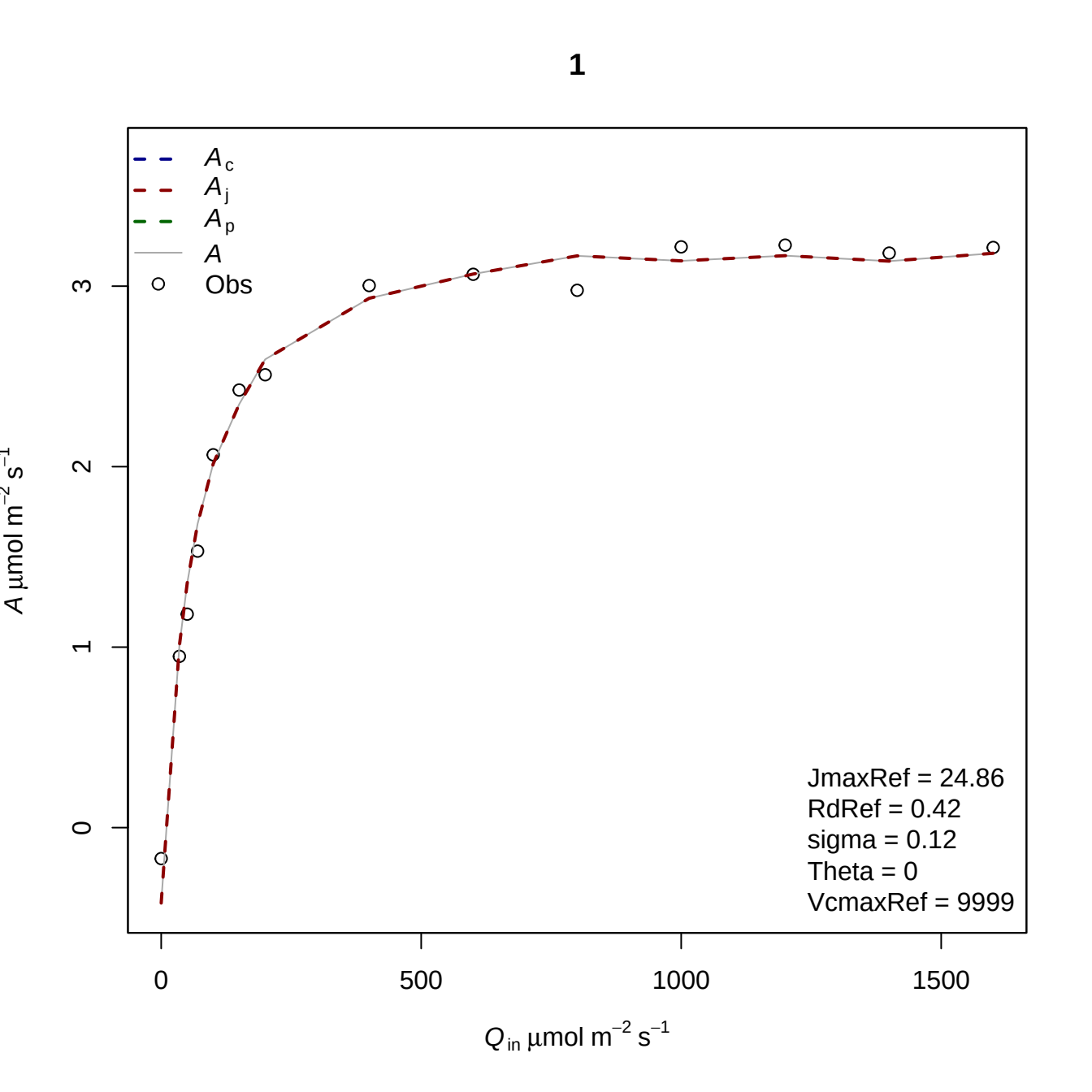
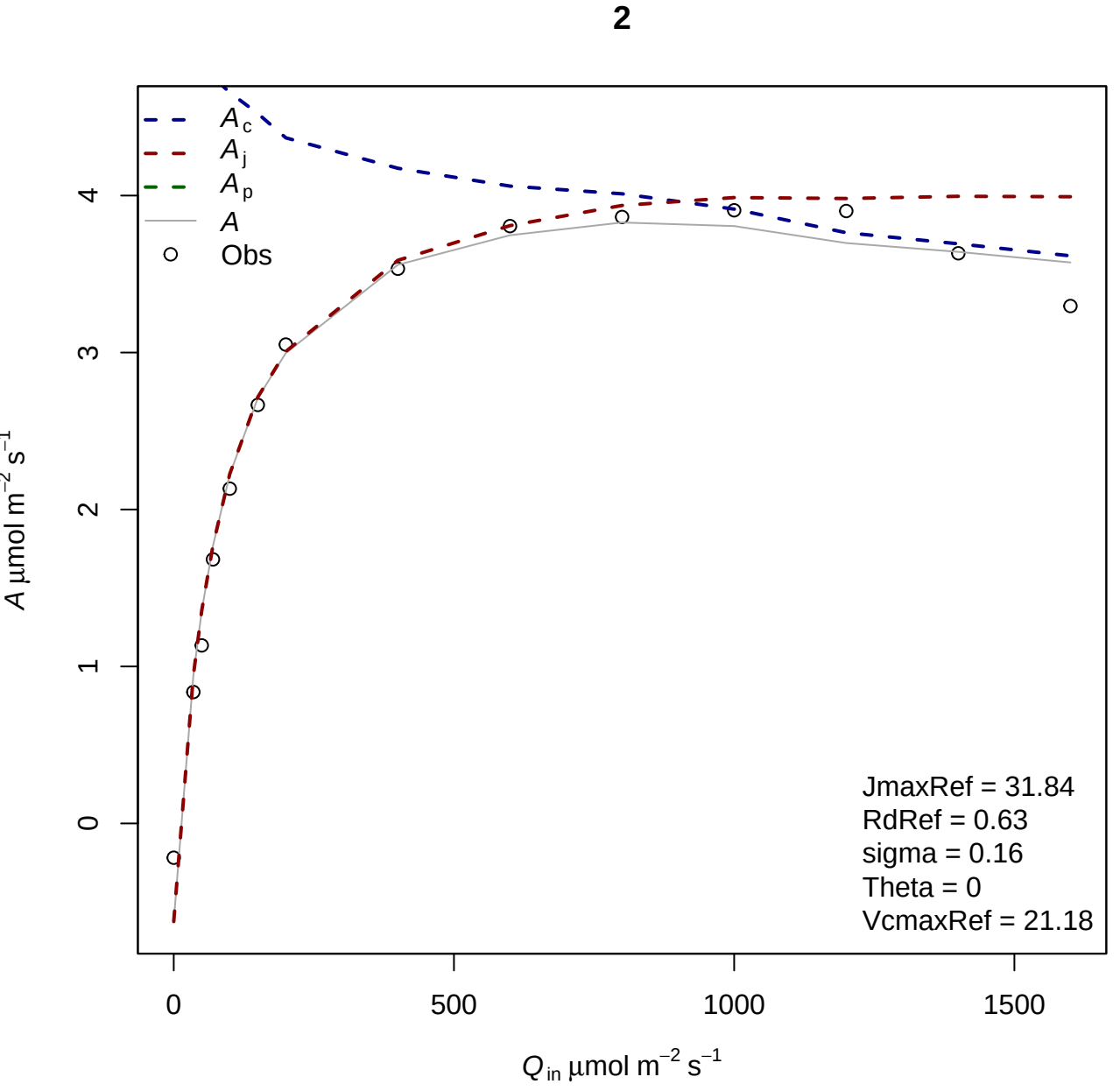
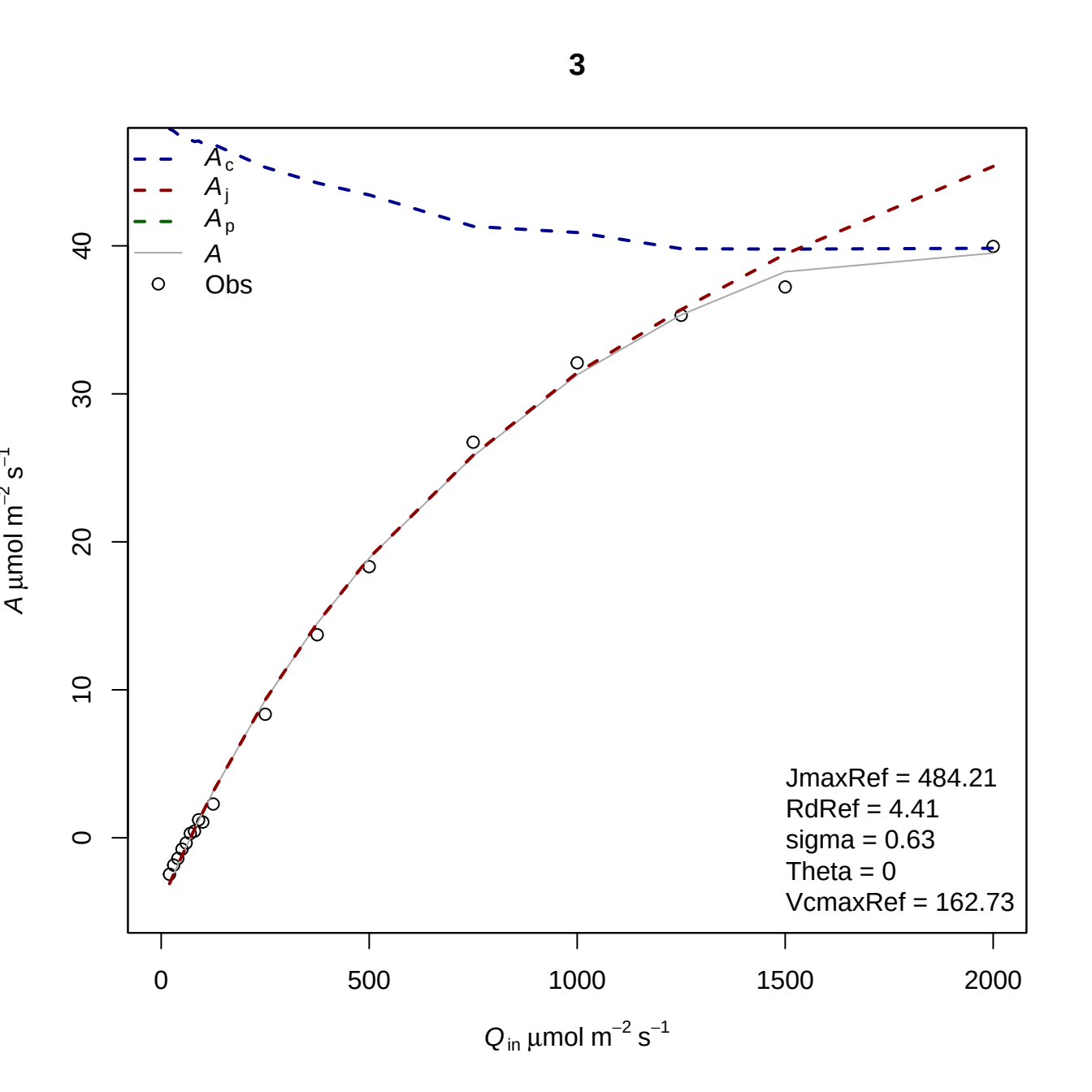
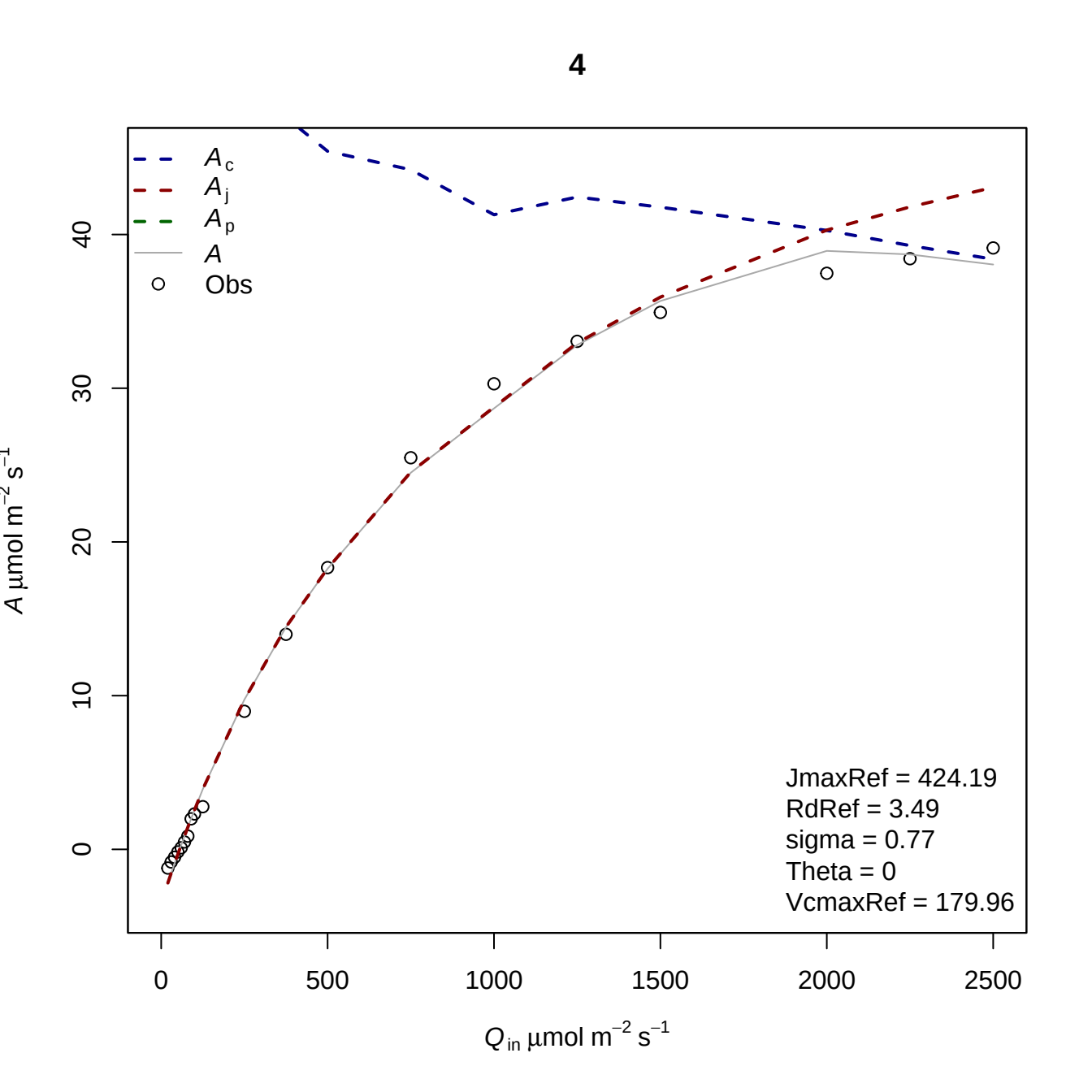
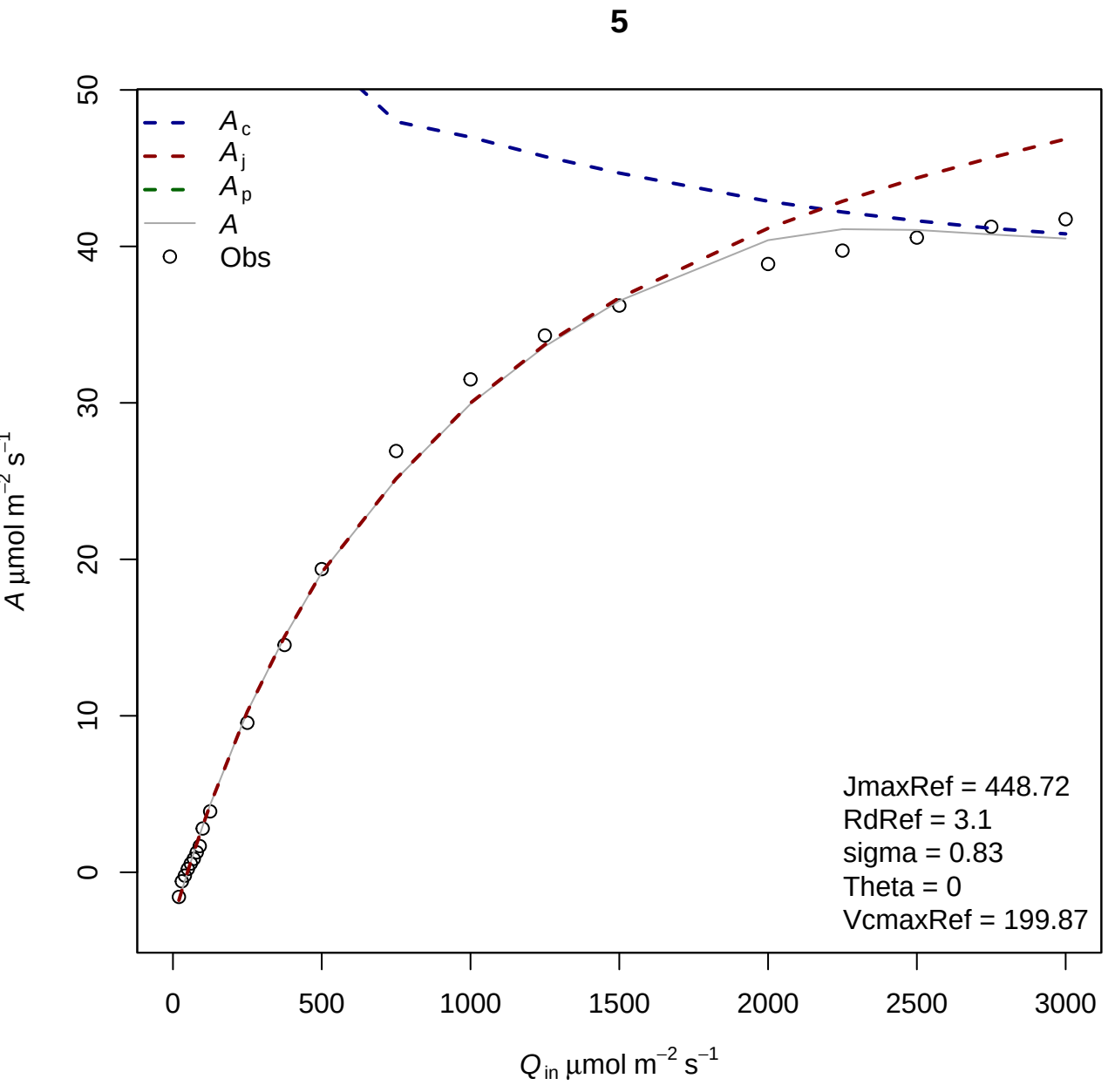
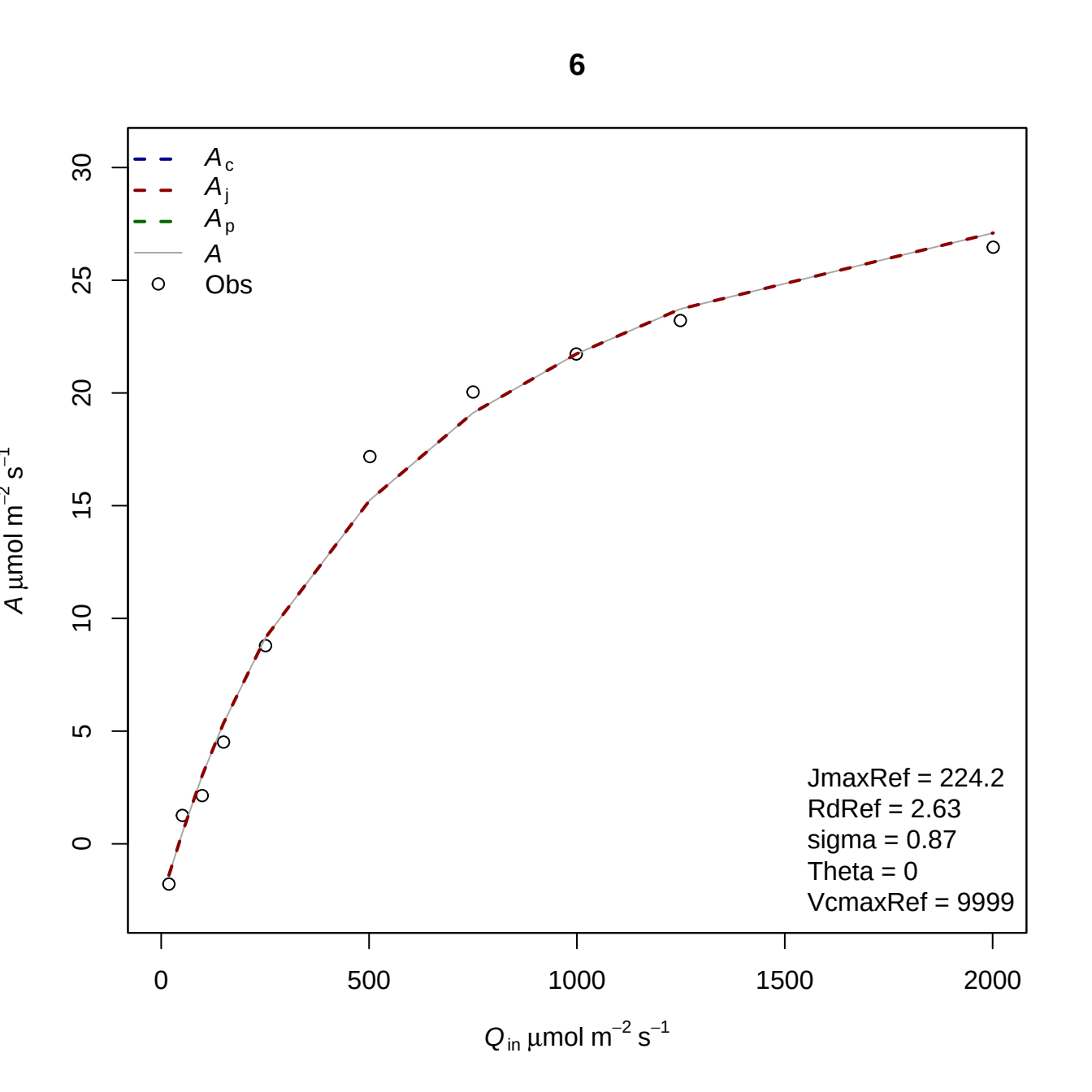

1
A_c
A_j
A_p
A
Obs
JmaxRef = 24.86
RdRef = 0.42
sigma = 0.12
Theta = 0
VcmaxRef = 9999
0
1
2
3
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

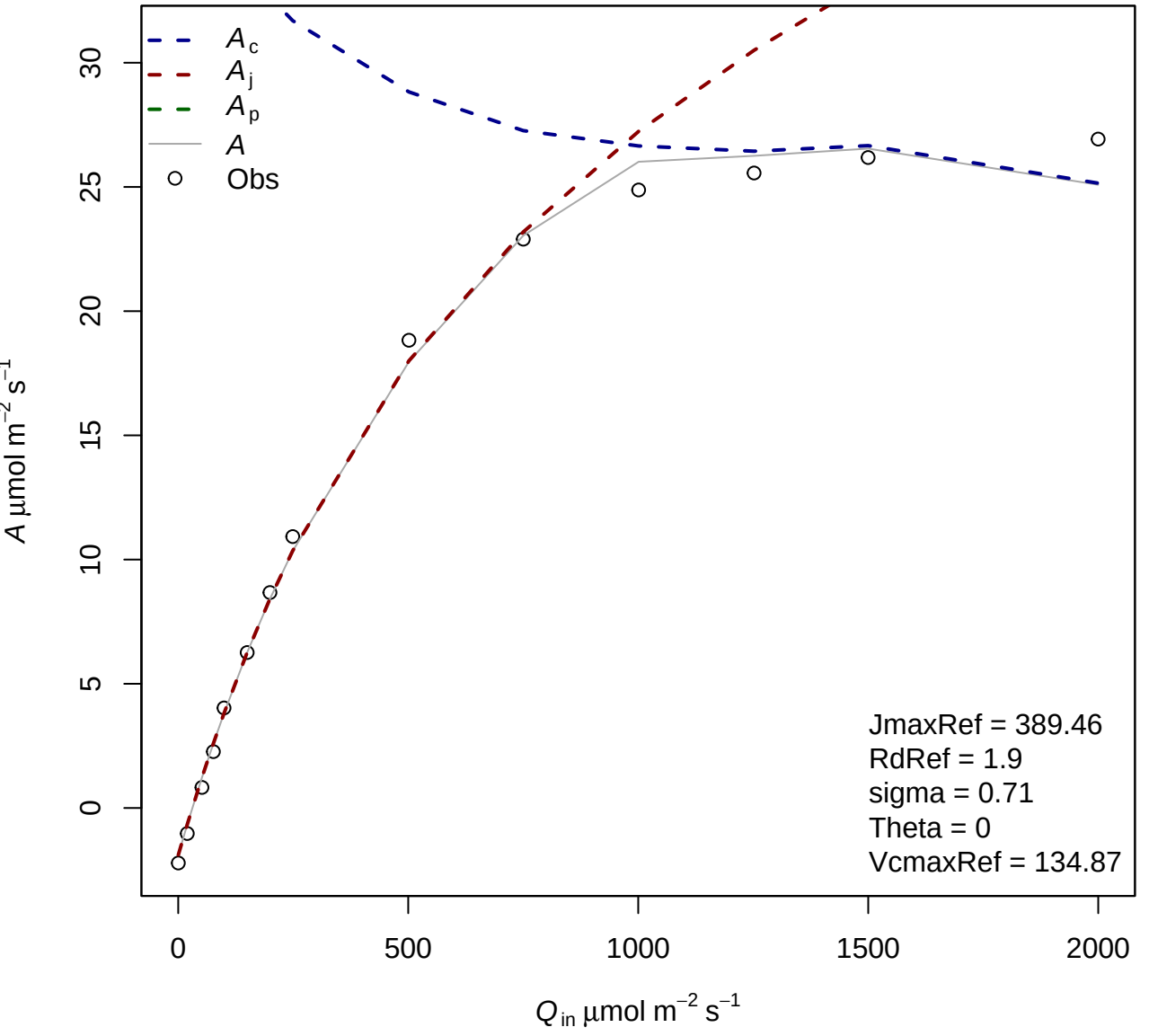

2
A_c
A_j
A_p
A
Obs
JmaxRef = 31.84
RdRef = 0.63
sigma = 0.16
Theta = 0
VcmaxRef = 21.18
0
1
2
3
4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

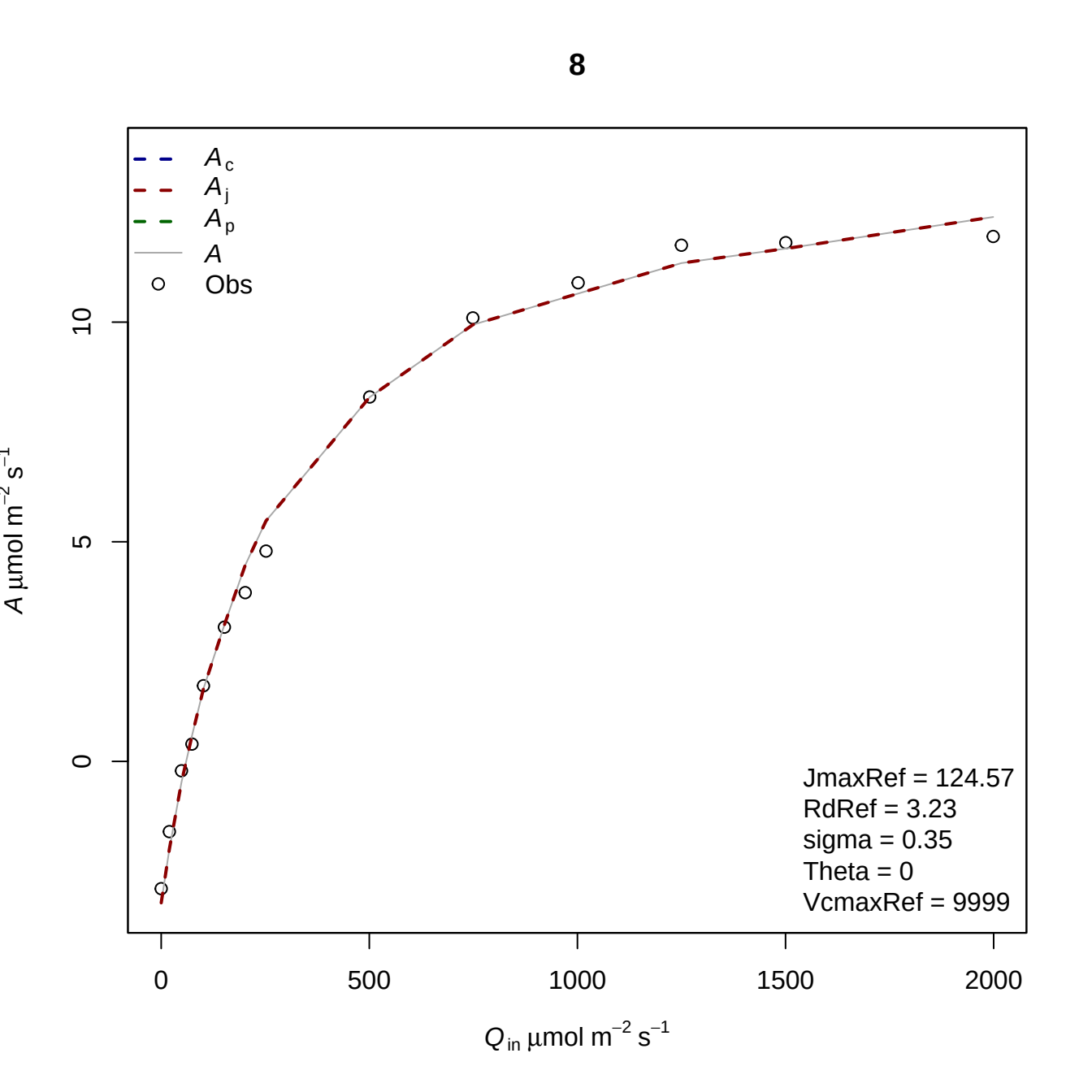

3
A_c
A_j
A_p
A
Obs
JmaxRef = 484.21
RdRef = 4.41
sigma = 0.63
Theta = 0
VcmaxRef = 162.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

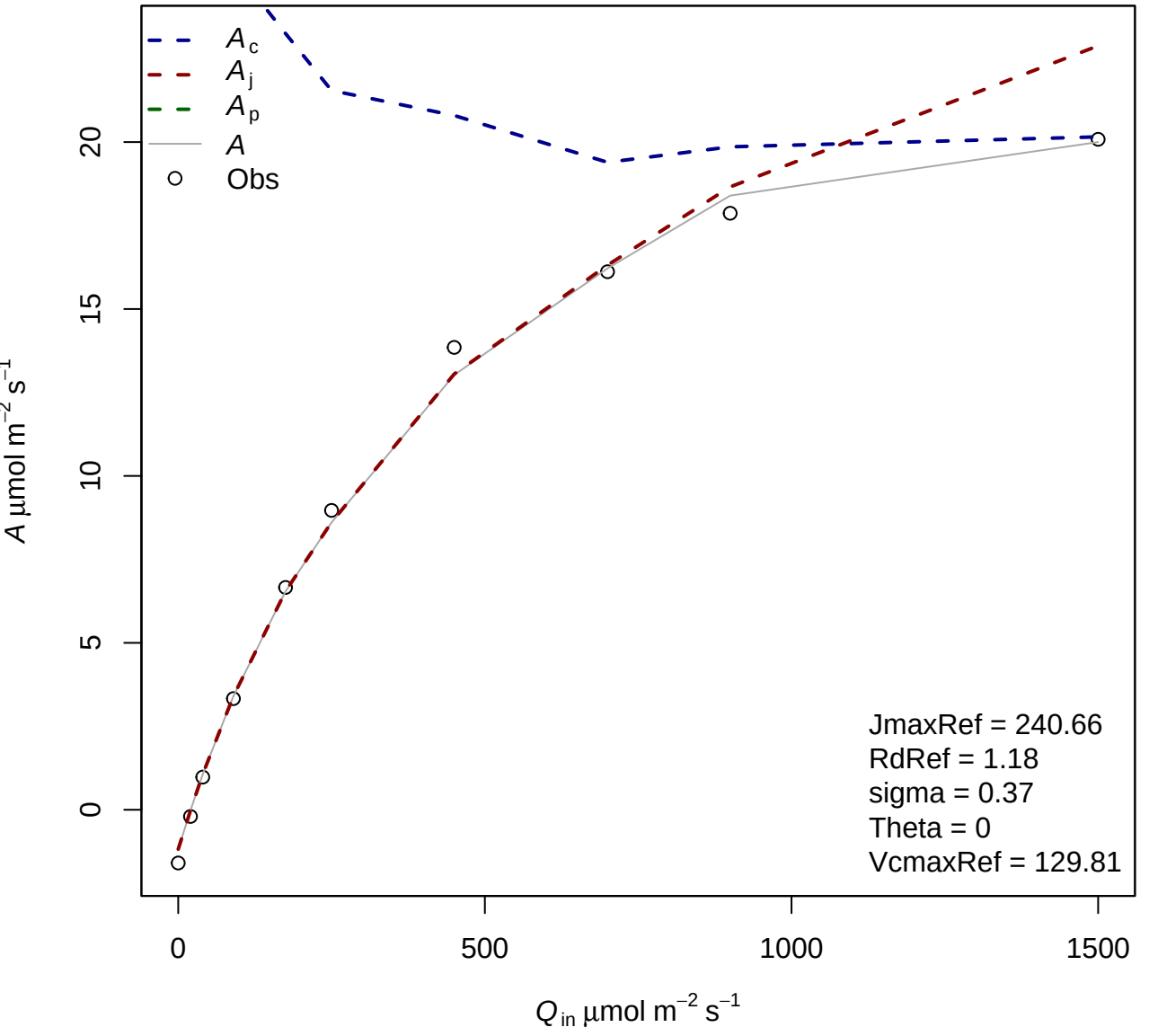

4
A_c
A_j
A_p
A
Obs
JmaxRef = 424.19
RdRef = 3.49
sigma = 0.77
Theta = 0
VcmaxRef = 179.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000
2500

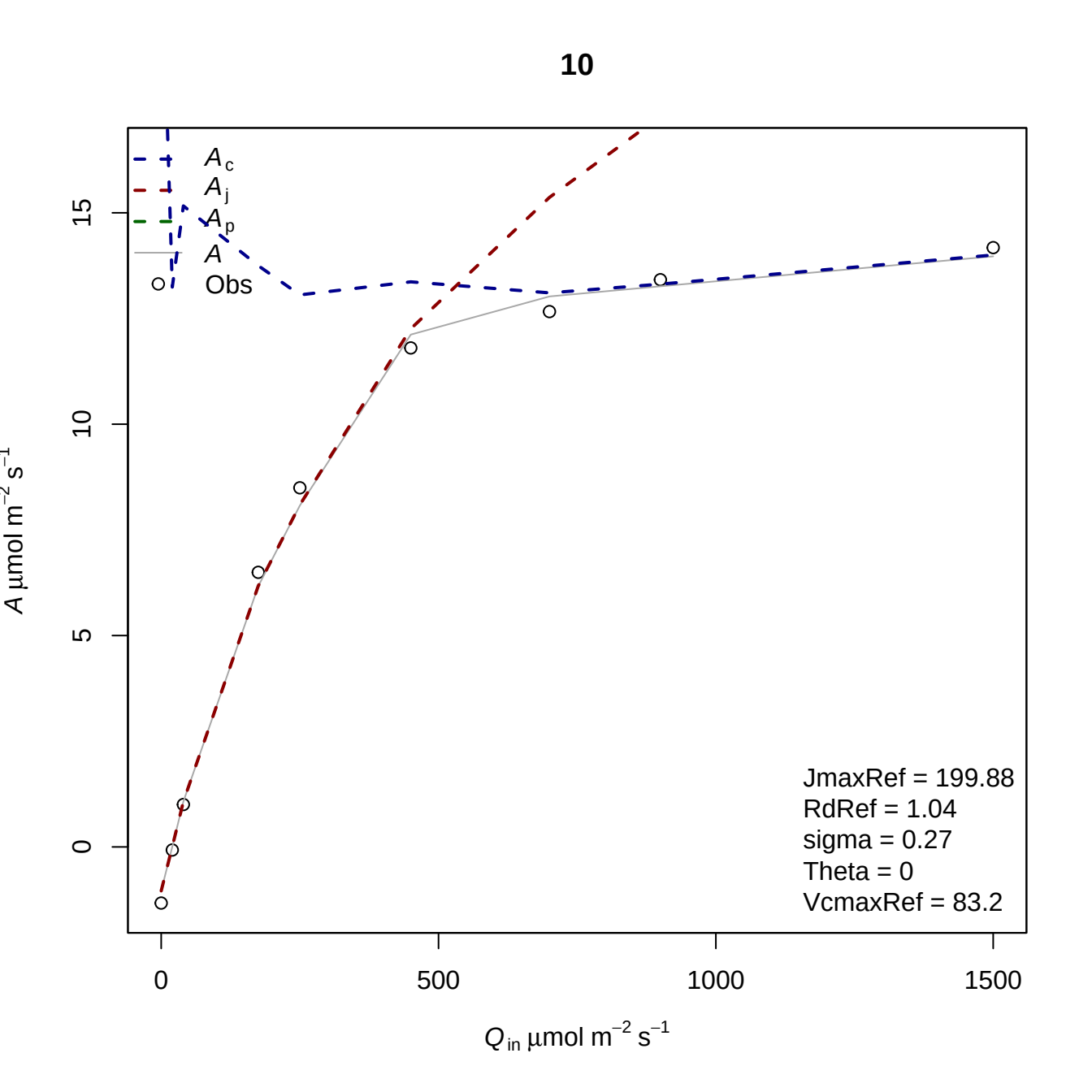

5
A_c
A_j
A_p
A
Obs
JmaxRef = 448.72
RdRef = 3.1
sigma = 0.83
Theta = 0
VcmaxRef = 199.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
50
0
500
1000
1500
2000
2500
3000

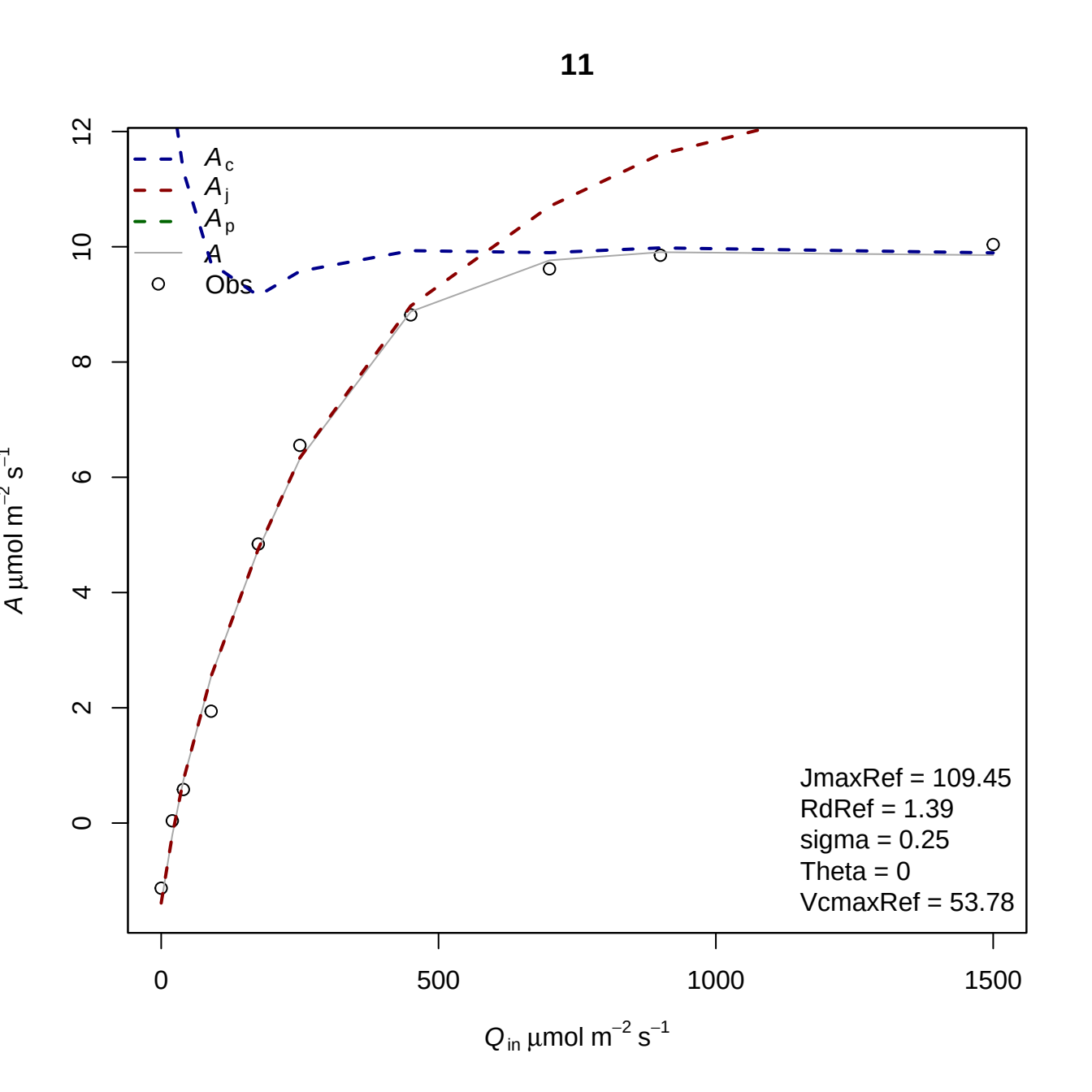

6
A_c
A_j
A_p
A
Obs
JmaxRef = 224.2
RdRef = 2.63
sigma = 0.87
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

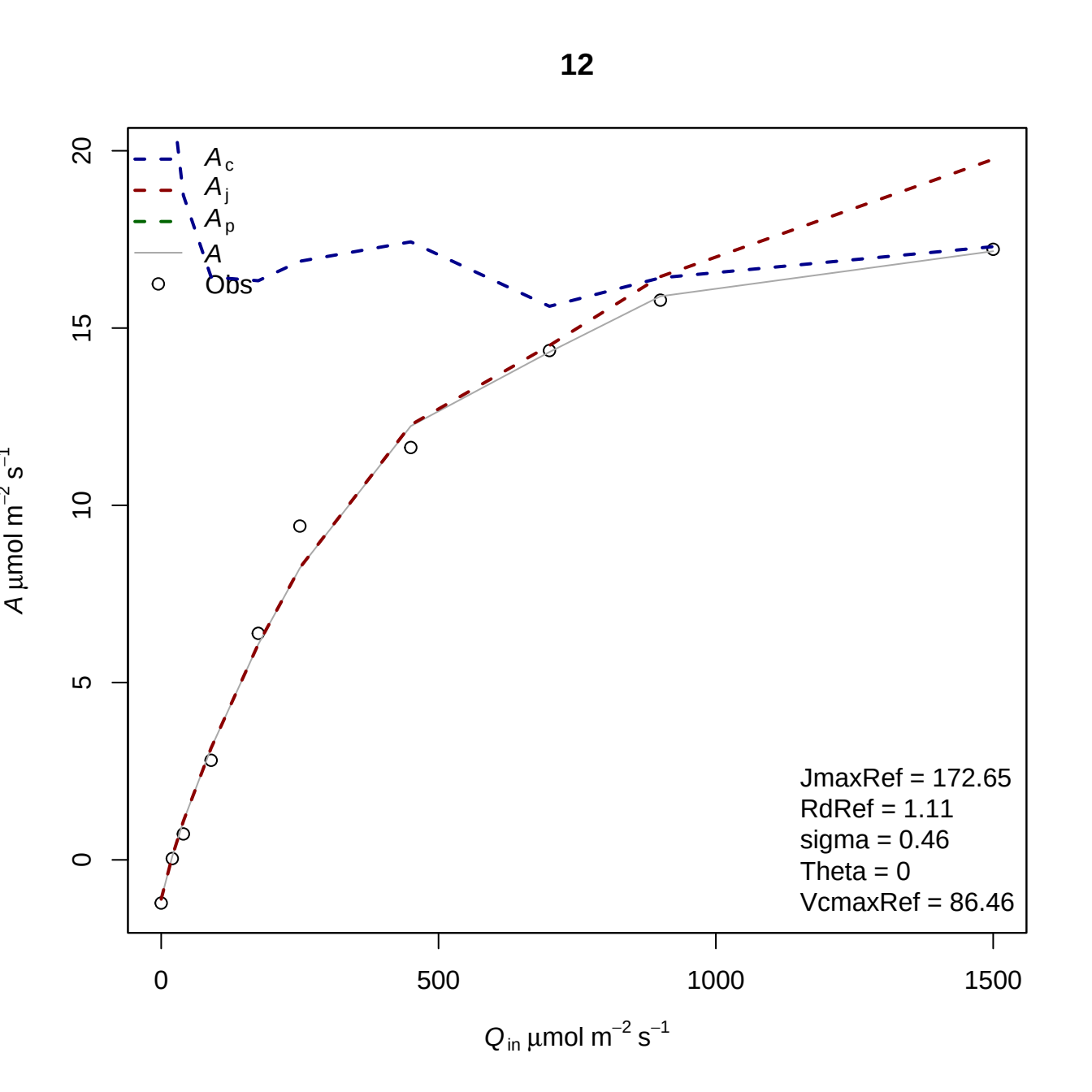

A_c
A_j
A_p
A
Obs
JmaxRef = 389.46
RdRef = 1.9
sigma = 0.71
Theta = 0
VcmaxRef = 134.87
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

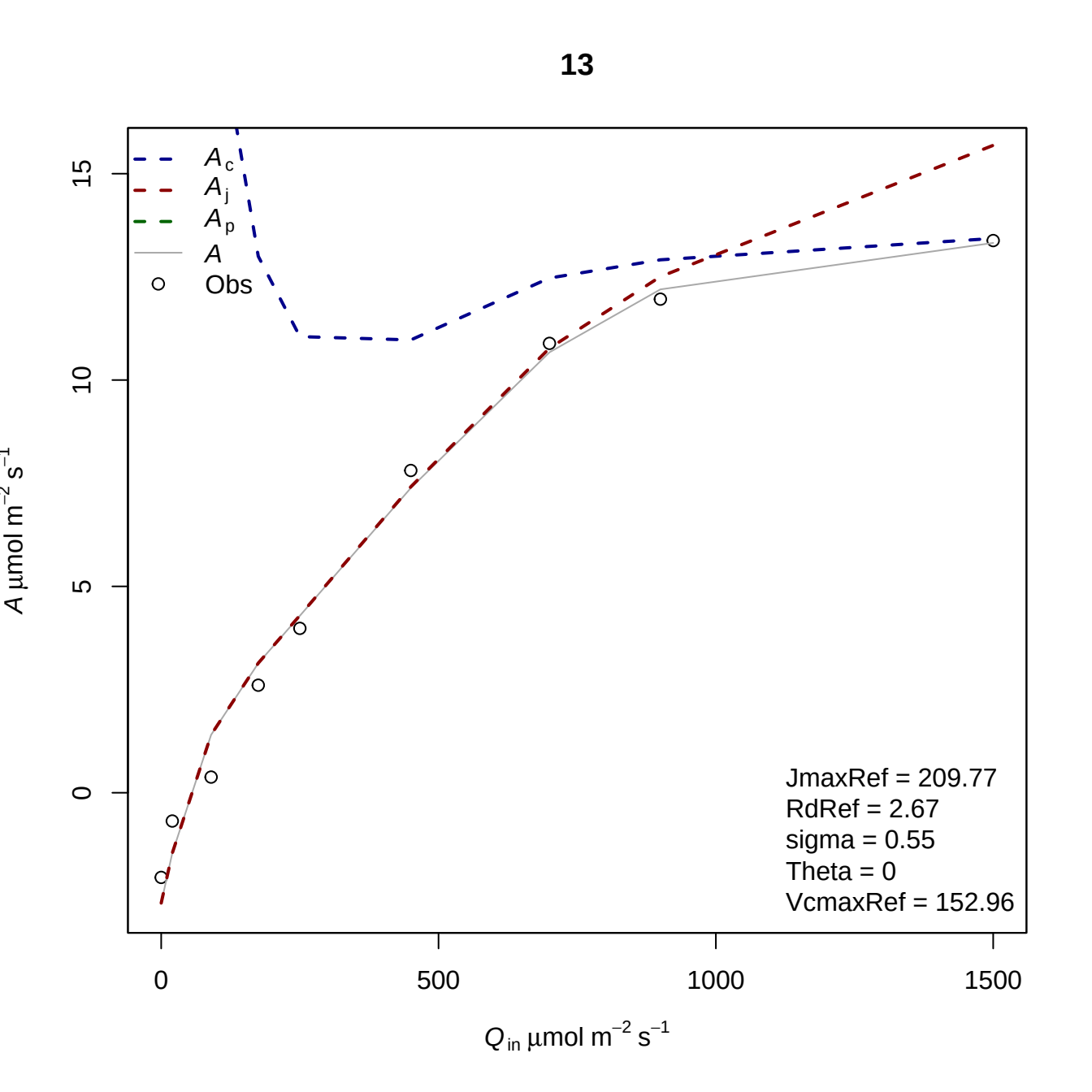

8
A_c
A_j
A_p
A
Obs
JmaxRef = 124.57
RdRef = 3.23
sigma = 0.35
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

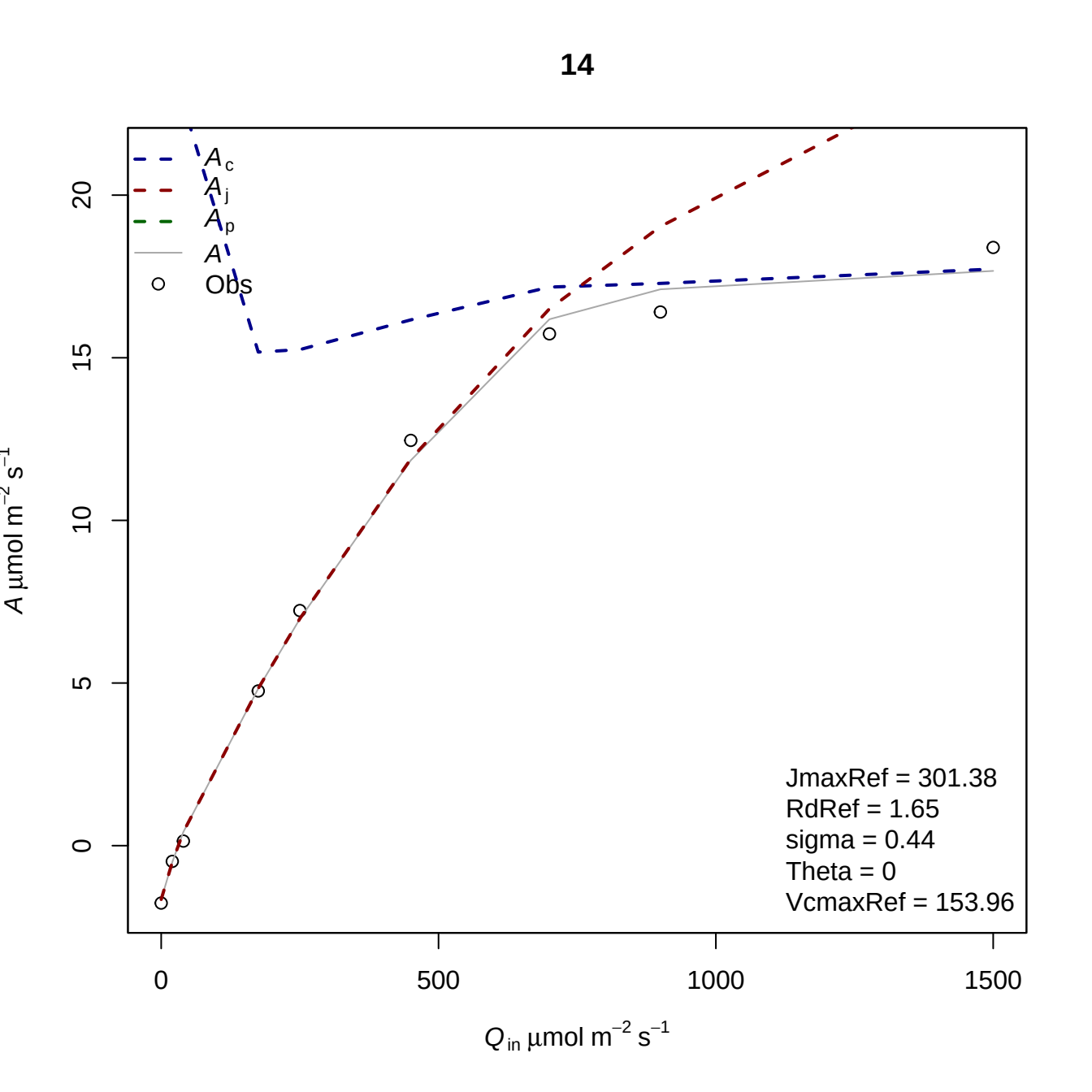

A_c
A_j
A_p
A
Obs
JmaxRef = 240.66
RdRef = 1.18
sigma = 0.37
Theta = 0
VcmaxRef = 129.81
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

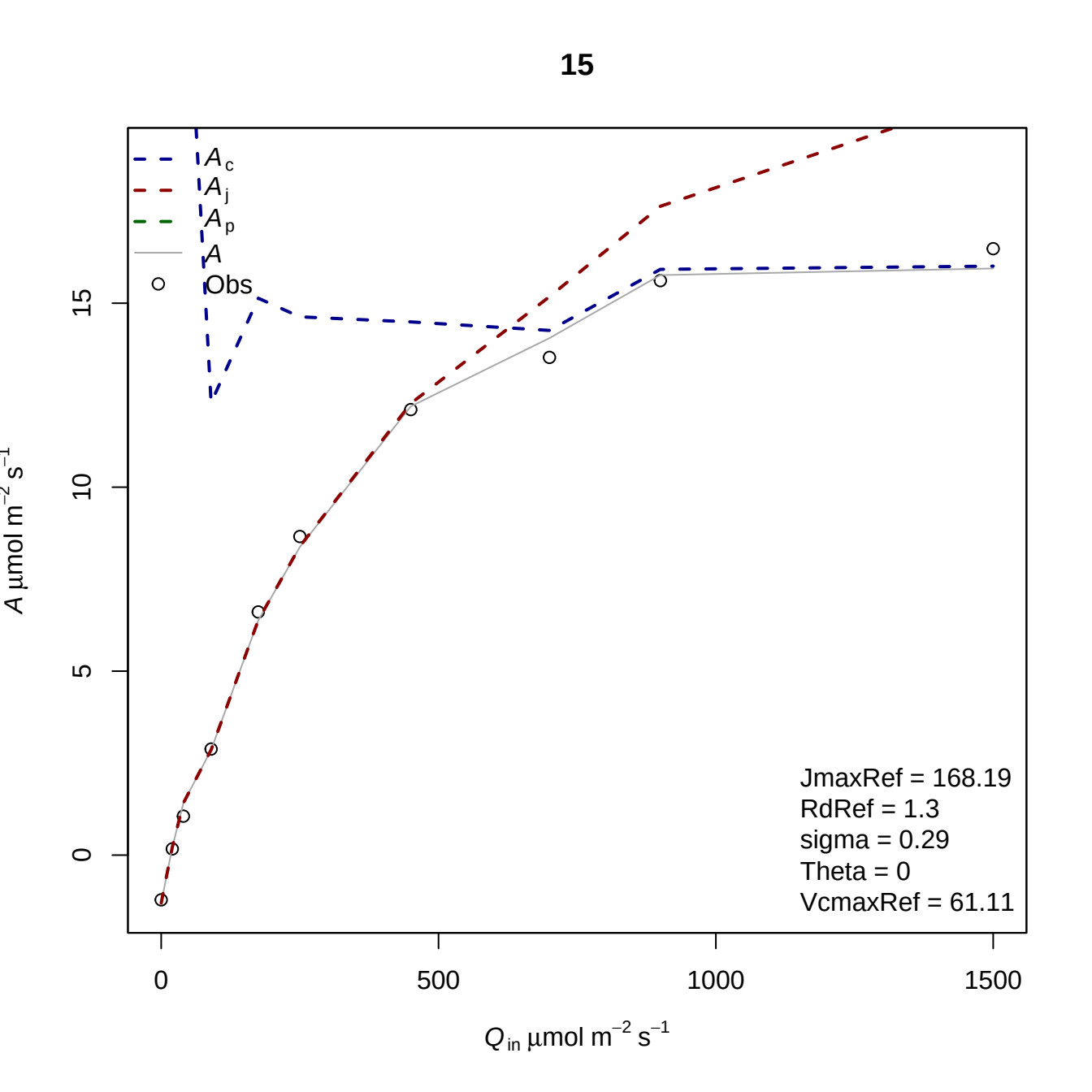

10
A_c
A_j
A_p
A
Obs
JmaxRef = 199.88
RdRef = 1.04
sigma = 0.27
Theta = 0
VcmaxRef = 83.2
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

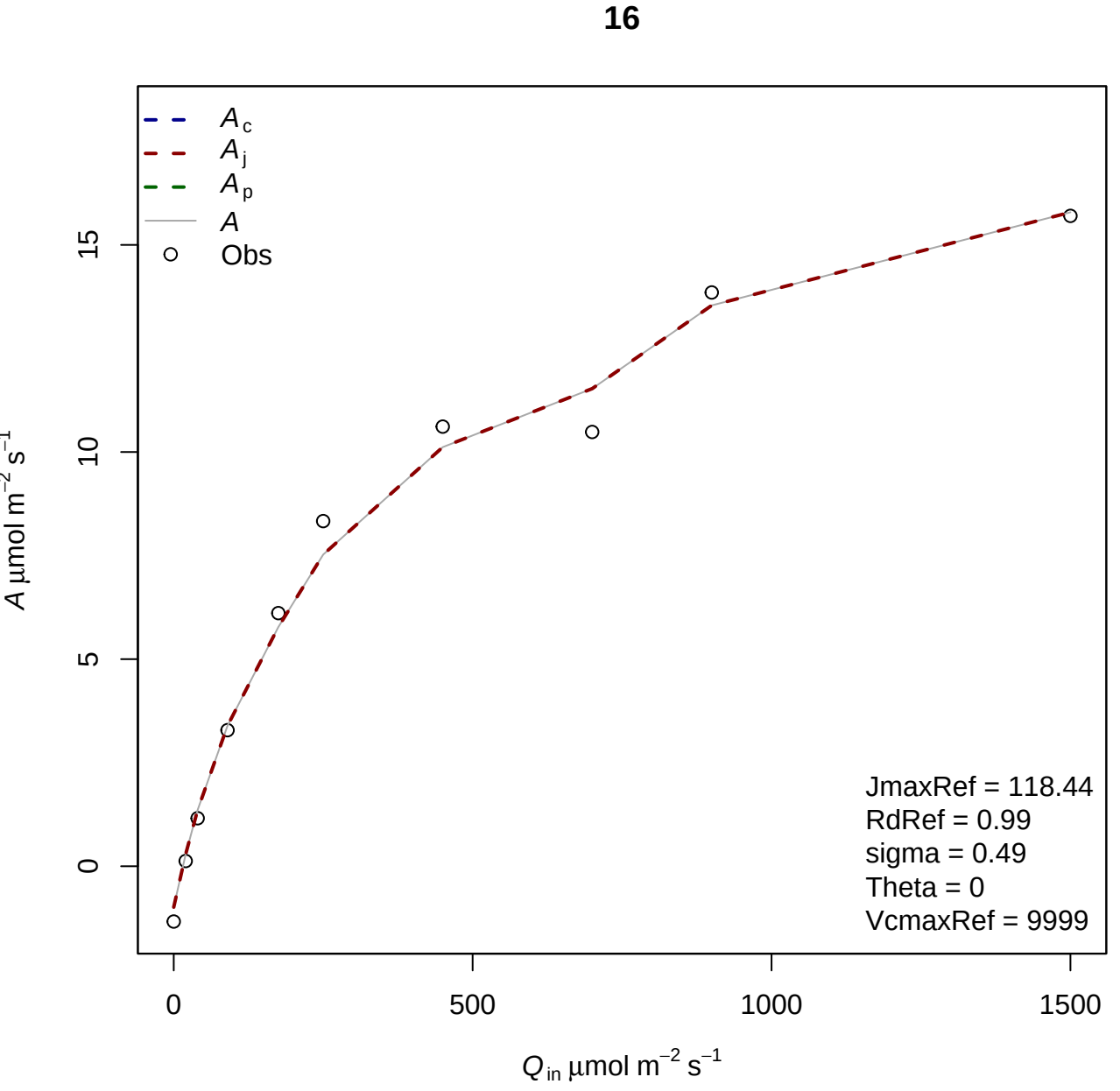

11
A_c
A_j
A_p
A
Obs
JmaxRef = 109.45
RdRef = 1.39
sigma = 0.25
Theta = 0
VcmaxRef = 53.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

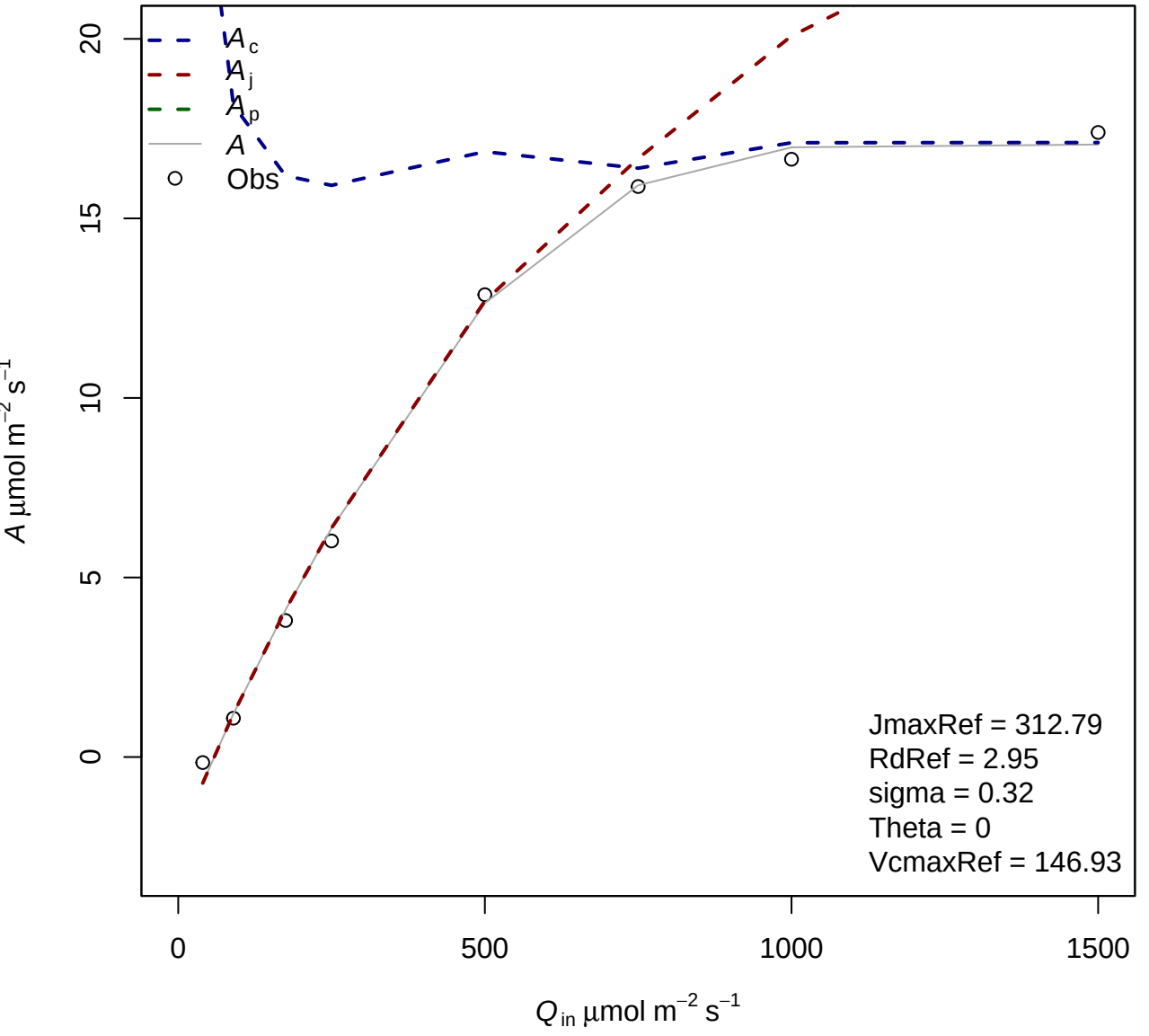

12
A_c
A_j
A_p
A
Obs
JmaxRef = 172.65
RdRef = 1.11
sigma = 0.46
Theta = 0
VcmaxRef = 86.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

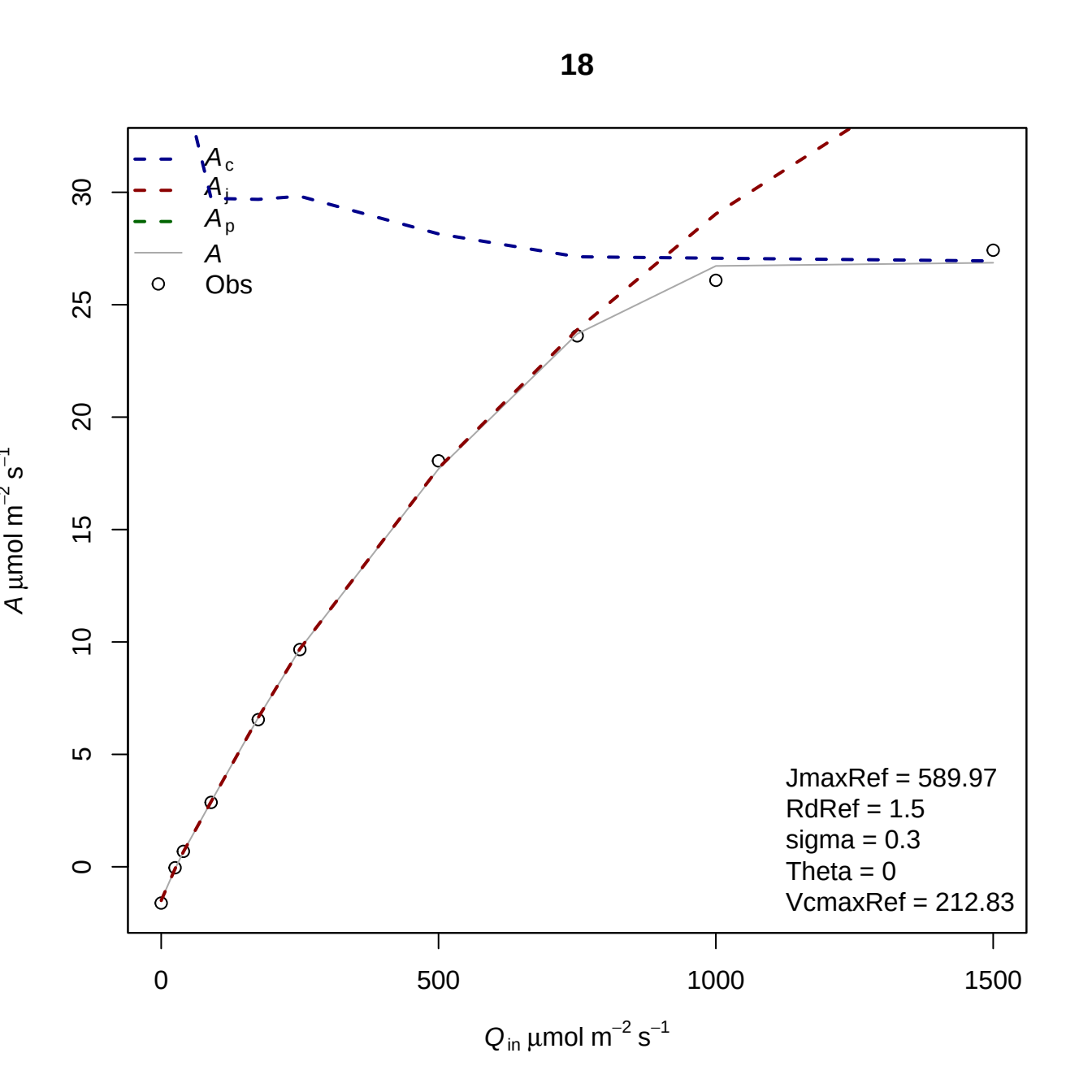

13
A_c
A_j
A_p
A
Obs
JmaxRef = 209.77
RdRef = 2.67
sigma = 0.55
Theta = 0
VcmaxRef = 152.96
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

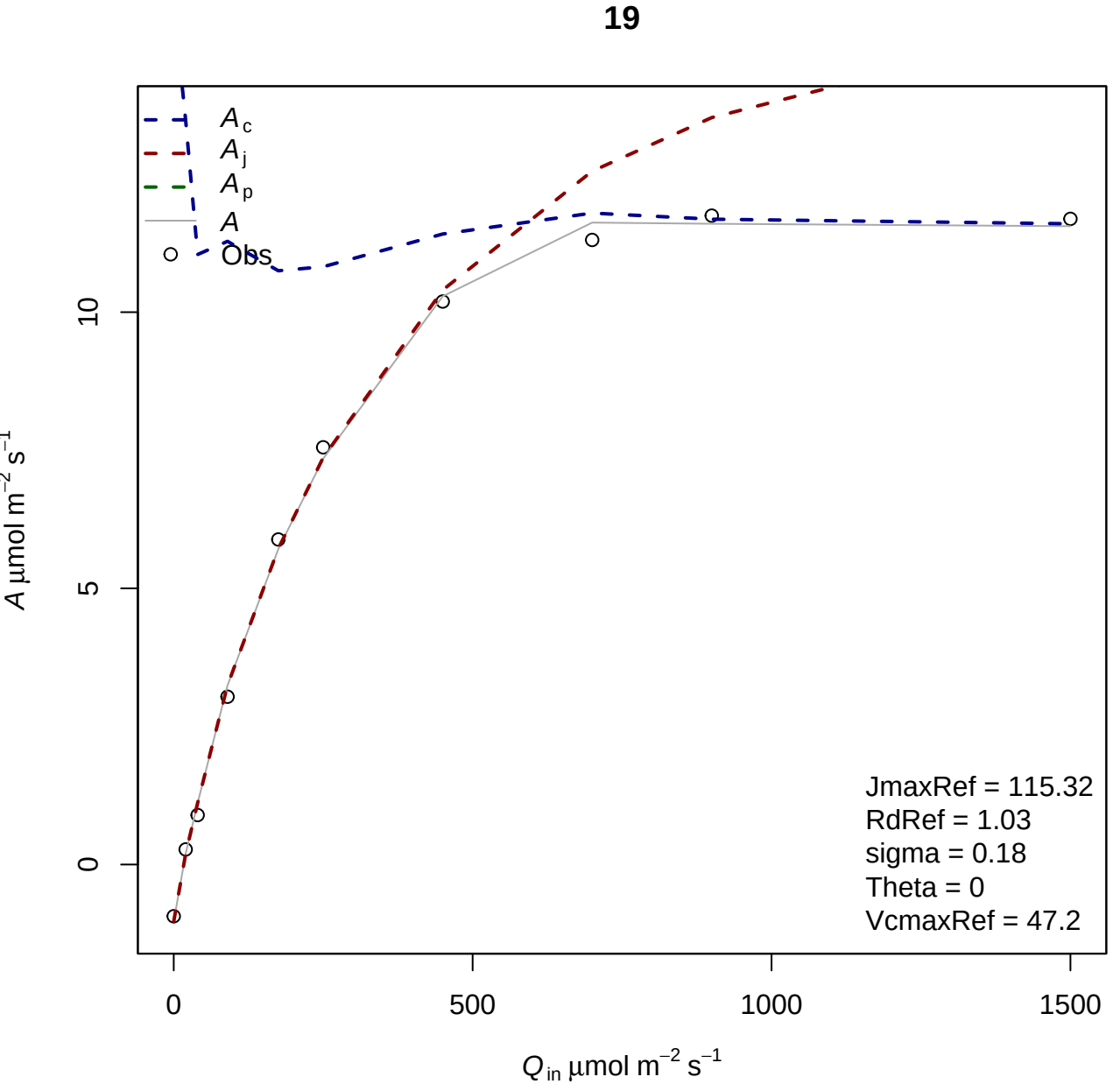

14
A_c
A_j
A_p
A
Obs
JmaxRef = 301.38
RdRef = 1.65
sigma = 0.44
Theta = 0
VcmaxRef = 153.96
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

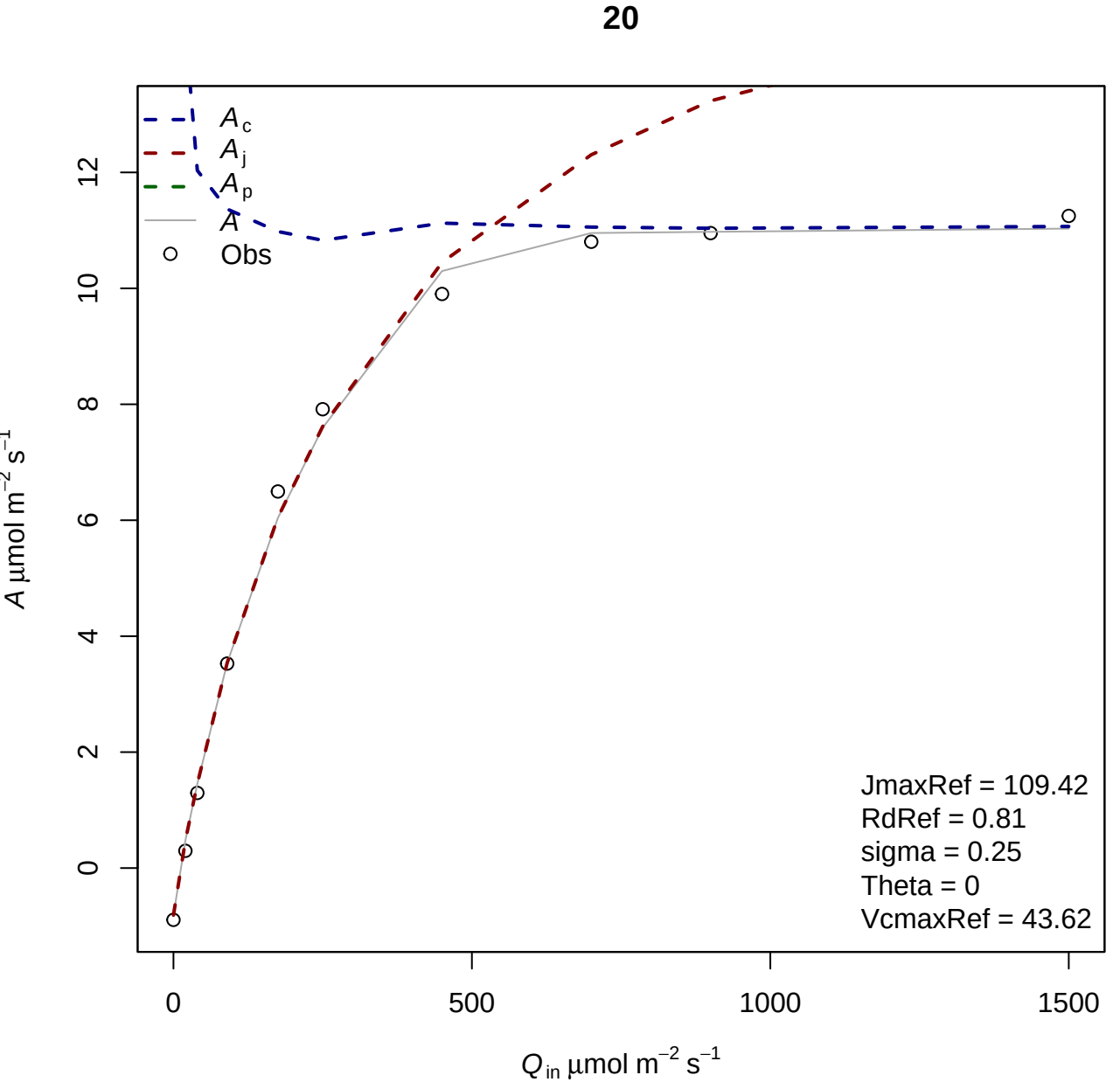

15
A_c
A_j
A_p
A
Obs
JmaxRef = 168.19
RdRef = 1.3
sigma = 0.29
Theta = 0
VcmaxRef = 61.11
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

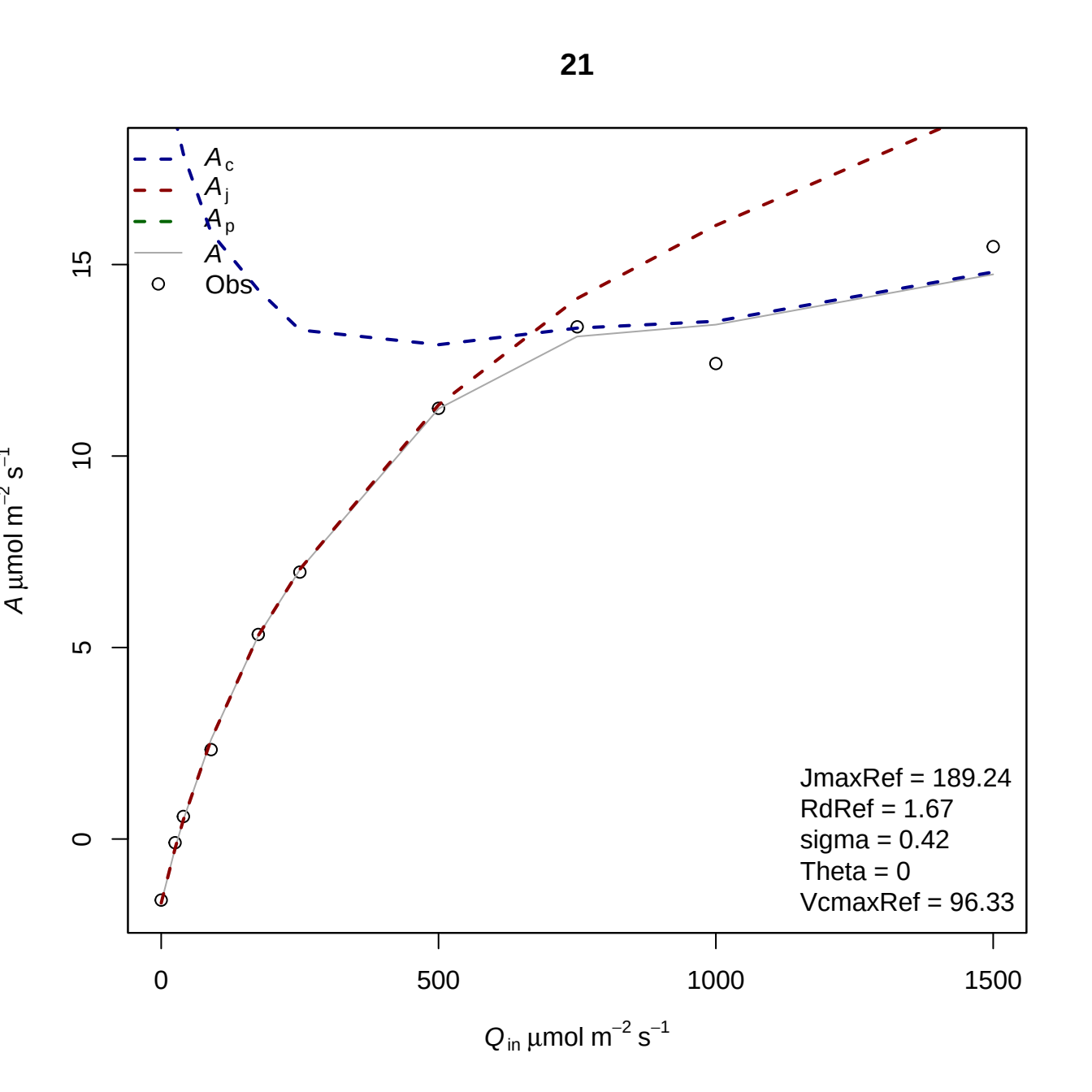

16
A_c
A_j
A_p
A
Obs
JmaxRef = 118.44
RdRef = 0.99
sigma = 0.49
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

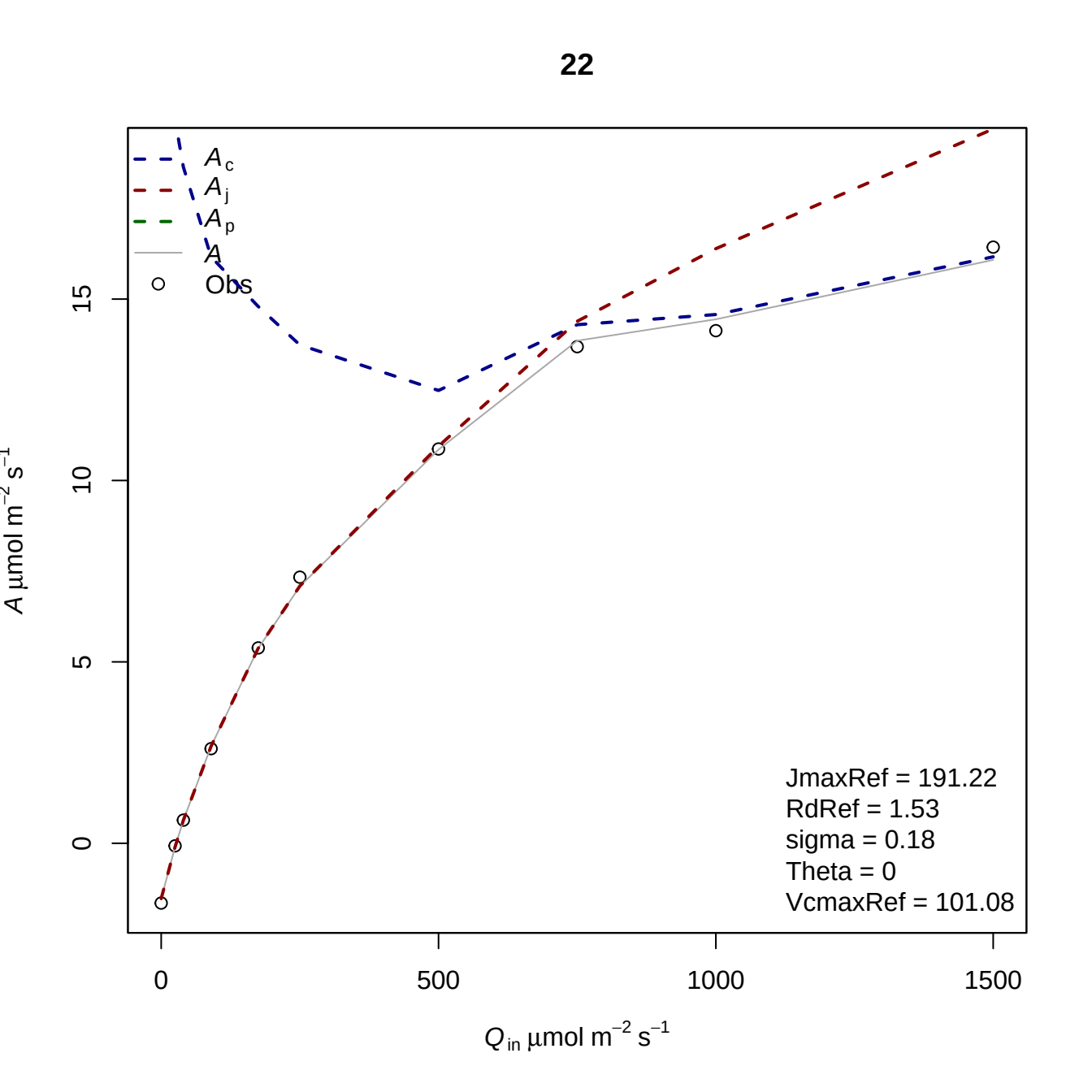

A_c
A_j
A_p
A
Obs
JmaxRef = 312.79
RdRef = 2.95
sigma = 0.32
Theta = 0
VcmaxRef = 146.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

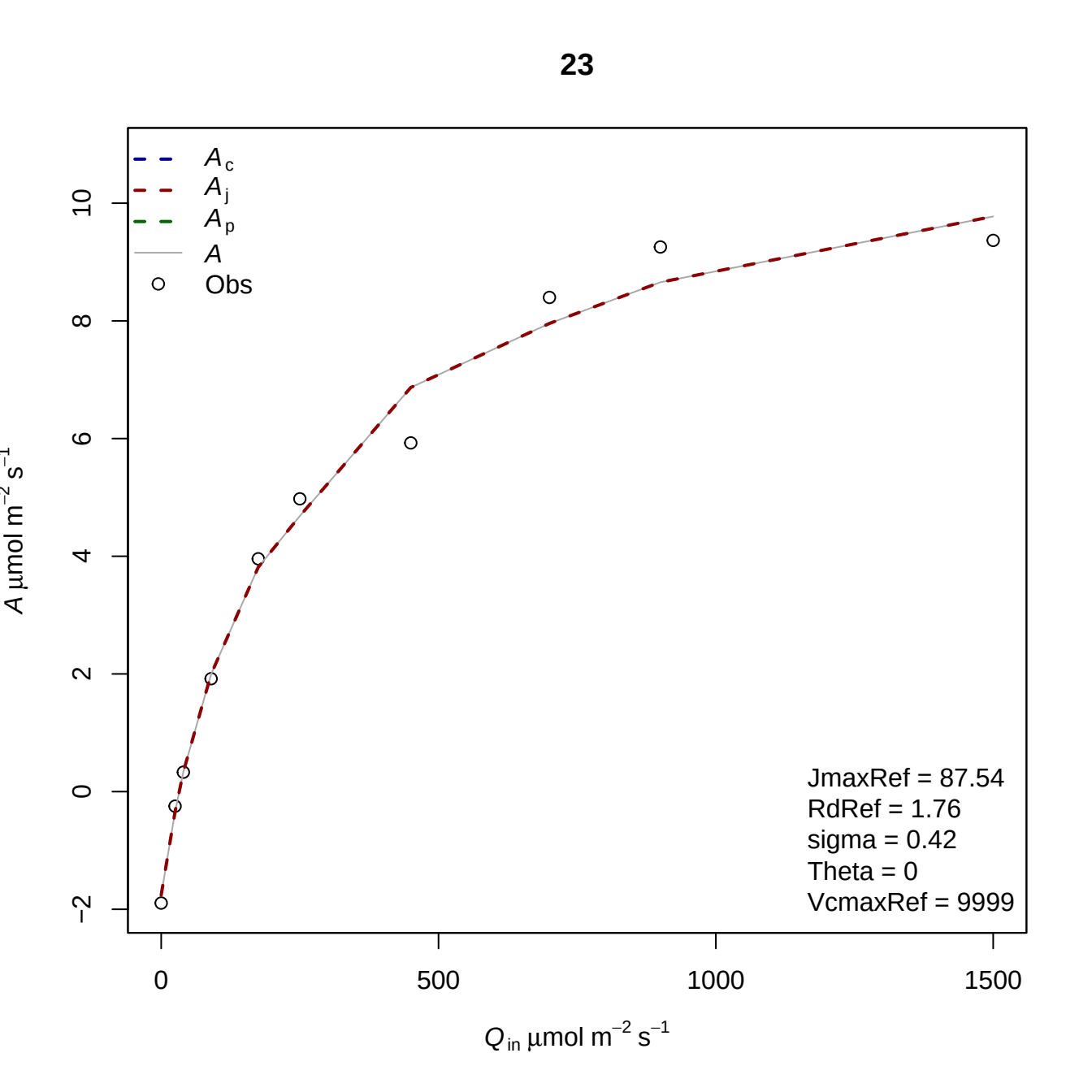

18
A_c
A_j
A_p
A
Obs
JmaxRef = 589.97
RdRef = 1.5
sigma = 0.3
Theta = 0
VcmaxRef = 212.83
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

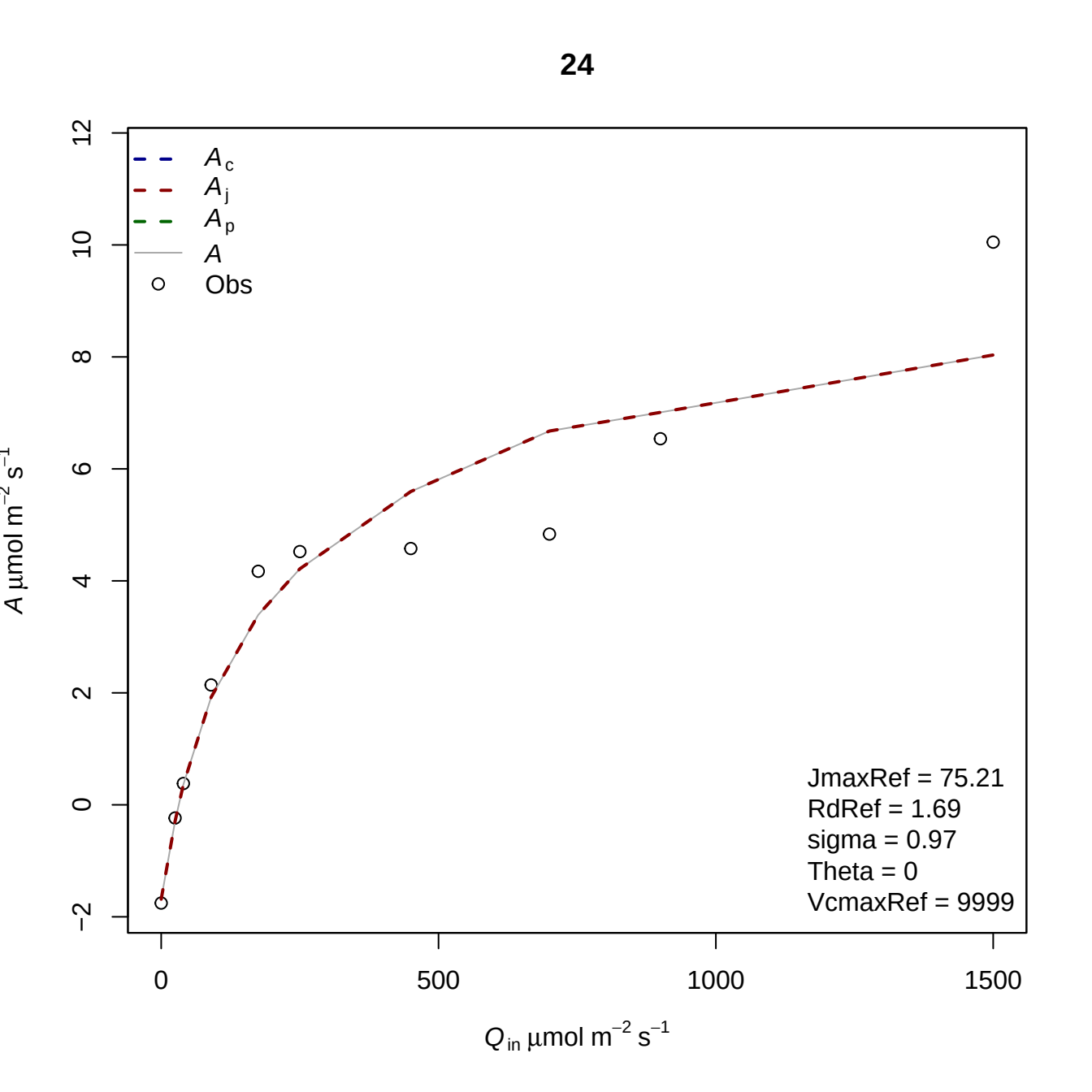

19
A_c
A_j
A_p
A
Obs
JmaxRef = 115.32
RdRef = 1.03
sigma = 0.18
Theta = 0
VcmaxRef = 47.2
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

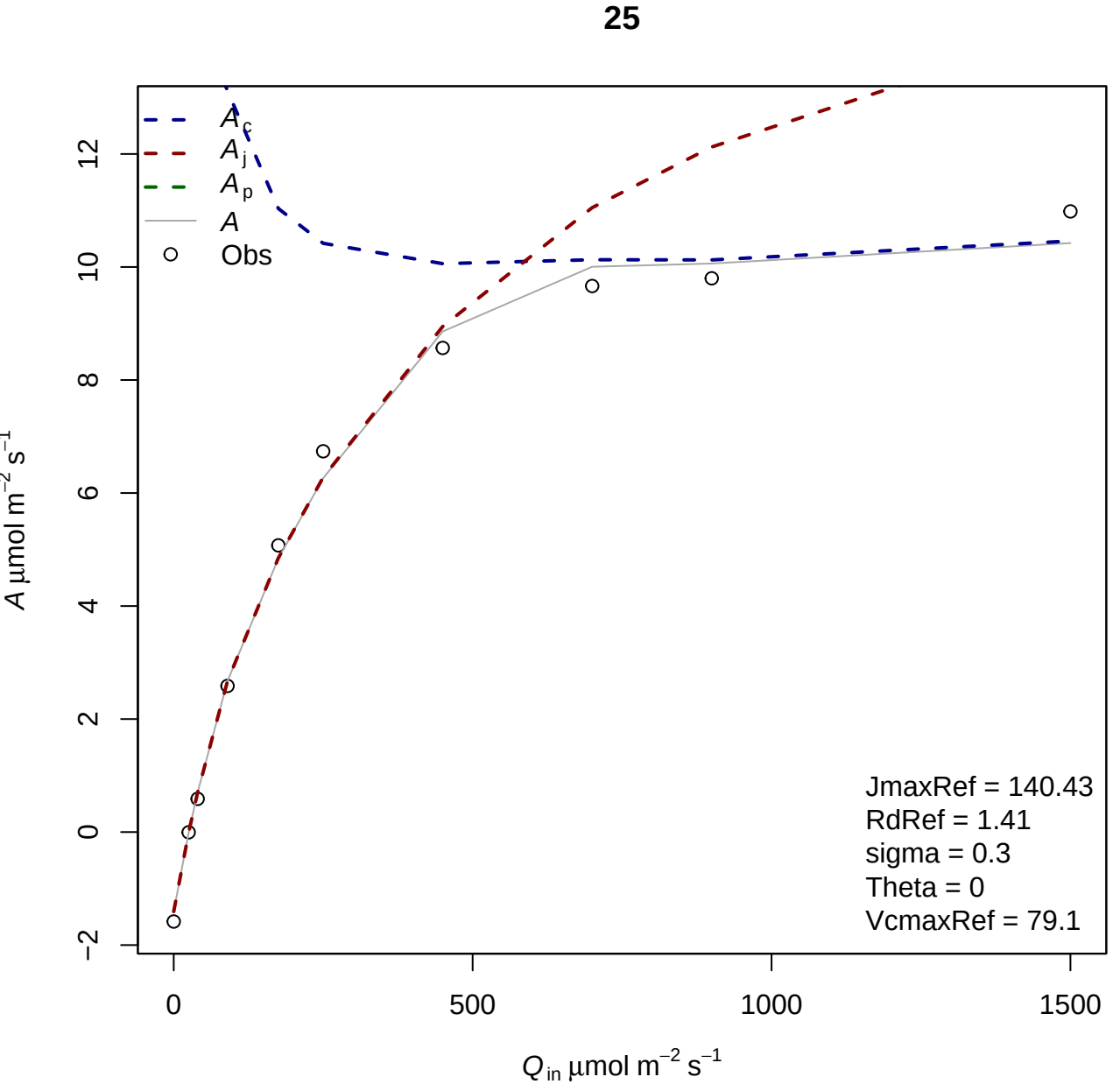

20
A_c
A_j
A_p
A
Obs
JmaxRef = 109.42
RdRef = 0.81
sigma = 0.25
Theta = 0
VcmaxRef = 43.62
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

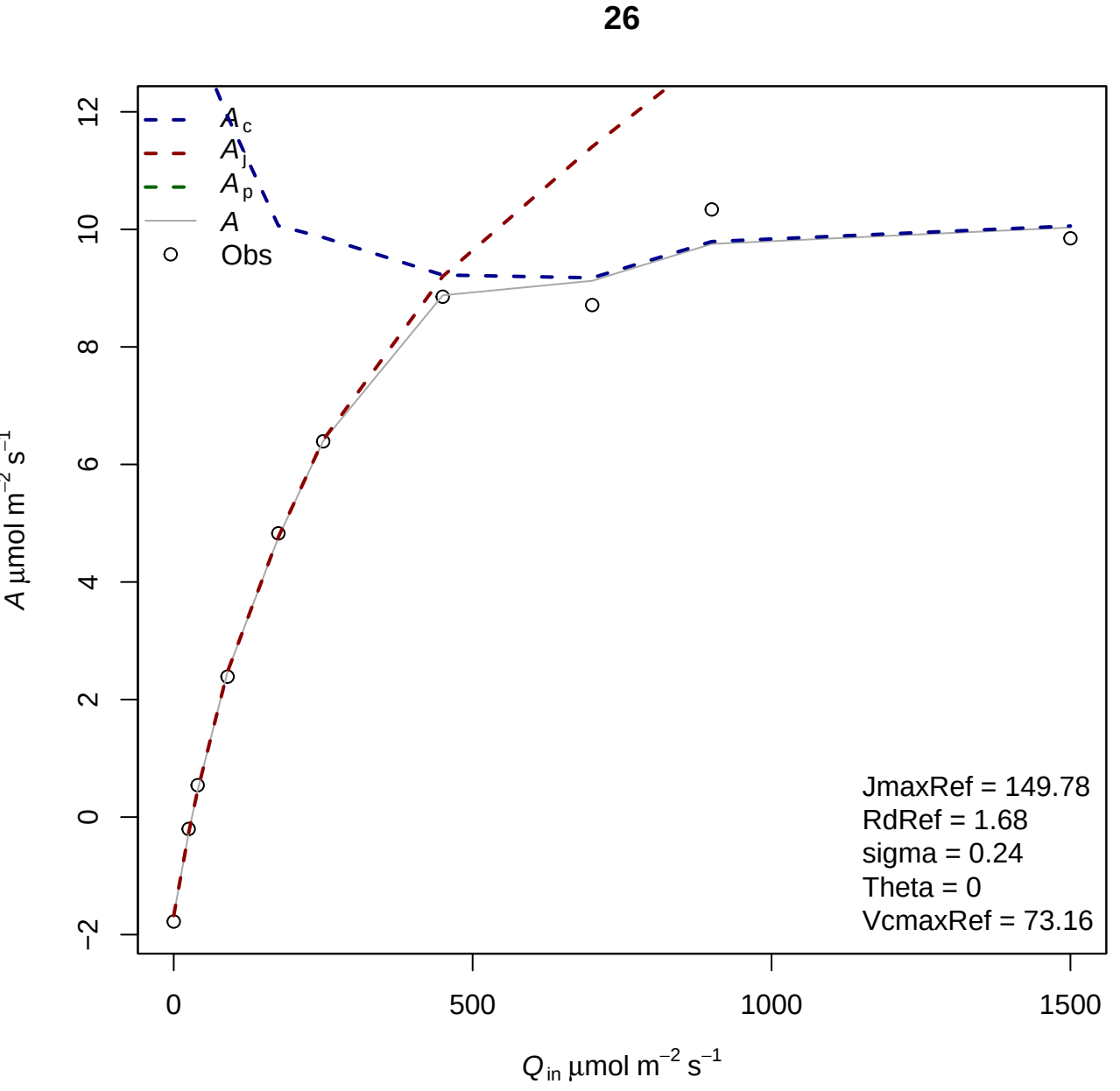

21
A_c
A_j
A_p
A
Obs
JmaxRef = 189.24
RdRef = 1.67
sigma = 0.42
Theta = 0
VcmaxRef = 96.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

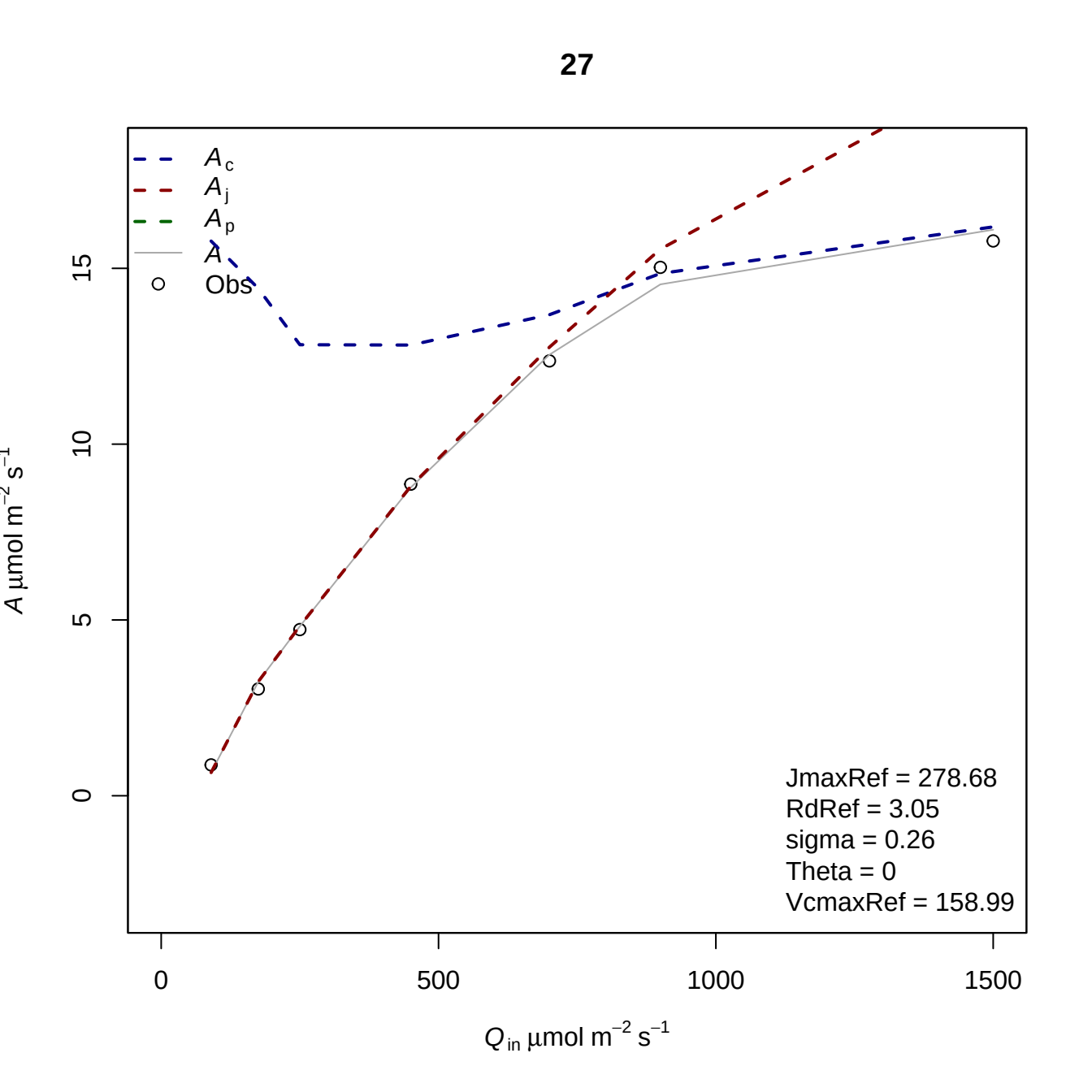

22
A_c
A_j
A_p
A
Obs
JmaxRef = 191.22
RdRef = 1.53
sigma = 0.18
Theta = 0
VcmaxRef = 101.08
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

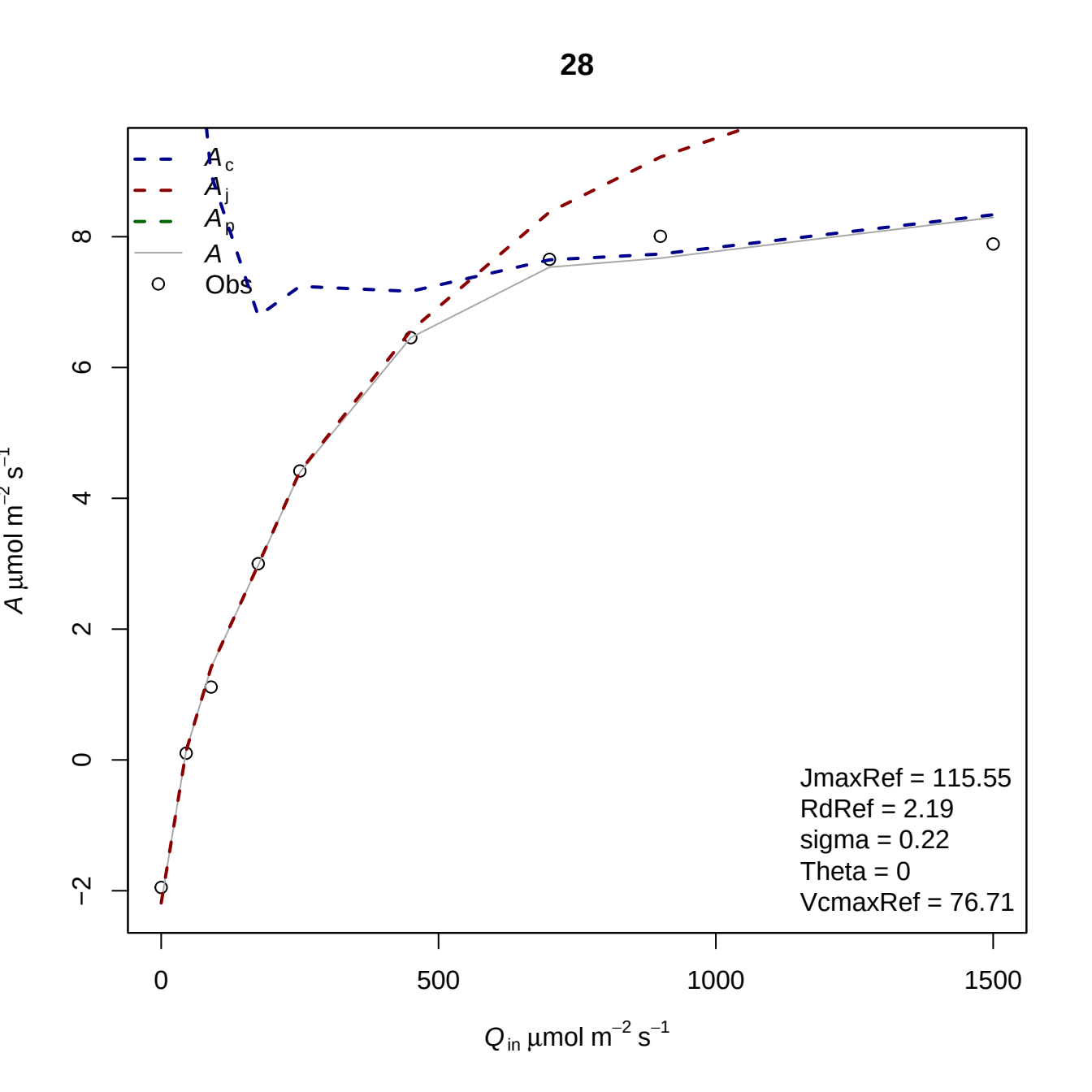

23
A_c
A_j
A_p
A
Obs
JmaxRef = 87.54
RdRef = 1.76
sigma = 0.42
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

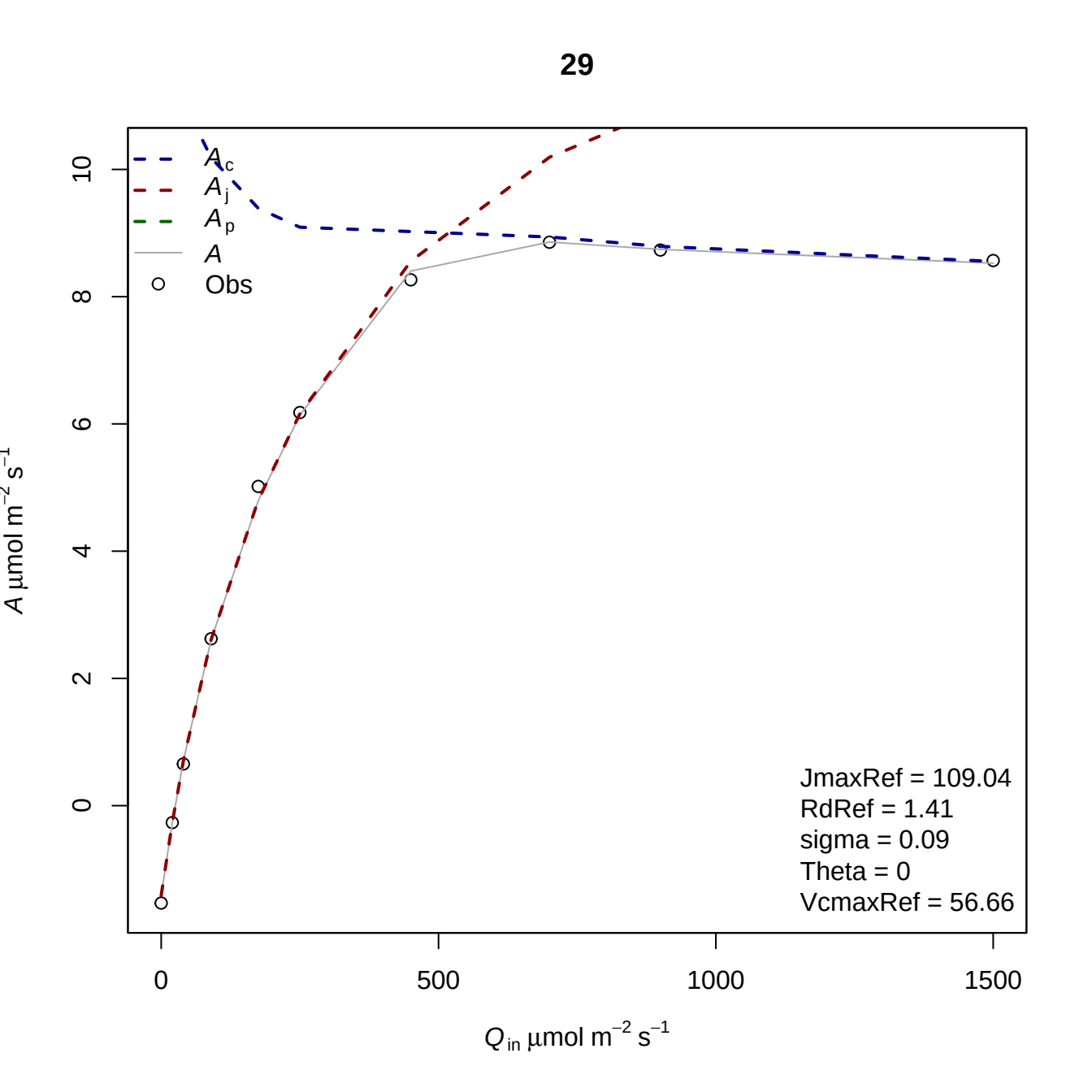

24
A_c
A_j
A_p
A
Obs
JmaxRef = 75.21
RdRef = 1.69
sigma = 0.97
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

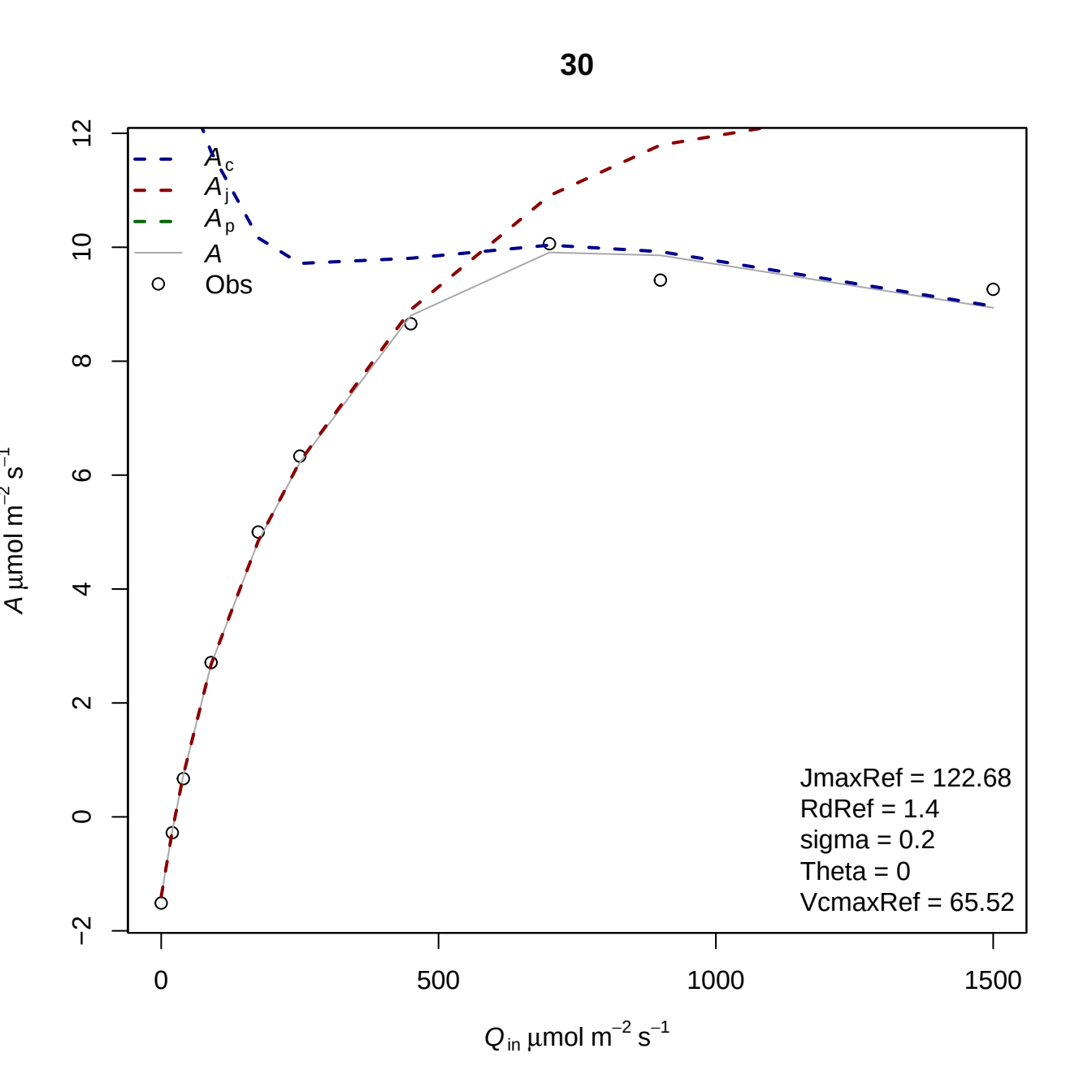

25
A_c
A_j
A_p
A
Obs
JmaxRef = 140.43
RdRef = 1.41
sigma = 0.3
Theta = 0
VcmaxRef = 79.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

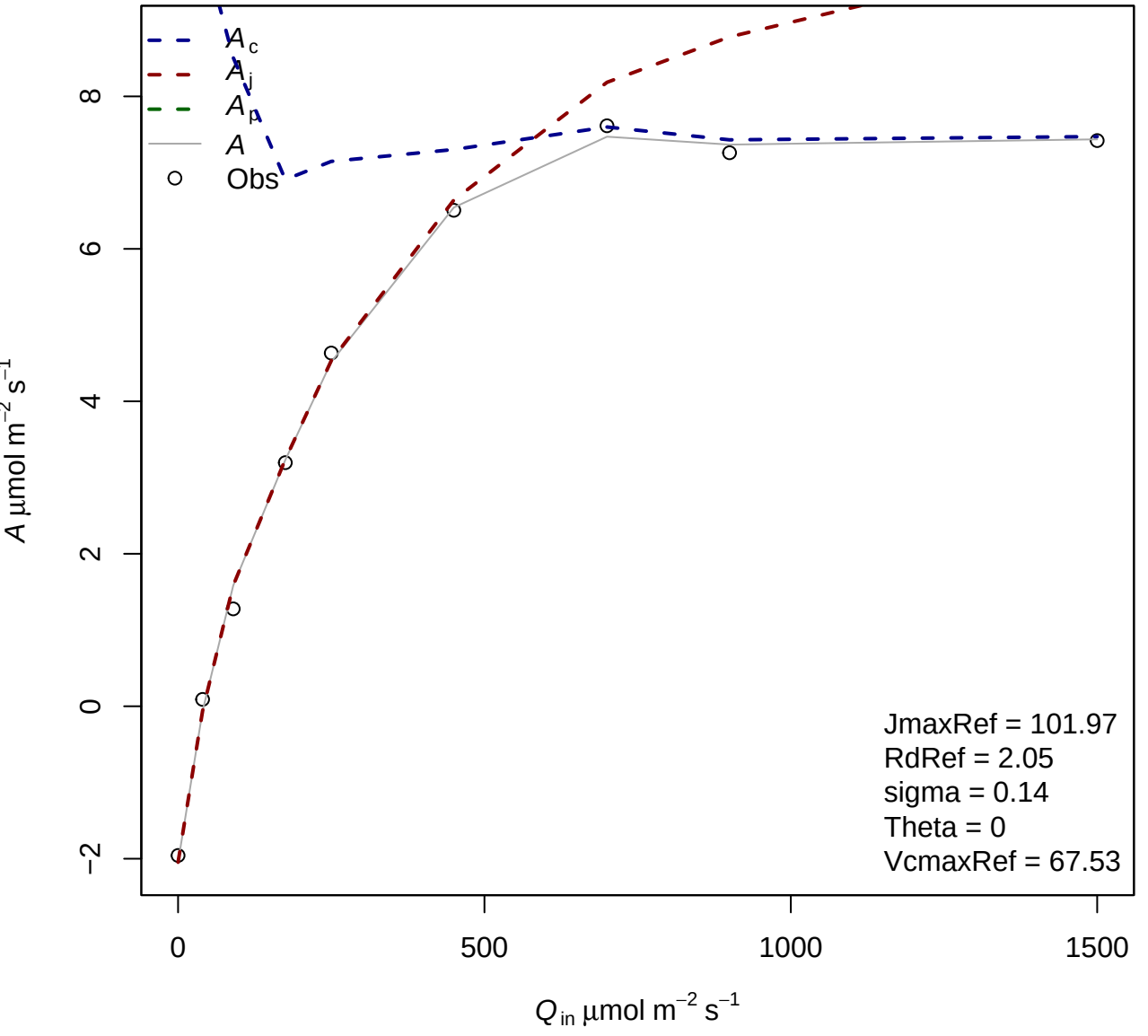

26
A_c
A_j
A_p
A
Obs
JmaxRef = 149.78
RdRef = 1.68
sigma = 0.24
Theta = 0
VcmaxRef = 73.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

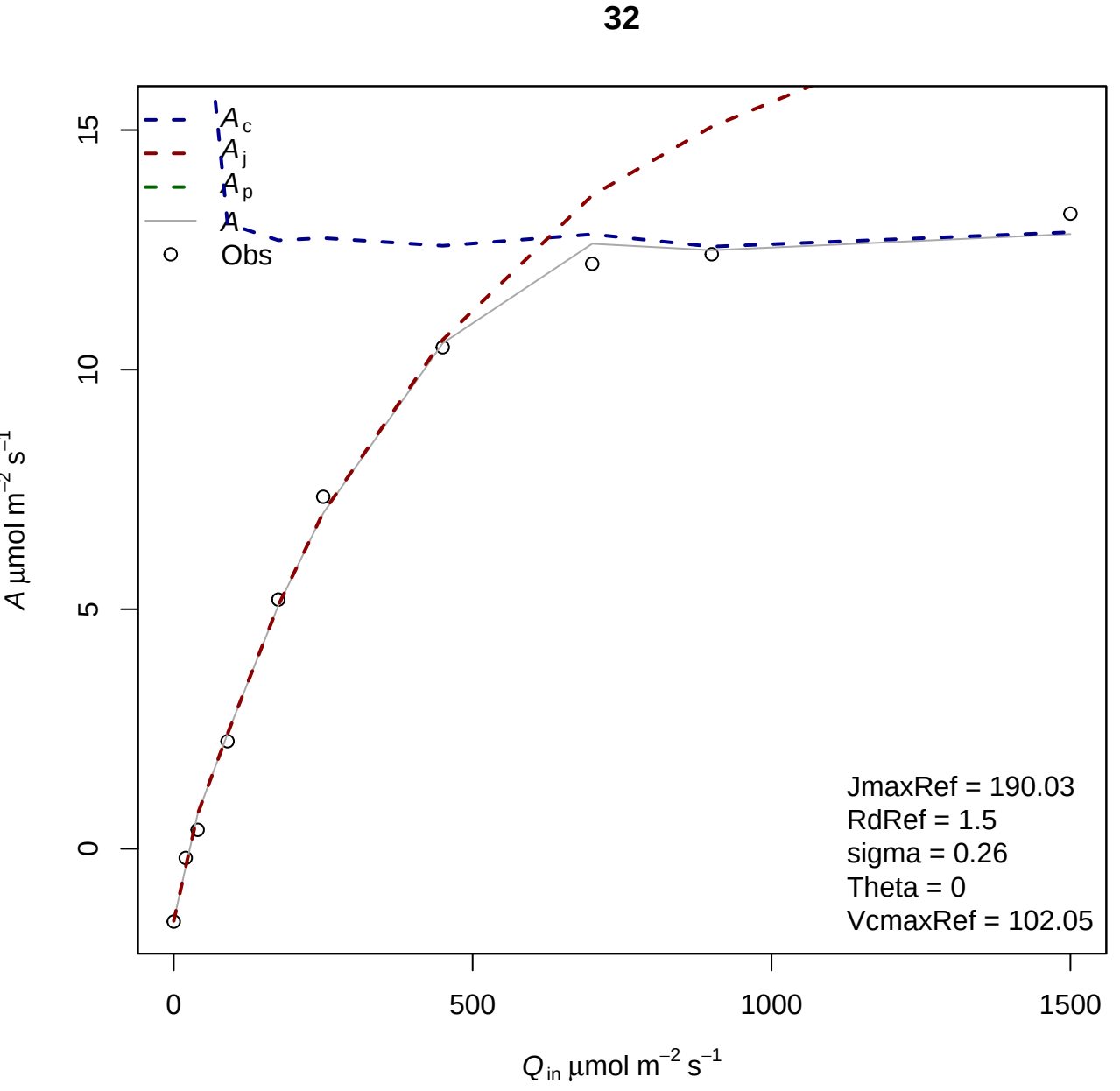

27
A_c
A_j
A_p
A
Obs
JmaxRef = 278.68
RdRef = 3.05
sigma = 0.26
Theta = 0
VcmaxRef = 158.99
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

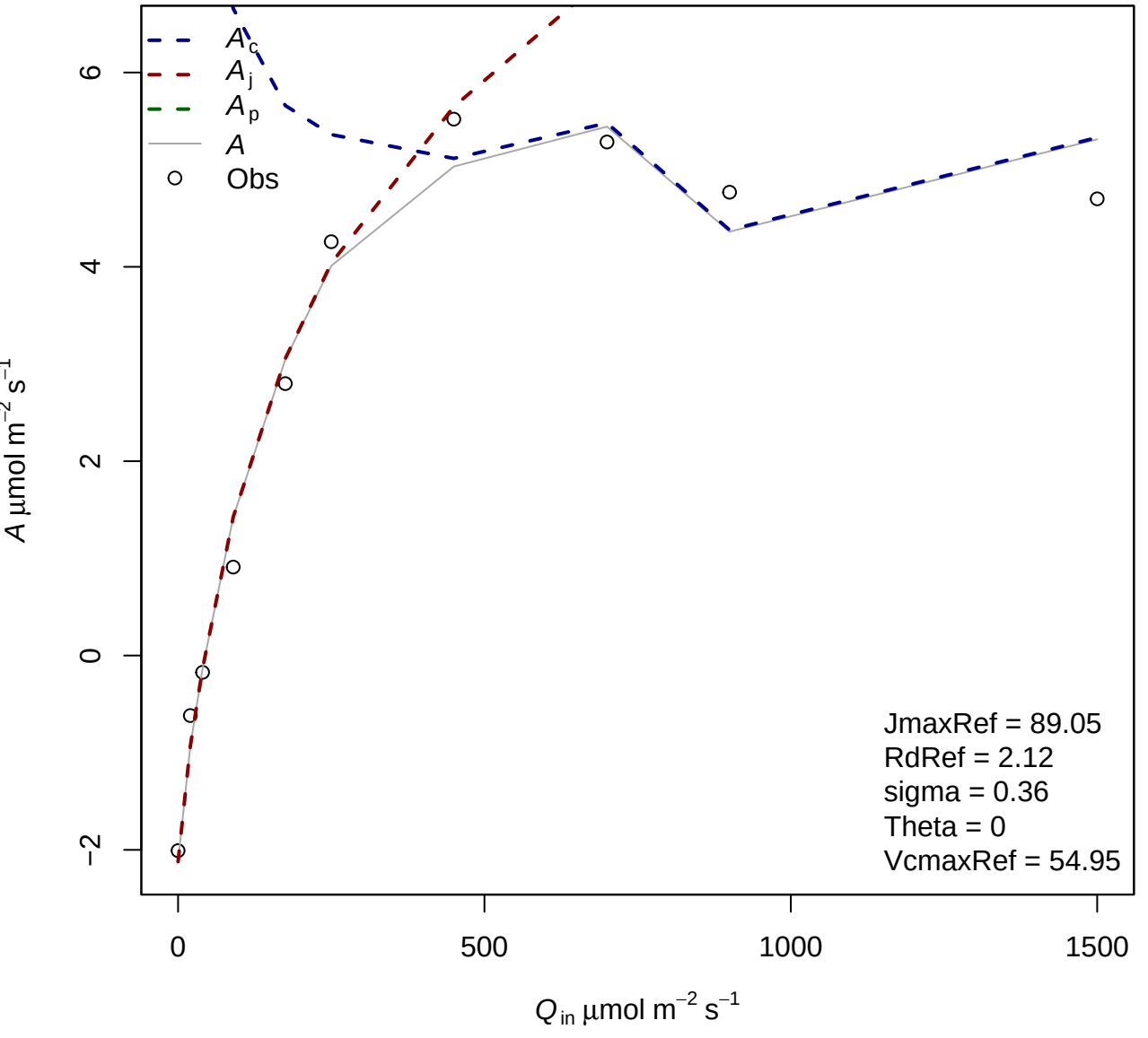

28
A_c
A_j
A_p
A
Obs
JmaxRef = 115.55
RdRef = 2.19
sigma = 0.22
Theta = 0
VcmaxRef = 76.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

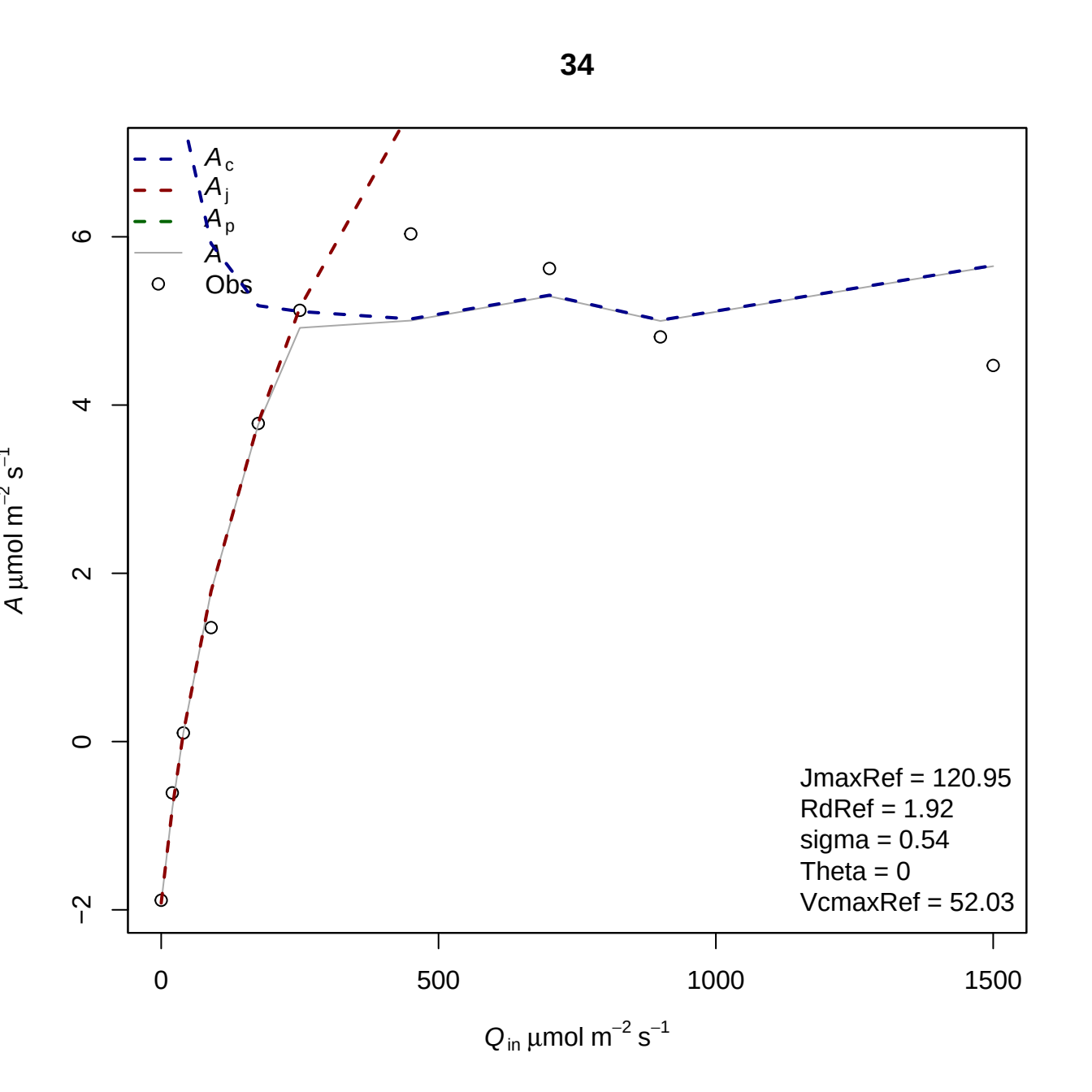

29
A_c
A_j
A_p
A
Obs
JmaxRef = 109.04
RdRef = 1.41
sigma = 0.09
Theta = 0
VcmaxRef = 56.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

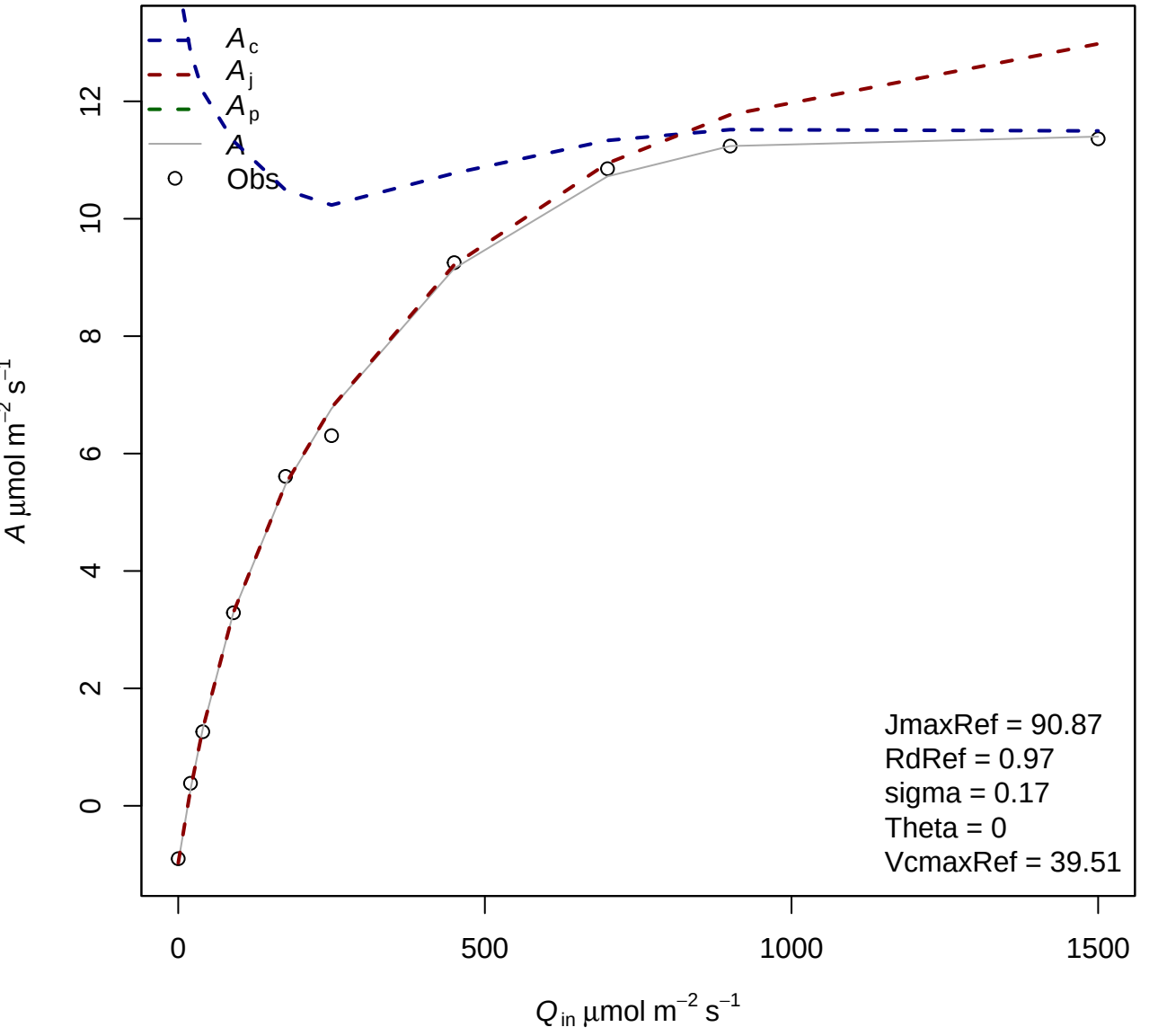

30
A_c
A_j
A_p
A
Obs
JmaxRef = 122.68
RdRef = 1.4
sigma = 0.2
Theta = 0
VcmaxRef = 65.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

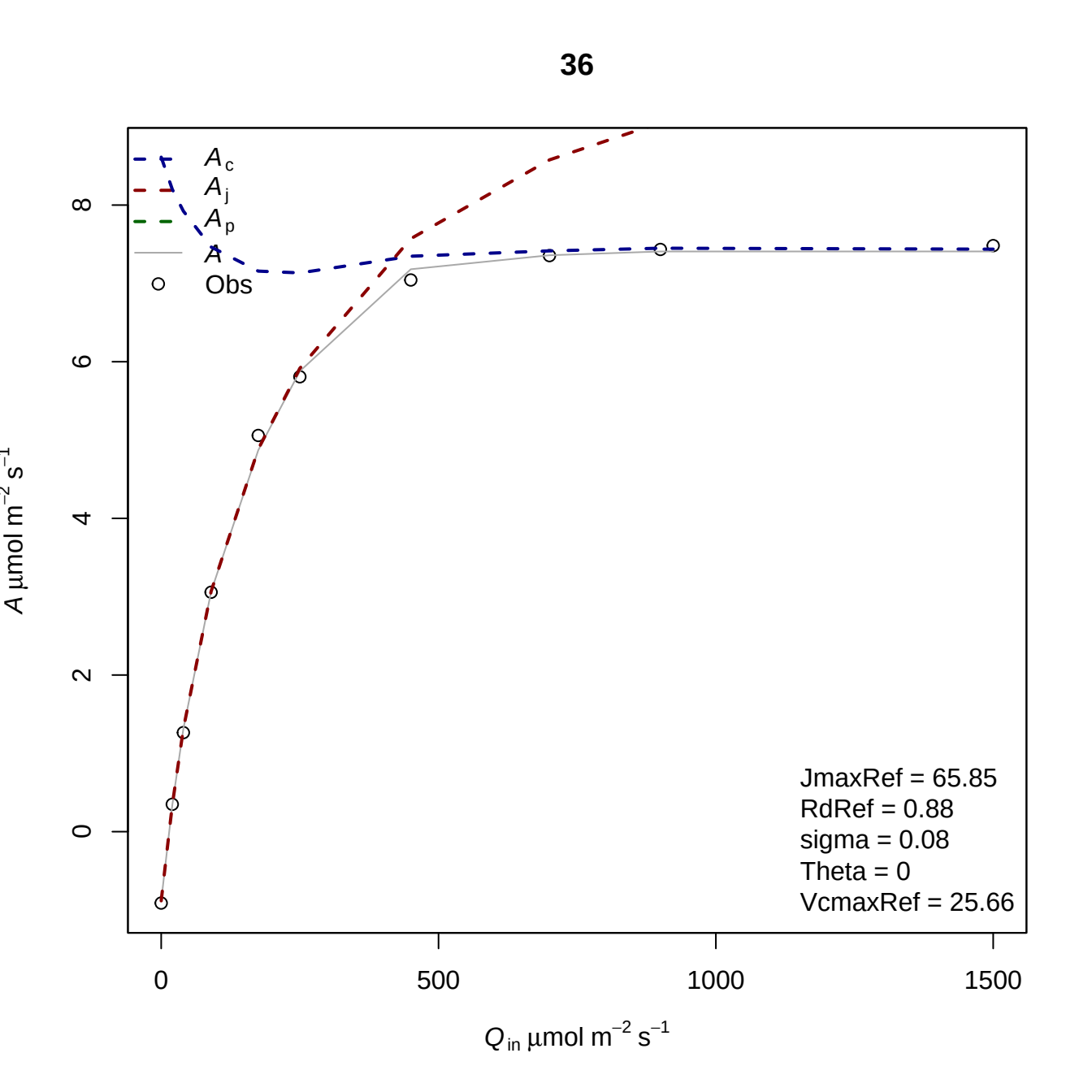

A_c
A_j
A_p
A
Obs
JmaxRef = 101.97
RdRef = 2.05
sigma = 0.14
Theta = 0
VcmaxRef = 67.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

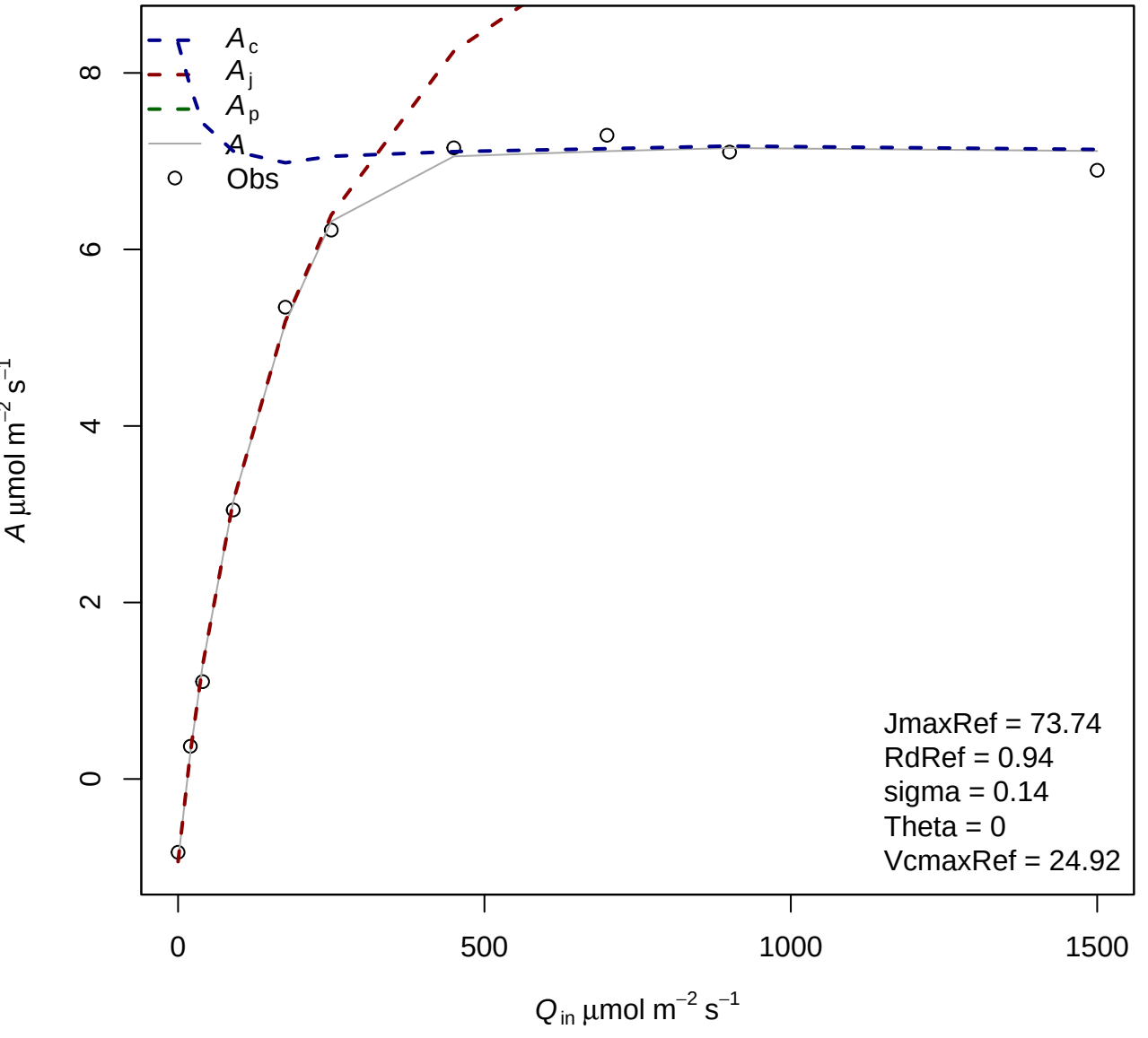

32
A_c
A_j
A_p
A
Obs
JmaxRef = 190.03
RdRef = 1.5
sigma = 0.26
Theta = 0
VcmaxRef = 102.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

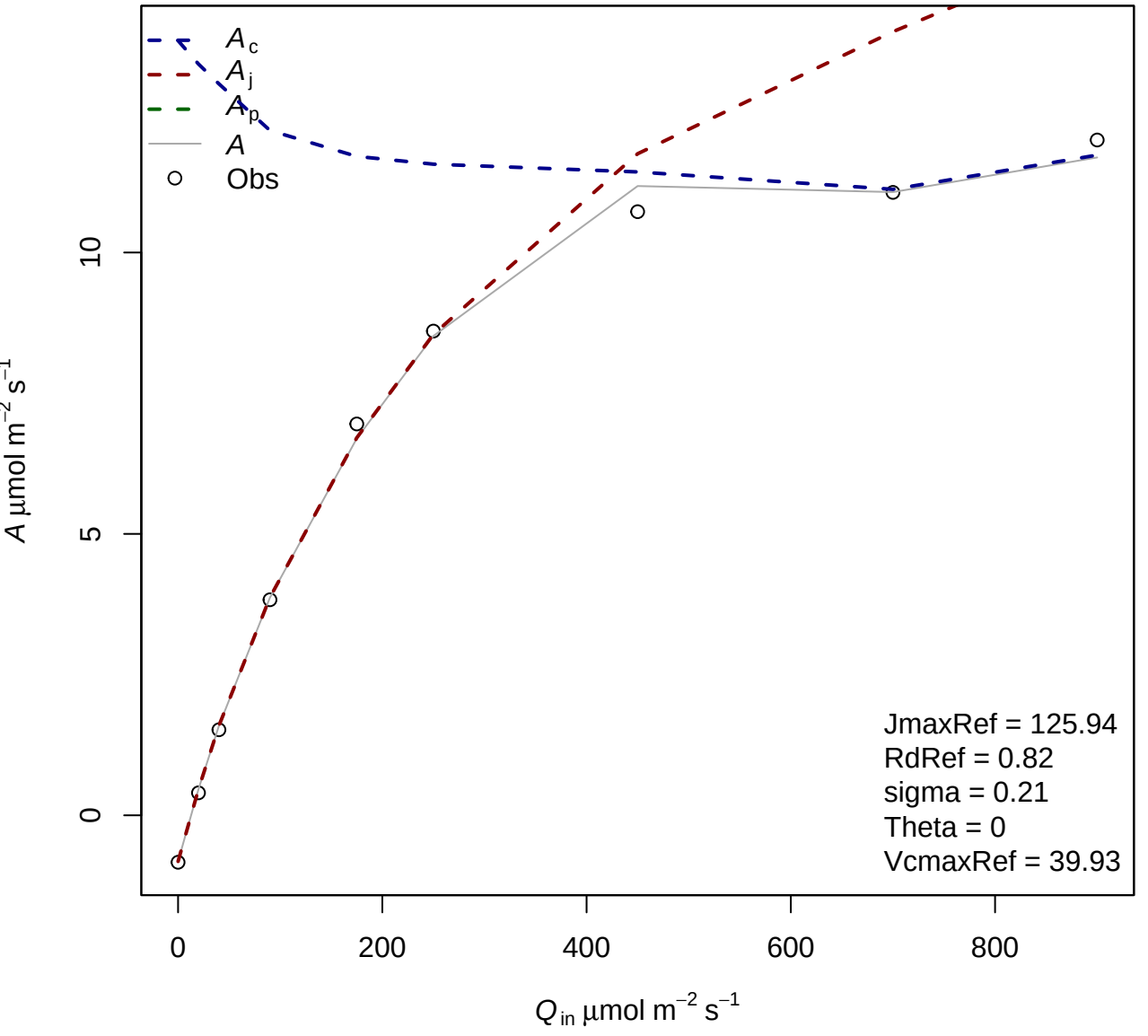

A_c
A_j
A_p
A
Obs
JmaxRef = 89.05
RdRef = 2.12
sigma = 0.36
Theta = 0
VcmaxRef = 54.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

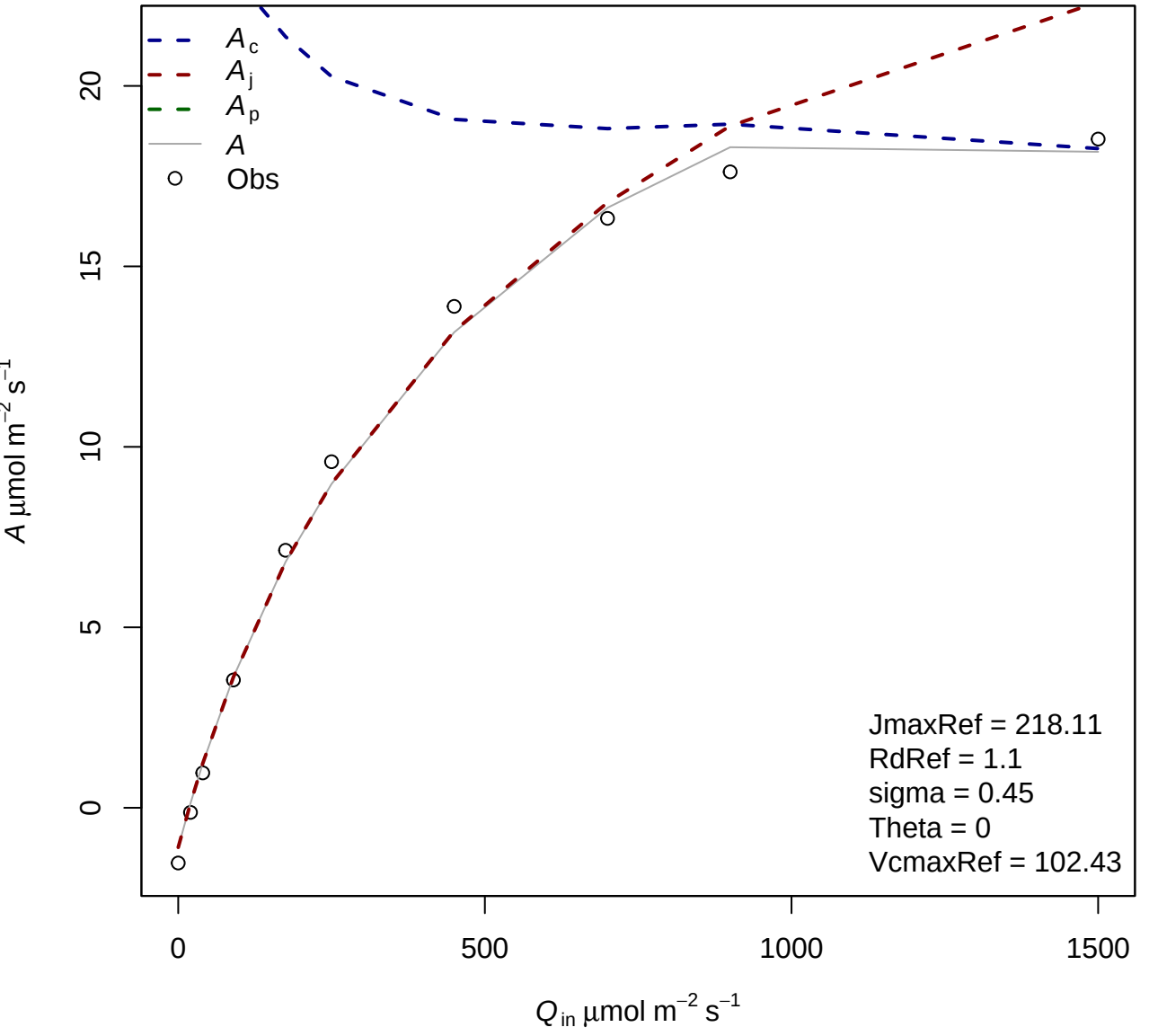

34
A_c
A_j
A_p
A
Obs
JmaxRef = 120.95
RdRef = 1.92
sigma = 0.54
Theta = 0
VcmaxRef = 52.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

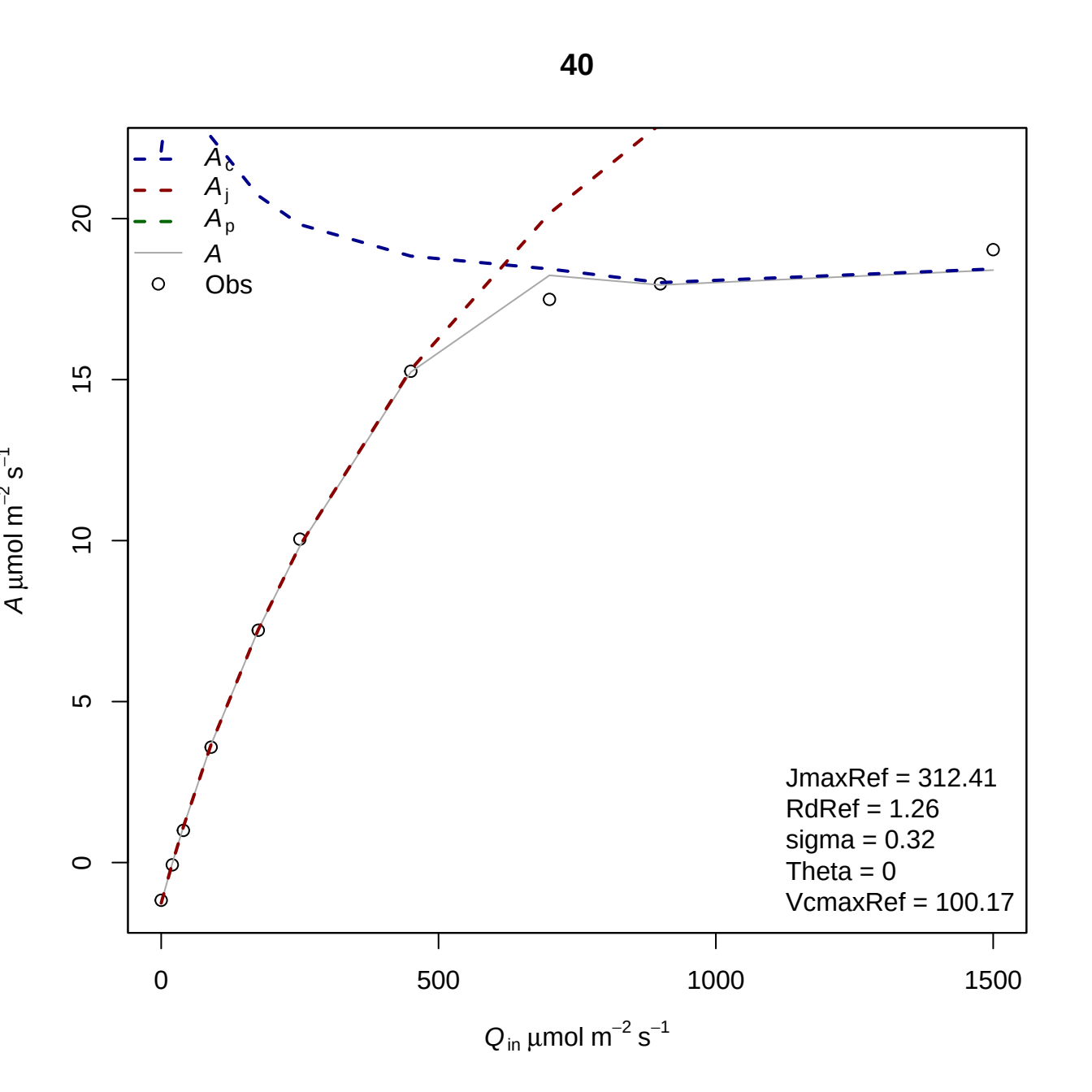

A_c
A_j
A_p
A
Obs
JmaxRef = 90.87
RdRef = 0.97
sigma = 0.17
Theta = 0
VcmaxRef = 39.51
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

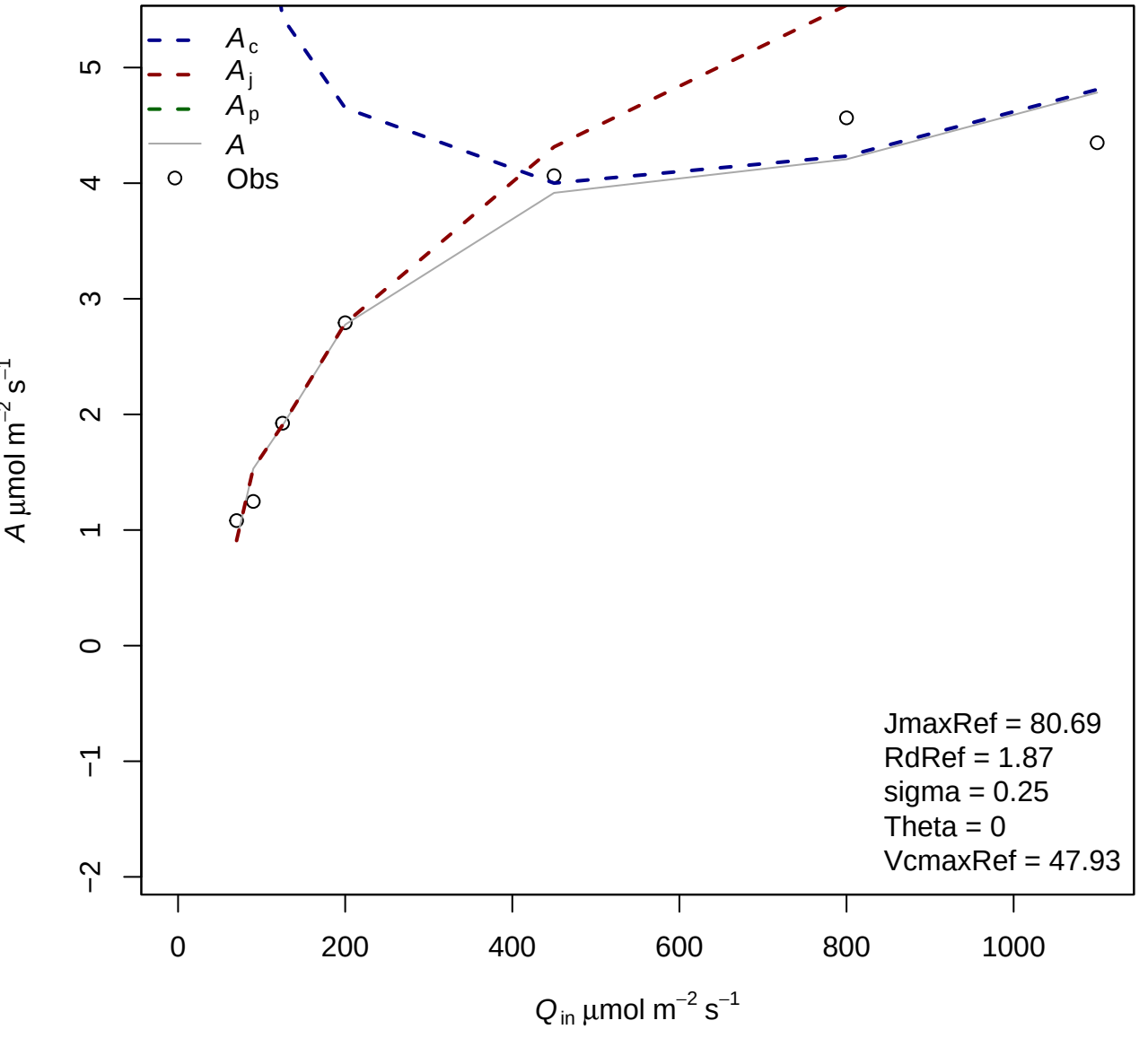

36
A_c
A_j
A_p
A
Obs
JmaxRef = 65.85
RdRef = 0.88
sigma = 0.08
Theta = 0
VcmaxRef = 25.66
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

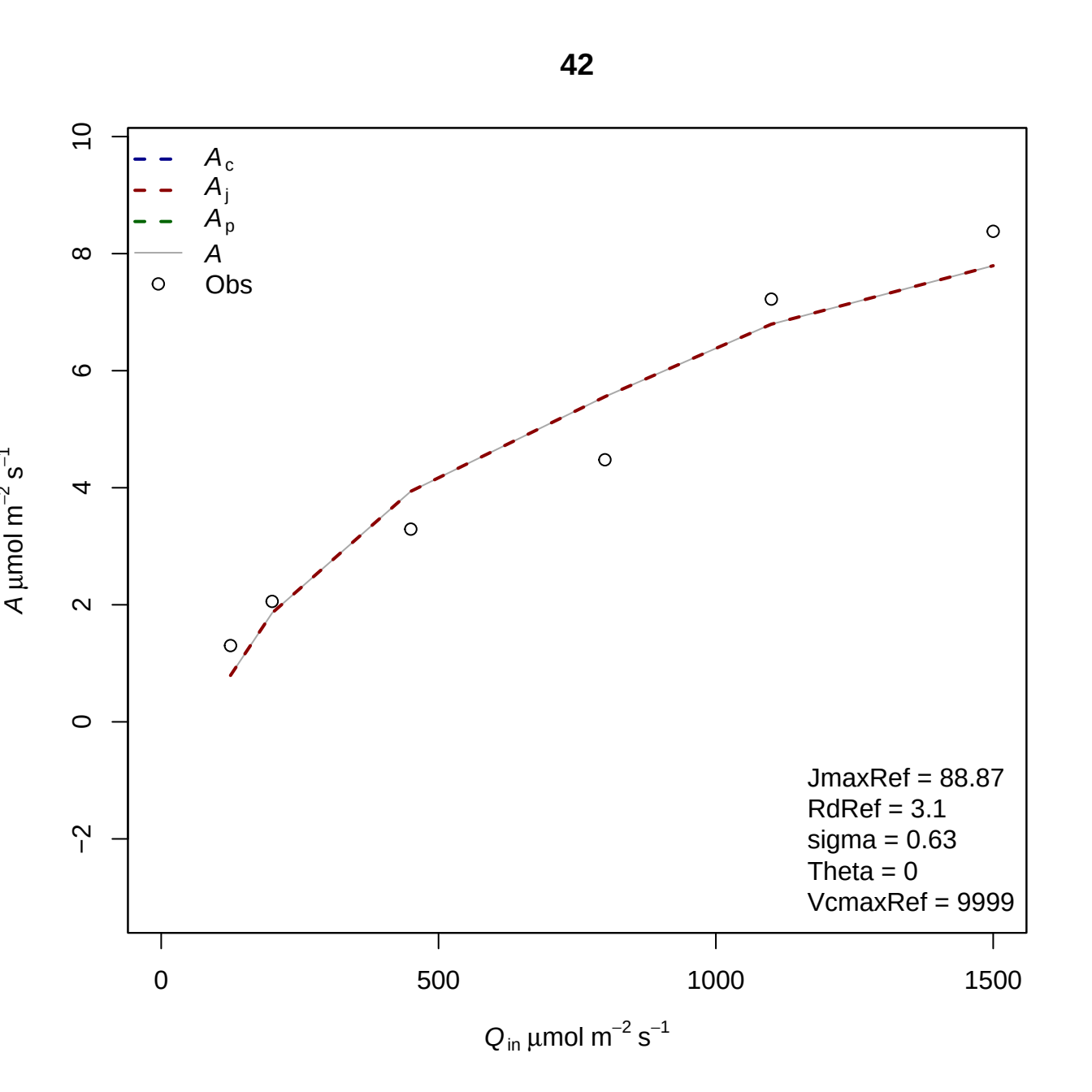

A_c
A_j
A_p
A
Obs
JmaxRef = 73.74
RdRef = 0.94
sigma = 0.14
Theta = 0
VcmaxRef = 24.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

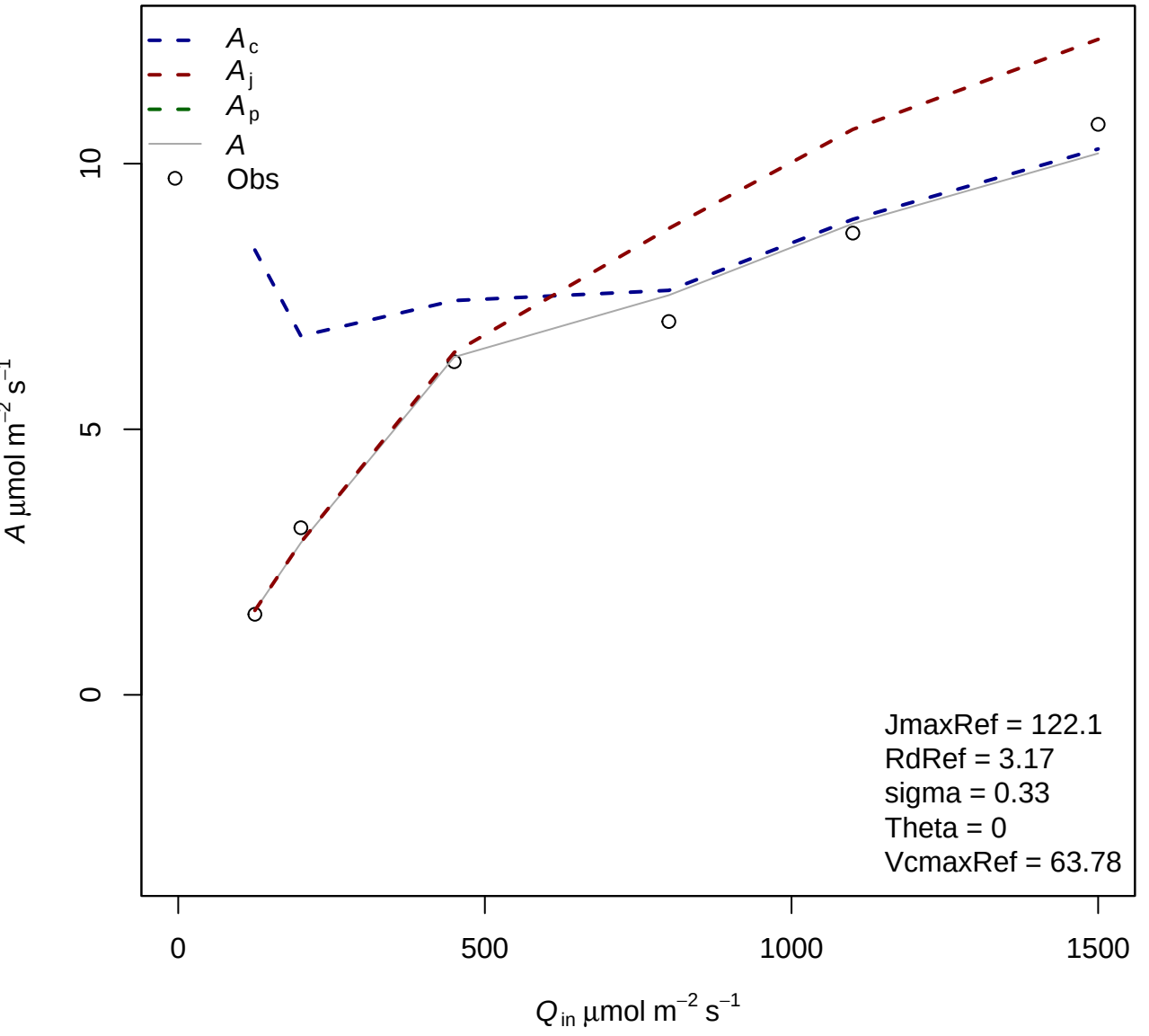

A_c
A_j
A_p
A
Obs
JmaxRef = 125.94
RdRef = 0.82
sigma = 0.21
Theta = 0
VcmaxRef = 39.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800

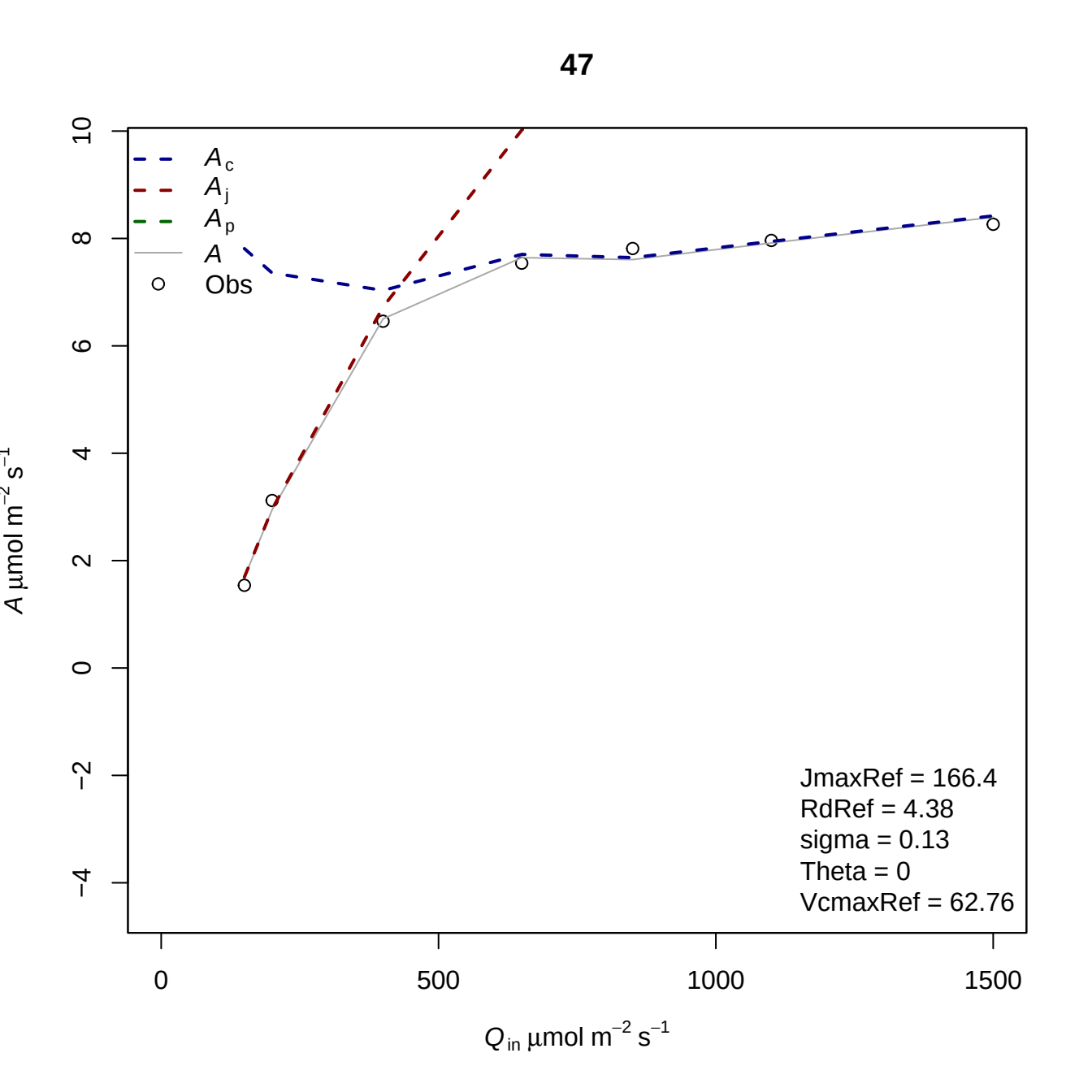

A_c
A_j
A_p
A
Obs
JmaxRef = 218.11
RdRef = 1.1
sigma = 0.45
Theta = 0
VcmaxRef = 102.43
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

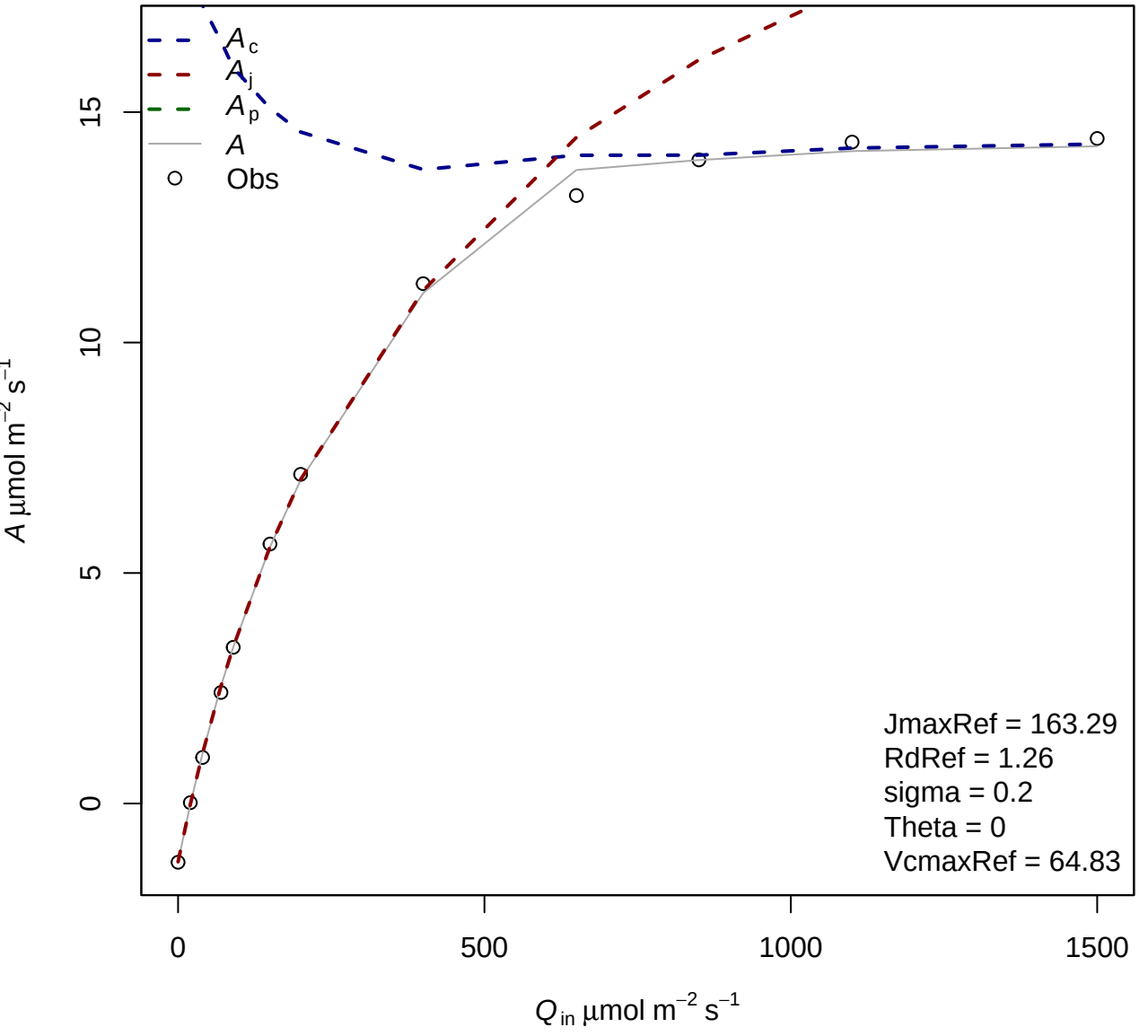

40
A_c
A_j
A_p
A
Obs
JmaxRef = 312.41
RdRef = 1.26
sigma = 0.32
Theta = 0
VcmaxRef = 100.17
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

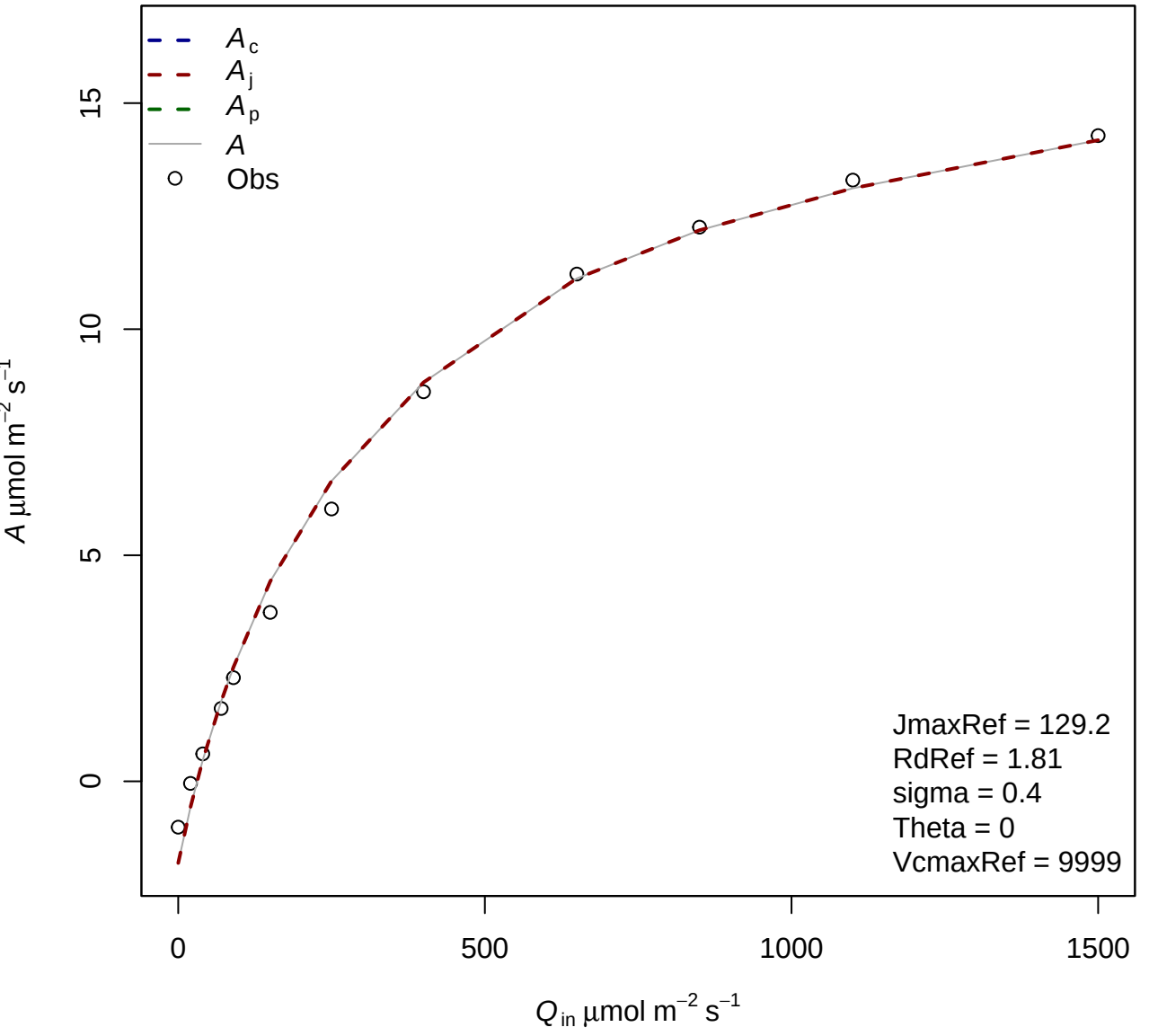

A_c
A_j
A_p
A
Obs
JmaxRef = 80.69
RdRef = 1.87
sigma = 0.25
Theta = 0
VcmaxRef = 47.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
5
0
200
400
600
800
1000

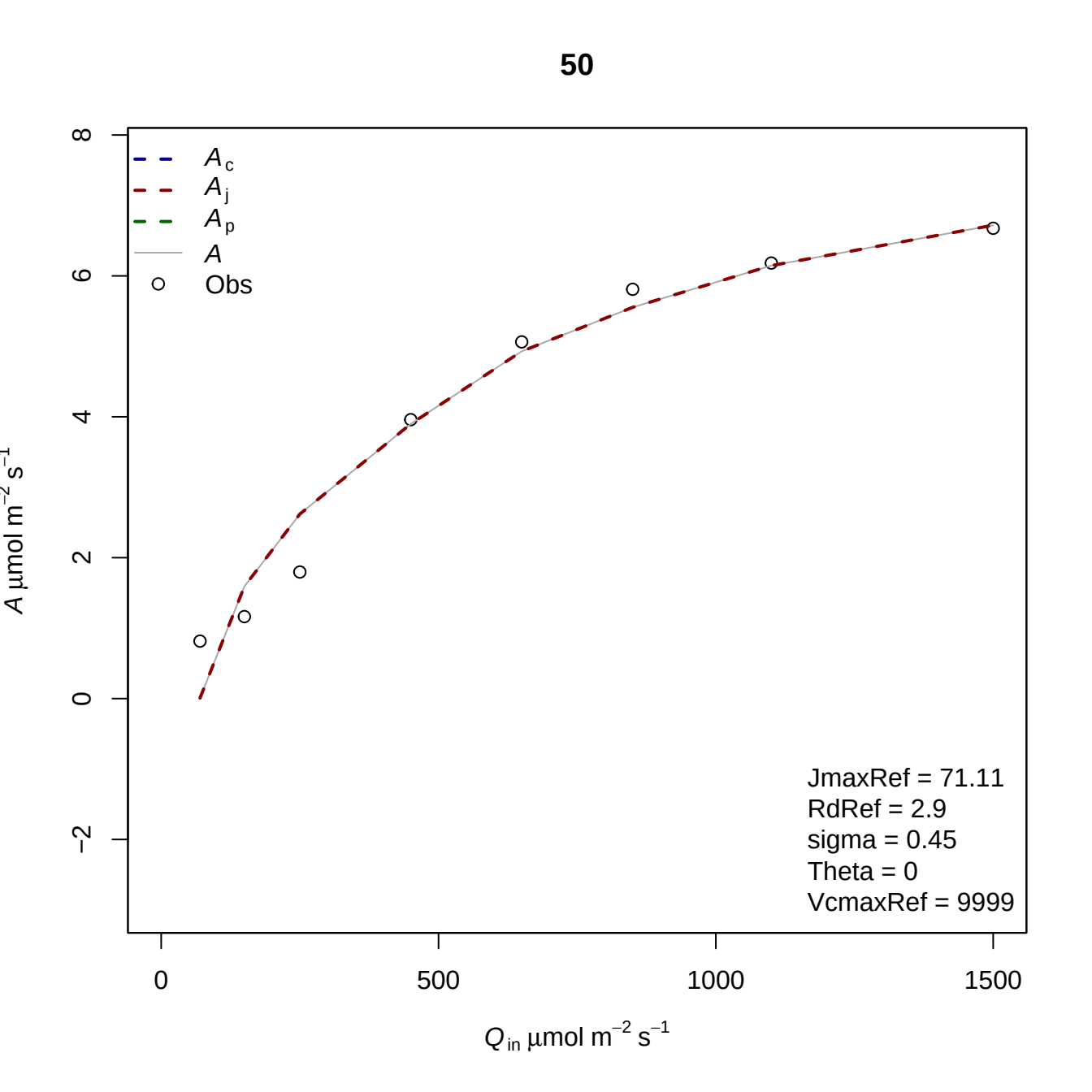

42
A_c
A_j
A_p
A
Obs
JmaxRef = 88.87
RdRef = 3.1
sigma = 0.63
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

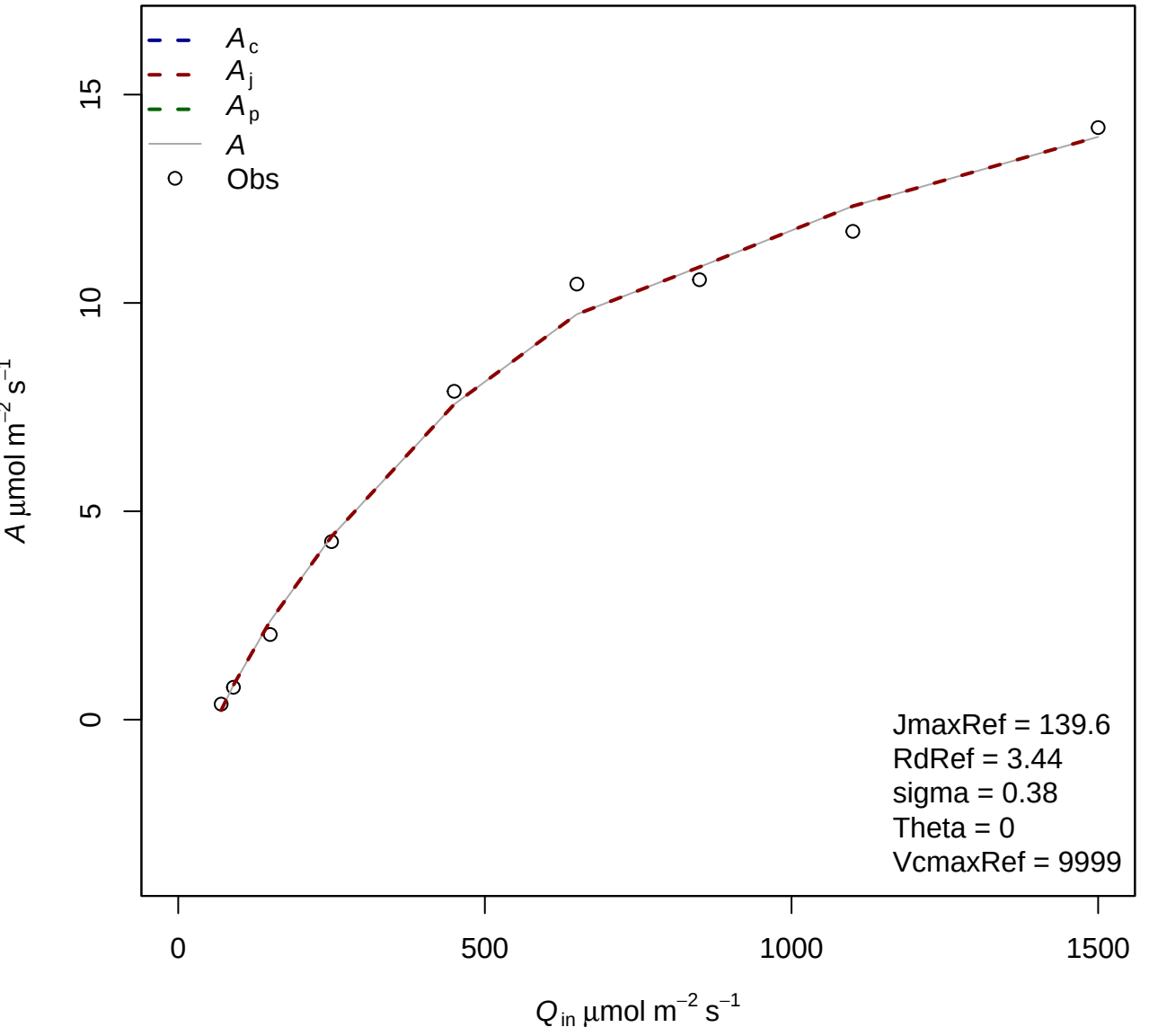

A_c
A_j
A_p
A
Obs
JmaxRef = 122.1
RdRef = 3.17
sigma = 0.33
Theta = 0
VcmaxRef = 63.78
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

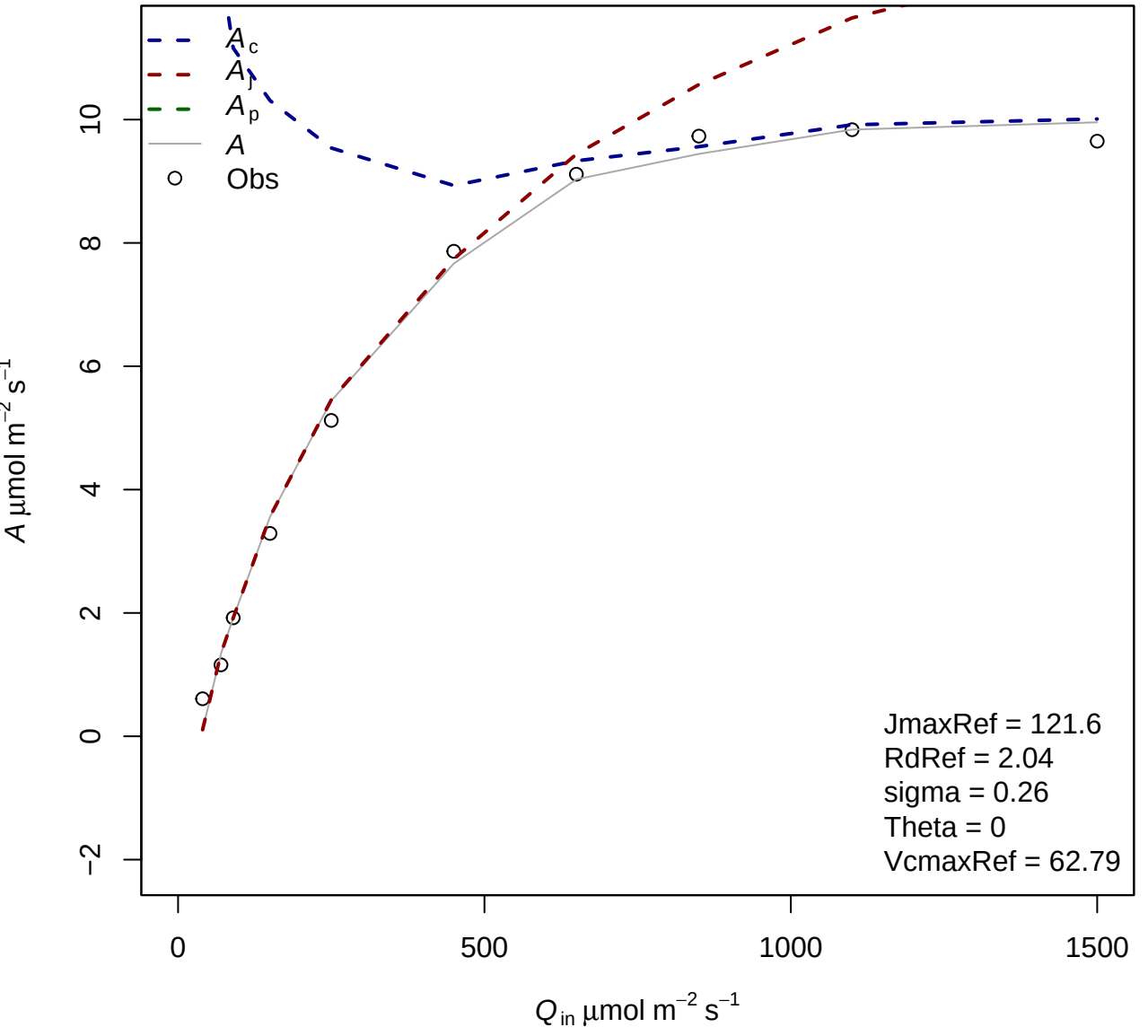

47
A_c
A_j
A_p
A
Obs
JmaxRef = 166.4
RdRef = 4.38
sigma = 0.13
Theta = 0
VcmaxRef = 62.76
10
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

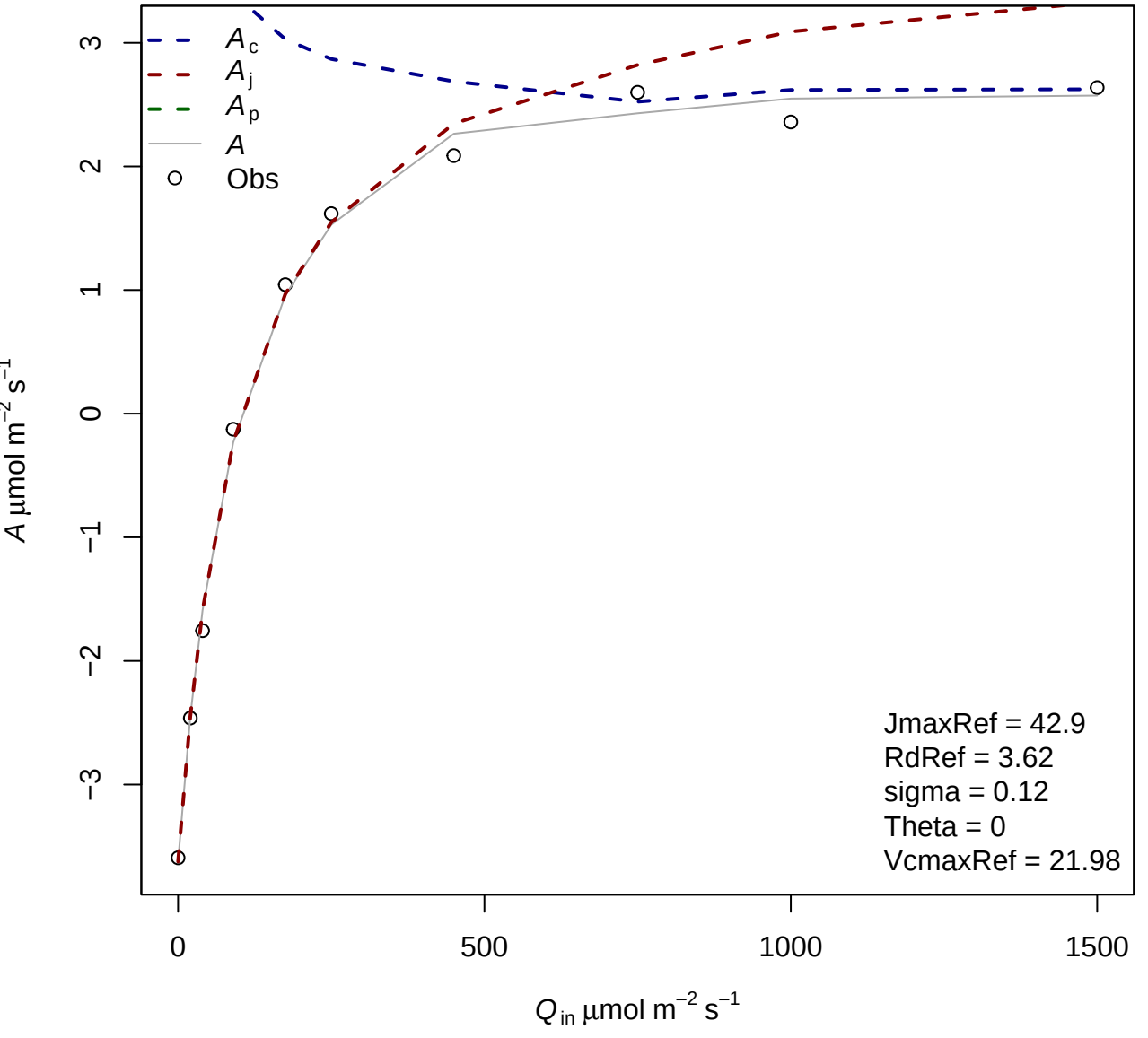

A_c
A_j
A_p
A
Obs
JmaxRef = 163.29
RdRef = 1.26
sigma = 0.2
Theta = 0
VcmaxRef = 64.83
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

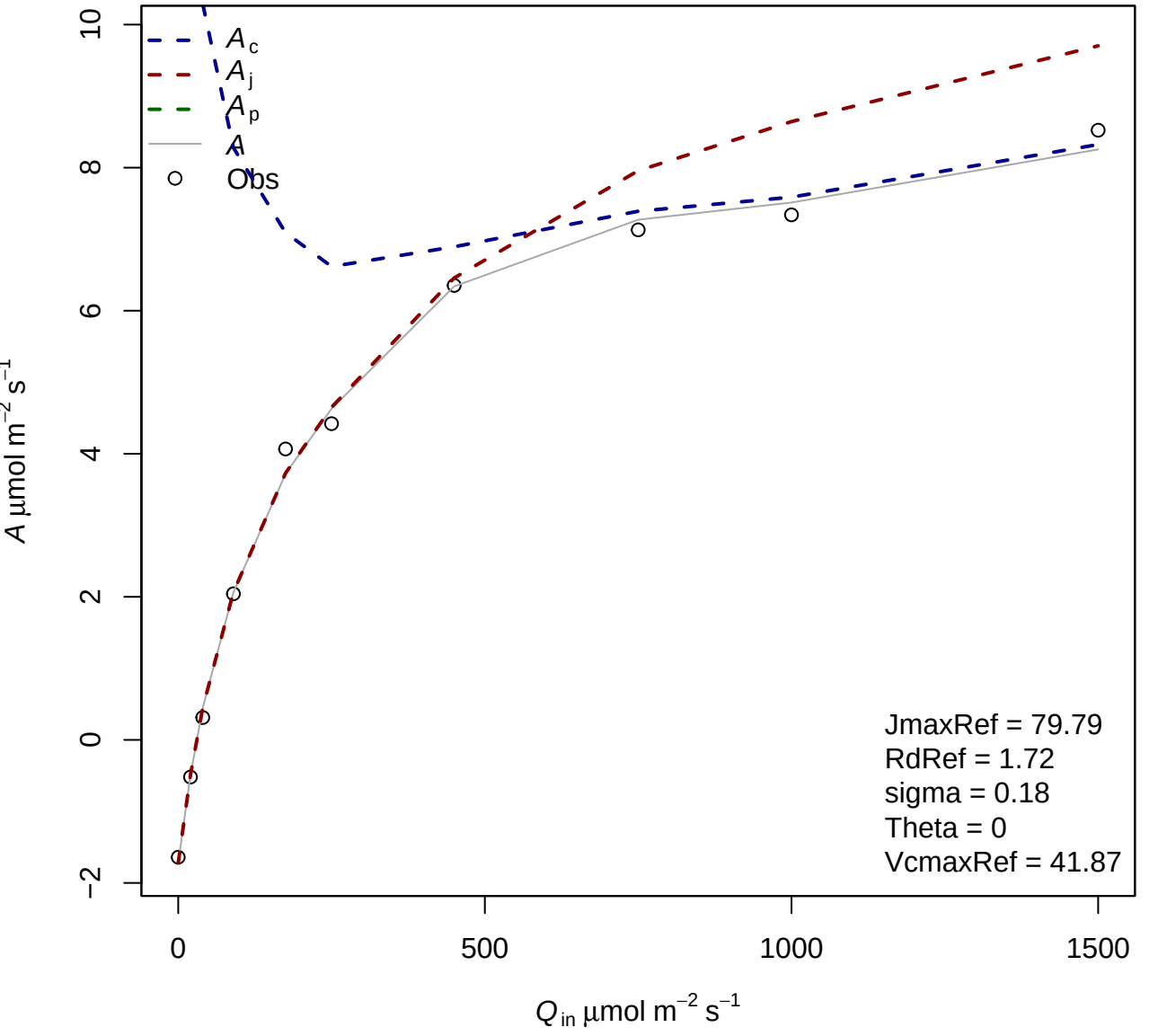

A_c
A_j
A_p
A
Obs
JmaxRef = 129.2
RdRef = 1.81
sigma = 0.4
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

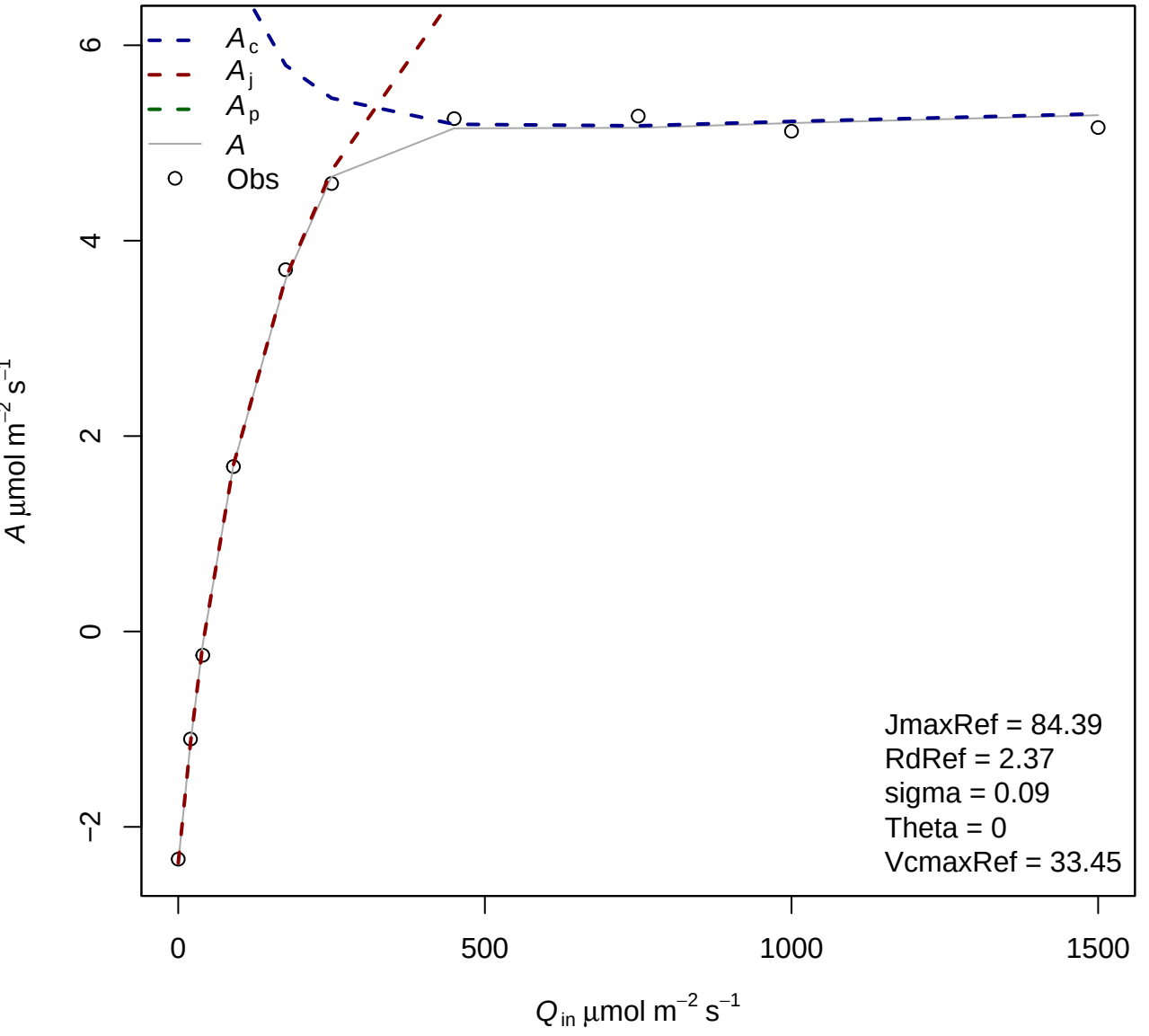

50
A_c
A_j
A_p
A
Obs
JmaxRef = 71.11
RdRef = 2.9
sigma = 0.45
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

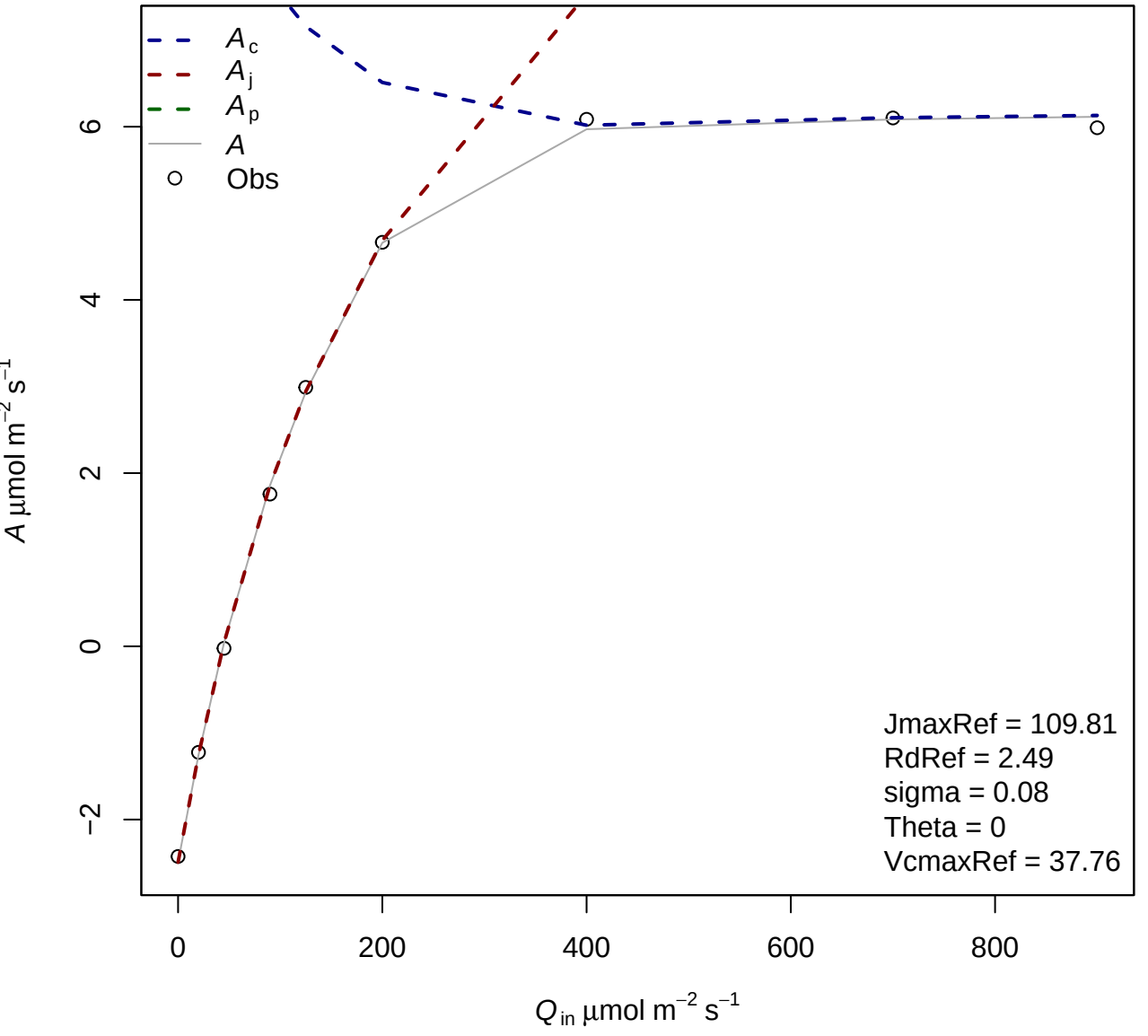

A_c
A_j
A_p
A
Obs
JmaxRef = 139.6
RdRef = 3.44
sigma = 0.38
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

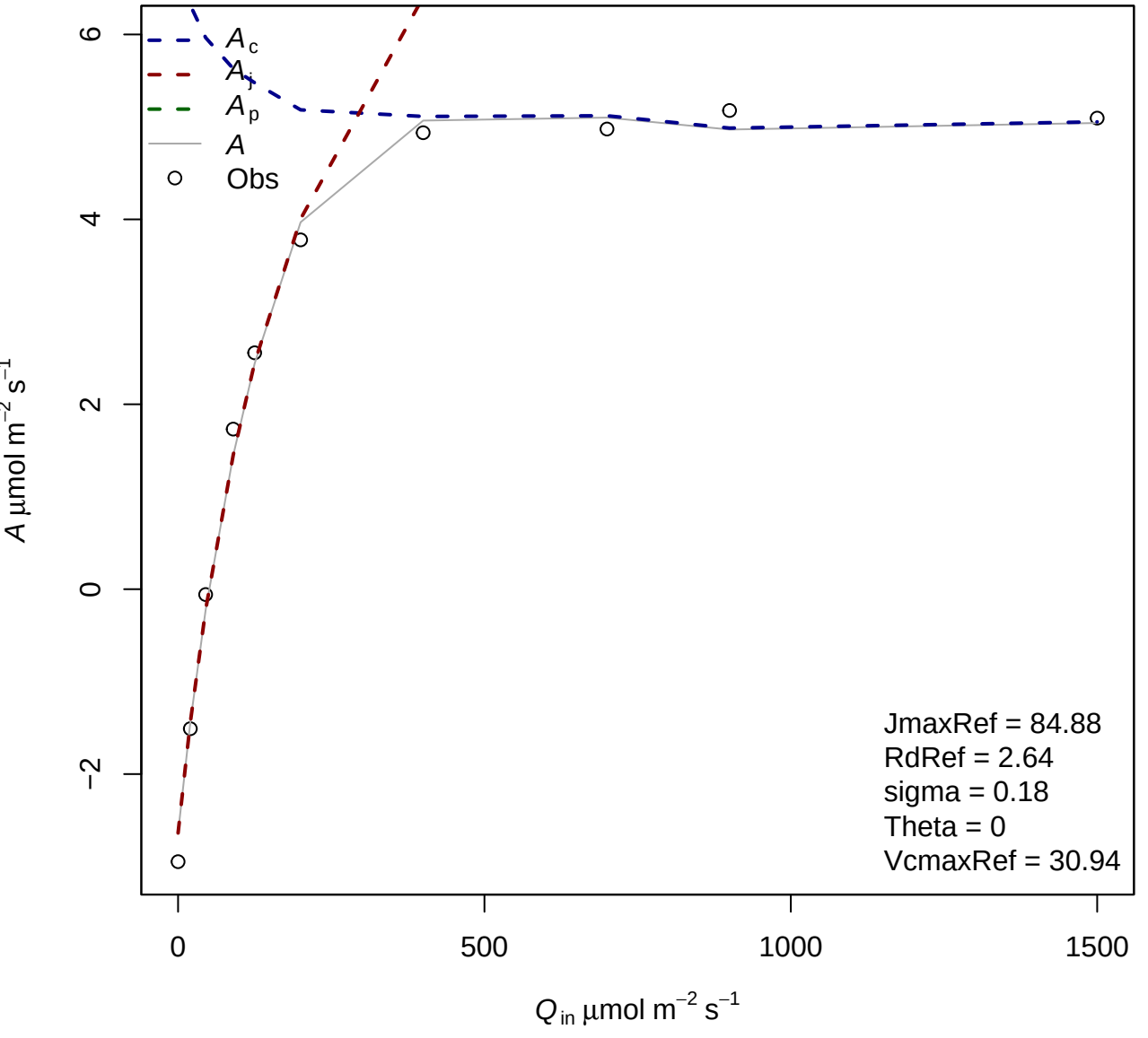

A_c
A_j
A_p
A
Obs
JmaxRef = 121.6
RdRef = 2.04
sigma = 0.26
Theta = 0
VcmaxRef = 62.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

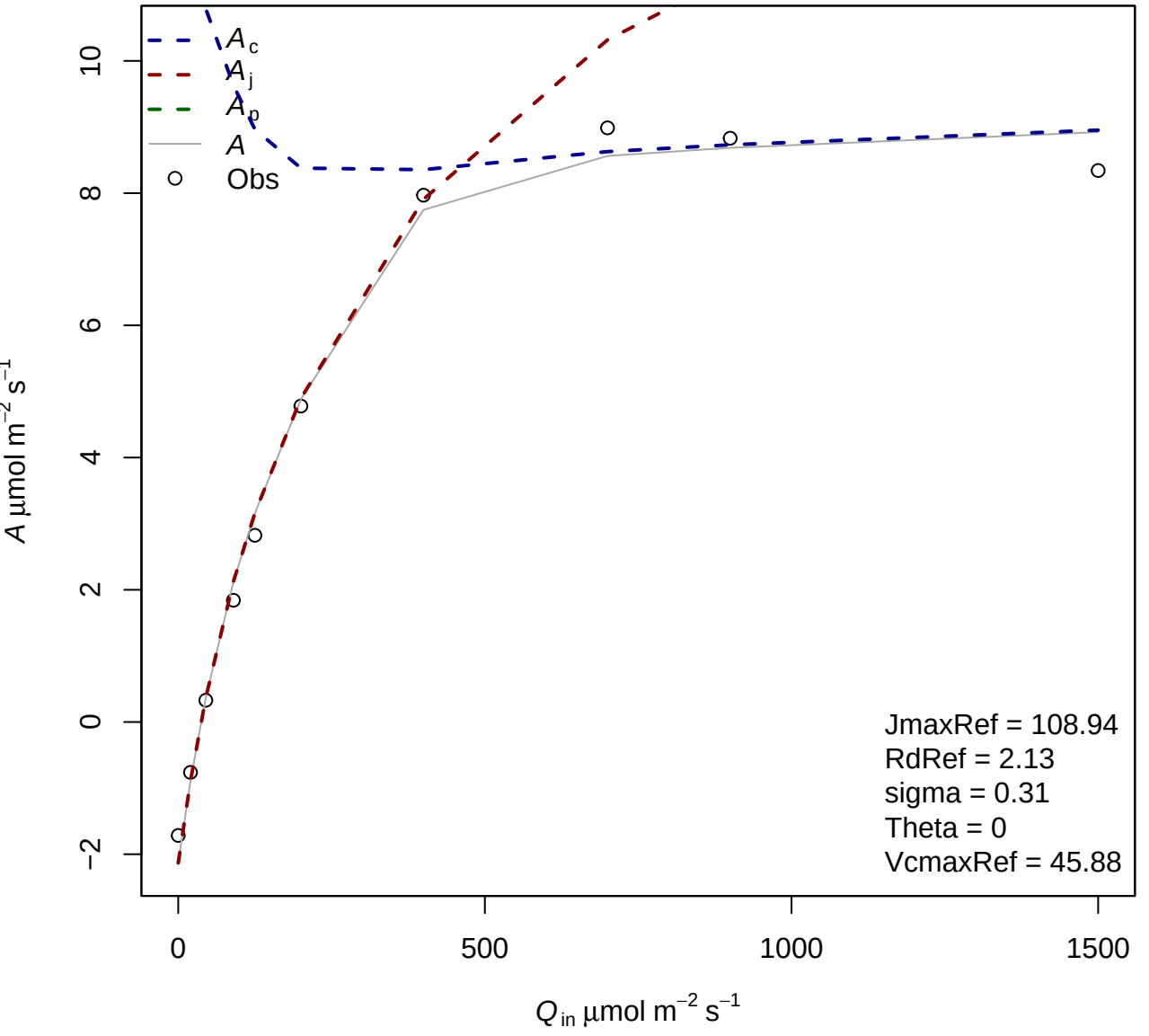

A_c
A_j
A_p
A
Obs
JmaxRef = 42.9
RdRef = 3.62
sigma = 0.12
Theta = 0
VcmaxRef = 21.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
−3
0
500
1000
1500

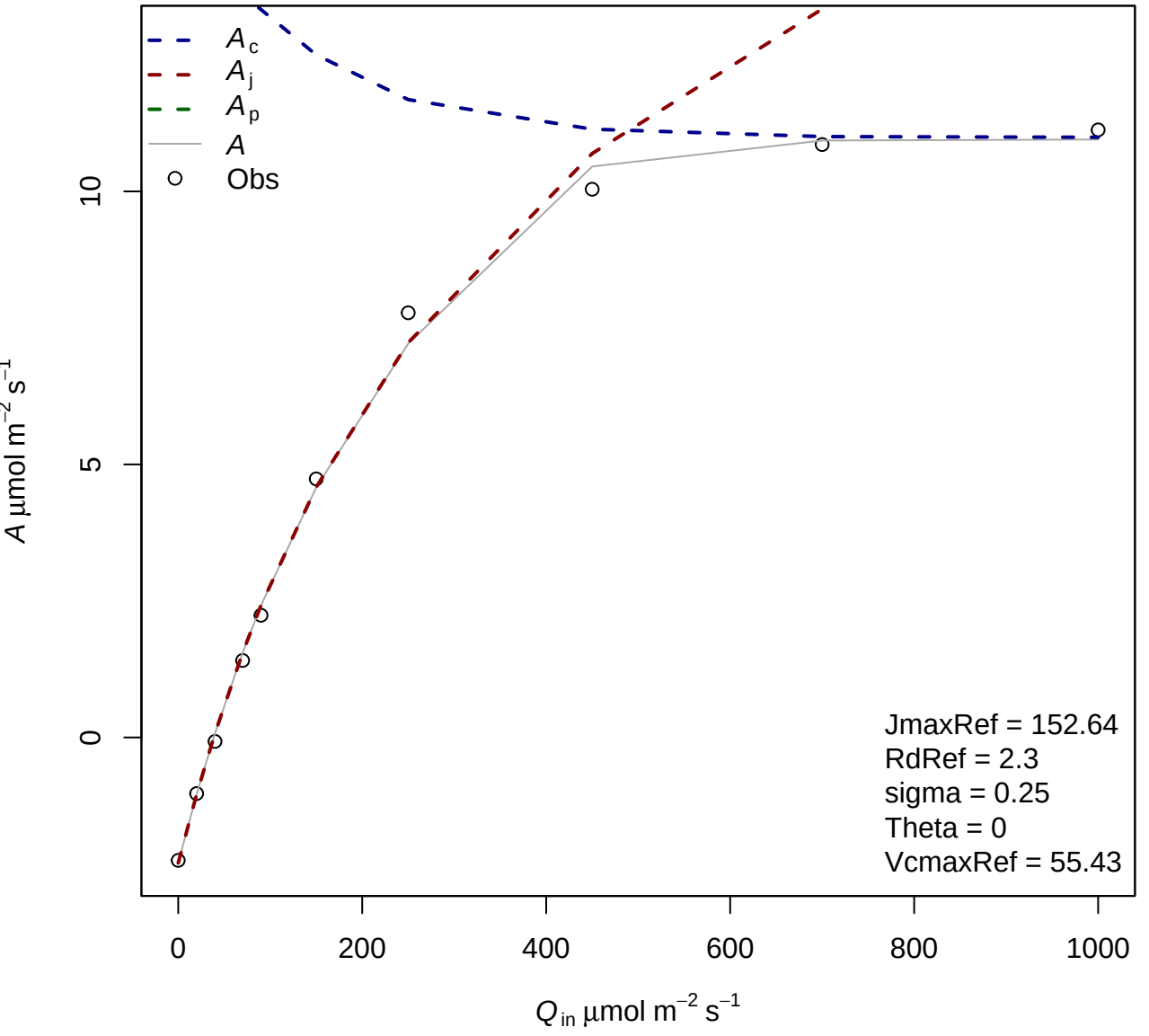

A_c
A_j
A_p
A
Obs
JmaxRef = 79.79
RdRef = 1.72
sigma = 0.18
Theta = 0
VcmaxRef = 41.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

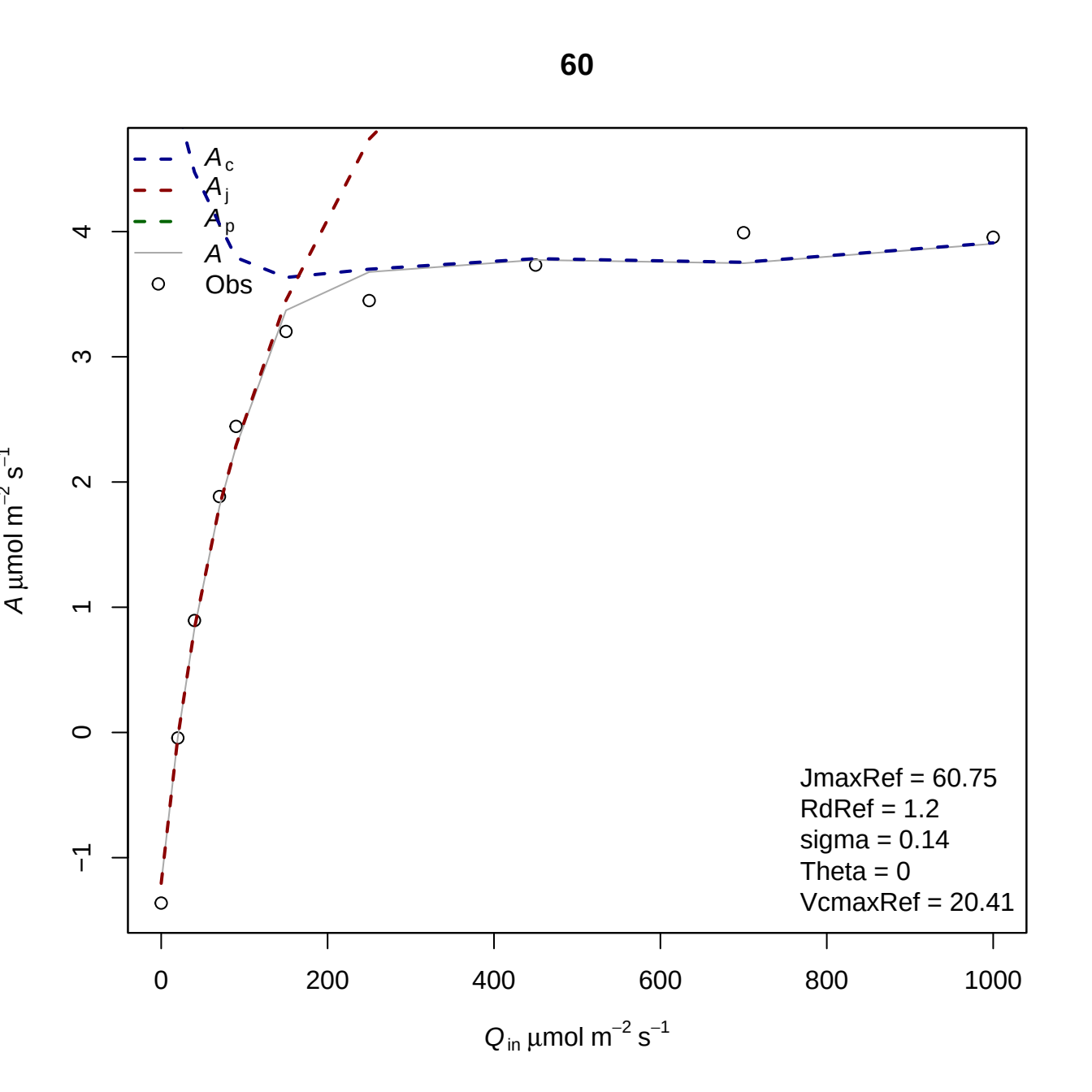

A_c
A_j
A_p
A
Obs
JmaxRef = 84.39
RdRef = 2.37
sigma = 0.09
Theta = 0
VcmaxRef = 33.45
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

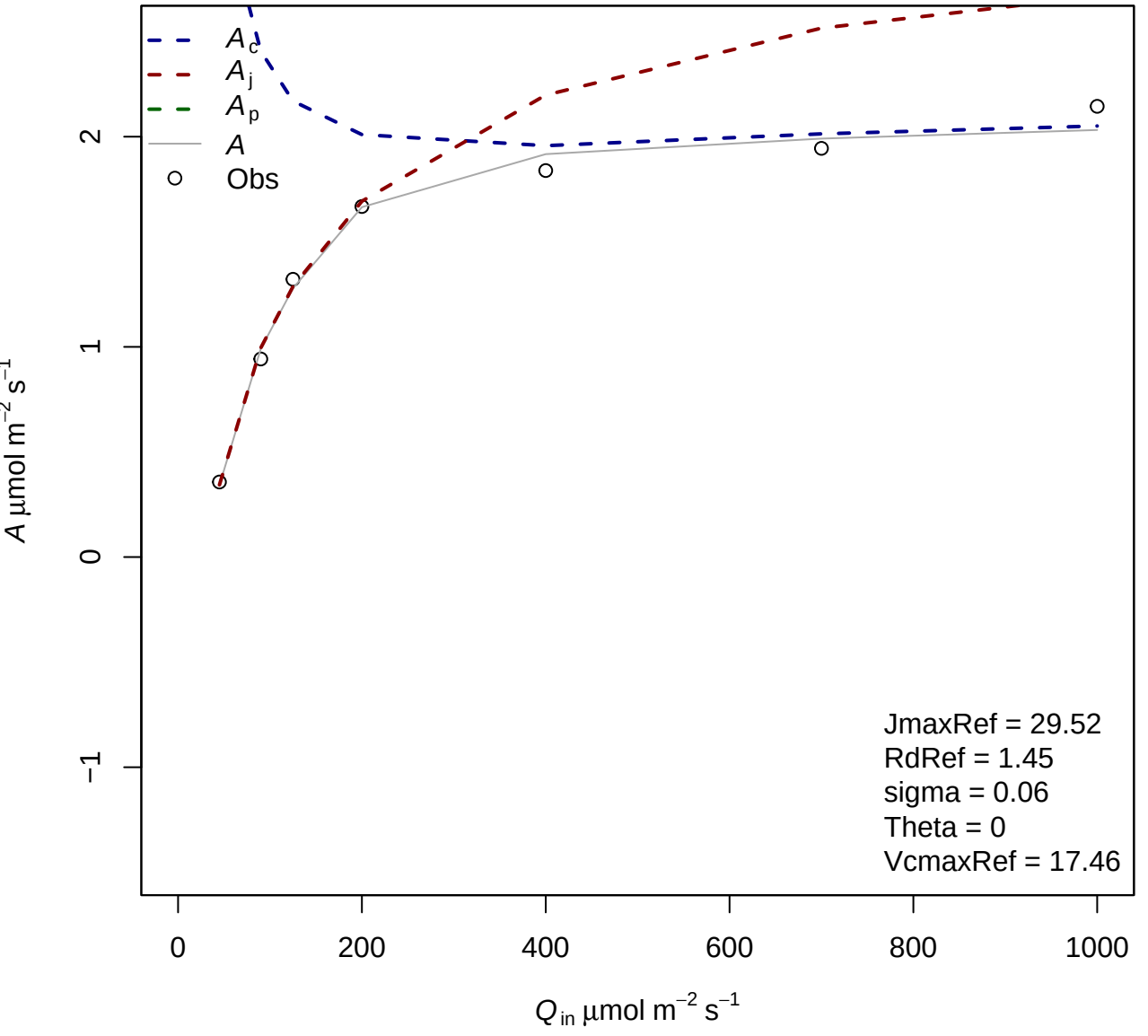

A_c
A_j
A_p
A
Obs
JmaxRef = 109.81
RdRef = 2.49
sigma = 0.08
Theta = 0
VcmaxRef = 37.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

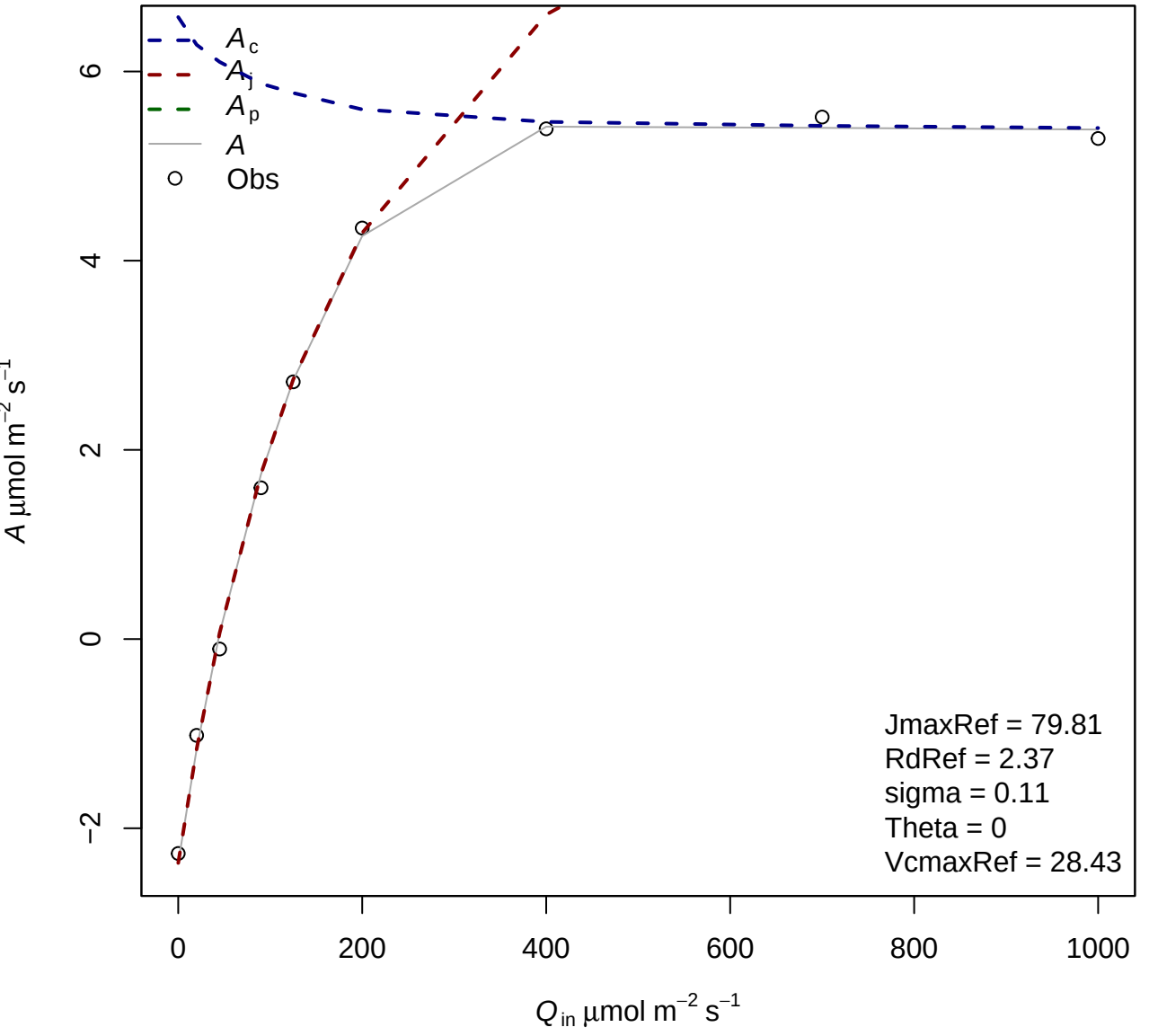

A_c
A_j
A_p
A
Obs
JmaxRef = 84.88
RdRef = 2.64
sigma = 0.18
Theta = 0
VcmaxRef = 30.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

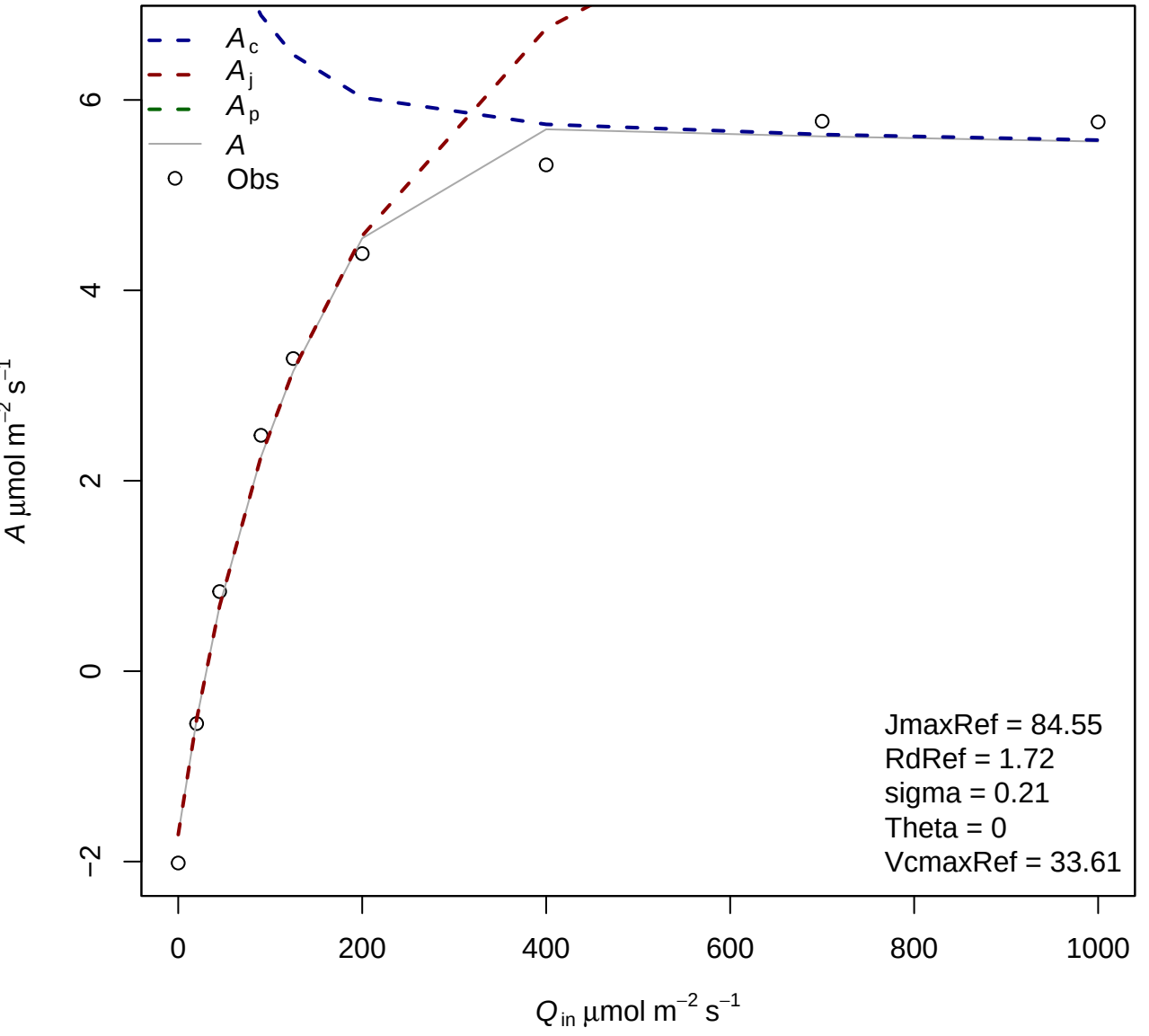

A_c
A_j
A_p
A
Obs
JmaxRef = 108.94
RdRef = 2.13
sigma = 0.31
Theta = 0
VcmaxRef = 45.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

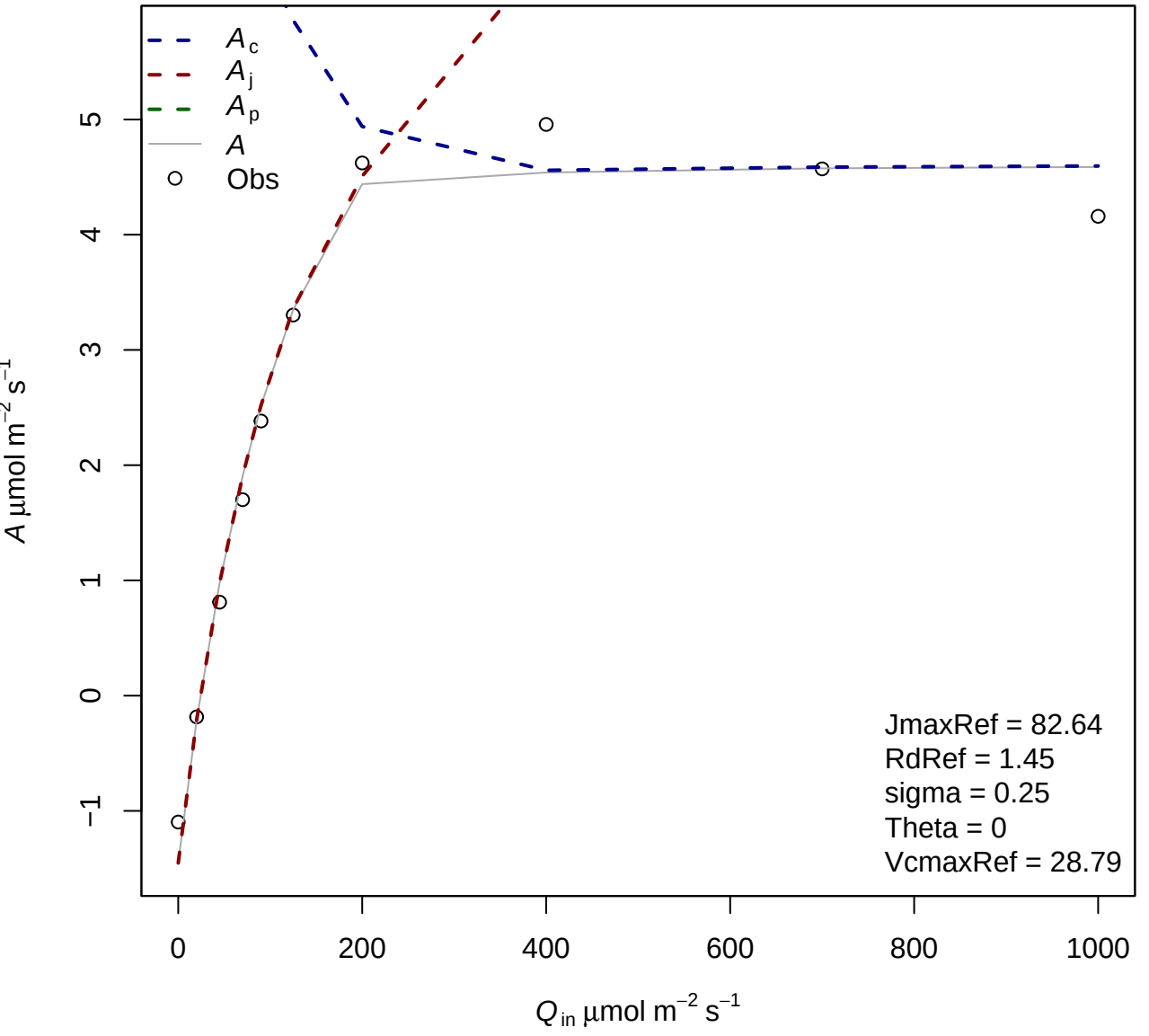

A_c
A_j
A_p
A
Obs
JmaxRef = 152.64
RdRef = 2.3
sigma = 0.25
Theta = 0
VcmaxRef = 55.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

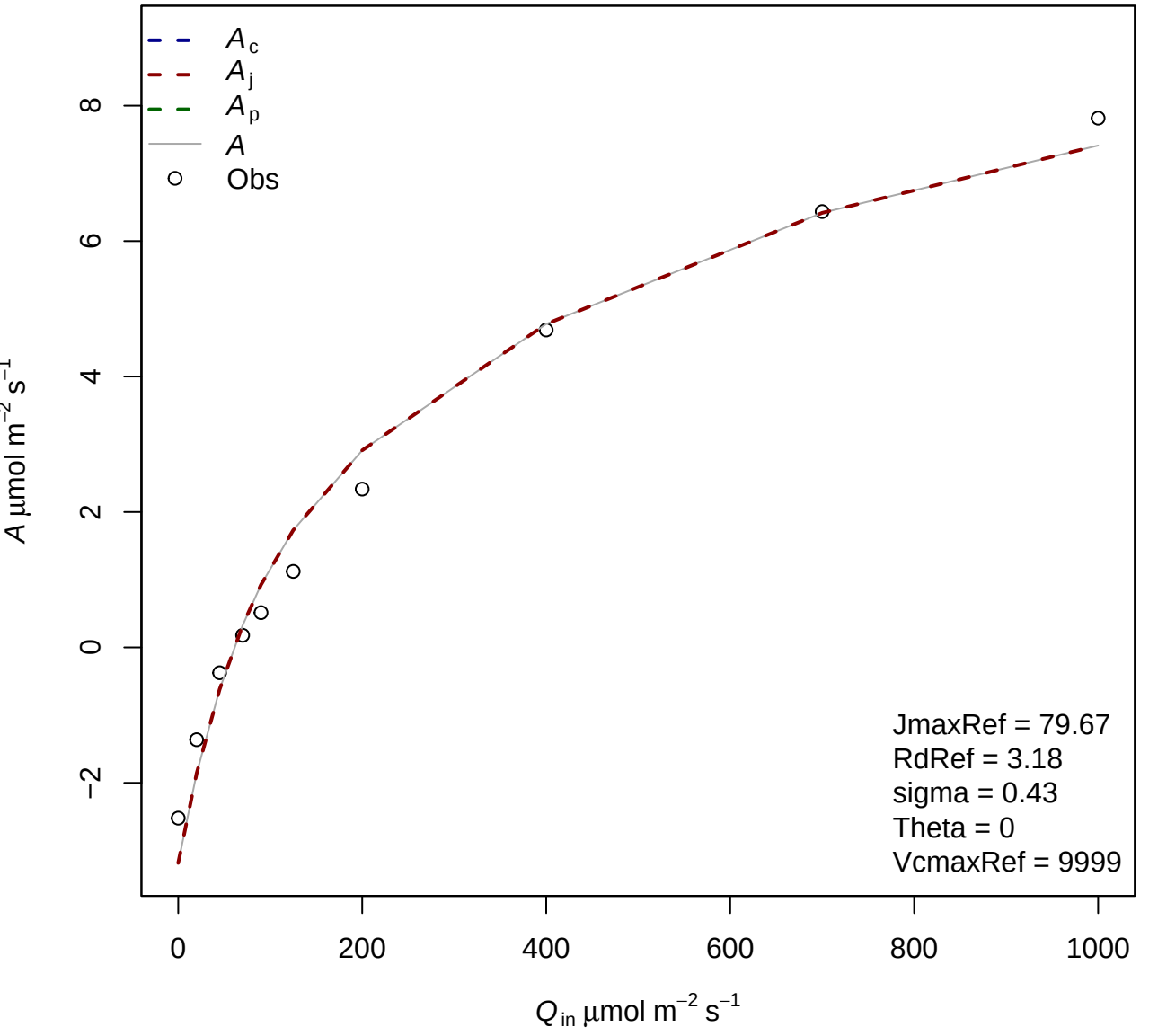

60
A_c
A_j
A_p
A
Obs
JmaxRef = 60.75
RdRef = 1.2
sigma = 0.14
Theta = 0
VcmaxRef = 20.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

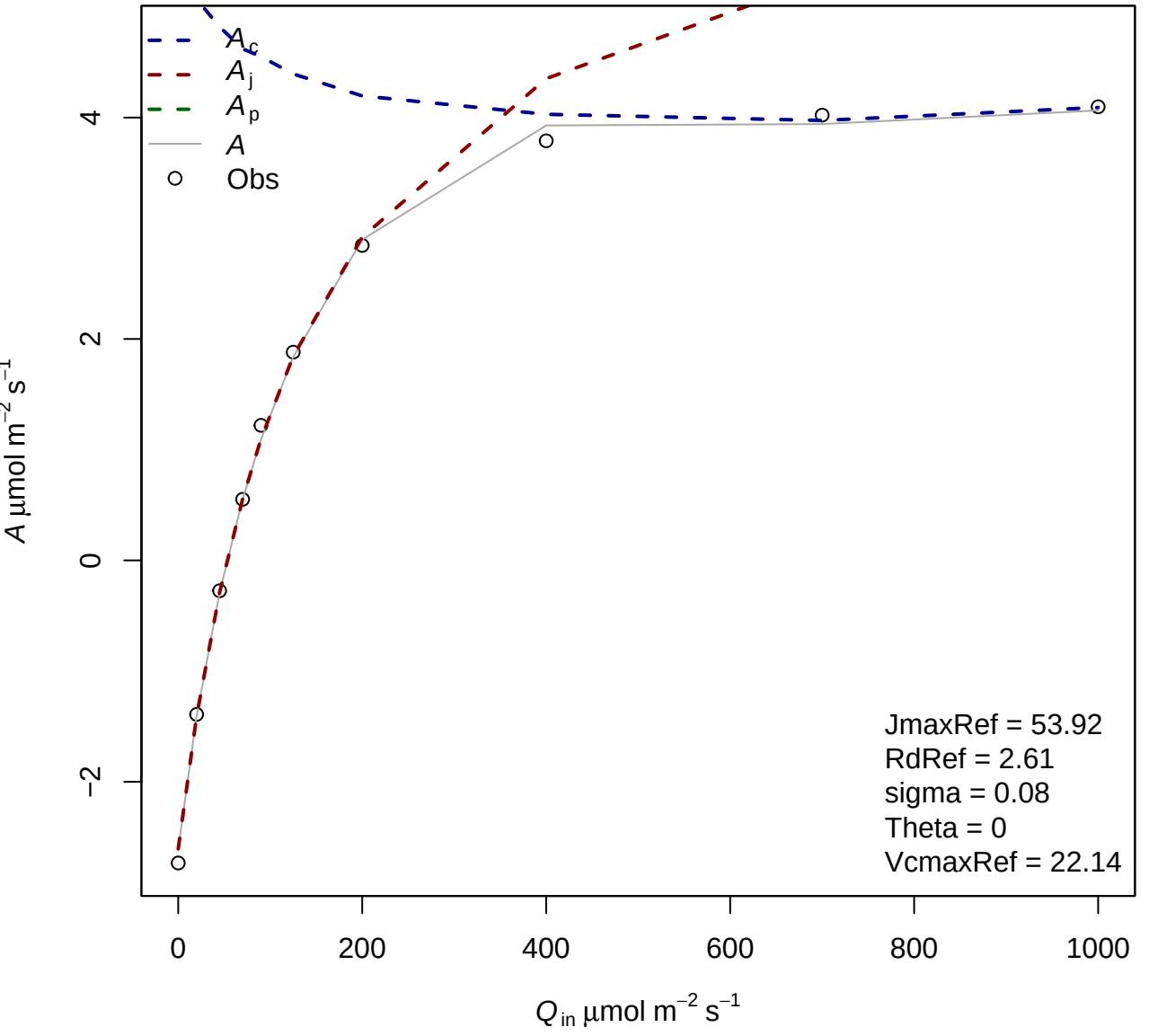

A_c
A_j
A_p
A
Obs
JmaxRef = 29.52
RdRef = 1.45
sigma = 0.06
Theta = 0
VcmaxRef = 17.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000

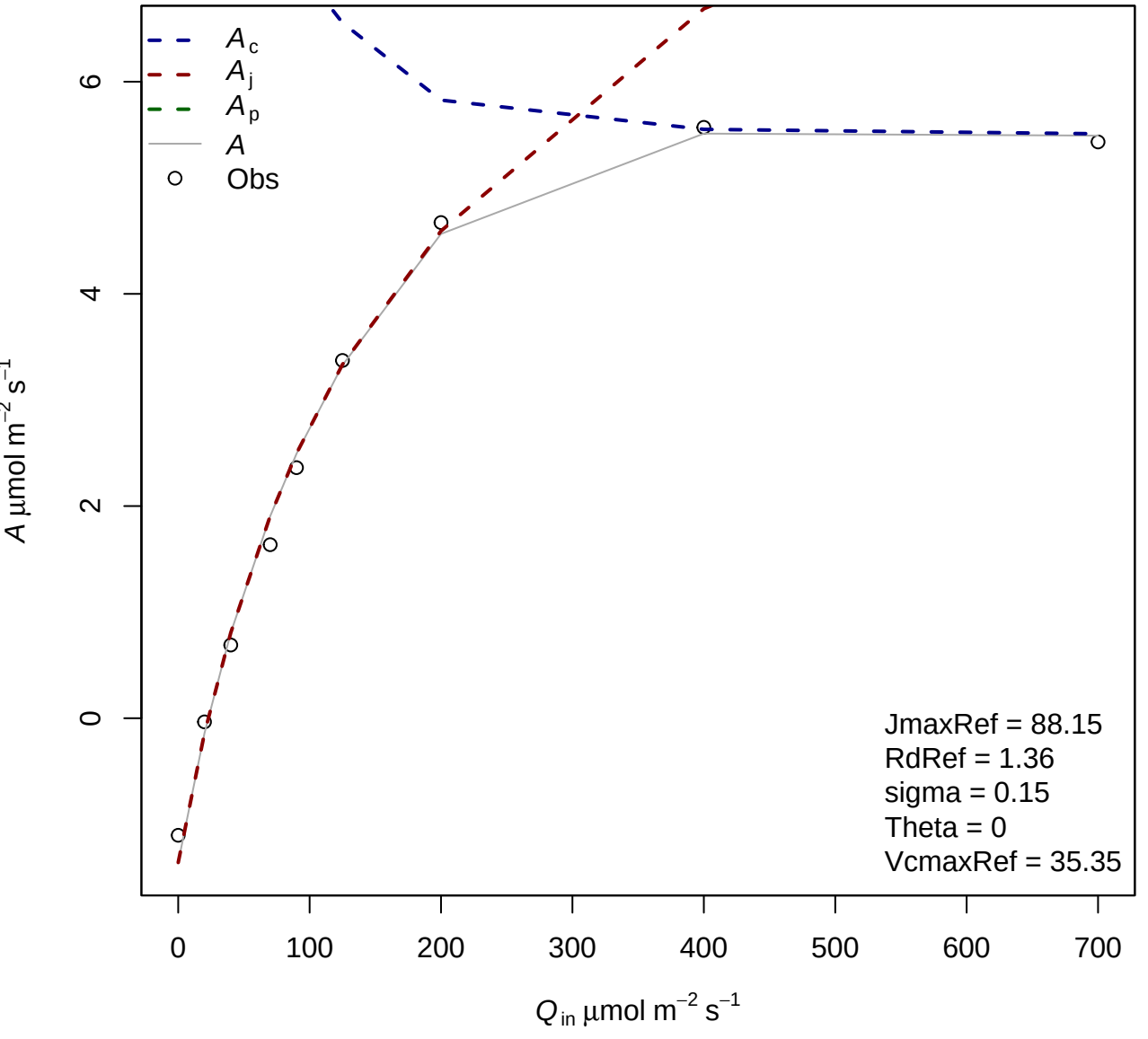

A_c
A_j
A_p
A
Obs
JmaxRef = 79.81
RdRef = 2.37
sigma = 0.11
Theta = 0
VcmaxRef = 28.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

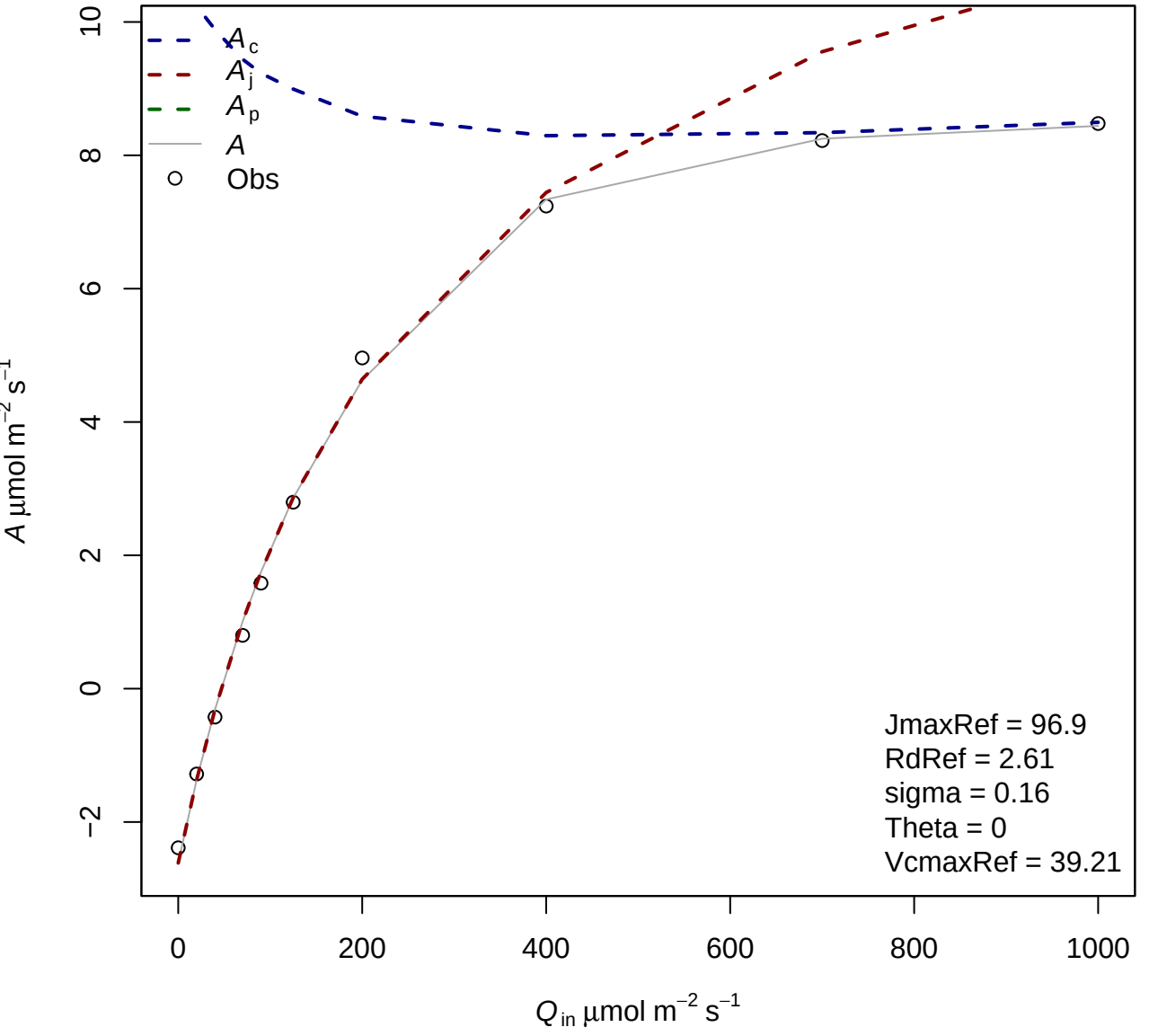

A_c
A_j
A_p
A
Obs
JmaxRef = 84.55
RdRef = 1.72
sigma = 0.21
Theta = 0
VcmaxRef = 33.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

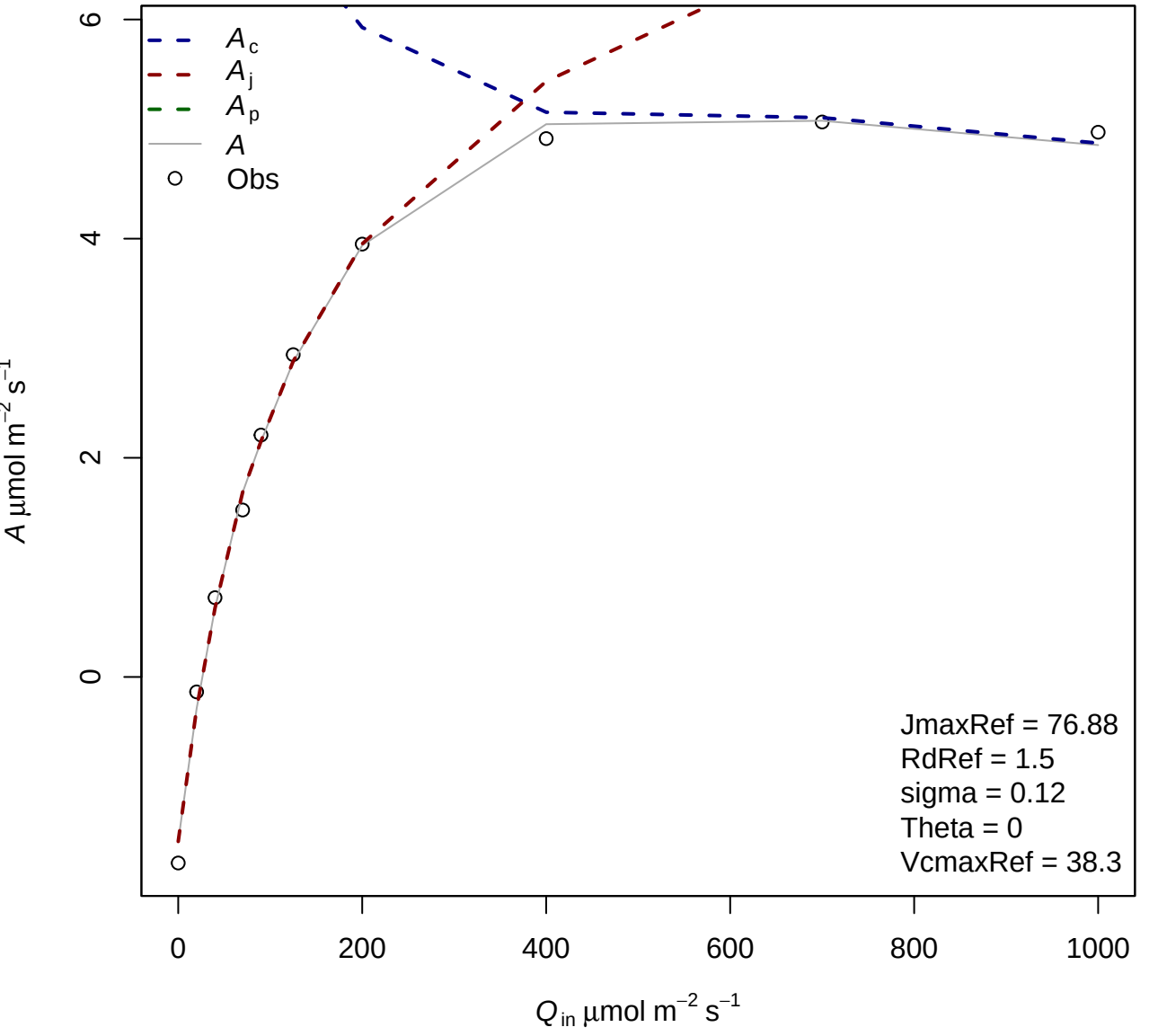

A_c
A_j
A_p
A
Obs
JmaxRef = 82.64
RdRef = 1.45
sigma = 0.25
Theta = 0
VcmaxRef = 28.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000

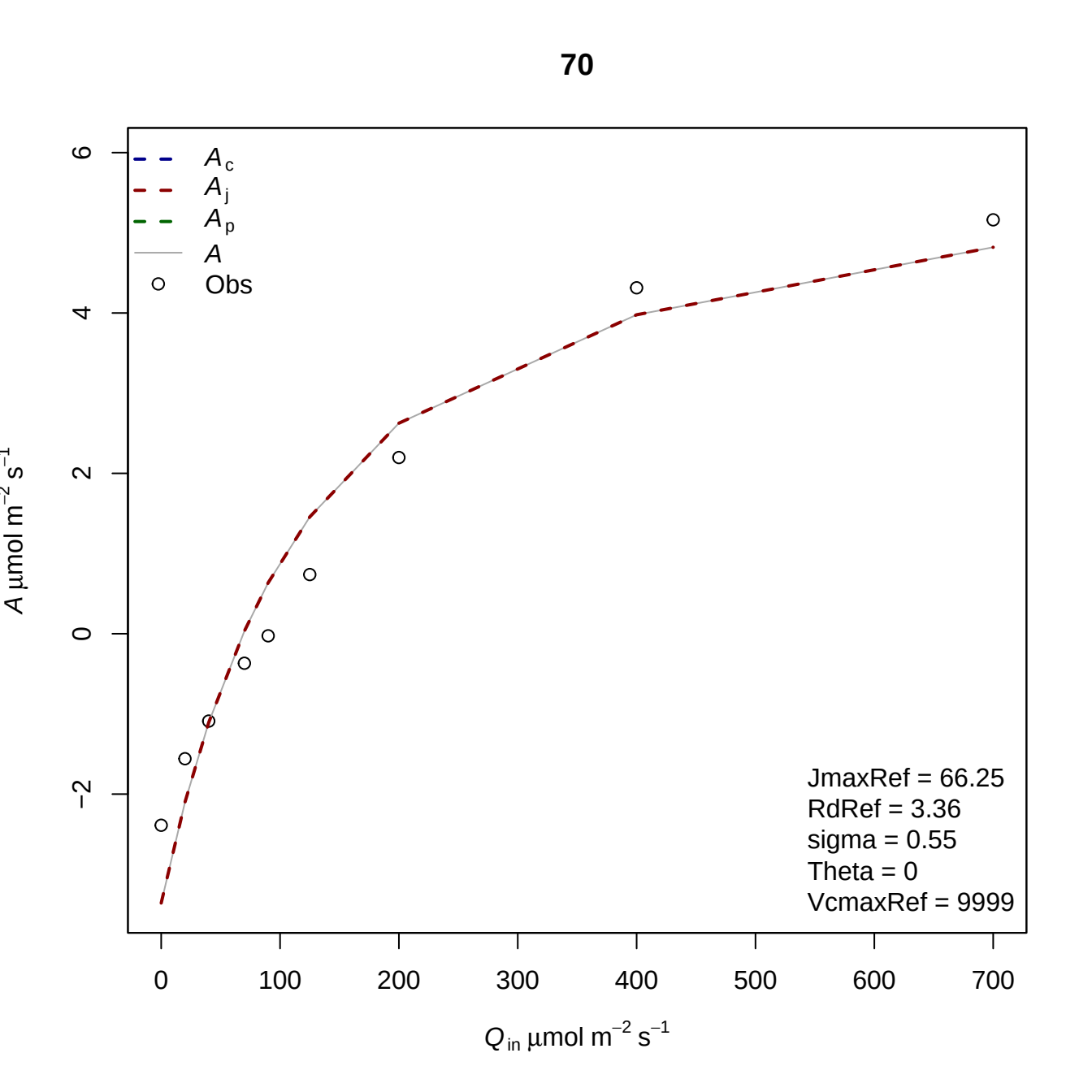

A_c
A_j
A_p
A
Obs
JmaxRef = 79.67
RdRef = 3.18
sigma = 0.43
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

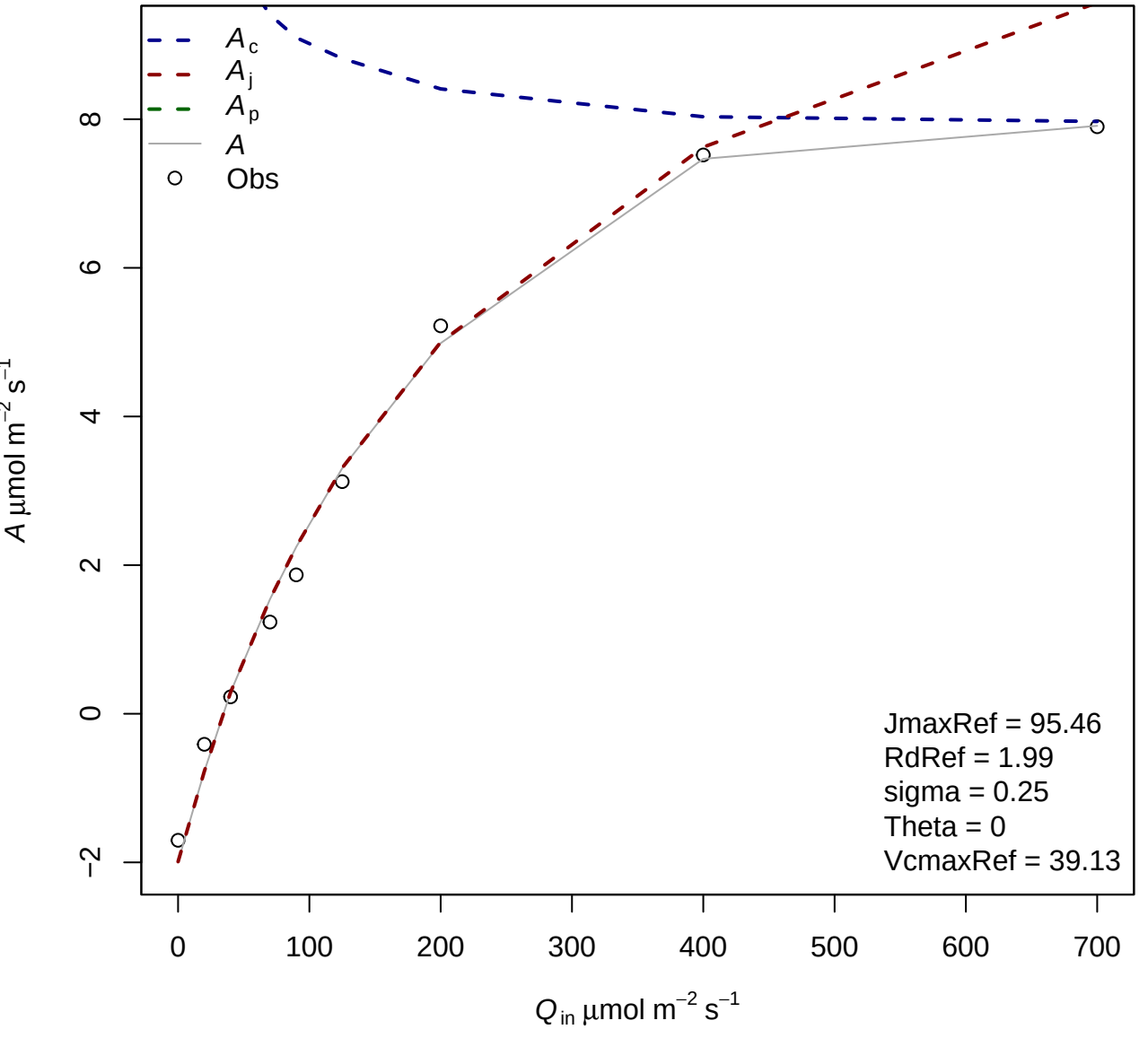

A_c
A_j
A_p
A
Obs
JmaxRef = 53.92
RdRef = 2.61
sigma = 0.08
Theta = 0
VcmaxRef = 22.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

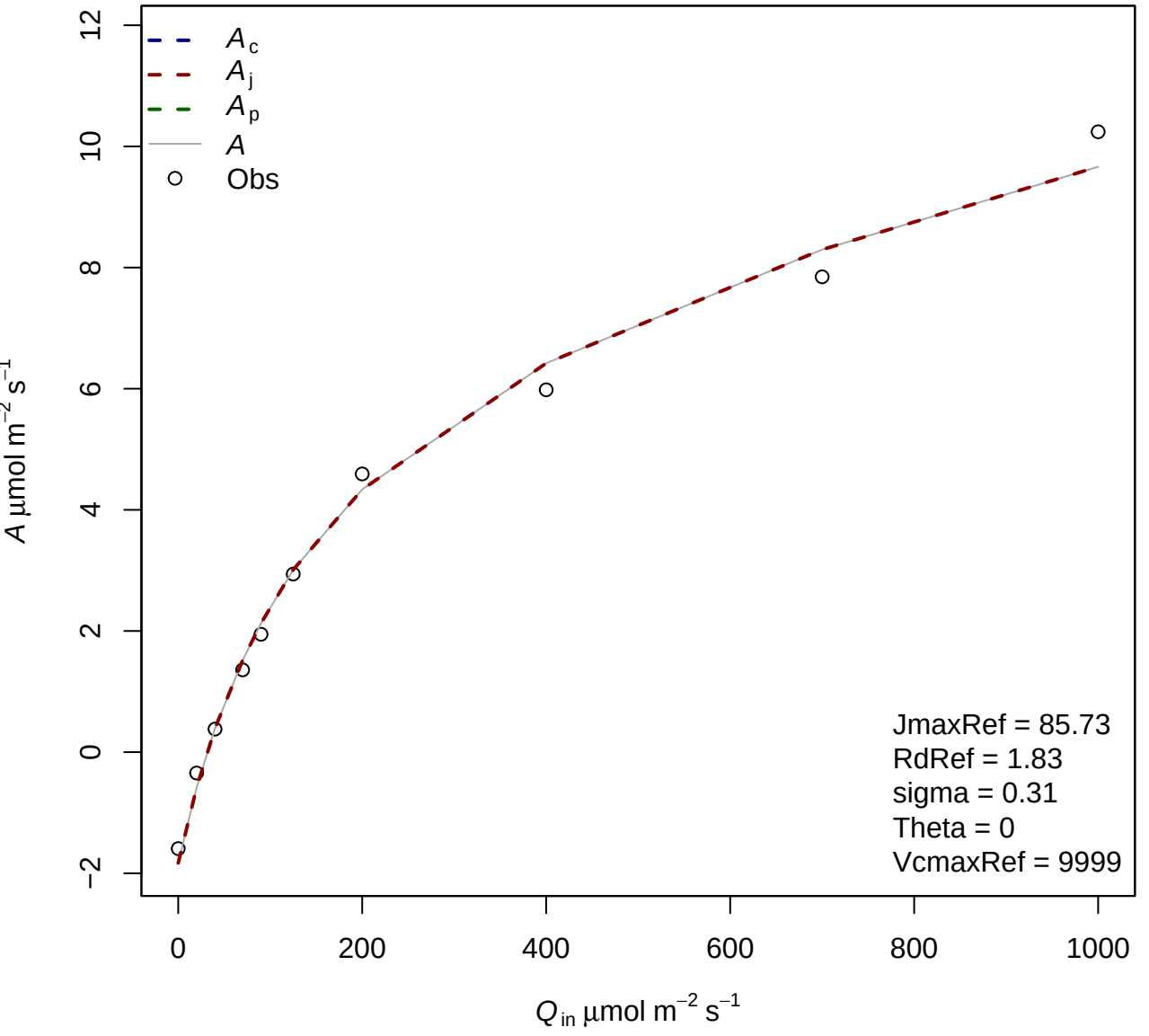

A_c
A_j
A_p
A
Obs
JmaxRef = 88.15
RdRef = 1.36
sigma = 0.15
Theta = 0
VcmaxRef = 35.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
100
200
300
400
500
600
700

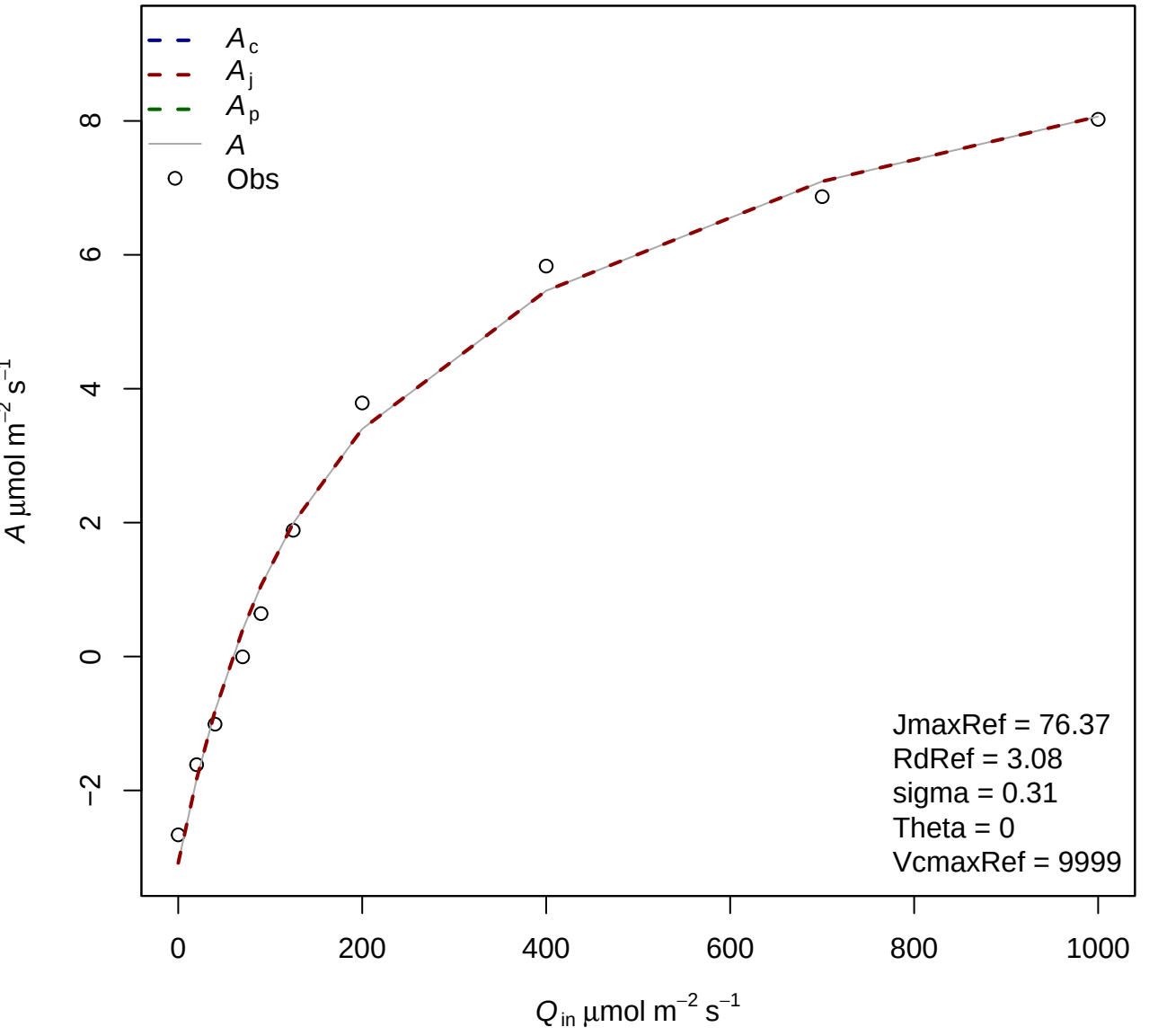

A_c
A_j
A_p
A
Obs
JmaxRef = 96.9
RdRef = 2.61
sigma = 0.16
Theta = 0
VcmaxRef = 39.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
200
400
600
800
1000

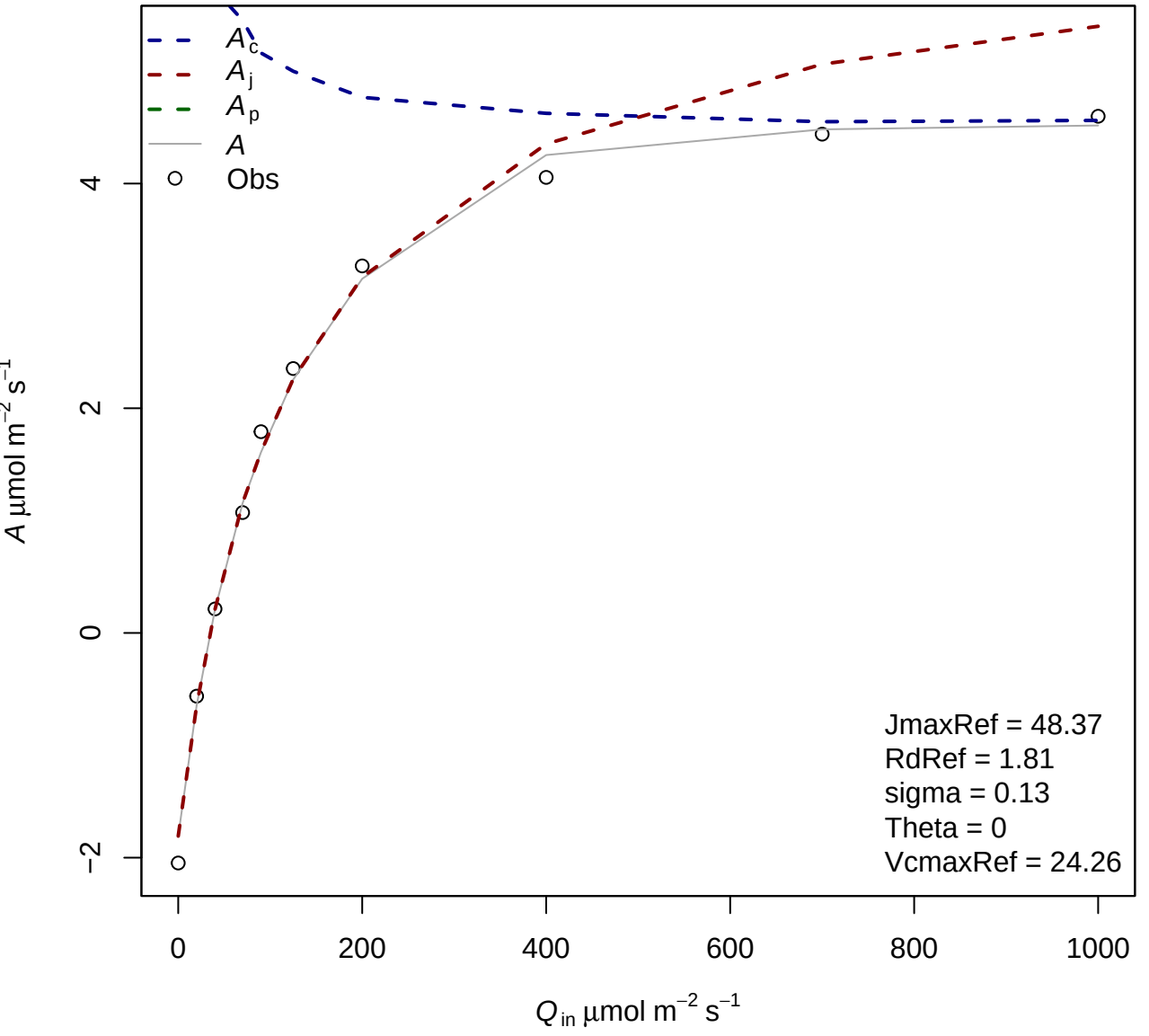

A_c
A_j
A_p
A
Obs
JmaxRef = 76.88
RdRef = 1.5
sigma = 0.12
Theta = 0
VcmaxRef = 38.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

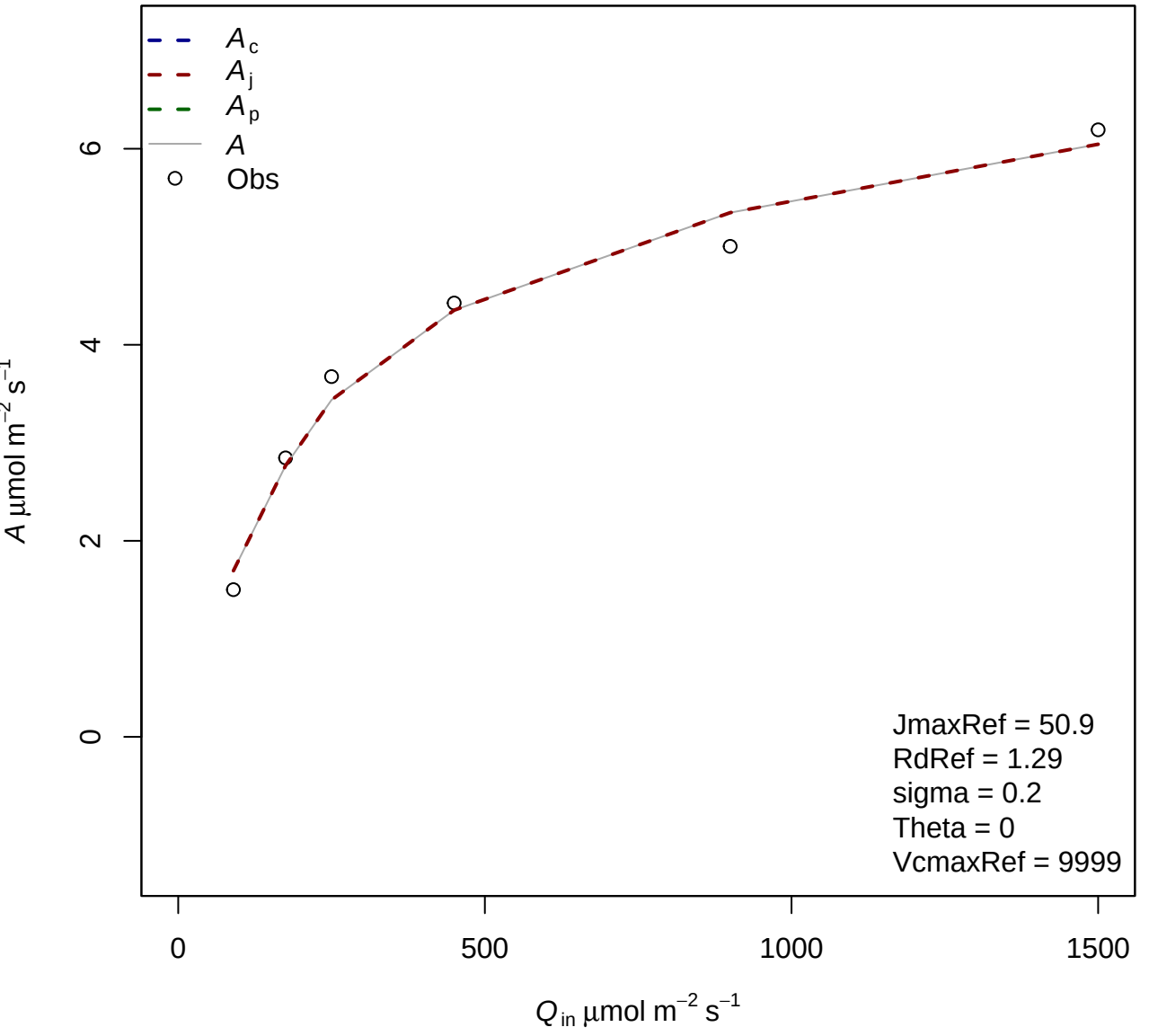

70
A_c
A_j
A_p
A
Obs
JmaxRef = 66.25
RdRef = 3.36
sigma = 0.55
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
100
200
300
400
500
600
700

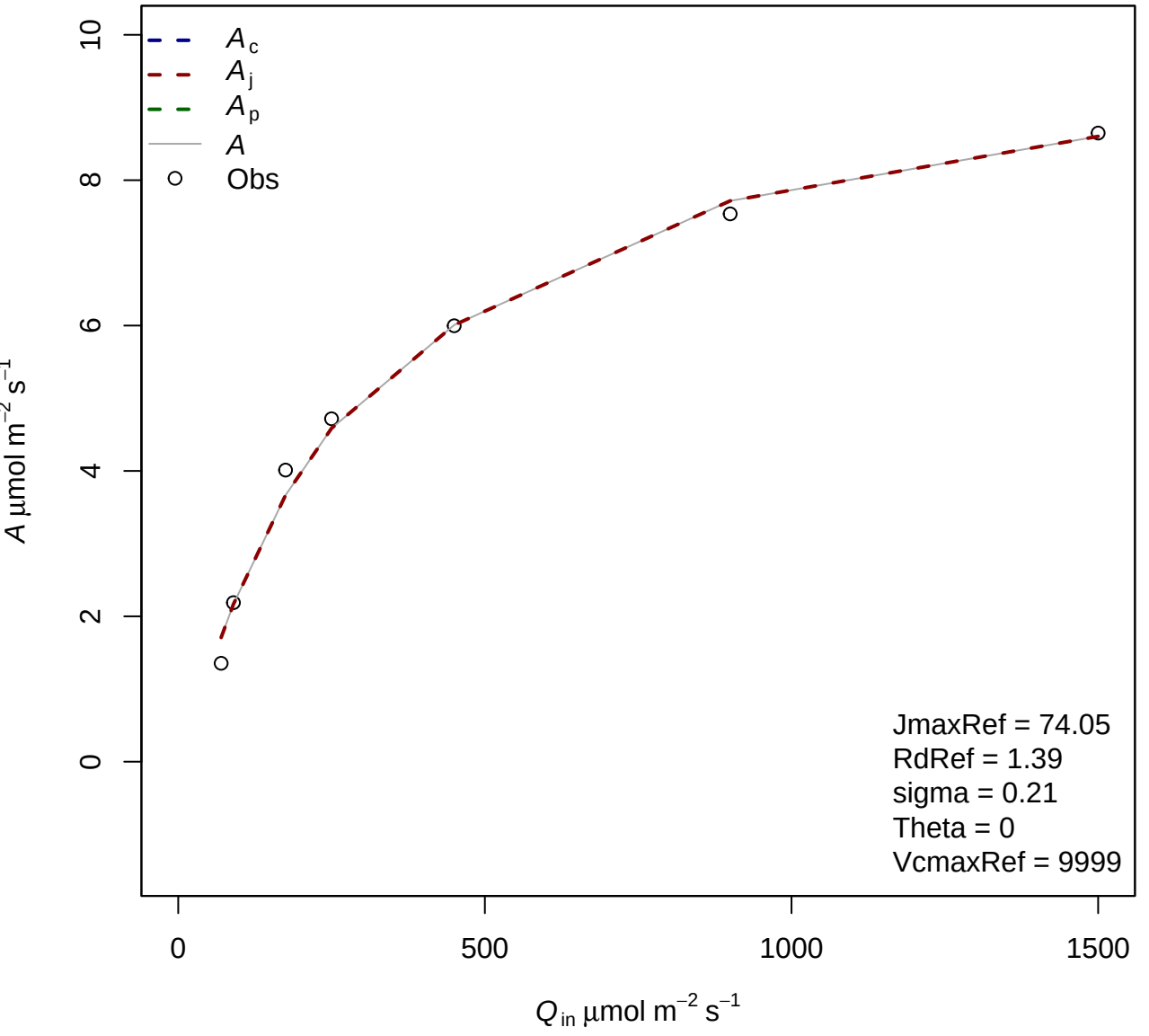

A_c
A_j
A_p
A
Obs
JmaxRef = 95.46
RdRef = 1.99
sigma = 0.25
Theta = 0
VcmaxRef = 39.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
100
200
300
400
500
600
700

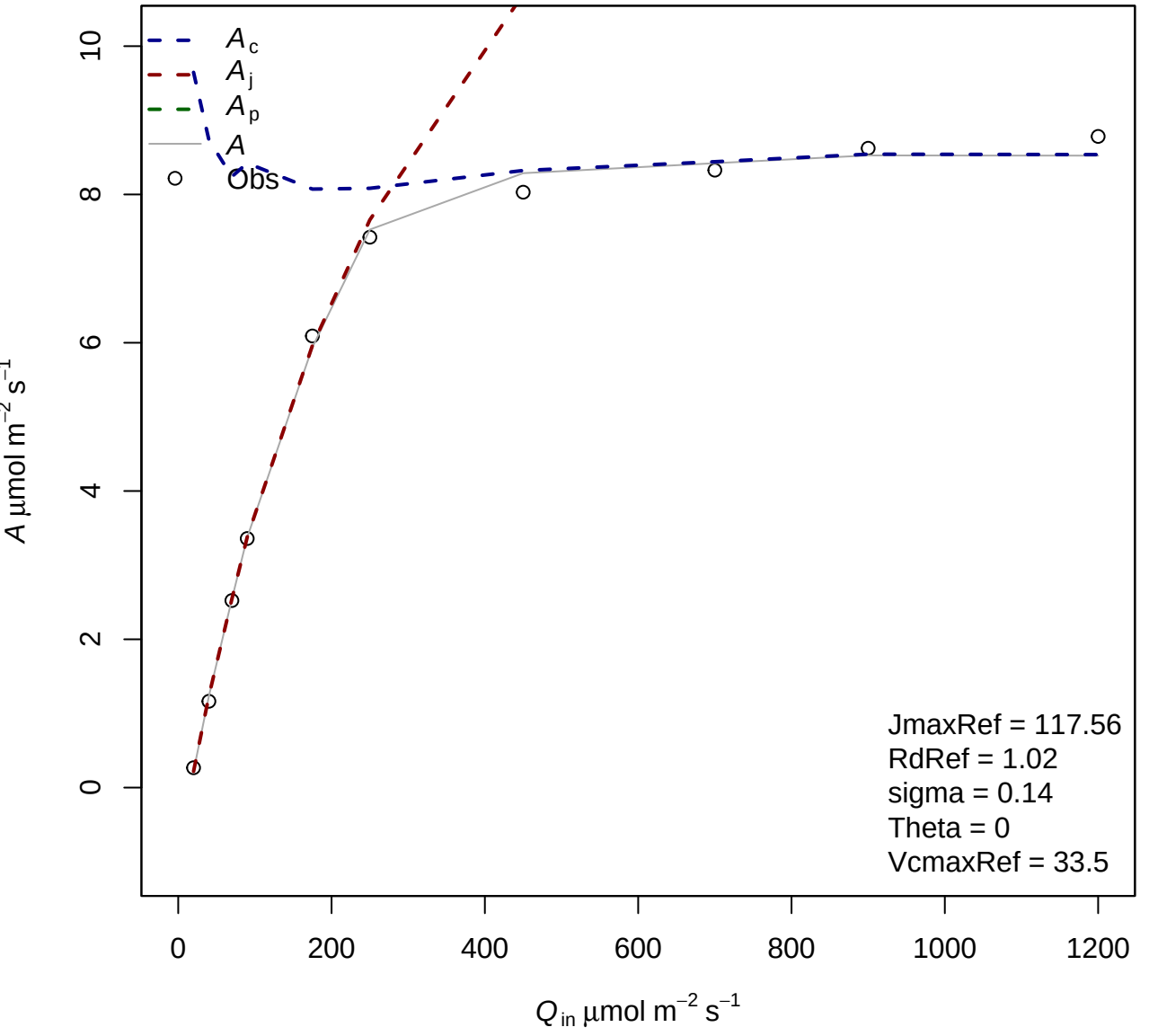

A_c
A_j
A_p
A
Obs
JmaxRef = 85.73
RdRef = 1.83
sigma = 0.31
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
200
400
600
800
1000

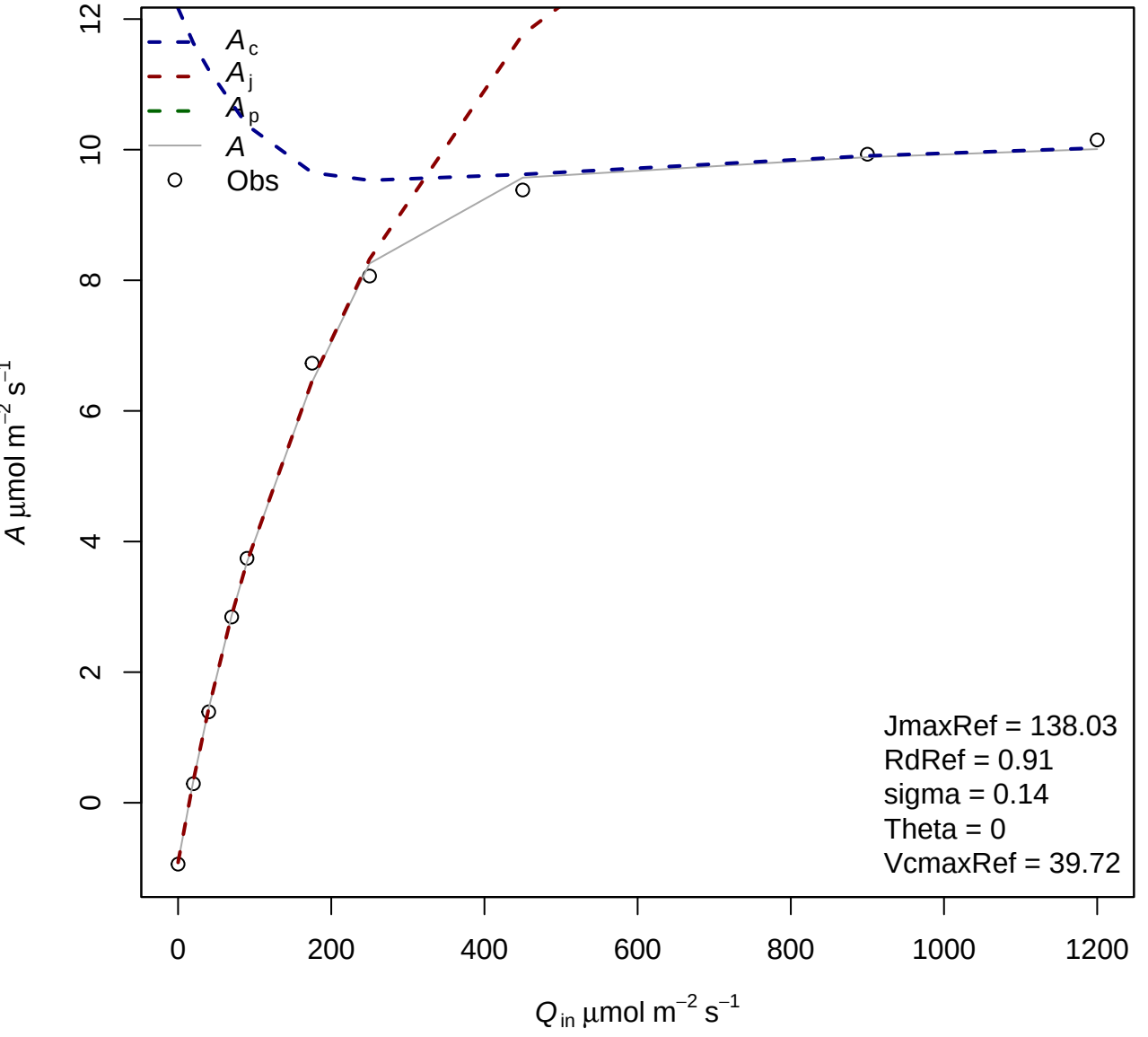

A_c
A_j
A_p
A
Obs
JmaxRef = 76.37
RdRef = 3.08
sigma = 0.31
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

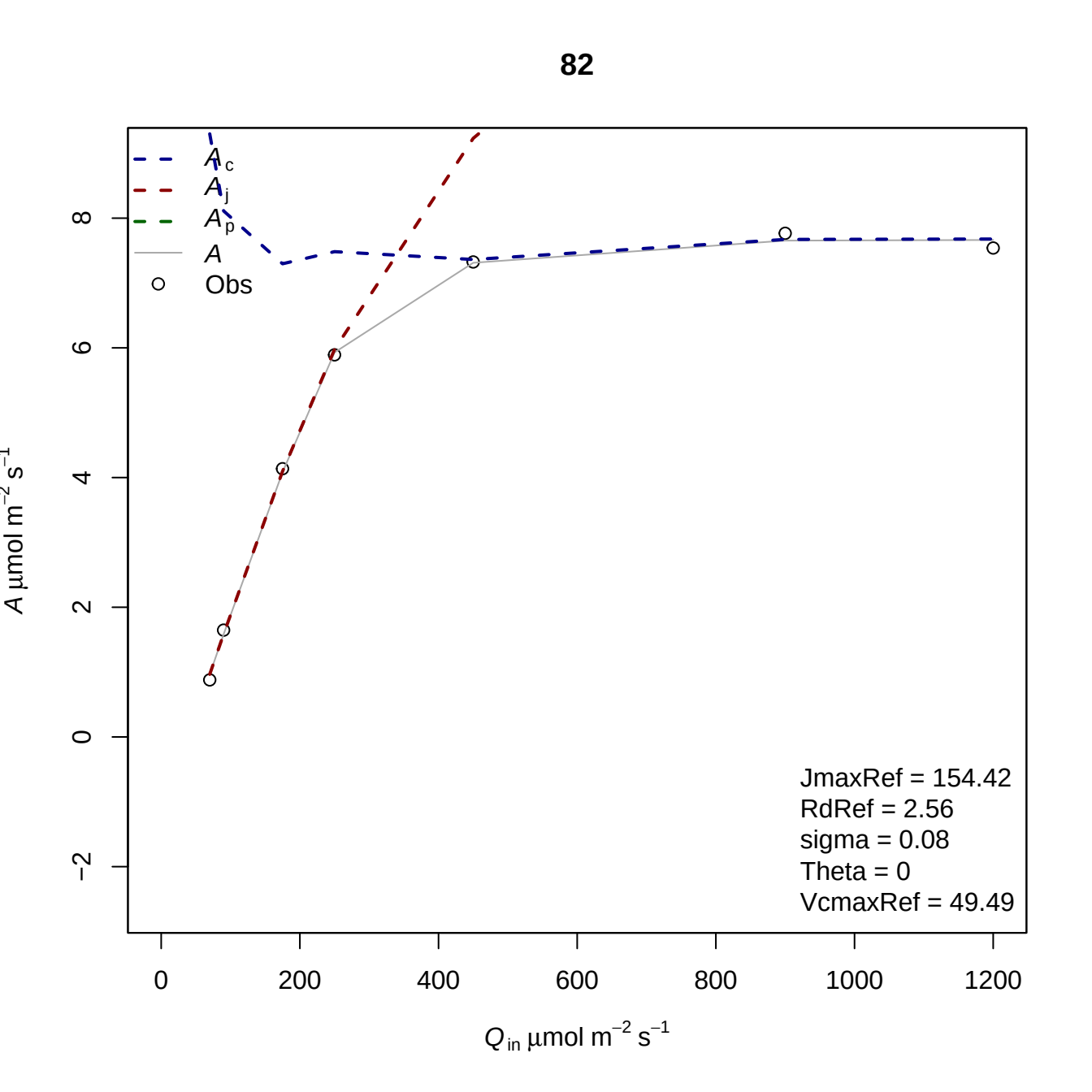

A_c
A_j
A_p
A
Obs
JmaxRef = 48.37
RdRef = 1.81
sigma = 0.13
Theta = 0
VcmaxRef = 24.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

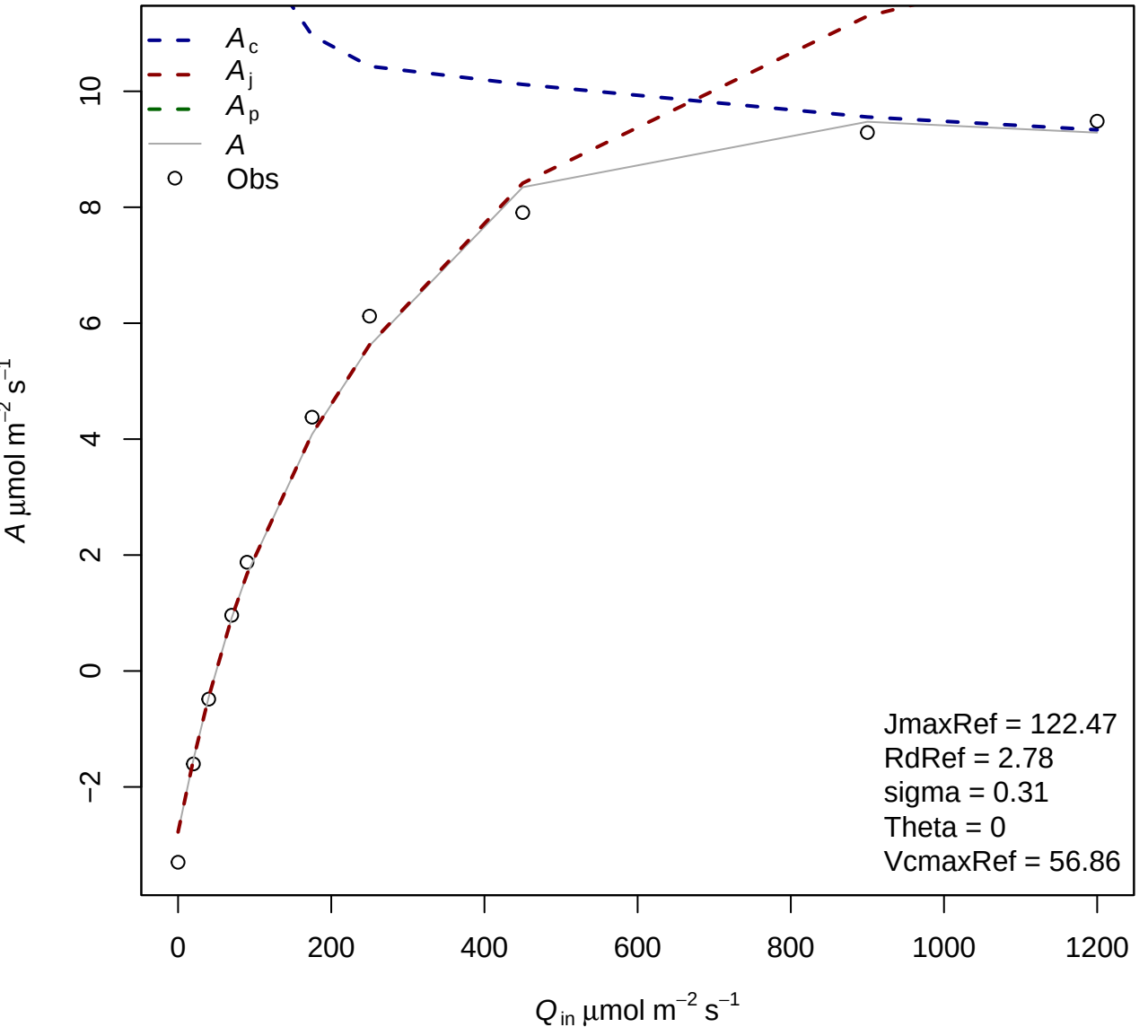

A_c
A_j
A_p
A
Obs
JmaxRef = 50.9
RdRef = 1.29
sigma = 0.2
Theta = 0
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

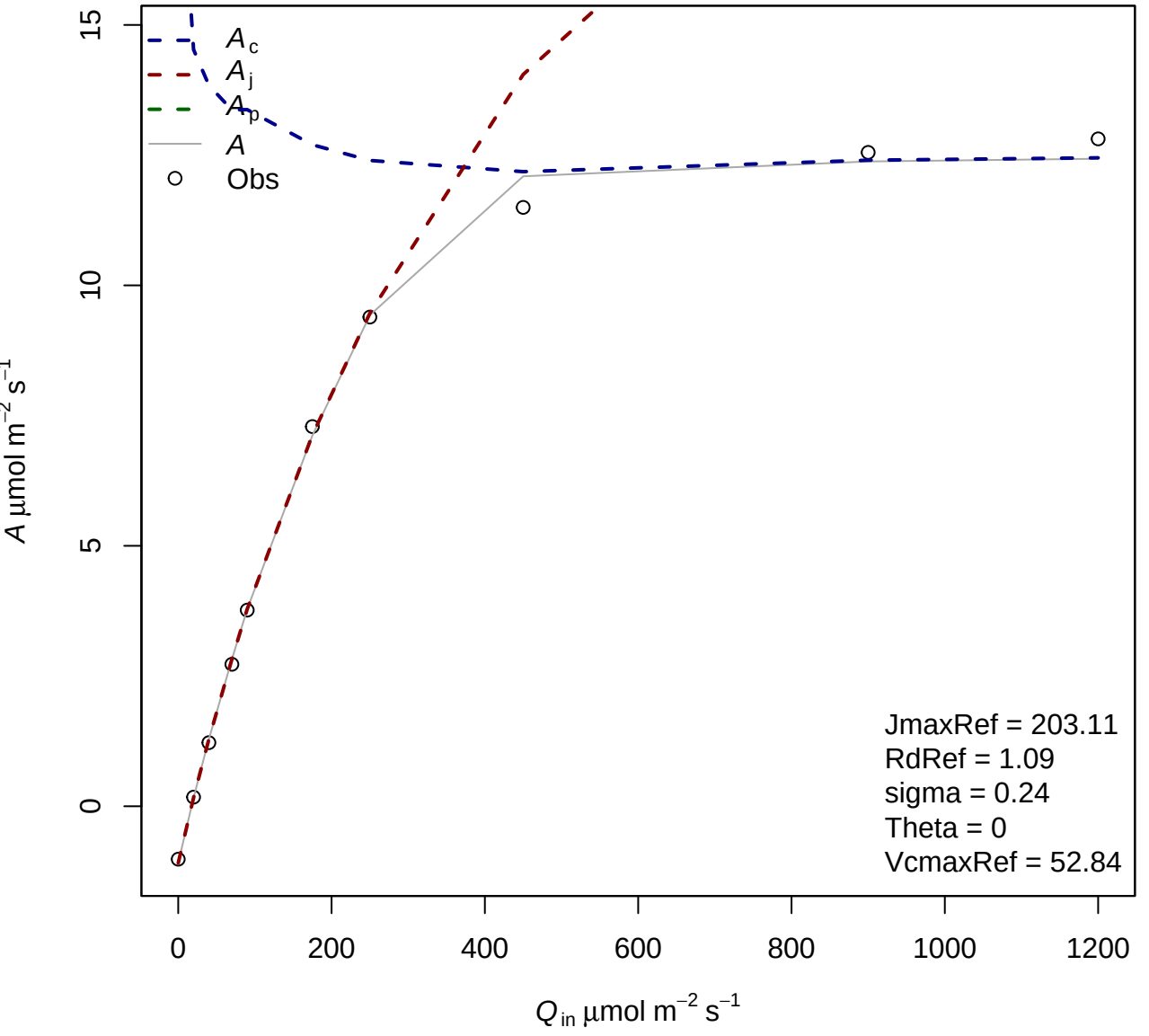

A_c
A_j
A_p
A
Obs
JmaxRef = 74.05
RdRef = 1.39
sigma = 0.21
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

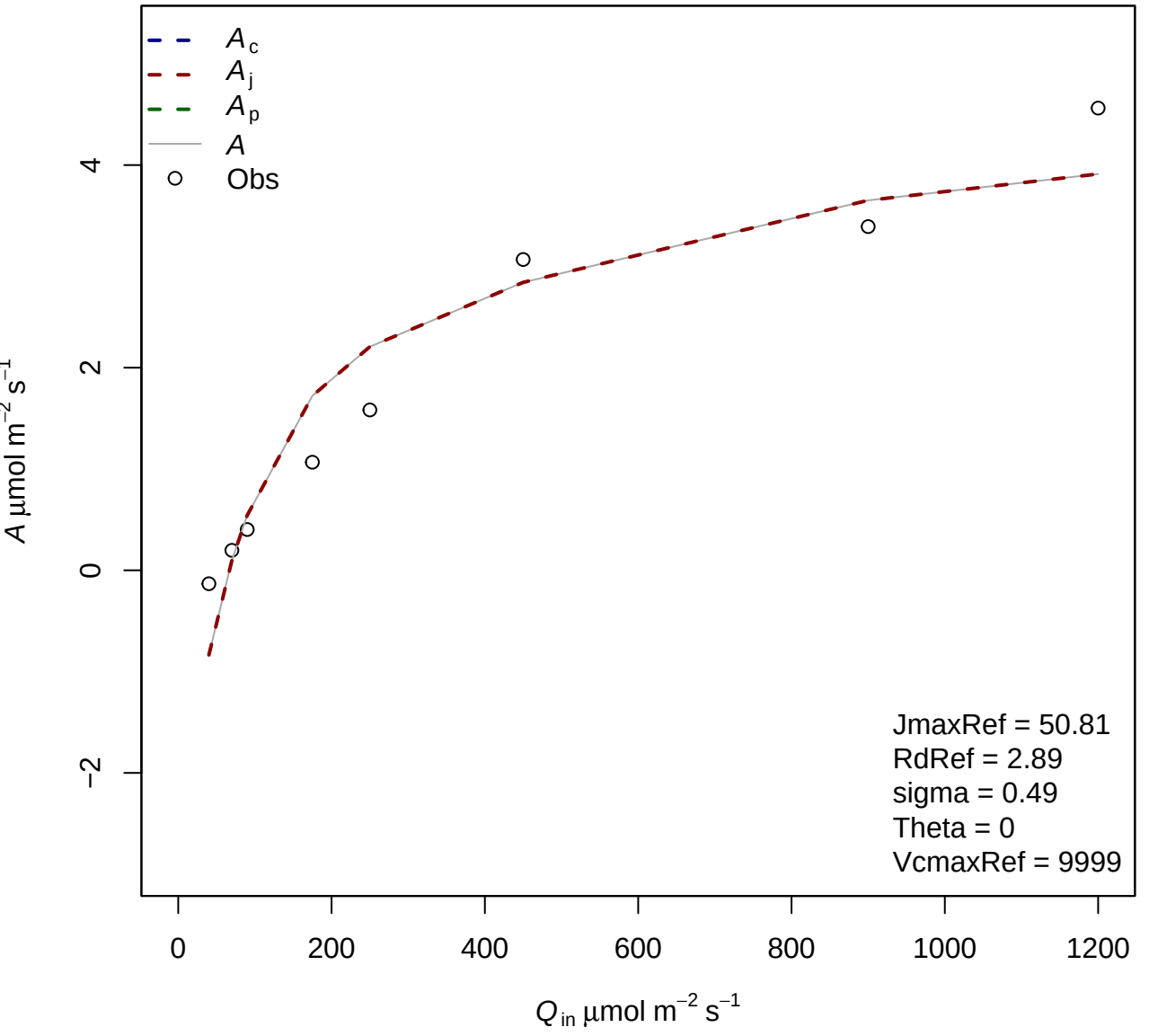

A_c
A_j
A_p
A
Obs
JmaxRef = 117.56
RdRef = 1.02
sigma = 0.14
Theta = 0
VcmaxRef = 33.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

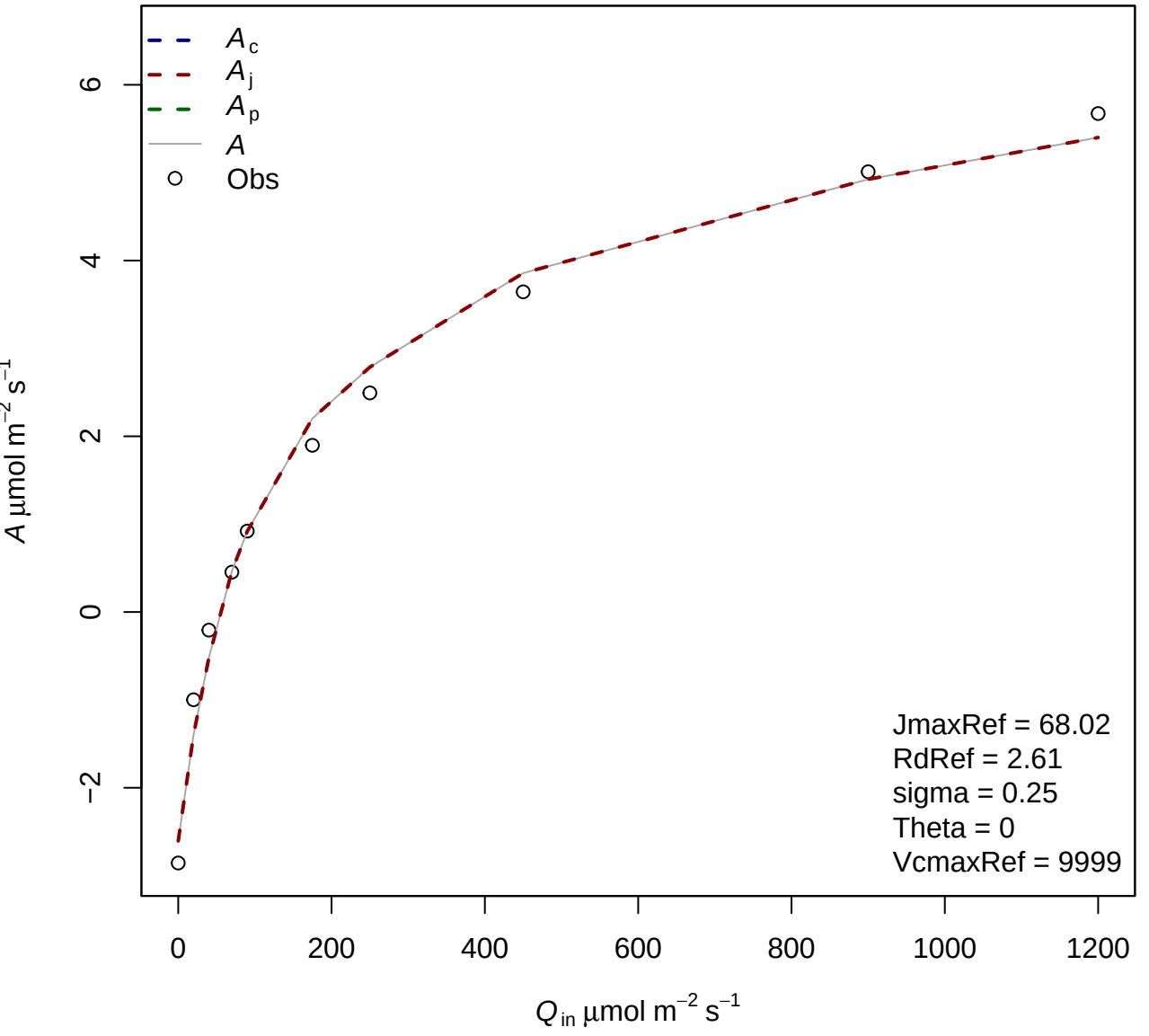

A_c
A_j
A_p
A
Obs
JmaxRef = 138.03
RdRef = 0.91
sigma = 0.14
Theta = 0
VcmaxRef = 39.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

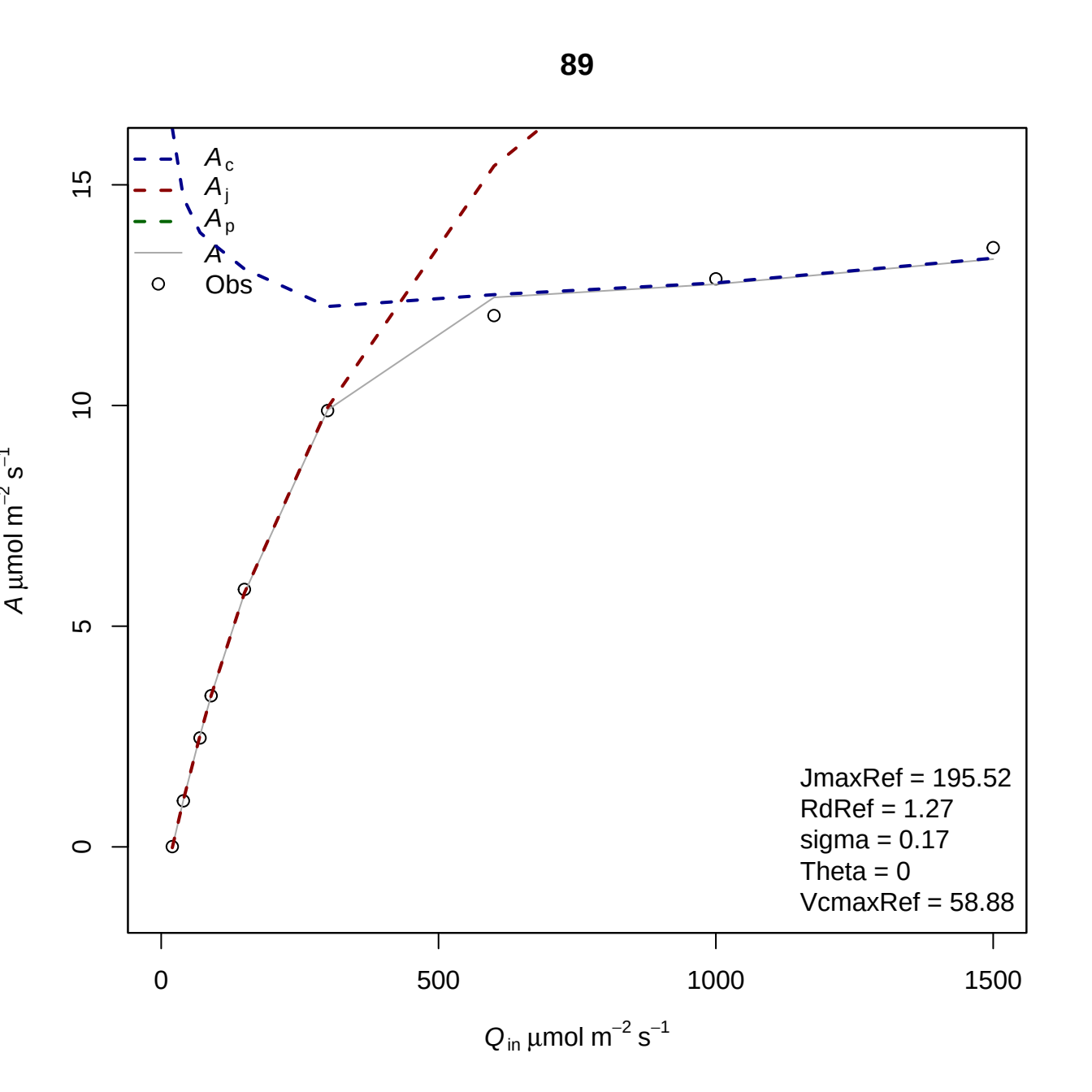

82
A_c
A_j
A_p
A
Obs
JmaxRef = 154.42
RdRef = 2.56
sigma = 0.08
Theta = 0
VcmaxRef = 49.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

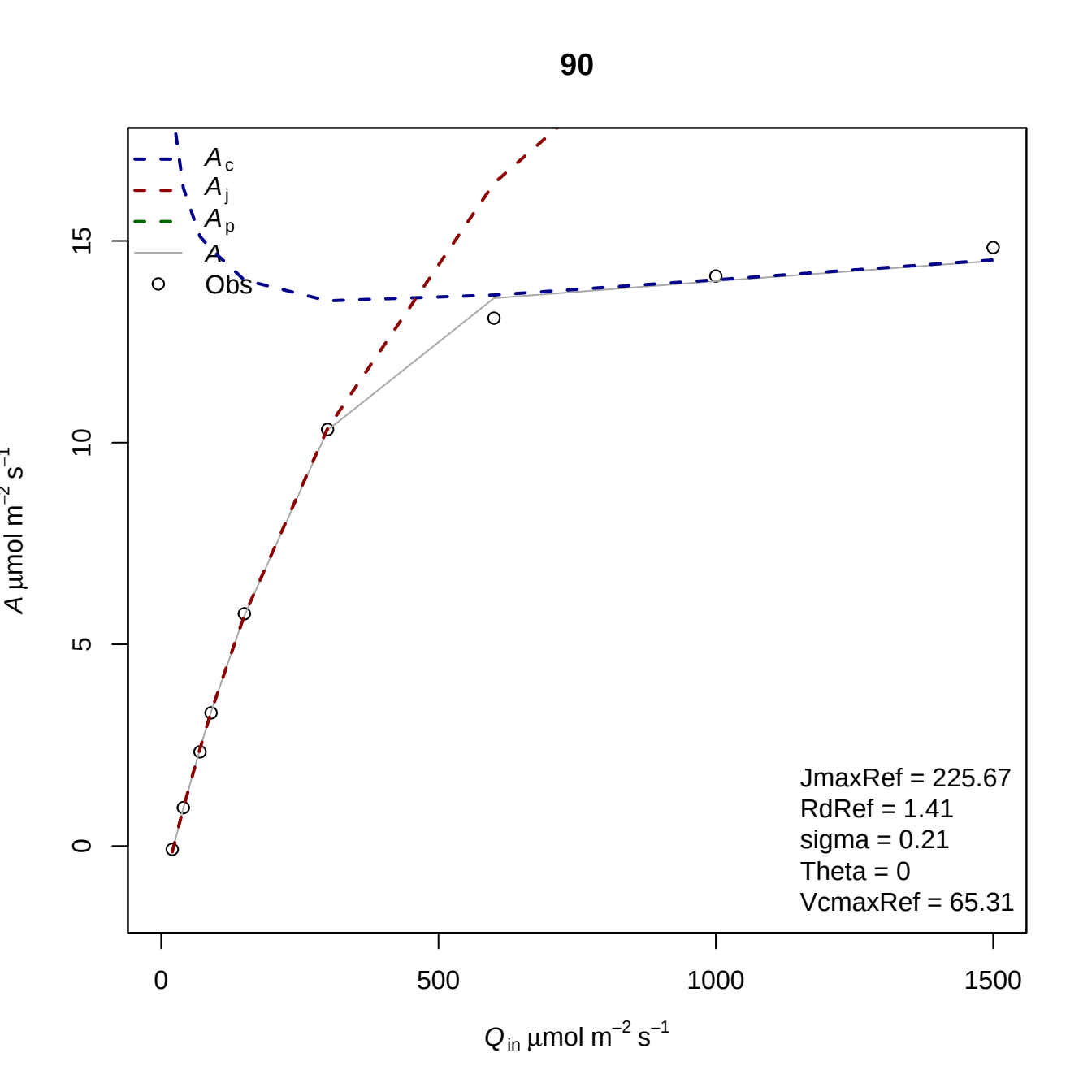

A_c
A_j
A_p
A
Obs
JmaxRef = 122.47
RdRef = 2.78
sigma = 0.31
Theta = 0
VcmaxRef = 56.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

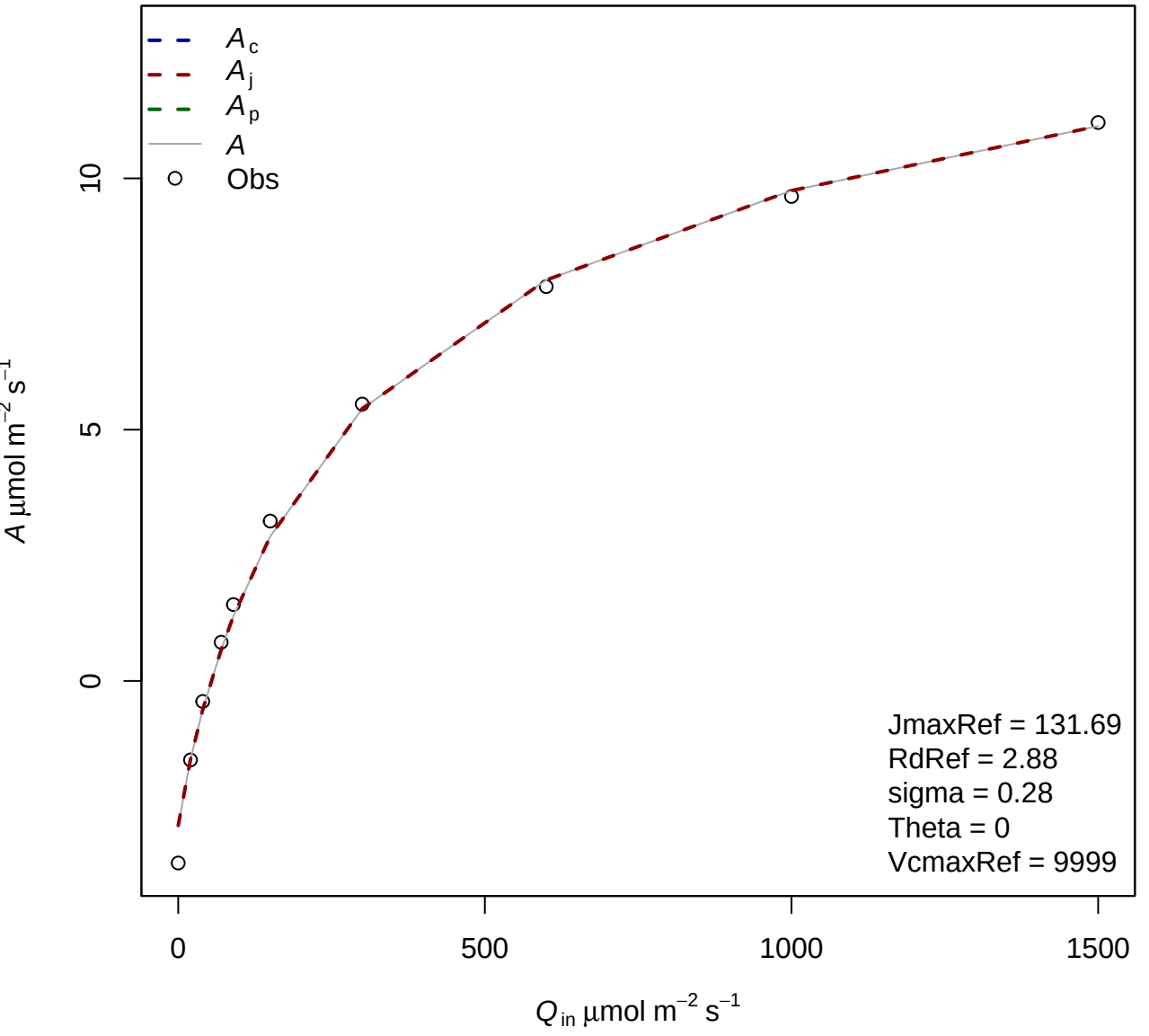

A_c
A_j
A_p
A
Obs
JmaxRef = 203.11
RdRef = 1.09
sigma = 0.24
Theta = 0
VcmaxRef = 52.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

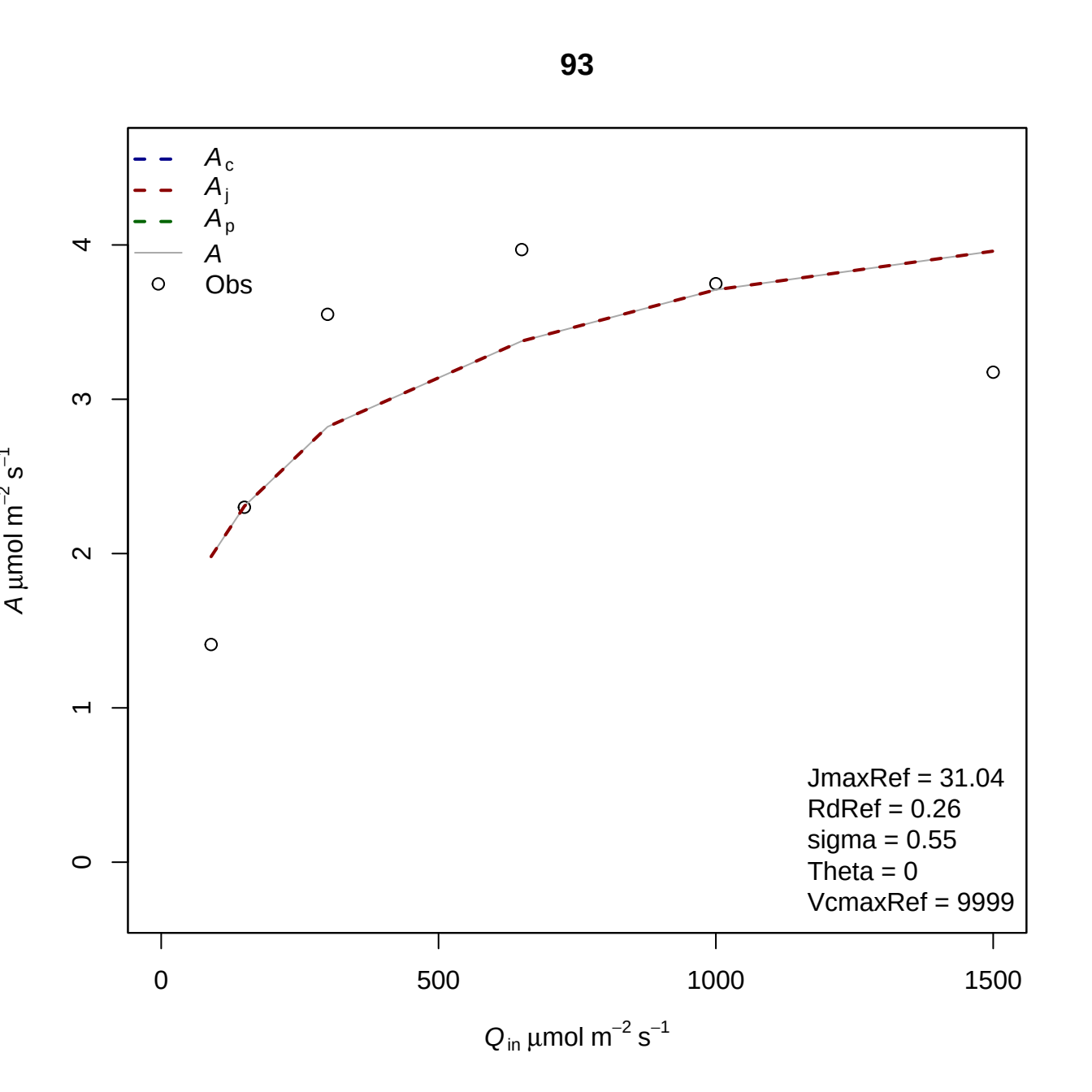

A_c
A_j
A_p
A
Obs
JmaxRef = 50.81
RdRef = 2.89
sigma = 0.49
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200

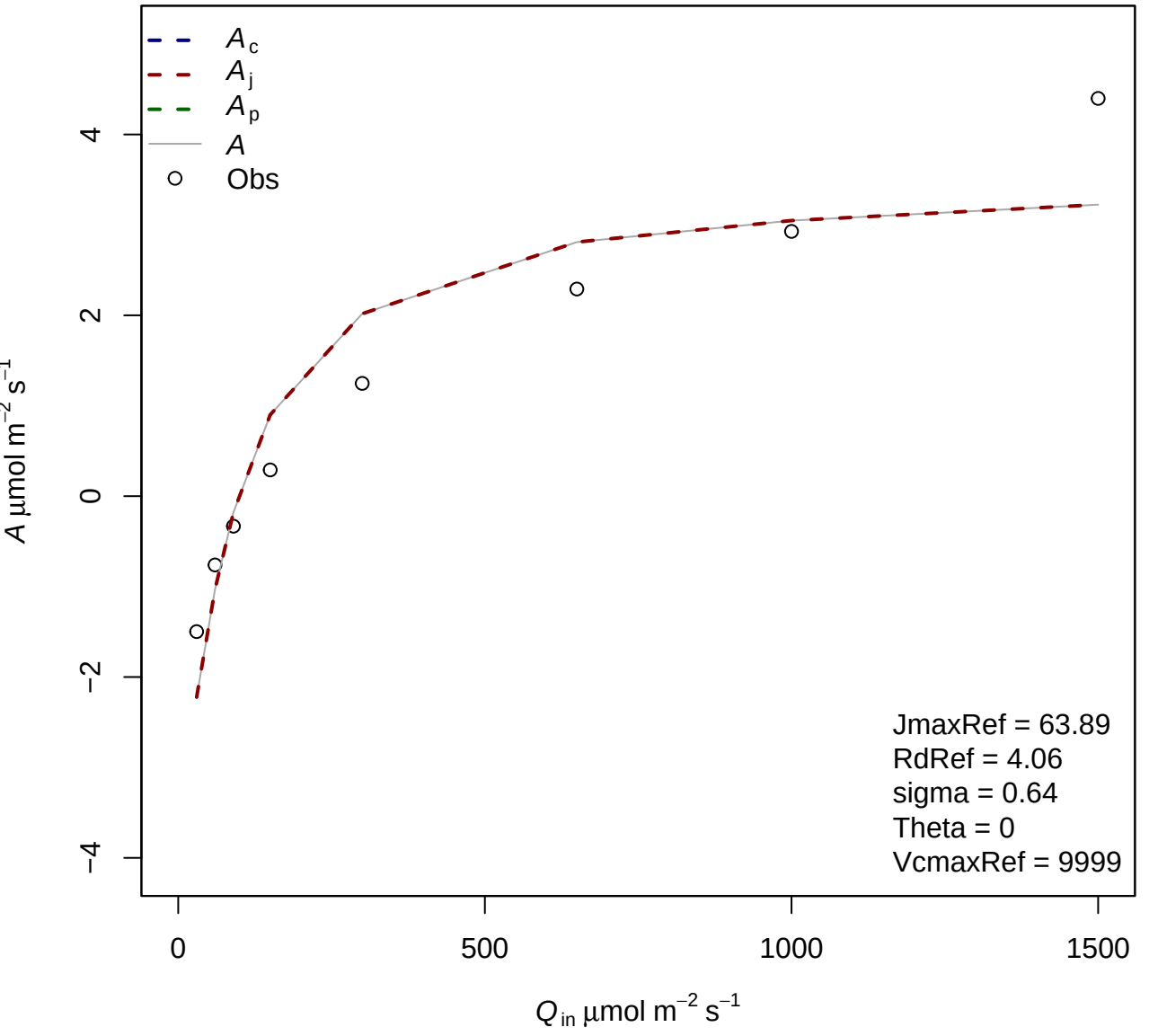

A_c
A_j
A_p
A
Obs
JmaxRef = 68.02
RdRef = 2.61
sigma = 0.25
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

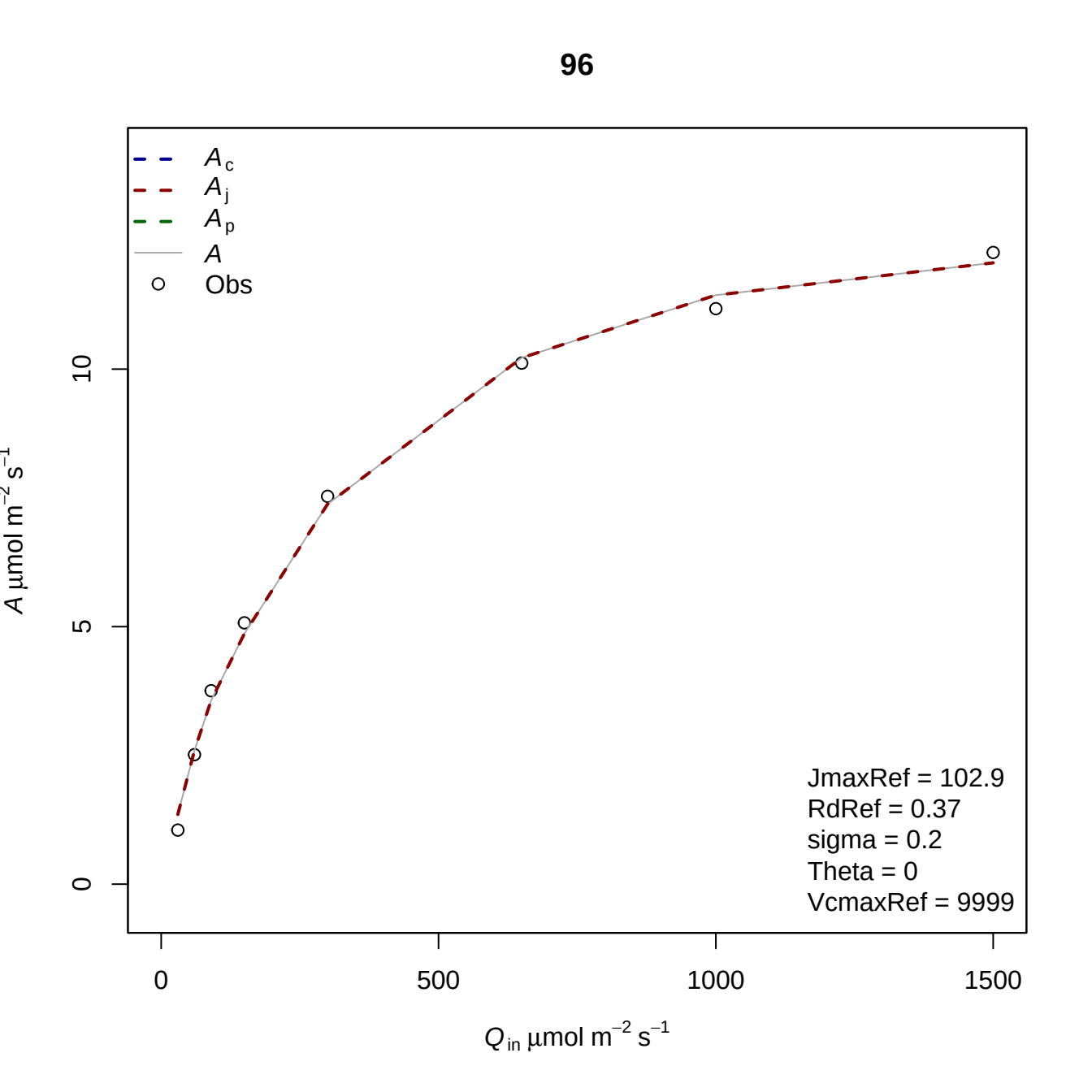

89
A_c
A_j
A_p
A
Obs
JmaxRef = 195.52
RdRef = 1.27
sigma = 0.17
Theta = 0
VcmaxRef = 58.88
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

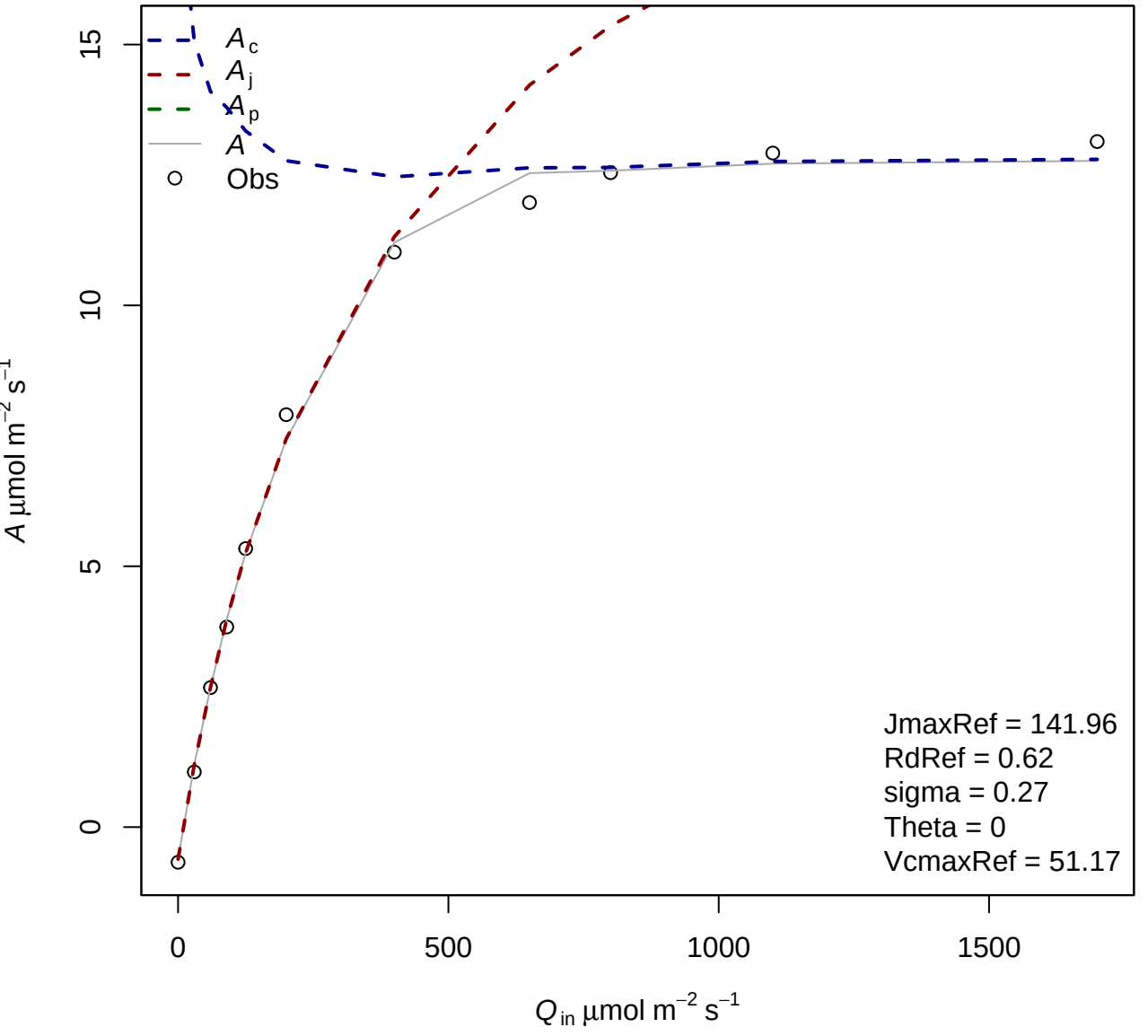

90
A_c
A_j
A_p
A
Obs
JmaxRef = 225.67
RdRef = 1.41
sigma = 0.21
Theta = 0
VcmaxRef = 65.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

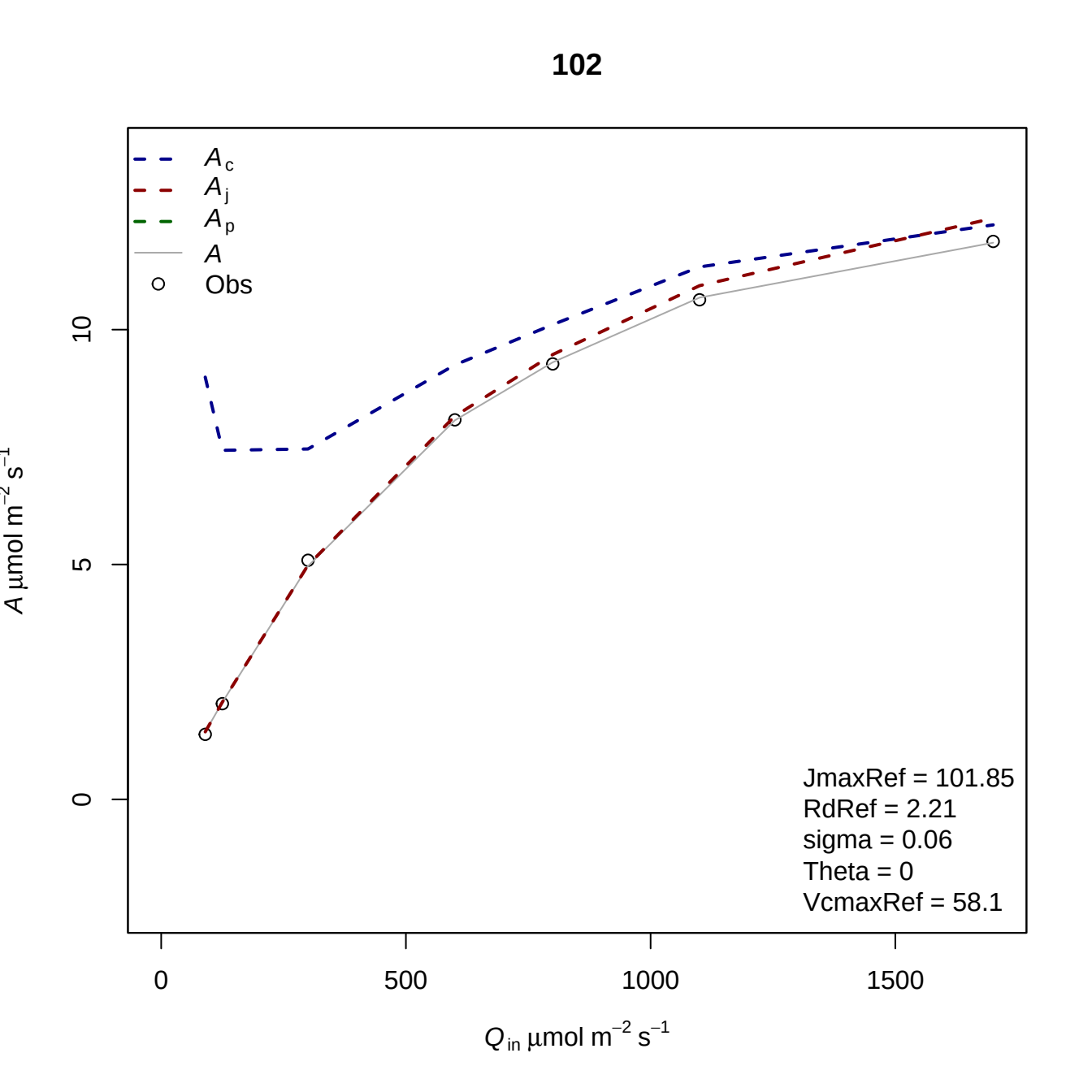

A_c
A_j
A_p
A
Obs
JmaxRef = 131.69
RdRef = 2.88
sigma = 0.28
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

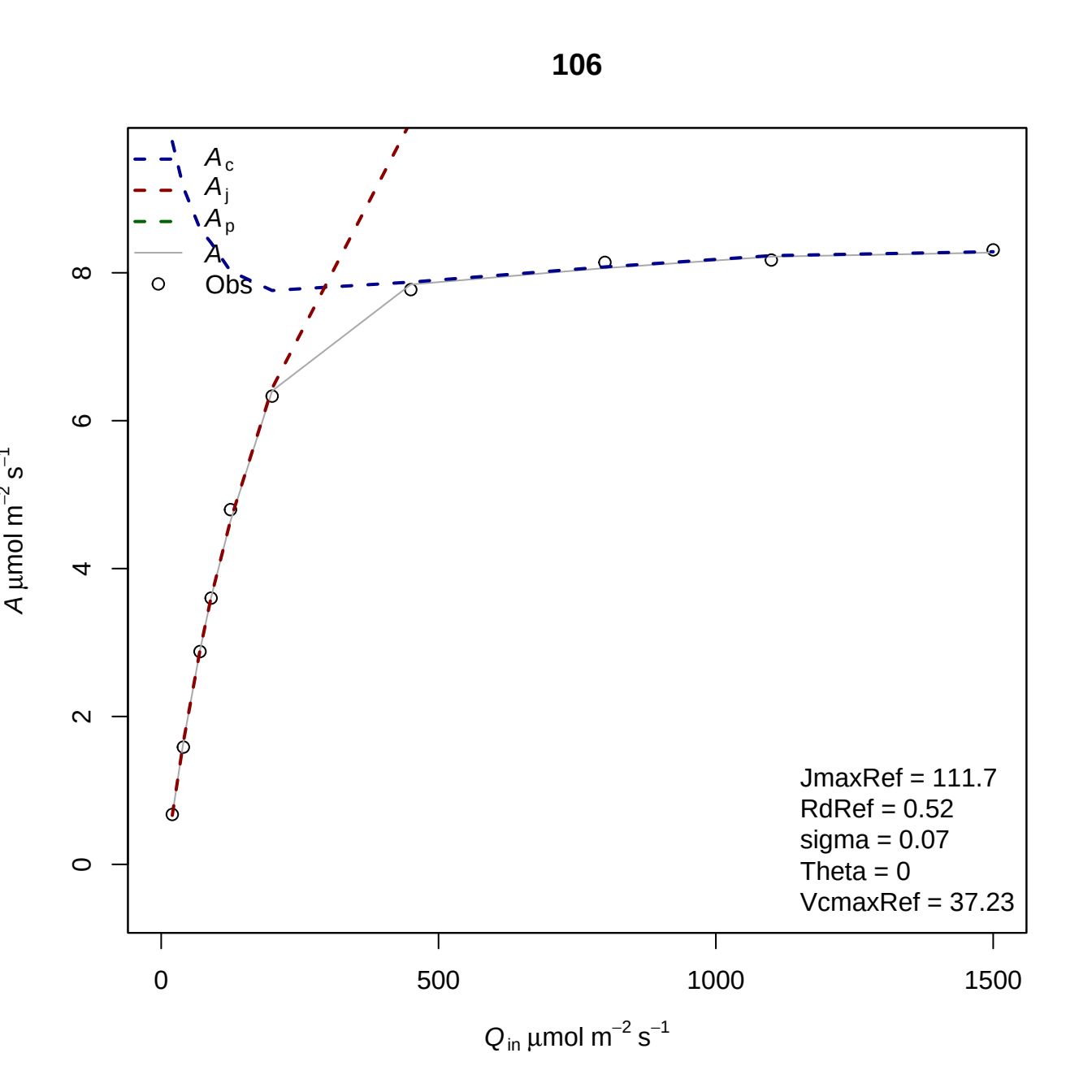

93
A_c
A_j
A_p
A
Obs
JmaxRef = 31.04
RdRef = 0.26
sigma = 0.55
Theta = 0
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

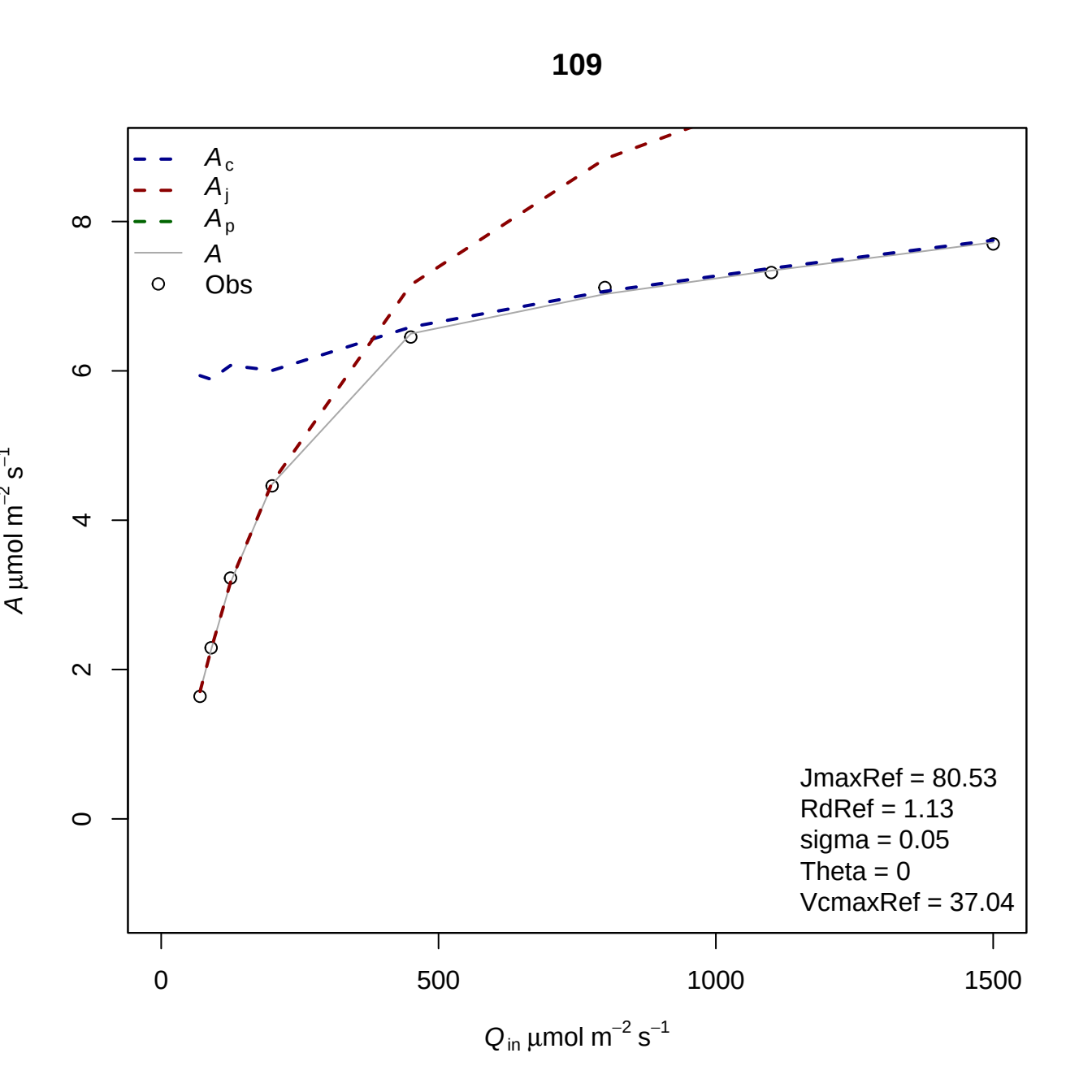

A_c
A_j
A_p
A
Obs
JmaxRef = 63.89
RdRef = 4.06
sigma = 0.64
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

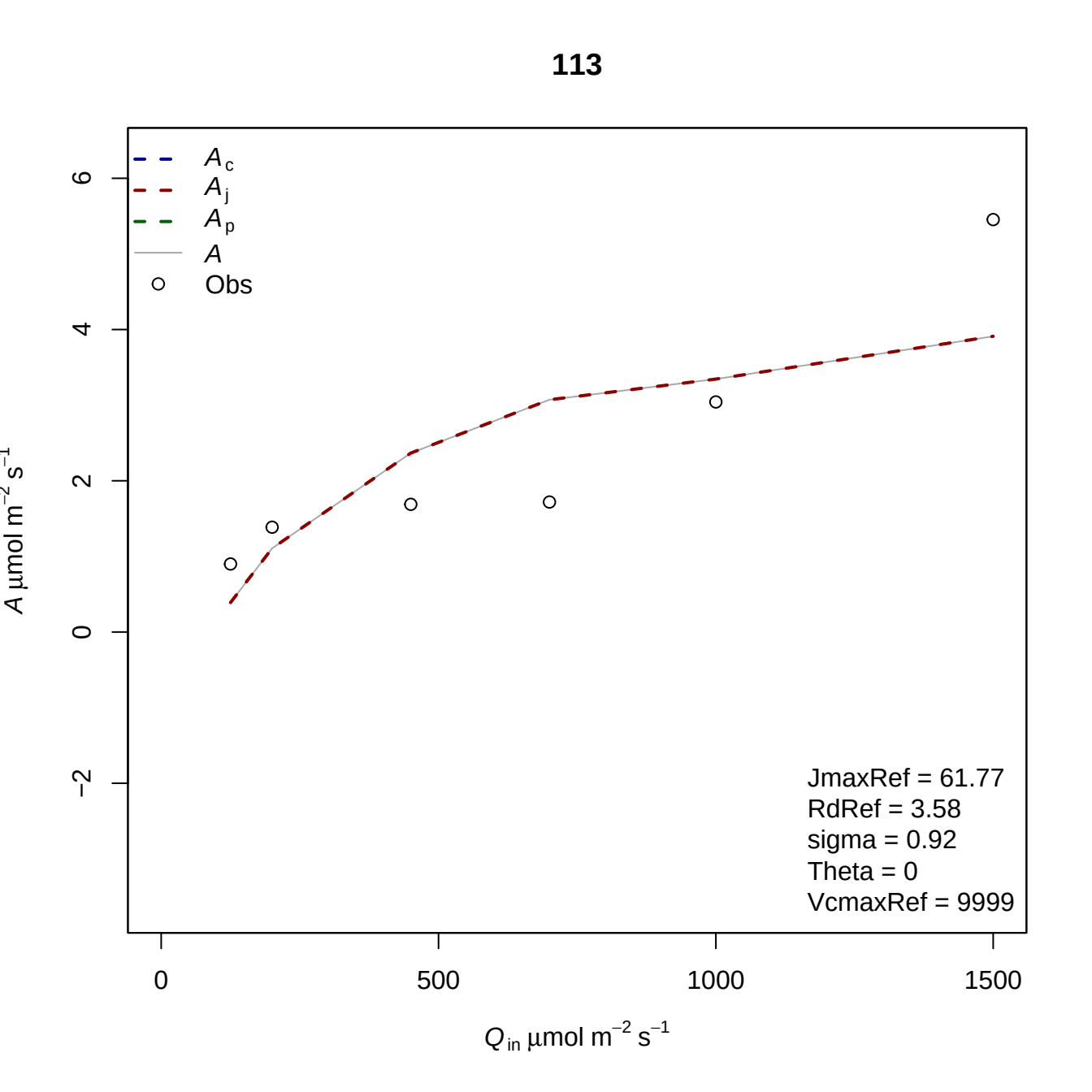

96
A_c
A_j
A_p
A
Obs
JmaxRef = 102.9
RdRef = 0.37
sigma = 0.2
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

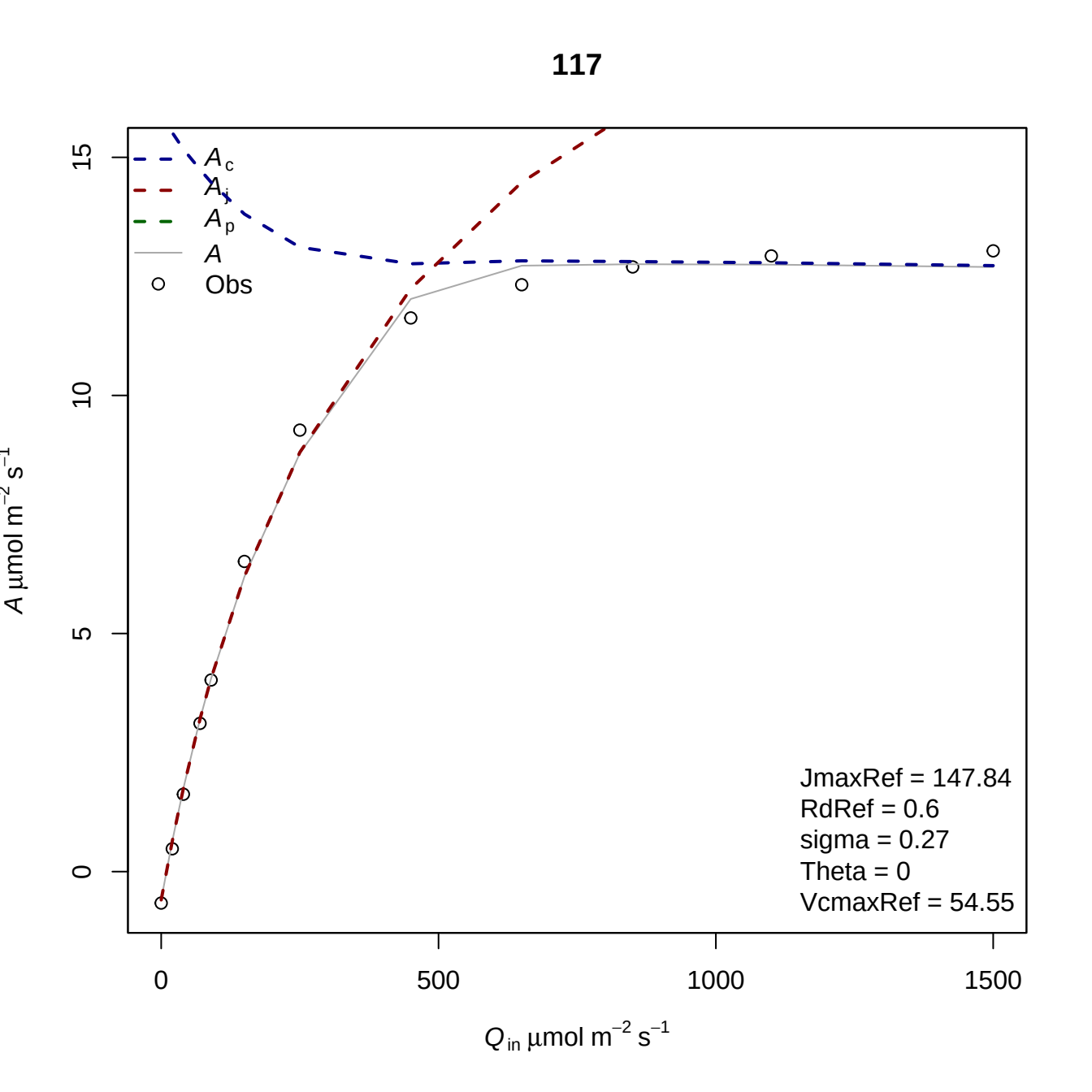

A_c
A_j
A_p
A
Obs
JmaxRef = 141.96
RdRef = 0.62
sigma = 0.27
Theta = 0
VcmaxRef = 51.17
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

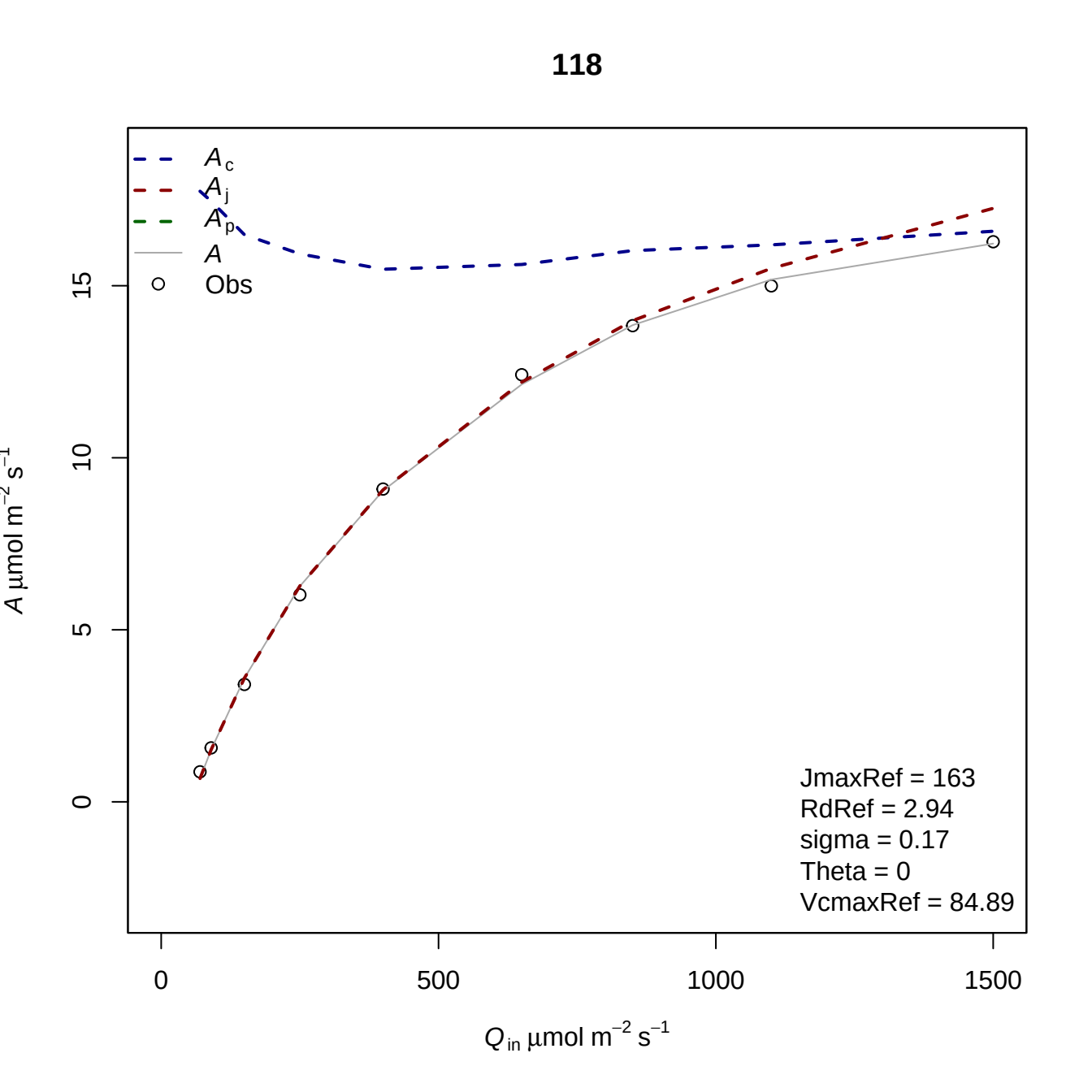

102
A_c
A_j
A_p
A
Obs
JmaxRef = 101.85
RdRef = 2.21
sigma = 0.06
Theta = 0
VcmaxRef = 58.1
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

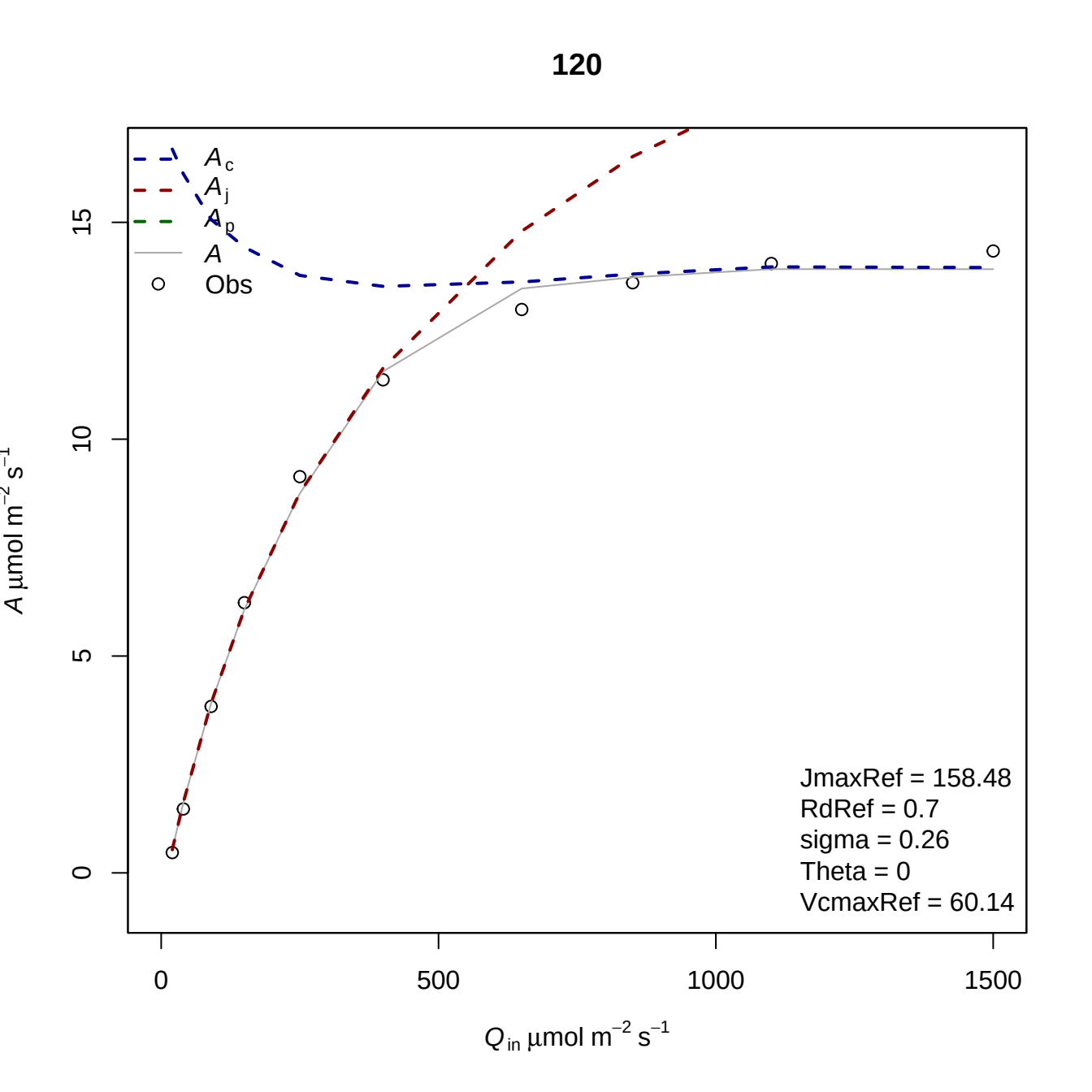

106
A_c
A_j
A_p
A
Obs
JmaxRef = 111.7
RdRef = 0.52
sigma = 0.07
Theta = 0
VcmaxRef = 37.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

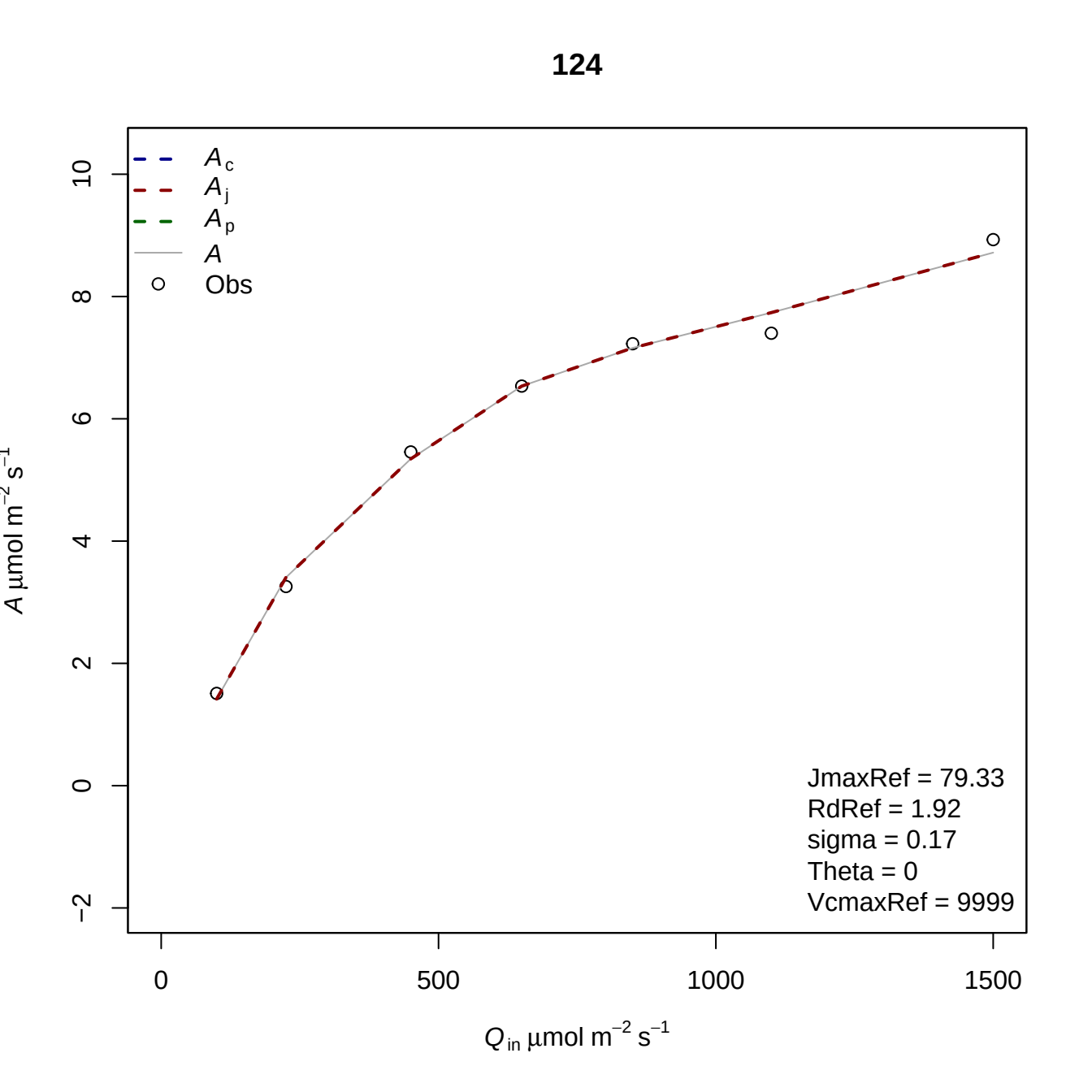

109
A_c
A_j
A_p
A
Obs
JmaxRef = 80.53
RdRef = 1.13
sigma = 0.05
Theta = 0
VcmaxRef = 37.04
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

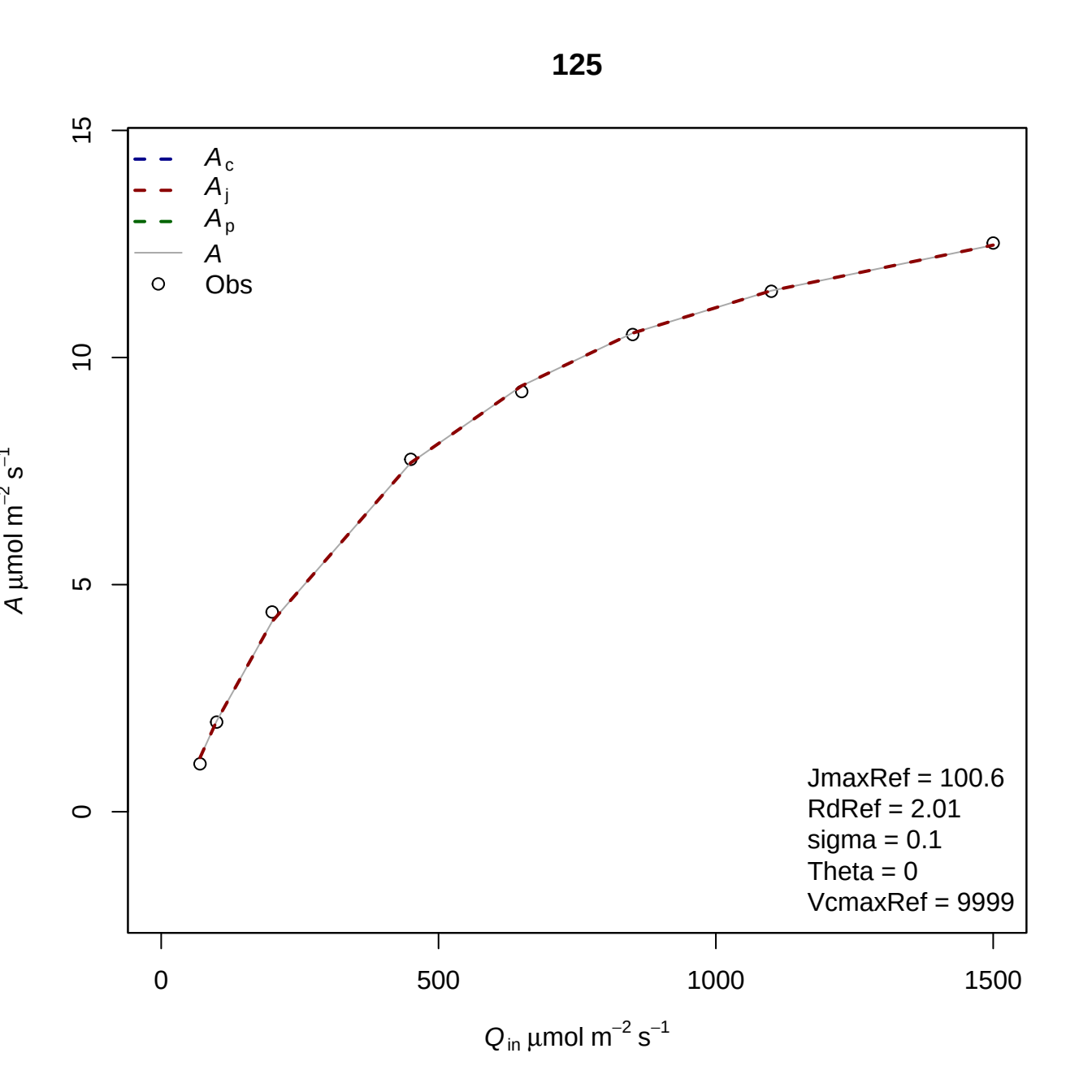

113
A_c
A_j
A_p
A
Obs
6
4
2
0
−2
A µmol m^{-2} s^{-1}
JmaxRef = 61.77
RdRef = 3.58
sigma = 0.92
Theta = 0
VcmaxRef = 9999
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

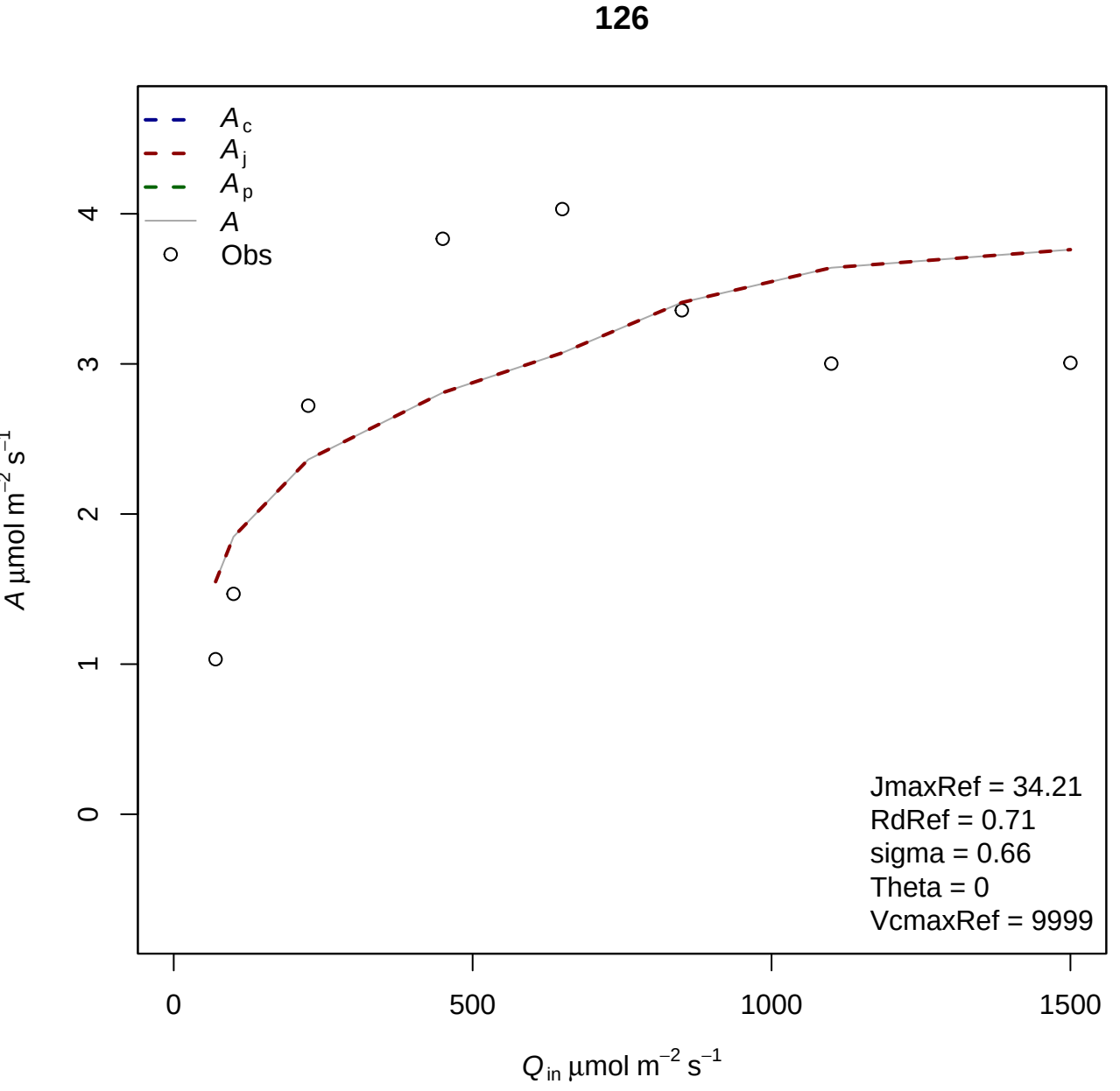

117
A_c
A_j
A_p
A
Obs
JmaxRef = 147.84
RdRef = 0.6
sigma = 0.27
Theta = 0
VcmaxRef = 54.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

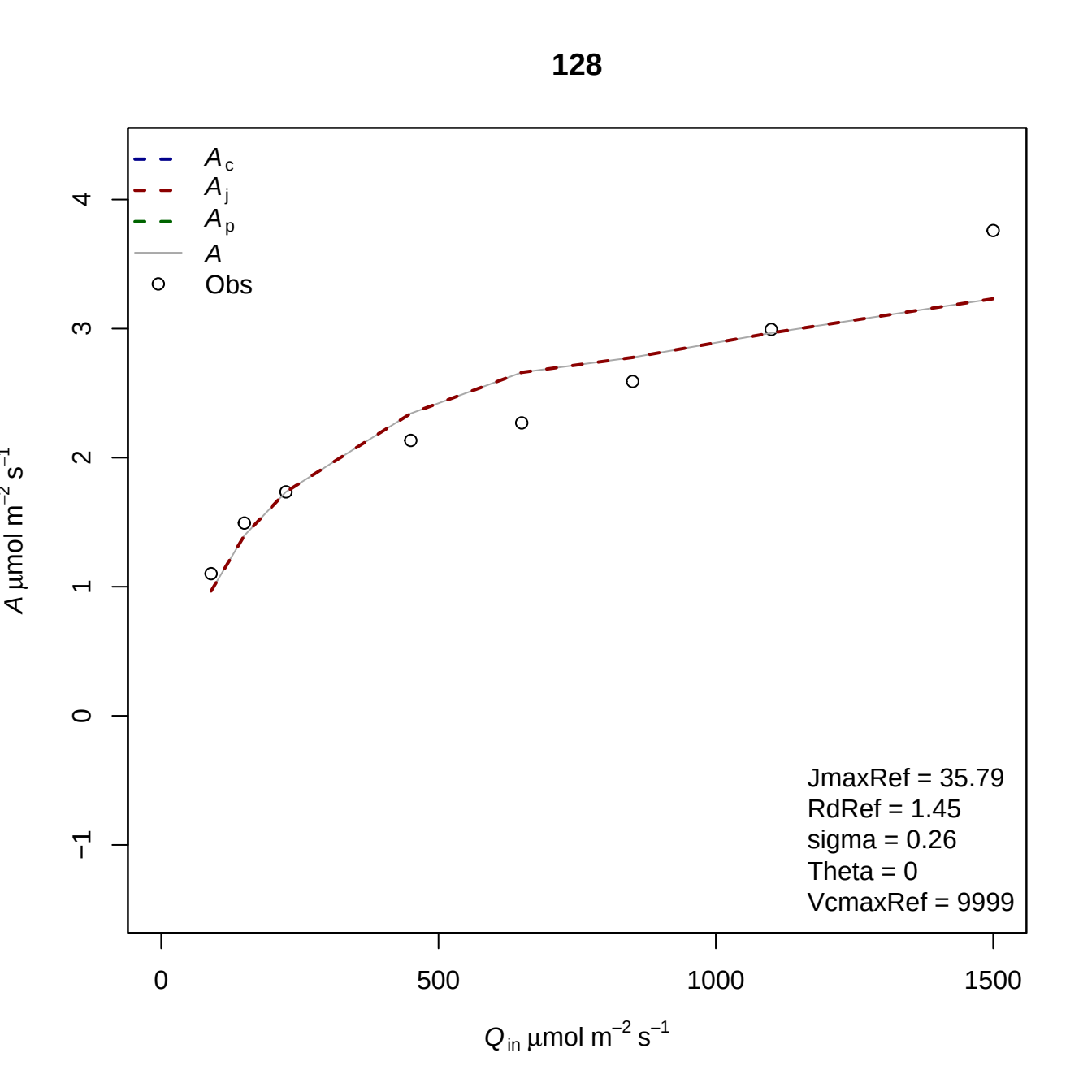

118
A_c
A_j
A_p
A
Obs
JmaxRef = 163
RdRef = 2.94
sigma = 0.17
Theta = 0
VcmaxRef = 84.89
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

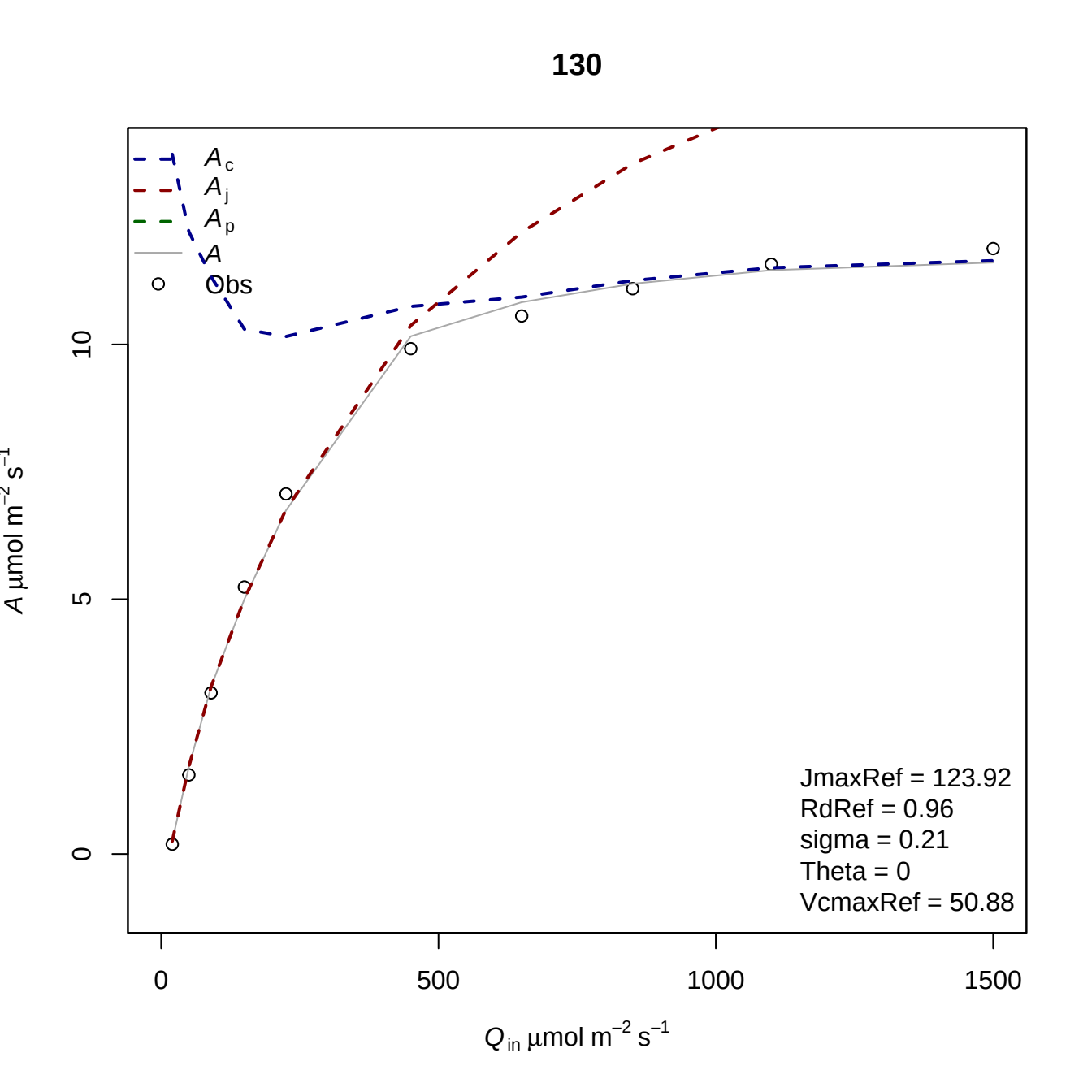

120
A_c
A_j
A_p
A
Obs
JmaxRef = 158.48
RdRef = 0.7
sigma = 0.26
Theta = 0
VcmaxRef = 60.14
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

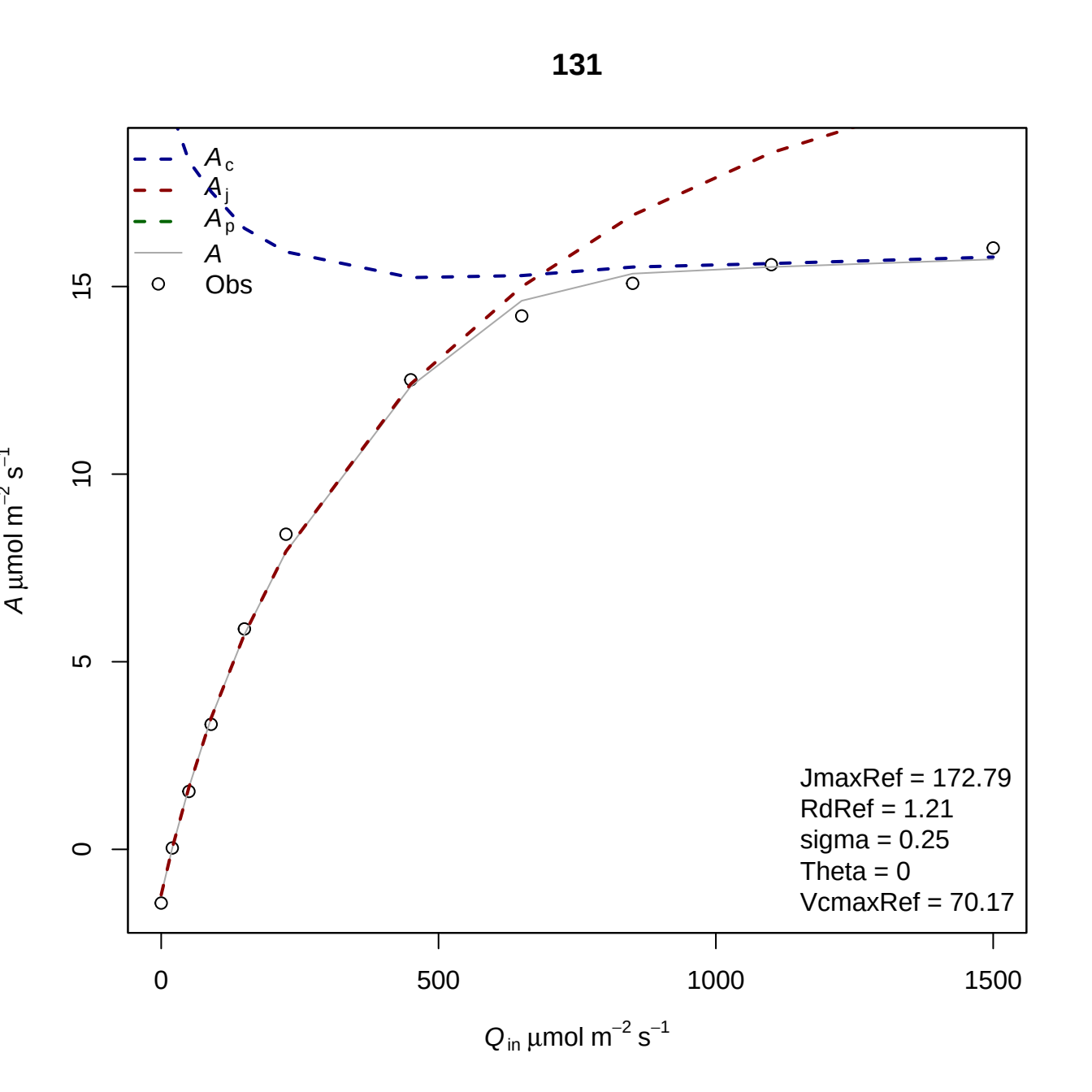

124
A_c
A_j
A_p
A
Obs
JmaxRef = 79.33
RdRef = 1.92
sigma = 0.17
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

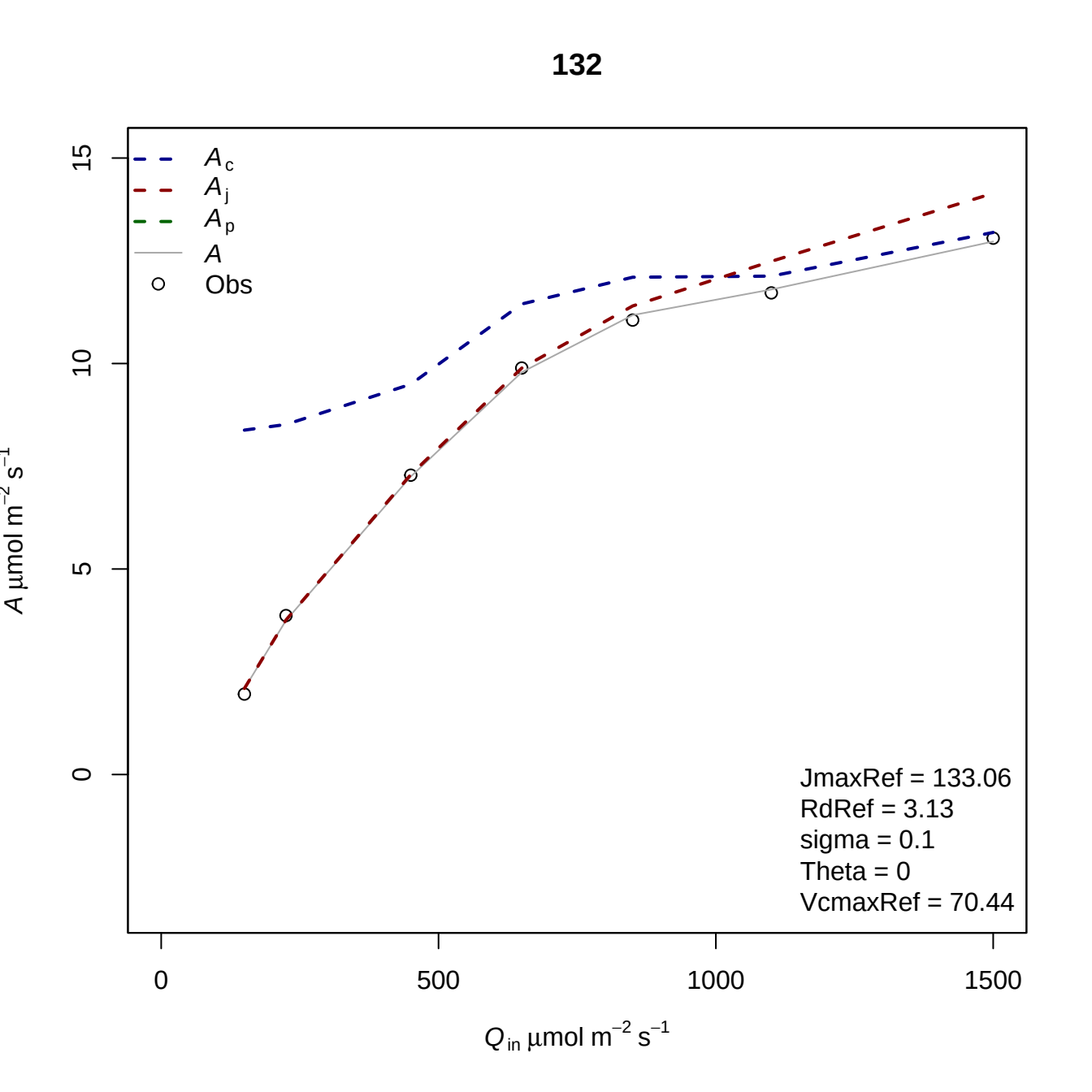

125
A_c
A_j
A_p
A
Obs
JmaxRef = 100.6
RdRef = 2.01
sigma = 0.1
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

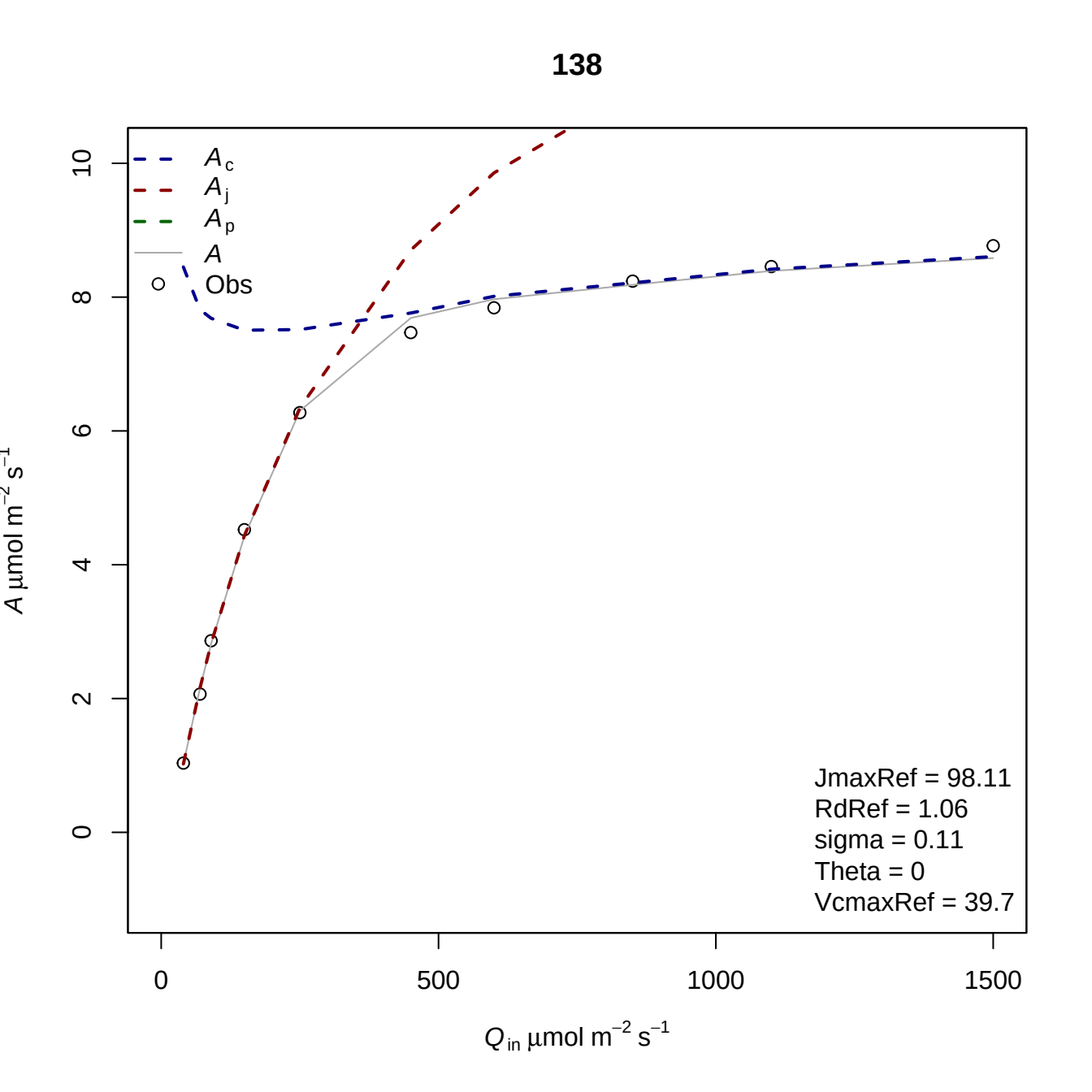

126
A_c
A_j
A_p
A
Obs
JmaxRef = 34.21
RdRef = 0.71
sigma = 0.66
Theta = 0
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

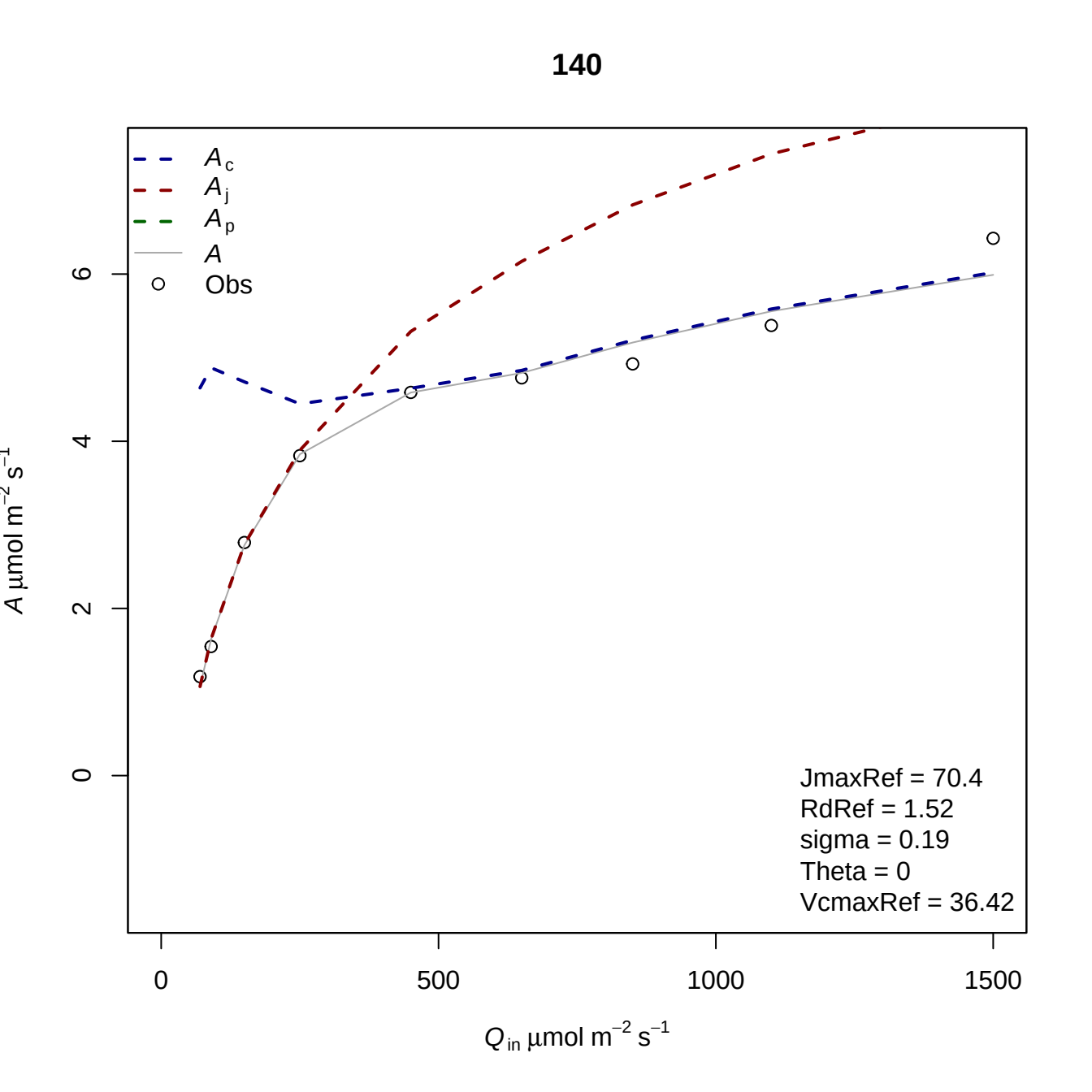

128
A_c
A_j
A_p
A
Obs
JmaxRef = 35.79
RdRef = 1.45
sigma = 0.26
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

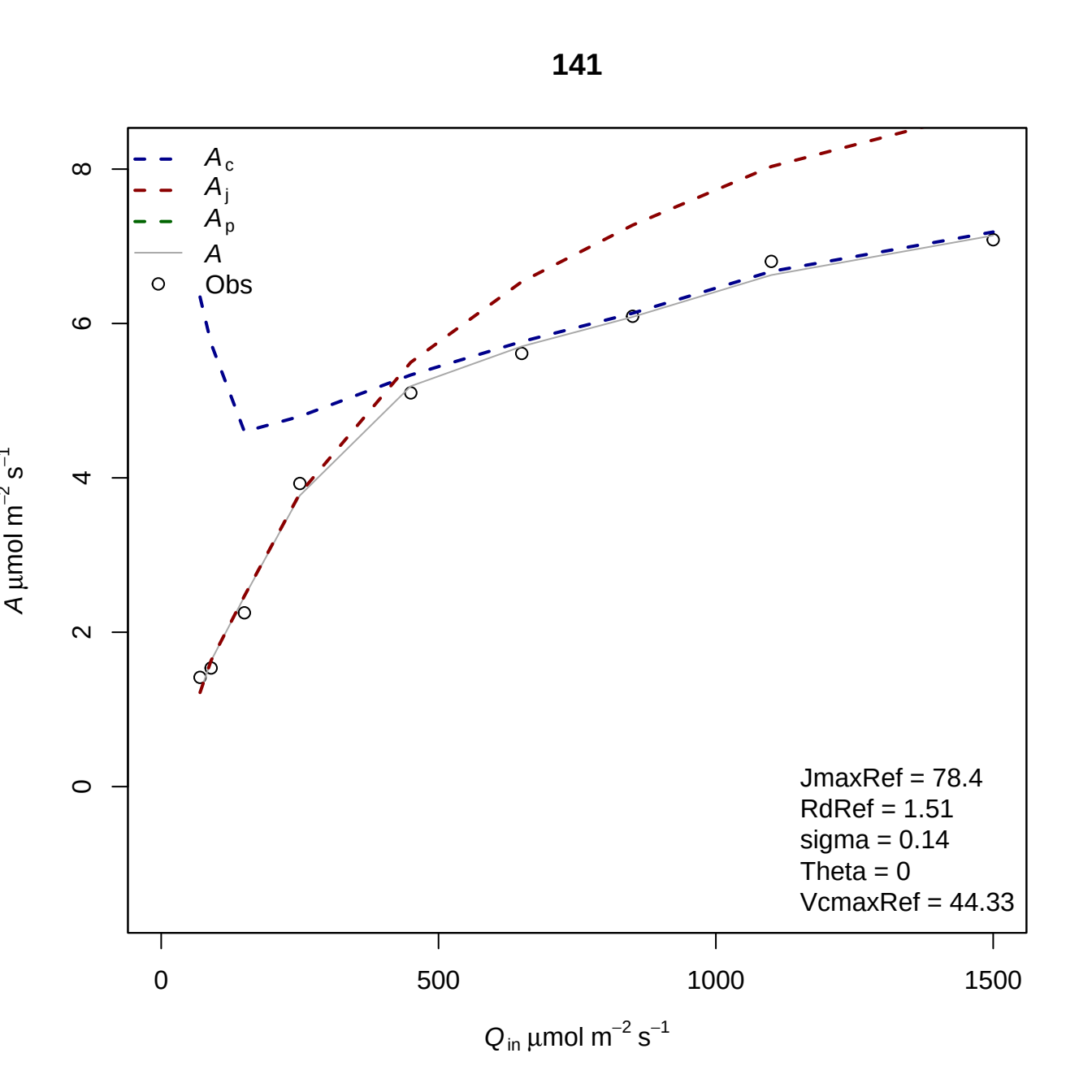

130
A_c
A_j
A_p
A
Obs
JmaxRef = 123.92
RdRef = 0.96
sigma = 0.21
Theta = 0
VcmaxRef = 50.88
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

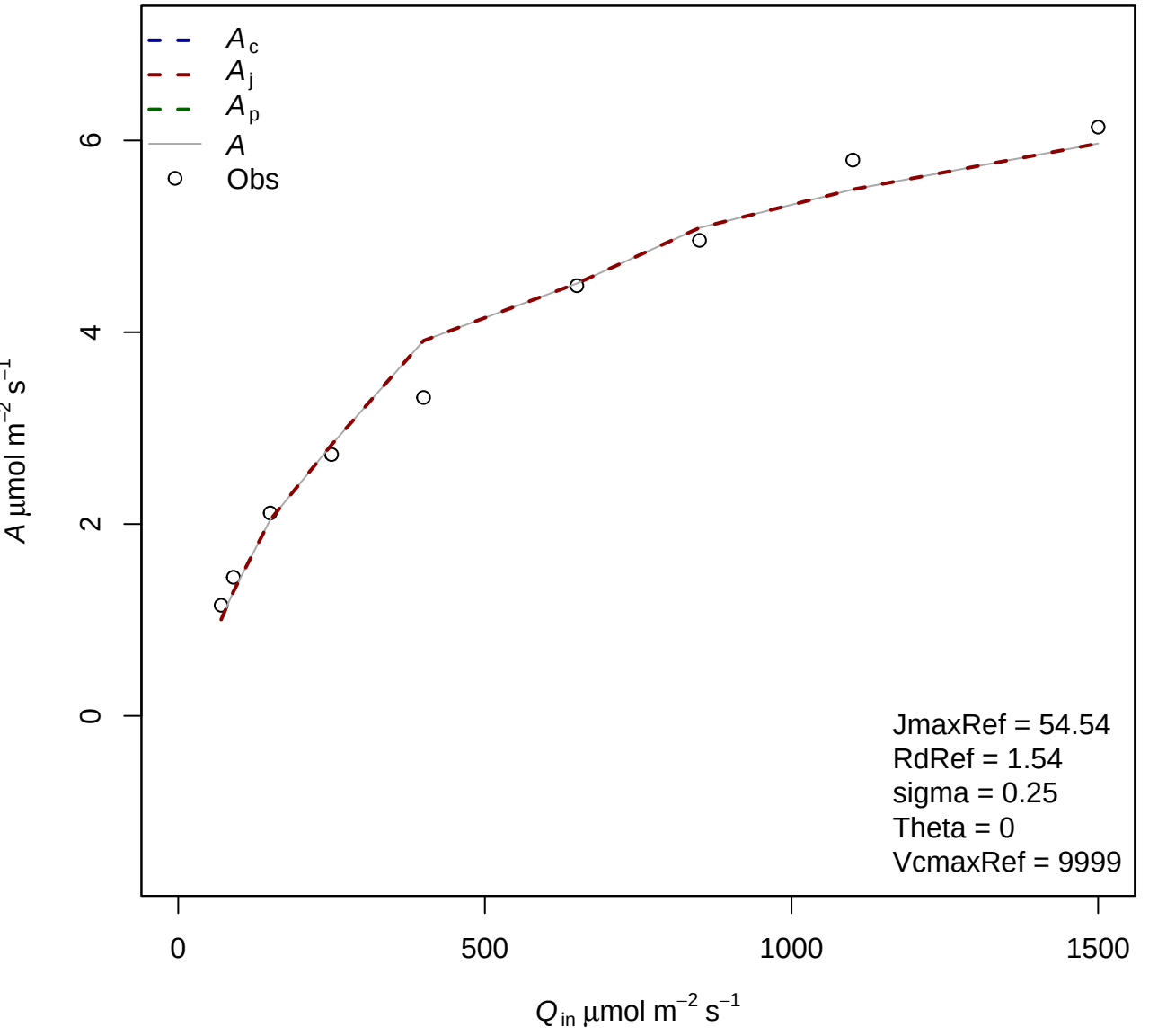

131
A_c
A_j
A_p
A
Obs
JmaxRef = 172.79
RdRef = 1.21
sigma = 0.25
Theta = 0
VcmaxRef = 70.17
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

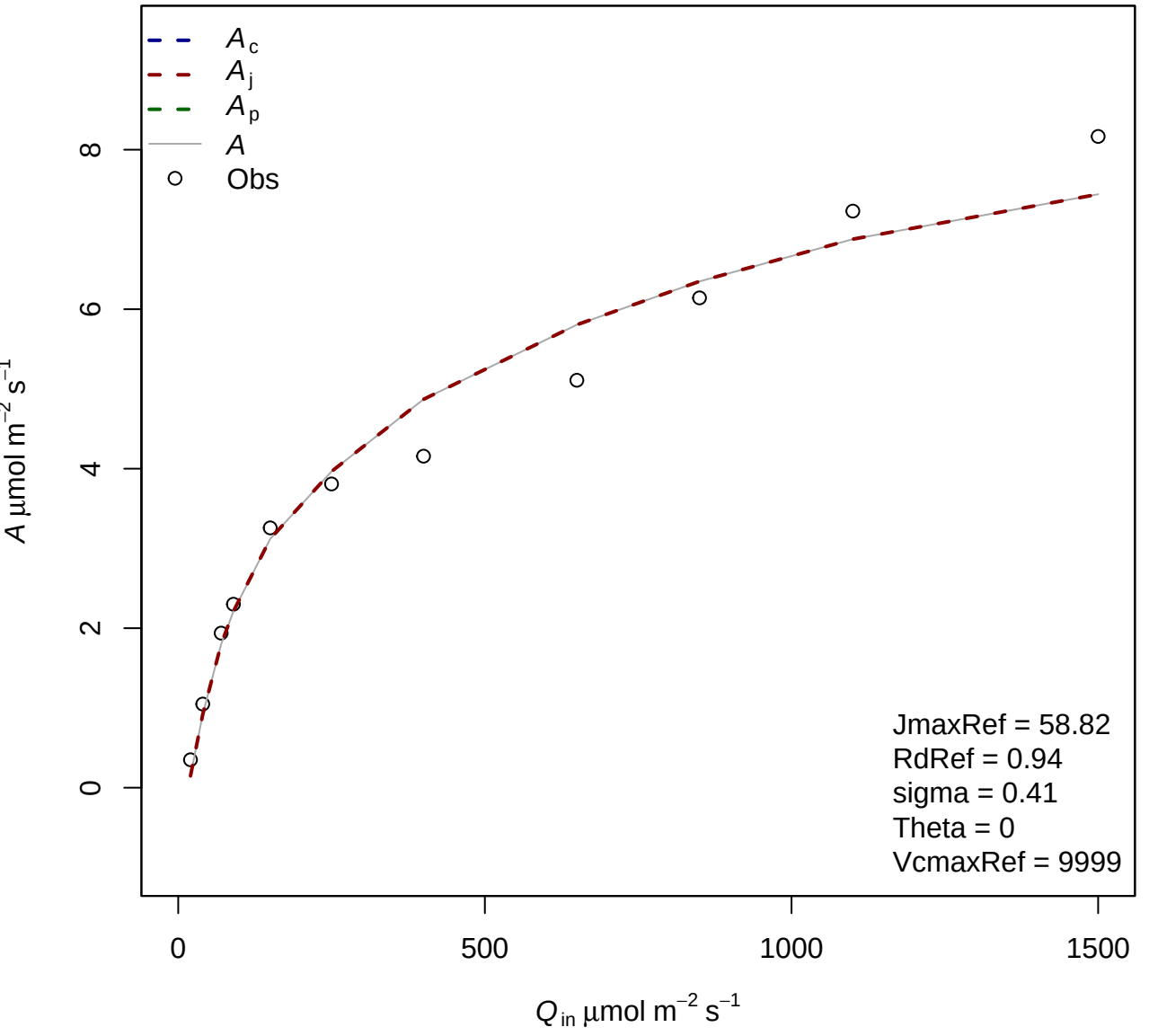

132
A_c
A_j
A_p
A
Obs
JmaxRef = 133.06
RdRef = 3.13
sigma = 0.1
Theta = 0
VcmaxRef = 70.44
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

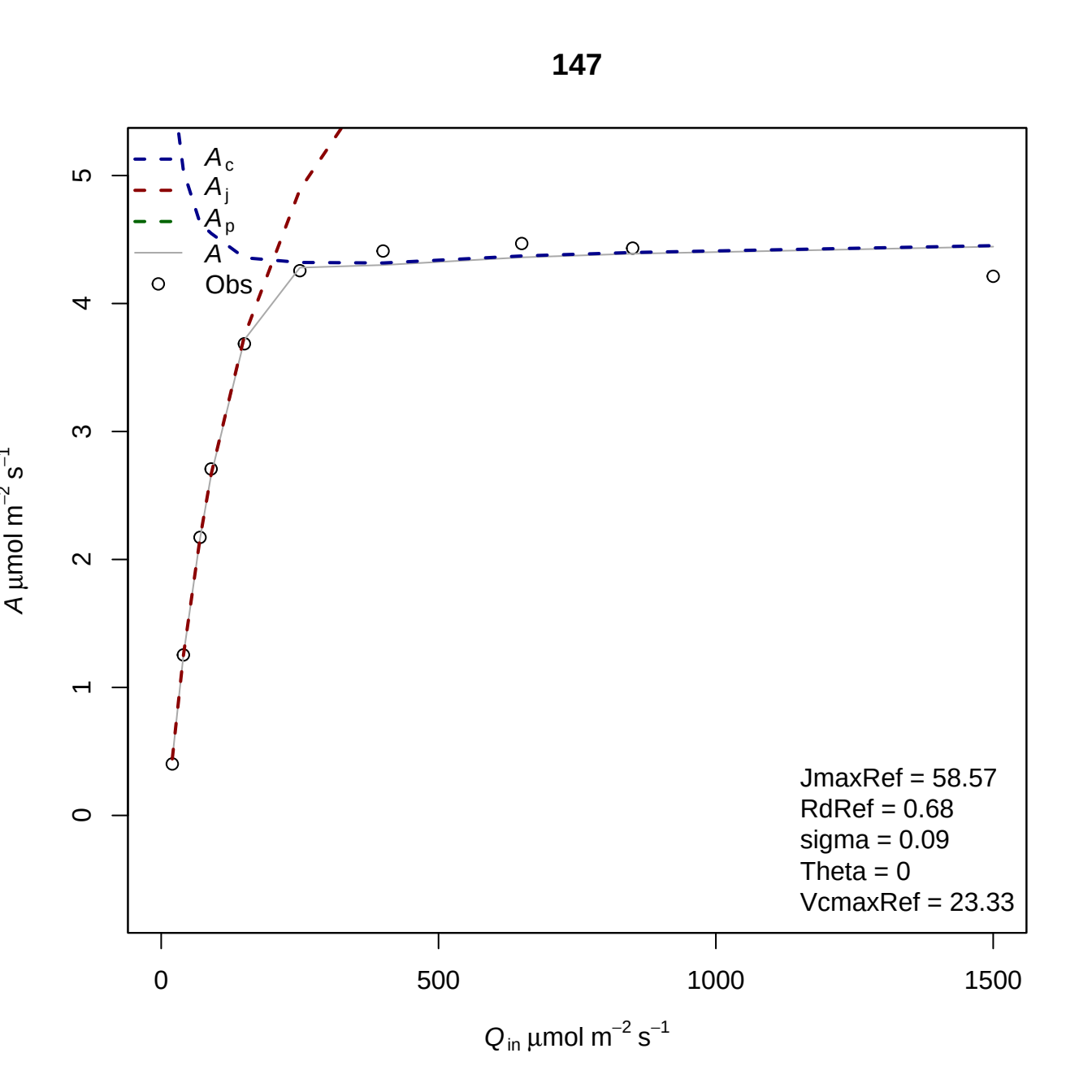

138
A_c
A_j
A_p
A
Obs
JmaxRef = 98.11
RdRef = 1.06
sigma = 0.11
Theta = 0
VcmaxRef = 39.7
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

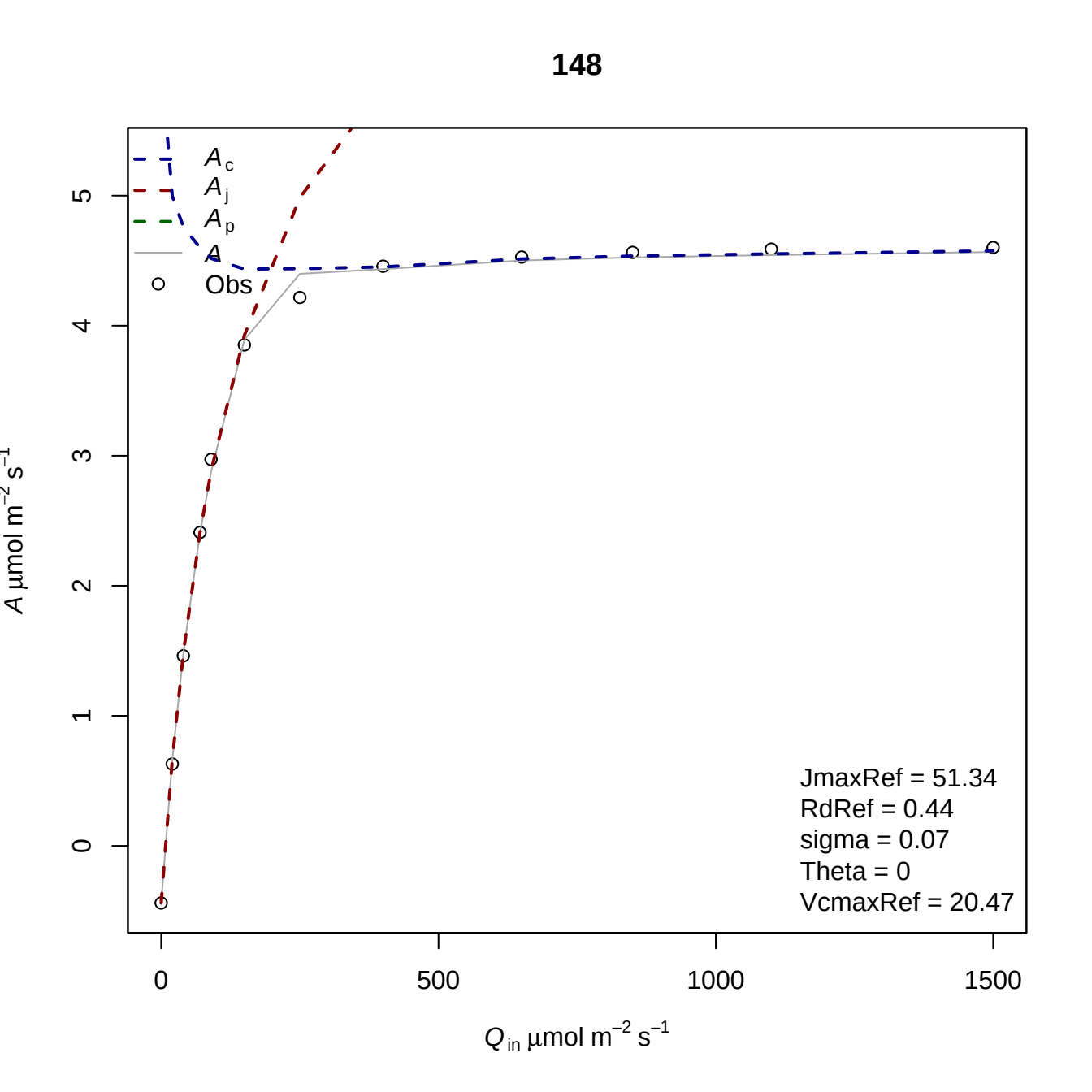

140
A_c
A_j
A_p
A
Obs
JmaxRef = 70.4
RdRef = 1.52
sigma = 0.19
Theta = 0
VcmaxRef = 36.42
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

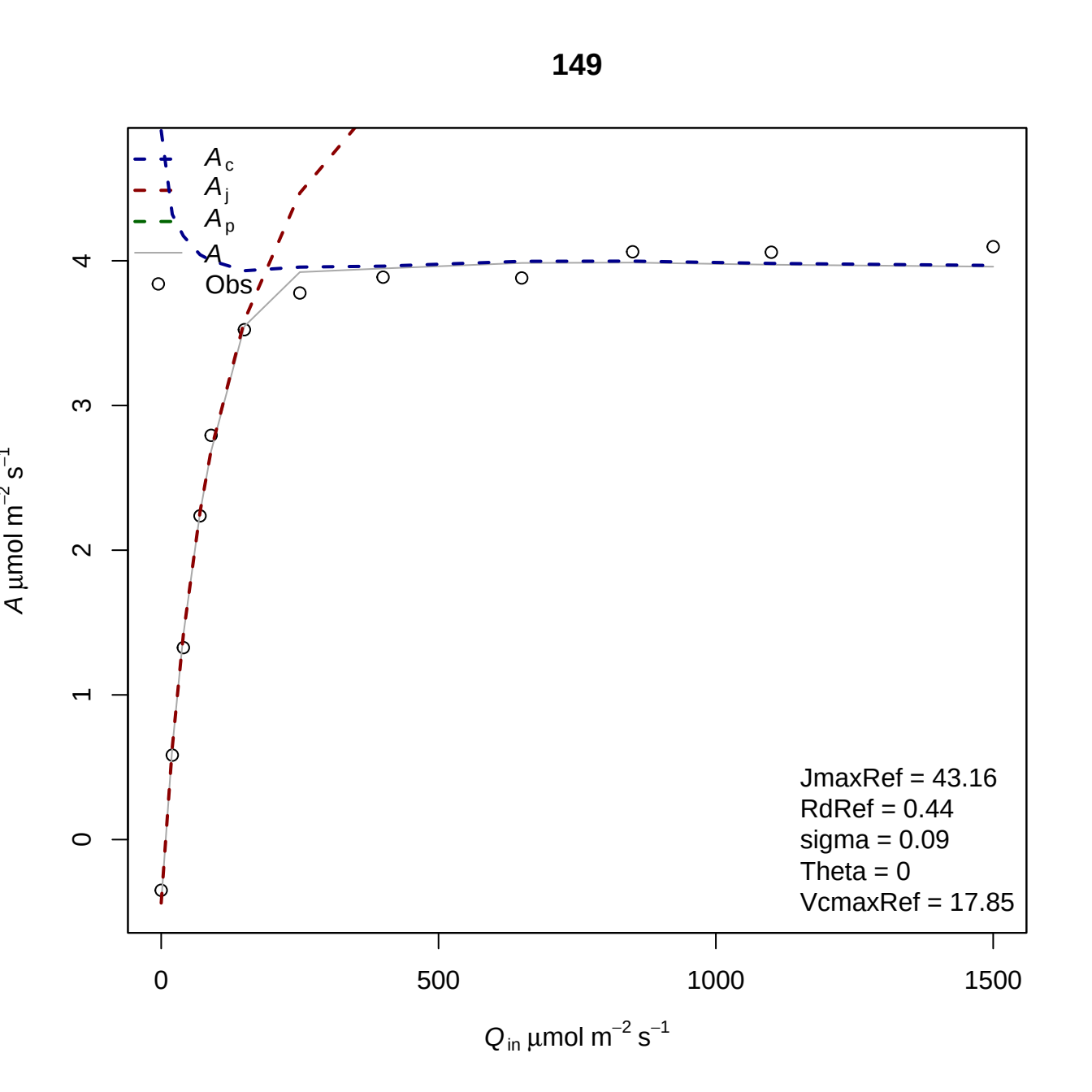

141
A_c
A_j
A_p
A
Obs
JmaxRef = 78.4
RdRef = 1.51
sigma = 0.14
Theta = 0
VcmaxRef = 44.33
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

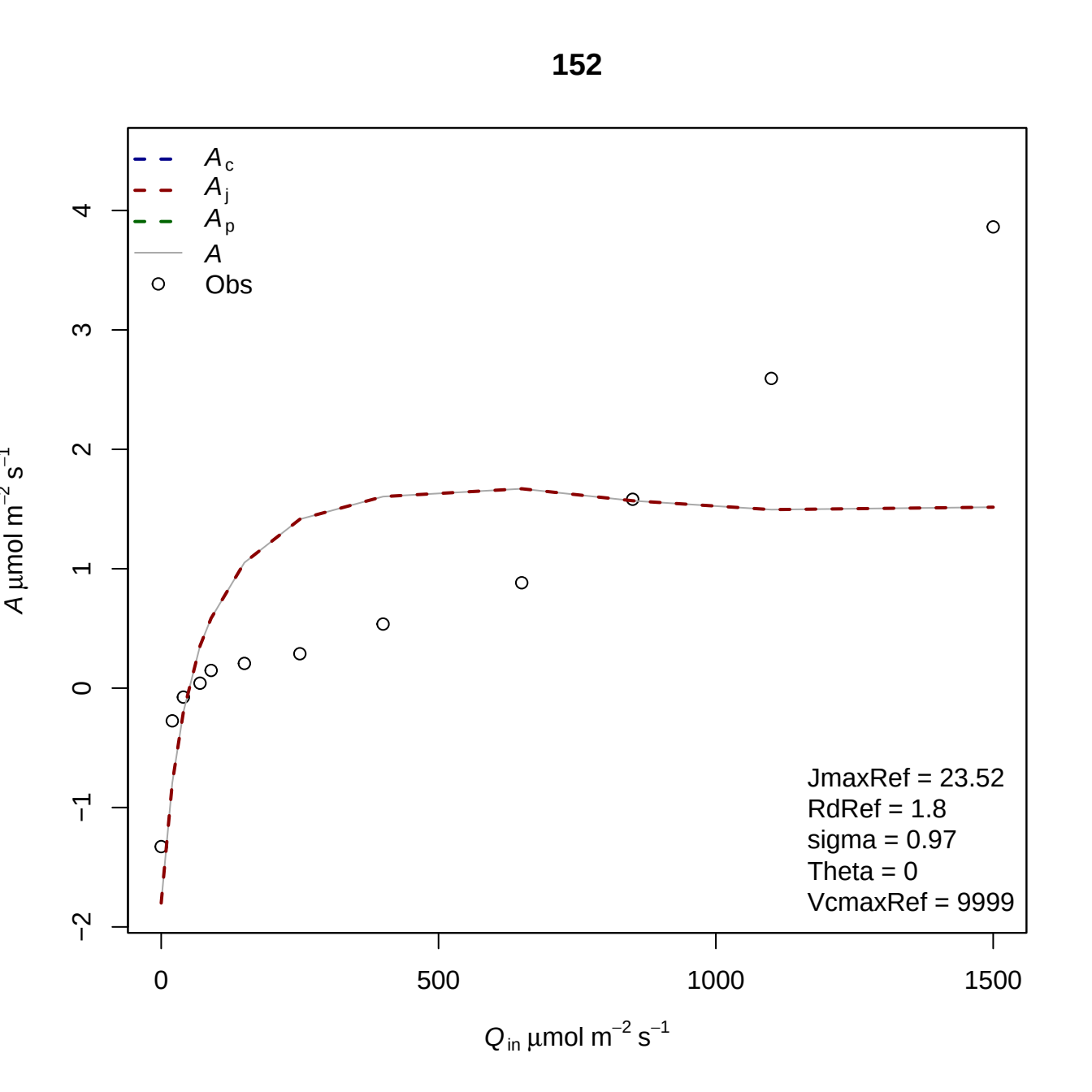

A_c
A_j
A_p
A
Obs
JmaxRef = 54.54
RdRef = 1.54
sigma = 0.25
Theta = 0
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

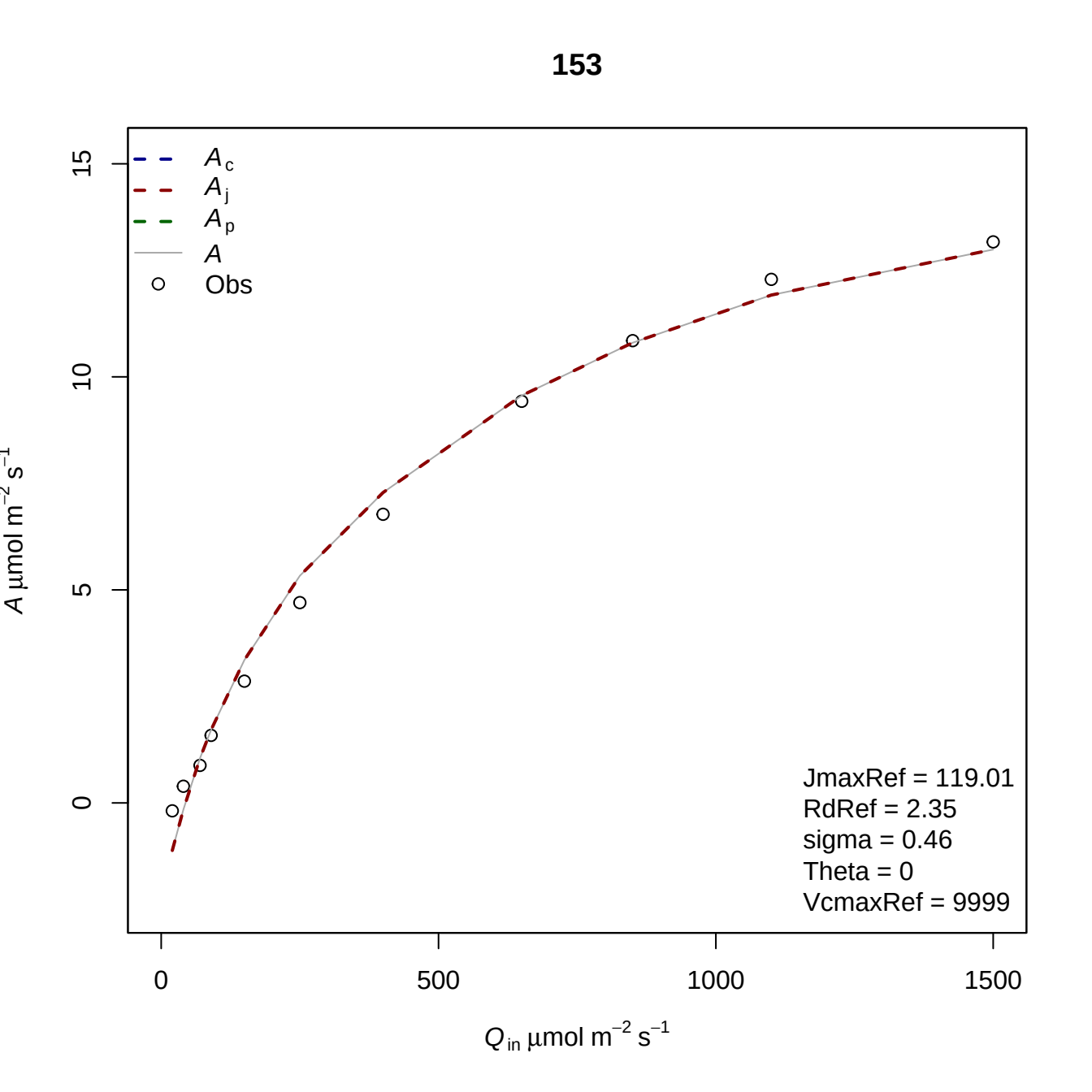

A_c
A_j
A_p
A
Obs
JmaxRef = 58.82
RdRef = 0.94
sigma = 0.41
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

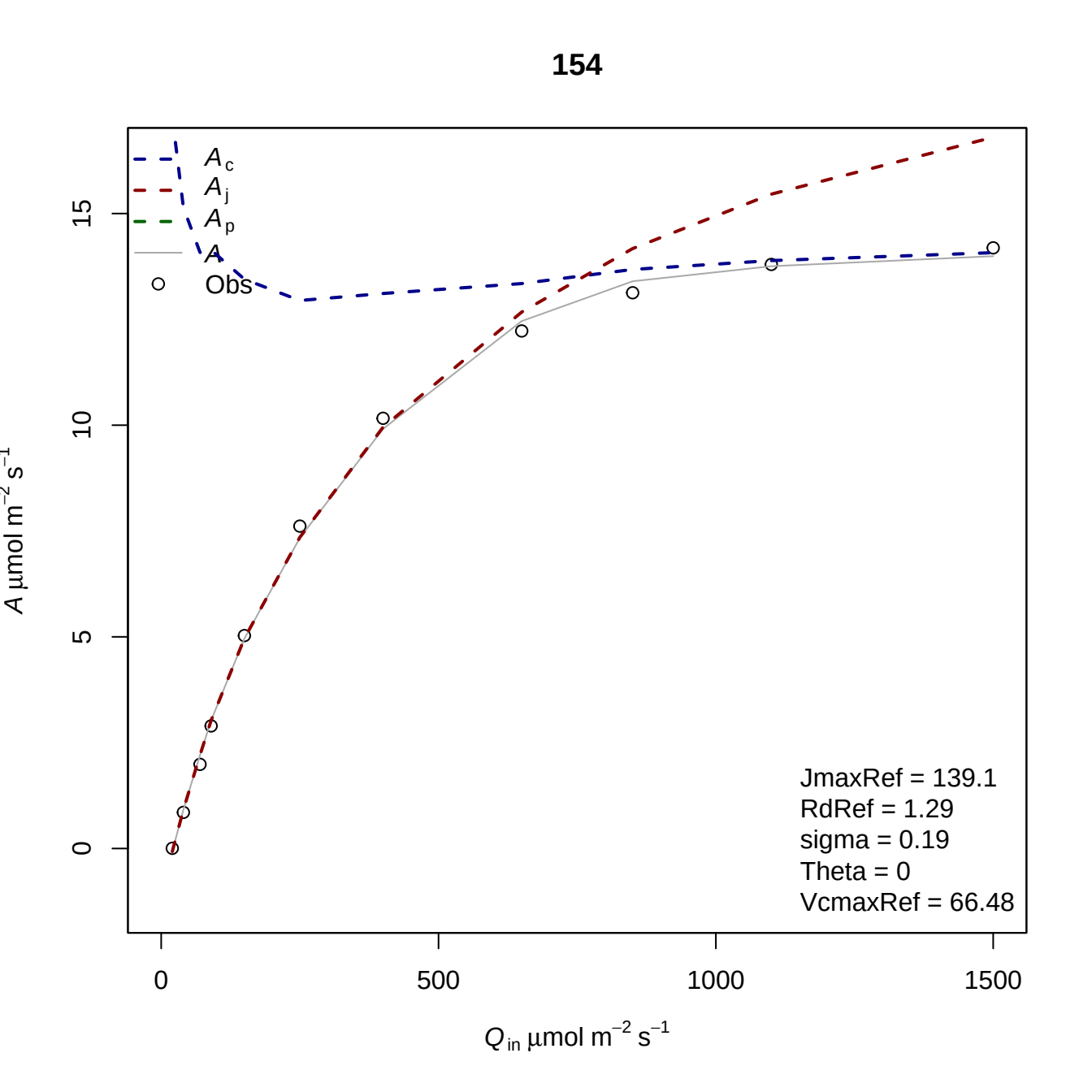

147
A_c
A_j
A_p
A
Obs
JmaxRef = 58.57
RdRef = 0.68
sigma = 0.09
Theta = 0
VcmaxRef = 23.33
0
1
2
3
4
5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

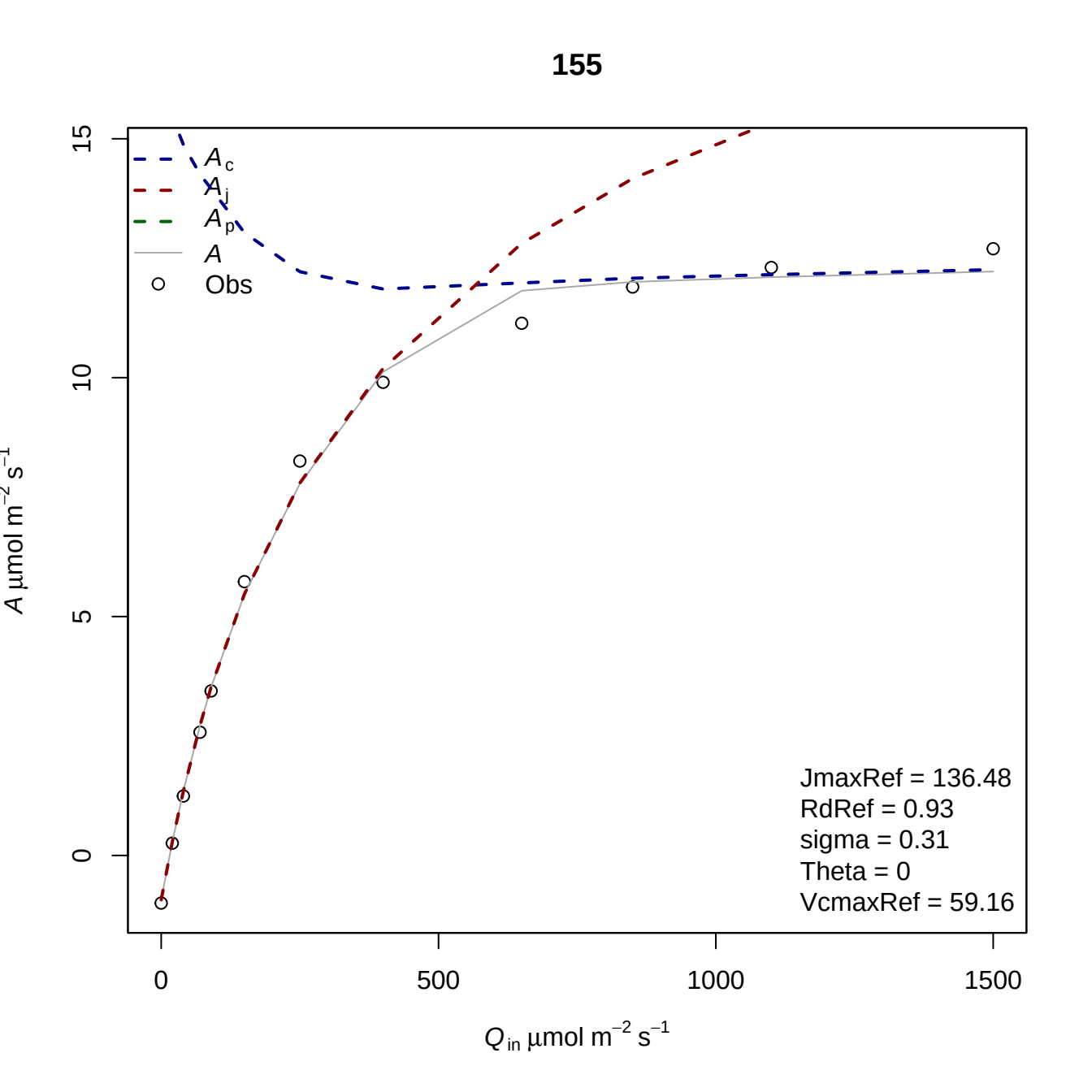

148
A_c
A_j
A_p
A
Obs
JmaxRef = 51.34
RdRef = 0.44
sigma = 0.07
Theta = 0
VcmaxRef = 20.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

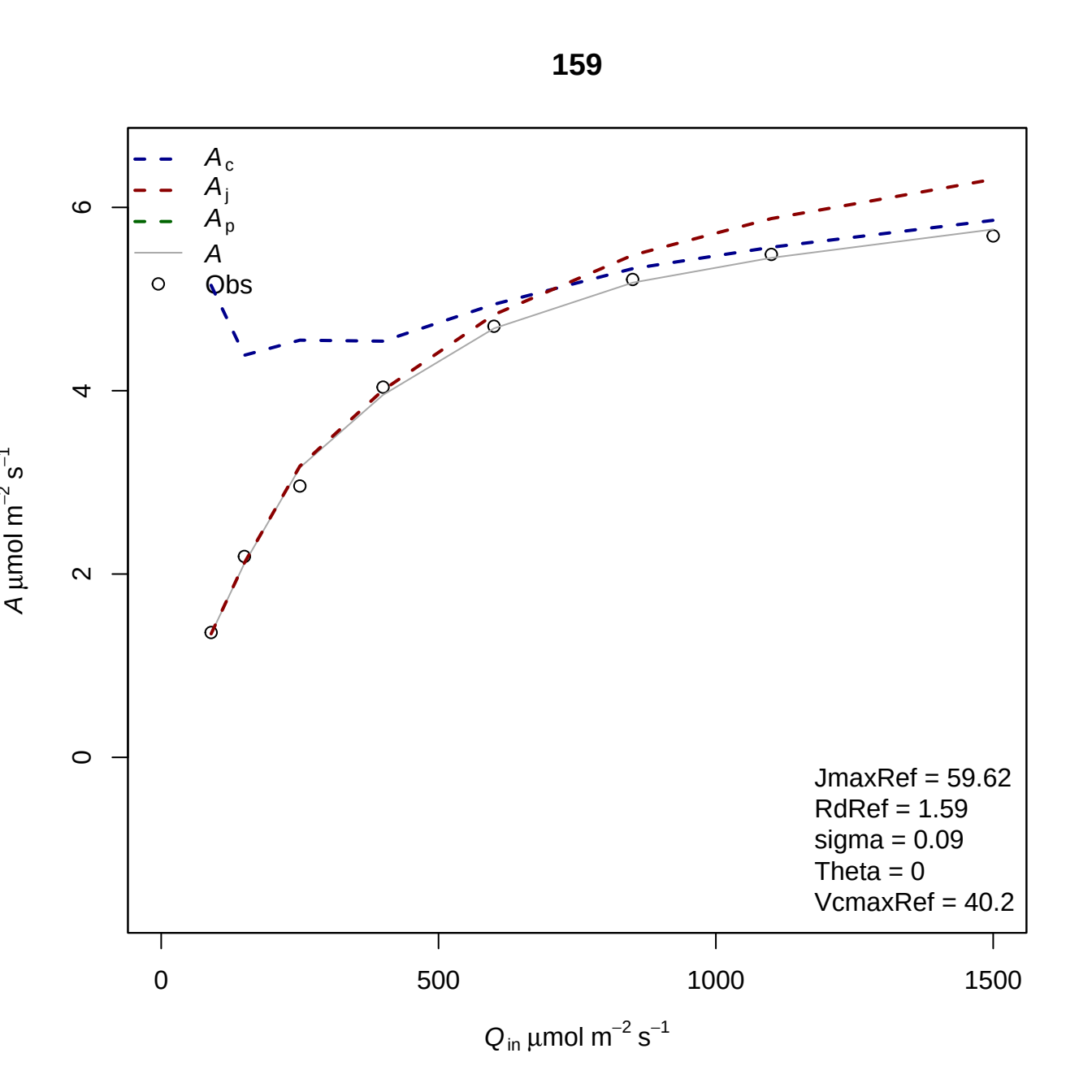

149
A_c
A_j
A_p
A
Obs
JmaxRef = 43.16
RdRef = 0.44
sigma = 0.09
Theta = 0
VcmaxRef = 17.85
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

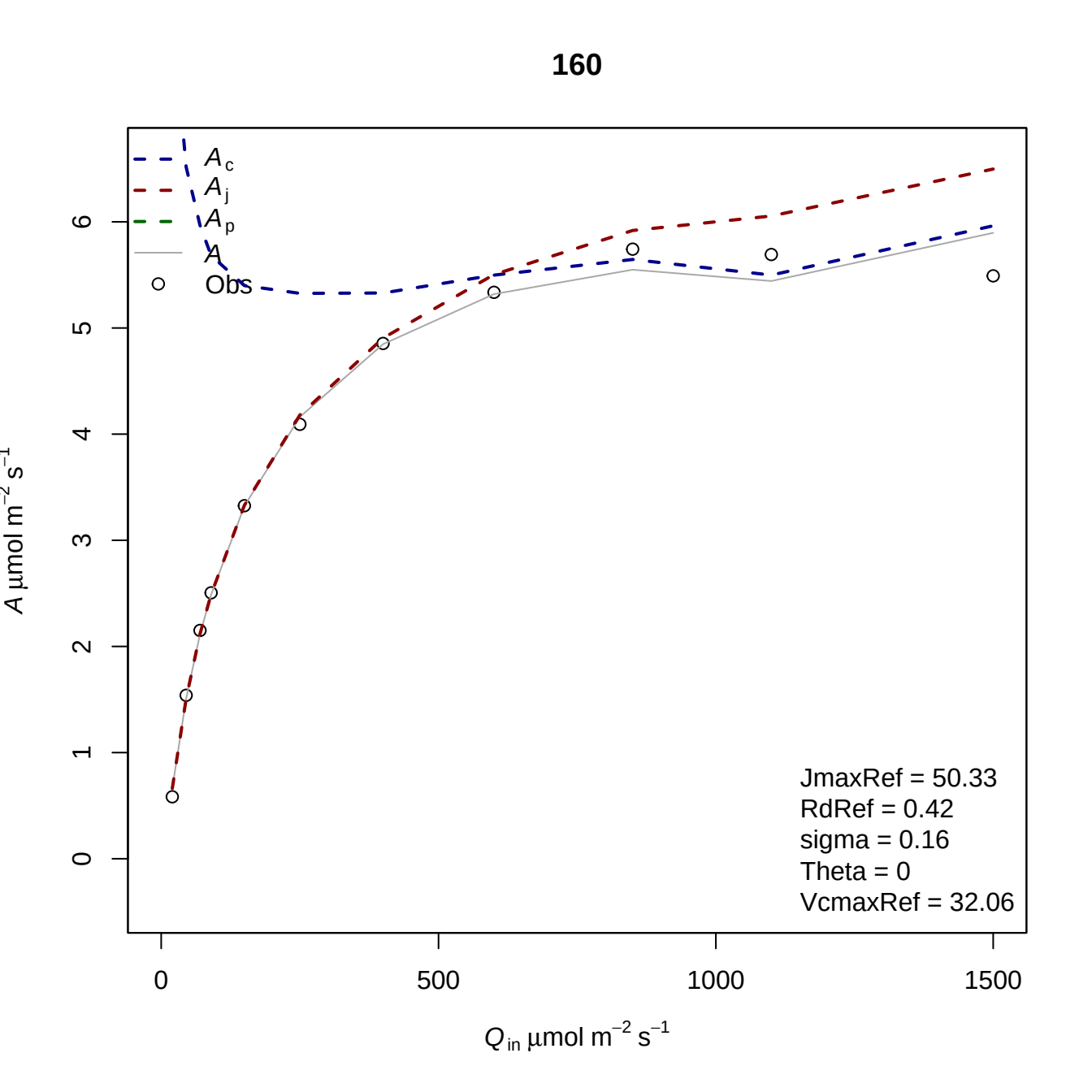

152
A_c
A_j
A_p
A
Obs
JmaxRef = 23.52
RdRef = 1.8
sigma = 0.97
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

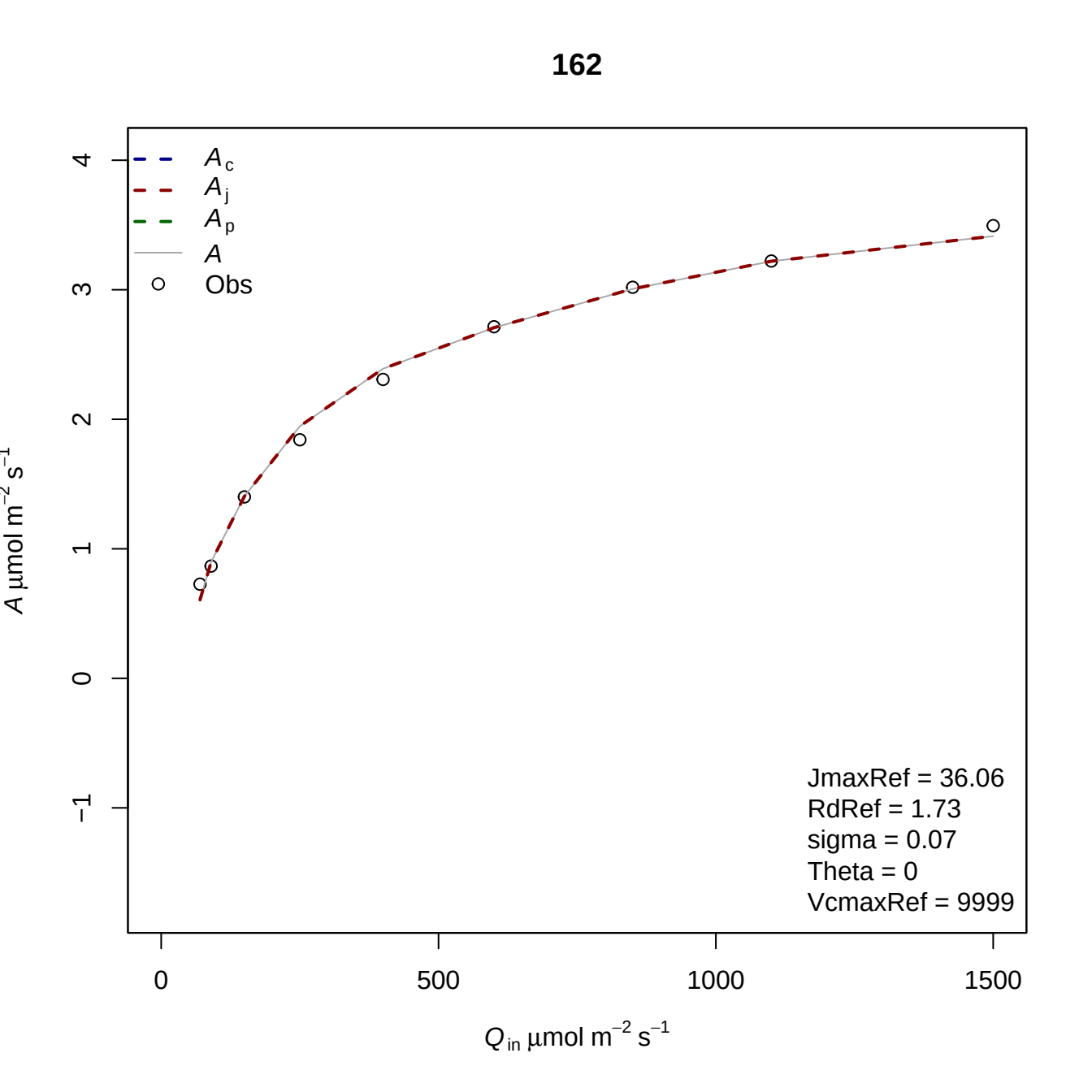

153
A_c
A_j
A_p
A
Obs
JmaxRef = 119.01
RdRef = 2.35
sigma = 0.46
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

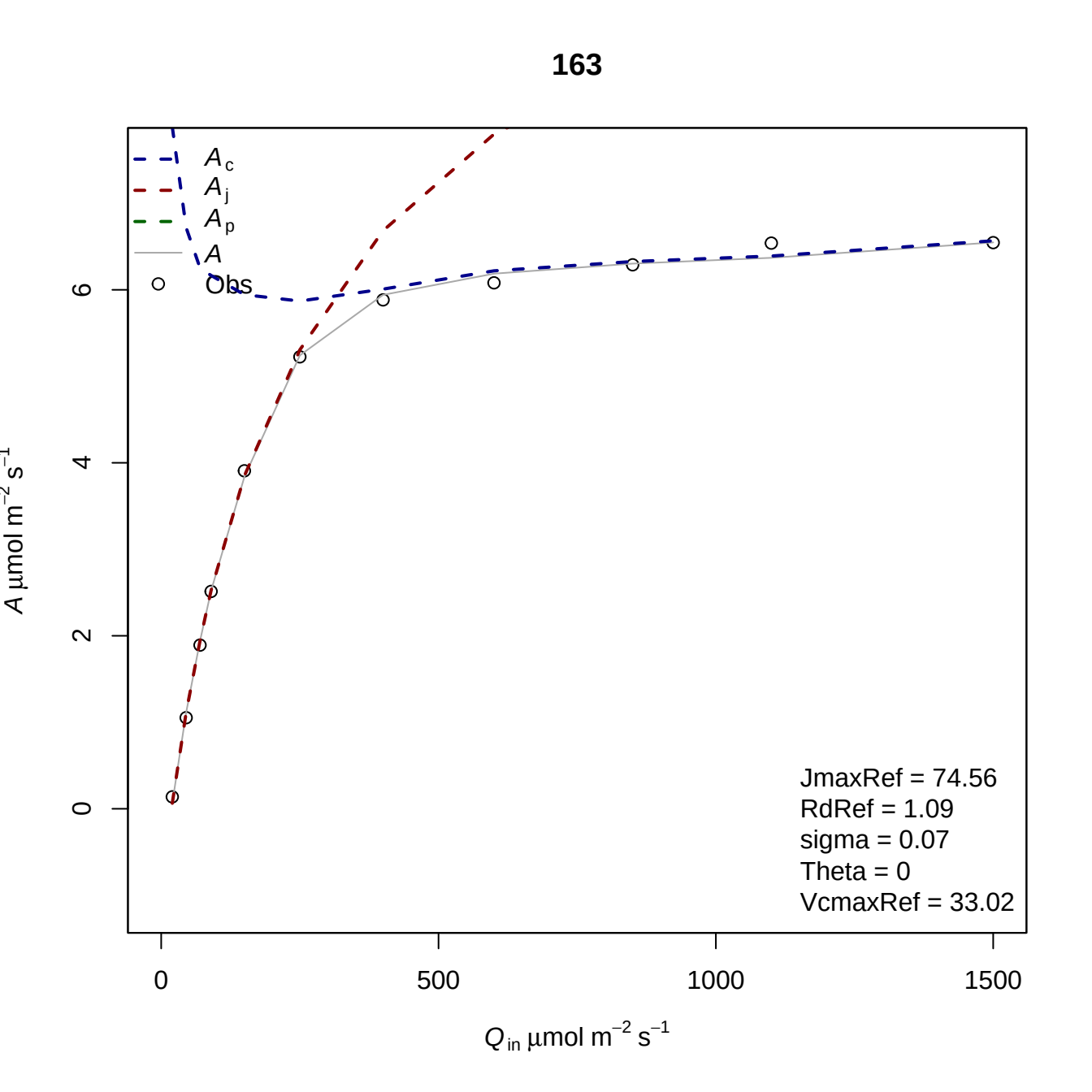

154
A_c
A_j
A_p
A
Obs
JmaxRef = 139.1
RdRef = 1.29
sigma = 0.19
Theta = 0
VcmaxRef = 66.48
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

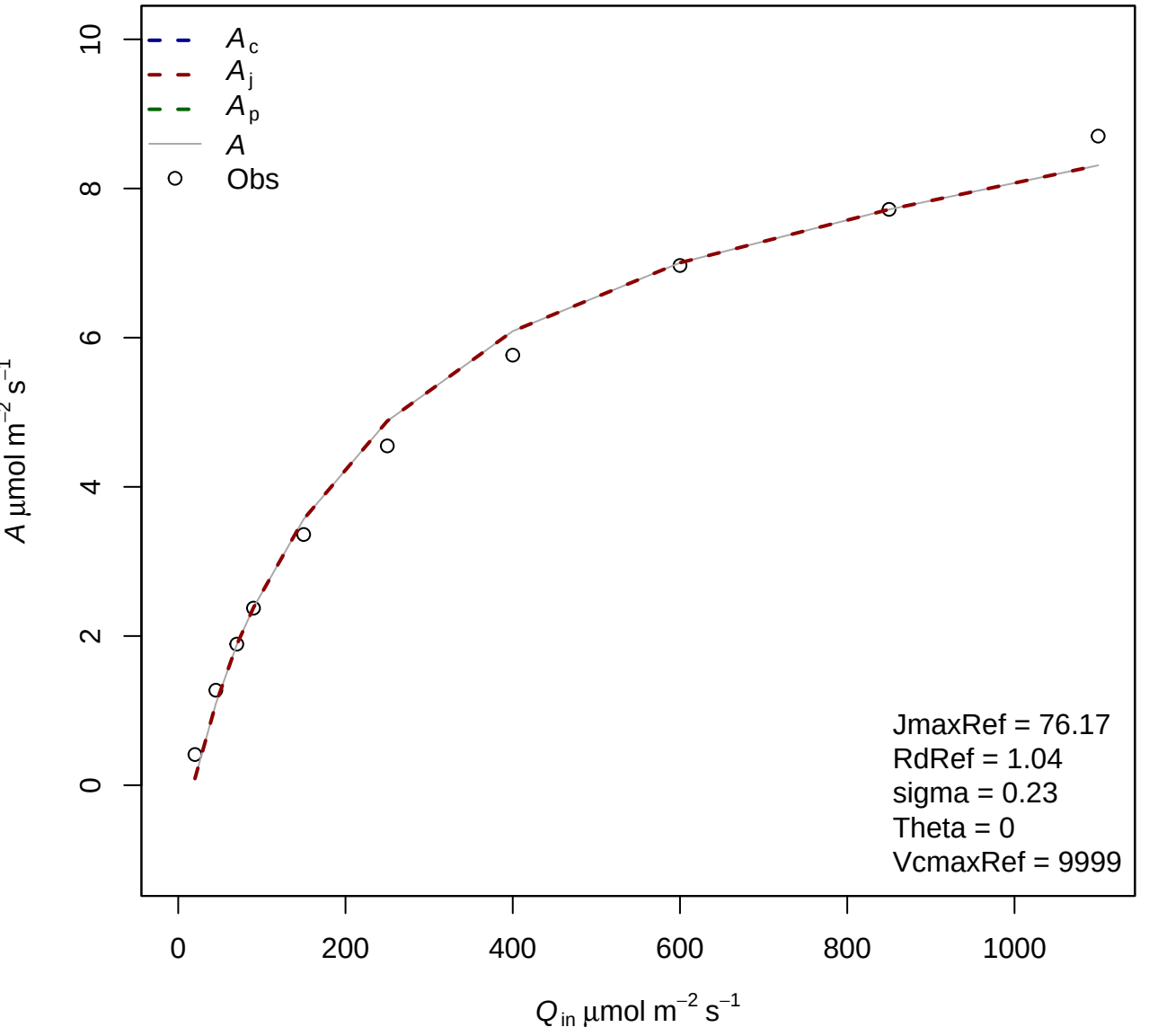

155
A_c
A_j
A_p
A
Obs
JmaxRef = 136.48
RdRef = 0.93
sigma = 0.31
Theta = 0
VcmaxRef = 59.16
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

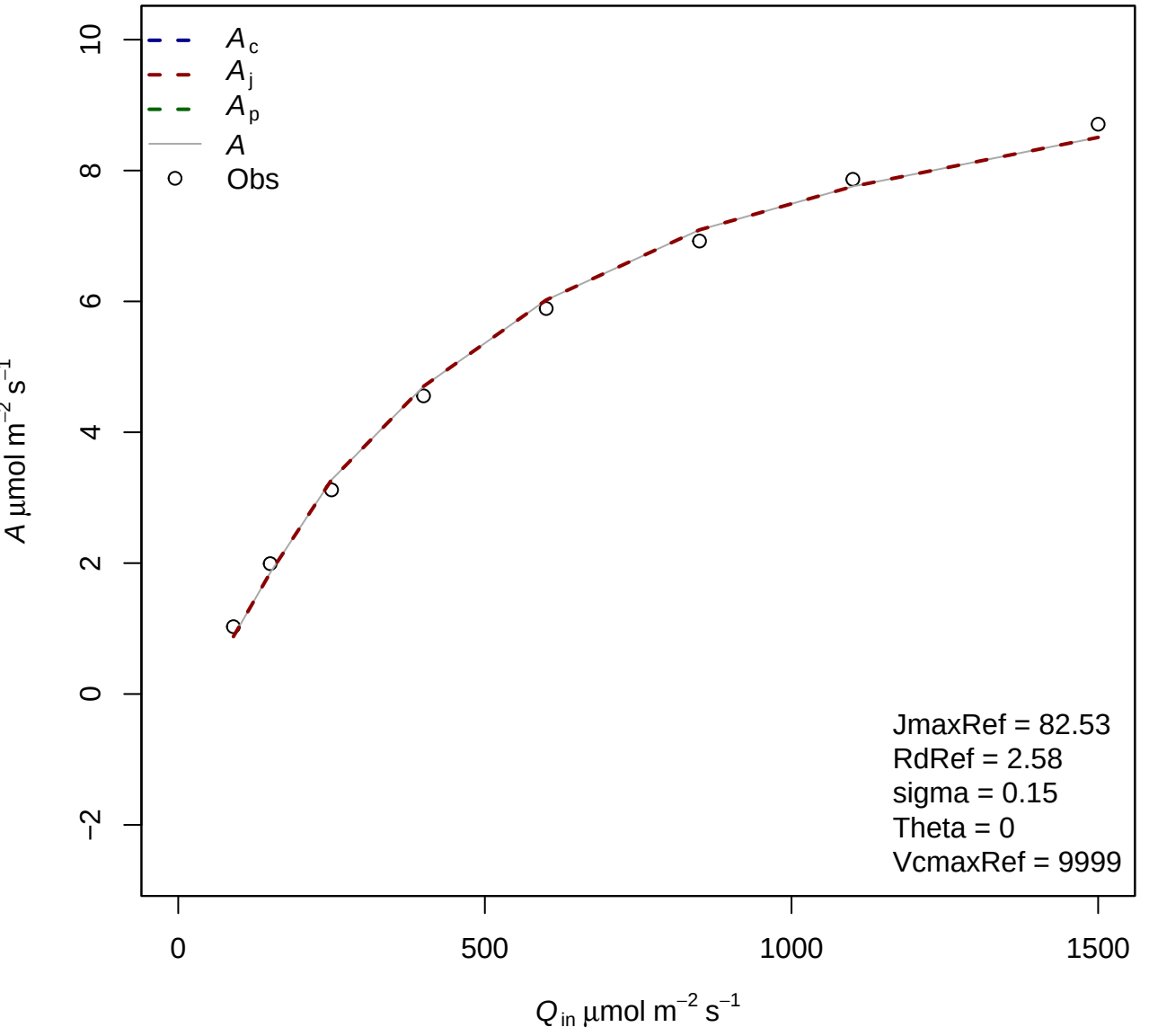

159
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 59.62
RdRef = 1.59
sigma = 0.09
Theta = 0
VcmaxRef = 40.2

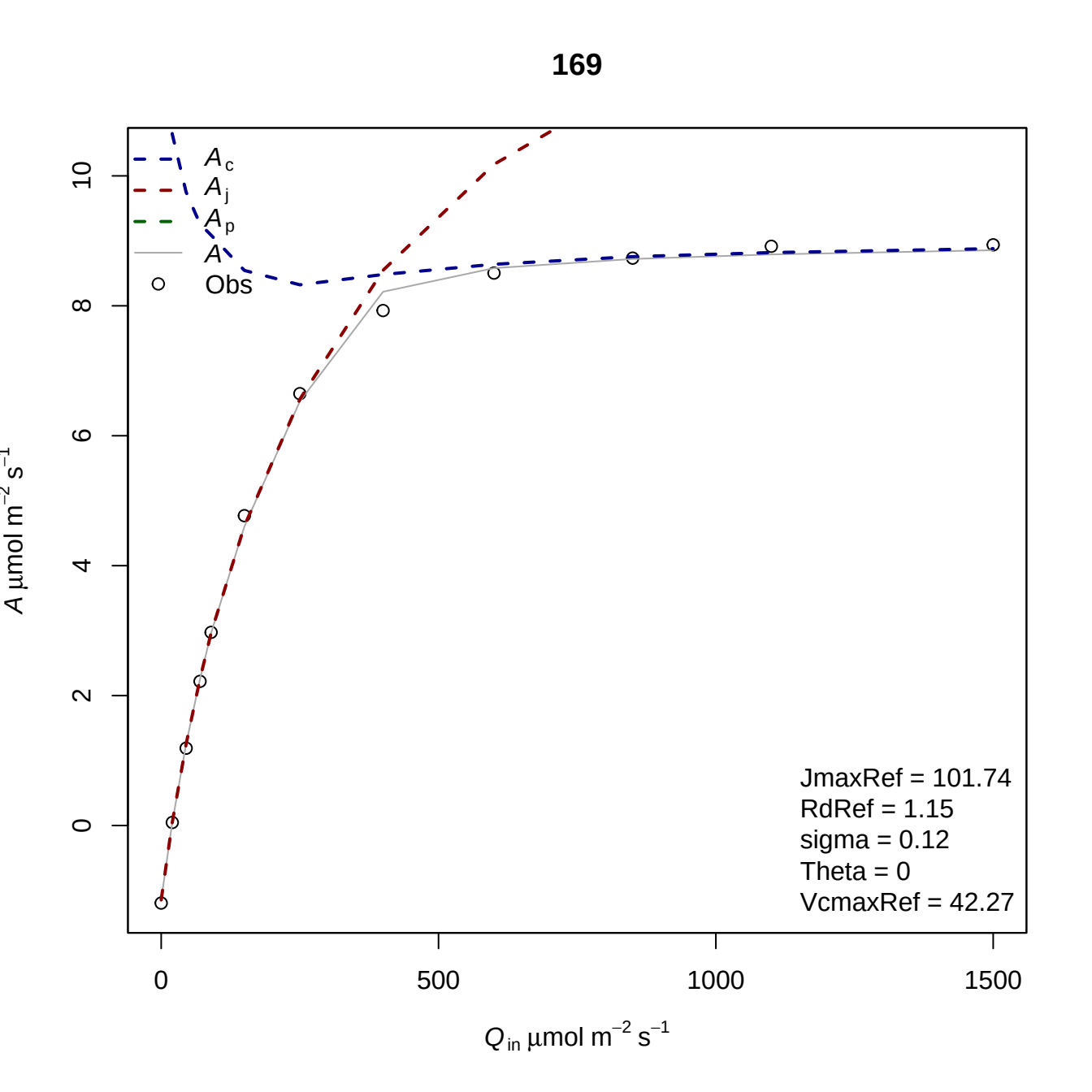

160
A_c
A_j
A_p
A
Obs
JmaxRef = 50.33
RdRef = 0.42
sigma = 0.16
Theta = 0
VcmaxRef = 32.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

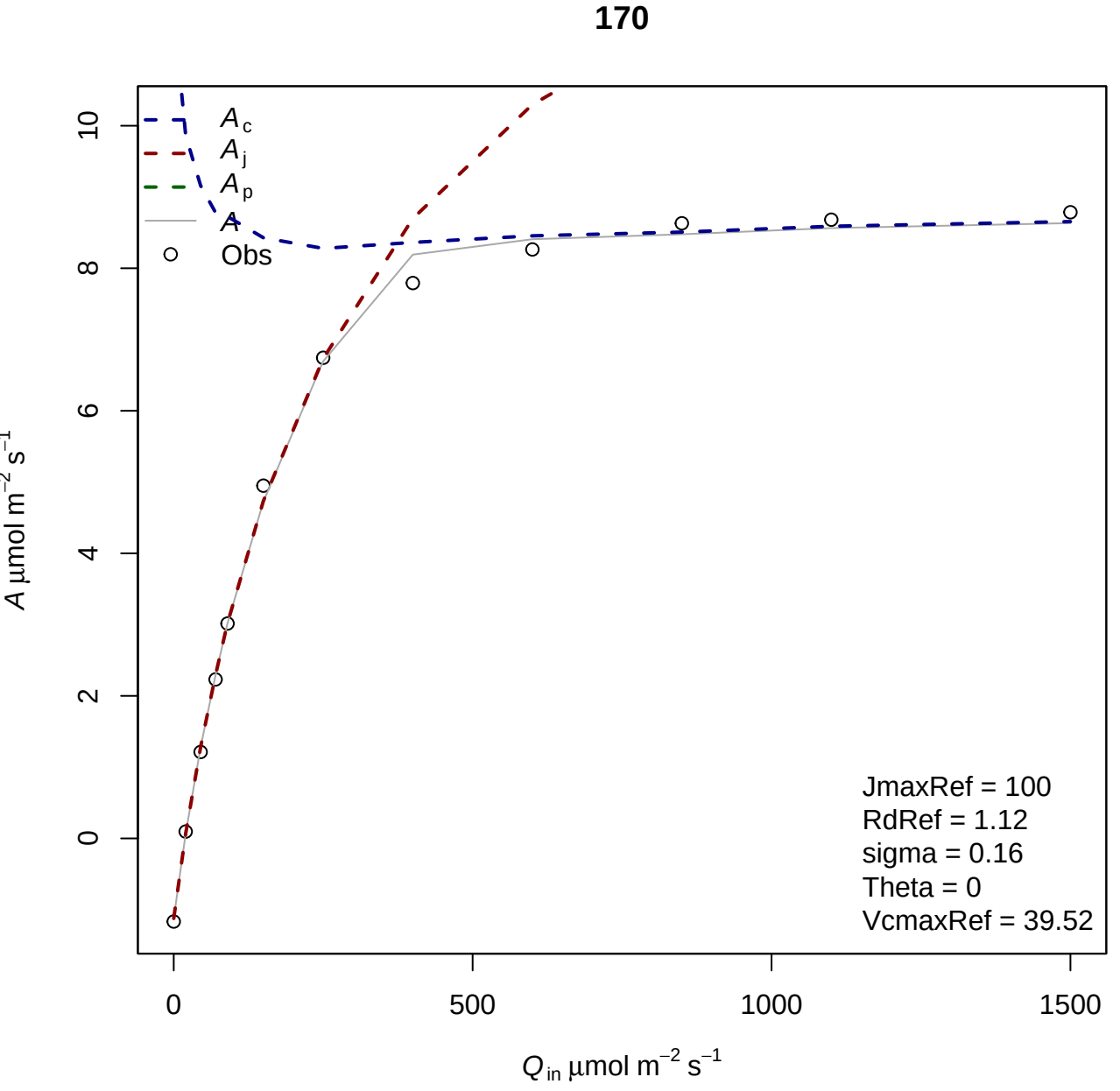

162
A_c
A_j
A_p
A
Obs
JmaxRef = 36.06
RdRef = 1.73
sigma = 0.07
Theta = 0
VcmaxRef = 9999
4
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

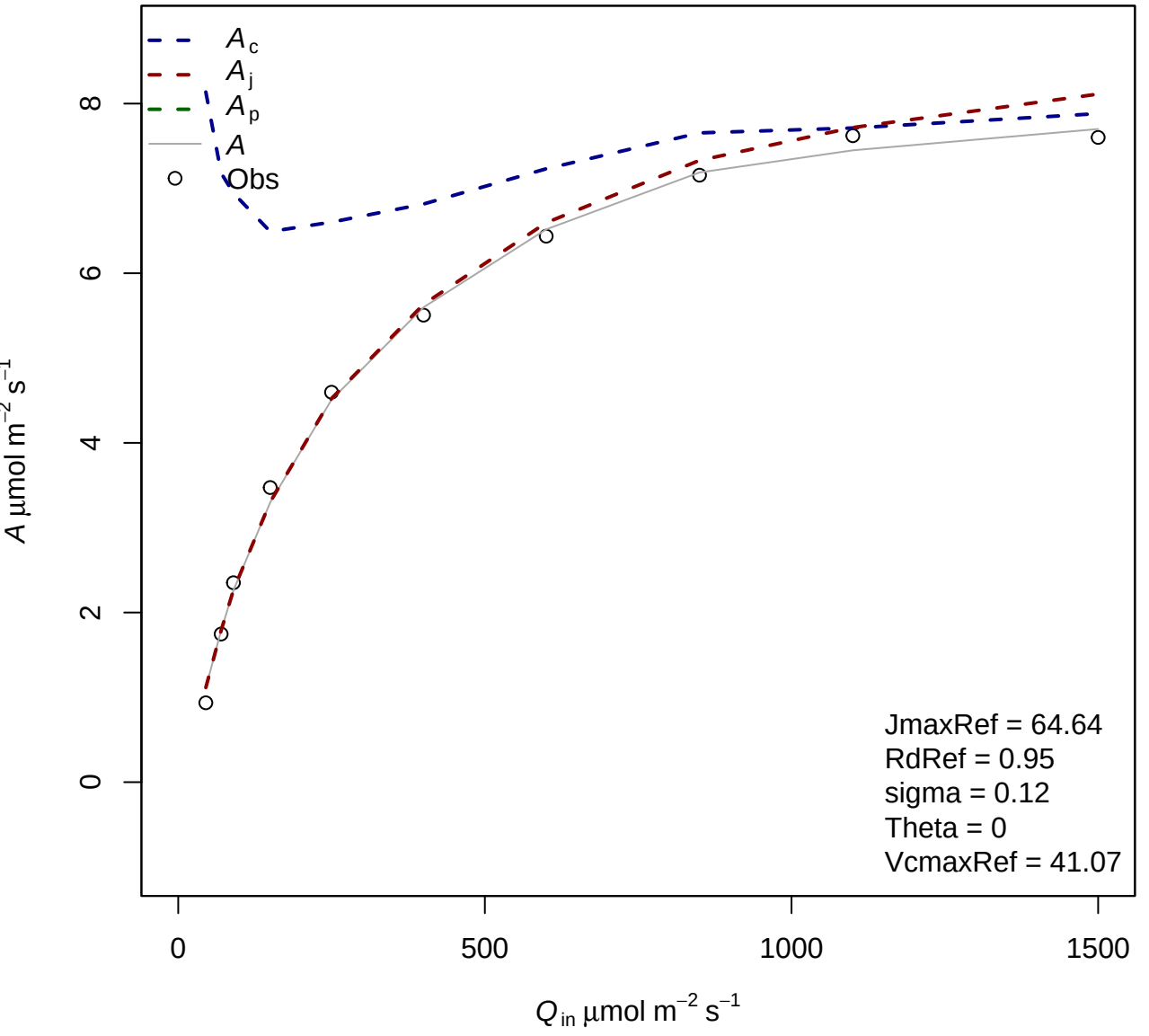

163
A_c
A_j
A_p
A
Obs
JmaxRef = 74.56
RdRef = 1.09
sigma = 0.07
Theta = 0
VcmaxRef = 33.02
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

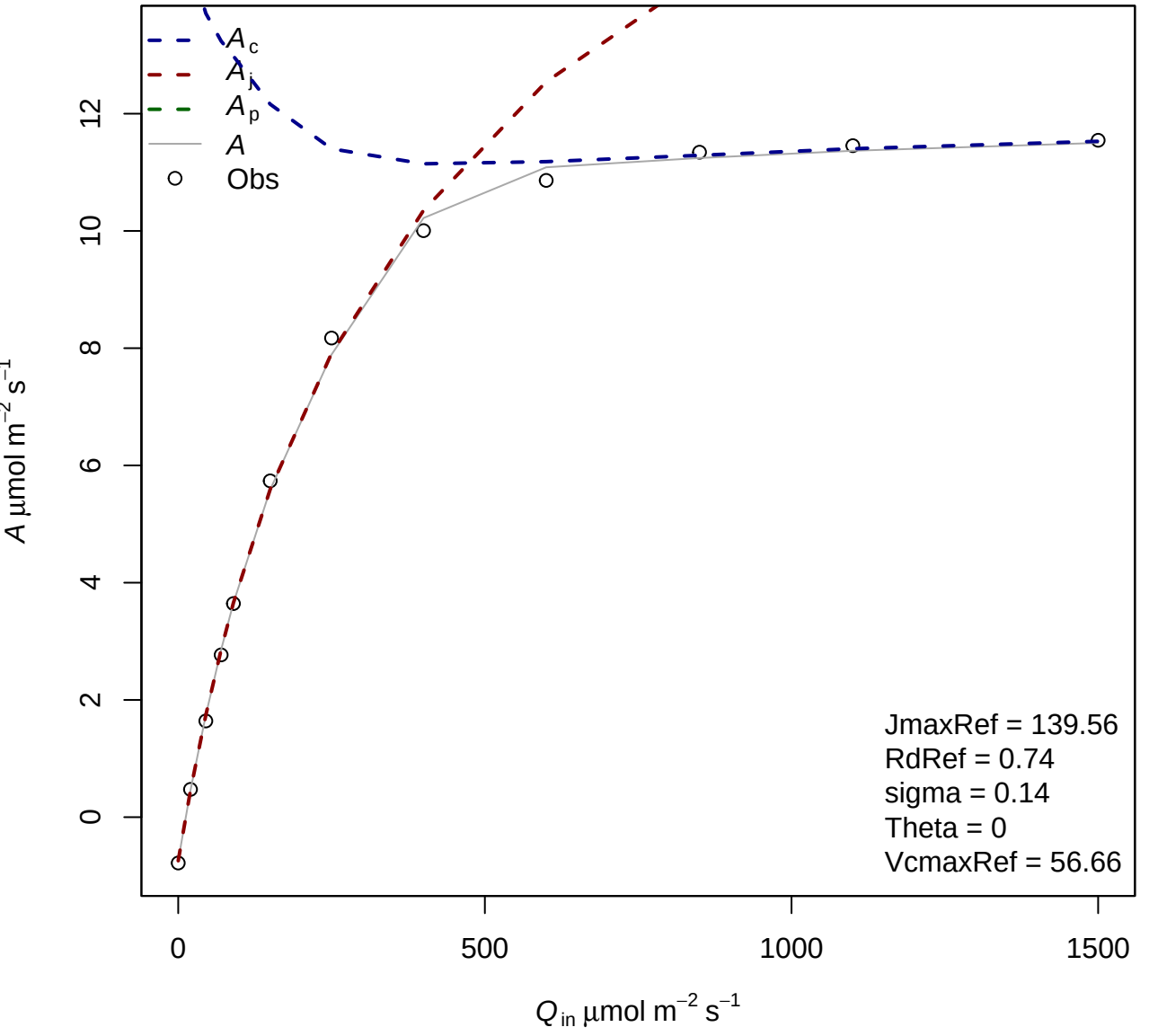

A_c
A_j
A_p
A
Obs
JmaxRef = 76.17
RdRef = 1.04
sigma = 0.23
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
10
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

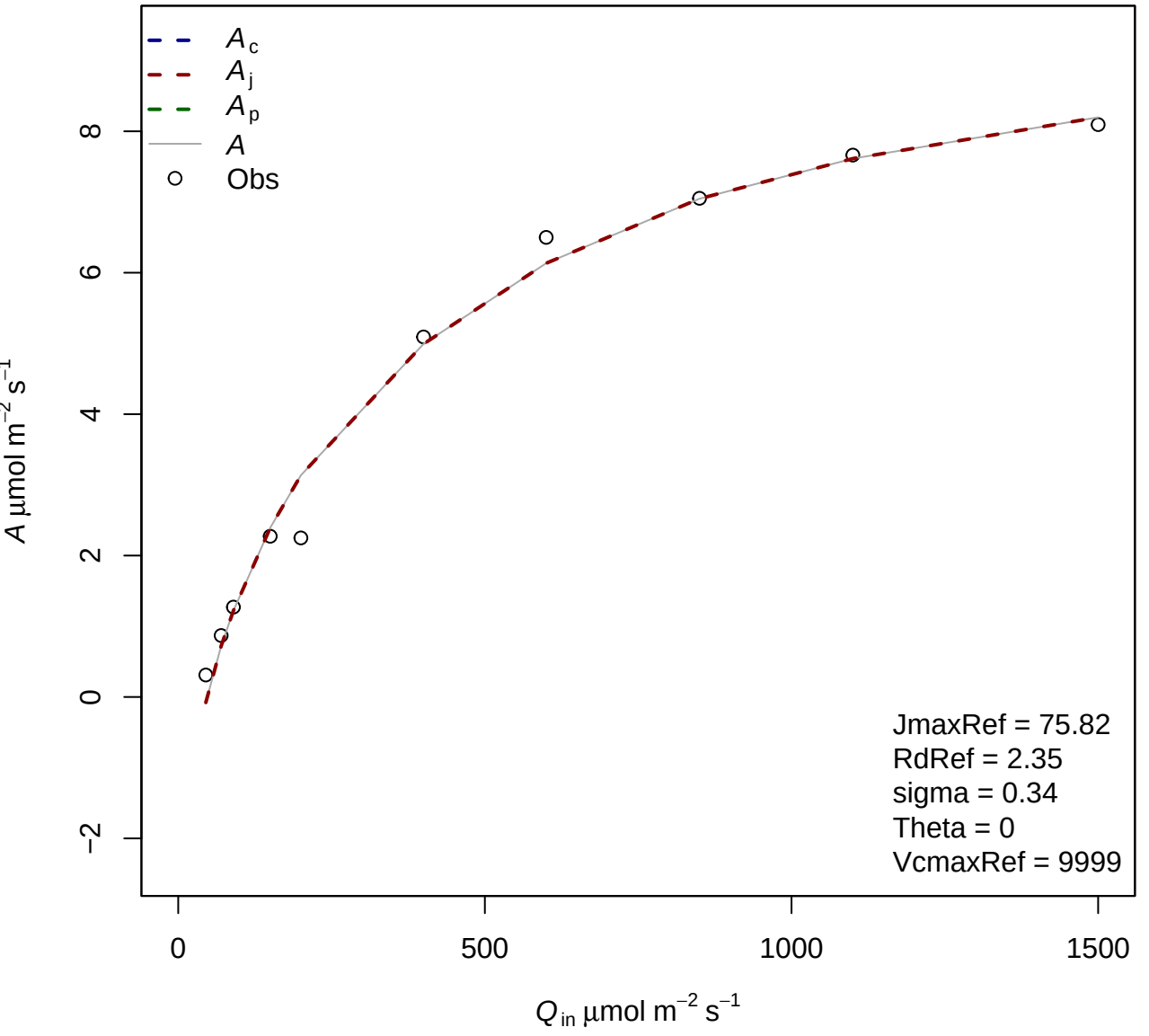

A_c
A_j
A_p
A
Obs
JmaxRef = 82.53
RdRef = 2.58
sigma = 0.15
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

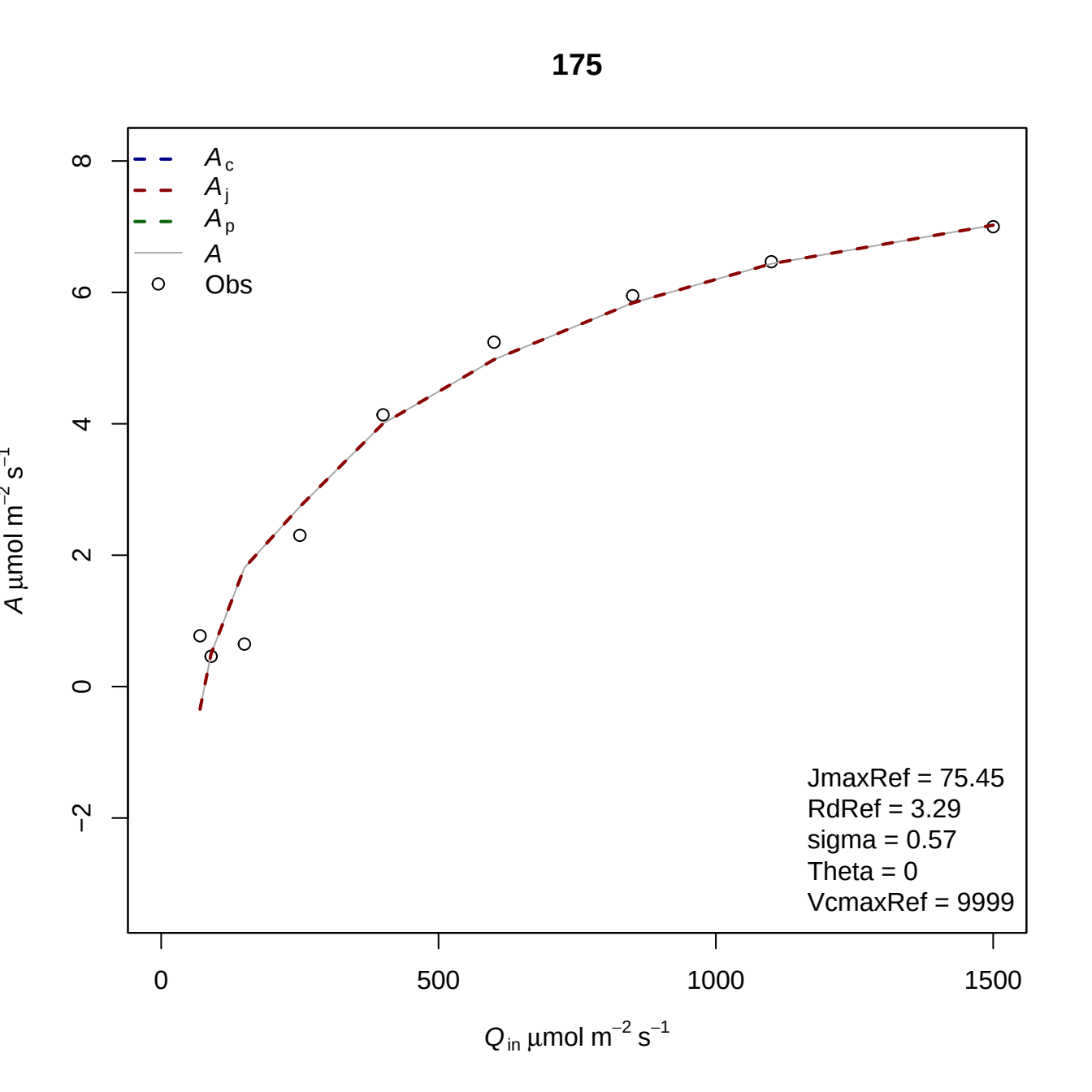

169
A_c
A_j
A_p
A
Obs
JmaxRef = 101.74
RdRef = 1.15
sigma = 0.12
Theta = 0
VcmaxRef = 42.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

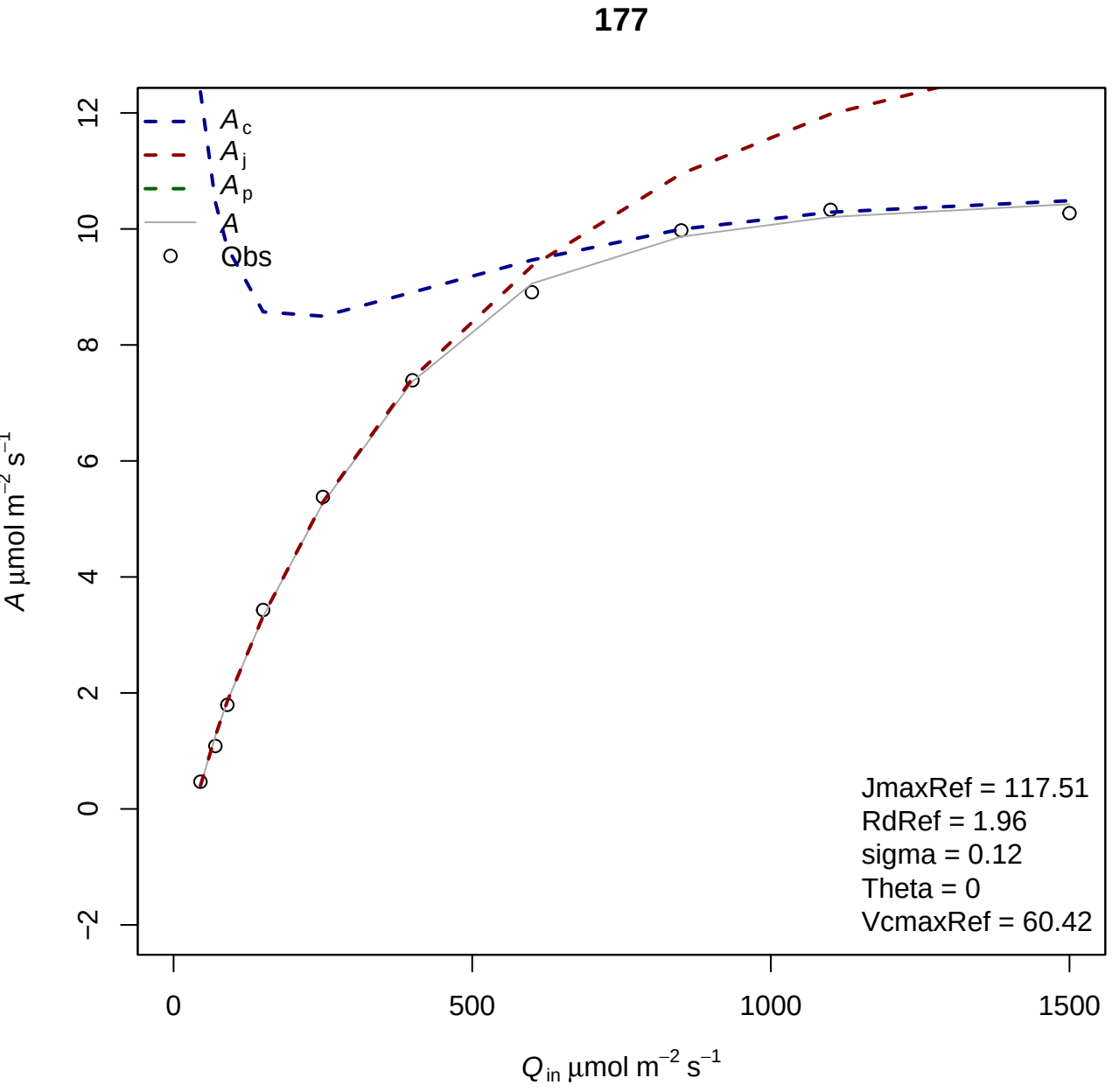

170
A_c
A_j
A_p
A
Obs
JmaxRef = 100
RdRef = 1.12
sigma = 0.16
Theta = 0
VcmaxRef = 39.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

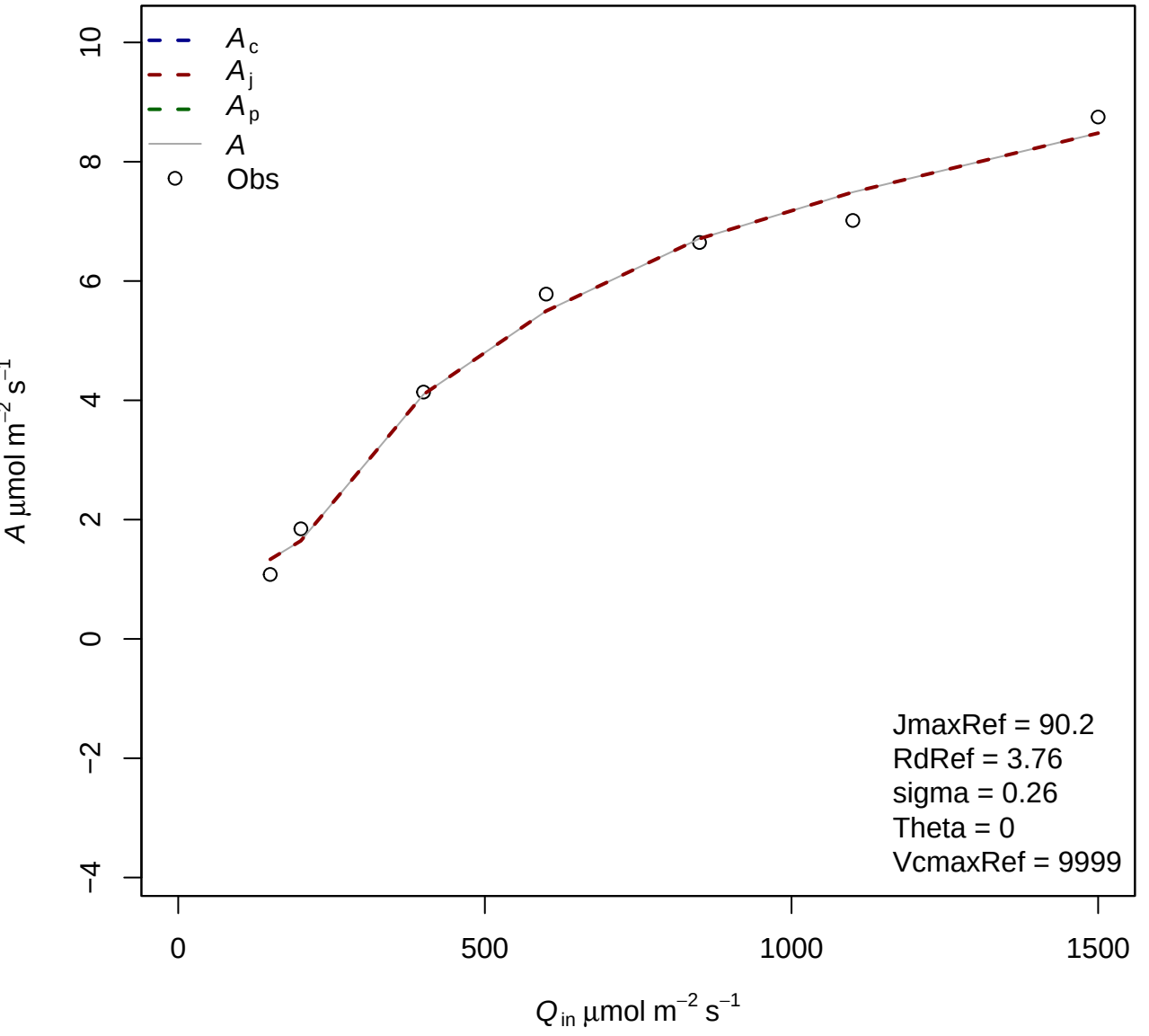

A_c
A_j
A_p
A
Obs
JmaxRef = 64.64
RdRef = 0.95
sigma = 0.12
Theta = 0
VcmaxRef = 41.07
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

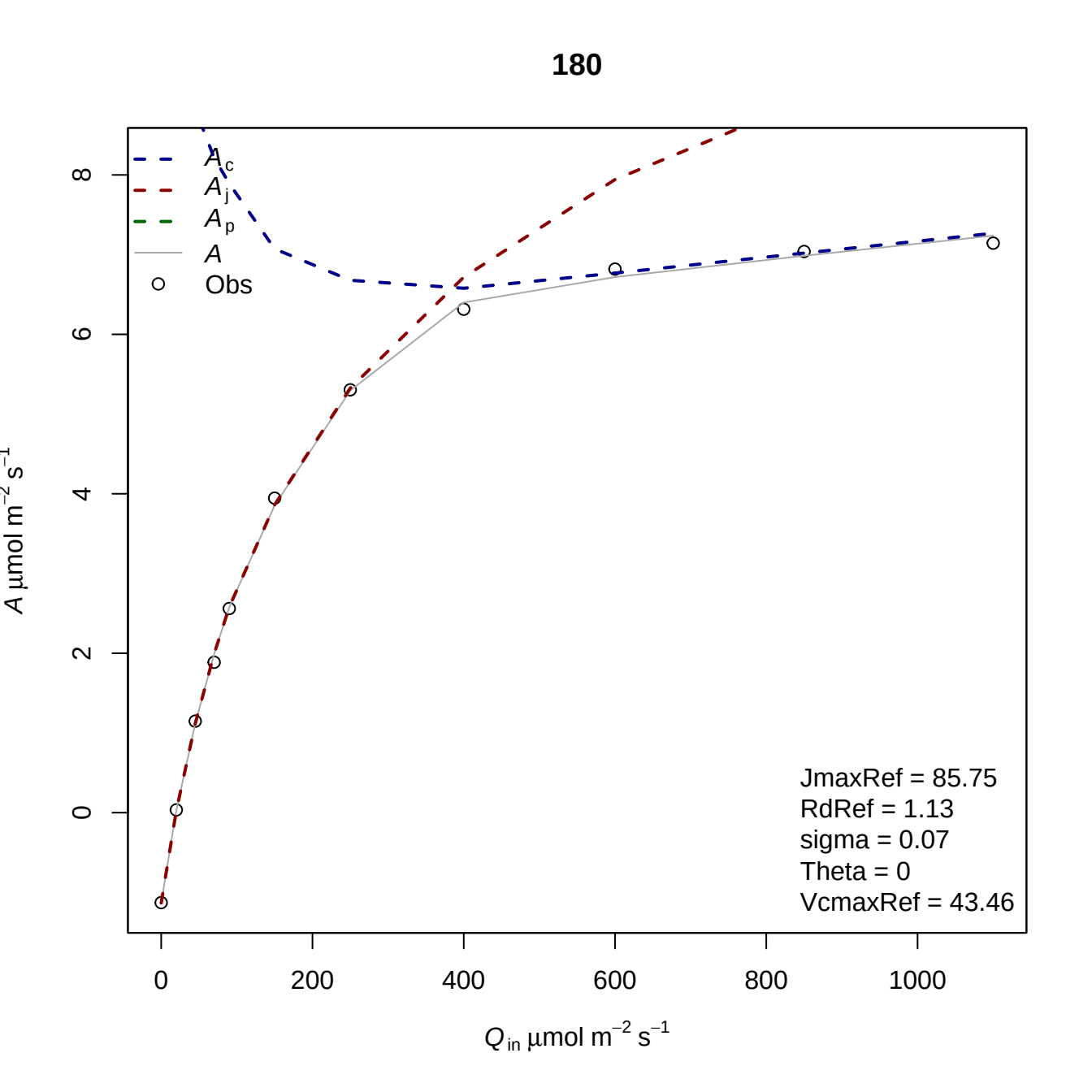

A_c
A_j
A_p
A
Obs
JmaxRef = 139.56
RdRef = 0.74
sigma = 0.14
Theta = 0
VcmaxRef = 56.66
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

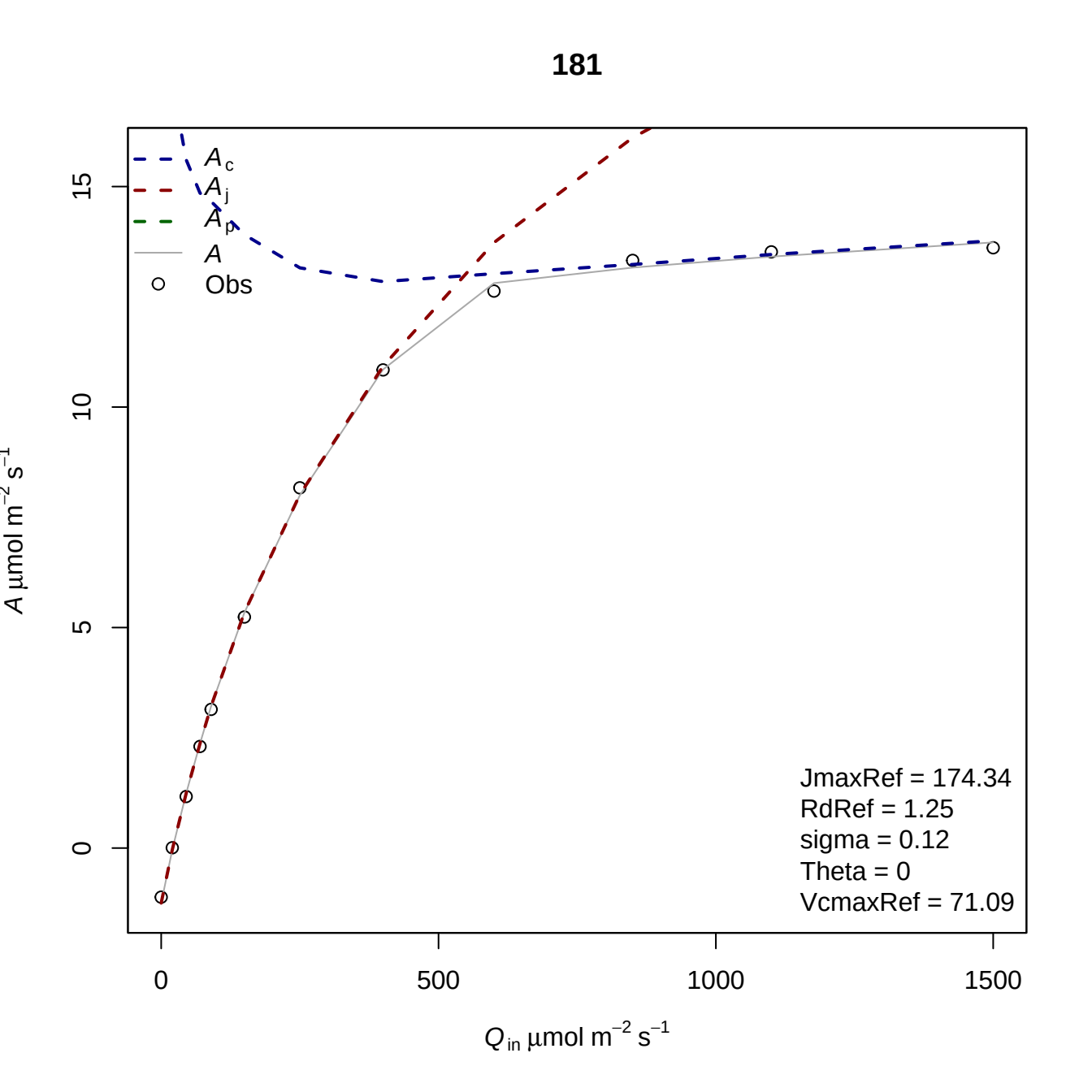

A_c
A_j
A_p
A
Obs
JmaxRef = 75.82
RdRef = 2.35
sigma = 0.34
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

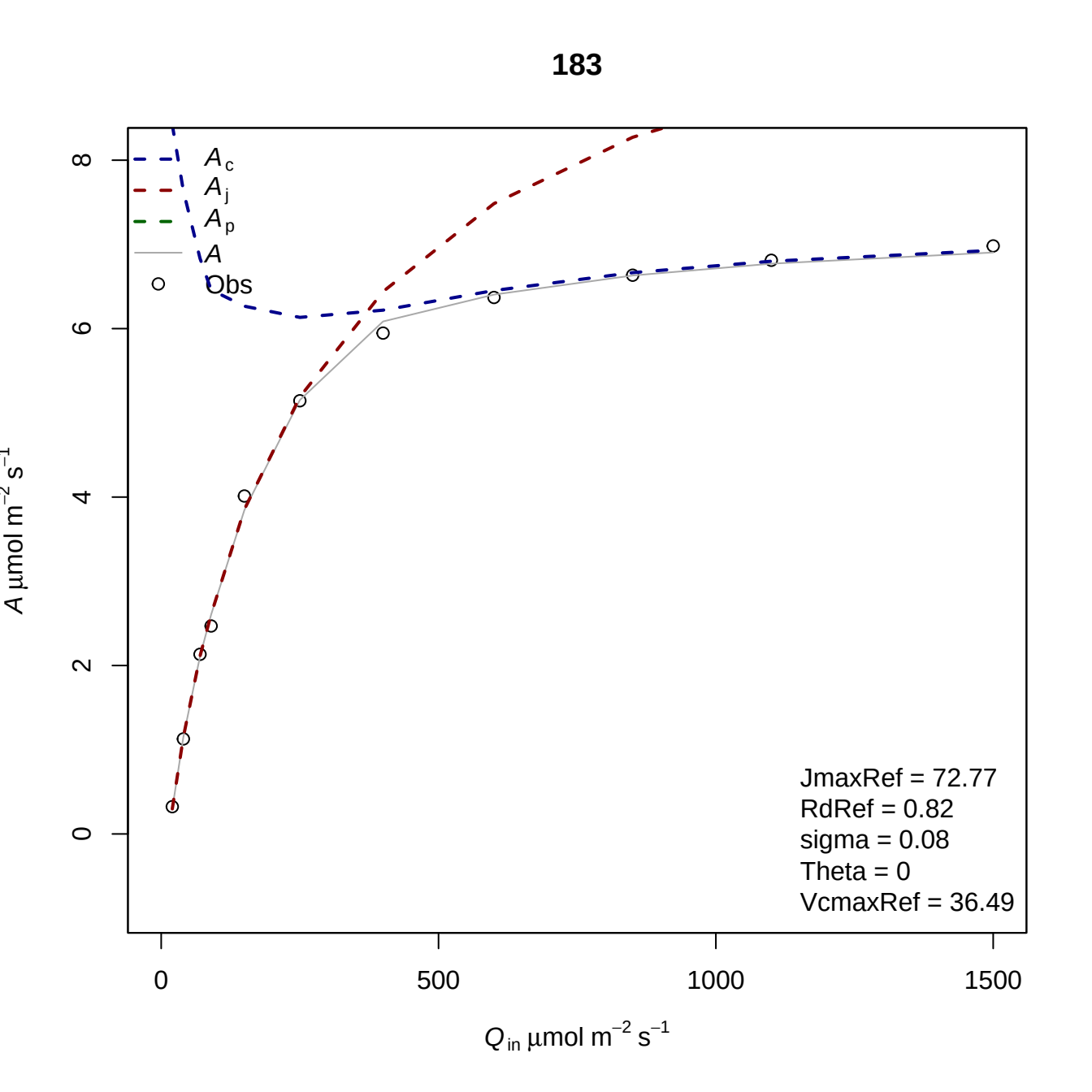

175
A_c
A_j
A_p
A
Obs
JmaxRef = 75.45
RdRef = 3.29
sigma = 0.57
Theta = 0
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

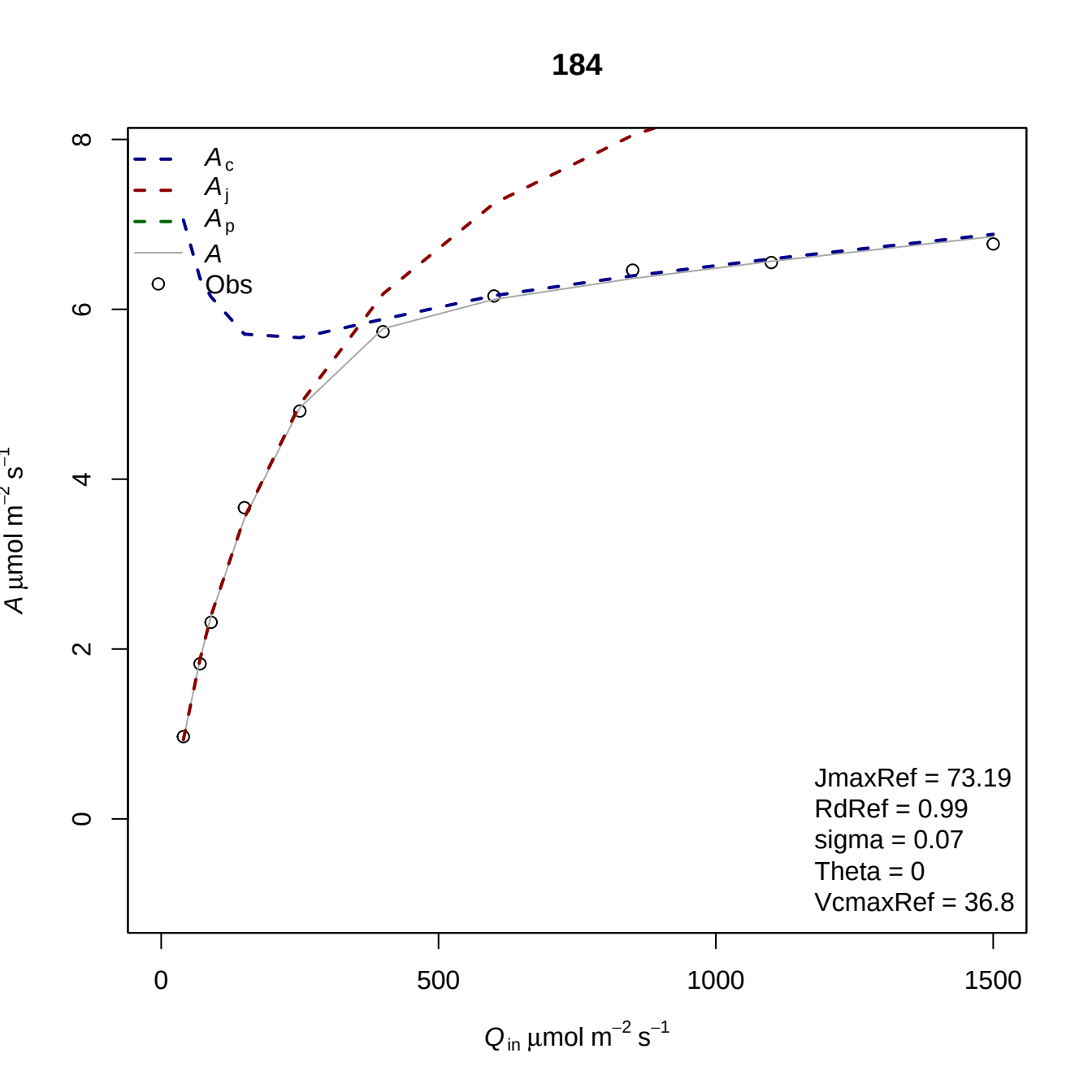

177
A_c
A_j
A_p
A
Obs
JmaxRef = 117.51
RdRef = 1.96
sigma = 0.12
Theta = 0
VcmaxRef = 60.42
12
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

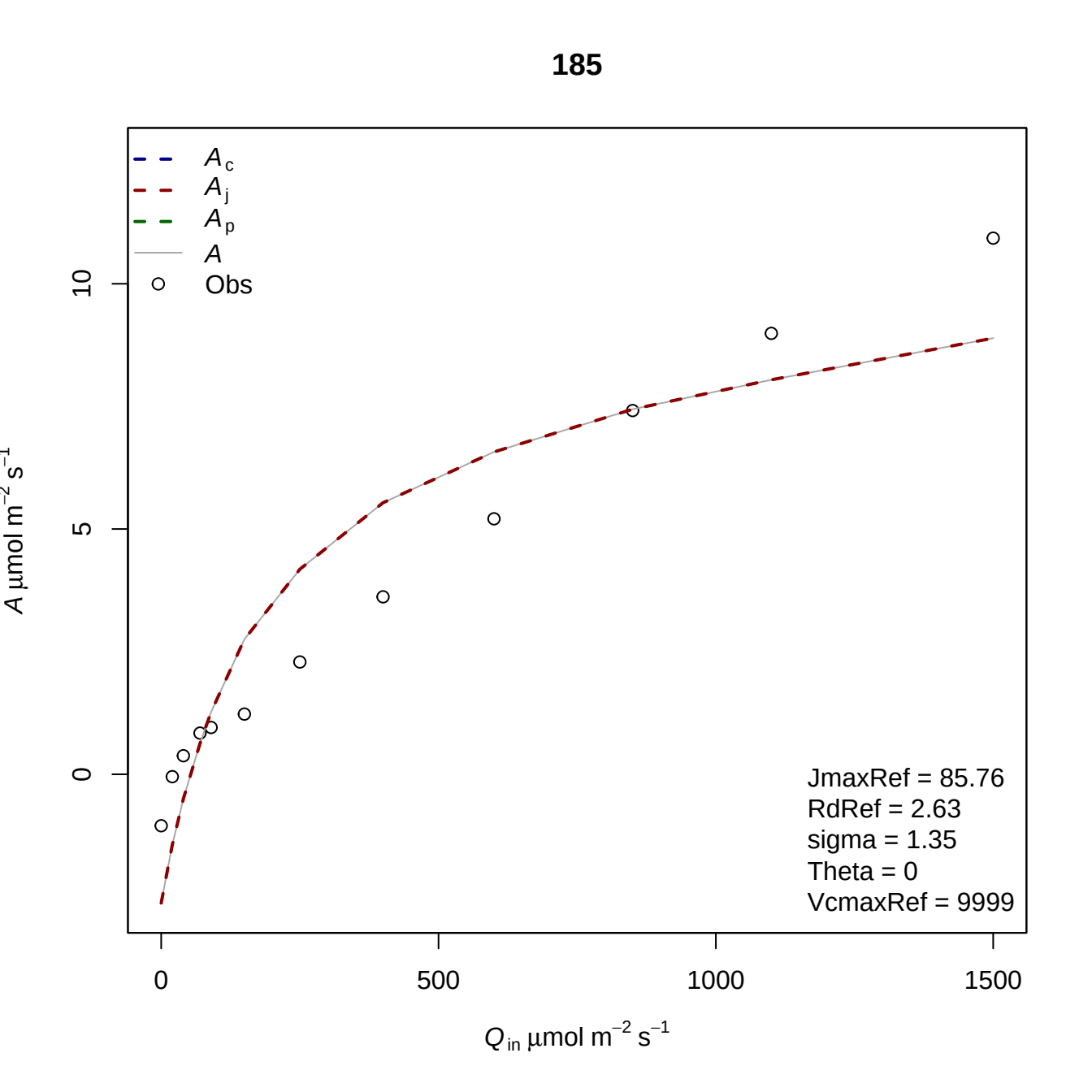

A_c
A_j
A_p
A
Obs
JmaxRef = 90.2
RdRef = 3.76
sigma = 0.26
Theta = 0
VcmaxRef = 9999
10
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

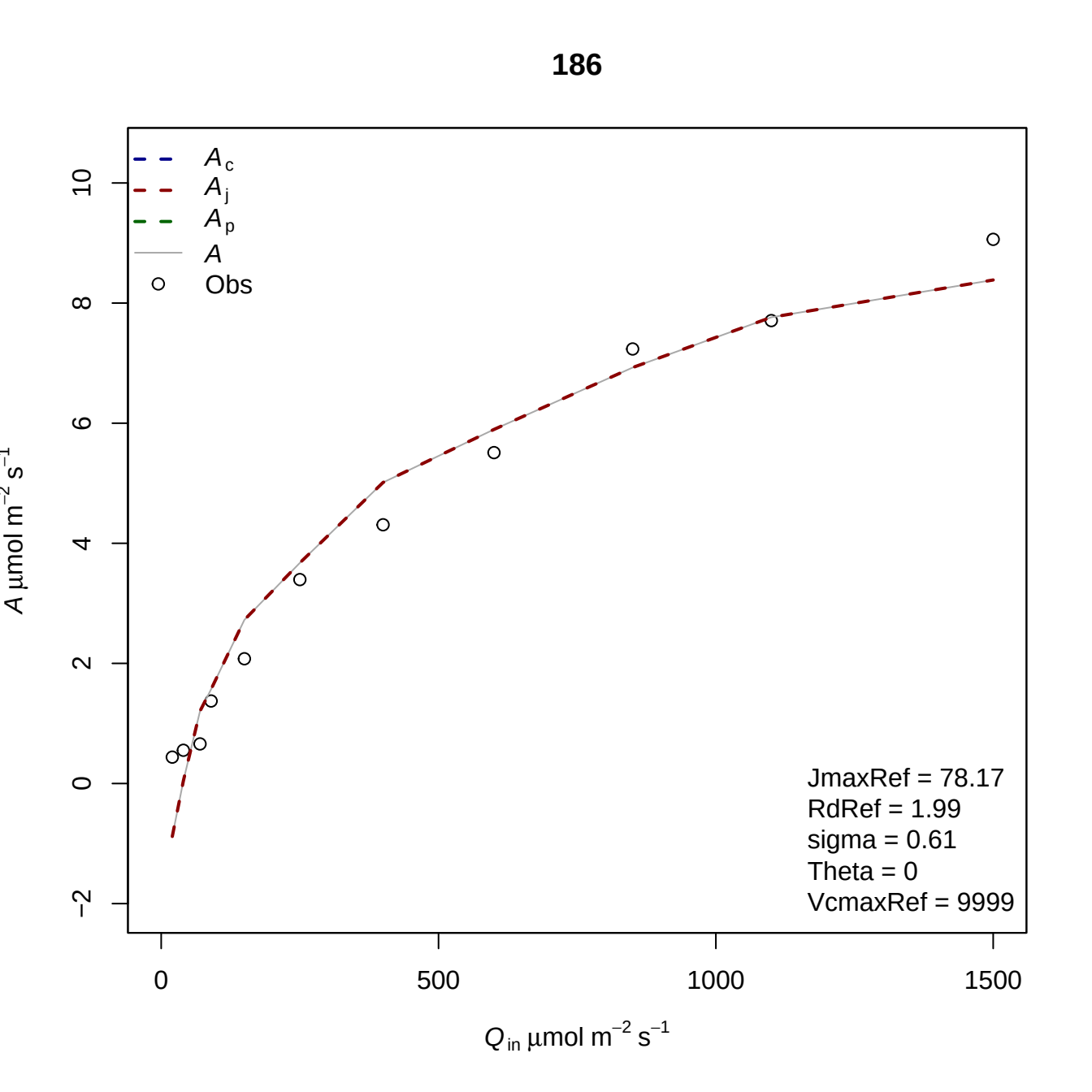

180
A_c
A_j
A_p
A
Obs
JmaxRef = 85.75
RdRef = 1.13
sigma = 0.07
Theta = 0
VcmaxRef = 43.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
200
400
600
800
1000

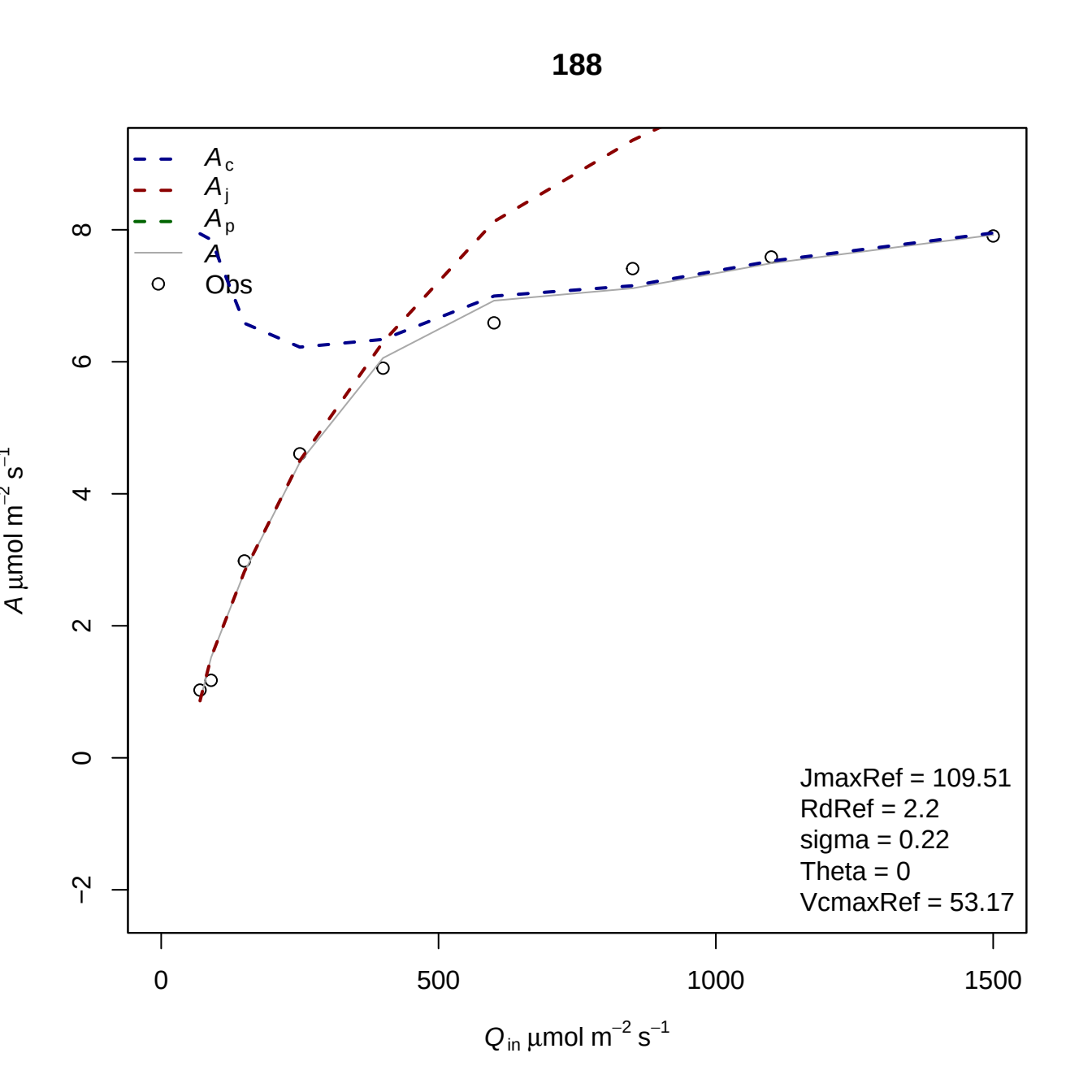

181
A_c
A_j
A_p
A
Obs
JmaxRef = 174.34
RdRef = 1.25
sigma = 0.12
Theta = 0
VcmaxRef = 71.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

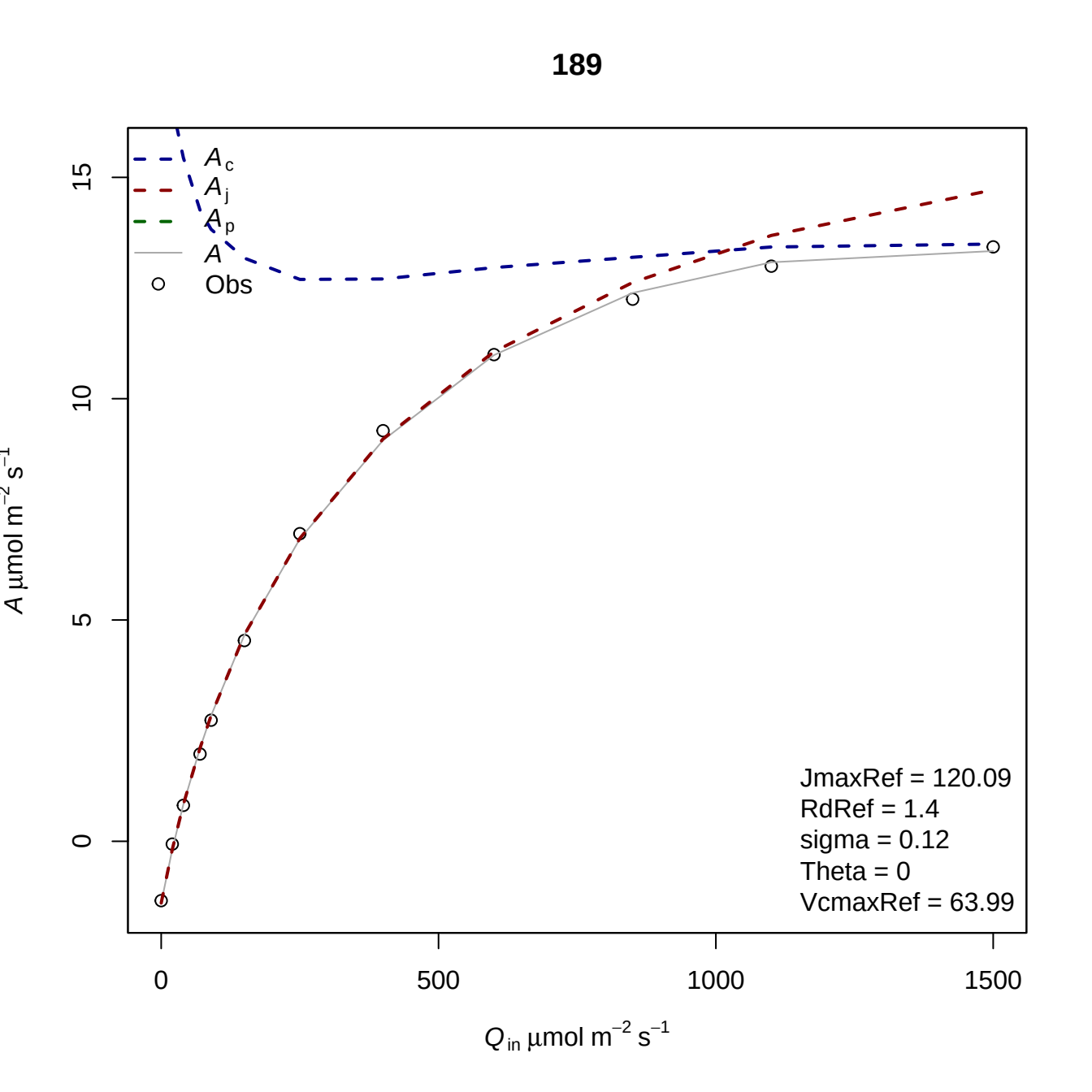

183
A_c
A_j
A_p
A
Obs
JmaxRef = 72.77
RdRef = 0.82
sigma = 0.08
Theta = 0
VcmaxRef = 36.49
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

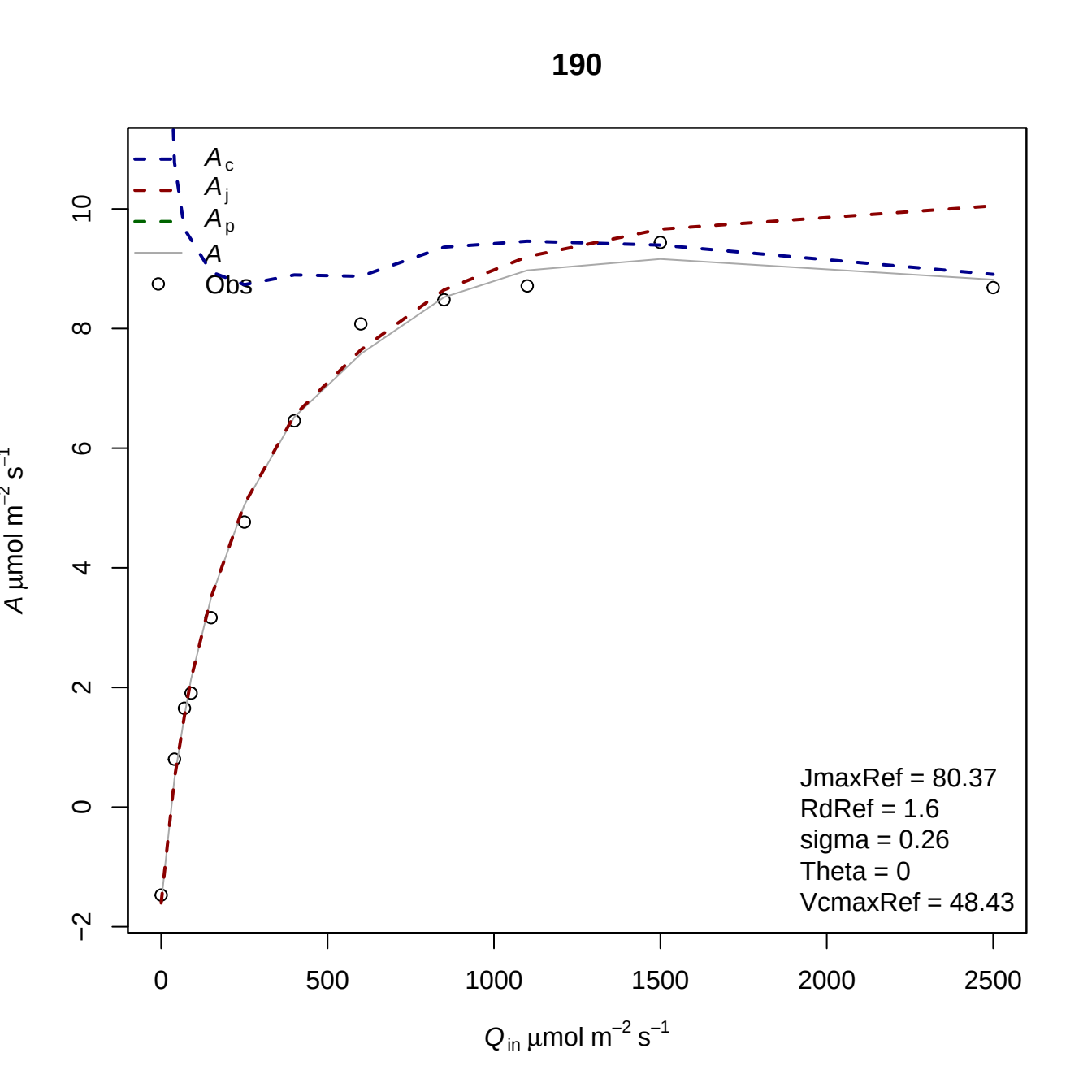

184
A_c
A_j
A_p
A
Obs
JmaxRef = 73.19
RdRef = 0.99
sigma = 0.07
Theta = 0
VcmaxRef = 36.8
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

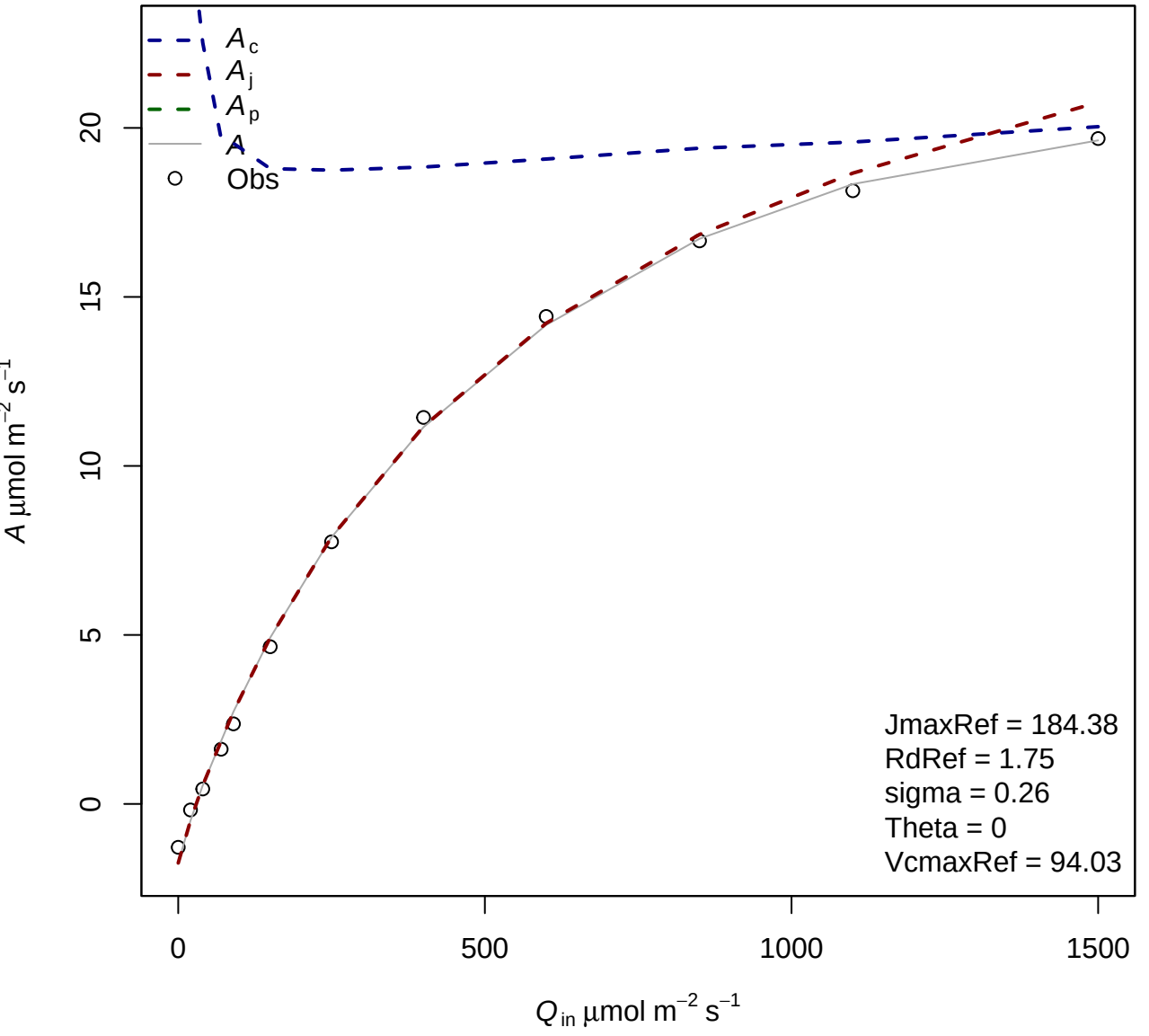

185
A_c
A_j
A_p
A
Obs
JmaxRef = 85.76
RdRef = 2.63
sigma = 1.35
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

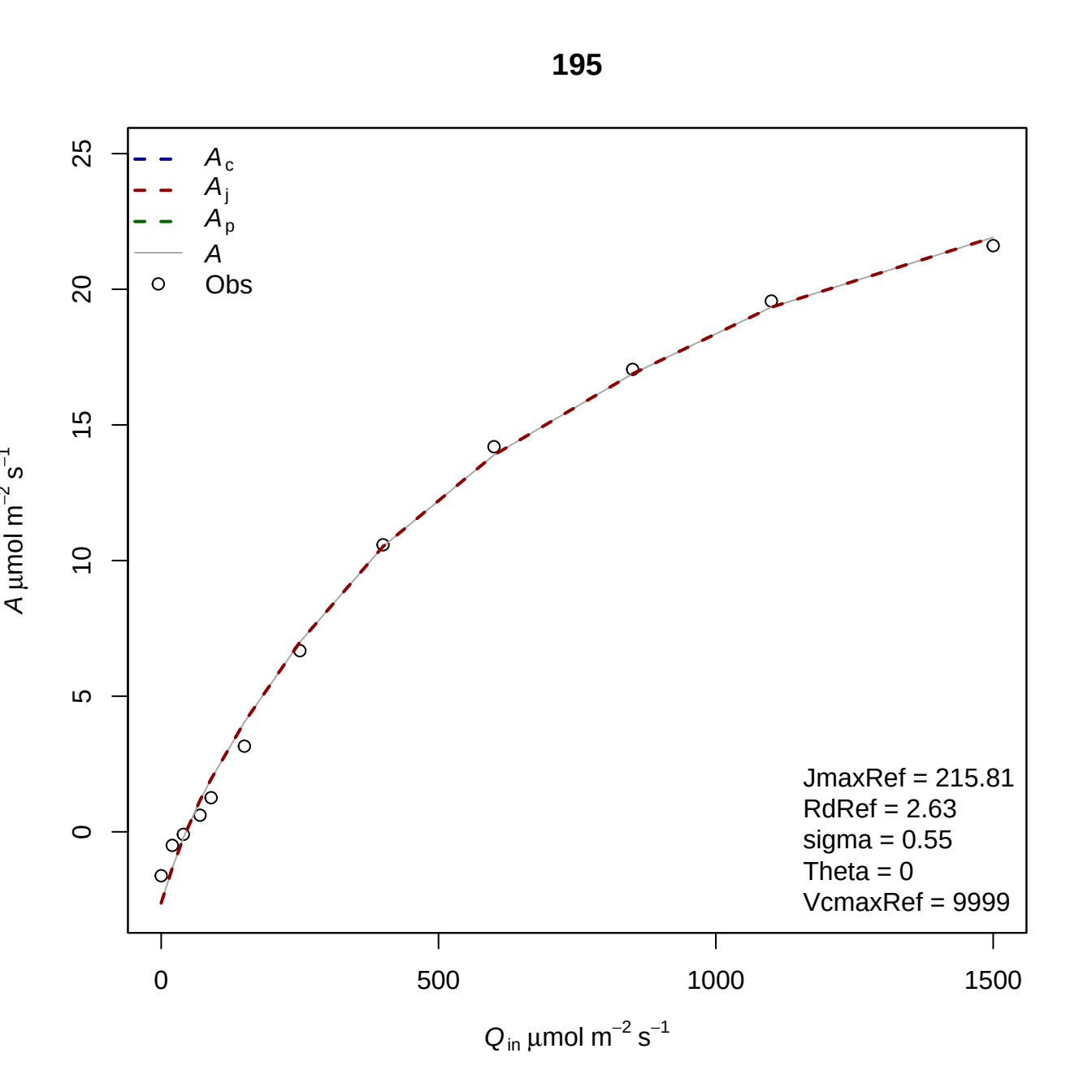

186
A_c
A_j
A_p
A
Obs
JmaxRef = 78.17
RdRef = 1.99
sigma = 0.61
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

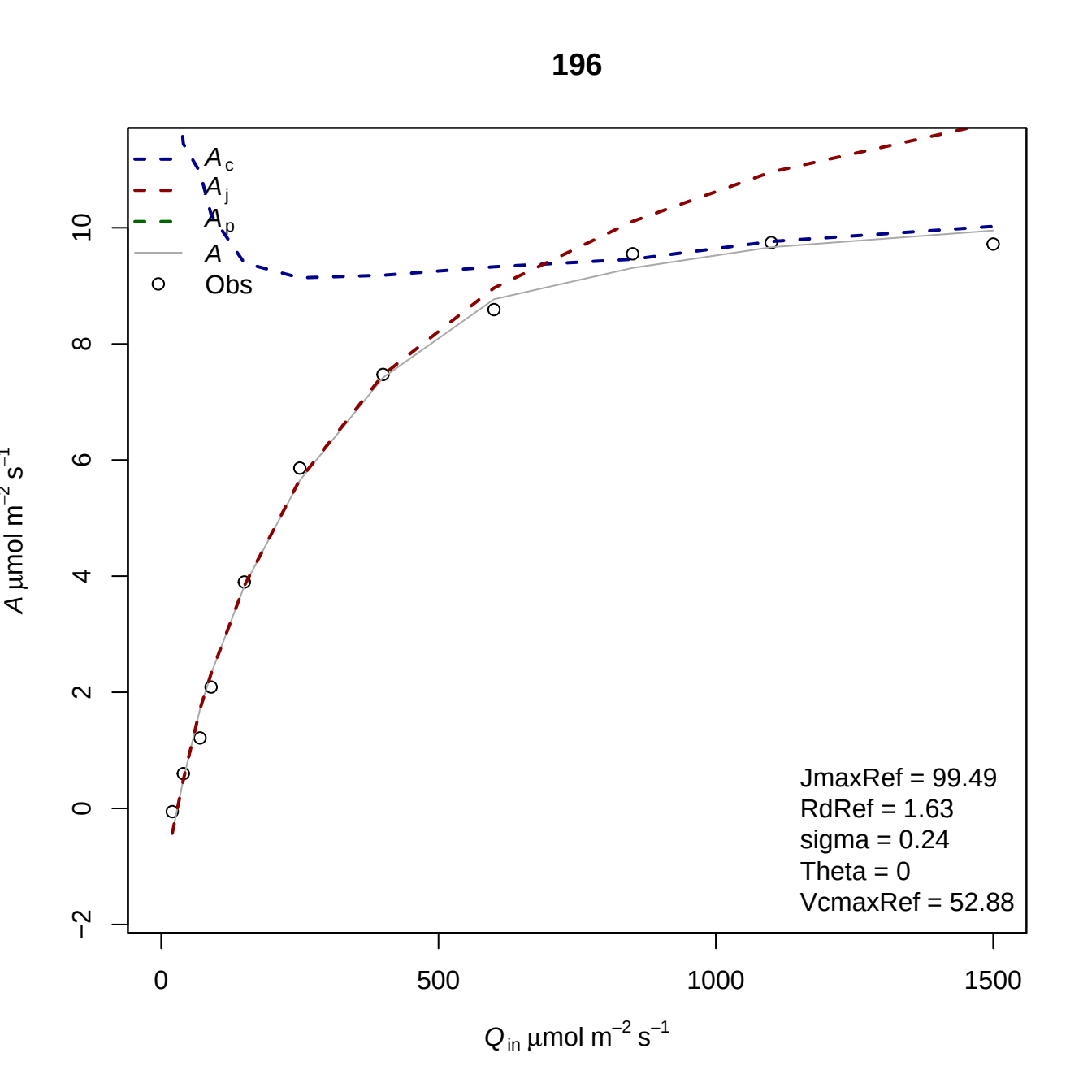

188
A_c
A_j
A_p
A
Obs
JmaxRef = 109.51
RdRef = 2.2
sigma = 0.22
Theta = 0
VcmaxRef = 53.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

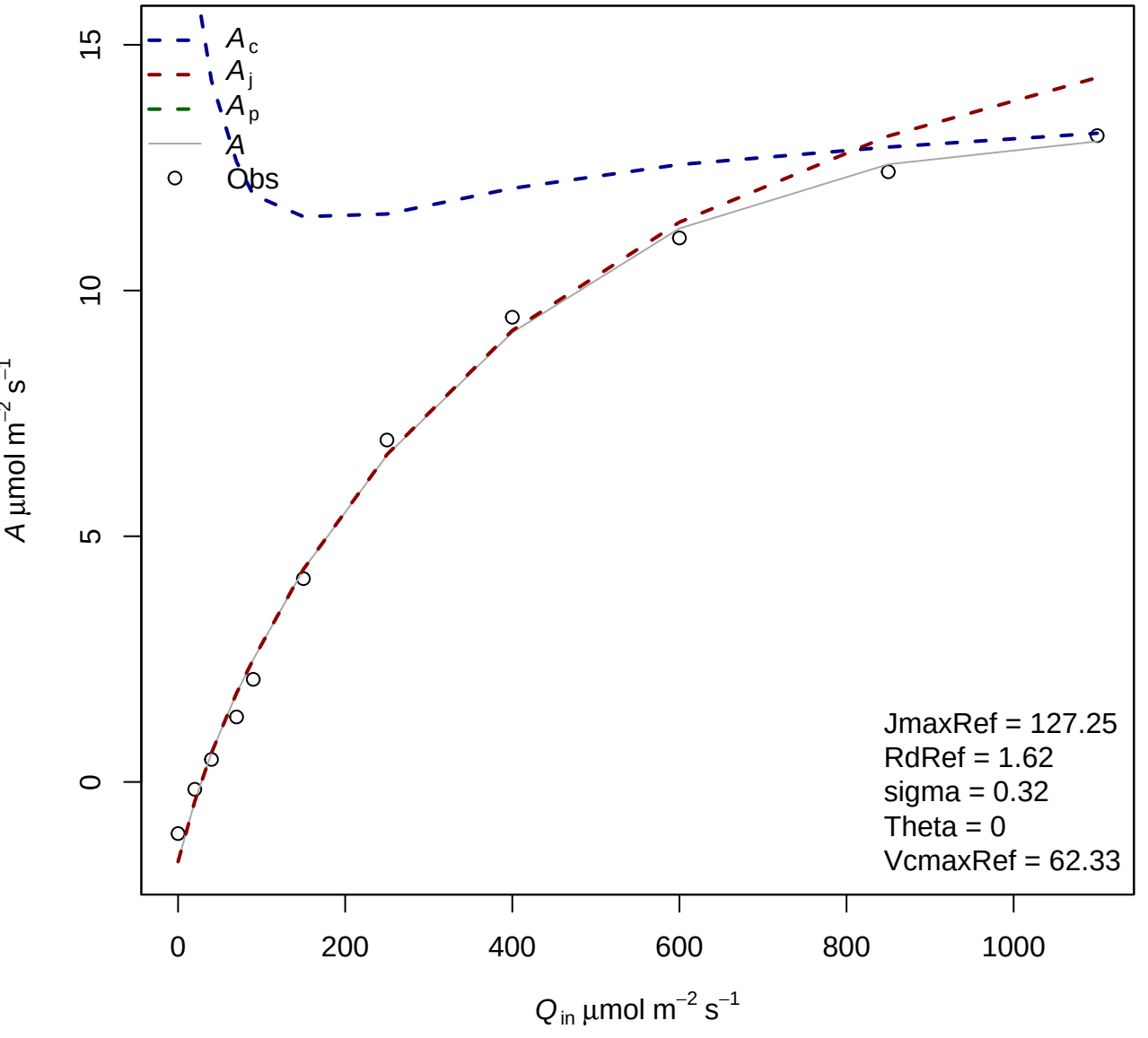

189
A_c
A_j
A_p
A
Obs
JmaxRef = 120.09
RdRef = 1.4
sigma = 0.12
Theta = 0
VcmaxRef = 63.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

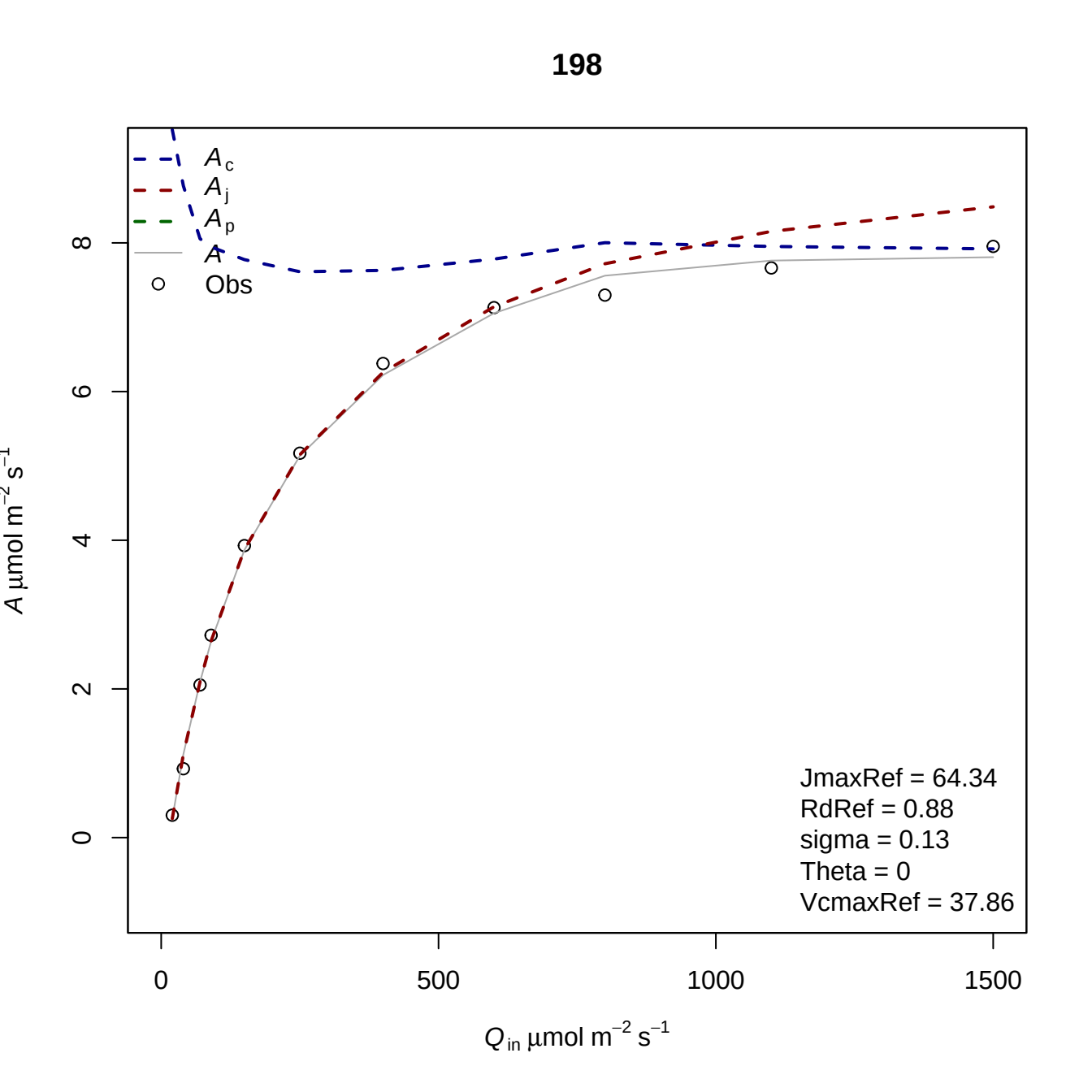

190
A_c
A_j
A_p
A
Obs
JmaxRef = 80.37
RdRef = 1.6
sigma = 0.26
Theta = 0
VcmaxRef = 48.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000
2500

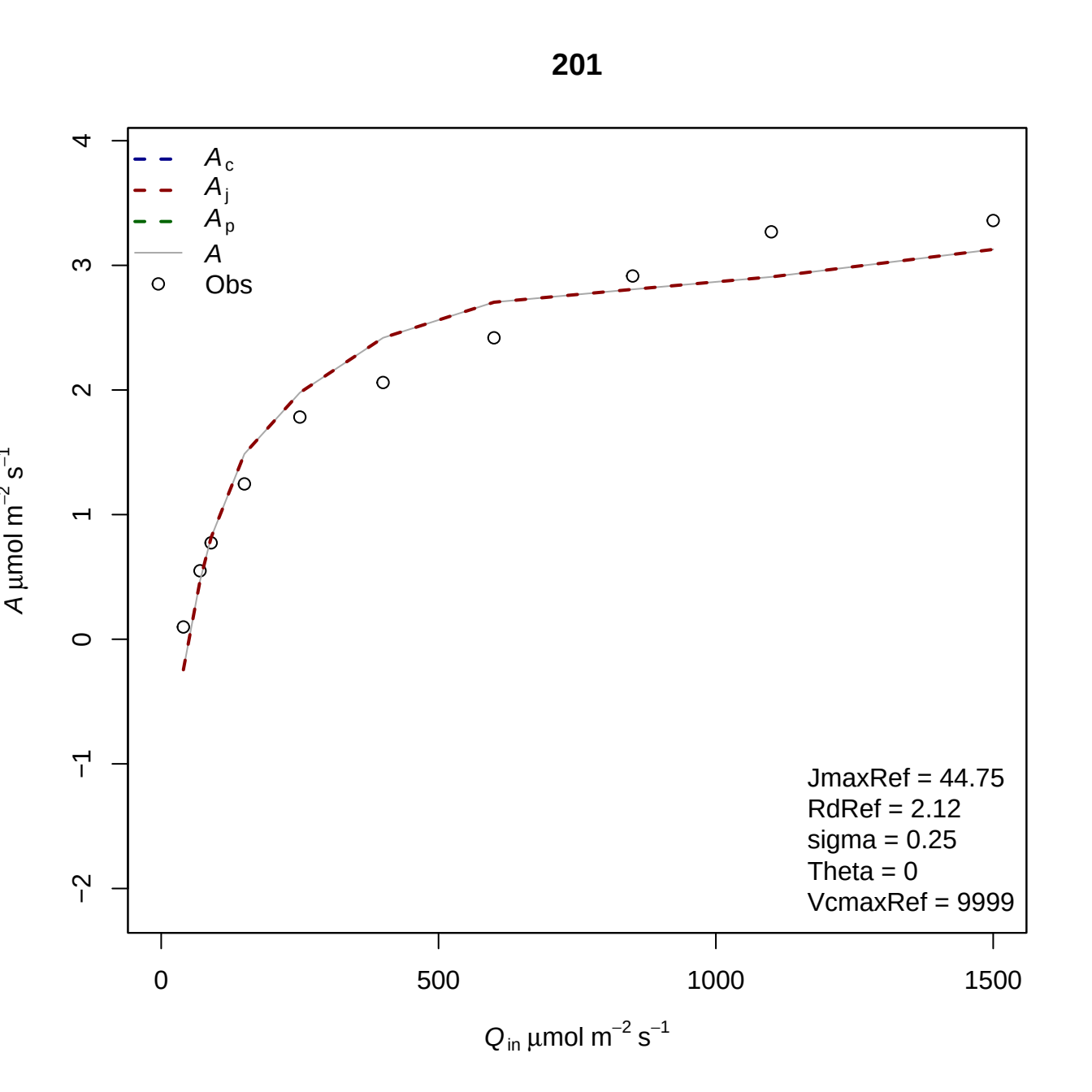

A_c
A_j
A_p
A
Obs
JmaxRef = 184.38
RdRef = 1.75
sigma = 0.26
Theta = 0
VcmaxRef = 94.03
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

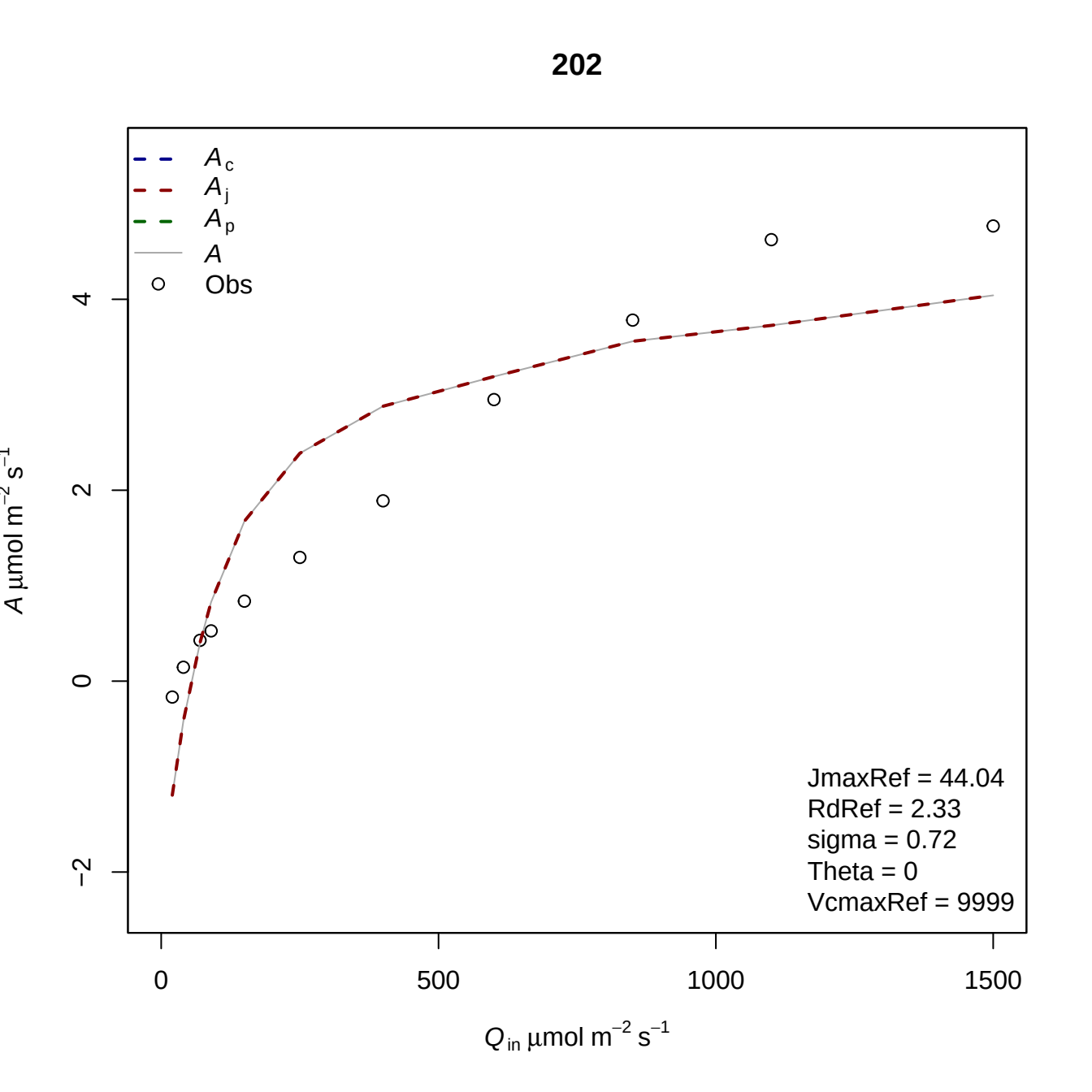

195
A_c
A_j
A_p
A
Obs
JmaxRef = 215.81
RdRef = 2.63
sigma = 0.55
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

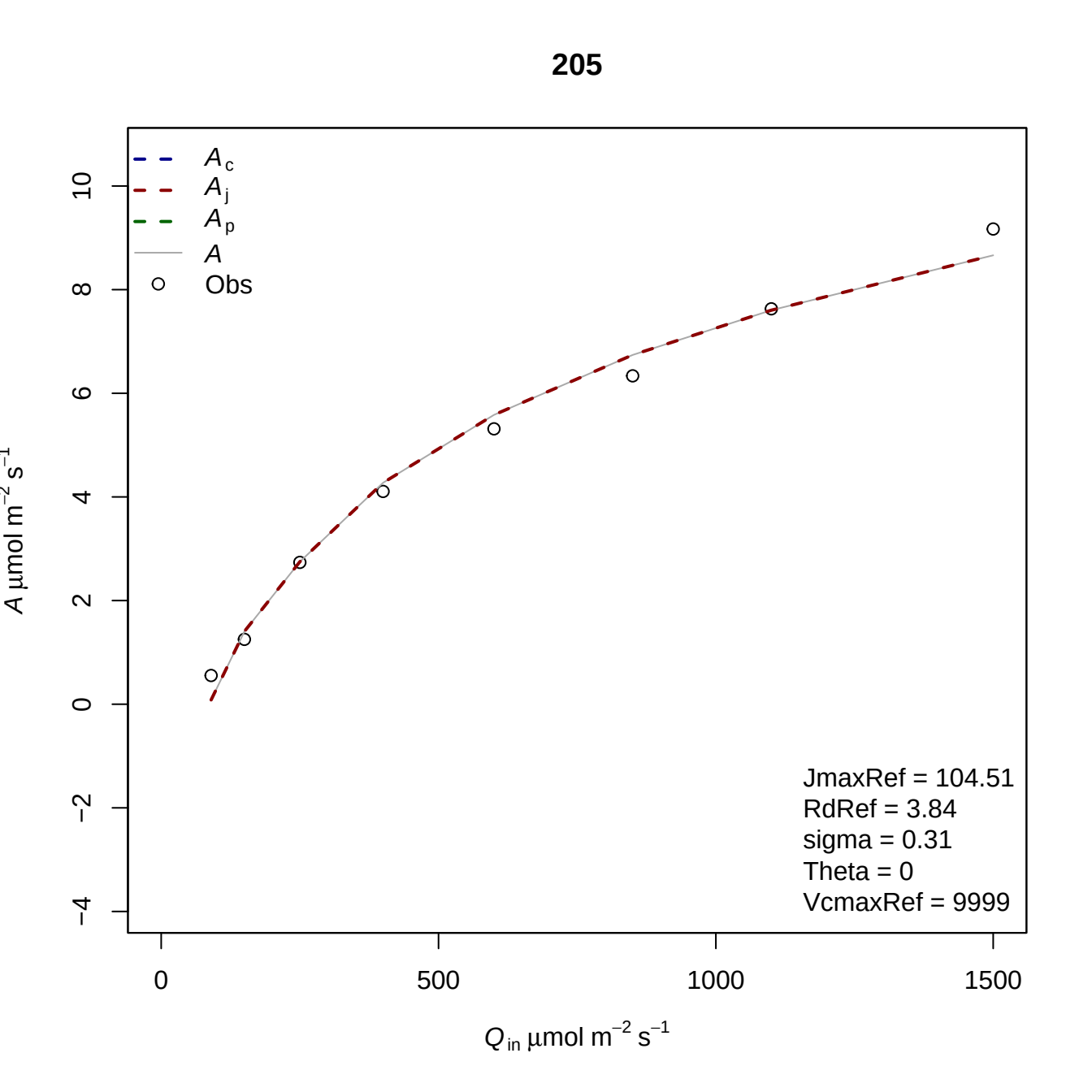

196
A_c
A_j
A_p
A
Obs
JmaxRef = 99.49
RdRef = 1.63
sigma = 0.24
Theta = 0
VcmaxRef = 52.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

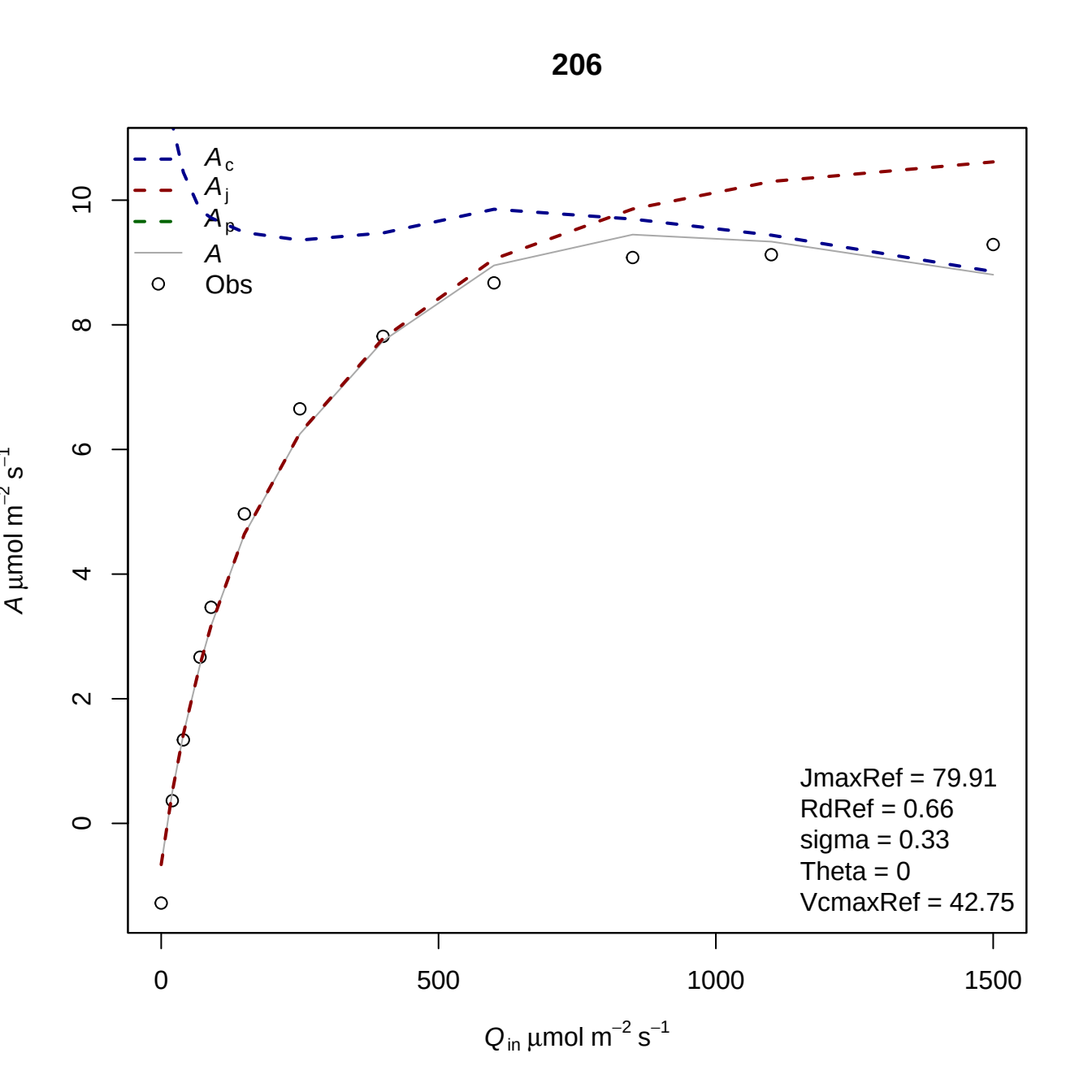

A_c
A_j
A_p
A
Obs
JmaxRef = 127.25
RdRef = 1.62
sigma = 0.32
Theta = 0
VcmaxRef = 62.33
0
5
10
15
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

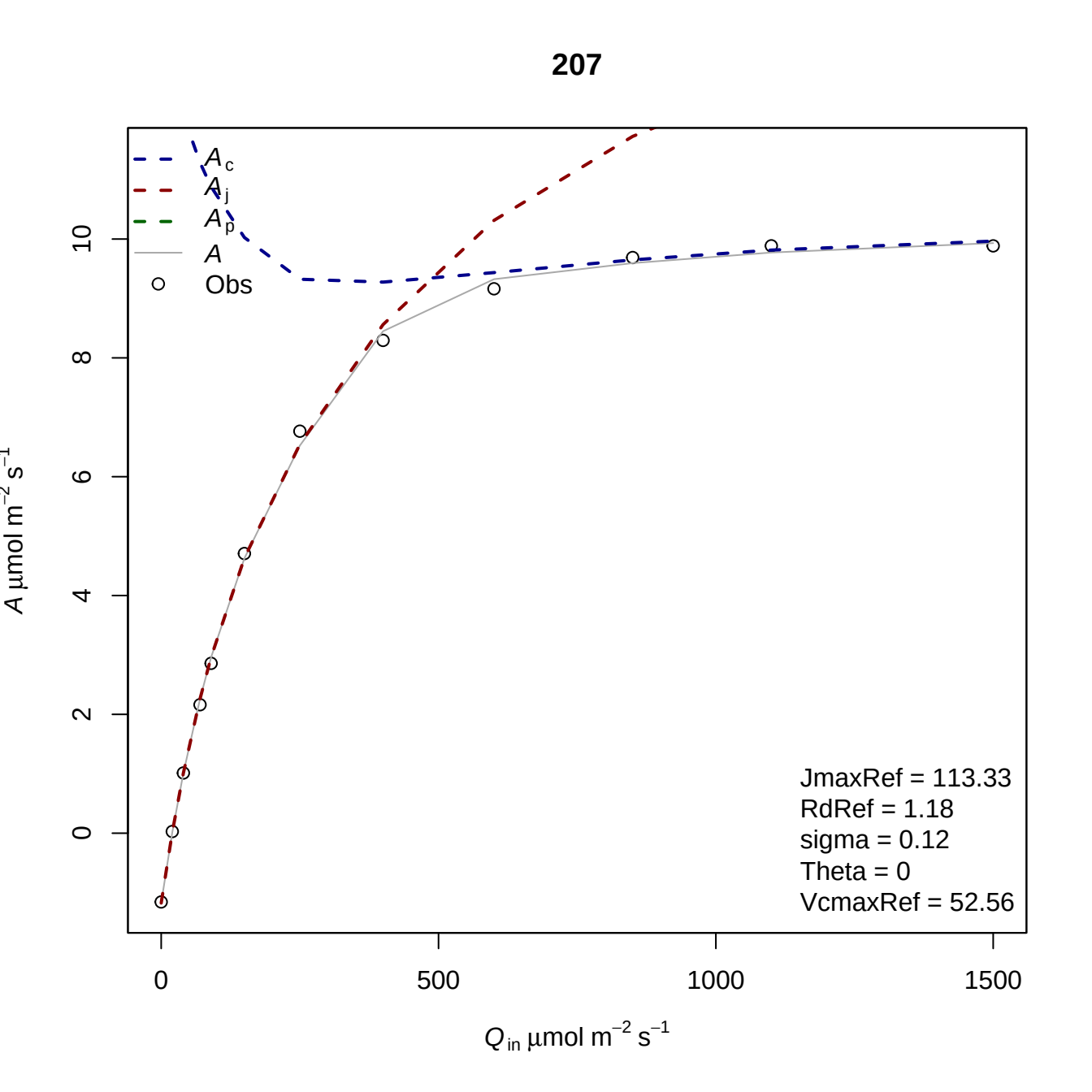

198
A_c
A_j
A_p
A
Obs
JmaxRef = 64.34
RdRef = 0.88
sigma = 0.13
Theta = 0
VcmaxRef = 37.86
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

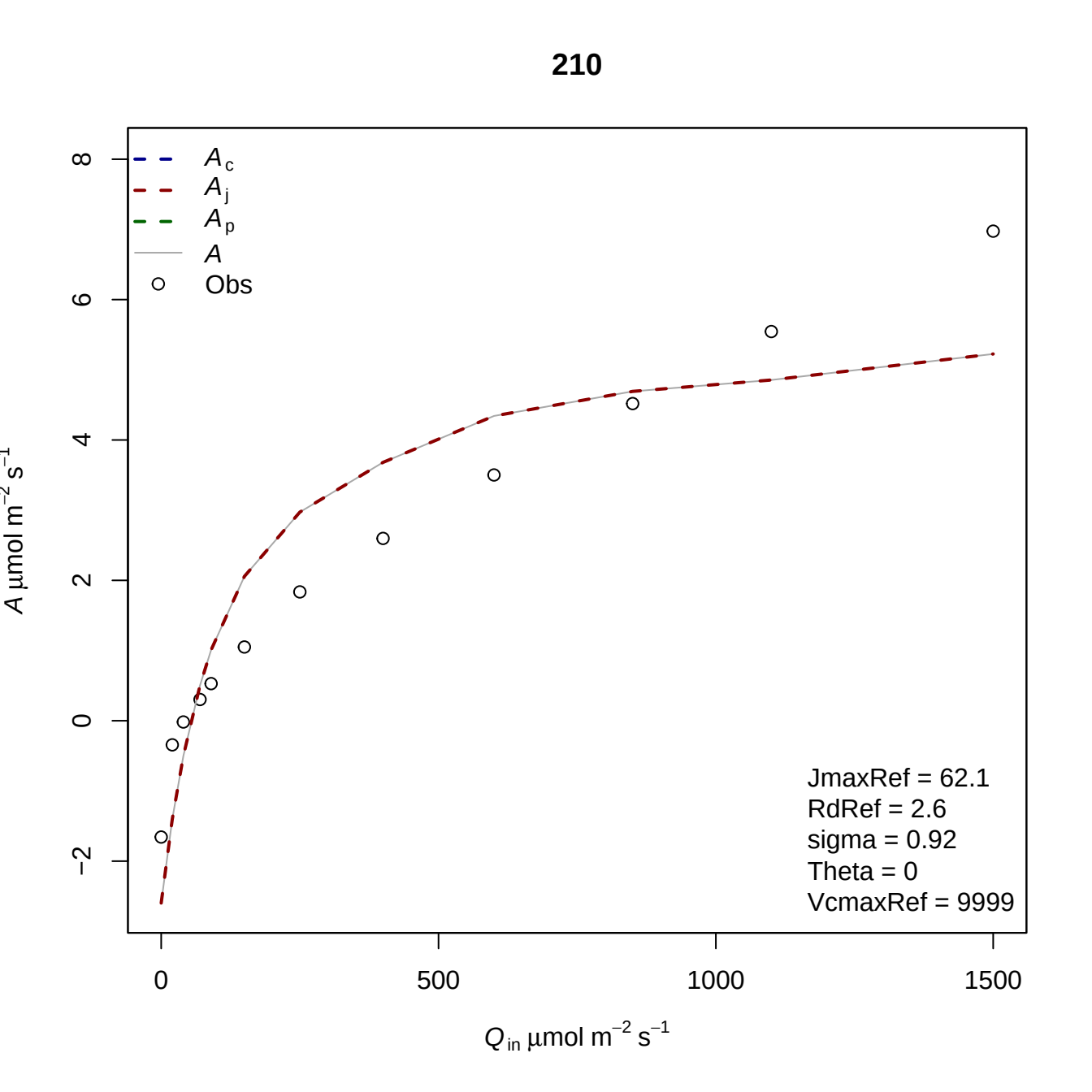

201
A_c
A_j
A_p
A
Obs
JmaxRef = 44.75
RdRef = 2.12
sigma = 0.25
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

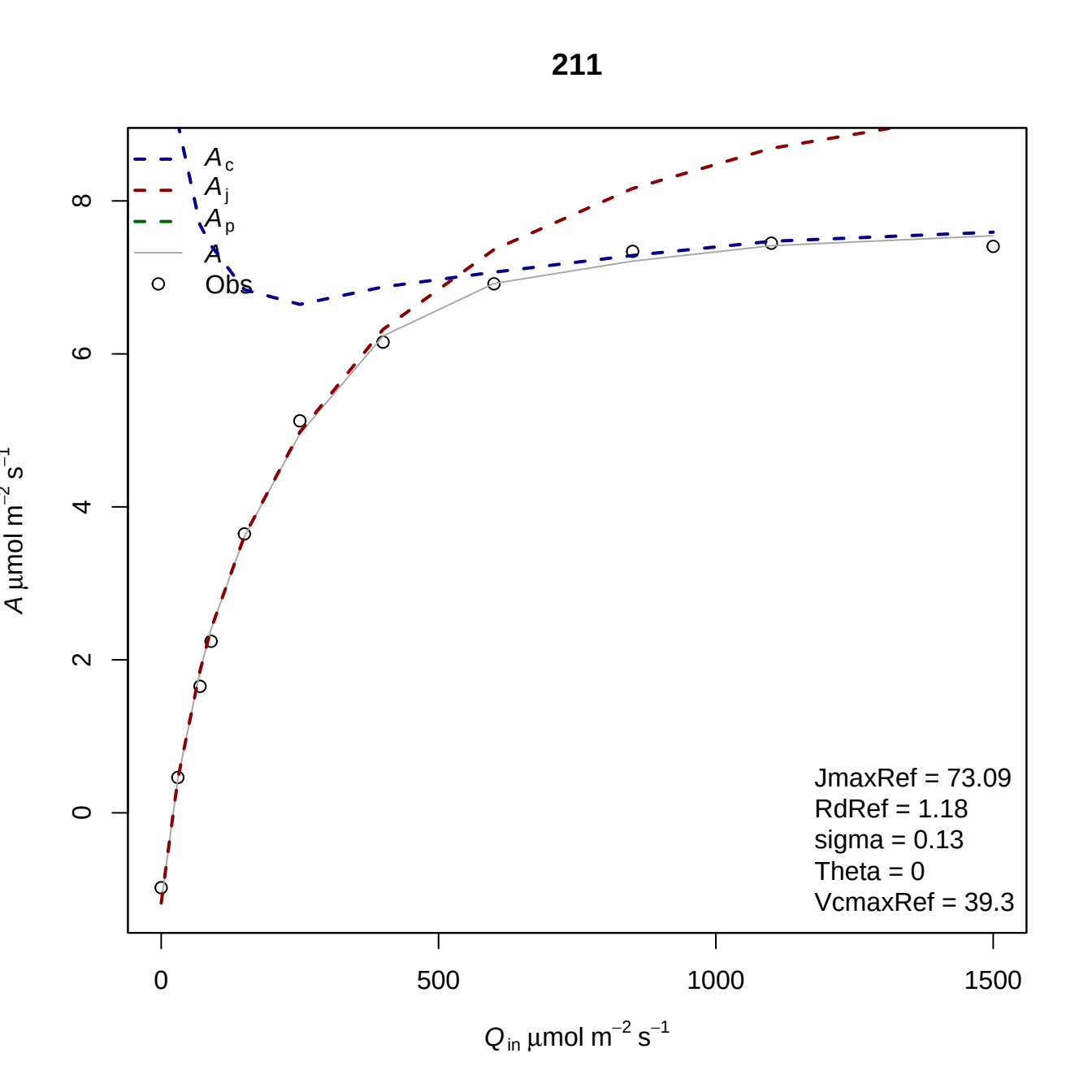

202
A_c
A_j
A_p
A
Obs
JmaxRef = 44.04
RdRef = 2.33
sigma = 0.72
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

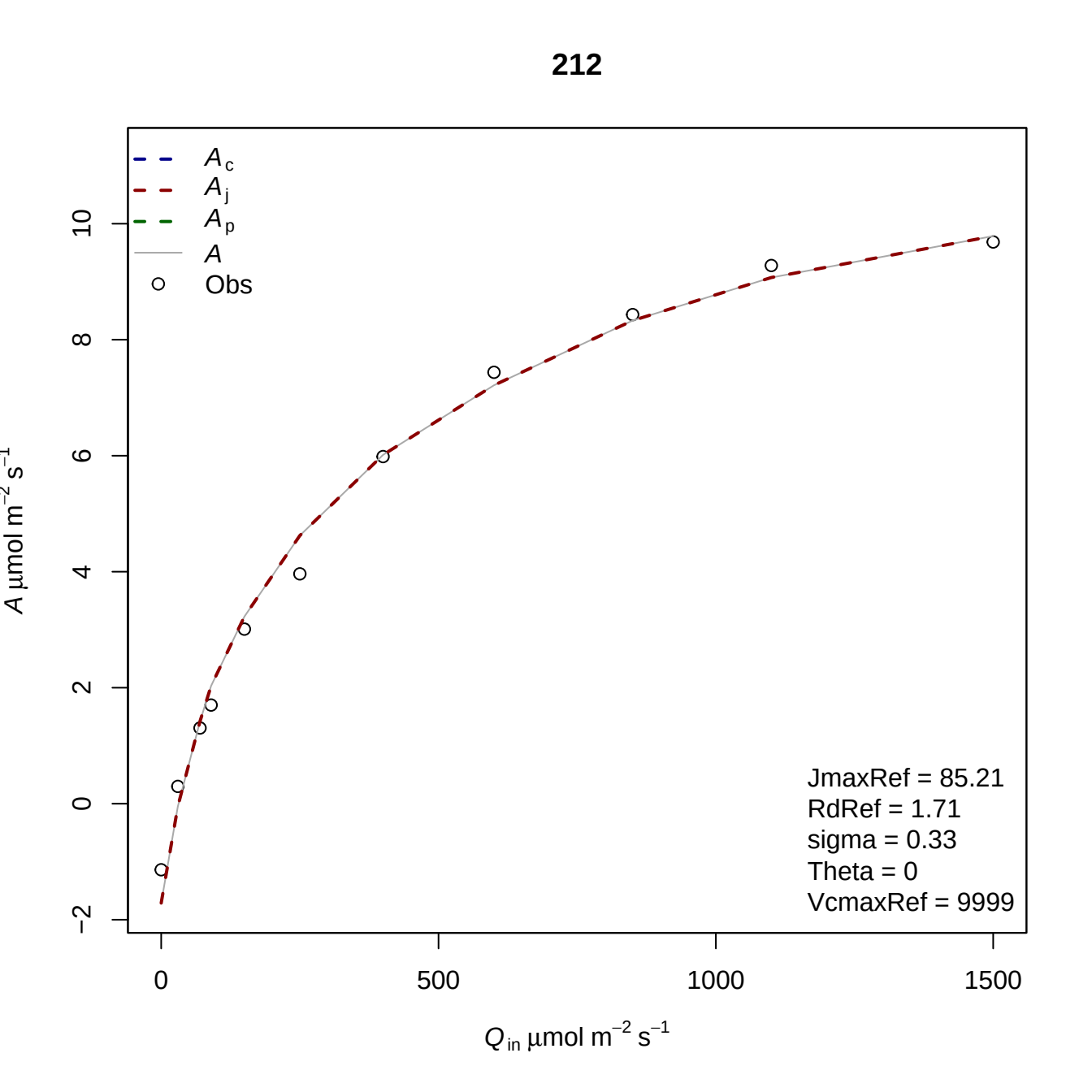

205
A_c
A_j
A_p
A
Obs
JmaxRef = 104.51
RdRef = 3.84
sigma = 0.31
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

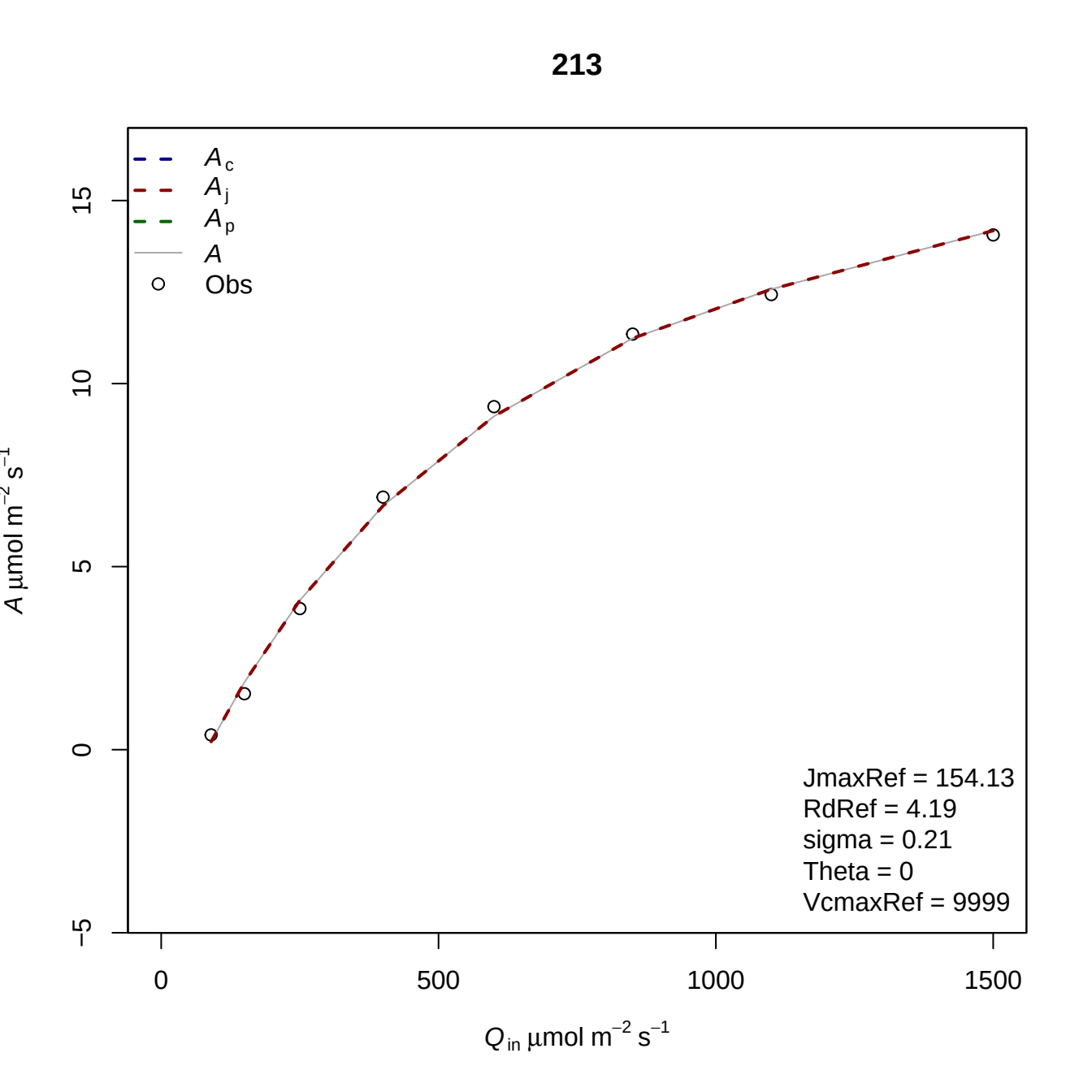

206
A_c
A_j
A_p
A
Obs
JmaxRef = 79.91
RdRef = 0.66
sigma = 0.33
Theta = 0
VcmaxRef = 42.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

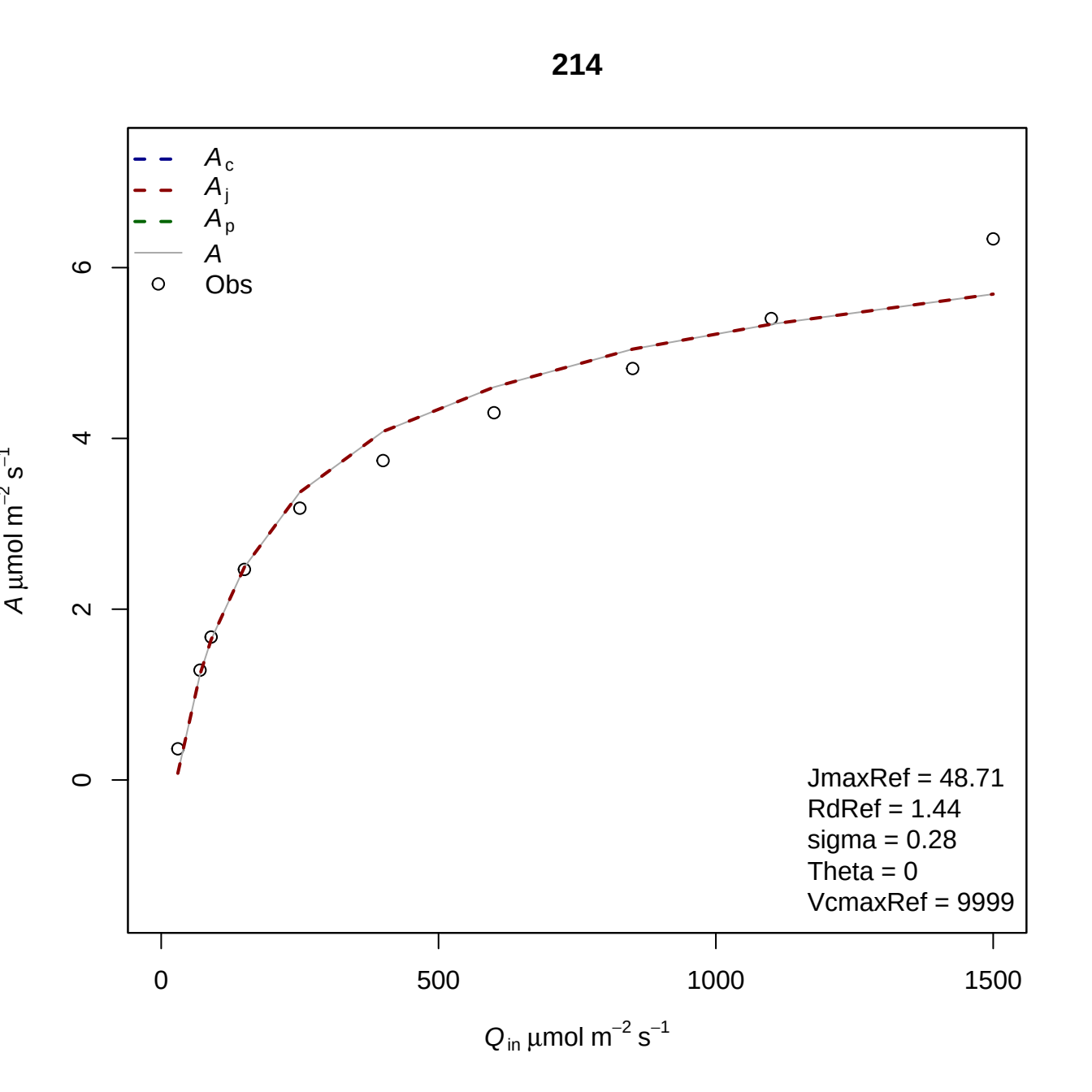

207
A_c
A_j
A_p
A
Obs
JmaxRef = 113.33
RdRef = 1.18
sigma = 0.12
Theta = 0
VcmaxRef = 52.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

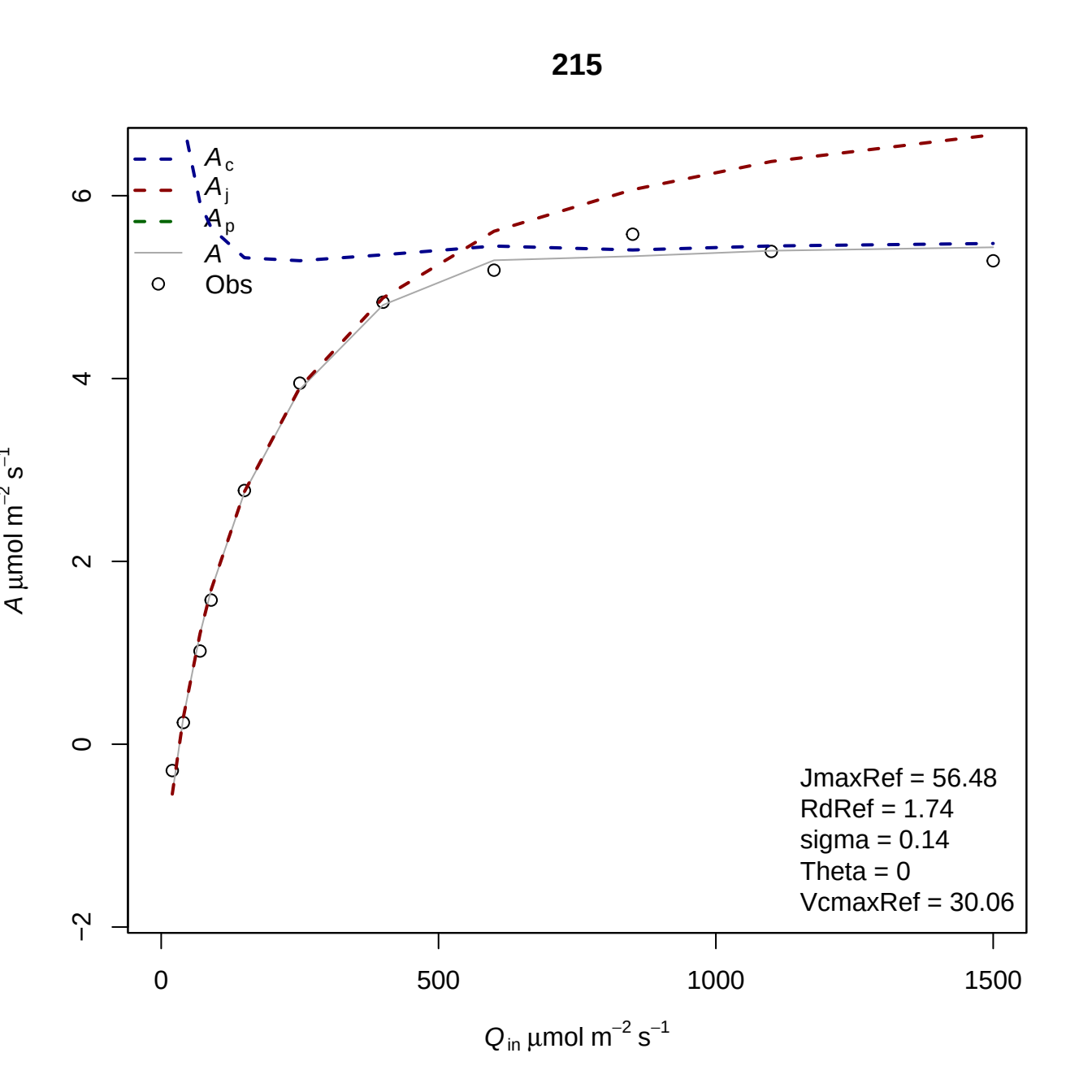

210
A_c
A_j
A_p
A
Obs
JmaxRef = 62.1
RdRef = 2.6
sigma = 0.92
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

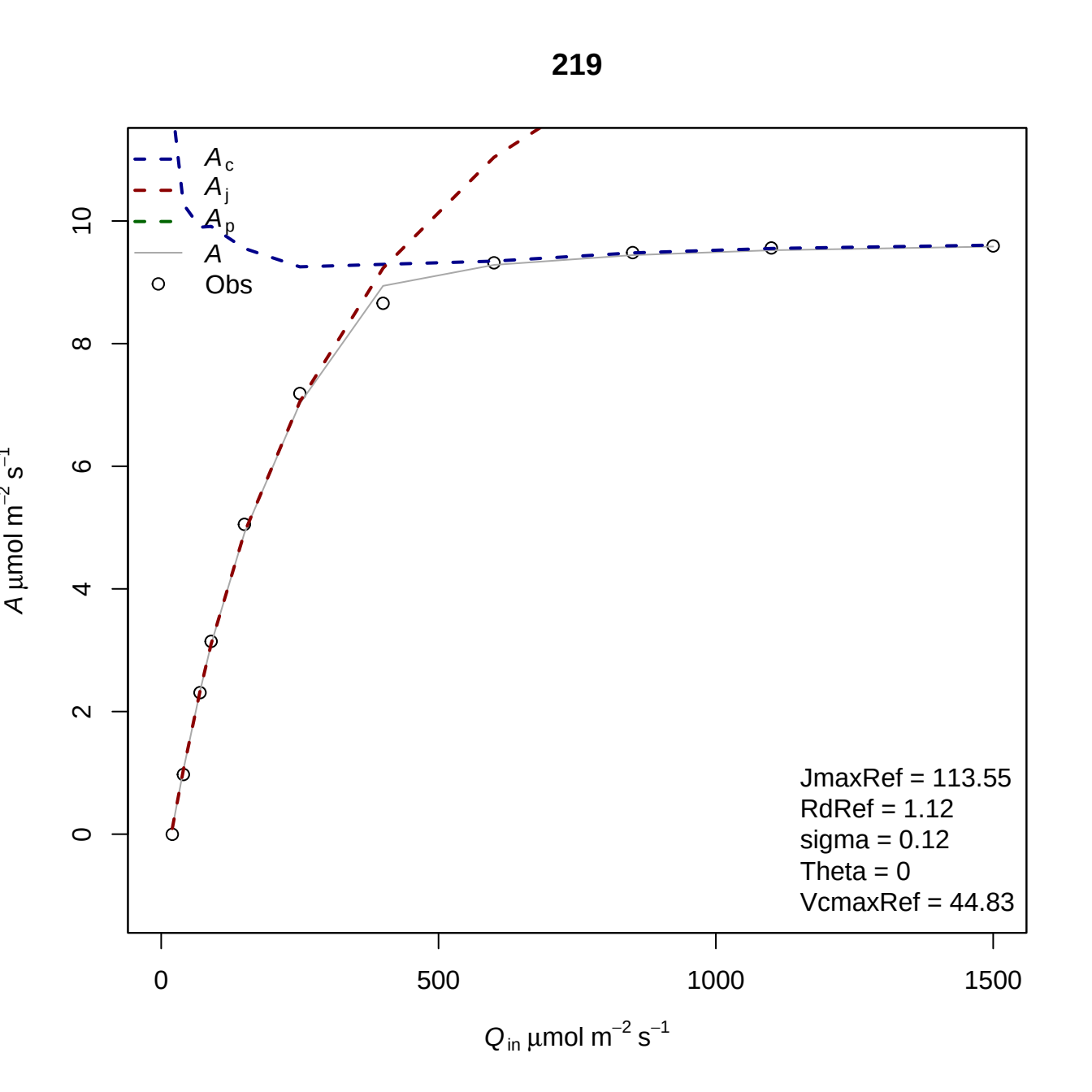

211
A_c
A_j
A_p
A
Obs
JmaxRef = 73.09
RdRef = 1.18
sigma = 0.13
Theta = 0
VcmaxRef = 39.3
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

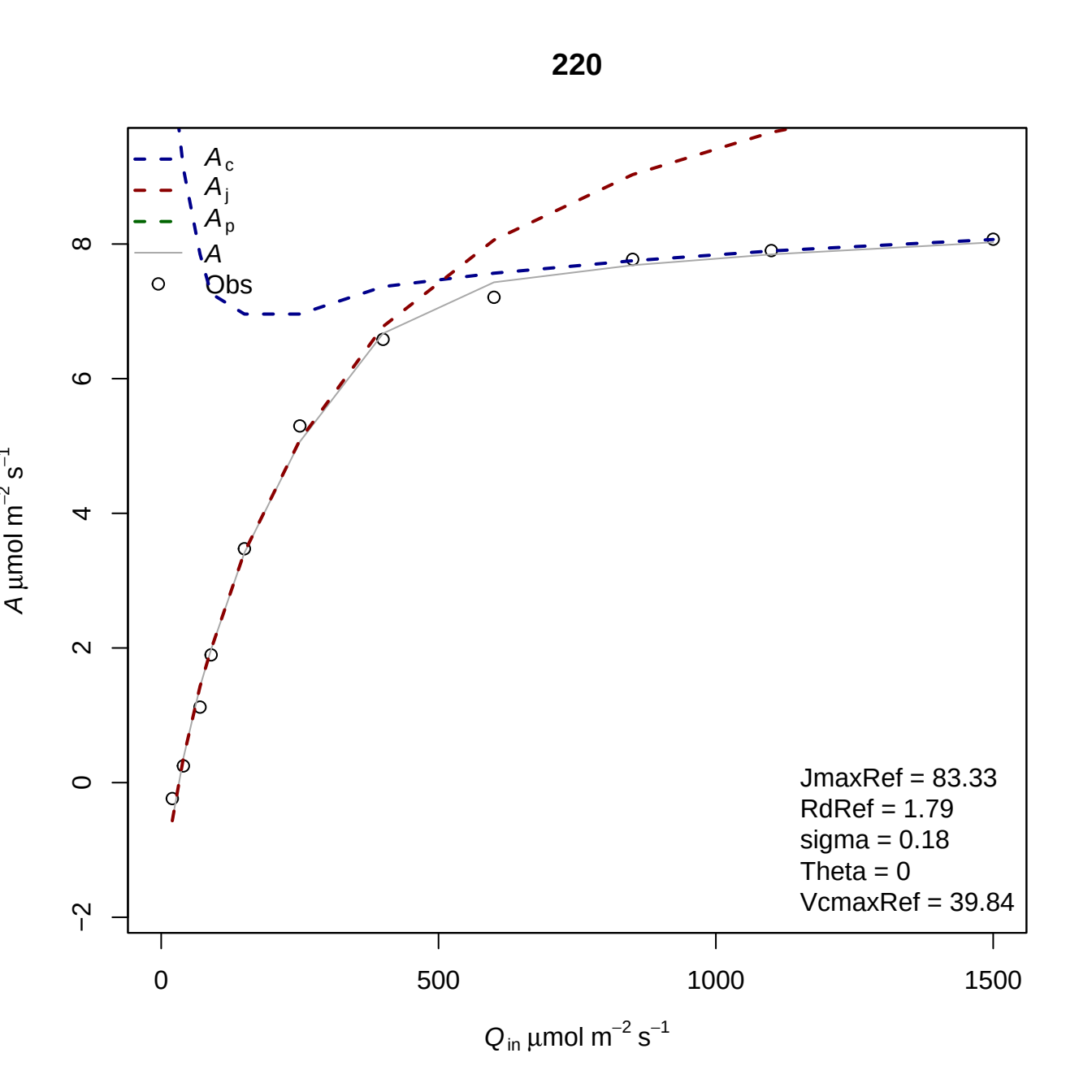

212
A_c
A_j
A_p
A
Obs
JmaxRef = 85.21
RdRef = 1.71
sigma = 0.33
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

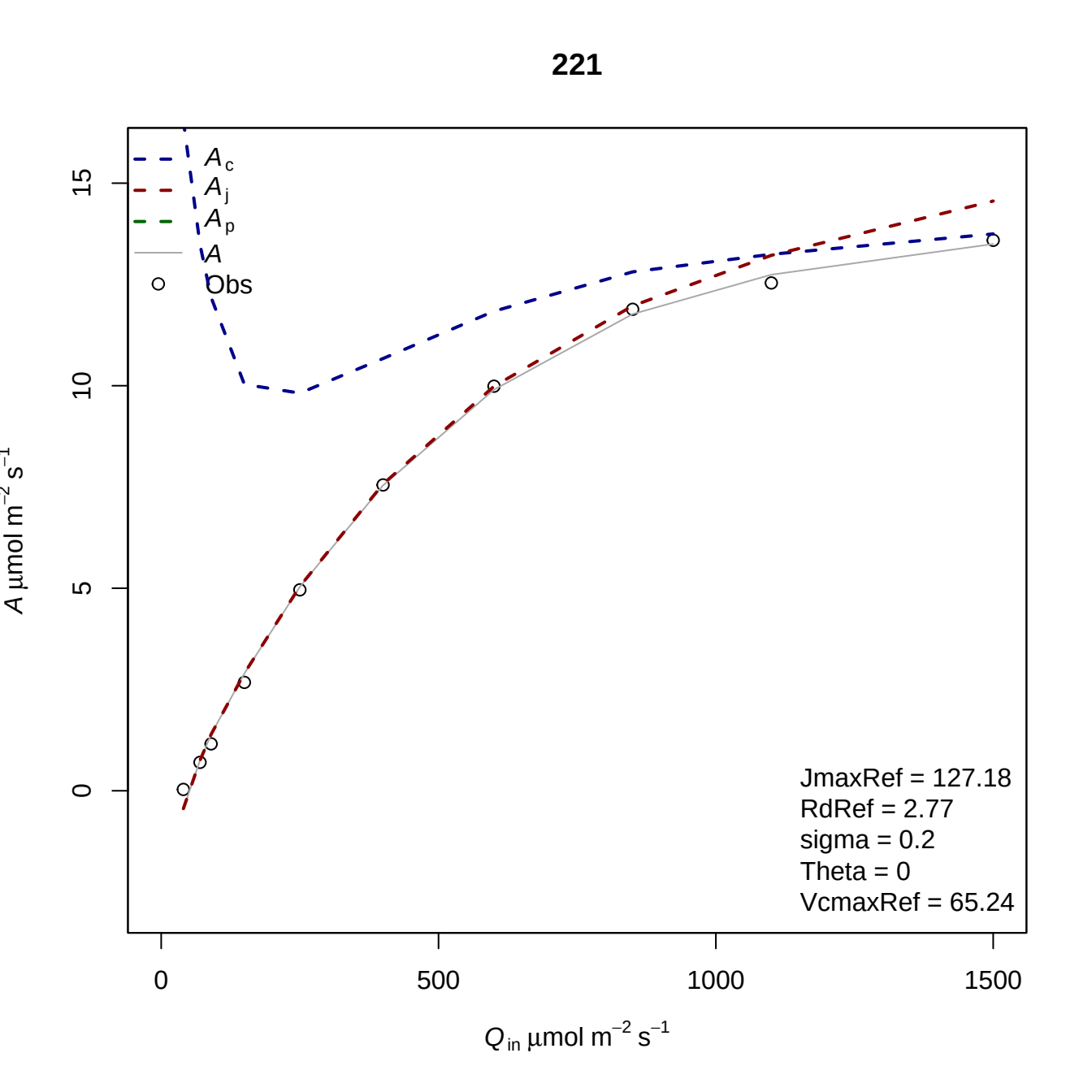

213
A_c
A_j
A_p
A
Obs
JmaxRef = 154.13
RdRef = 4.19
sigma = 0.21
Theta = 0
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

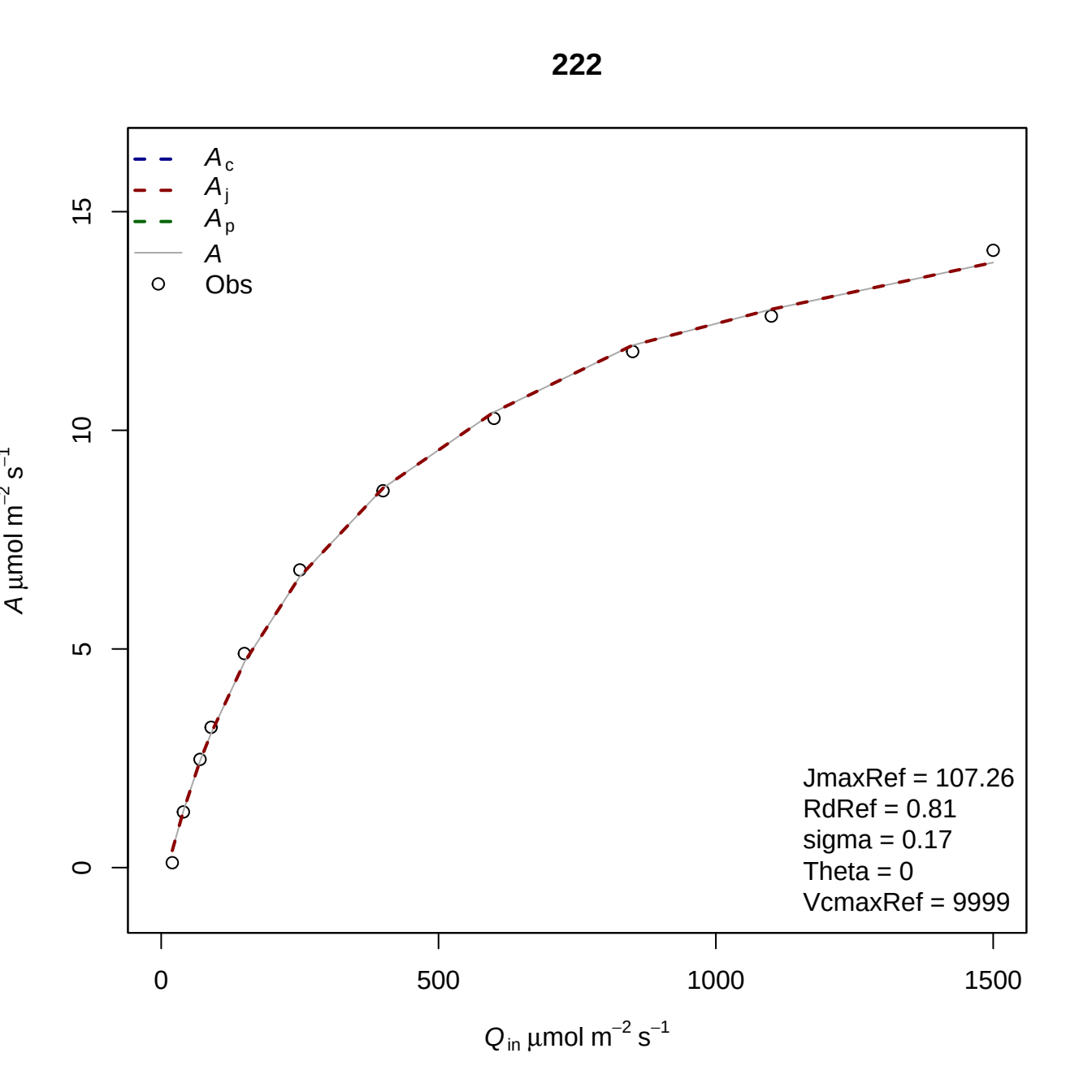

214
A_c
A_j
A_p
A
Obs
JmaxRef = 48.71
RdRef = 1.44
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

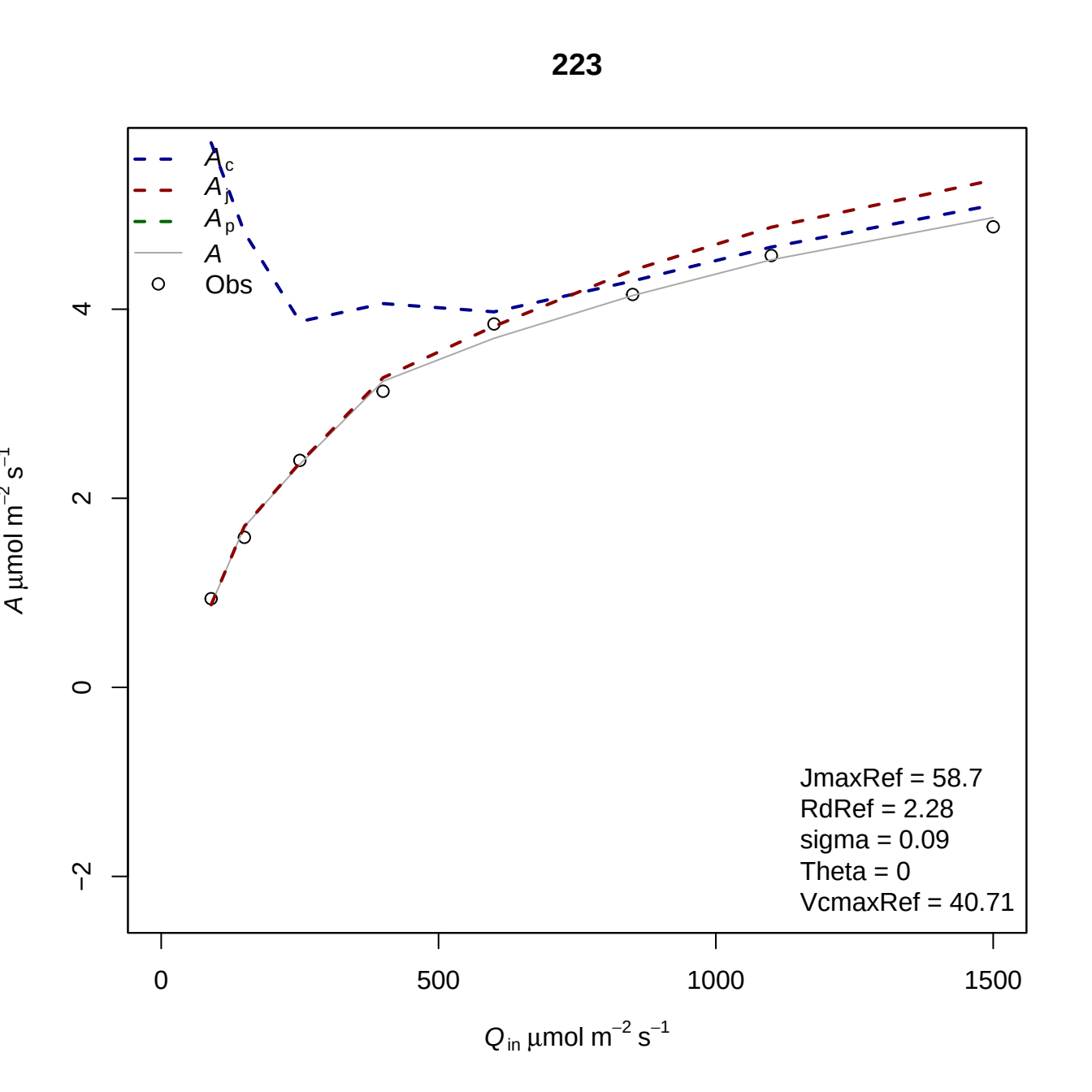

215
A_c
A_j
A_p
A
Obs
JmaxRef = 56.48
RdRef = 1.74
sigma = 0.14
Theta = 0
VcmaxRef = 30.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

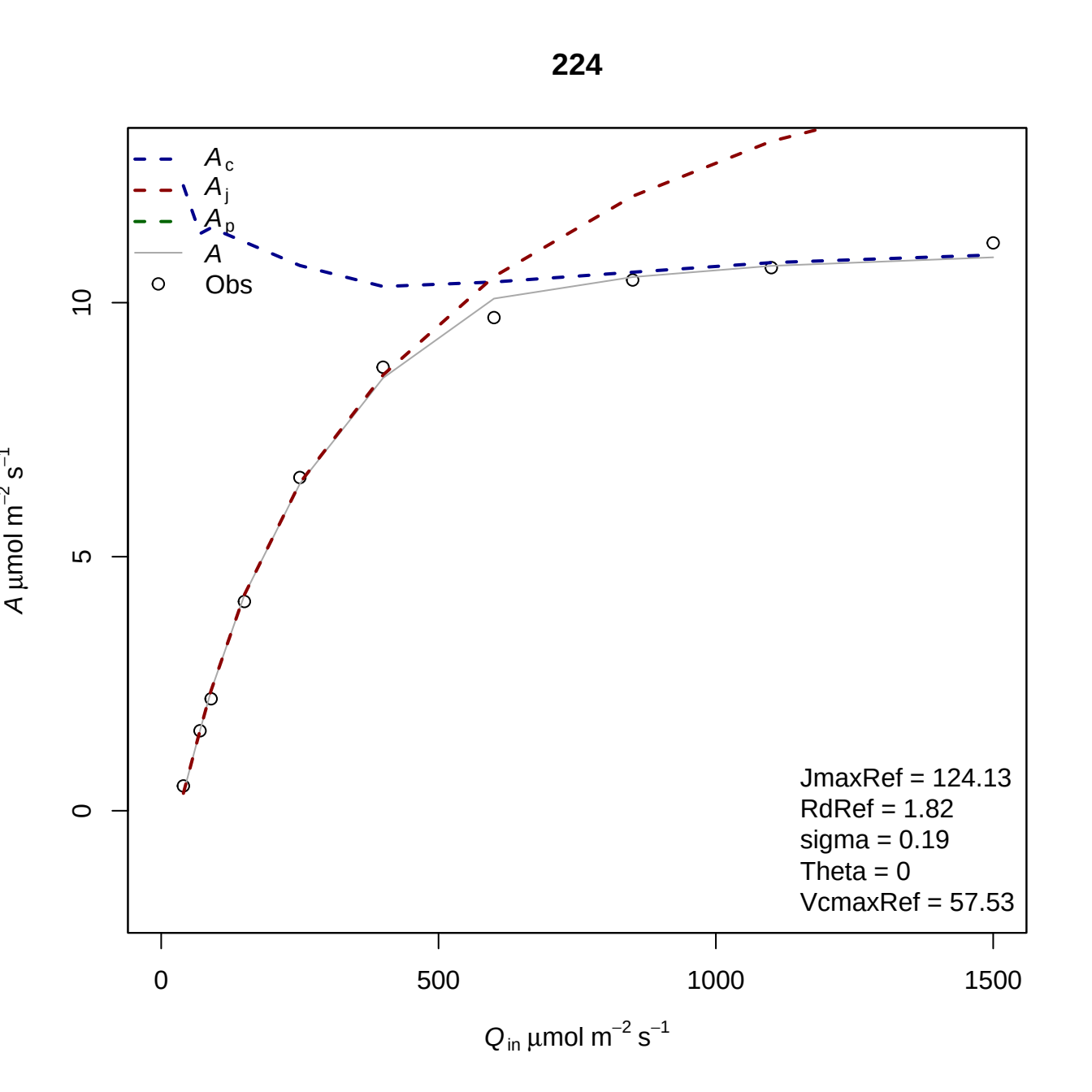

219
A_c
A_j
A_p
A
Obs
JmaxRef = 113.55
RdRef = 1.12
sigma = 0.12
Theta = 0
VcmaxRef = 44.83
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

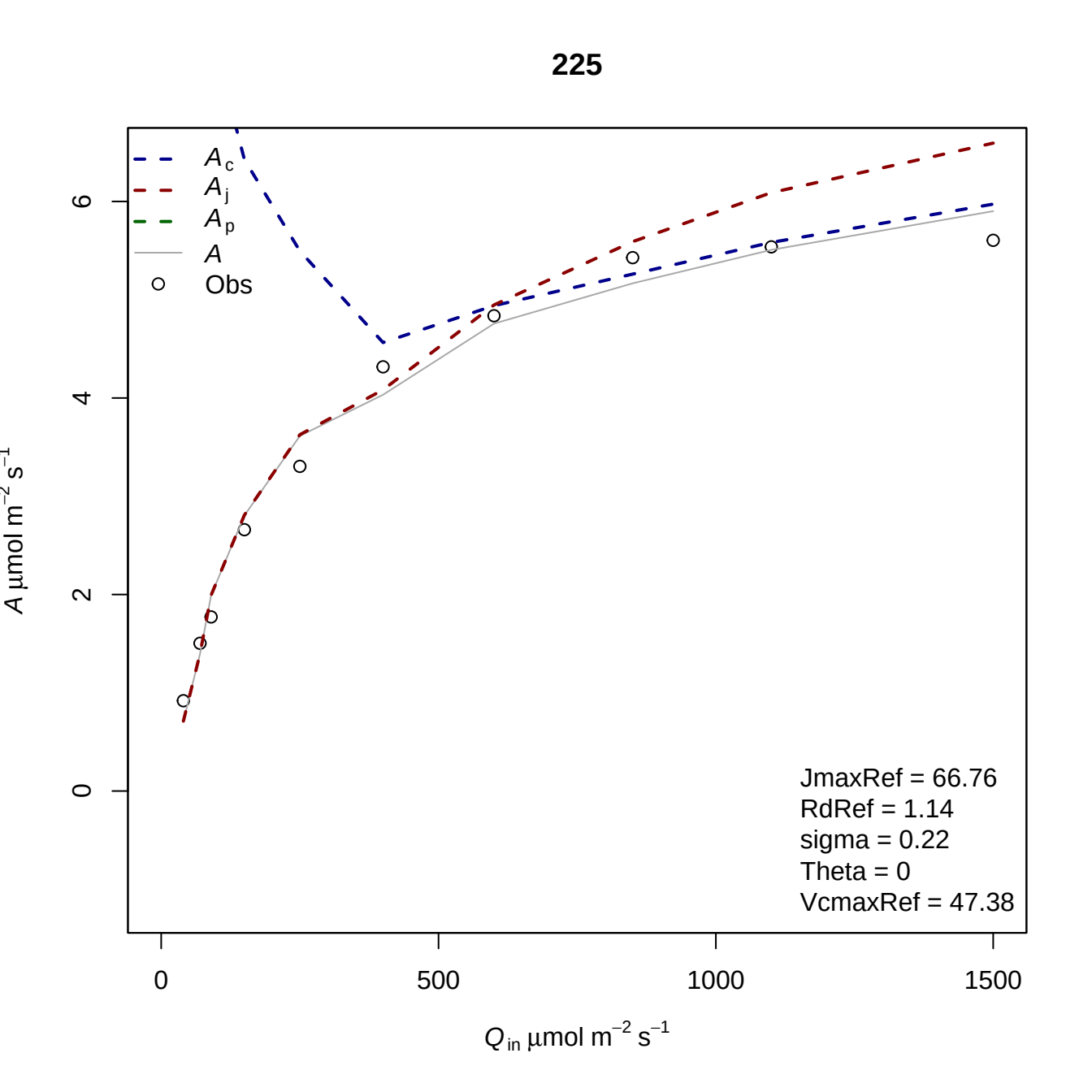

220
A_c
A_j
A_p
A
Obs
JmaxRef = 83.33
RdRef = 1.79
sigma = 0.18
Theta = 0
VcmaxRef = 39.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

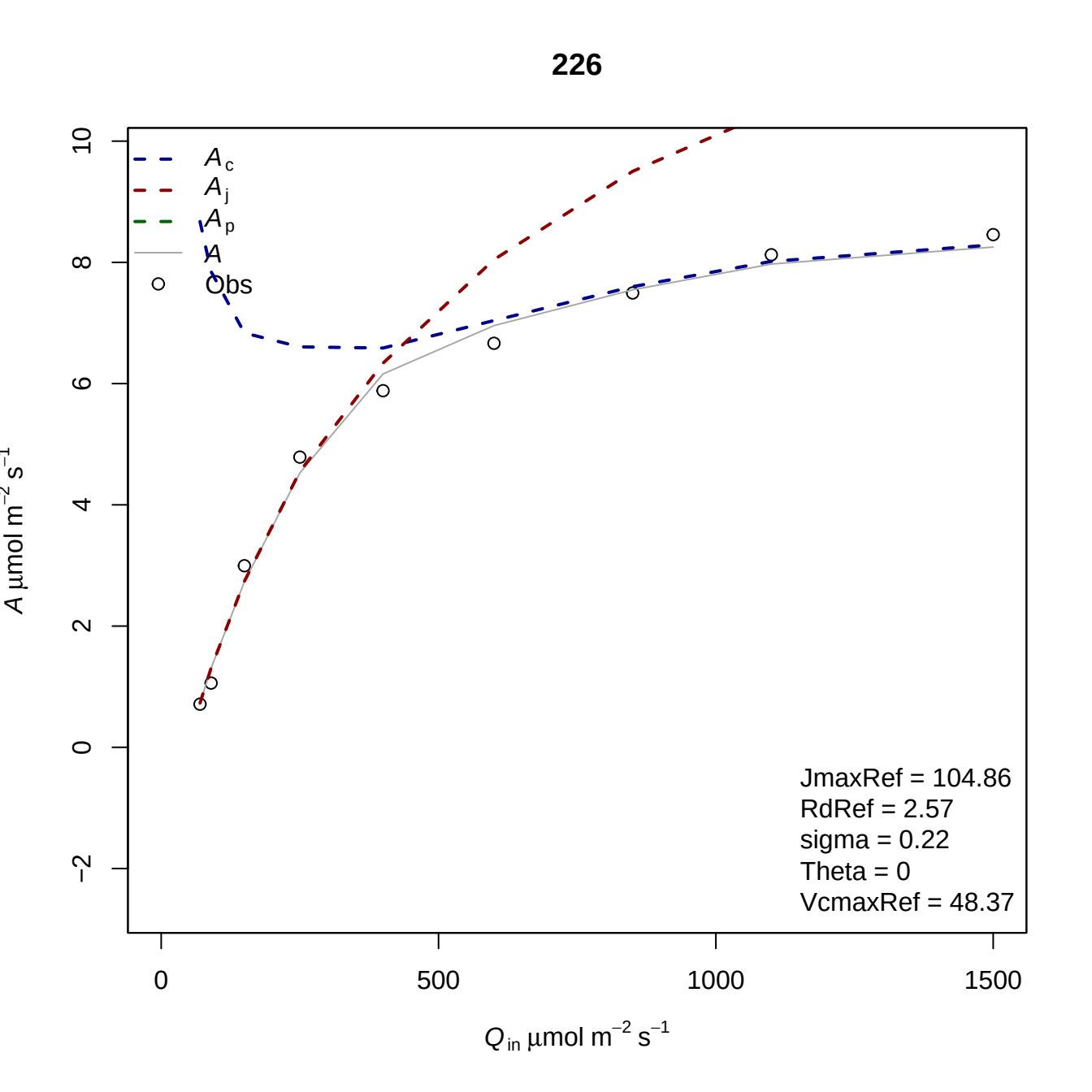

221
A_c
A_j
A_p
A
Obs
JmaxRef = 127.18
RdRef = 2.77
sigma = 0.2
Theta = 0
VcmaxRef = 65.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

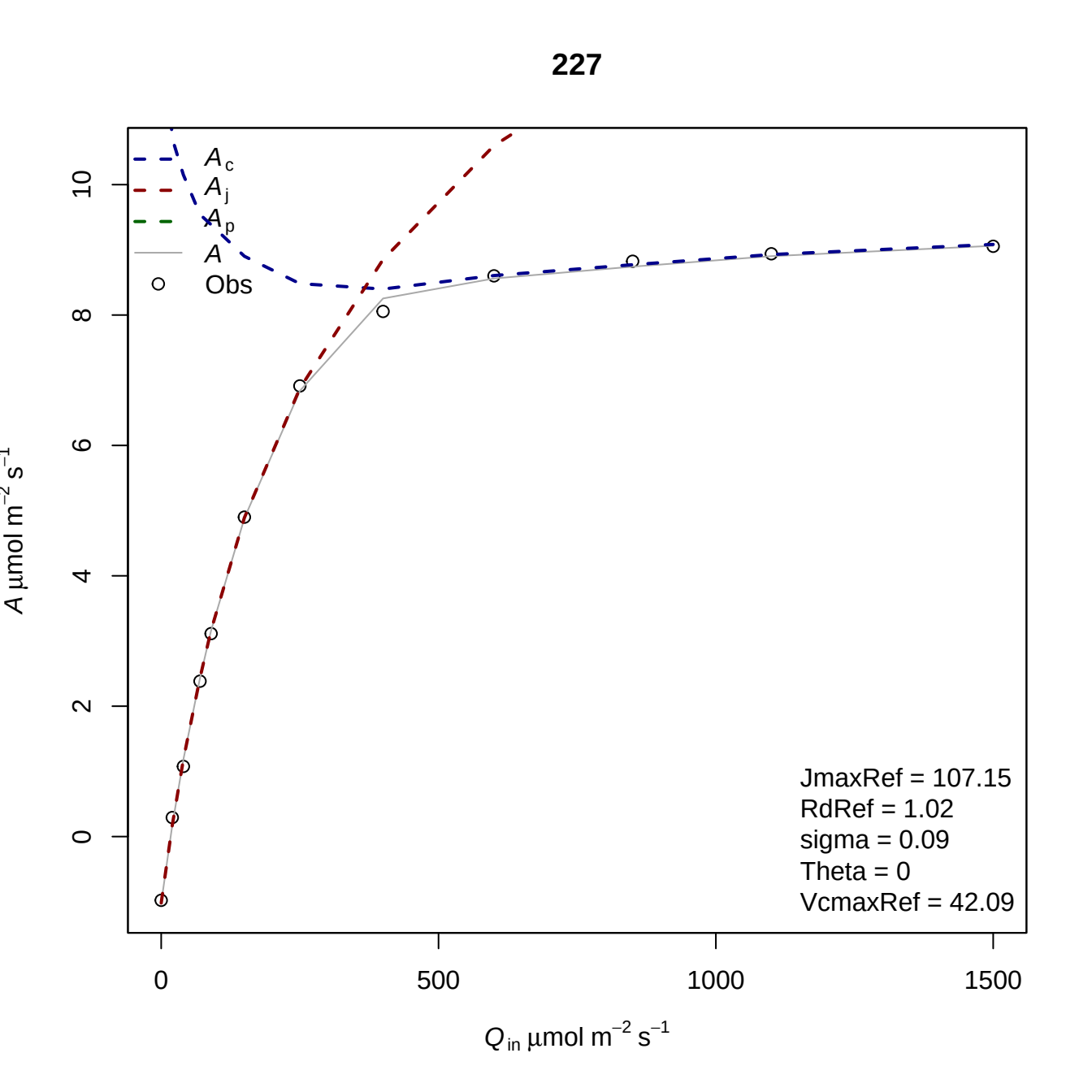

222
A_c
A_j
A_p
A
Obs
JmaxRef = 107.26
RdRef = 0.81
sigma = 0.17
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

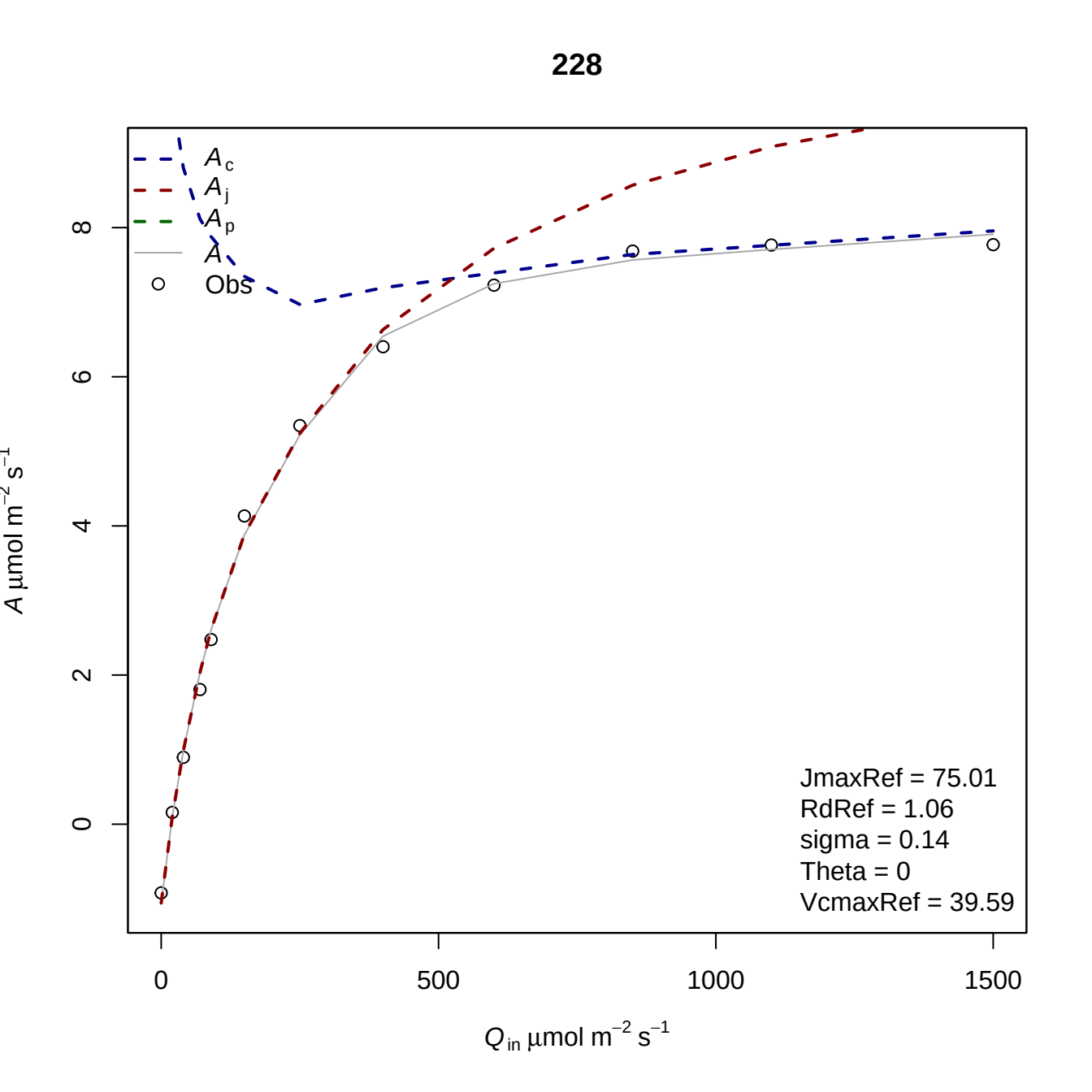

223
A_c
A_j
A_p
A
Obs
JmaxRef = 58.7
RdRef = 2.28
sigma = 0.09
Theta = 0
VcmaxRef = 40.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

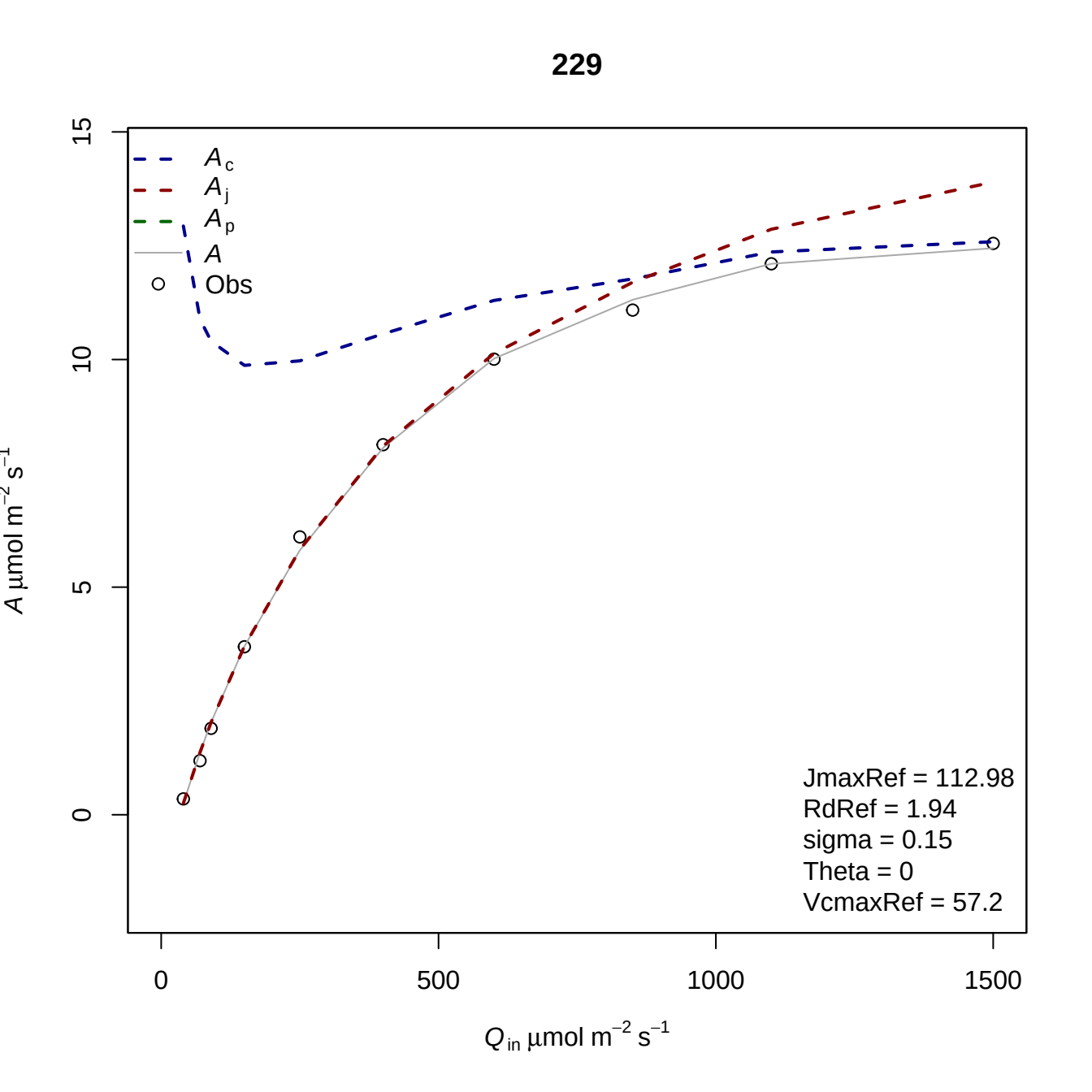

224
A_c
A_j
A_p
A
Obs
JmaxRef = 124.13
RdRef = 1.82
sigma = 0.19
Theta = 0
VcmaxRef = 57.53
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

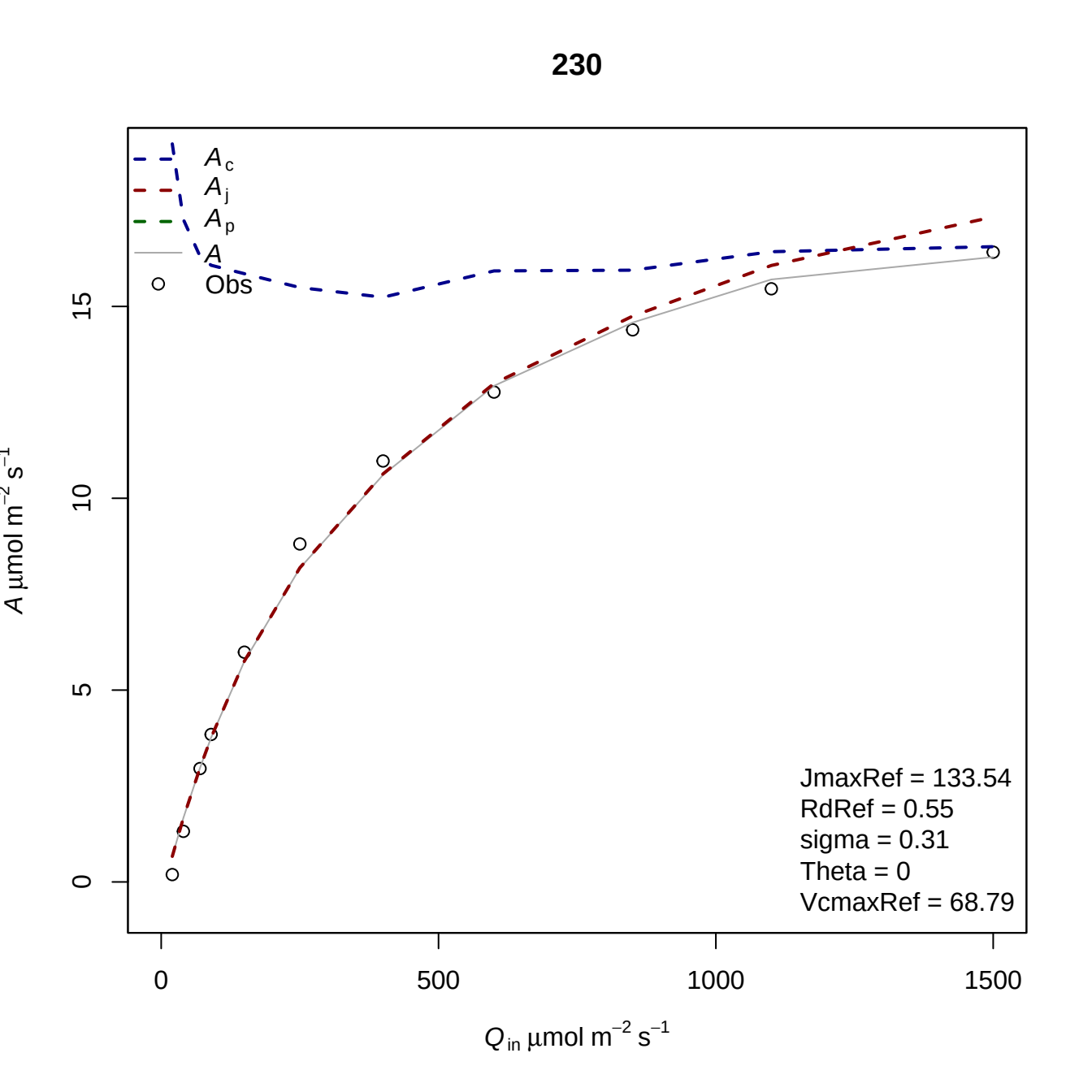

225
A_c
A_j
A_p
A
Obs
JmaxRef = 66.76
RdRef = 1.14
sigma = 0.22
Theta = 0
VcmaxRef = 47.38
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

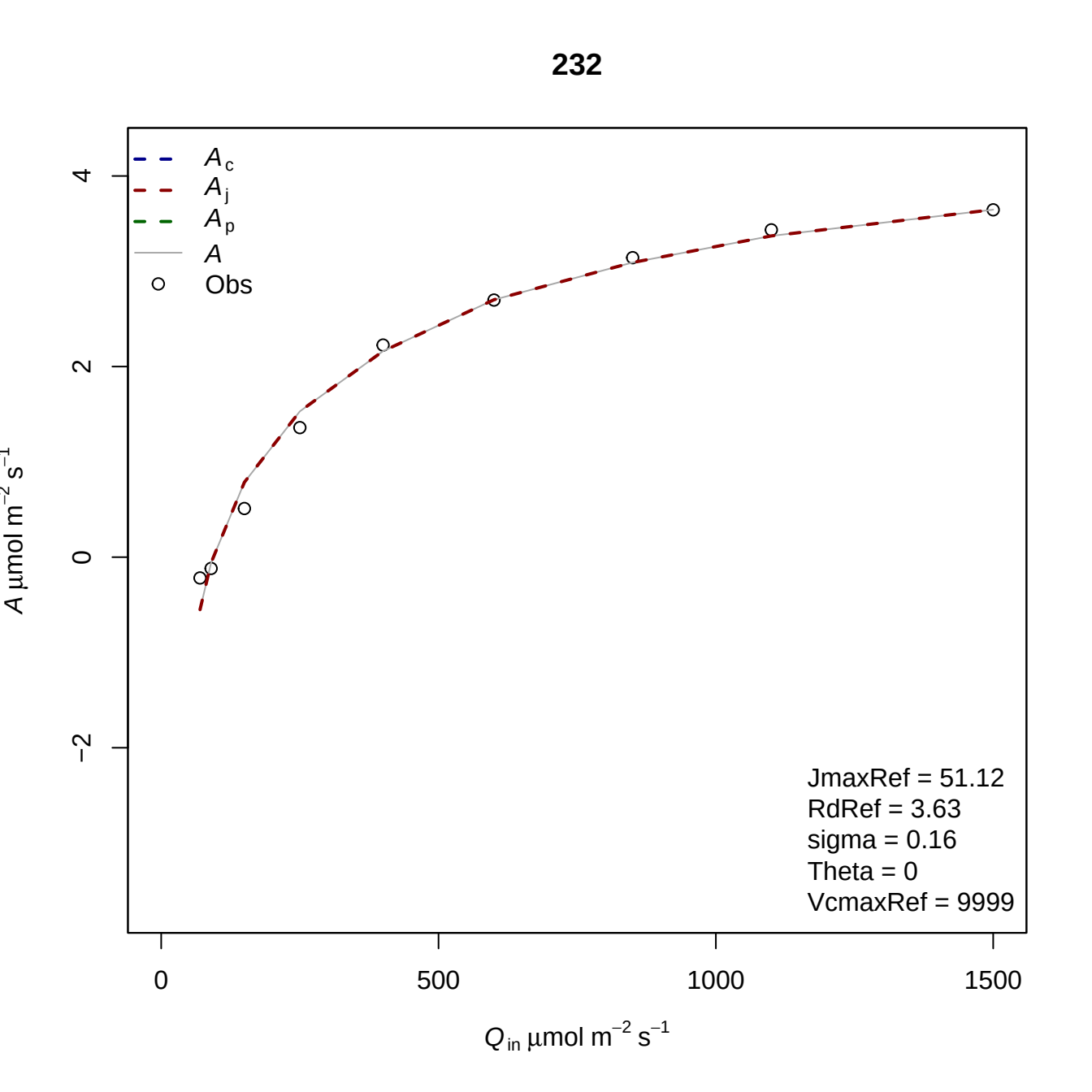

226
A_c
A_j
A_p
A
Obs
JmaxRef = 104.86
RdRef = 2.57
sigma = 0.22
Theta = 0
VcmaxRef = 48.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

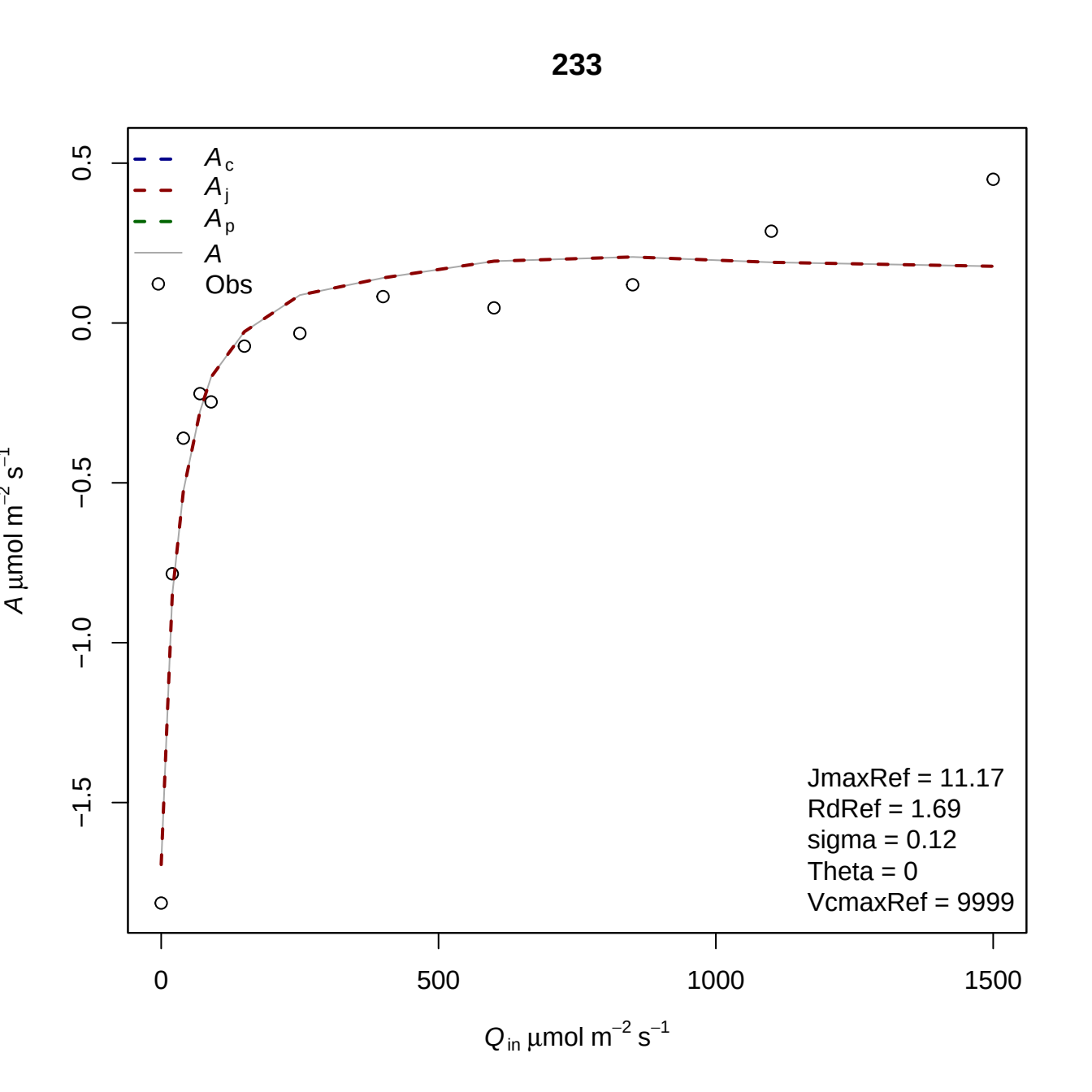

227
A_c
A_j
A_p
A
Obs
JmaxRef = 107.15
RdRef = 1.02
sigma = 0.09
Theta = 0
VcmaxRef = 42.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

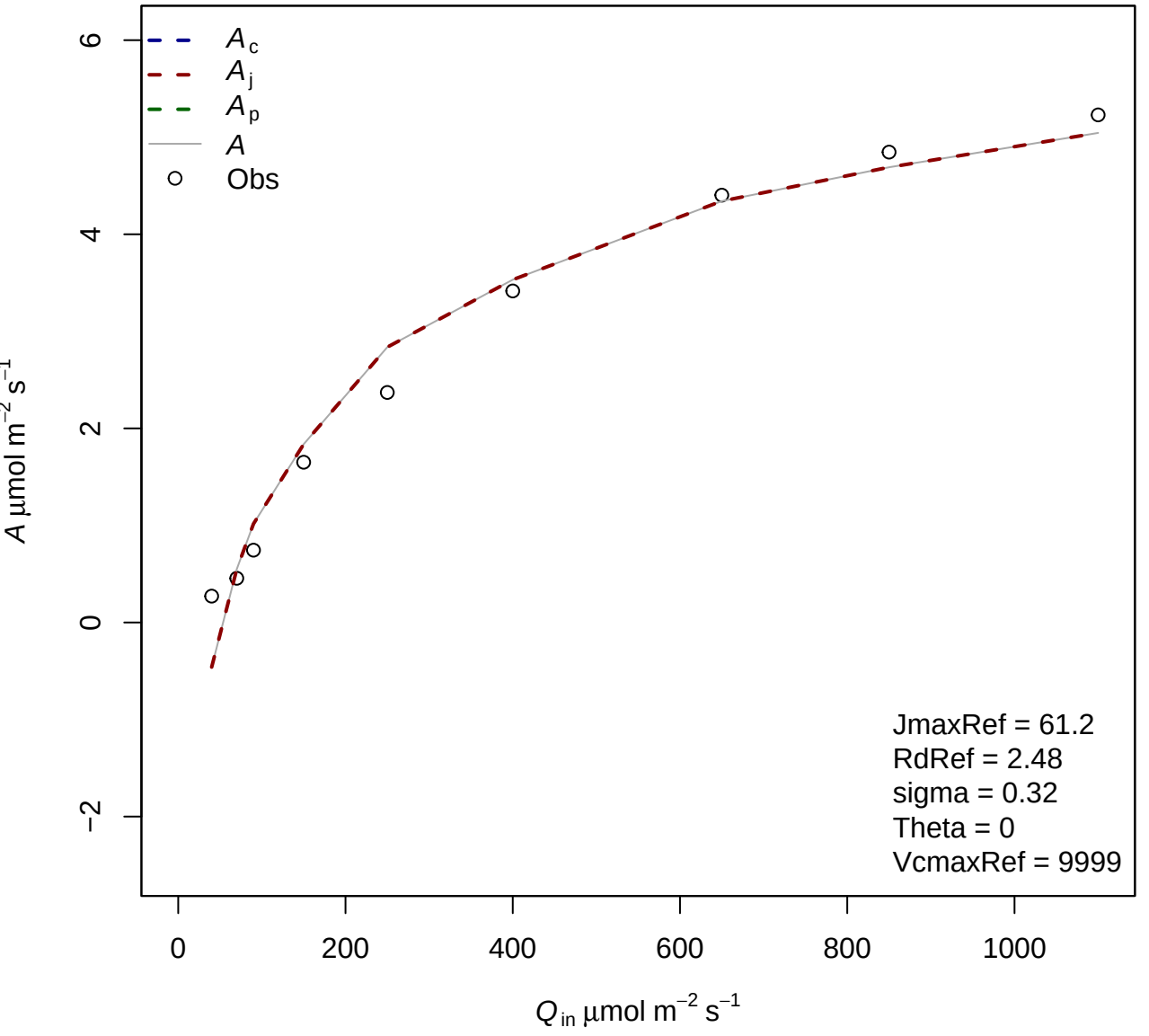

228
A_c
A_j
A_p
A
Obs
JmaxRef = 75.01
RdRef = 1.06
sigma = 0.14
Theta = 0
VcmaxRef = 39.59
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

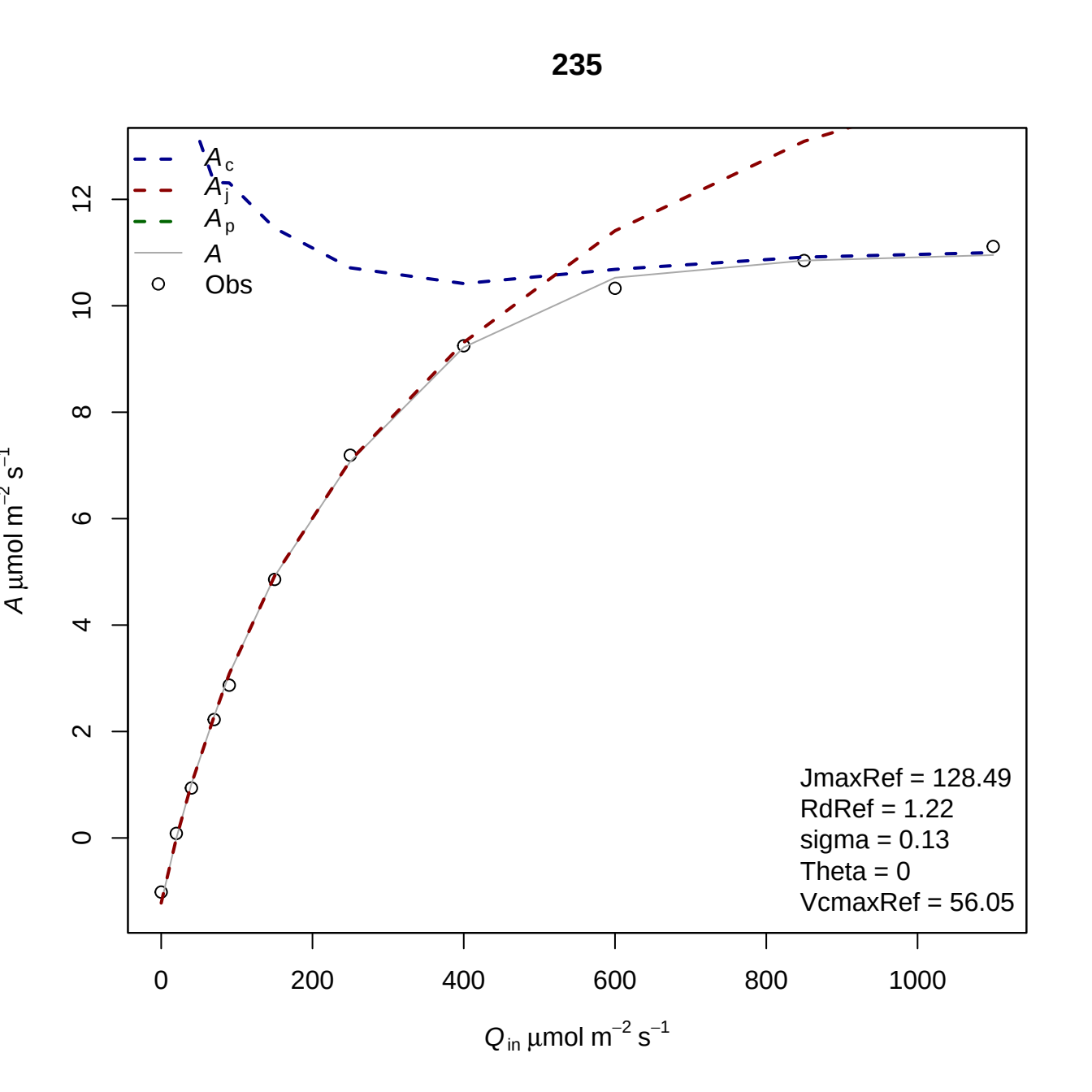

229
A_c
A_j
A_p
A
Obs
JmaxRef = 112.98
RdRef = 1.94
sigma = 0.15
Theta = 0
VcmaxRef = 57.2
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

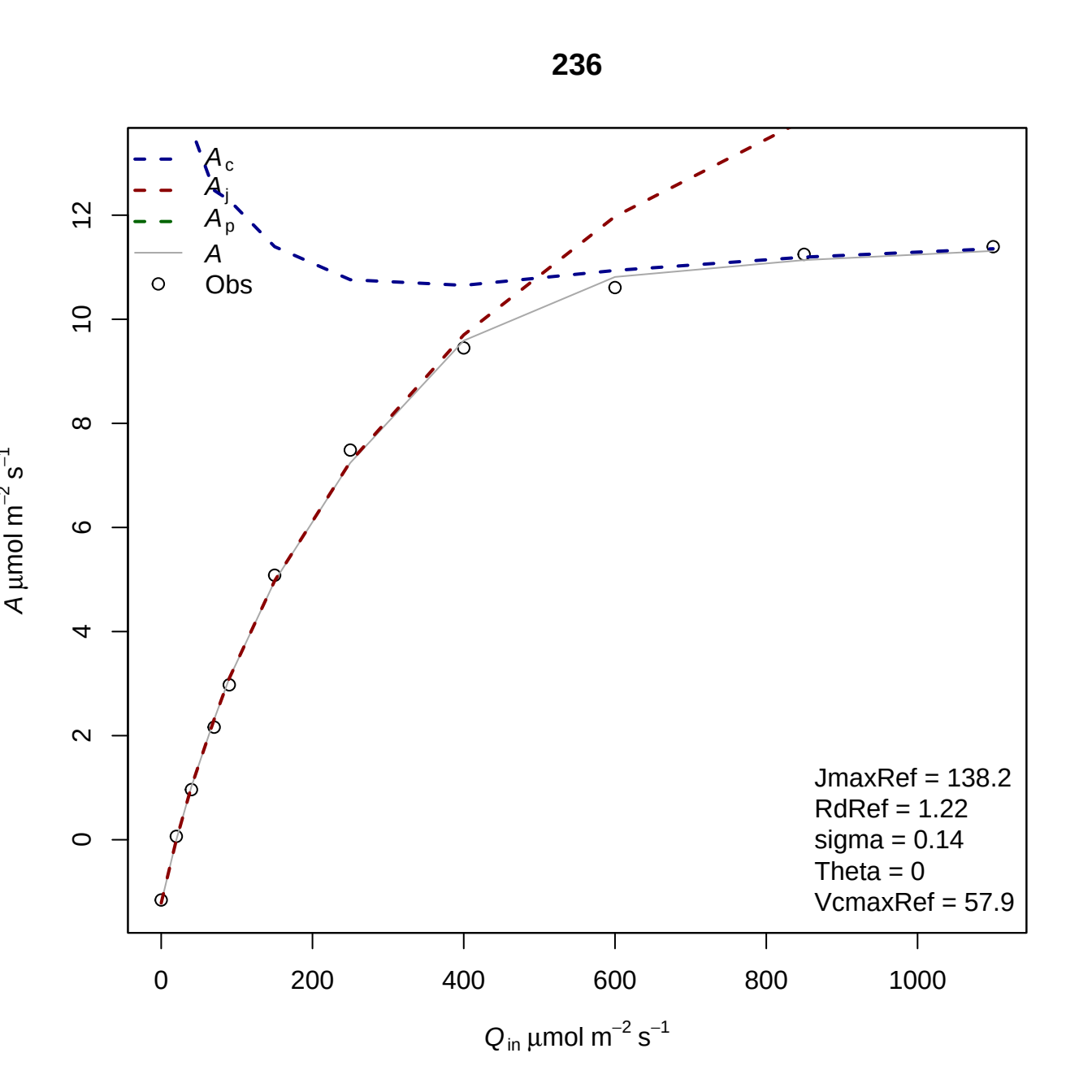

230
A_c
A_j
A_p
A
Obs
JmaxRef = 133.54
RdRef = 0.55
sigma = 0.31
Theta = 0
VcmaxRef = 68.79
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

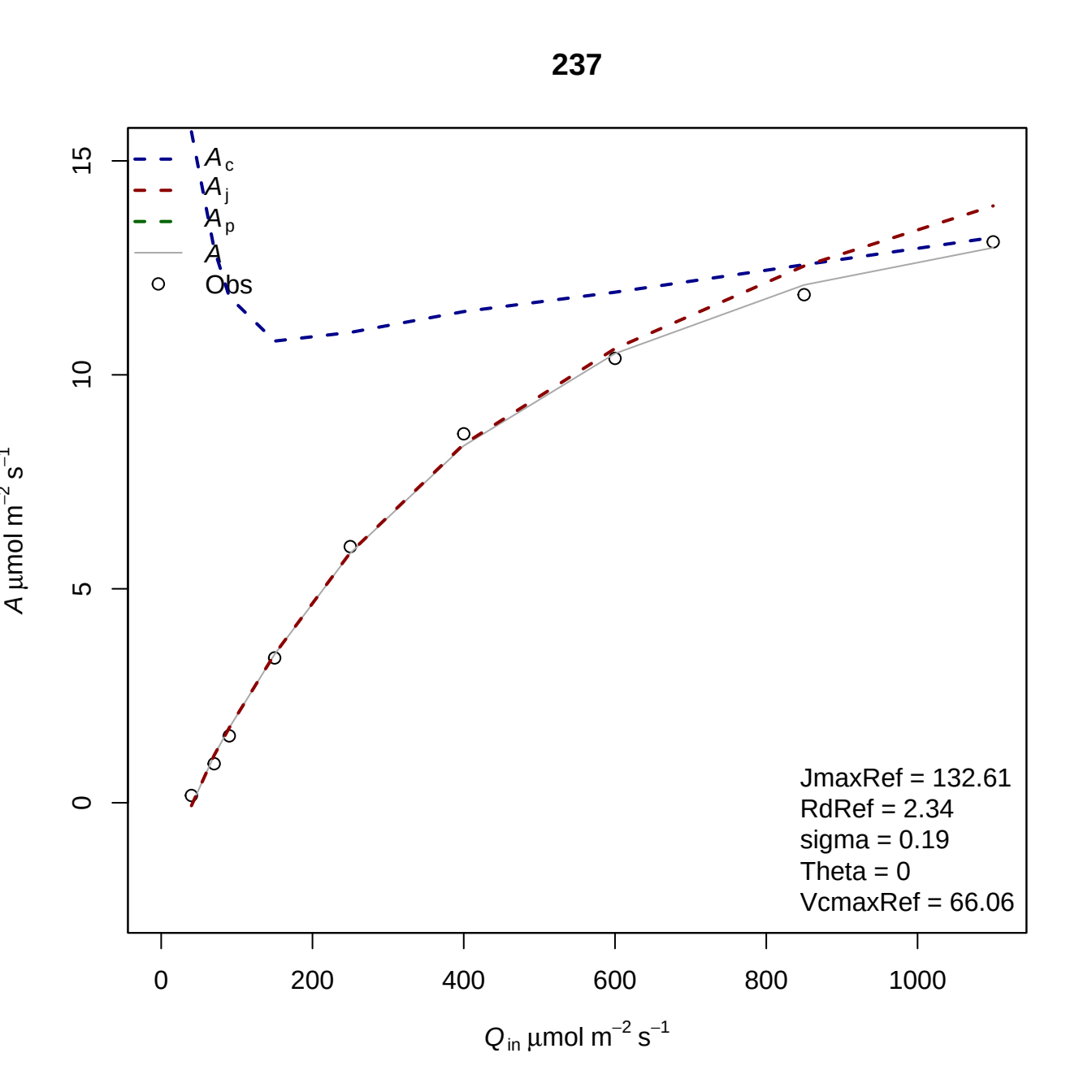

232
A_c
A_j
A_p
A
Obs
4
2
0
−2
A µmol m^{-2} s^{-1}
JmaxRef = 51.12
RdRef = 3.63
sigma = 0.16
Theta = 0
VcmaxRef = 9999
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

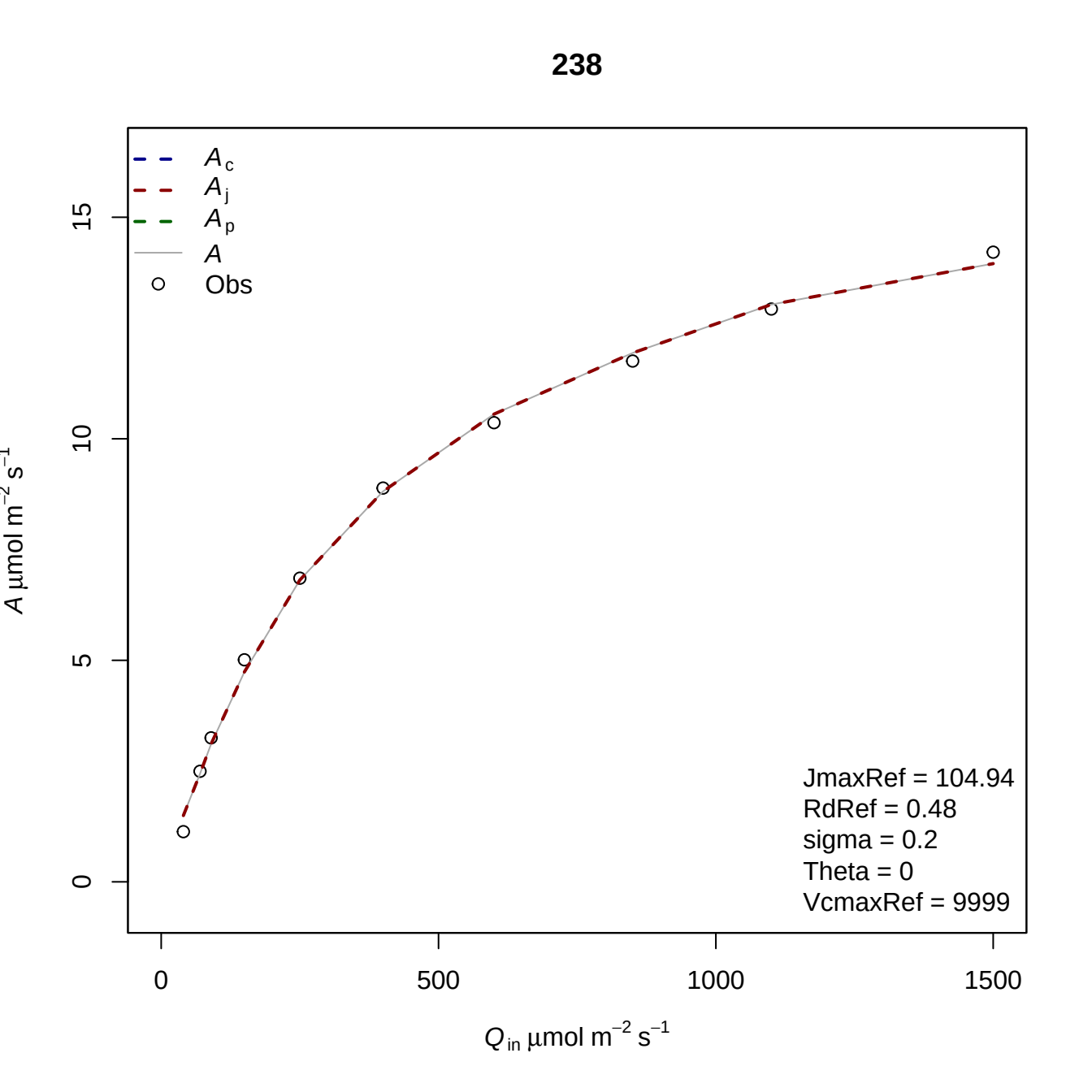

233
A_c
A_j
A_p
A
Obs
JmaxRef = 11.17
RdRef = 1.69
sigma = 0.12
Theta = 0
VcmaxRef = 9999
0.5
0.0
−0.5
−1.0
−1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

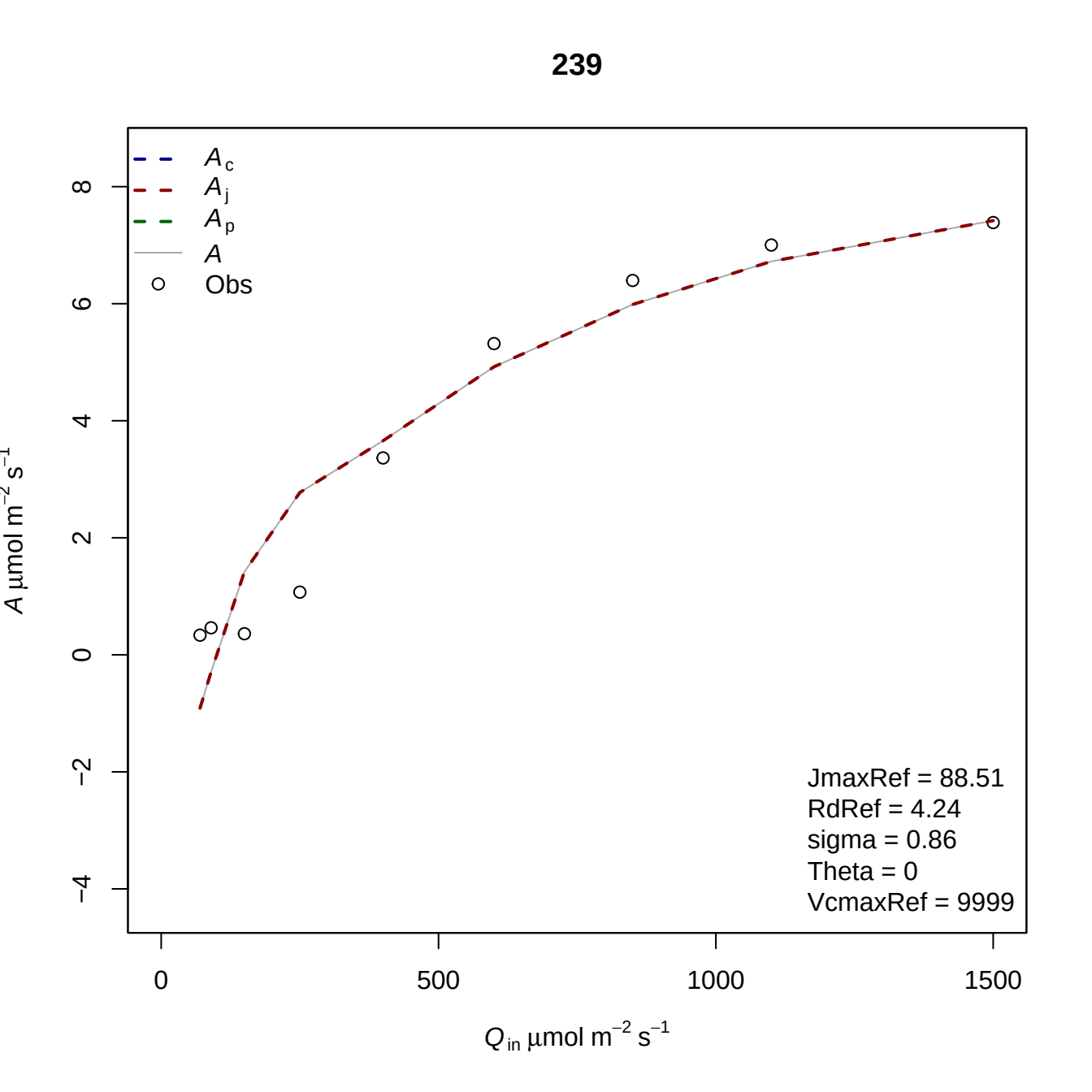

A_c
A_j
A_p
A
Obs
JmaxRef = 61.2
RdRef = 2.48
sigma = 0.32
Theta = 0
VcmaxRef = 9999
6
4
2
0
−2
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

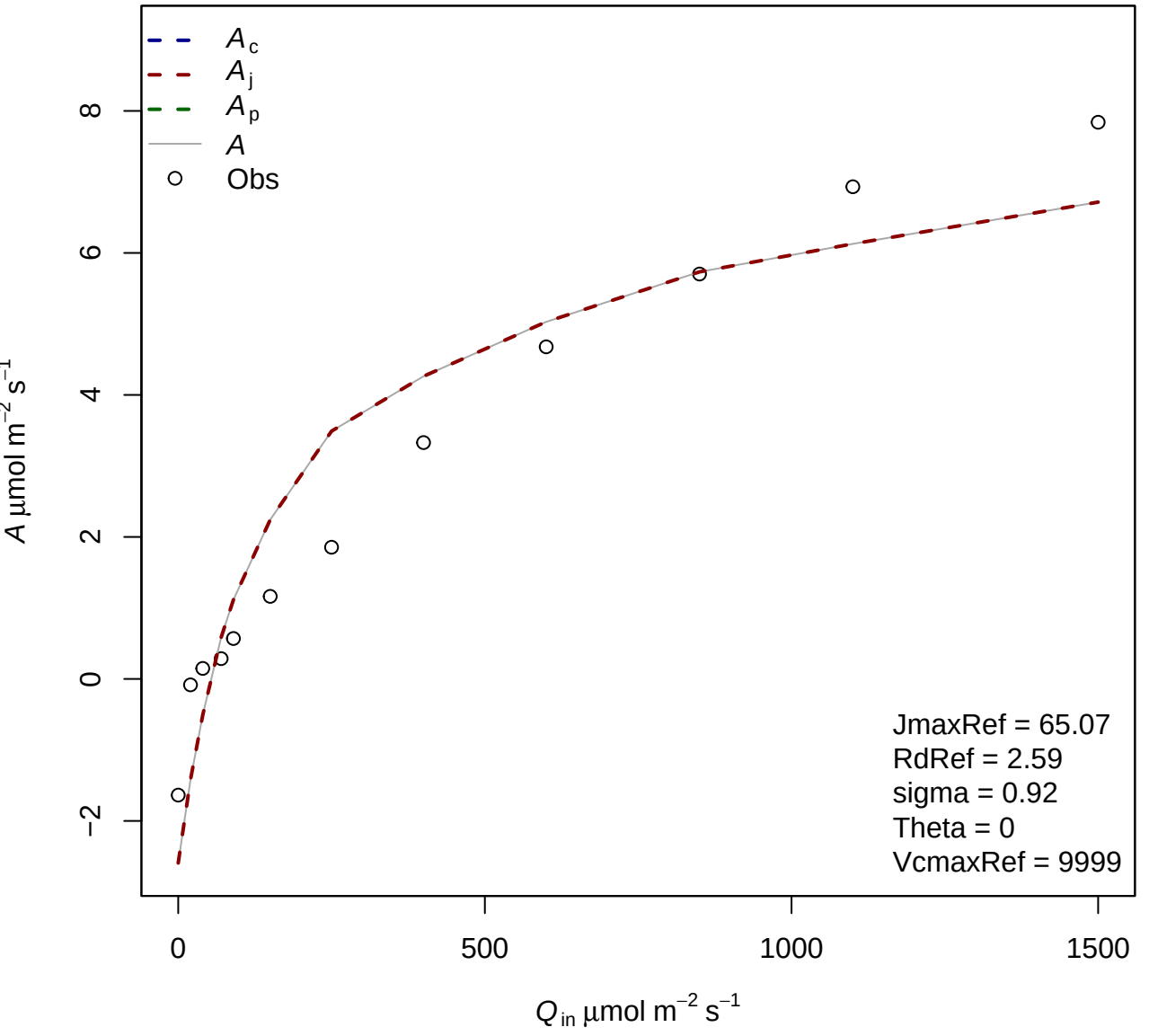

235
A_c
A_j
A_p
A
Obs
JmaxRef = 128.49
RdRef = 1.22
sigma = 0.13
Theta = 0
VcmaxRef = 56.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000

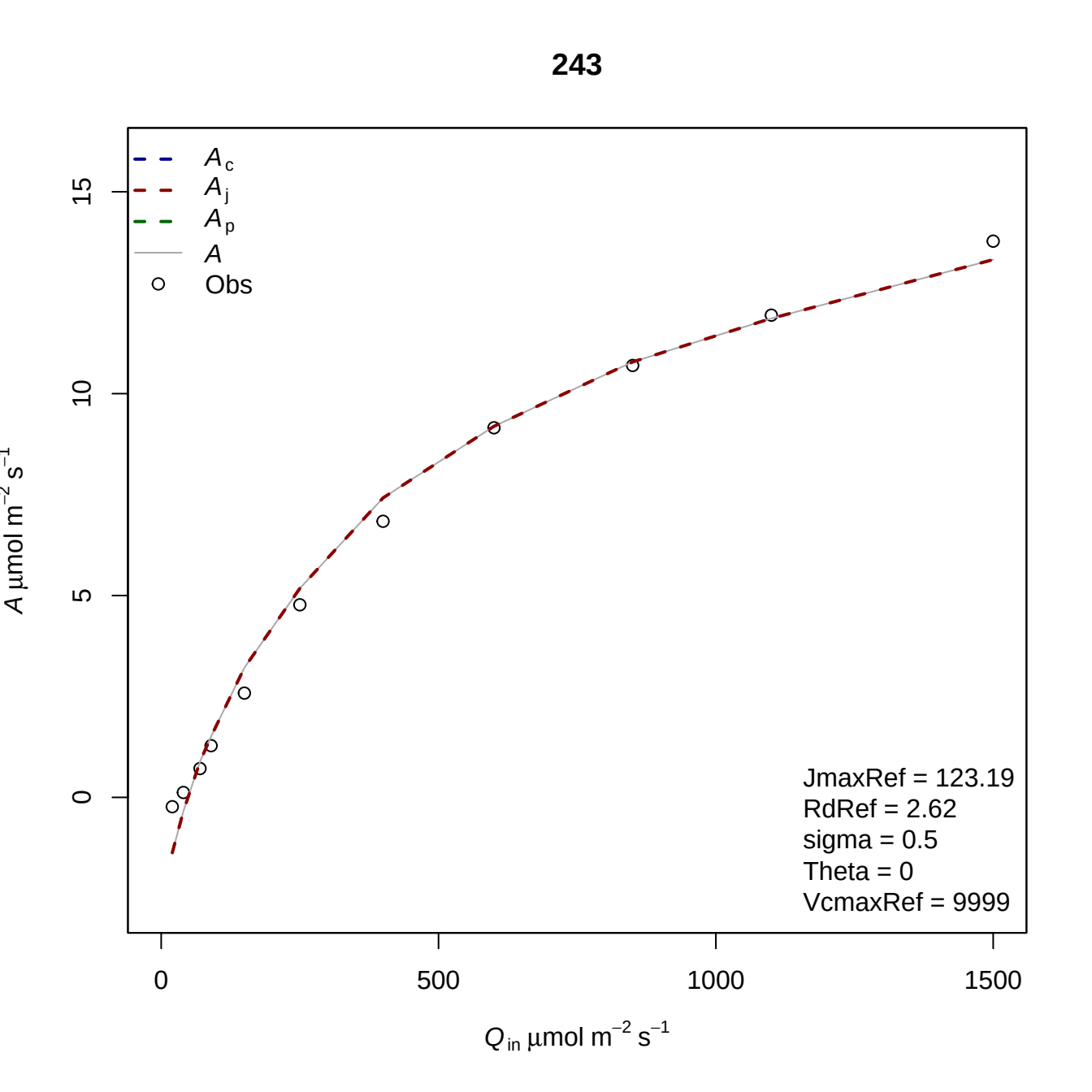

236
A_c
A_j
A_p
A
Obs
JmaxRef = 138.2
RdRef = 1.22
sigma = 0.14
Theta = 0
VcmaxRef = 57.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000

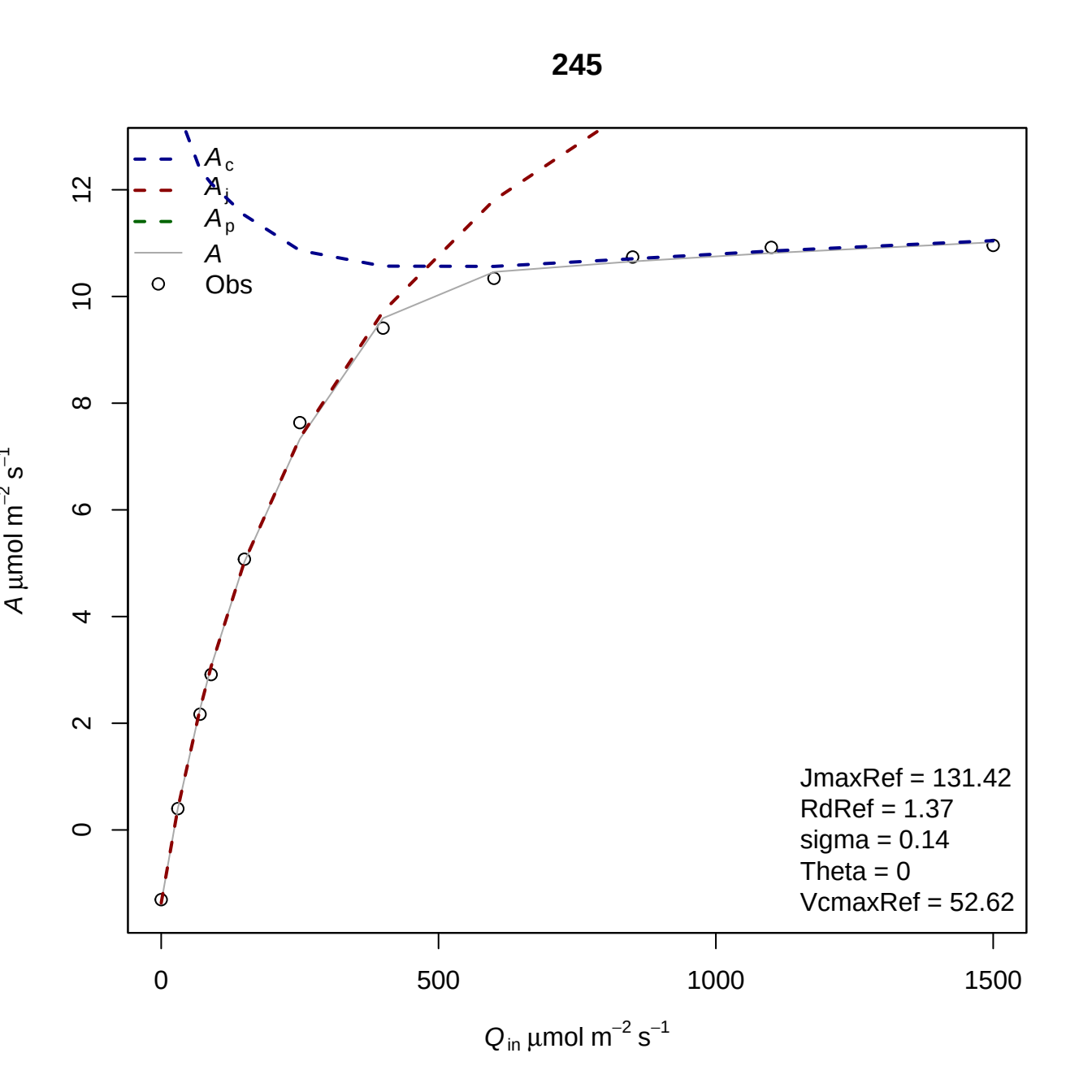

237
A_c
A_j
A_p
A
Obs
JmaxRef = 132.61
RdRef = 2.34
sigma = 0.19
Theta = 0
VcmaxRef = 66.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000

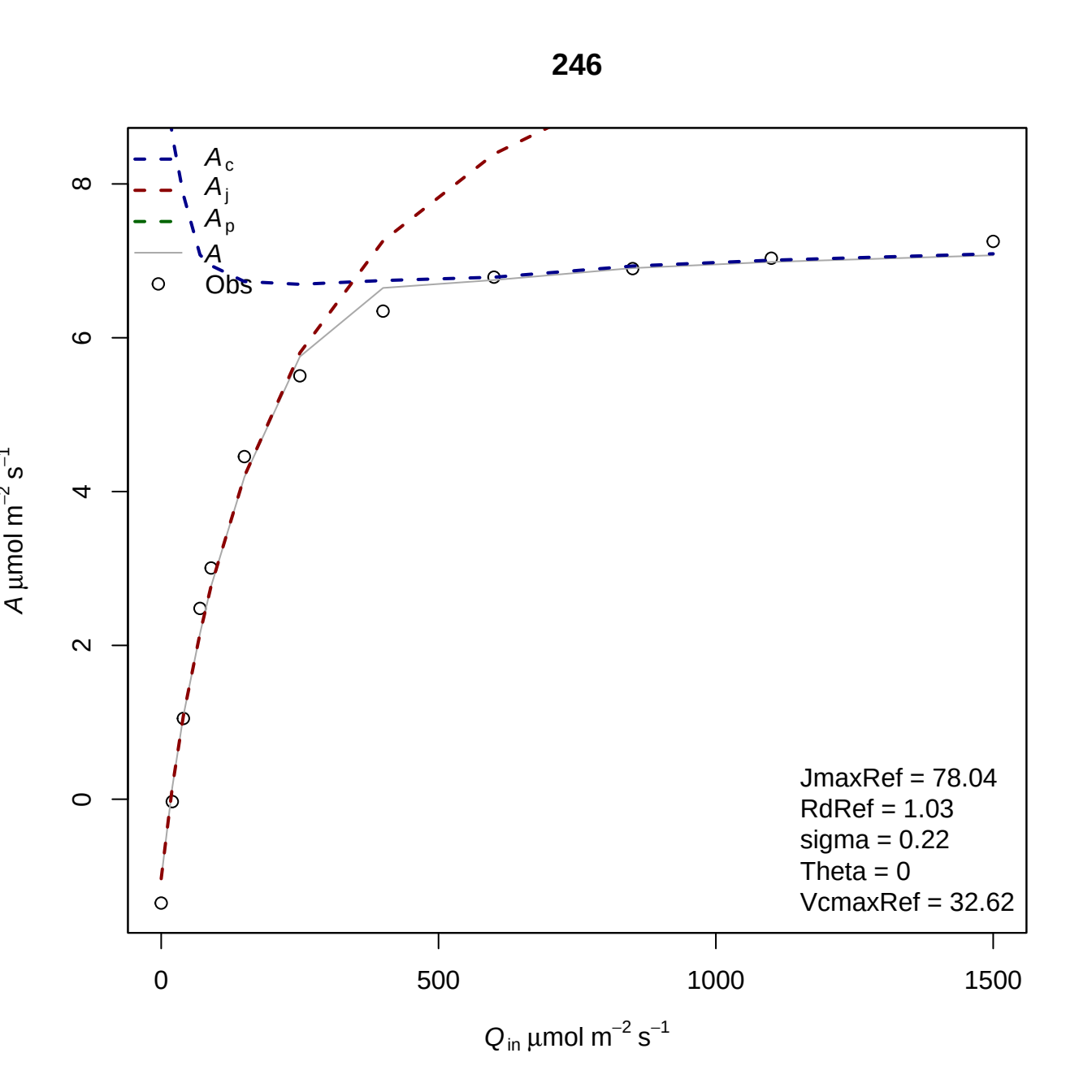

238
A_c
A_j
A_p
A
Obs
JmaxRef = 104.94
RdRef = 0.48
sigma = 0.2
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

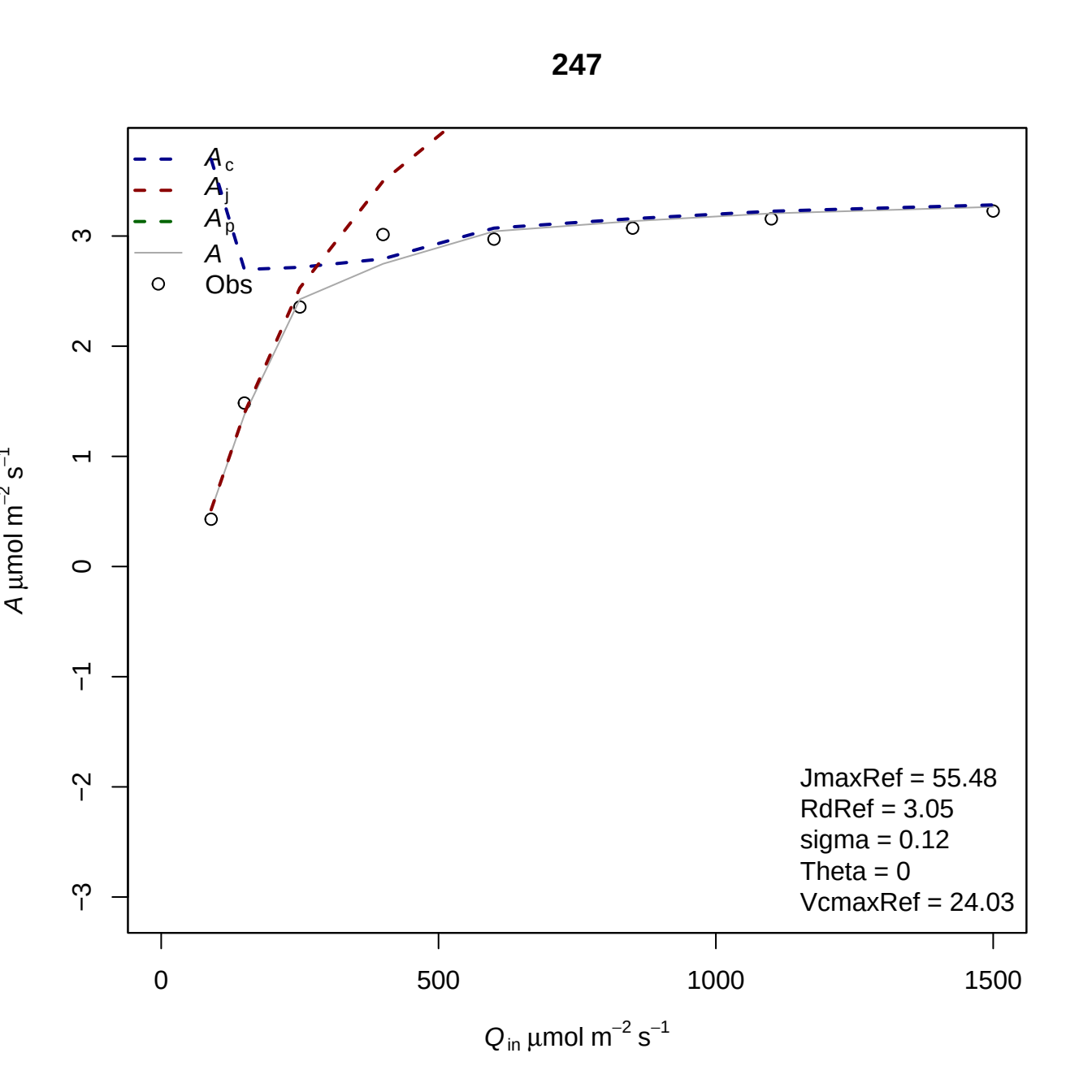

239
A_c
A_j
A_p
A
Obs
JmaxRef = 88.51
RdRef = 4.24
sigma = 0.86
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

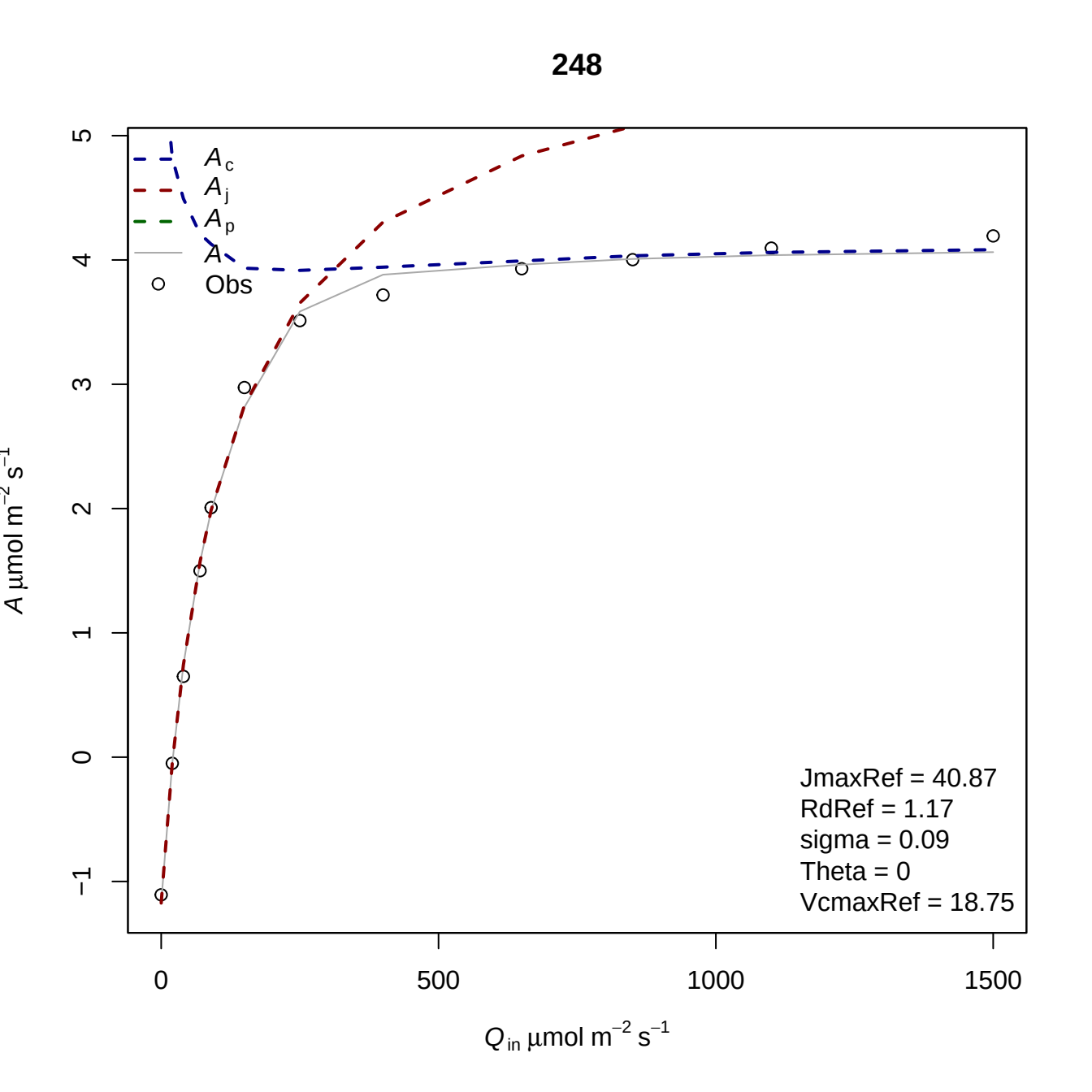

A_c
A_j
A_p
A
Obs
JmaxRef = 65.07
RdRef = 2.59
sigma = 0.92
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

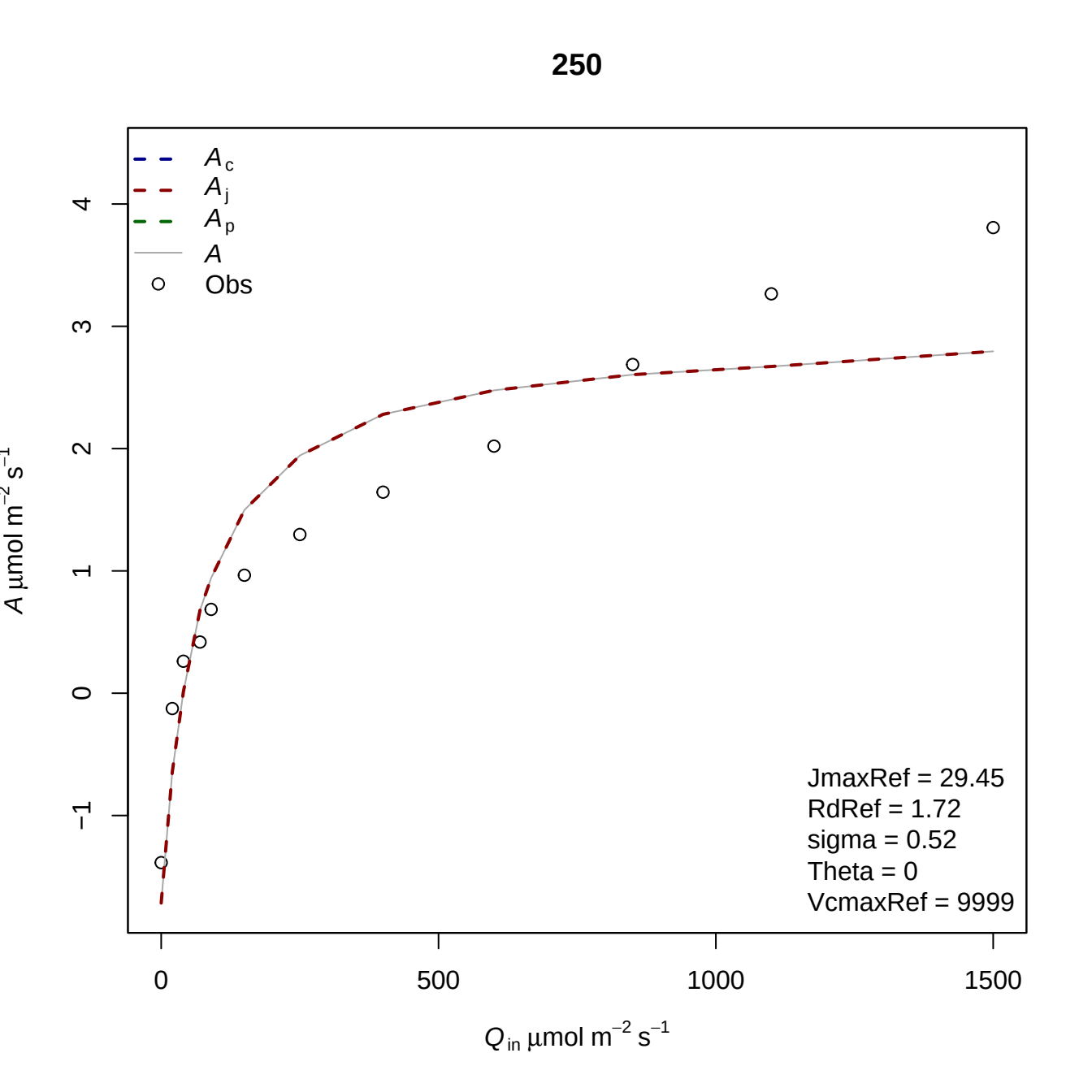

243
A_c
A_j
A_p
A
Obs
JmaxRef = 123.19
RdRef = 2.62
sigma = 0.5
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

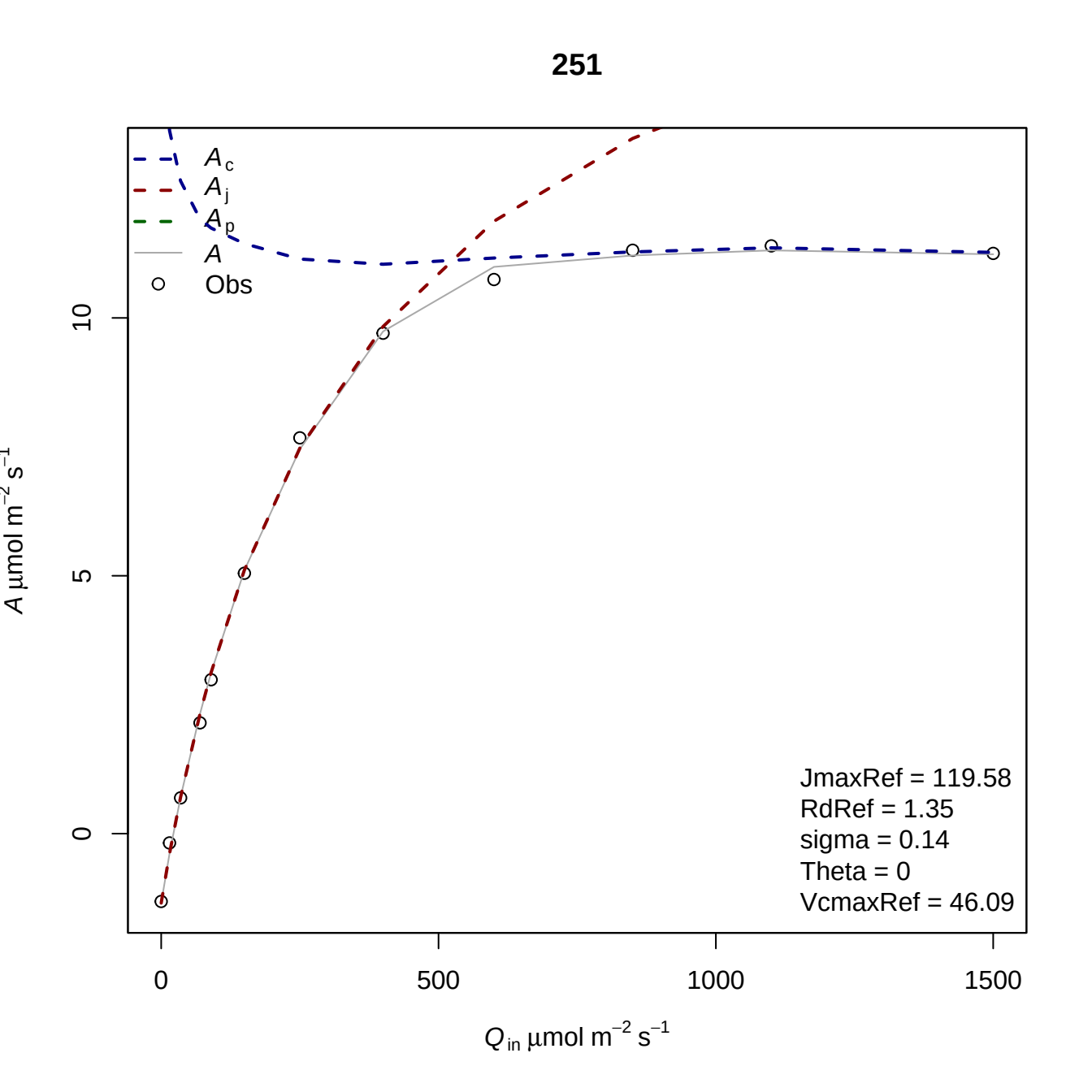

245
A_c
A_j
A_p
A
Obs
JmaxRef = 131.42
RdRef = 1.37
sigma = 0.14
Theta = 0
VcmaxRef = 52.62
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

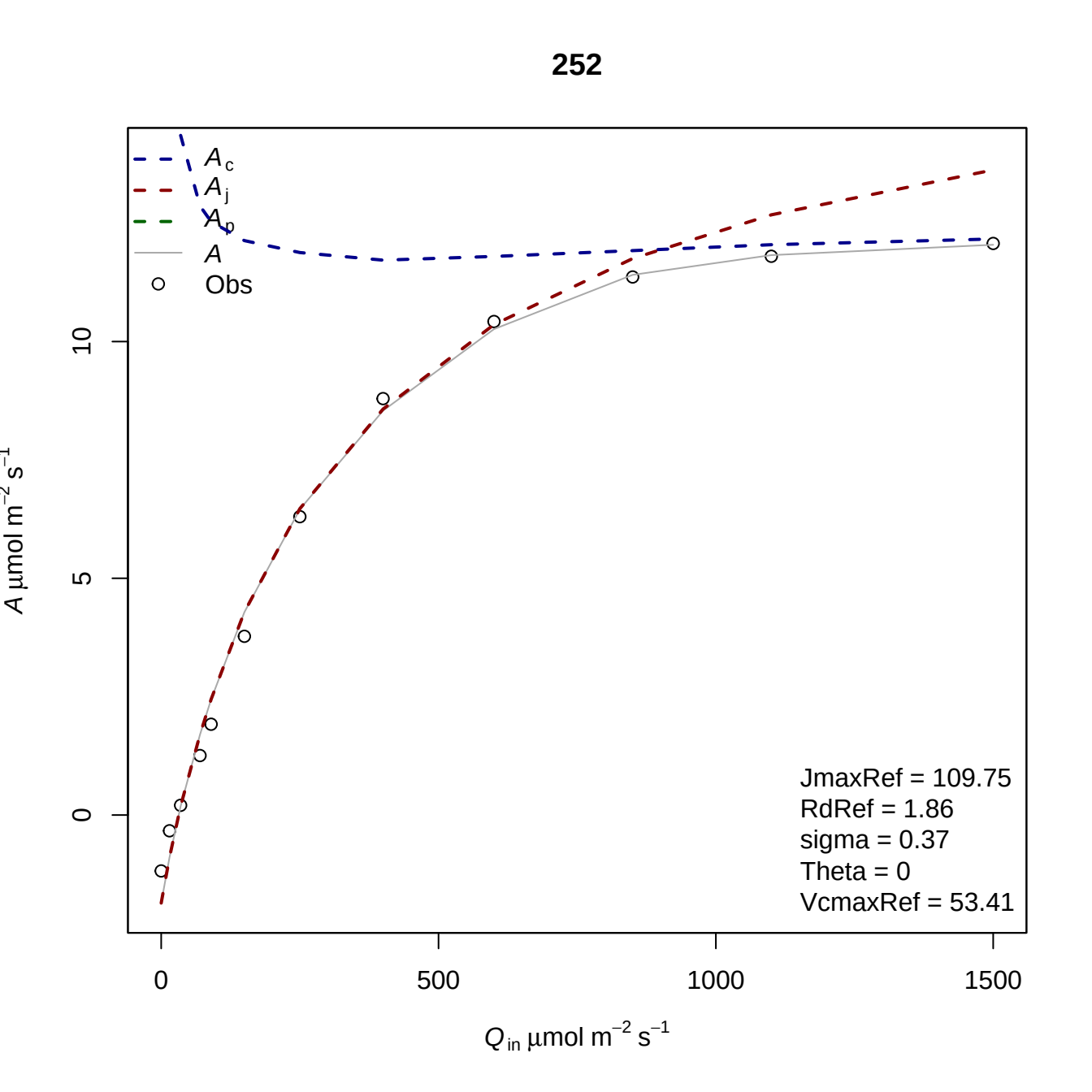

246
A_c
A_j
A_p
A
Obs
JmaxRef = 78.04
RdRef = 1.03
sigma = 0.22
Theta = 0
VcmaxRef = 32.62
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

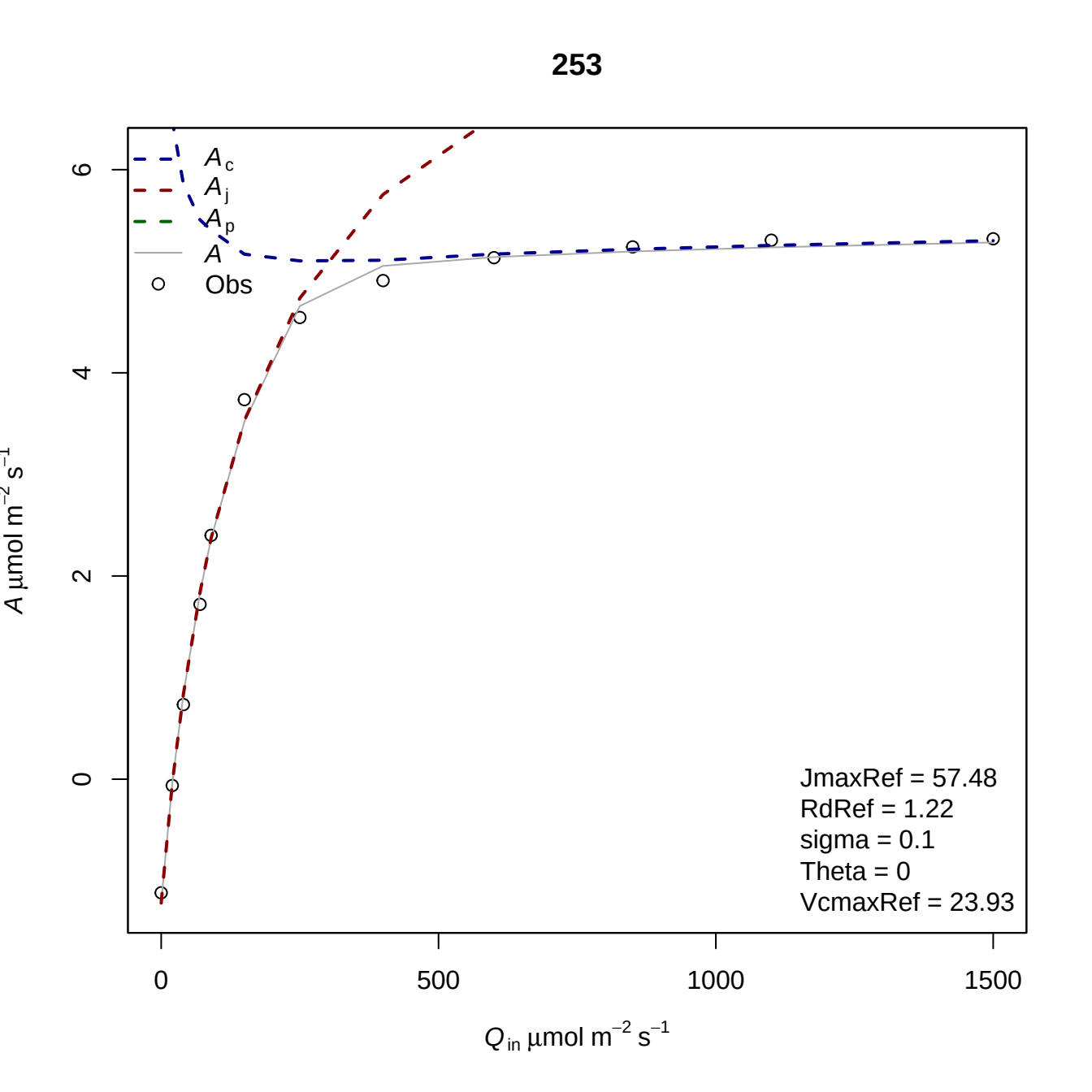

247
A_c
A_j
A_p
A
Obs
JmaxRef = 55.48
RdRef = 3.05
sigma = 0.12
Theta = 0
VcmaxRef = 24.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
−3
0
500
1000
1500

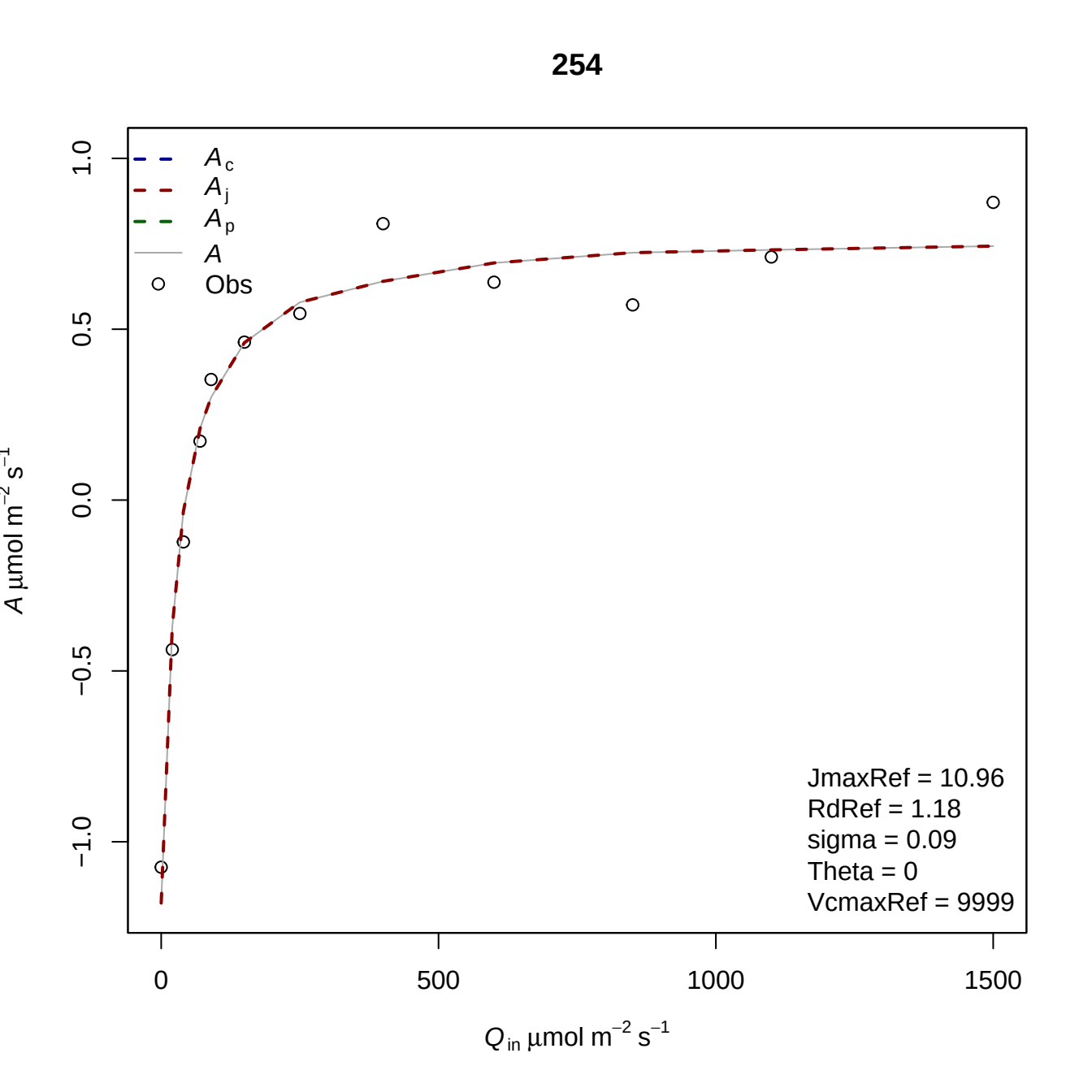

248
A_c
A_j
A_p
A
Obs
JmaxRef = 40.87
RdRef = 1.17
sigma = 0.09
Theta = 0
VcmaxRef = 18.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

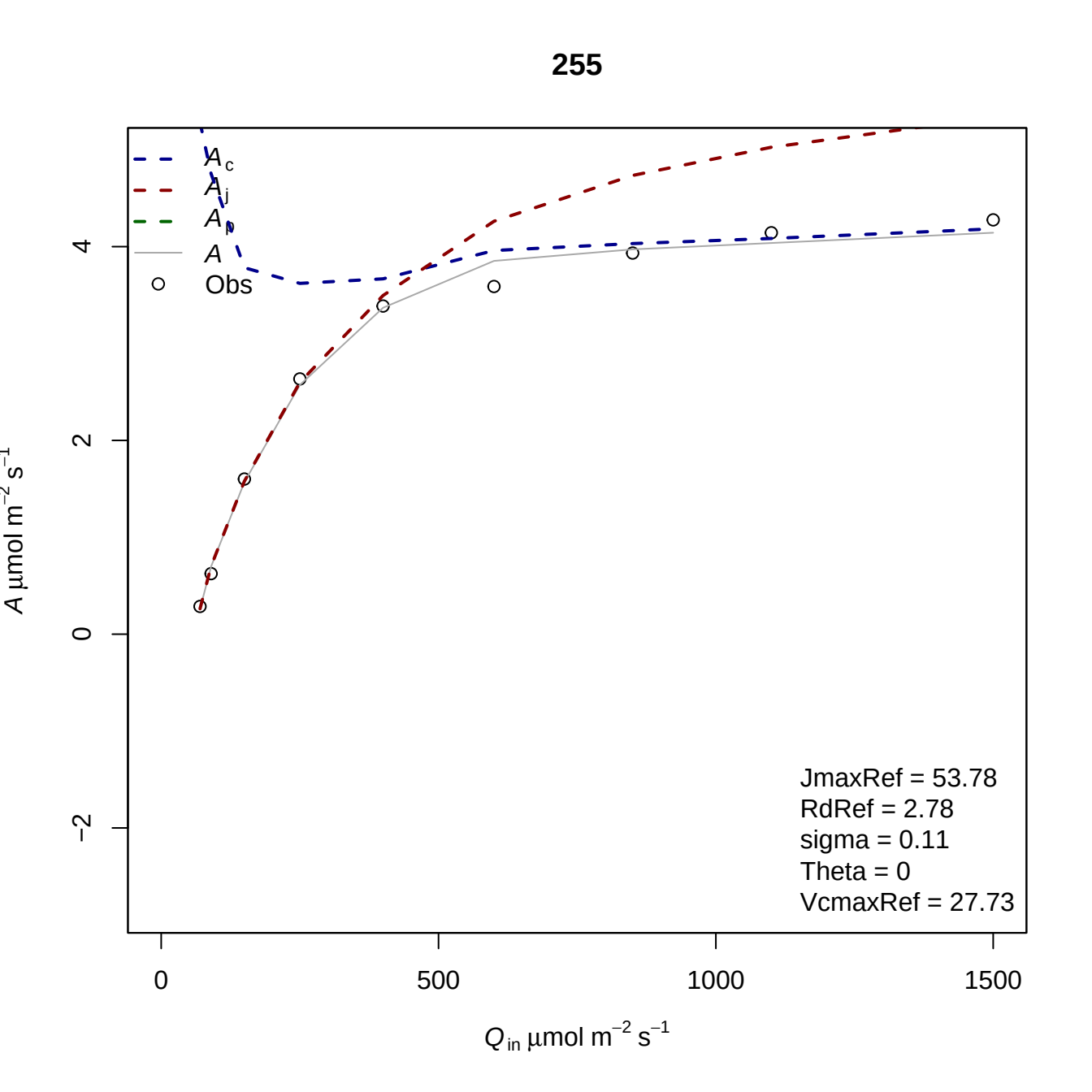

250
A_c
A_j
A_p
A
Obs
JmaxRef = 29.45
RdRef = 1.72
sigma = 0.52
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

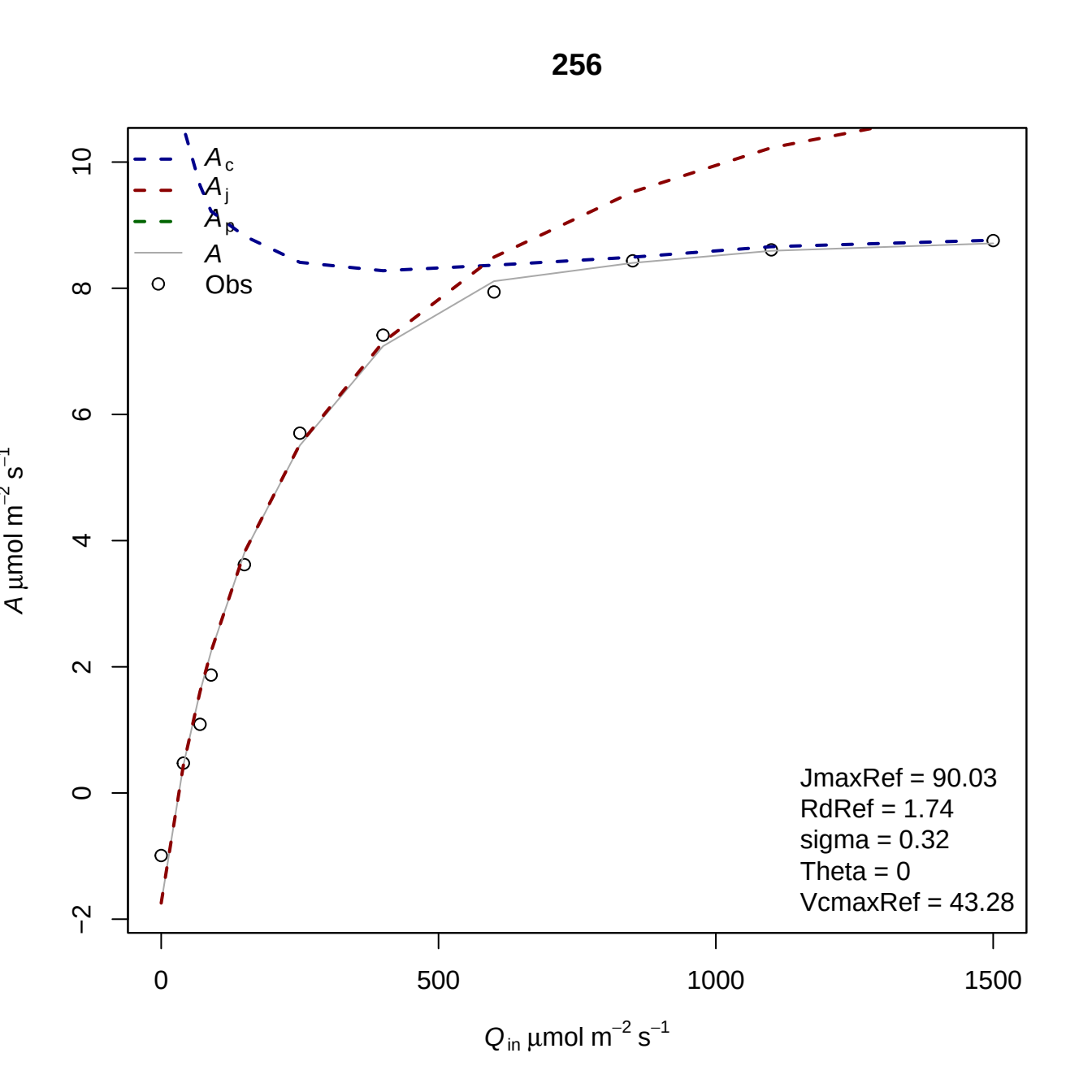

251
A_c
A_j
A_p
A
Obs
JmaxRef = 119.58
RdRef = 1.35
sigma = 0.14
Theta = 0
VcmaxRef = 46.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

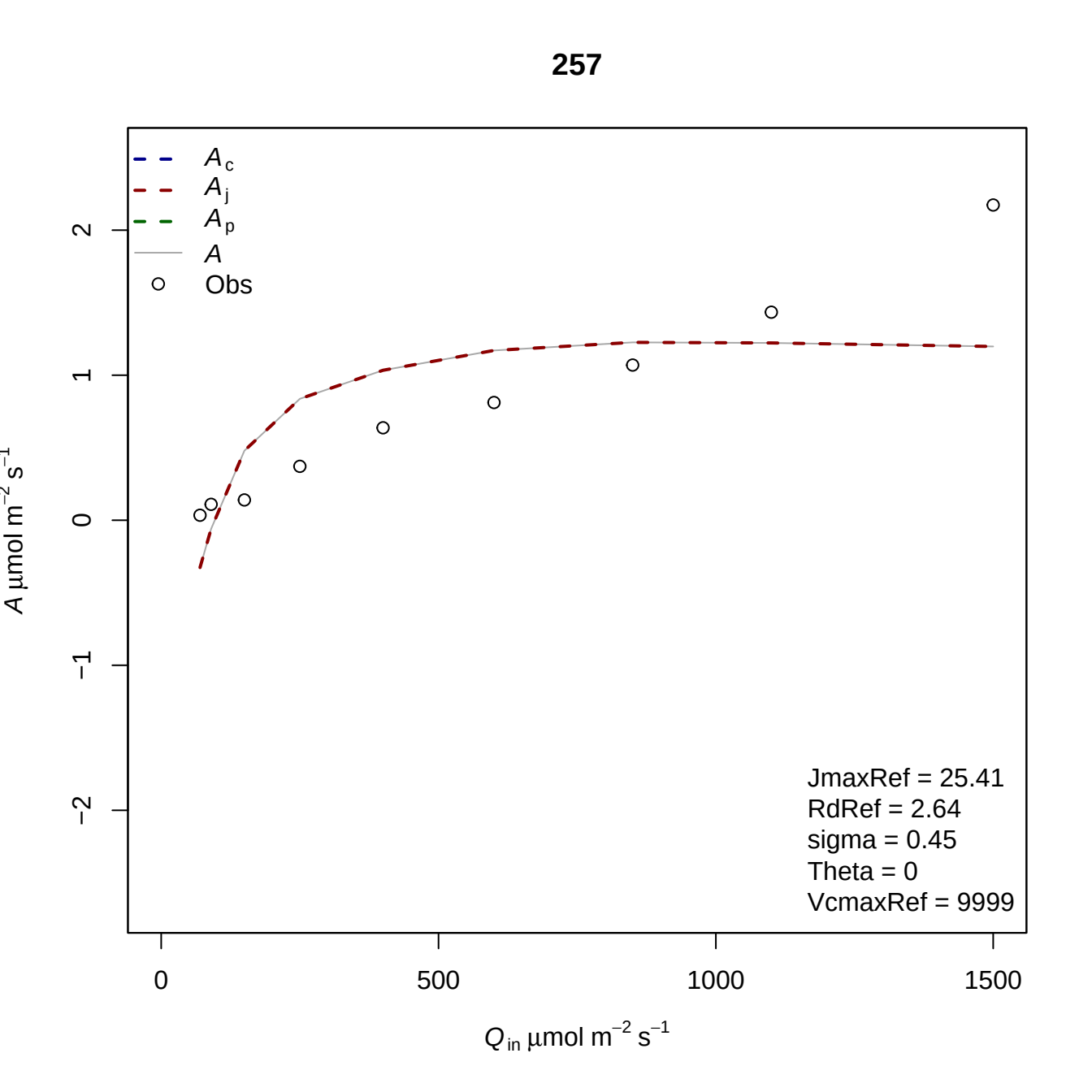

252
A_c
A_j
A_p
A
Obs
JmaxRef = 109.75
RdRef = 1.86
sigma = 0.37
Theta = 0
VcmaxRef = 53.41
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

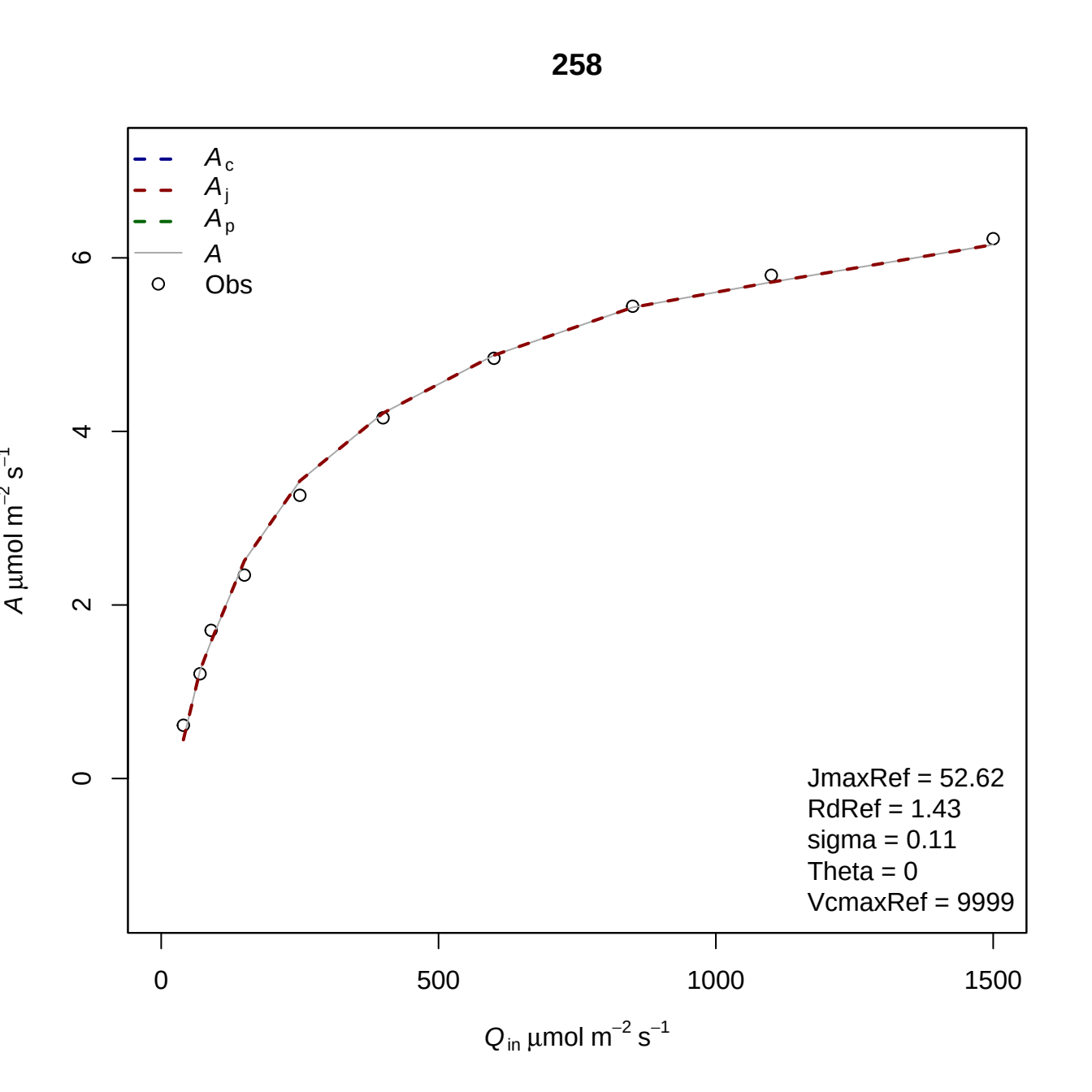

253
A_c
A_j
A_p
A
Obs
JmaxRef = 57.48
RdRef = 1.22
sigma = 0.1
Theta = 0
VcmaxRef = 23.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

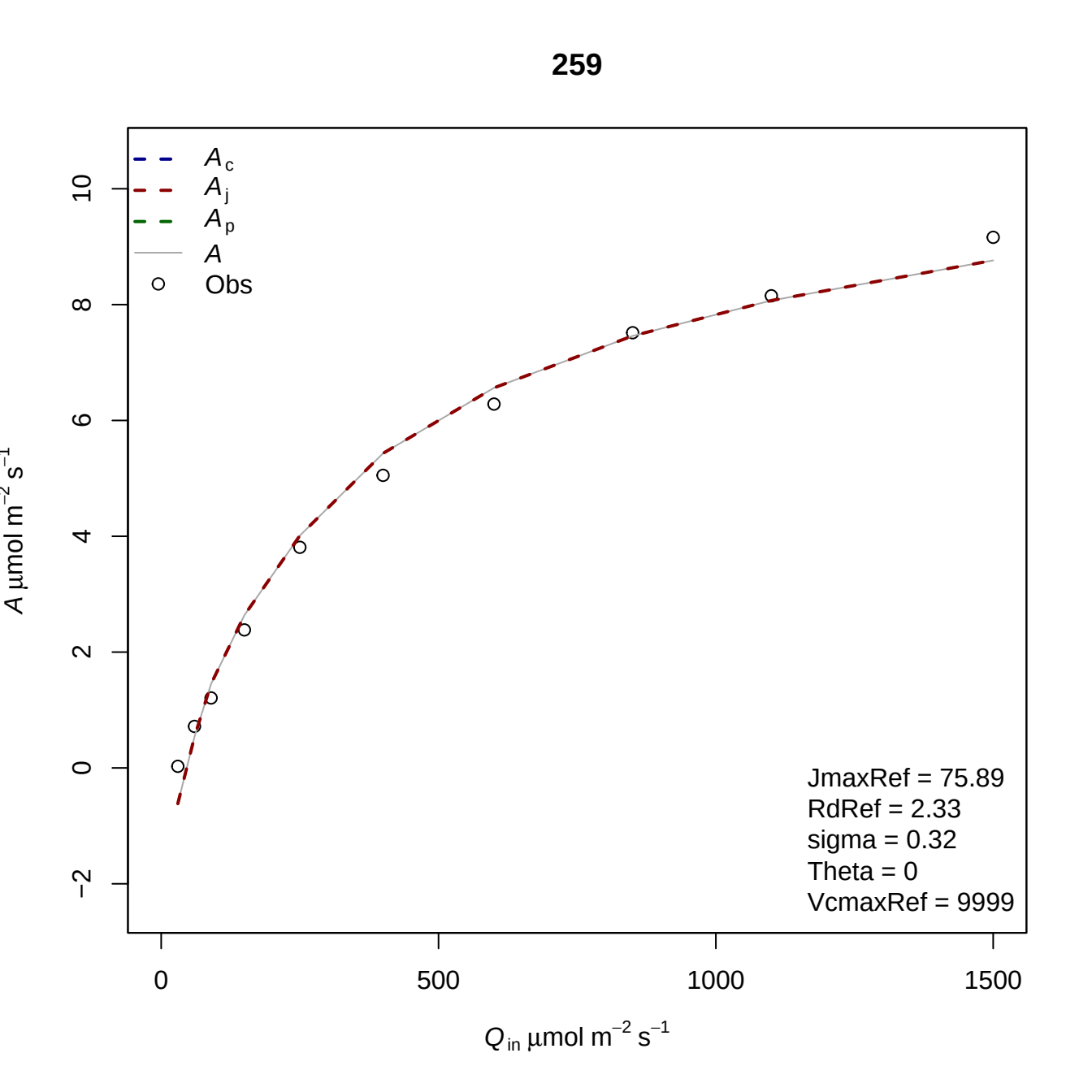

254
A_c
A_j
A_p
A
Obs
JmaxRef = 10.96
RdRef = 1.18
sigma = 0.09
Theta = 0
VcmaxRef = 9999
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

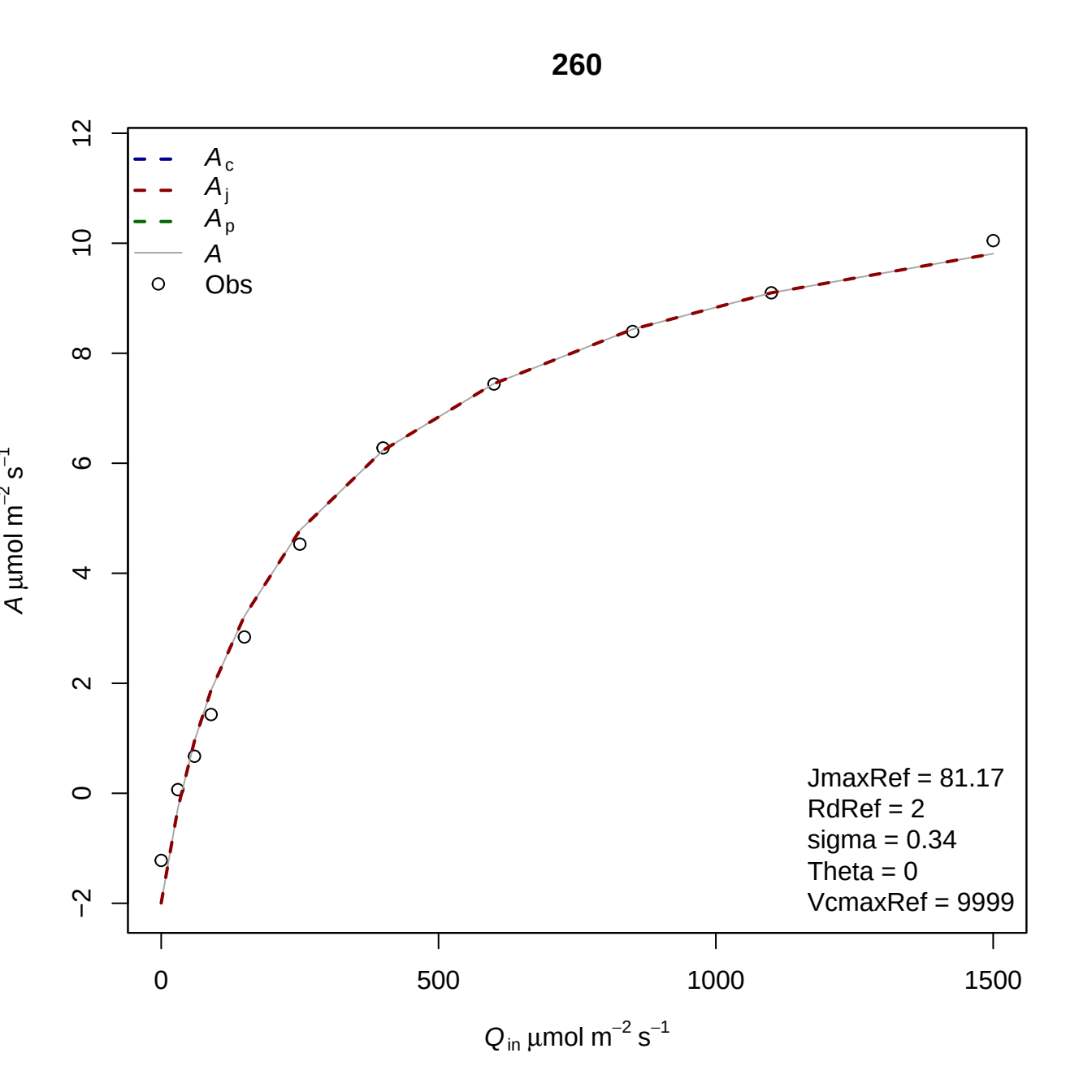

255
A_c
A_j
A_p
A
Obs
JmaxRef = 53.78
RdRef = 2.78
sigma = 0.11
Theta = 0
VcmaxRef = 27.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

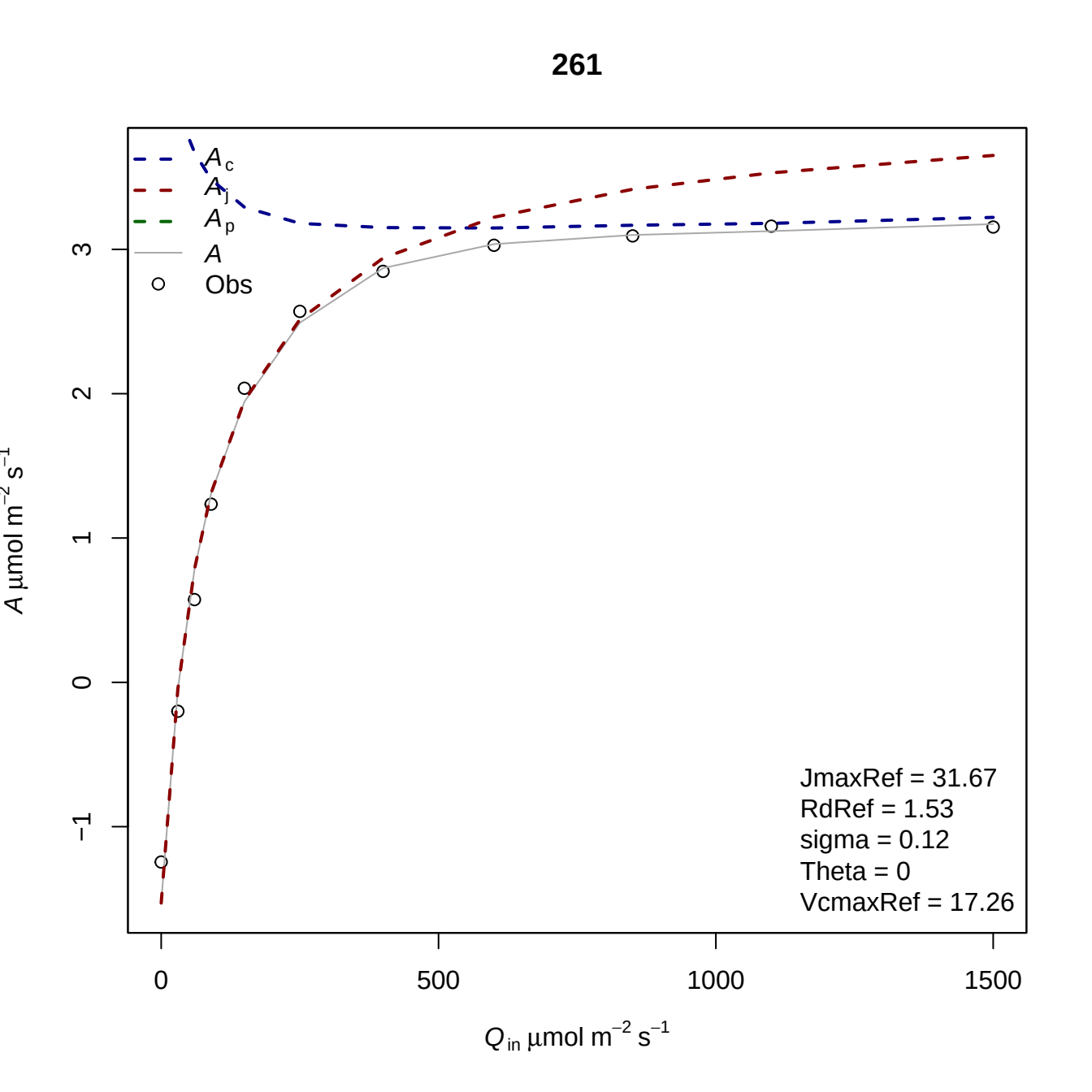

256
A_c
A_j
A_p
A
Obs
JmaxRef = 90.03
RdRef = 1.74
sigma = 0.32
Theta = 0
VcmaxRef = 43.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

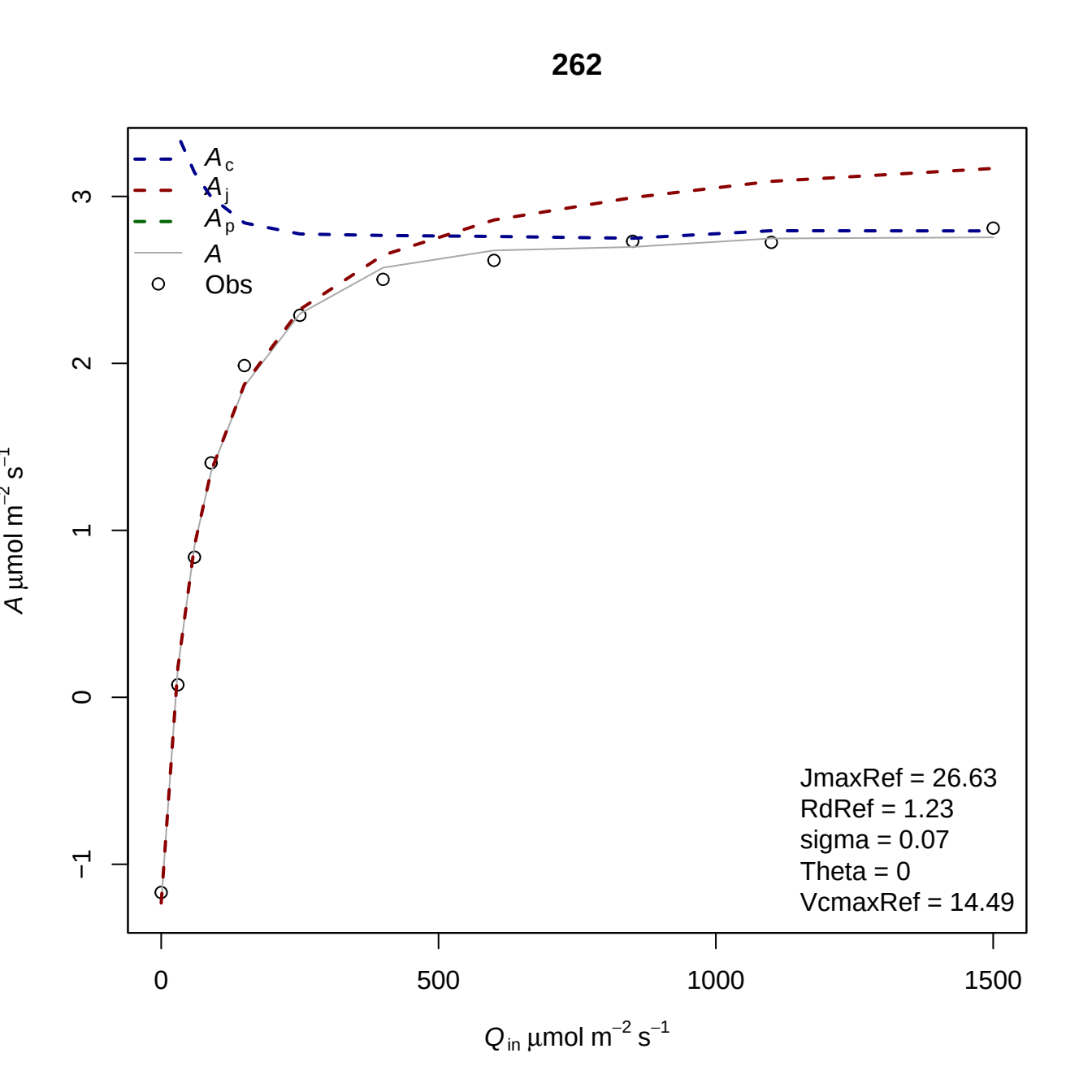

257
A_c
A_j
A_p
A
Obs
JmaxRef = 25.41
RdRef = 2.64
sigma = 0.45
Theta = 0
VcmaxRef = 9999
2
1
0
−1
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

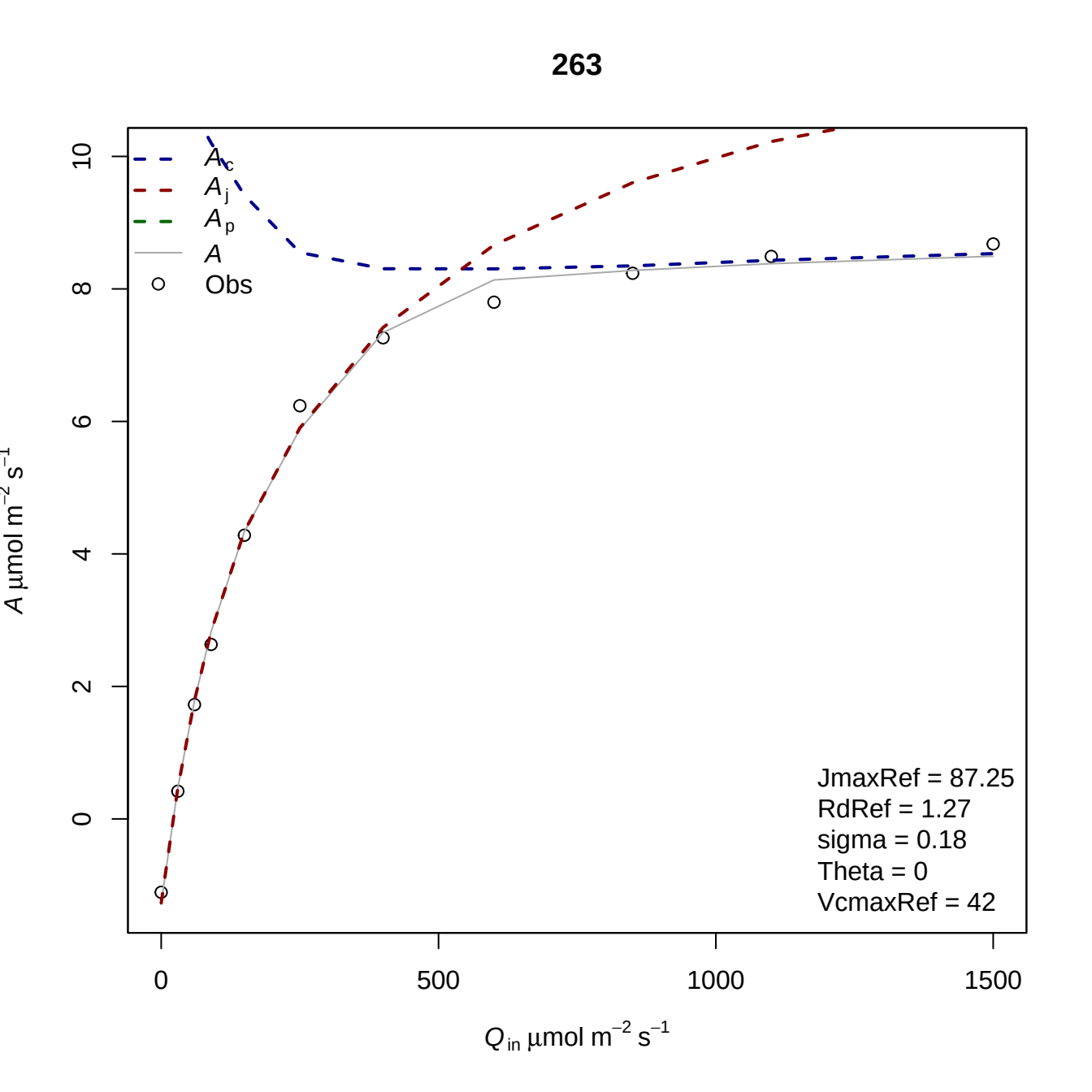

258
A_c
A_j
A_p
A
Obs
JmaxRef = 52.62
RdRef = 1.43
sigma = 0.11
Theta = 0
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

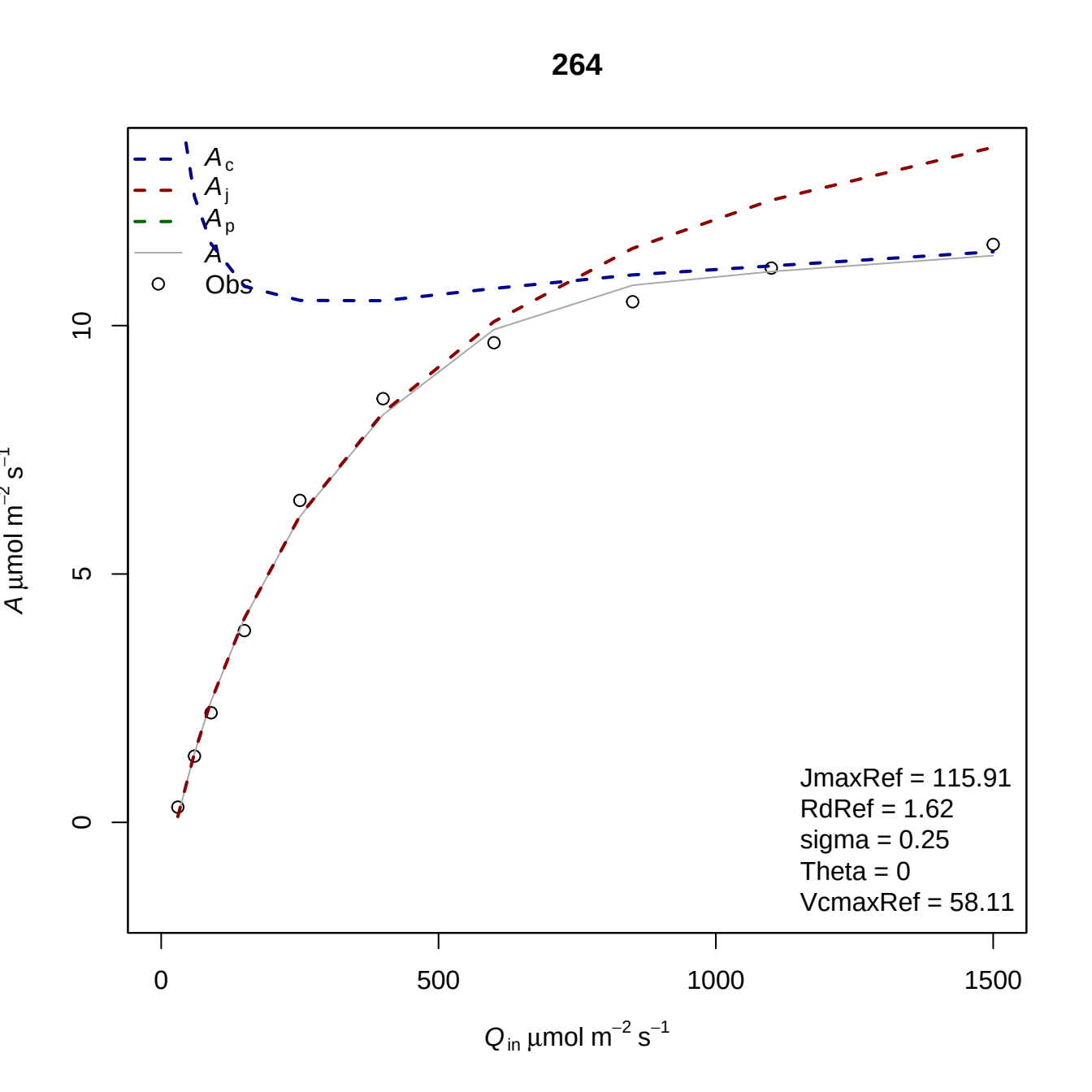

259
A_c
A_j
A_p
A
Obs
JmaxRef = 75.89
RdRef = 2.33
sigma = 0.32
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

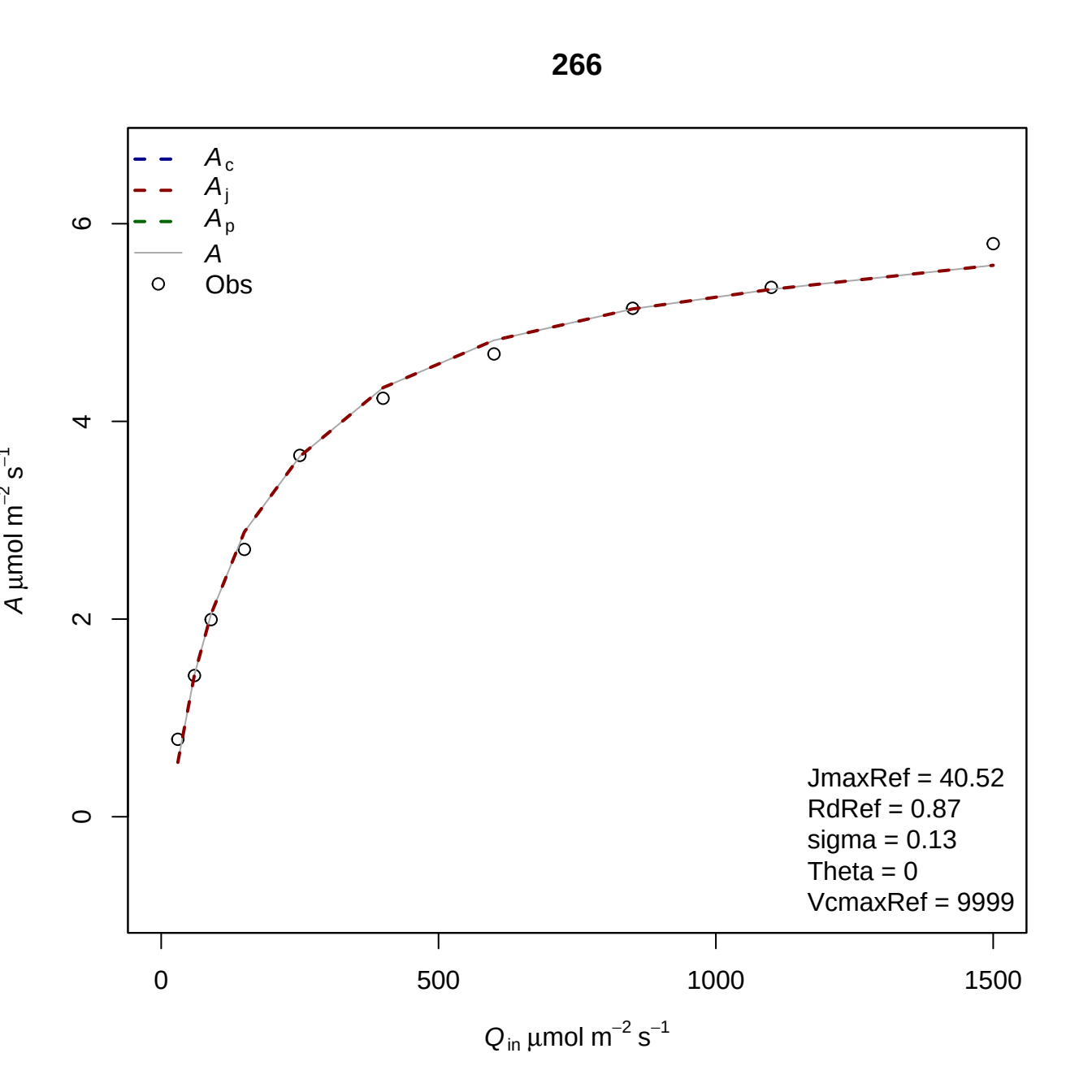

260
A_c
A_j
A_p
A
Obs
JmaxRef = 81.17
RdRef = 2
sigma = 0.34
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

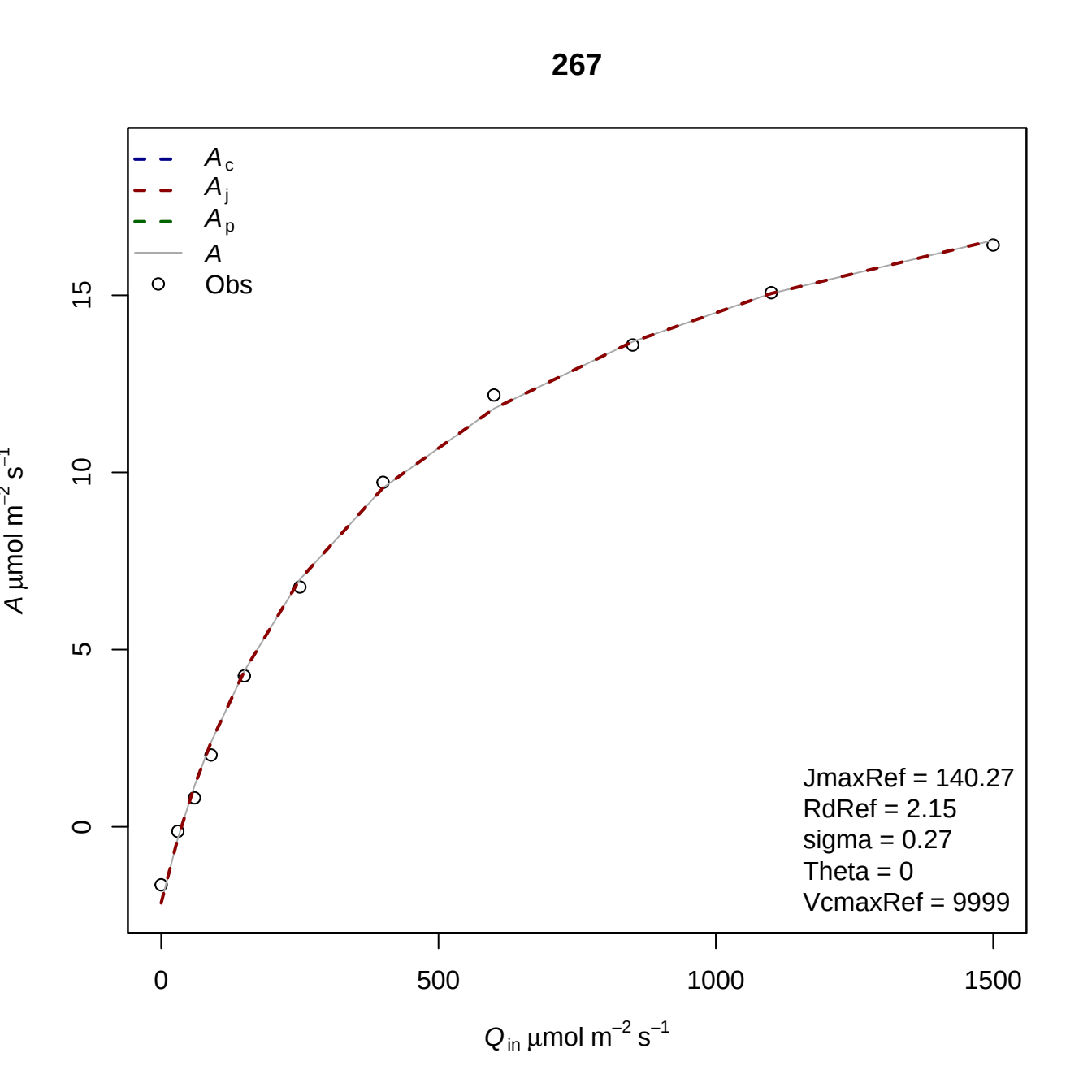

261
A_c
A_j
A_p
A
Obs
JmaxRef = 31.67
RdRef = 1.53
sigma = 0.12
Theta = 0
VcmaxRef = 17.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

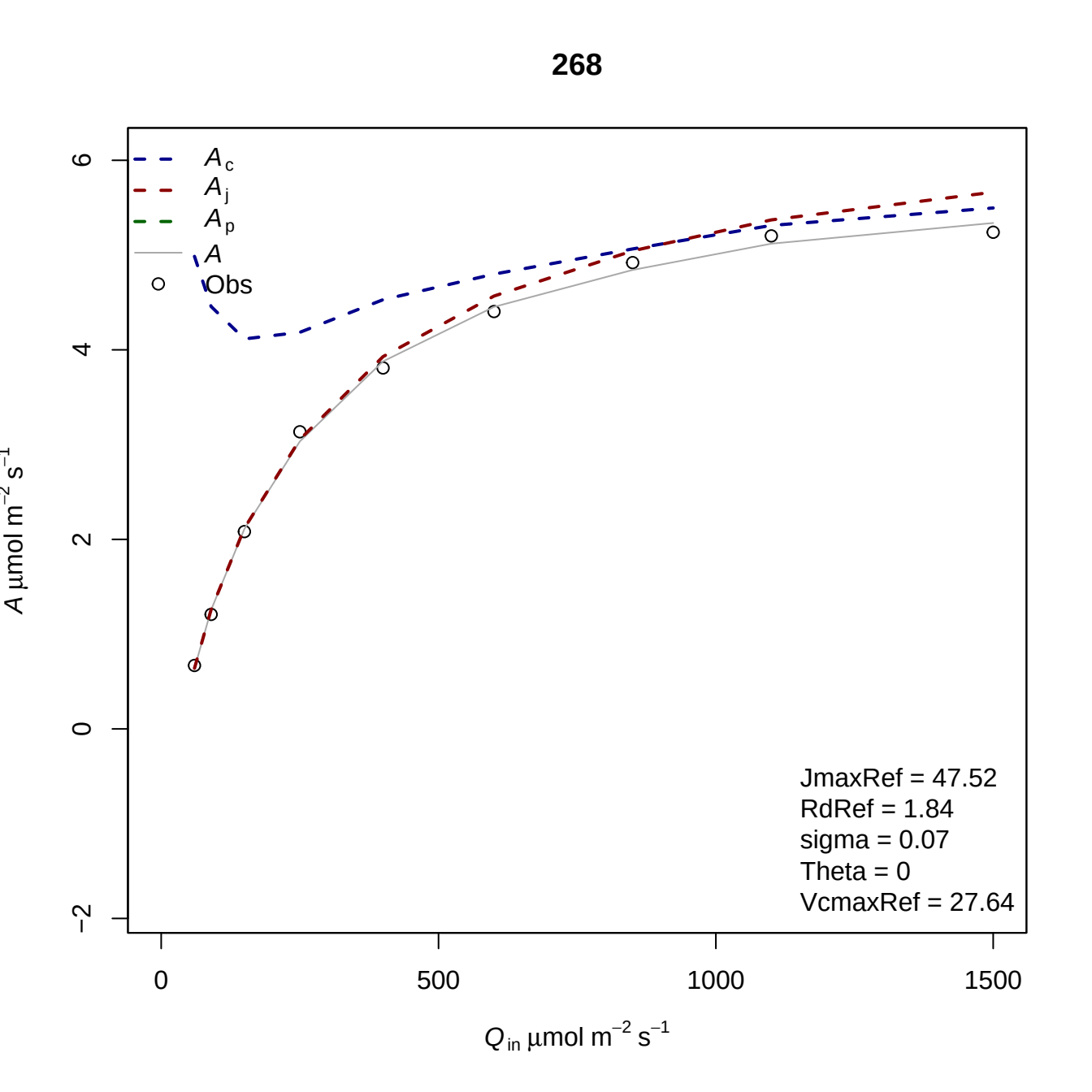

262
A_c
A_j
A_p
A
Obs
JmaxRef = 26.63
RdRef = 1.23
sigma = 0.07
Theta = 0
VcmaxRef = 14.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

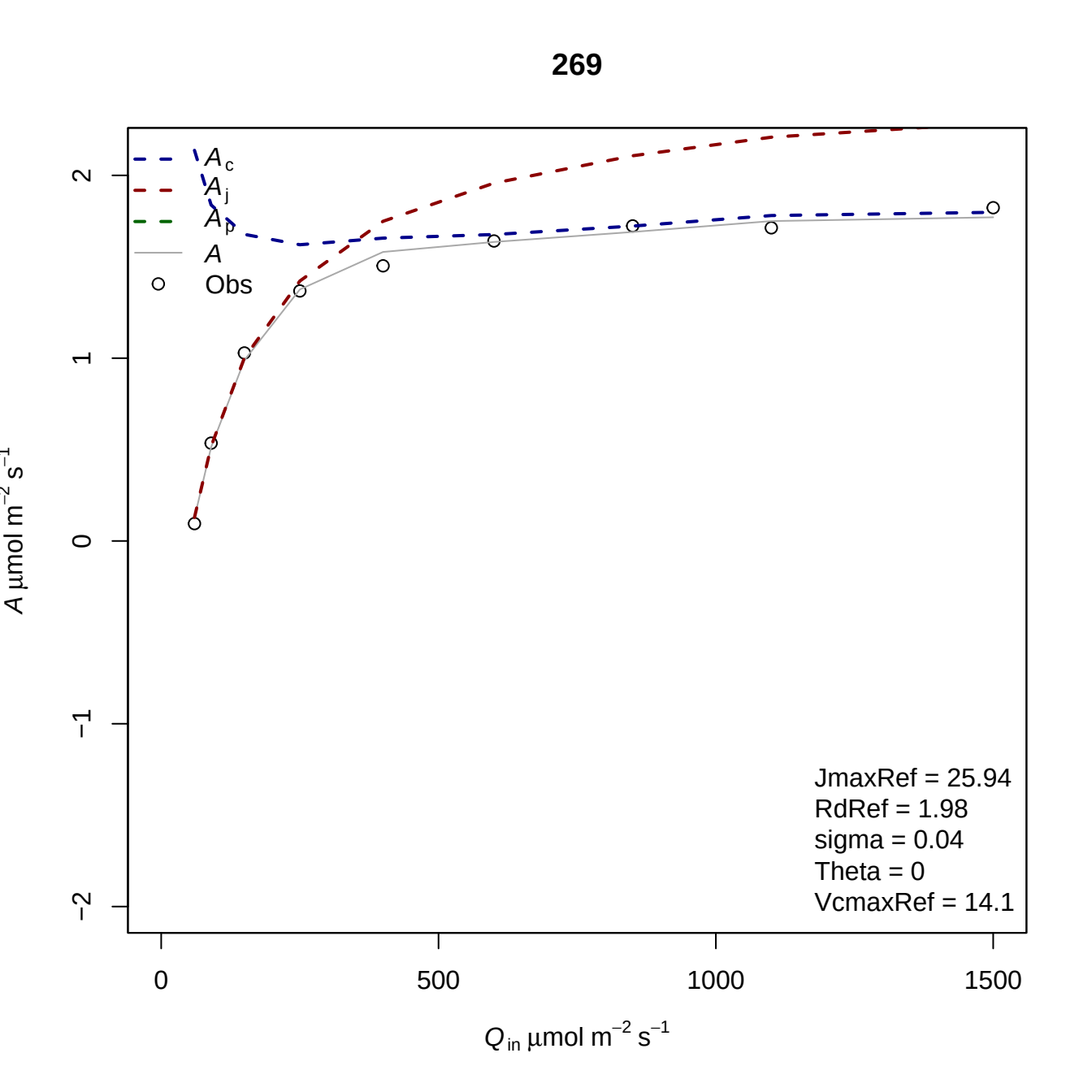

263
A_c
A_j
A_p
A
Obs
JmaxRef = 87.25
RdRef = 1.27
sigma = 0.18
Theta = 0
VcmaxRef = 42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

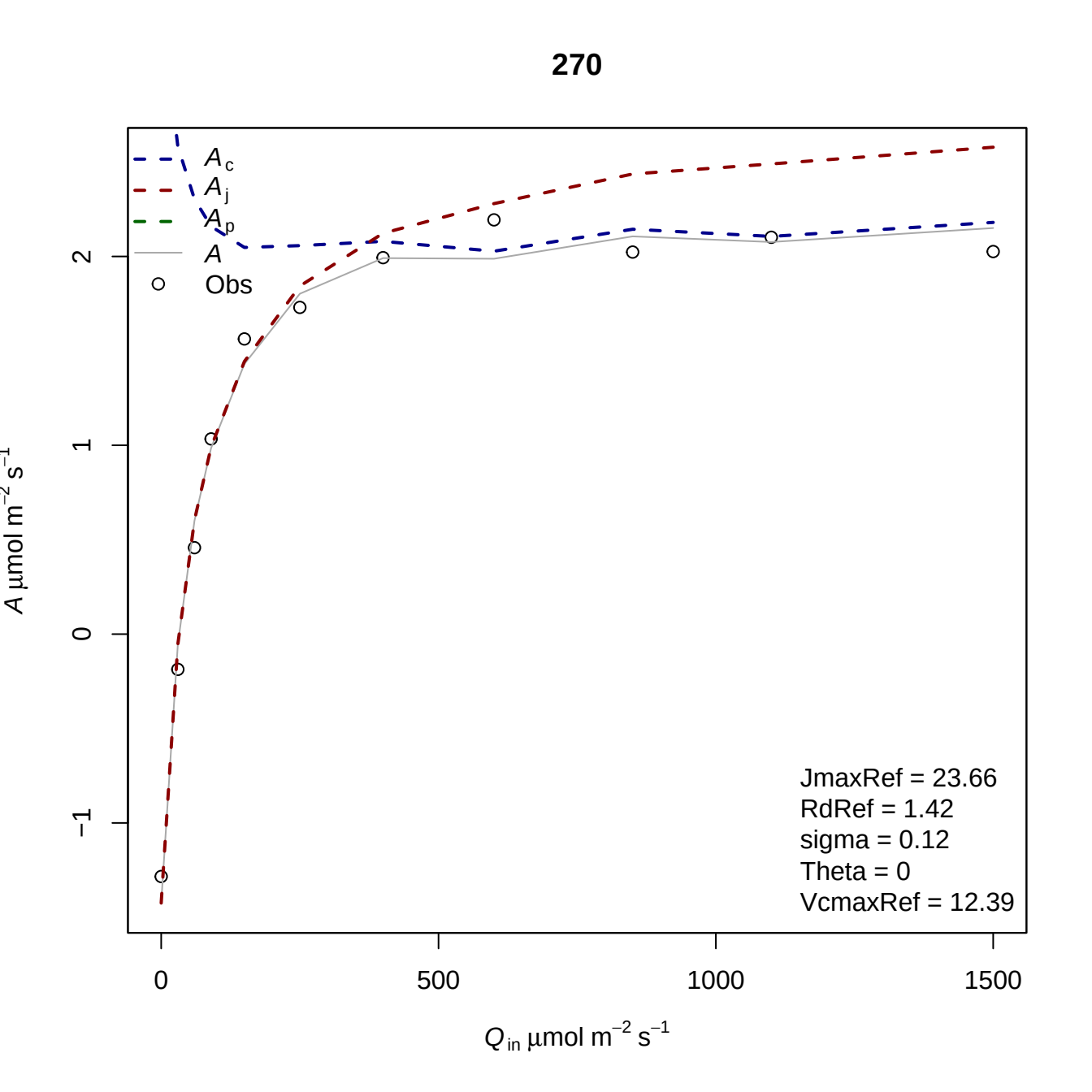

264
A_c
A_j
A_p
A
Obs
JmaxRef = 115.91
RdRef = 1.62
sigma = 0.25
Theta = 0
VcmaxRef = 58.11
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

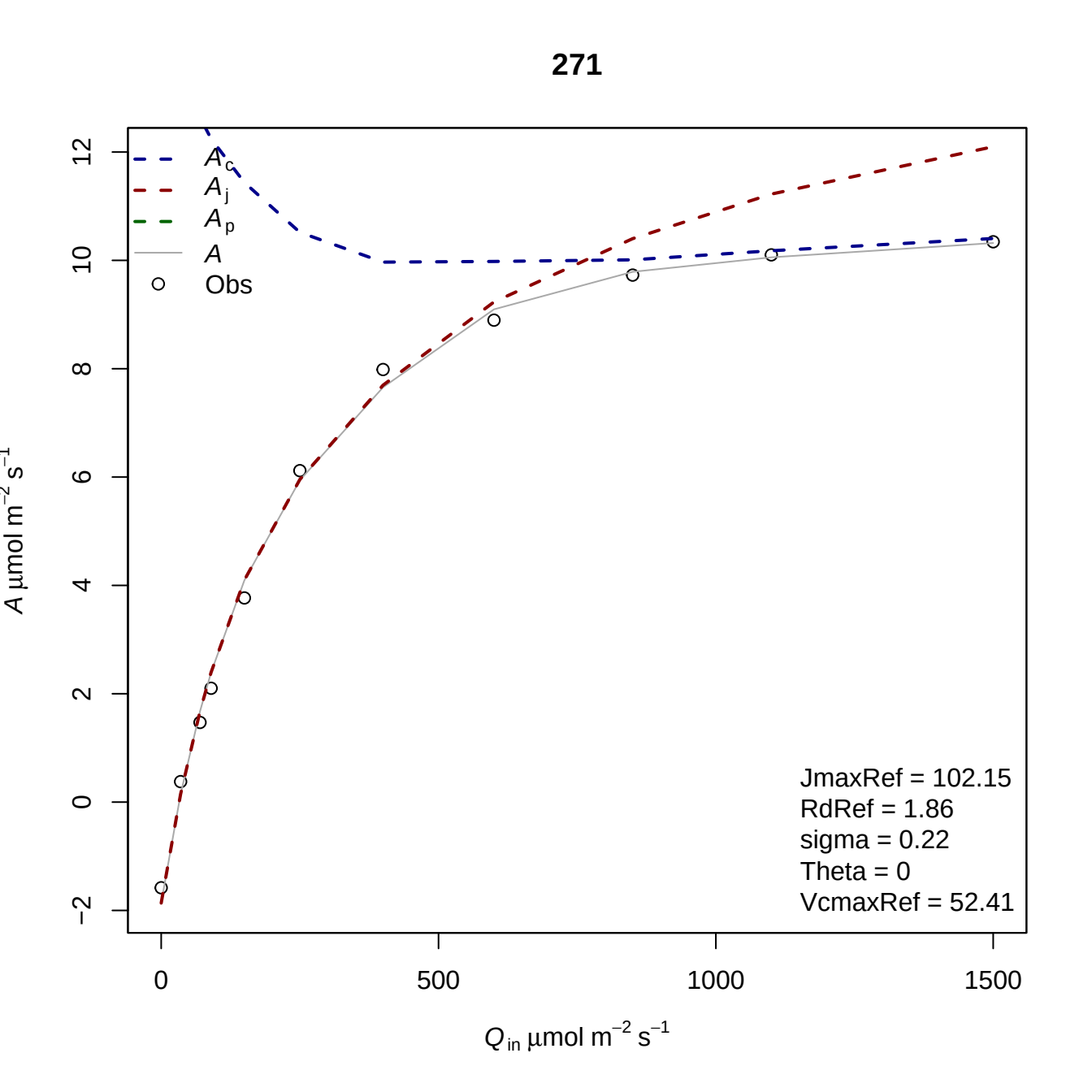

266
A_c
A_j
A_p
A
Obs
JmaxRef = 40.52
RdRef = 0.87
sigma = 0.13
Theta = 0
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

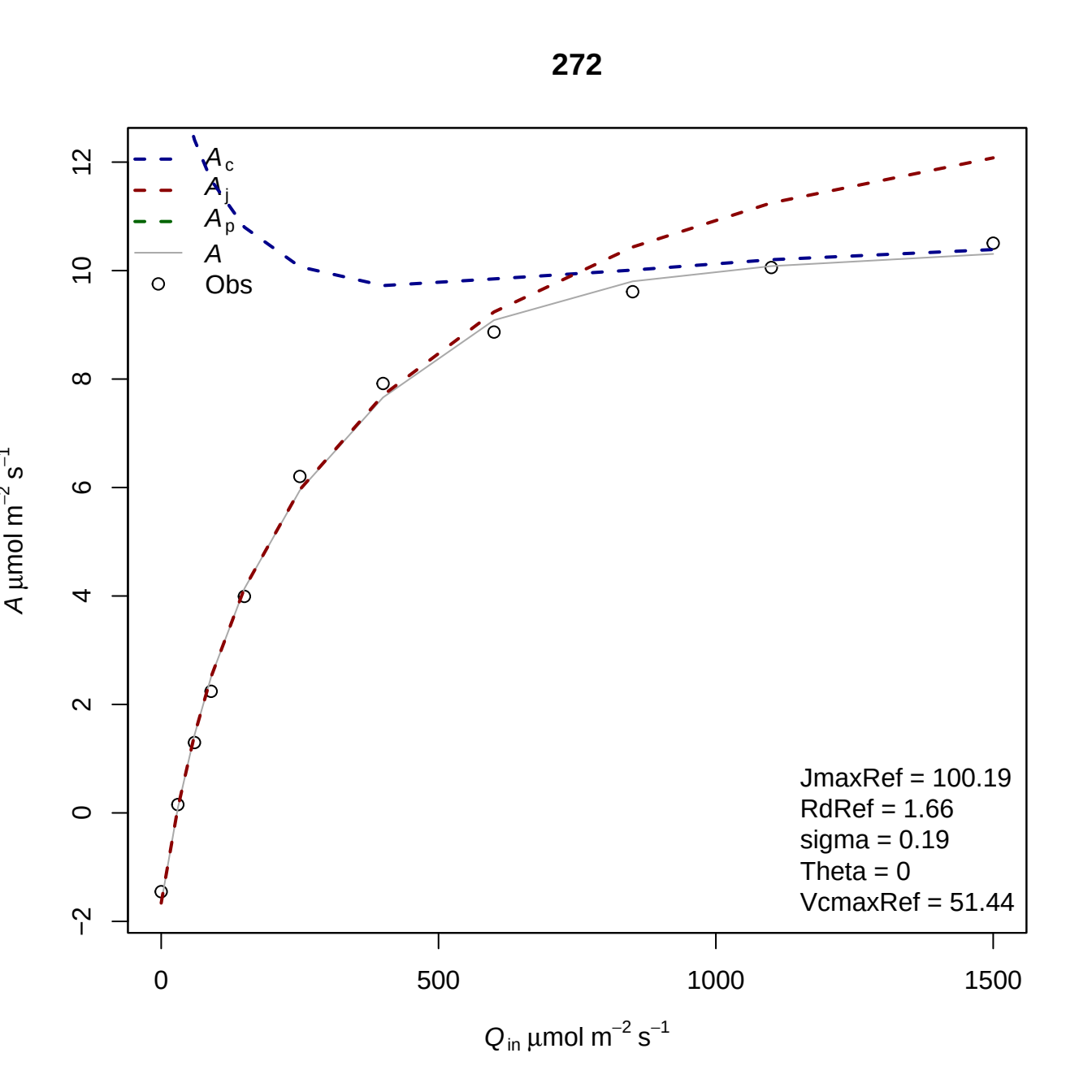

267
A_c
A_j
A_p
A
Obs
JmaxRef = 140.27
RdRef = 2.15
sigma = 0.27
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

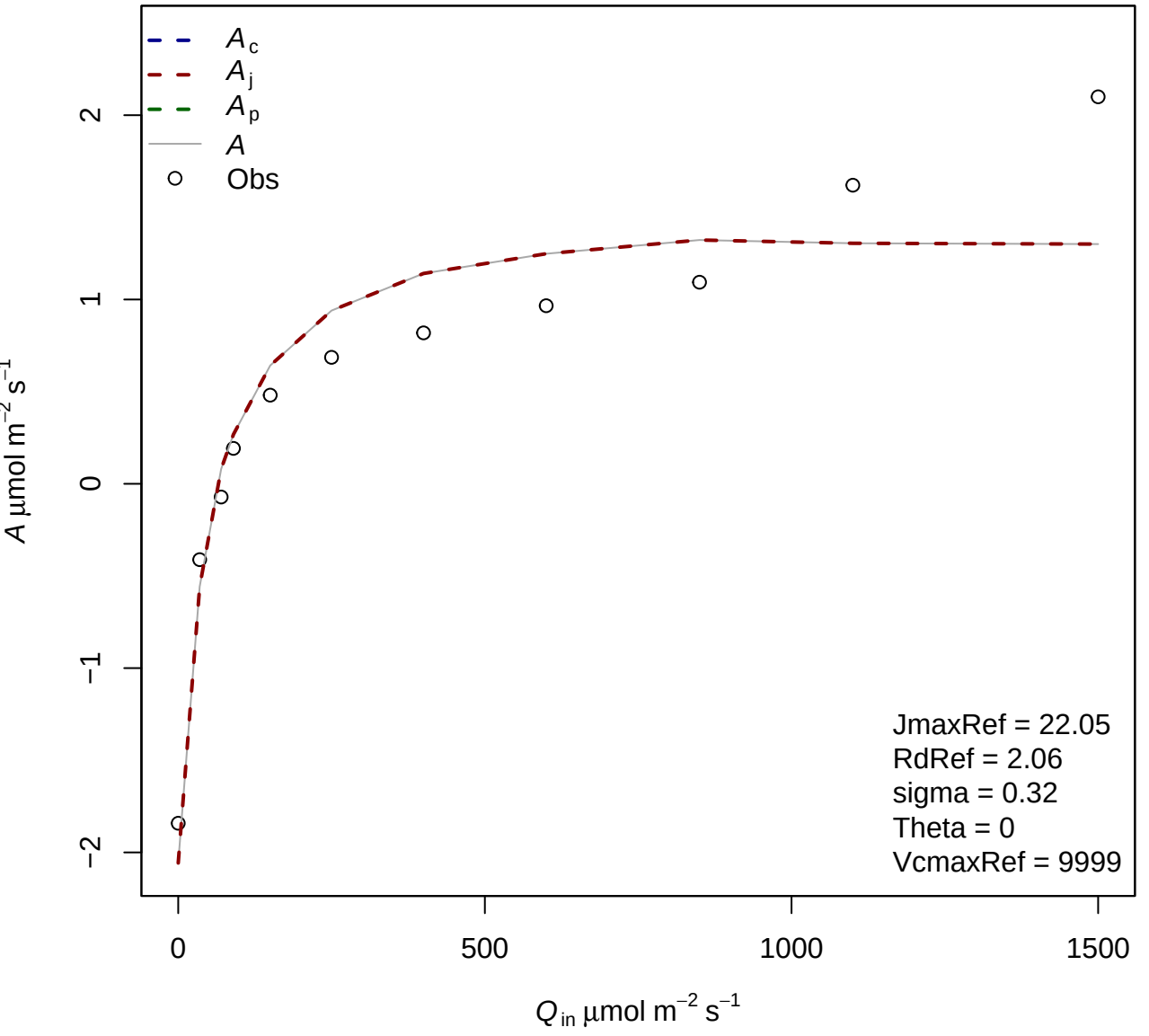

268
A_c
A_j
A_p
A
Obs
JmaxRef = 47.52
RdRef = 1.84
sigma = 0.07
Theta = 0
VcmaxRef = 27.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

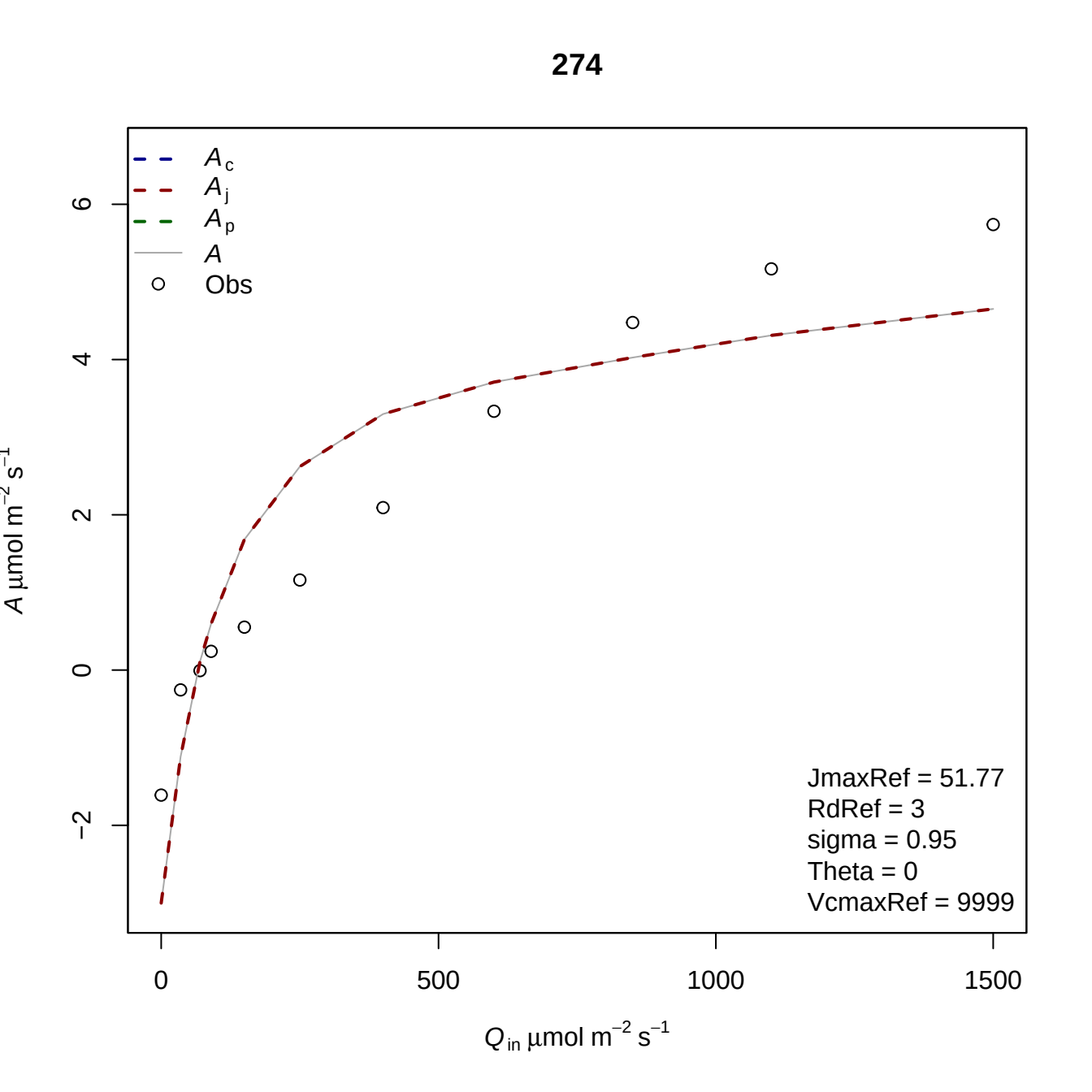

269
A_c
A_j
A_p
A
Obs
JmaxRef = 25.94
RdRef = 1.98
sigma = 0.04
Theta = 0
VcmaxRef = 14.1
2
1
0
−1
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

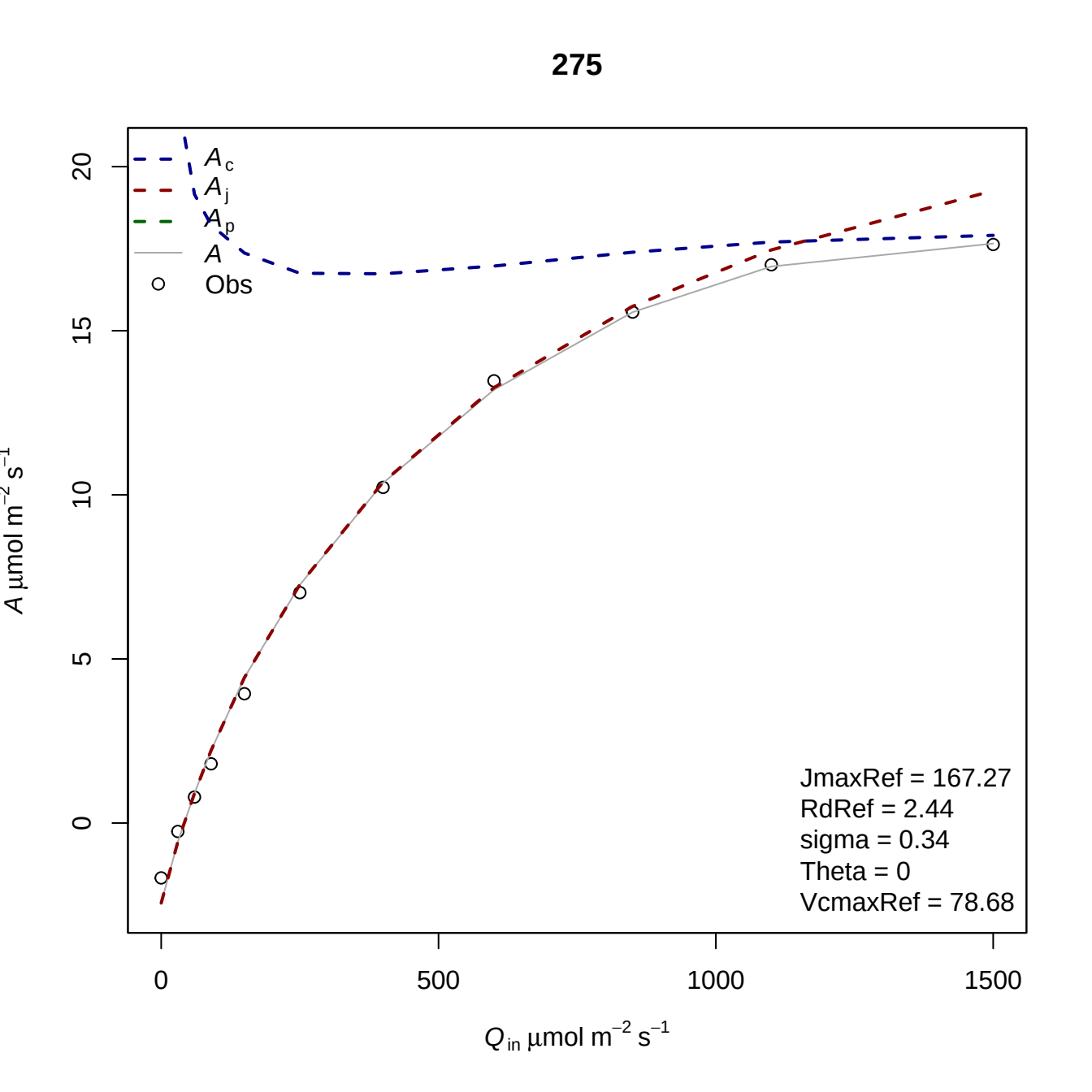

270
A_c
A_j
A_p
A
Obs
JmaxRef = 23.66
RdRef = 1.42
sigma = 0.12
Theta = 0
VcmaxRef = 12.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

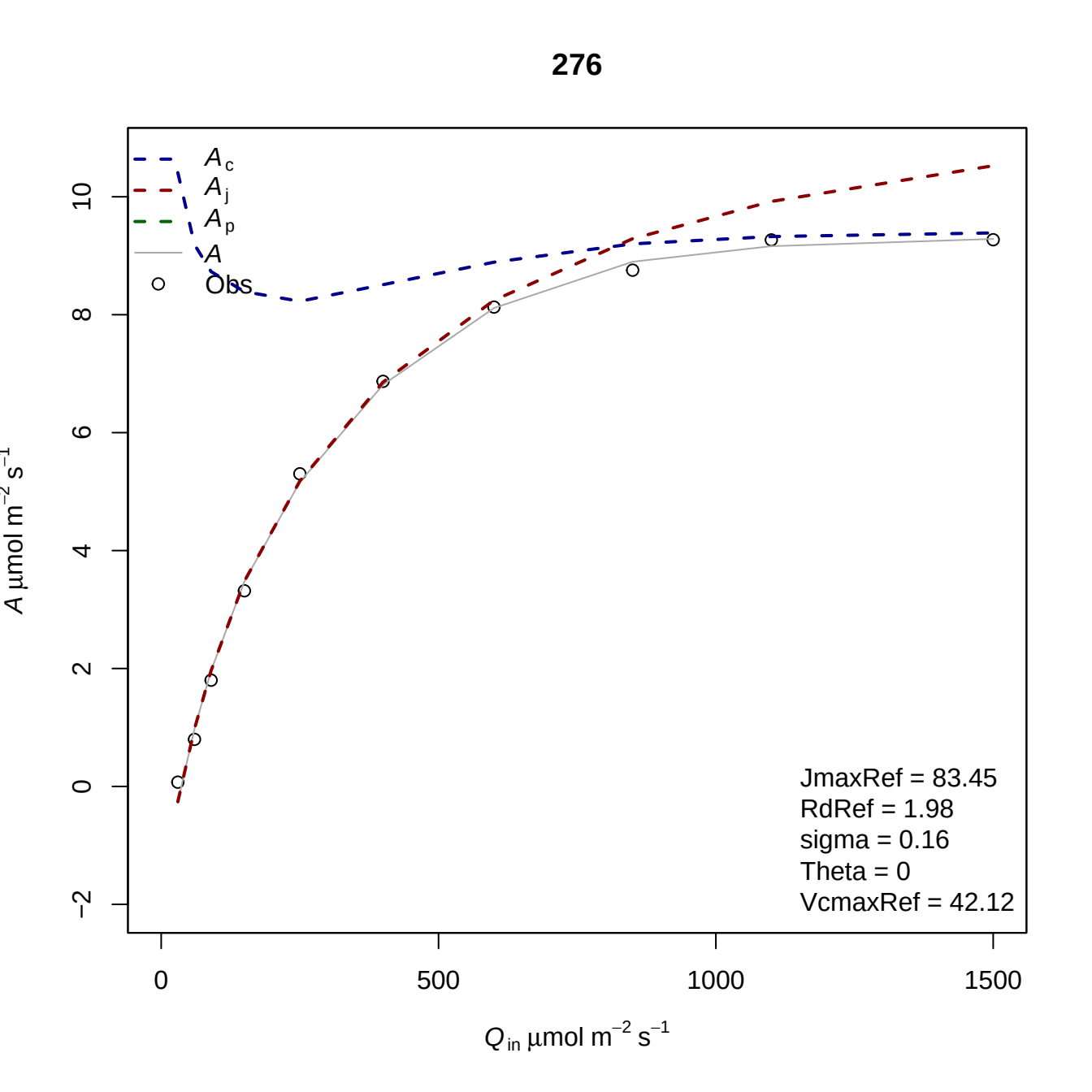

271
A_c
A_j
A_p
A
Obs
JmaxRef = 102.15
RdRef = 1.86
sigma = 0.22
Theta = 0
VcmaxRef = 52.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

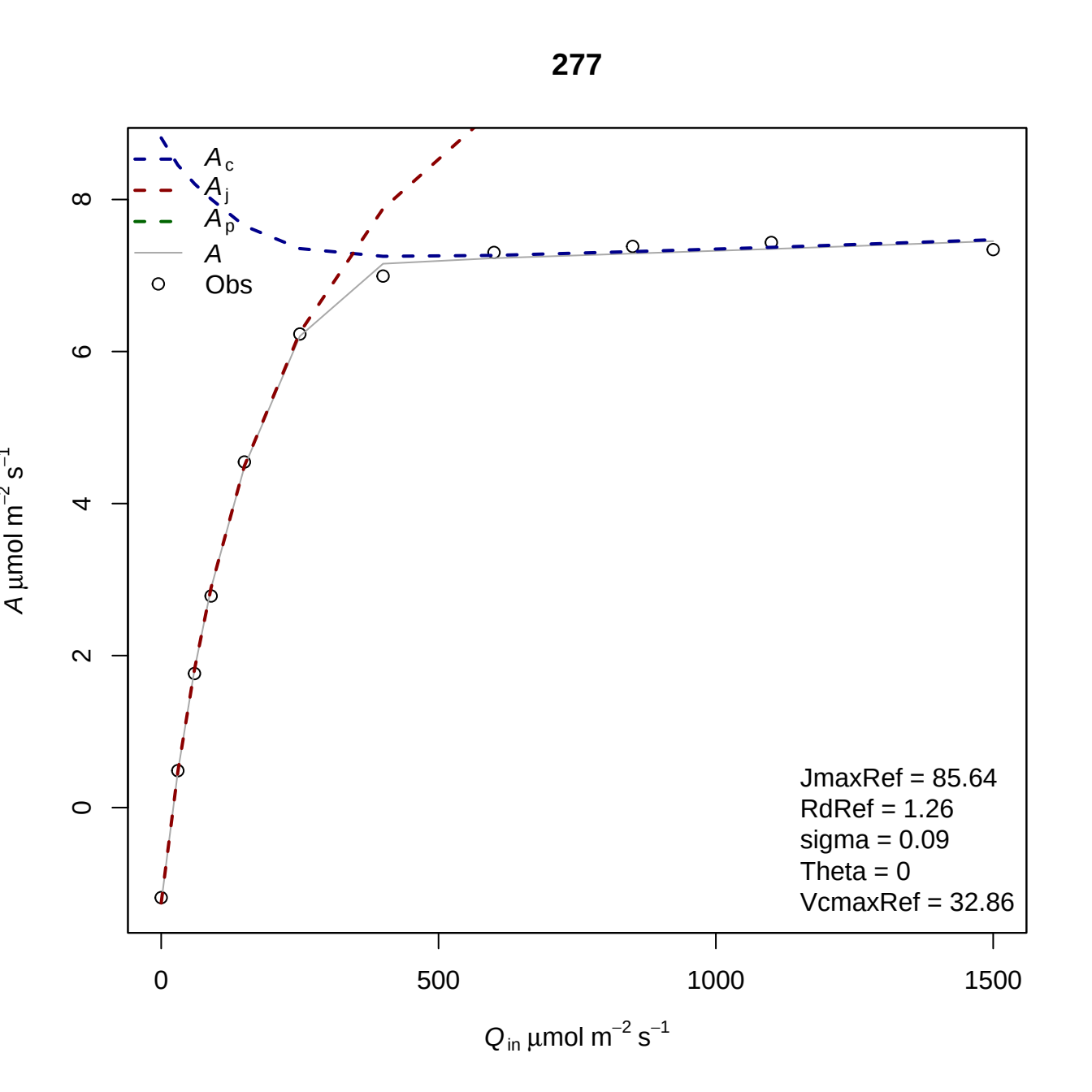

272
A_c
A_j
A_p
A
Obs
JmaxRef = 100.19
RdRef = 1.66
sigma = 0.19
Theta = 0
VcmaxRef = 51.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

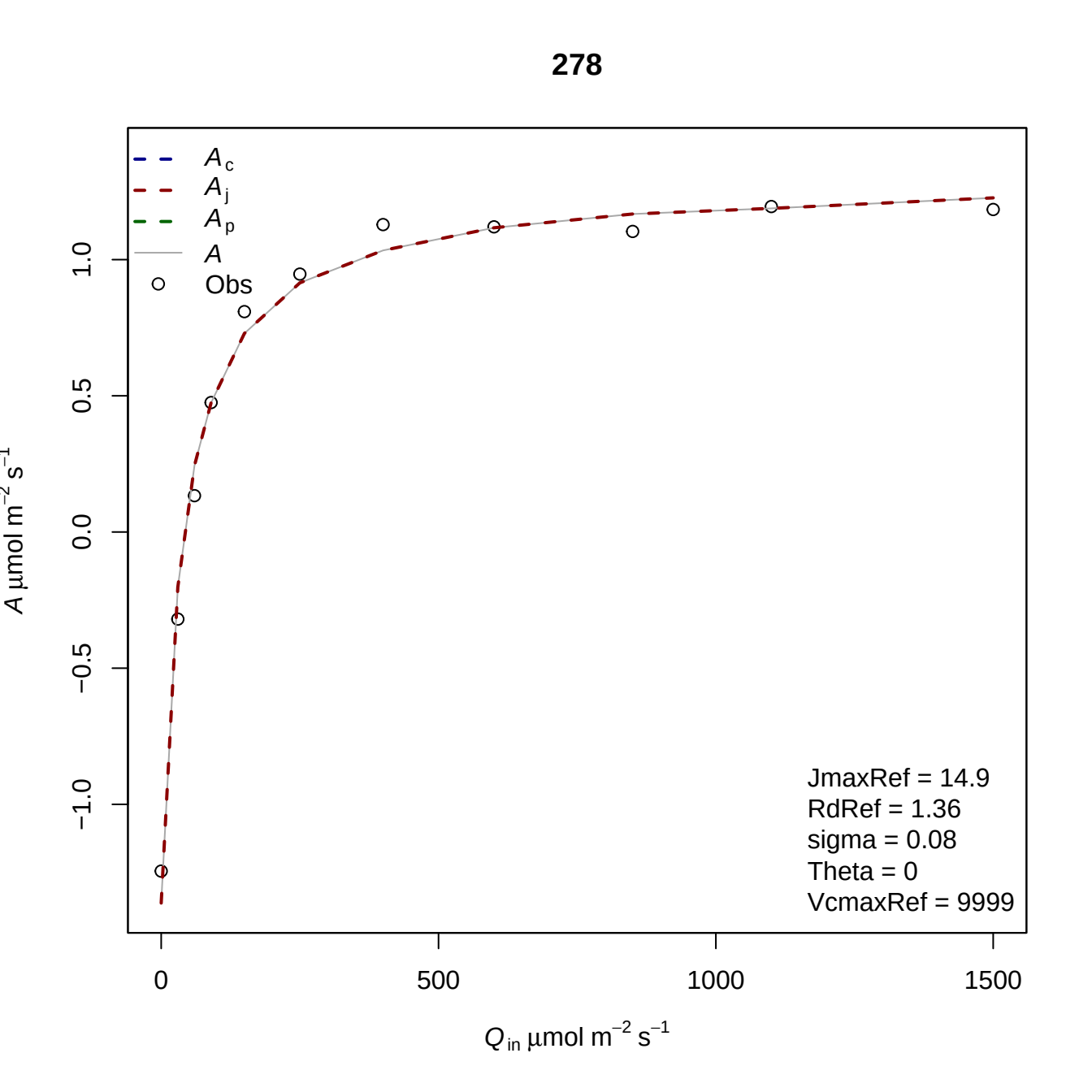

A_c
A_j
A_p
A
Obs
JmaxRef = 22.05
RdRef = 2.06
sigma = 0.32
Theta = 0
VcmaxRef = 9999
2
1
0
−1
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

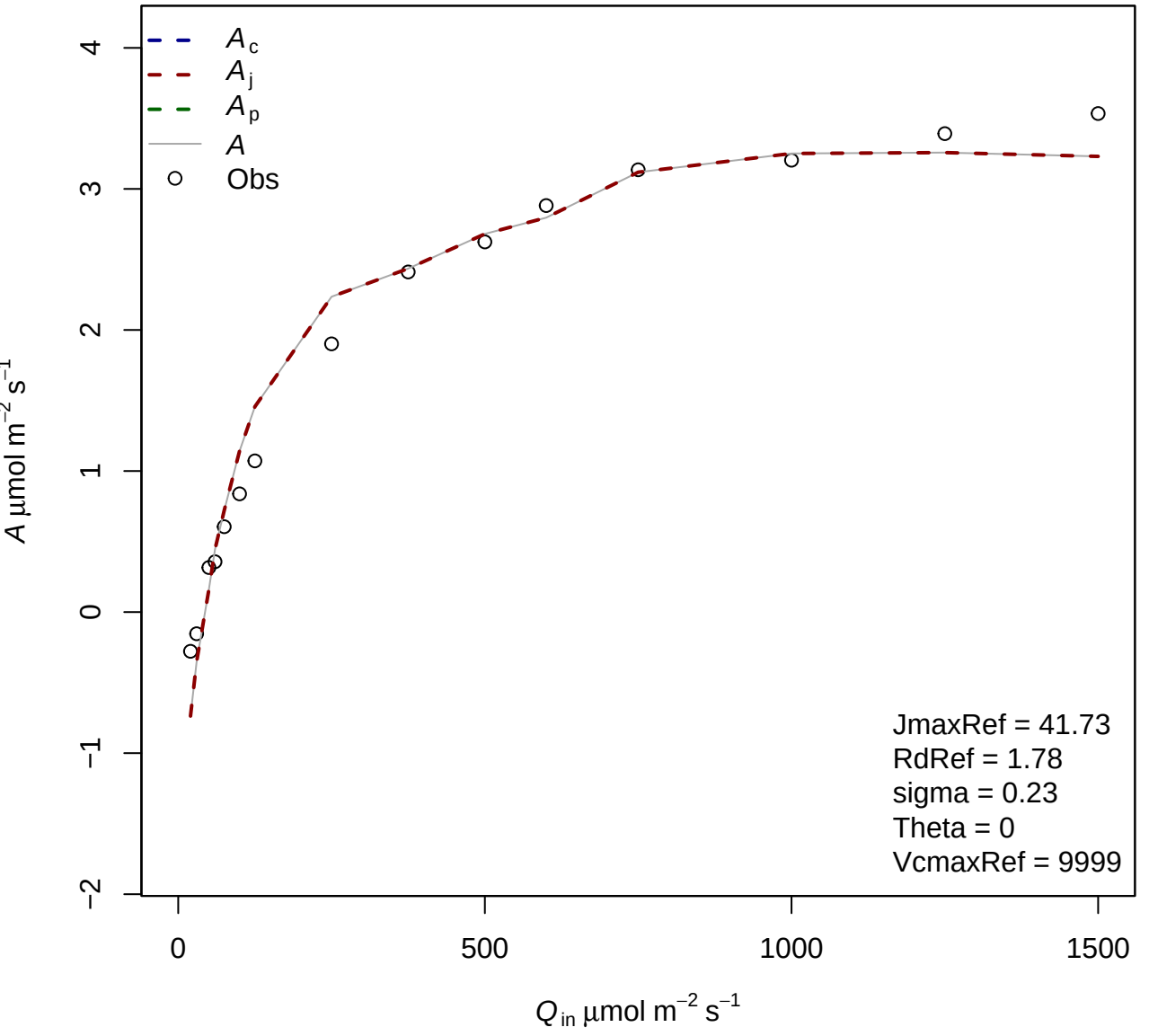

274
A_c
A_j
A_p
A
Obs
JmaxRef = 51.77
RdRef = 3
sigma = 0.95
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

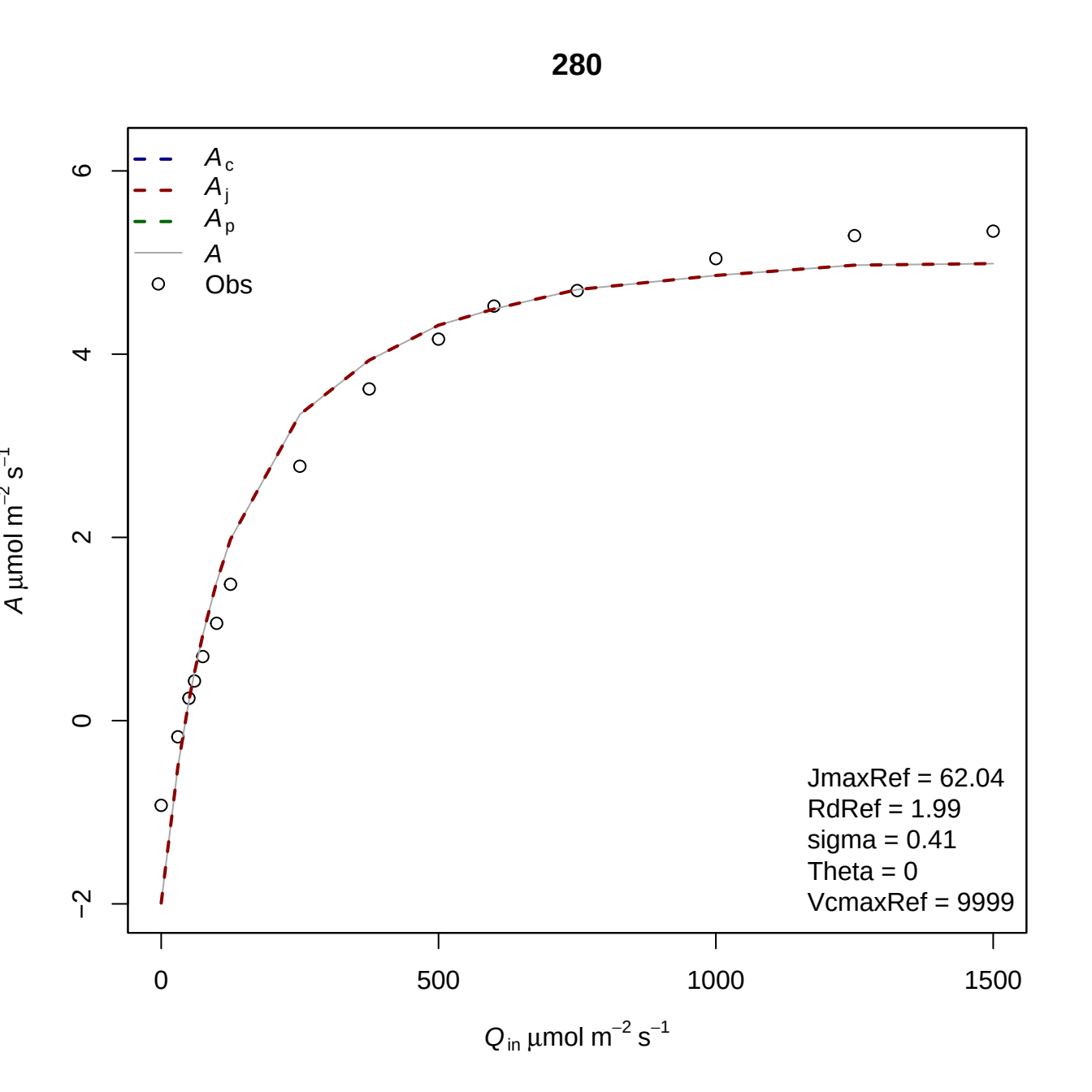

275
A_c
A_j
A_p
A
Obs
JmaxRef = 167.27
RdRef = 2.44
sigma = 0.34
Theta = 0
VcmaxRef = 78.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

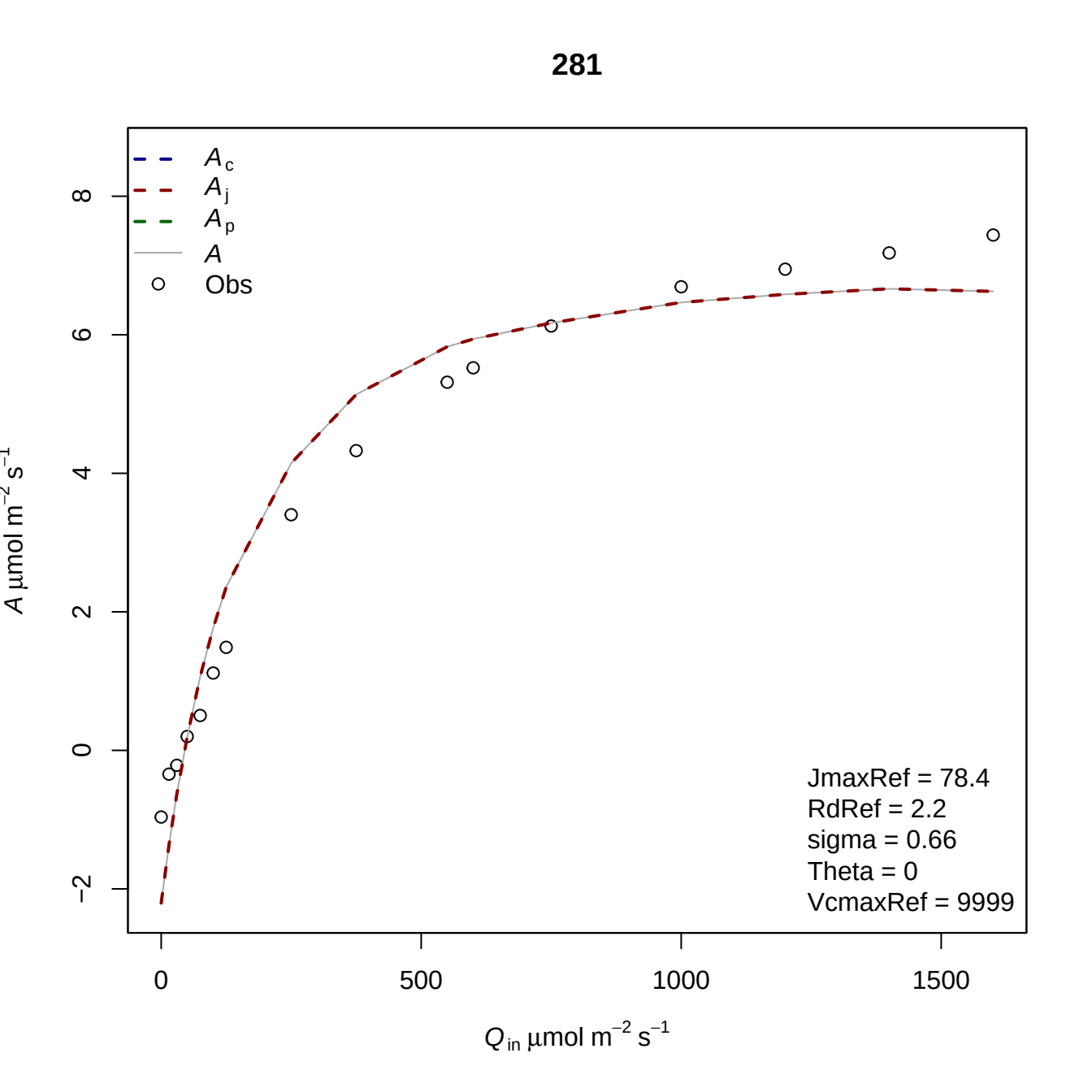

276
A_c
A_j
A_p
A
Obs
JmaxRef = 83.45
RdRef = 1.98
sigma = 0.16
Theta = 0
VcmaxRef = 42.12
−2
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

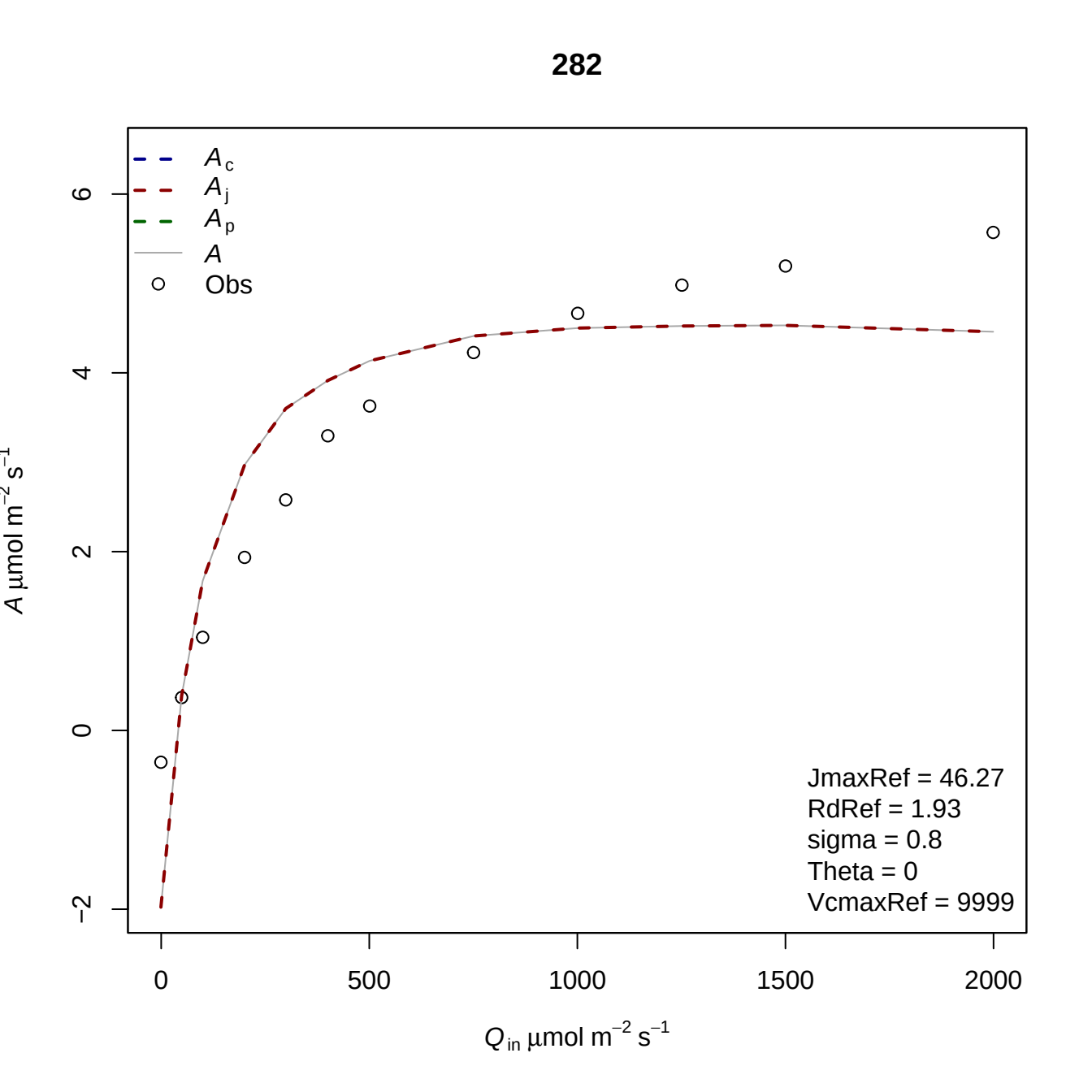

277
A_c
A_j
A_p
A
Obs
JmaxRef = 85.64
RdRef = 1.26
sigma = 0.09
Theta = 0
VcmaxRef = 32.86
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

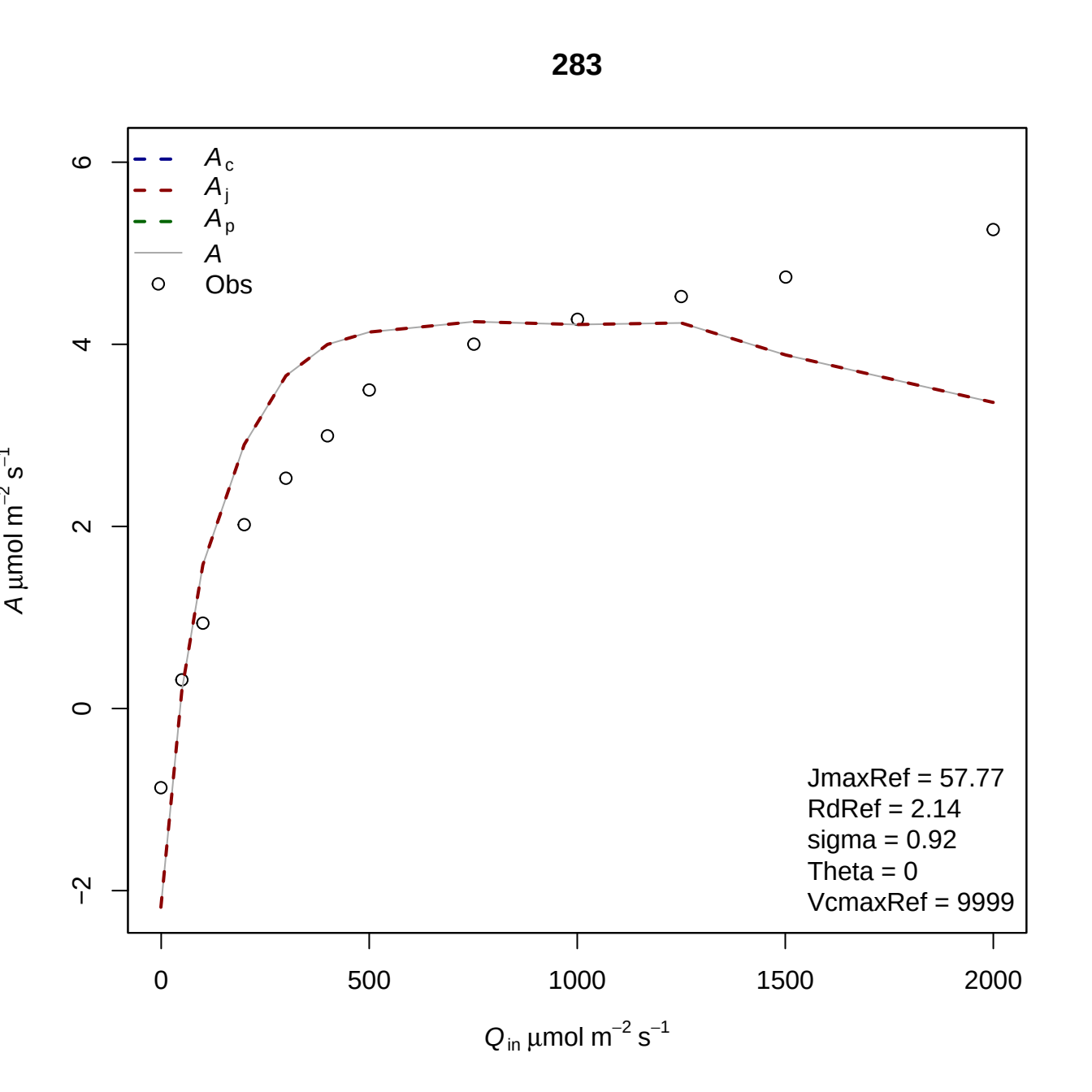

278
A_c
A_j
A_p
A
Obs
JmaxRef = 14.9
RdRef = 1.36
sigma = 0.08
Theta = 0
VcmaxRef = 9999
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

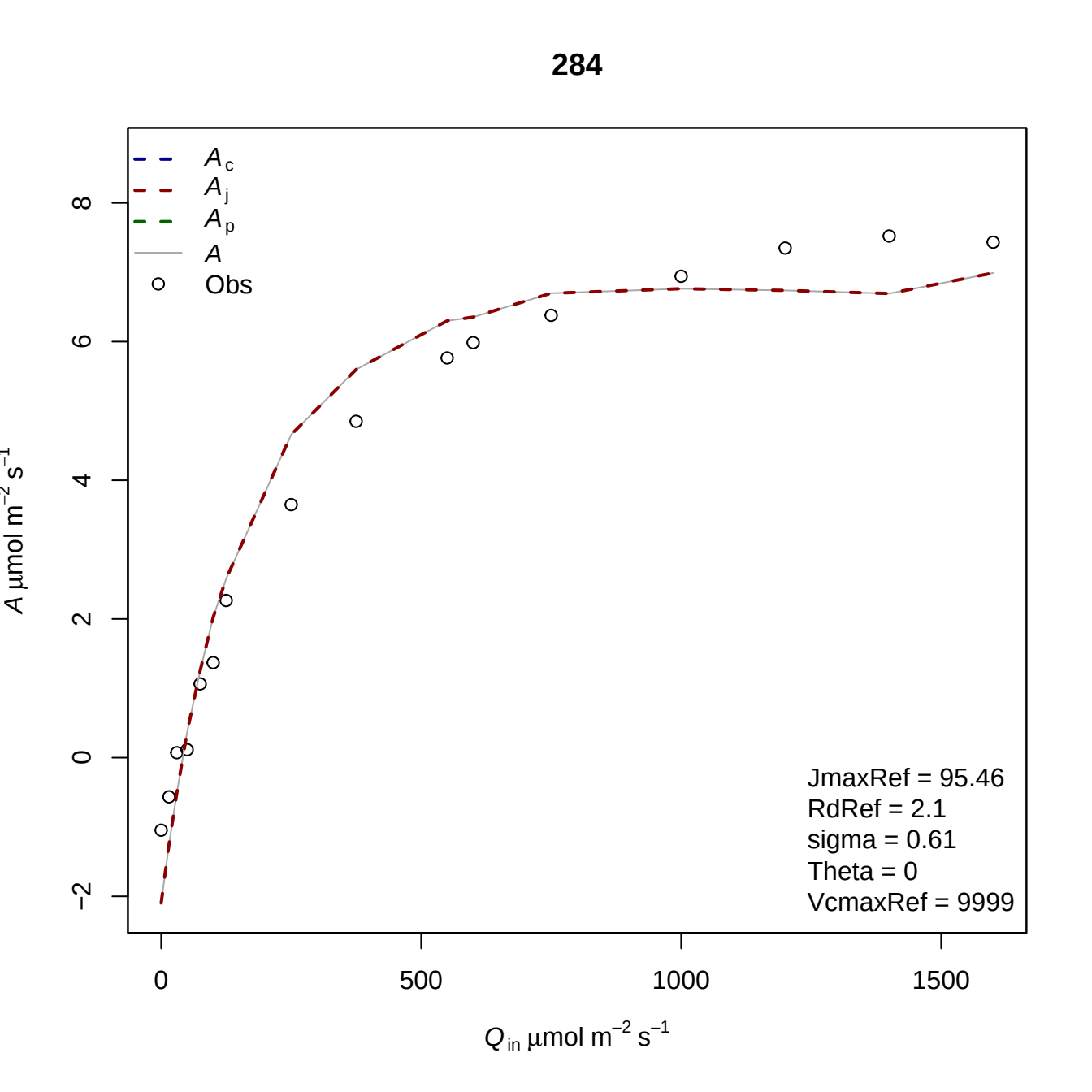

A_c
A_j
A_p
A
Obs
JmaxRef = 41.73
RdRef = 1.78
sigma = 0.23
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

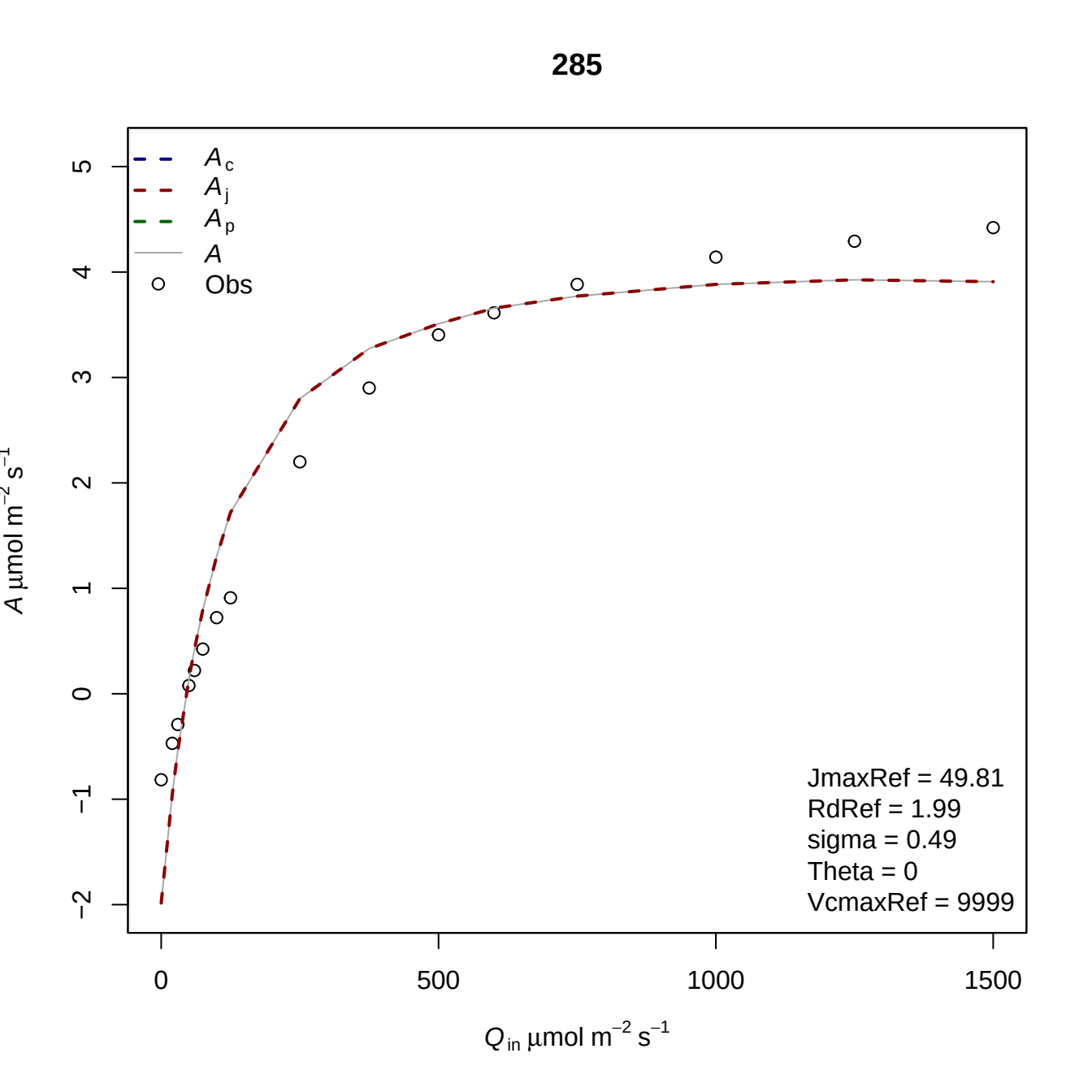

280
A_c
A_j
A_p
A
Obs
JmaxRef = 62.04
RdRef = 1.99
sigma = 0.41
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

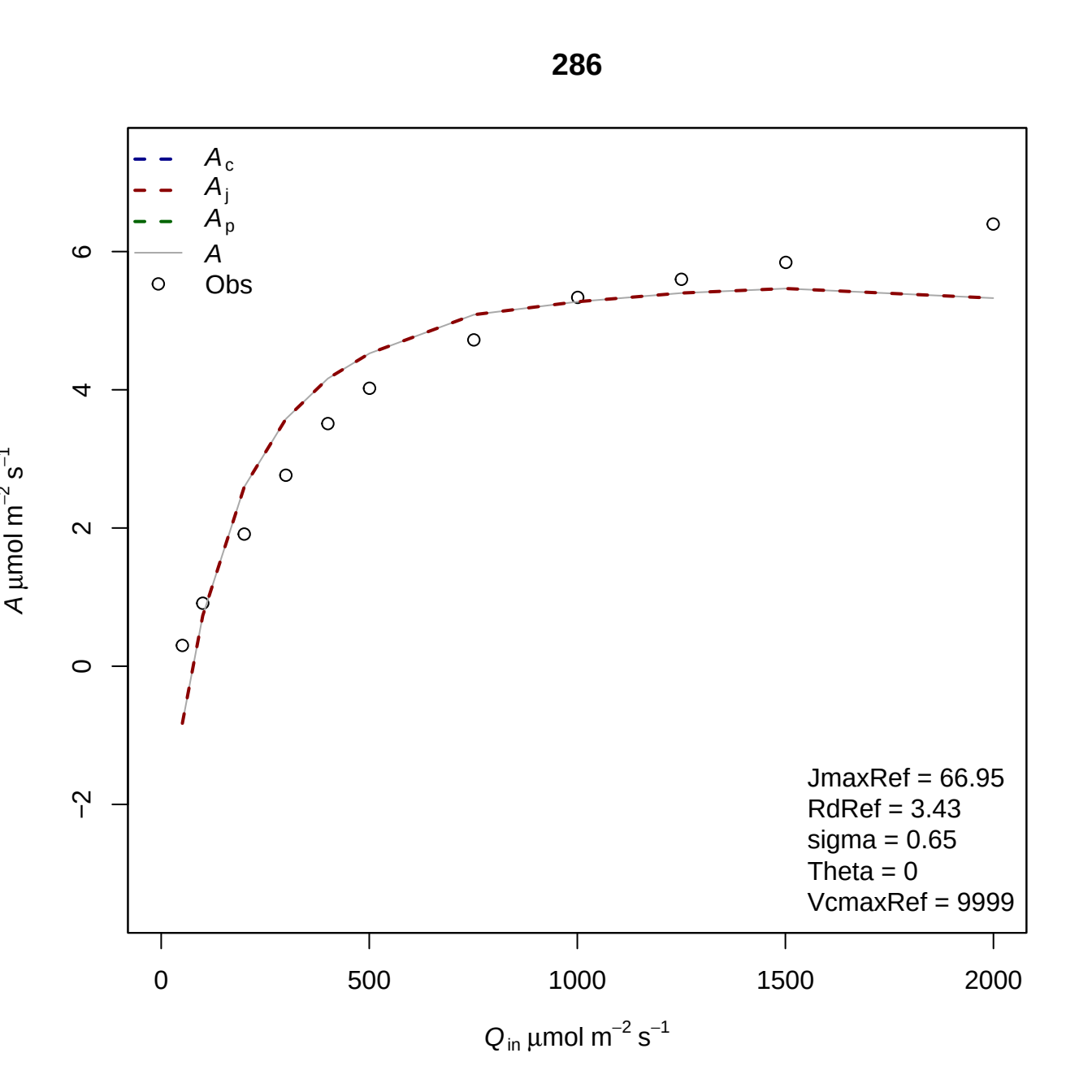

281
A_c
A_j
A_p
A
Obs
JmaxRef = 78.4
RdRef = 2.2
sigma = 0.66
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

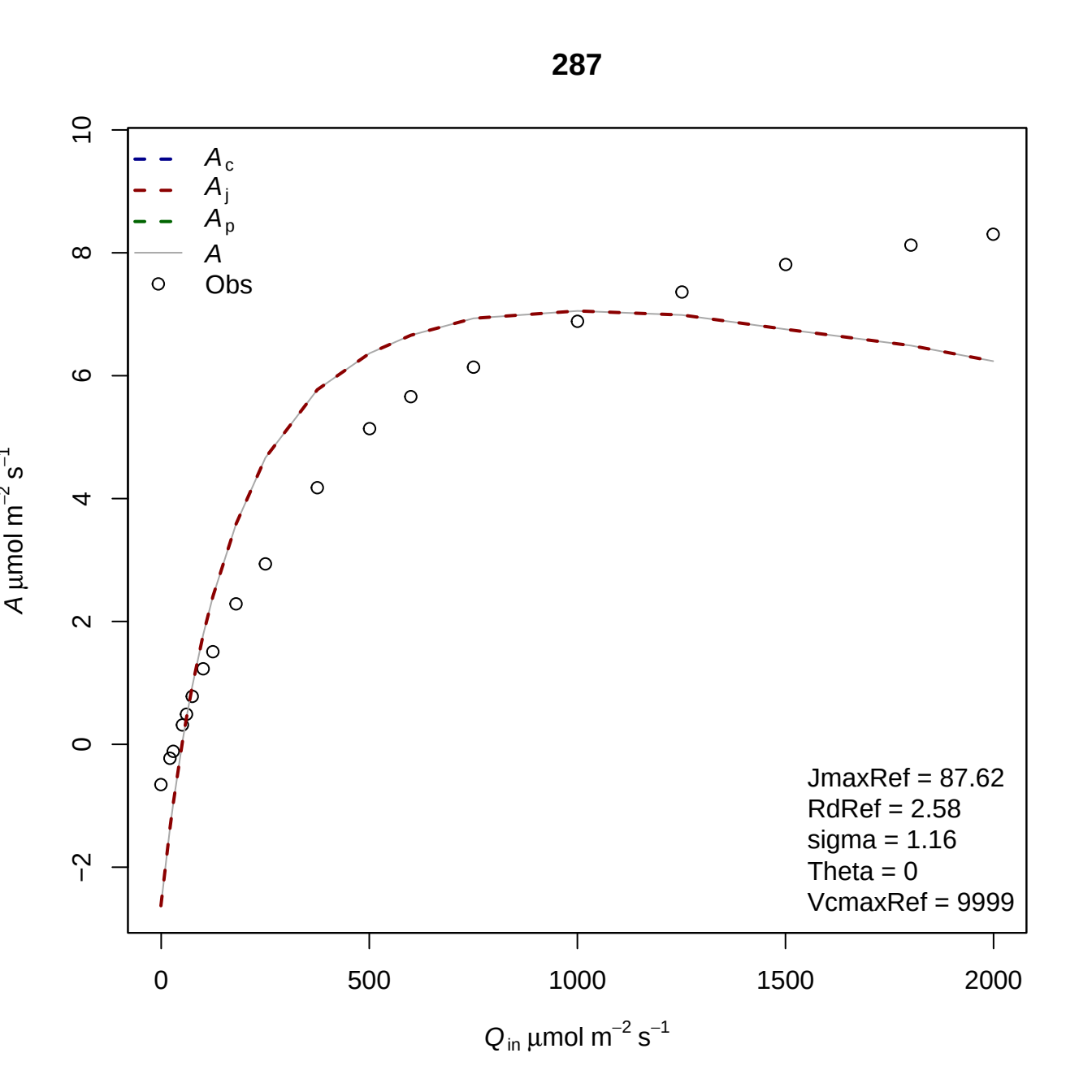

282
A_c
A_j
A_p
A
Obs
JmaxRef = 46.27
RdRef = 1.93
sigma = 0.8
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

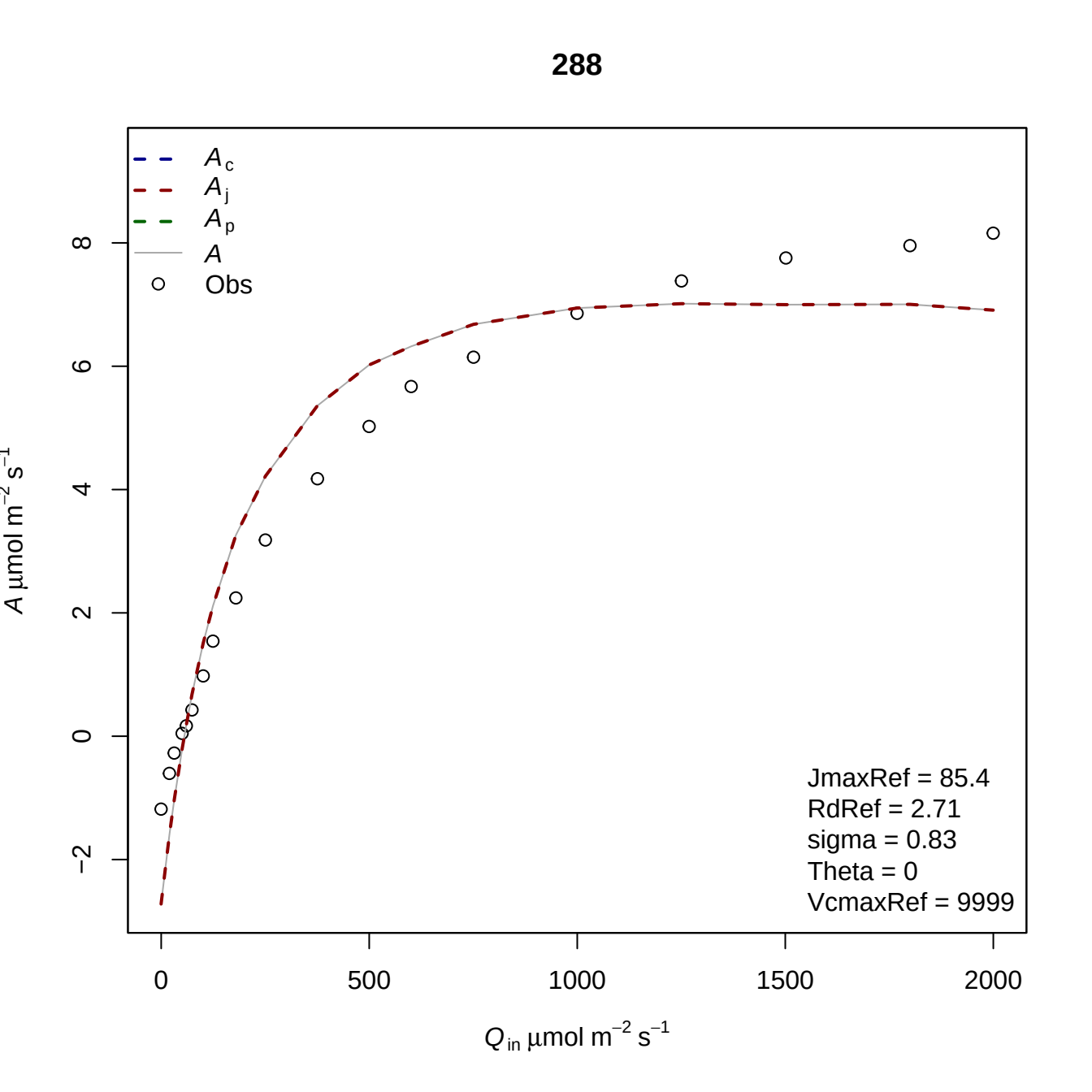

283
A_c
A_j
A_p
A
Obs
JmaxRef = 57.77
RdRef = 2.14
sigma = 0.92
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

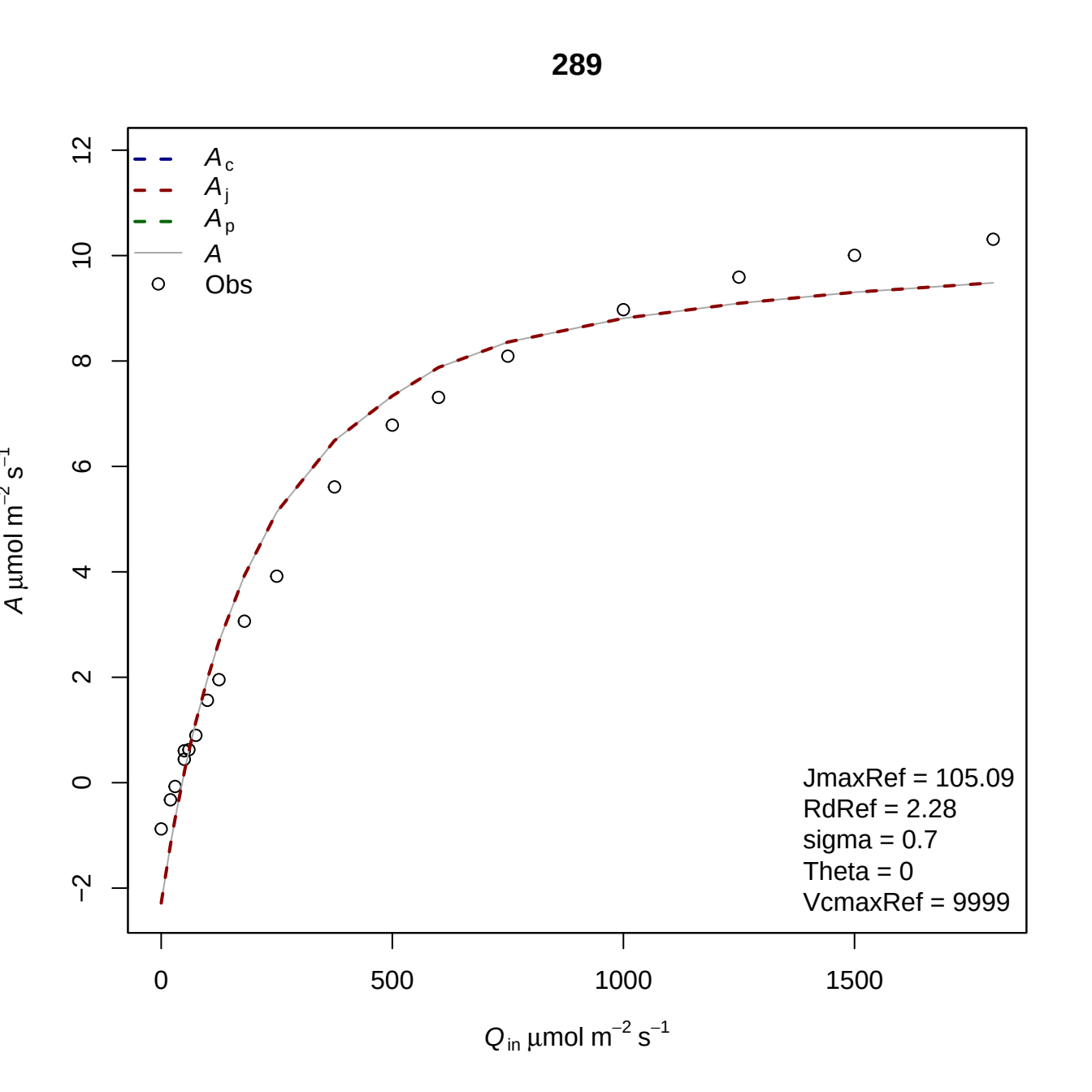

284
A_c
A_j
A_p
A
Obs
JmaxRef = 95.46
RdRef = 2.1
sigma = 0.61
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

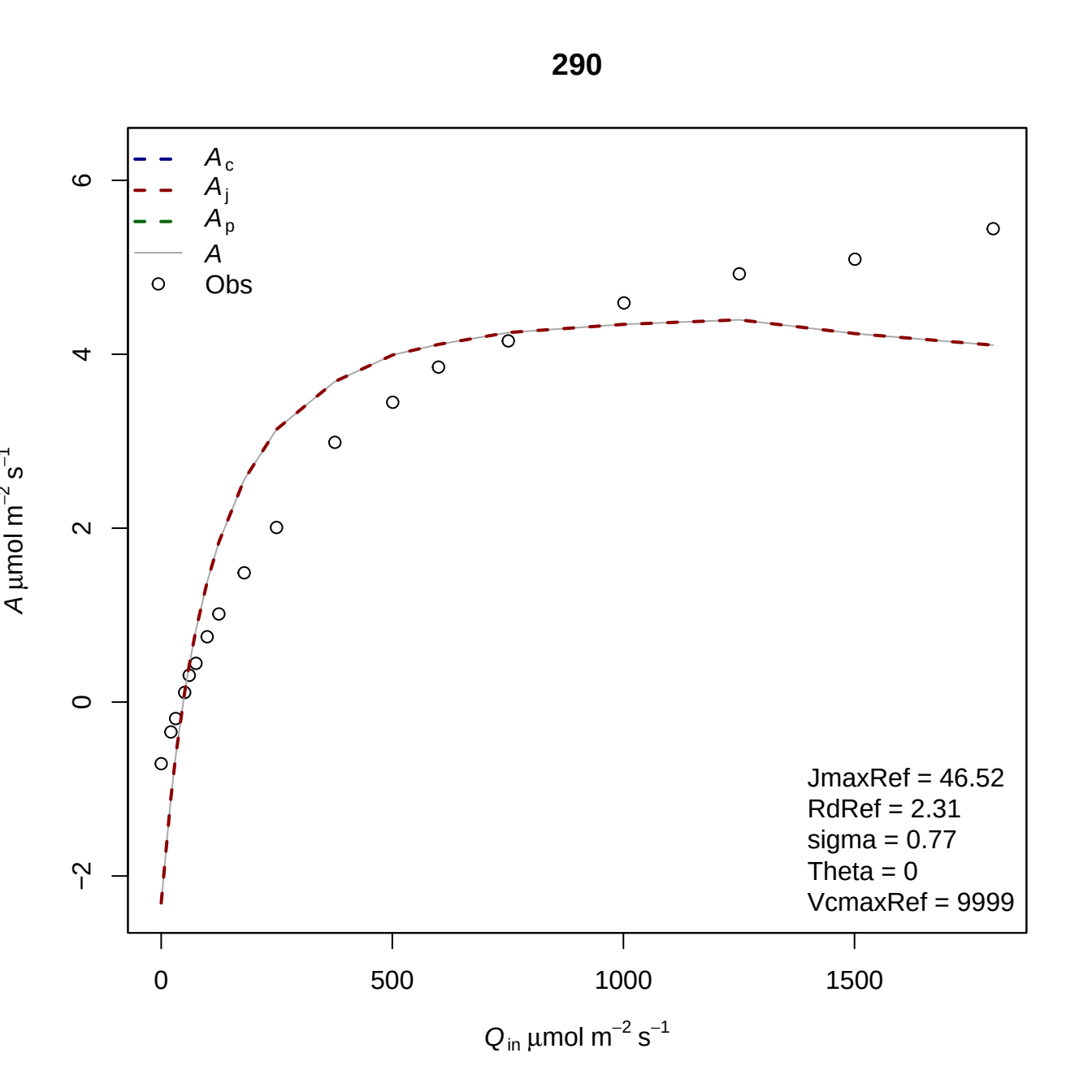

285
A_c
A_j
A_p
A
Obs
JmaxRef = 49.81
RdRef = 1.99
sigma = 0.49
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
−2
0
500
1000
1500

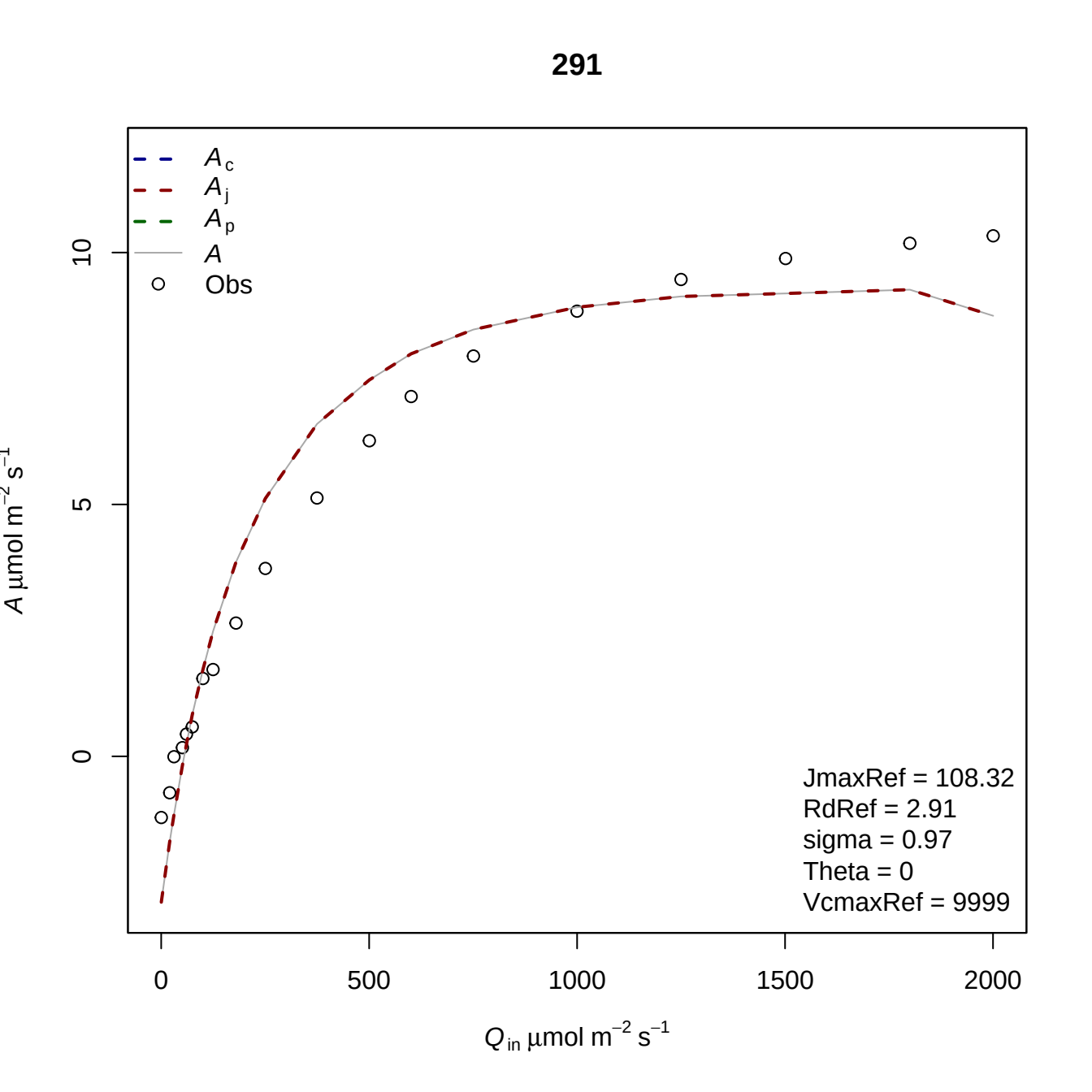

286
A_c
A_j
A_p
A
Obs
JmaxRef = 66.95
RdRef = 3.43
sigma = 0.65
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

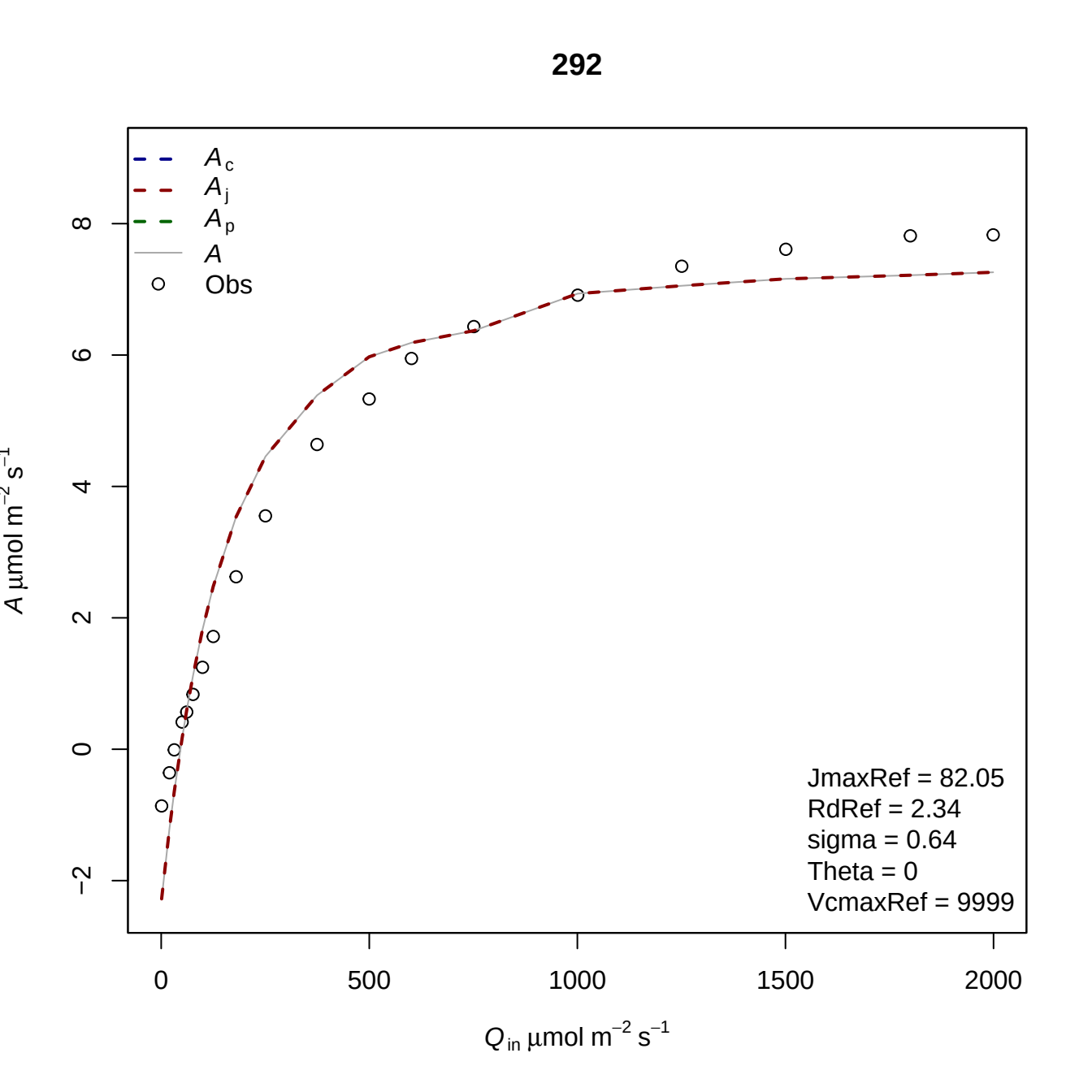

287
A_c
A_j
A_p
A
Obs
JmaxRef = 87.62
RdRef = 2.58
sigma = 1.16
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

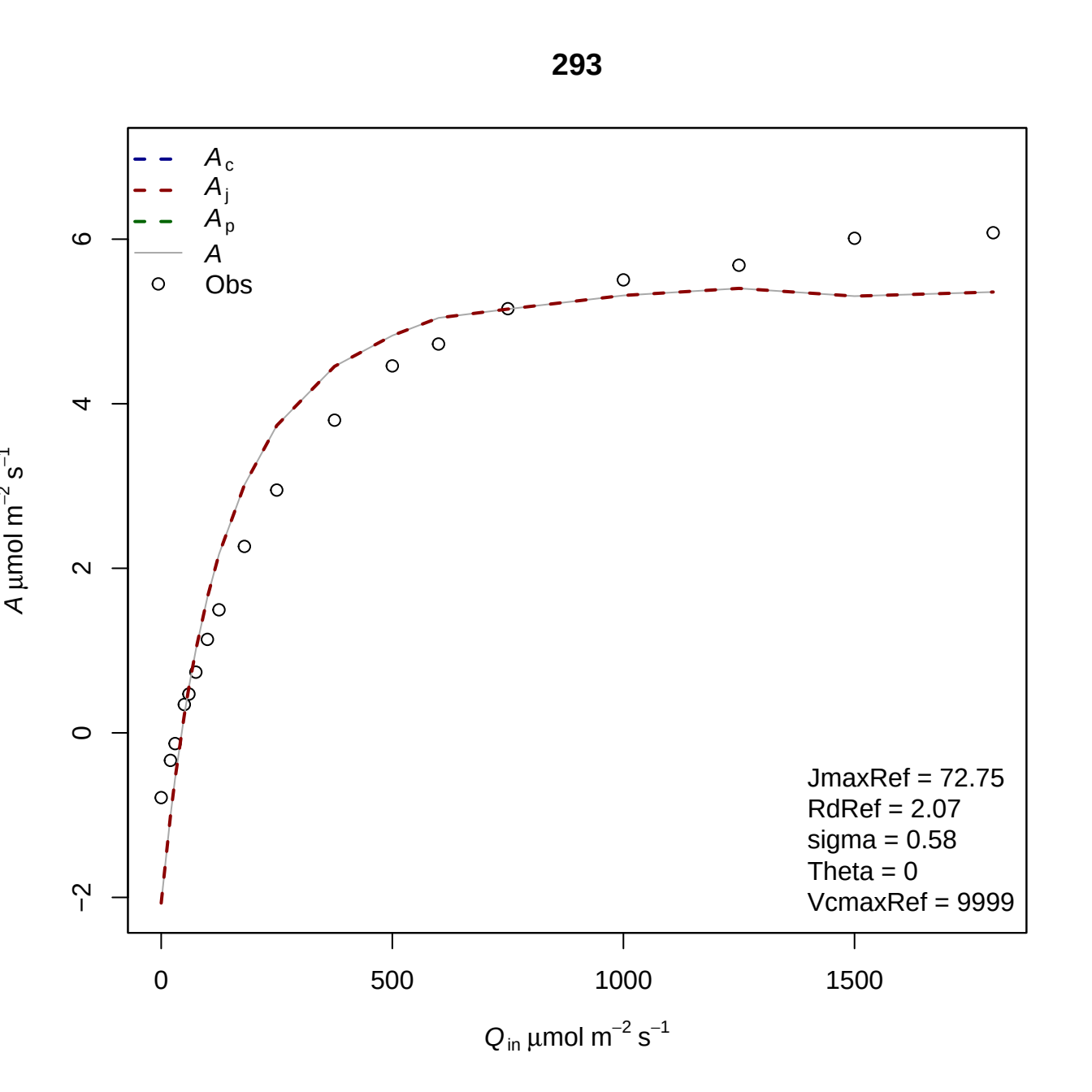

288
A_c
A_j
A_p
A
Obs
JmaxRef = 85.4
RdRef = 2.71
sigma = 0.83
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

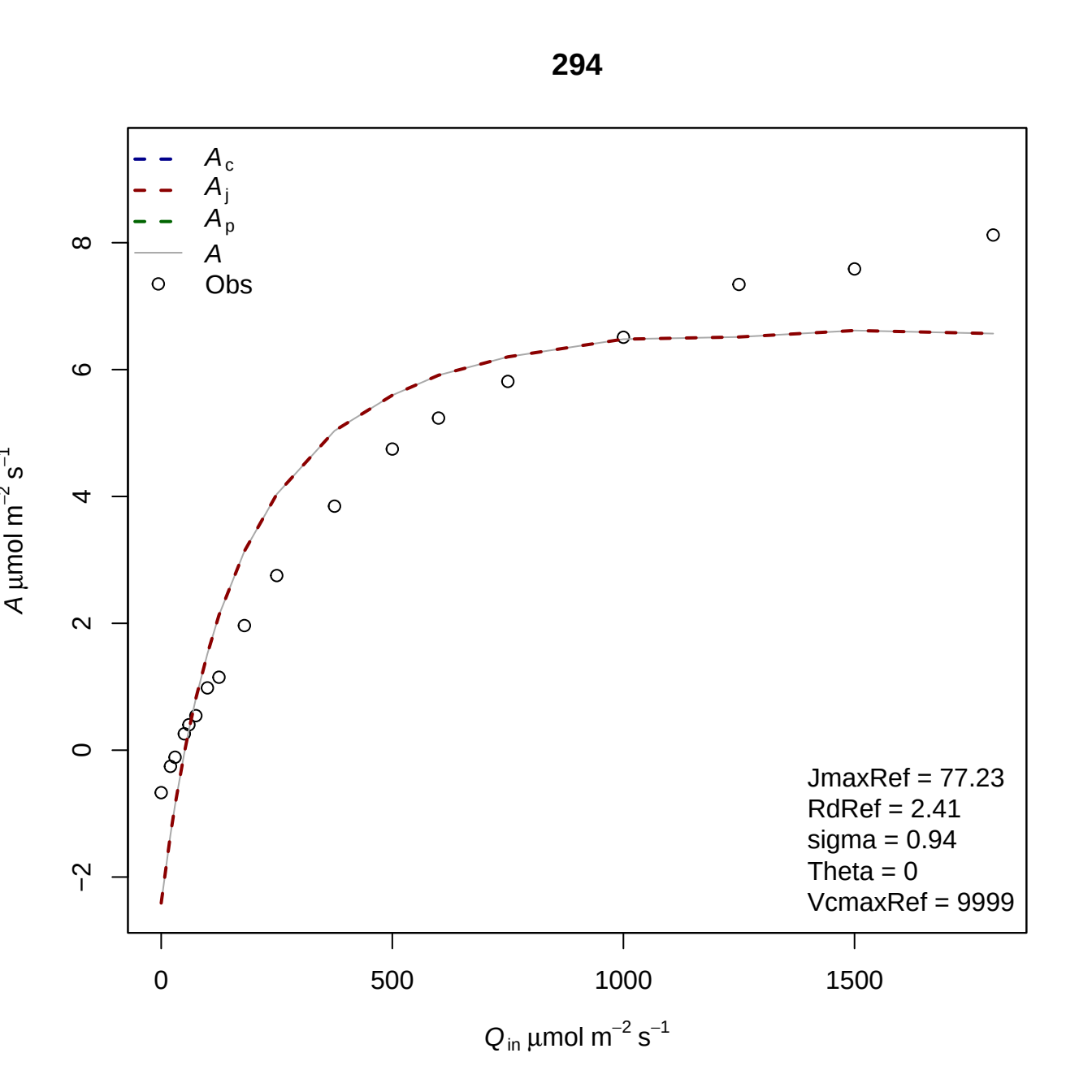

289
A_c
A_j
A_p
A
Obs
JmaxRef = 105.09
RdRef = 2.28
sigma = 0.7
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

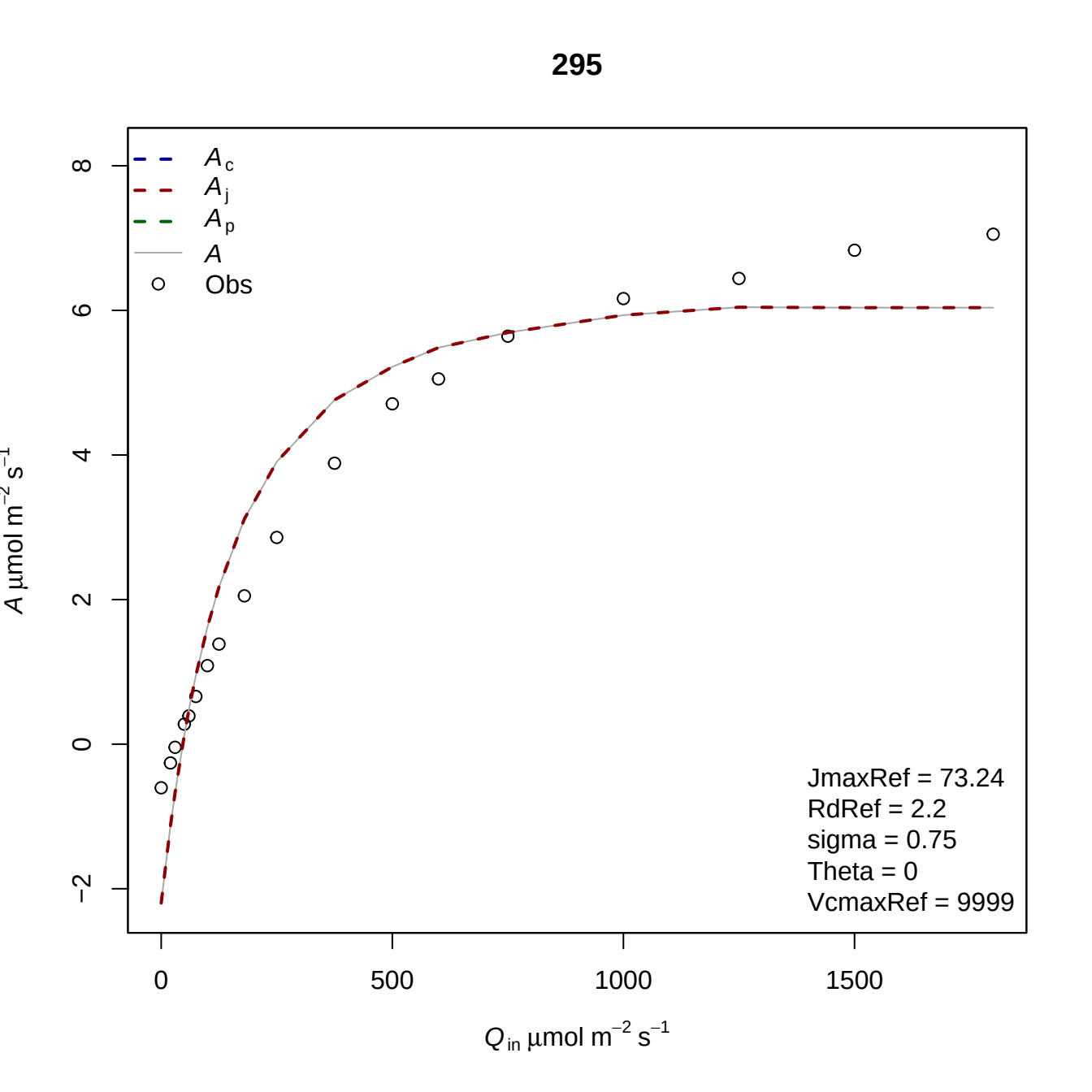

290
A_c
A_j
A_p
A
Obs
JmaxRef = 46.52
RdRef = 2.31
sigma = 0.77
Theta = 0
VcmaxRef = 9999
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

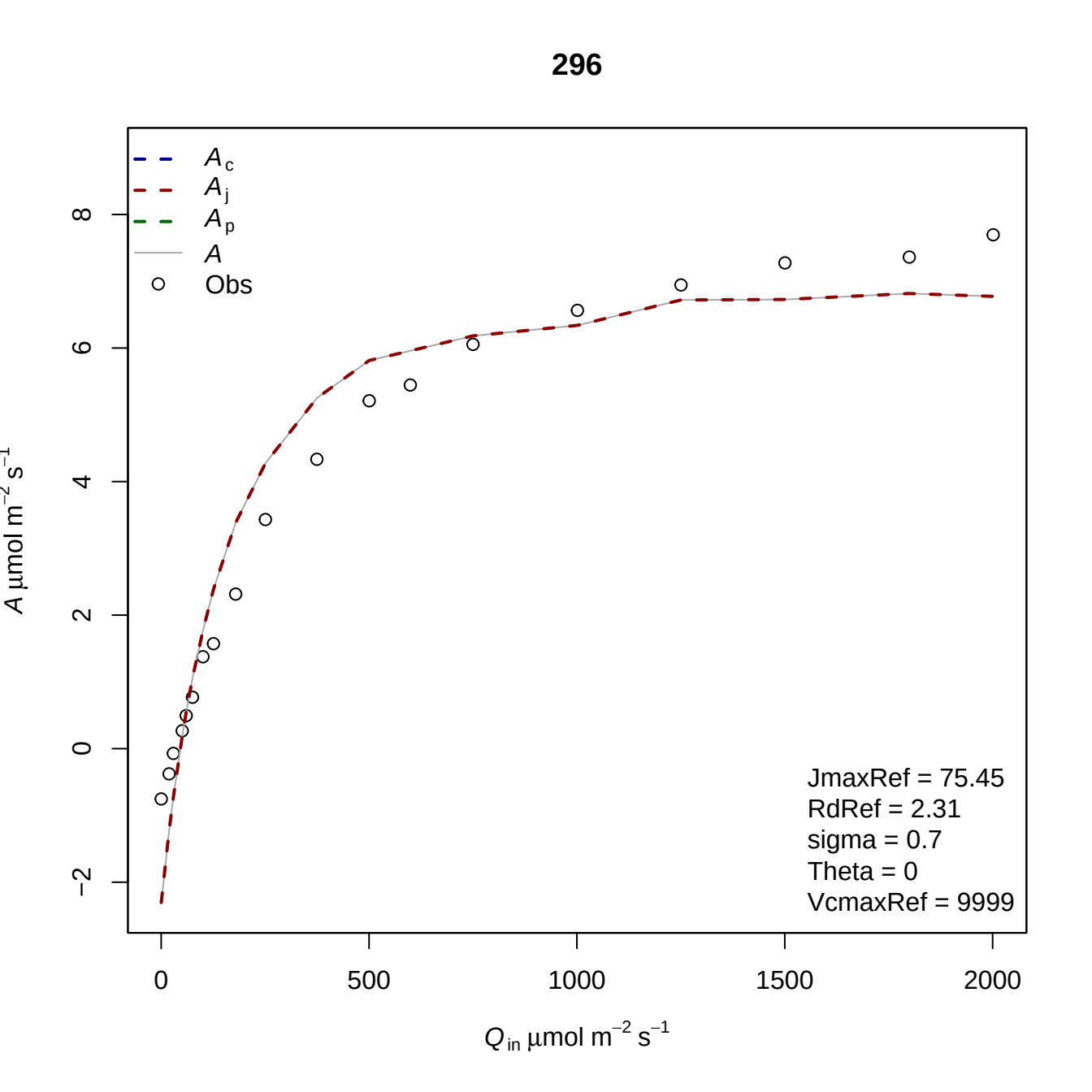

291
A_c
A_j
A_p
A
Obs
JmaxRef = 108.32
RdRef = 2.91
sigma = 0.97
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

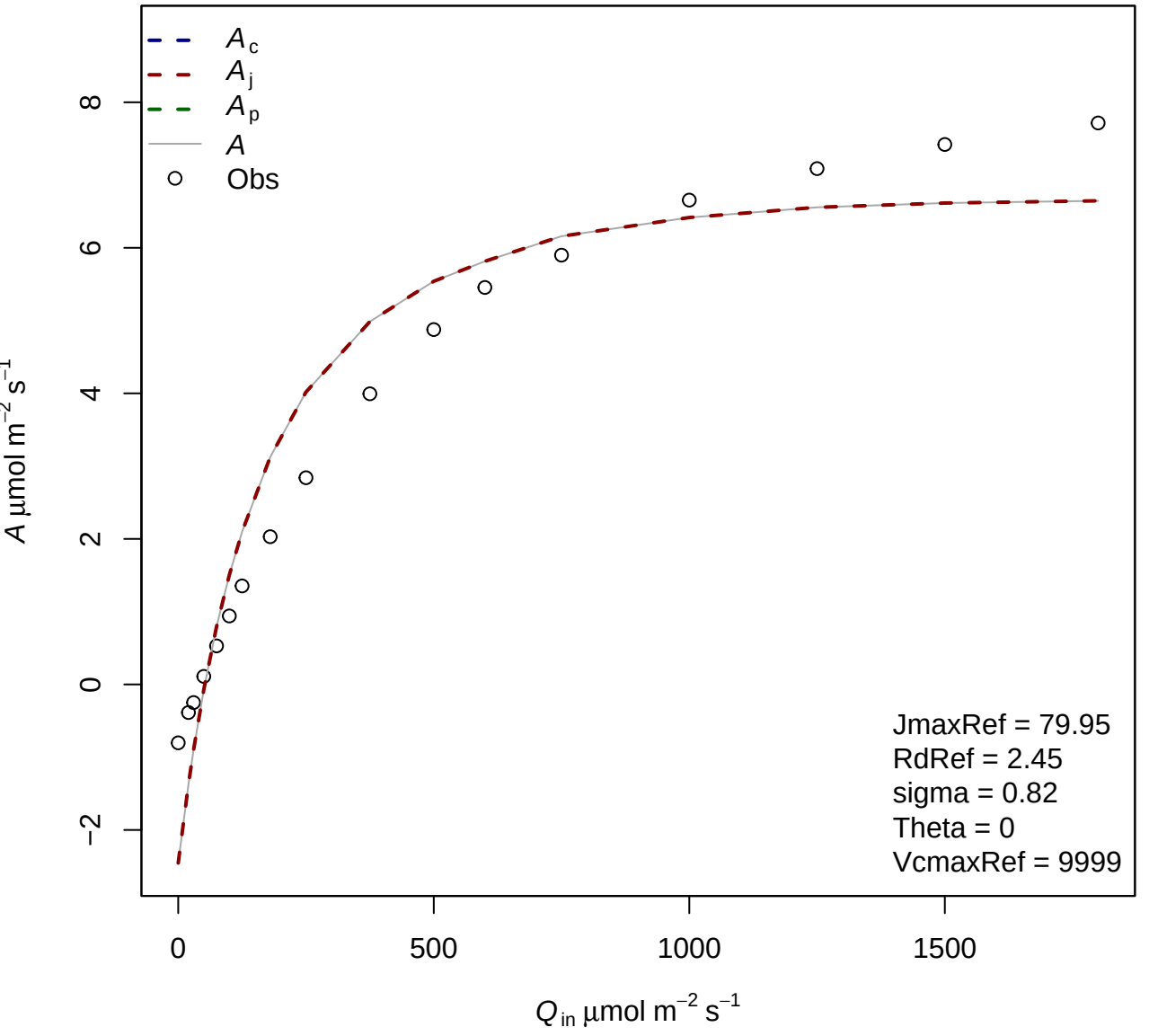

292
A_c
A_j
A_p
A
Obs
JmaxRef = 82.05
RdRef = 2.34
sigma = 0.64
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

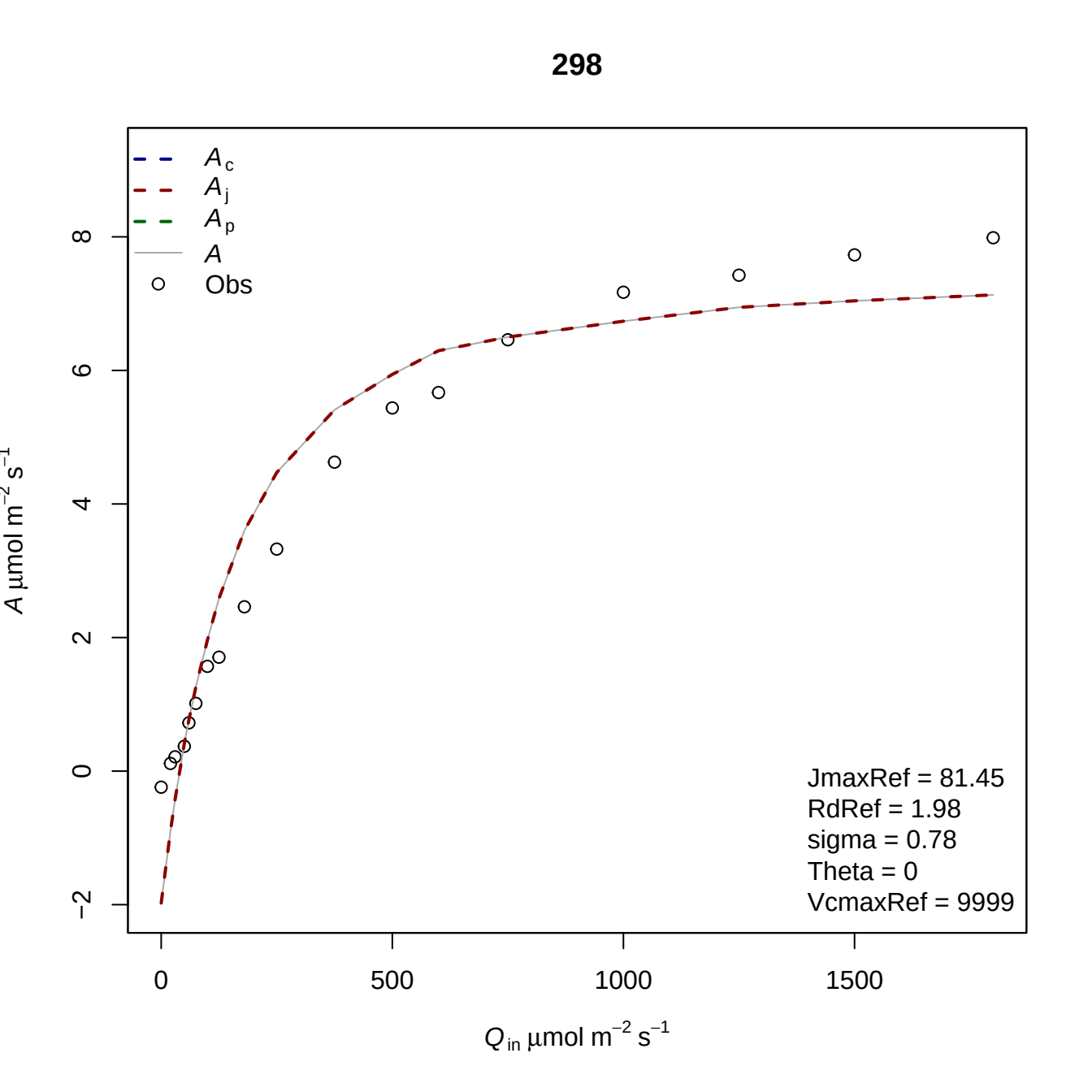

293
A_c
A_j
A_p
A
Obs
JmaxRef = 72.75
RdRef = 2.07
sigma = 0.58
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

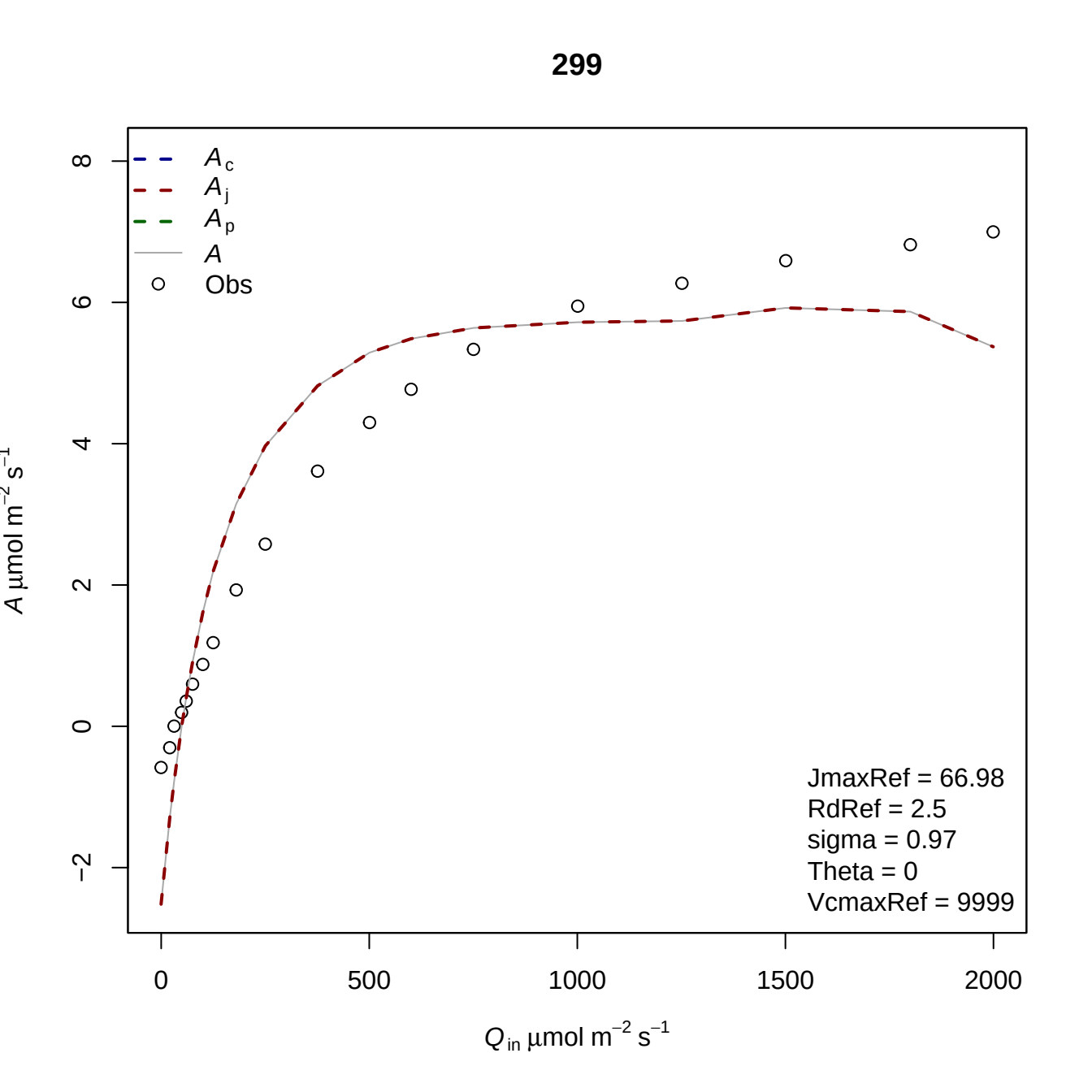

294
A_c
A_j
A_p
A
Obs
JmaxRef = 77.23
RdRef = 2.41
sigma = 0.94
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

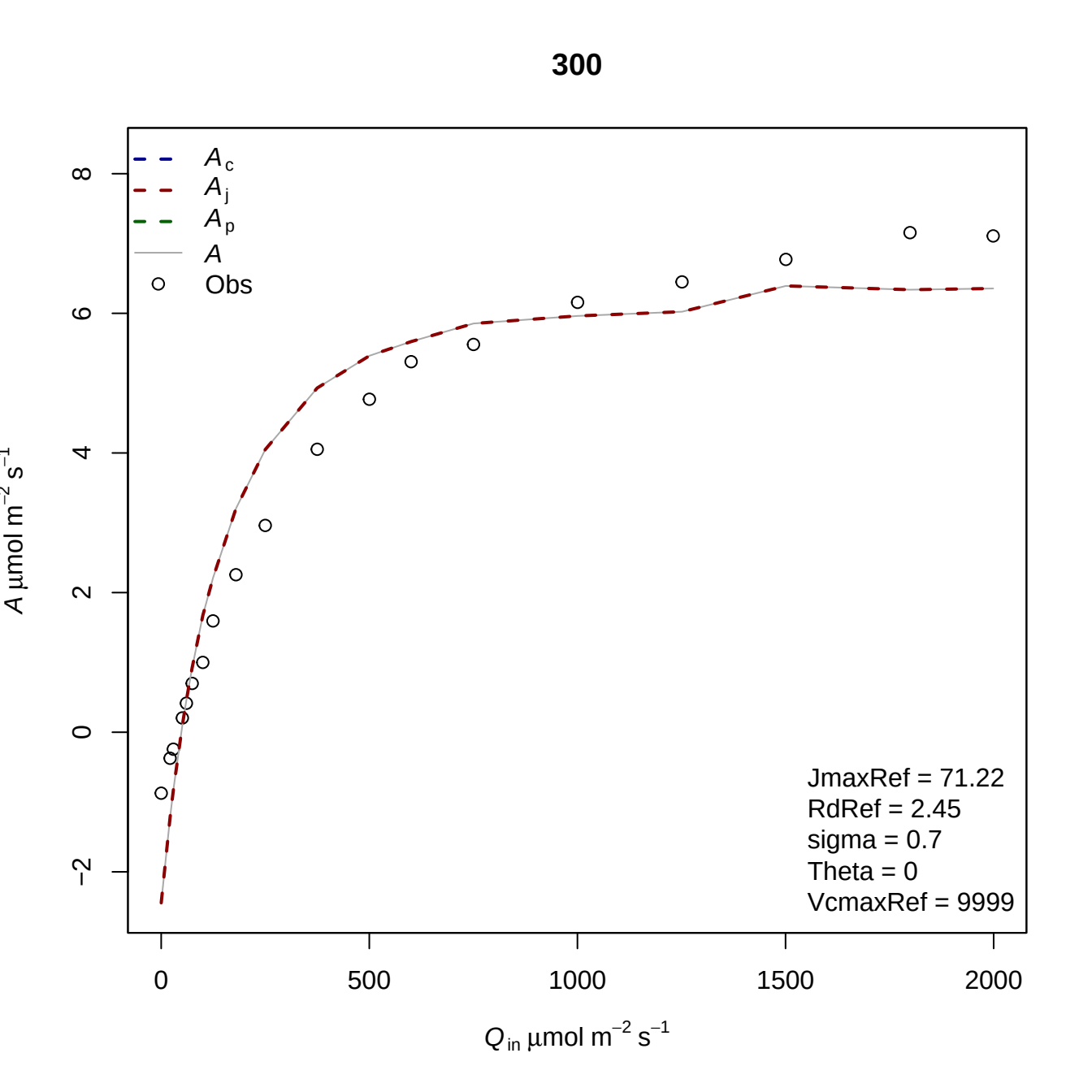

295
A_c
A_j
A_p
A
Obs
JmaxRef = 73.24
RdRef = 2.2
sigma = 0.75
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

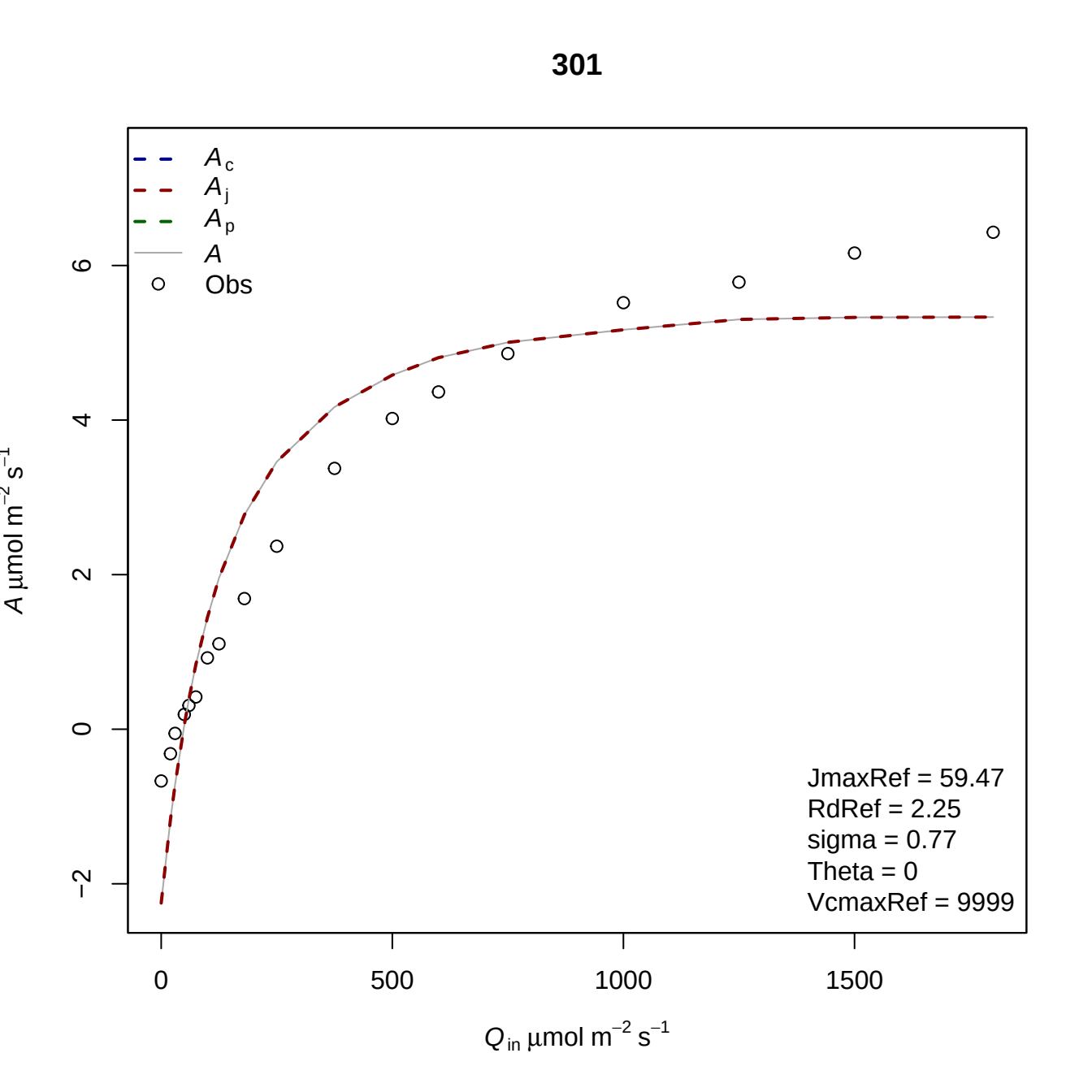

296
A_c
A_j
A_p
A
Obs
JmaxRef = 75.45
RdRef = 2.31
sigma = 0.7
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

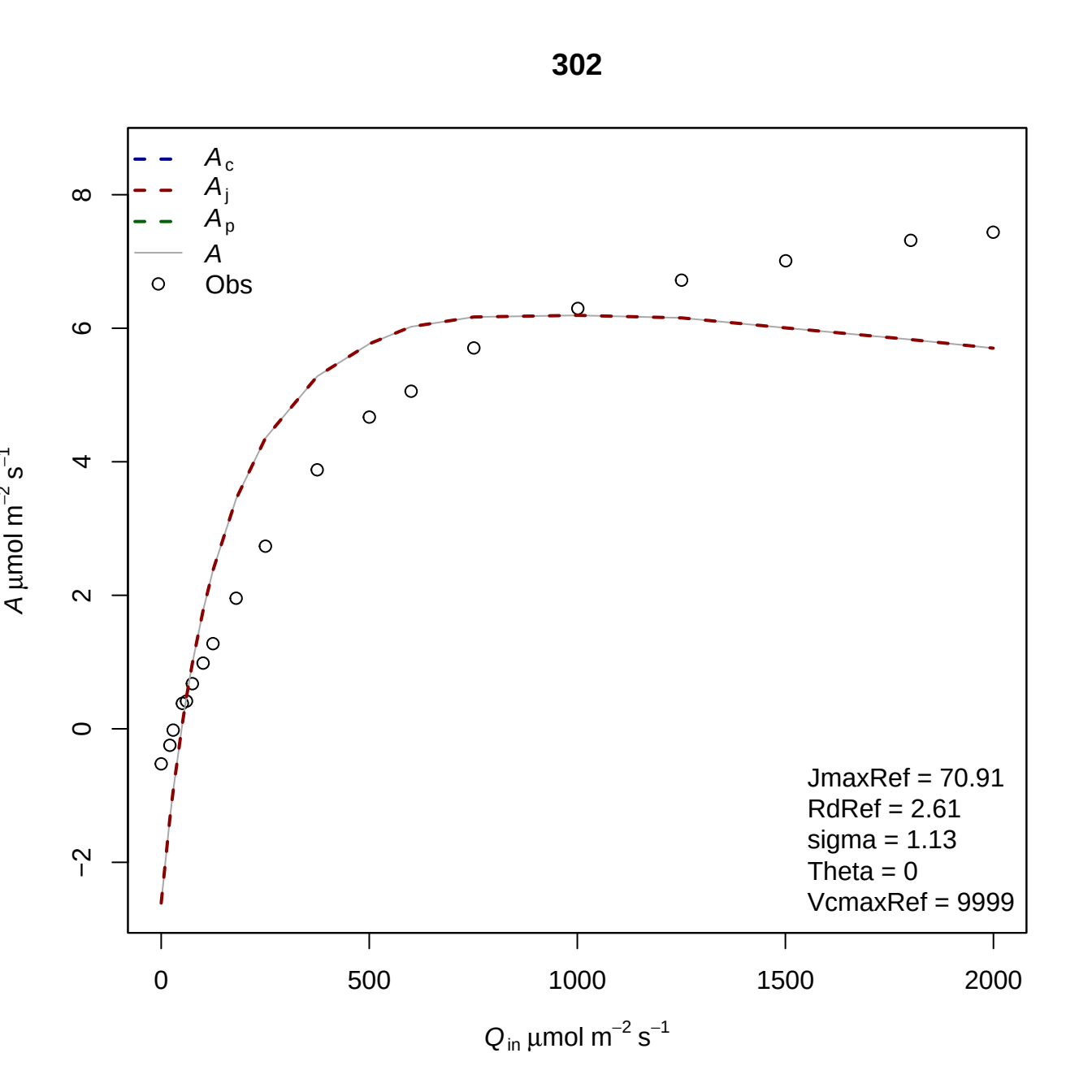

A_c
A_j
A_p
A
Obs
JmaxRef = 79.95
RdRef = 2.45
sigma = 0.82
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

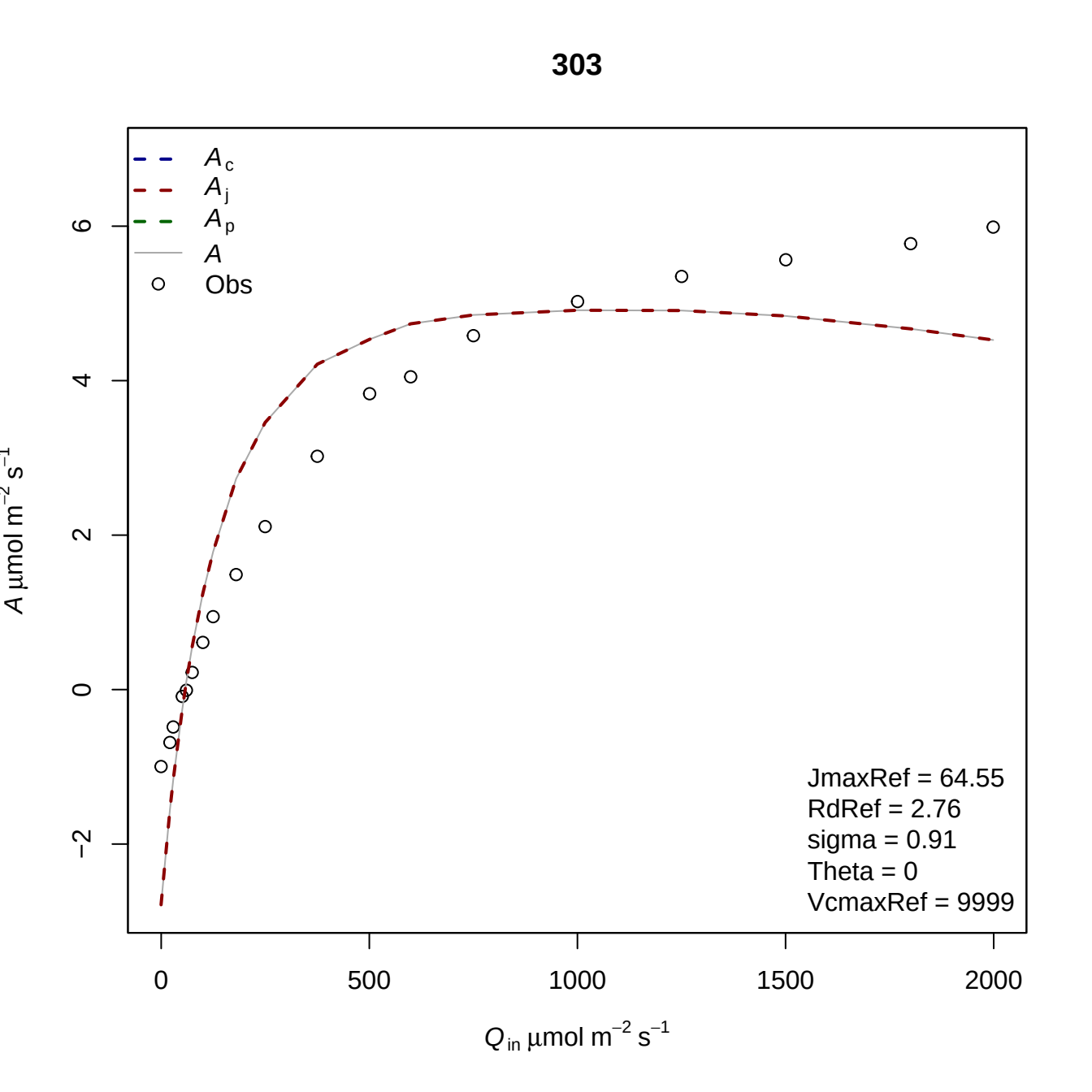

298
A_c
A_j
A_p
A
Obs
JmaxRef = 81.45
RdRef = 1.98
sigma = 0.78
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

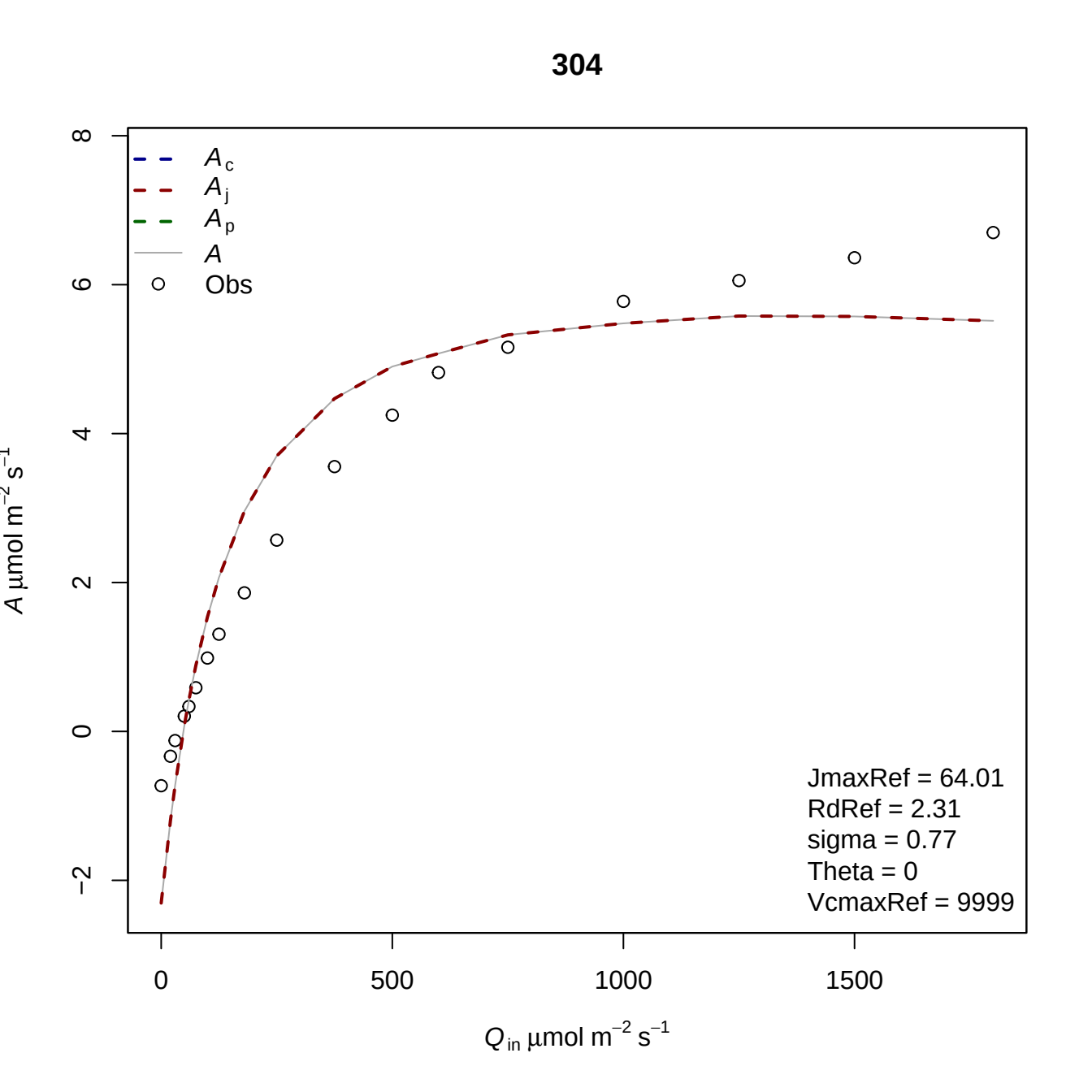

299
A_c
A_j
A_p
A
Obs
JmaxRef = 66.98
RdRef = 2.5
sigma = 0.97
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

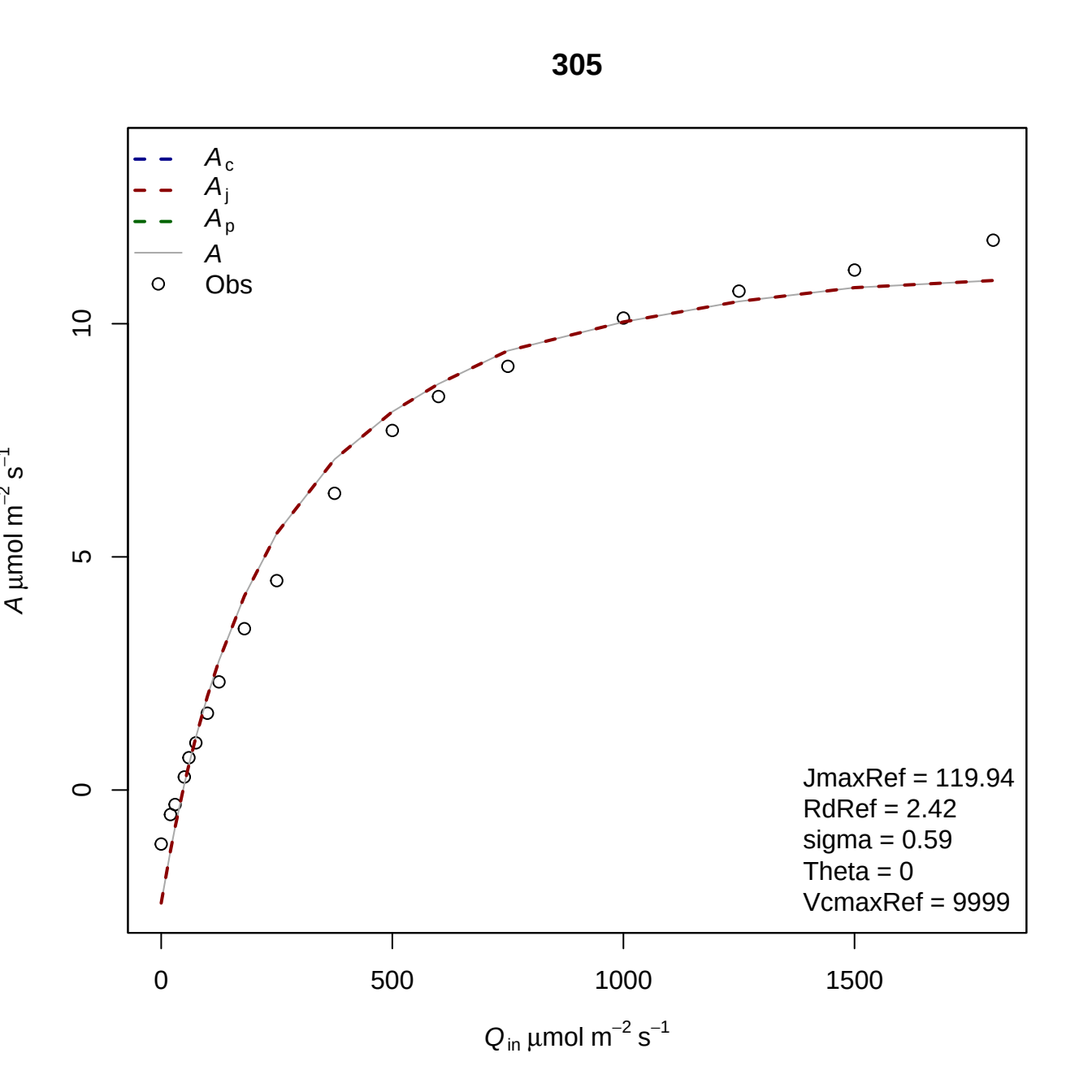

300
A_c
A_j
A_p
A
Obs
JmaxRef = 71.22
RdRef = 2.45
sigma = 0.7
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

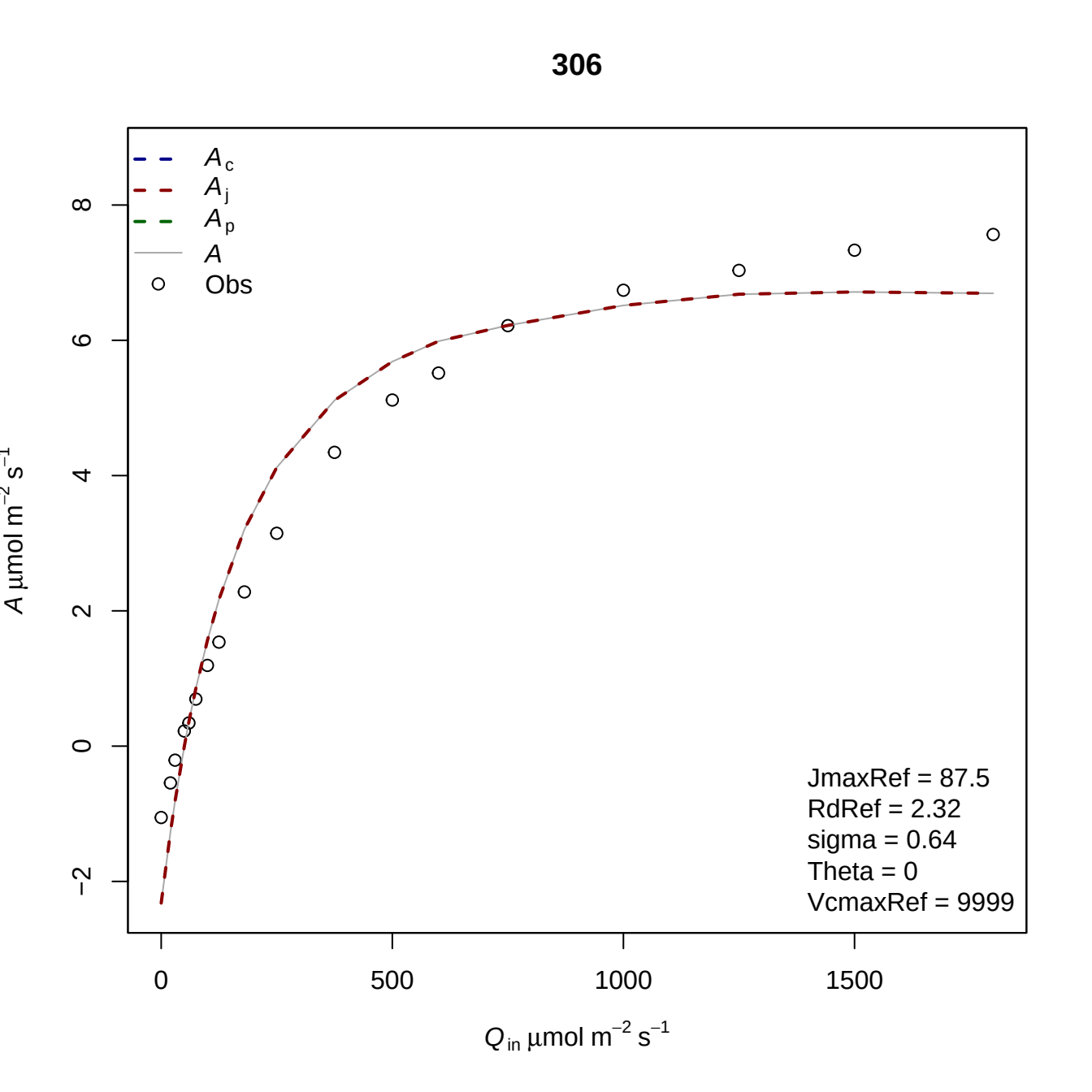

301
A_c
A_j
A_p
A
Obs
JmaxRef = 59.47
RdRef = 2.25
sigma = 0.77
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

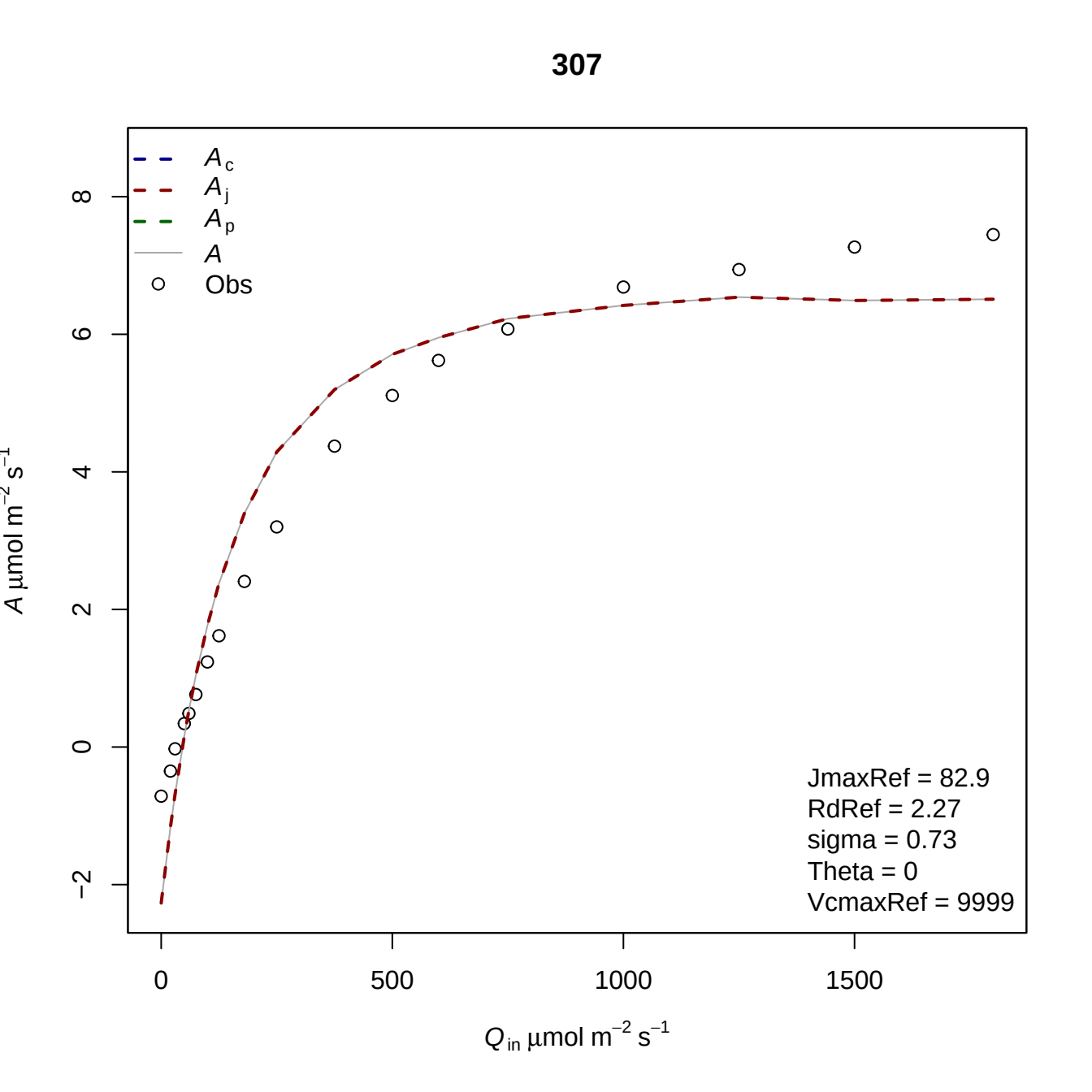

302
A_c
A_j
A_p
A
Obs
JmaxRef = 70.91
RdRef = 2.61
sigma = 1.13
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

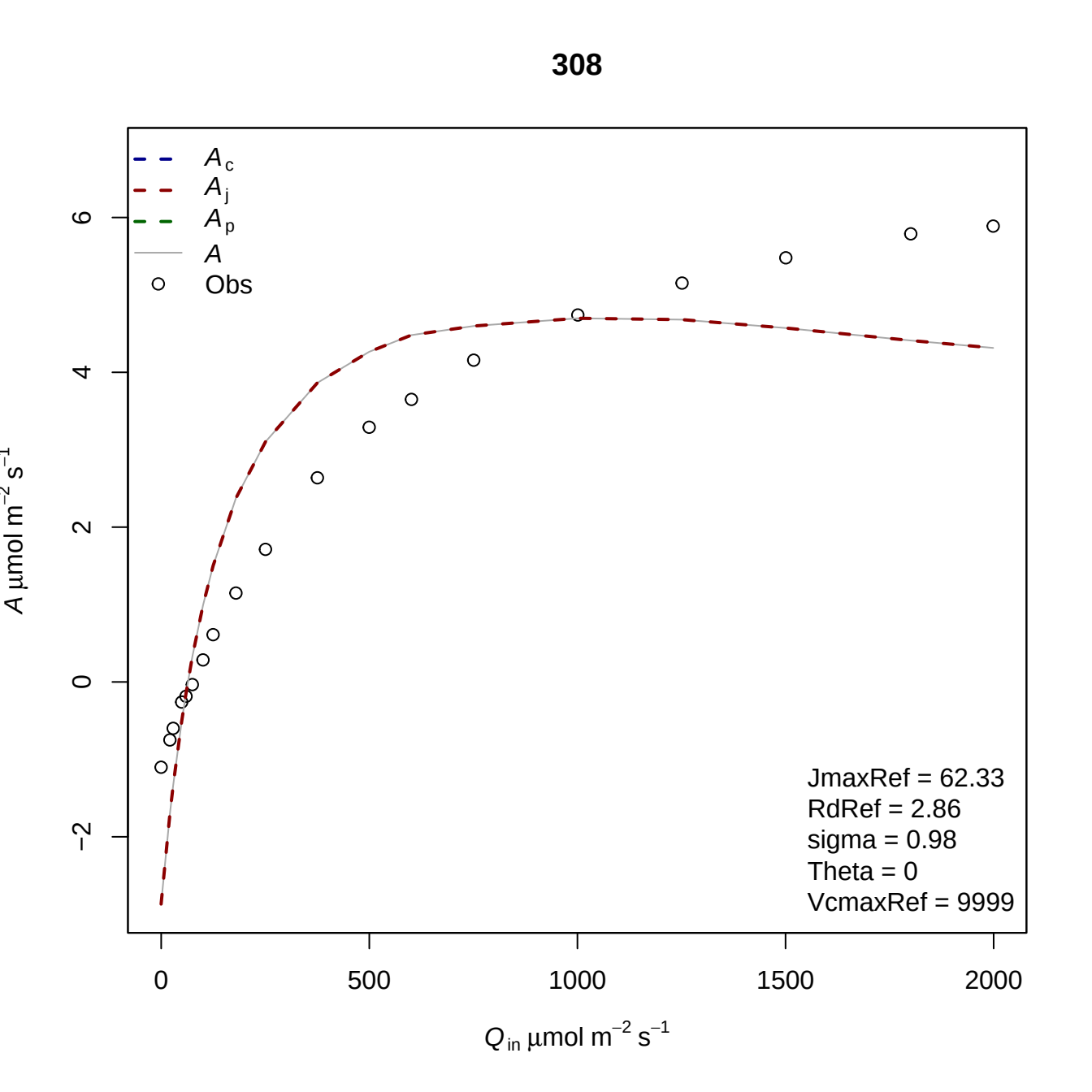

303
A_c
A_j
A_p
A
Obs
JmaxRef = 64.55
RdRef = 2.76
sigma = 0.91
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

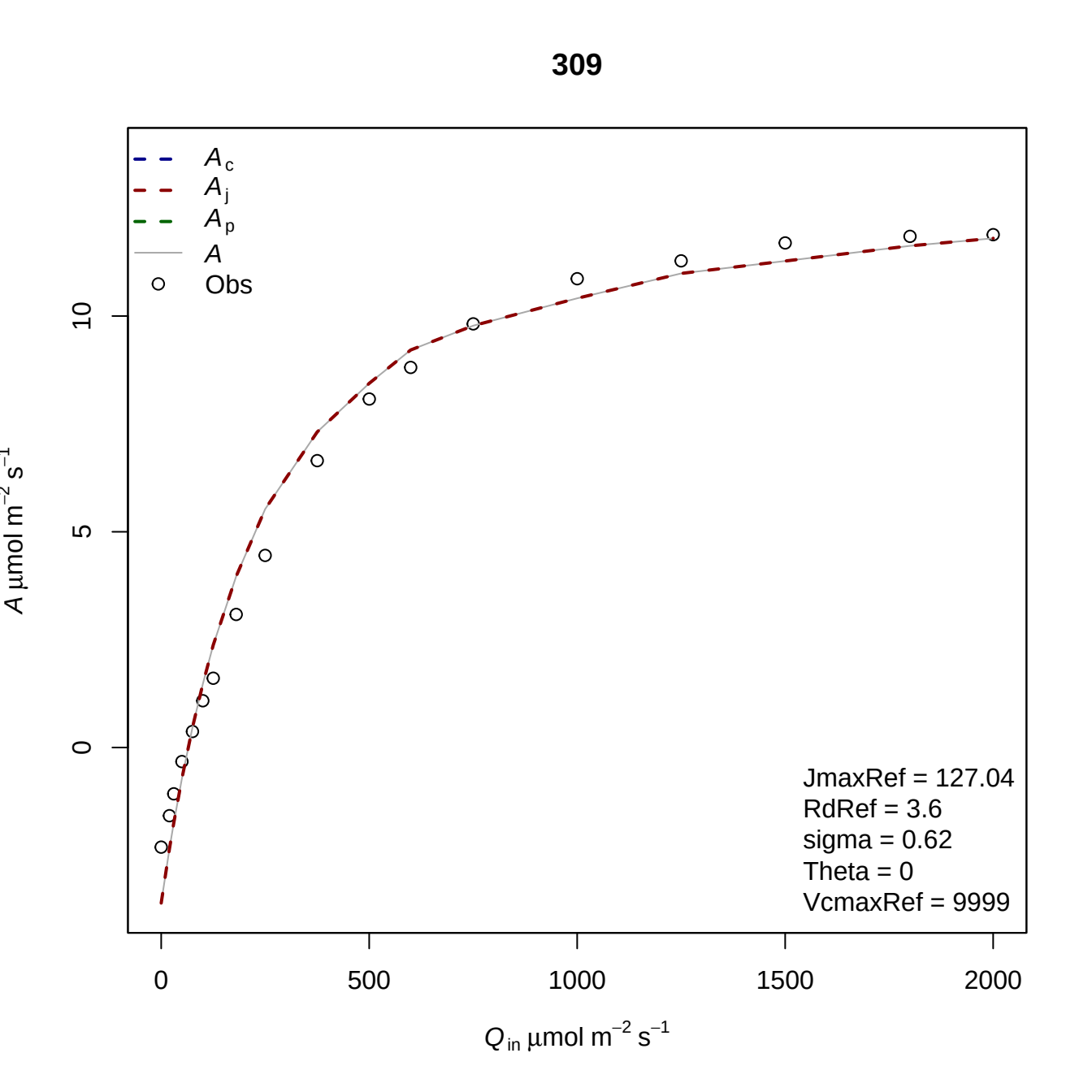

304
A_c
A_j
A_p
A
Obs
JmaxRef = 64.01
RdRef = 2.31
sigma = 0.77
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

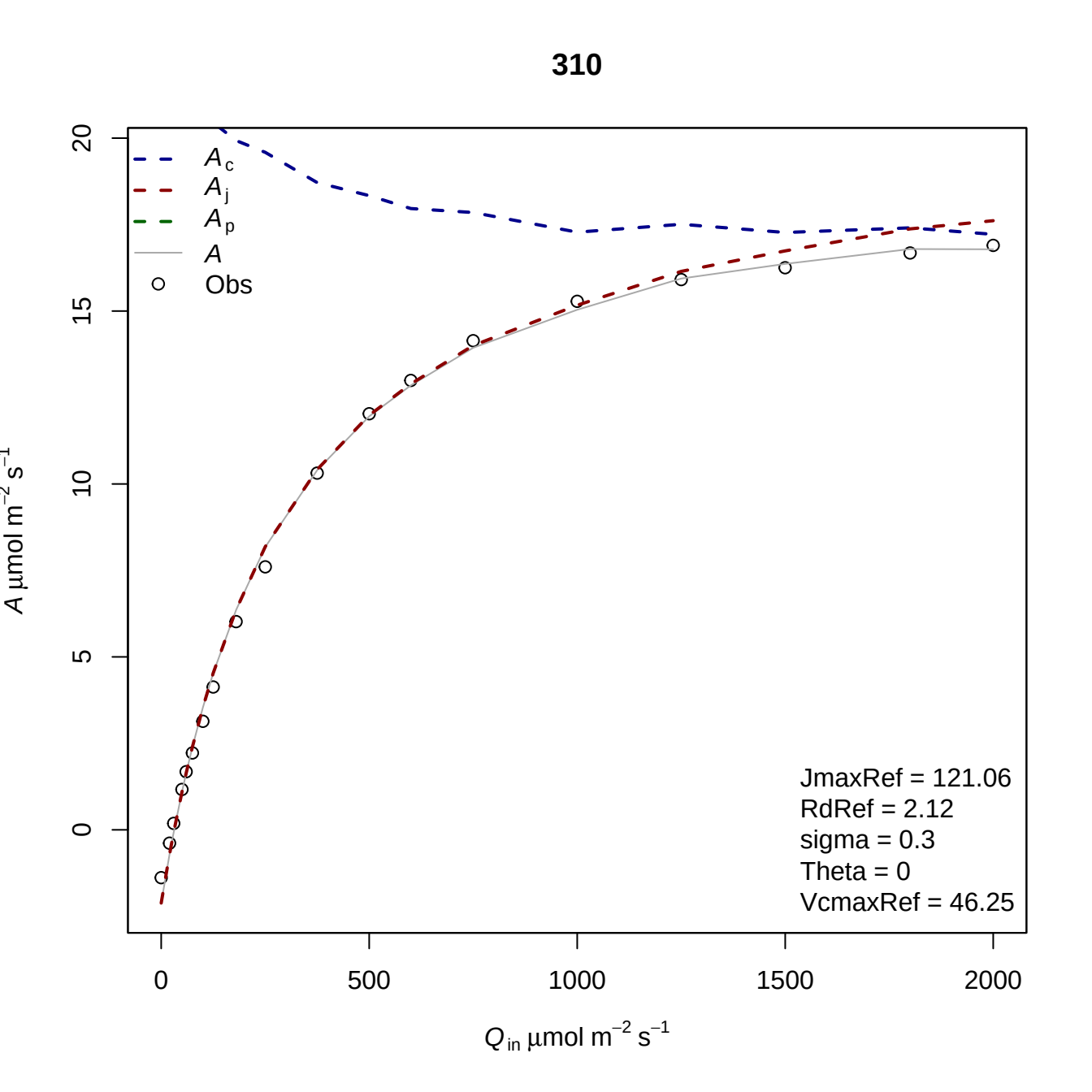

305
A_c
A_j
A_p
A
Obs
JmaxRef = 119.94
RdRef = 2.42
sigma = 0.59
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

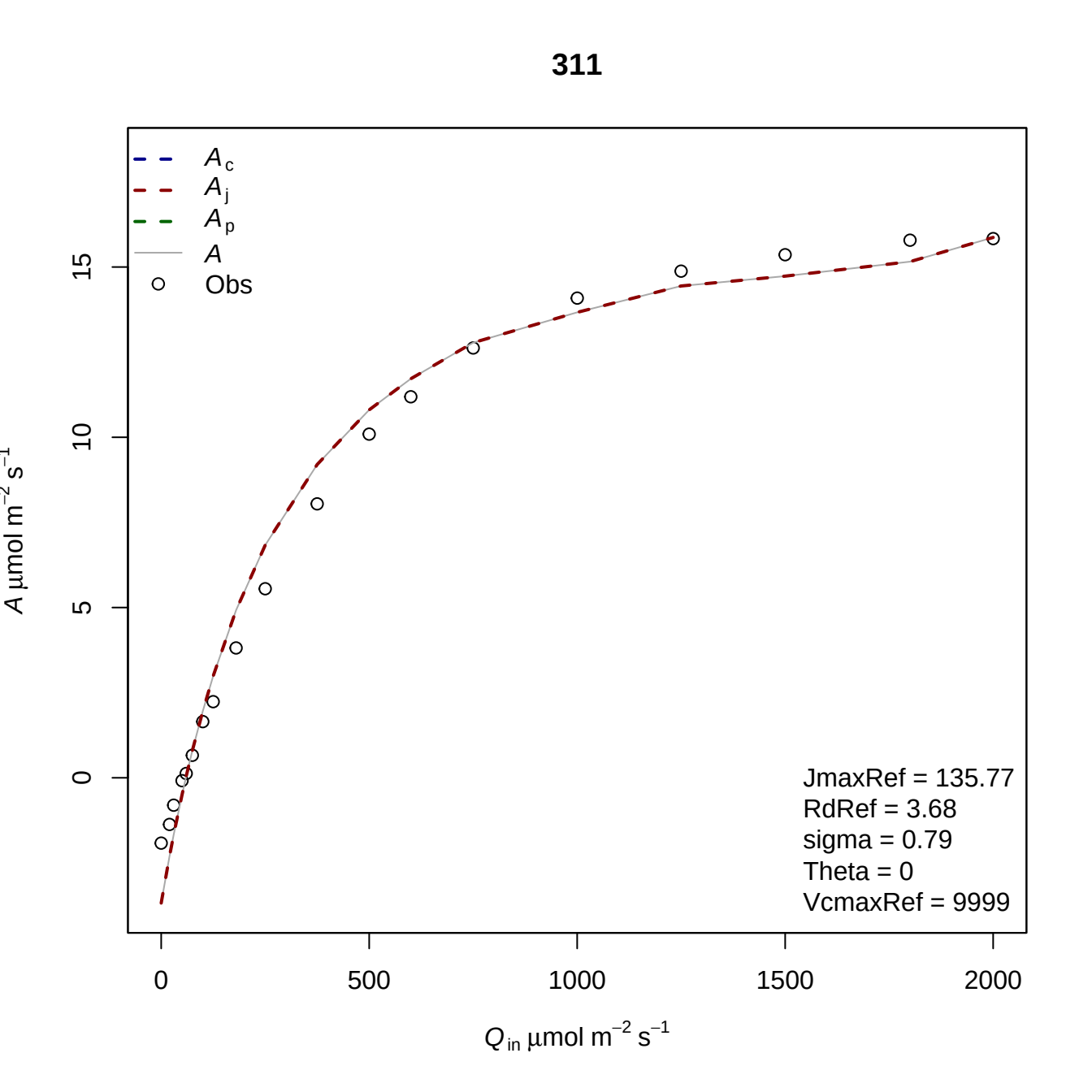

306
A_c
A_j
A_p
A
Obs
JmaxRef = 87.5
RdRef = 2.32
sigma = 0.64
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

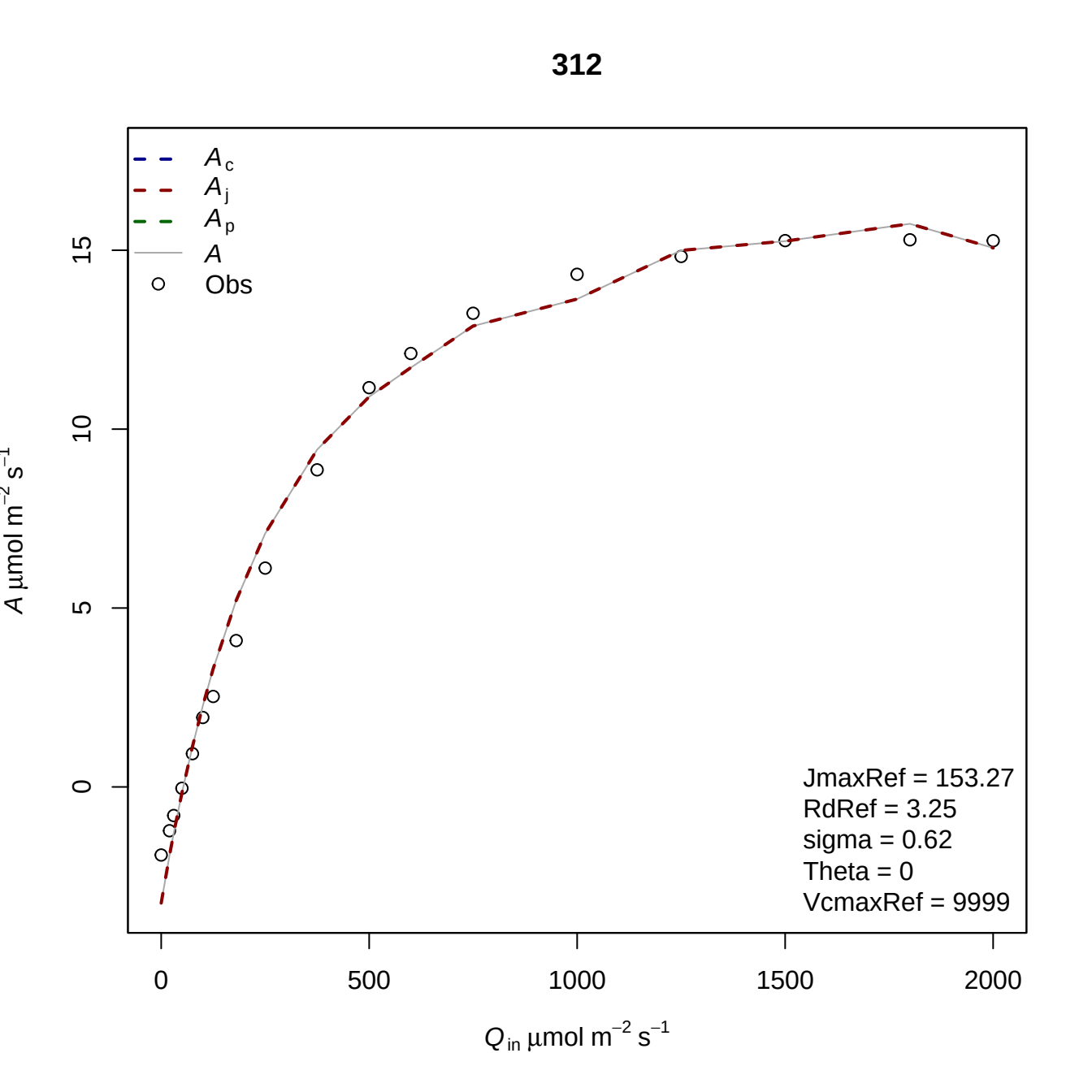

307
A_c
A_j
A_p
A
Obs
JmaxRef = 82.9
RdRef = 2.27
sigma = 0.73
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

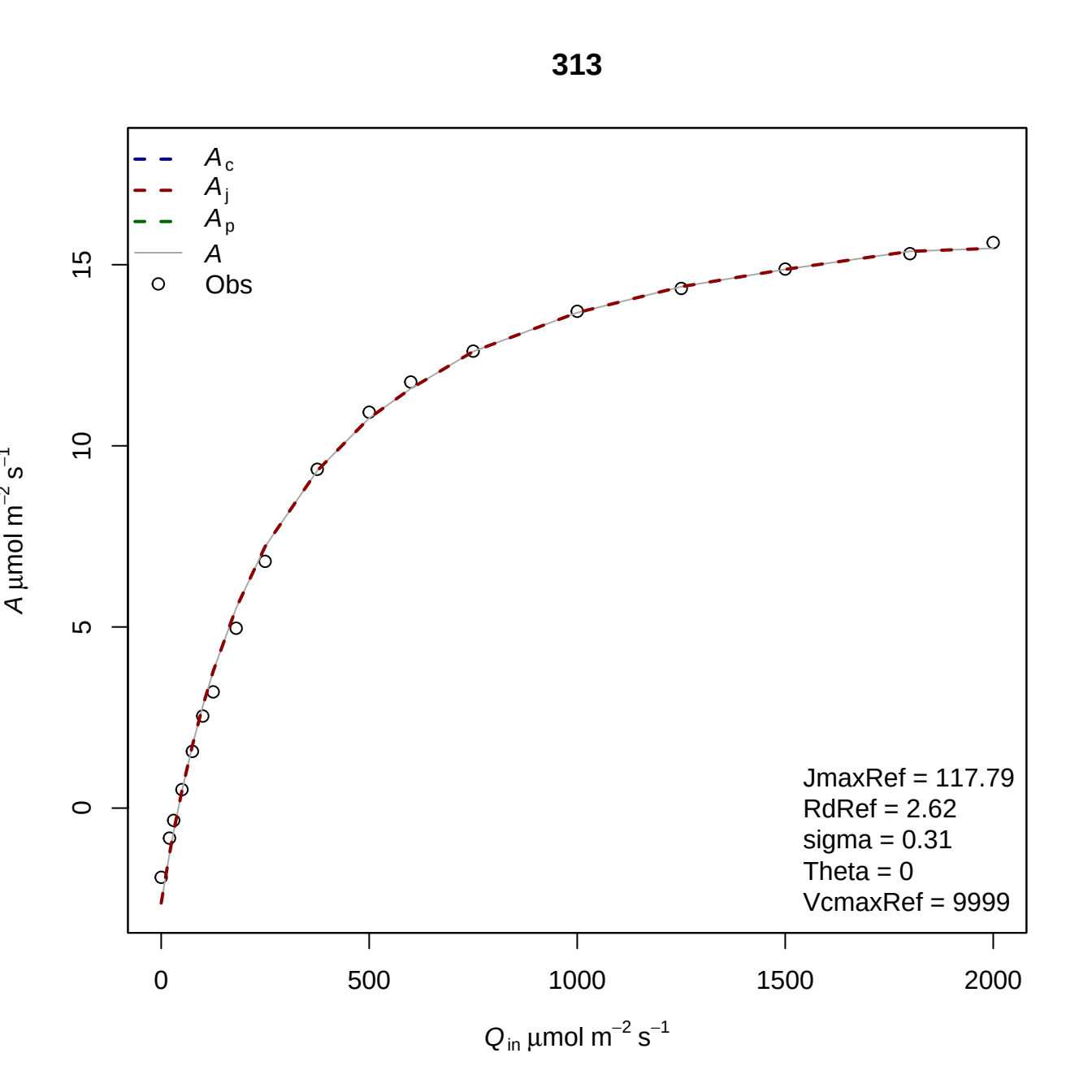

308
A_c
A_j
A_p
A
Obs
JmaxRef = 62.33
RdRef = 2.86
sigma = 0.98
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

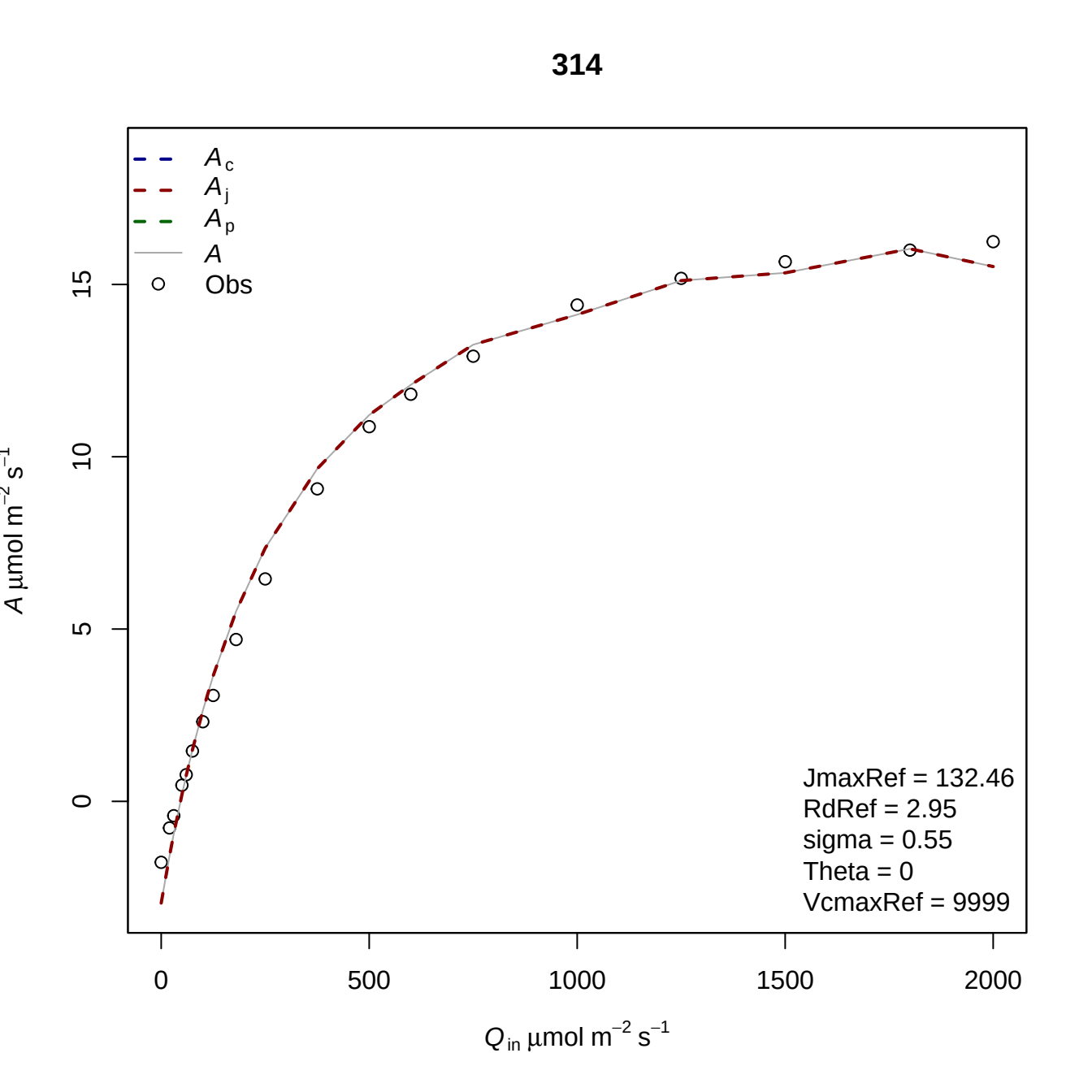

309
A_c
A_j
A_p
A
Obs
JmaxRef = 127.04
RdRef = 3.6
sigma = 0.62
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

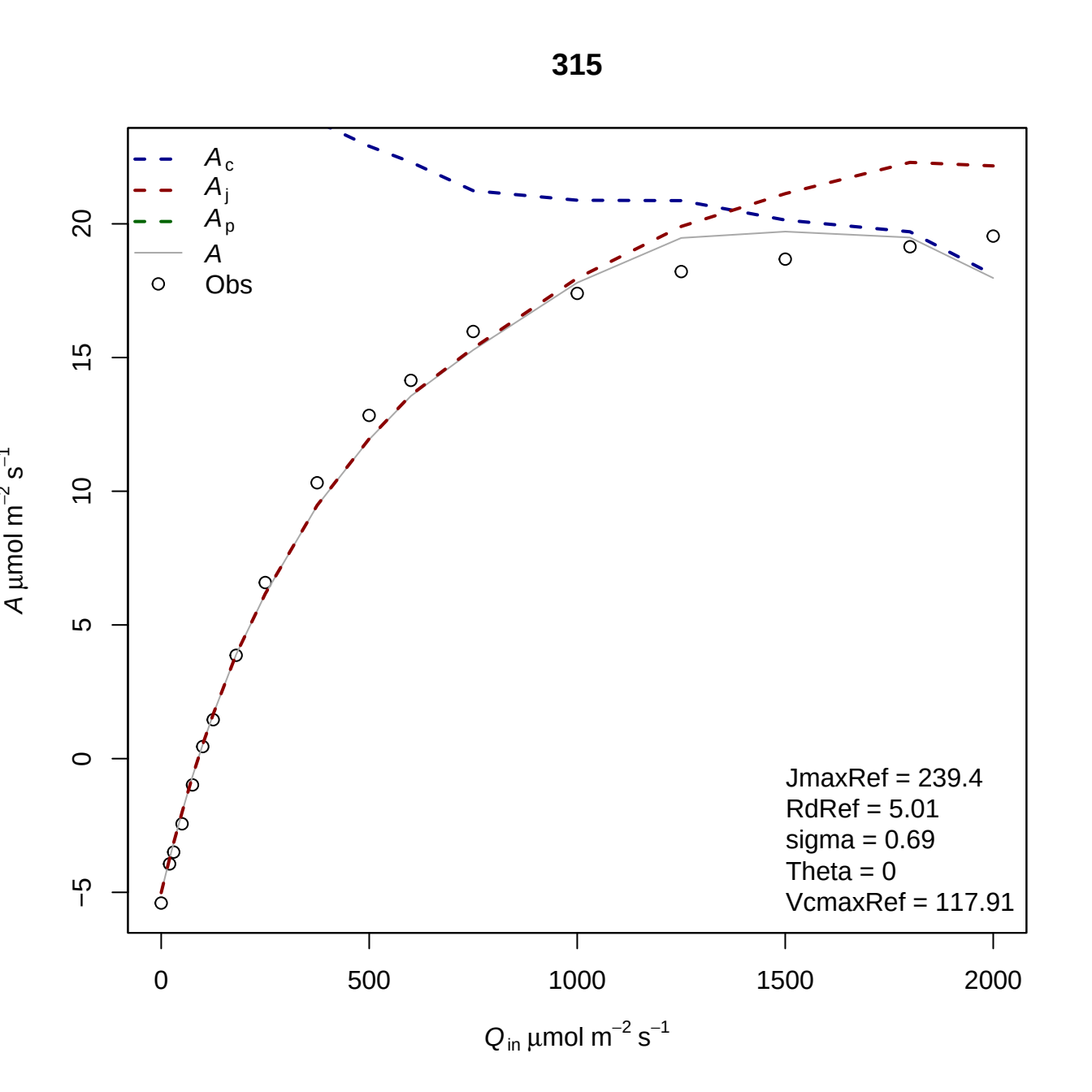

310
A_c
A_j
A_p
A
Obs
JmaxRef = 121.06
RdRef = 2.12
sigma = 0.3
Theta = 0
VcmaxRef = 46.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

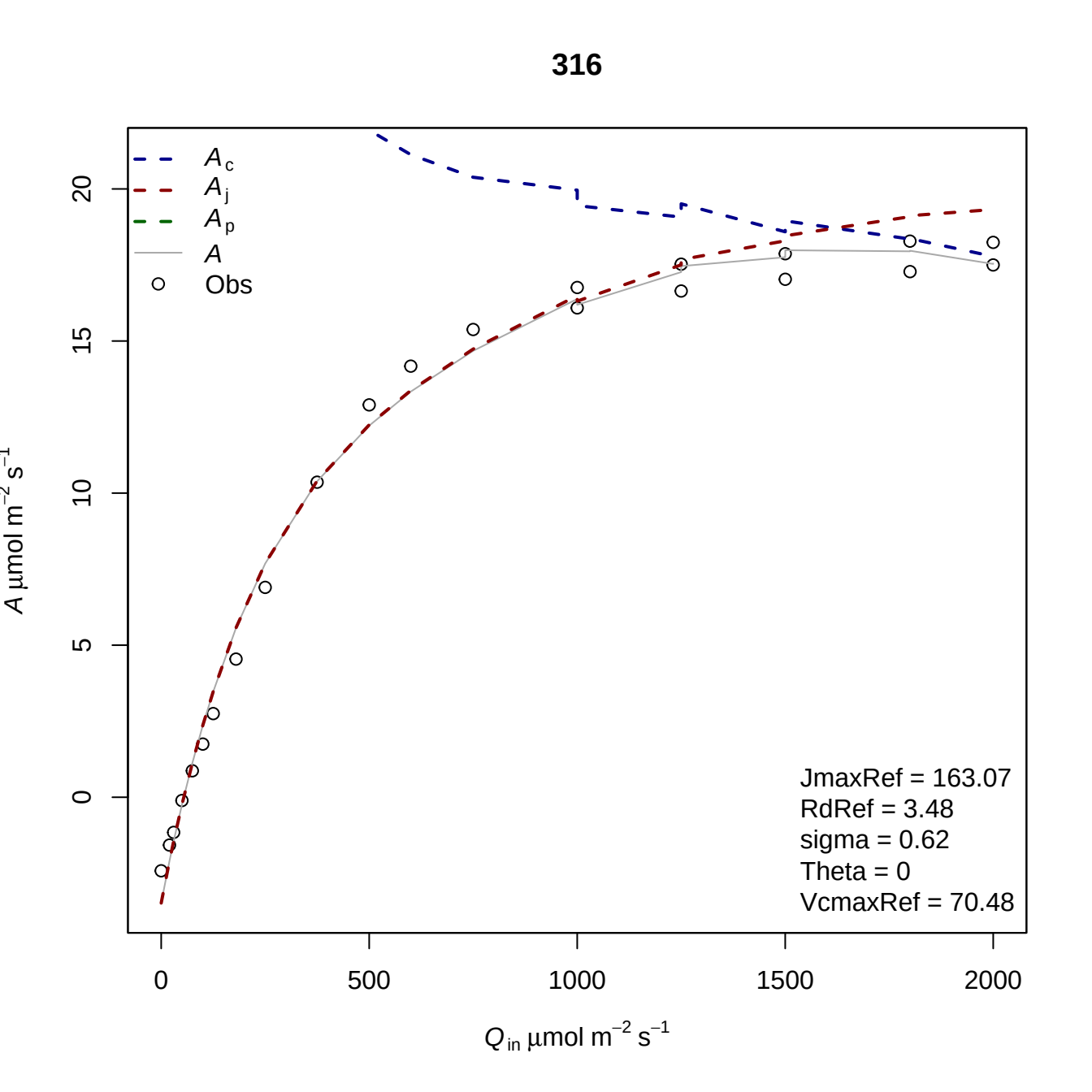

311
A_c
A_j
A_p
A
Obs
JmaxRef = 135.77
RdRef = 3.68
sigma = 0.79
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

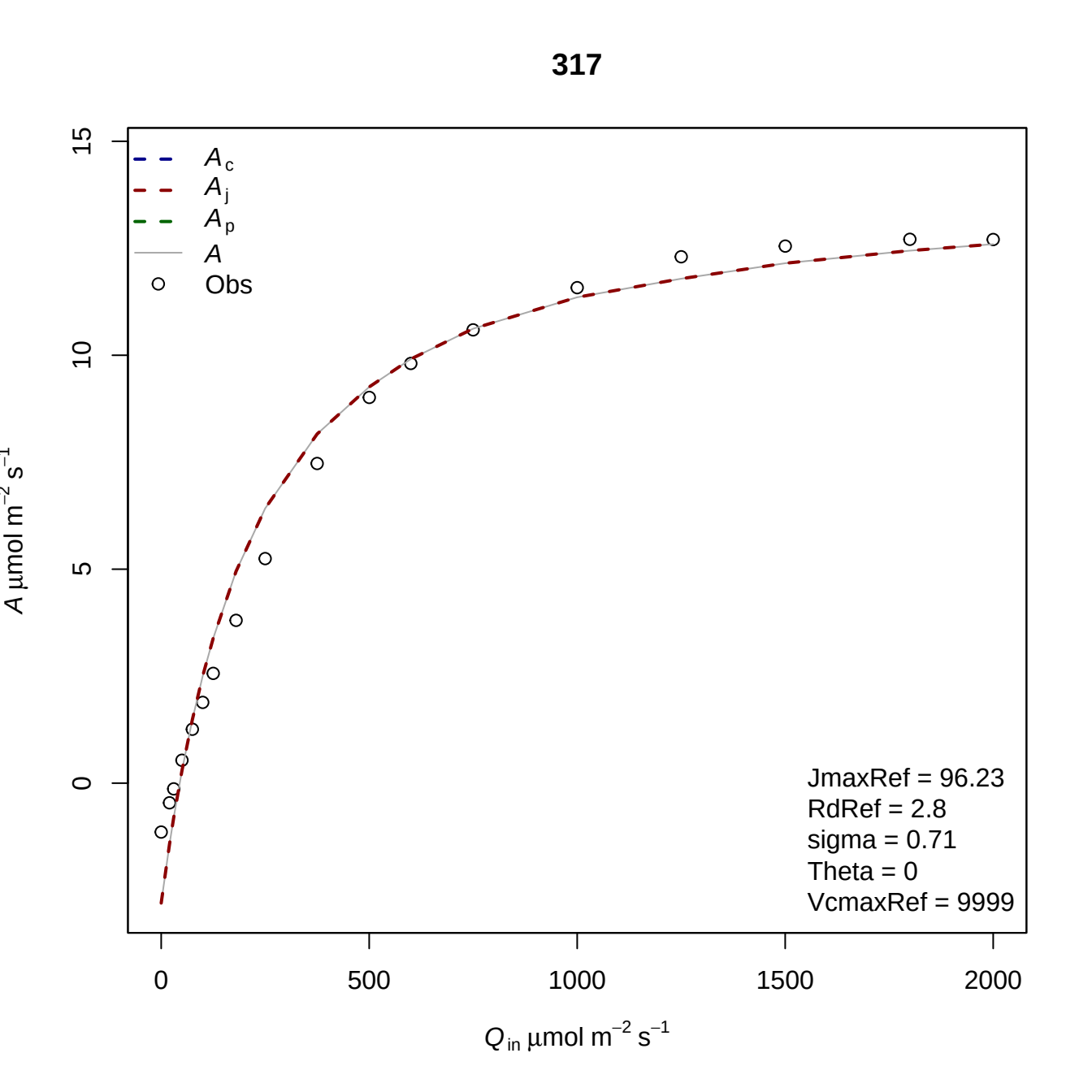

312
A_c
A_j
A_p
A
Obs
JmaxRef = 153.27
RdRef = 3.25
sigma = 0.62
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

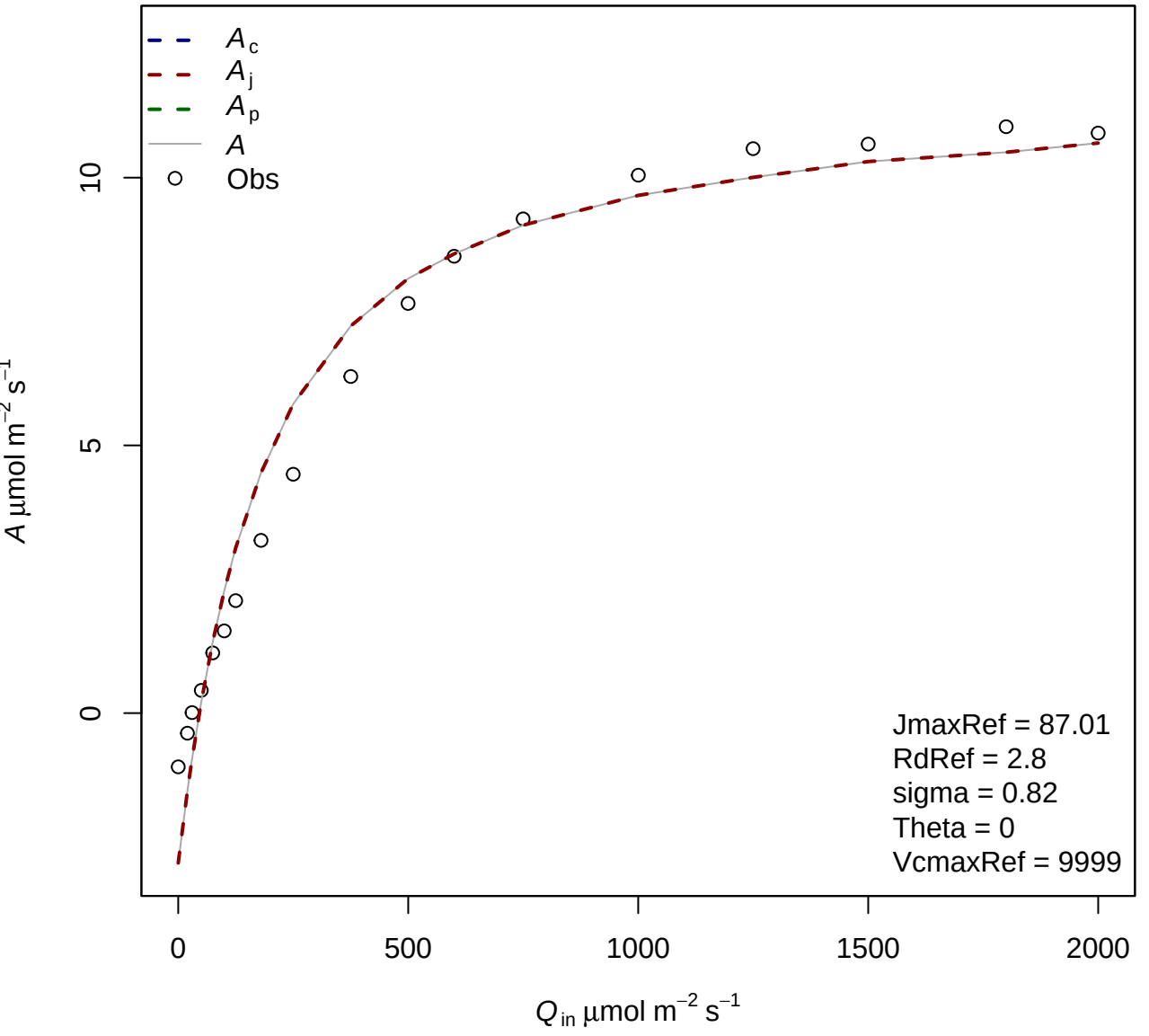

313
A_c
A_j
A_p
A
Obs
JmaxRef = 117.79
RdRef = 2.62
sigma = 0.31
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

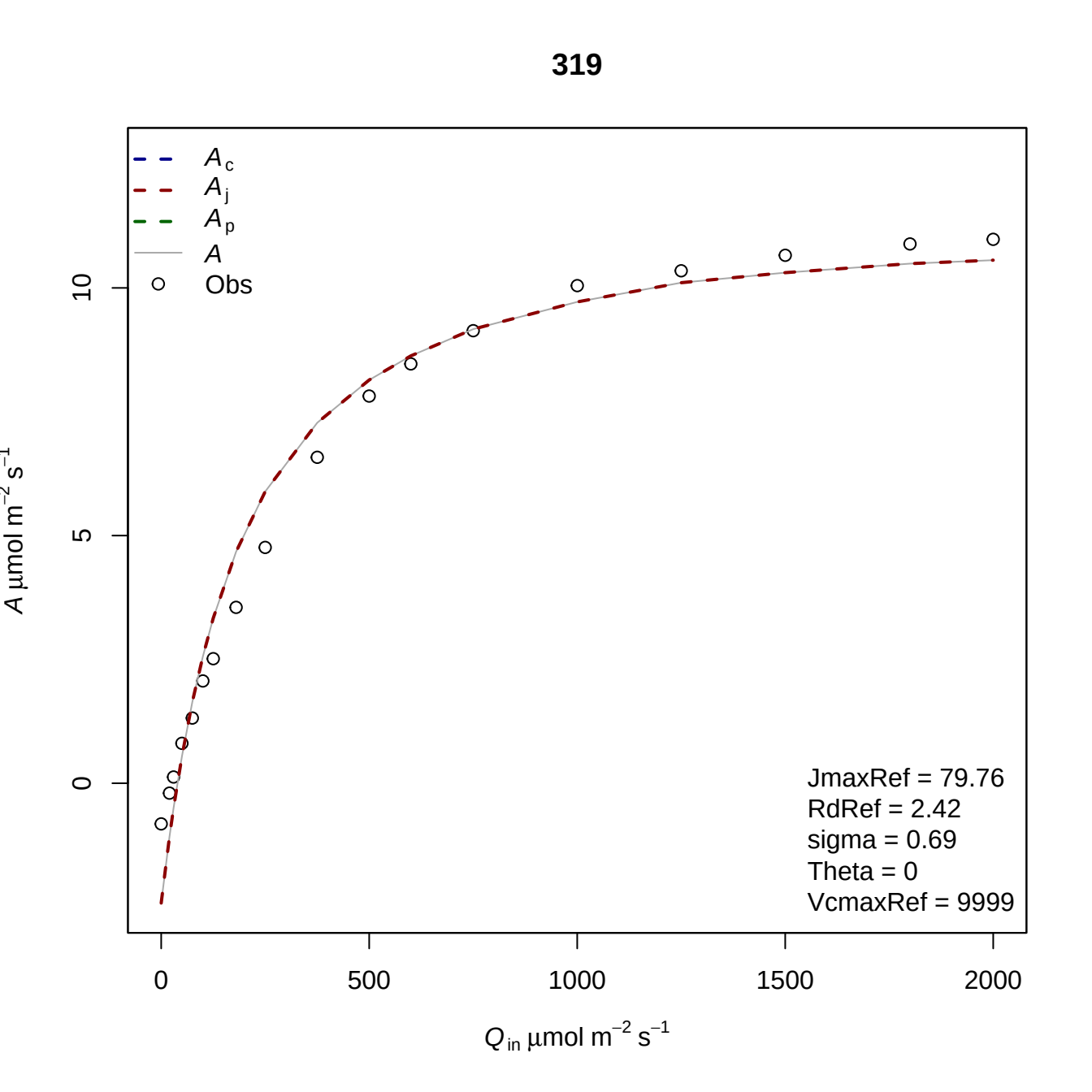

314
A_c
A_j
A_p
A
Obs
JmaxRef = 132.46
RdRef = 2.95
sigma = 0.55
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

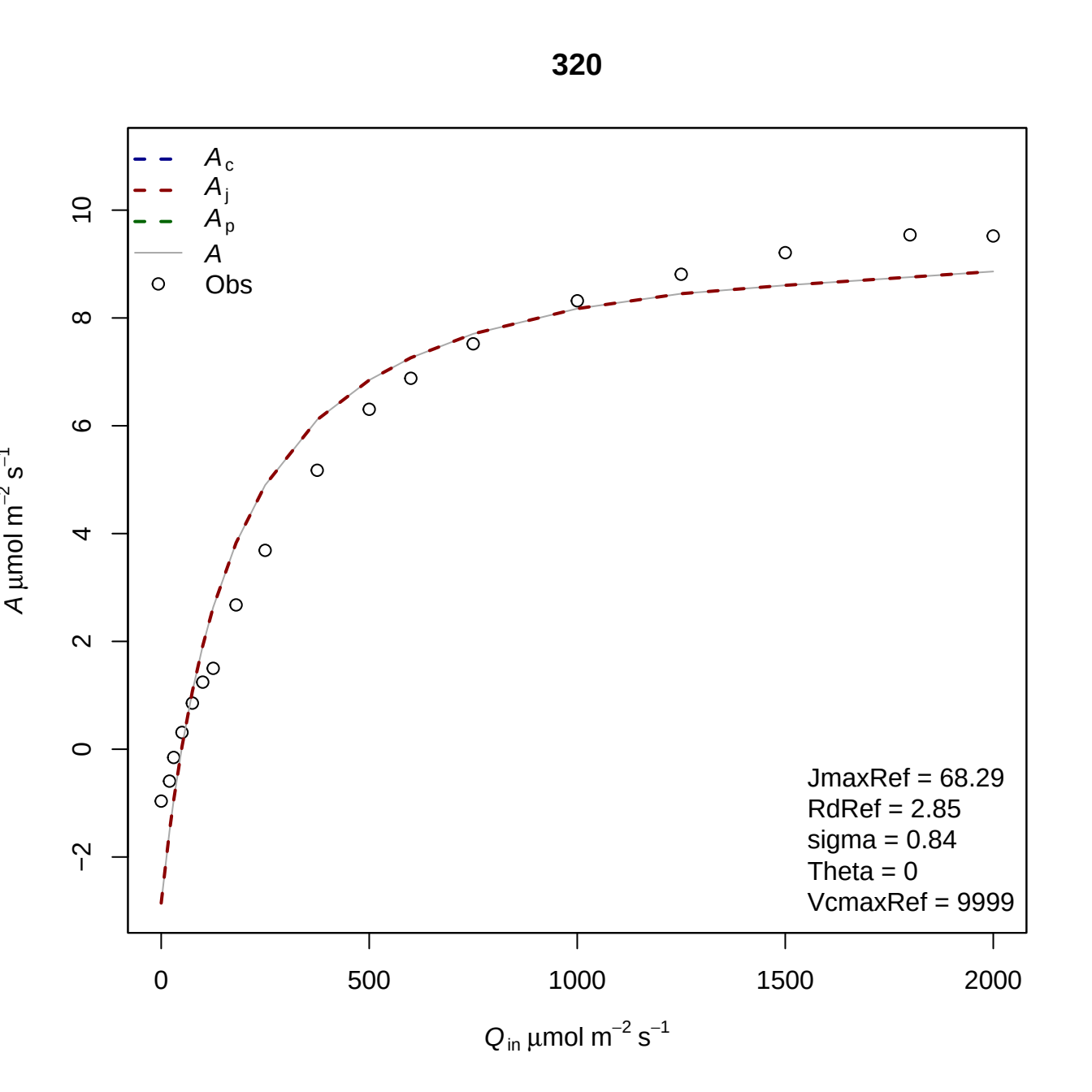

315
A_c
A_j
A_p
A
Obs
JmaxRef = 239.4
RdRef = 5.01
sigma = 0.69
Theta = 0
VcmaxRef = 117.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

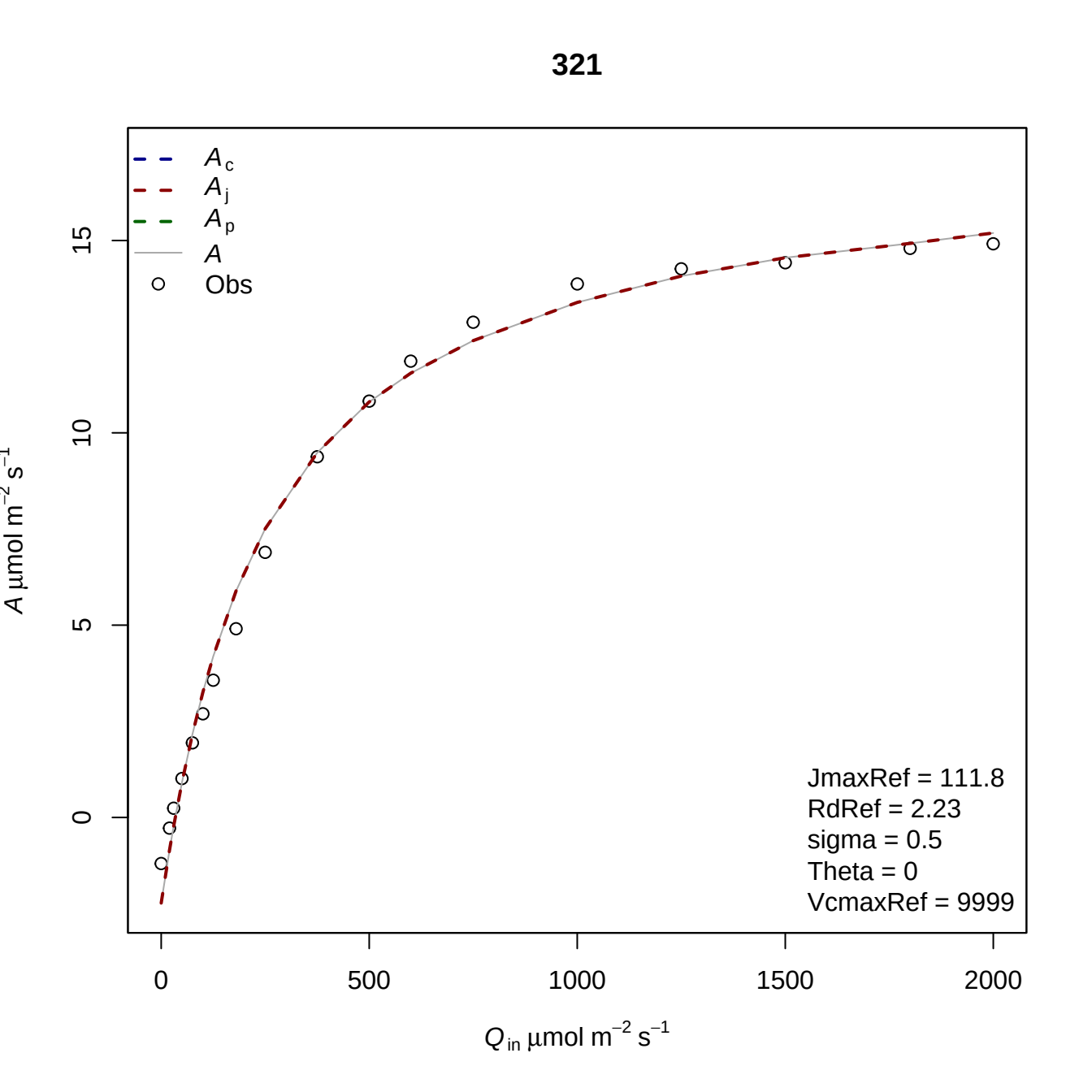

316
A_c
A_j
A_p
A
Obs
JmaxRef = 163.07
RdRef = 3.48
sigma = 0.62
Theta = 0
VcmaxRef = 70.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

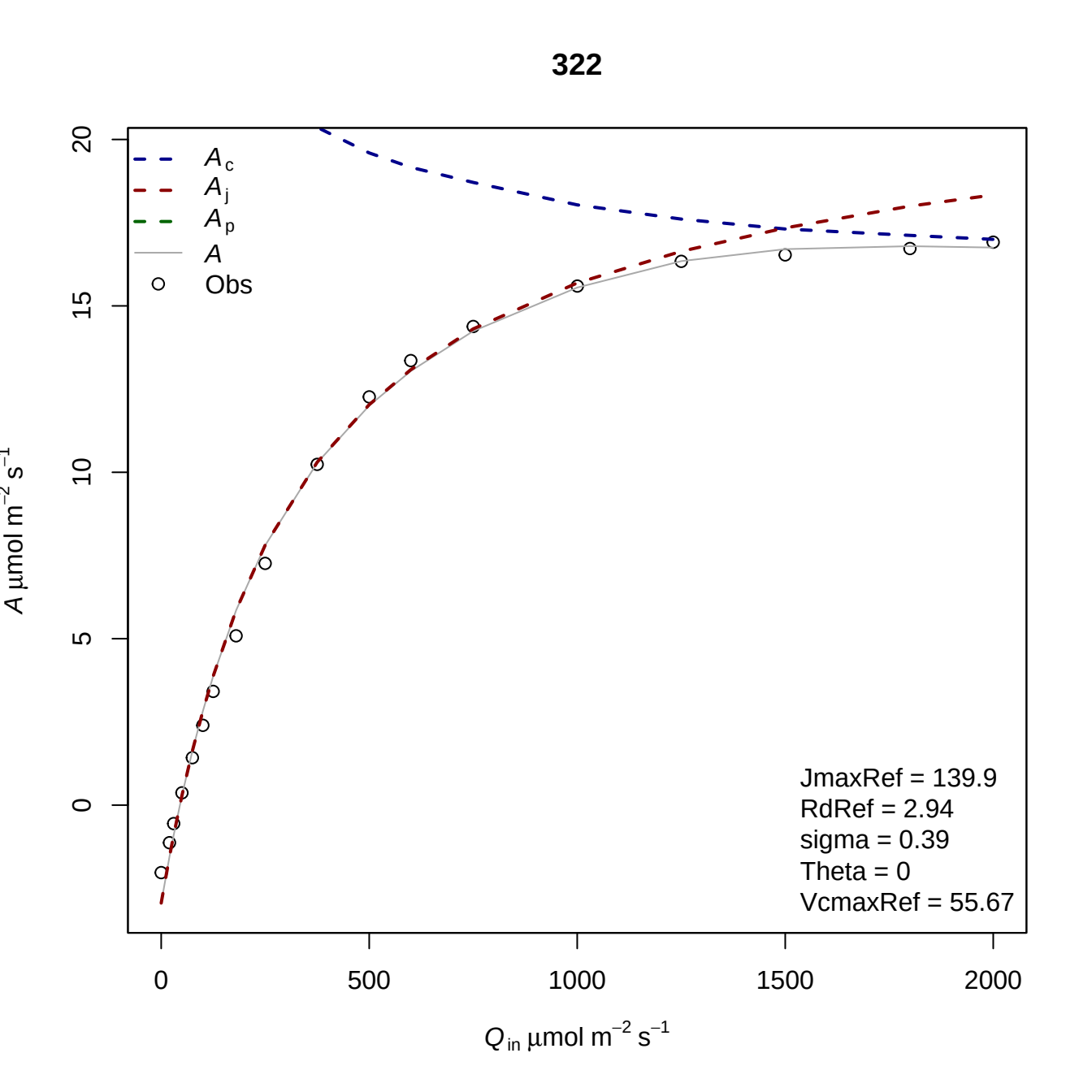

317
A_c
A_j
A_p
A
Obs
JmaxRef = 96.23
RdRef = 2.8
sigma = 0.71
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

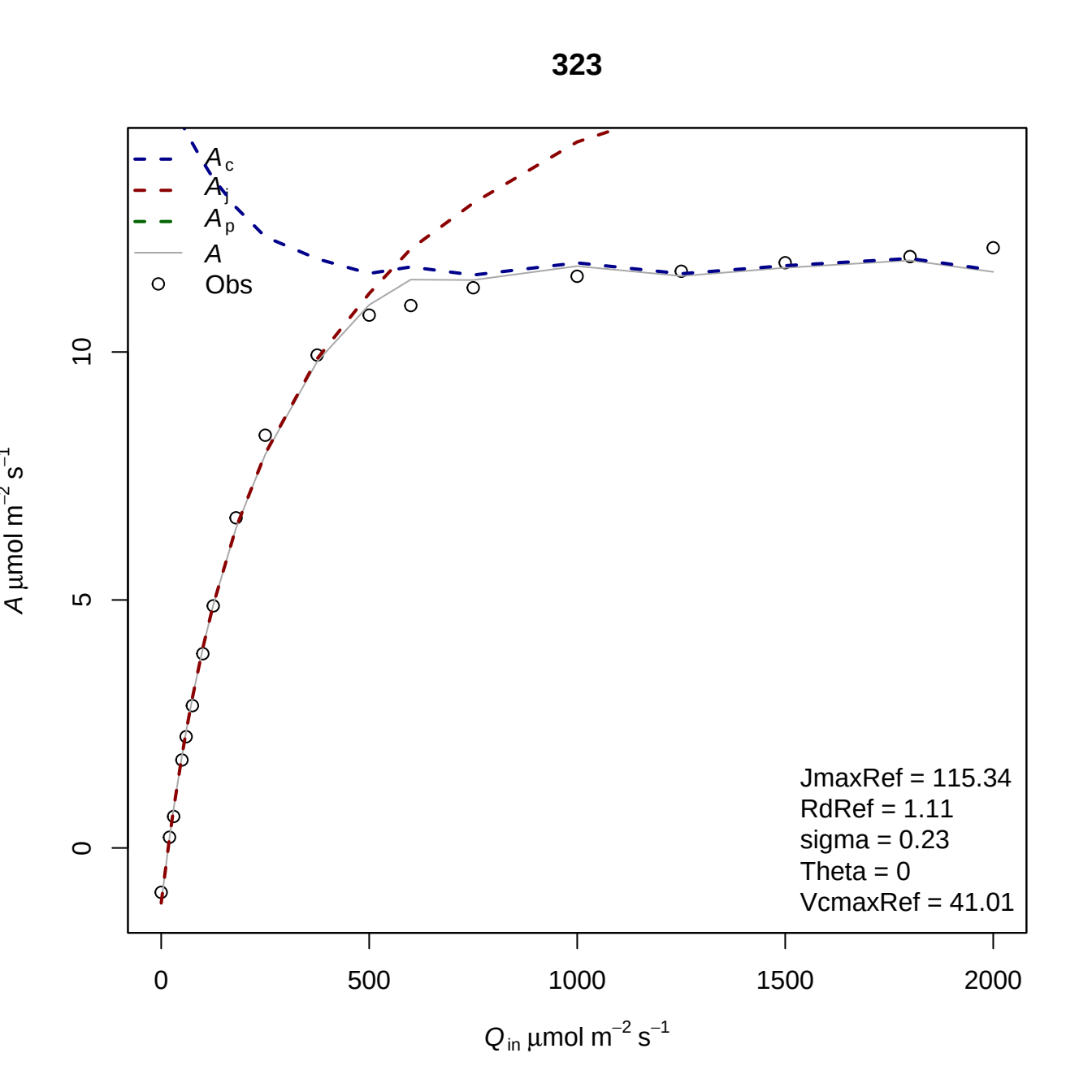

A_c
A_j
A_p
A
Obs
JmaxRef = 87.01
RdRef = 2.8
sigma = 0.82
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

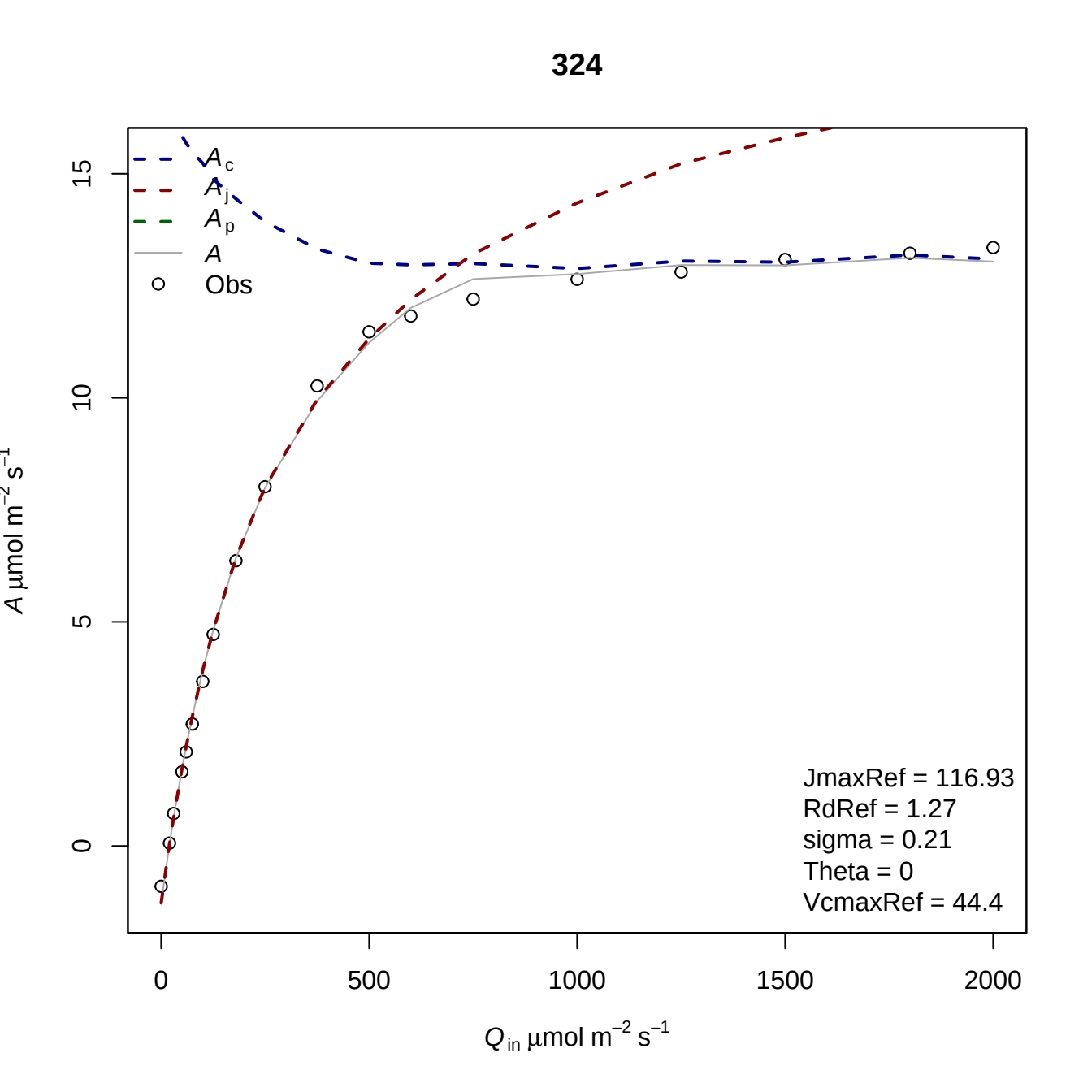

319
A_c
A_j
A_p
A
Obs
JmaxRef = 79.76
RdRef = 2.42
sigma = 0.69
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

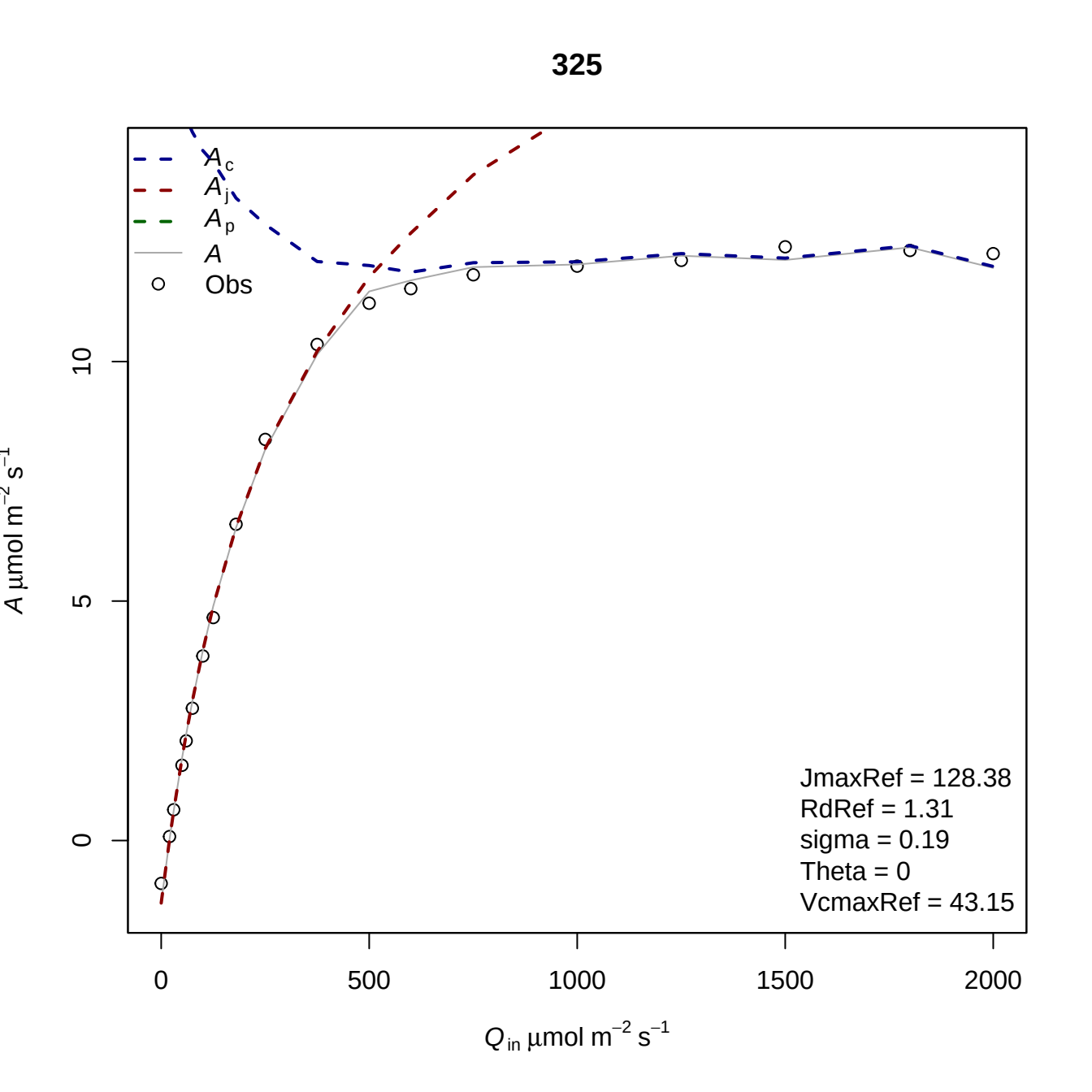

320
A_c
A_j
A_p
A
Obs
JmaxRef = 68.29
RdRef = 2.85
sigma = 0.84
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

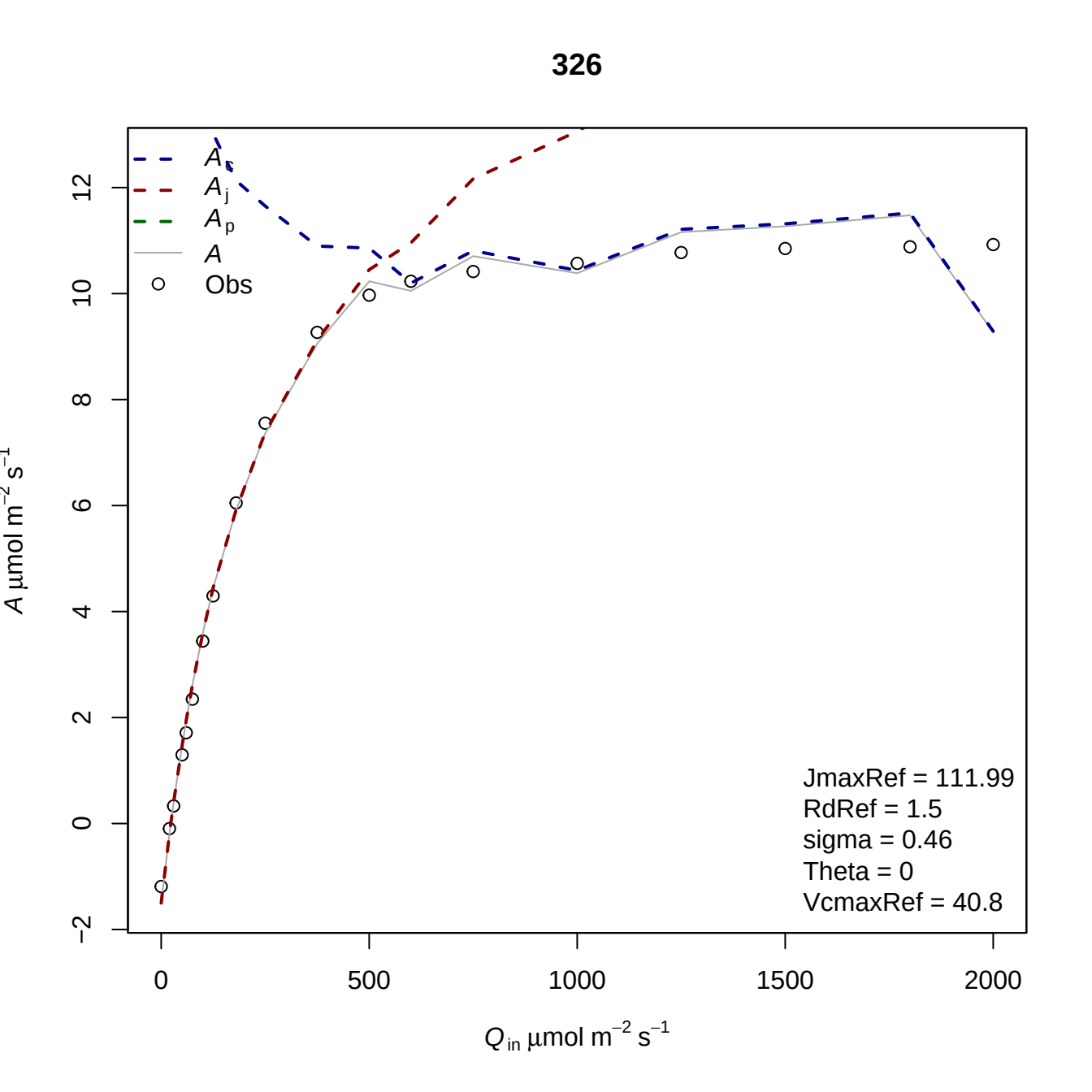

321
A_c
A_j
A_p
A
Obs
JmaxRef = 111.8
RdRef = 2.23
sigma = 0.5
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

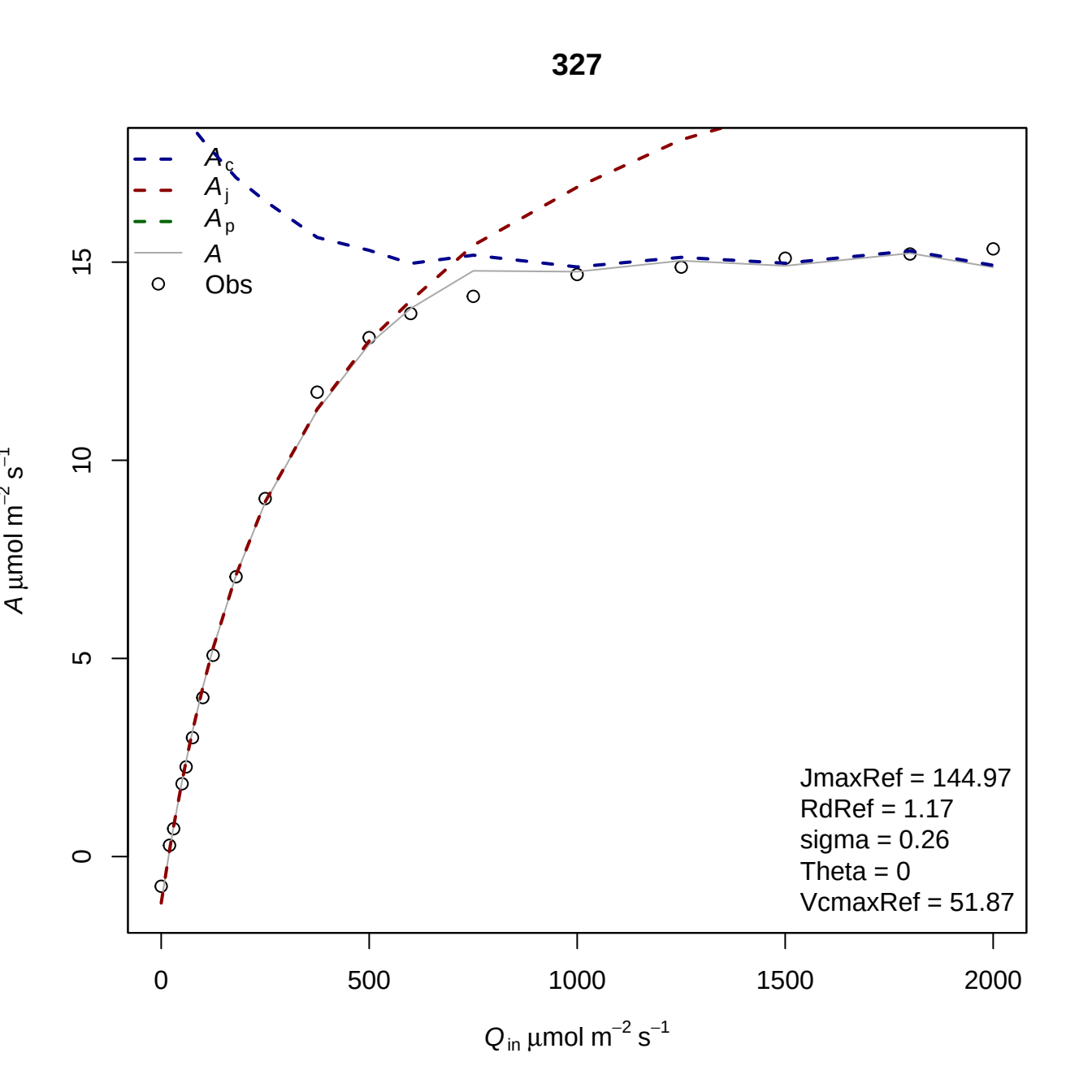

322
A_c
A_j
A_p
A
Obs
JmaxRef = 139.9
RdRef = 2.94
sigma = 0.39
Theta = 0
VcmaxRef = 55.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

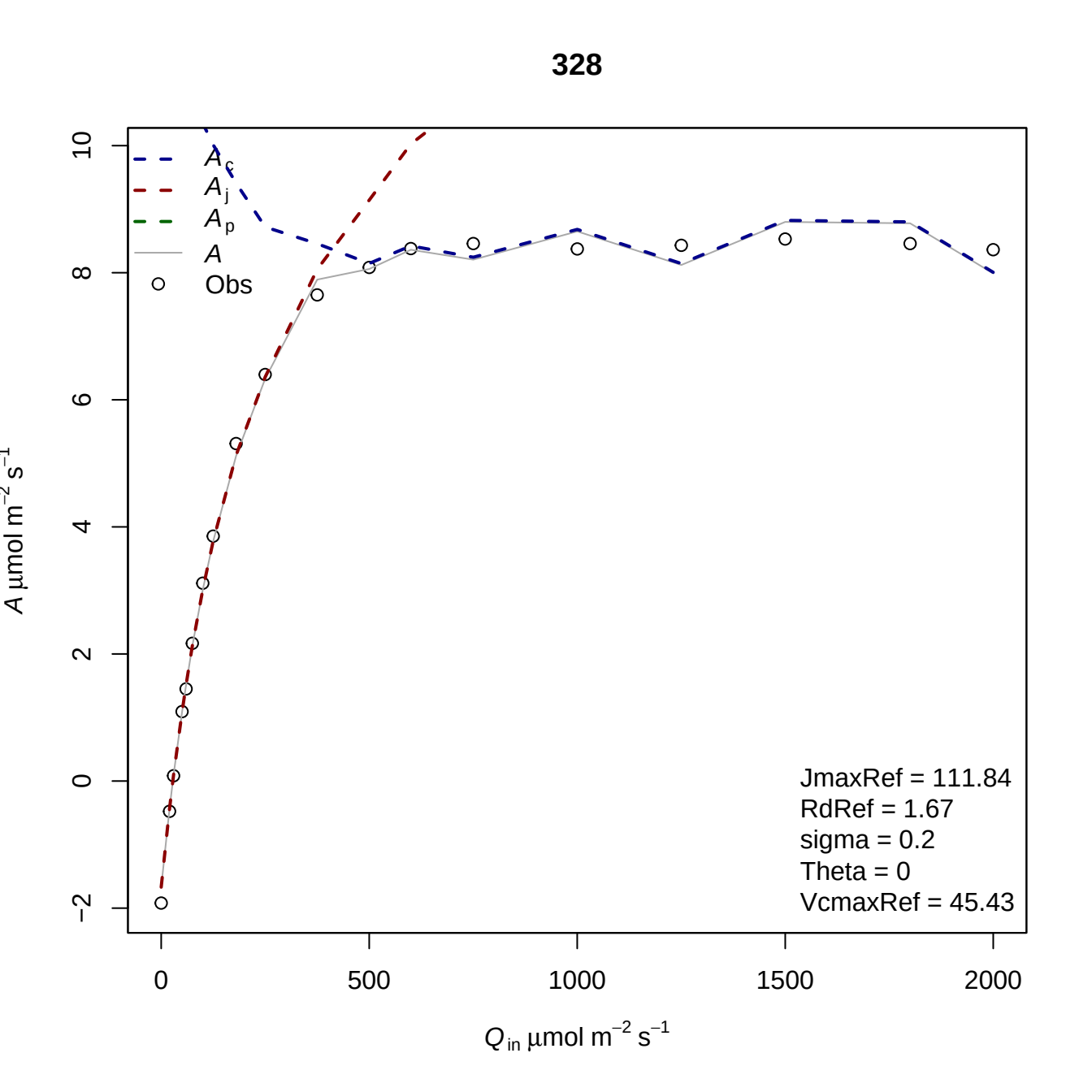

323
A_c
A_j
A_p
A
Obs
JmaxRef = 115.34
RdRef = 1.11
sigma = 0.23
Theta = 0
VcmaxRef = 41.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

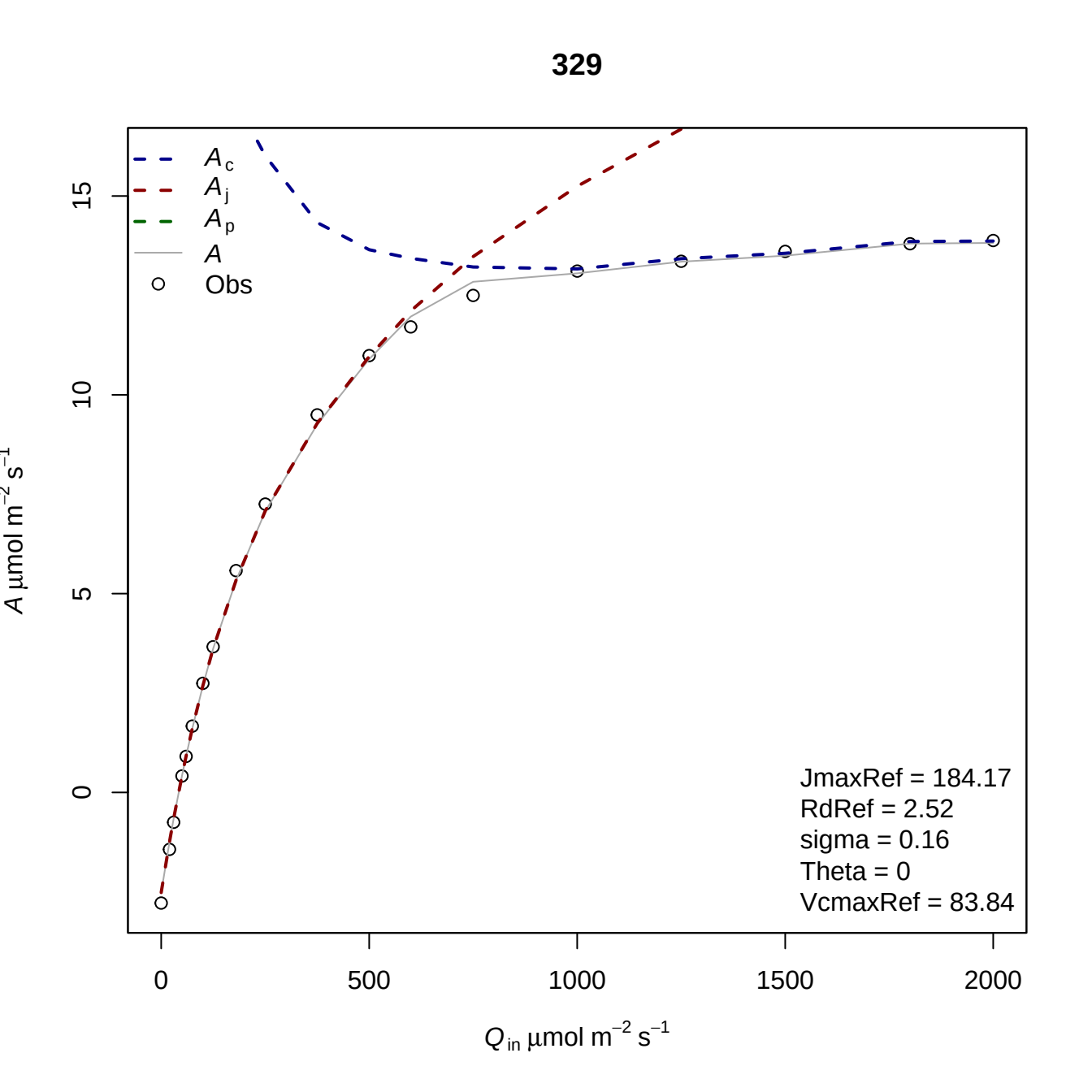

324
A_c
A_j
A_p
A
Obs
JmaxRef = 116.93
RdRef = 1.27
sigma = 0.21
Theta = 0
VcmaxRef = 44.4
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

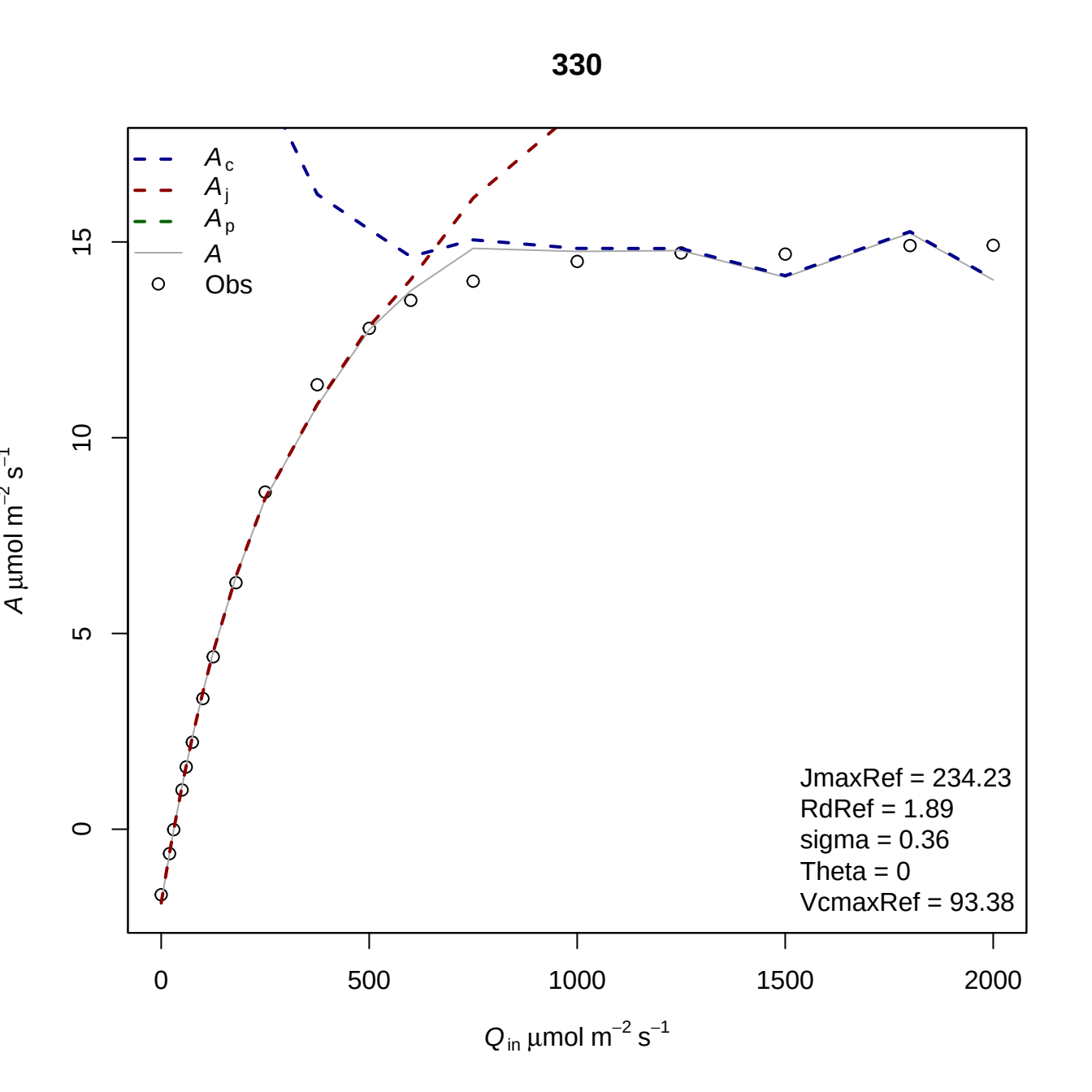

325
A_c
A_j
A_p
A
Obs
JmaxRef = 128.38
RdRef = 1.31
sigma = 0.19
Theta = 0
VcmaxRef = 43.15
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

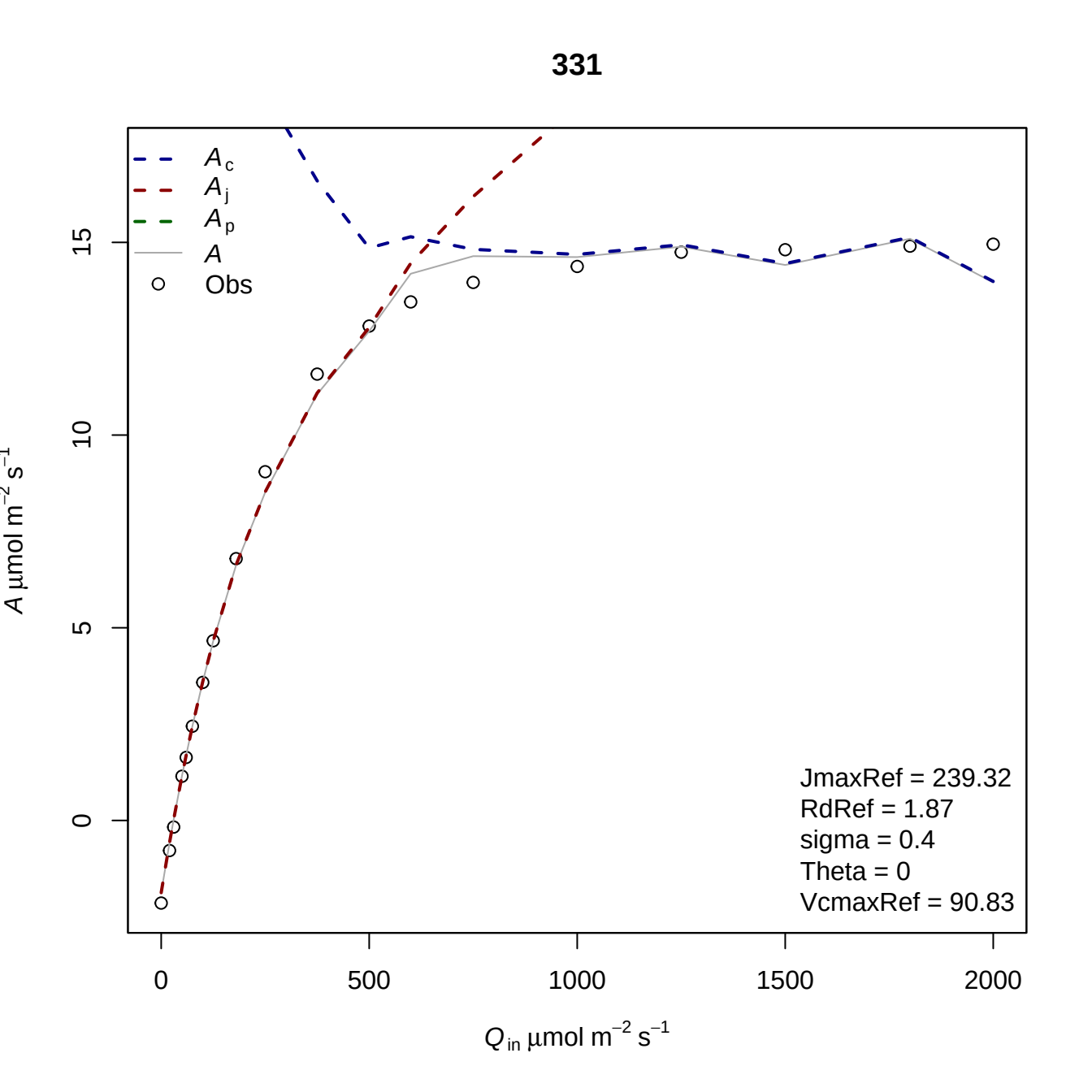

326
A_c
A_j
A_p
A
Obs
JmaxRef = 111.99
RdRef = 1.5
sigma = 0.46
Theta = 0
VcmaxRef = 40.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

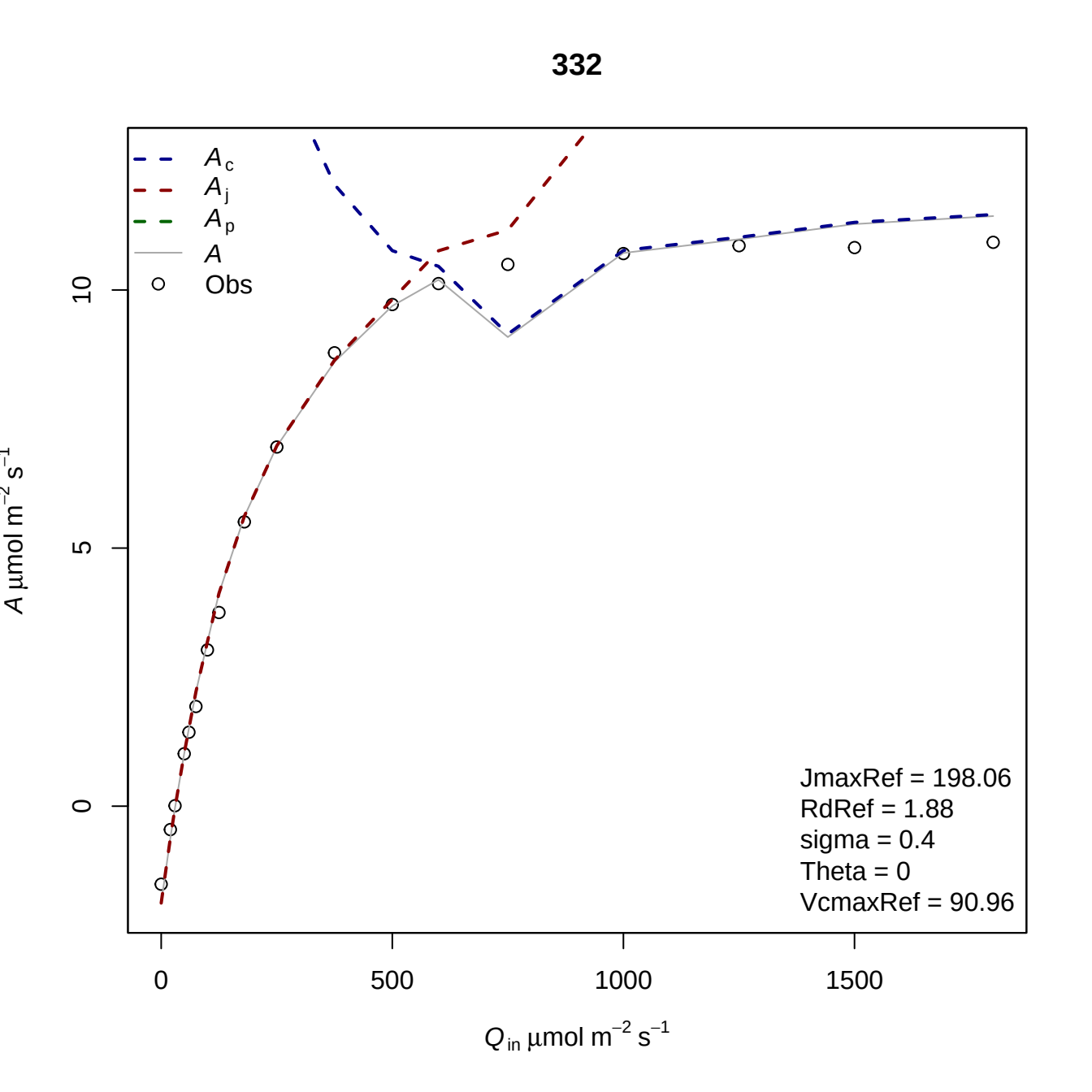

327
A_c
A_j
A_p
A
Obs
JmaxRef = 144.97
RdRef = 1.17
sigma = 0.26
Theta = 0
VcmaxRef = 51.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

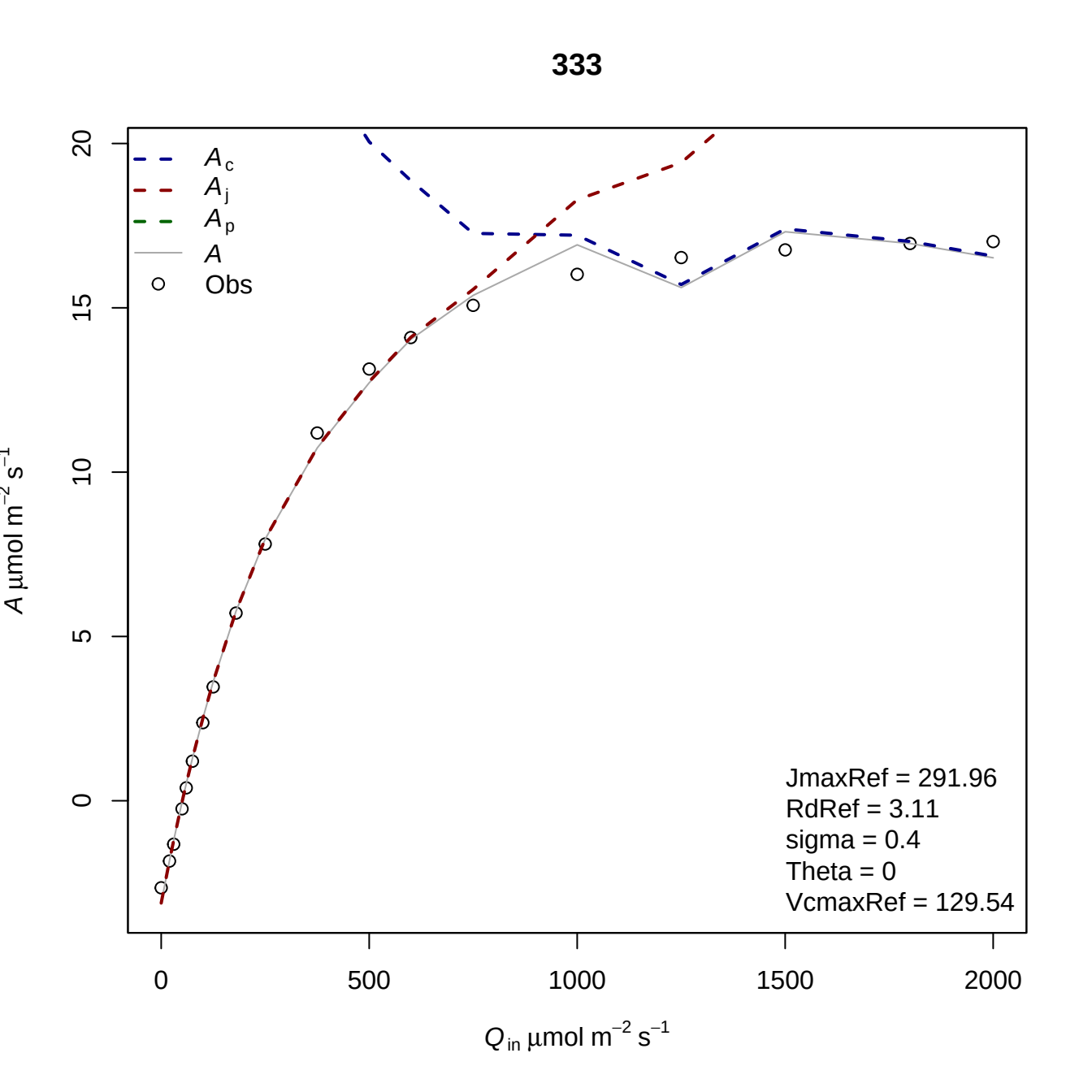

328
A_c
A_j
A_p
A
Obs
JmaxRef = 111.84
RdRef = 1.67
sigma = 0.2
Theta = 0
VcmaxRef = 45.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

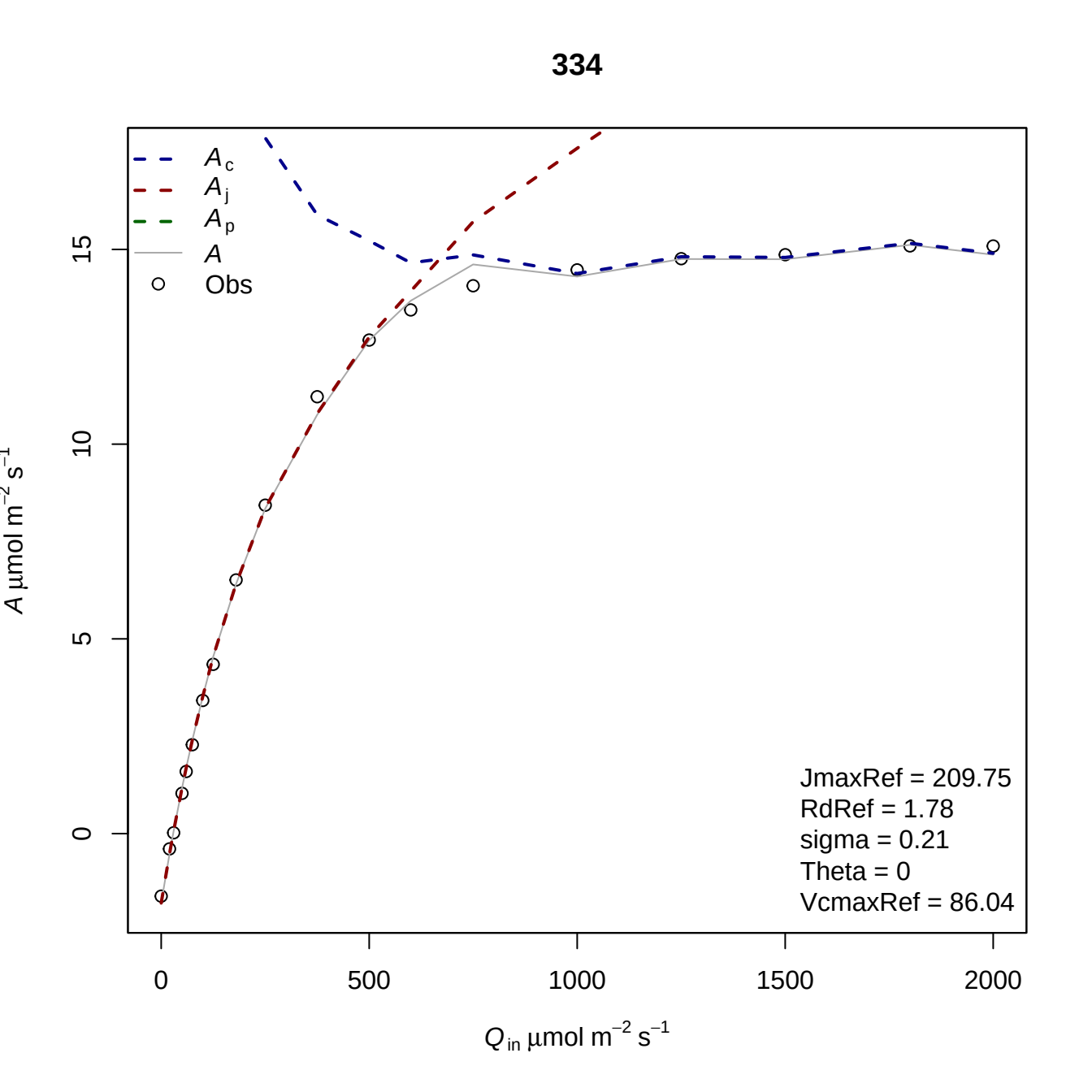

329
A_c
A_j
A_p
A
Obs
JmaxRef = 184.17
RdRef = 2.52
sigma = 0.16
Theta = 0
VcmaxRef = 83.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

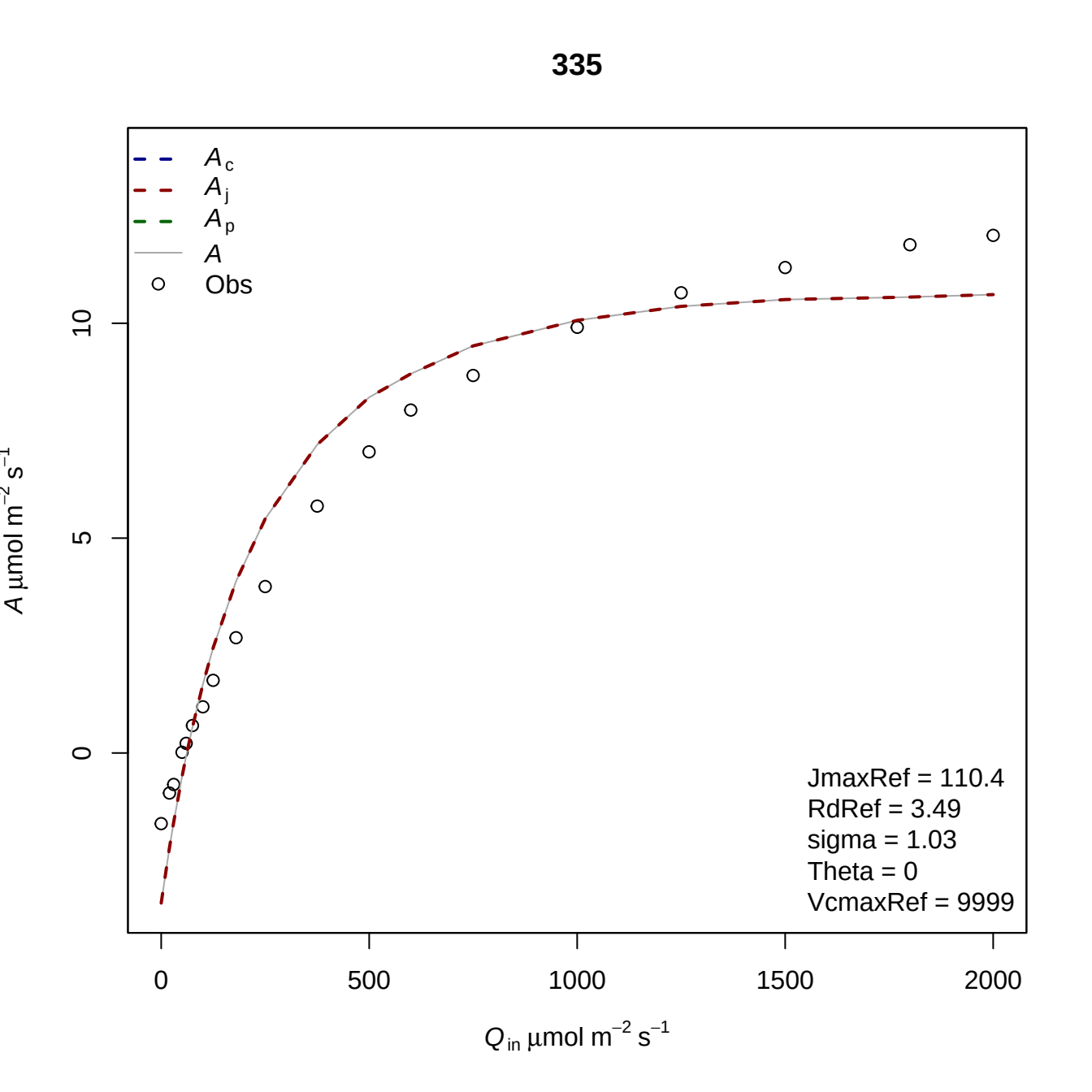

330
A_c
A_j
A_p
A
Obs
JmaxRef = 234.23
RdRef = 1.89
sigma = 0.36
Theta = 0
VcmaxRef = 93.38
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

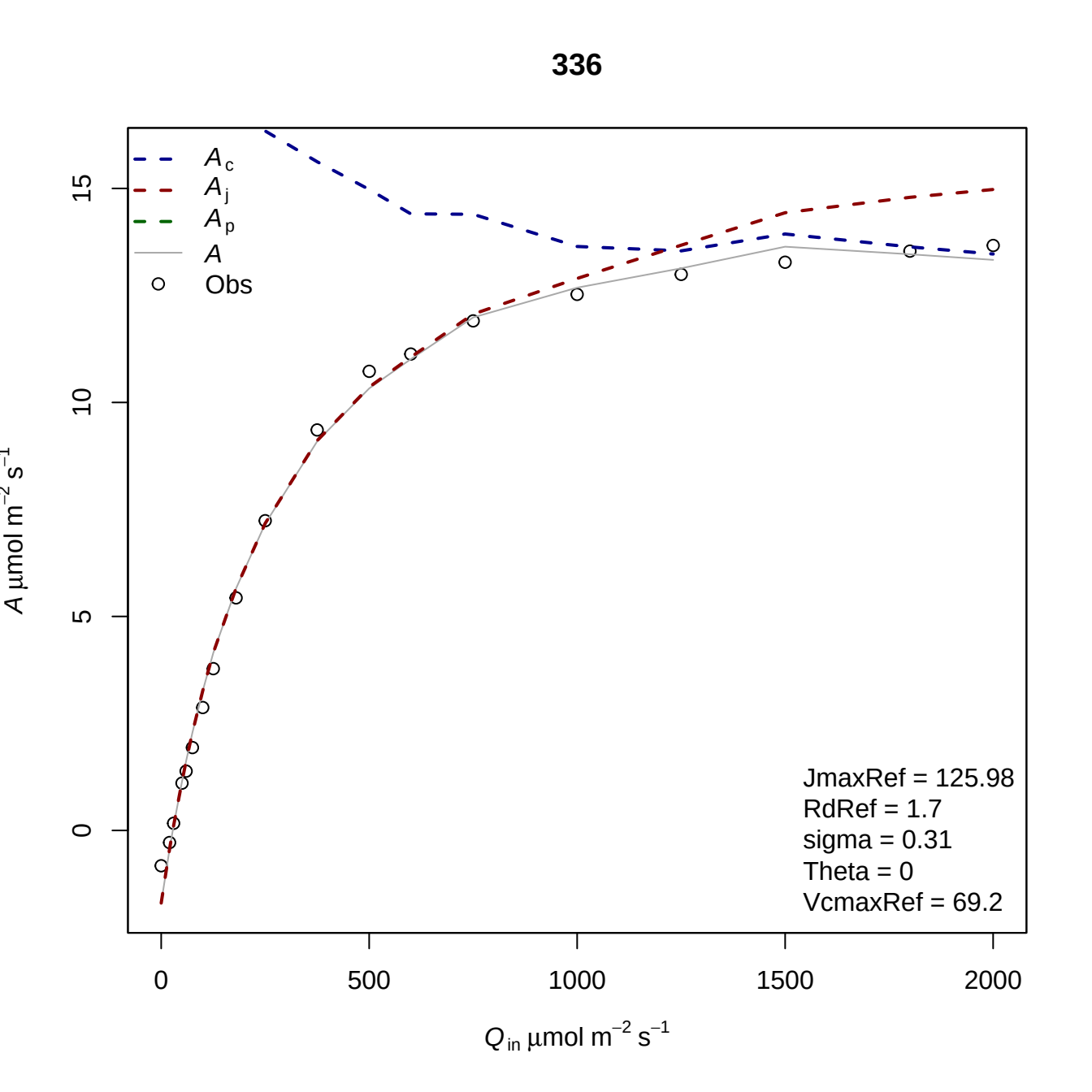

331
A_c
A_j
A_p
A
Obs
JmaxRef = 239.32
RdRef = 1.87
sigma = 0.4
Theta = 0
VcmaxRef = 90.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

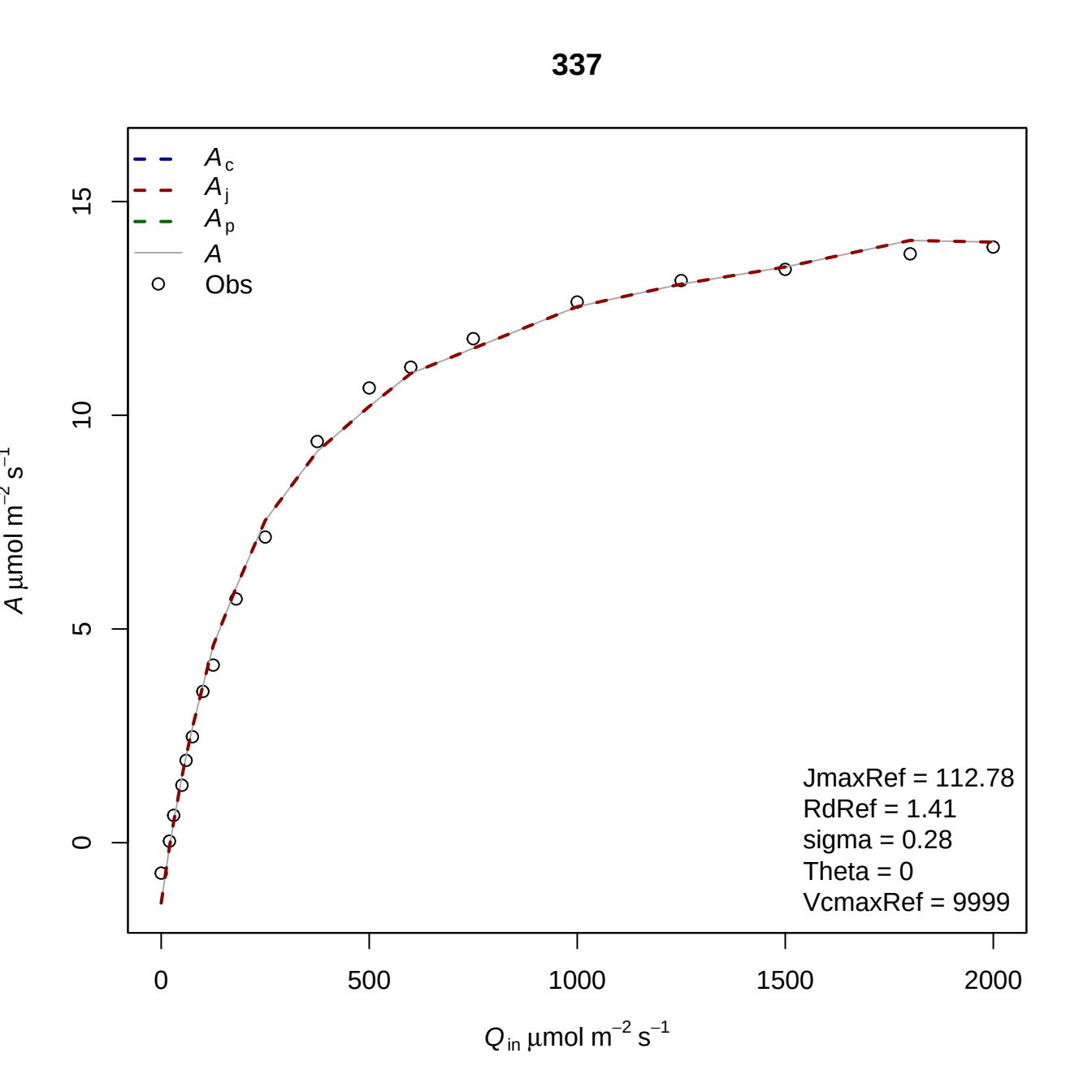

332
A_c
A_j
A_p
A
Obs
JmaxRef = 198.06
RdRef = 1.88
sigma = 0.4
Theta = 0
VcmaxRef = 90.96
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

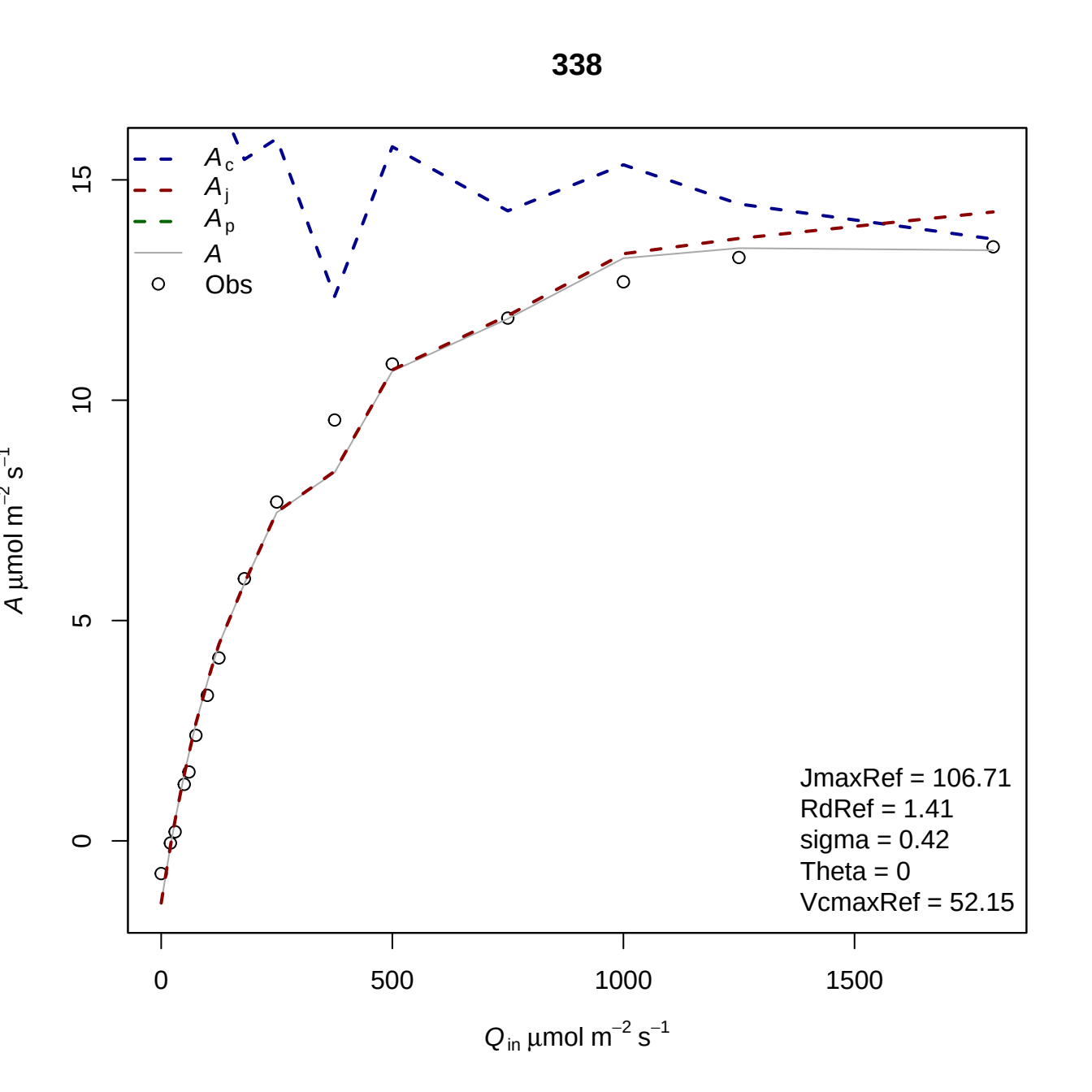

333
A_c
A_j
A_p
A
Obs
JmaxRef = 291.96
RdRef = 3.11
sigma = 0.4
Theta = 0
VcmaxRef = 129.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

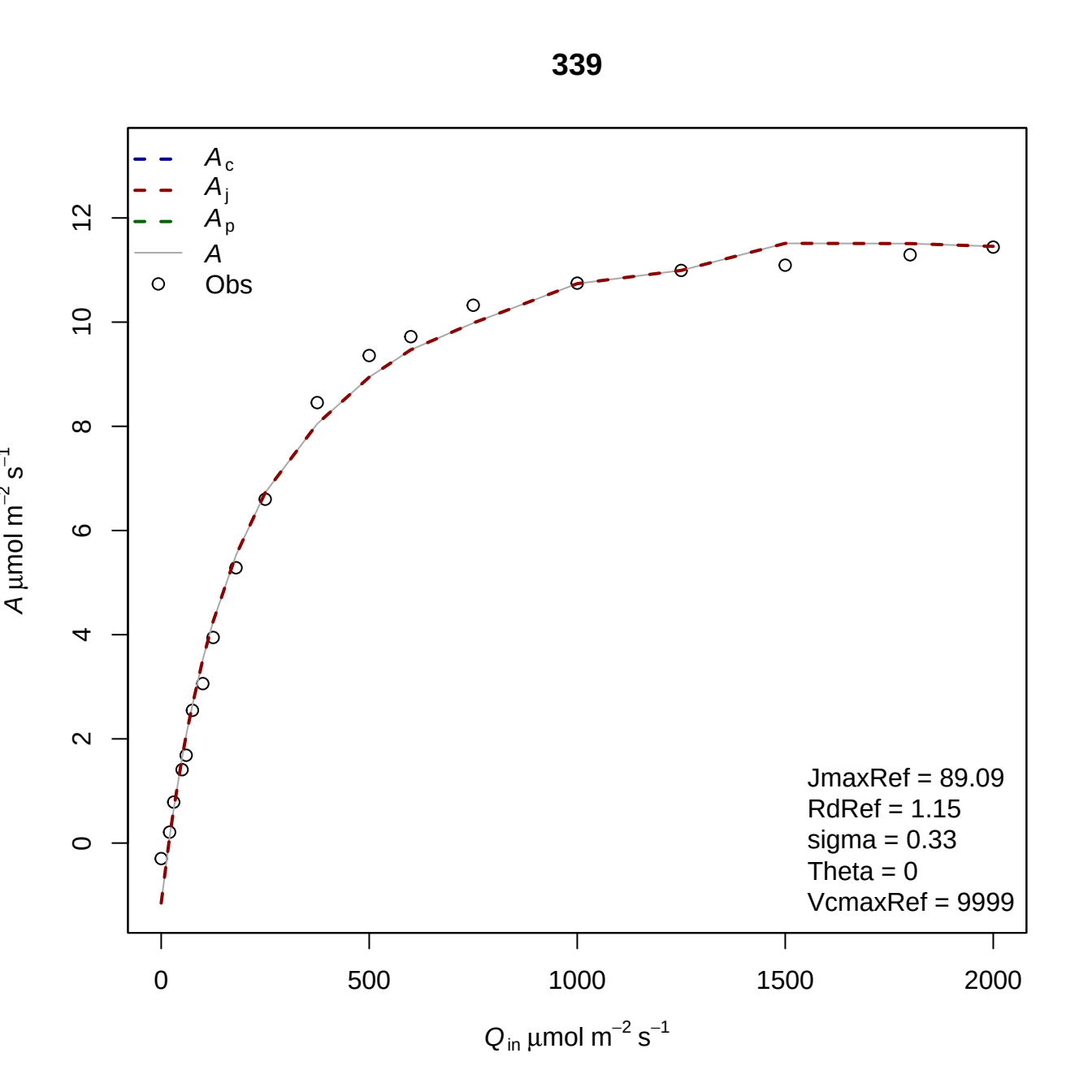

334
A_c
A_j
A_p
A
Obs
JmaxRef = 209.75
RdRef = 1.78
sigma = 0.21
Theta = 0
VcmaxRef = 86.04
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

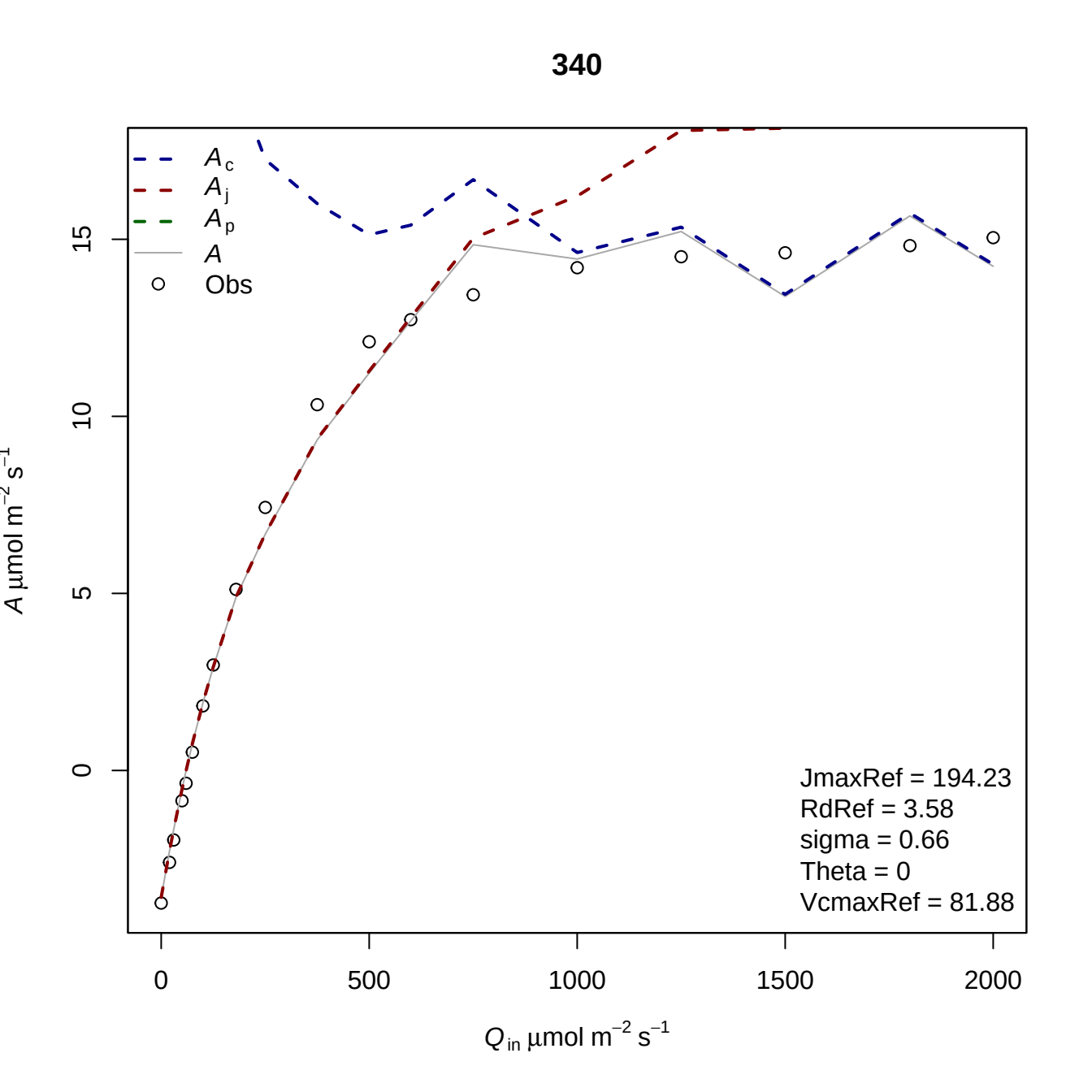

335
A_c
A_j
A_p
A
Obs
JmaxRef = 110.4
RdRef = 3.49
sigma = 1.03
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

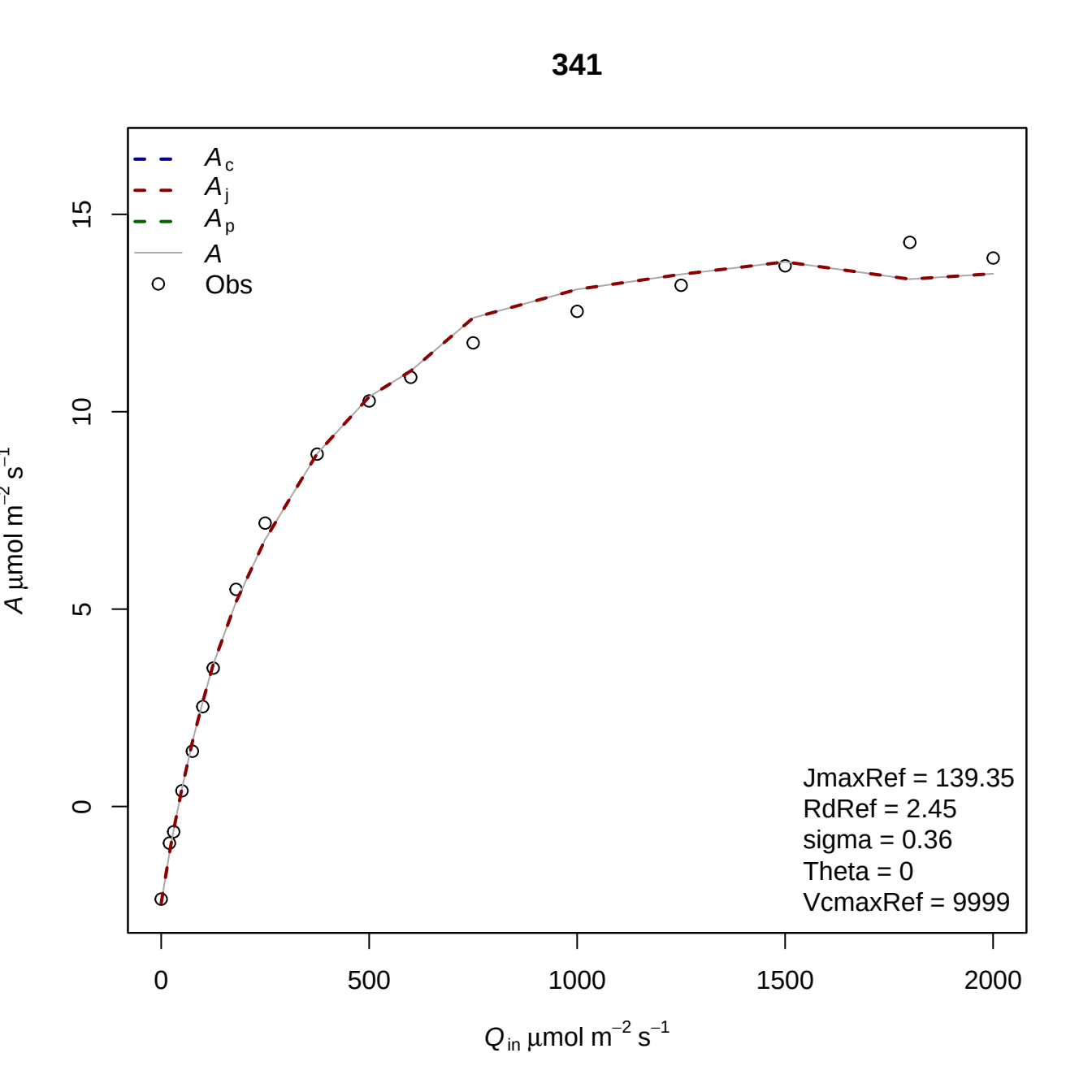

336
A_c
A_j
A_p
A
Obs
JmaxRef = 125.98
RdRef = 1.7
sigma = 0.31
Theta = 0
VcmaxRef = 69.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

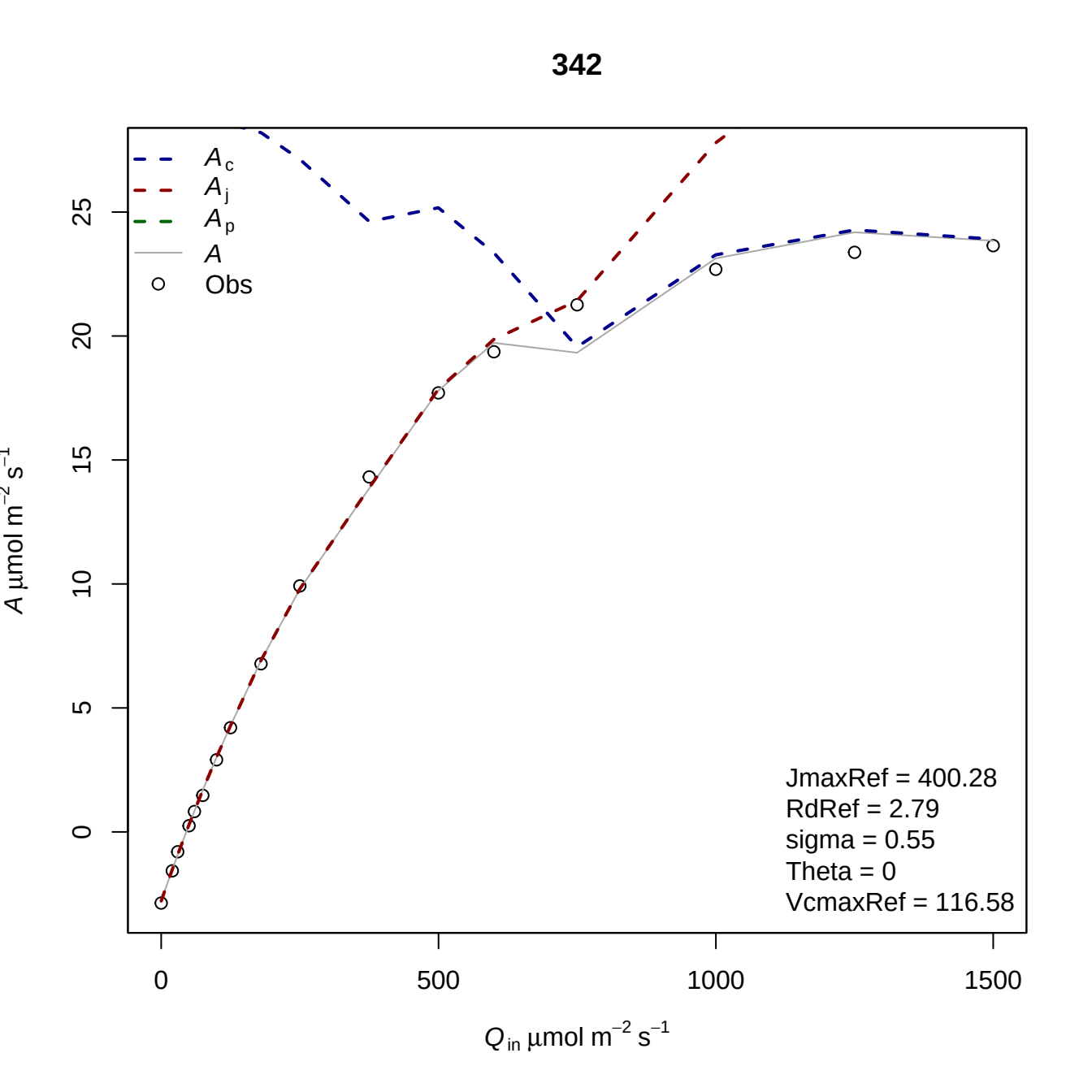

337
A_c
A_j
A_p
A
Obs
JmaxRef = 112.78
RdRef = 1.41
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

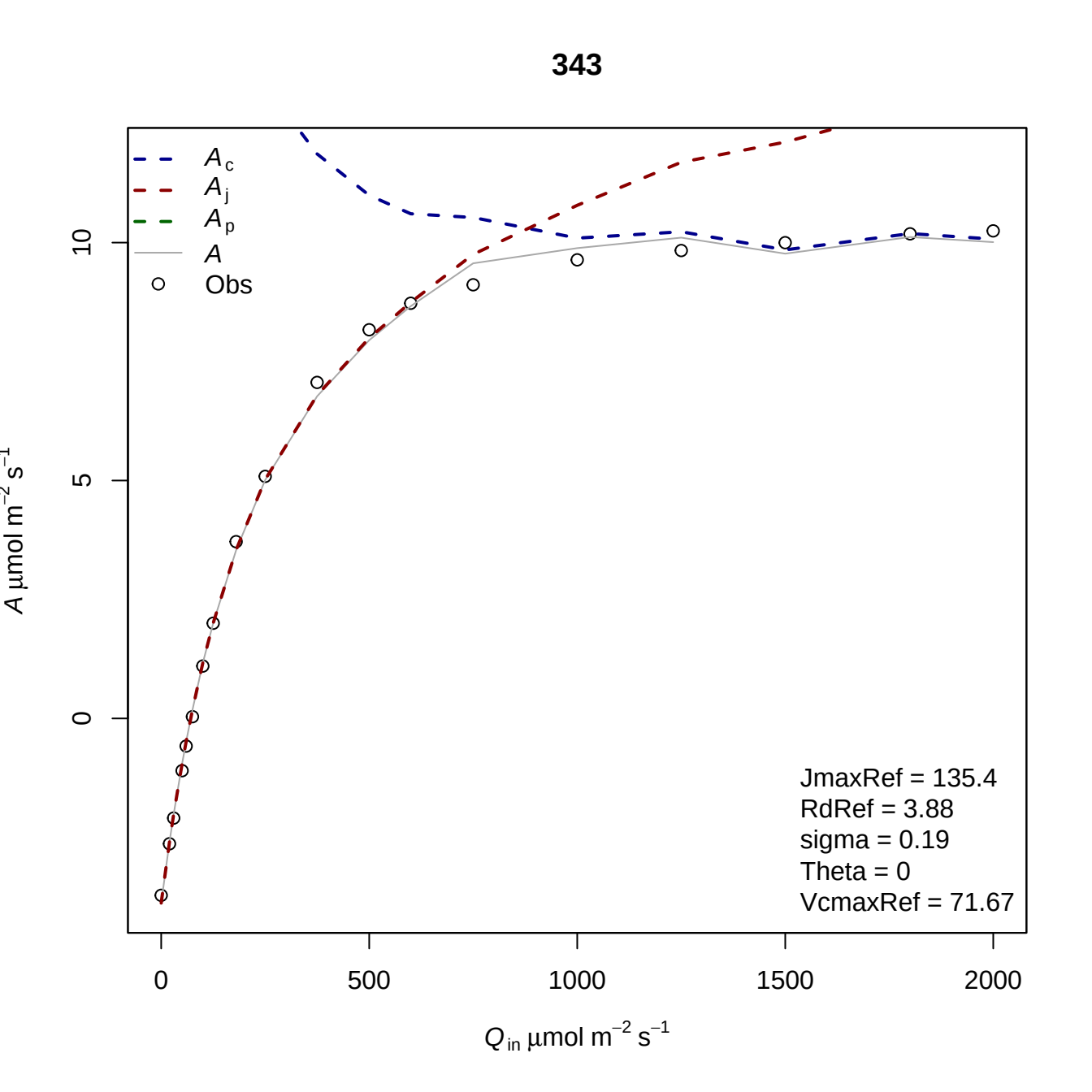

338
A_c
A_j
A_p
A
Obs
JmaxRef = 106.71
RdRef = 1.41
sigma = 0.42
Theta = 0
VcmaxRef = 52.15
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

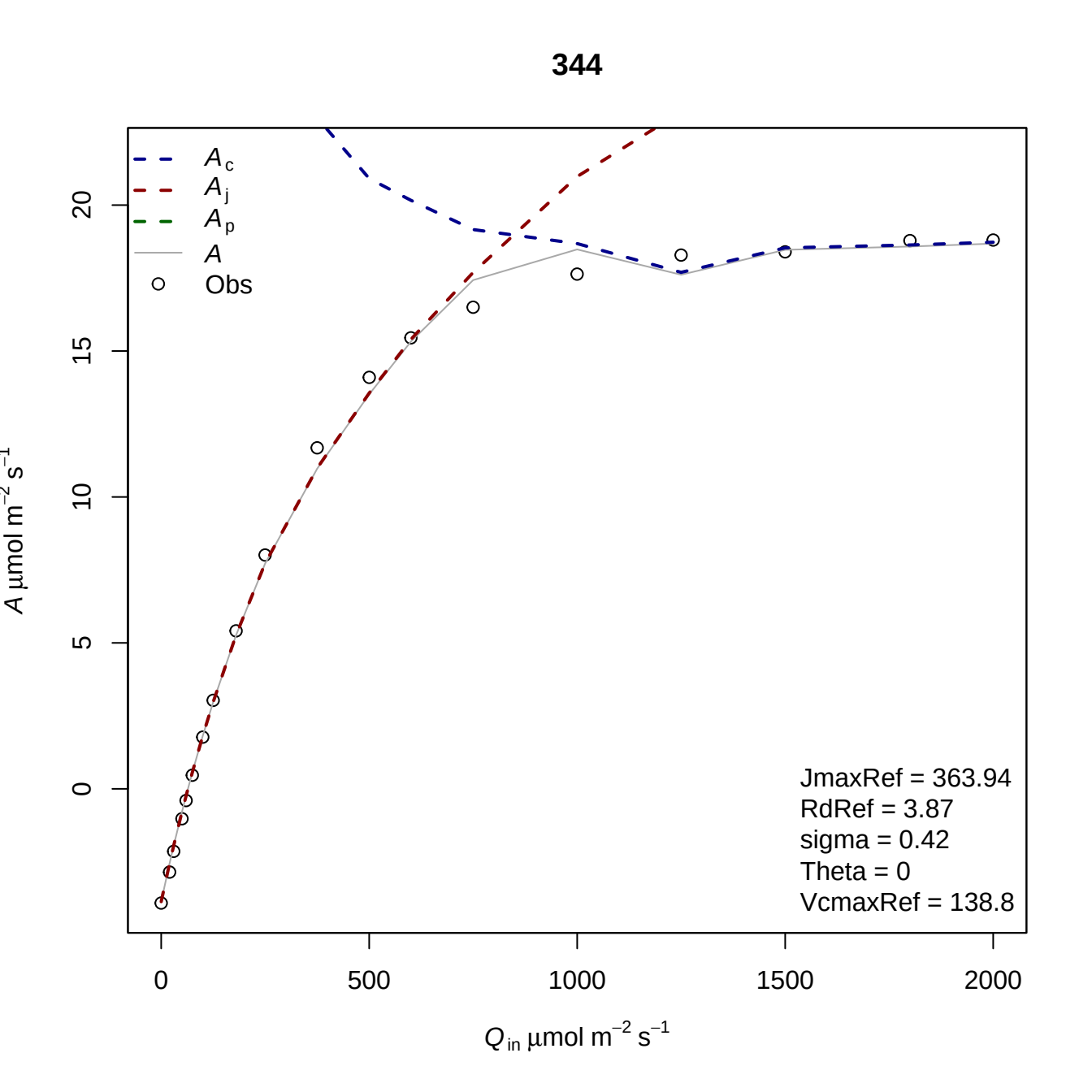

339
A_c
A_j
A_p
A
Obs
JmaxRef = 89.09
RdRef = 1.15
sigma = 0.33
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

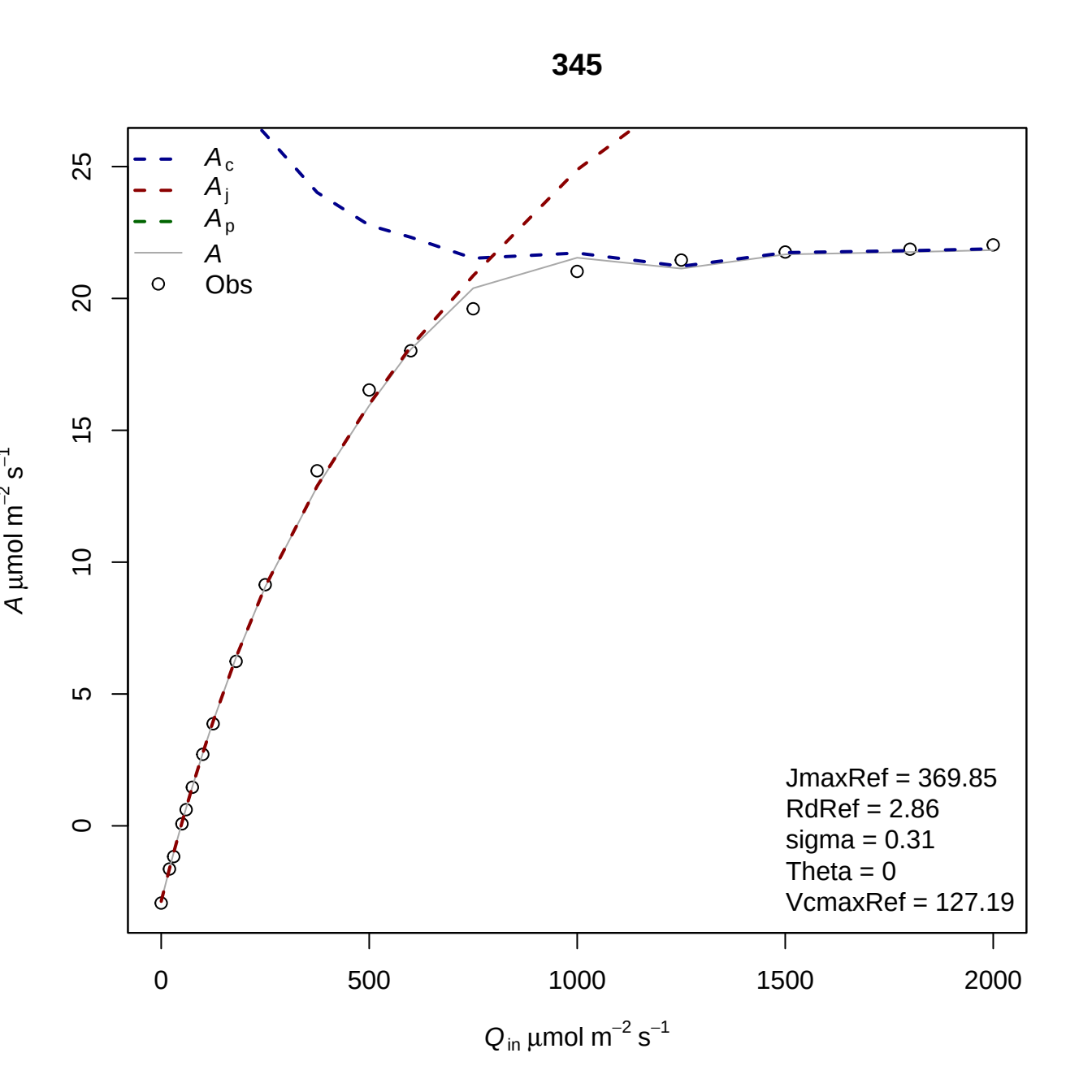

340
A_c
A_j
A_p
A
Obs
JmaxRef = 194.23
RdRef = 3.58
sigma = 0.66
Theta = 0
VcmaxRef = 81.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

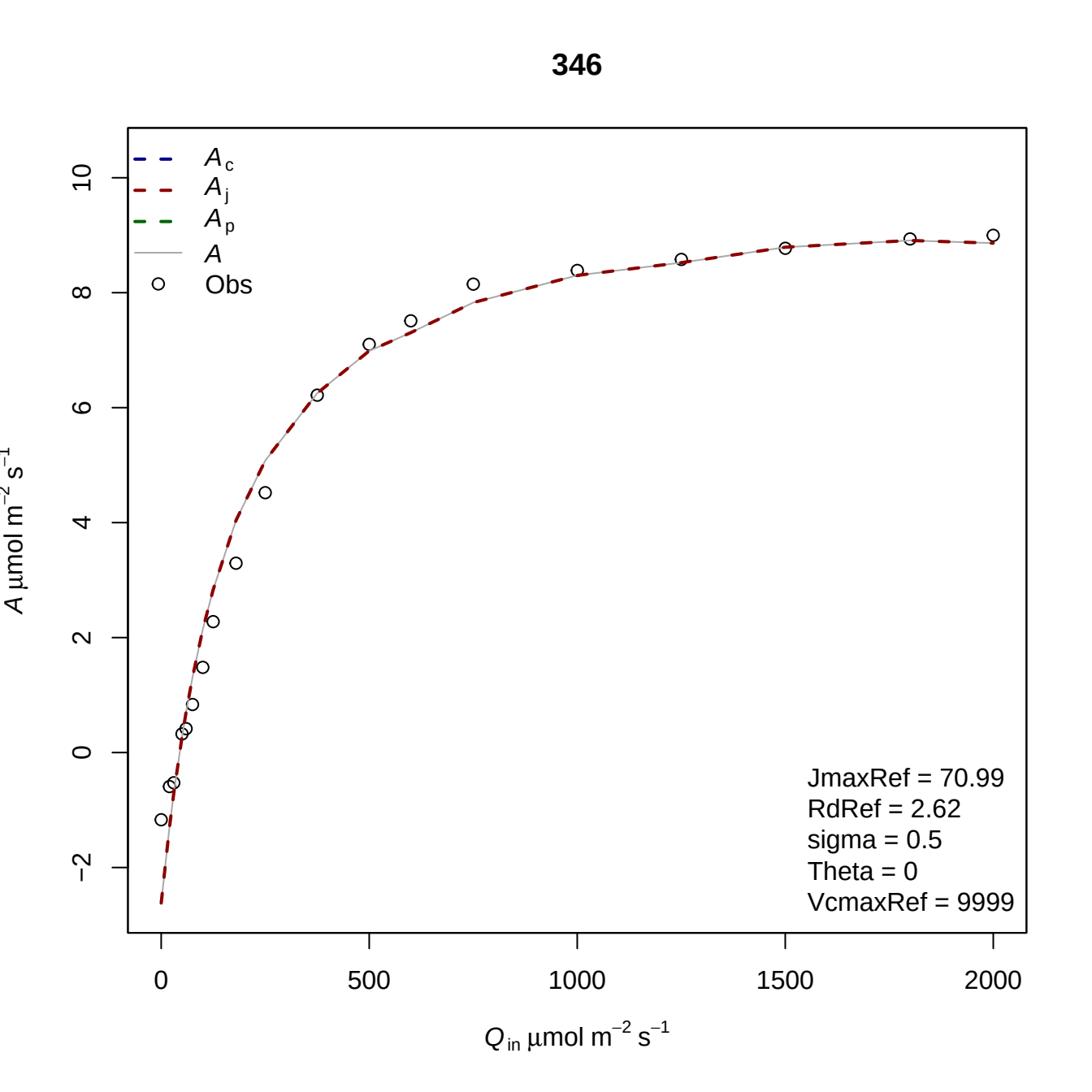

341
A_c
A_j
A_p
A
Obs
JmaxRef = 139.35
RdRef = 2.45
sigma = 0.36
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

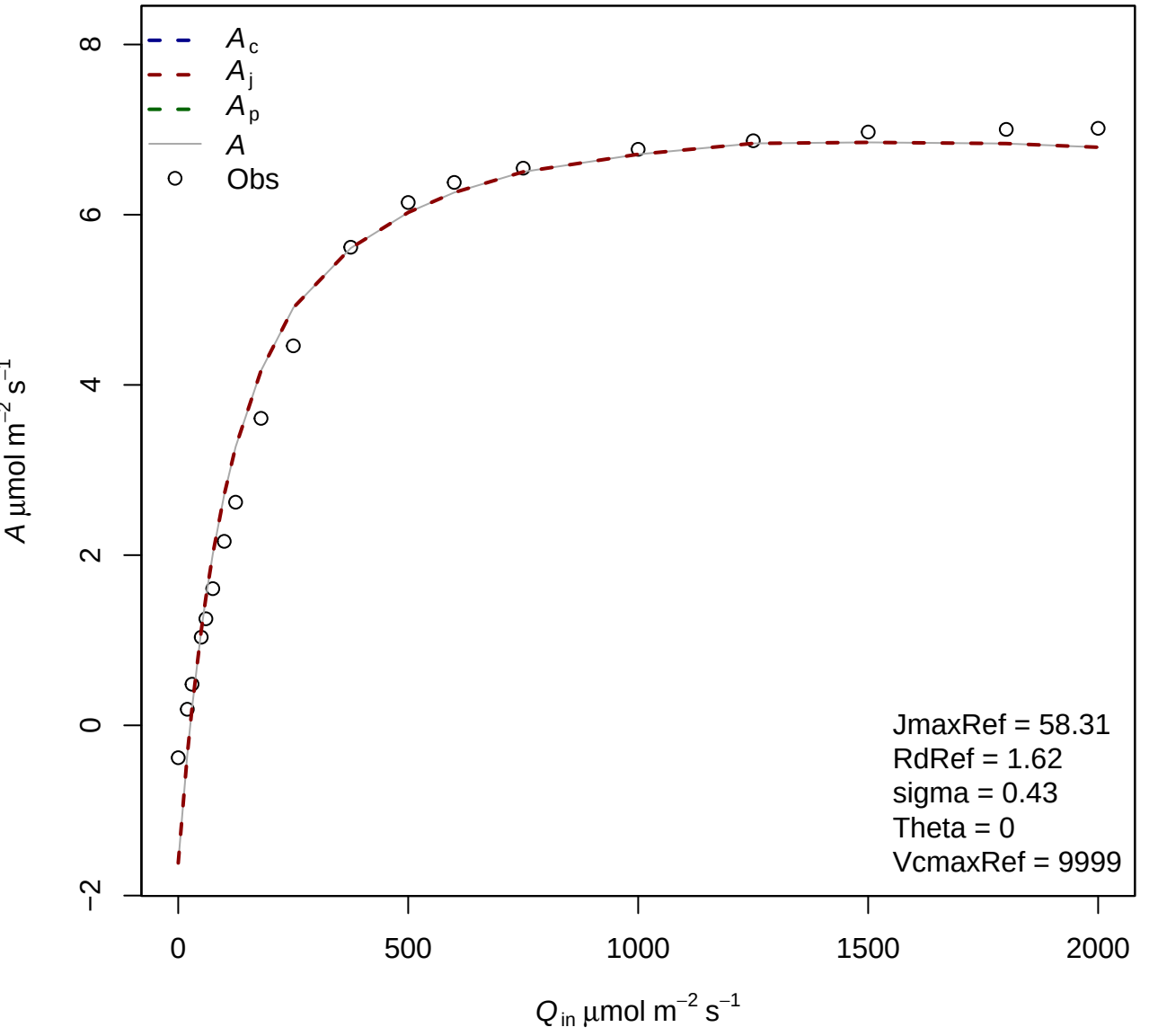

342
A_c
A_j
A_p
A
Obs
JmaxRef = 400.28
RdRef = 2.79
sigma = 0.55
Theta = 0
VcmaxRef = 116.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

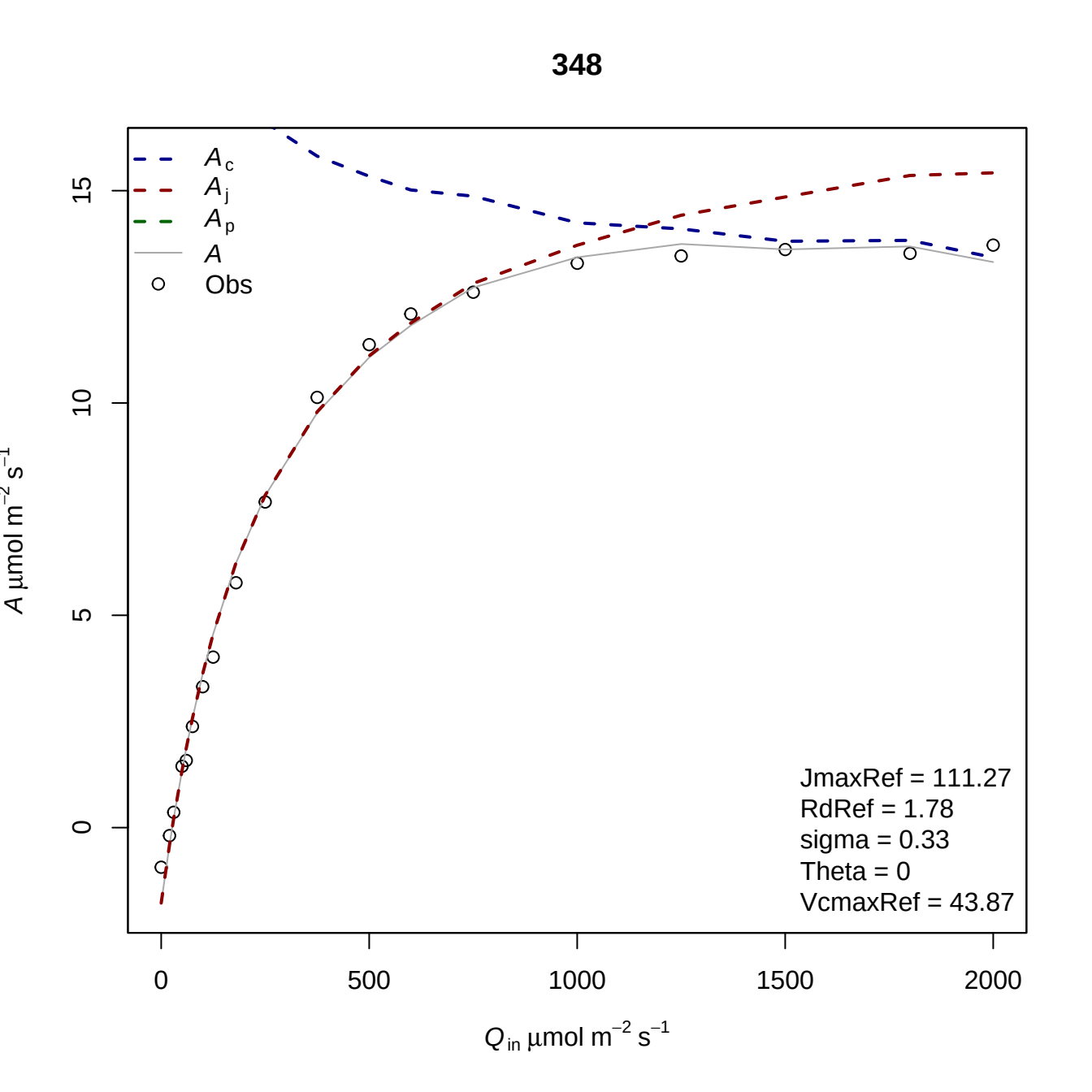

343
A_c
A_j
A_p
A
Obs
JmaxRef = 135.4
RdRef = 3.88
sigma = 0.19
Theta = 0
VcmaxRef = 71.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

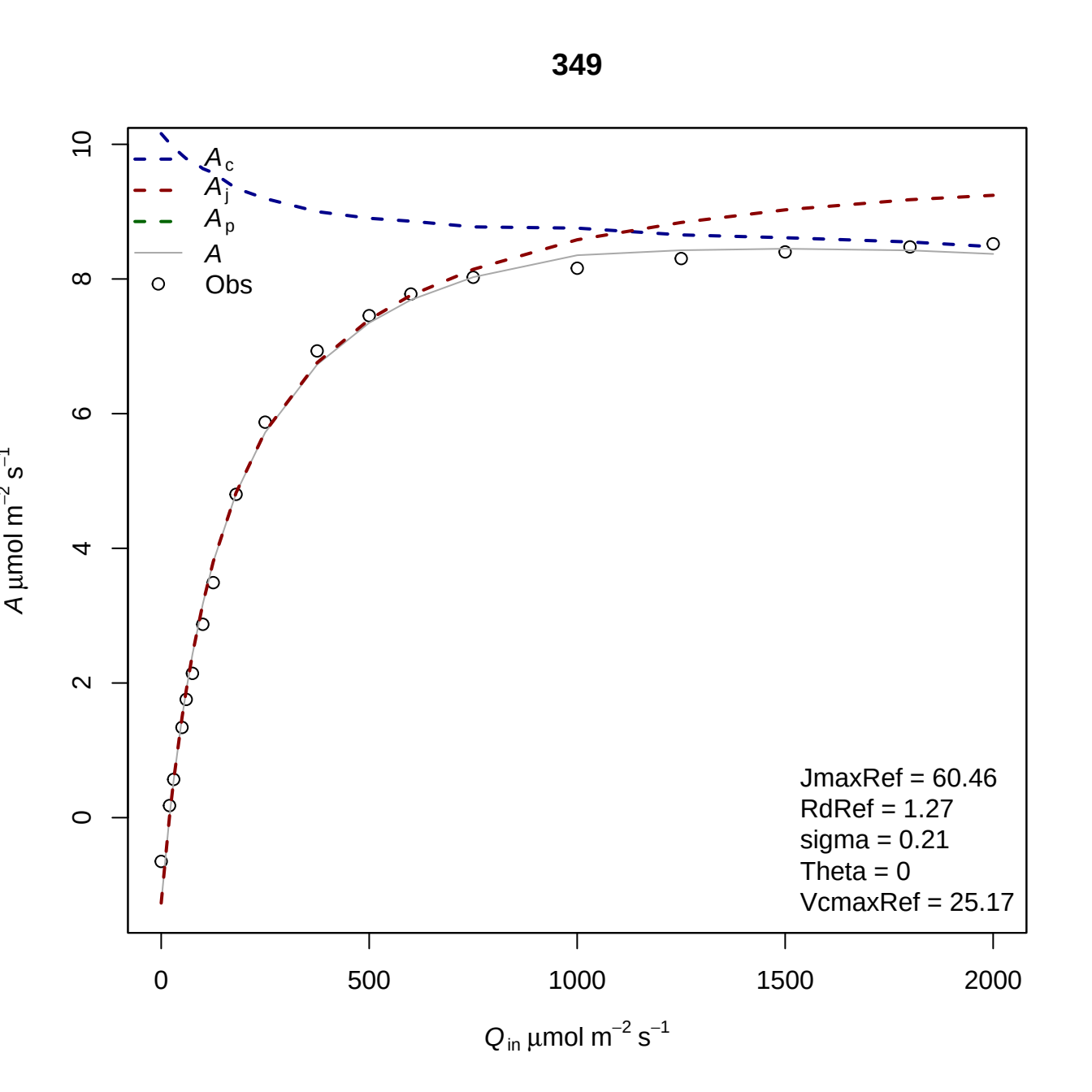

344
A_c
A_j
A_p
A
Obs
JmaxRef = 363.94
RdRef = 3.87
sigma = 0.42
Theta = 0
VcmaxRef = 138.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

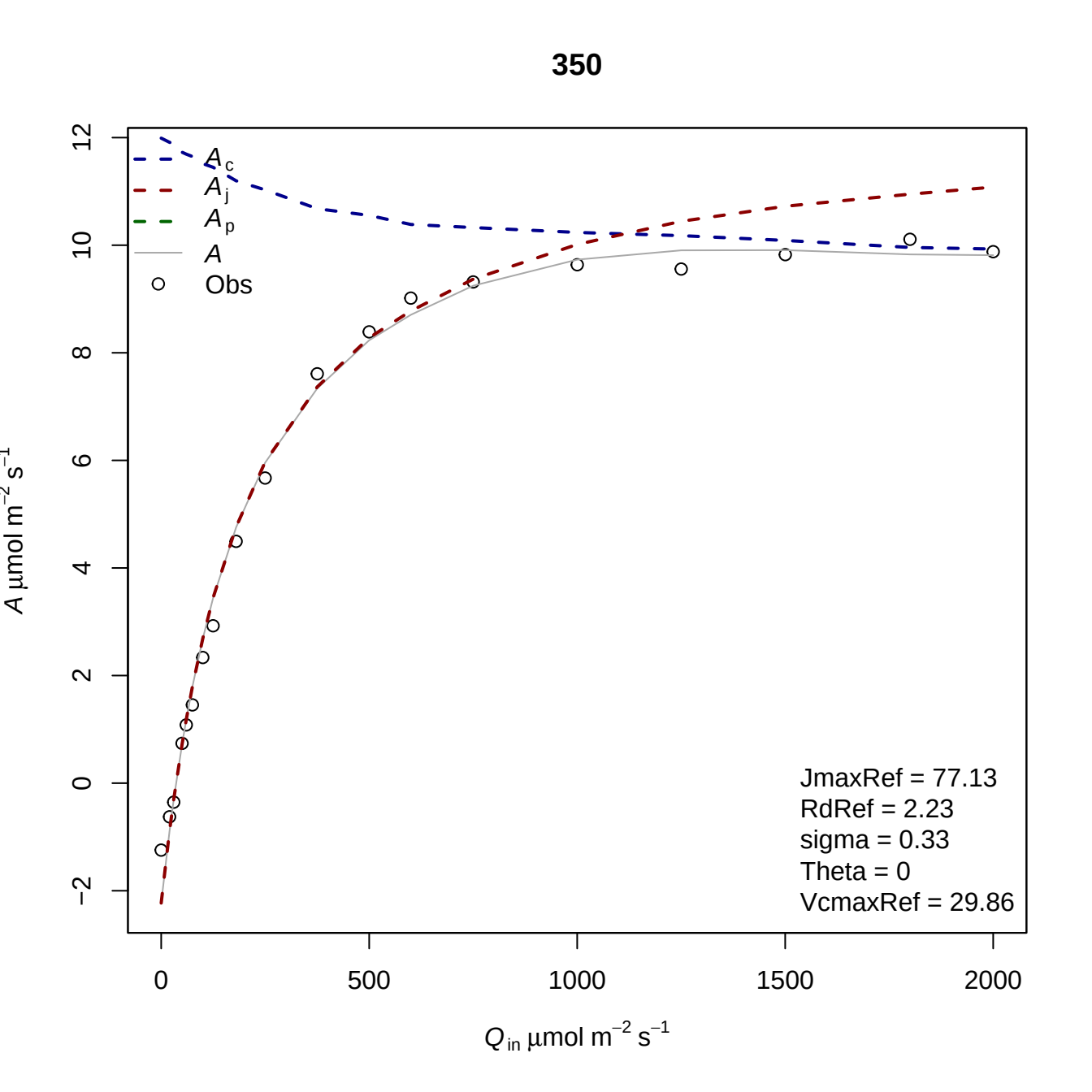

345
A_c
A_j
A_p
A
Obs
JmaxRef = 369.85
RdRef = 2.86
sigma = 0.31
Theta = 0
VcmaxRef = 127.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

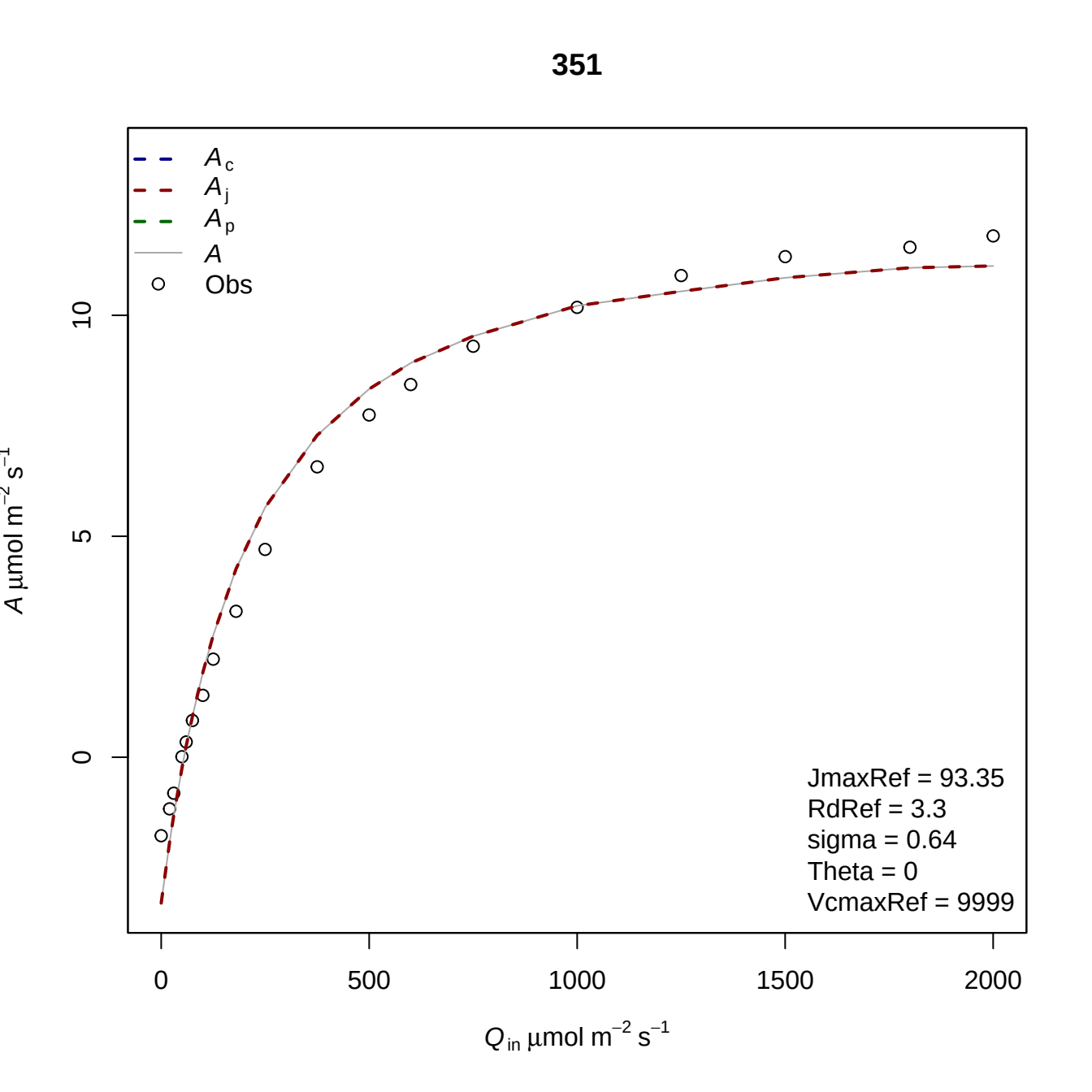

346
A_c
A_j
A_p
A
Obs
JmaxRef = 70.99
RdRef = 2.62
sigma = 0.5
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

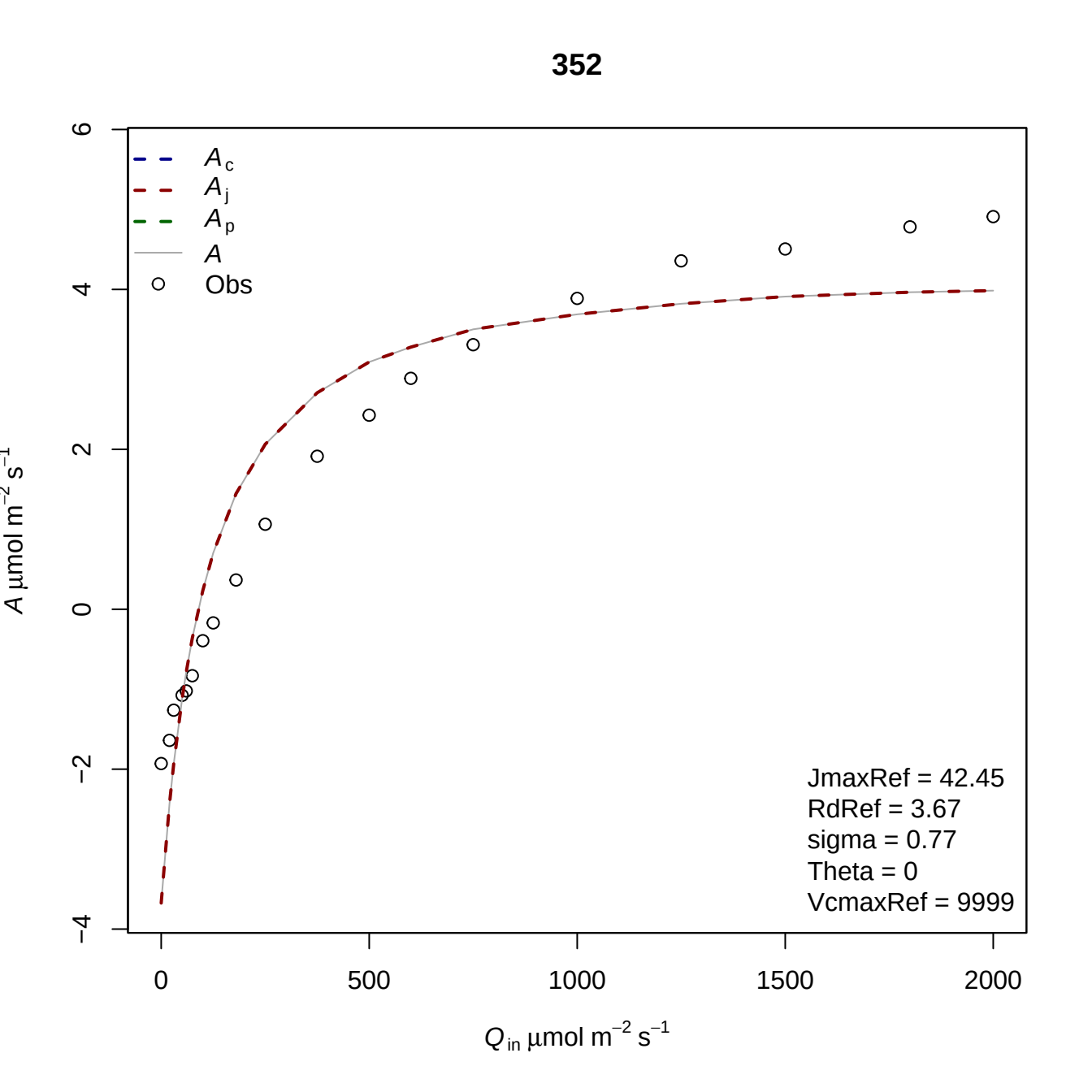

A_c
A_j
A_p
A
Obs
JmaxRef = 58.31
RdRef = 1.62
sigma = 0.43
Theta = 0
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

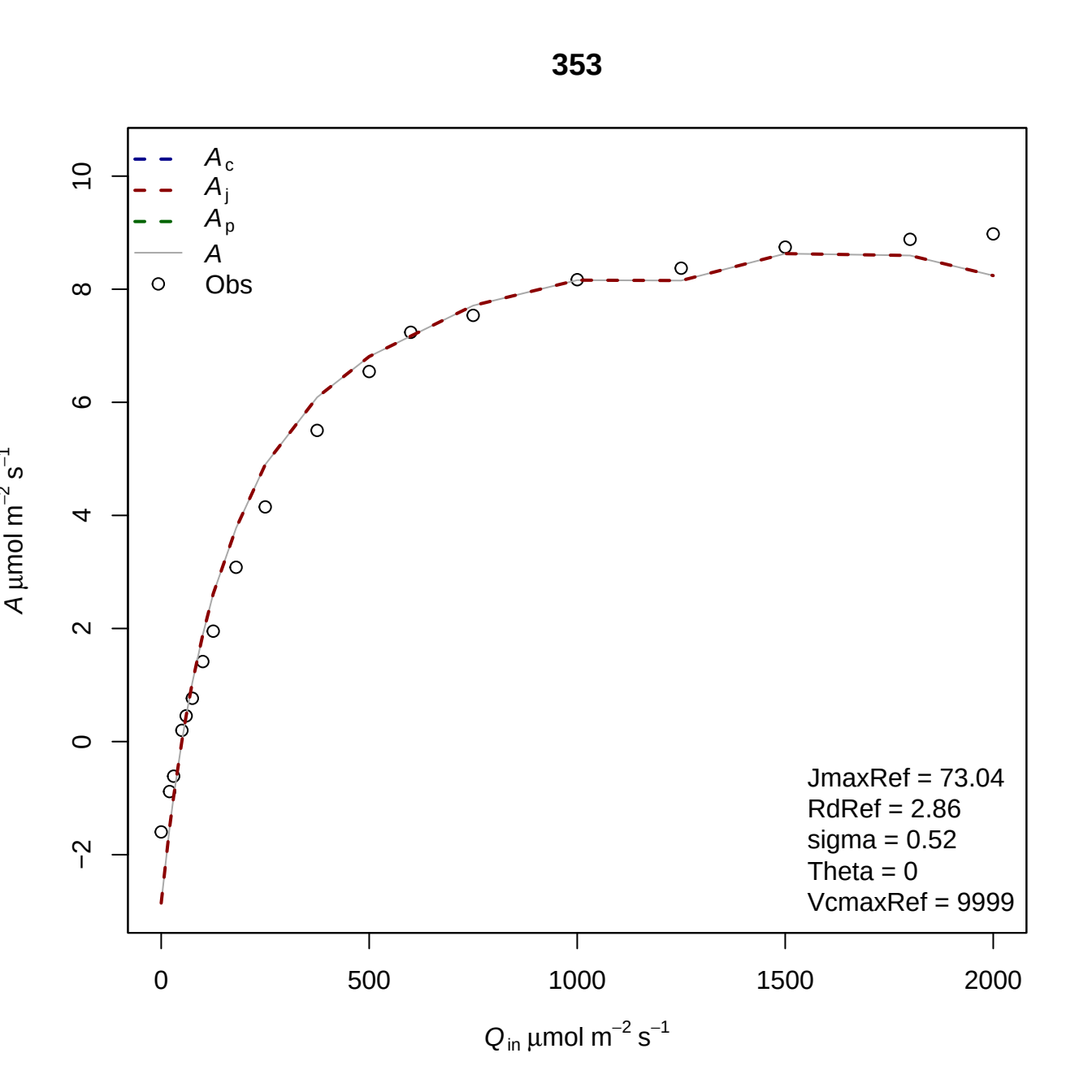

348
A_c
A_j
A_p
A
Obs
JmaxRef = 111.27
RdRef = 1.78
sigma = 0.33
Theta = 0
VcmaxRef = 43.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

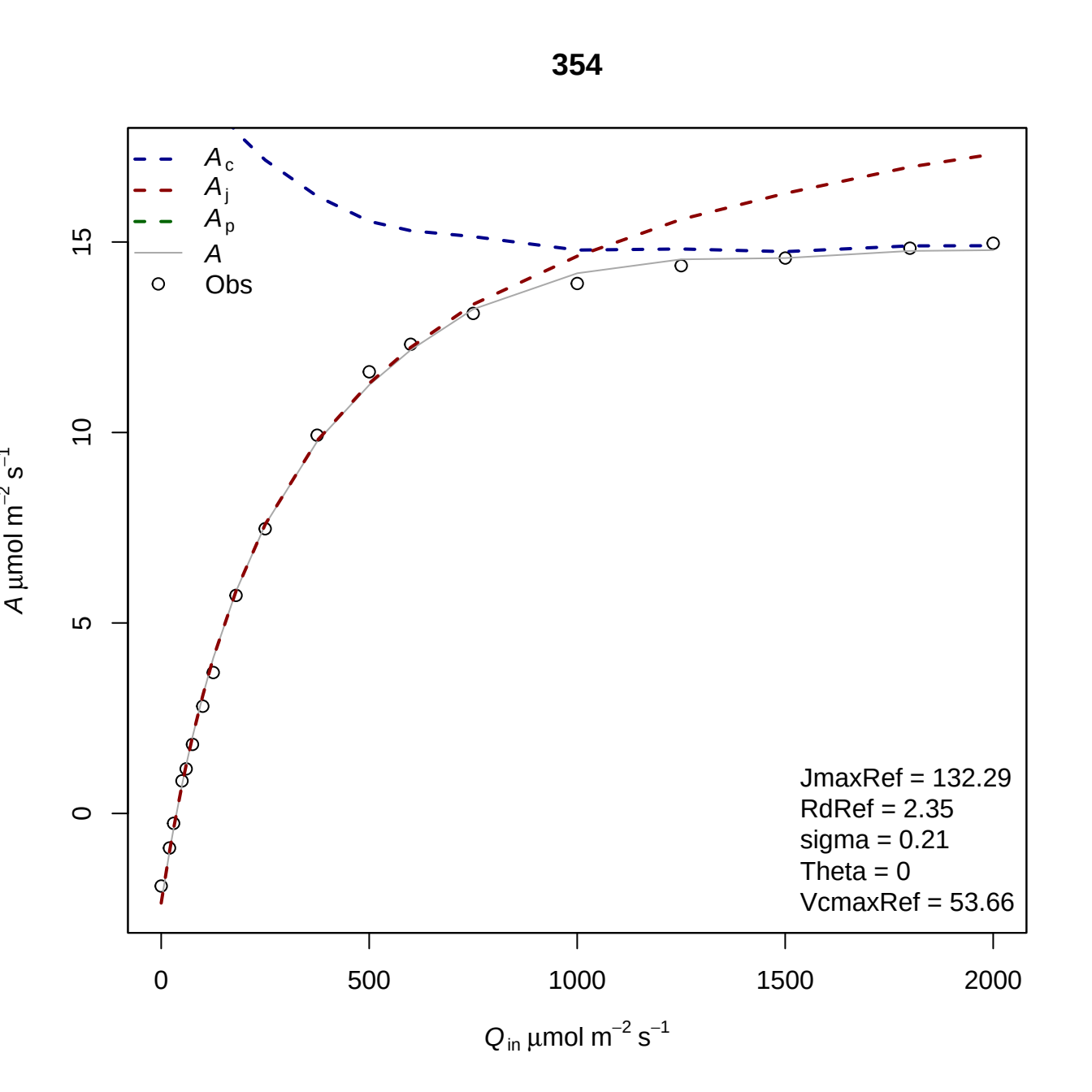

349
A_c
A_j
A_p
A
Obs
JmaxRef = 60.46
RdRef = 1.27
sigma = 0.21
Theta = 0
VcmaxRef = 25.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

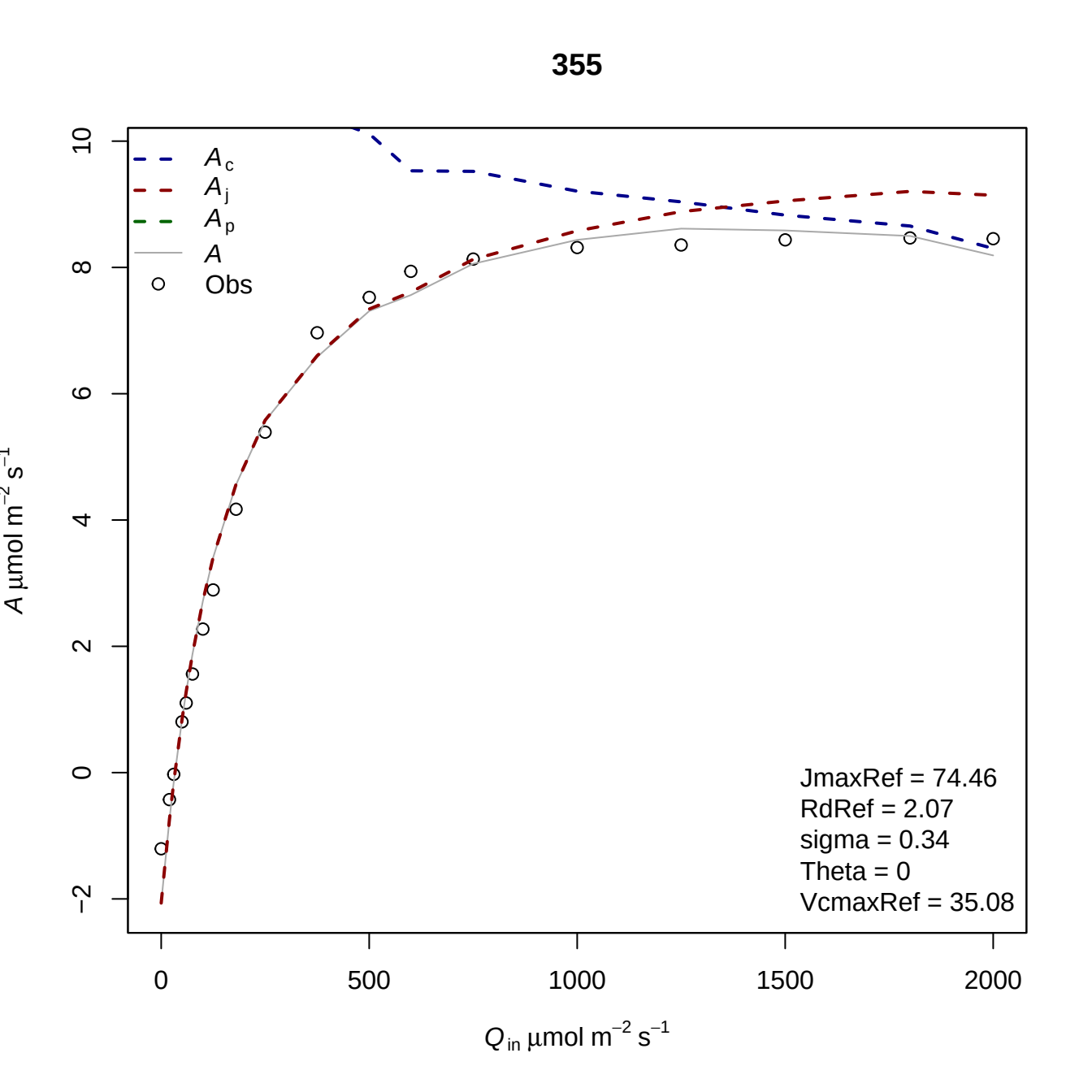

350
A_c
A_j
A_p
A
Obs
JmaxRef = 77.13
RdRef = 2.23
sigma = 0.33
Theta = 0
VcmaxRef = 29.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

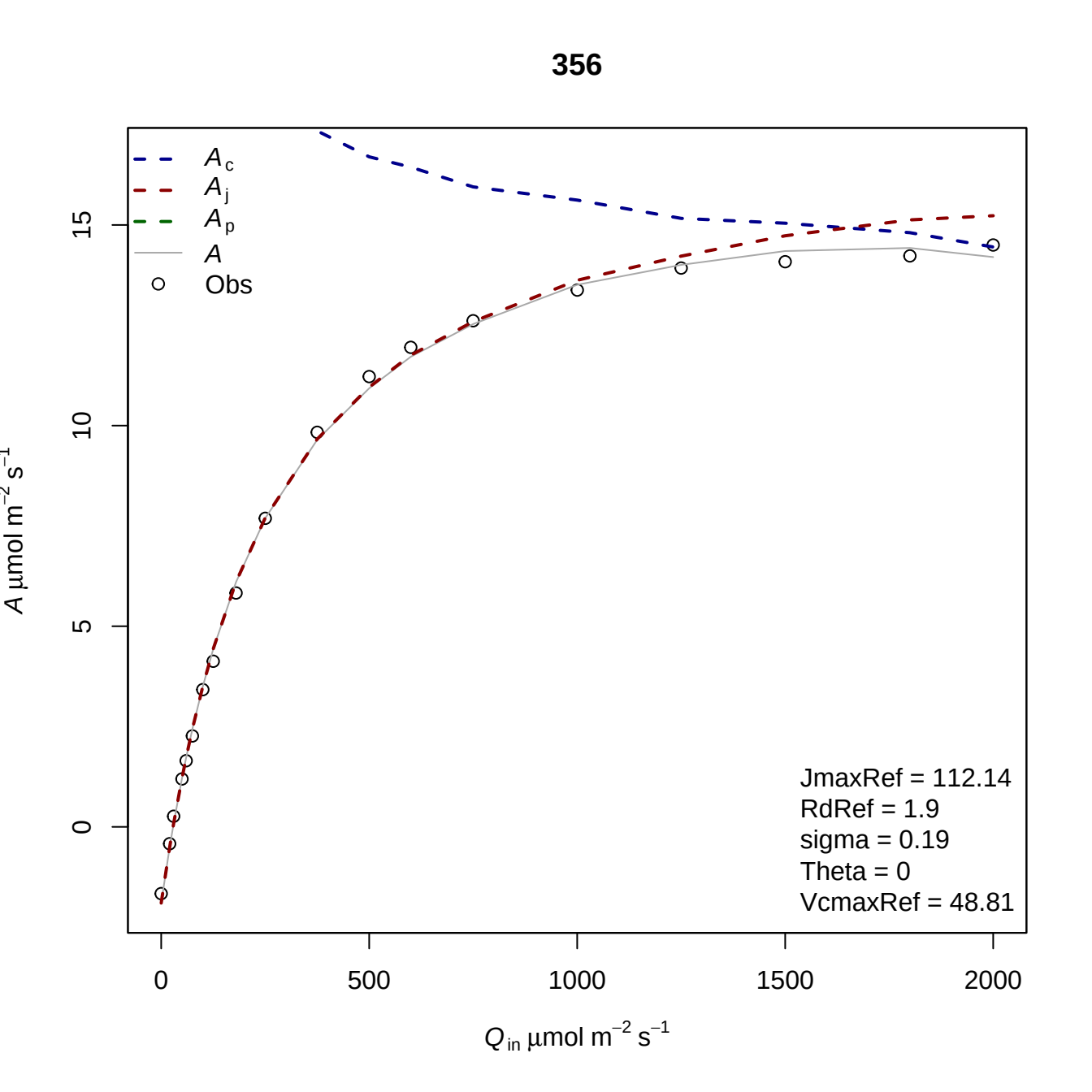

351
A_c
A_j
A_p
A
Obs
JmaxRef = 93.35
RdRef = 3.3
sigma = 0.64
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

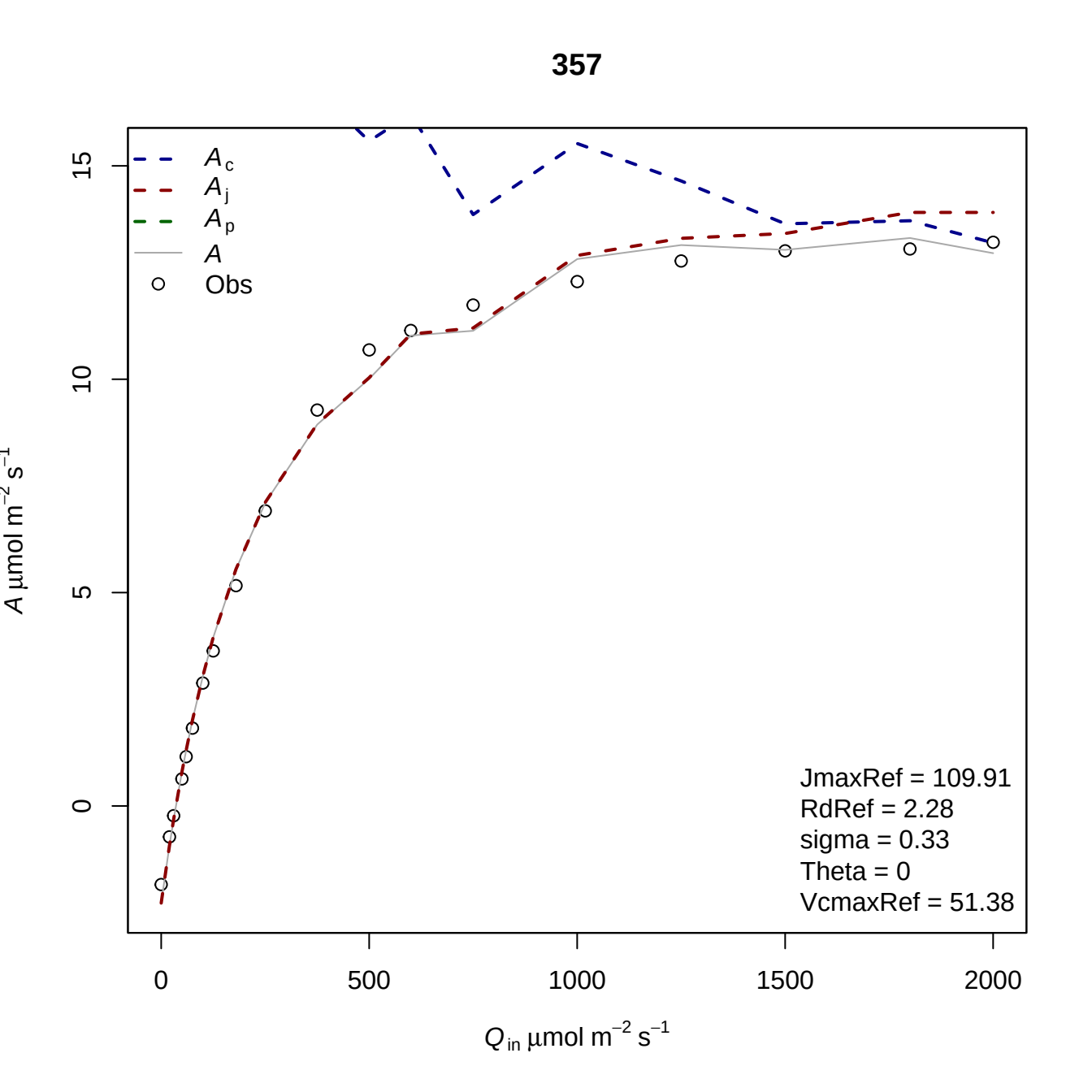

352
A_c
A_j
A_p
A
Obs
JmaxRef = 42.45
RdRef = 3.67
sigma = 0.77
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

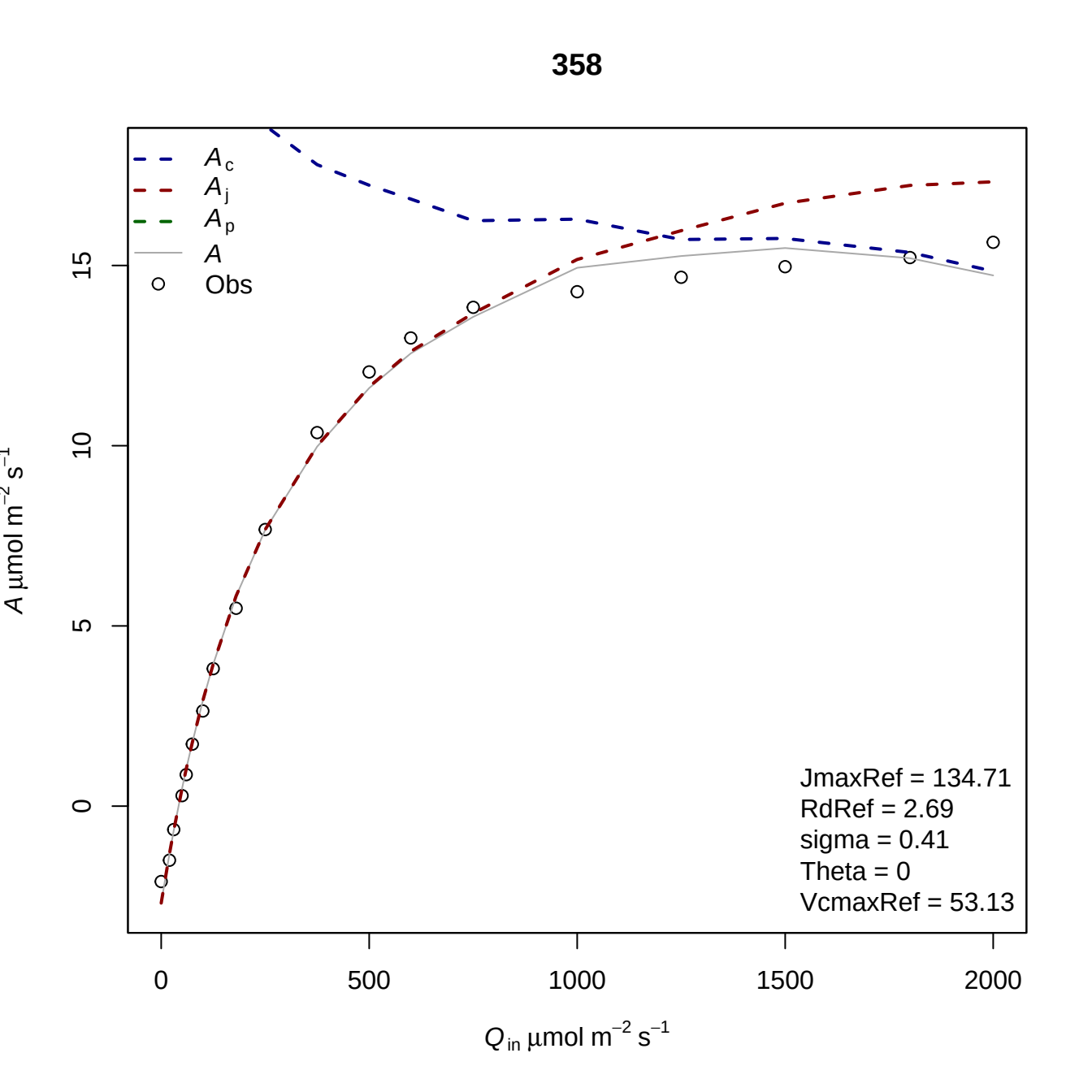

353
A_c
A_j
A_p
A
Obs
JmaxRef = 73.04
RdRef = 2.86
sigma = 0.52
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

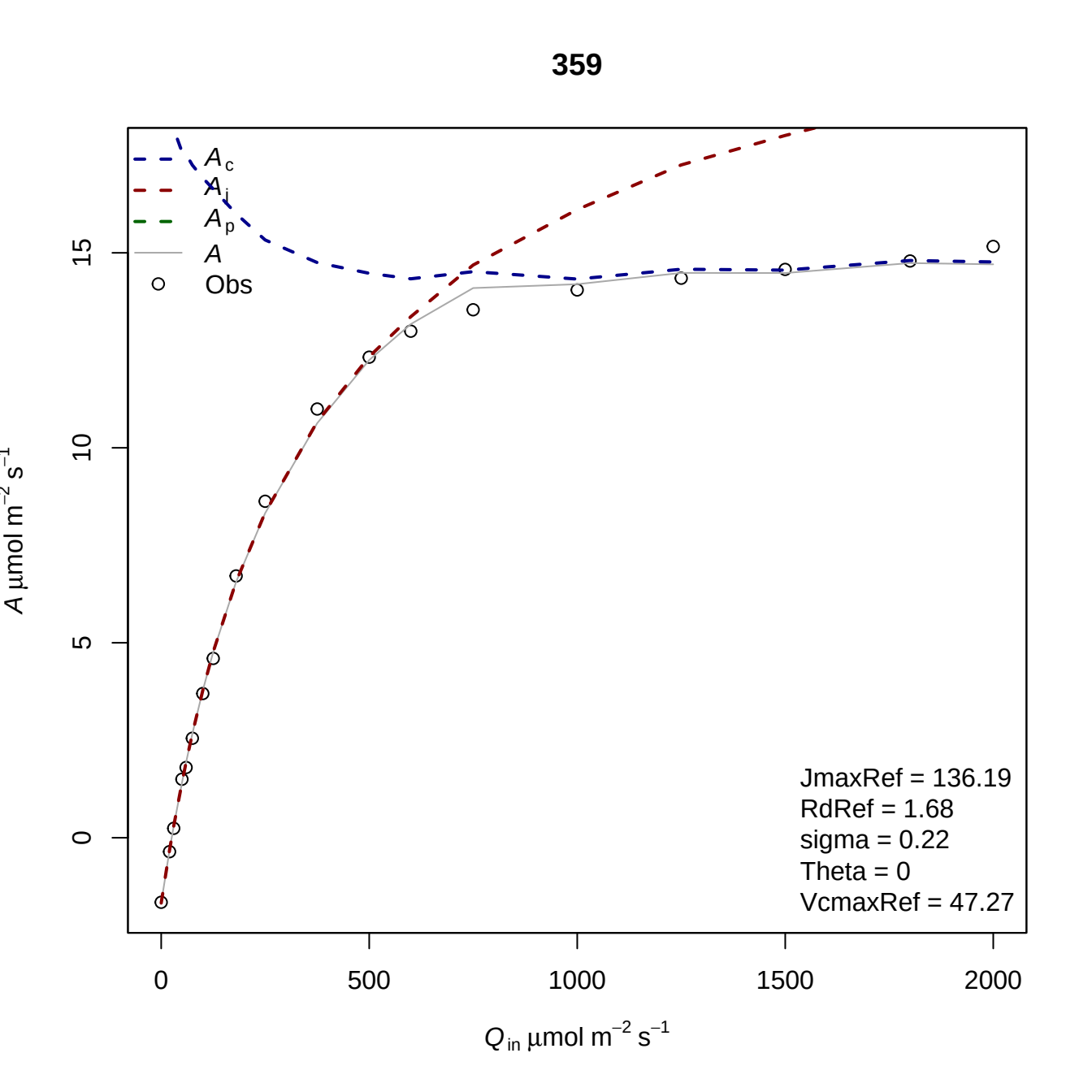

354
A_c
A_j
A_p
A
Obs
JmaxRef = 132.29
RdRef = 2.35
sigma = 0.21
Theta = 0
VcmaxRef = 53.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

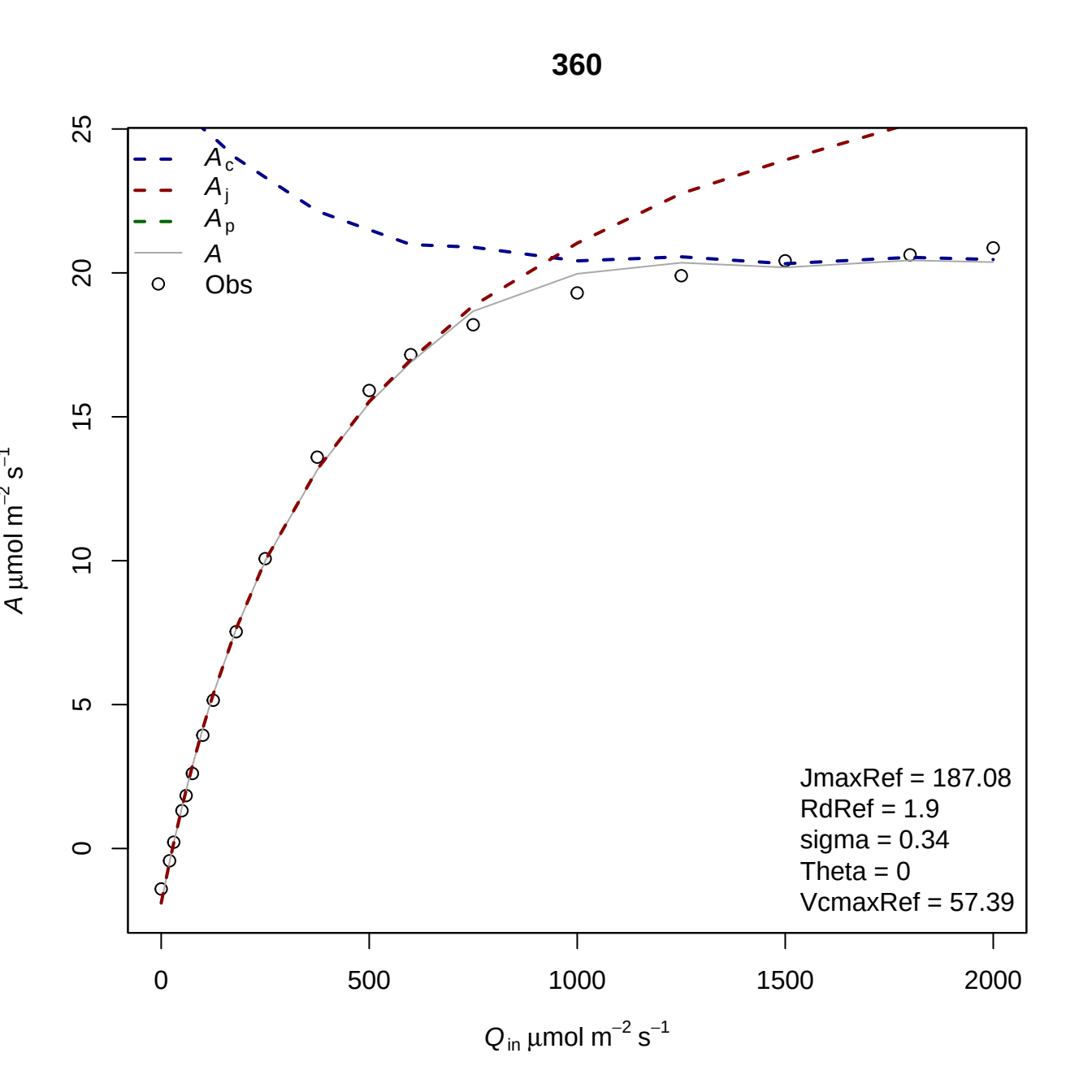

355
A_c
A_j
A_p
A
Obs
JmaxRef = 74.46
RdRef = 2.07
sigma = 0.34
Theta = 0
VcmaxRef = 35.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

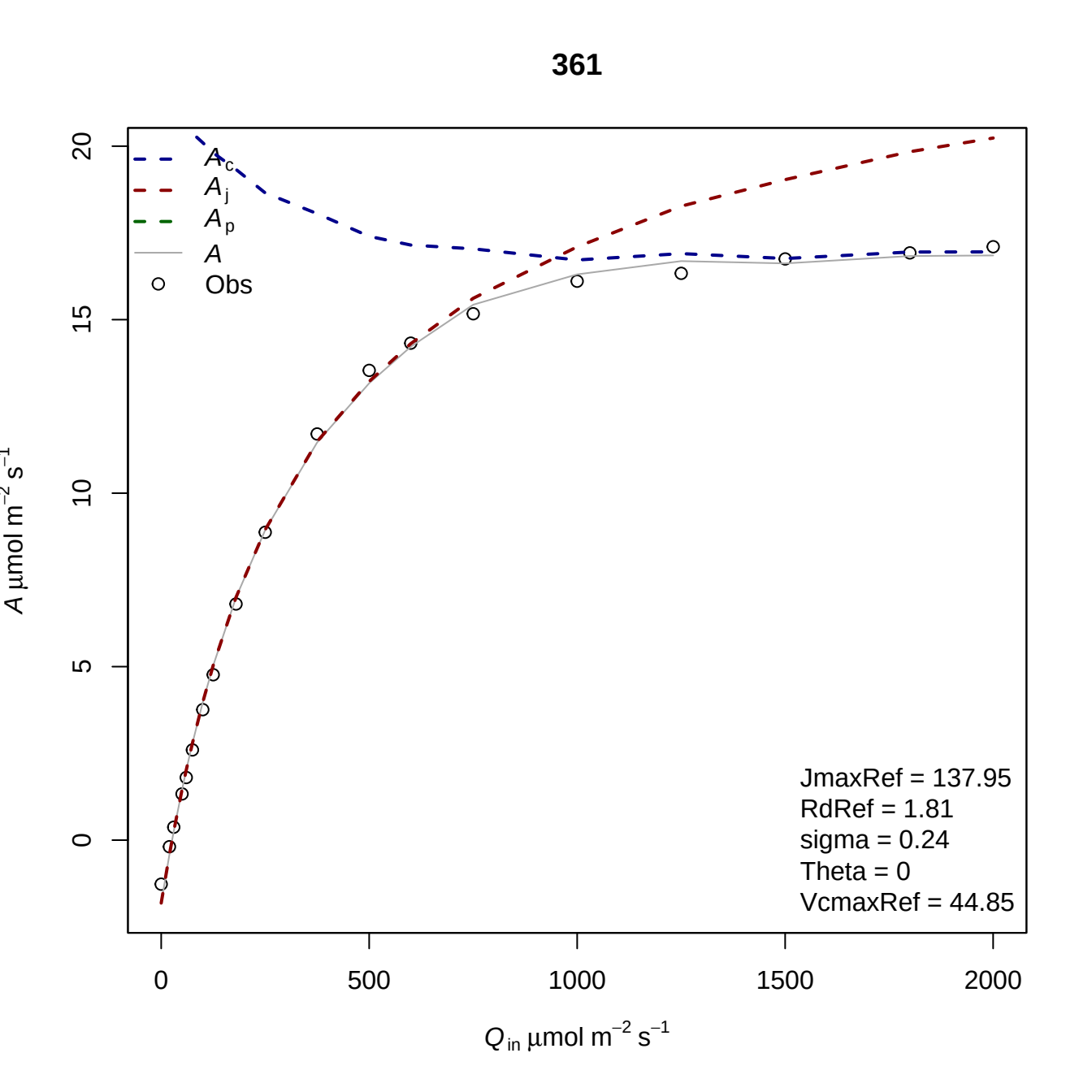

356
A_c
A_j
A_p
A
Obs
JmaxRef = 112.14
RdRef = 1.9
sigma = 0.19
Theta = 0
VcmaxRef = 48.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

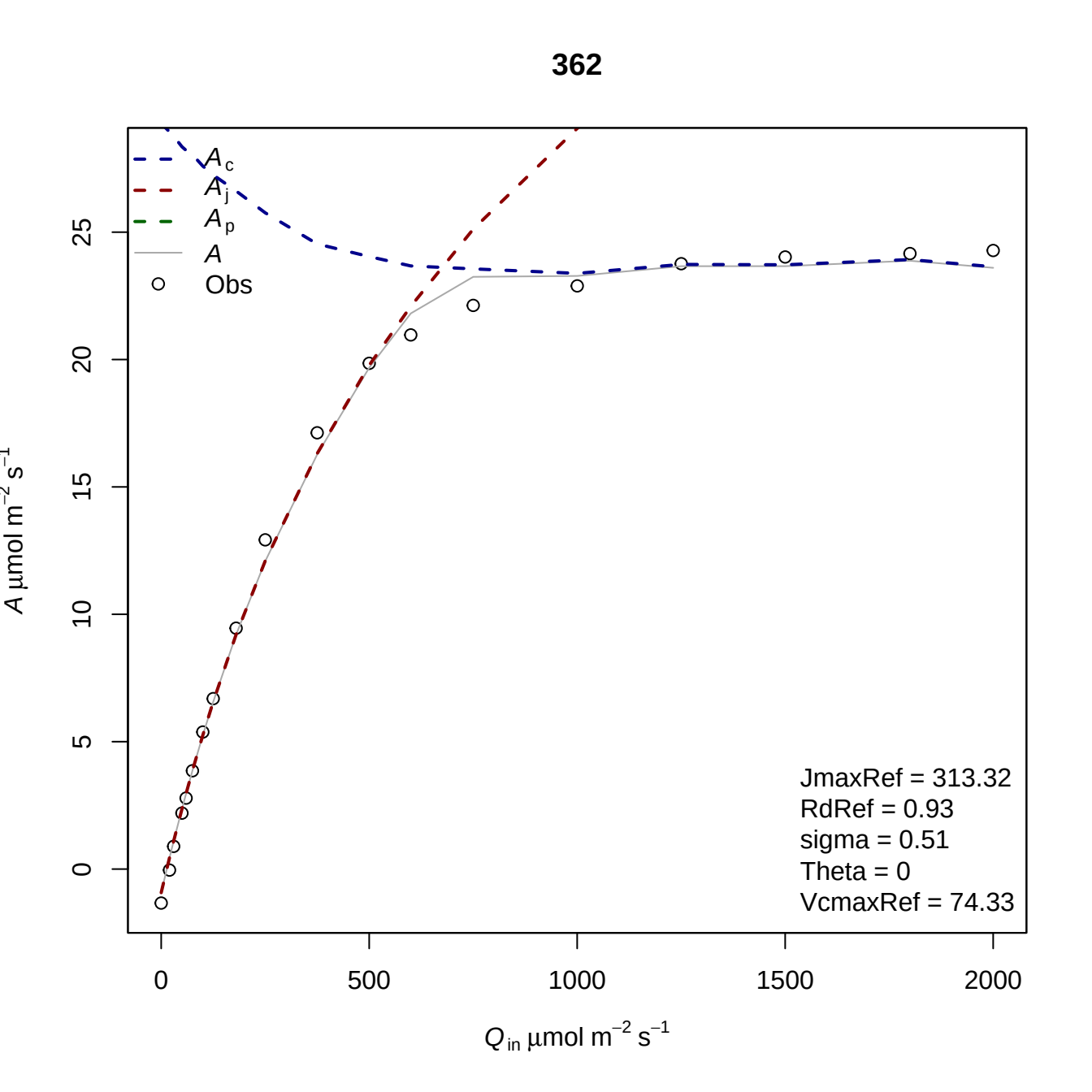

357
A_c
A_j
A_p
A
Obs
JmaxRef = 109.91
RdRef = 2.28
sigma = 0.33
Theta = 0
VcmaxRef = 51.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

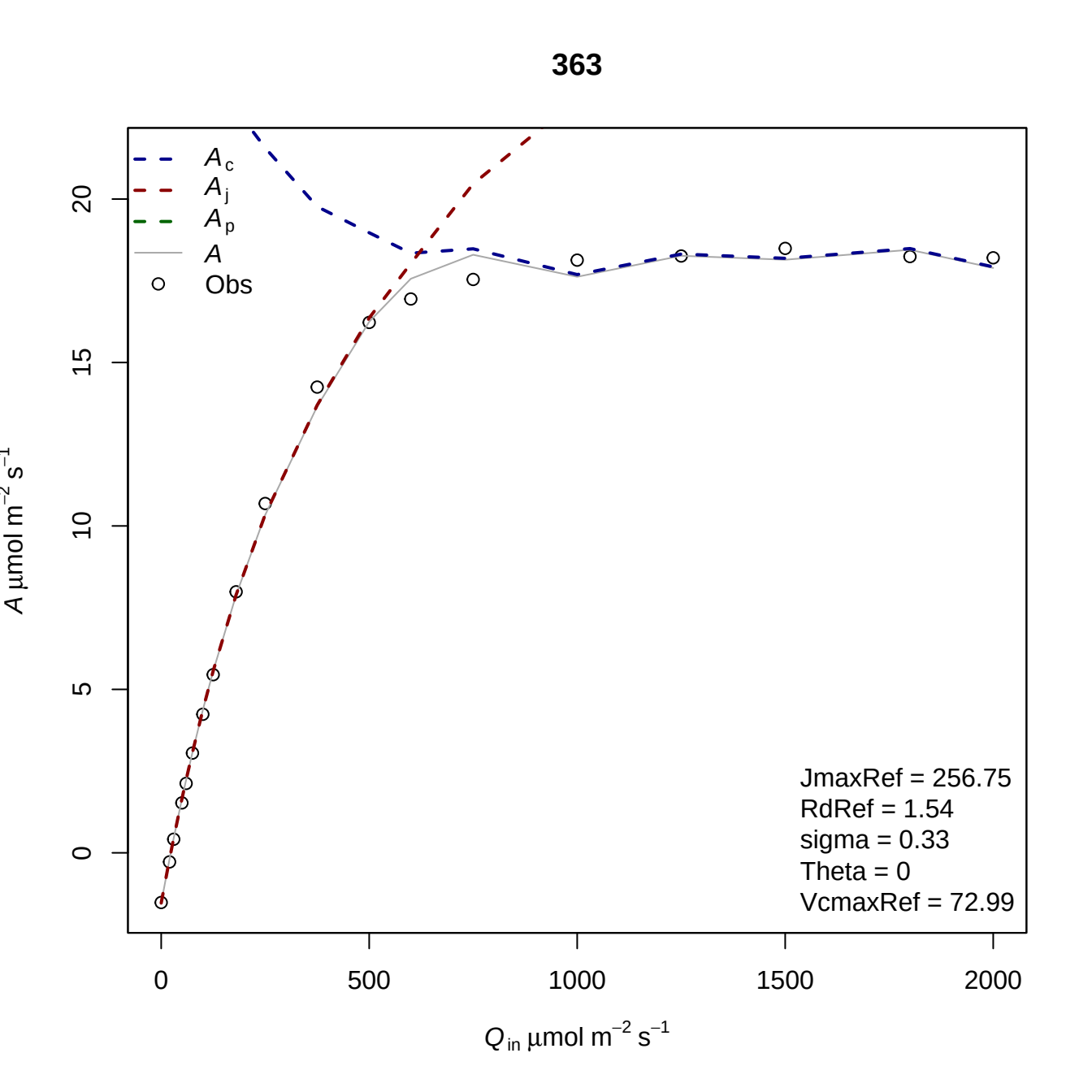

358
A_c
A_j
A_p
A
Obs
JmaxRef = 134.71
RdRef = 2.69
sigma = 0.41
Theta = 0
VcmaxRef = 53.13
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

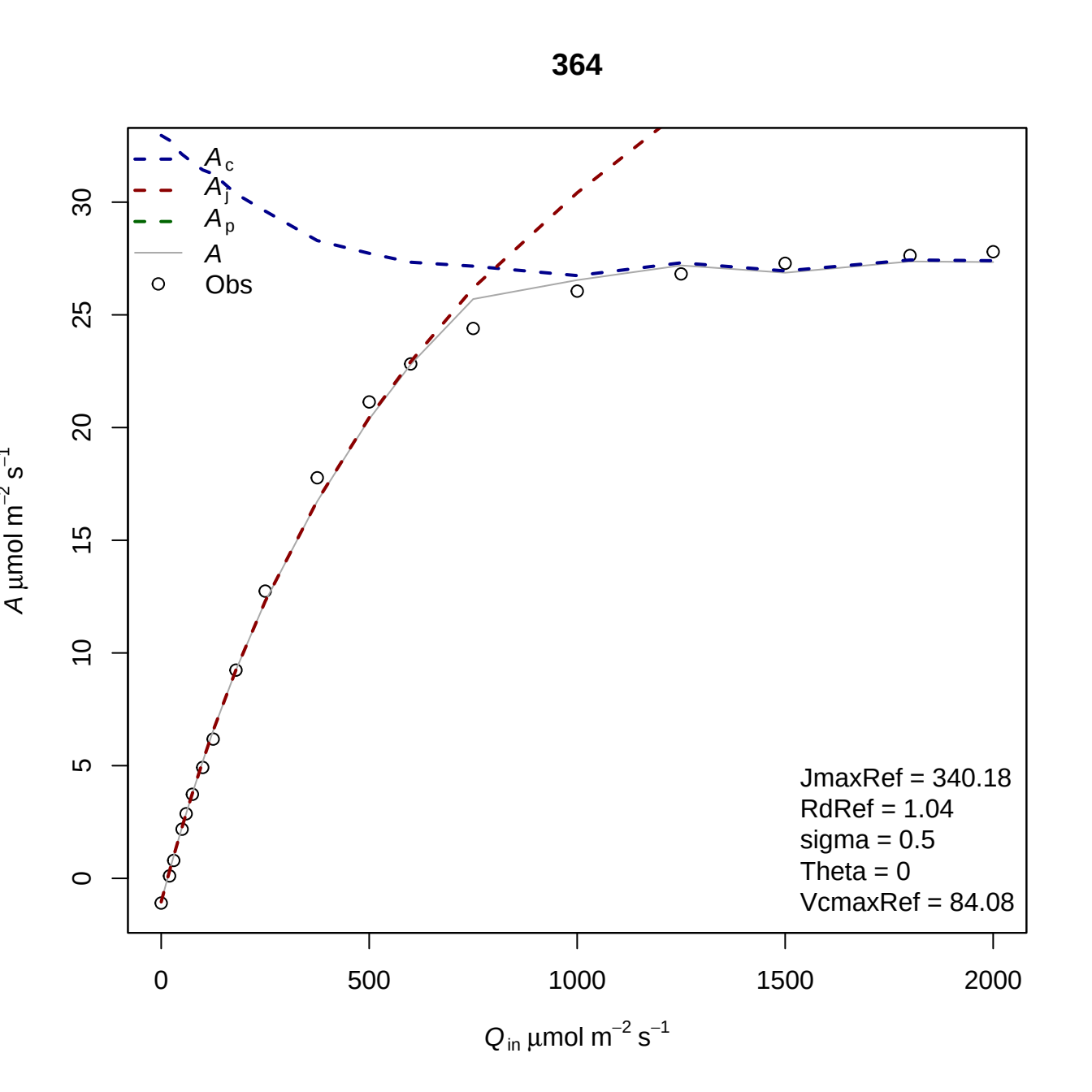

359
A_c
A_j
A_p
A
Obs
JmaxRef = 136.19
RdRef = 1.68
sigma = 0.22
Theta = 0
VcmaxRef = 47.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

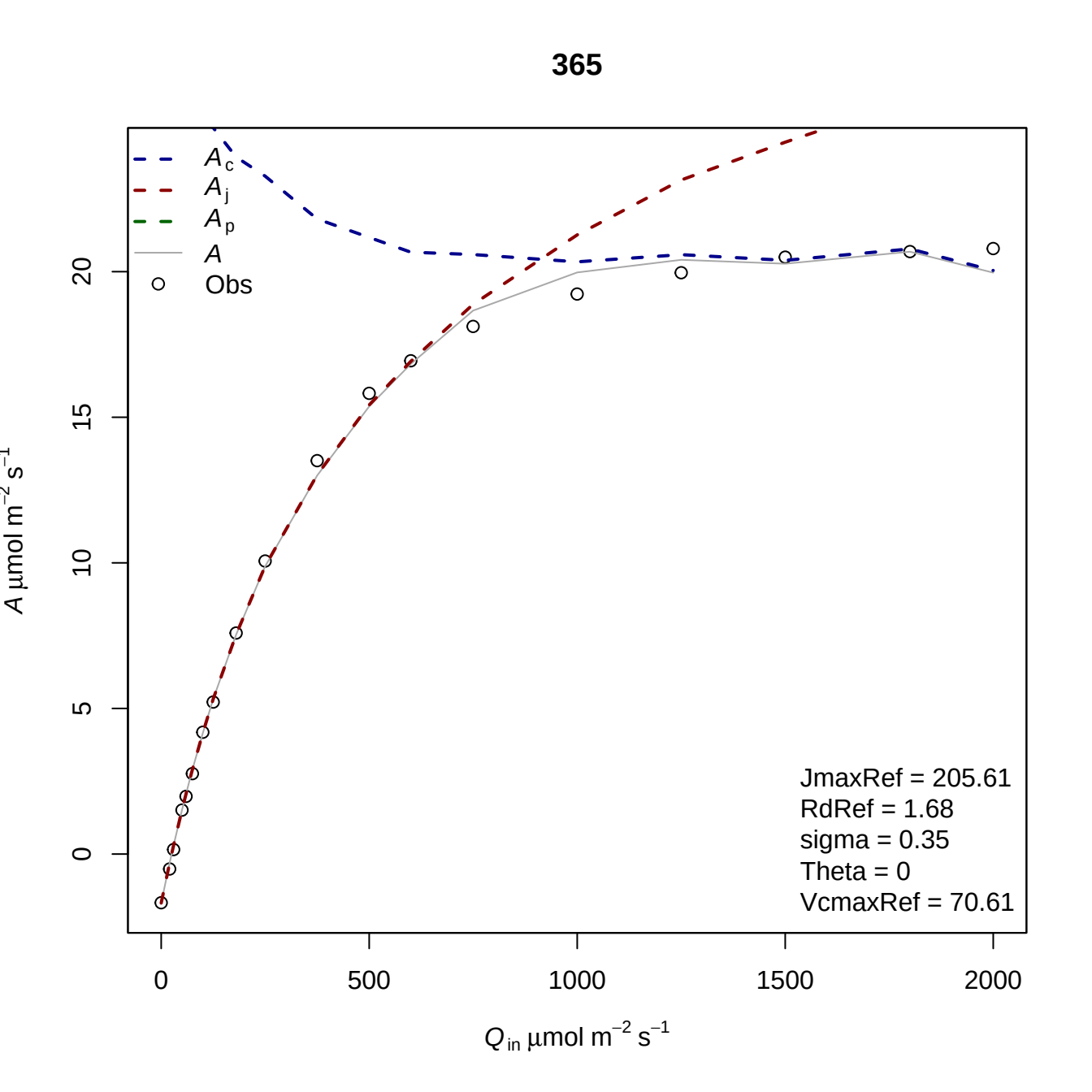

360
A_c
A_j
A_p
A
Obs
JmaxRef = 187.08
RdRef = 1.9
sigma = 0.34
Theta = 0
VcmaxRef = 57.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

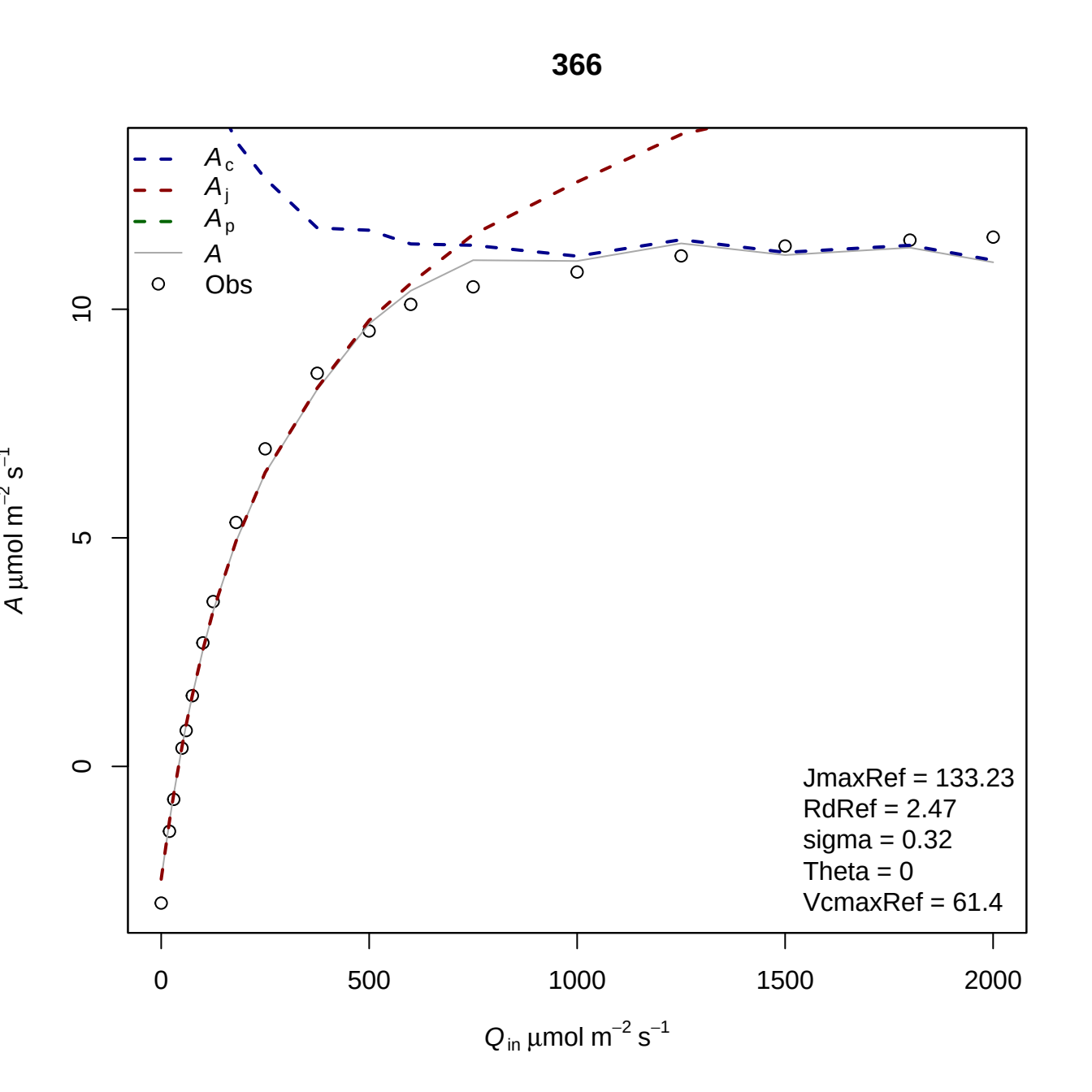

361
A_c
A_j
A_p
A
Obs
JmaxRef = 137.95
RdRef = 1.81
sigma = 0.24
Theta = 0
VcmaxRef = 44.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

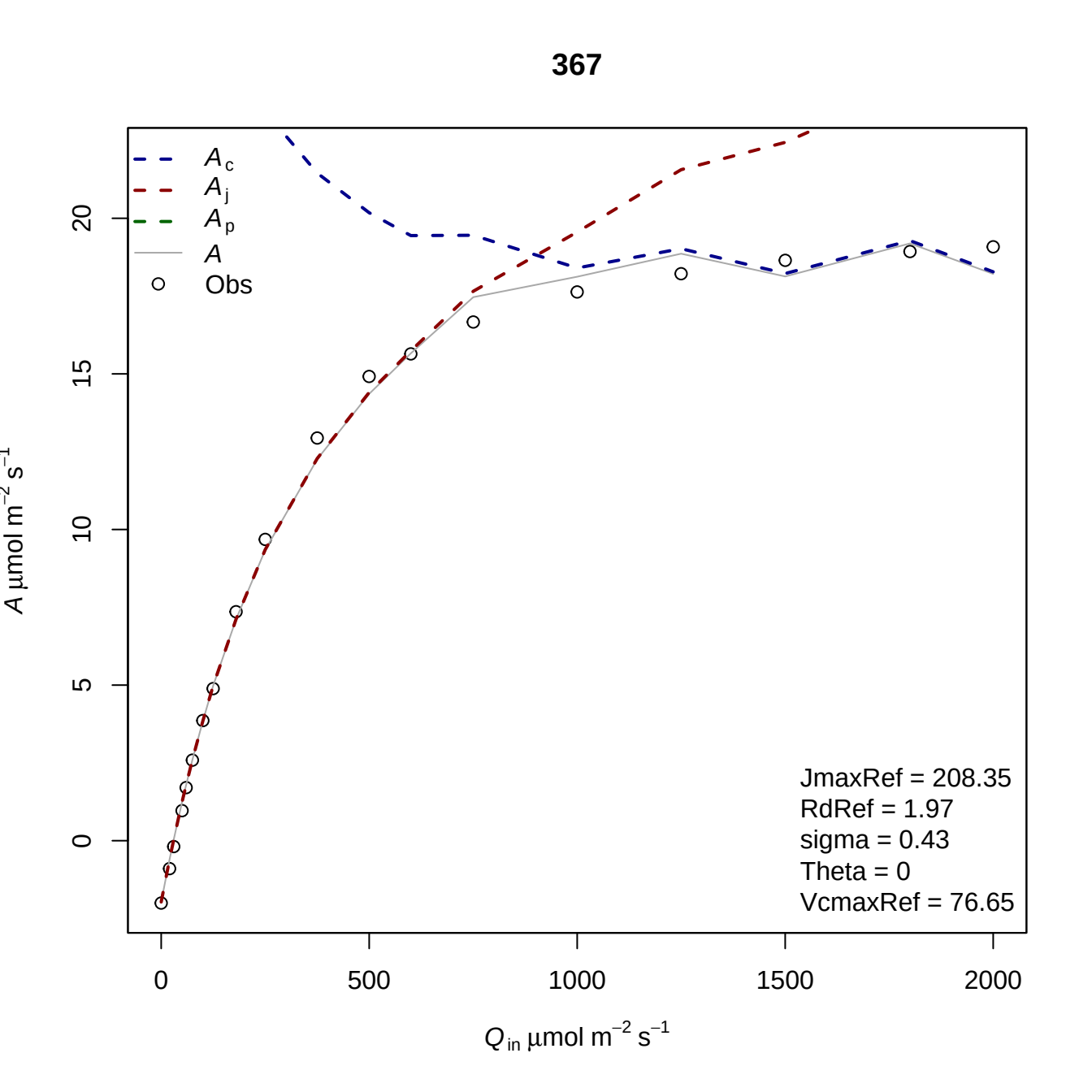

362
A_c
A_j
A_p
A
Obs
JmaxRef = 313.32
RdRef = 0.93
sigma = 0.51
Theta = 0
VcmaxRef = 74.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

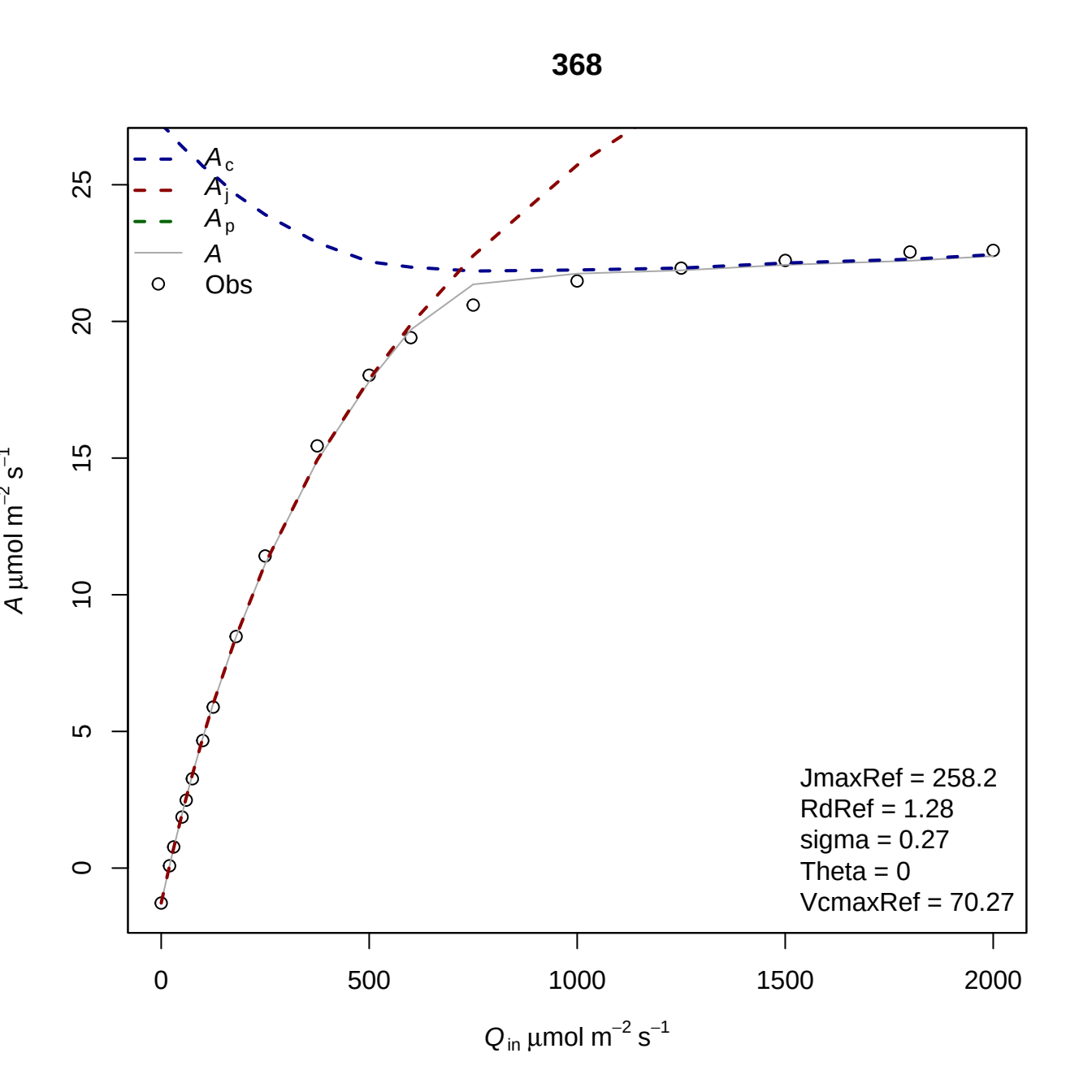

363
A_c
A_j
A_p
A
Obs
JmaxRef = 256.75
RdRef = 1.54
sigma = 0.33
Theta = 0
VcmaxRef = 72.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

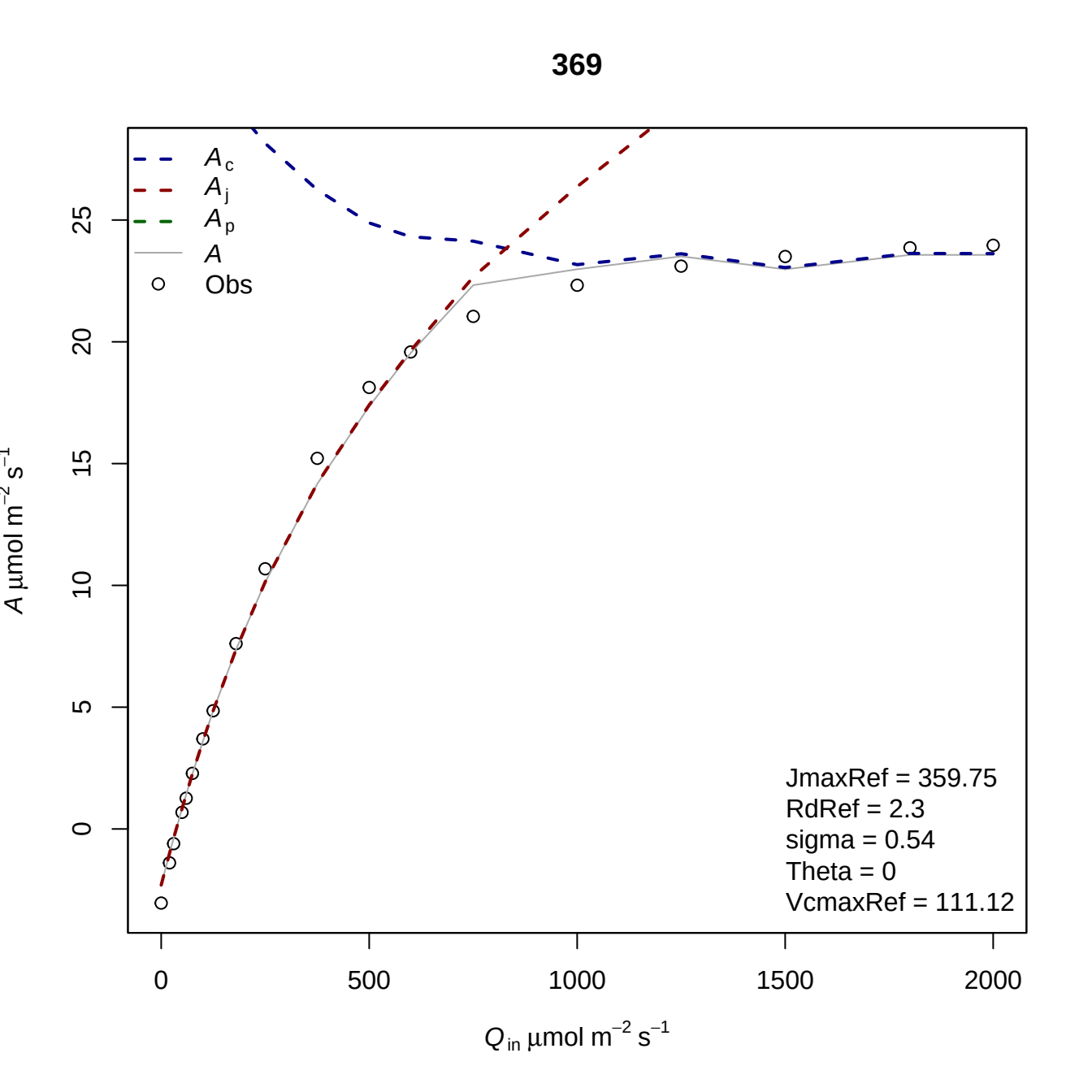

364
A_c
A_j
A_p
A
Obs
JmaxRef = 340.18
RdRef = 1.04
sigma = 0.5
Theta = 0
VcmaxRef = 84.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

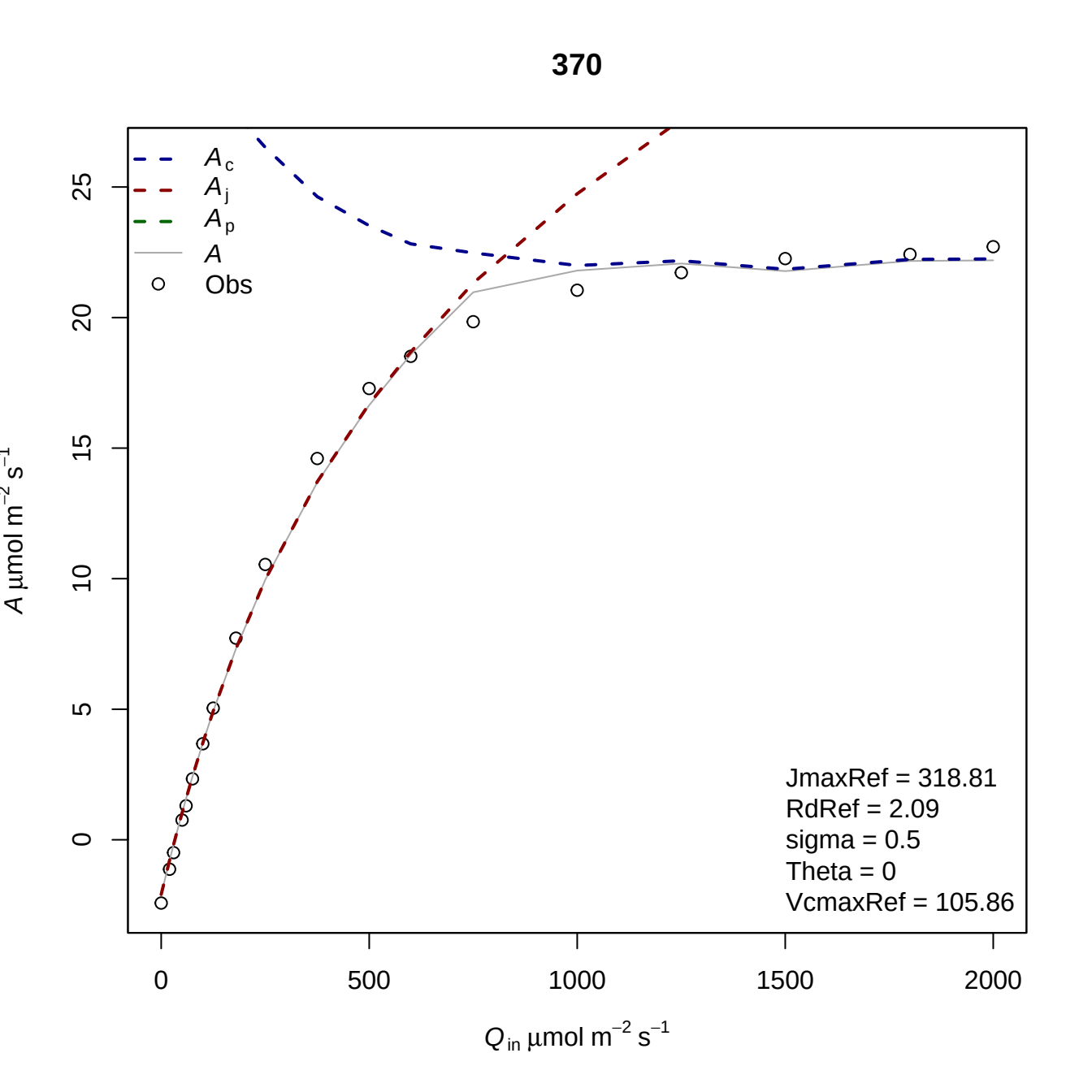

365
A_c
A_j
A_p
A
Obs
JmaxRef = 205.61
RdRef = 1.68
sigma = 0.35
Theta = 0
VcmaxRef = 70.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

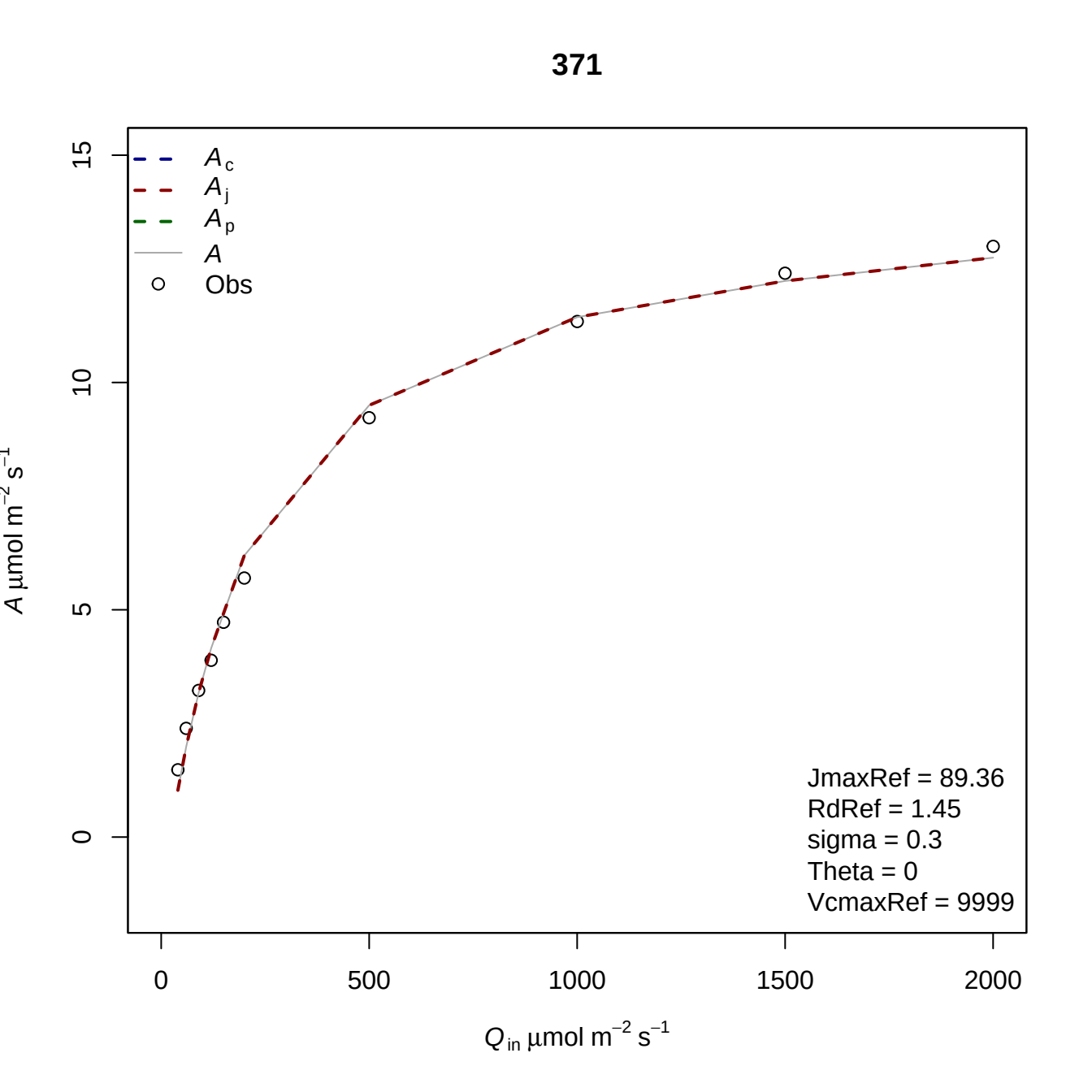

366
A_c
A_j
A_p
A
Obs
JmaxRef = 133.23
RdRef = 2.47
sigma = 0.32
Theta = 0
VcmaxRef = 61.4
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

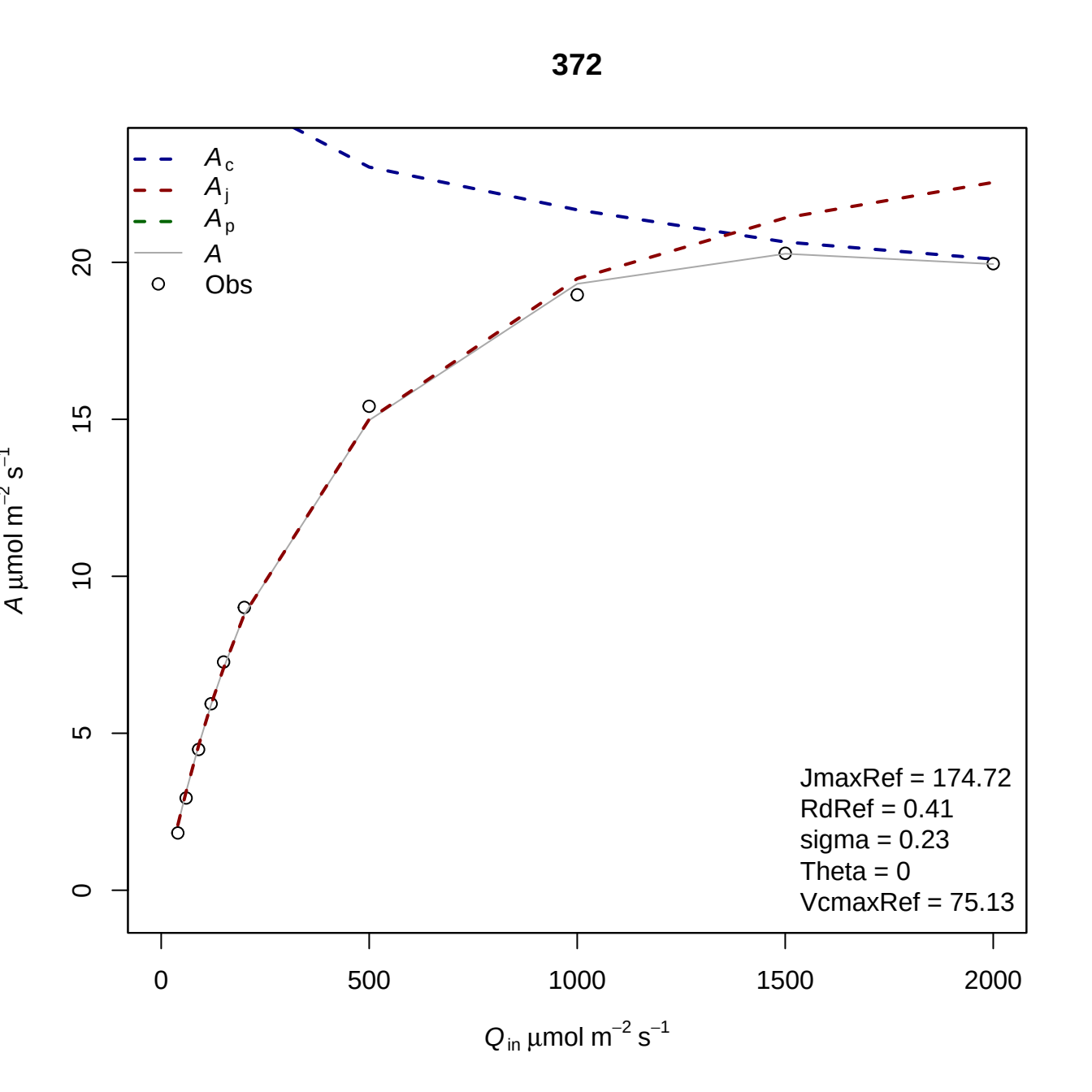

367
A_c
A_j
A_p
A
Obs
JmaxRef = 208.35
RdRef = 1.97
sigma = 0.43
Theta = 0
VcmaxRef = 76.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

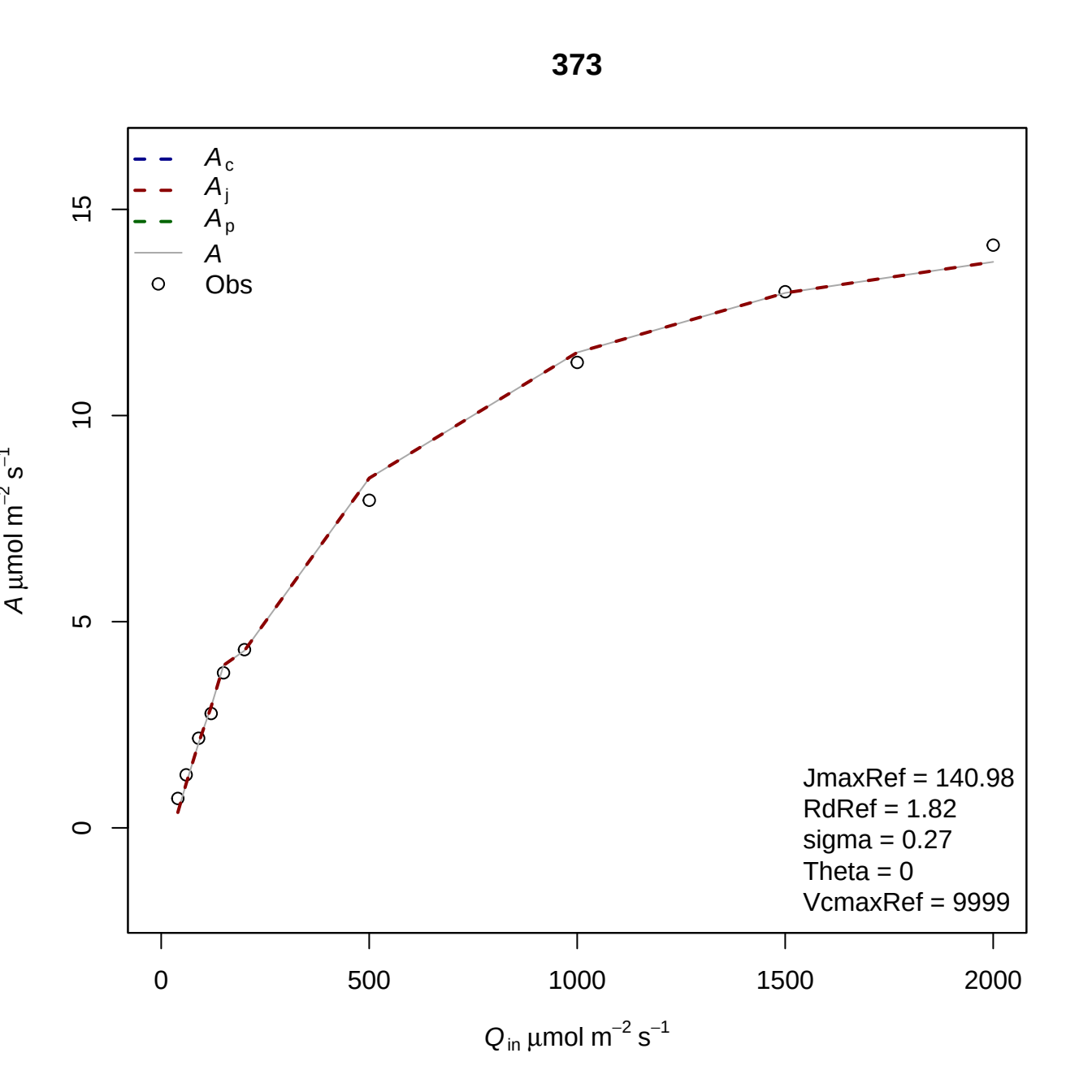

368
A_c
A_j
A_p
A
Obs
JmaxRef = 258.2
RdRef = 1.28
sigma = 0.27
Theta = 0
VcmaxRef = 70.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

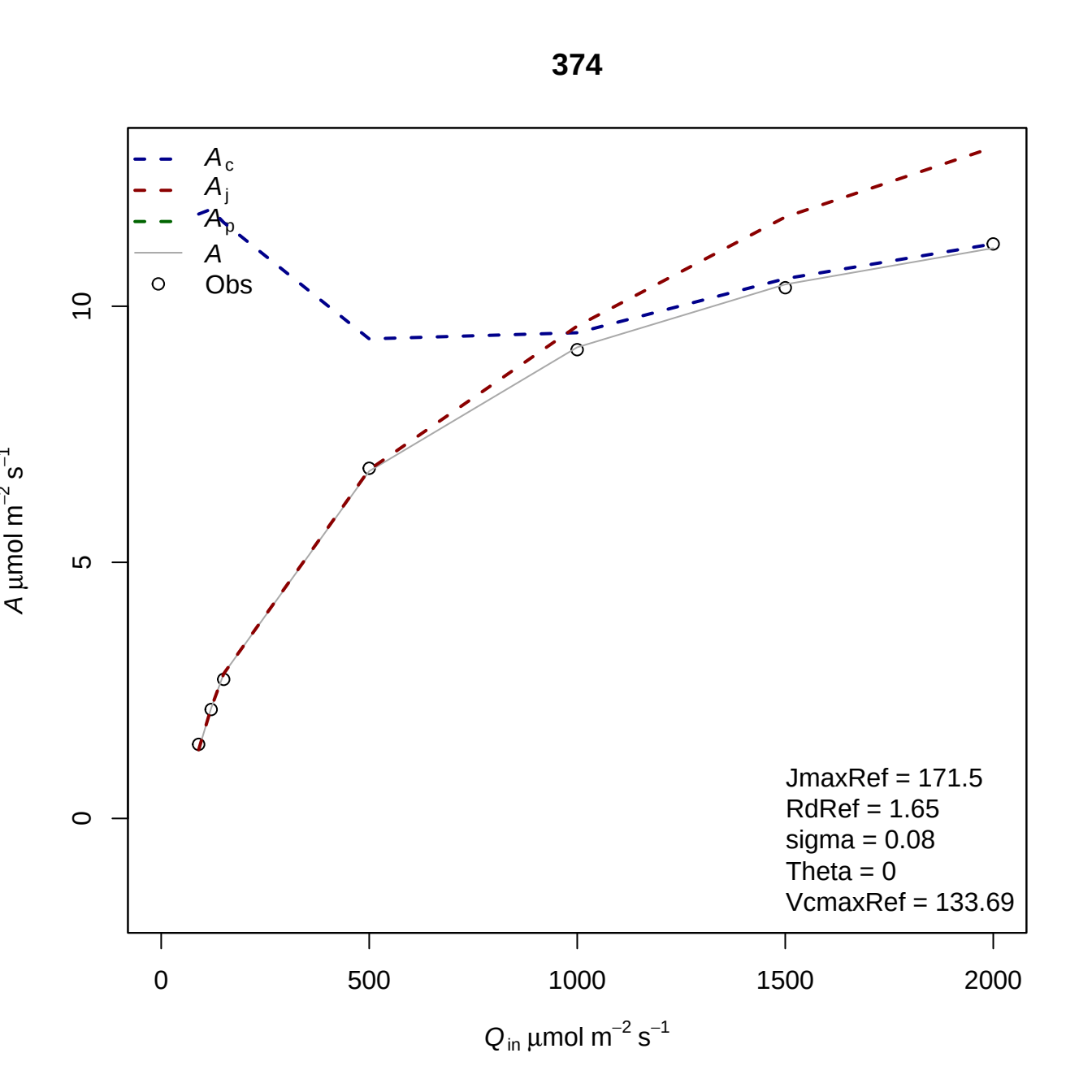

369
A_c
A_j
A_p
A
Obs
JmaxRef = 359.75
RdRef = 2.3
sigma = 0.54
Theta = 0
VcmaxRef = 111.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

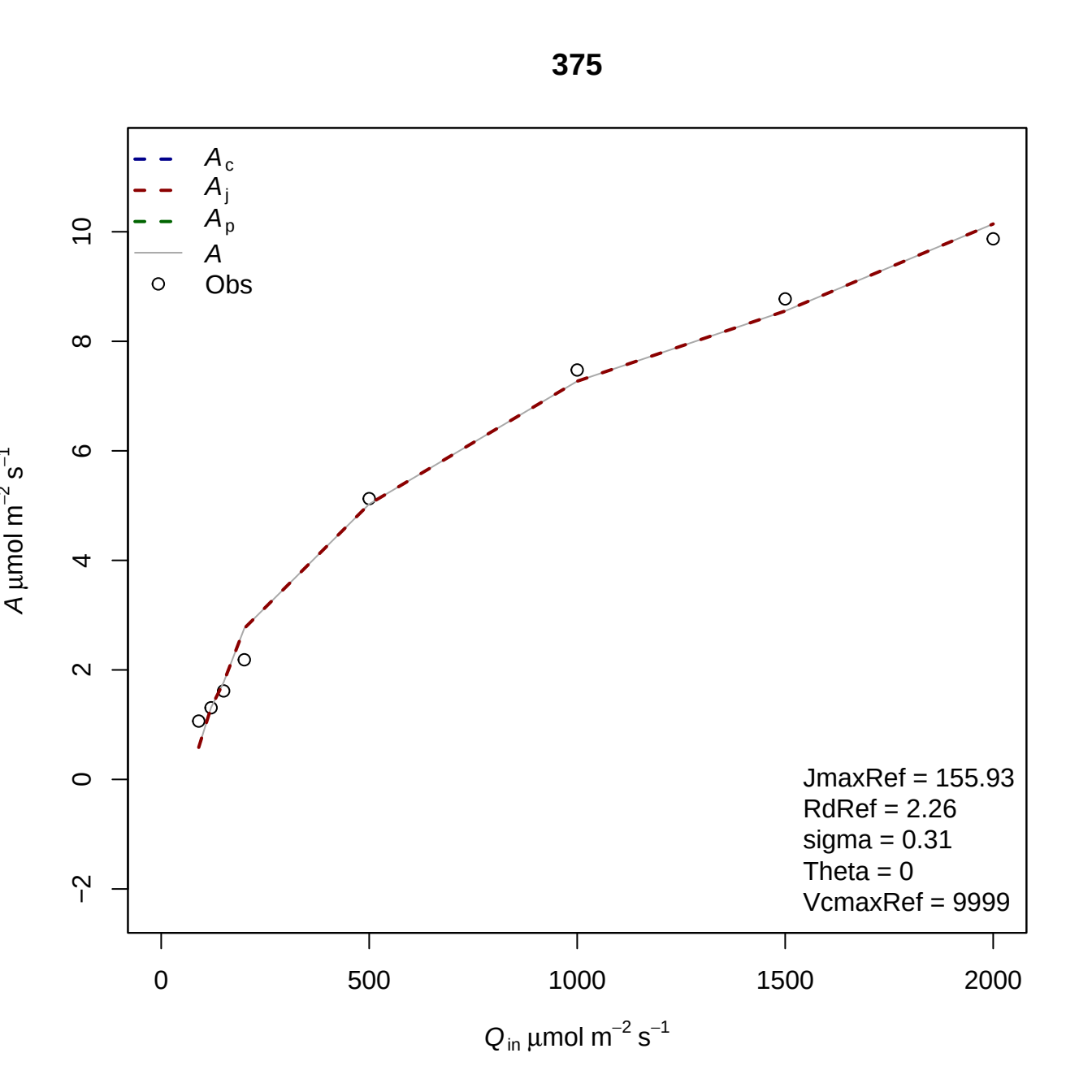

370
A_c
A_j
A_p
A
Obs
JmaxRef = 318.81
RdRef = 2.09
sigma = 0.5
Theta = 0
VcmaxRef = 105.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

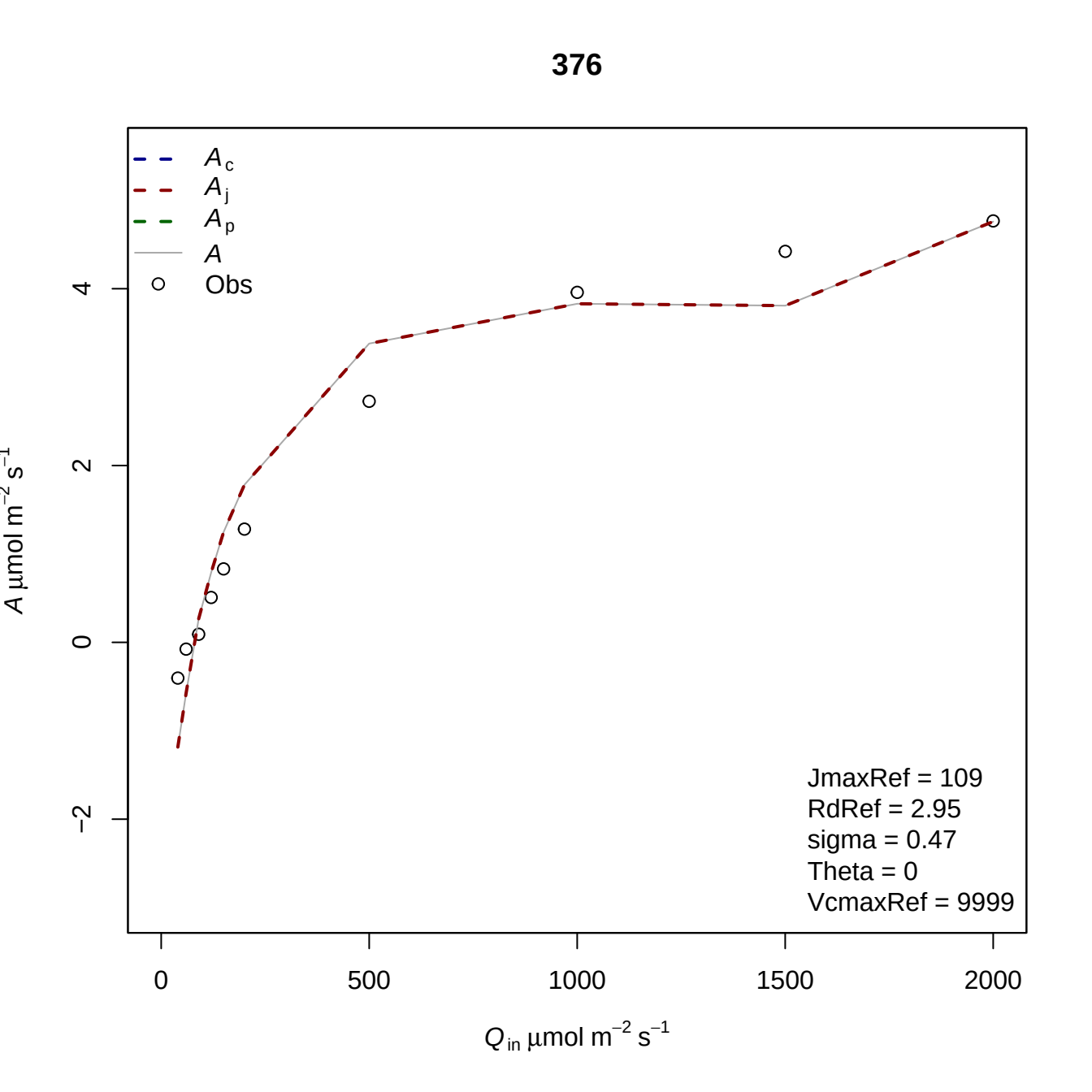

371
A_c
A_j
A_p
A
Obs
JmaxRef = 89.36
RdRef = 1.45
sigma = 0.3
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

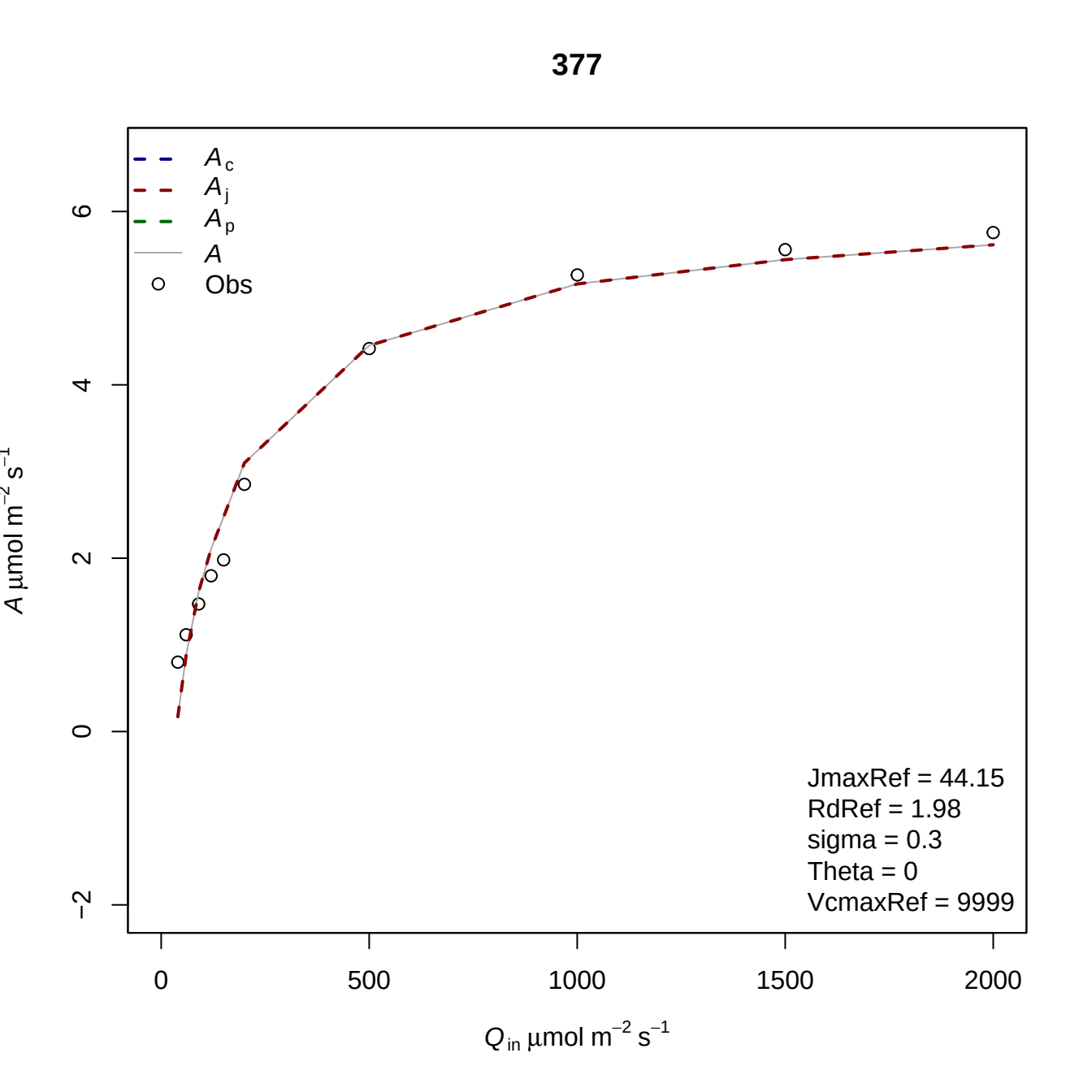

372
A_c
A_j
A_p
A
Obs
JmaxRef = 174.72
RdRef = 0.41
sigma = 0.23
Theta = 0
VcmaxRef = 75.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

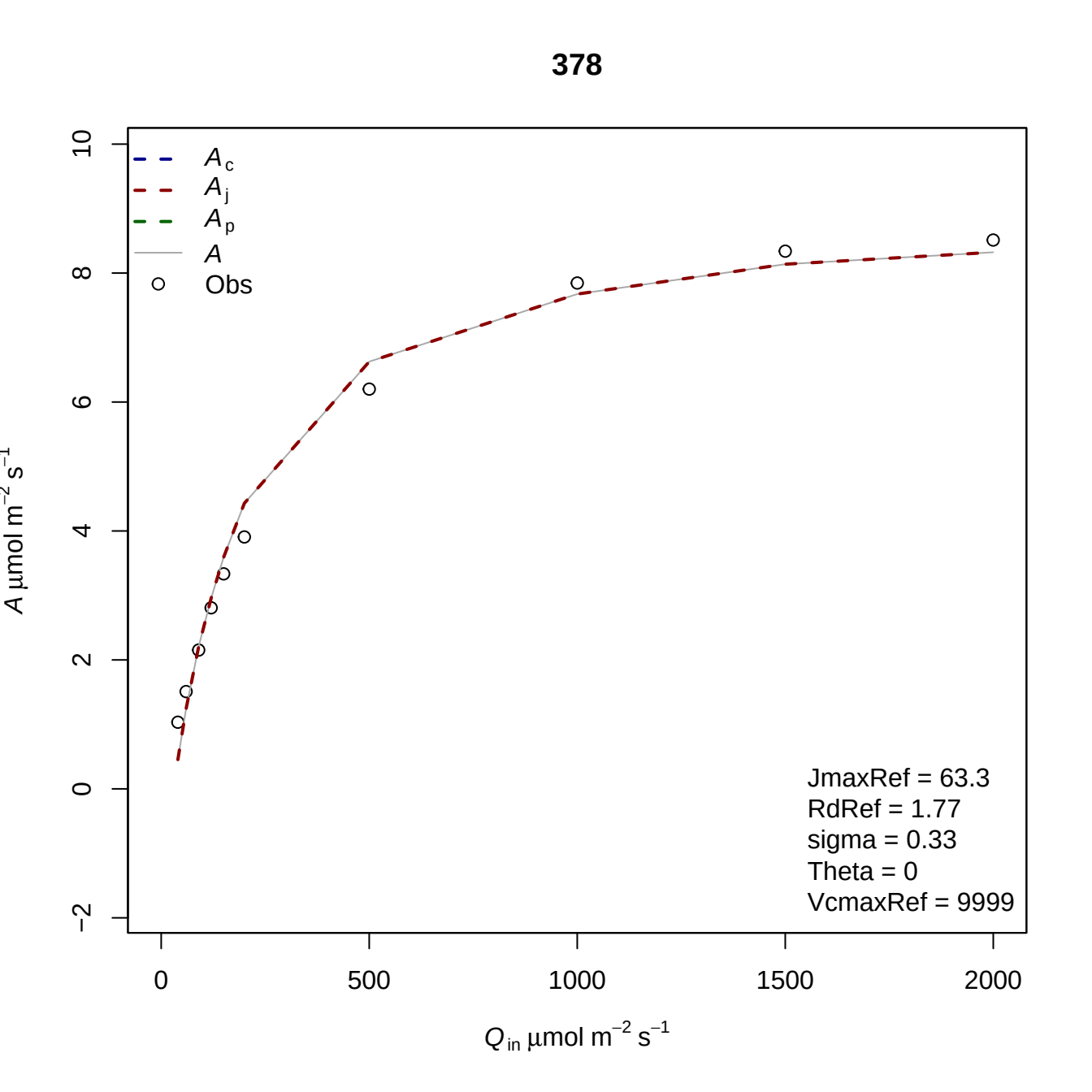

373
A_c
A_j
A_p
A
Obs
JmaxRef = 140.98
RdRef = 1.82
sigma = 0.27
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

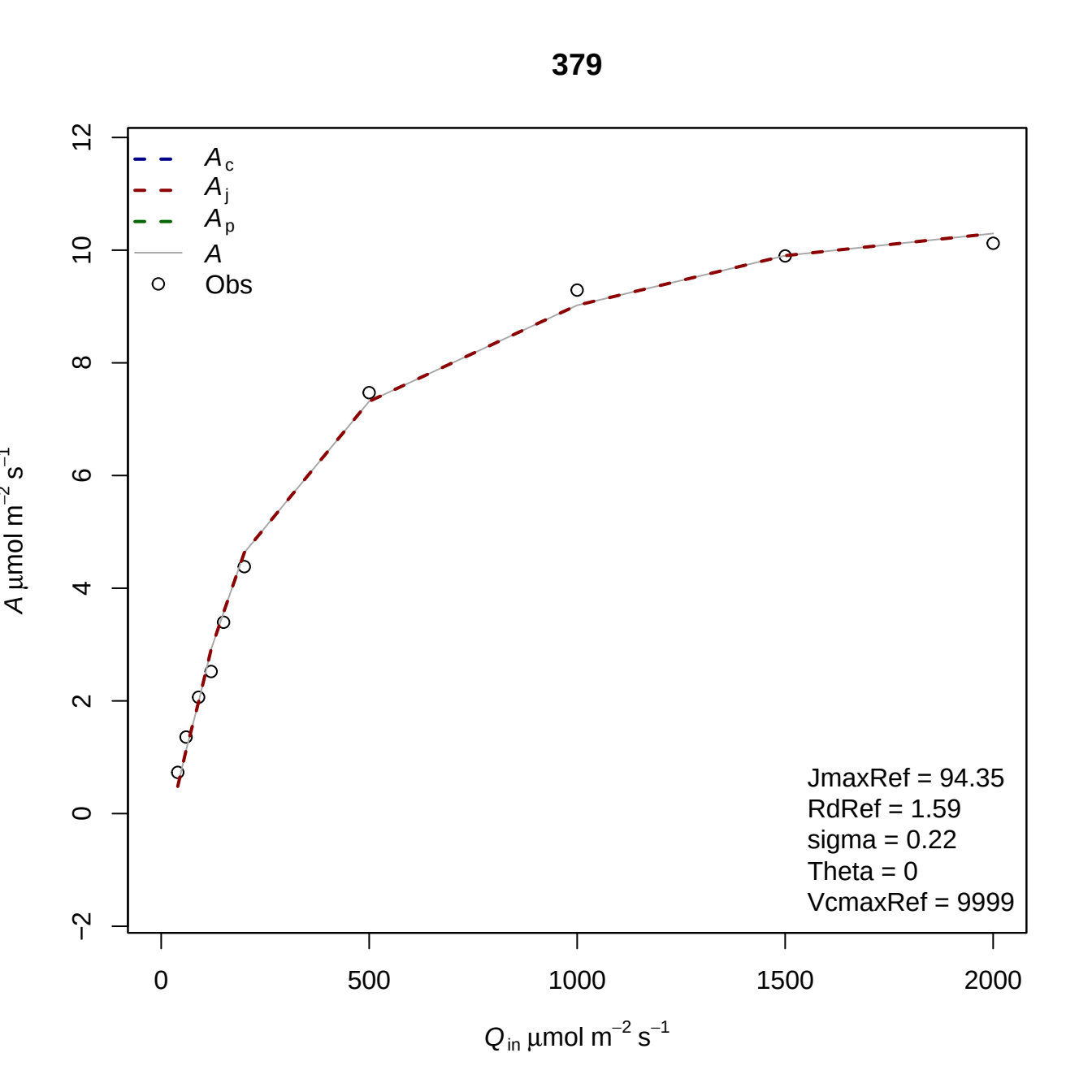

374
A_c
A_j
A_p
A
Obs
JmaxRef = 171.5
RdRef = 1.65
sigma = 0.08
Theta = 0
VcmaxRef = 133.69
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

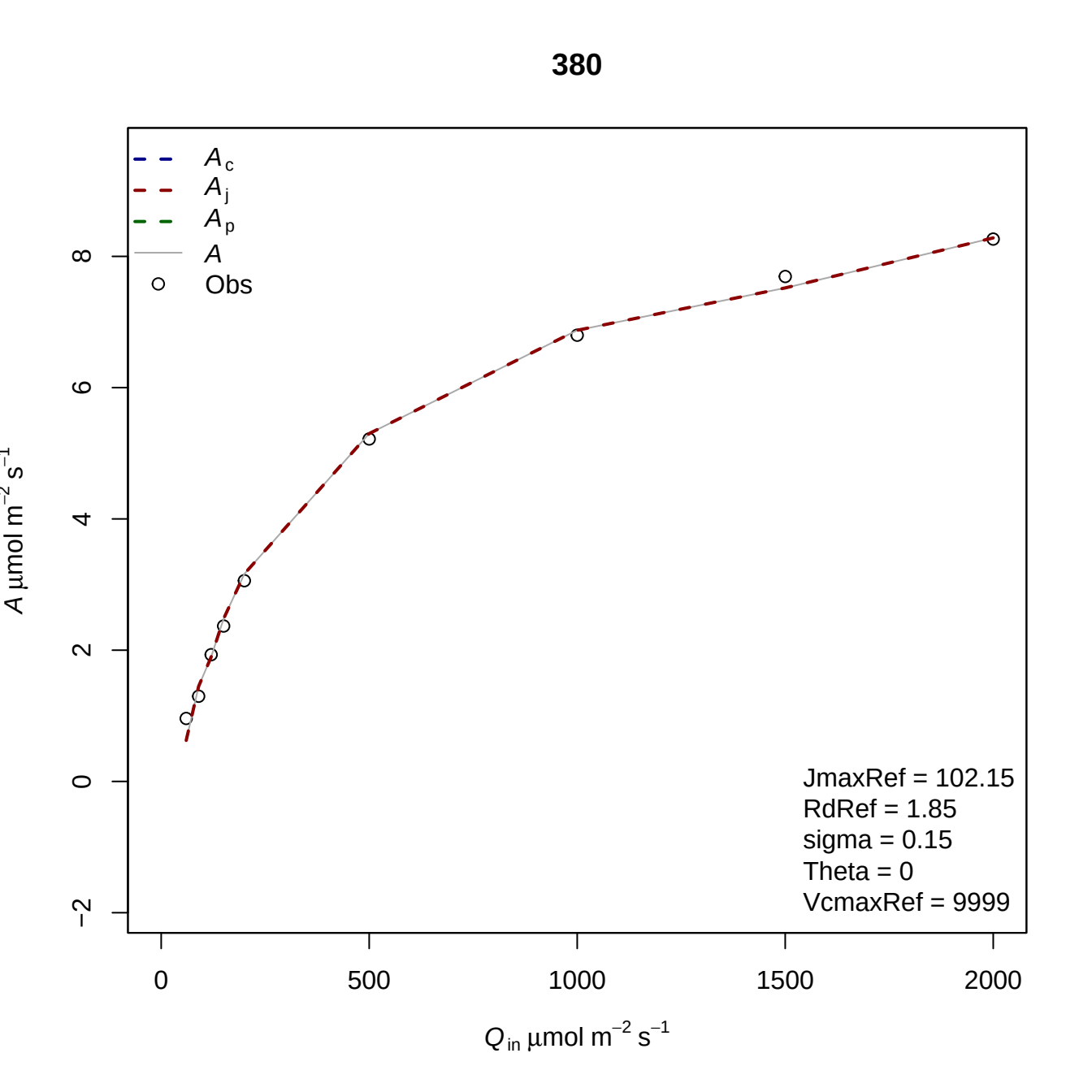

375
A_c
A_j
A_p
A
Obs
JmaxRef = 155.93
RdRef = 2.26
sigma = 0.31
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

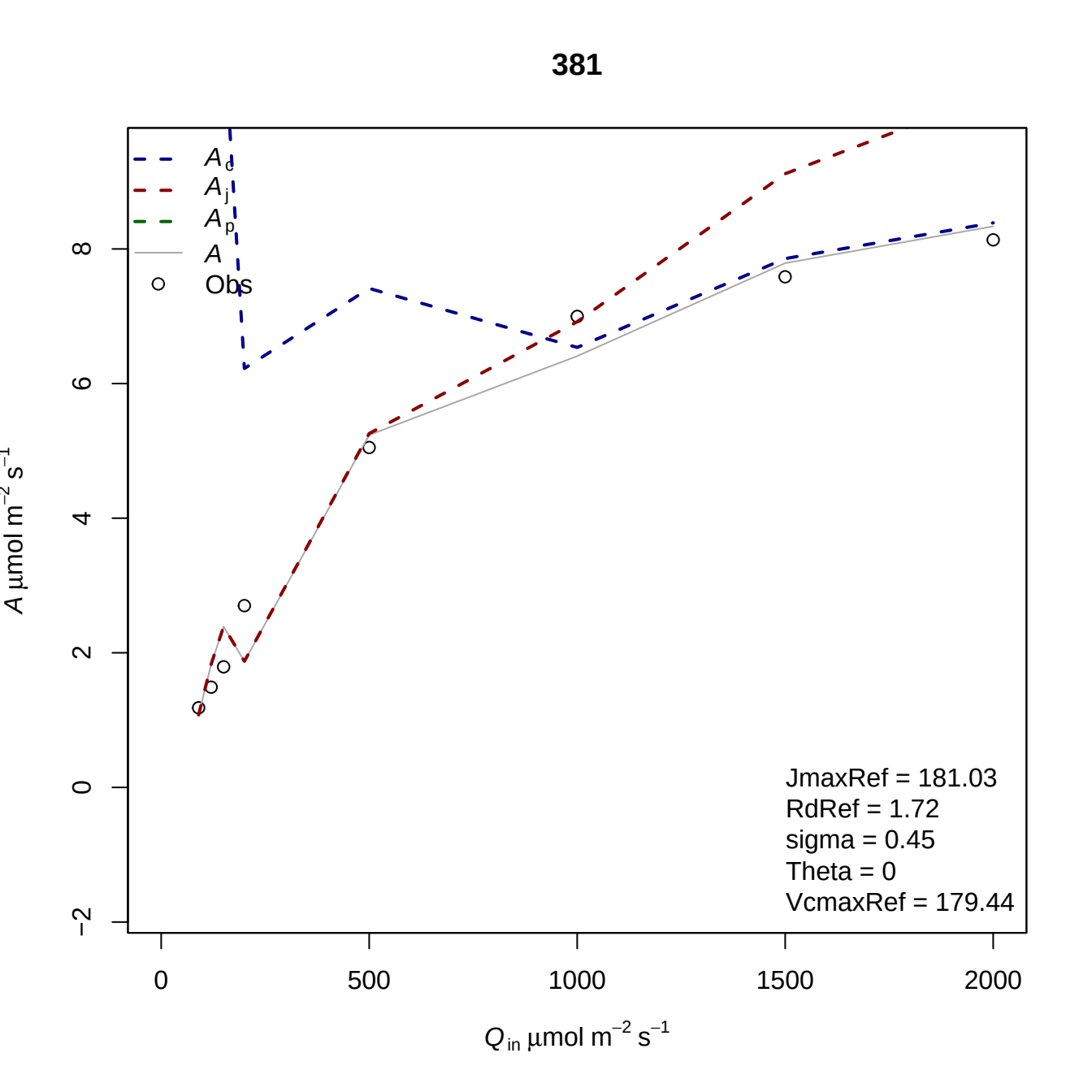

376
A_c
A_j
A_p
A
Obs
JmaxRef = 109
RdRef = 2.95
sigma = 0.47
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

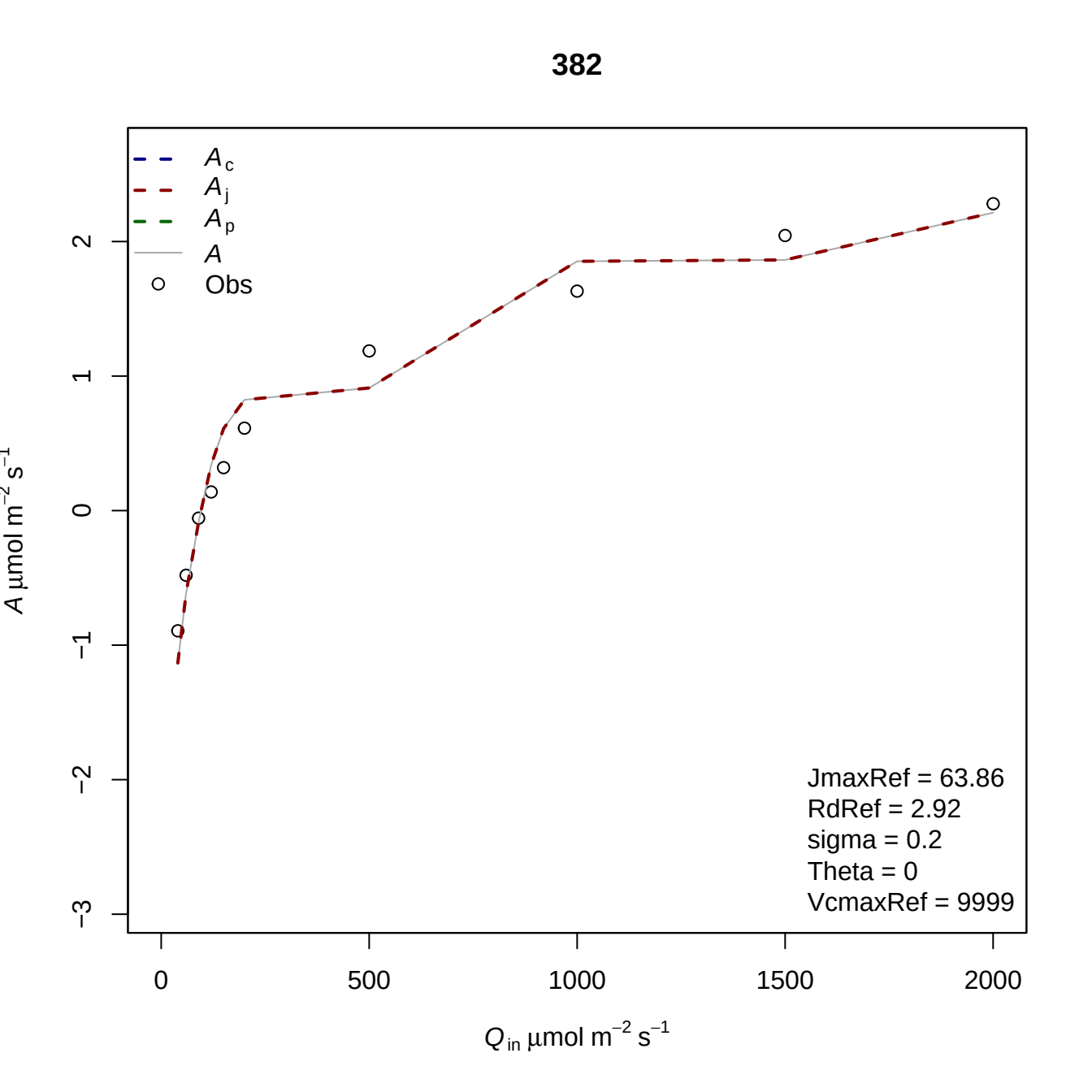

377
A_c
A_j
A_p
A
Obs
JmaxRef = 44.15
RdRef = 1.98
sigma = 0.3
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

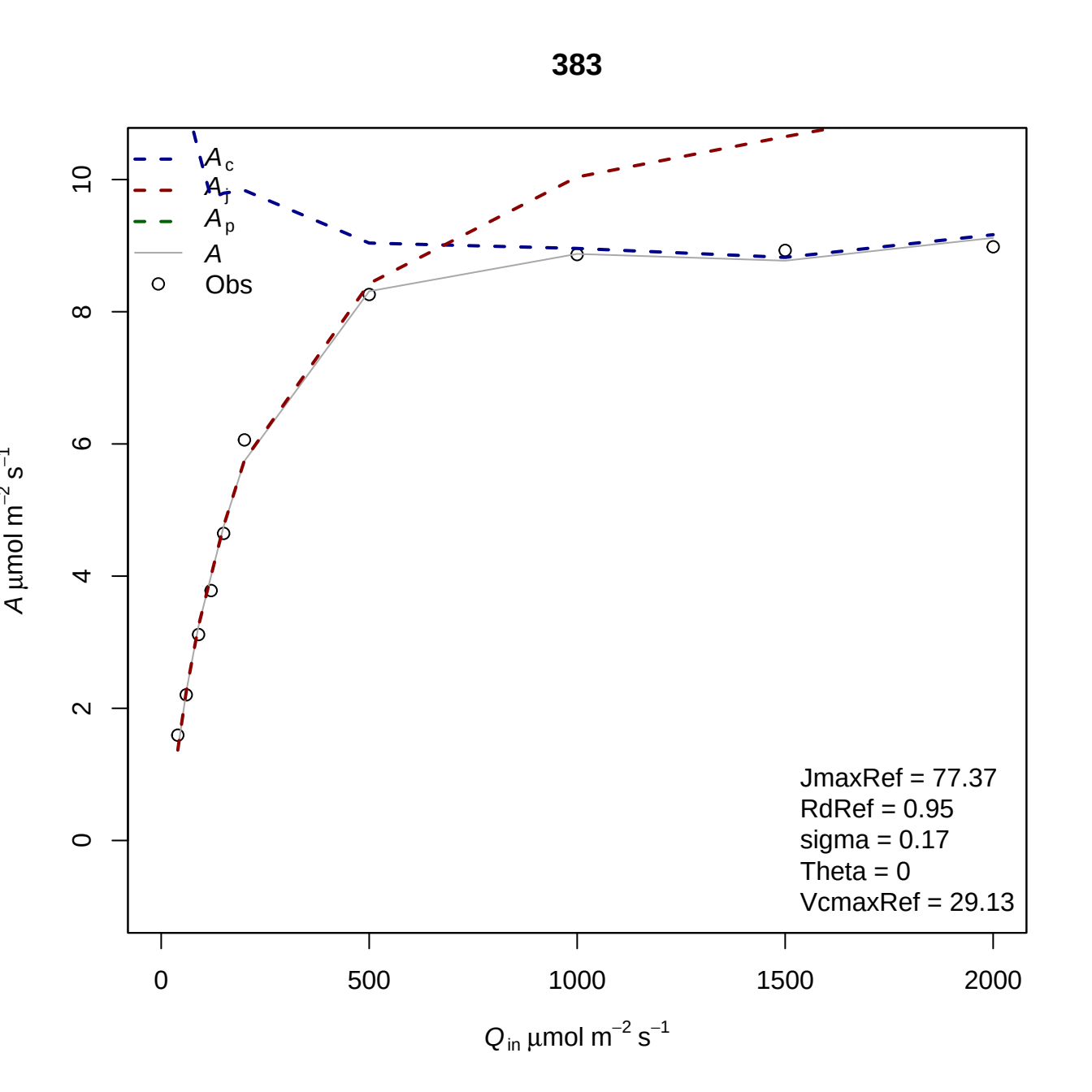

378
A_c
A_j
A_p
A
Obs
JmaxRef = 63.3
RdRef = 1.77
sigma = 0.33
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

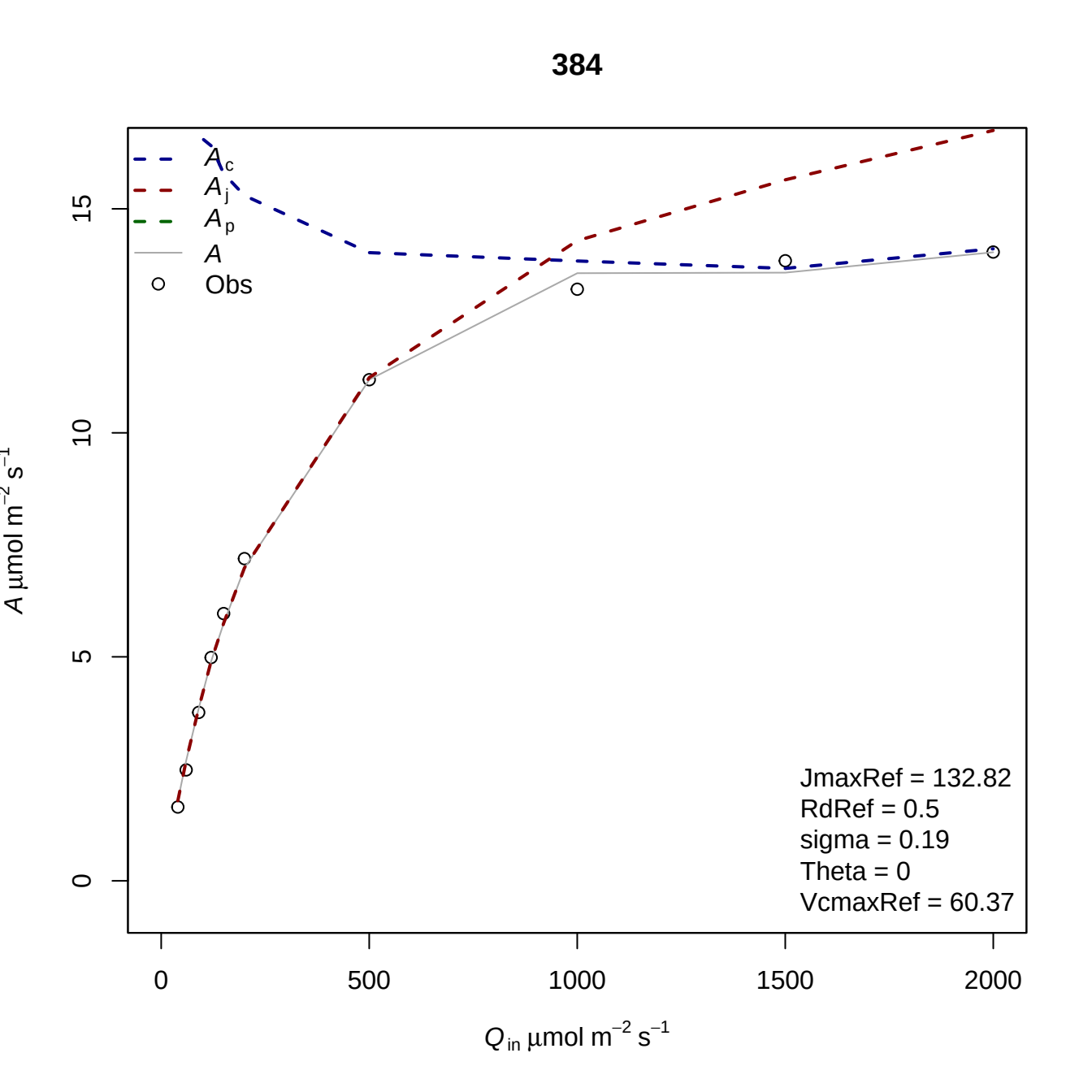

379
A_c
A_j
A_p
A
Obs
JmaxRef = 94.35
RdRef = 1.59
sigma = 0.22
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

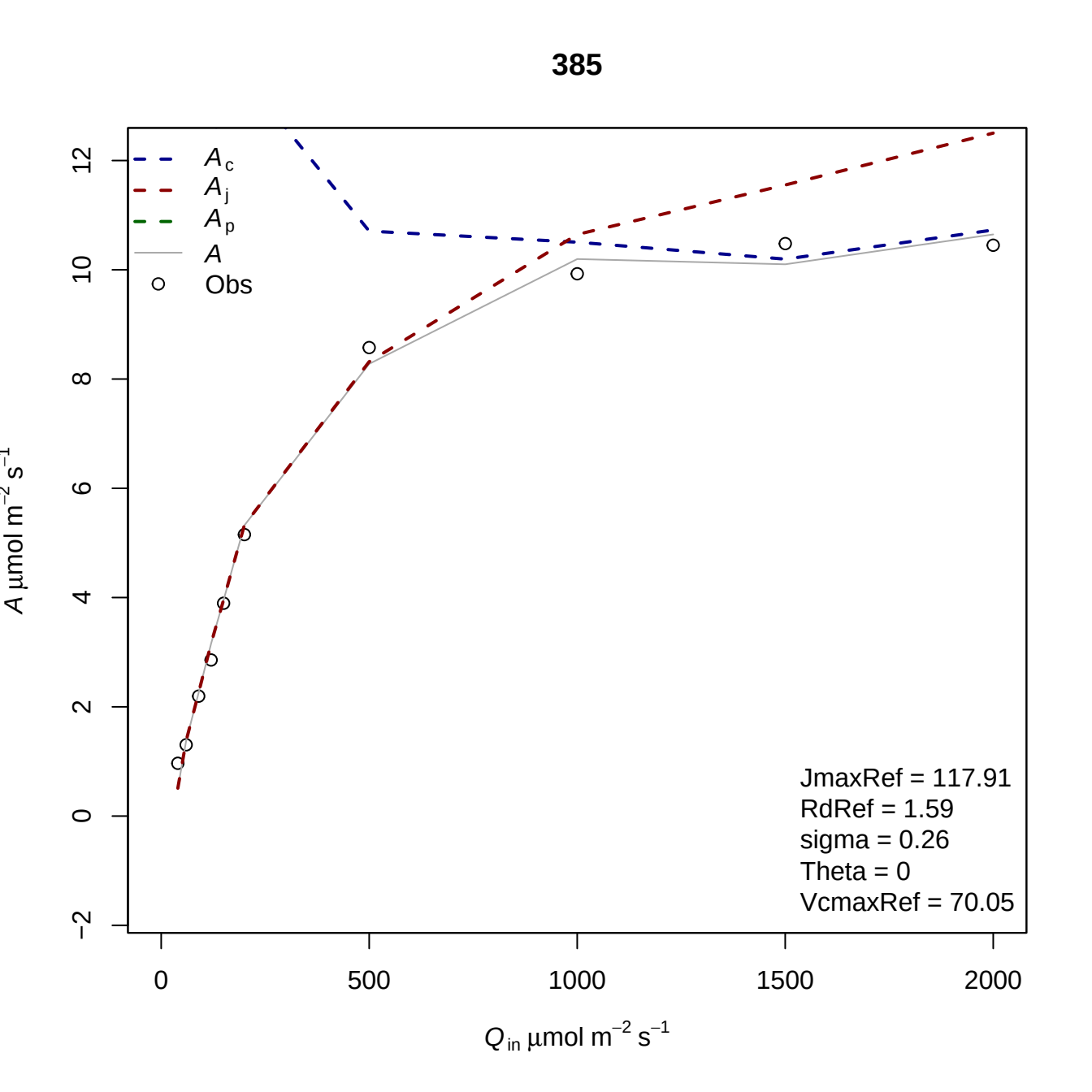

380
A_c
A_j
A_p
A
Obs
JmaxRef = 102.15
RdRef = 1.85
sigma = 0.15
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

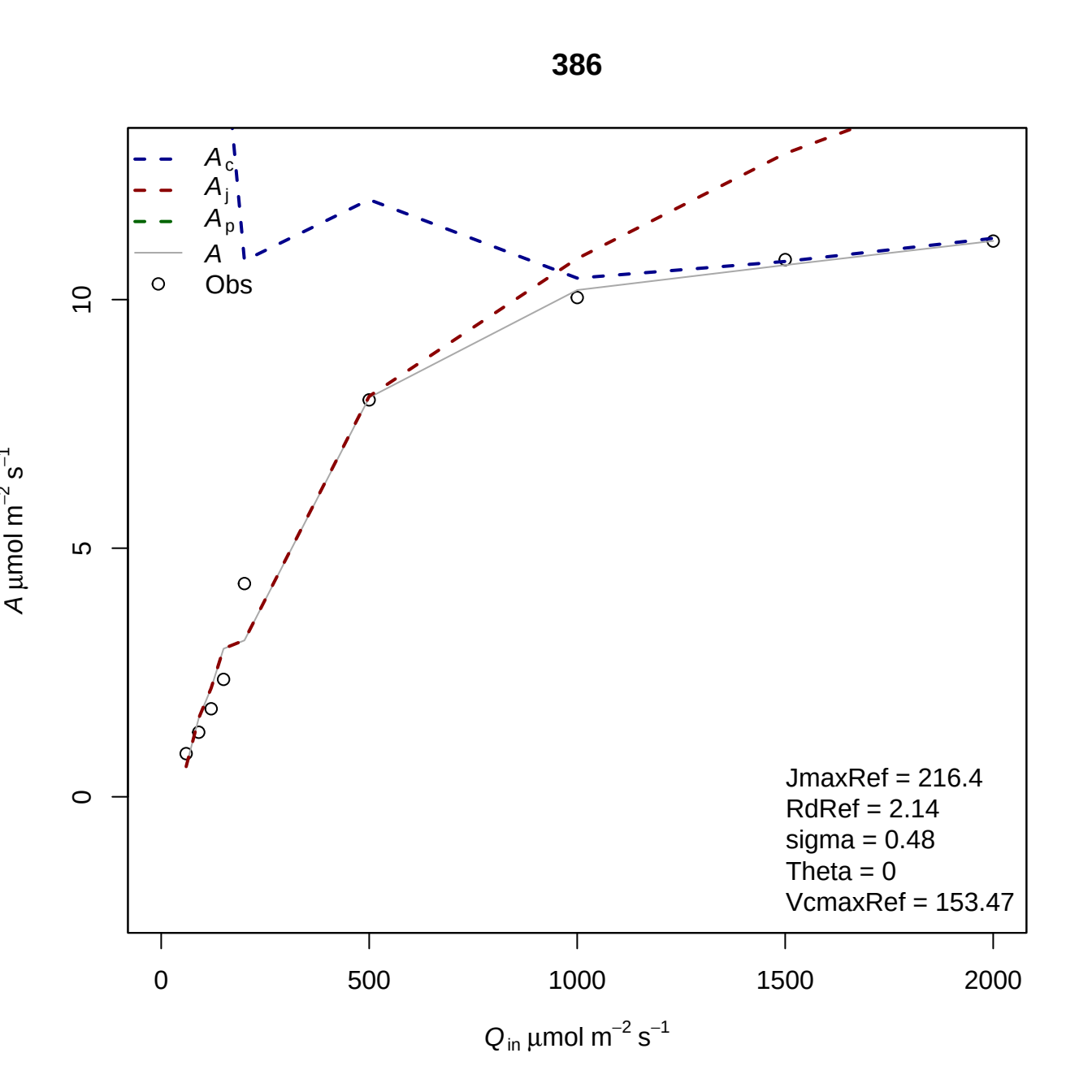

381
A_c
A_j
A_p
A
Obs
JmaxRef = 181.03
RdRef = 1.72
sigma = 0.45
Theta = 0
VcmaxRef = 179.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

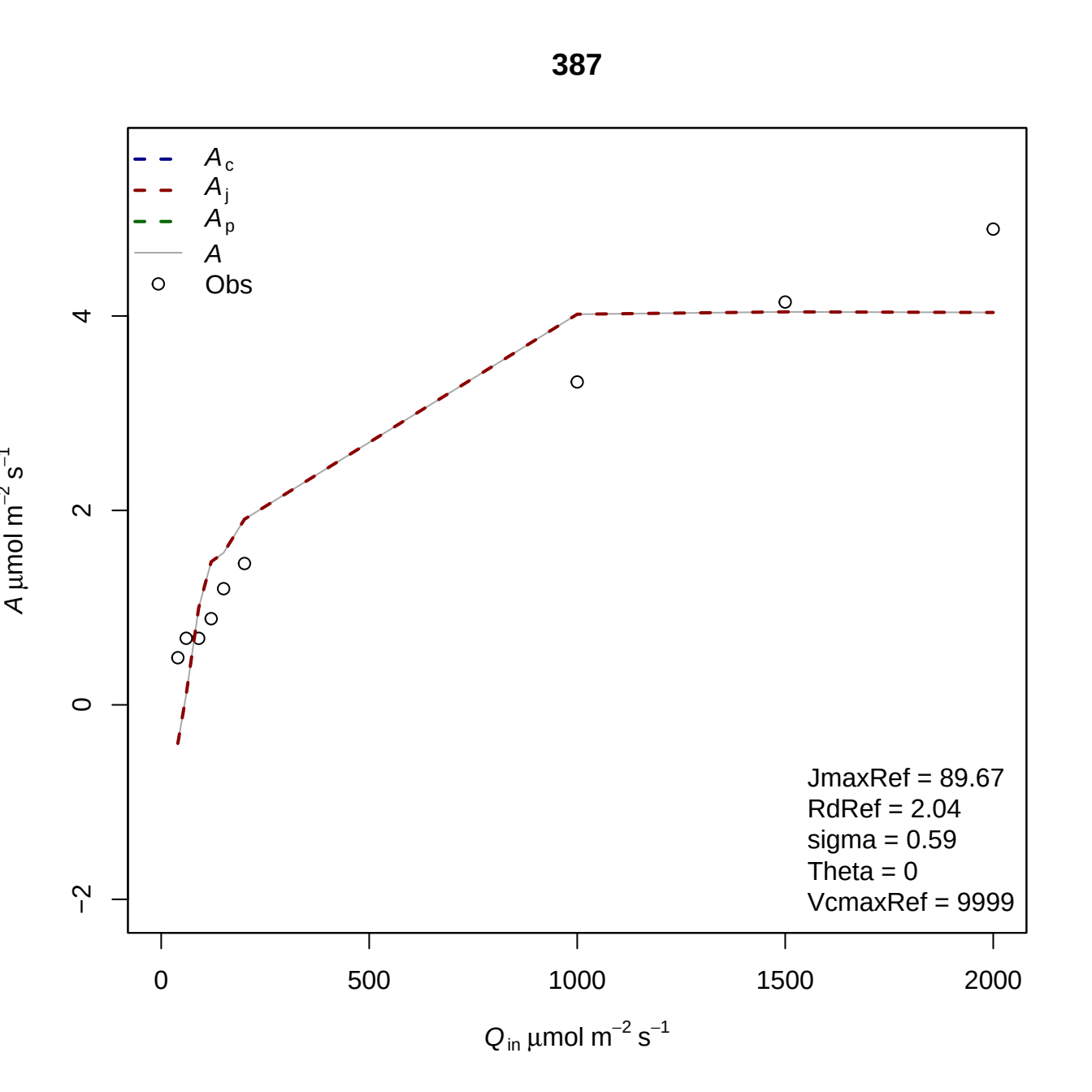

382
A_c
A_j
A_p
A
Obs
JmaxRef = 63.86
RdRef = 2.92
sigma = 0.2
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
−3
0
500
1000
1500
2000

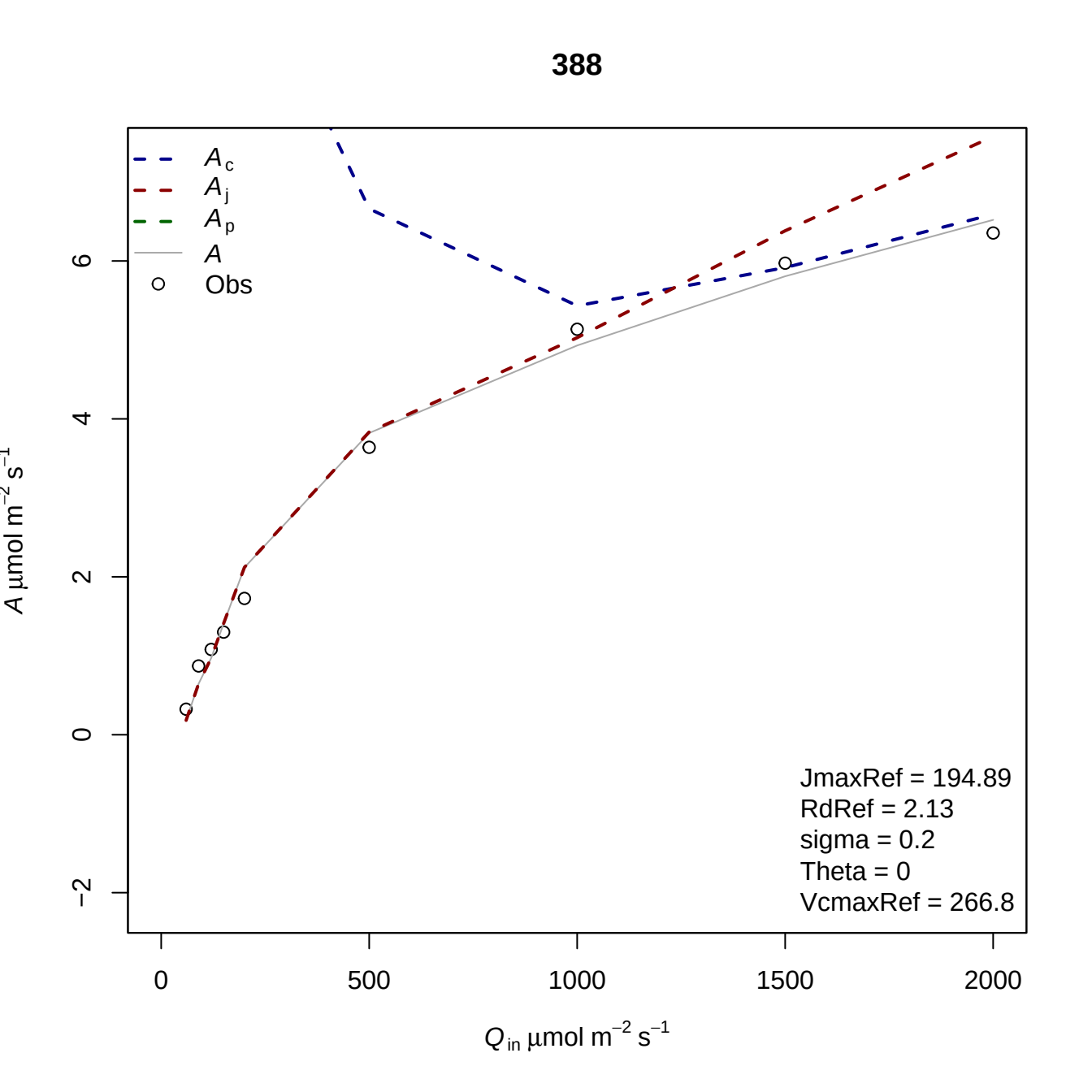

383
A_c
A_j
A_p
A
Obs
JmaxRef = 77.37
RdRef = 0.95
sigma = 0.17
Theta = 0
VcmaxRef = 29.13
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

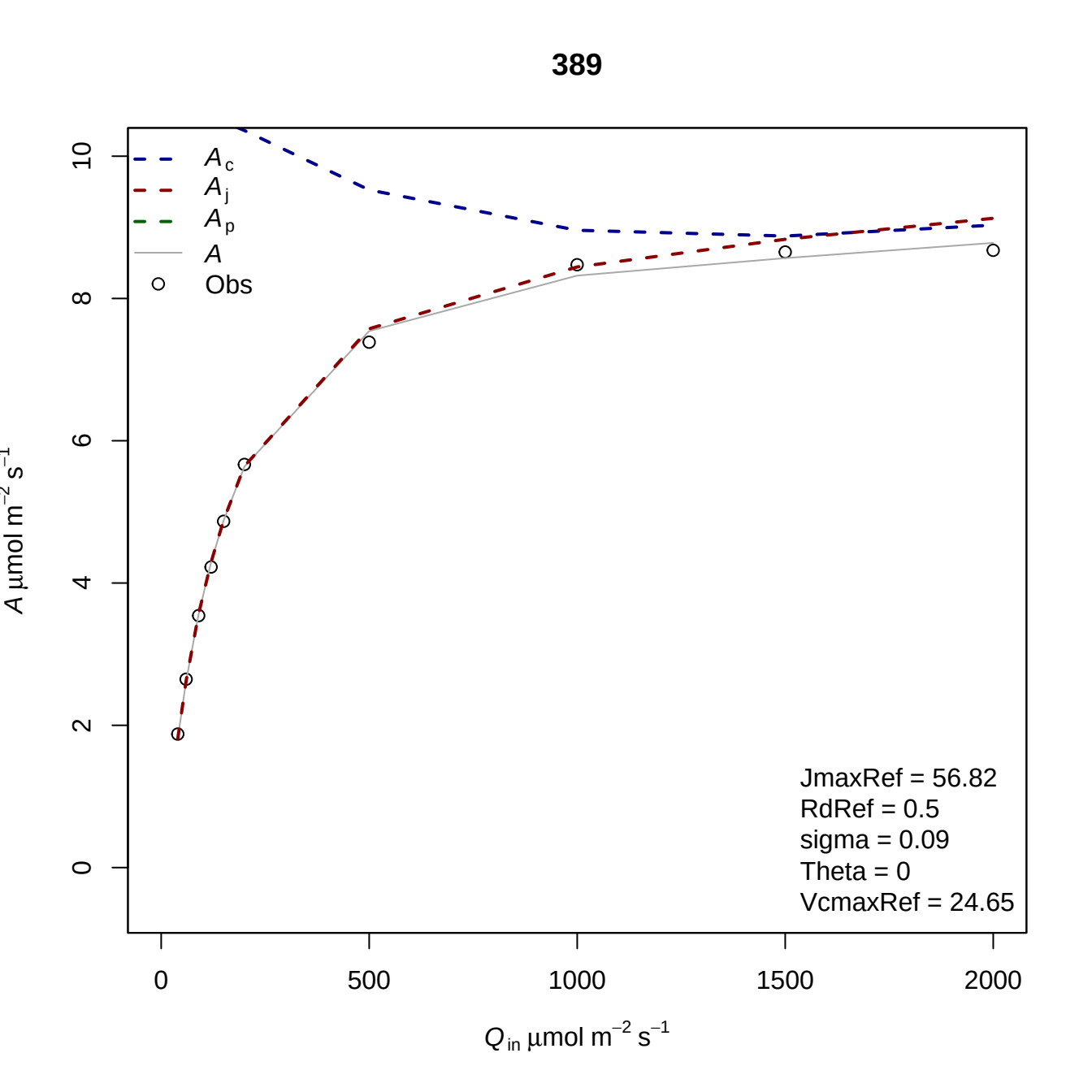

384
A_c
A_j
A_p
A
Obs
JmaxRef = 132.82
RdRef = 0.5
sigma = 0.19
Theta = 0
VcmaxRef = 60.37
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

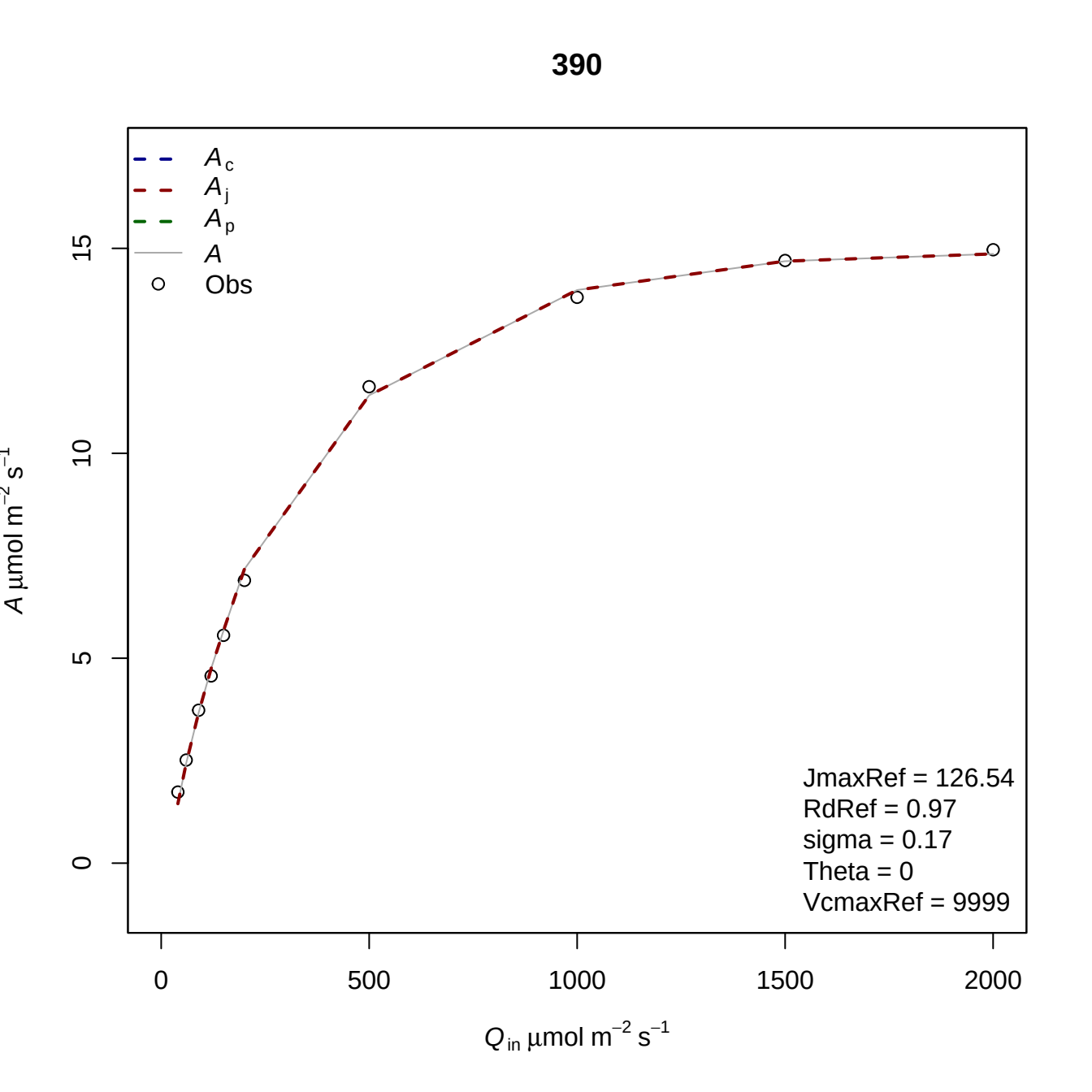

385
A_c
A_j
A_p
A
Obs
JmaxRef = 117.91
RdRef = 1.59
sigma = 0.26
Theta = 0
VcmaxRef = 70.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

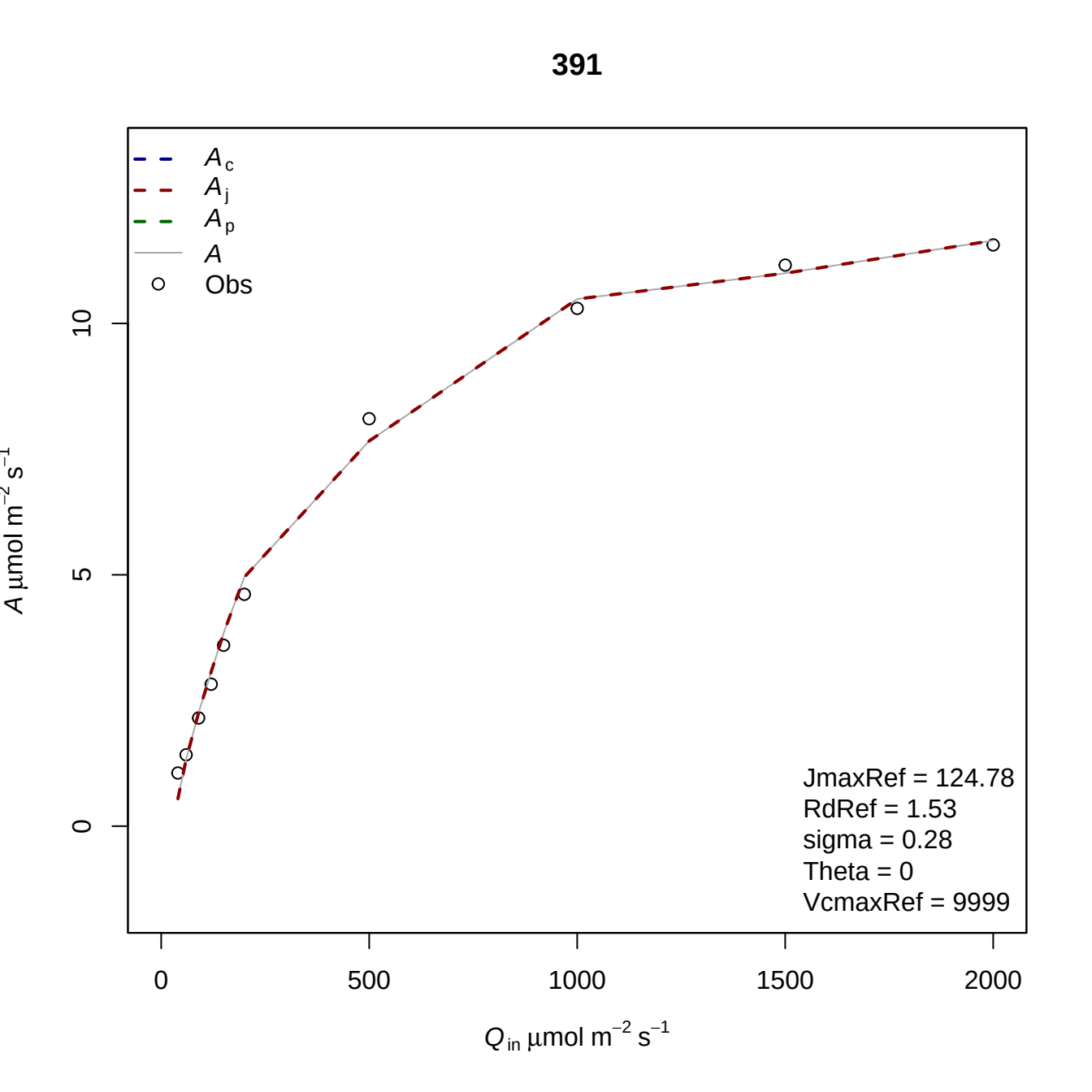

386
A_c
A_j
A_p
A
Obs
JmaxRef = 216.4
RdRef = 2.14
sigma = 0.48
Theta = 0
VcmaxRef = 153.47
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

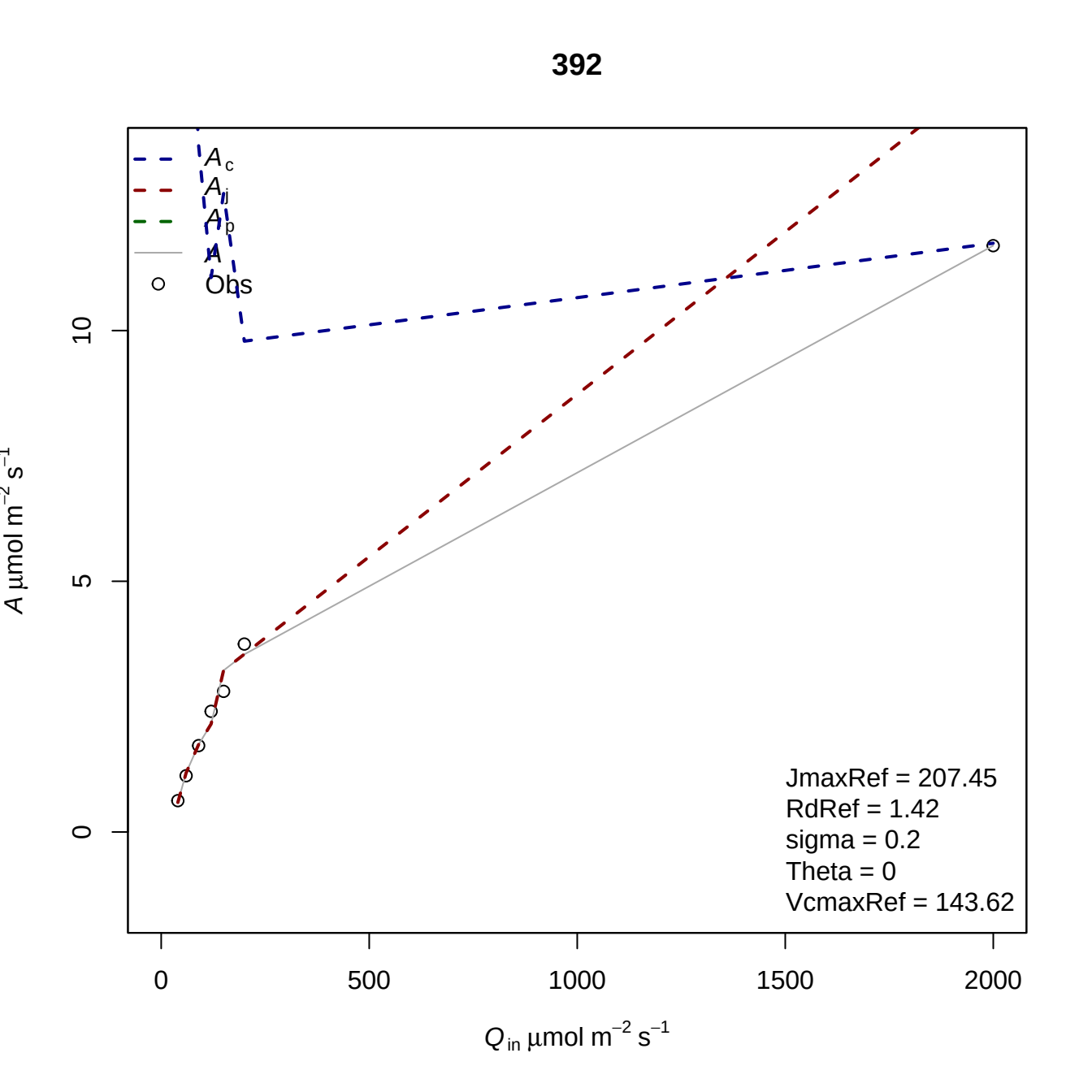

387
A_c
A_j
A_p
A
Obs
JmaxRef = 89.67
RdRef = 2.04
sigma = 0.59
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

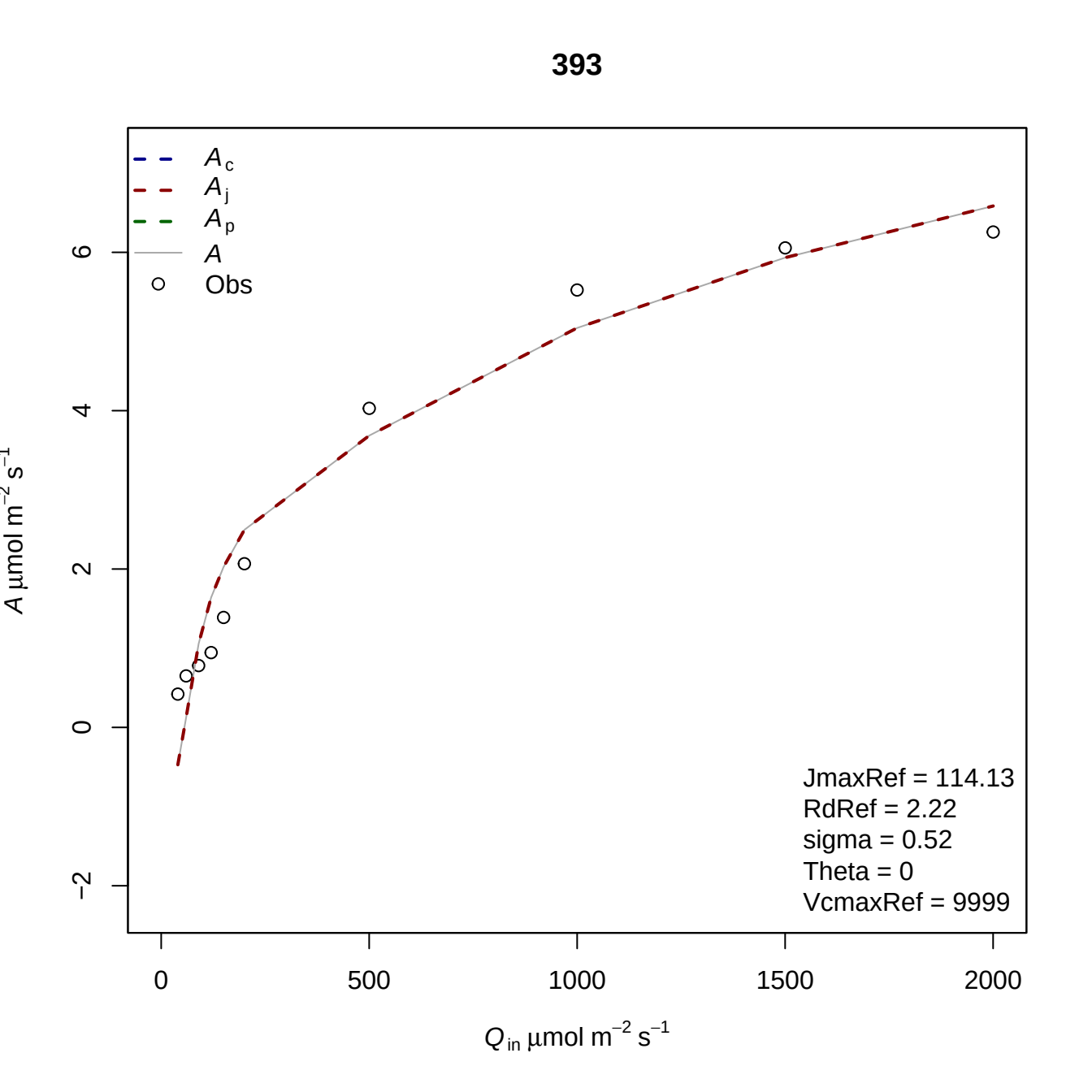

388
A_c
A_j
A_p
A
Obs
JmaxRef = 194.89
RdRef = 2.13
sigma = 0.2
Theta = 0
VcmaxRef = 266.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

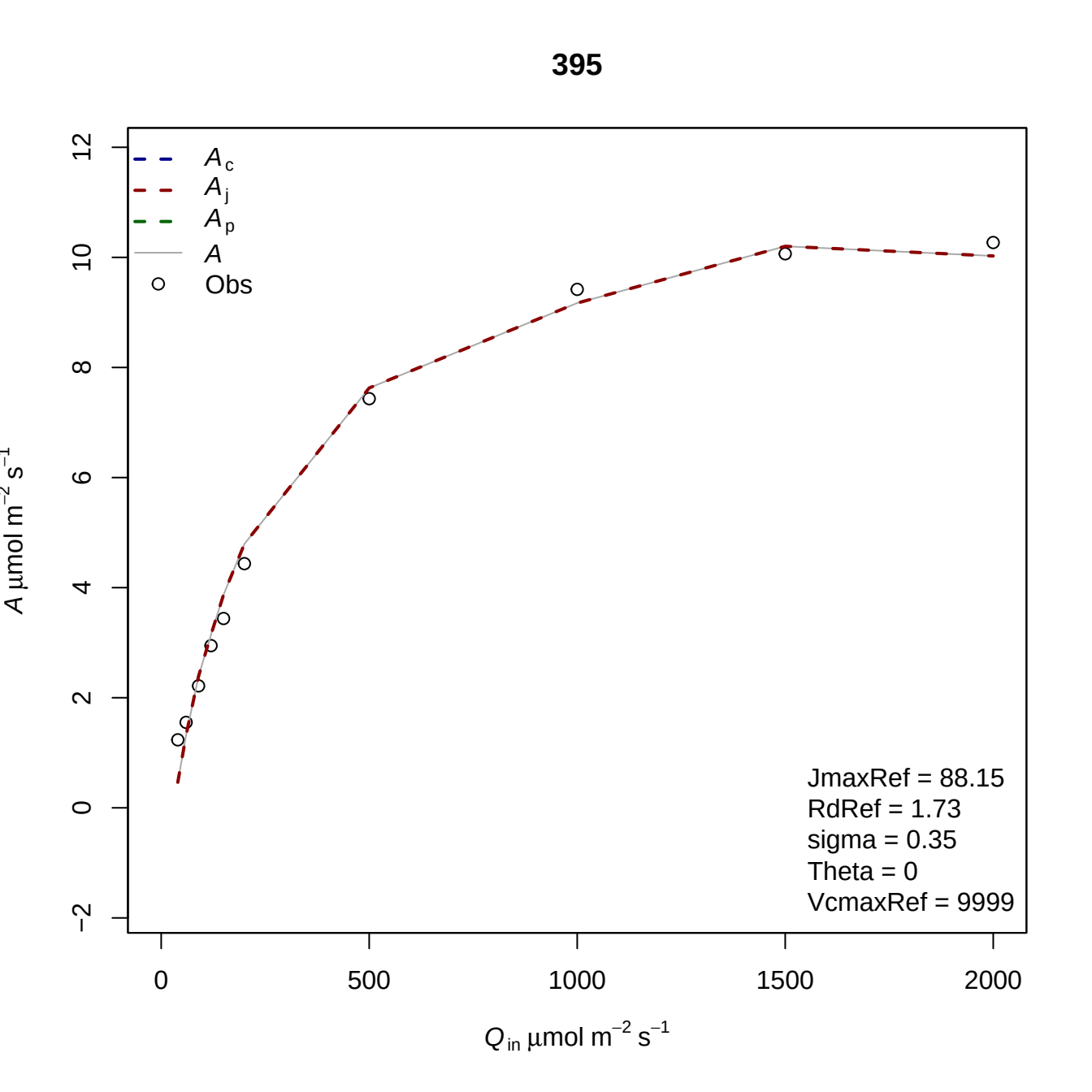

389
A_c
A_j
A_p
A
Obs
JmaxRef = 56.82
RdRef = 0.5
sigma = 0.09
Theta = 0
VcmaxRef = 24.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

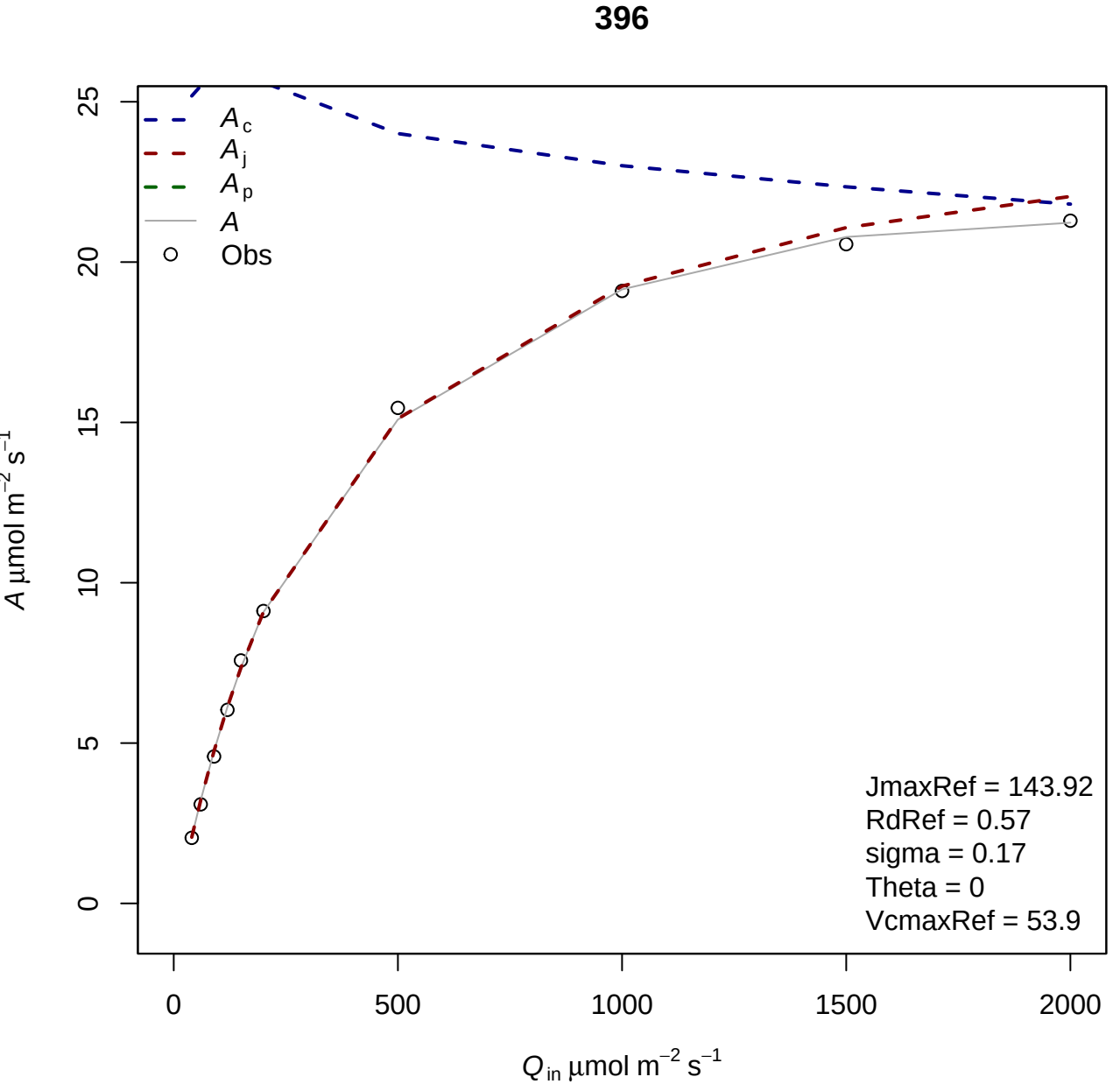

390
A_c
A_j
A_p
A
Obs
JmaxRef = 126.54
RdRef = 0.97
sigma = 0.17
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

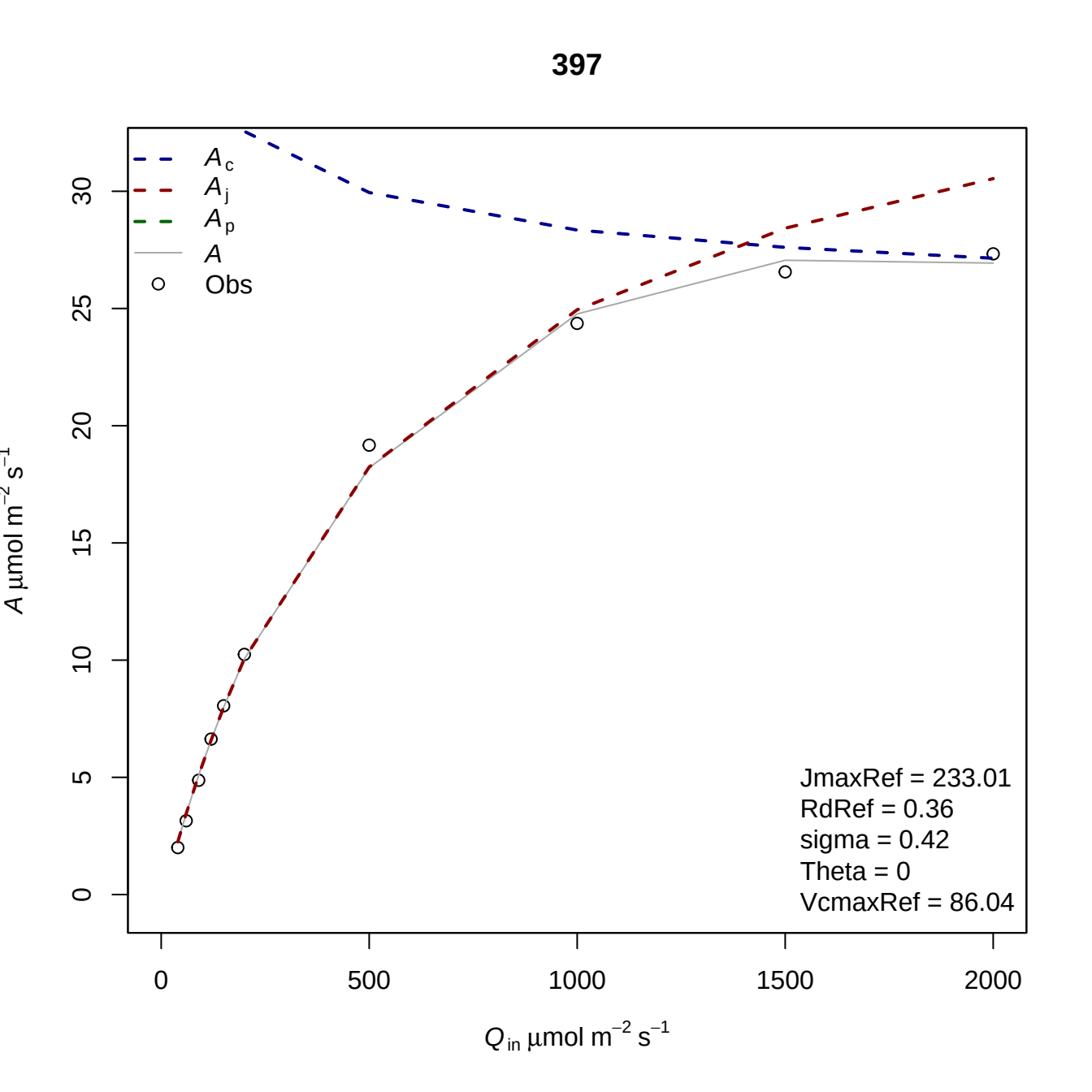

391
A_c
A_j
A_p
A
Obs
JmaxRef = 124.78
RdRef = 1.53
sigma = 0.28
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

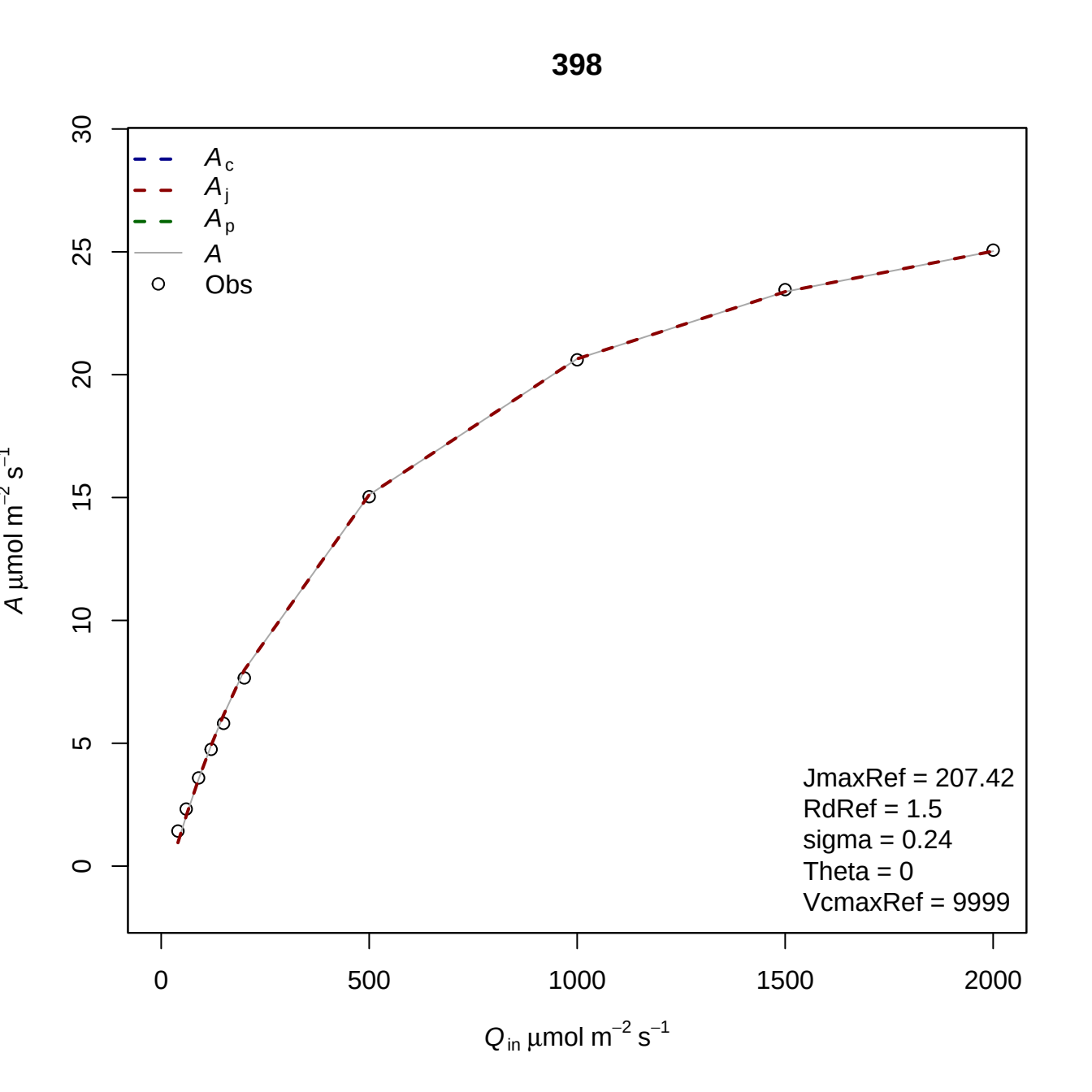

392
A_c
A_j
A_p
A
Obs
JmaxRef = 207.45
RdRef = 1.42
sigma = 0.2
Theta = 0
VcmaxRef = 143.62
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

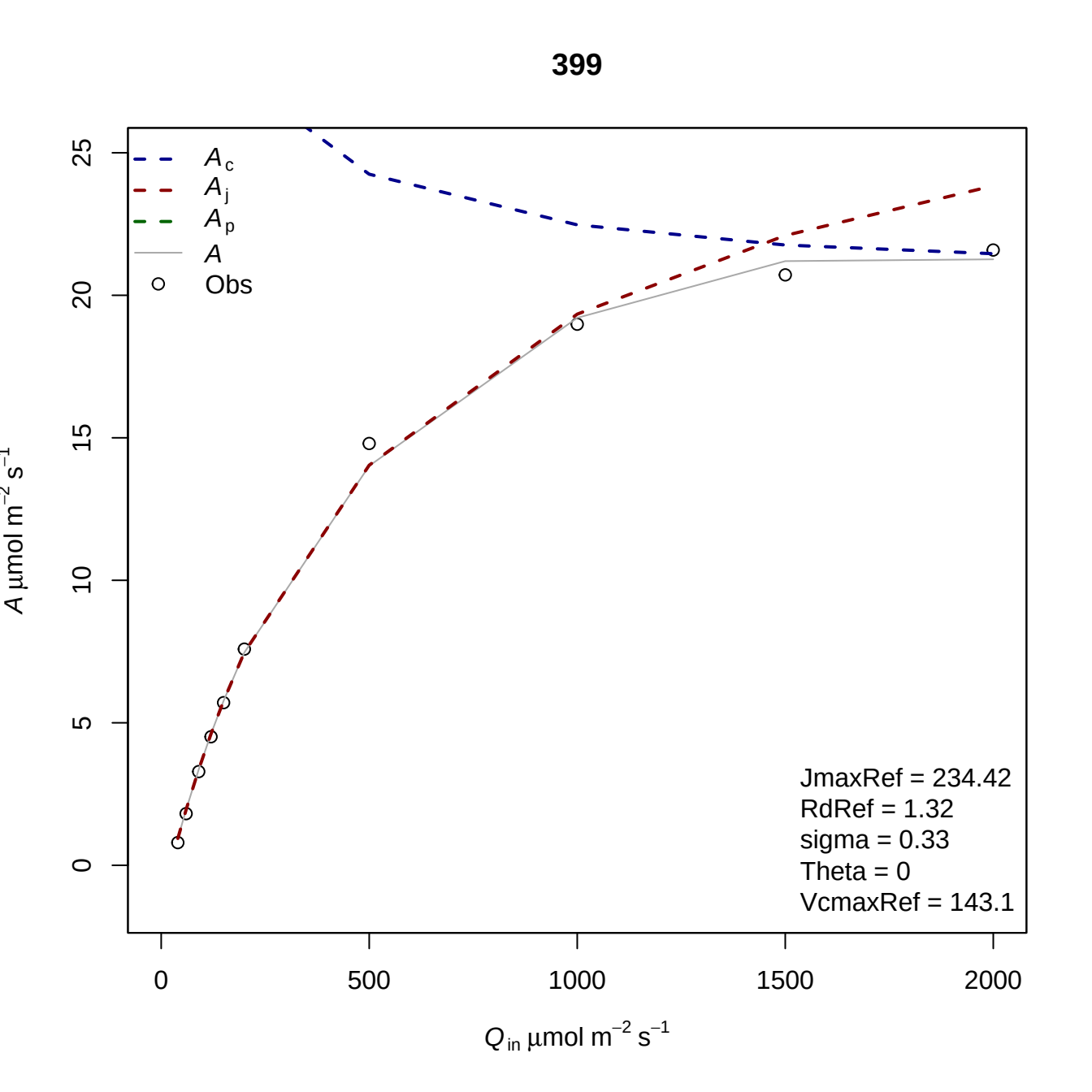

393
A_c
A_j
A_p
A
Obs
JmaxRef = 114.13
RdRef = 2.22
sigma = 0.52
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

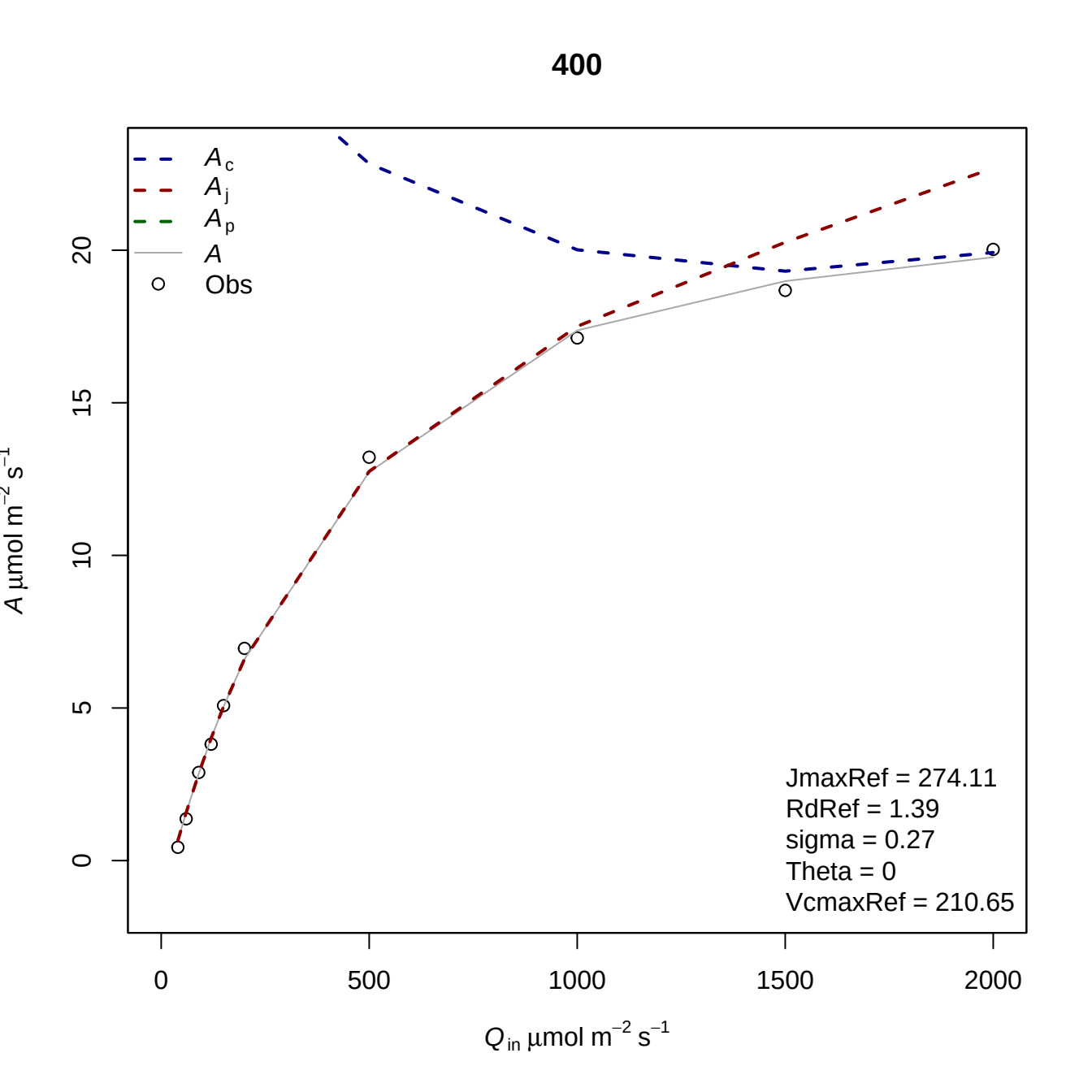

395
A_c
A_j
A_p
A
Obs
JmaxRef = 88.15
RdRef = 1.73
sigma = 0.35
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

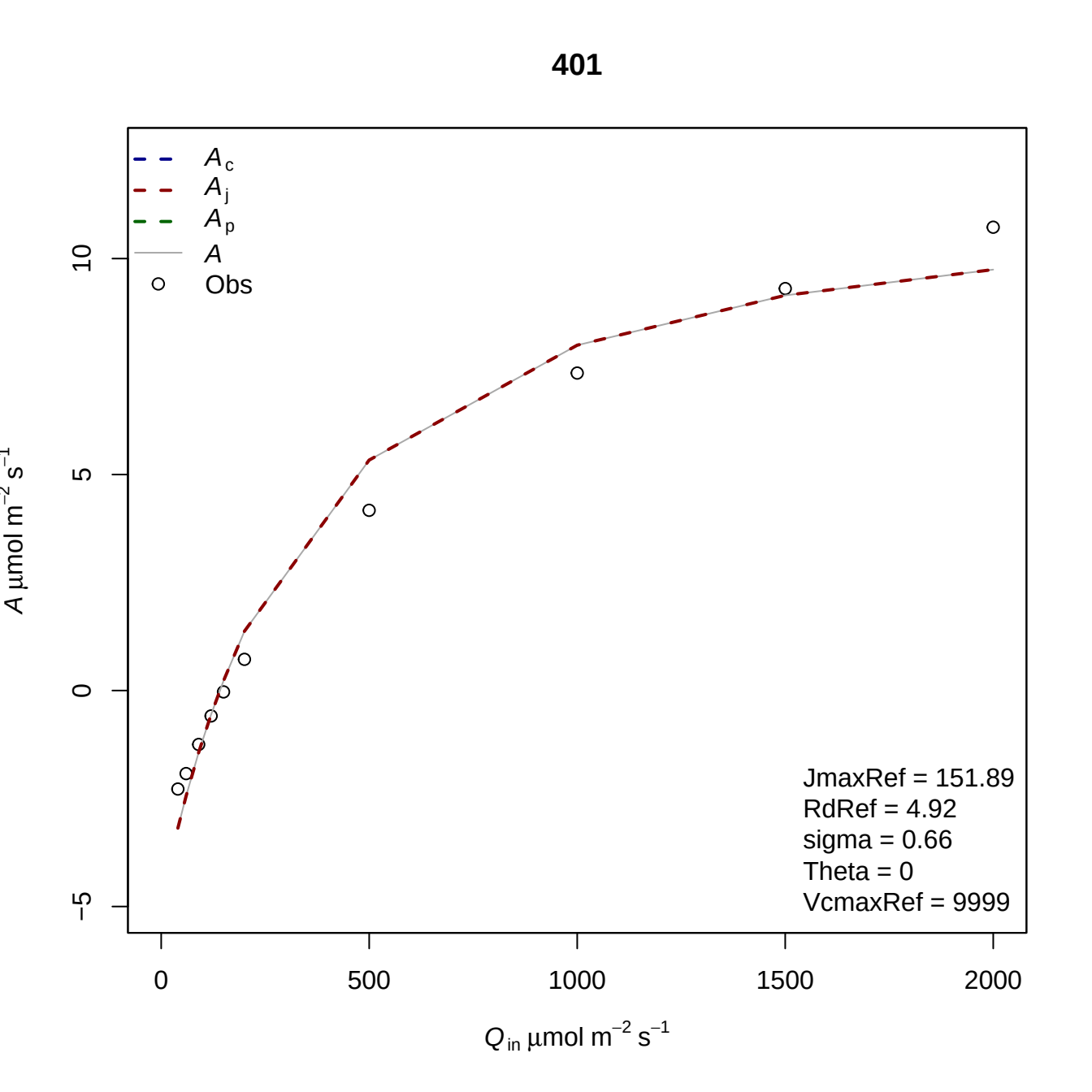

396
A_c
A_j
A_p
A
Obs
JmaxRef = 143.92
RdRef = 0.57
sigma = 0.17
Theta = 0
VcmaxRef = 53.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

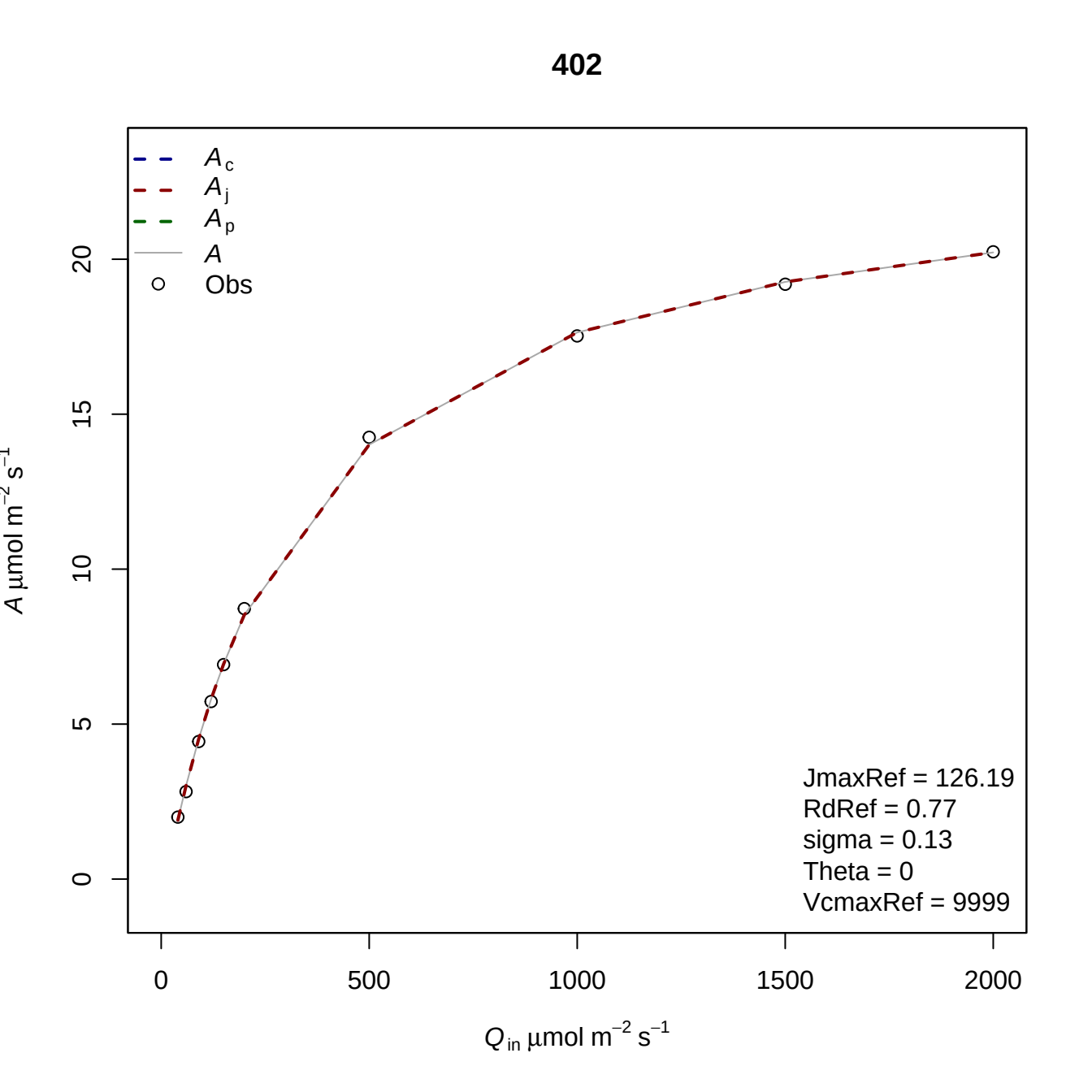

397
A_c
A_j
A_p
A
Obs
JmaxRef = 233.01
RdRef = 0.36
sigma = 0.42
Theta = 0
VcmaxRef = 86.04
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

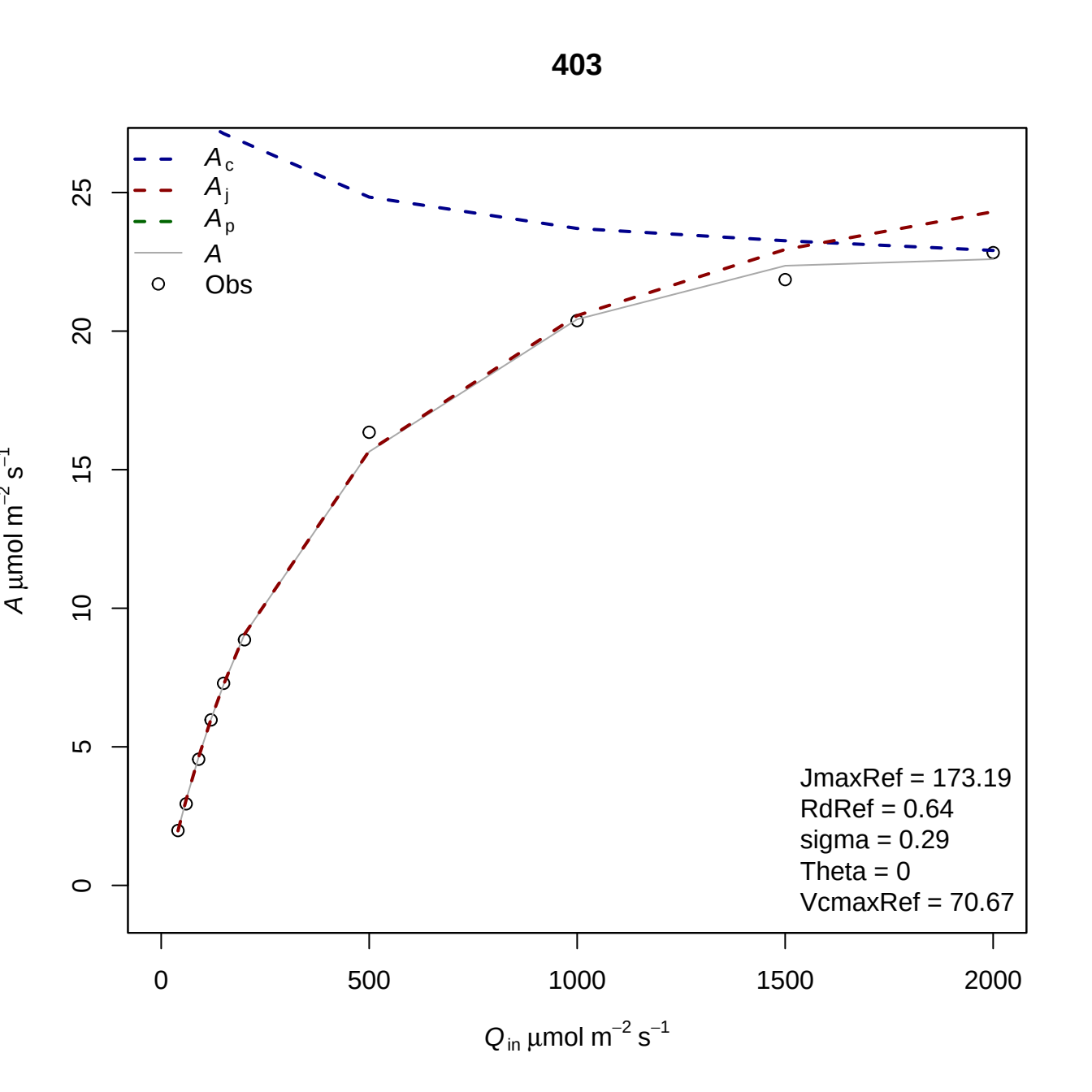

398
A_c
A_j
A_p
A
Obs
JmaxRef = 207.42
RdRef = 1.5
sigma = 0.24
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

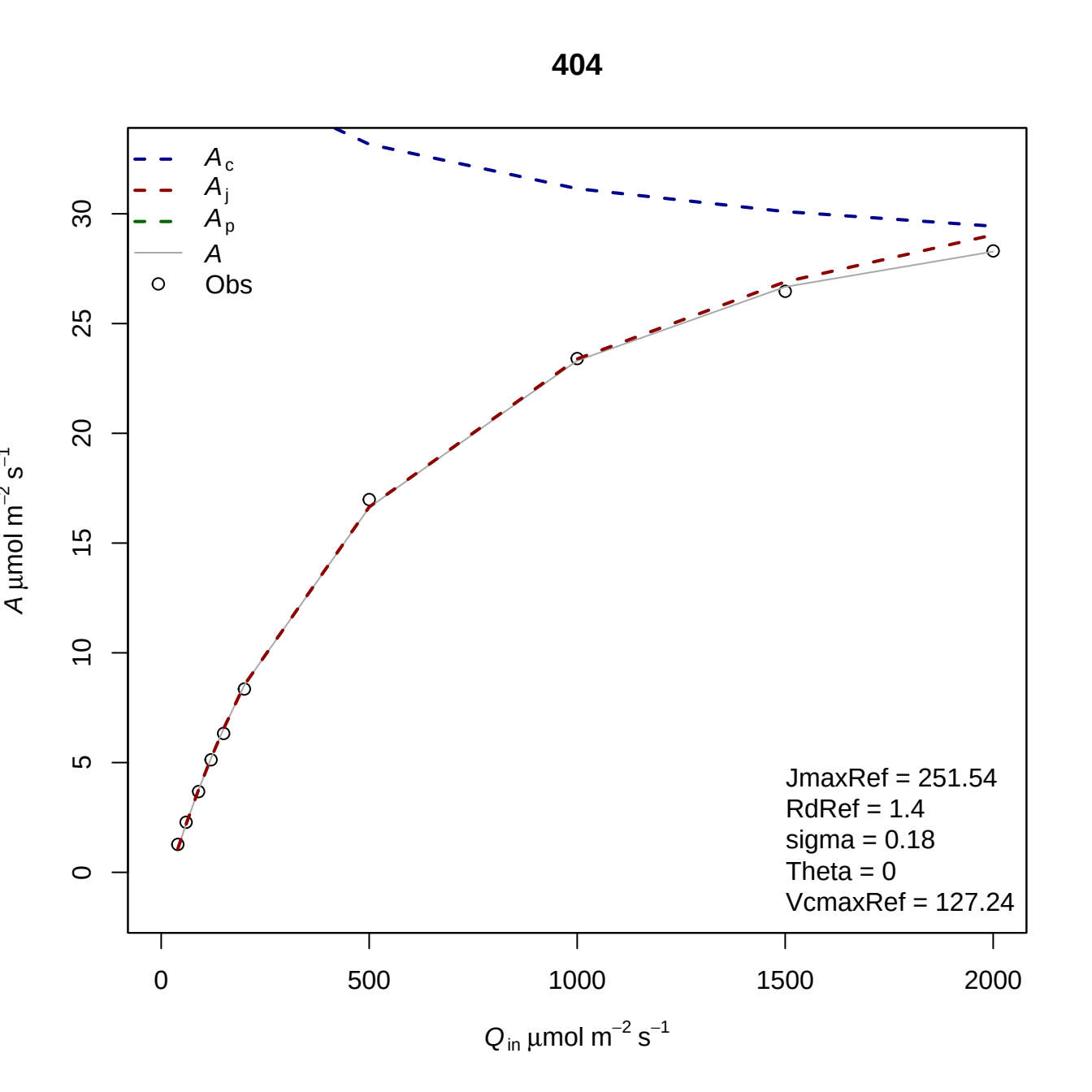

399
A_c
A_j
A_p
A
Obs
JmaxRef = 234.42
RdRef = 1.32
sigma = 0.33
Theta = 0
VcmaxRef = 143.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

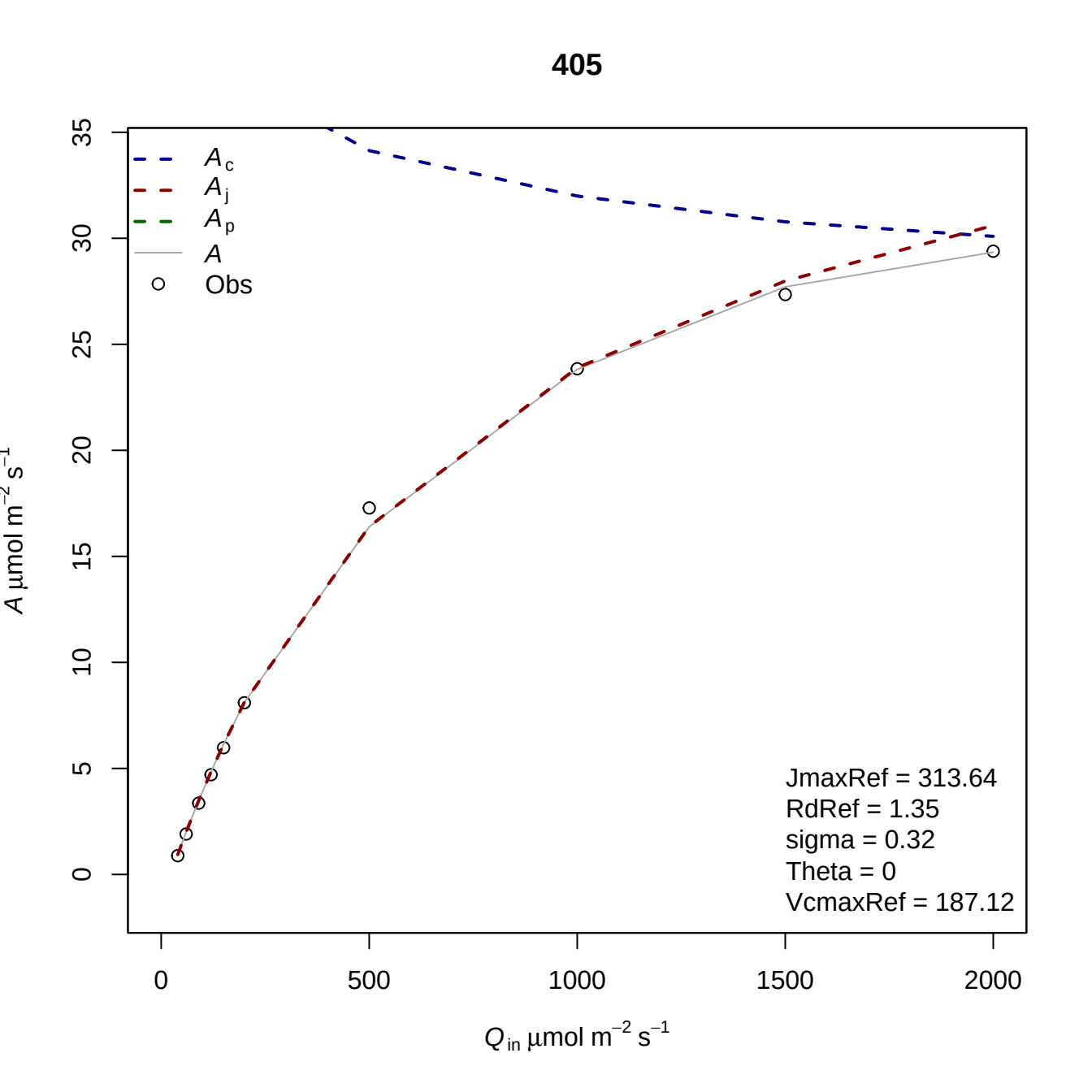

400
A_c
A_j
A_p
A
Obs
JmaxRef = 274.11
RdRef = 1.39
sigma = 0.27
Theta = 0
VcmaxRef = 210.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

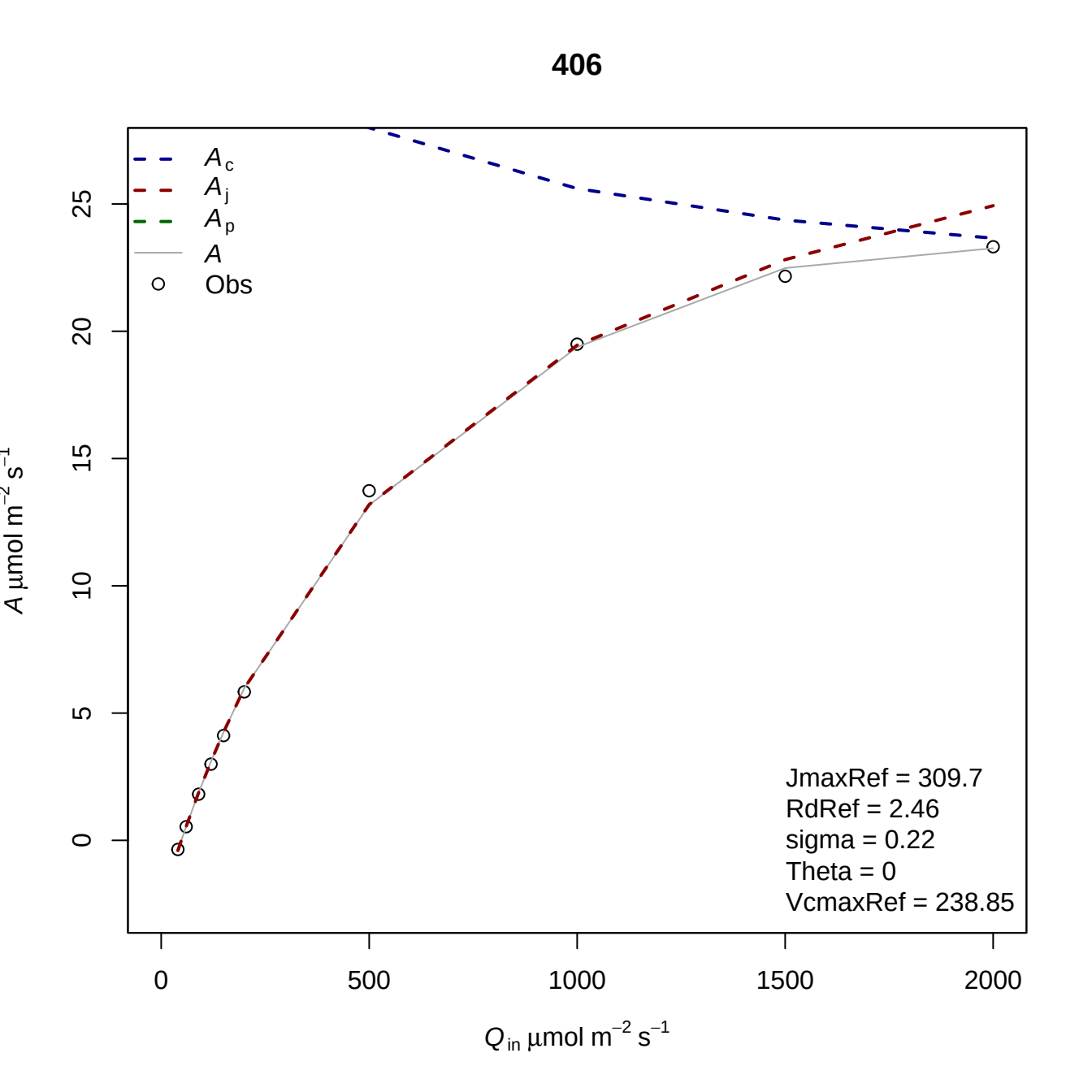

401
A_c
A_j
A_p
A
Obs
JmaxRef = 151.89
RdRef = 4.92
sigma = 0.66
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

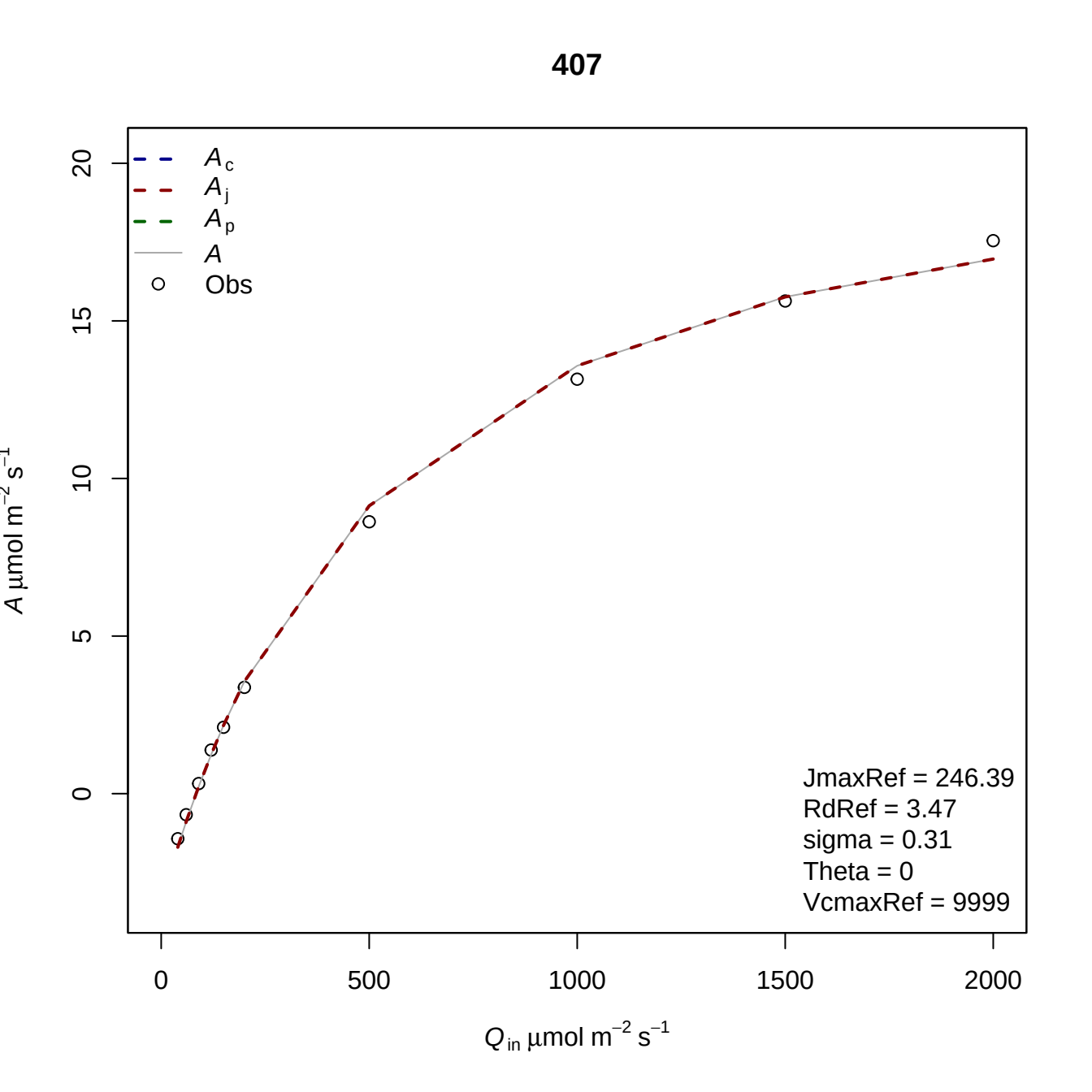

402
A_c
A_j
A_p
A
Obs
JmaxRef = 126.19
RdRef = 0.77
sigma = 0.13
Theta = 0
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

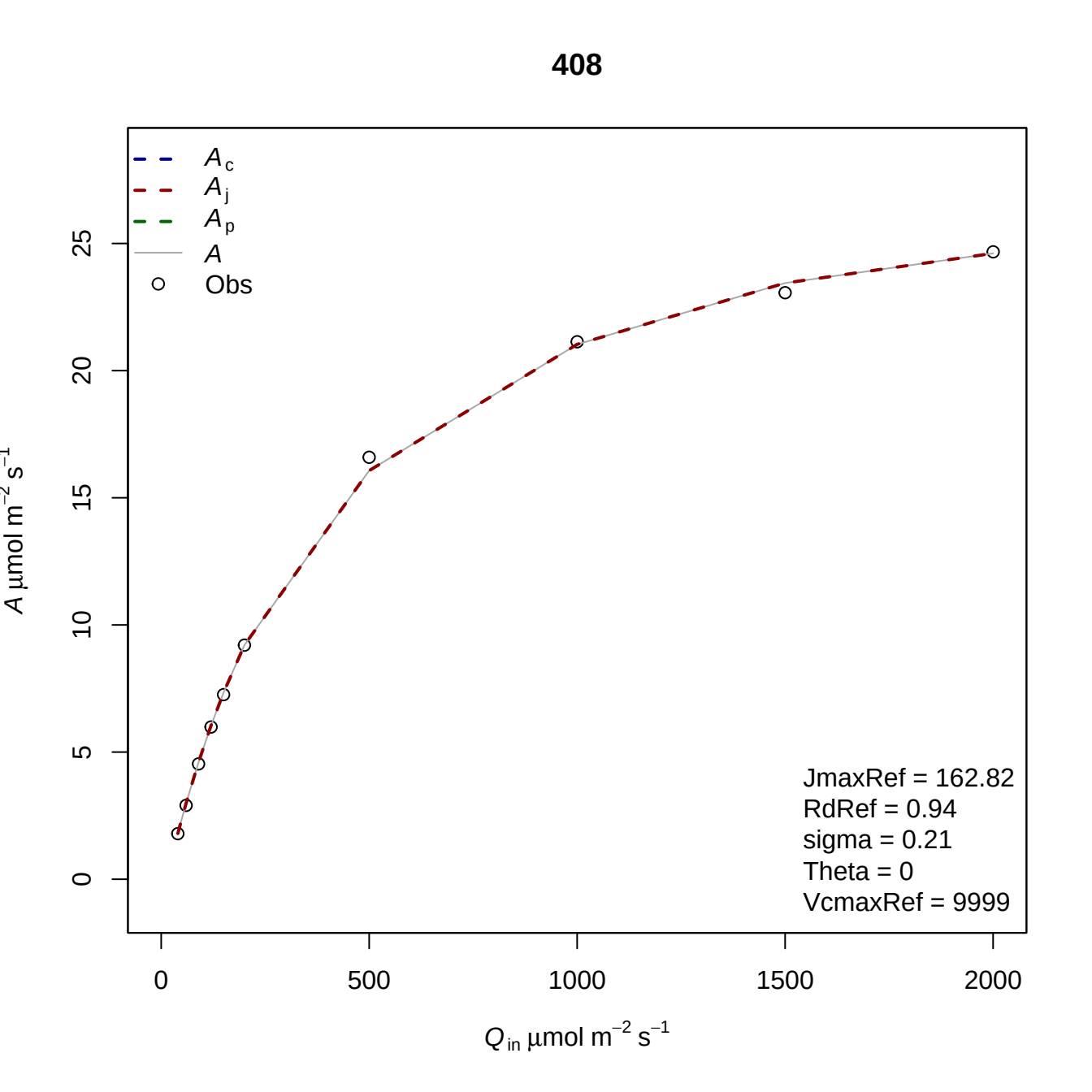

403
A_c
A_j
A_p
A
Obs
JmaxRef = 173.19
RdRef = 0.64
sigma = 0.29
Theta = 0
VcmaxRef = 70.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

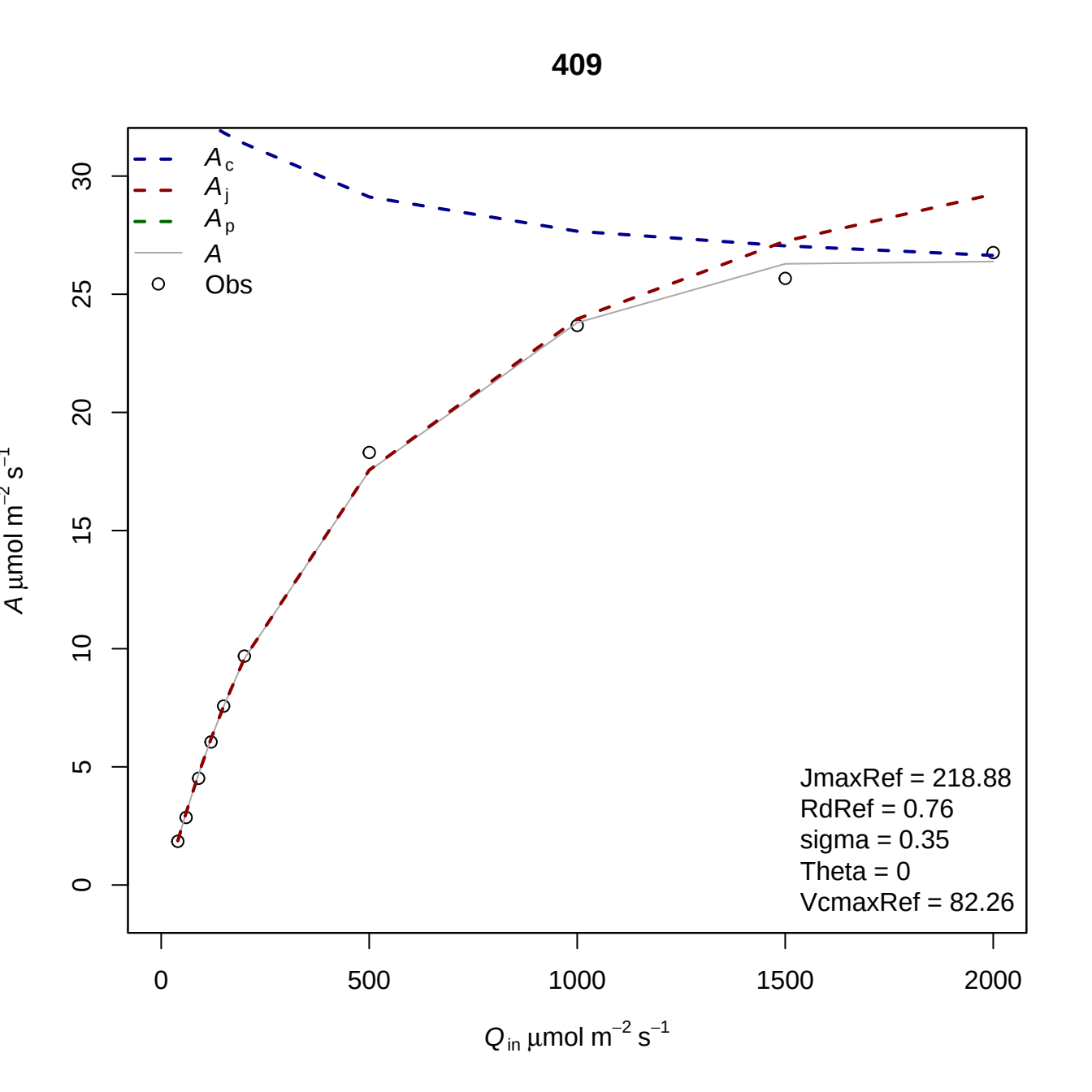

404
A_c
A_j
A_p
A
Obs
JmaxRef = 251.54
RdRef = 1.4
sigma = 0.18
Theta = 0
VcmaxRef = 127.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

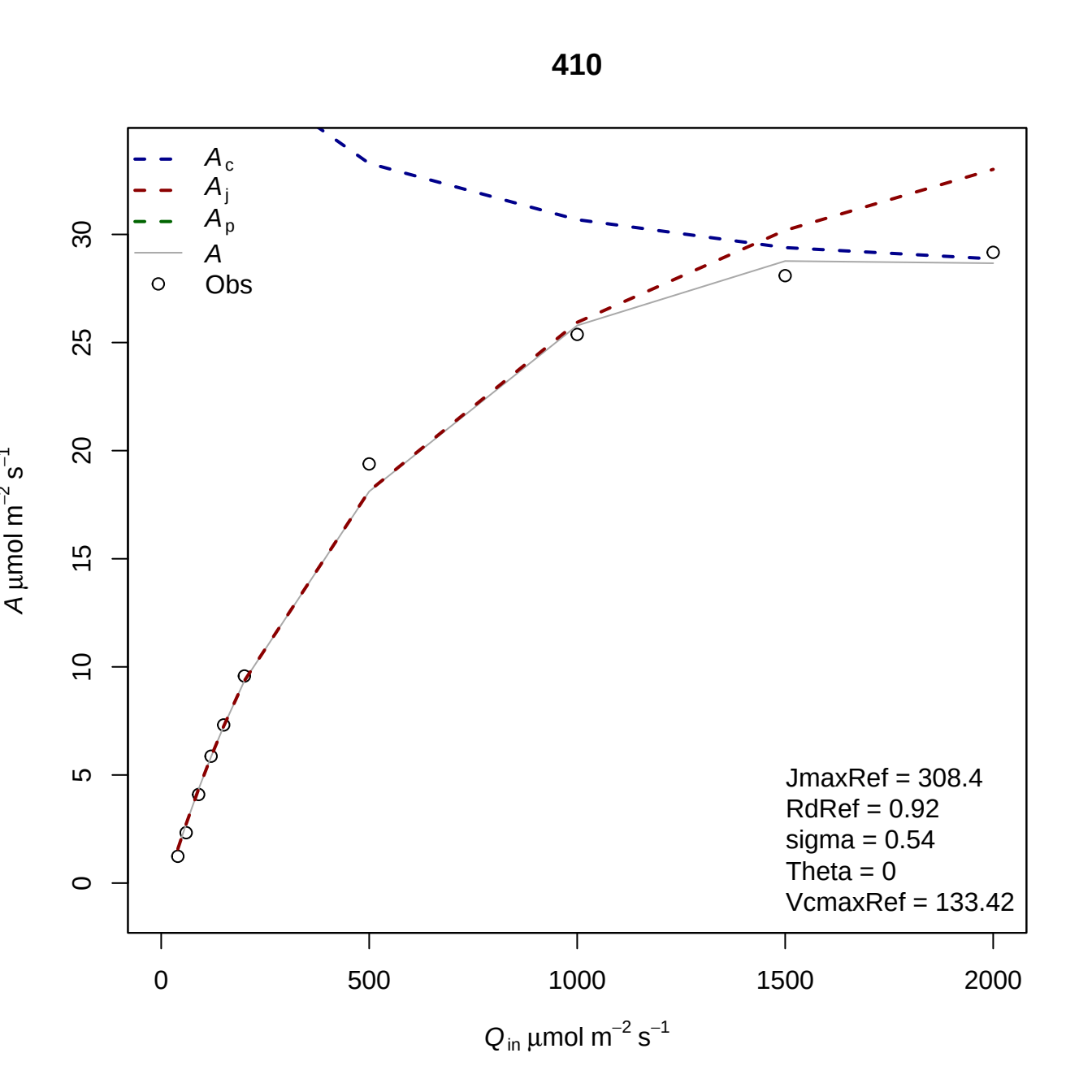

405
A_c
A_j
A_p
A
Obs
JmaxRef = 313.64
RdRef = 1.35
sigma = 0.32
Theta = 0
VcmaxRef = 187.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

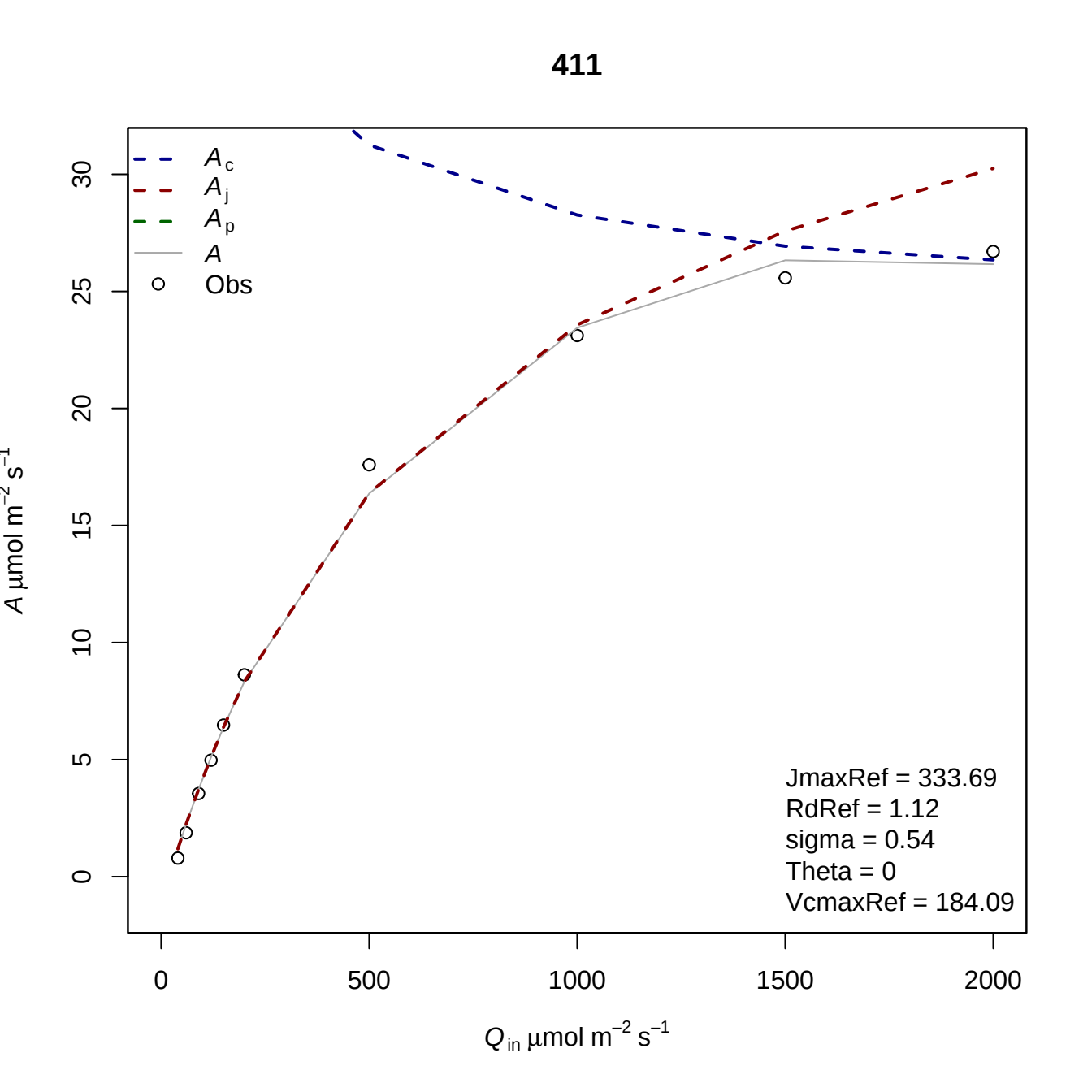

406
A_c
A_j
A_p
A
Obs
JmaxRef = 309.7
RdRef = 2.46
sigma = 0.22
Theta = 0
VcmaxRef = 238.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

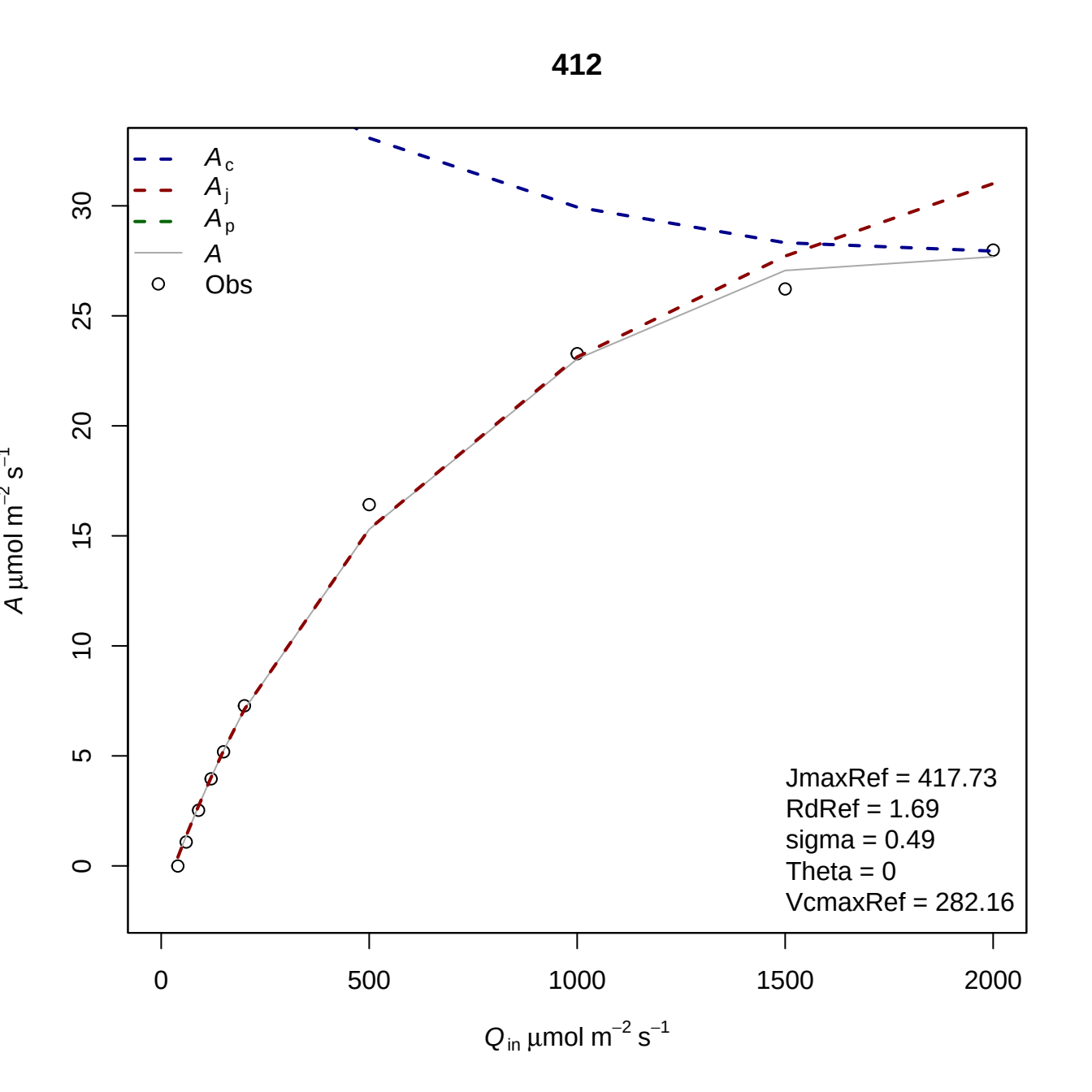

407
A_c
A_j
A_p
A
Obs
JmaxRef = 246.39
RdRef = 3.47
sigma = 0.31
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

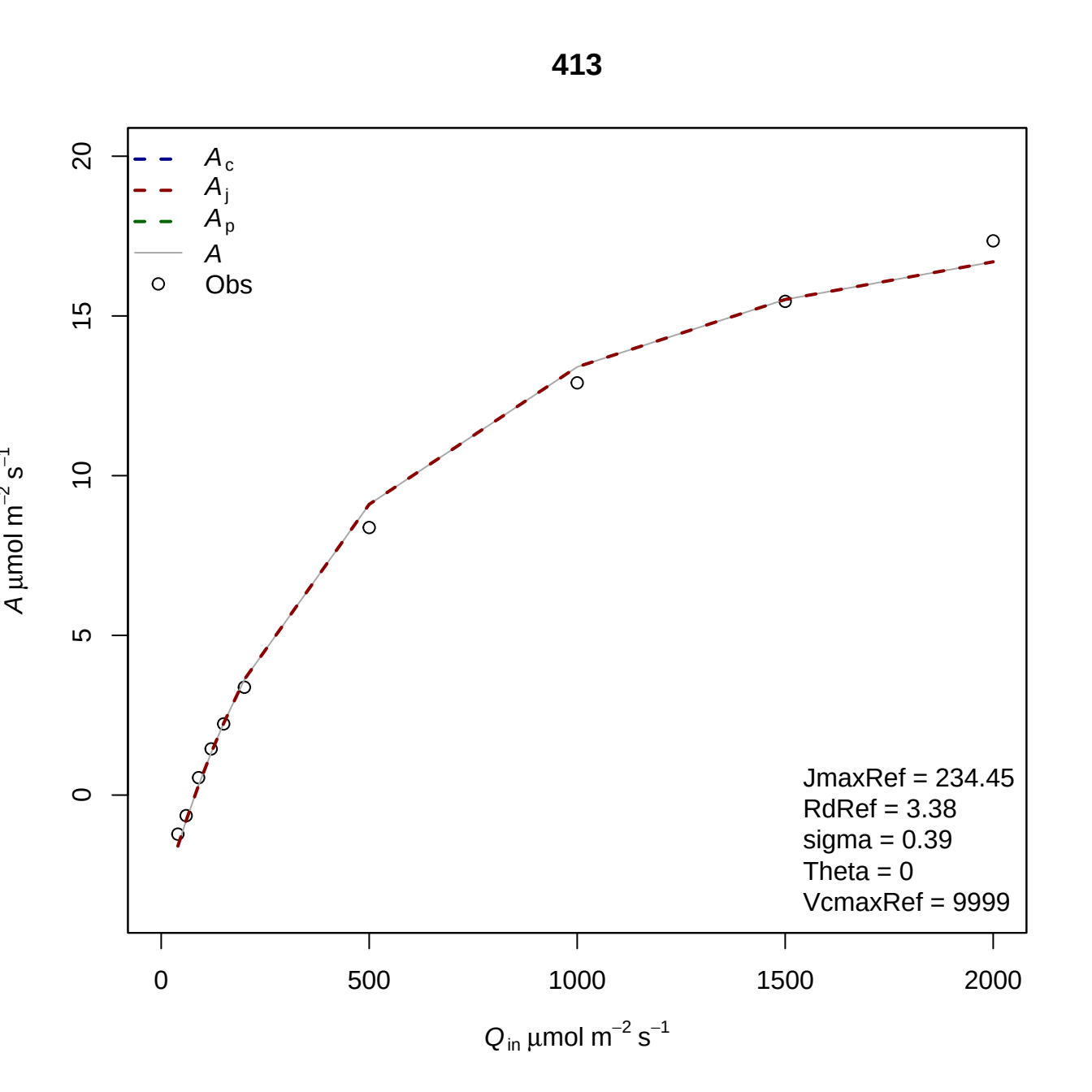

408
A_c
A_j
A_p
A
Obs
JmaxRef = 162.82
RdRef = 0.94
sigma = 0.21
Theta = 0
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

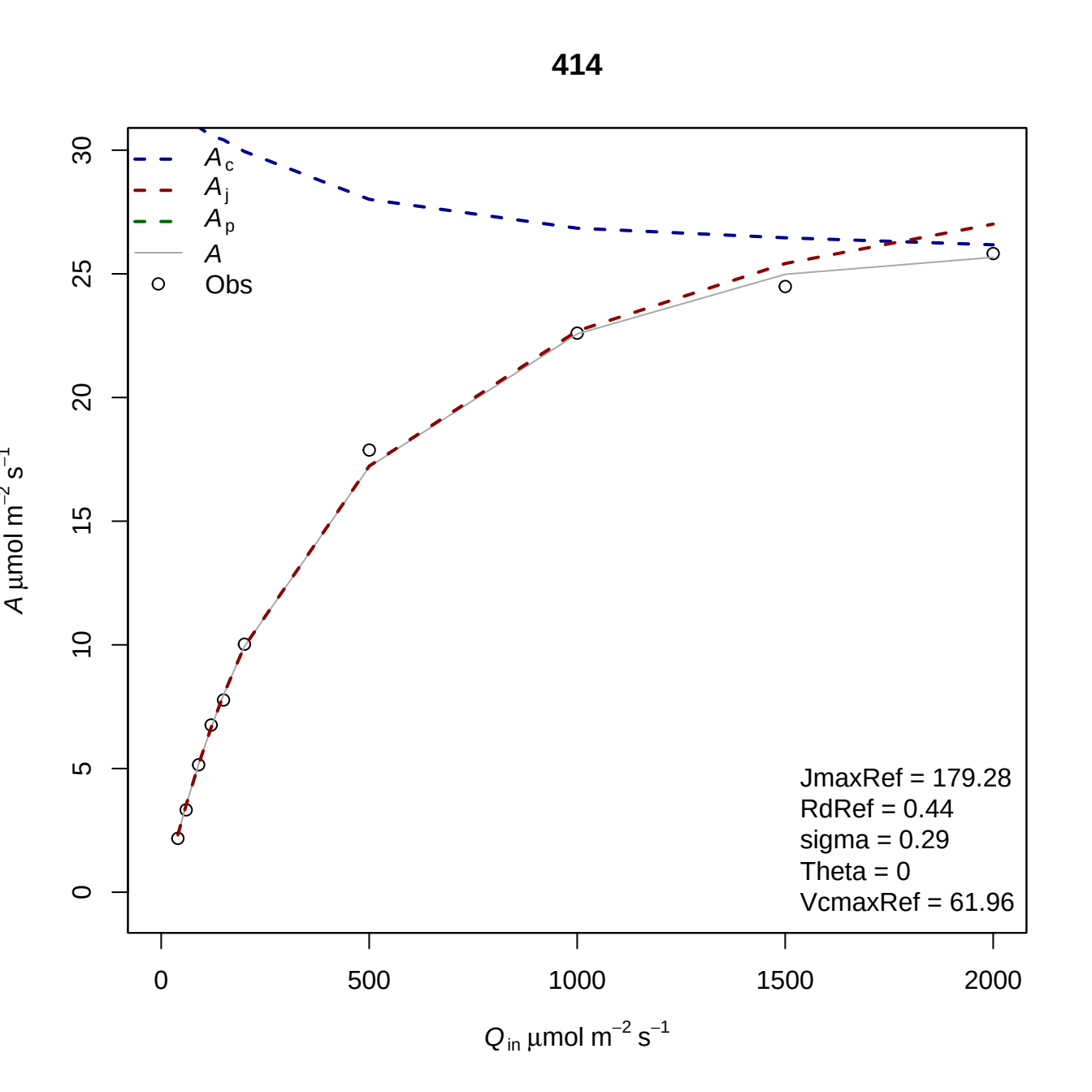

409
A_c
A_j
A_p
A
Obs
JmaxRef = 218.88
RdRef = 0.76
sigma = 0.35
Theta = 0
VcmaxRef = 82.26
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

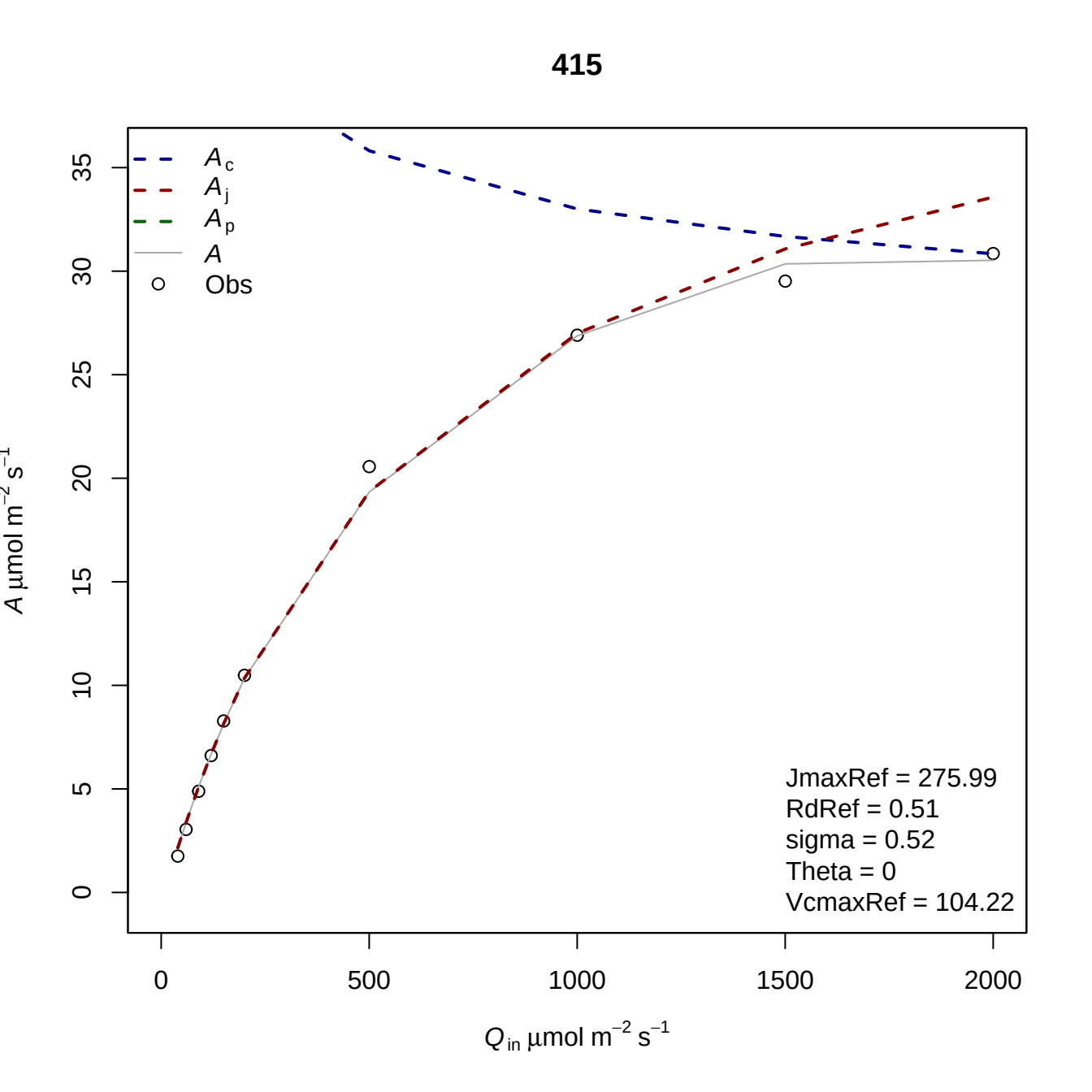

410
A_c
A_j
A_p
A
Obs
JmaxRef = 308.4
RdRef = 0.92
sigma = 0.54
Theta = 0
VcmaxRef = 133.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

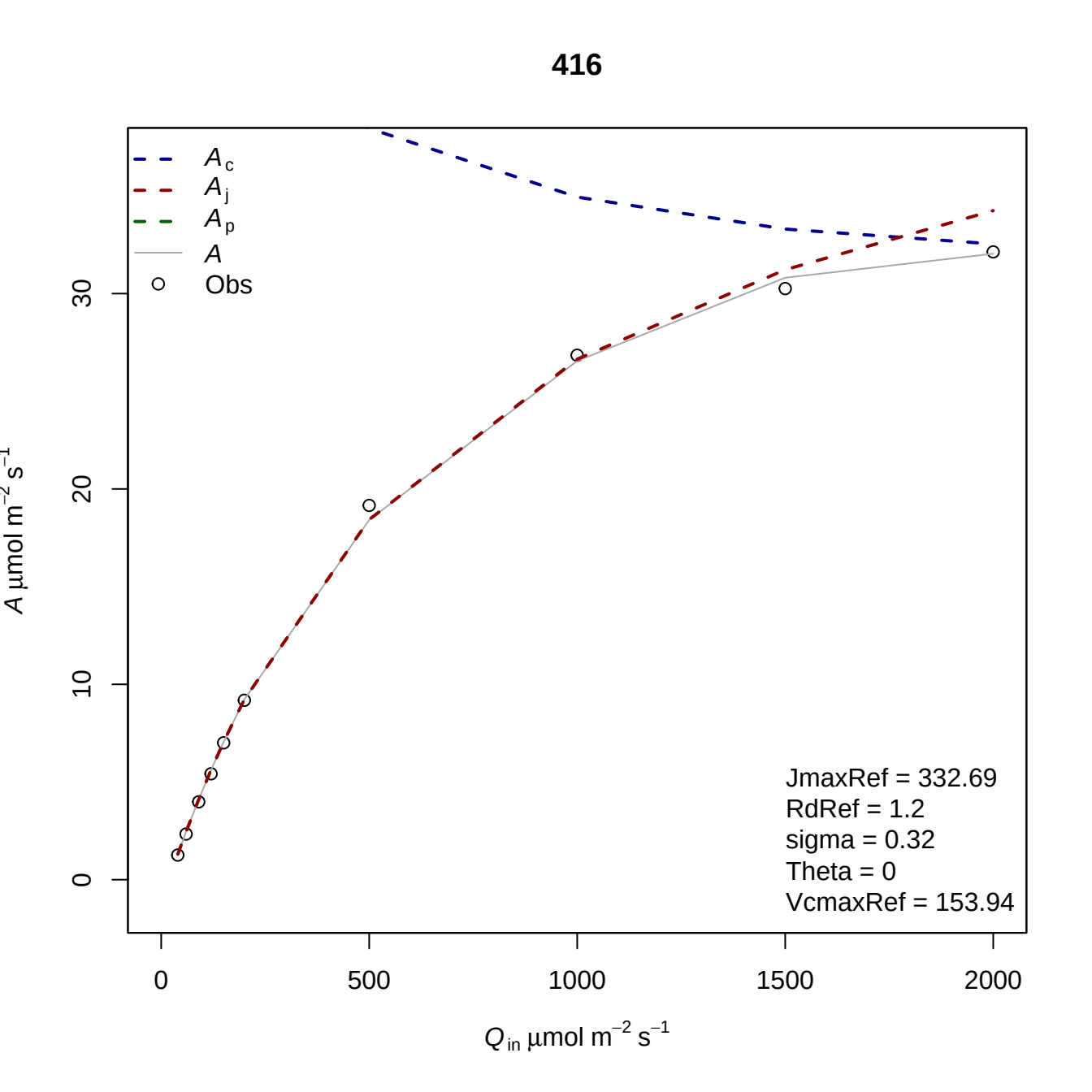

411
A_c
A_j
A_p
A
Obs
JmaxRef = 333.69
RdRef = 1.12
sigma = 0.54
Theta = 0
VcmaxRef = 184.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

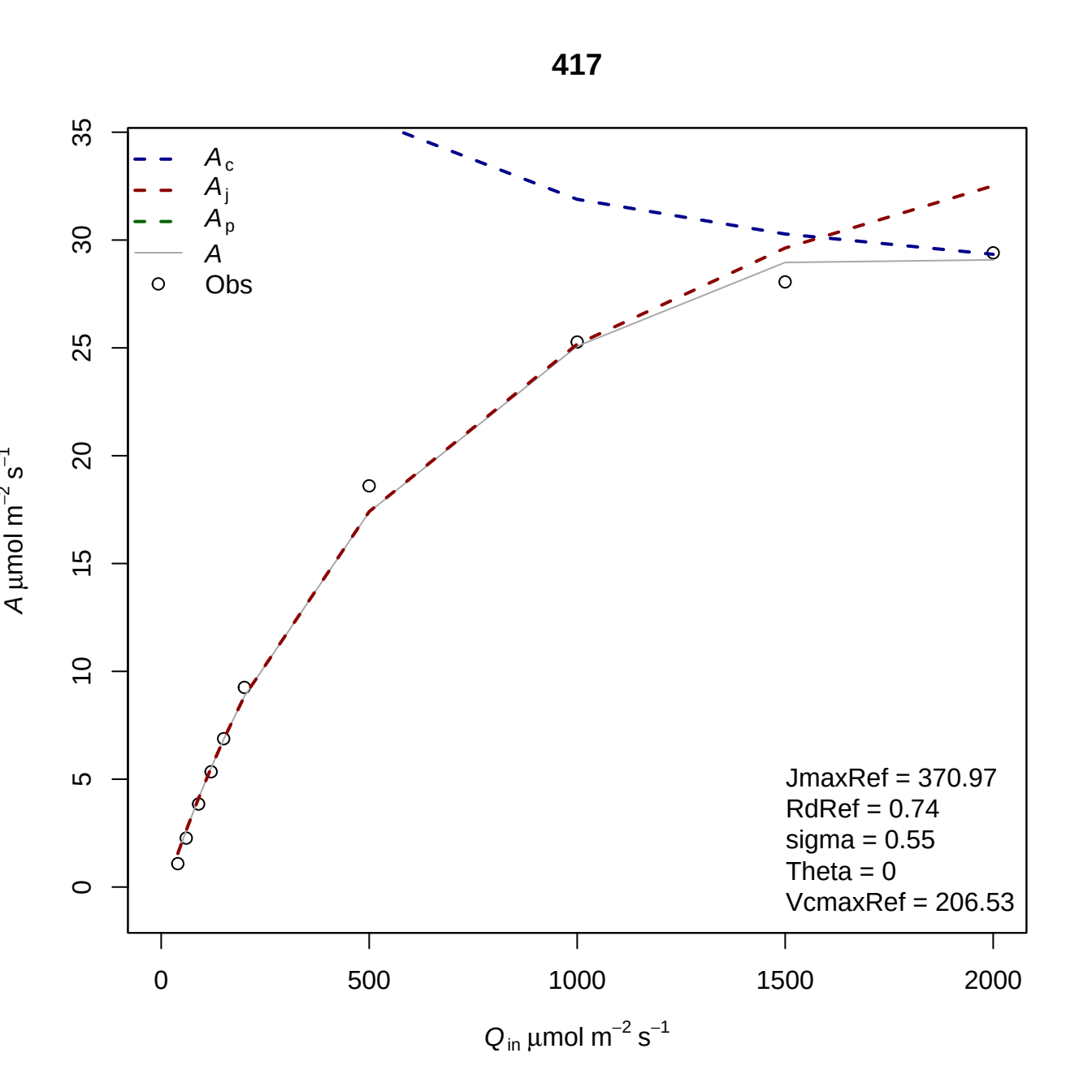

412
A_c
A_j
A_p
A
Obs
JmaxRef = 417.73
RdRef = 1.69
sigma = 0.49
Theta = 0
VcmaxRef = 282.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

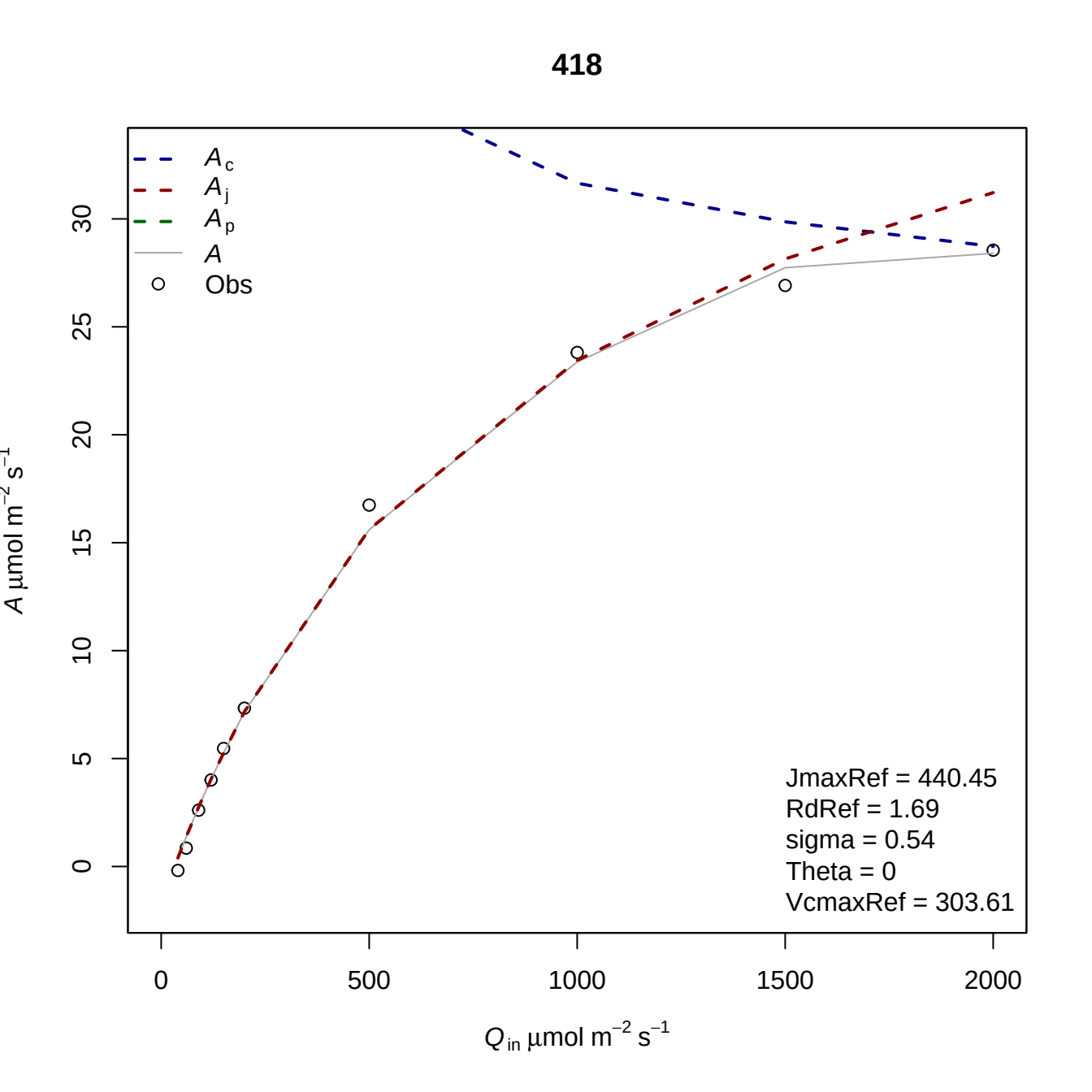

413
A_c
A_j
A_p
A
Obs
JmaxRef = 234.45
RdRef = 3.38
sigma = 0.39
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

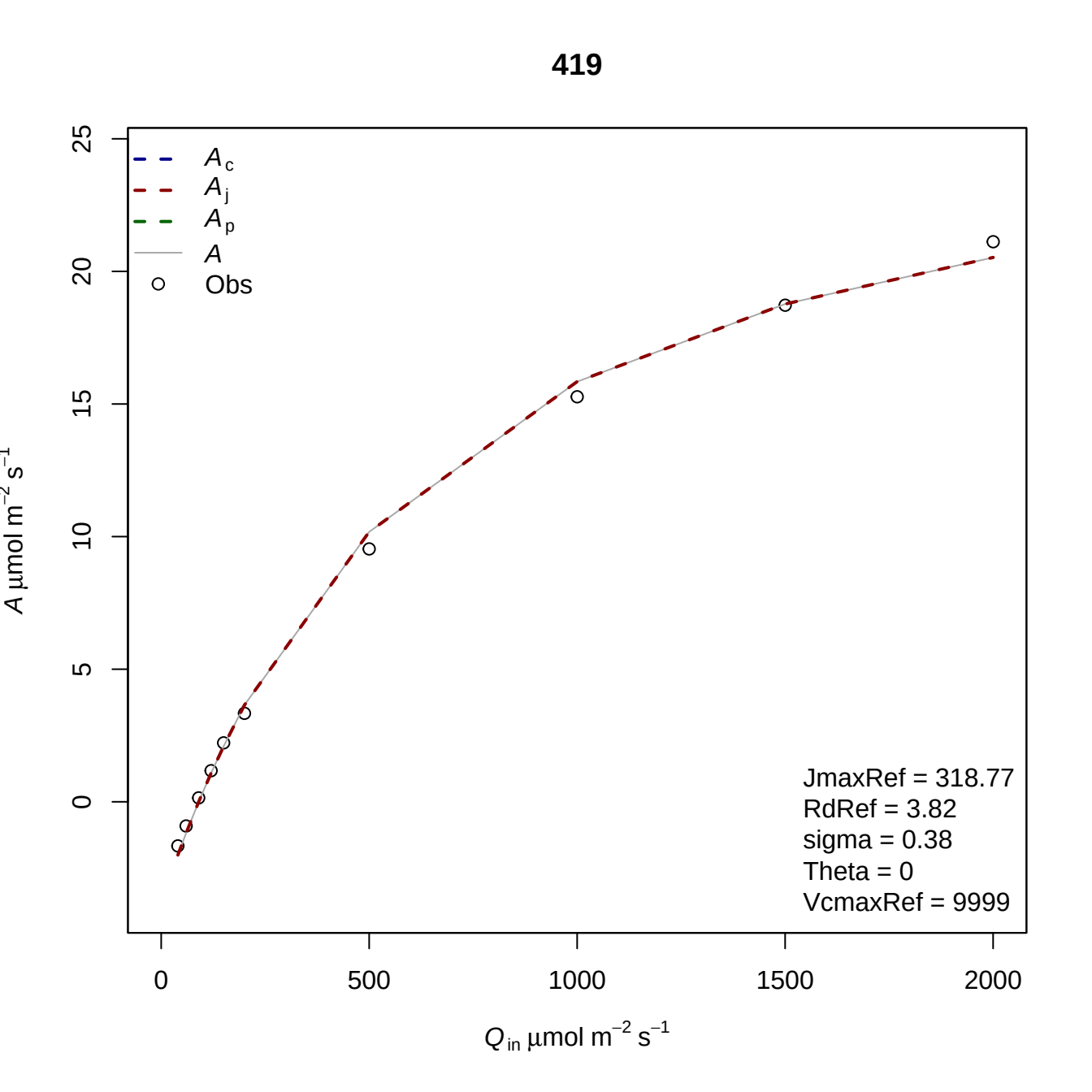

414
A_c
A_j
A_p
A
Obs
JmaxRef = 179.28
RdRef = 0.44
sigma = 0.29
Theta = 0
VcmaxRef = 61.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

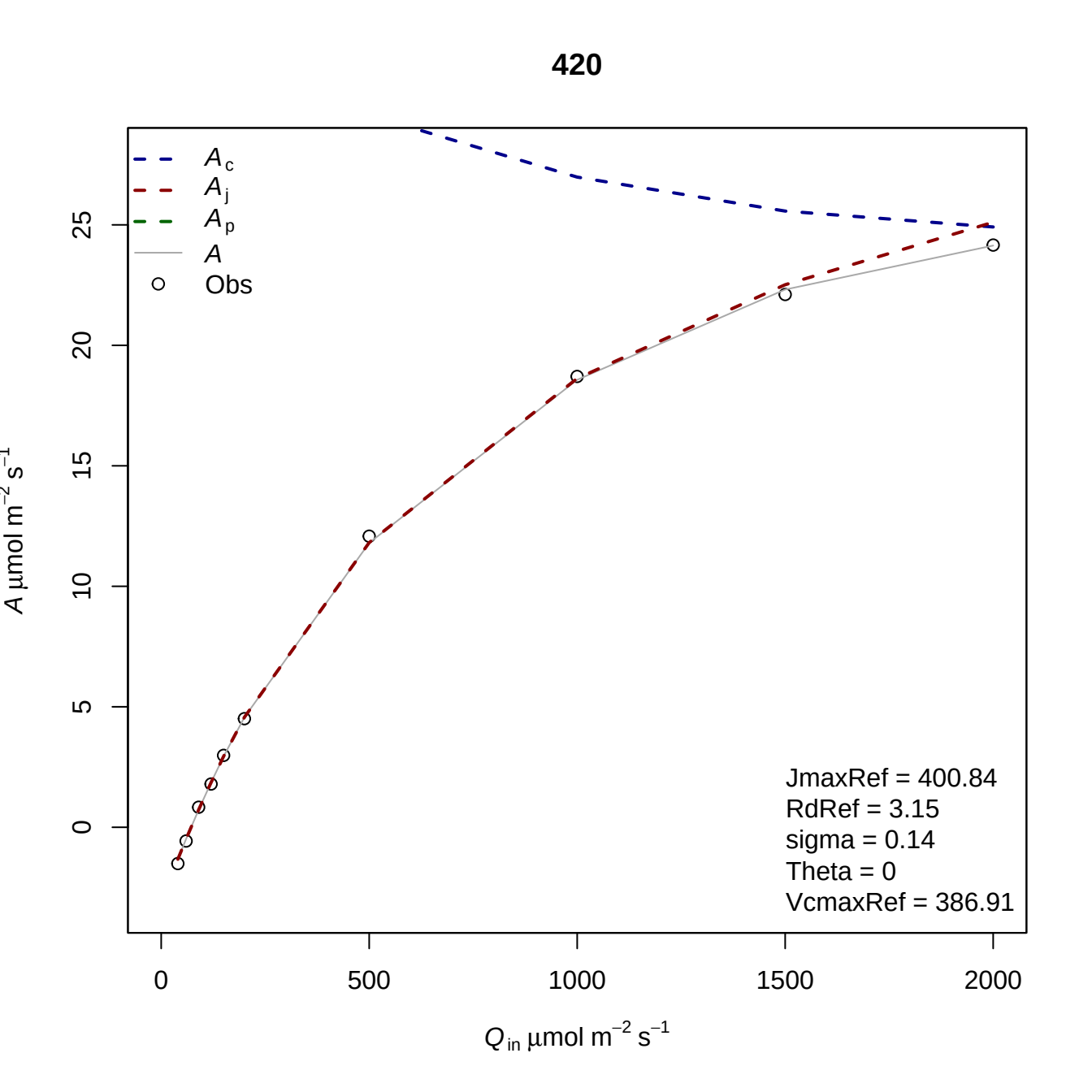

415
A_c
A_j
A_p
A
Obs
JmaxRef = 275.99
RdRef = 0.51
sigma = 0.52
Theta = 0
VcmaxRef = 104.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

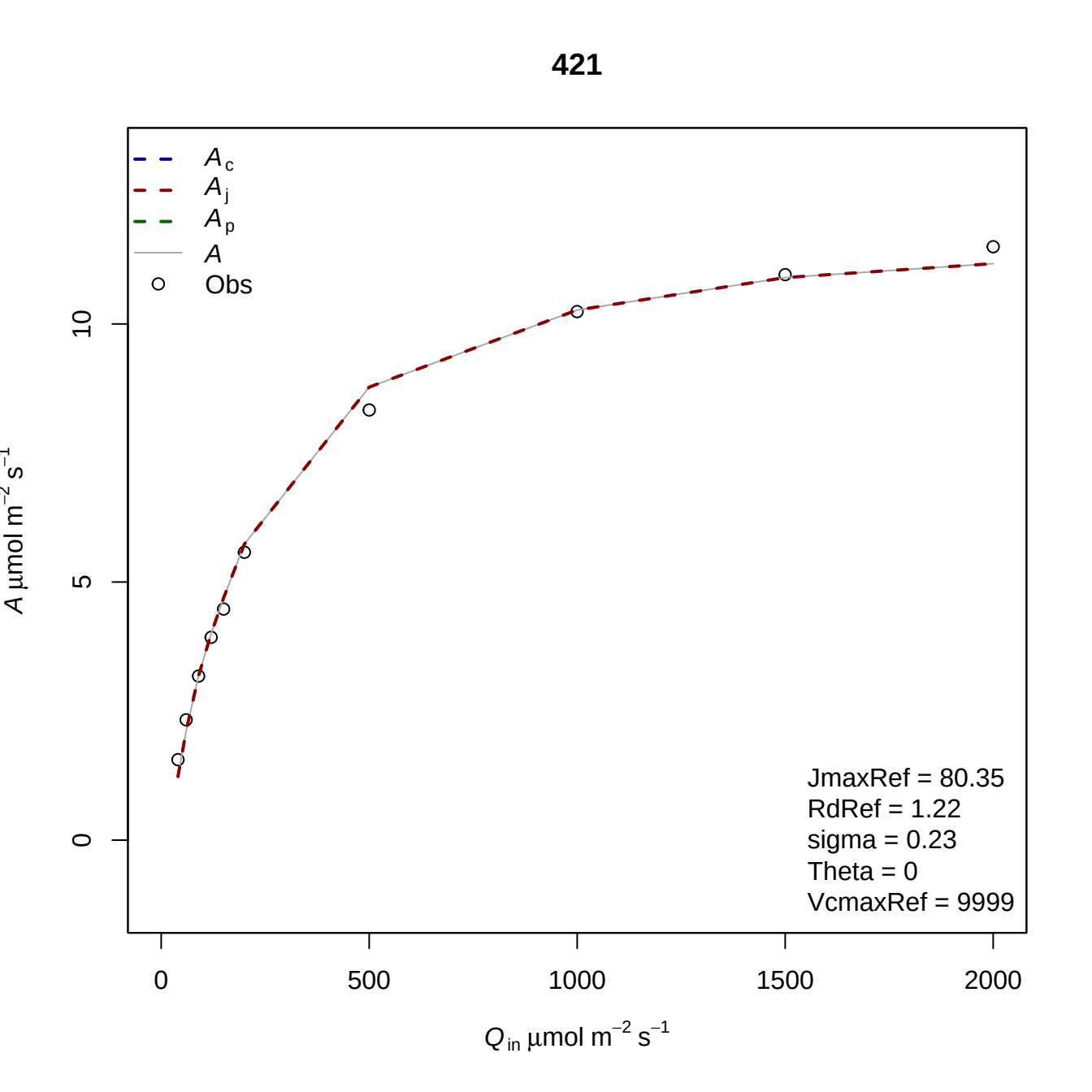

416
A_c
A_j
A_p
A
Obs
JmaxRef = 332.69
RdRef = 1.2
sigma = 0.32
Theta = 0
VcmaxRef = 153.94
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

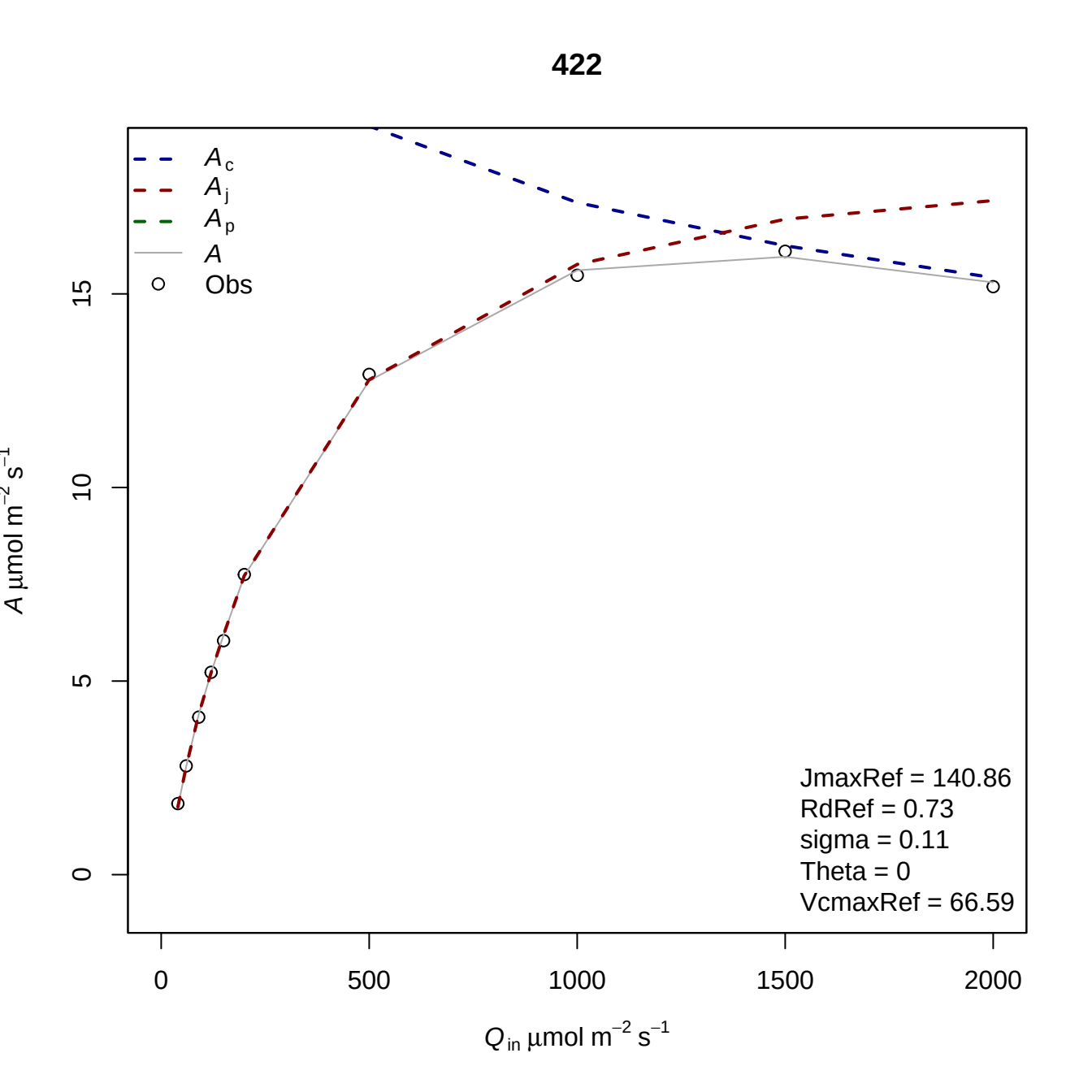

417
A_c
A_j
A_p
A
Obs
JmaxRef = 370.97
RdRef = 0.74
sigma = 0.55
Theta = 0
VcmaxRef = 206.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

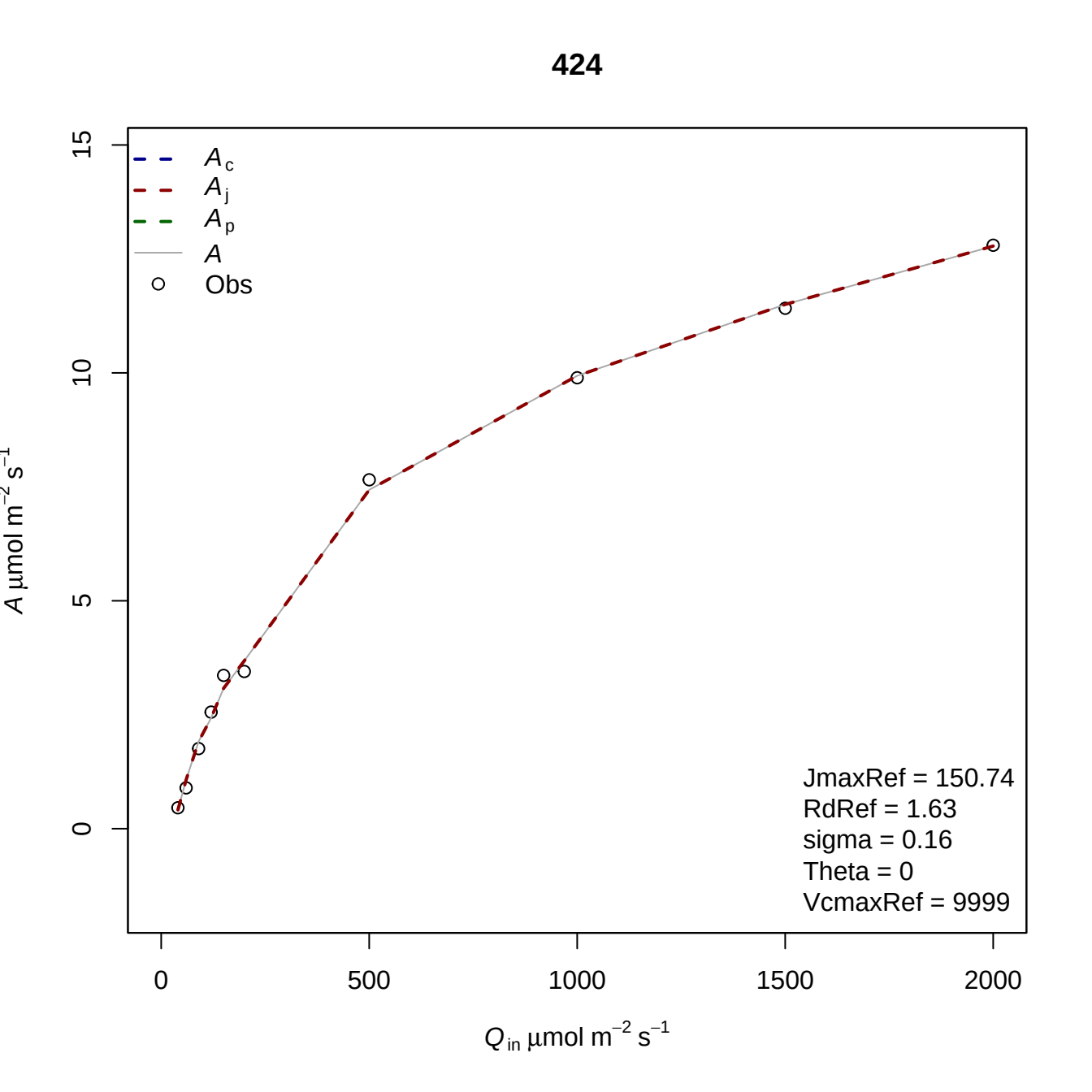

418
A_c
A_j
A_p
A
Obs
JmaxRef = 440.45
RdRef = 1.69
sigma = 0.54
Theta = 0
VcmaxRef = 303.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

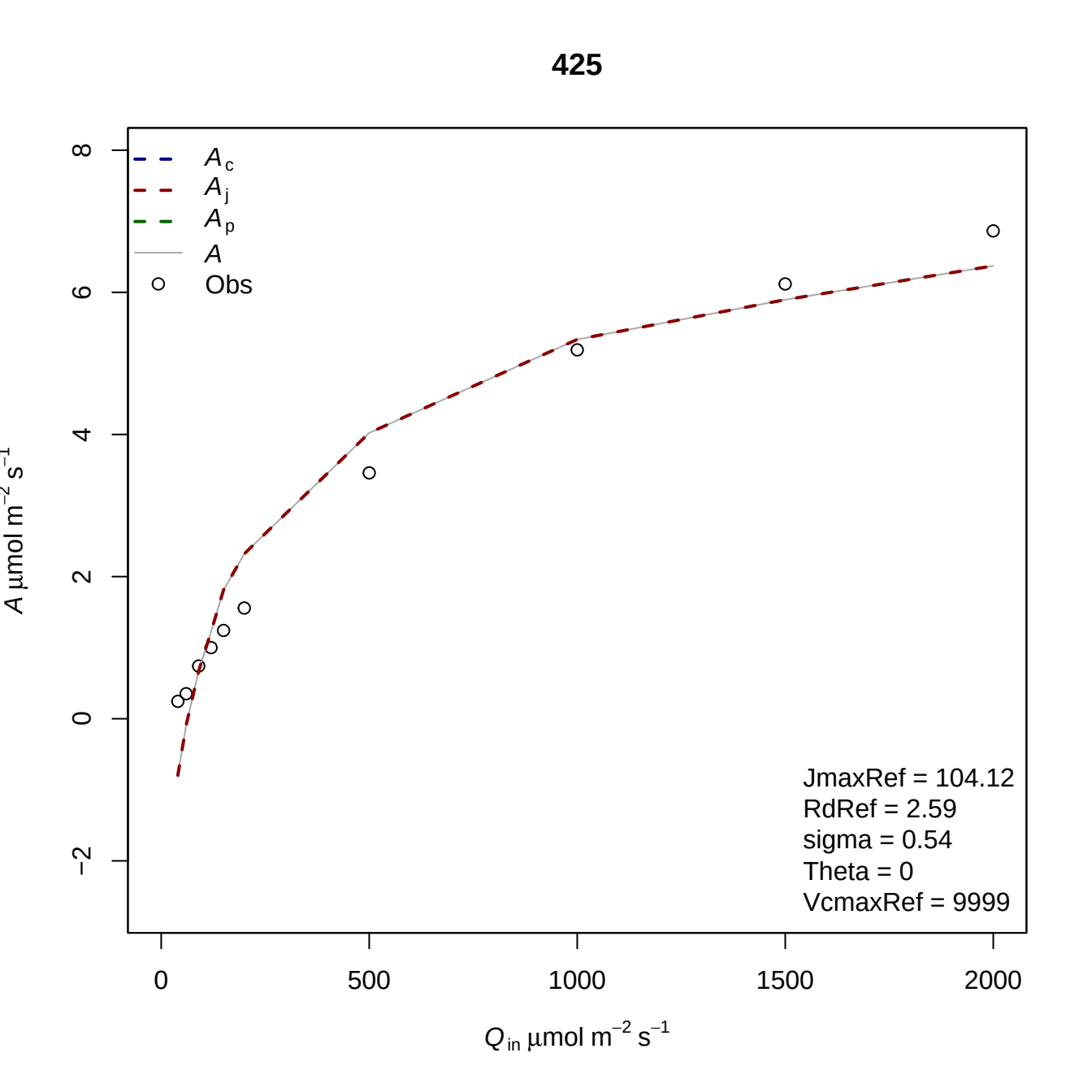

419
A_c
A_j
A_p
A
Obs
JmaxRef = 318.77
RdRef = 3.82
sigma = 0.38
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

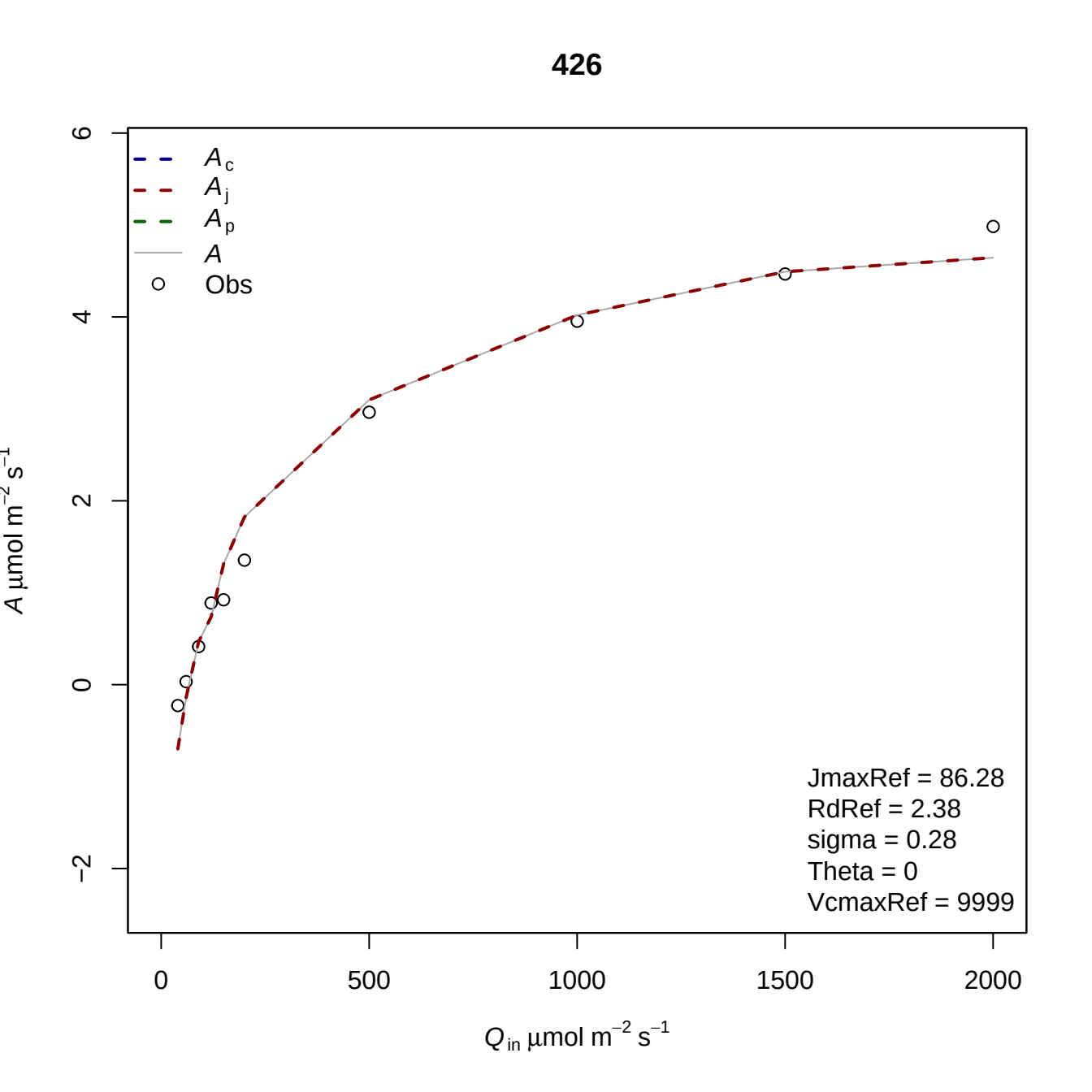

420
A_c
A_j
A_p
A
Obs
JmaxRef = 400.84
RdRef = 3.15
sigma = 0.14
Theta = 0
VcmaxRef = 386.91
A µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

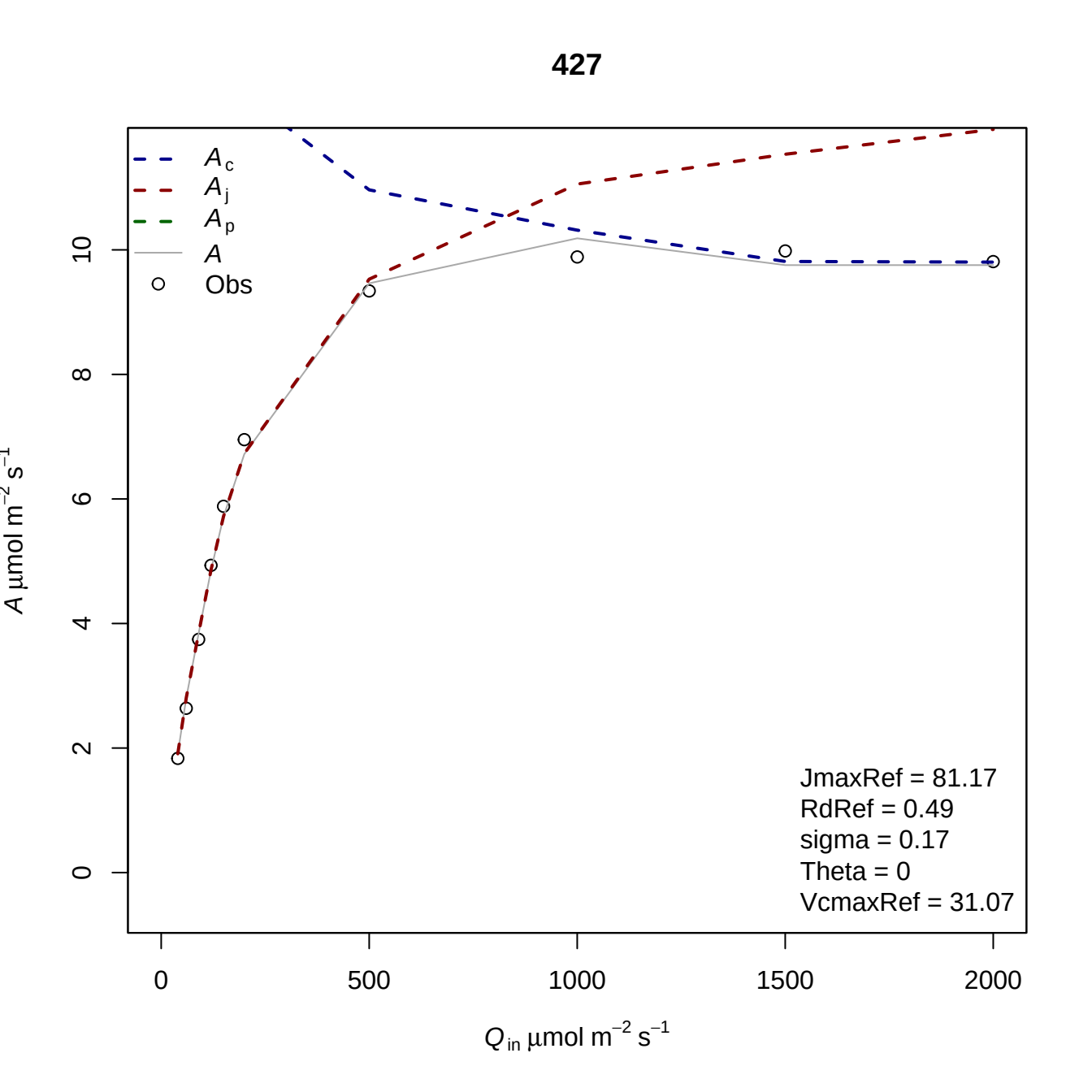

421
A_c
A_j
A_p
A
Obs
JmaxRef = 80.35
RdRef = 1.22
sigma = 0.23
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

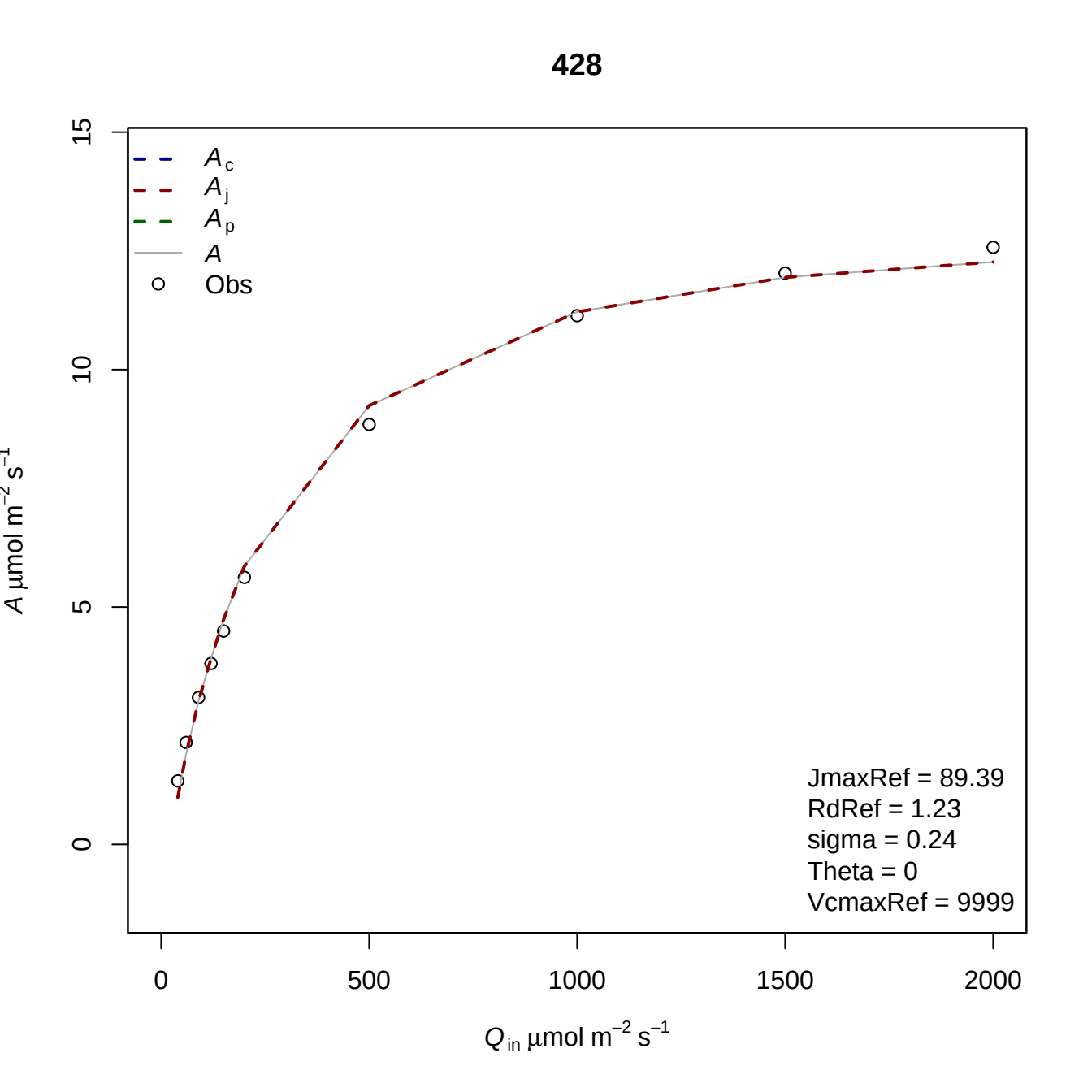

422
A_c
A_j
A_p
A
Obs
JmaxRef = 140.86
RdRef = 0.73
sigma = 0.11
Theta = 0
VcmaxRef = 66.59
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

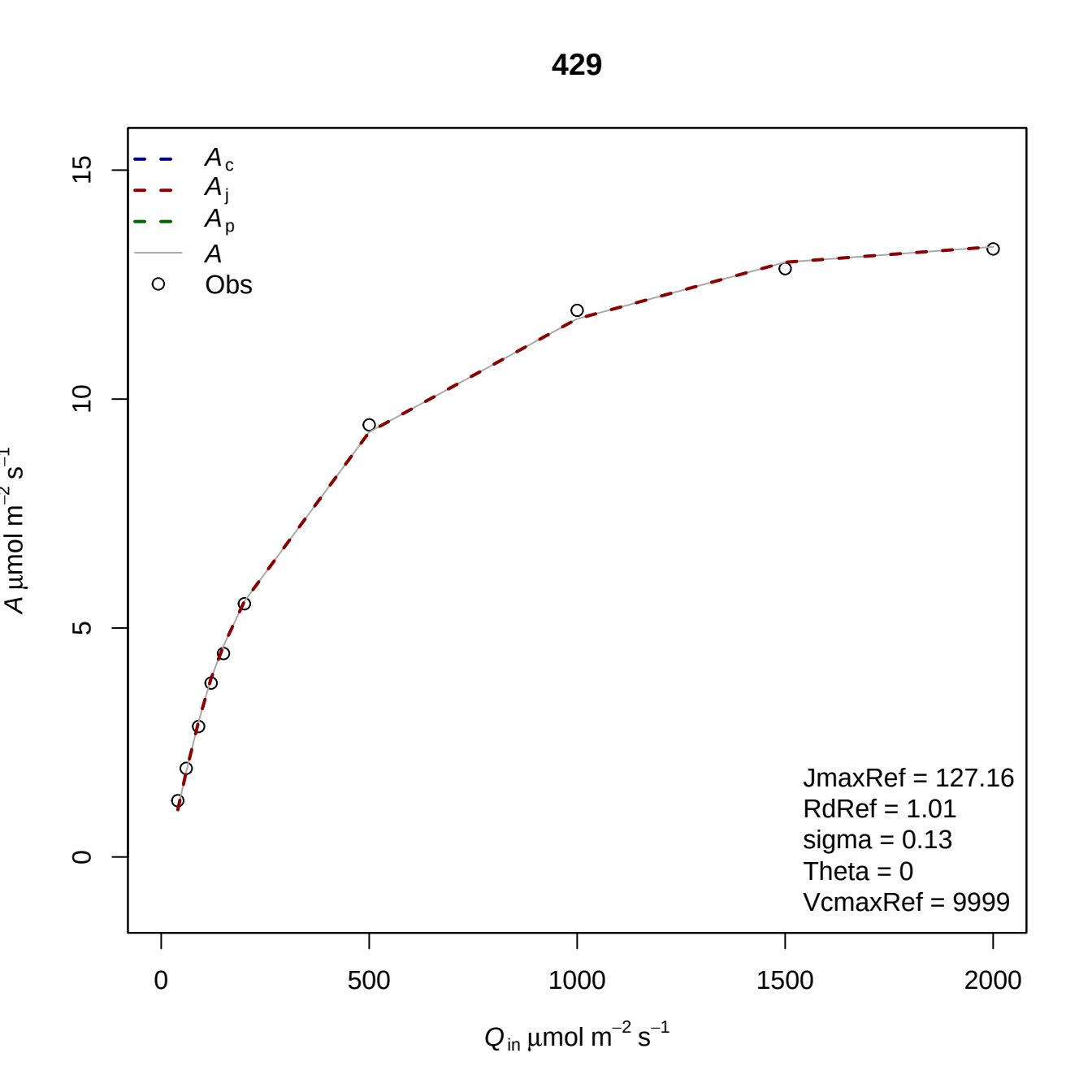

424
A_c
A_j
A_p
A
Obs
JmaxRef = 150.74
RdRef = 1.63
sigma = 0.16
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

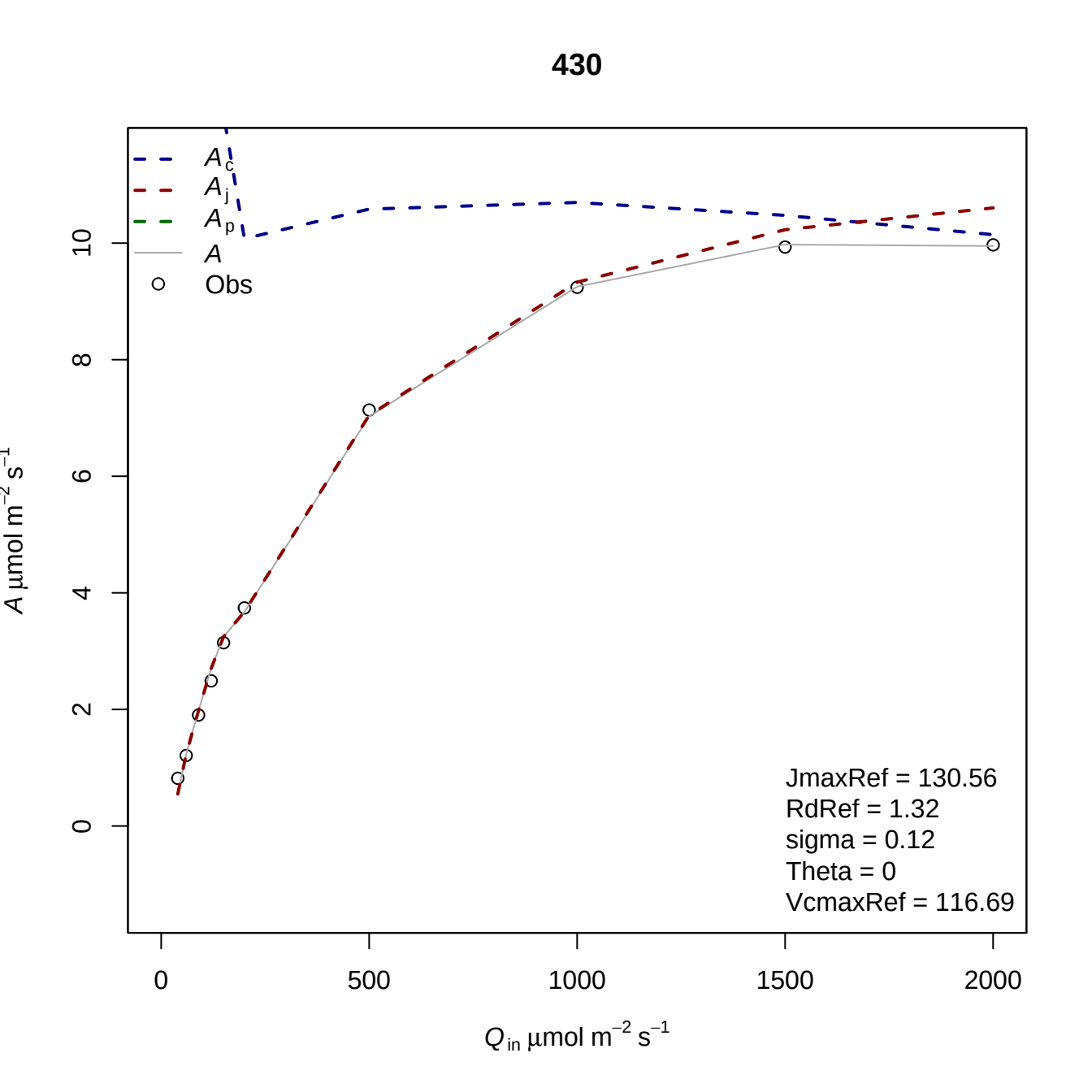

425
A_c
A_j
A_p
A
Obs
JmaxRef = 104.12
RdRef = 2.59
sigma = 0.54
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

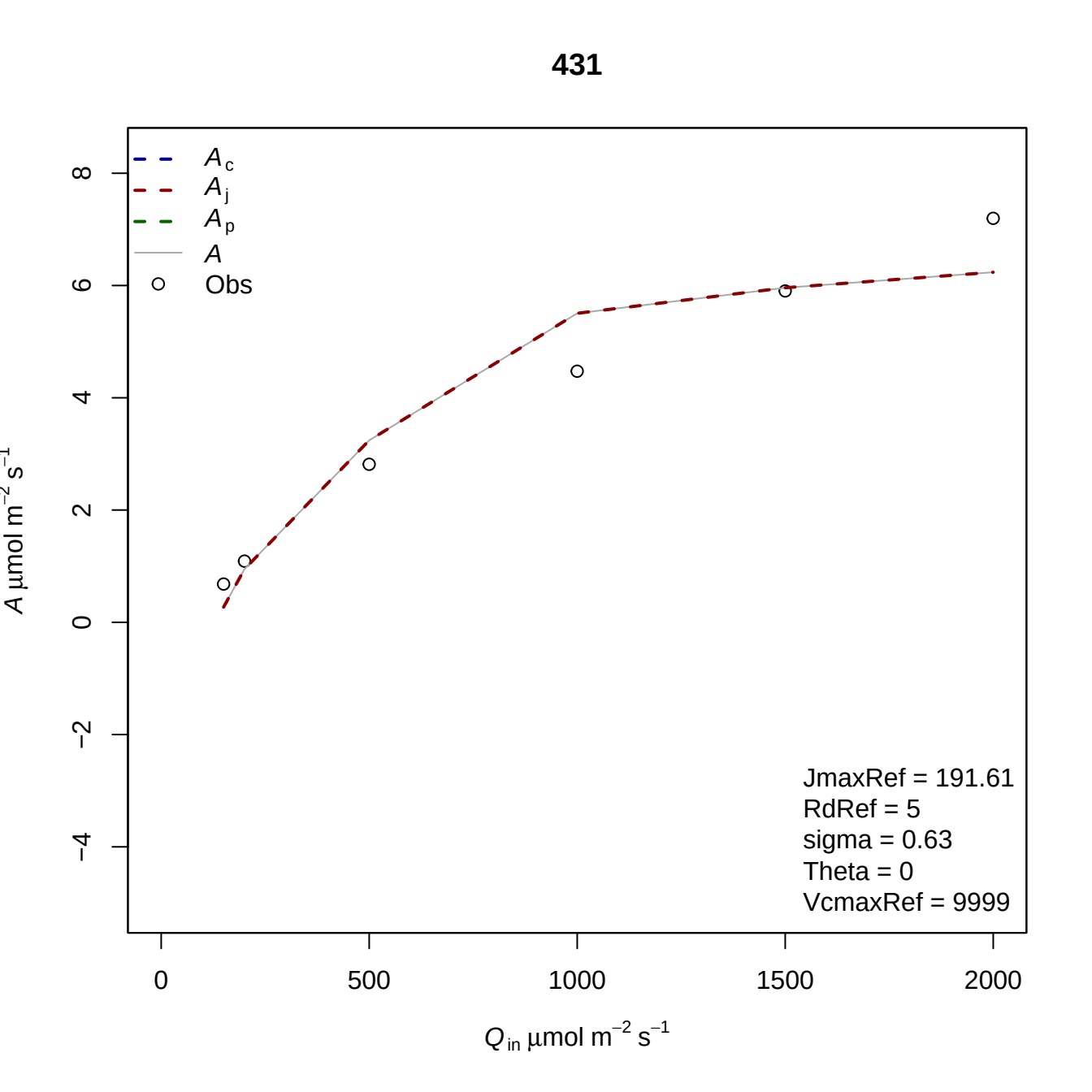

426
A_c
A_j
A_p
A
Obs
JmaxRef = 86.28
RdRef = 2.38
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

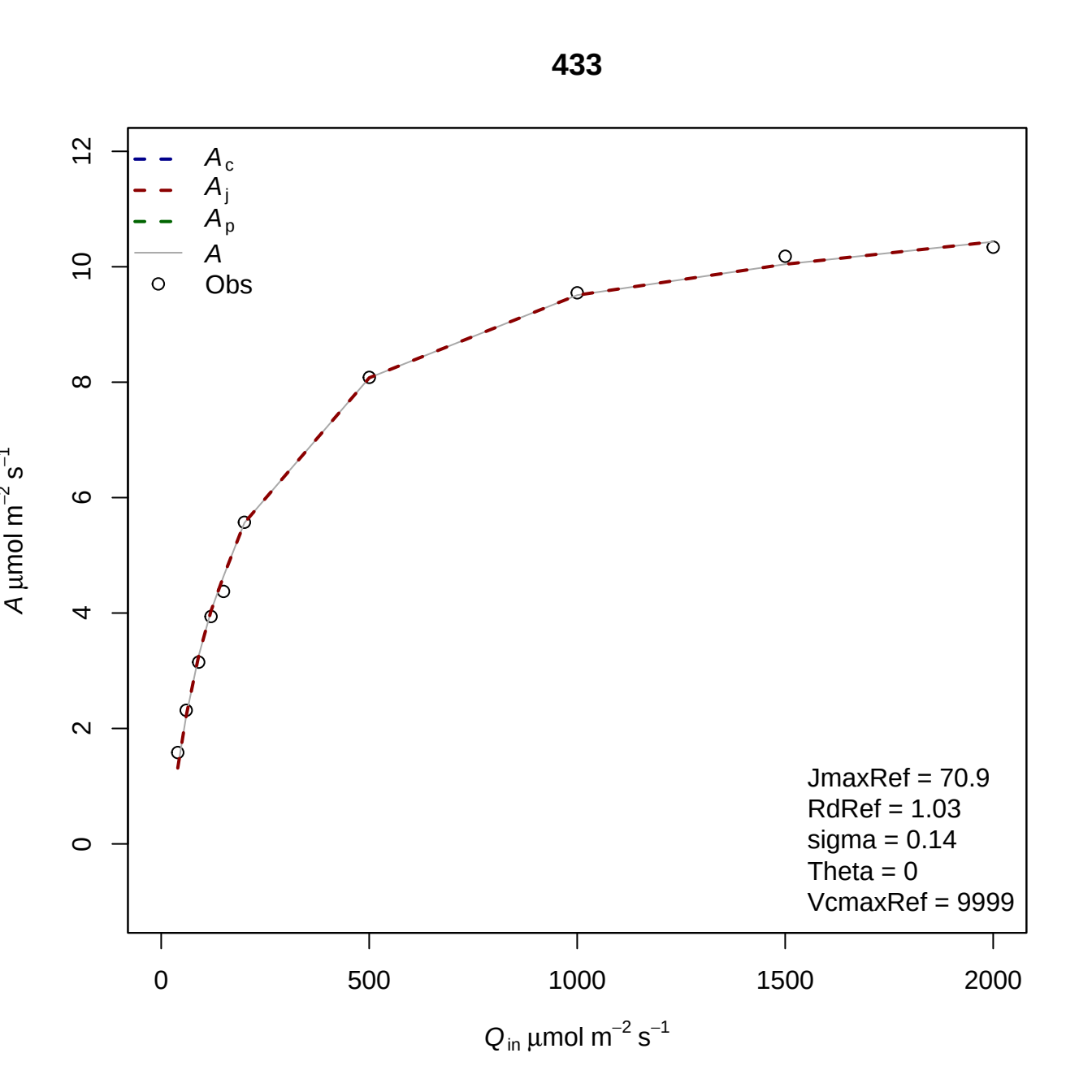

427
A_c
A_j
A_p
A
Obs
JmaxRef = 81.17
RdRef = 0.49
sigma = 0.17
Theta = 0
VcmaxRef = 31.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

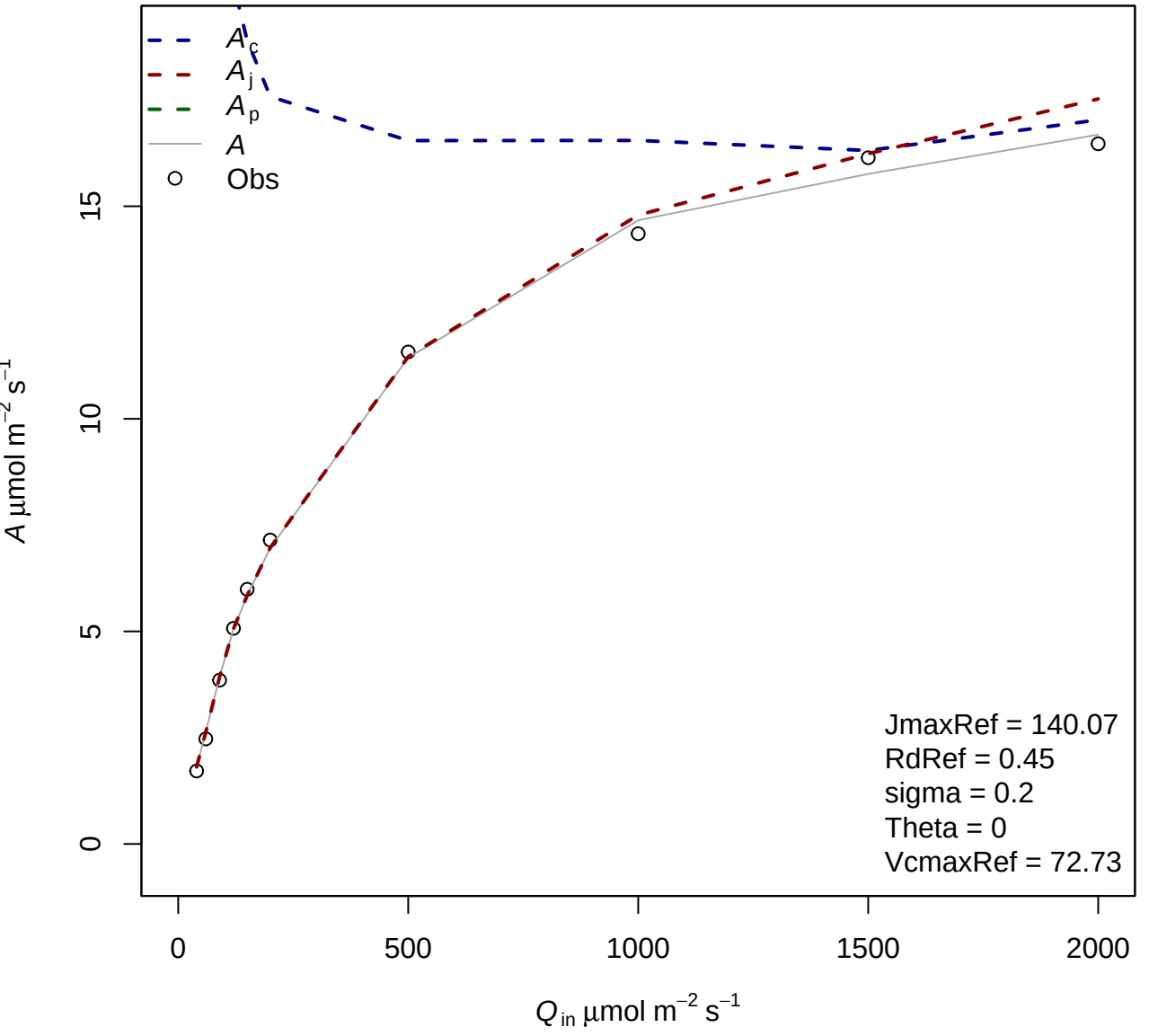

428
A_c
A_j
A_p
A
Obs
JmaxRef = 89.39
RdRef = 1.23
sigma = 0.24
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

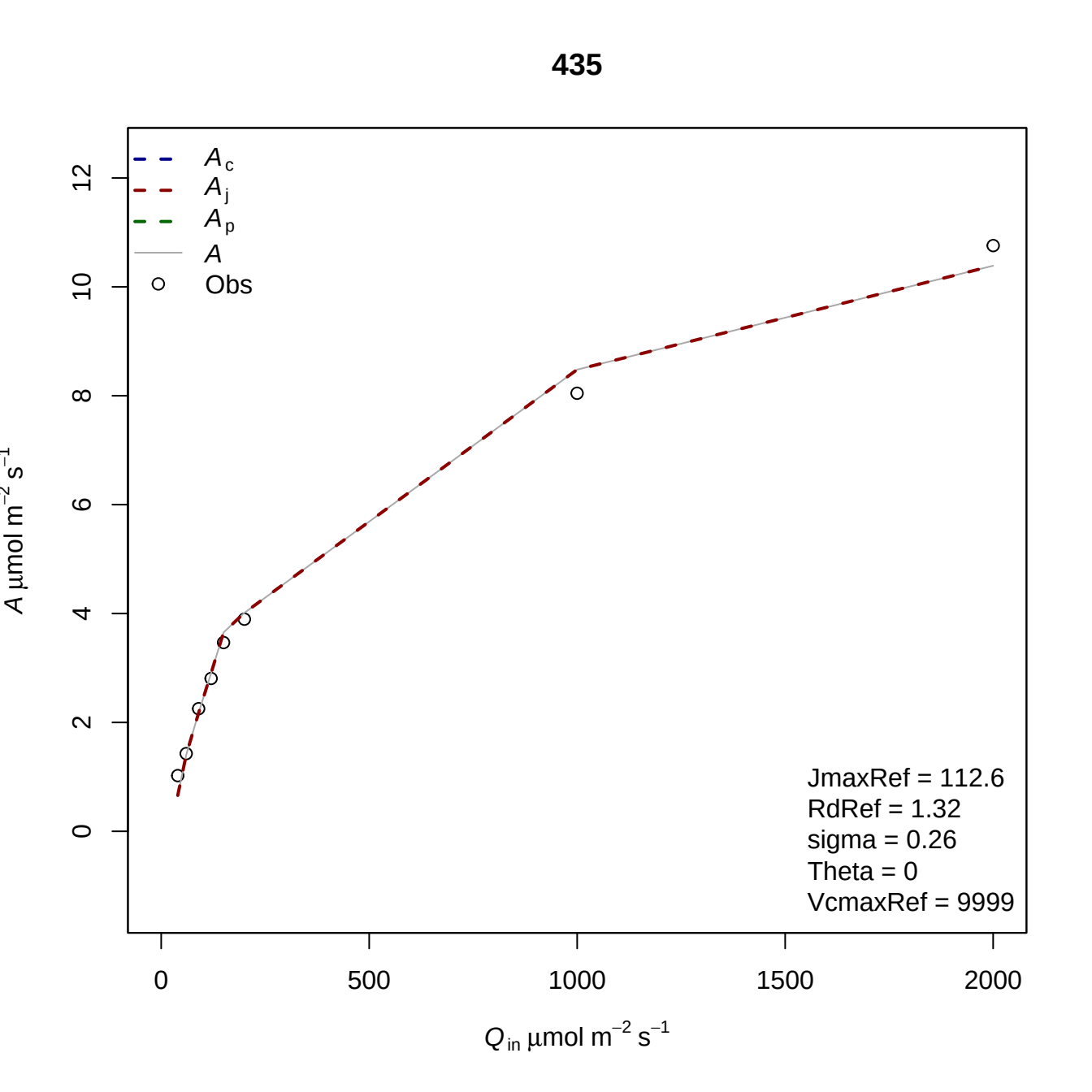

429
A_c
A_j
A_p
A
Obs
JmaxRef = 127.16
RdRef = 1.01
sigma = 0.13
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

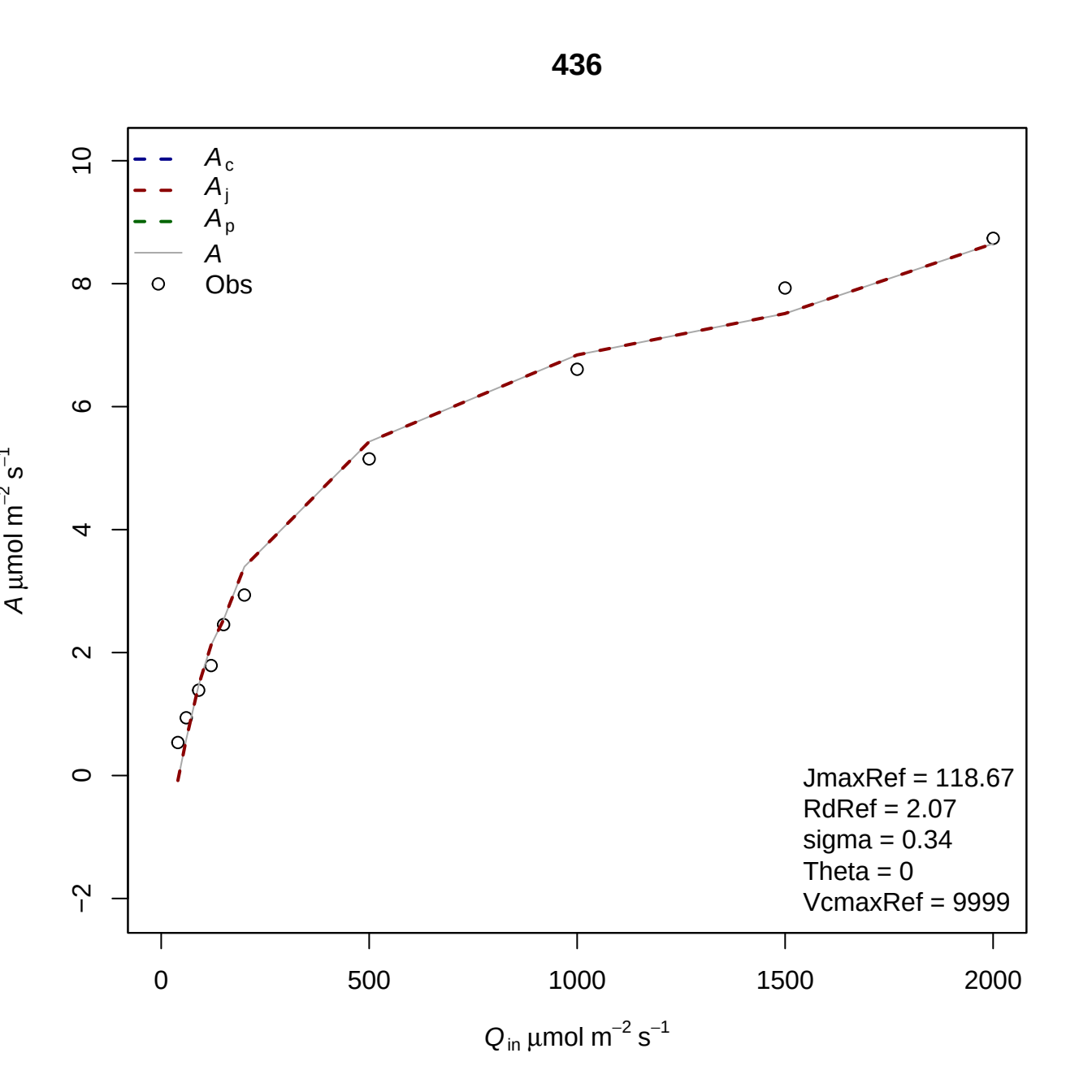

430
A_c
A_j
A_p
A
Obs
JmaxRef = 130.56
RdRef = 1.32
sigma = 0.12
Theta = 0
VcmaxRef = 116.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

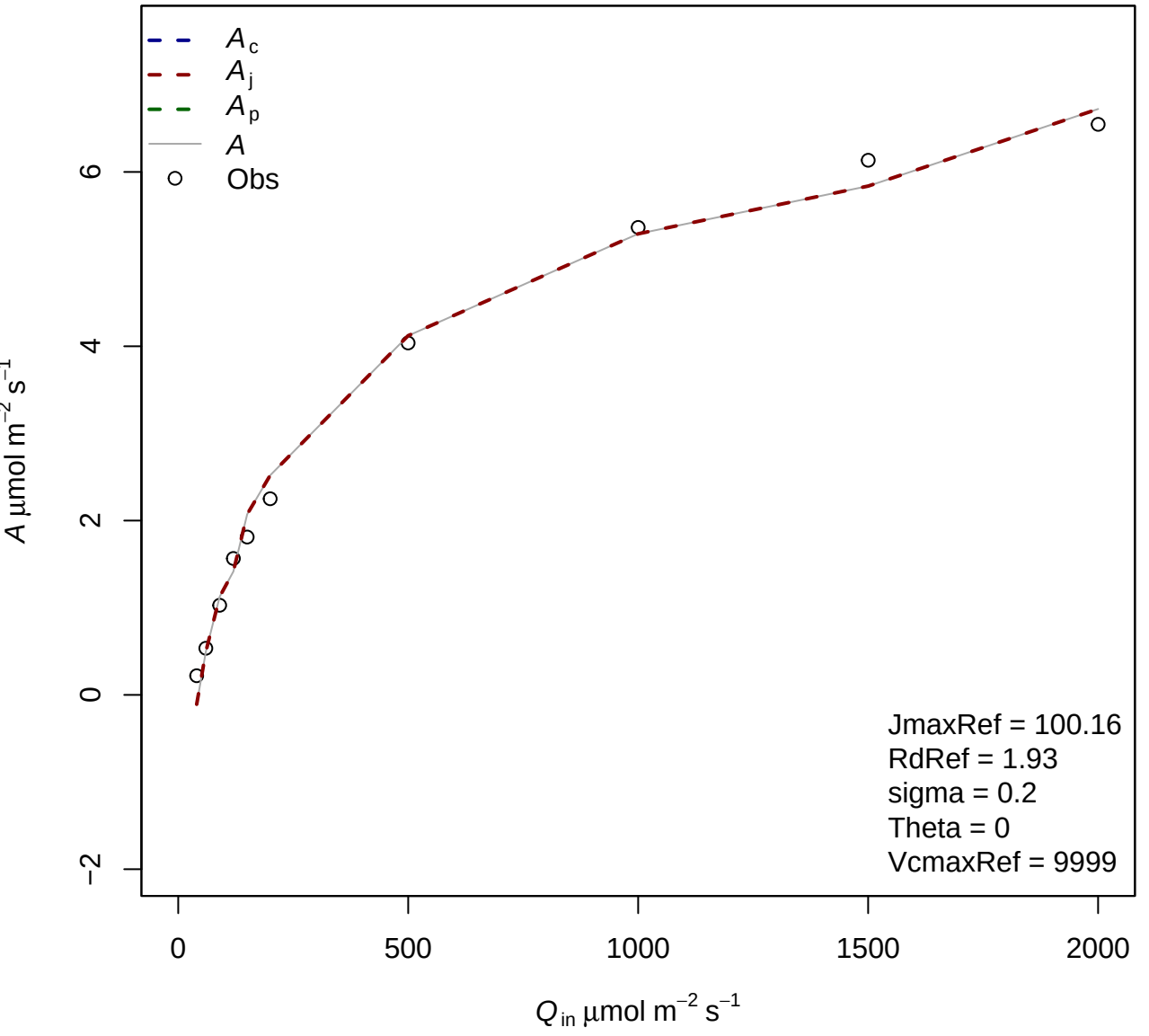

431
A_c
A_j
A_p
A
Obs
JmaxRef = 191.61
RdRef = 5
sigma = 0.63
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

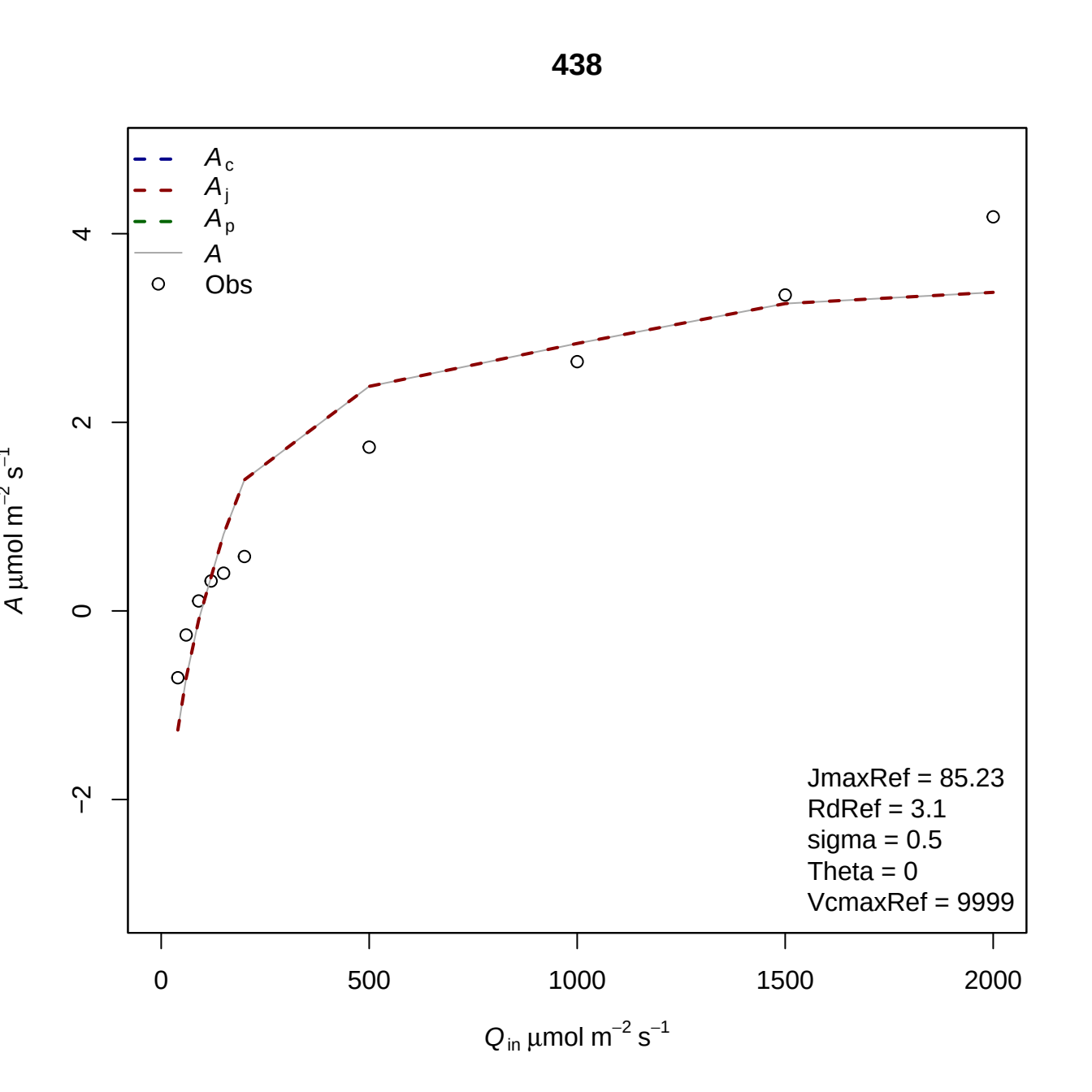

433
A_c
A_j
A_p
A
Obs
JmaxRef = 70.9
RdRef = 1.03
sigma = 0.14
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

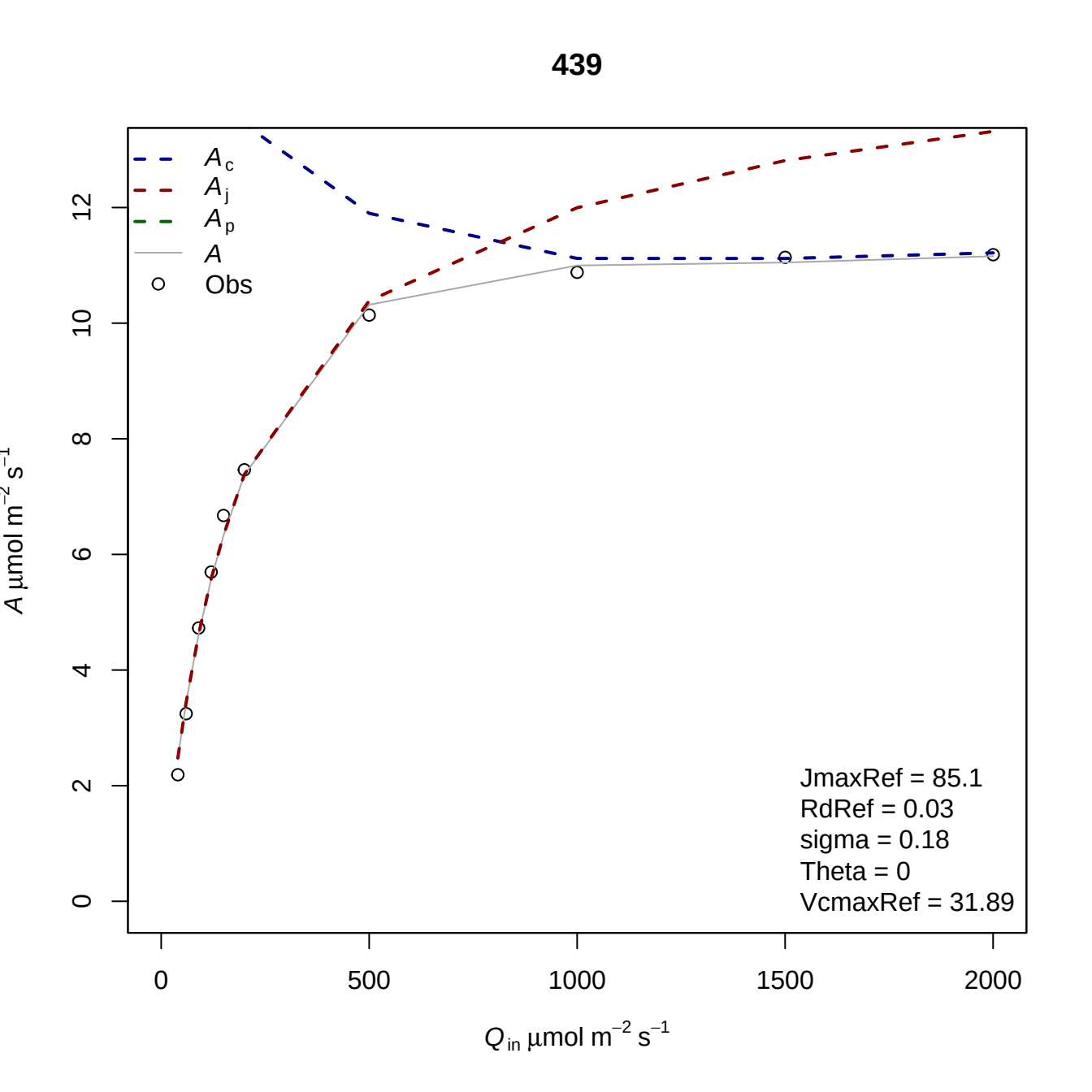

A_c
A_j
A_p
A
Obs
JmaxRef = 140.07
RdRef = 0.45
sigma = 0.2
Theta = 0
VcmaxRef = 72.73
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

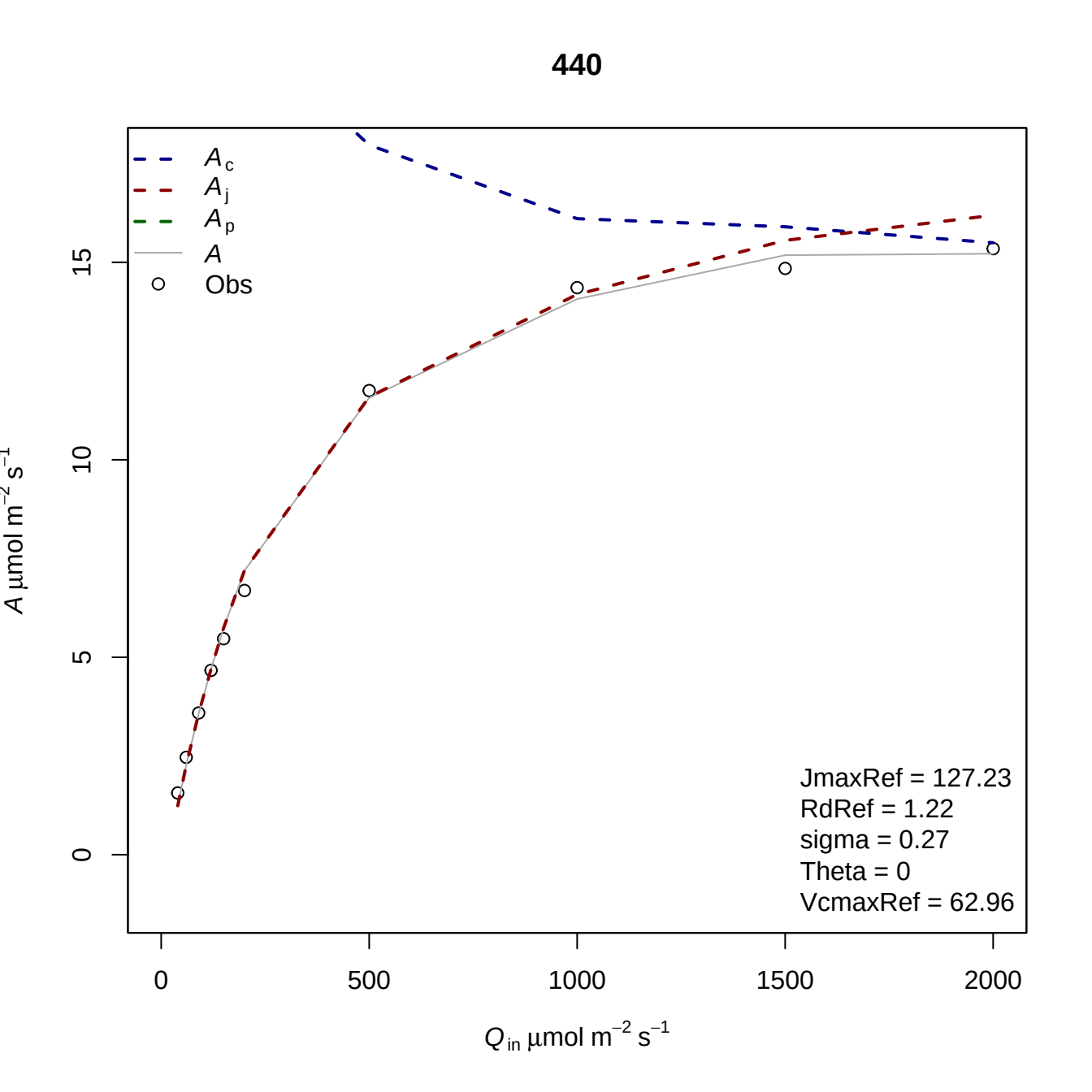

435
A_c
A_j
A_p
A
Obs
JmaxRef = 112.6
RdRef = 1.32
sigma = 0.26
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

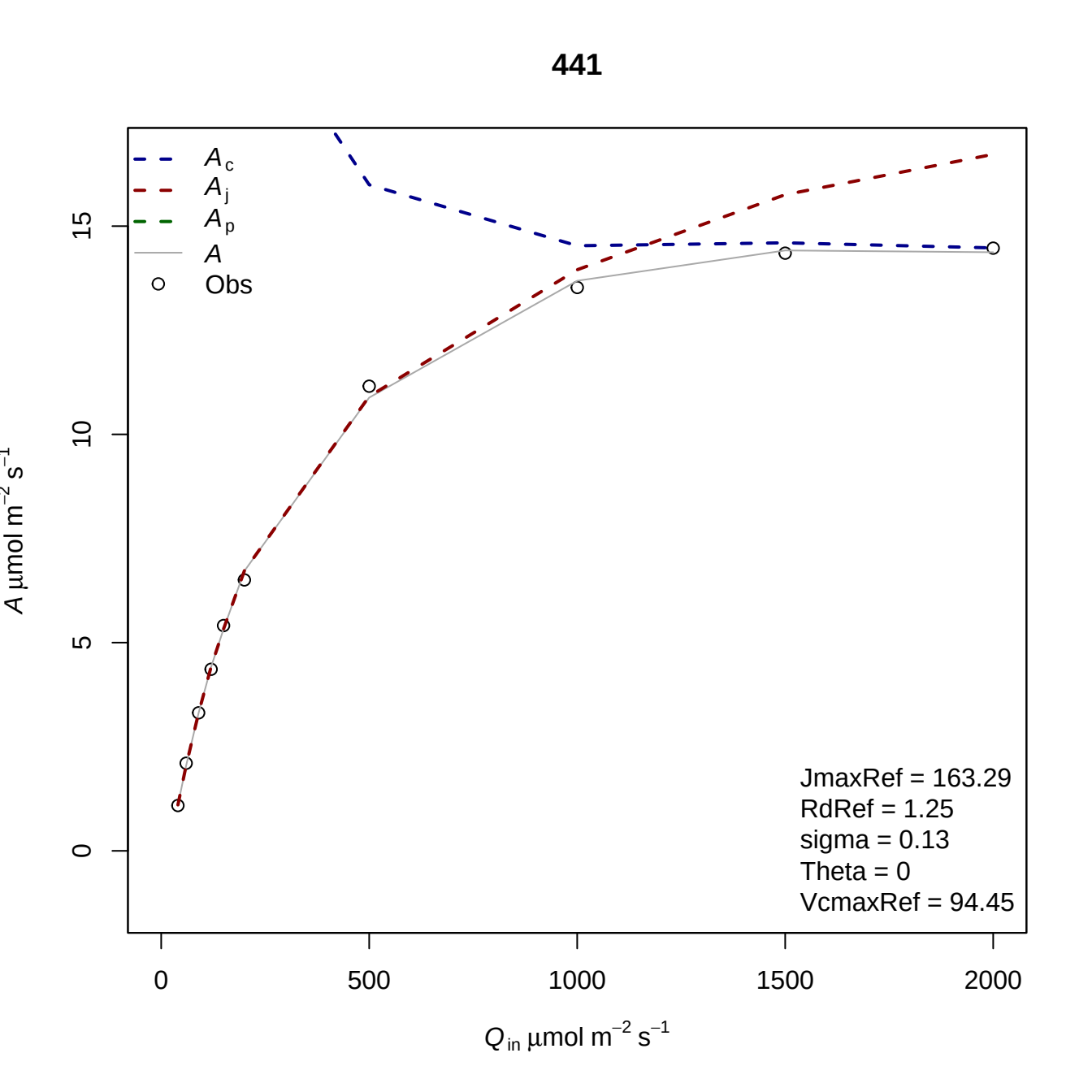

436
A_c
A_j
A_p
A
Obs
JmaxRef = 118.67
RdRef = 2.07
sigma = 0.34
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

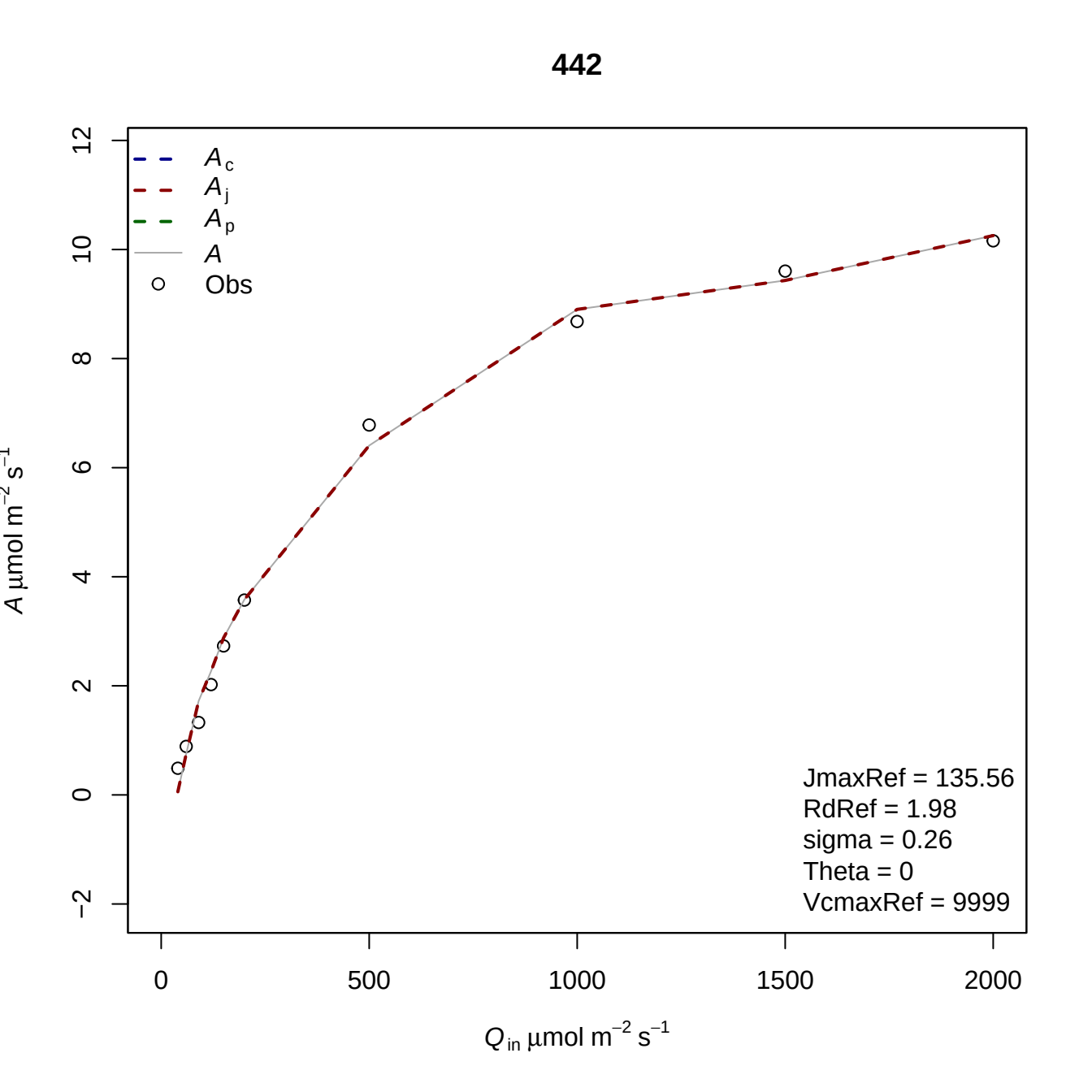

A_c
A_j
A_p
A
Obs
JmaxRef = 100.16
RdRef = 1.93
sigma = 0.2
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

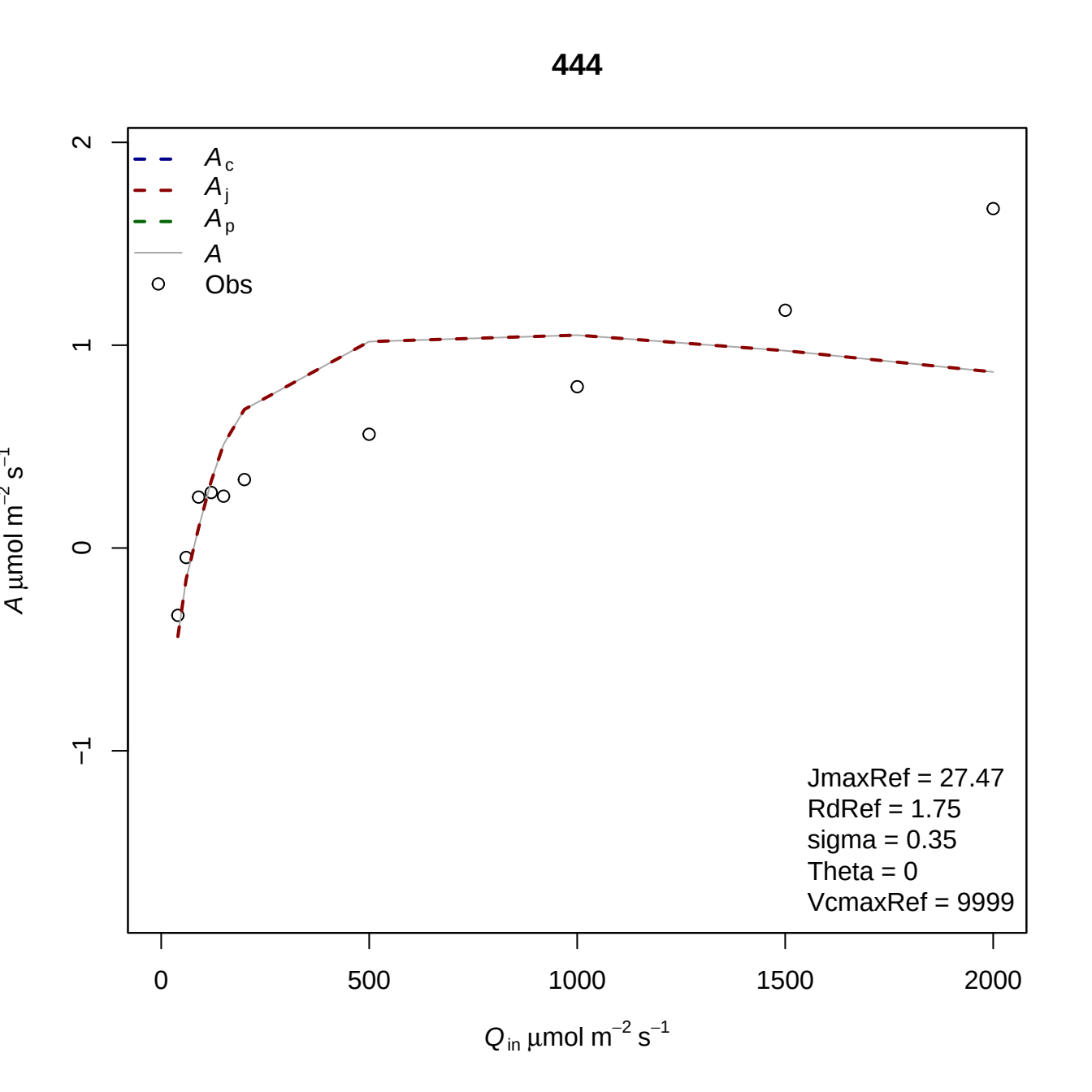

438
A_c
A_j
A_p
A
Obs
JmaxRef = 85.23
RdRef = 3.1
sigma = 0.5
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

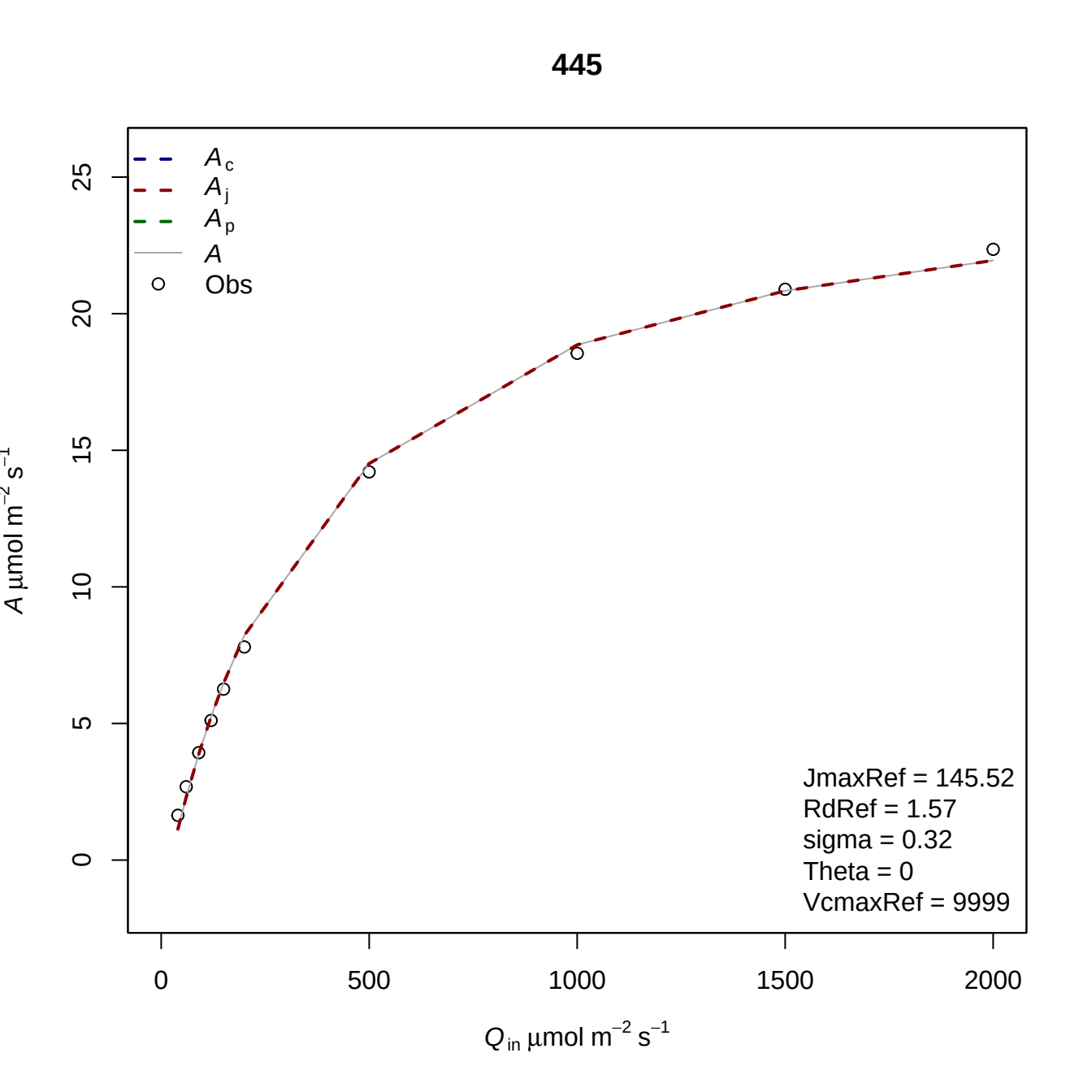

439
A_c
A_j
A_p
A
Obs
JmaxRef = 85.1
RdRef = 0.03
sigma = 0.18
Theta = 0
VcmaxRef = 31.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

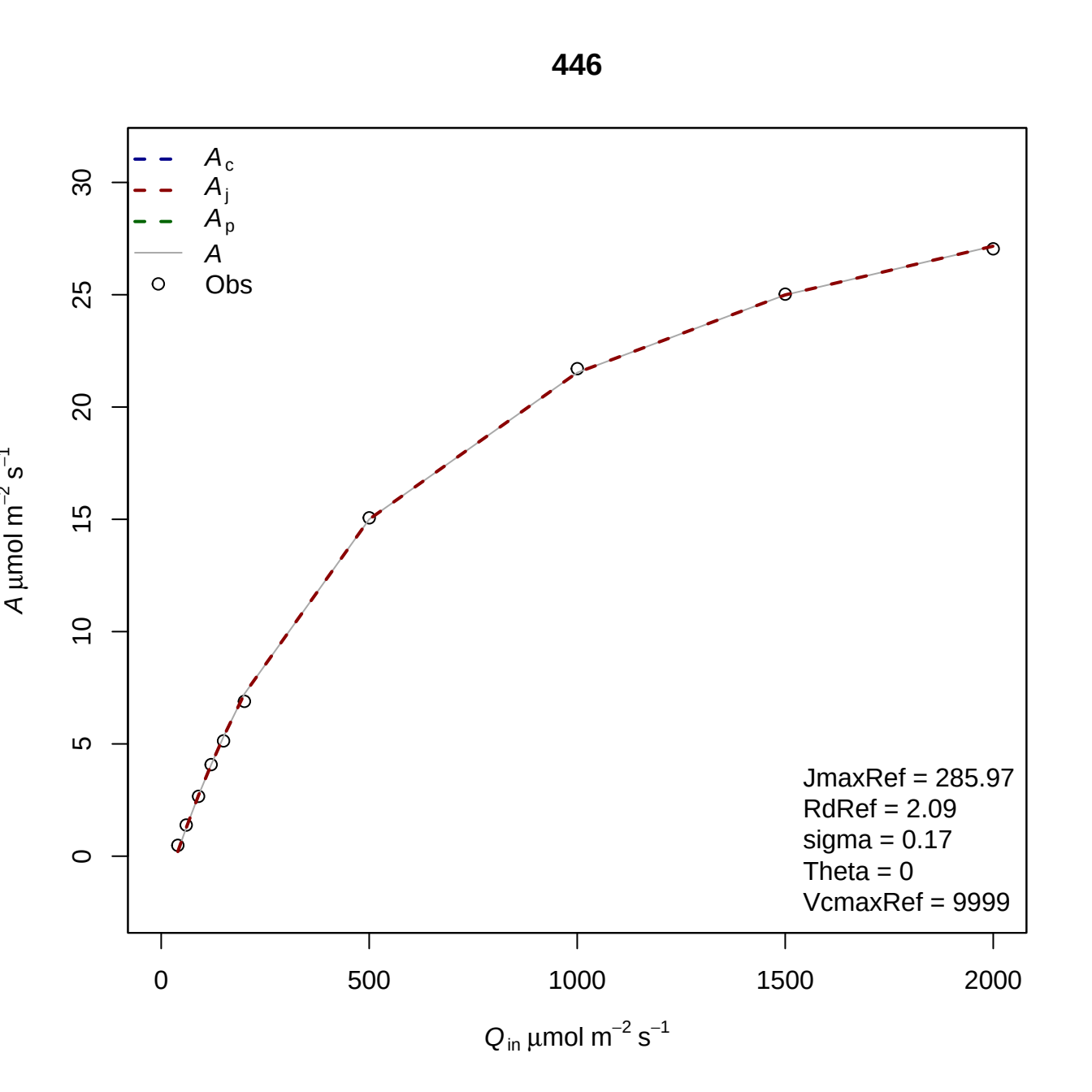

440
A_c
A_j
A_p
A
Obs
JmaxRef = 127.23
RdRef = 1.22
sigma = 0.27
Theta = 0
VcmaxRef = 62.96
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

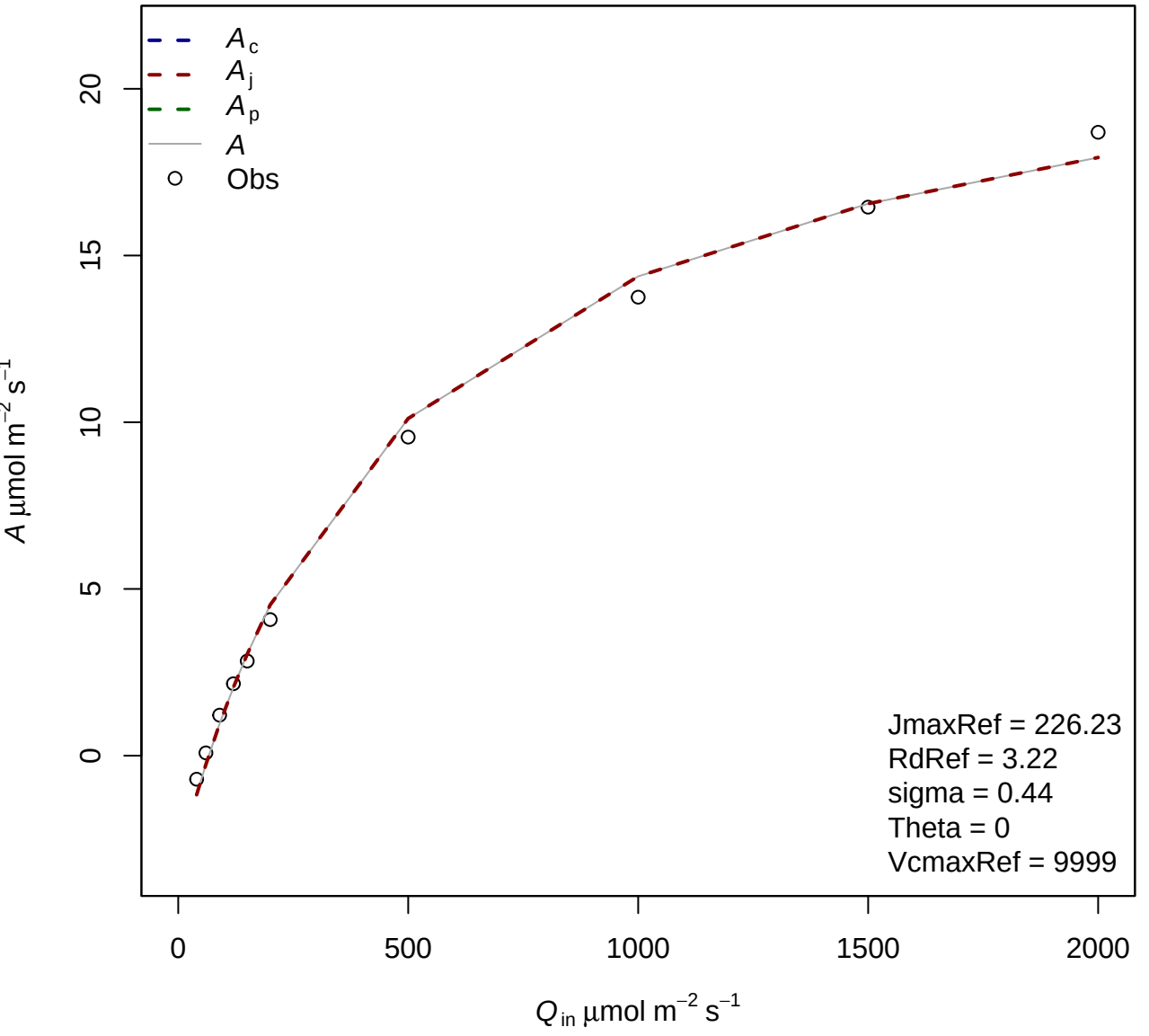

441
A_c
A_j
A_p
A
Obs
JmaxRef = 163.29
RdRef = 1.25
sigma = 0.13
Theta = 0
VcmaxRef = 94.45
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

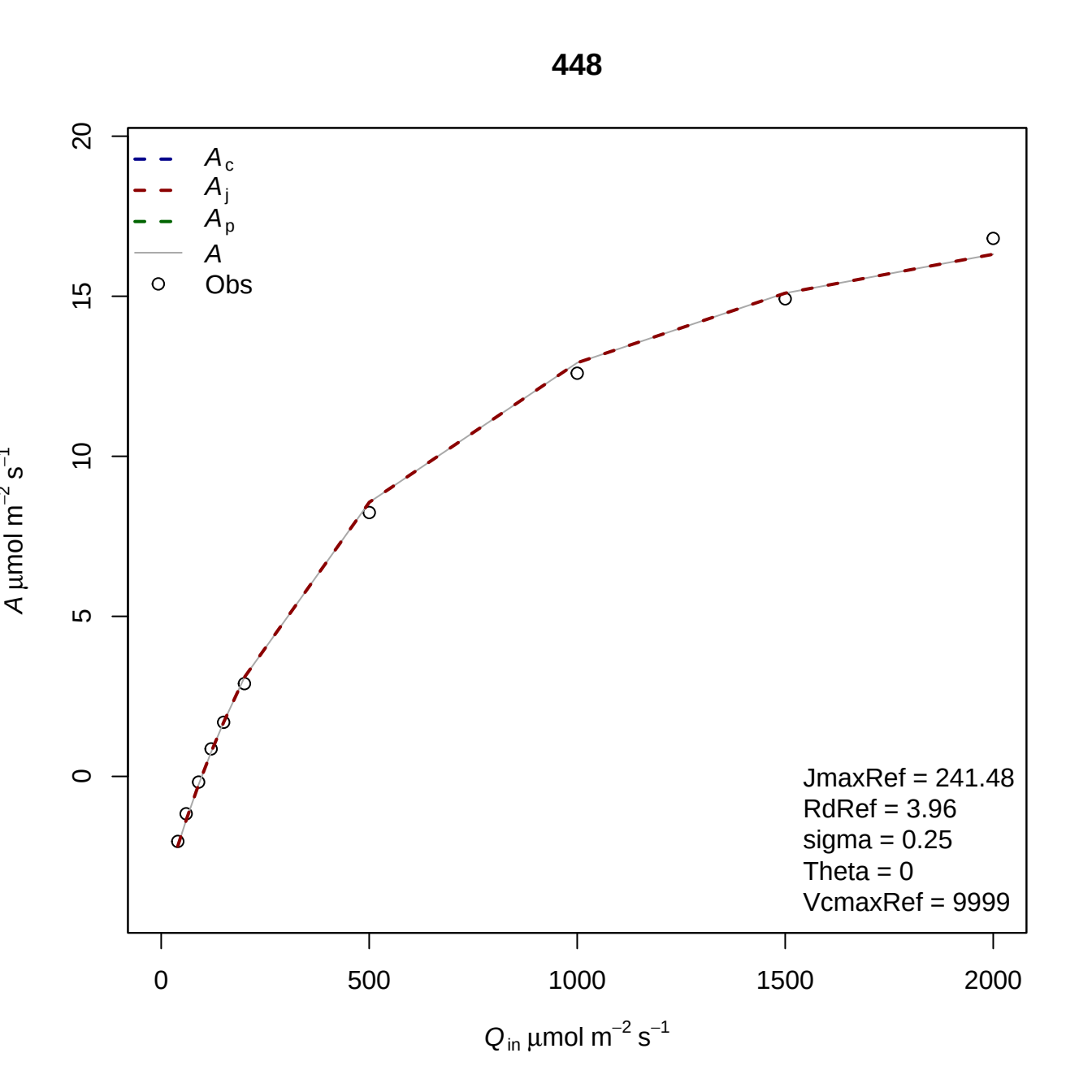

442
A_c
A_j
A_p
A
Obs
JmaxRef = 135.56
RdRef = 1.98
sigma = 0.26
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

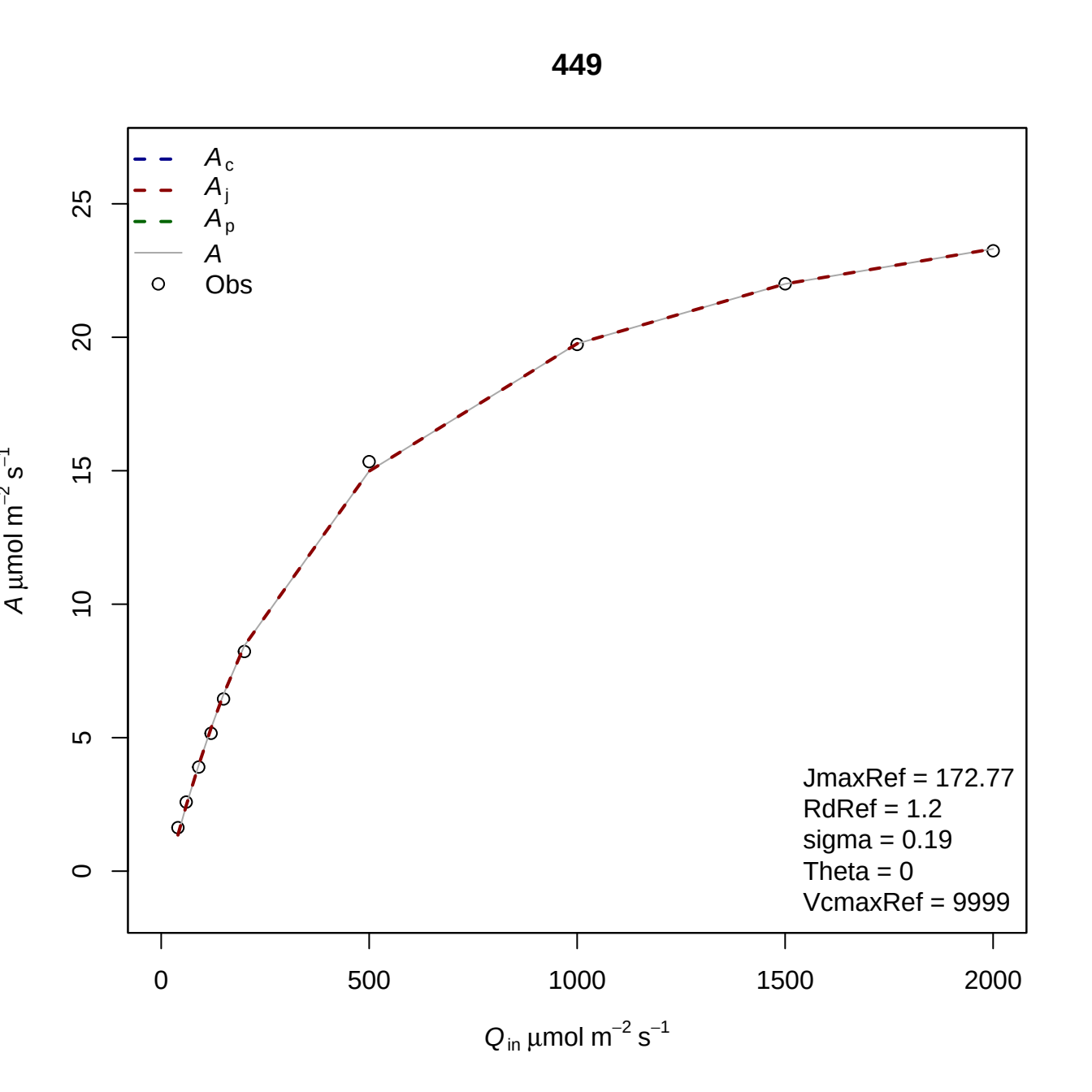

444
A_c
A_j
A_p
A
Obs
JmaxRef = 27.47
RdRef = 1.75
sigma = 0.35
Theta = 0
VcmaxRef = 9999
2
1
0
−1
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

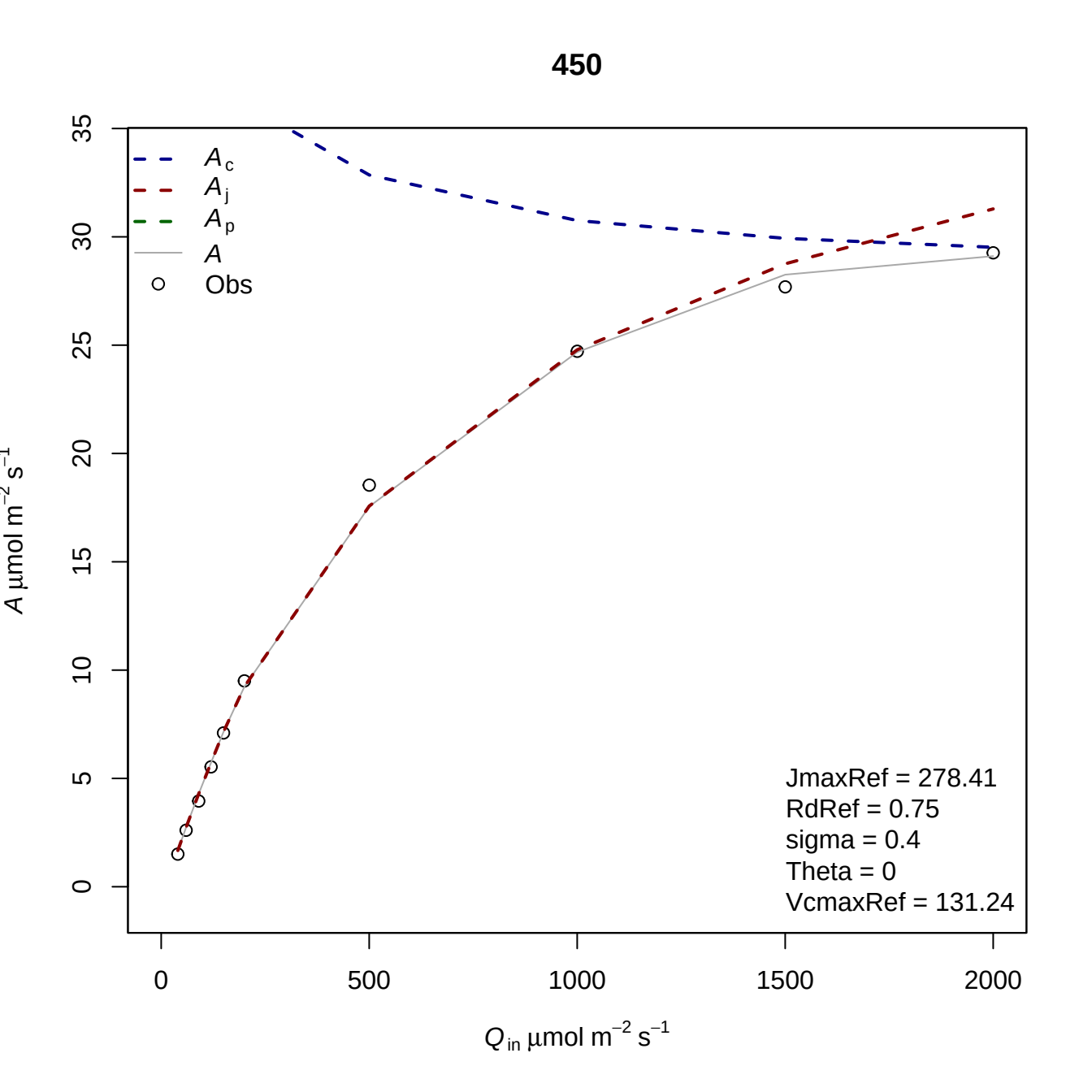

445
A_c
A_j
A_p
A
Obs
JmaxRef = 145.52
RdRef = 1.57
sigma = 0.32
Theta = 0
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

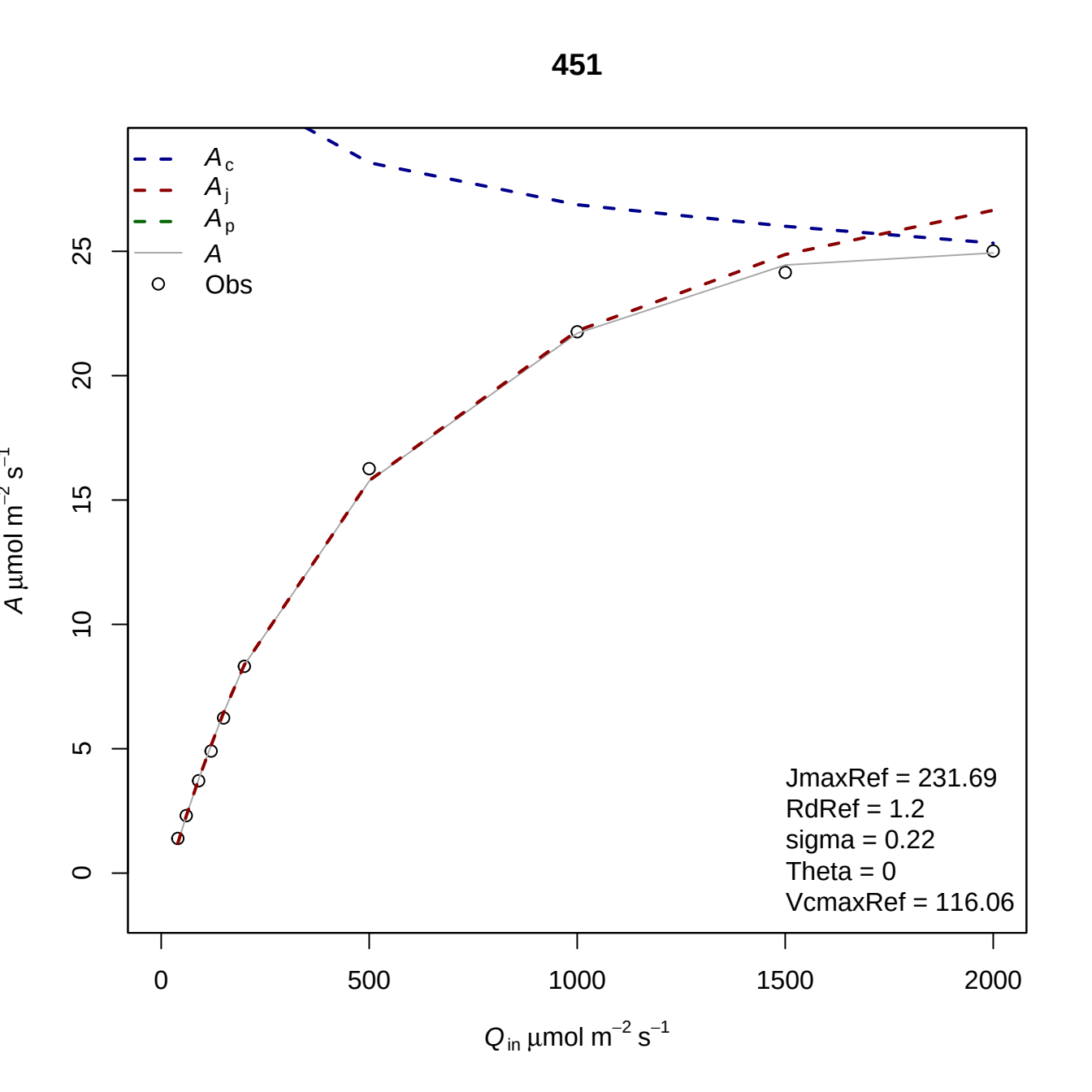

446
A_c
A_j
A_p
A
Obs
JmaxRef = 285.97
RdRef = 2.09
sigma = 0.17
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

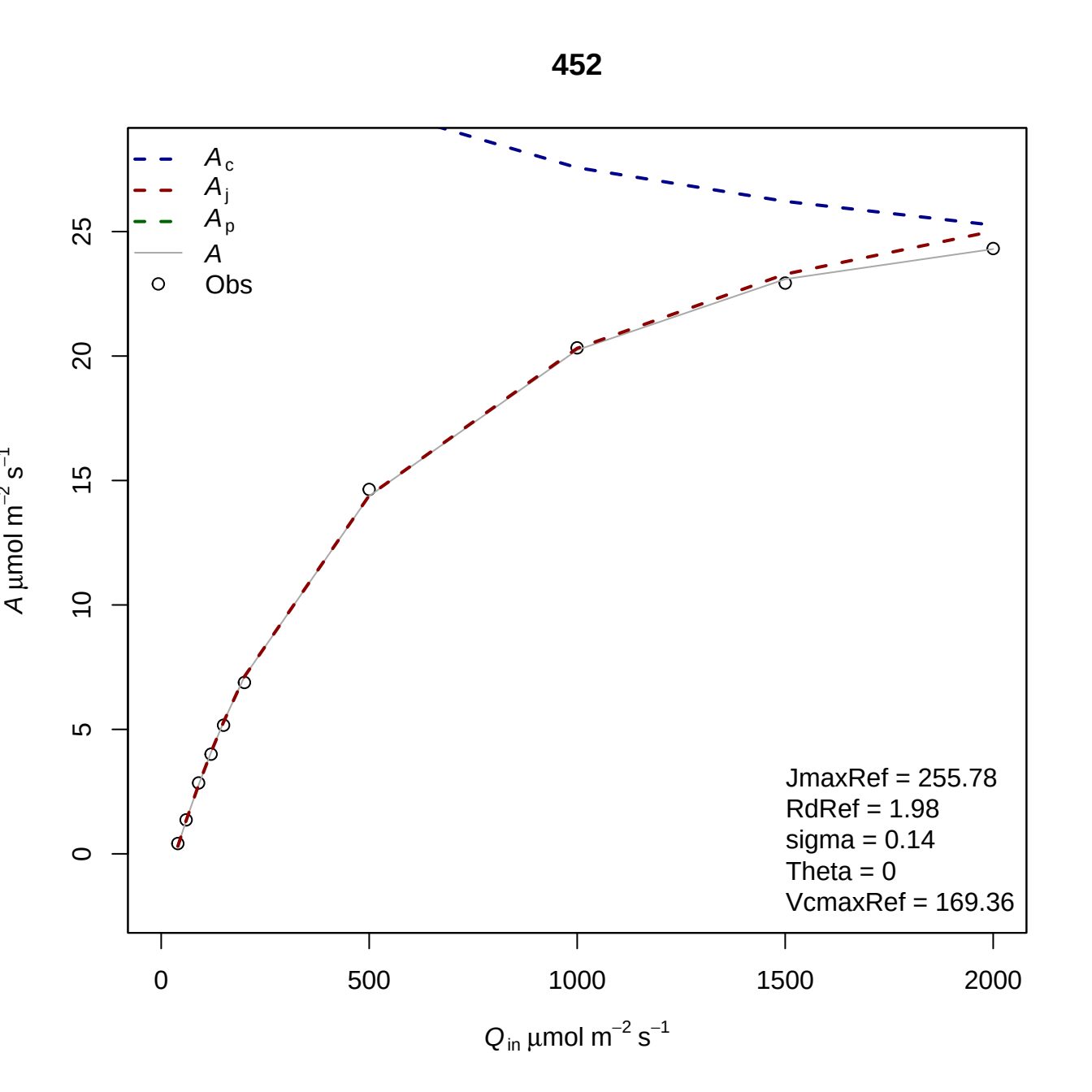

A_c
A_j
A_p
A
Obs
JmaxRef = 226.23
RdRef = 3.22
sigma = 0.44
Theta = 0
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

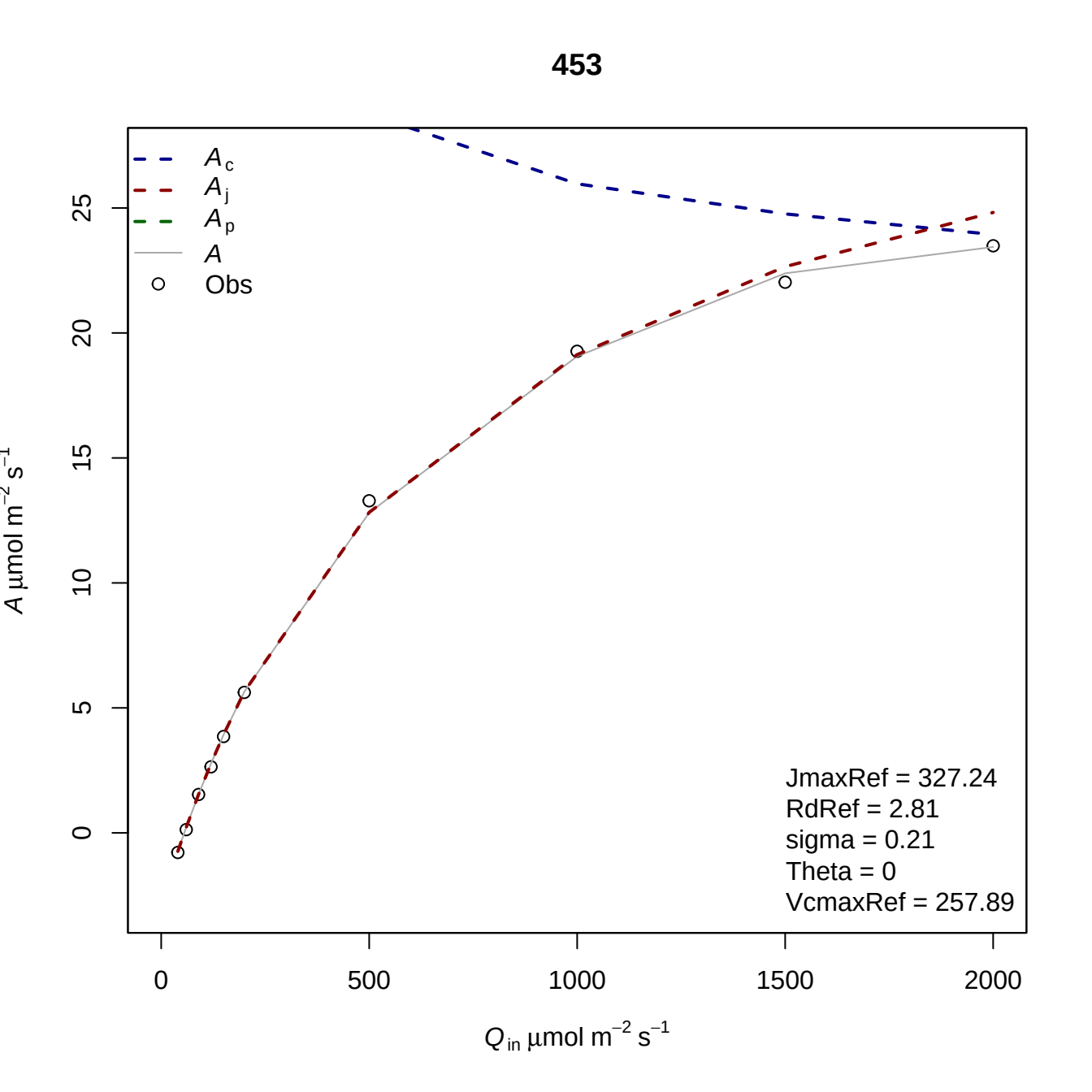

448
A_c
A_j
A_p
A
Obs
JmaxRef = 241.48
RdRef = 3.96
sigma = 0.25
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

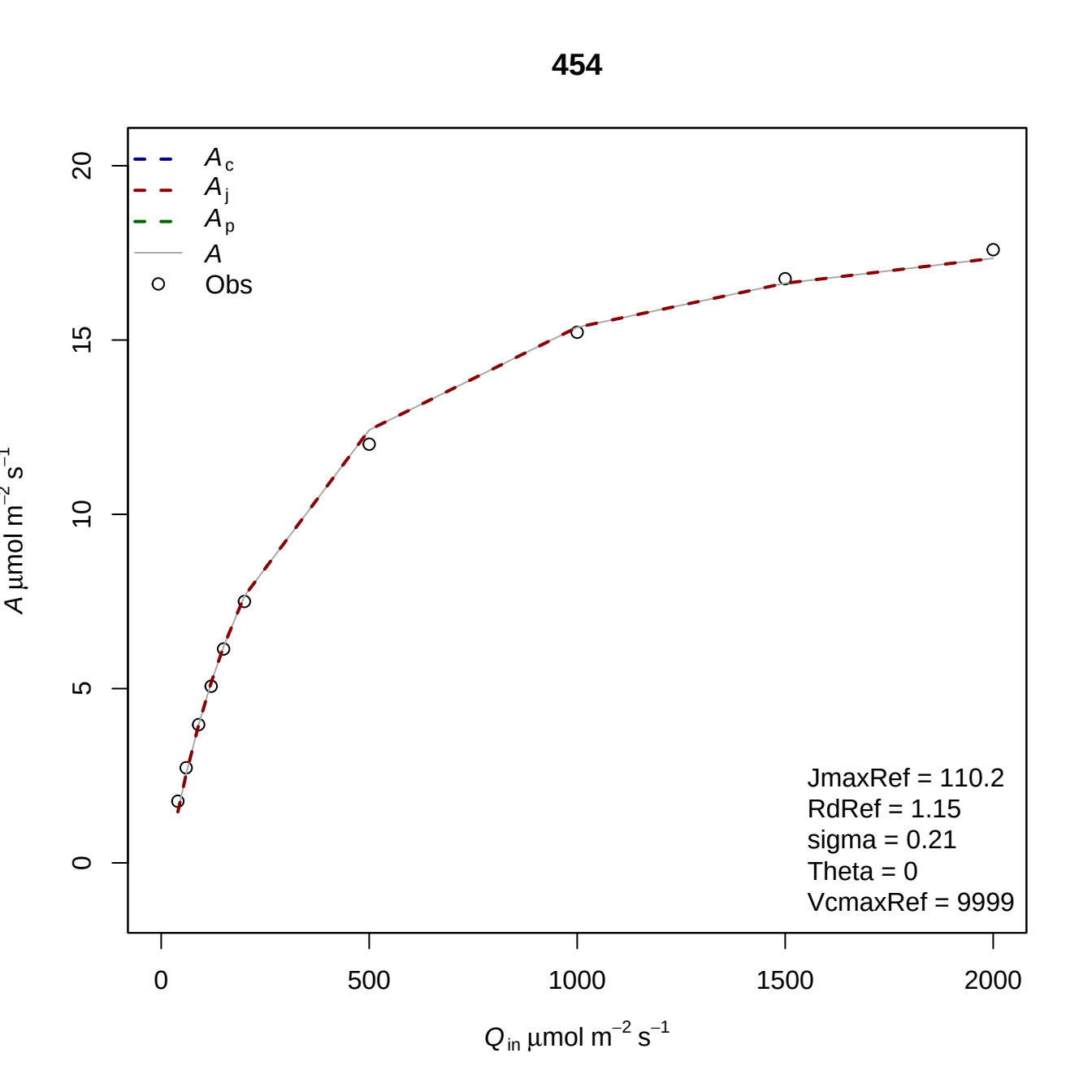

449
A_c
A_j
A_p
A
Obs
JmaxRef = 172.77
RdRef = 1.2
sigma = 0.19
Theta = 0
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

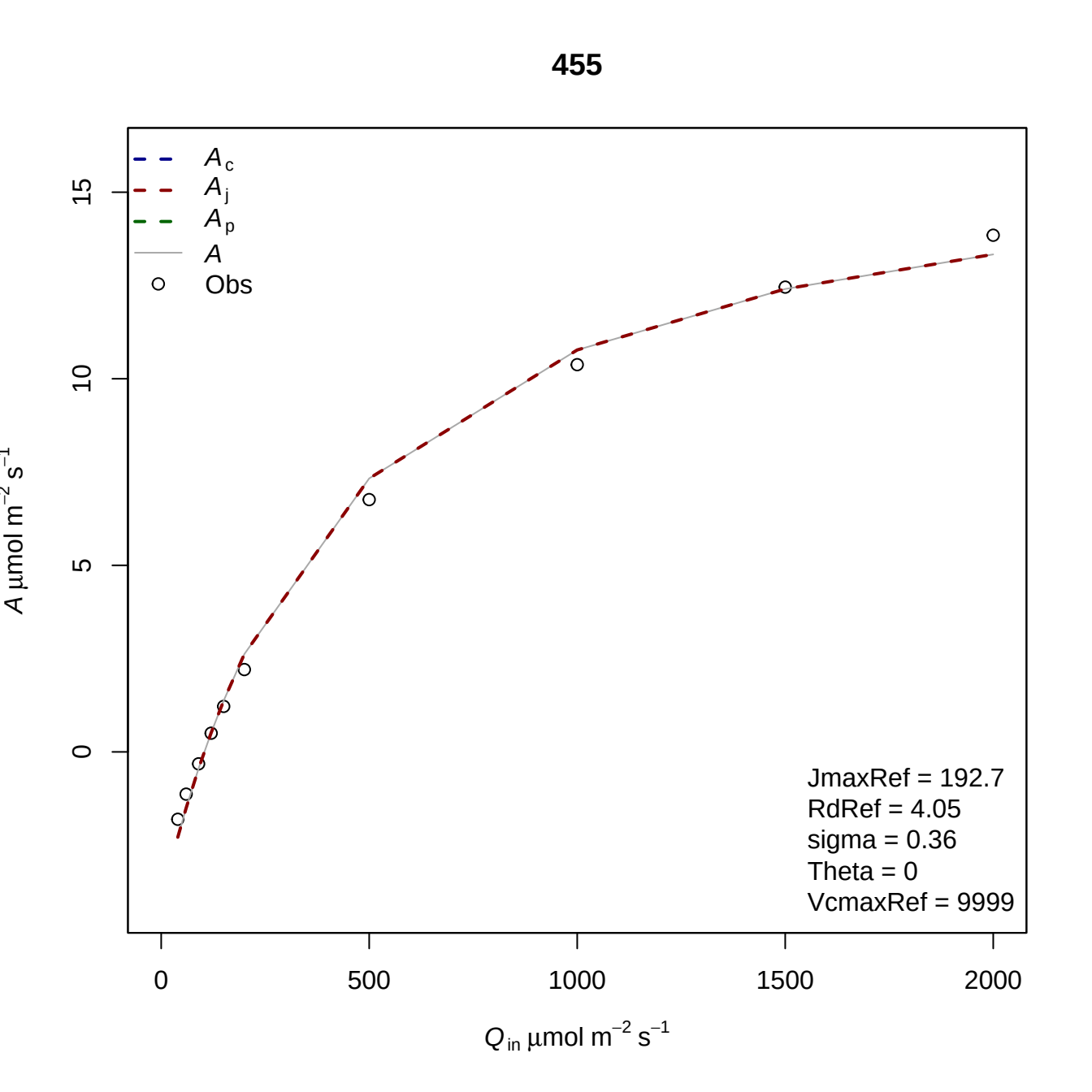

450
A_c
A_j
A_p
A
Obs
JmaxRef = 278.41
RdRef = 0.75
sigma = 0.4
Theta = 0
VcmaxRef = 131.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

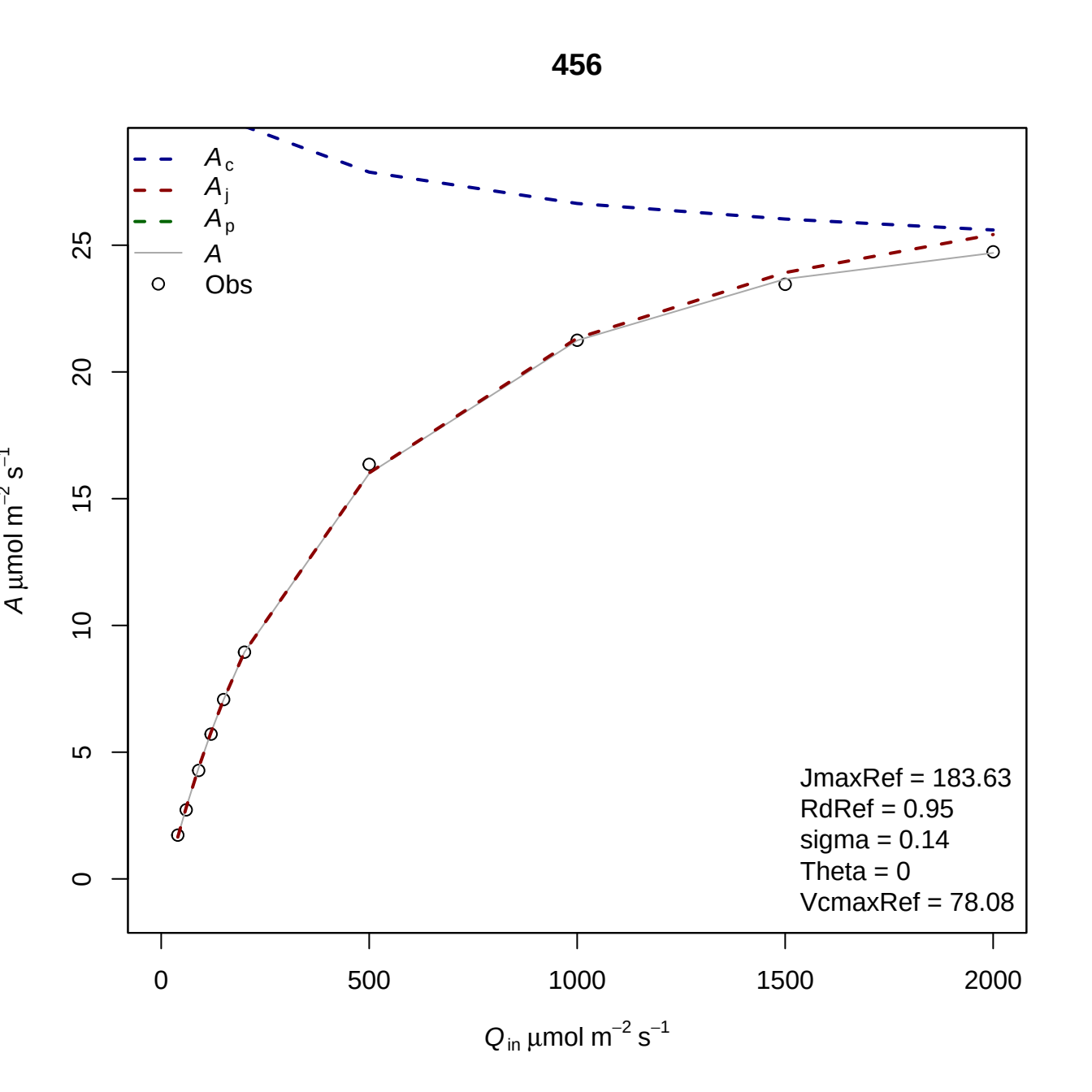

451
A_c
A_j
A_p
A
Obs
JmaxRef = 231.69
RdRef = 1.2
sigma = 0.22
Theta = 0
VcmaxRef = 116.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

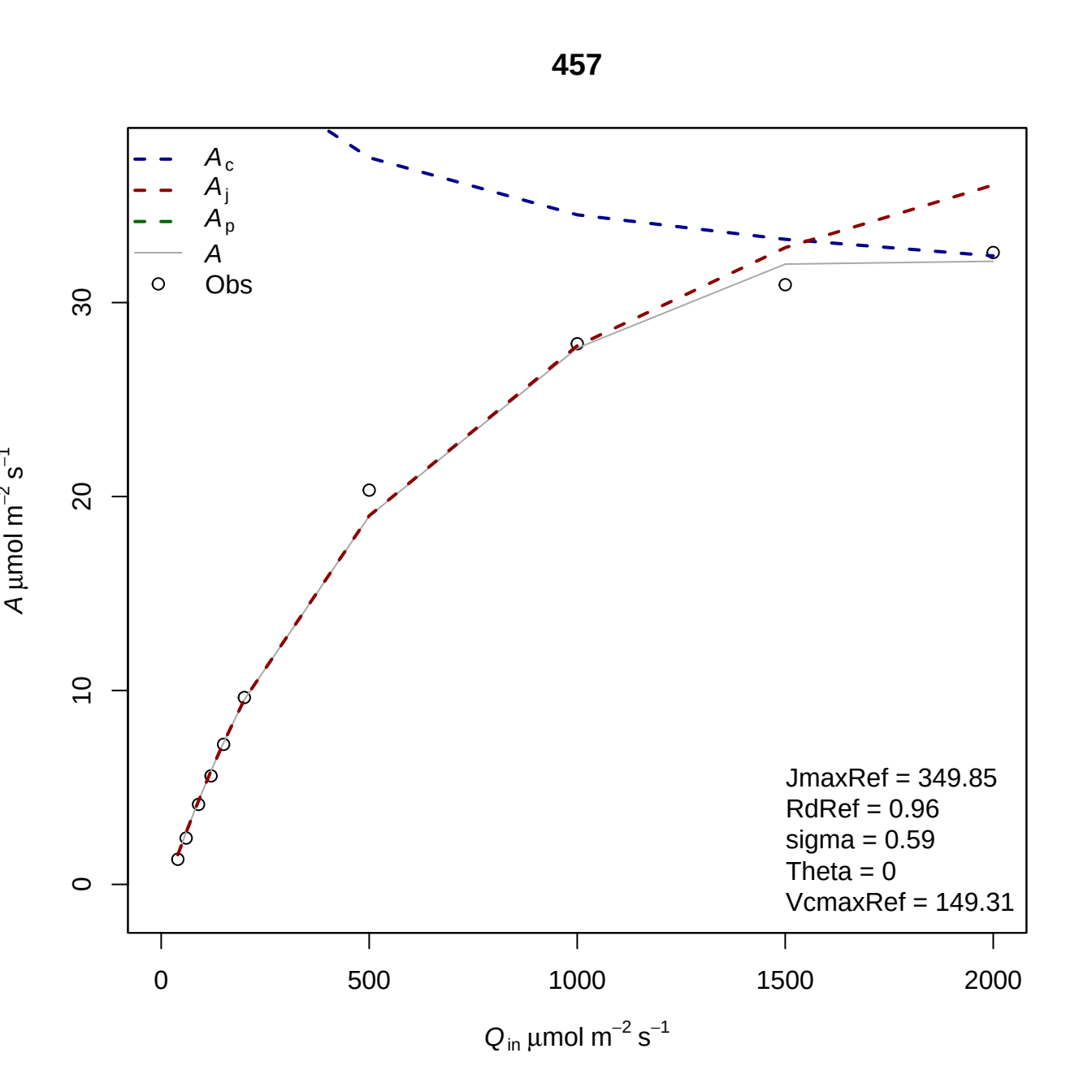

452
A_c
A_j
A_p
A
Obs
JmaxRef = 255.78
RdRef = 1.98
sigma = 0.14
Theta = 0
VcmaxRef = 169.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

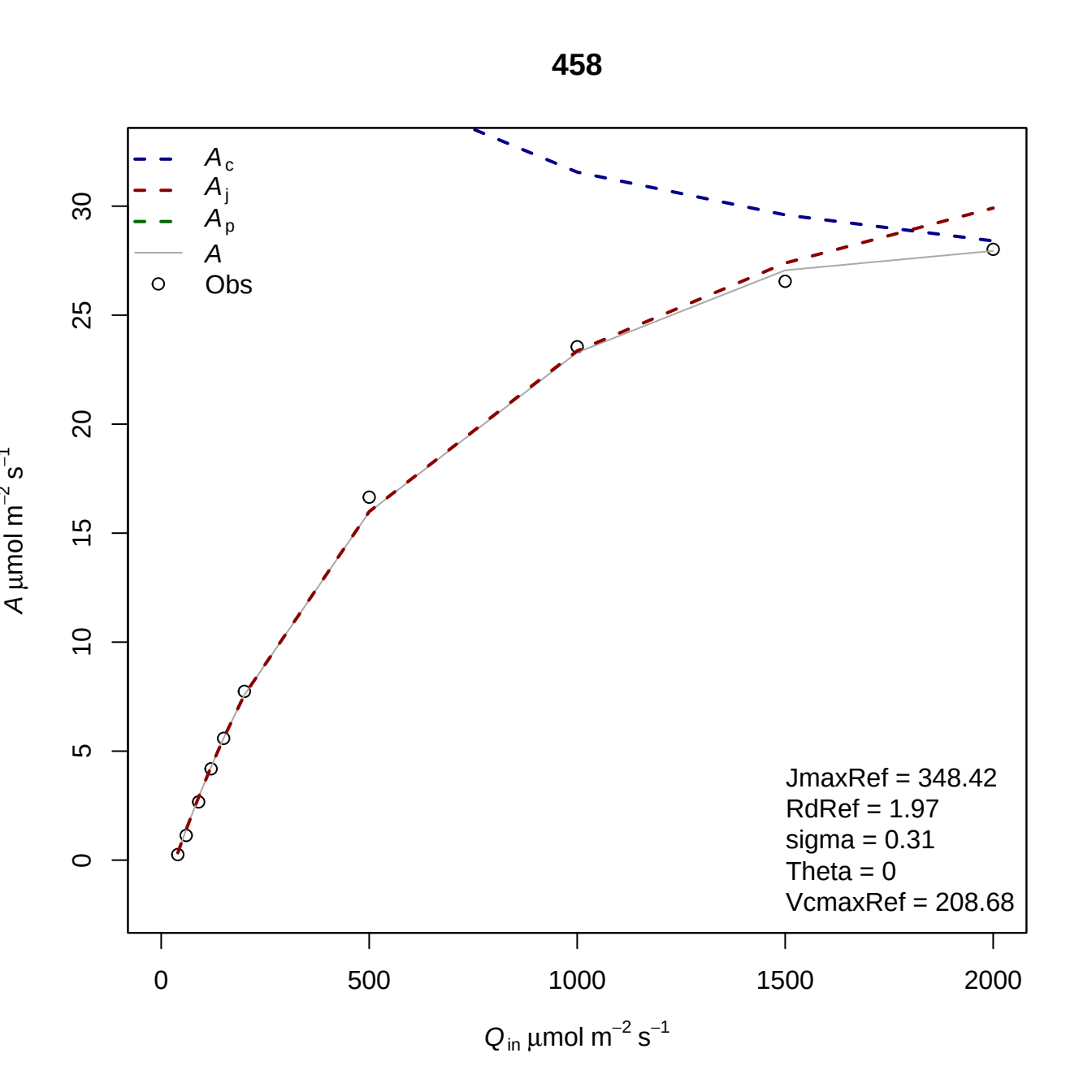

453
A_c
A_j
A_p
A
Obs
JmaxRef = 327.24
RdRef = 2.81
sigma = 0.21
Theta = 0
VcmaxRef = 257.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

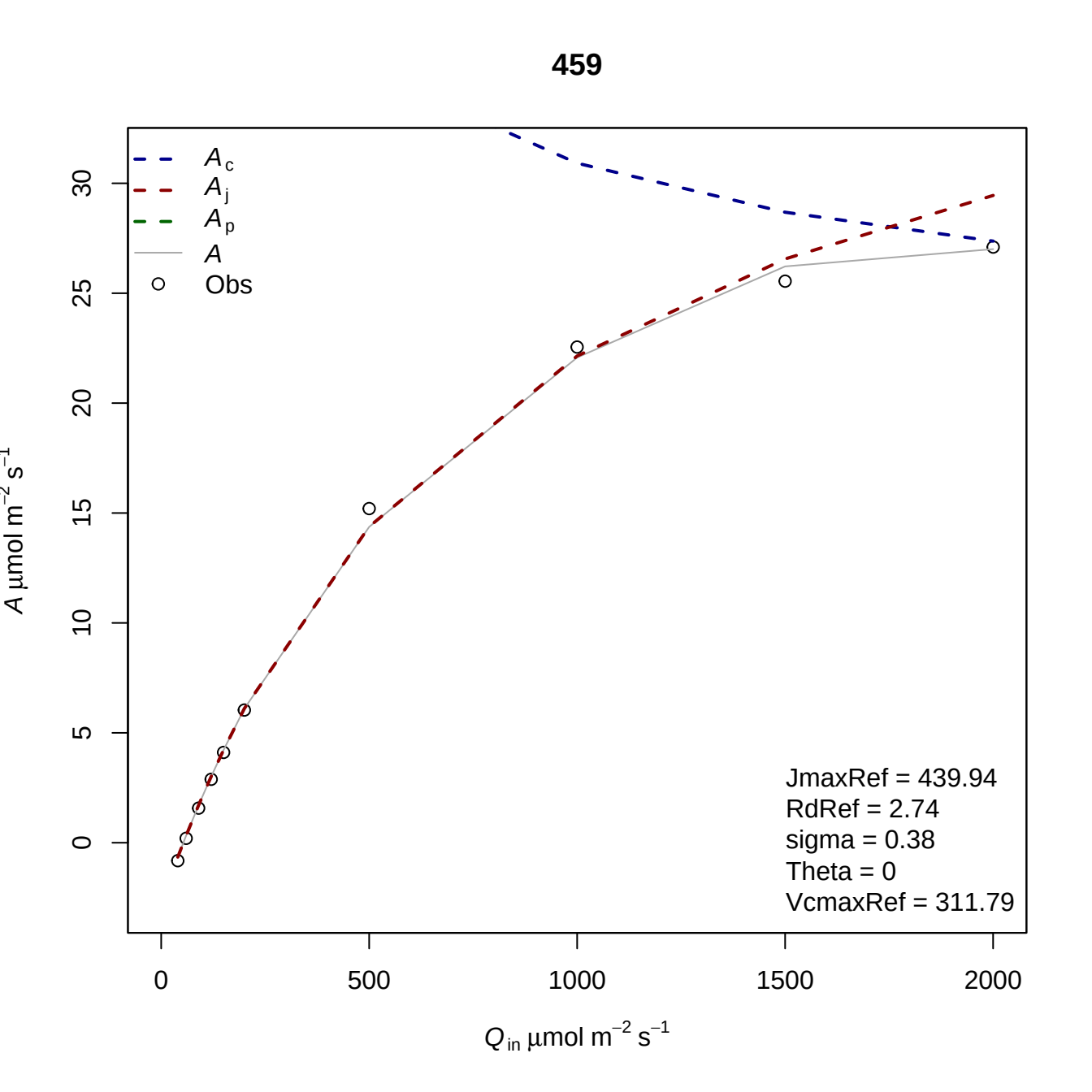

454
A_c
A_j
A_p
A
Obs
JmaxRef = 110.2
RdRef = 1.15
sigma = 0.21
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

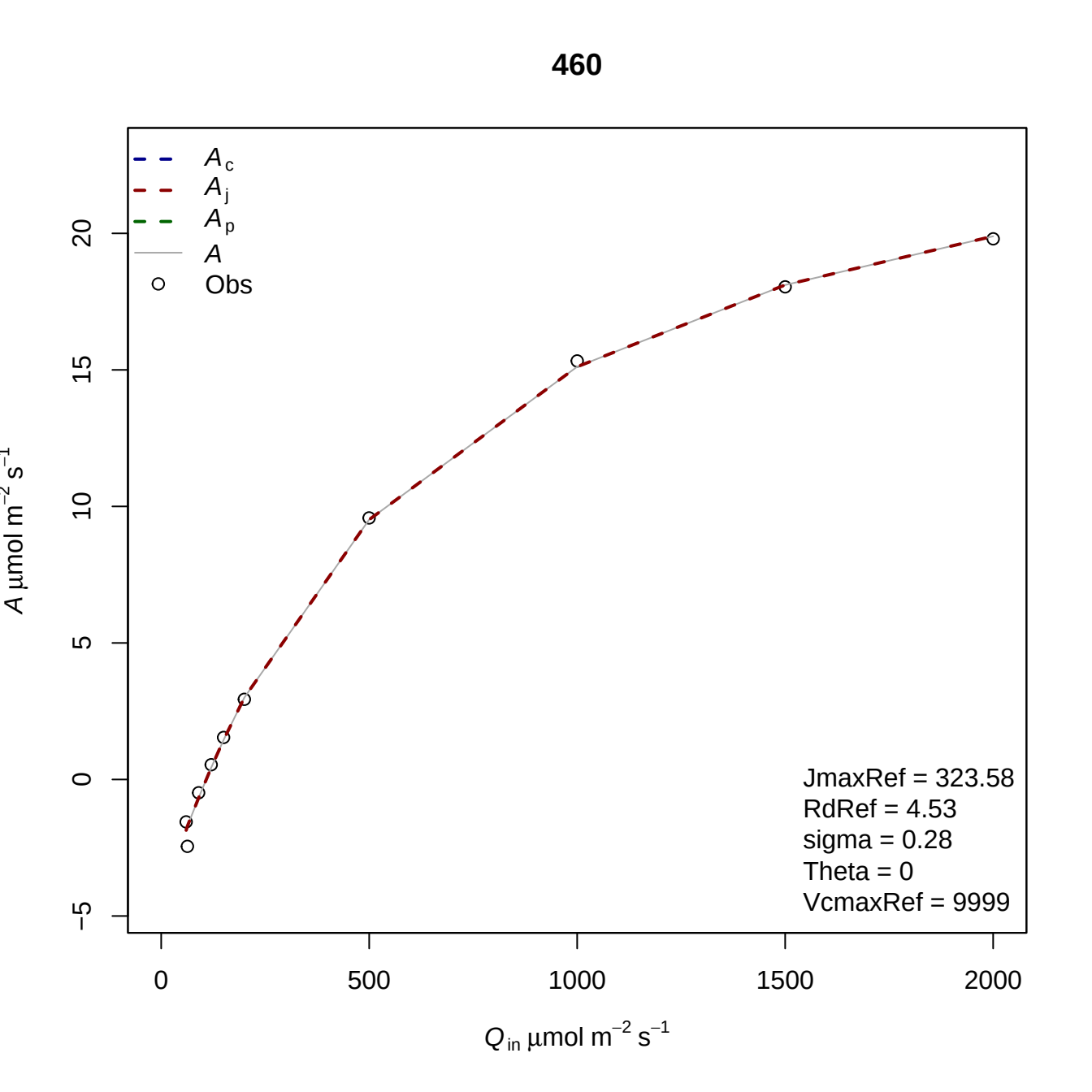

455
A_c
A_j
A_p
A
Obs
JmaxRef = 192.7
RdRef = 4.05
sigma = 0.36
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

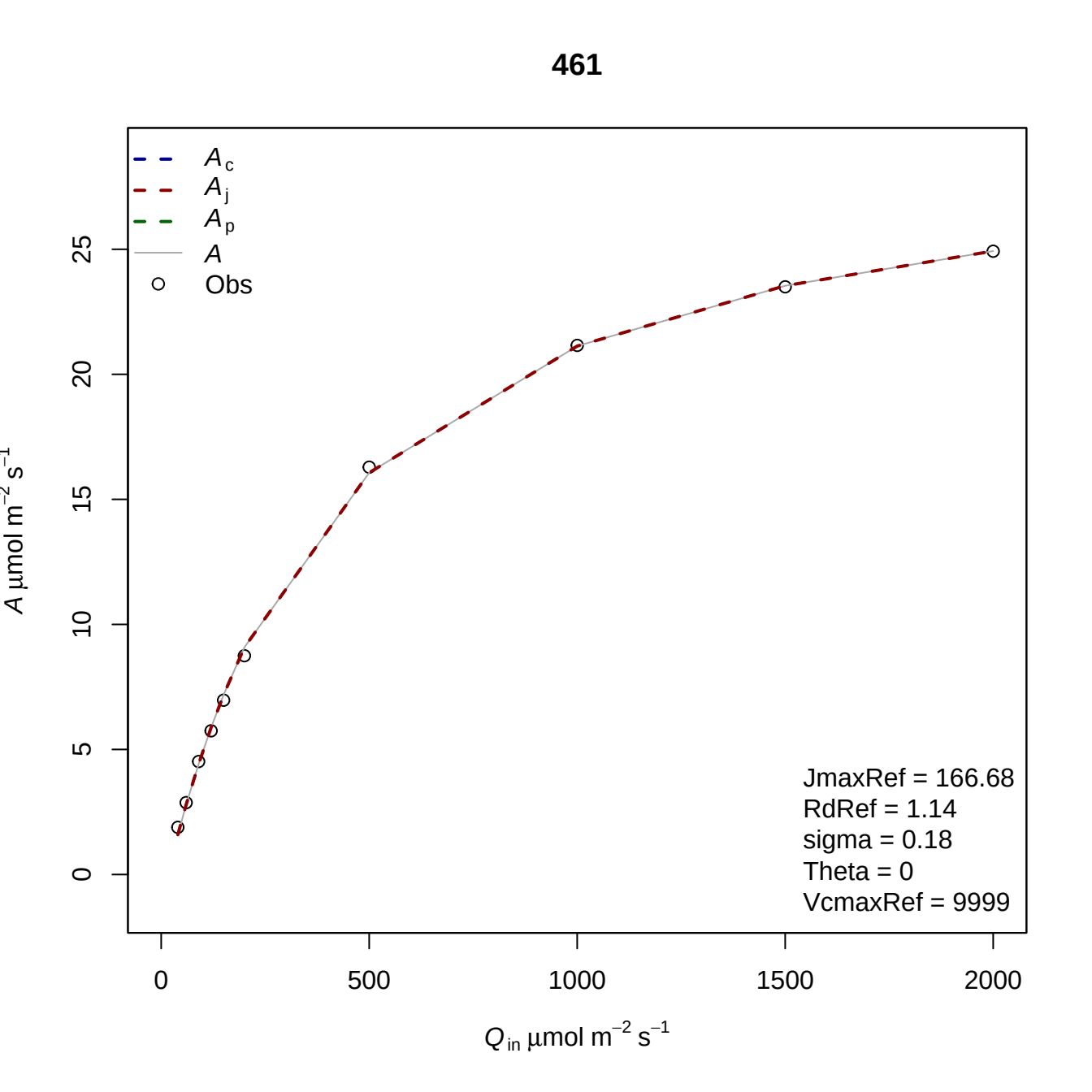

456
A_c
A_j
A_p
A
Obs
JmaxRef = 183.63
RdRef = 0.95
sigma = 0.14
Theta = 0
VcmaxRef = 78.08
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

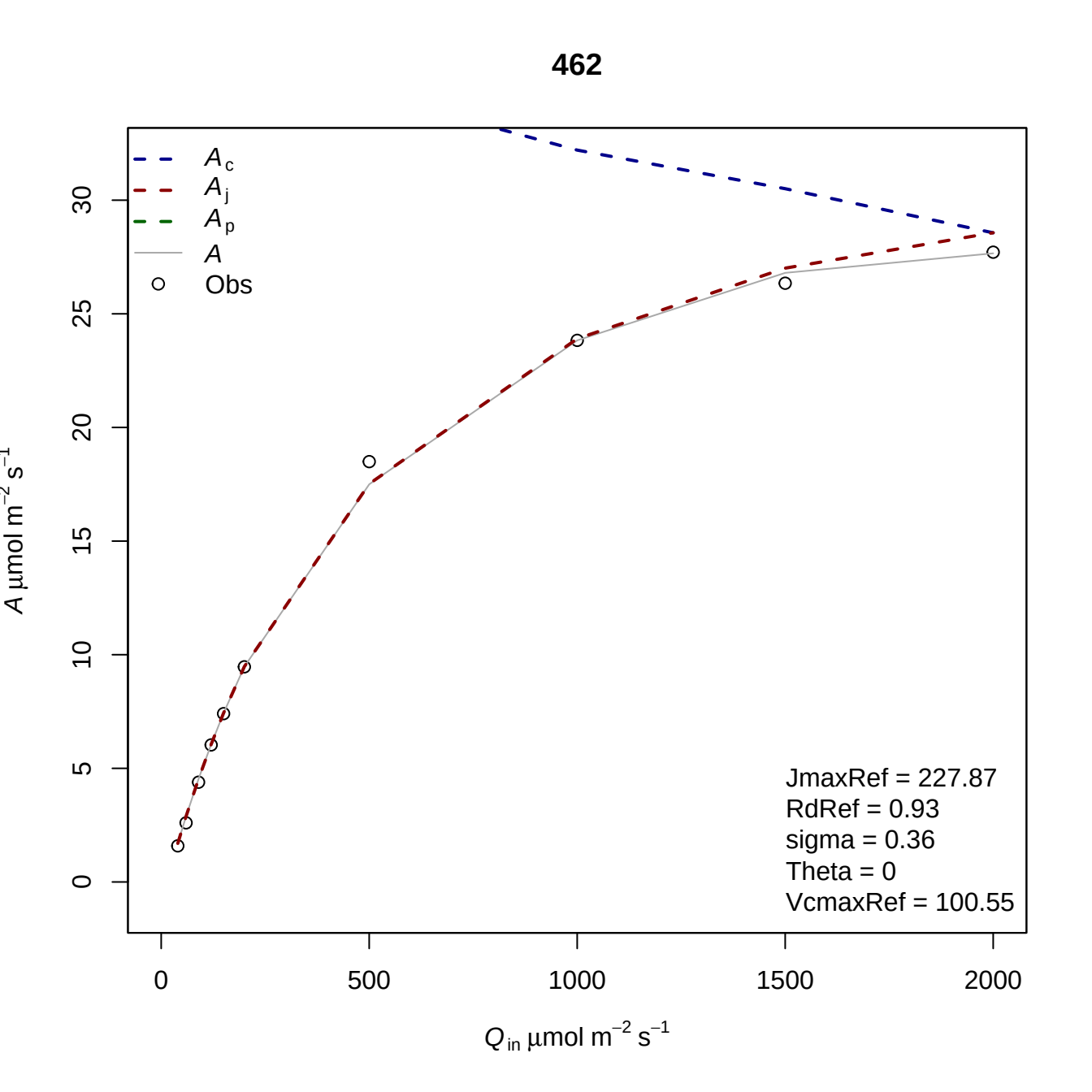

457
A_c
A_j
A_p
A
Obs
JmaxRef = 349.85
RdRef = 0.96
sigma = 0.59
Theta = 0
VcmaxRef = 149.31
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

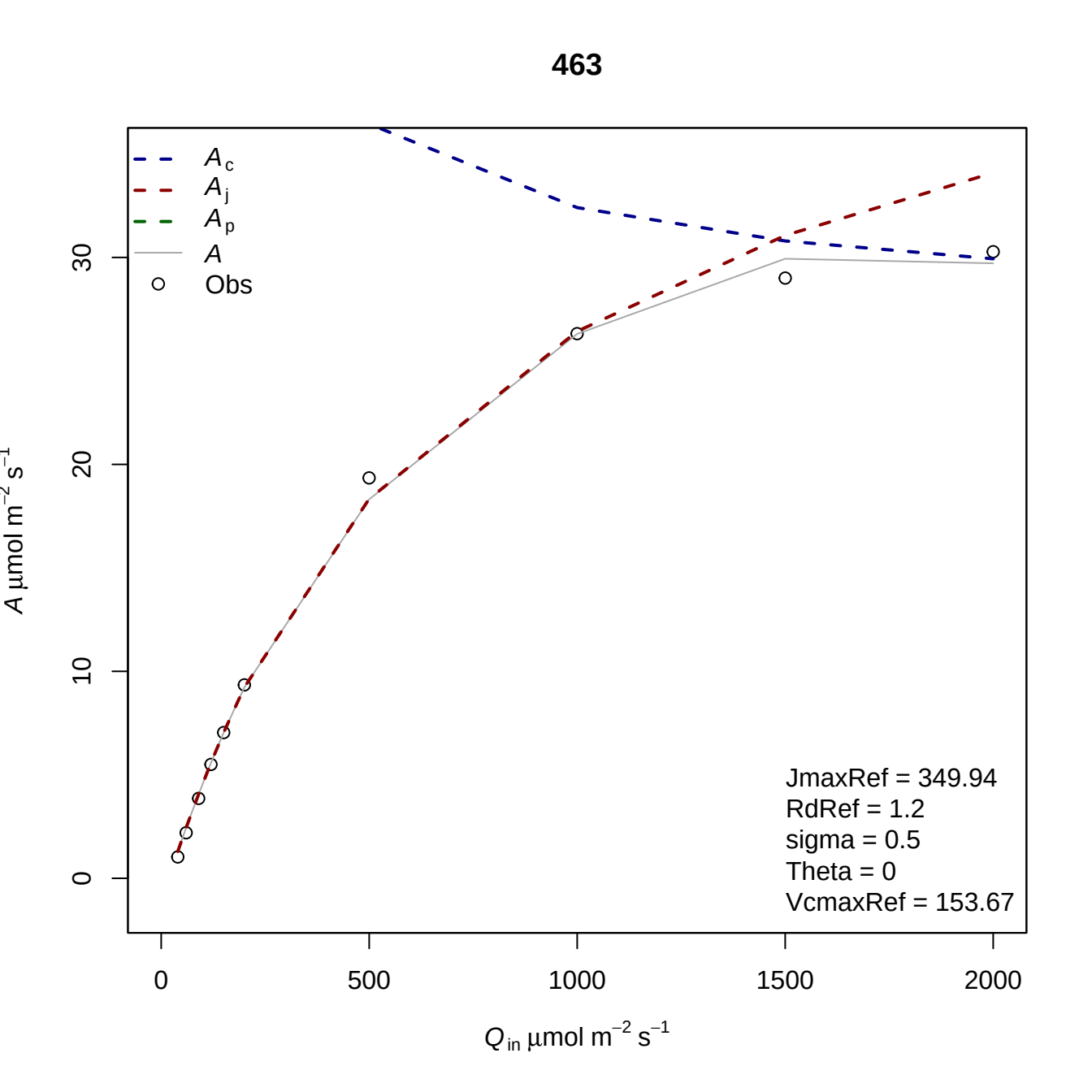

458
A_c
A_j
A_p
A
Obs
JmaxRef = 348.42
RdRef = 1.97
sigma = 0.31
Theta = 0
VcmaxRef = 208.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

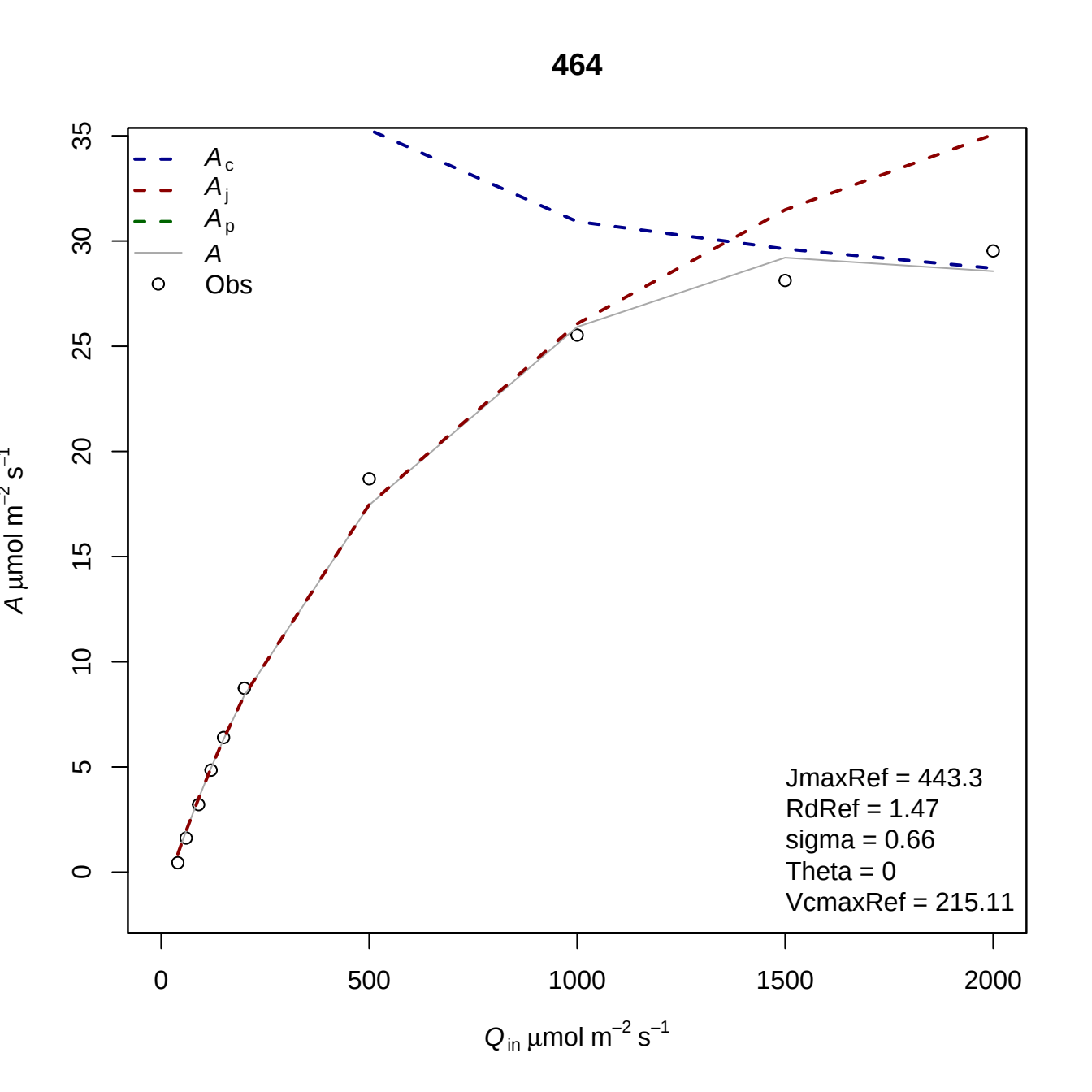

459
A_c
A_j
A_p
A
Obs
JmaxRef = 439.94
RdRef = 2.74
sigma = 0.38
Theta = 0
VcmaxRef = 311.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

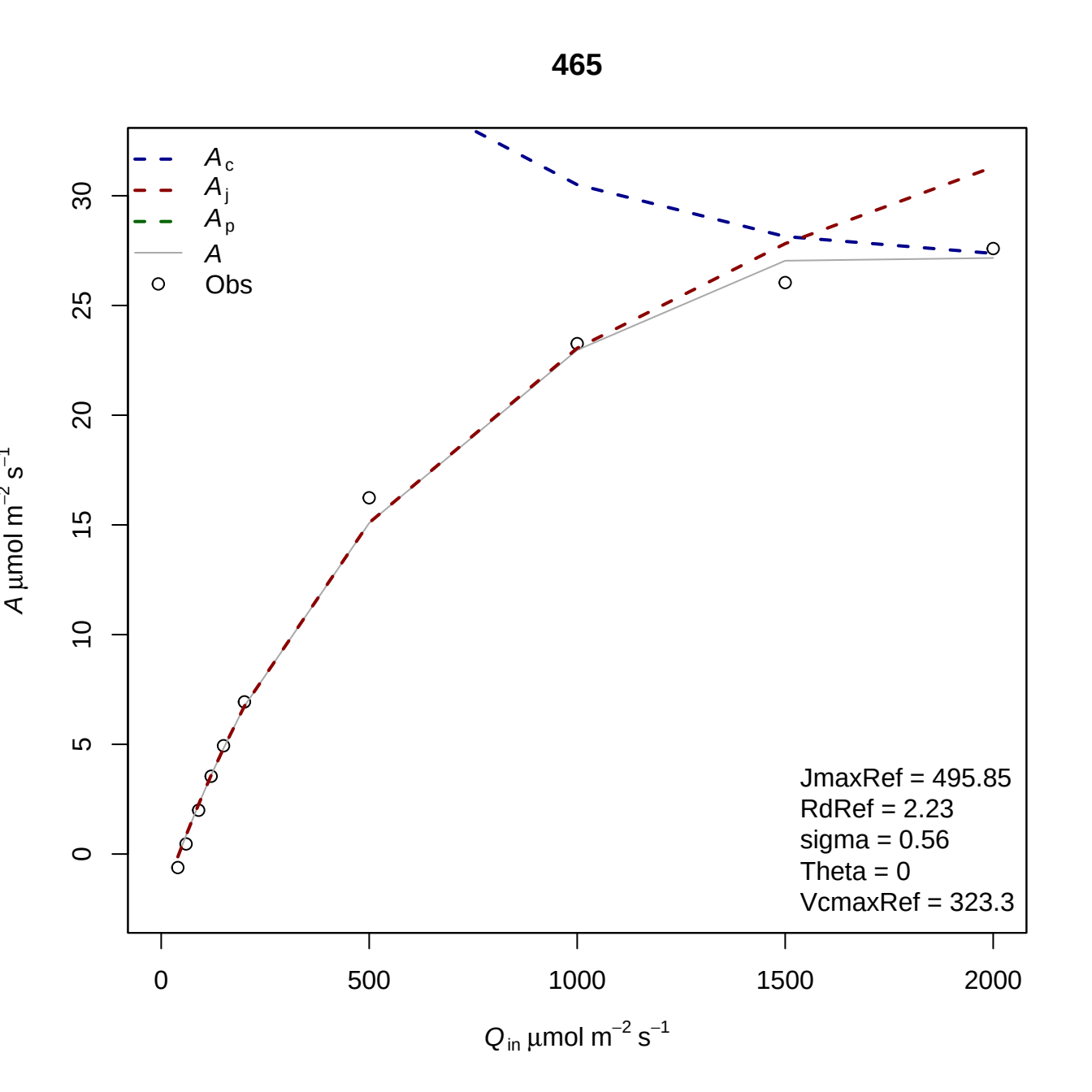

460
A_c
A_j
A_p
A
Obs
JmaxRef = 323.58
RdRef = 4.53
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

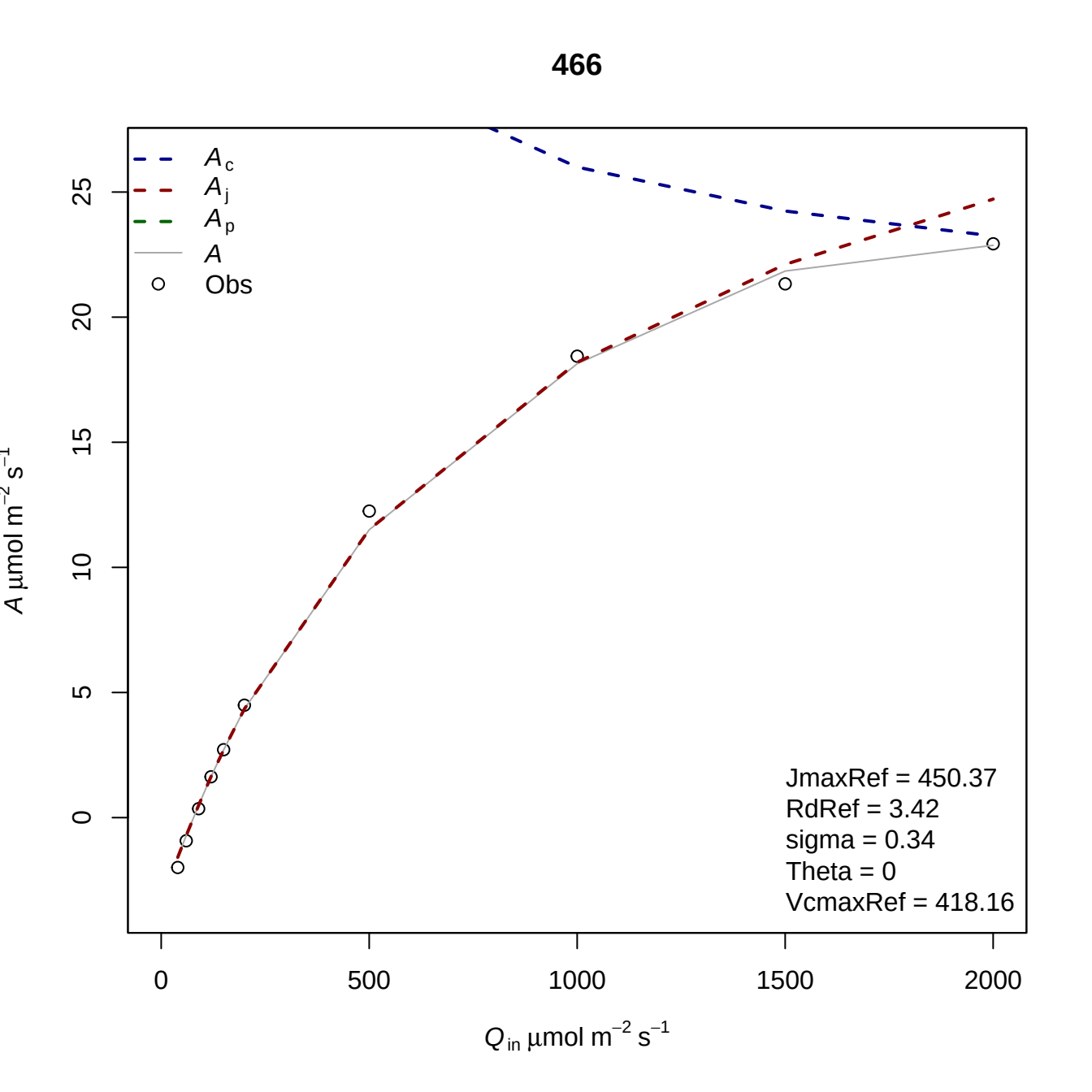

461
A_c
A_j
A_p
A
Obs
JmaxRef = 166.68
RdRef = 1.14
sigma = 0.18
Theta = 0
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

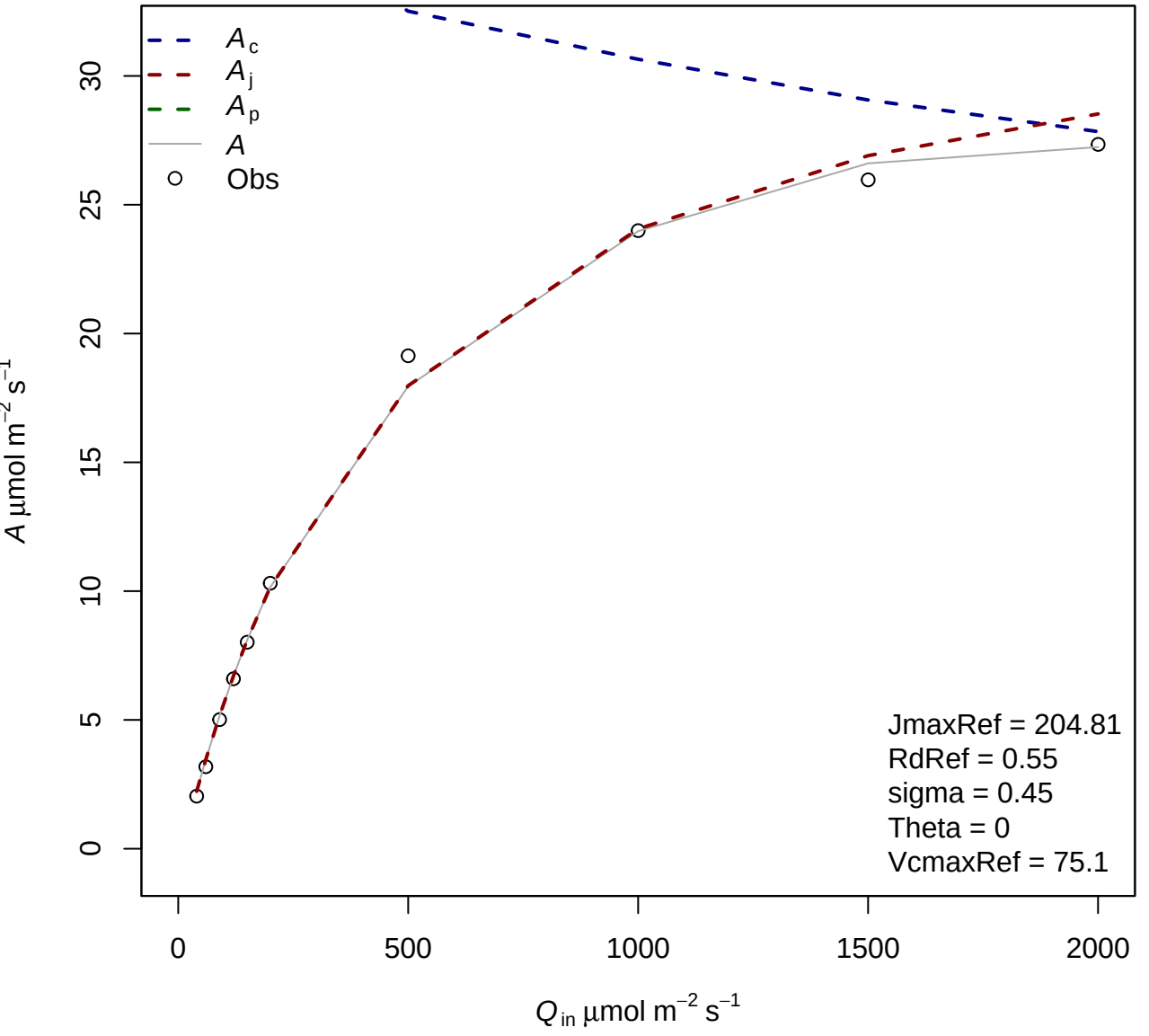

462
A_c
A_j
A_p
A
Obs
JmaxRef = 227.87
RdRef = 0.93
sigma = 0.36
Theta = 0
VcmaxRef = 100.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

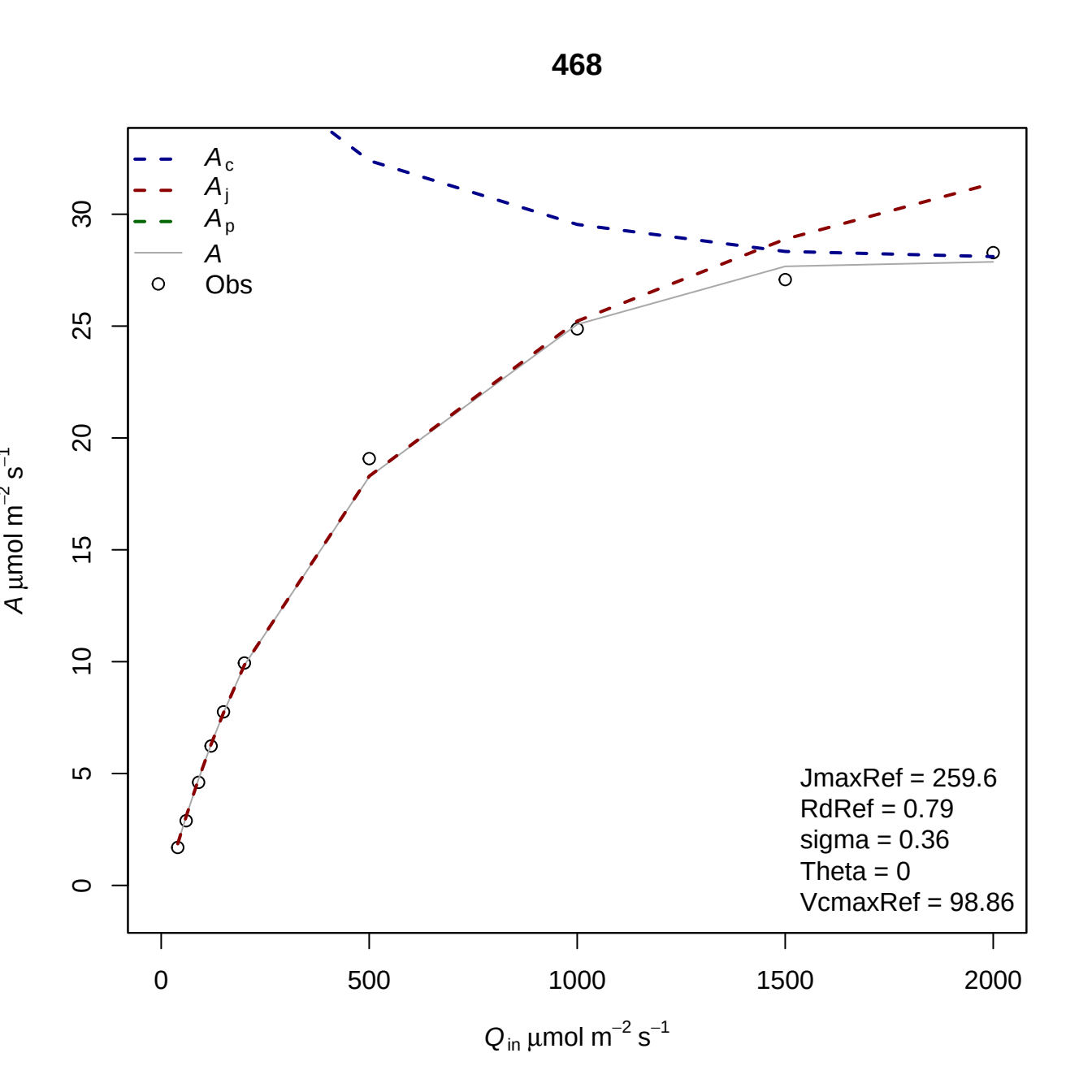

463
A_c
A_j
A_p
A
Obs
JmaxRef = 349.94
RdRef = 1.2
sigma = 0.5
Theta = 0
VcmaxRef = 153.67
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

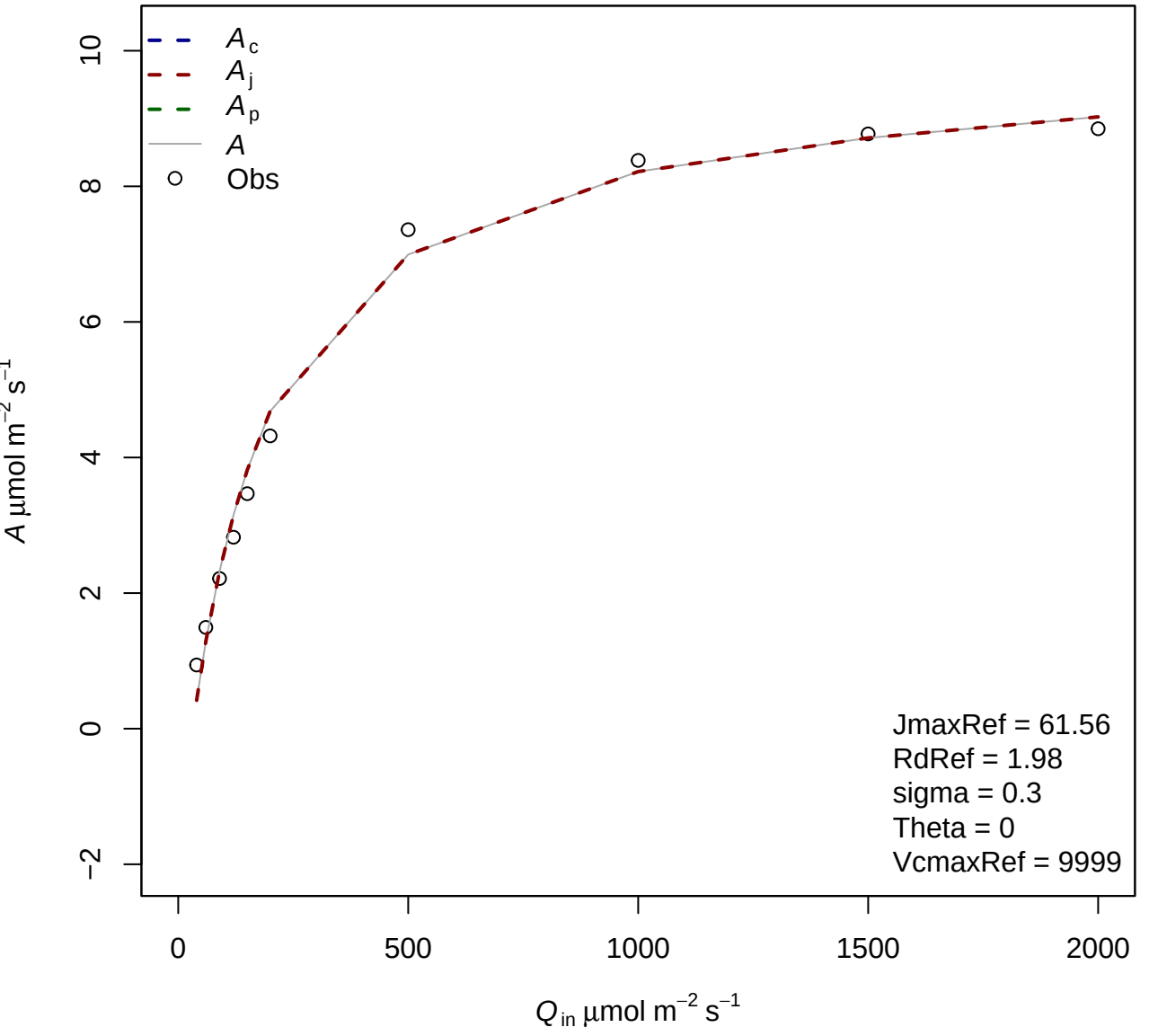

464
A_c
A_j
A_p
A
Obs
JmaxRef = 443.3
RdRef = 1.47
sigma = 0.66
Theta = 0
VcmaxRef = 215.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

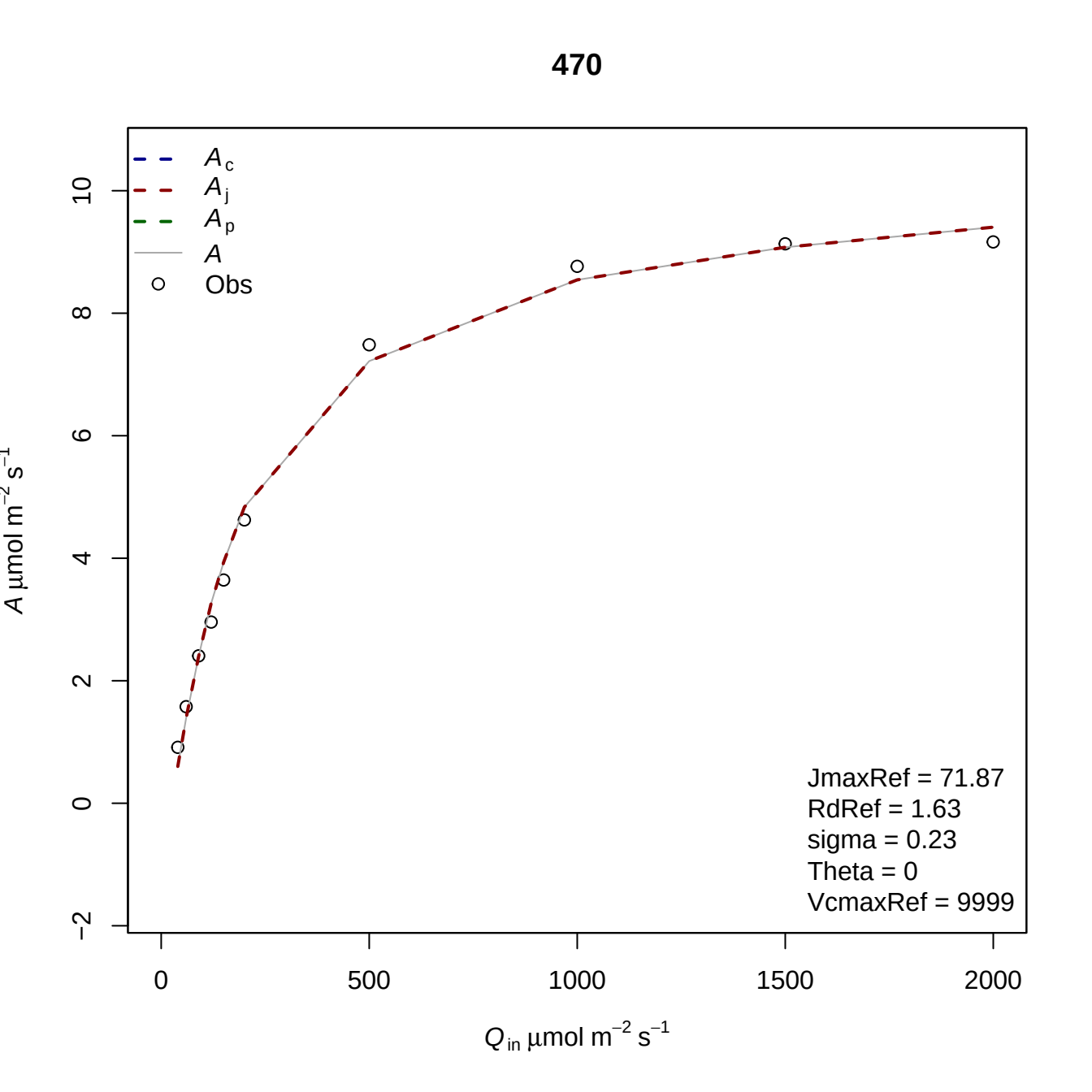

465
A_c
A_j
A_p
A
Obs
JmaxRef = 495.85
RdRef = 2.23
sigma = 0.56
Theta = 0
VcmaxRef = 323.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

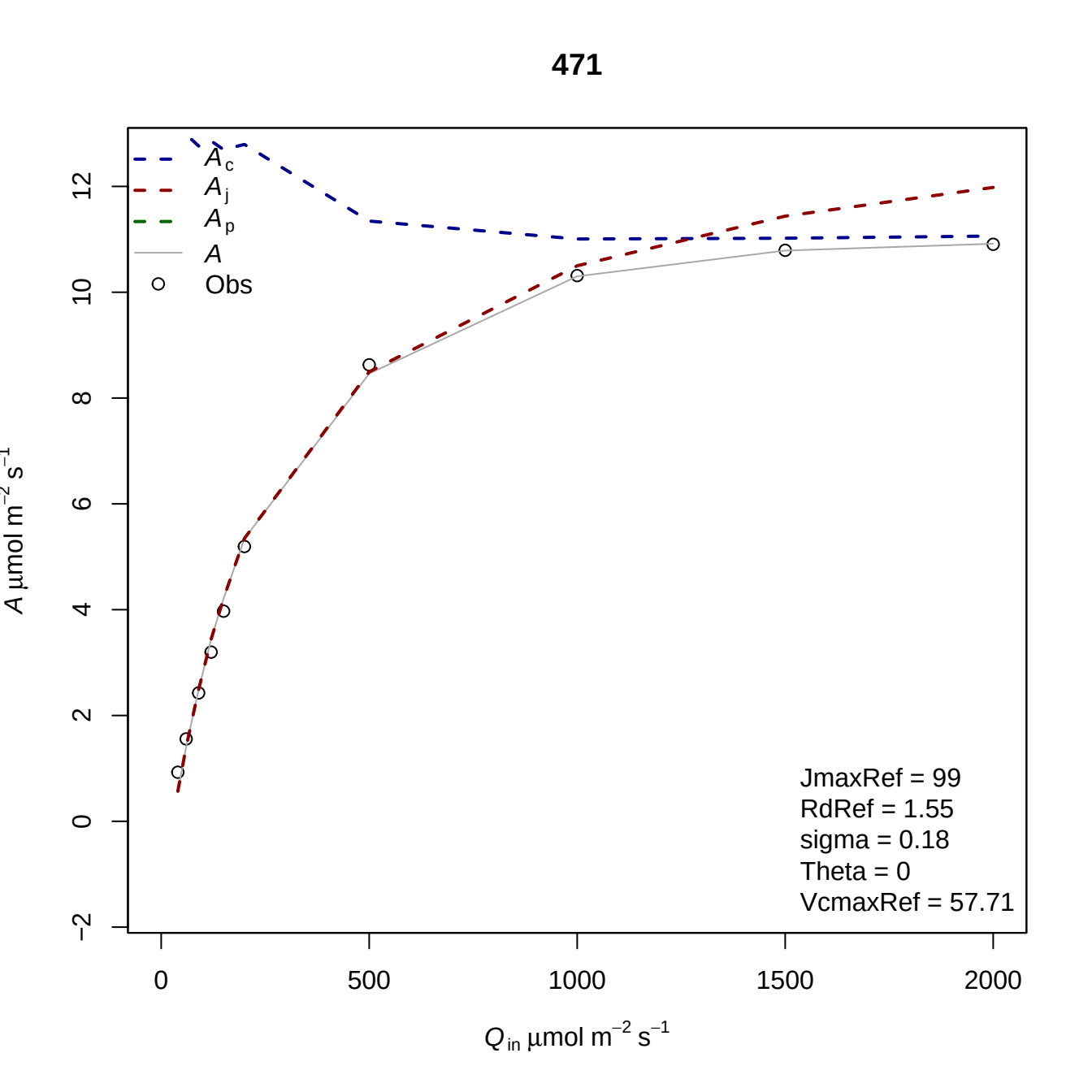

466
A_c
A_j
A_p
A
Obs
JmaxRef = 450.37
RdRef = 3.42
sigma = 0.34
Theta = 0
VcmaxRef = 418.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

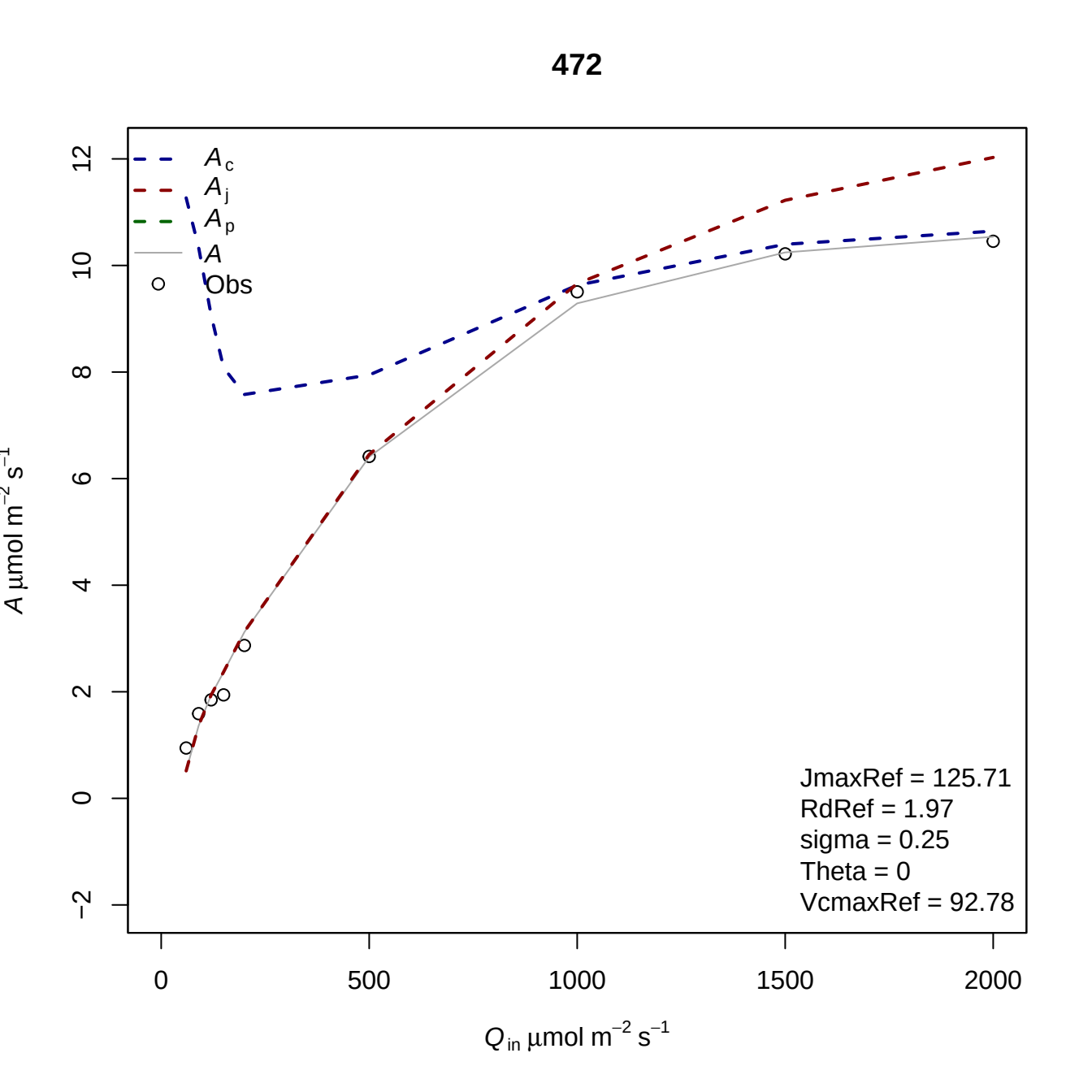

A_c
A_j
A_p
A
Obs
JmaxRef = 204.81
RdRef = 0.55
sigma = 0.45
Theta = 0
VcmaxRef = 75.1
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

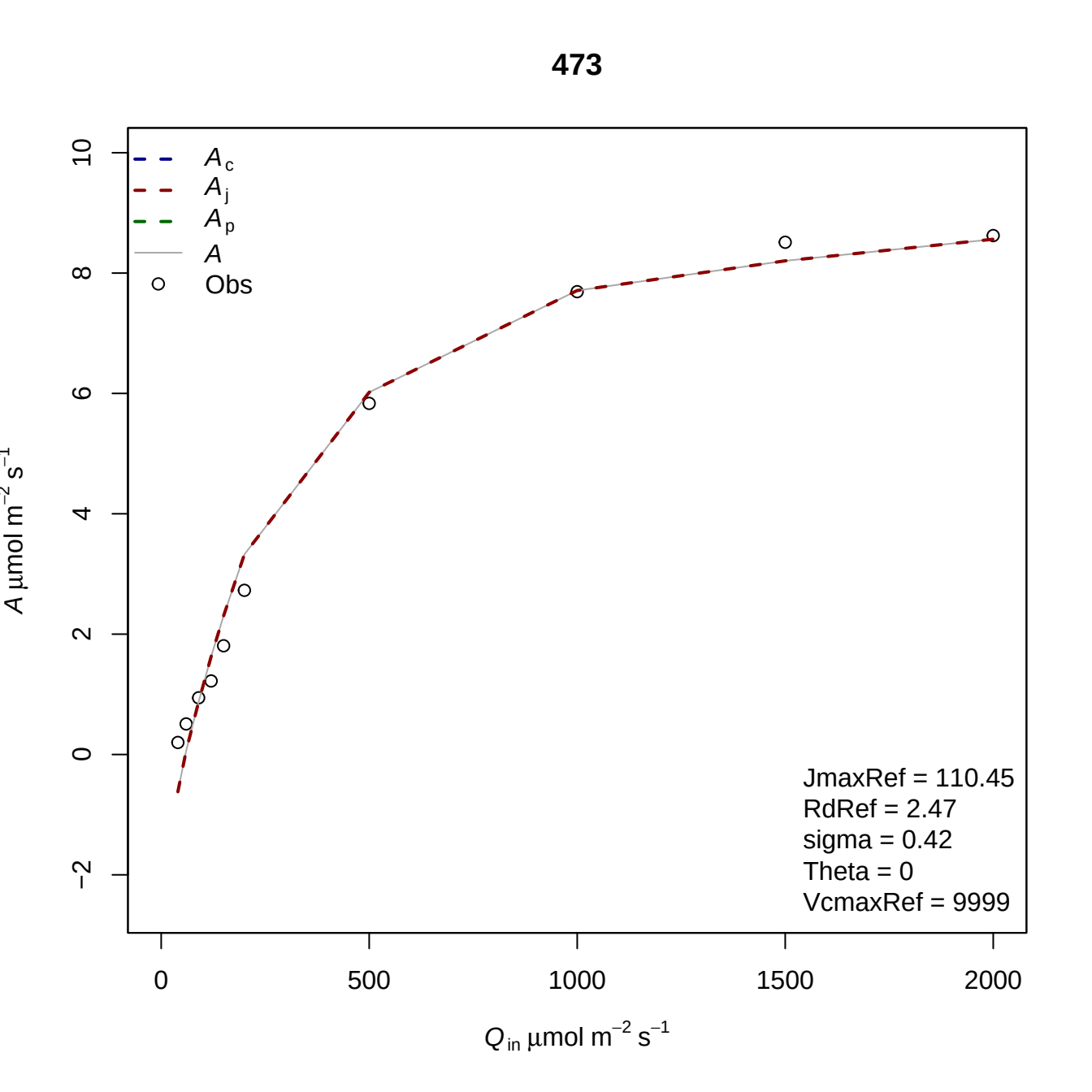

468
A_c
A_j
A_p
A
Obs
JmaxRef = 259.6
RdRef = 0.79
sigma = 0.36
Theta = 0
VcmaxRef = 98.86
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

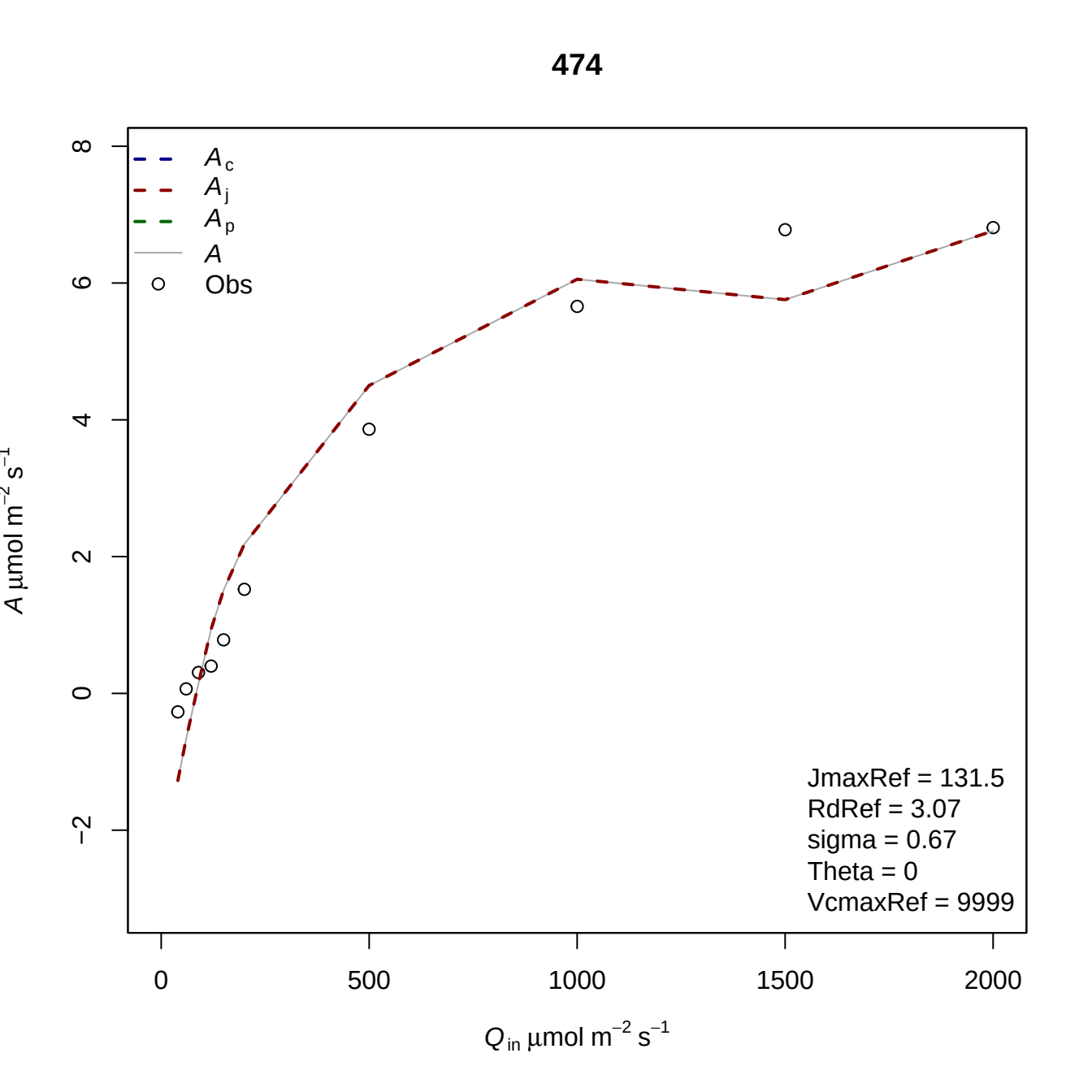

A_c
A_j
A_p
A
Obs
JmaxRef = 61.56
RdRef = 1.98
sigma = 0.3
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

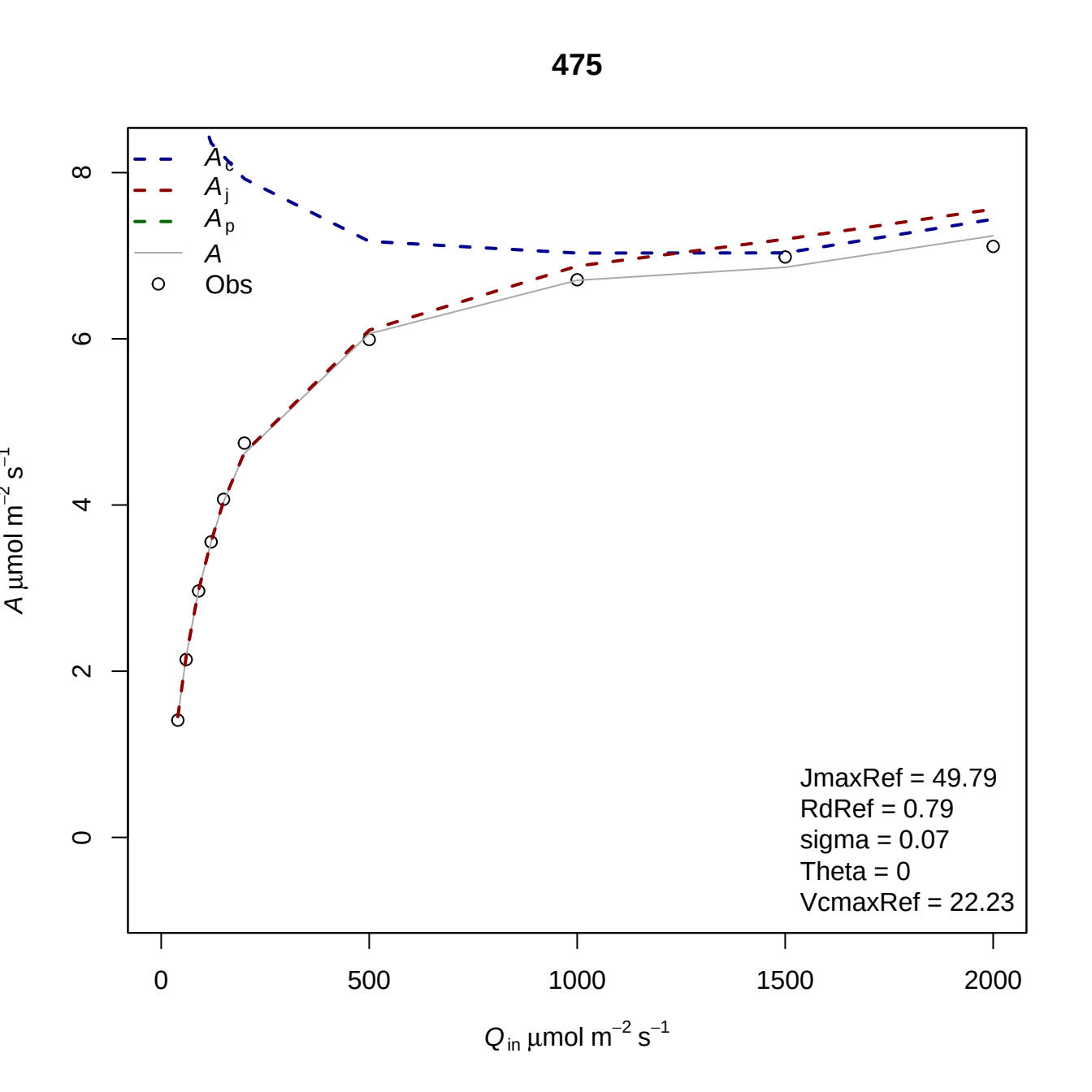

470
A_c
A_j
A_p
A
Obs
JmaxRef = 71.87
RdRef = 1.63
sigma = 0.23
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

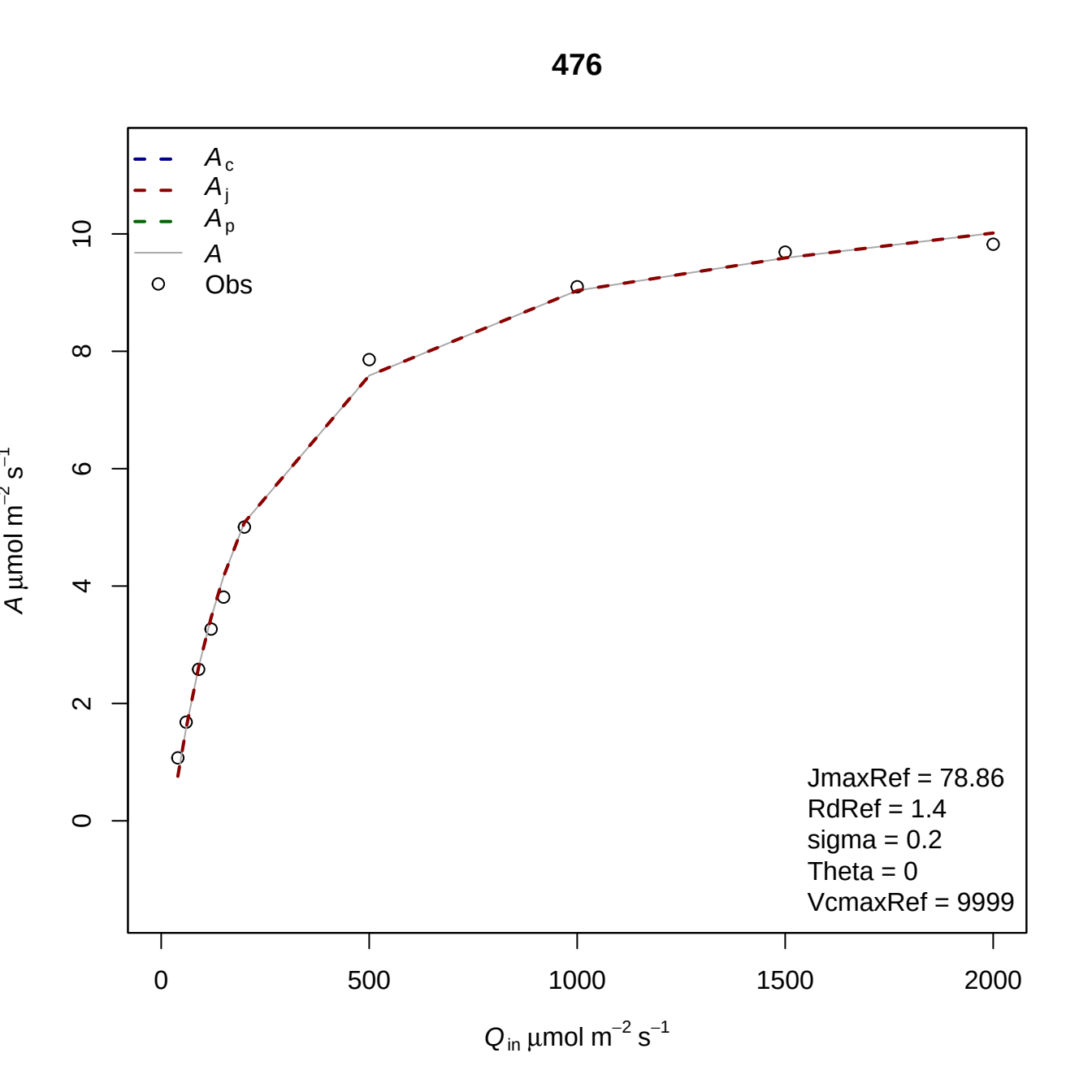

471
A_c
A_j
A_p
A
Obs
JmaxRef = 99
RdRef = 1.55
sigma = 0.18
Theta = 0
VcmaxRef = 57.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

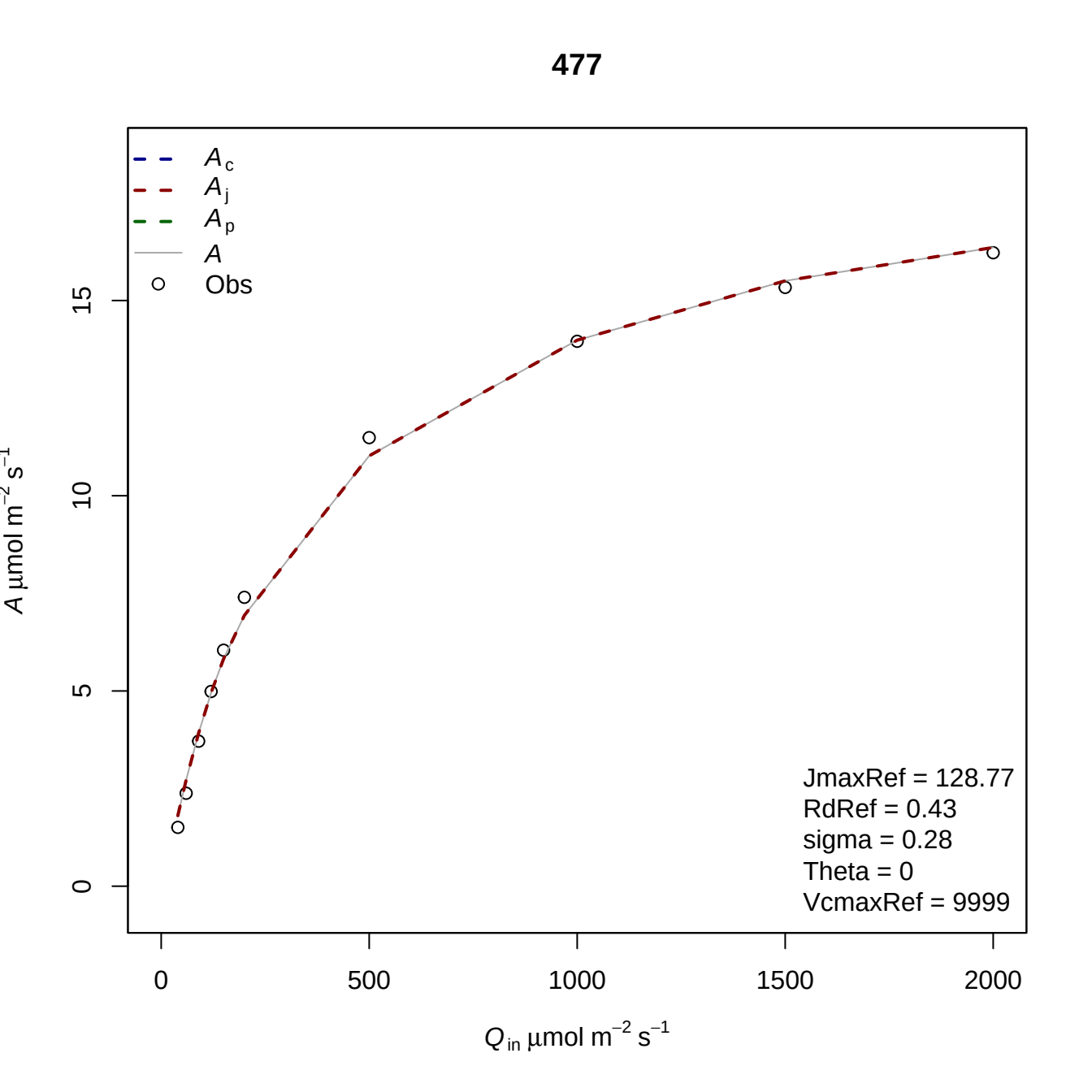

472
A_c
A_j
A_p
A
Obs
JmaxRef = 125.71
RdRef = 1.97
sigma = 0.25
Theta = 0
VcmaxRef = 92.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

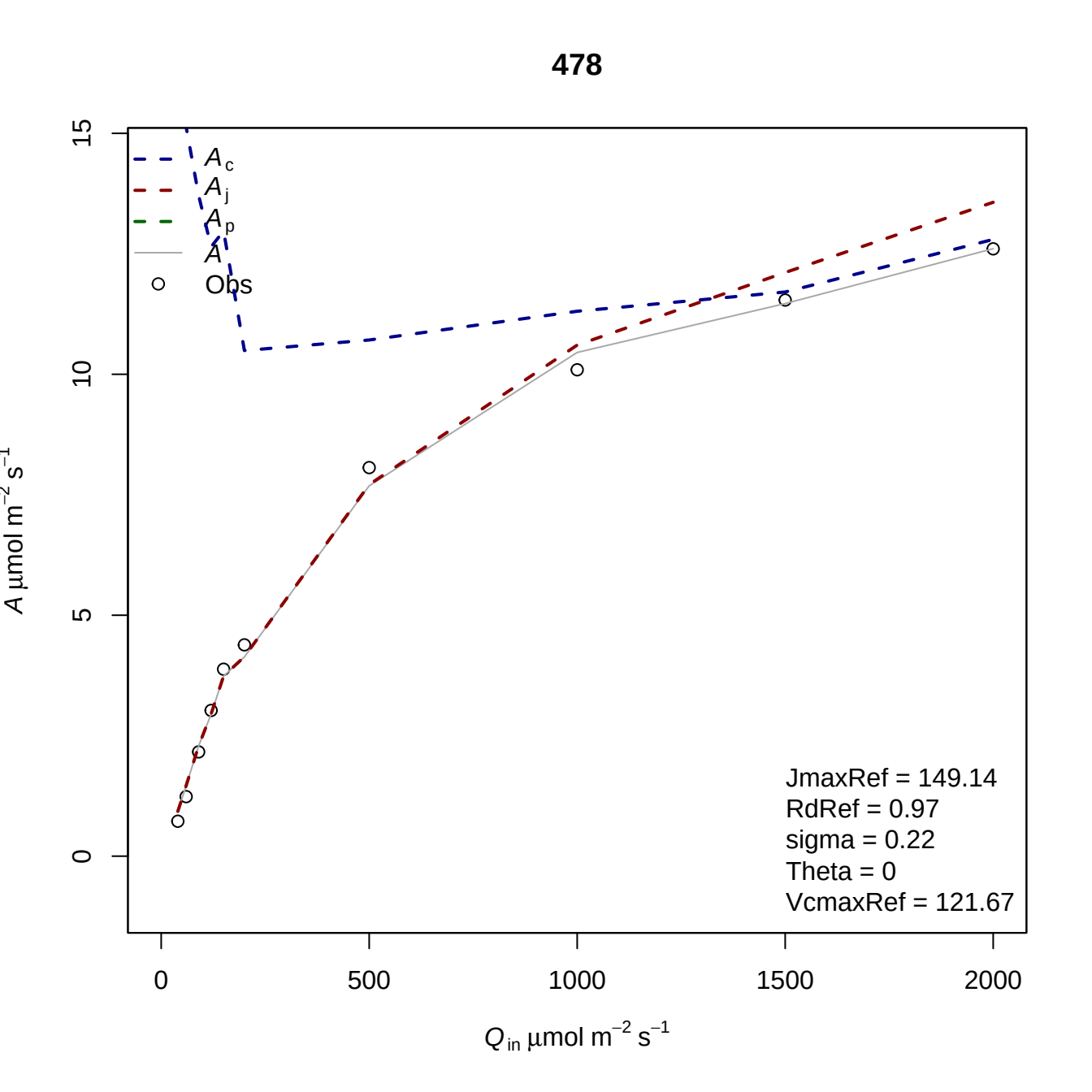

473
A_c
A_j
A_p
A
Obs
JmaxRef = 110.45
RdRef = 2.47
sigma = 0.42
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

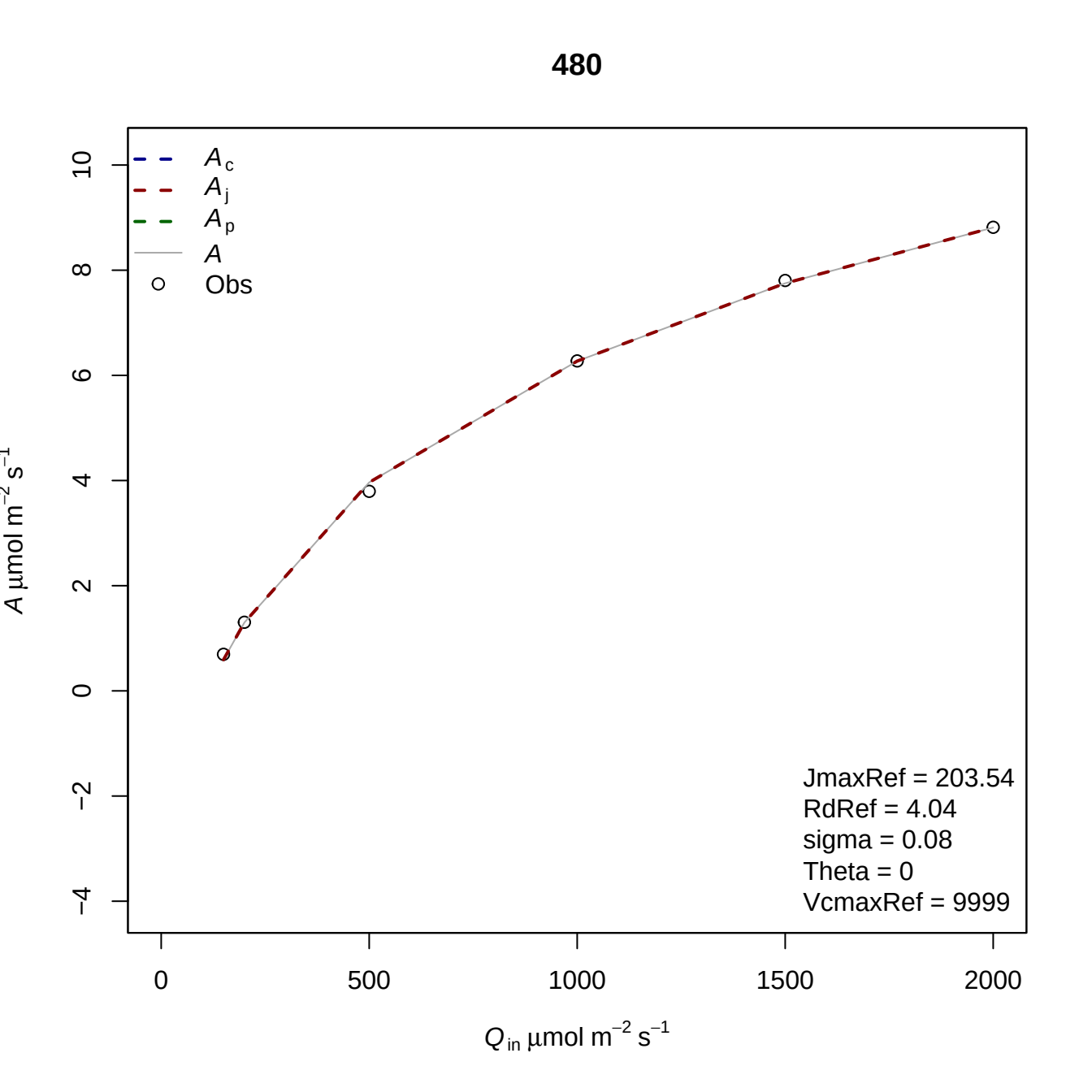

474
A_c
A_j
A_p
A
Obs
JmaxRef = 131.5
RdRef = 3.07
sigma = 0.67
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

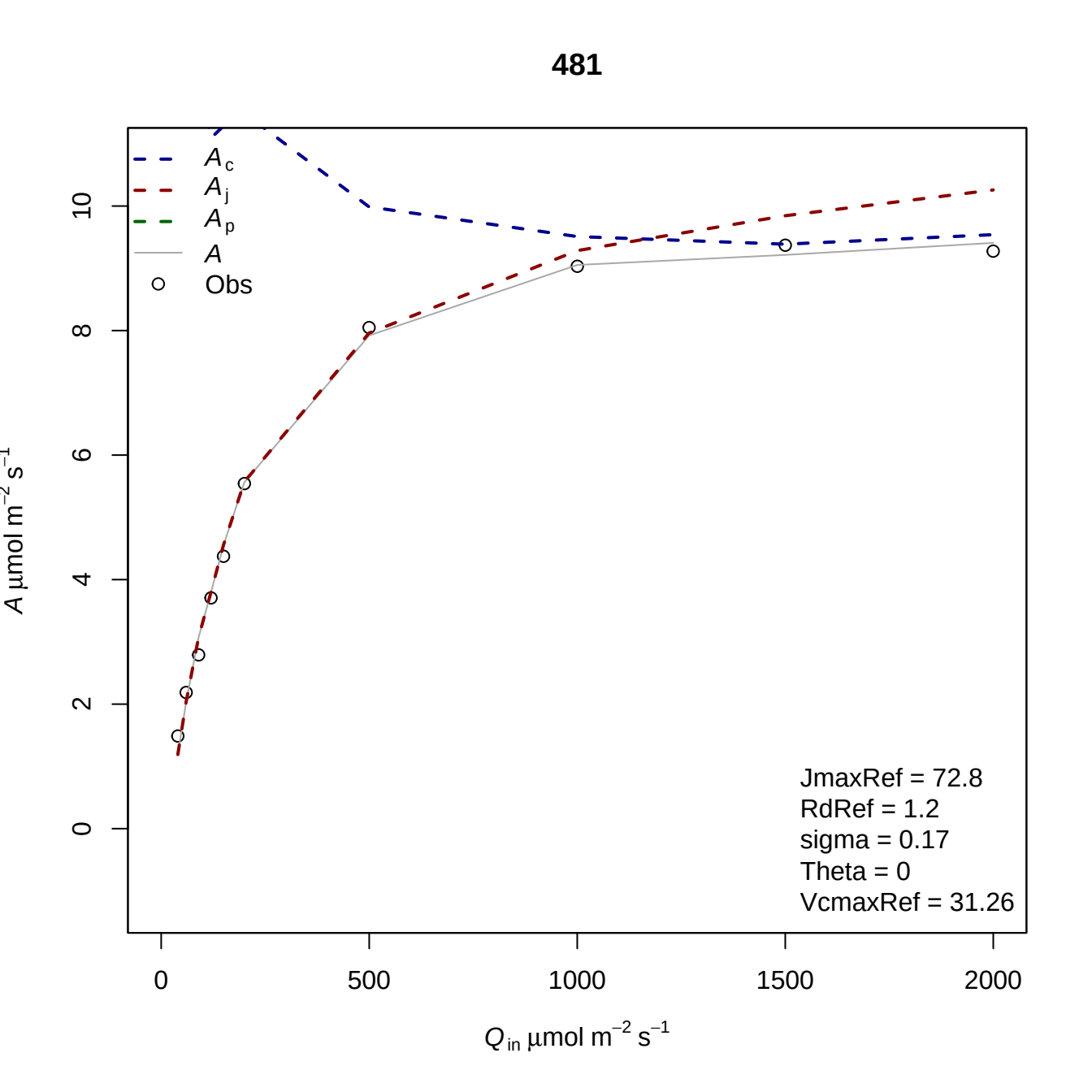

475
A_c
A_j
A_p
A
Obs
JmaxRef = 49.79
RdRef = 0.79
sigma = 0.07
Theta = 0
VcmaxRef = 22.23
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

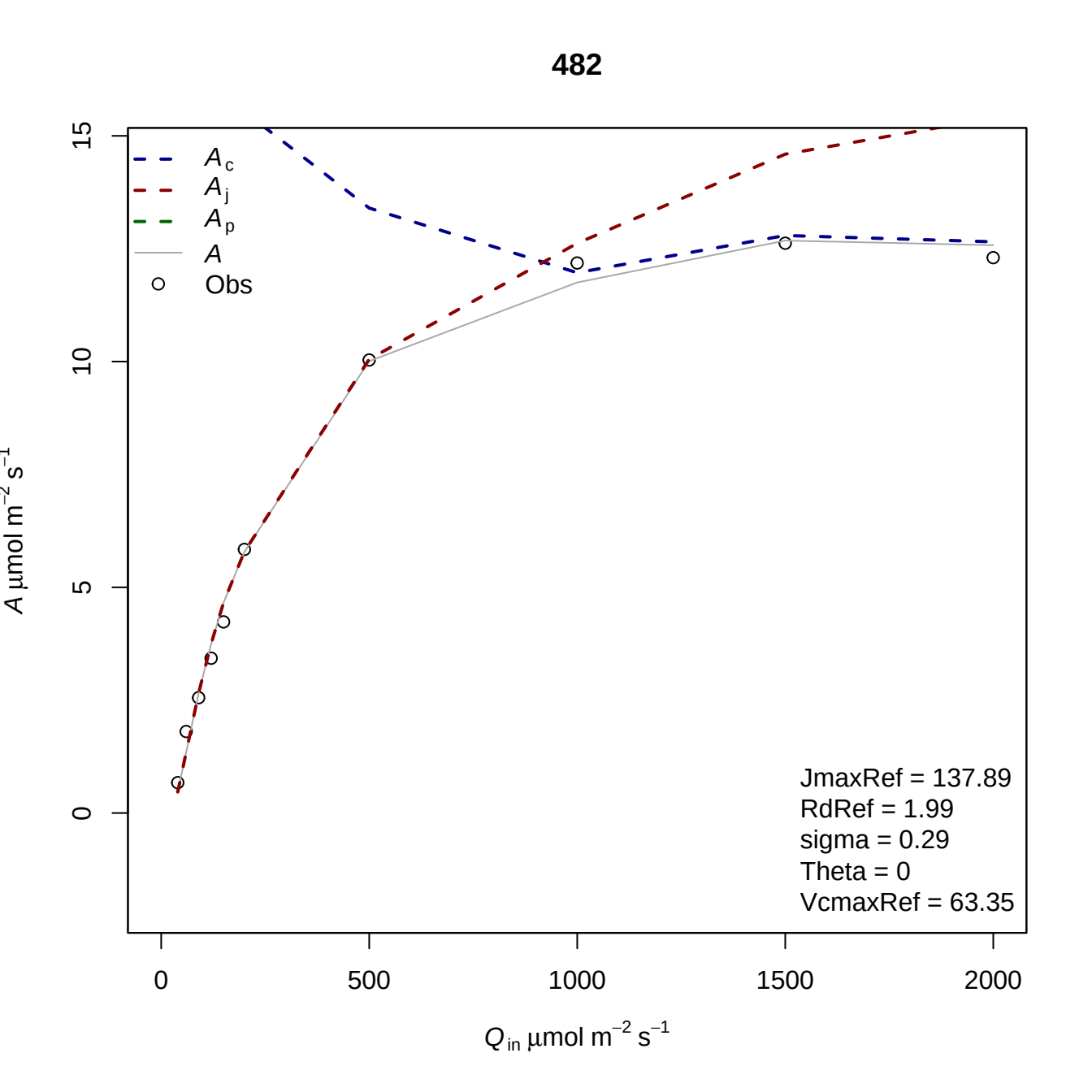

476
A_c
A_j
A_p
A
Obs
JmaxRef = 78.86
RdRef = 1.4
sigma = 0.2
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

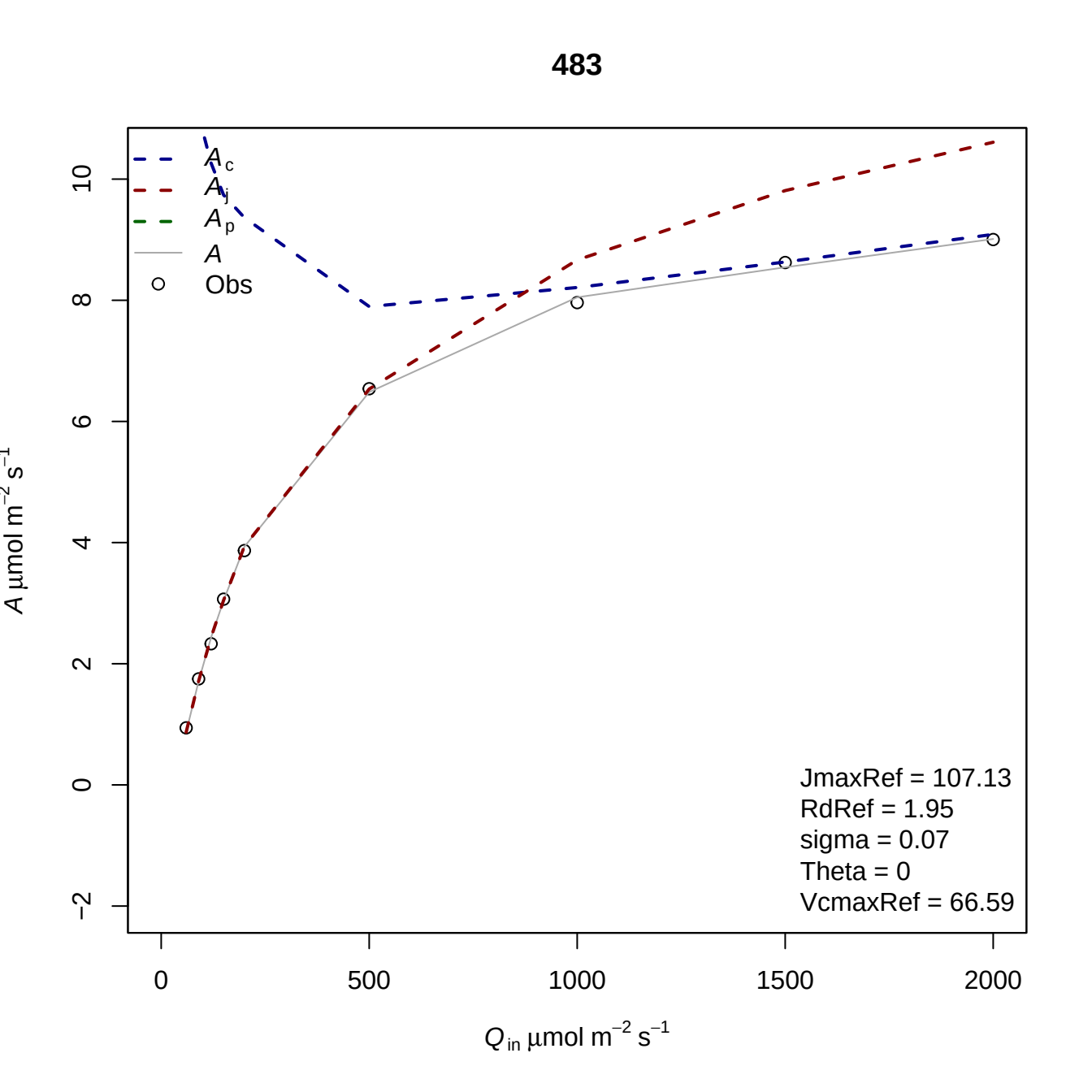

477
A_c
A_j
A_p
A
Obs
JmaxRef = 128.77
RdRef = 0.43
sigma = 0.28
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

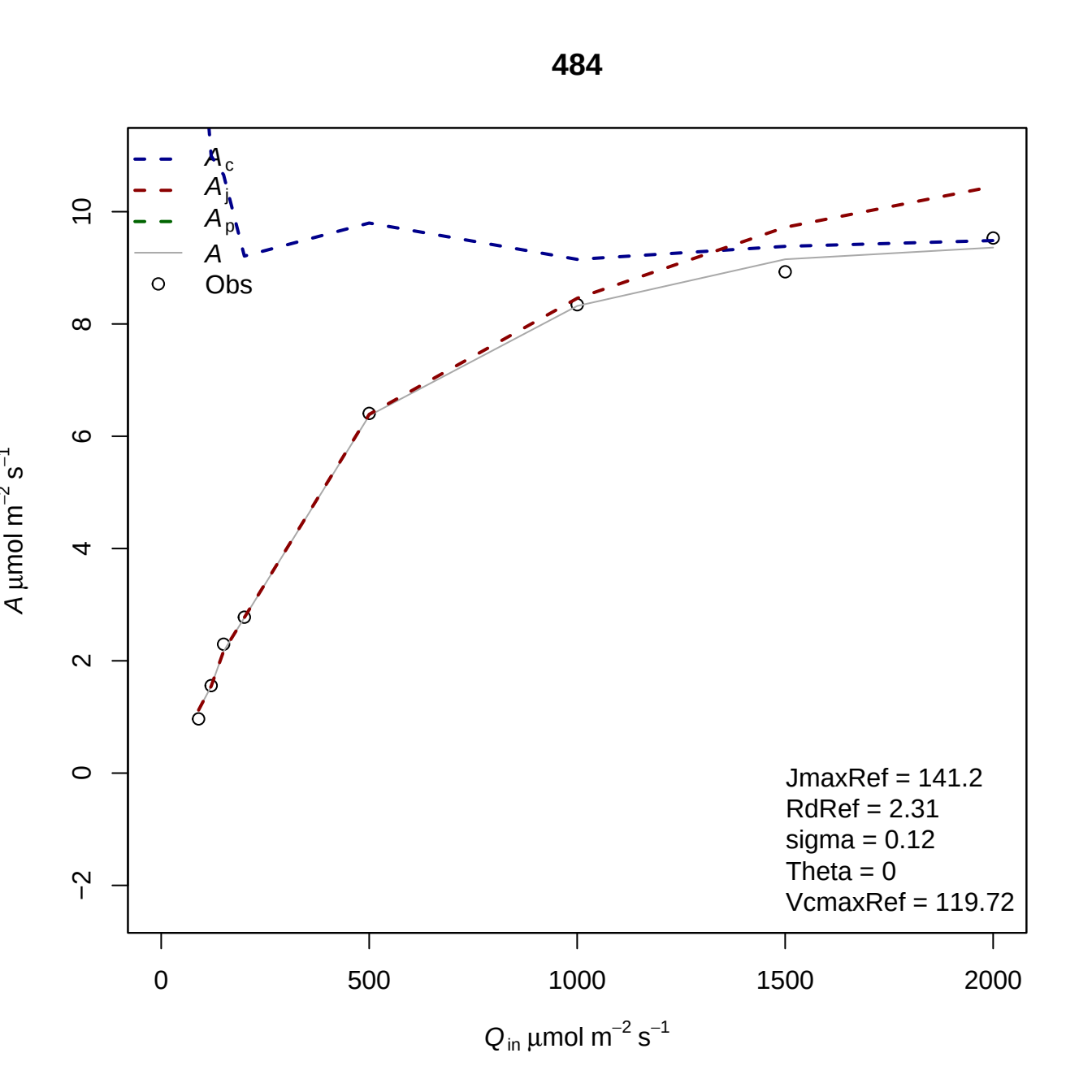

478
A_c
A_j
A_p
A
Obs
JmaxRef = 149.14
RdRef = 0.97
sigma = 0.22
Theta = 0
VcmaxRef = 121.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

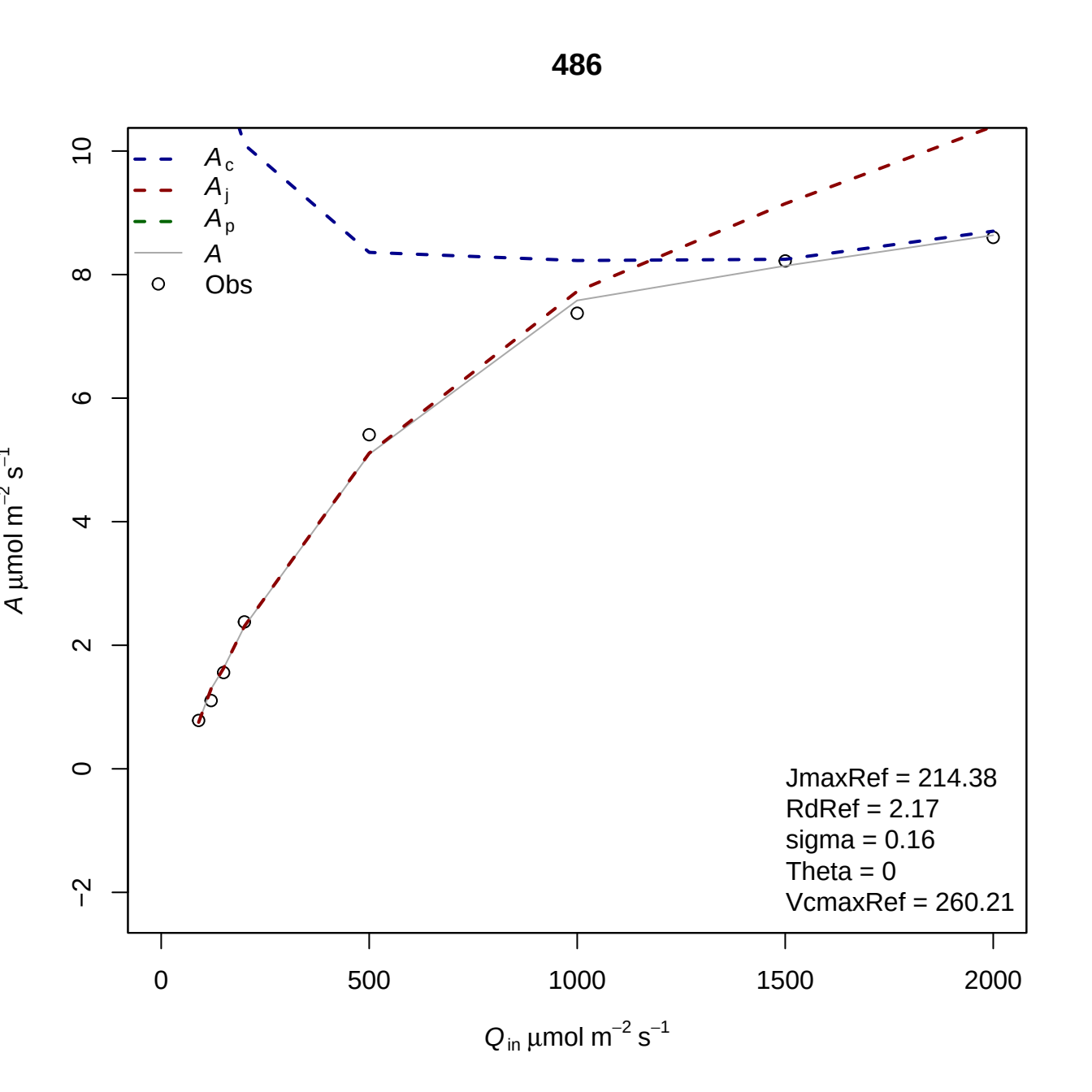

480
A_c
A_j
A_p
A
Obs
JmaxRef = 203.54
RdRef = 4.04
sigma = 0.08
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

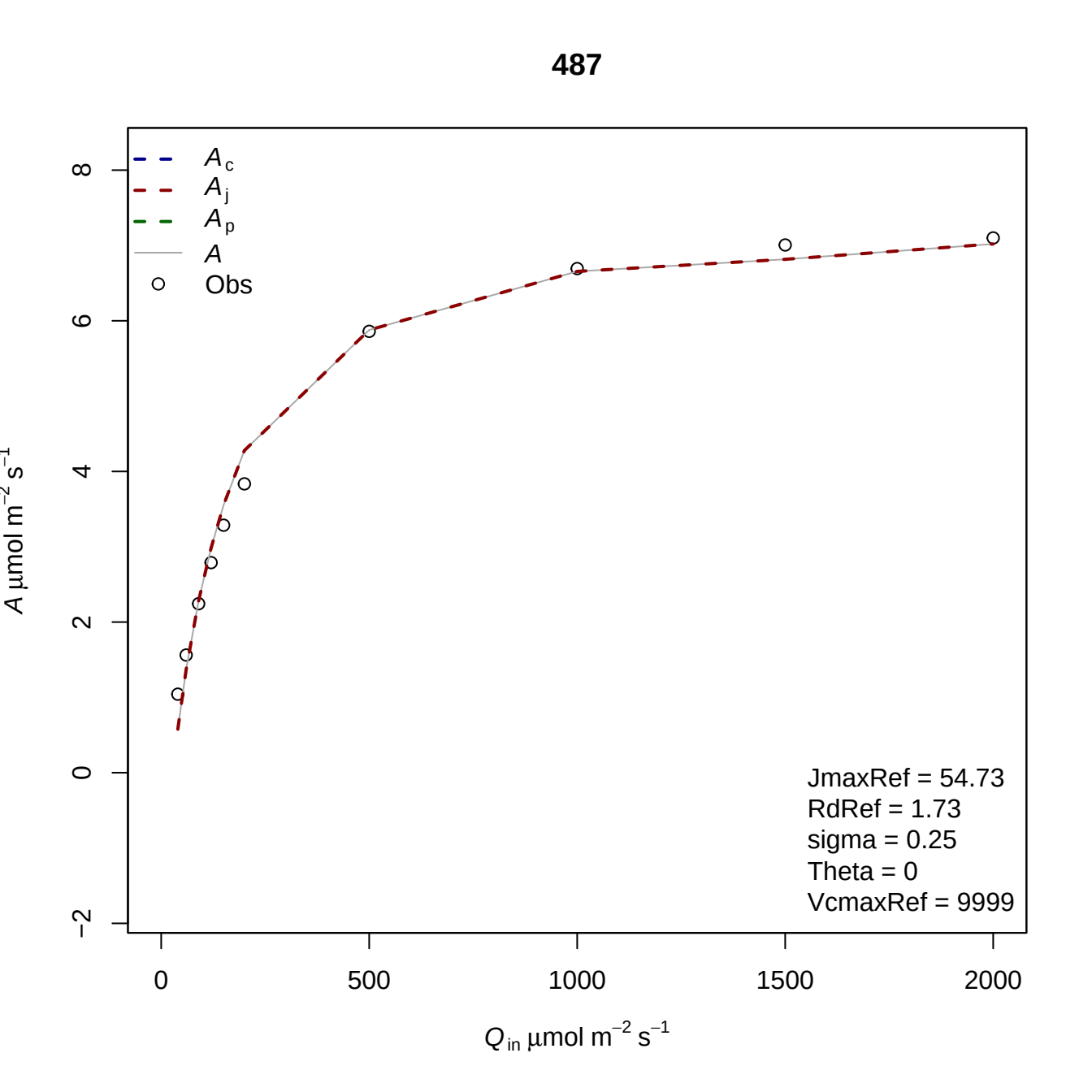

481
A_c
A_j
A_p
A
Obs
JmaxRef = 72.8
RdRef = 1.2
sigma = 0.17
Theta = 0
VcmaxRef = 31.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

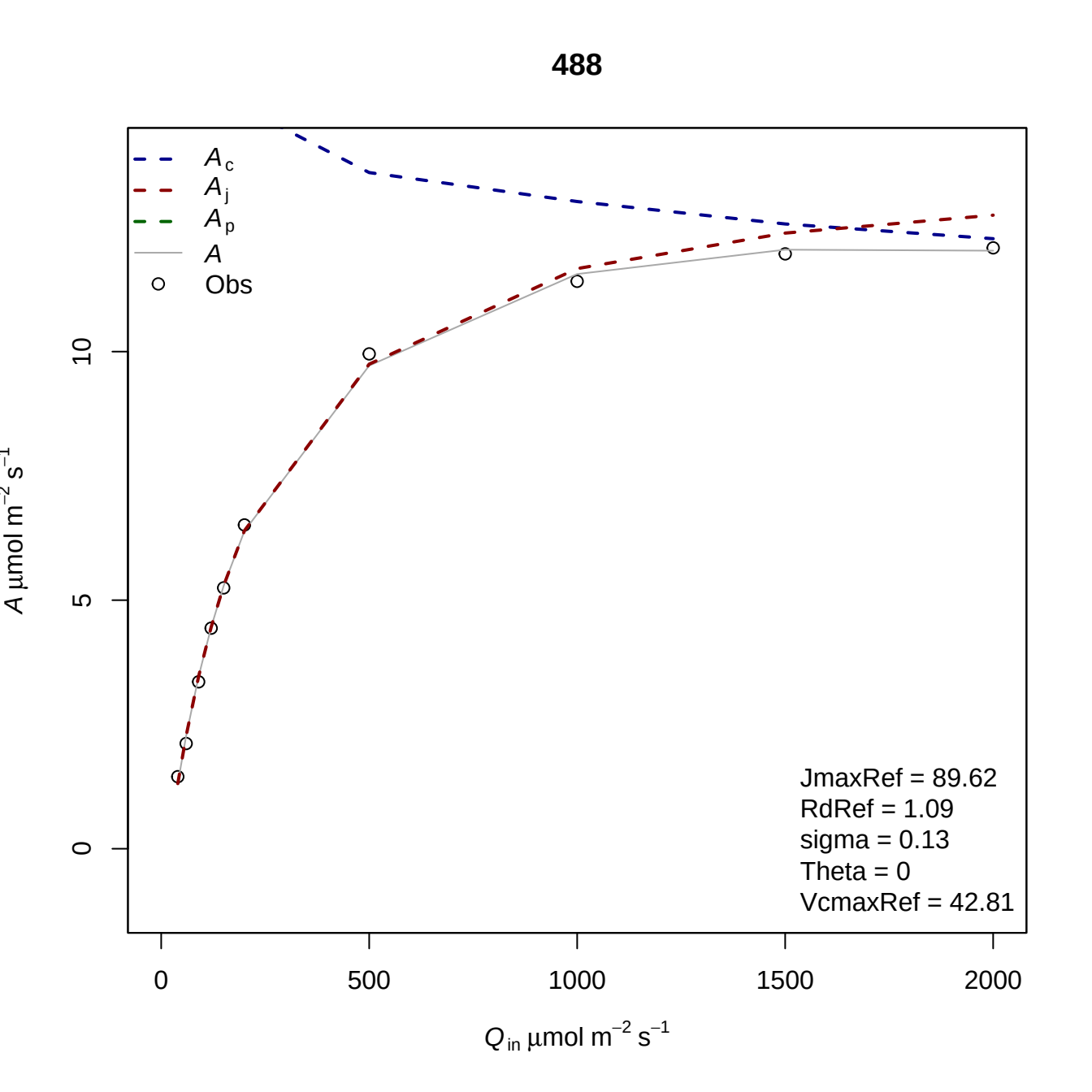

482
A_c
A_j
A_p
A
Obs
JmaxRef = 137.89
RdRef = 1.99
sigma = 0.29
Theta = 0
VcmaxRef = 63.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

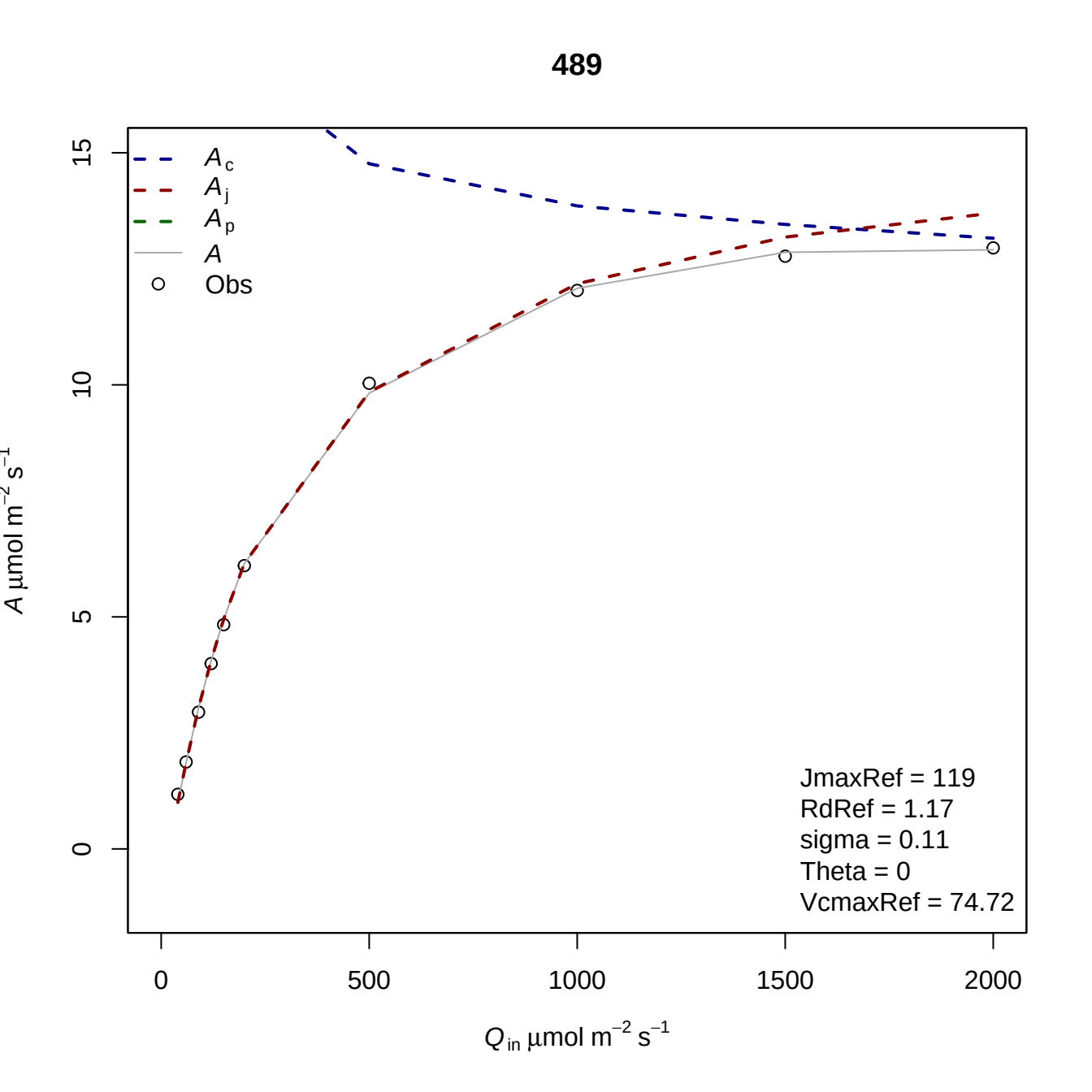

483
A_c
A_j
A_p
A
Obs
JmaxRef = 107.13
RdRef = 1.95
sigma = 0.07
Theta = 0
VcmaxRef = 66.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

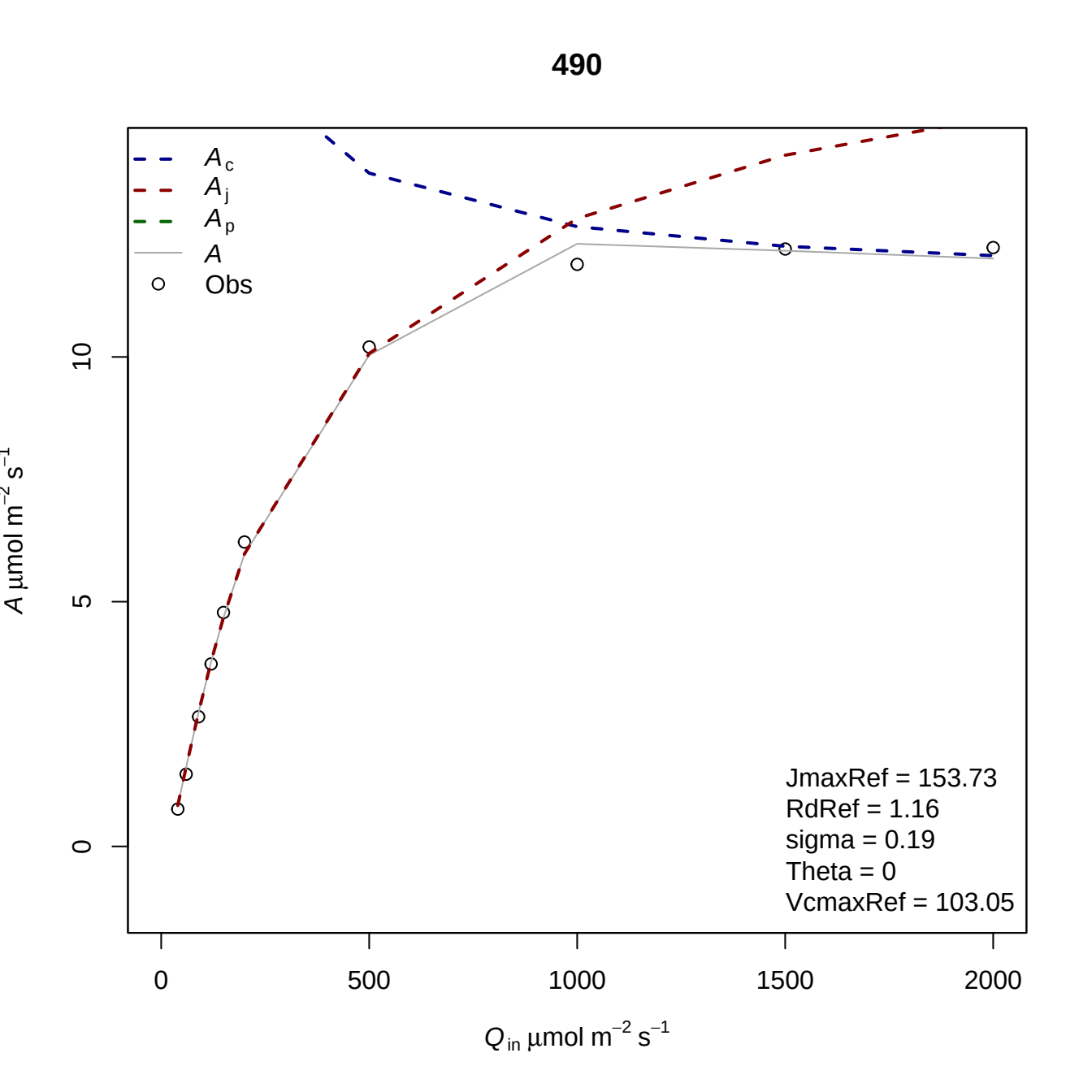

484
A_c
A_j
A_p
A
Obs
JmaxRef = 141.2
RdRef = 2.31
sigma = 0.12
Theta = 0
VcmaxRef = 119.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

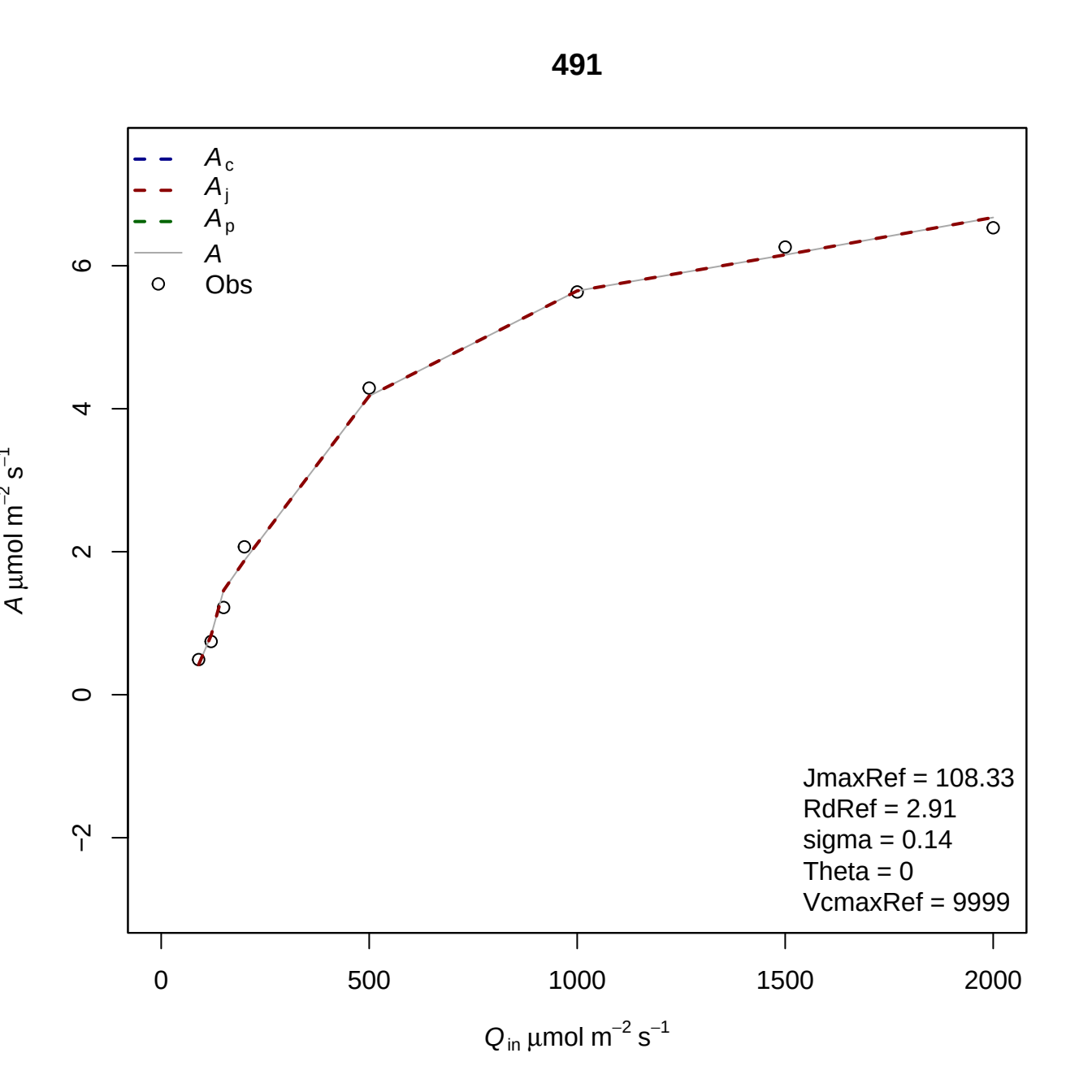

486
A_c
A_j
A_p
A
Obs
JmaxRef = 214.38
RdRef = 2.17
sigma = 0.16
Theta = 0
VcmaxRef = 260.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

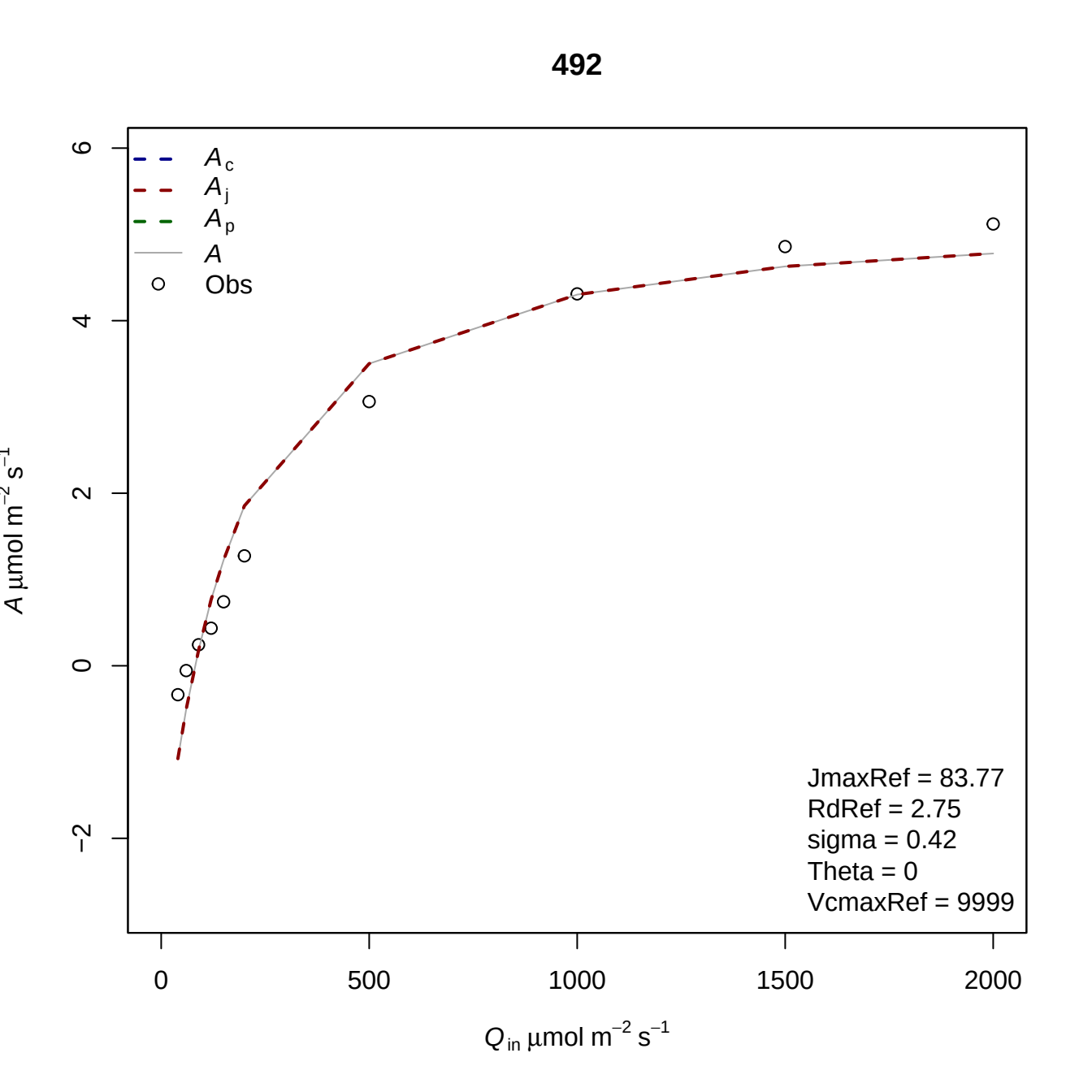

487
A_c
A_j
A_p
A
Obs
JmaxRef = 54.73
RdRef = 1.73
sigma = 0.25
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

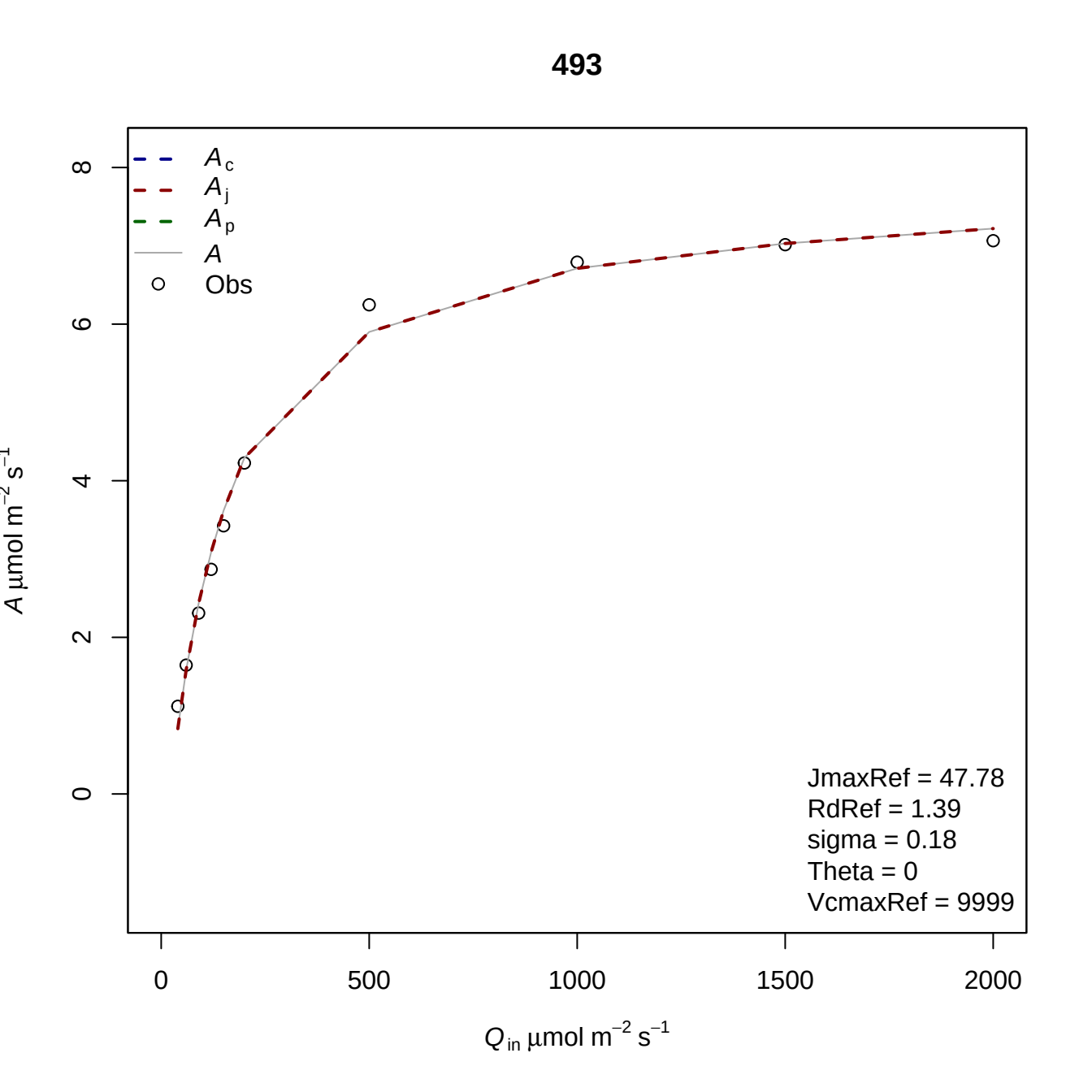

488
A_c
A_j
A_p
A
Obs
JmaxRef = 89.62
RdRef = 1.09
sigma = 0.13
Theta = 0
VcmaxRef = 42.81
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

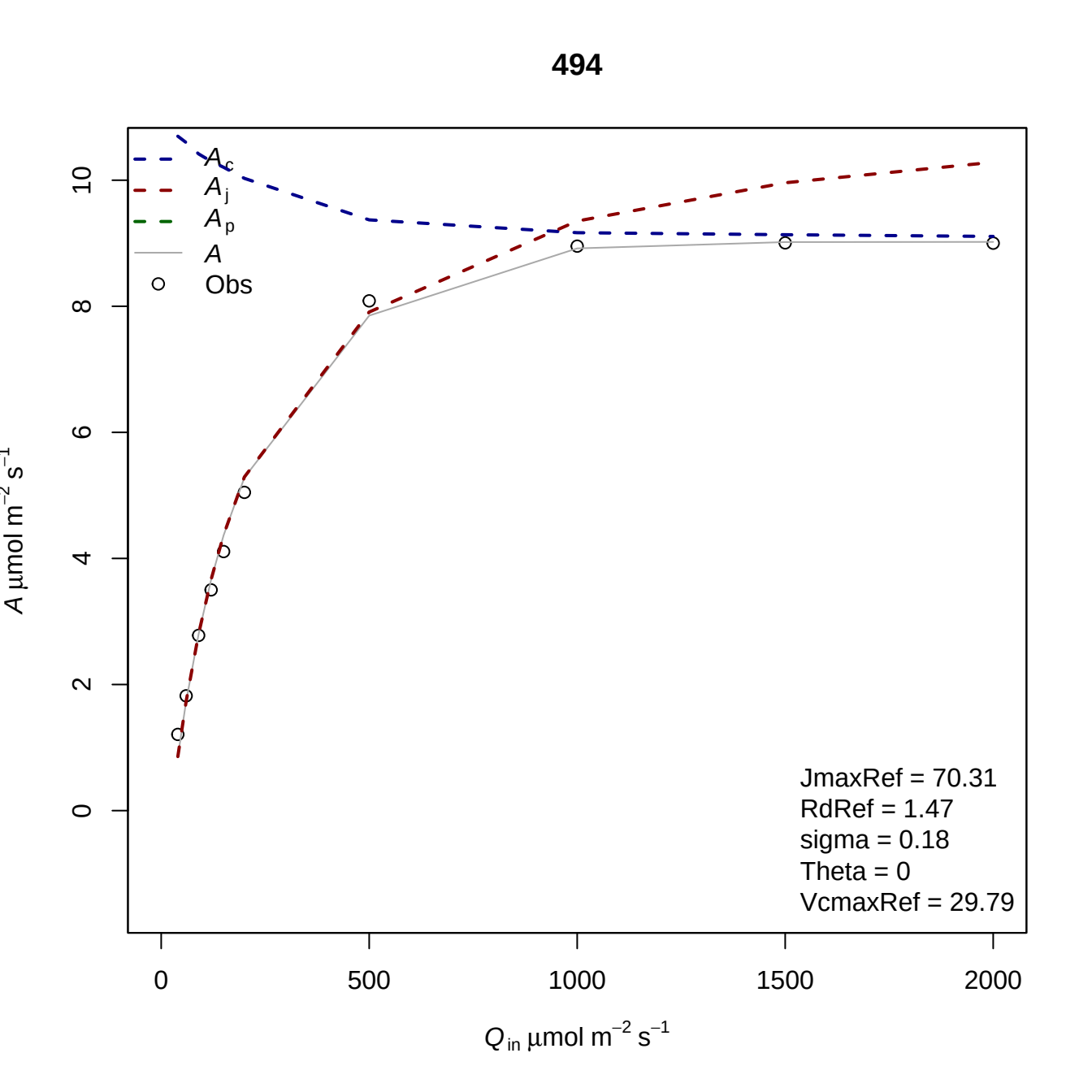

489
A_c
A_j
A_p
A
Obs
JmaxRef = 119
RdRef = 1.17
sigma = 0.11
Theta = 0
VcmaxRef = 74.72
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

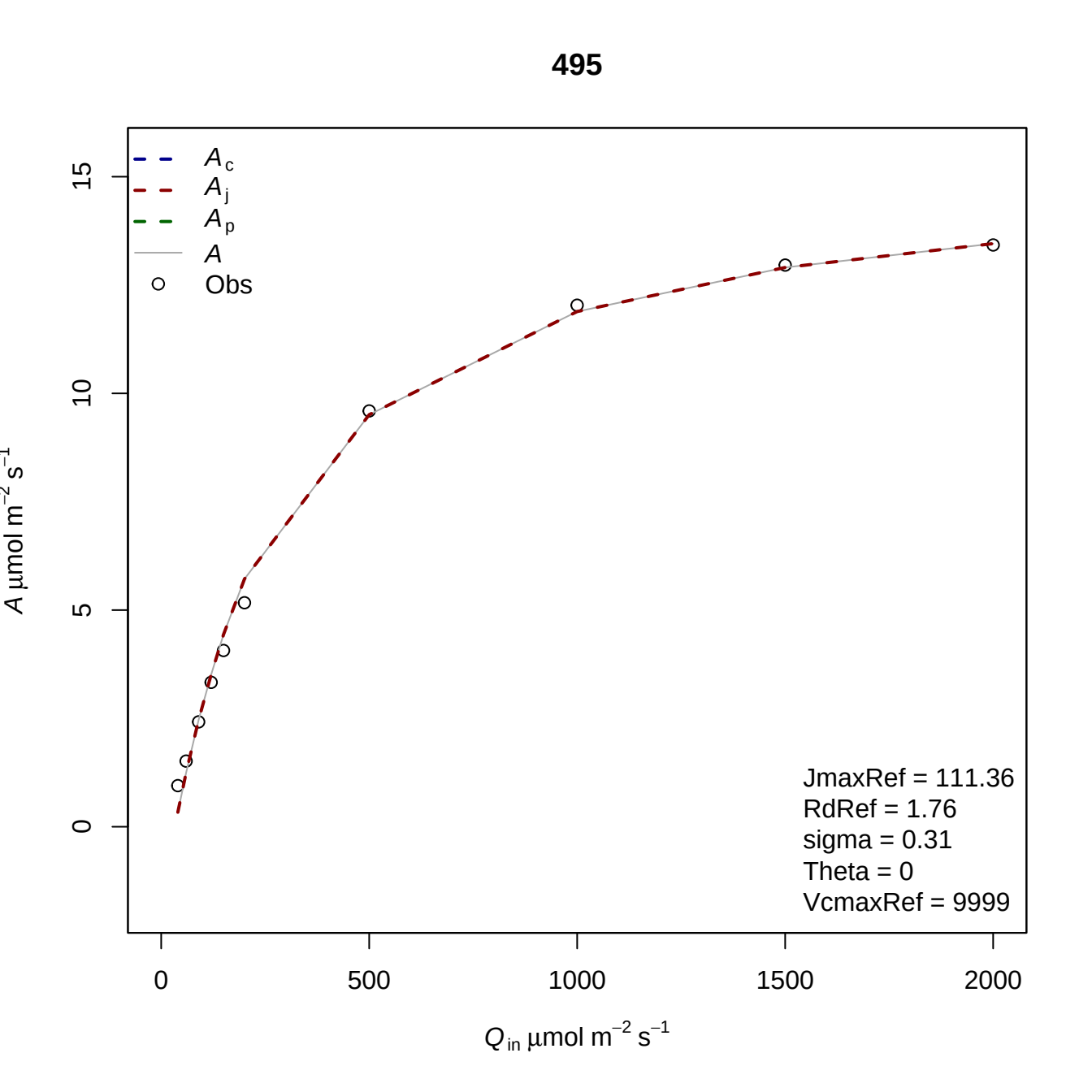

490
A_c
A_j
A_p
A
Obs
JmaxRef = 153.73
RdRef = 1.16
sigma = 0.19
Theta = 0
VcmaxRef = 103.05
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

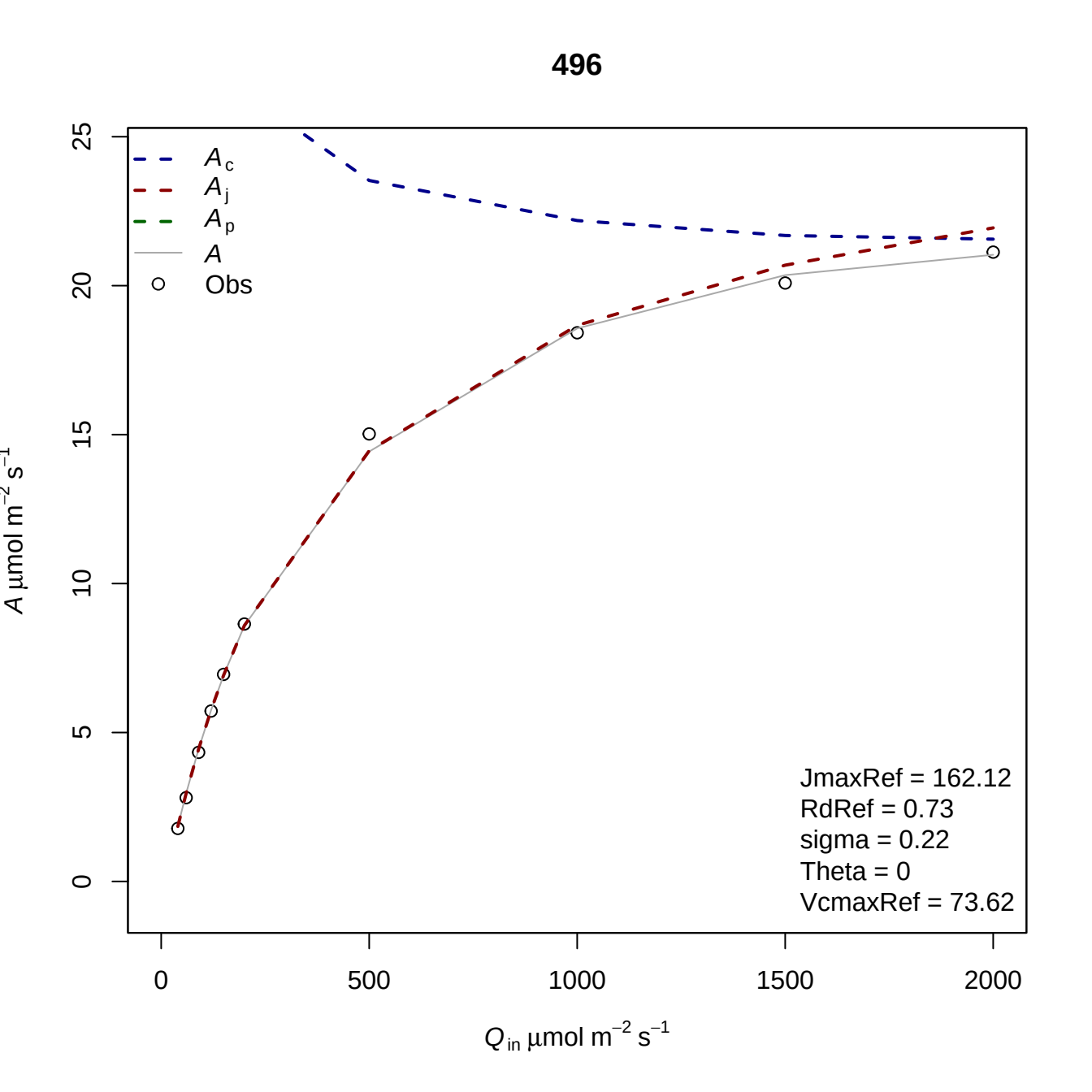

491
A_c
A_j
A_p
A
Obs
JmaxRef = 108.33
RdRef = 2.91
sigma = 0.14
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

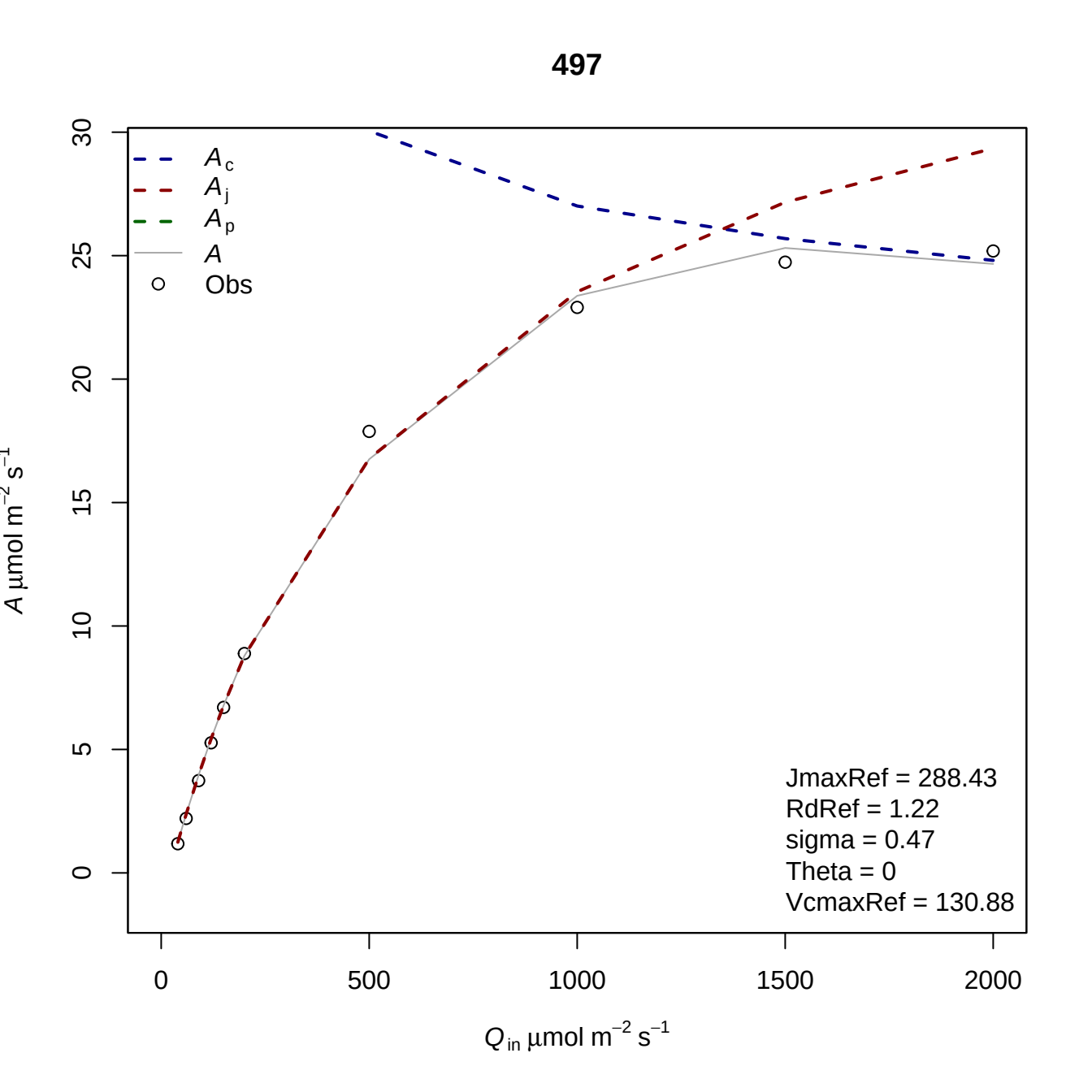

492
A_c
A_j
A_p
A
Obs
JmaxRef = 83.77
RdRef = 2.75
sigma = 0.42
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

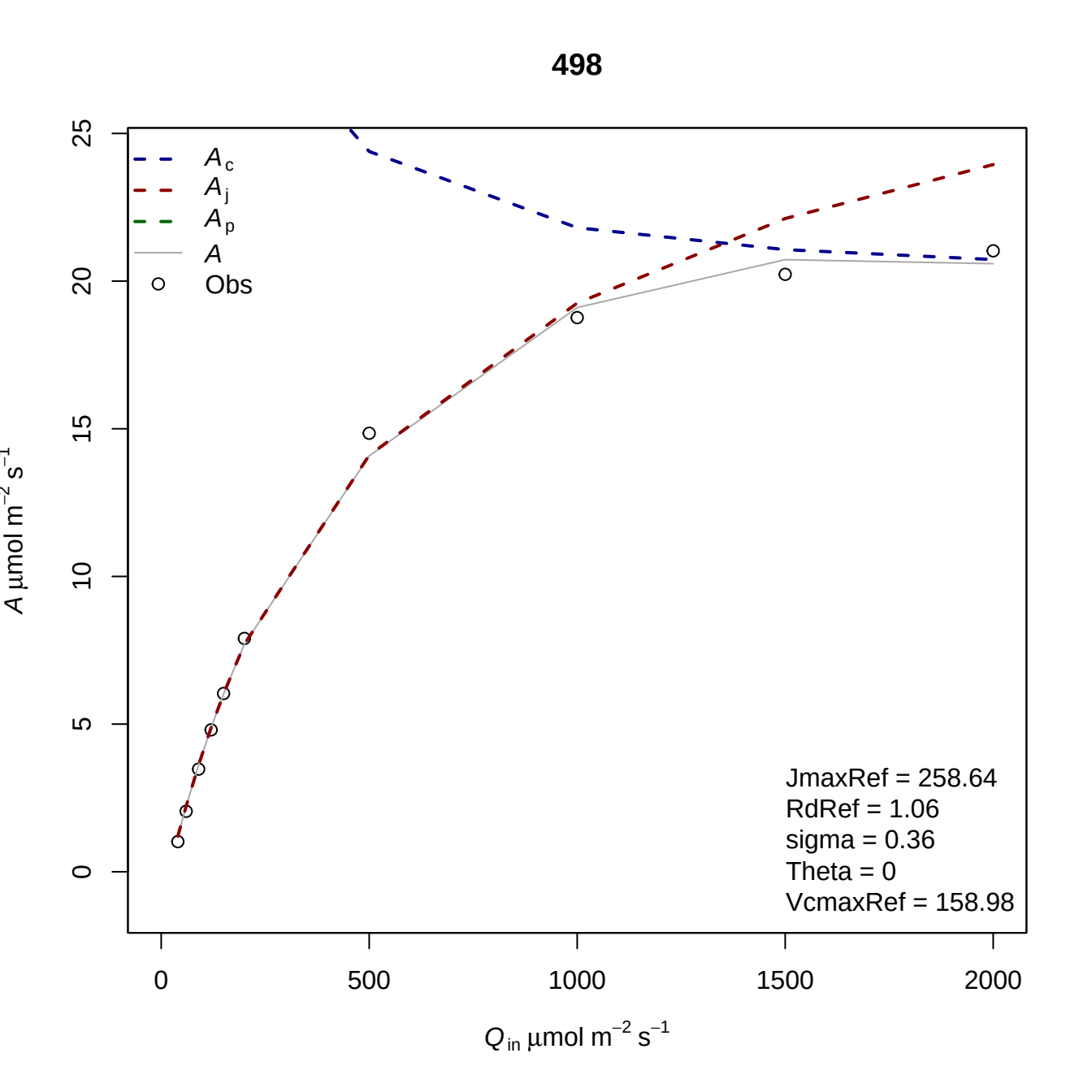

493
A_c
A_j
A_p
A
Obs
JmaxRef = 47.78
RdRef = 1.39
sigma = 0.18
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

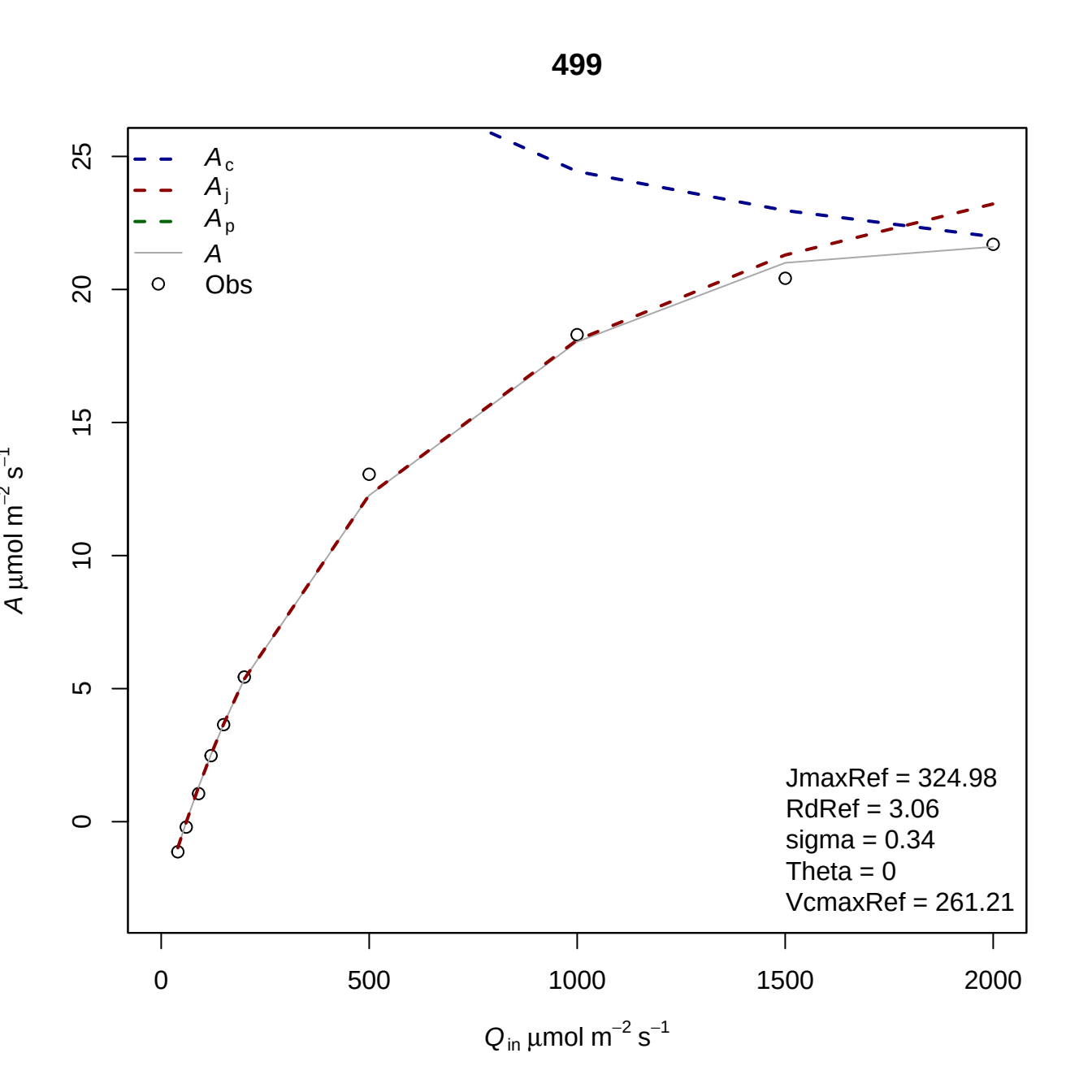

494
A_c
A_j
A_p
A
Obs
JmaxRef = 70.31
RdRef = 1.47
sigma = 0.18
Theta = 0
VcmaxRef = 29.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

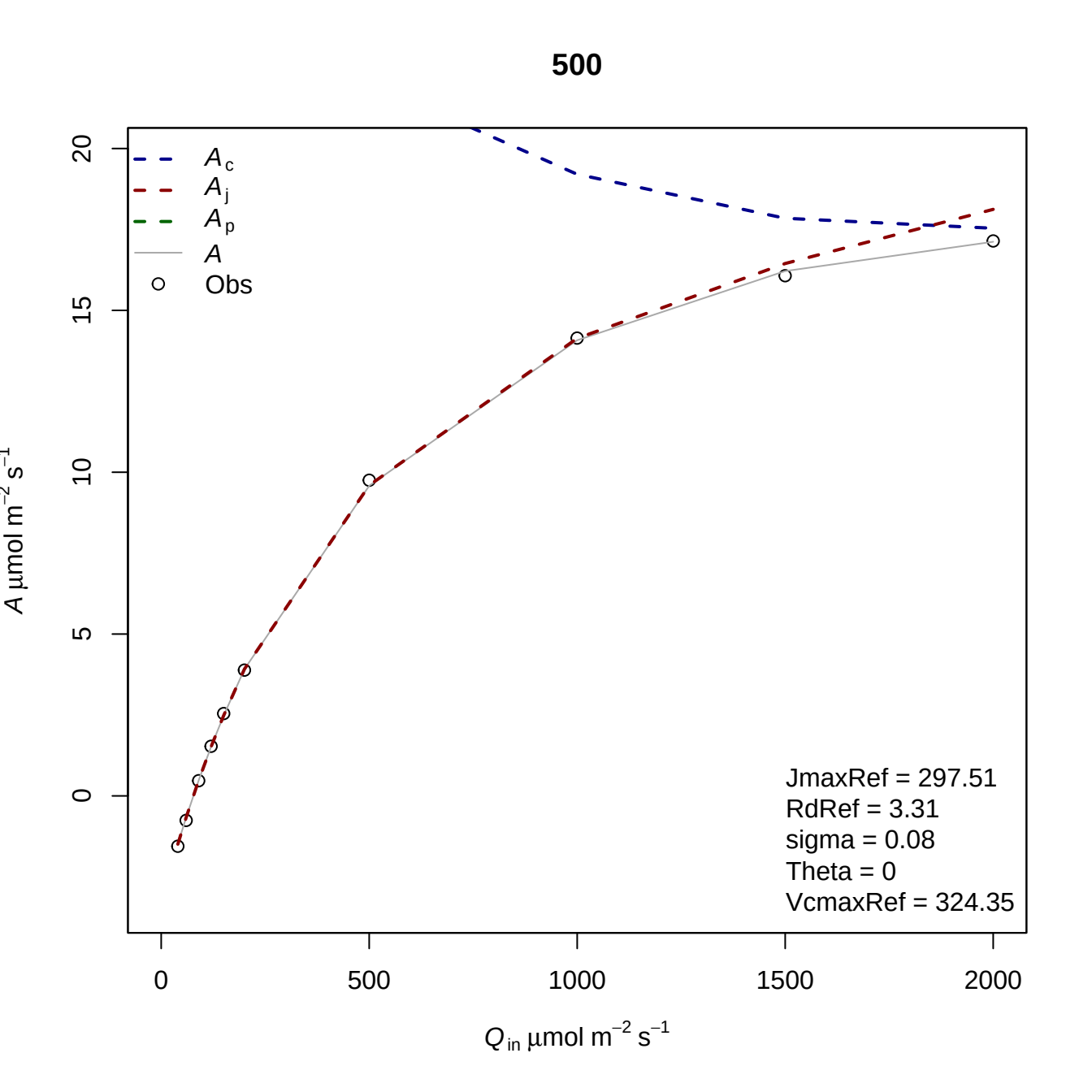

495
A_c
A_j
A_p
A
Obs
JmaxRef = 111.36
RdRef = 1.76
sigma = 0.31
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

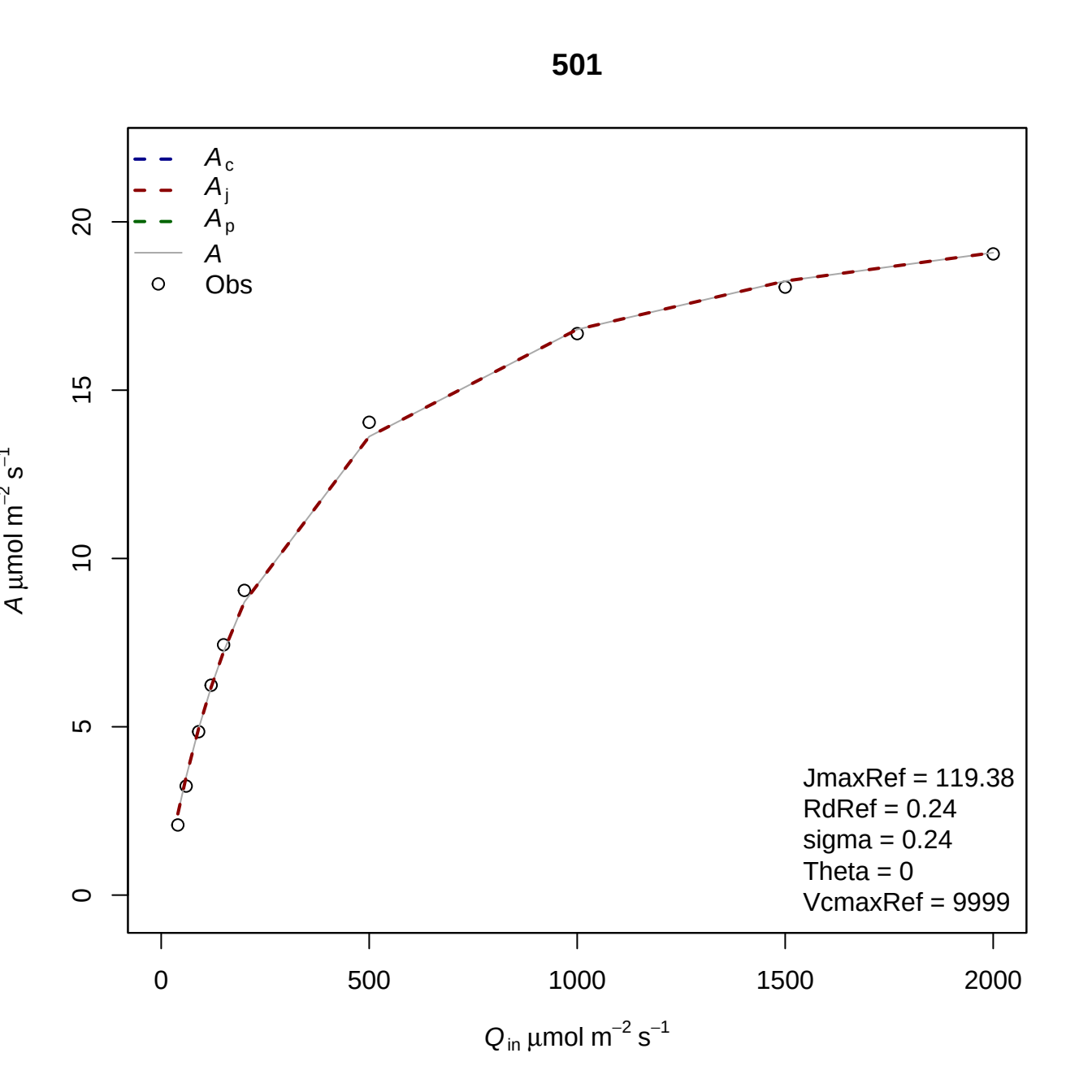

496
A_c
A_j
A_p
A
Obs
JmaxRef = 162.12
RdRef = 0.73
sigma = 0.22
Theta = 0
VcmaxRef = 73.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

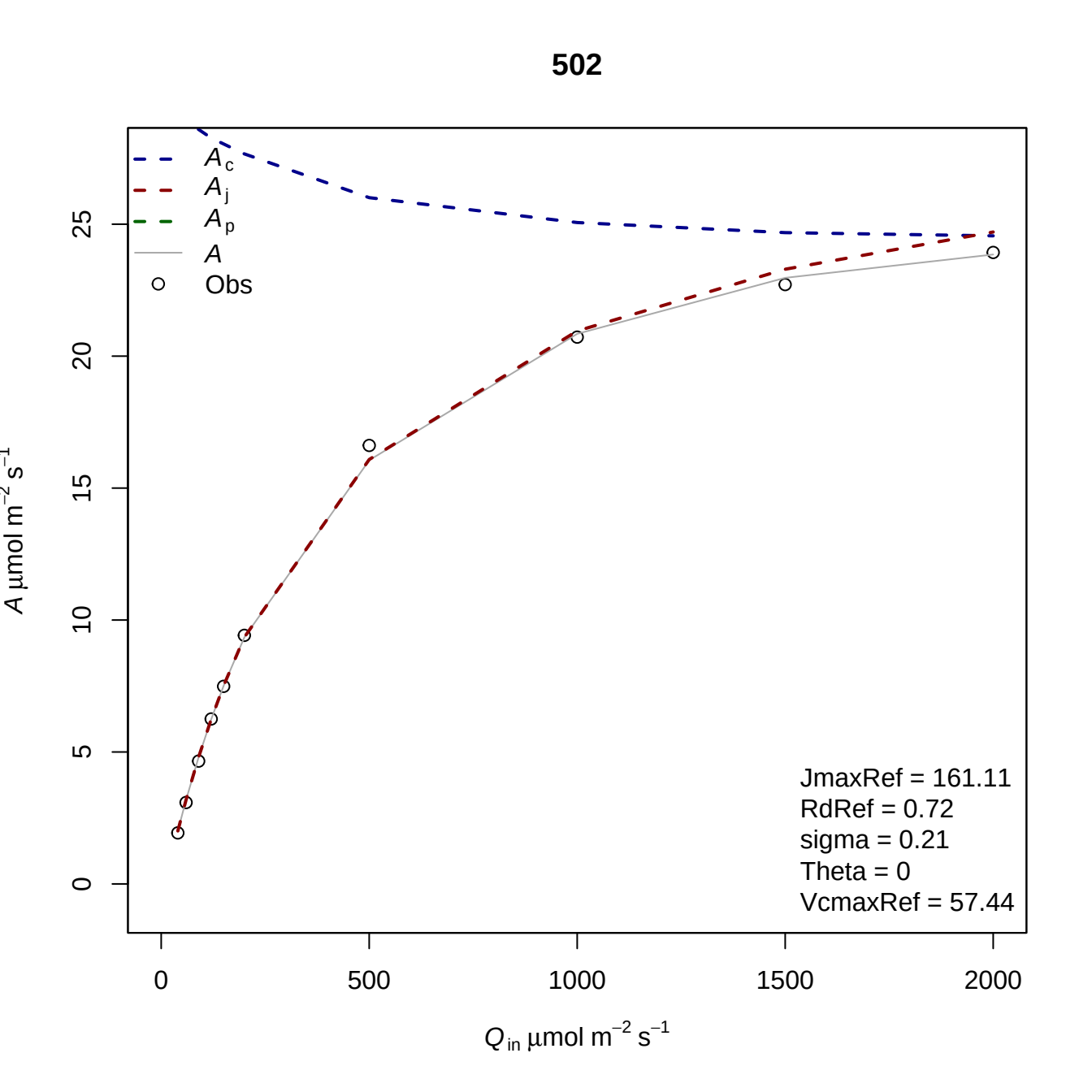

497
A_c
A_j
A_p
A
Obs
JmaxRef = 288.43
RdRef = 1.22
sigma = 0.47
Theta = 0
VcmaxRef = 130.88
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

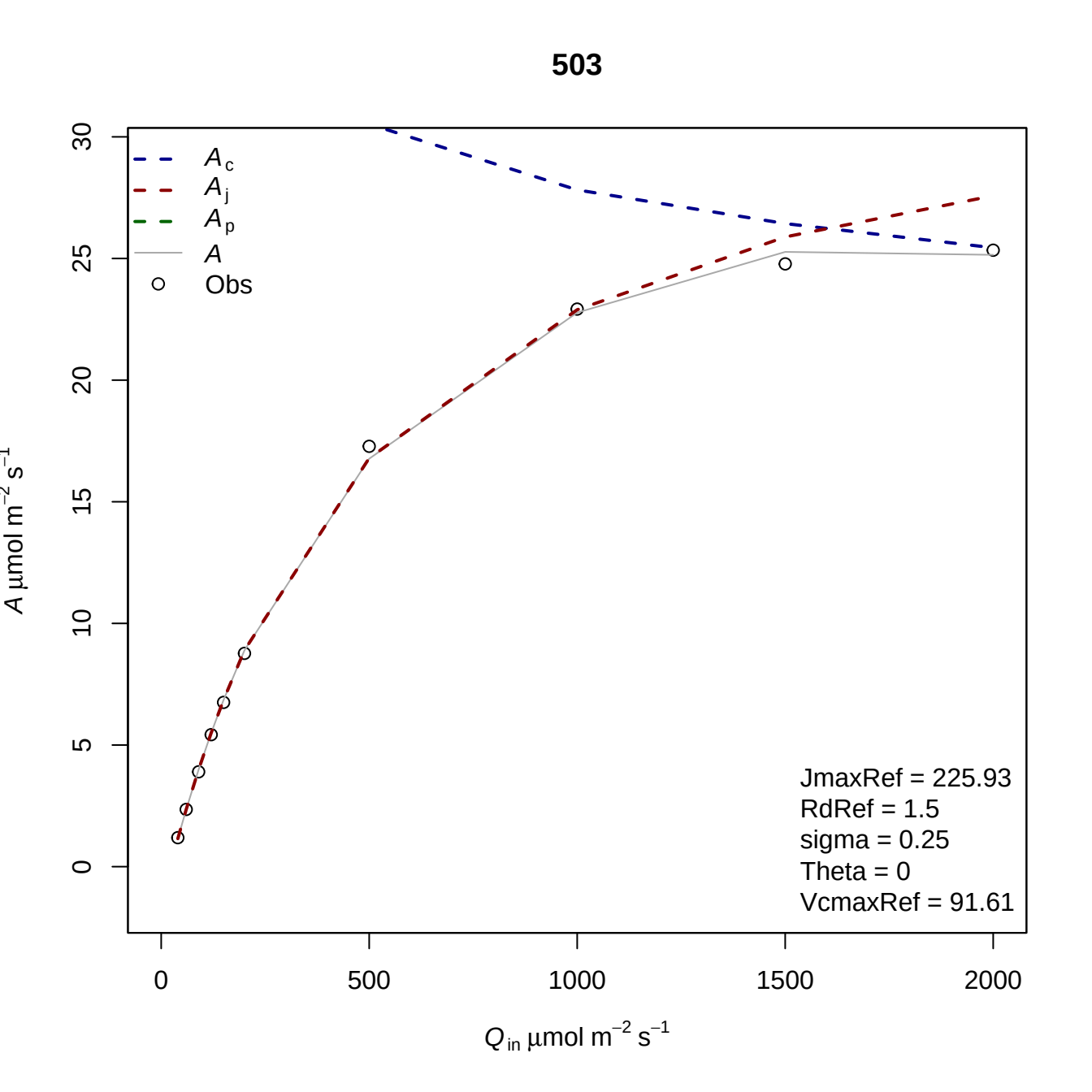

498
A_c
A_j
A_p
A
Obs
JmaxRef = 258.64
RdRef = 1.06
sigma = 0.36
Theta = 0
VcmaxRef = 158.98
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

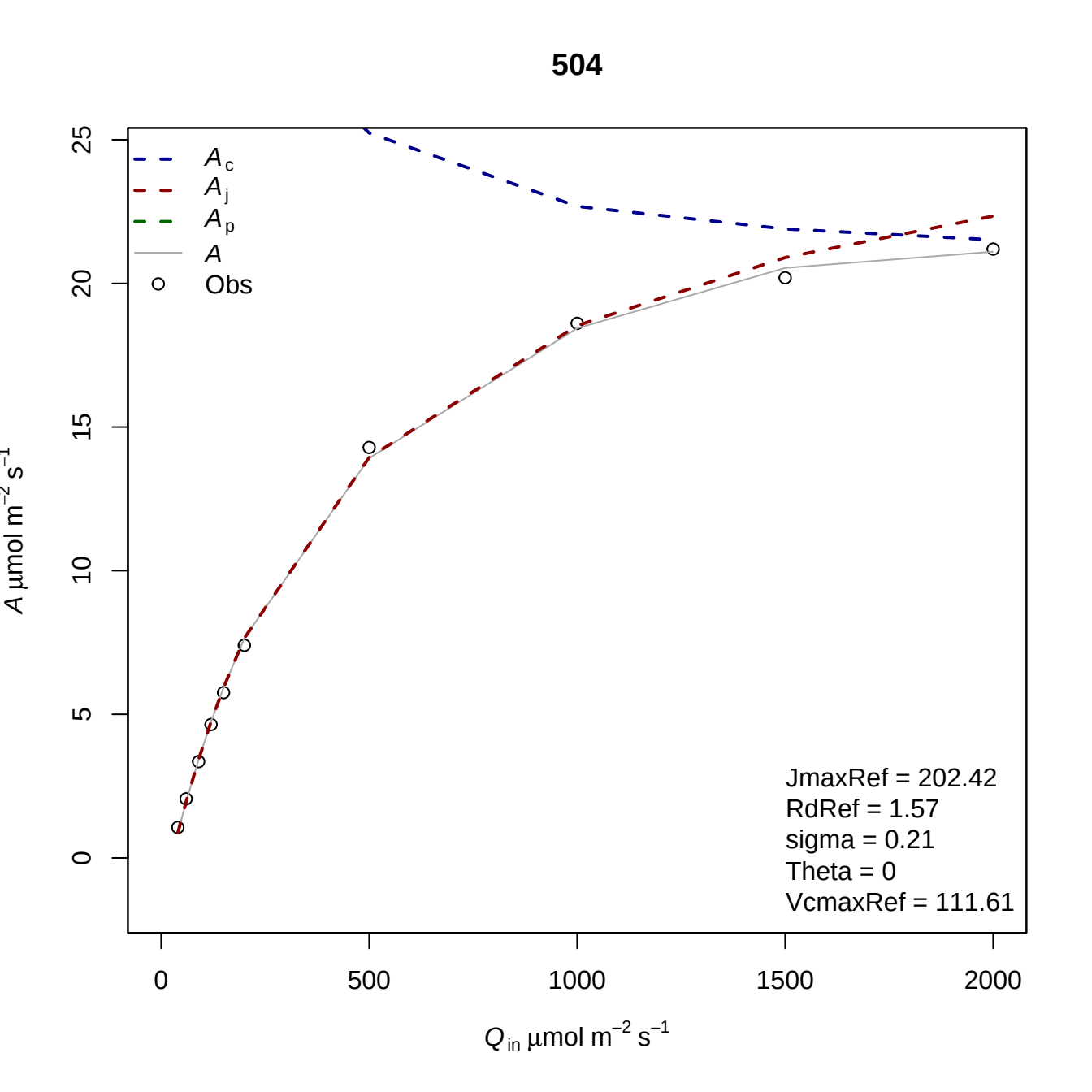

499
A_c
A_j
A_p
A
Obs
JmaxRef = 324.98
RdRef = 3.06
sigma = 0.34
Theta = 0
VcmaxRef = 261.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

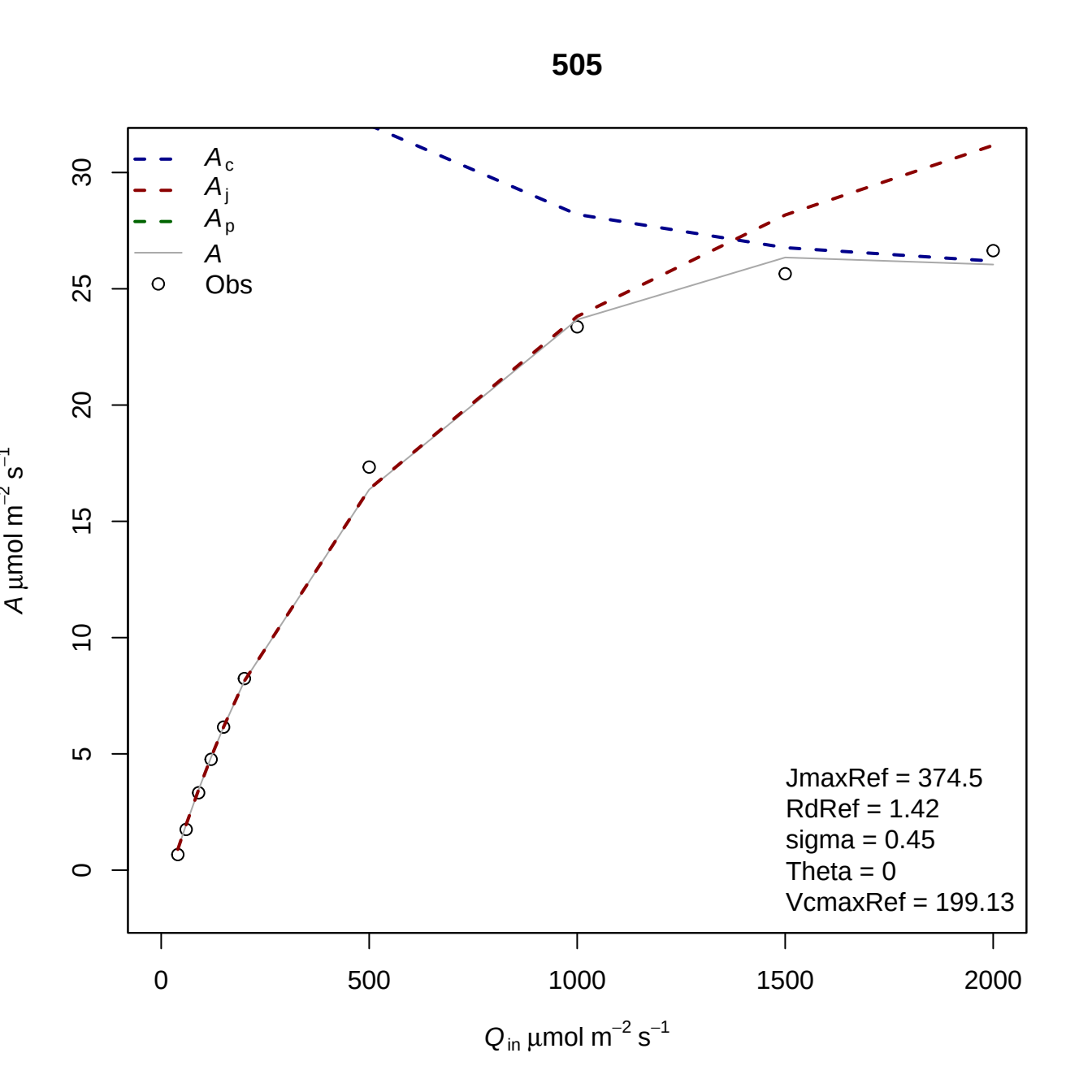

500
A_c
A_j
A_p
A
Obs
JmaxRef = 297.51
RdRef = 3.31
sigma = 0.08
Theta = 0
VcmaxRef = 324.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

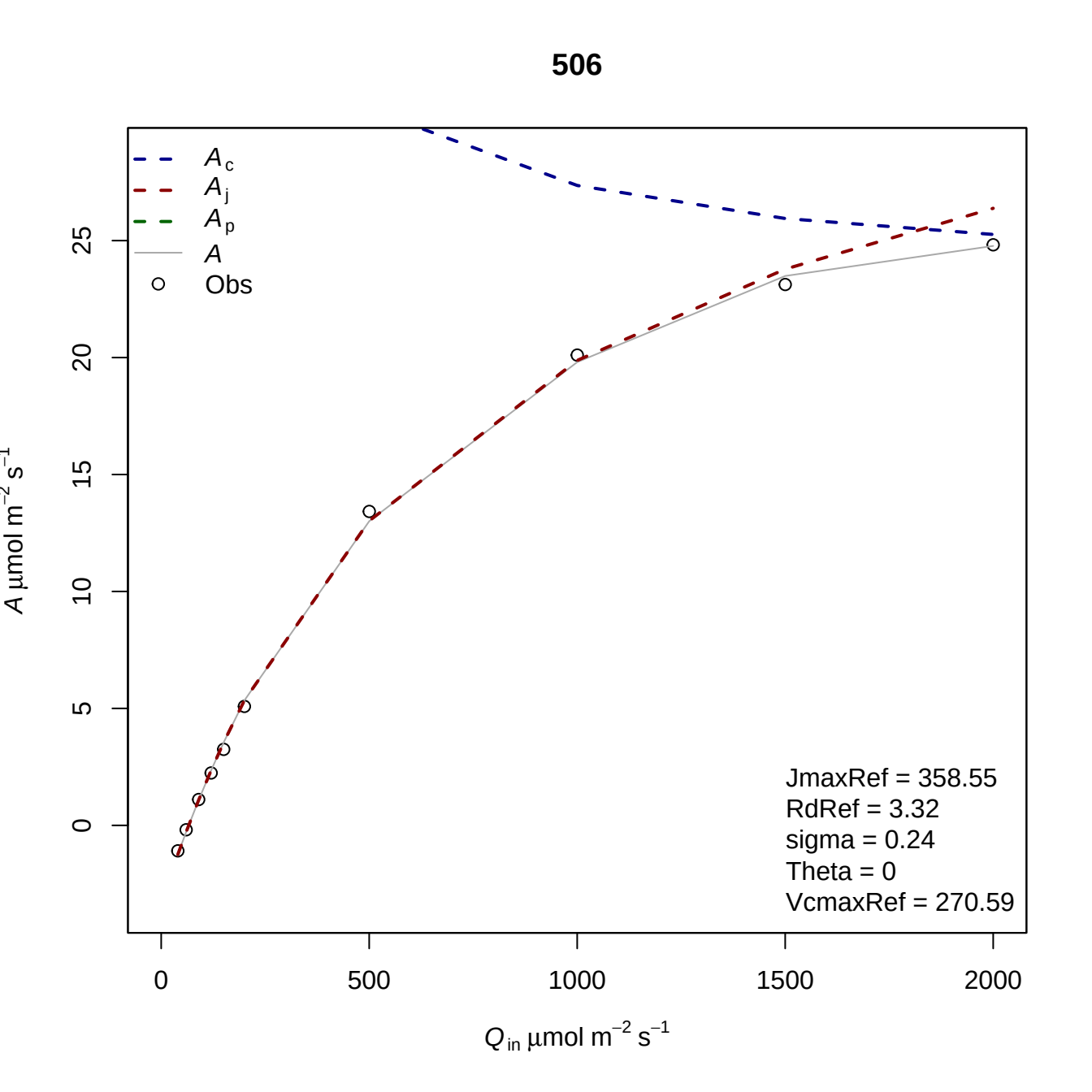

501
A_c
A_j
A_p
A
Obs
JmaxRef = 119.38
RdRef = 0.24
sigma = 0.24
Theta = 0
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

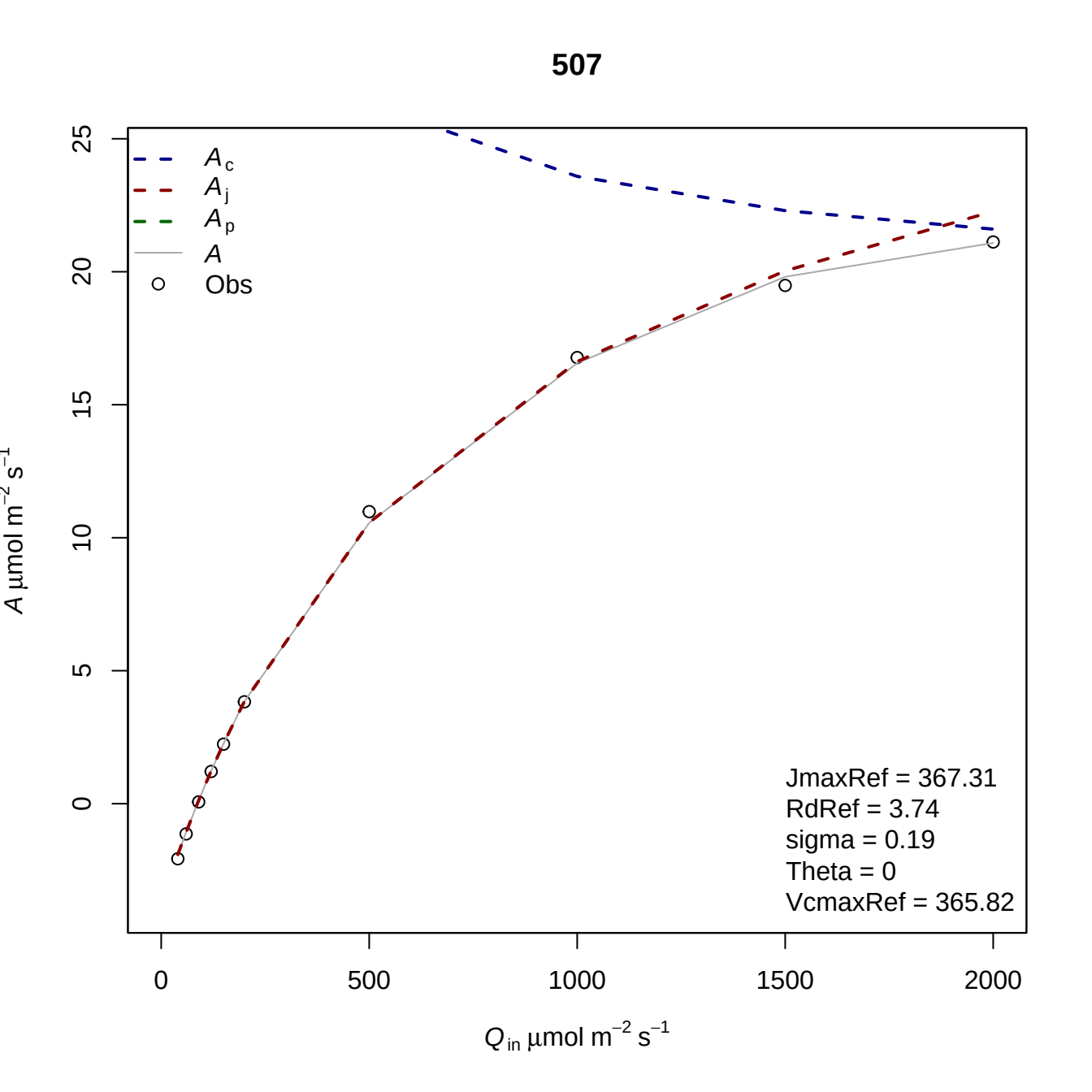

502
A_c
A_j
A_p
A
Obs
JmaxRef = 161.11
RdRef = 0.72
sigma = 0.21
Theta = 0
VcmaxRef = 57.44
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

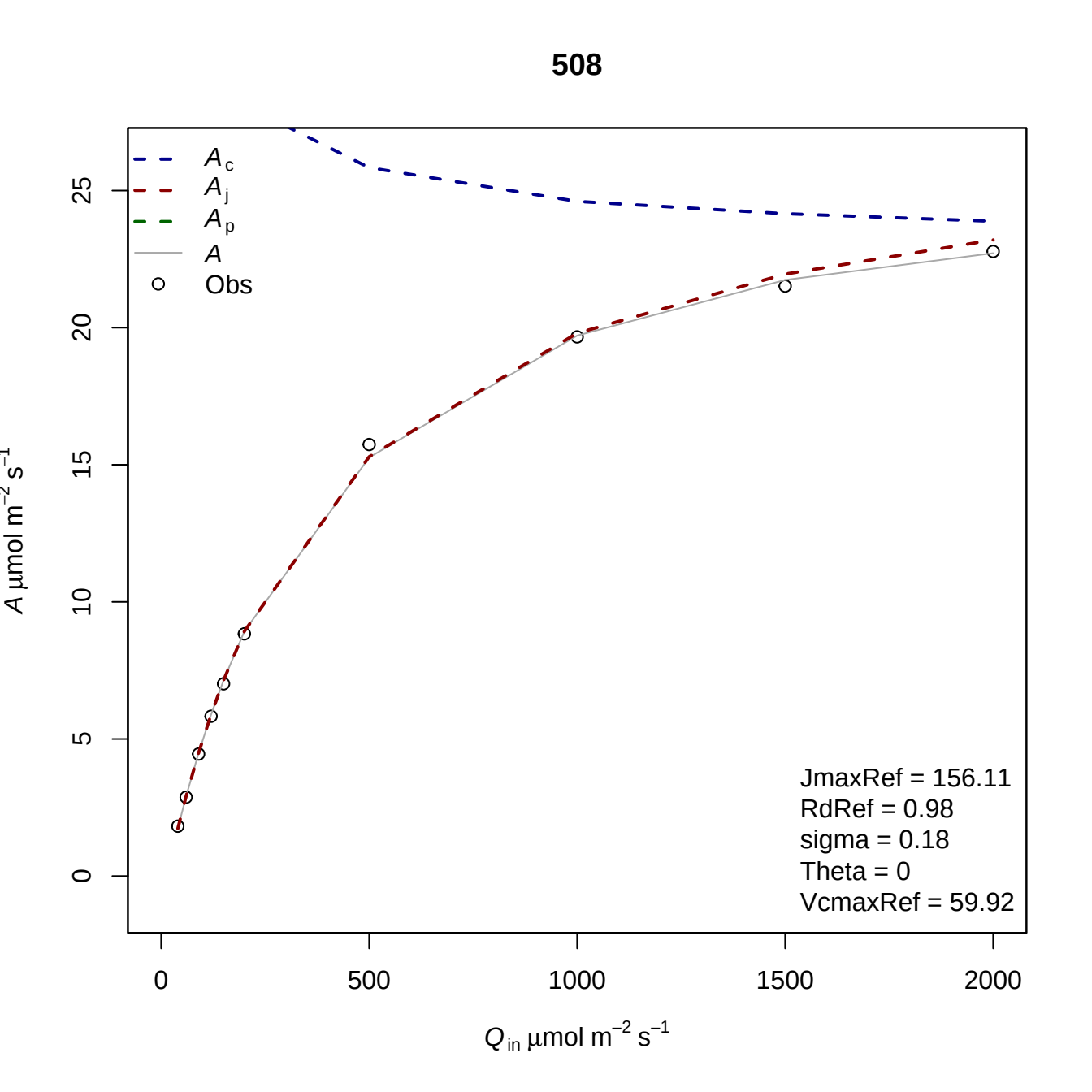

503
A_c
A_j
A_p
A
Obs
JmaxRef = 225.93
RdRef = 1.5
sigma = 0.25
Theta = 0
VcmaxRef = 91.61
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

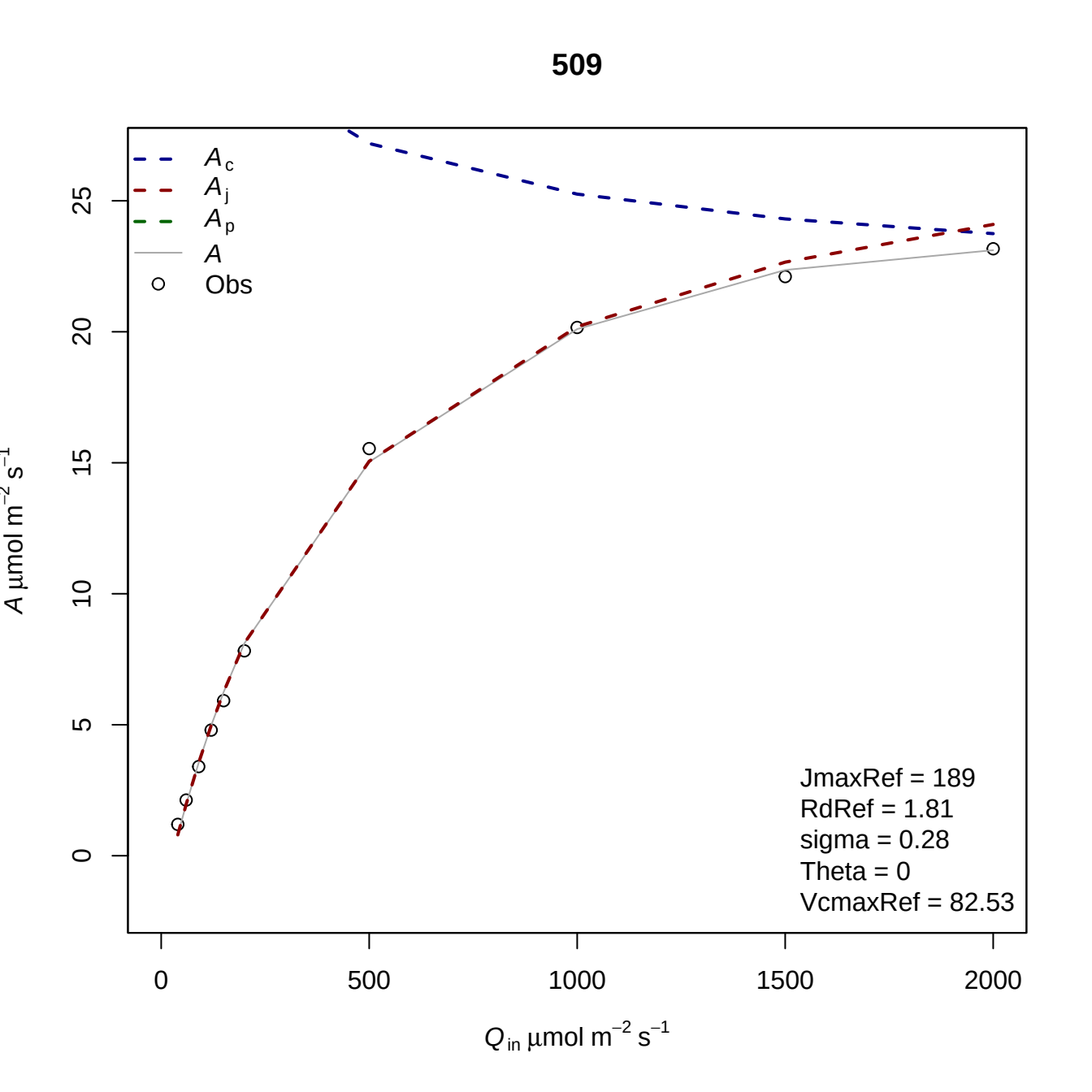

504
A_c
A_j
A_p
A
Obs
JmaxRef = 202.42
RdRef = 1.57
sigma = 0.21
Theta = 0
VcmaxRef = 111.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

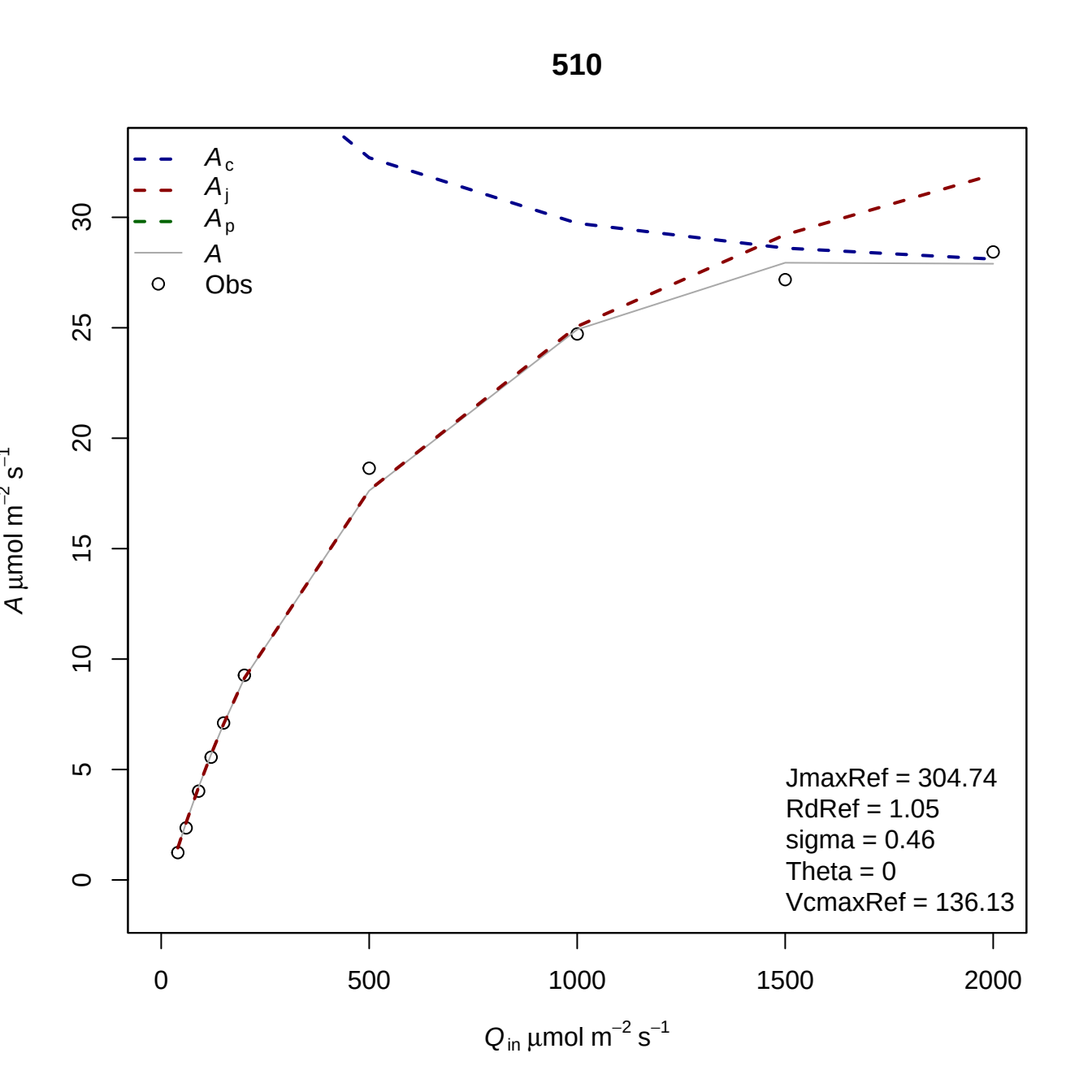

505
A_c
A_j
A_p
A
Obs
JmaxRef = 374.5
RdRef = 1.42
sigma = 0.45
Theta = 0
VcmaxRef = 199.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

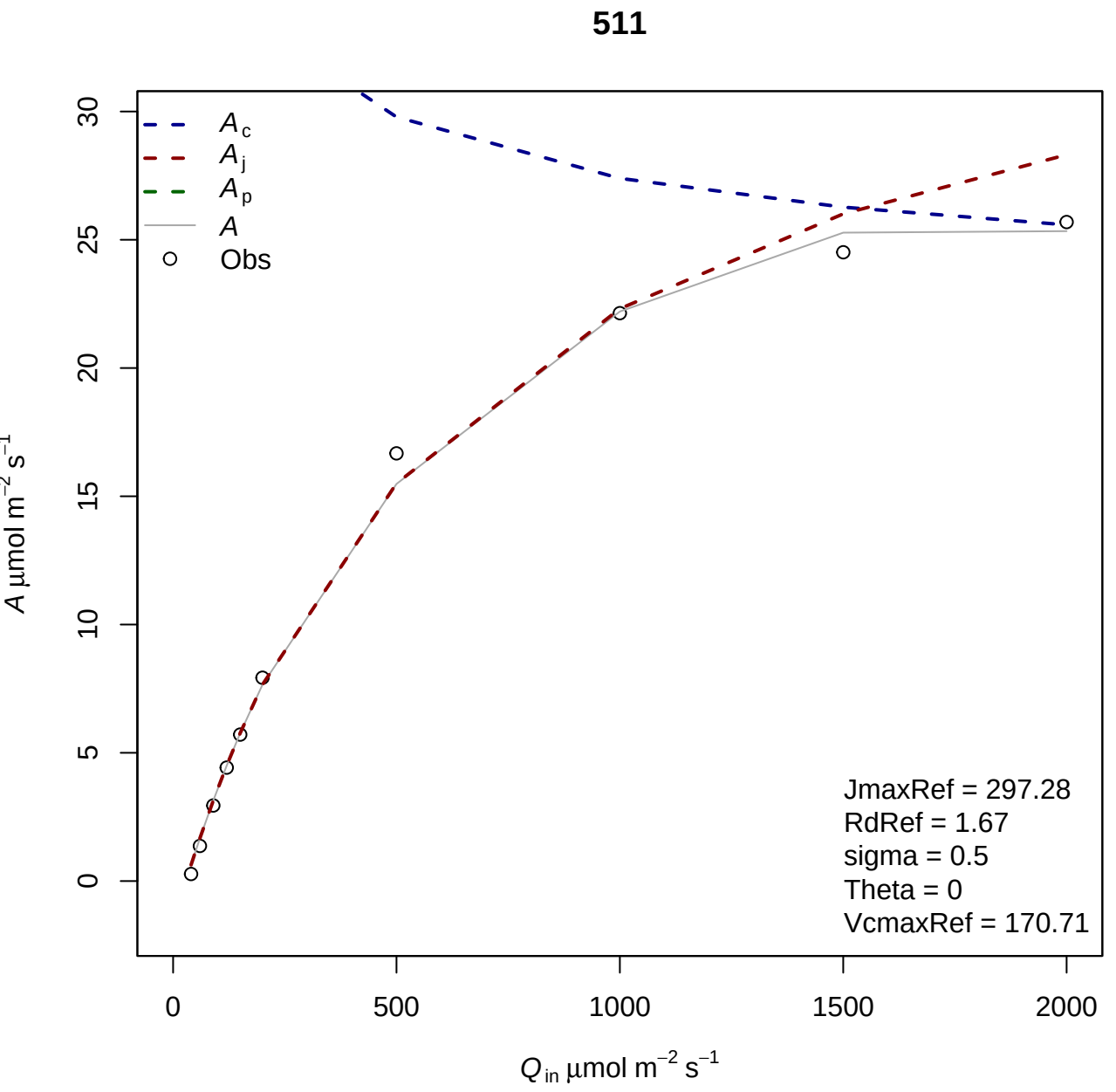

506
A_c
A_j
A_p
A
Obs
JmaxRef = 358.55
RdRef = 3.32
sigma = 0.24
Theta = 0
VcmaxRef = 270.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

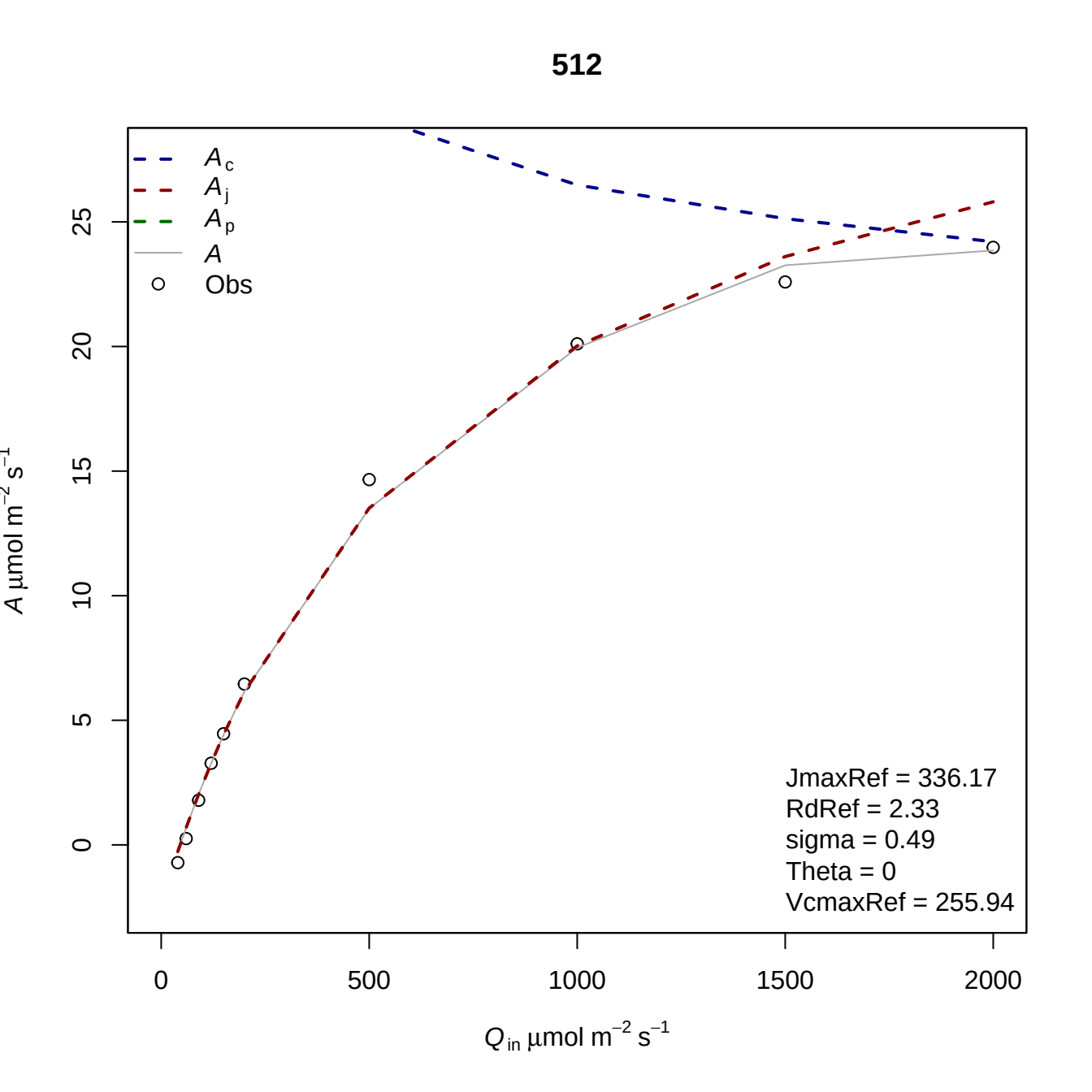

507
A_c
A_j
A_p
A
Obs
JmaxRef = 367.31
RdRef = 3.74
sigma = 0.19
Theta = 0
VcmaxRef = 365.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

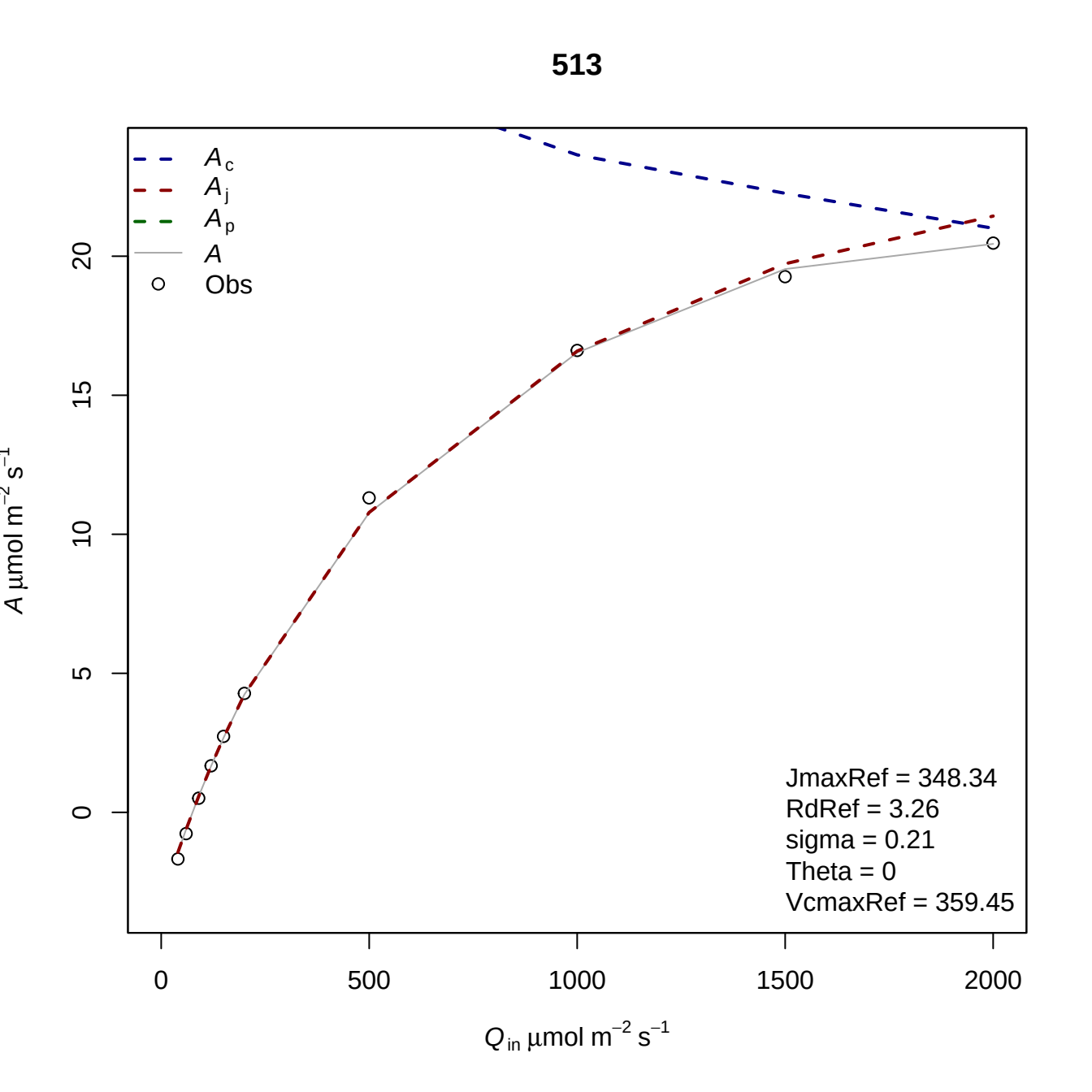

508
A_c
A_j
A_p
A
Obs
JmaxRef = 156.11
RdRef = 0.98
sigma = 0.18
Theta = 0
VcmaxRef = 59.92
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

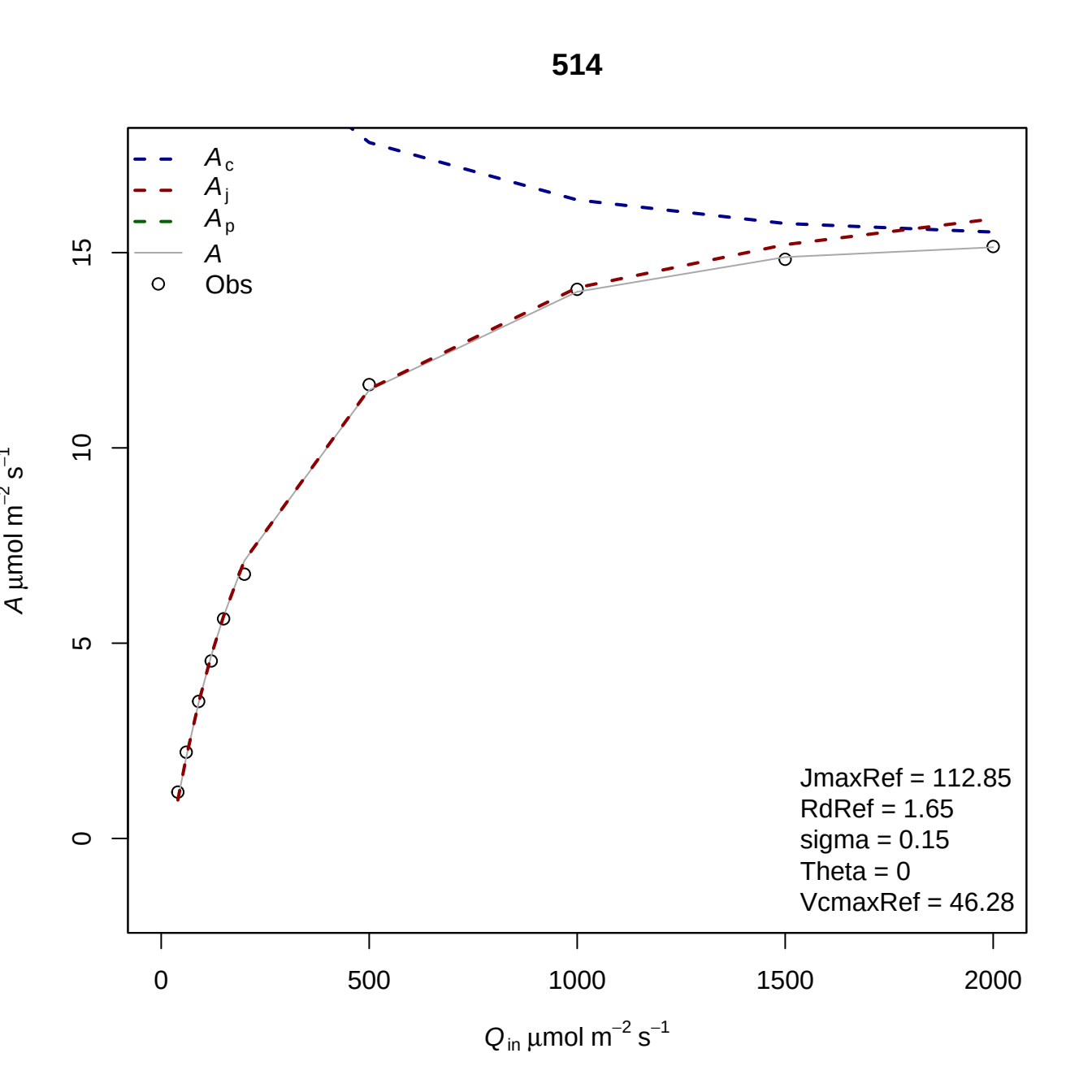

509
A_c
A_j
A_p
A
Obs
JmaxRef = 189
RdRef = 1.81
sigma = 0.28
Theta = 0
VcmaxRef = 82.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

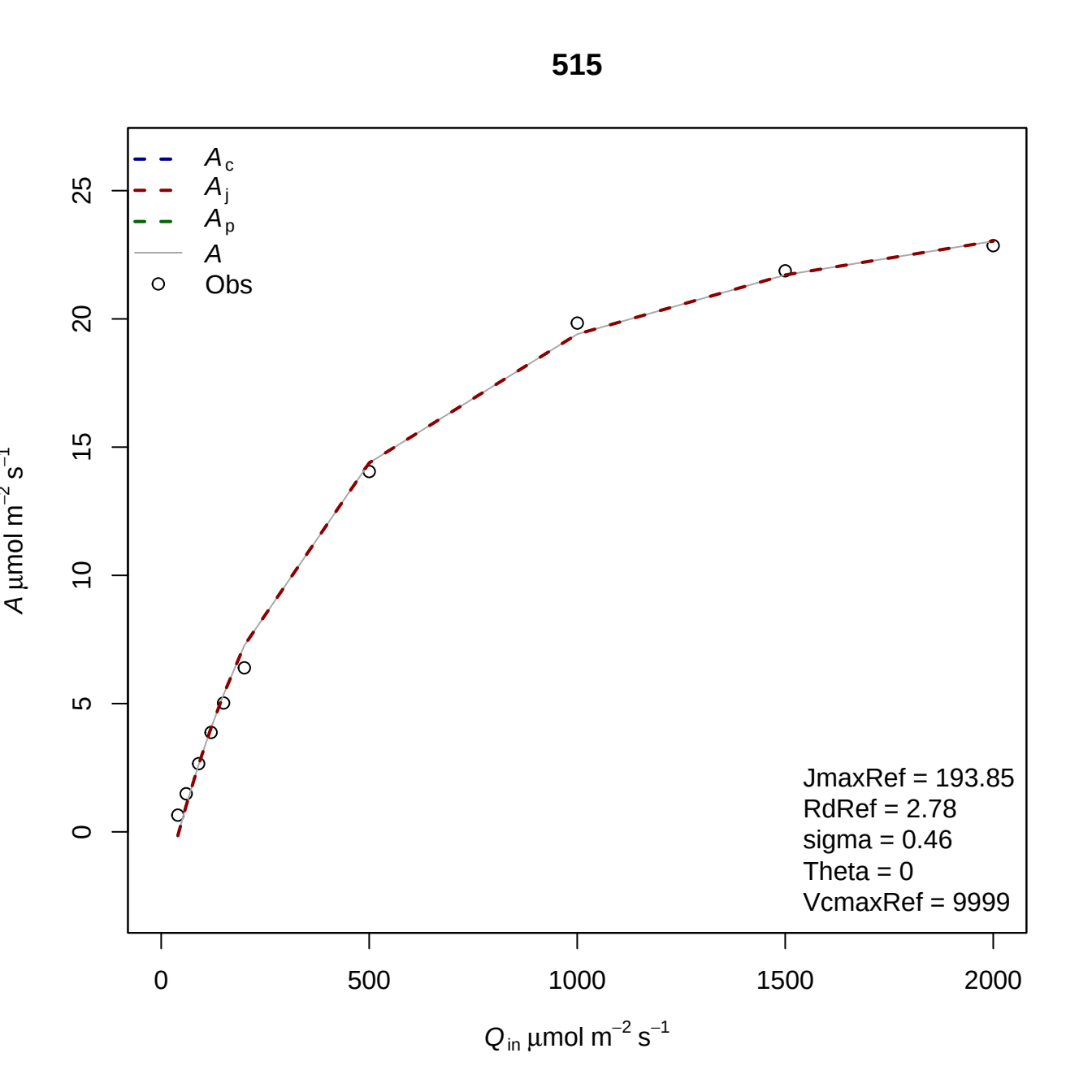

510
A_c
A_j
A_p
A
Obs
JmaxRef = 304.74
RdRef = 1.05
sigma = 0.46
Theta = 0
VcmaxRef = 136.13
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

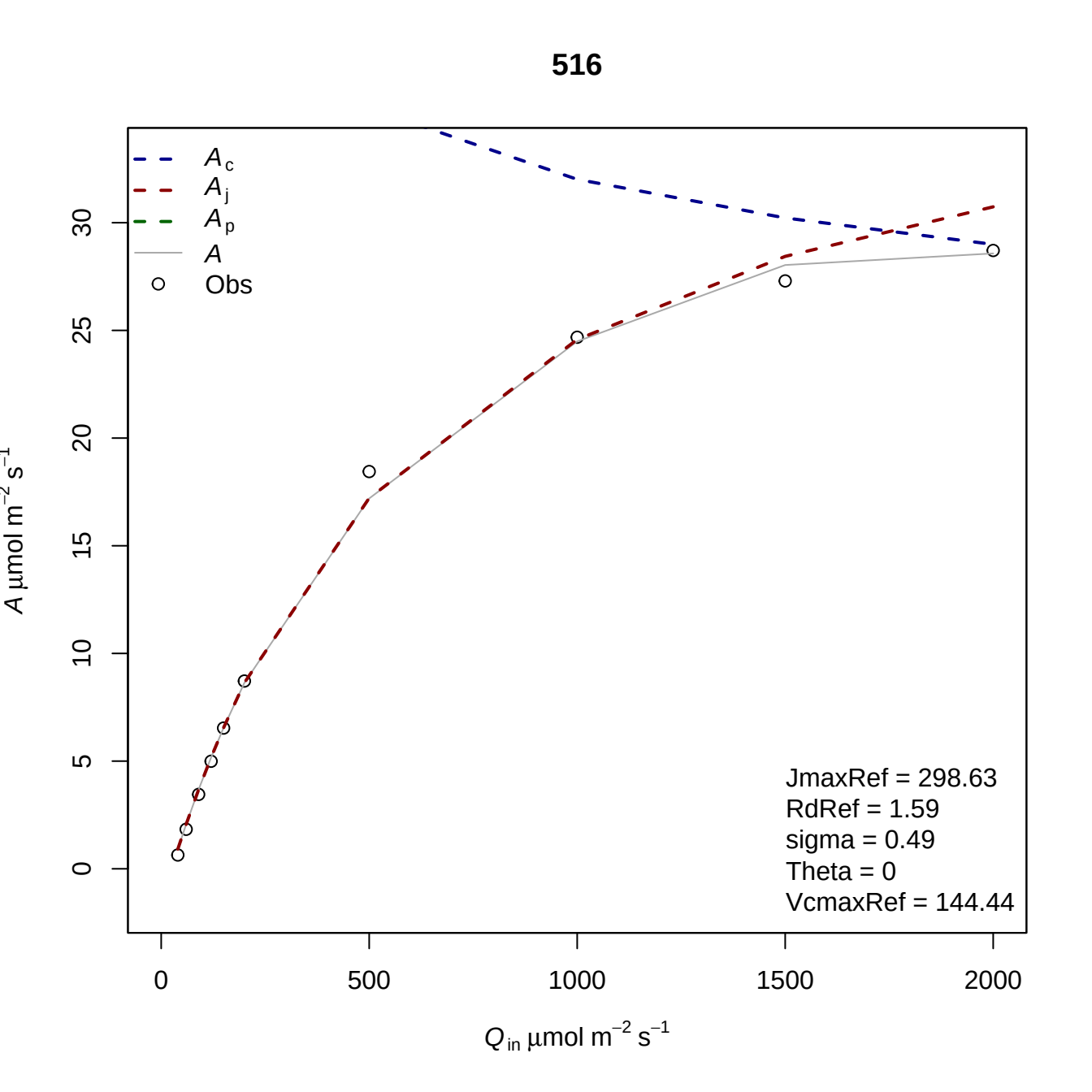

511
A_c
A_j
A_p
A
Obs
JmaxRef = 297.28
RdRef = 1.67
sigma = 0.5
Theta = 0
VcmaxRef = 170.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

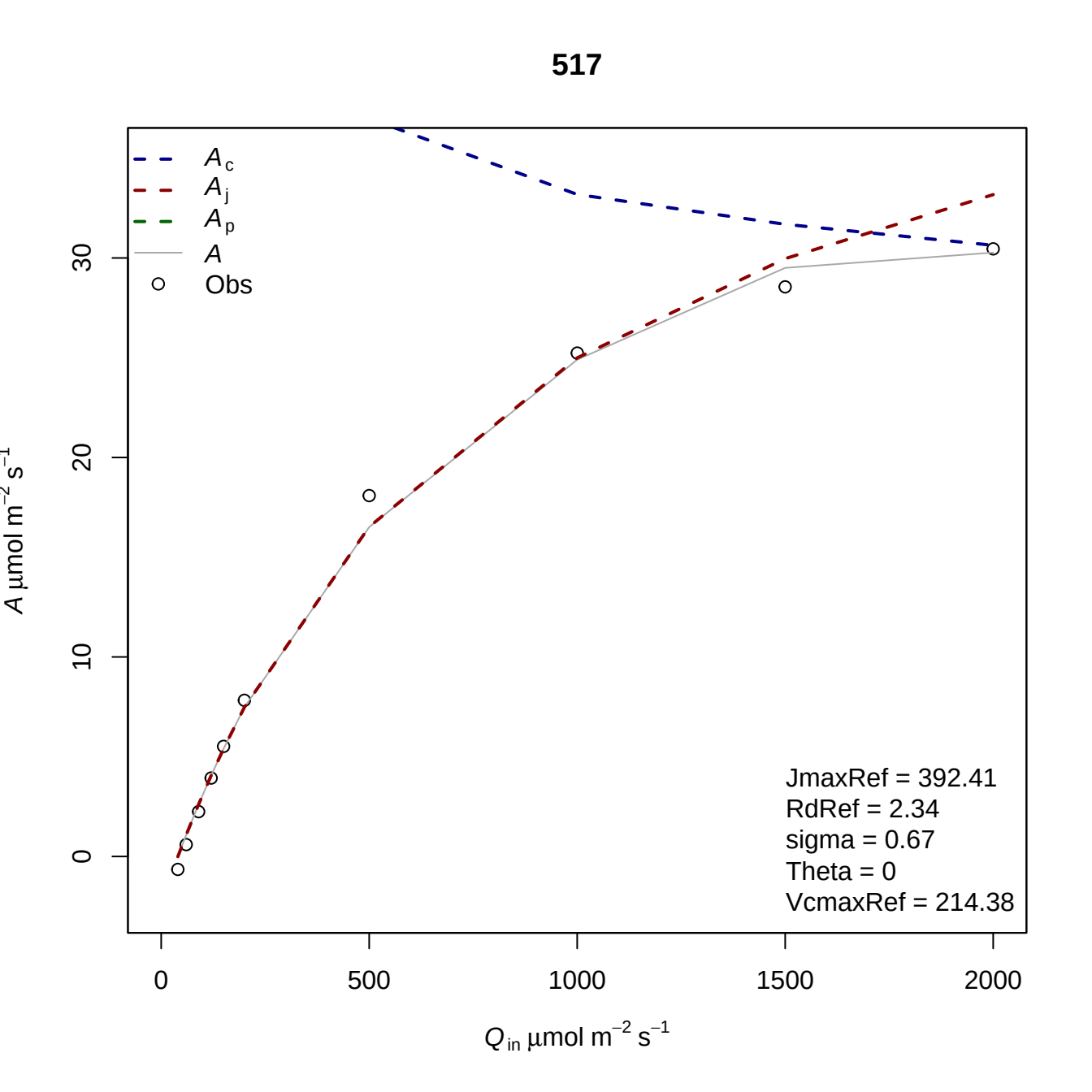

512
A_c
A_j
A_p
A
Obs
JmaxRef = 336.17
RdRef = 2.33
sigma = 0.49
Theta = 0
VcmaxRef = 255.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

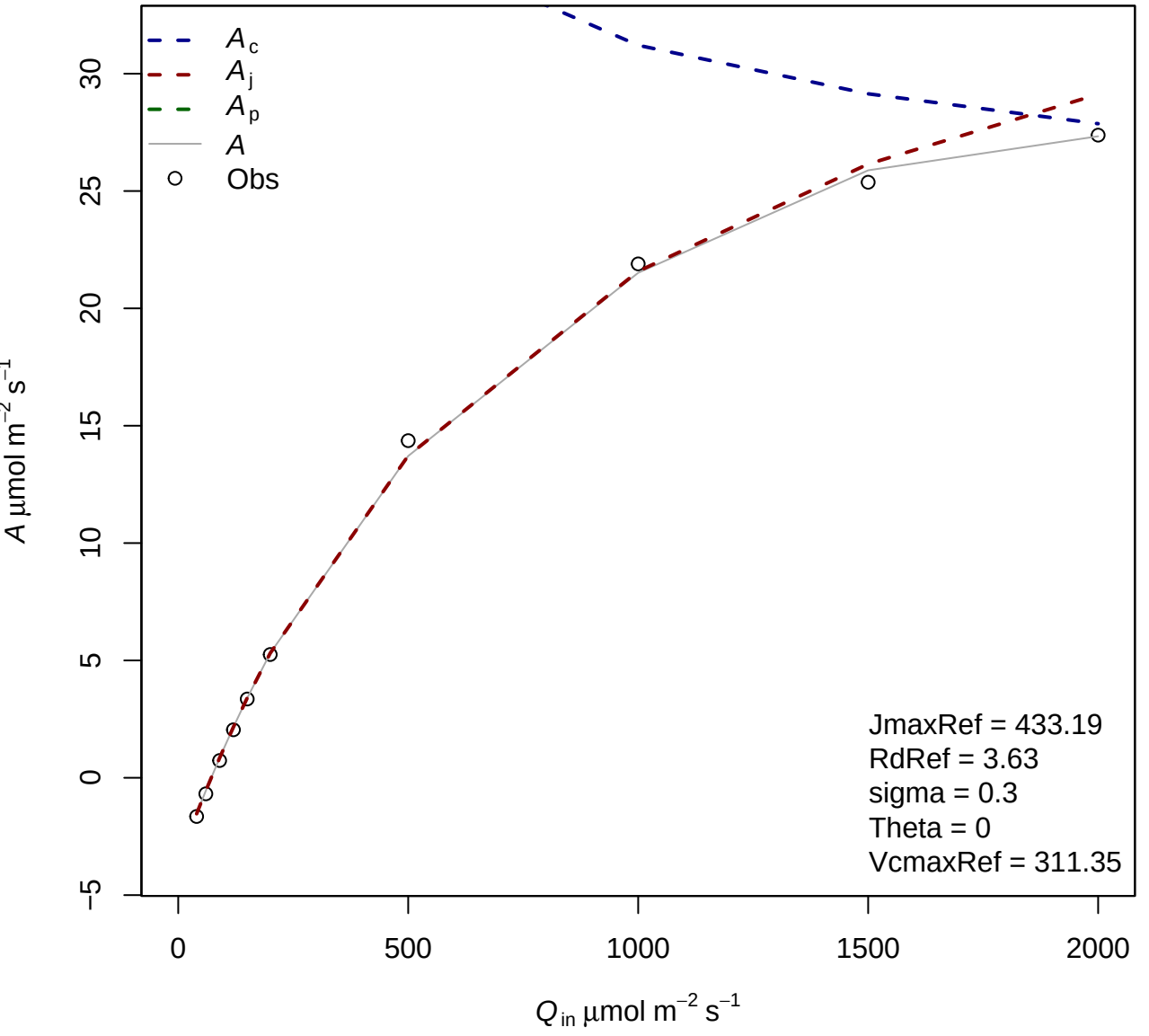

513
A_c
A_j
A_p
A
Obs
JmaxRef = 348.34
RdRef = 3.26
sigma = 0.21
Theta = 0
VcmaxRef = 359.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

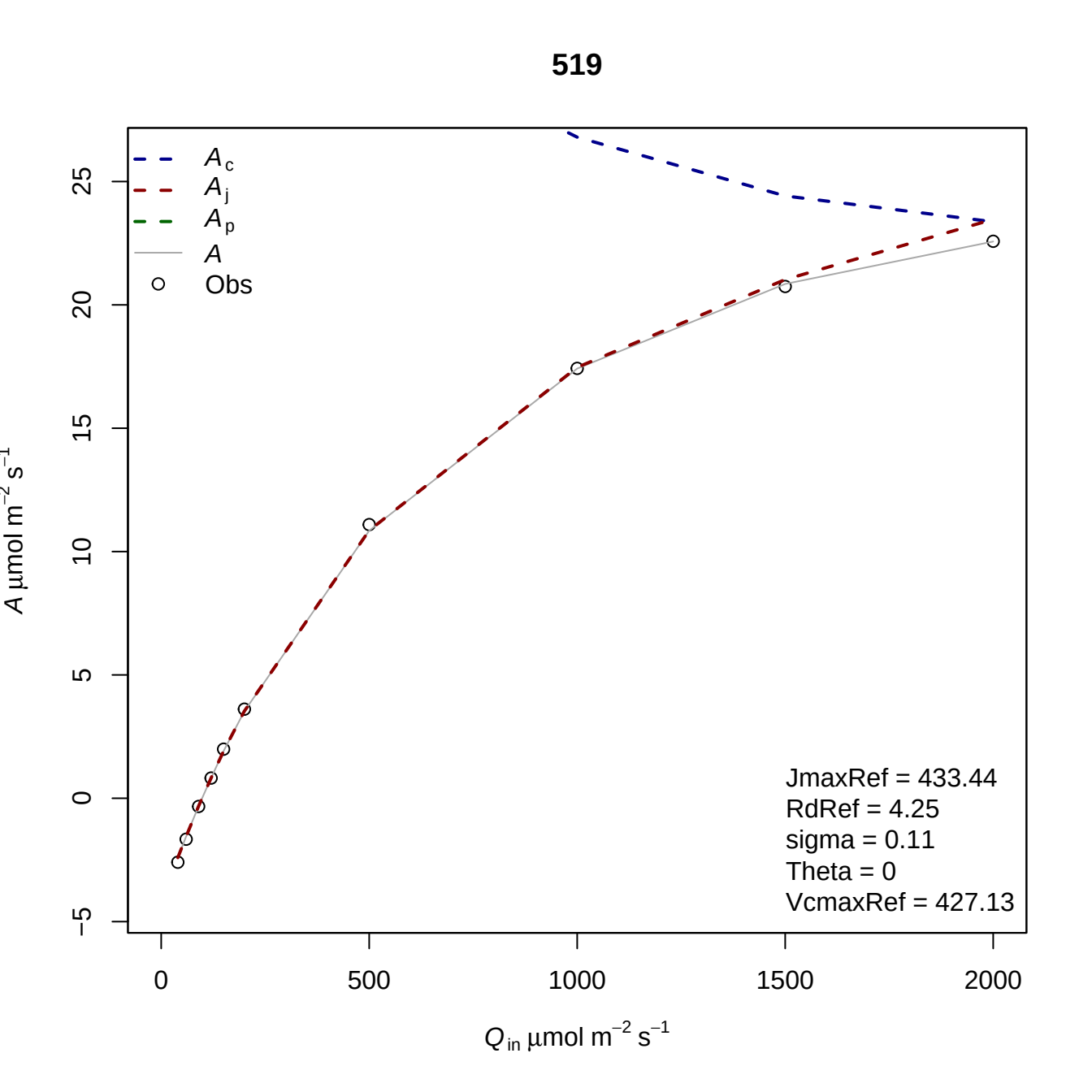

514
A_c
A_j
A_p
A
Obs
JmaxRef = 112.85
RdRef = 1.65
sigma = 0.15
Theta = 0
VcmaxRef = 46.28
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

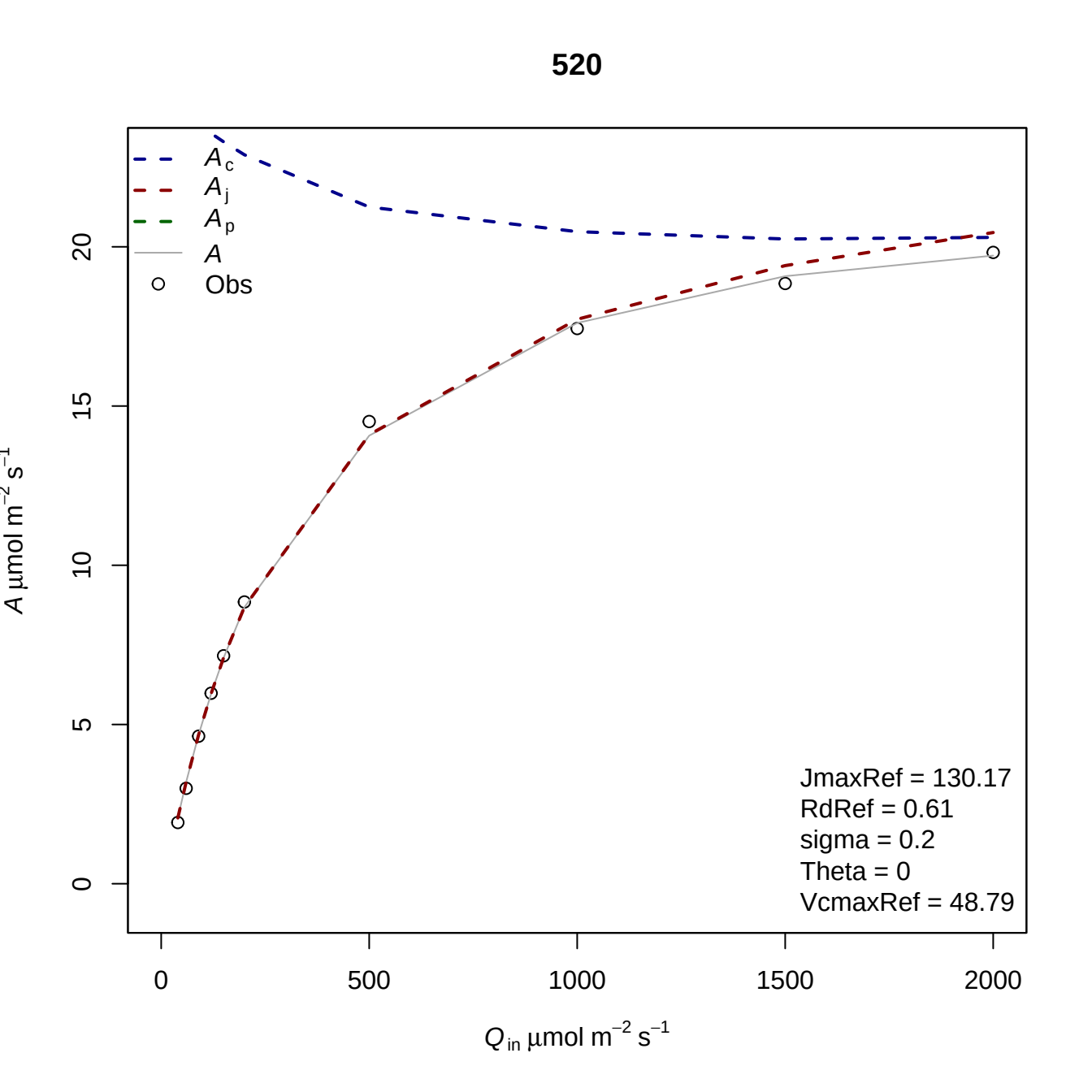

515
A_c
A_j
A_p
A
Obs
JmaxRef = 193.85
RdRef = 2.78
sigma = 0.46
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

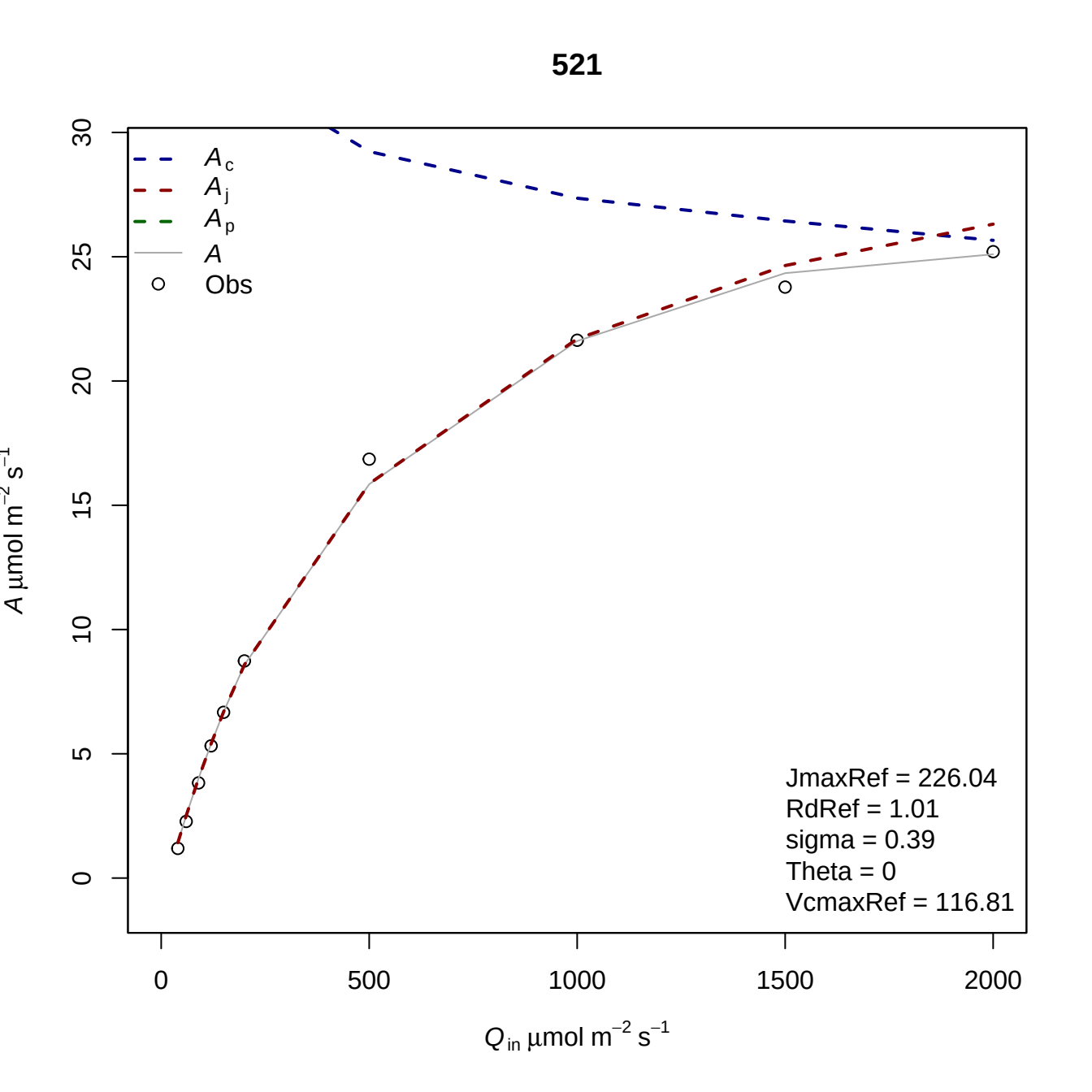

516
A_c
A_j
A_p
A
Obs
JmaxRef = 298.63
RdRef = 1.59
sigma = 0.49
Theta = 0
VcmaxRef = 144.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

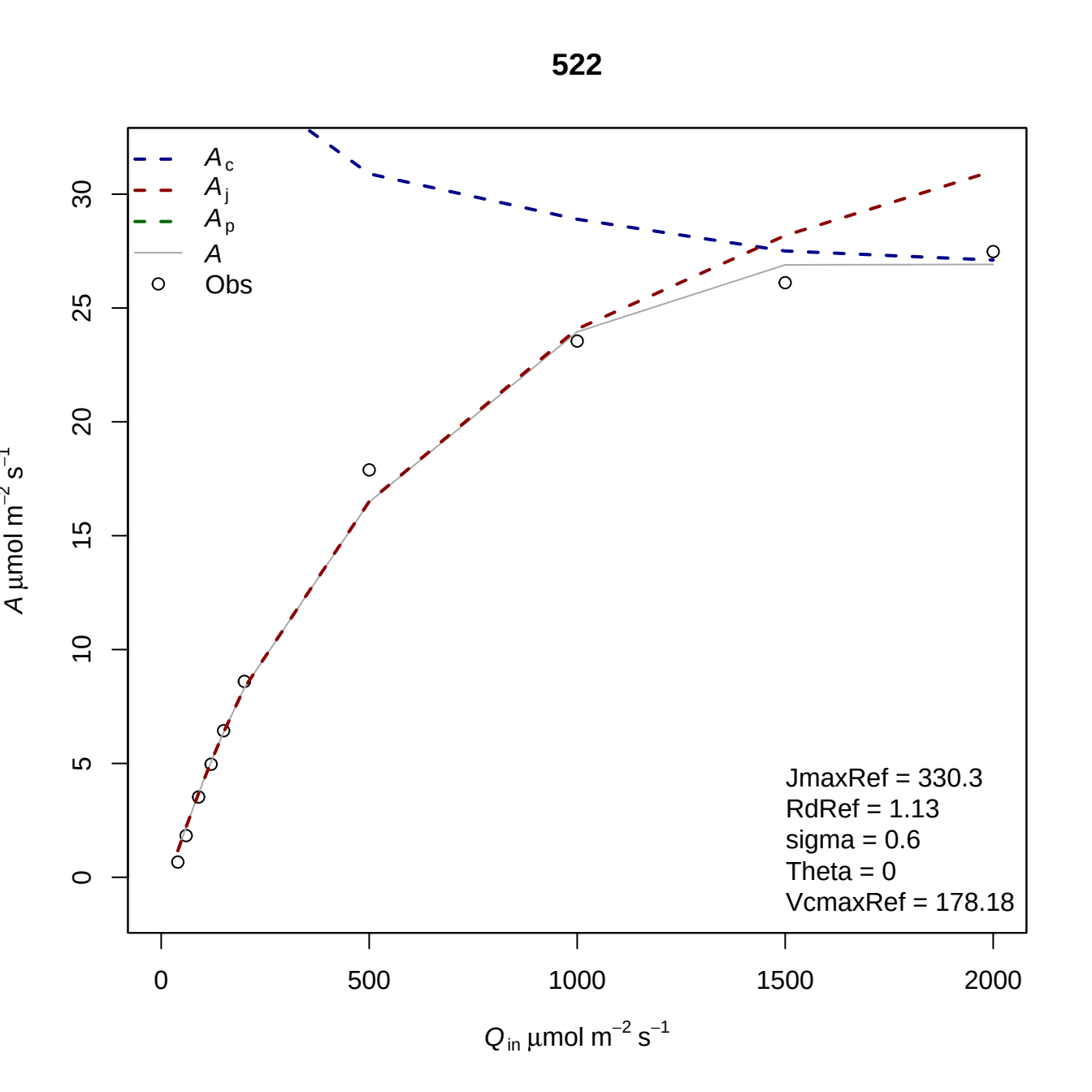

517
A_c
A_j
A_p
A
Obs
JmaxRef = 392.41
RdRef = 2.34
sigma = 0.67
Theta = 0
VcmaxRef = 214.38
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

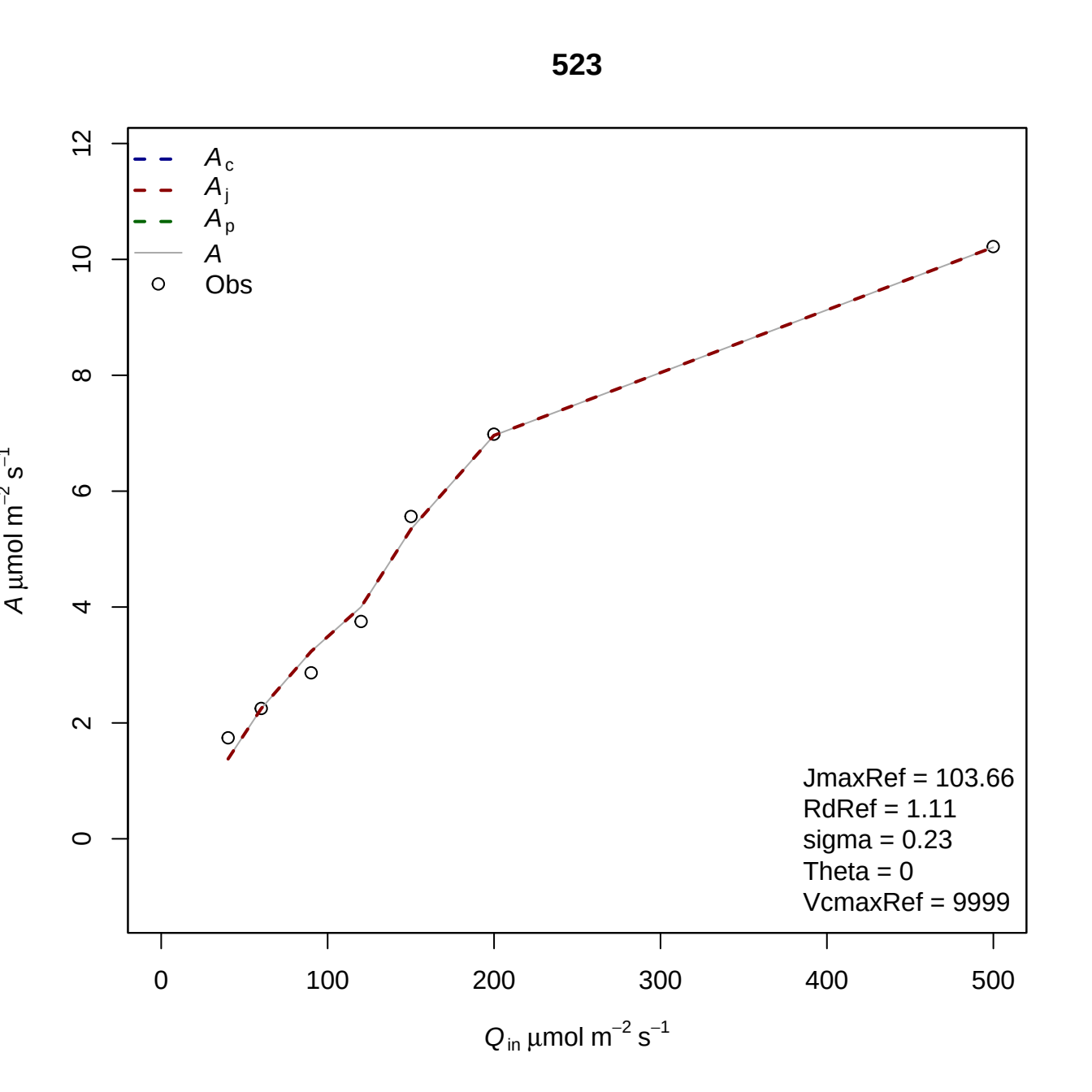

A_c
A_j
A_p
A
Obs
JmaxRef = 433.19
RdRef = 3.63
sigma = 0.3
Theta = 0
VcmaxRef = 311.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
30
0
500
1000
1500
2000

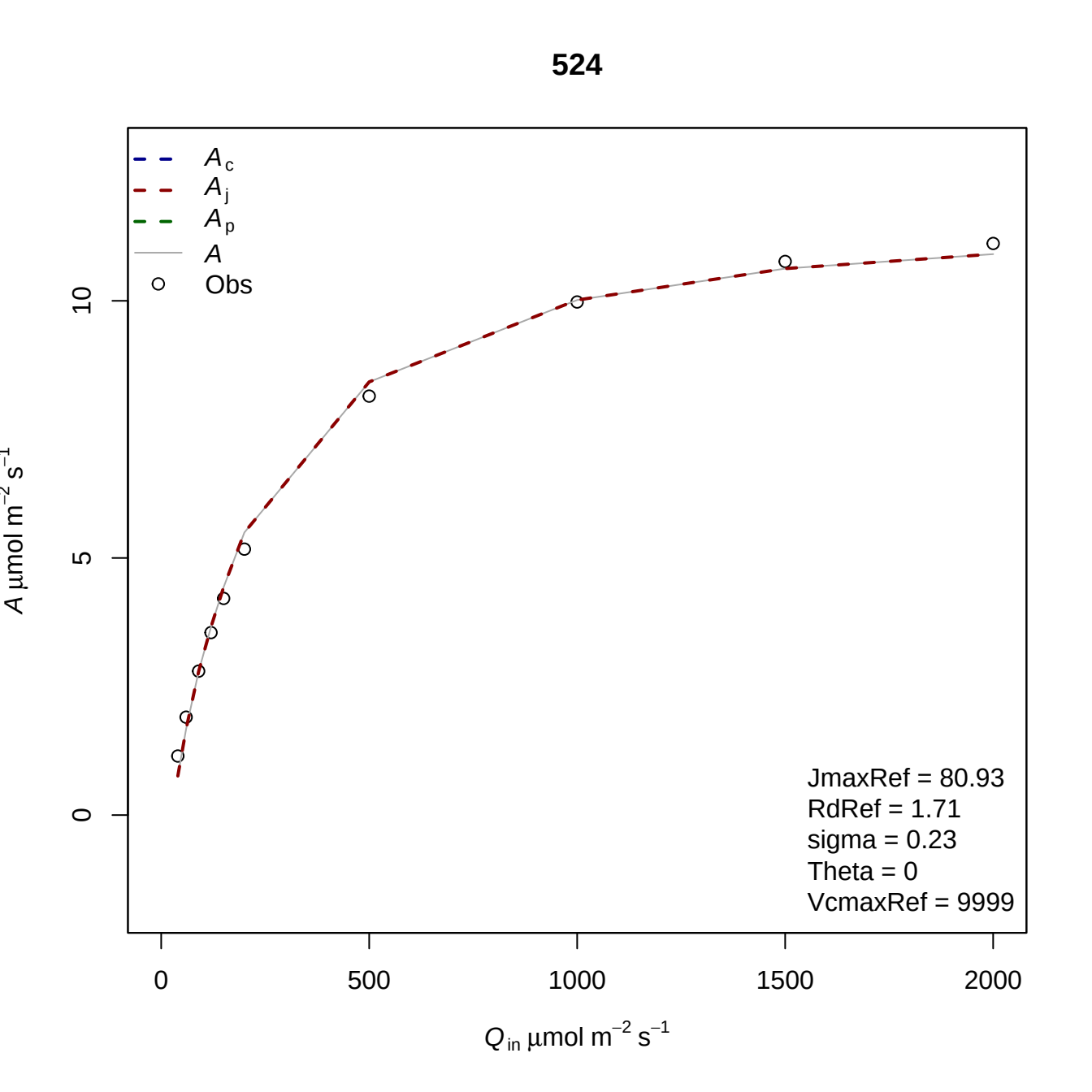

519
A_c
A_j
A_p
A
Obs
JmaxRef = 433.44
RdRef = 4.25
sigma = 0.11
Theta = 0
VcmaxRef = 427.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500
2000

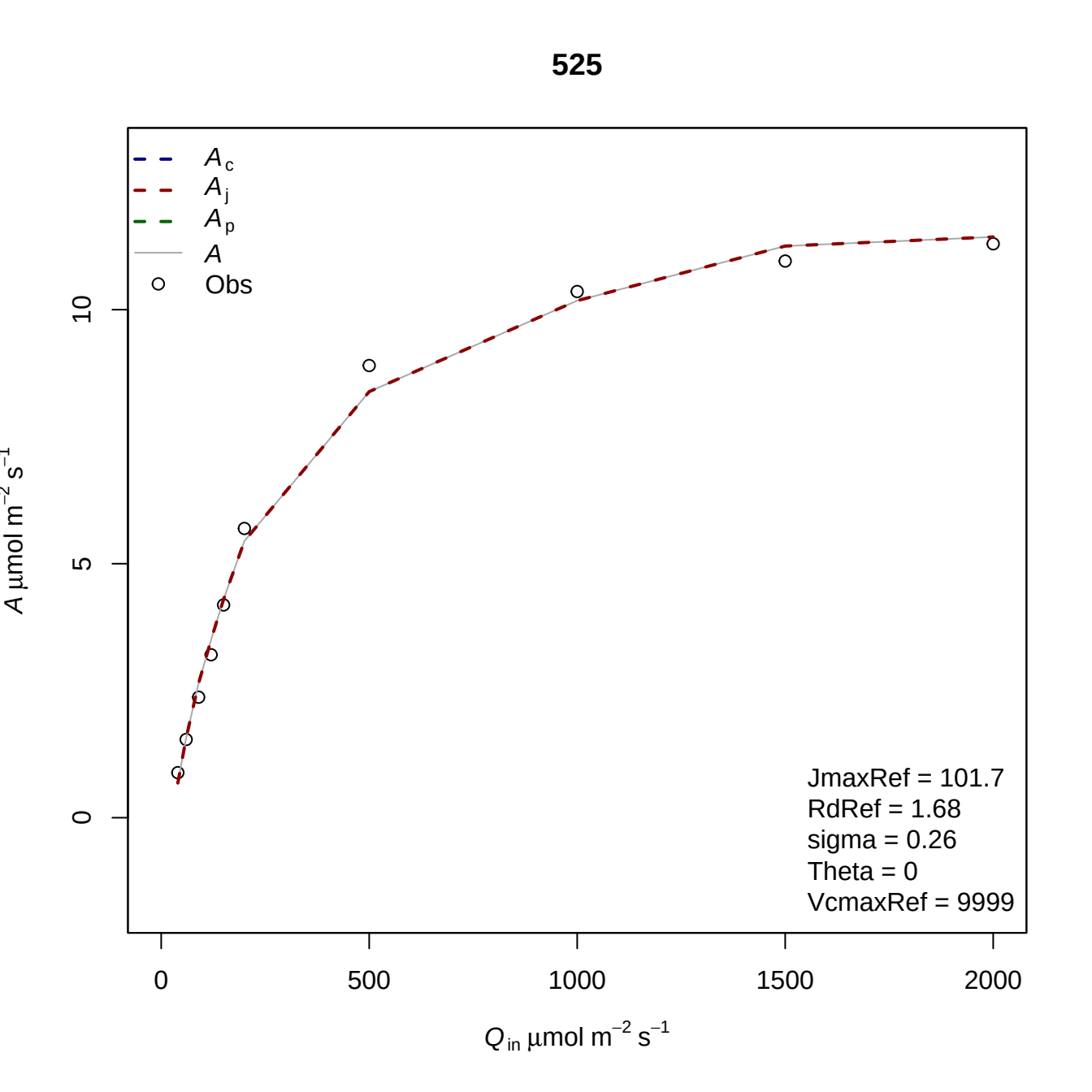

520
A_c
A_j
A_p
A
Obs
JmaxRef = 130.17
RdRef = 0.61
sigma = 0.2
Theta = 0
VcmaxRef = 48.79
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

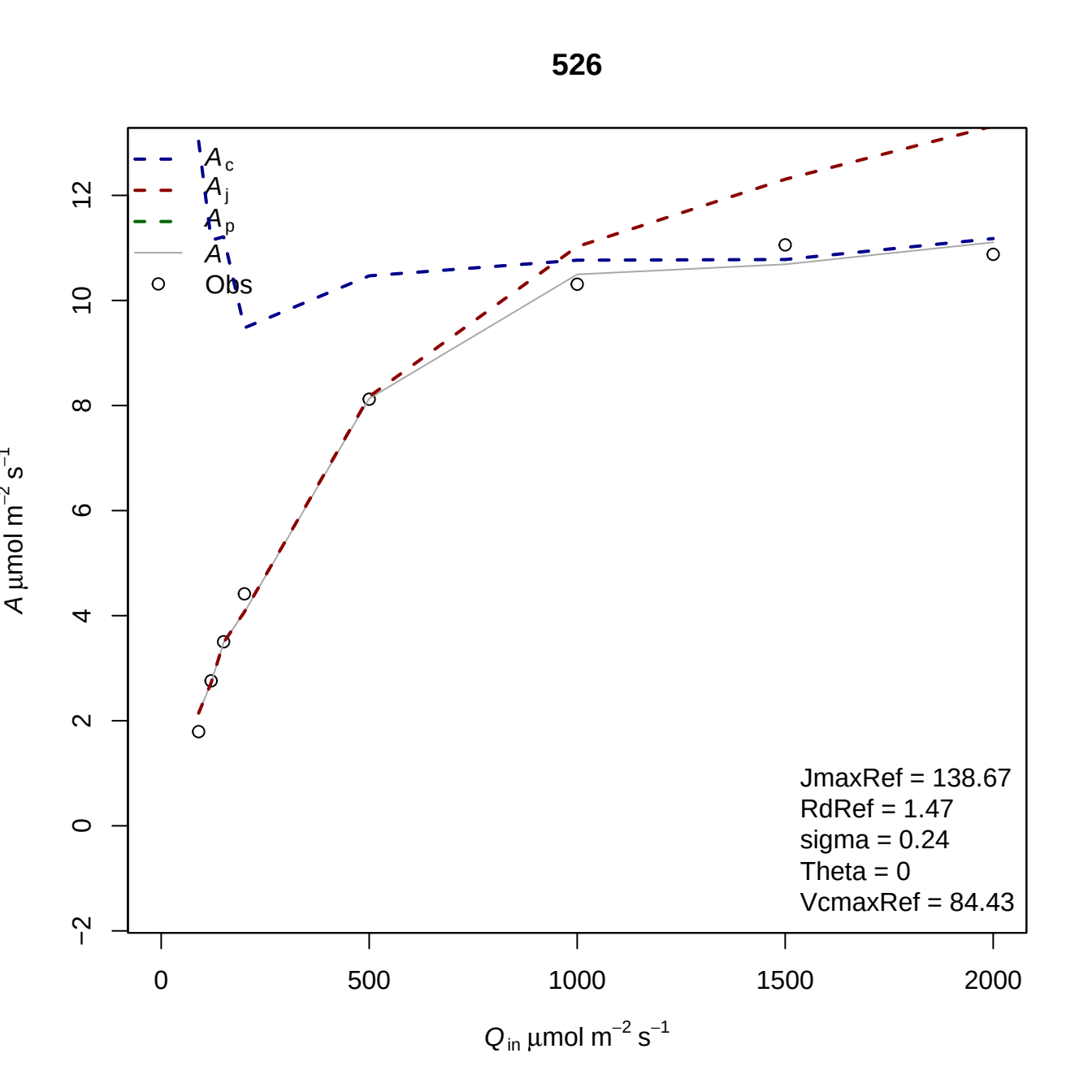

521
A_c
A_j
A_p
A
Obs
JmaxRef = 226.04
RdRef = 1.01
sigma = 0.39
Theta = 0
VcmaxRef = 116.81
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

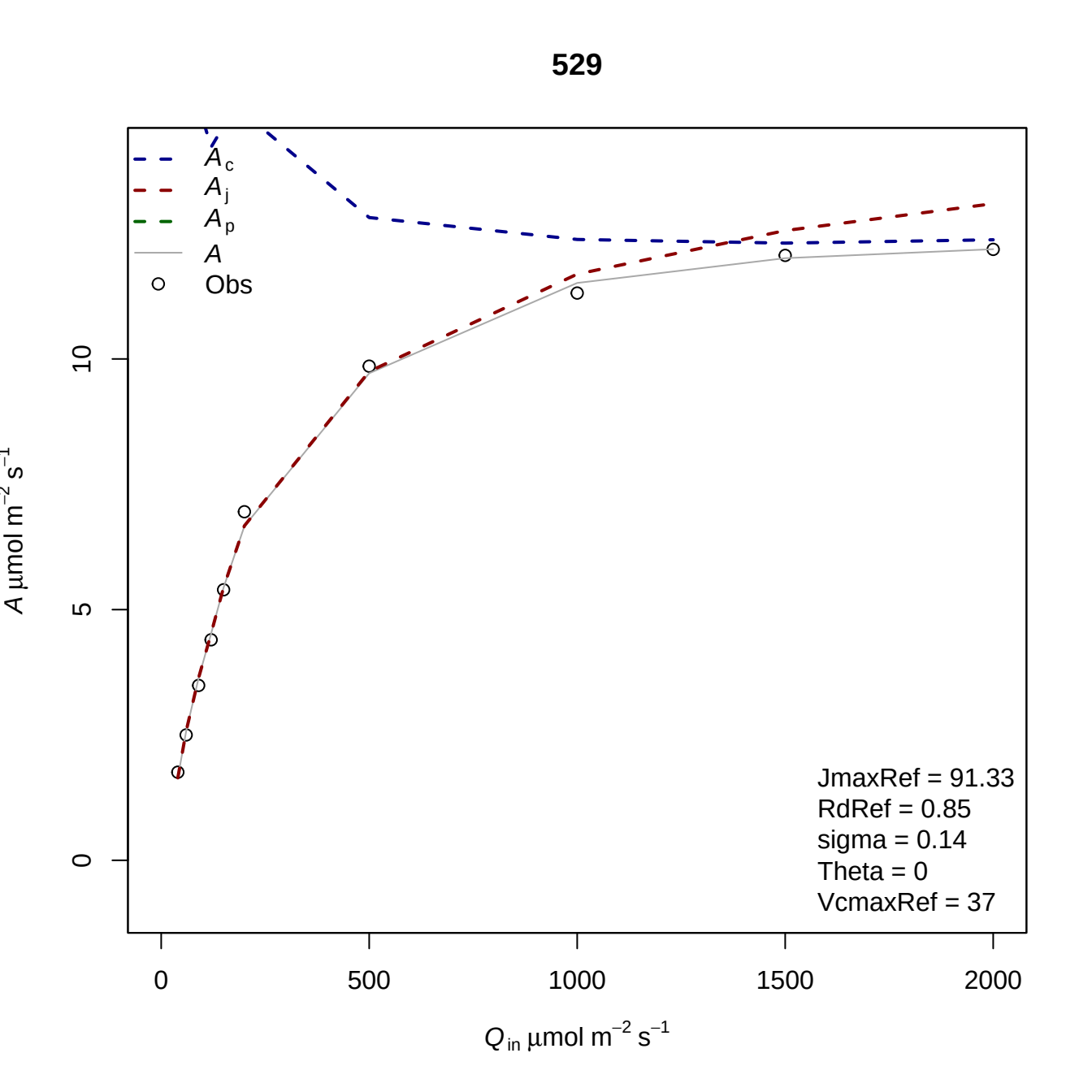

522
A_c
A_j
A_p
A
Obs
JmaxRef = 330.3
RdRef = 1.13
sigma = 0.6
Theta = 0
VcmaxRef = 178.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

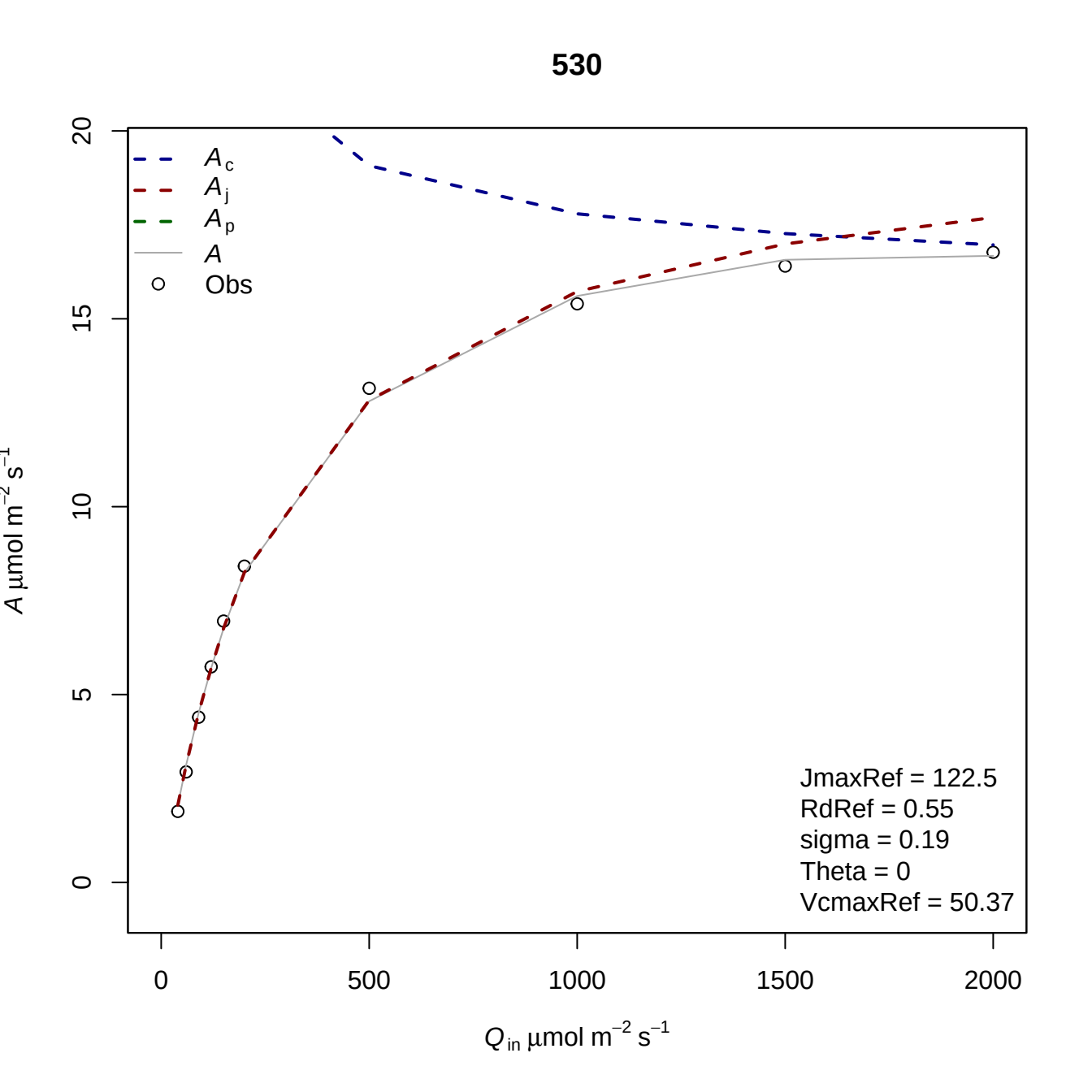

523
A_c
A_j
A_p
A
Obs
JmaxRef = 103.66
RdRef = 1.11
sigma = 0.23
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
100
200
300
400
500

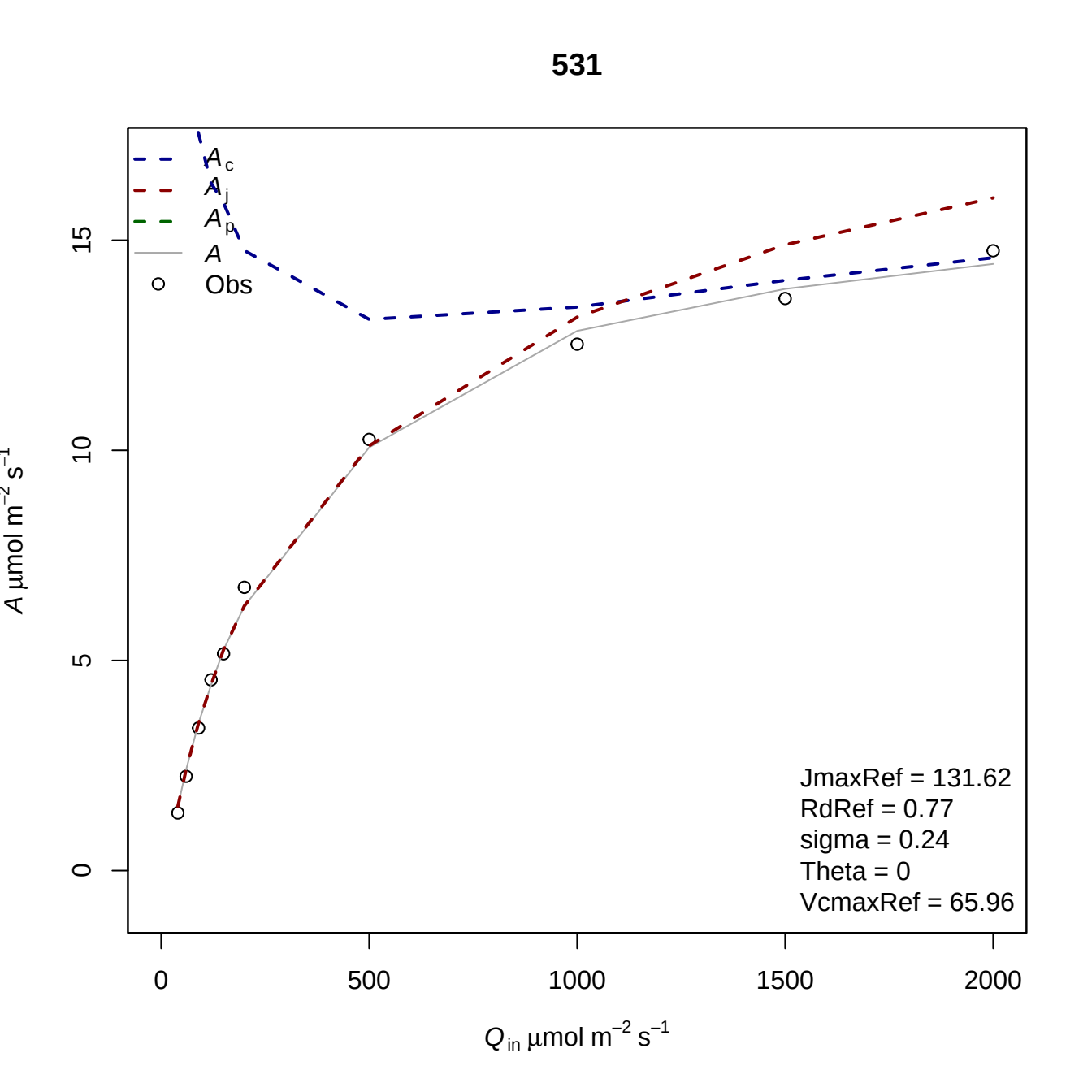

524
A_c
A_j
A_p
A
Obs
JmaxRef = 80.93
RdRef = 1.71
sigma = 0.23
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

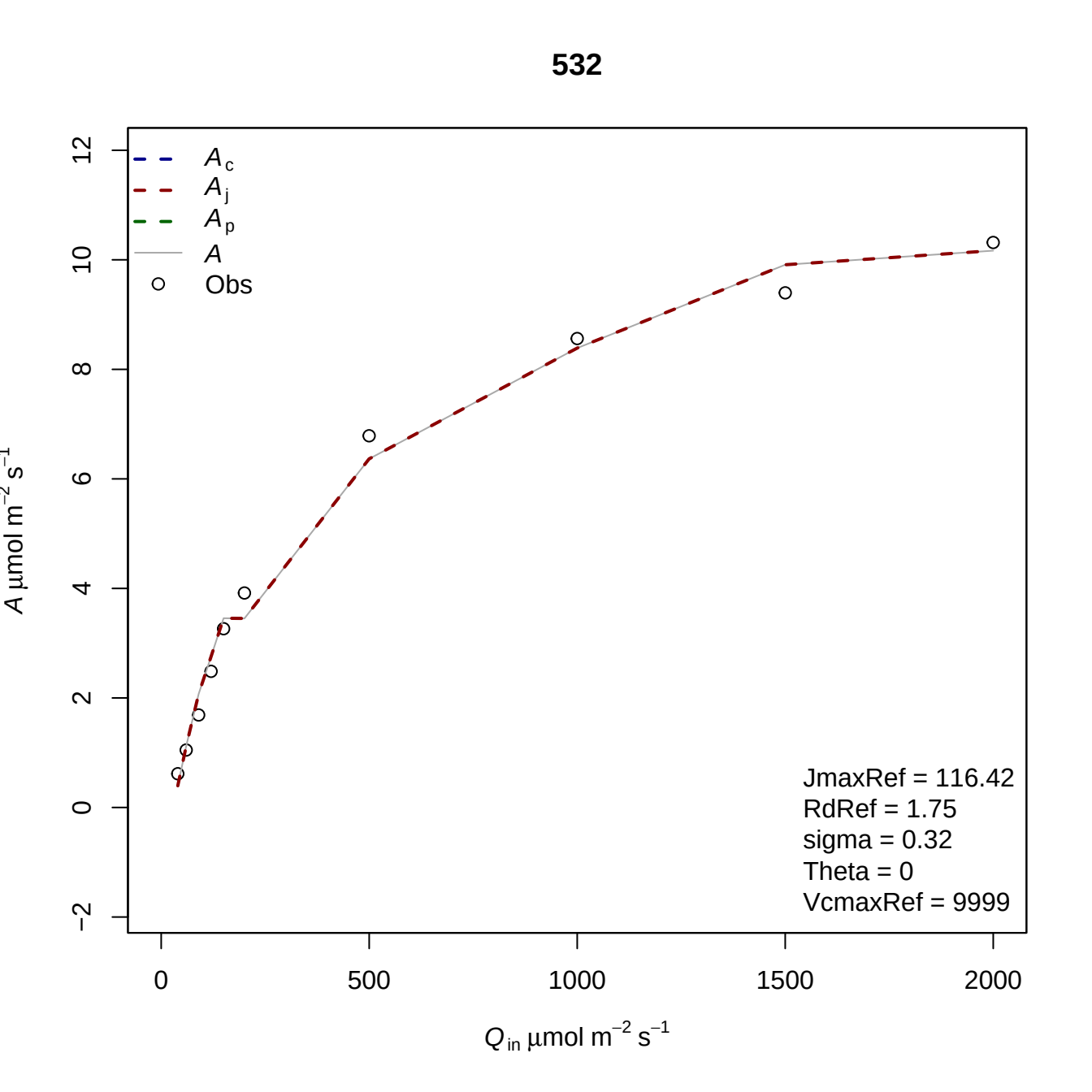

525
A_c
A_j
A_p
A
Obs
JmaxRef = 101.7
RdRef = 1.68
sigma = 0.26
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

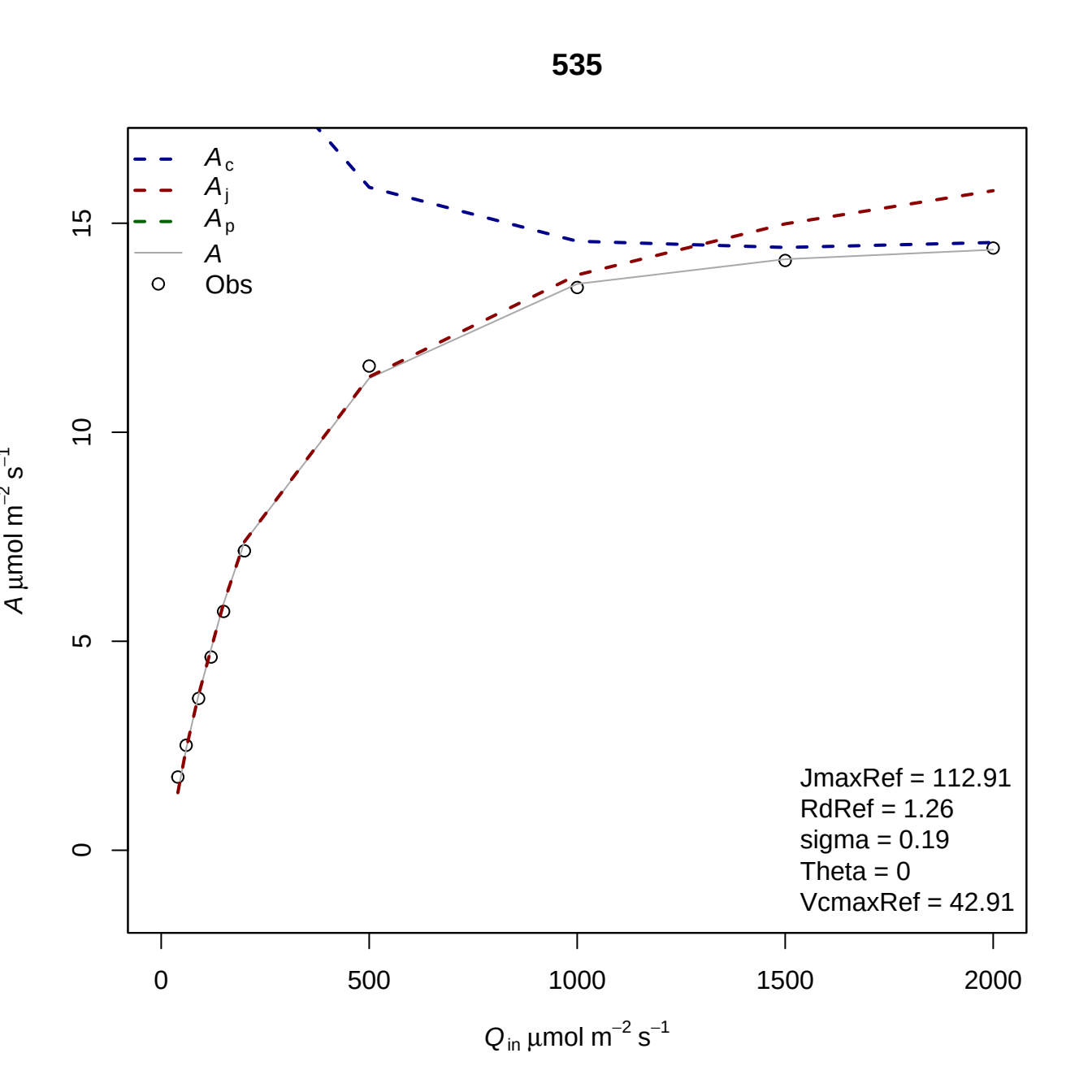

526
A_c
A_j
A_p
A
Obs
JmaxRef = 138.67
RdRef = 1.47
sigma = 0.24
Theta = 0
VcmaxRef = 84.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

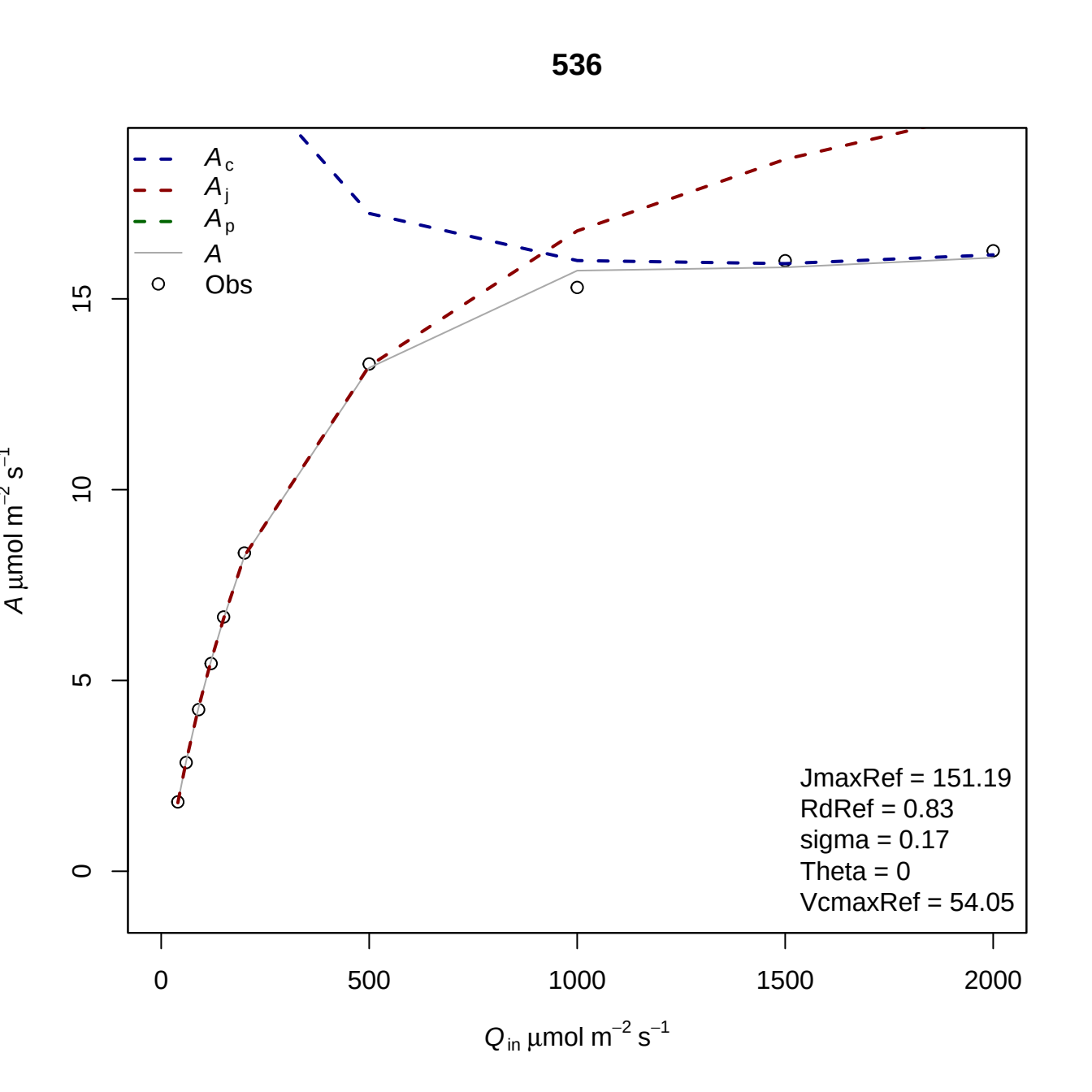

529
A_c
A_j
A_p
A
Obs
JmaxRef = 91.33
RdRef = 0.85
sigma = 0.14
Theta = 0
VcmaxRef = 37
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

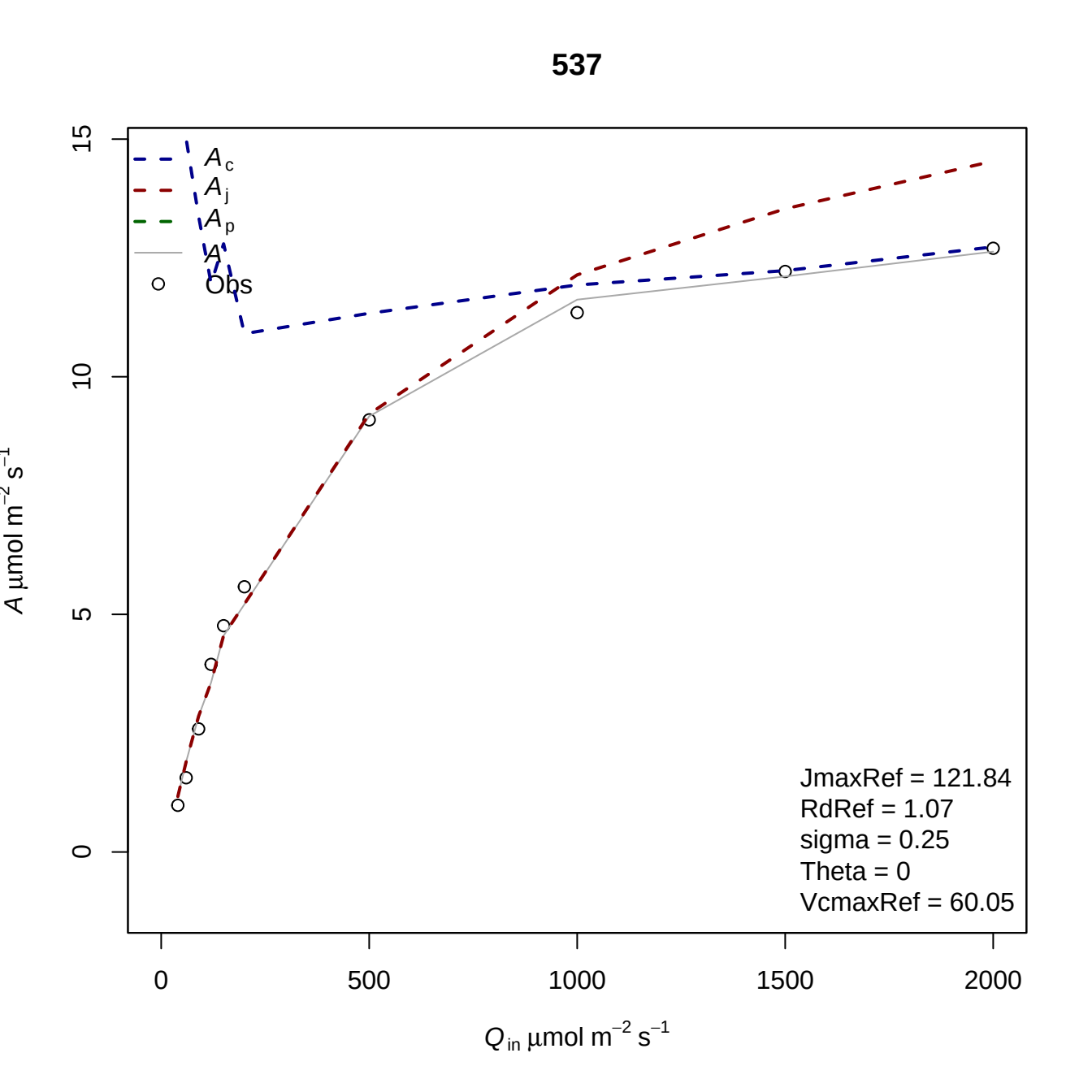

530
A_c
A_j
A_p
A
Obs
JmaxRef = 122.5
RdRef = 0.55
sigma = 0.19
Theta = 0
VcmaxRef = 50.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

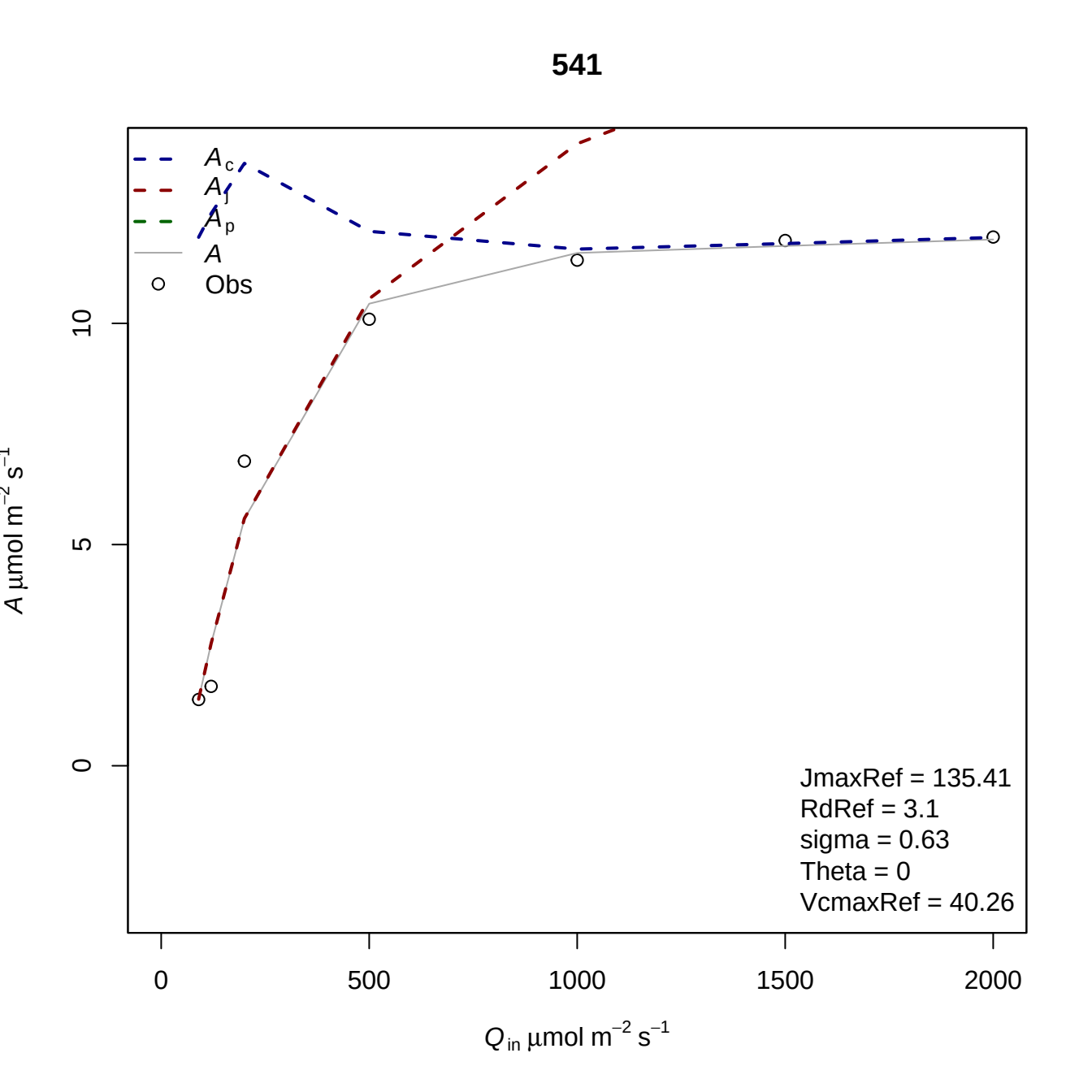

531
A_c
A_j
A_p
A
Obs
JmaxRef = 131.62
RdRef = 0.77
sigma = 0.24
Theta = 0
VcmaxRef = 65.96
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

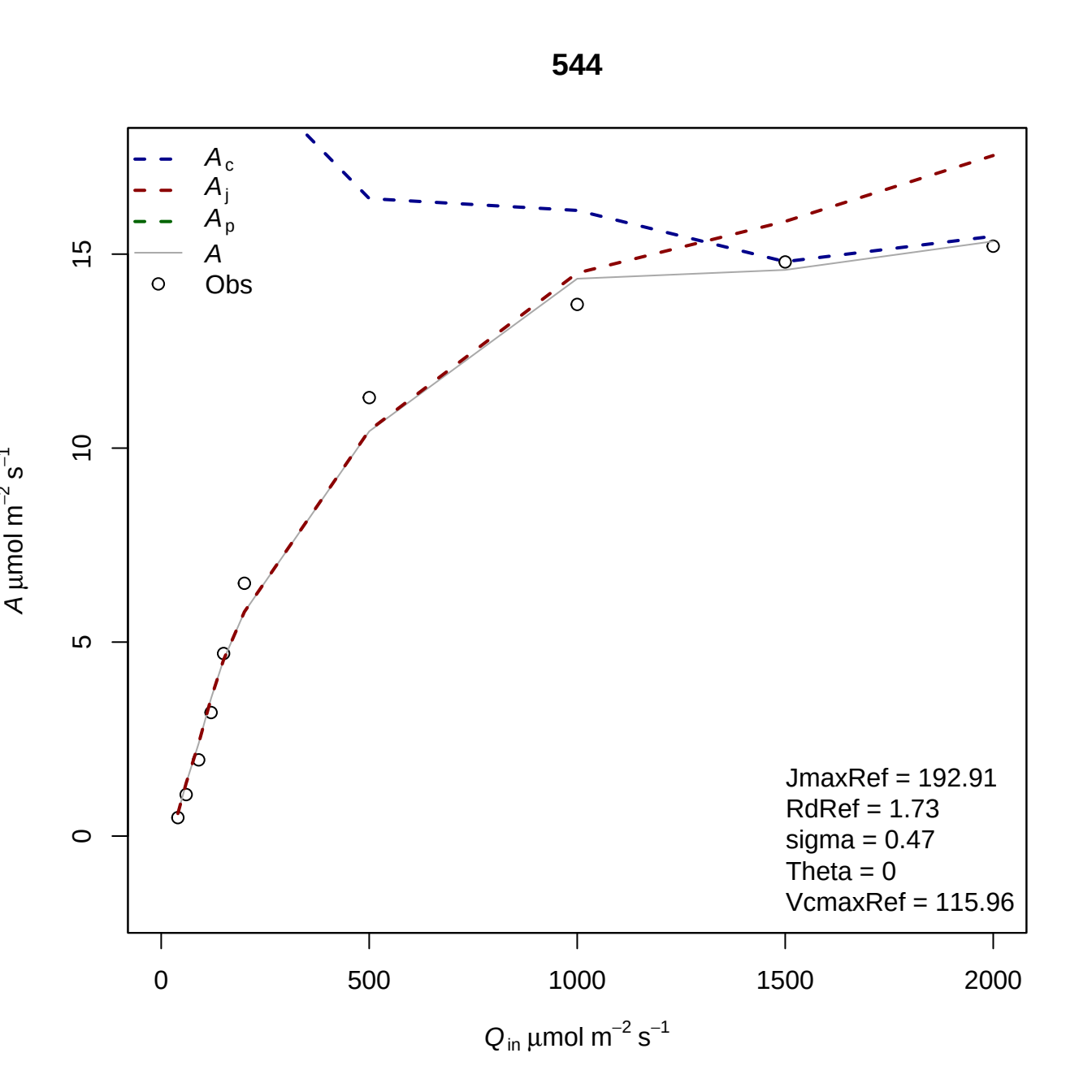

532
A_c
A_j
A_p
A
Obs
JmaxRef = 116.42
RdRef = 1.75
sigma = 0.32
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

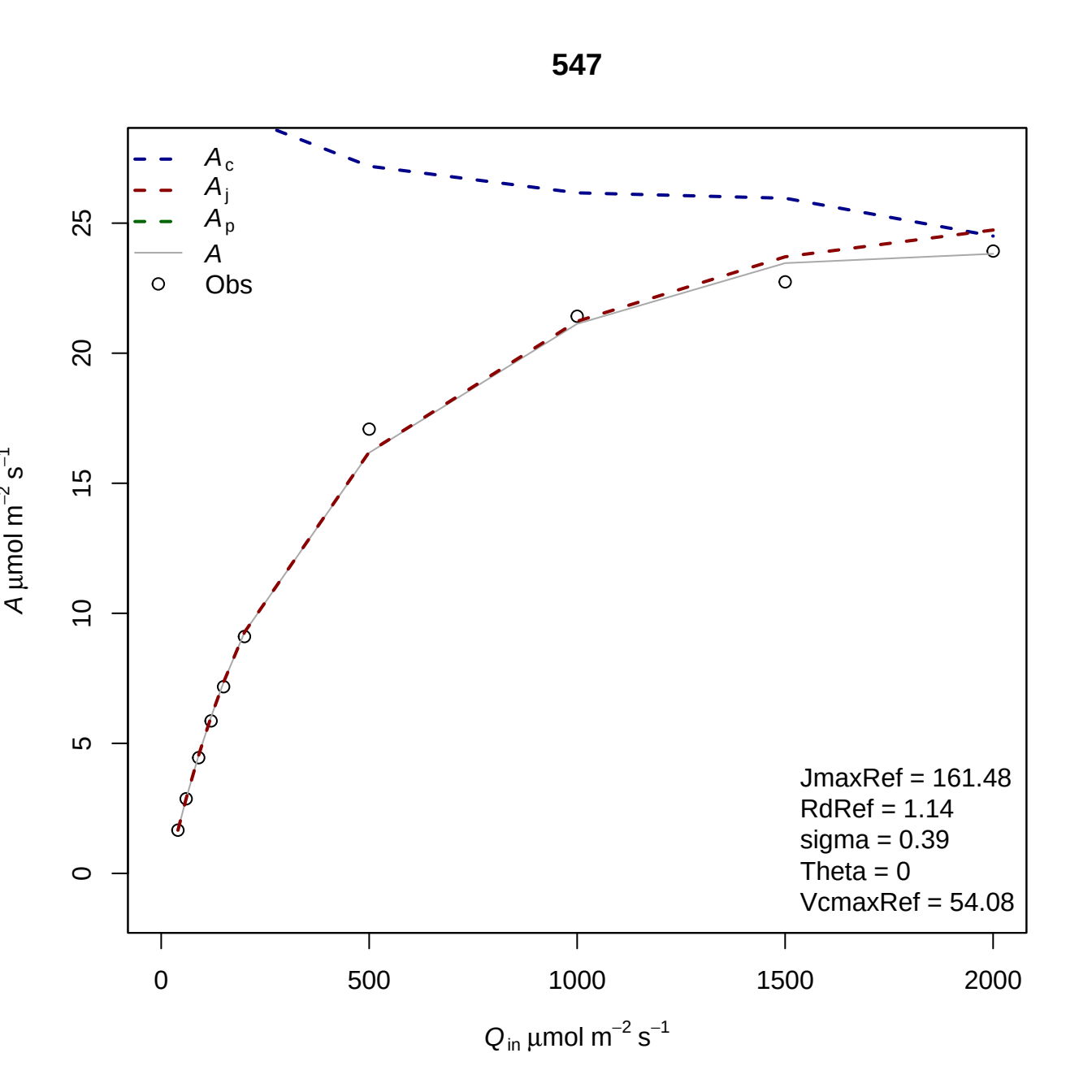

535
A_c
A_j
A_p
A
Obs
JmaxRef = 112.91
RdRef = 1.26
sigma = 0.19
Theta = 0
VcmaxRef = 42.91
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

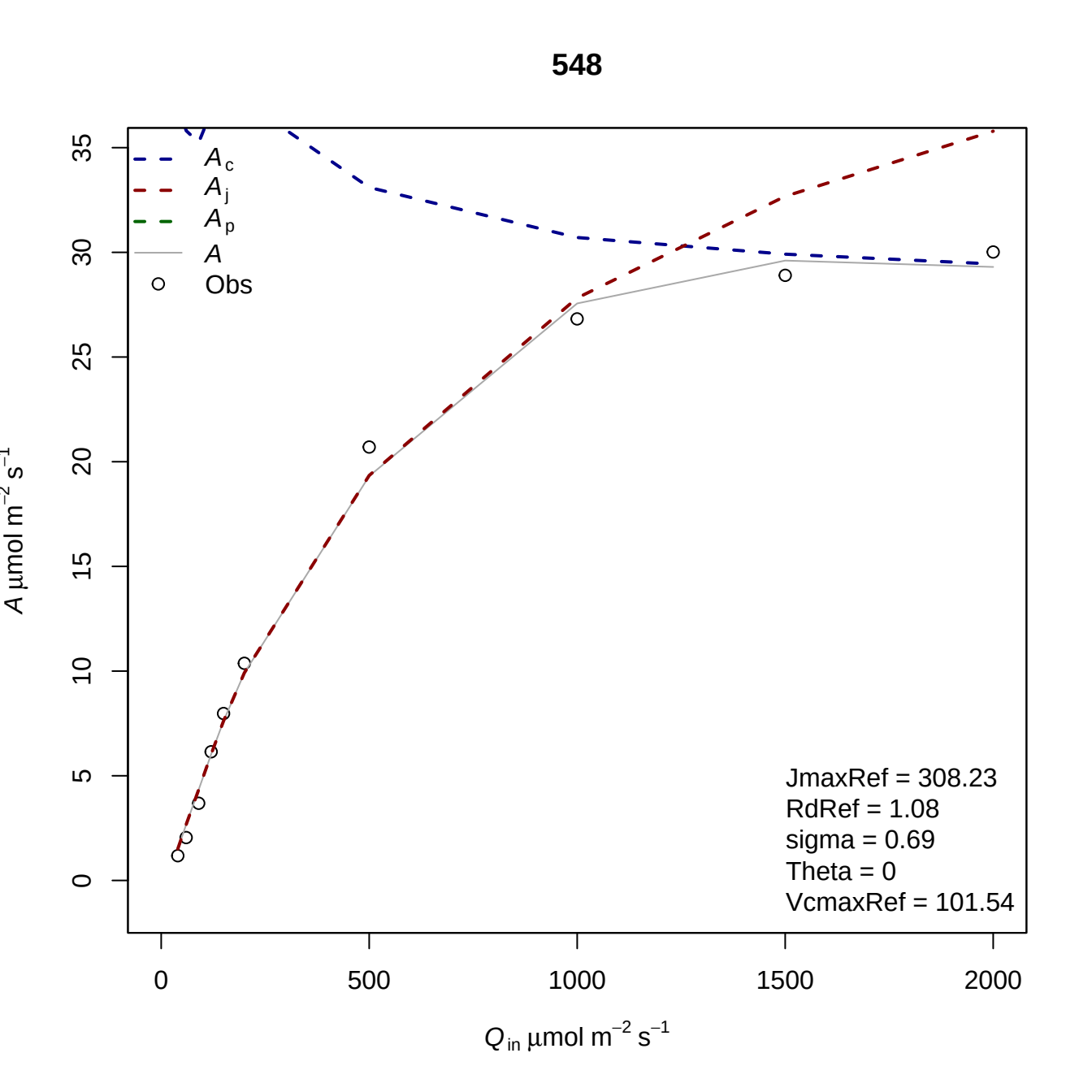

536
A_c
A_j
A_p
A
Obs
JmaxRef = 151.19
RdRef = 0.83
sigma = 0.17
Theta = 0
VcmaxRef = 54.05
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

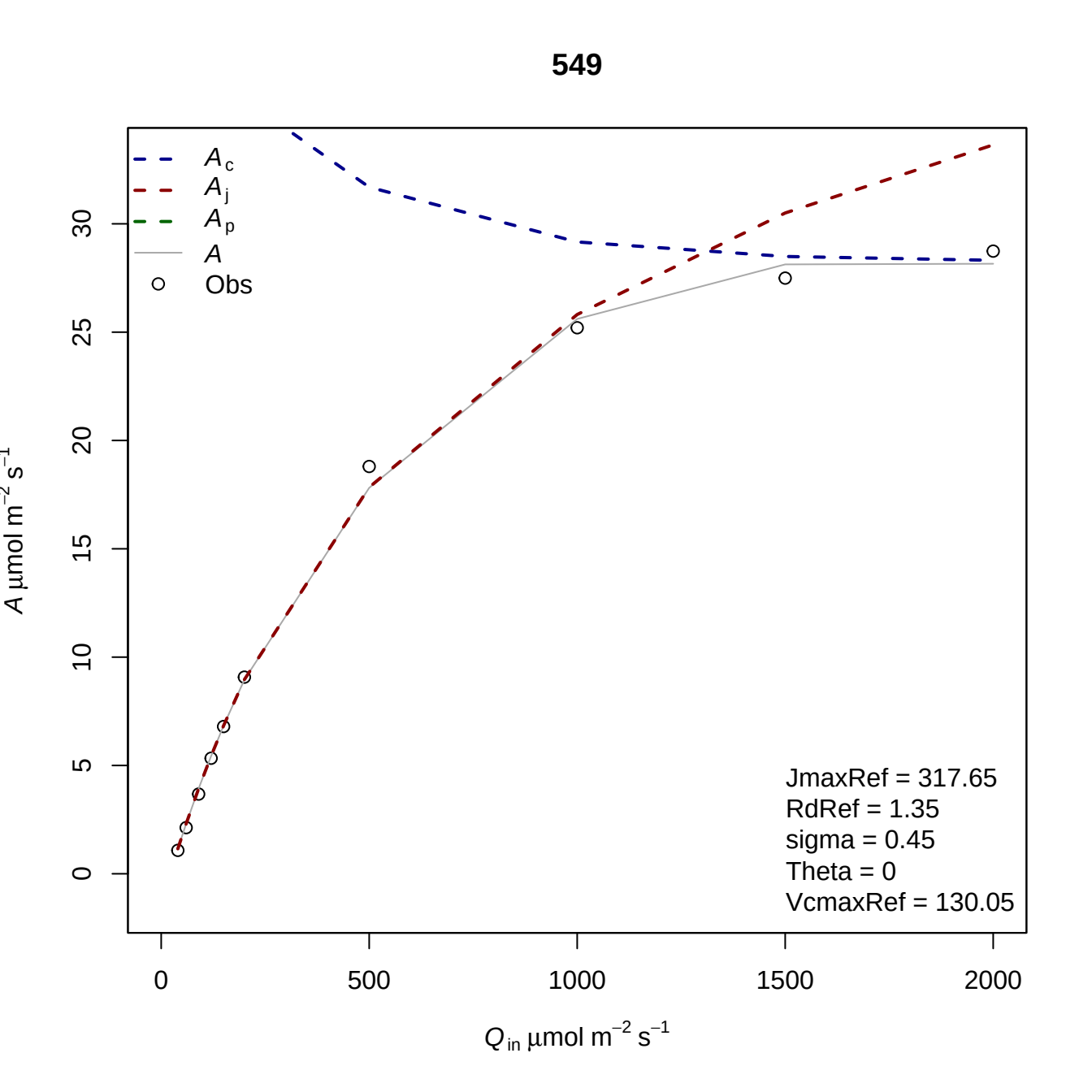

537
A_c
A_j
A_p
A
Obs
JmaxRef = 121.84
RdRef = 1.07
sigma = 0.25
Theta = 0
VcmaxRef = 60.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

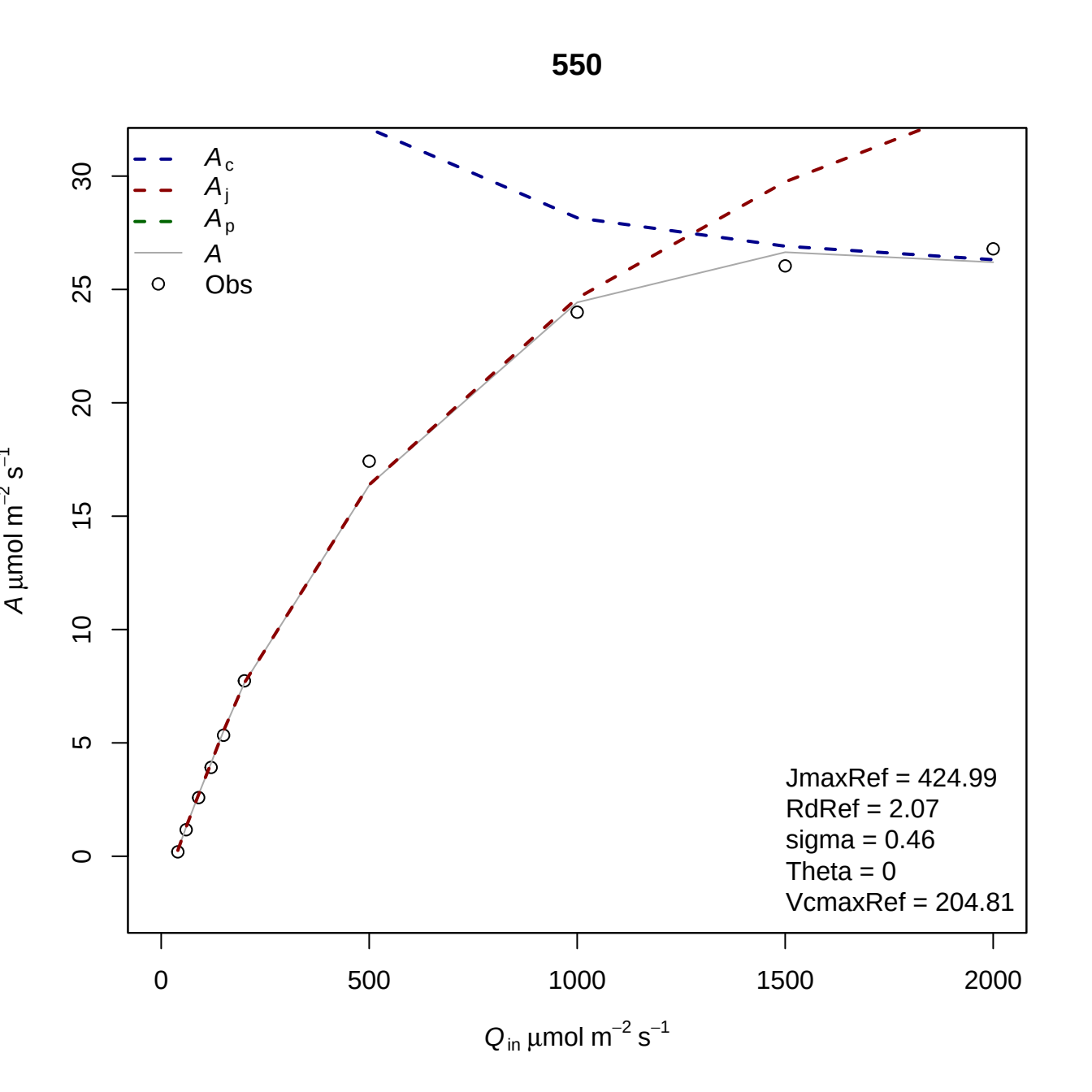

541
A_c
A_j
A_p
A
Obs
JmaxRef = 135.41
RdRef = 3.1
sigma = 0.63
Theta = 0
VcmaxRef = 40.26
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

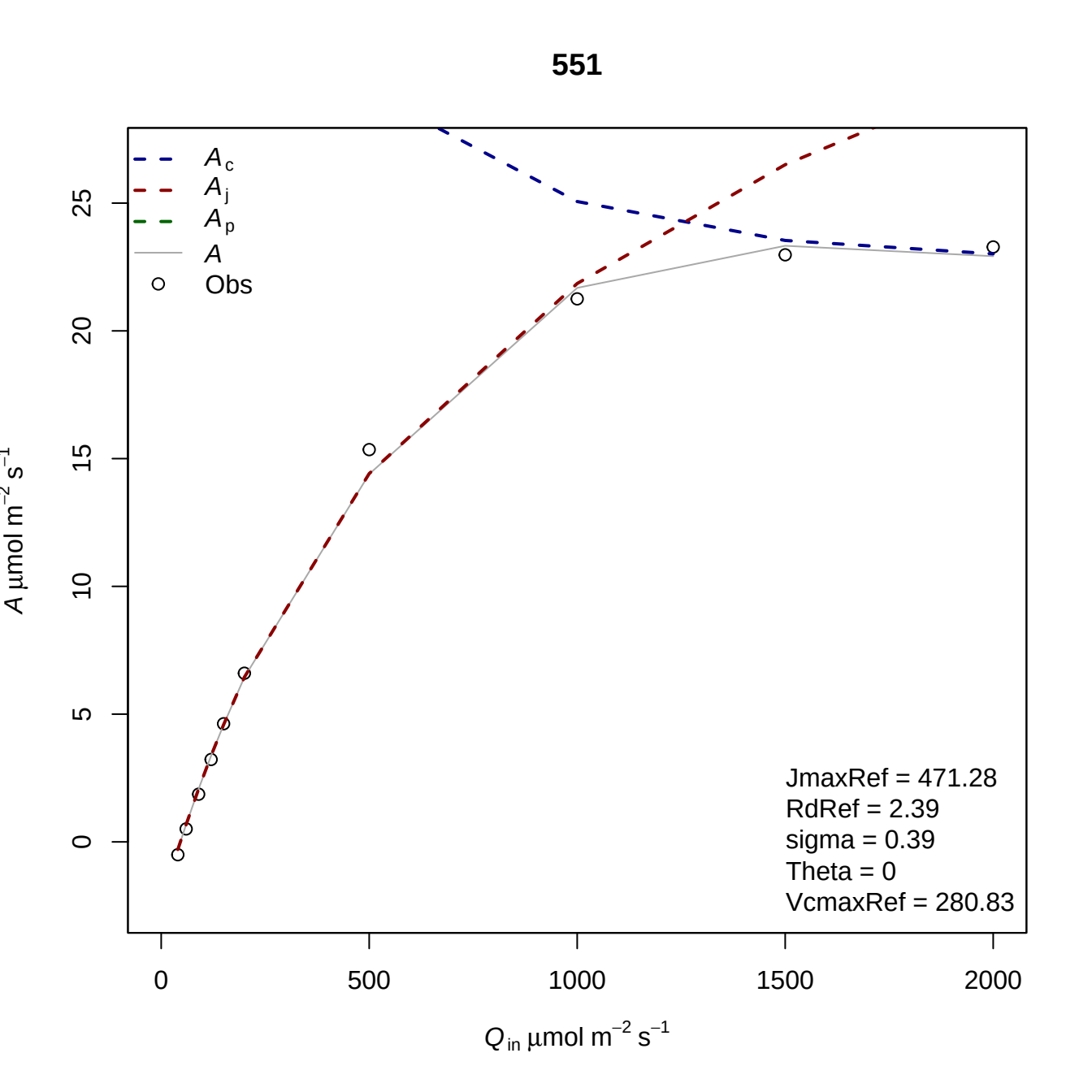

544
A_c
A_j
A_p
A
Obs
JmaxRef = 192.91
RdRef = 1.73
sigma = 0.47
Theta = 0
VcmaxRef = 115.96
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

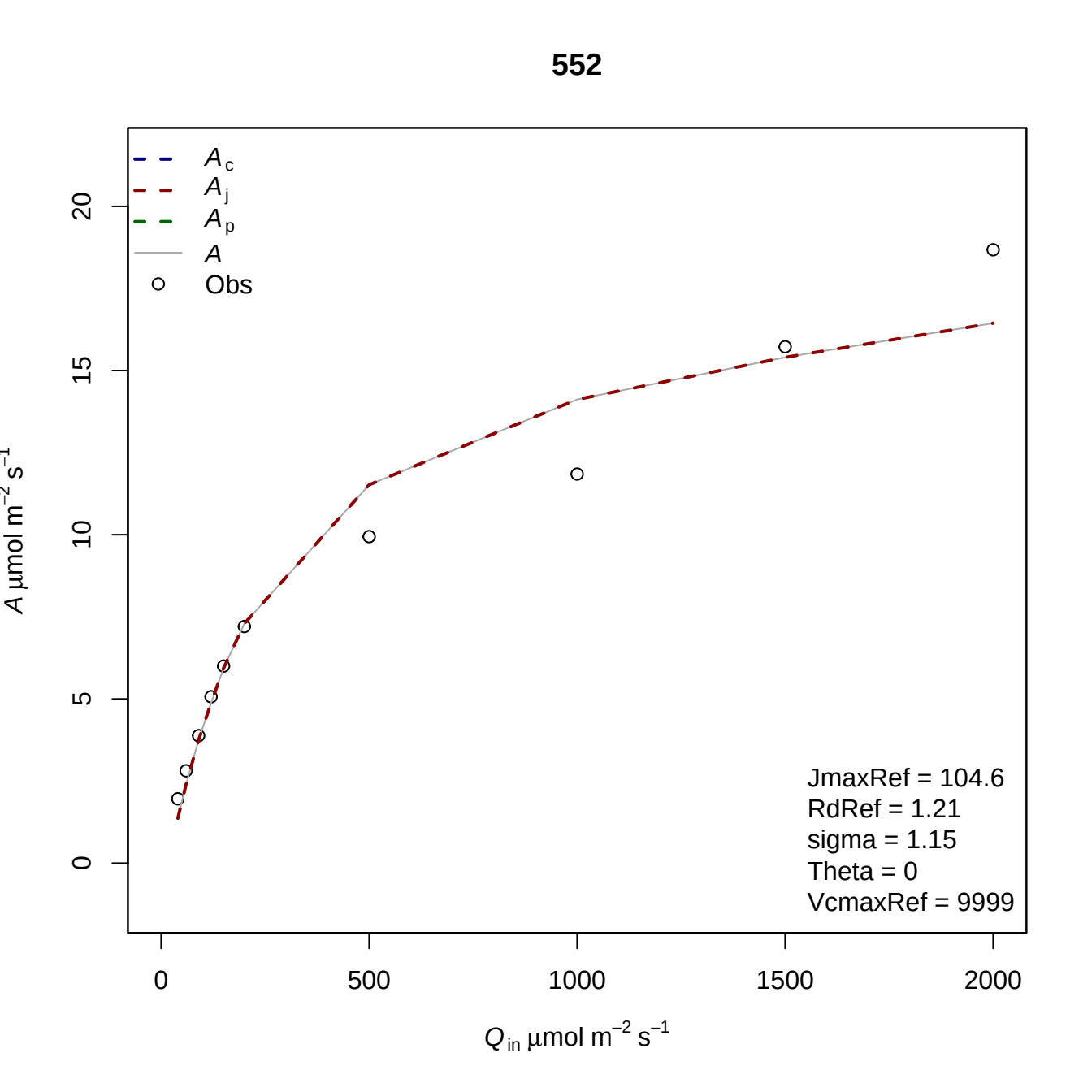

547
A_c
A_j
A_p
A
Obs
JmaxRef = 161.48
RdRef = 1.14
sigma = 0.39
Theta = 0
VcmaxRef = 54.08
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

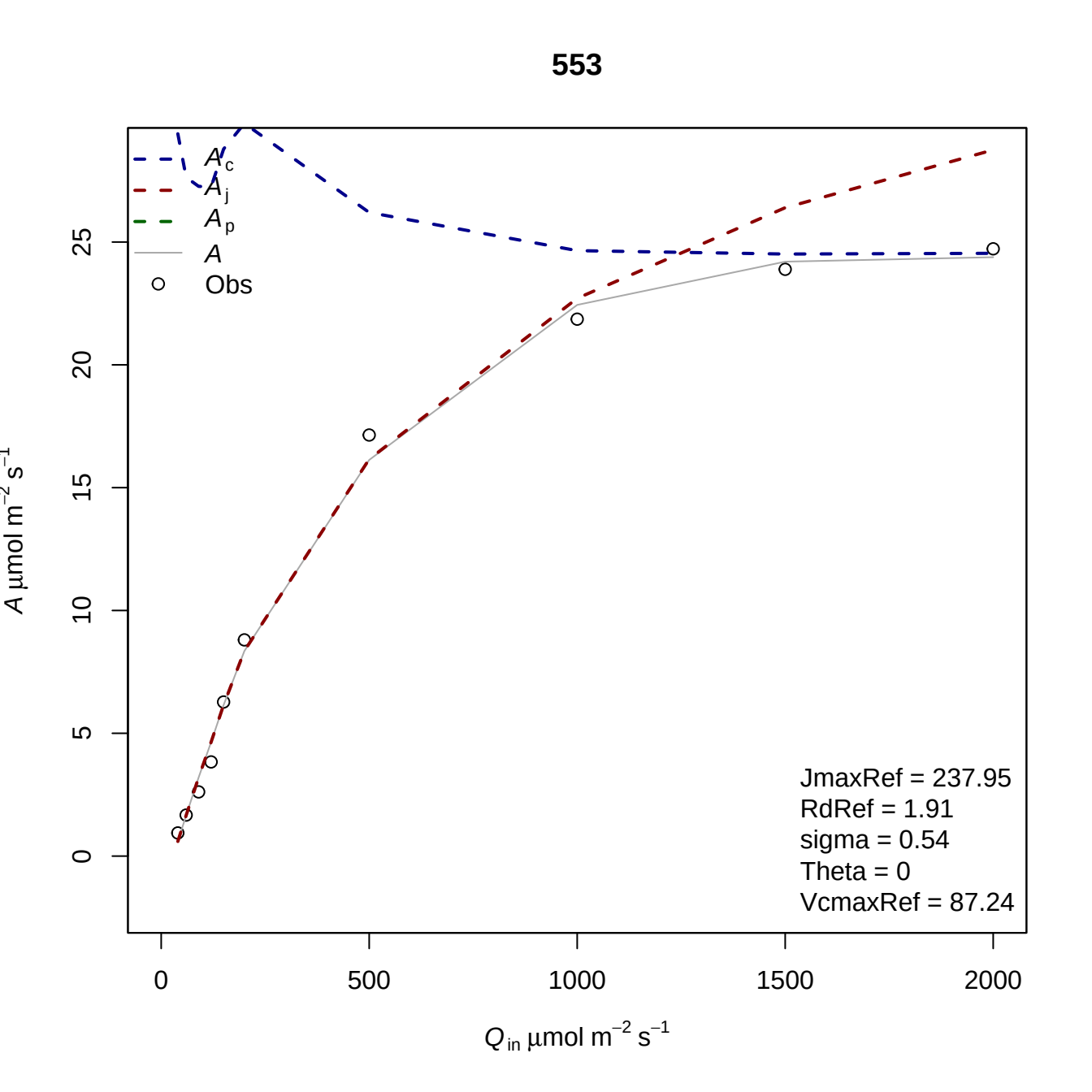

548
A_c
A_j
A_p
A
Obs
JmaxRef = 308.23
RdRef = 1.08
sigma = 0.69
Theta = 0
VcmaxRef = 101.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

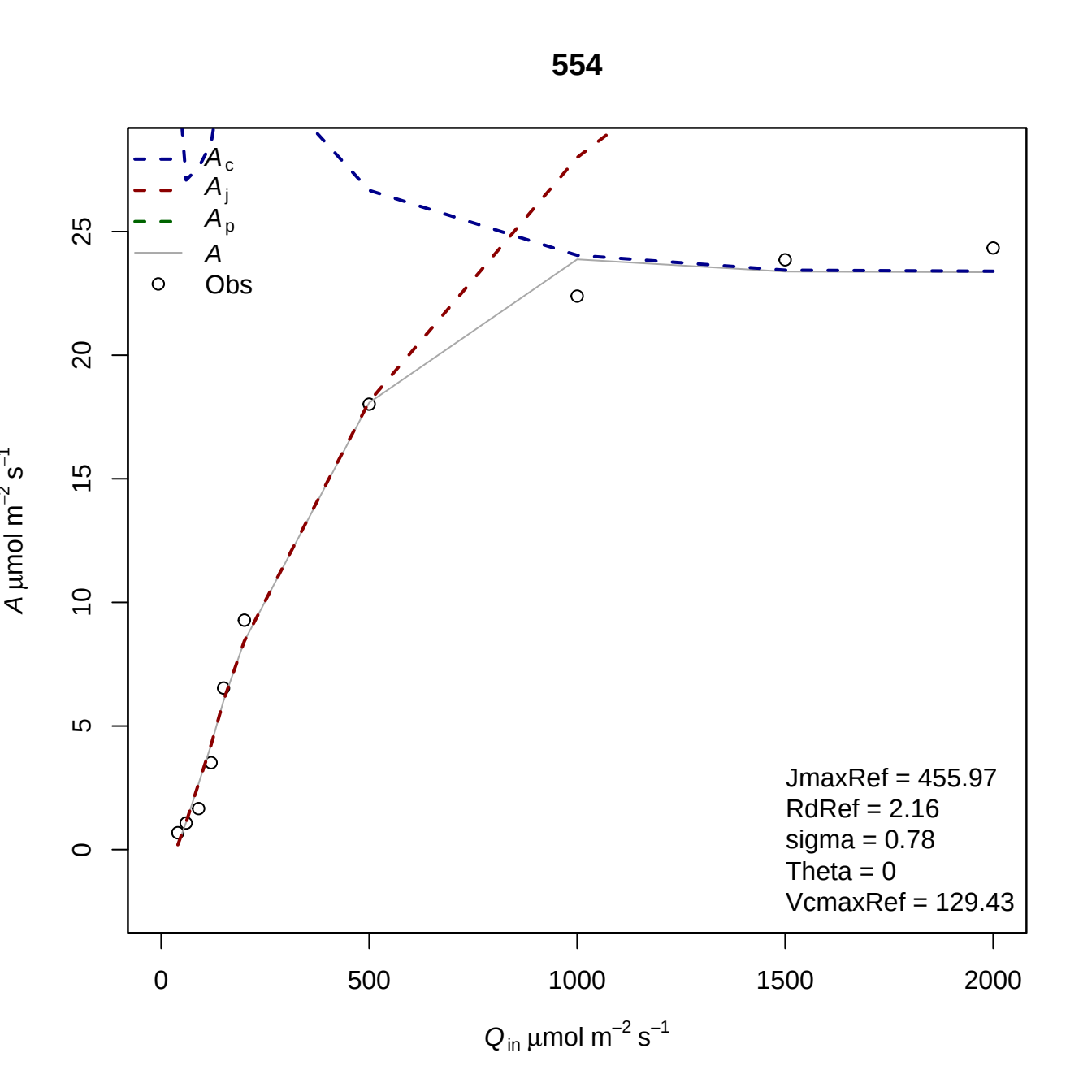

549
A_c
A_j
A_p
A
Obs
JmaxRef = 317.65
RdRef = 1.35
sigma = 0.45
Theta = 0
VcmaxRef = 130.05
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

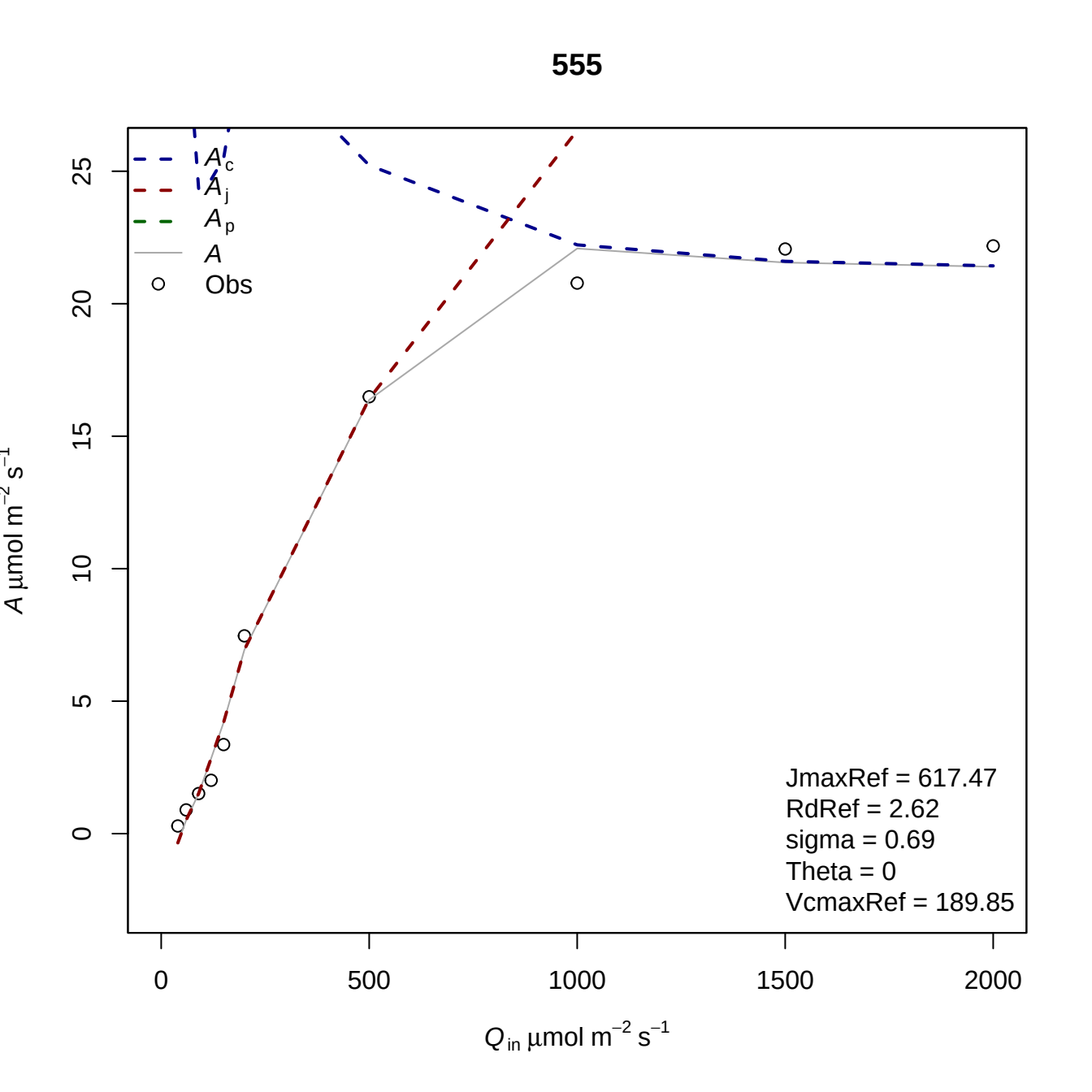

550
A_c
A_j
A_p
A
Obs
JmaxRef = 424.99
RdRef = 2.07
sigma = 0.46
Theta = 0
VcmaxRef = 204.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

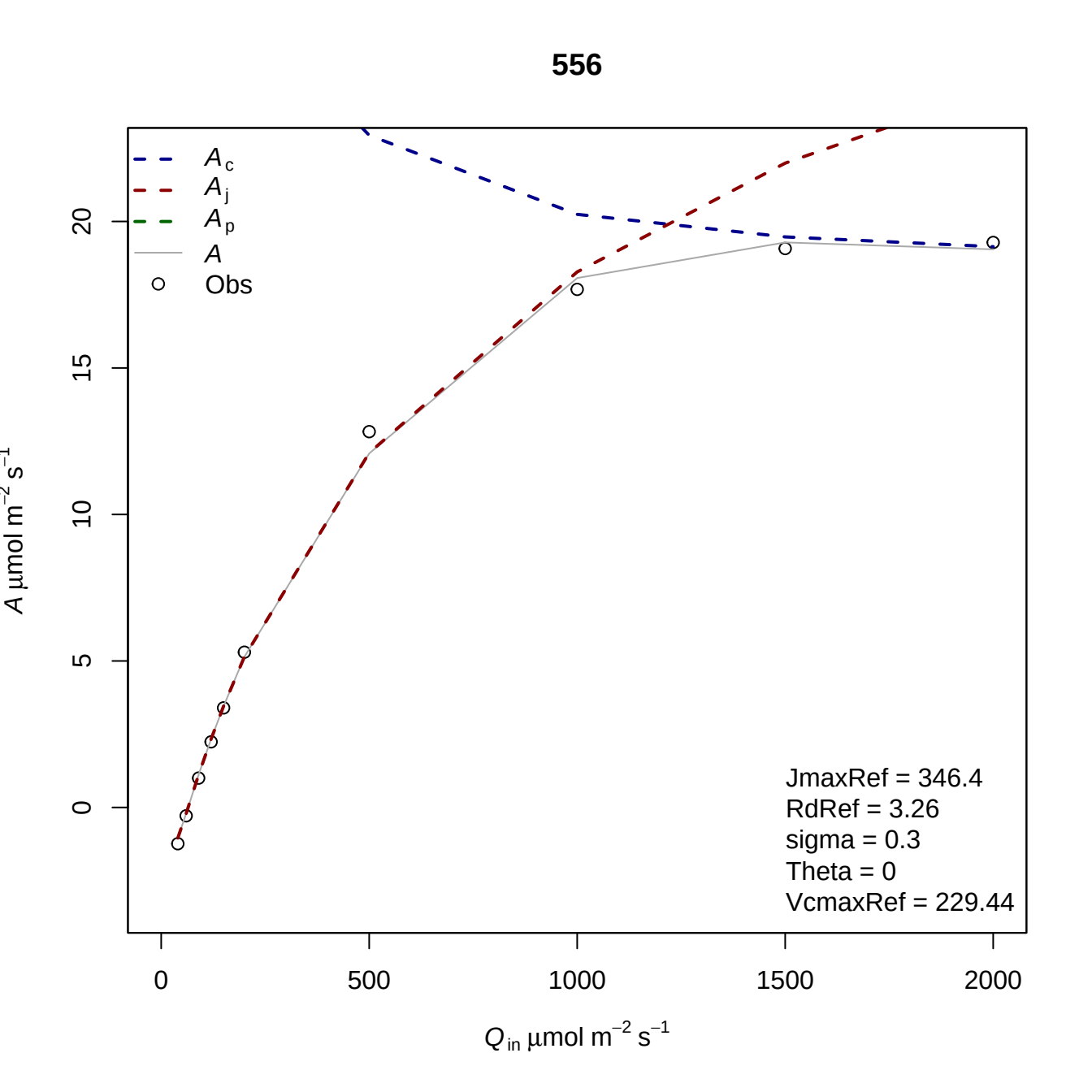

551
A_c
A_j
A_p
A
Obs
JmaxRef = 471.28
RdRef = 2.39
sigma = 0.39
Theta = 0
VcmaxRef = 280.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

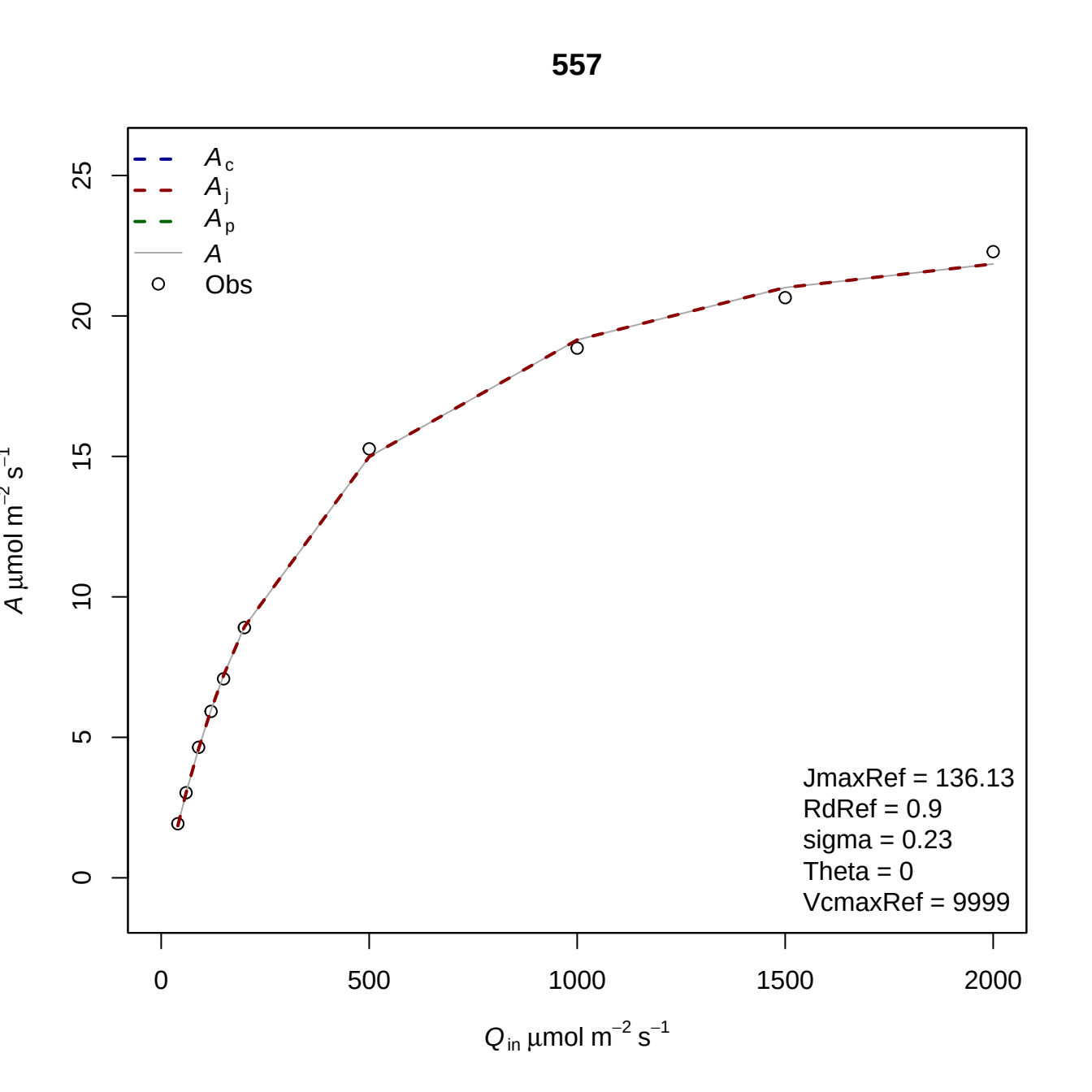

552
A_c
A_j
A_p
A
Obs
JmaxRef = 104.6
RdRef = 1.21
sigma = 1.15
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

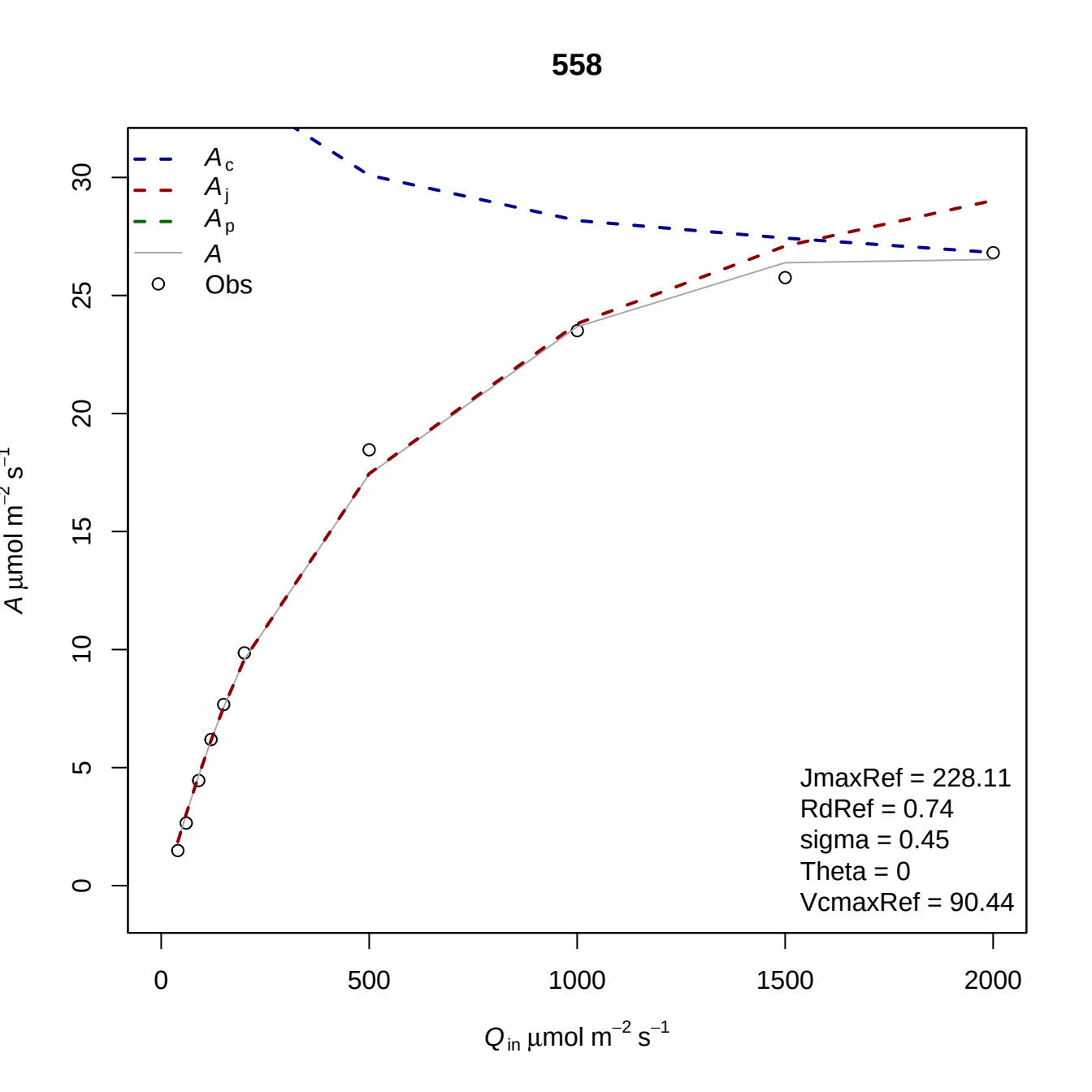

553
A_c
A_j
A_p
A
Obs
JmaxRef = 237.95
RdRef = 1.91
sigma = 0.54
Theta = 0
VcmaxRef = 87.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

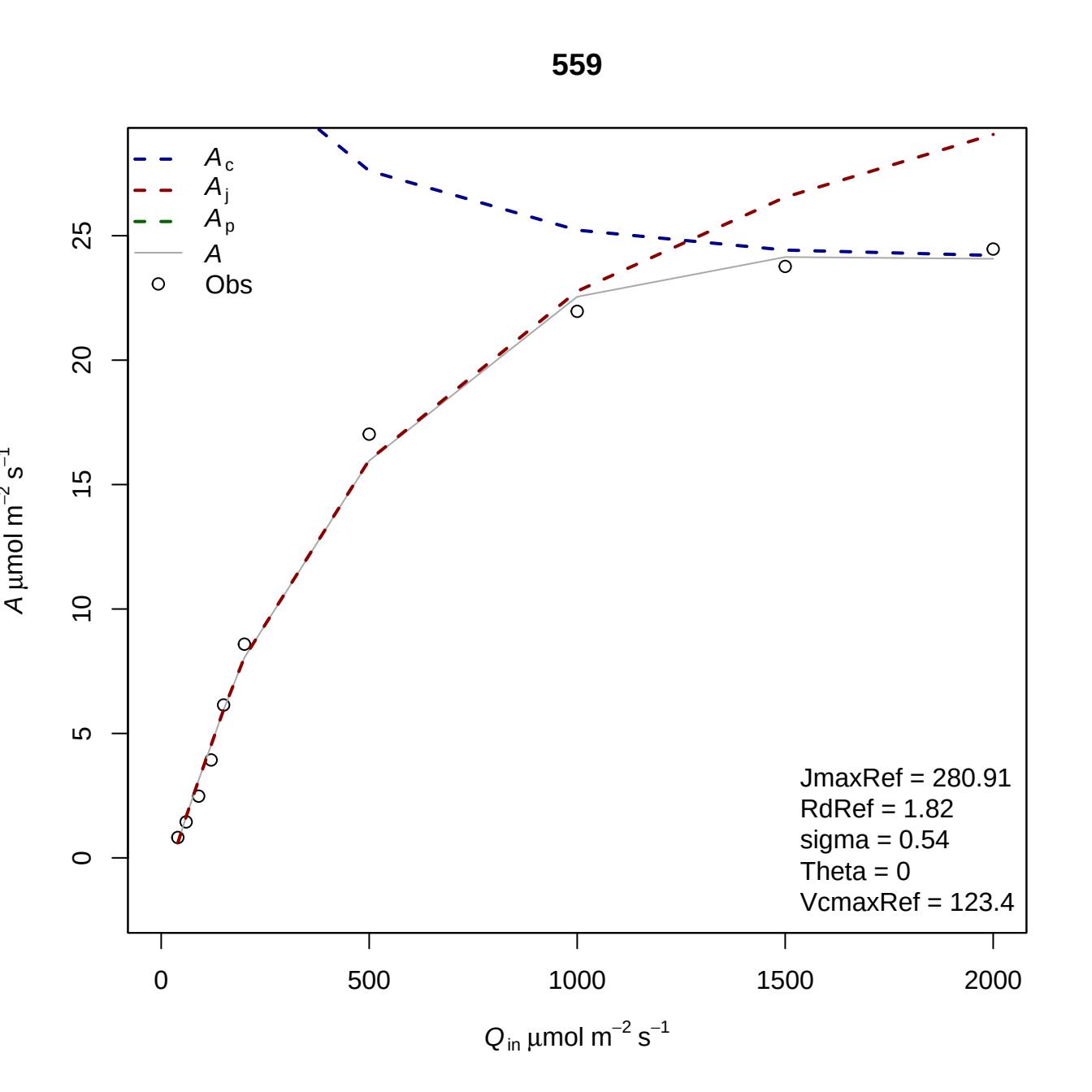

554
A_c
A_j
A_p
A
Obs
JmaxRef = 455.97
RdRef = 2.16
sigma = 0.78
Theta = 0
VcmaxRef = 129.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

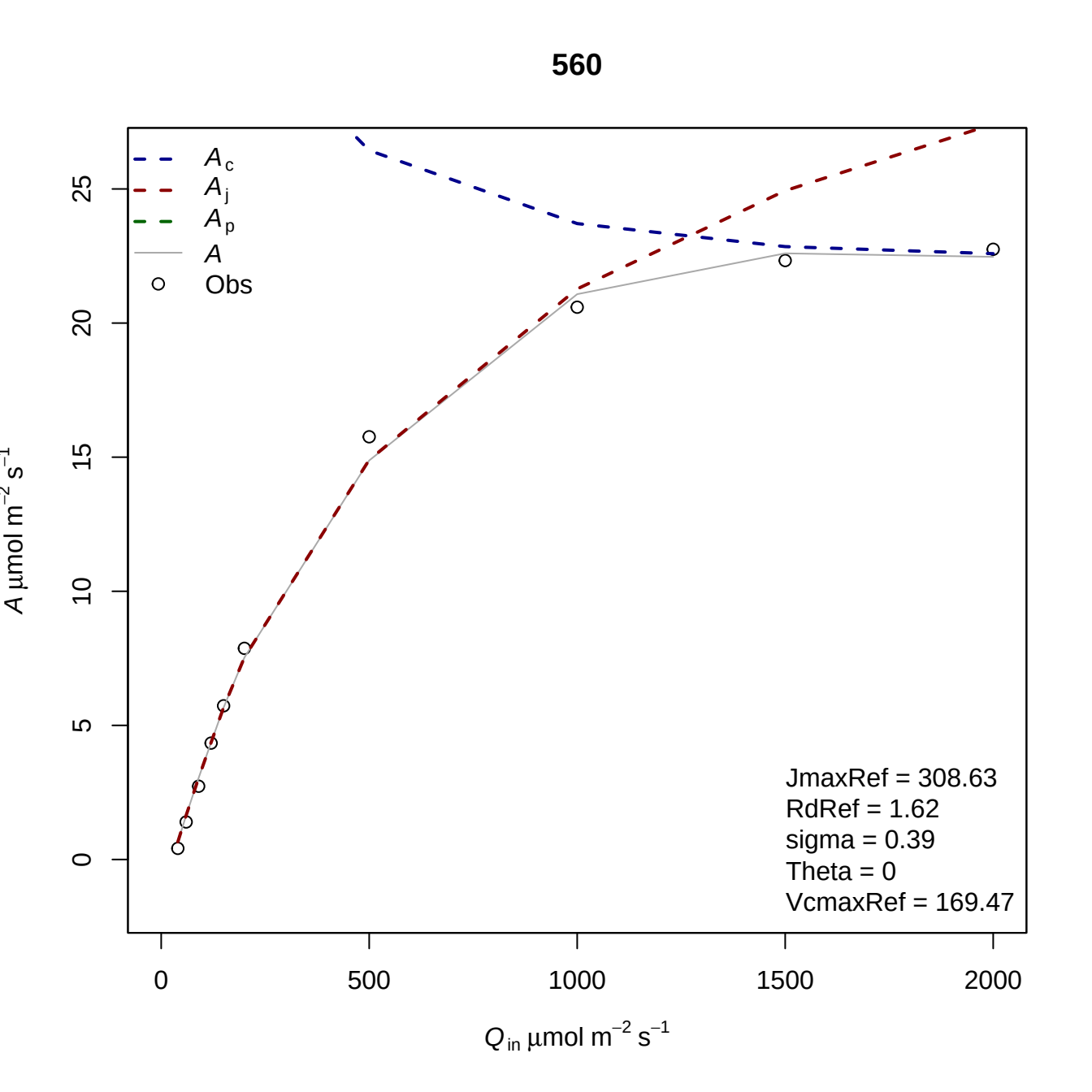

555
A_c
A_j
A_p
A
Obs
JmaxRef = 617.47
RdRef = 2.62
sigma = 0.69
Theta = 0
VcmaxRef = 189.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

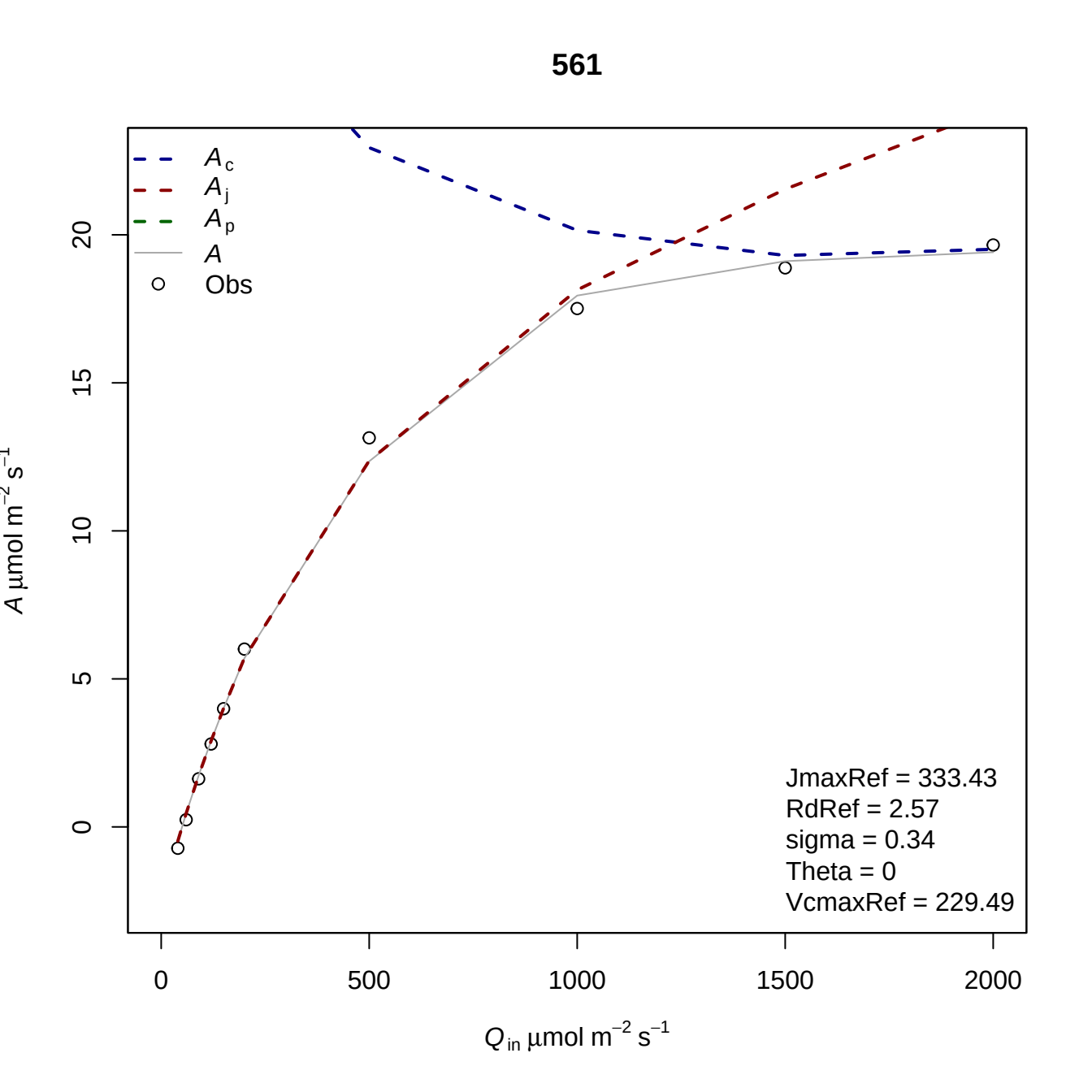

556
A_c
A_j
A_p
A
Obs
JmaxRef = 346.4
RdRef = 3.26
sigma = 0.3
Theta = 0
VcmaxRef = 229.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

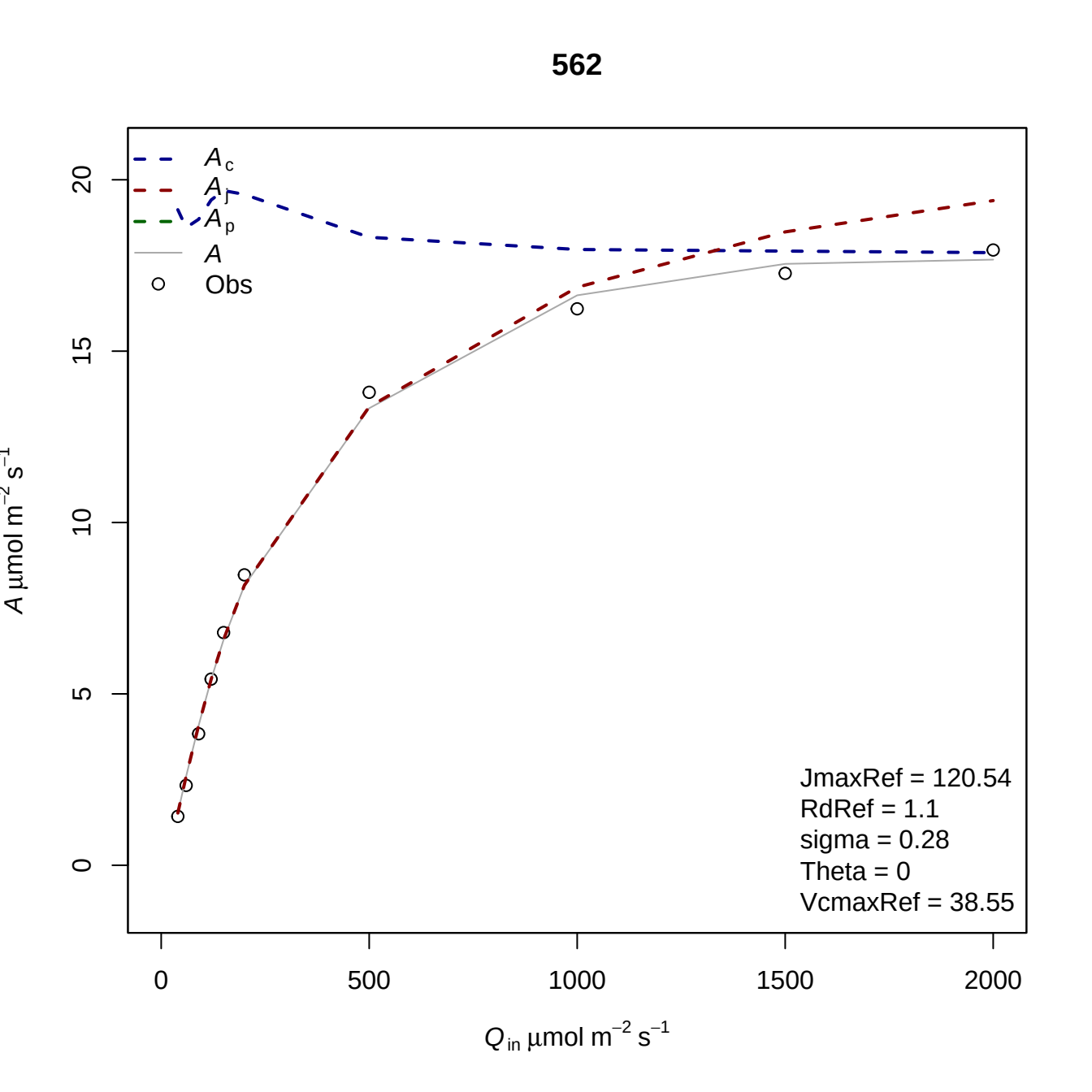

557
A_c
A_j
A_p
A
Obs
JmaxRef = 136.13
RdRef = 0.9
sigma = 0.23
Theta = 0
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

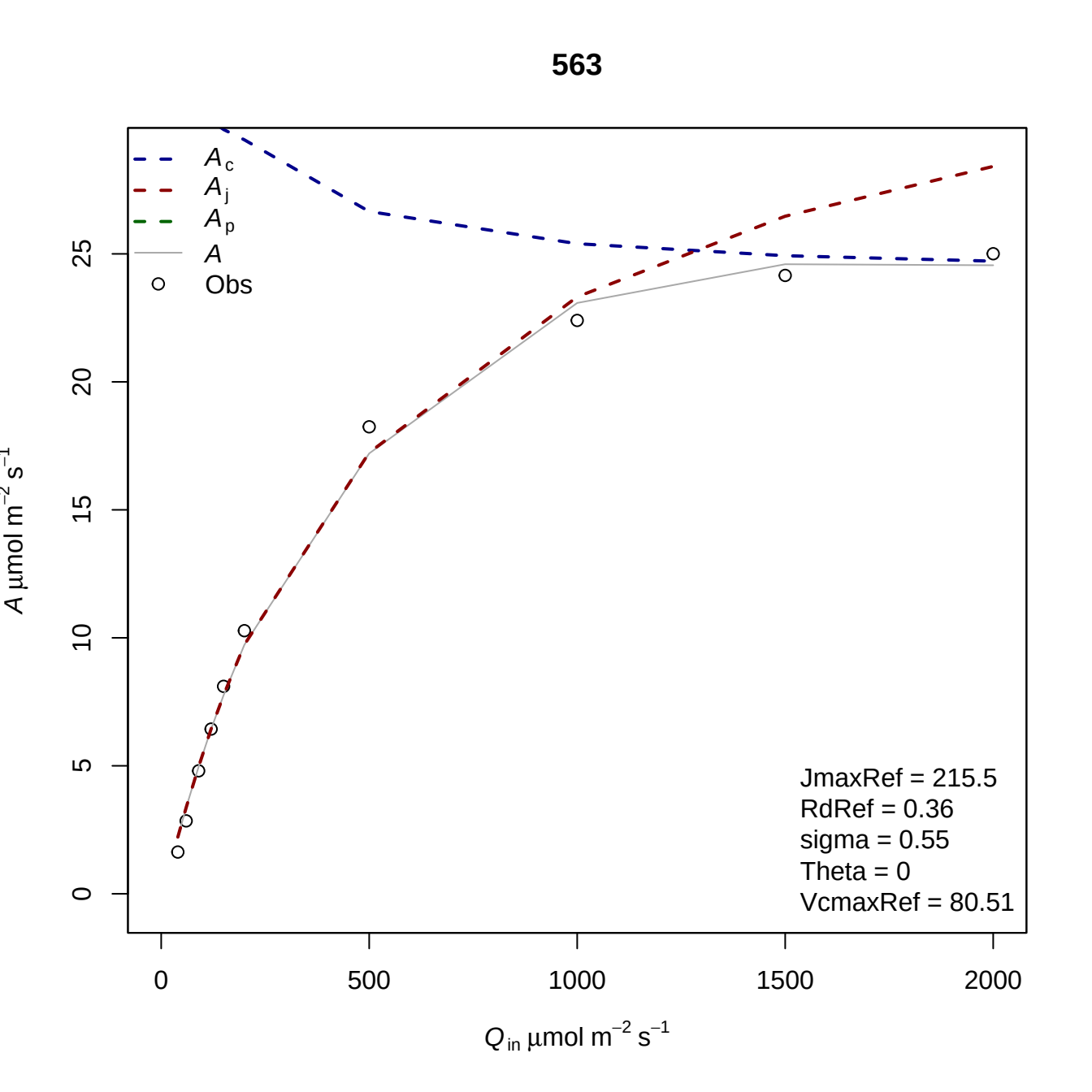

558
A_c
A_j
A_p
A
Obs
JmaxRef = 228.11
RdRef = 0.74
sigma = 0.45
Theta = 0
VcmaxRef = 90.44
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

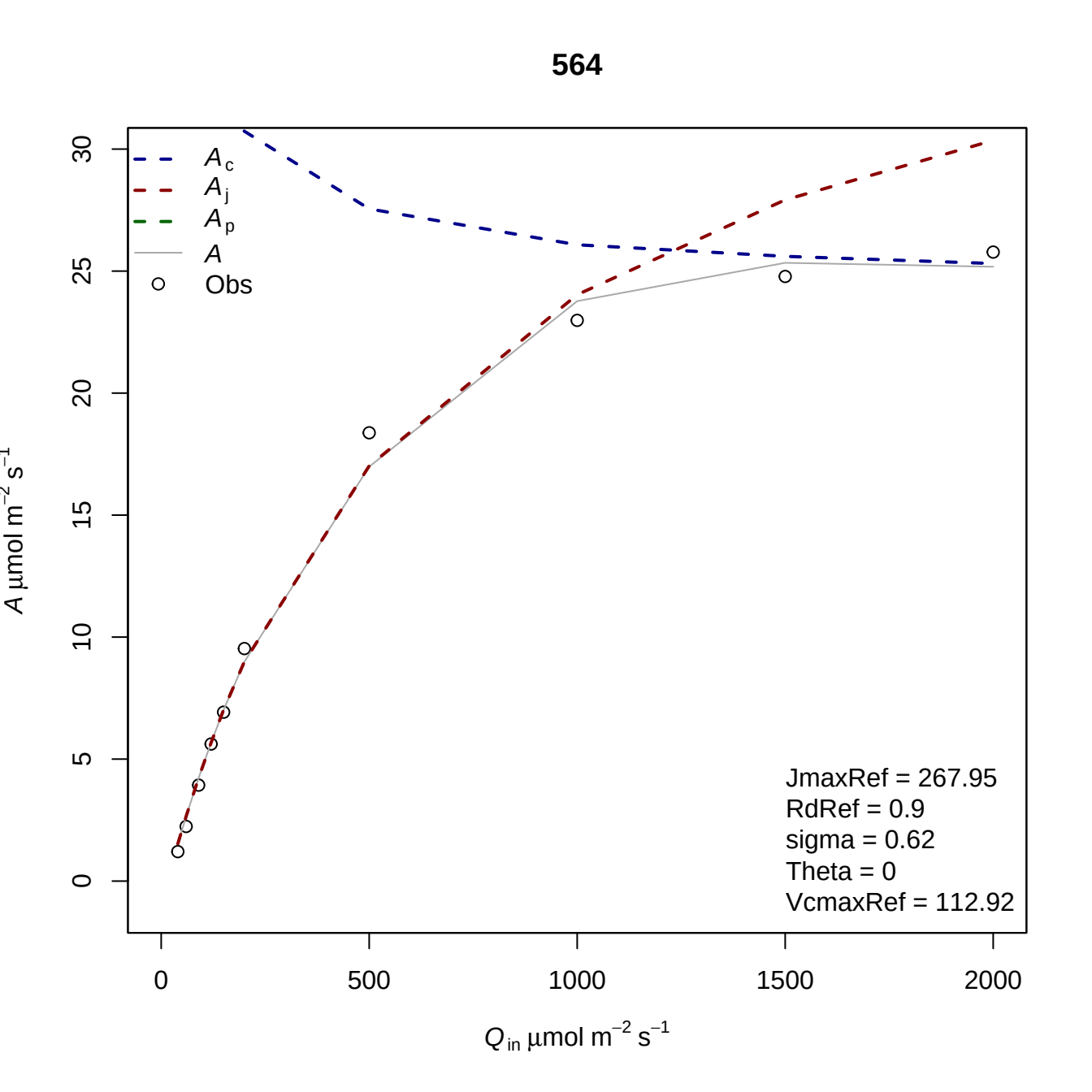

559
A_c
A_j
A_p
A
Obs
JmaxRef = 280.91
RdRef = 1.82
sigma = 0.54
Theta = 0
VcmaxRef = 123.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

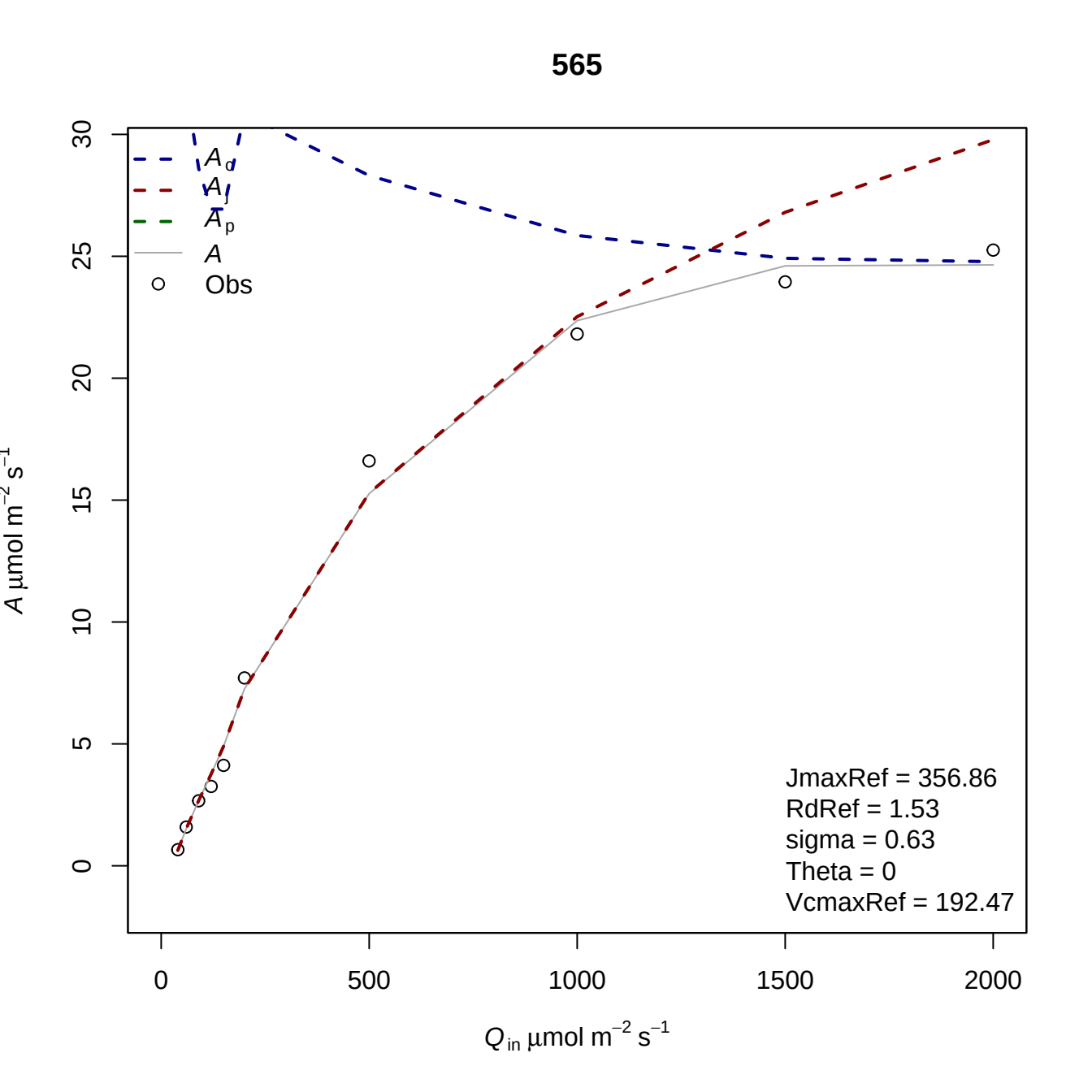

560
A_c
A_j
A_p
A
Obs
JmaxRef = 308.63
RdRef = 1.62
sigma = 0.39
Theta = 0
VcmaxRef = 169.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

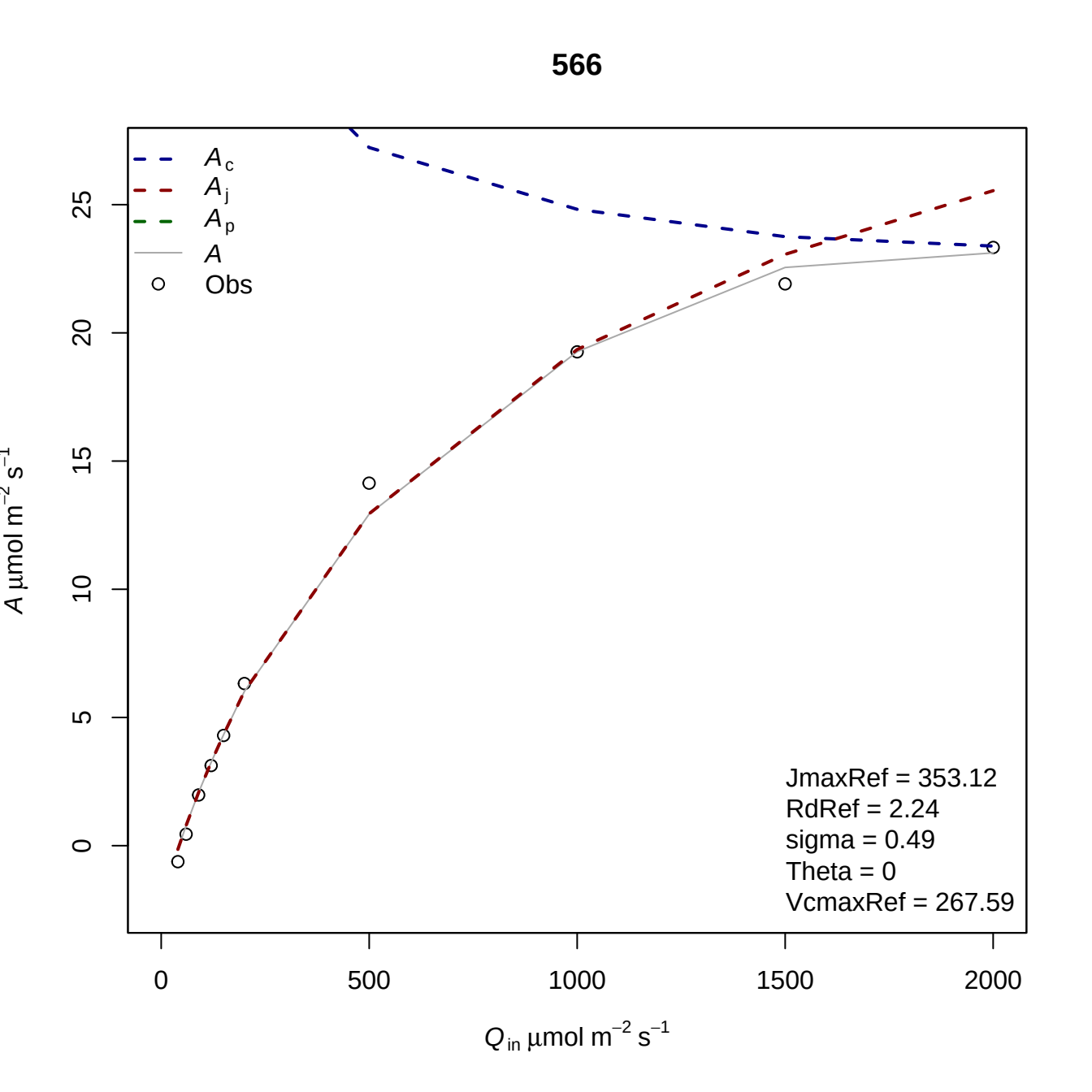

561
A_c
A_j
A_p
A
Obs
JmaxRef = 333.43
RdRef = 2.57
sigma = 0.34
Theta = 0
VcmaxRef = 229.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

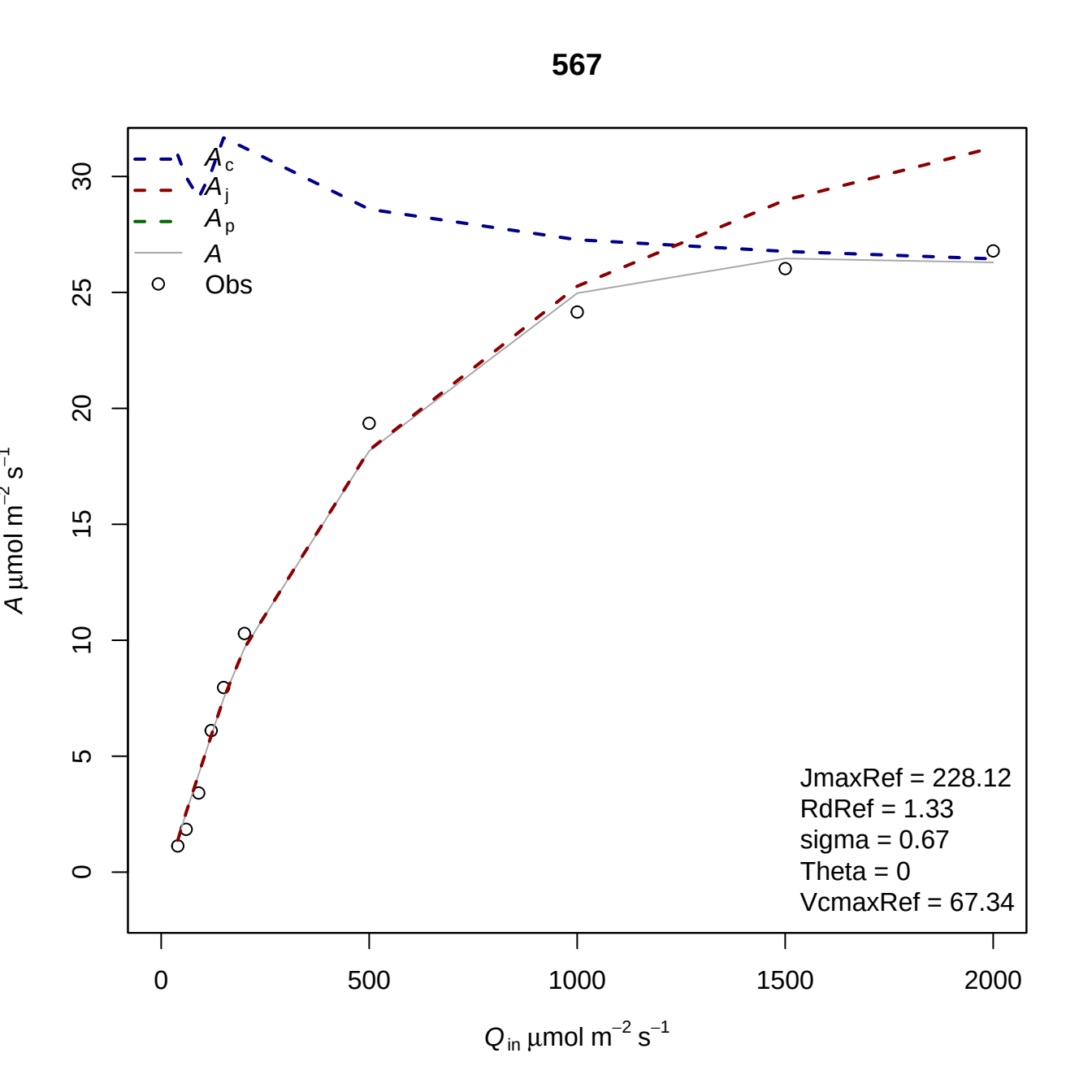

562
A_c
A_j
A_p
A
Obs
JmaxRef = 120.54
RdRef = 1.1
sigma = 0.28
Theta = 0
VcmaxRef = 38.55
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

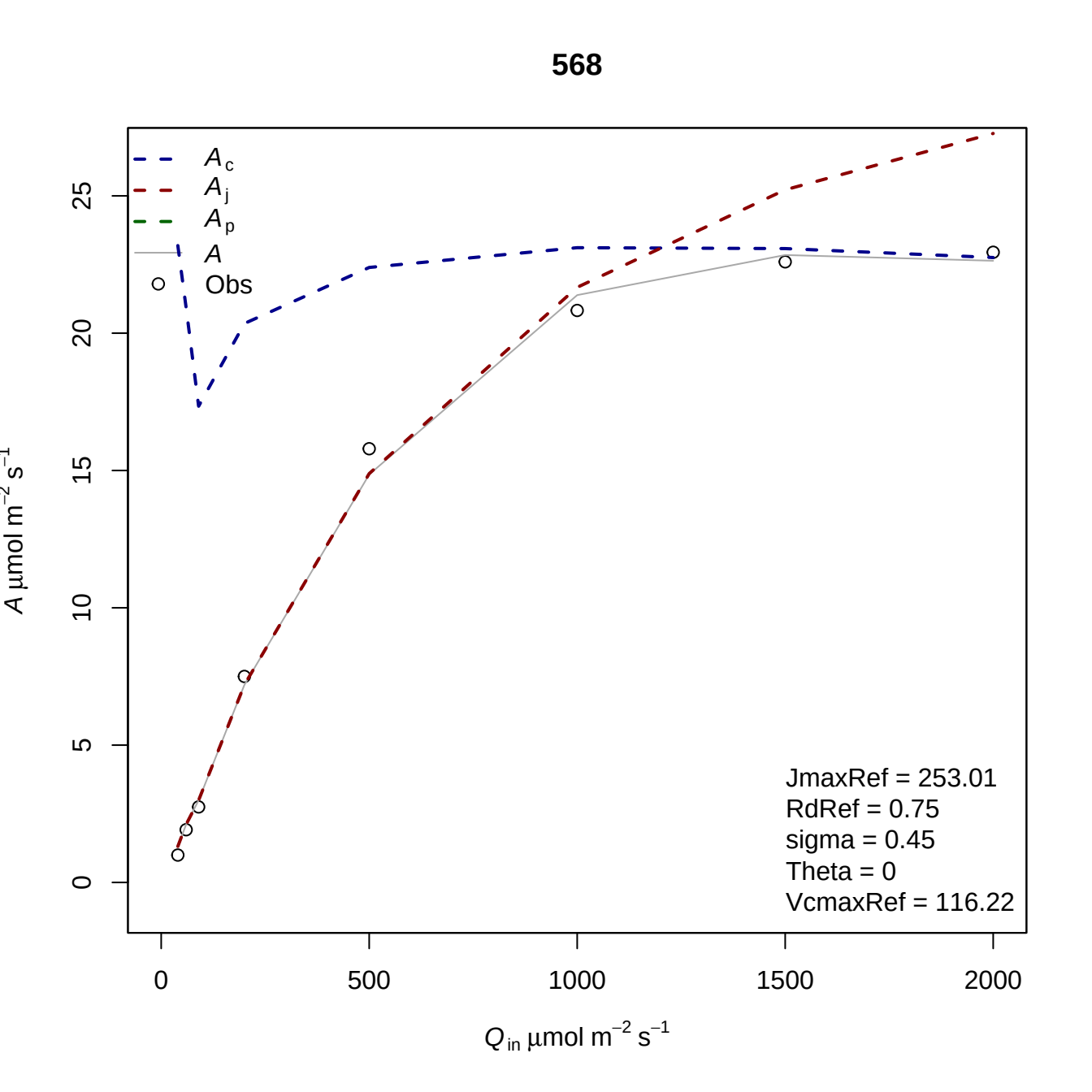

563
A_c
A_j
A_p
A
Obs
JmaxRef = 215.5
RdRef = 0.36
sigma = 0.55
Theta = 0
VcmaxRef = 80.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

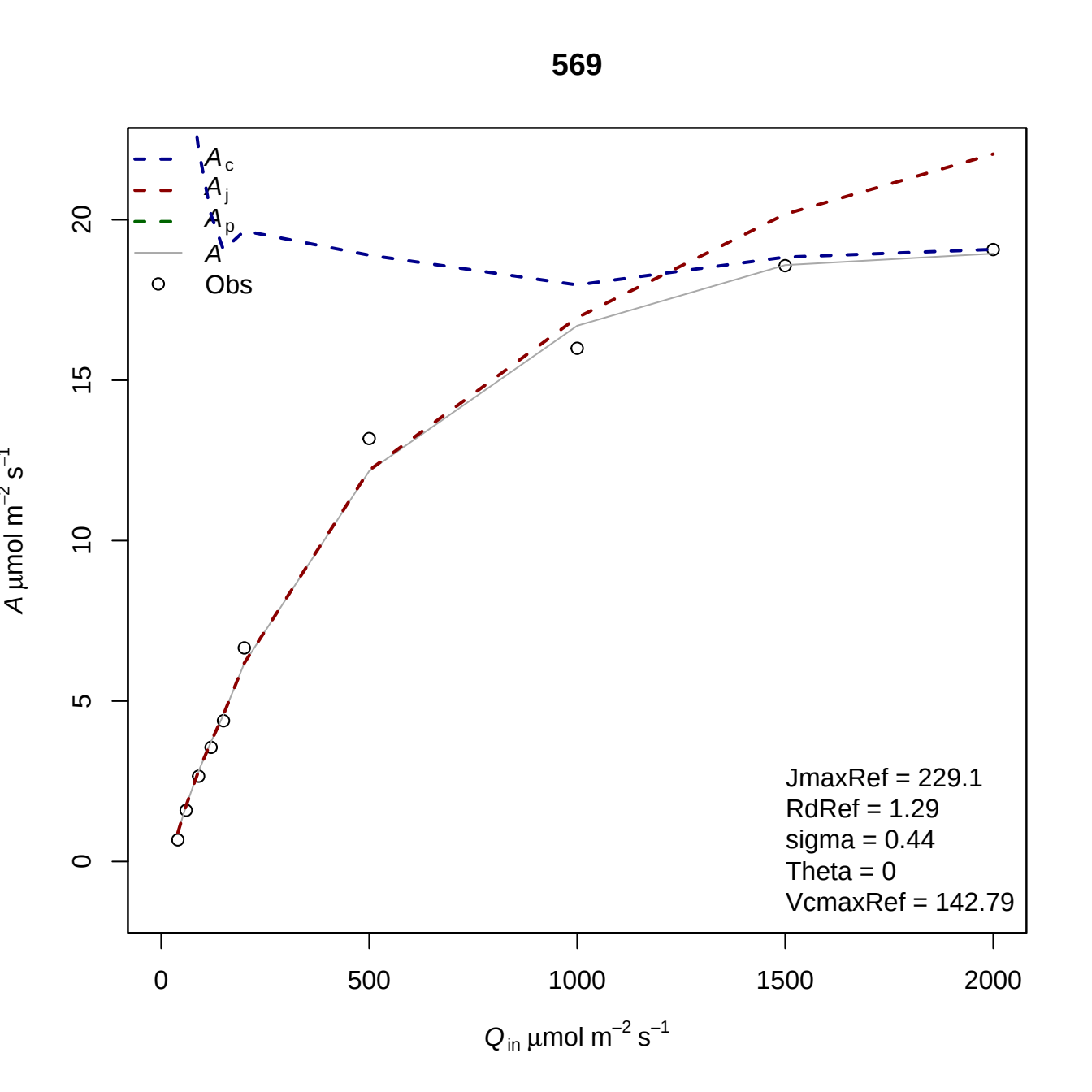

564
A_c
A_j
A_p
A
Obs
JmaxRef = 267.95
RdRef = 0.9
sigma = 0.62
Theta = 0
VcmaxRef = 112.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

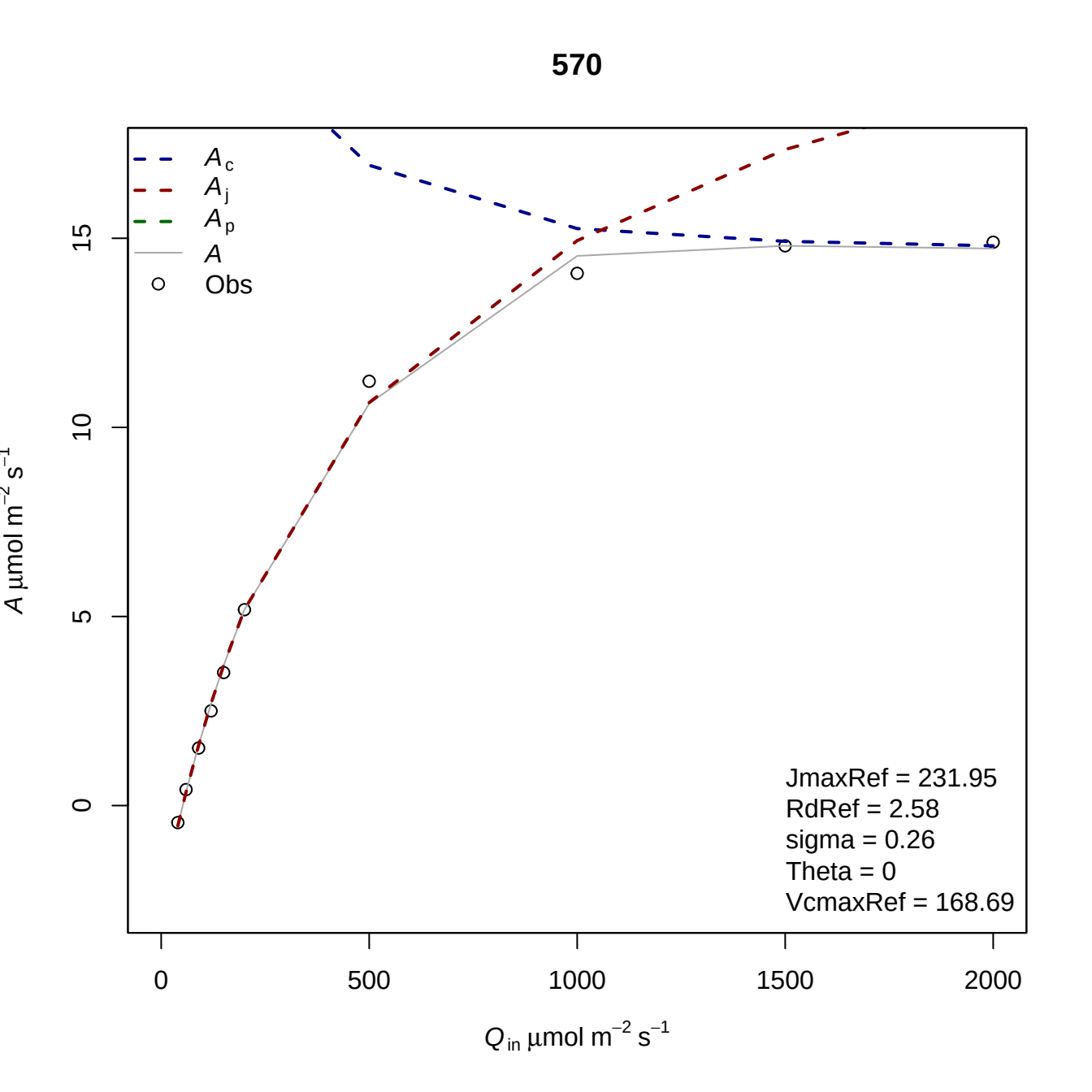

565
A_d
A_j
A_p
A
Obs
JmaxRef = 356.86
RdRef = 1.53
sigma = 0.63
Theta = 0
VcmaxRef = 192.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

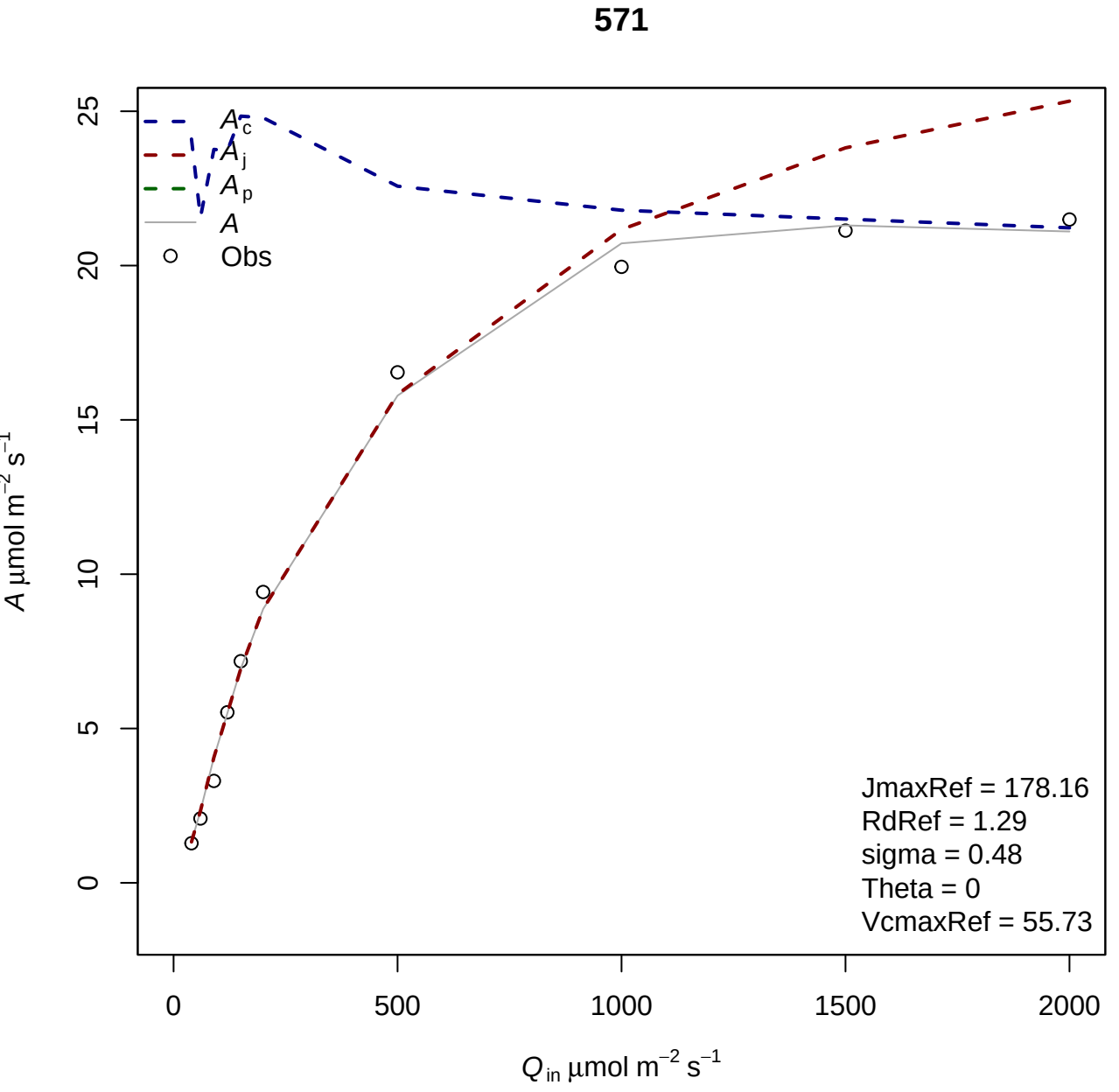

566
A_c
A_j
A_p
A
Obs
JmaxRef = 353.12
RdRef = 2.24
sigma = 0.49
Theta = 0
VcmaxRef = 267.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

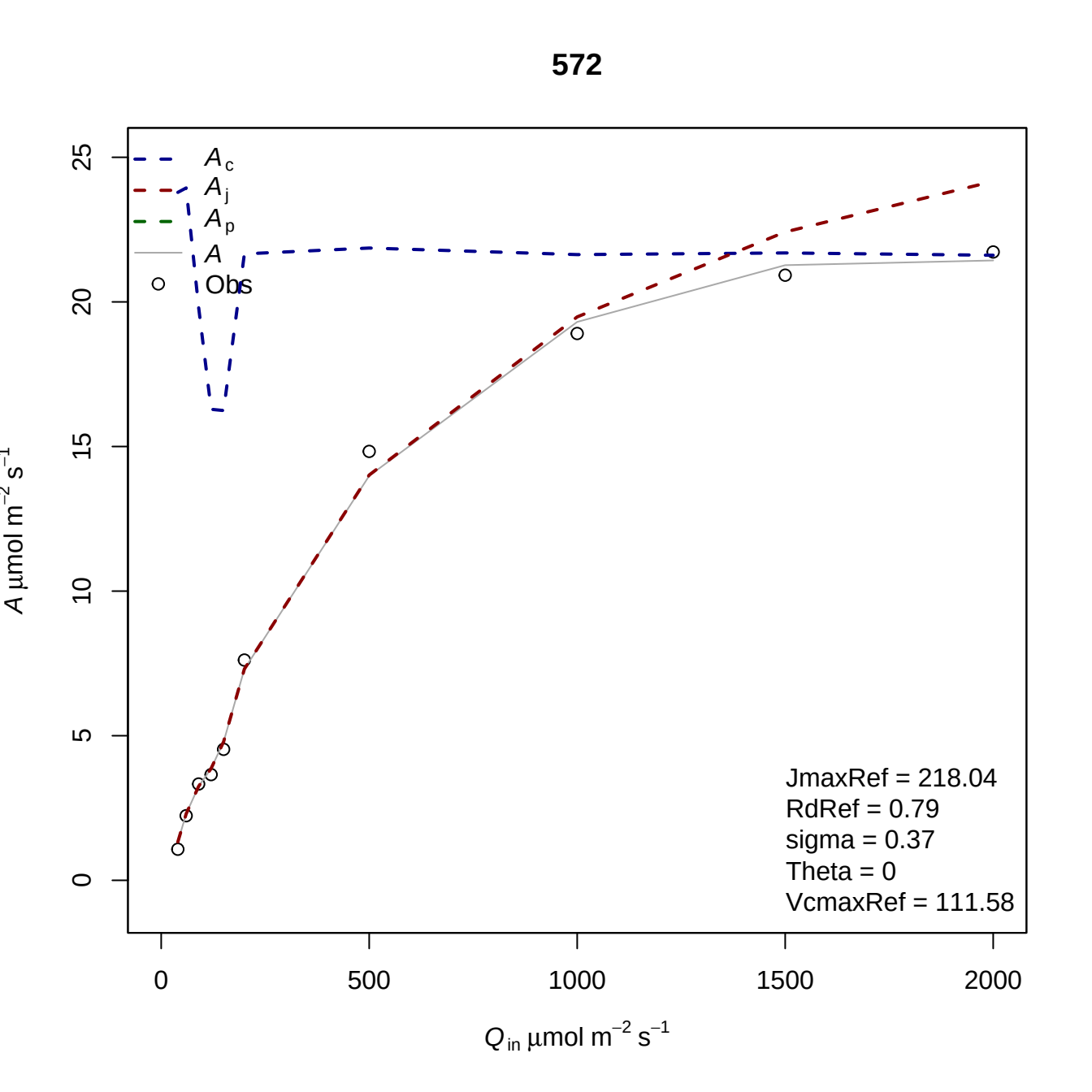

567
A_c
A_j
A_p
A
Obs
JmaxRef = 228.12
RdRef = 1.33
sigma = 0.67
Theta = 0
VcmaxRef = 67.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

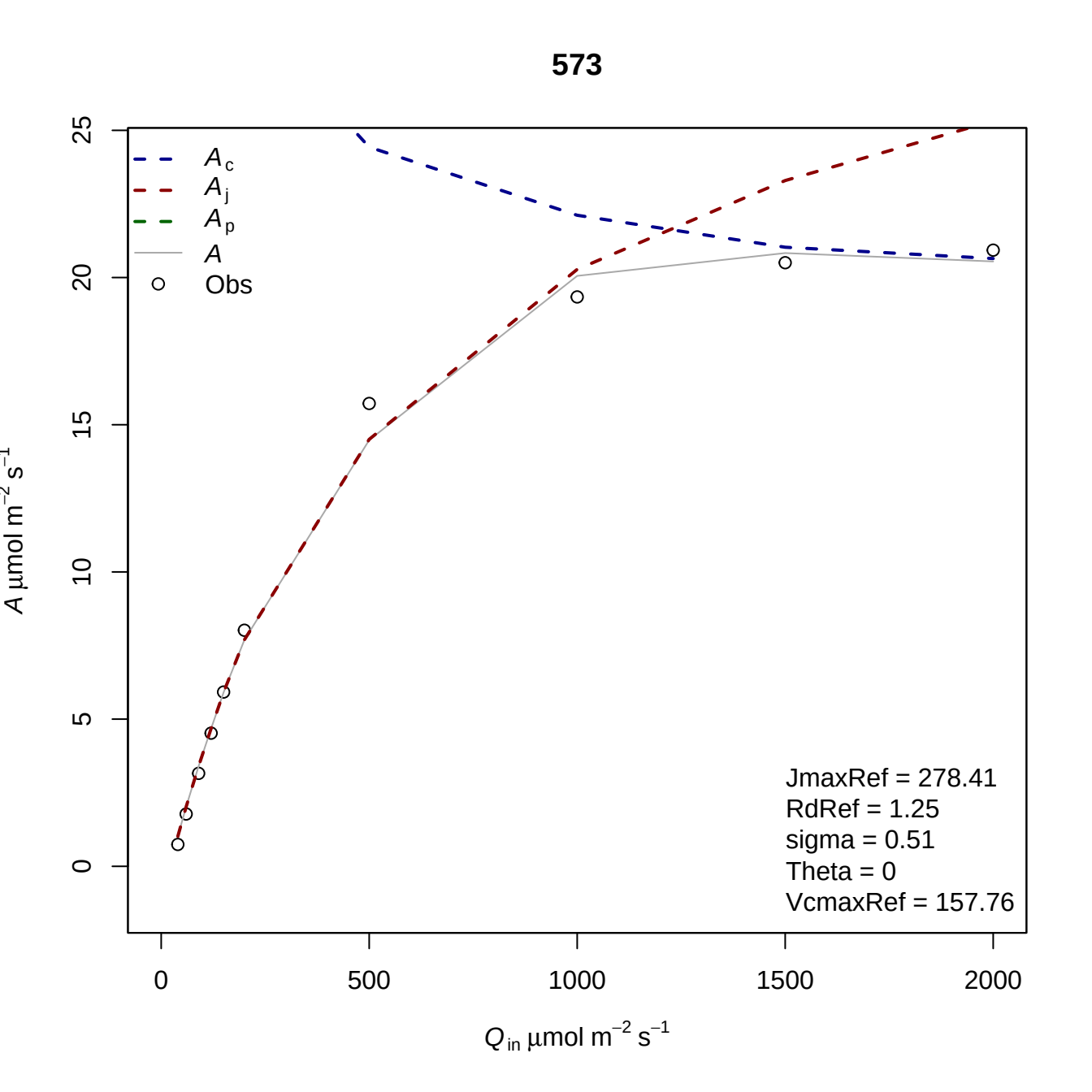

568
A_c
A_j
A_p
A
Obs
JmaxRef = 253.01
RdRef = 0.75
sigma = 0.45
Theta = 0
VcmaxRef = 116.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

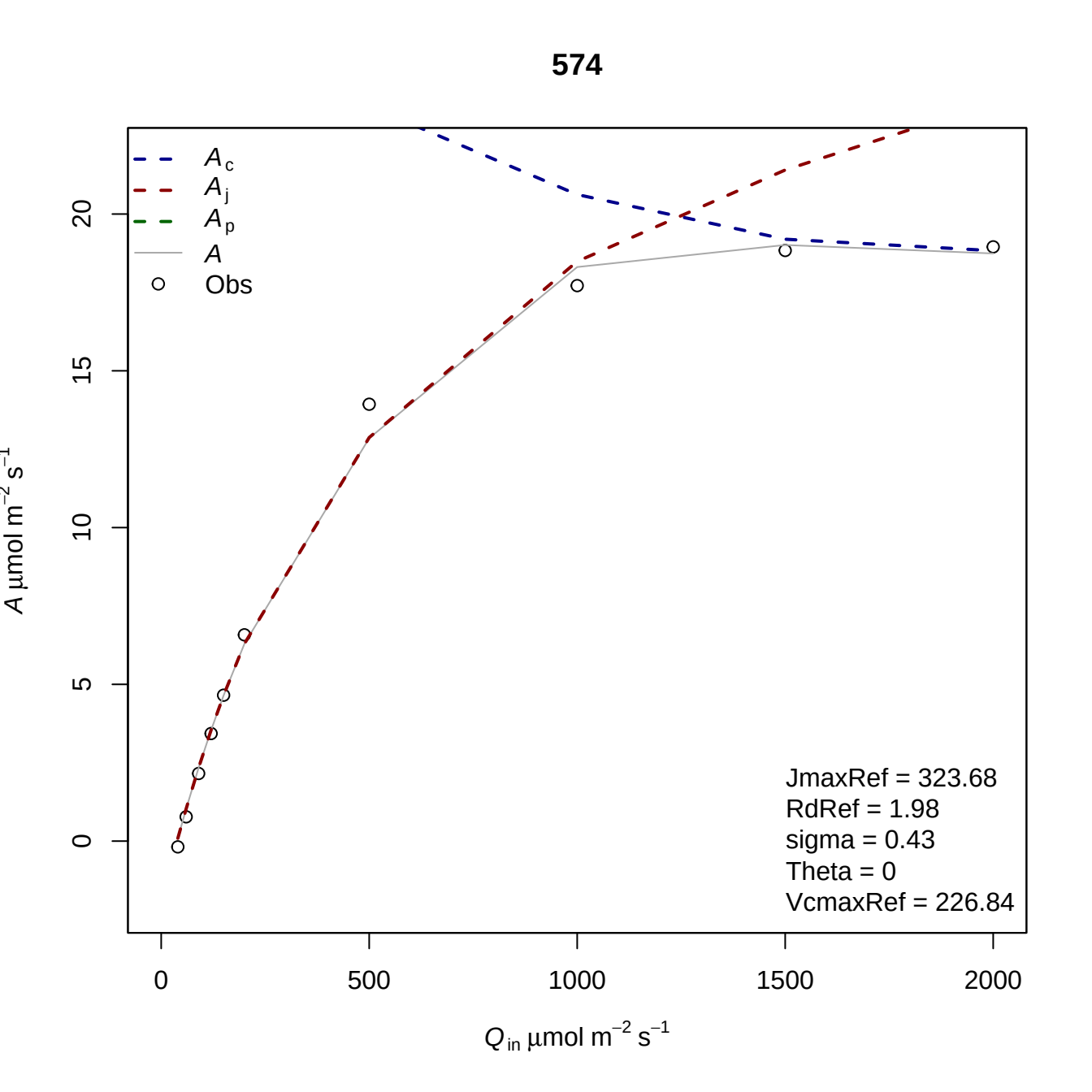

569
A_c
A_j
A_p
A
Obs
JmaxRef = 229.1
RdRef = 1.29
sigma = 0.44
Theta = 0
VcmaxRef = 142.79
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

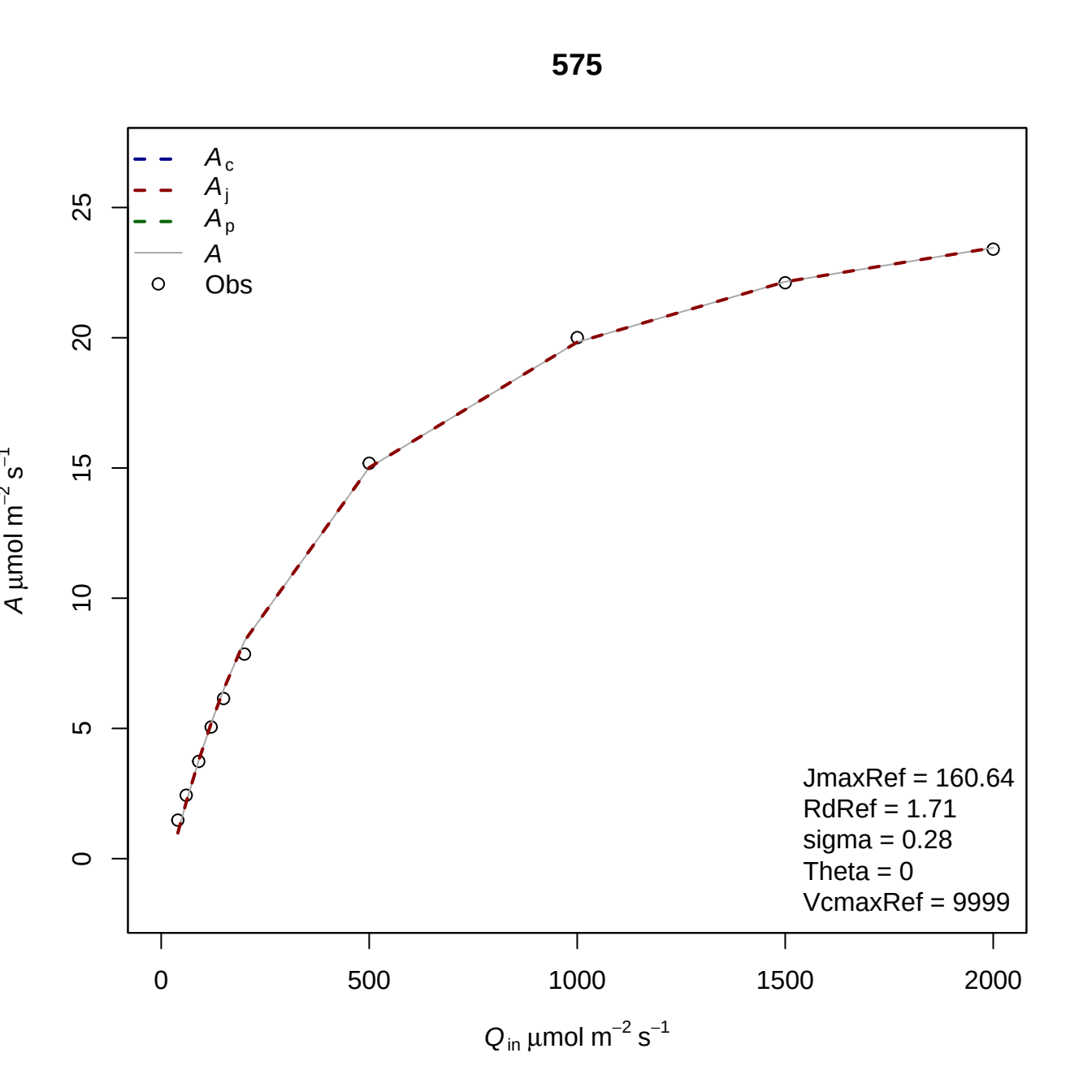

570
A_c
A_j
A_p
A
Obs
JmaxRef = 231.95
RdRef = 2.58
sigma = 0.26
Theta = 0
VcmaxRef = 168.69
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

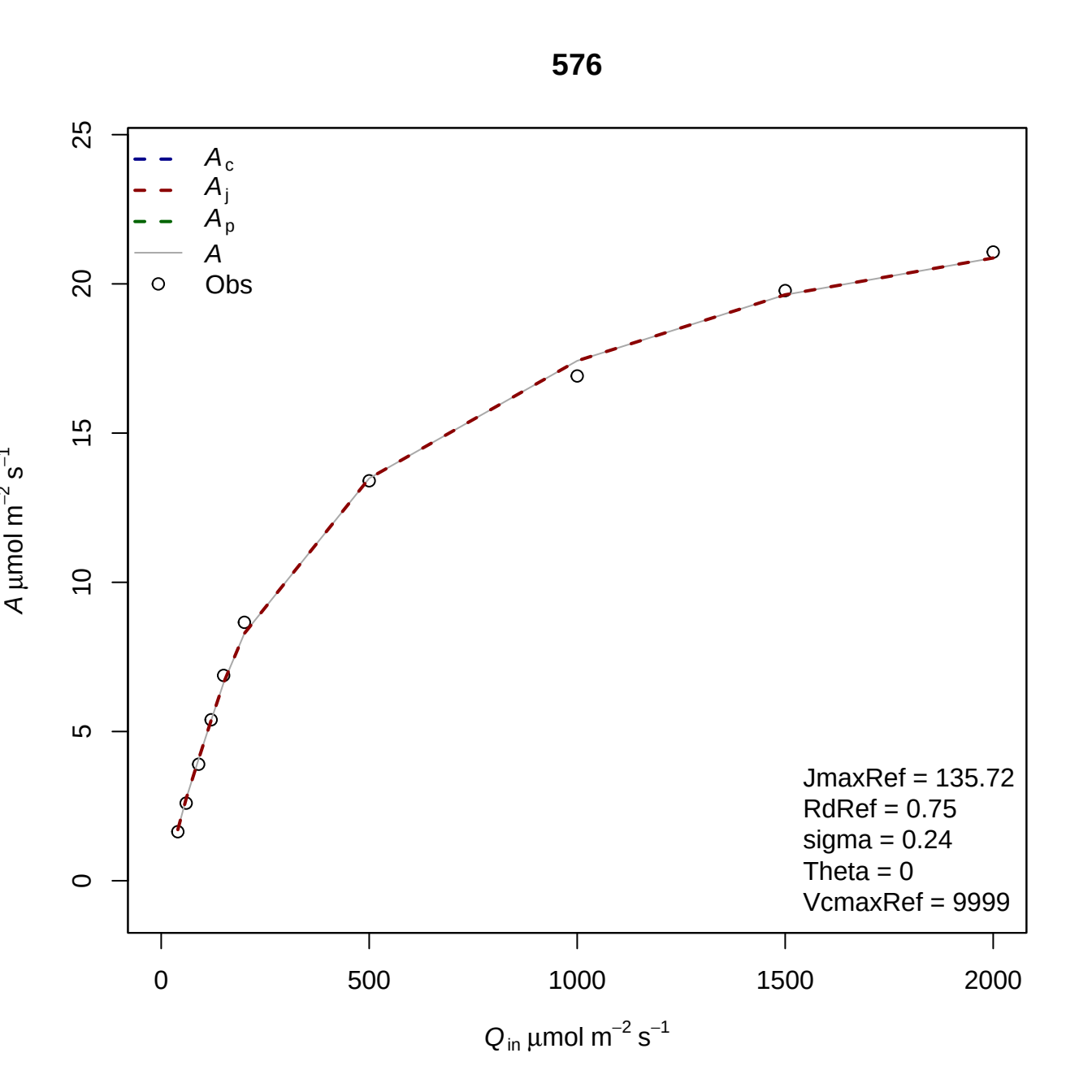

571
A_c
A_j
A_p
A
Obs
JmaxRef = 178.16
RdRef = 1.29
sigma = 0.48
Theta = 0
VcmaxRef = 55.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

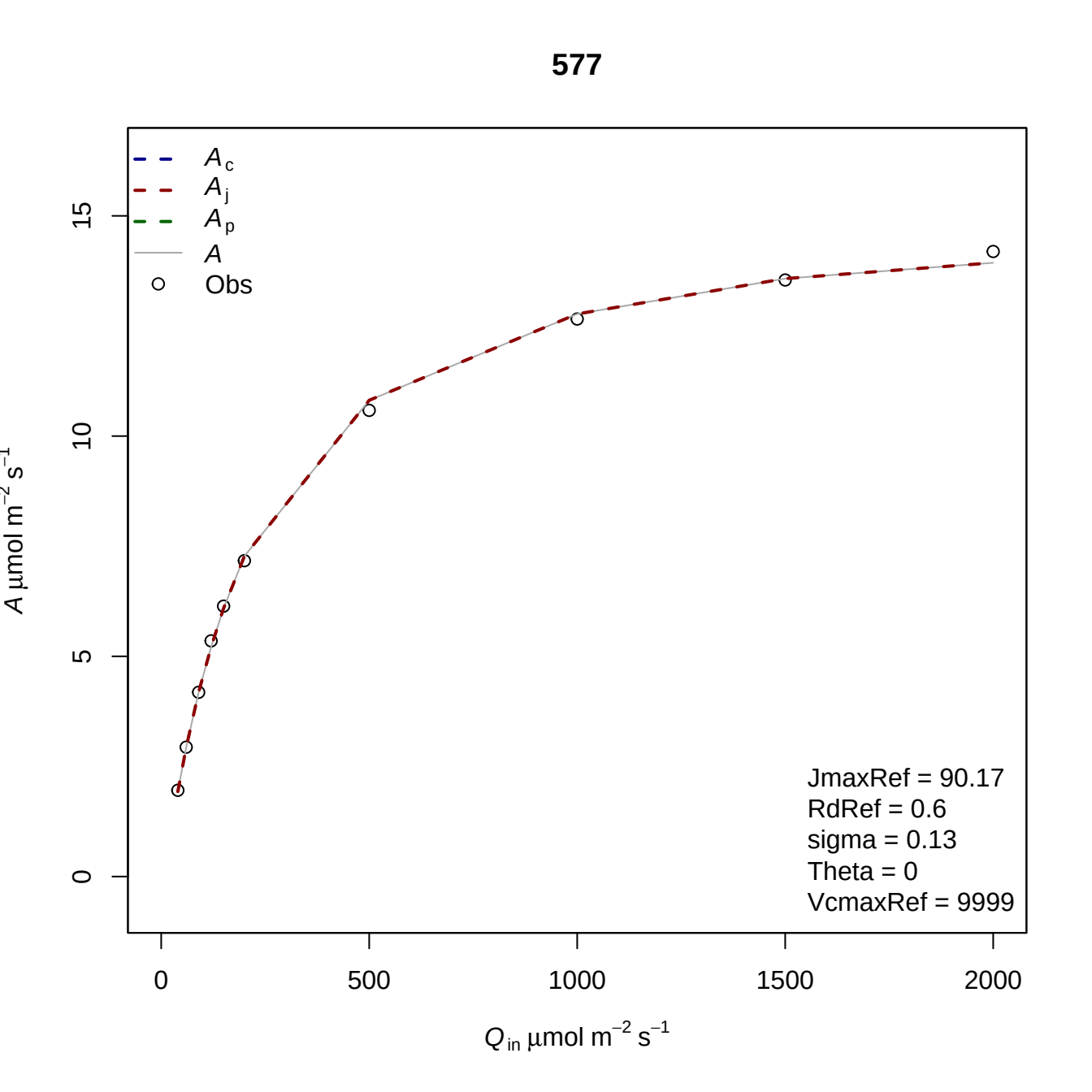

572
A_c
A_j
A_p
A
Obs
JmaxRef = 218.04
RdRef = 0.79
sigma = 0.37
Theta = 0
VcmaxRef = 111.58
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

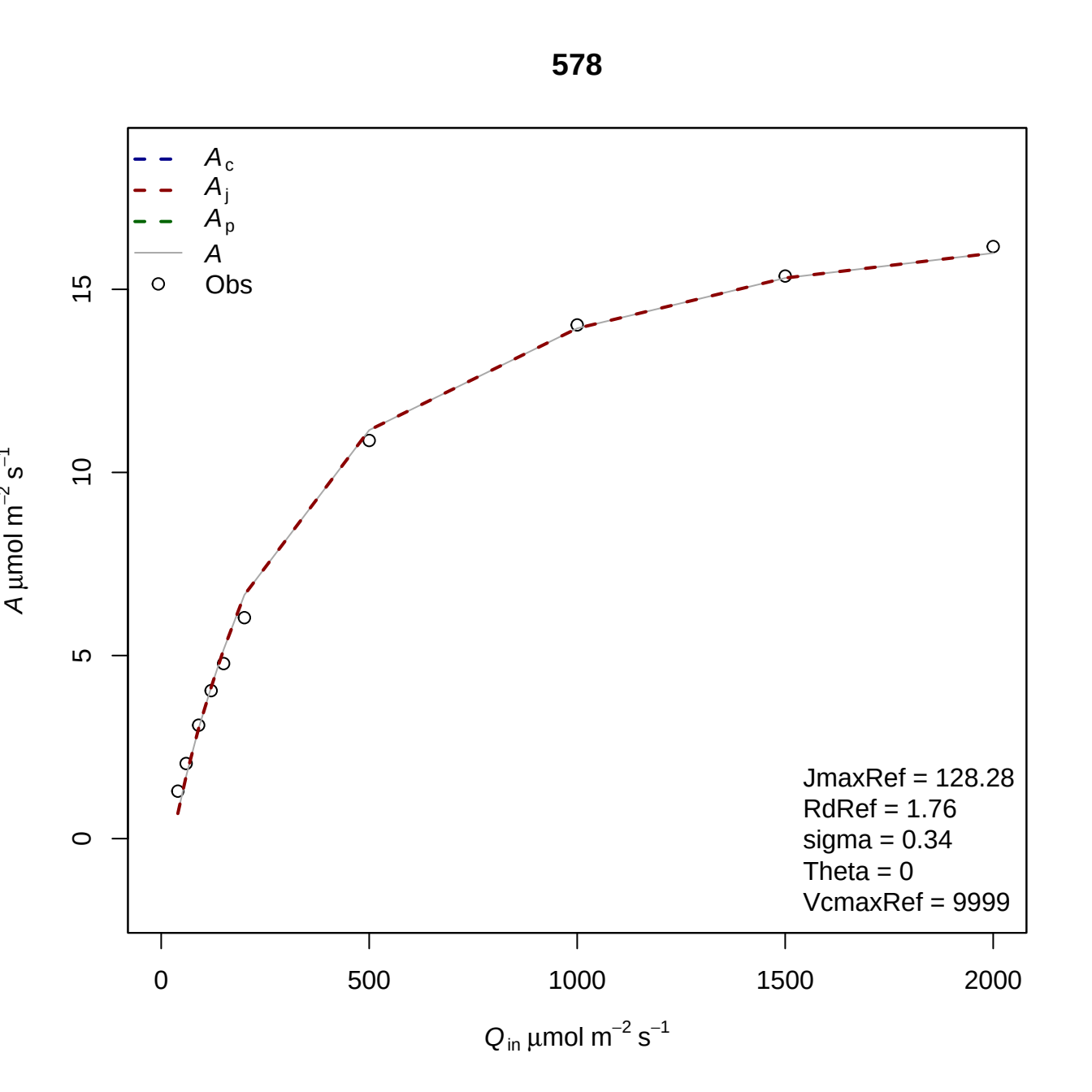

573
A_c
A_j
A_p
A
Obs
JmaxRef = 278.41
RdRef = 1.25
sigma = 0.51
Theta = 0
VcmaxRef = 157.76
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

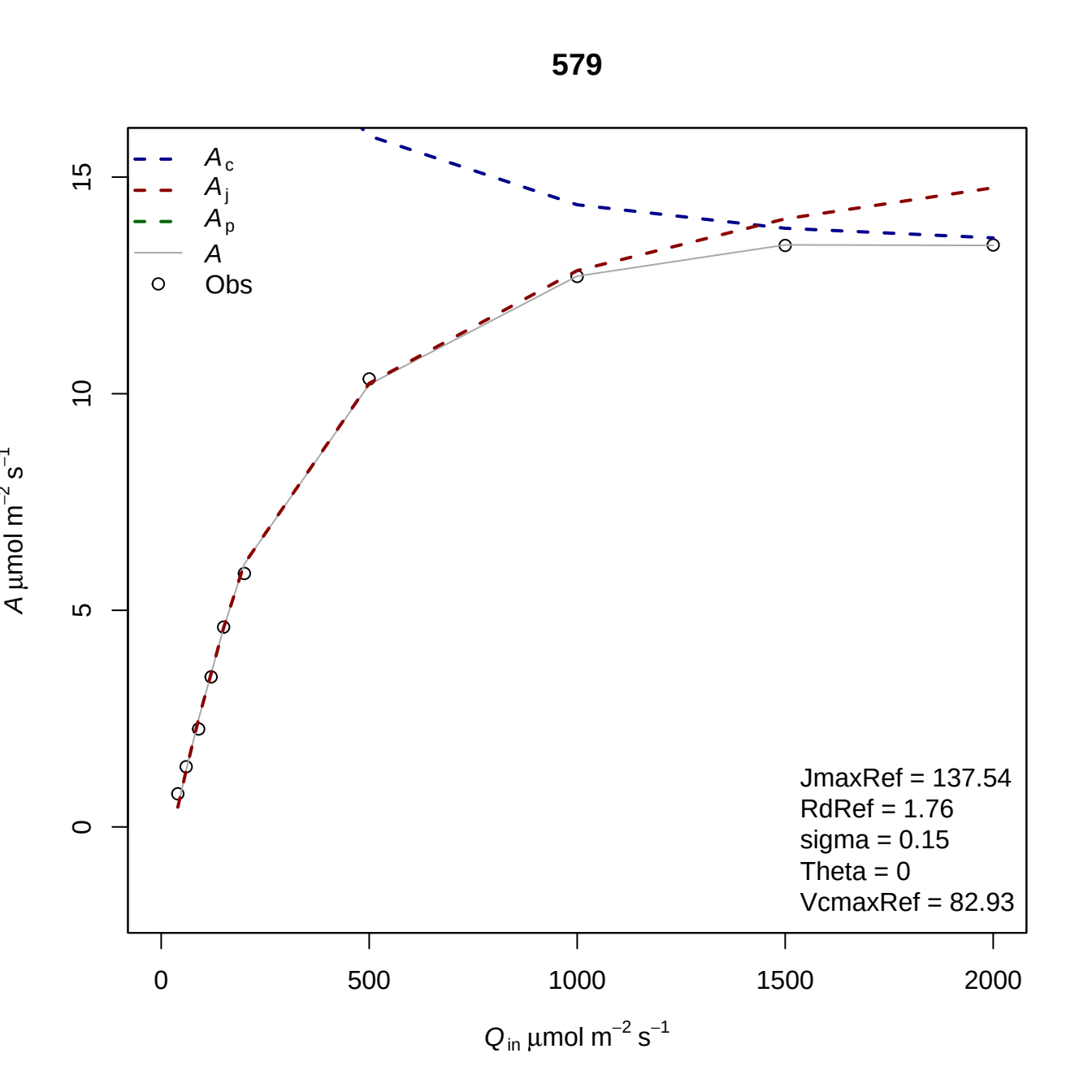

574
A_c
A_j
A_p
A
Obs
JmaxRef = 323.68
RdRef = 1.98
sigma = 0.43
Theta = 0
VcmaxRef = 226.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

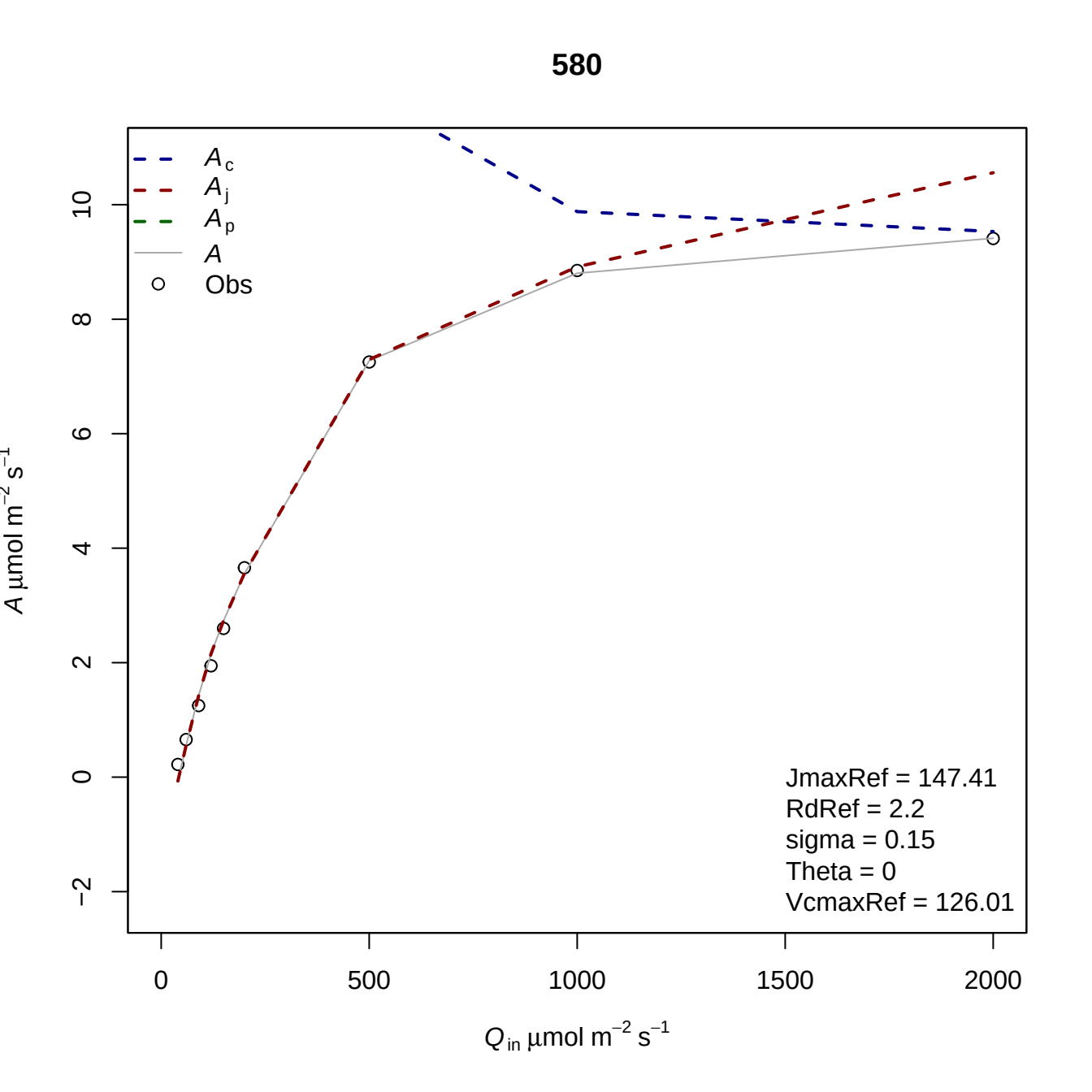

575
A_c
A_j
A_p
A
Obs
JmaxRef = 160.64
RdRef = 1.71
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

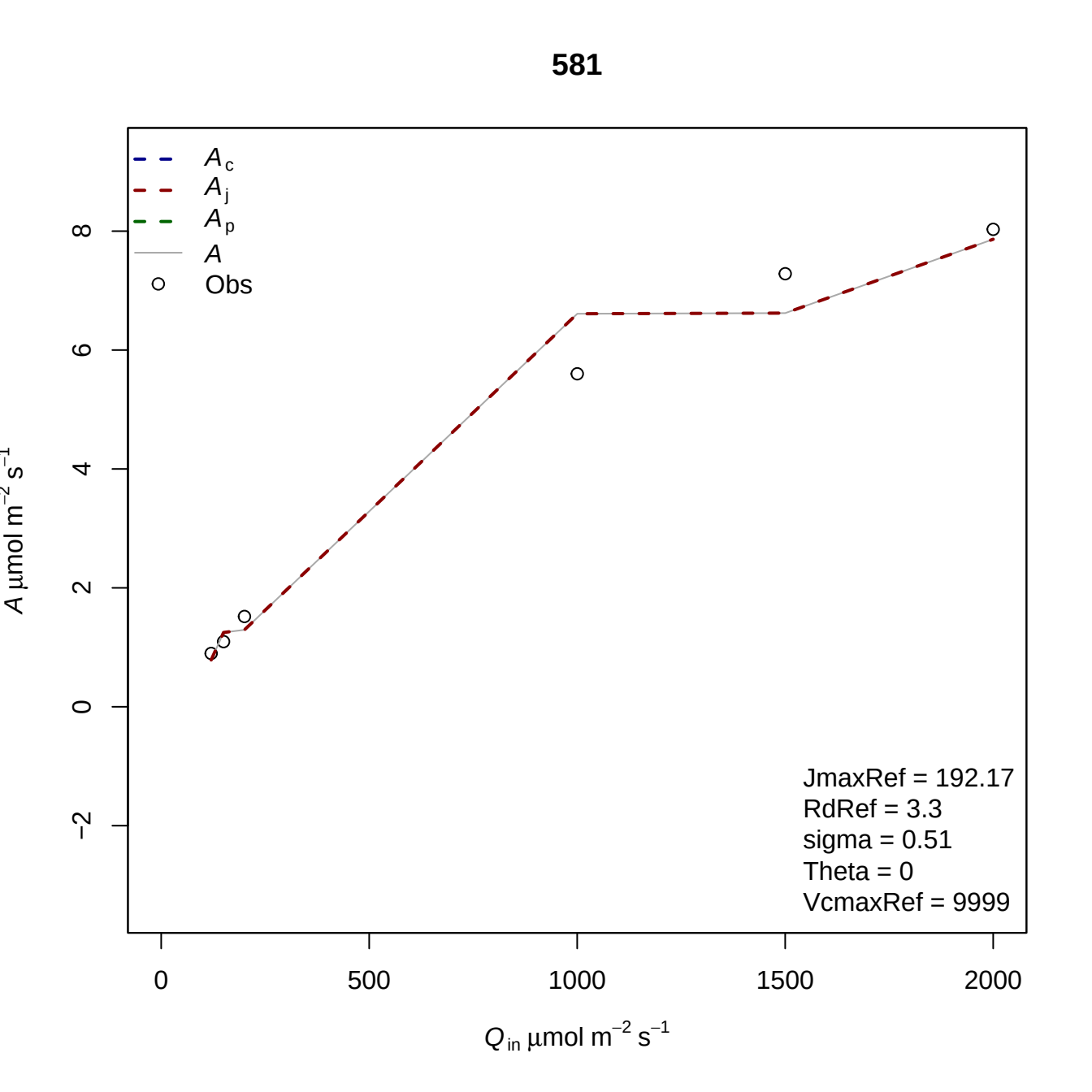

576
A_c
A_j
A_p
A
Obs
JmaxRef = 135.72
RdRef = 0.75
sigma = 0.24
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

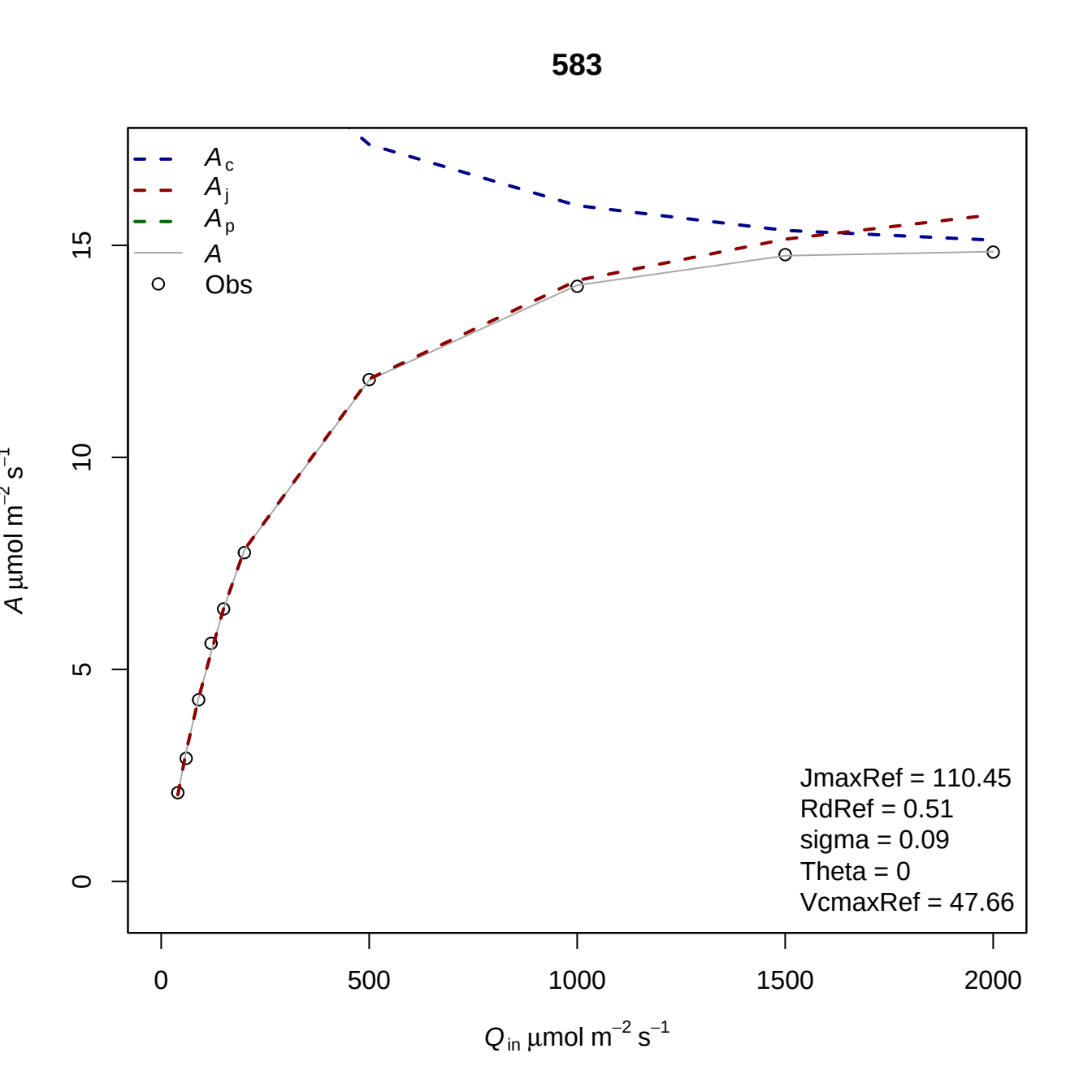

577
A_c
A_j
A_p
A
Obs
JmaxRef = 90.17
RdRef = 0.6
sigma = 0.13
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

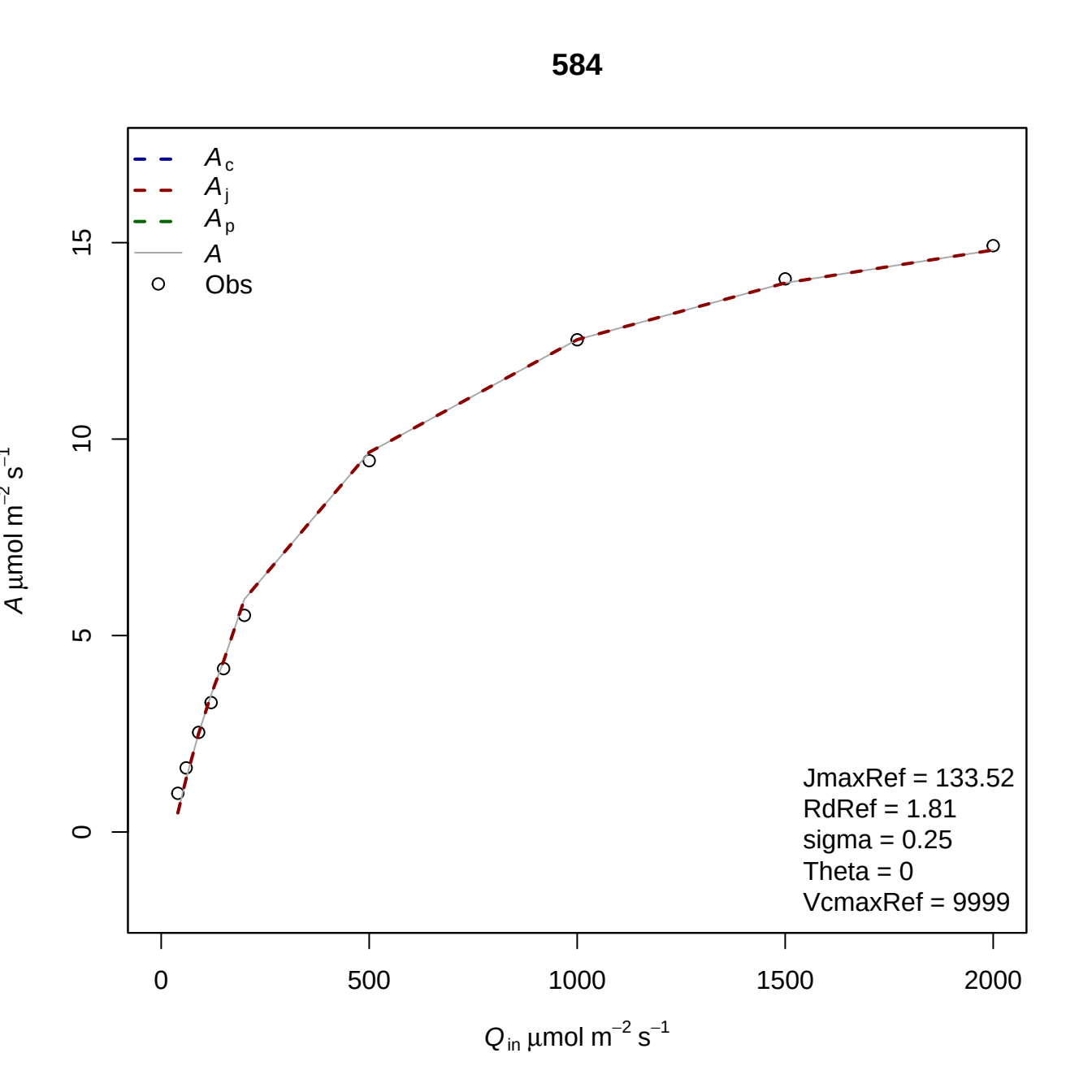

578
A_c
A_j
A_p
A
Obs
JmaxRef = 128.28
RdRef = 1.76
sigma = 0.34
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

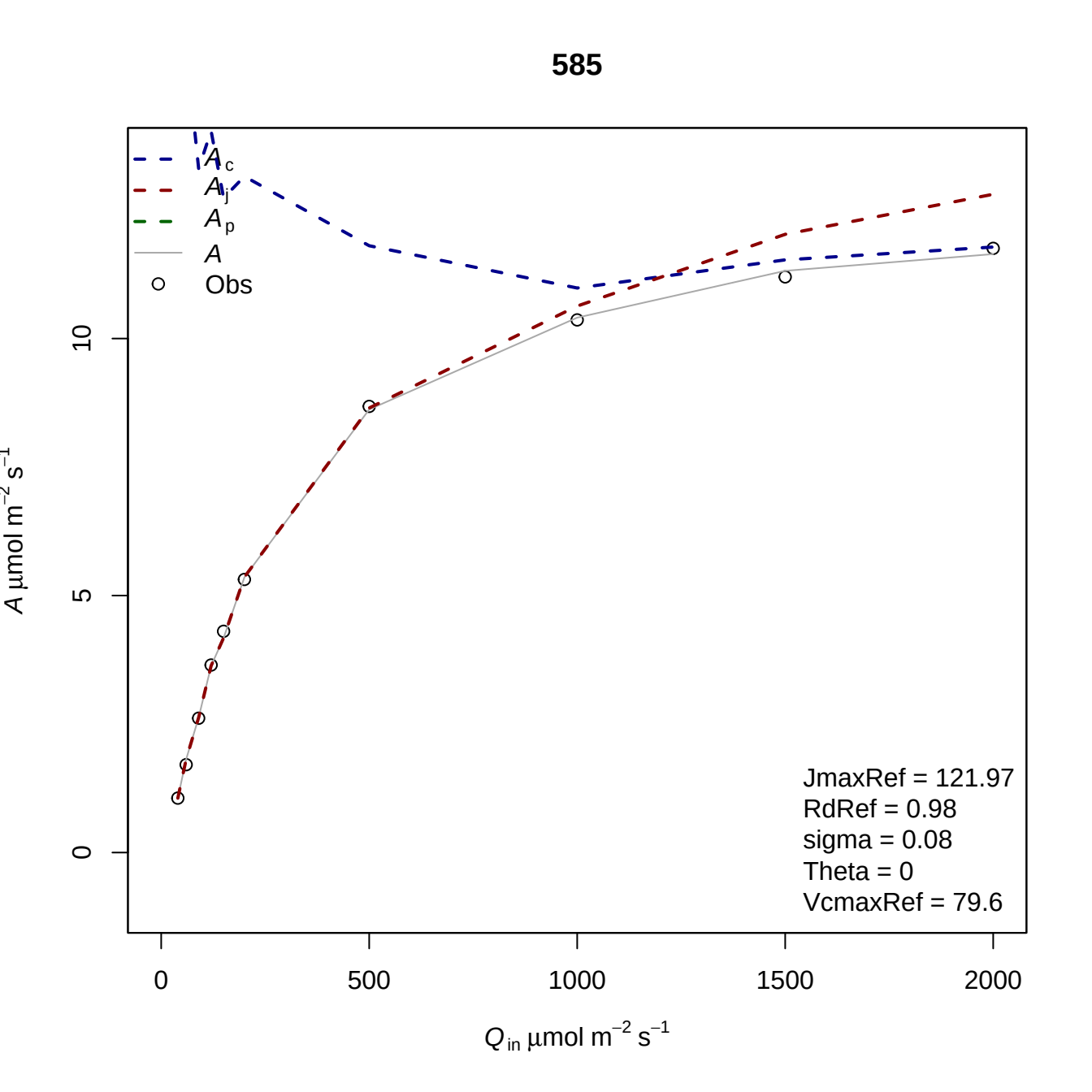

579
A_c
A_j
A_p
A
Obs
JmaxRef = 137.54
RdRef = 1.76
sigma = 0.15
Theta = 0
VcmaxRef = 82.93
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

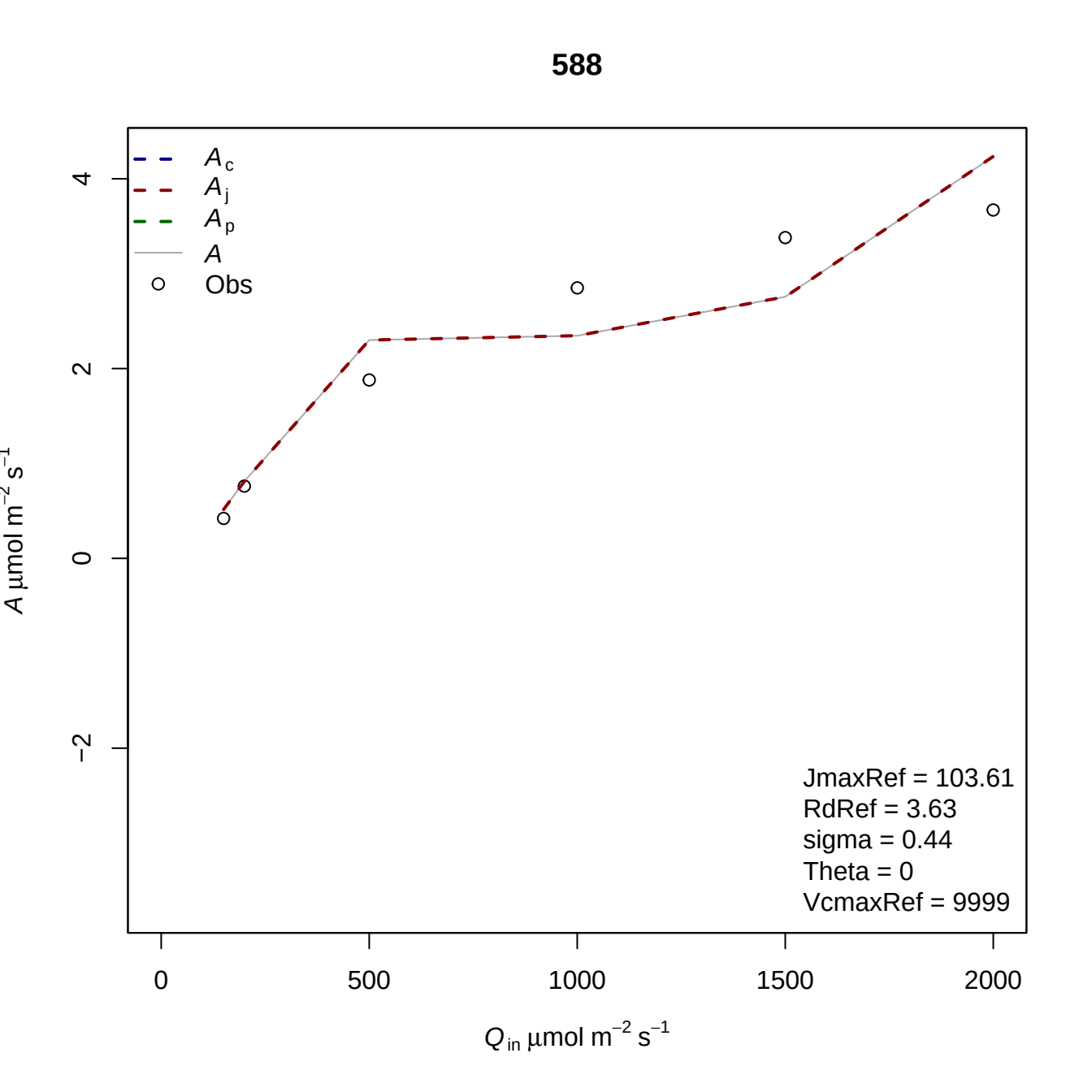

580
A_c
A_j
A_p
A
Obs
JmaxRef = 147.41
RdRef = 2.2
sigma = 0.15
Theta = 0
VcmaxRef = 126.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

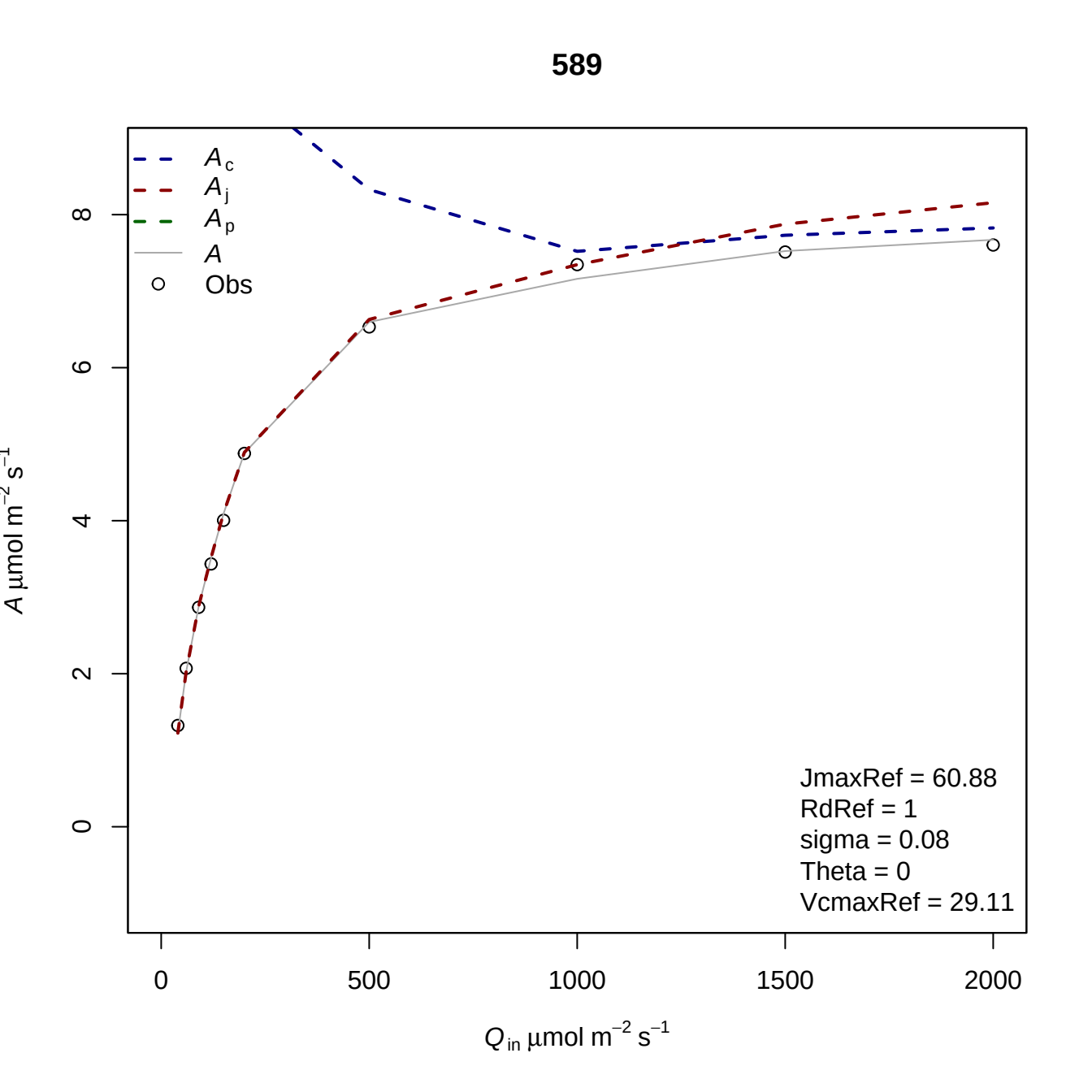

581
A_c
A_j
A_p
A
Obs
JmaxRef = 192.17
RdRef = 3.3
sigma = 0.51
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

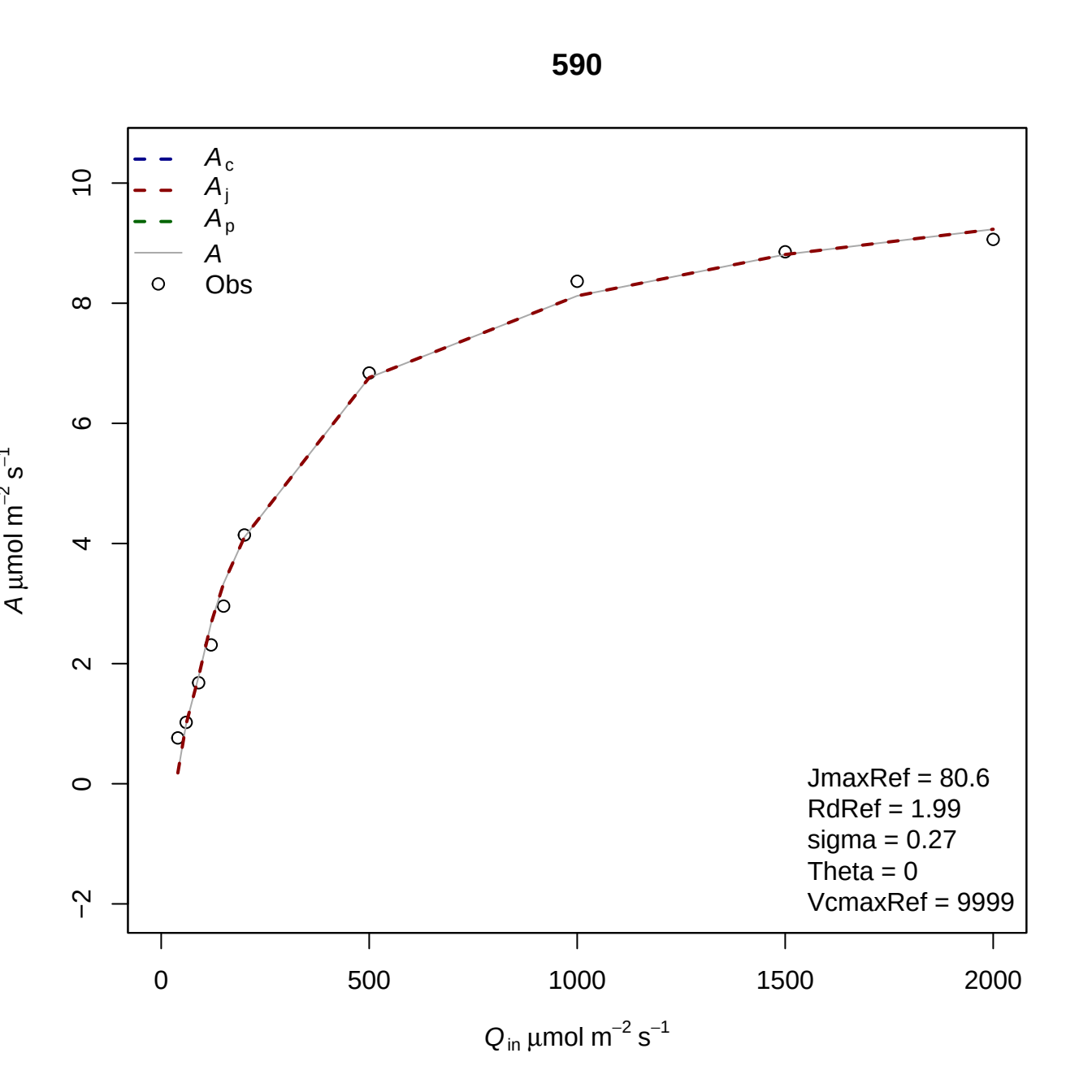

583
A_c
A_j
A_p
A
Obs
JmaxRef = 110.45
RdRef = 0.51
sigma = 0.09
Theta = 0
VcmaxRef = 47.66
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

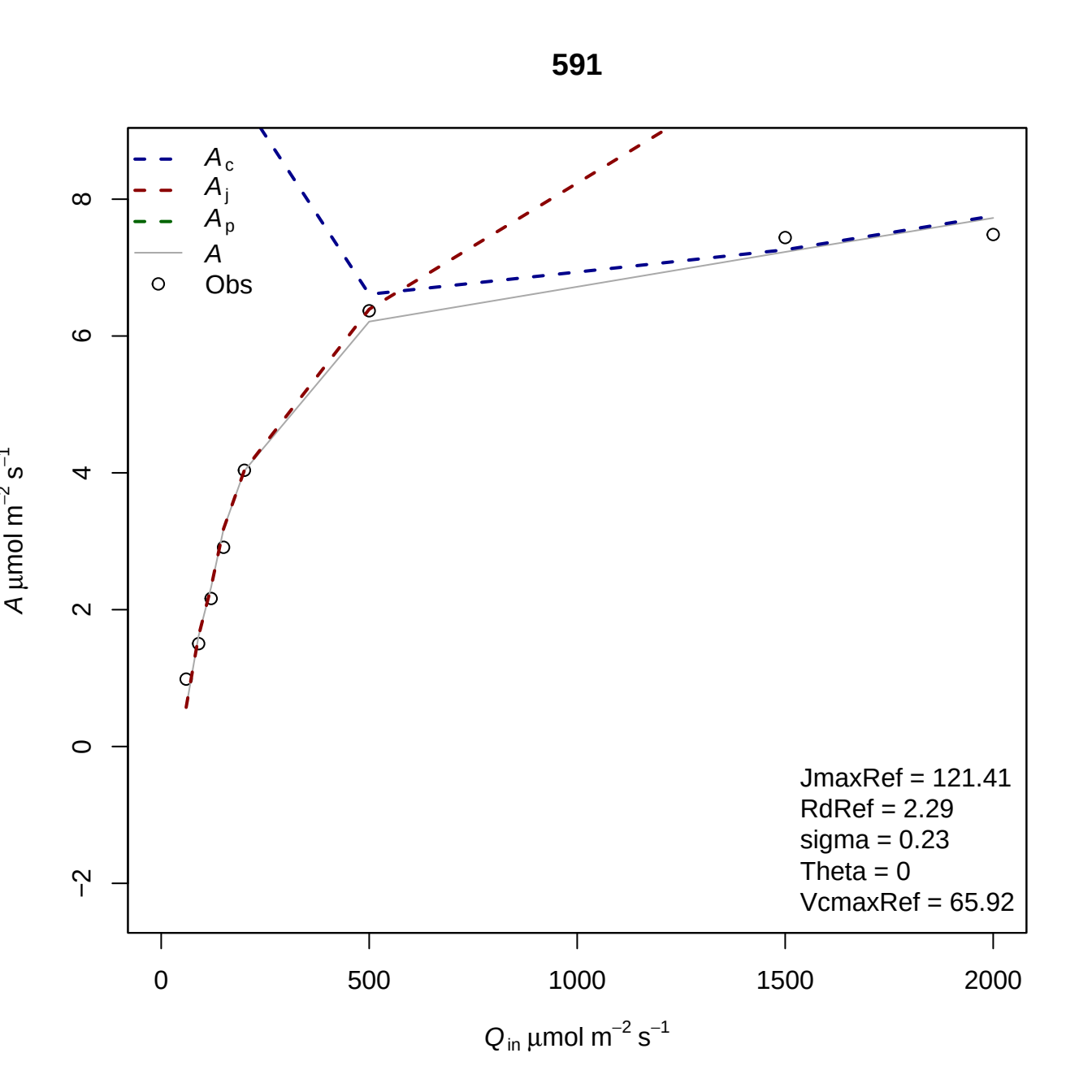

584
A_c
A_j
A_p
A
Obs
JmaxRef = 133.52
RdRef = 1.81
sigma = 0.25
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

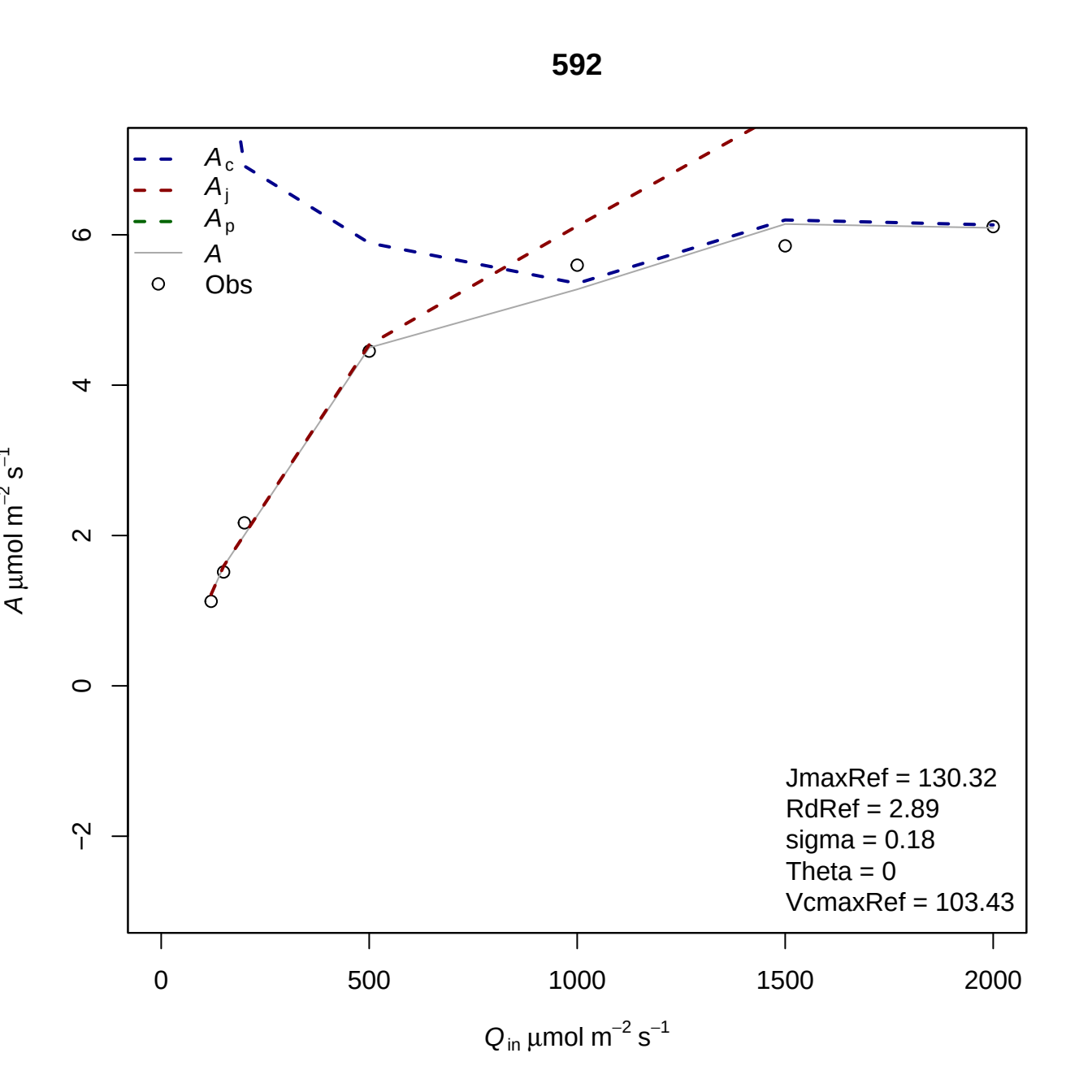

585
A_c
A_j
A_p
A
Obs
JmaxRef = 121.97
RdRef = 0.98
sigma = 0.08
Theta = 0
VcmaxRef = 79.6
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

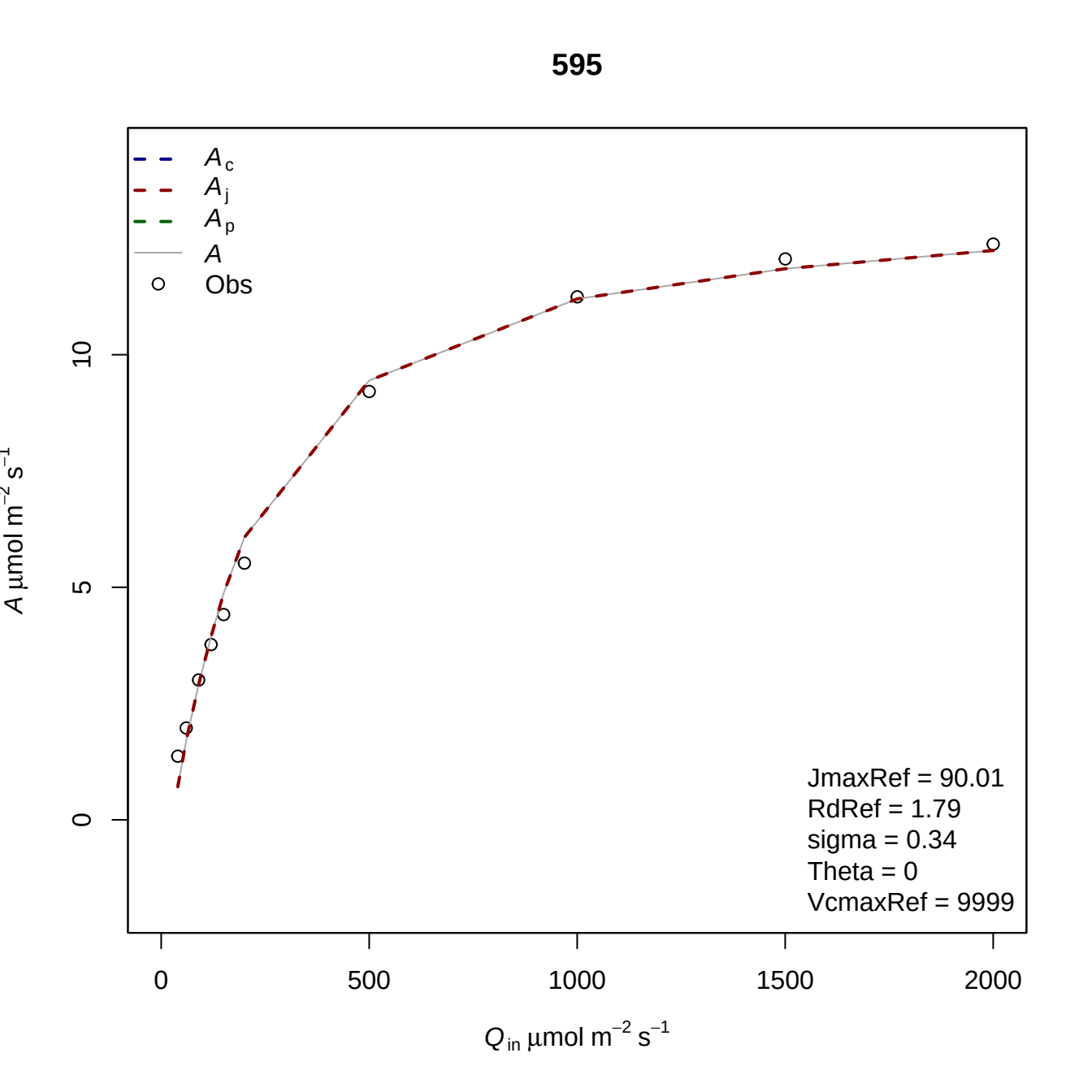

588
A_c
A_j
A_p
A
Obs
JmaxRef = 103.61
RdRef = 3.63
sigma = 0.44
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
4
2
0
−2
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

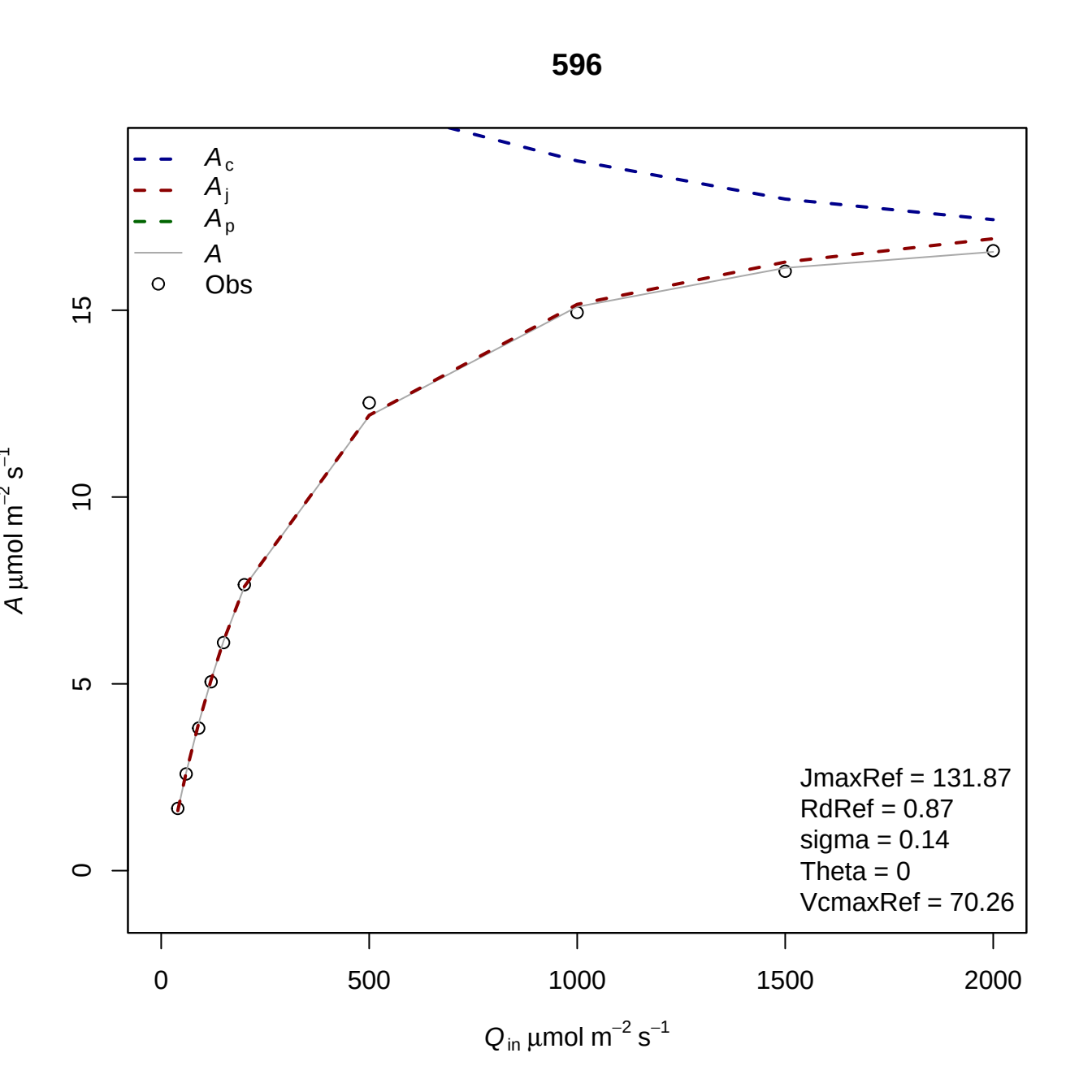

589
A_c
A_j
A_p
A
Obs
JmaxRef = 60.88
RdRef = 1
sigma = 0.08
Theta = 0
VcmaxRef = 29.11
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

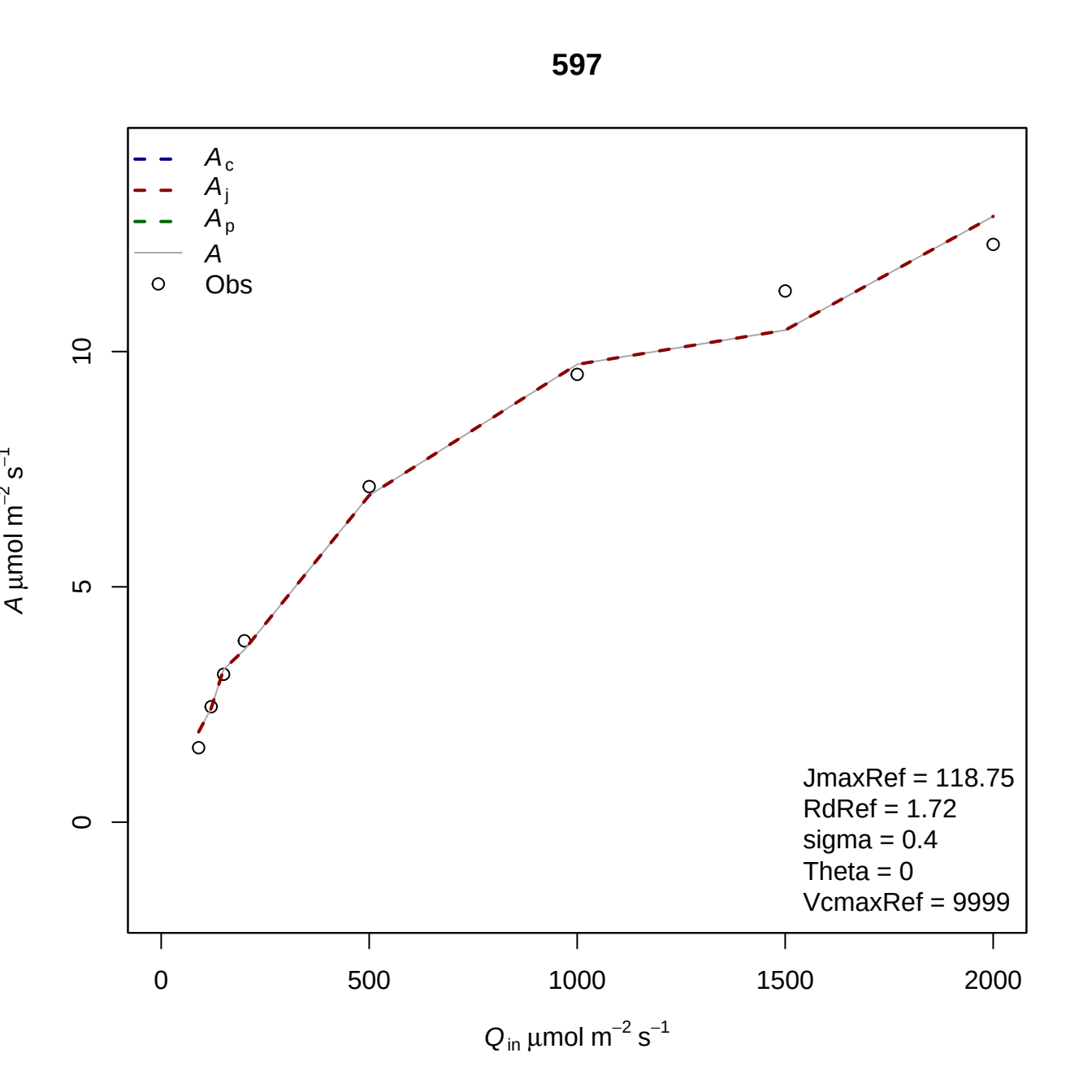

590
A_c
A_j
A_p
A
Obs
JmaxRef = 80.6
RdRef = 1.99
sigma = 0.27
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

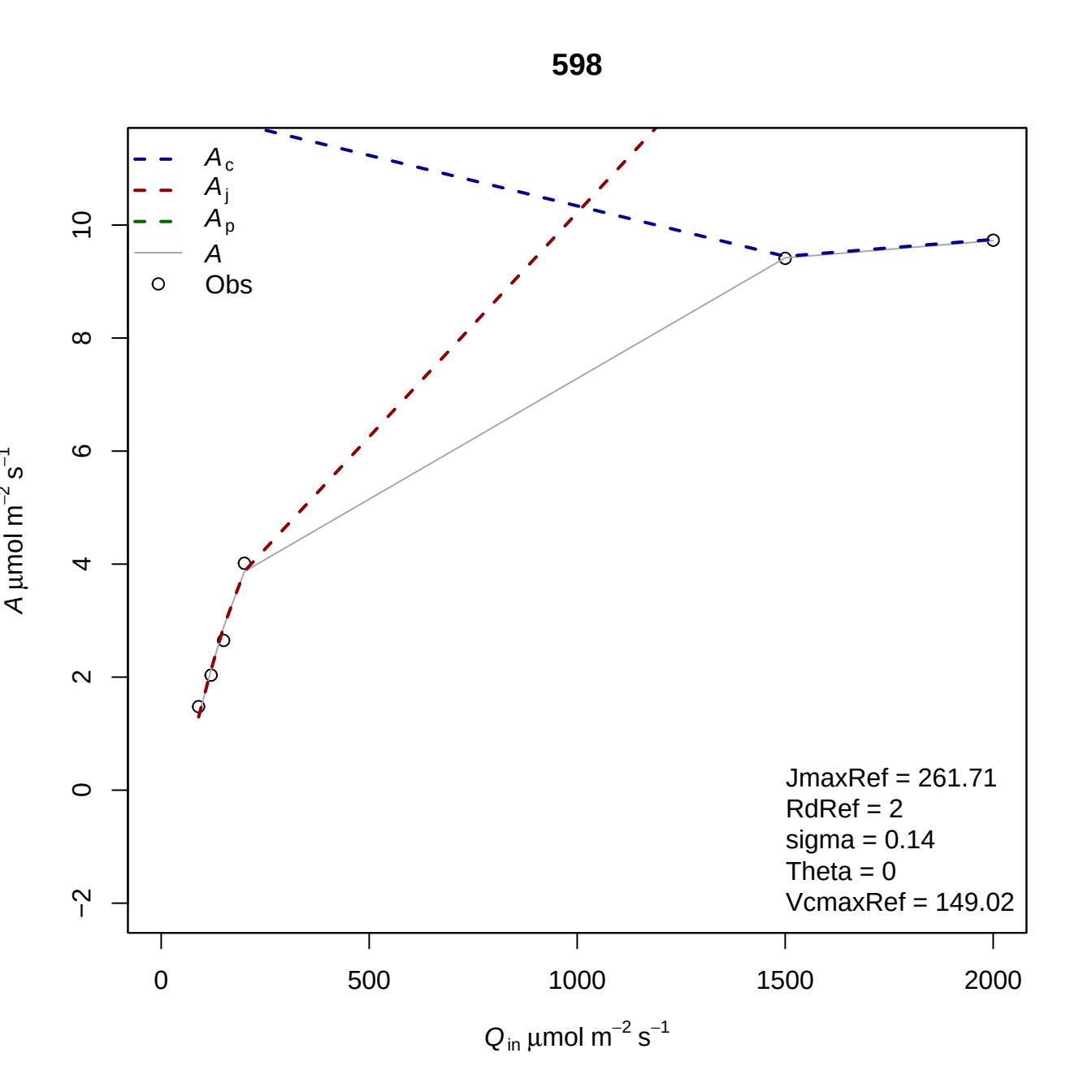

591
A_c
A_j
A_p
A
Obs
JmaxRef = 121.41
RdRef = 2.29
sigma = 0.23
Theta = 0
VcmaxRef = 65.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

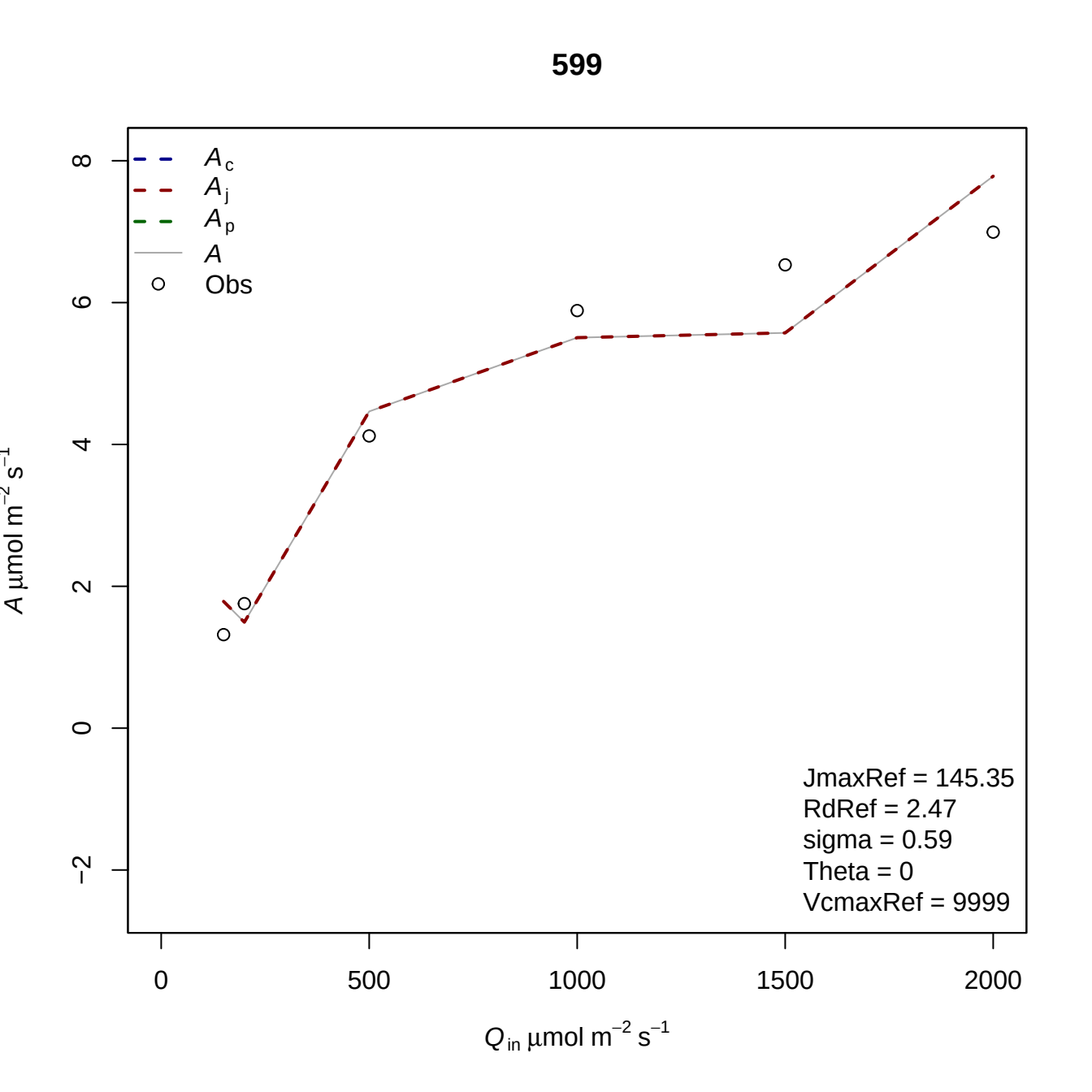

592
A_c
A_j
A_p
A
Obs
JmaxRef = 130.32
RdRef = 2.89
sigma = 0.18
Theta = 0
VcmaxRef = 103.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

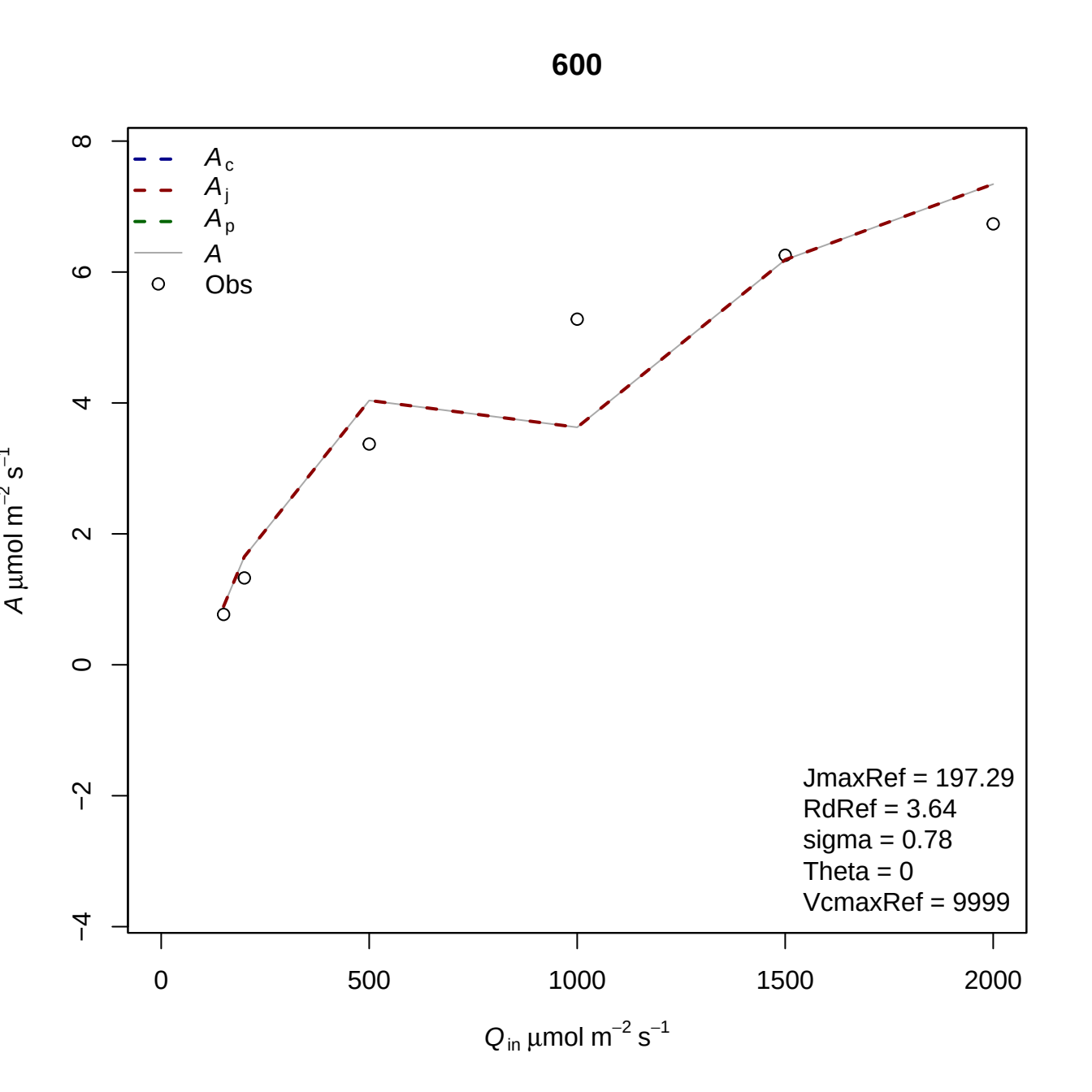

595
A_c
A_j
A_p
A
Obs
JmaxRef = 90.01
RdRef = 1.79
sigma = 0.34
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

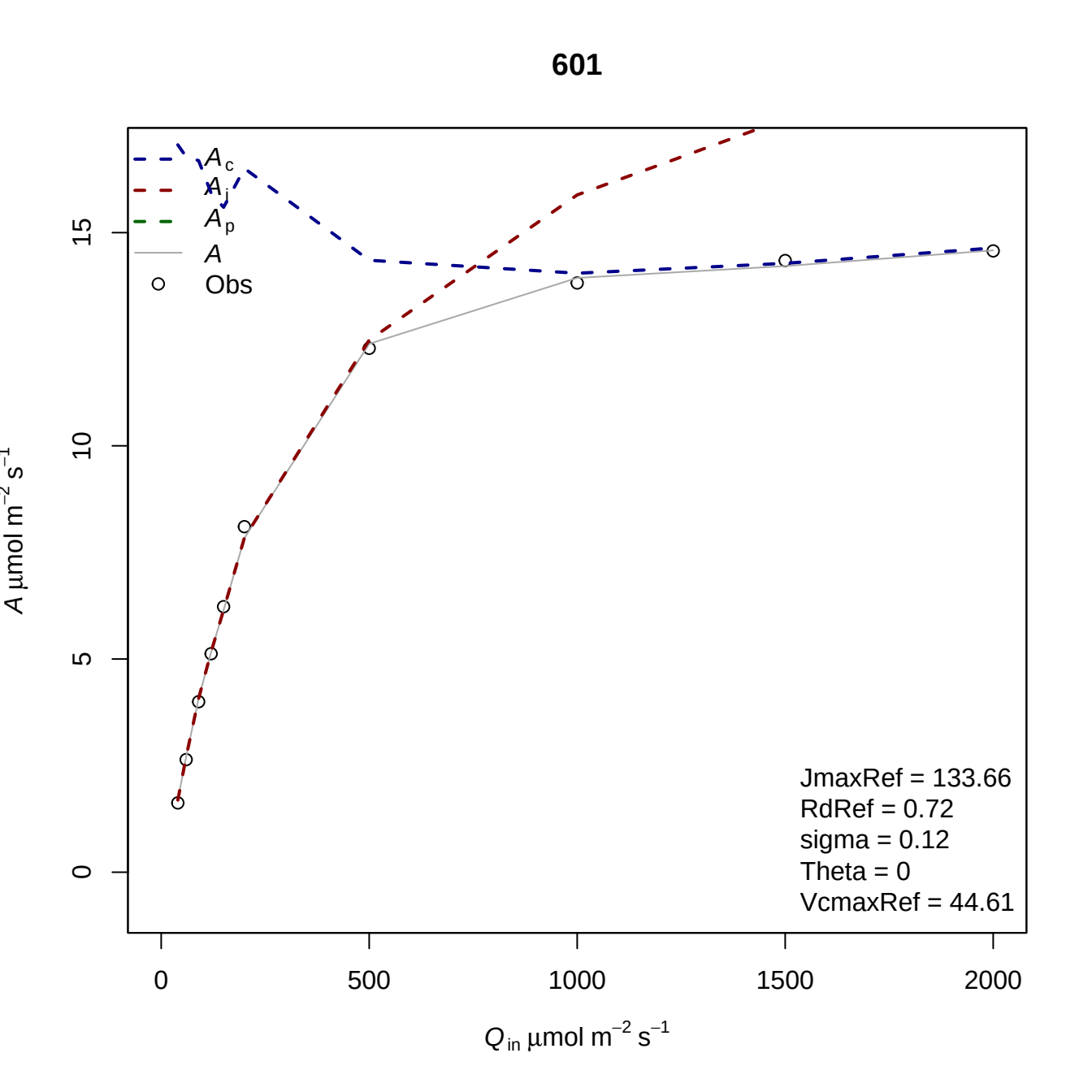

596
A_c
A_j
A_p
A
Obs
JmaxRef = 131.87
RdRef = 0.87
sigma = 0.14
Theta = 0
VcmaxRef = 70.26
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

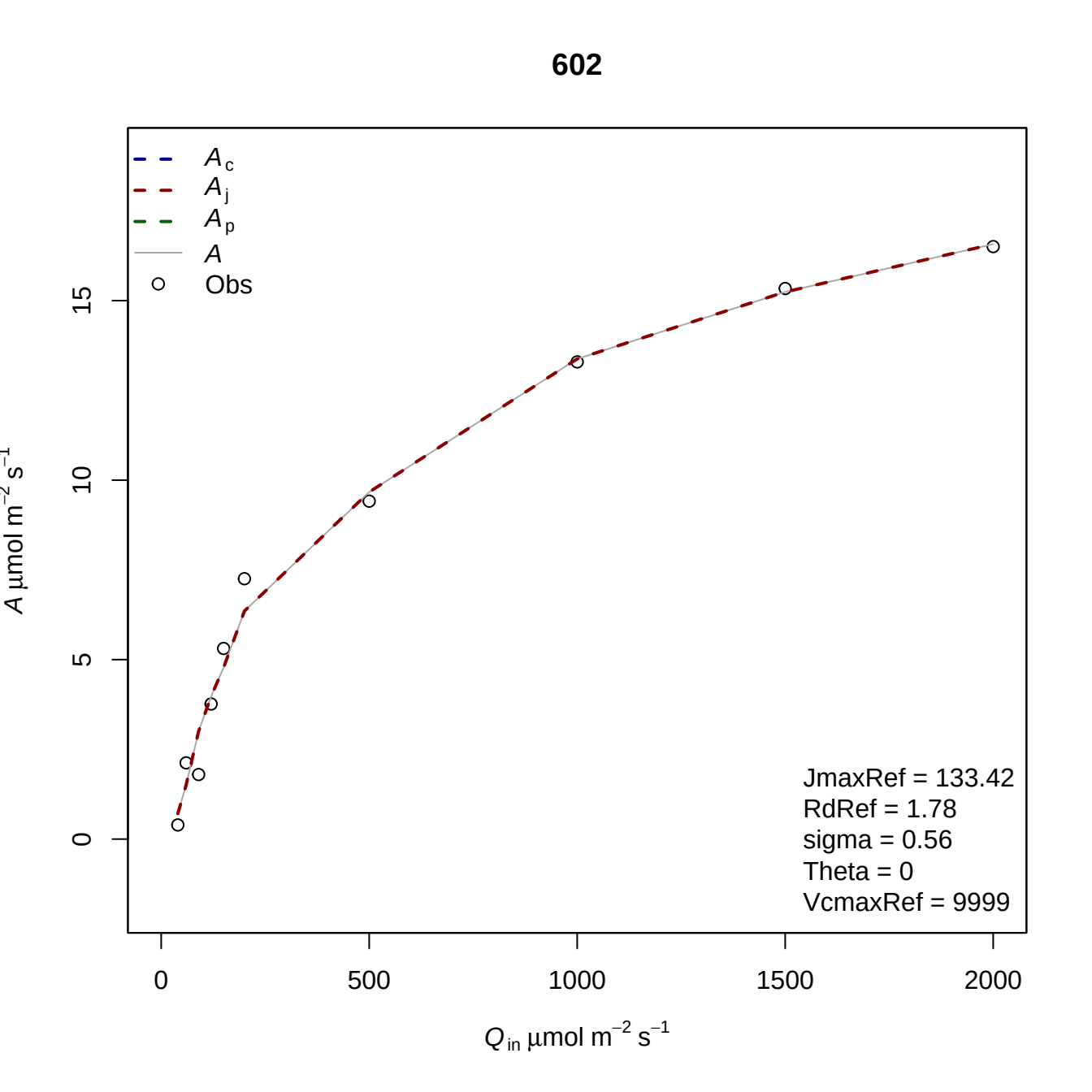

597
A_c
A_j
A_p
A
Obs
JmaxRef = 118.75
RdRef = 1.72
sigma = 0.4
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

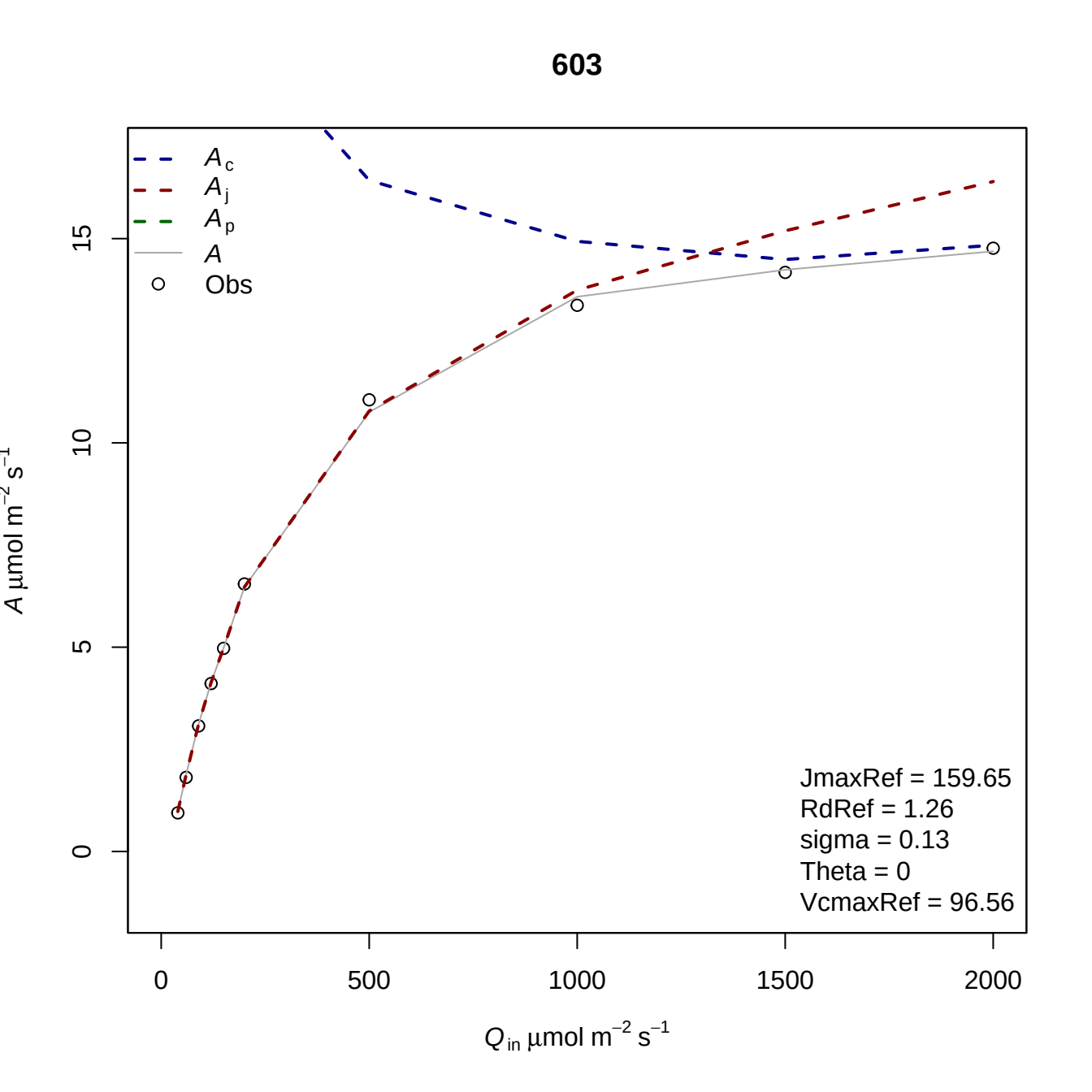

598
A_c
A_j
A_p
A
Obs
JmaxRef = 261.71
RdRef = 2
sigma = 0.14
Theta = 0
VcmaxRef = 149.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

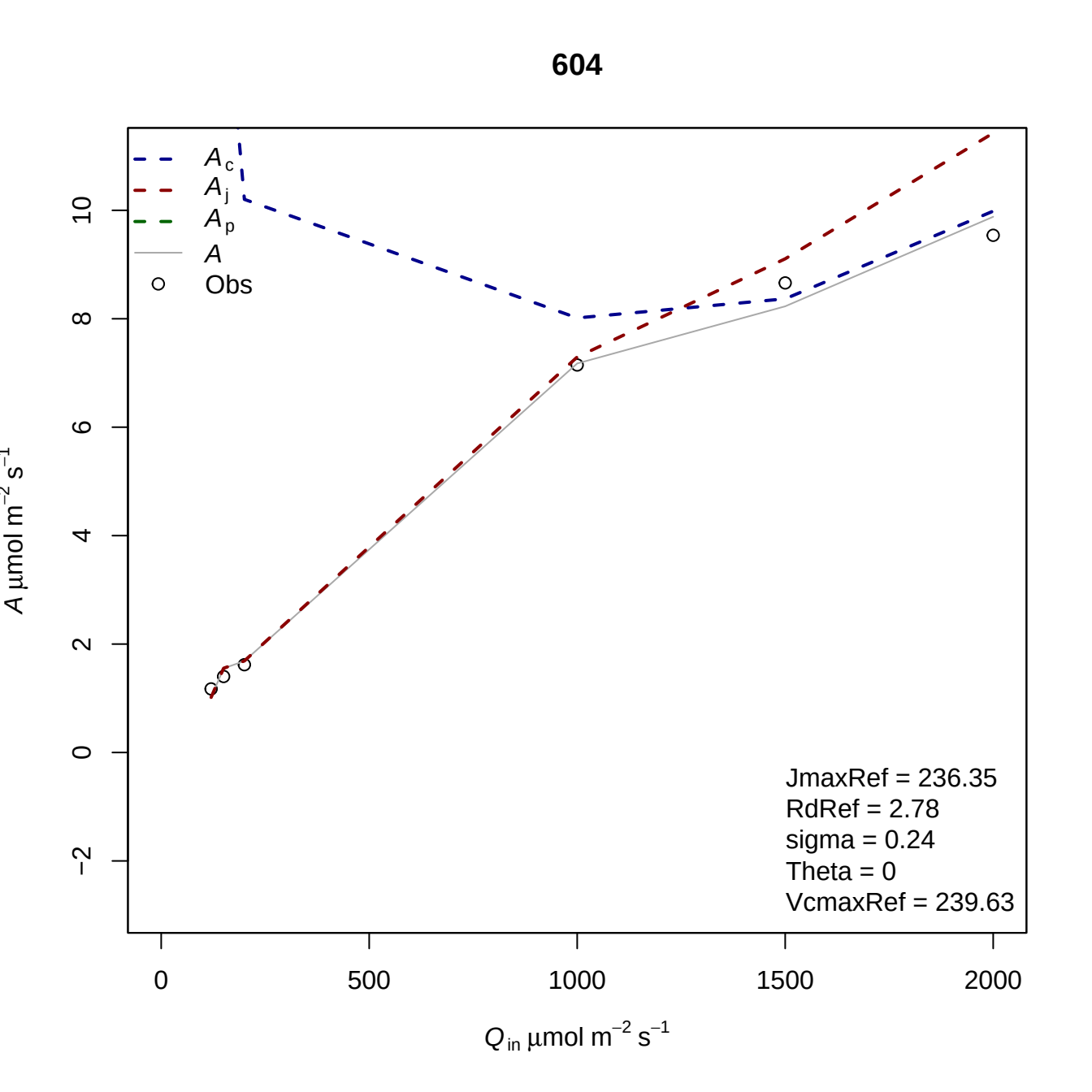

599
A_c
A_j
A_p
A
Obs
JmaxRef = 145.35
RdRef = 2.47
sigma = 0.59
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

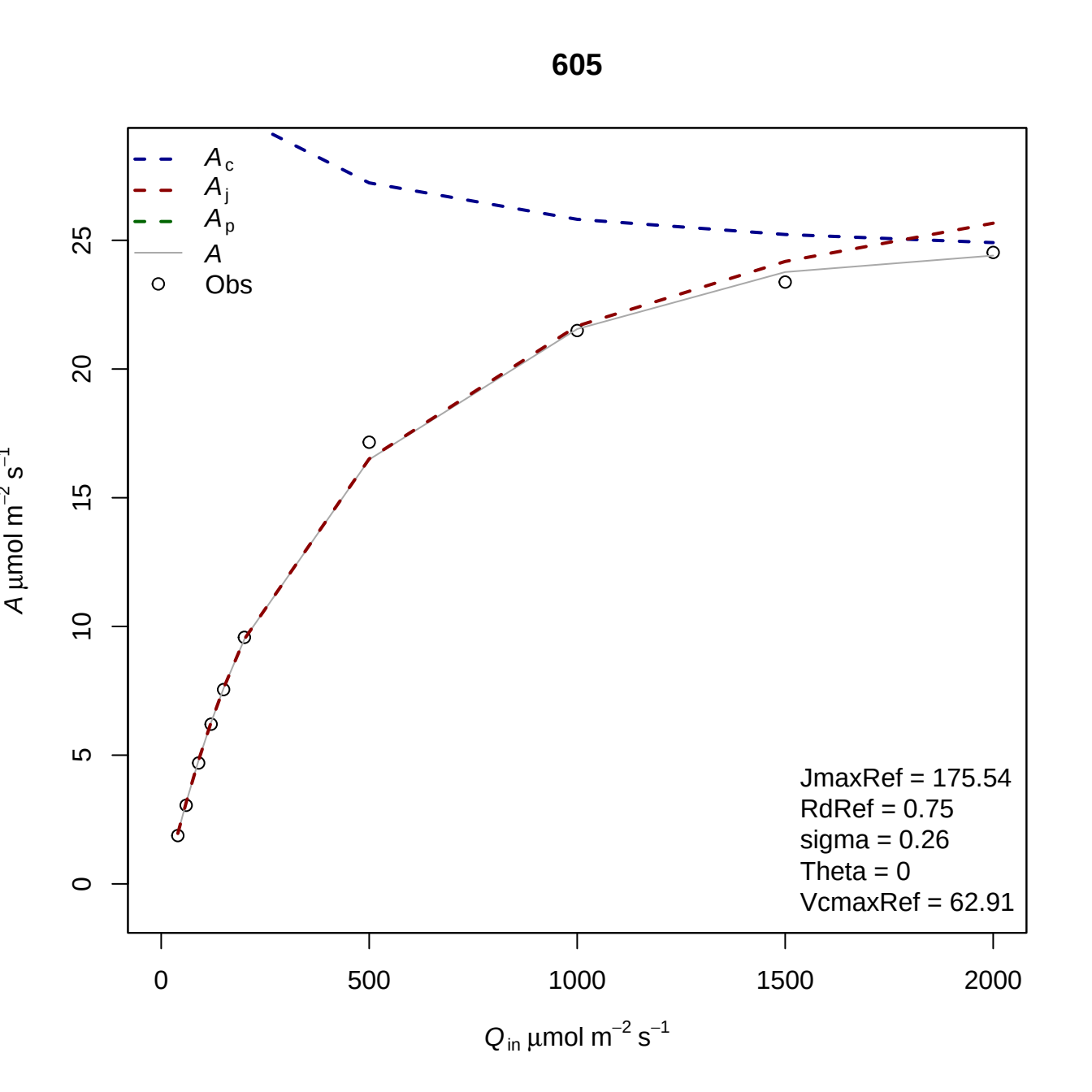

600
A_c
A_j
A_p
A
Obs
JmaxRef = 197.29
RdRef = 3.64
sigma = 0.78
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

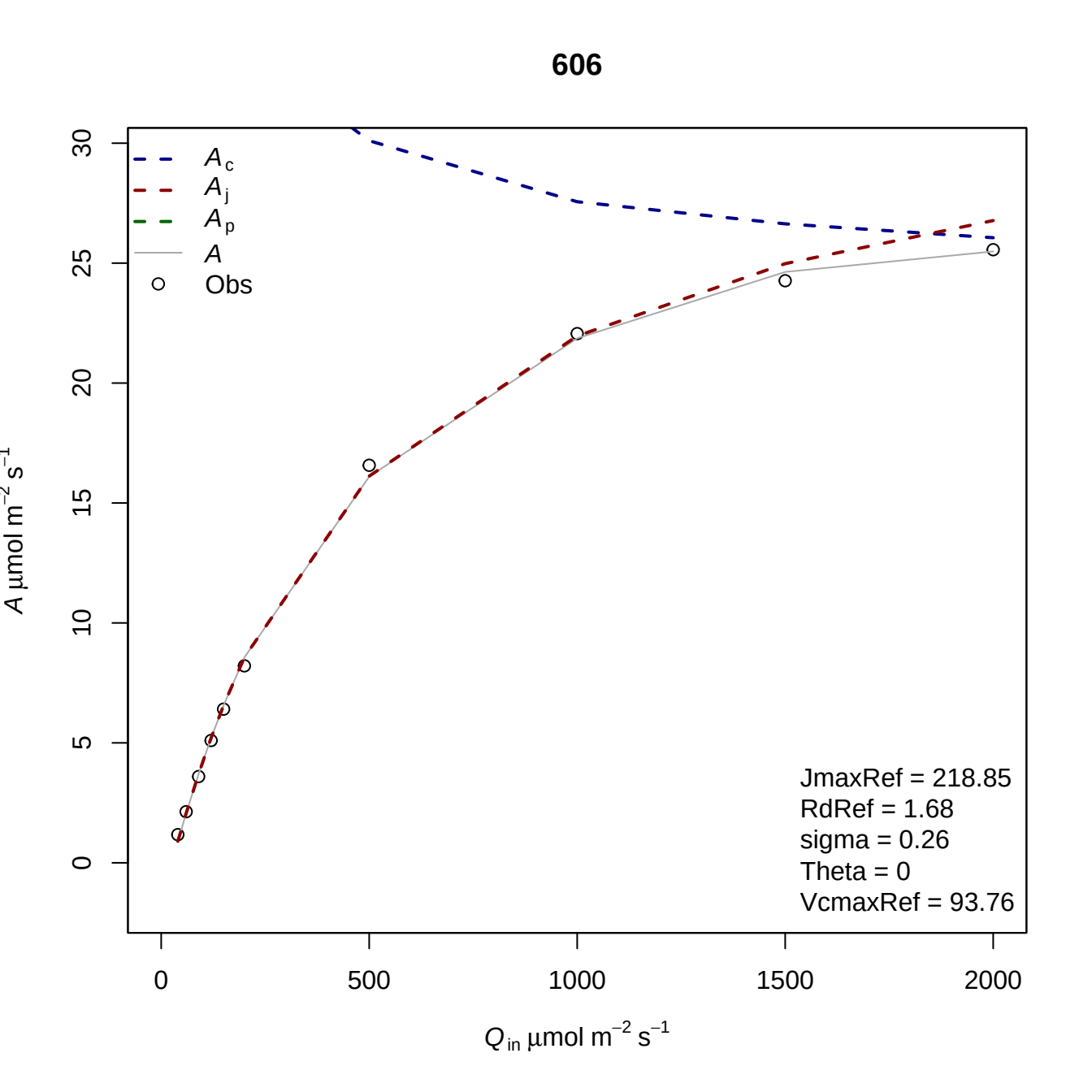

601
A_c
A_j
A_p
A
Obs
JmaxRef = 133.66
RdRef = 0.72
sigma = 0.12
Theta = 0
VcmaxRef = 44.61
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

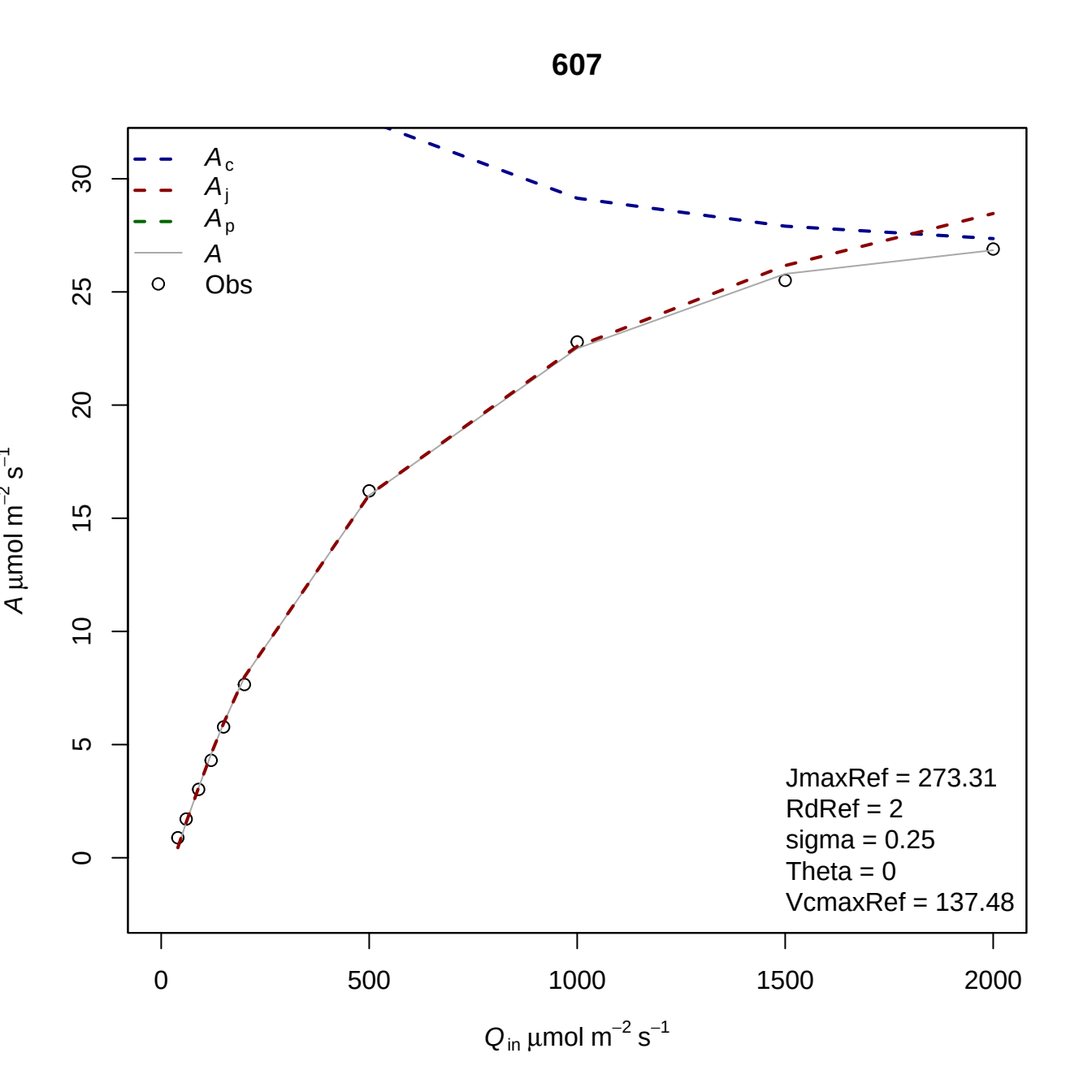

602
A_c
A_j
A_p
A
Obs
JmaxRef = 133.42
RdRef = 1.78
sigma = 0.56
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

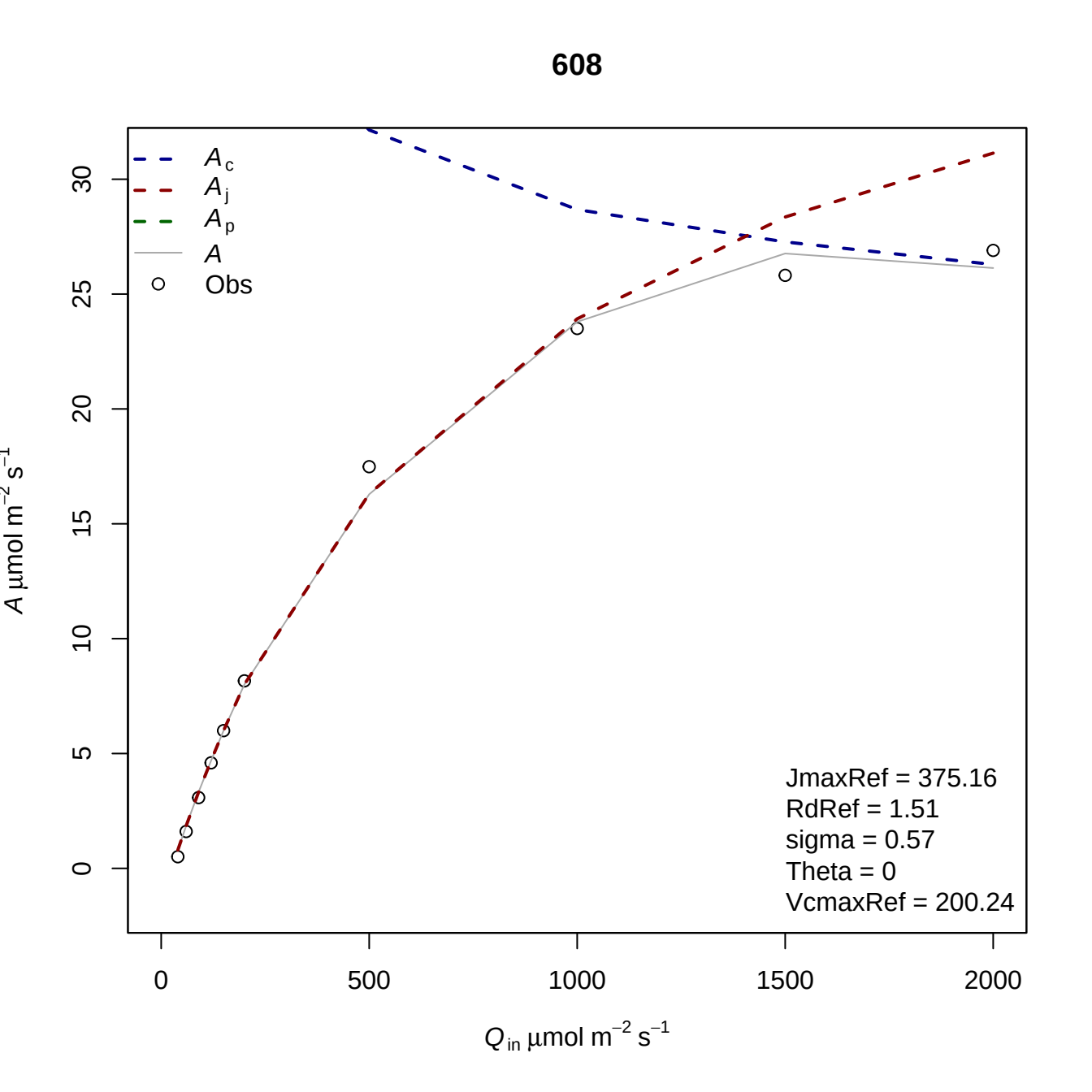

603
A_c
A_j
A_p
A
Obs
JmaxRef = 159.65
RdRef = 1.26
sigma = 0.13
Theta = 0
VcmaxRef = 96.56
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

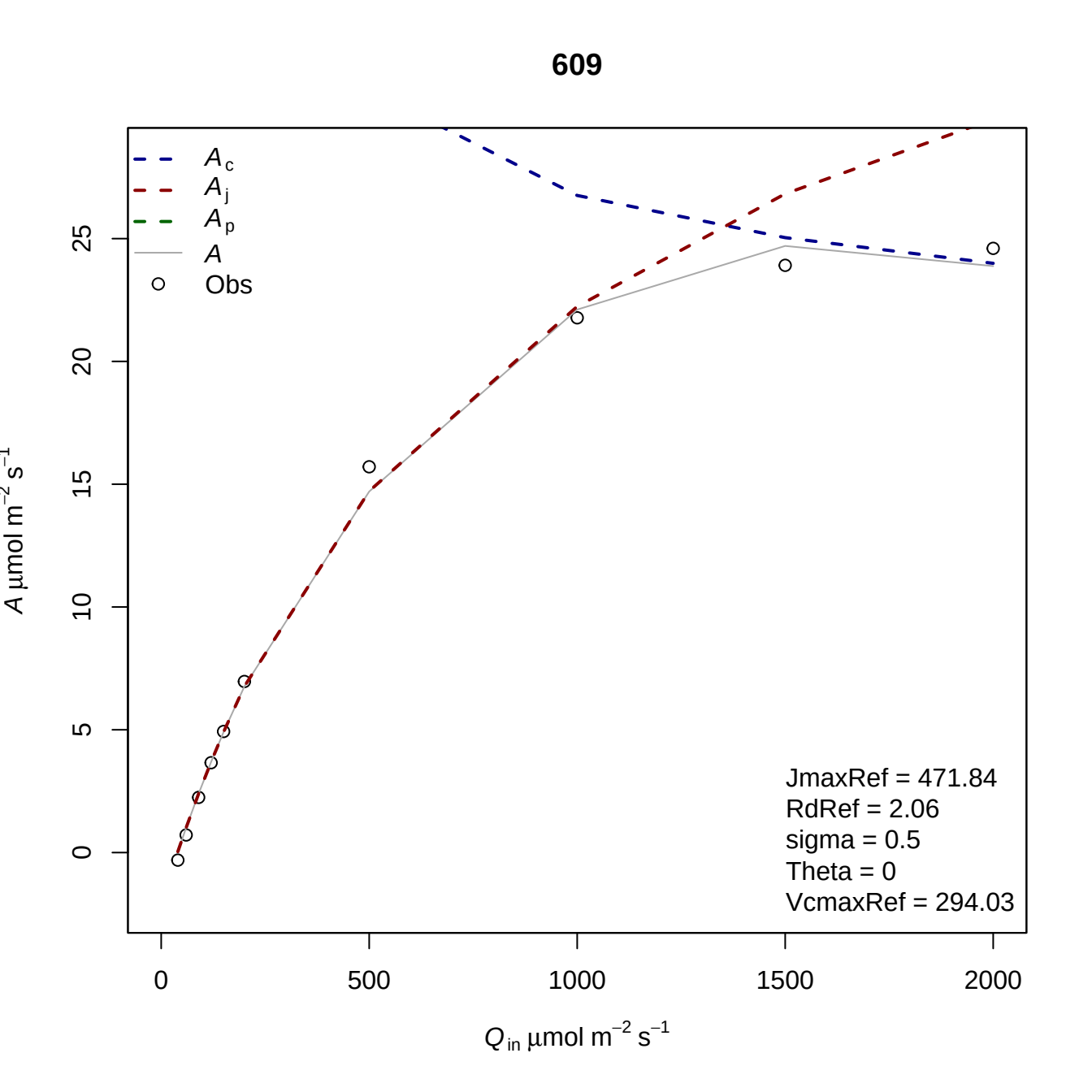

604
A_c
A_j
A_p
A
Obs
JmaxRef = 236.35
RdRef = 2.78
sigma = 0.24
Theta = 0
VcmaxRef = 239.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

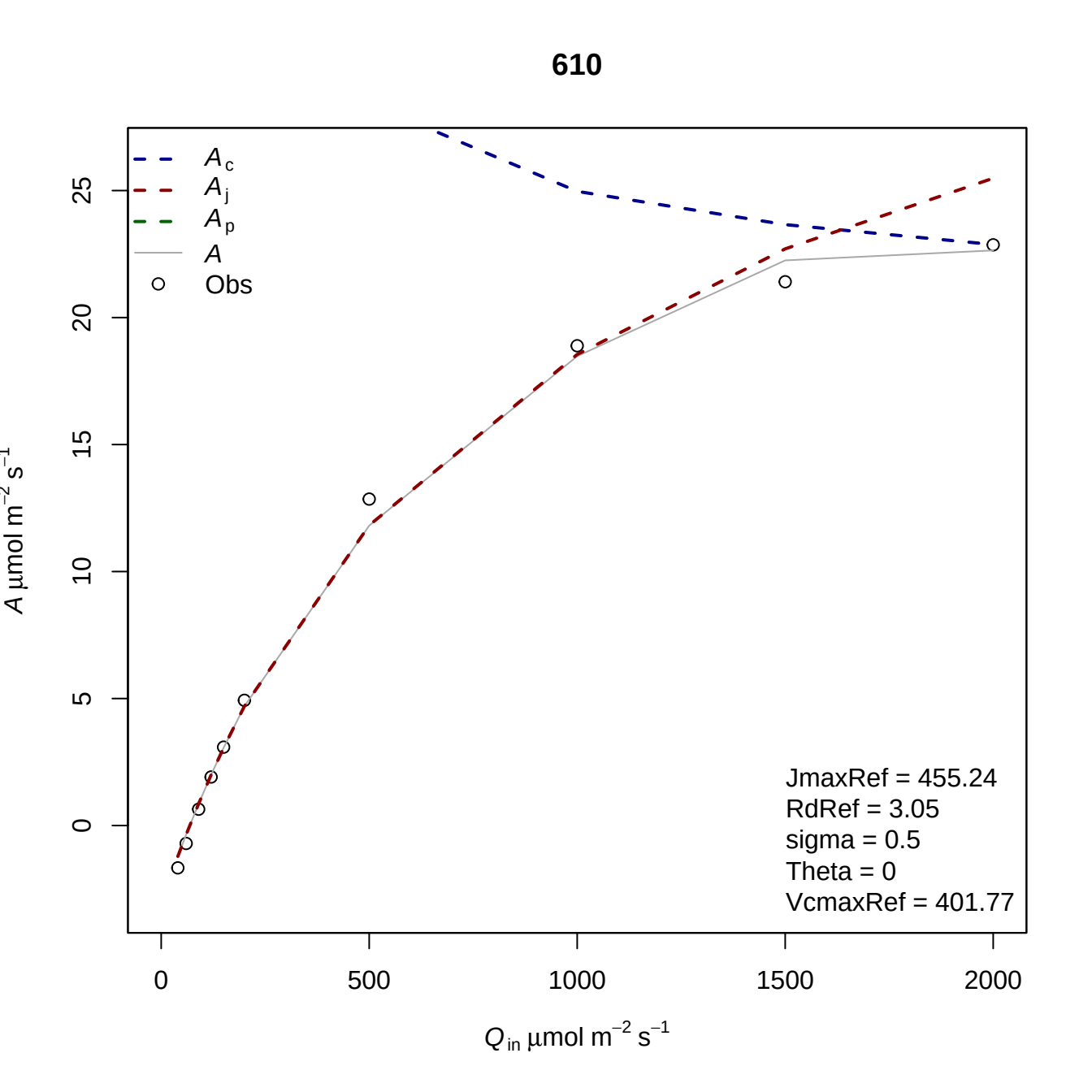

605
A_c
A_j
A_p
A
Obs
JmaxRef = 175.54
RdRef = 0.75
sigma = 0.26
Theta = 0
VcmaxRef = 62.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

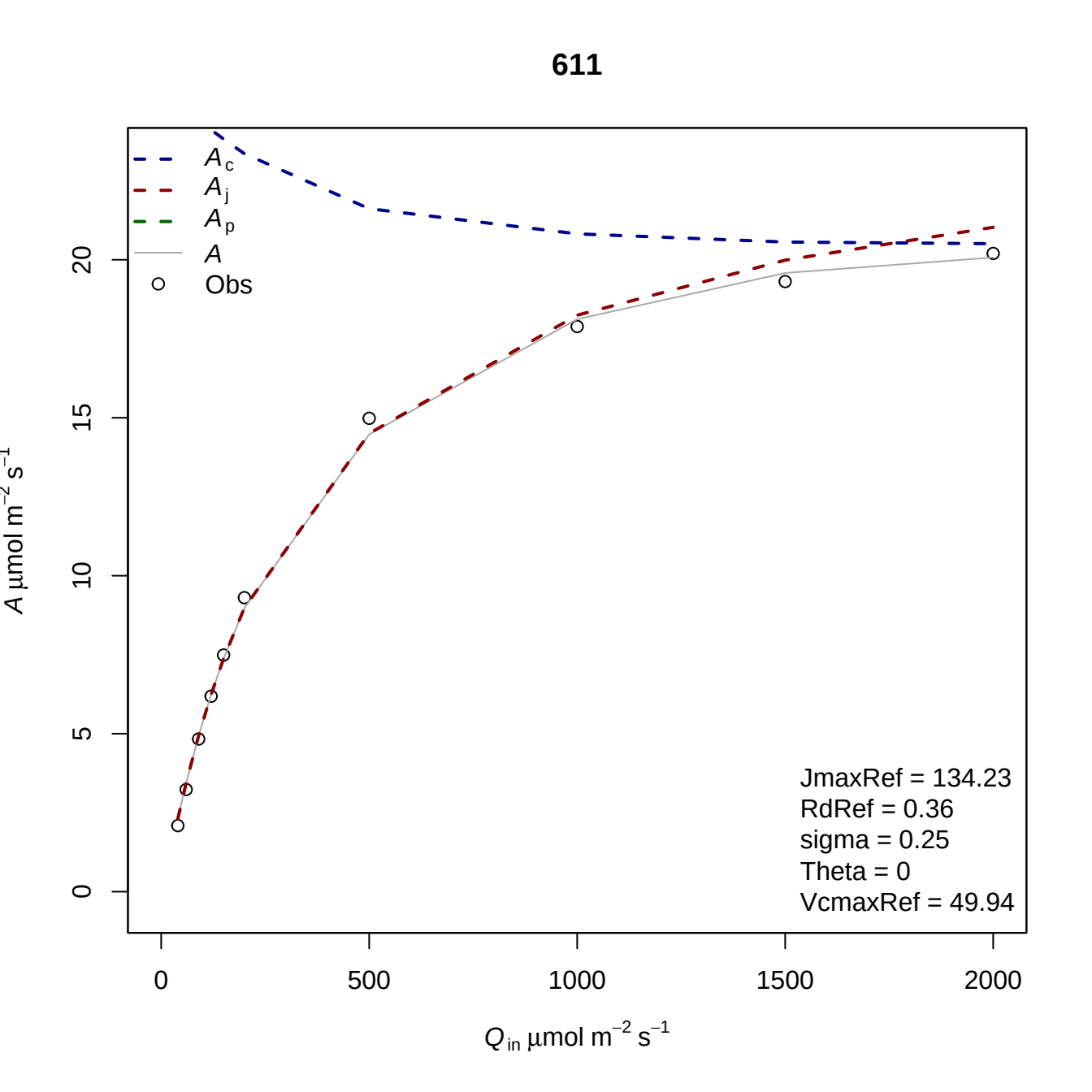

606
A_c
A_j
A_p
A
Obs
JmaxRef = 218.85
RdRef = 1.68
sigma = 0.26
Theta = 0
VcmaxRef = 93.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

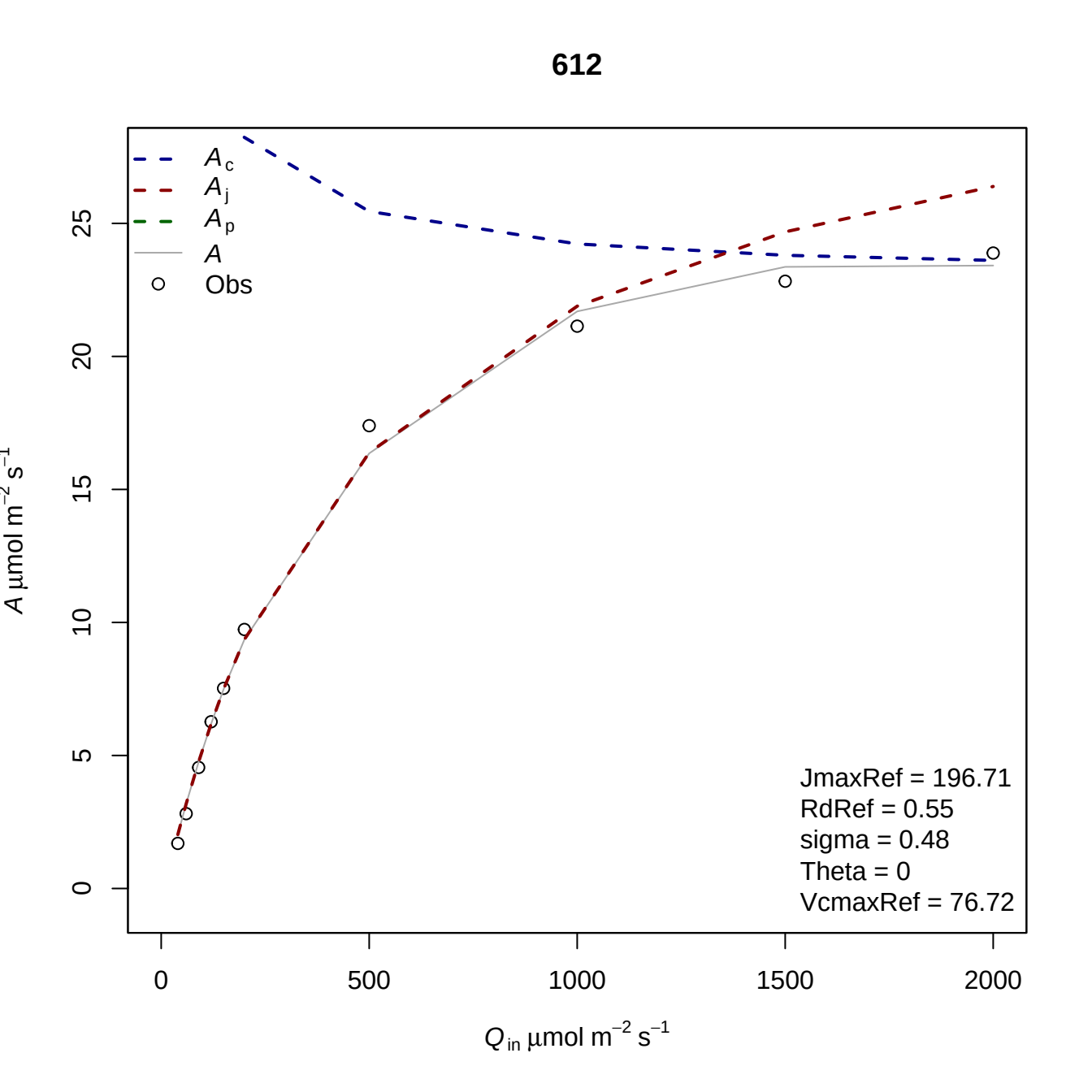

607
A_c
A_j
A_p
A
Obs
JmaxRef = 273.31
RdRef = 2
sigma = 0.25
Theta = 0
VcmaxRef = 137.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

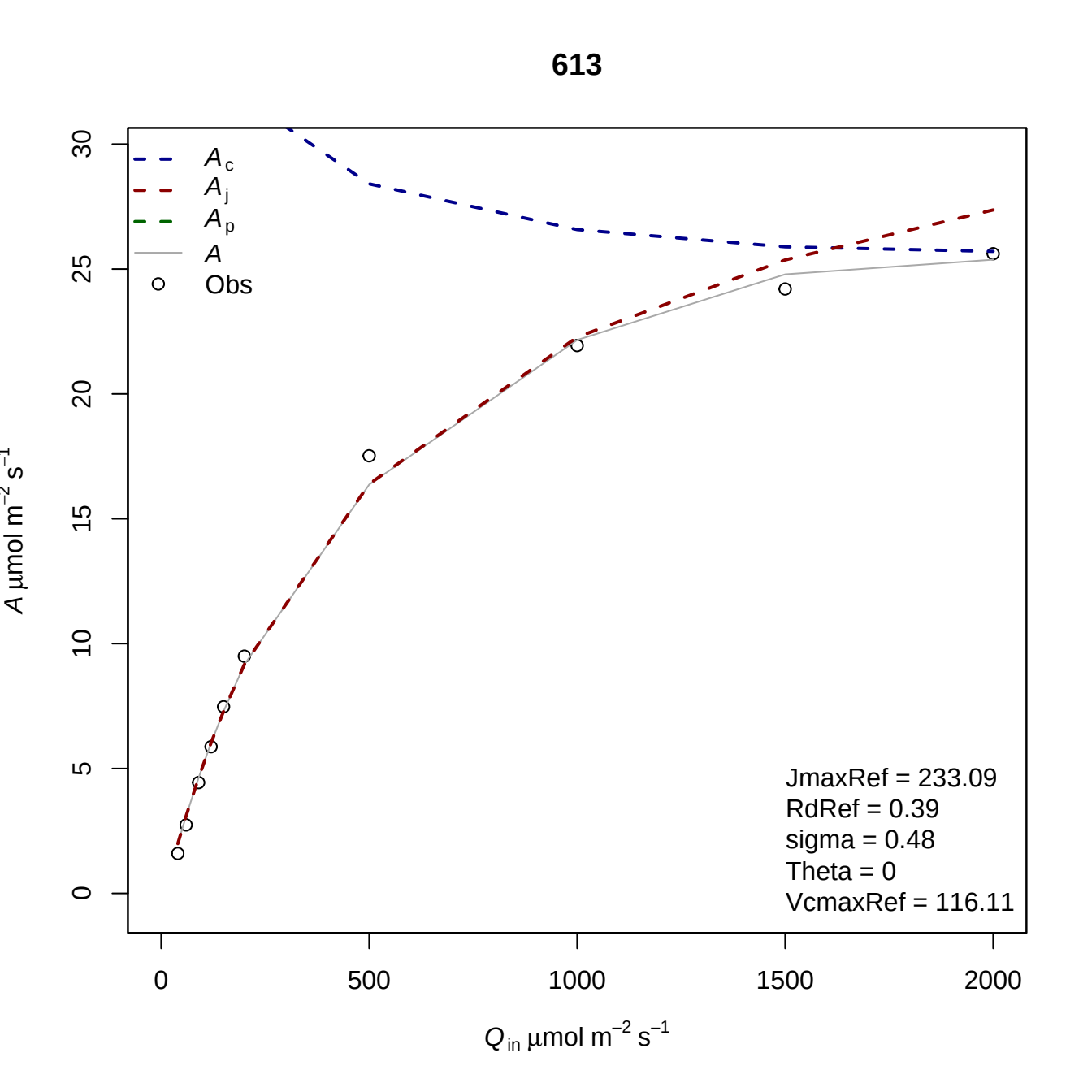

608
A_c
A_j
A_p
A
Obs
JmaxRef = 375.16
RdRef = 1.51
sigma = 0.57
Theta = 0
VcmaxRef = 200.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

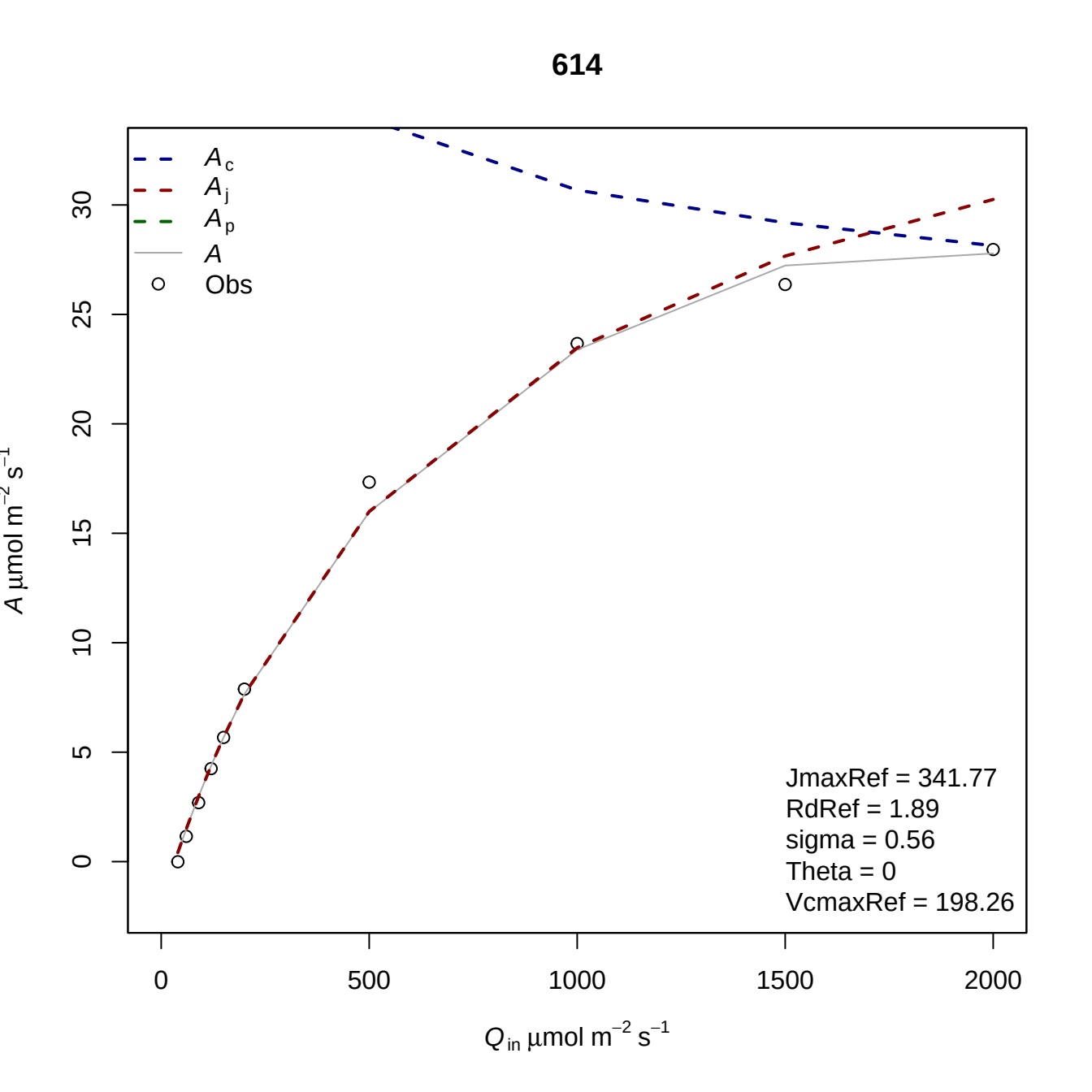

609
A_c
A_j
A_p
A
Obs
JmaxRef = 471.84
RdRef = 2.06
sigma = 0.5
Theta = 0
VcmaxRef = 294.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

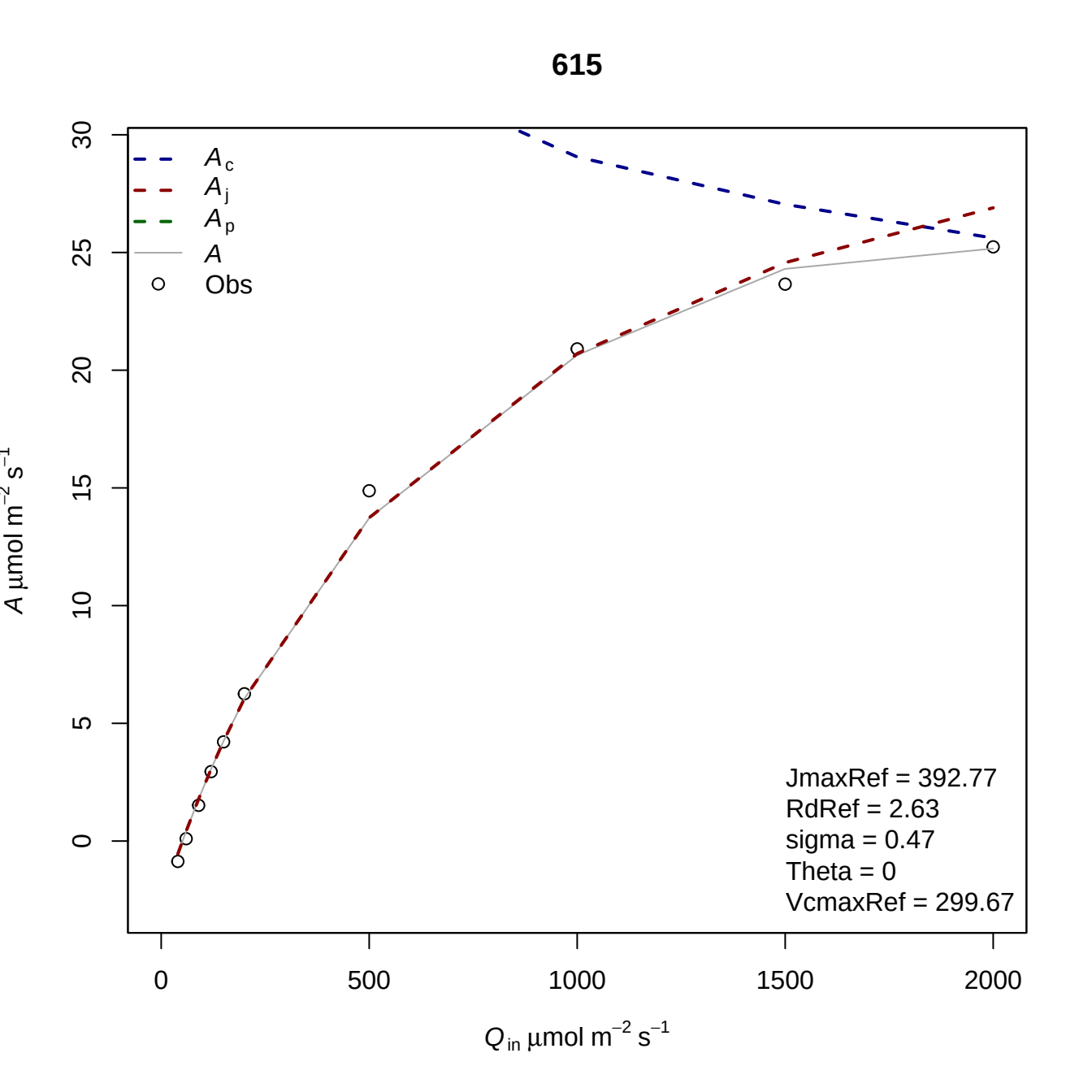

610
A_c
A_j
A_p
A
Obs
JmaxRef = 455.24
RdRef = 3.05
sigma = 0.5
Theta = 0
VcmaxRef = 401.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

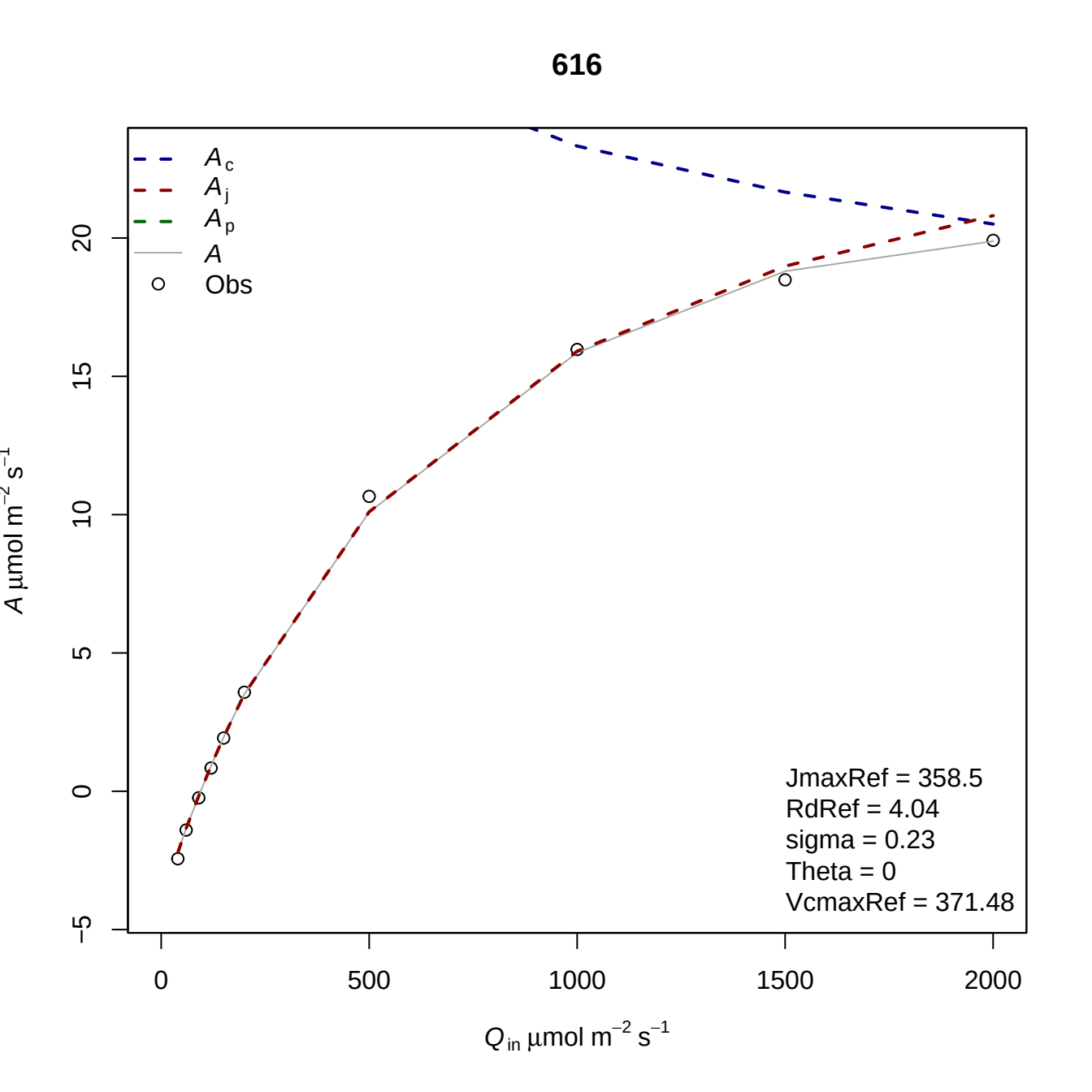

611
A_c
A_j
A_p
A
Obs
JmaxRef = 134.23
RdRef = 0.36
sigma = 0.25
Theta = 0
VcmaxRef = 49.94
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

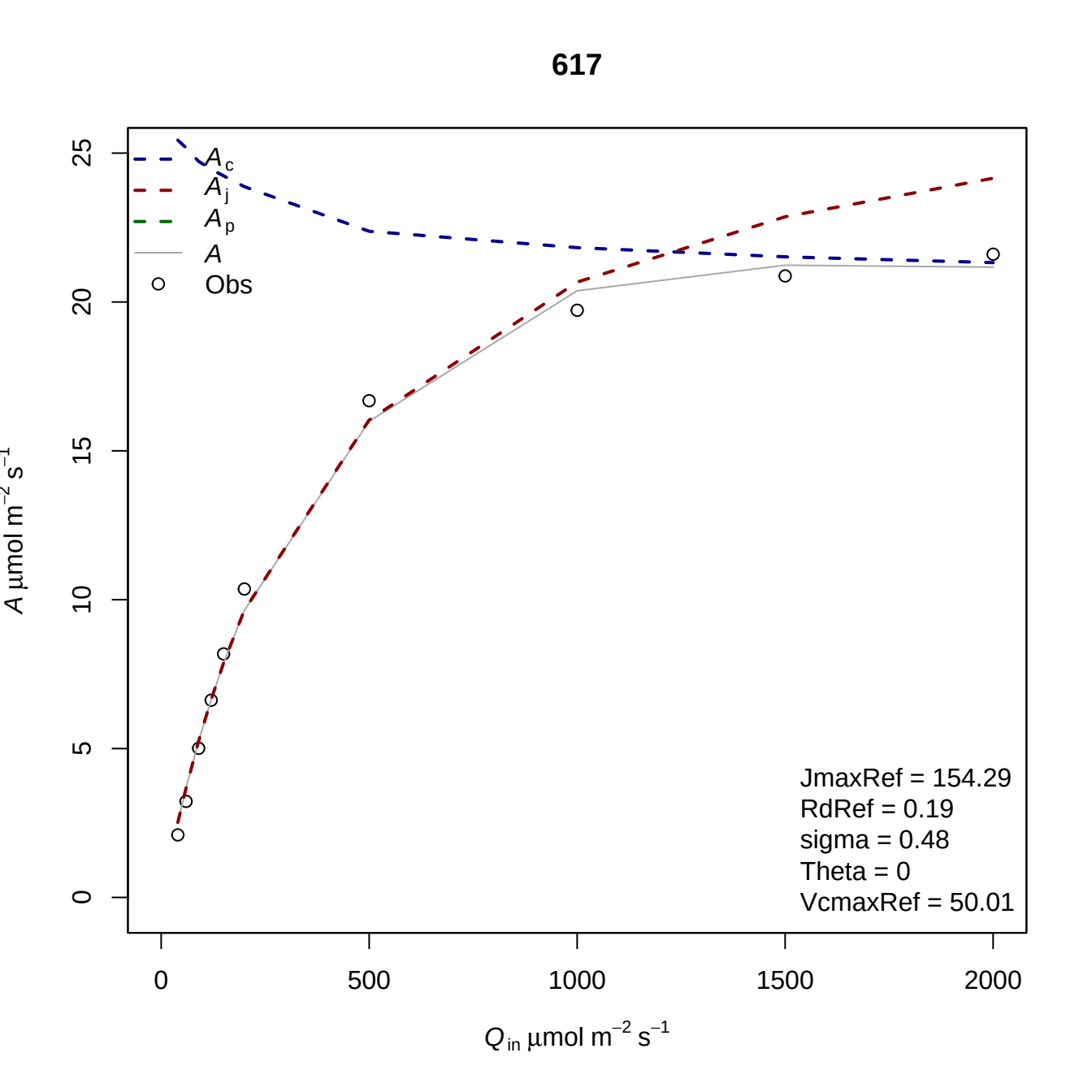

612
A_c
A_j
A_p
A
Obs
JmaxRef = 196.71
RdRef = 0.55
sigma = 0.48
Theta = 0
VcmaxRef = 76.72
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

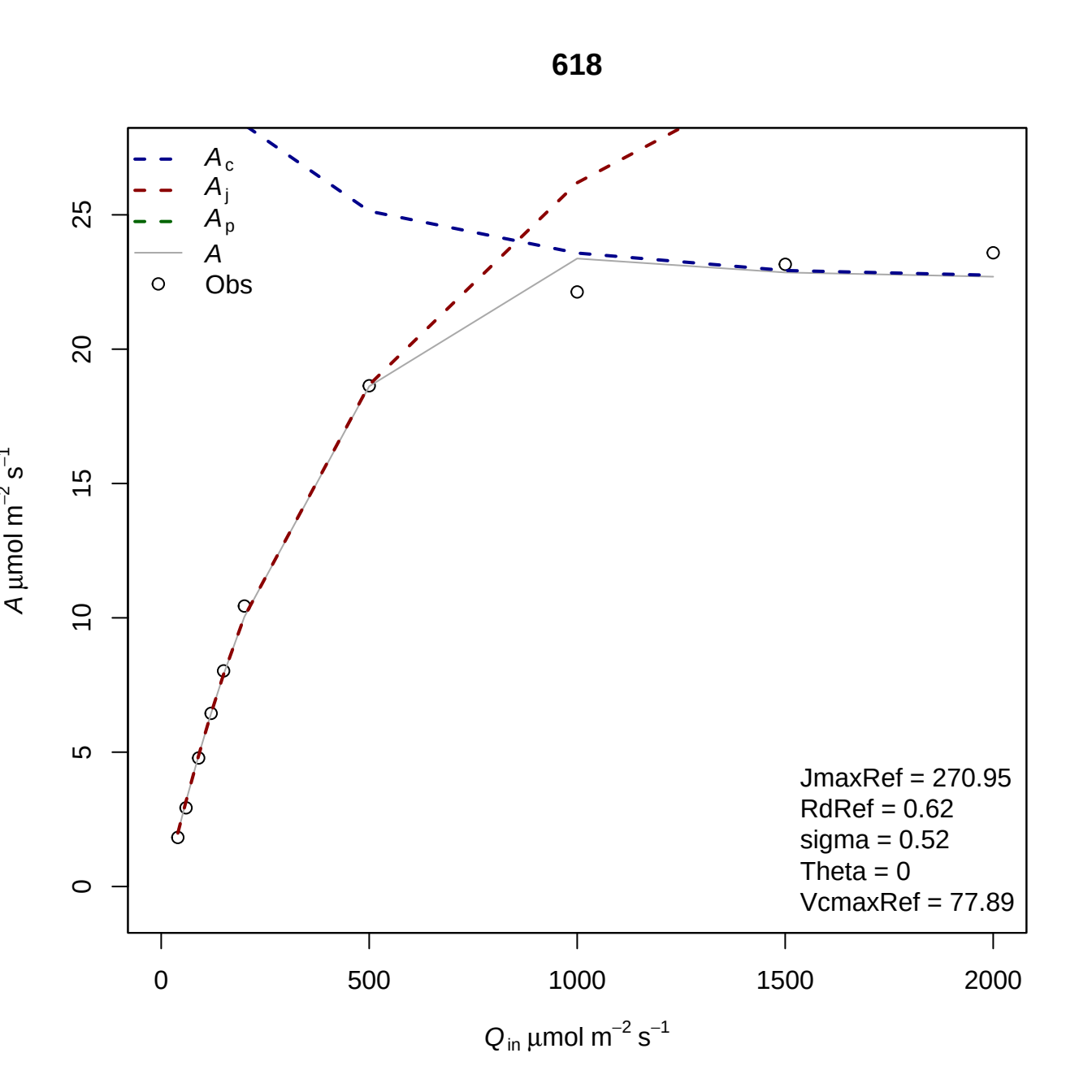

613
A_c
A_j
A_p
A
Obs
JmaxRef = 233.09
RdRef = 0.39
sigma = 0.48
Theta = 0
VcmaxRef = 116.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

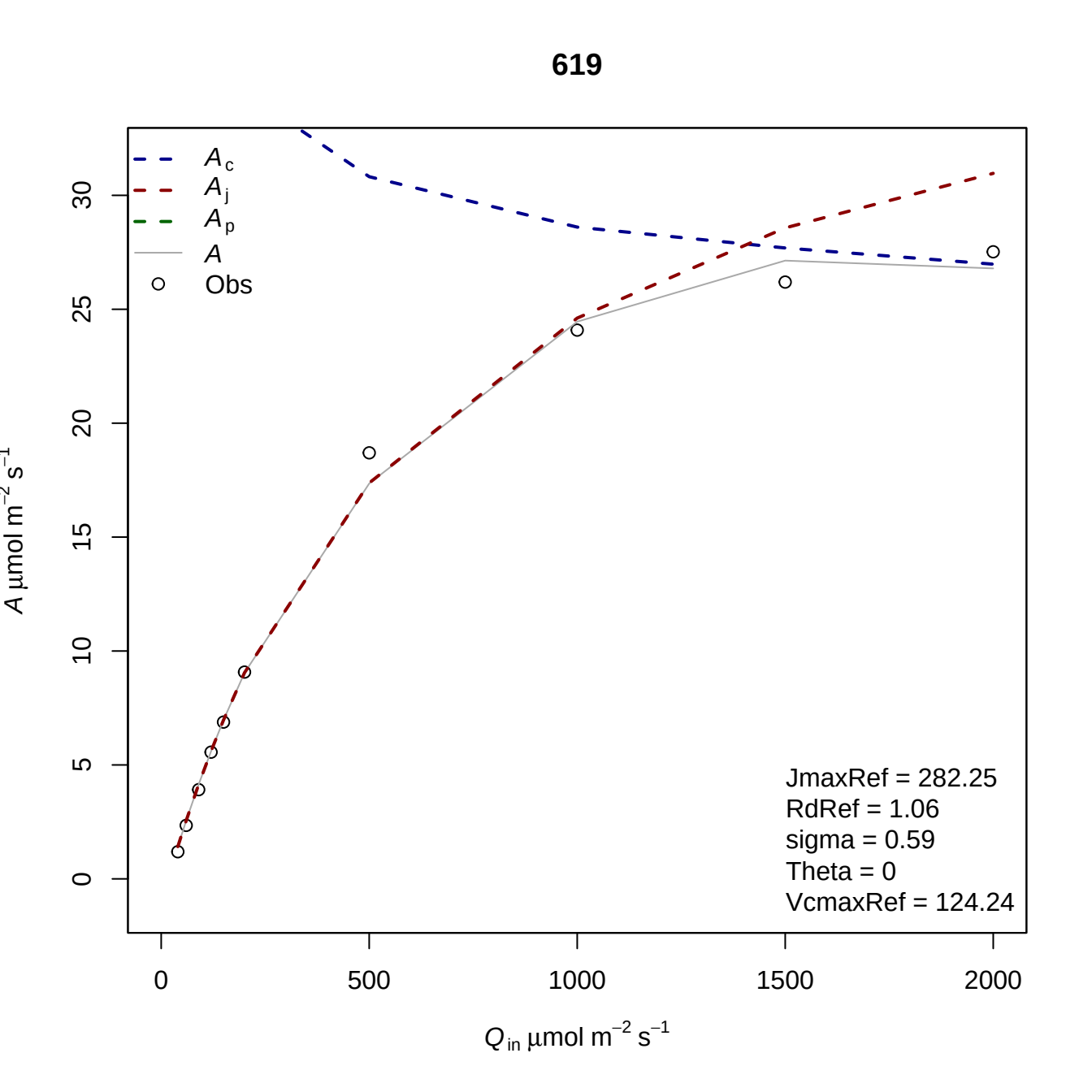

614
A_c
A_j
A_p
A
Obs
JmaxRef = 341.77
RdRef = 1.89
sigma = 0.56
Theta = 0
VcmaxRef = 198.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

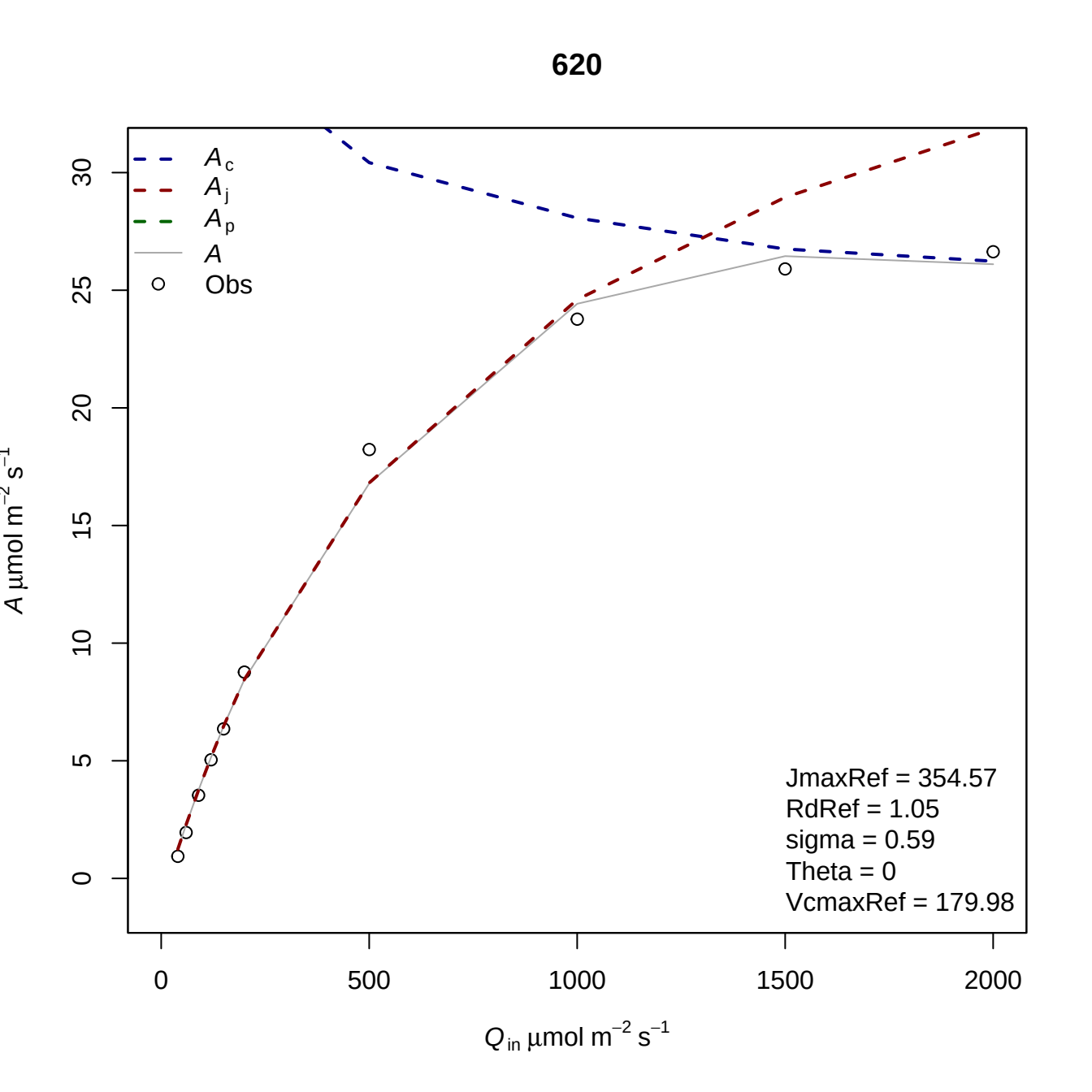

615
A_c
A_j
A_p
A
Obs
JmaxRef = 392.77
RdRef = 2.63
sigma = 0.47
Theta = 0
VcmaxRef = 299.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

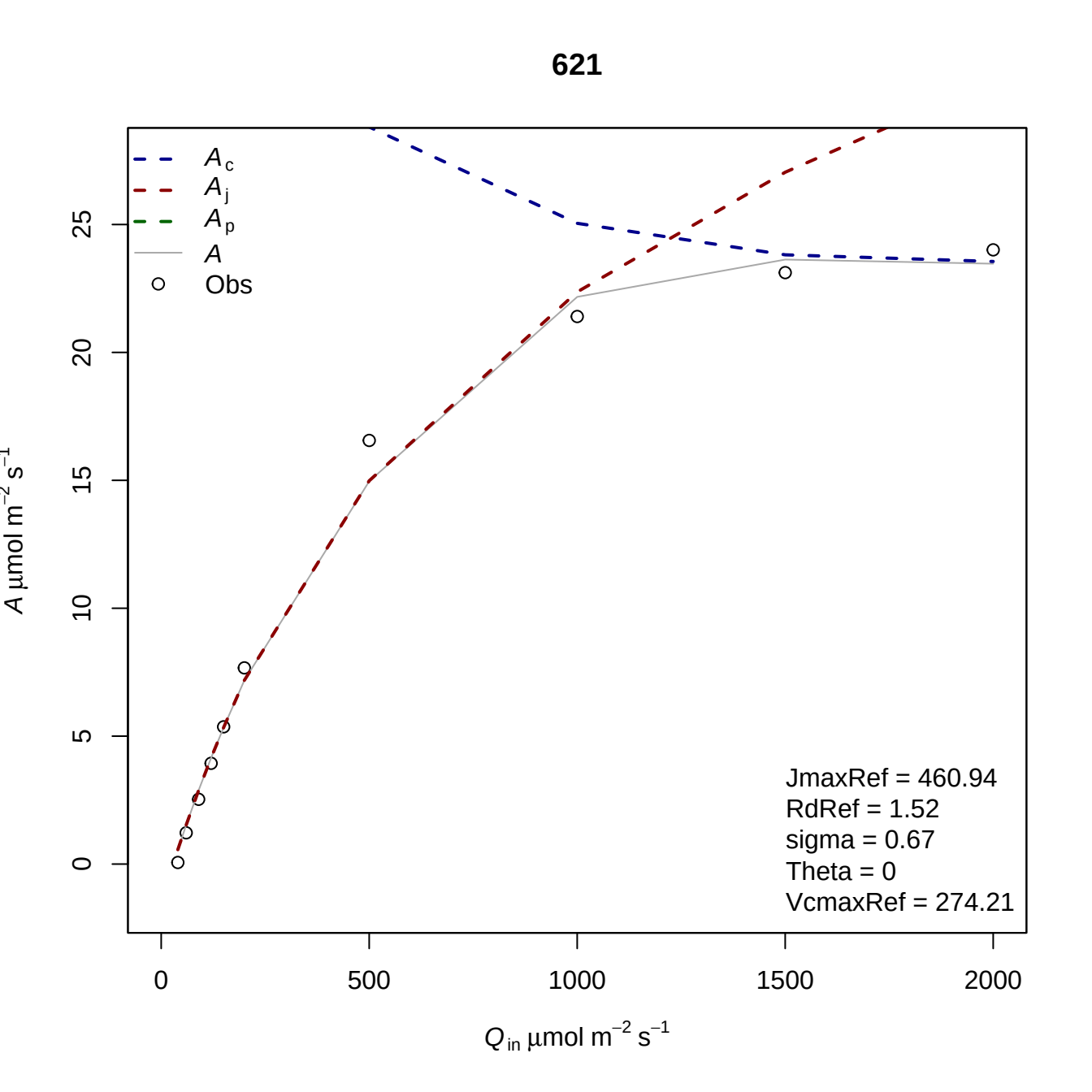

616
A_c
A_j
A_p
A
Obs
JmaxRef = 358.5
RdRef = 4.04
sigma = 0.23
Theta = 0
VcmaxRef = 371.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

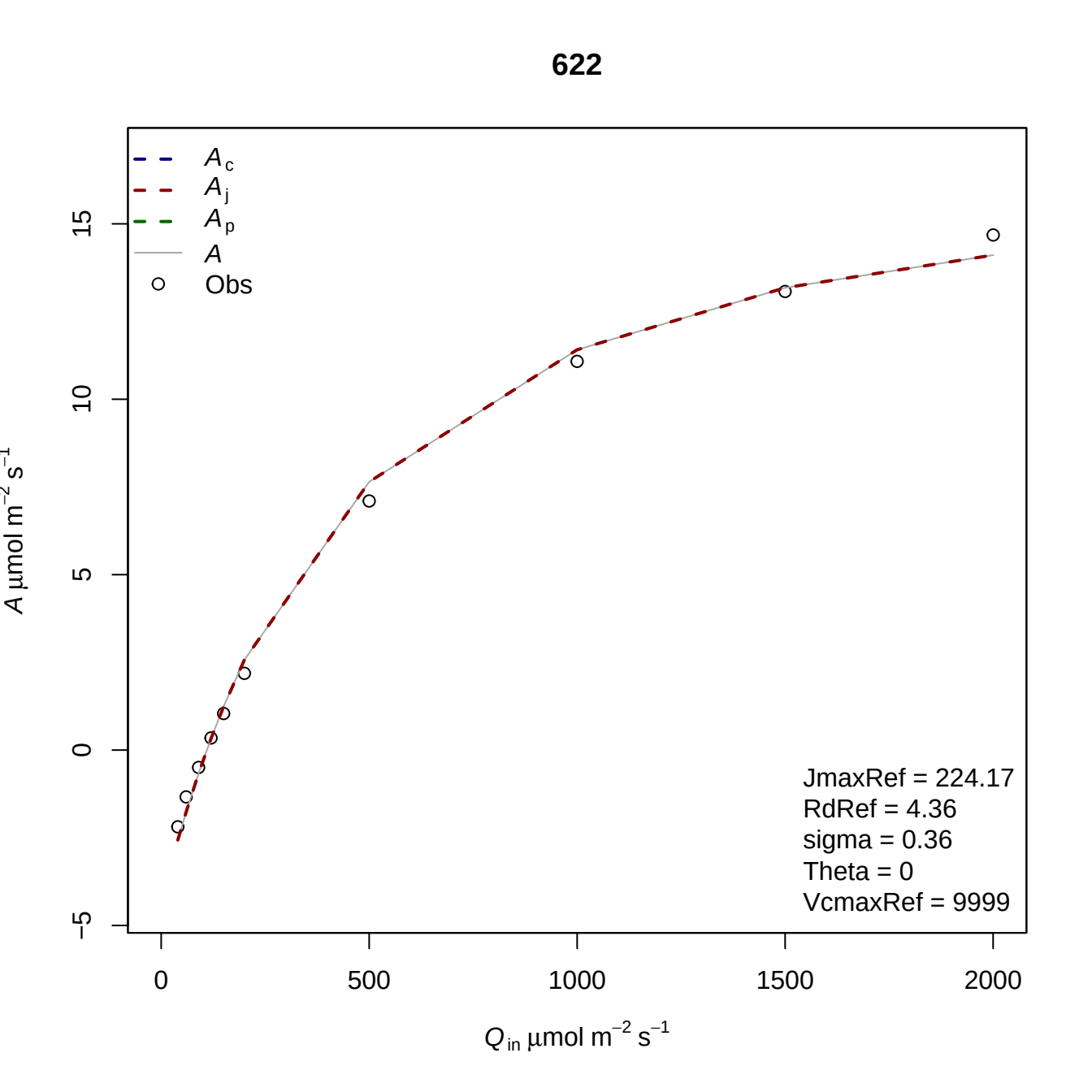

617
A_c
A_j
A_p
A
Obs
JmaxRef = 154.29
RdRef = 0.19
sigma = 0.48
Theta = 0
VcmaxRef = 50.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

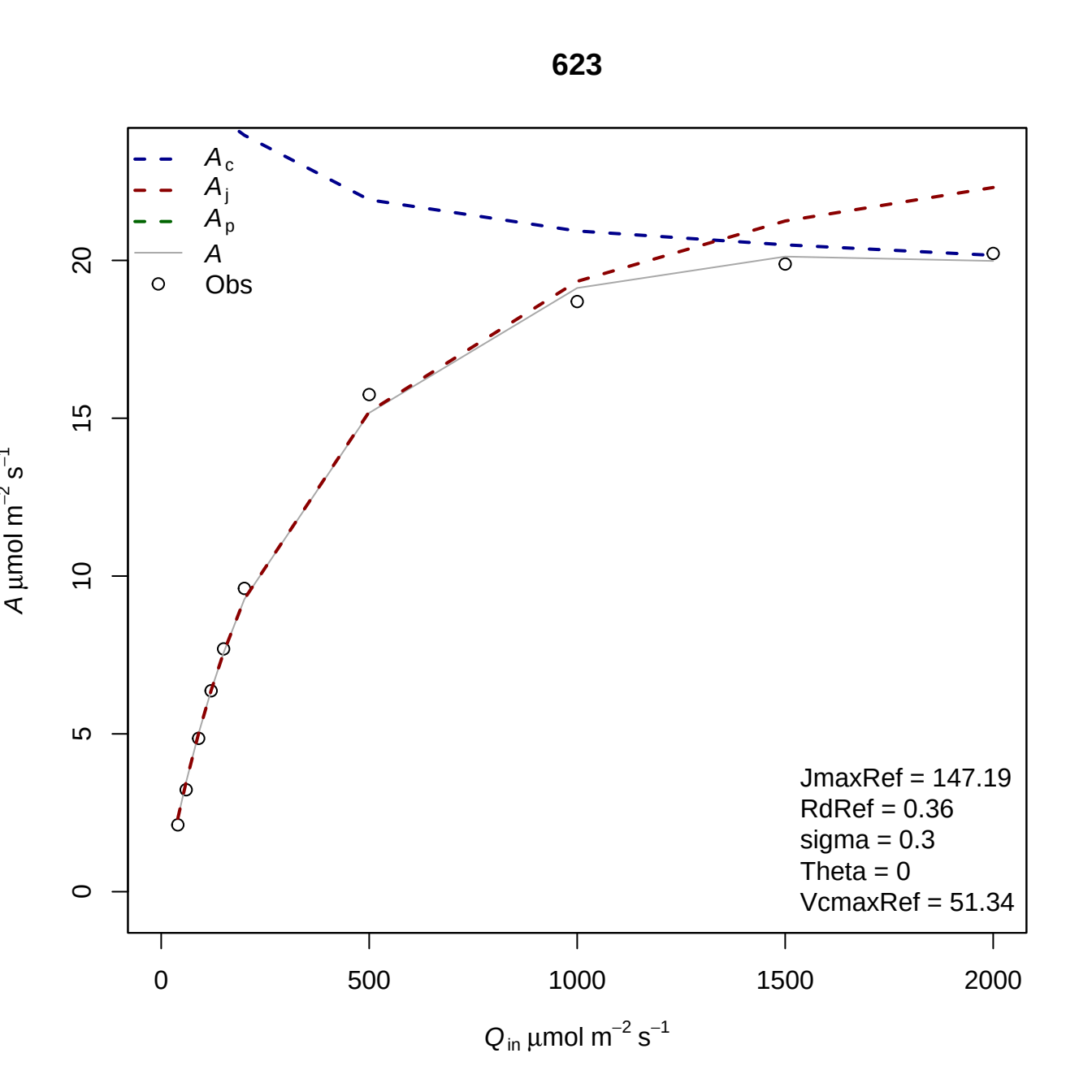

618
A_c
A_j
A_p
A
Obs
JmaxRef = 270.95
RdRef = 0.62
sigma = 0.52
Theta = 0
VcmaxRef = 77.89
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

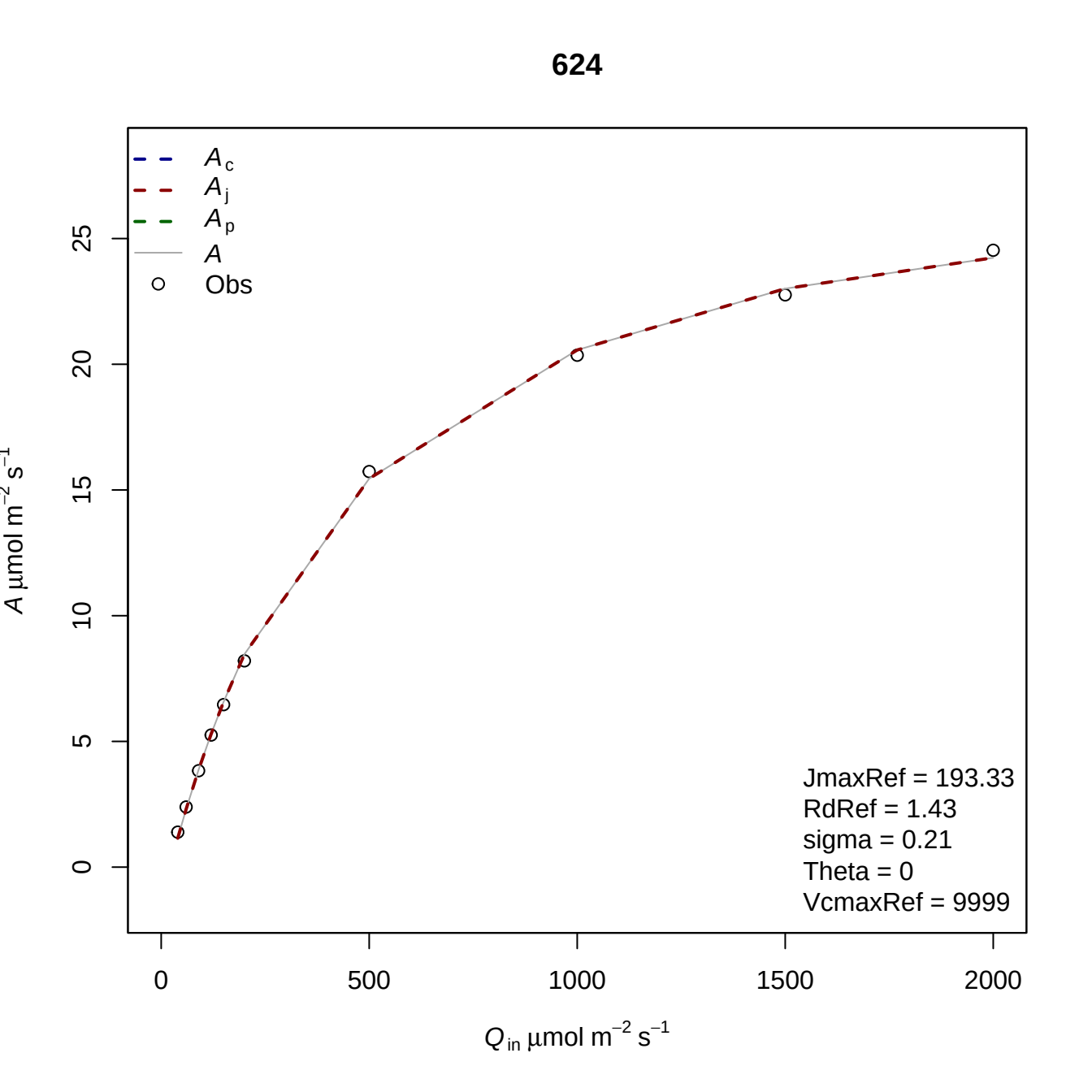

619
A_c
A_j
A_p
A
Obs
JmaxRef = 282.25
RdRef = 1.06
sigma = 0.59
Theta = 0
VcmaxRef = 124.24
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

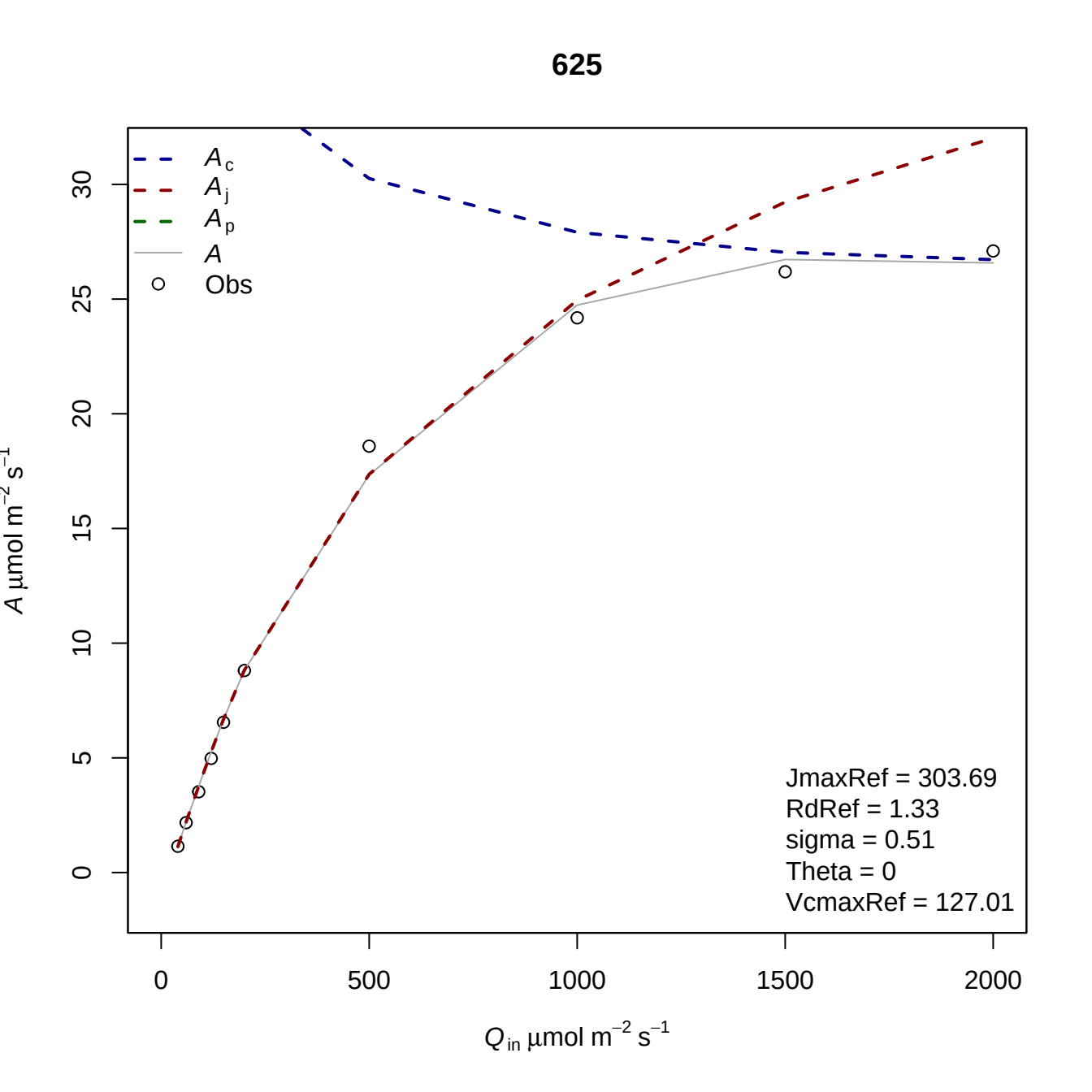

620
A_c
A_j
A_p
A
Obs
JmaxRef = 354.57
RdRef = 1.05
sigma = 0.59
Theta = 0
VcmaxRef = 179.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

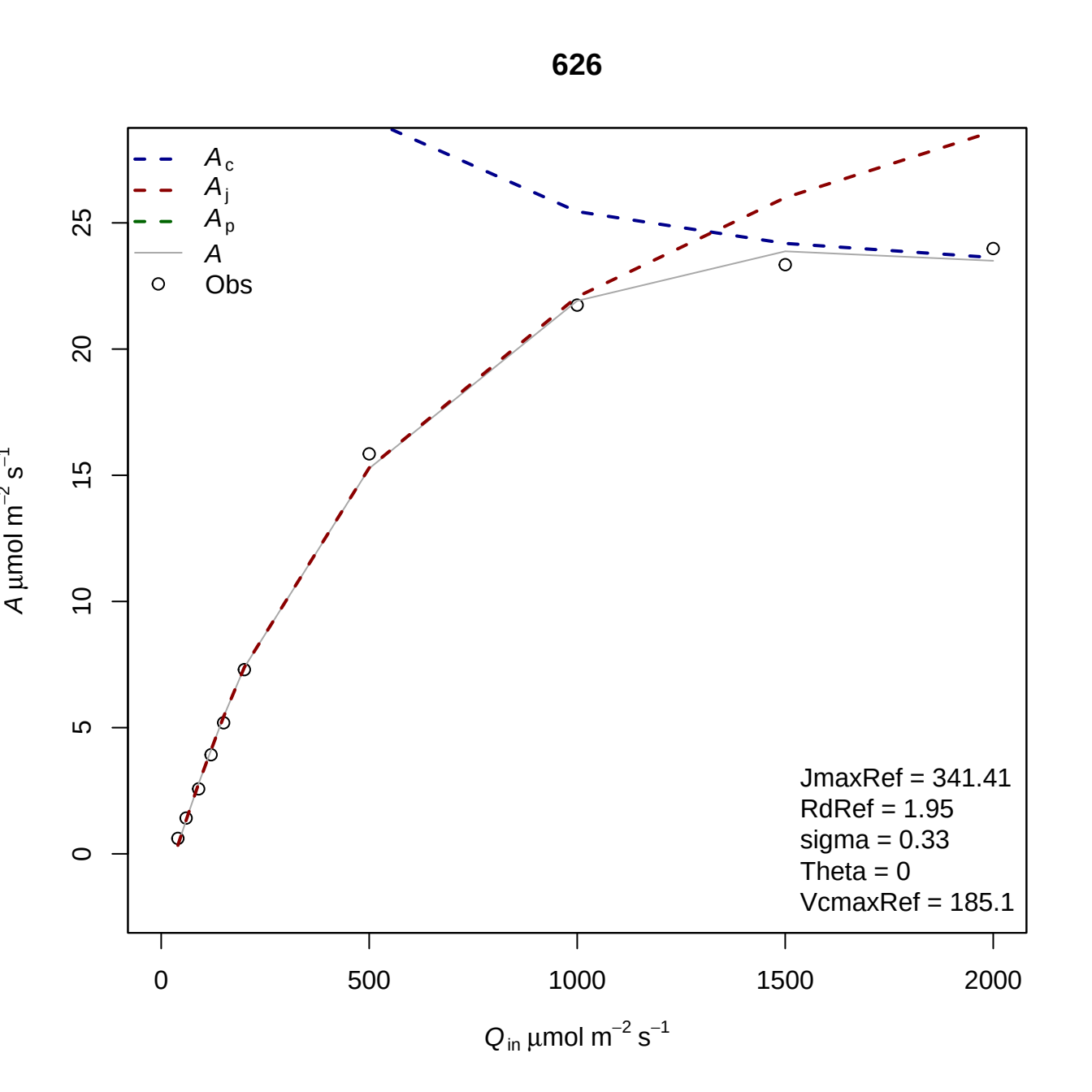

621
A_c
A_j
A_p
A
Obs
JmaxRef = 460.94
RdRef = 1.52
sigma = 0.67
Theta = 0
VcmaxRef = 274.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

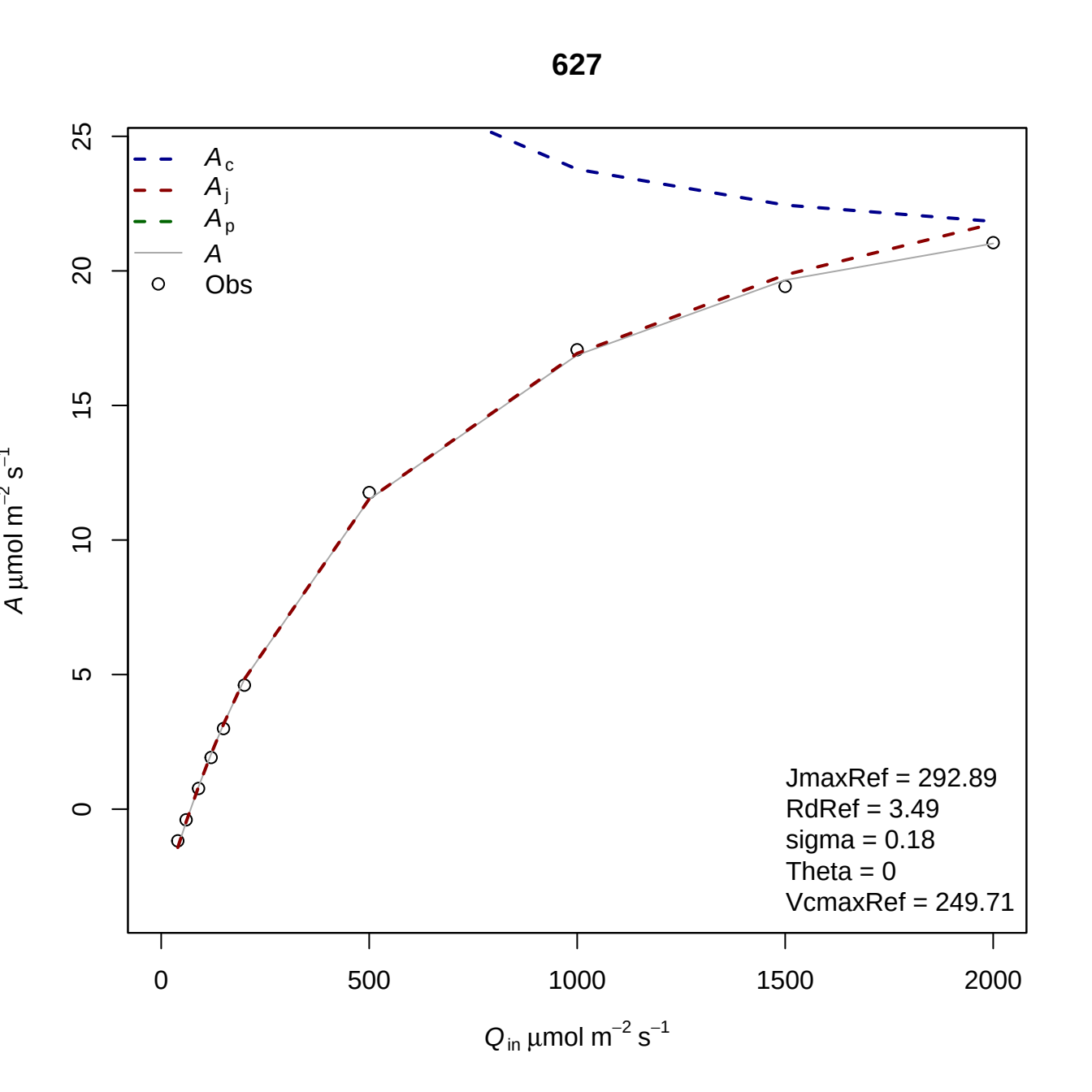

622
A_c
A_j
A_p
A
Obs
JmaxRef = 224.17
RdRef = 4.36
sigma = 0.36
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

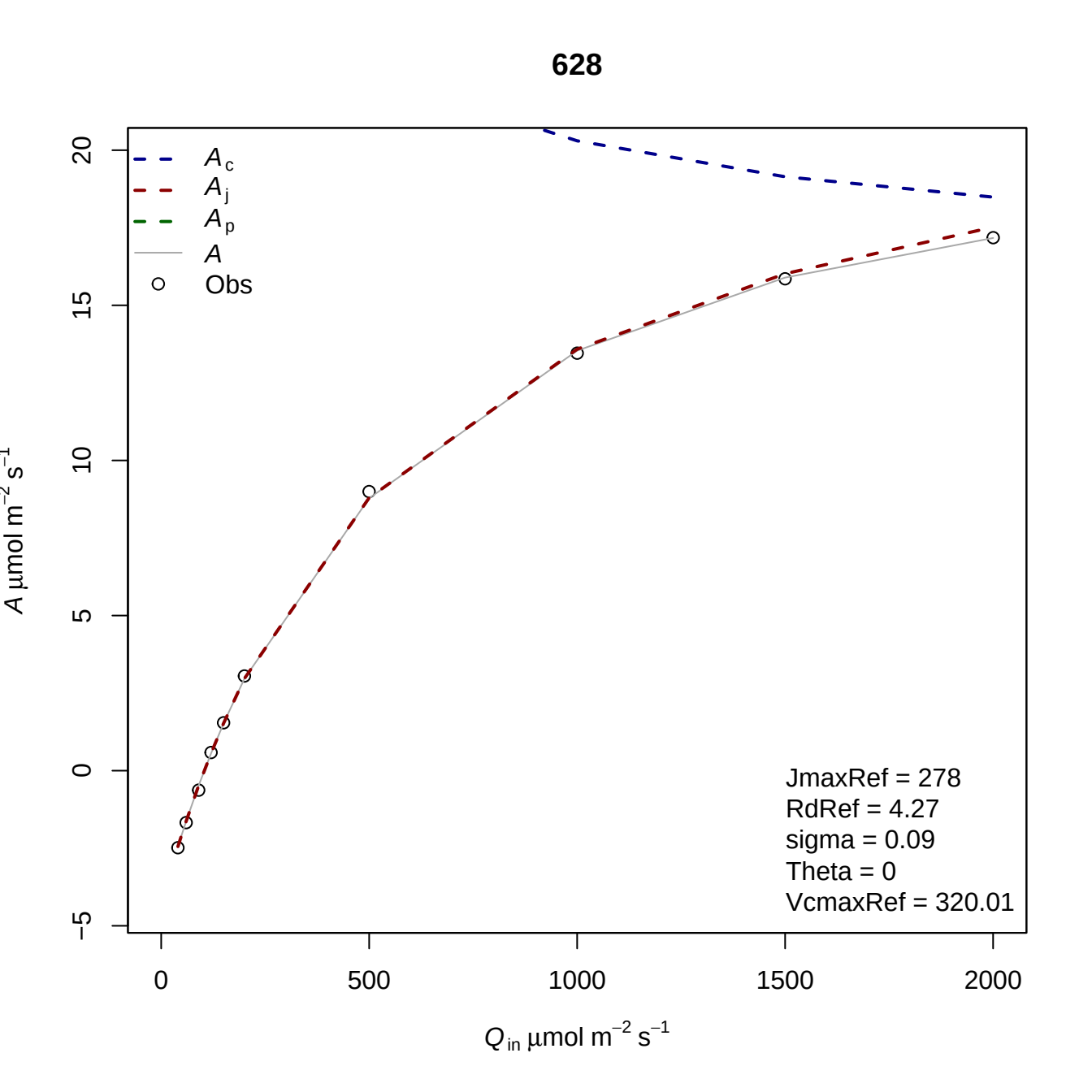

623
A_c
A_j
A_p
A
Obs
JmaxRef = 147.19
RdRef = 0.36
sigma = 0.3
Theta = 0
VcmaxRef = 51.34
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

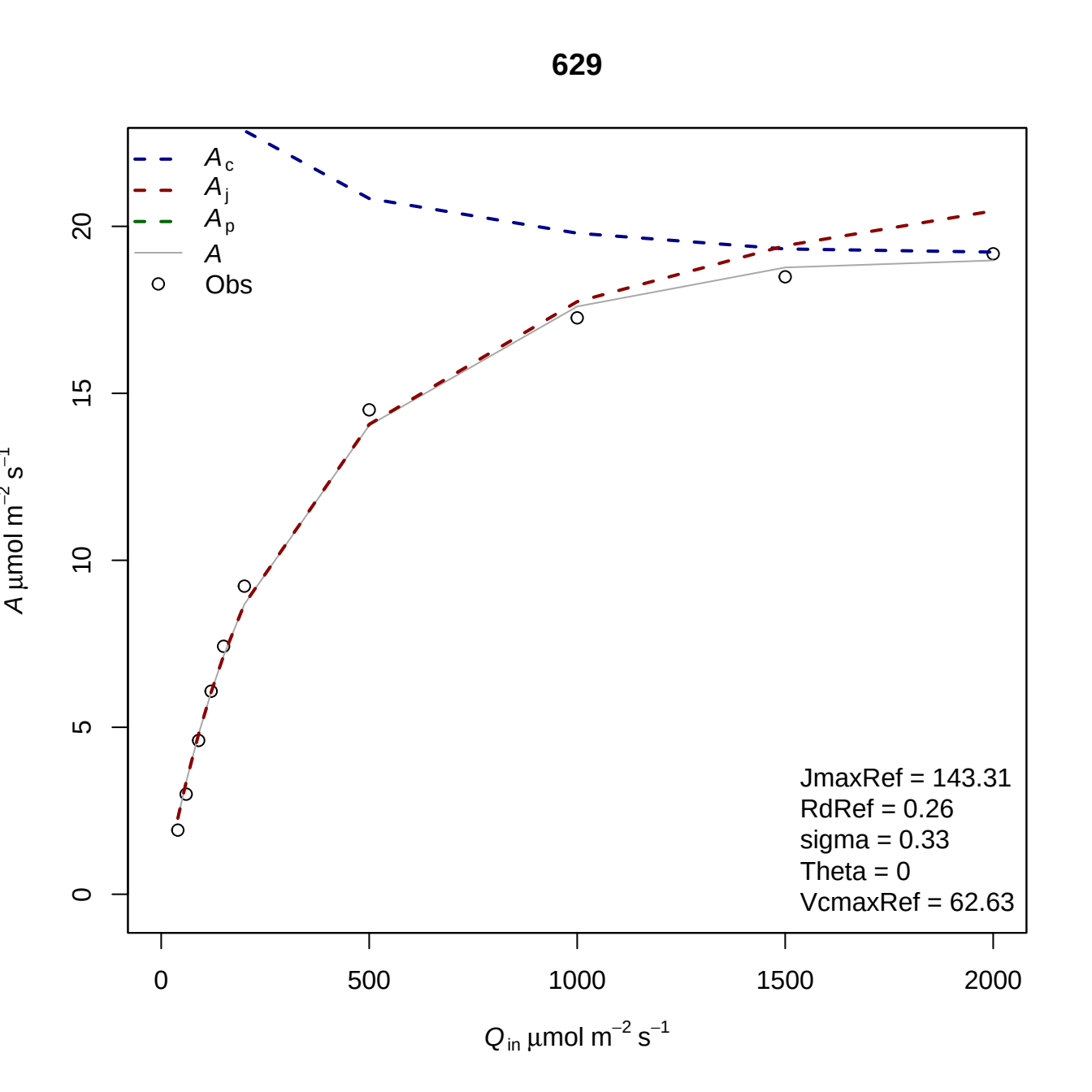

624
A_c
A_j
A_p
A
Obs
JmaxRef = 193.33
RdRef = 1.43
sigma = 0.21
Theta = 0
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

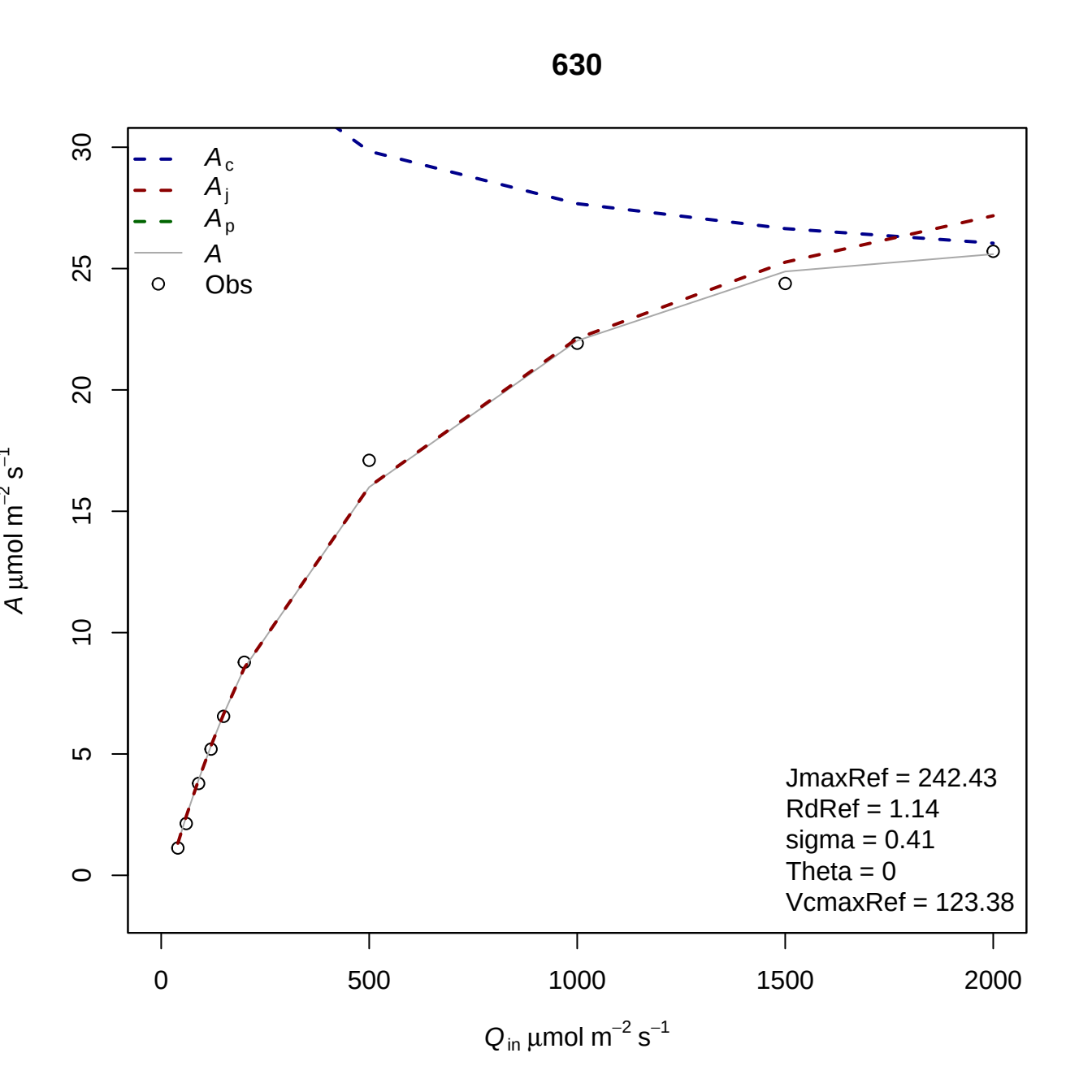

625
A_c
A_j
A_p
A
Obs
JmaxRef = 303.69
RdRef = 1.33
sigma = 0.51
Theta = 0
VcmaxRef = 127.01
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

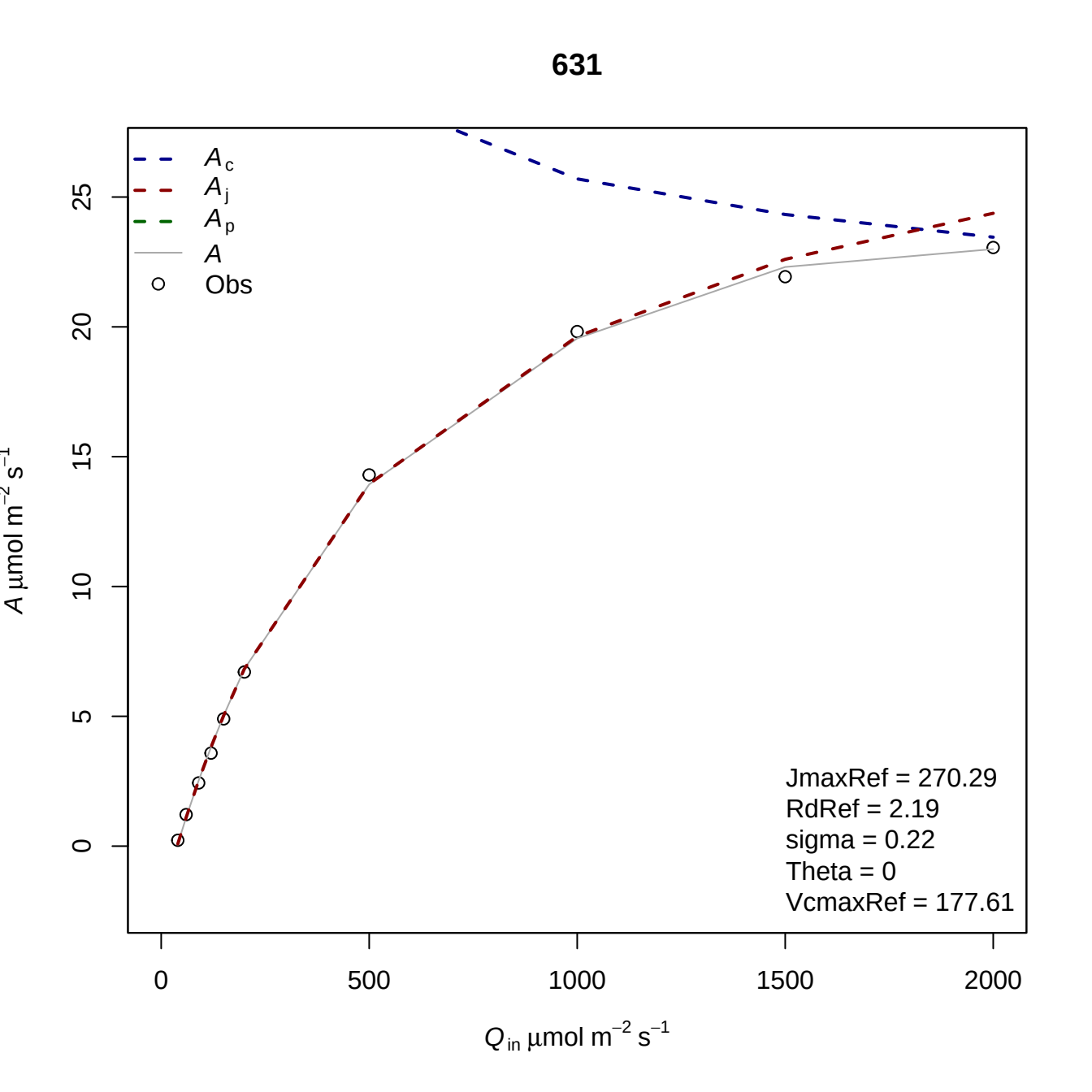

626
A_c
A_j
A_p
A
Obs
JmaxRef = 341.41
RdRef = 1.95
sigma = 0.33
Theta = 0
VcmaxRef = 185.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

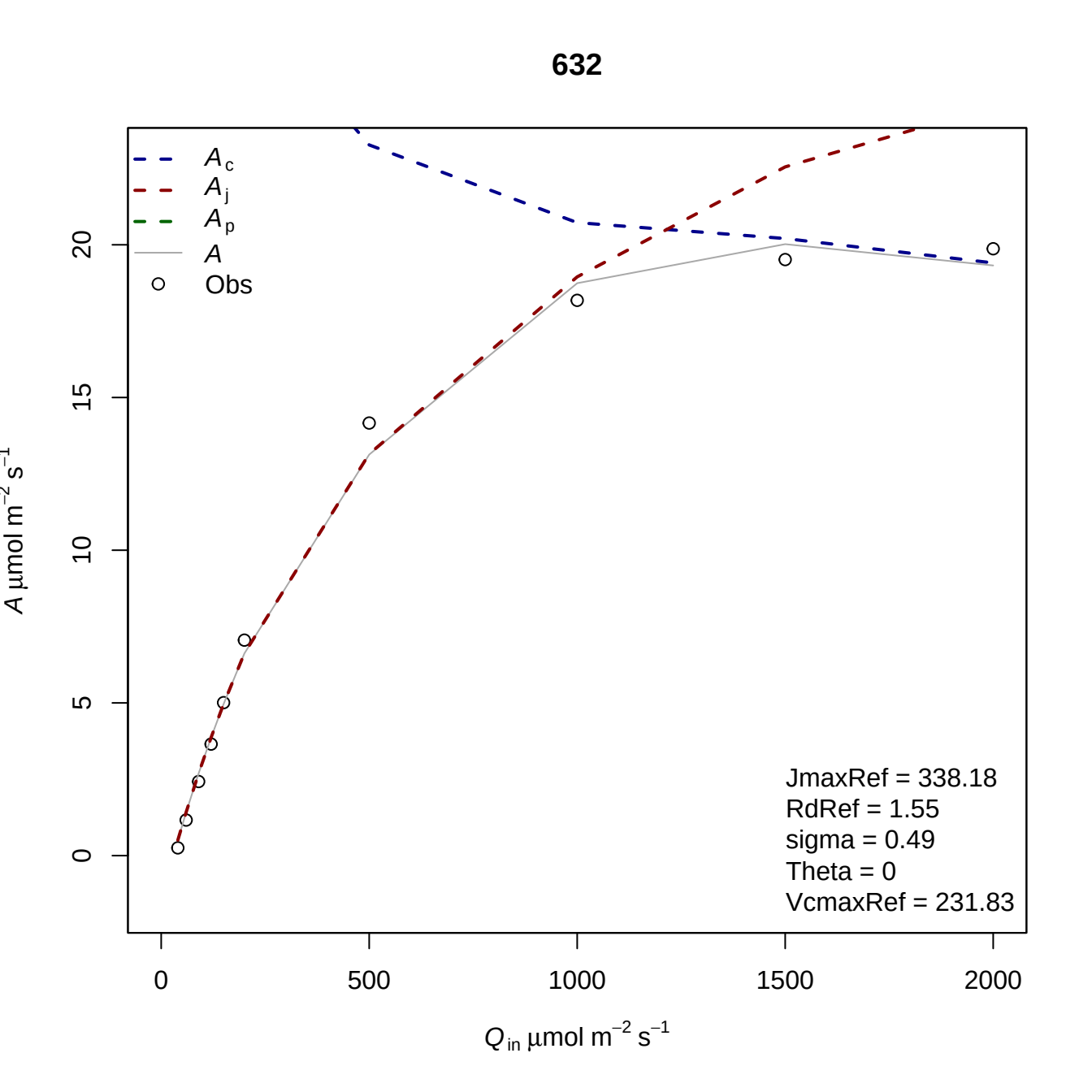

627
A_c
A_j
A_p
A
Obs
JmaxRef = 292.89
RdRef = 3.49
sigma = 0.18
Theta = 0
VcmaxRef = 249.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

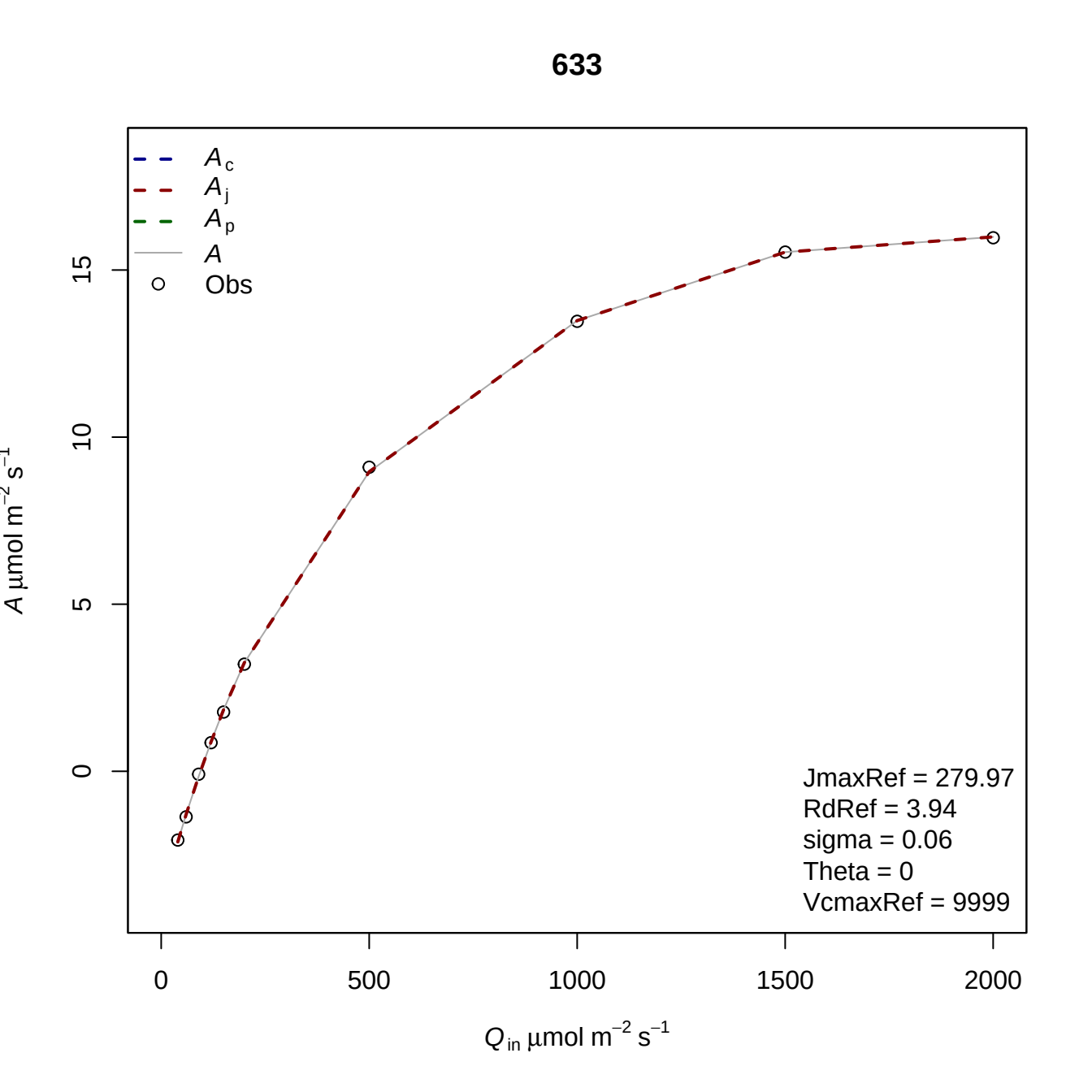

628
A_c
A_j
A_p
A
Obs
JmaxRef = 278
RdRef = 4.27
sigma = 0.09
Theta = 0
VcmaxRef = 320.01
20
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

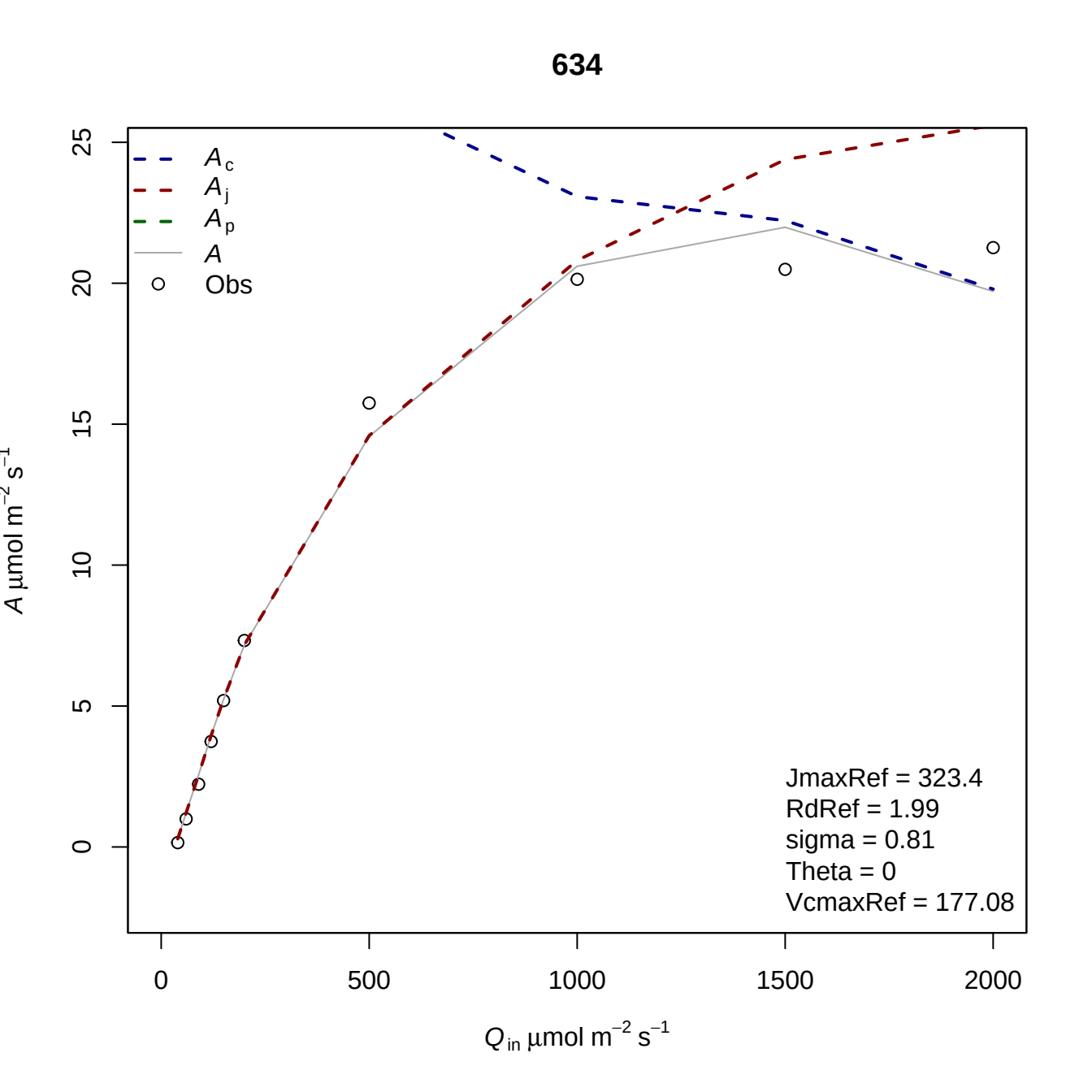

629
A_c
A_j
A_p
A
Obs
JmaxRef = 143.31
RdRef = 0.26
sigma = 0.33
Theta = 0
VcmaxRef = 62.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

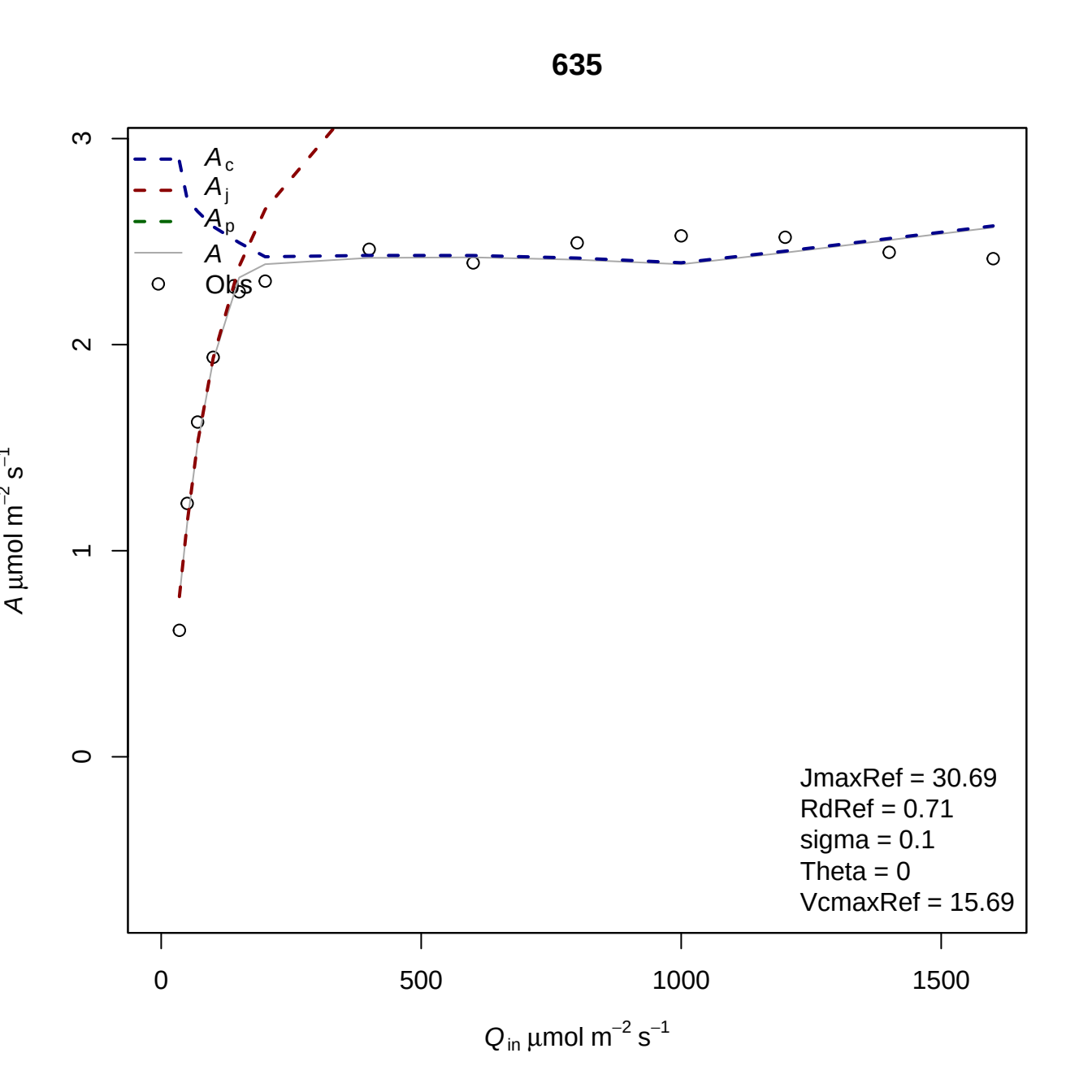

630
A_c
A_j
A_p
A
Obs
JmaxRef = 242.43
RdRef = 1.14
sigma = 0.41
Theta = 0
VcmaxRef = 123.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

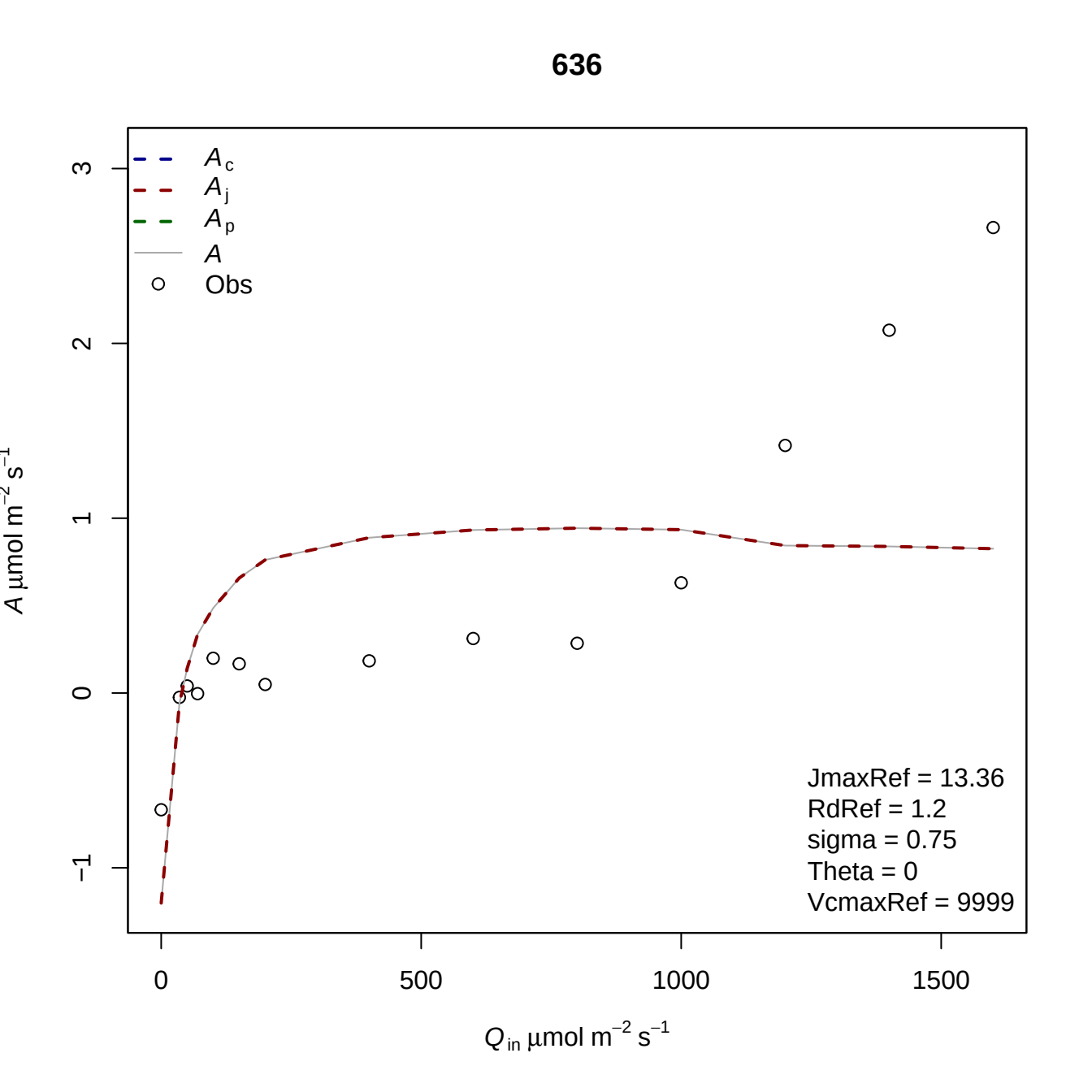

631
A_c
A_j
A_p
A
Obs
JmaxRef = 270.29
RdRef = 2.19
sigma = 0.22
Theta = 0
VcmaxRef = 177.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

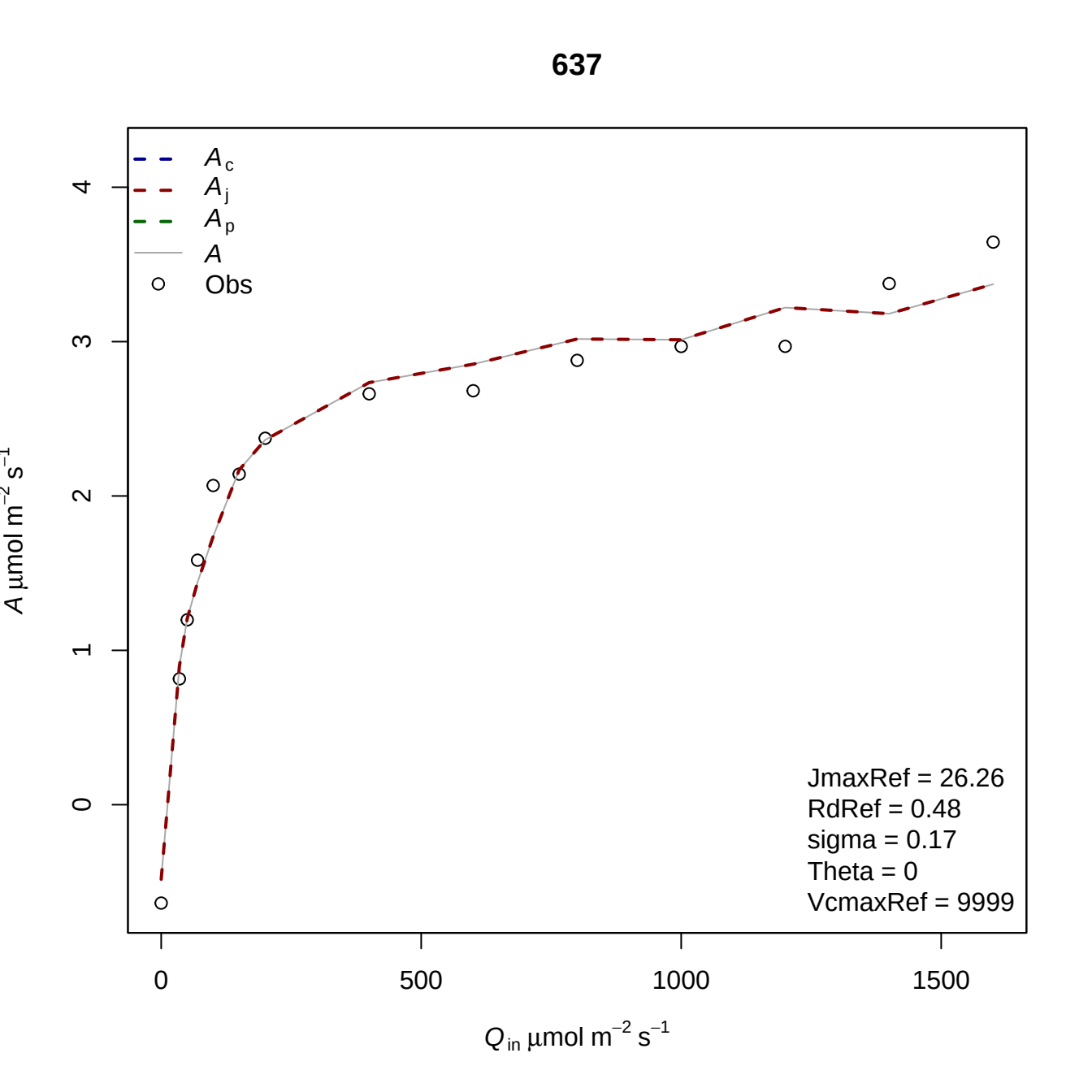

632
A_c
A_j
A_p
A
Obs
JmaxRef = 338.18
RdRef = 1.55
sigma = 0.49
Theta = 0
VcmaxRef = 231.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

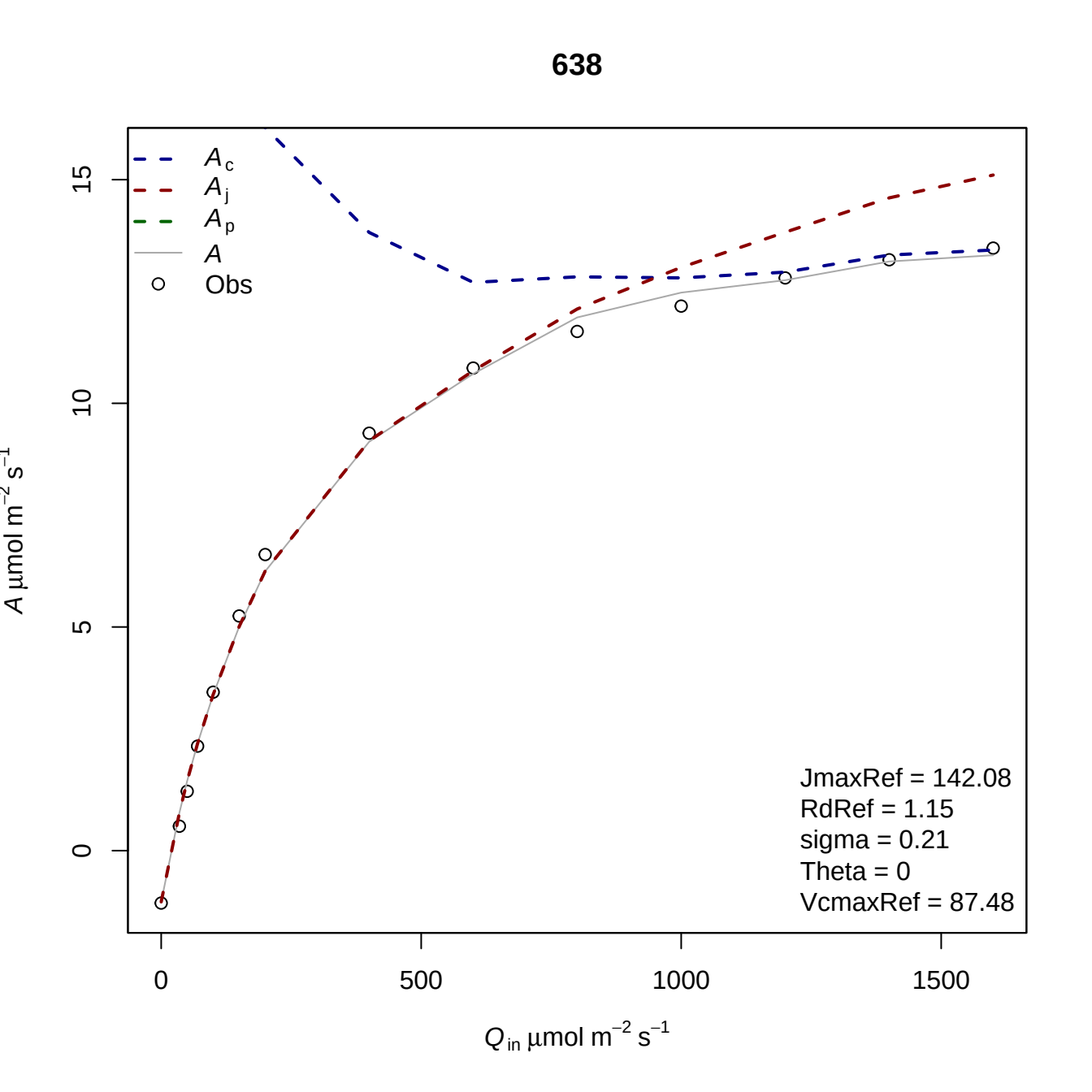

633
A_c
A_j
A_p
A
Obs
JmaxRef = 279.97
RdRef = 3.94
sigma = 0.06
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

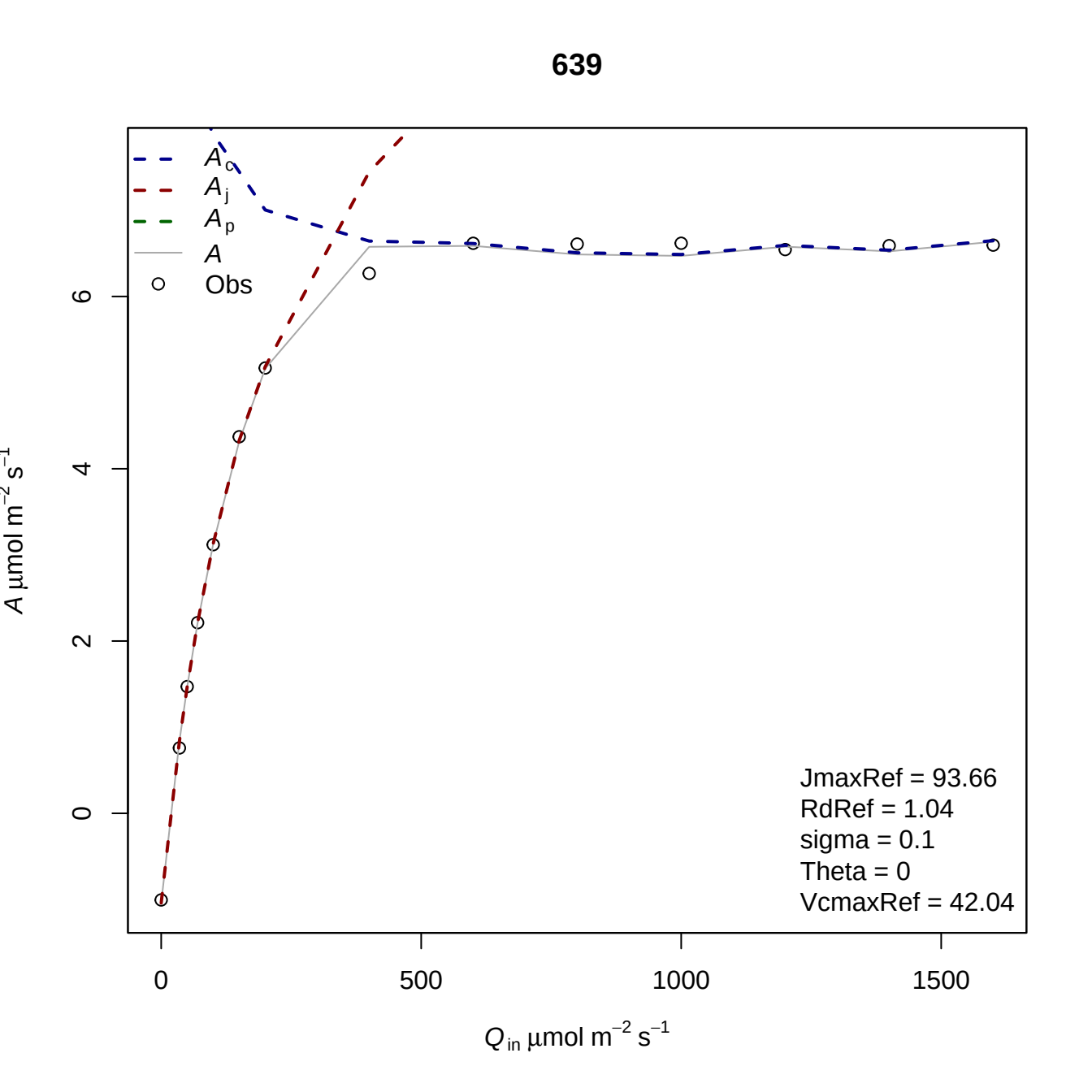

634
A_c
A_j
A_p
A
Obs
JmaxRef = 323.4
RdRef = 1.99
sigma = 0.81
Theta = 0
VcmaxRef = 177.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

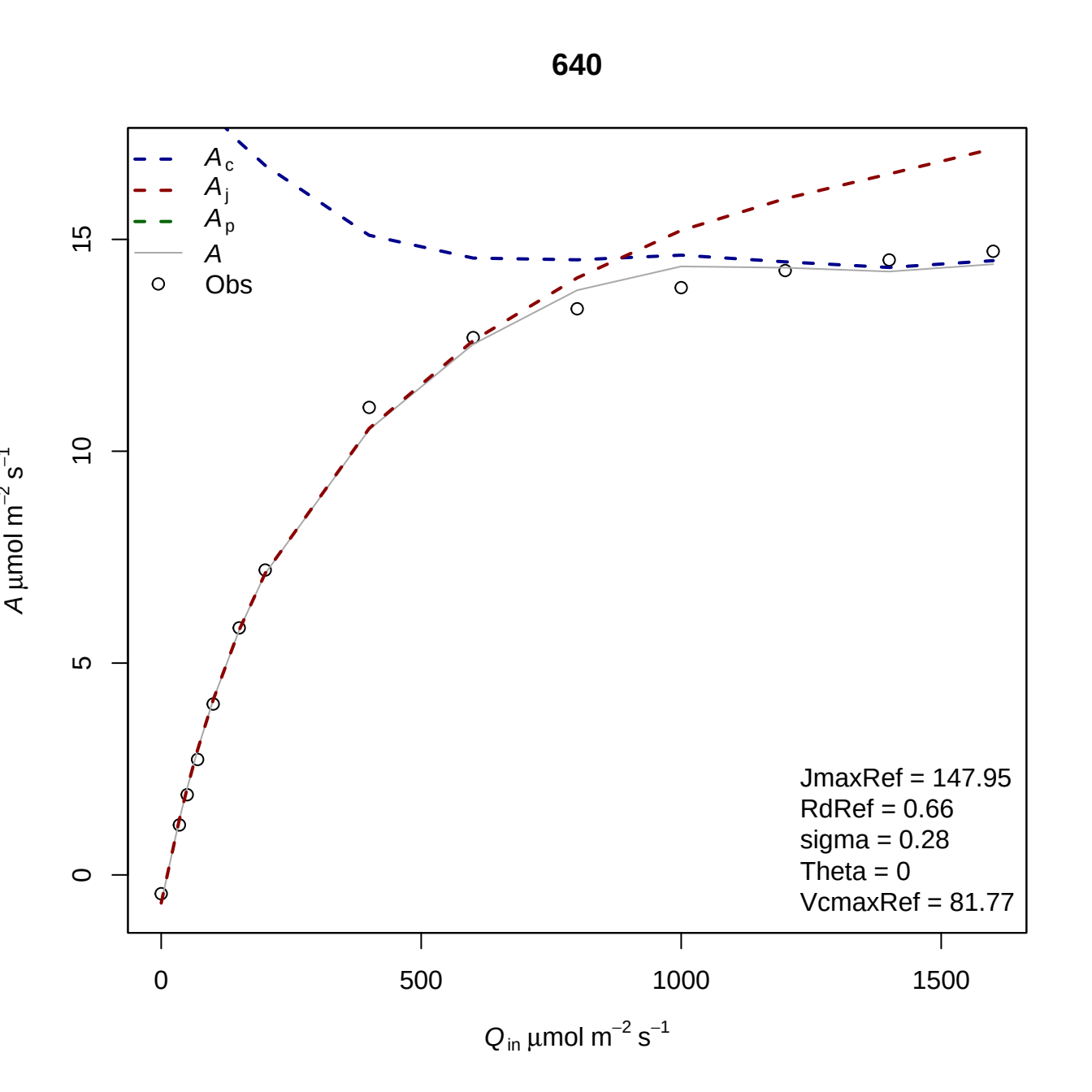

635
A_c
A_j
A_p
A
Obs
JmaxRef = 30.69
RdRef = 0.71
sigma = 0.1
Theta = 0
VcmaxRef = 15.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

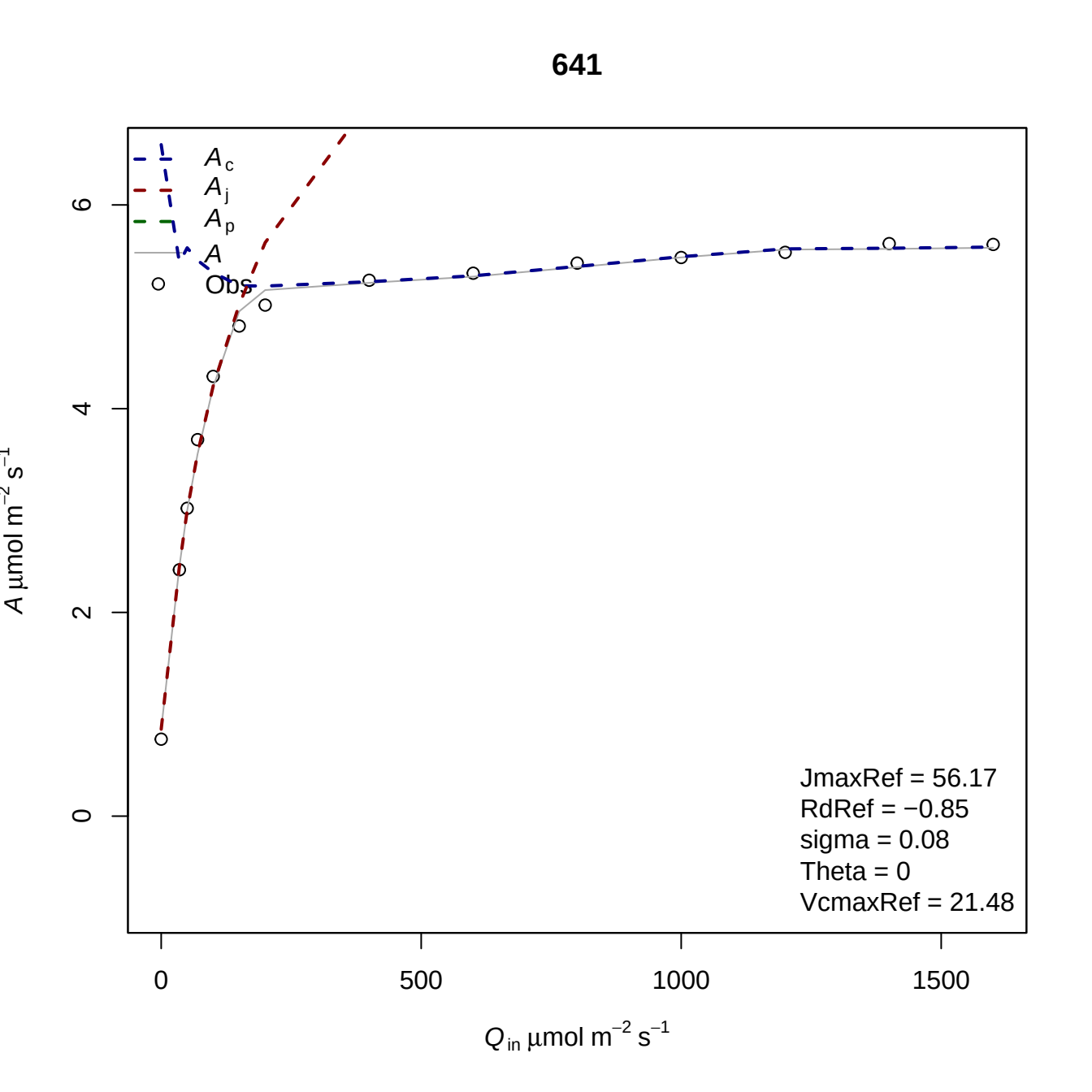

636
A_c
A_j
A_p
A
Obs
JmaxRef = 13.36
RdRef = 1.2
sigma = 0.75
Theta = 0
VcmaxRef = 9999
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

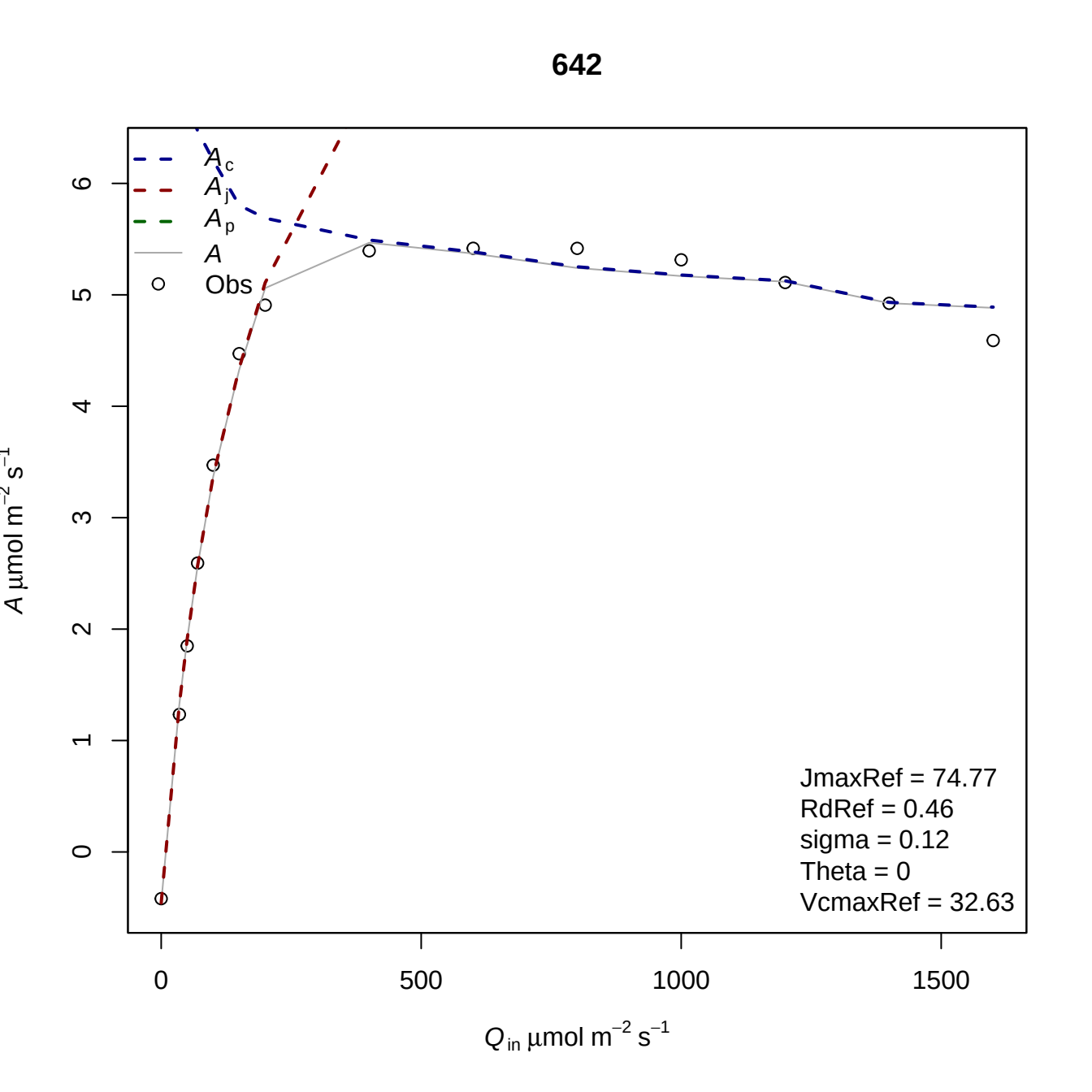

637
A_c
A_j
A_p
A
Obs
JmaxRef = 26.26
RdRef = 0.48
sigma = 0.17
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

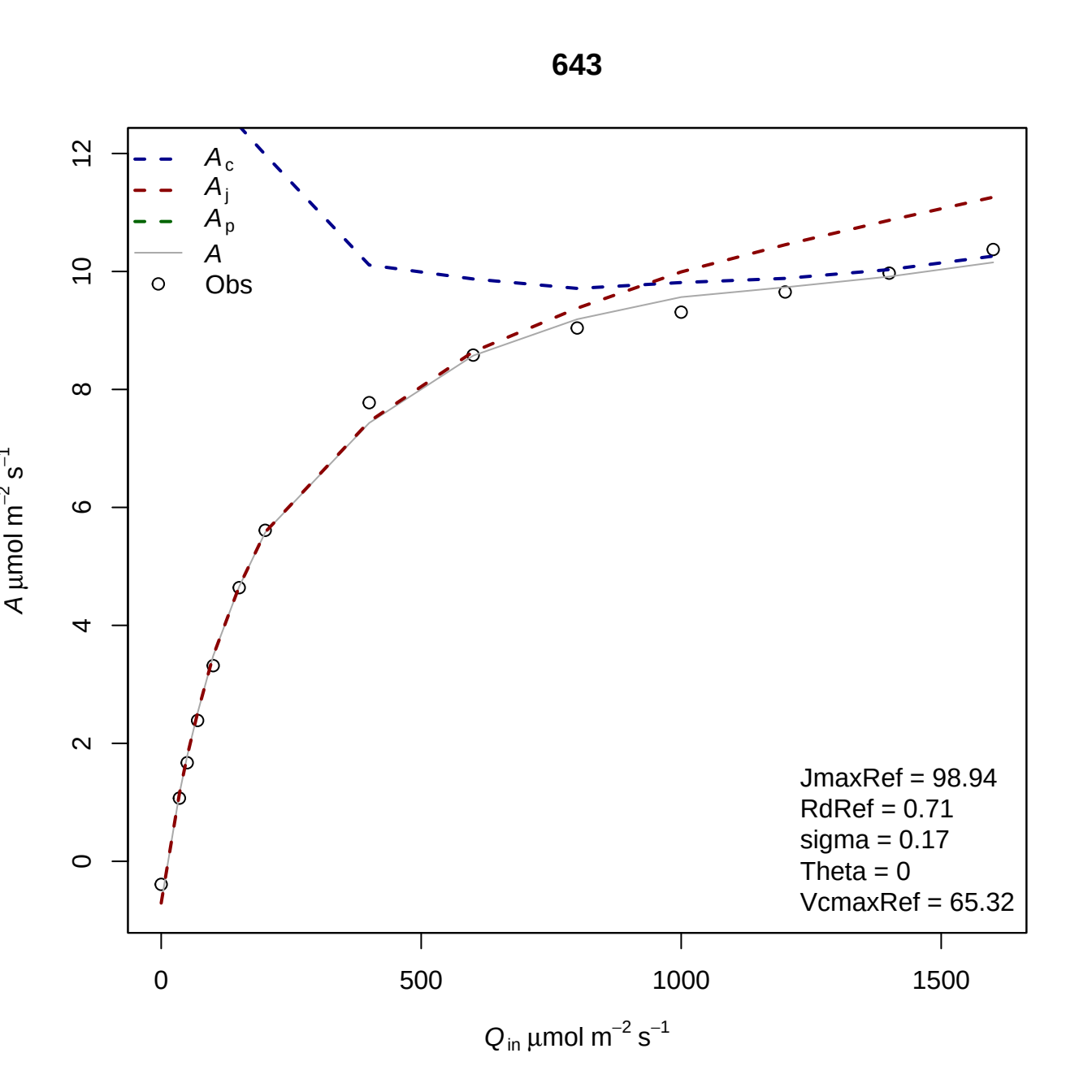

638
A_c
A_j
A_p
A
Obs
JmaxRef = 142.08
RdRef = 1.15
sigma = 0.21
Theta = 0
VcmaxRef = 87.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

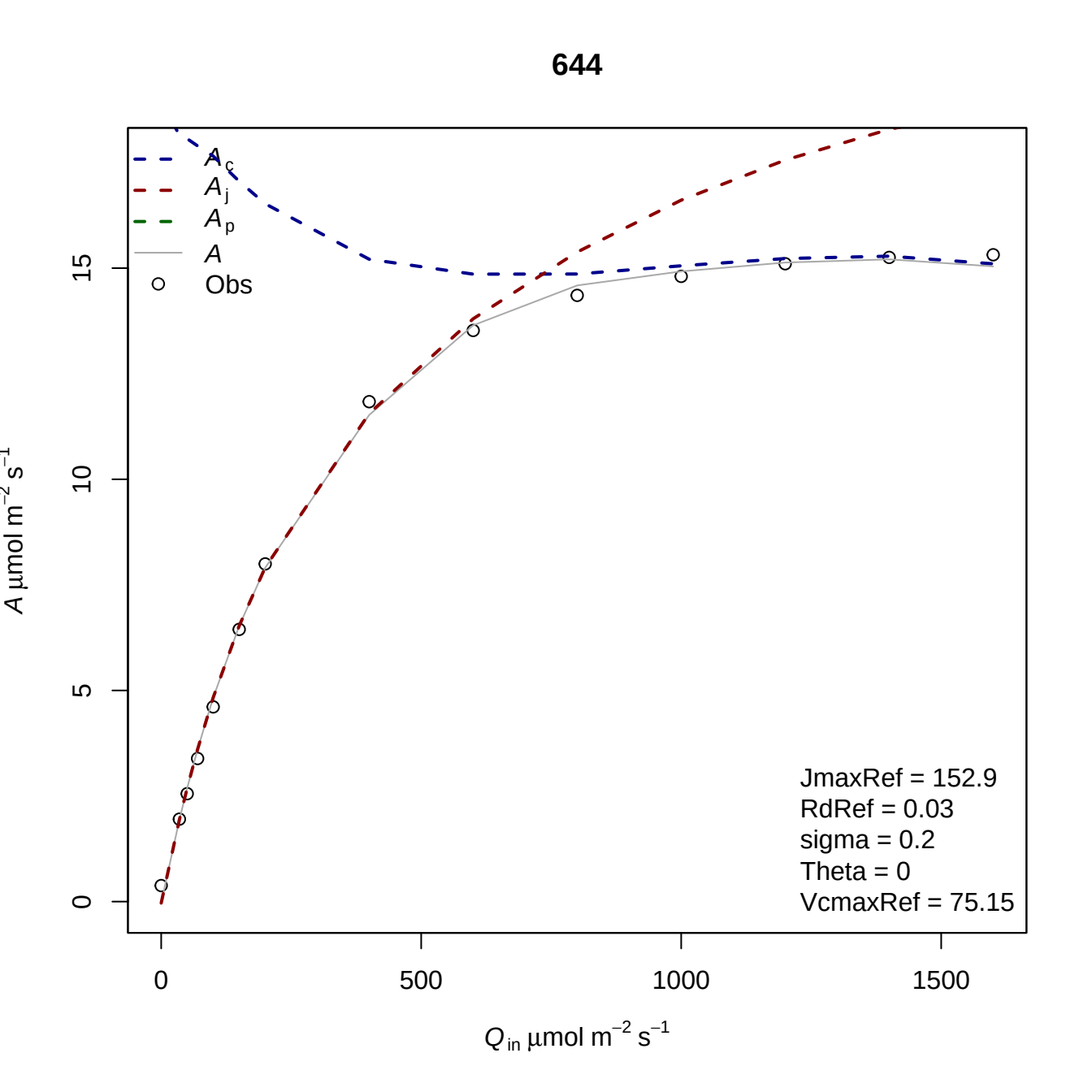

639
A_c
A_j
A_p
A
Obs
JmaxRef = 93.66
RdRef = 1.04
sigma = 0.1
Theta = 0
VcmaxRef = 42.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

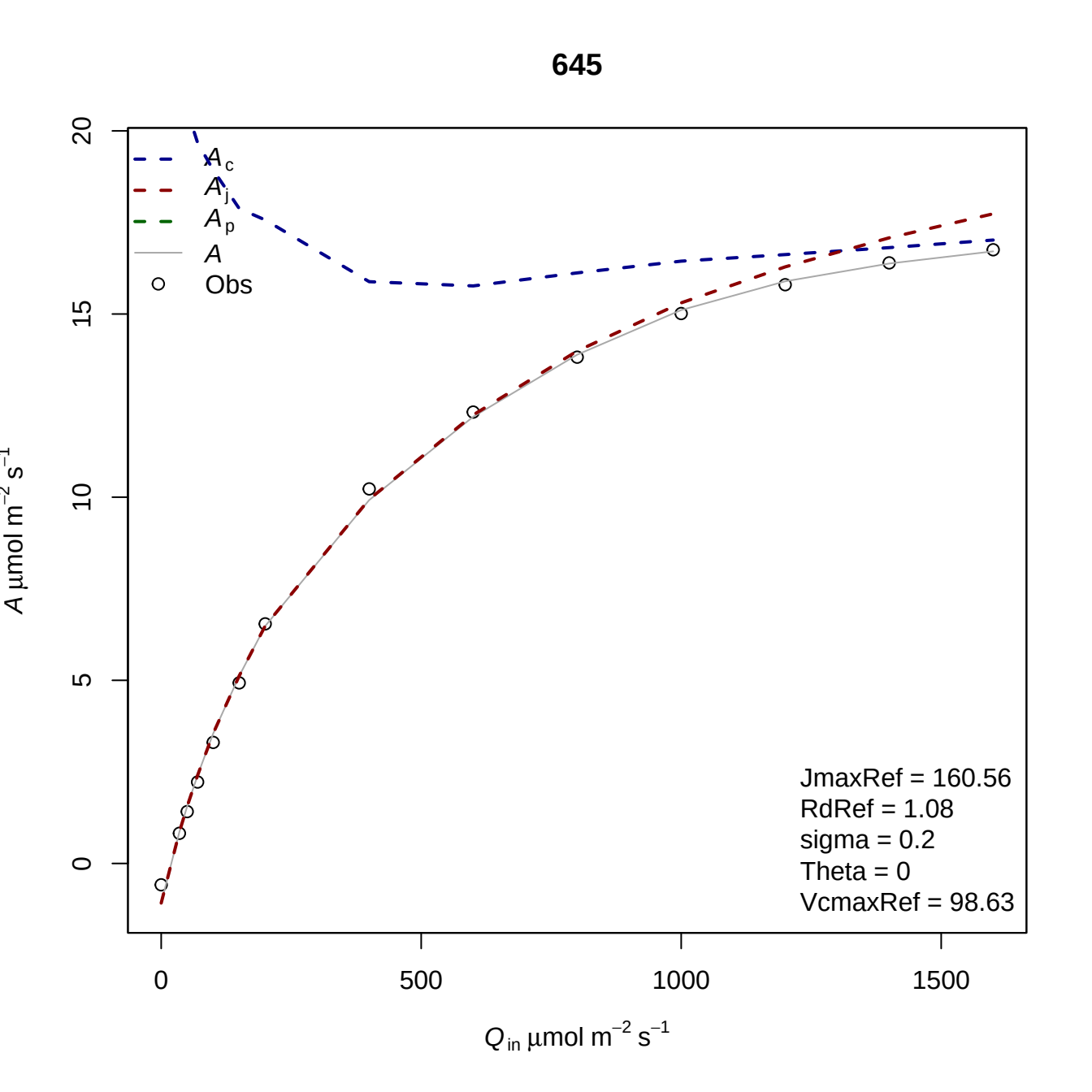

640
A_c
A_j
A_p
A
Obs
JmaxRef = 147.95
RdRef = 0.66
sigma = 0.28
Theta = 0
VcmaxRef = 81.77
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

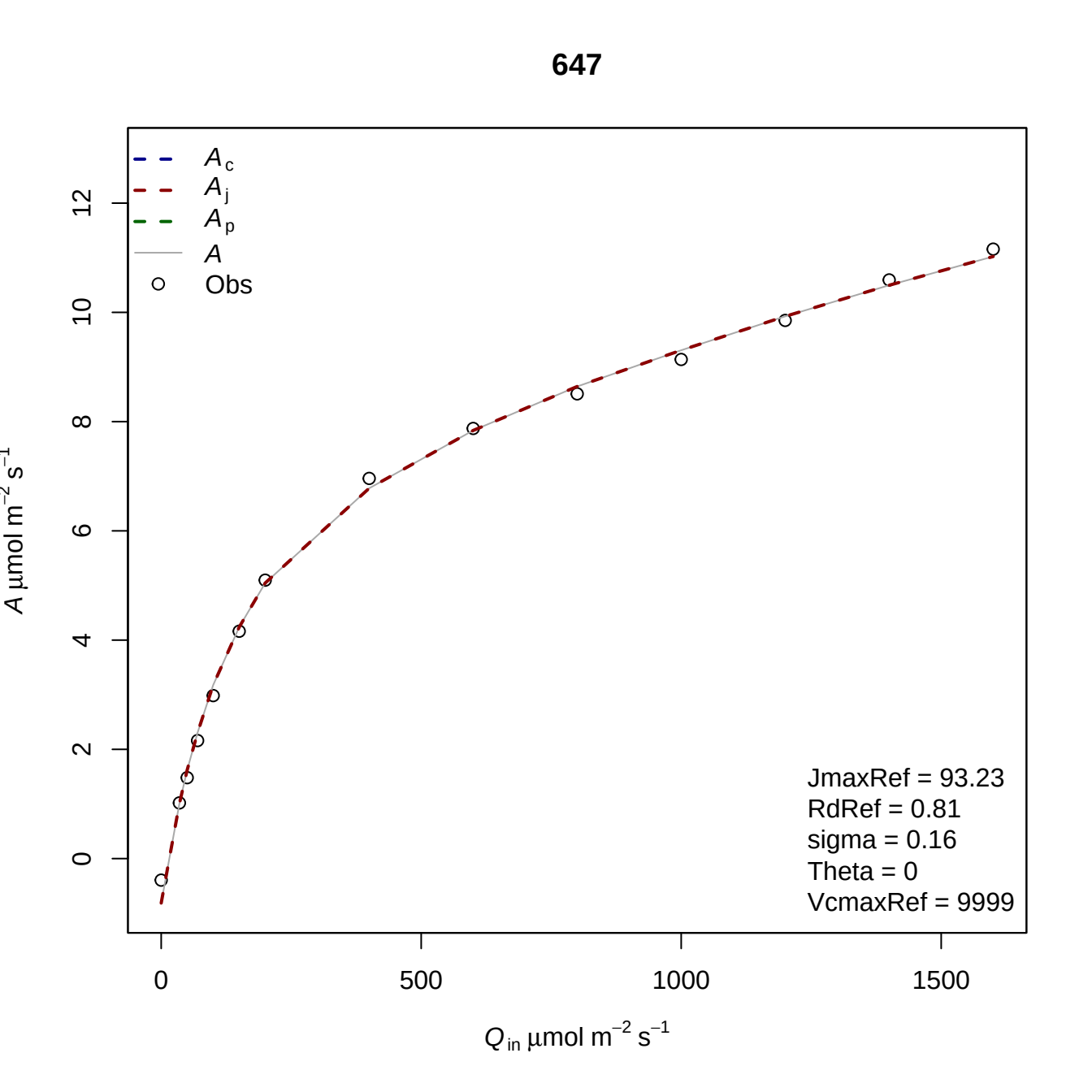

641
A_c
A_j
A_p
A
Obs
JmaxRef = 56.17
RdRef = −0.85
sigma = 0.08
Theta = 0
VcmaxRef = 21.48
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

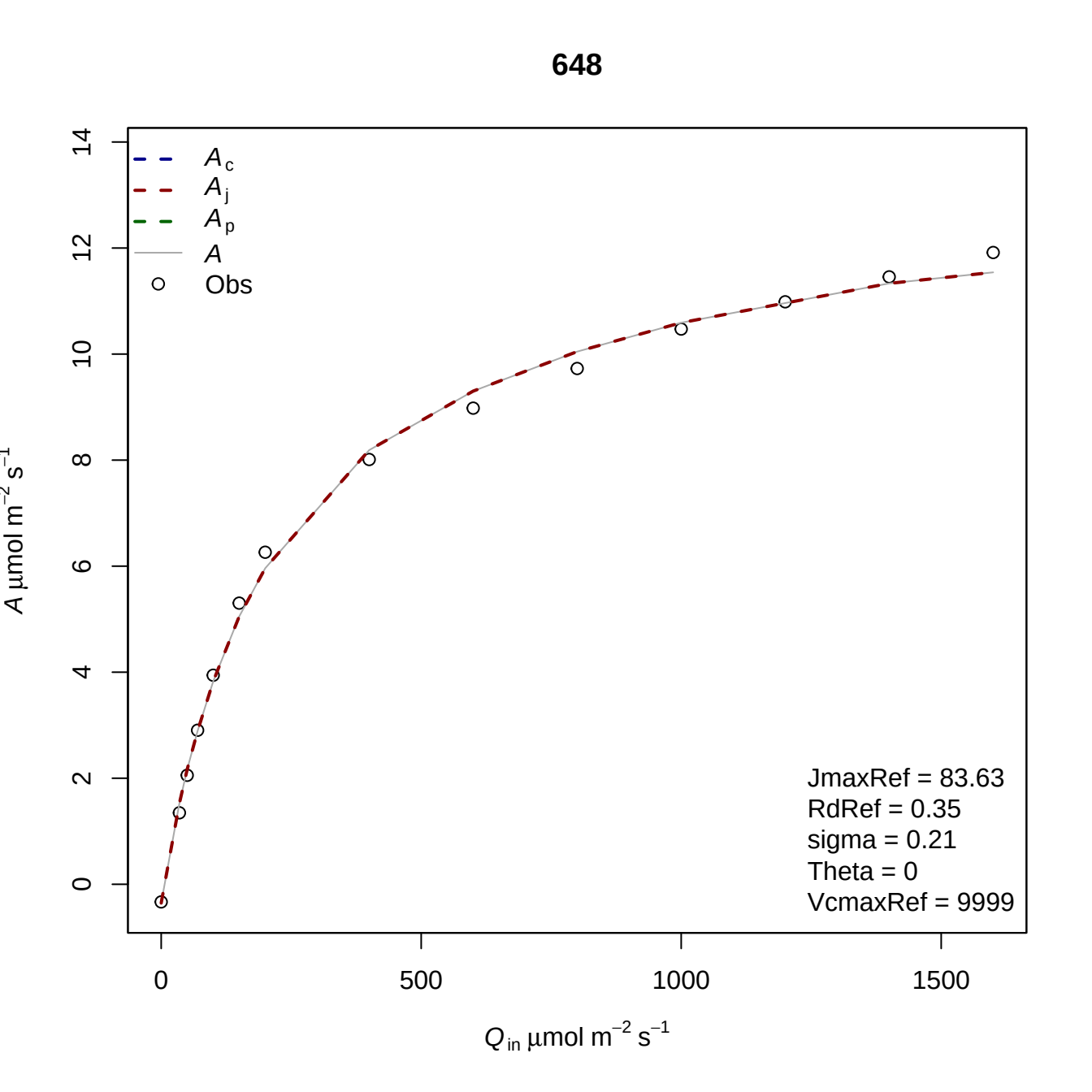

642
A_c
A_j
A_p
A
Obs
JmaxRef = 74.77
RdRef = 0.46
sigma = 0.12
Theta = 0
VcmaxRef = 32.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

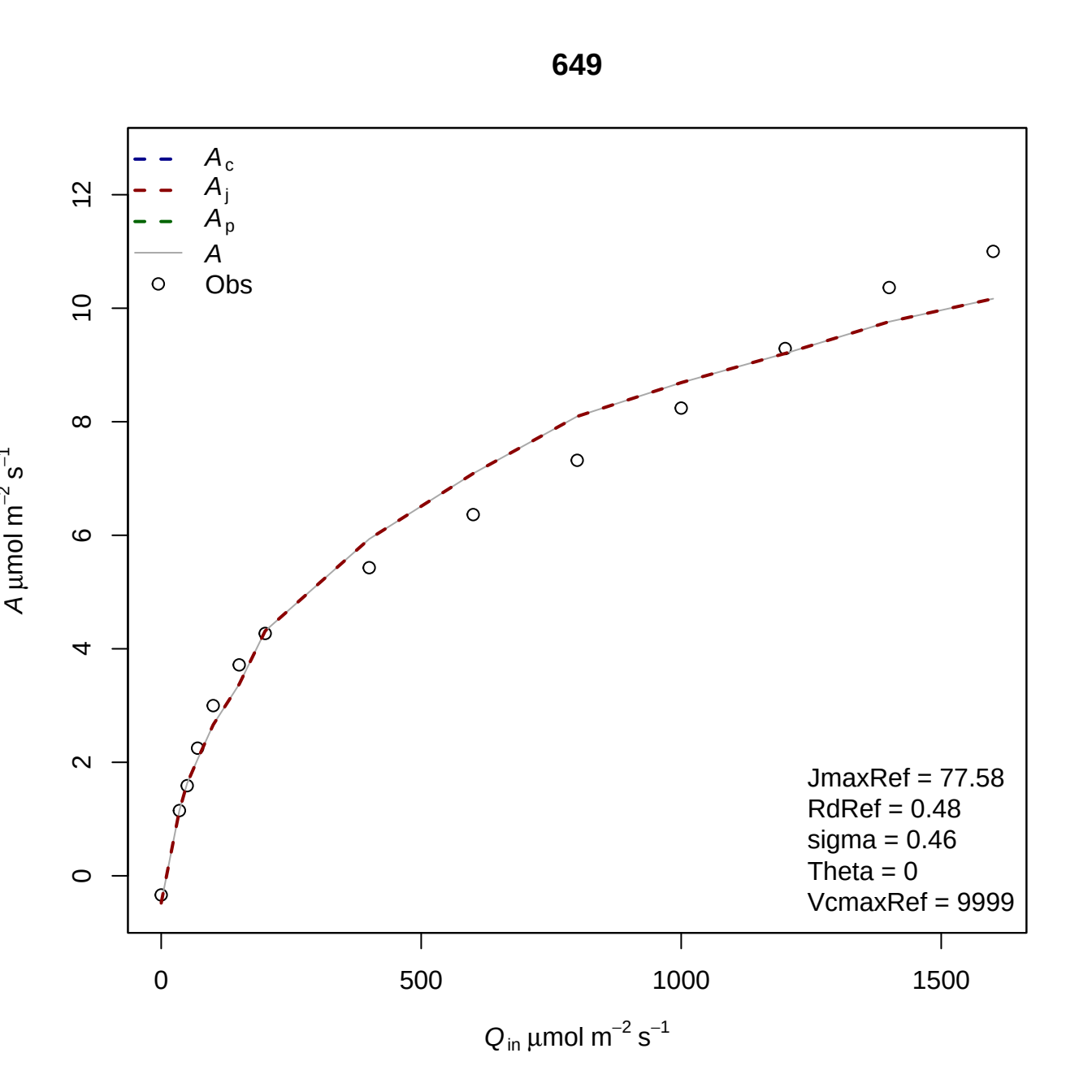

643
A_c
A_j
A_p
A
Obs
JmaxRef = 98.94
RdRef = 0.71
sigma = 0.17
Theta = 0
VcmaxRef = 65.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

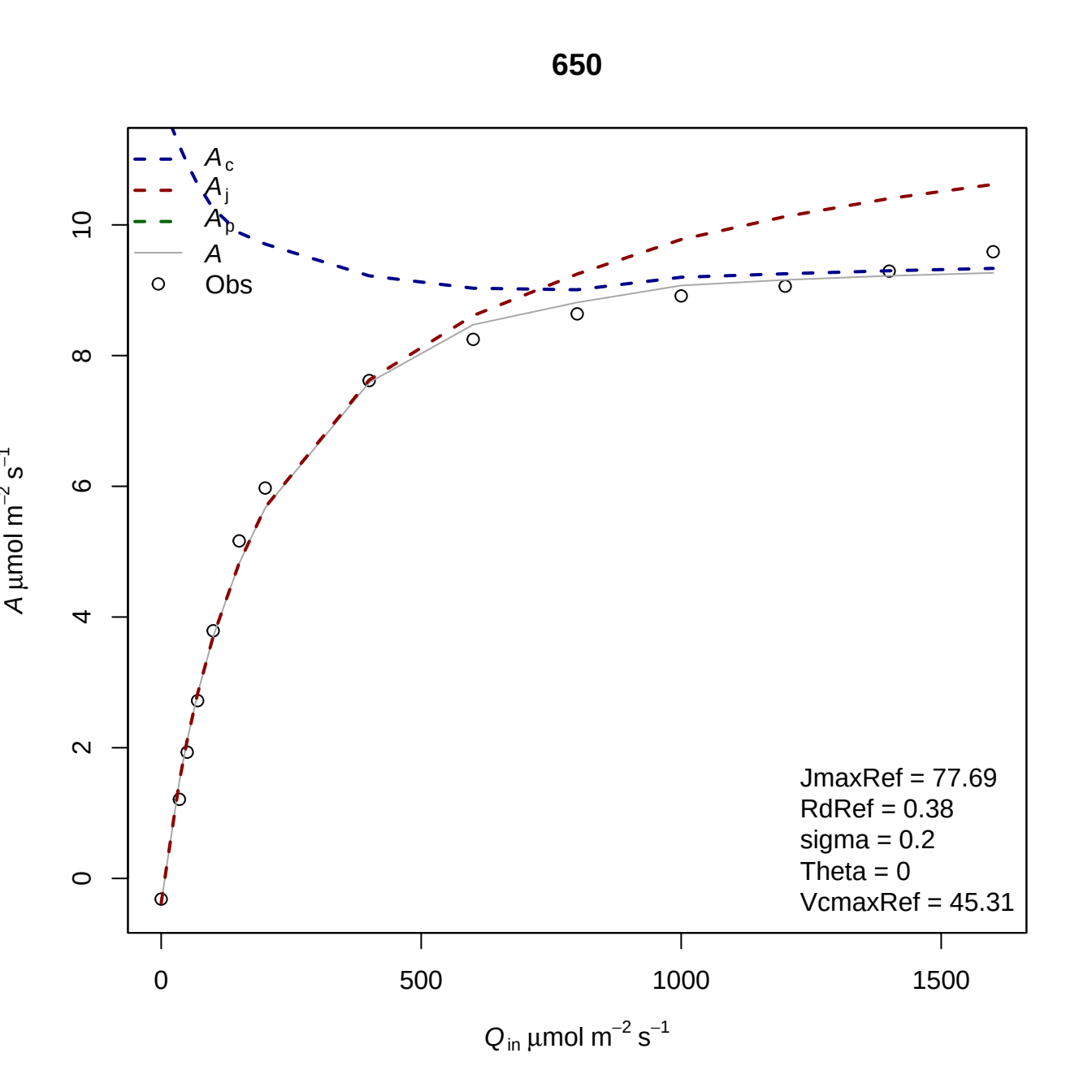

644
A_c
A_j
A_p
A
Obs
JmaxRef = 152.9
RdRef = 0.03
sigma = 0.2
Theta = 0
VcmaxRef = 75.15
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

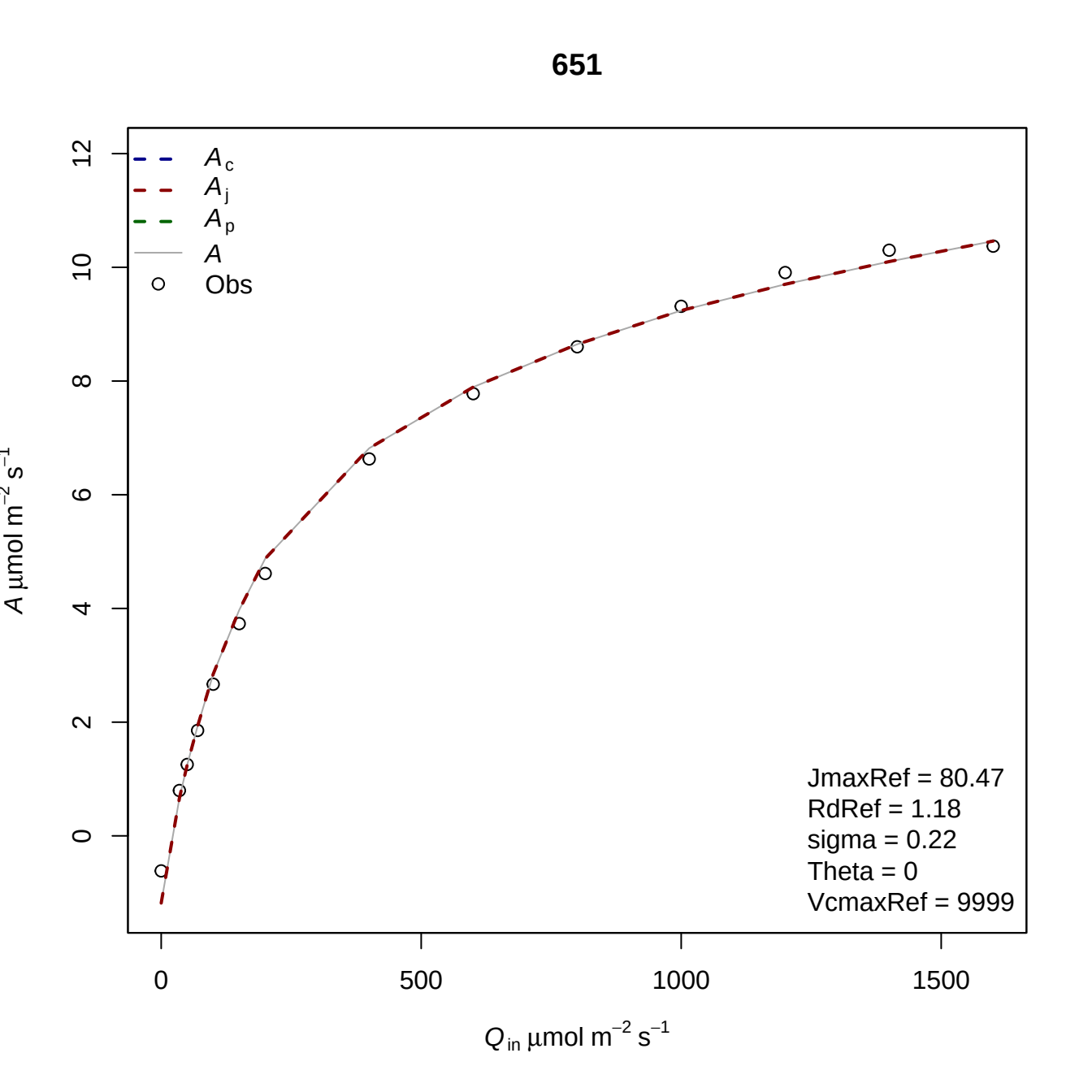

645
A_c
A_j
A_p
A
Obs
JmaxRef = 160.56
RdRef = 1.08
sigma = 0.2
Theta = 0
VcmaxRef = 98.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

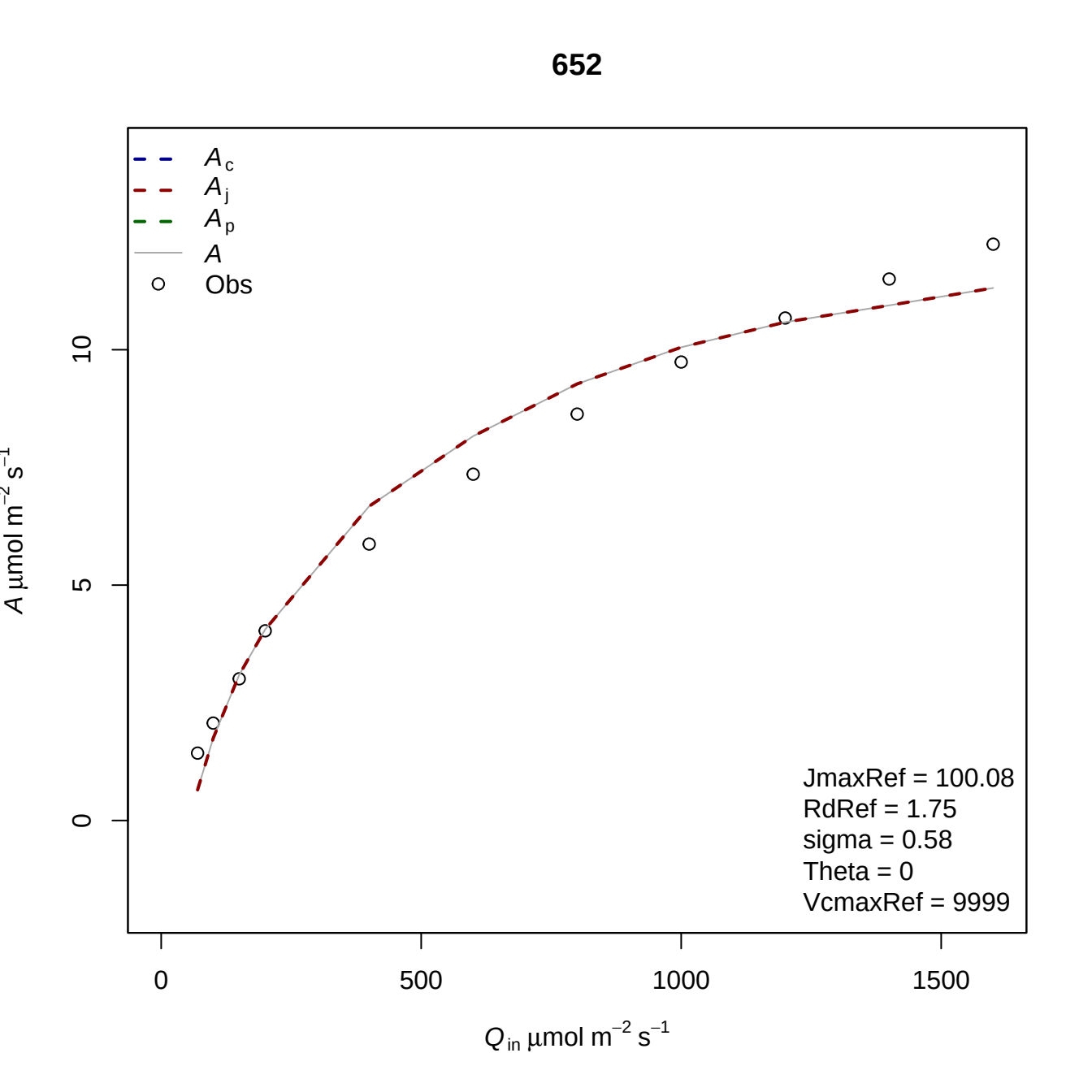

647
A_c
A_j
A_p
A
Obs
JmaxRef = 93.23
RdRef = 0.81
sigma = 0.16
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

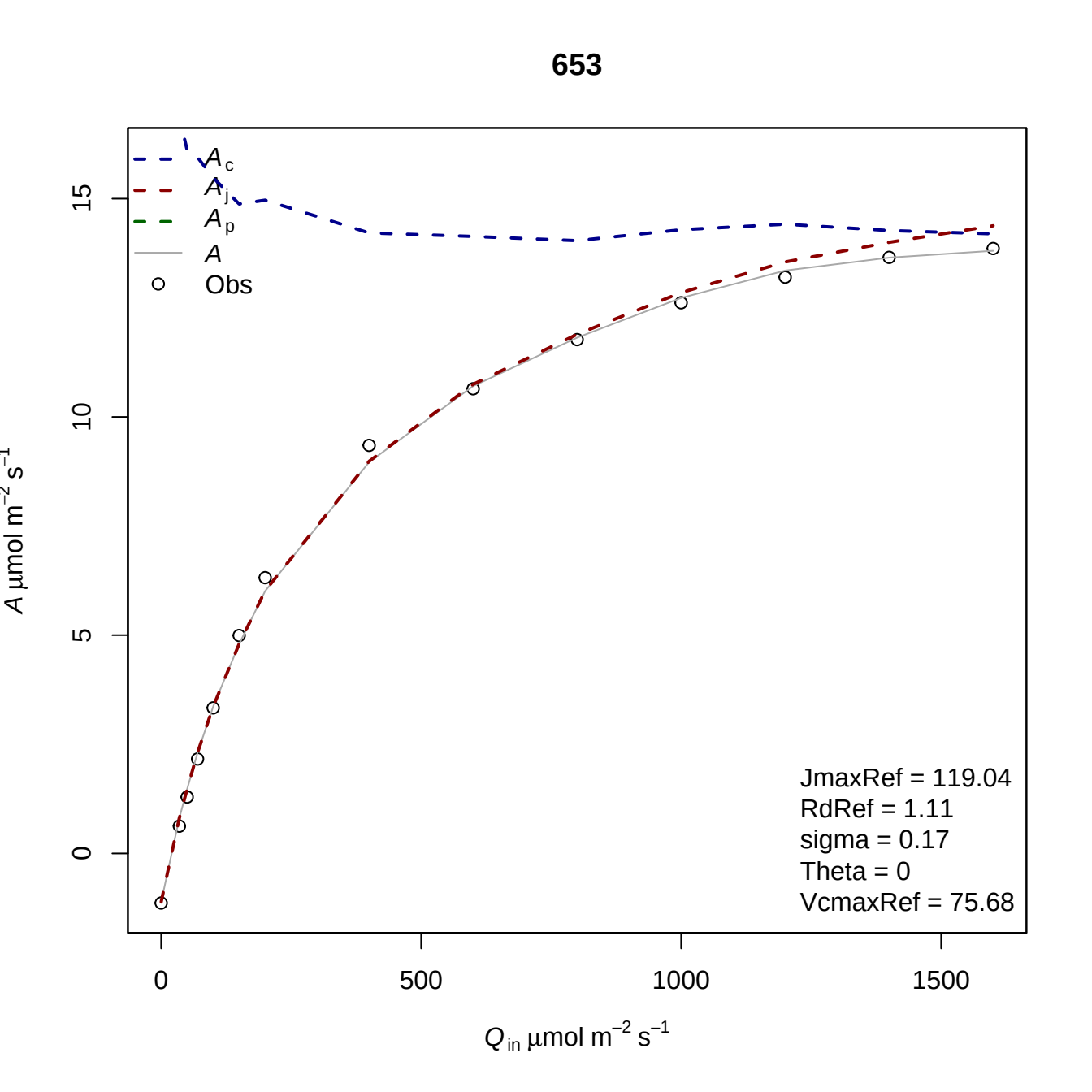

648
A_c
A_j
A_p
A
Obs
JmaxRef = 83.63
RdRef = 0.35
sigma = 0.21
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

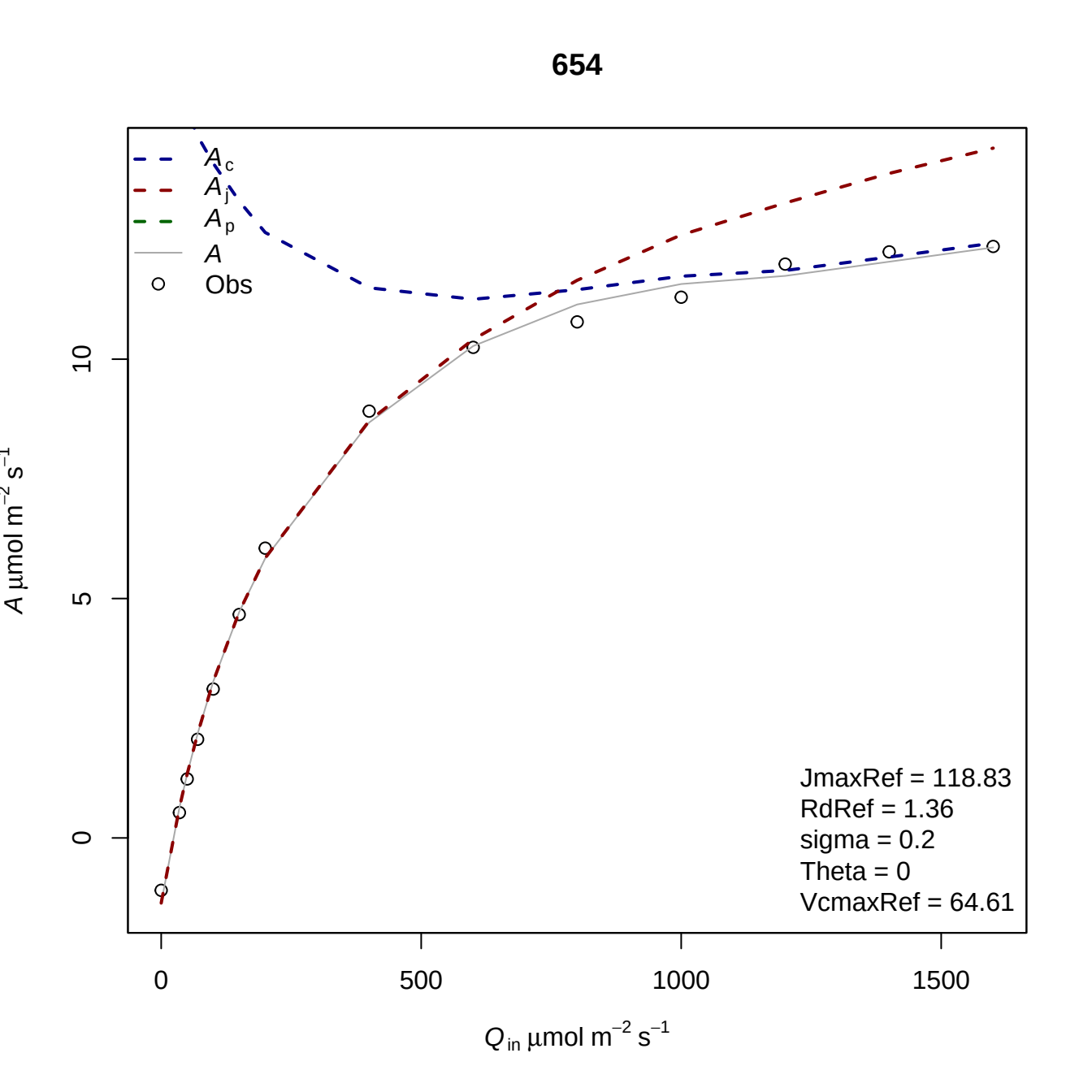

649
A_c
A_j
A_p
A
Obs
JmaxRef = 77.58
RdRef = 0.48
sigma = 0.46
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

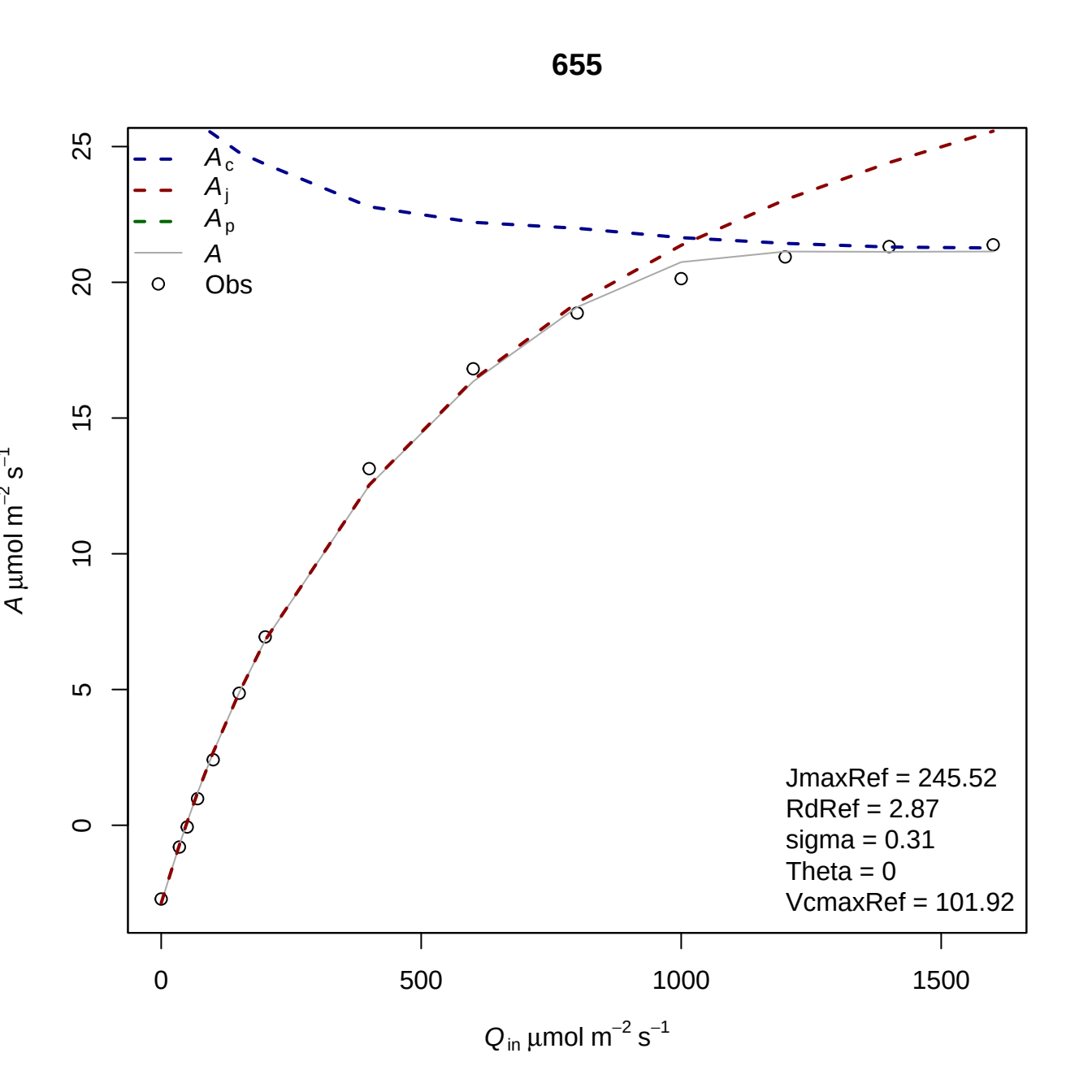

650
A_c
A_j
A_p
A
Obs
JmaxRef = 77.69
RdRef = 0.38
sigma = 0.2
Theta = 0
VcmaxRef = 45.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

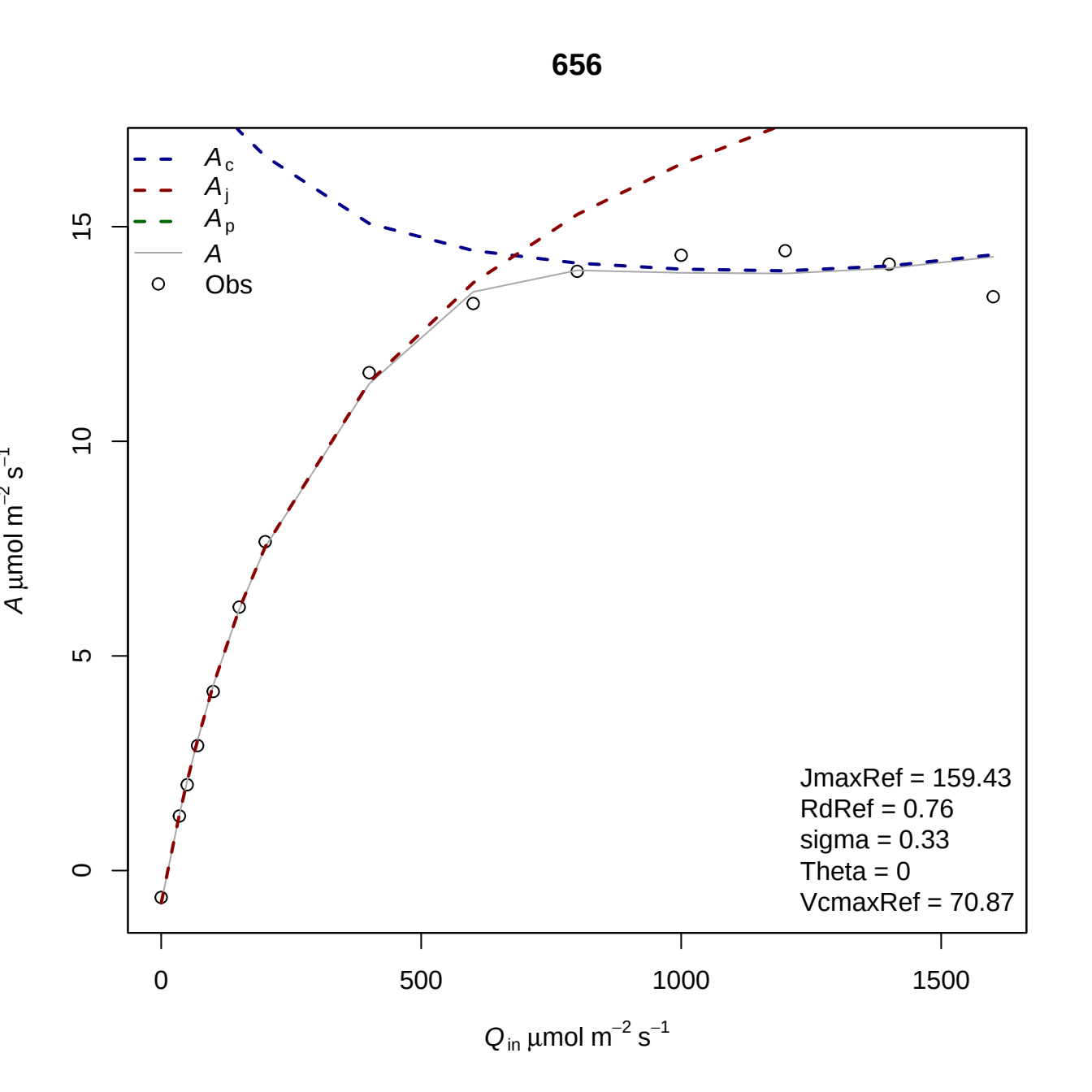

651
A_c
A_j
A_p
A
Obs
JmaxRef = 80.47
RdRef = 1.18
sigma = 0.22
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

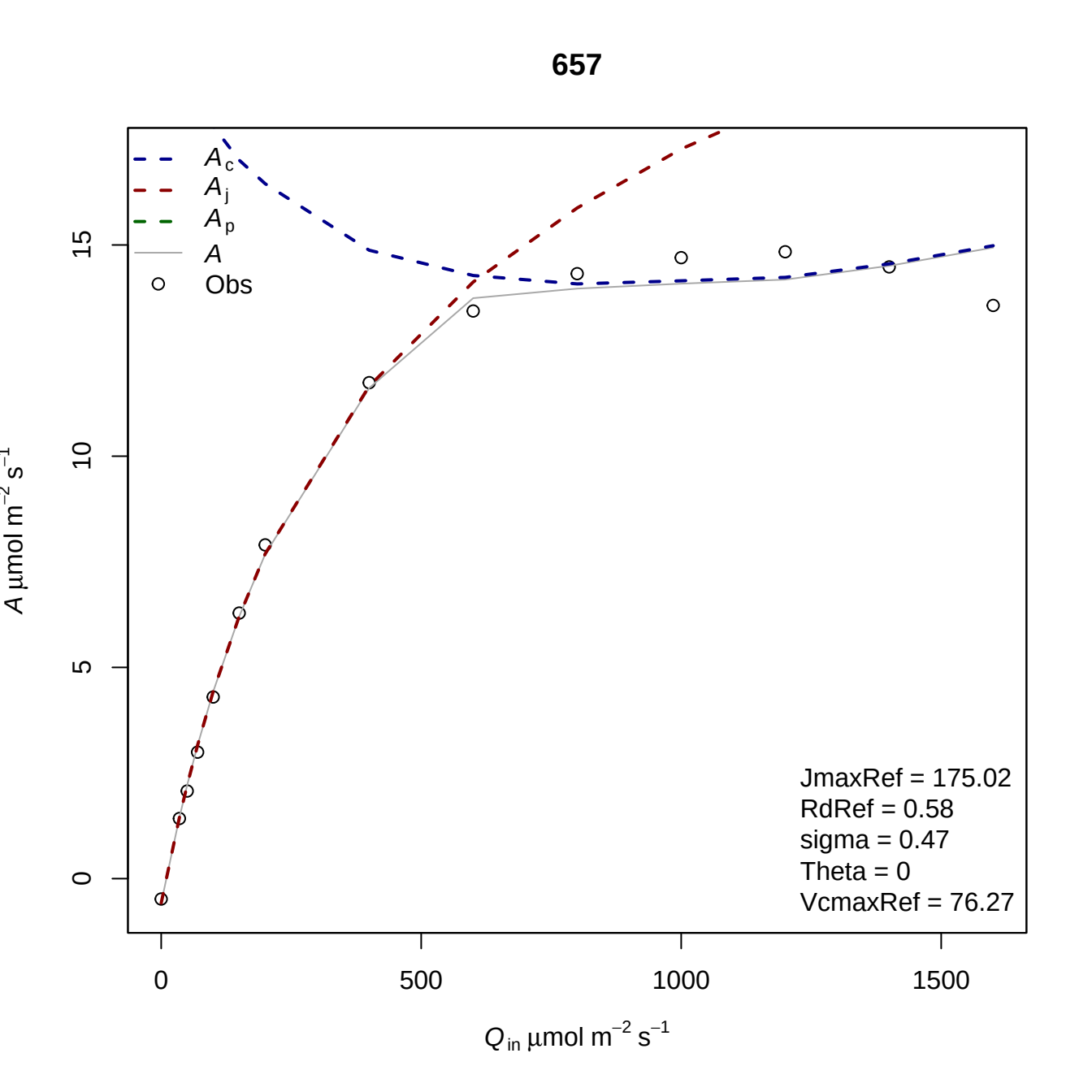

652
A_c
A_j
A_p
A
Obs
JmaxRef = 100.08
RdRef = 1.75
sigma = 0.58
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

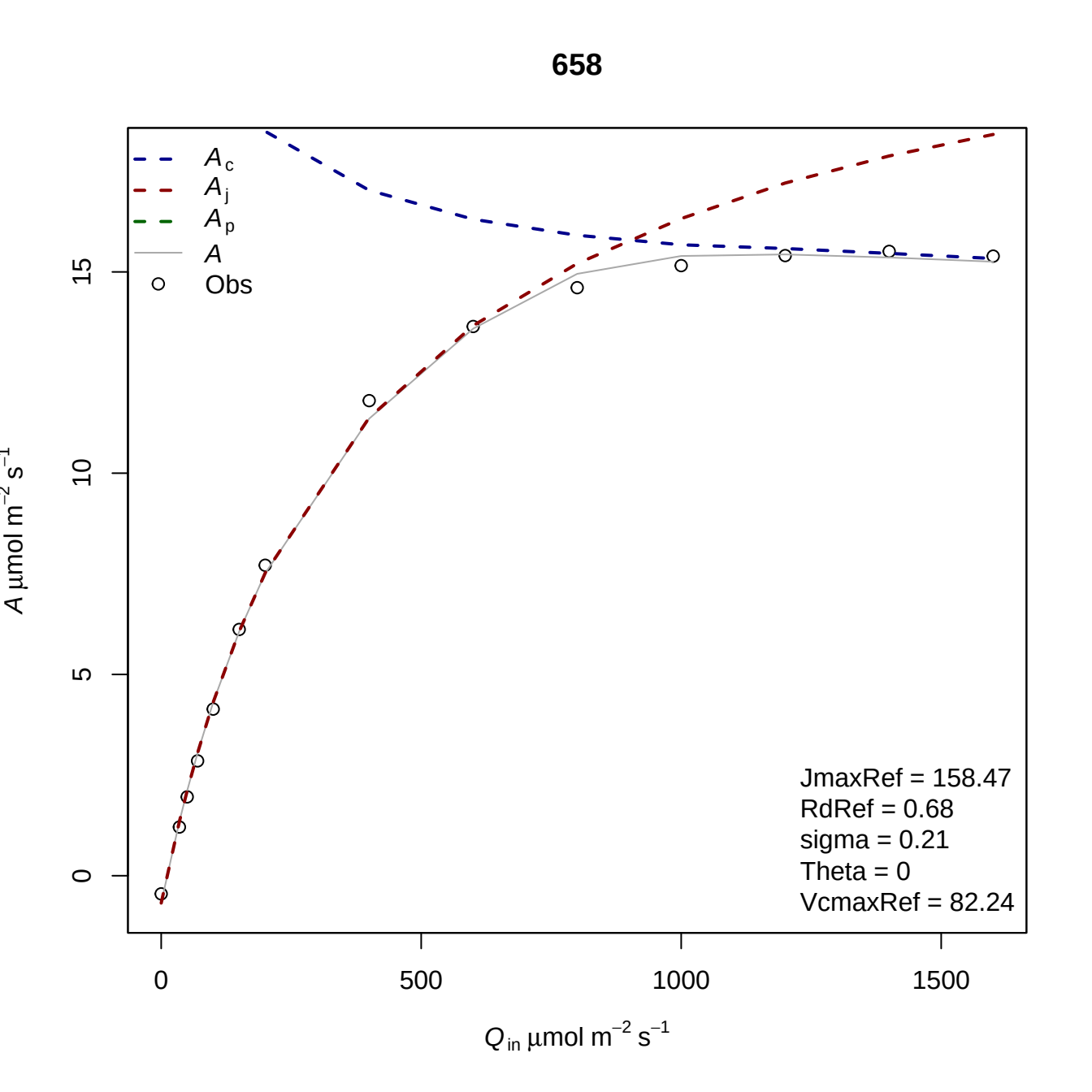

653
A_c
A_j
A_p
A
Obs
JmaxRef = 119.04
RdRef = 1.11
sigma = 0.17
Theta = 0
VcmaxRef = 75.68
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

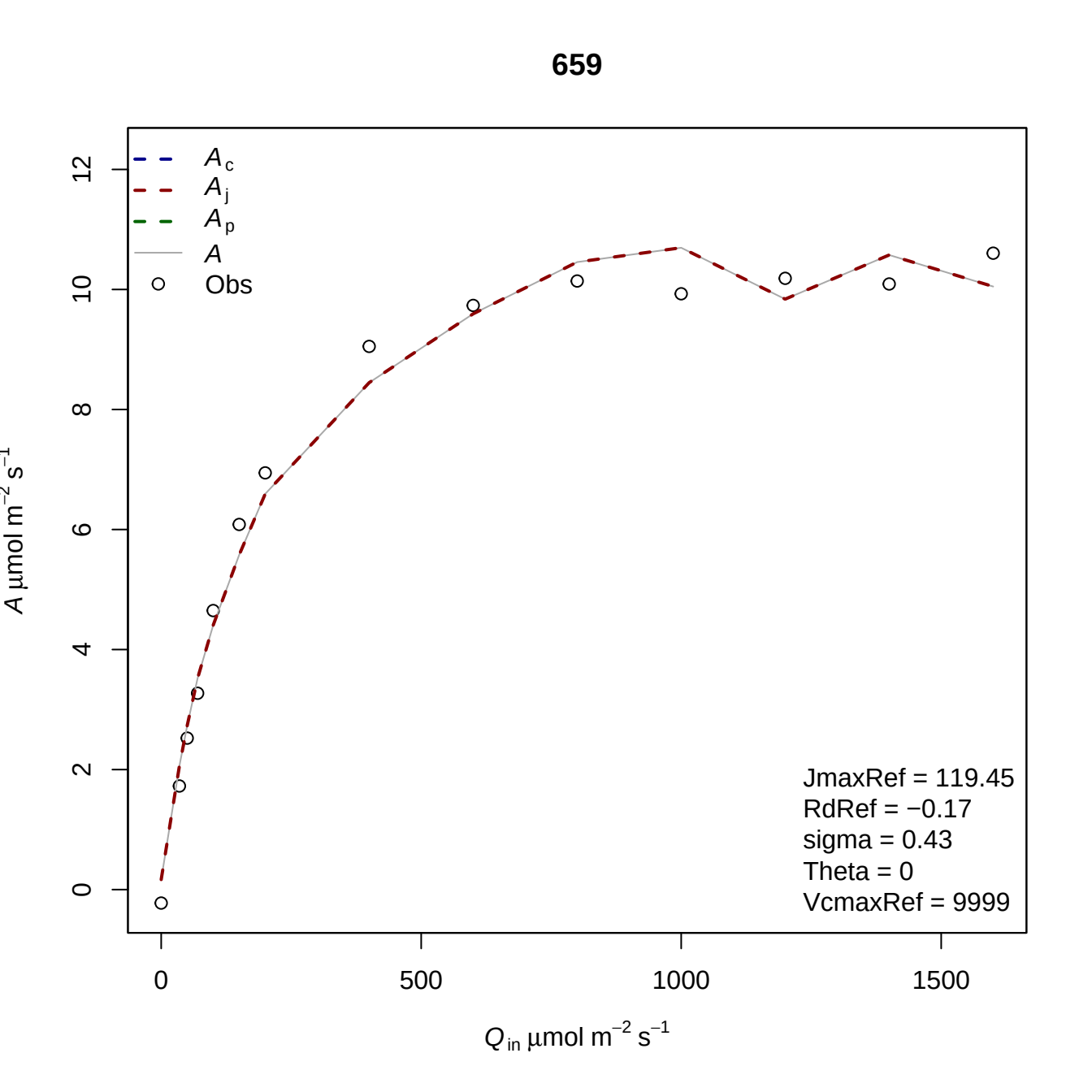

654
A_c
A_j
A_p
A
Obs
JmaxRef = 118.83
RdRef = 1.36
sigma = 0.2
Theta = 0
VcmaxRef = 64.61
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

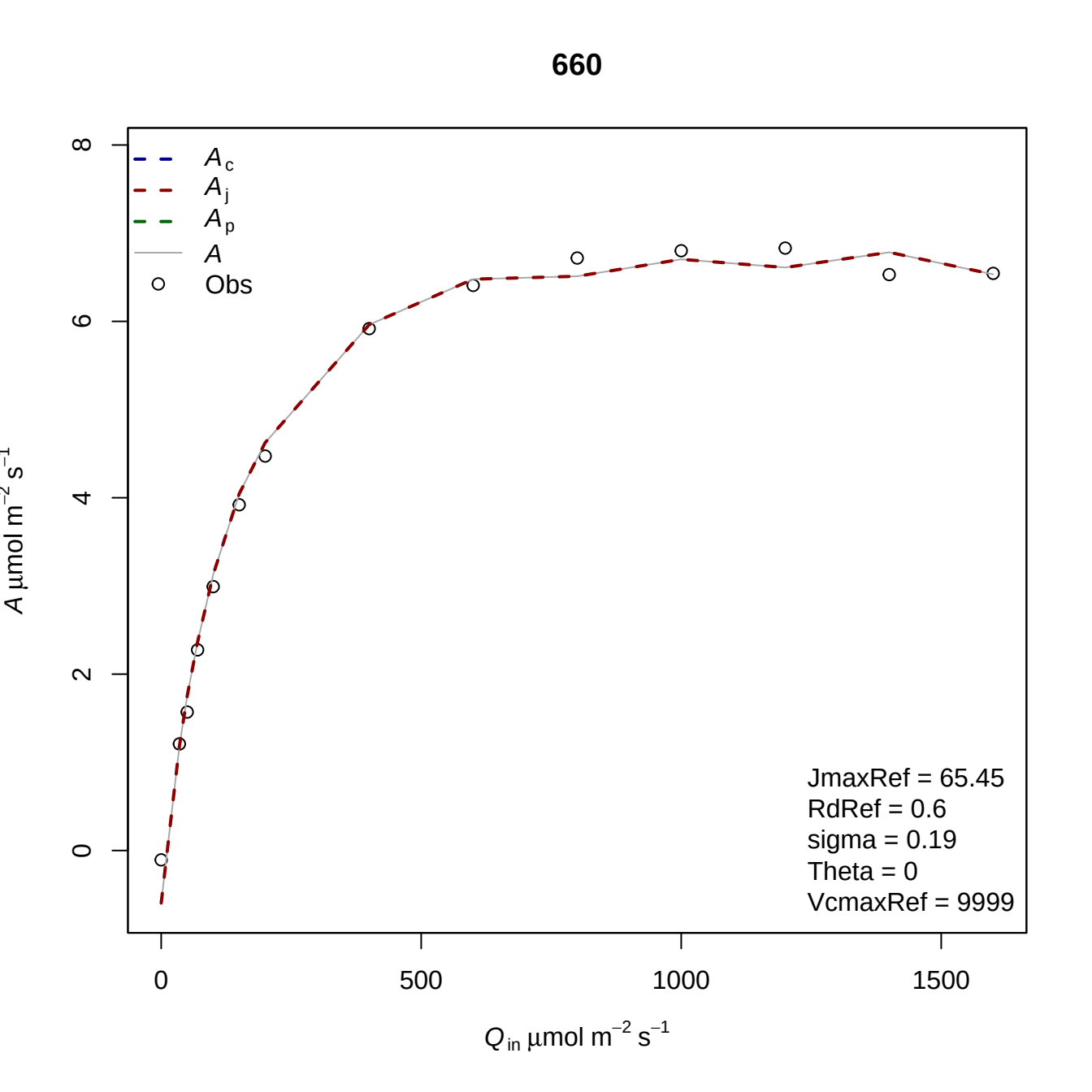

655
A_c
A_j
A_p
A
Obs
JmaxRef = 245.52
RdRef = 2.87
sigma = 0.31
Theta = 0
VcmaxRef = 101.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

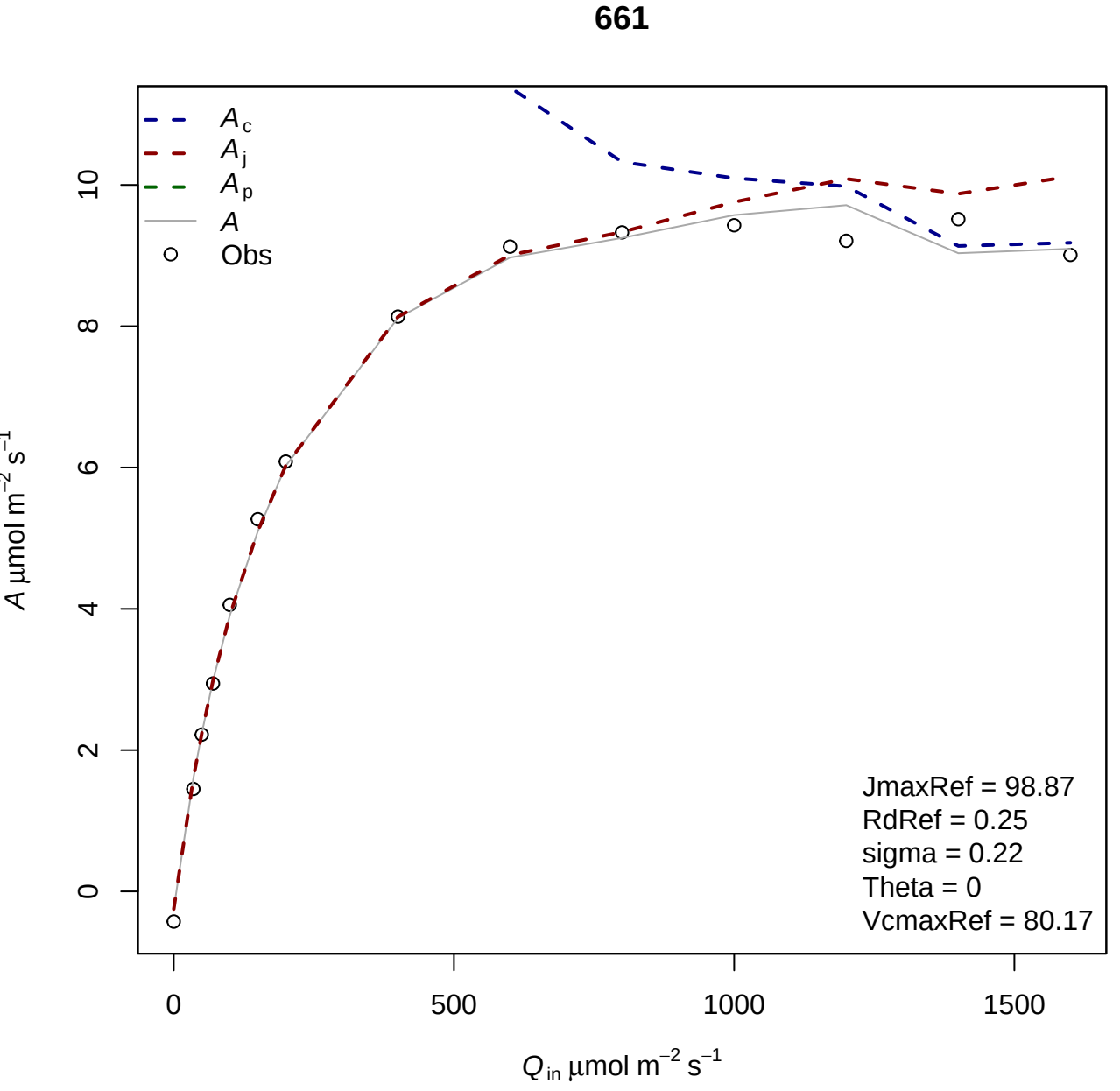

656
A_c
A_j
A_p
A
Obs
JmaxRef = 159.43
RdRef = 0.76
sigma = 0.33
Theta = 0
VcmaxRef = 70.87
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

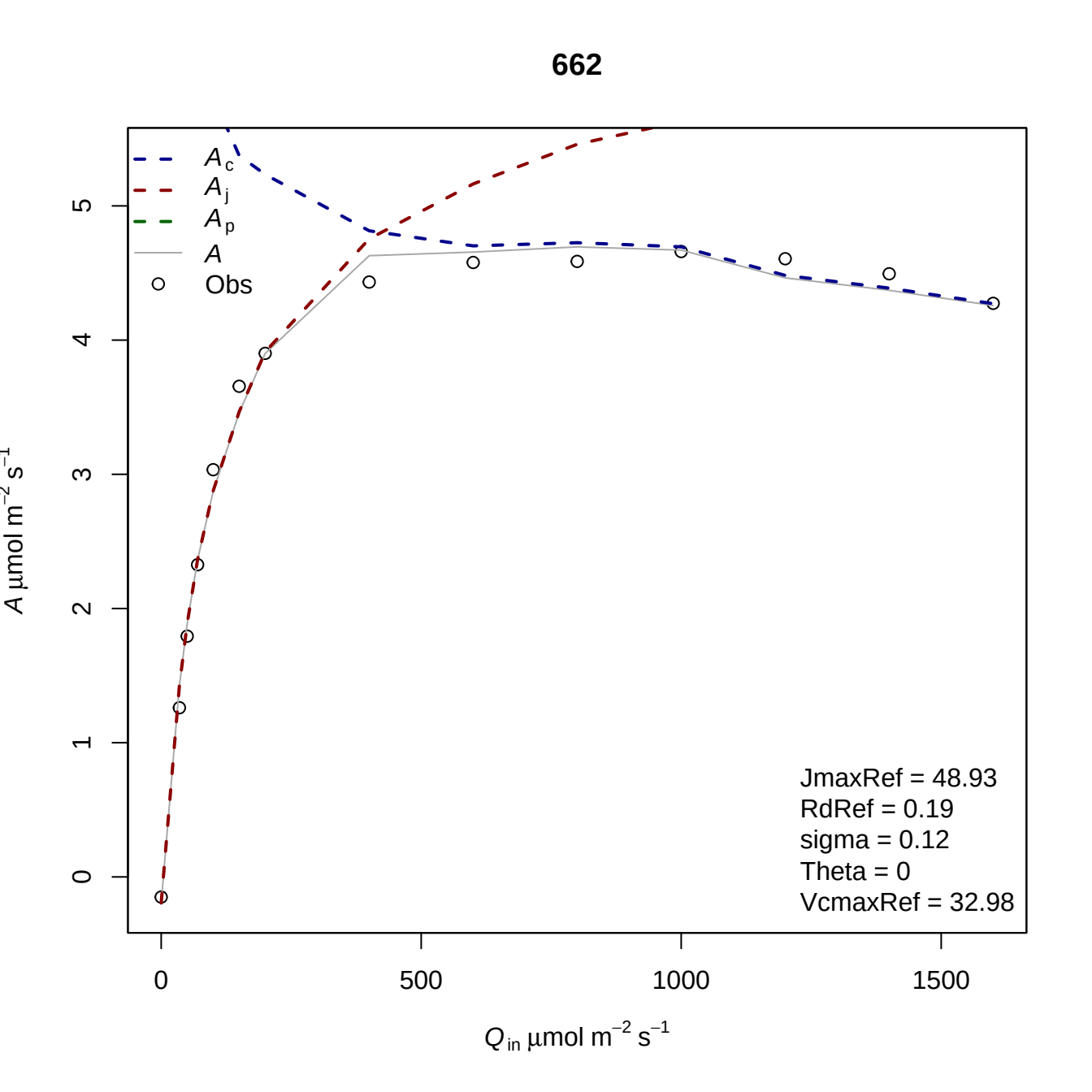

657
A_c
A_j
A_p
A
Obs
JmaxRef = 175.02
RdRef = 0.58
sigma = 0.47
Theta = 0
VcmaxRef = 76.27
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

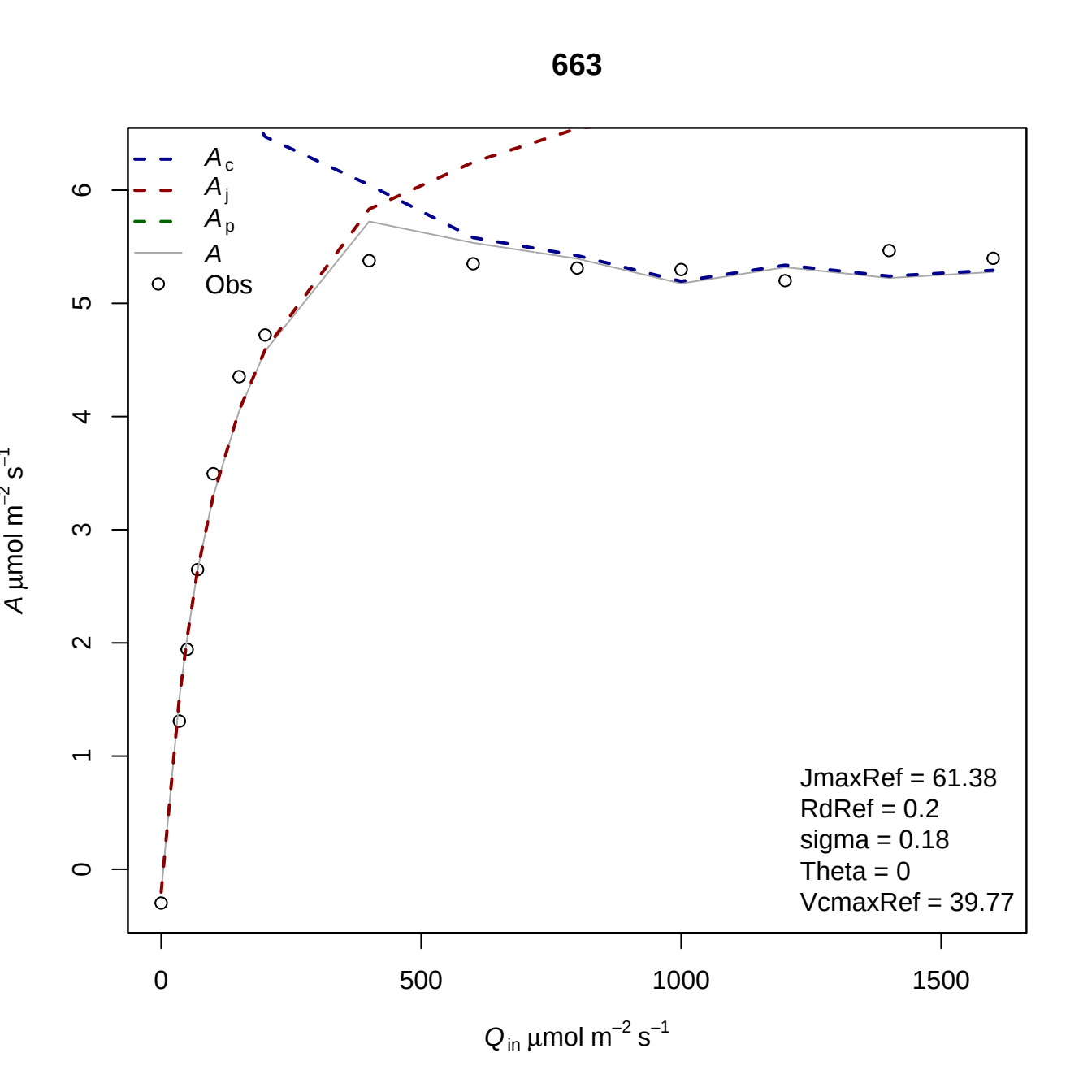

658
A_c
A_j
A_p
A
Obs
JmaxRef = 158.47
RdRef = 0.68
sigma = 0.21
Theta = 0
VcmaxRef = 82.24
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

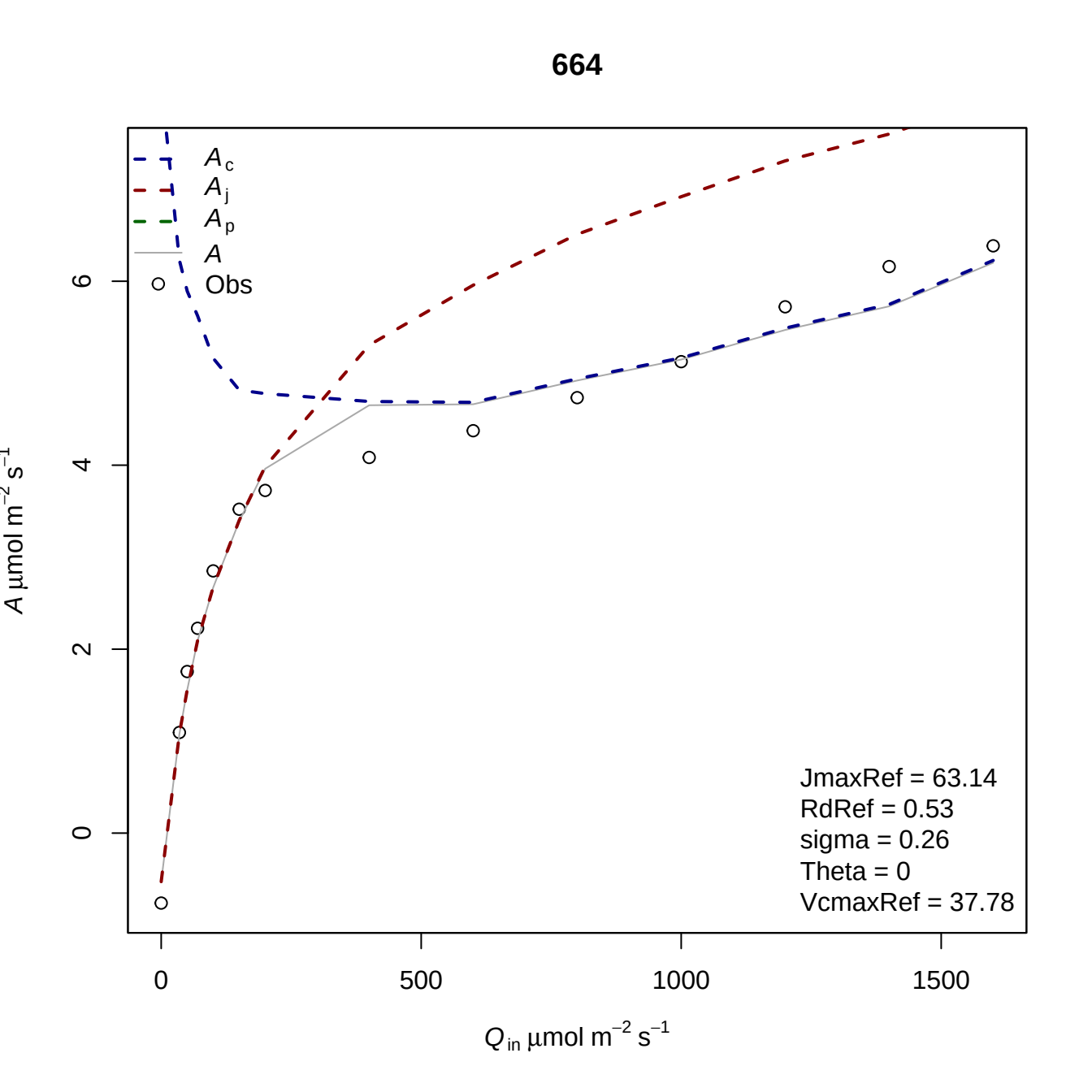

659
A_c
A_j
A_p
A
Obs
JmaxRef = 119.45
RdRef = −0.17
sigma = 0.43
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

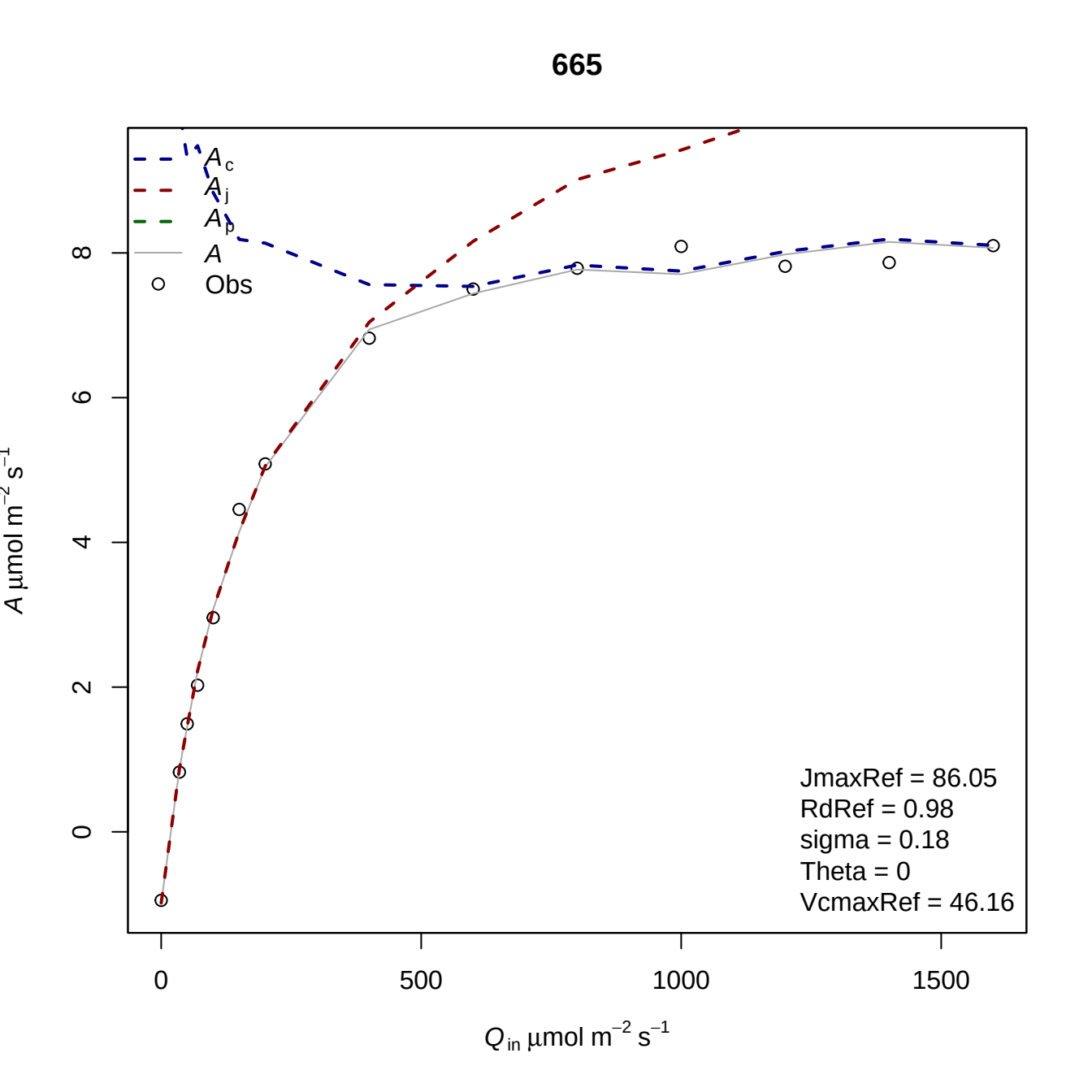

660
A_c
A_j
A_p
A
Obs
JmaxRef = 65.45
RdRef = 0.6
sigma = 0.19
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

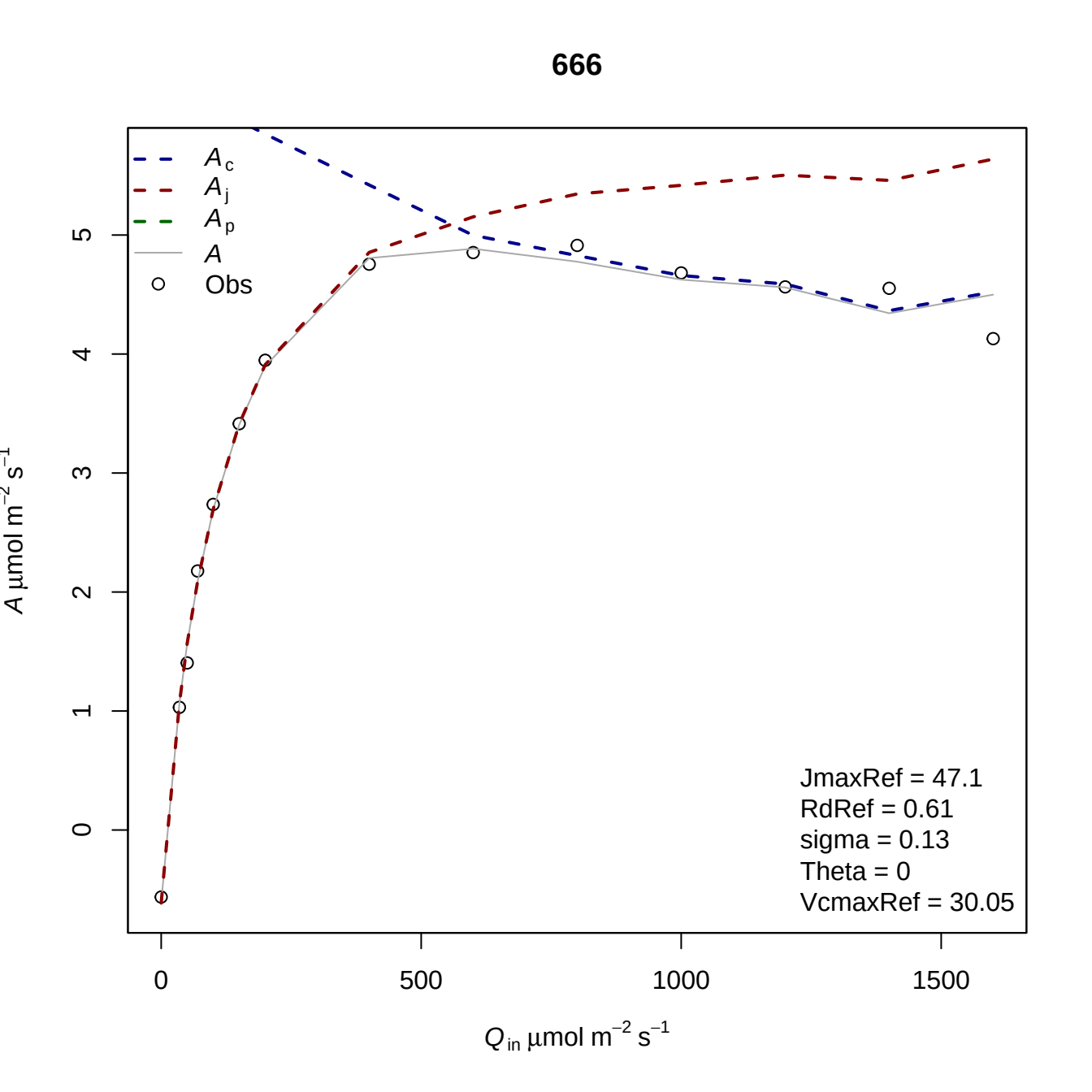

661
A_c
A_j
A_p
A
Obs
JmaxRef = 98.87
RdRef = 0.25
sigma = 0.22
Theta = 0
VcmaxRef = 80.17
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

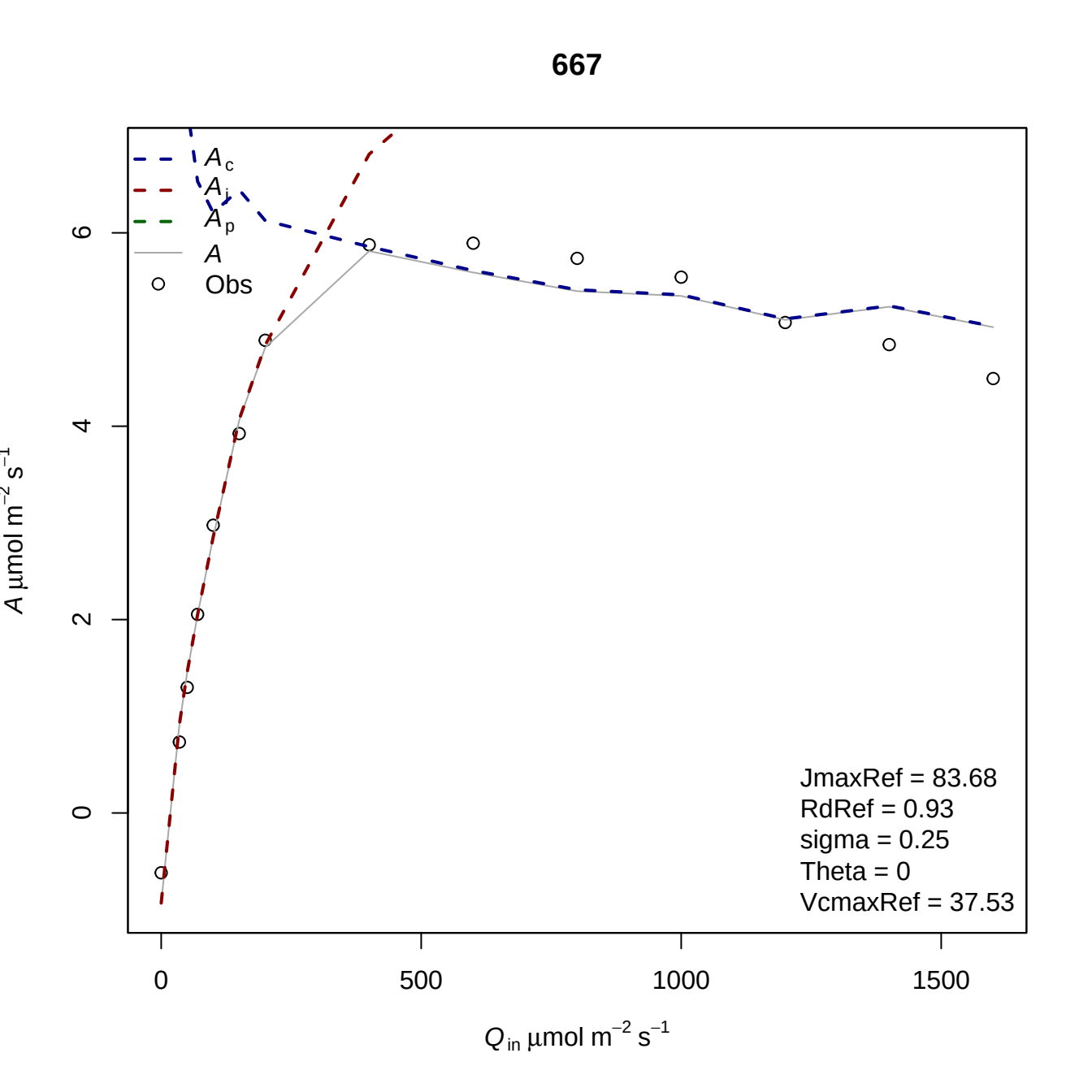

662
A_c
A_j
A_p
A
Obs
JmaxRef = 48.93
RdRef = 0.19
sigma = 0.12
Theta = 0
VcmaxRef = 32.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

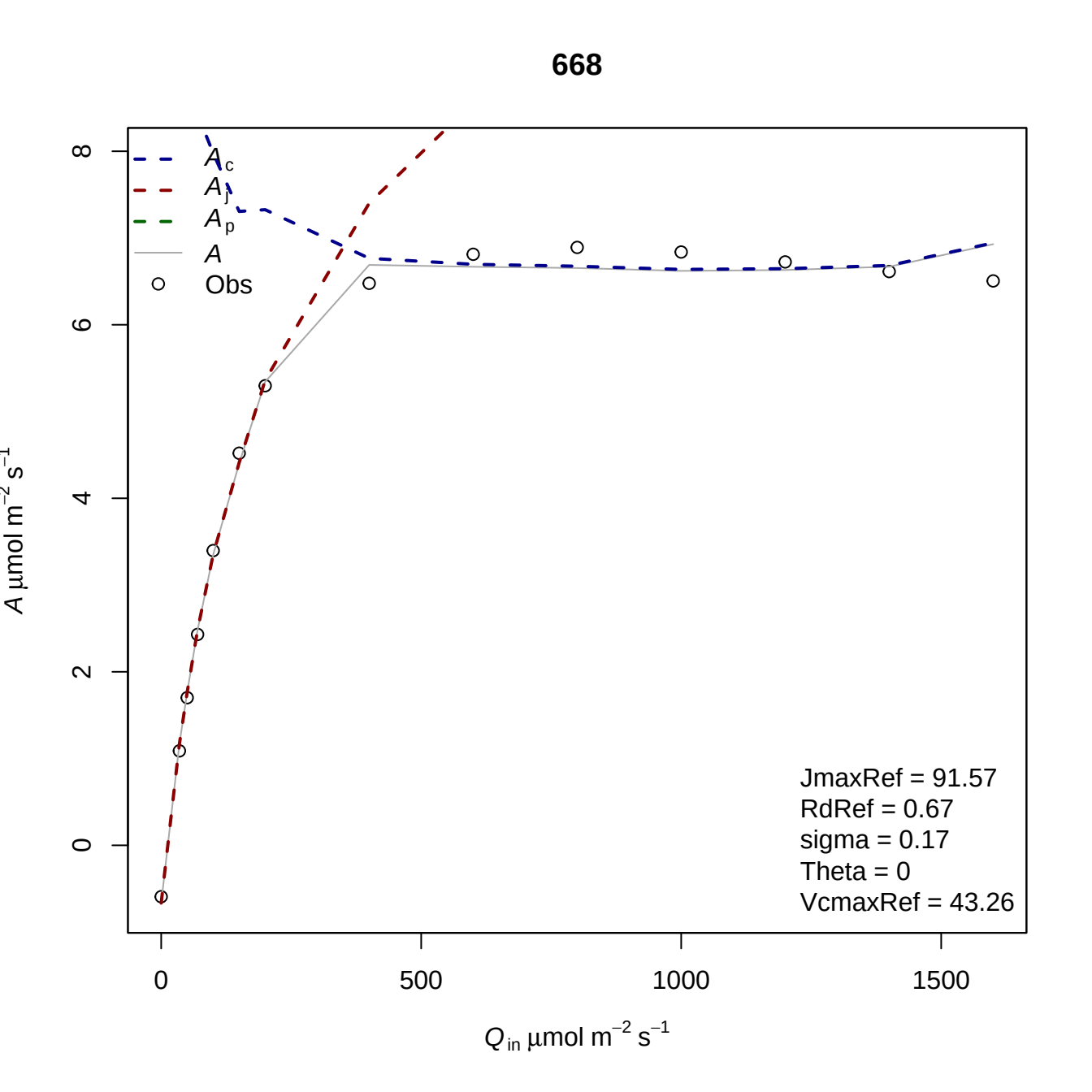

663
A_c
A_j
A_p
A
Obs
JmaxRef = 61.38
RdRef = 0.2
sigma = 0.18
Theta = 0
VcmaxRef = 39.77
0
1
2
3
4
5
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

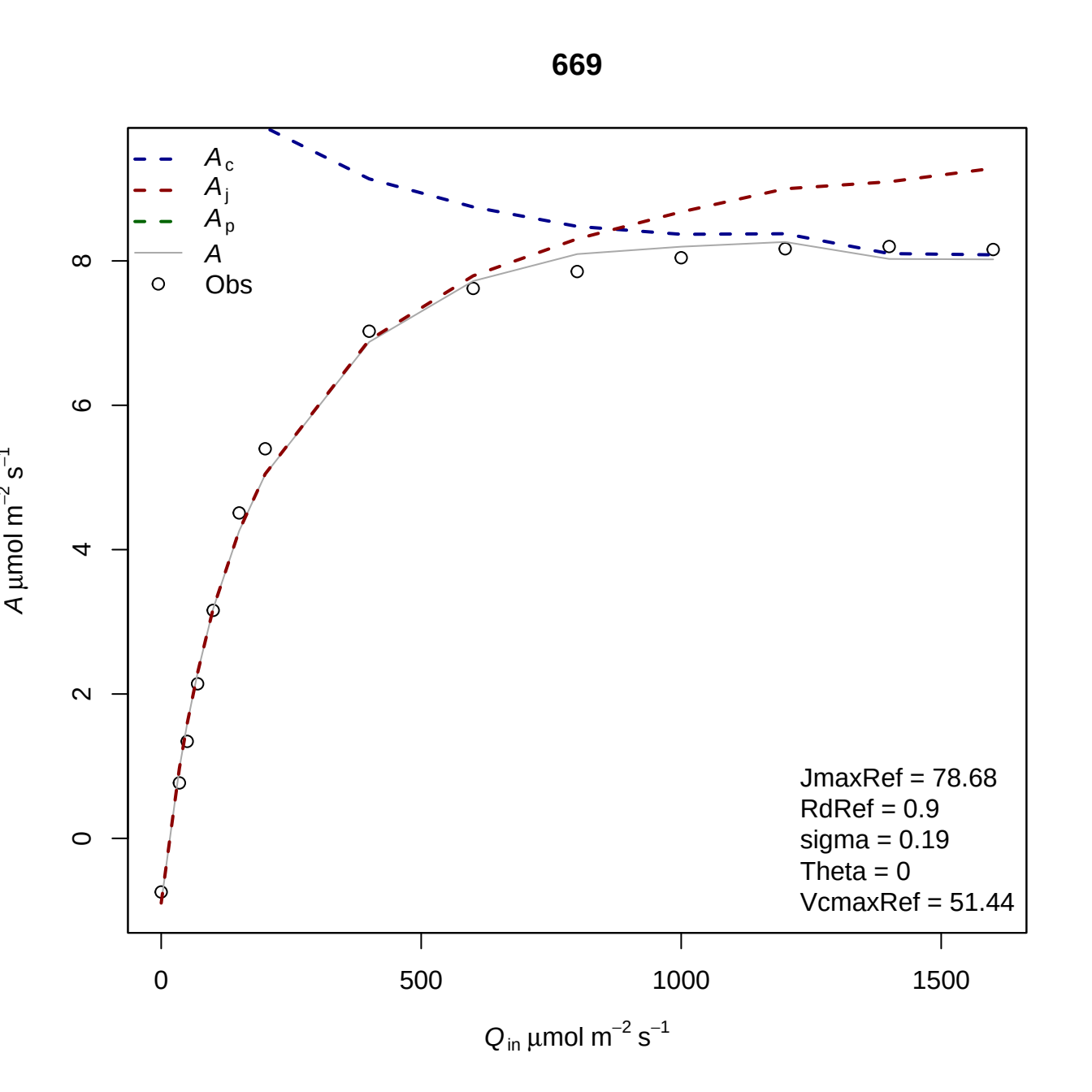

664
A_c
A_j
A_p
A
Obs
JmaxRef = 63.14
RdRef = 0.53
sigma = 0.26
Theta = 0
VcmaxRef = 37.78
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

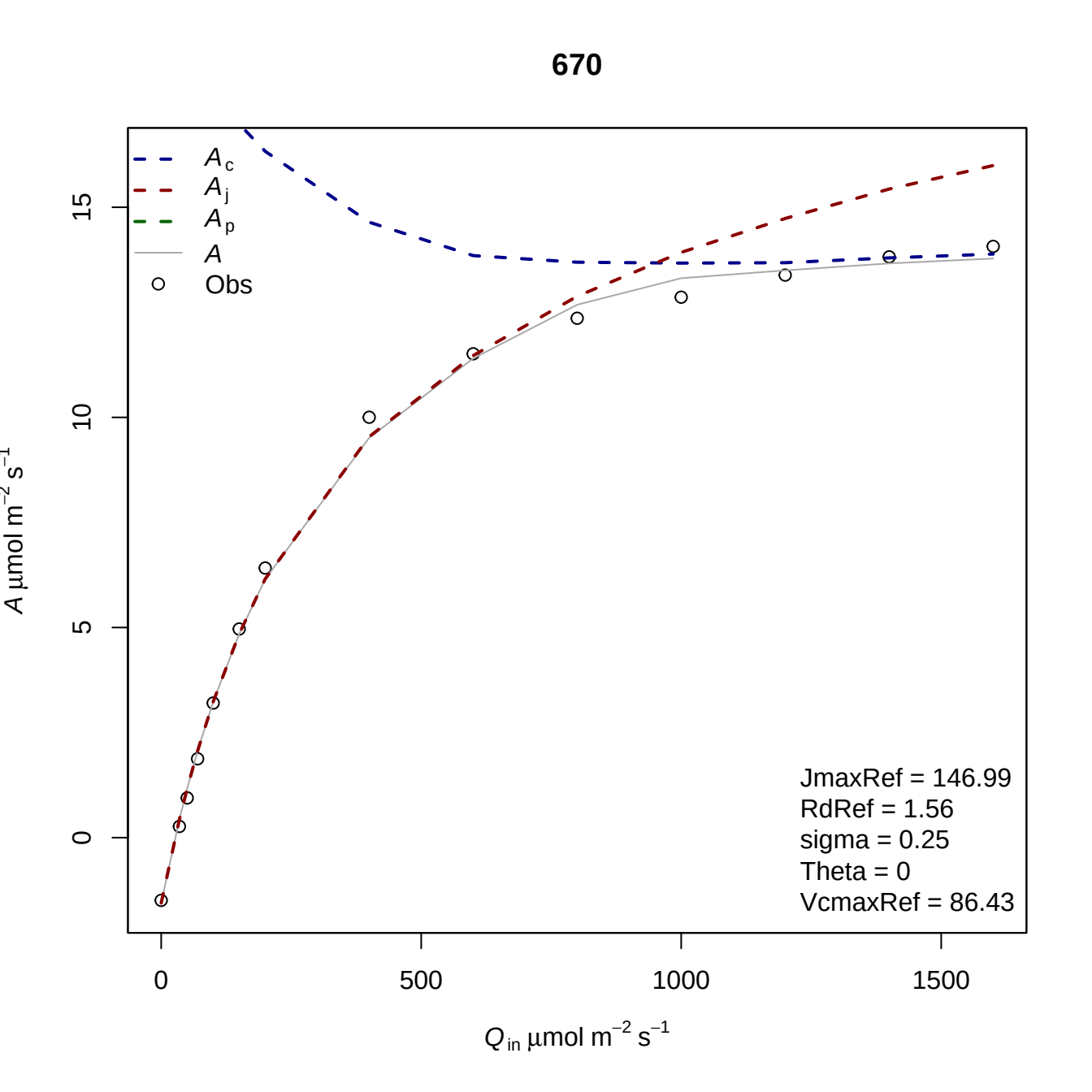

665
A_c
A_j
A_p
A
Obs
JmaxRef = 86.05
RdRef = 0.98
sigma = 0.18
Theta = 0
VcmaxRef = 46.16
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

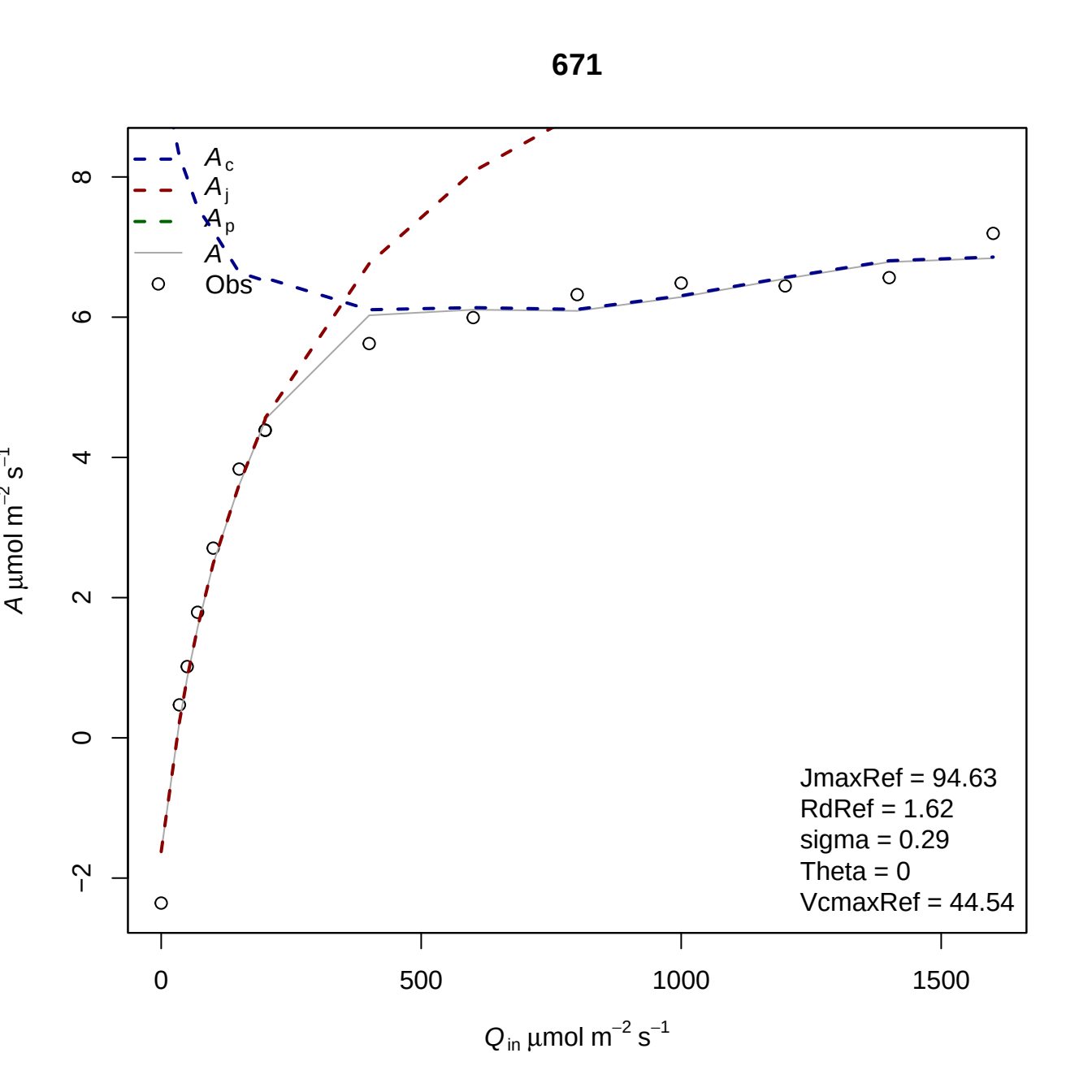

666
A_c
A_j
A_p
A
Obs
JmaxRef = 47.1
RdRef = 0.61
sigma = 0.13
Theta = 0
VcmaxRef = 30.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

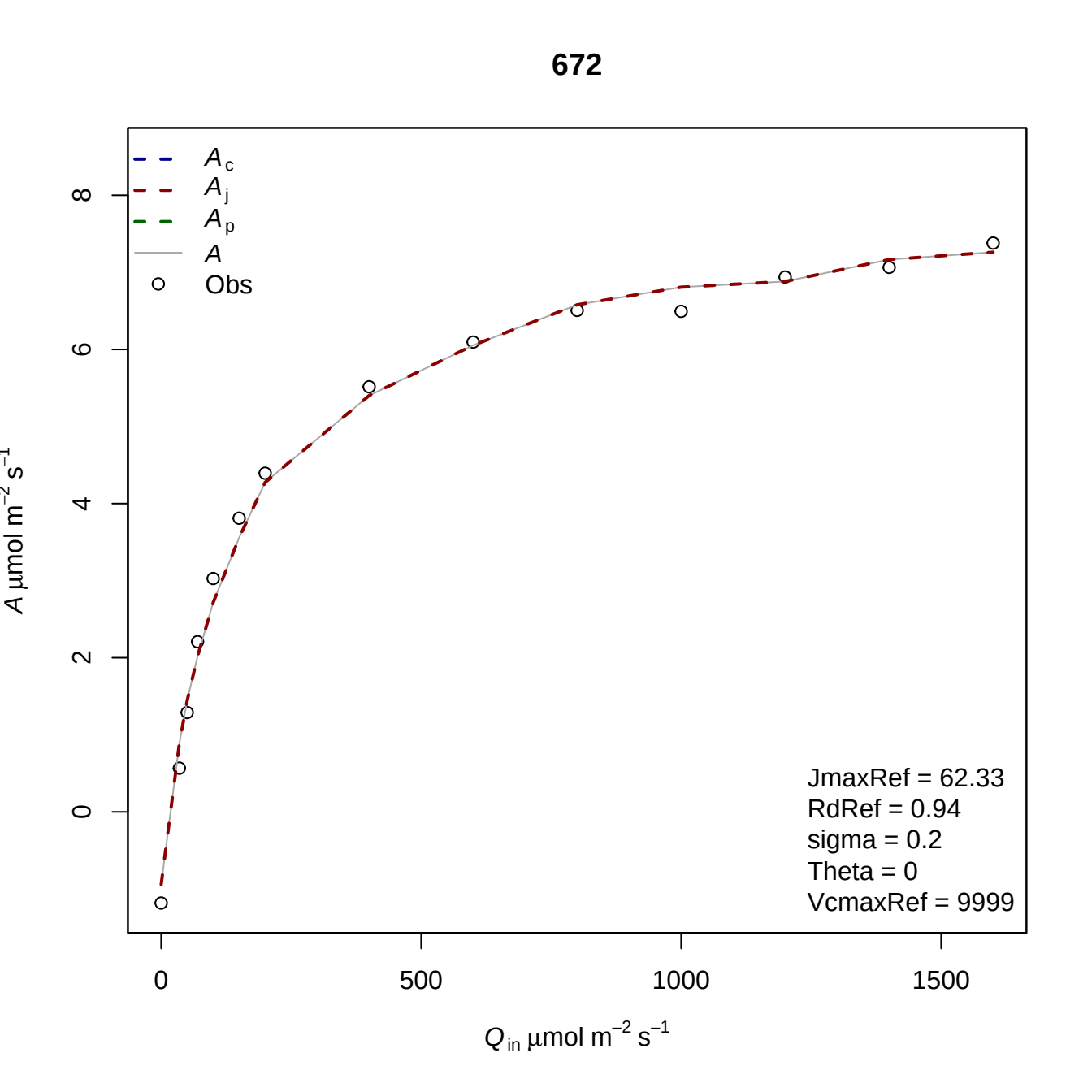

667
A_c
A_j
A_p
A
Obs
JmaxRef = 83.68
RdRef = 0.93
sigma = 0.25
Theta = 0
VcmaxRef = 37.53
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

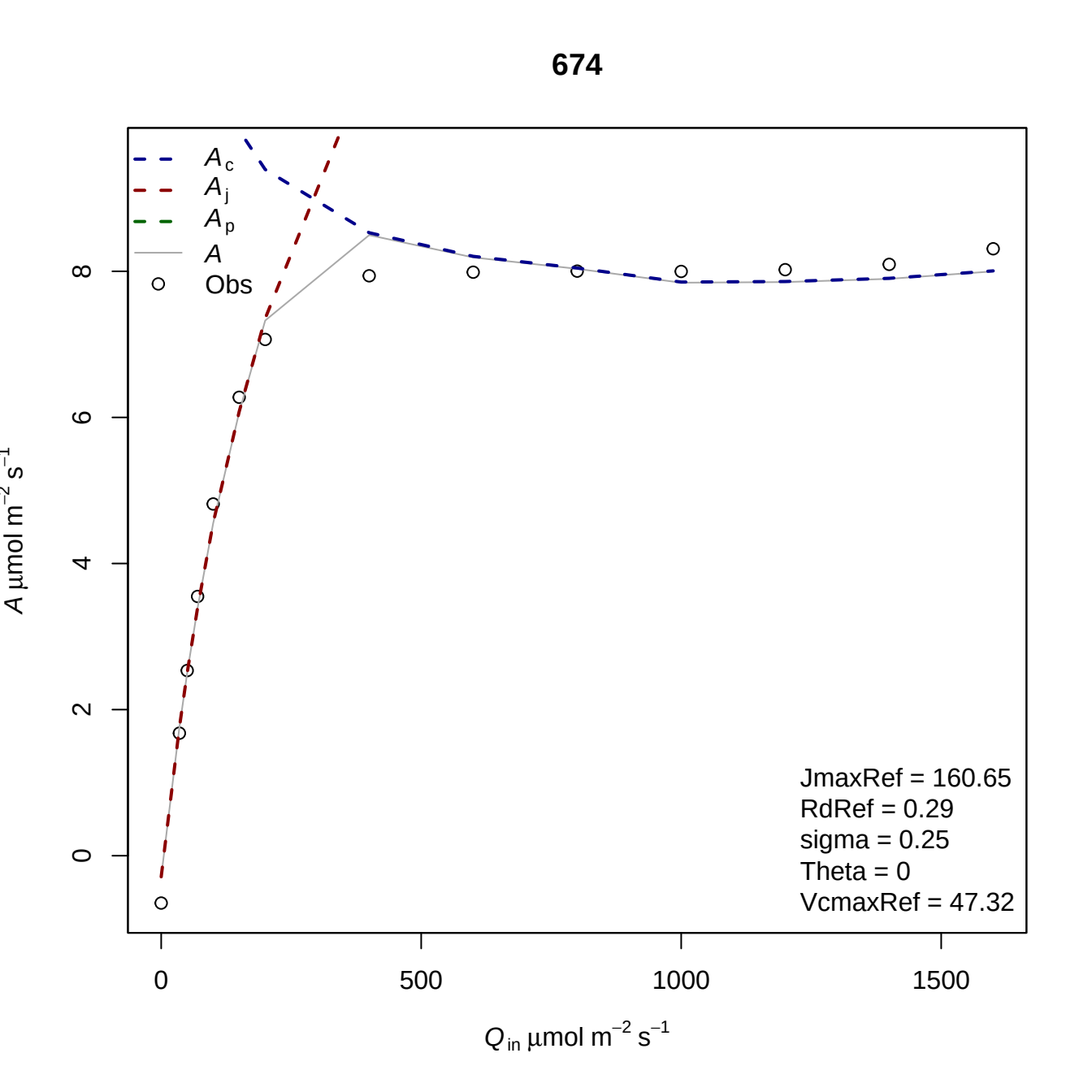

668
A_c
A_j
A_p
A
Obs
JmaxRef = 91.57
RdRef = 0.67
sigma = 0.17
Theta = 0
VcmaxRef = 43.26
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

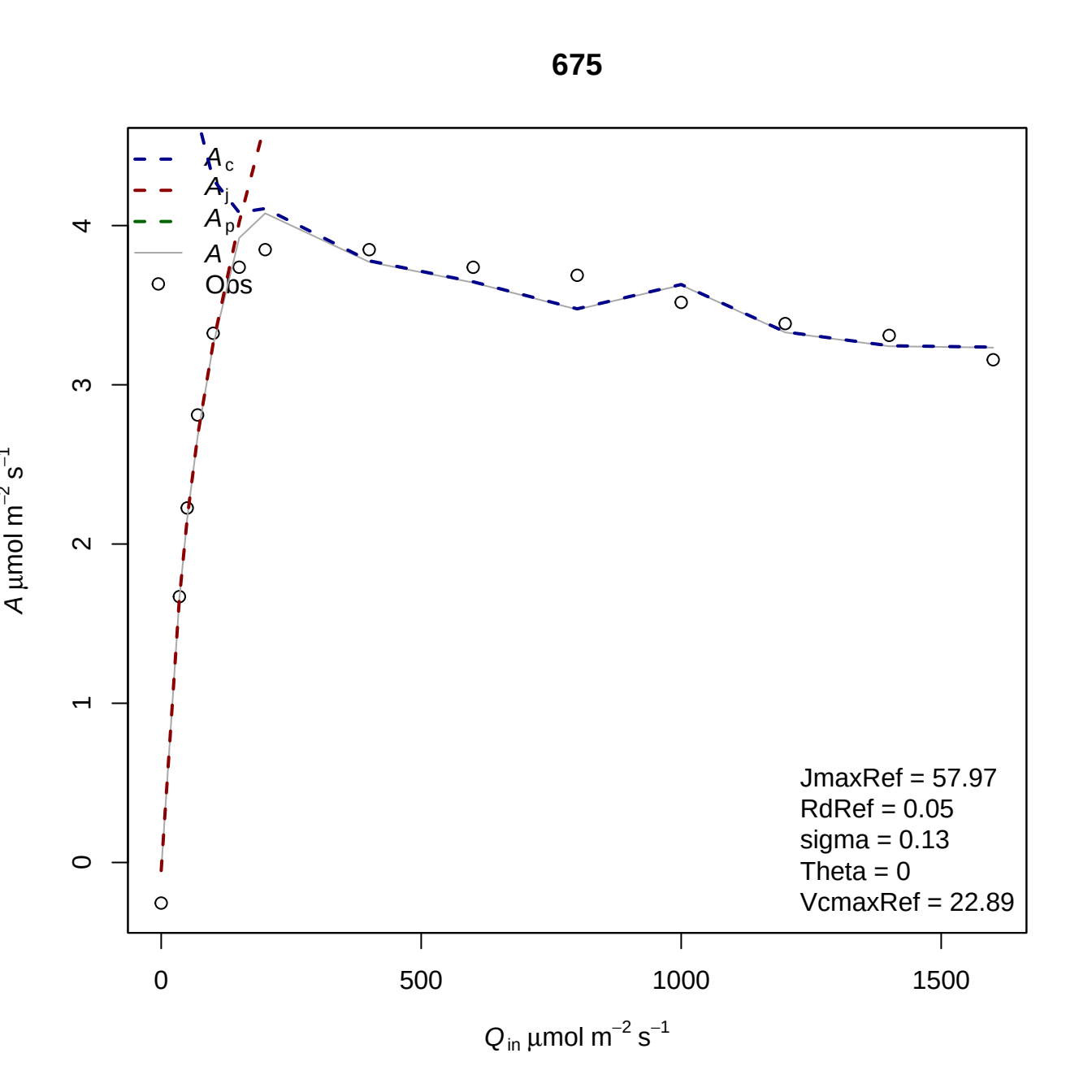

669
A_c
A_j
A_p
A
Obs
JmaxRef = 78.68
RdRef = 0.9
sigma = 0.19
Theta = 0
VcmaxRef = 51.44
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

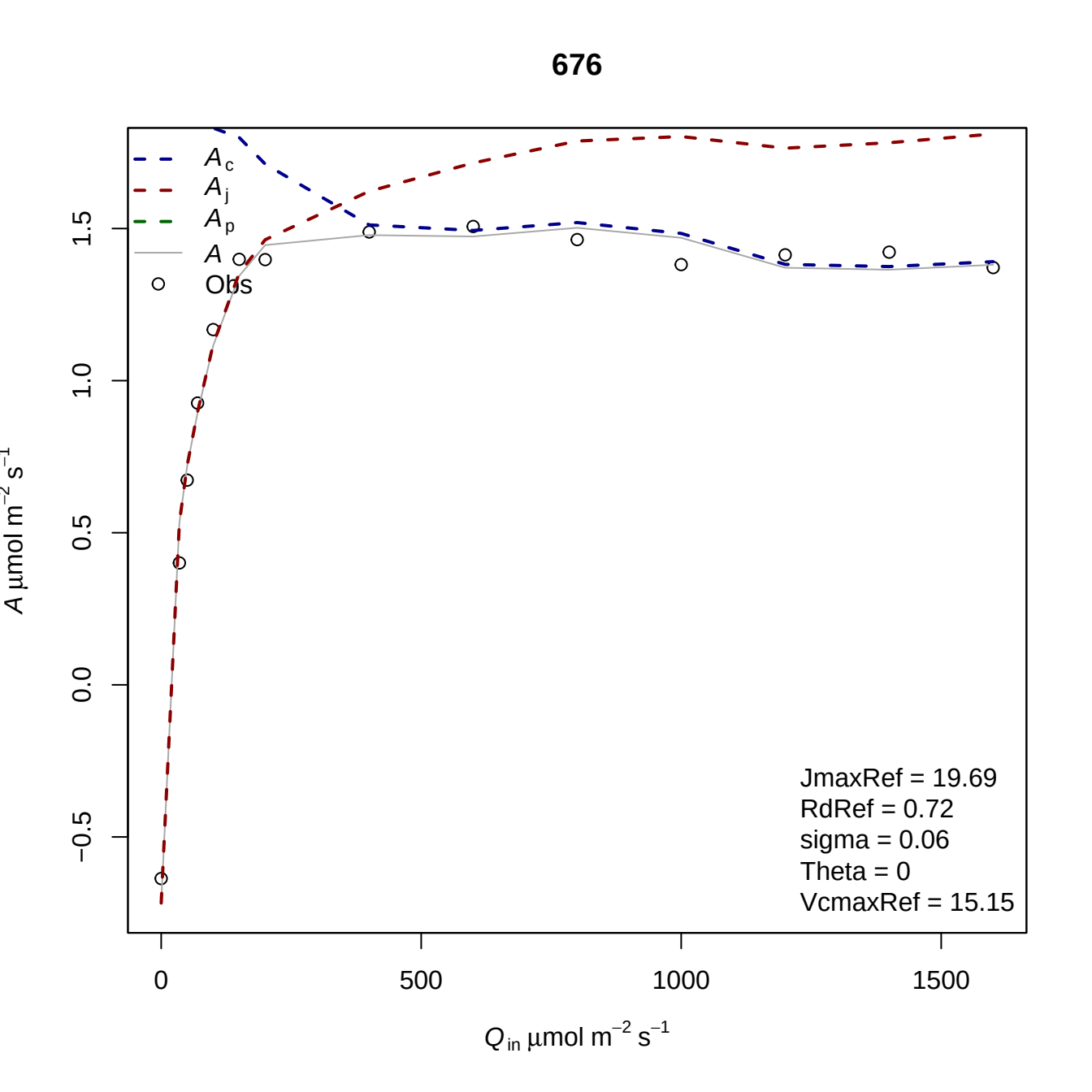

670
A_c
A_j
A_p
A
Obs
JmaxRef = 146.99
RdRef = 1.56
sigma = 0.25
Theta = 0
VcmaxRef = 86.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

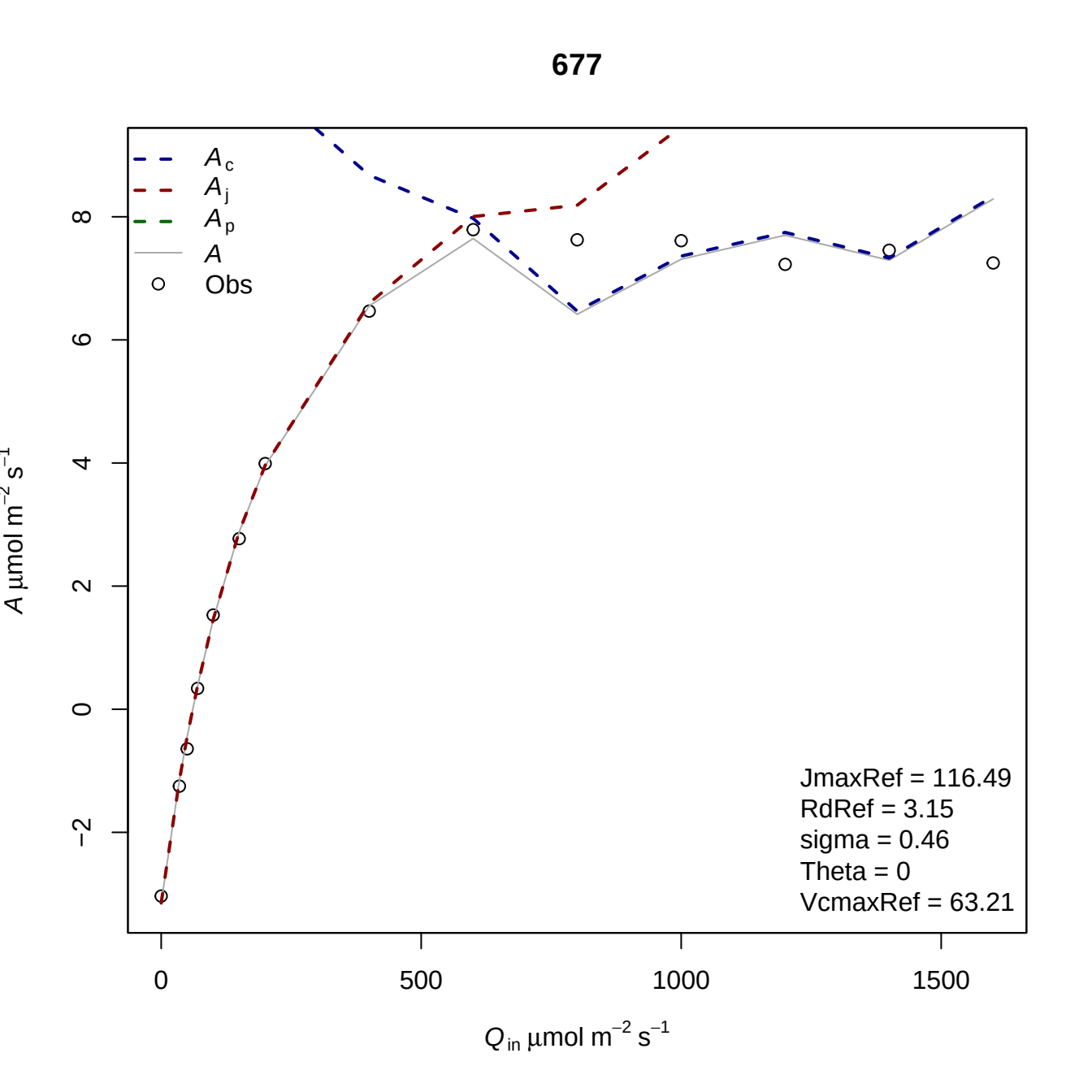

671
A_c
A_j
A_p
A
Obs
JmaxRef = 94.63
RdRef = 1.62
sigma = 0.29
Theta = 0
VcmaxRef = 44.54
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

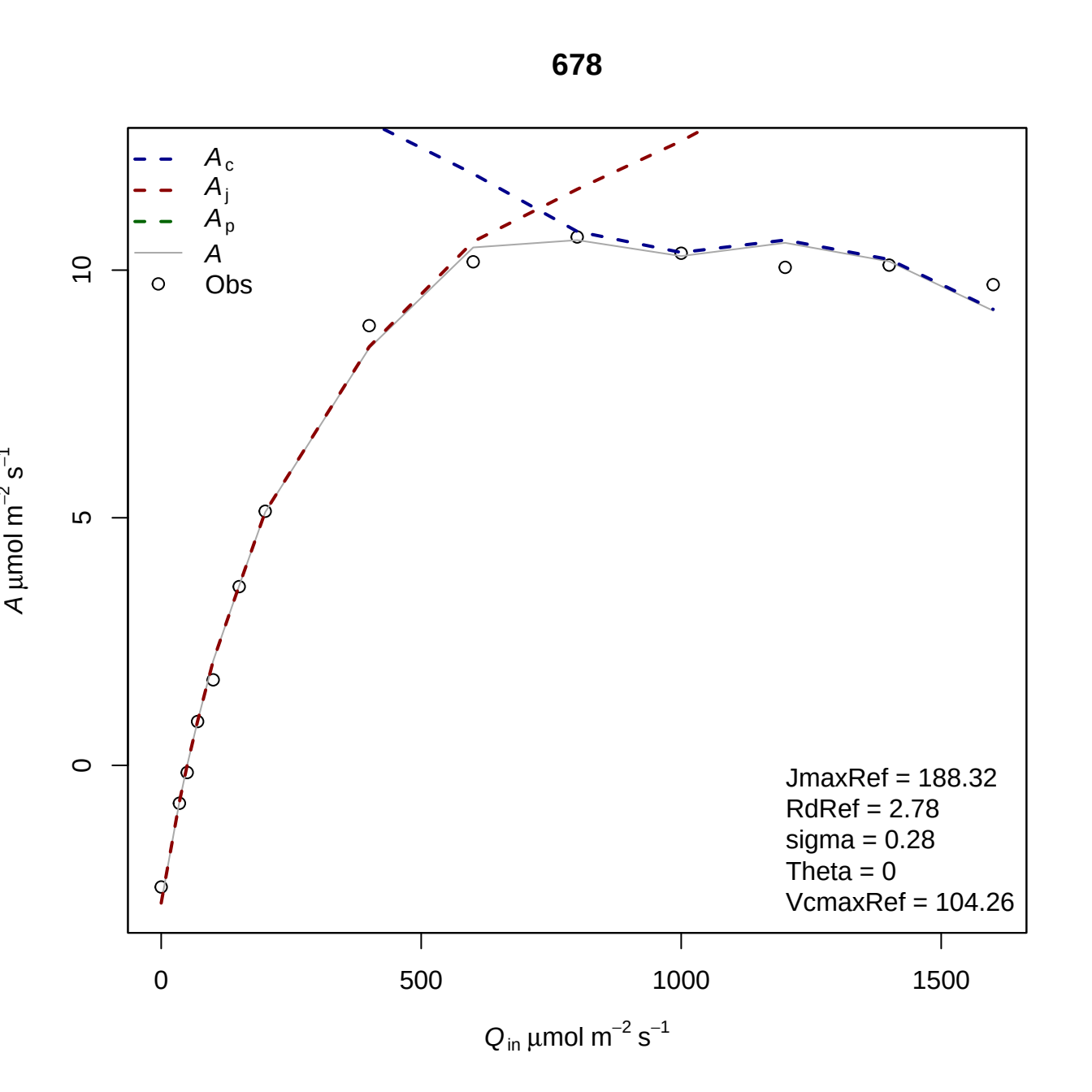

672
A_c
A_j
A_p
A
Obs
JmaxRef = 62.33
RdRef = 0.94
sigma = 0.2
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

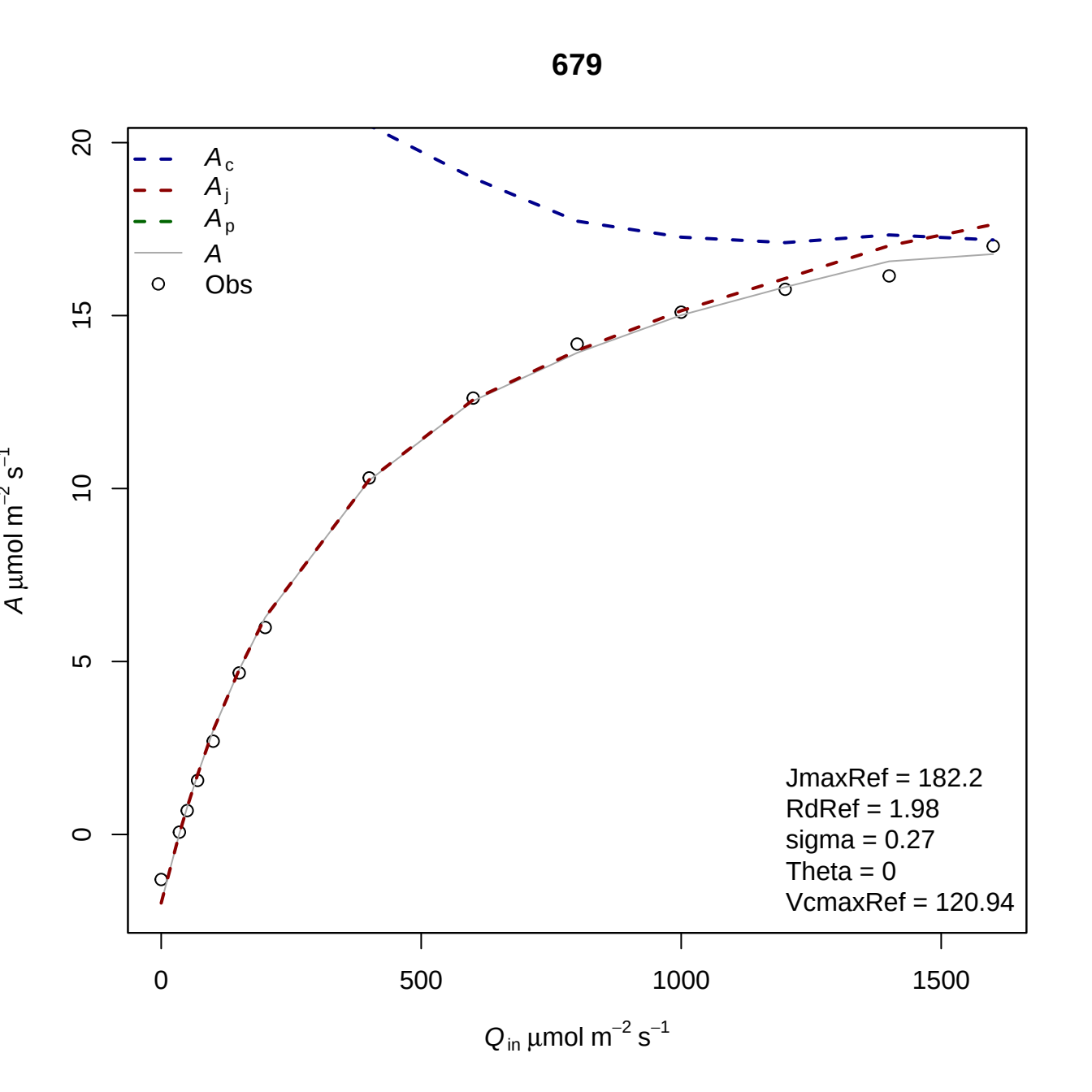

674
A_c
A_j
A_p
A
Obs
JmaxRef = 160.65
RdRef = 0.29
sigma = 0.25
Theta = 0
VcmaxRef = 47.32
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

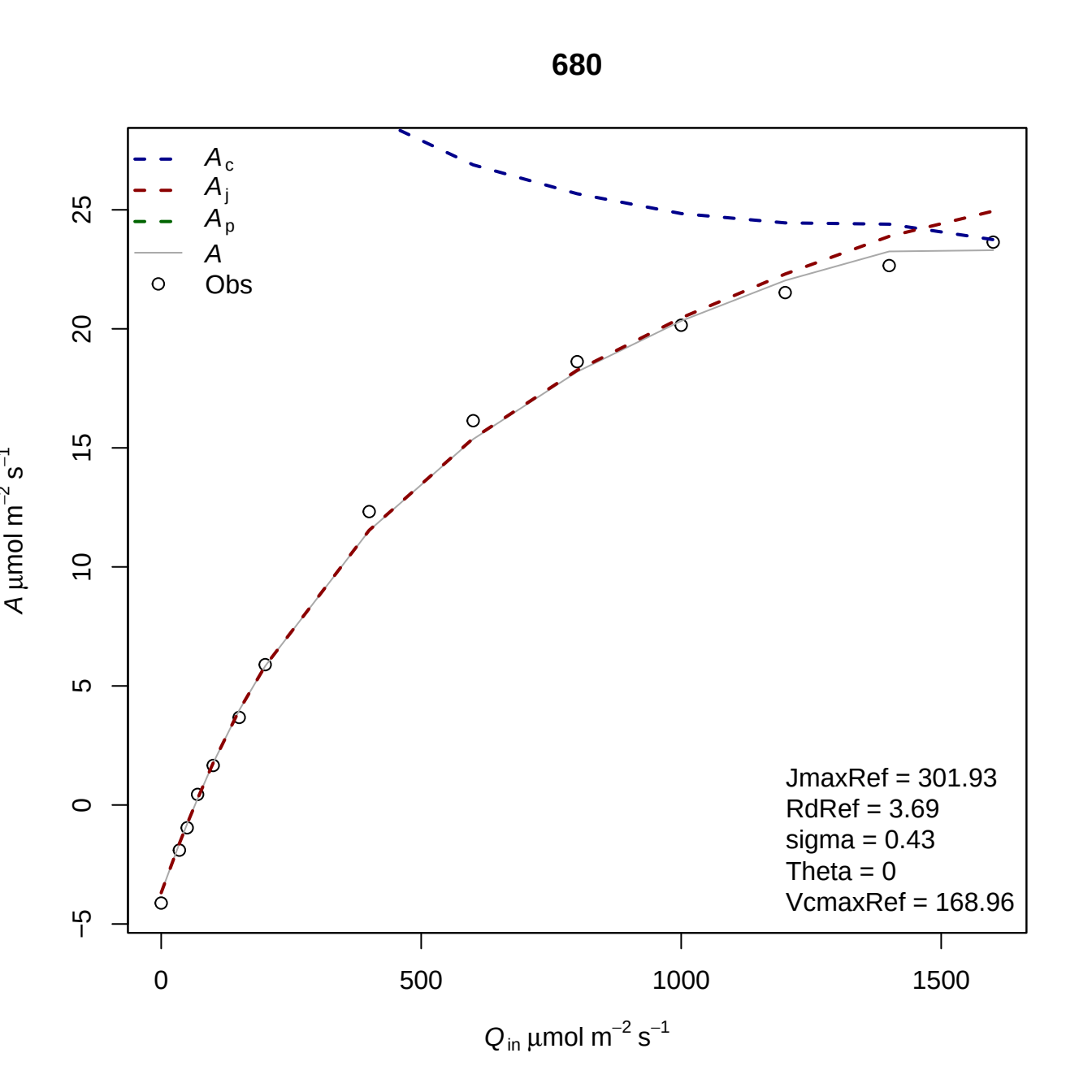

675
A_c
A_j
A_p
A
Obs
JmaxRef = 57.97
RdRef = 0.05
sigma = 0.13
Theta = 0
VcmaxRef = 22.89
0
1
2
3
4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

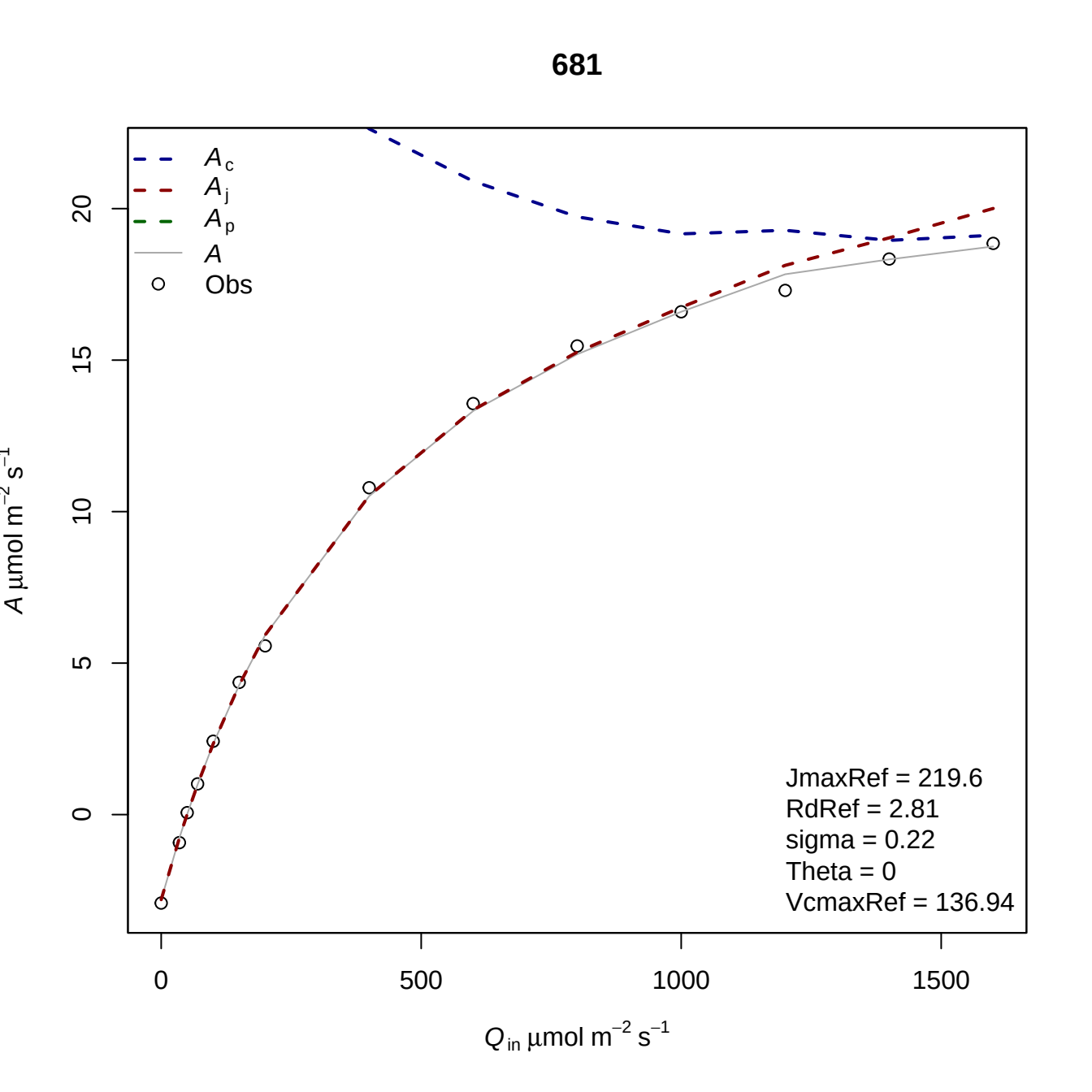

676
A_c
A_j
A_p
A
Obs
JmaxRef = 19.69
RdRef = 0.72
sigma = 0.06
Theta = 0
VcmaxRef = 15.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
0
500
1000
1500

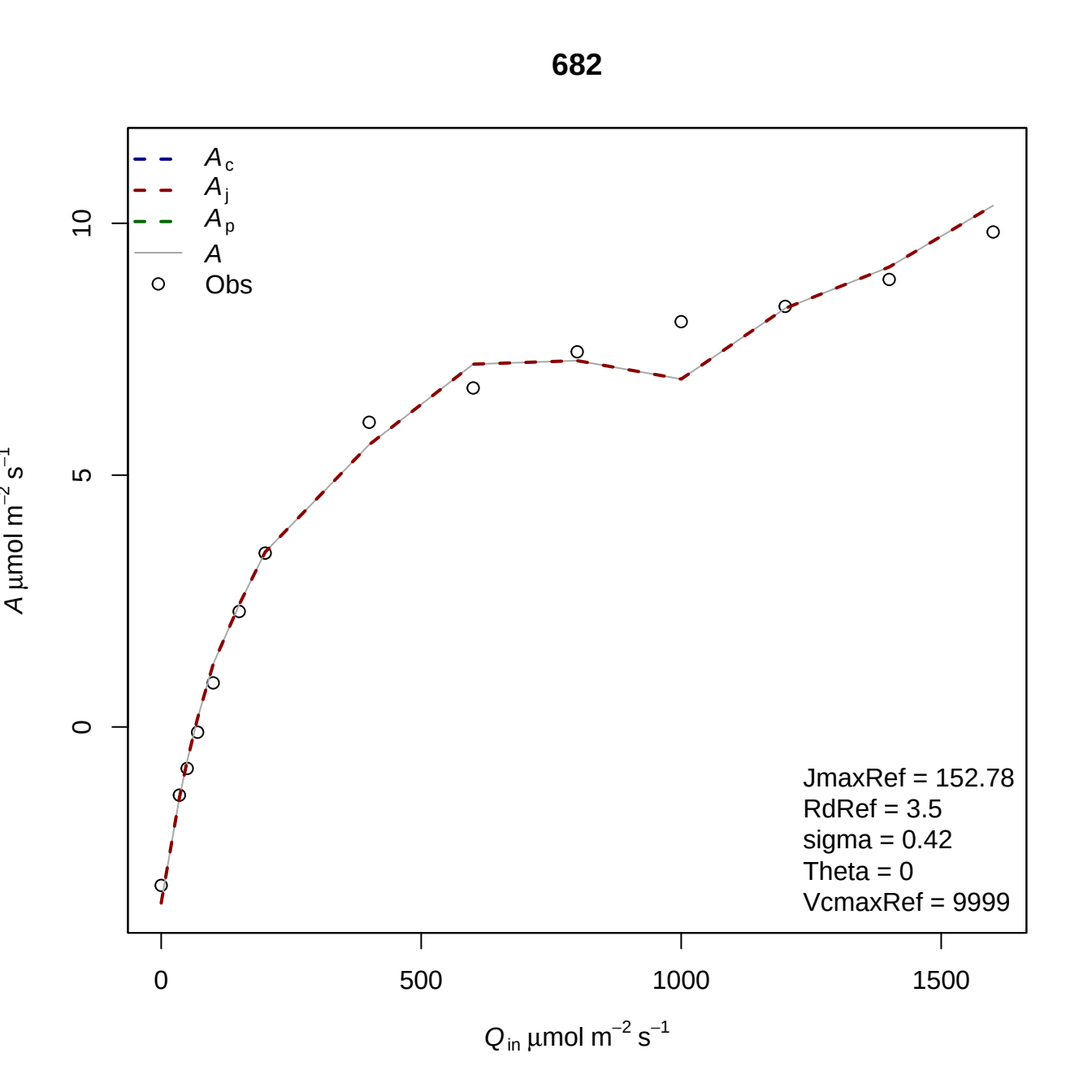

677
A_c
A_j
A_p
A
Obs
JmaxRef = 116.49
RdRef = 3.15
sigma = 0.46
Theta = 0
VcmaxRef = 63.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

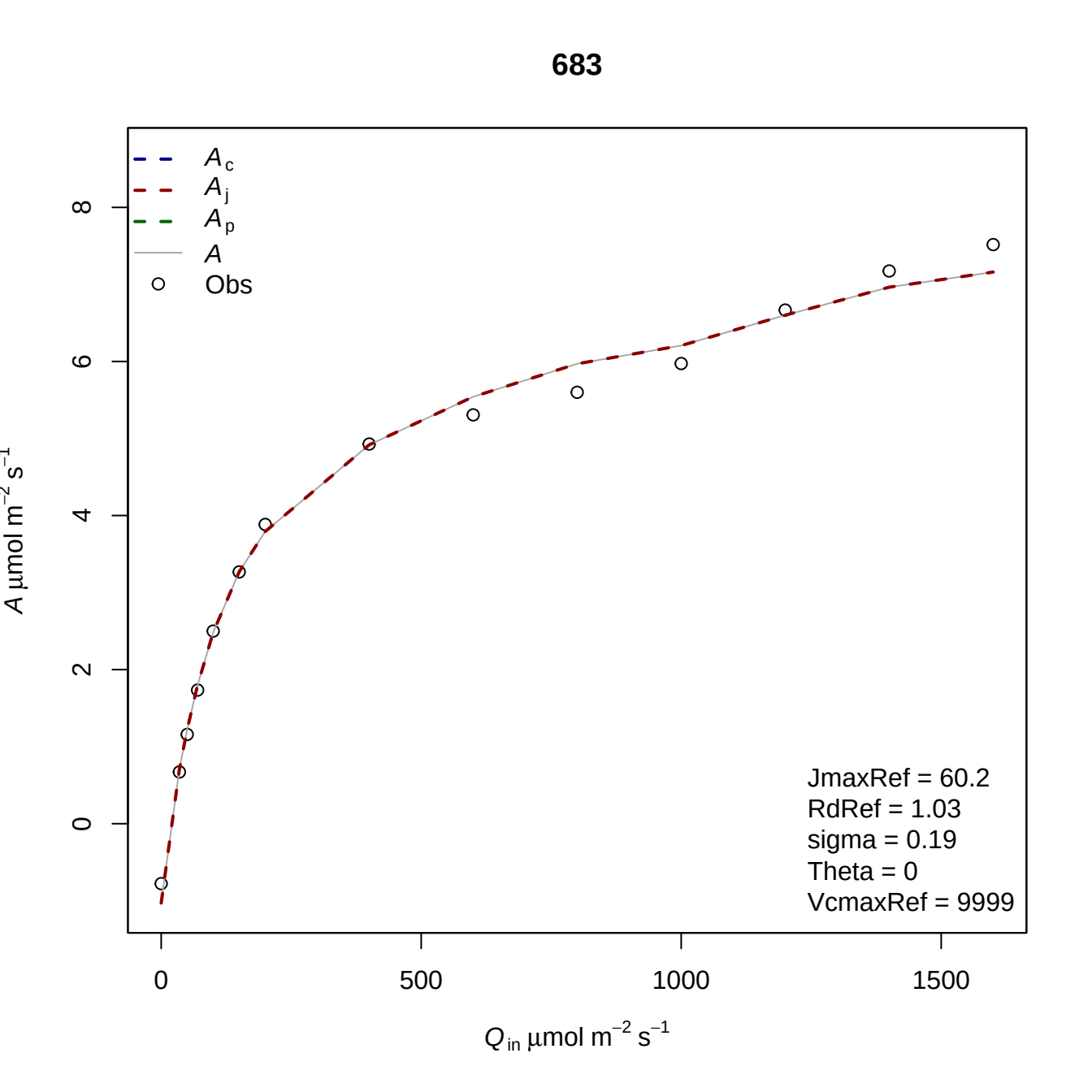

678
A_c
A_j
A_p
A
Obs
JmaxRef = 188.32
RdRef = 2.78
sigma = 0.28
Theta = 0
VcmaxRef = 104.26
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

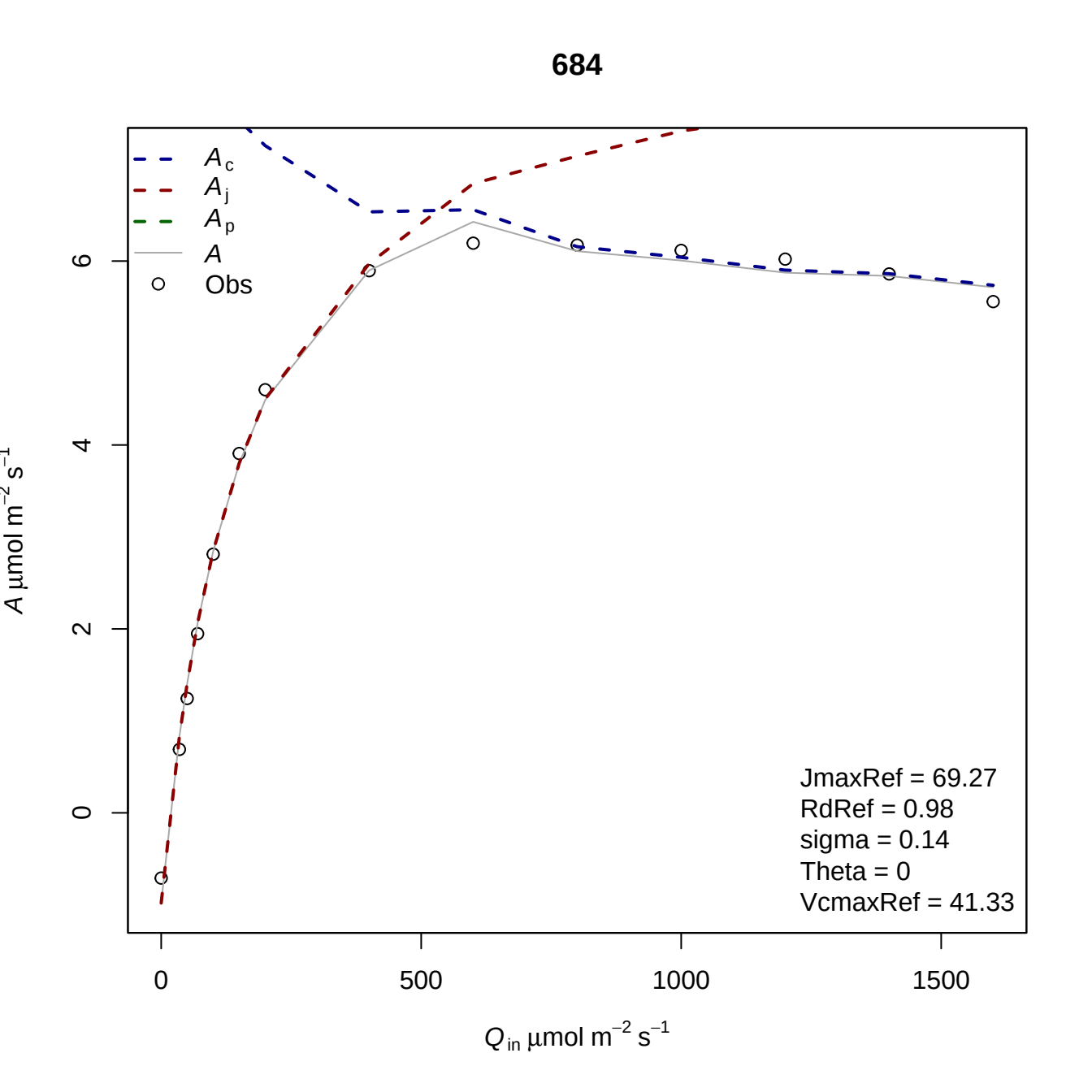

679
A_c
A_j
A_p
A
Obs
JmaxRef = 182.2
RdRef = 1.98
sigma = 0.27
Theta = 0
VcmaxRef = 120.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

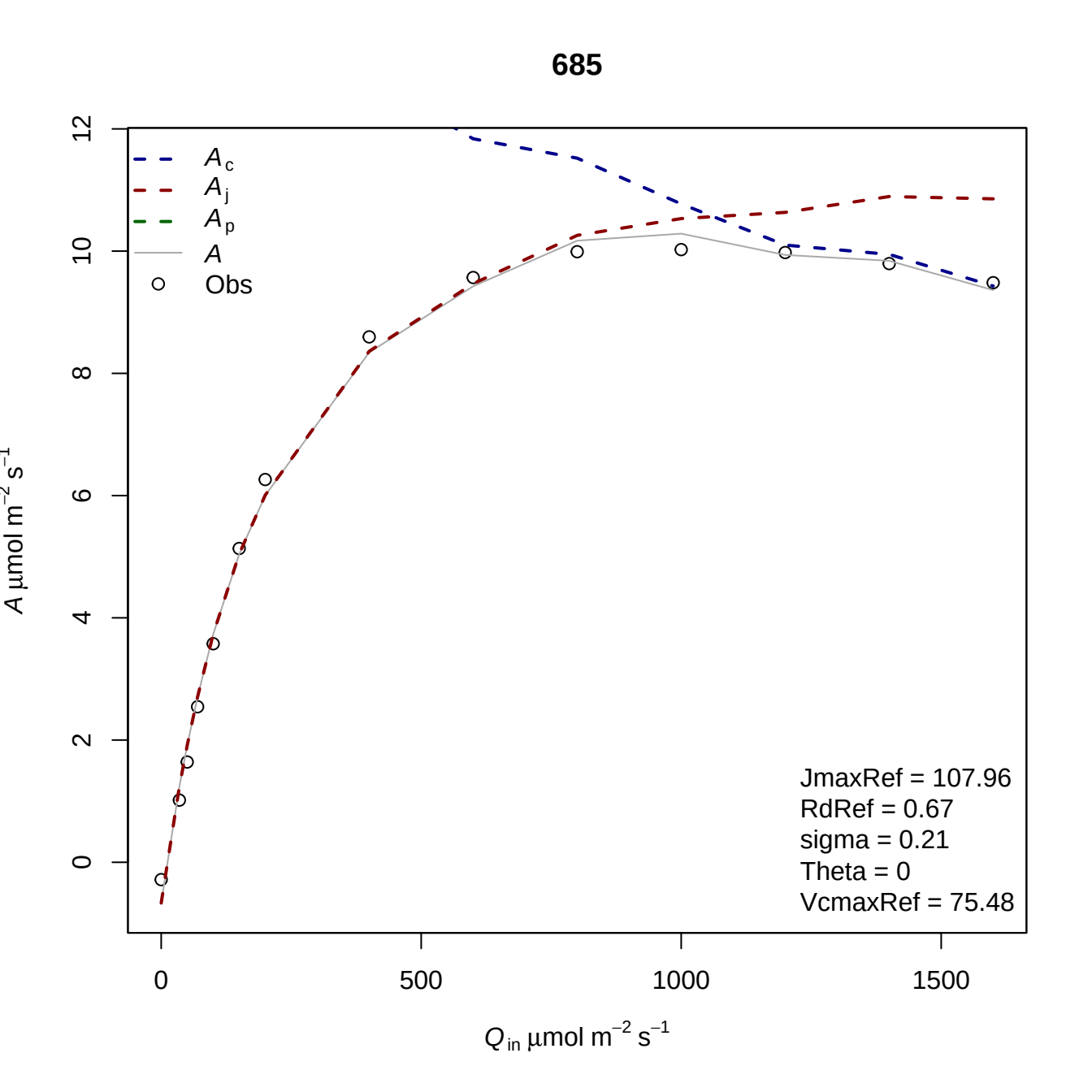

680
A_c
A_j
A_p
A
Obs
JmaxRef = 301.93
RdRef = 3.69
sigma = 0.43
Theta = 0
VcmaxRef = 168.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500

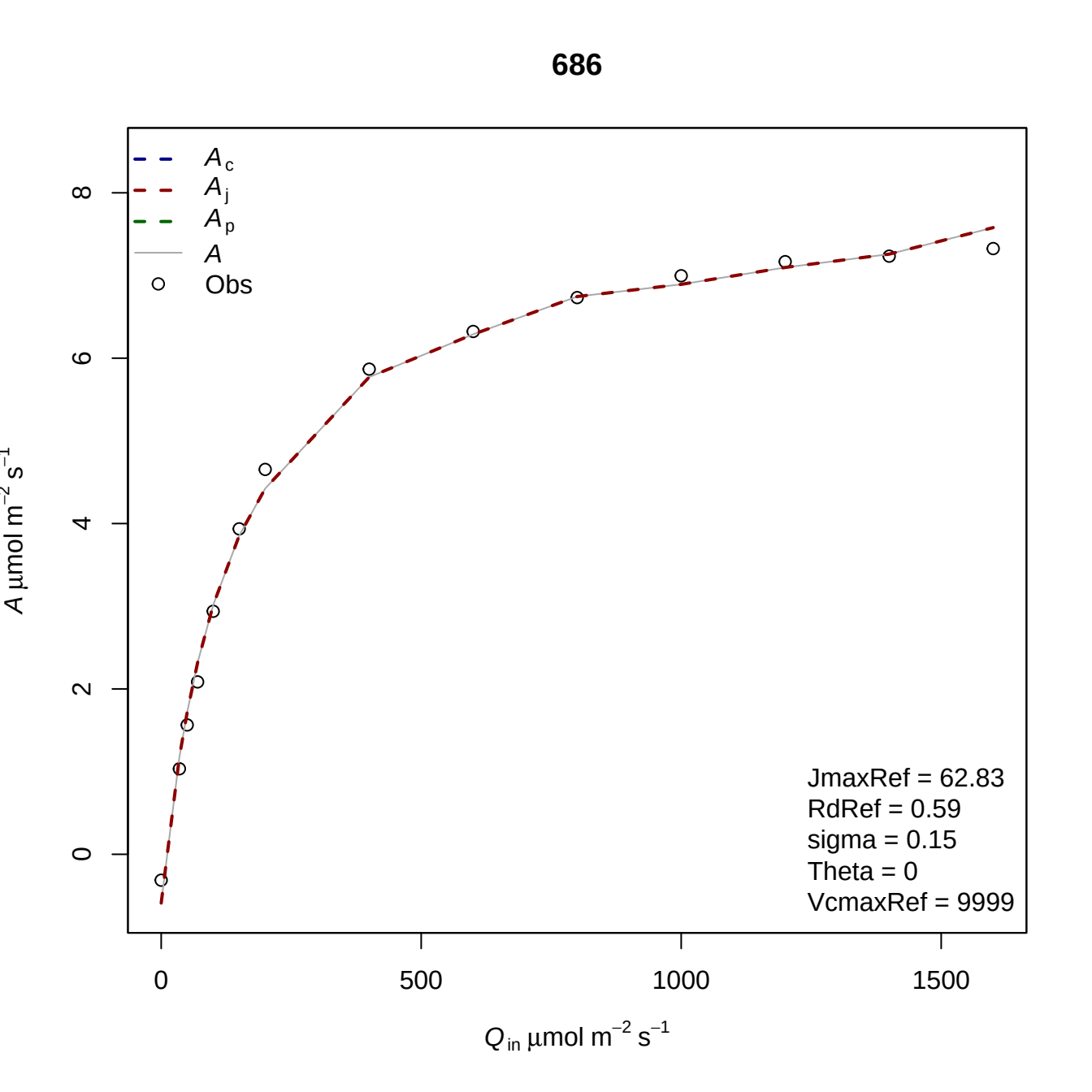

681
A_c
A_j
A_p
A
Obs
JmaxRef = 219.6
RdRef = 2.81
sigma = 0.22
Theta = 0
VcmaxRef = 136.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

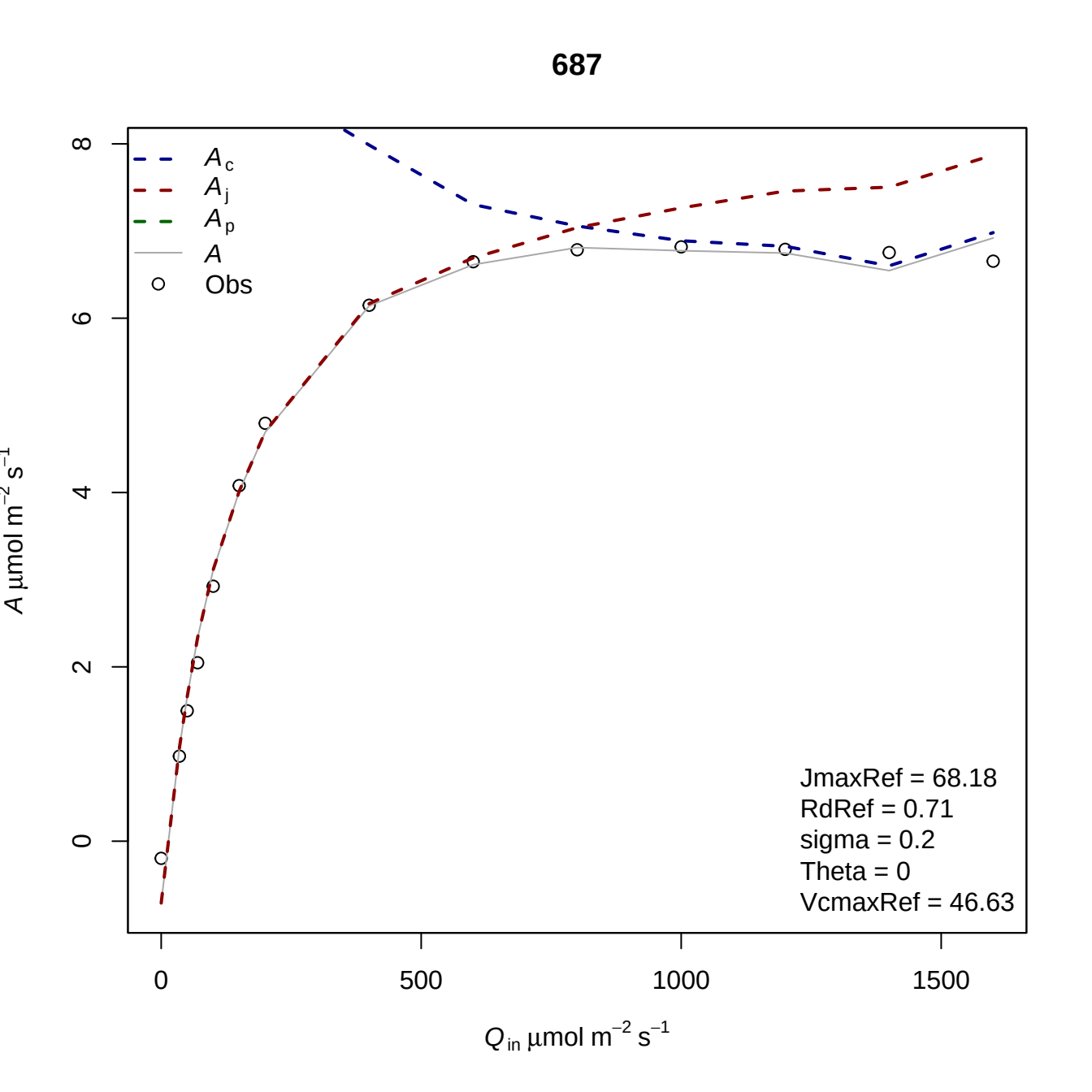

682
A_c
A_j
A_p
A
Obs
JmaxRef = 152.78
RdRef = 3.5
sigma = 0.42
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

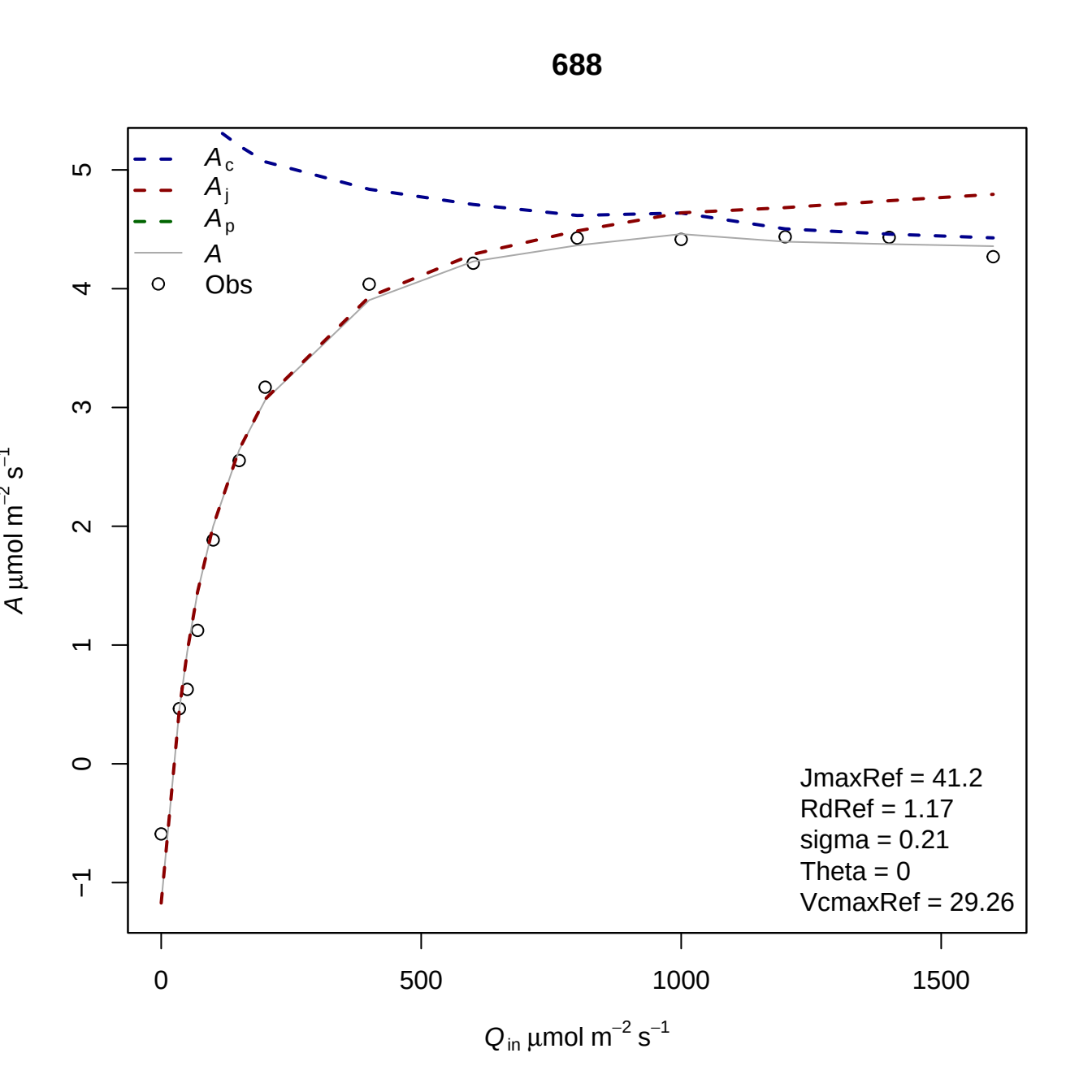

683
A_c
A_j
A_p
A
Obs
JmaxRef = 60.2
RdRef = 1.03
sigma = 0.19
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

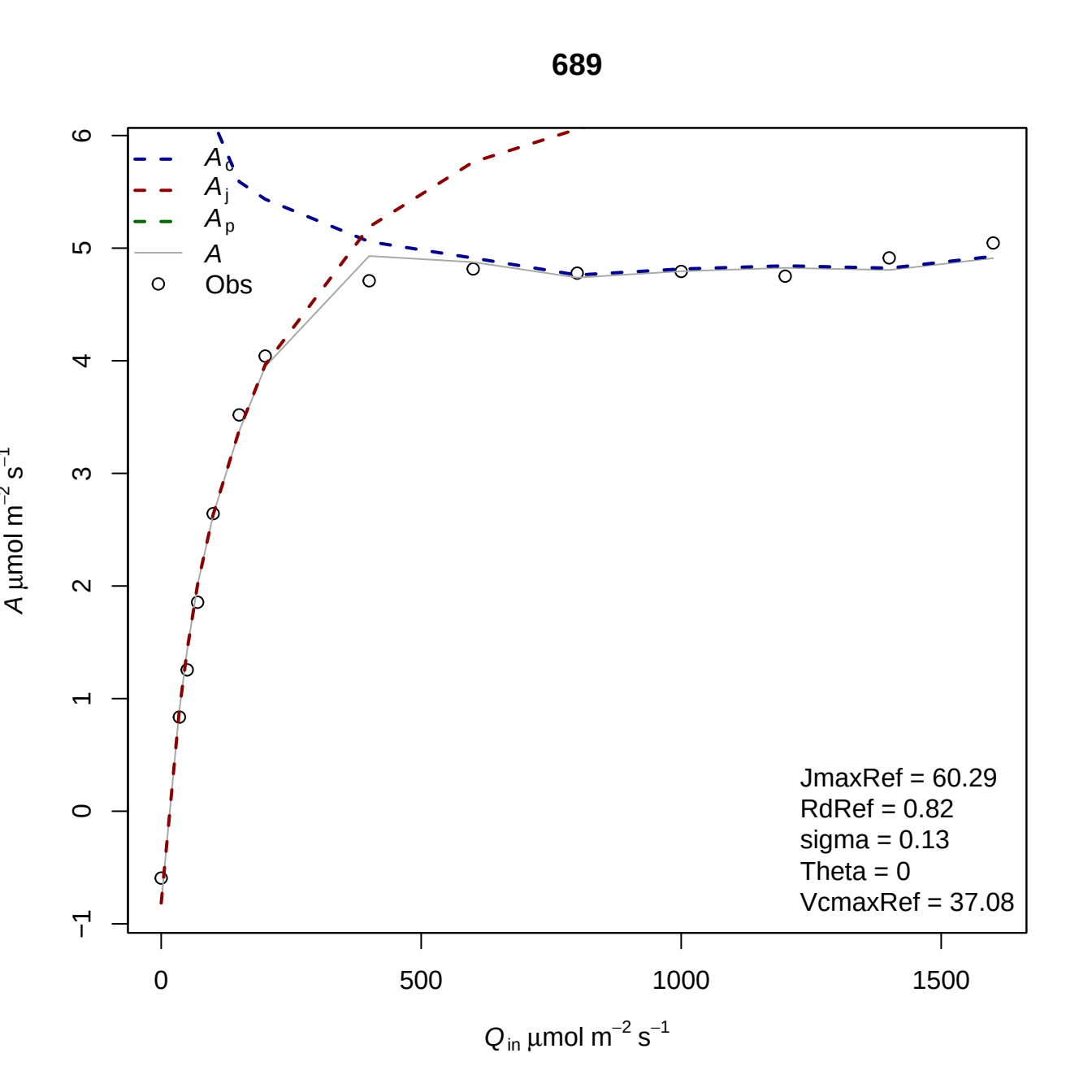

684
A_c
A_j
A_p
A
Obs
JmaxRef = 69.27
RdRef = 0.98
sigma = 0.14
Theta = 0
VcmaxRef = 41.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

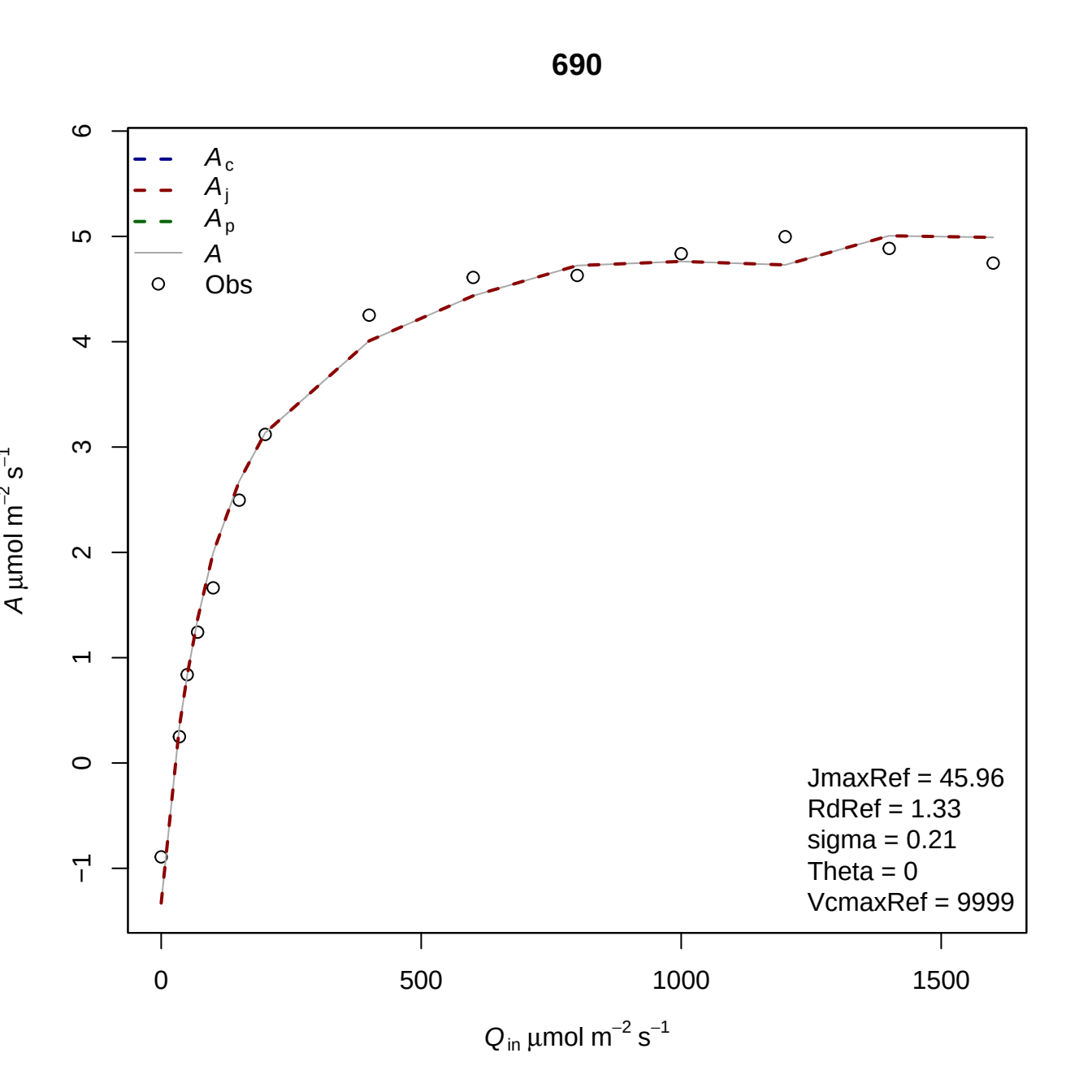

685
A_c
A_j
A_p
A
Obs
JmaxRef = 107.96
RdRef = 0.67
sigma = 0.21
Theta = 0
VcmaxRef = 75.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

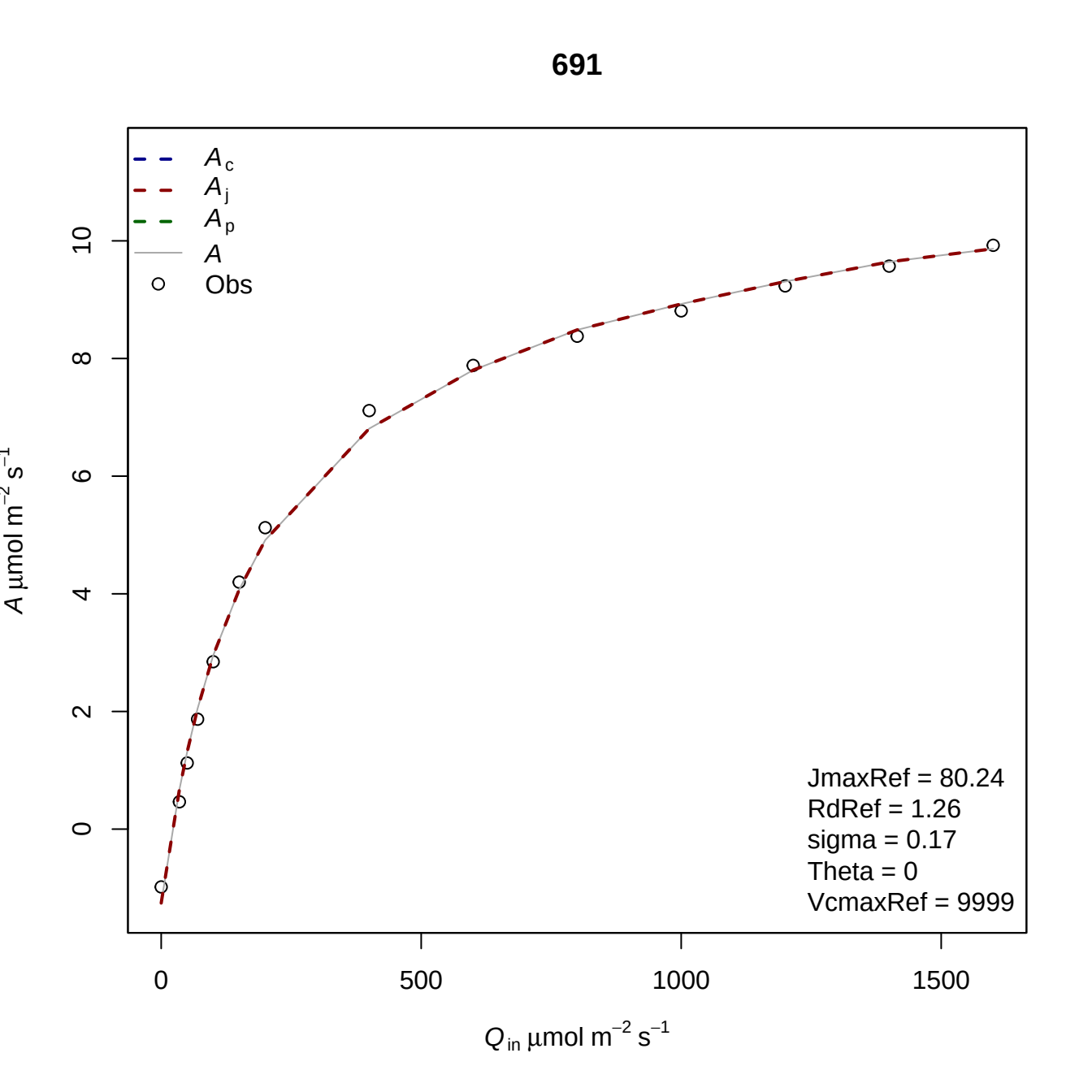

686
A_c
A_j
A_p
A
Obs
JmaxRef = 62.83
RdRef = 0.59
sigma = 0.15
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

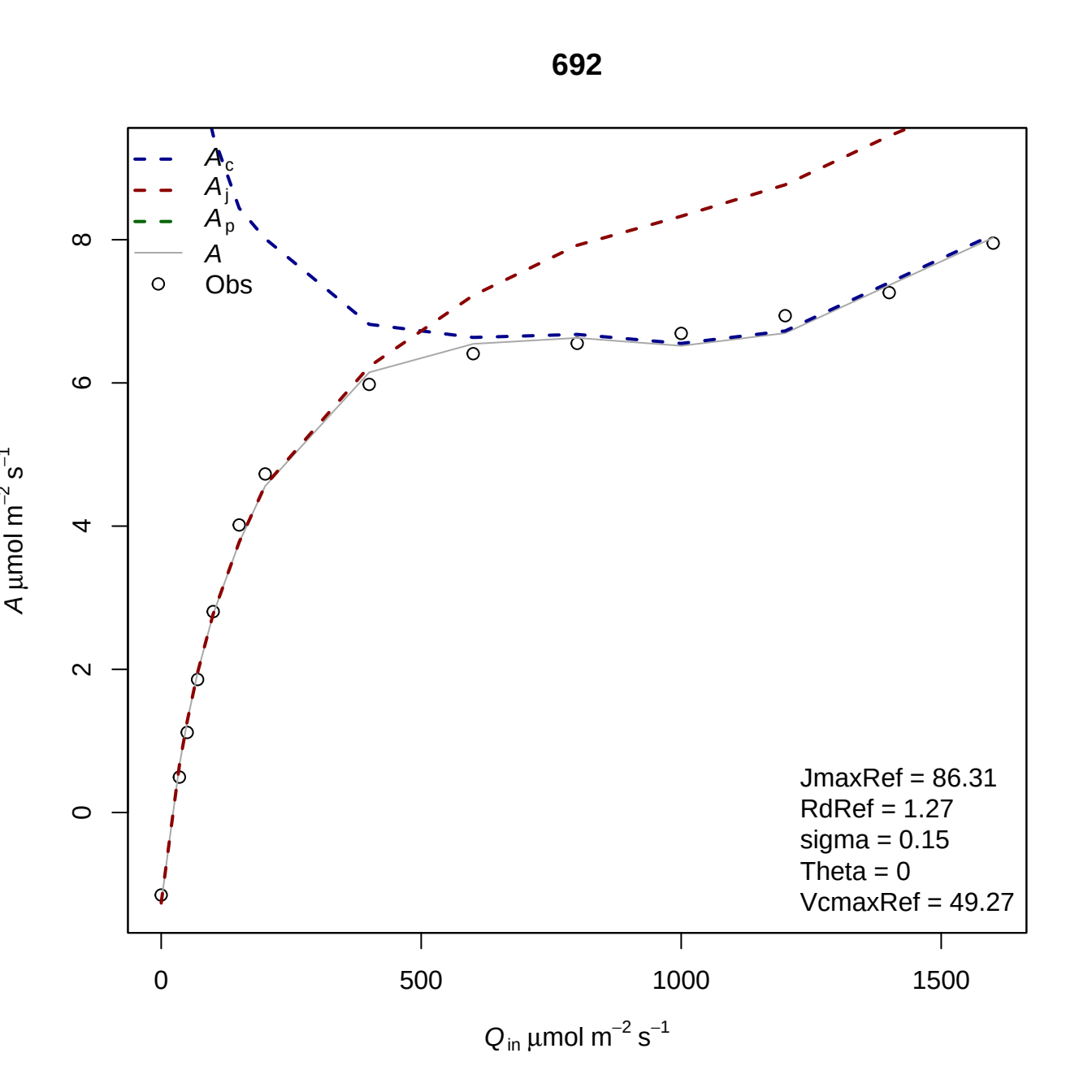

687
A_c
A_j
A_p
A
Obs
JmaxRef = 68.18
RdRef = 0.71
sigma = 0.2
Theta = 0
VcmaxRef = 46.63
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

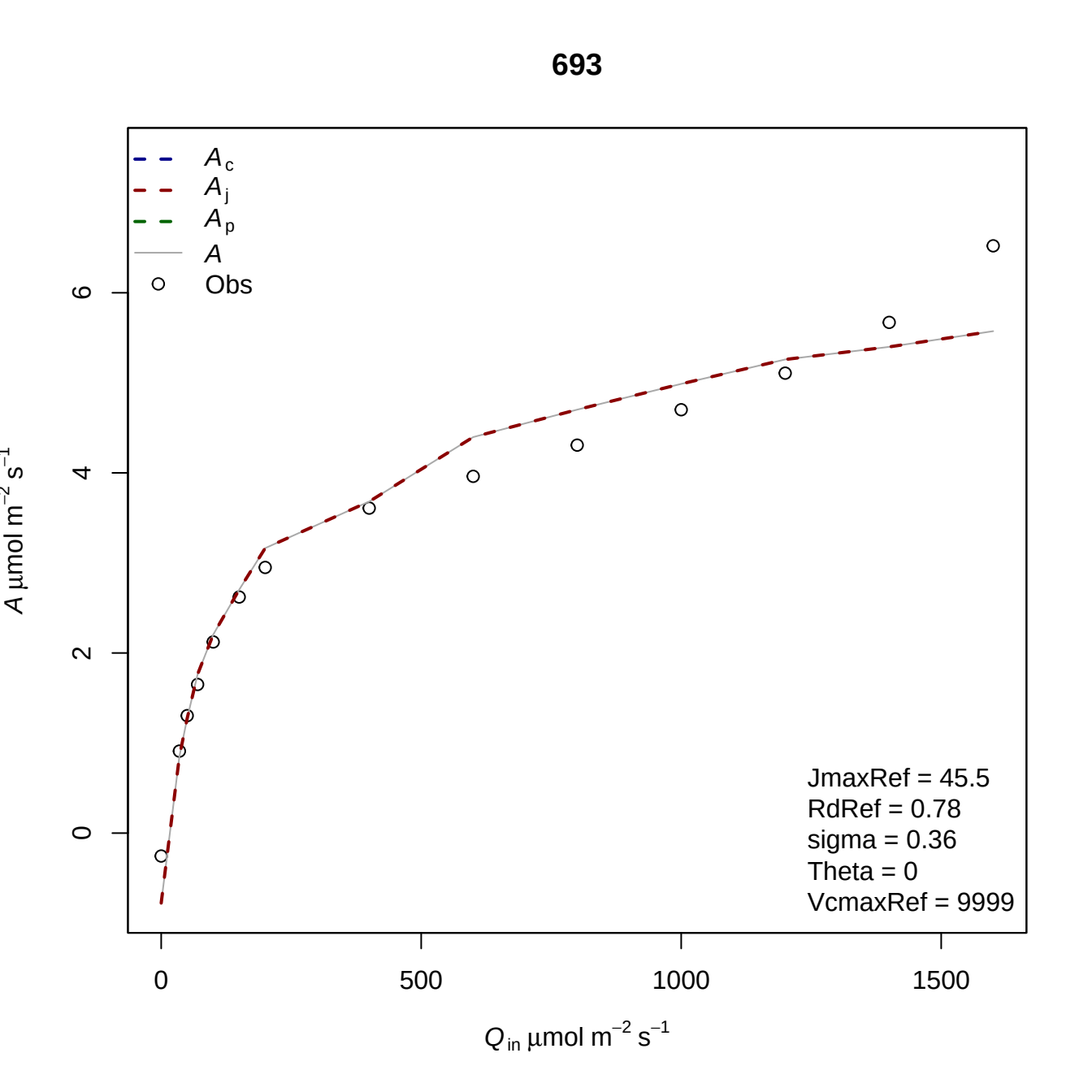

688
A_c
A_j
A_p
A
Obs
JmaxRef = 41.2
RdRef = 1.17
sigma = 0.21
Theta = 0
VcmaxRef = 29.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

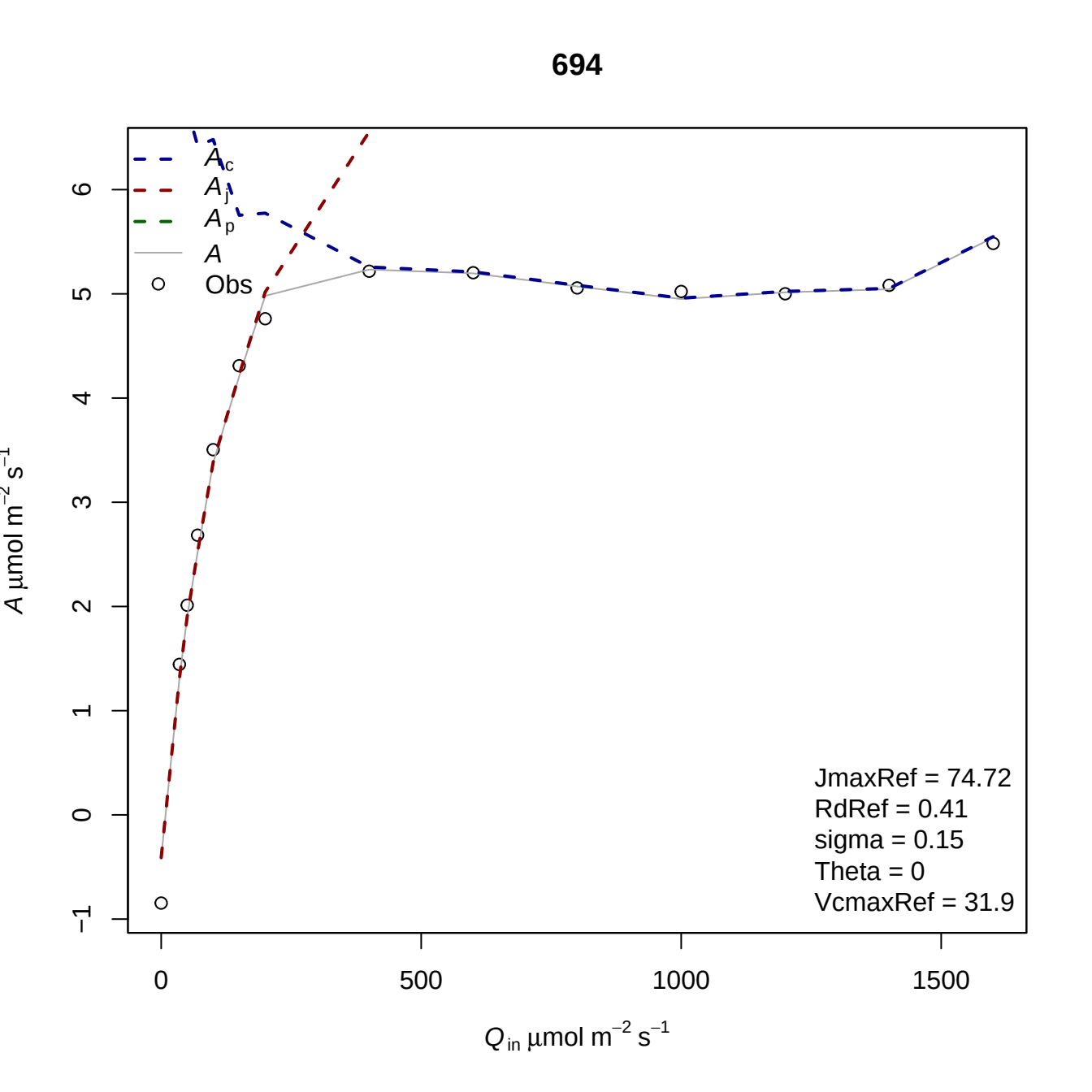

689
A_v
A_j
A_p
A
Obs
JmaxRef = 60.29
RdRef = 0.82
sigma = 0.13
Theta = 0
VcmaxRef = 37.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

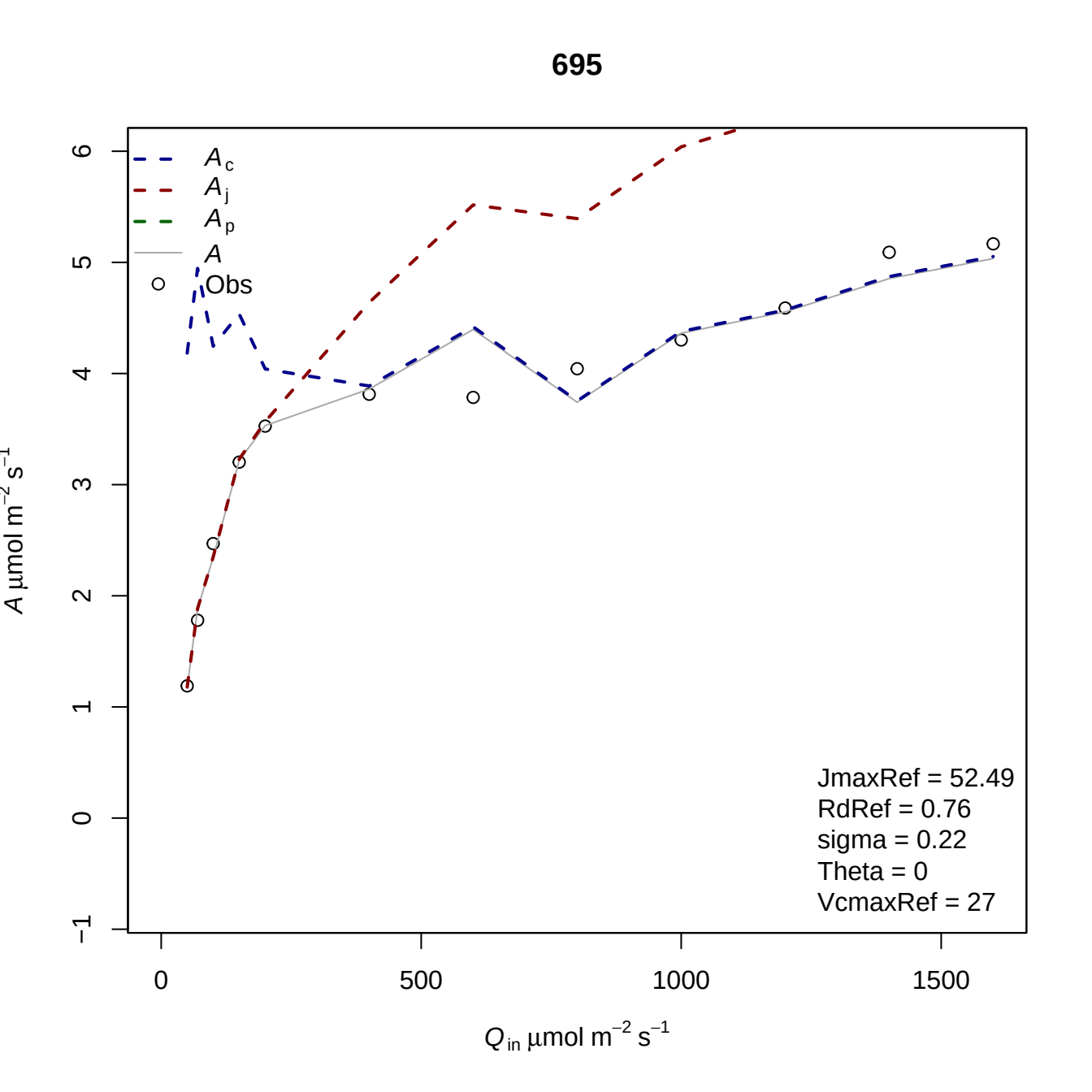

690
A_c
A_j
A_p
A
Obs
JmaxRef = 45.96
RdRef = 1.33
sigma = 0.21
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

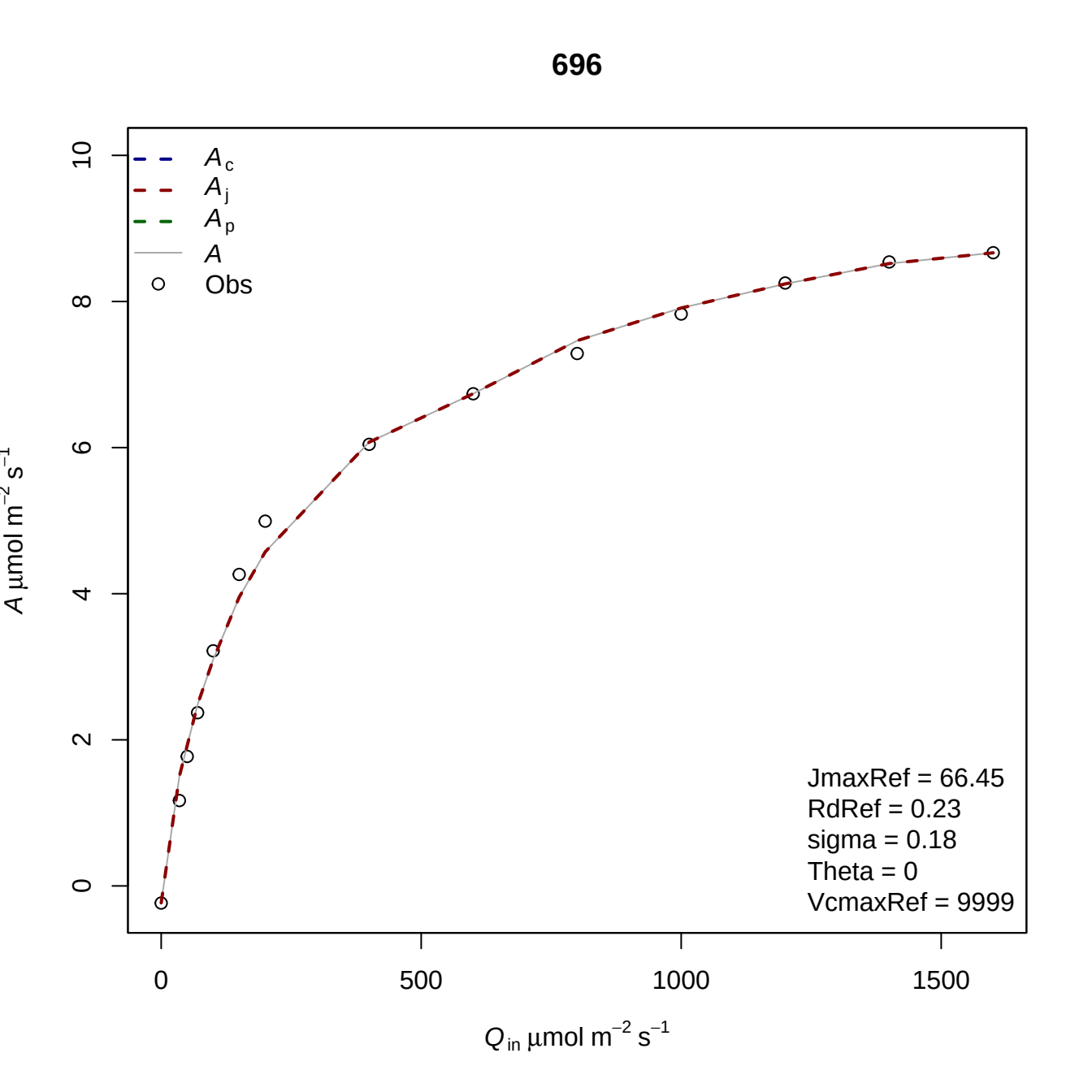

691
A_c
A_j
A_p
A
Obs
JmaxRef = 80.24
RdRef = 1.26
sigma = 0.17
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

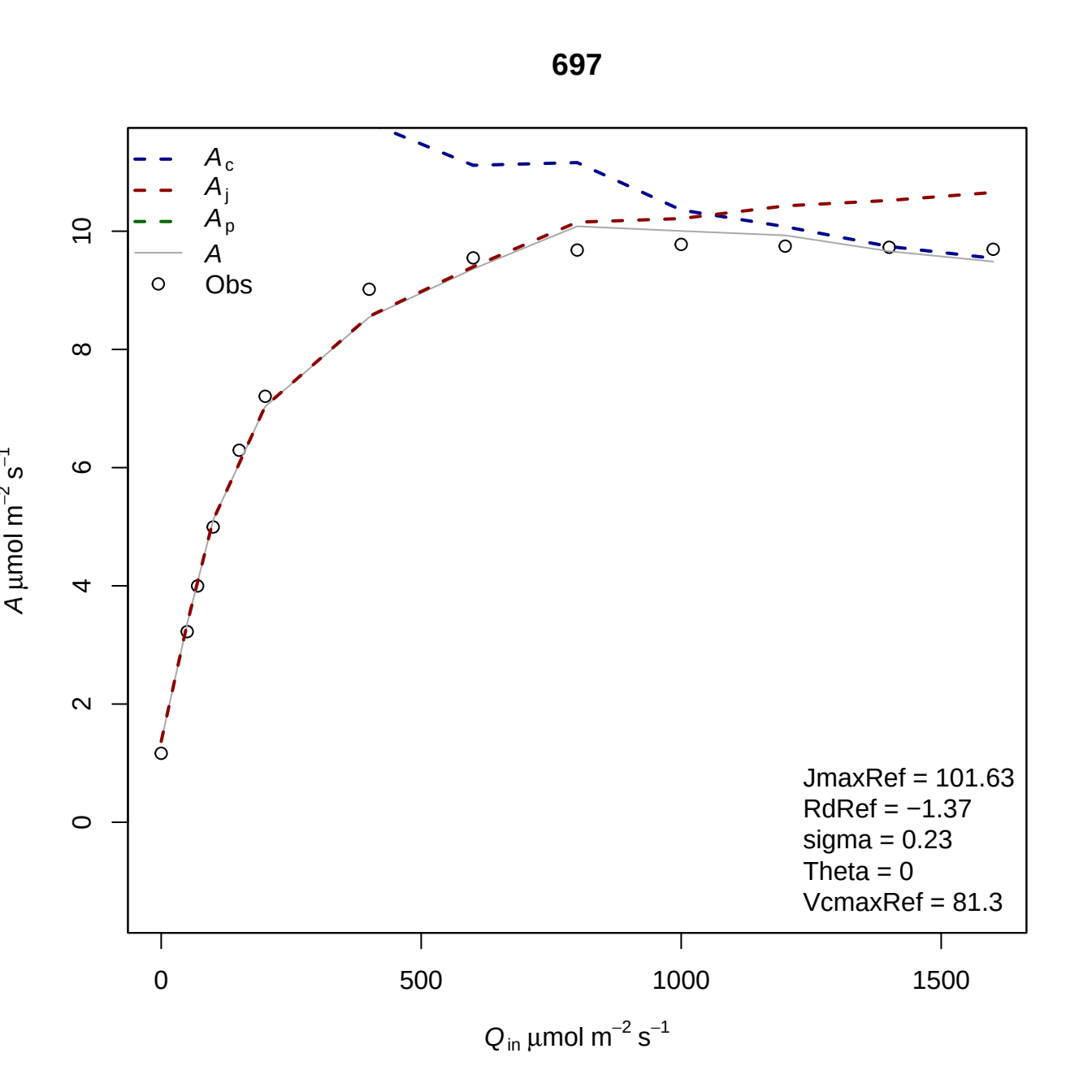

692
A_c
A_j
A_p
A
Obs
JmaxRef = 86.31
RdRef = 1.27
sigma = 0.15
Theta = 0
VcmaxRef = 49.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

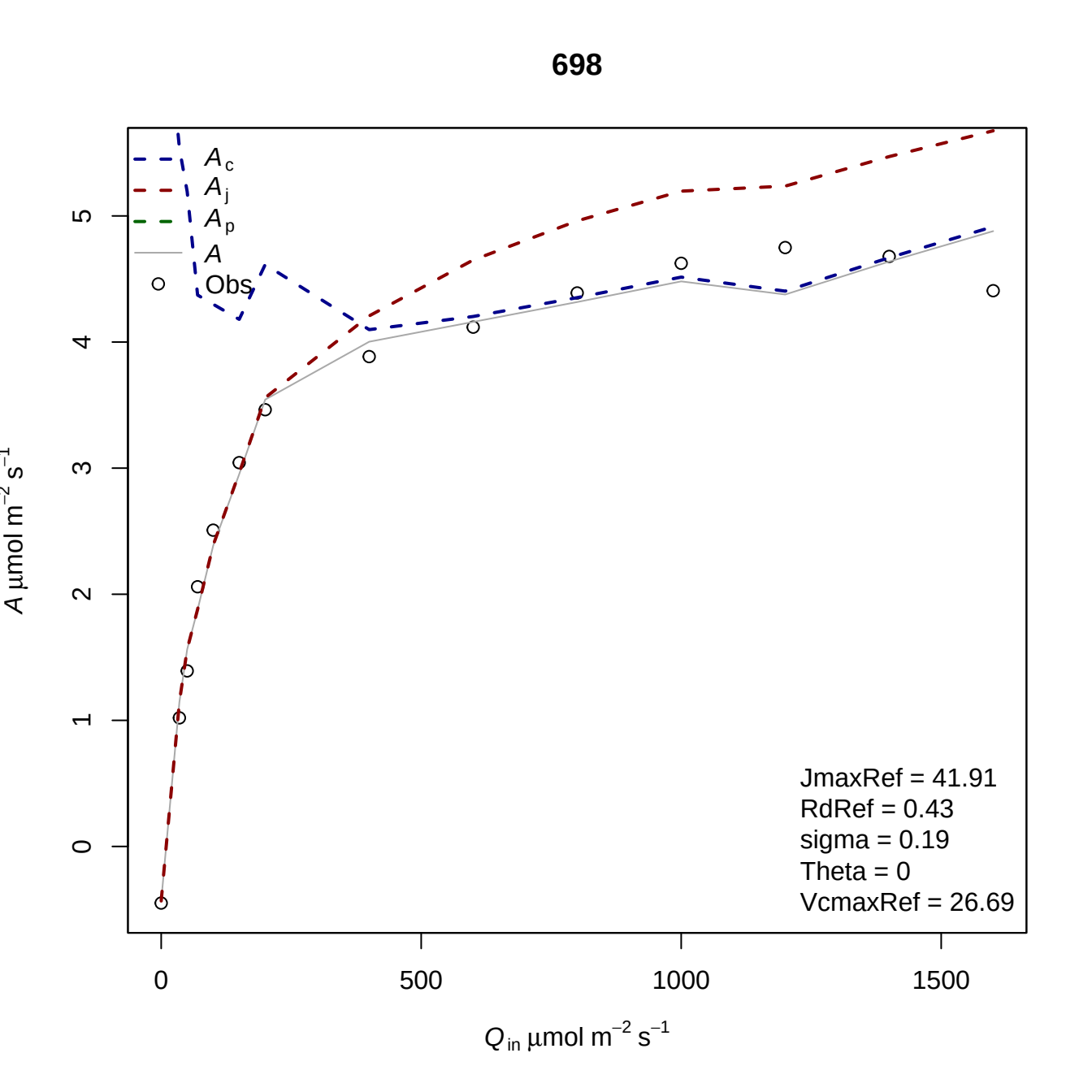

693
A_c
A_j
A_p
A
Obs
JmaxRef = 45.5
RdRef = 0.78
sigma = 0.36
Theta = 0
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

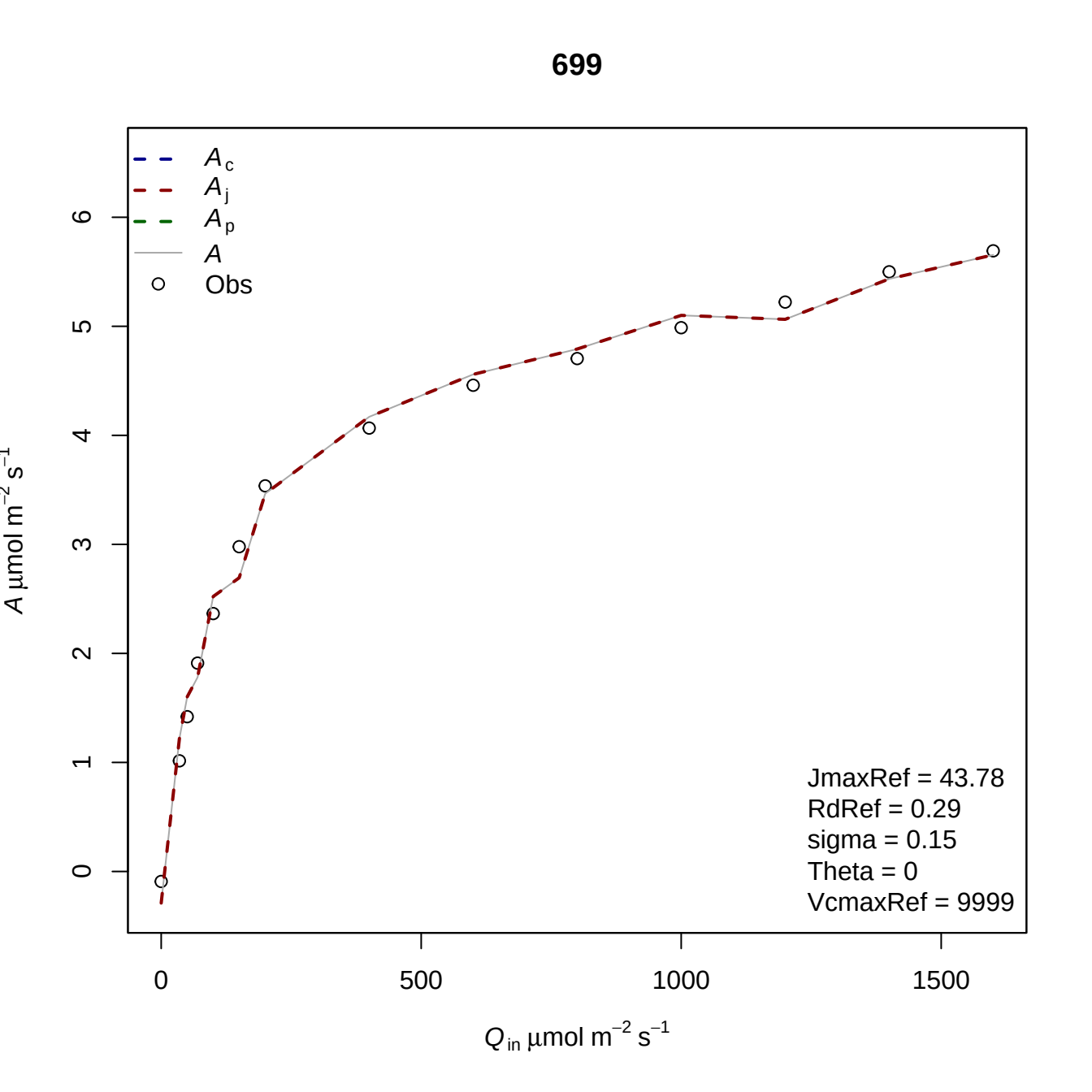

694
A_c
A_j
A_p
A
Obs
JmaxRef = 74.72
RdRef = 0.41
sigma = 0.15
Theta = 0
VcmaxRef = 31.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

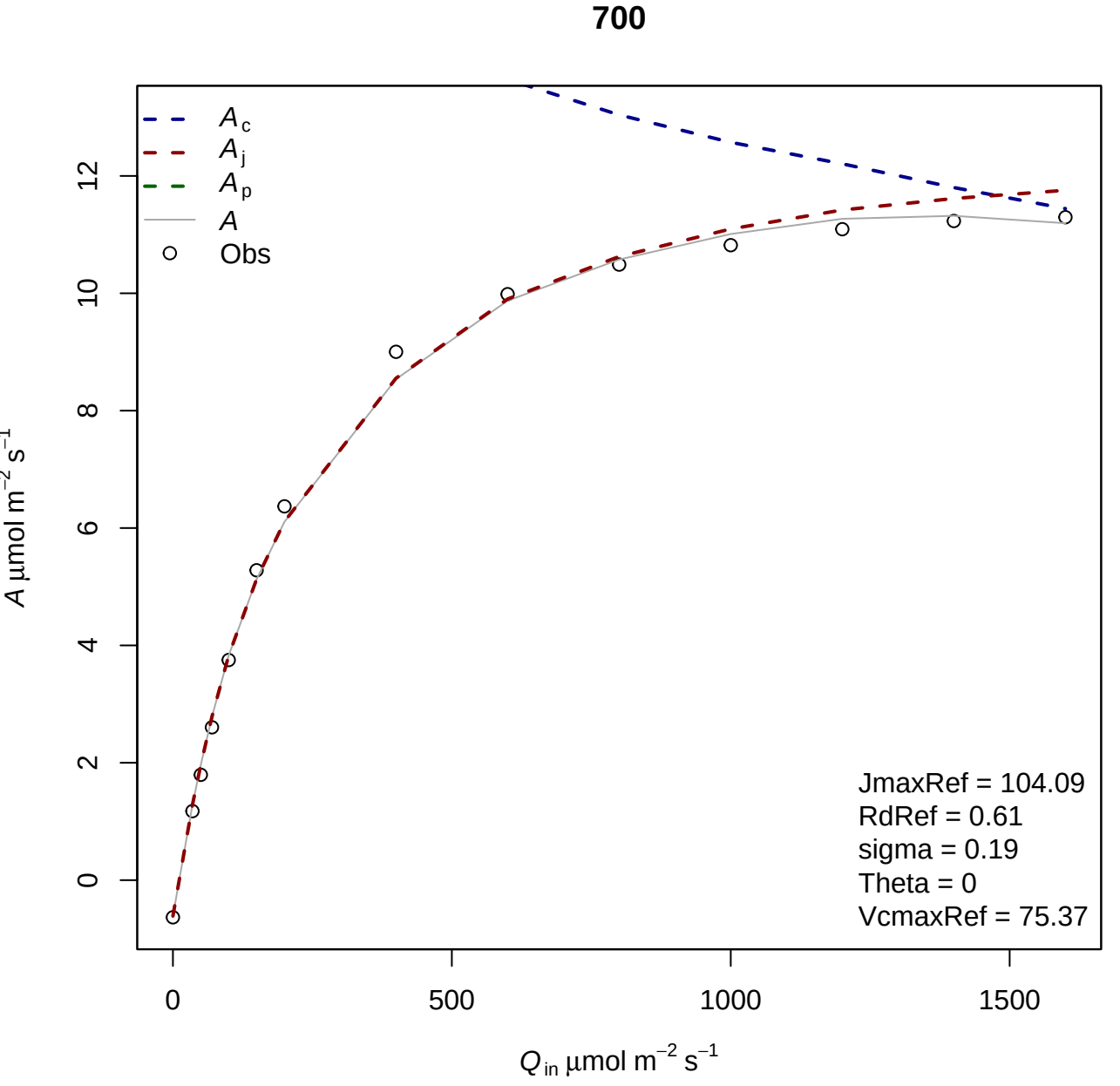

695
A_c
A_j
A_p
A
Obs
JmaxRef = 52.49
RdRef = 0.76
sigma = 0.22
Theta = 0
VcmaxRef = 27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

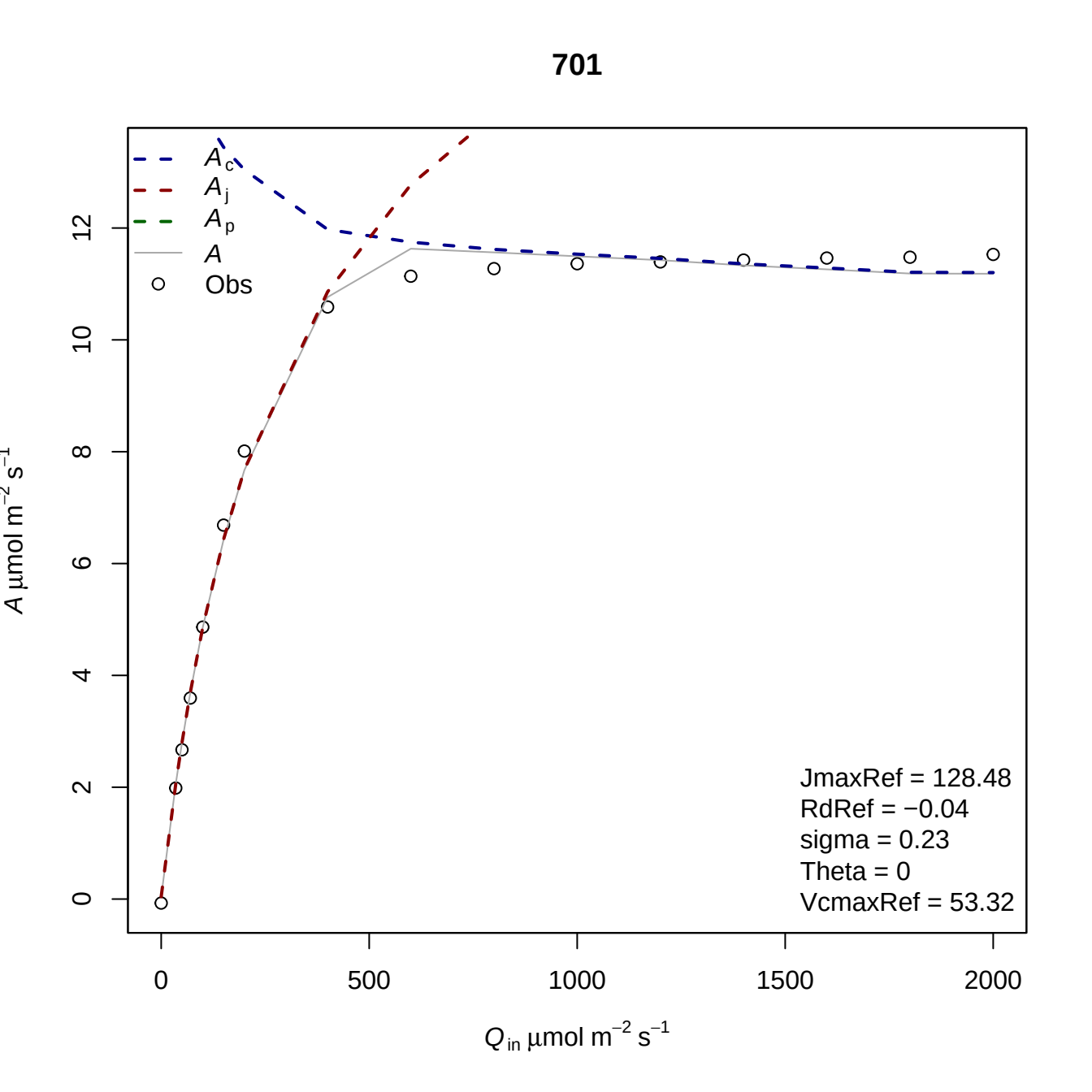

696
A_c
A_j
A_p
A
Obs
JmaxRef = 66.45
RdRef = 0.23
sigma = 0.18
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

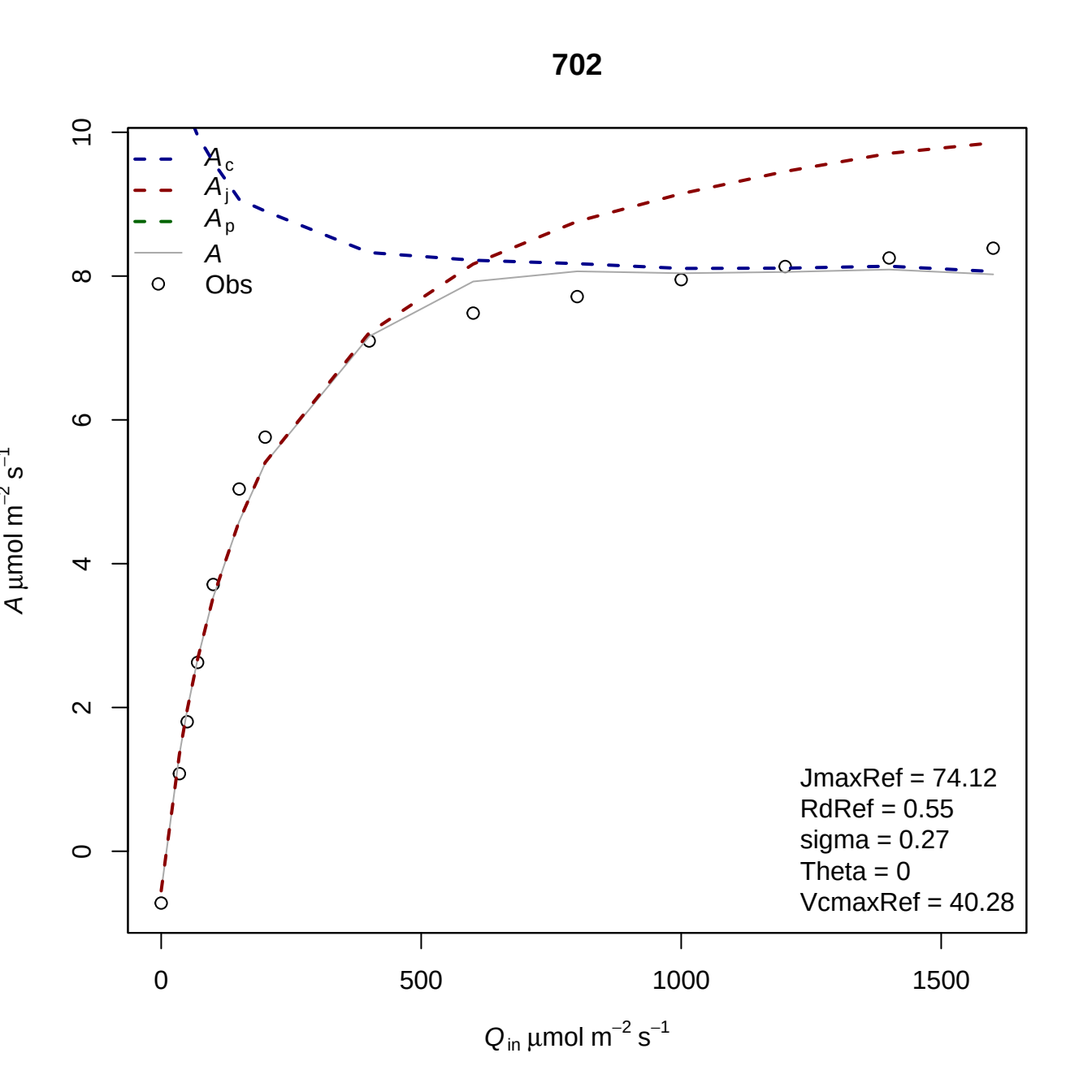

697
A_c
A_j
A_p
A
Obs
JmaxRef = 101.63
RdRef = −1.37
sigma = 0.23
Theta = 0
VcmaxRef = 81.3
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

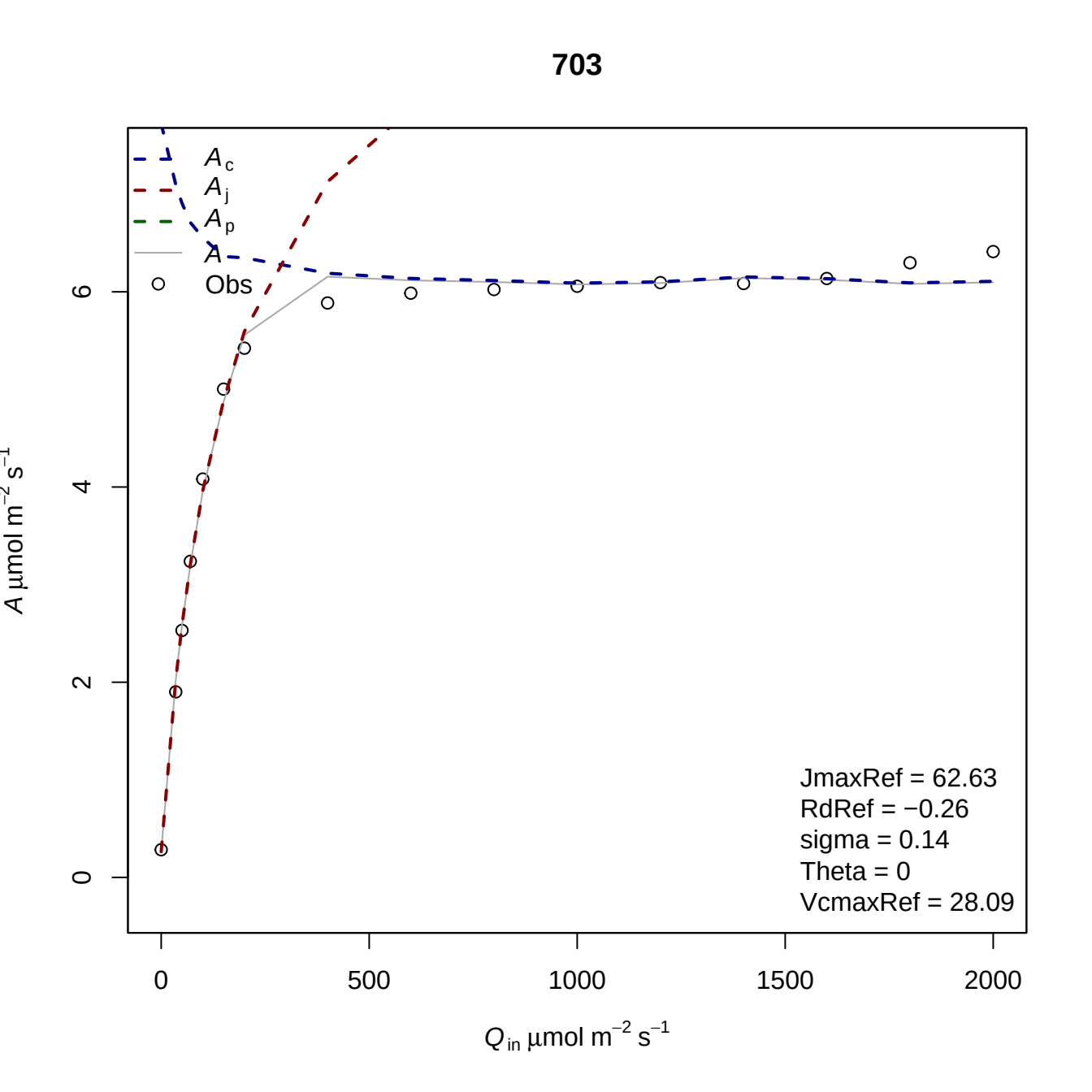

698
A_c
A_j
A_p
A
Obs
JmaxRef = 41.91
RdRef = 0.43
sigma = 0.19
Theta = 0
VcmaxRef = 26.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

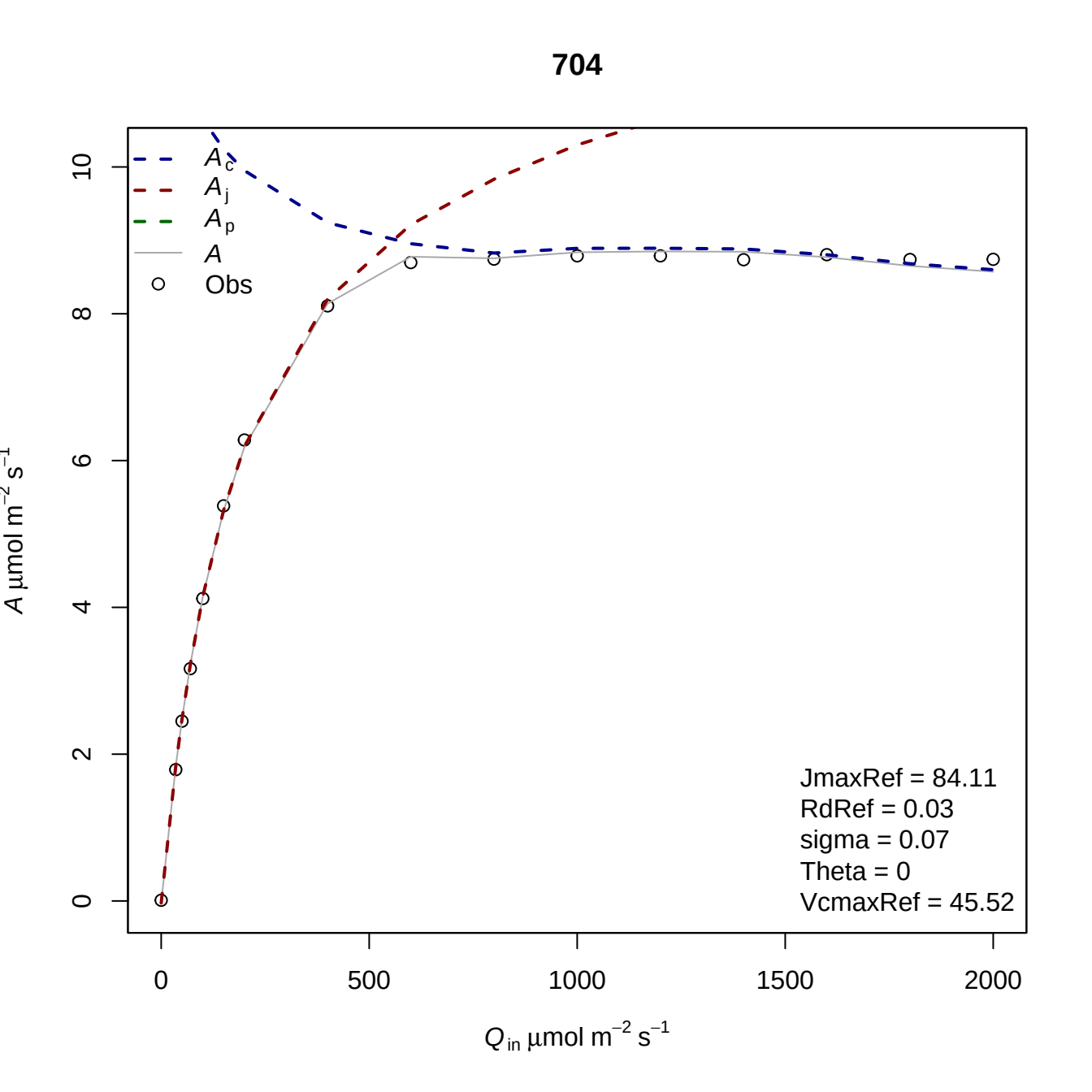

699
A_c
A_j
A_p
A
Obs
JmaxRef = 43.78
RdRef = 0.29
sigma = 0.15
Theta = 0
VcmaxRef = 9999
0
1
2
3
4
5
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

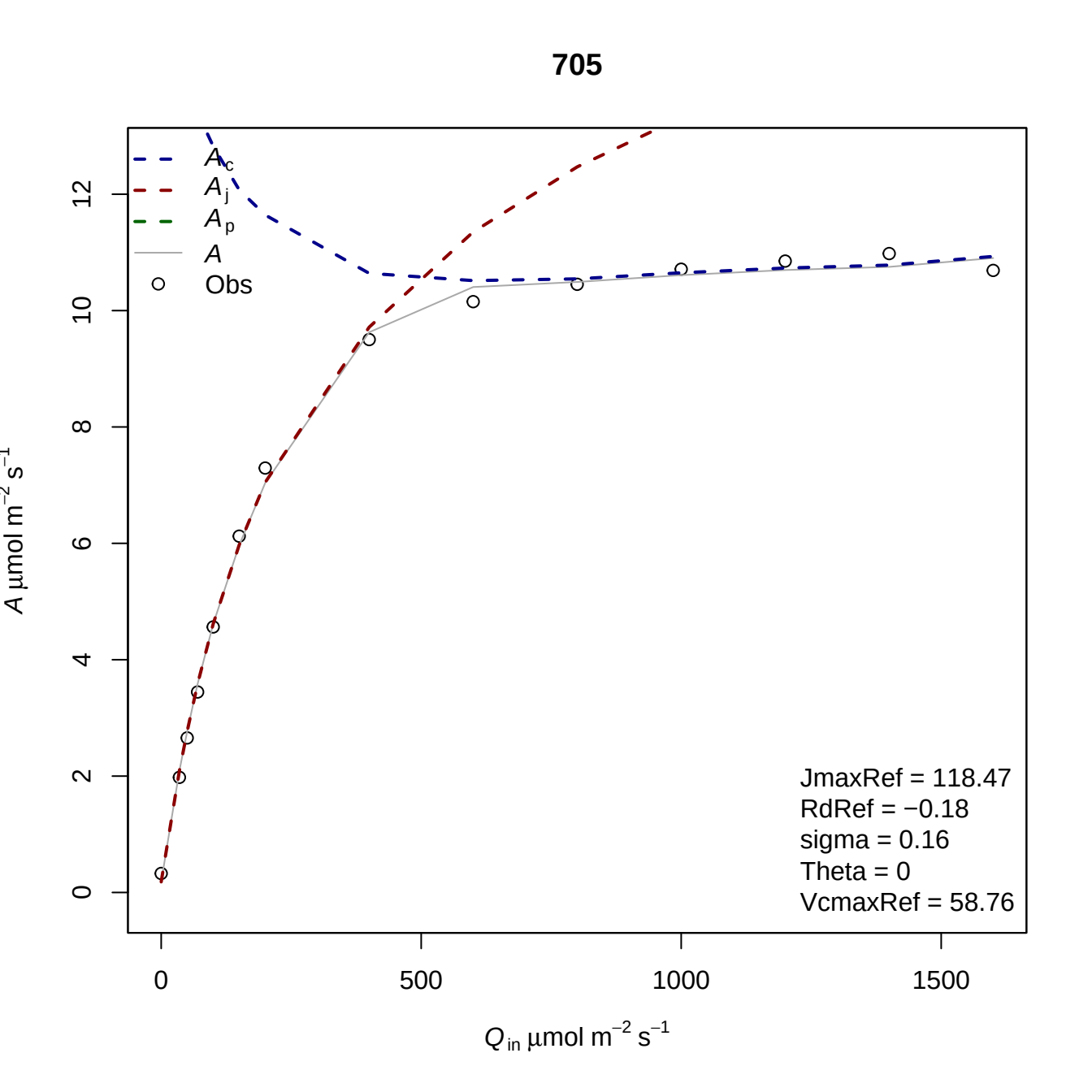

700
A_c
A_j
A_p
A
Obs
JmaxRef = 104.09
RdRef = 0.61
sigma = 0.19
Theta = 0
VcmaxRef = 75.37
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

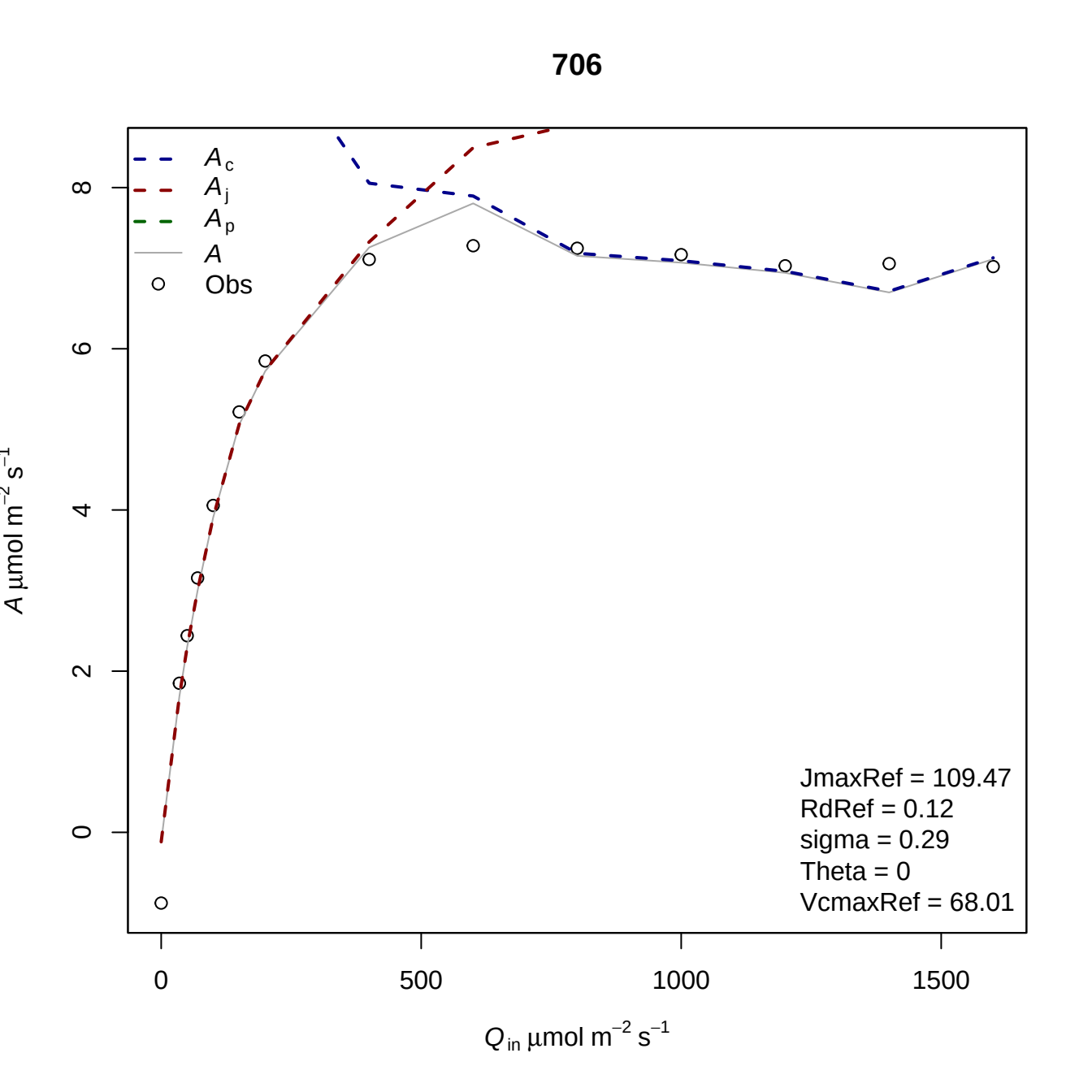

701
A_c
A_j
A_p
A
Obs
JmaxRef = 128.48
RdRef = −0.04
sigma = 0.23
Theta = 0
VcmaxRef = 53.32
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

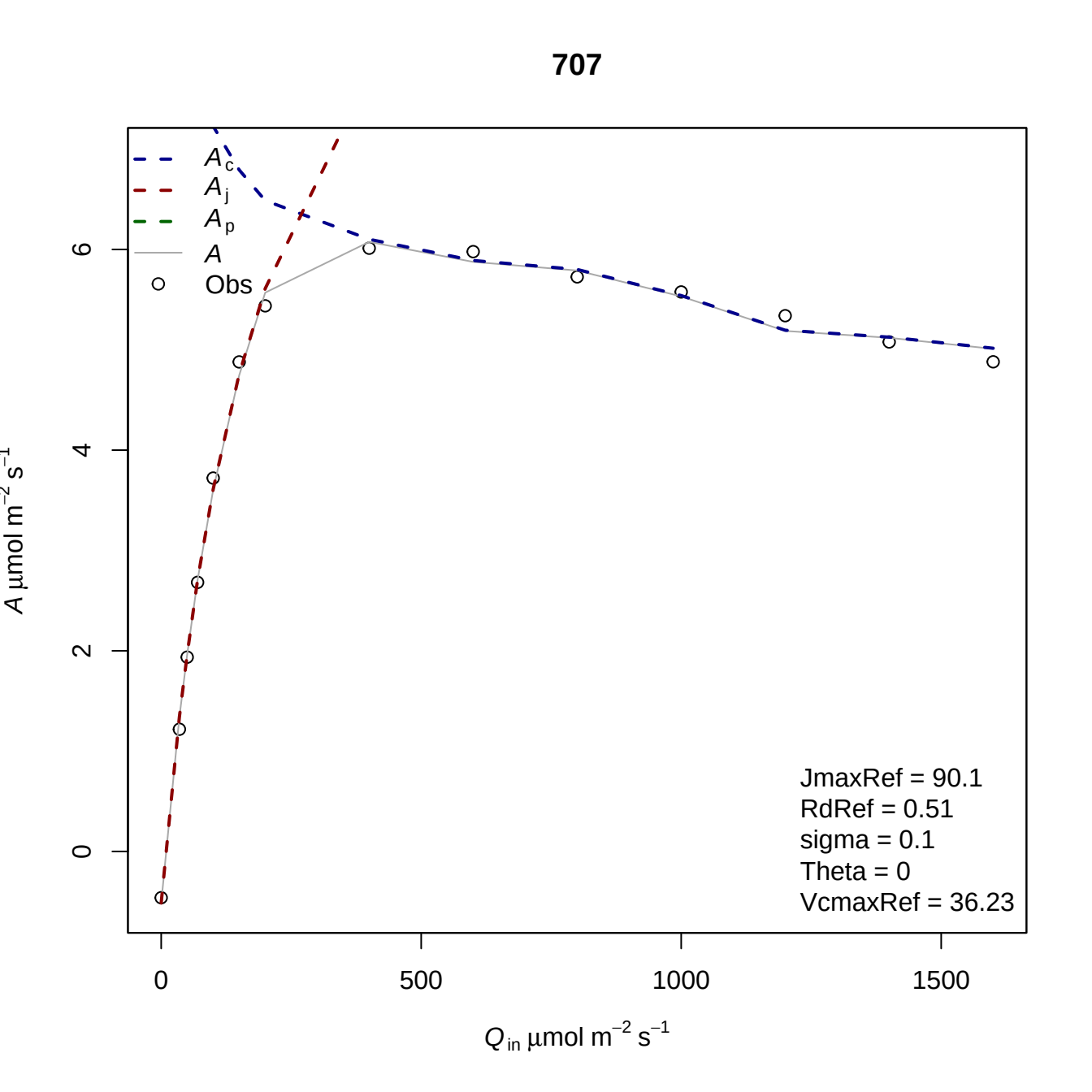

702
A_c
A_j
A_p
A
Obs
JmaxRef = 74.12
RdRef = 0.55
sigma = 0.27
Theta = 0
VcmaxRef = 40.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

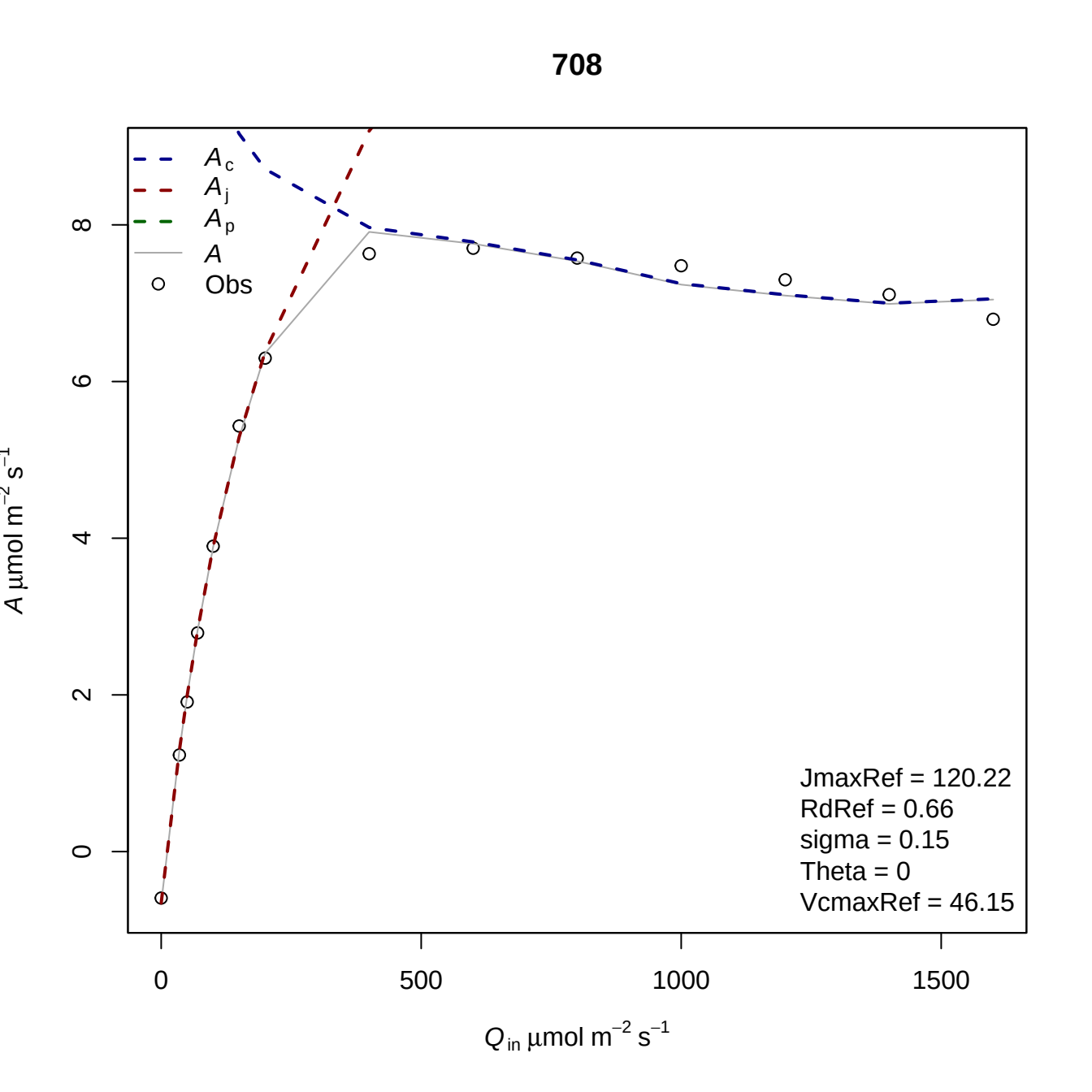

703
A_c
A_j
A_p
A
Obs
JmaxRef = 62.63
RdRef = −0.26
sigma = 0.14
Theta = 0
VcmaxRef = 28.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

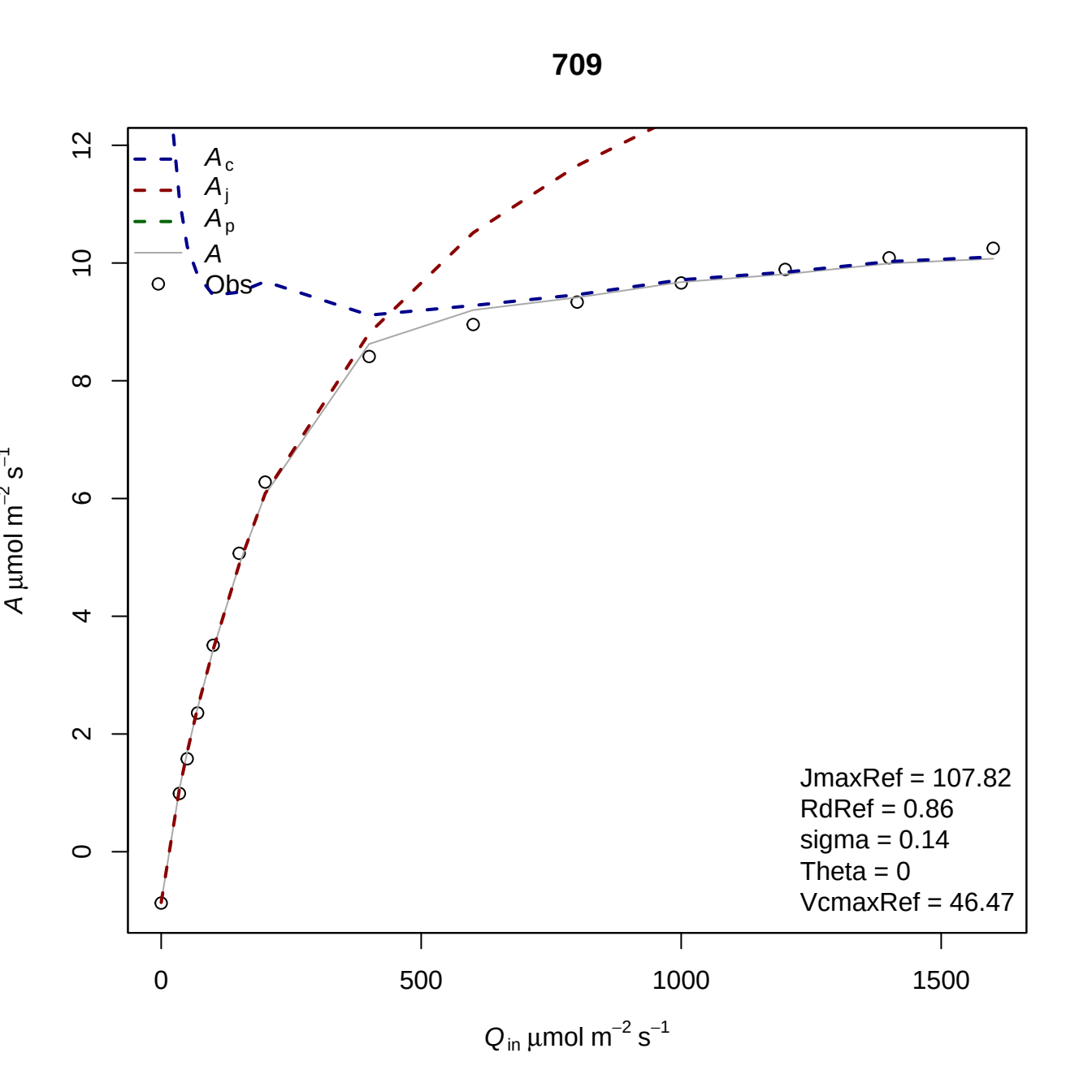

704
A_c
A_j
A_p
A
Obs
JmaxRef = 84.11
RdRef = 0.03
sigma = 0.07
Theta = 0
VcmaxRef = 45.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

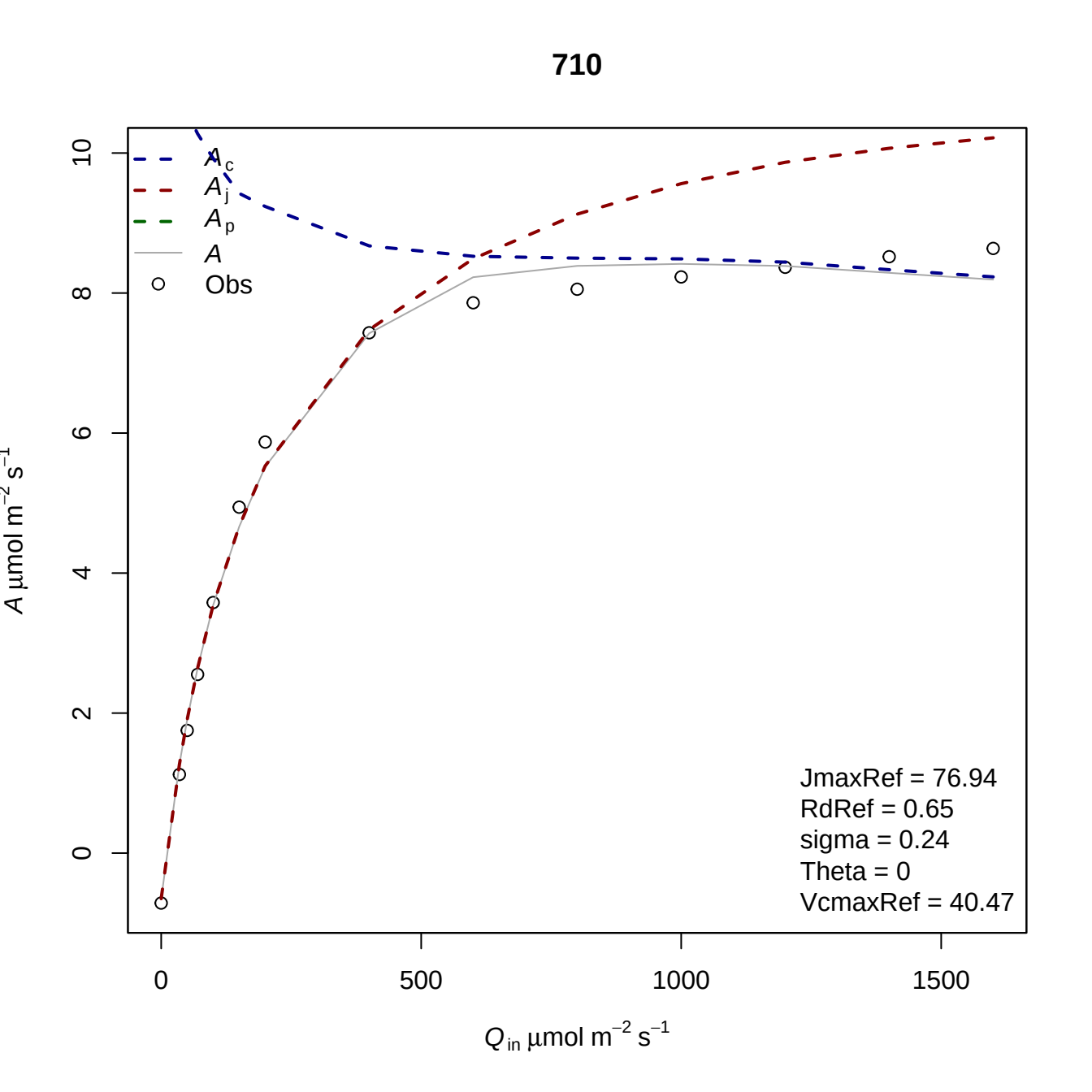

705
A_c
A_j
A_p
A
Obs
JmaxRef = 118.47
RdRef = −0.18
sigma = 0.16
Theta = 0
VcmaxRef = 58.76
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

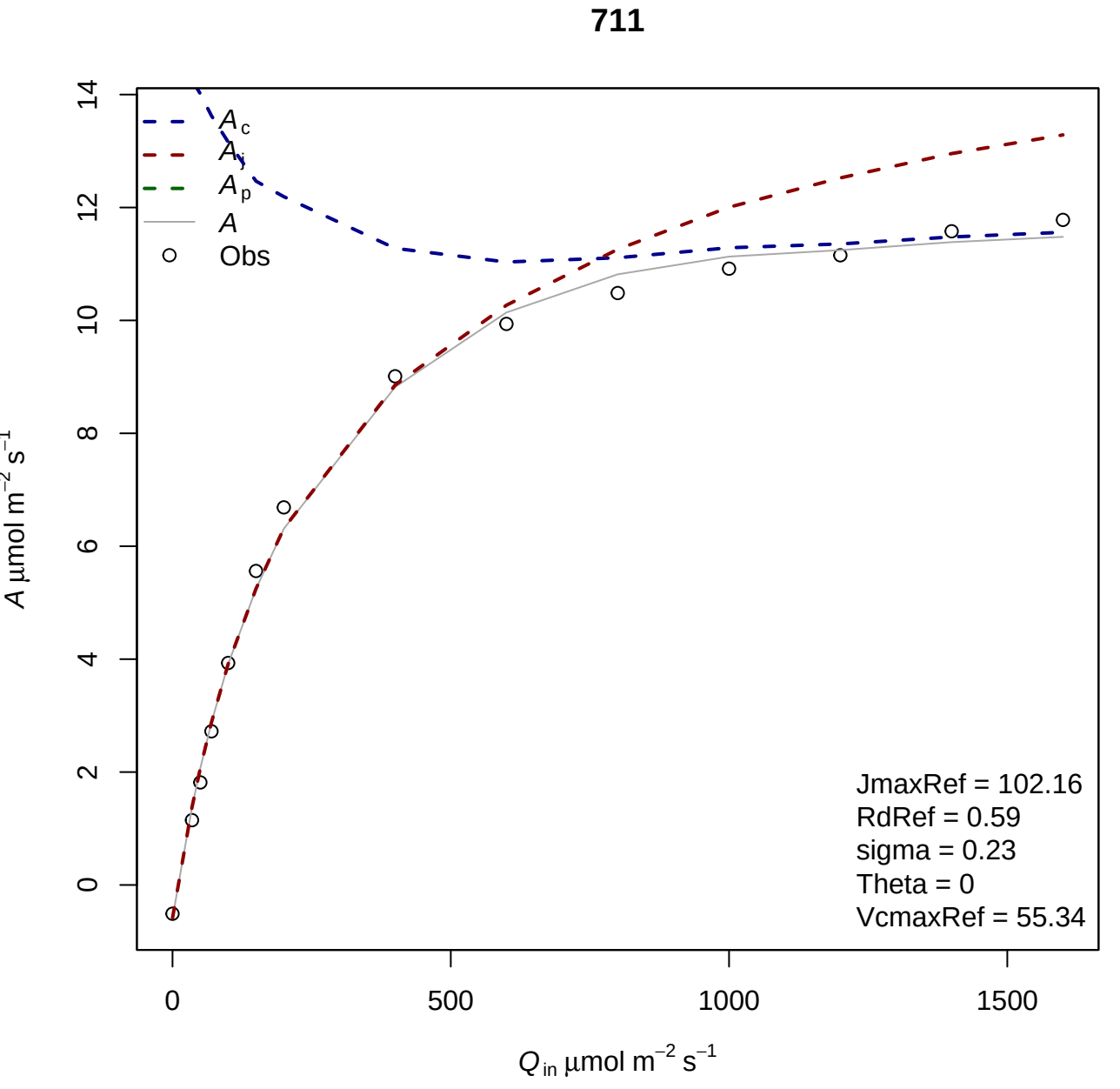

706
A_c
A_j
A_p
A
Obs
JmaxRef = 109.47
RdRef = 0.12
sigma = 0.29
Theta = 0
VcmaxRef = 68.01
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

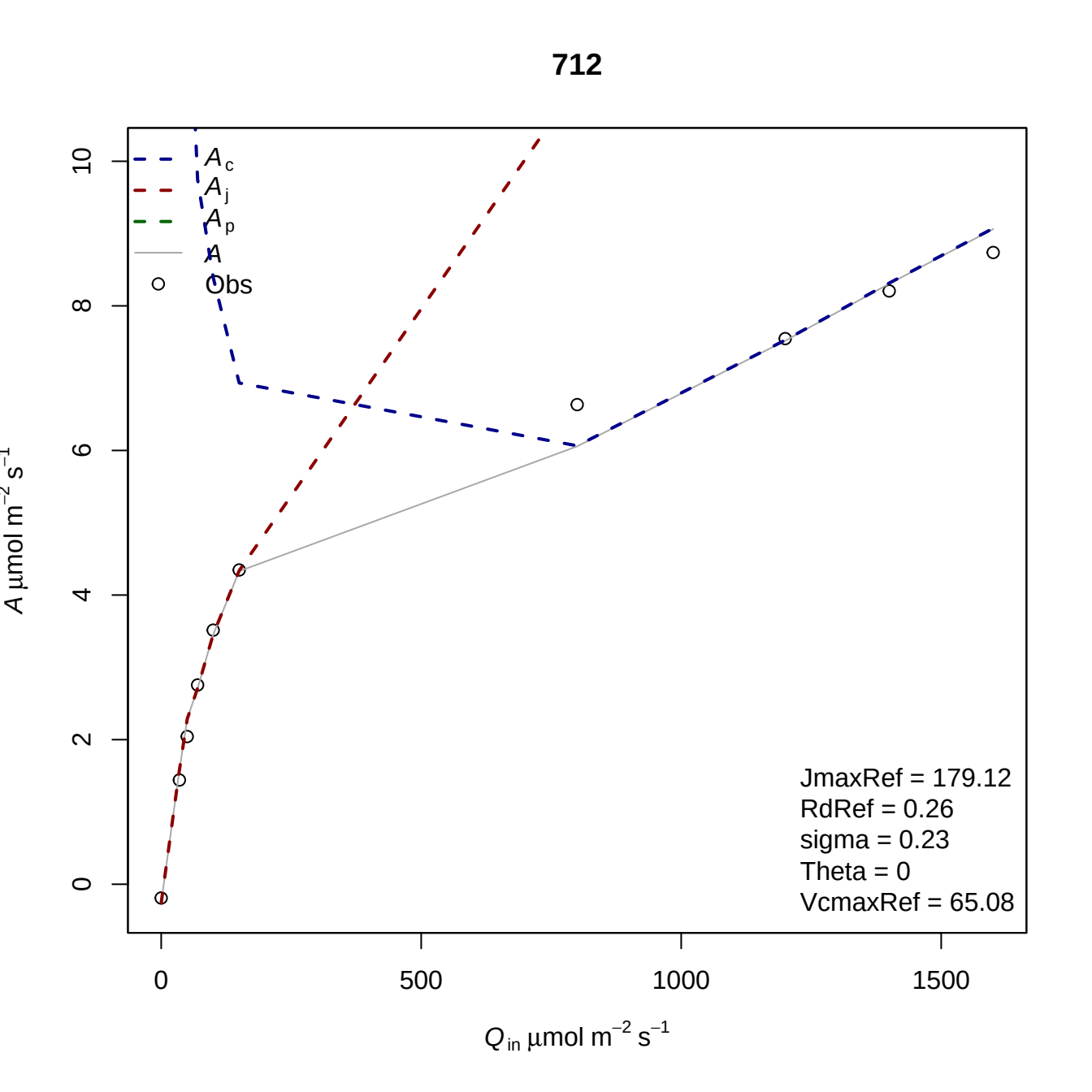

707
A_c
A_j
A_p
A
Obs
JmaxRef = 90.1
RdRef = 0.51
sigma = 0.1
Theta = 0
VcmaxRef = 36.23
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

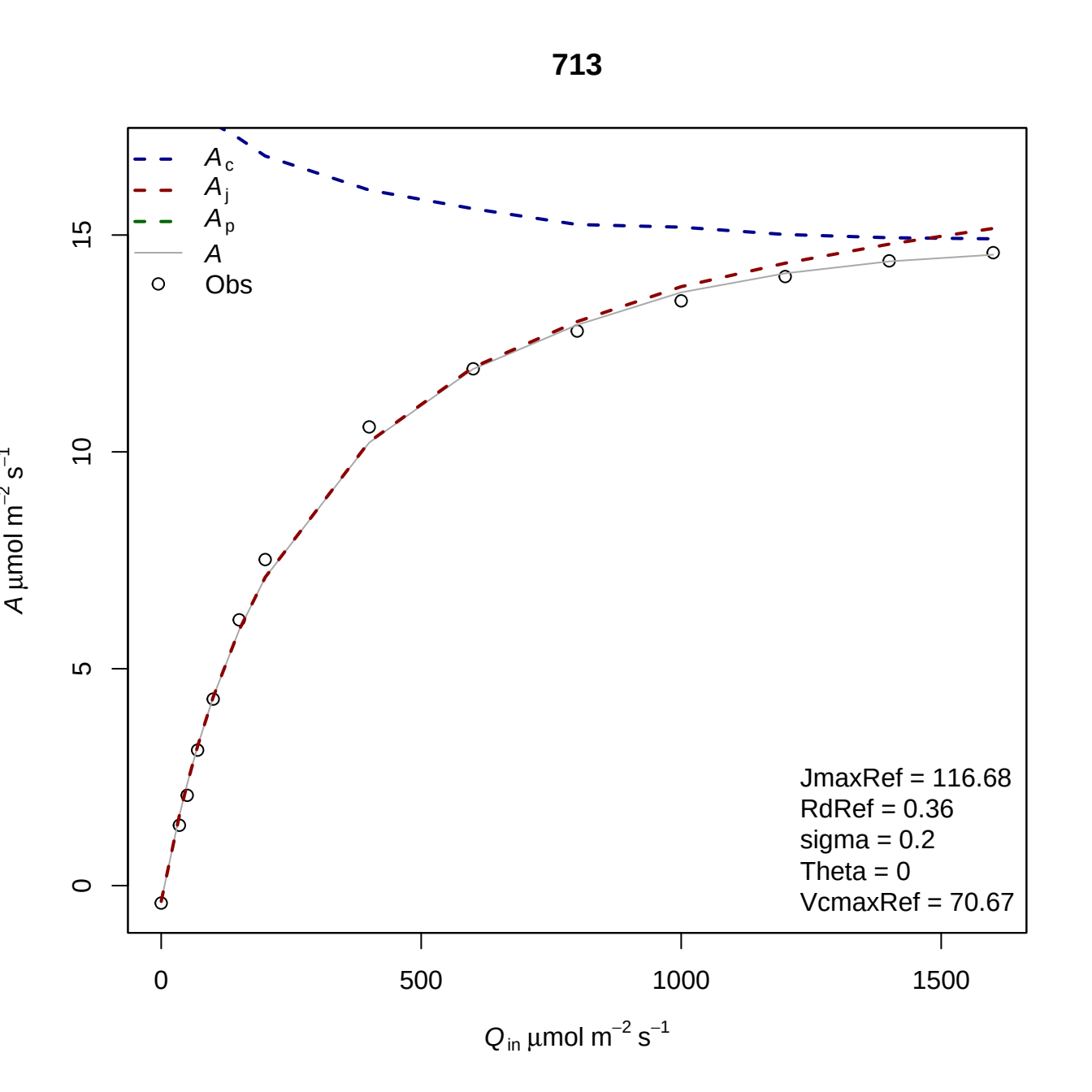

708
A_c
A_j
A_p
A
Obs
JmaxRef = 120.22
RdRef = 0.66
sigma = 0.15
Theta = 0
VcmaxRef = 46.15
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

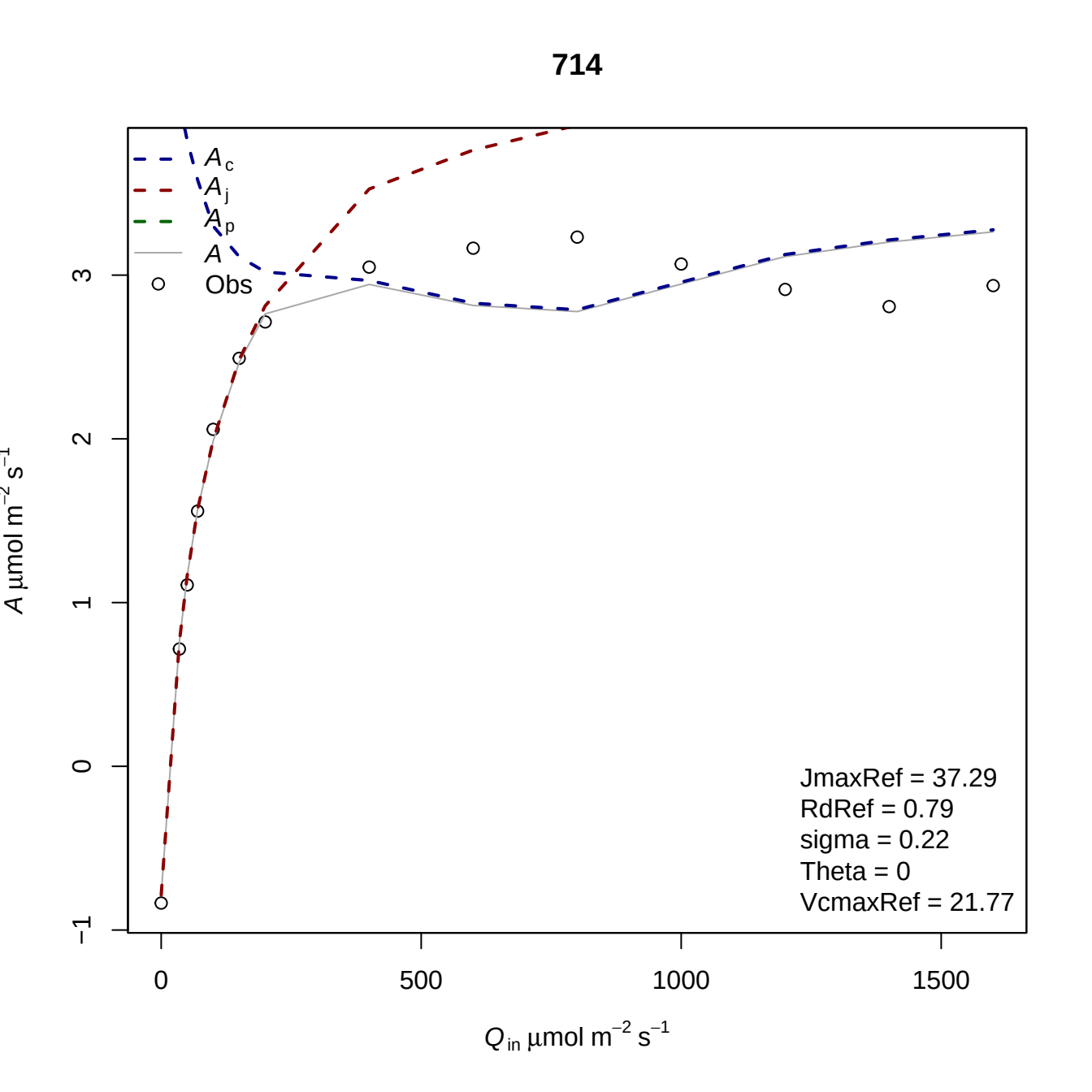

709
A_c
A_j
A_p
A
Obs
JmaxRef = 107.82
RdRef = 0.86
sigma = 0.14
Theta = 0
VcmaxRef = 46.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

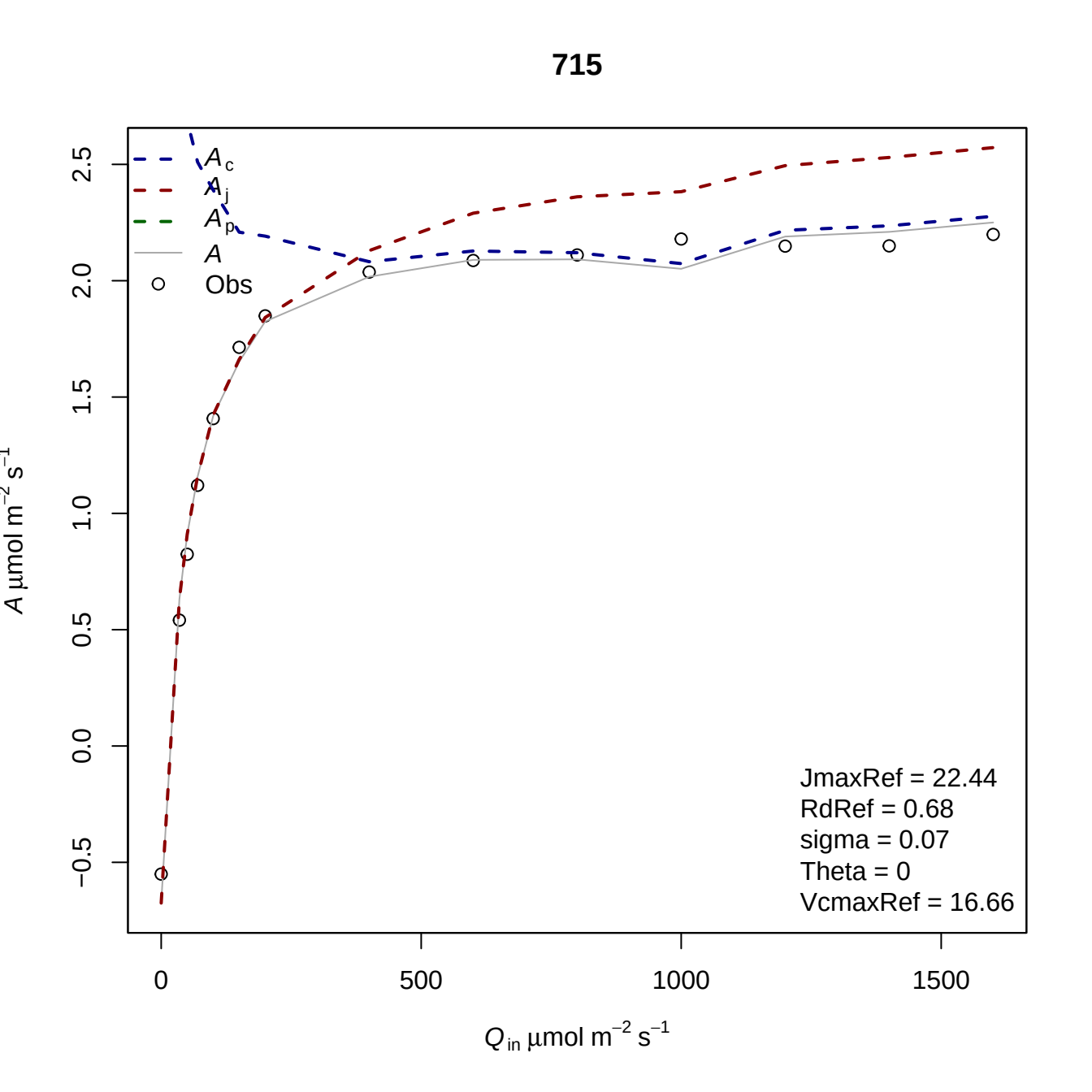

710
A_c
A_j
A_p
A
Obs
JmaxRef = 76.94
RdRef = 0.65
sigma = 0.24
Theta = 0
VcmaxRef = 40.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

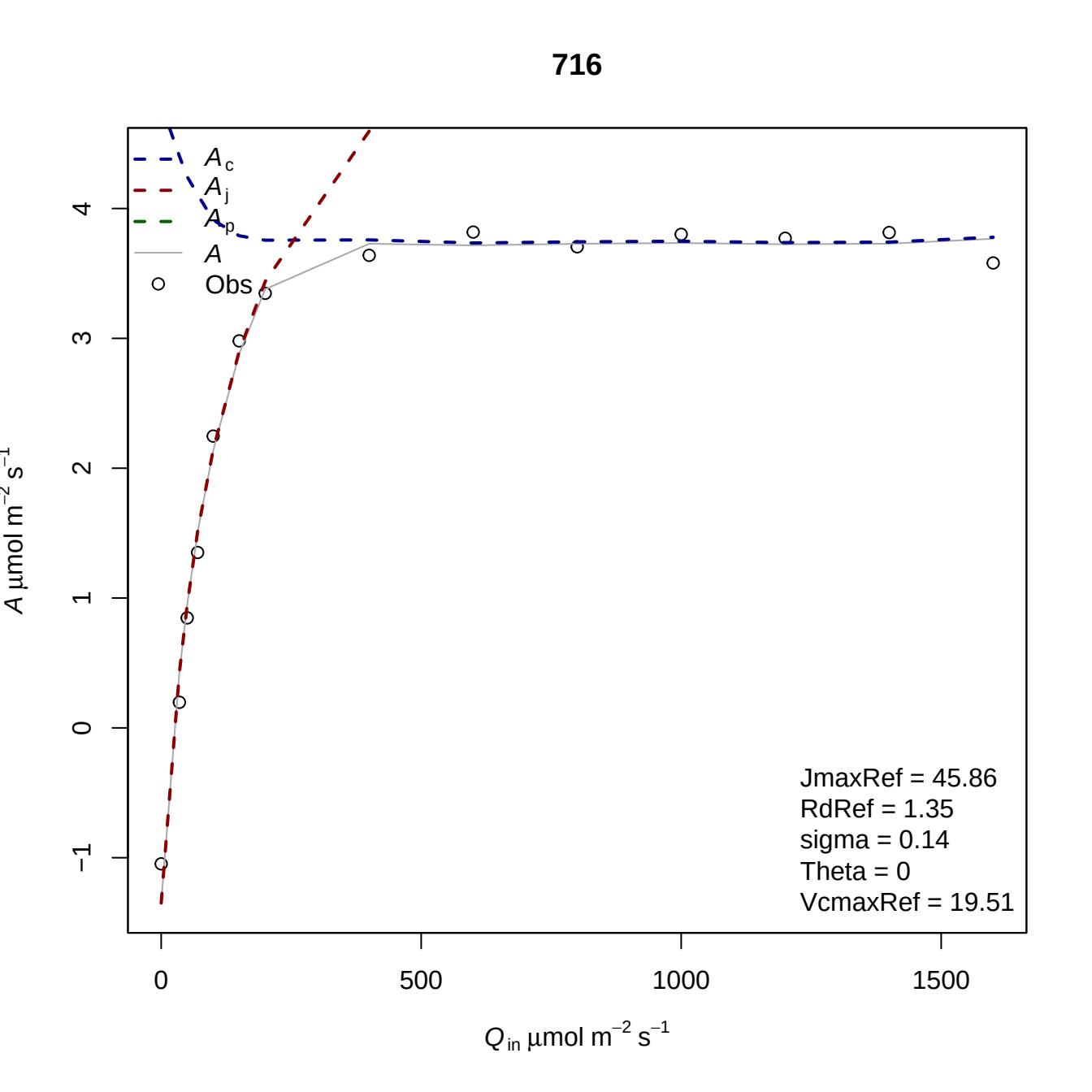

711
A_c
A_j
A_p
A
Obs
JmaxRef = 102.16
RdRef = 0.59
sigma = 0.23
Theta = 0
VcmaxRef = 55.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

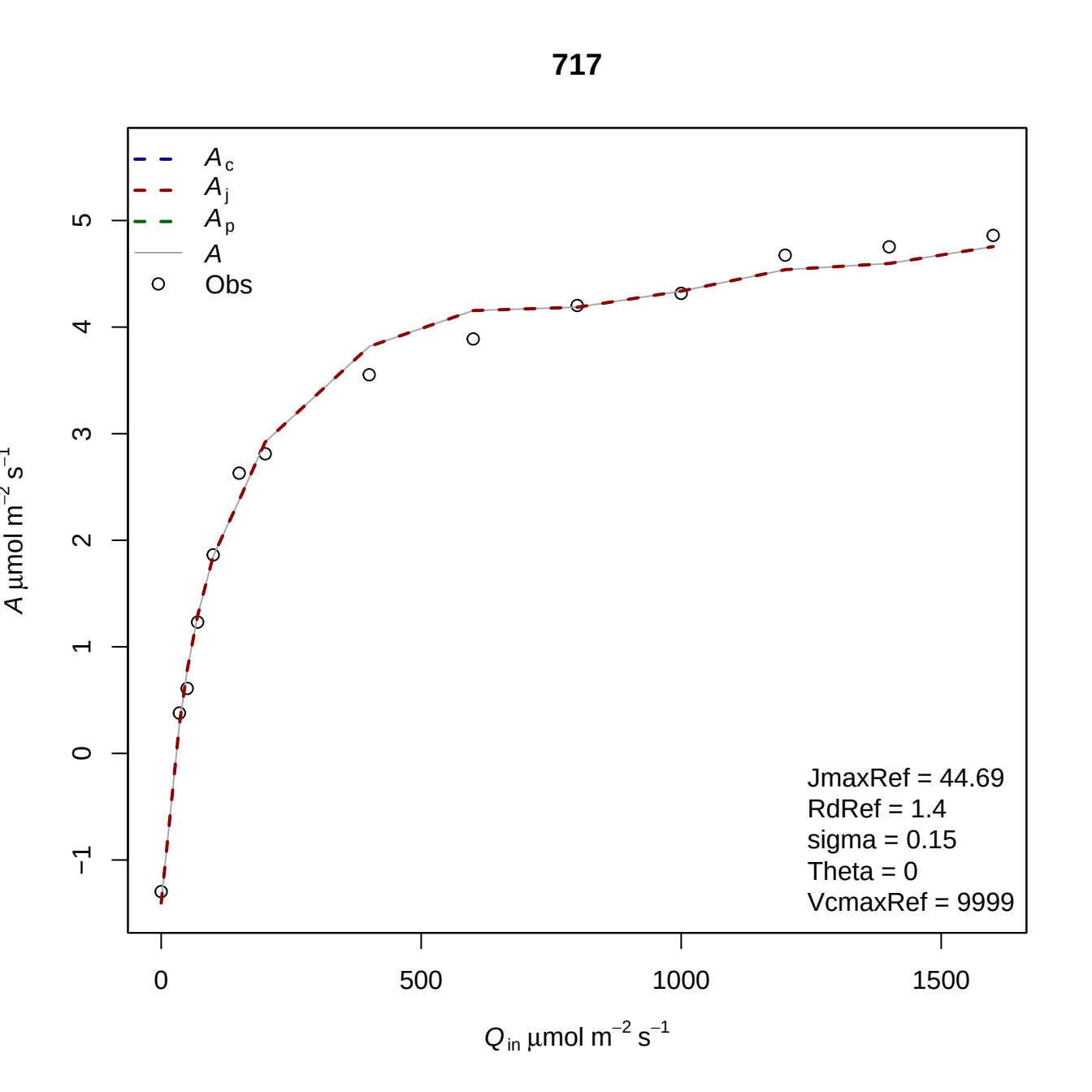

712
A_c
A_j
A_p
A
Obs
JmaxRef = 179.12
RdRef = 0.26
sigma = 0.23
Theta = 0
VcmaxRef = 65.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

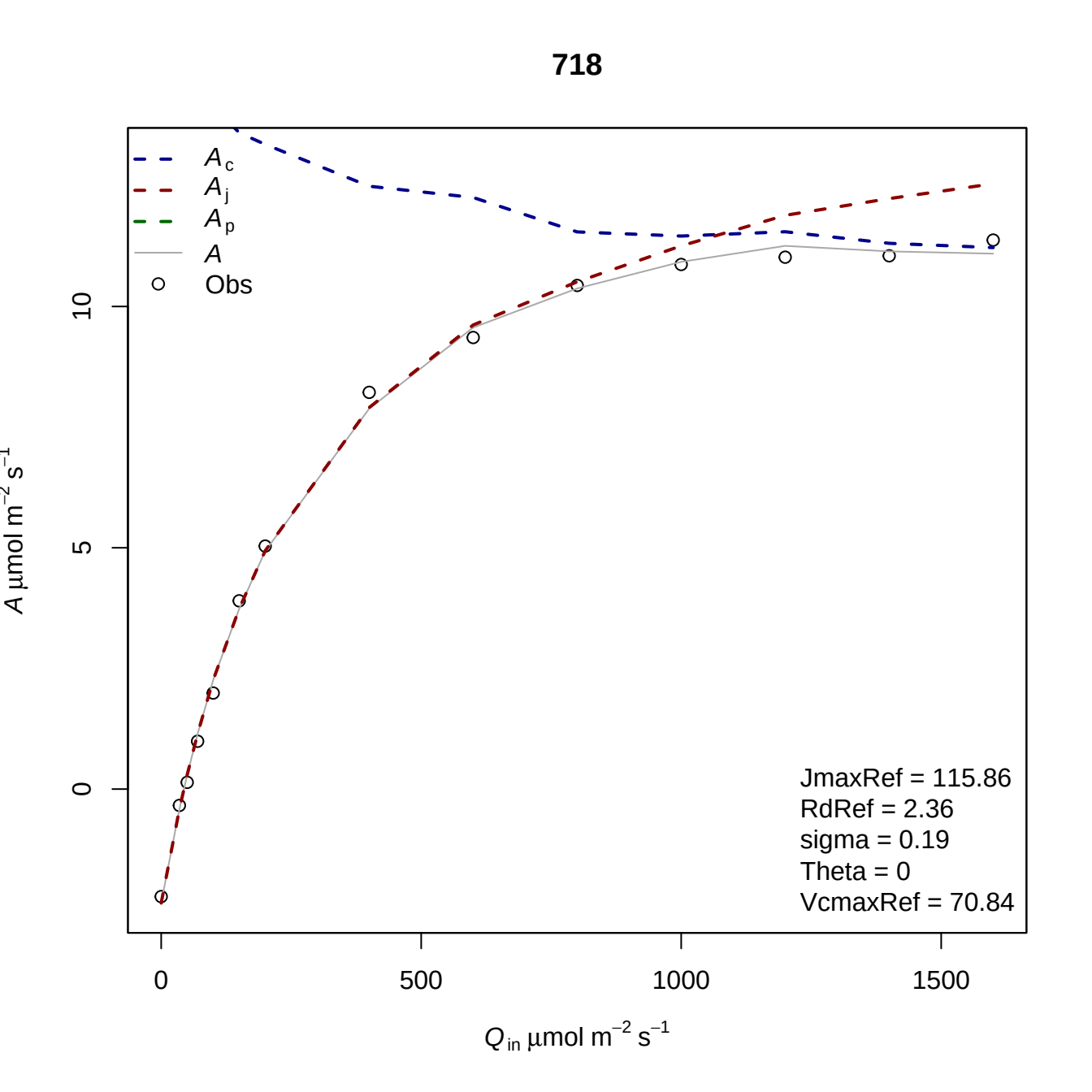

713
A_c
A_j
A_p
A
Obs
JmaxRef = 116.68
RdRef = 0.36
sigma = 0.2
Theta = 0
VcmaxRef = 70.67
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

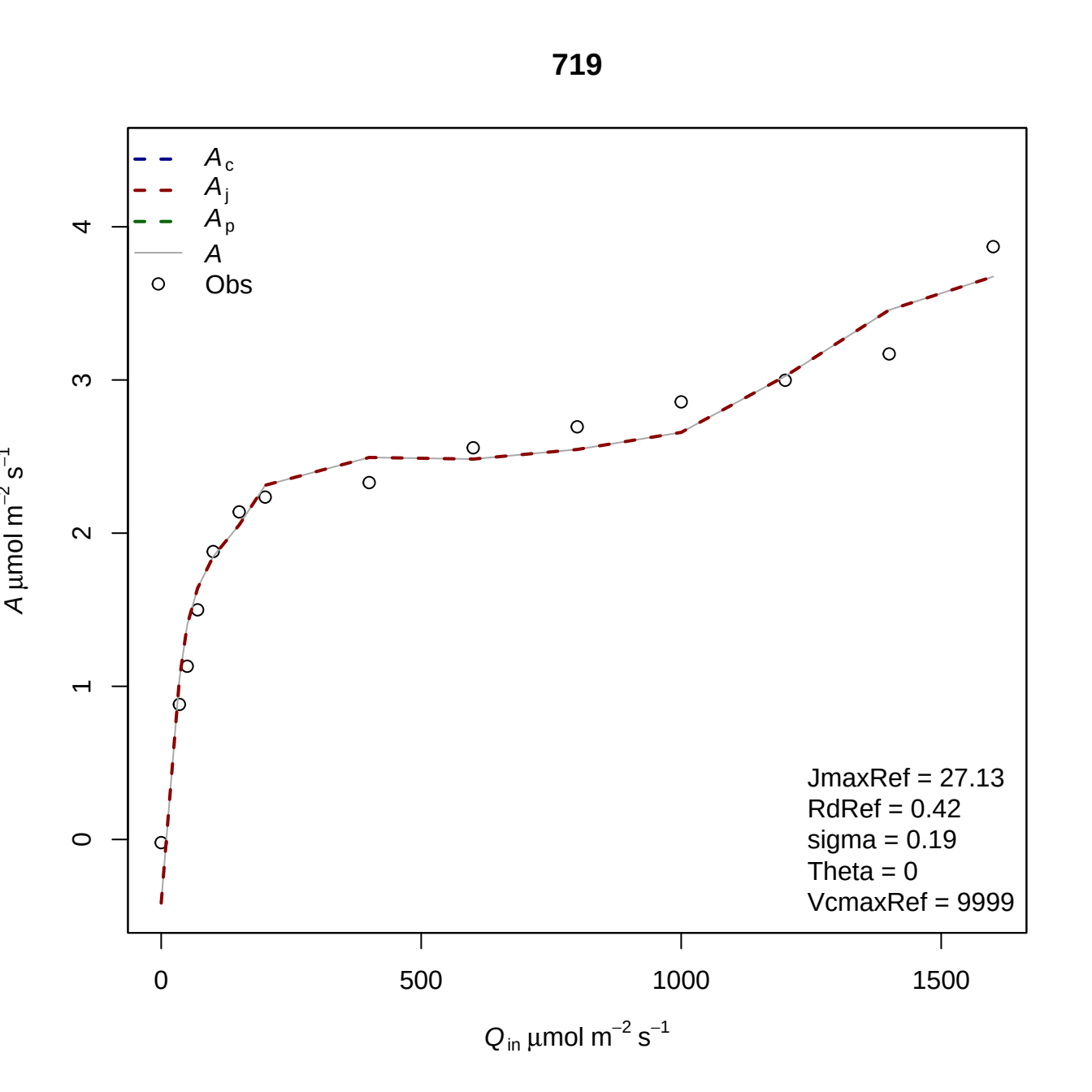

714
A_c
A_j
A_p
A
Obs
JmaxRef = 37.29
RdRef = 0.79
sigma = 0.22
Theta = 0
VcmaxRef = 21.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

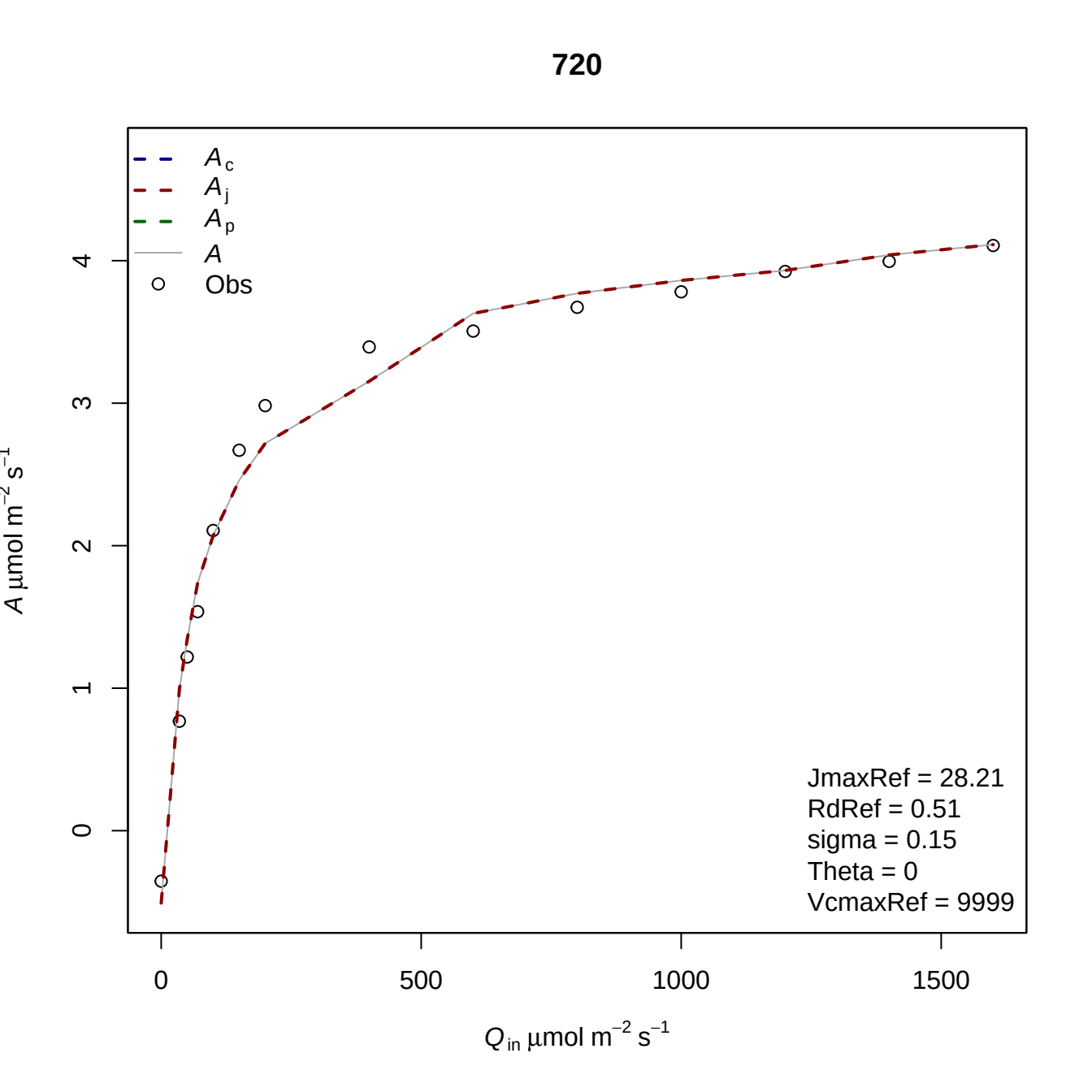

715
A_c
A_j
A_p
A
Obs
JmaxRef = 22.44
RdRef = 0.68
sigma = 0.07
Theta = 0
VcmaxRef = 16.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500

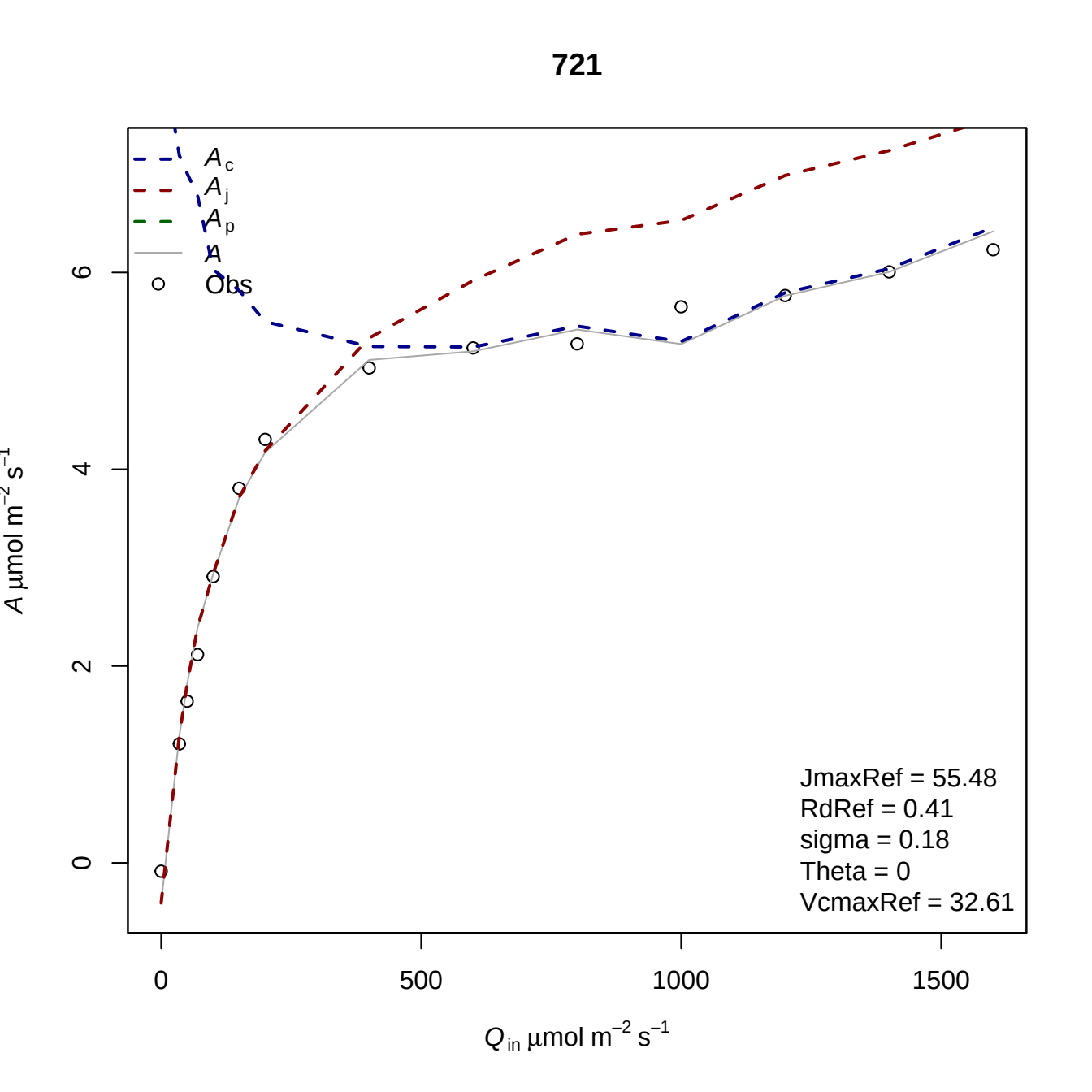

716
A_c
A_j
A_p
A
Obs
JmaxRef = 45.86
RdRef = 1.35
sigma = 0.14
Theta = 0
VcmaxRef = 19.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

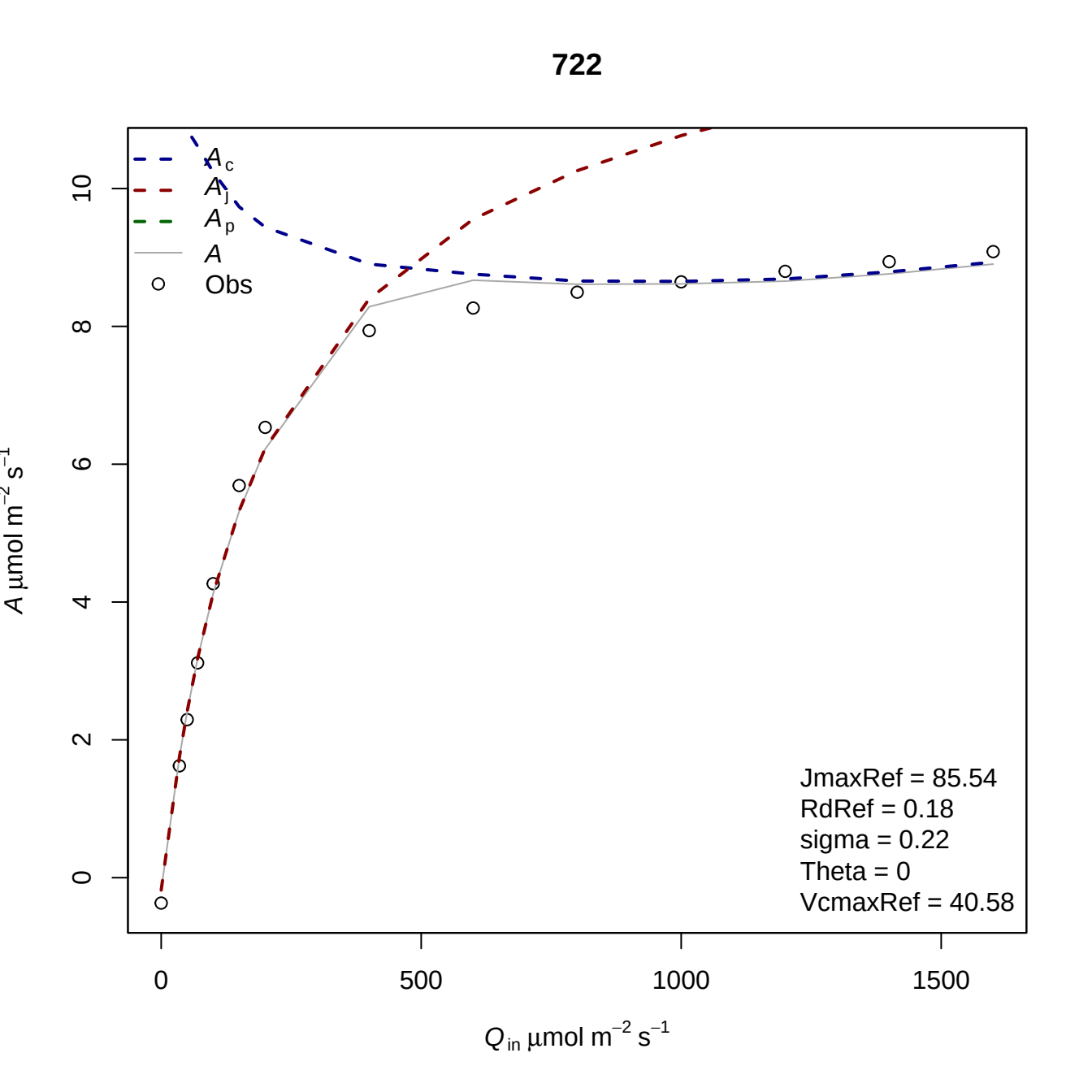

717
A_c
A_j
A_p
A
Obs
JmaxRef = 44.69
RdRef = 1.4
sigma = 0.15
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

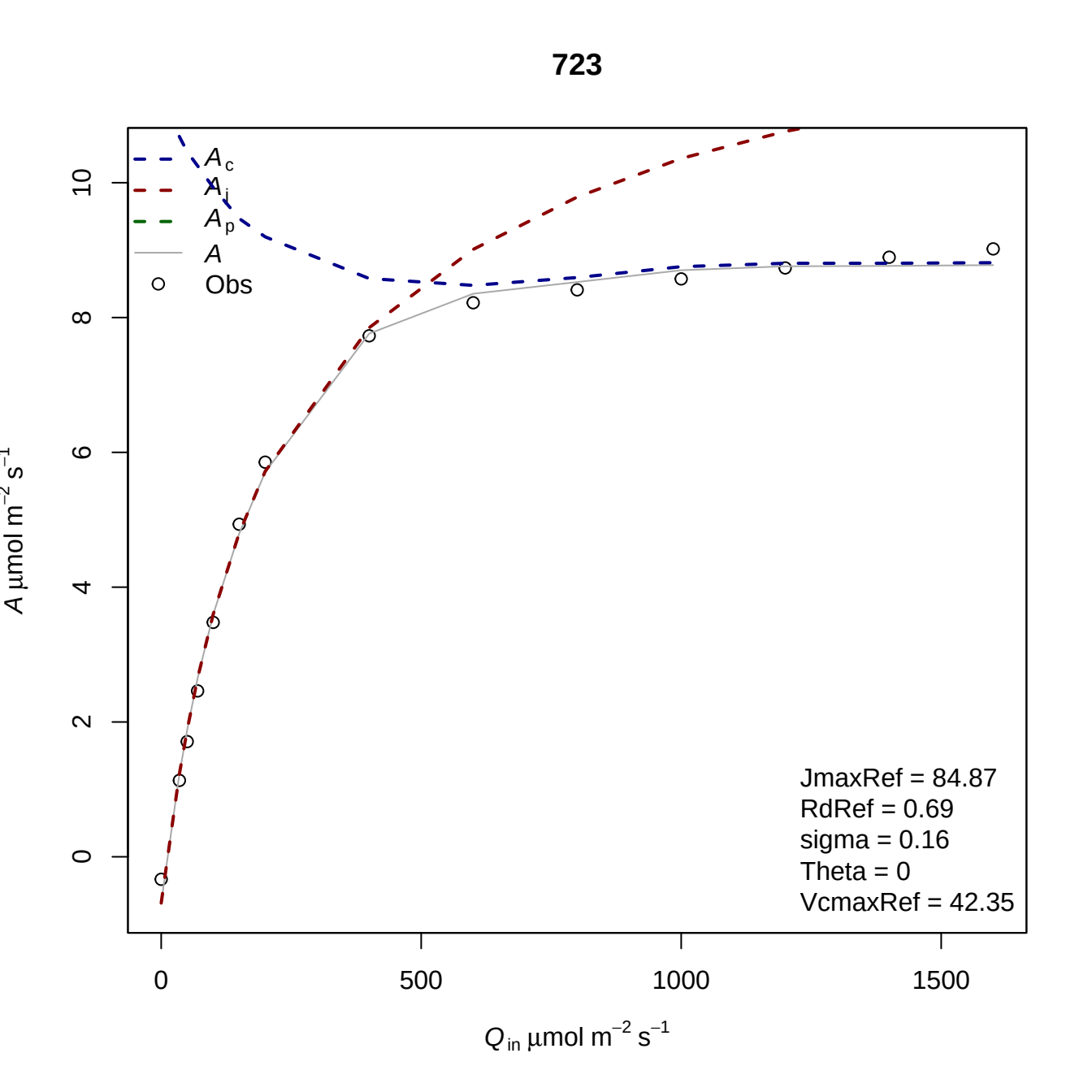

718
A_c
A_j
A_p
A
Obs
JmaxRef = 115.86
RdRef = 2.36
sigma = 0.19
Theta = 0
VcmaxRef = 70.84
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

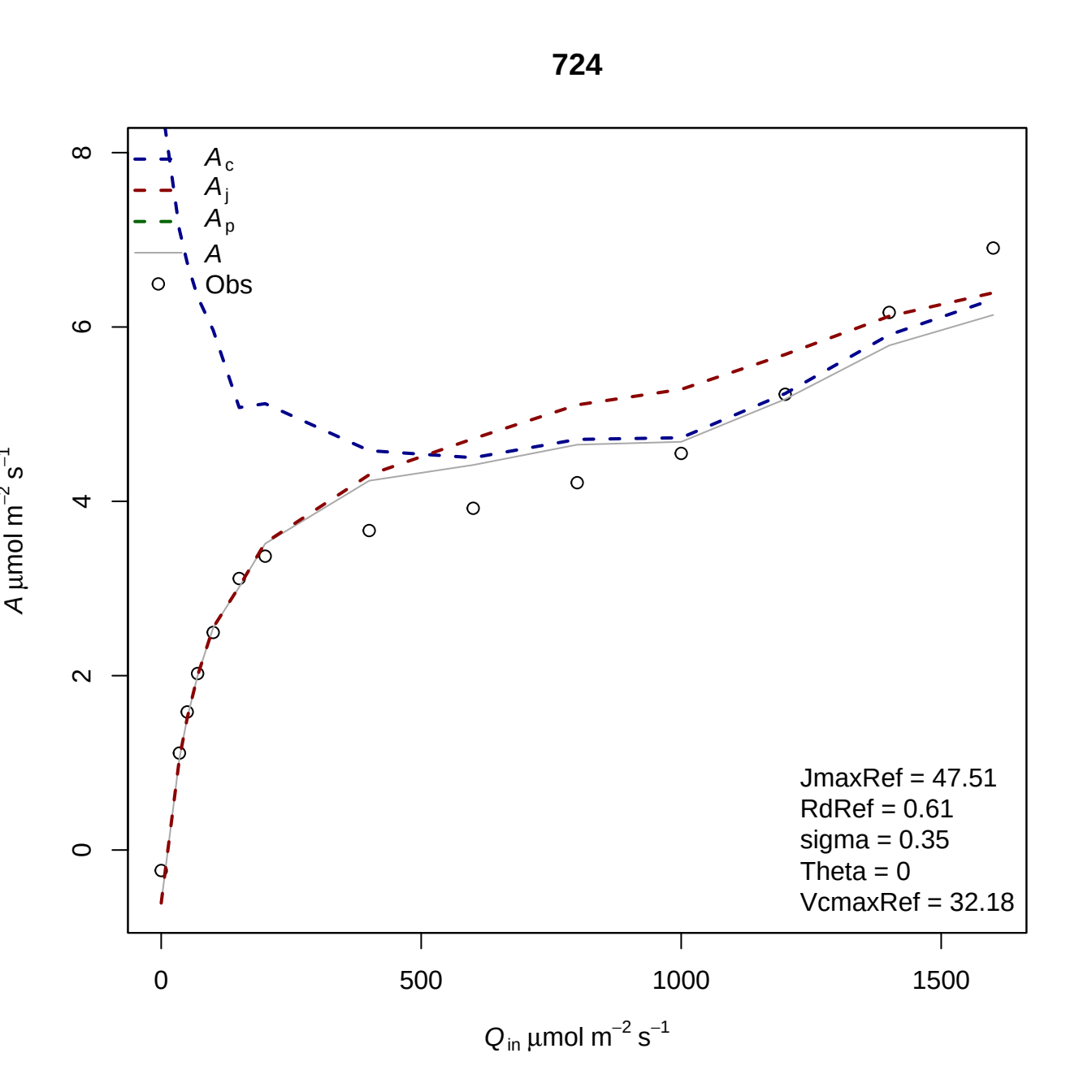

719
A_c
A_j
A_p
A
Obs
JmaxRef = 27.13
RdRef = 0.42
sigma = 0.19
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

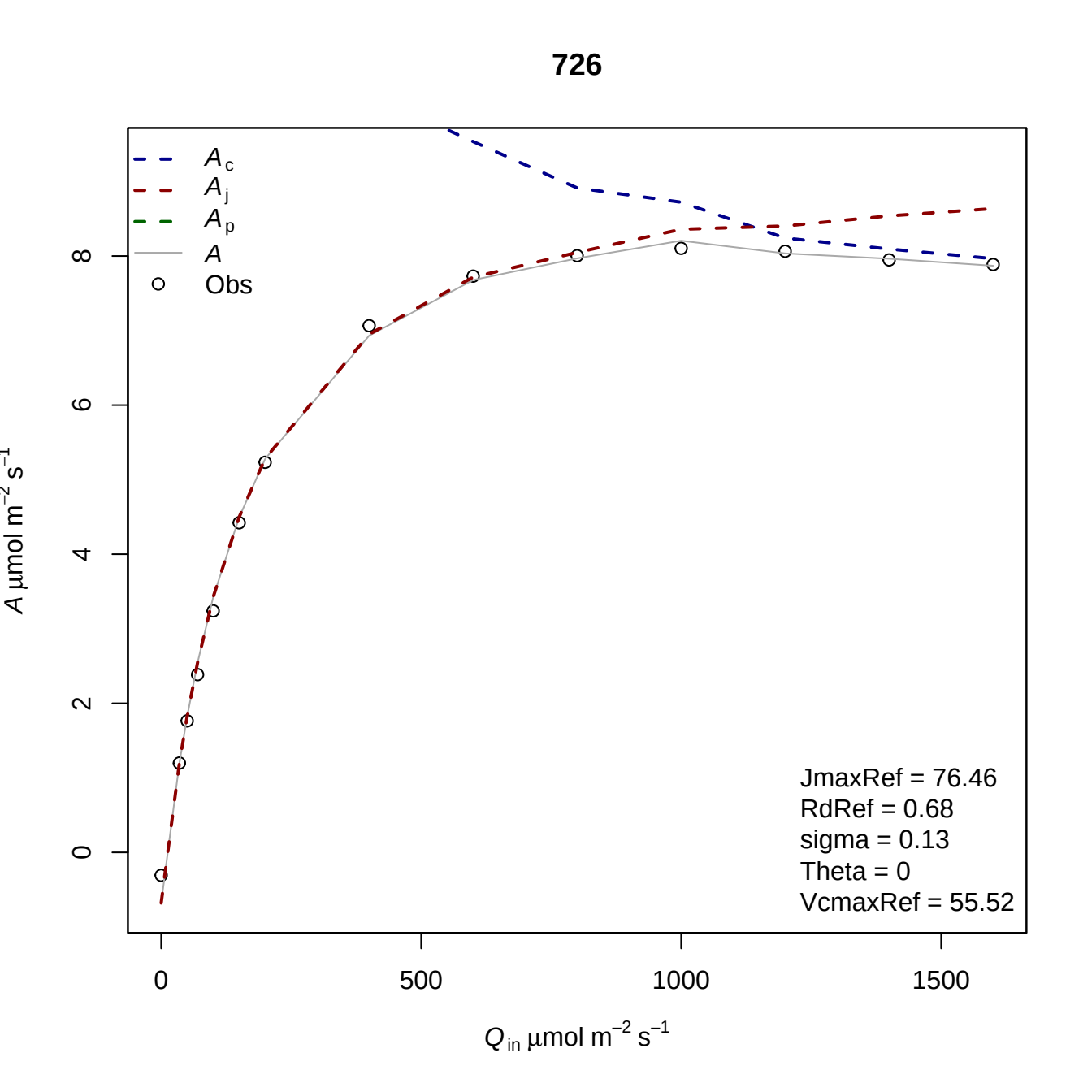

720
A_c
A_j
A_p
A
Obs
JmaxRef = 28.21
RdRef = 0.51
sigma = 0.15
Theta = 0
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

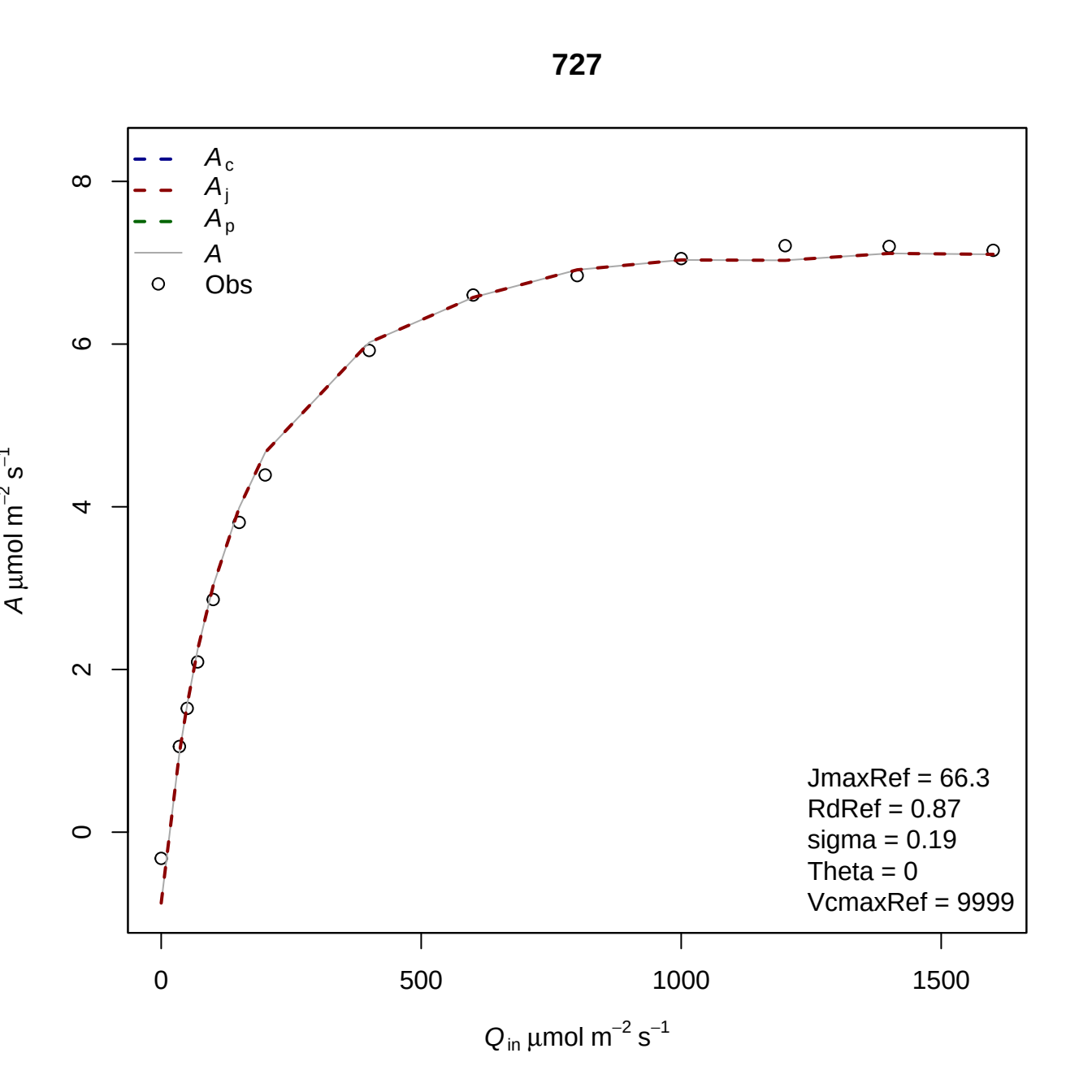

721
A_c
A_j
A_p
A
Obs
JmaxRef = 55.48
RdRef = 0.41
sigma = 0.18
Theta = 0
VcmaxRef = 32.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

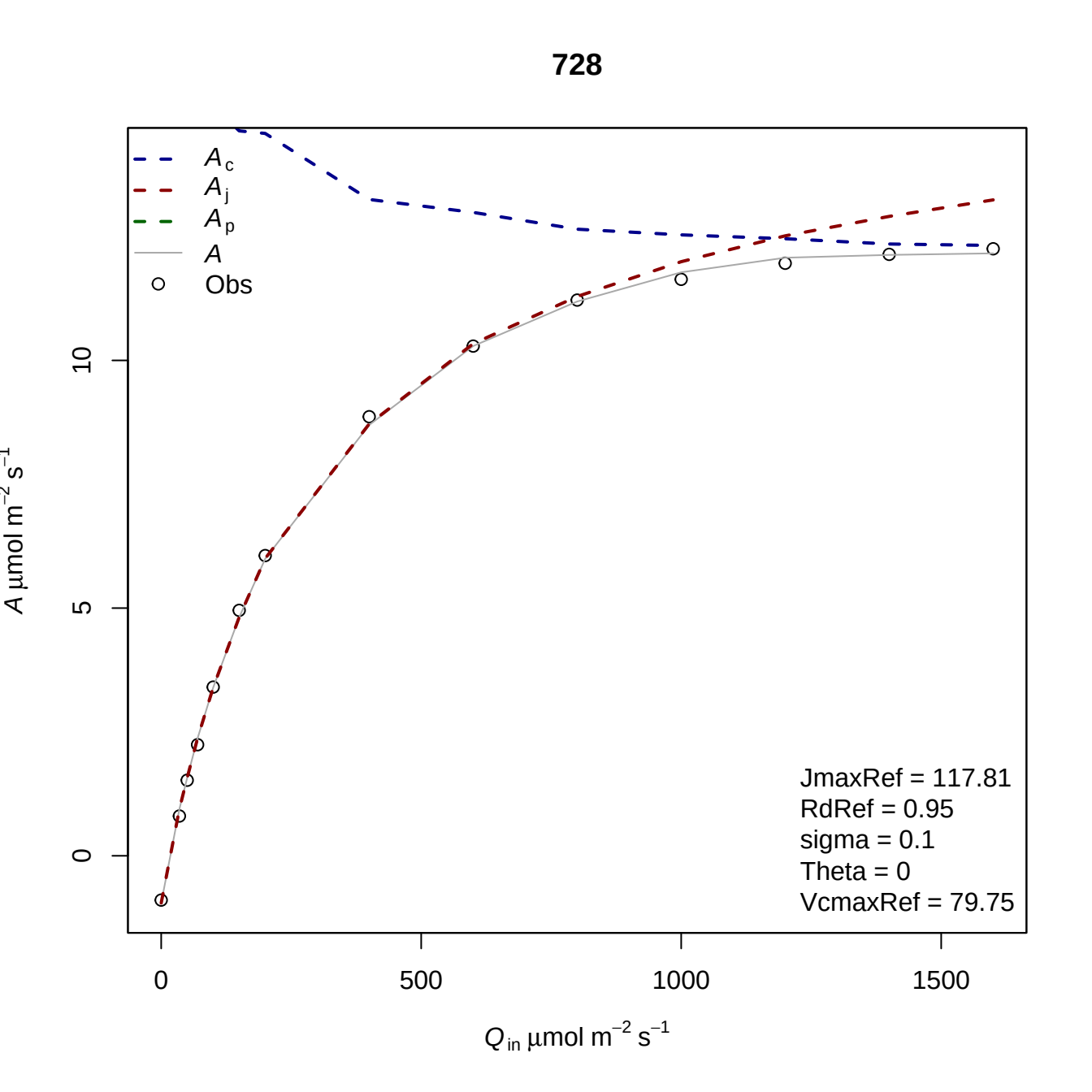

722
A_c
A_j
A_p
A
Obs
JmaxRef = 85.54
RdRef = 0.18
sigma = 0.22
Theta = 0
VcmaxRef = 40.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

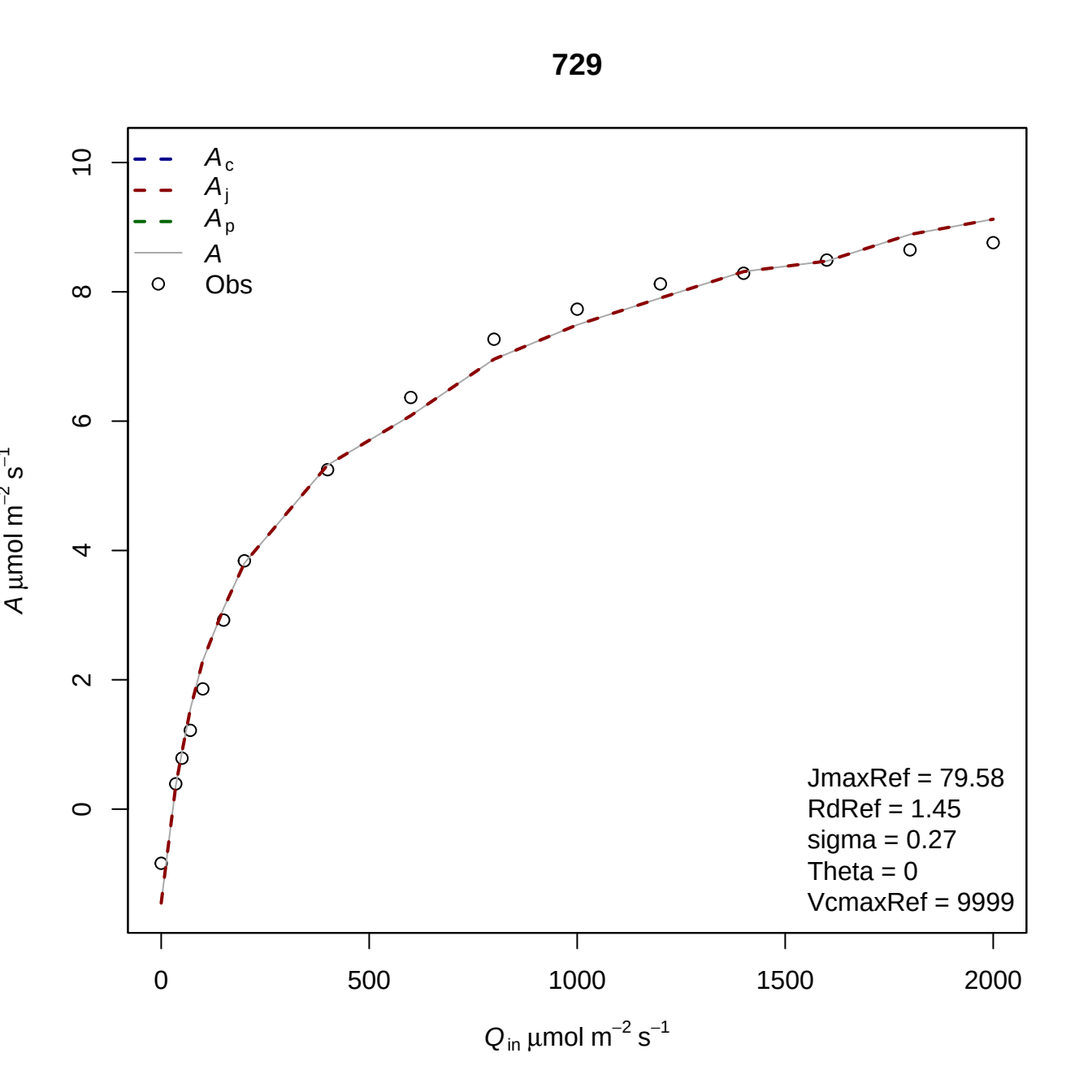

723
A_c
A_j
A_p
A
Obs
JmaxRef = 84.87
RdRef = 0.69
sigma = 0.16
Theta = 0
VcmaxRef = 42.35
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

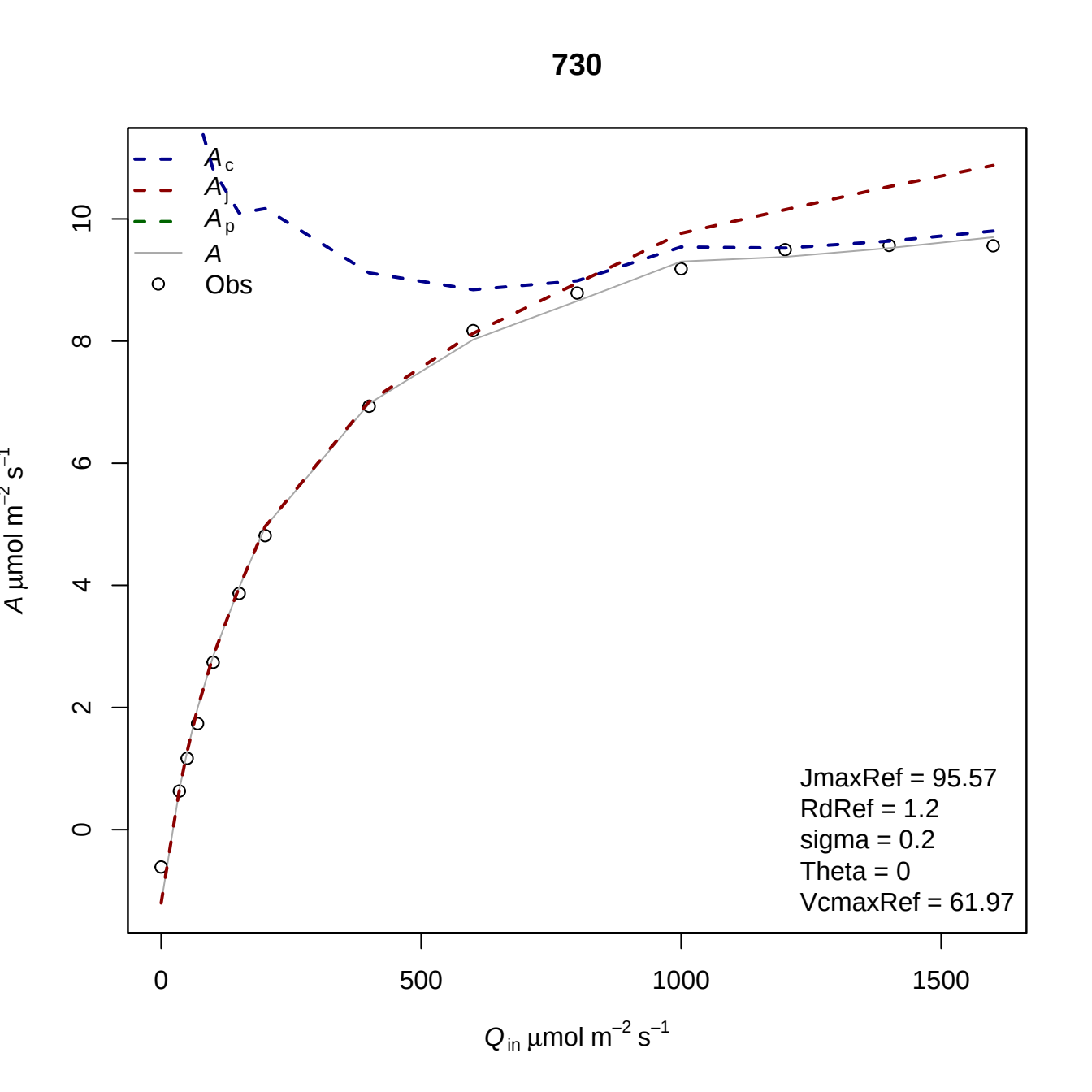

724
A_c
A_j
A_p
A
Obs
JmaxRef = 47.51
RdRef = 0.61
sigma = 0.35
Theta = 0
VcmaxRef = 32.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

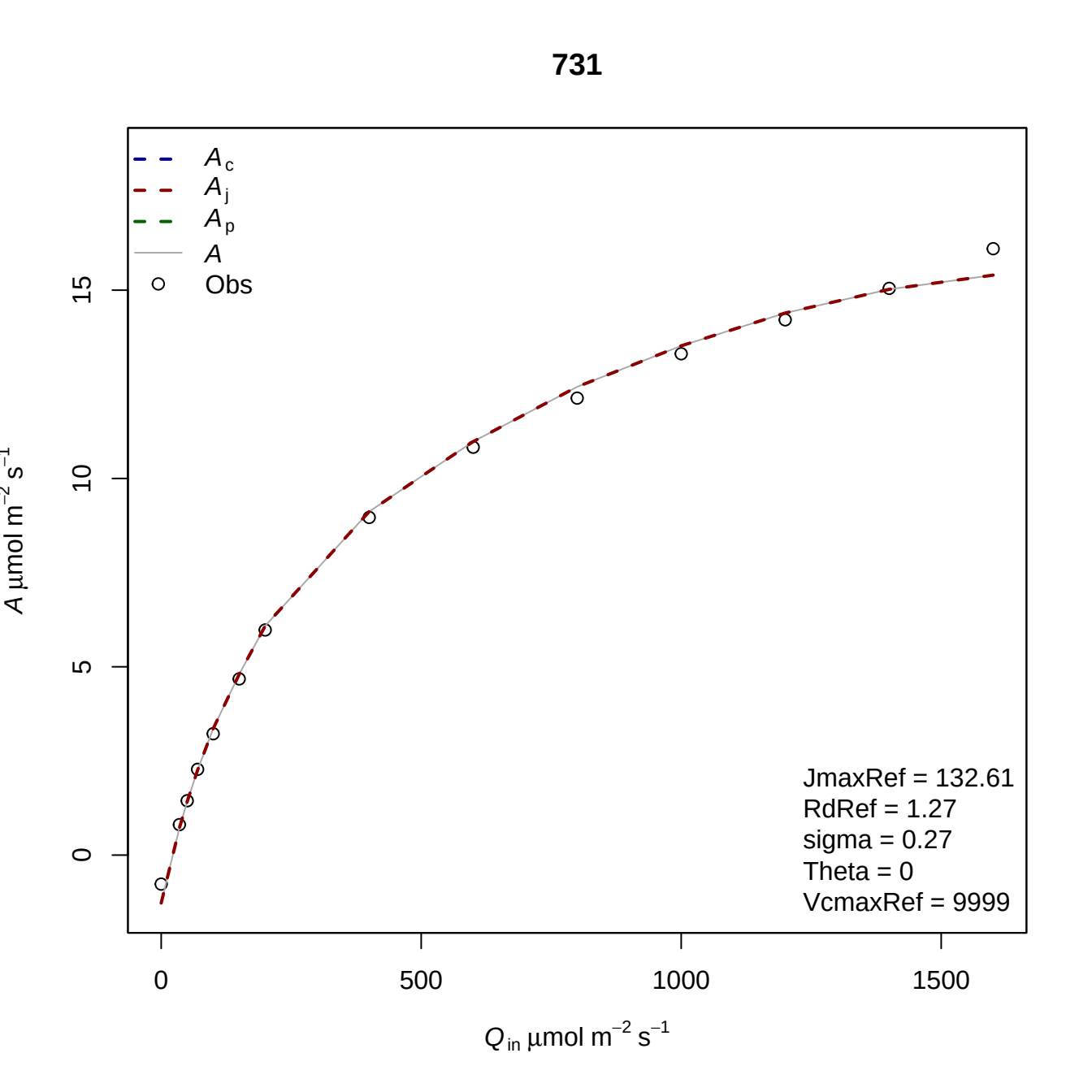

726
A_c
A_j
A_p
A
Obs
JmaxRef = 76.46
RdRef = 0.68
sigma = 0.13
Theta = 0
VcmaxRef = 55.52
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

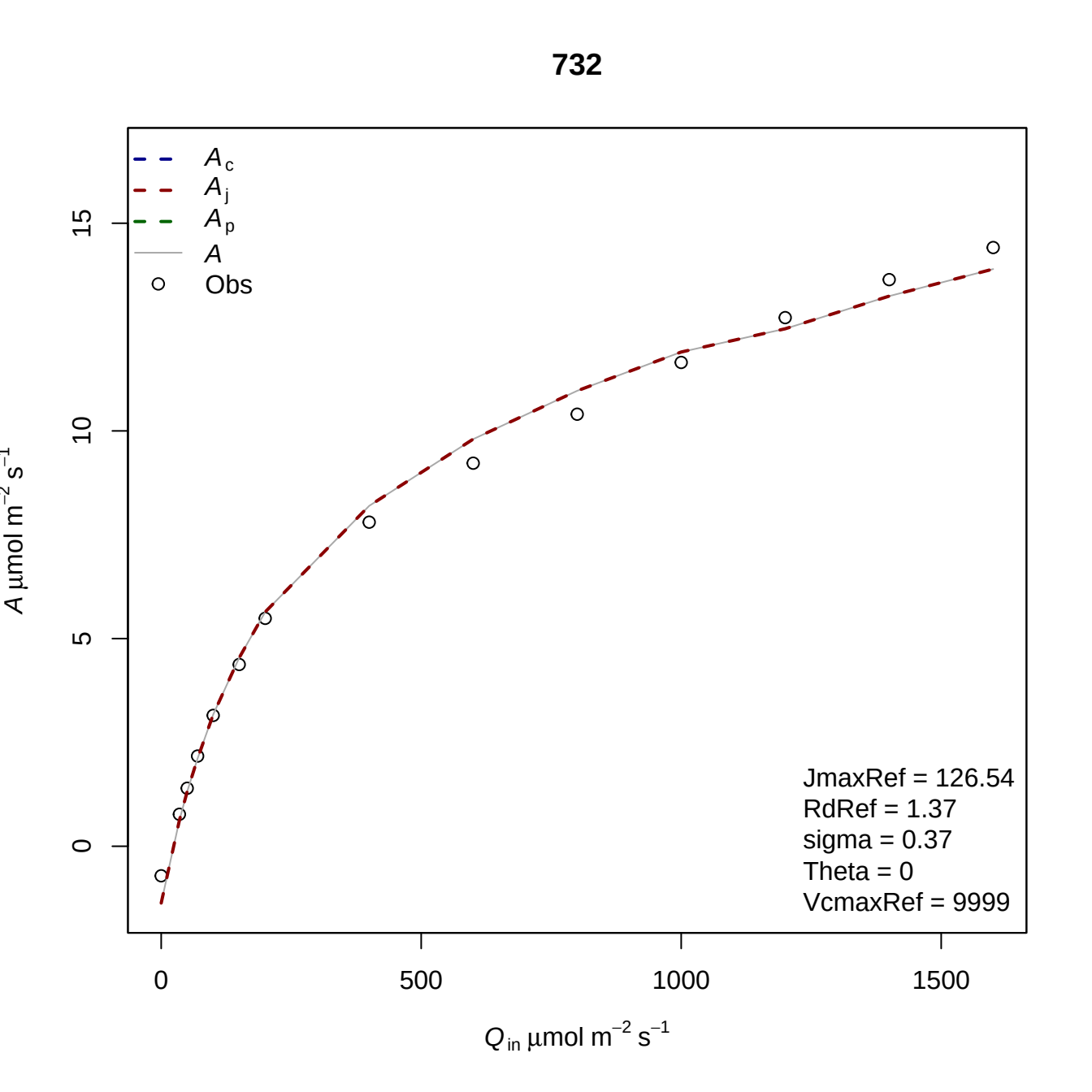

727
A_c
A_j
A_p
A
Obs
JmaxRef = 66.3
RdRef = 0.87
sigma = 0.19
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

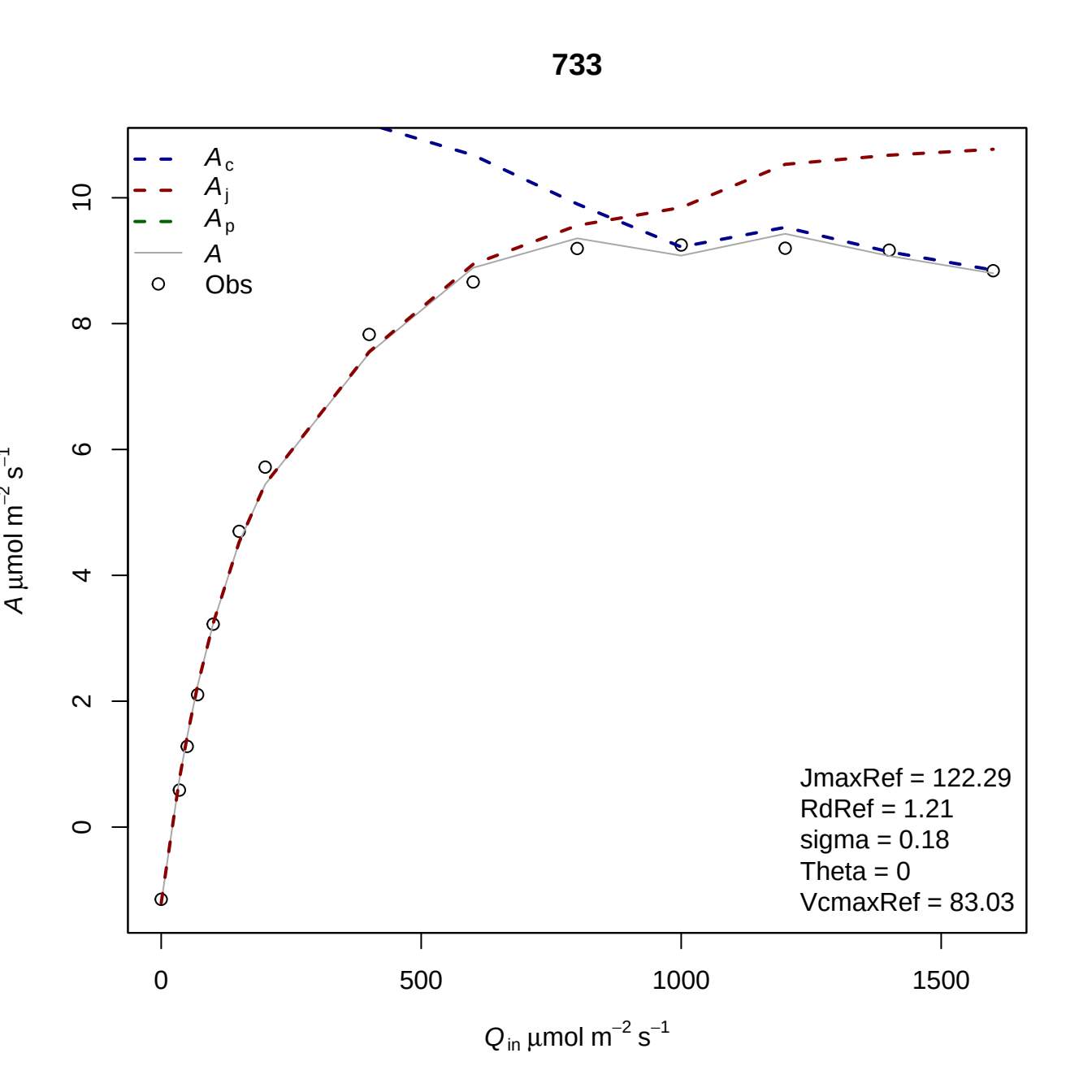

728
A_c
A_j
A_p
A
Obs
JmaxRef = 117.81
RdRef = 0.95
sigma = 0.1
Theta = 0
VcmaxRef = 79.75
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

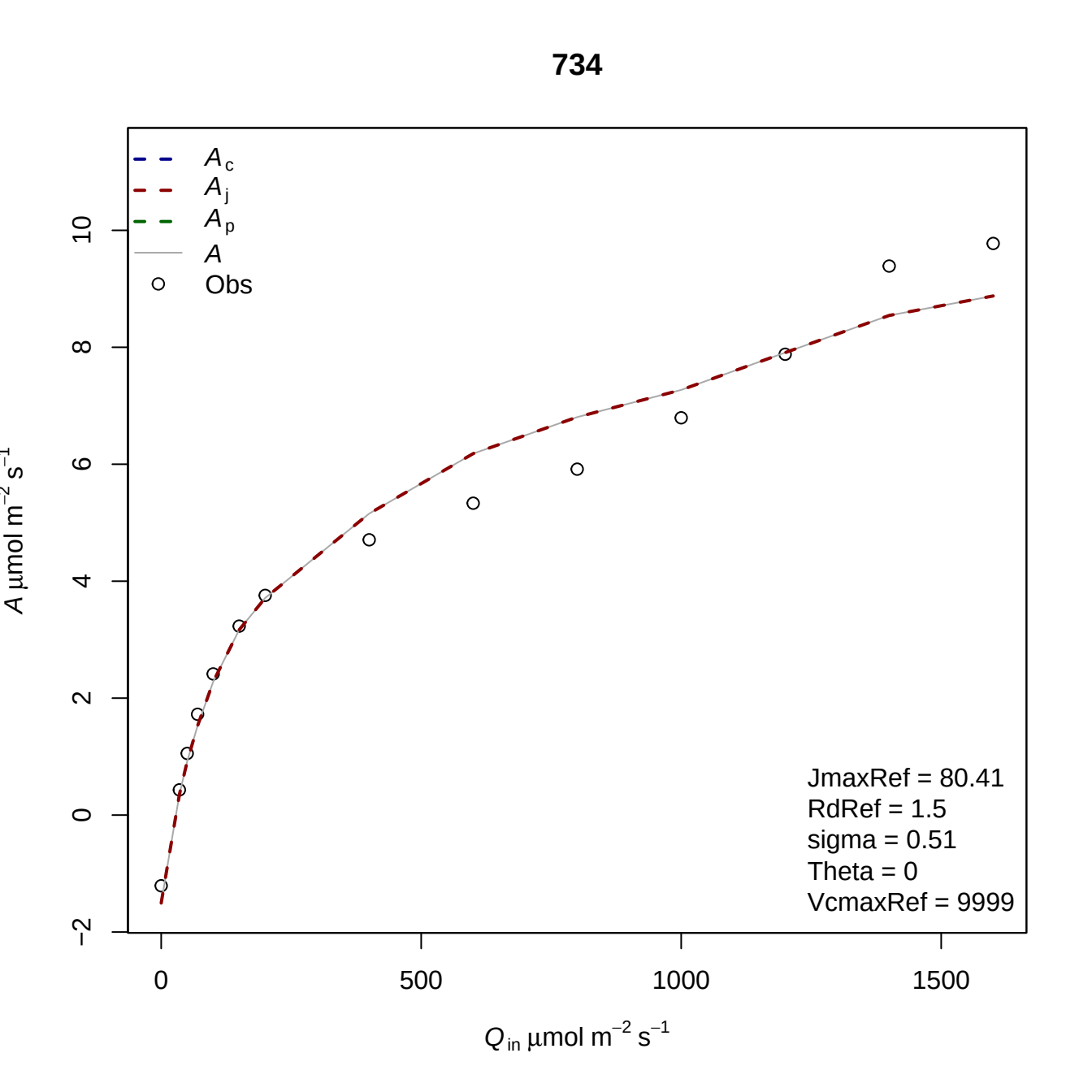

729
A_c
A_j
A_p
A
Obs
JmaxRef = 79.58
RdRef = 1.45
sigma = 0.27
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

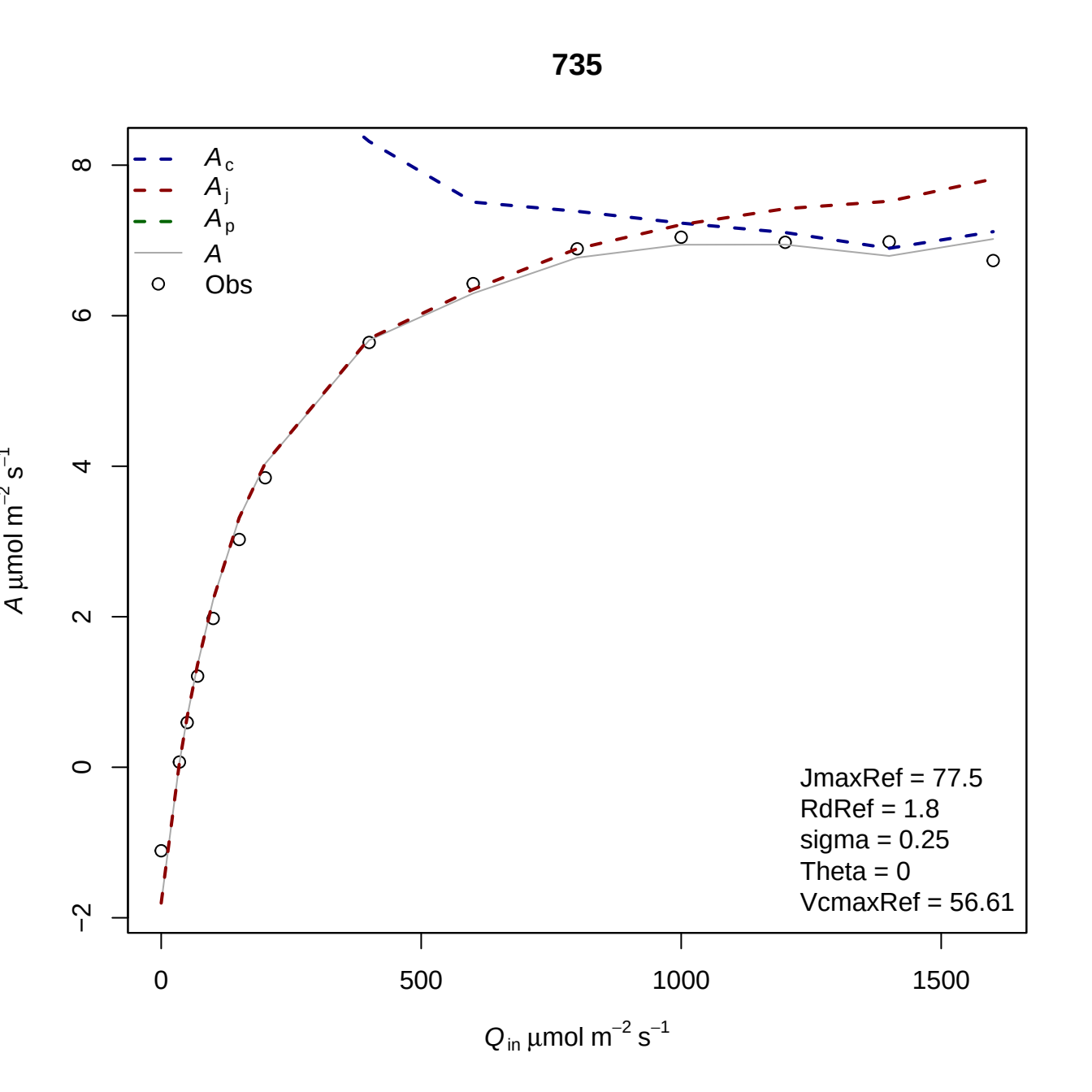

730
A_c
A_j
A_p
A
Obs
JmaxRef = 95.57
RdRef = 1.2
sigma = 0.2
Theta = 0
VcmaxRef = 61.97
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

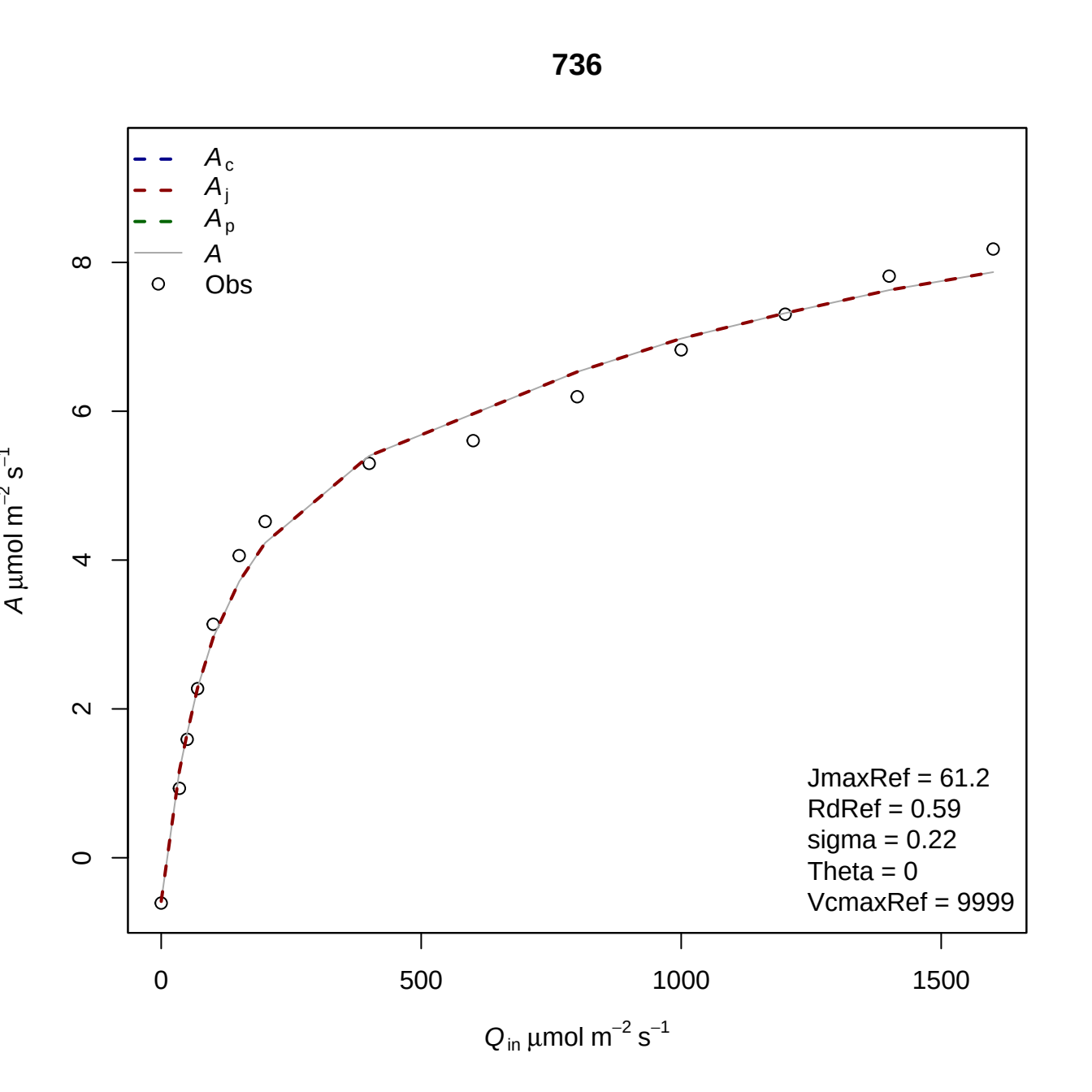

731
A_c
A_j
A_p
A
Obs
JmaxRef = 132.61
RdRef = 1.27
sigma = 0.27
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

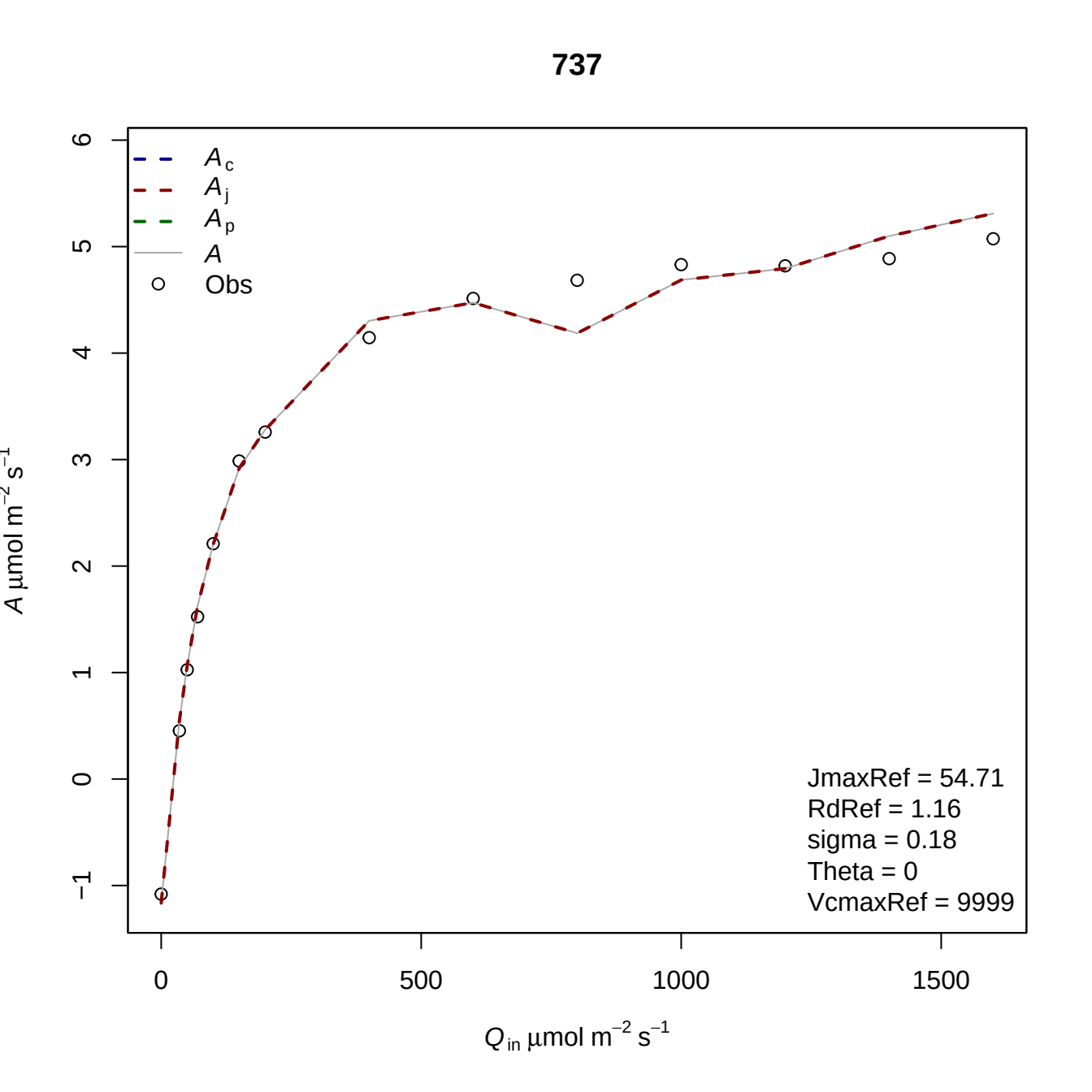

732
A_c
A_j
A_p
A
Obs
JmaxRef = 126.54
RdRef = 1.37
sigma = 0.37
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

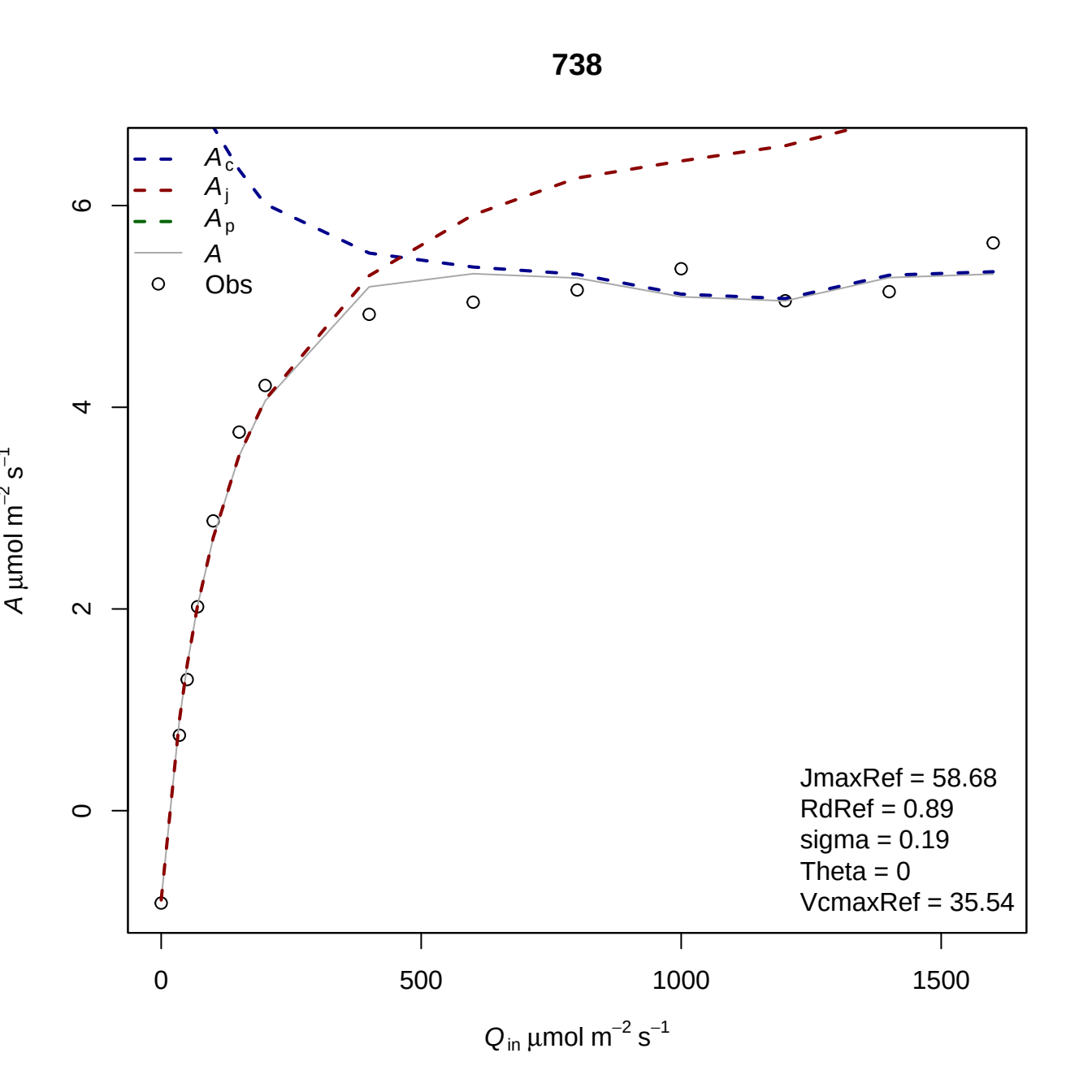

733
A_c
A_j
A_p
A
Obs
JmaxRef = 122.29
RdRef = 1.21
sigma = 0.18
Theta = 0
VcmaxRef = 83.03
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

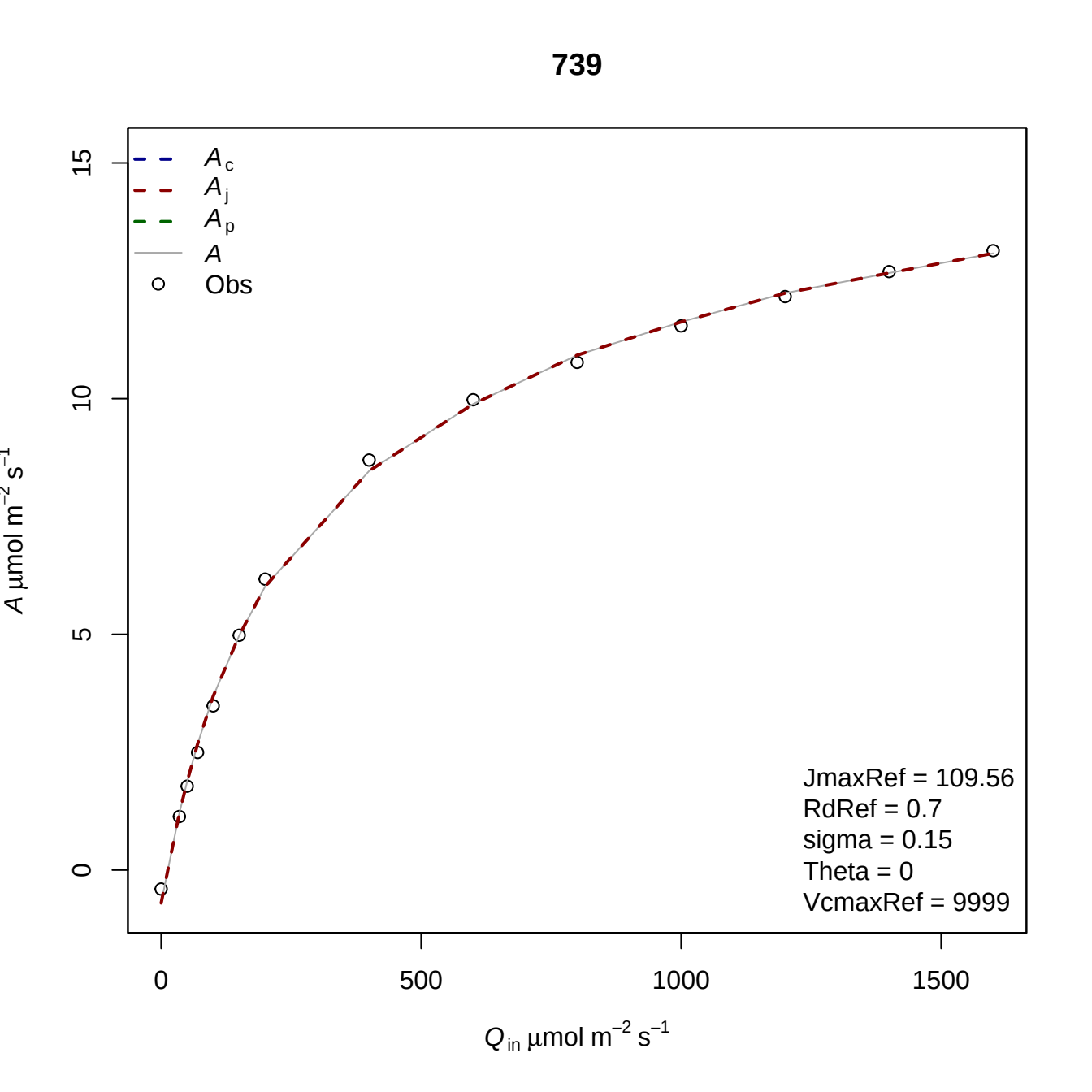

734
A_c
A_j
A_p
A
Obs
JmaxRef = 80.41
RdRef = 1.5
sigma = 0.51
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

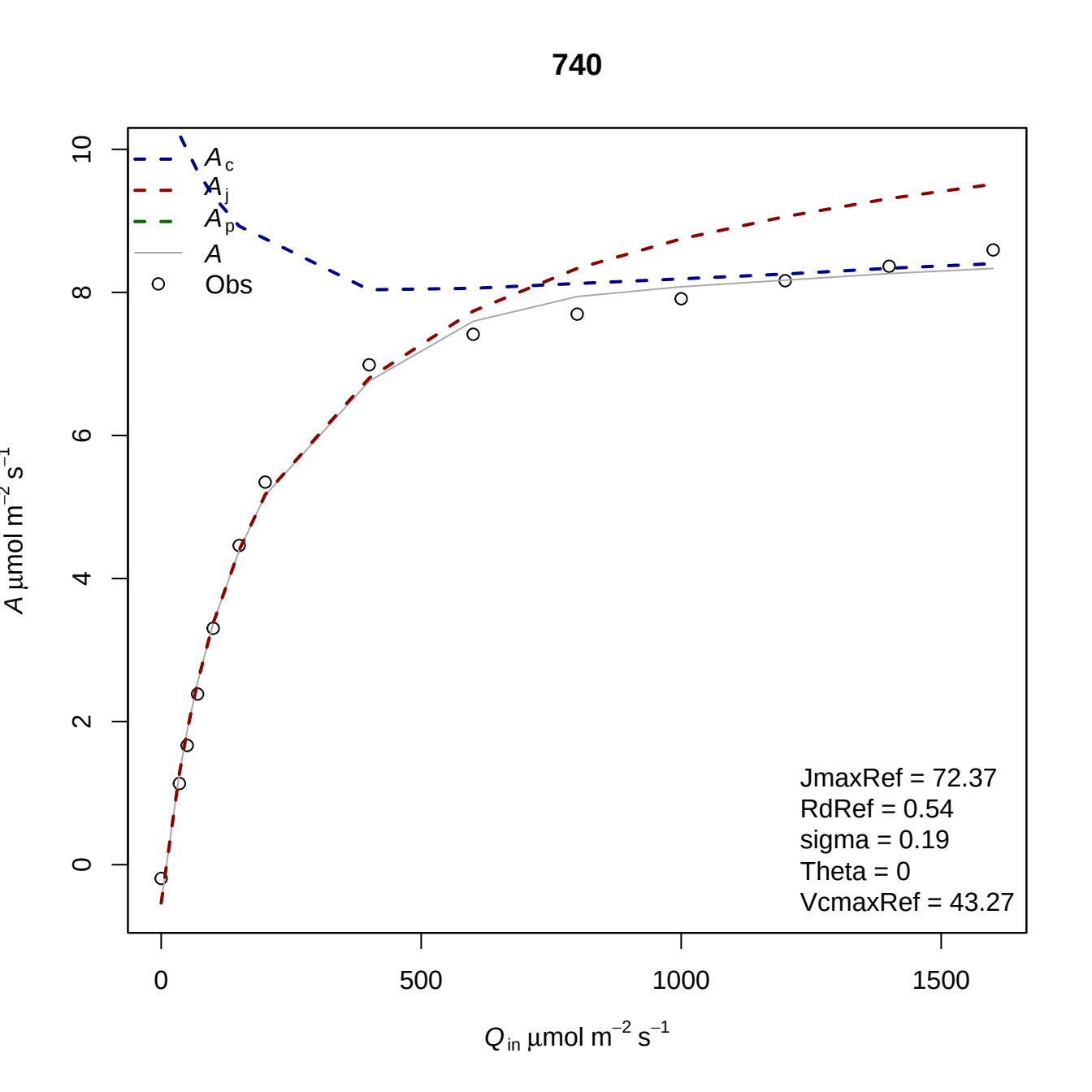

735
A_c
A_j
A_p
A
Obs
JmaxRef = 77.5
RdRef = 1.8
sigma = 0.25
Theta = 0
VcmaxRef = 56.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

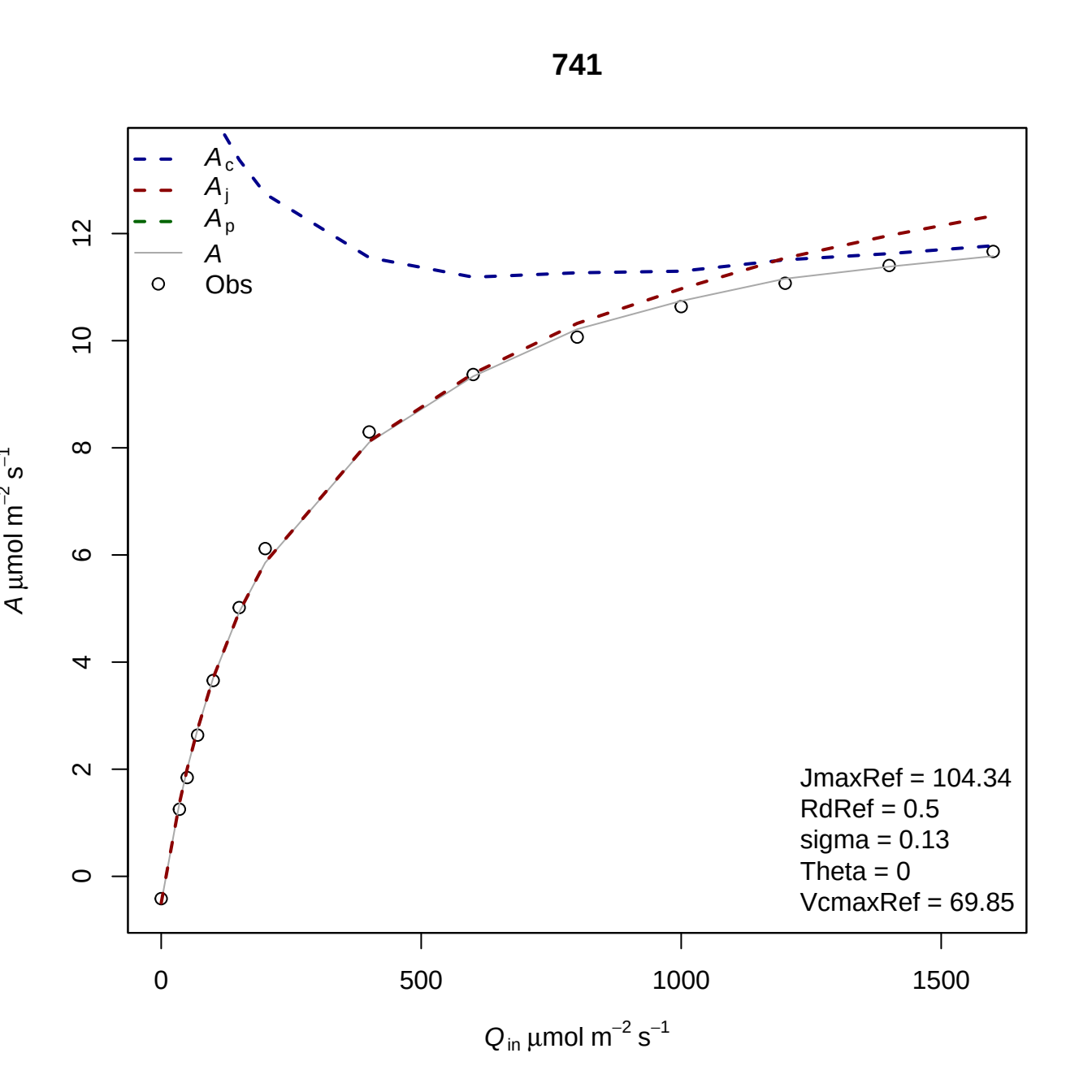

736
A_c
A_j
A_p
A
Obs
JmaxRef = 61.2
RdRef = 0.59
sigma = 0.22
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

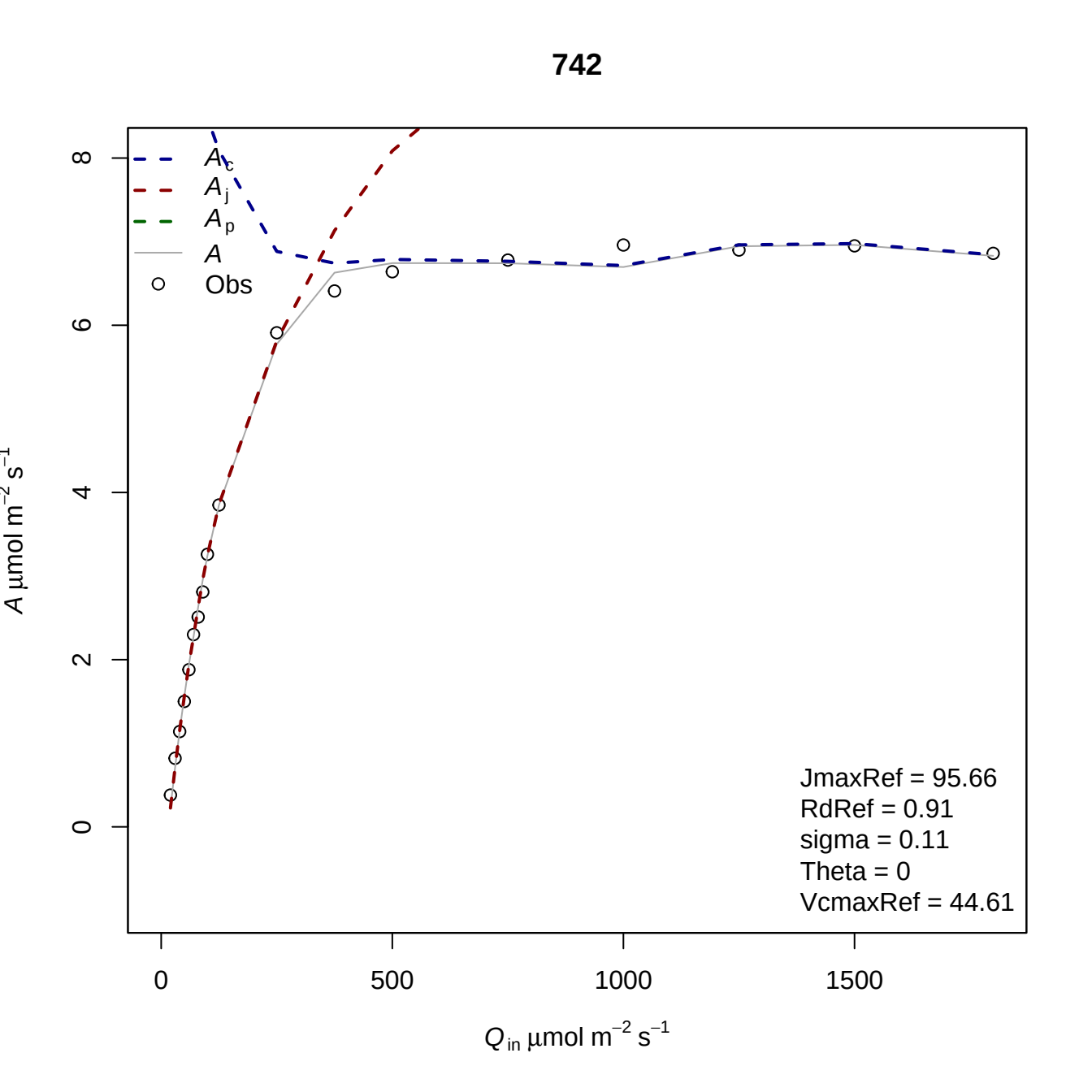

737
A_c
A_j
A_p
A
Obs
JmaxRef = 54.71
RdRef = 1.16
sigma = 0.18
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

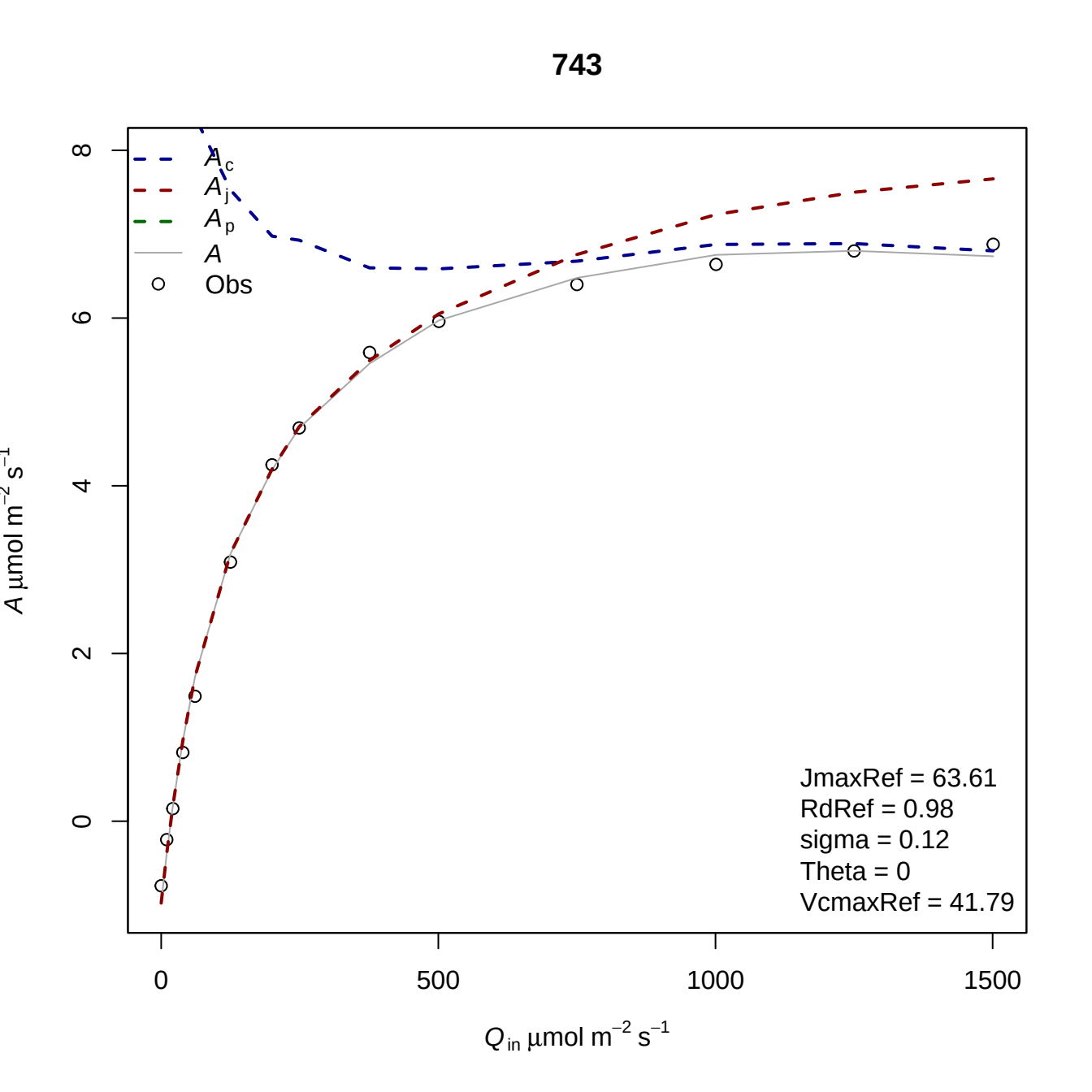

738
A_c
A_j
A_p
A
Obs
JmaxRef = 58.68
RdRef = 0.89
sigma = 0.19
Theta = 0
VcmaxRef = 35.54
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

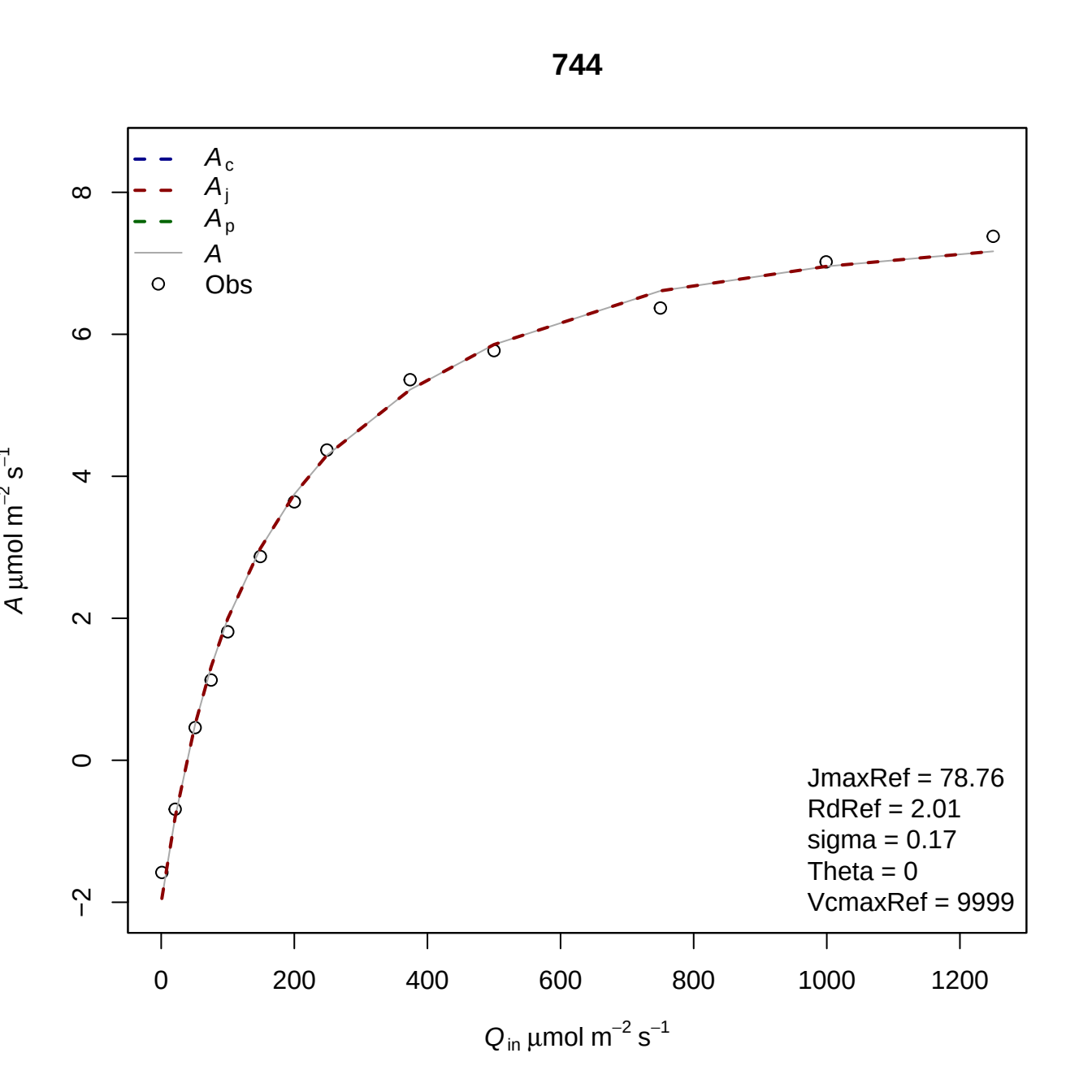

739
A_c
A_j
A_p
A
Obs
JmaxRef = 109.56
RdRef = 0.7
sigma = 0.15
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

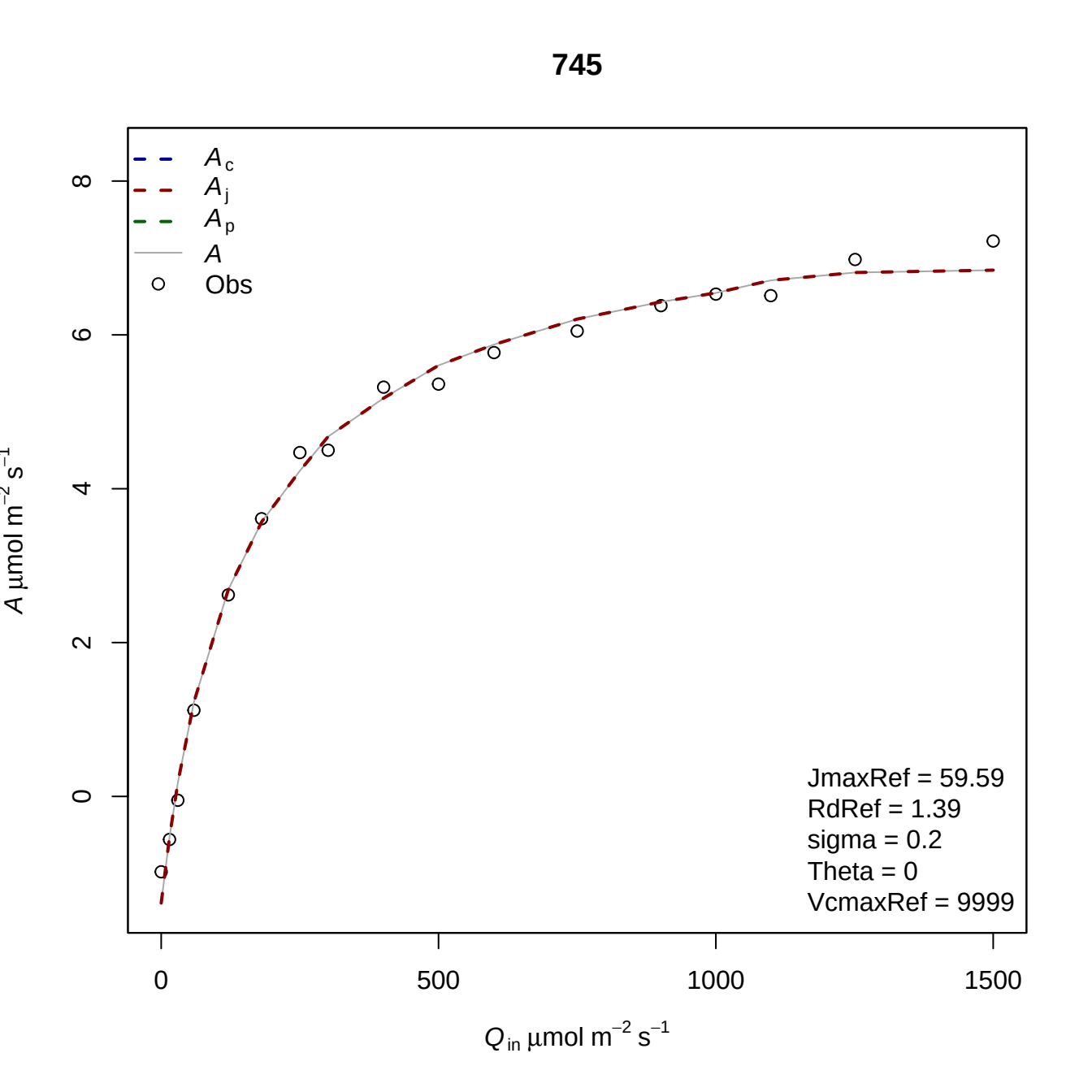

740
A_c
A_j
A_p
A
Obs
JmaxRef = 72.37
RdRef = 0.54
sigma = 0.19
Theta = 0
VcmaxRef = 43.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

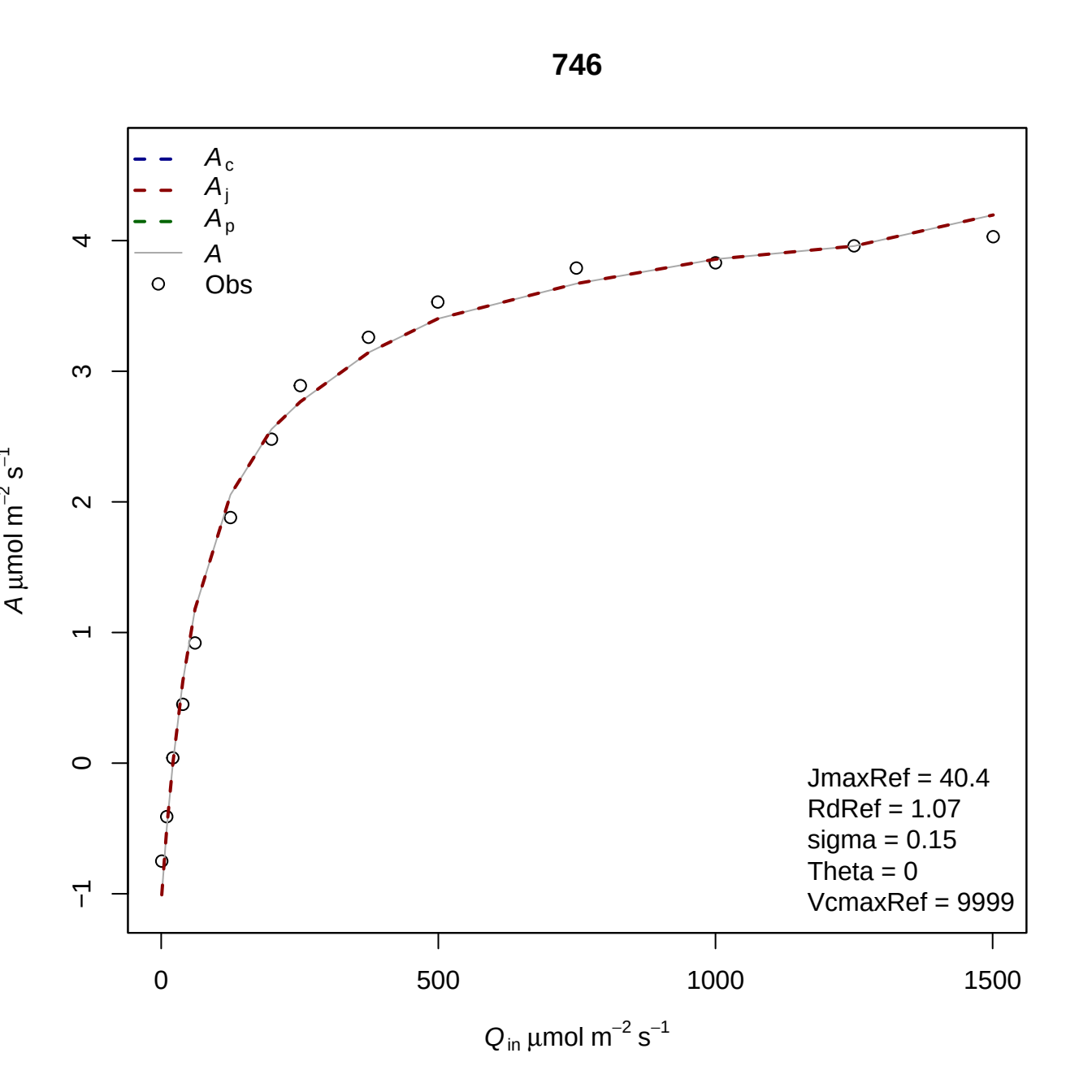

741
A_c
A_j
A_p
A
Obs
JmaxRef = 104.34
RdRef = 0.5
sigma = 0.13
Theta = 0
VcmaxRef = 69.85
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

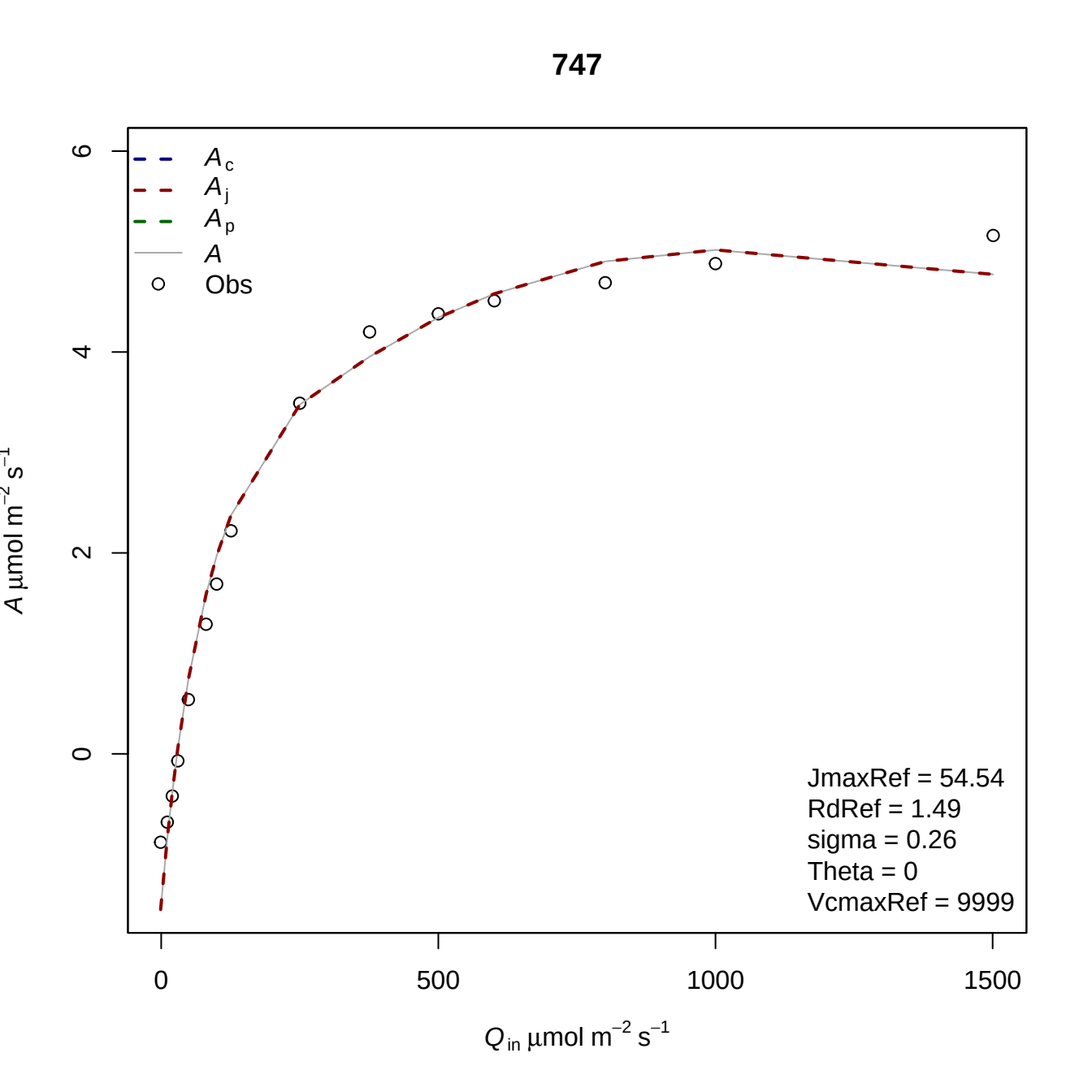

742
A_c
A_j
A_p
A
Obs
JmaxRef = 95.66
RdRef = 0.91
sigma = 0.11
Theta = 0
VcmaxRef = 44.61
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

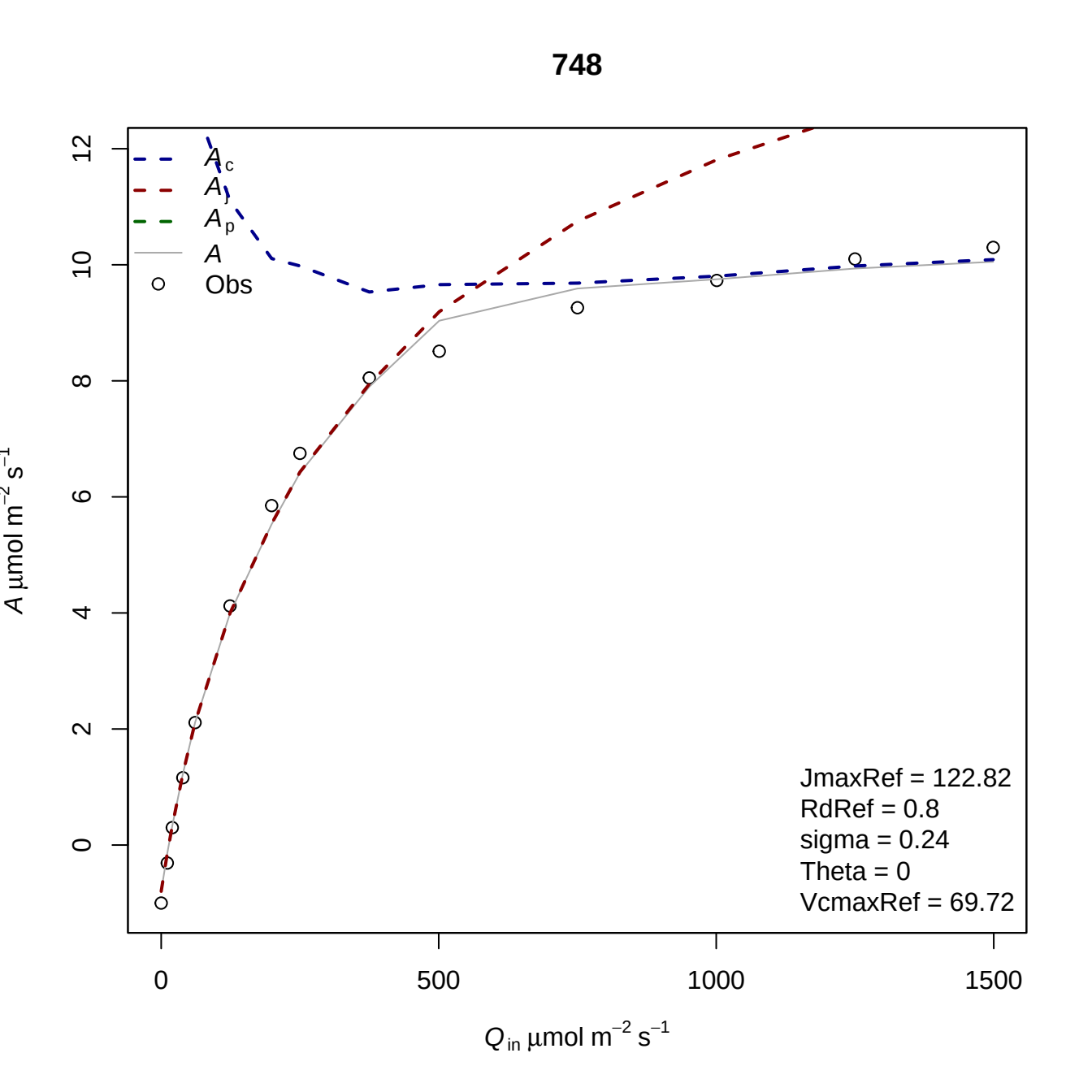

743
A_c
A_j
A_p
A
Obs
JmaxRef = 63.61
RdRef = 0.98
sigma = 0.12
Theta = 0
VcmaxRef = 41.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

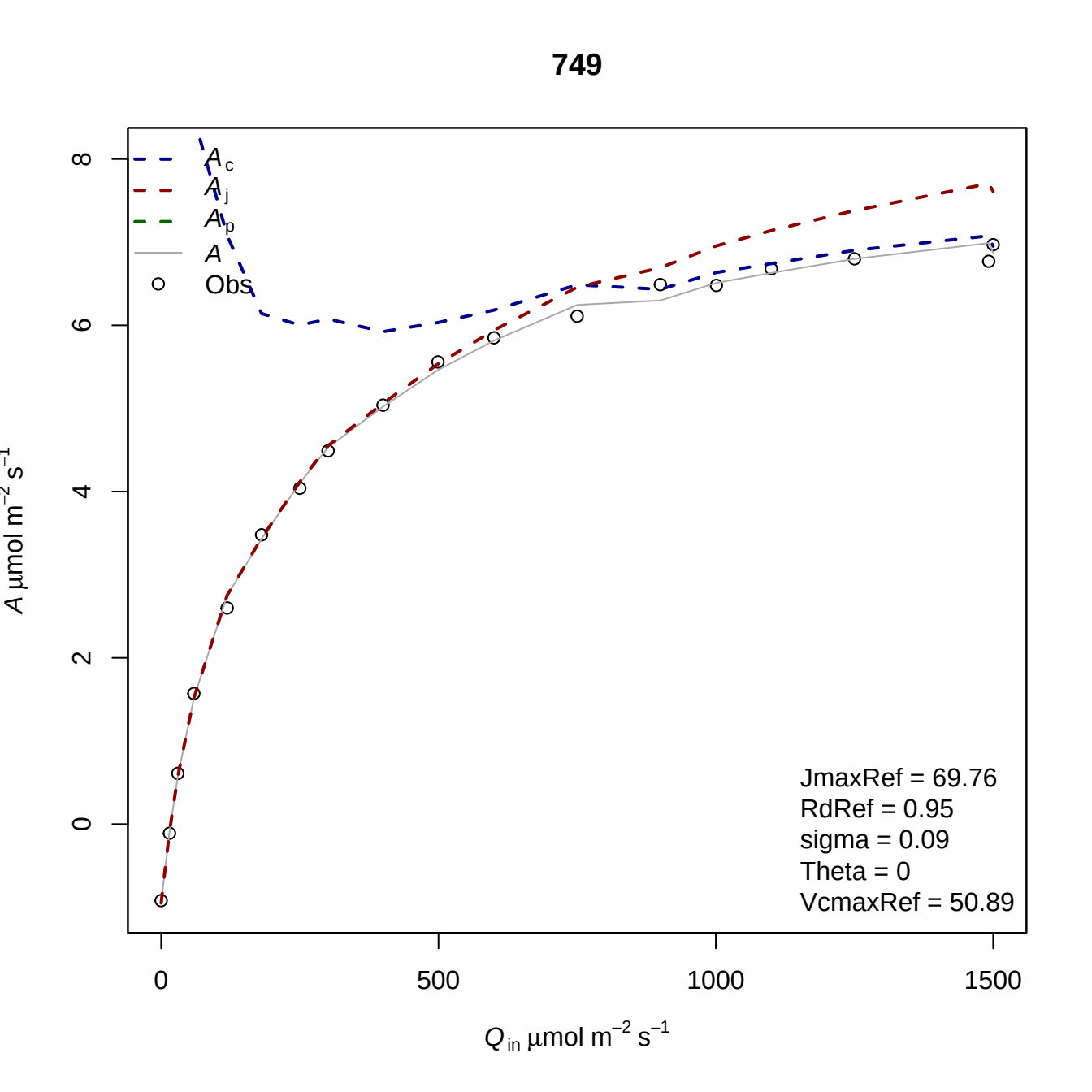

744
A_c
A_j
A_p
A
Obs
JmaxRef = 78.76
RdRef = 2.01
sigma = 0.17
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

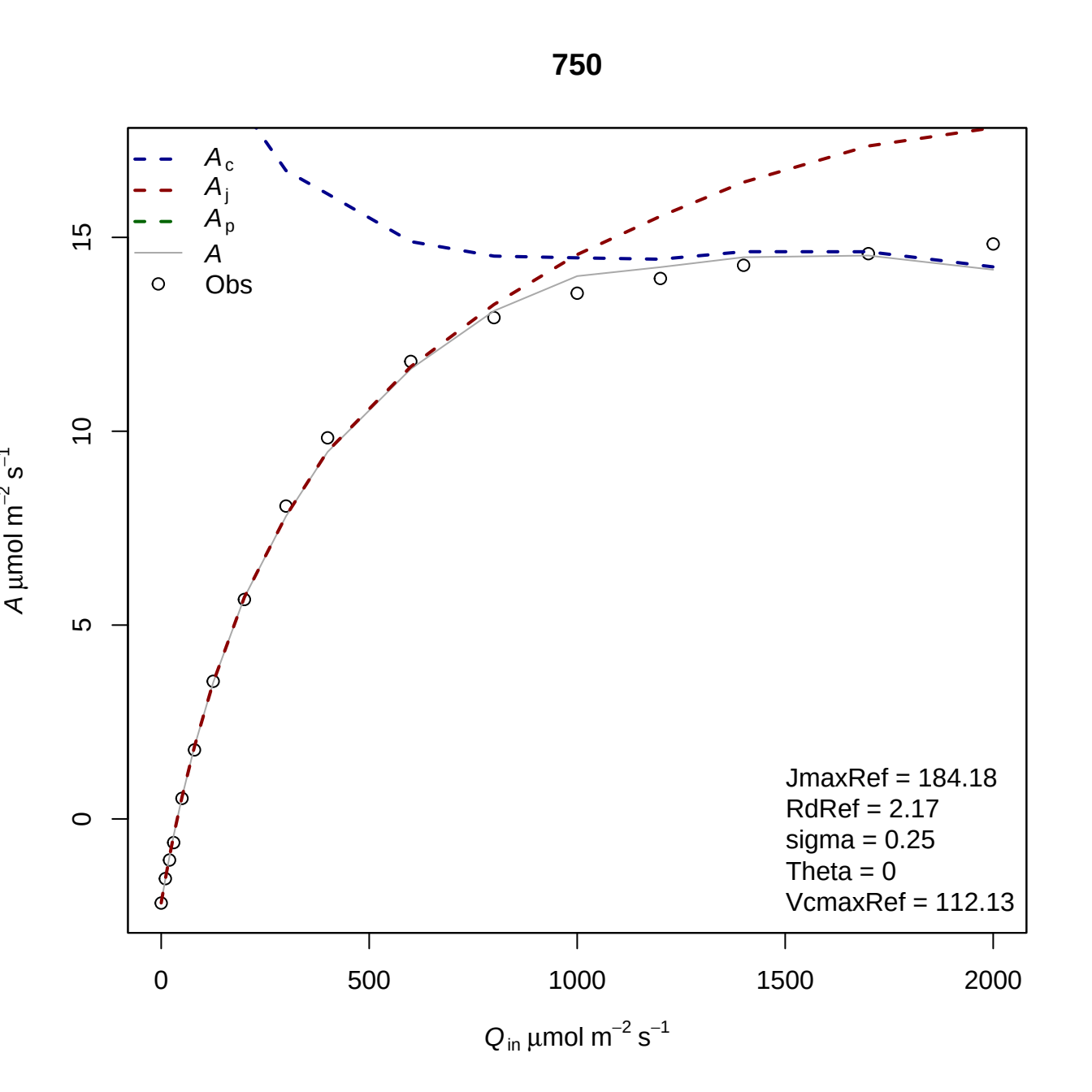

745
A_c
A_j
A_p
A
Obs
JmaxRef = 59.59
RdRef = 1.39
sigma = 0.2
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

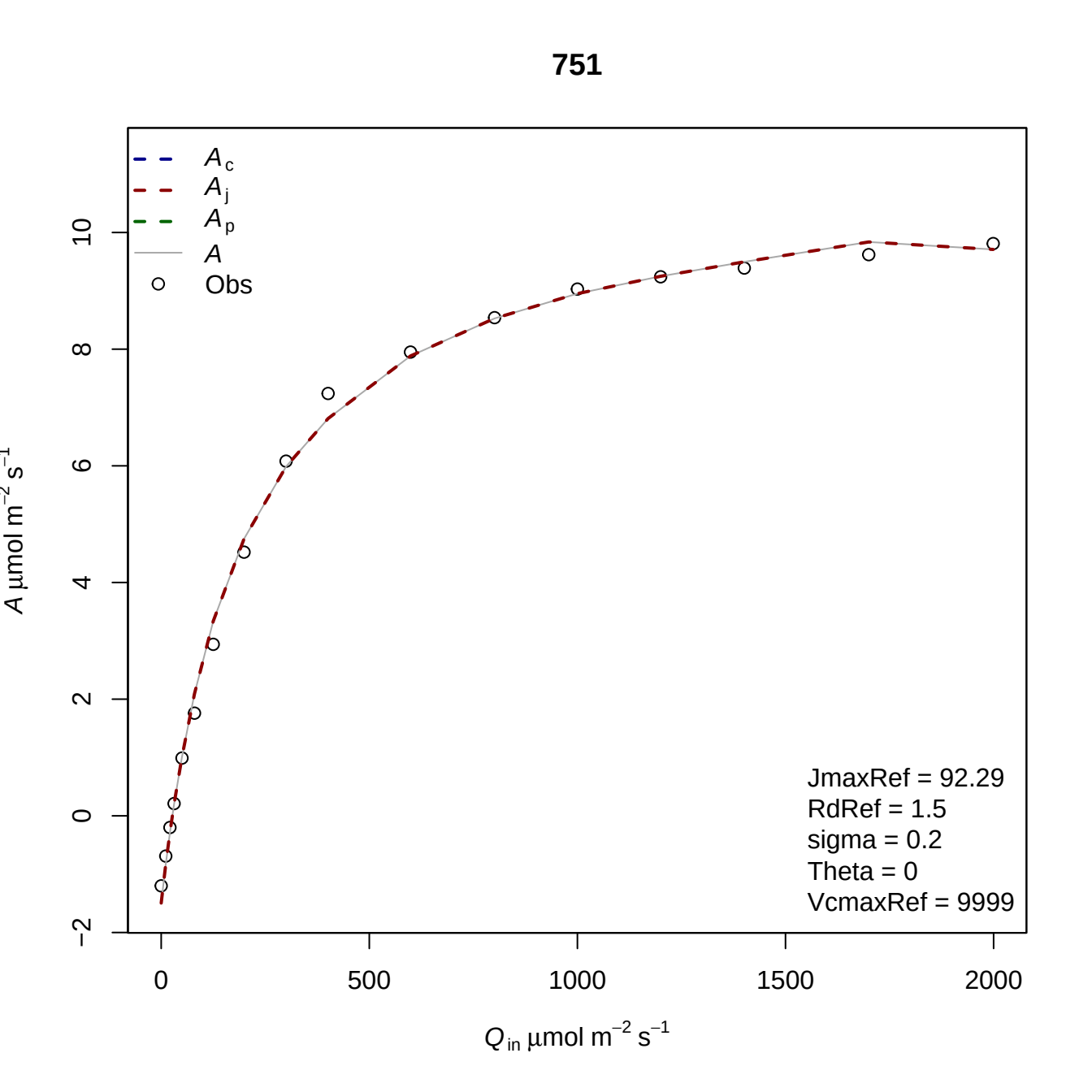

746
A_c
A_j
A_p
A
Obs
JmaxRef = 40.4
RdRef = 1.07
sigma = 0.15
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

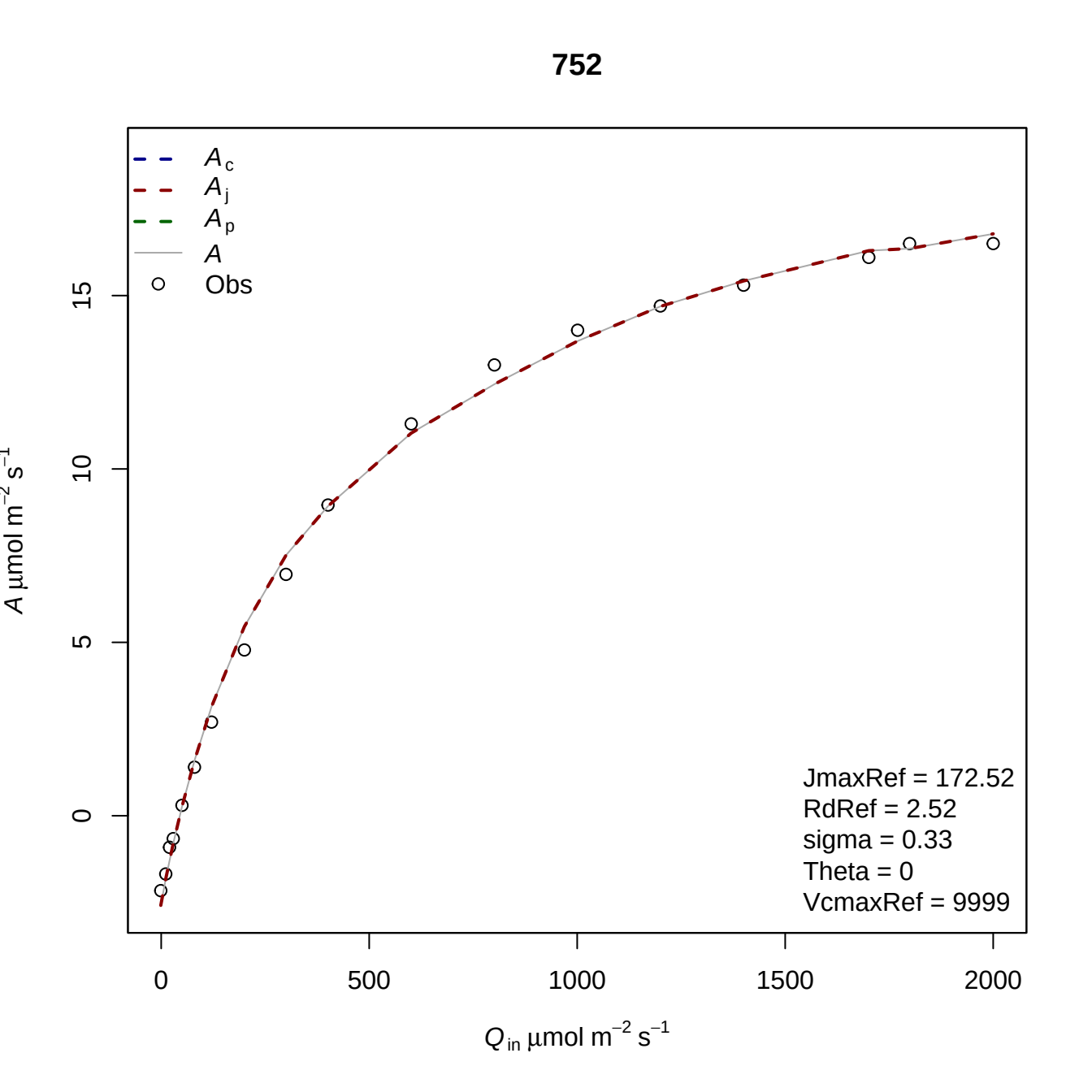

747
A_c
A_j
A_p
A
Obs
JmaxRef = 54.54
RdRef = 1.49
sigma = 0.26
Theta = 0
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

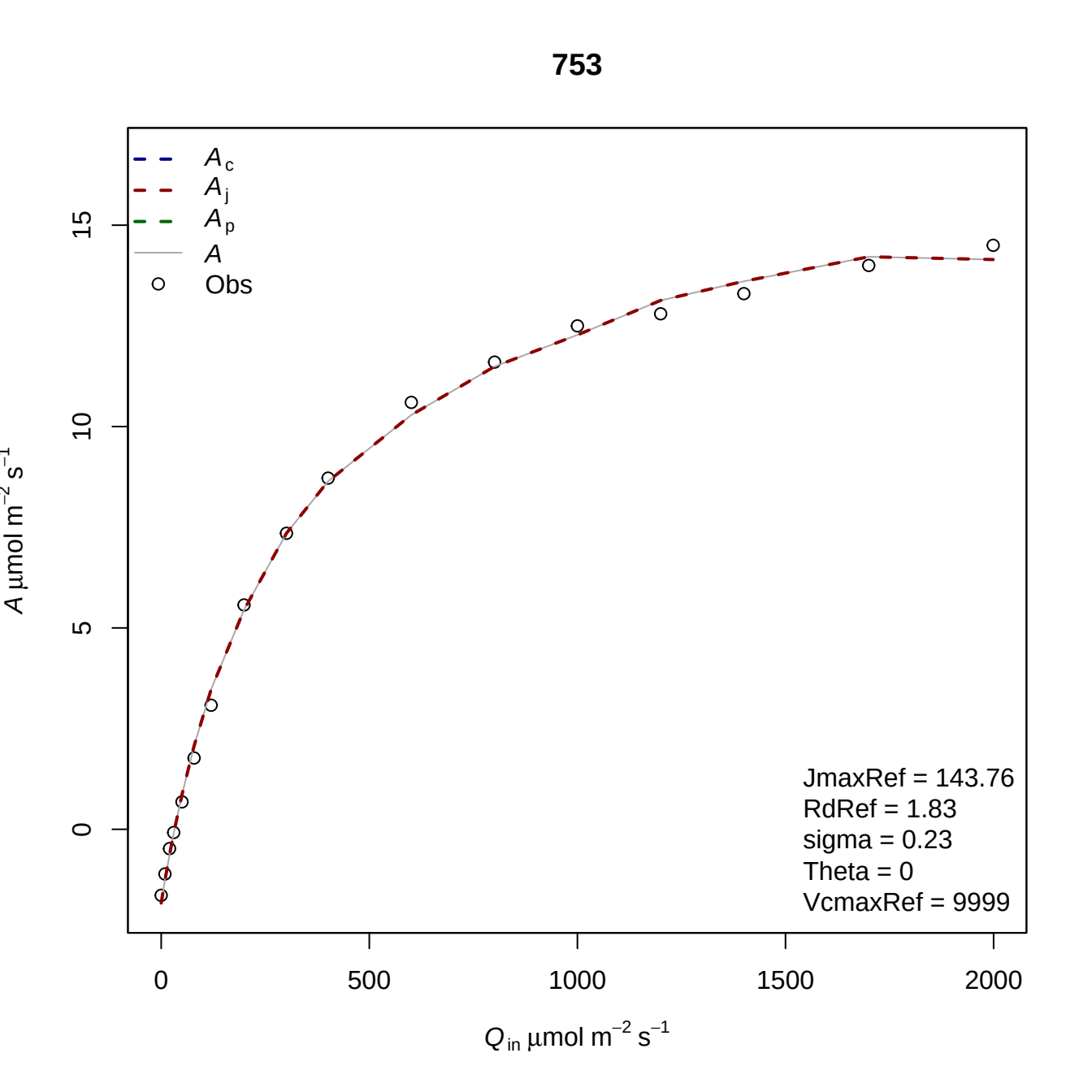

748
A_c
A_j
A_p
A
Obs
JmaxRef = 122.82
RdRef = 0.8
sigma = 0.24
Theta = 0
VcmaxRef = 69.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

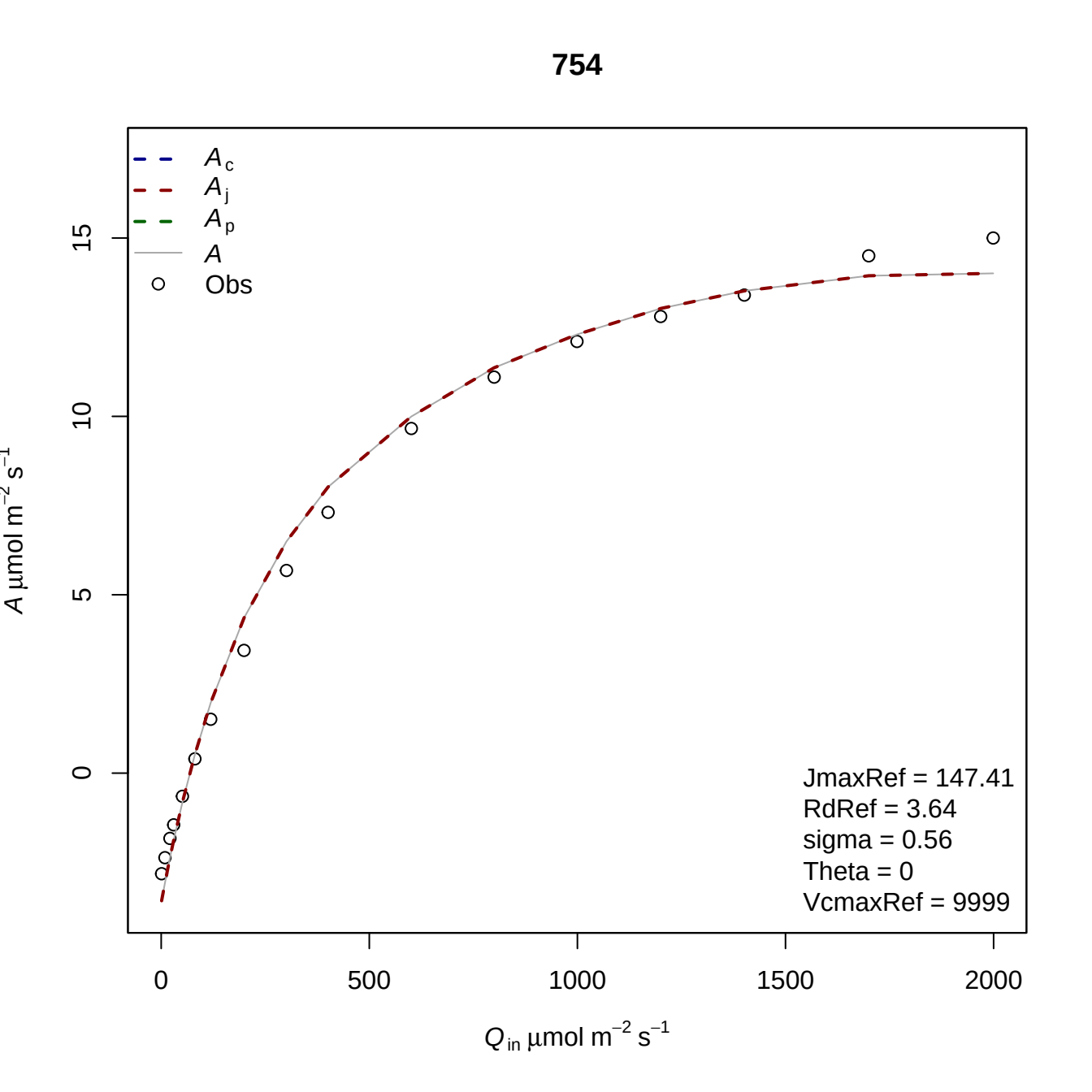

749
A_c
A_j
A_p
A
Obs
JmaxRef = 69.76
RdRef = 0.95
sigma = 0.09
Theta = 0
VcmaxRef = 50.89
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

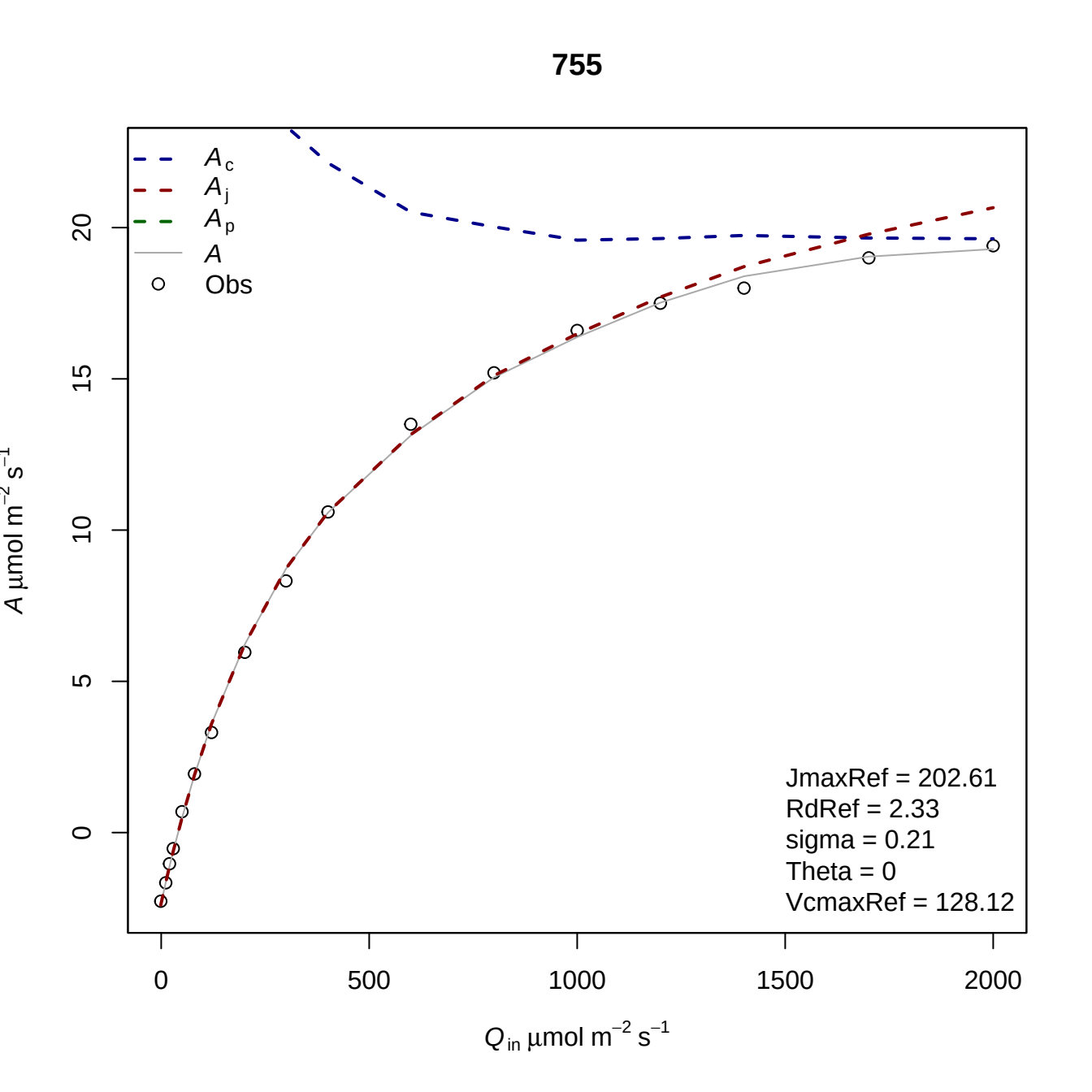

750
A_c
A_j
A_p
A
Obs
JmaxRef = 184.18
RdRef = 2.17
sigma = 0.25
Theta = 0
VcmaxRef = 112.13
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

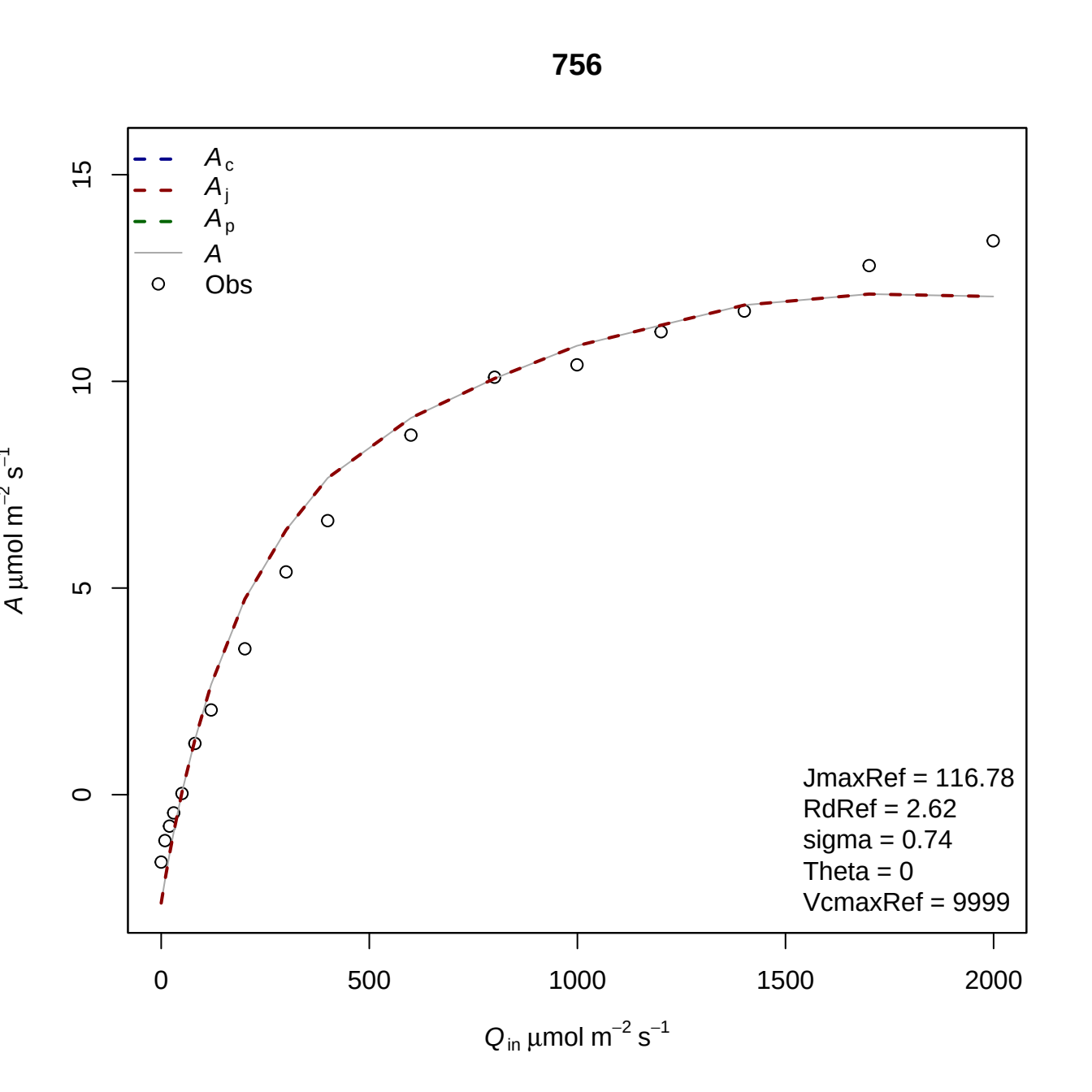

751
A_c
A_j
A_p
A
Obs
JmaxRef = 92.29
RdRef = 1.5
sigma = 0.2
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

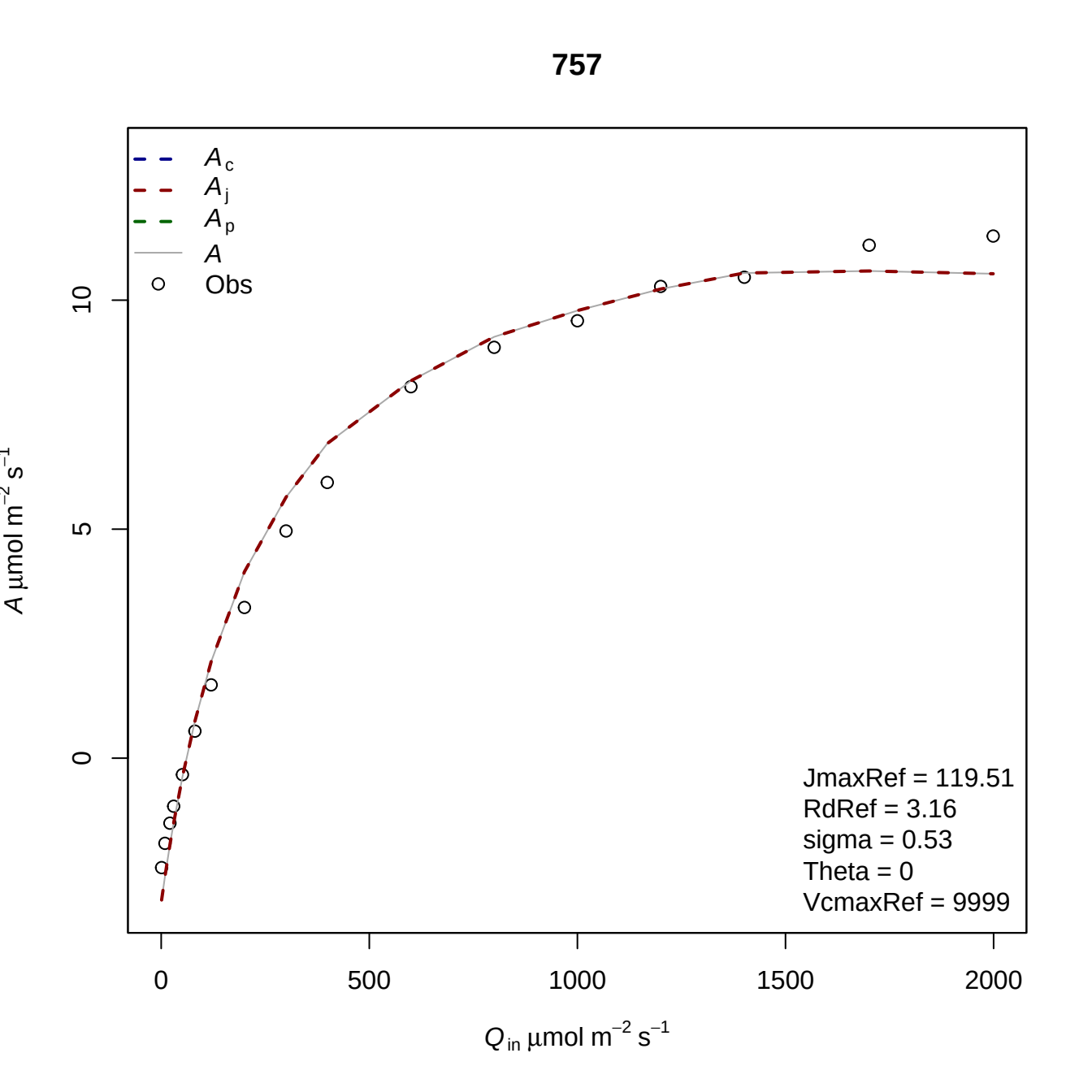

752
A_c
A_j
A_p
A
Obs
JmaxRef = 172.52
RdRef = 2.52
sigma = 0.33
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

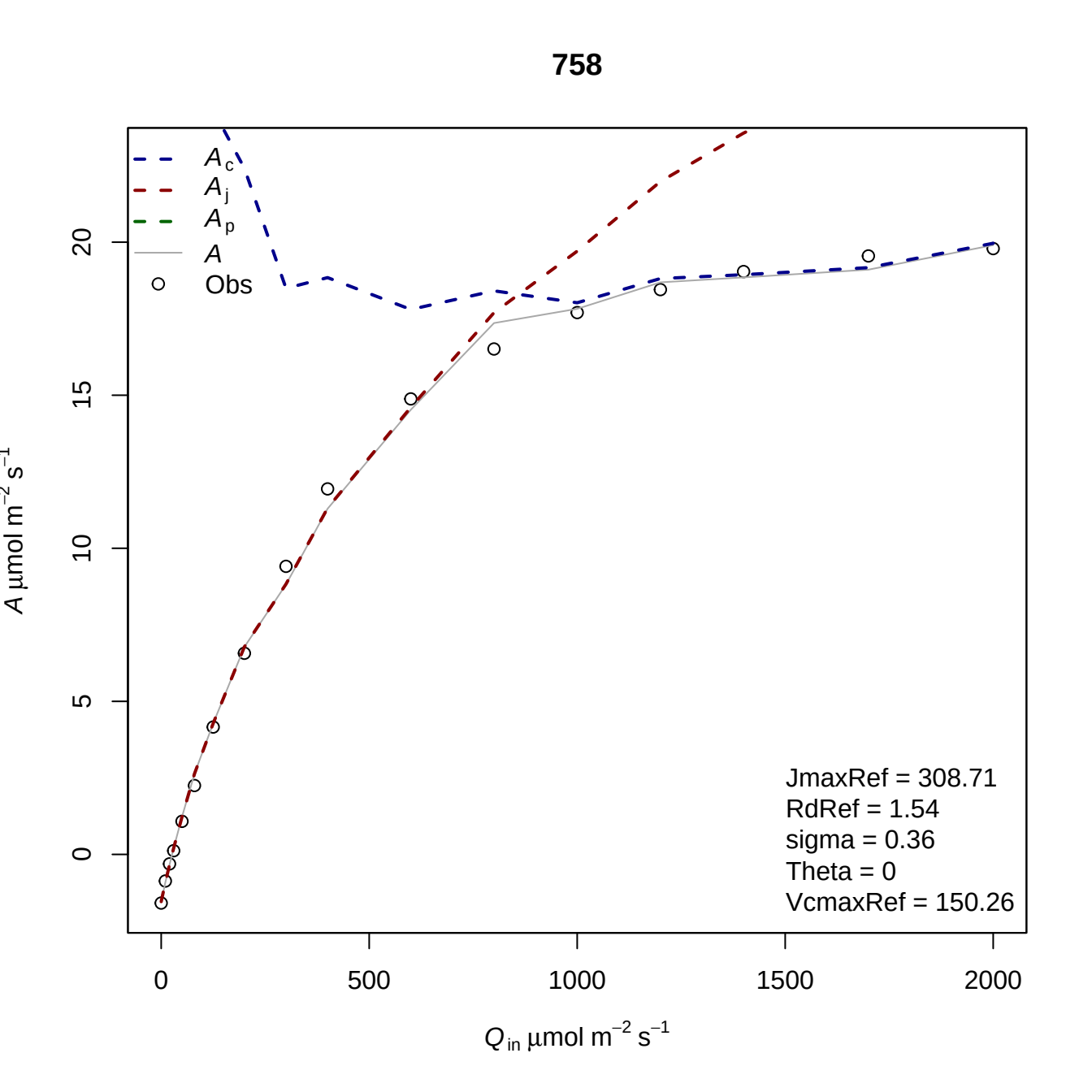

753
A_c
A_j
A_p
A
Obs
JmaxRef = 143.76
RdRef = 1.83
sigma = 0.23
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

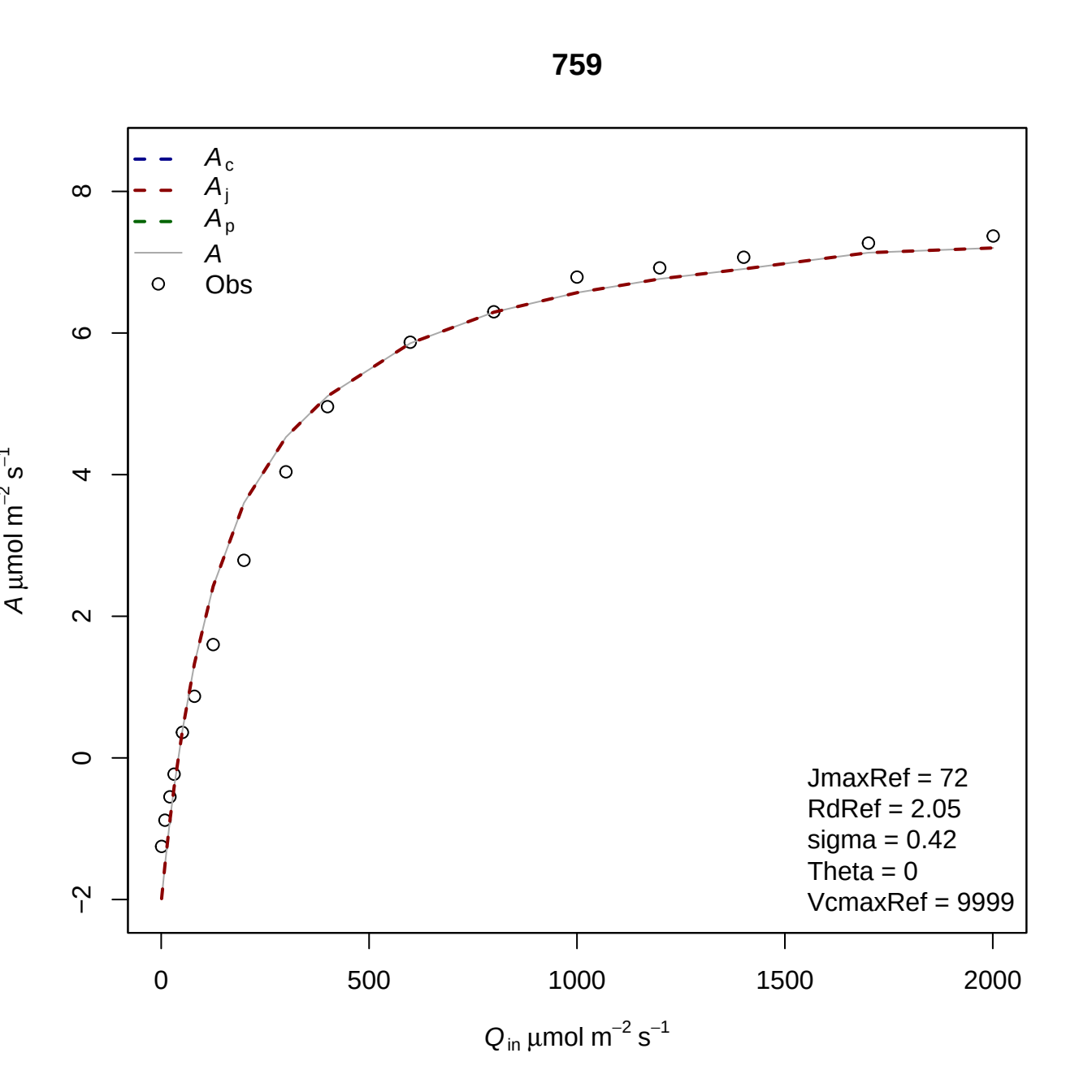

754
A_c
A_j
A_p
A
Obs
JmaxRef = 147.41
RdRef = 3.64
sigma = 0.56
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

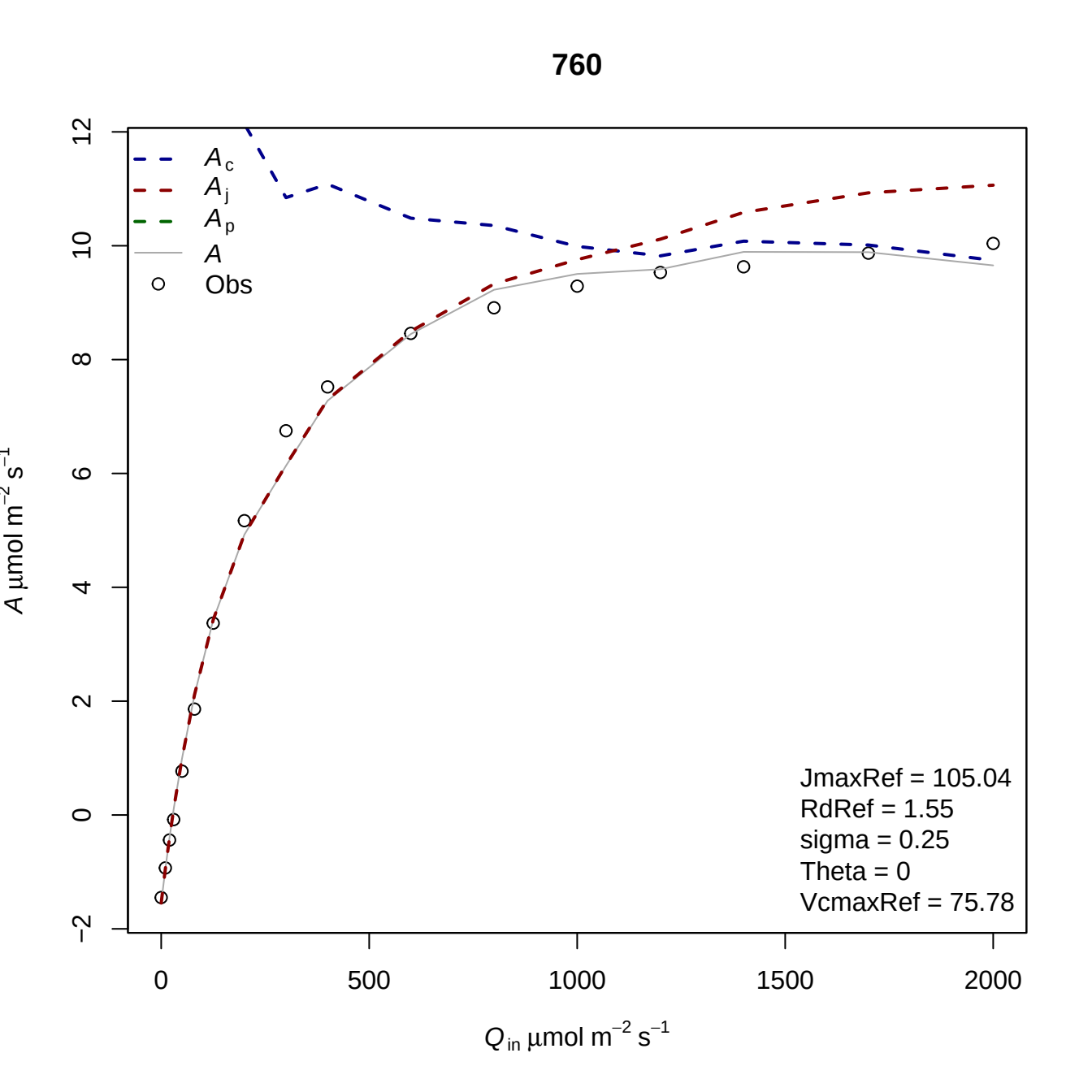

755
A_c
A_j
A_p
A
Obs
JmaxRef = 202.61
RdRef = 2.33
sigma = 0.21
Theta = 0
VcmaxRef = 128.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

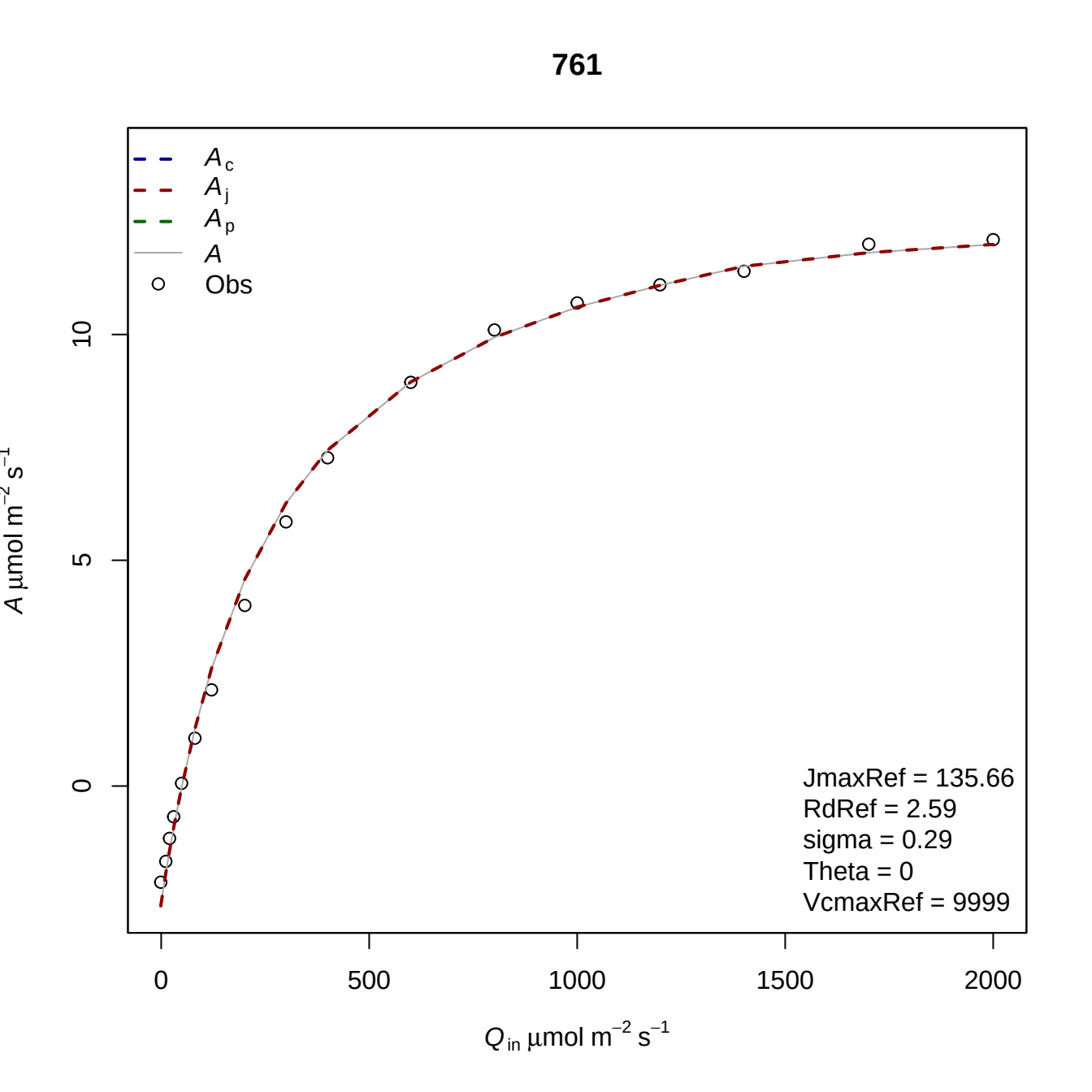

756
A_c
A_j
A_p
A
Obs
JmaxRef = 116.78
RdRef = 2.62
sigma = 0.74
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

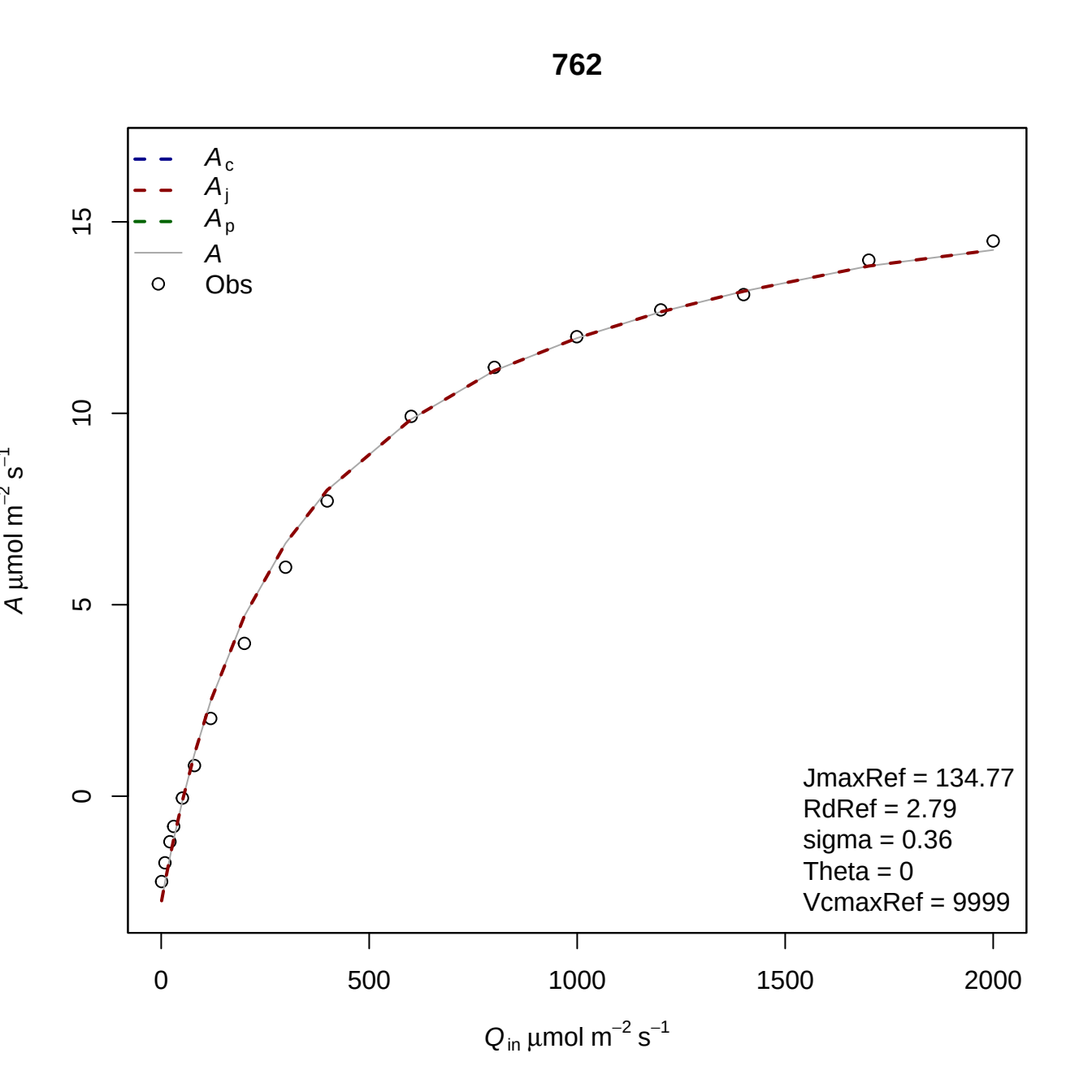

757
A_c
A_j
A_p
A
Obs
JmaxRef = 119.51
RdRef = 3.16
sigma = 0.53
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

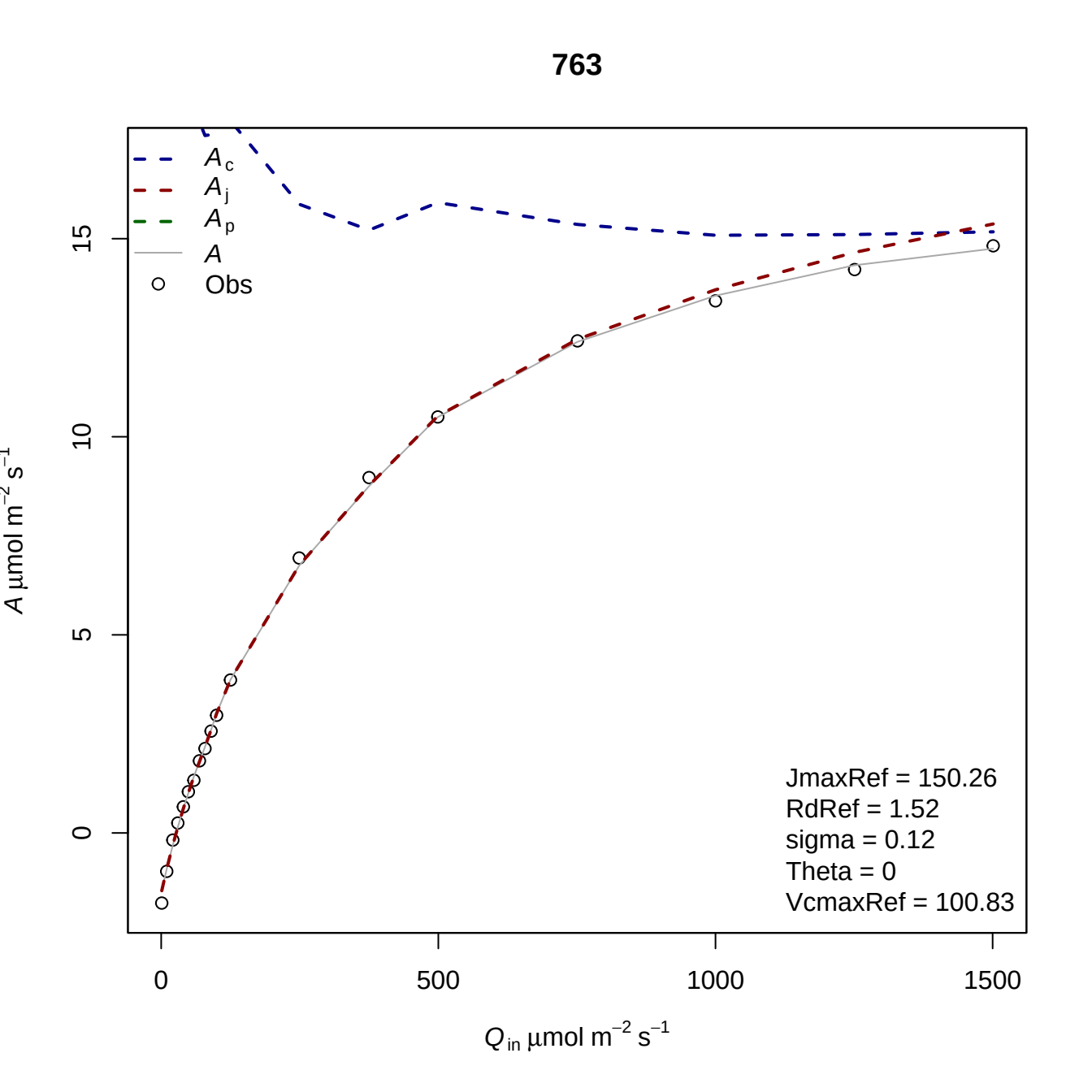

758
A_c
A_j
A_p
A
Obs
JmaxRef = 308.71
RdRef = 1.54
sigma = 0.36
Theta = 0
VcmaxRef = 150.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

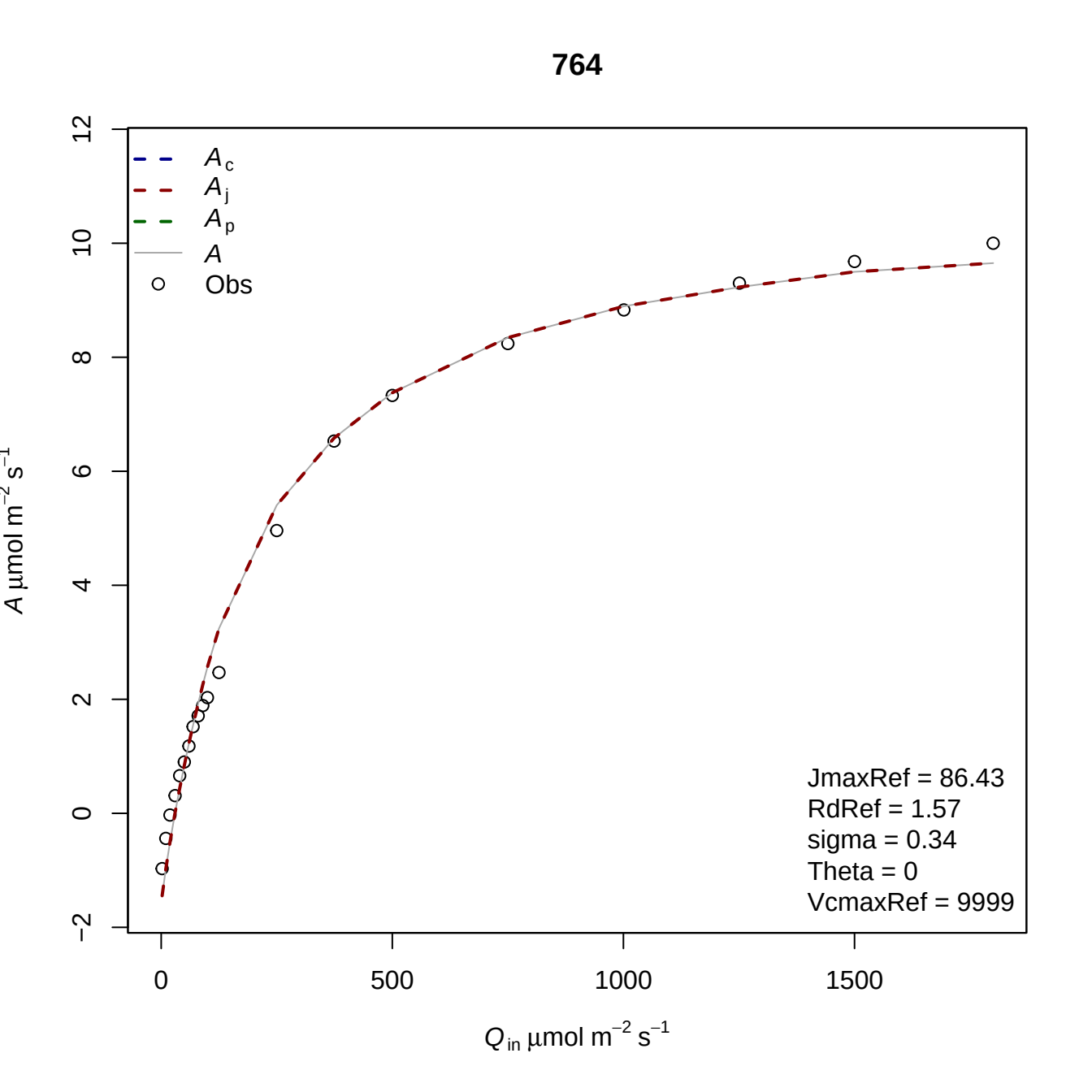

759
A_c
A_j
A_p
A
Obs
JmaxRef = 72
RdRef = 2.05
sigma = 0.42
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

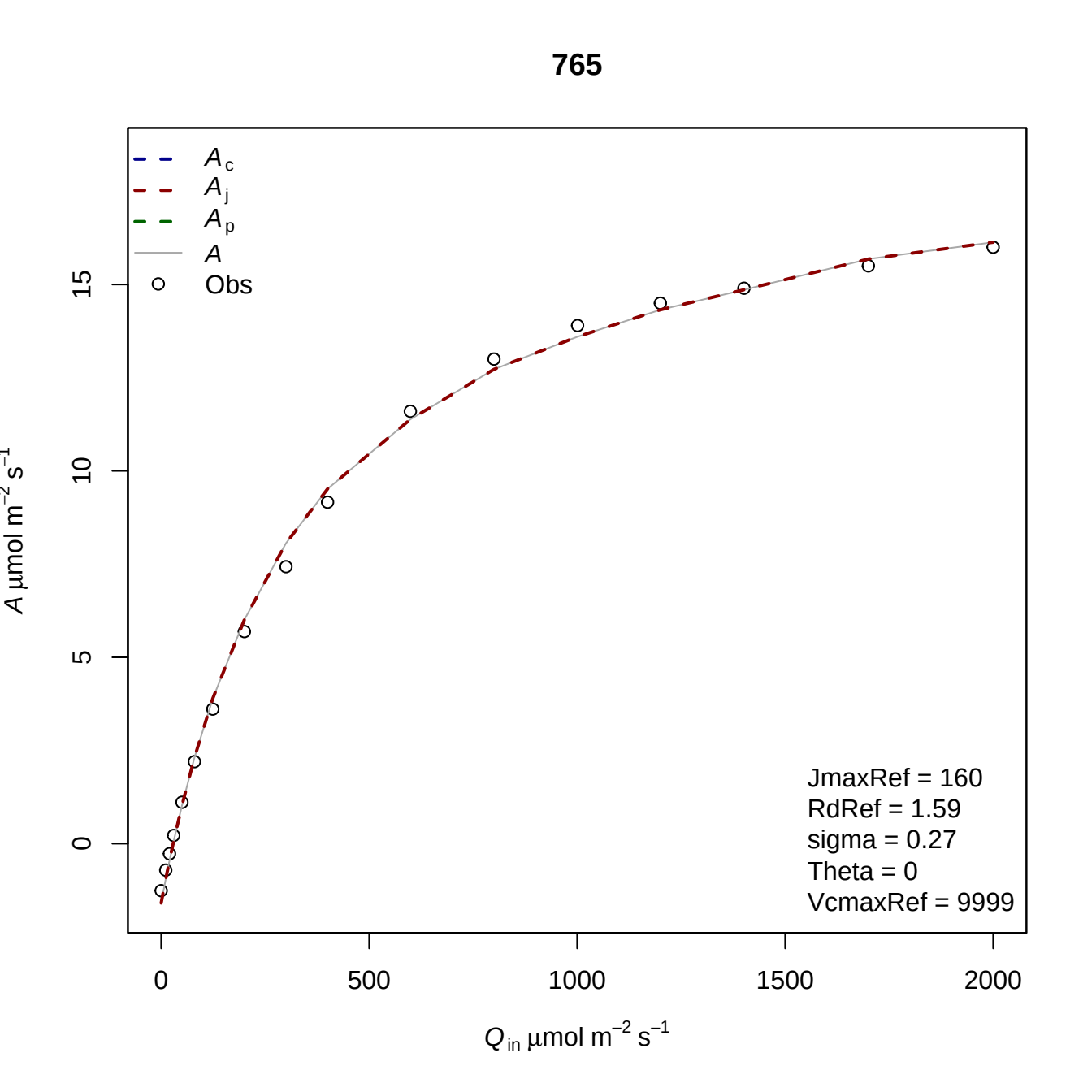

760
A_c
A_j
A_p
A
Obs
JmaxRef = 105.04
RdRef = 1.55
sigma = 0.25
Theta = 0
VcmaxRef = 75.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

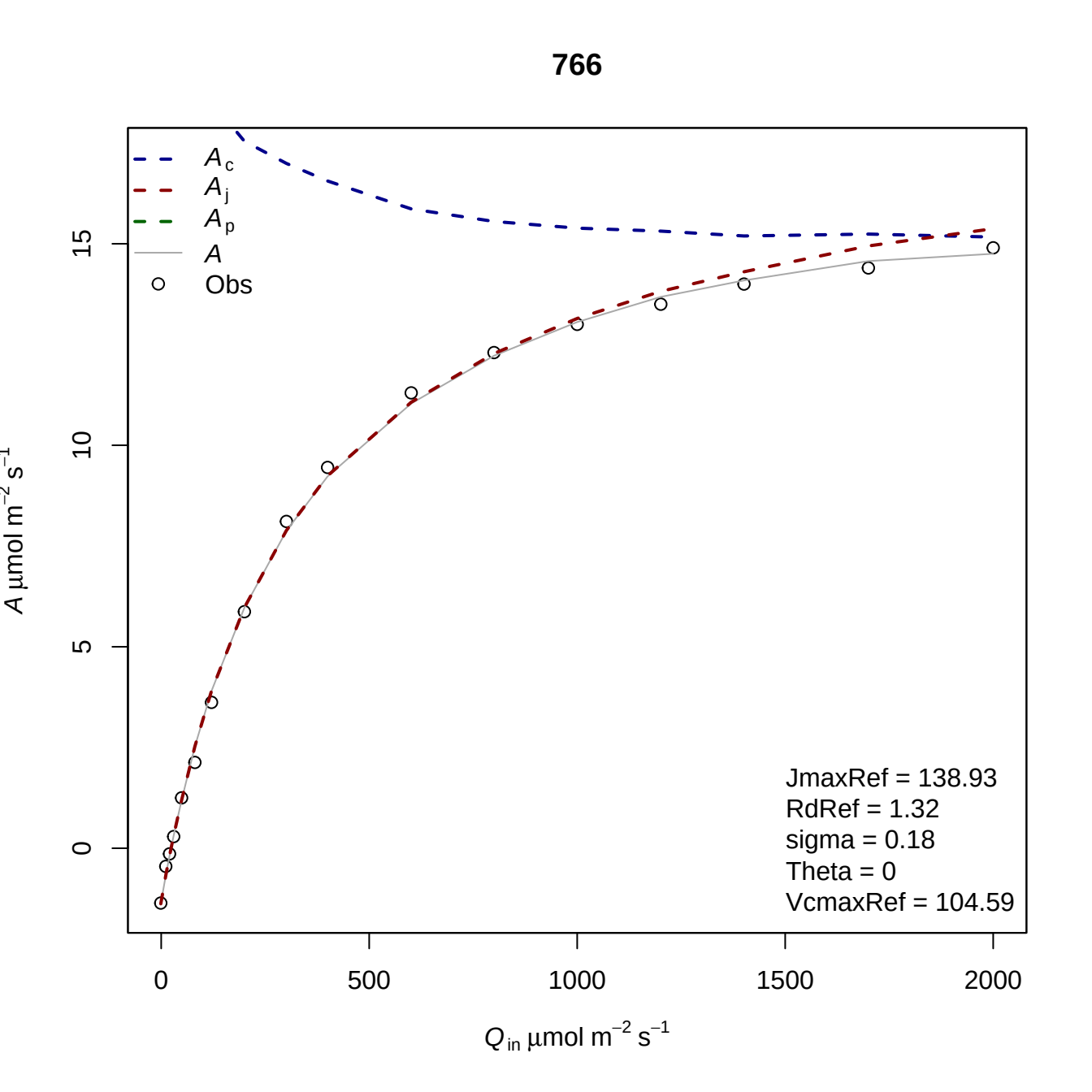

761
A_c
A_j
A_p
A
Obs
JmaxRef = 135.66
RdRef = 2.59
sigma = 0.29
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

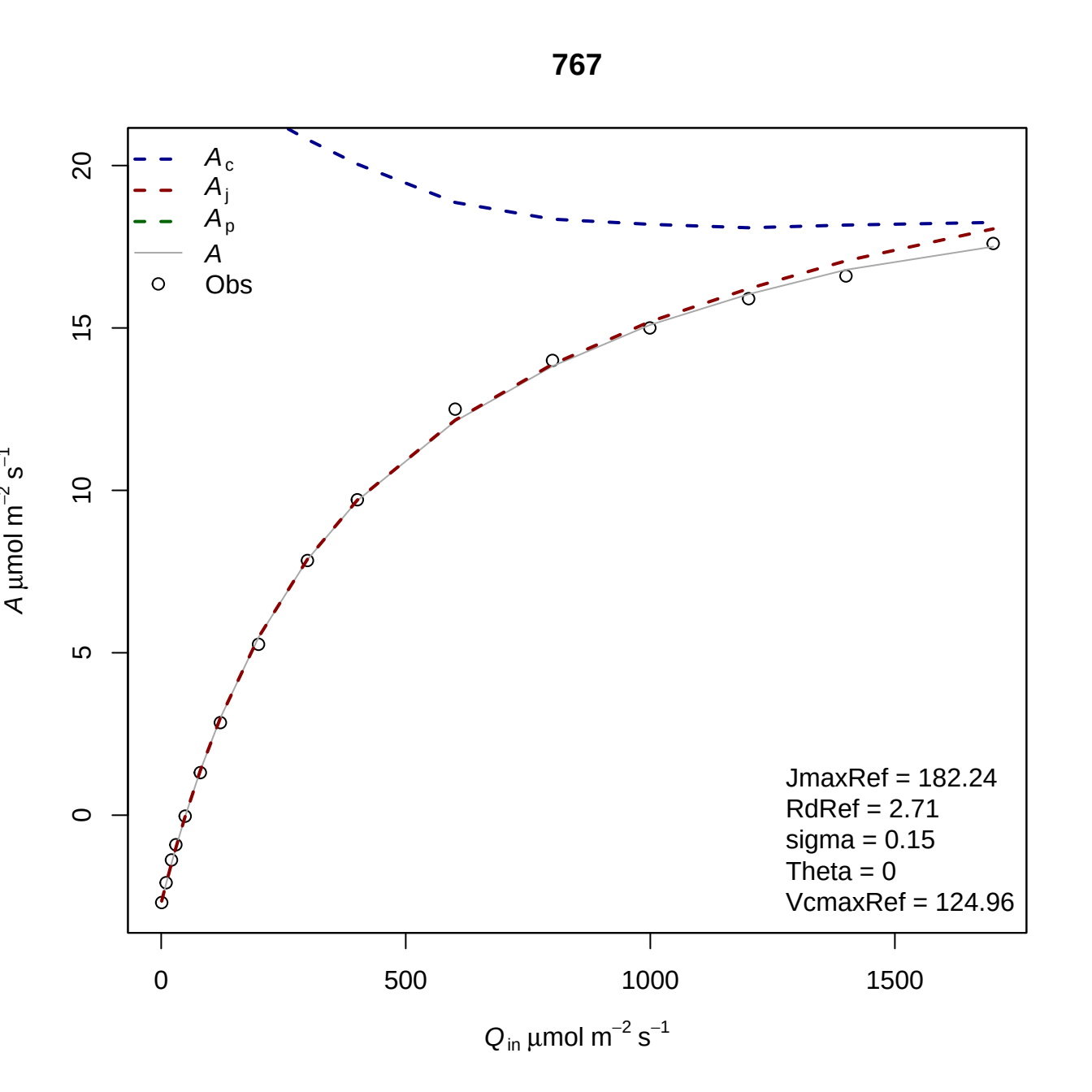

762
A_c
A_j
A_p
A
Obs
JmaxRef = 134.77
RdRef = 2.79
sigma = 0.36
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

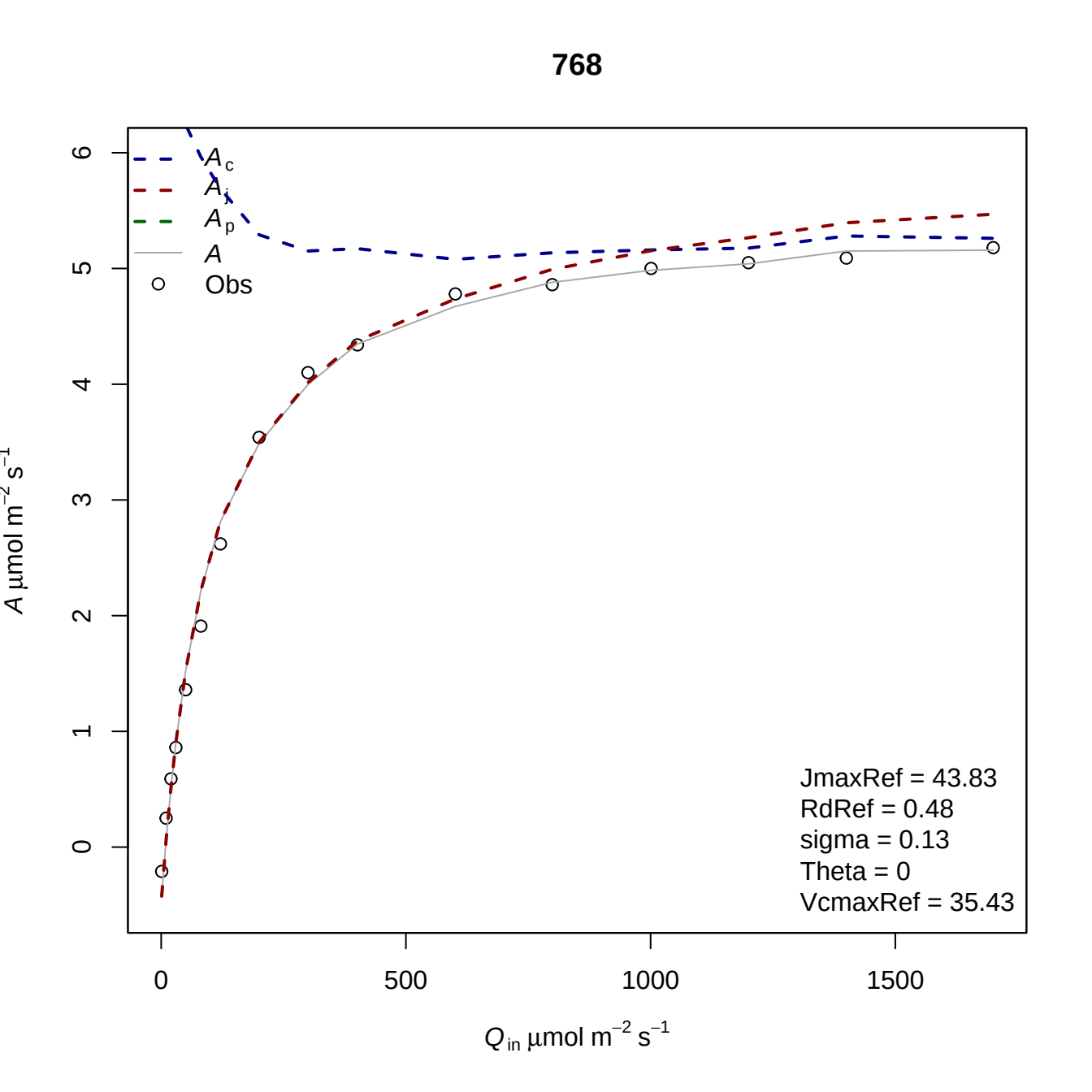

763
A_c
A_j
A_p
A
Obs
JmaxRef = 150.26
RdRef = 1.52
sigma = 0.12
Theta = 0
VcmaxRef = 100.83
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

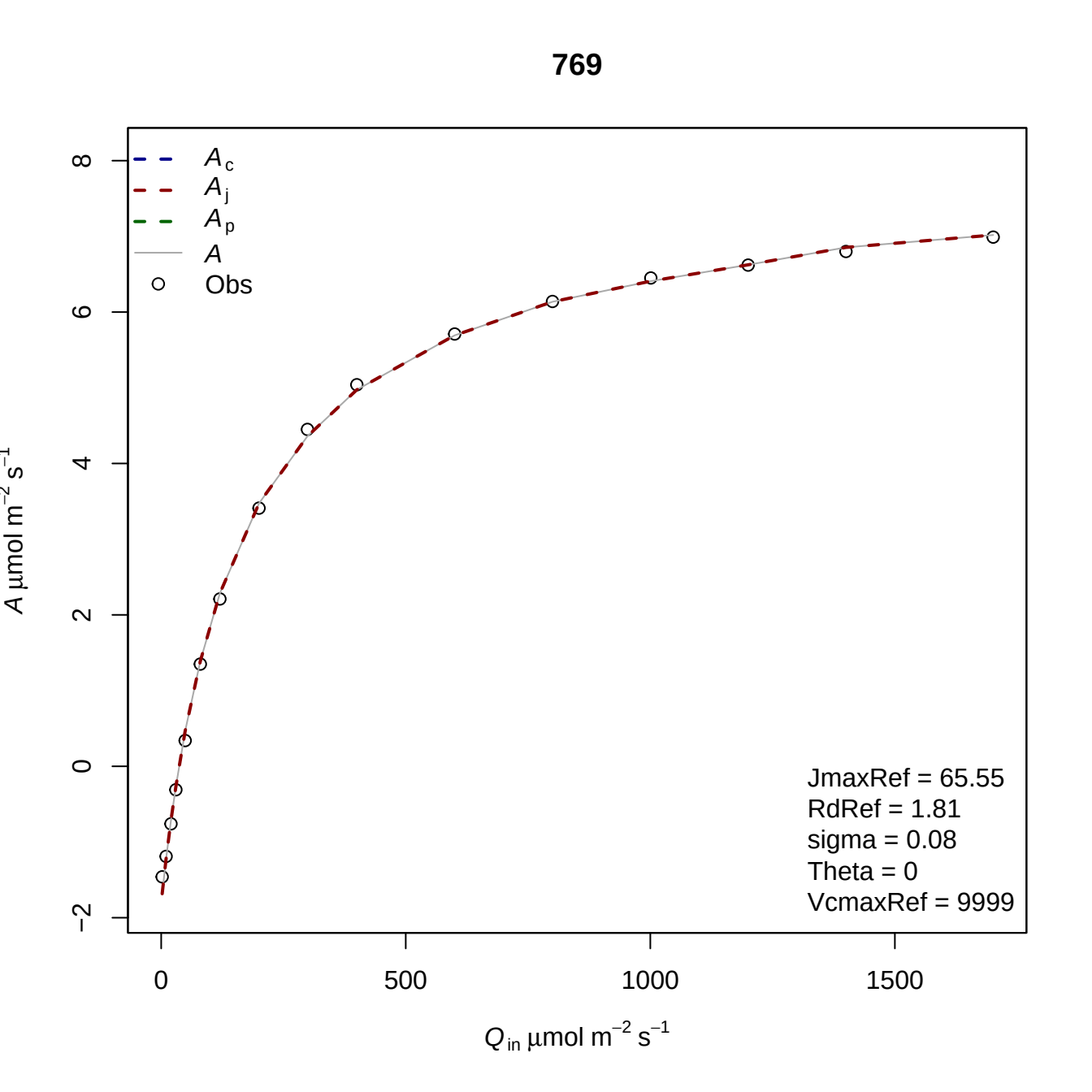

764
A_c
A_j
A_p
A
Obs
JmaxRef = 86.43
RdRef = 1.57
sigma = 0.34
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

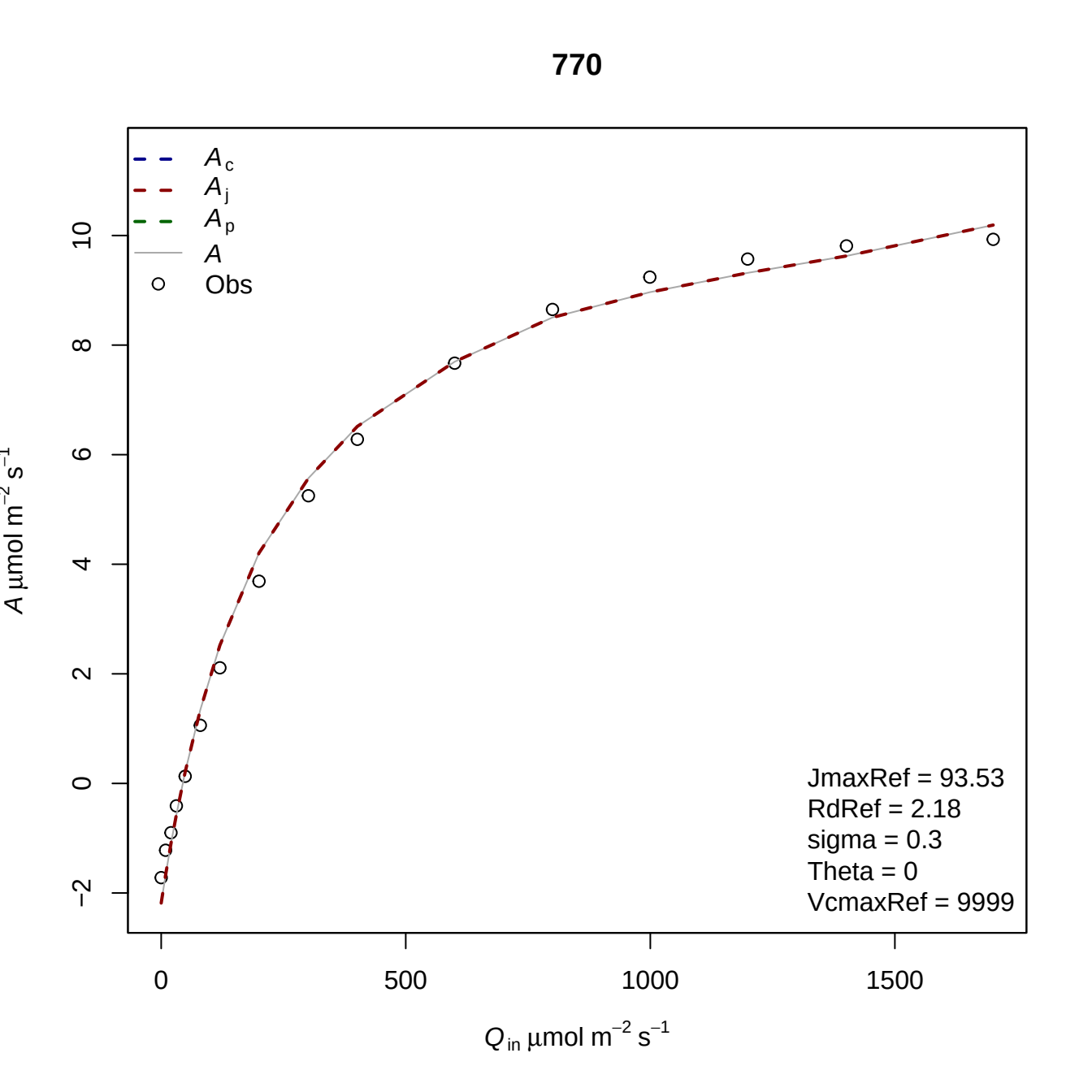

765
A_c
A_j
A_p
A
Obs
JmaxRef = 160
RdRef = 1.59
sigma = 0.27
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

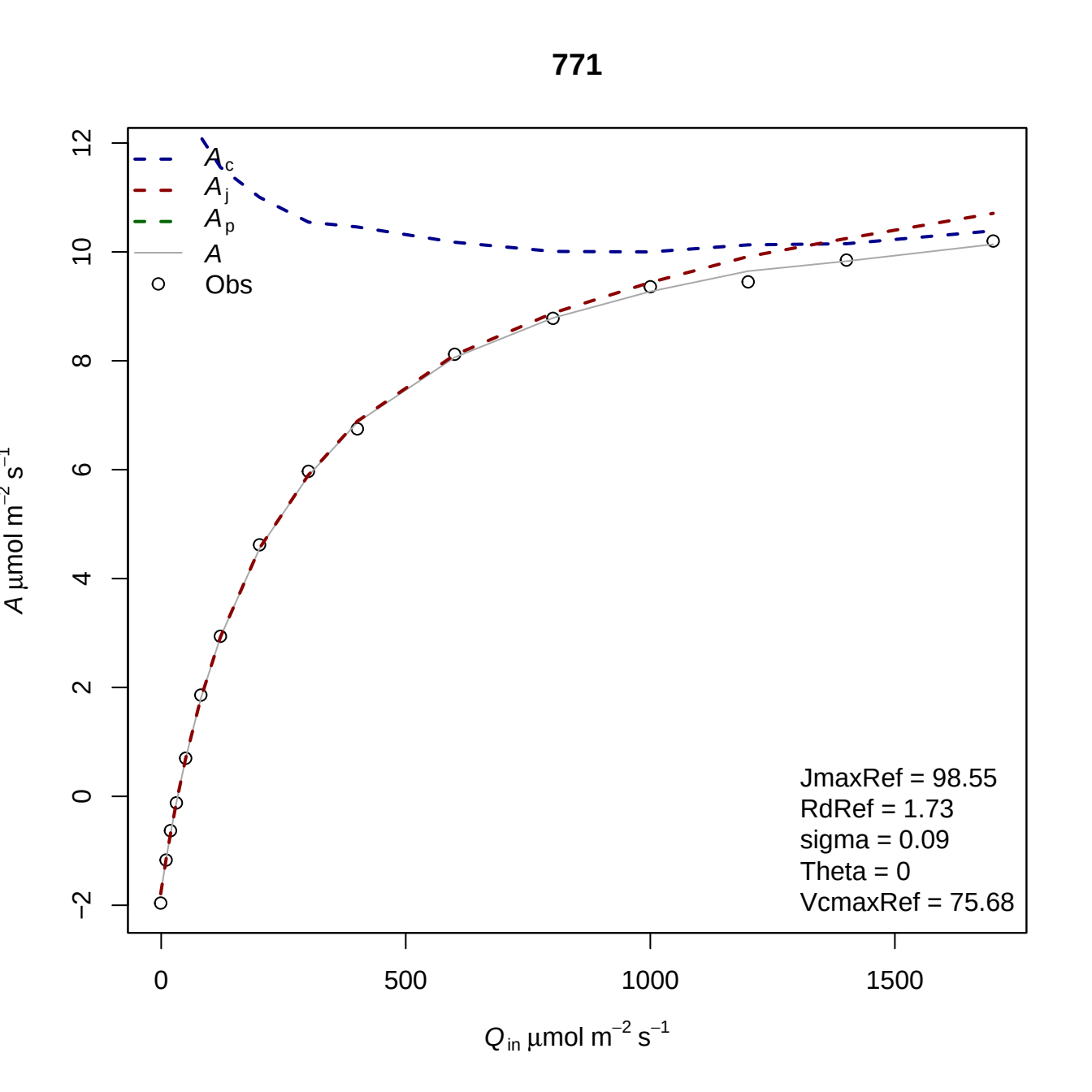

766
A_c
A_j
A_p
A
Obs
JmaxRef = 138.93
RdRef = 1.32
sigma = 0.18
Theta = 0
VcmaxRef = 104.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

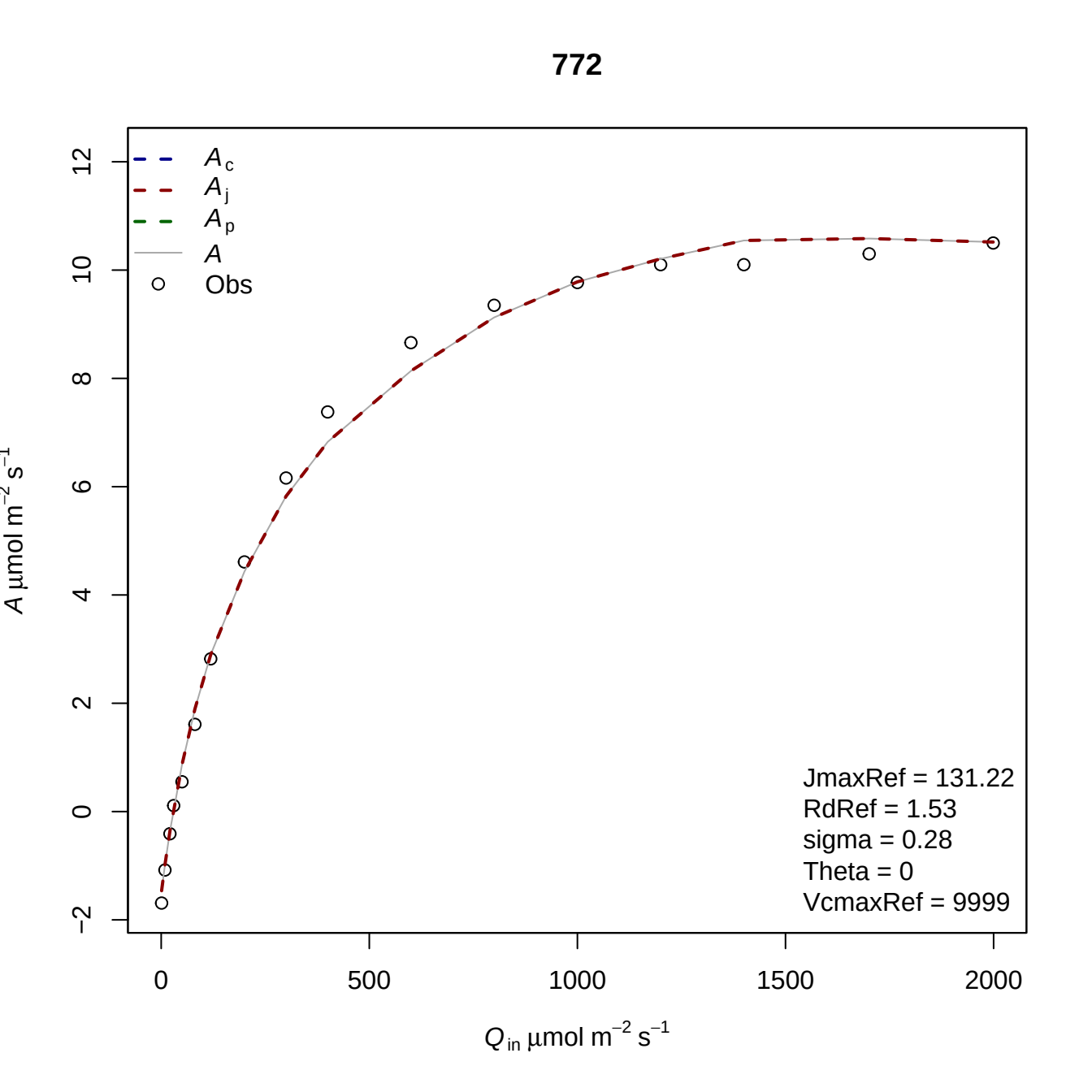

767
A_c
A_j
A_p
A
Obs
JmaxRef = 182.24
RdRef = 2.71
sigma = 0.15
Theta = 0
VcmaxRef = 124.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

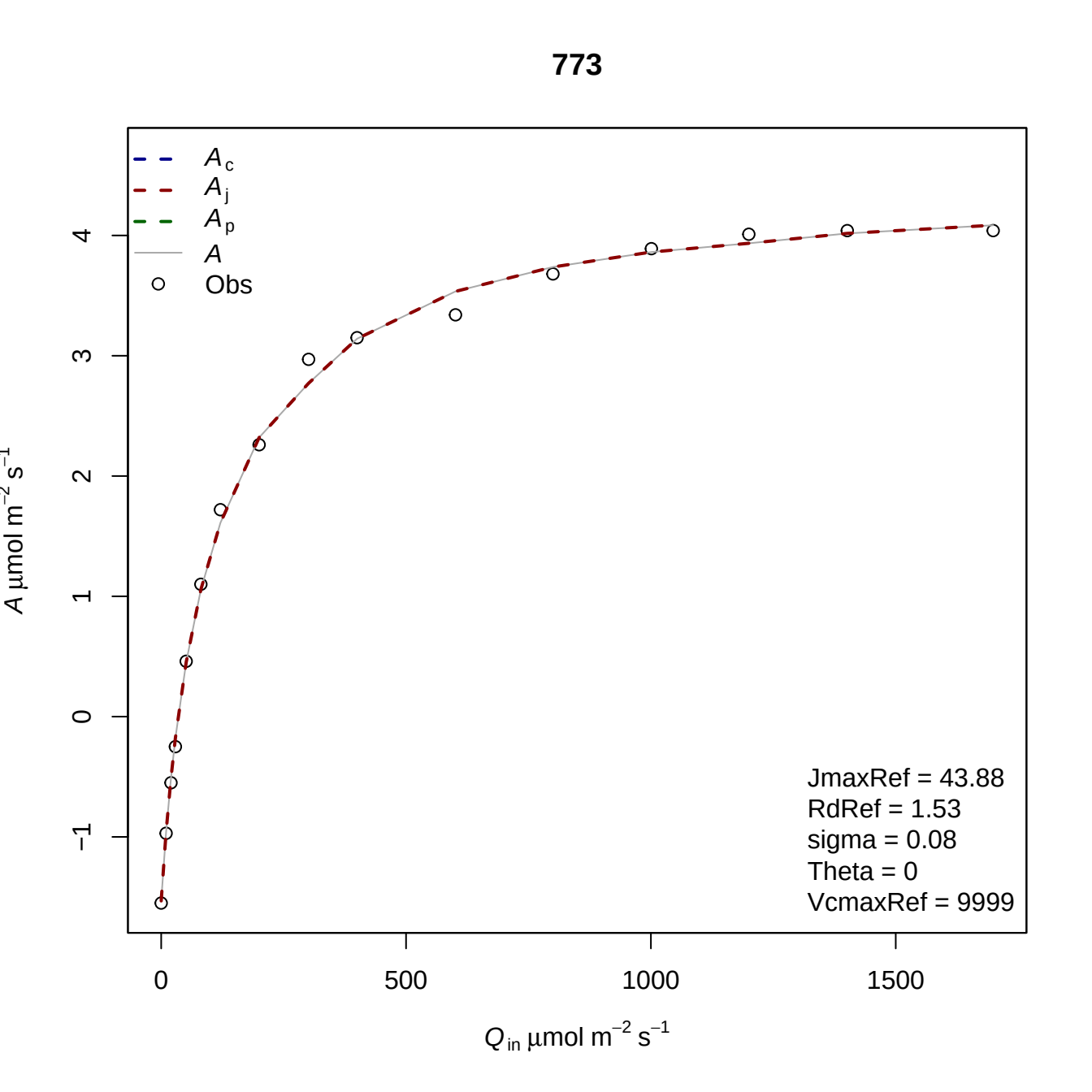

768
A_c
A_j
A_p
A
Obs
JmaxRef = 43.83
RdRef = 0.48
sigma = 0.13
Theta = 0
VcmaxRef = 35.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

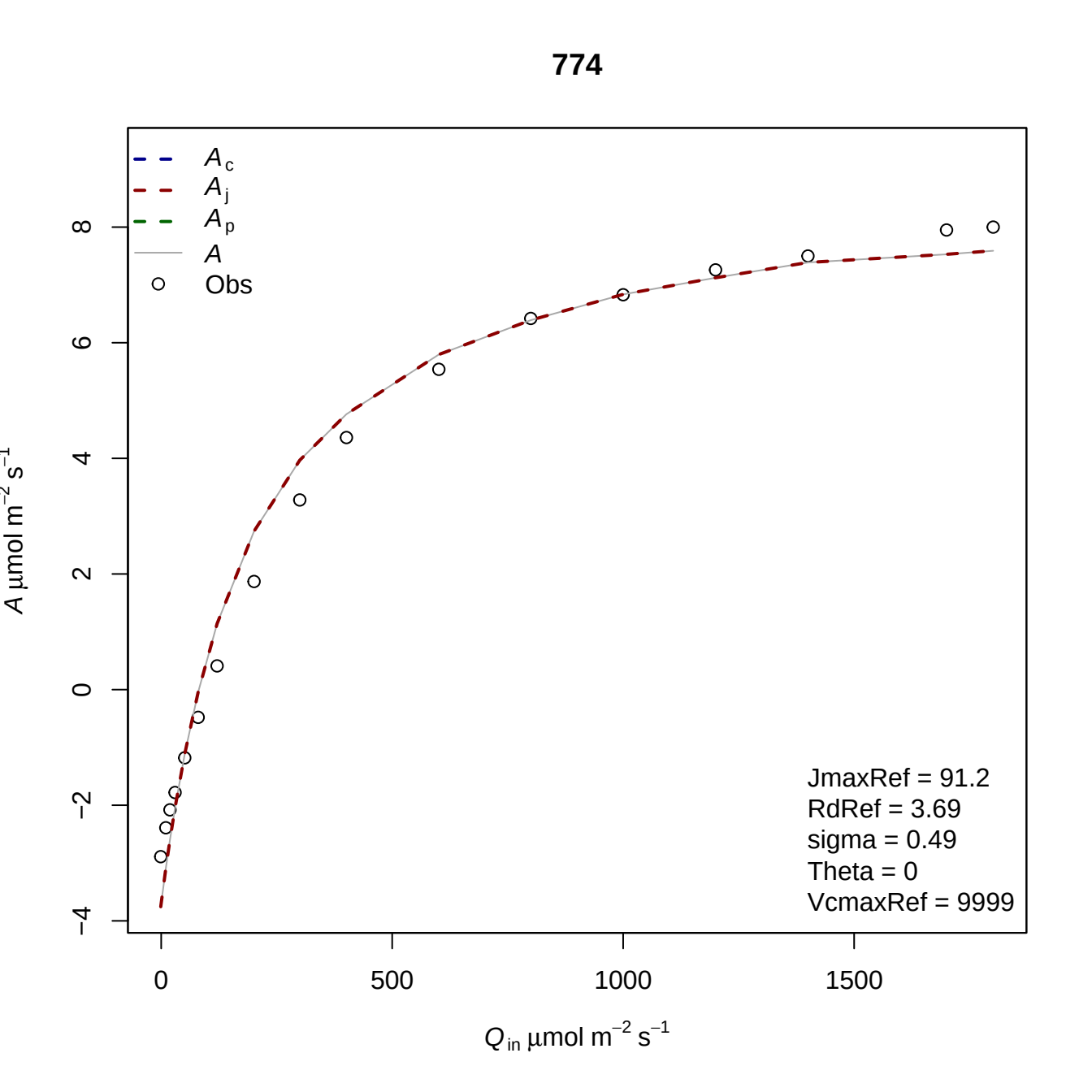

769
A_c
A_j
A_p
A
Obs
JmaxRef = 65.55
RdRef = 1.81
sigma = 0.08
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

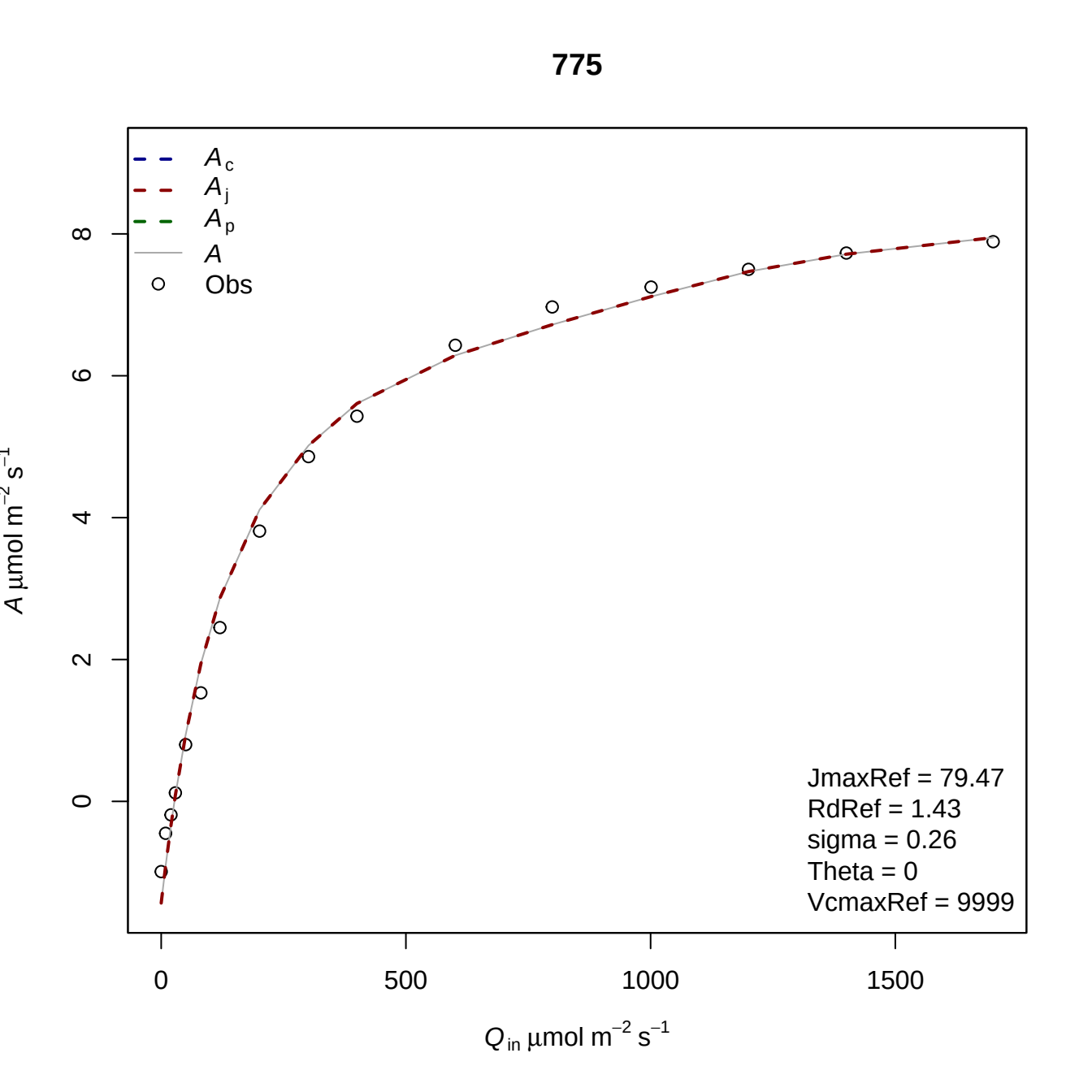

770
A_c
A_j
A_p
A
Obs
JmaxRef = 93.53
RdRef = 2.18
sigma = 0.3
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

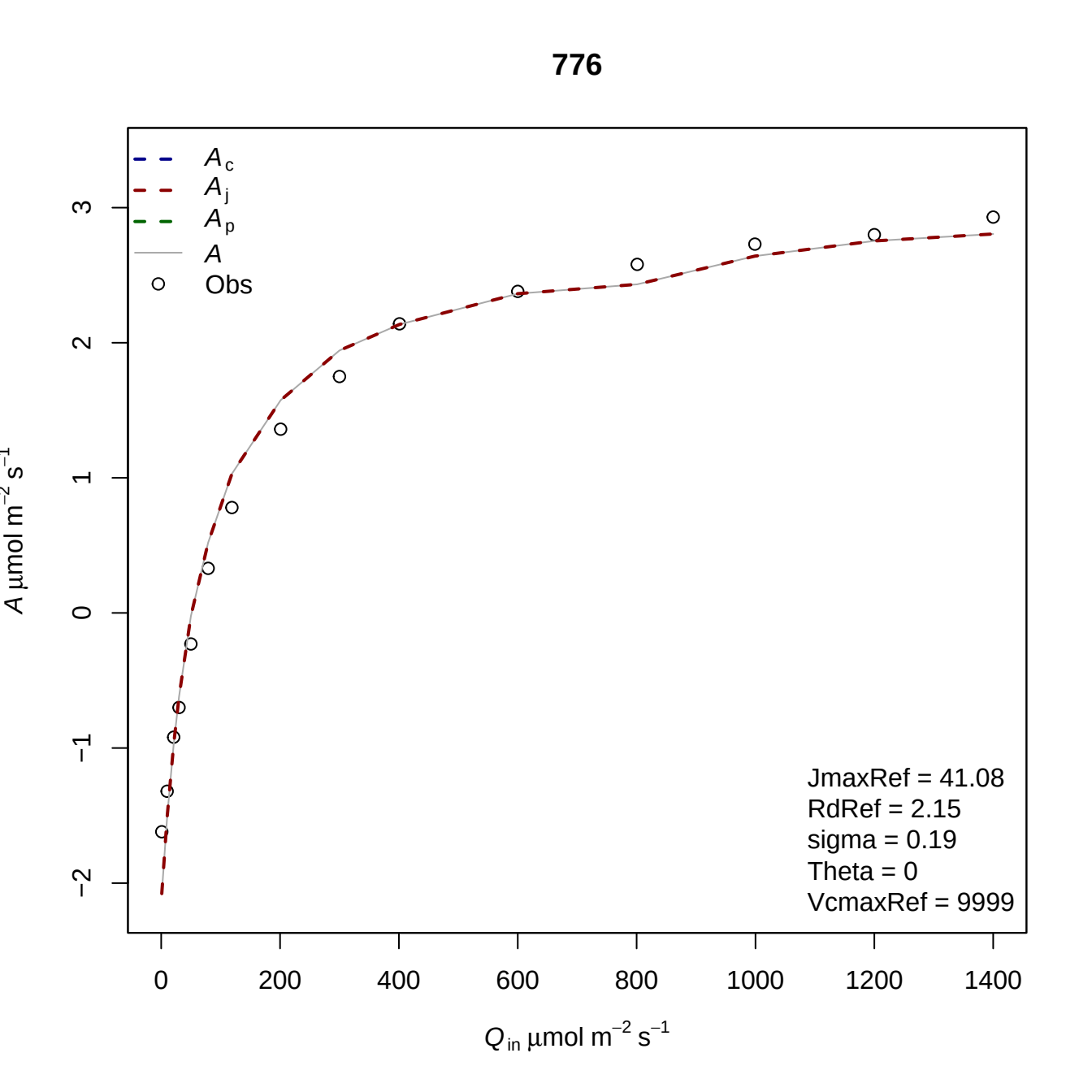

771
A_c
A_j
A_p
A
Obs
JmaxRef = 98.55
RdRef = 1.73
sigma = 0.09
Theta = 0
VcmaxRef = 75.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

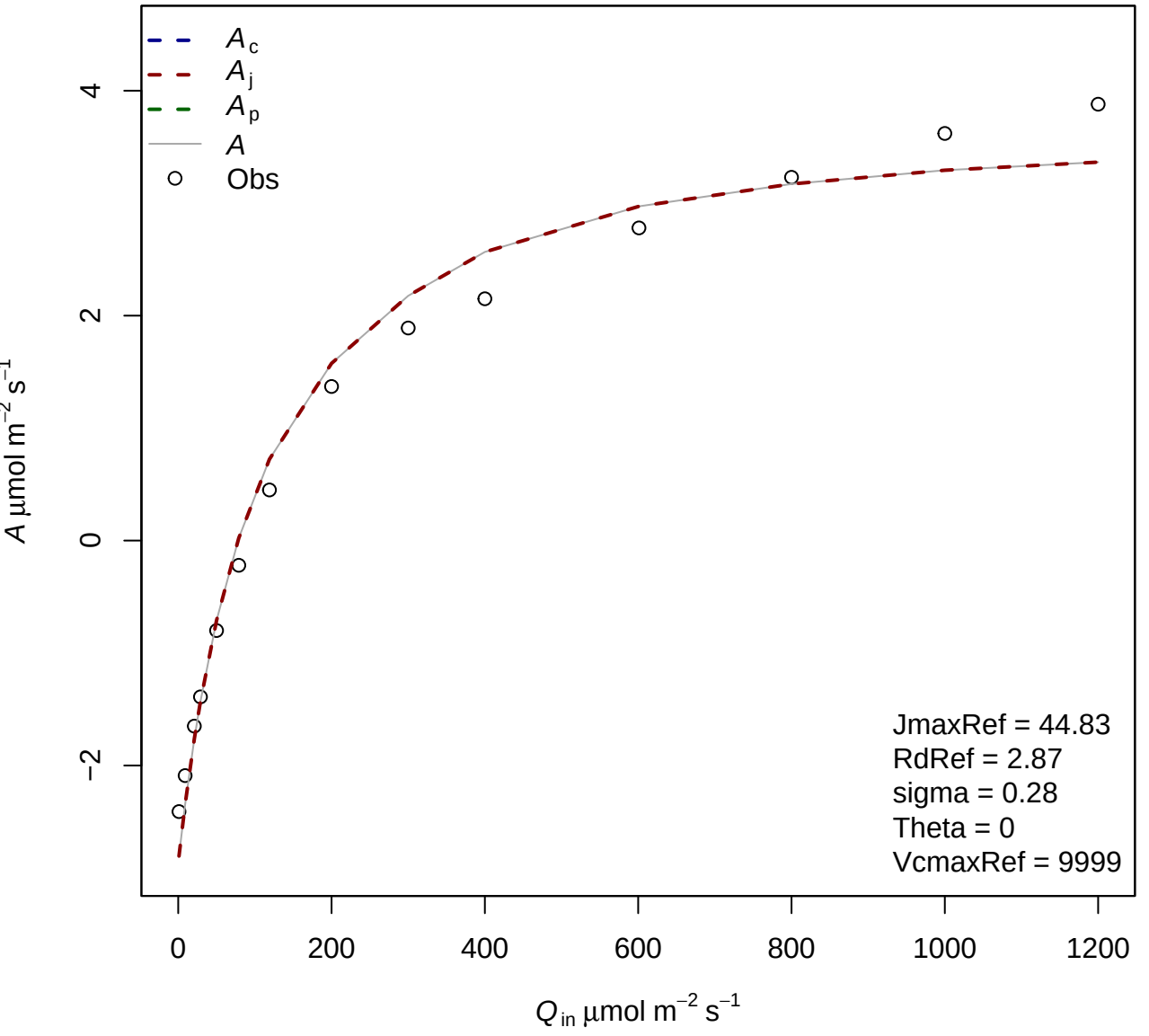

772
A_c
A_j
A_p
A
Obs
JmaxRef = 131.22
RdRef = 1.53
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

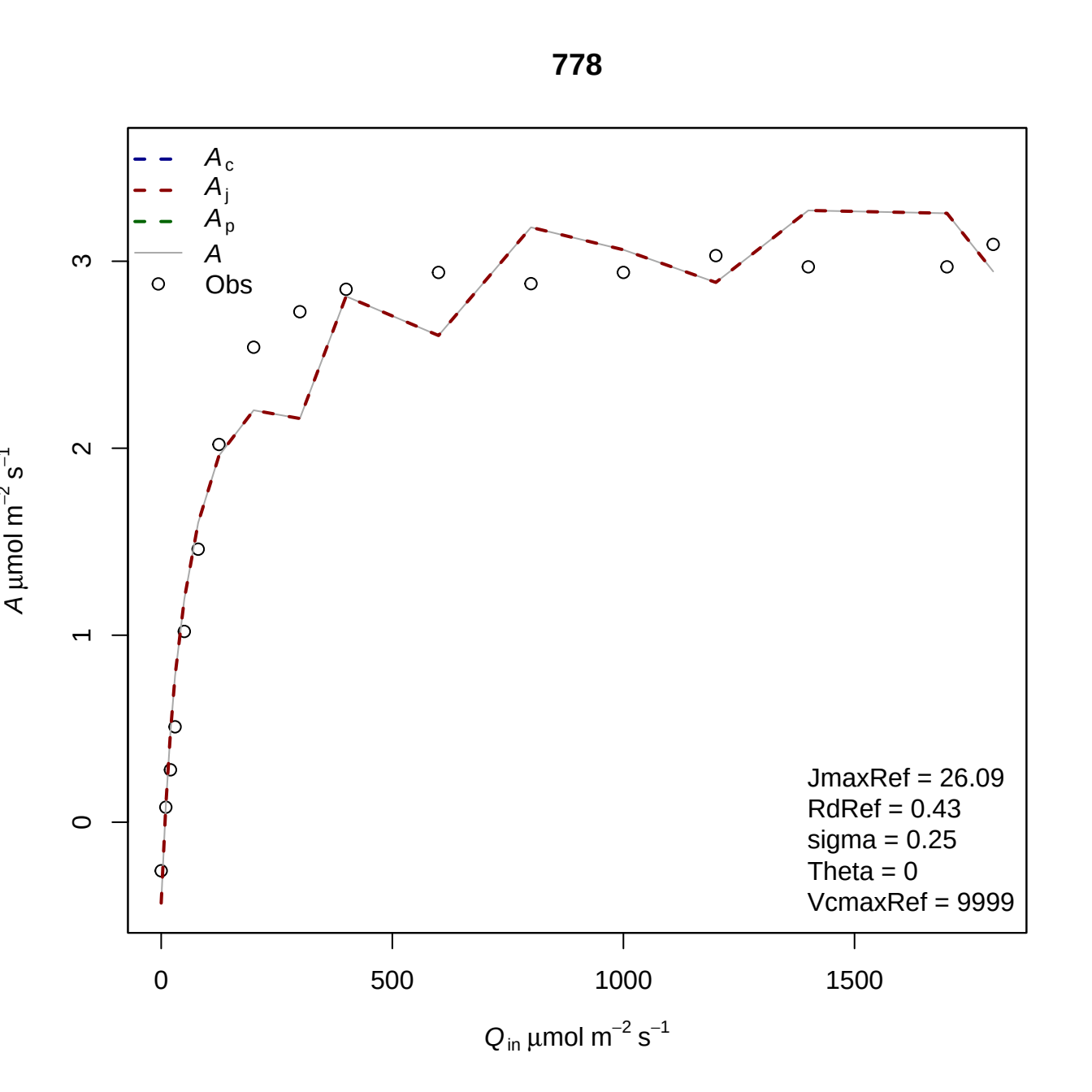

773
A_c
A_j
A_p
A
Obs
JmaxRef = 43.88
RdRef = 1.53
sigma = 0.08
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

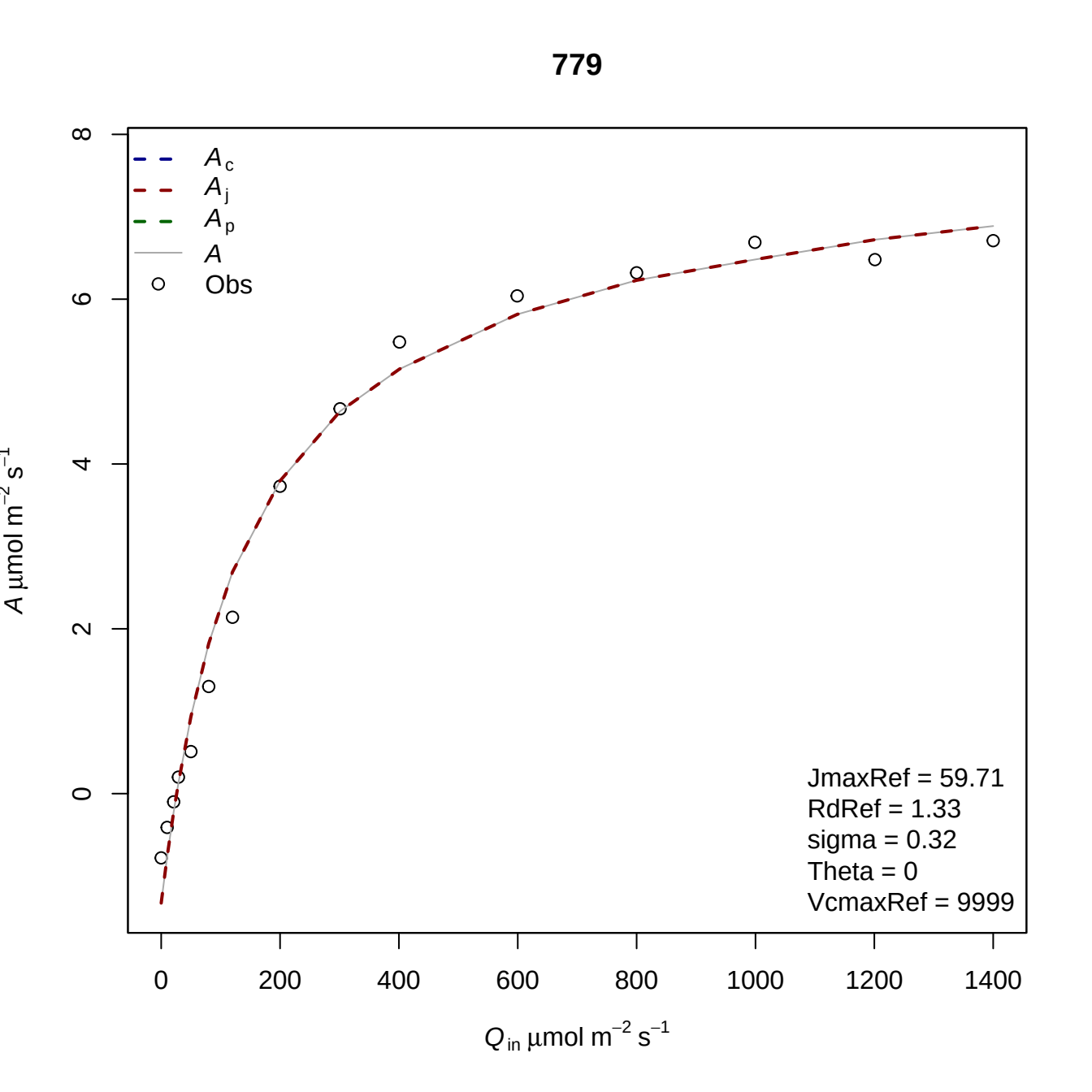

774
A_c
A_j
A_p
A
Obs
JmaxRef = 91.2
RdRef = 3.69
sigma = 0.49
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
8
0
500
1000
1500

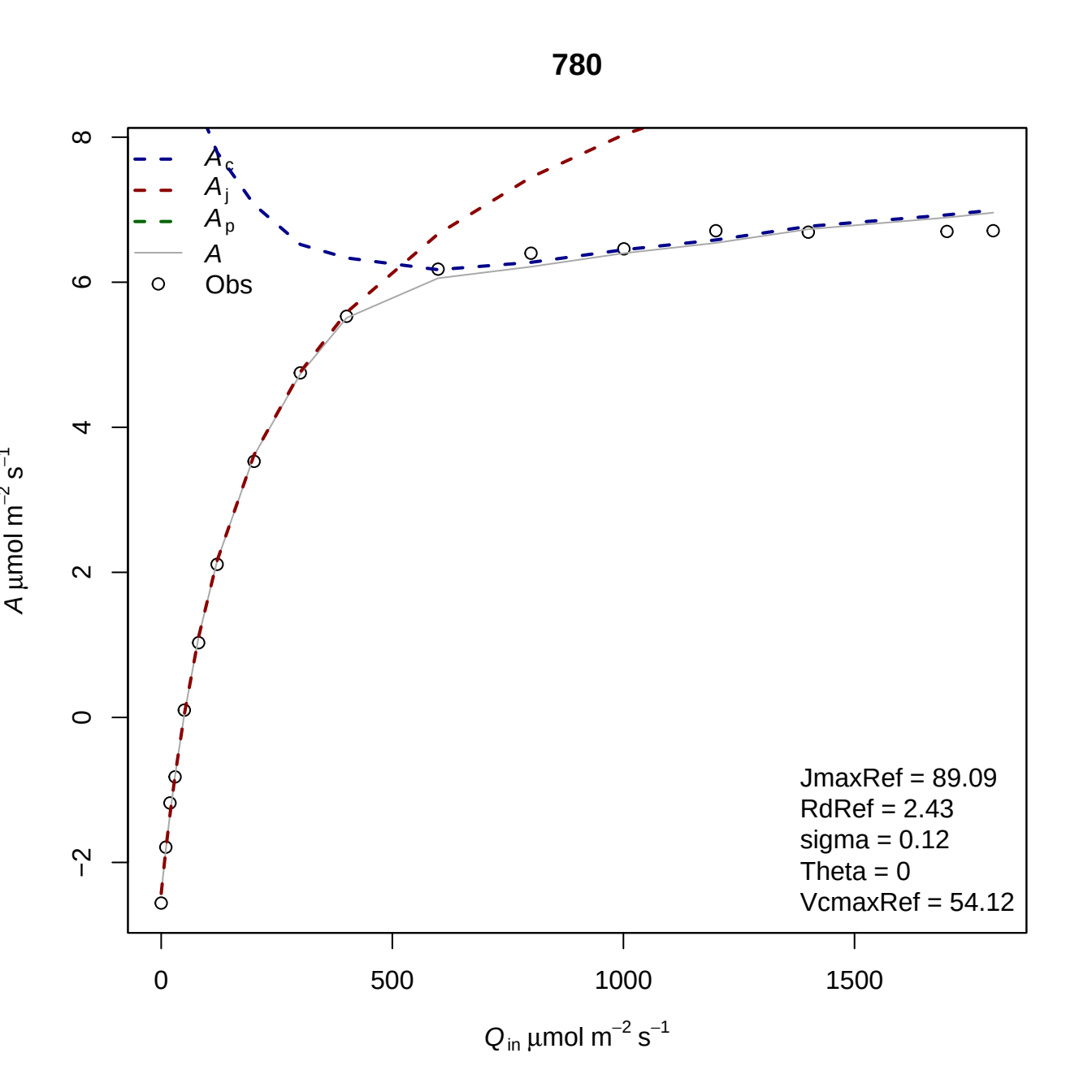

775
A_c
A_j
A_p
A
Obs
JmaxRef = 79.47
RdRef = 1.43
sigma = 0.26
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

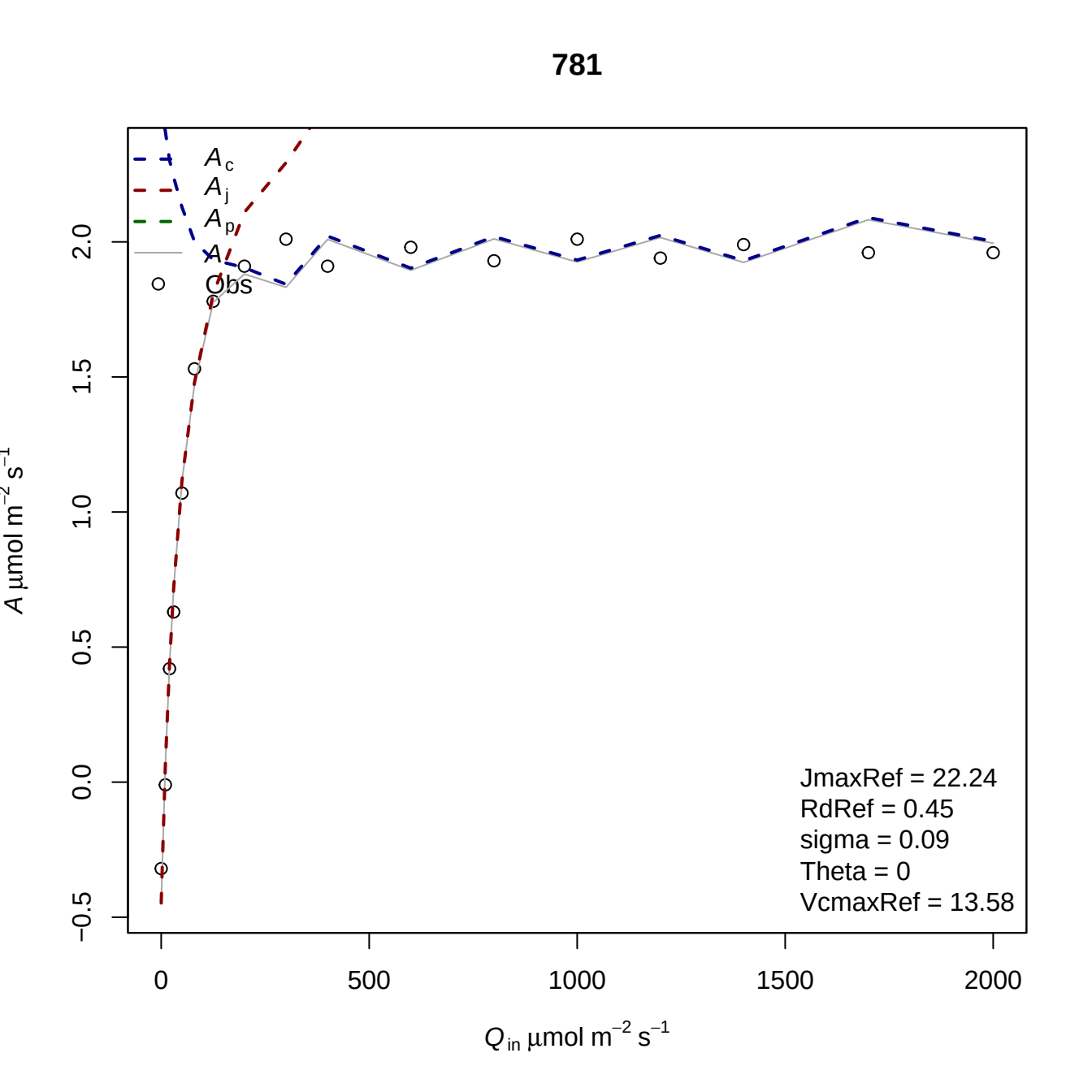

776
A_c
A_j
A_p
A
Obs
JmaxRef = 41.08
RdRef = 2.15
sigma = 0.19
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

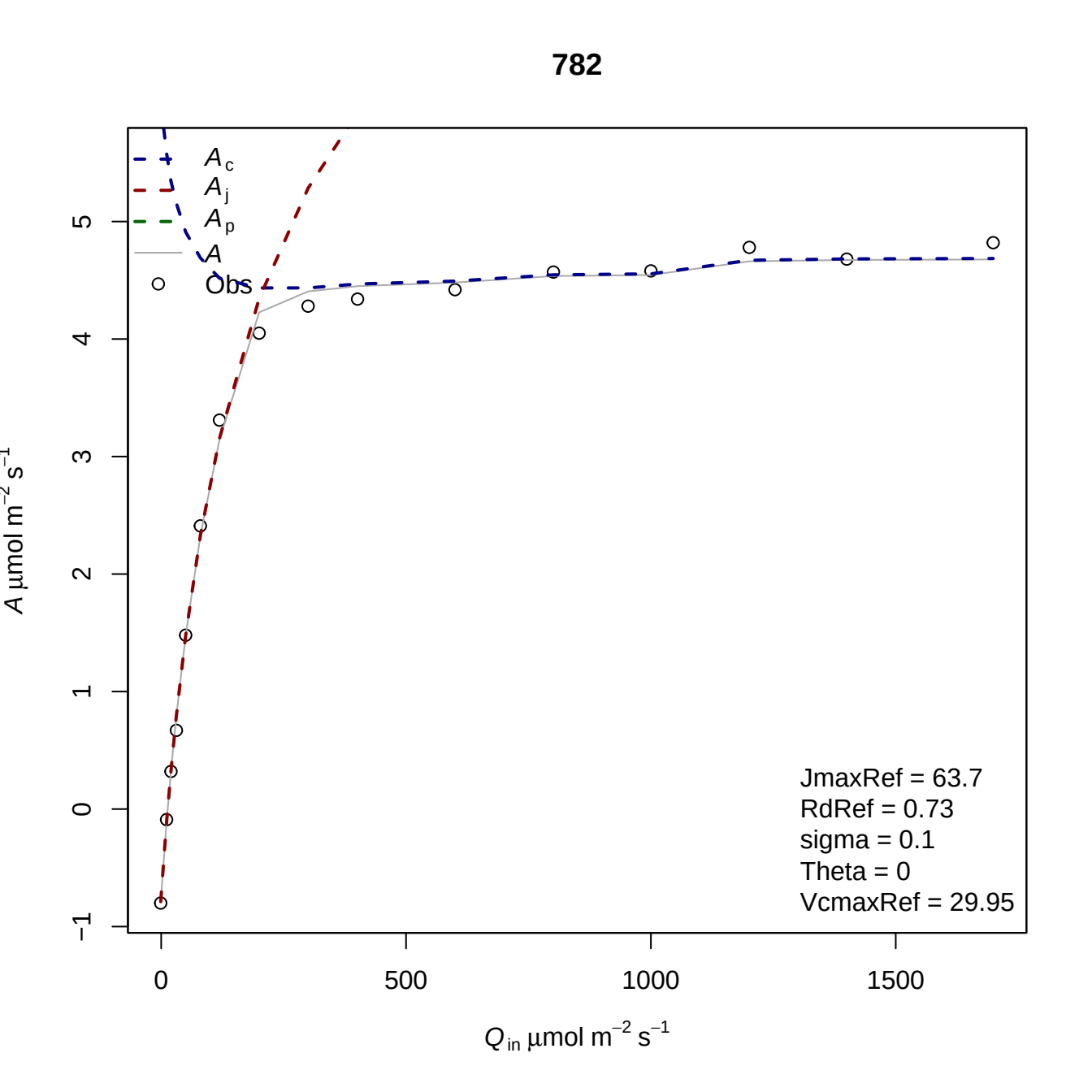

A_c
A_j
A_p
A
Obs
JmaxRef = 44.83
RdRef = 2.87
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200

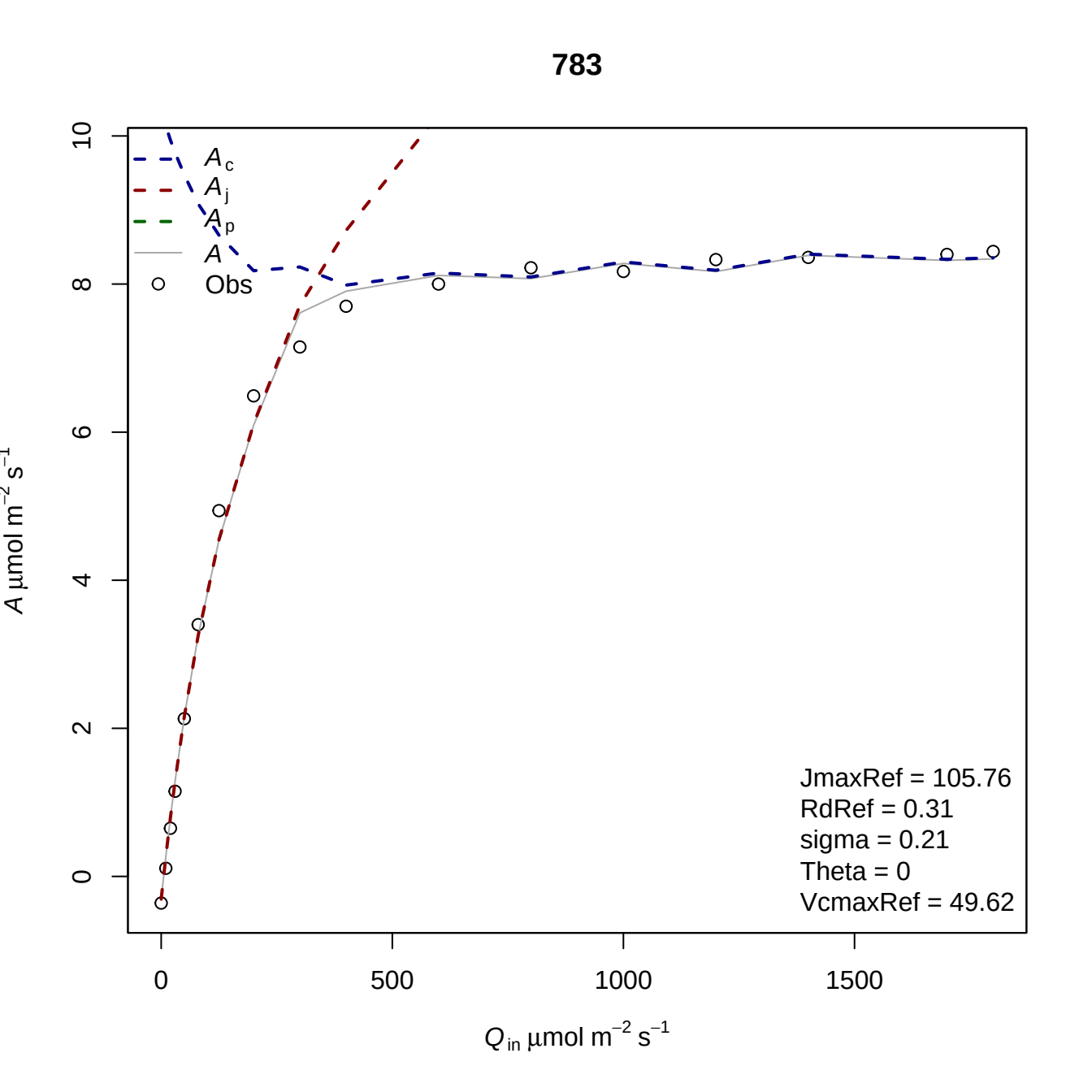

778
A_c
A_j
A_p
A
Obs
JmaxRef = 26.09
RdRef = 0.43
sigma = 0.25
Theta = 0
VcmaxRef = 9999
0
1
2
3
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

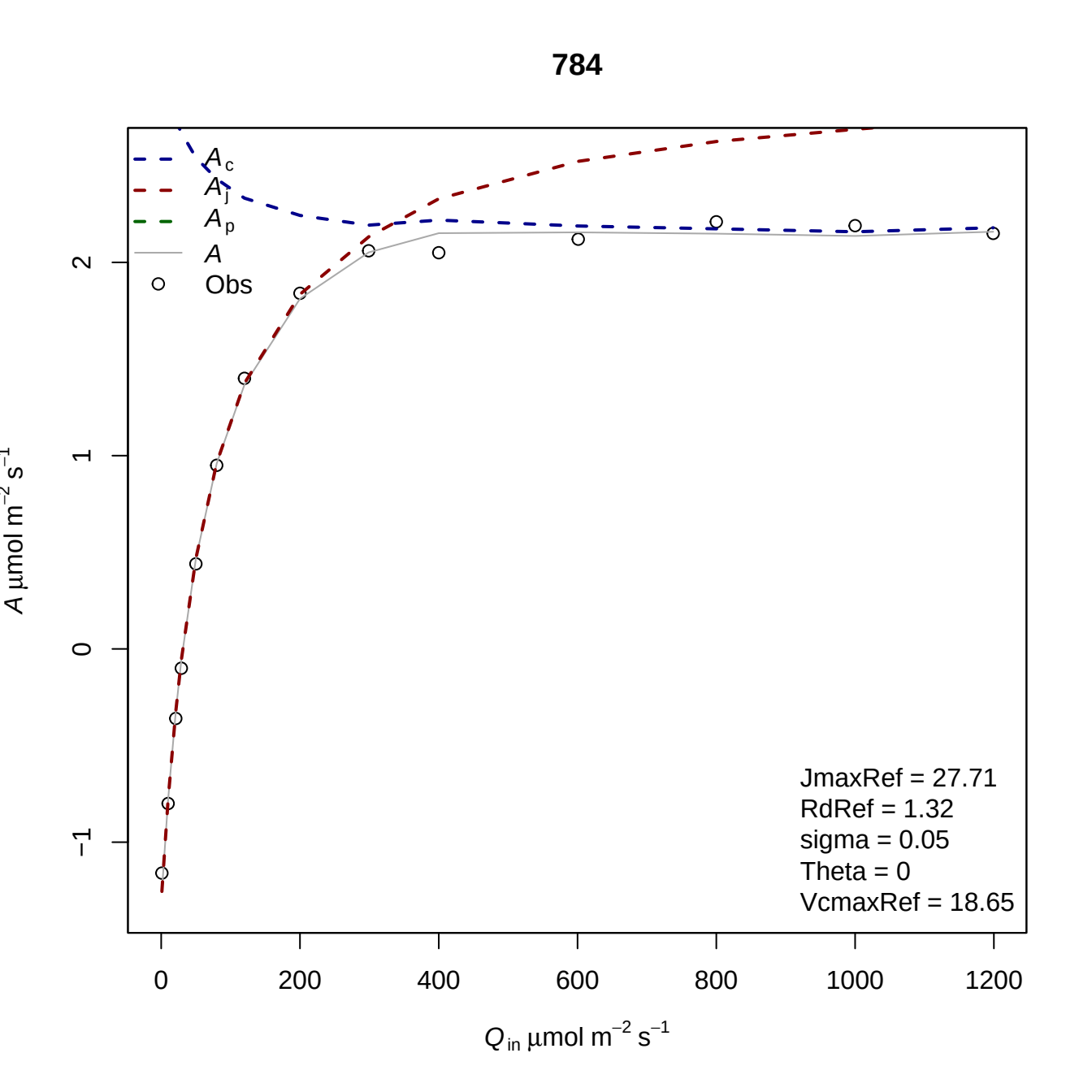

779
A_c
A_j
A_p
A
Obs
JmaxRef = 59.71
RdRef = 1.33
sigma = 0.32
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

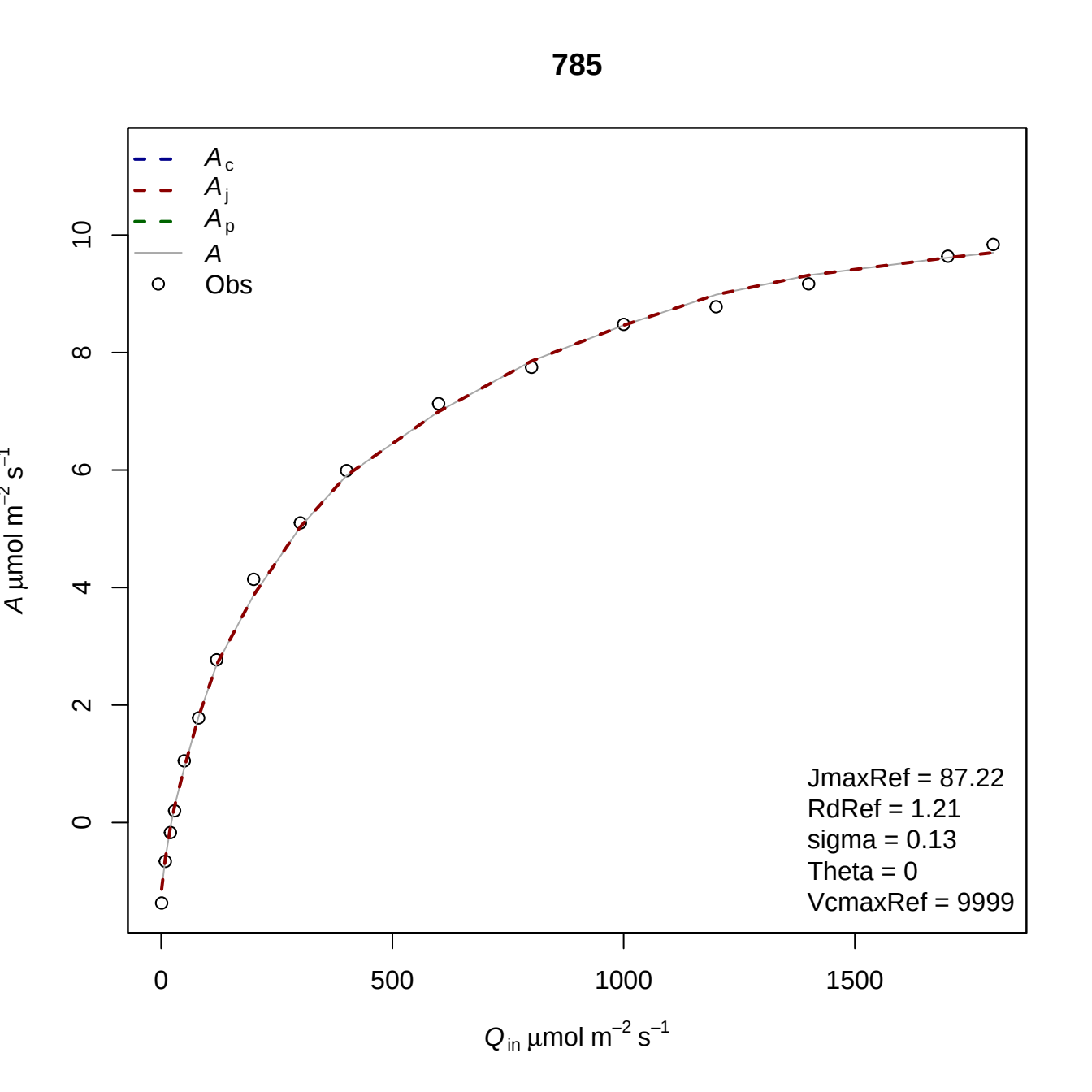

780
A_c
A_j
A_p
A
Obs
JmaxRef = 89.09
RdRef = 2.43
sigma = 0.12
Theta = 0
VcmaxRef = 54.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

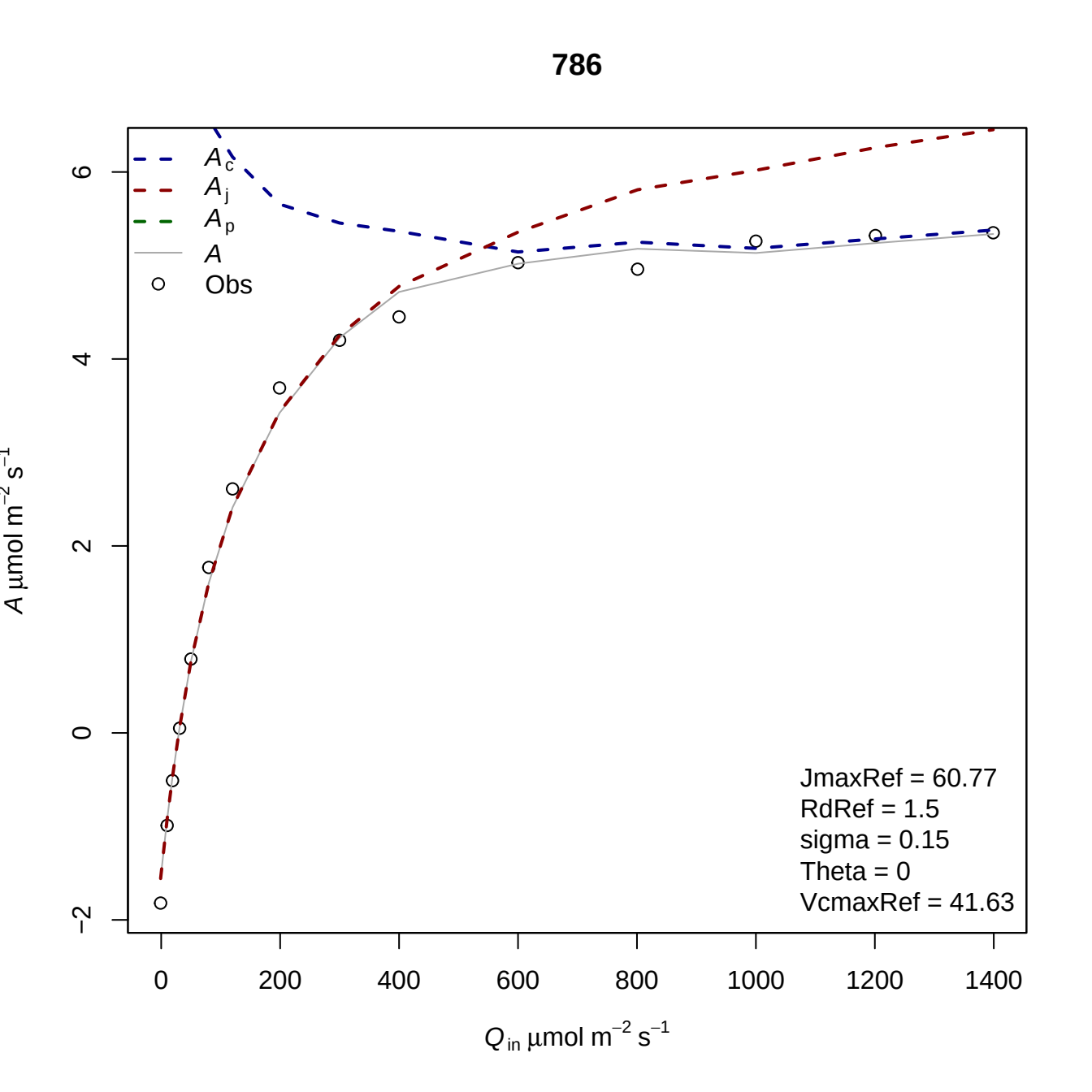

781
A_c
A_j
A_p
A
Obs
JmaxRef = 22.24
RdRef = 0.45
sigma = 0.09
Theta = 0
VcmaxRef = 13.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
2000

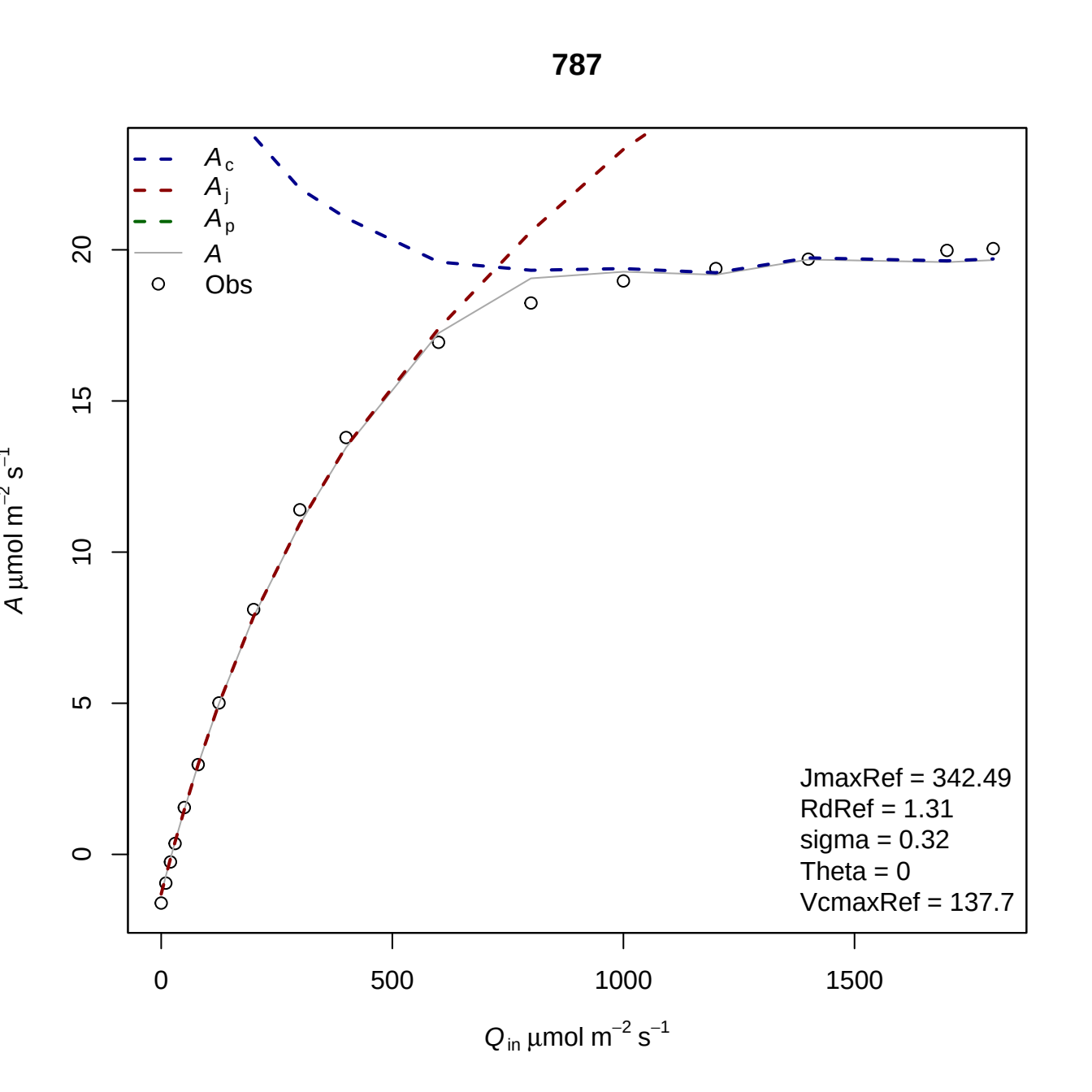

782
A_c
A_j
A_p
A
Obs
JmaxRef = 63.7
RdRef = 0.73
sigma = 0.1
Theta = 0
VcmaxRef = 29.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

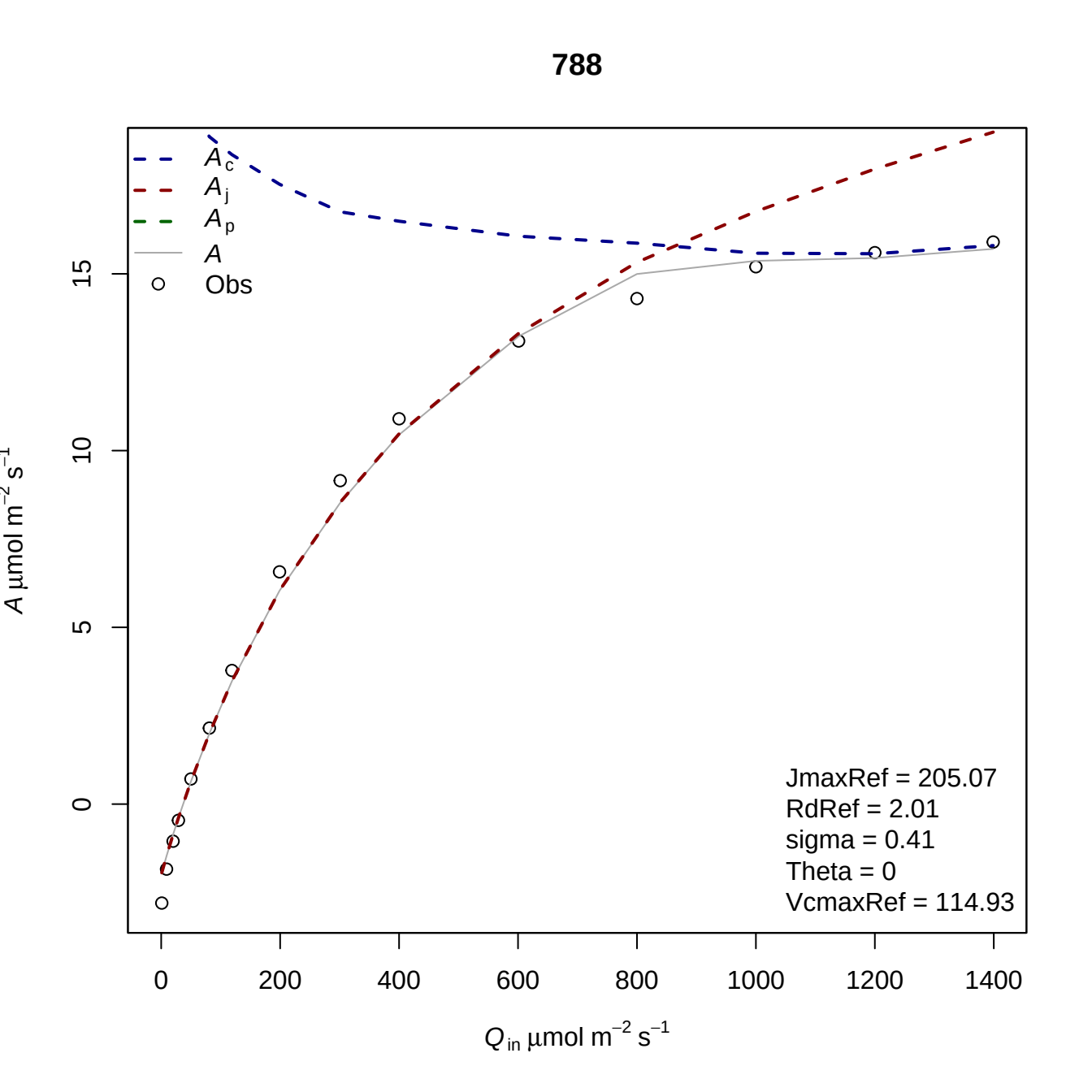

783
A_c
A_j
A_p
A
Obs
JmaxRef = 105.76
RdRef = 0.31
sigma = 0.21
Theta = 0
VcmaxRef = 49.62
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

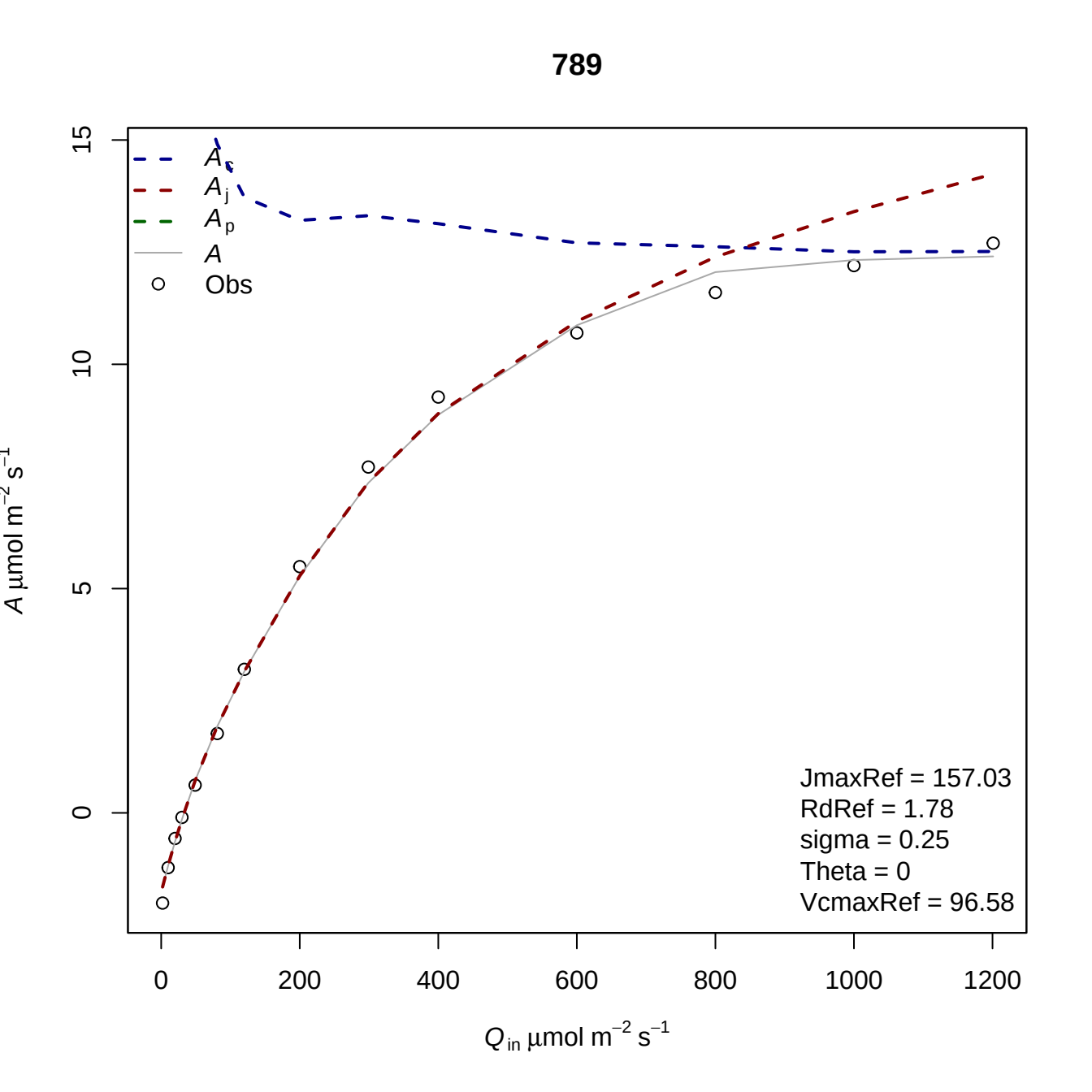

784
A_c
A_j
A_p
A
Obs
JmaxRef = 27.71
RdRef = 1.32
sigma = 0.05
Theta = 0
VcmaxRef = 18.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000
1200

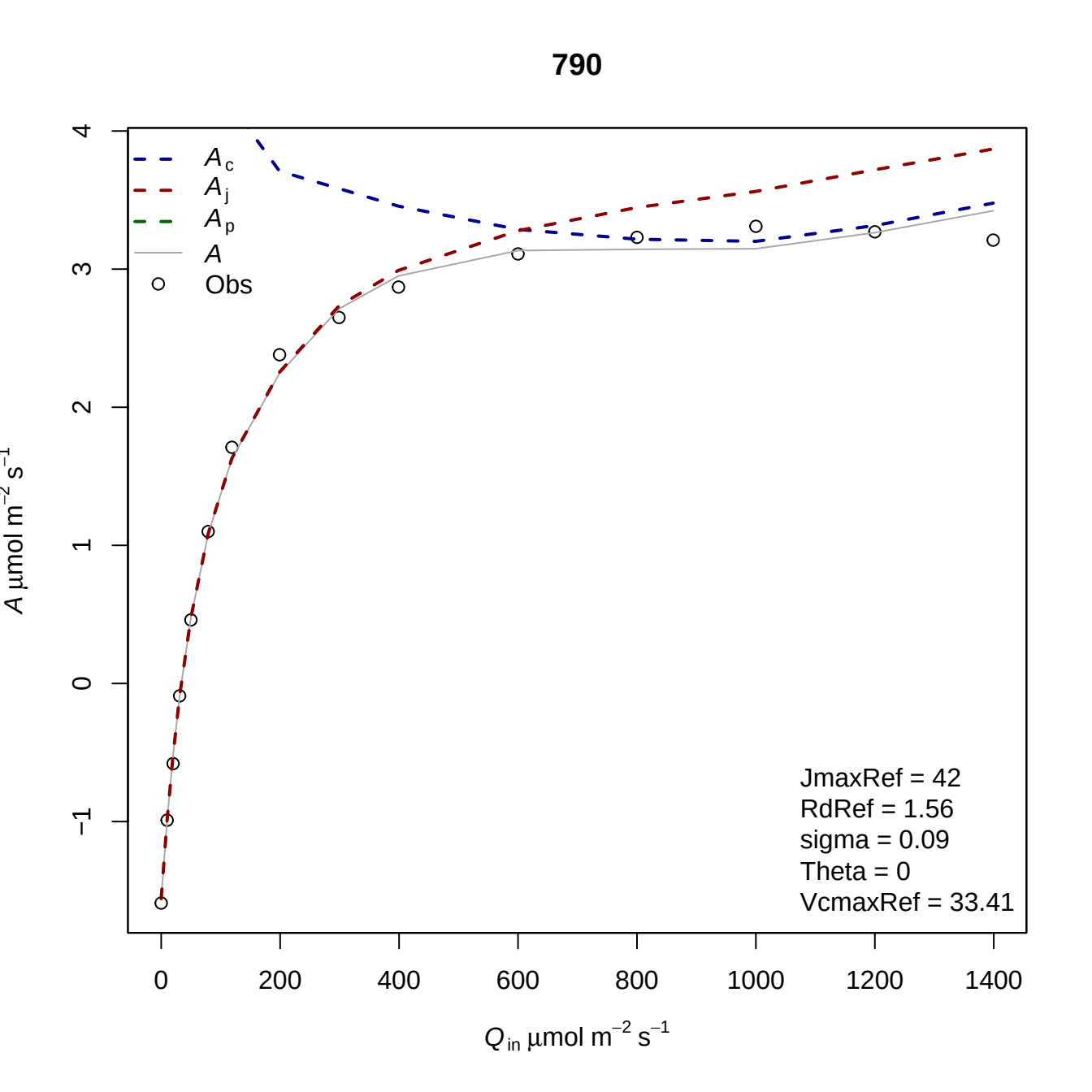

785
A_c
A_j
A_p
A
Obs
JmaxRef = 87.22
RdRef = 1.21
sigma = 0.13
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

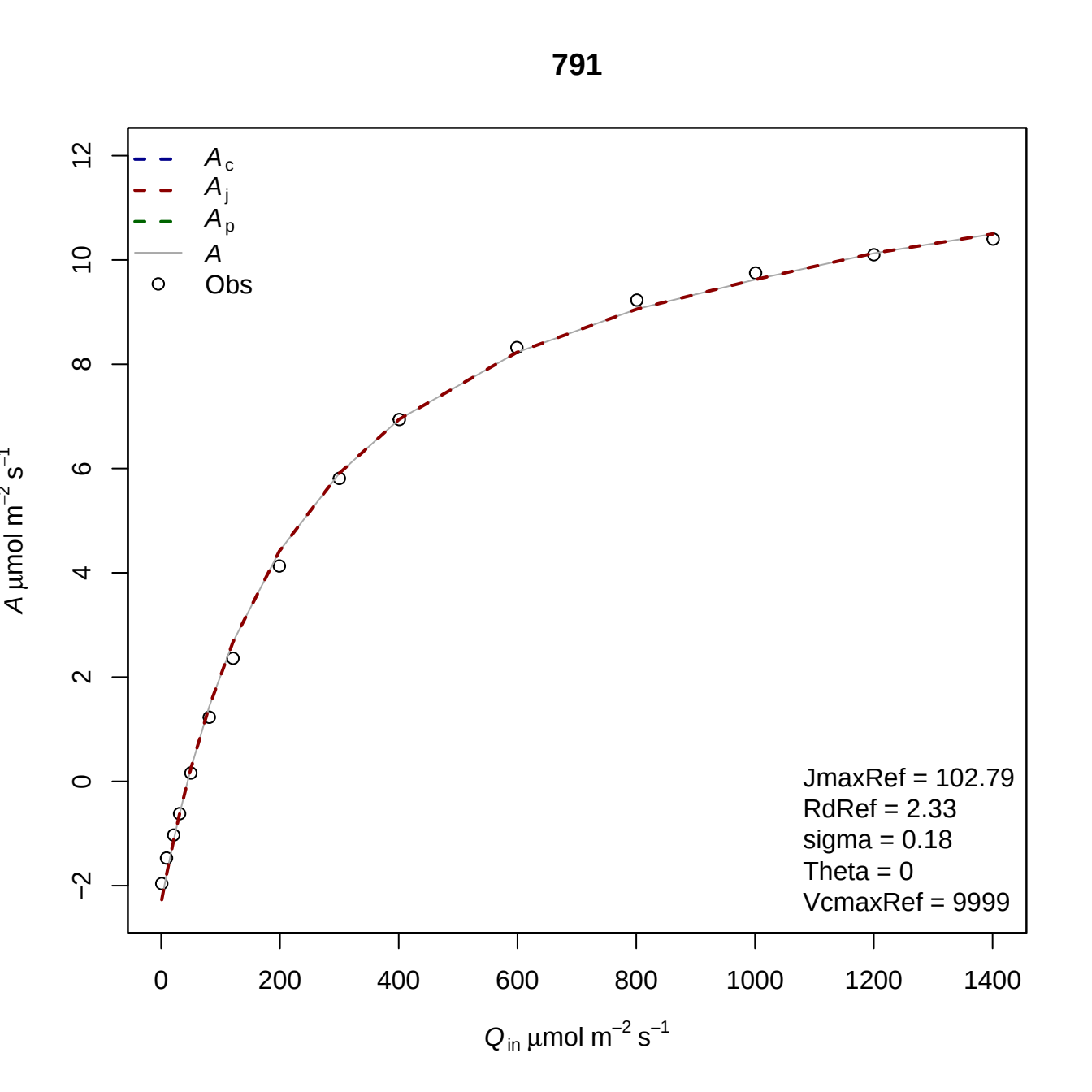

786
A_c
A_j
A_p
A
Obs
JmaxRef = 60.77
RdRef = 1.5
sigma = 0.15
Theta = 0
VcmaxRef = 41.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

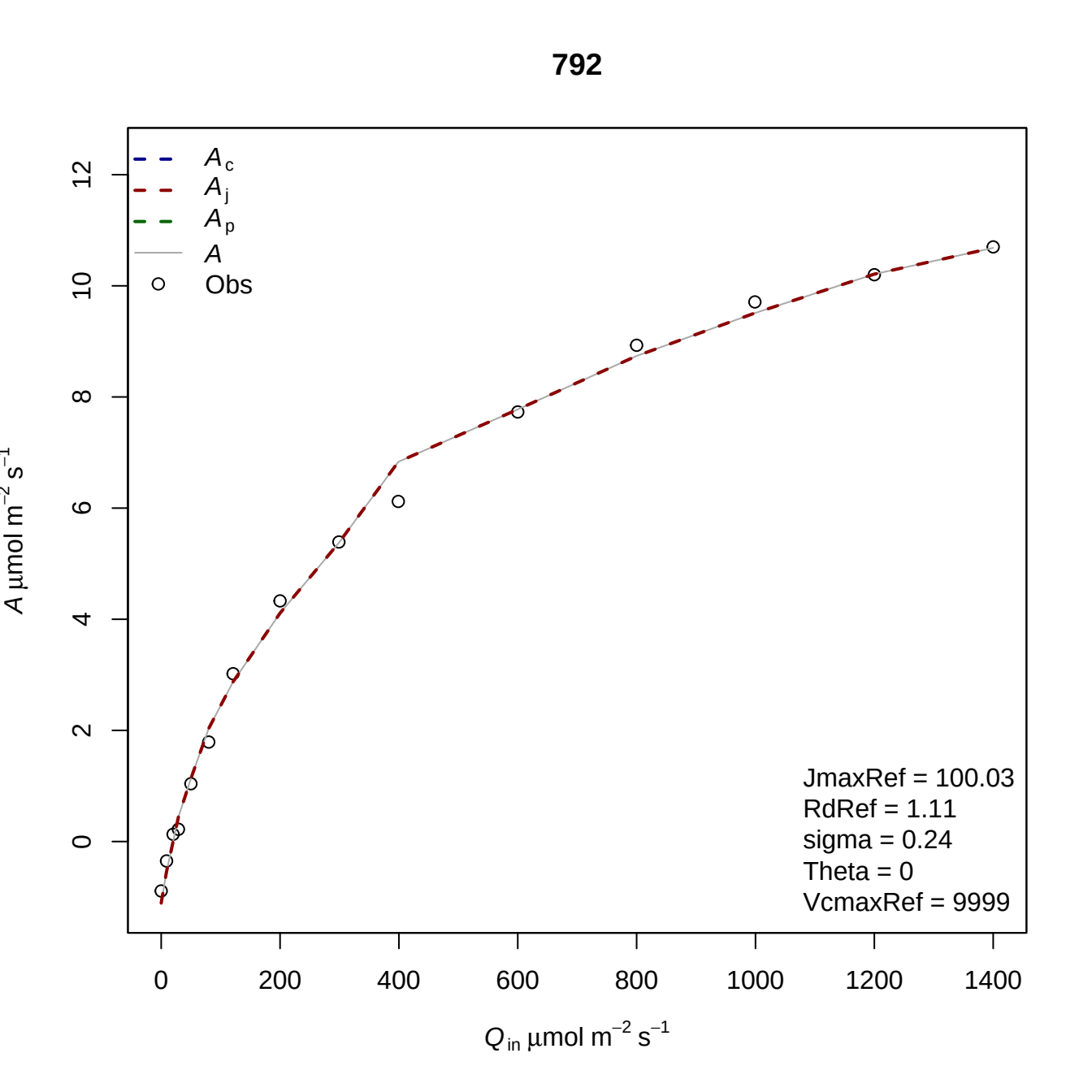

787
A_c
A_j
A_p
A
Obs
JmaxRef = 342.49
RdRef = 1.31
sigma = 0.32
Theta = 0
VcmaxRef = 137.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

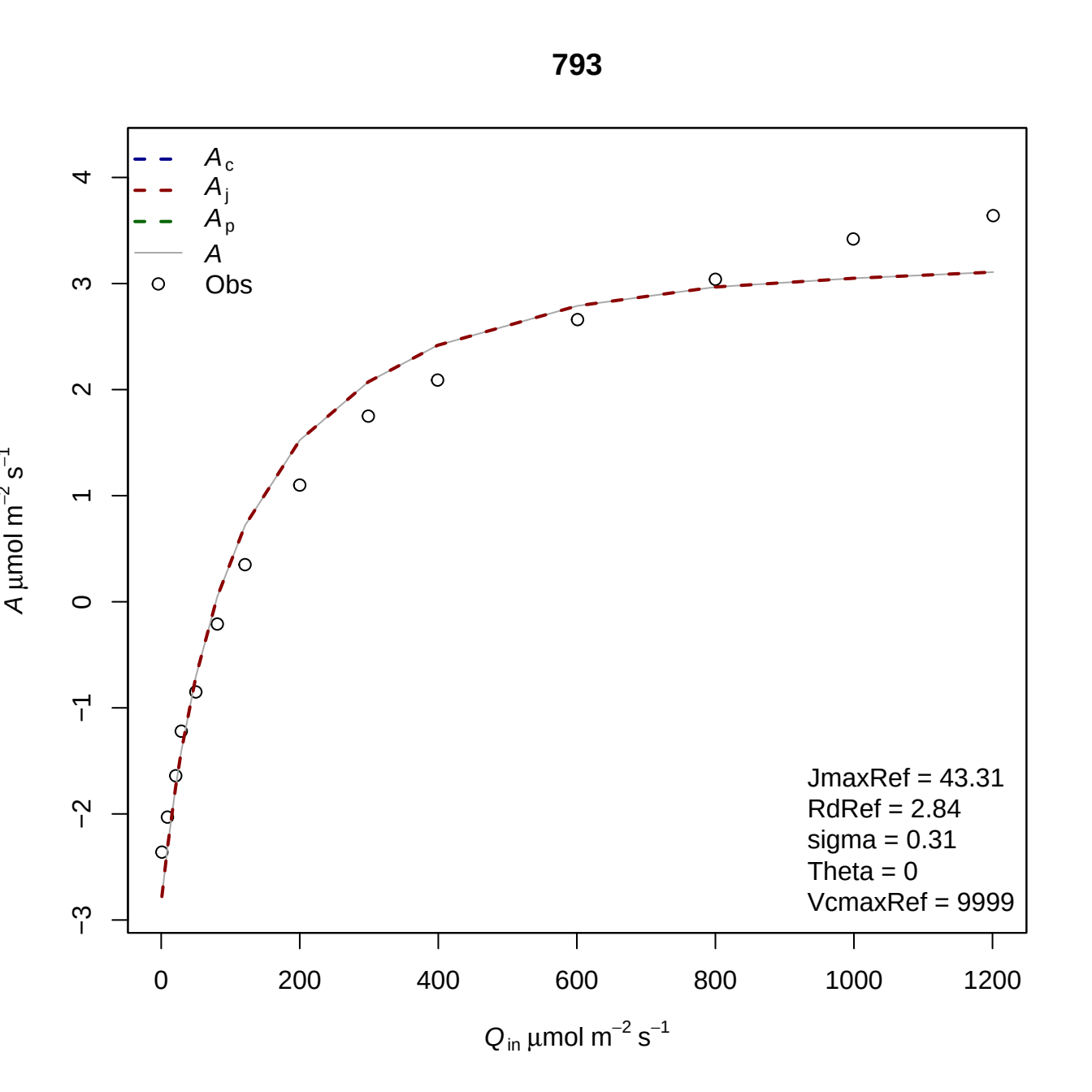

788
A_c
A_j
A_p
A
Obs
JmaxRef = 205.07
RdRef = 2.01
sigma = 0.41
Theta = 0
VcmaxRef = 114.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

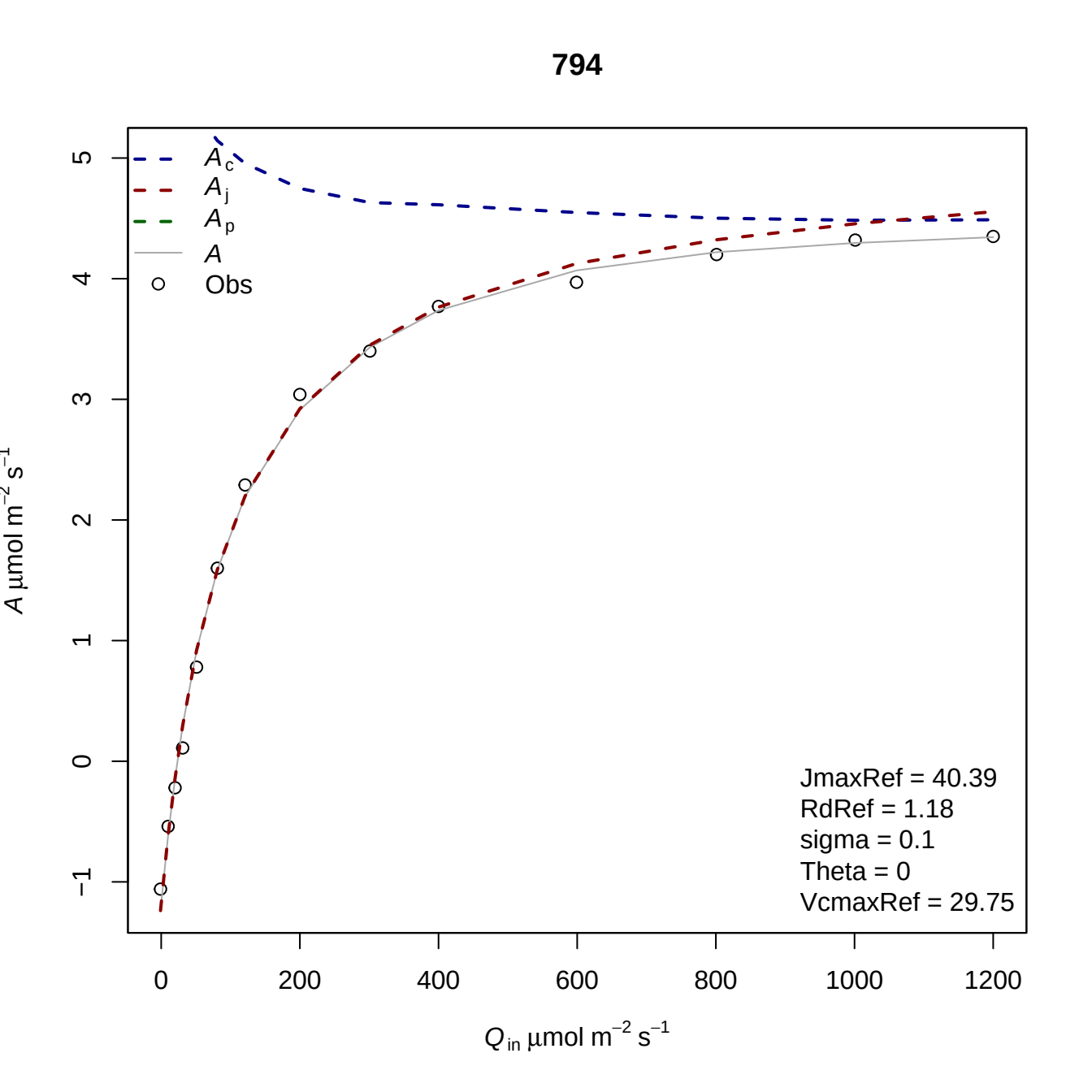

789
A_c
A_j
A_p
A
Obs
JmaxRef = 157.03
RdRef = 1.78
sigma = 0.25
Theta = 0
VcmaxRef = 96.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

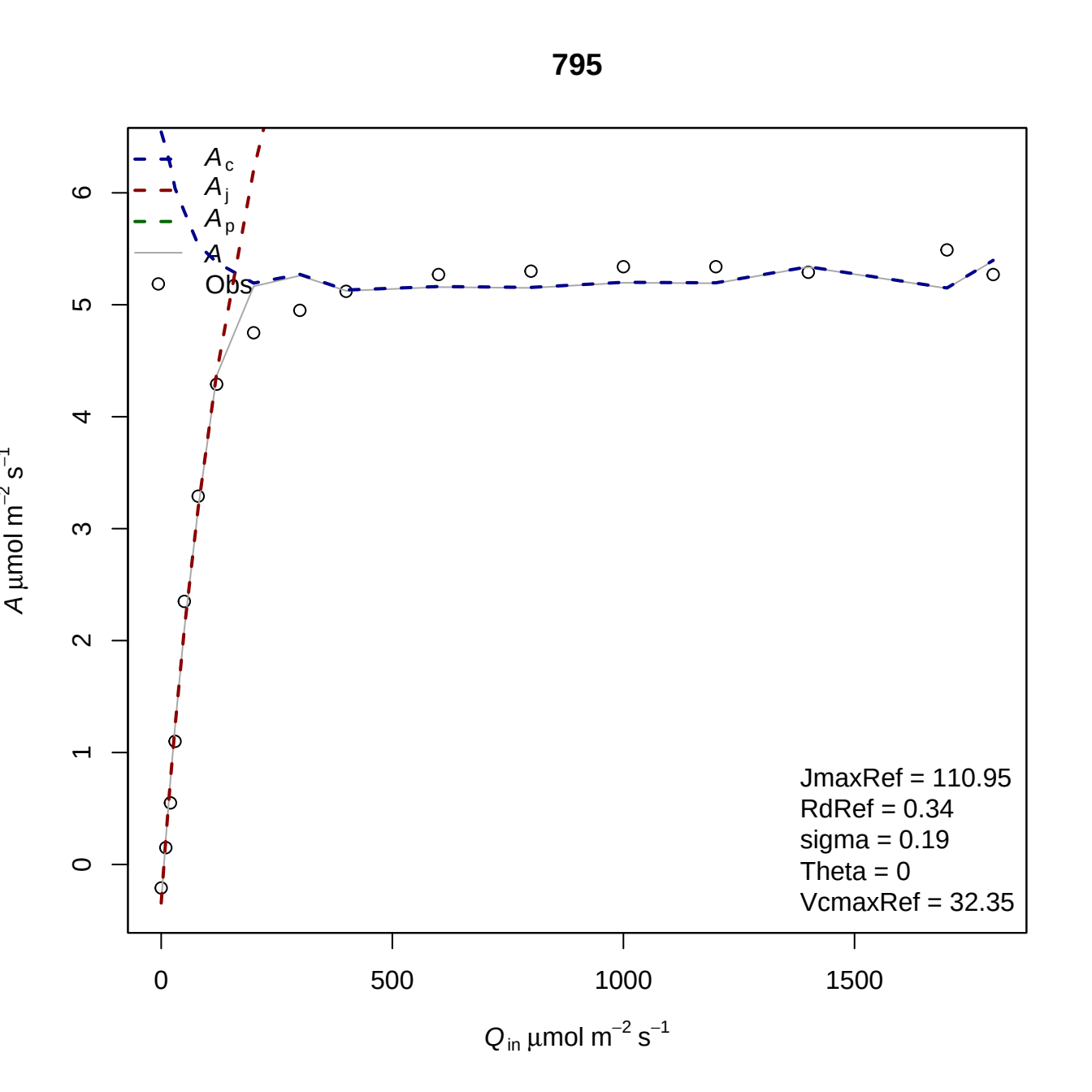

790
A_c
A_j
A_p
A
Obs
JmaxRef = 42
RdRef = 1.56
sigma = 0.09
Theta = 0
VcmaxRef = 33.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

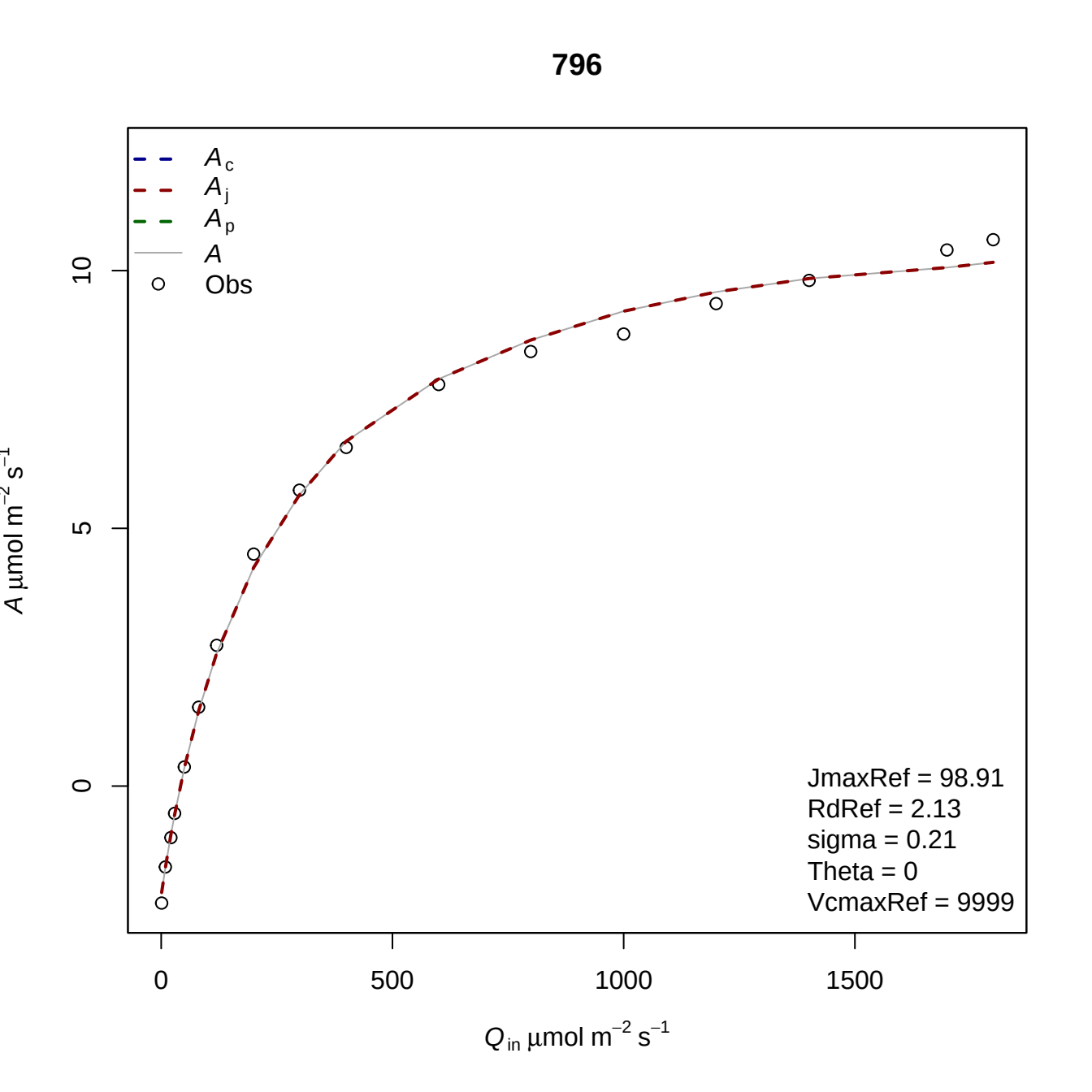

791
A_c
A_j
A_p
A
Obs
JmaxRef = 102.79
RdRef = 2.33
sigma = 0.18
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
200
400
600
800
1000
1200
1400

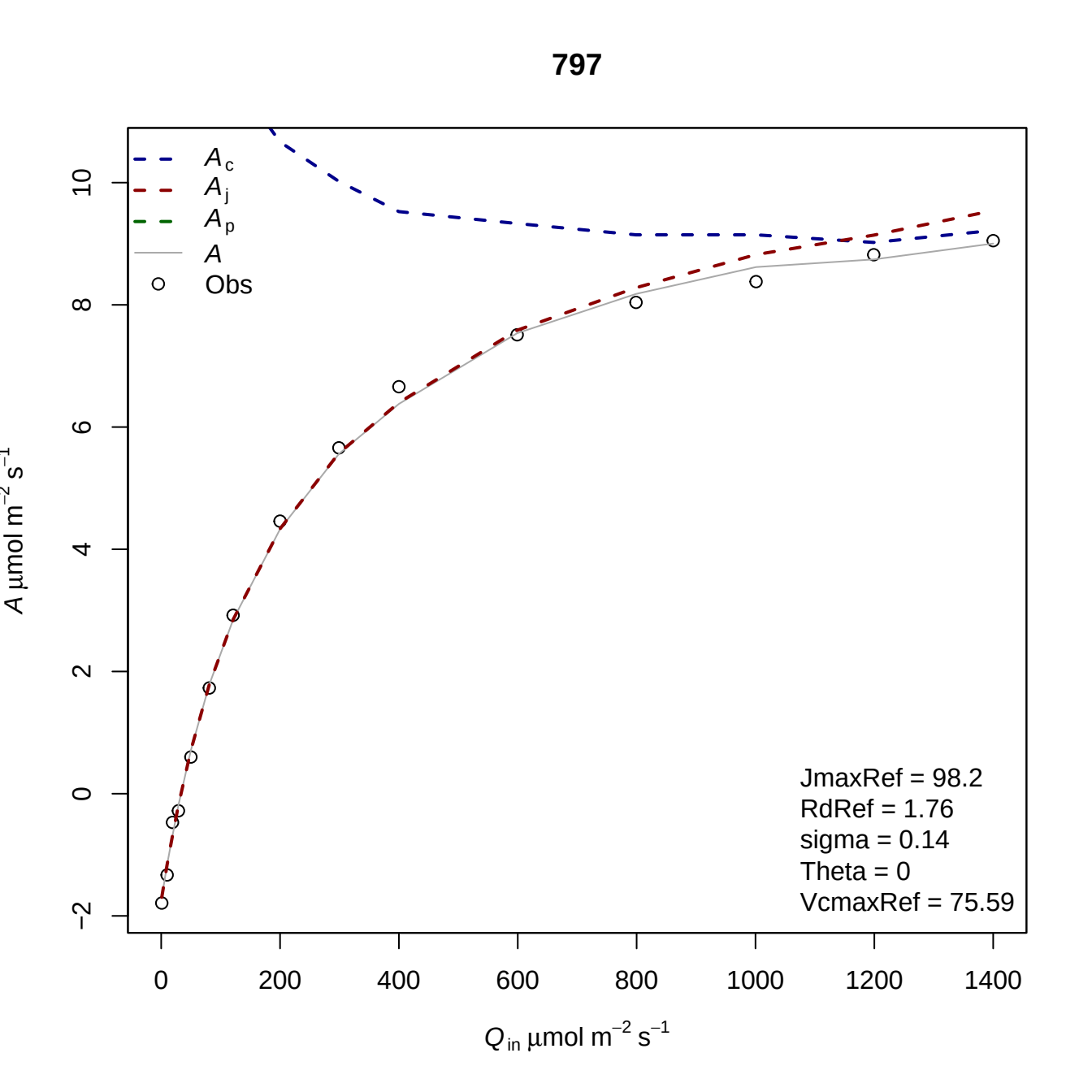

792
A_c
A_j
A_p
A
Obs
JmaxRef = 100.03
RdRef = 1.11
sigma = 0.24
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

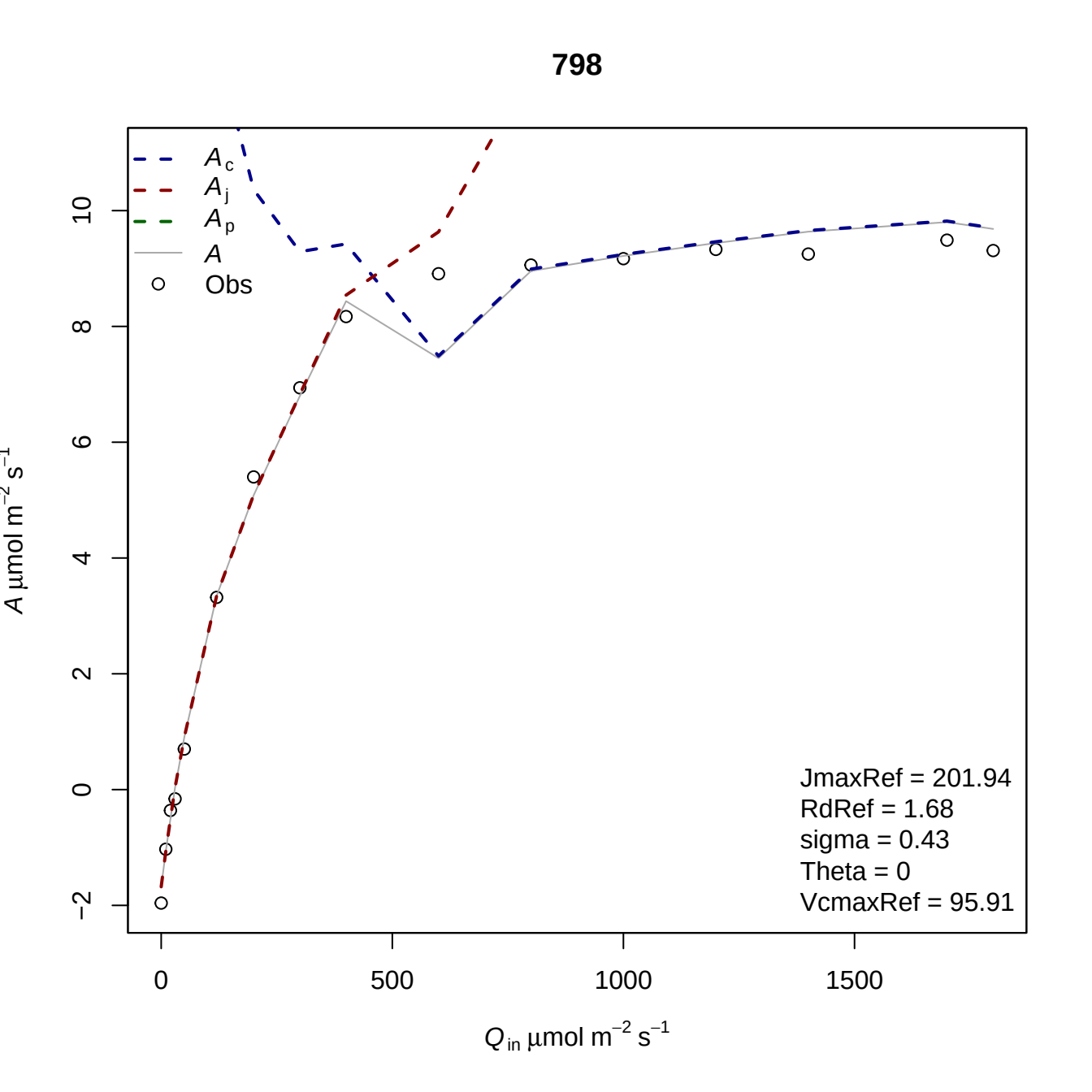

793
A_c
A_j
A_p
A
Obs
JmaxRef = 43.31
RdRef = 2.84
sigma = 0.31
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−3
−2
−1
0
1
2
3
4
0
200
400
600
800
1000
1200

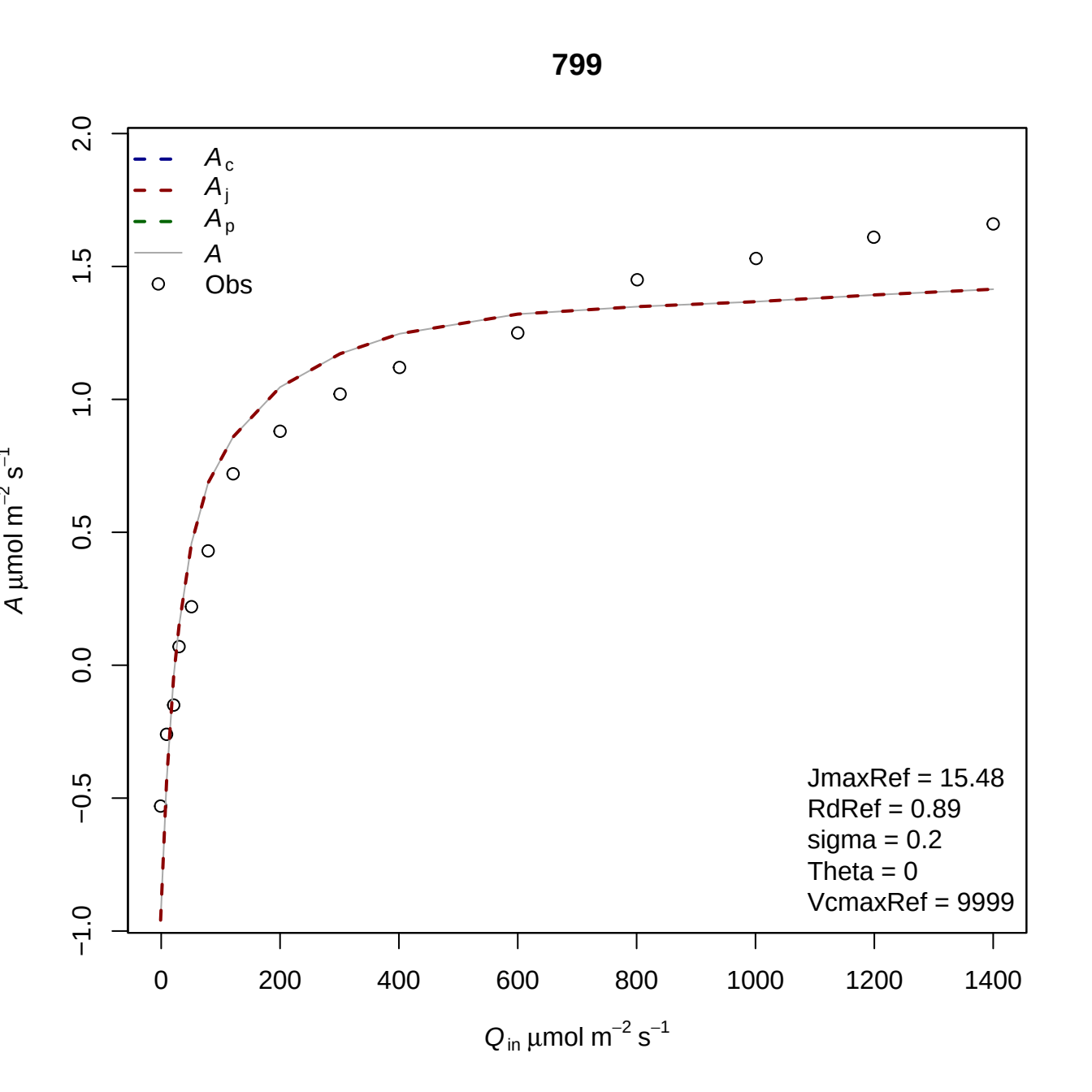

794
A_c
A_j
A_p
A
Obs
JmaxRef = 40.39
RdRef = 1.18
sigma = 0.1
Theta = 0
VcmaxRef = 29.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200

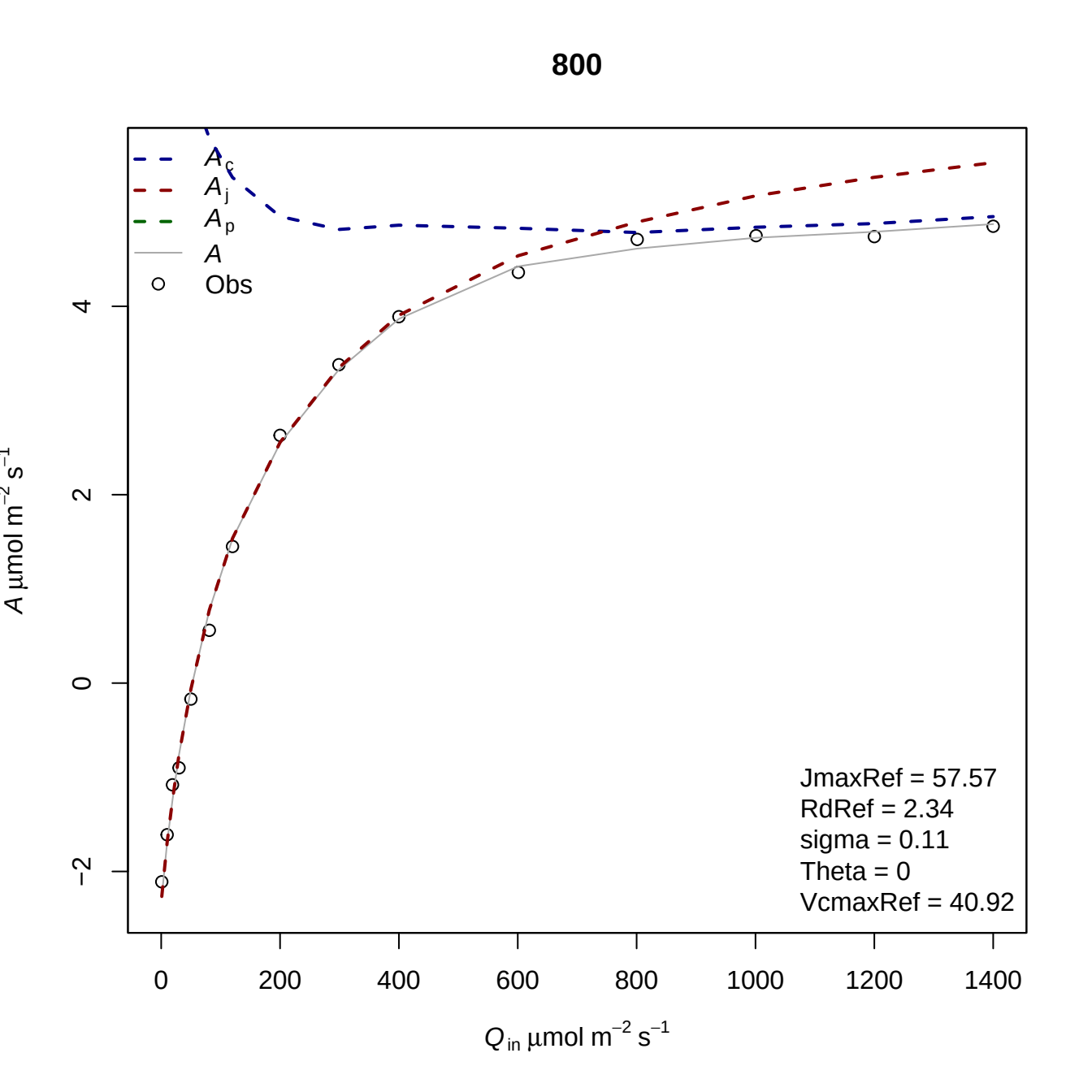

795
A_c
A_j
A_p
A
Obs
JmaxRef = 110.95
RdRef = 0.34
sigma = 0.19
Theta = 0
VcmaxRef = 32.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

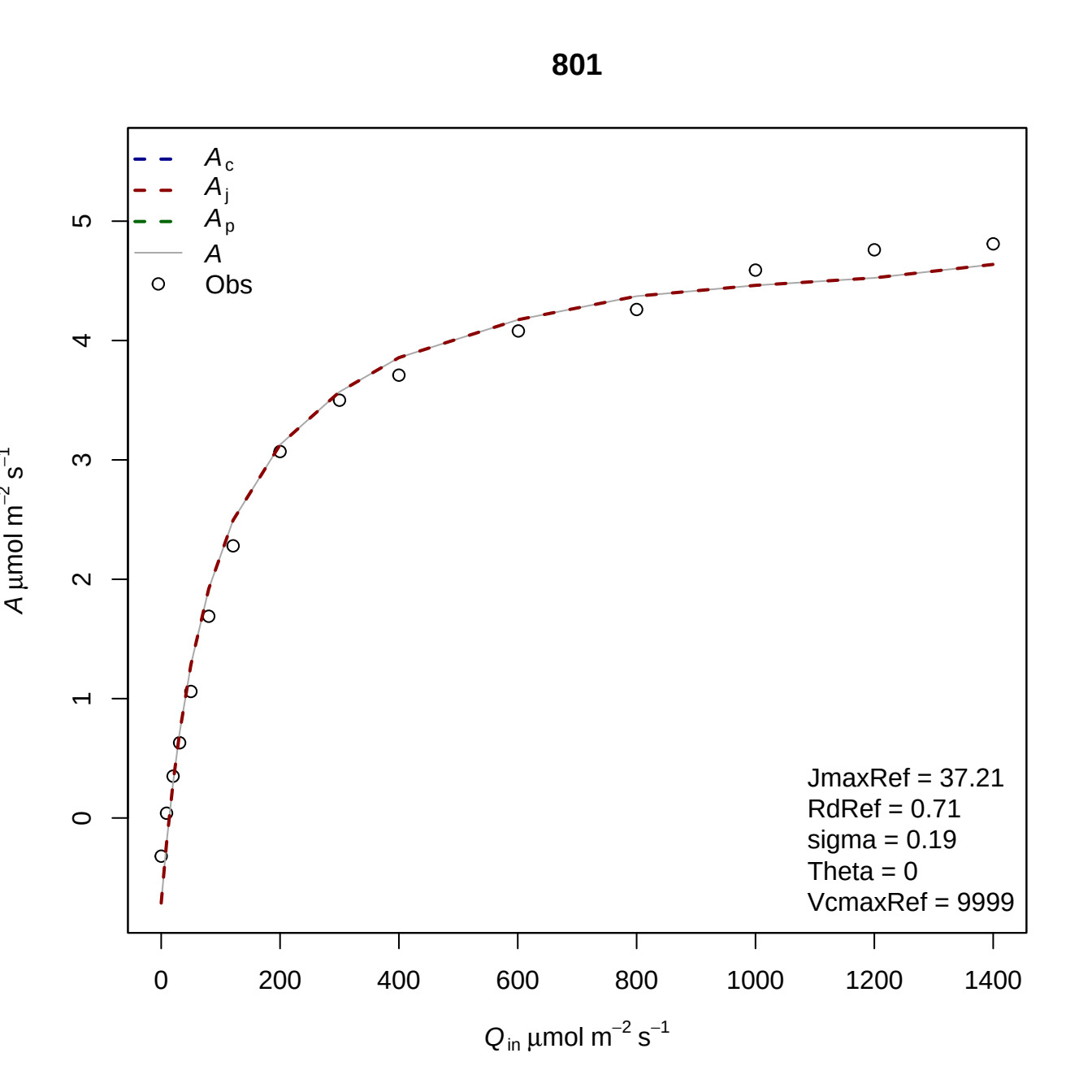

796
A_c
A_j
A_p
A
Obs
JmaxRef = 98.91
RdRef = 2.13
sigma = 0.21
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

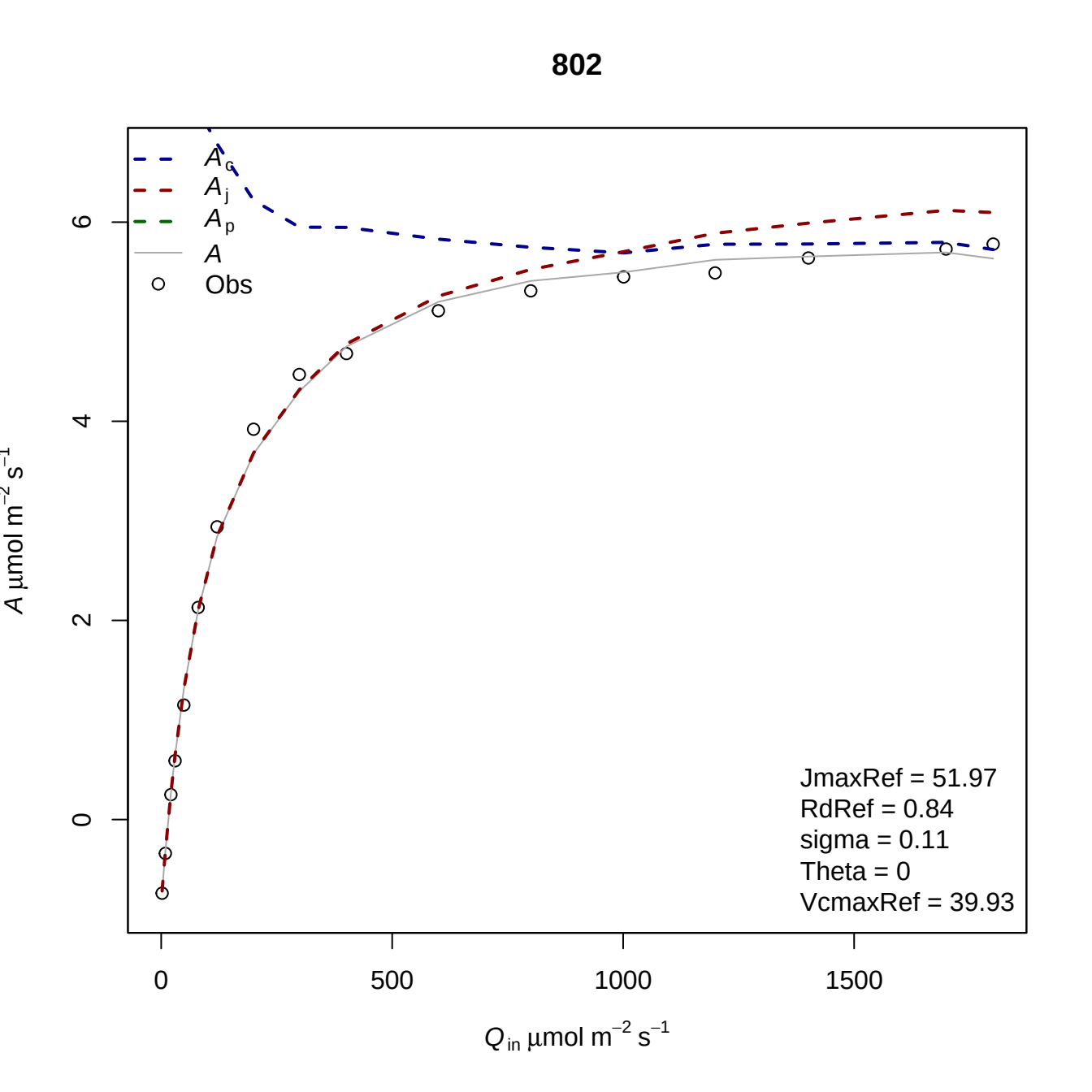

797
A_c
A_j
A_p
A
Obs
JmaxRef = 98.2
RdRef = 1.76
sigma = 0.14
Theta = 0
VcmaxRef = 75.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

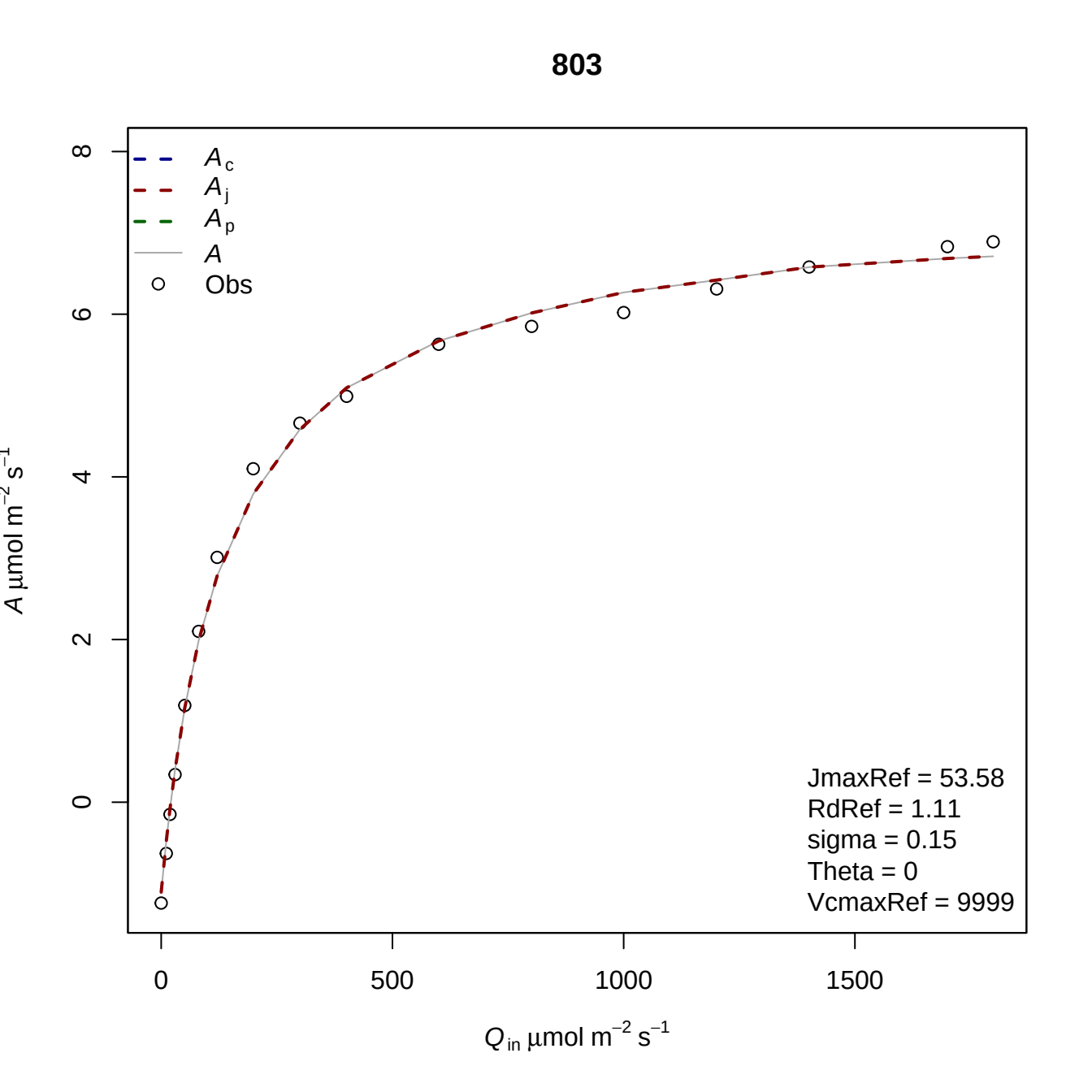

798
A_c
A_j
A_p
A
Obs
JmaxRef = 201.94
RdRef = 1.68
sigma = 0.43
Theta = 0
VcmaxRef = 95.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

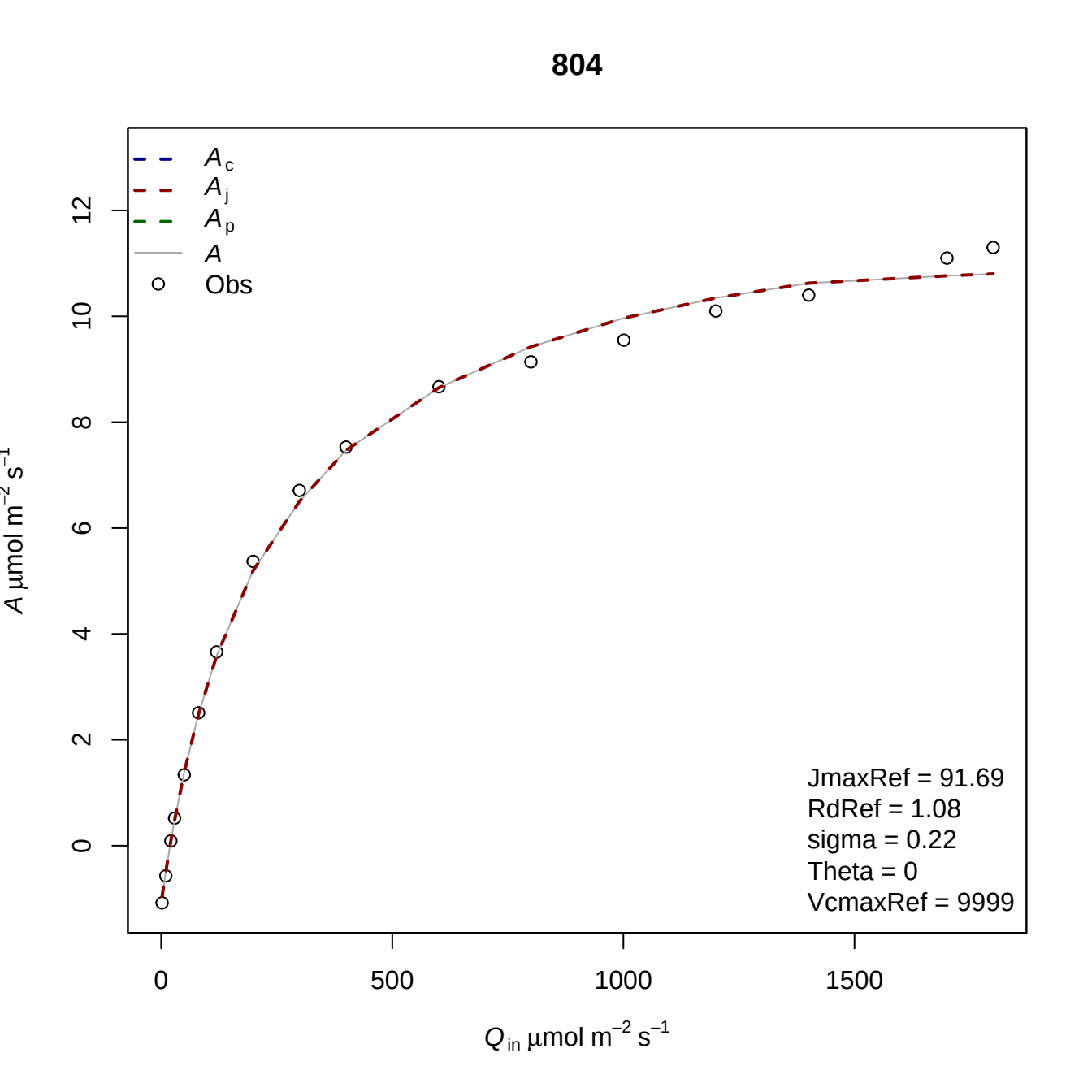

799
A_c
A_j
A_p
A
Obs
JmaxRef = 15.48
RdRef = 0.89
sigma = 0.2
Theta = 0
VcmaxRef = 9999
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

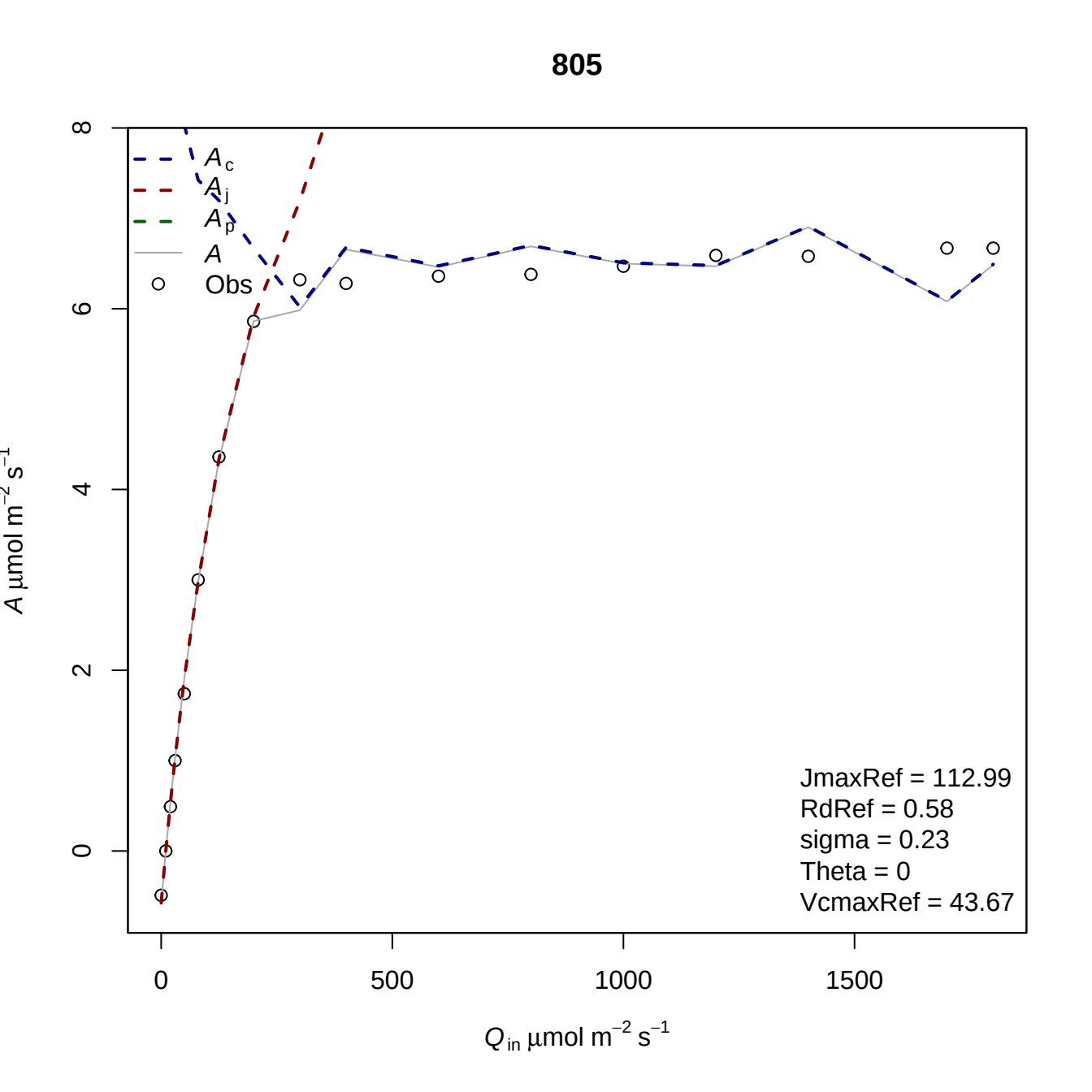

800
A_c
A_j
A_p
A
Obs
JmaxRef = 57.57
RdRef = 2.34
sigma = 0.11
Theta = 0
VcmaxRef = 40.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200
1400

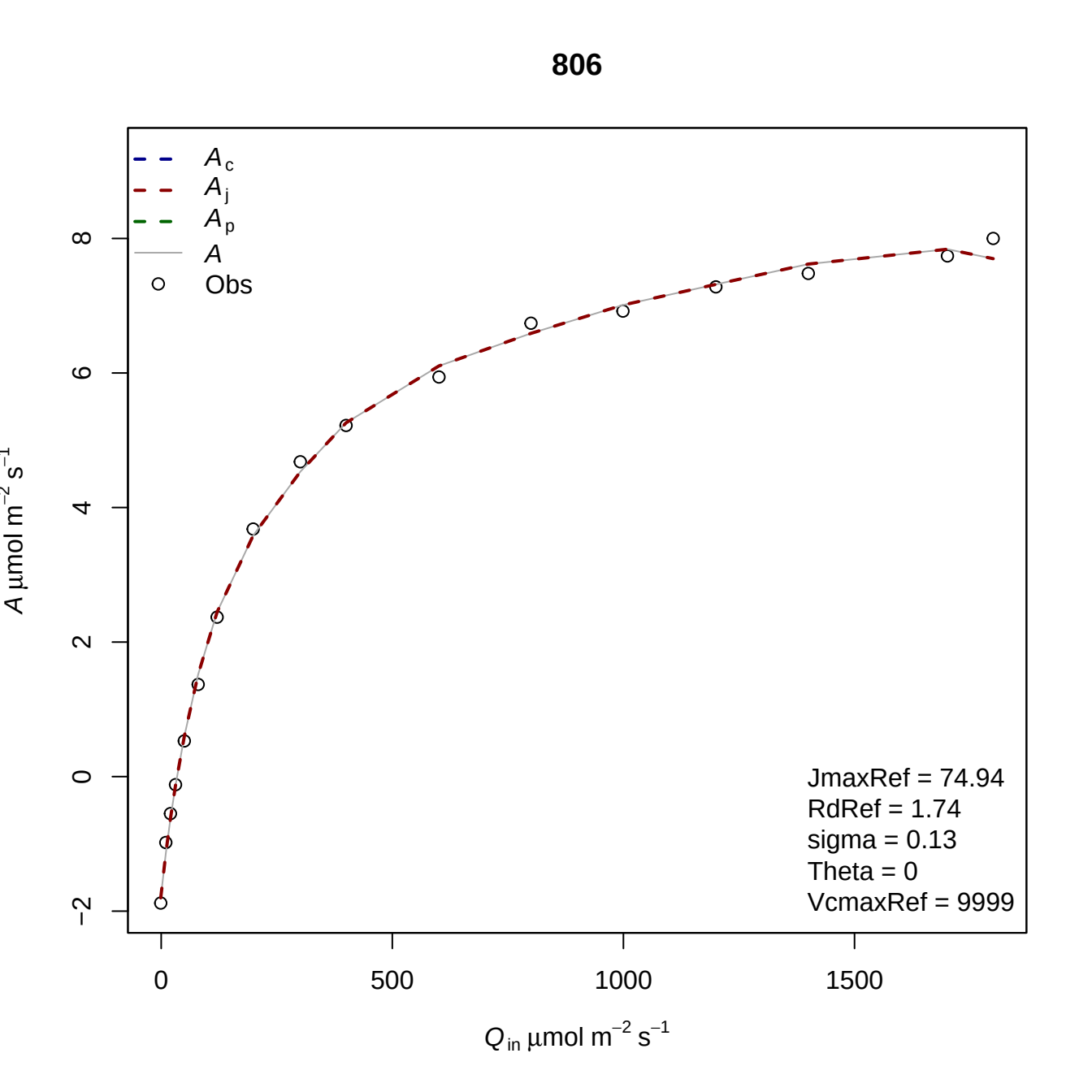

801
A_c
A_j
A_p
A
Obs
JmaxRef = 37.21
RdRef = 0.71
sigma = 0.19
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000
1200
1400

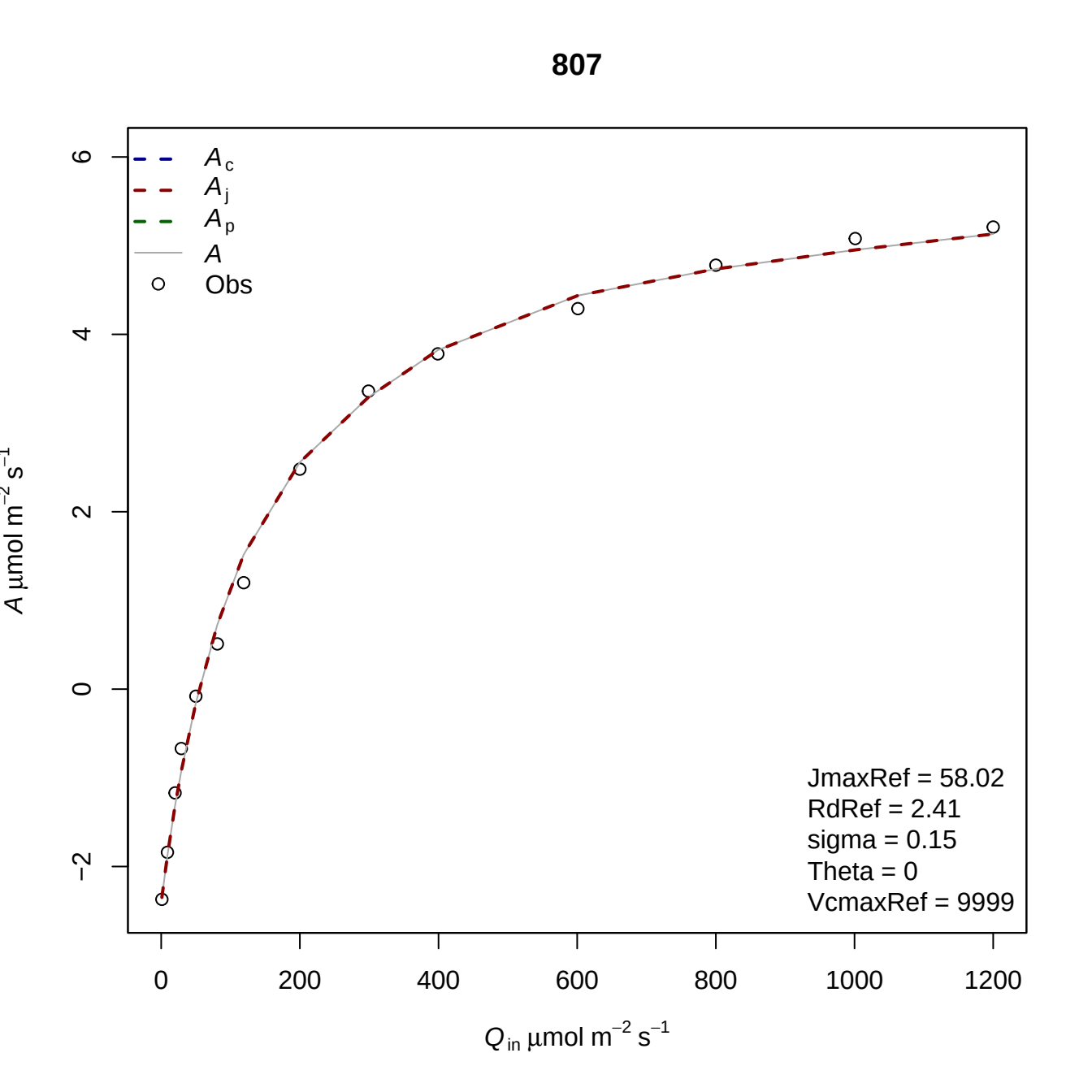

802
A_c
A_j
A_p
A
Obs
JmaxRef = 51.97
RdRef = 0.84
sigma = 0.11
Theta = 0
VcmaxRef = 39.93
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

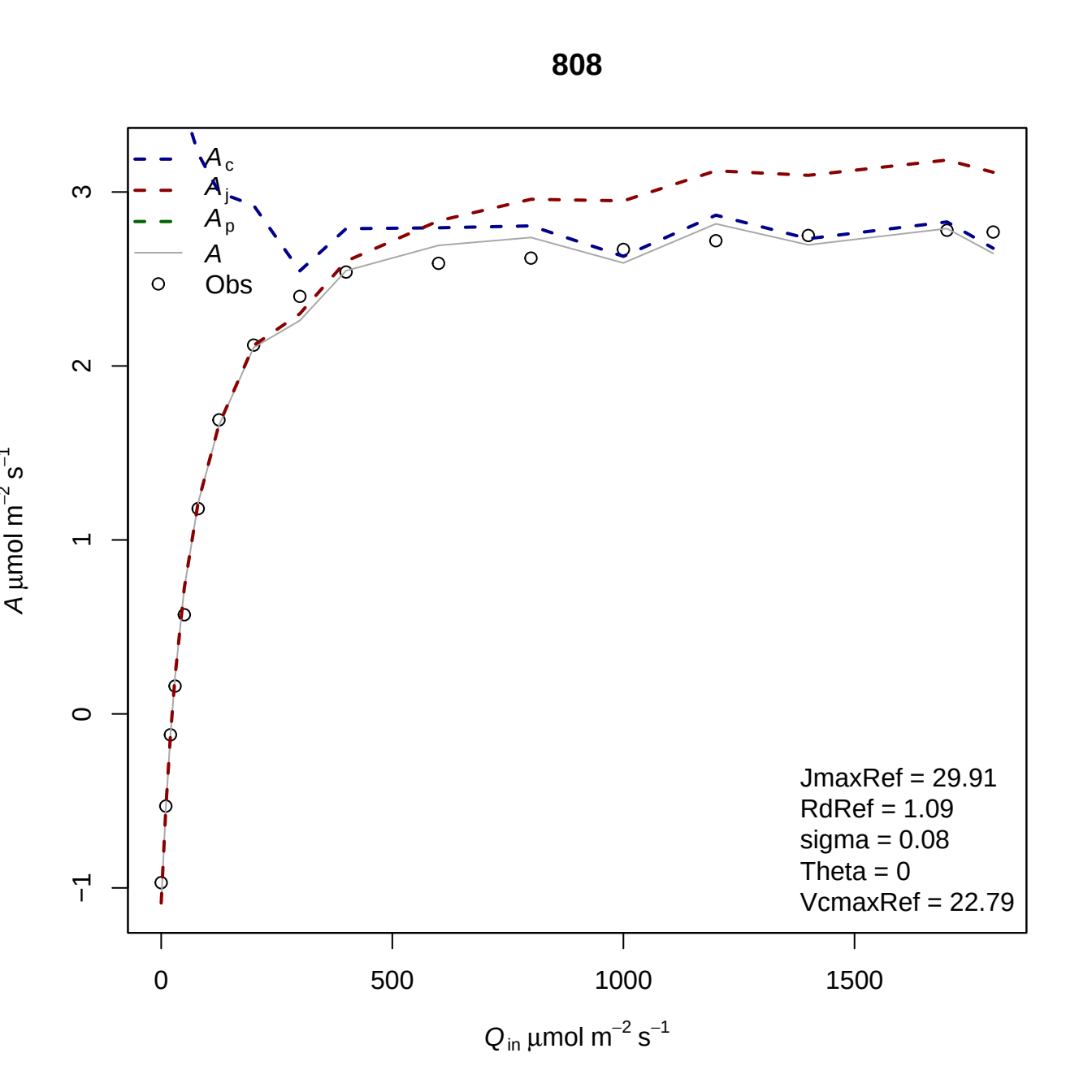

803
A_c
A_j
A_p
A
Obs
JmaxRef = 53.58
RdRef = 1.11
sigma = 0.15
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

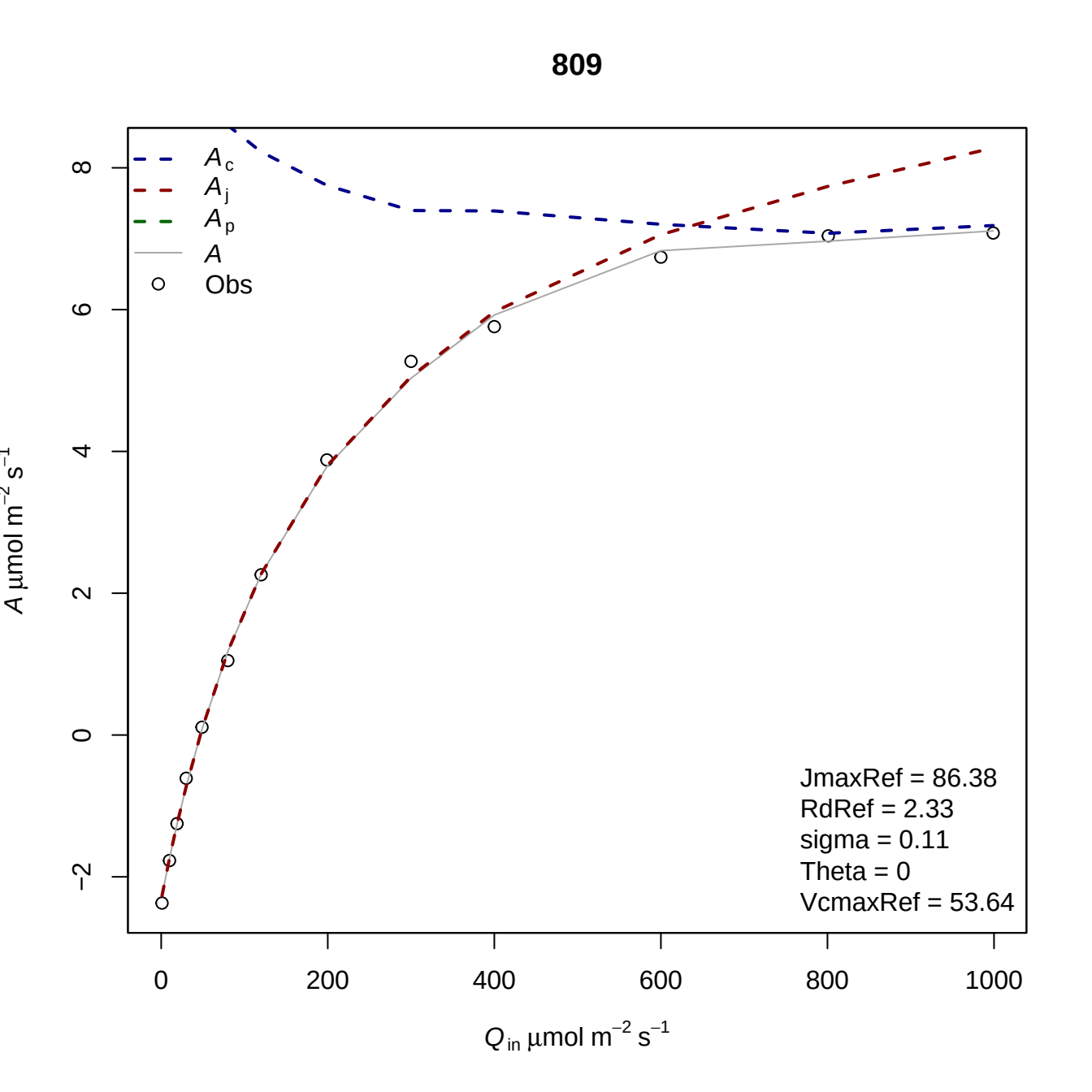

804
A_c
A_j
A_p
A
Obs
JmaxRef = 91.69
RdRef = 1.08
sigma = 0.22
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

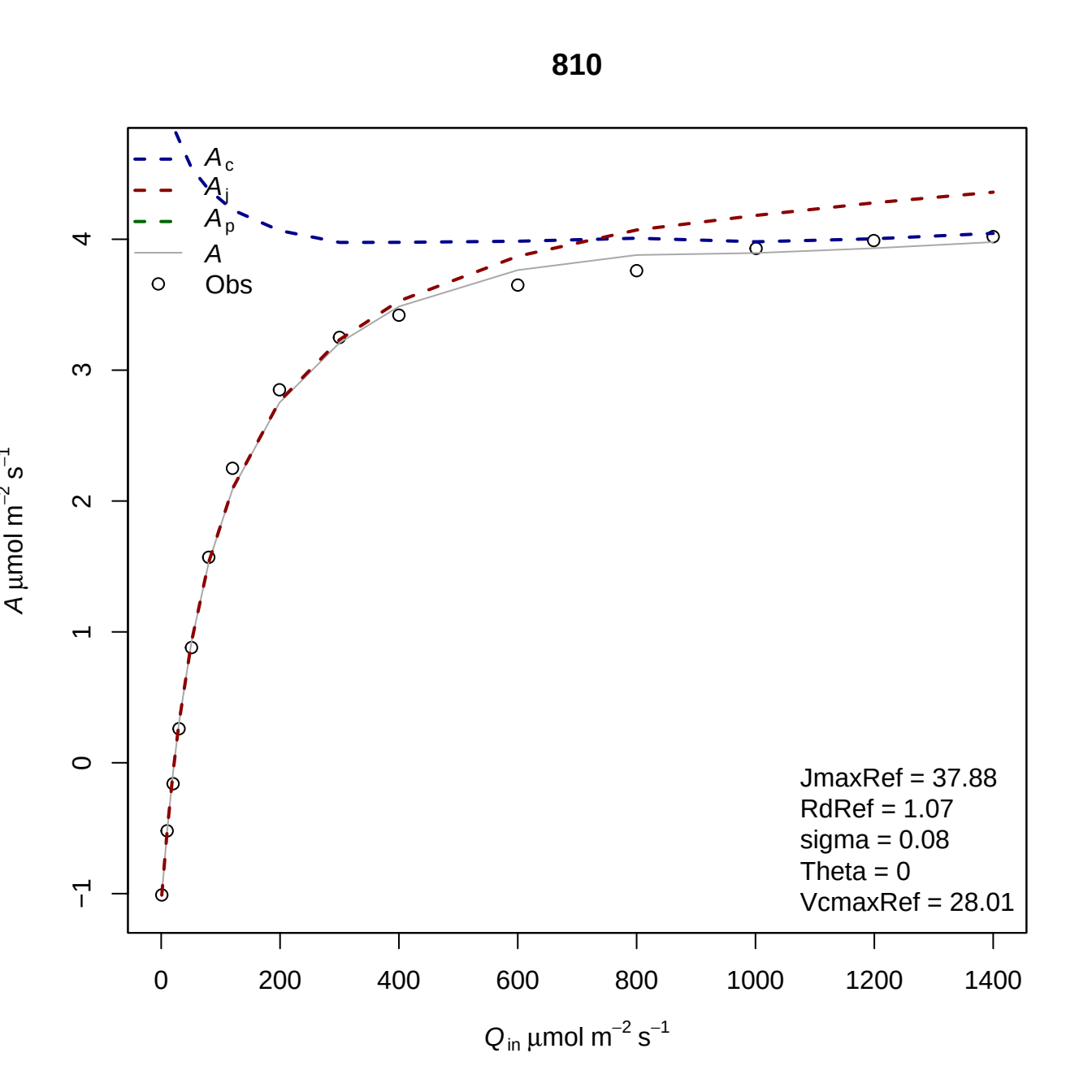

805
A_c
A_j
A_p
A
Obs
JmaxRef = 112.99
RdRef = 0.58
sigma = 0.23
Theta = 0
VcmaxRef = 43.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

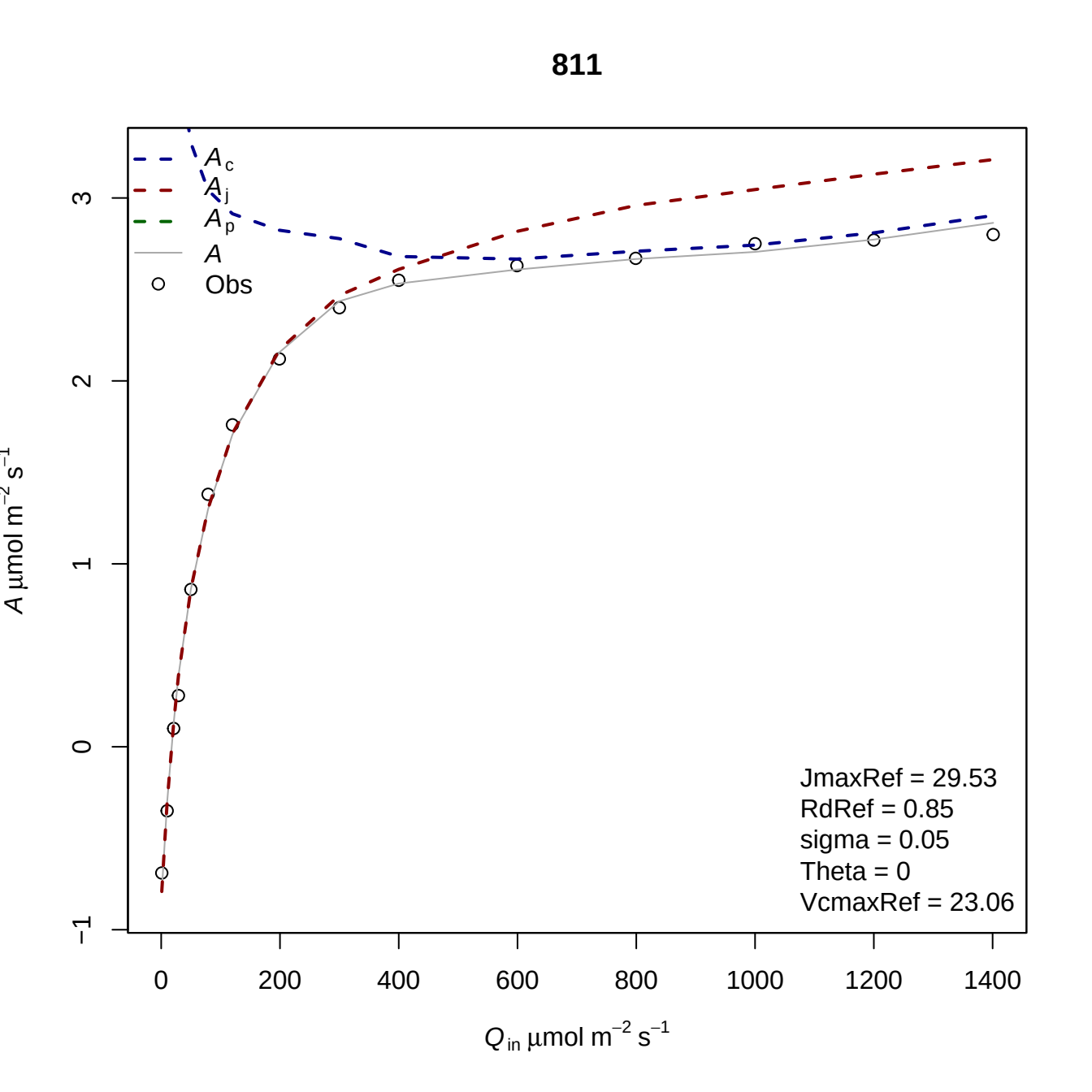

806
A_c
A_j
A_p
A
Obs
JmaxRef = 74.94
RdRef = 1.74
sigma = 0.13
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

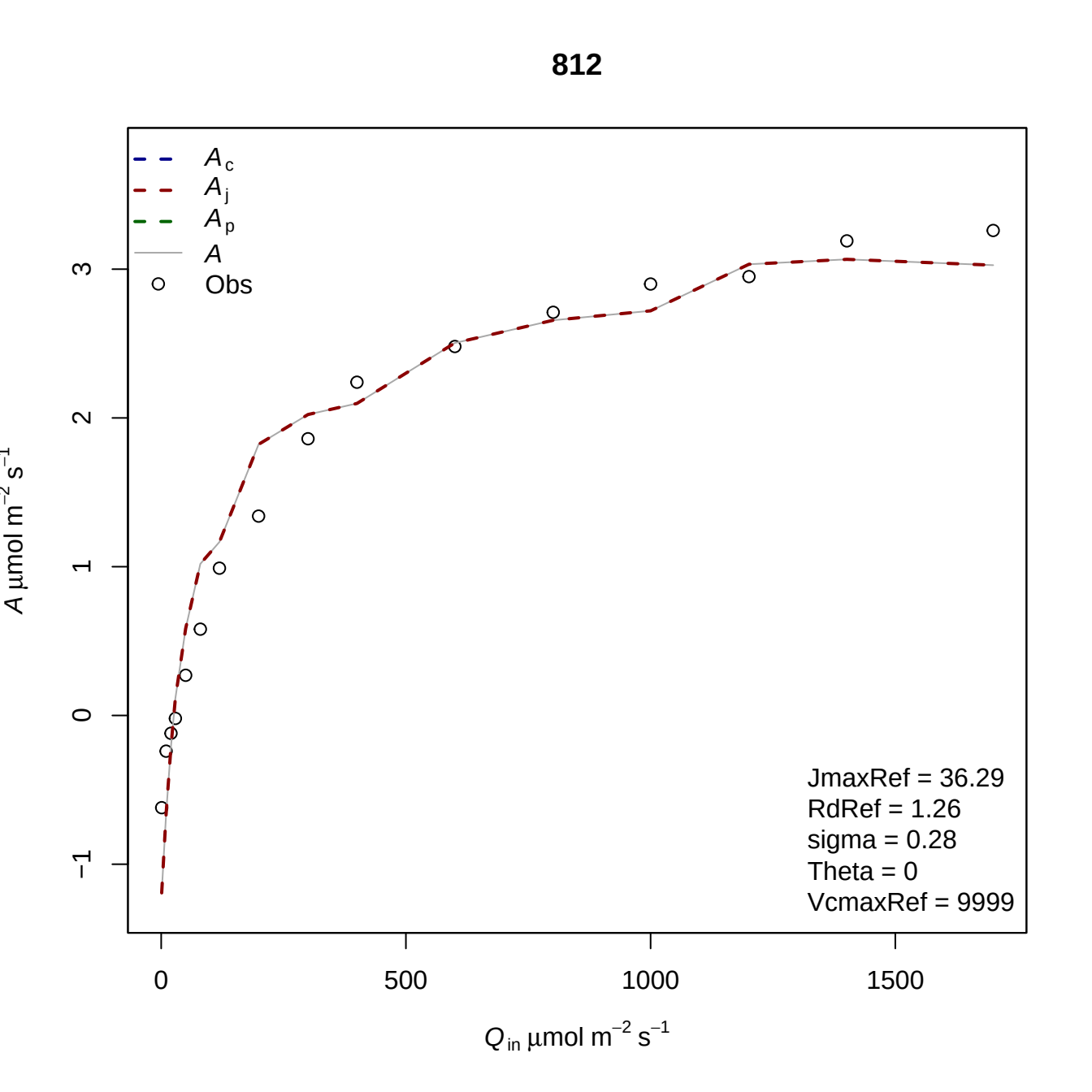

807
A_c
A_j
A_p
A
Obs
JmaxRef = 58.02
RdRef = 2.41
sigma = 0.15
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

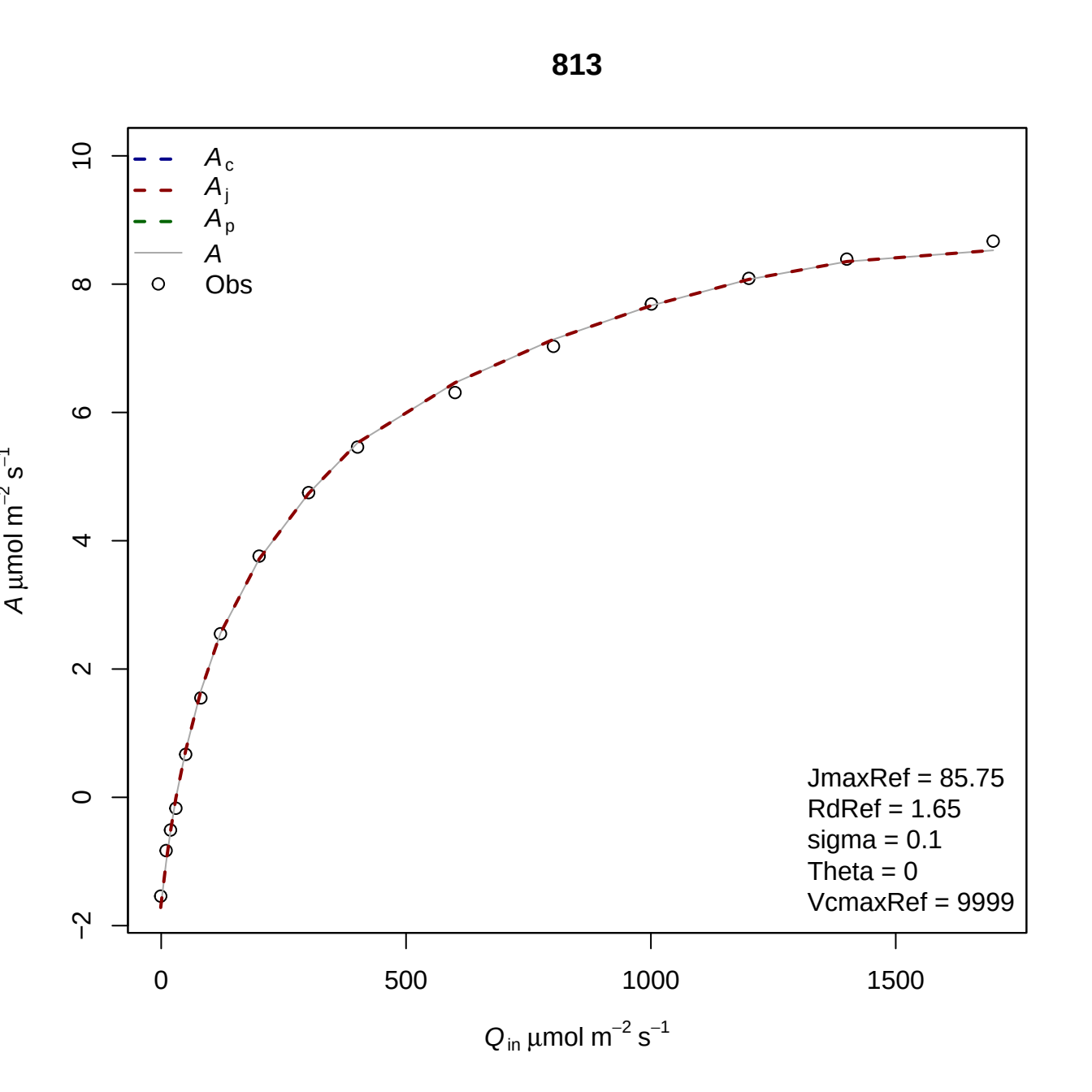

808
A_c
A_j
A_p
A
Obs
JmaxRef = 29.91
RdRef = 1.09
sigma = 0.08
Theta = 0
VcmaxRef = 22.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

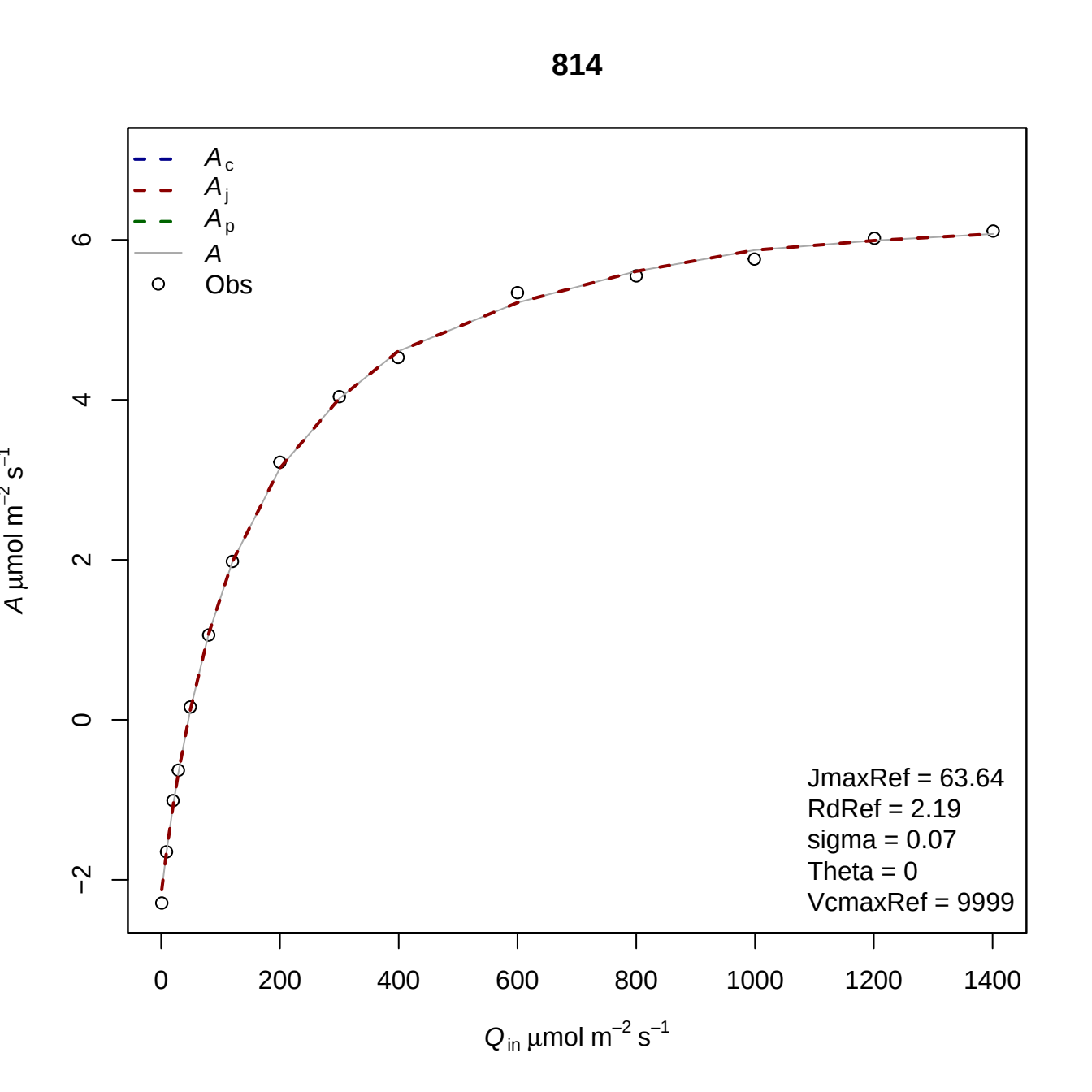

809
A_c
A_j
A_p
A
Obs
JmaxRef = 86.38
RdRef = 2.33
sigma = 0.11
Theta = 0
VcmaxRef = 53.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

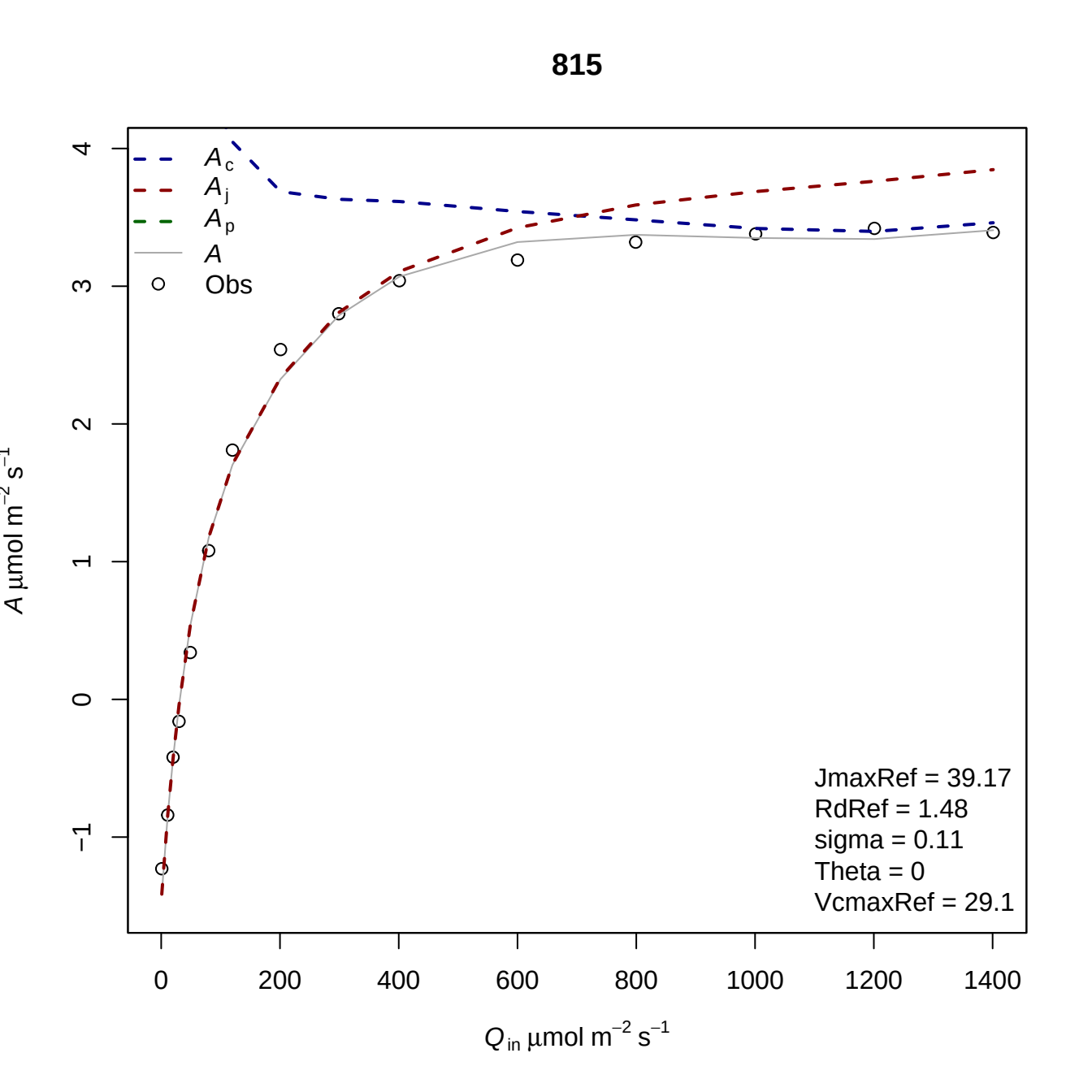

810
A_c
A_j
A_p
A
Obs
JmaxRef = 37.88
RdRef = 1.07
sigma = 0.08
Theta = 0
VcmaxRef = 28.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

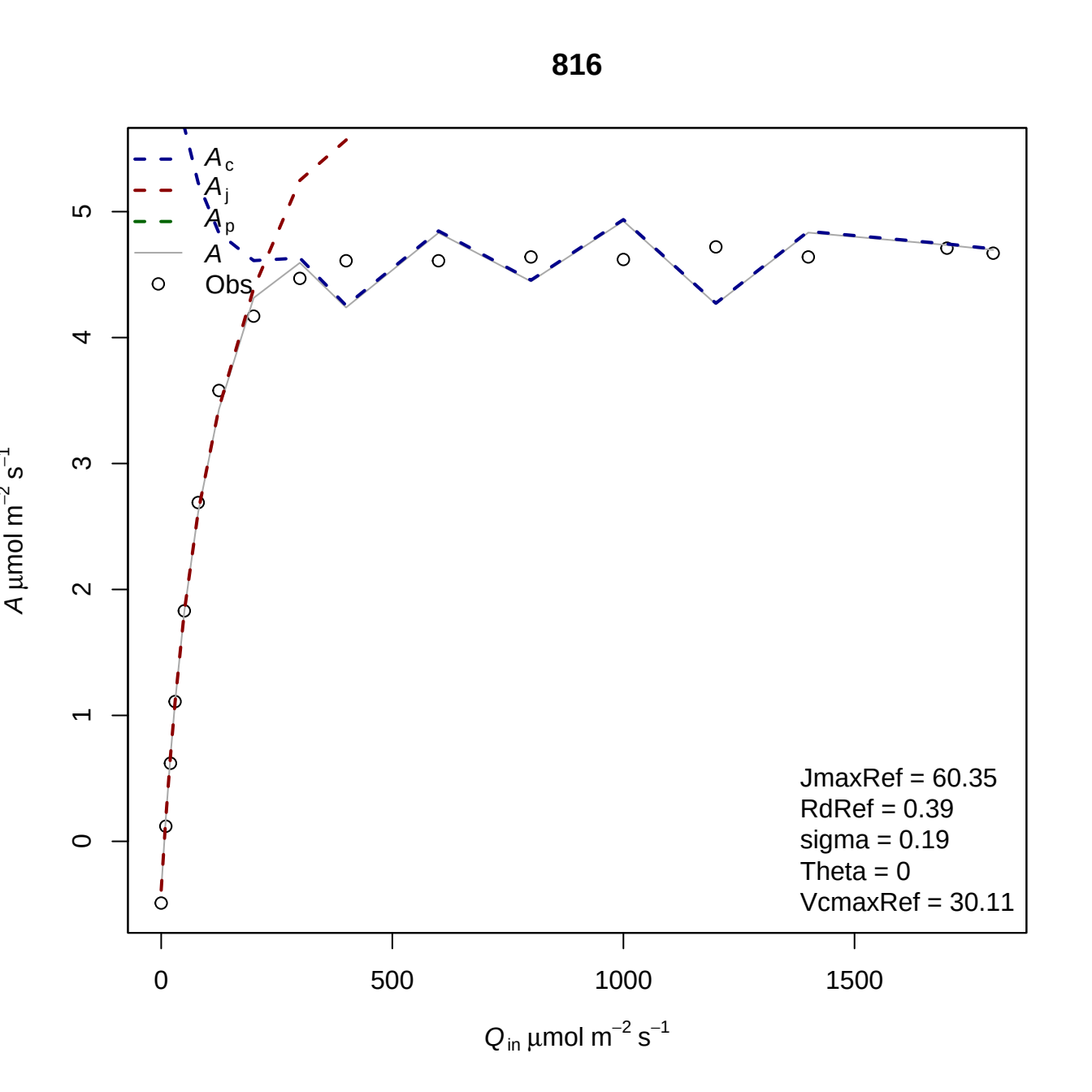

811
A_c
A_j
A_p
A
Obs
JmaxRef = 29.53
RdRef = 0.85
sigma = 0.05
Theta = 0
VcmaxRef = 23.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

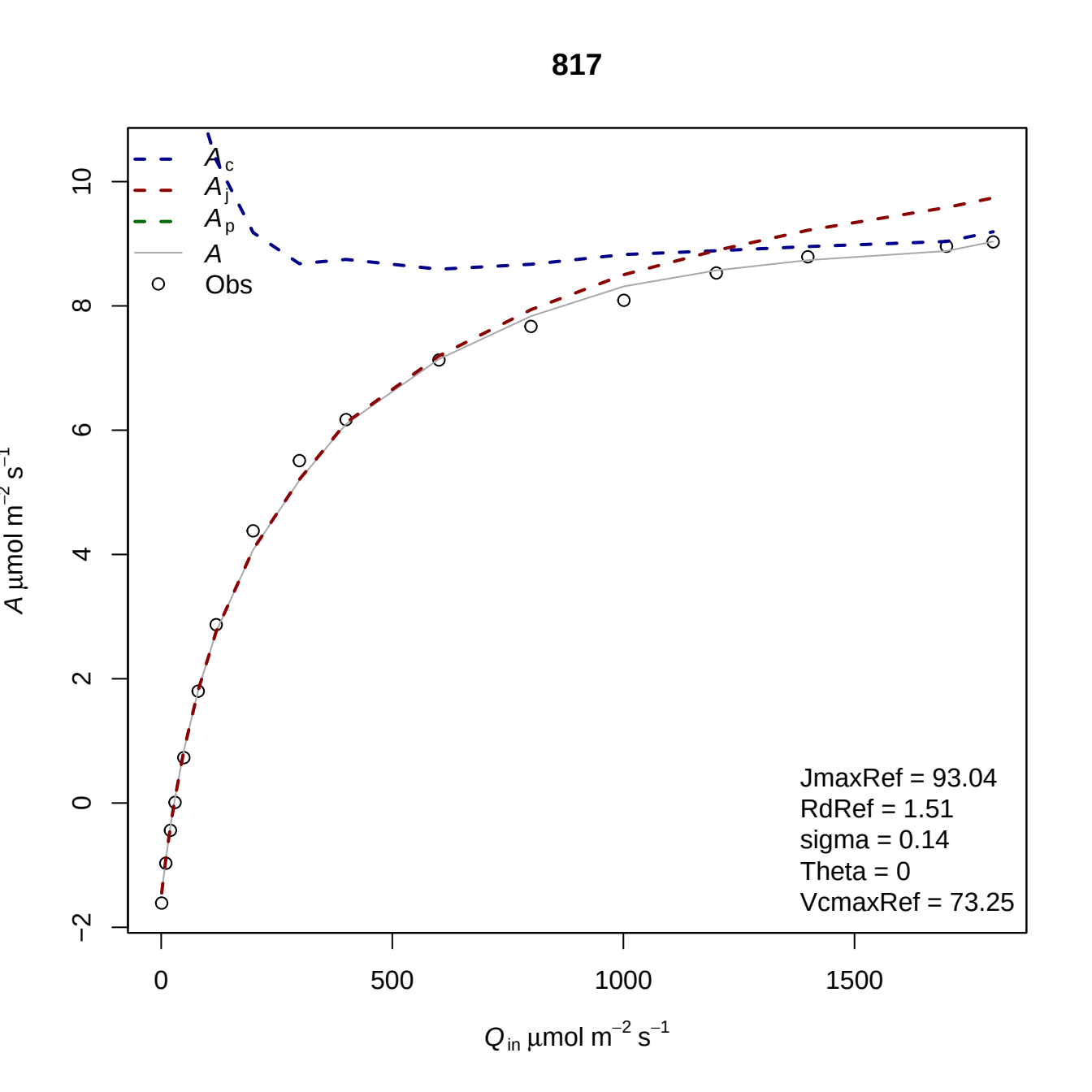

812
A_c
A_j
A_p
A
Obs
JmaxRef = 36.29
RdRef = 1.26
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

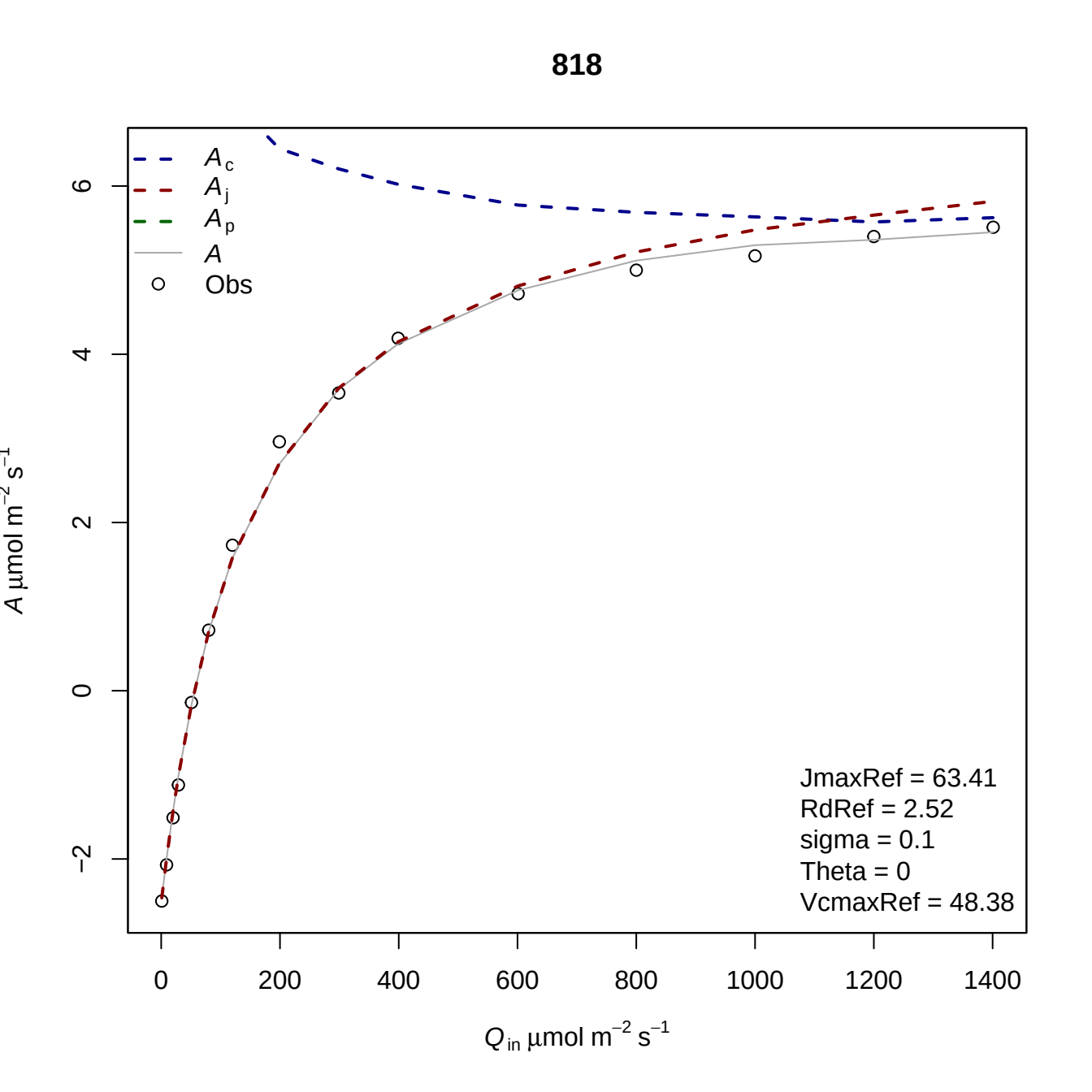

813
A_c
A_j
A_p
A
Obs
JmaxRef = 85.75
RdRef = 1.65
sigma = 0.1
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

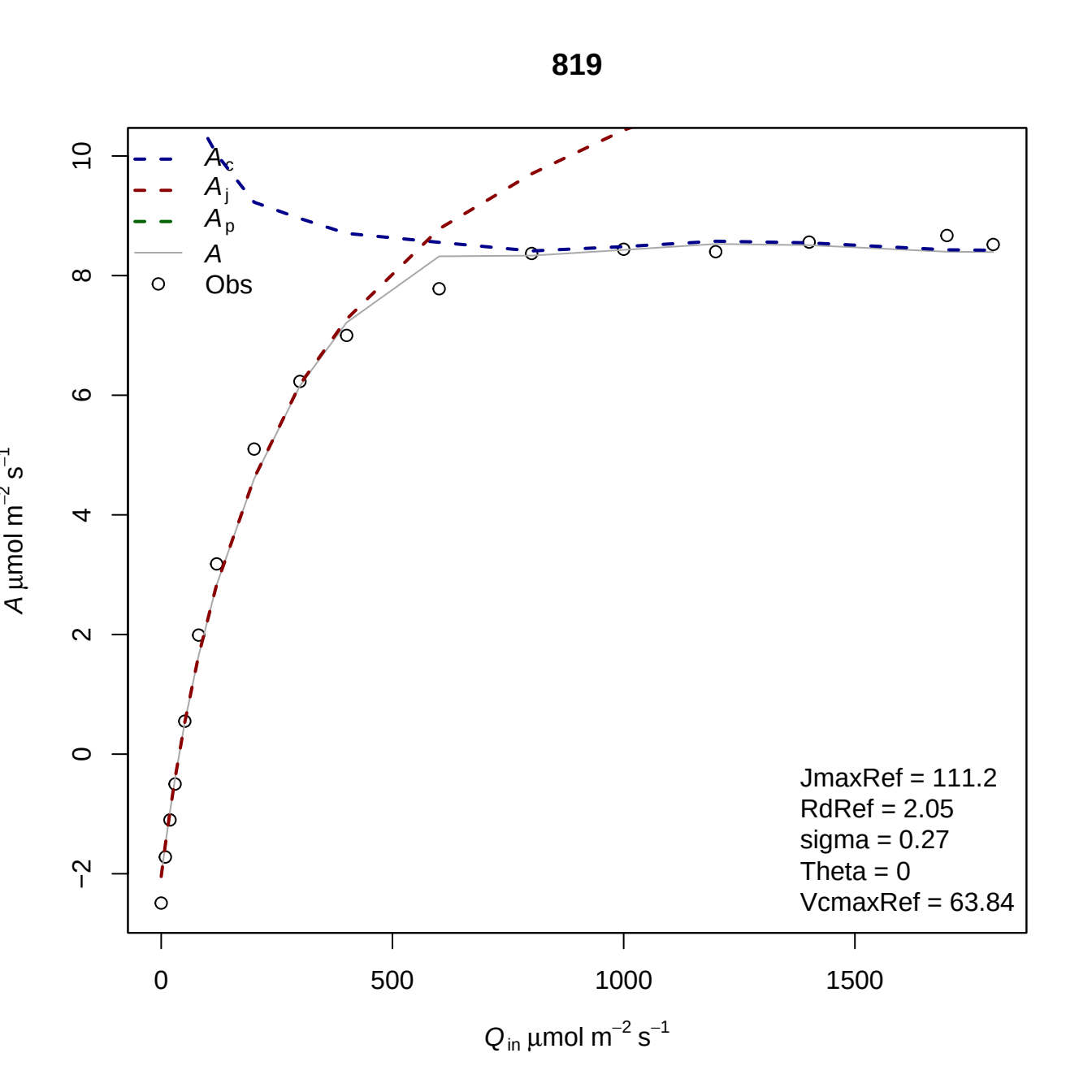

814
A_c
A_j
A_p
A
Obs
JmaxRef = 63.64
RdRef = 2.19
sigma = 0.07
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

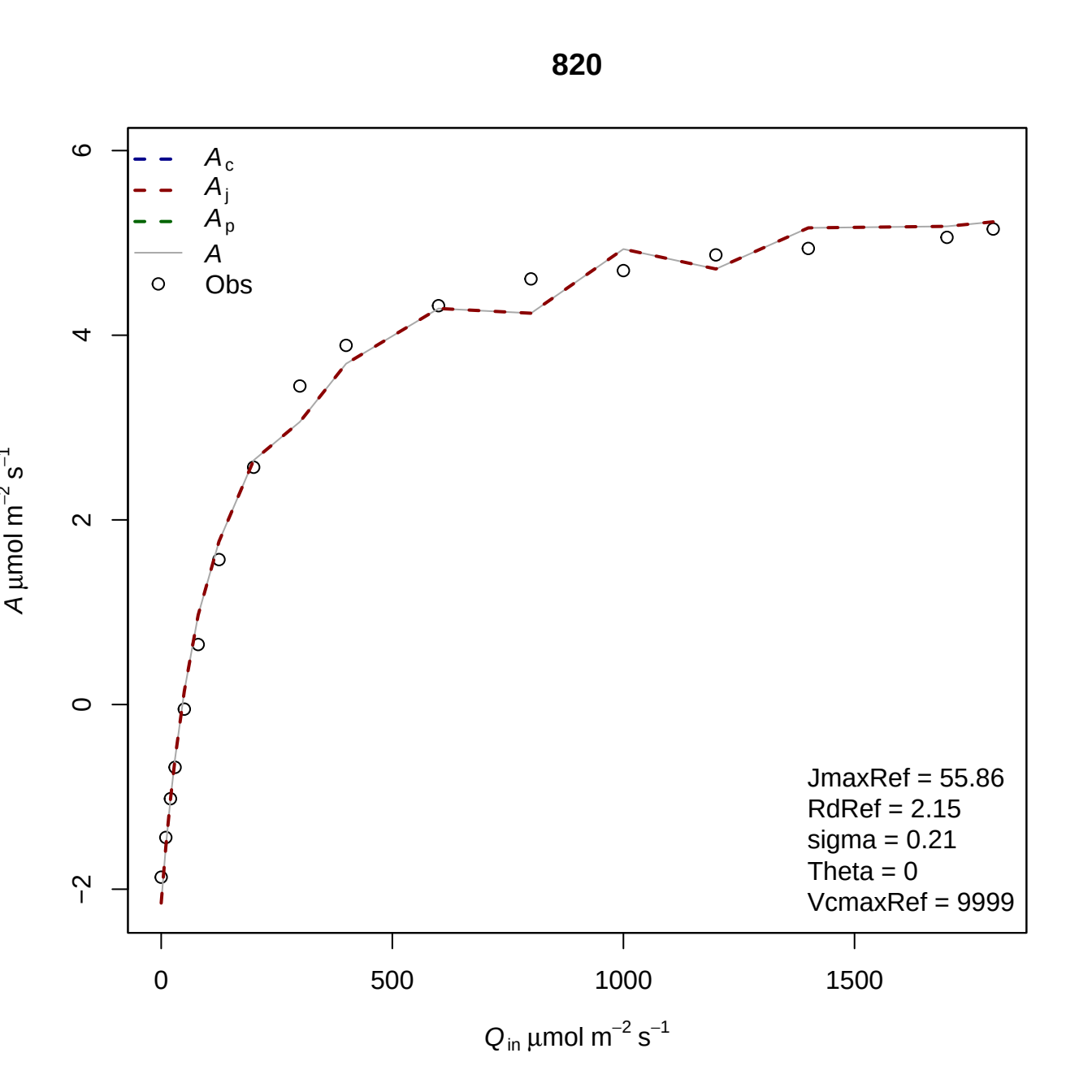

815
A_c
A_j
A_p
A
Obs
JmaxRef = 39.17
RdRef = 1.48
sigma = 0.11
Theta = 0
VcmaxRef = 29.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

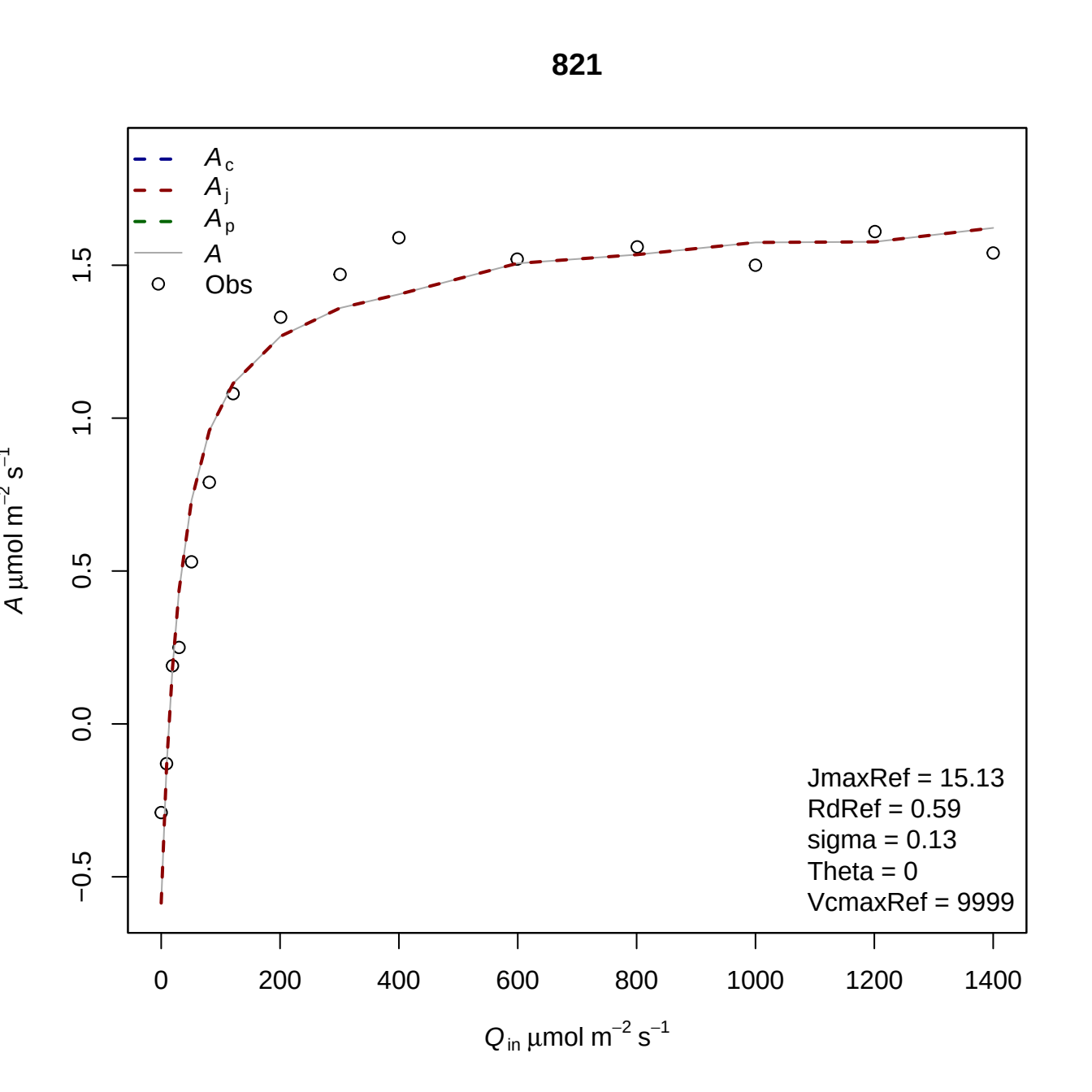

816
A_c
A_j
A_p
A
Obs
JmaxRef = 60.35
RdRef = 0.39
sigma = 0.19
Theta = 0
VcmaxRef = 30.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

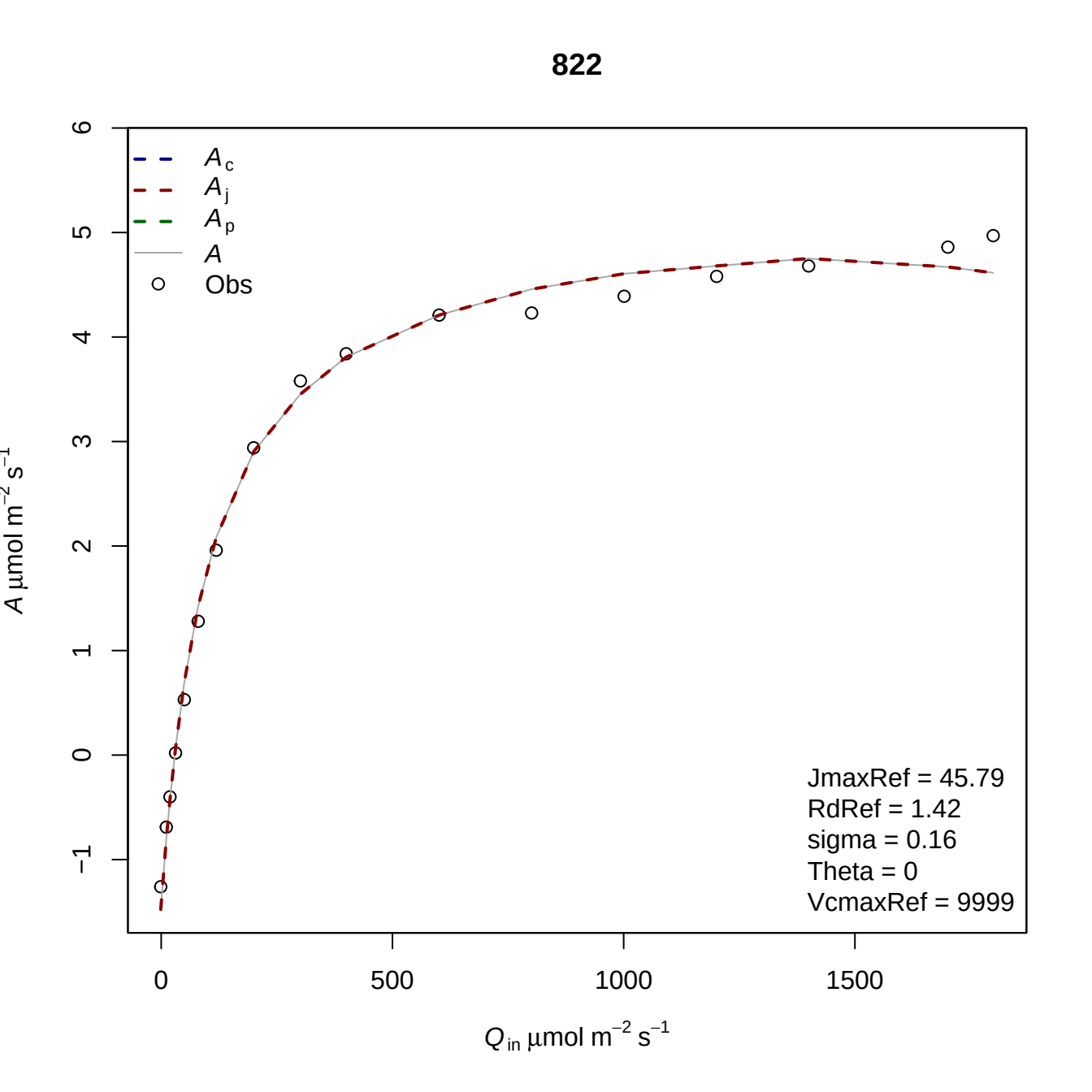

817
A_c
A_j
A_p
A
Obs
JmaxRef = 93.04
RdRef = 1.51
sigma = 0.14
Theta = 0
VcmaxRef = 73.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

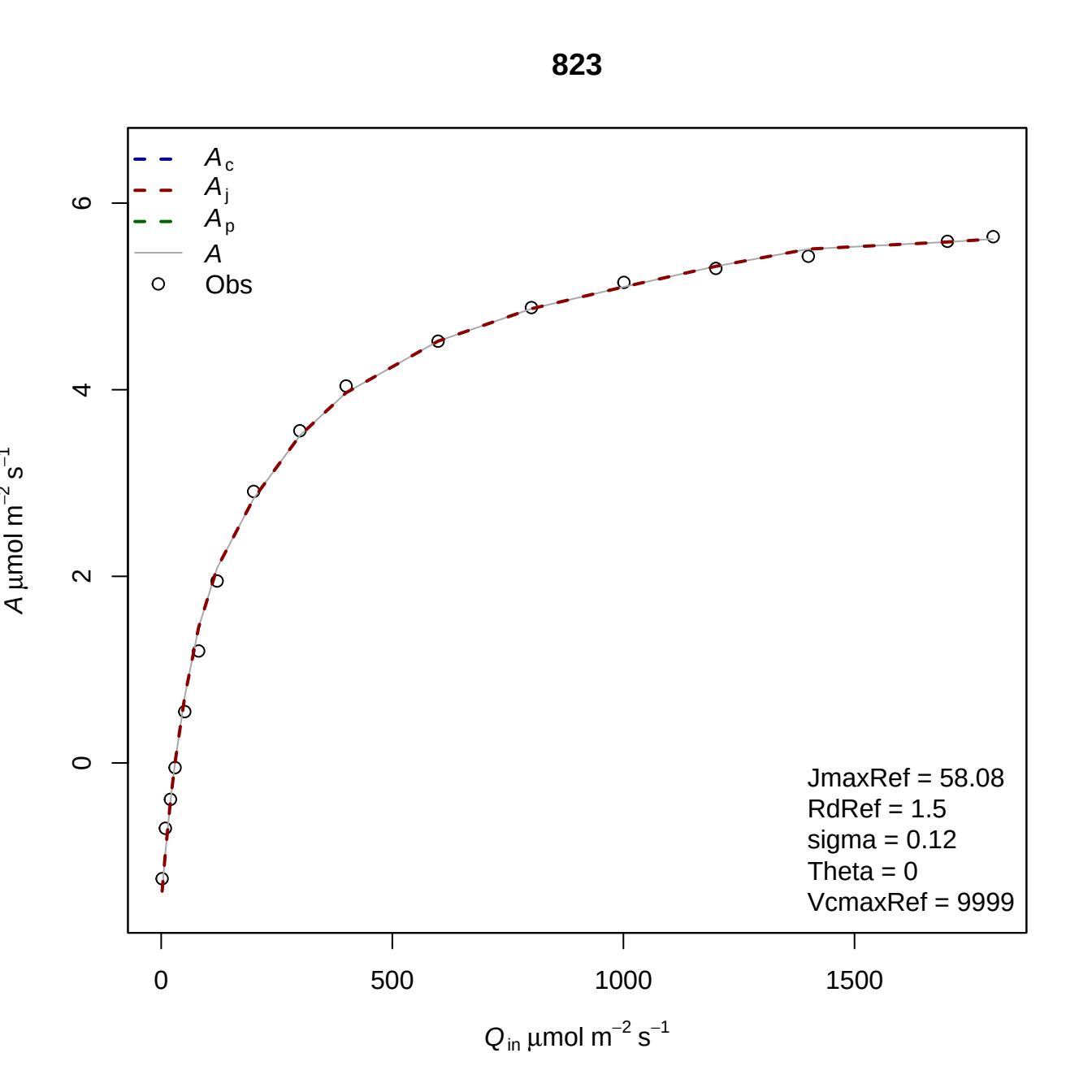

818
A_c
A_j
A_p
A
Obs
JmaxRef = 63.41
RdRef = 2.52
sigma = 0.1
Theta = 0
VcmaxRef = 48.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

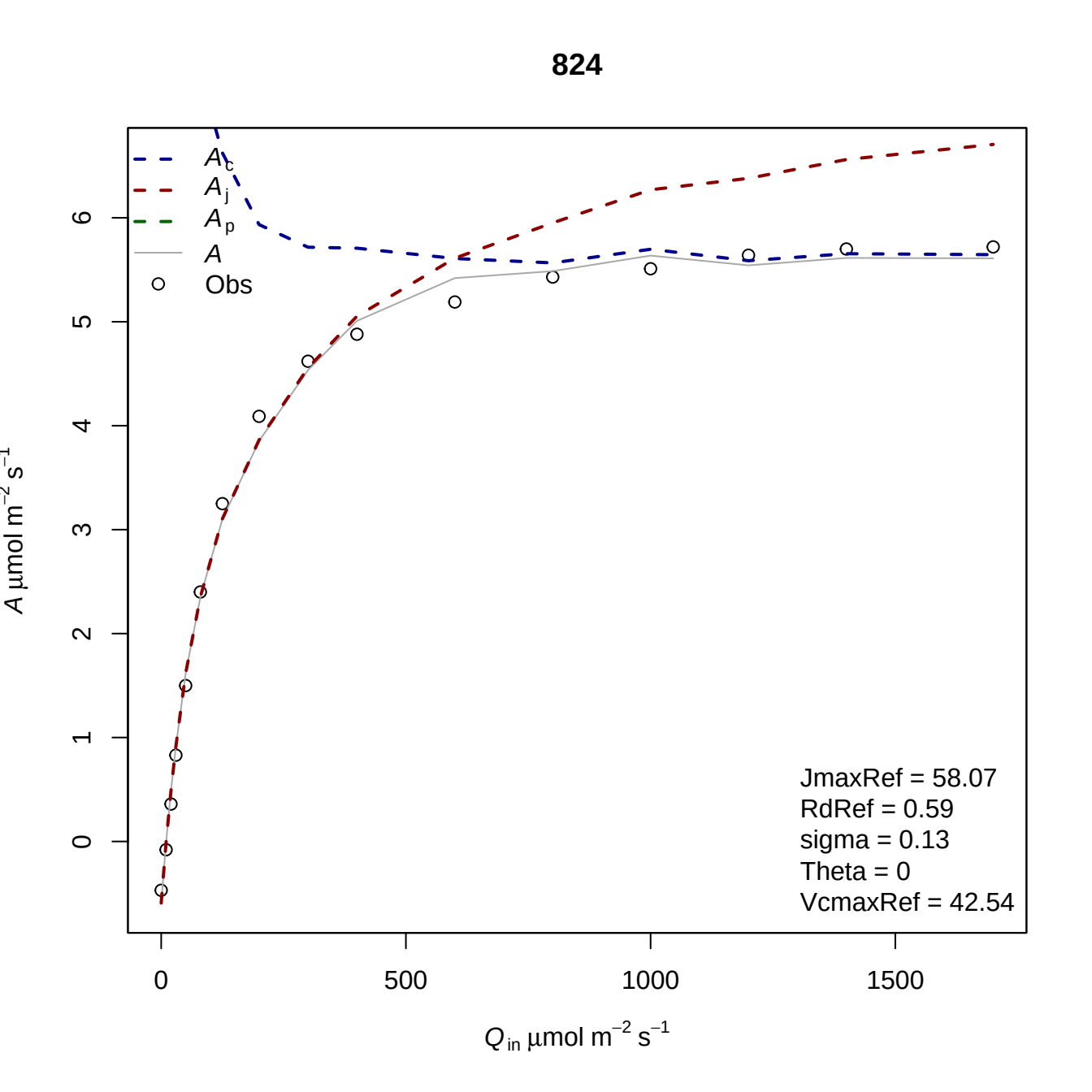

819
A_c
A_j
A_p
A
Obs
JmaxRef = 111.2
RdRef = 2.05
sigma = 0.27
Theta = 0
VcmaxRef = 63.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

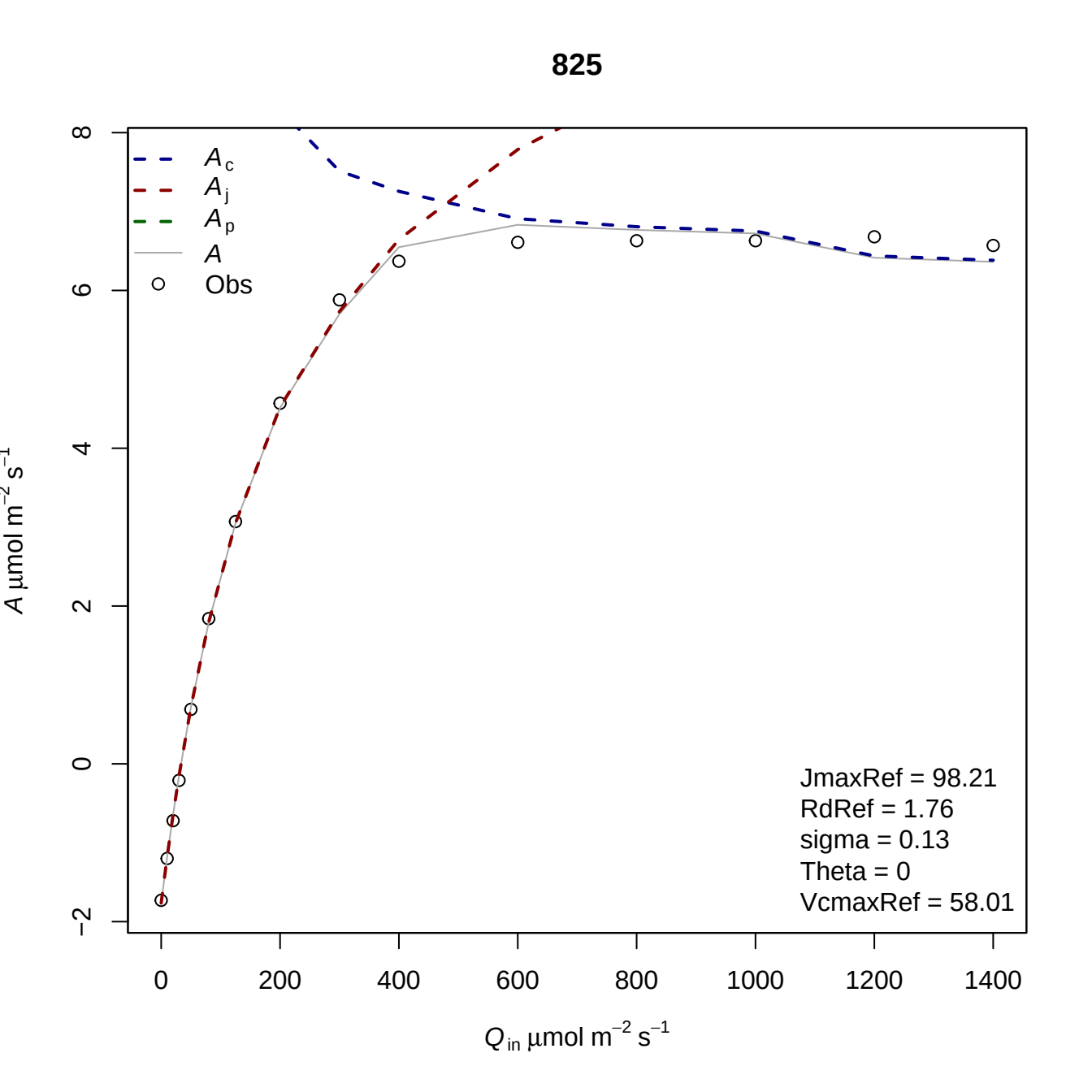

820
A_c
A_j
A_p
A
Obs
JmaxRef = 55.86
RdRef = 2.15
sigma = 0.21
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

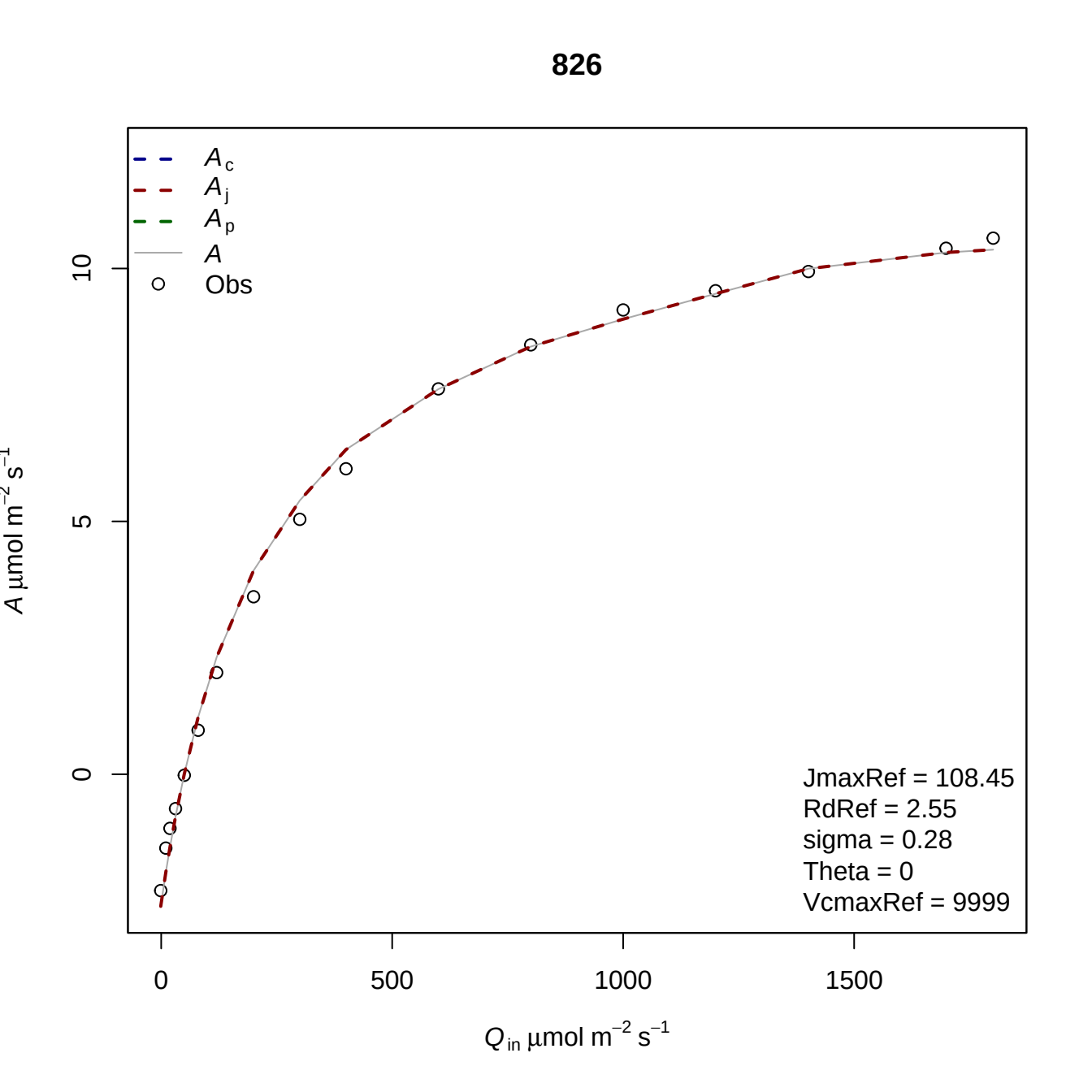

821
A_c
A_j
A_p
A
Obs
JmaxRef = 15.13
RdRef = 0.59
sigma = 0.13
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
0
200
400
600
800
1000
1200
1400

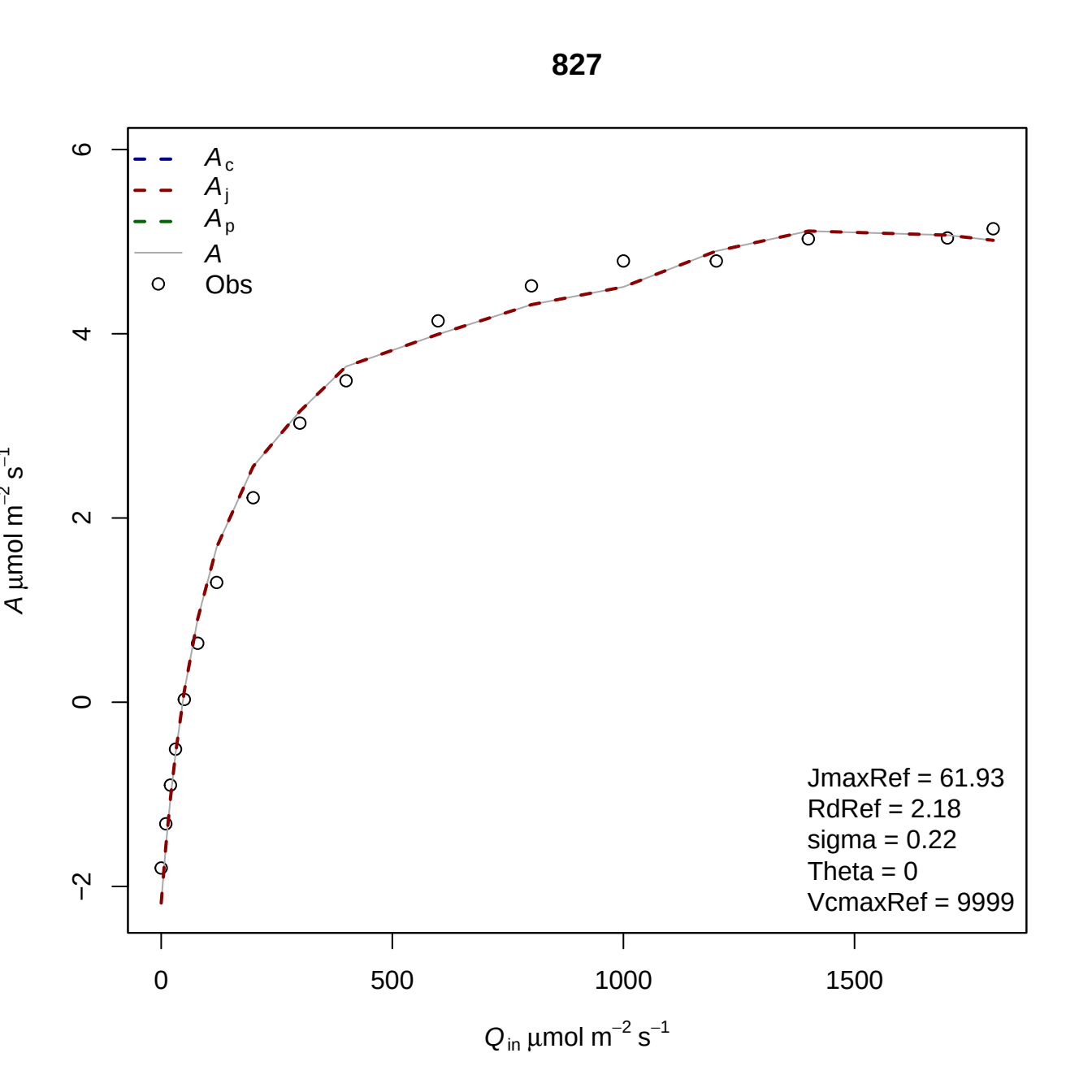

822
A_c
A_j
A_p
A
Obs
JmaxRef = 45.79
RdRef = 1.42
sigma = 0.16
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

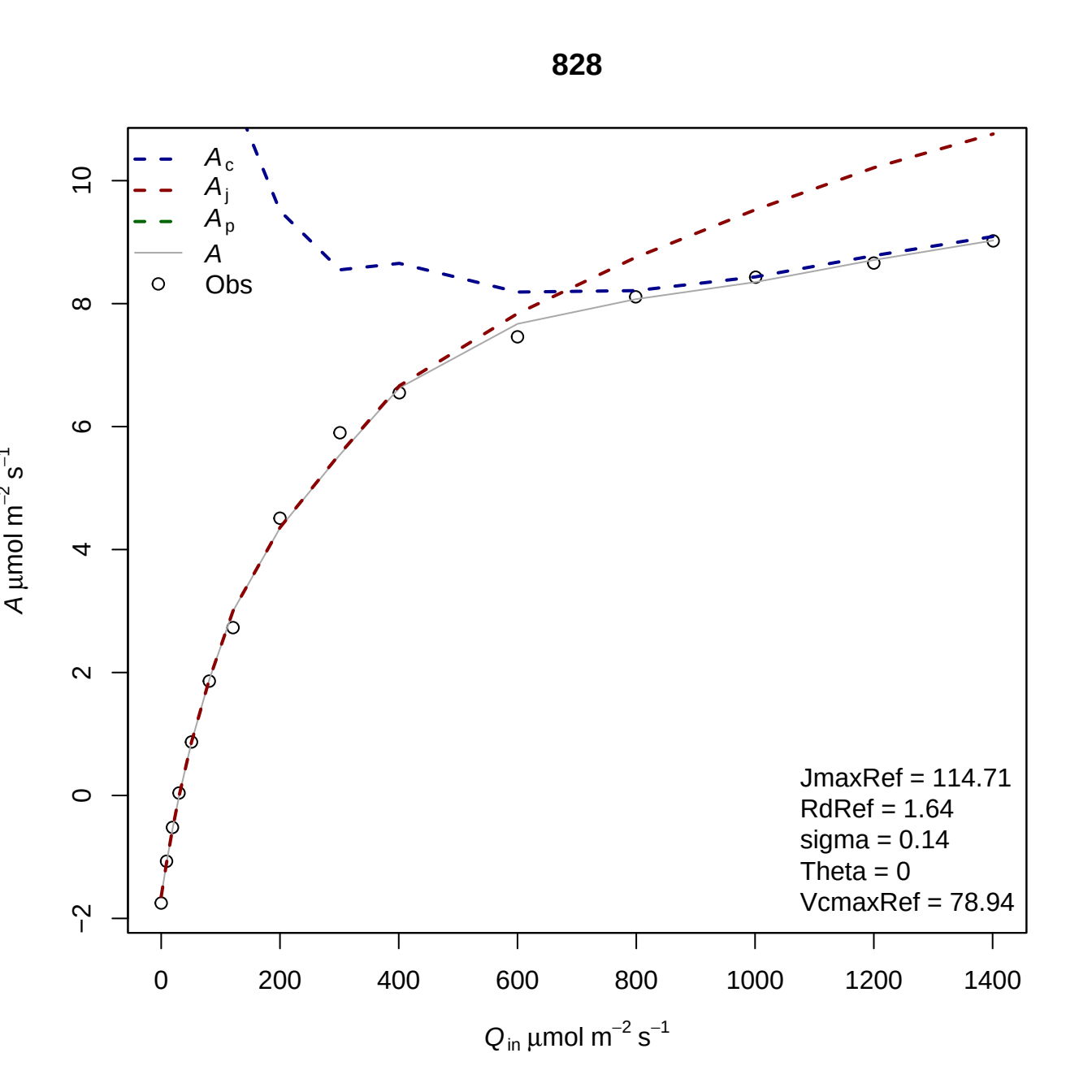

823
A_c
A_j
A_p
A
Obs
JmaxRef = 58.08
RdRef = 1.5
sigma = 0.12
Theta = 0
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

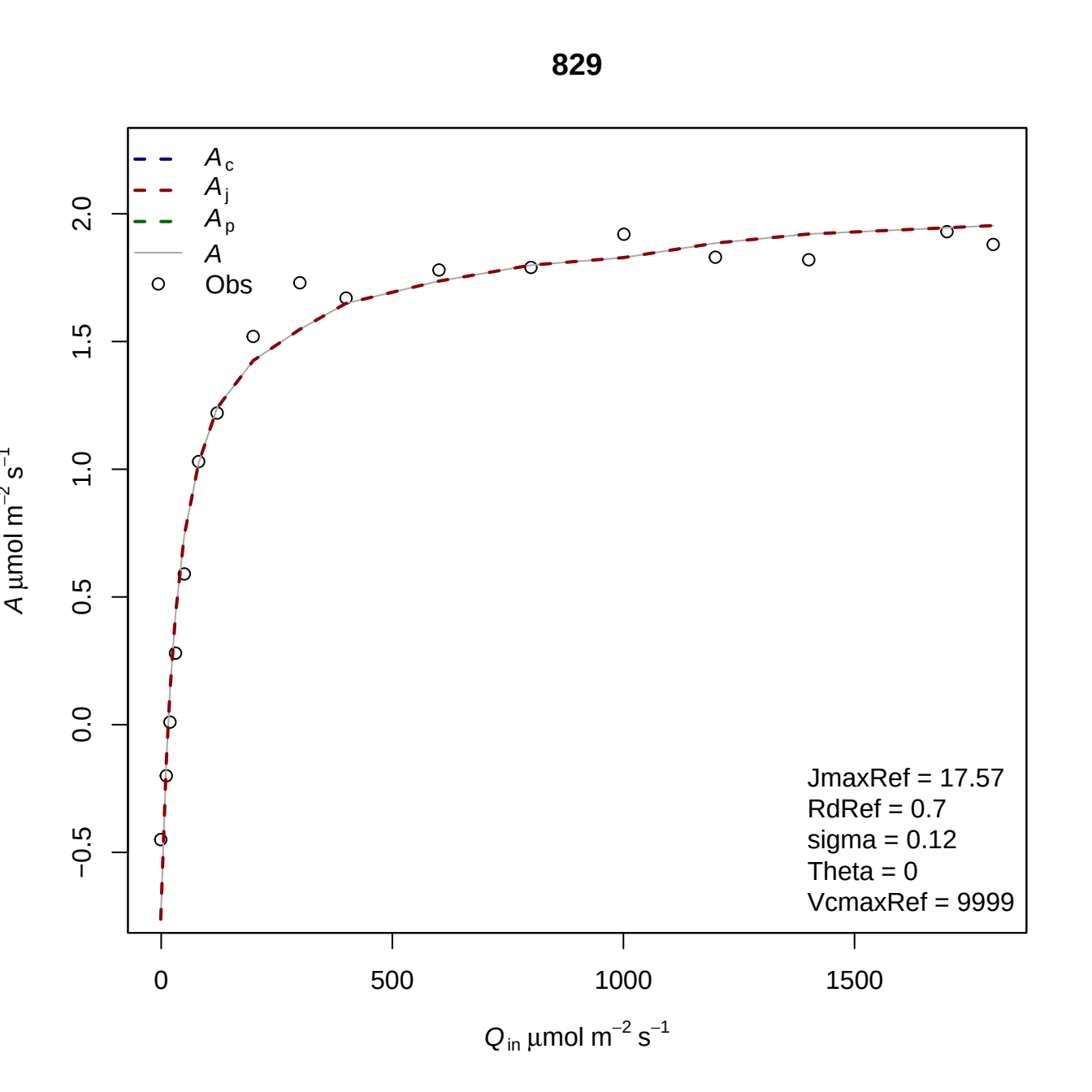

824
A_c
A_j
A_p
A
Obs
JmaxRef = 58.07
RdRef = 0.59
sigma = 0.13
Theta = 0
VcmaxRef = 42.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

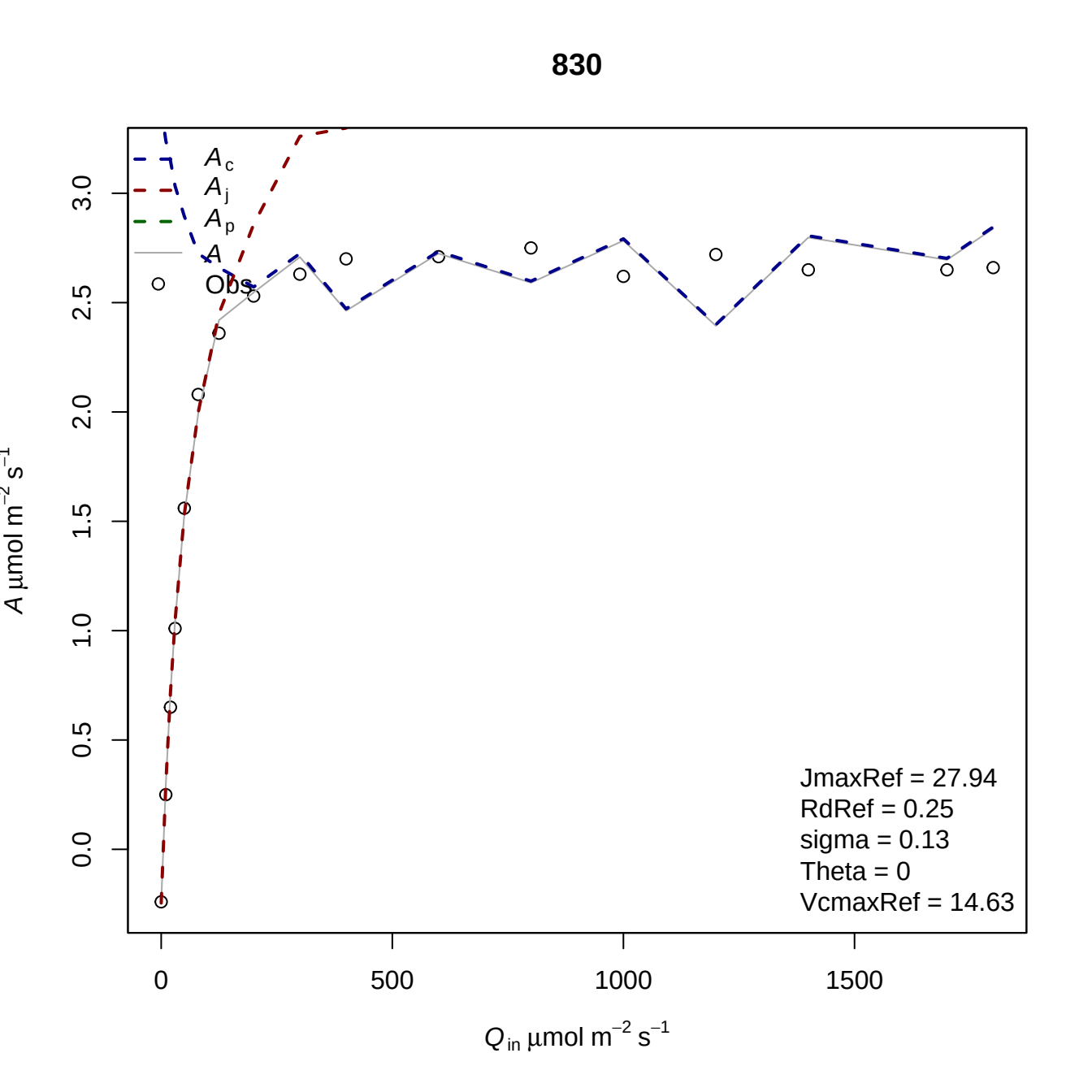

825
A_c
A_j
A_p
A
Obs
JmaxRef = 98.21
RdRef = 1.76
sigma = 0.13
Theta = 0
VcmaxRef = 58.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

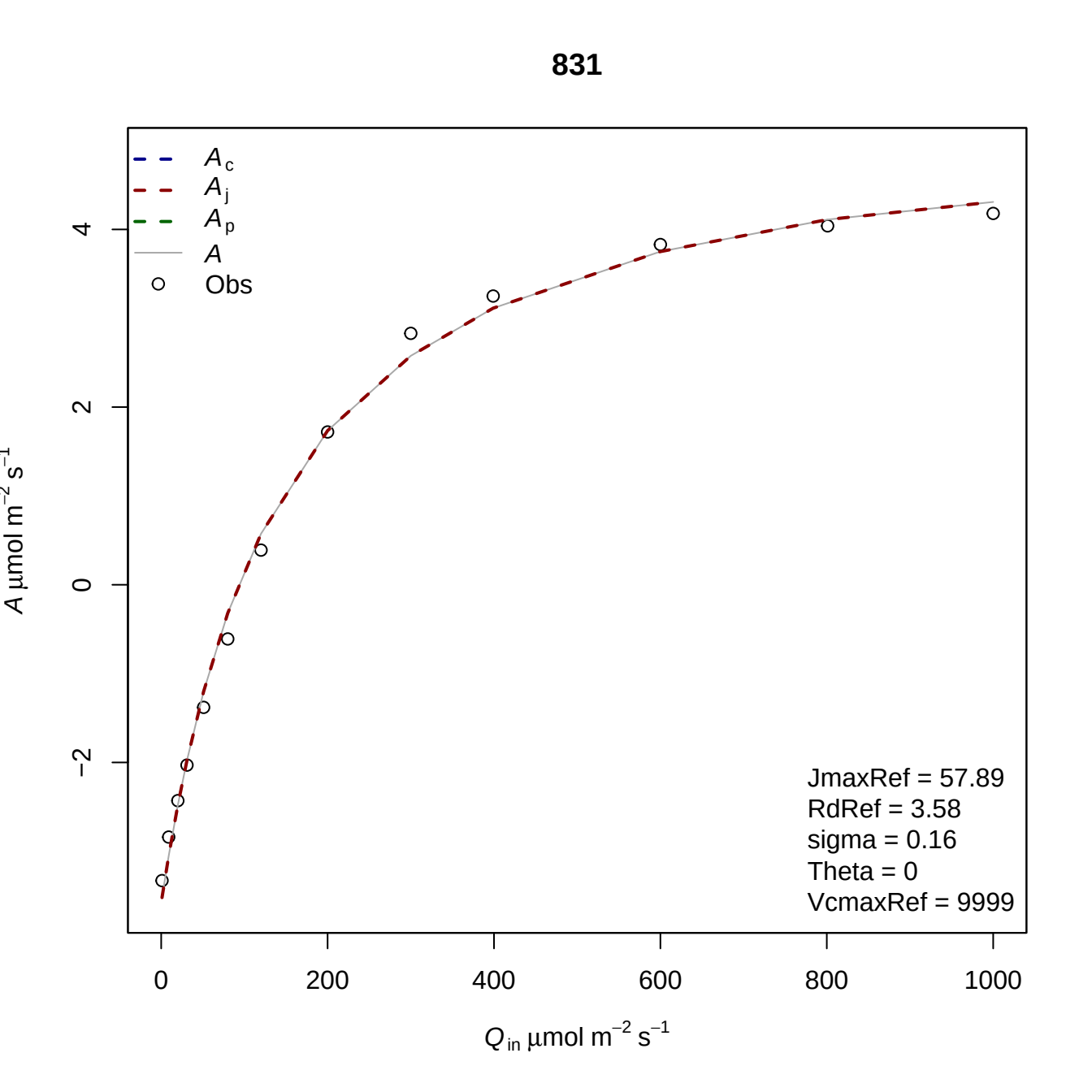

826
A_c
A_j
A_p
A
Obs
JmaxRef = 108.45
RdRef = 2.55
sigma = 0.28
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

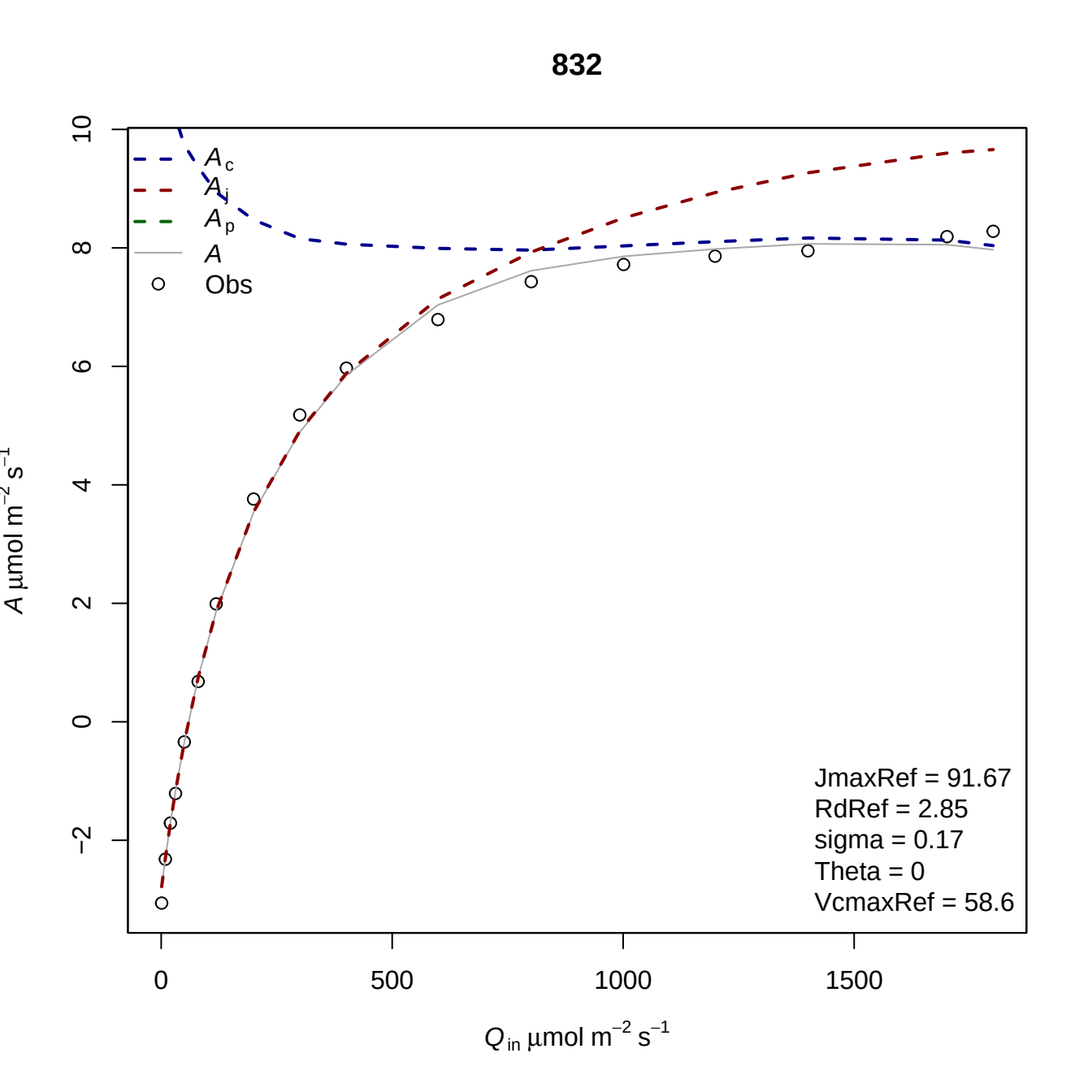

827
A_c
A_j
A_p
A
Obs
JmaxRef = 61.93
RdRef = 2.18
sigma = 0.22
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

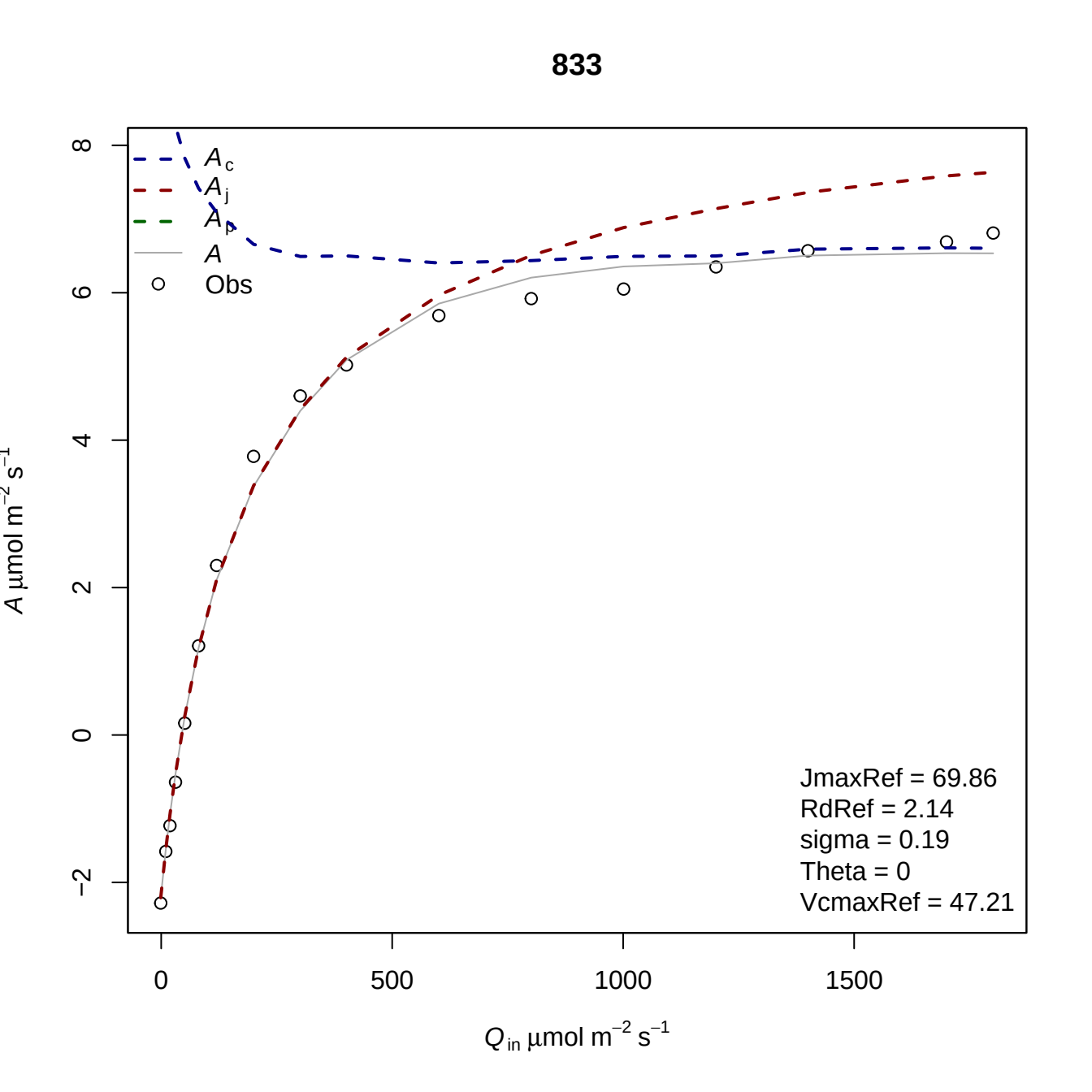

828
A_c
A_j
A_p
A
Obs
JmaxRef = 114.71
RdRef = 1.64
sigma = 0.14
Theta = 0
VcmaxRef = 78.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

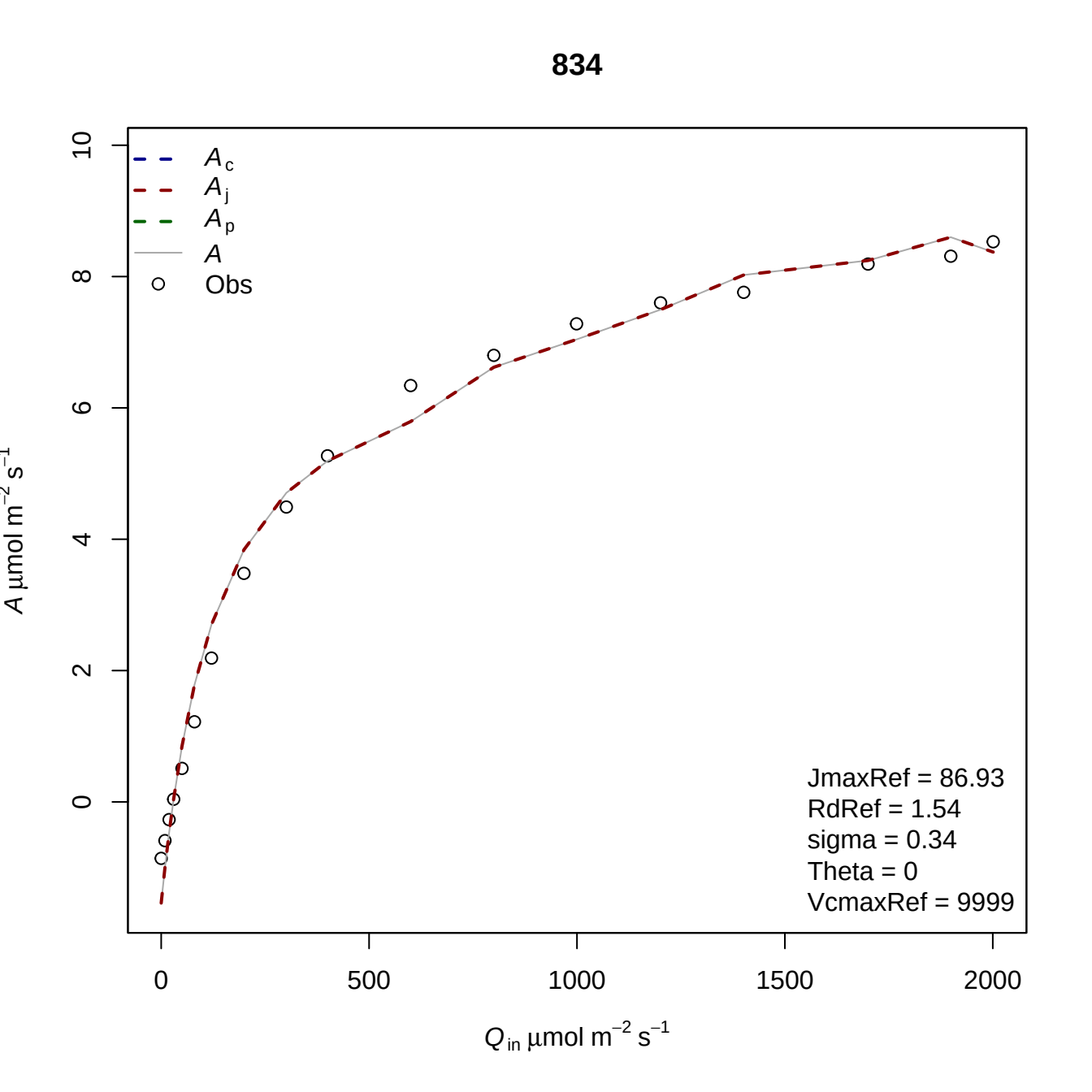

829
A_c
A_j
A_p
A
Obs
JmaxRef = 17.57
RdRef = 0.7
sigma = 0.12
Theta = 0
VcmaxRef = 9999
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

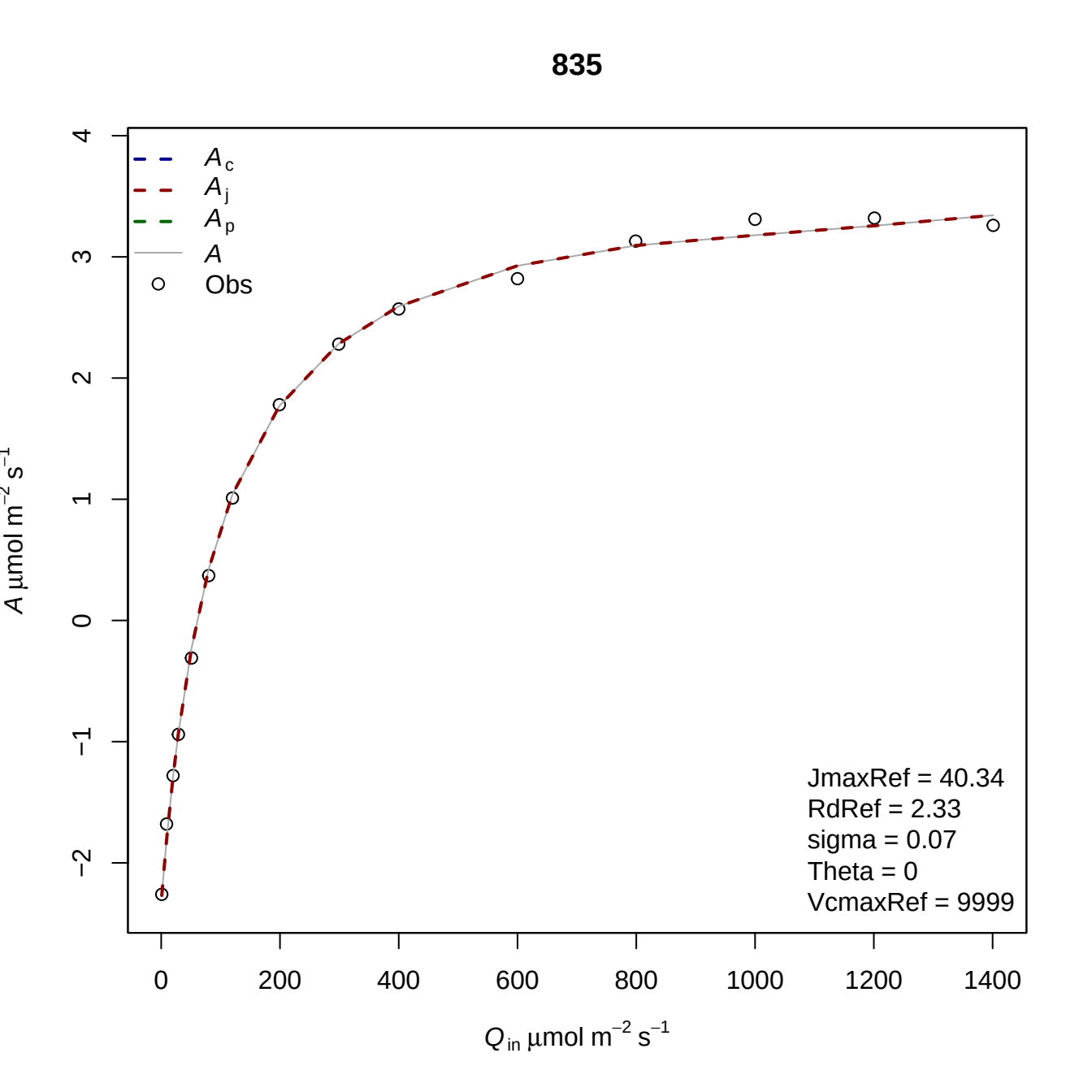

830
A_c
A_j
A_p
A
Obs
JmaxRef = 27.94
RdRef = 0.25
sigma = 0.13
Theta = 0
VcmaxRef = 14.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

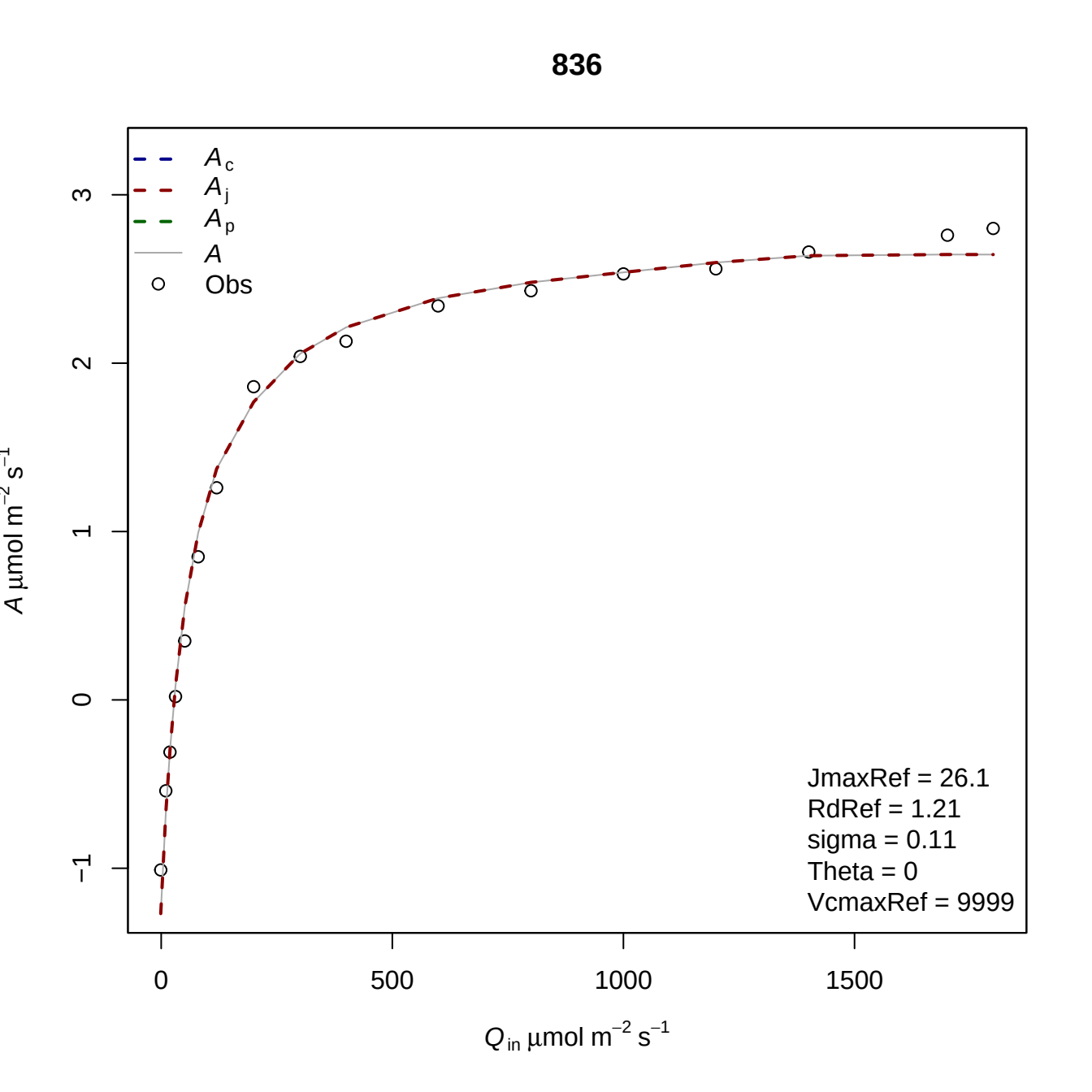

831
A_c
A_j
A_p
A
Obs
JmaxRef = 57.89
RdRef = 3.58
sigma = 0.16
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
200
400
600
800
1000

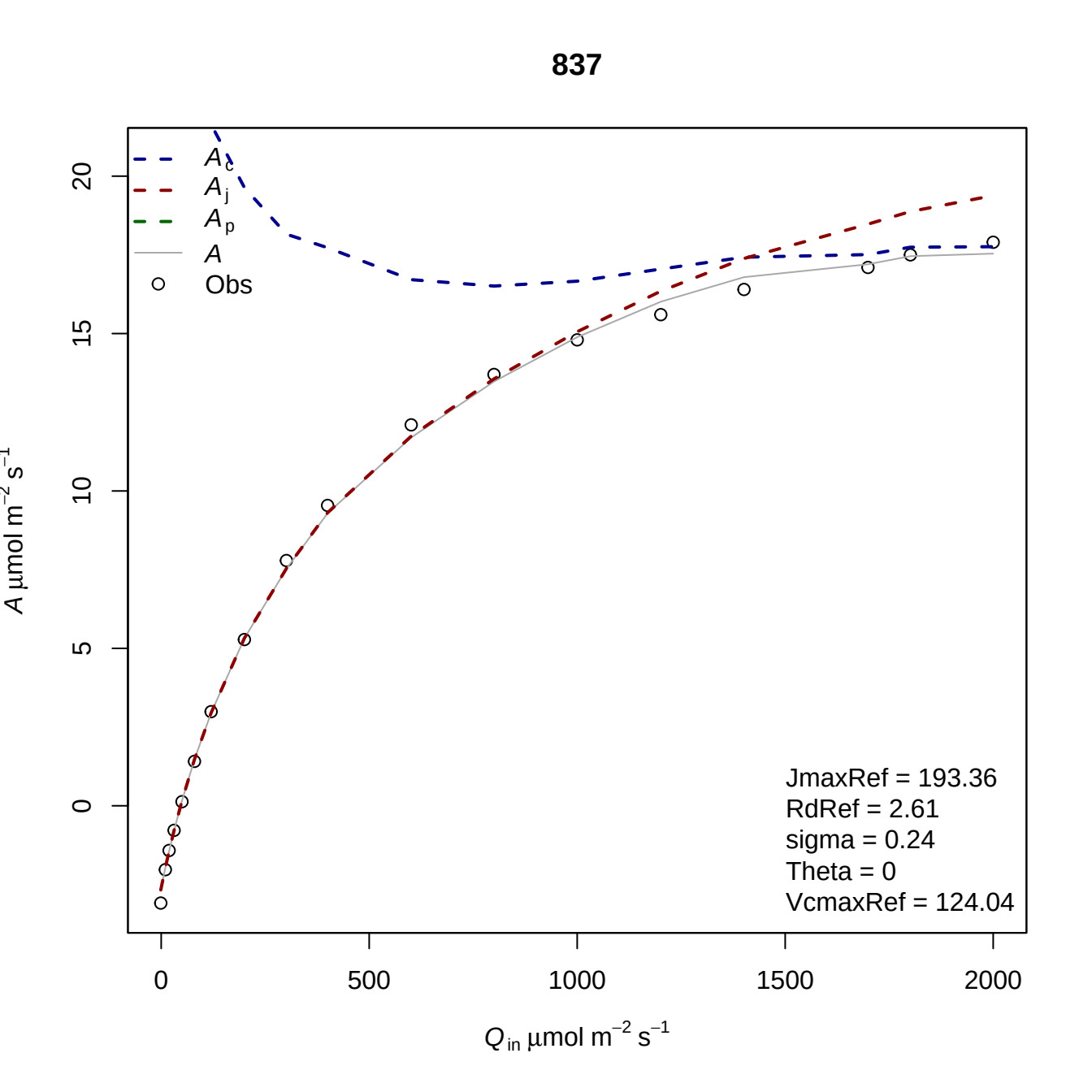

832
A_c
A_j
A_p
A
Obs
JmaxRef = 91.67
RdRef = 2.85
sigma = 0.17
Theta = 0
VcmaxRef = 58.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

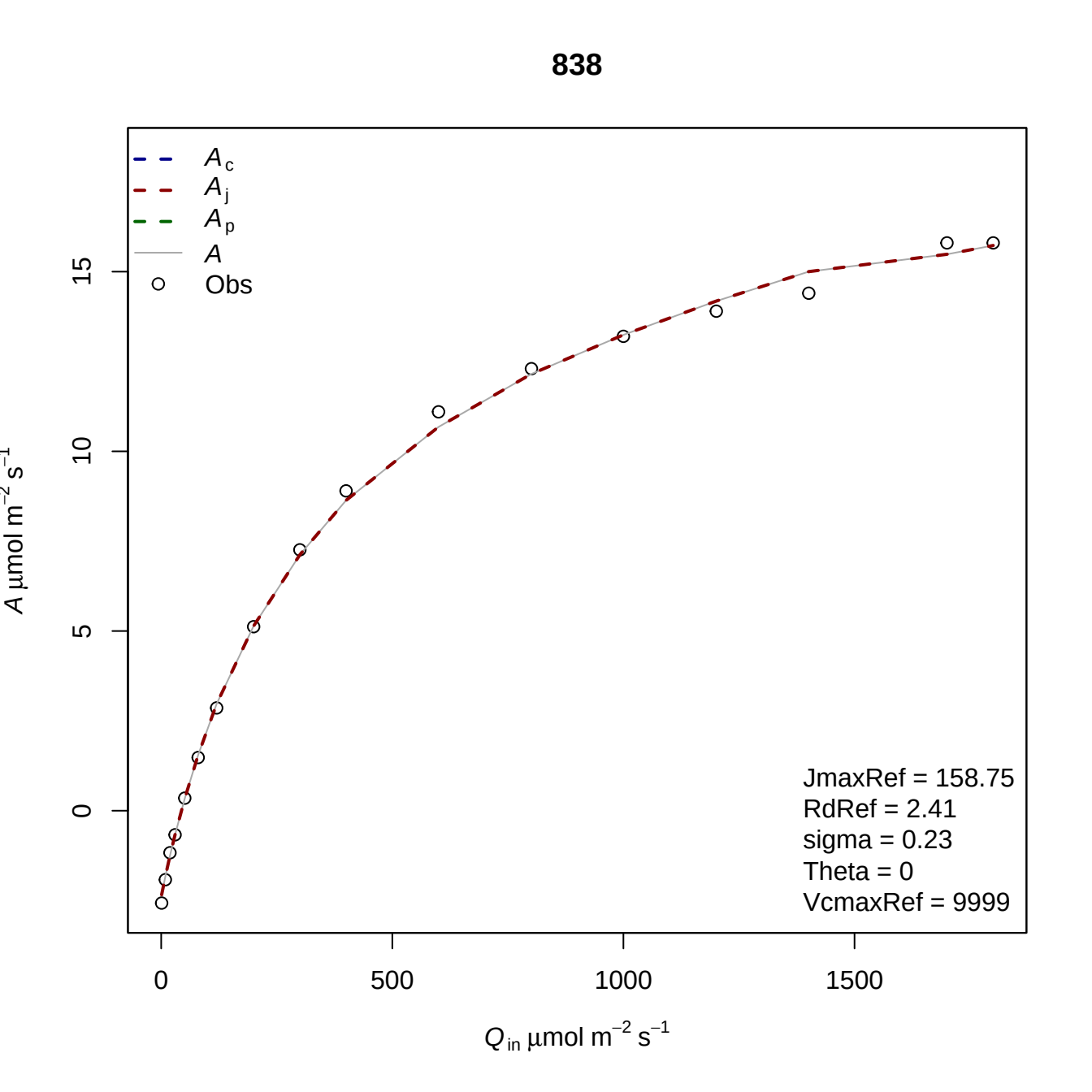

833
A_c
A_j
A_p
A
Obs
JmaxRef = 69.86
RdRef = 2.14
sigma = 0.19
Theta = 0
VcmaxRef = 47.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

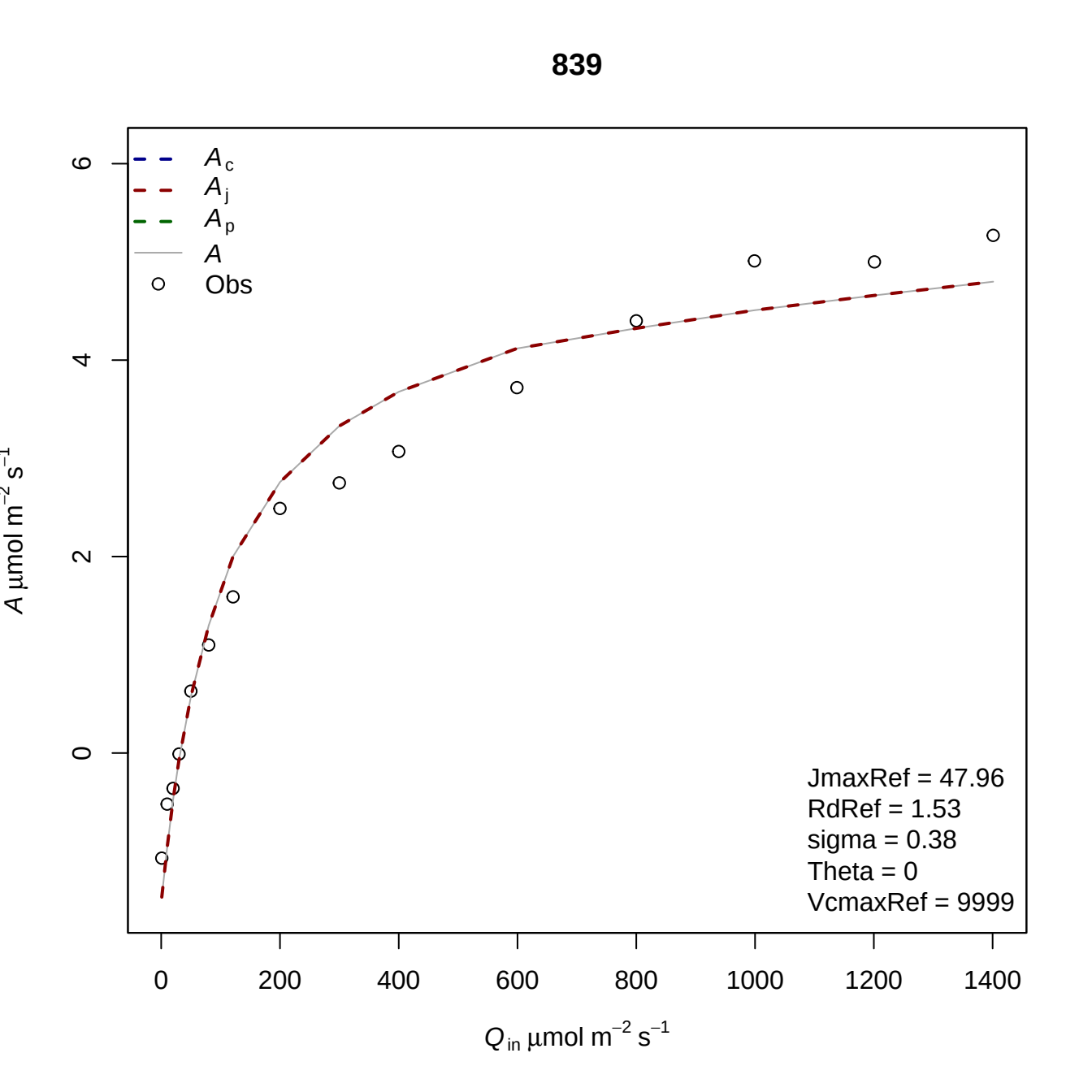

834
A_c
A_j
A_p
A
Obs
JmaxRef = 86.93
RdRef = 1.54
sigma = 0.34
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

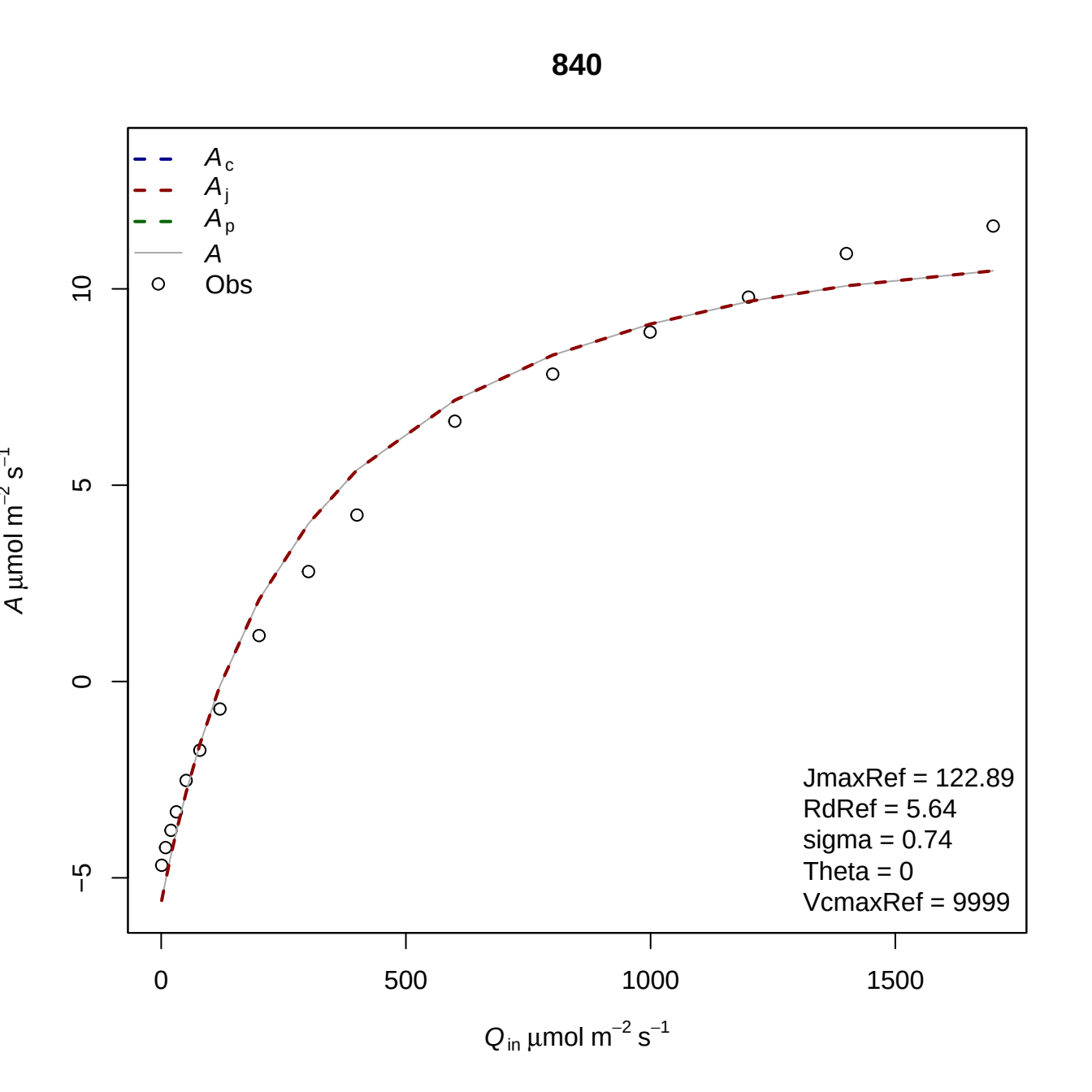

835
A_c
A_j
A_p
A
Obs
JmaxRef = 40.34
RdRef = 2.33
sigma = 0.07
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

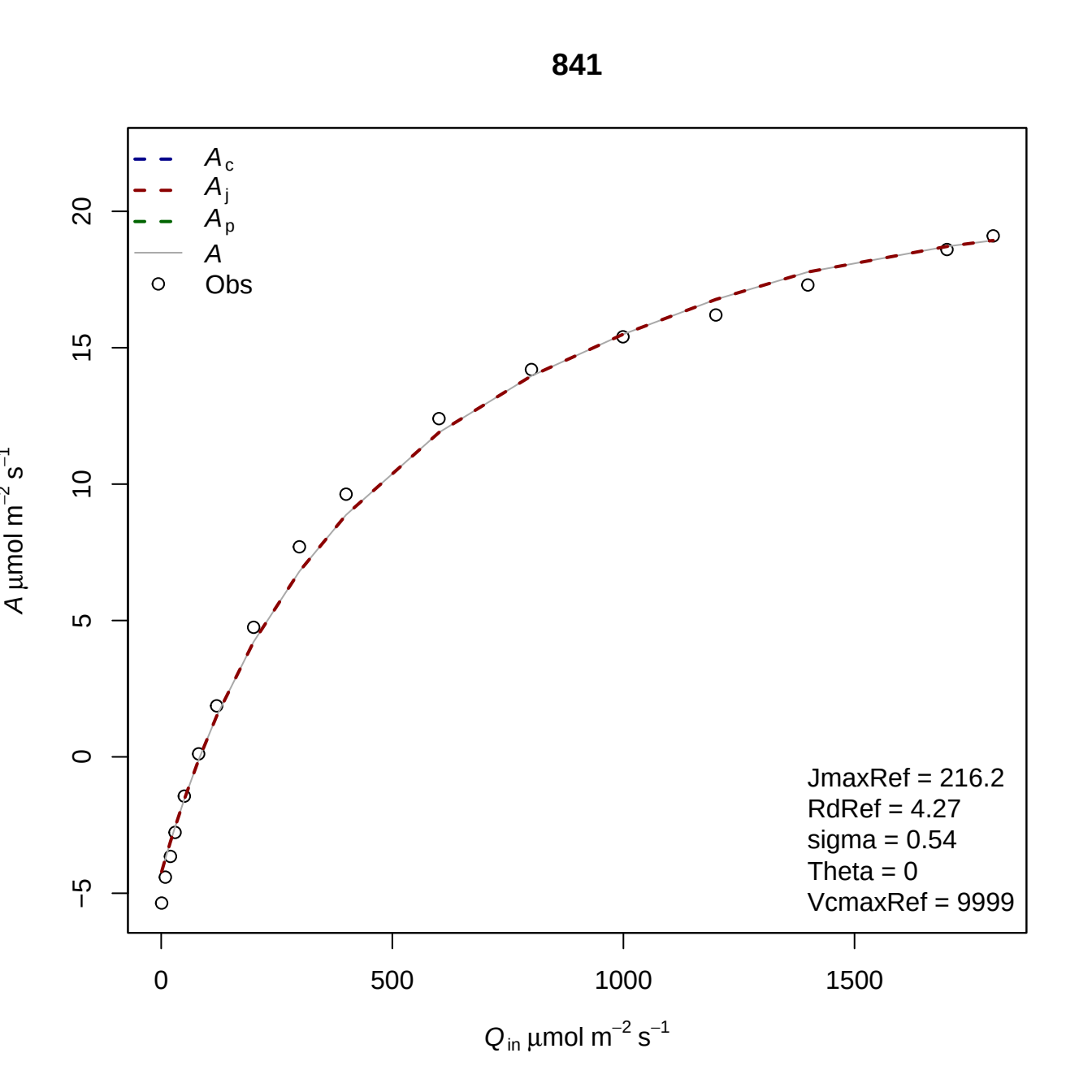

836
A_c
A_j
A_p
A
Obs
JmaxRef = 26.1
RdRef = 1.21
sigma = 0.11
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

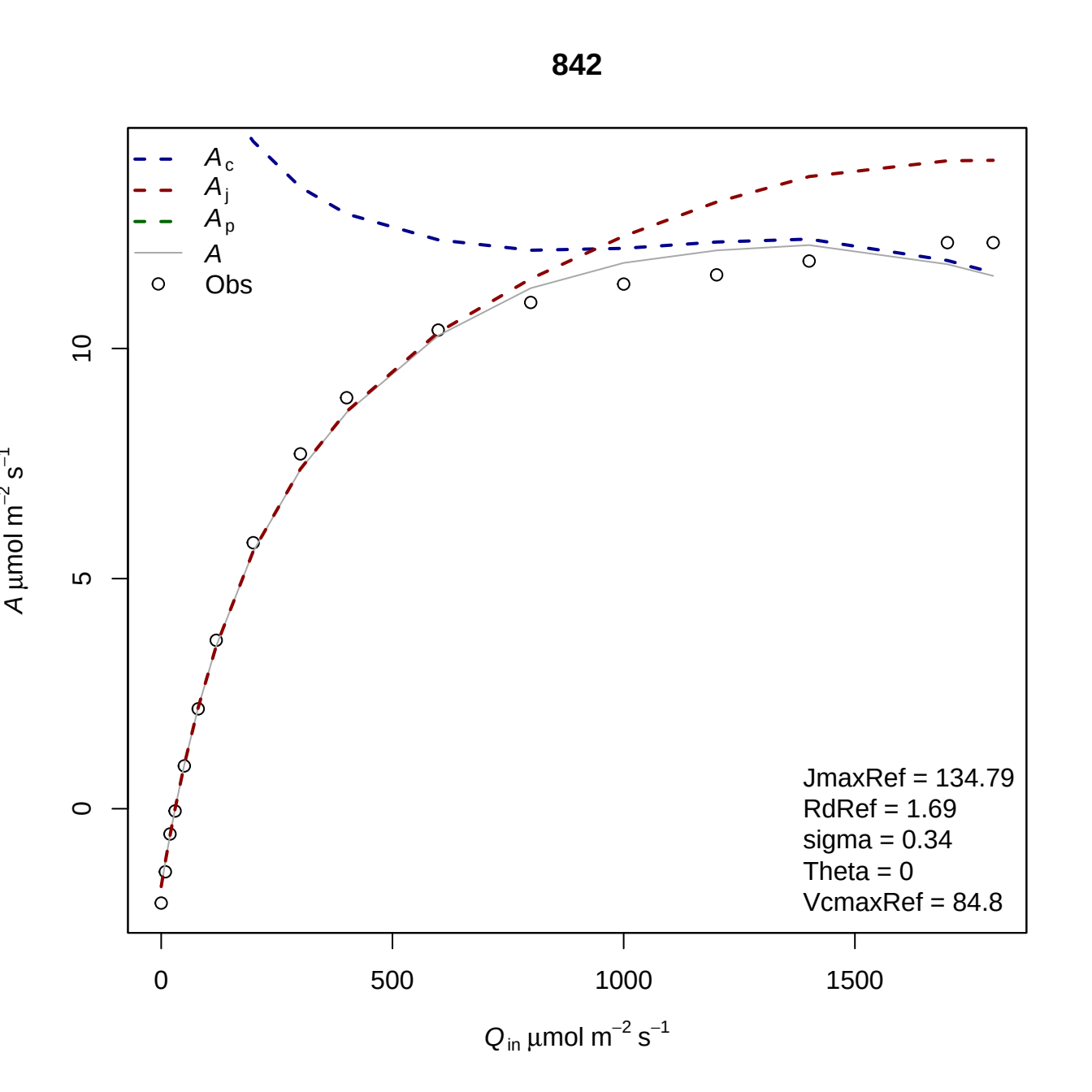

837
A_c
A_j
A_p
A
Obs
JmaxRef = 193.36
RdRef = 2.61
sigma = 0.24
Theta = 0
VcmaxRef = 124.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

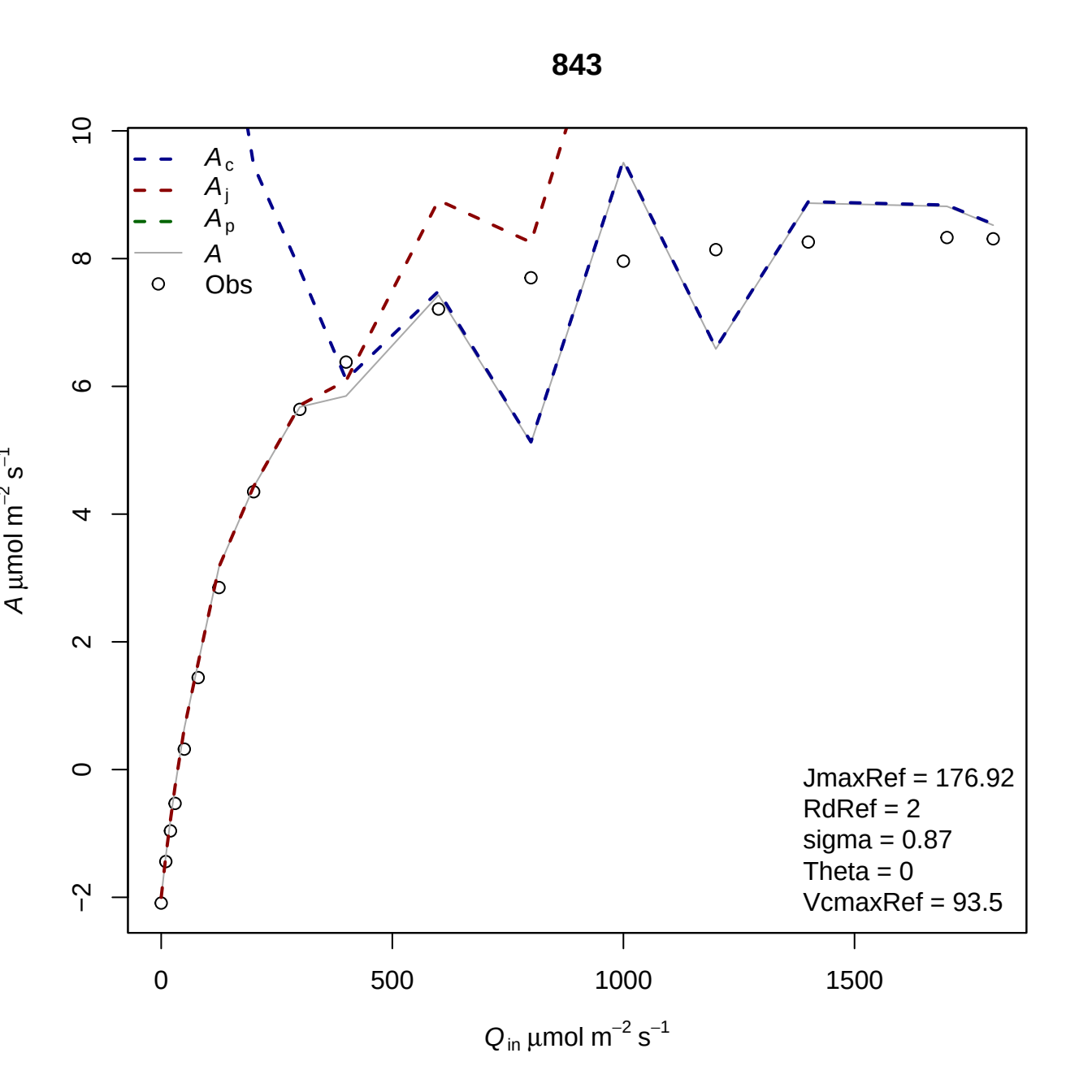

838
A_c
A_j
A_p
A
Obs
JmaxRef = 158.75
RdRef = 2.41
sigma = 0.23
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

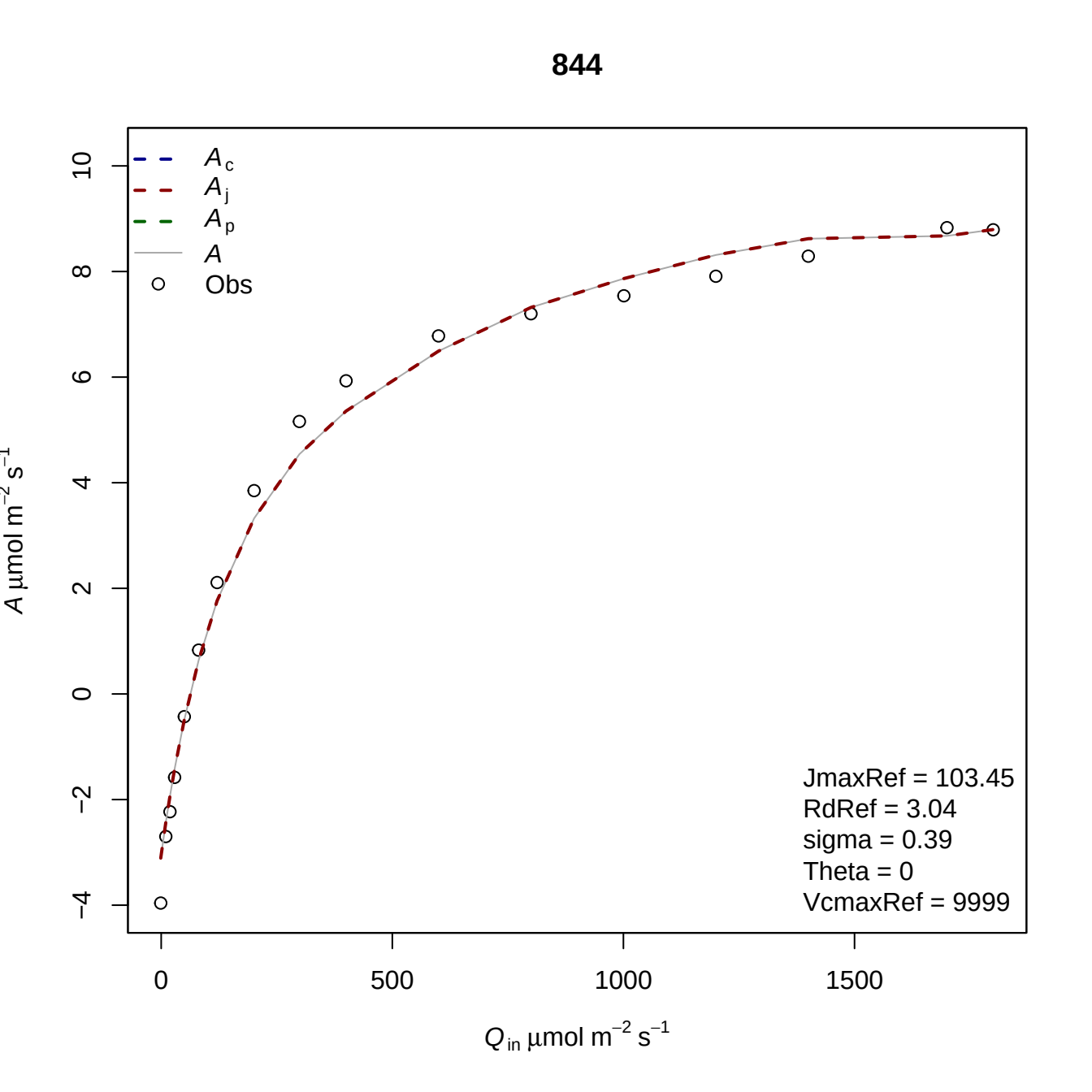

839
A_c
A_j
A_p
A
Obs
JmaxRef = 47.96
RdRef = 1.53
sigma = 0.38
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000
1200
1400

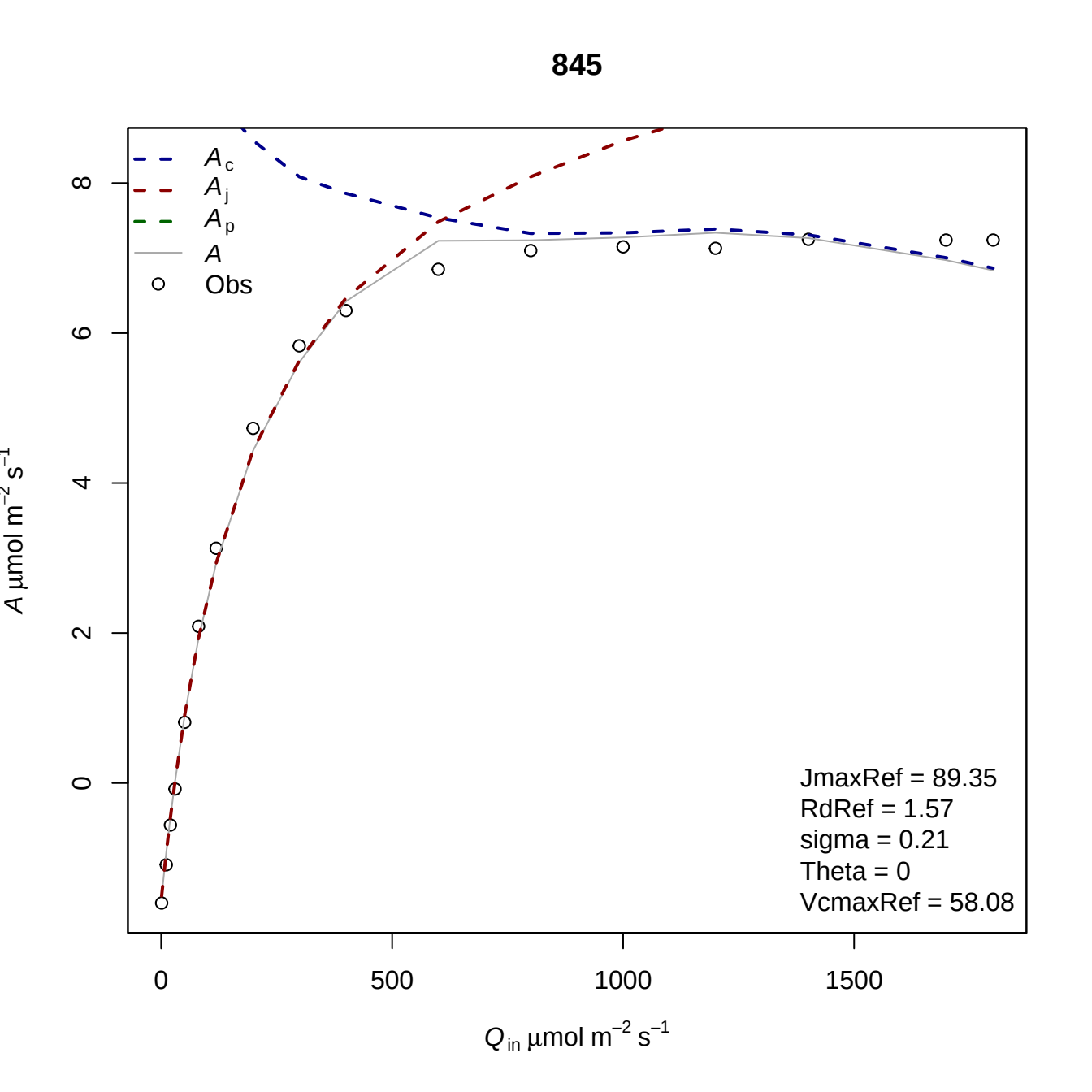

840
A_c
A_j
A_p
A
Obs
JmaxRef = 122.89
RdRef = 5.64
sigma = 0.74
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

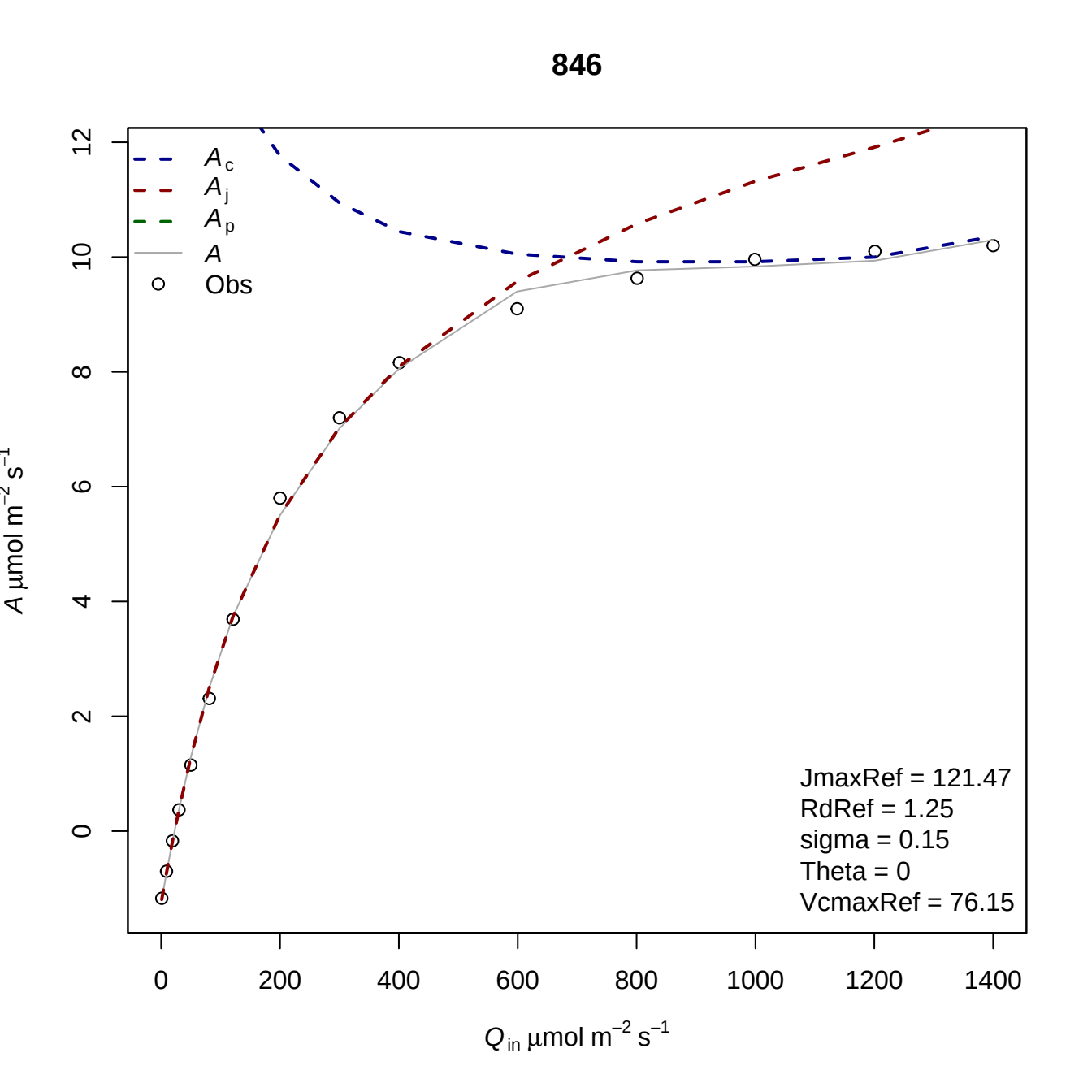

841
A_c
A_j
A_p
A
Obs
JmaxRef = 216.2
RdRef = 4.27
sigma = 0.54
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500

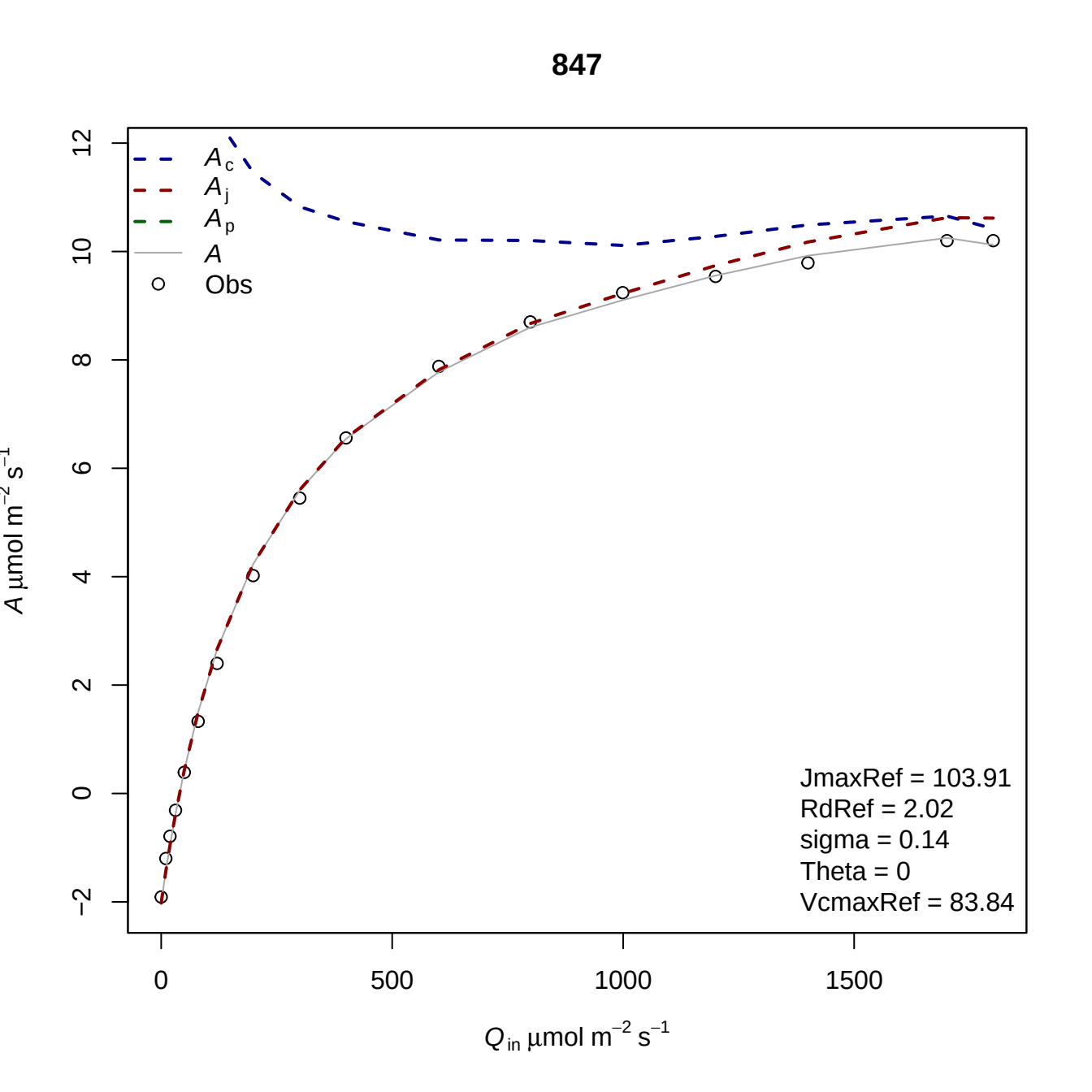

842
A_c
A_j
A_p
A
Obs
JmaxRef = 134.79
RdRef = 1.69
sigma = 0.34
Theta = 0
VcmaxRef = 84.8
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

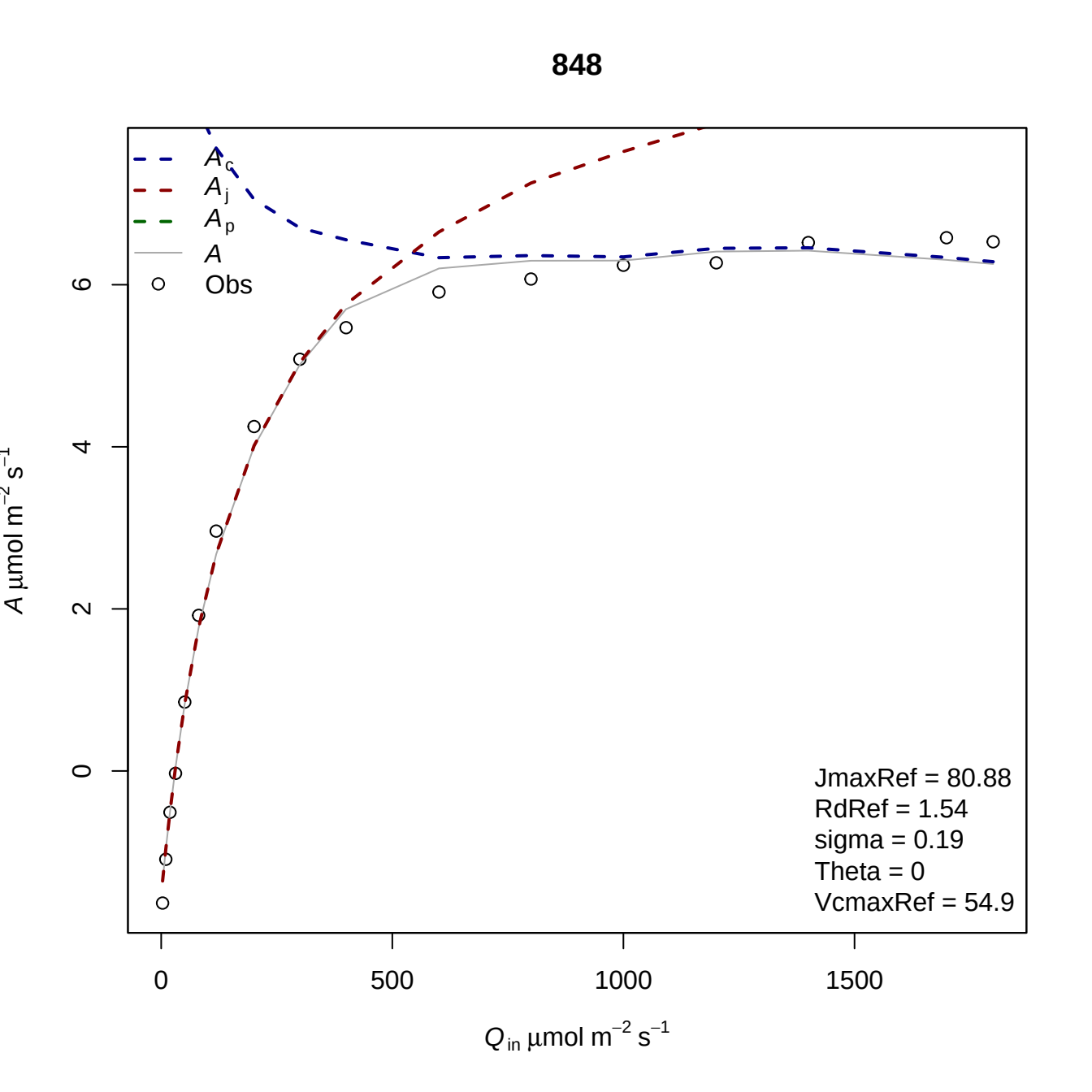

843
A_c
A_j
A_p
A
Obs
JmaxRef = 176.92
RdRef = 2
sigma = 0.87
Theta = 0
VcmaxRef = 93.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

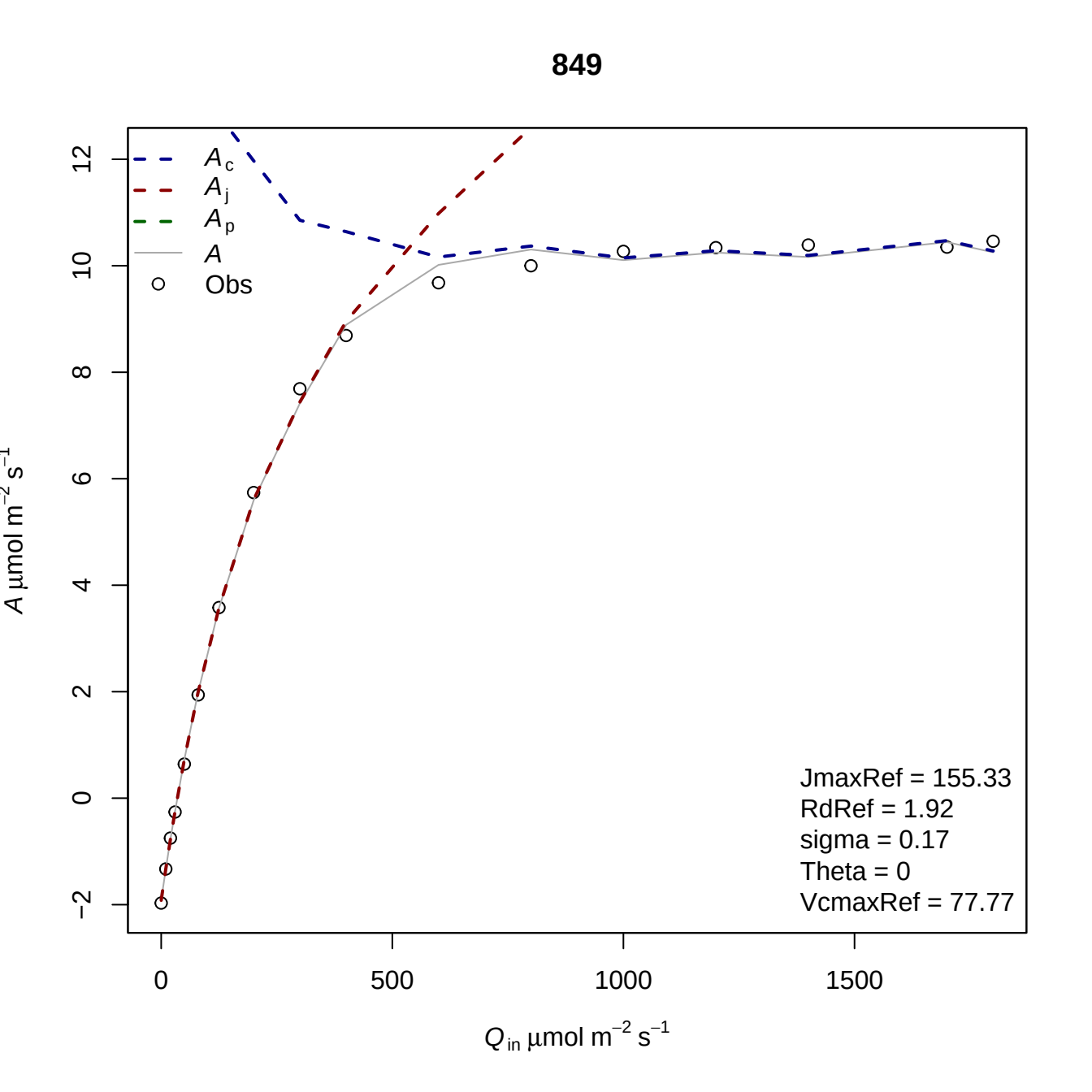

844
A_c
A_j
A_p
A
Obs
JmaxRef = 103.45
RdRef = 3.04
sigma = 0.39
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

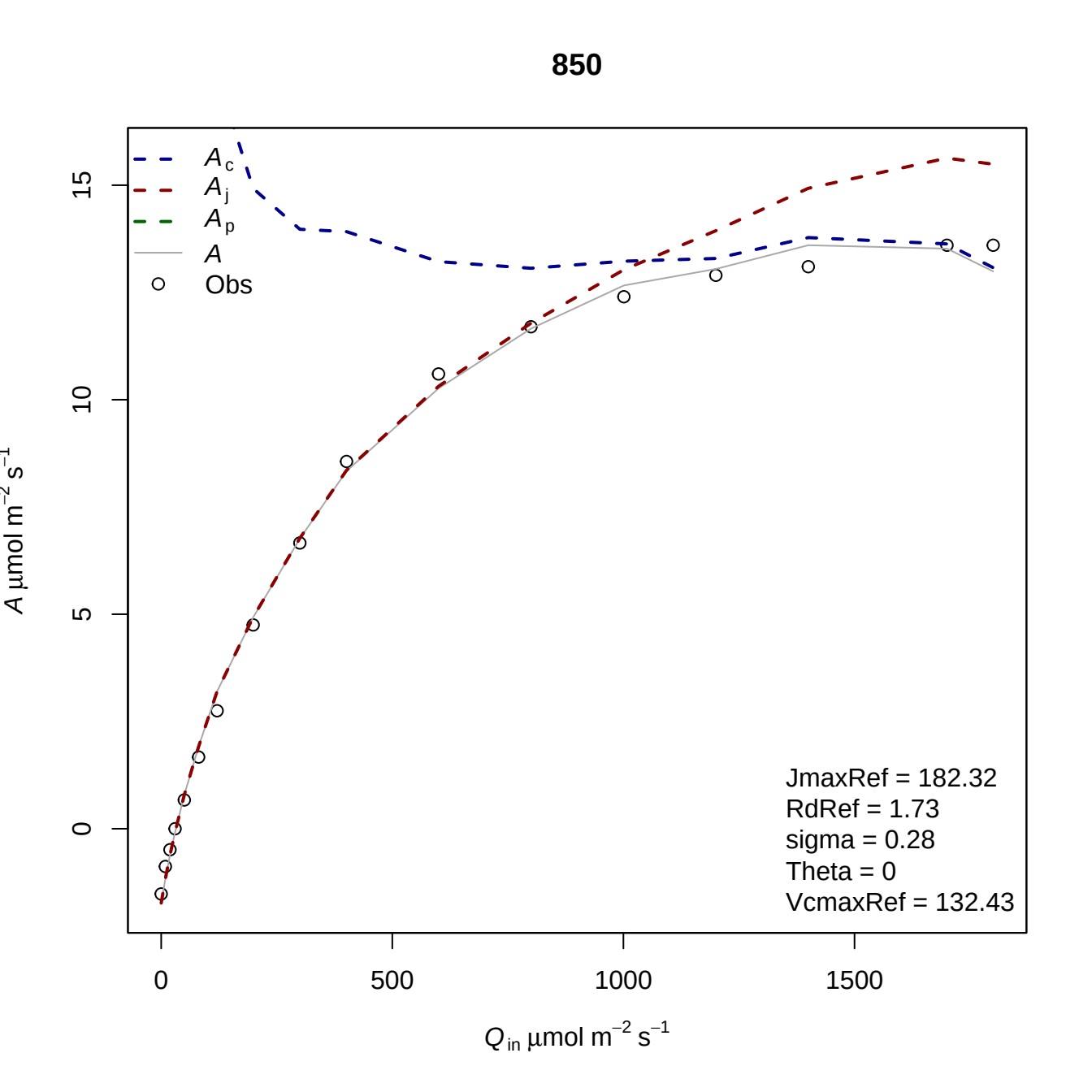

845
A_c
A_j
A_p
A
Obs
JmaxRef = 89.35
RdRef = 1.57
sigma = 0.21
Theta = 0
VcmaxRef = 58.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

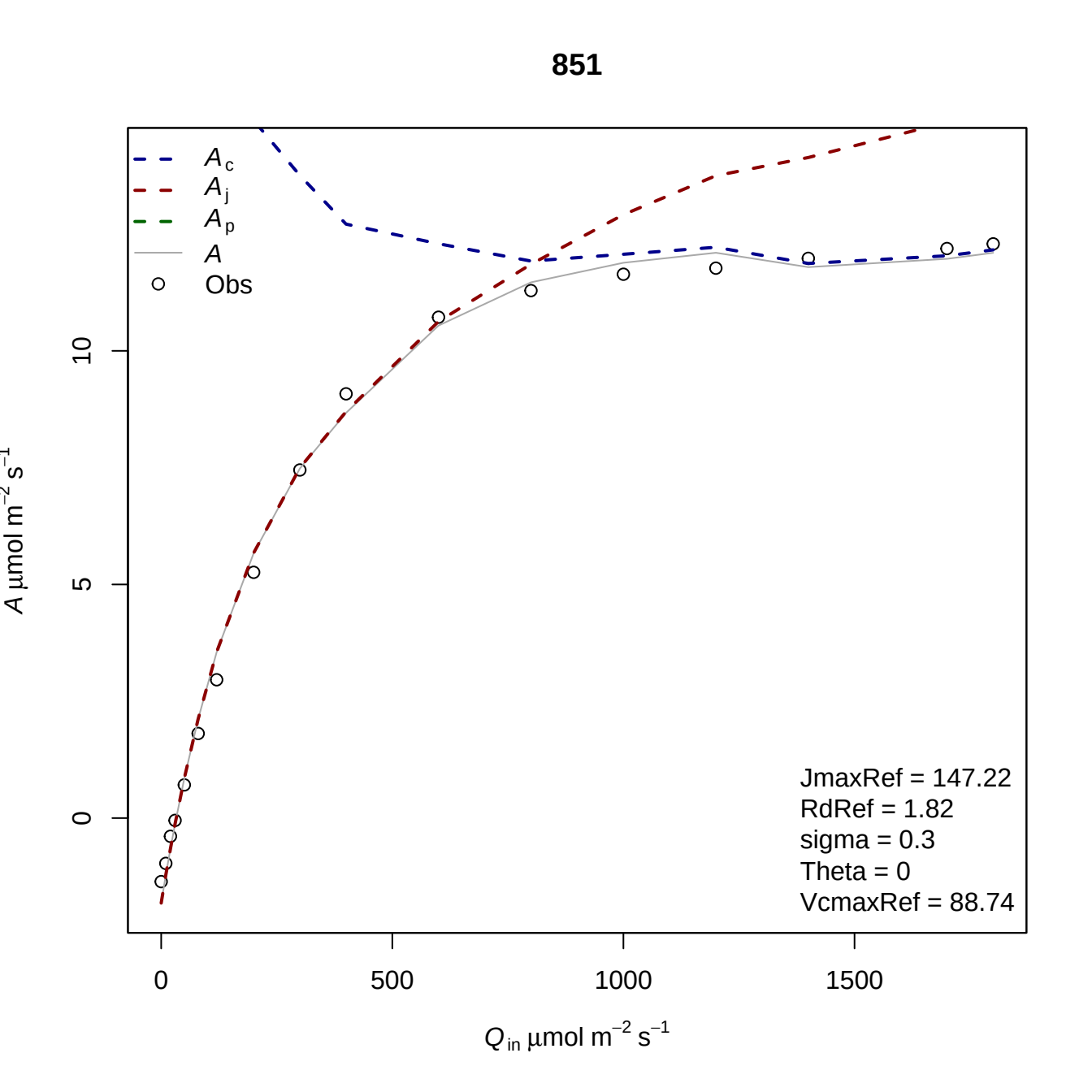

846
A_c
A_j
A_p
A
Obs
JmaxRef = 121.47
RdRef = 1.25
sigma = 0.15
Theta = 0
VcmaxRef = 76.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

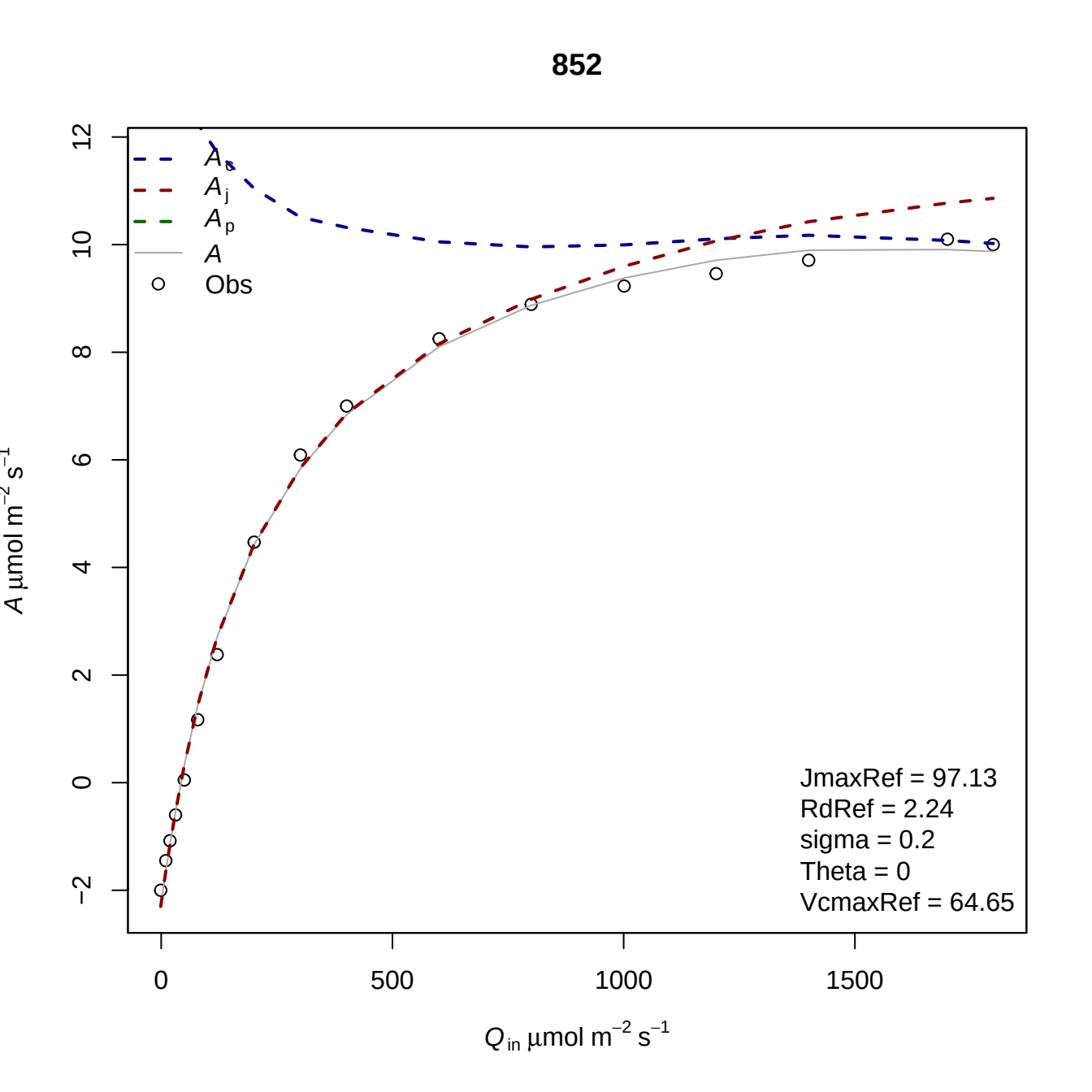

847
A_c
A_j
A_p
A
Obs
JmaxRef = 103.91
RdRef = 2.02
sigma = 0.14
Theta = 0
VcmaxRef = 83.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

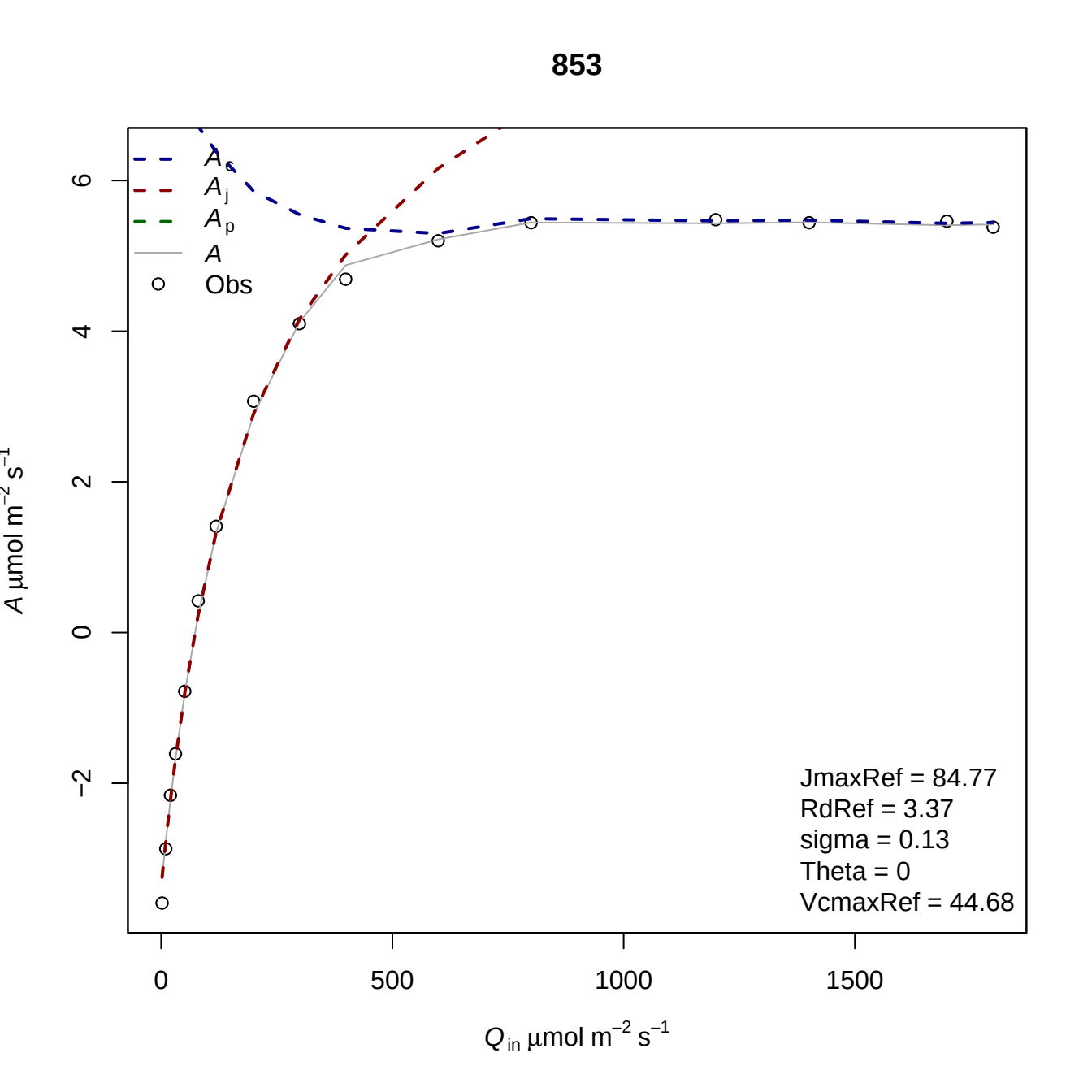

848
A_c
A_j
A_p
A
Obs
JmaxRef = 80.88
RdRef = 1.54
sigma = 0.19
Theta = 0
VcmaxRef = 54.9
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

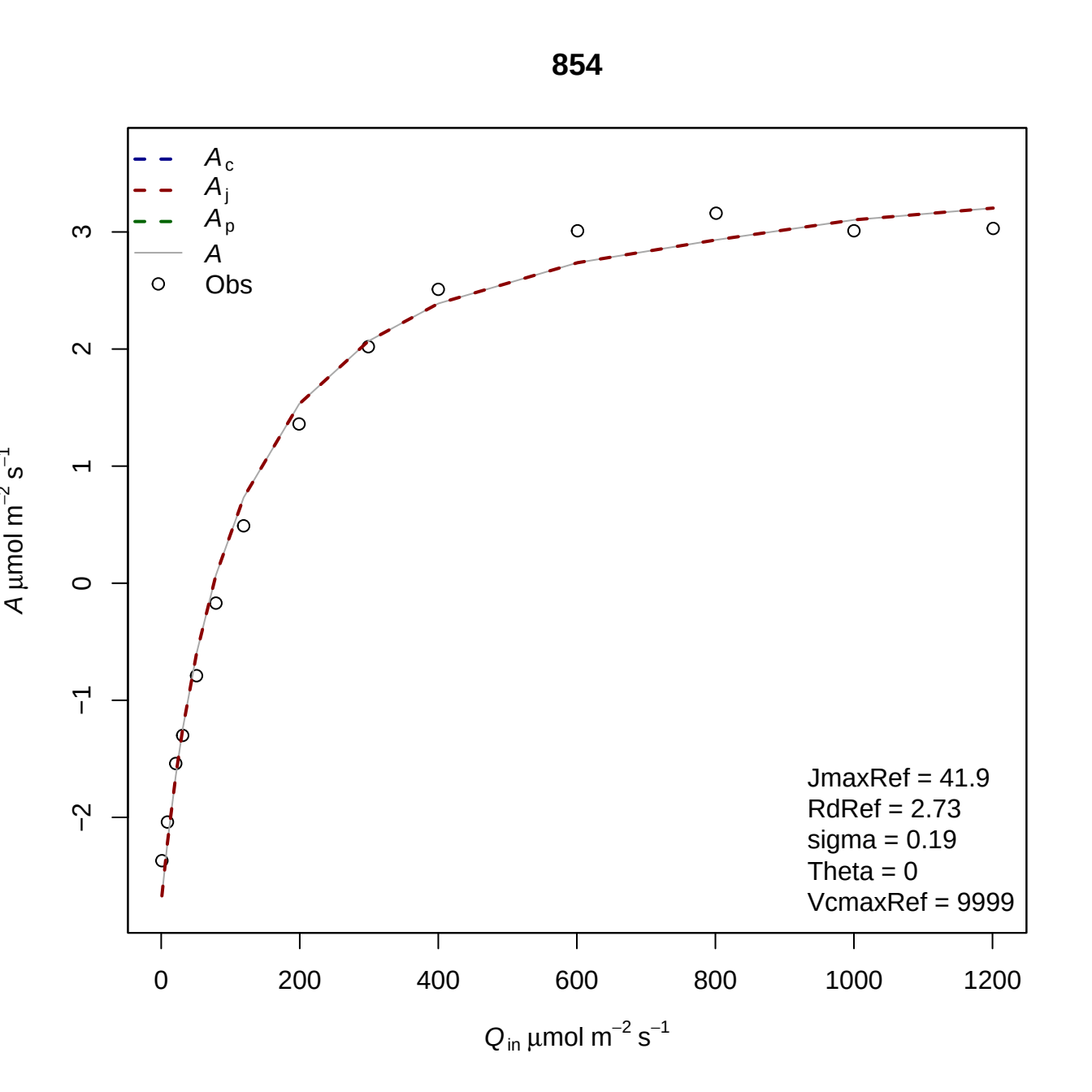

849
A_c
A_j
A_p
A
Obs
JmaxRef = 155.33
RdRef = 1.92
sigma = 0.17
Theta = 0
VcmaxRef = 77.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

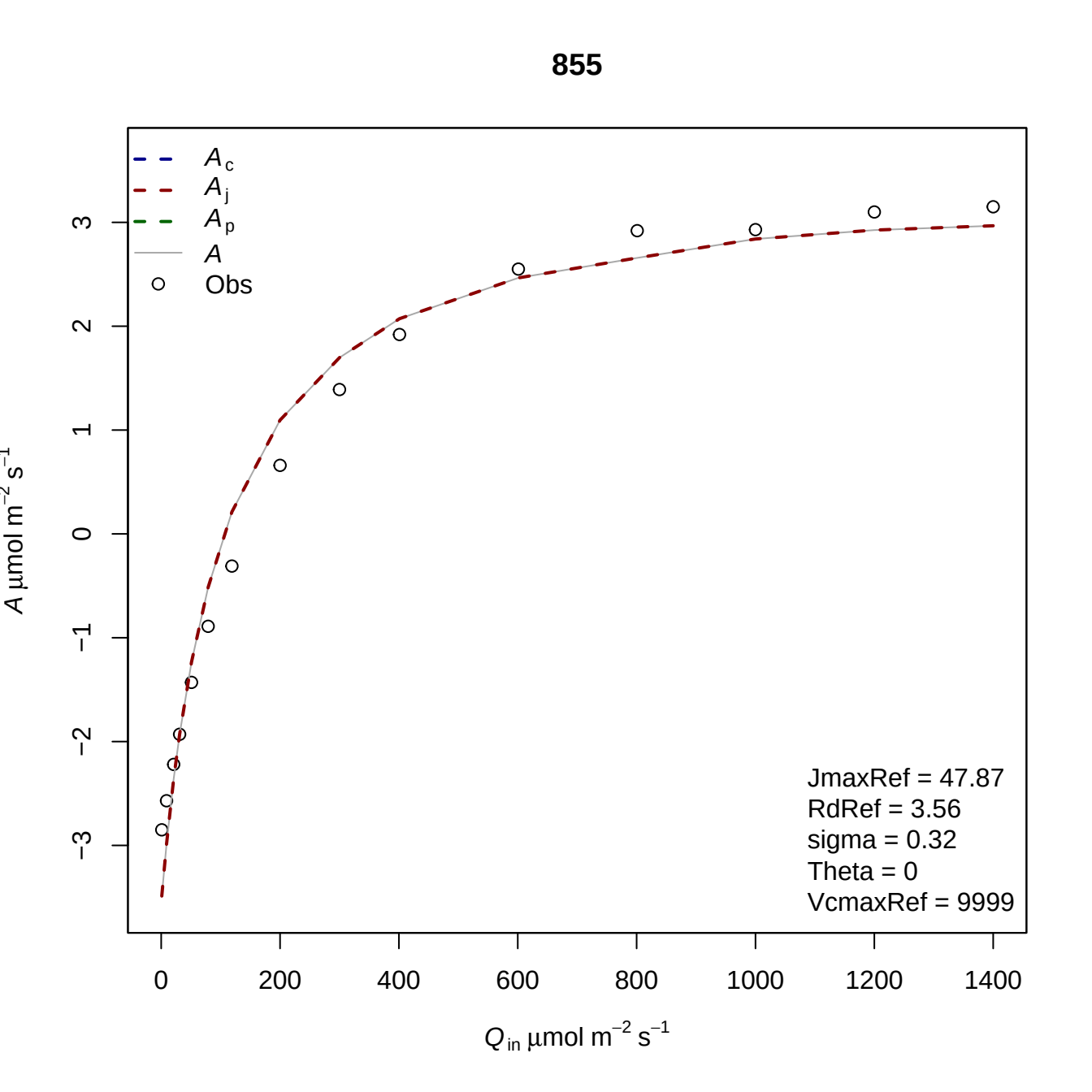

850
A_c
A_j
A_p
A
Obs
JmaxRef = 182.32
RdRef = 1.73
sigma = 0.28
Theta = 0
VcmaxRef = 132.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

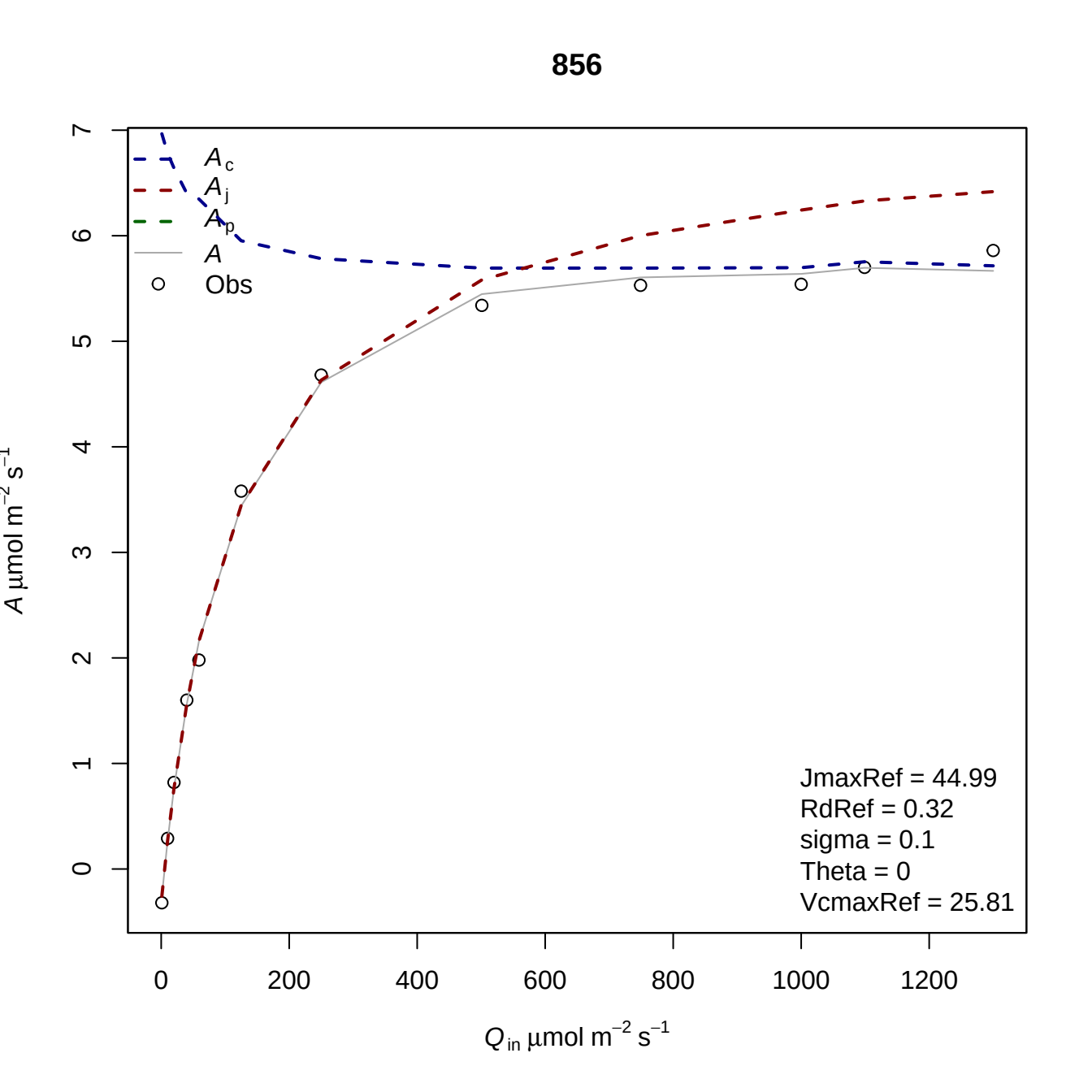

851
A_c
A_j
A_p
A
Obs
JmaxRef = 147.22
RdRef = 1.82
sigma = 0.3
Theta = 0
VcmaxRef = 88.74
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

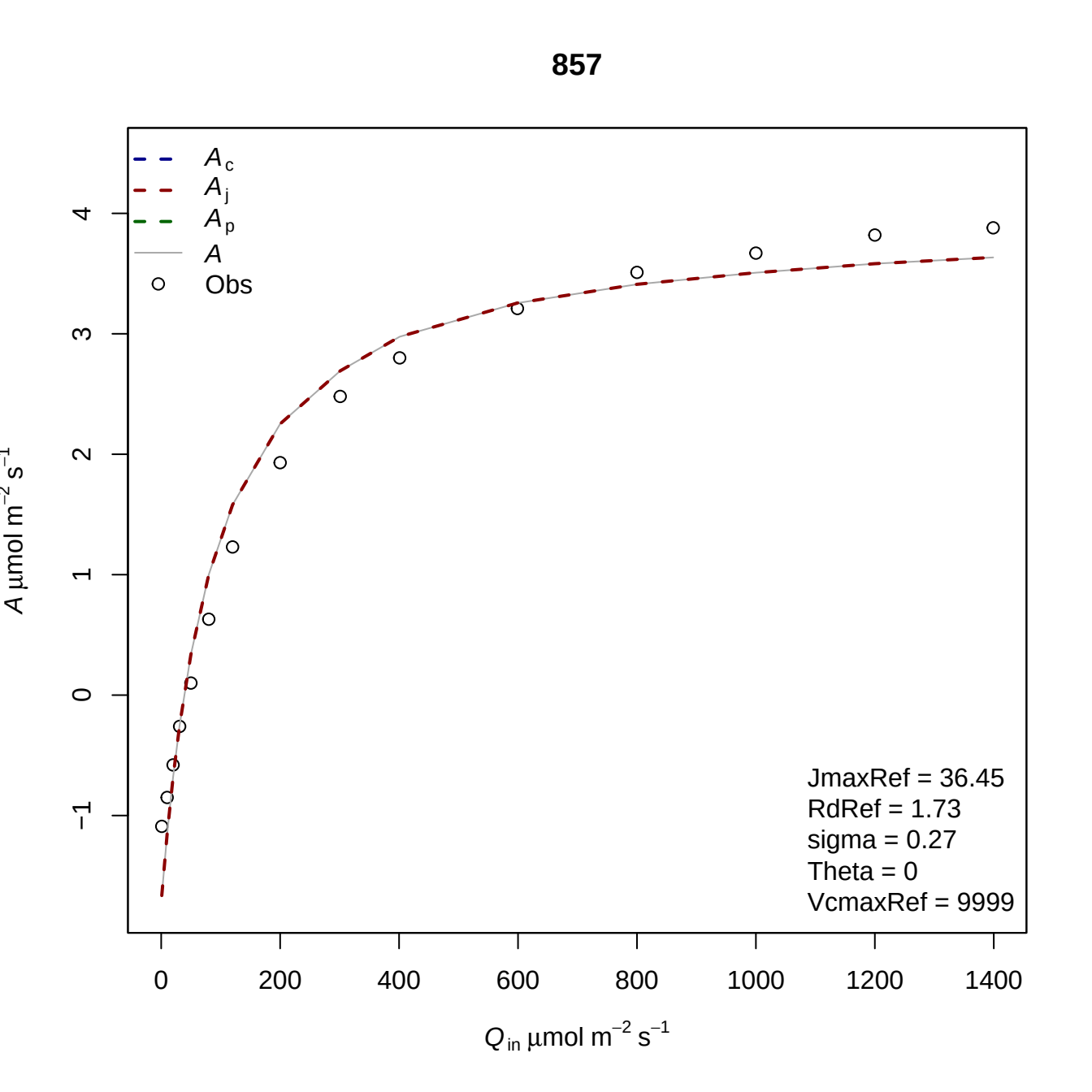

852
A_c
A_j
A_p
A
Obs
JmaxRef = 97.13
RdRef = 2.24
sigma = 0.2
Theta = 0
VcmaxRef = 64.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

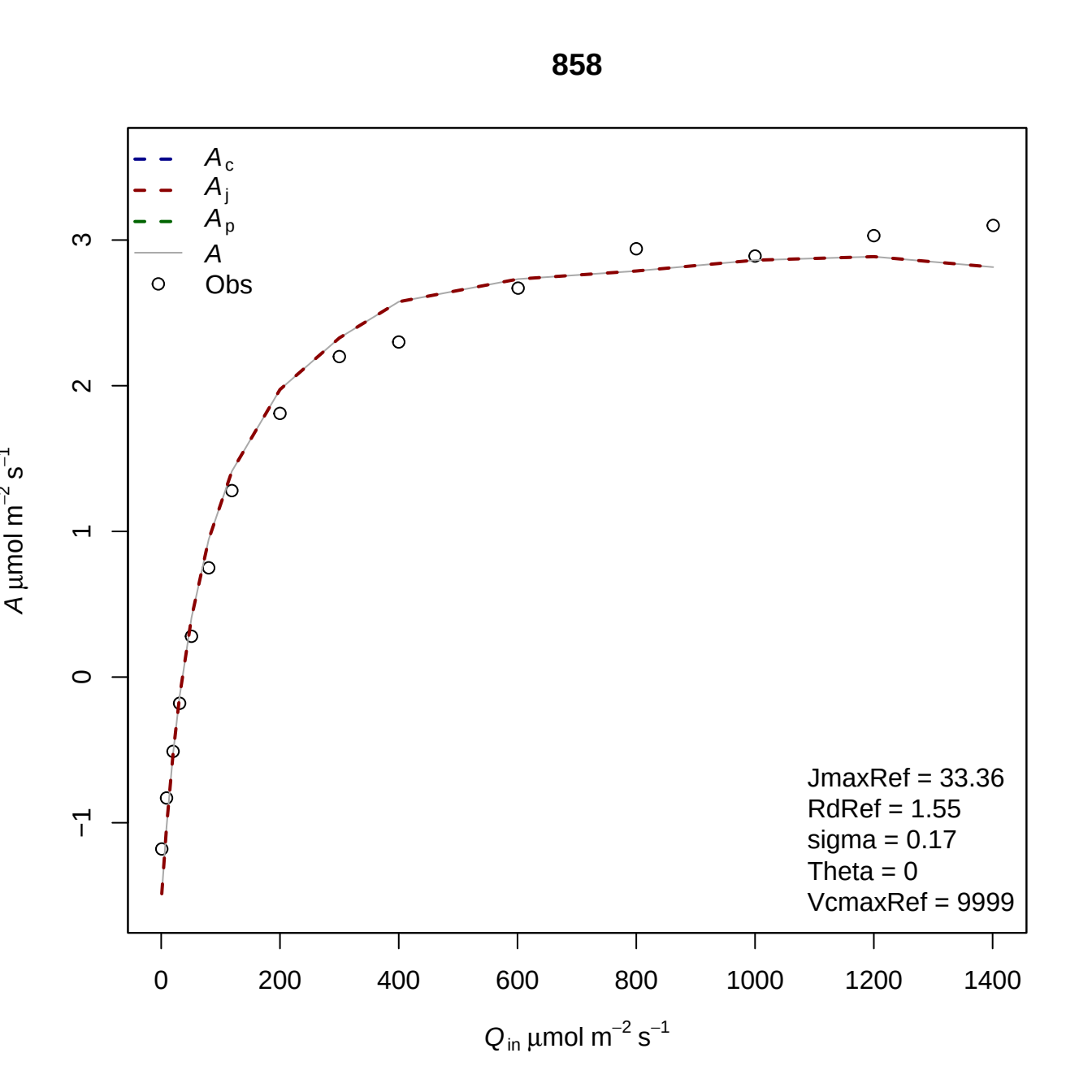

853
A_c
A_j
A_p
A
Obs
JmaxRef = 84.77
RdRef = 3.37
sigma = 0.13
Theta = 0
VcmaxRef = 44.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

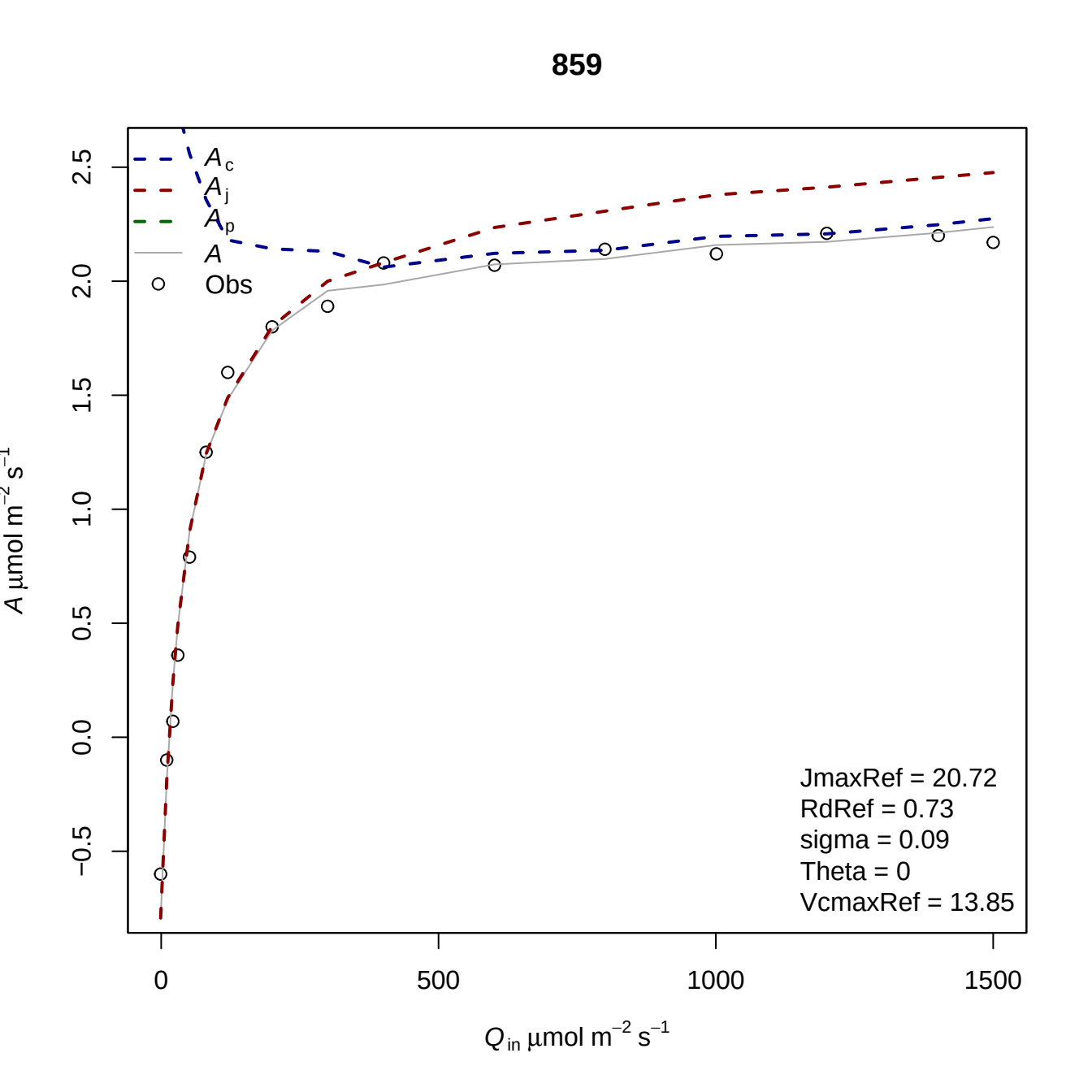

854
A_c
A_j
A_p
A
Obs
JmaxRef = 41.9
RdRef = 2.73
sigma = 0.19
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200

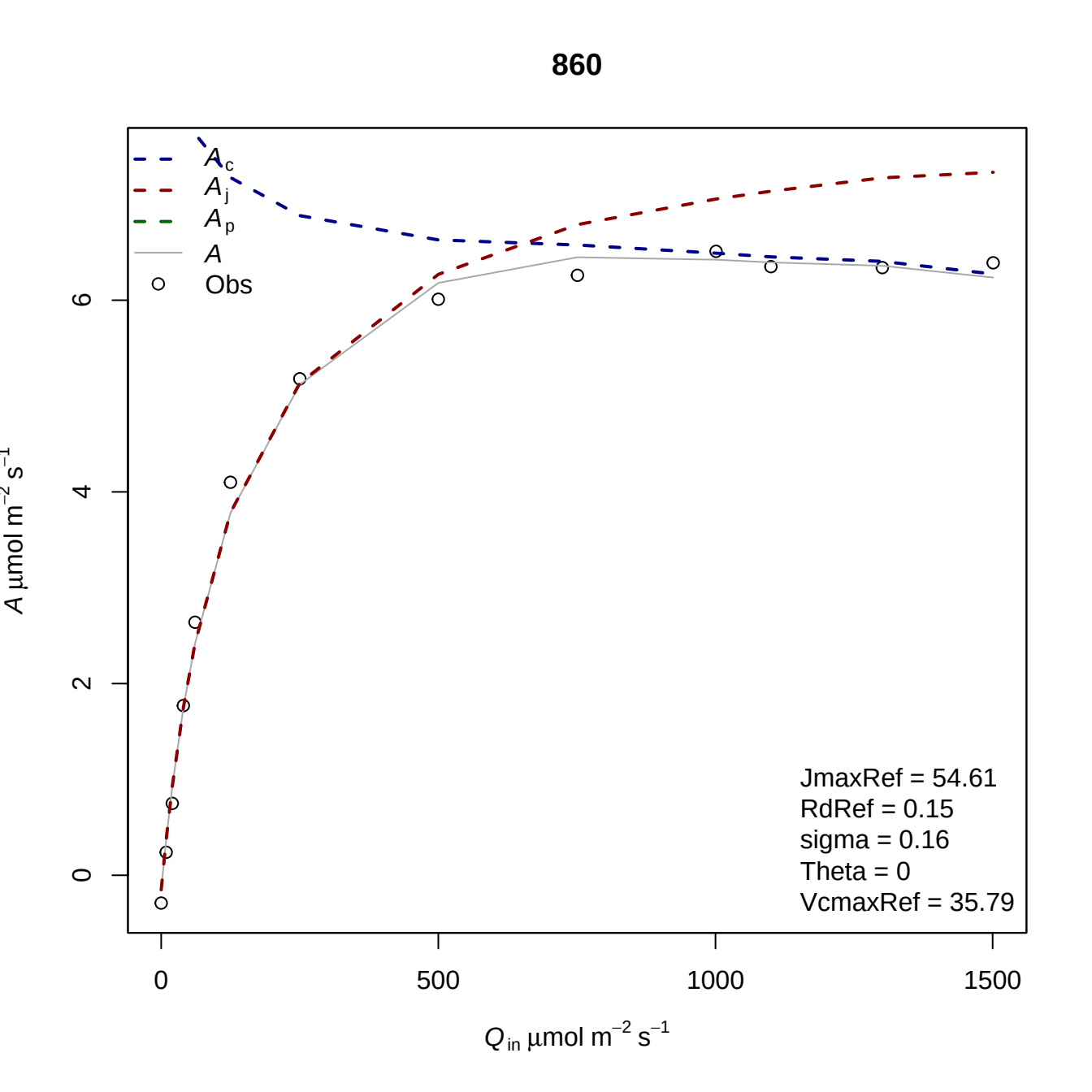

855
A_c
A_j
A_p
A
Obs
JmaxRef = 47.87
RdRef = 3.56
sigma = 0.32
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−3
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

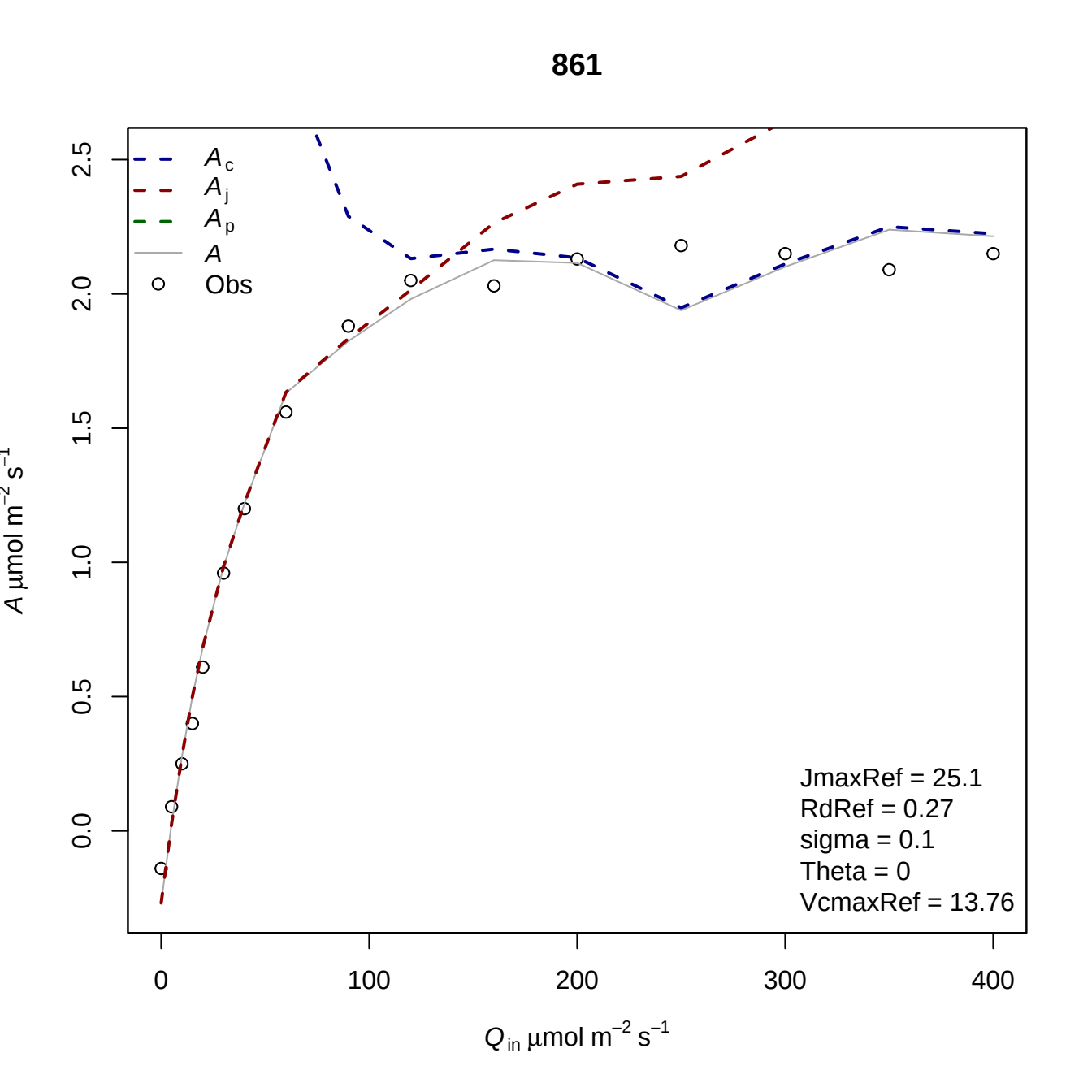

856
A_c
A_j
A_p
A
Obs
JmaxRef = 44.99
RdRef = 0.32
sigma = 0.1
Theta = 0
VcmaxRef = 25.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200

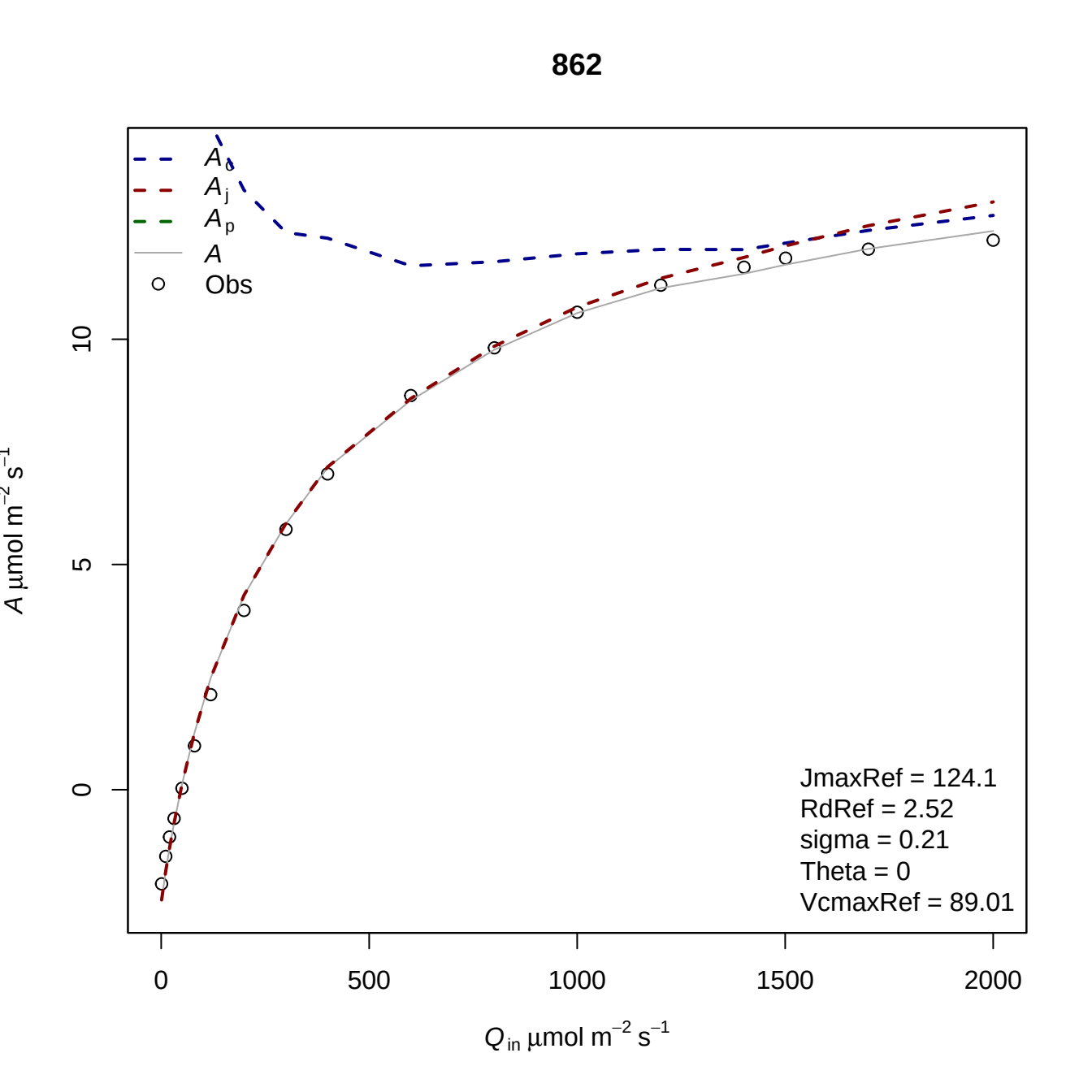

857
A_c
A_j
A_p
A
Obs
JmaxRef = 36.45
RdRef = 1.73
sigma = 0.27
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

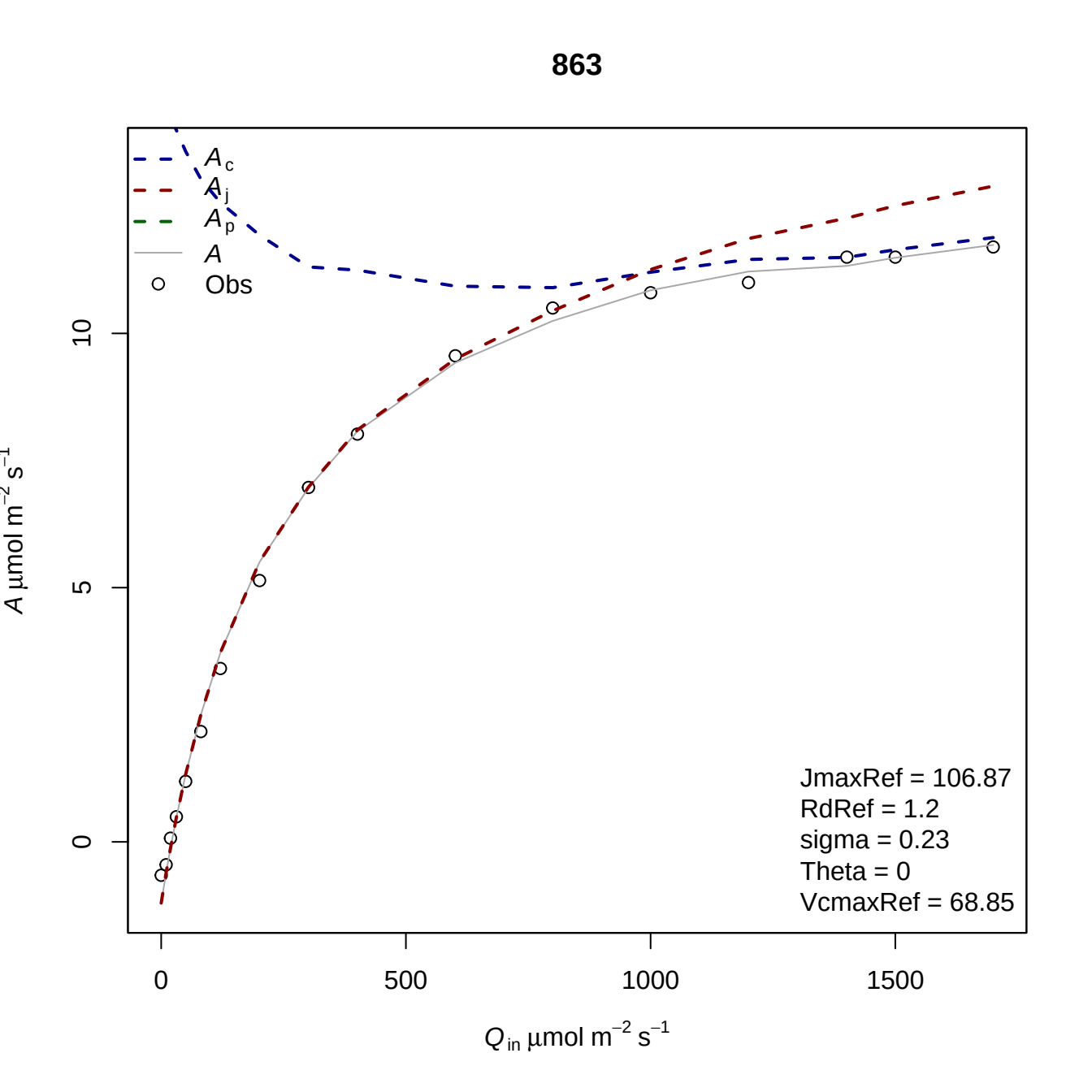

858
A_c
A_j
A_p
A
Obs
JmaxRef = 33.36
RdRef = 1.55
sigma = 0.17
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

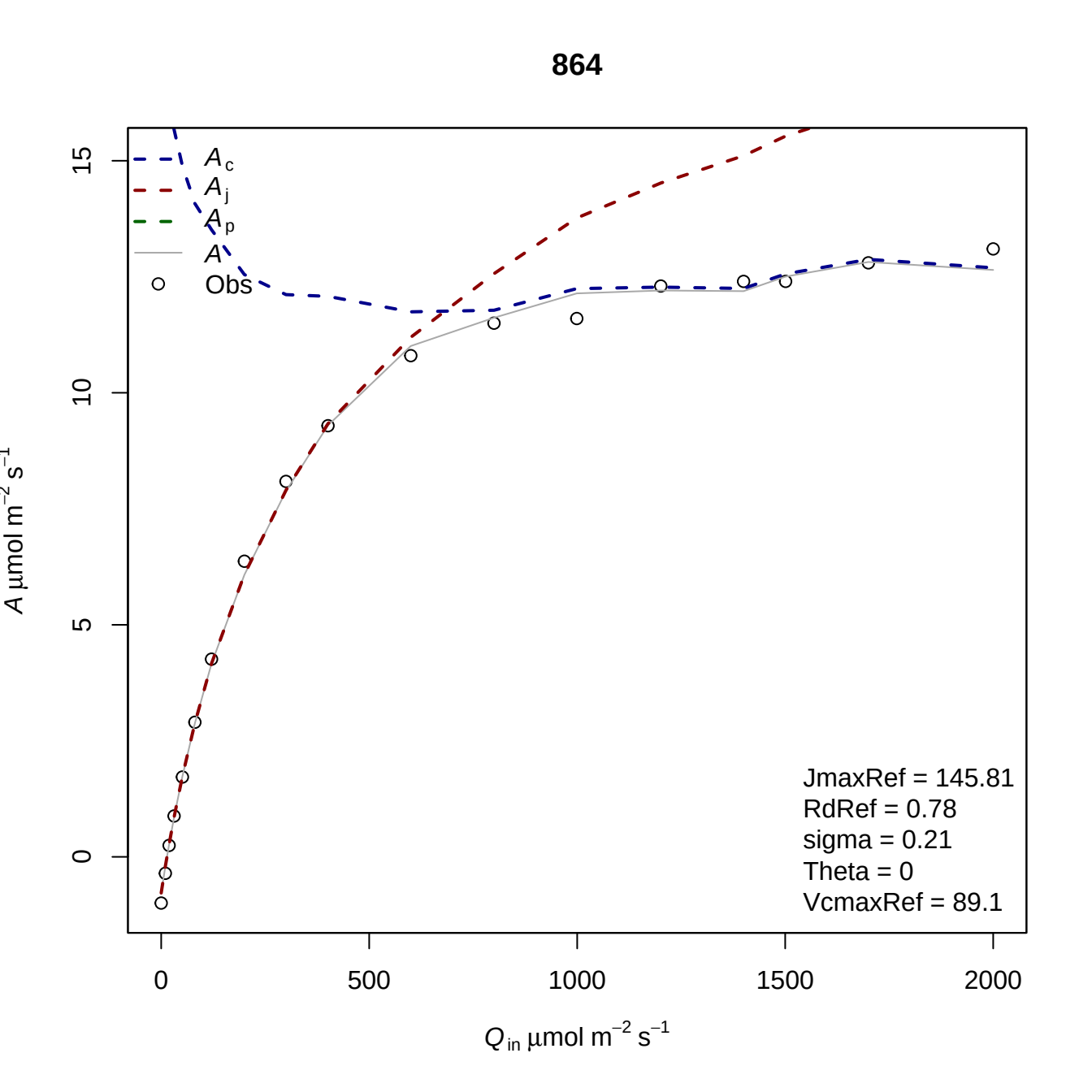

859
A_c
A_j
A_p
A
Obs
JmaxRef = 20.72
RdRef = 0.73
sigma = 0.09
Theta = 0
VcmaxRef = 13.85
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

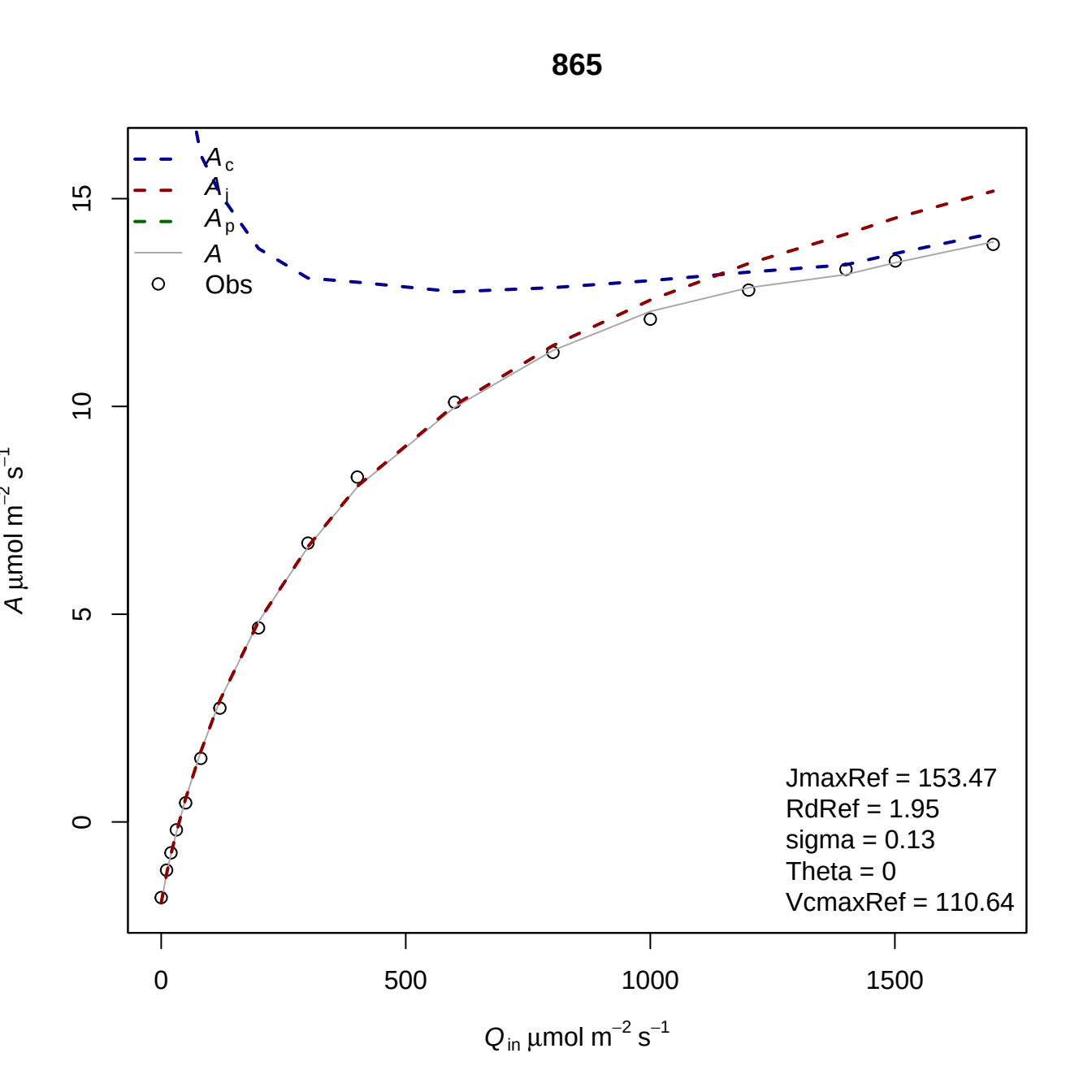

860
A_c
A_j
A_p
A
Obs
JmaxRef = 54.61
RdRef = 0.15
sigma = 0.16
Theta = 0
VcmaxRef = 35.79
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

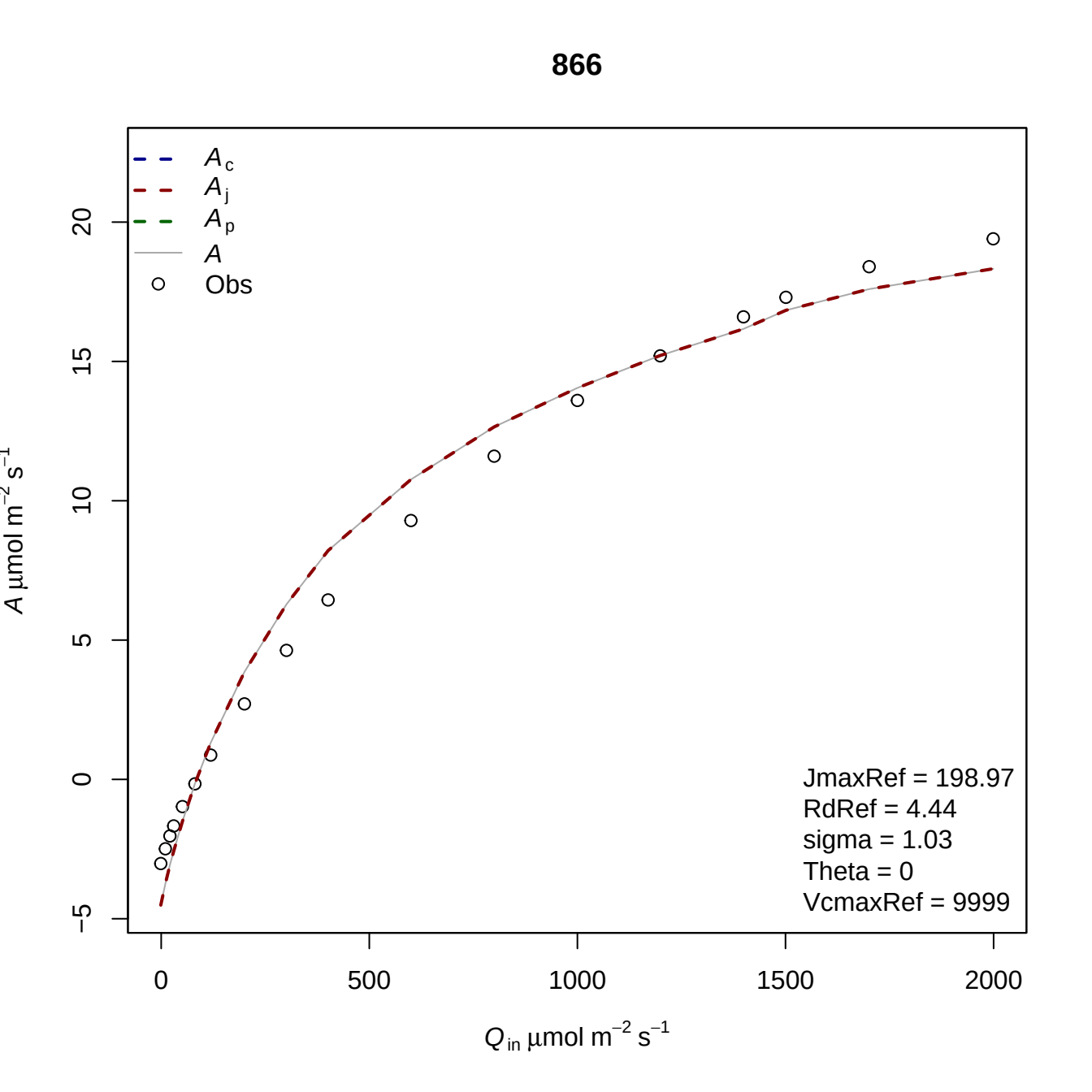

861
A_c
A_j
A_p
A
Obs
JmaxRef = 25.1
RdRef = 0.27
sigma = 0.1
Theta = 0
VcmaxRef = 13.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
0
100
200
300
400

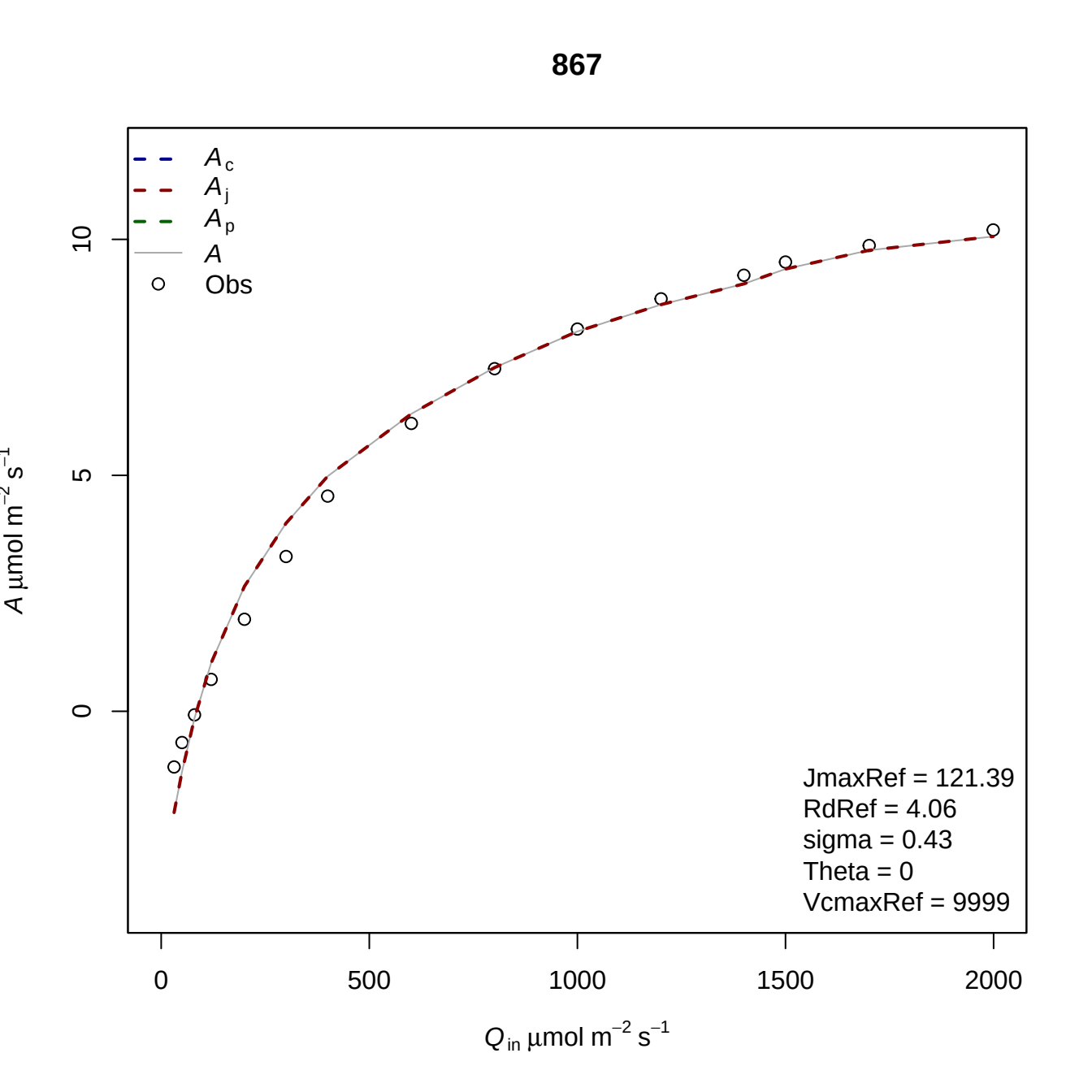

862
A_c
A_j
A_p
A
Obs
JmaxRef = 124.1
RdRef = 2.52
sigma = 0.21
Theta = 0
VcmaxRef = 89.01
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

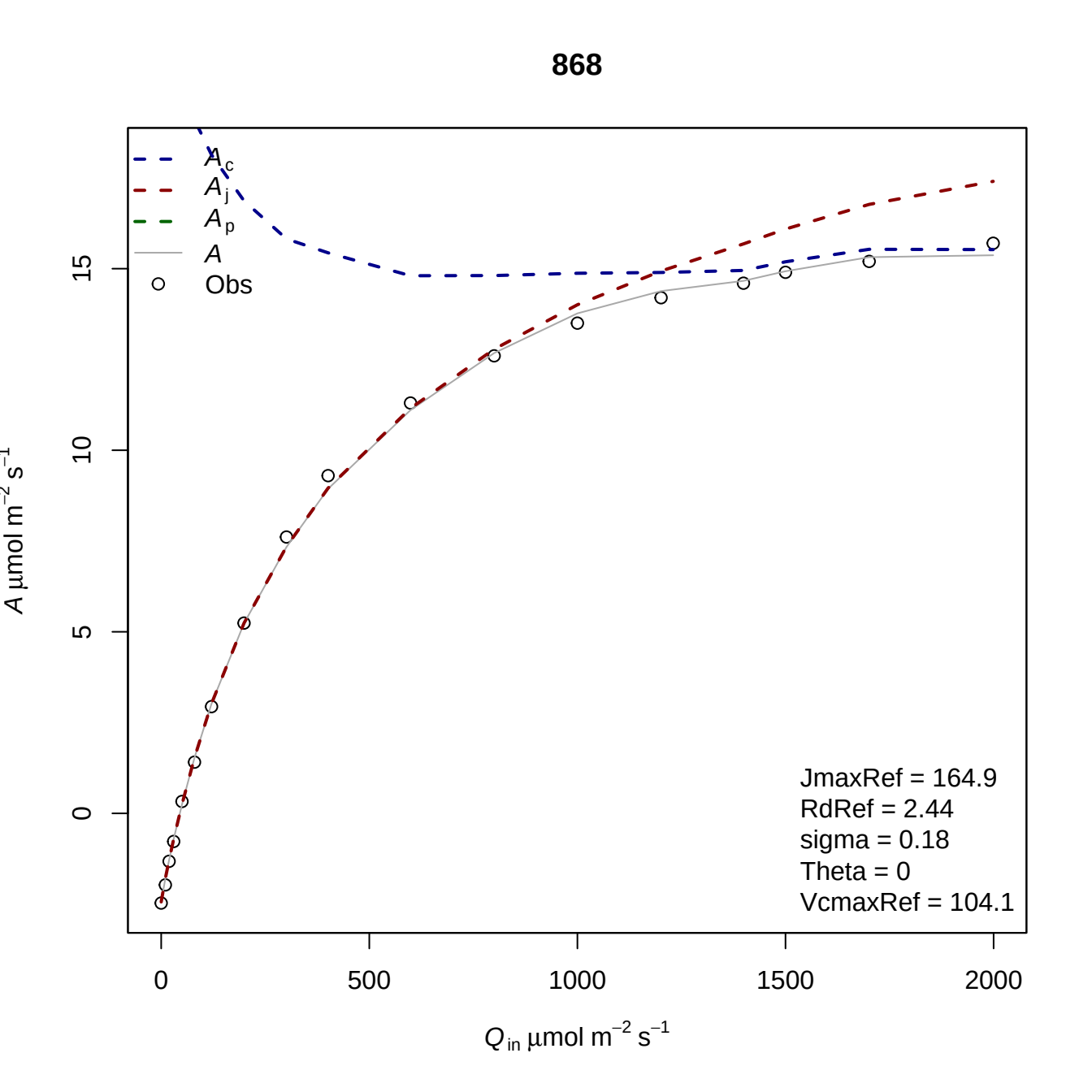

863
A_c
A_j
A_p
A
Obs
JmaxRef = 106.87
RdRef = 1.2
sigma = 0.23
Theta = 0
VcmaxRef = 68.85
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

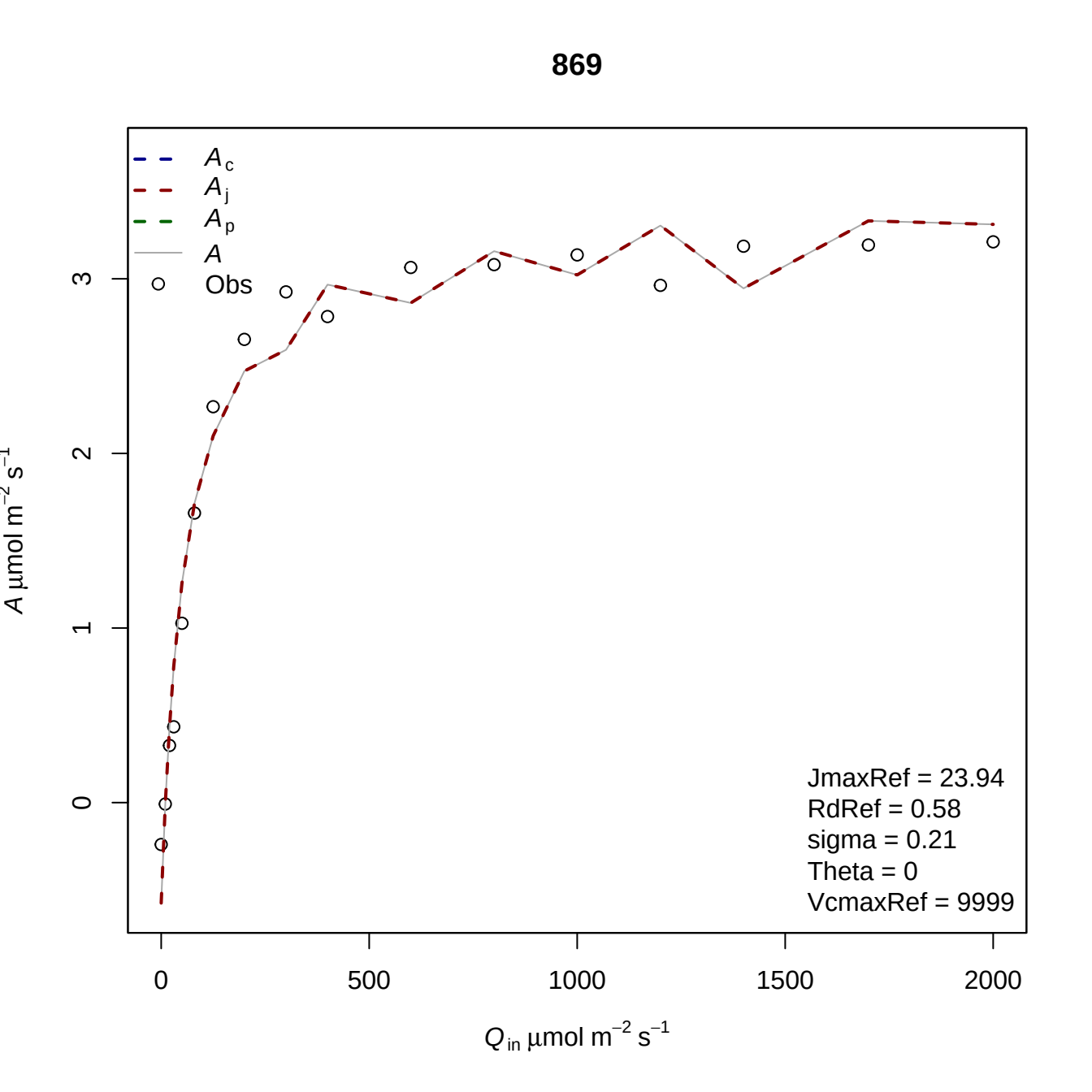

864
A_c
A_j
A_p
A
Obs
JmaxRef = 145.81
RdRef = 0.78
sigma = 0.21
Theta = 0
VcmaxRef = 89.1
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

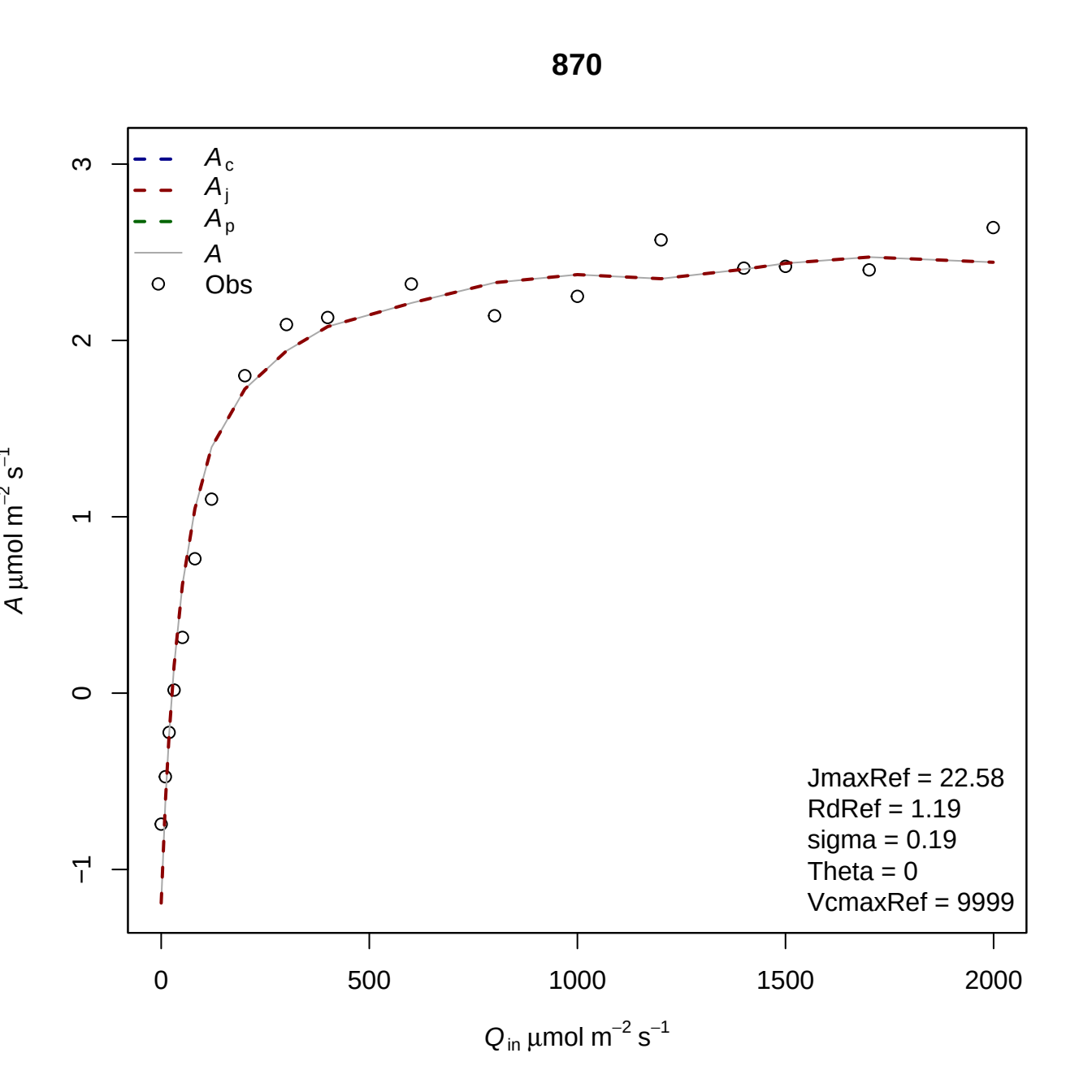

865
A_c
A_j
A_p
A
Obs
JmaxRef = 153.47
RdRef = 1.95
sigma = 0.13
Theta = 0
VcmaxRef = 110.64
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

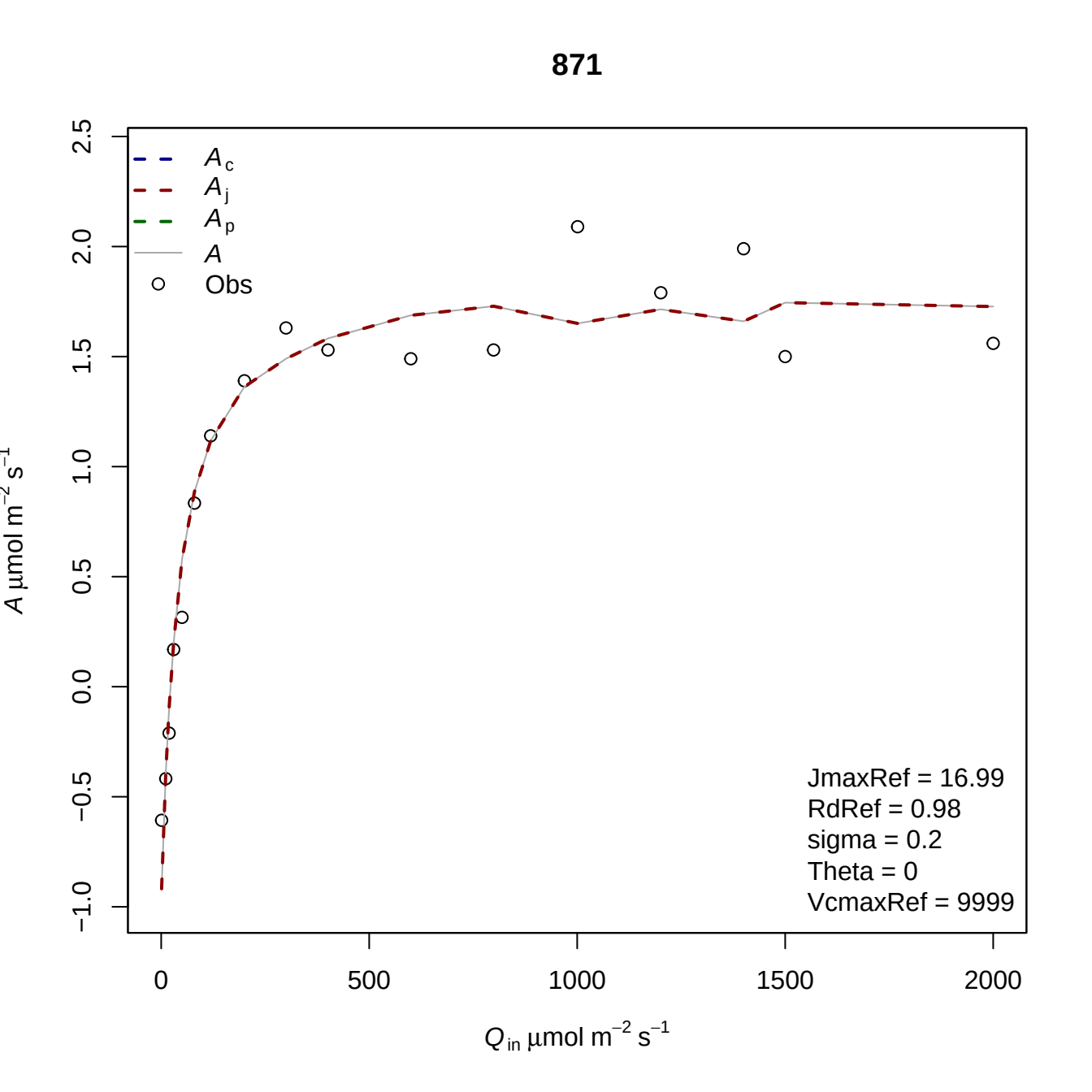

866
A_c
A_j
A_p
A
Obs
JmaxRef = 198.97
RdRef = 4.44
sigma = 1.03
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

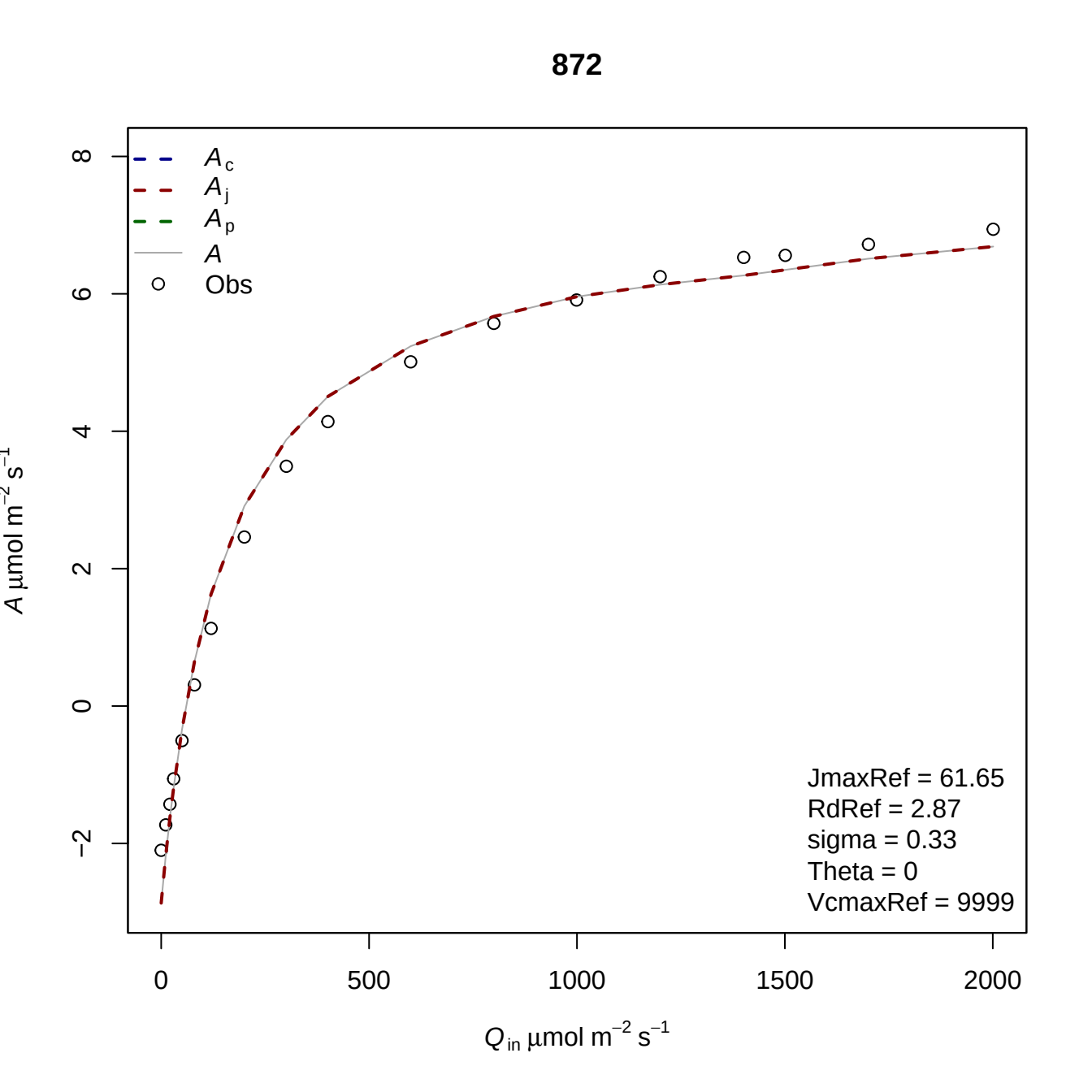

867
A_c
A_j
A_p
A
Obs
JmaxRef = 121.39
RdRef = 4.06
sigma = 0.43
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

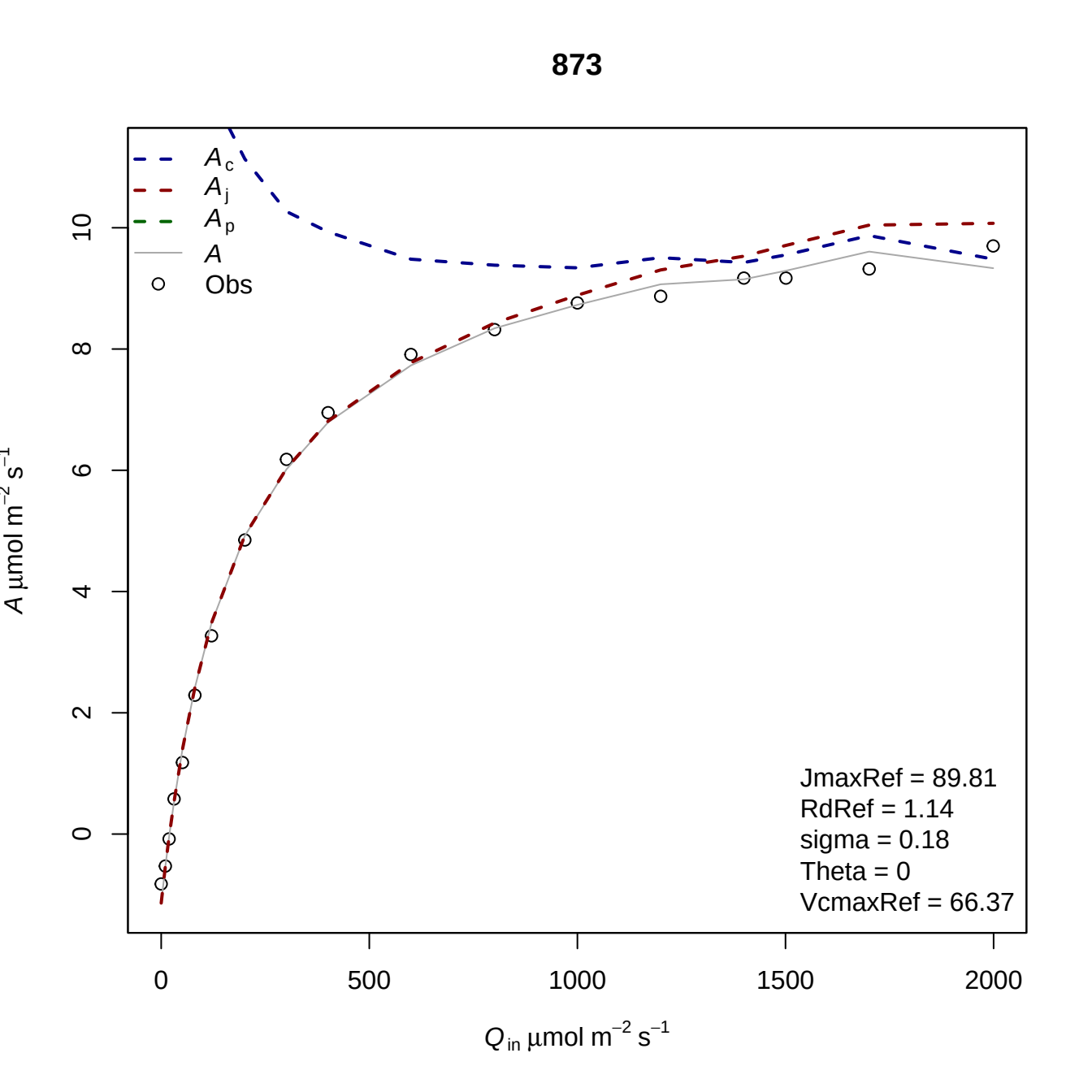

868
A_c
A_j
A_p
A
Obs
JmaxRef = 164.9
RdRef = 2.44
sigma = 0.18
Theta = 0
VcmaxRef = 104.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

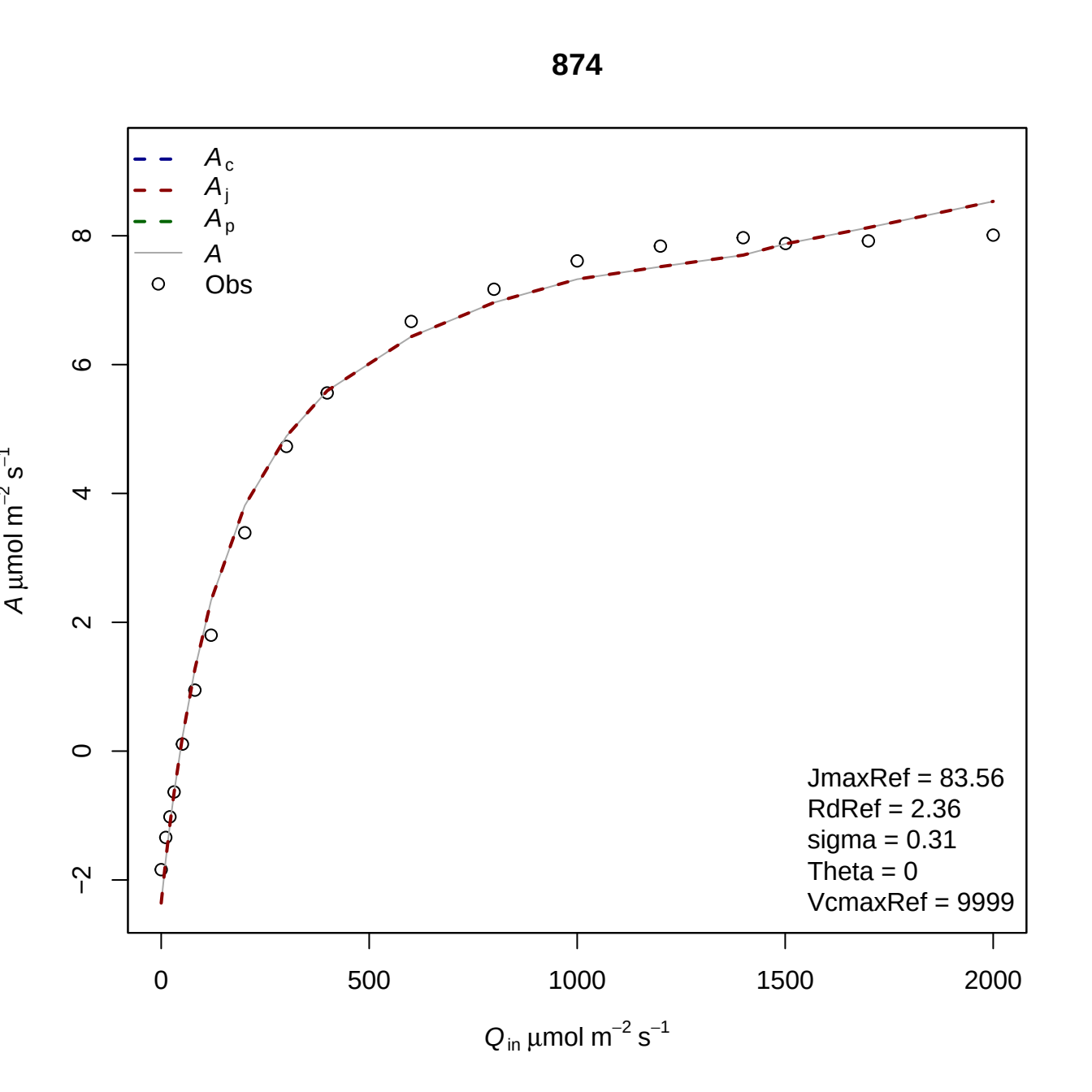

869
A_c
A_j
A_p
A
Obs
JmaxRef = 23.94
RdRef = 0.58
sigma = 0.21
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

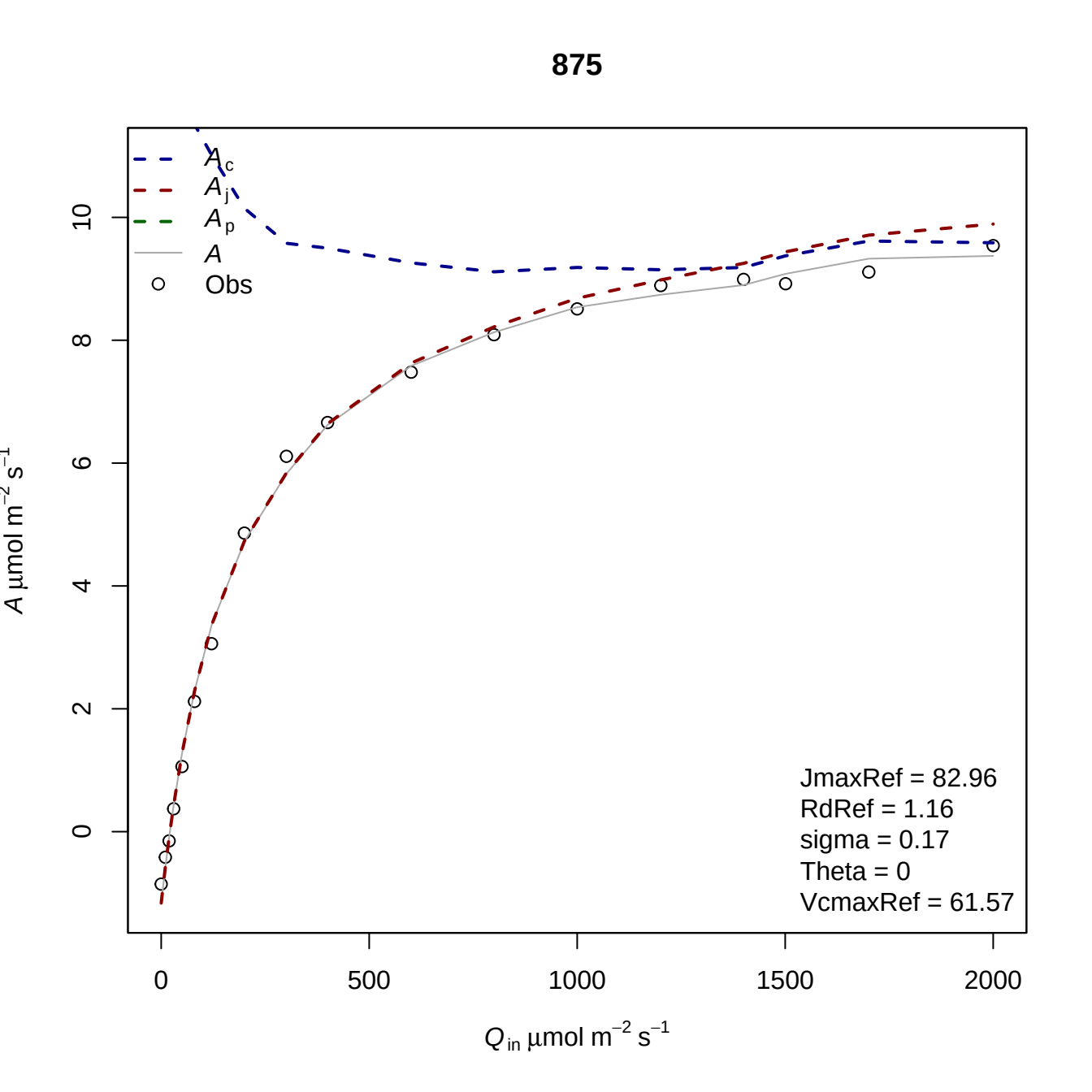

870
A_c
A_j
A_p
A
Obs
JmaxRef = 22.58
RdRef = 1.19
sigma = 0.19
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500
2000

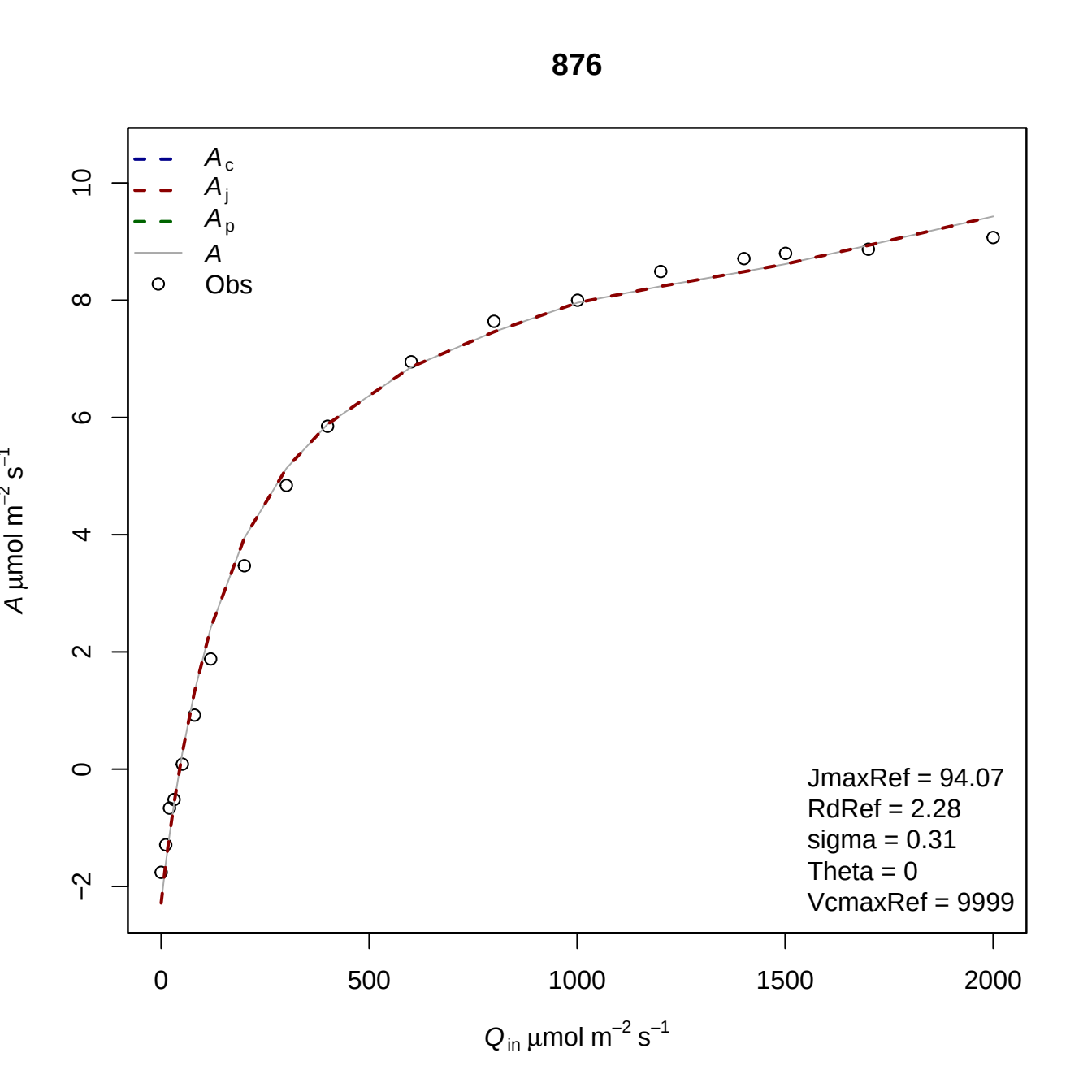

871
A_c
A_j
A_p
A
Obs
JmaxRef = 16.99
RdRef = 0.98
sigma = 0.2
Theta = 0
VcmaxRef = 9999
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

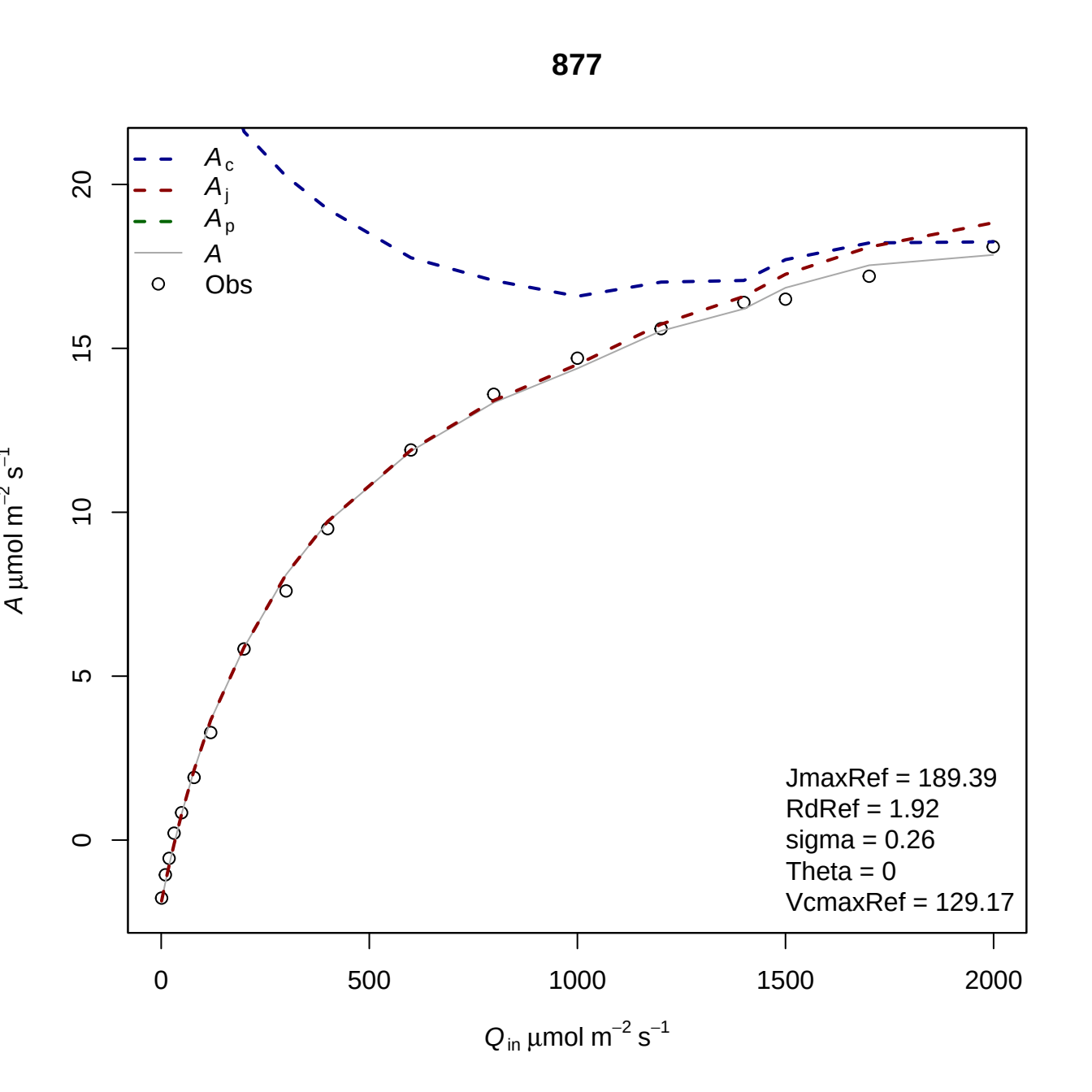

872
A_c
A_j
A_p
A
Obs
JmaxRef = 61.65
RdRef = 2.87
sigma = 0.33
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

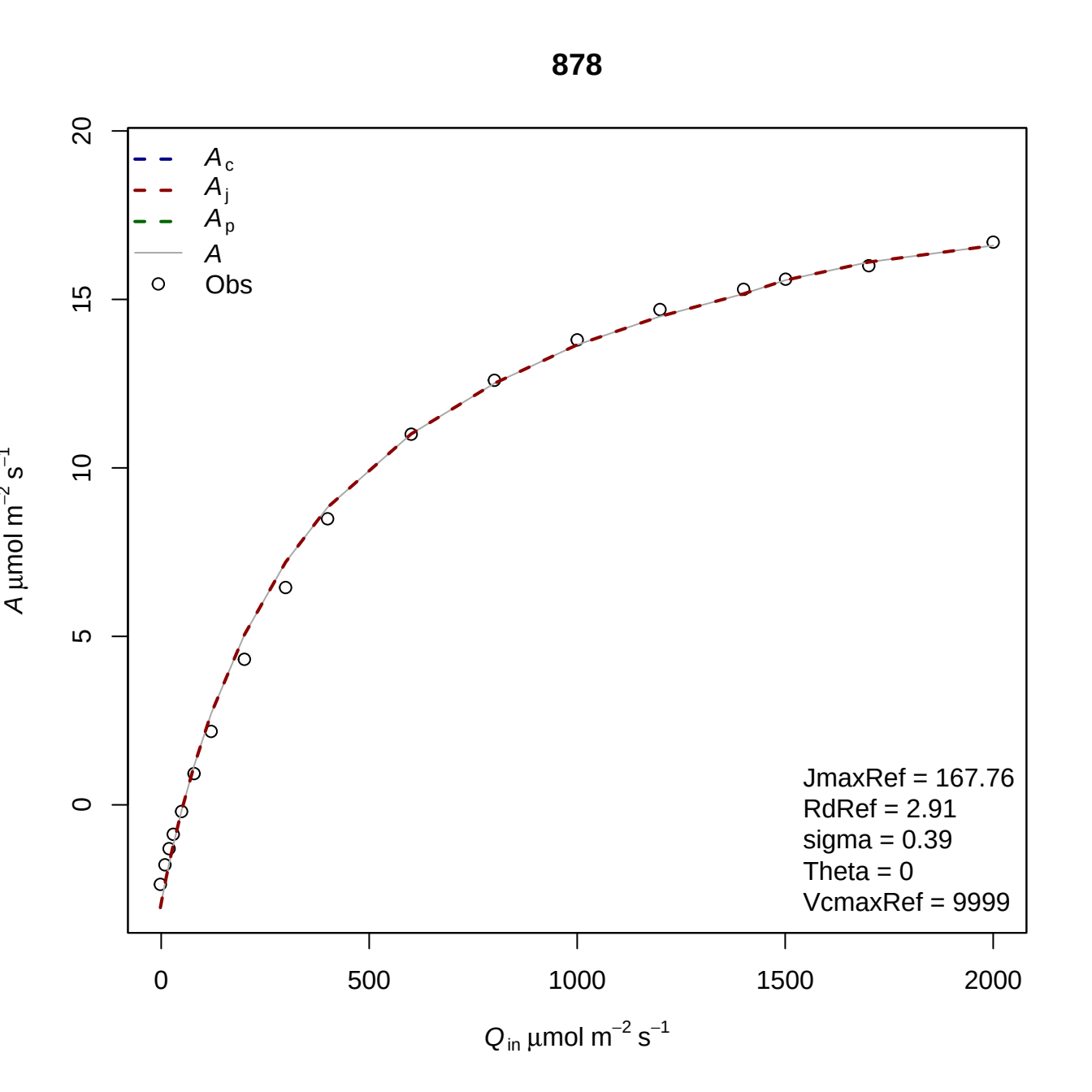

873
A_c
A_j
A_p
A
Obs
JmaxRef = 89.81
RdRef = 1.14
sigma = 0.18
Theta = 0
VcmaxRef = 66.37
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

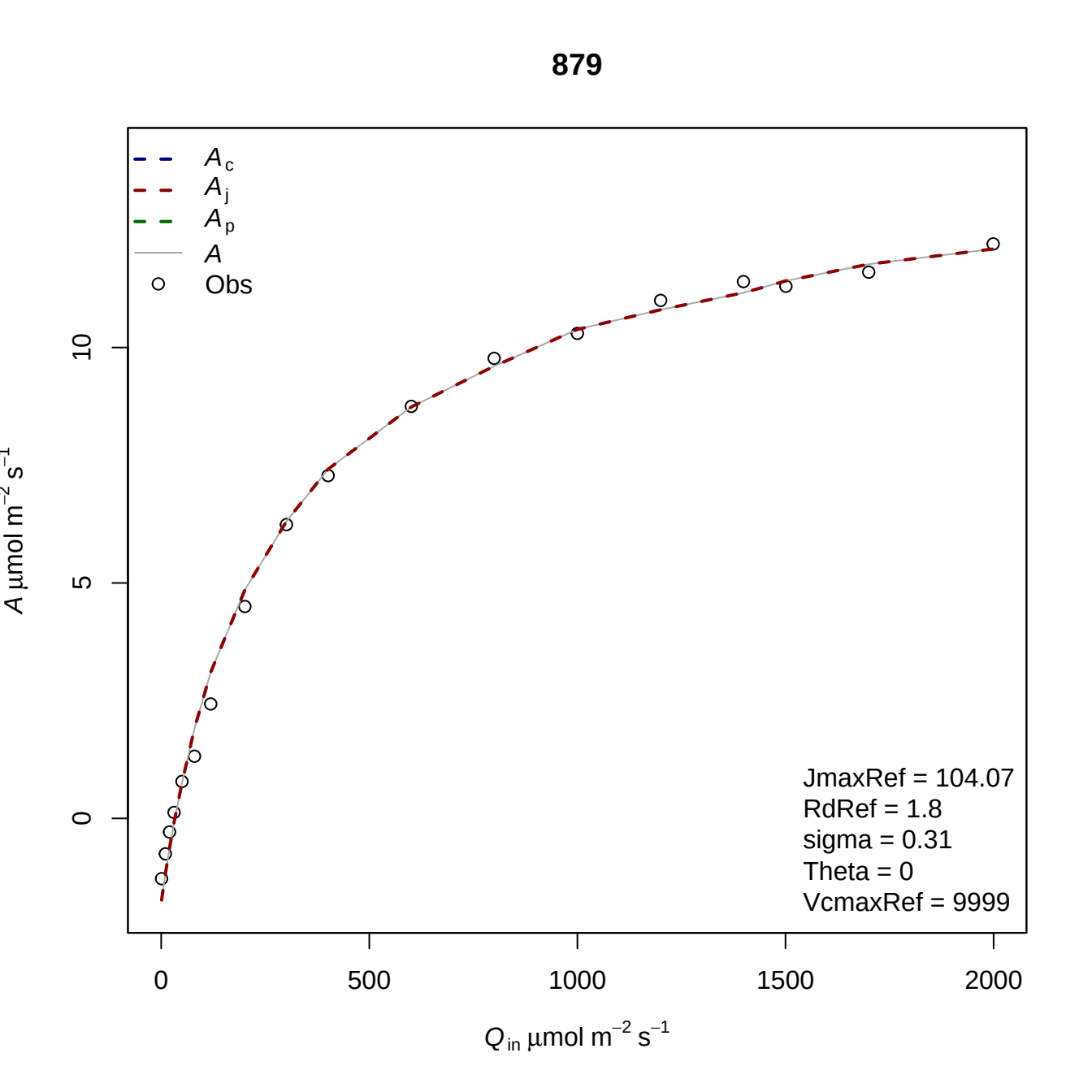

874
A_c
A_j
A_p
A
Obs
JmaxRef = 83.56
RdRef = 2.36
sigma = 0.31
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

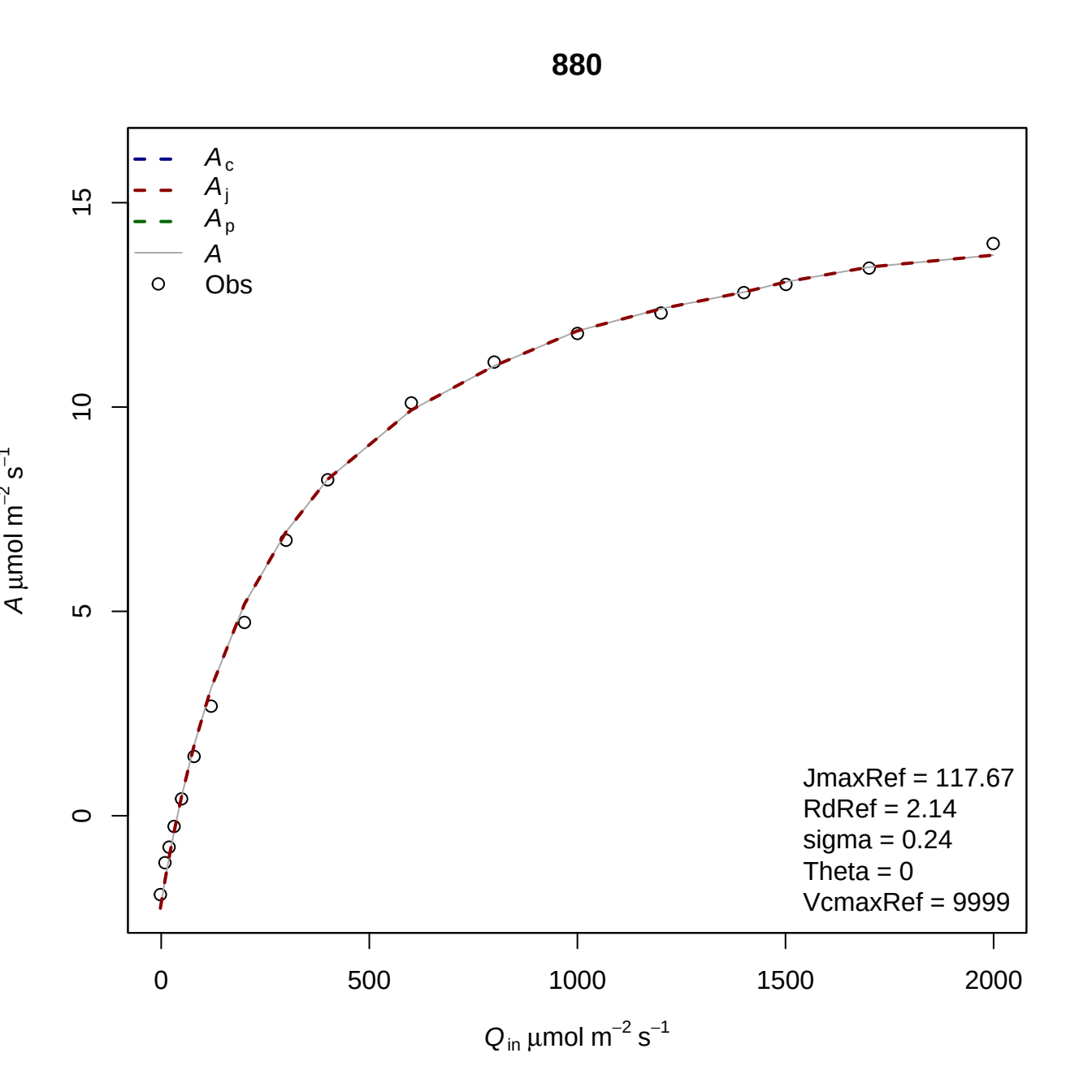

875
A_c
A_j
A_p
A
Obs
JmaxRef = 82.96
RdRef = 1.16
sigma = 0.17
Theta = 0
VcmaxRef = 61.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

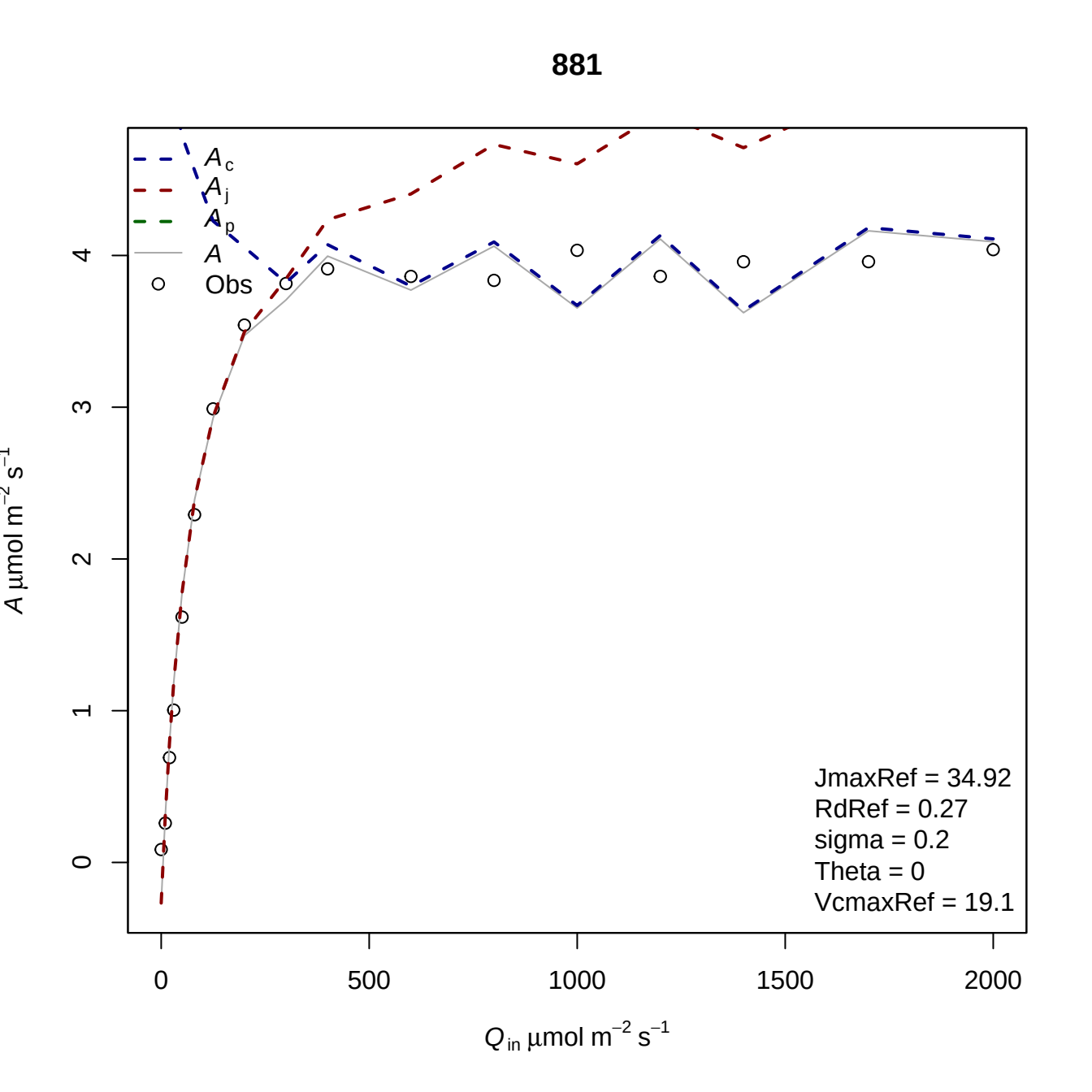

876
A_c
A_j
A_p
A
Obs
JmaxRef = 94.07
RdRef = 2.28
sigma = 0.31
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

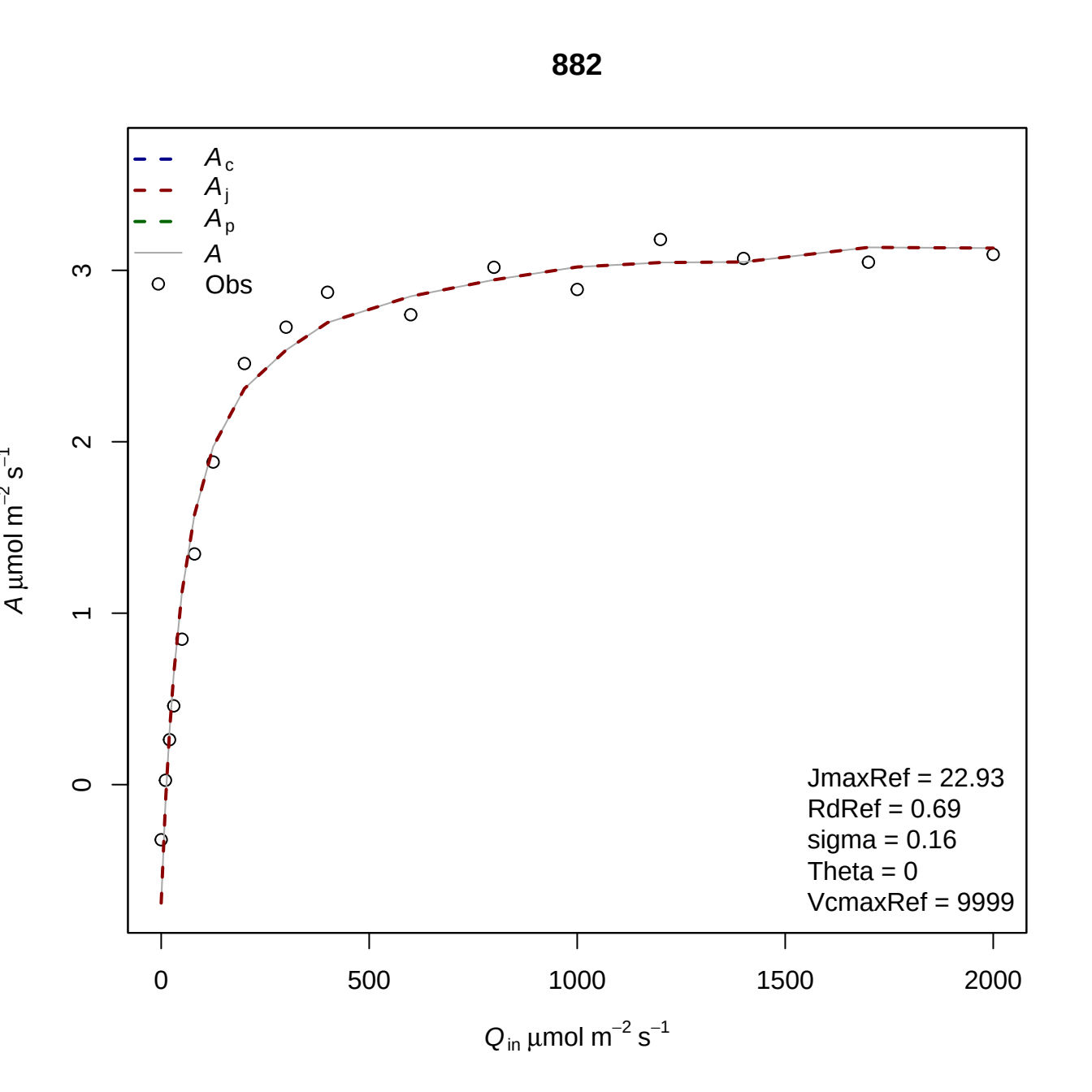

877
A_c
A_j
A_p
A
Obs
JmaxRef = 189.39
RdRef = 1.92
sigma = 0.26
Theta = 0
VcmaxRef = 129.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

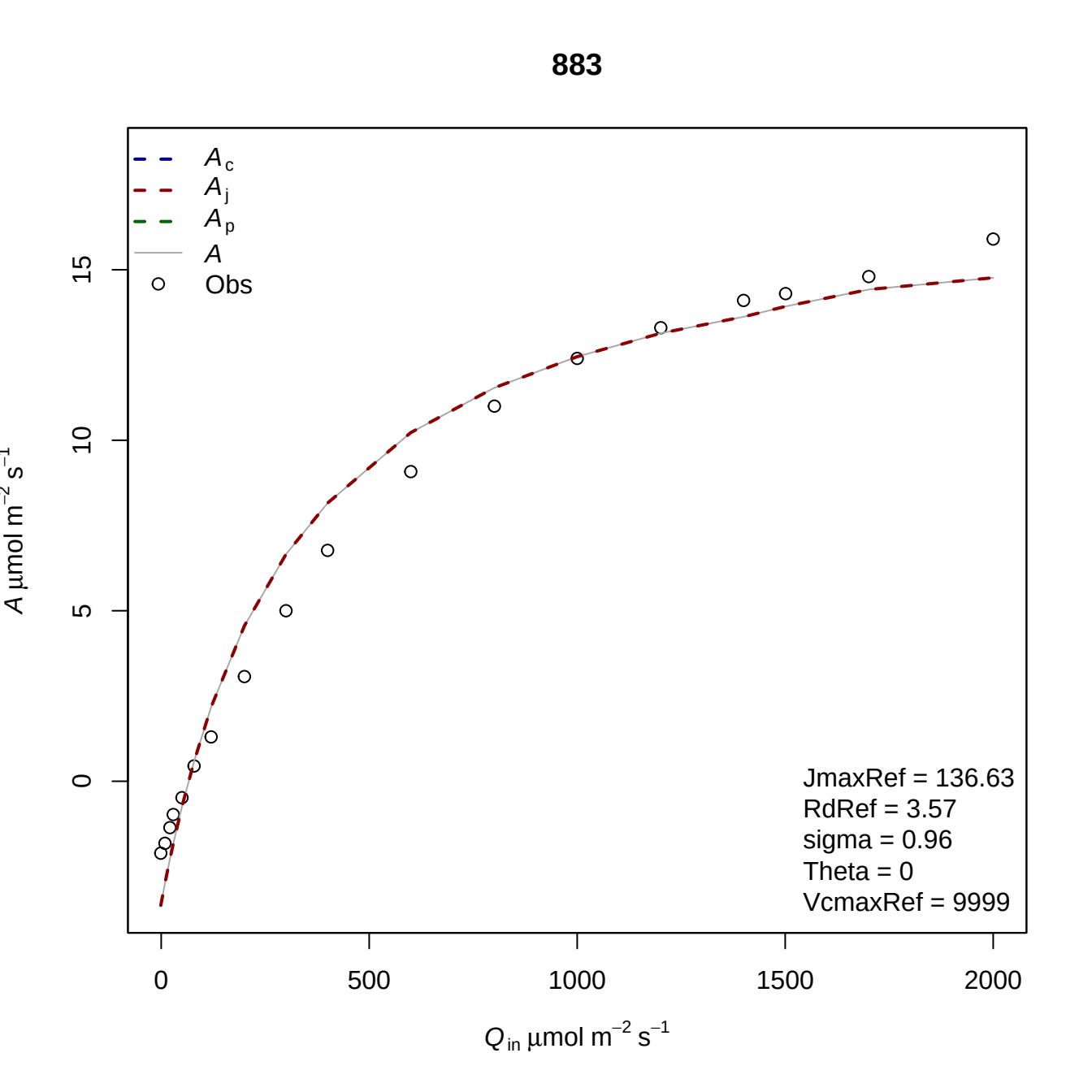

878
A_c
A_j
A_p
A
Obs
JmaxRef = 167.76
RdRef = 2.91
sigma = 0.39
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

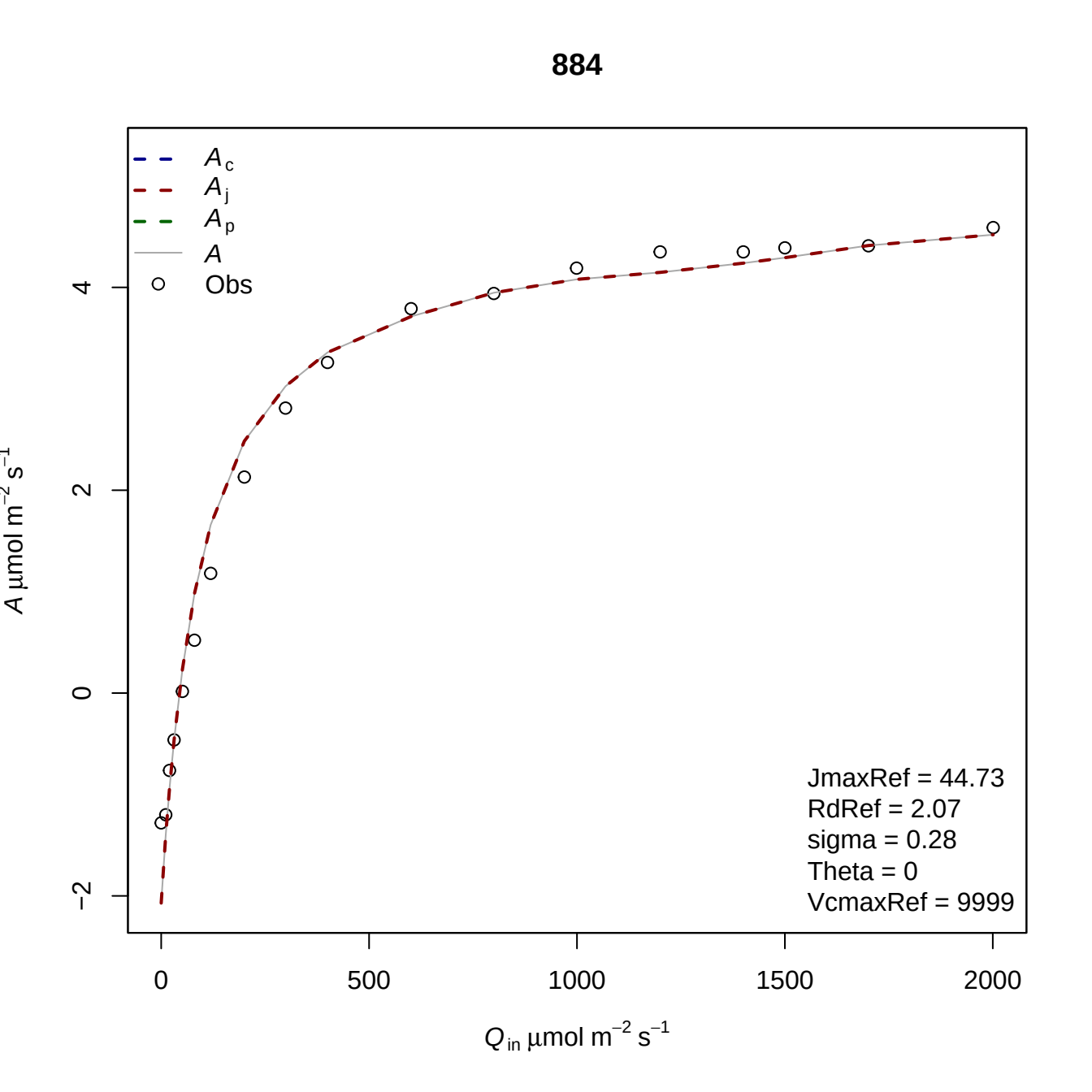

879
A_c
A_j
A_p
A
Obs
JmaxRef = 104.07
RdRef = 1.8
sigma = 0.31
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

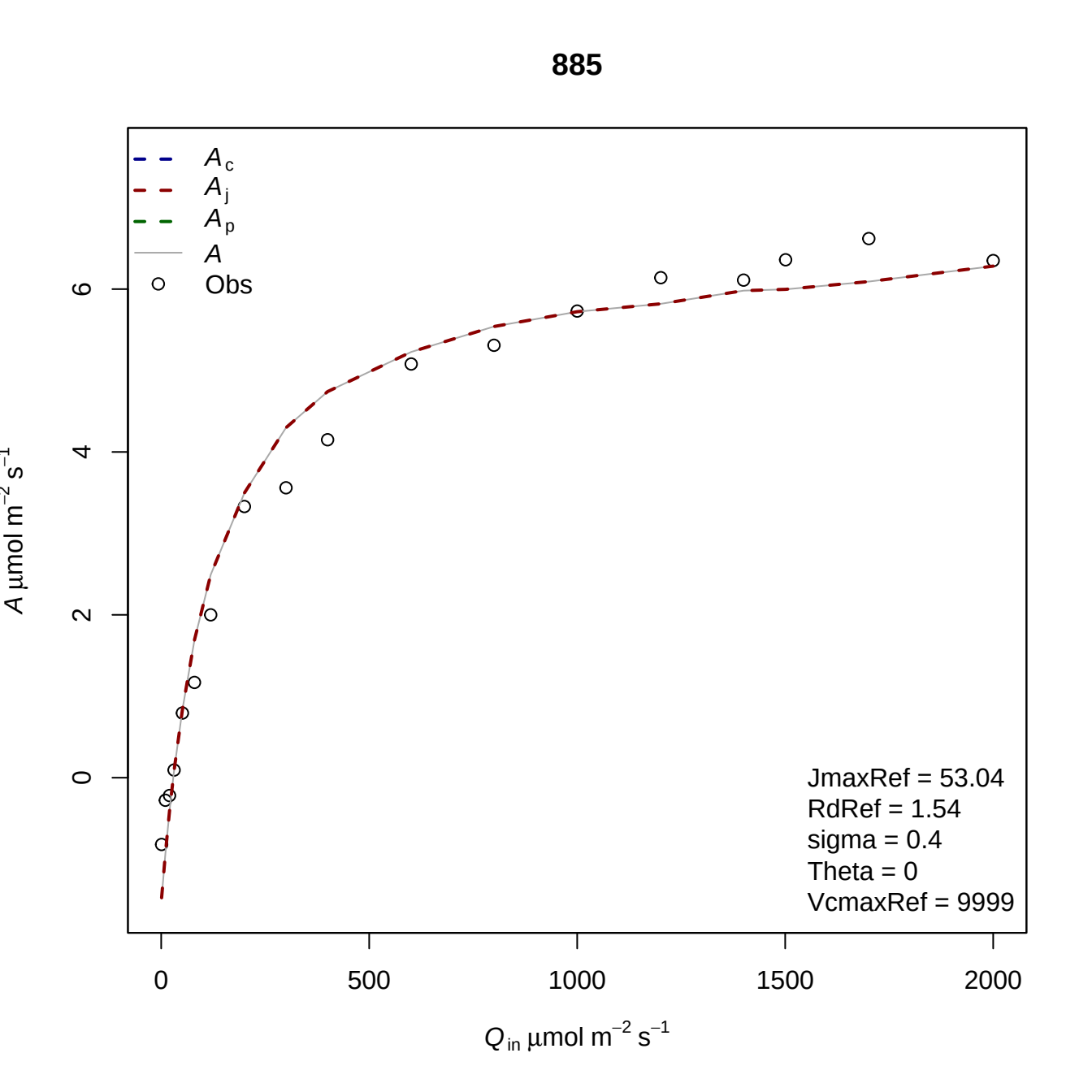

880
A_c
A_j
A_p
A
Obs
JmaxRef = 117.67
RdRef = 2.14
sigma = 0.24
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

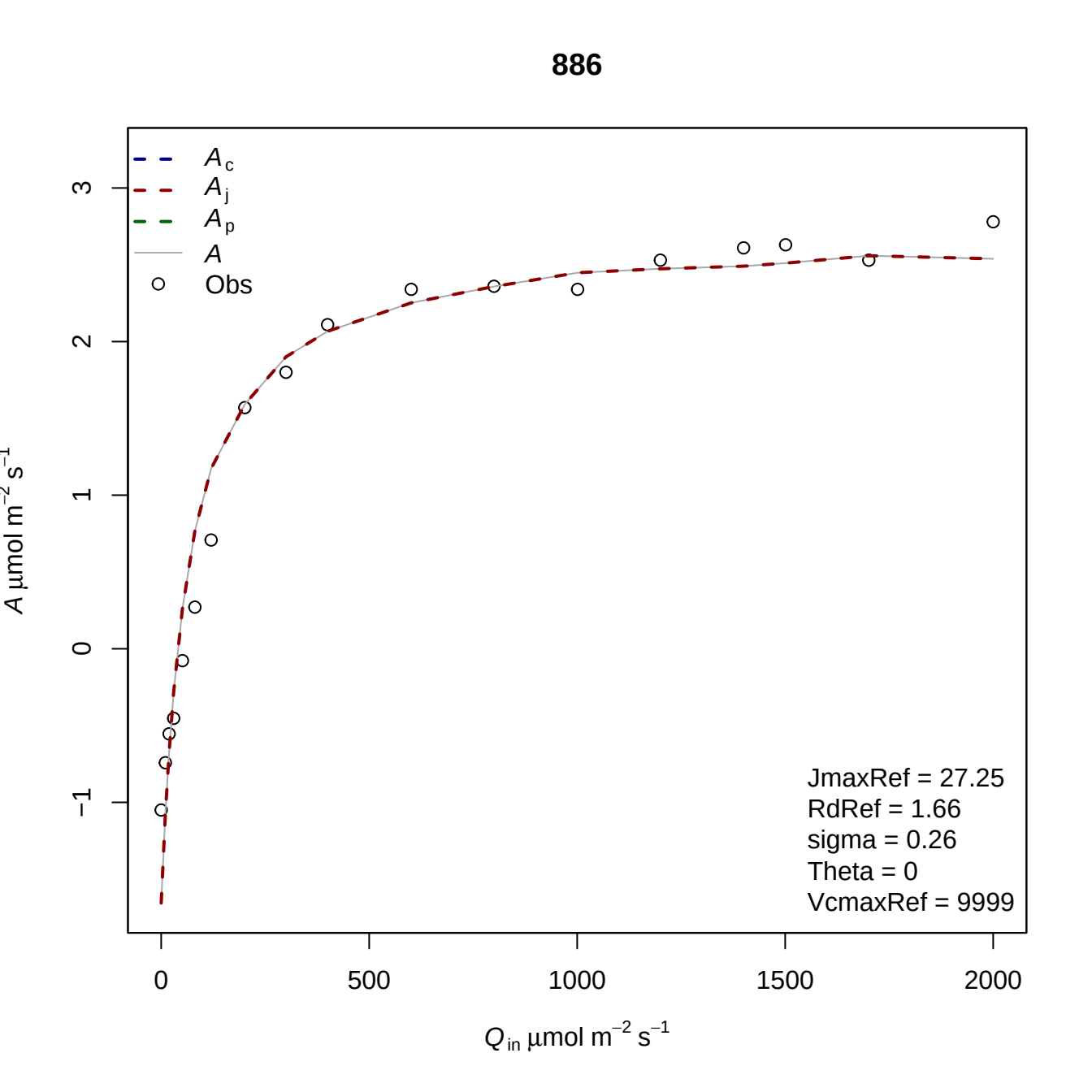

881
A_c
A_j
A_p
A
Obs
JmaxRef = 34.92
RdRef = 0.27
sigma = 0.2
Theta = 0
VcmaxRef = 19.1
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

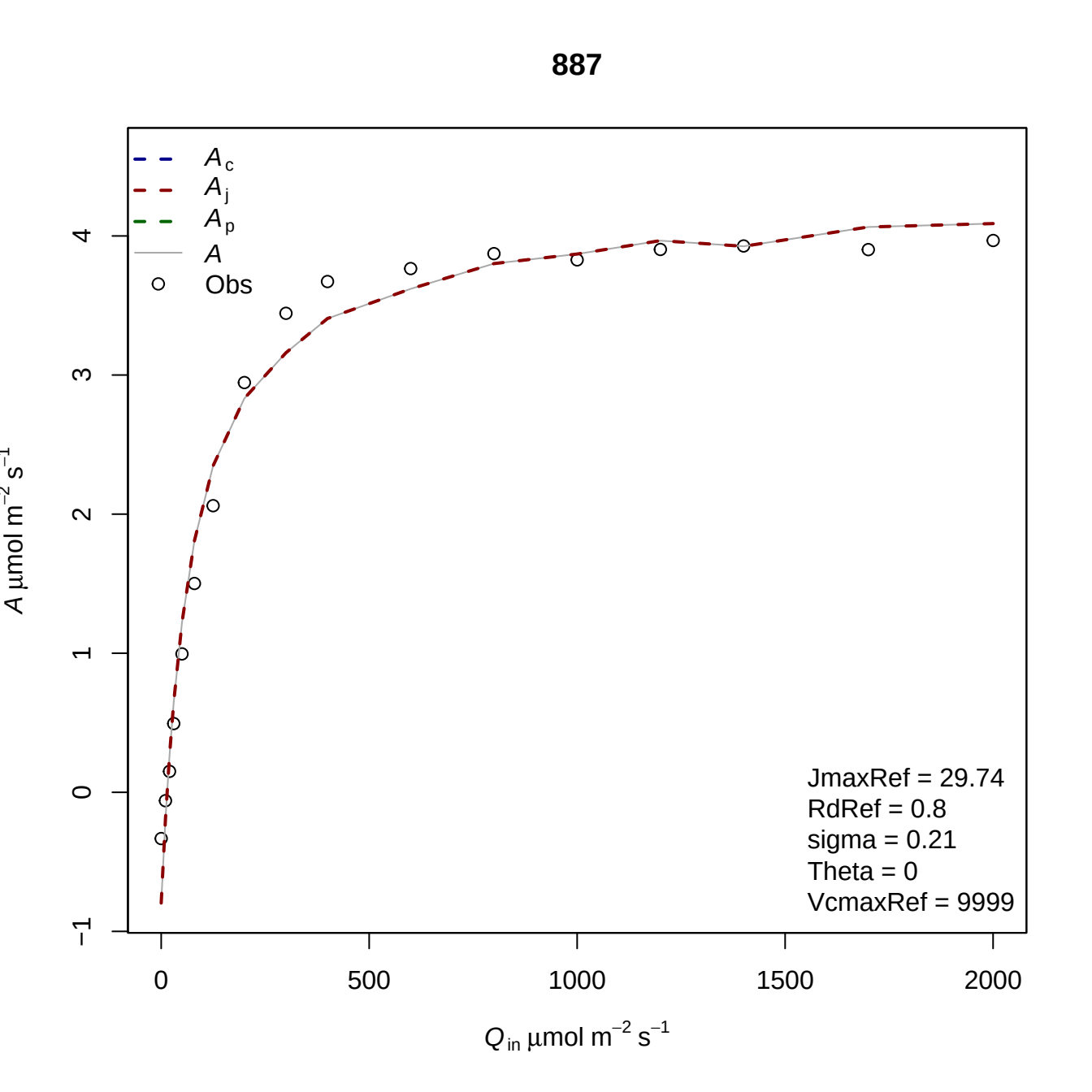

882
A_c
A_j
A_p
A
Obs
JmaxRef = 22.93
RdRef = 0.69
sigma = 0.16
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

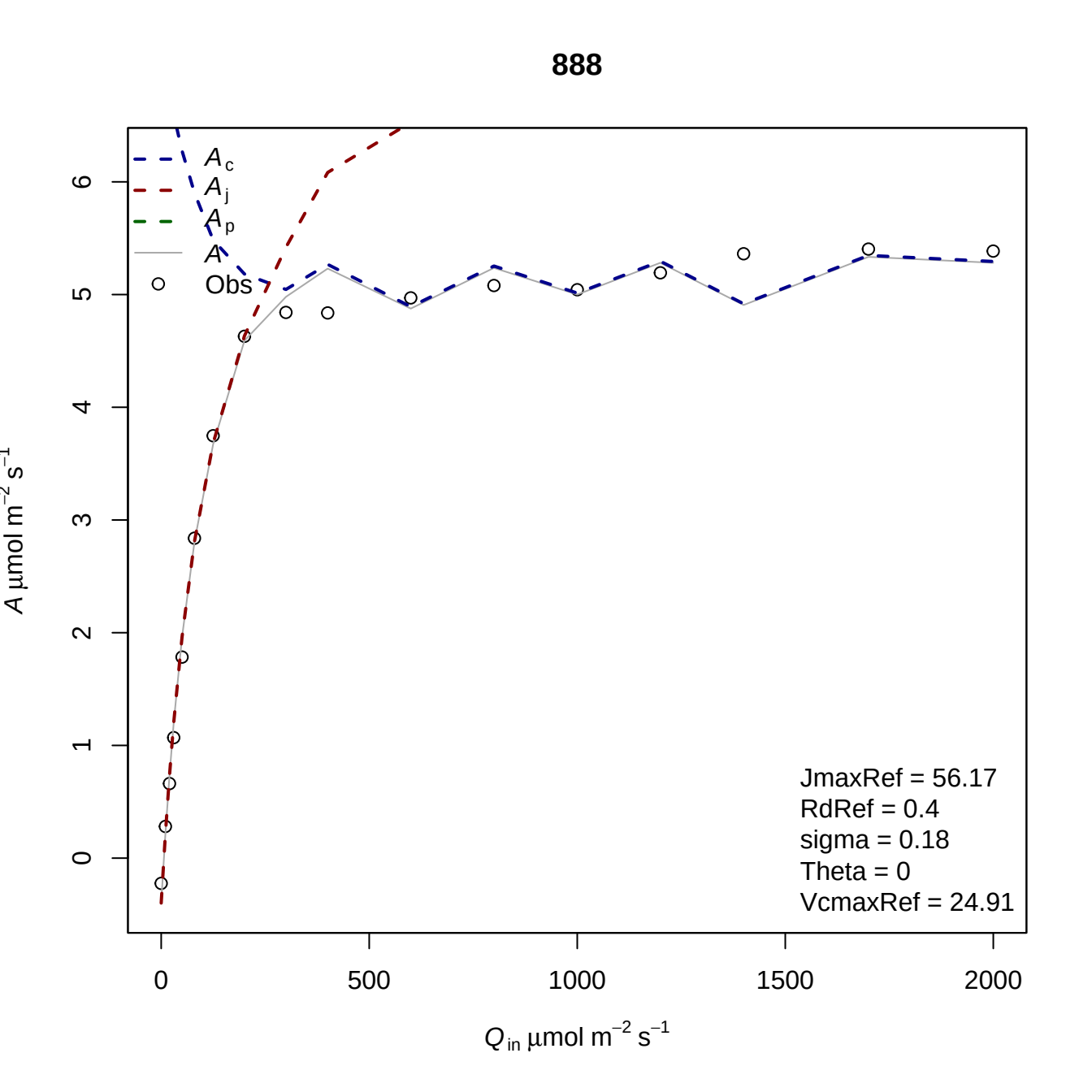

883
A_c
A_j
A_p
A
Obs
JmaxRef = 136.63
RdRef = 3.57
sigma = 0.96
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

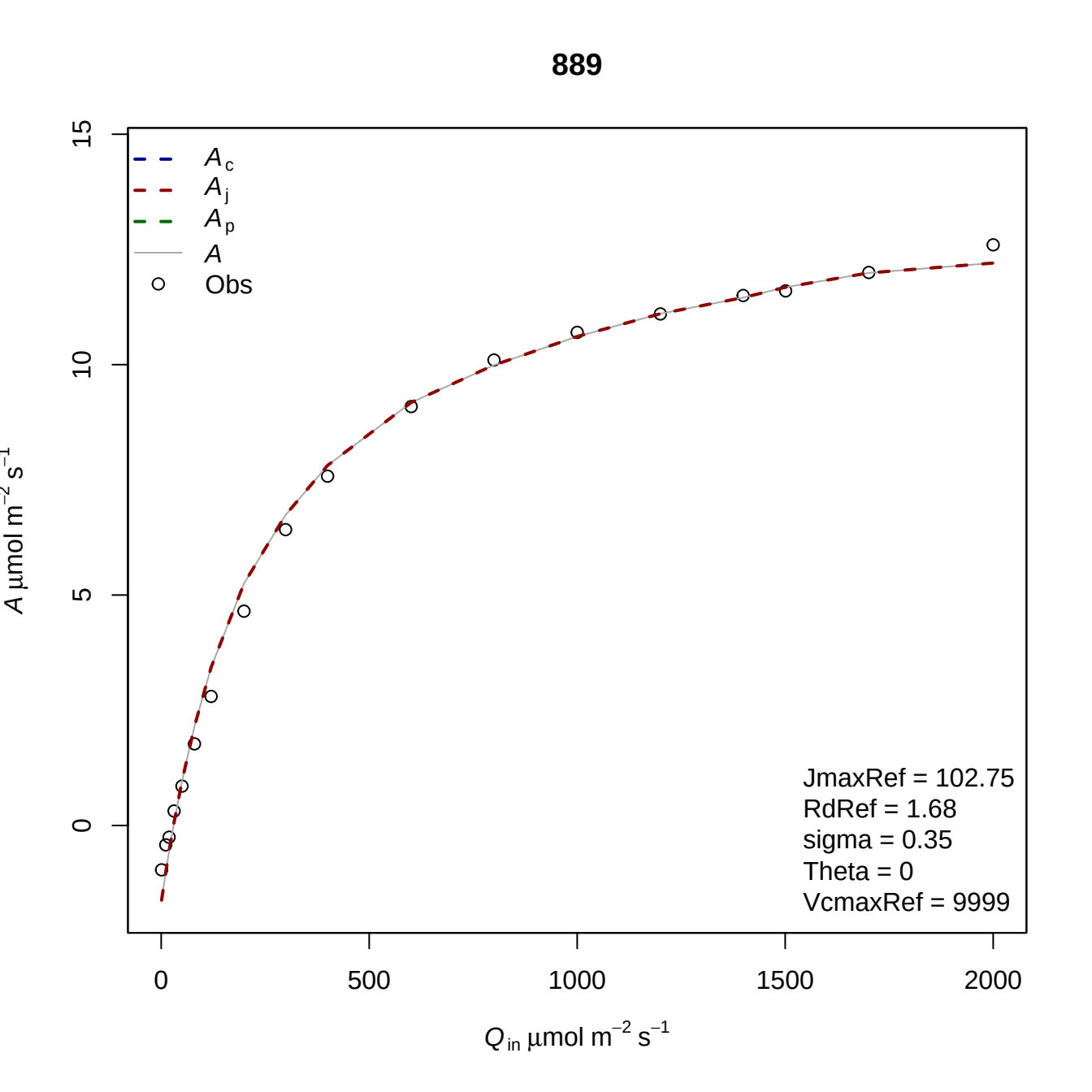

884
A_c
A_j
A_p
A
Obs
JmaxRef = 44.73
RdRef = 2.07
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

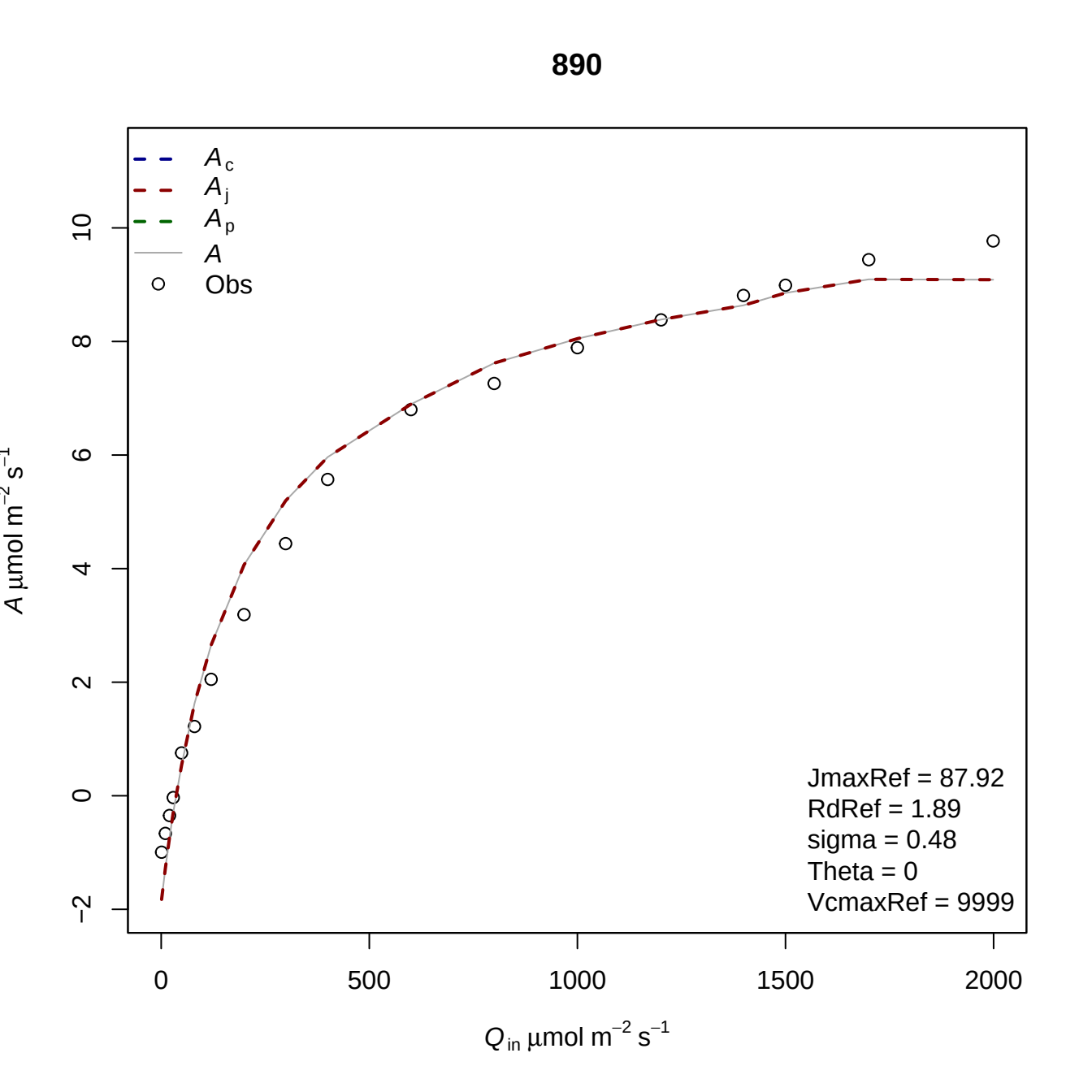

885
A_c
A_j
A_p
A
Obs
JmaxRef = 53.04
RdRef = 1.54
sigma = 0.4
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

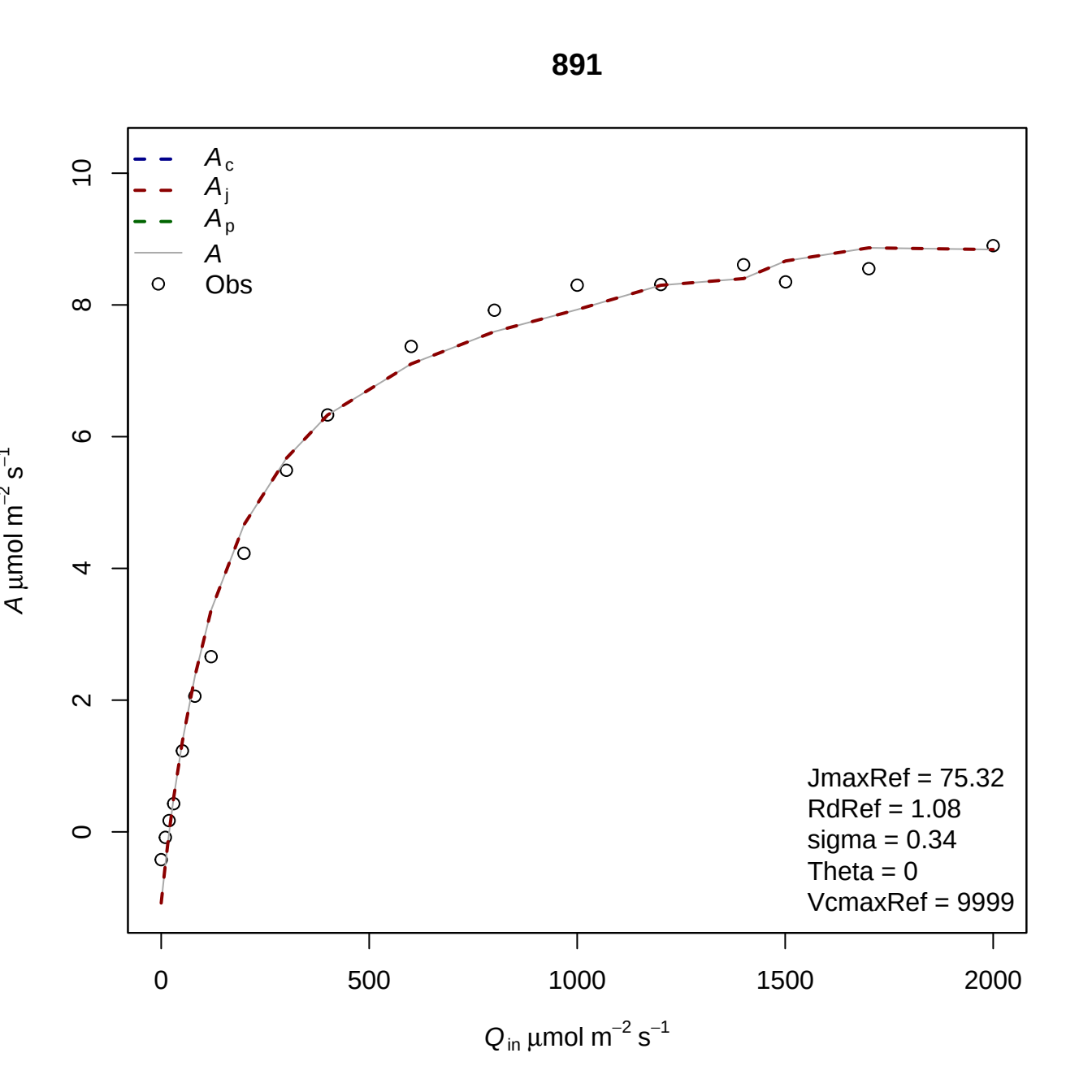

886
A_c
A_j
A_p
A
Obs
JmaxRef = 27.25
RdRef = 1.66
sigma = 0.26
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

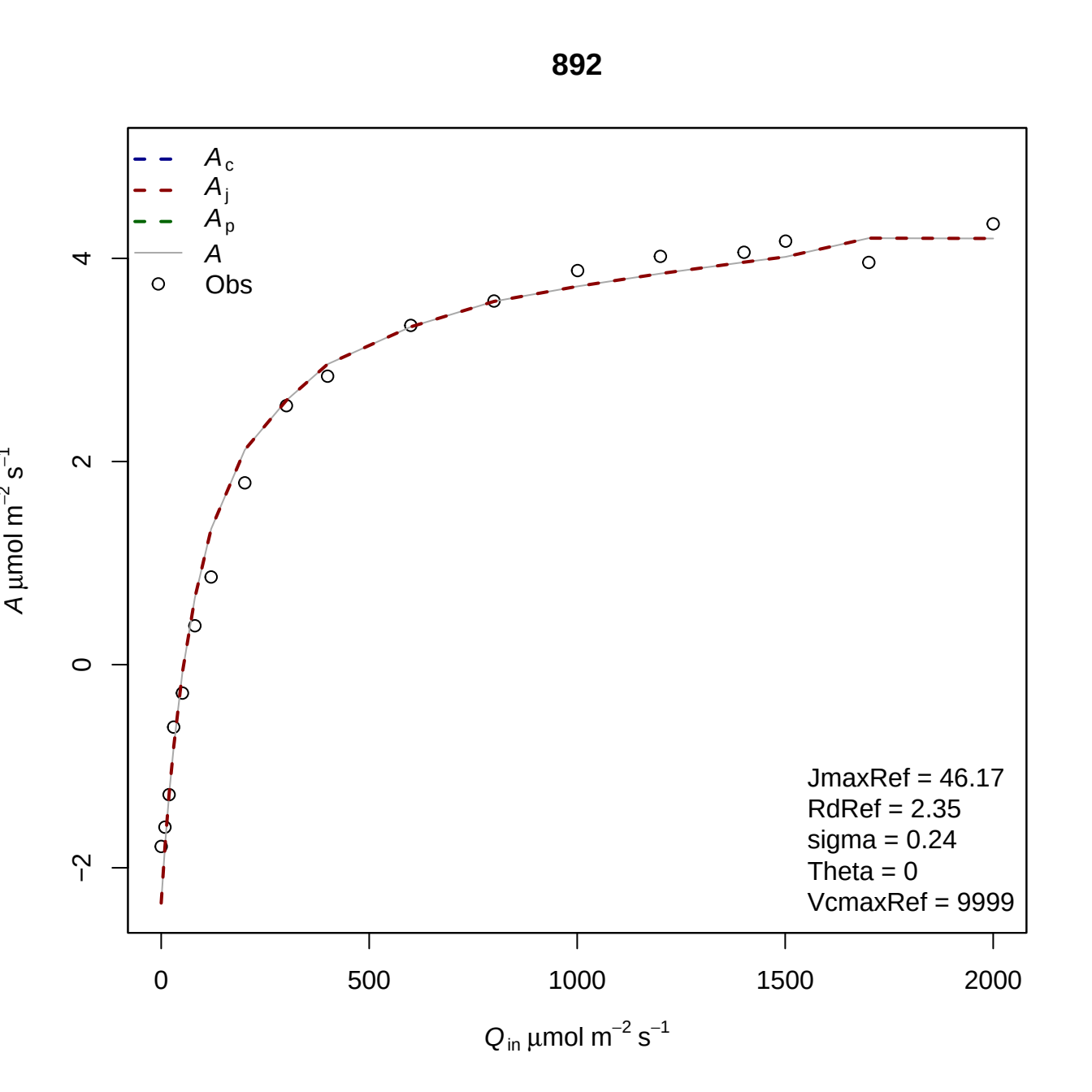

887
A_c
A_j
A_p
A
Obs
JmaxRef = 29.74
RdRef = 0.8
sigma = 0.21
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

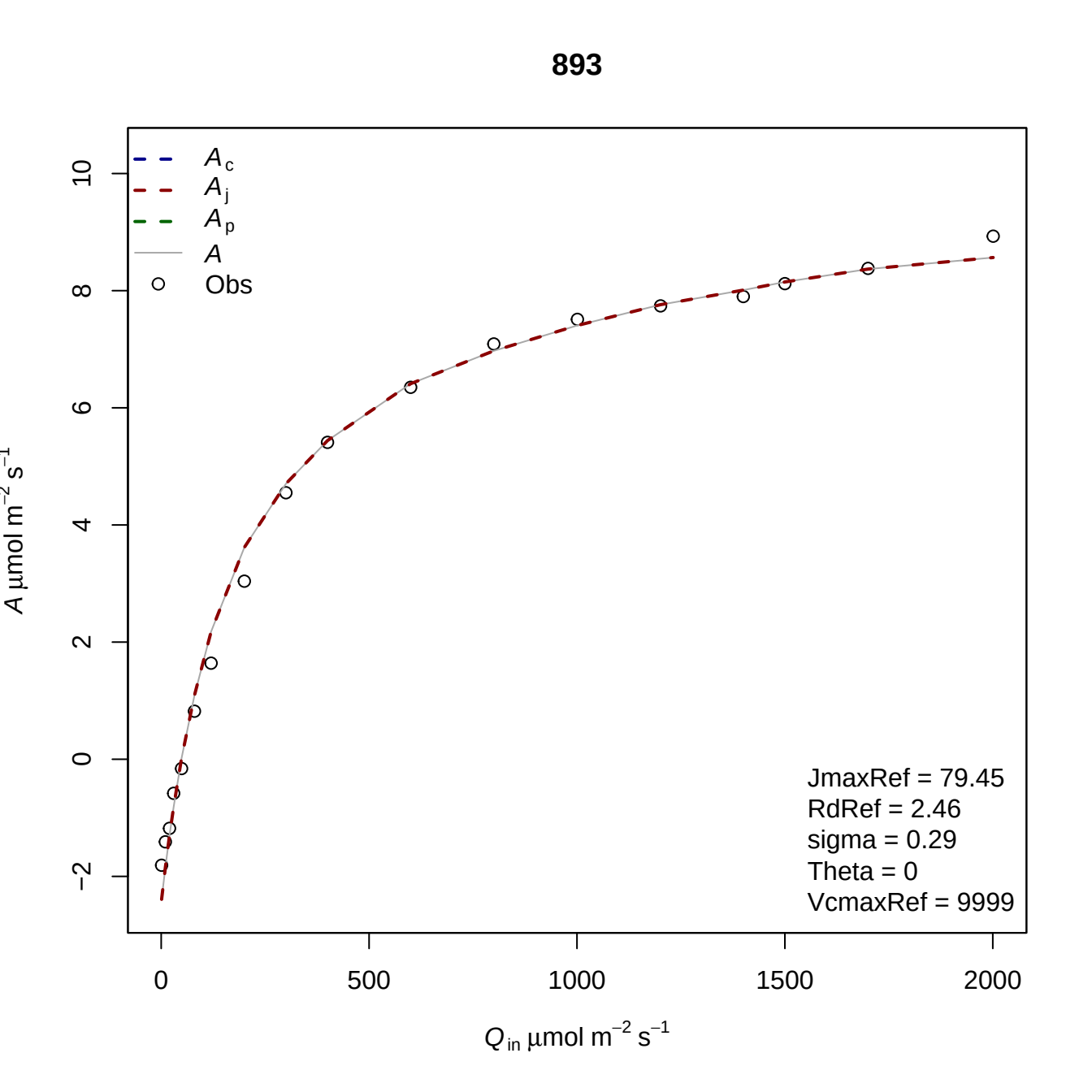

888
A_c
A_j
A_p
A
Obs
JmaxRef = 56.17
RdRef = 0.4
sigma = 0.18
Theta = 0
VcmaxRef = 24.91
0
1
2
3
4
5
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

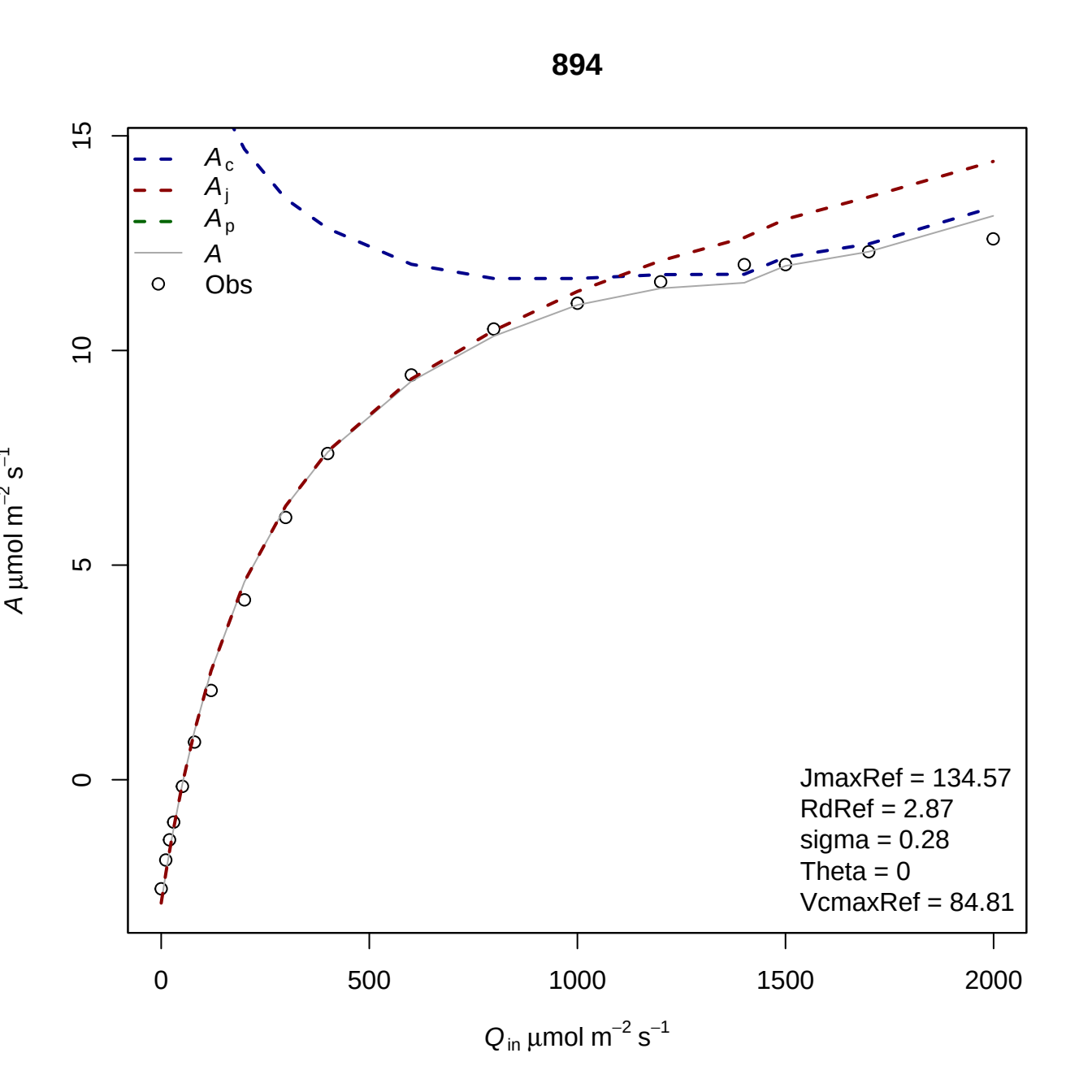

889
A_c
A_j
A_p
A
Obs
JmaxRef = 102.75
RdRef = 1.68
sigma = 0.35
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

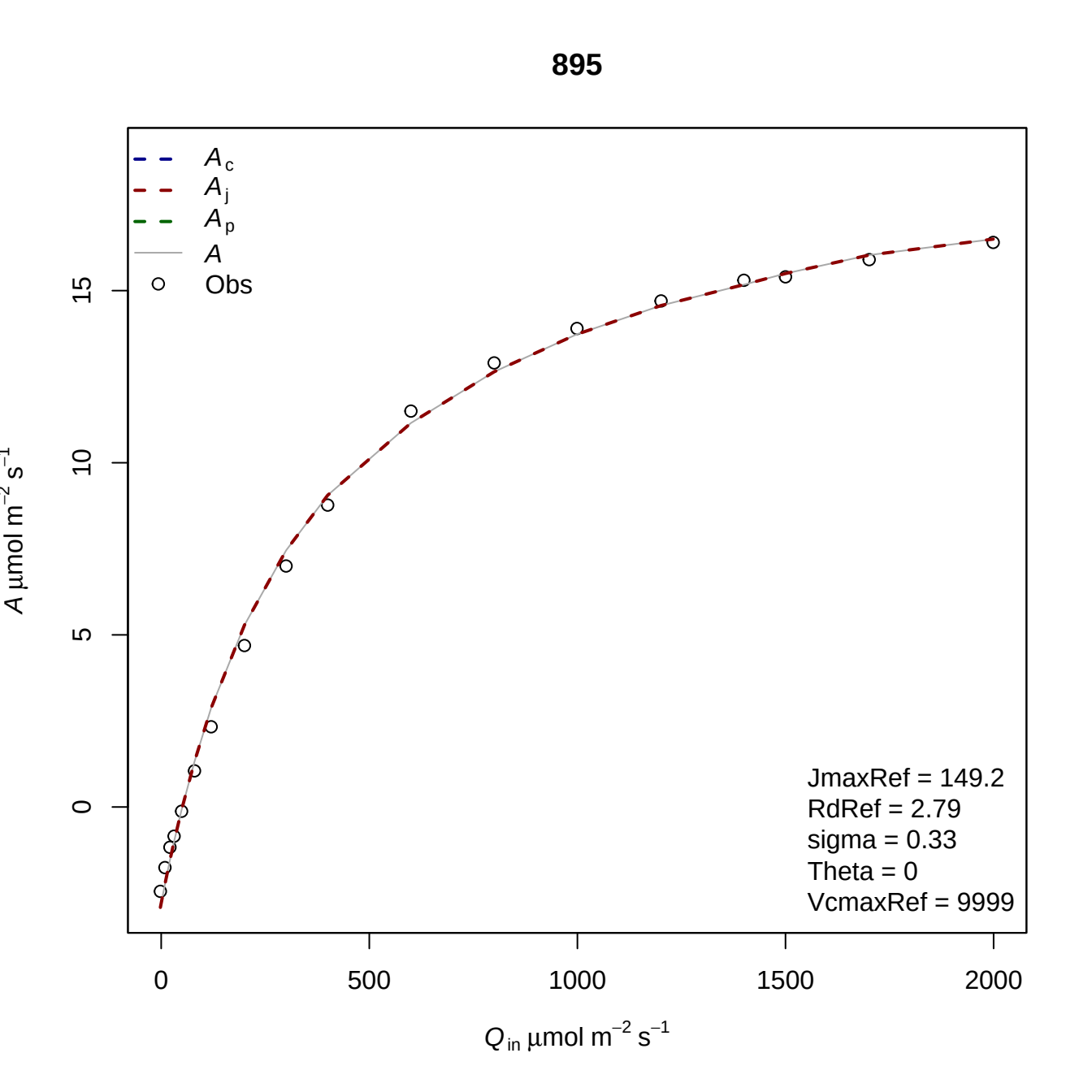

890
A_c
A_j
A_p
A
Obs
JmaxRef = 87.92
RdRef = 1.89
sigma = 0.48
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

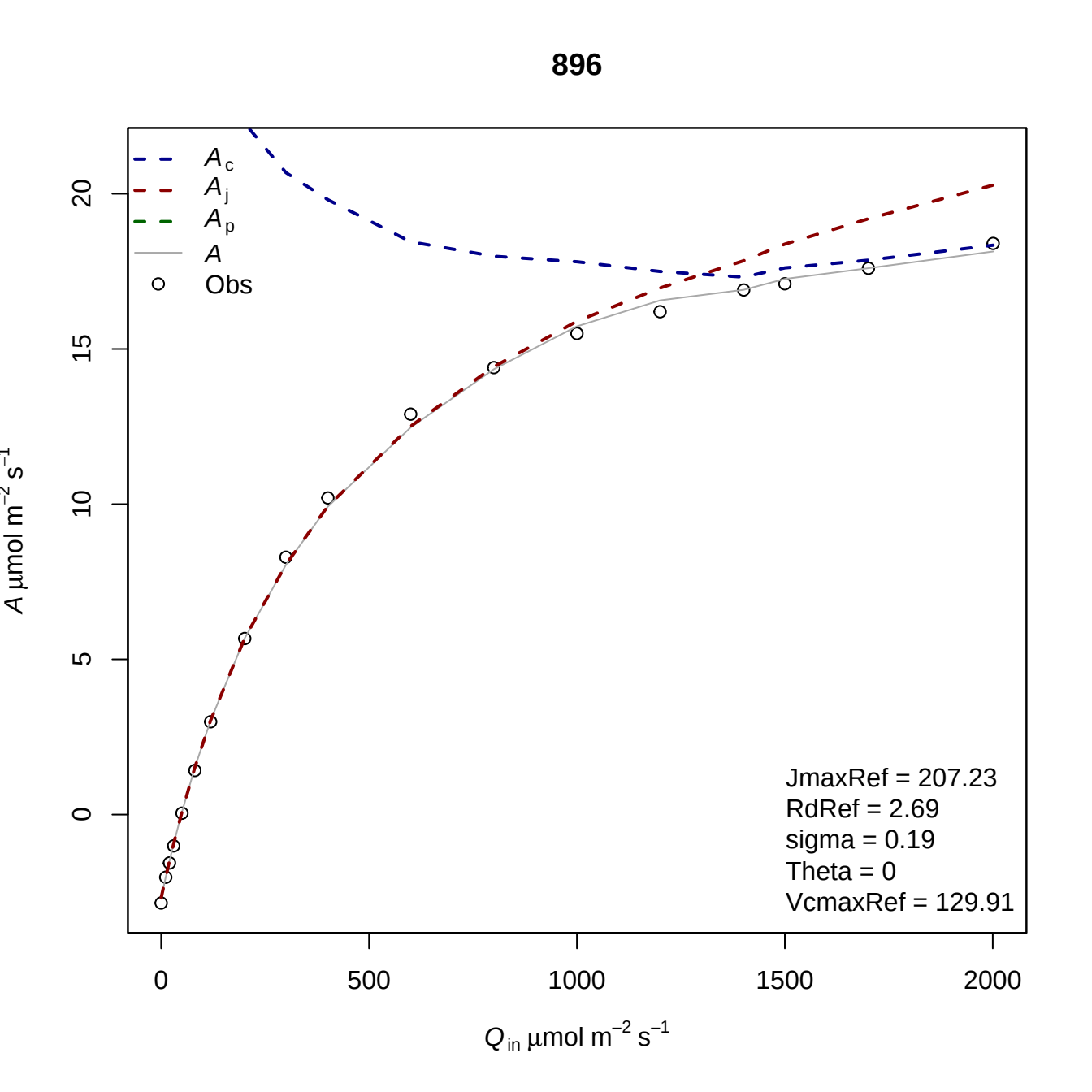

891
A_c
A_j
A_p
A
Obs
JmaxRef = 75.32
RdRef = 1.08
sigma = 0.34
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

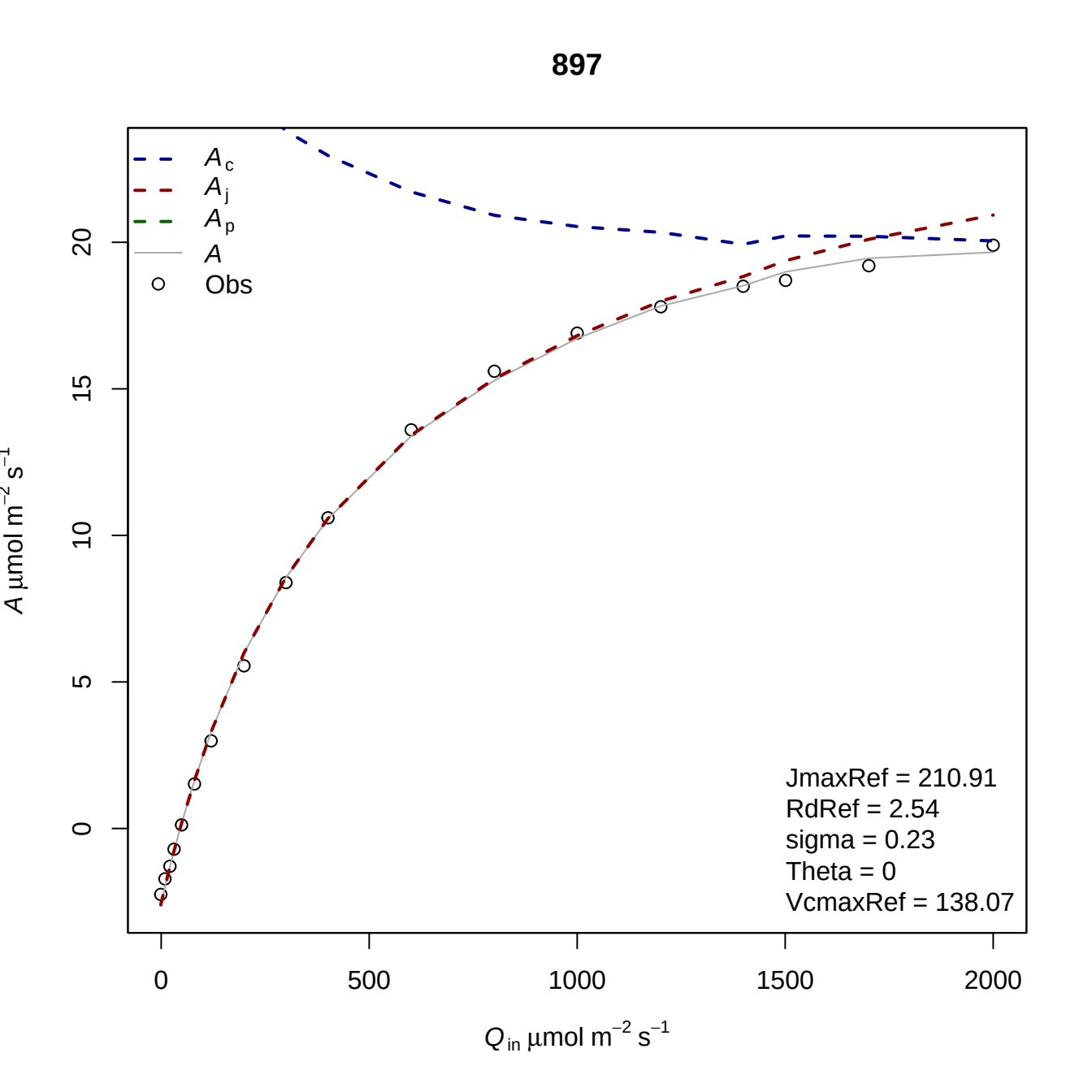

892
A_c
A_j
A_p
A
Obs
JmaxRef = 46.17
RdRef = 2.35
sigma = 0.24
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

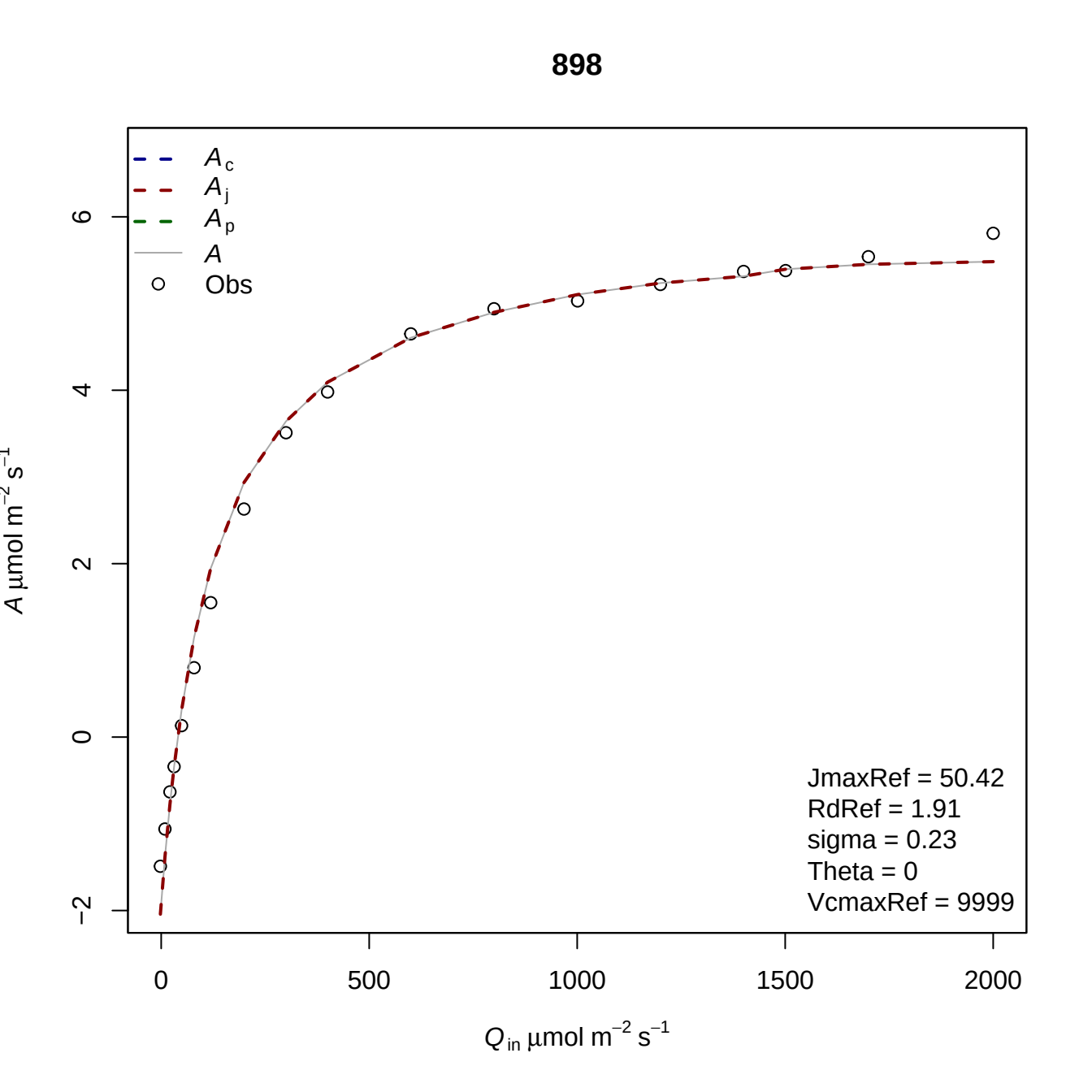

893
A_c
A_j
A_p
A
Obs
JmaxRef = 79.45
RdRef = 2.46
sigma = 0.29
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

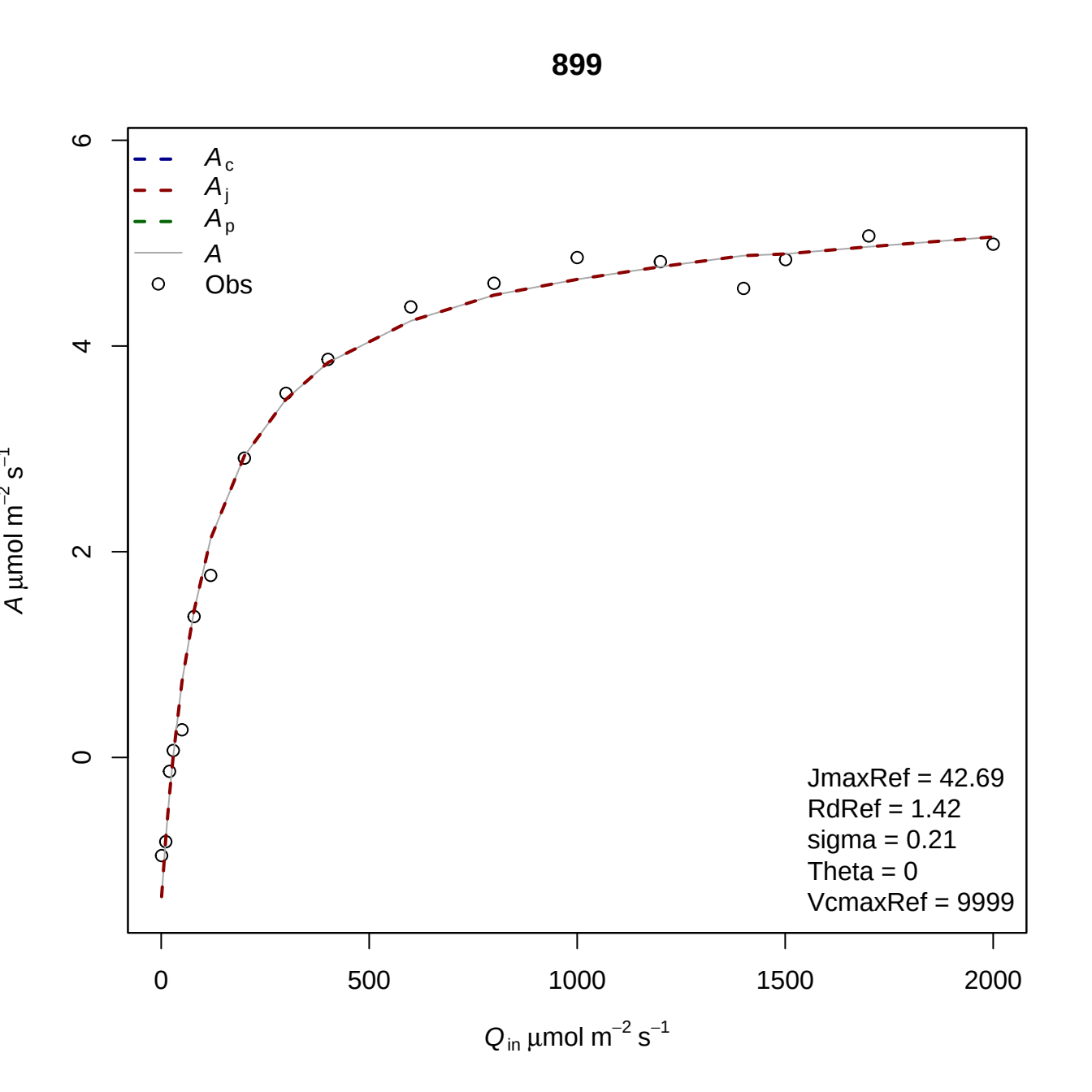

894
A_c
A_j
A_p
A
Obs
JmaxRef = 134.57
RdRef = 2.87
sigma = 0.28
Theta = 0
VcmaxRef = 84.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

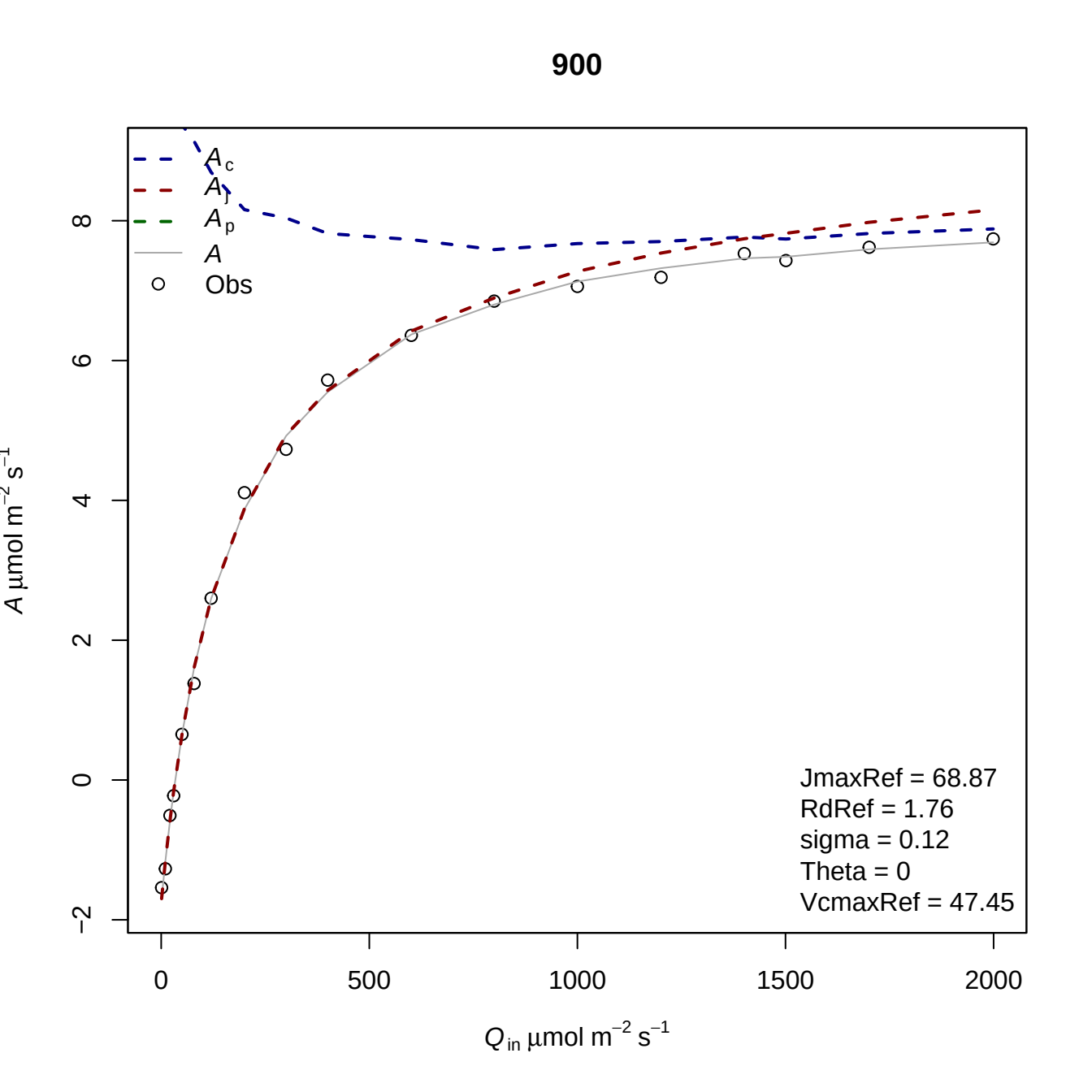

895
A_c
A_j
A_p
A
Obs
JmaxRef = 149.2
RdRef = 2.79
sigma = 0.33
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

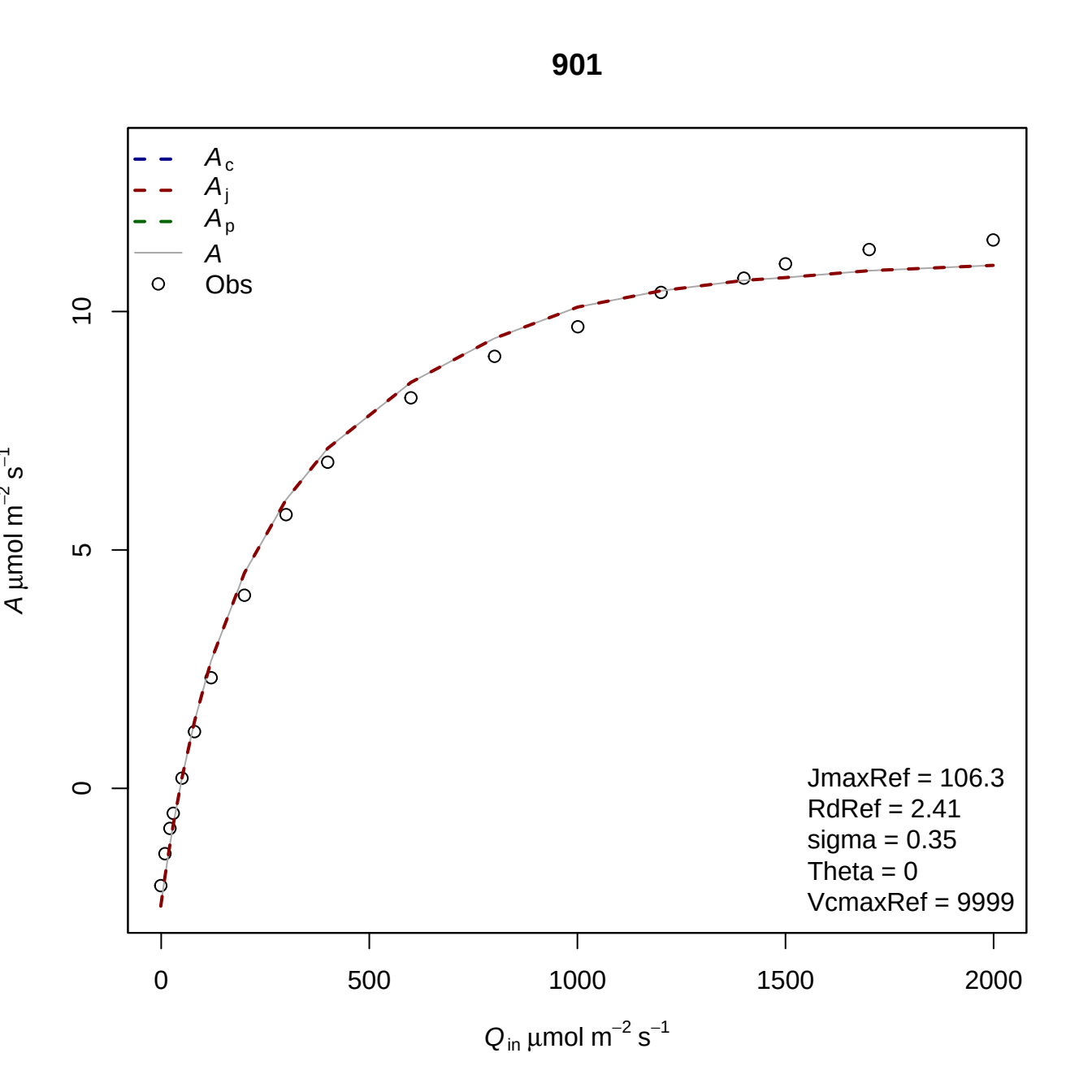

896
A_c
A_j
A_p
A
Obs
JmaxRef = 207.23
RdRef = 2.69
sigma = 0.19
Theta = 0
VcmaxRef = 129.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

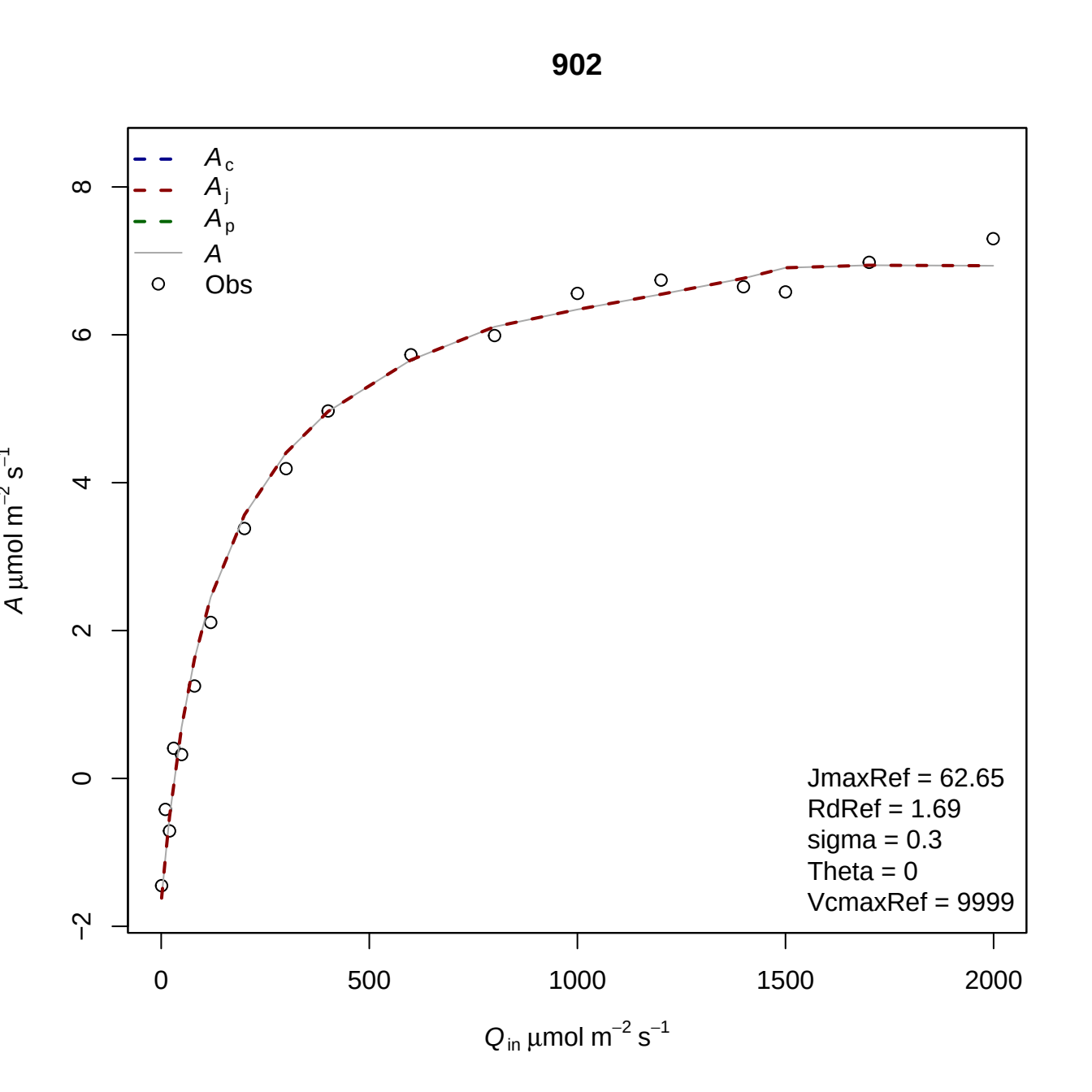

897
A_c
A_j
A_p
A
Obs
JmaxRef = 210.91
RdRef = 2.54
sigma = 0.23
Theta = 0
VcmaxRef = 138.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

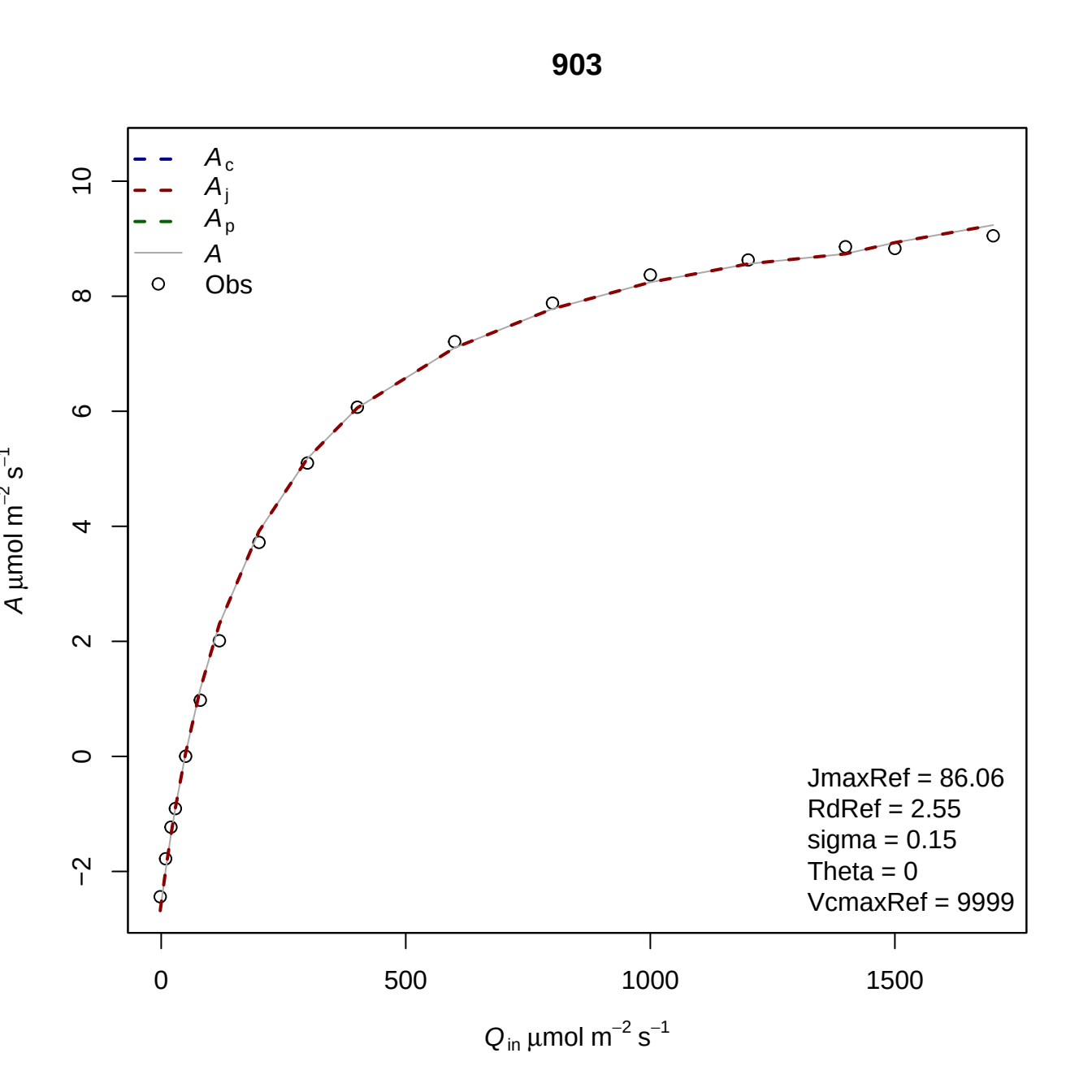

898
A_c
A_j
A_p
A
Obs
JmaxRef = 50.42
RdRef = 1.91
sigma = 0.23
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

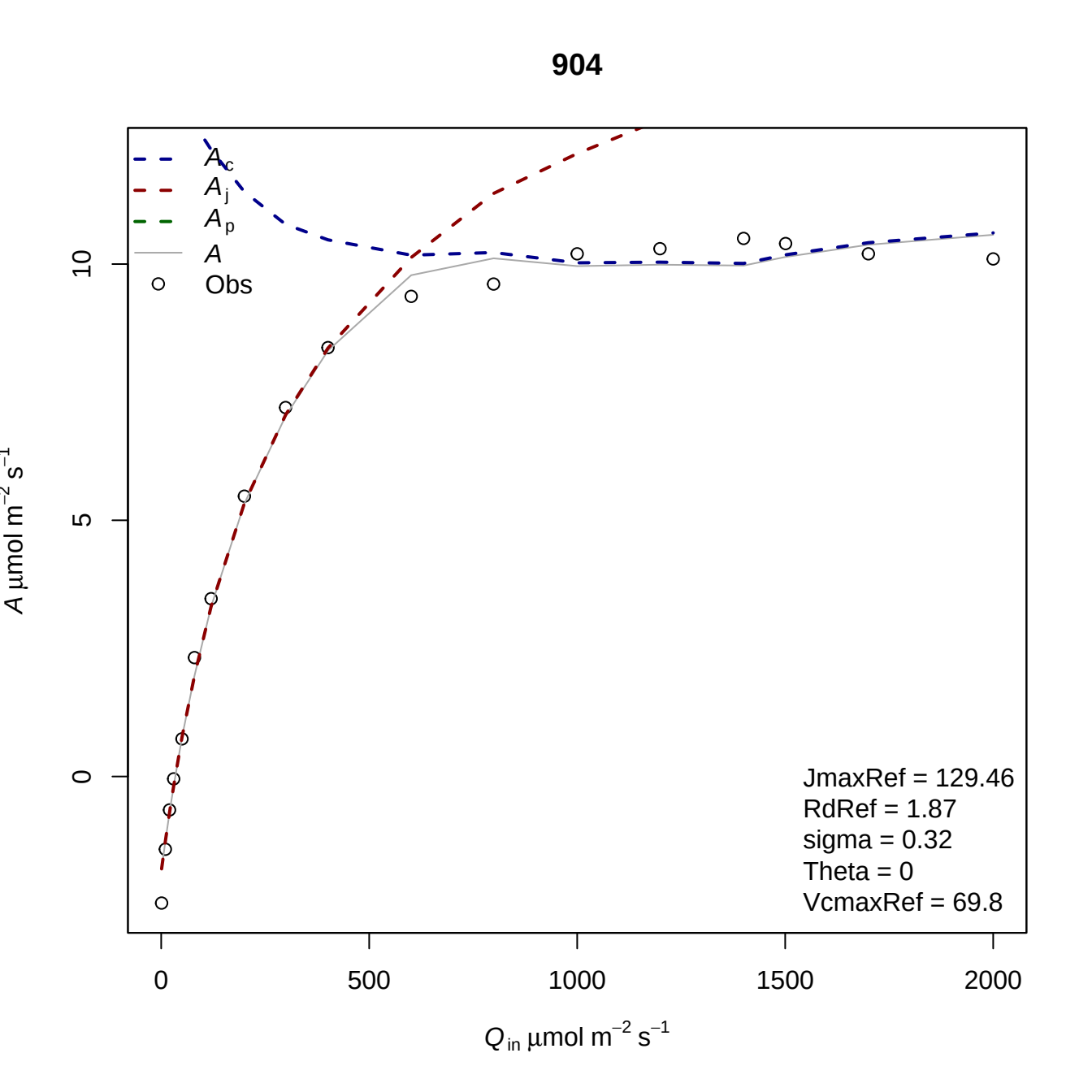

899
A_c
A_j
A_p
A
Obs
JmaxRef = 42.69
RdRef = 1.42
sigma = 0.21
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

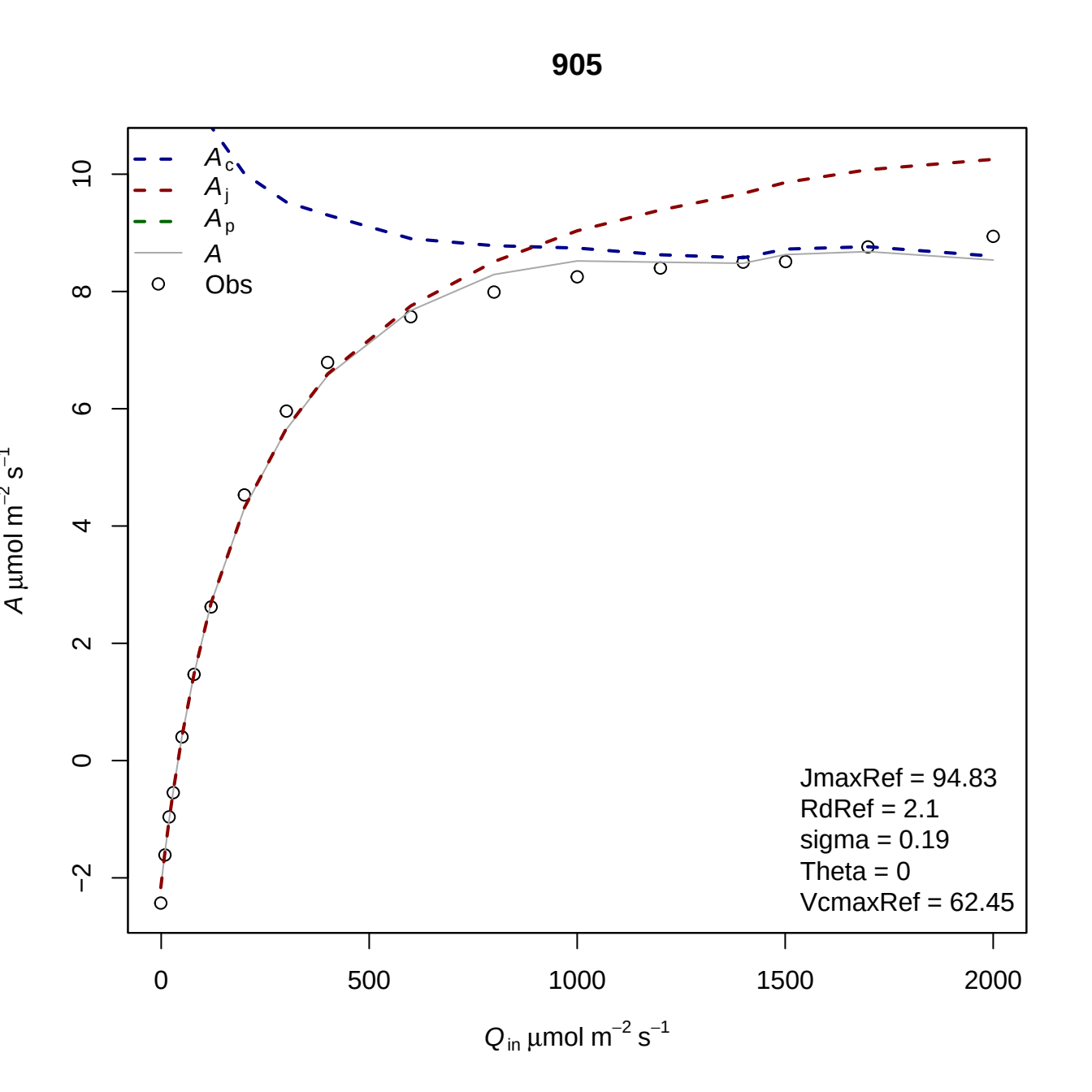

900
A_c
A_j
A_p
A
Obs
JmaxRef = 68.87
RdRef = 1.76
sigma = 0.12
Theta = 0
VcmaxRef = 47.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

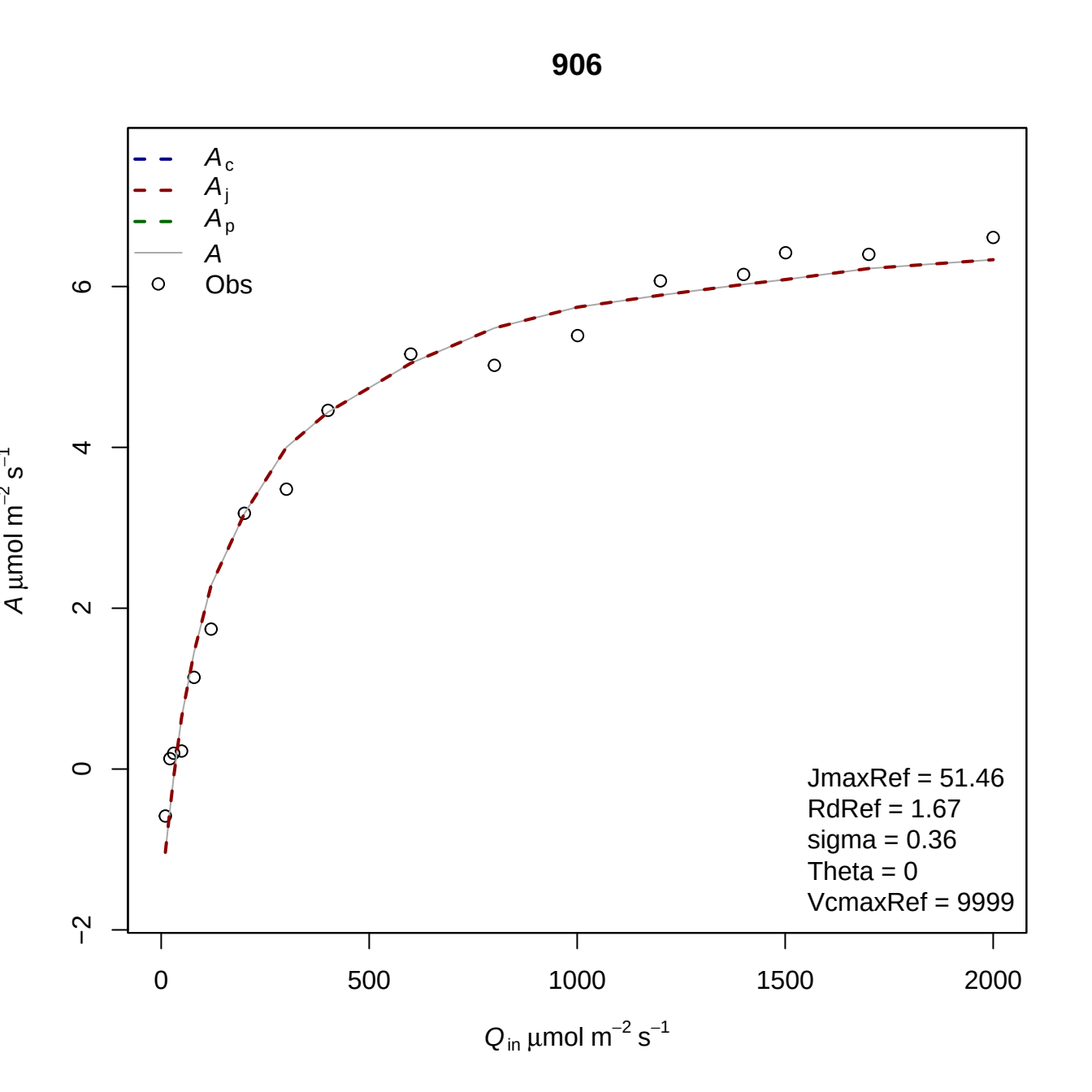

901
A_c
A_j
A_p
A
Obs
JmaxRef = 106.3
RdRef = 2.41
sigma = 0.35
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

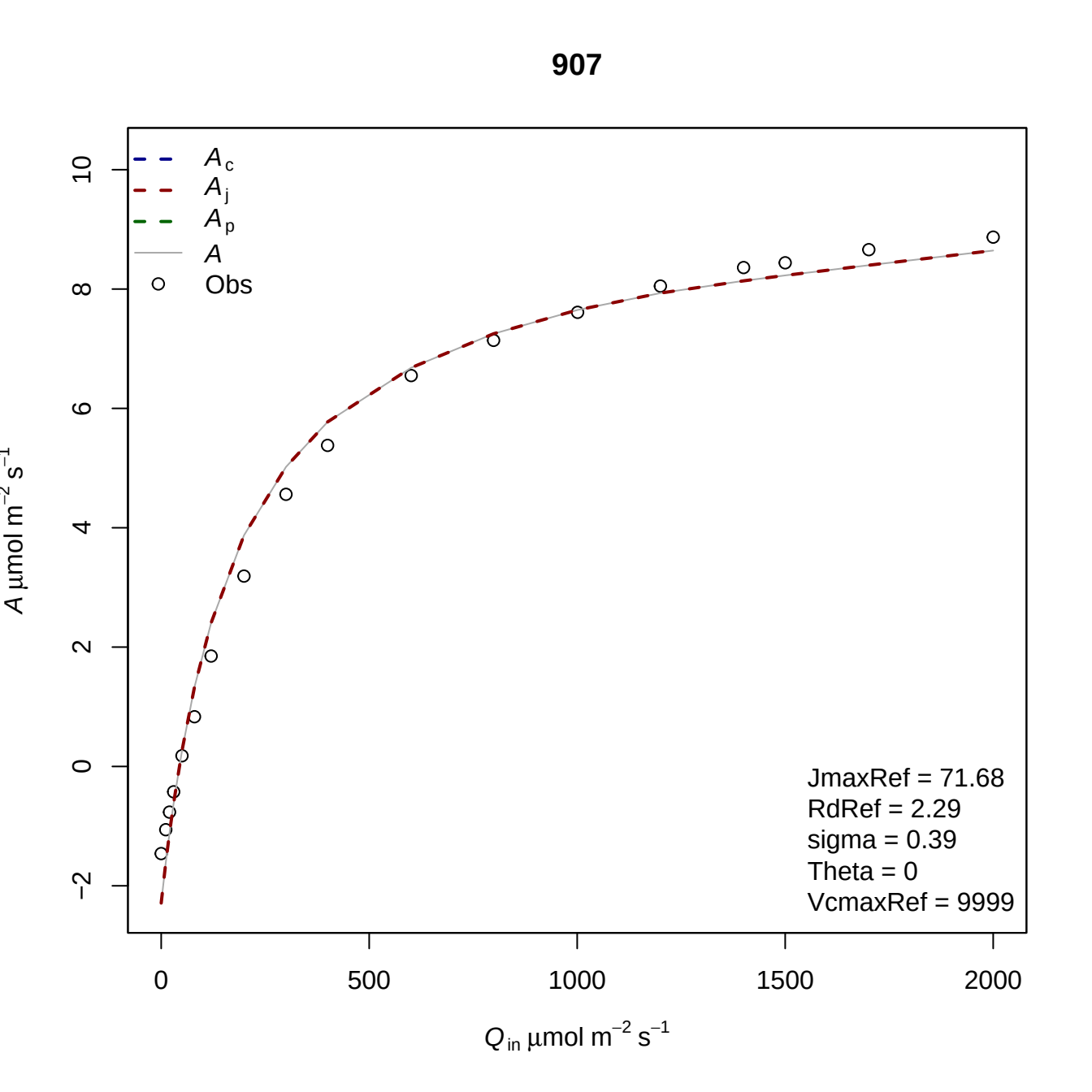

902
A_c
A_j
A_p
A
Obs
JmaxRef = 62.65
RdRef = 1.69
sigma = 0.3
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

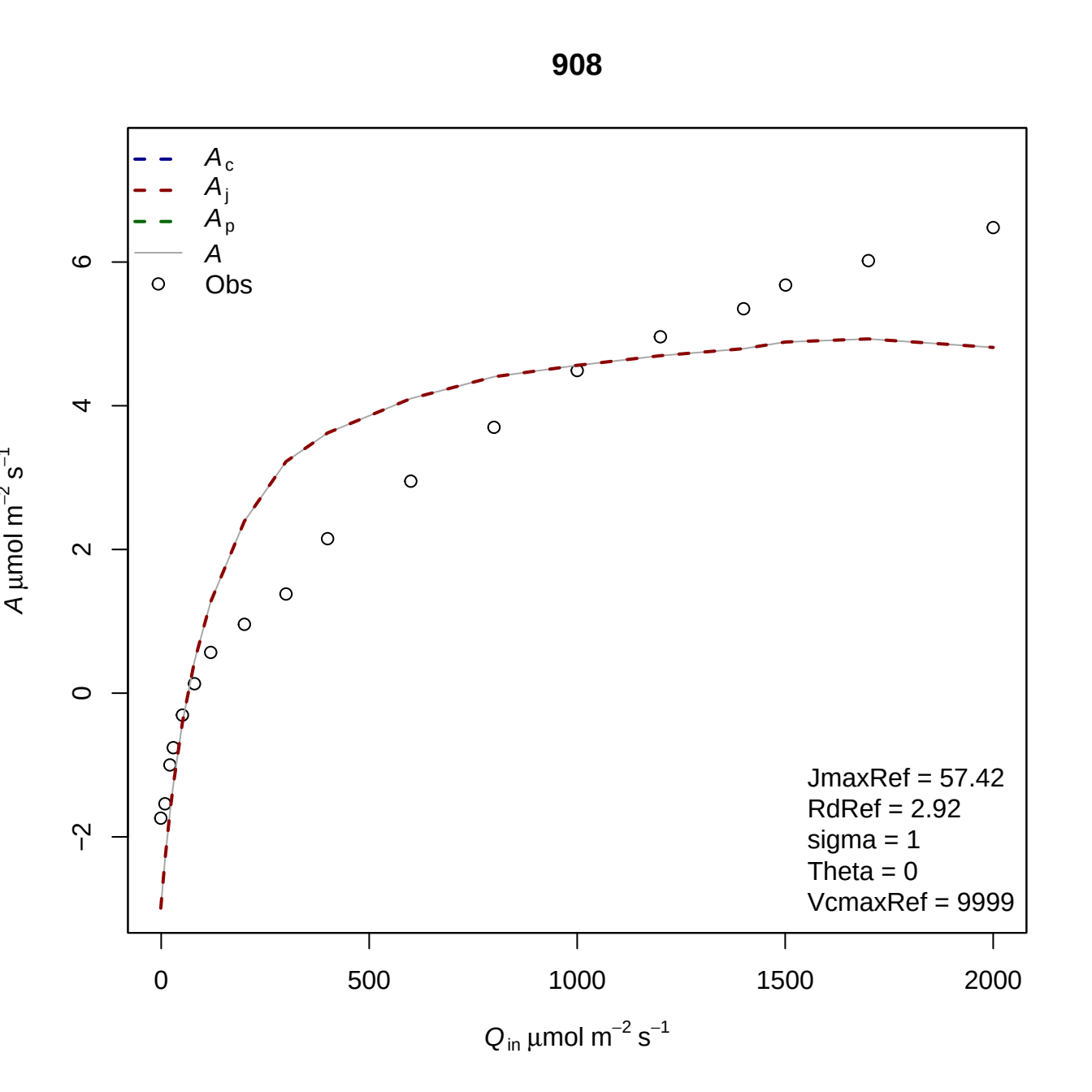

903
A_c
A_j
A_p
A
Obs
JmaxRef = 86.06
RdRef = 2.55
sigma = 0.15
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

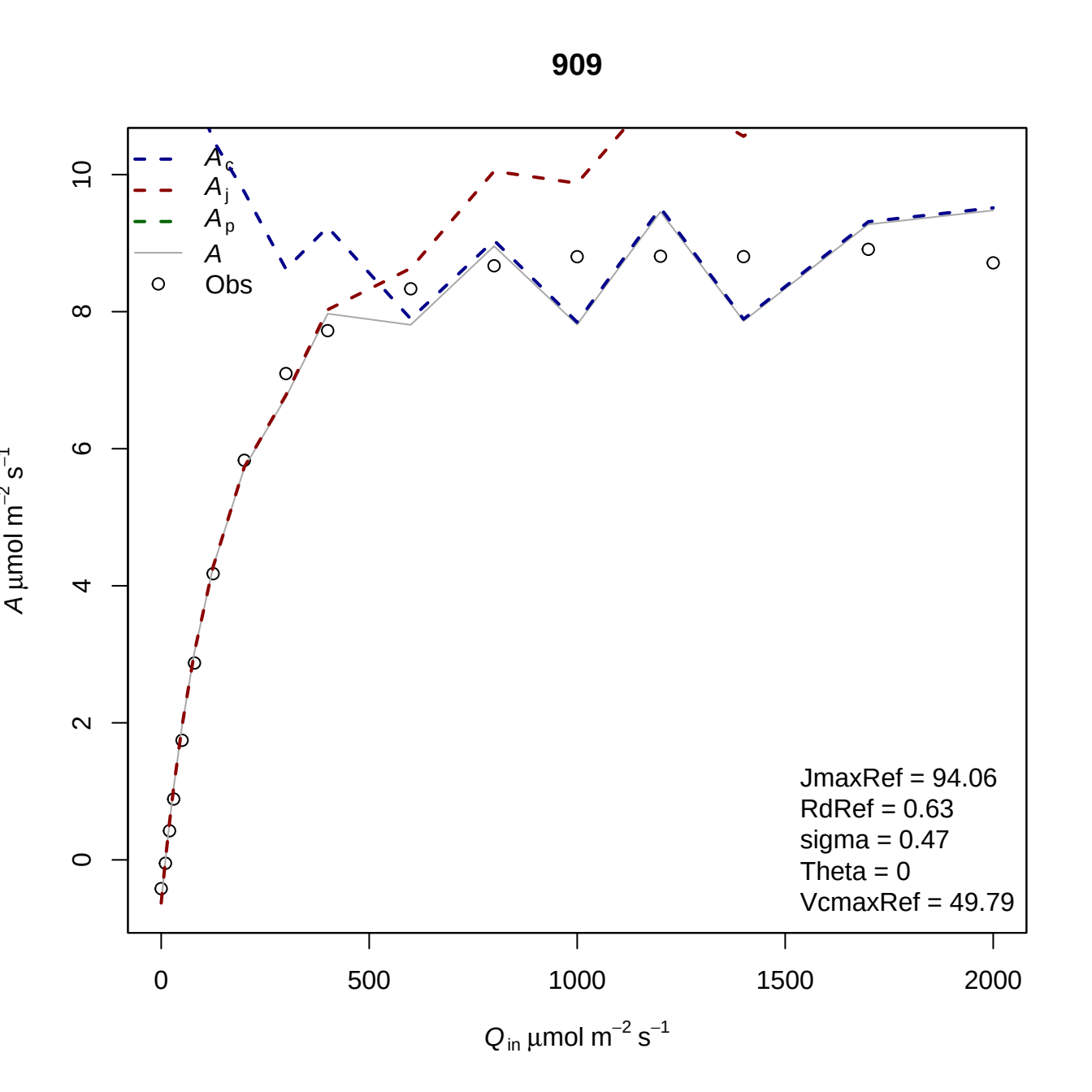

904
A_c
A_j
A_p
A
Obs
JmaxRef = 129.46
RdRef = 1.87
sigma = 0.32
Theta = 0
VcmaxRef = 69.8
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

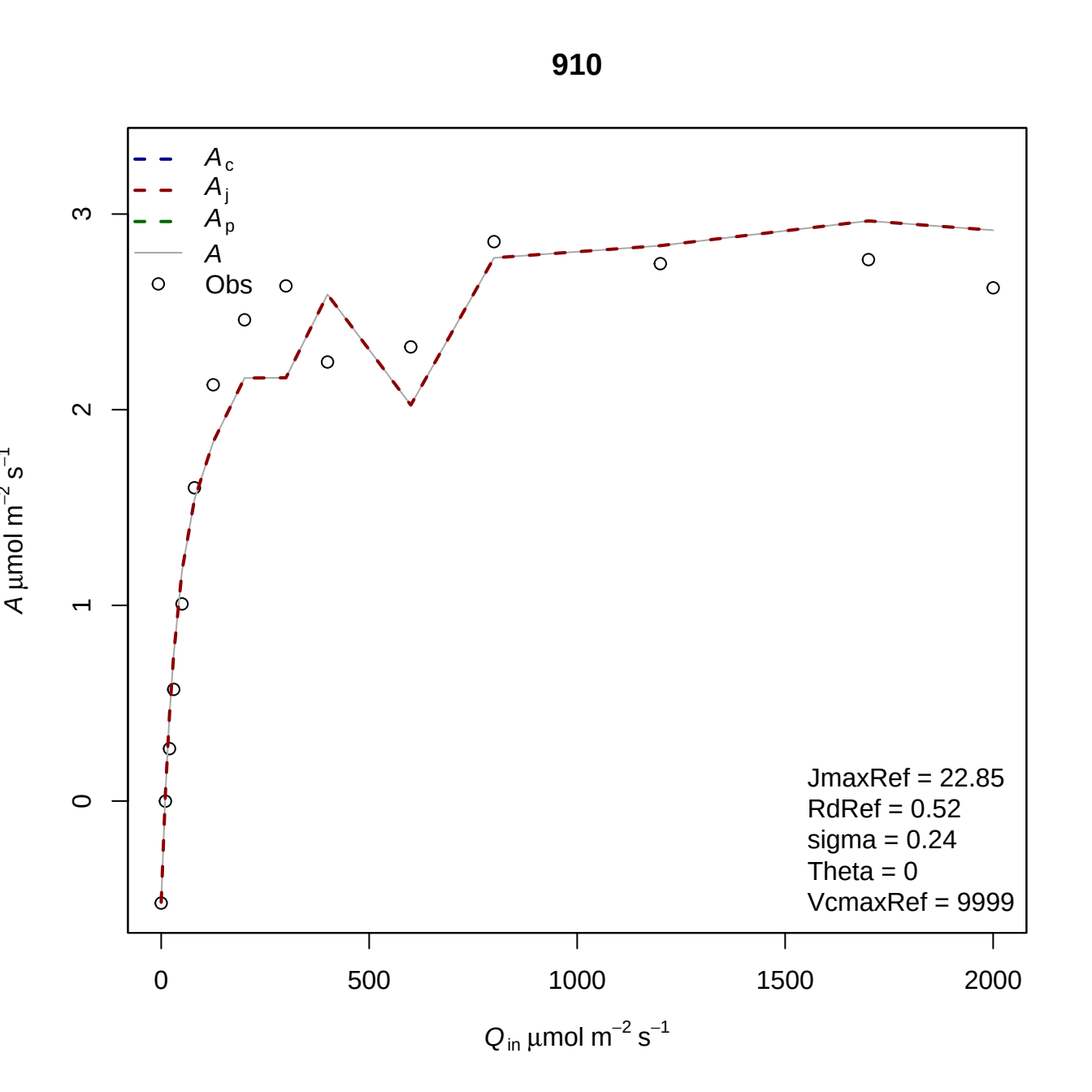

905
A_c
A_j
A_p
A
Obs
JmaxRef = 94.83
RdRef = 2.1
sigma = 0.19
Theta = 0
VcmaxRef = 62.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

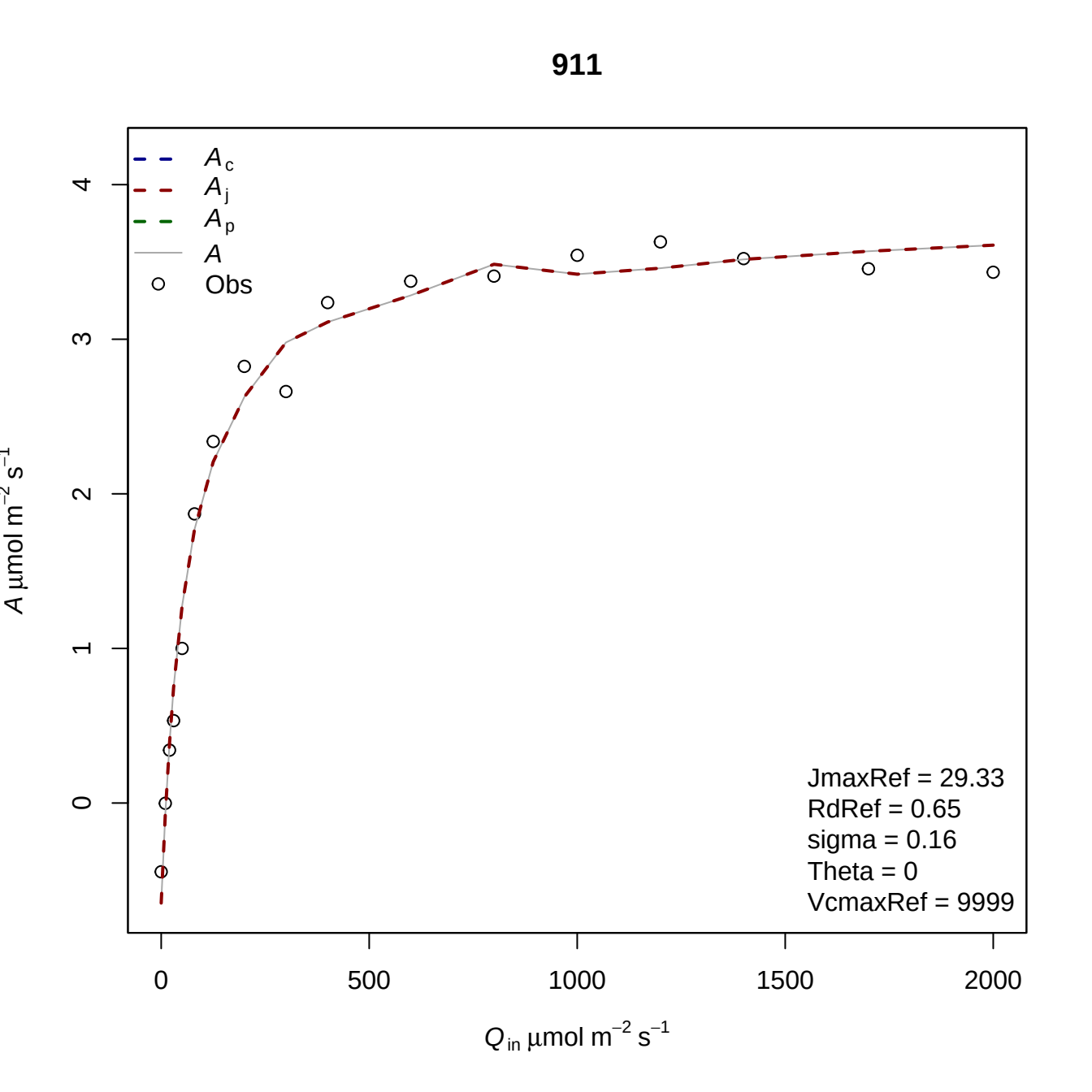

906
A_c
A_j
A_p
A
Obs
JmaxRef = 51.46
RdRef = 1.67
sigma = 0.36
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

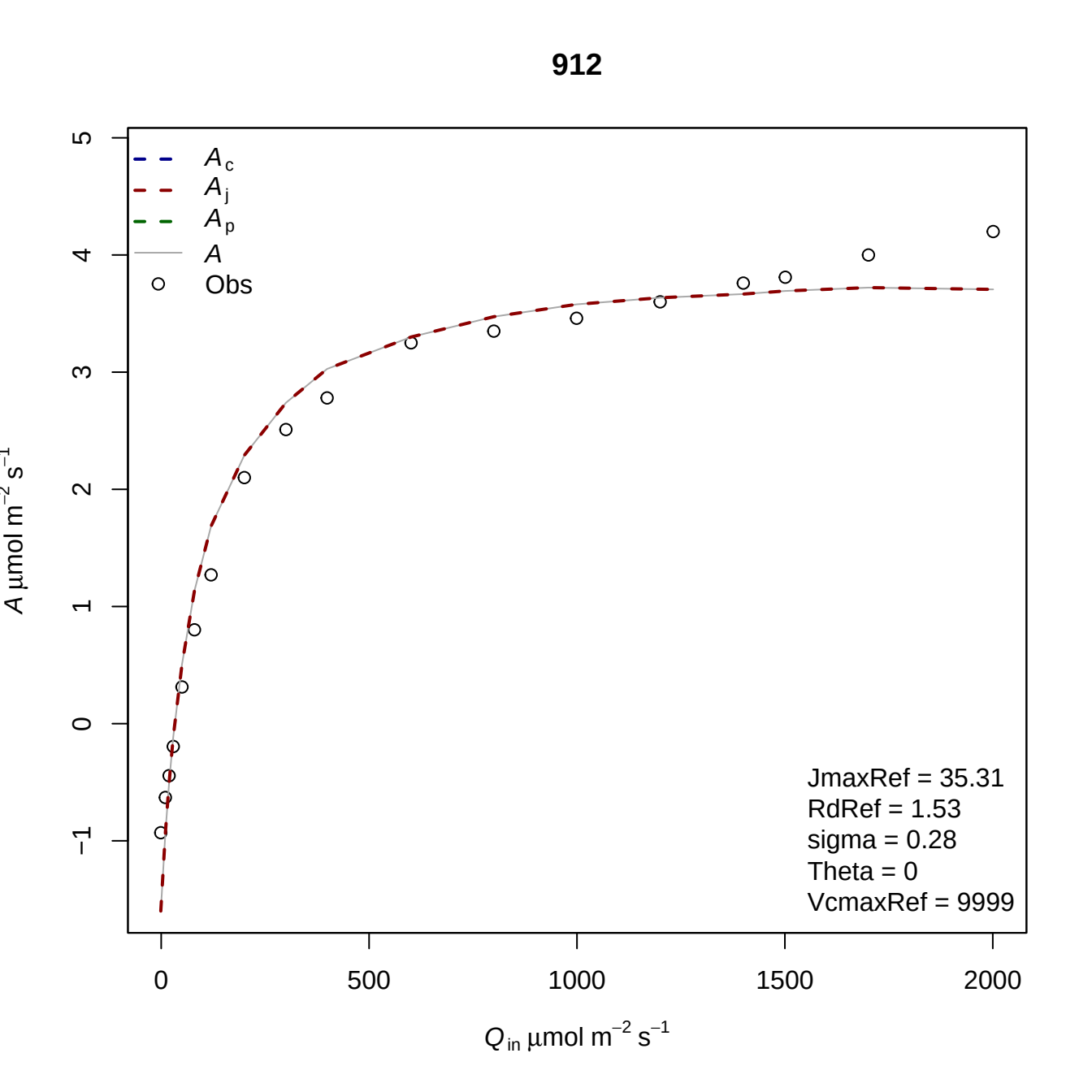

907
A_c
A_j
A_p
A
Obs
JmaxRef = 71.68
RdRef = 2.29
sigma = 0.39
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

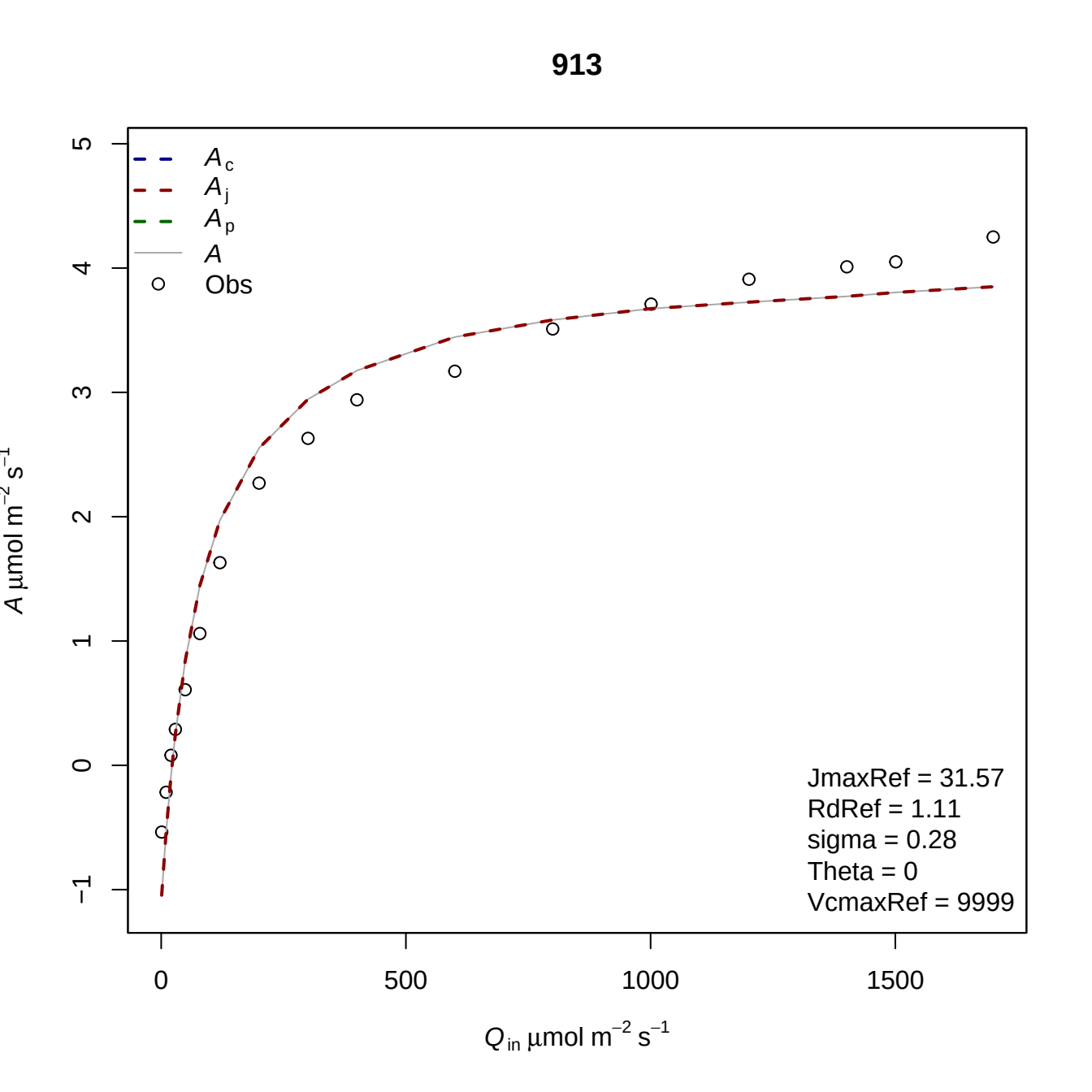

908
A_c
A_j
A_p
A
Obs
JmaxRef = 57.42
RdRef = 2.92
sigma = 1
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

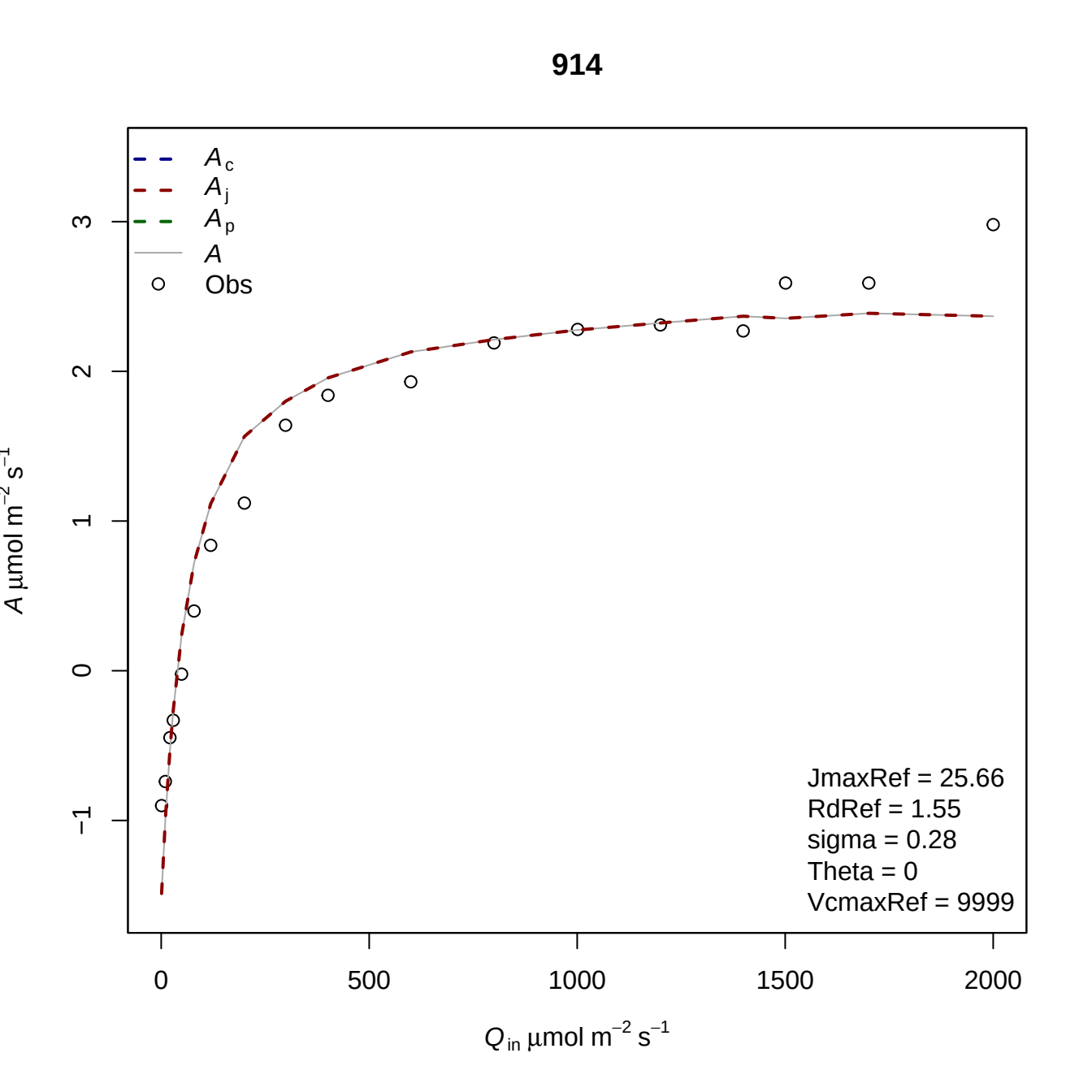

909
A_c
A_j
A_p
A
Obs
JmaxRef = 94.06
RdRef = 0.63
sigma = 0.47
Theta = 0
VcmaxRef = 49.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

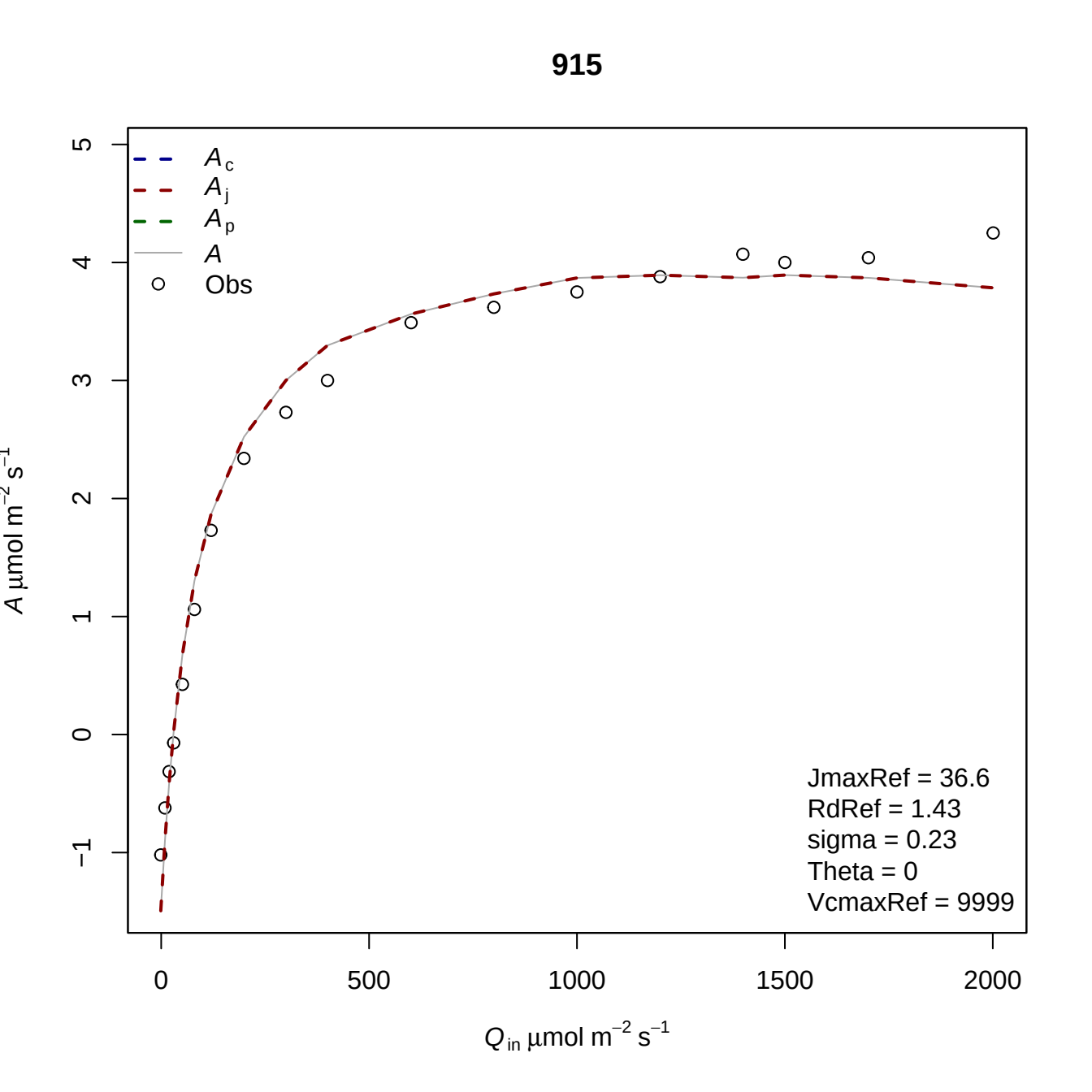

910
A_c
A_j
A_p
A
Obs
JmaxRef = 22.85
RdRef = 0.52
sigma = 0.24
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

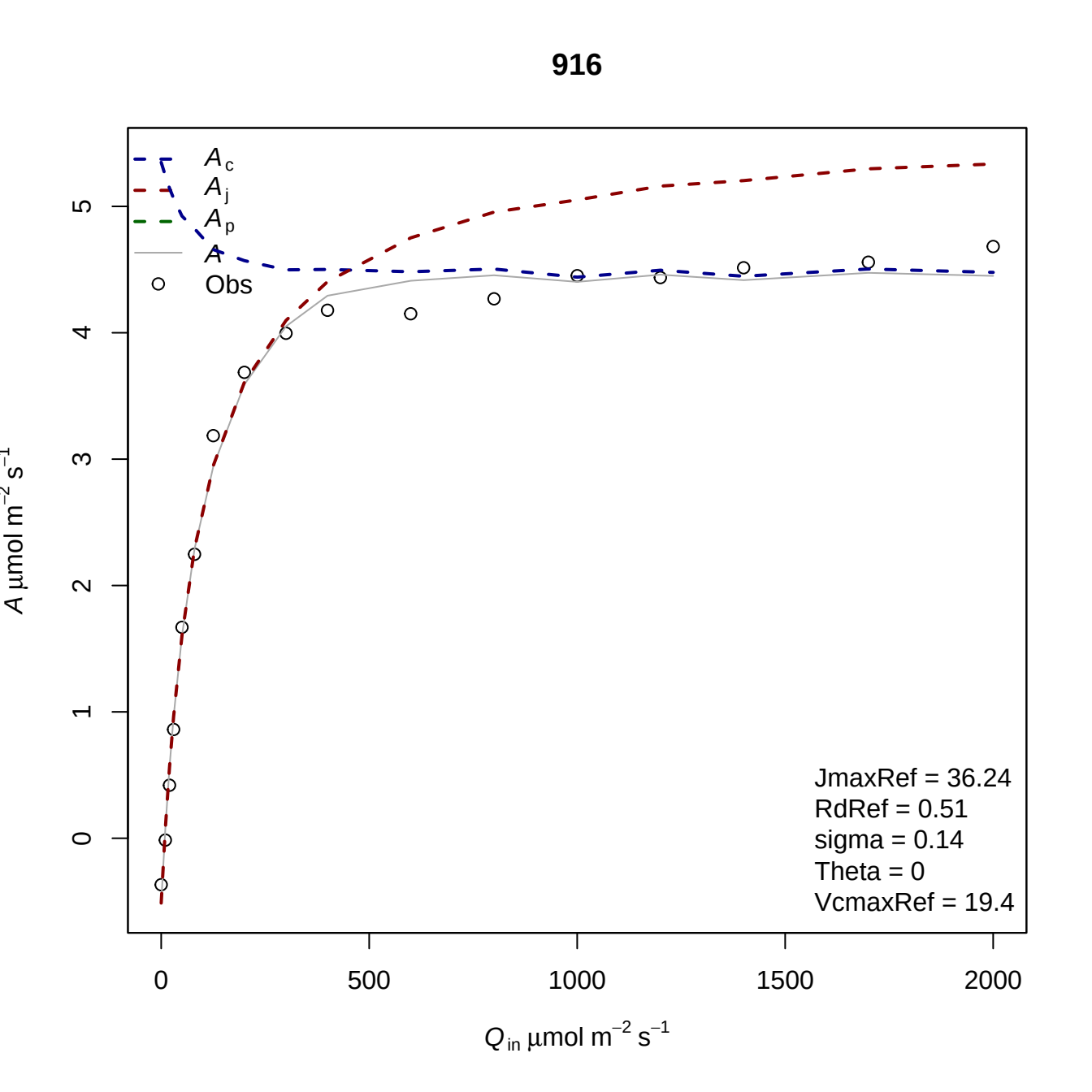

911
A_c
A_j
A_p
A
Obs
JmaxRef = 29.33
RdRef = 0.65
sigma = 0.16
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

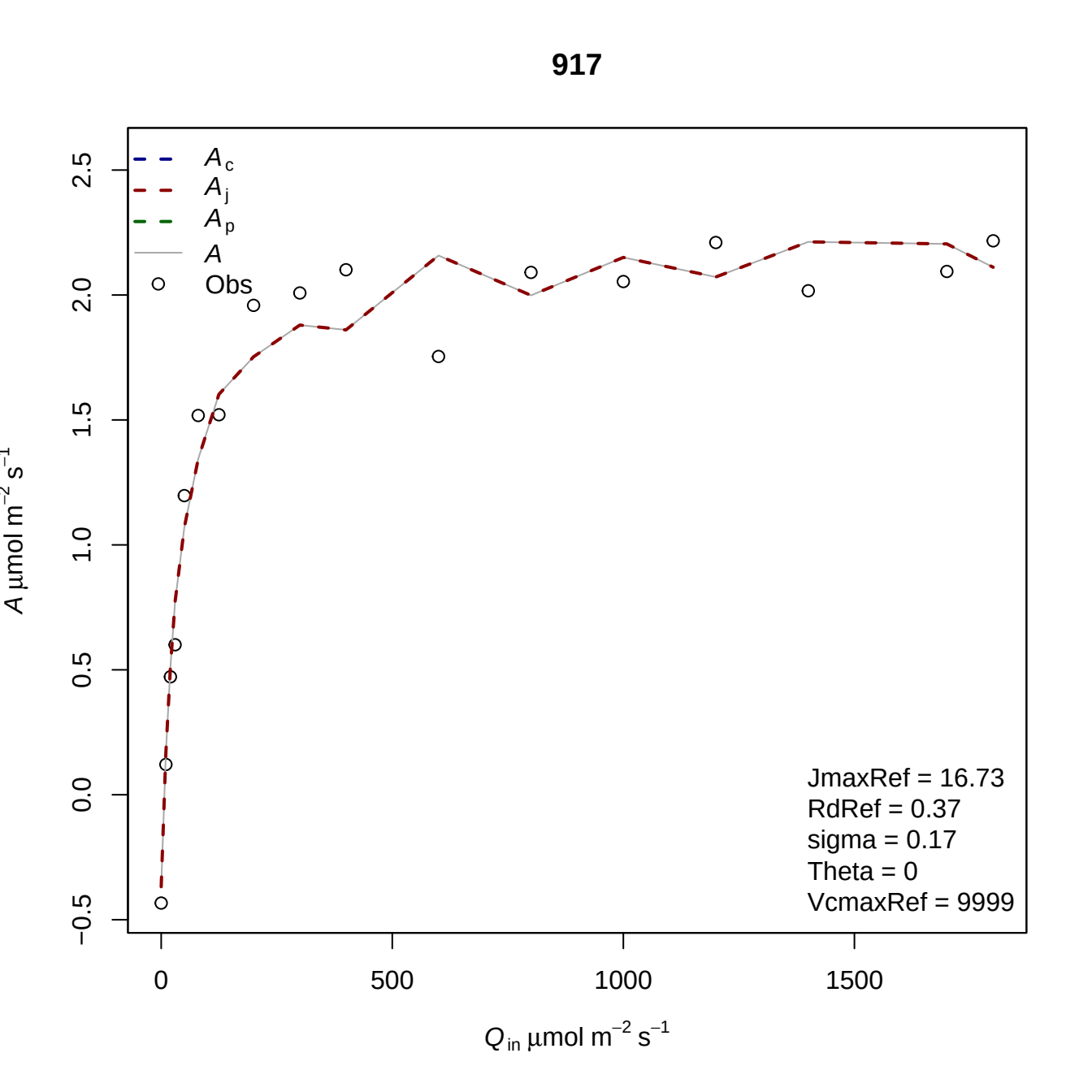

912
A_c
A_j
A_p
A
Obs
JmaxRef = 35.31
RdRef = 1.53
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
500
1000
1500
2000

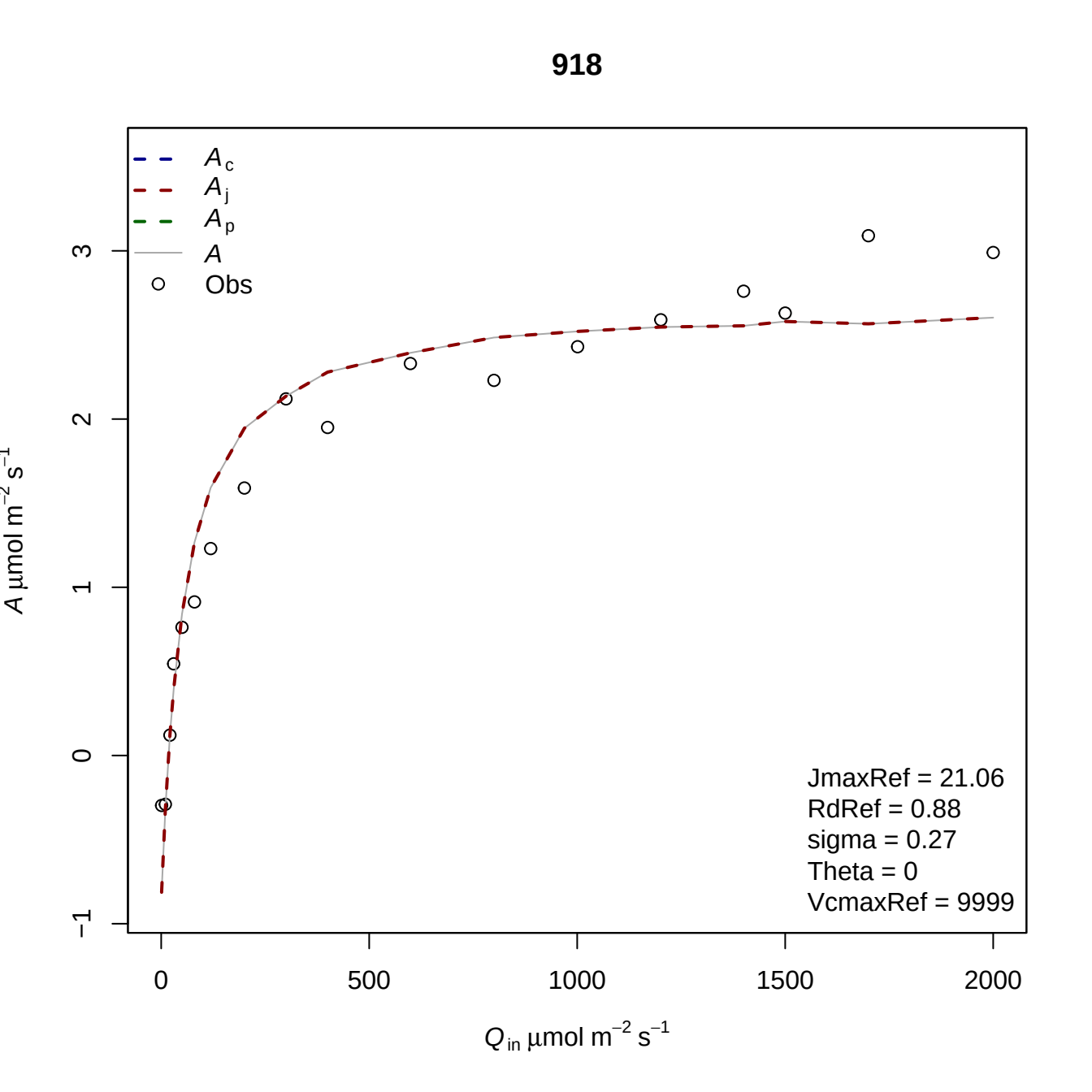

913
A_c
A_j
A_p
A
Obs
JmaxRef = 31.57
RdRef = 1.11
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

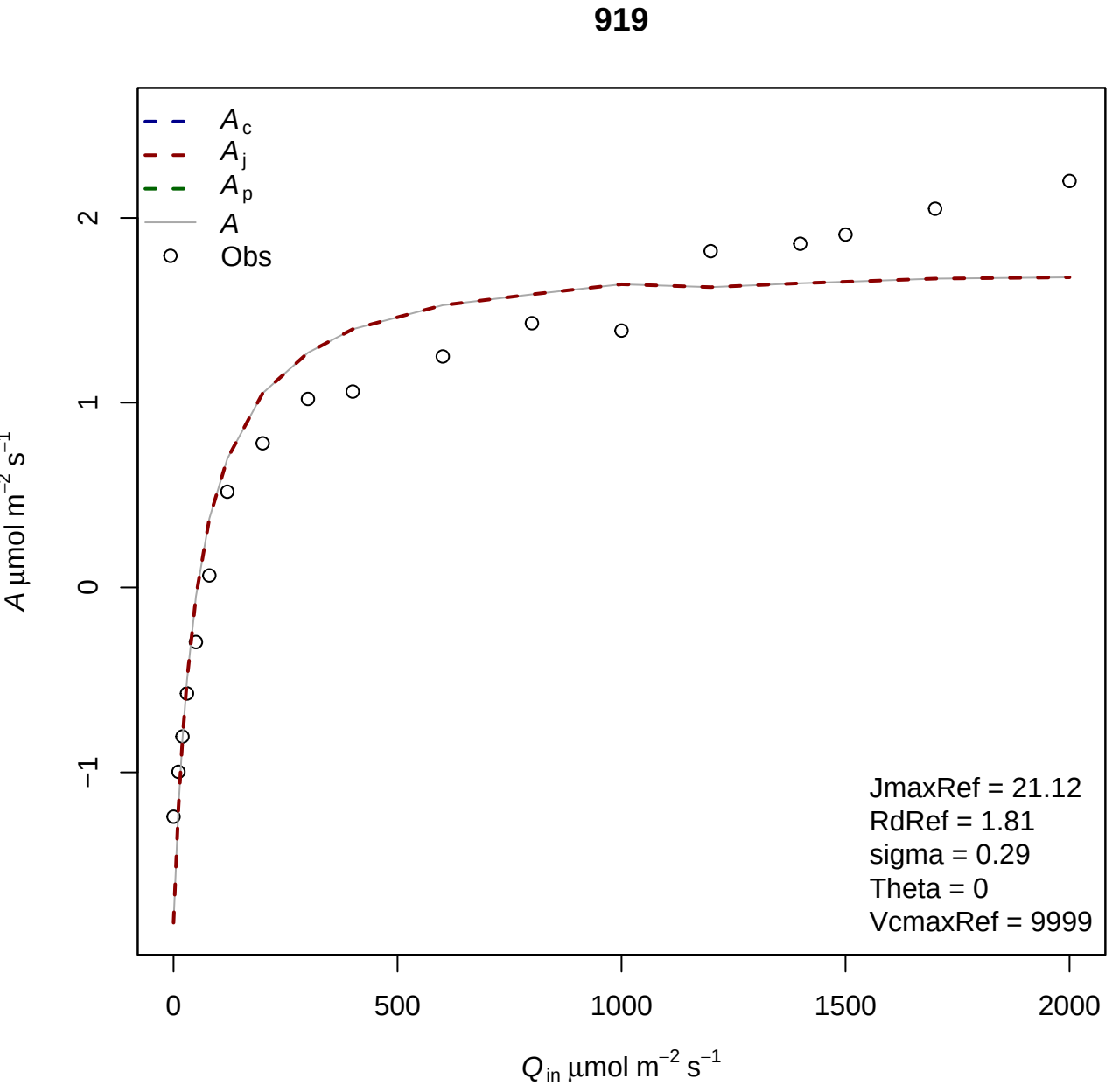

914
A_c
A_j
A_p
A
Obs
JmaxRef = 25.66
RdRef = 1.55
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

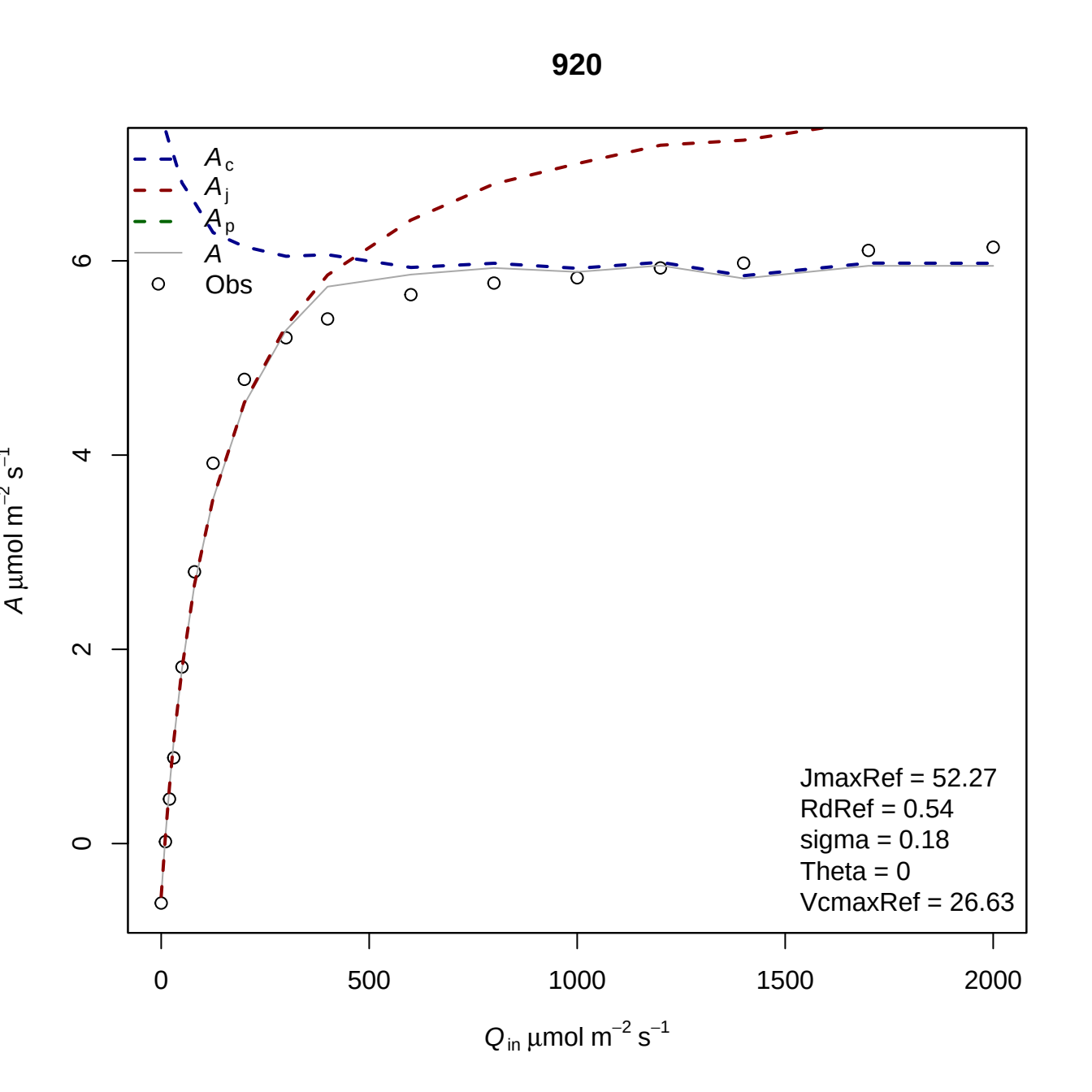

915
A_c
A_j
A_p
A
Obs
JmaxRef = 36.6
RdRef = 1.43
sigma = 0.23
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
500
1000
1500
2000

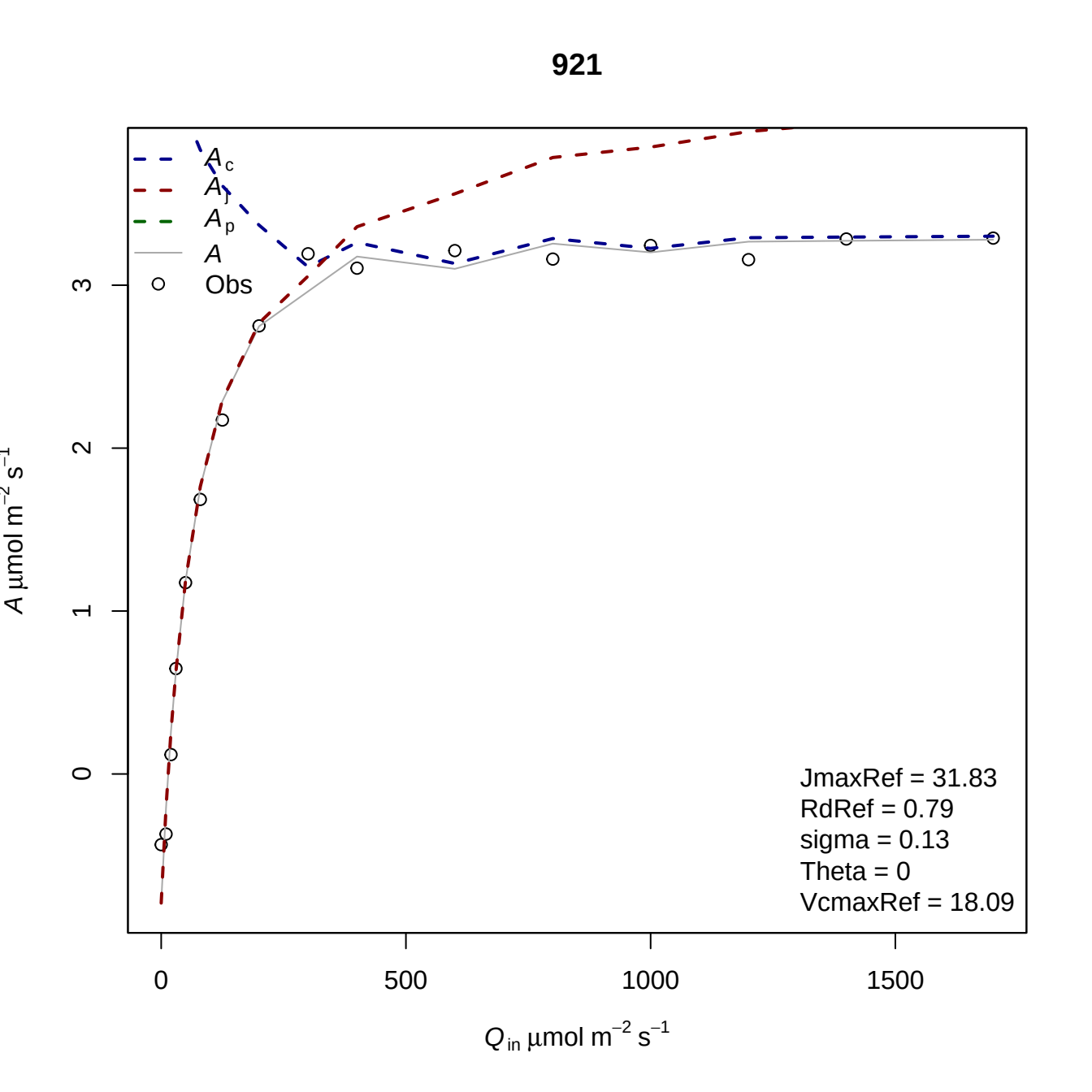

916
A_c
A_j
A_p
A
Obs
JmaxRef = 36.24
RdRef = 0.51
sigma = 0.14
Theta = 0
VcmaxRef = 19.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500
2000

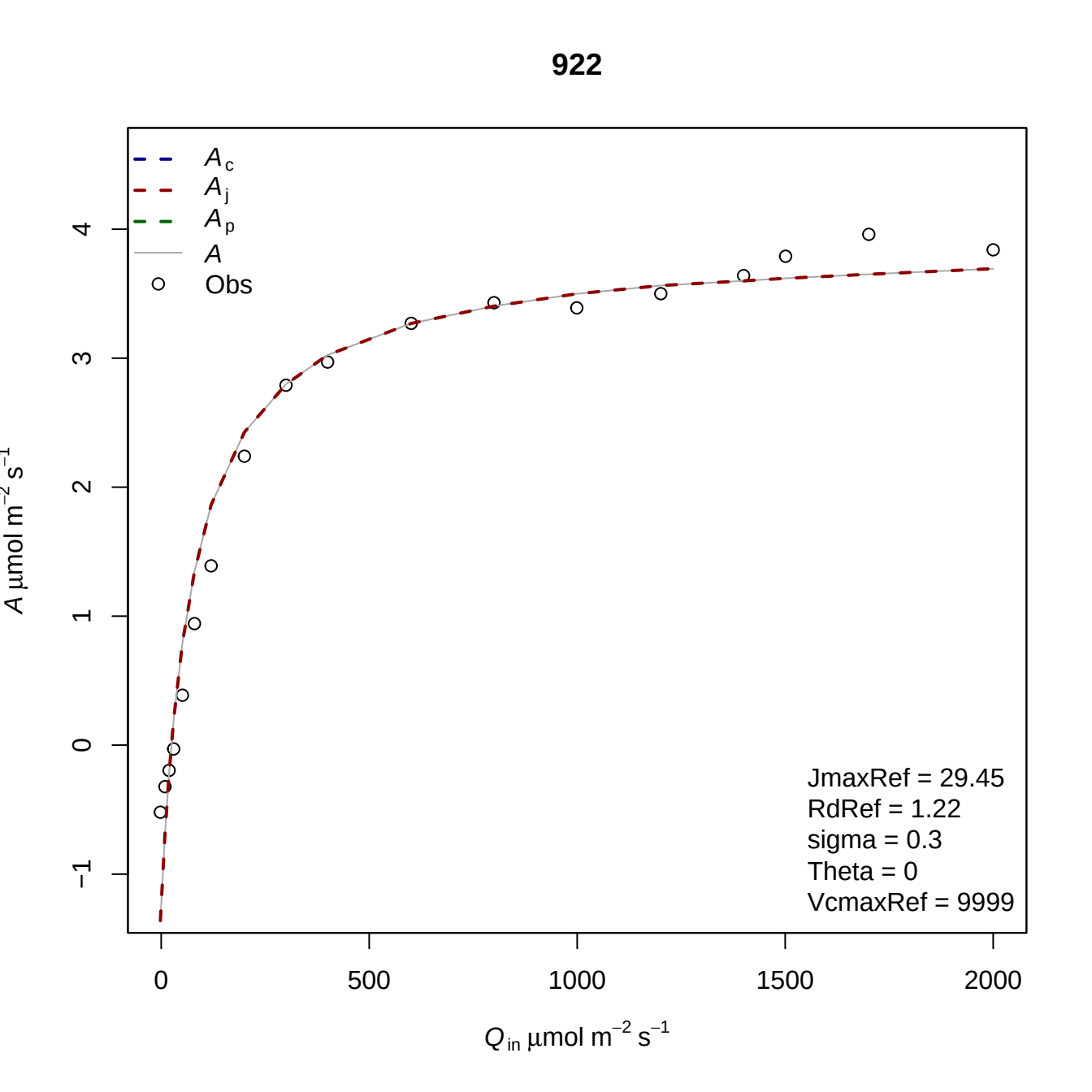

917
A_c
A_j
A_p
A
Obs
JmaxRef = 16.73
RdRef = 0.37
sigma = 0.17
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

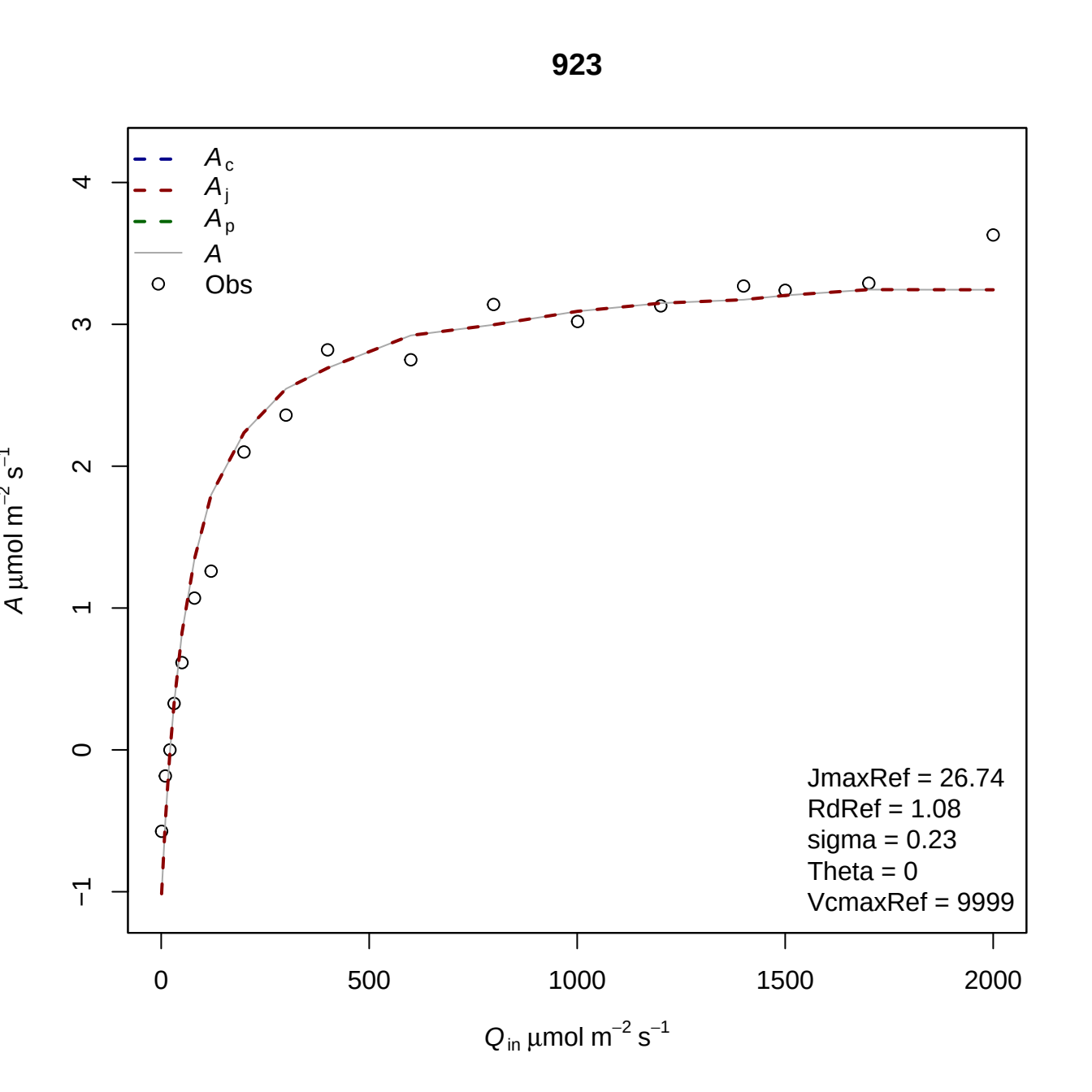

918
A_c
A_j
A_p
A
Obs
JmaxRef = 21.06
RdRef = 0.88
sigma = 0.27
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

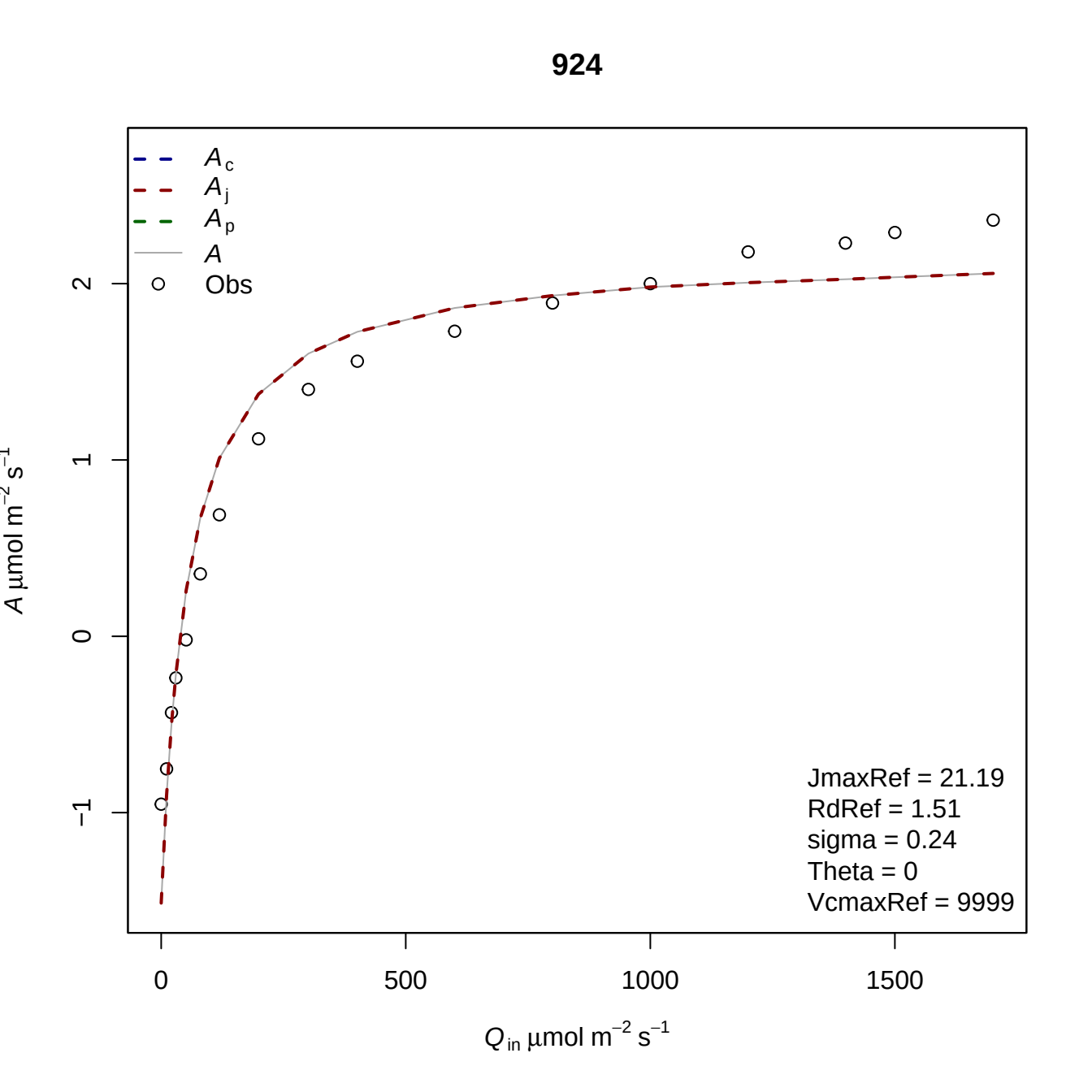

919
A_c
A_j
A_p
A
Obs
JmaxRef = 21.12
RdRef = 1.81
sigma = 0.29
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500
2000

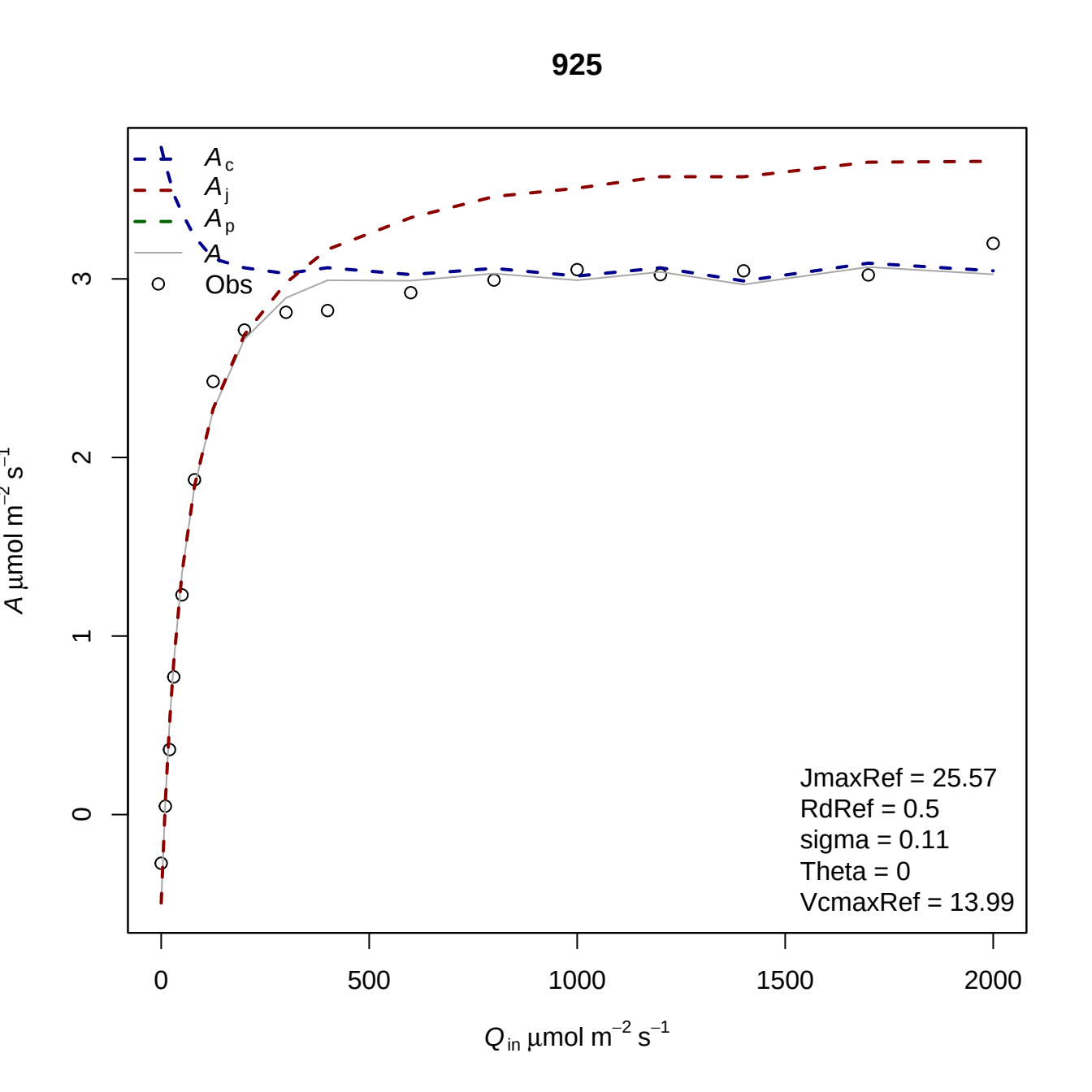

920
A_c
A_j
A_p
A
Obs
JmaxRef = 52.27
RdRef = 0.54
sigma = 0.18
Theta = 0
VcmaxRef = 26.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

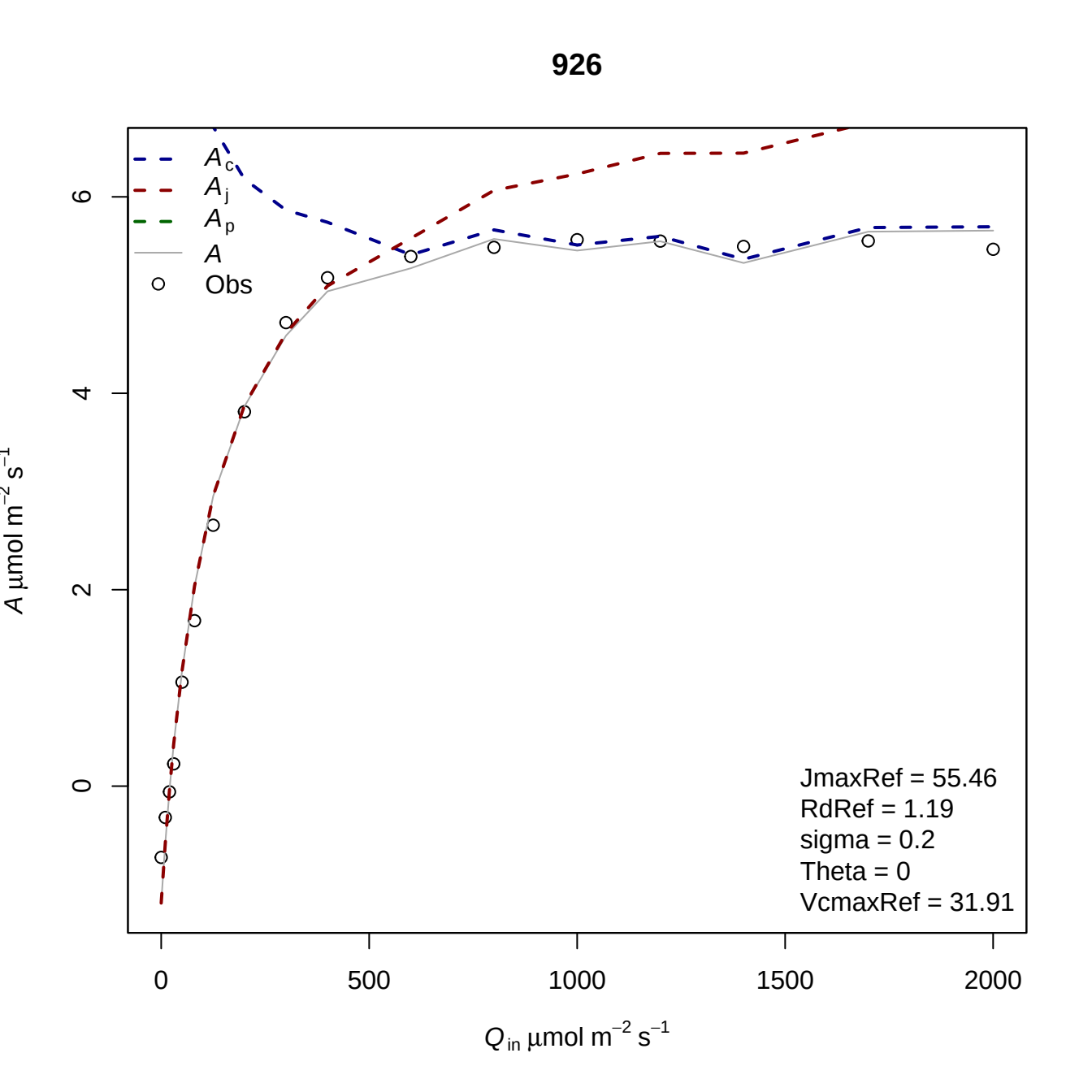

921
A_c
A_j
A_p
A
Obs
JmaxRef = 31.83
RdRef = 0.79
sigma = 0.13
Theta = 0
VcmaxRef = 18.09
0
1
2
3
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

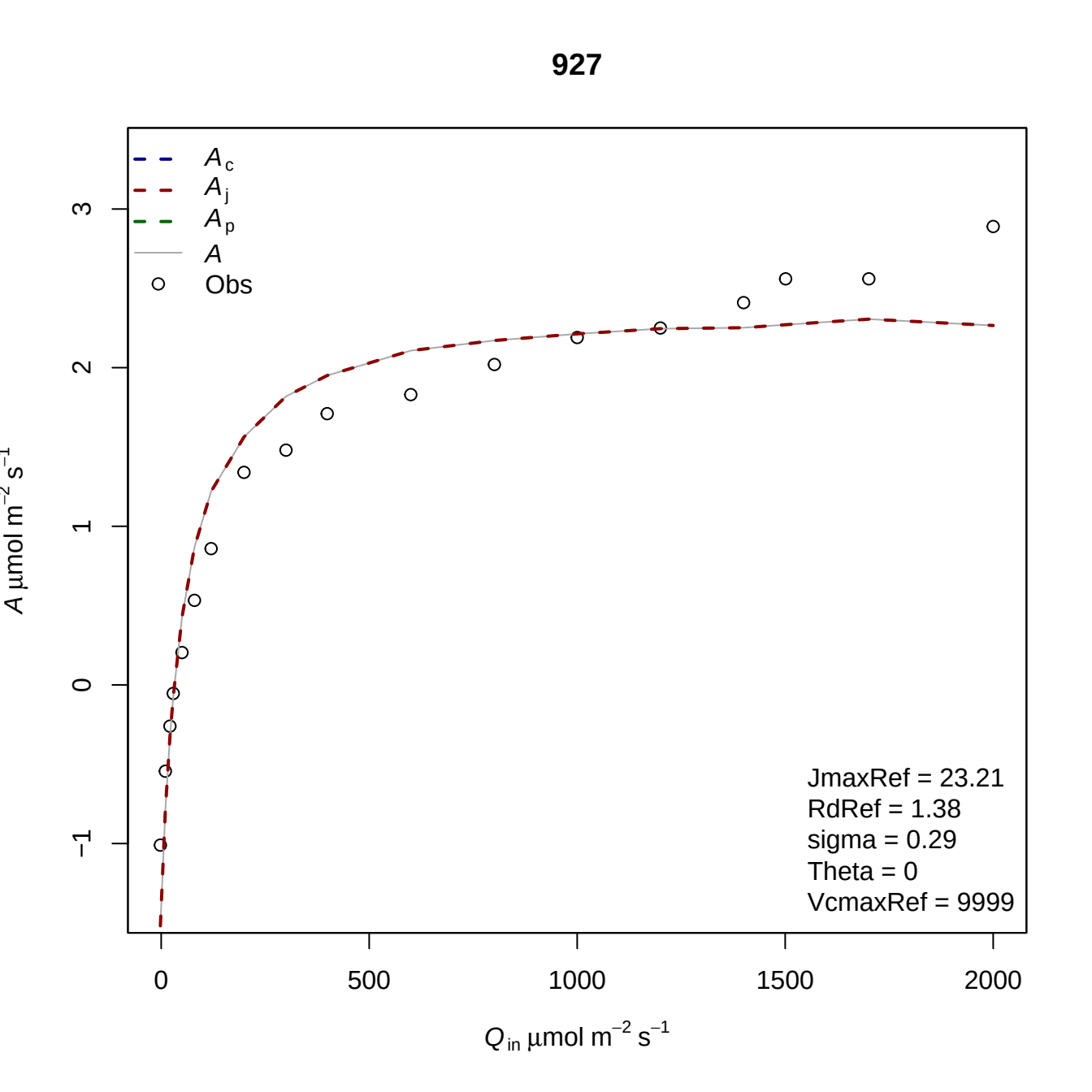

922
A_c
A_j
A_p
A
Obs
JmaxRef = 29.45
RdRef = 1.22
sigma = 0.3
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

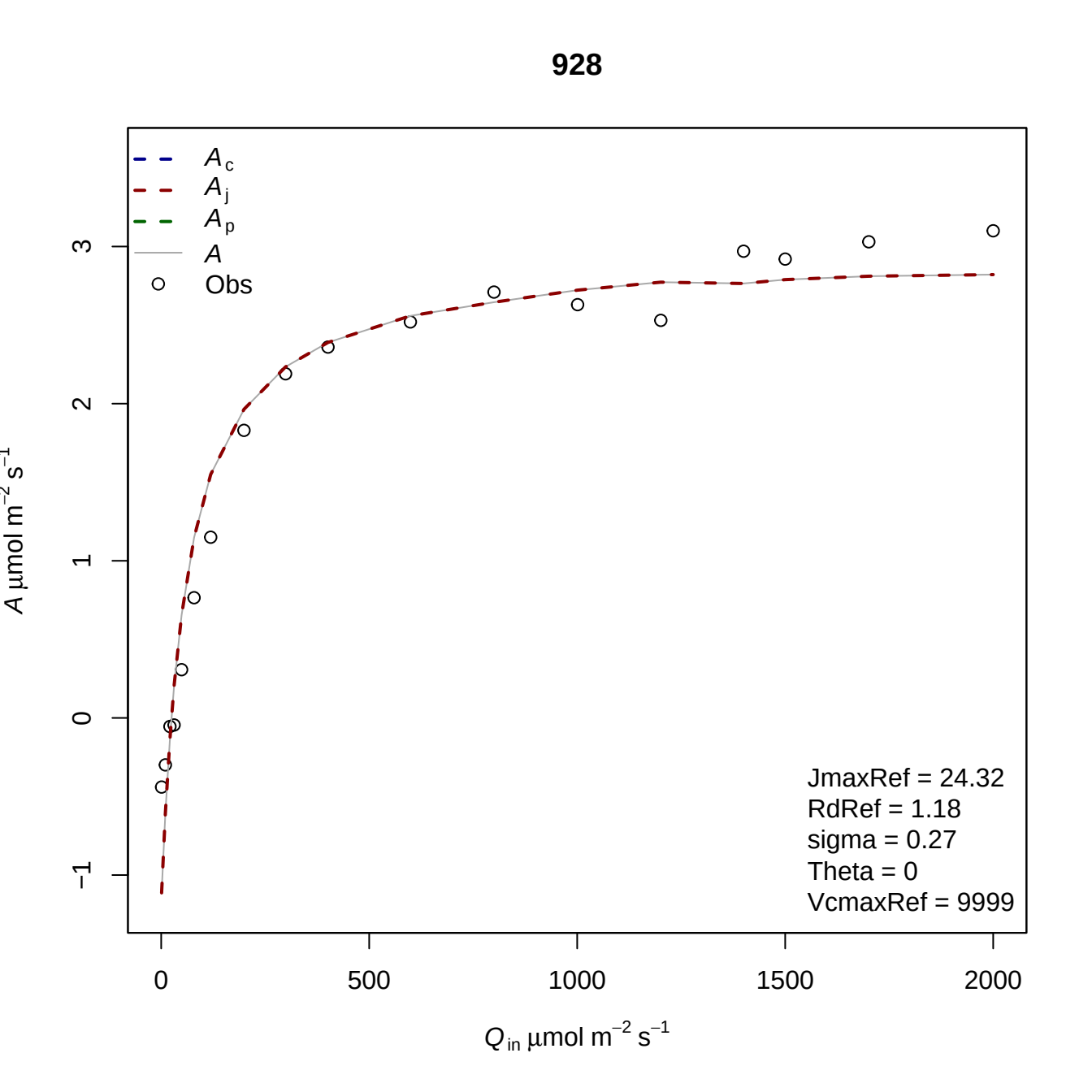

923
A_c
A_j
A_p
A
Obs
JmaxRef = 26.74
RdRef = 1.08
sigma = 0.23
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

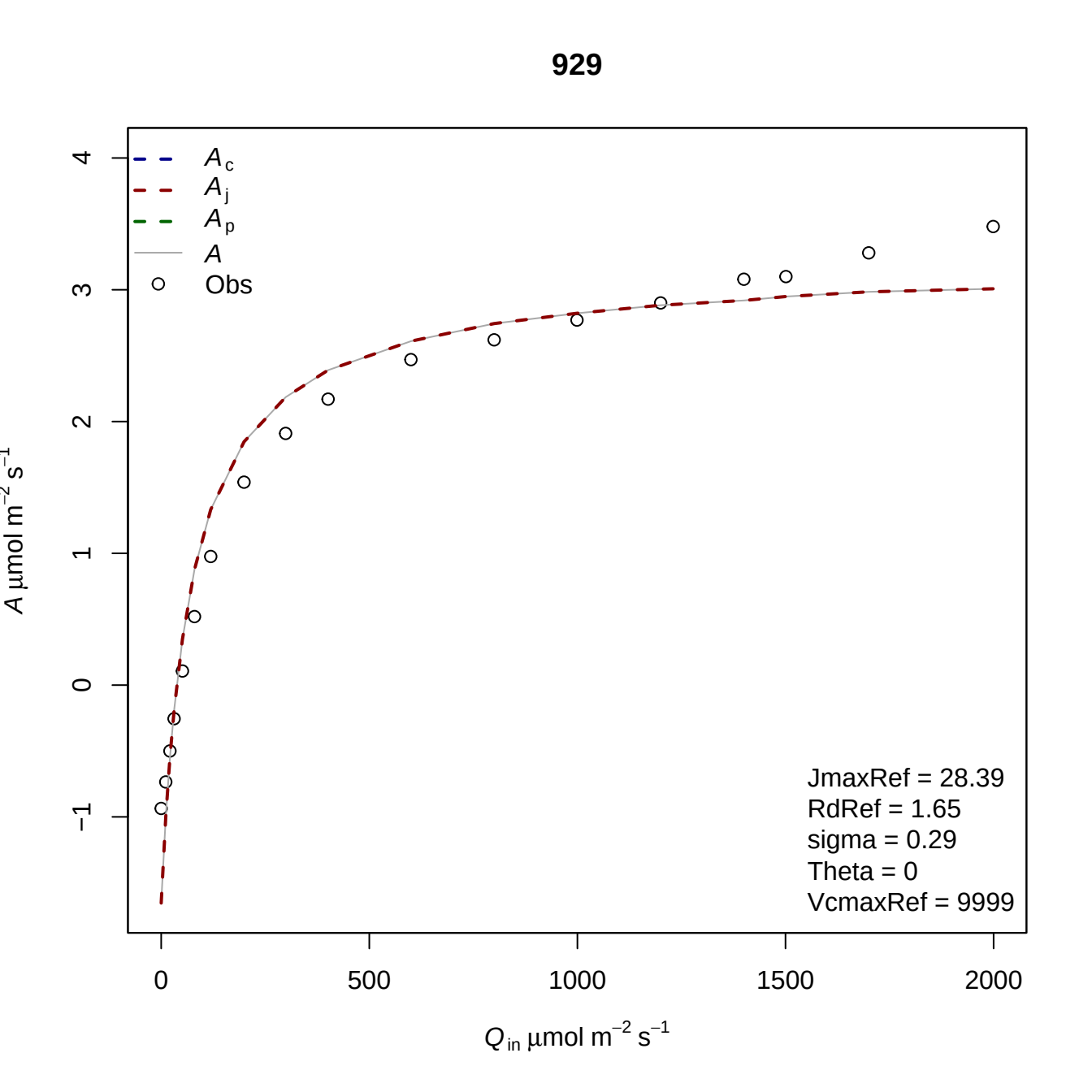

924
A_c
A_j
A_p
A
Obs
JmaxRef = 21.19
RdRef = 1.51
sigma = 0.24
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

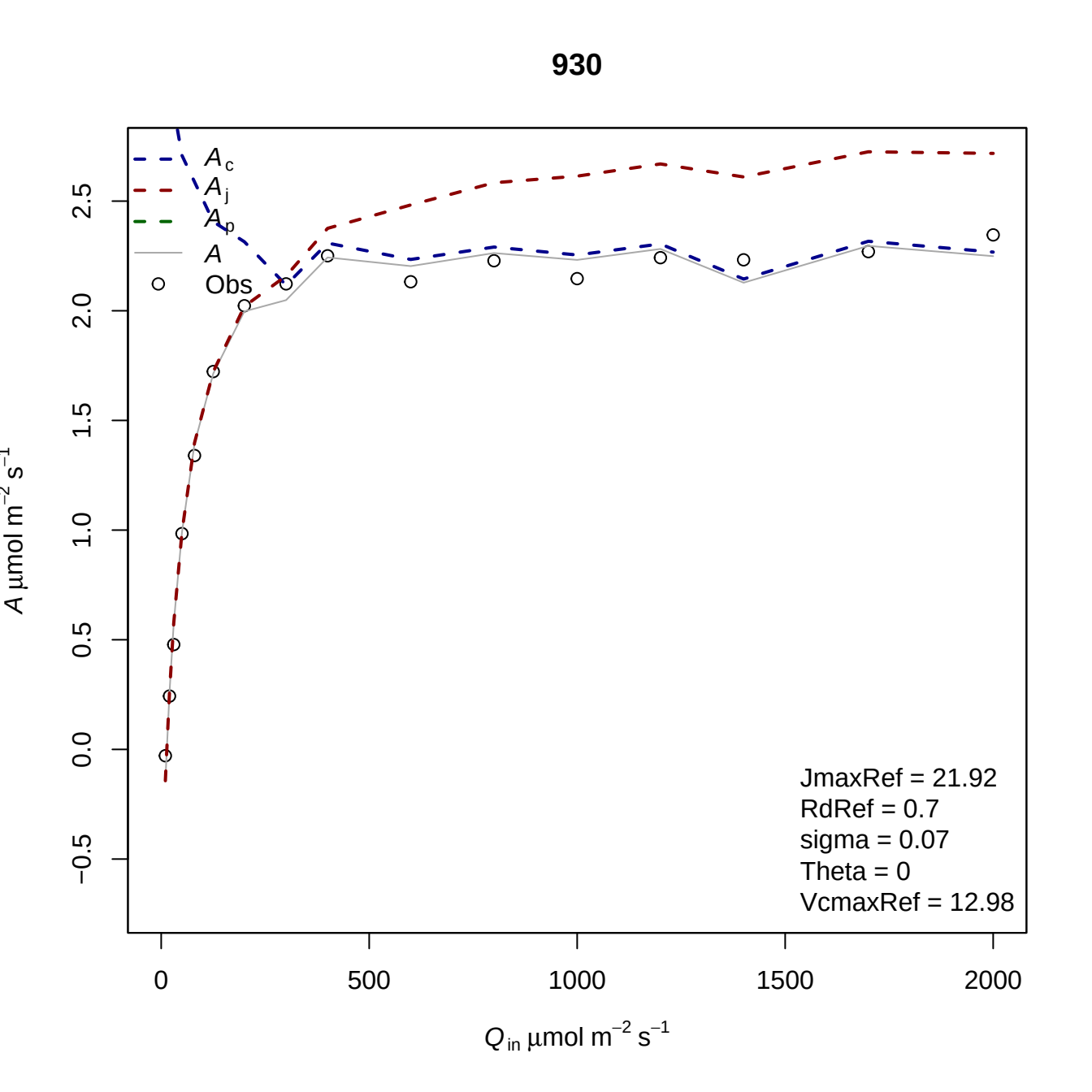

925
A_c
A_j
A_p
A
Obs
JmaxRef = 25.57
RdRef = 0.5
sigma = 0.11
Theta = 0
VcmaxRef = 13.99
0
1
2
3
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

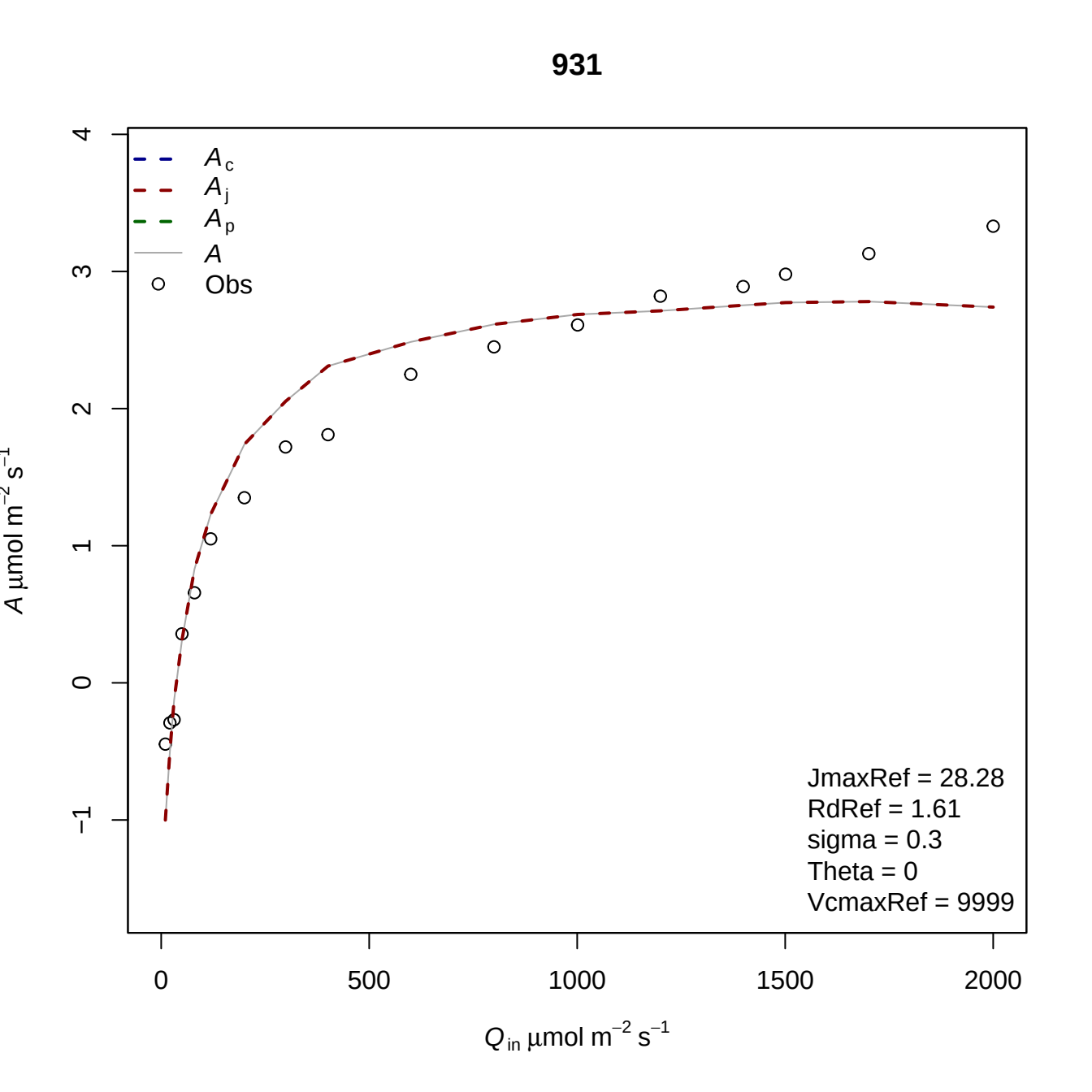

926
A_c
A_j
A_p
A
Obs
JmaxRef = 55.46
RdRef = 1.19
sigma = 0.2
Theta = 0
VcmaxRef = 31.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

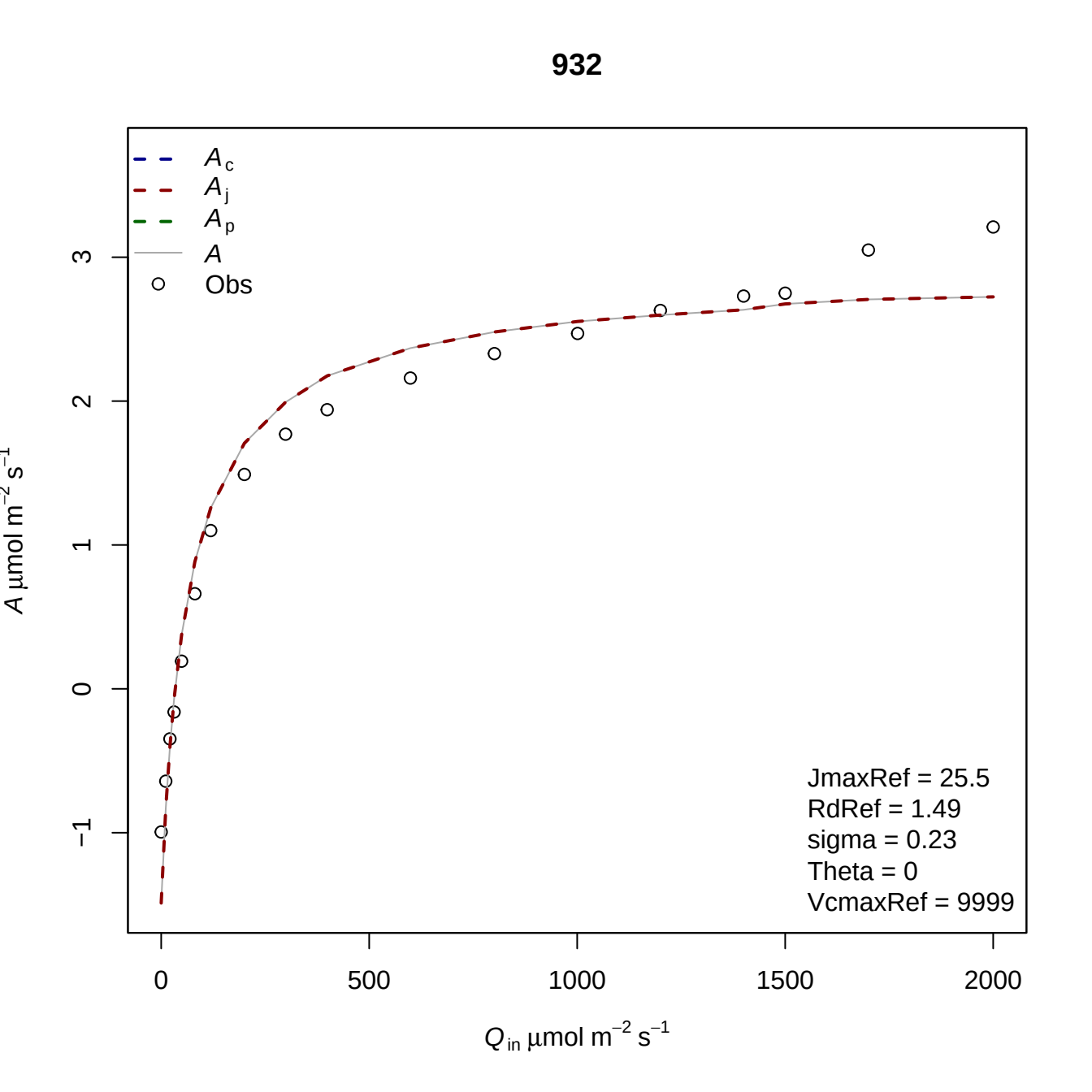

927
A_c
A_j
A_p
A
Obs
JmaxRef = 23.21
RdRef = 1.38
sigma = 0.29
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

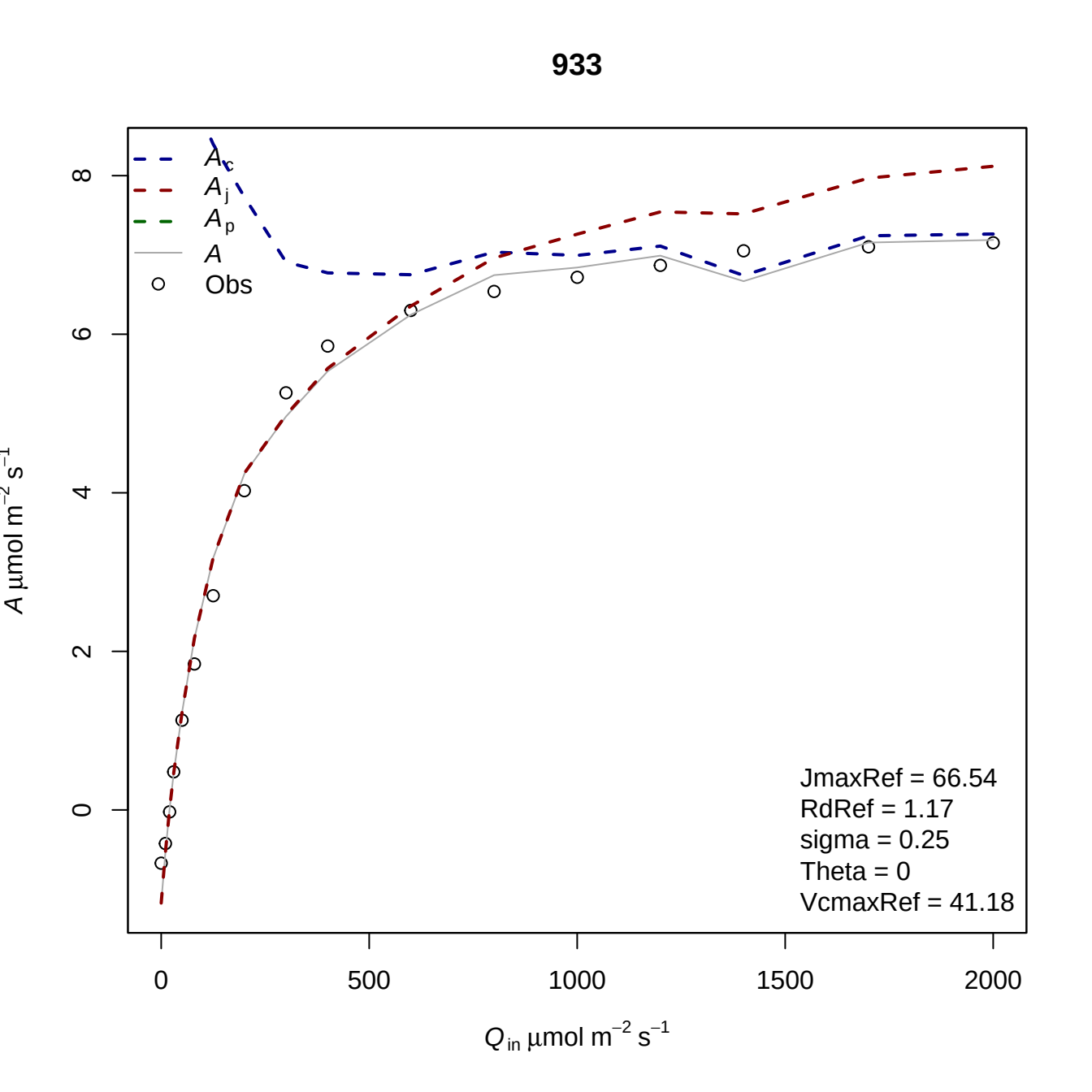

928
A_c
A_j
A_p
A
Obs
JmaxRef = 24.32
RdRef = 1.18
sigma = 0.27
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

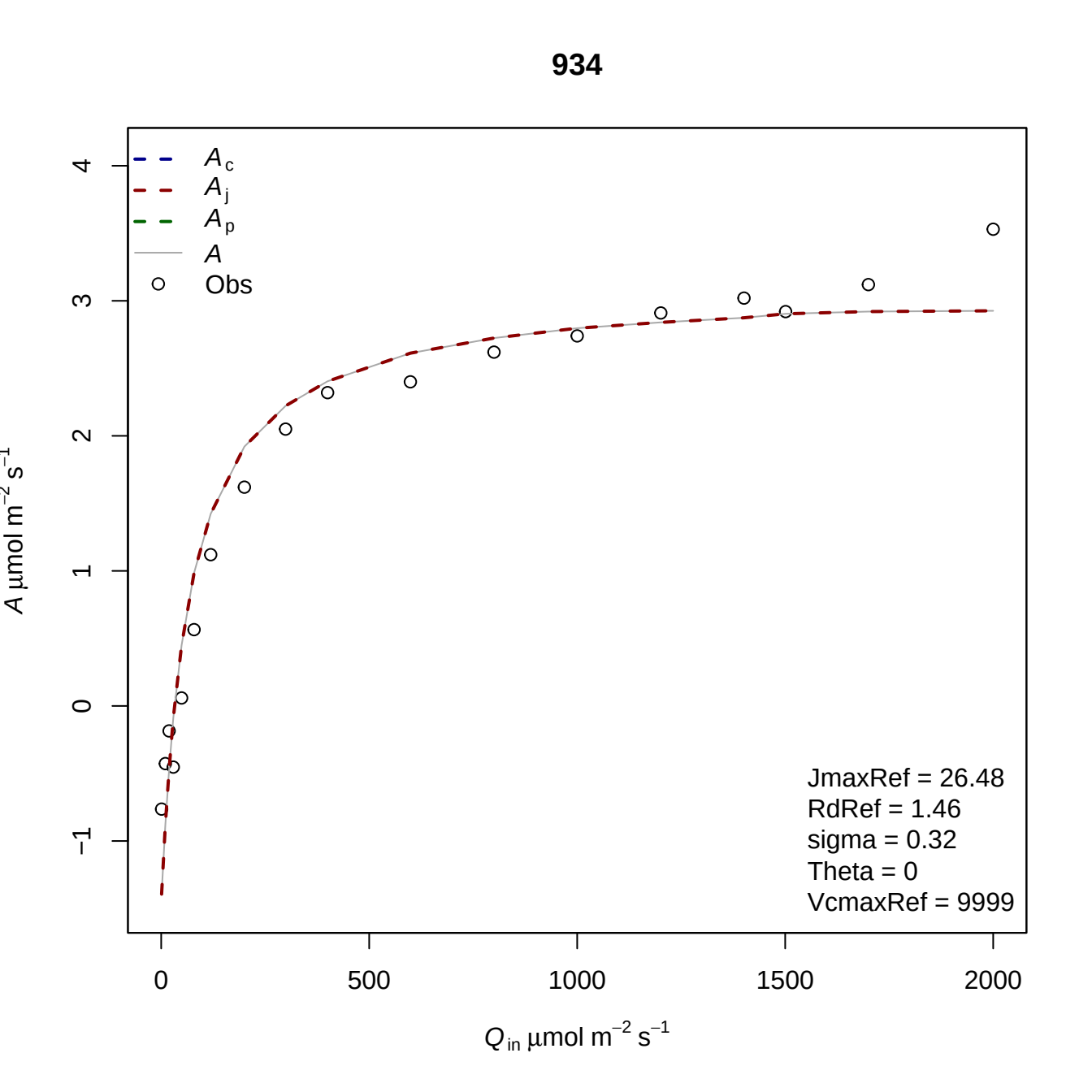

929
A_c
A_j
A_p
A
Obs
JmaxRef = 28.39
RdRef = 1.65
sigma = 0.29
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500
2000

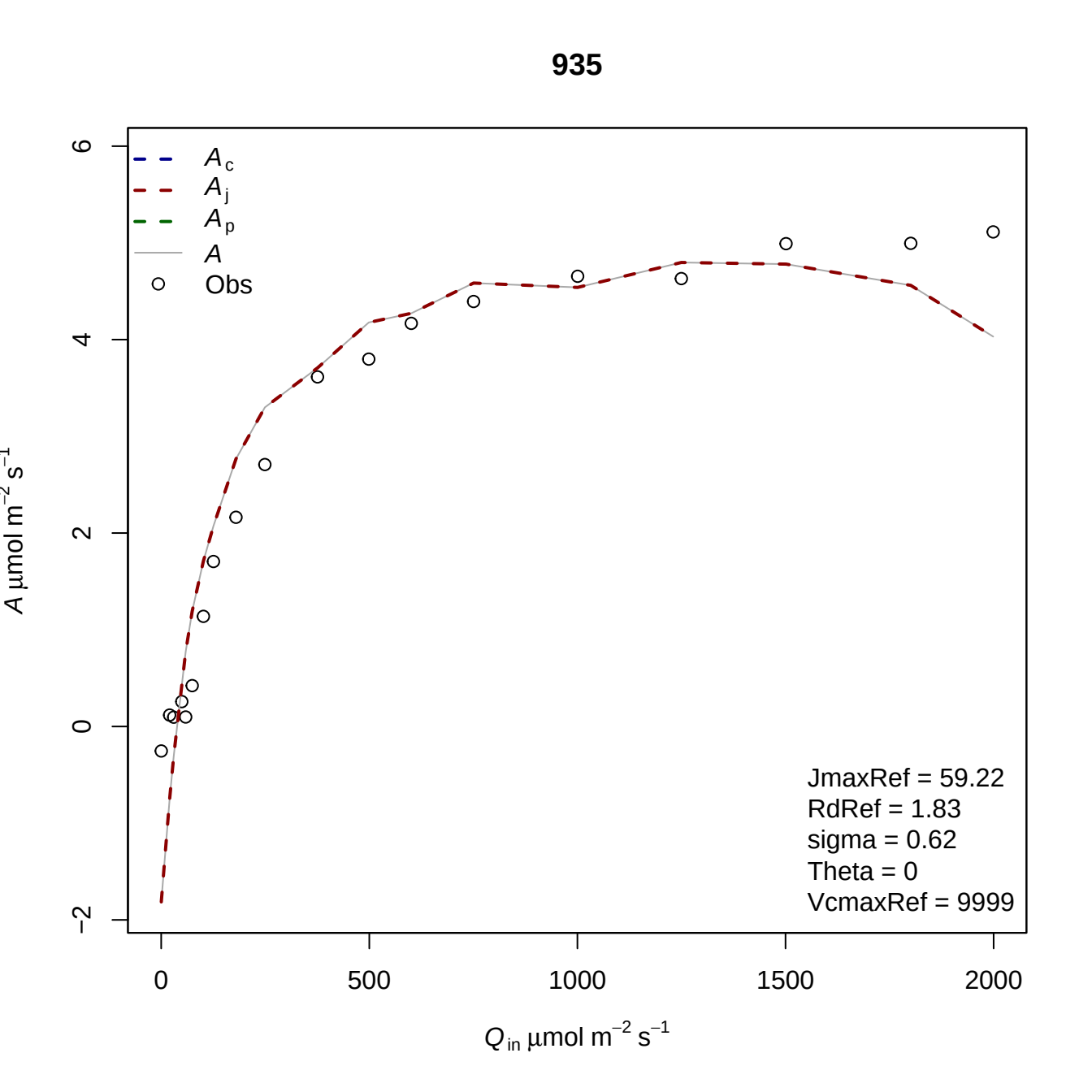

930
A_c
A_j
A_p
A
Obs
JmaxRef = 21.92
RdRef = 0.7
sigma = 0.07
Theta = 0
VcmaxRef = 12.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500
2000

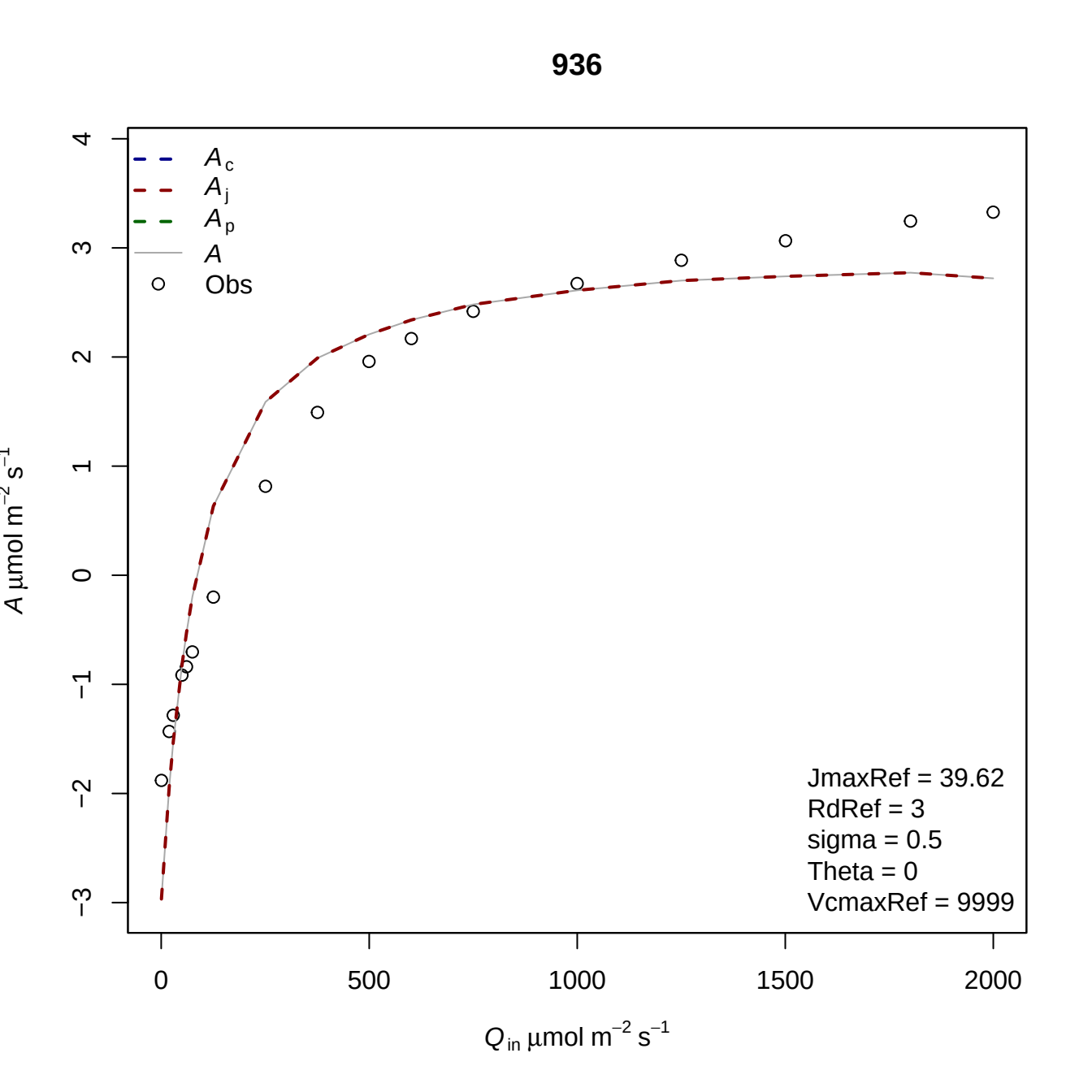

931
A_c
A_j
A_p
A
Obs
JmaxRef = 28.28
RdRef = 1.61
sigma = 0.3
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

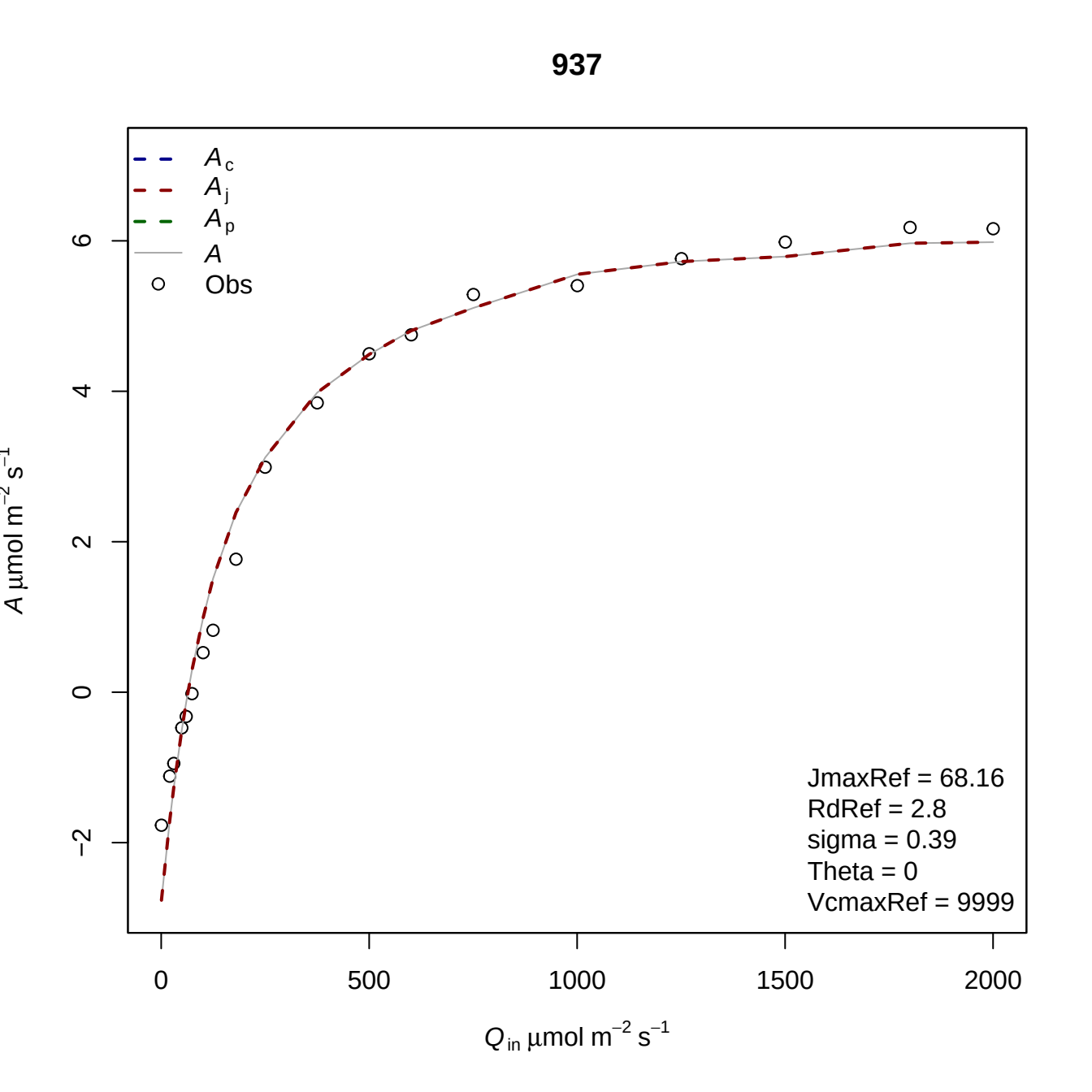

932
A_c
A_j
A_p
A
Obs
JmaxRef = 25.5
RdRef = 1.49
sigma = 0.23
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

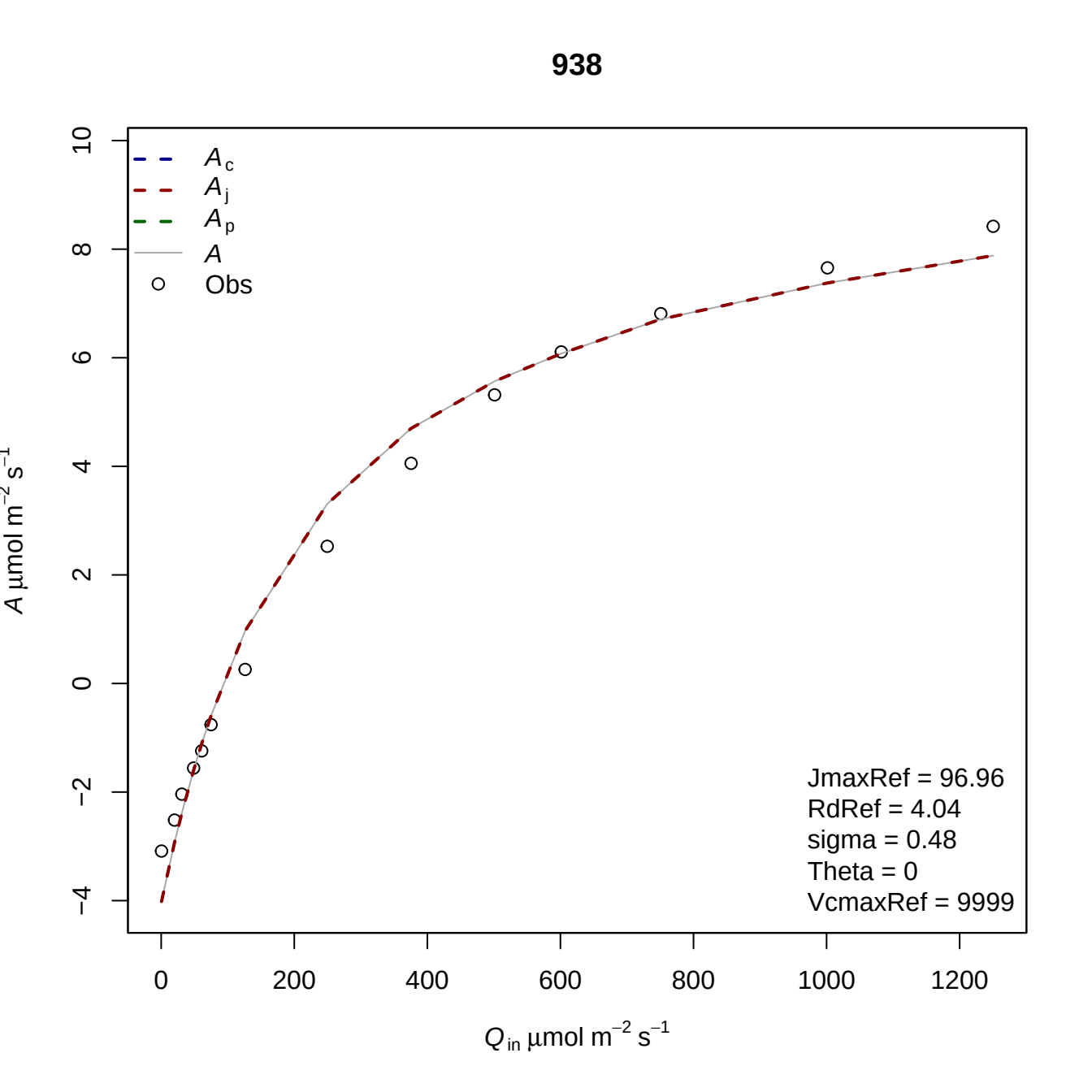

933
A_c
A_j
A_p
A
Obs
JmaxRef = 66.54
RdRef = 1.17
sigma = 0.25
Theta = 0
VcmaxRef = 41.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

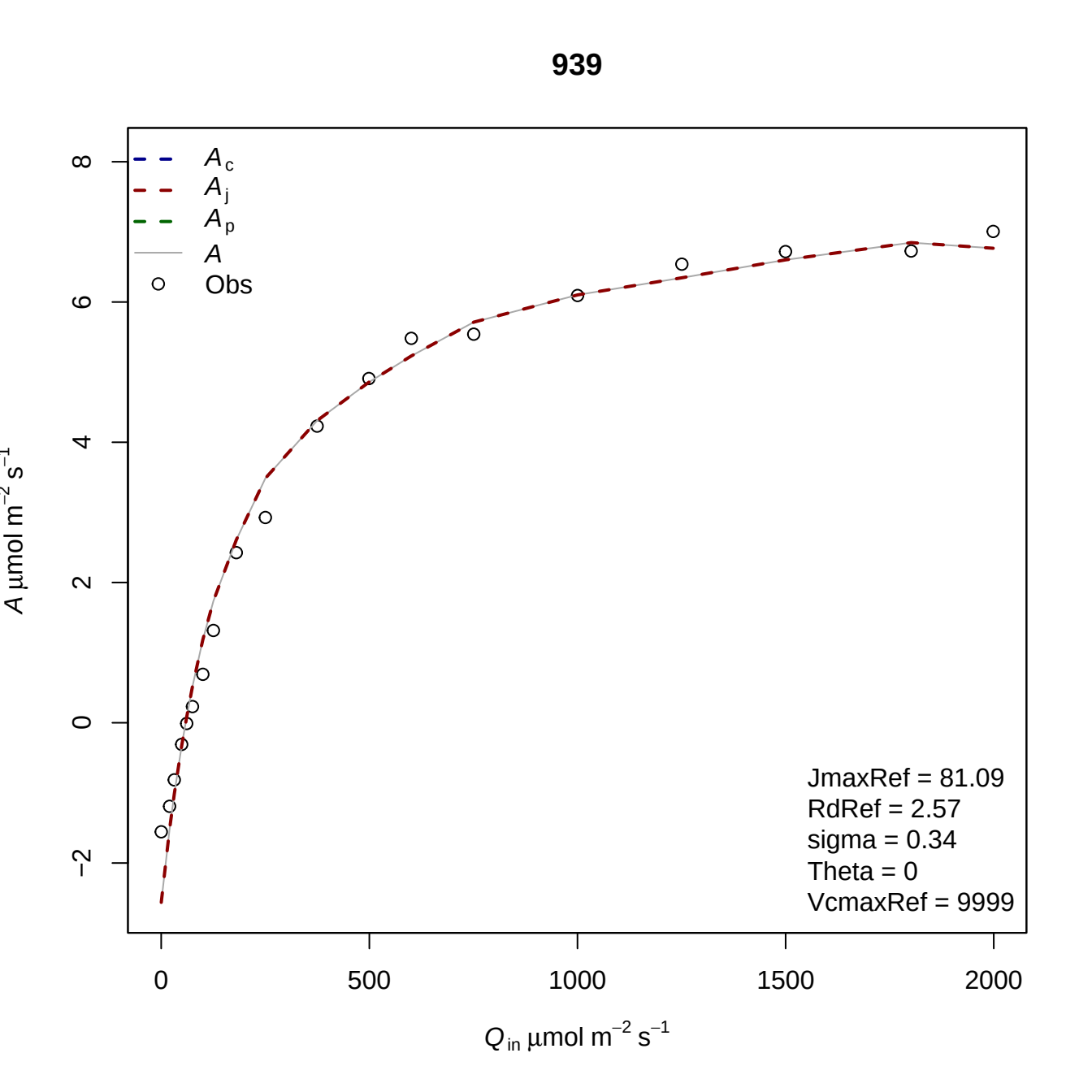

934
A_c
A_j
A_p
A
Obs
JmaxRef = 26.48
RdRef = 1.46
sigma = 0.32
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

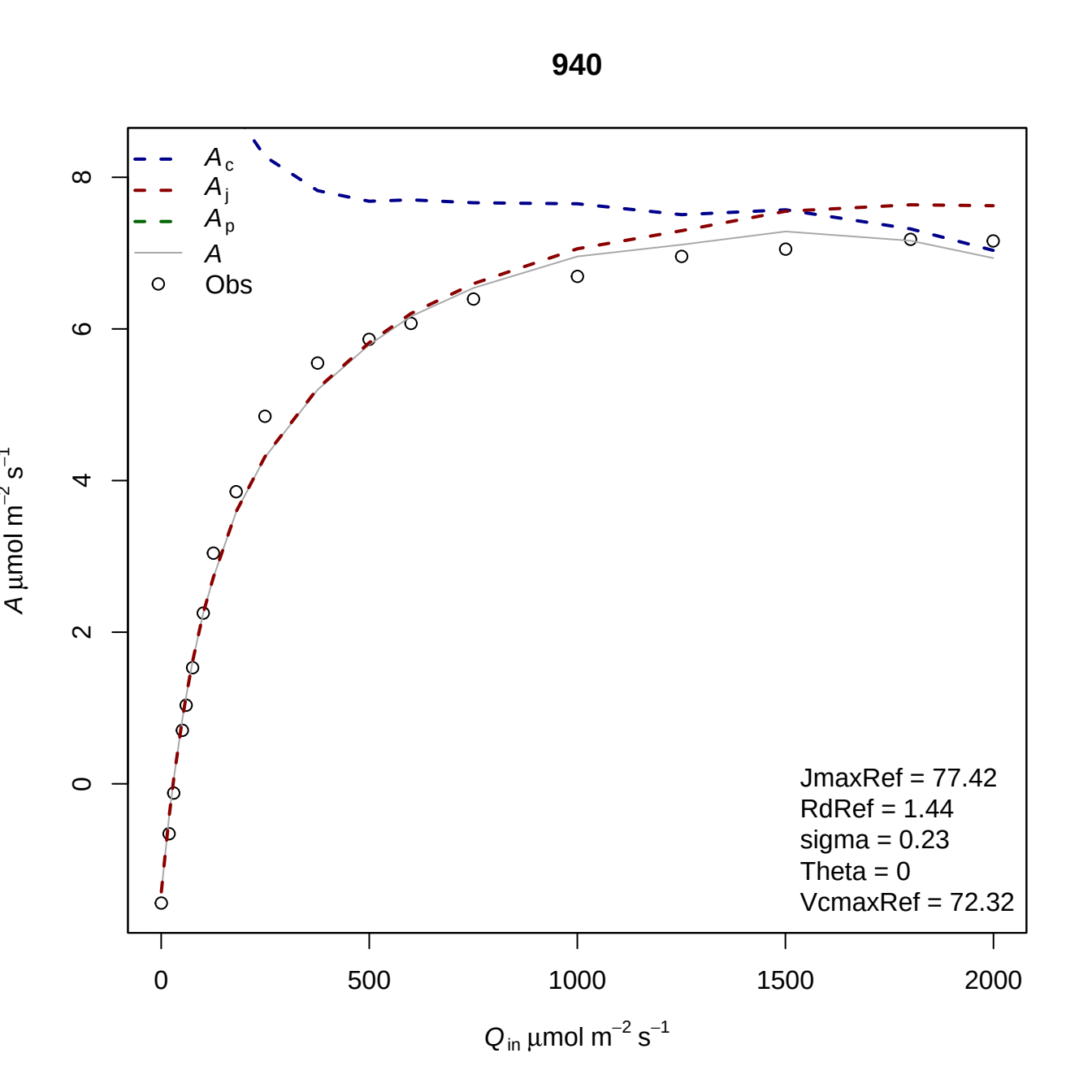

935
A_c
A_j
A_p
A
Obs
JmaxRef = 59.22
RdRef = 1.83
sigma = 0.62
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

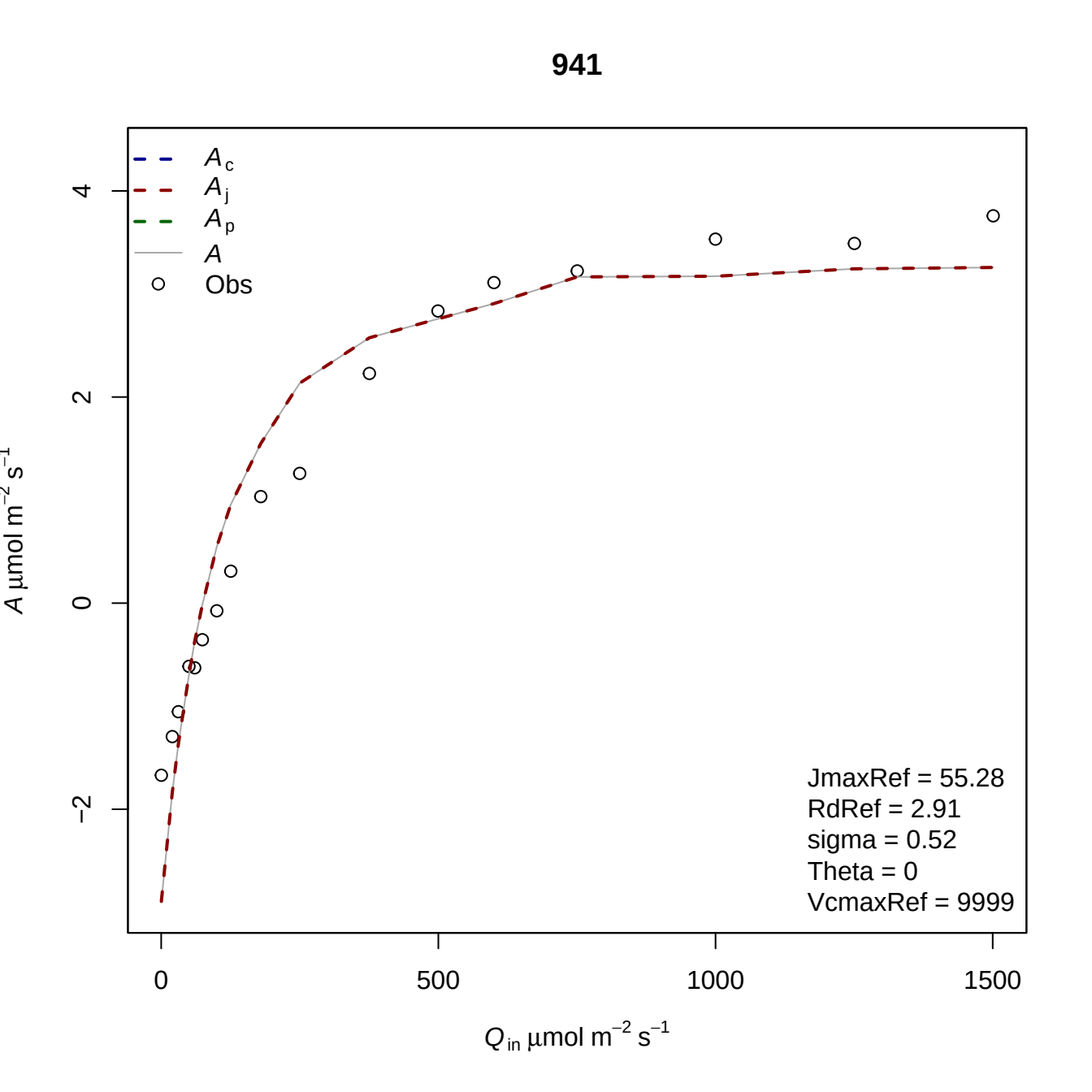

936
A_c
A_j
A_p
A
Obs
JmaxRef = 39.62
RdRef = 3
sigma = 0.5
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500
2000

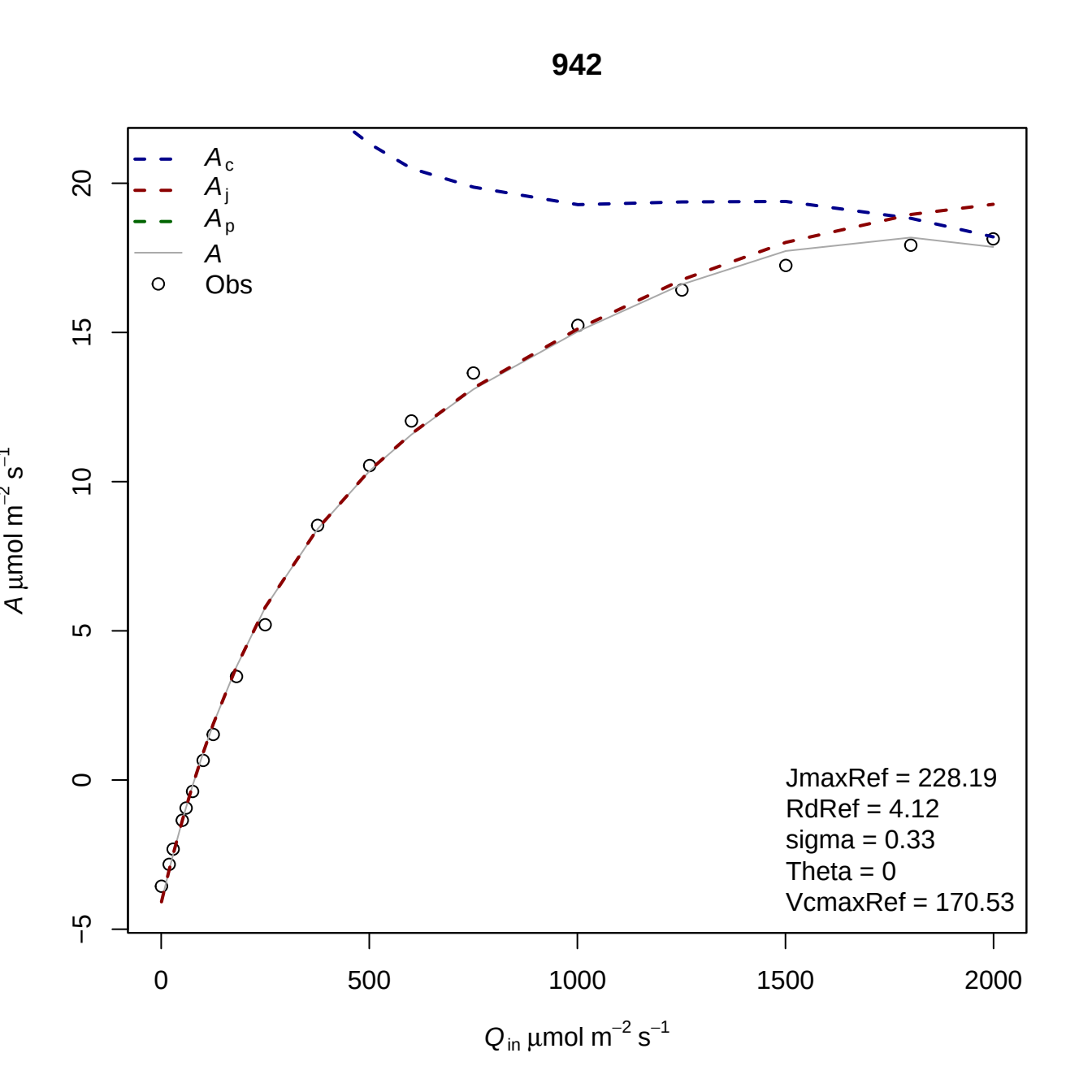

937
A_c
A_j
A_p
A
Obs
JmaxRef = 68.16
RdRef = 2.8
sigma = 0.39
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

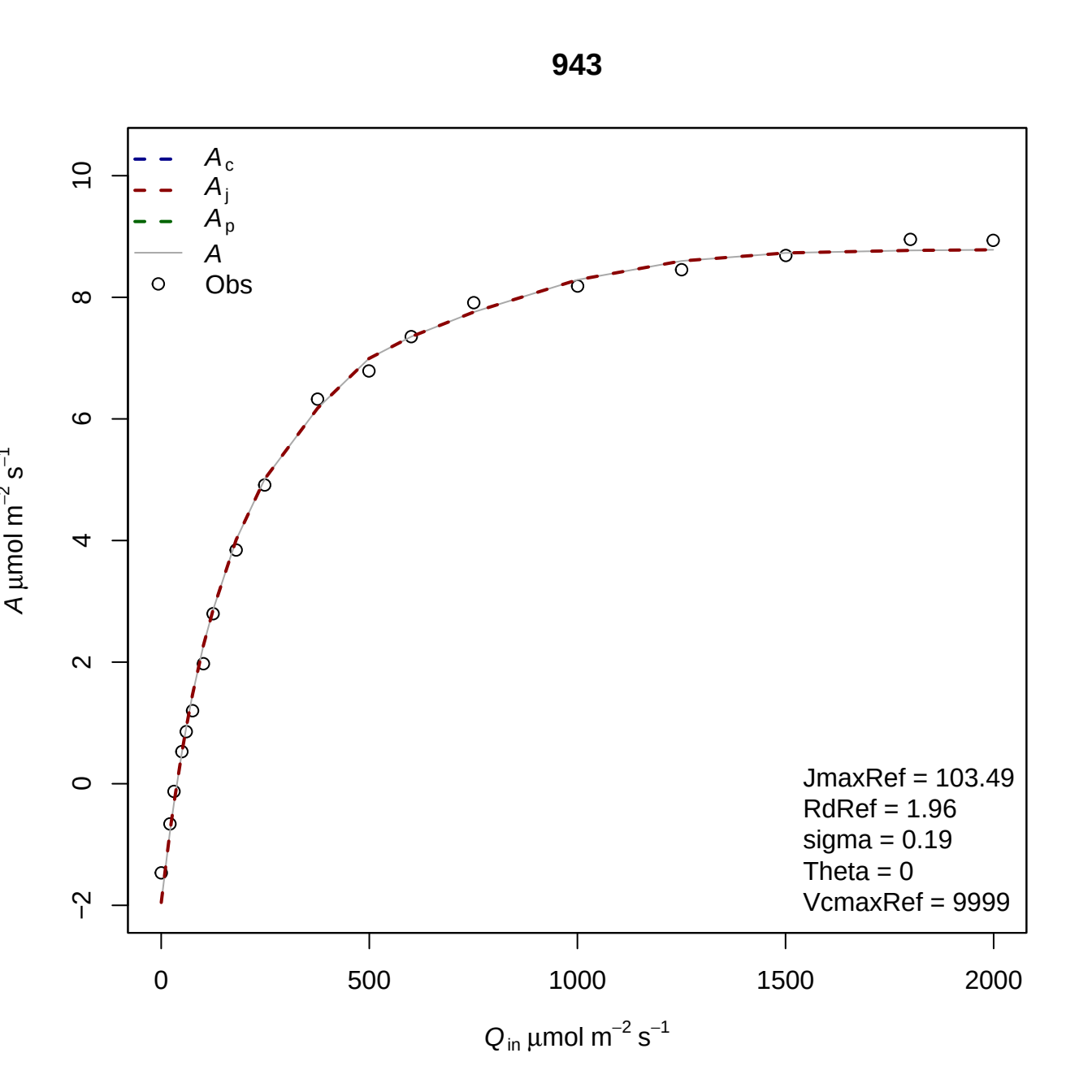

938
A_c
A_j
A_p
A
Obs
JmaxRef = 96.96
RdRef = 4.04
sigma = 0.48
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

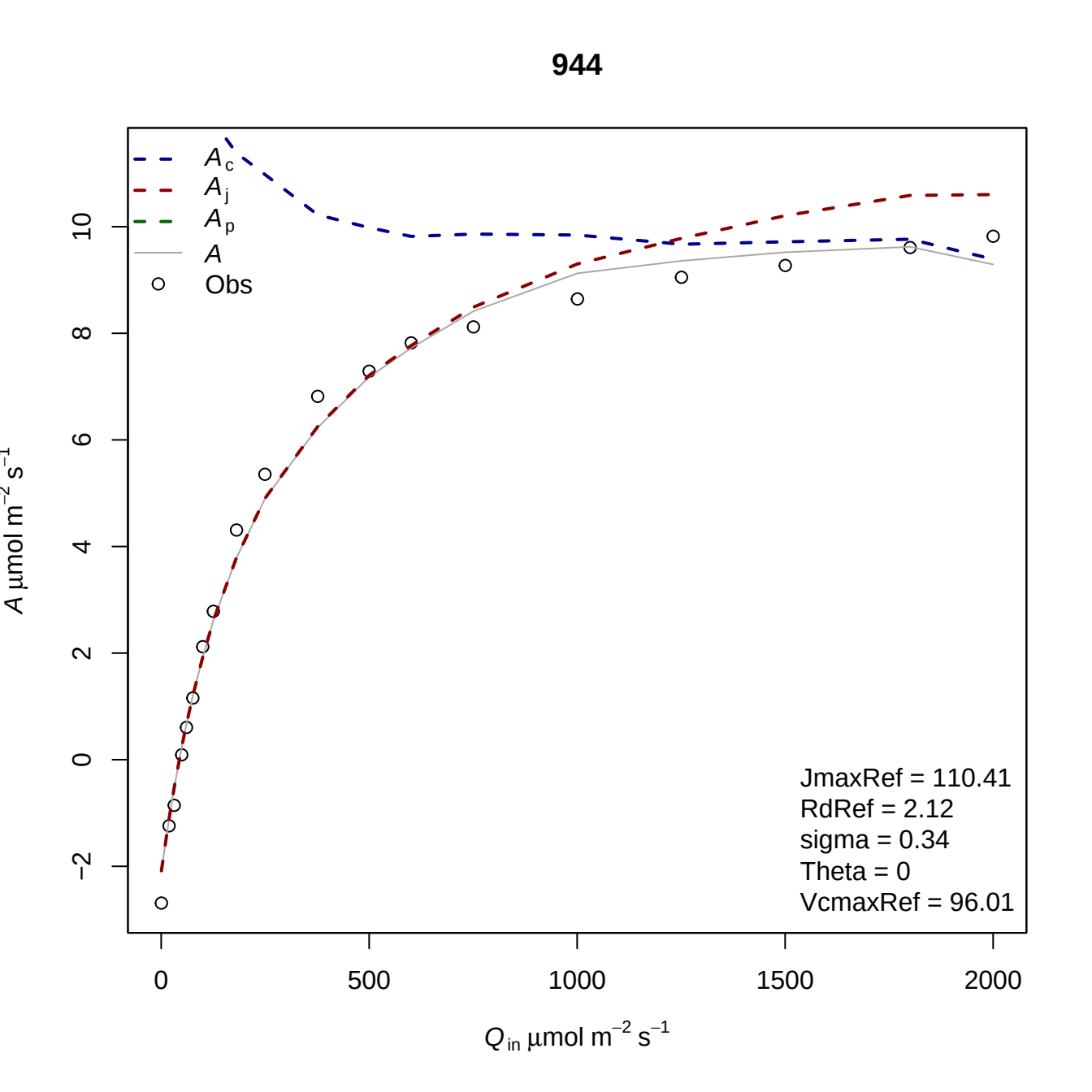

939
A_c
A_j
A_p
A
Obs
JmaxRef = 81.09
RdRef = 2.57
sigma = 0.34
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

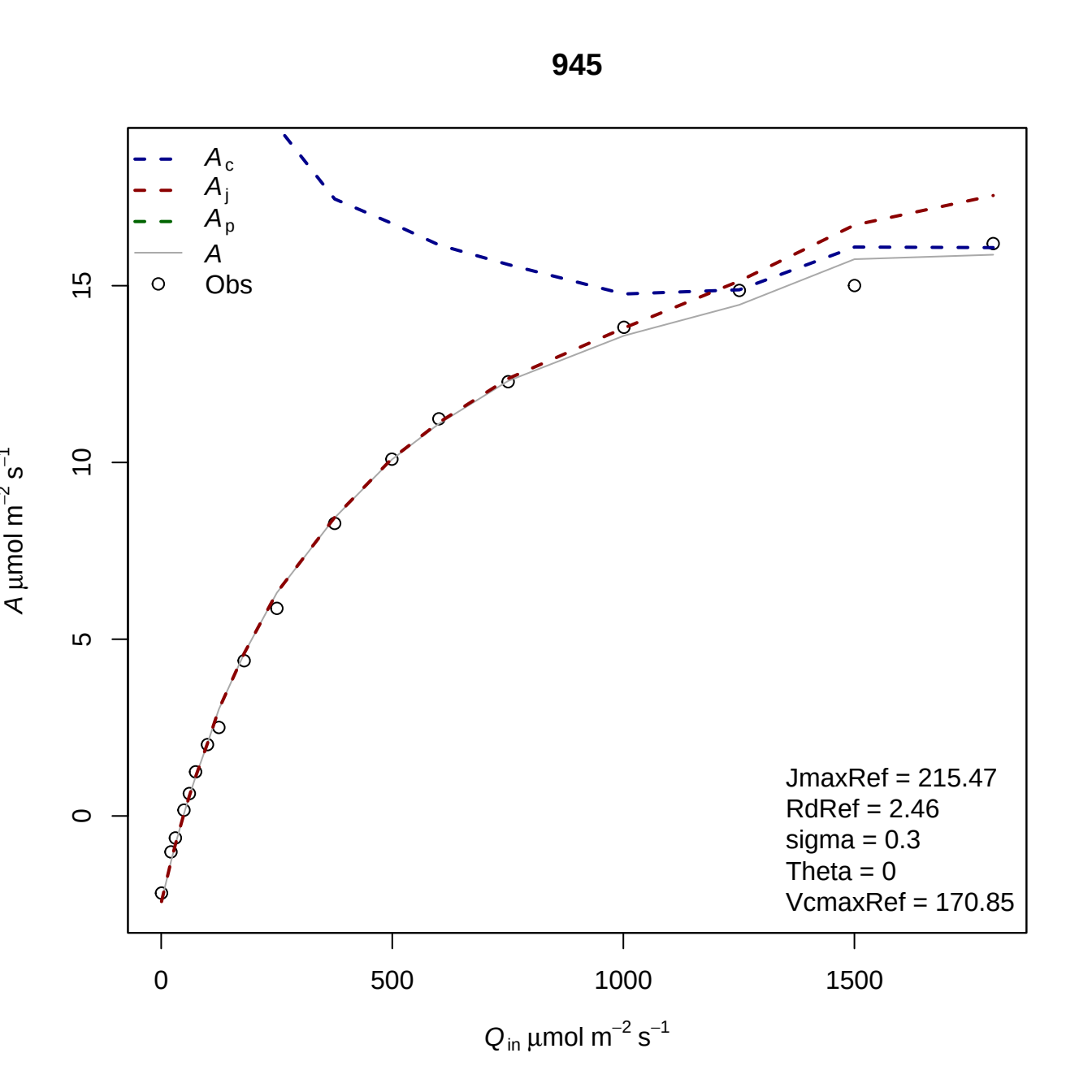

940
A_c
A_j
A_p
A
Obs
JmaxRef = 77.42
RdRef = 1.44
sigma = 0.23
Theta = 0
VcmaxRef = 72.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

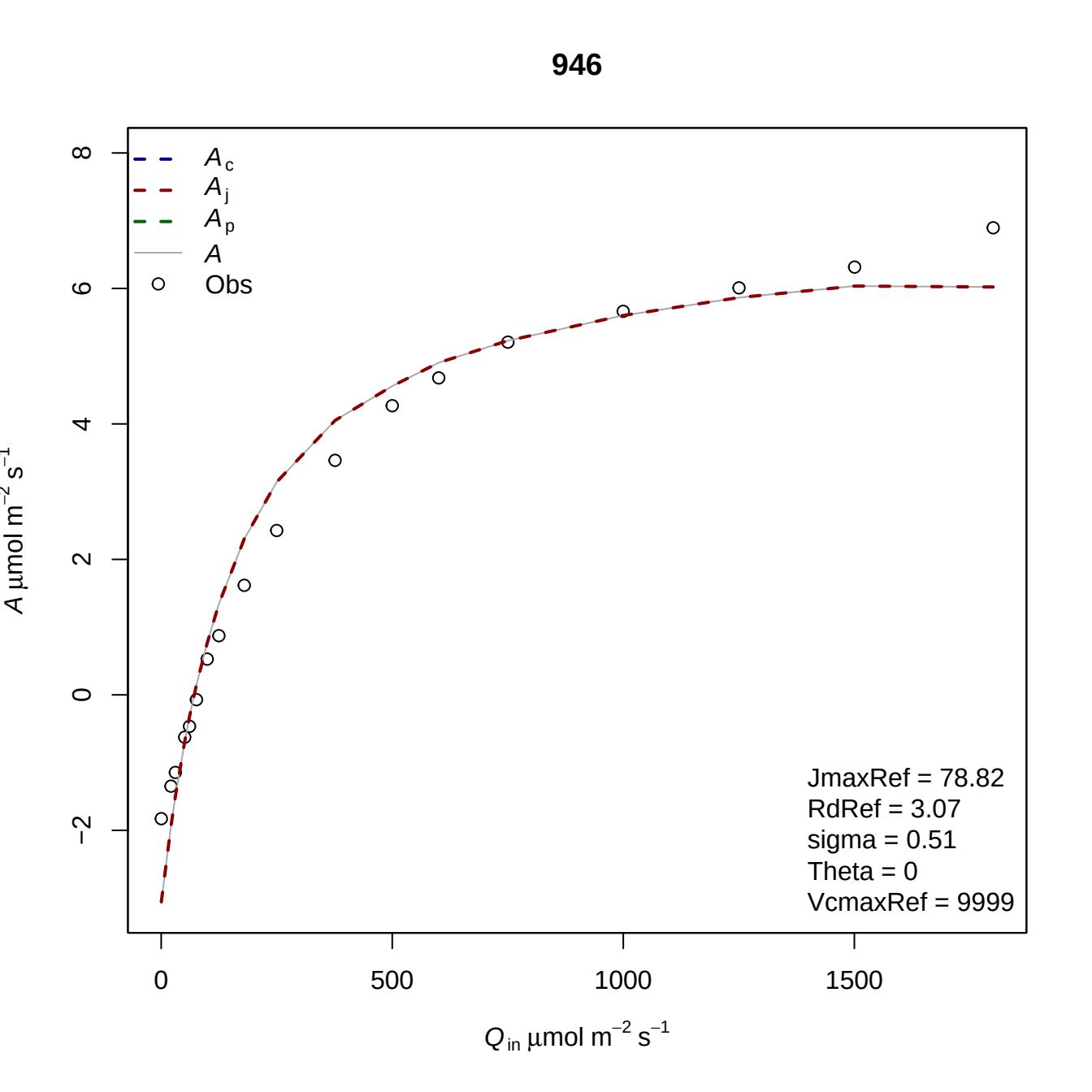

941
A_c
A_j
A_p
A
Obs
JmaxRef = 55.28
RdRef = 2.91
sigma = 0.52
Theta = 0
VcmaxRef = 9999
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

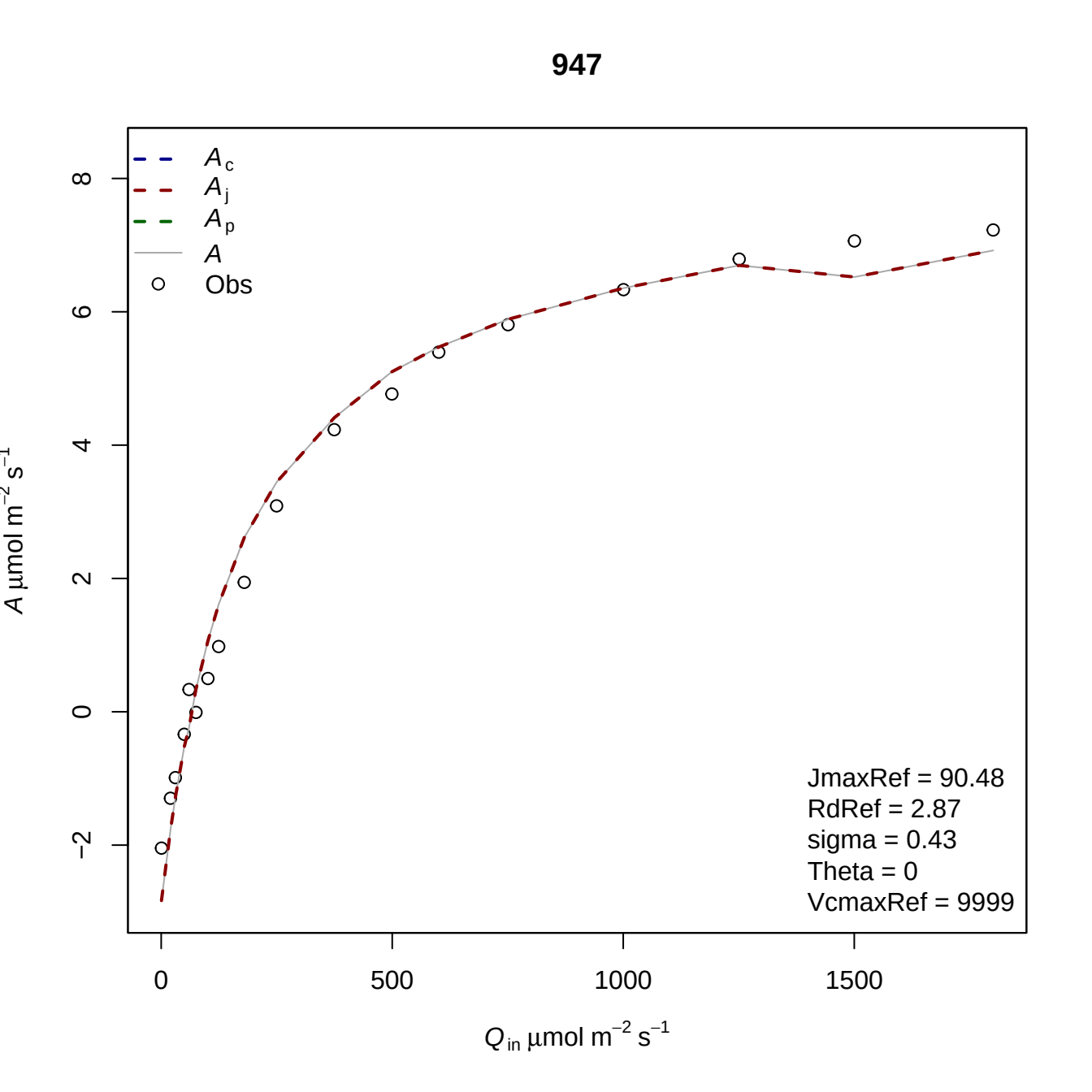

942
A_c
A_j
A_p
A
Obs
JmaxRef = 228.19
RdRef = 4.12
sigma = 0.33
Theta = 0
VcmaxRef = 170.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

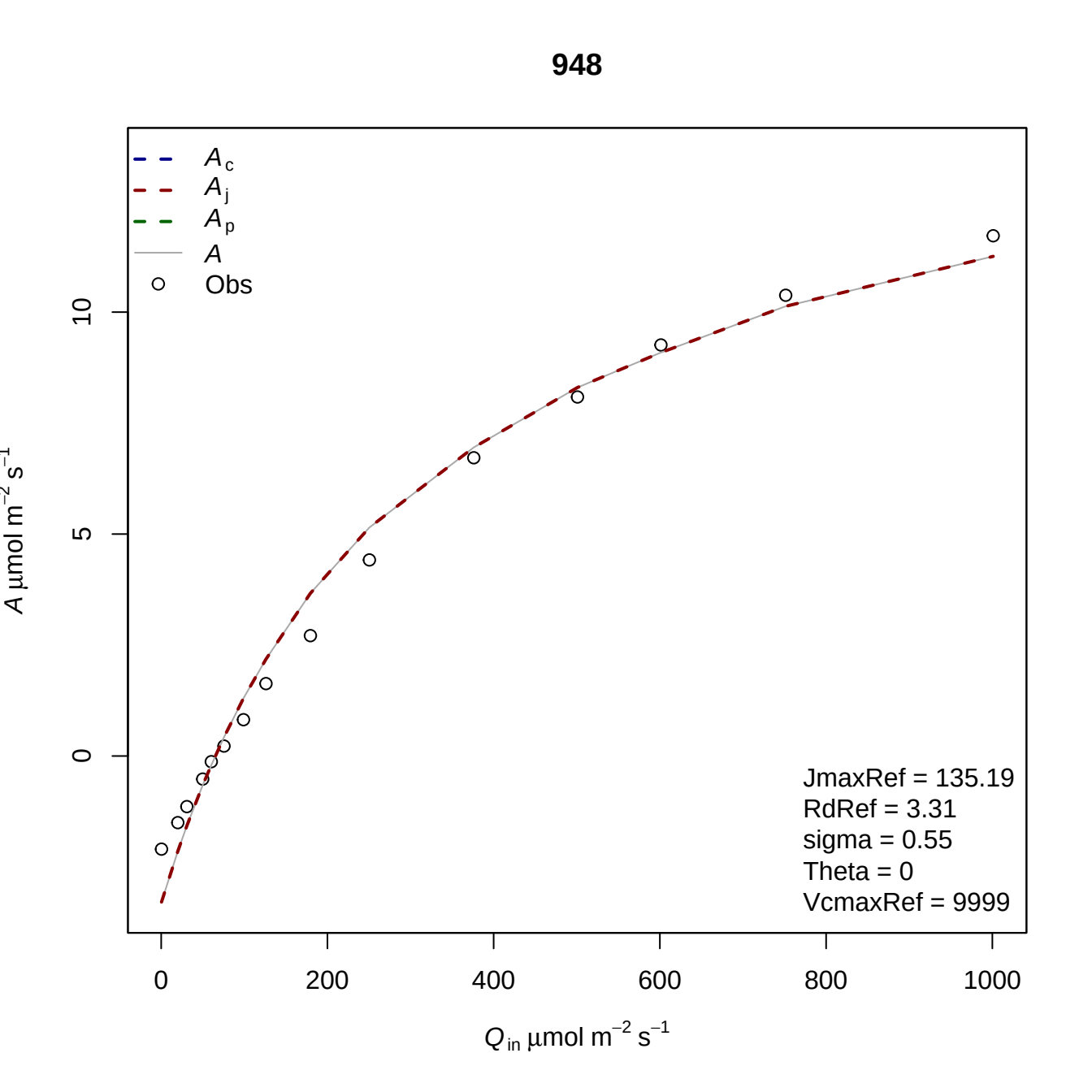

943
A_c
A_j
A_p
A
Obs
JmaxRef = 103.49
RdRef = 1.96
sigma = 0.19
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

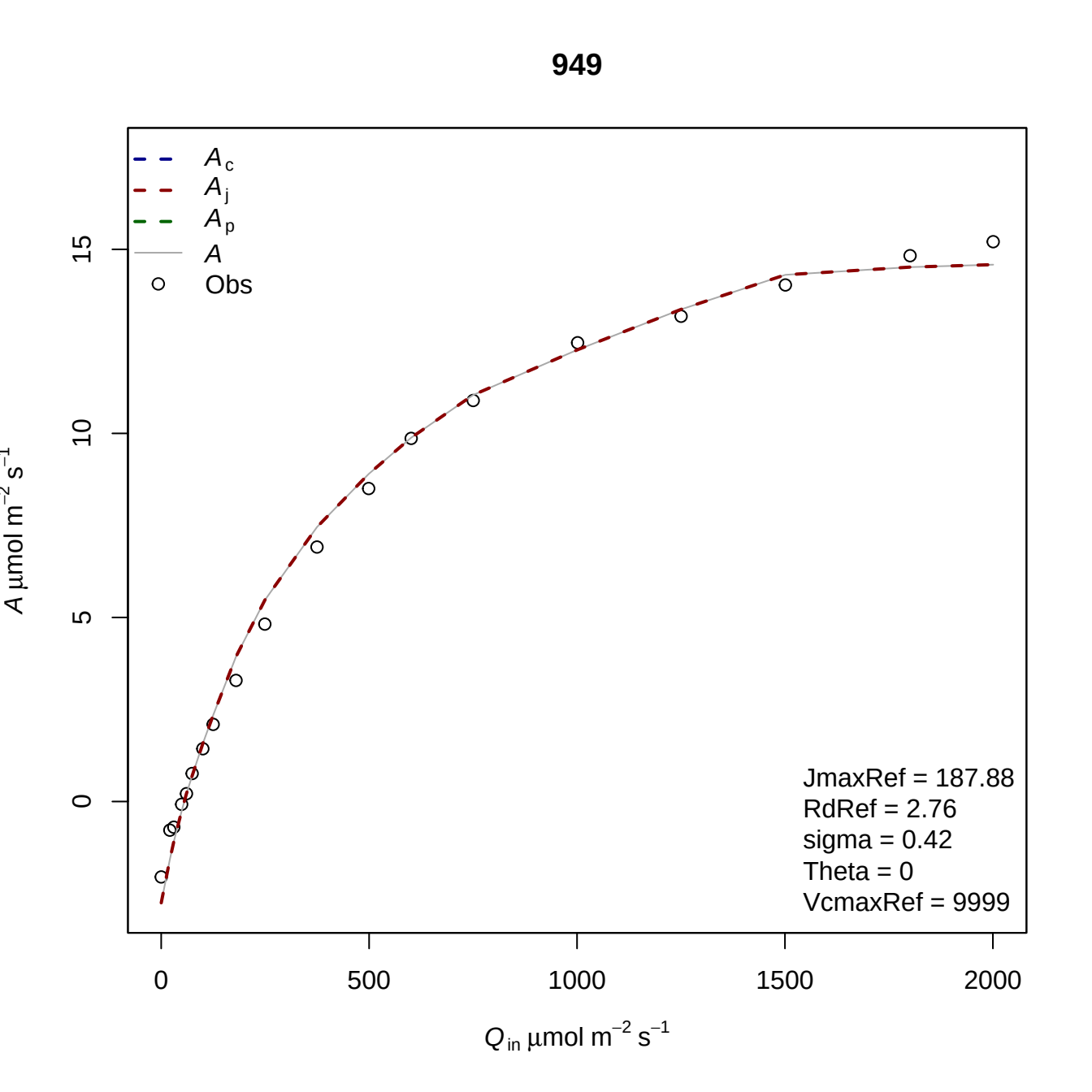

944
A_c
A_j
A_p
A
Obs
JmaxRef = 110.41
RdRef = 2.12
sigma = 0.34
Theta = 0
VcmaxRef = 96.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

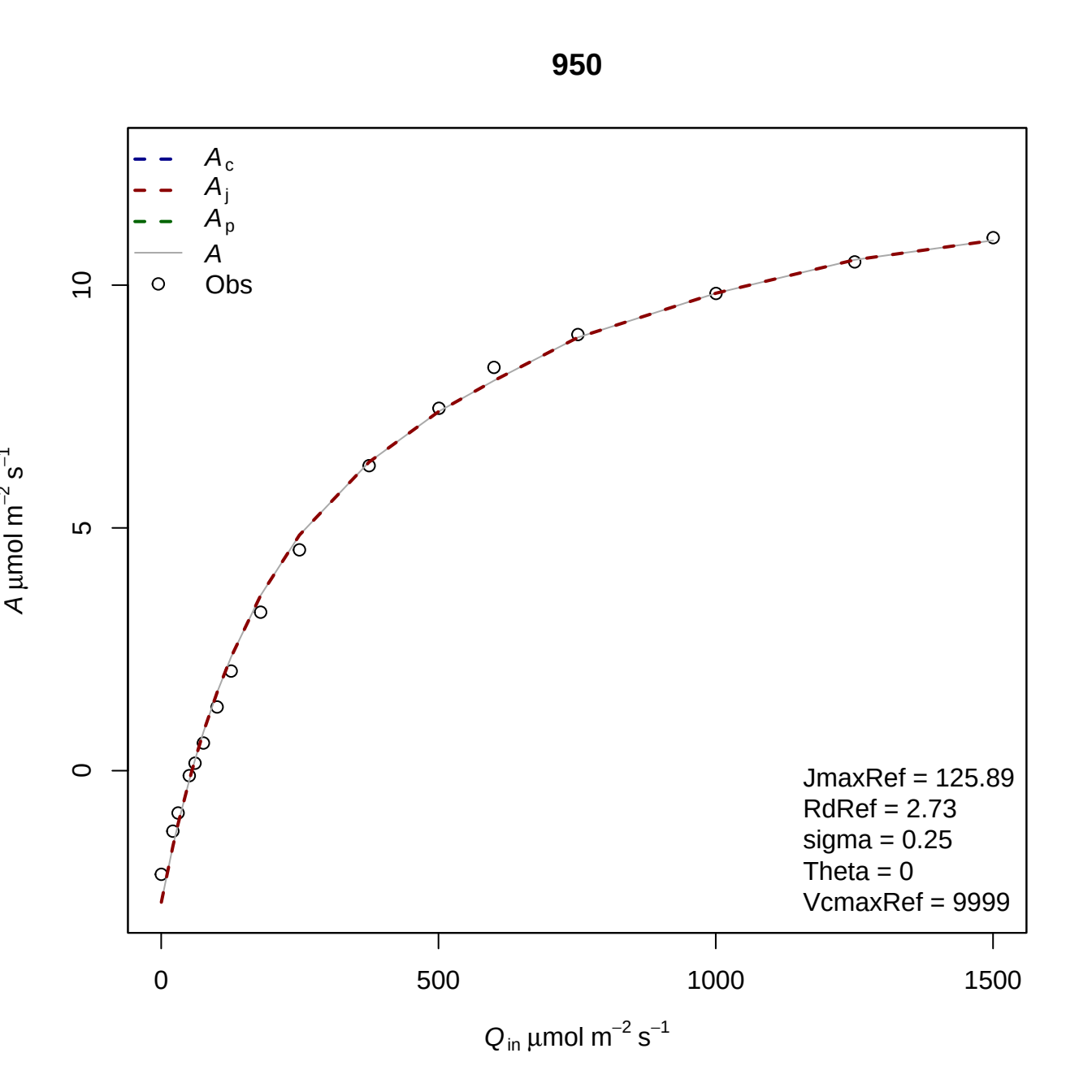

945
A_c
A_j
A_p
A
Obs
JmaxRef = 215.47
RdRef = 2.46
sigma = 0.3
Theta = 0
VcmaxRef = 170.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

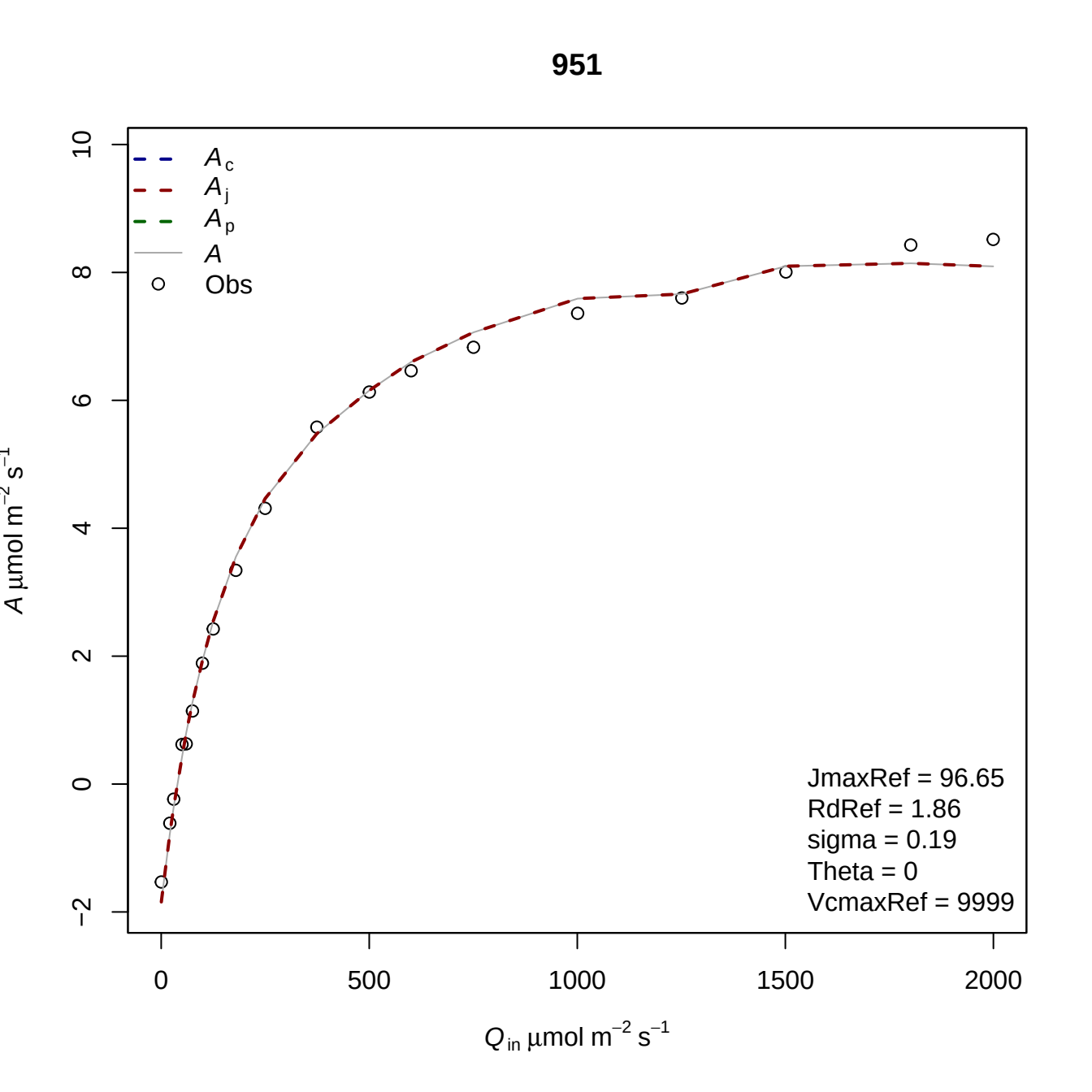

946
A_c
A_j
A_p
A
Obs
JmaxRef = 78.82
RdRef = 3.07
sigma = 0.51
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

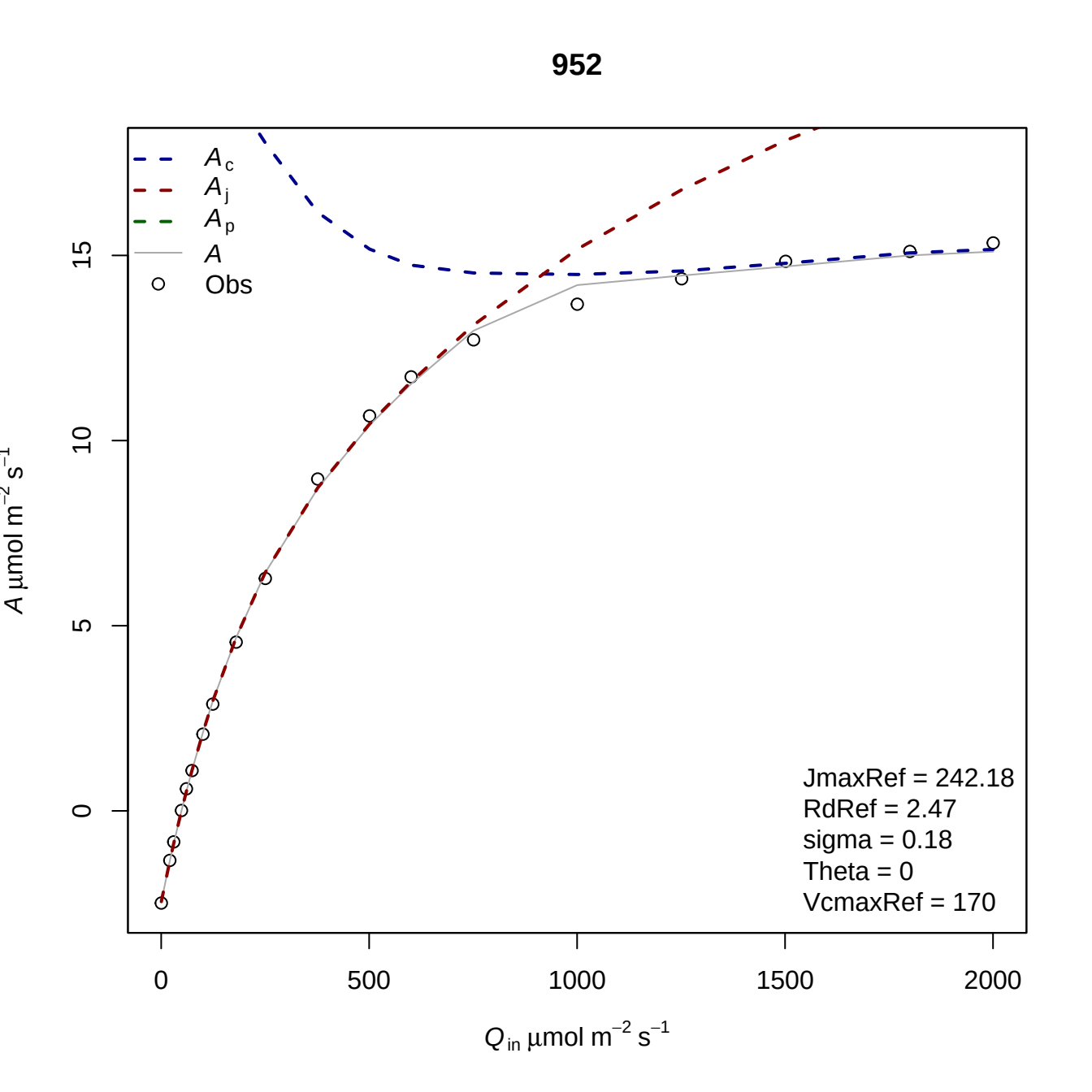

947
A_c
A_j
A_p
A
Obs
JmaxRef = 90.48
RdRef = 2.87
sigma = 0.43
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

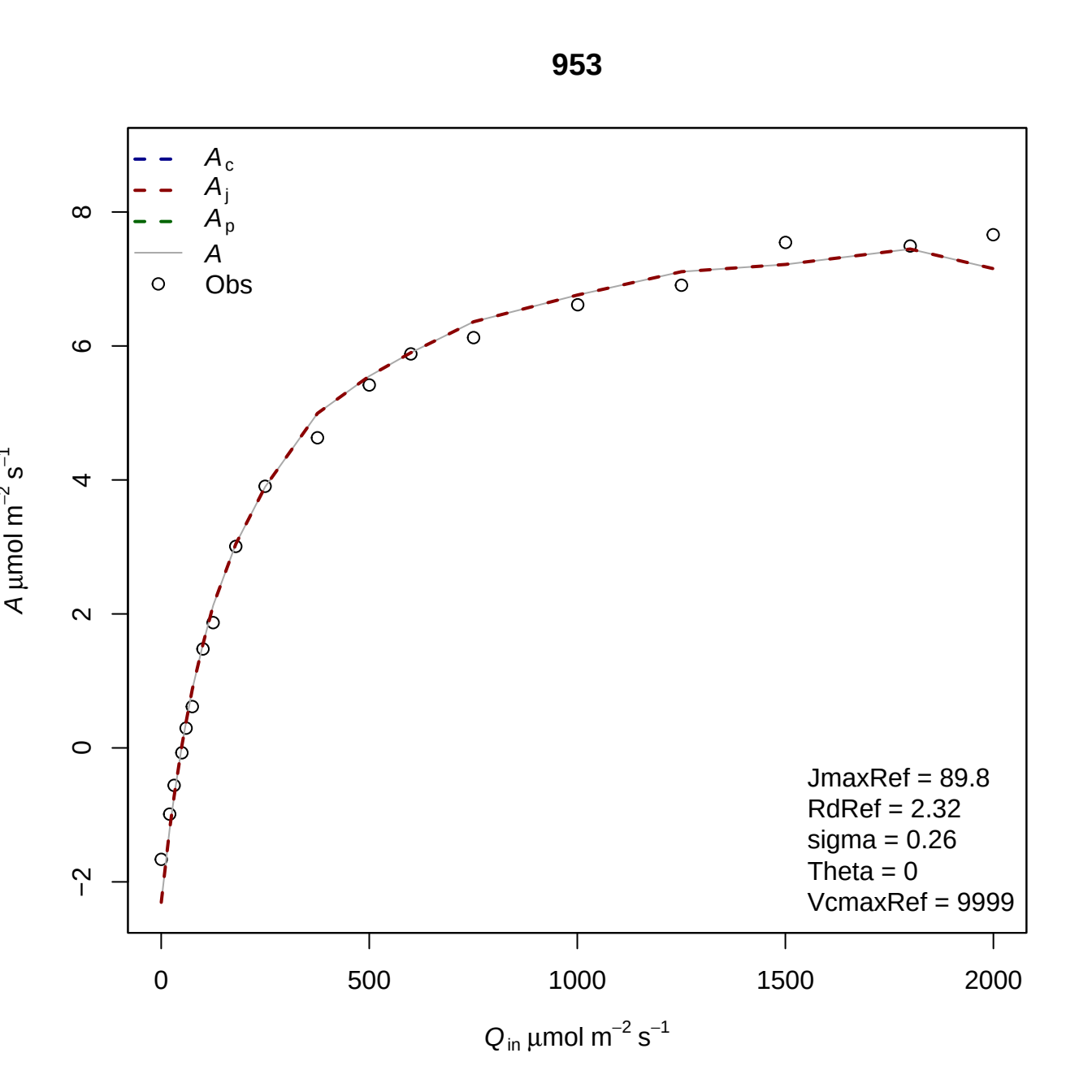

948
A_c
A_j
A_p
A
Obs
JmaxRef = 135.19
RdRef = 3.31
sigma = 0.55
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

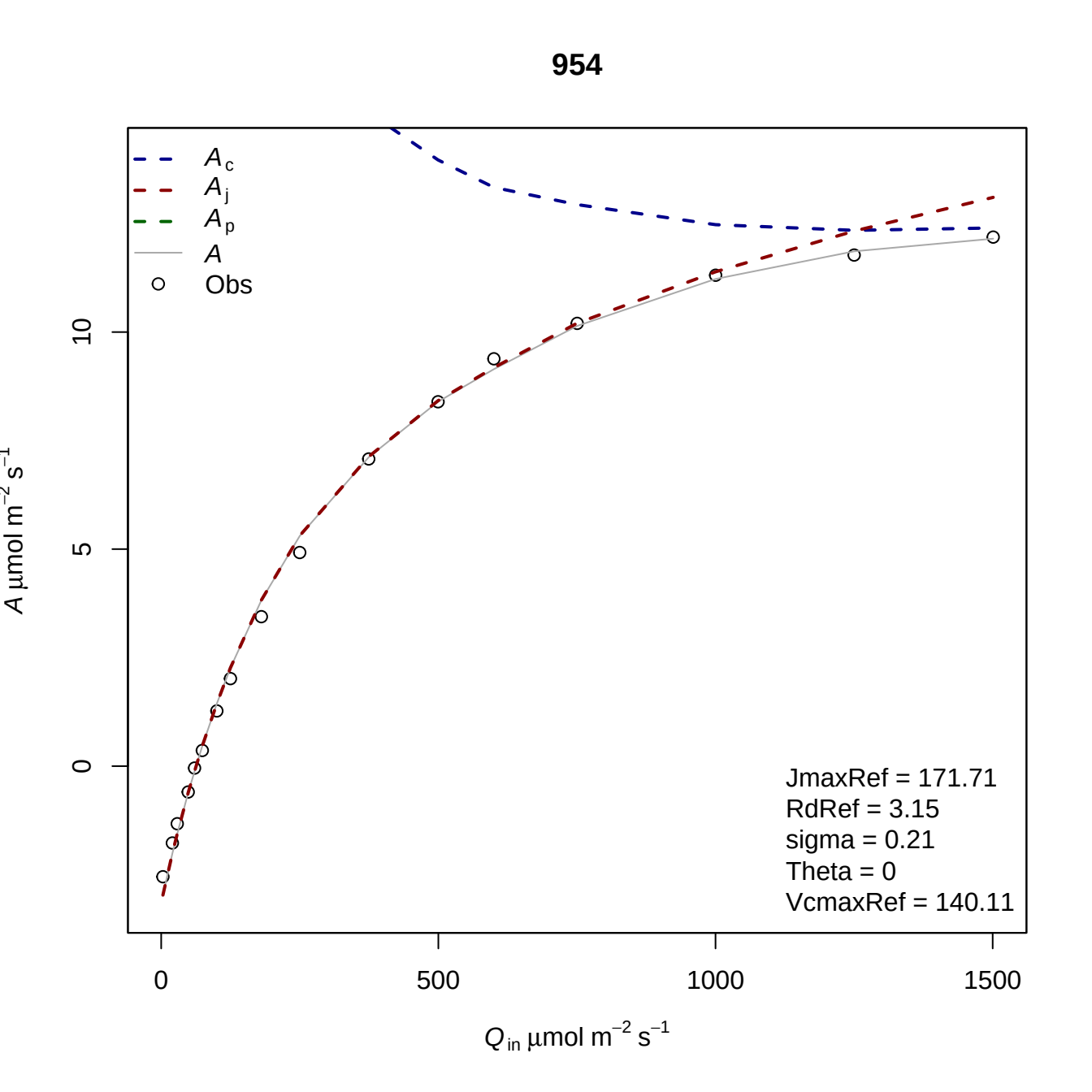

949
A_c
A_j
A_p
A
Obs
JmaxRef = 187.88
RdRef = 2.76
sigma = 0.42
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

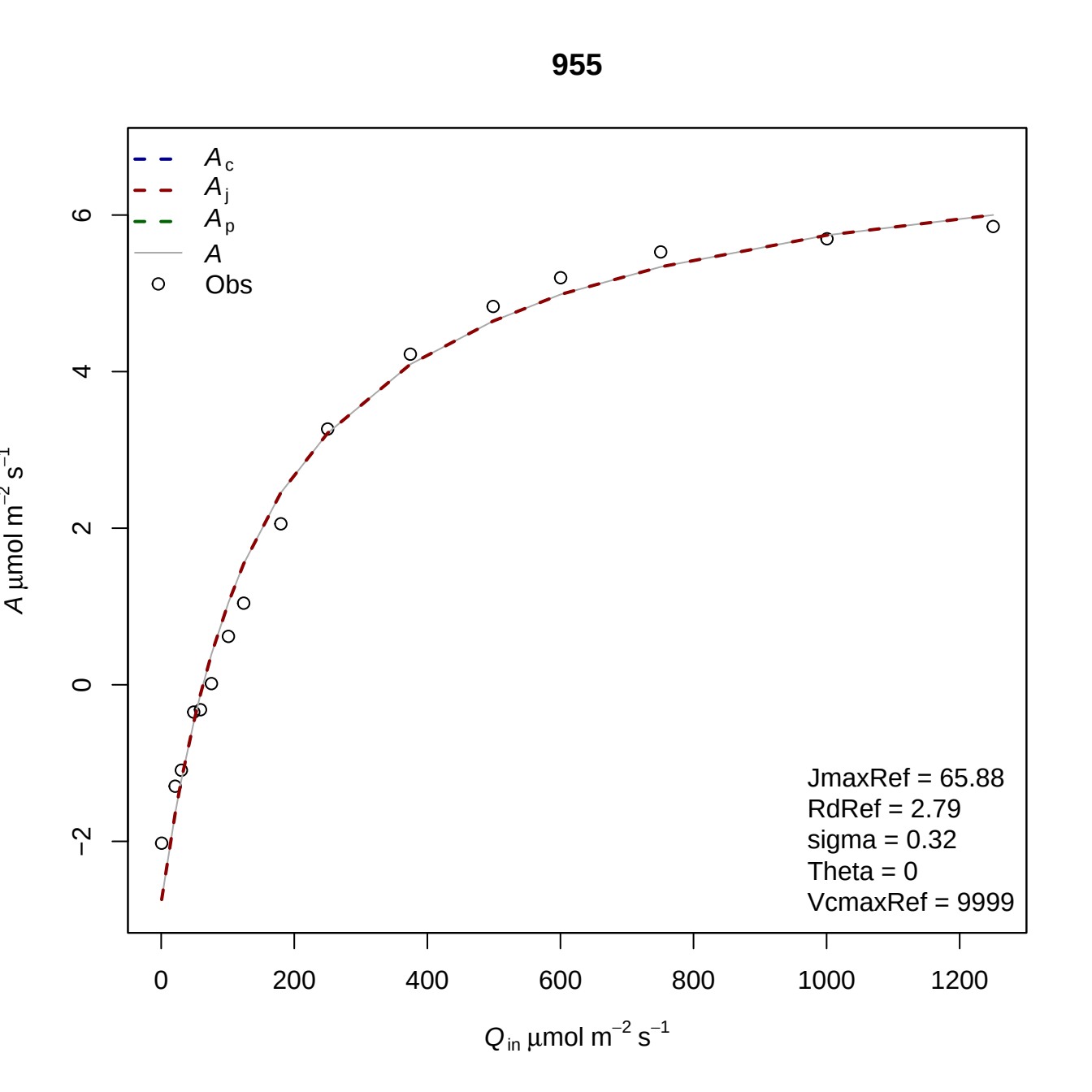

950
A_c
A_j
A_p
A
Obs
JmaxRef = 125.89
RdRef = 2.73
sigma = 0.25
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

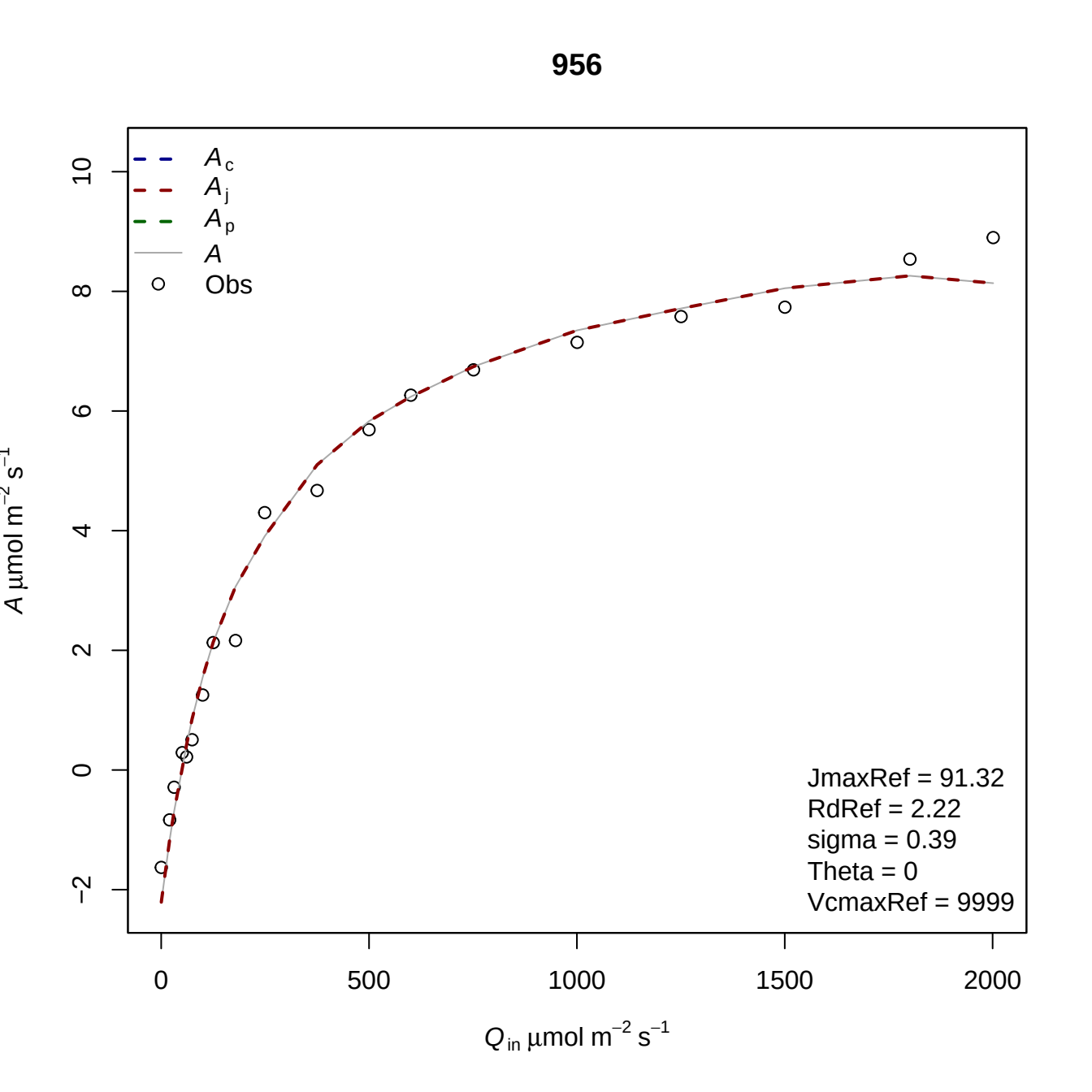

951
A_c
A_j
A_p
A
Obs
JmaxRef = 96.65
RdRef = 1.86
sigma = 0.19
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

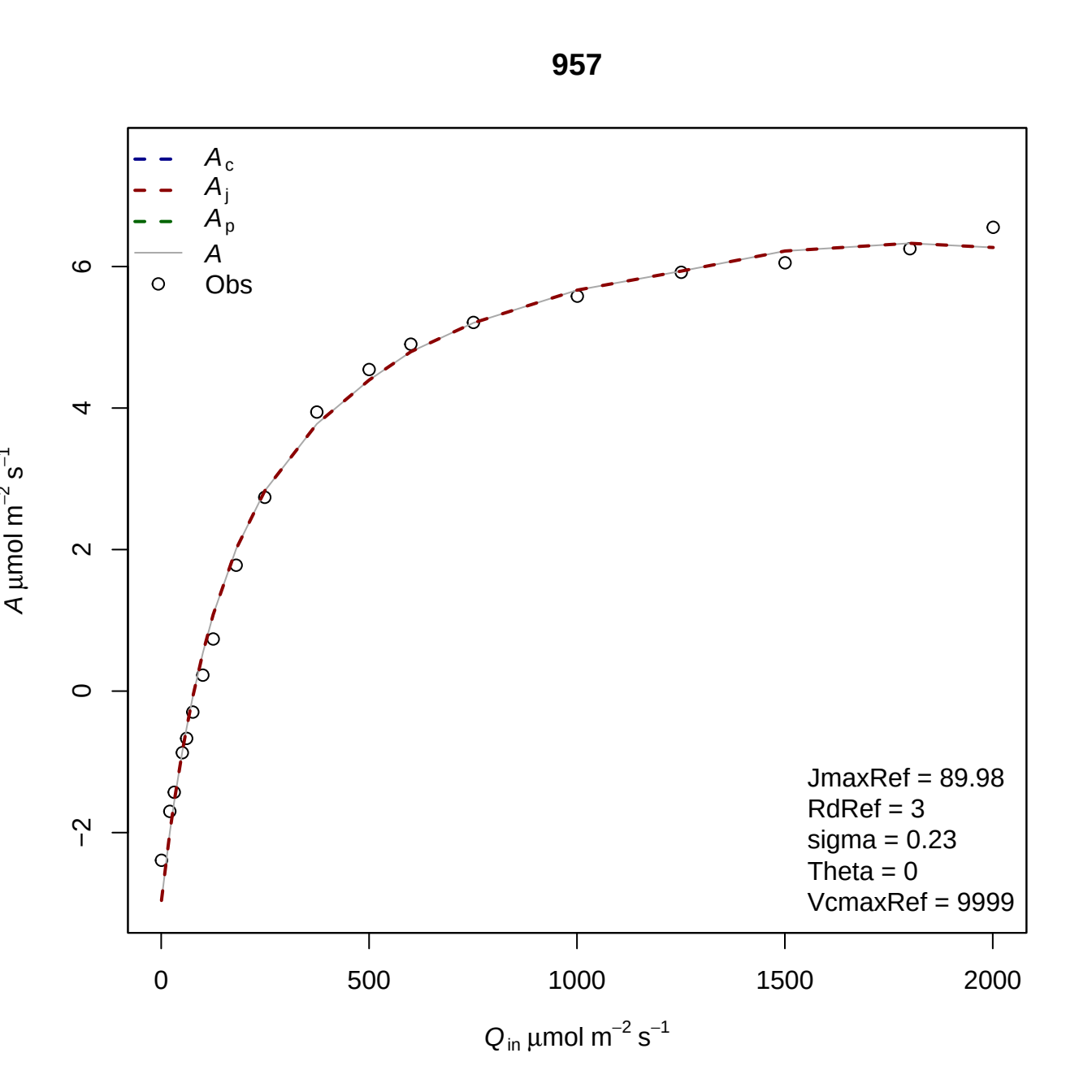

952
A_c
A_j
A_p
A
Obs
JmaxRef = 242.18
RdRef = 2.47
sigma = 0.18
Theta = 0
VcmaxRef = 170
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

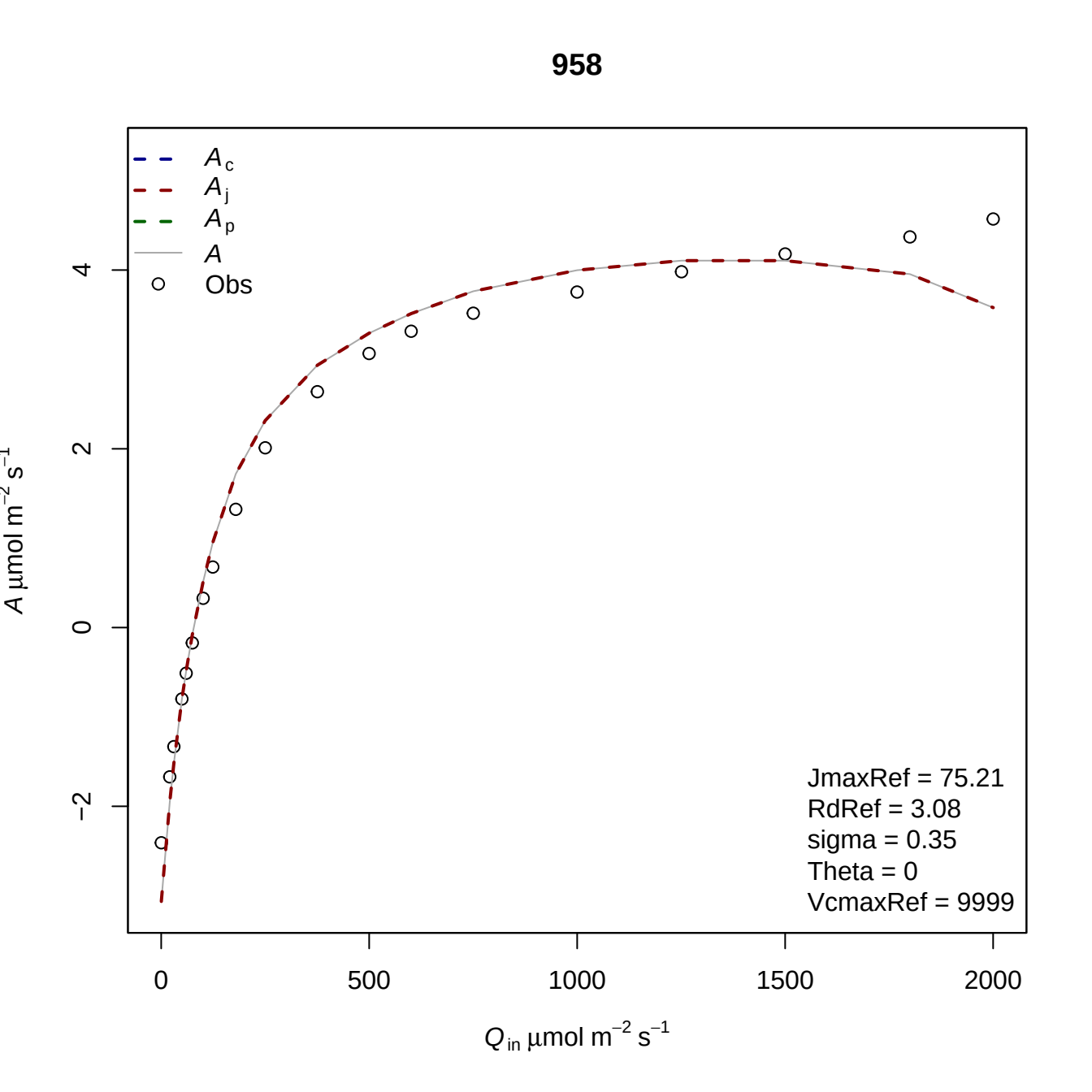

953
A_c
A_j
A_p
A
Obs
JmaxRef = 89.8
RdRef = 2.32
sigma = 0.26
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

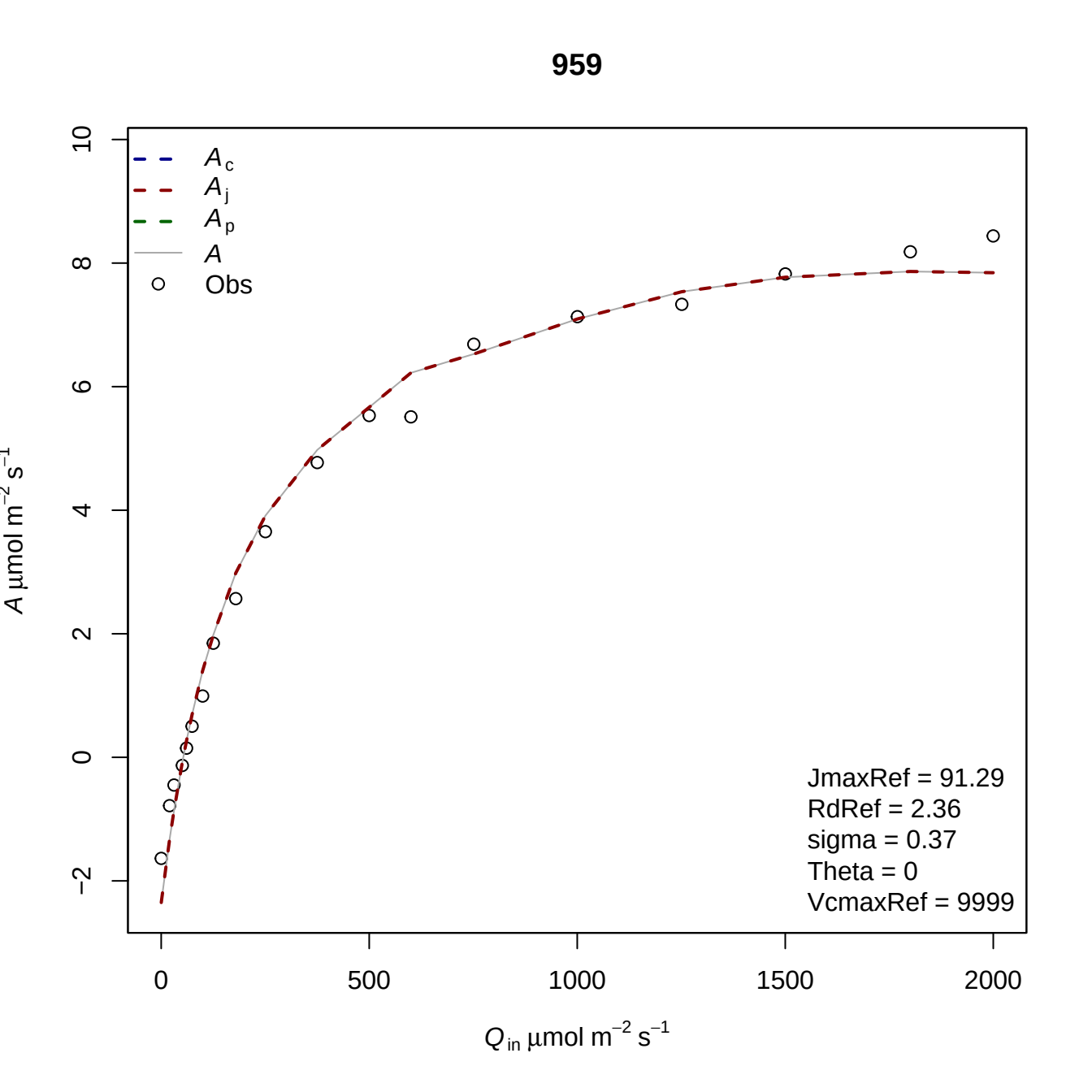

954
A_c
A_j
A_p
A
Obs
JmaxRef = 171.71
RdRef = 3.15
sigma = 0.21
Theta = 0
VcmaxRef = 140.11
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

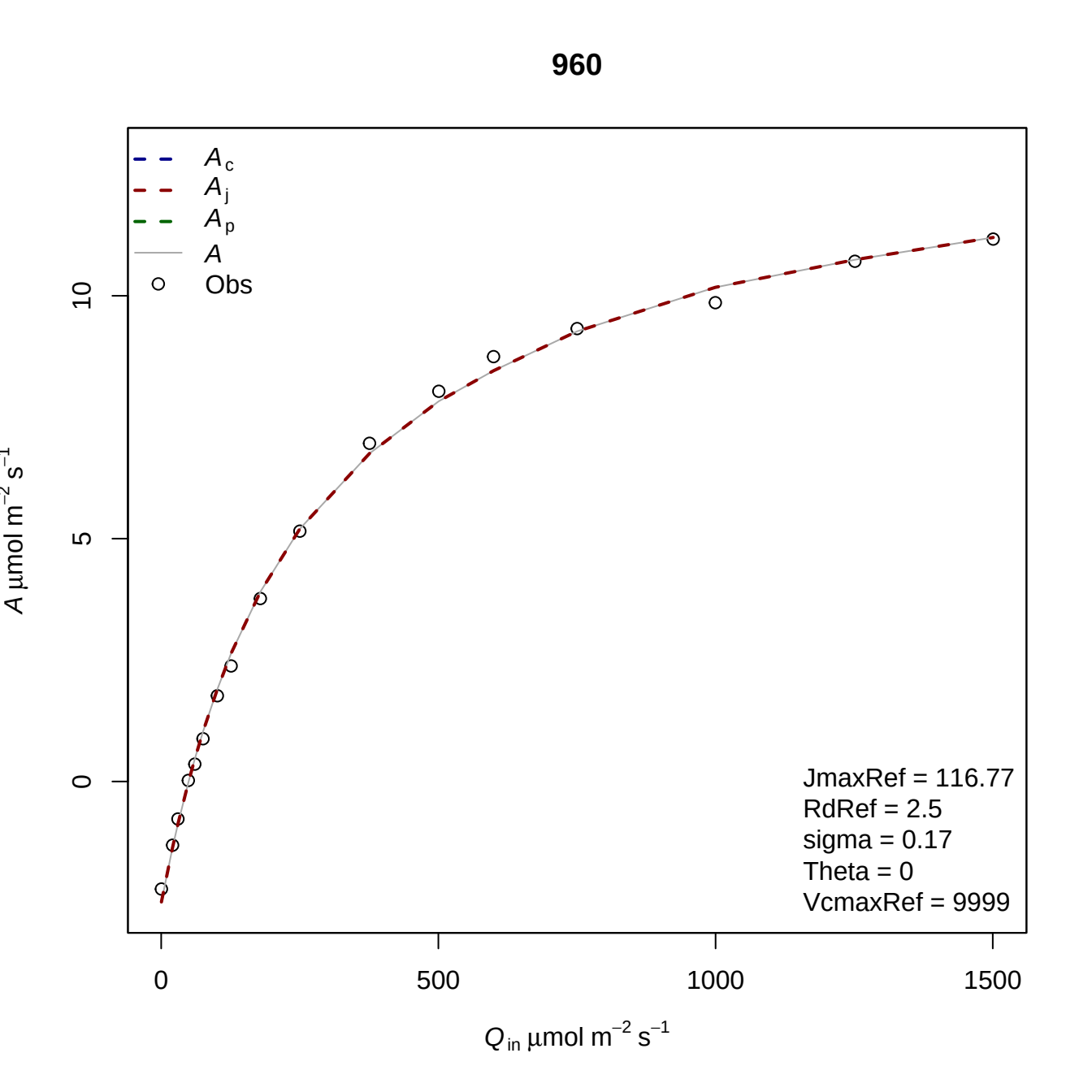

955
A_c
A_j
A_p
A
Obs
JmaxRef = 65.88
RdRef = 2.79
sigma = 0.32
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

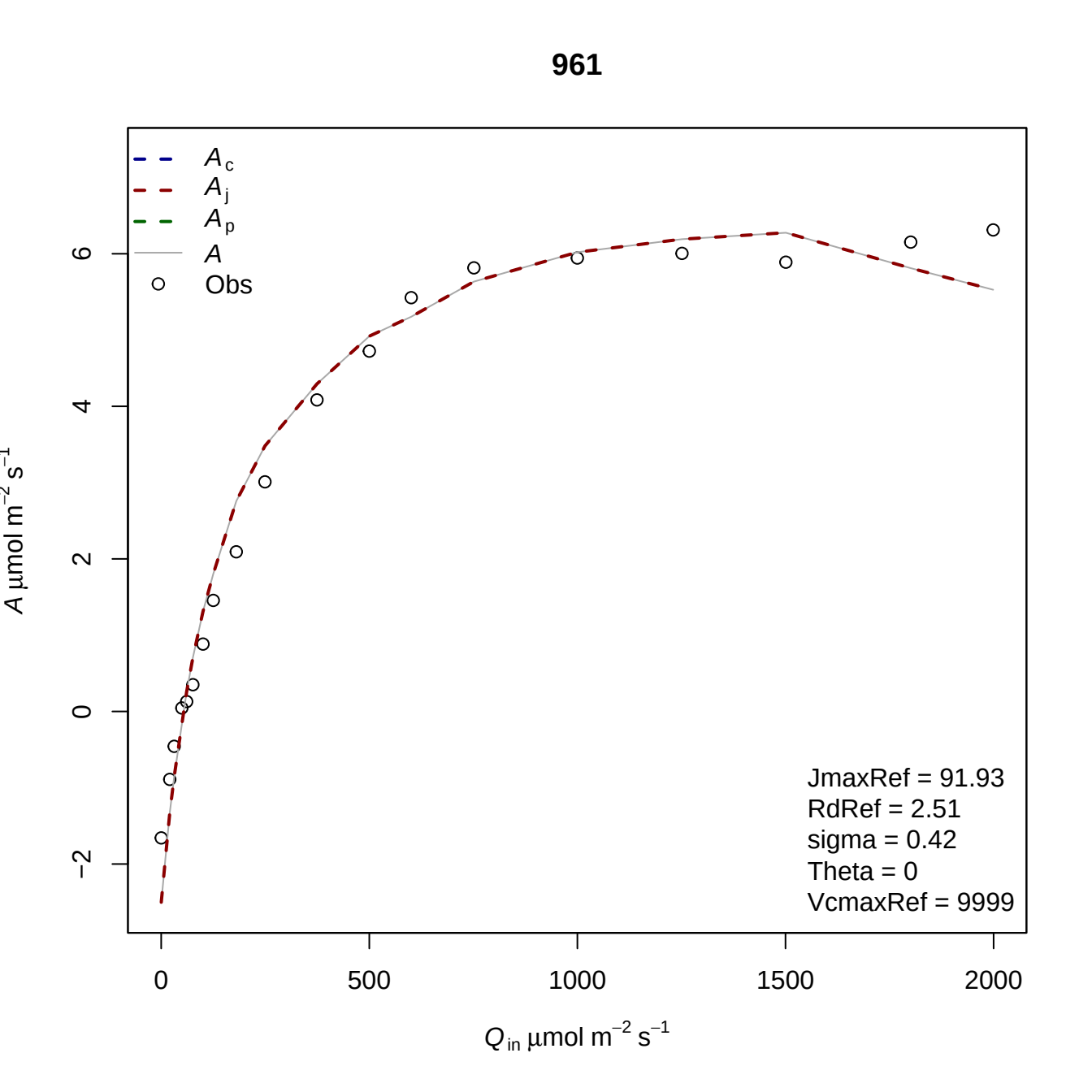

956
A_c
A_j
A_p
A
Obs
JmaxRef = 91.32
RdRef = 2.22
sigma = 0.39
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

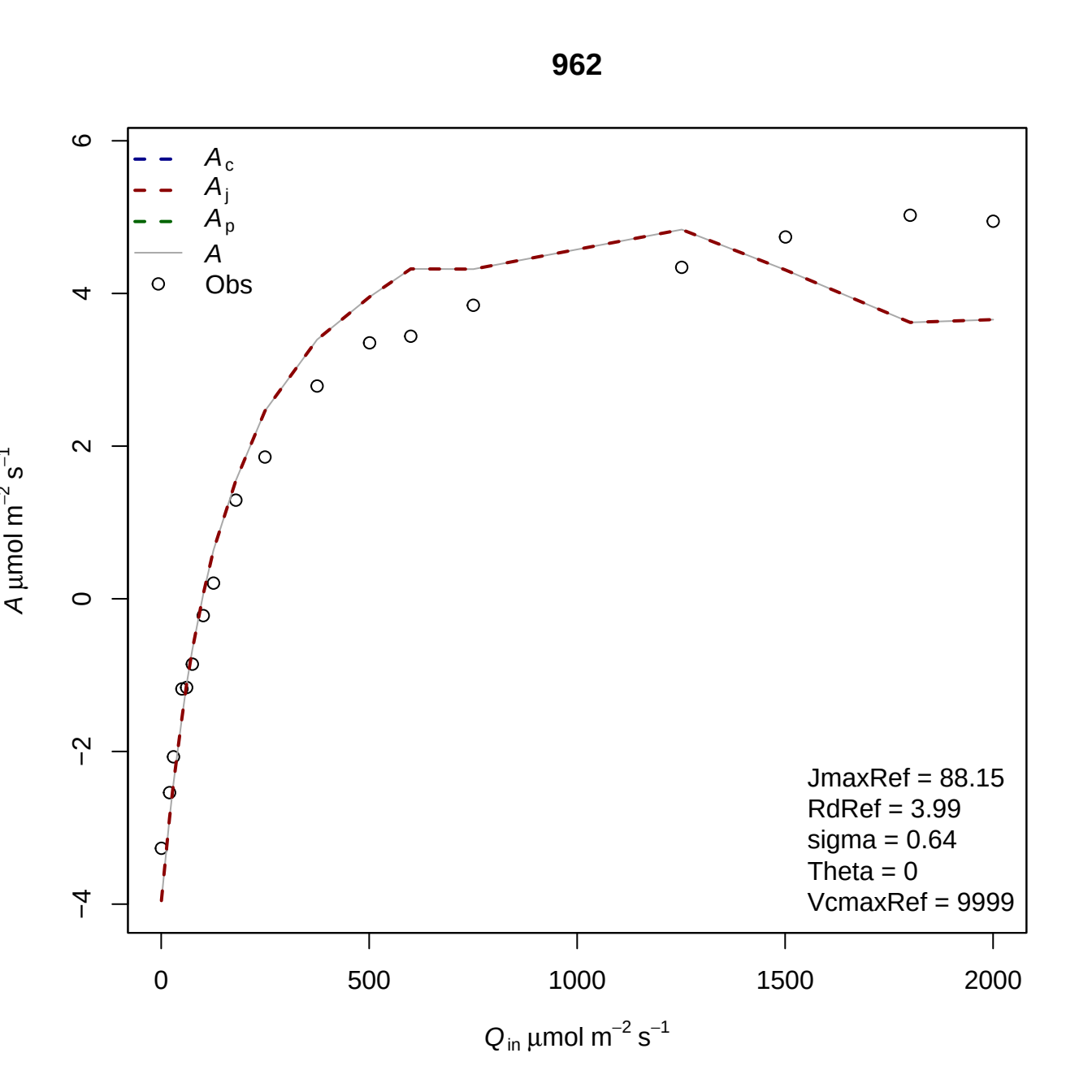

957
A_c
A_j
A_p
A
Obs
JmaxRef = 89.98
RdRef = 3
sigma = 0.23
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

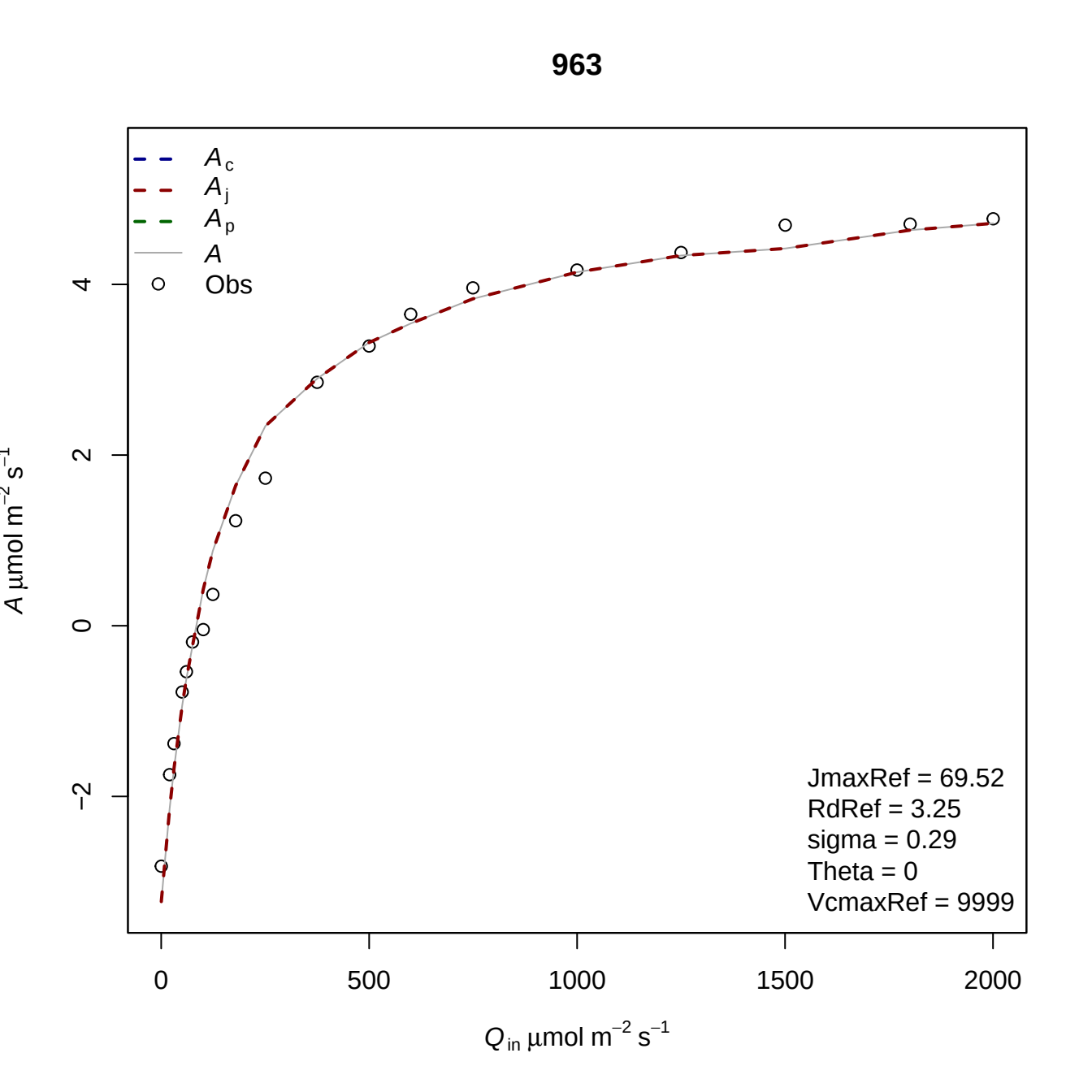

958
A_c
A_j
A_p
A
Obs
JmaxRef = 75.21
RdRef = 3.08
sigma = 0.35
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

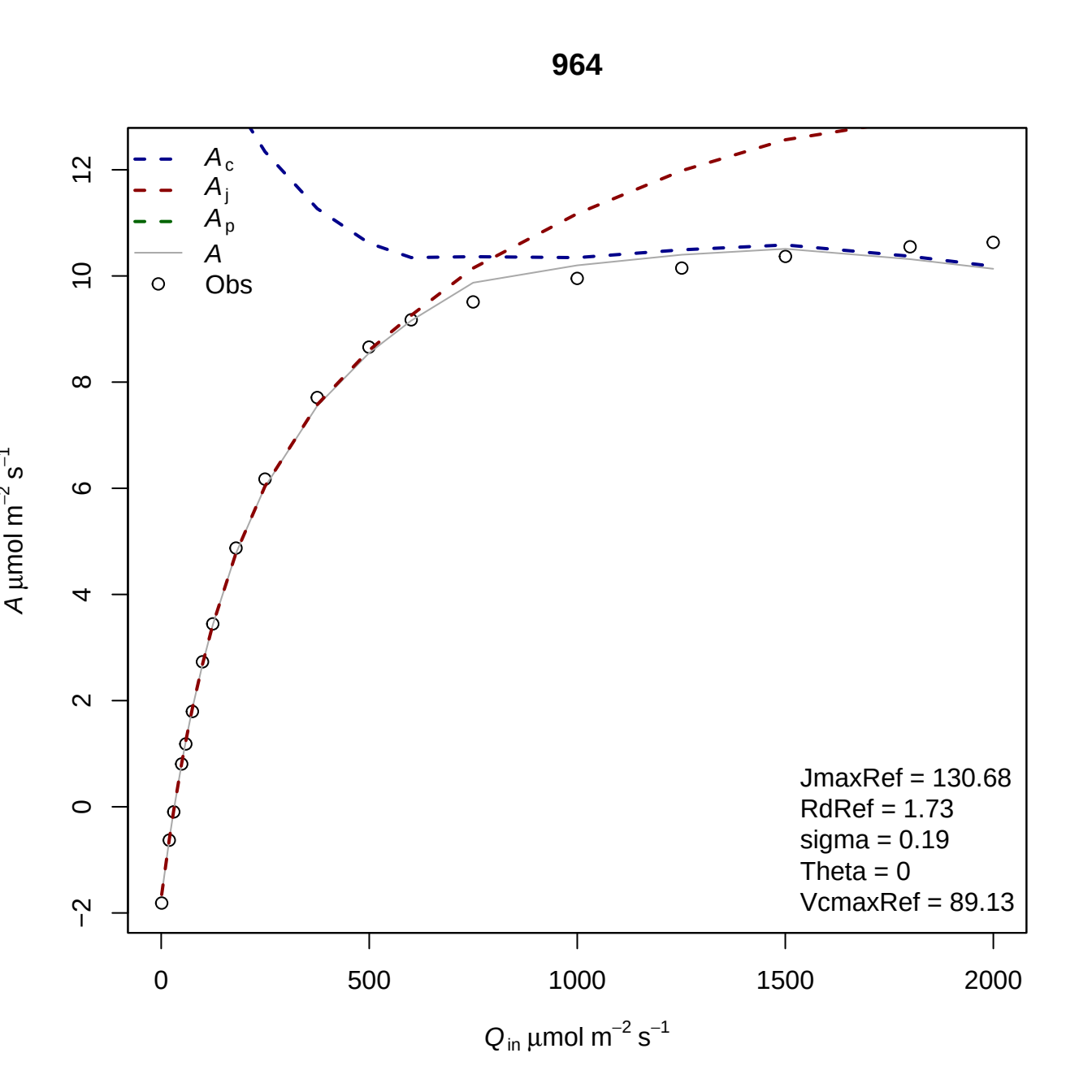

959
A_c
A_j
A_p
A
Obs
JmaxRef = 91.29
RdRef = 2.36
sigma = 0.37
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

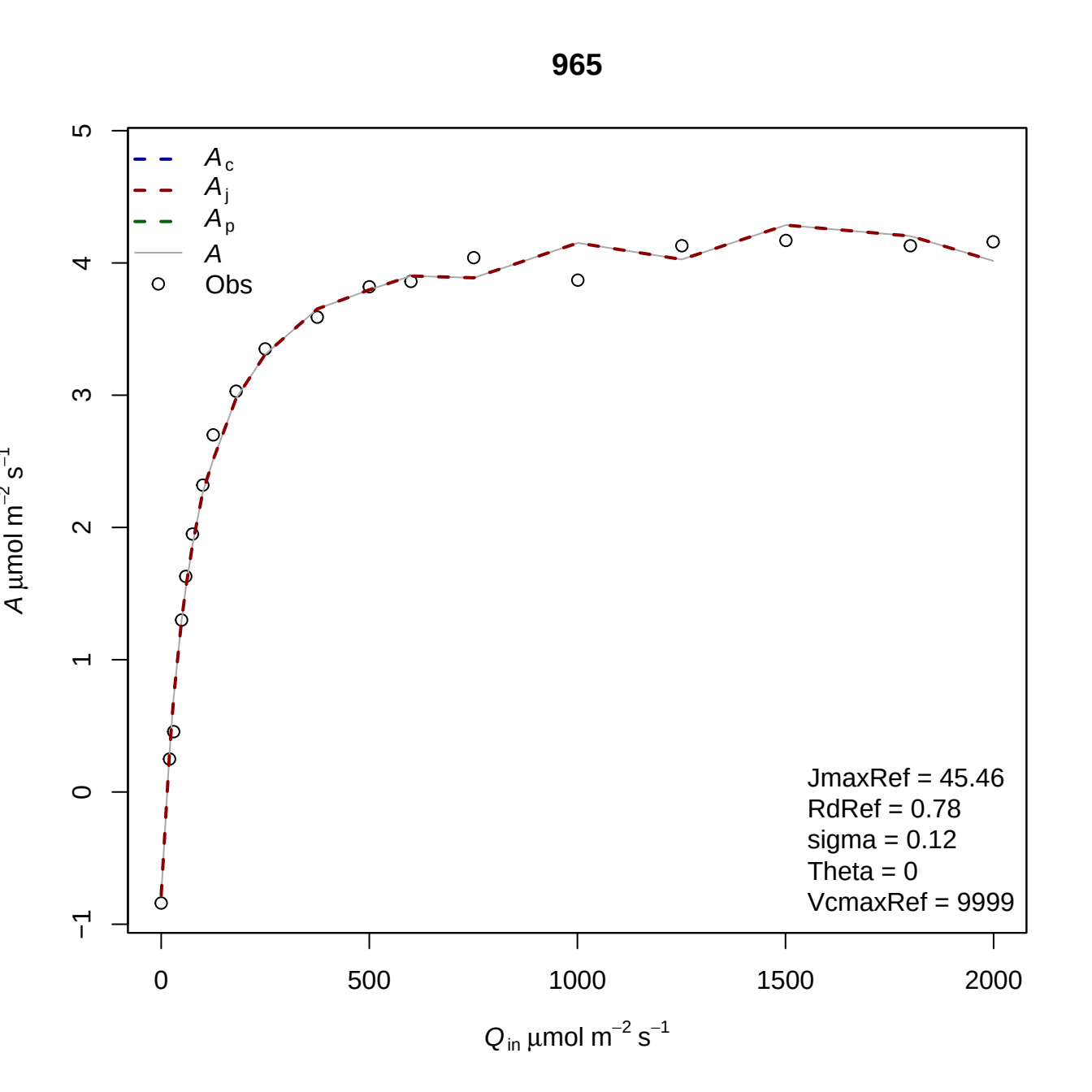

960
A_c
A_j
A_p
A
Obs
JmaxRef = 116.77
RdRef = 2.5
sigma = 0.17
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

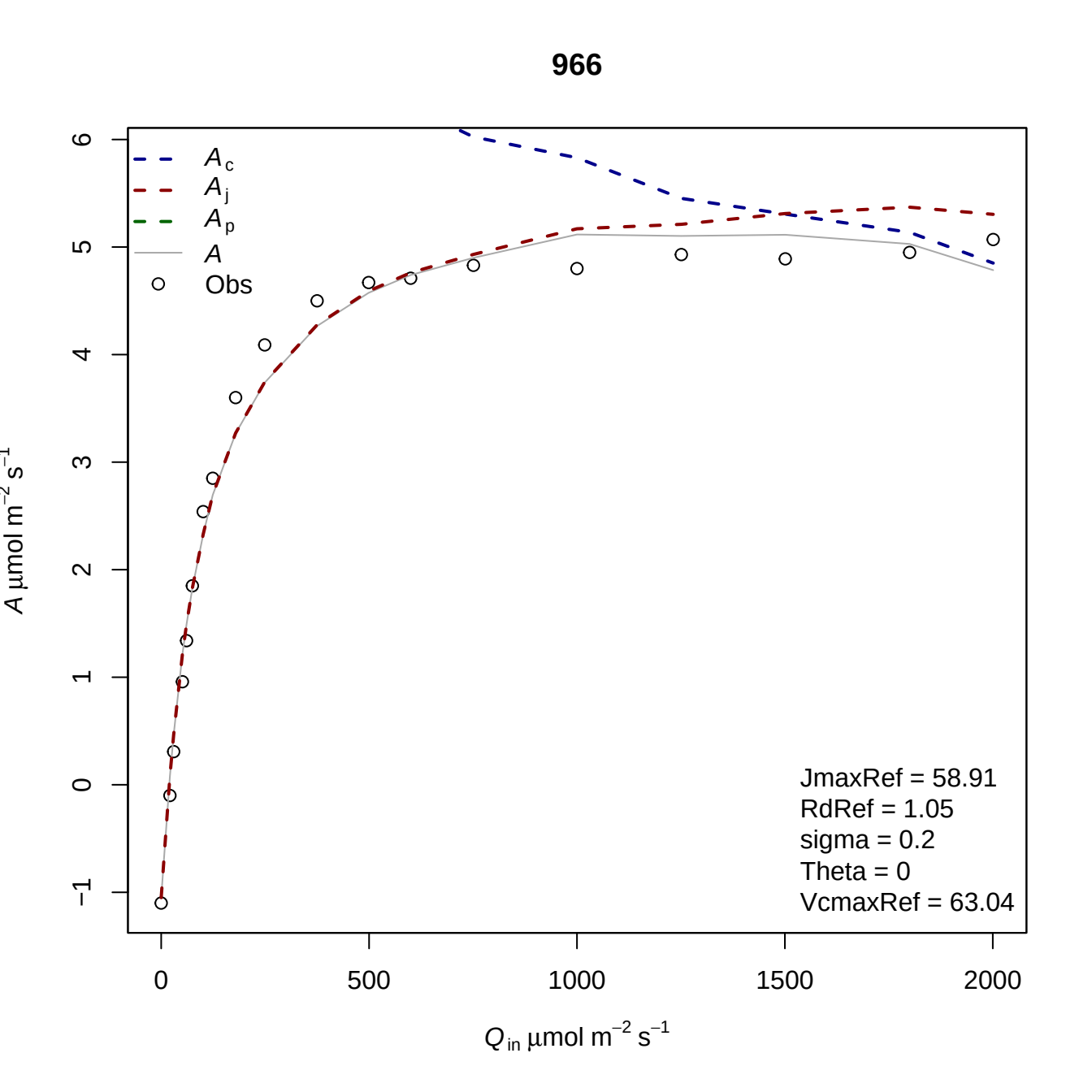

961
A_c
A_j
A_p
A
Obs
JmaxRef = 91.93
RdRef = 2.51
sigma = 0.42
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

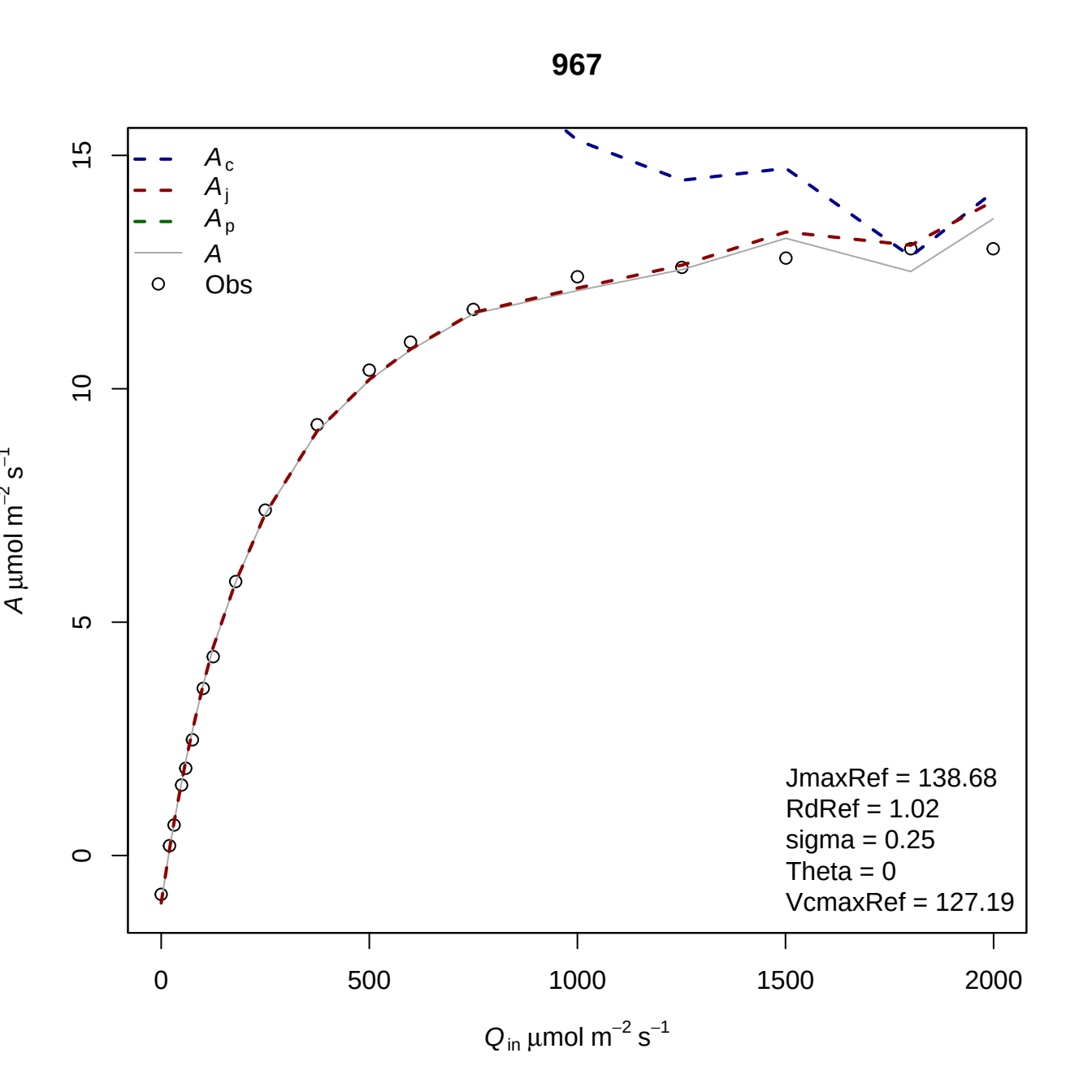

962
A_c
A_j
A_p
A
Obs
JmaxRef = 88.15
RdRef = 3.99
sigma = 0.64
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

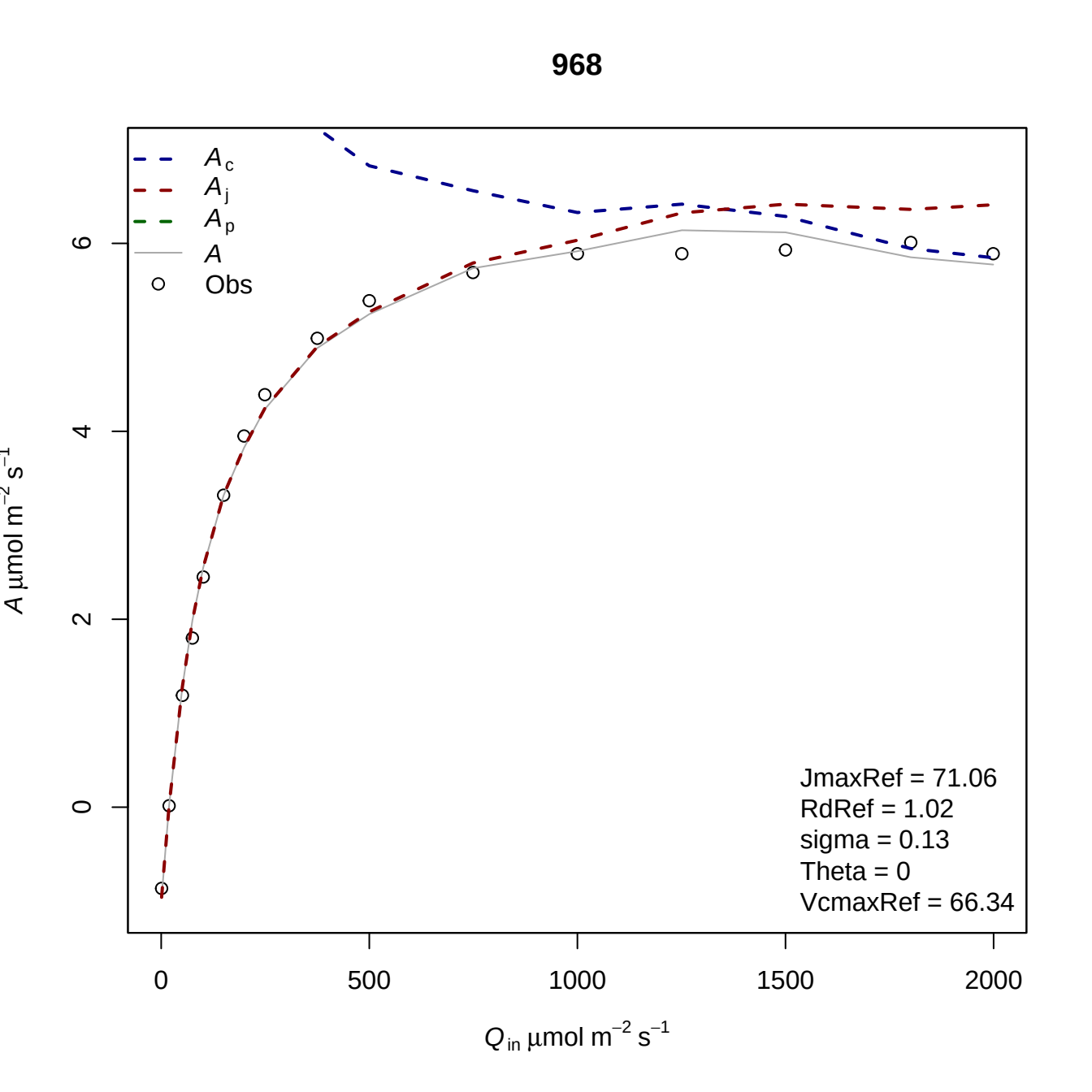

963
A_c
A_j
A_p
A
Obs
JmaxRef = 69.52
RdRef = 3.25
sigma = 0.29
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

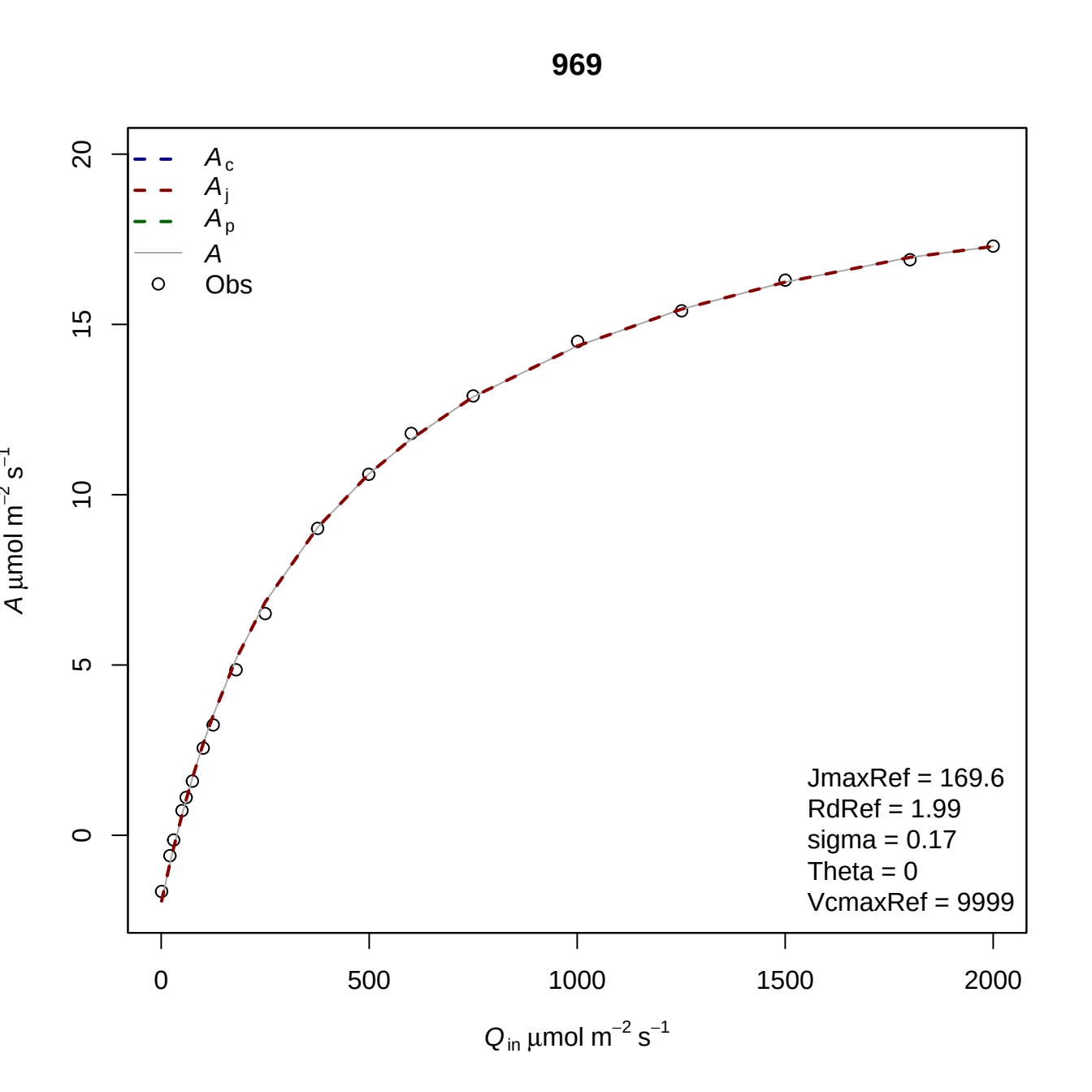

964
A_c
A_j
A_p
A
Obs
JmaxRef = 130.68
RdRef = 1.73
sigma = 0.19
Theta = 0
VcmaxRef = 89.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

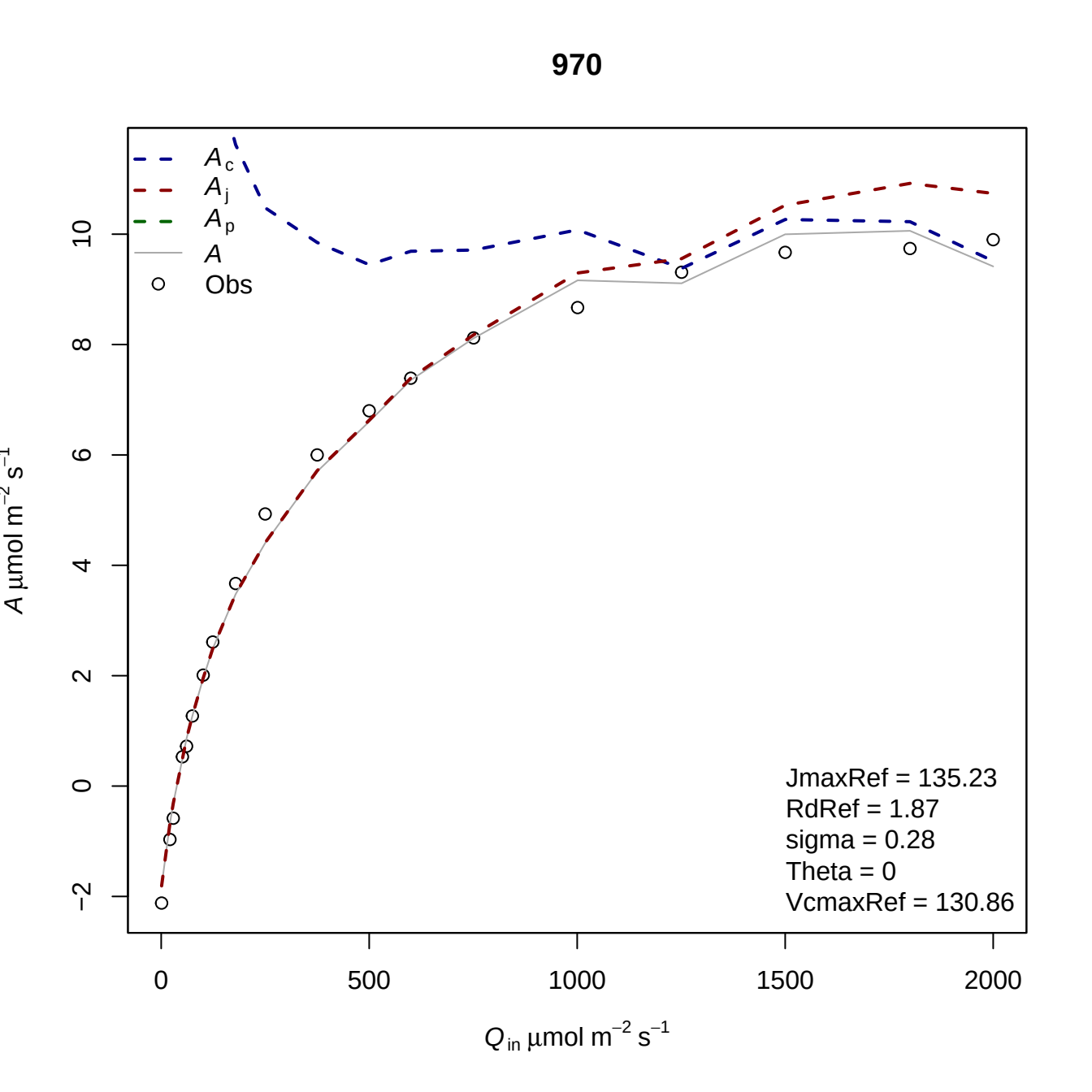

965
A_c
A_j
A_p
A
Obs
JmaxRef = 45.46
RdRef = 0.78
sigma = 0.12
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

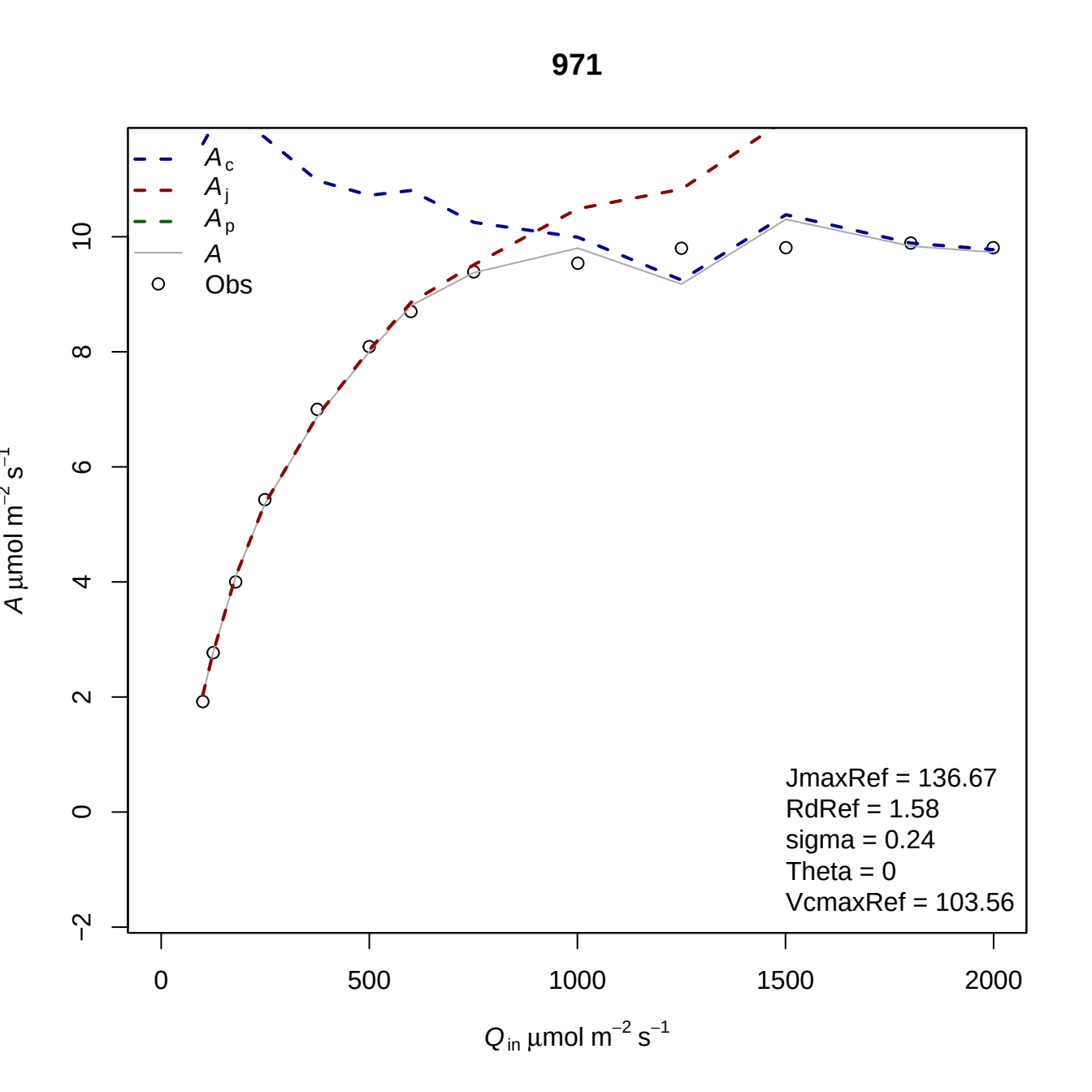

966
A_c
A_j
A_p
A
Obs
JmaxRef = 58.91
RdRef = 1.05
sigma = 0.2
Theta = 0
VcmaxRef = 63.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

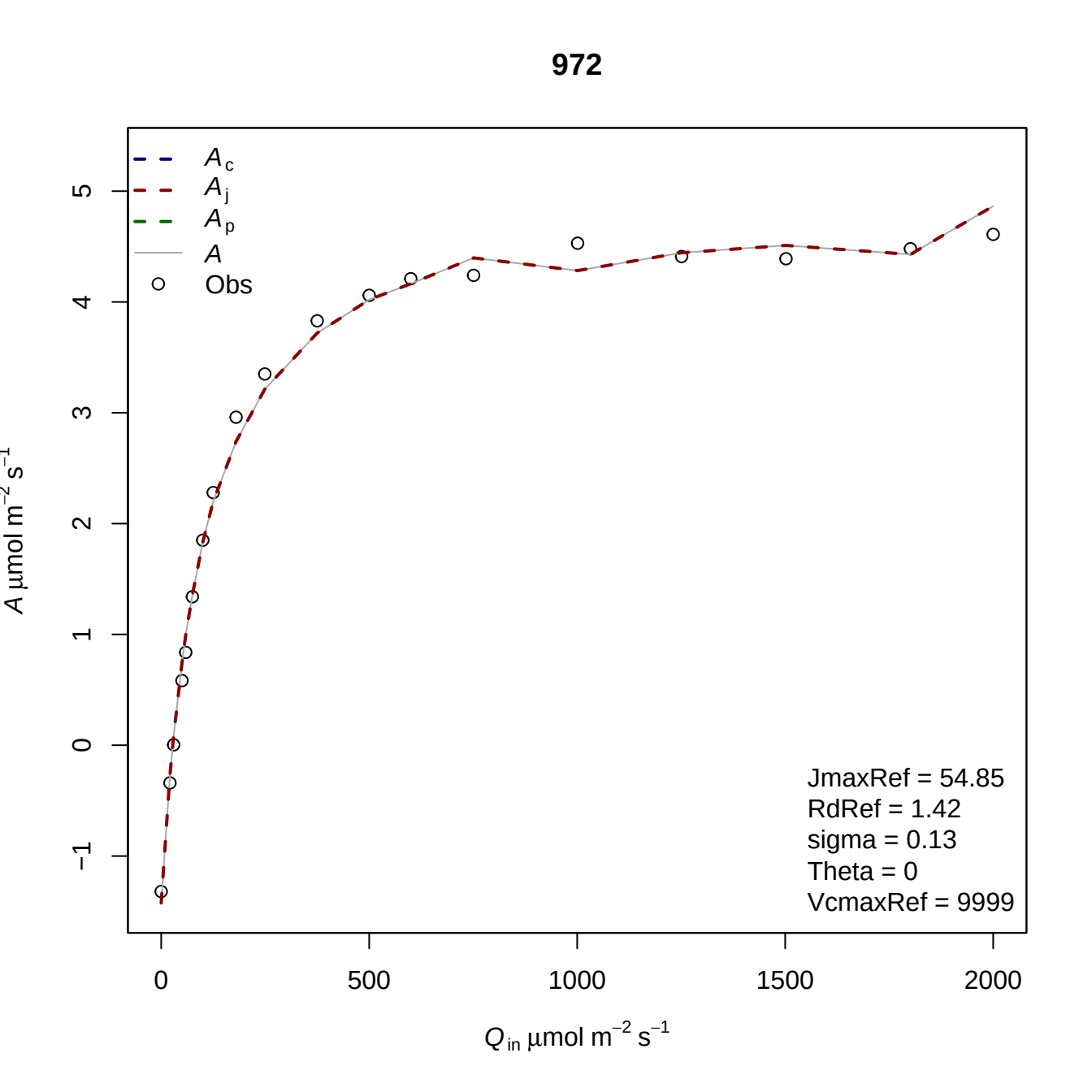

967
A_c
A_j
A_p
A
Obs
JmaxRef = 138.68
RdRef = 1.02
sigma = 0.25
Theta = 0
VcmaxRef = 127.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

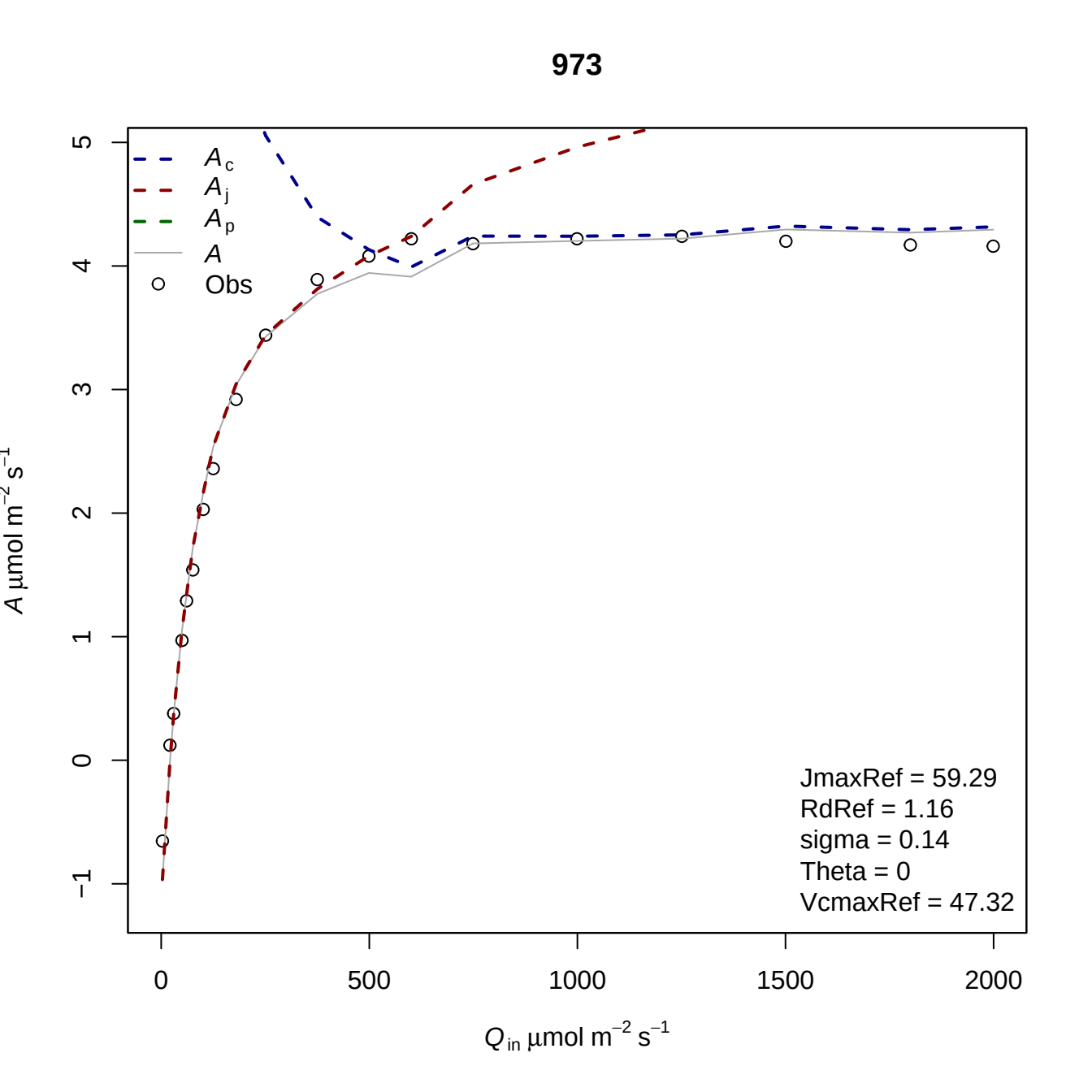

968
A_c
A_j
A_p
A
Obs
JmaxRef = 71.06
RdRef = 1.02
sigma = 0.13
Theta = 0
VcmaxRef = 66.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

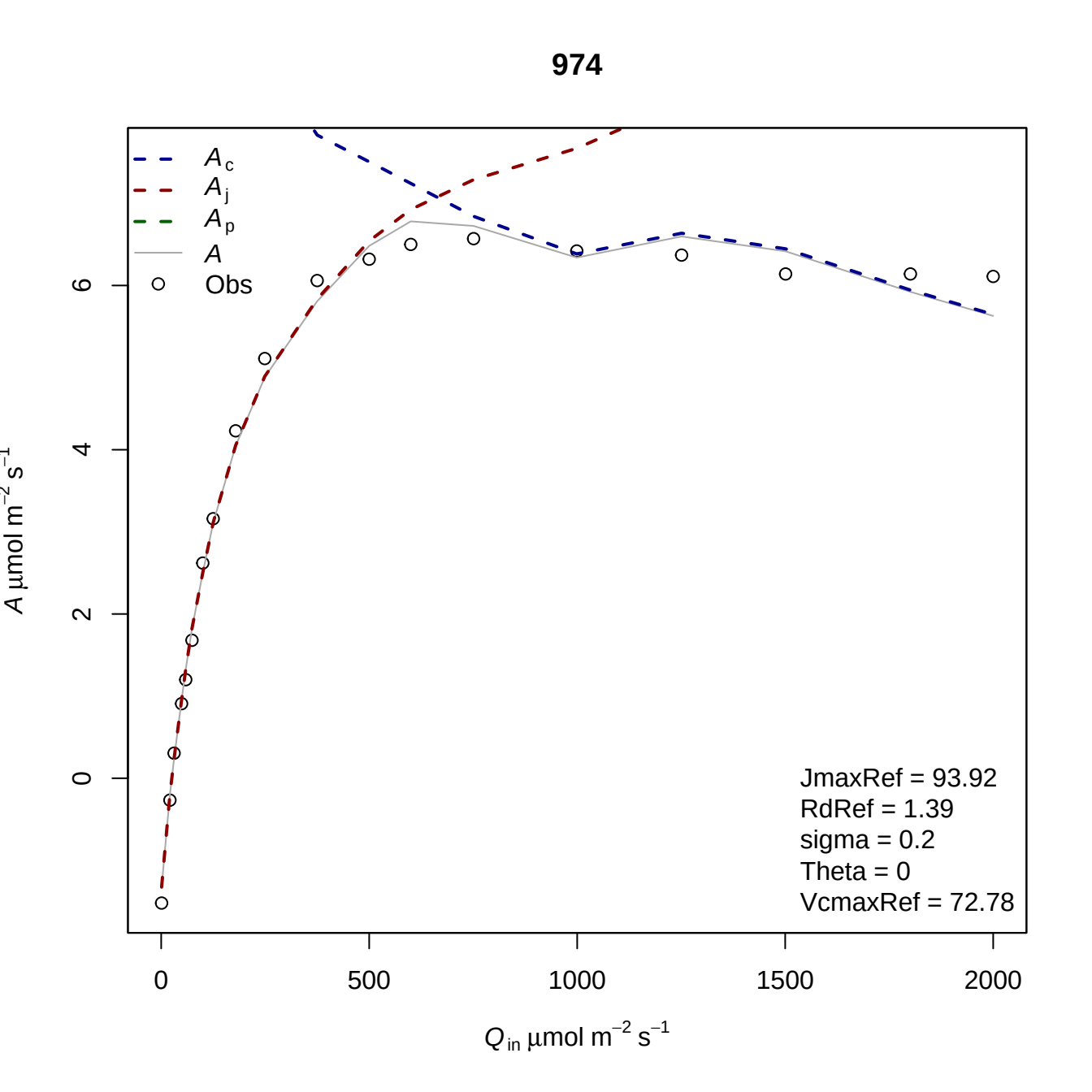

969
A_c
A_j
A_p
A
Obs
JmaxRef = 169.6
RdRef = 1.99
sigma = 0.17
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

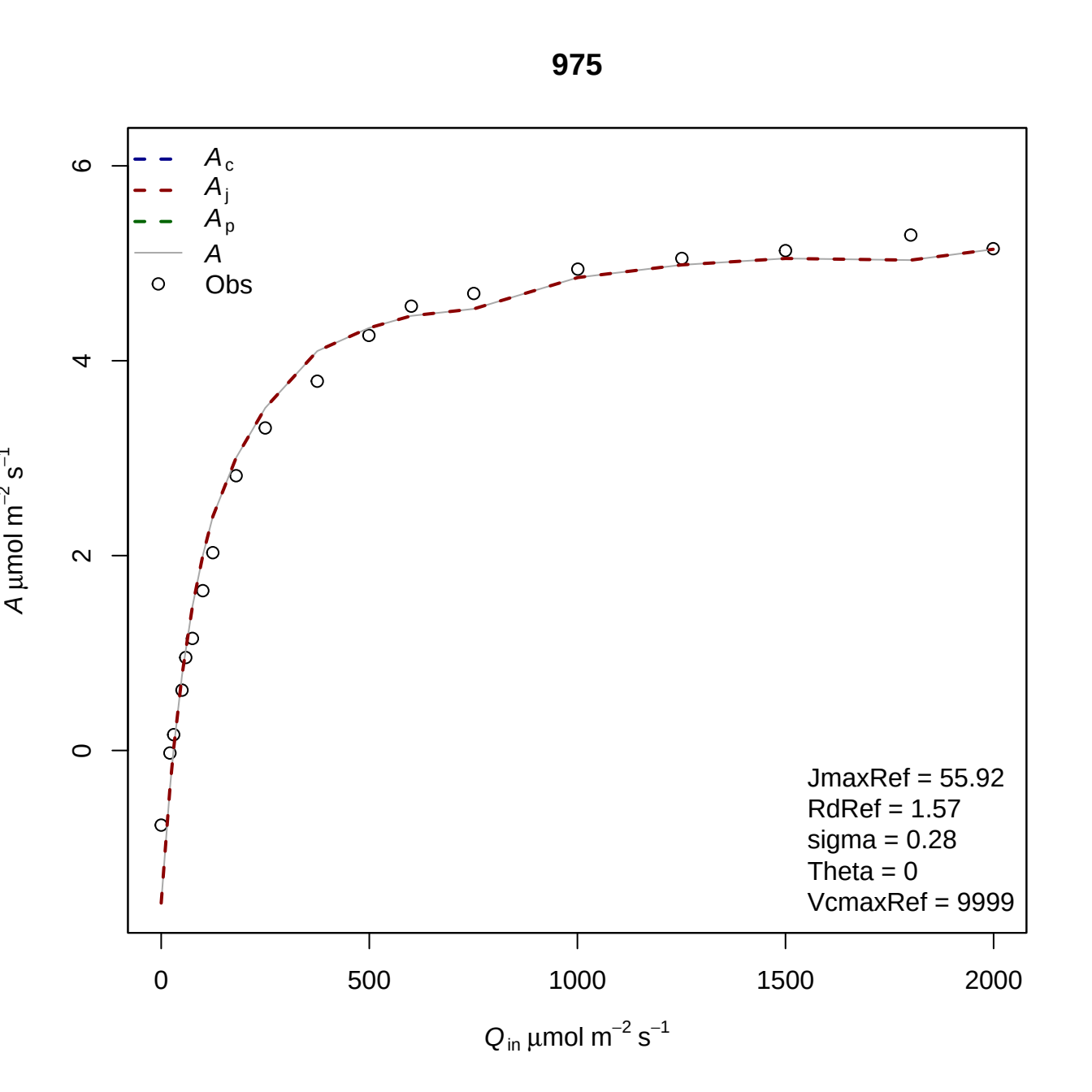

970
A_c
A_j
A_p
A
Obs
JmaxRef = 135.23
RdRef = 1.87
sigma = 0.28
Theta = 0
VcmaxRef = 130.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

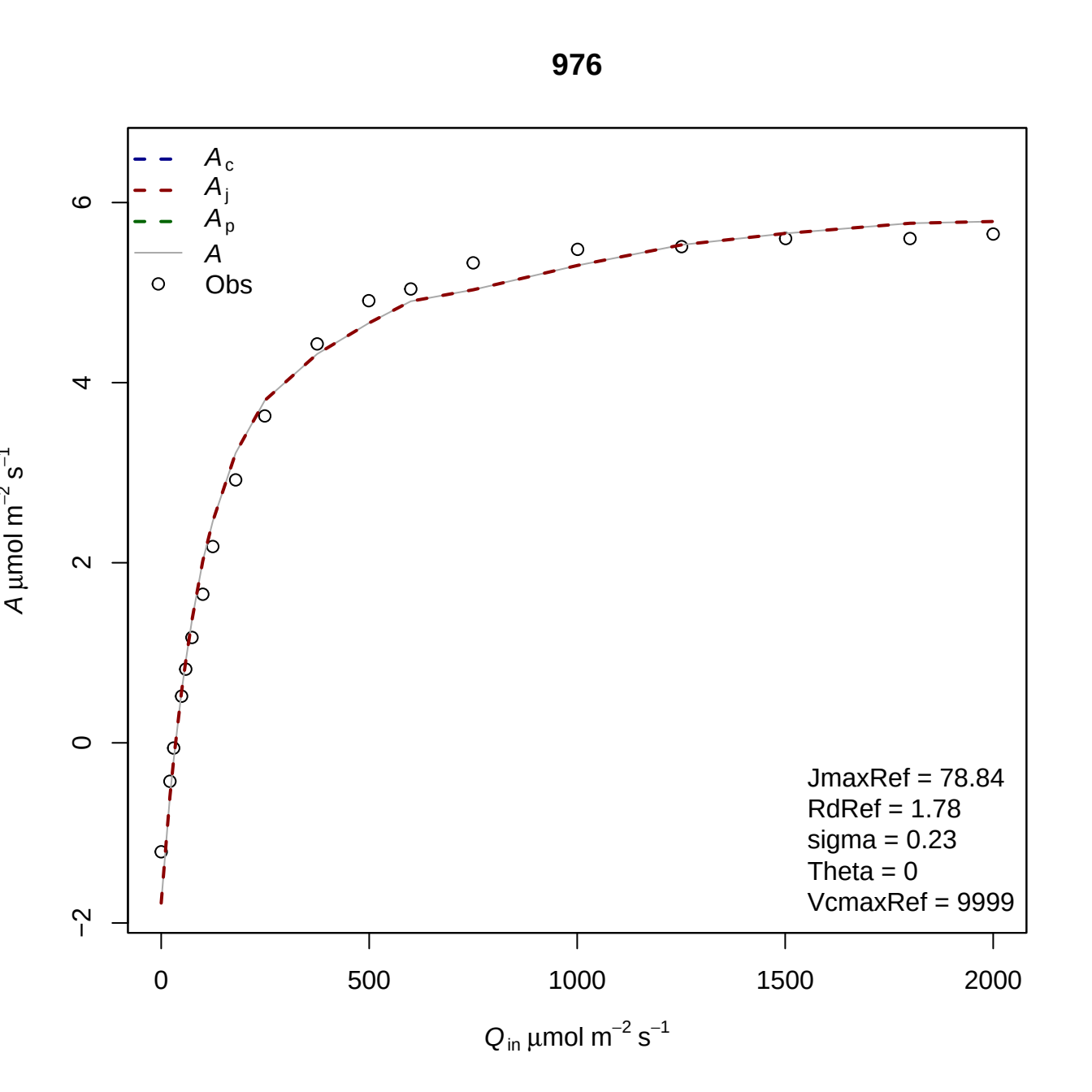

971
A_c
A_j
A_p
A
Obs
JmaxRef = 136.67
RdRef = 1.58
sigma = 0.24
Theta = 0
VcmaxRef = 103.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

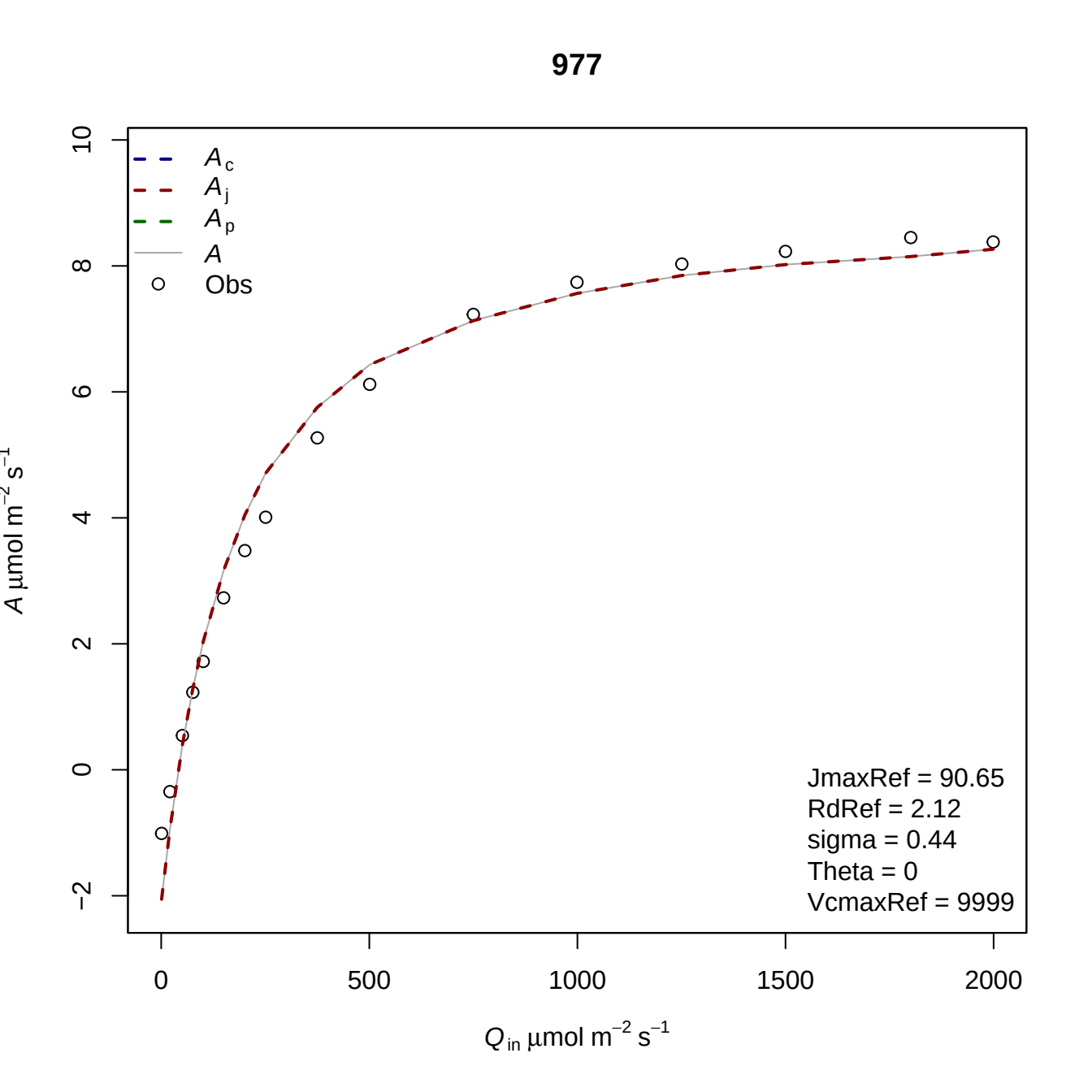

972
A_c
A_j
A_p
A
Obs
JmaxRef = 54.85
RdRef = 1.42
sigma = 0.13
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

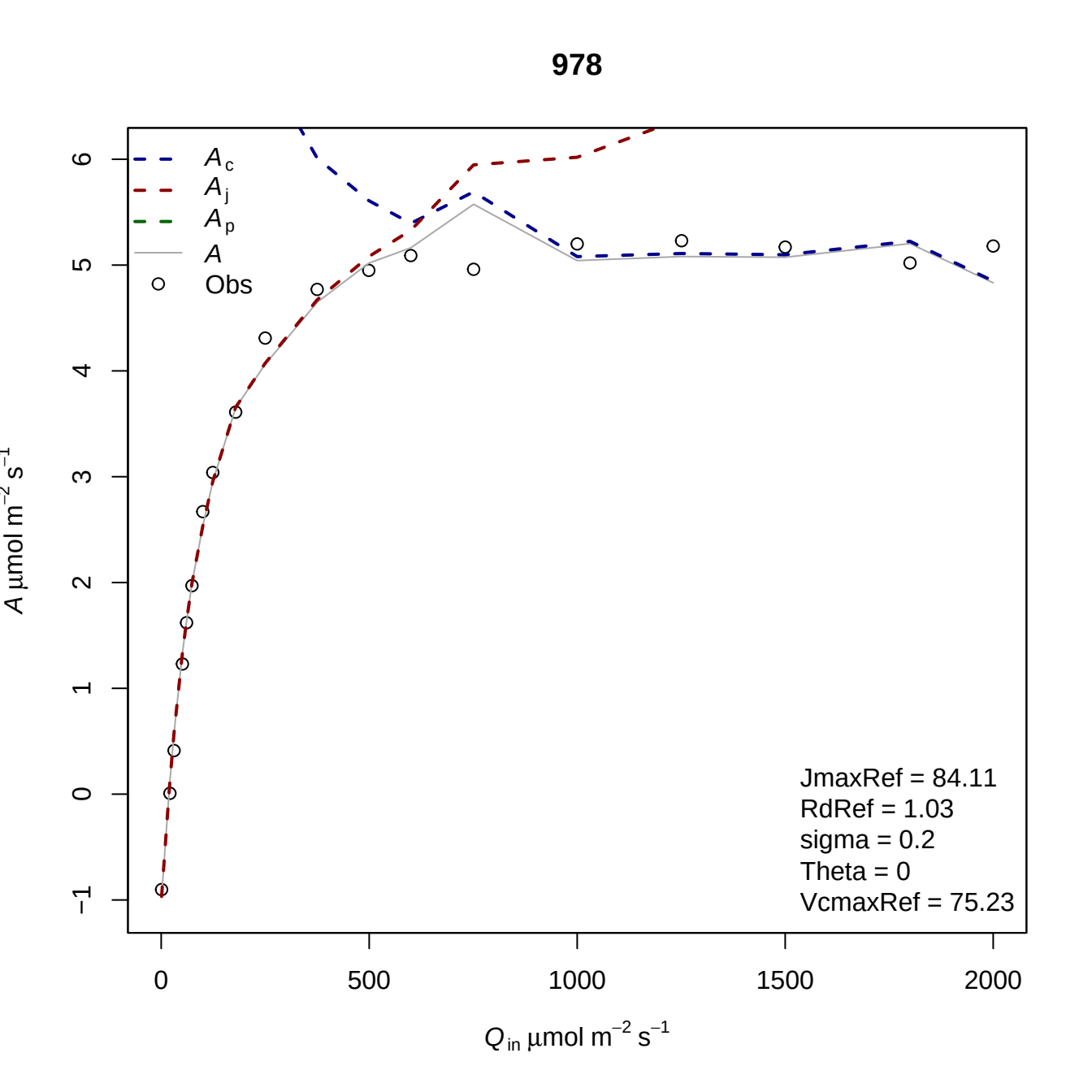

973
A_c
A_j
A_p
A
Obs
JmaxRef = 59.29
RdRef = 1.16
sigma = 0.14
Theta = 0
VcmaxRef = 47.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

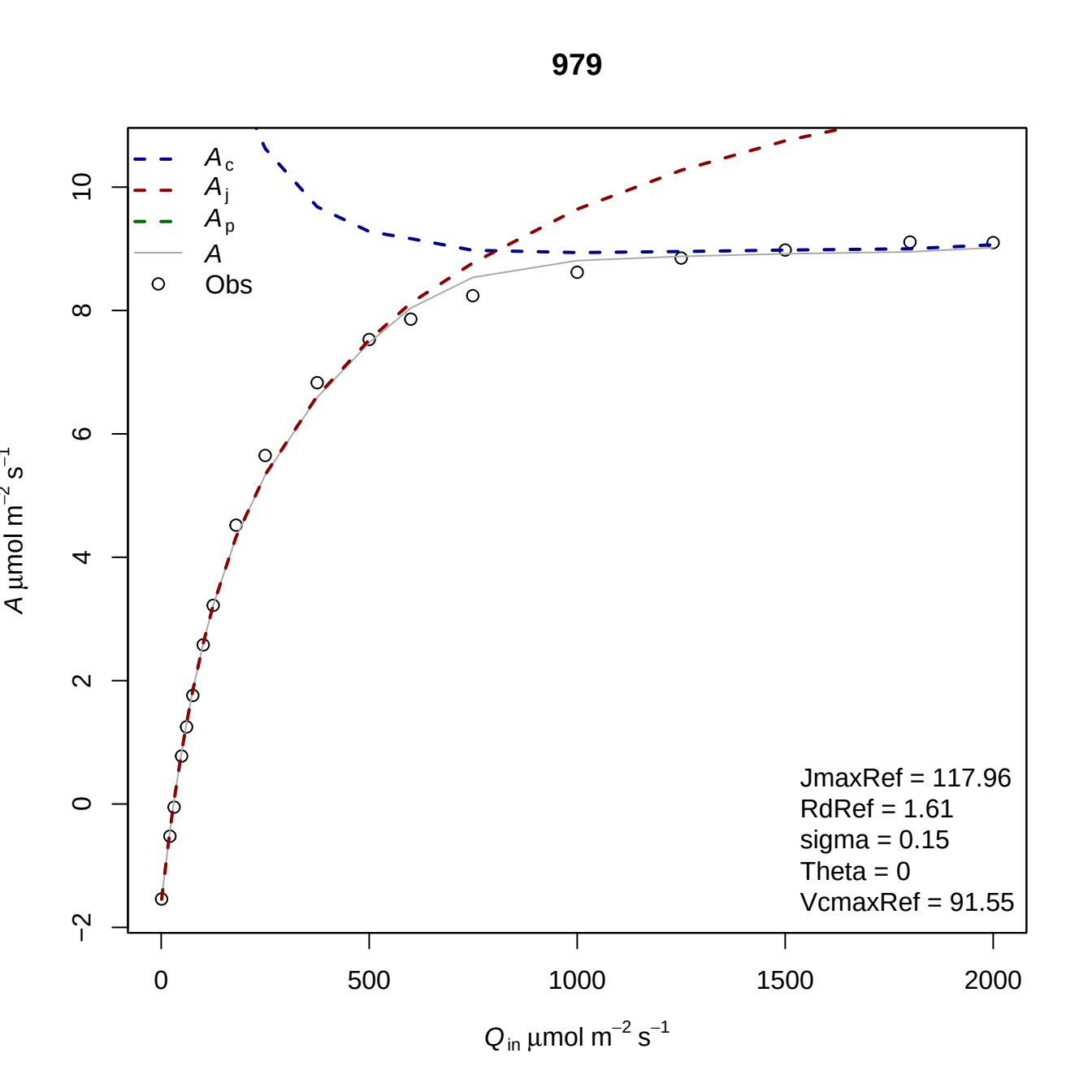

974
A_c
A_j
A_p
A
Obs
JmaxRef = 93.92
RdRef = 1.39
sigma = 0.2
Theta = 0
VcmaxRef = 72.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

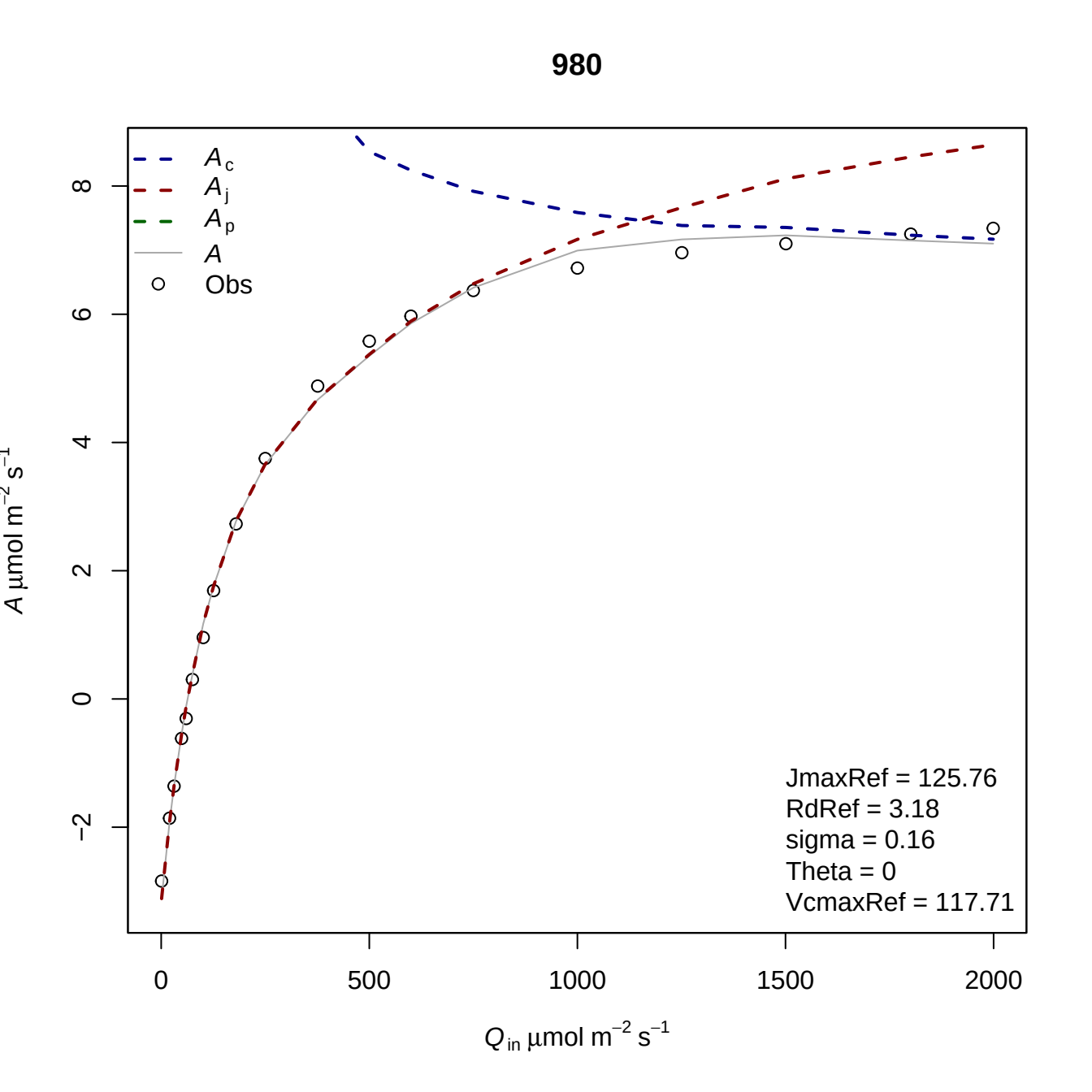

975
A_c
A_j
A_p
A
Obs
JmaxRef = 55.92
RdRef = 1.57
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

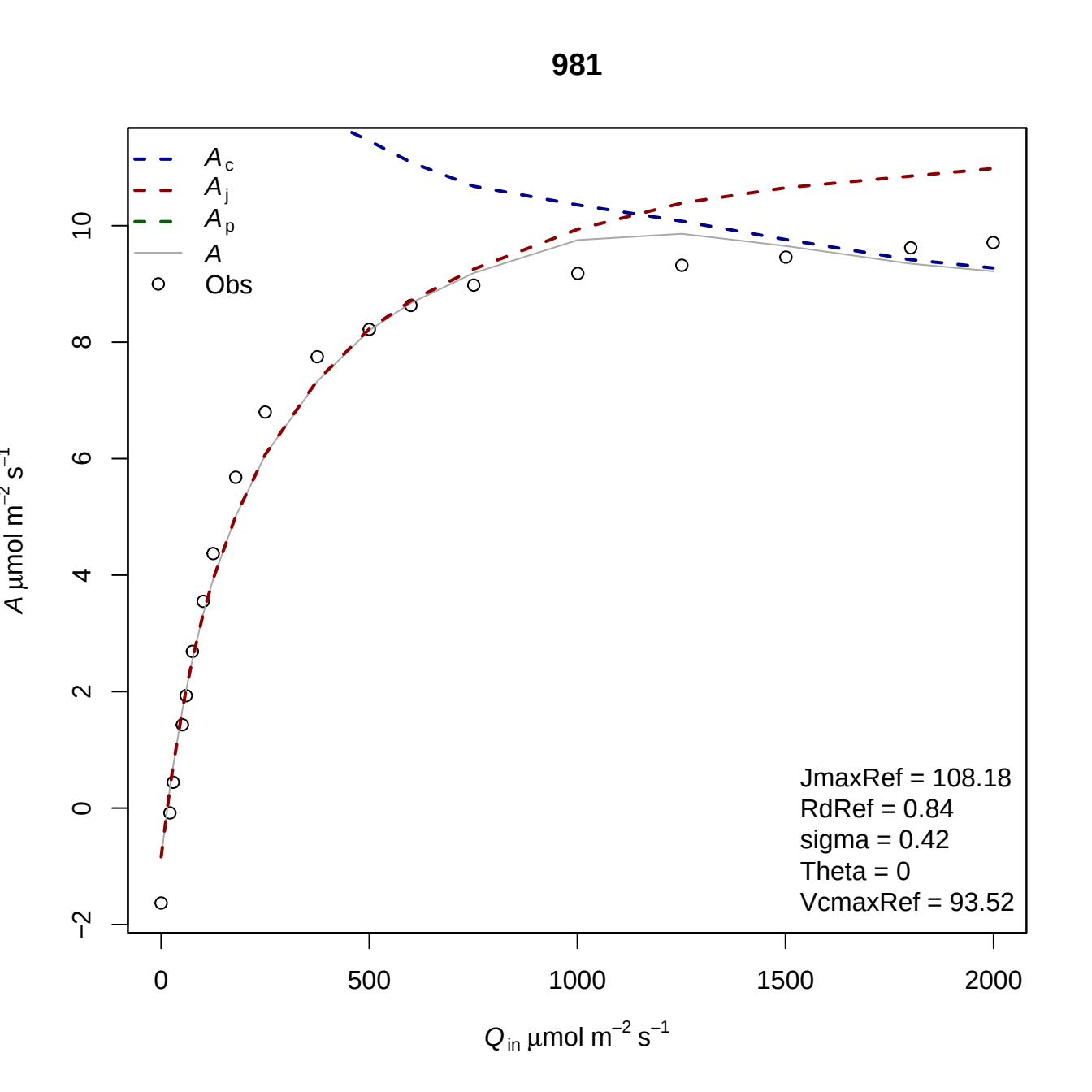

976
A_c
A_j
A_p
A
Obs
JmaxRef = 78.84
RdRef = 1.78
sigma = 0.23
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

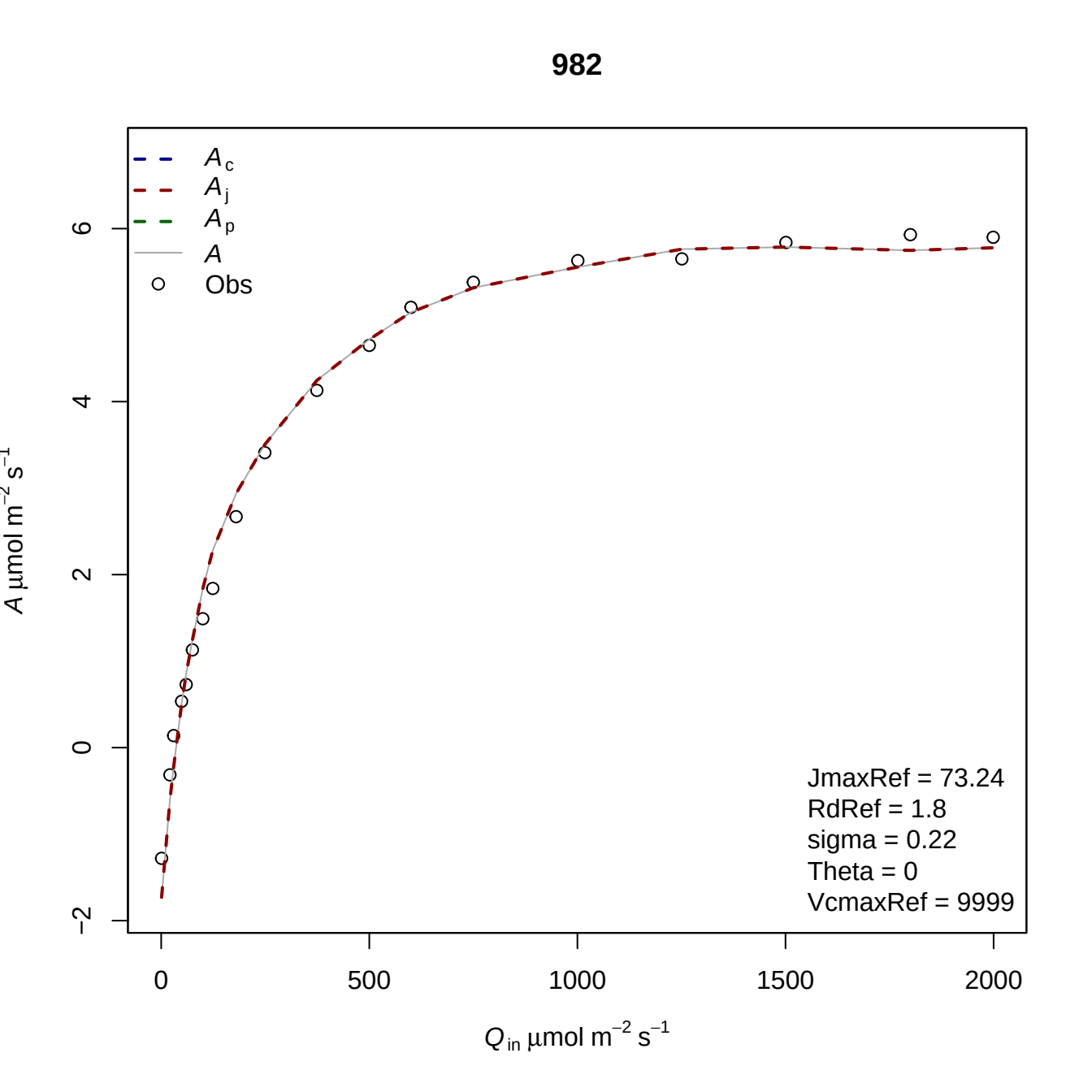

977
A_c
A_j
A_p
A
Obs
JmaxRef = 90.65
RdRef = 2.12
sigma = 0.44
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

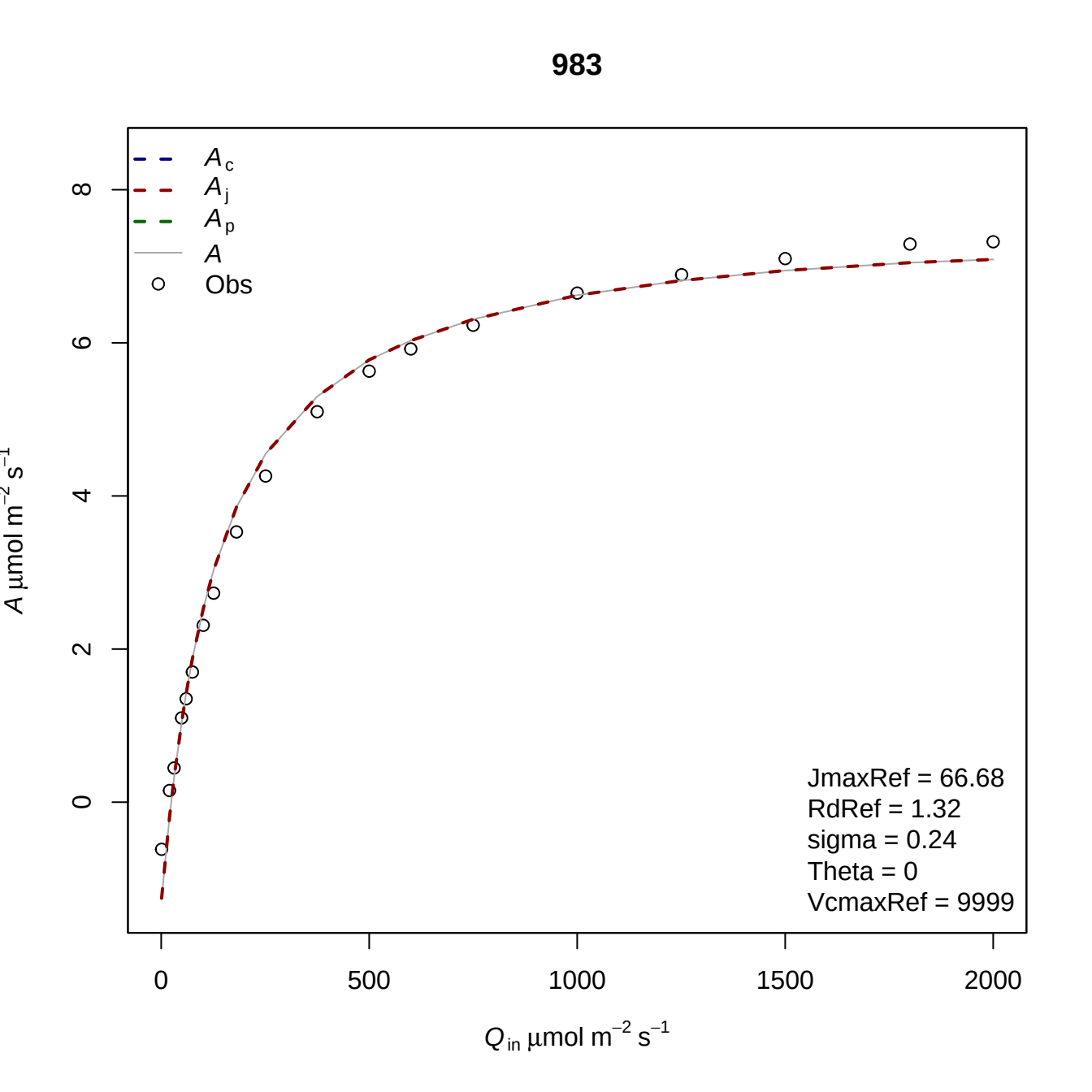

978
A_c
A_j
A_p
A
Obs
JmaxRef = 84.11
RdRef = 1.03
sigma = 0.2
Theta = 0
VcmaxRef = 75.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

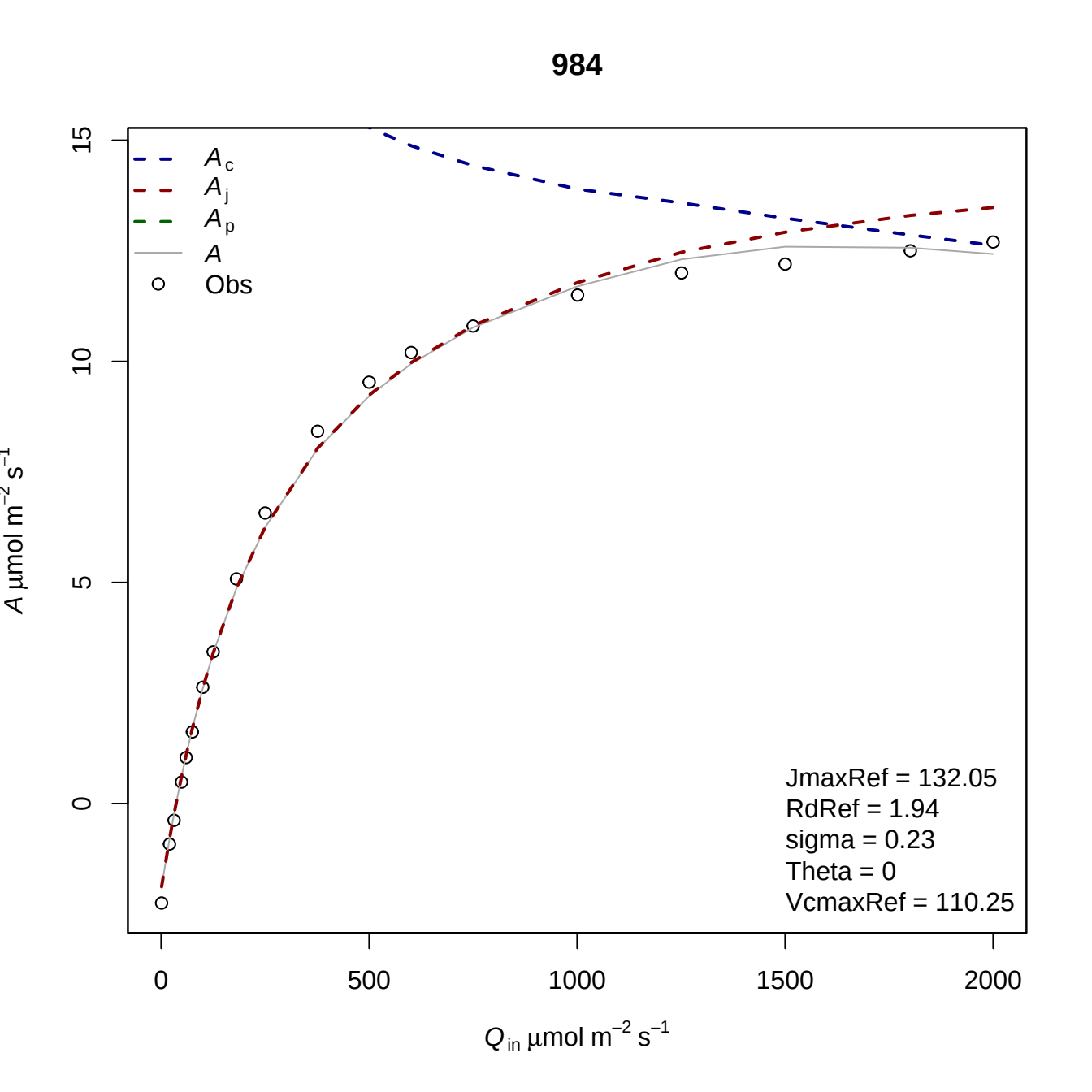

979
A_c
A_j
A_p
A
Obs
JmaxRef = 117.96
RdRef = 1.61
sigma = 0.15
Theta = 0
VcmaxRef = 91.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

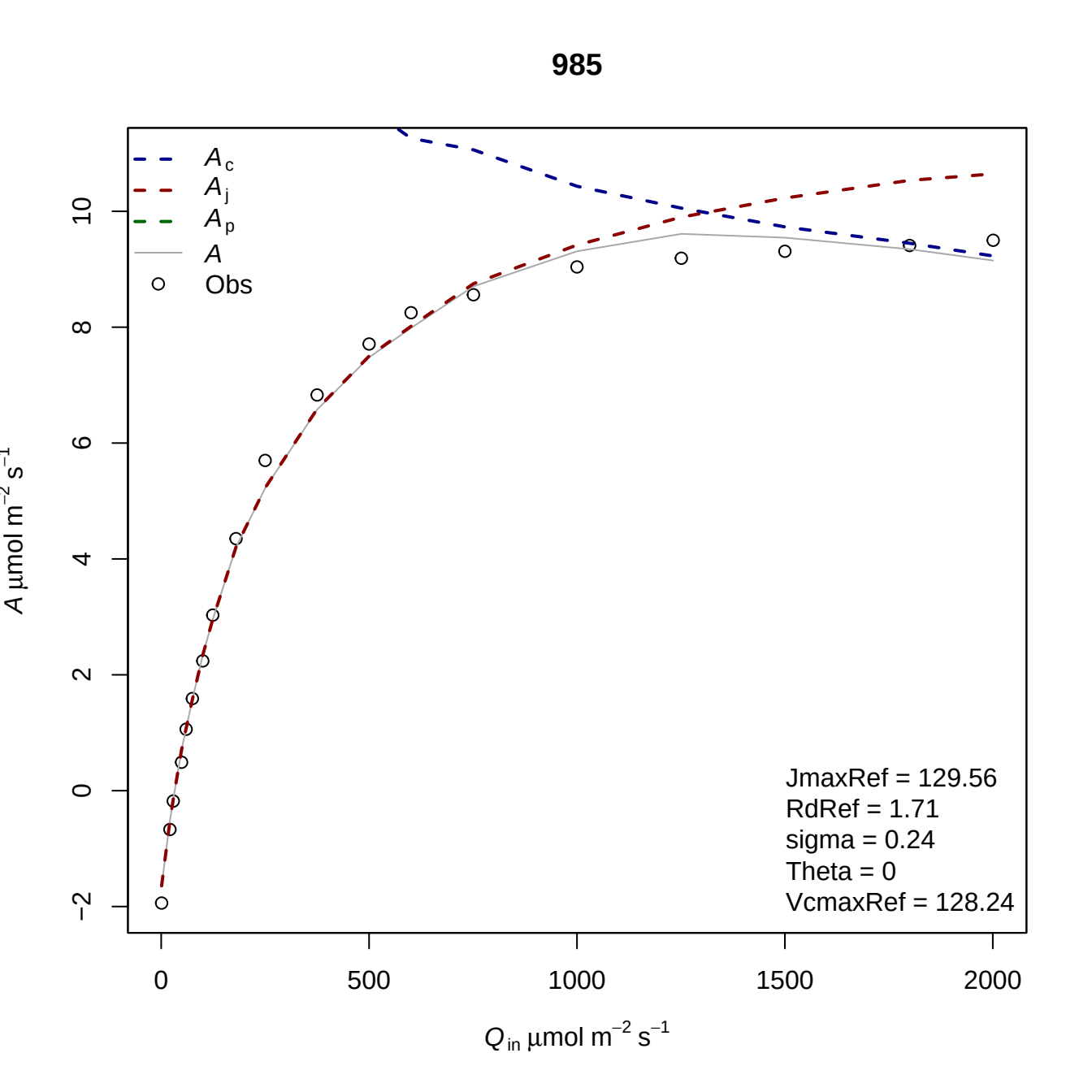

980
A_c
A_j
A_p
A
Obs
JmaxRef = 125.76
RdRef = 3.18
sigma = 0.16
Theta = 0
VcmaxRef = 117.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

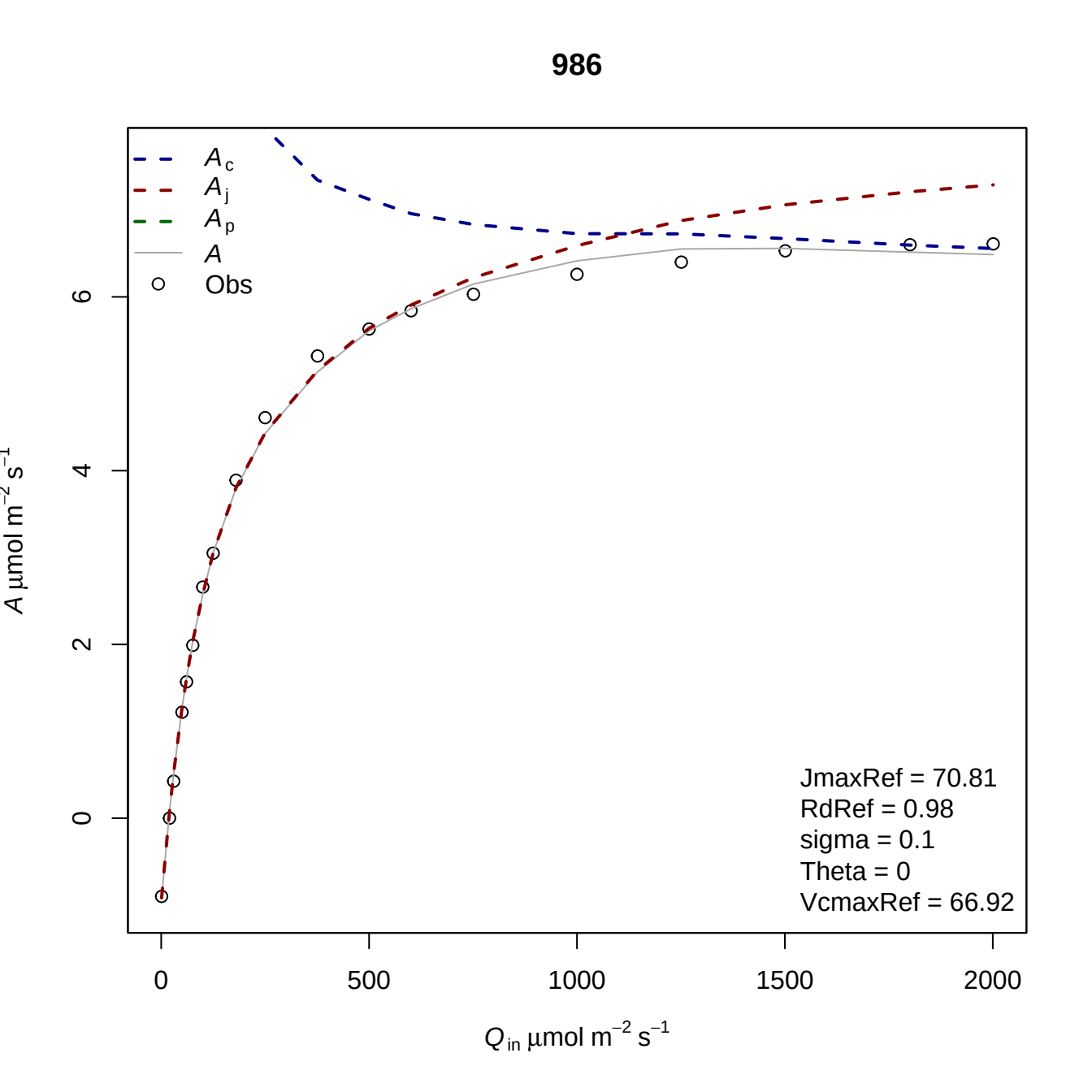

981
A_c
A_j
A_p
A
Obs
JmaxRef = 108.18
RdRef = 0.84
sigma = 0.42
Theta = 0
VcmaxRef = 93.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

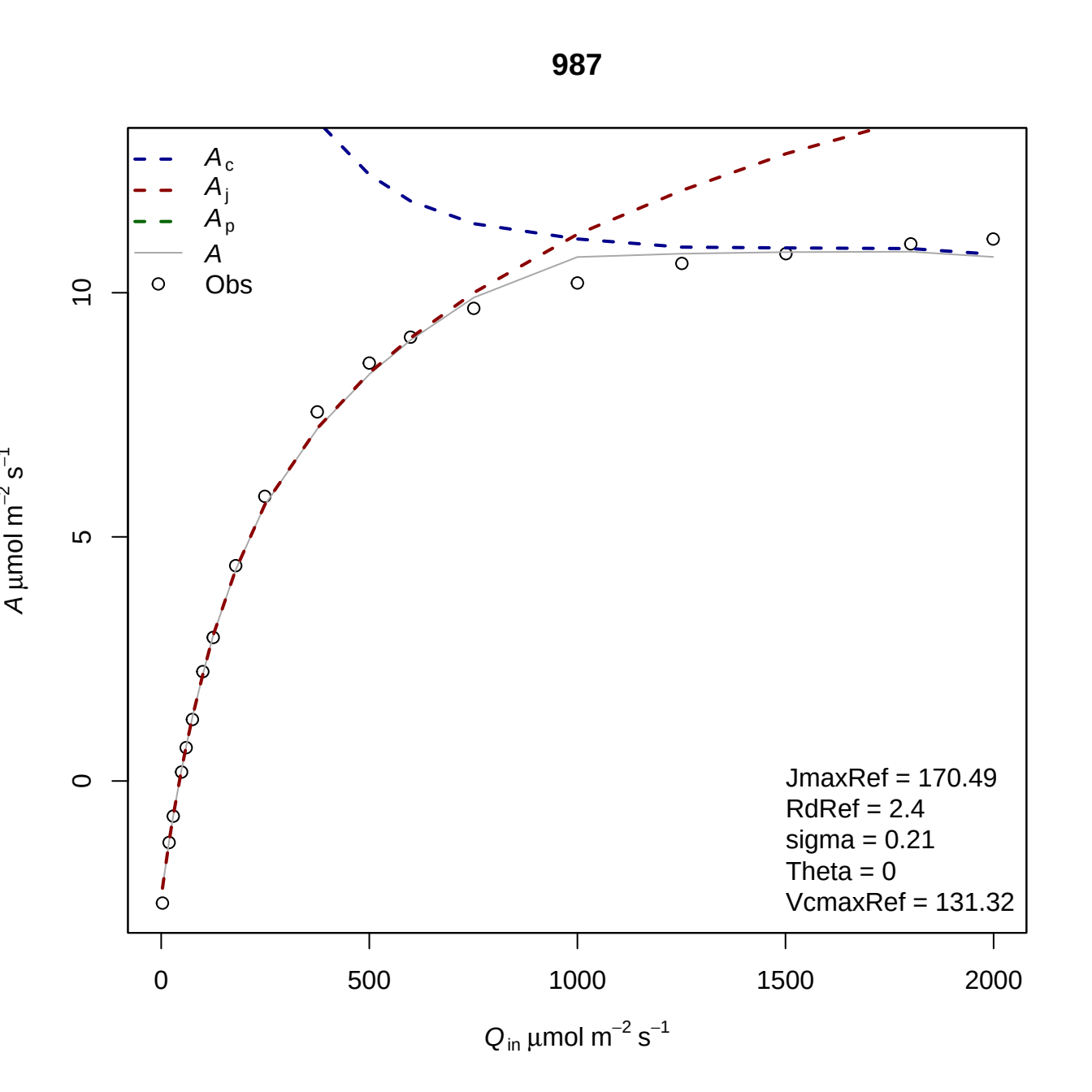

982
A_c
A_j
A_p
A
Obs
JmaxRef = 73.24
RdRef = 1.8
sigma = 0.22
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

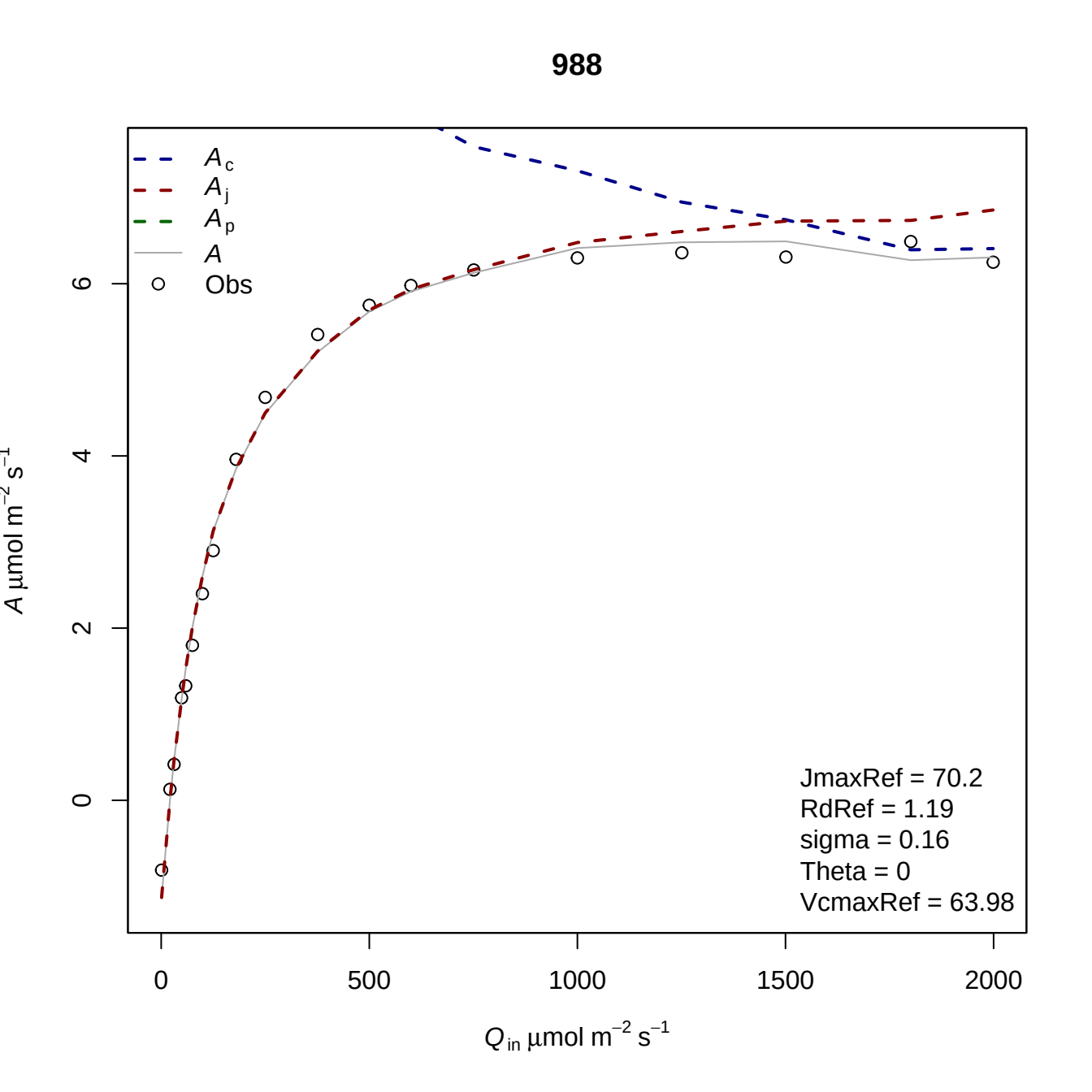

983
A_c
A_j
A_p
A
Obs
JmaxRef = 66.68
RdRef = 1.32
sigma = 0.24
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

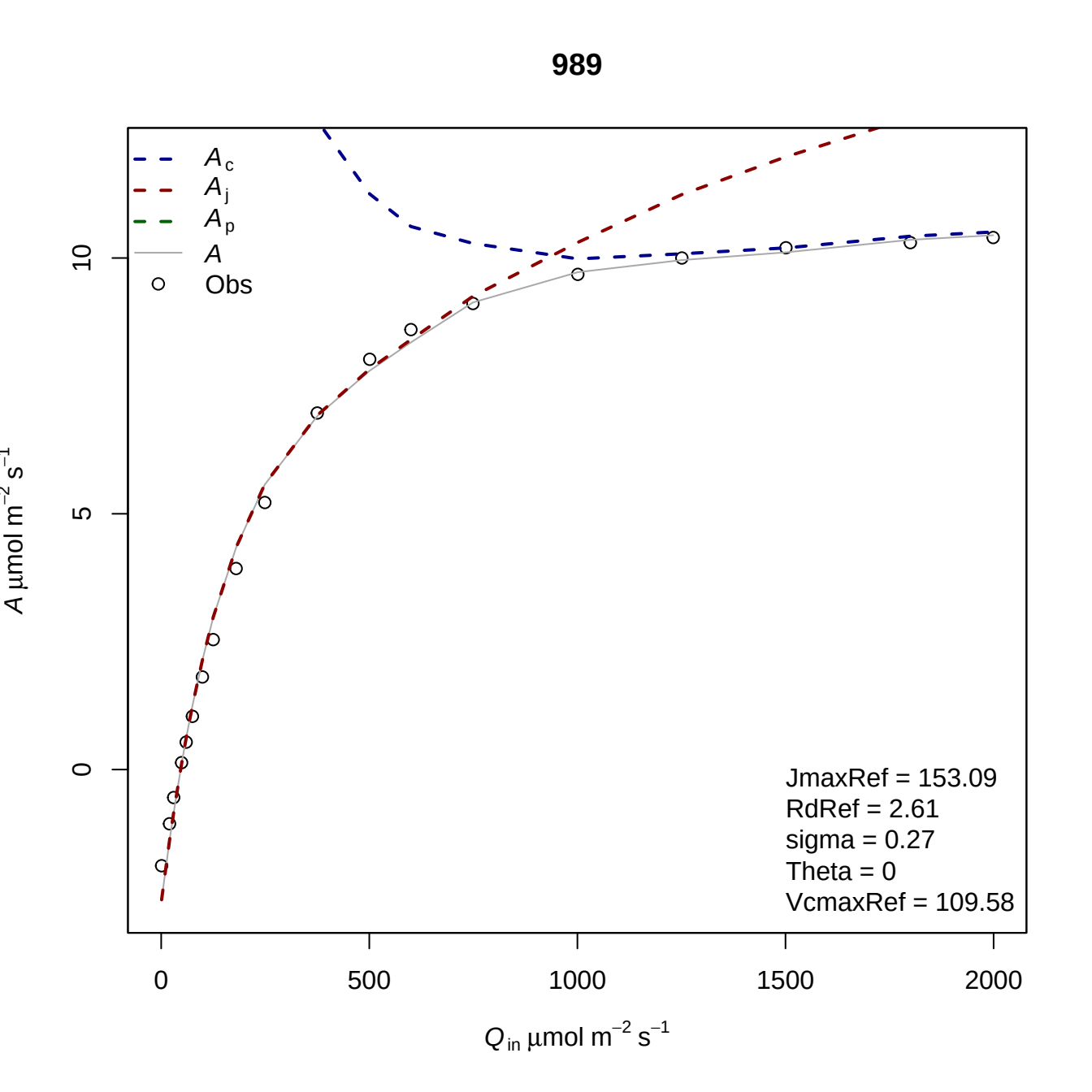

984
A_c
A_j
A_p
A
Obs
JmaxRef = 132.05
RdRef = 1.94
sigma = 0.23
Theta = 0
VcmaxRef = 110.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

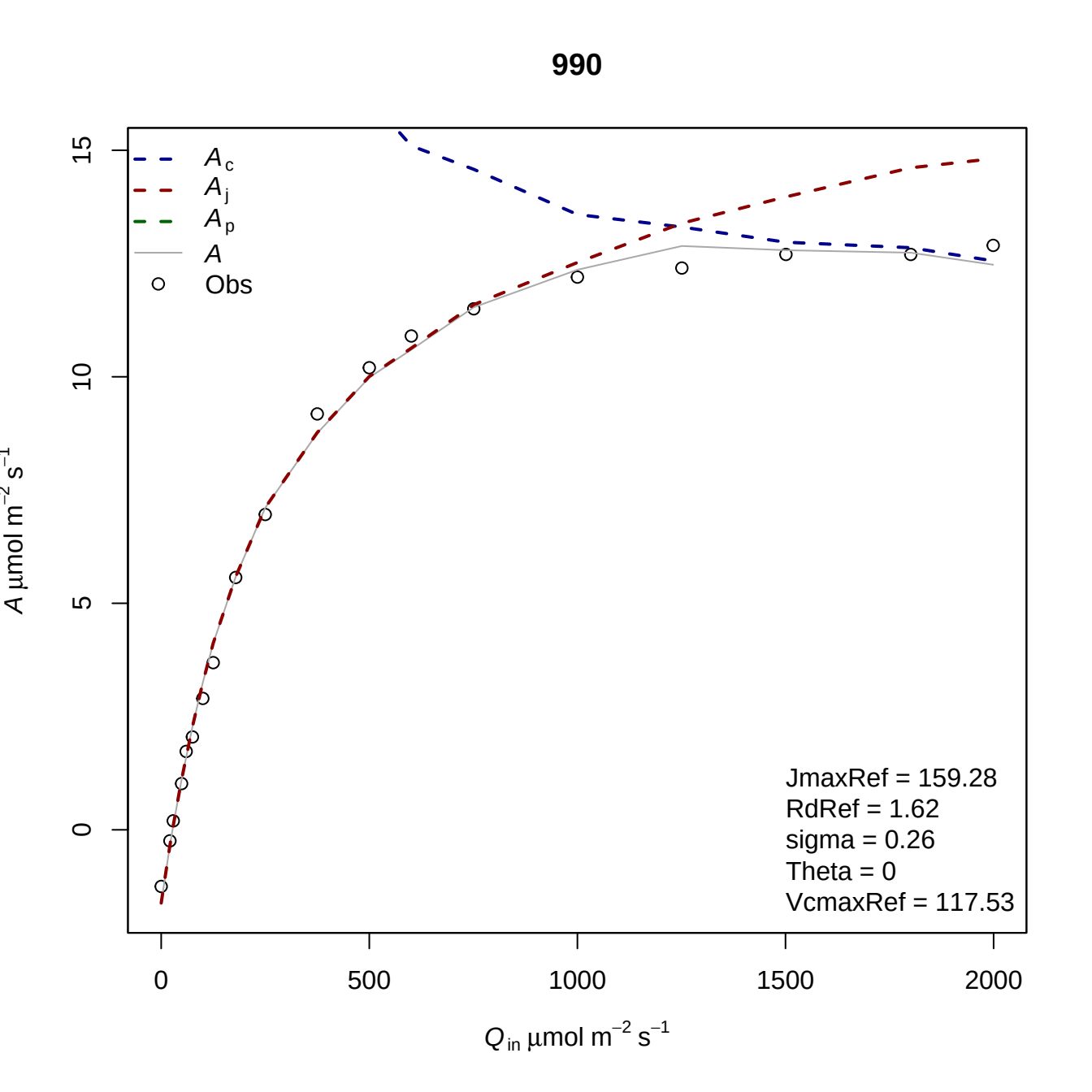

985
A_c
A_j
A_p
A
Obs
JmaxRef = 129.56
RdRef = 1.71
sigma = 0.24
Theta = 0
VcmaxRef = 128.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

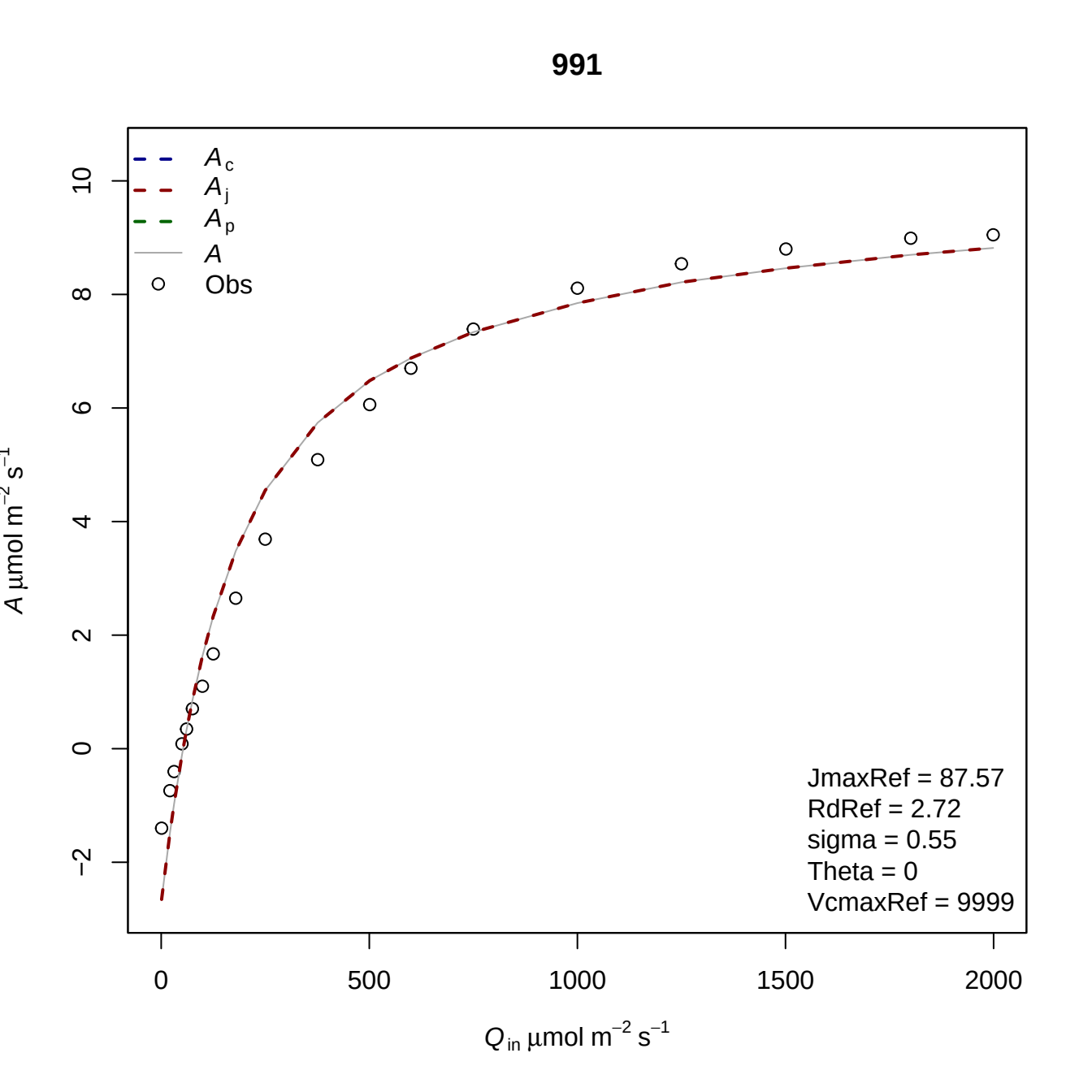

986
A_c
A_j
A_p
A
Obs
JmaxRef = 70.81
RdRef = 0.98
sigma = 0.1
Theta = 0
VcmaxRef = 66.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

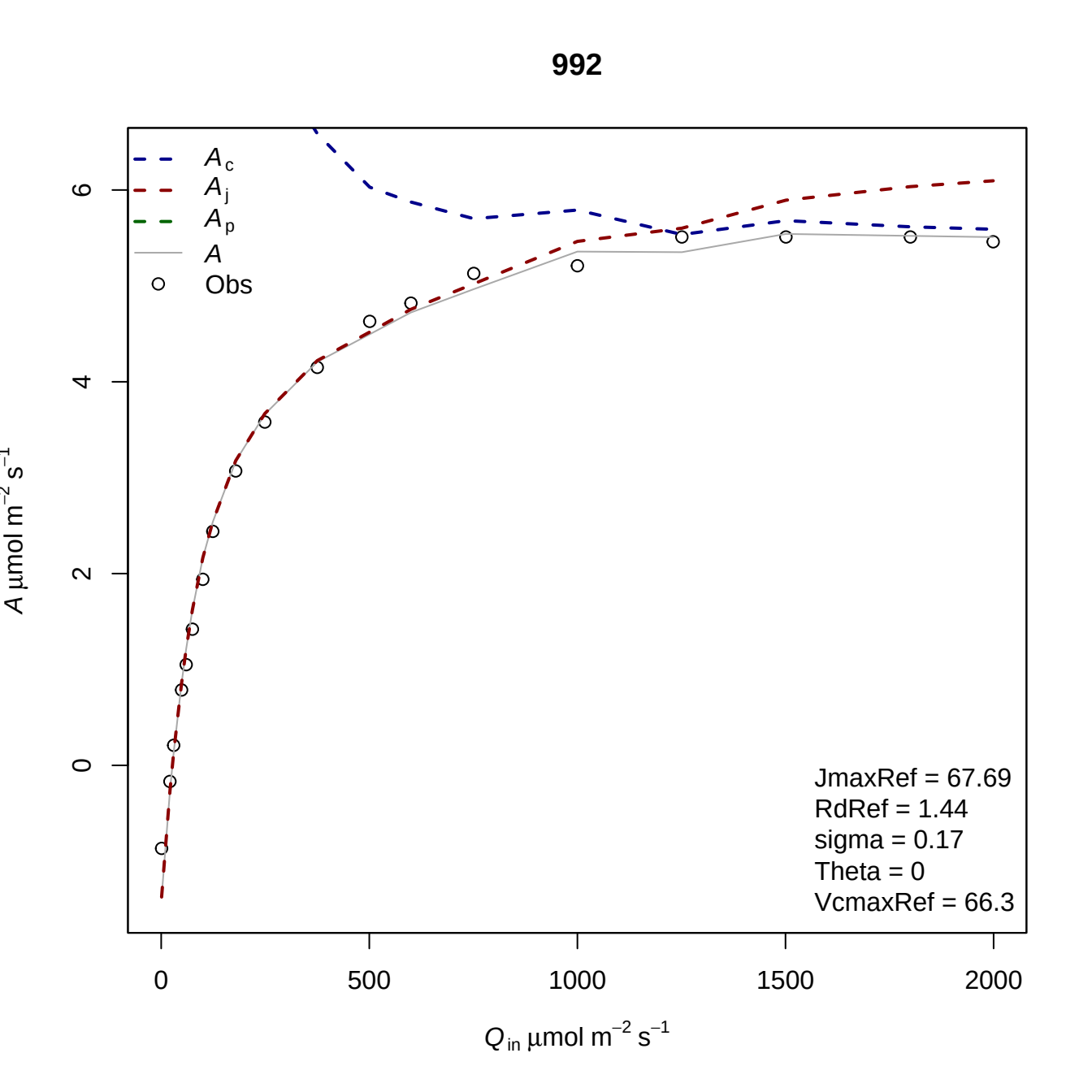

987
A_c
A_j
A_p
A
Obs
JmaxRef = 170.49
RdRef = 2.4
sigma = 0.21
Theta = 0
VcmaxRef = 131.32
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

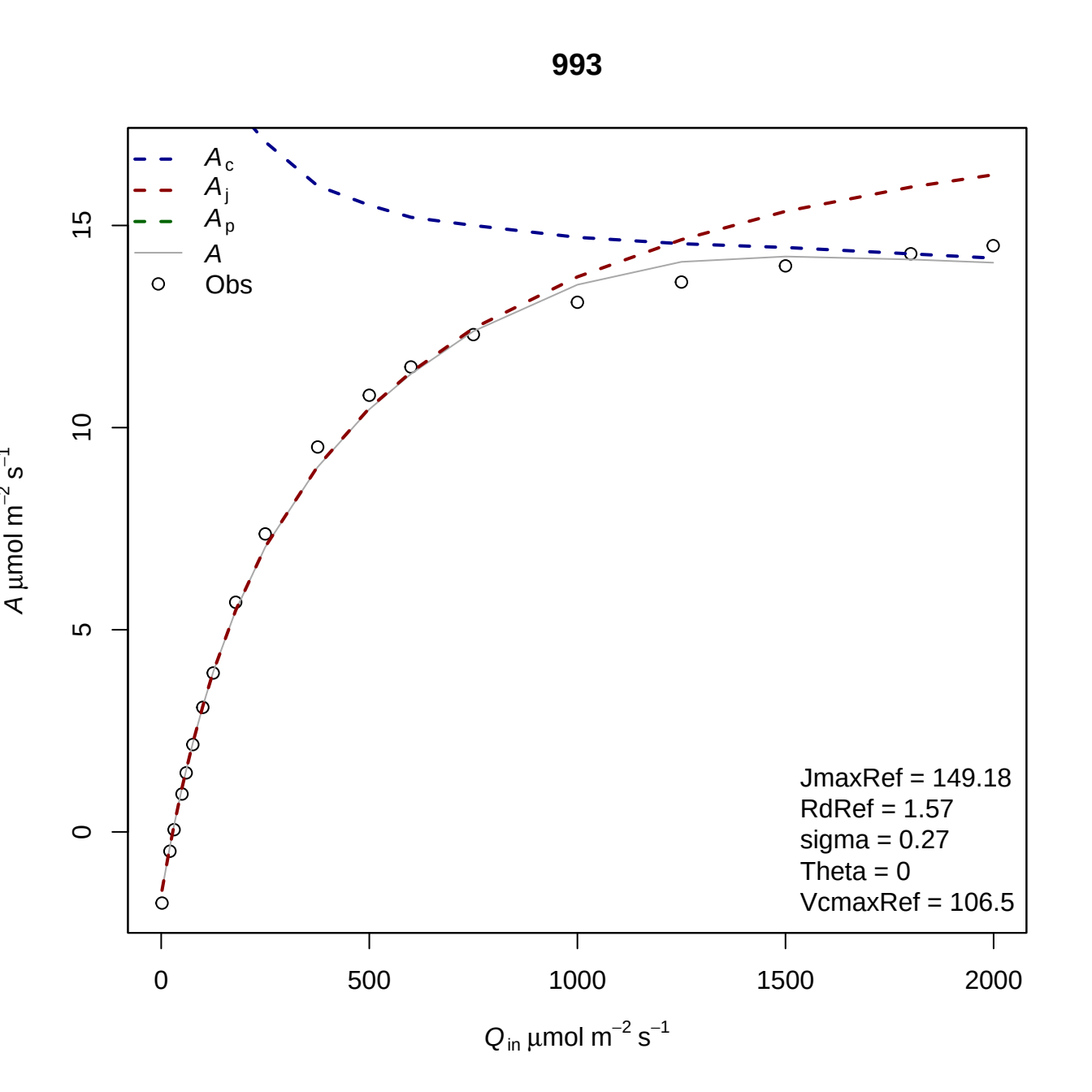

988
A_c
A_j
A_p
A
Obs
JmaxRef = 70.2
RdRef = 1.19
sigma = 0.16
Theta = 0
VcmaxRef = 63.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

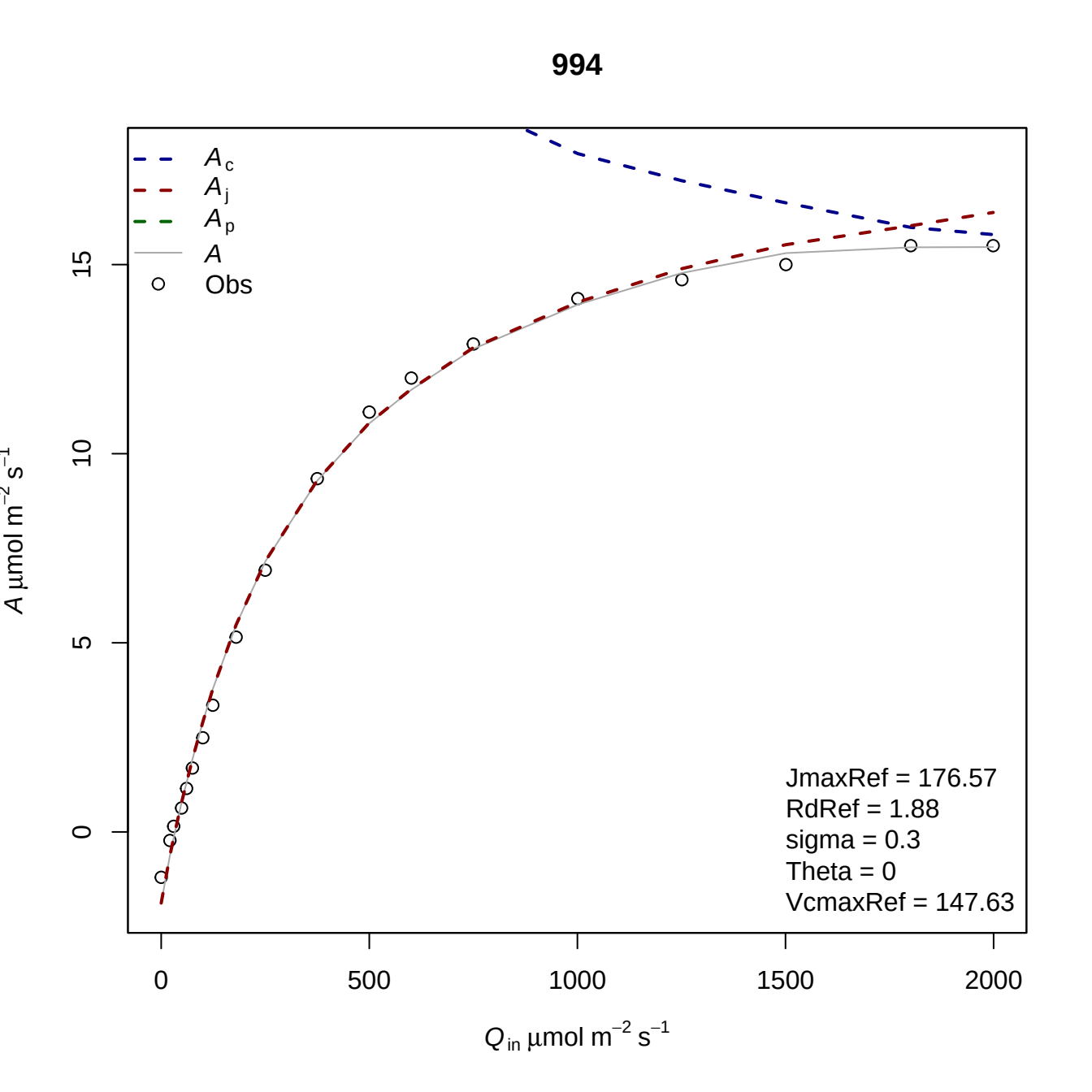

989
A_c
A_j
A_p
A
Obs
JmaxRef = 153.09
RdRef = 2.61
sigma = 0.27
Theta = 0
VcmaxRef = 109.58
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

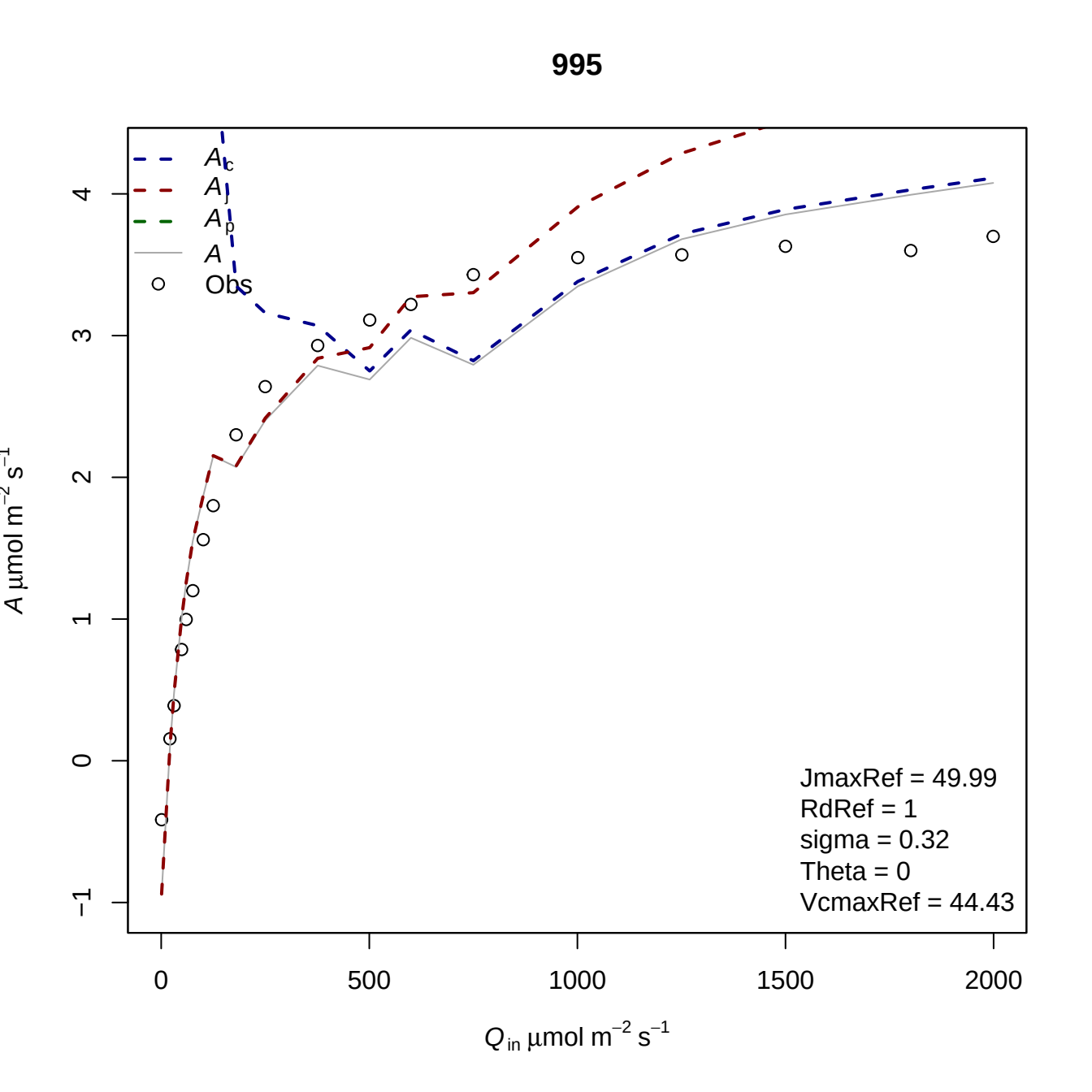

990
A_c
A_j
A_p
A
Obs
JmaxRef = 159.28
RdRef = 1.62
sigma = 0.26
Theta = 0
VcmaxRef = 117.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

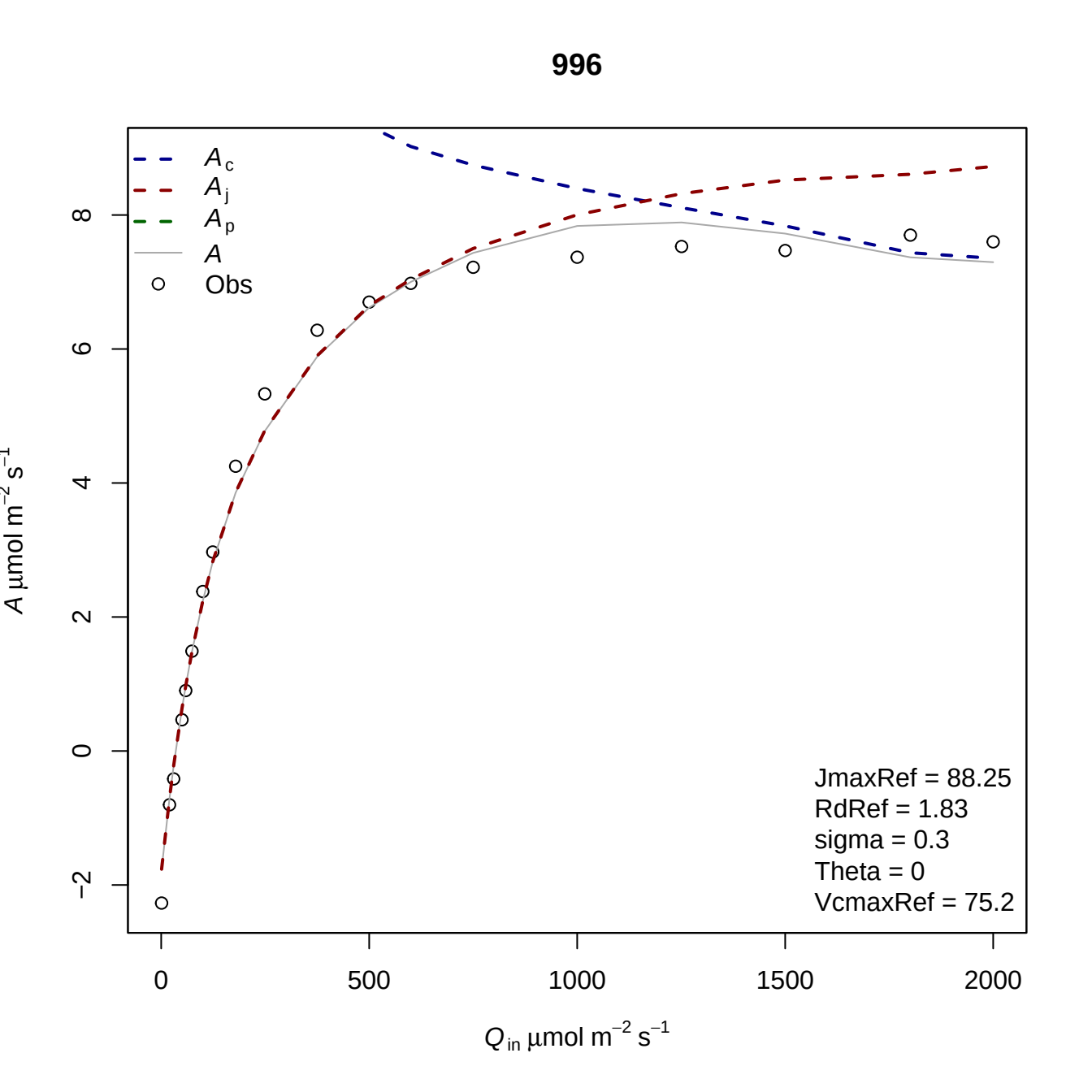

991
A_c
A_j
A_p
A
Obs
JmaxRef = 87.57
RdRef = 2.72
sigma = 0.55
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

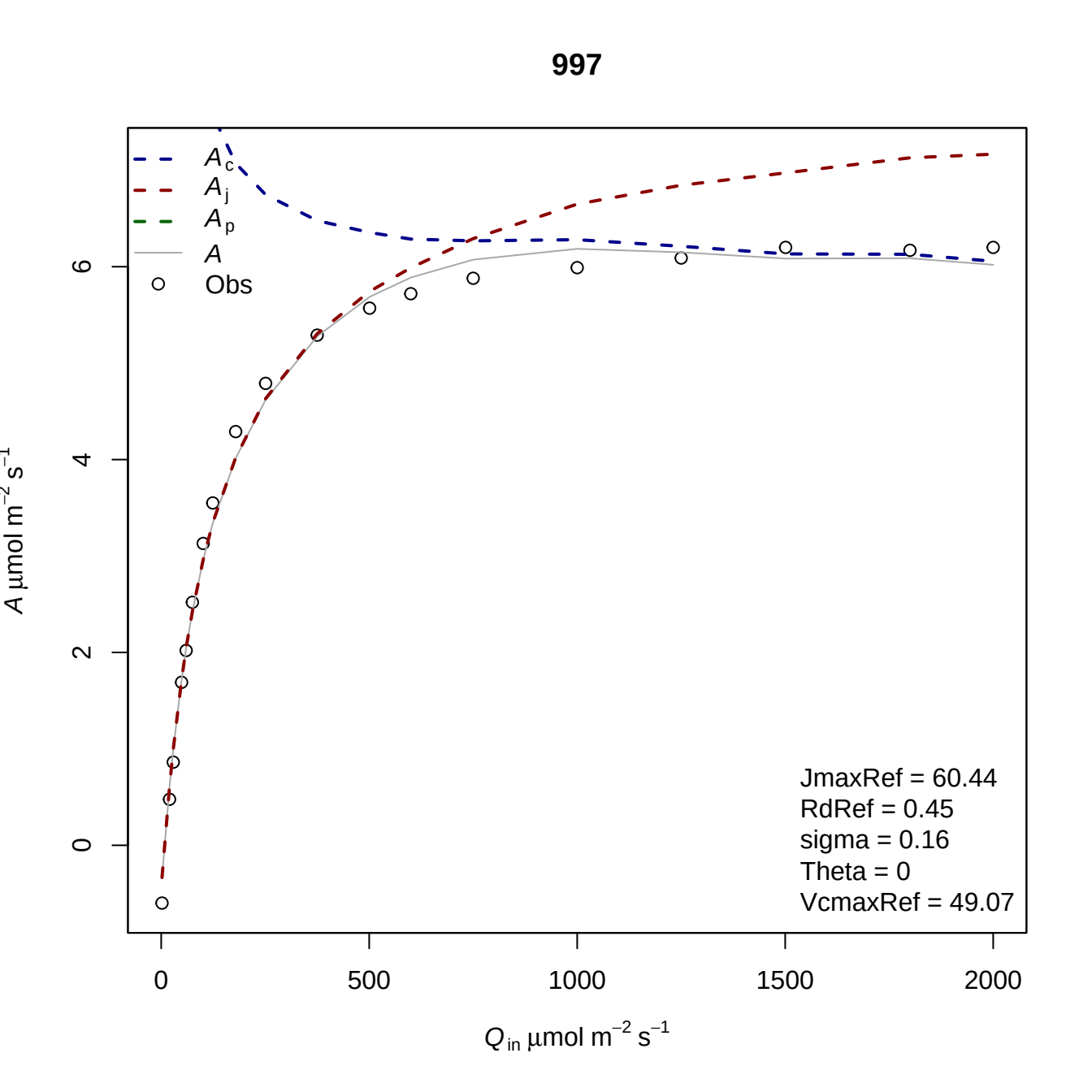

992
A_c
A_j
A_p
A
Obs
JmaxRef = 67.69
RdRef = 1.44
sigma = 0.17
Theta = 0
VcmaxRef = 66.3
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

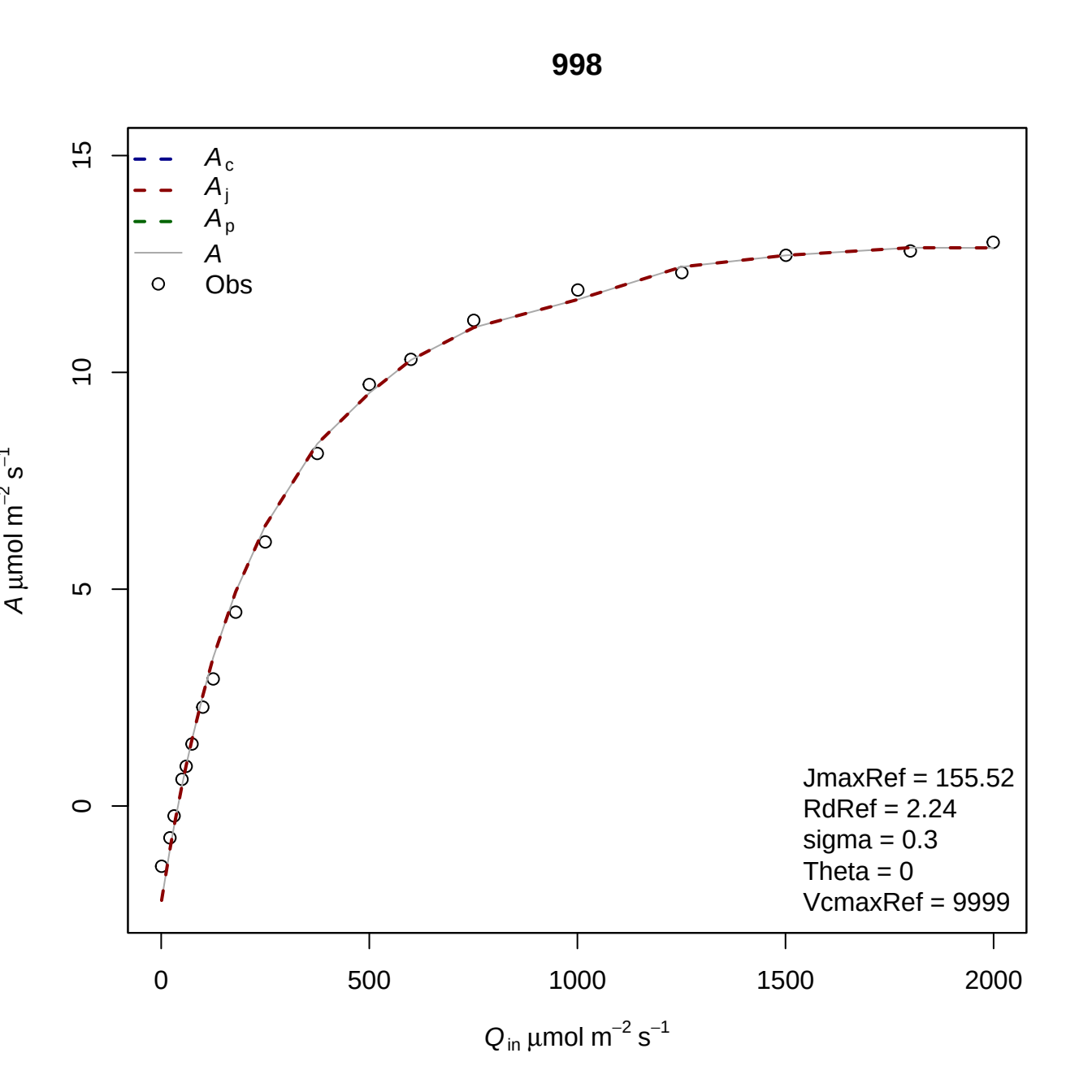

993
A_c
A_j
A_p
A
Obs
JmaxRef = 149.18
RdRef = 1.57
sigma = 0.27
Theta = 0
VcmaxRef = 106.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

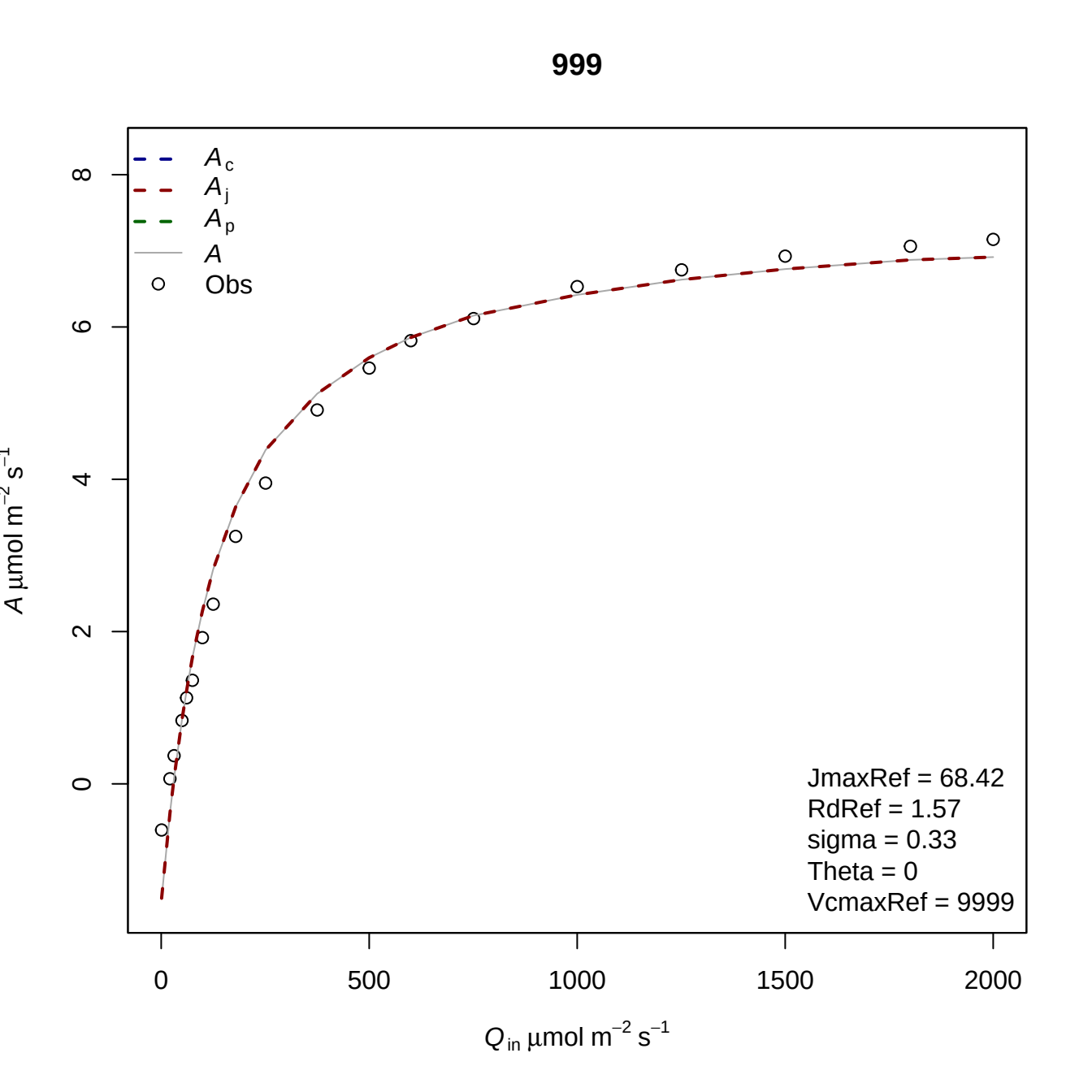

994
A_c
A_j
A_p
A
Obs
JmaxRef = 176.57
RdRef = 1.88
sigma = 0.3
Theta = 0
VcmaxRef = 147.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

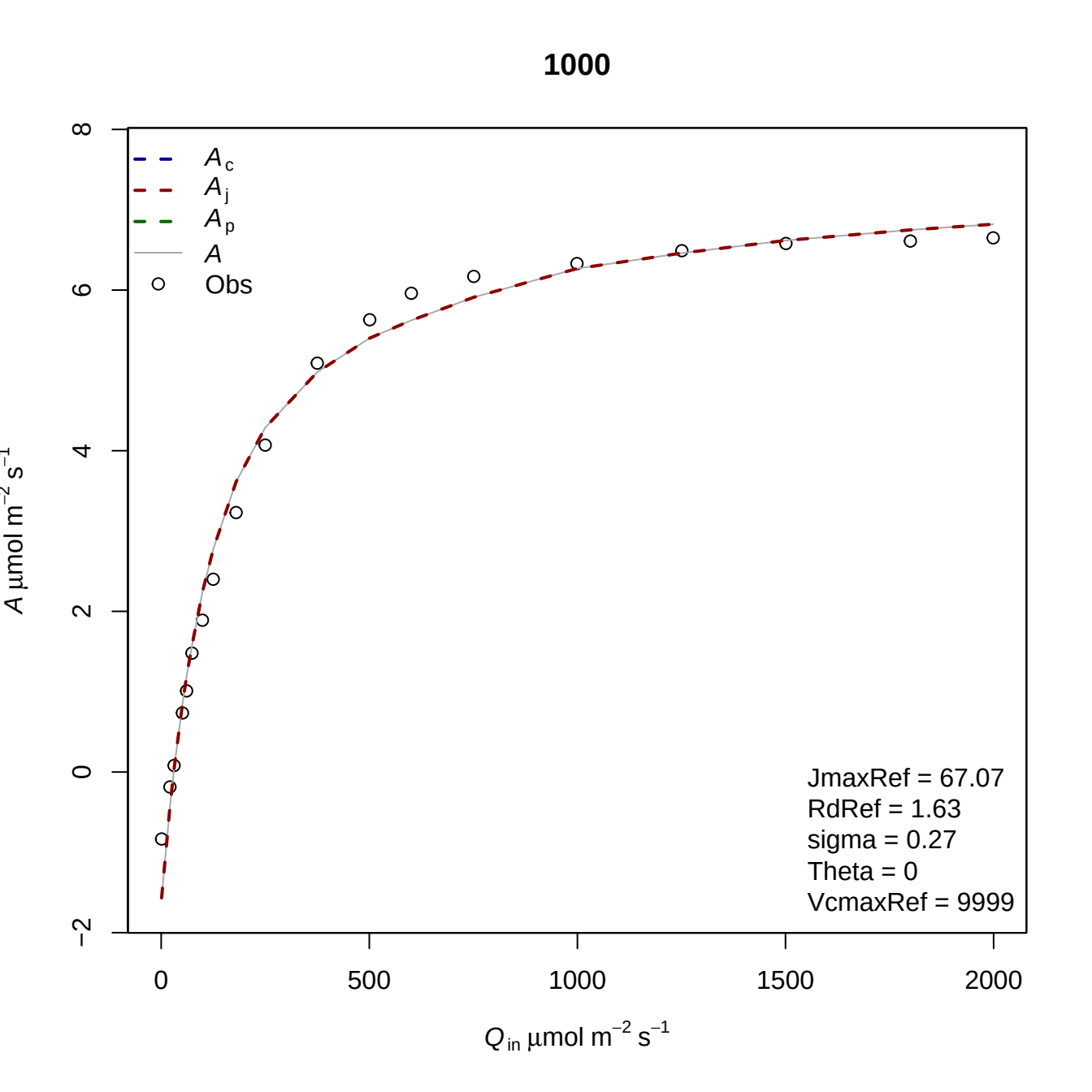

995
A_c
A_j
A_p
A
Obs
JmaxRef = 49.99
RdRef = 1
sigma = 0.32
Theta = 0
VcmaxRef = 44.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

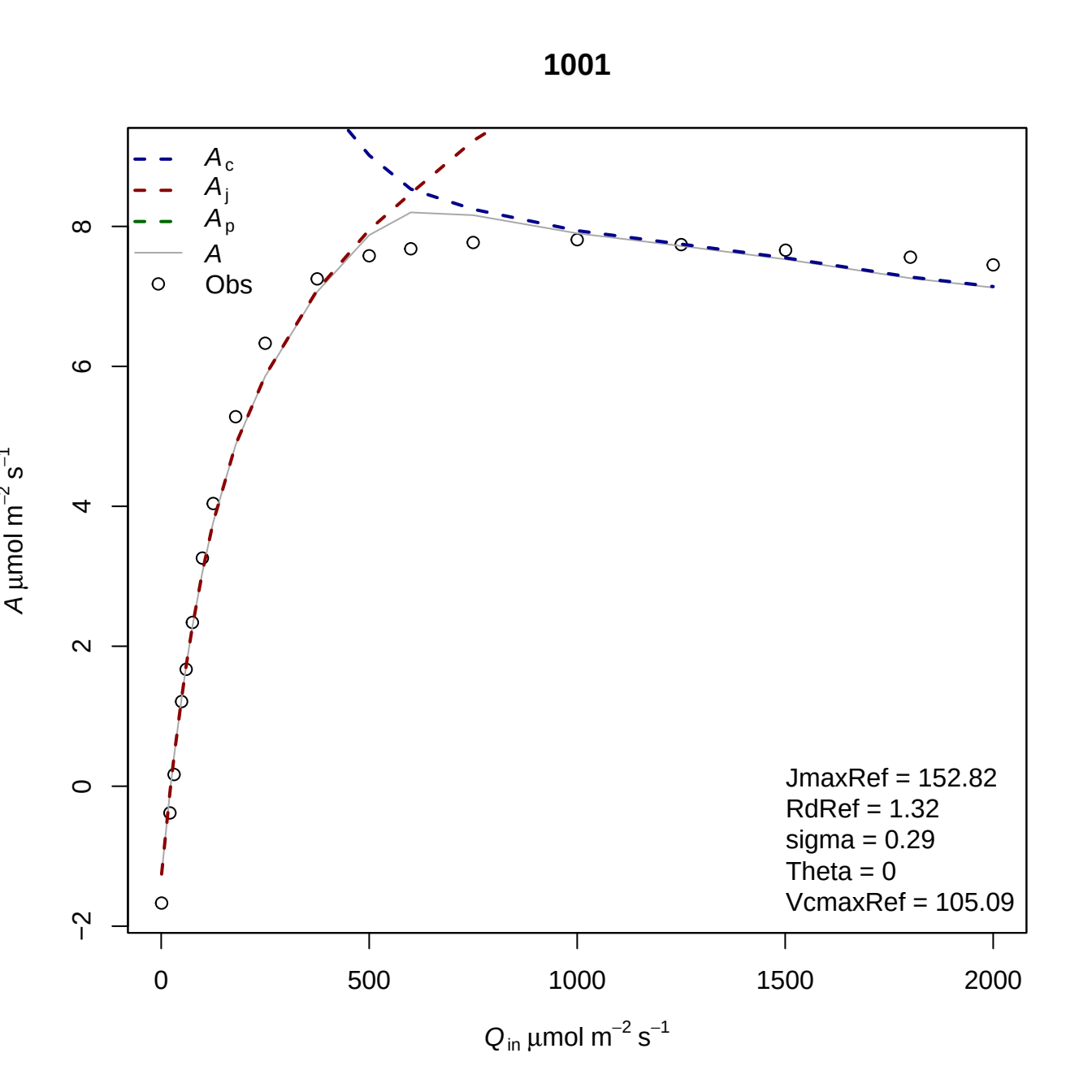

996
A_c
A_j
A_p
A
Obs
JmaxRef = 88.25
RdRef = 1.83
sigma = 0.3
Theta = 0
VcmaxRef = 75.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

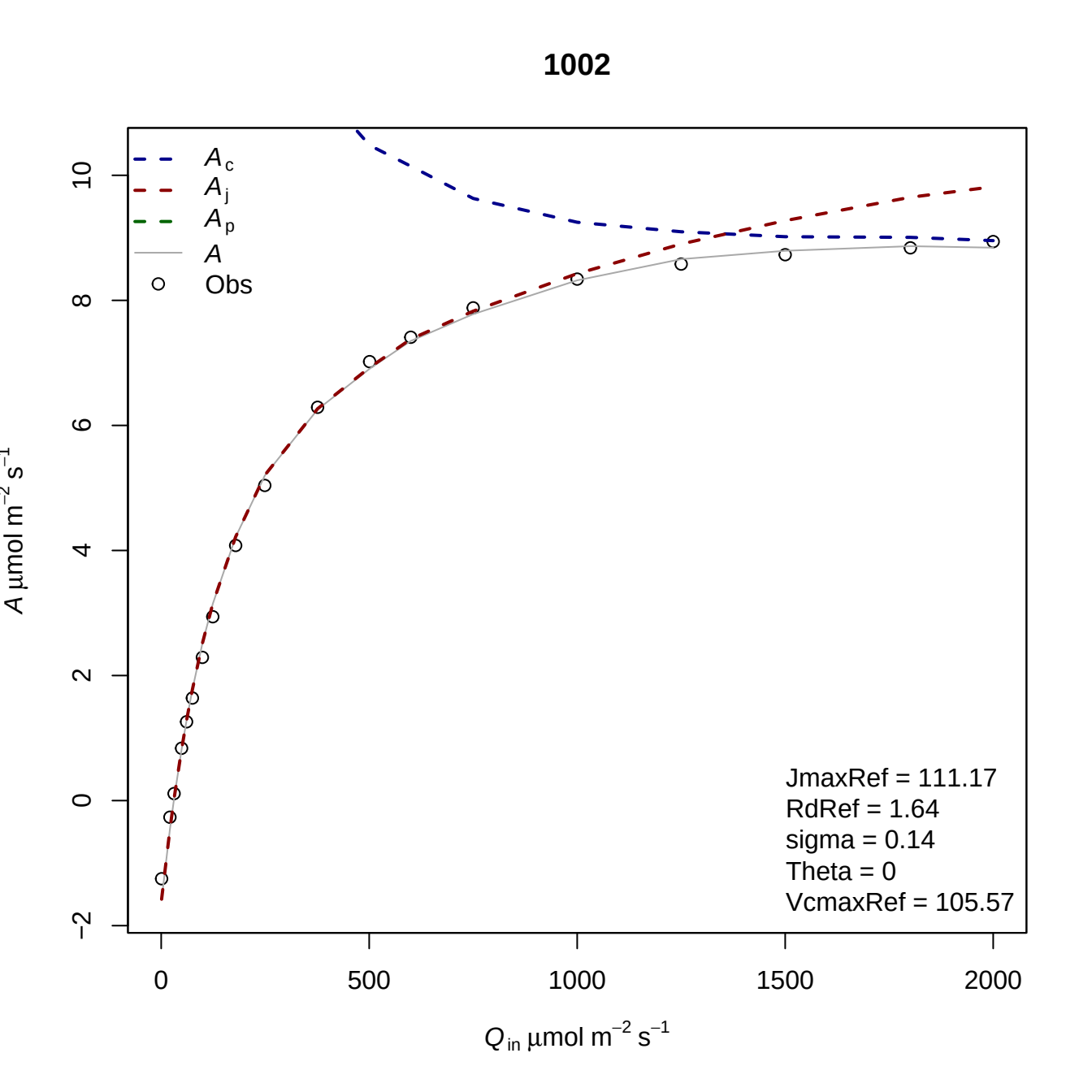

997
A_c
A_j
A_p
A
Obs
JmaxRef = 60.44
RdRef = 0.45
sigma = 0.16
Theta = 0
VcmaxRef = 49.07
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

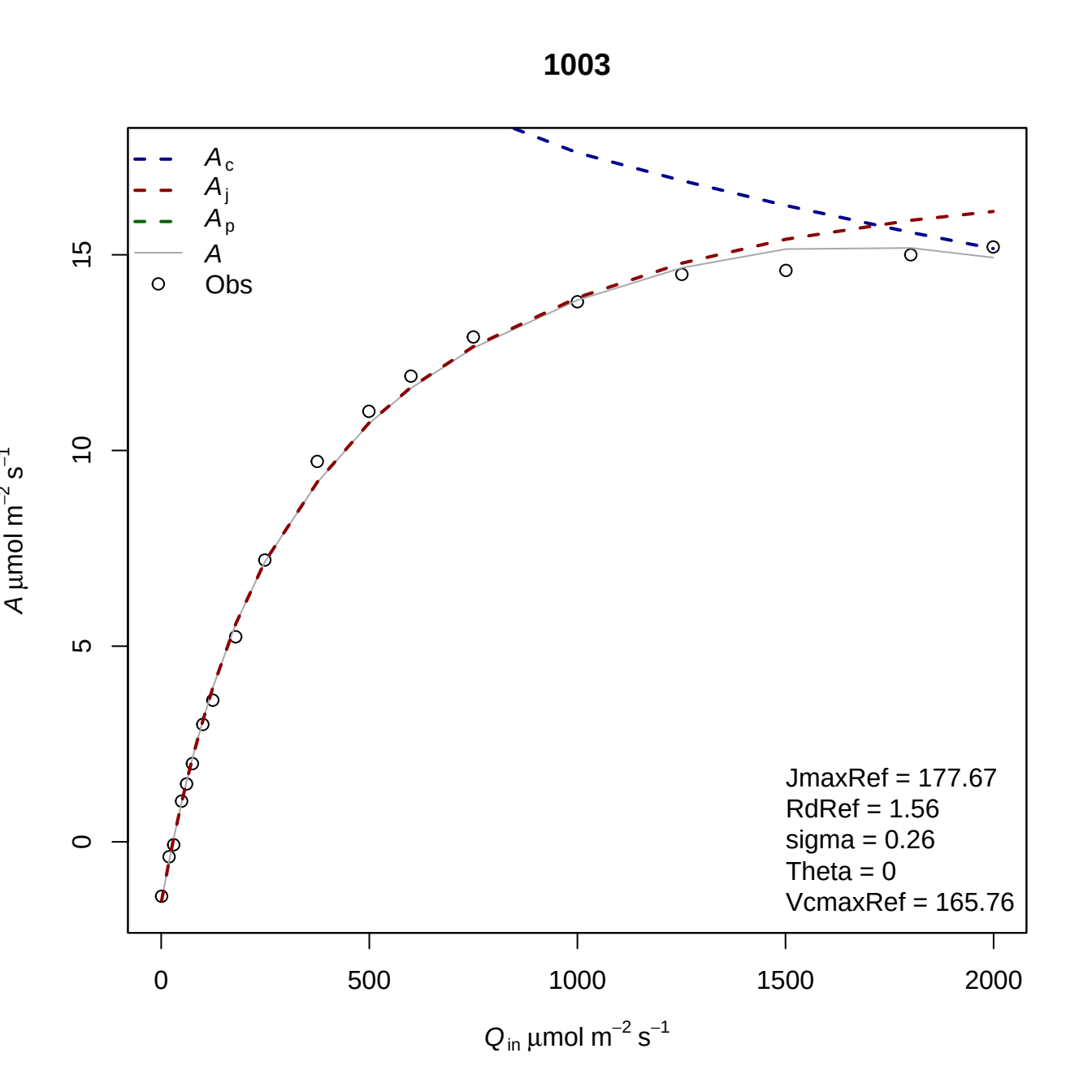

998
A_c
A_j
A_p
A
Obs
JmaxRef = 155.52
RdRef = 2.24
sigma = 0.3
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

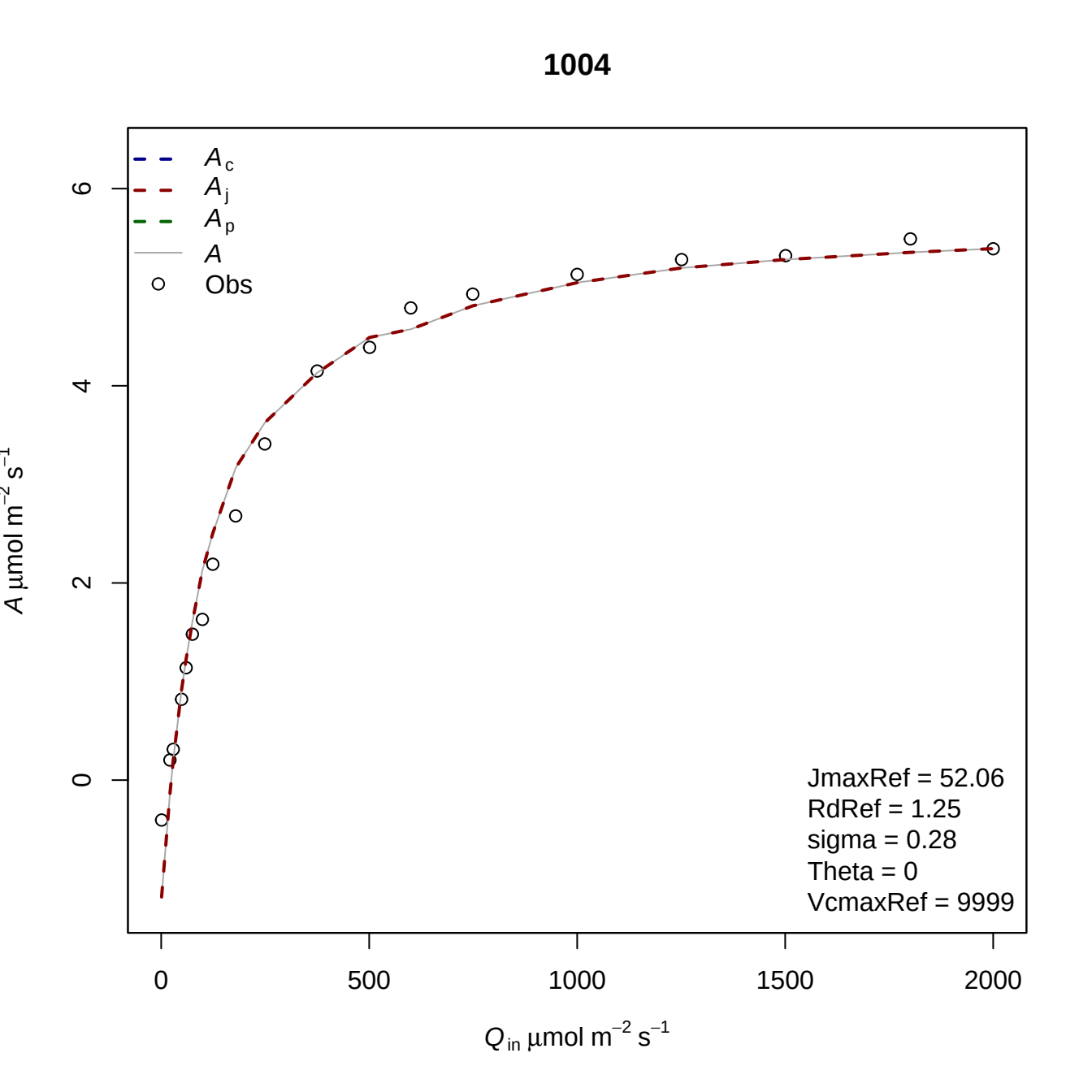

999
A_c
A_j
A_p
A
Obs
JmaxRef = 68.42
RdRef = 1.57
sigma = 0.33
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

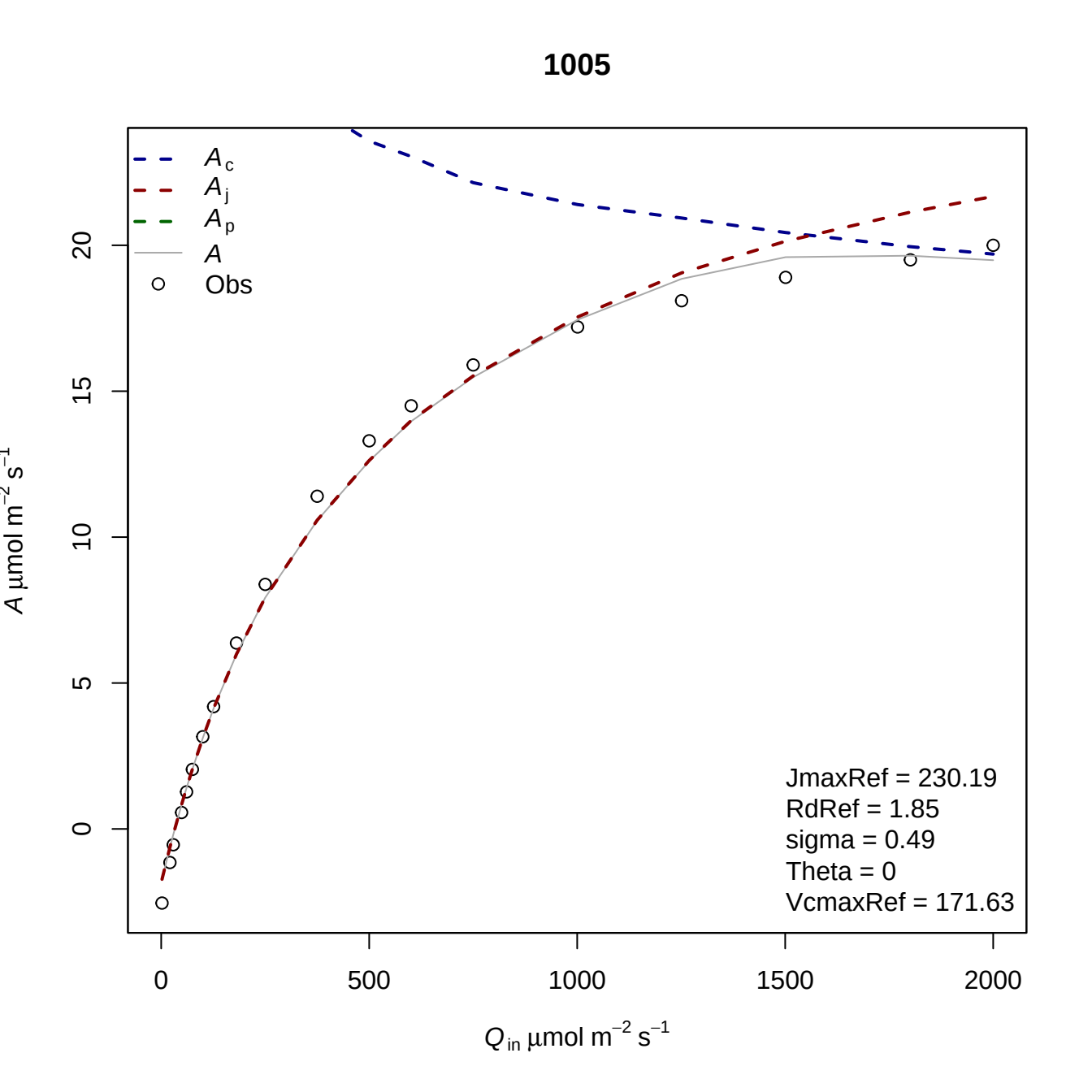

1000
A_c
A_j
A_p
A
Obs
JmaxRef = 67.07
RdRef = 1.63
sigma = 0.27
Theta = 0
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

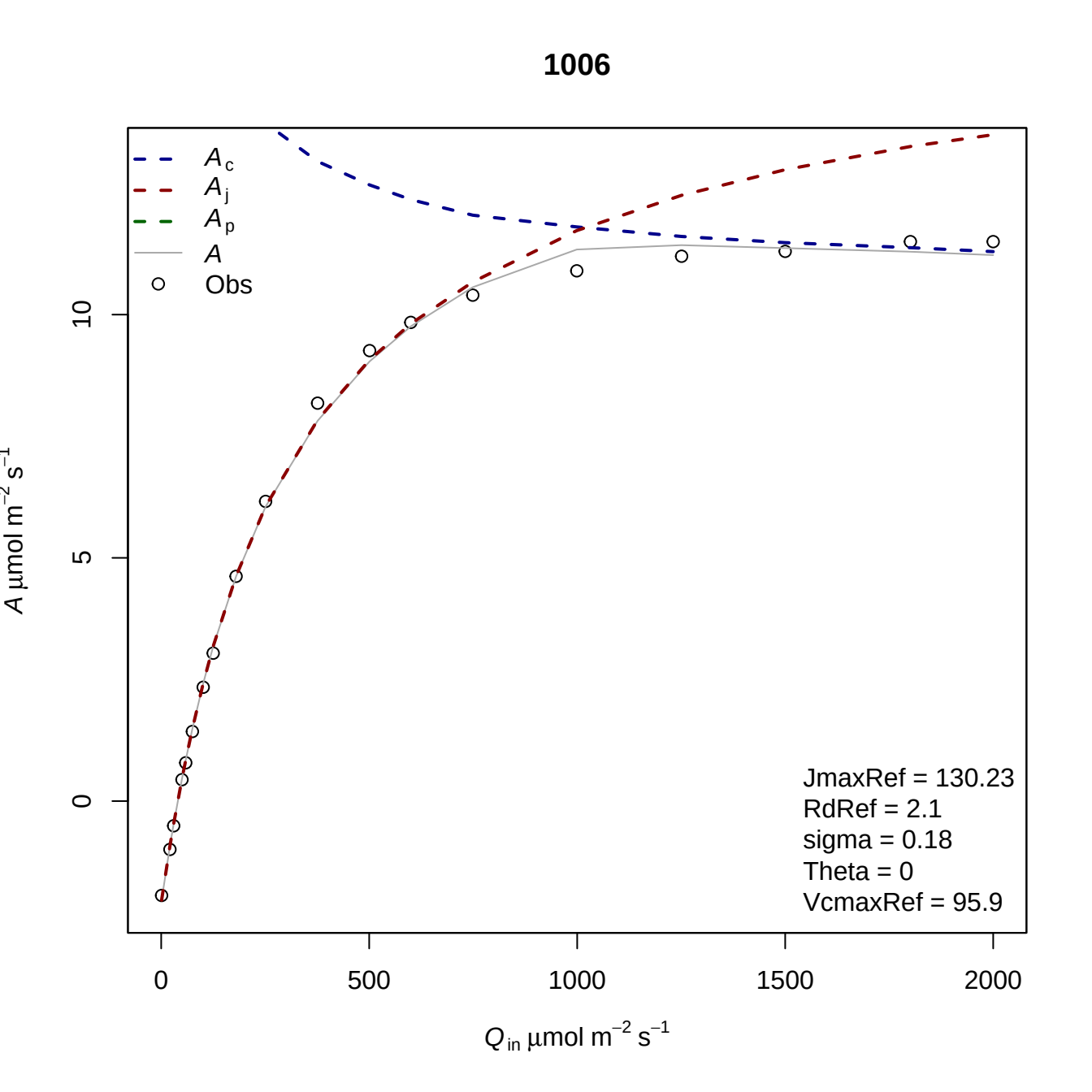

1001
A_c
A_j
A_p
A
Obs
JmaxRef = 152.82
RdRef = 1.32
sigma = 0.29
Theta = 0
VcmaxRef = 105.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

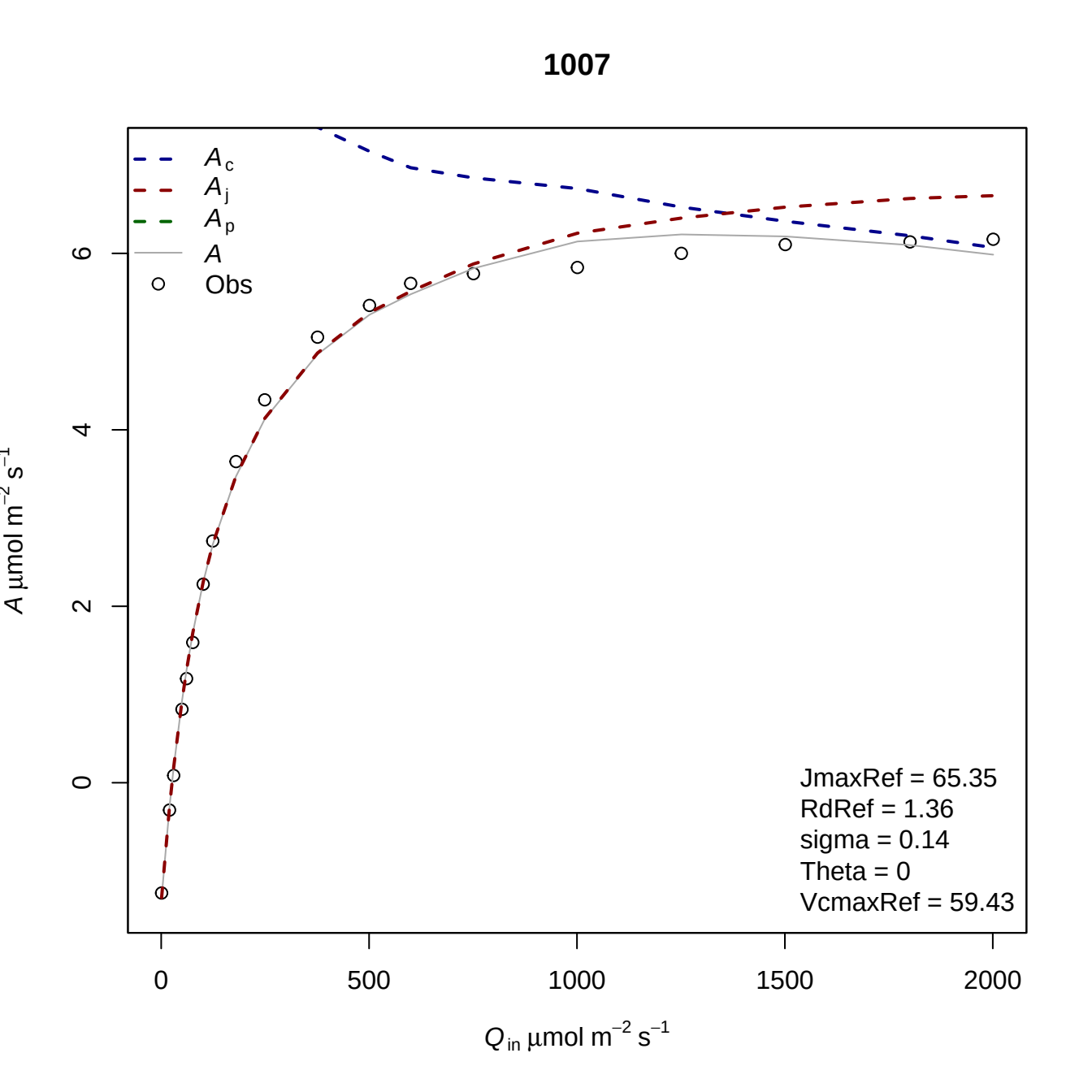

1002
A_c
A_j
A_p
A
Obs
JmaxRef = 111.17
RdRef = 1.64
sigma = 0.14
Theta = 0
VcmaxRef = 105.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

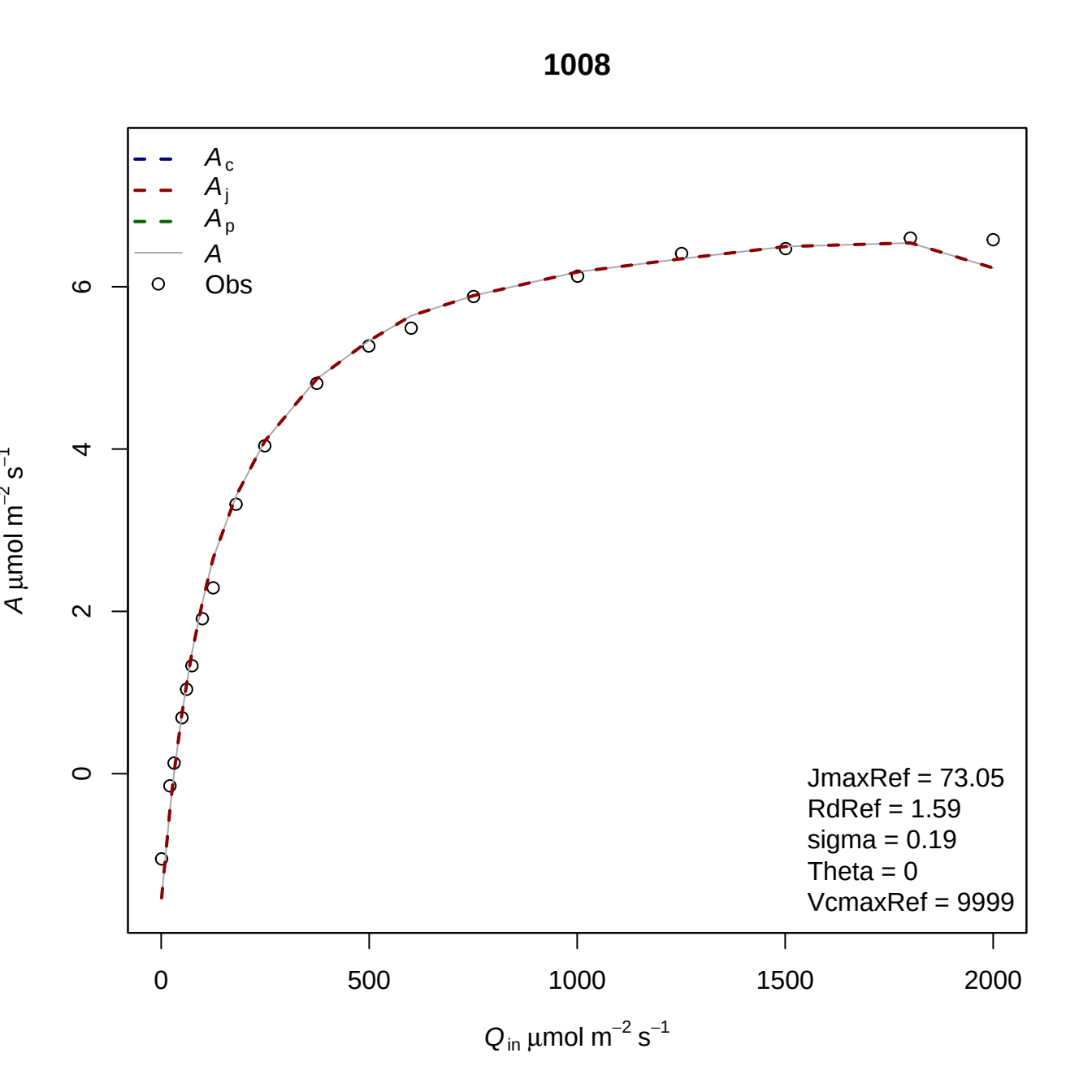

1003
A_c
A_j
A_p
A
Obs
JmaxRef = 177.67
RdRef = 1.56
sigma = 0.26
Theta = 0
VcmaxRef = 165.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

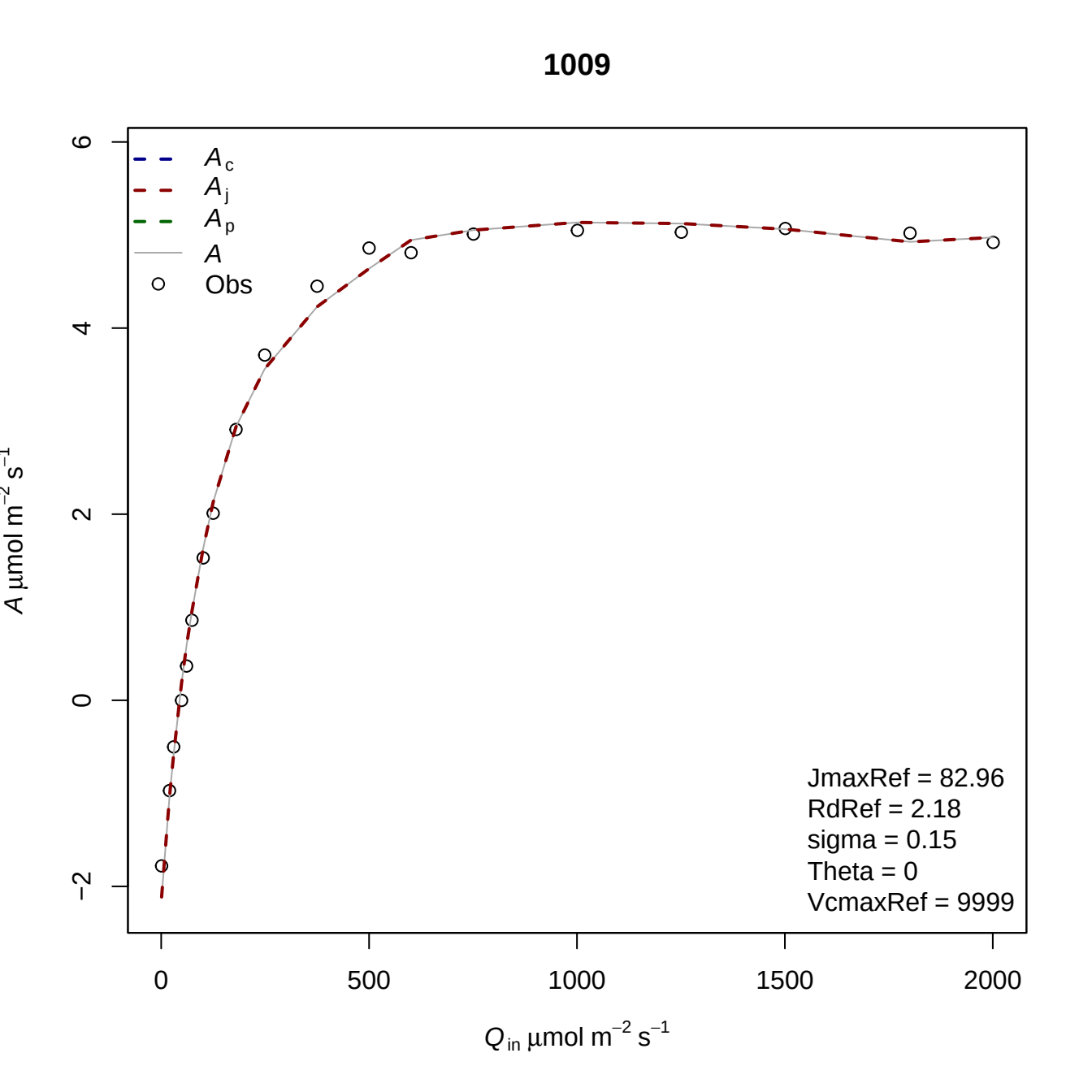

1004
A_c
A_j
A_p
A
Obs
JmaxRef = 52.06
RdRef = 1.25
sigma = 0.28
Theta = 0
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

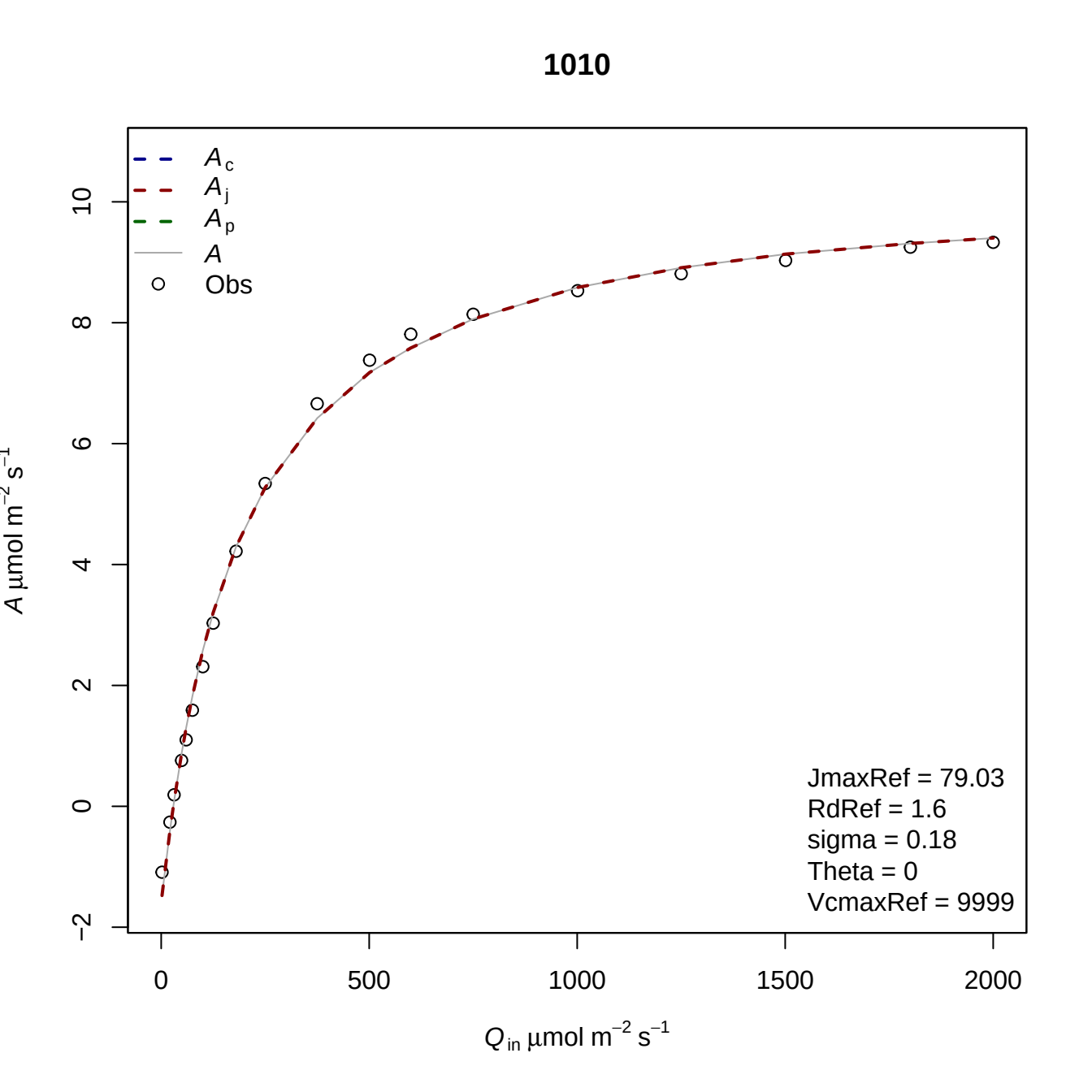

1005
A_c
A_j
A_p
A
Obs
JmaxRef = 230.19
RdRef = 1.85
sigma = 0.49
Theta = 0
VcmaxRef = 171.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

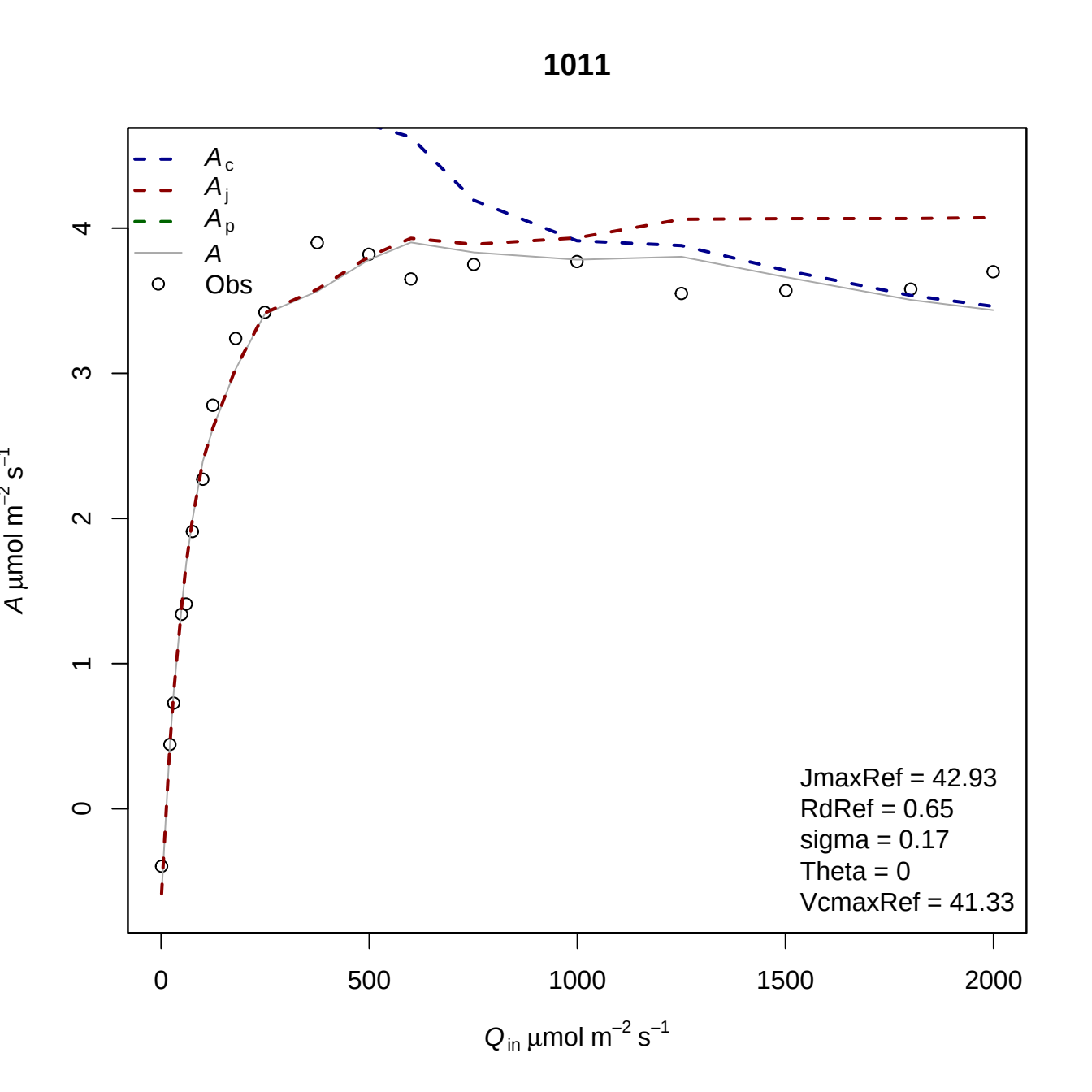

1006
A_c
A_j
A_p
A
Obs
JmaxRef = 130.23
RdRef = 2.1
sigma = 0.18
Theta = 0
VcmaxRef = 95.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

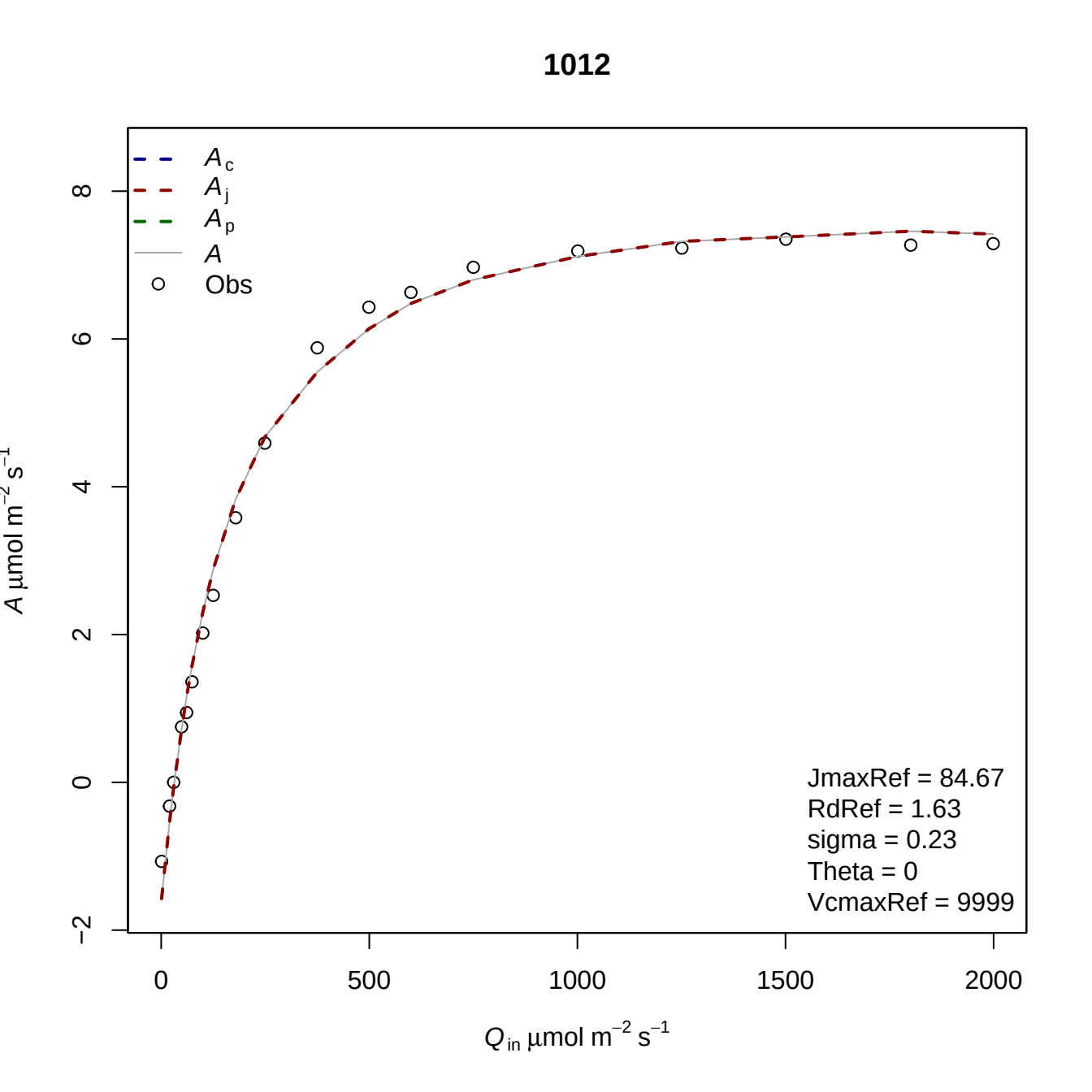

1007
A_c
A_j
A_p
A
Obs
JmaxRef = 65.35
RdRef = 1.36
sigma = 0.14
Theta = 0
VcmaxRef = 59.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

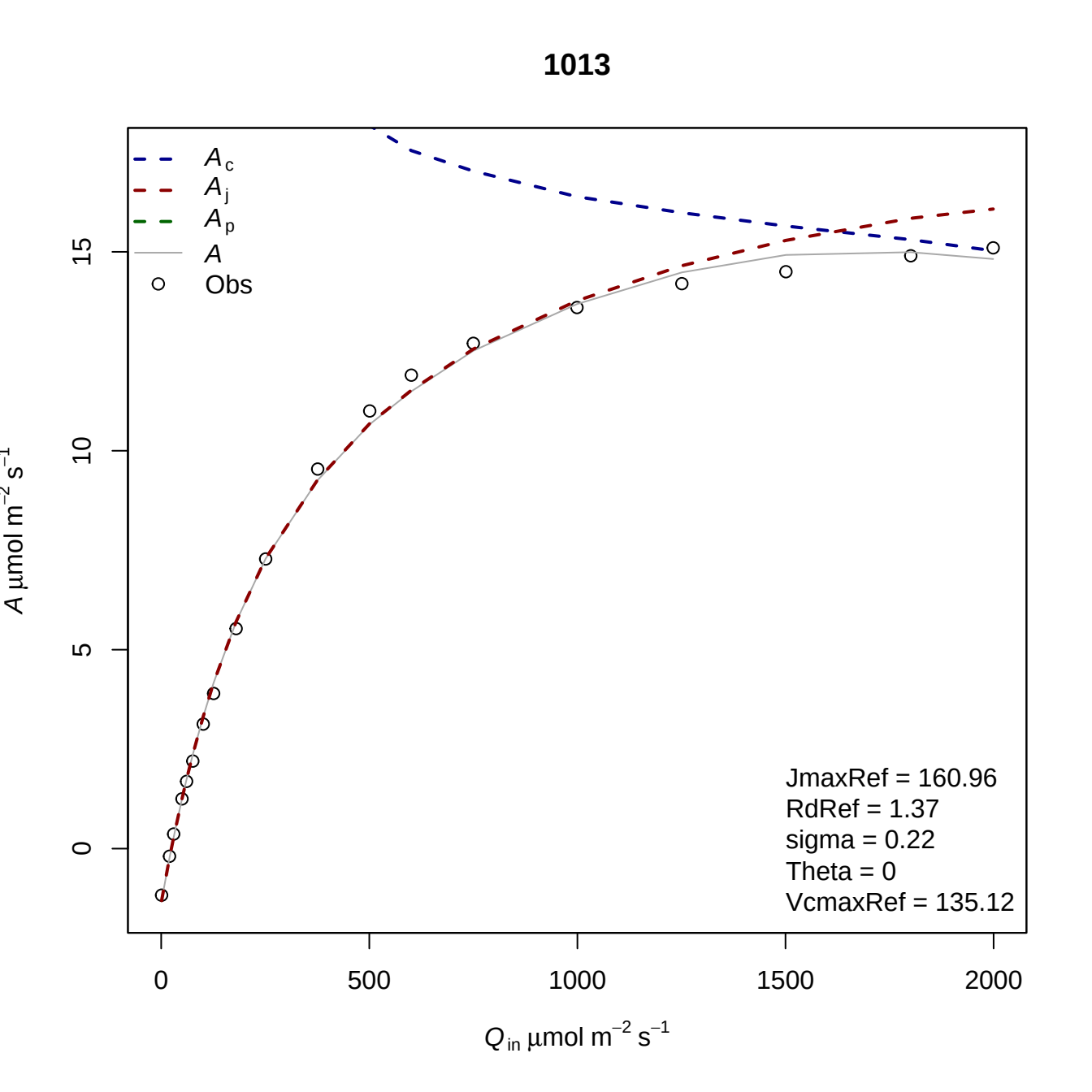

1008
A_c
A_j
A_p
A
Obs
JmaxRef = 73.05
RdRef = 1.59
sigma = 0.19
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

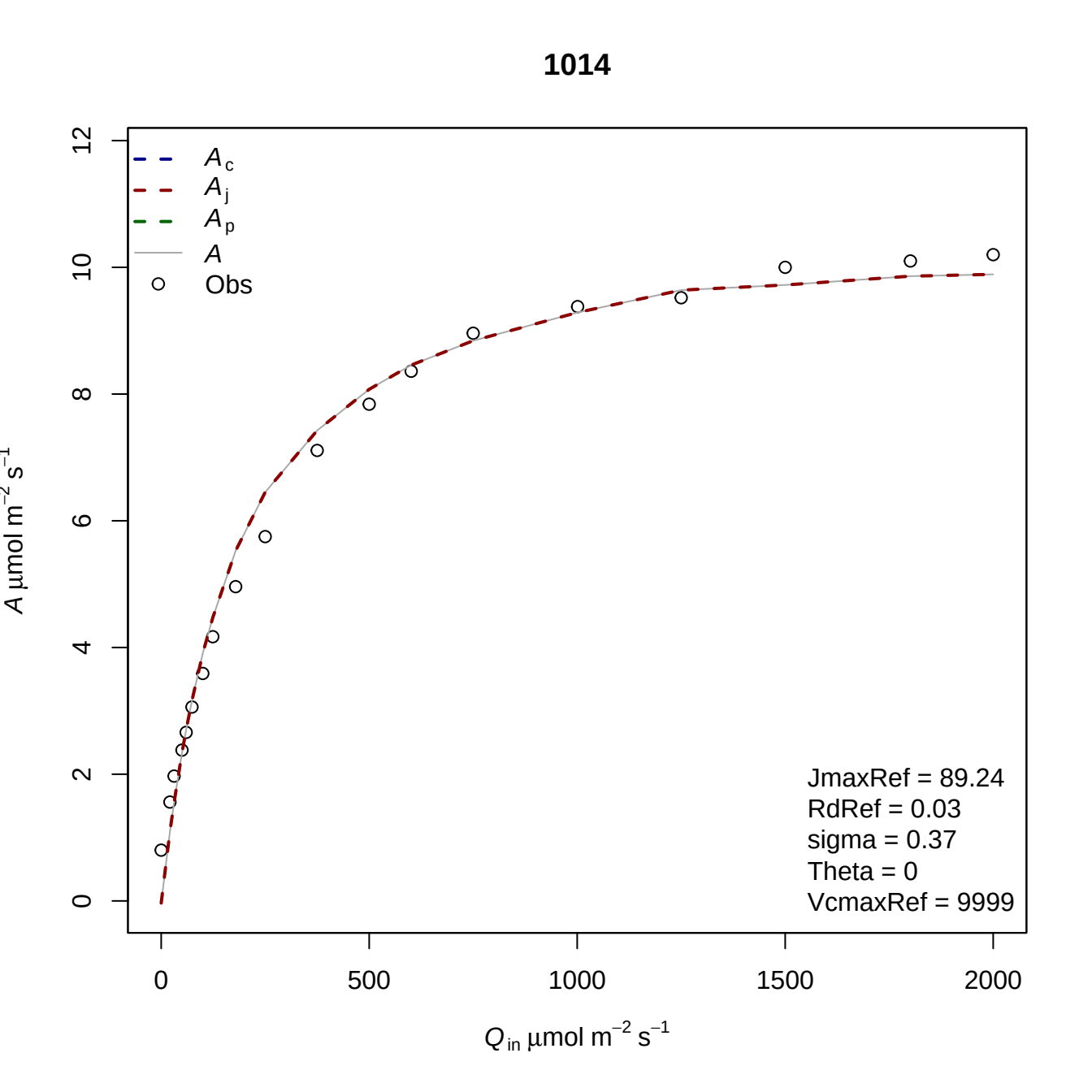

1009
A_c
A_j
A_p
A
Obs
JmaxRef = 82.96
RdRef = 2.18
sigma = 0.15
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

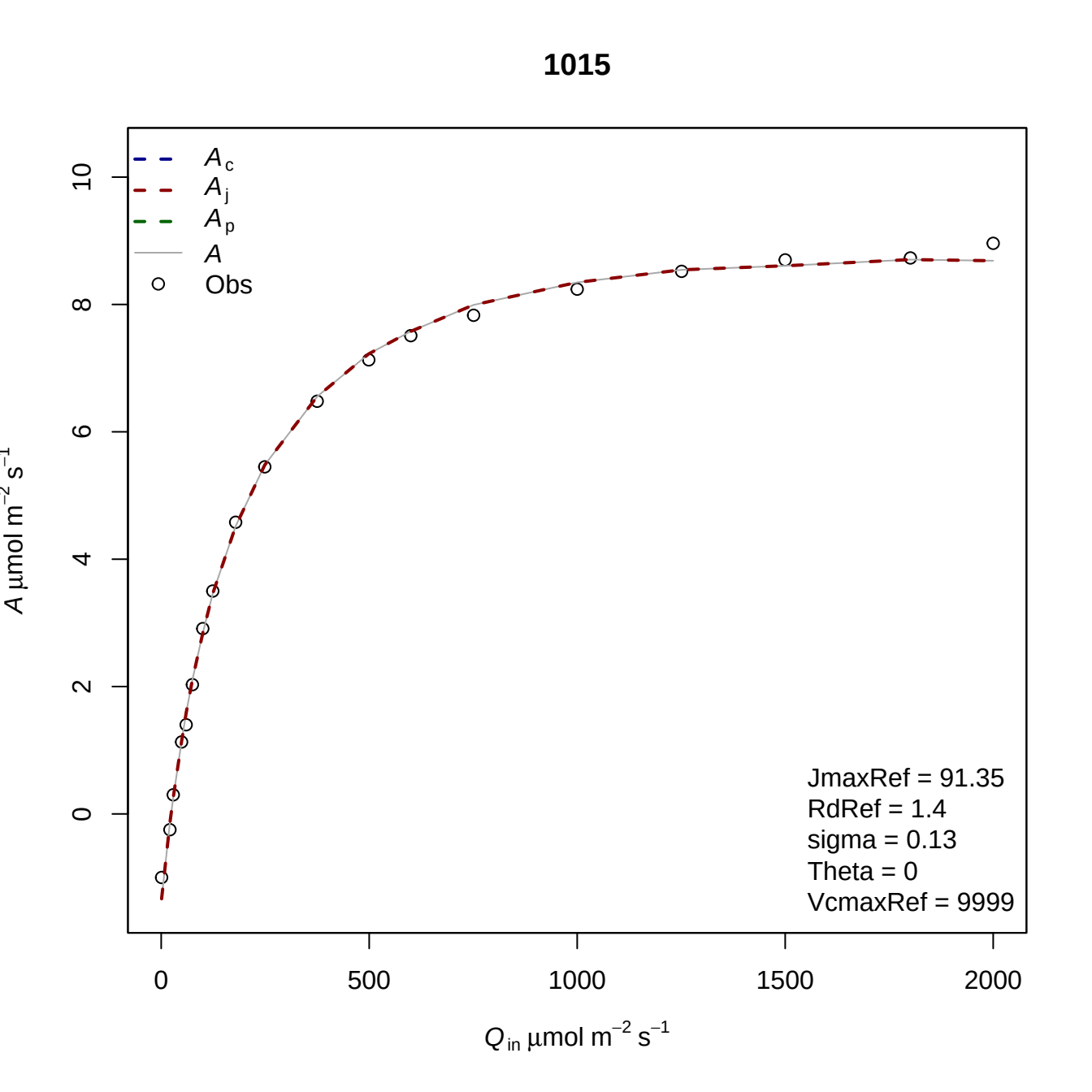

1010
A_c
A_j
A_p
A
Obs
JmaxRef = 79.03
RdRef = 1.6
sigma = 0.18
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

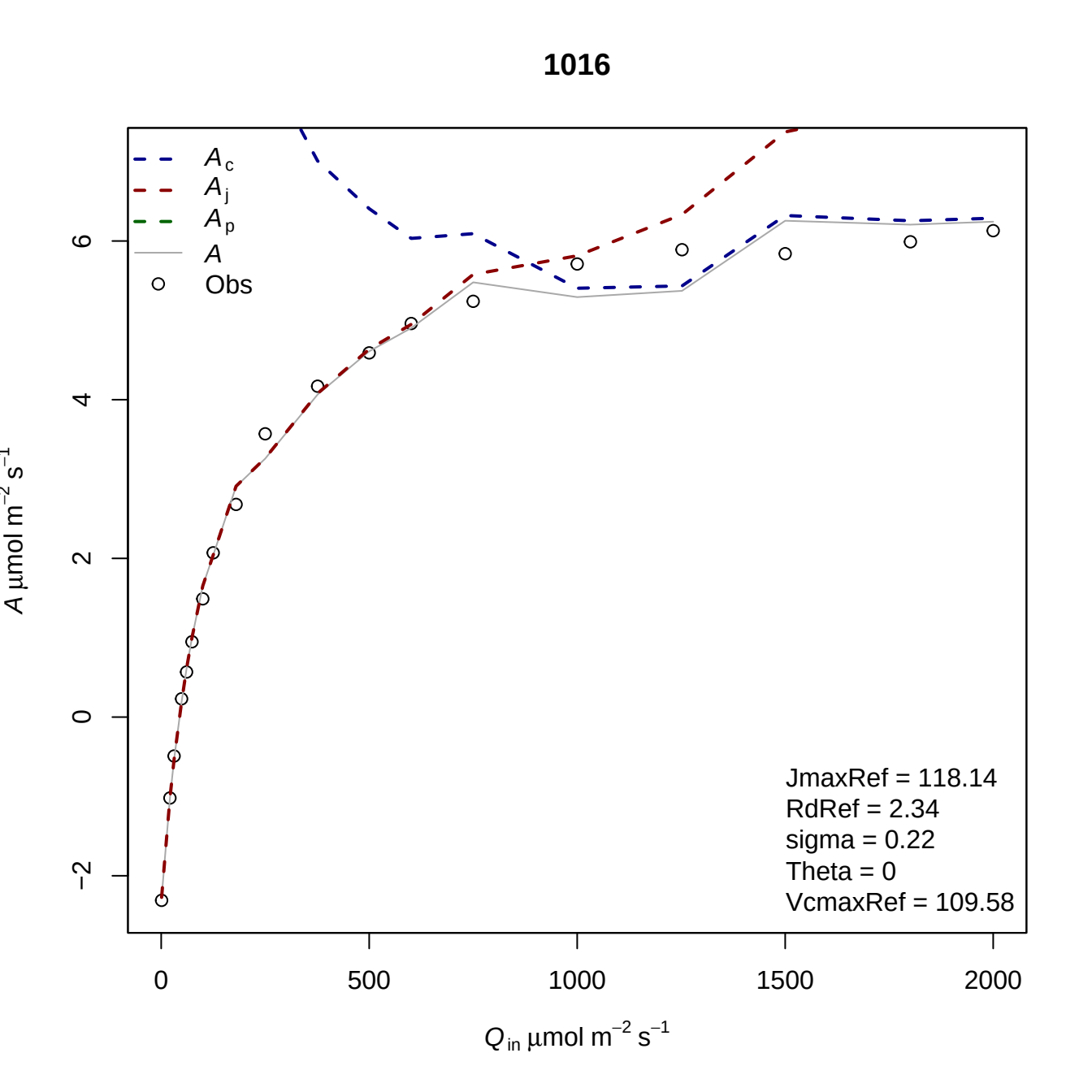

1011
A_c
A_j
A_p
A
Obs
JmaxRef = 42.93
RdRef = 0.65
sigma = 0.17
Theta = 0
VcmaxRef = 41.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

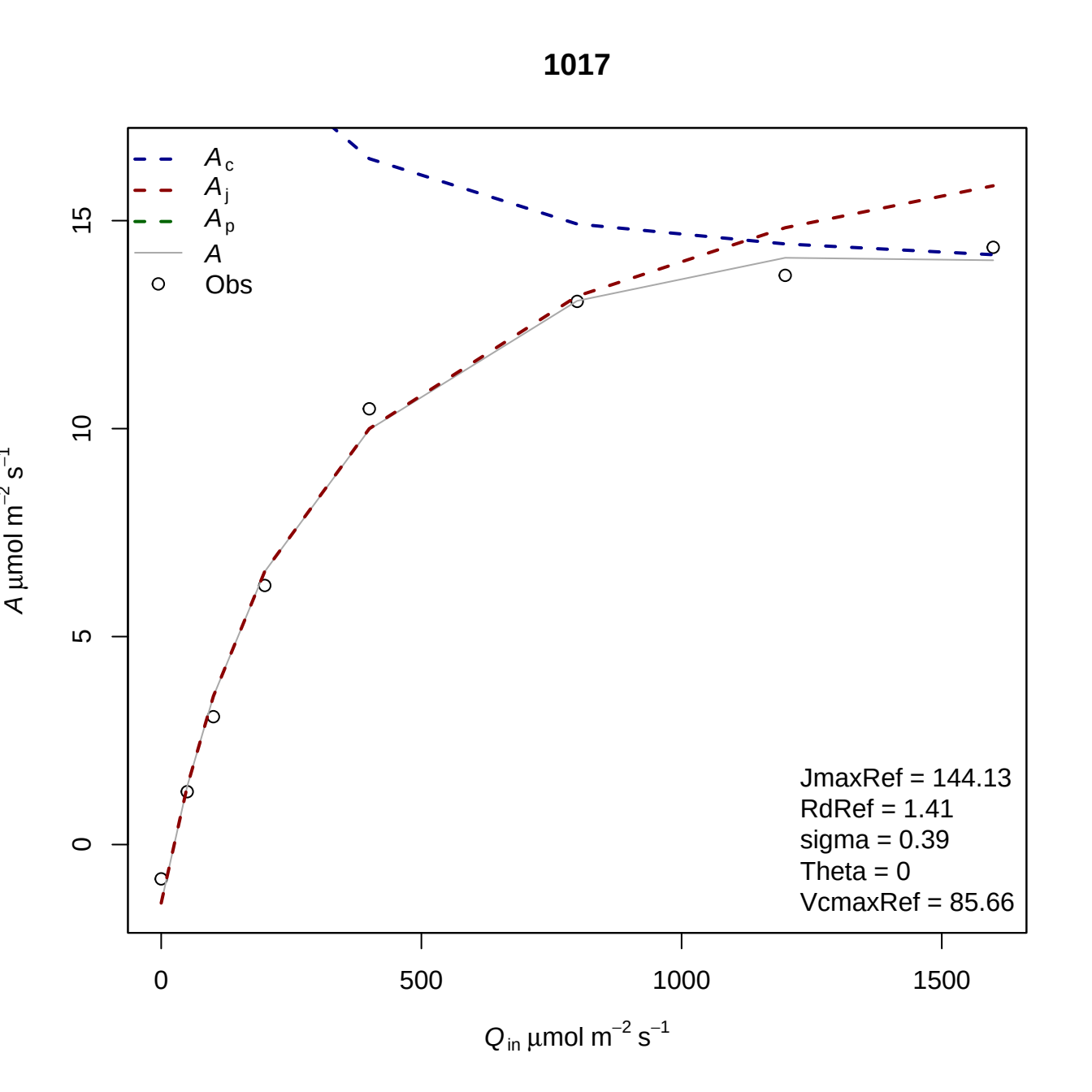

1012
A_c
A_j
A_p
A
Obs
JmaxRef = 84.67
RdRef = 1.63
sigma = 0.23
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

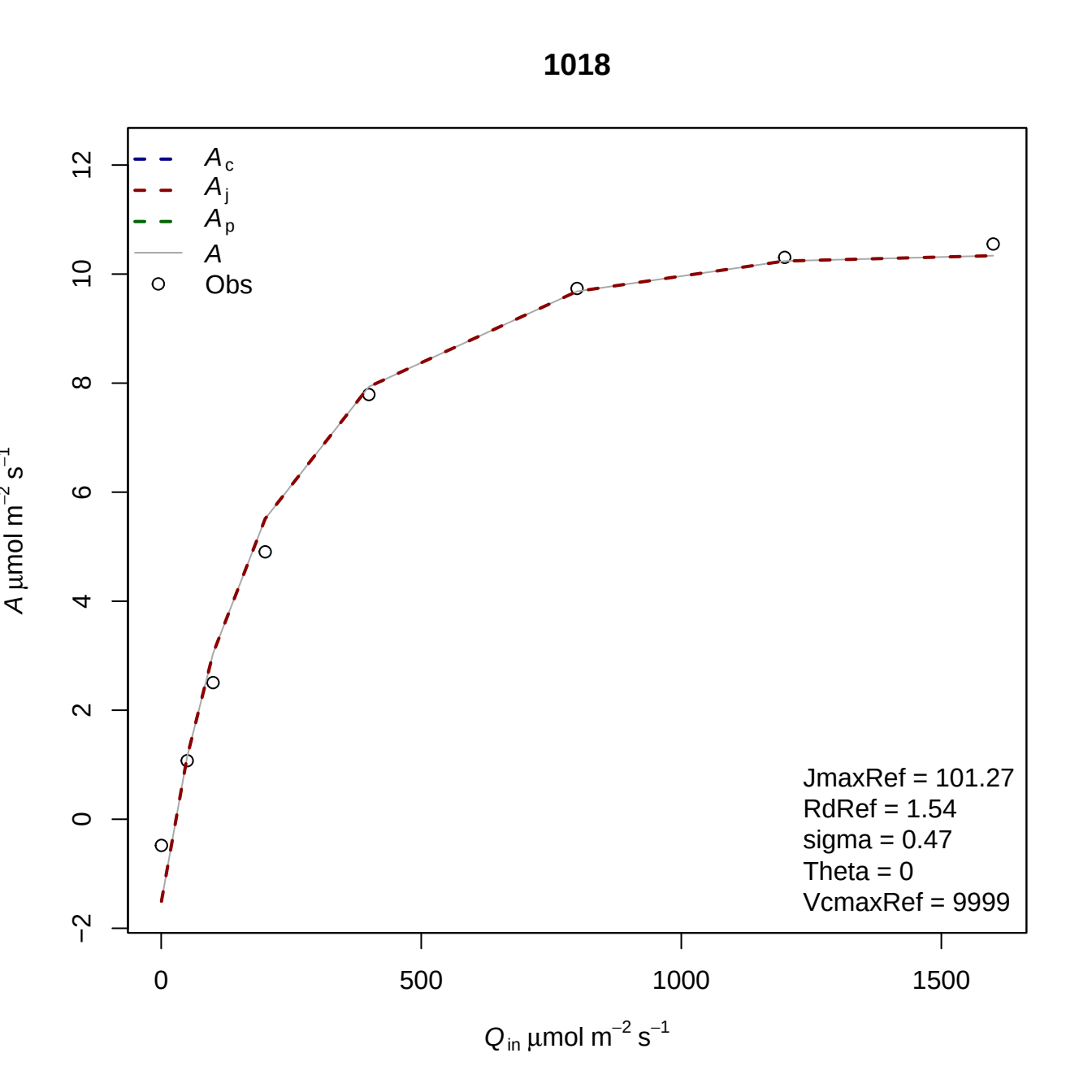

1013
A_c
A_j
A_p
A
Obs
JmaxRef = 160.96
RdRef = 1.37
sigma = 0.22
Theta = 0
VcmaxRef = 135.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

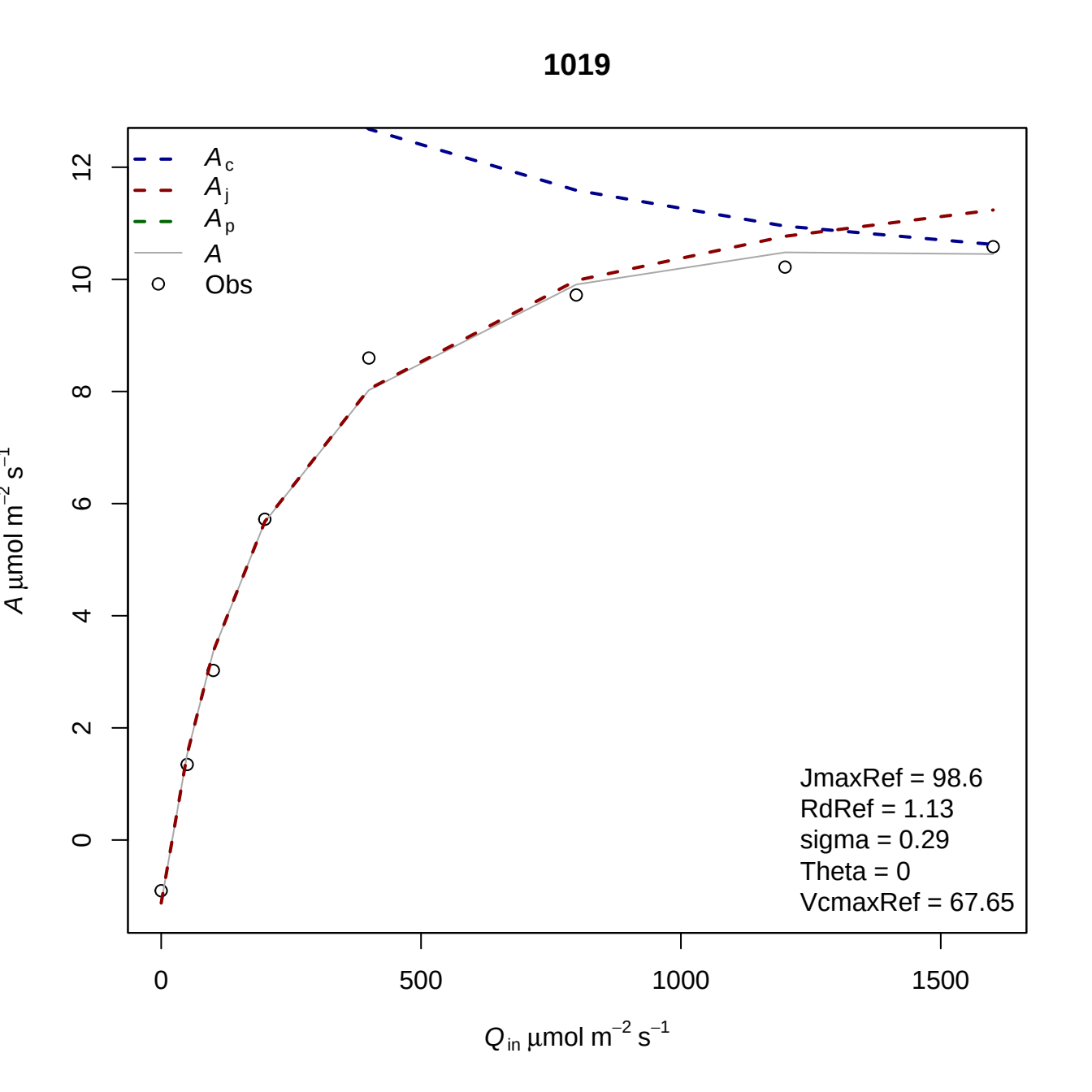

1014
A_c
A_j
A_p
A
Obs
JmaxRef = 89.24
RdRef = 0.03
sigma = 0.37
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

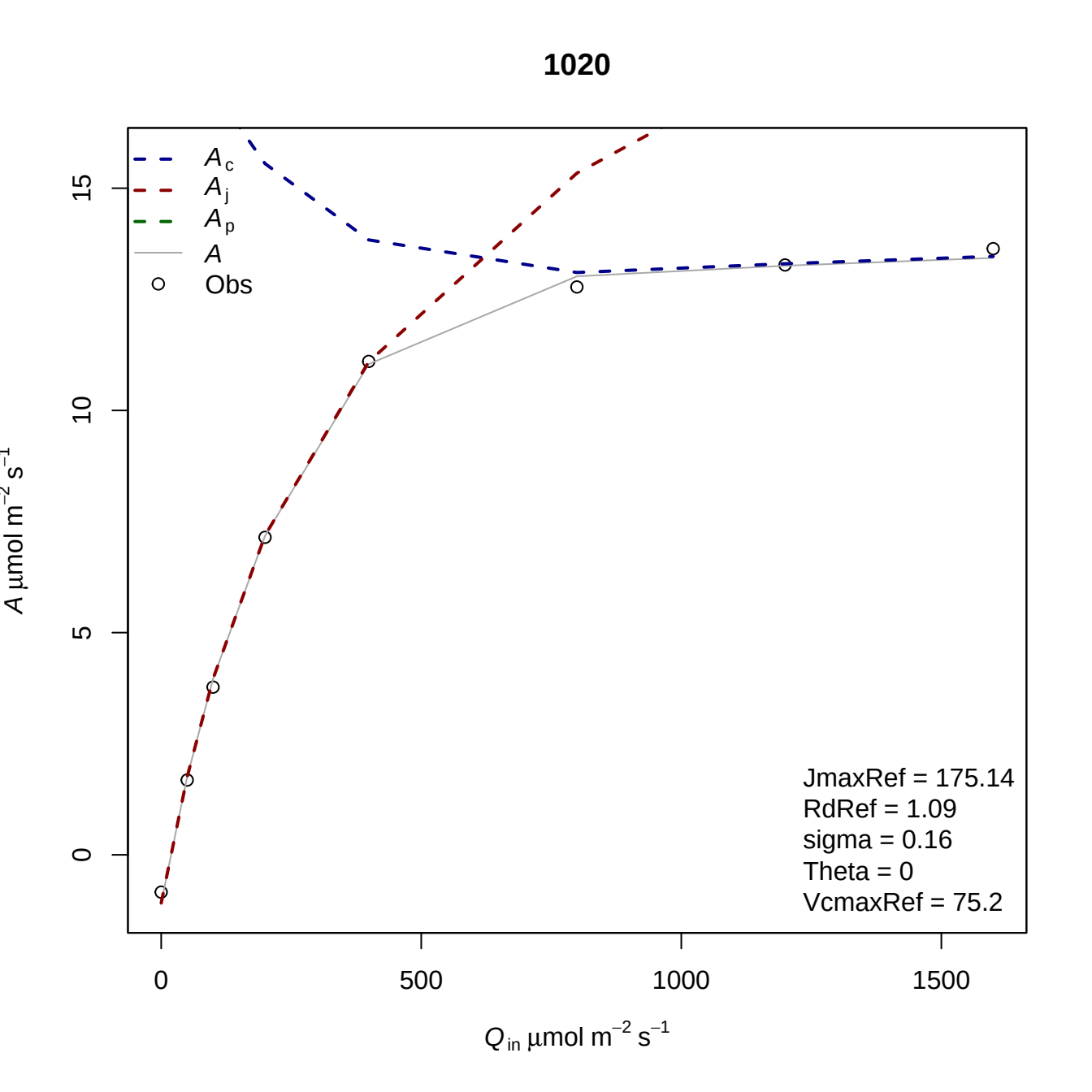

1015
A_c
A_j
A_p
A
Obs
JmaxRef = 91.35
RdRef = 1.4
sigma = 0.13
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

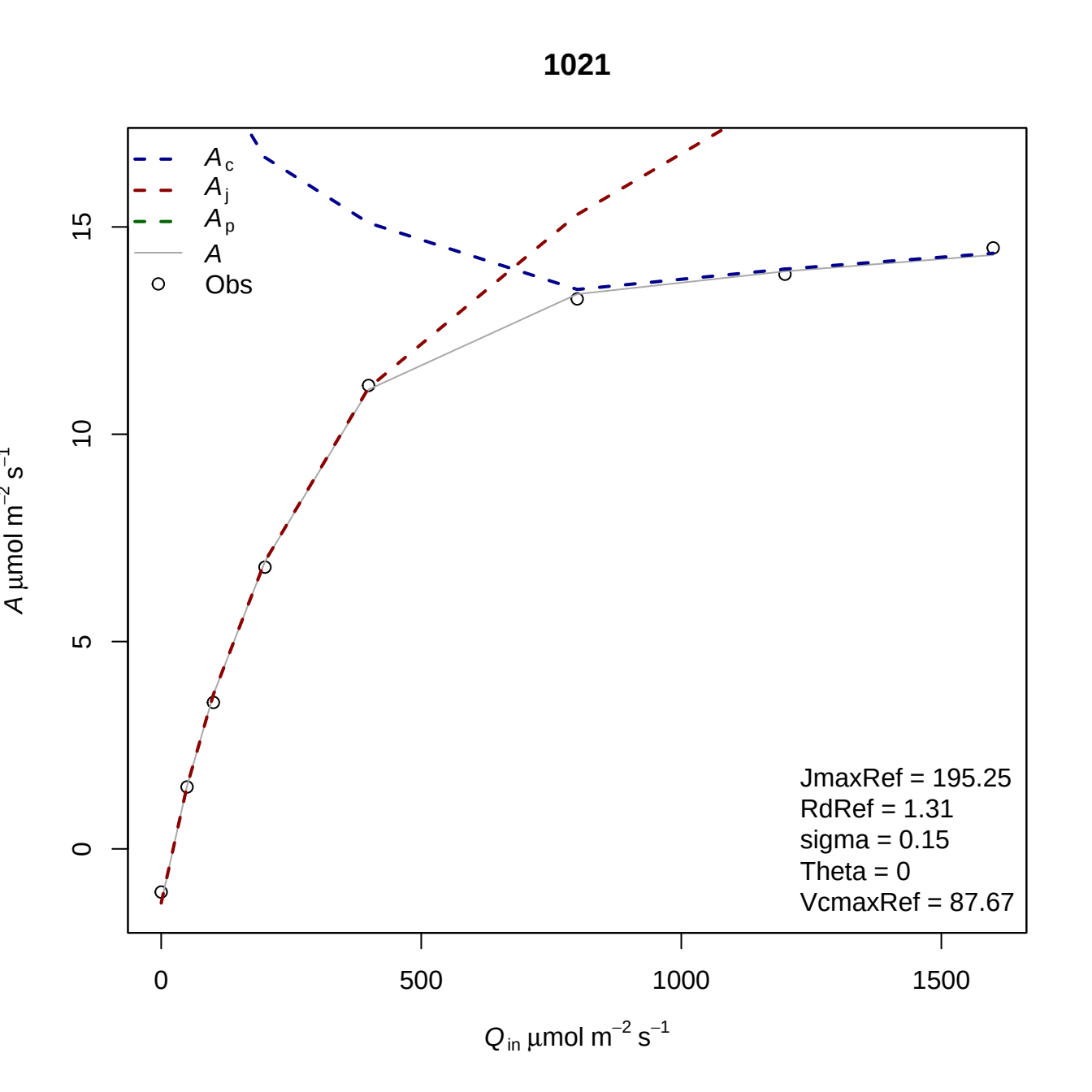

1016
A_c
A_j
A_p
A
Obs
JmaxRef = 118.14
RdRef = 2.34
sigma = 0.22
Theta = 0
VcmaxRef = 109.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

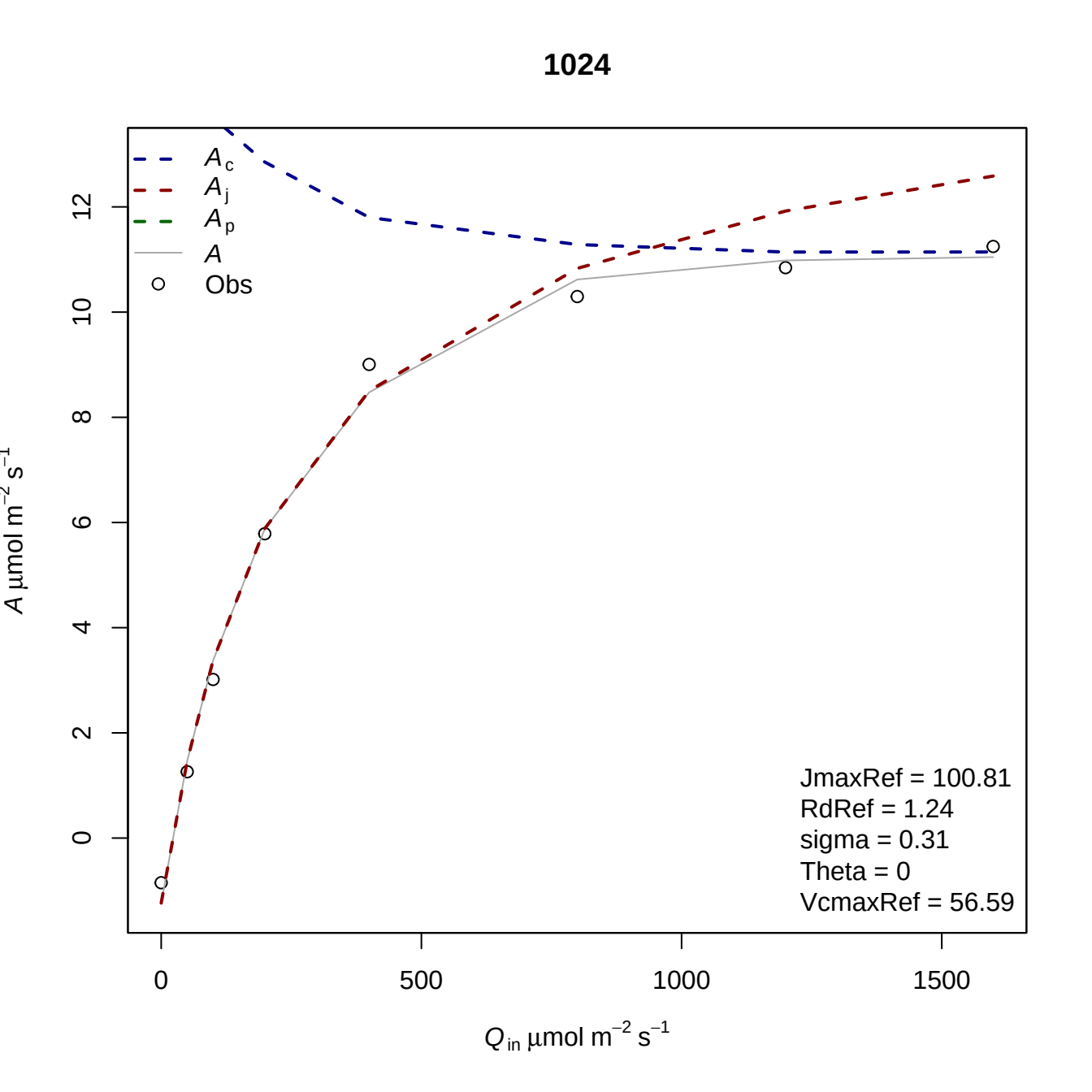

1017
A_c
A_j
A_p
A
Obs
JmaxRef = 144.13
RdRef = 1.41
sigma = 0.39
Theta = 0
VcmaxRef = 85.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

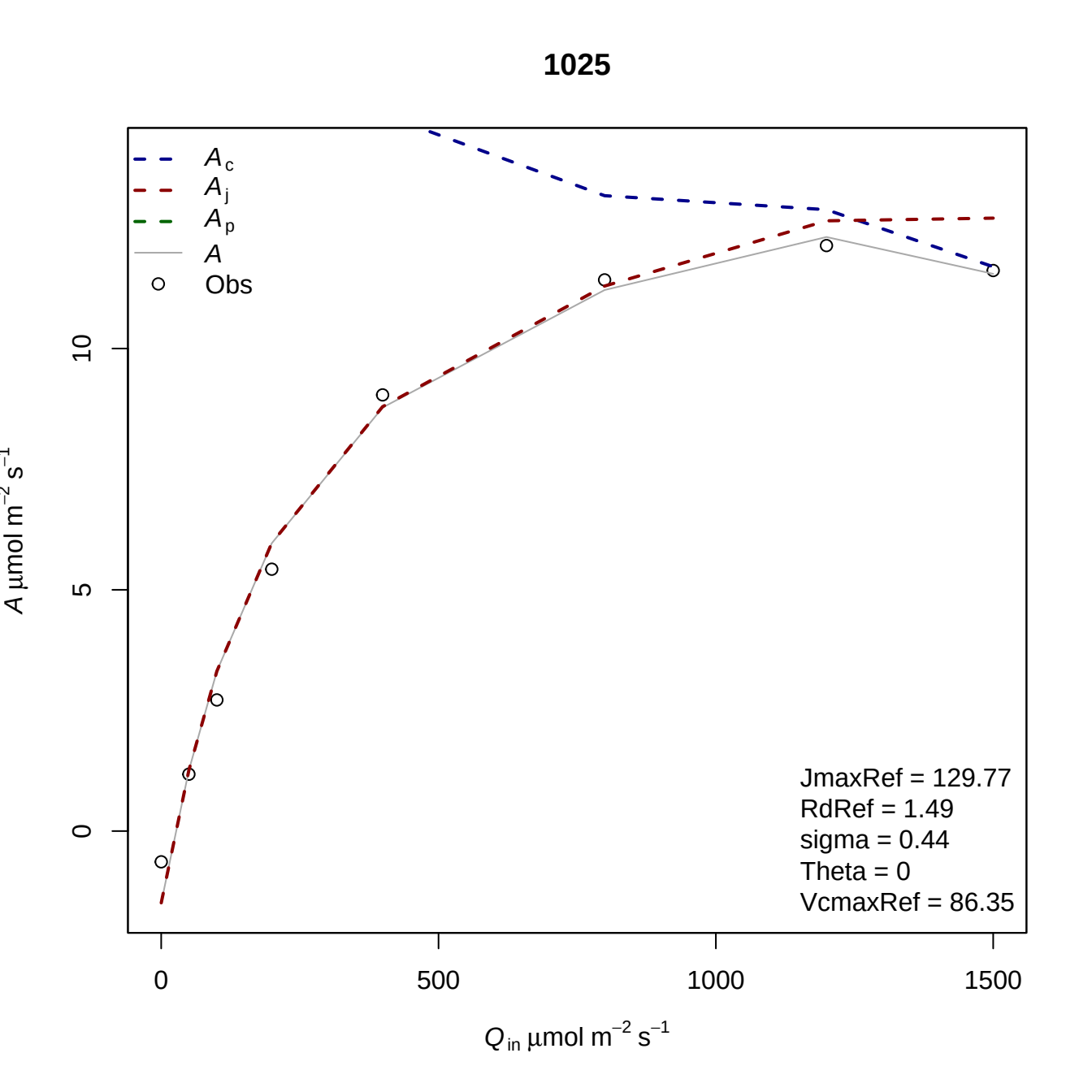

1018
A_c
A_j
A_p
A
Obs
JmaxRef = 101.27
RdRef = 1.54
sigma = 0.47
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

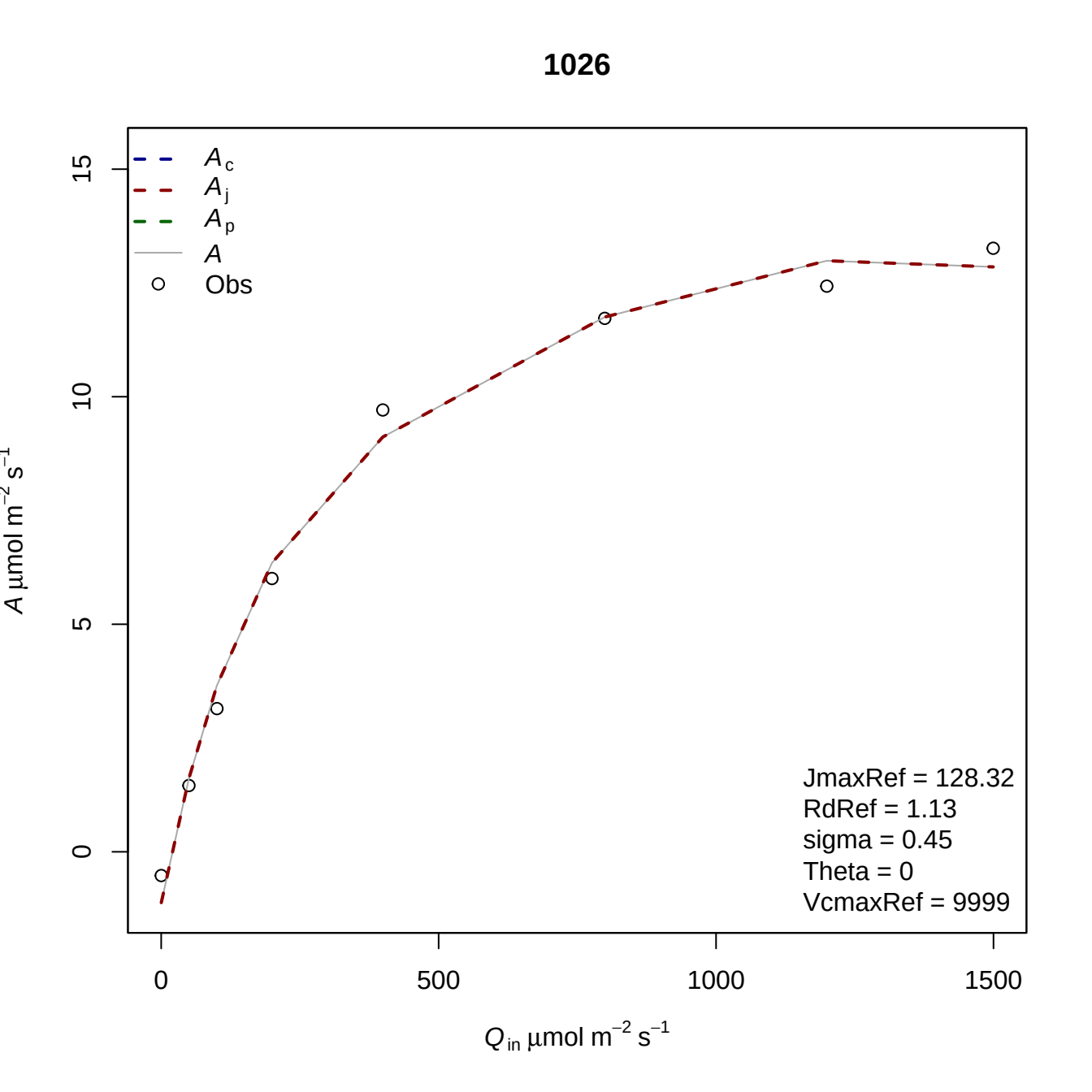

1019
A_c
A_j
A_p
A
Obs
JmaxRef = 98.6
RdRef = 1.13
sigma = 0.29
Theta = 0
VcmaxRef = 67.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

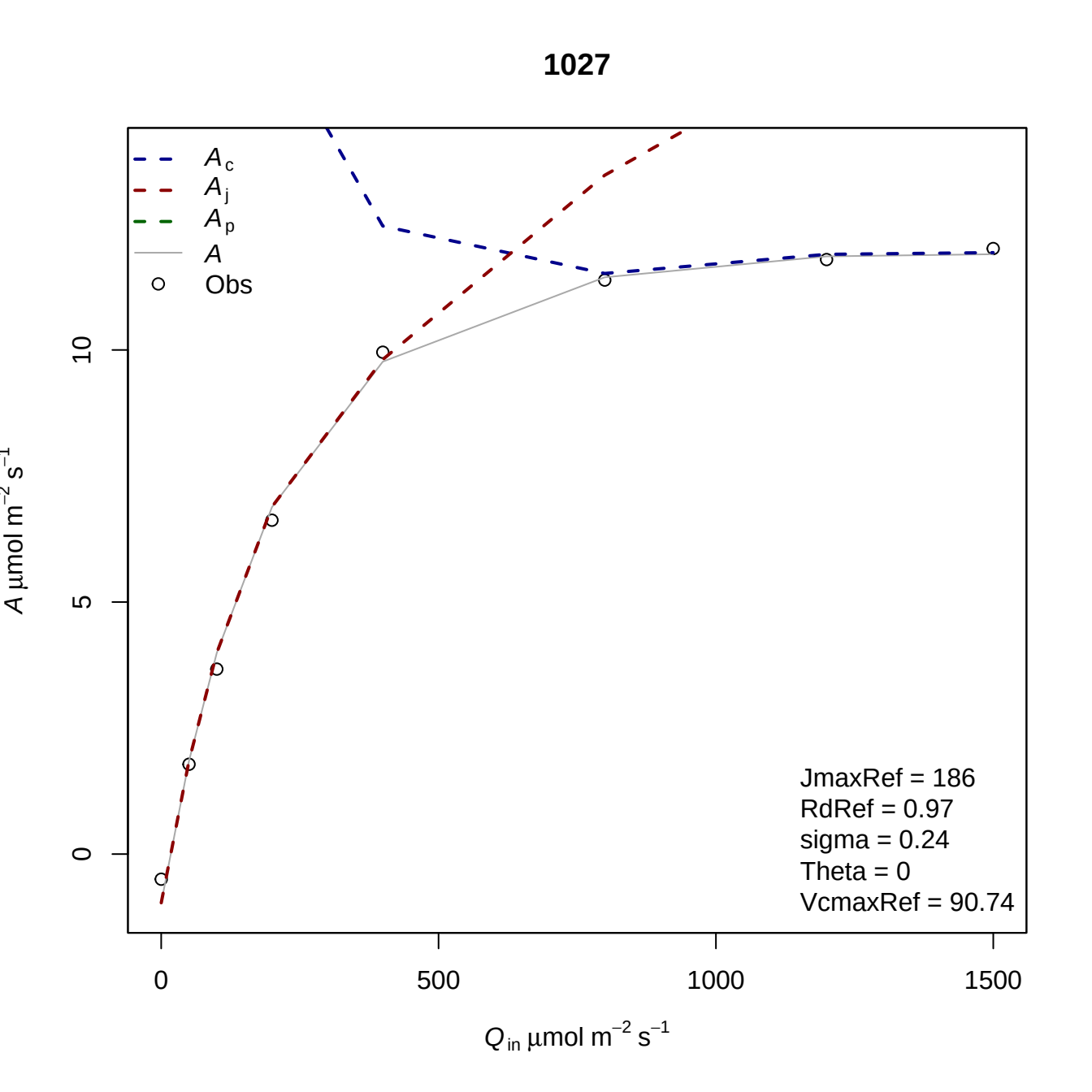

1020
A_c
A_j
A_p
A
Obs
JmaxRef = 175.14
RdRef = 1.09
sigma = 0.16
Theta = 0
VcmaxRef = 75.2
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

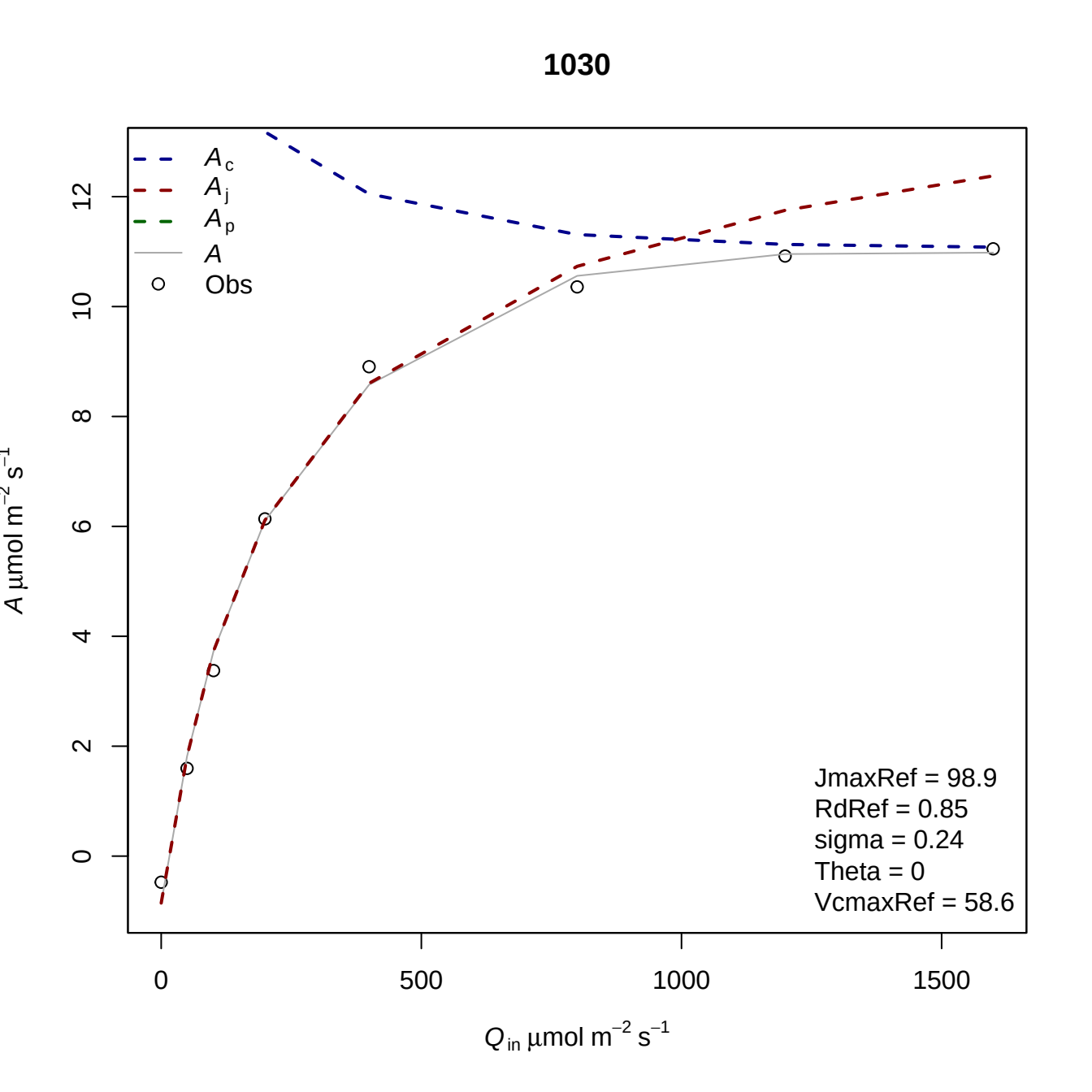

1021
A_c
A_j
A_p
A
Obs
JmaxRef = 195.25
RdRef = 1.31
sigma = 0.15
Theta = 0
VcmaxRef = 87.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

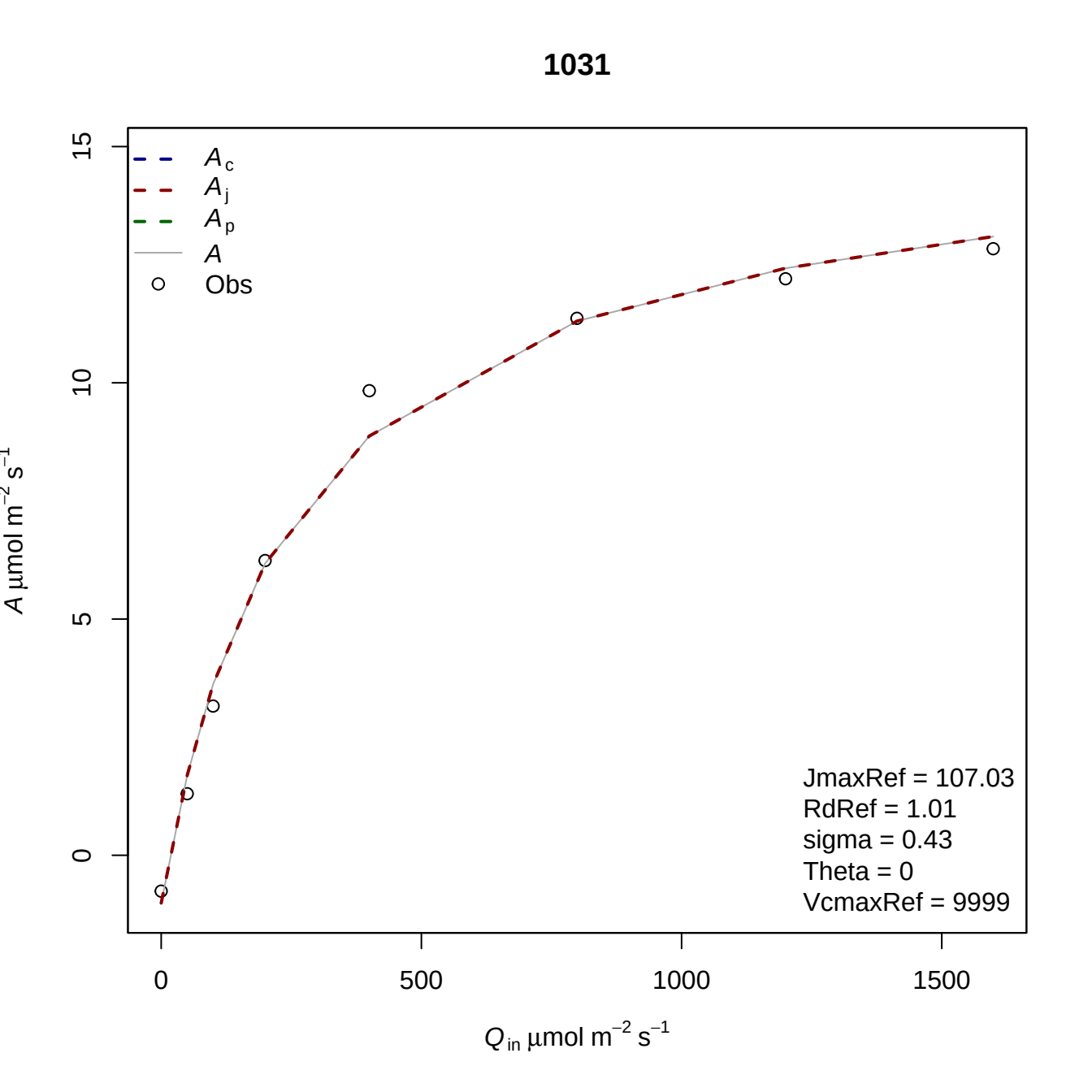

1024
A_c
A_j
A_p
A
Obs
JmaxRef = 100.81
RdRef = 1.24
sigma = 0.31
Theta = 0
VcmaxRef = 56.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

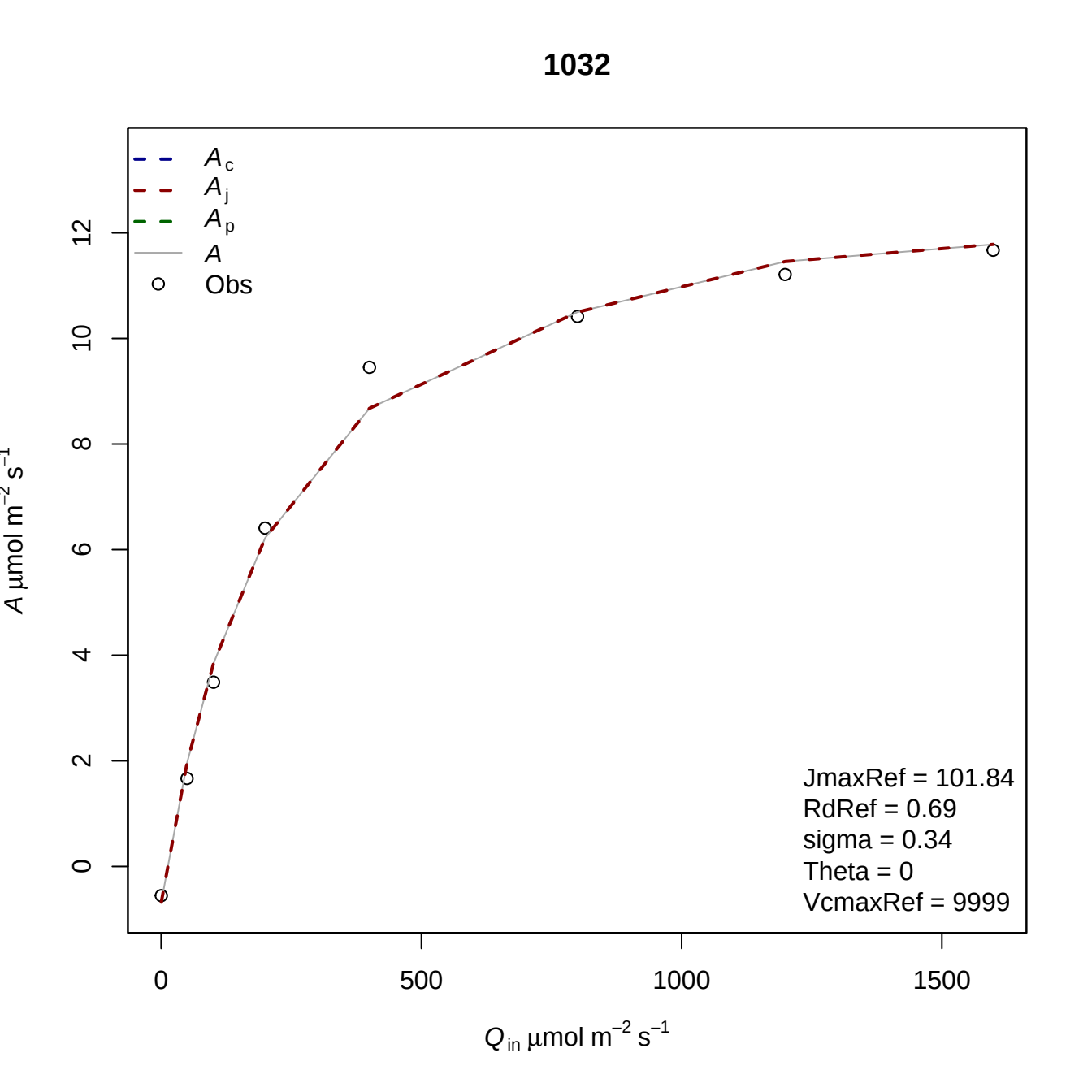

1025
A_c
A_j
A_p
A
Obs
JmaxRef = 129.77
RdRef = 1.49
sigma = 0.44
Theta = 0
VcmaxRef = 86.35
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

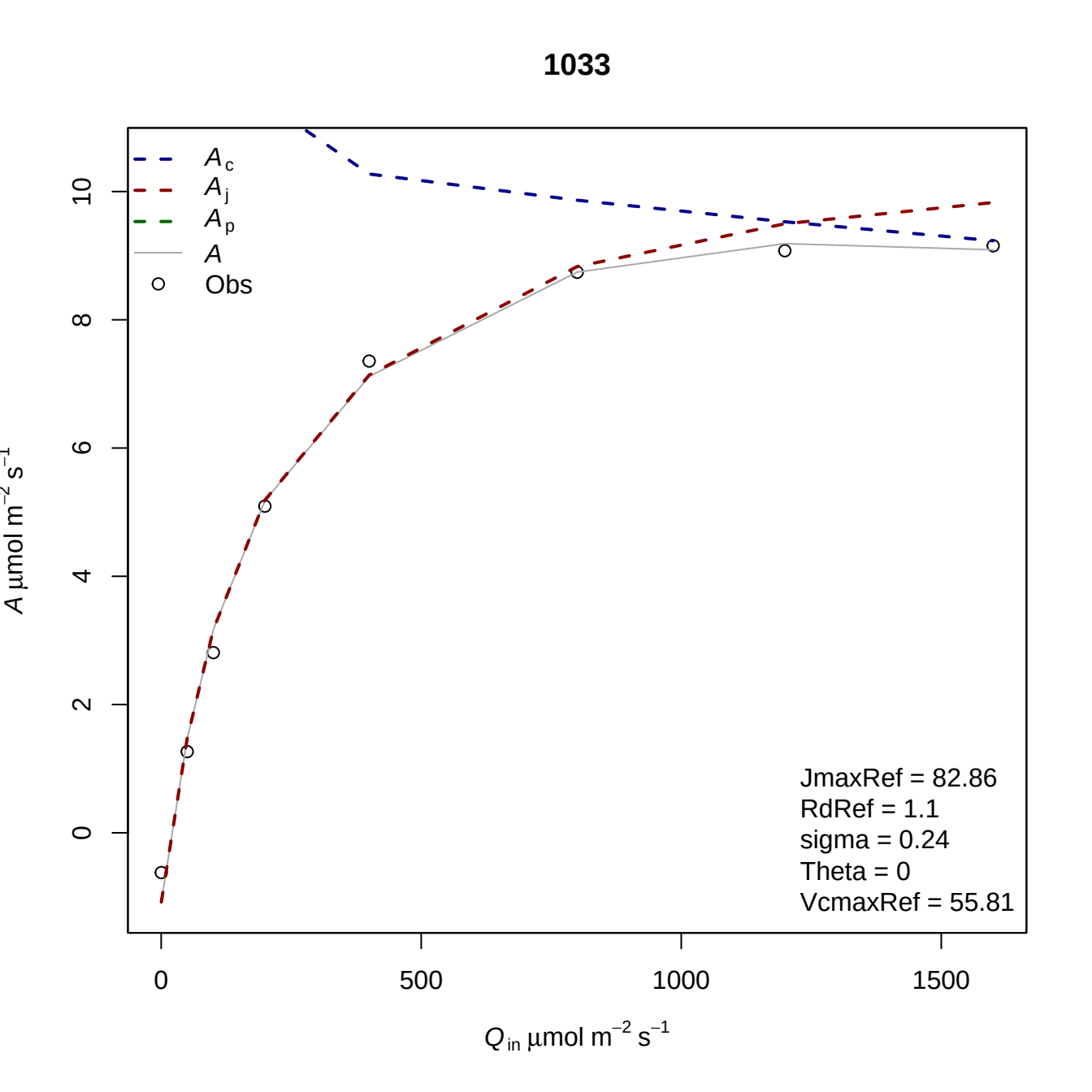

1026
A_c
A_j
A_p
A
Obs
JmaxRef = 128.32
RdRef = 1.13
sigma = 0.45
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

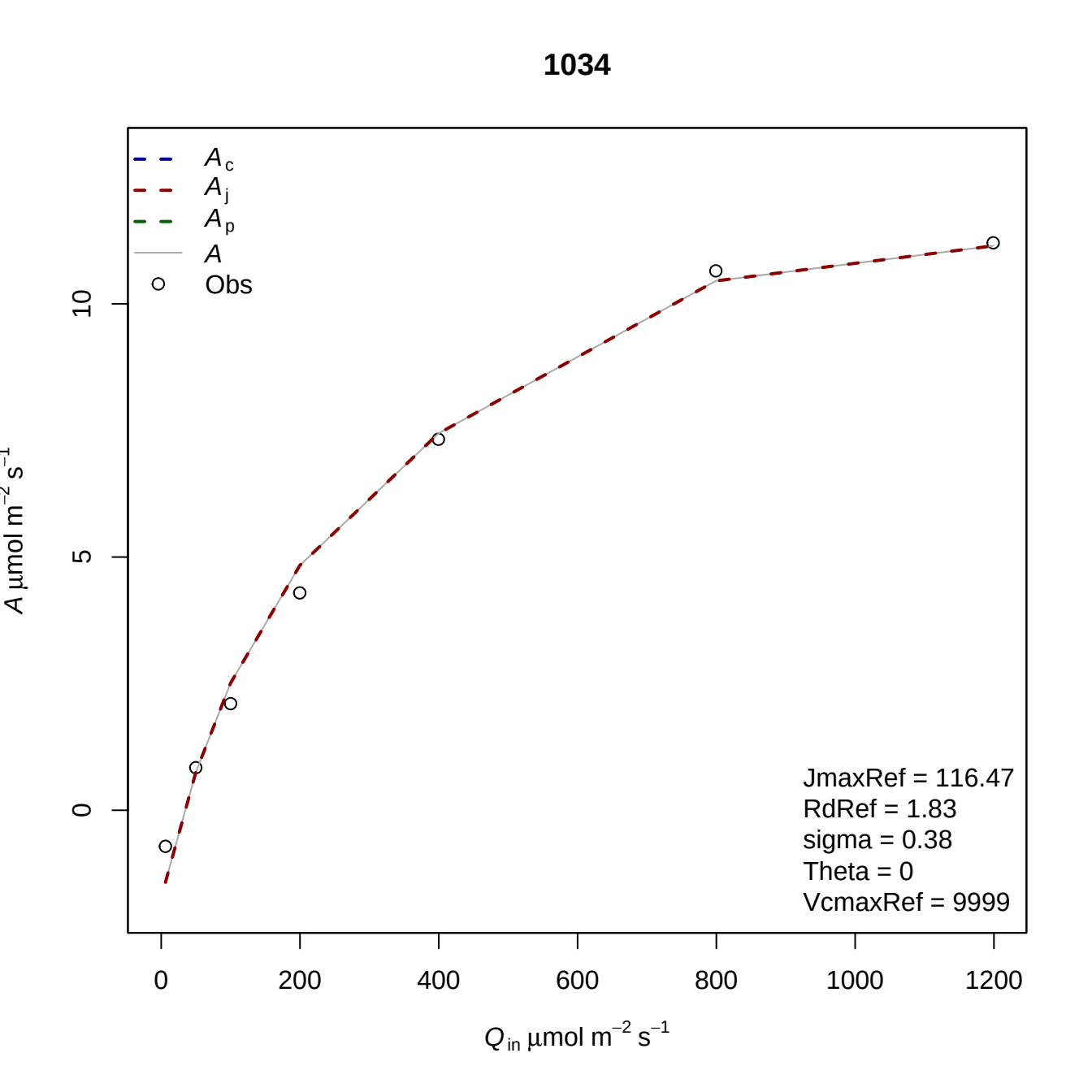

1027
A_c
A_j
A_p
A
Obs
JmaxRef = 186
RdRef = 0.97
sigma = 0.24
Theta = 0
VcmaxRef = 90.74
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

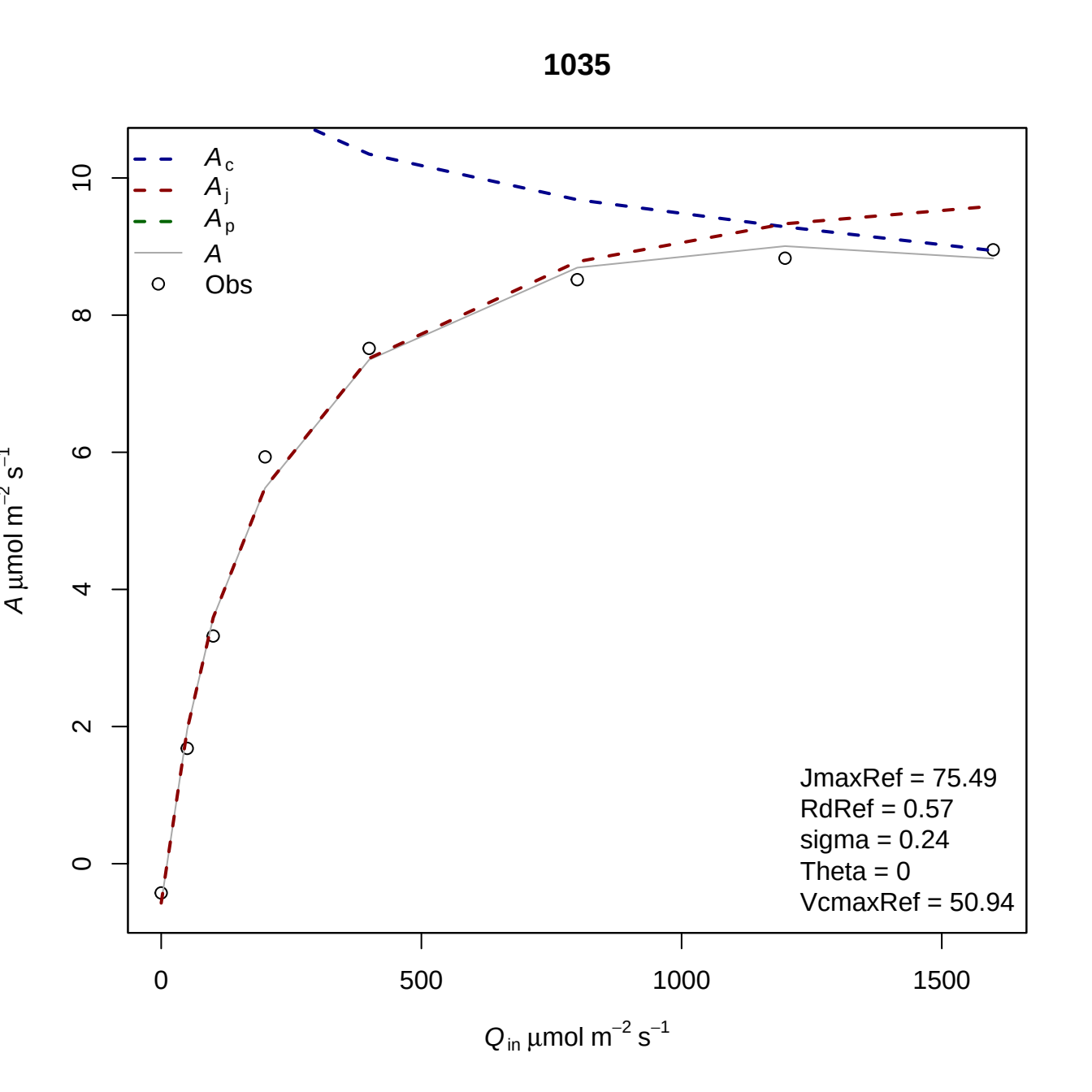

1030
A_c
A_j
A_p
A
Obs
JmaxRef = 98.9
RdRef = 0.85
sigma = 0.24
Theta = 0
VcmaxRef = 58.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

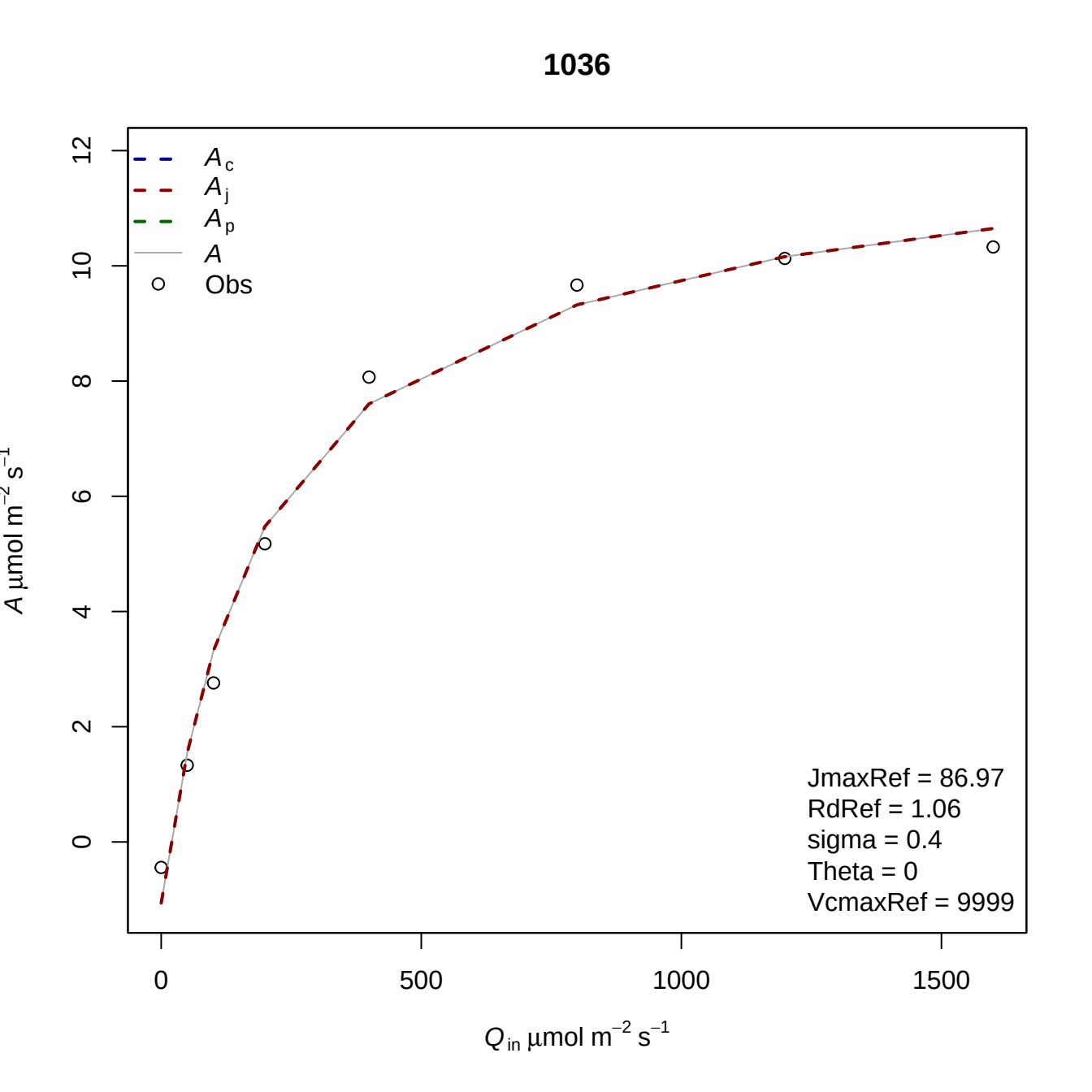

1031
A_c
A_j
A_p
A
Obs
JmaxRef = 107.03
RdRef = 1.01
sigma = 0.43
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

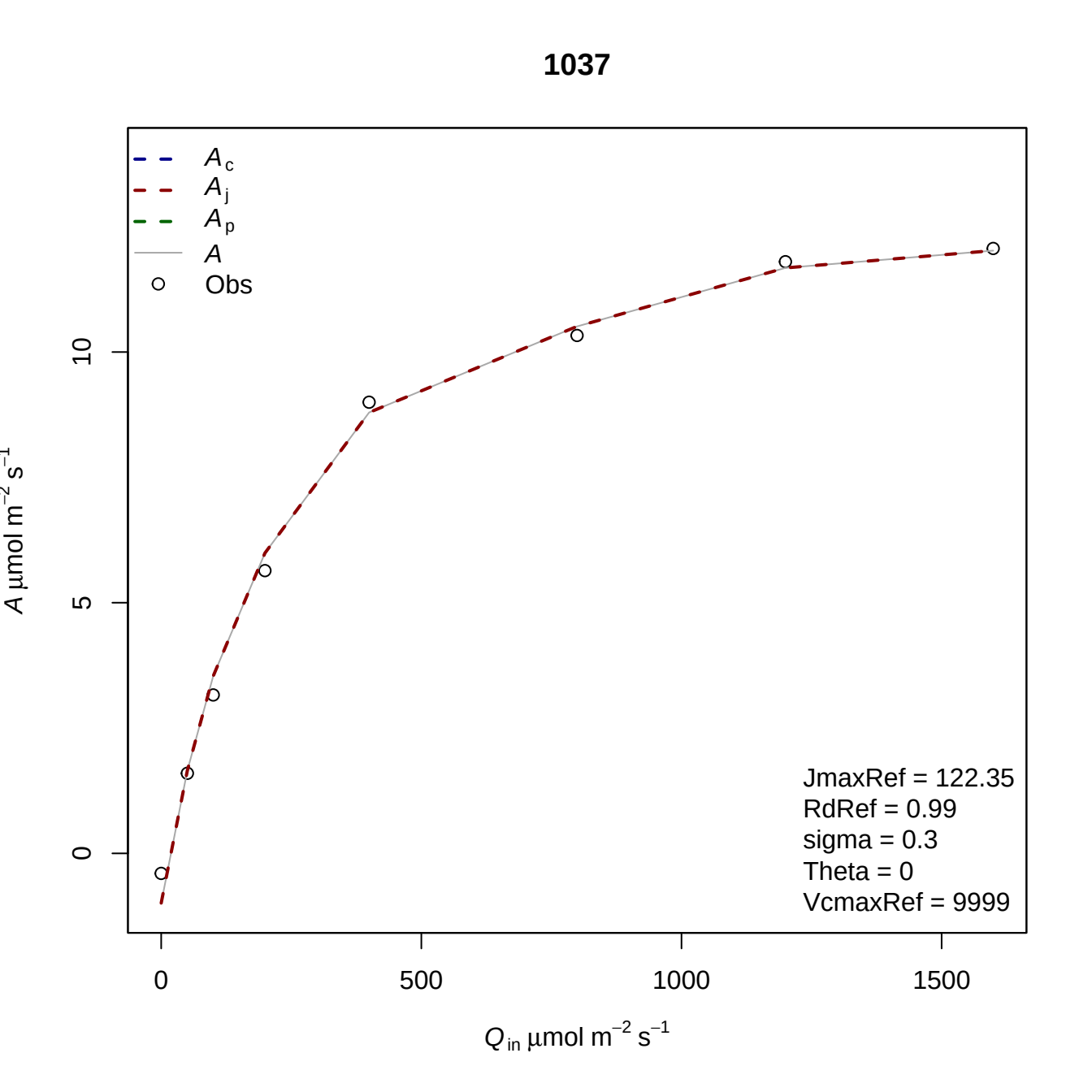

1032
A_c
A_j
A_p
A
Obs
JmaxRef = 101.84
RdRef = 0.69
sigma = 0.34
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

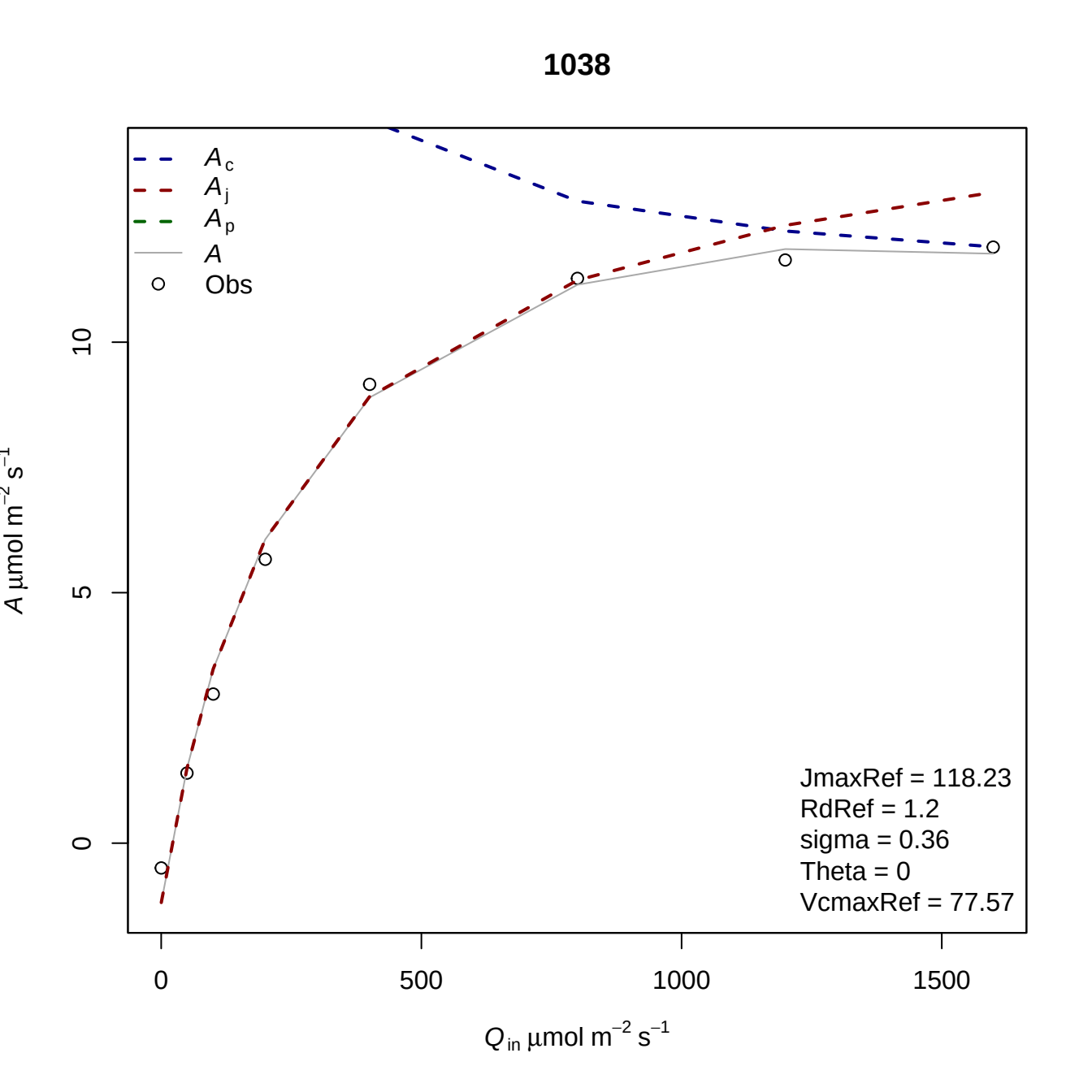

1033
A_c
A_j
A_p
A
Obs
JmaxRef = 82.86
RdRef = 1.1
sigma = 0.24
Theta = 0
VcmaxRef = 55.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

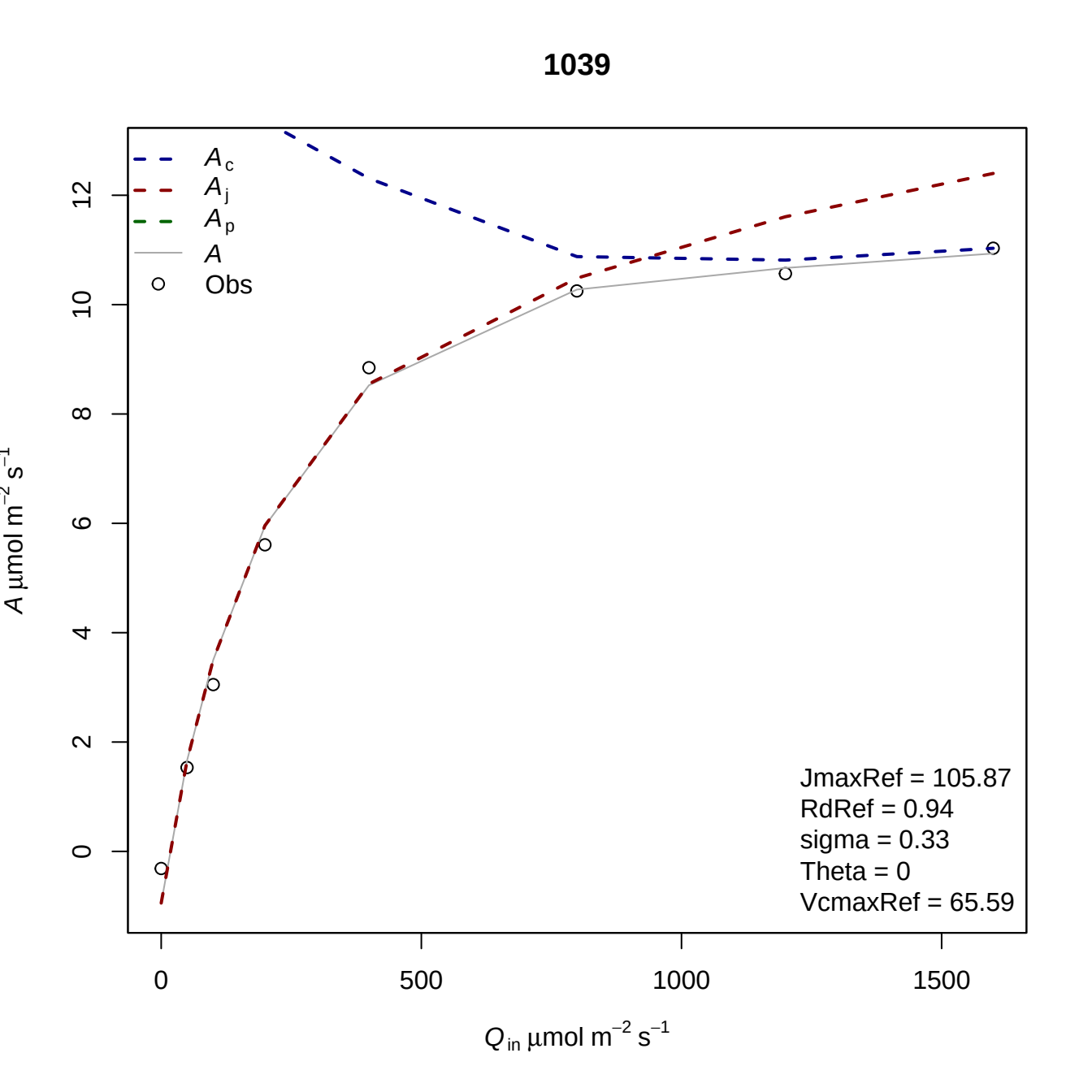

1034
A_c
A_j
A_p
A
Obs
JmaxRef = 116.47
RdRef = 1.83
sigma = 0.38
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

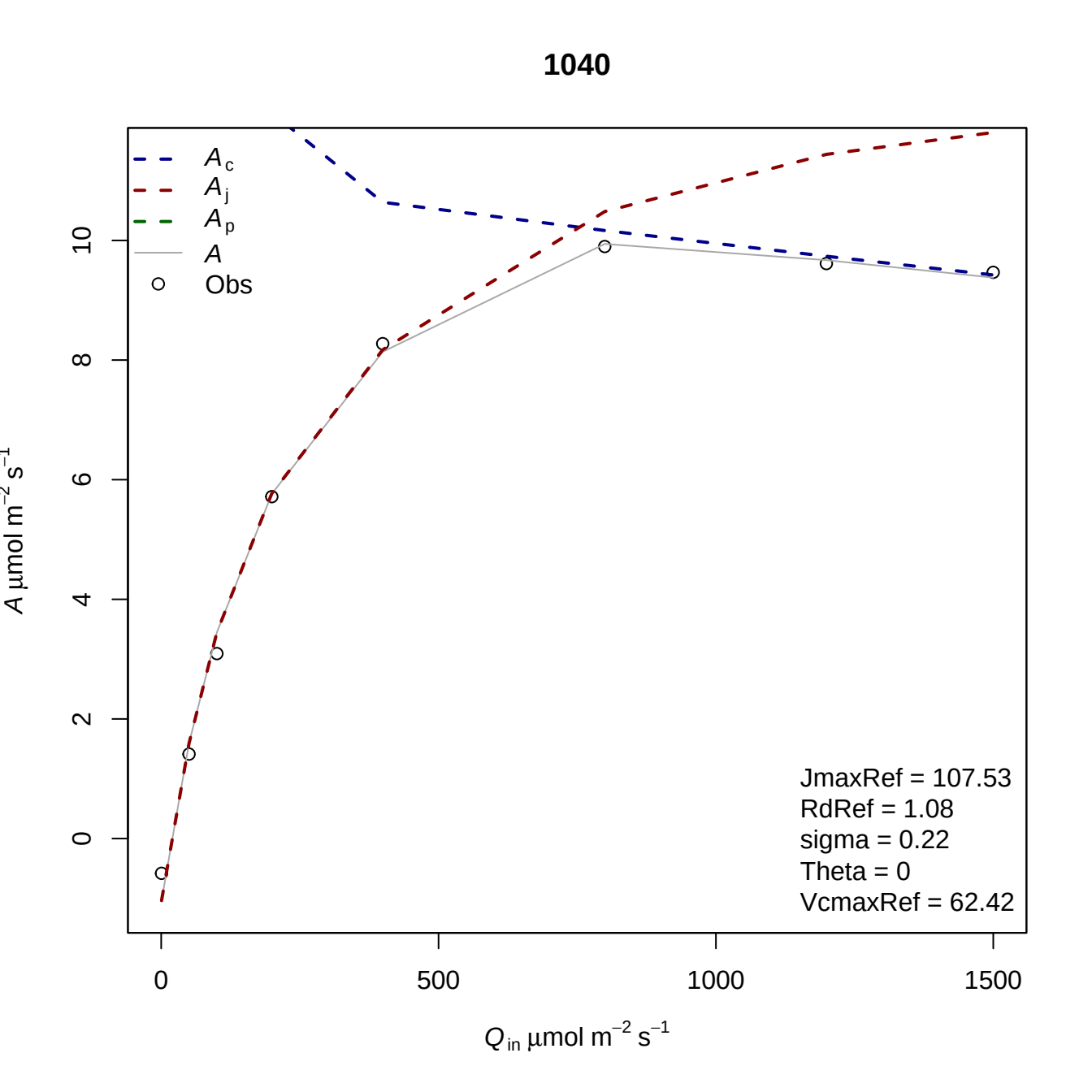

1035
A_c
A_j
A_p
A
Obs
JmaxRef = 75.49
RdRef = 0.57
sigma = 0.24
Theta = 0
VcmaxRef = 50.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

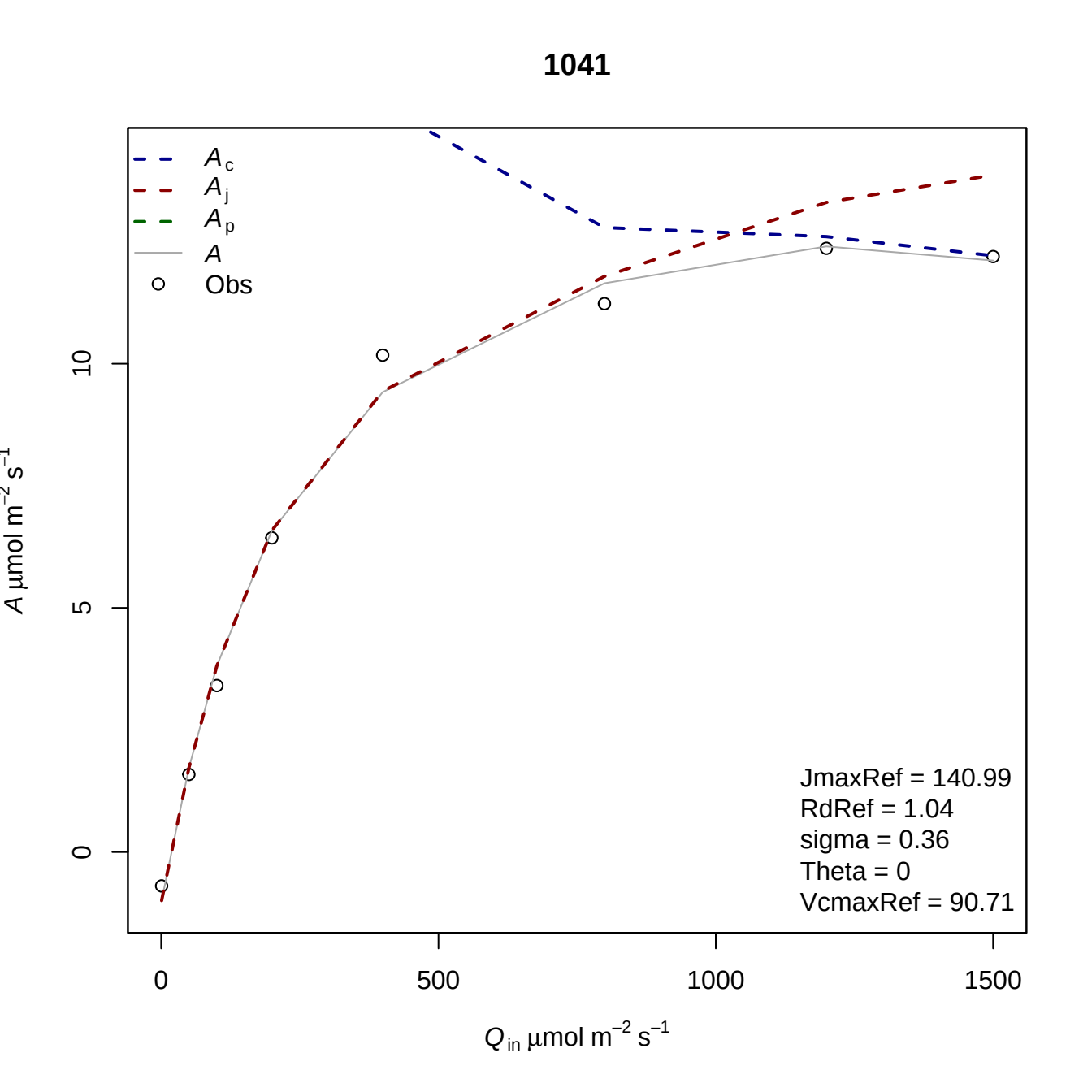

1036
A_c
A_j
A_p
A
Obs
JmaxRef = 86.97
RdRef = 1.06
sigma = 0.4
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

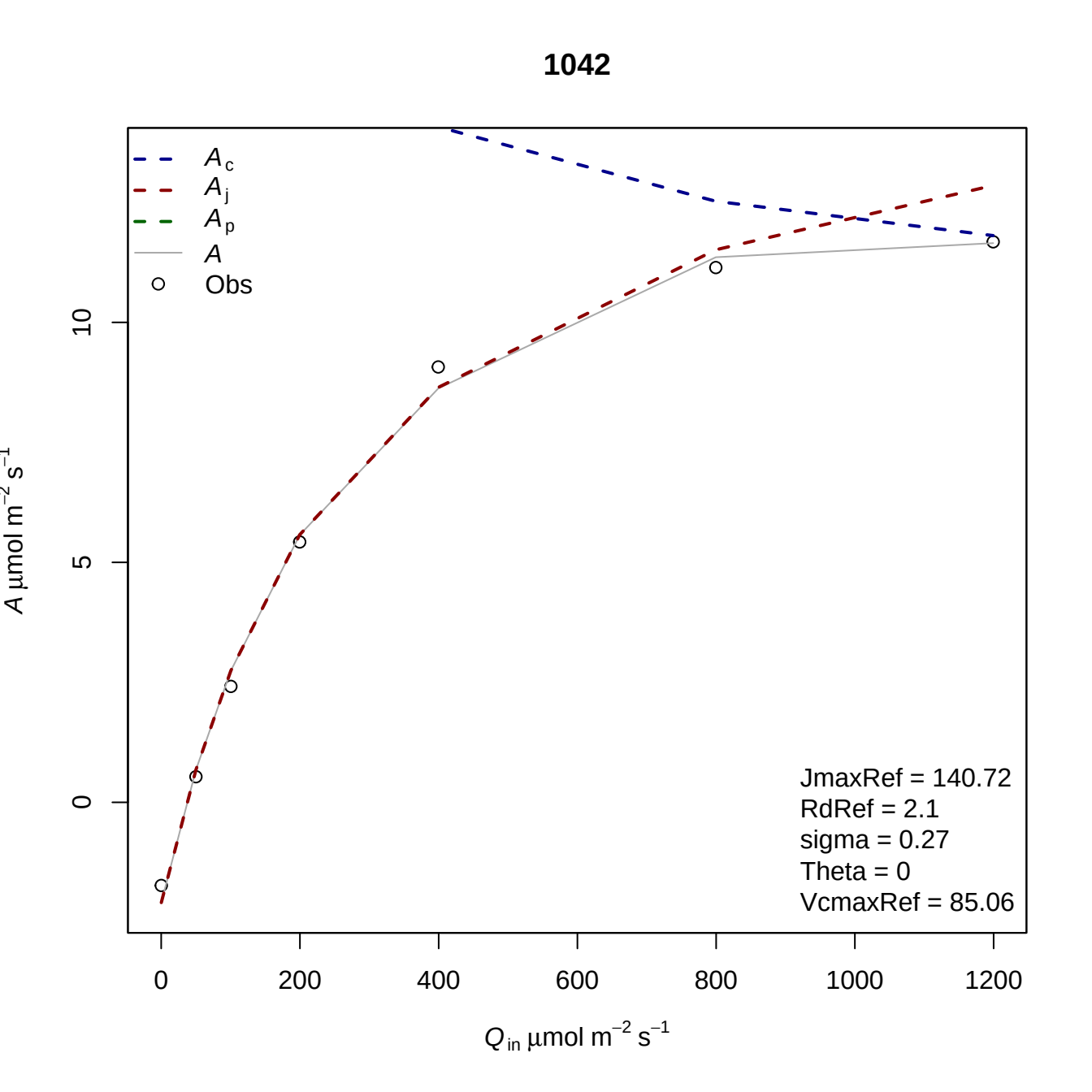

1037
A_c
A_j
A_p
A
Obs
JmaxRef = 122.35
RdRef = 0.99
sigma = 0.3
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

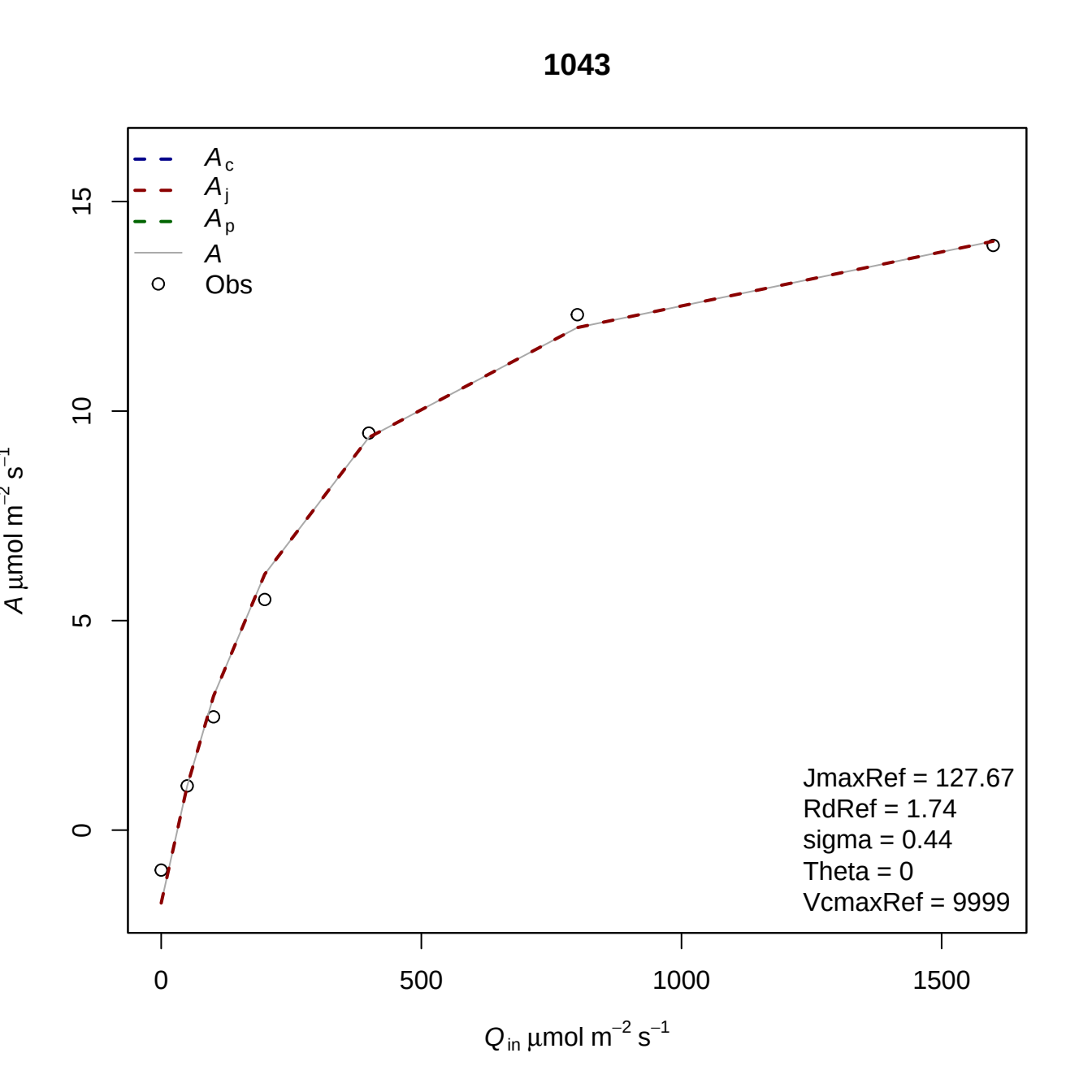

1038
A_c
A_j
A_p
A
Obs
JmaxRef = 118.23
RdRef = 1.2
sigma = 0.36
Theta = 0
VcmaxRef = 77.57
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

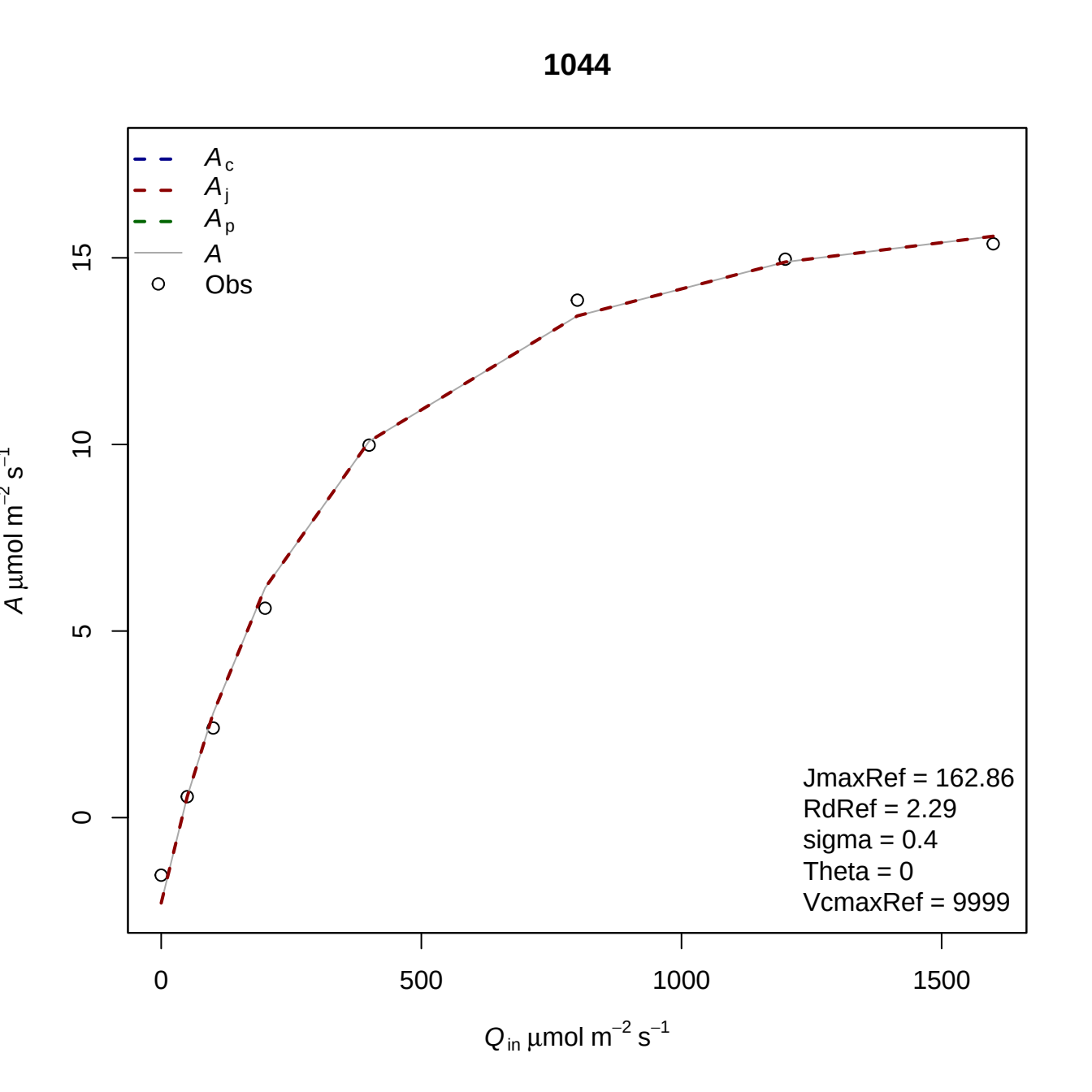

1039
A_c
A_j
A_p
A
Obs
JmaxRef = 105.87
RdRef = 0.94
sigma = 0.33
Theta = 0
VcmaxRef = 65.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

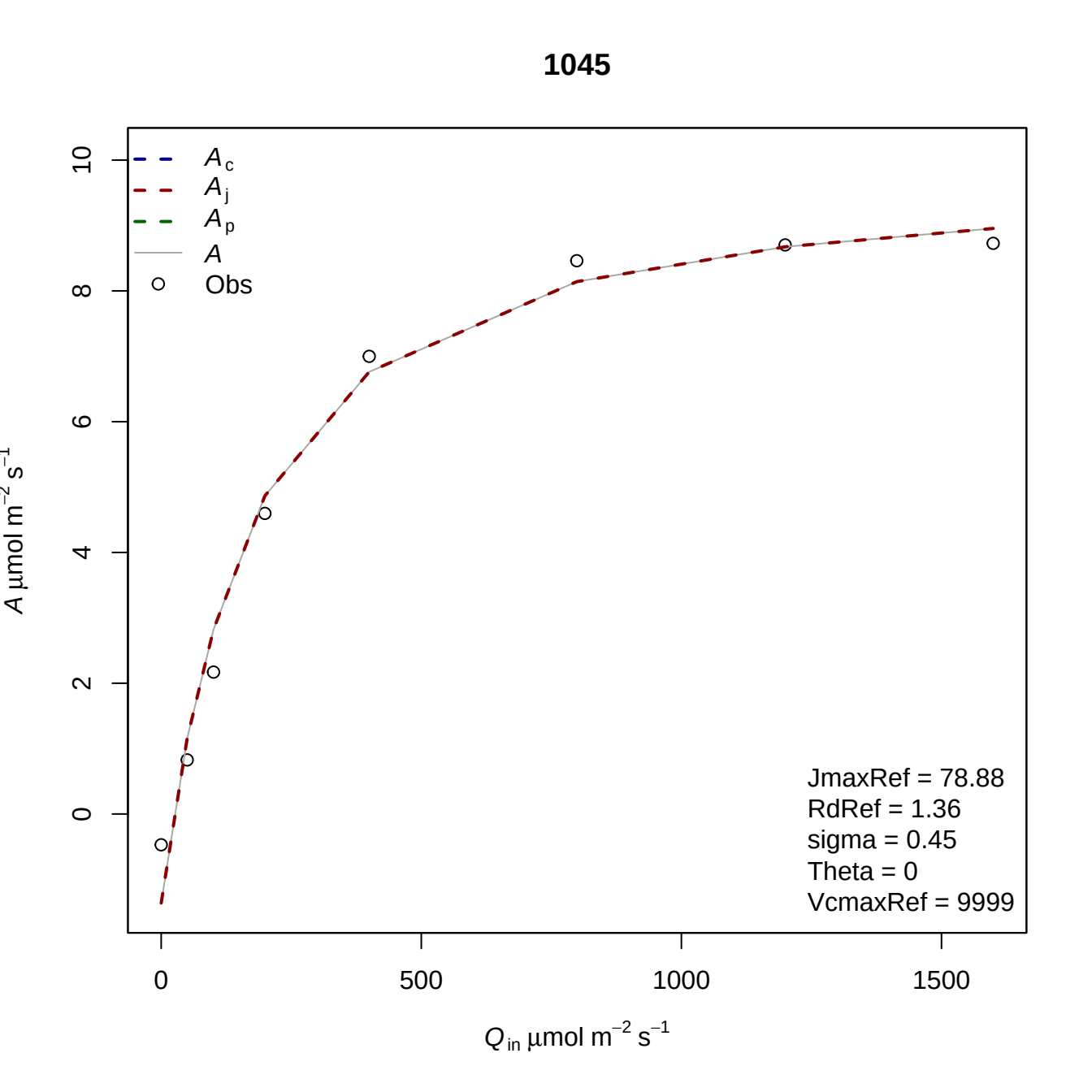

1040
A_c
A_j
A_p
A
Obs
JmaxRef = 107.53
RdRef = 1.08
sigma = 0.22
Theta = 0
VcmaxRef = 62.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

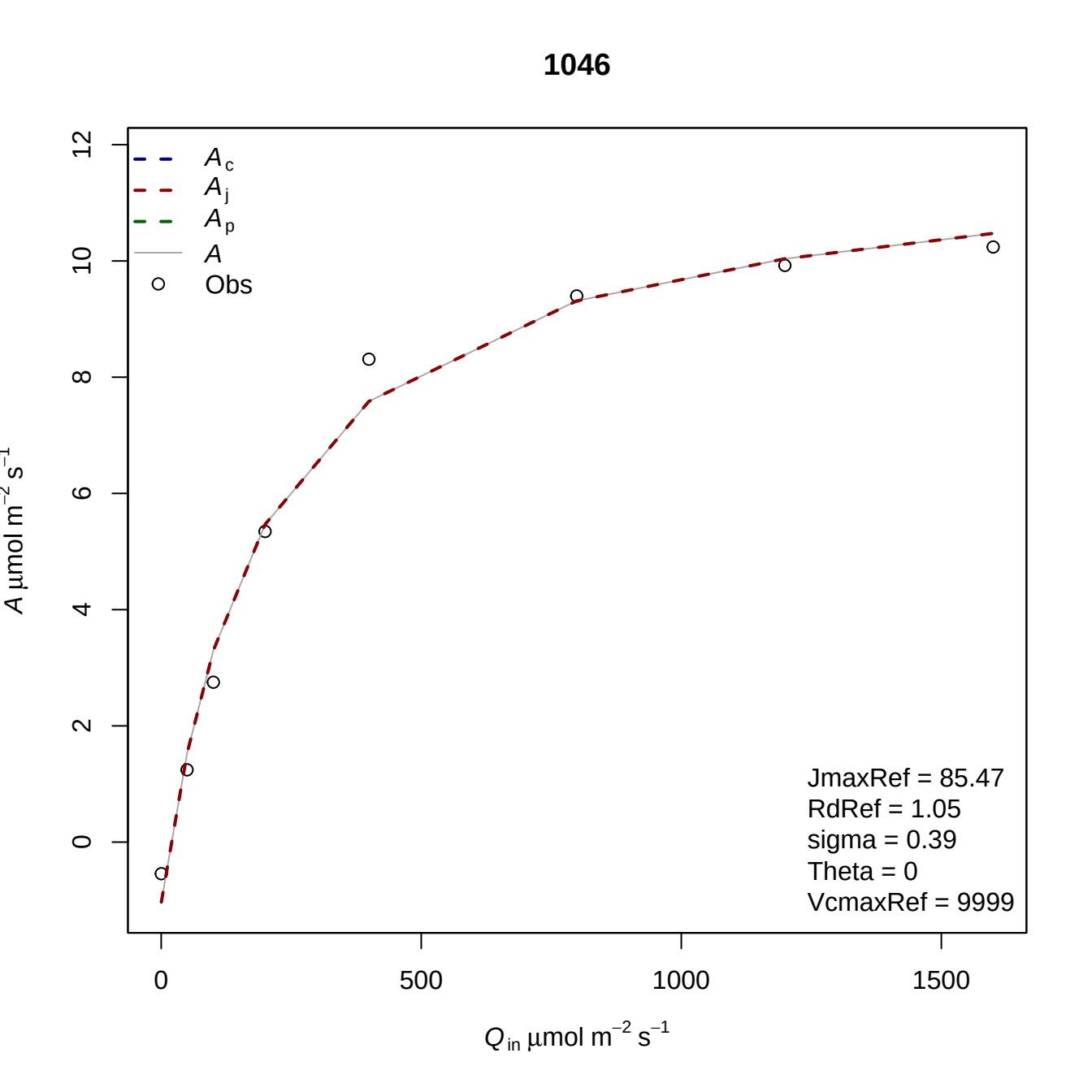

1041
A_c
A_j
A_p
A
Obs
JmaxRef = 140.99
RdRef = 1.04
sigma = 0.36
Theta = 0
VcmaxRef = 90.71
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

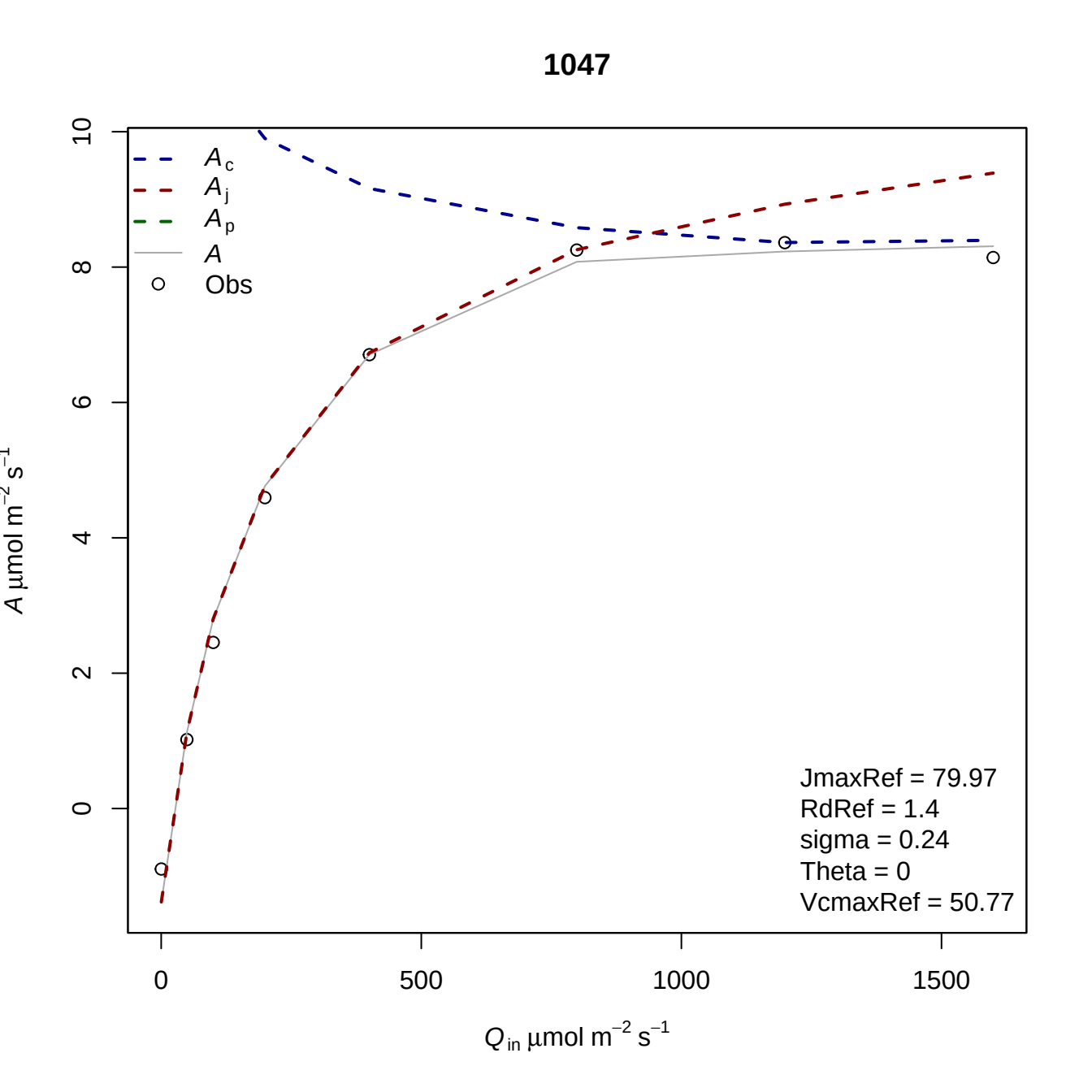

1042
A_c
A_j
A_p
A
Obs
JmaxRef = 140.72
RdRef = 2.1
sigma = 0.27
Theta = 0
VcmaxRef = 85.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

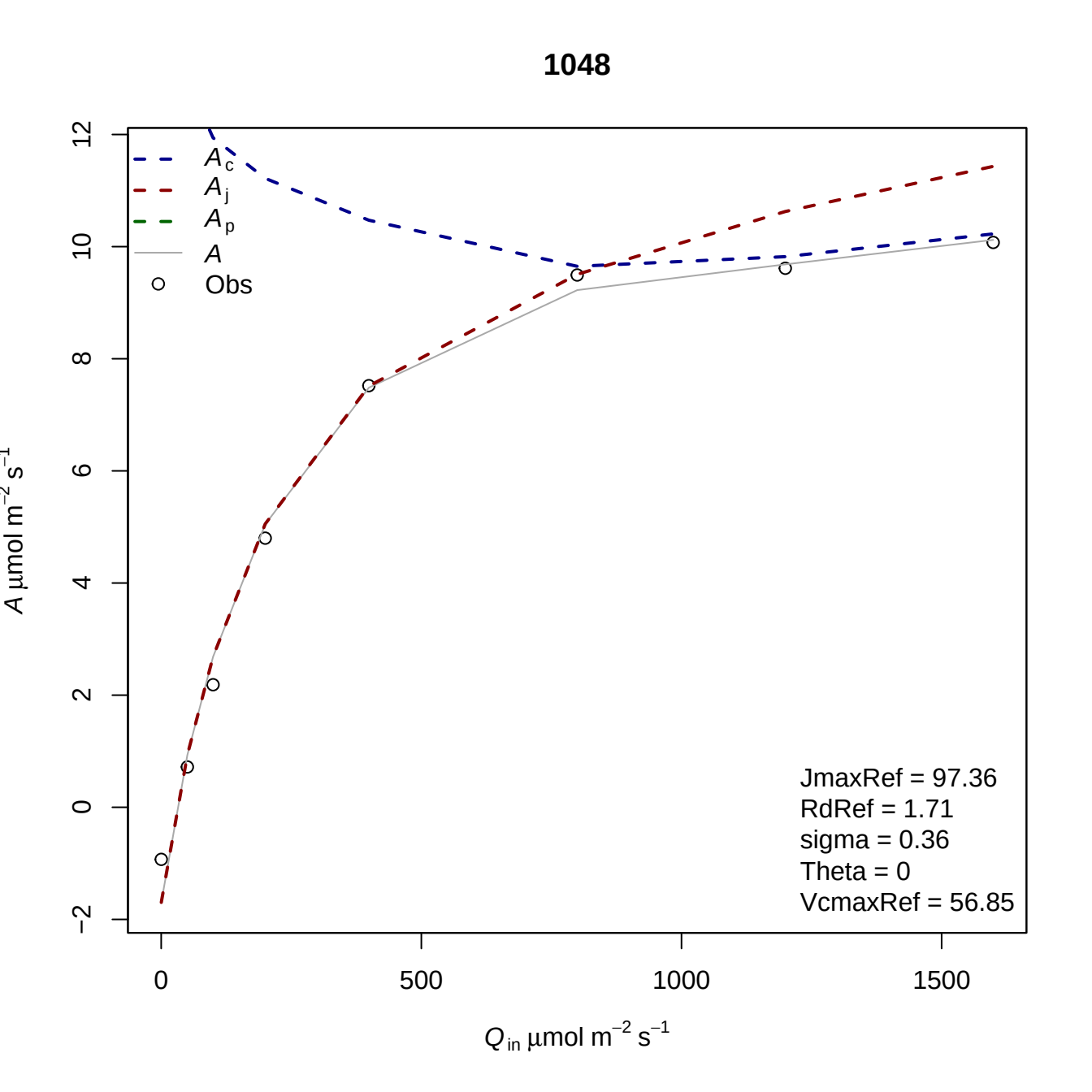

1043
A_c
A_j
A_p
A
Obs
JmaxRef = 127.67
RdRef = 1.74
sigma = 0.44
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

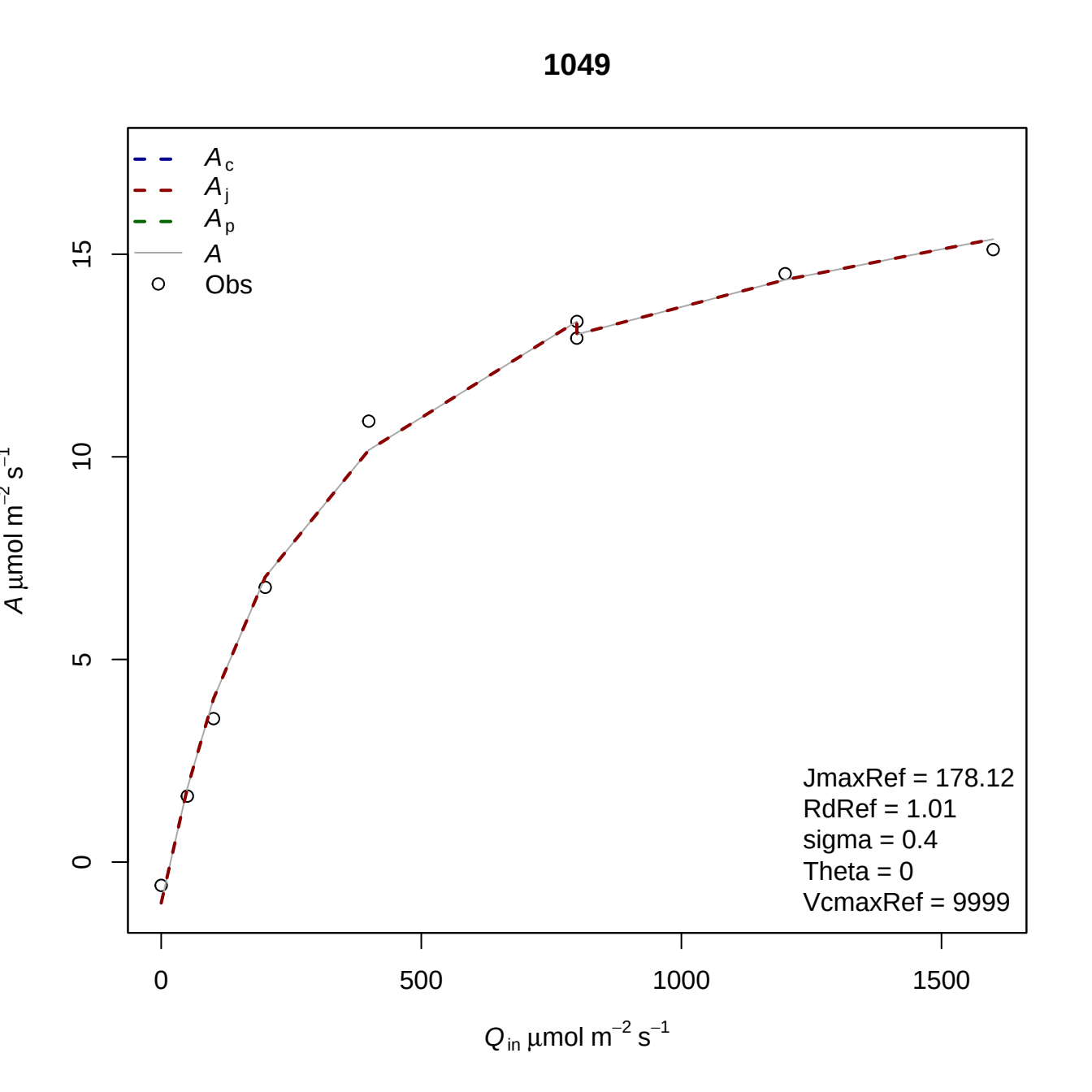

1044
A_c
A_j
A_p
A
Obs
JmaxRef = 162.86
RdRef = 2.29
sigma = 0.4
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

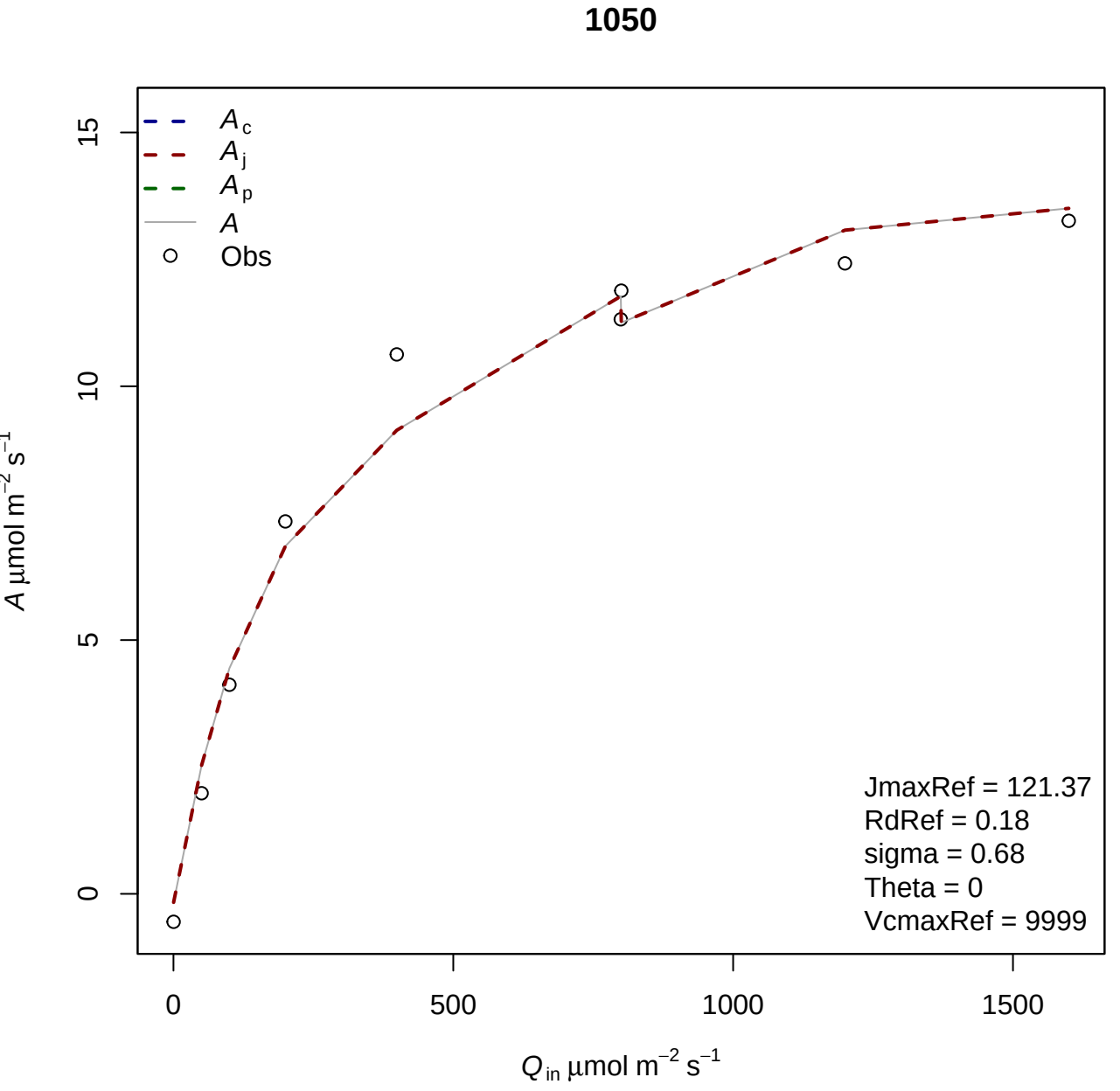

1045
A_c
A_j
A_p
A
Obs
JmaxRef = 78.88
RdRef = 1.36
sigma = 0.45
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

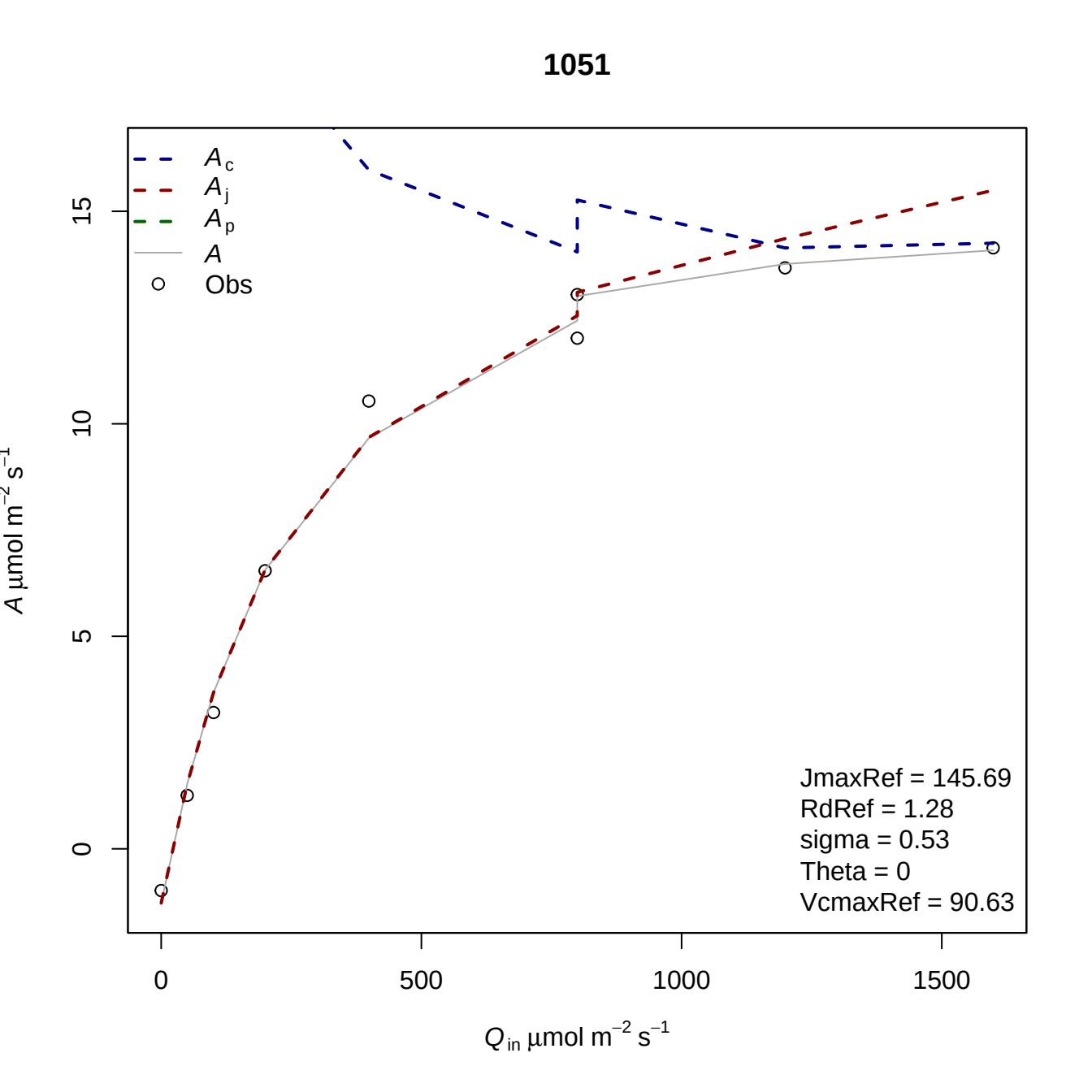

1046
A_c
A_j
A_p
A
Obs
JmaxRef = 85.47
RdRef = 1.05
sigma = 0.39
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

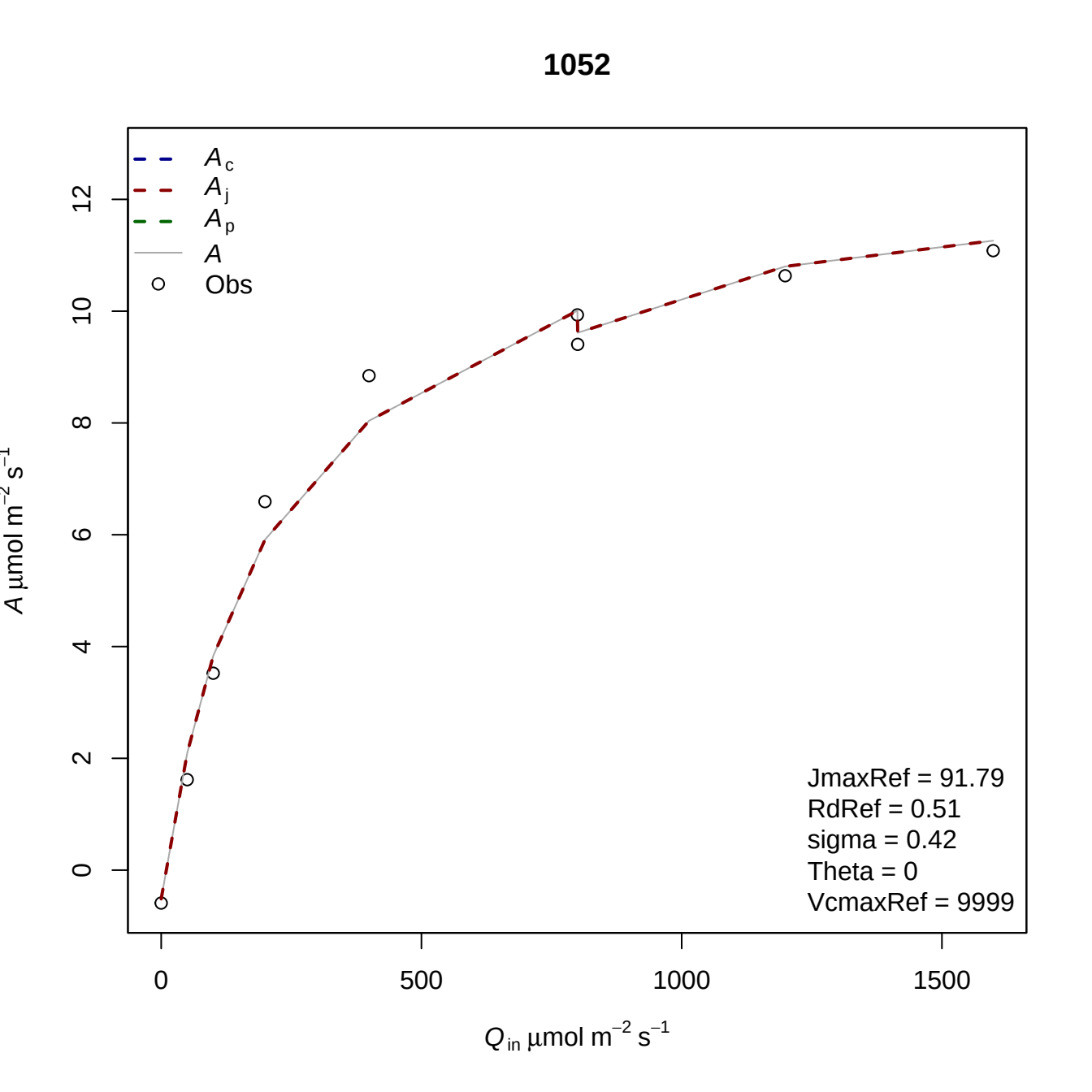

1047
A_c
A_j
A_p
A
Obs
JmaxRef = 79.97
RdRef = 1.4
sigma = 0.24
Theta = 0
VcmaxRef = 50.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

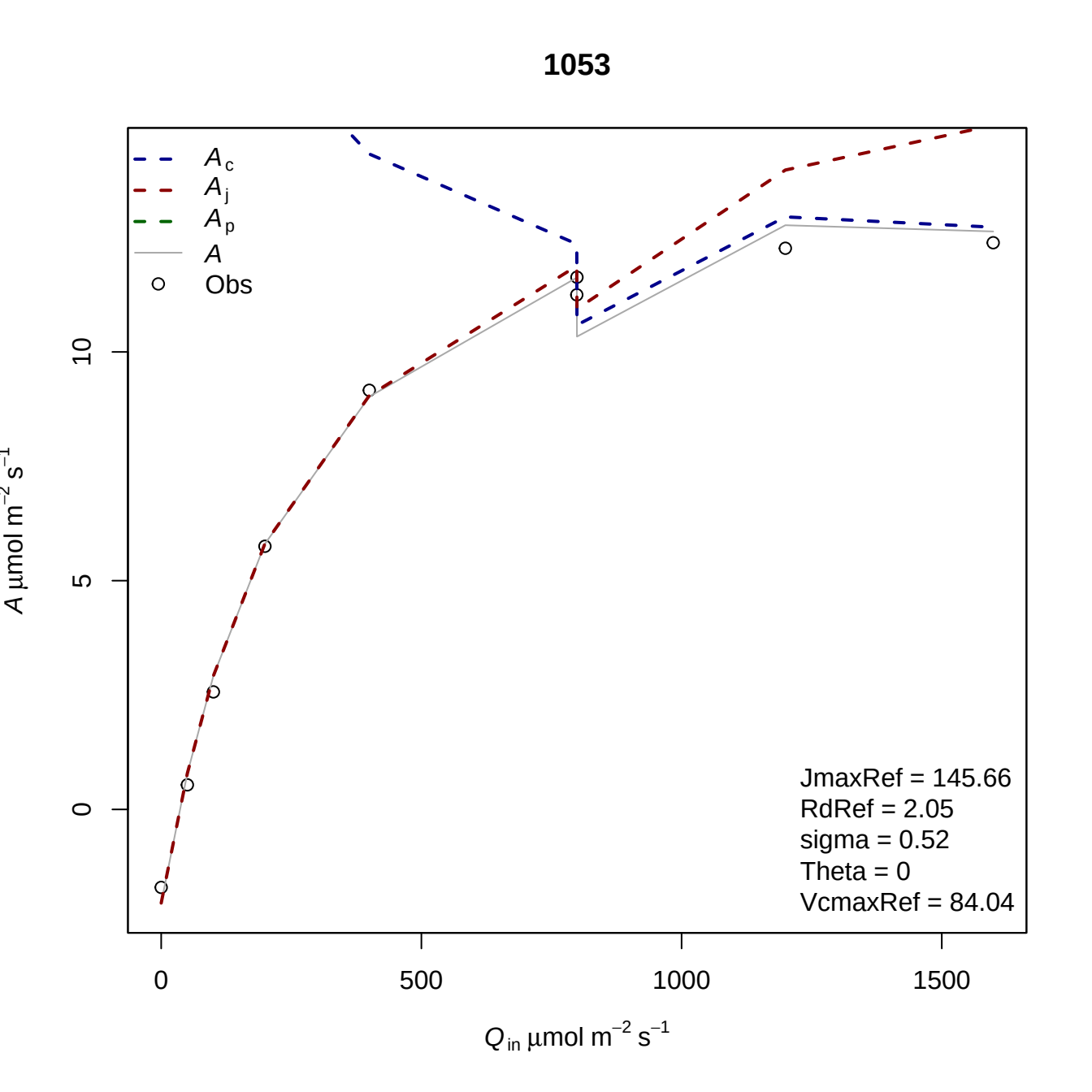

1048
A_c
A_j
A_p
A
Obs
JmaxRef = 97.36
RdRef = 1.71
sigma = 0.36
Theta = 0
VcmaxRef = 56.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

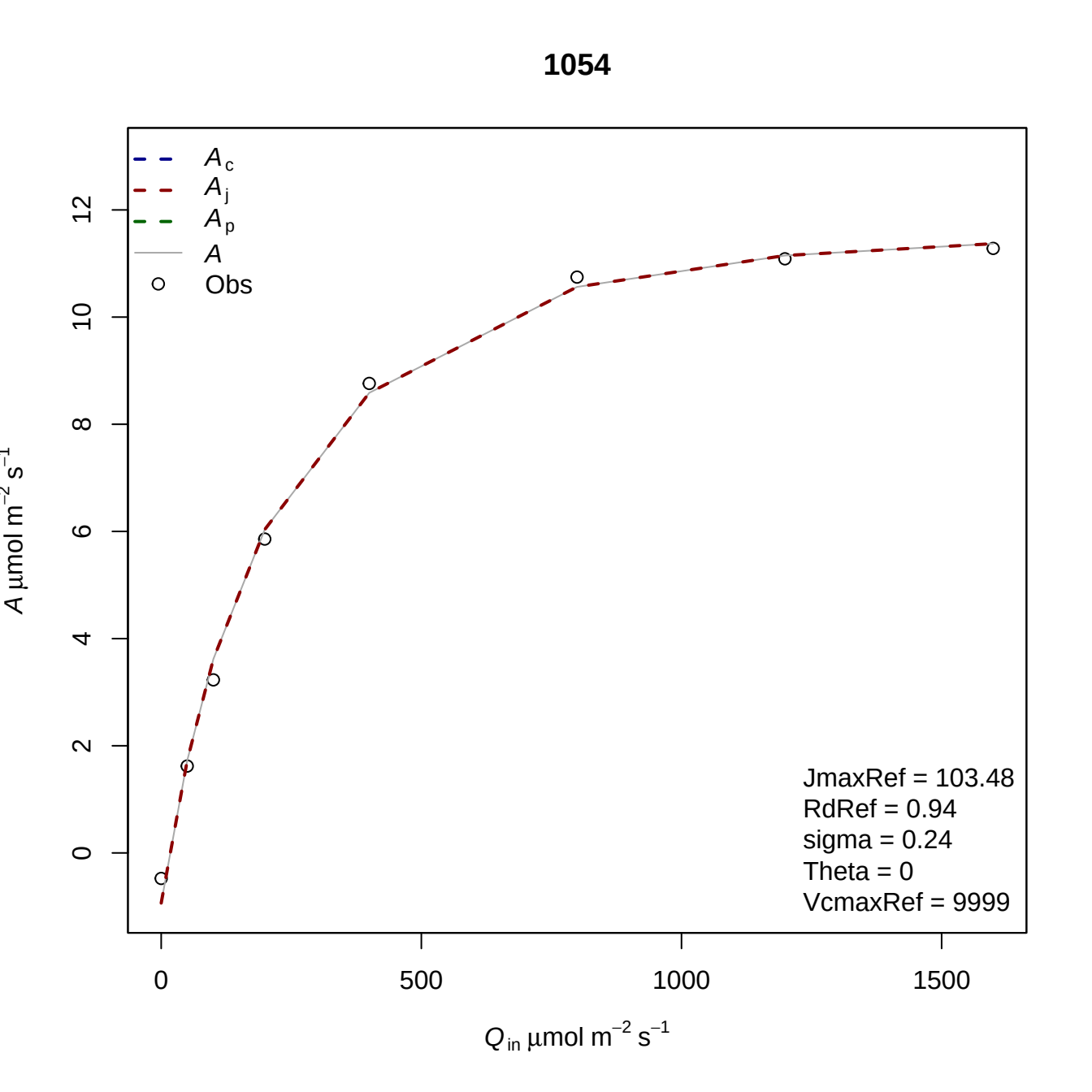

1049
A_c
A_j
A_p
A
Obs
JmaxRef = 178.12
RdRef = 1.01
sigma = 0.4
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

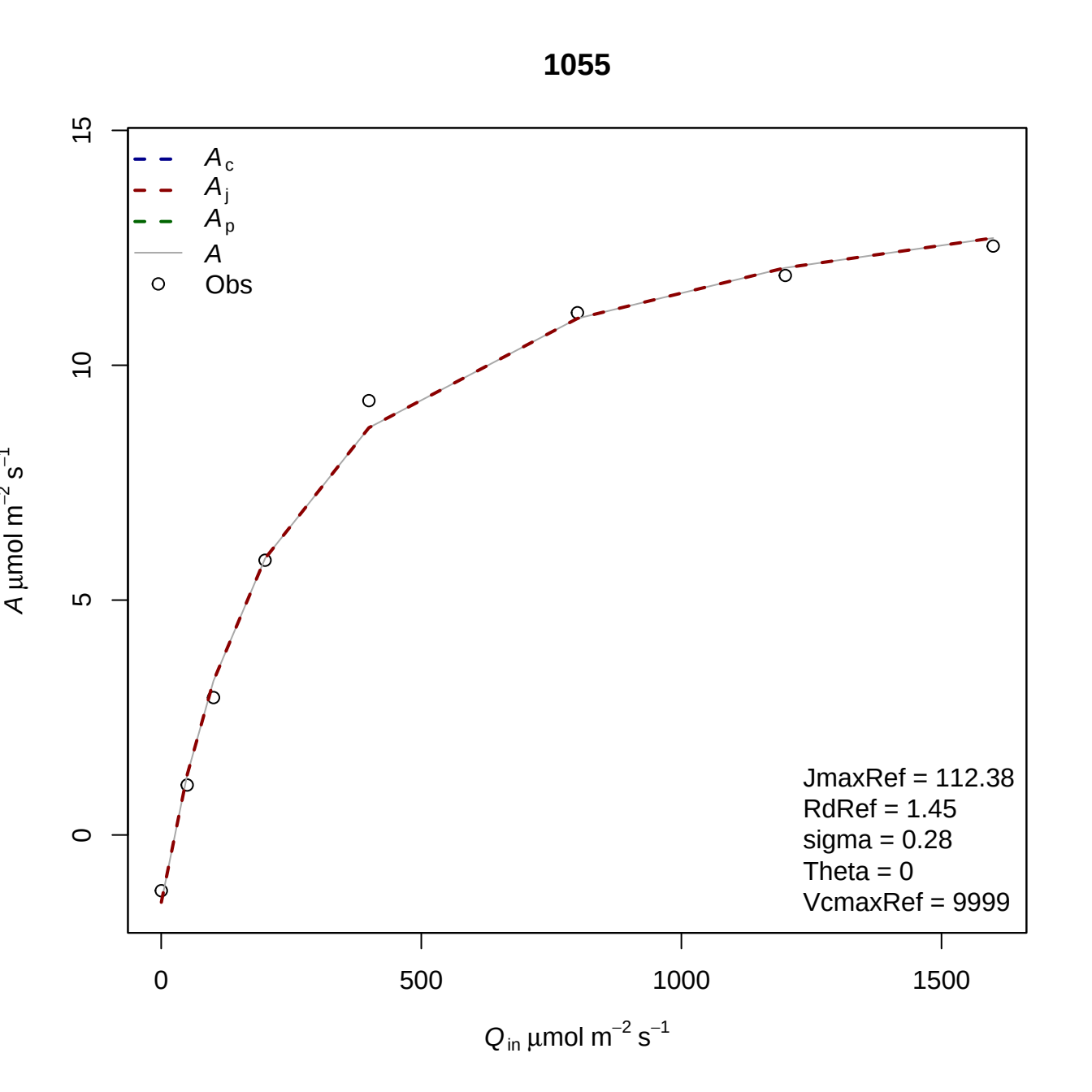

1050
A_c
A_j
A_p
A
Obs
JmaxRef = 121.37
RdRef = 0.18
sigma = 0.68
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

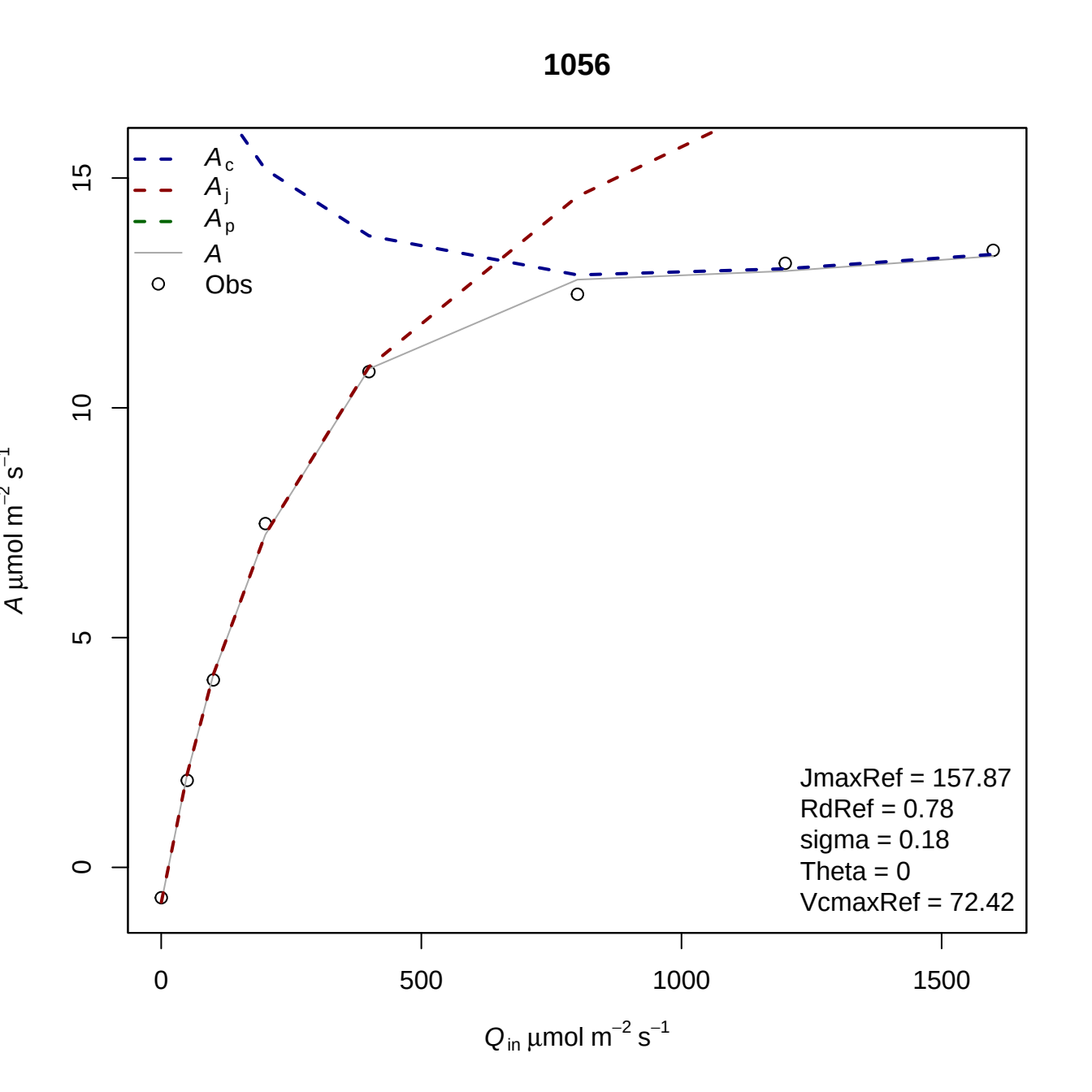

1051
A_c
A_j
A_p
A
Obs
JmaxRef = 145.69
RdRef = 1.28
sigma = 0.53
Theta = 0
VcmaxRef = 90.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

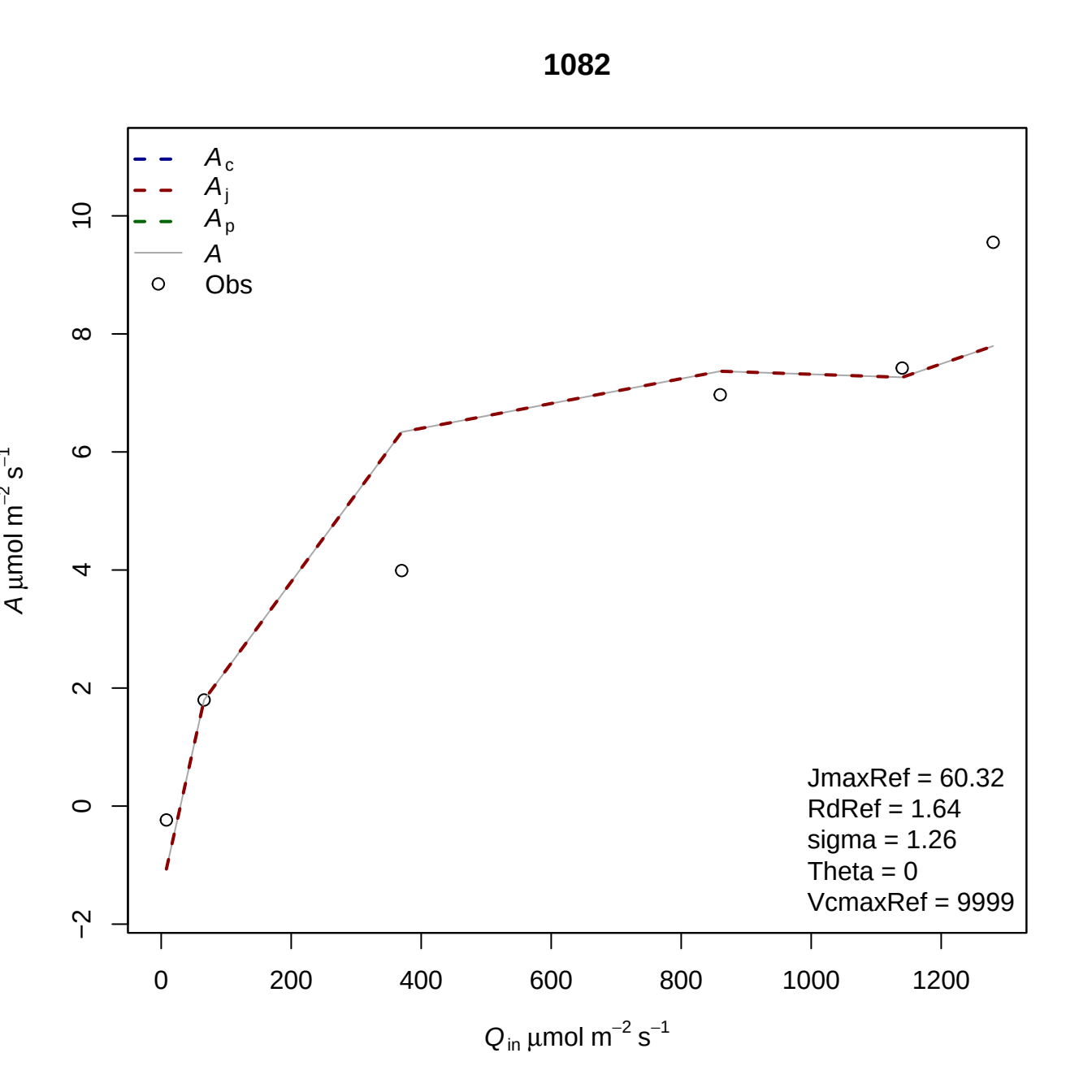

1052
A_c
A_j
A_p
A
Obs
JmaxRef = 91.79
RdRef = 0.51
sigma = 0.42
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

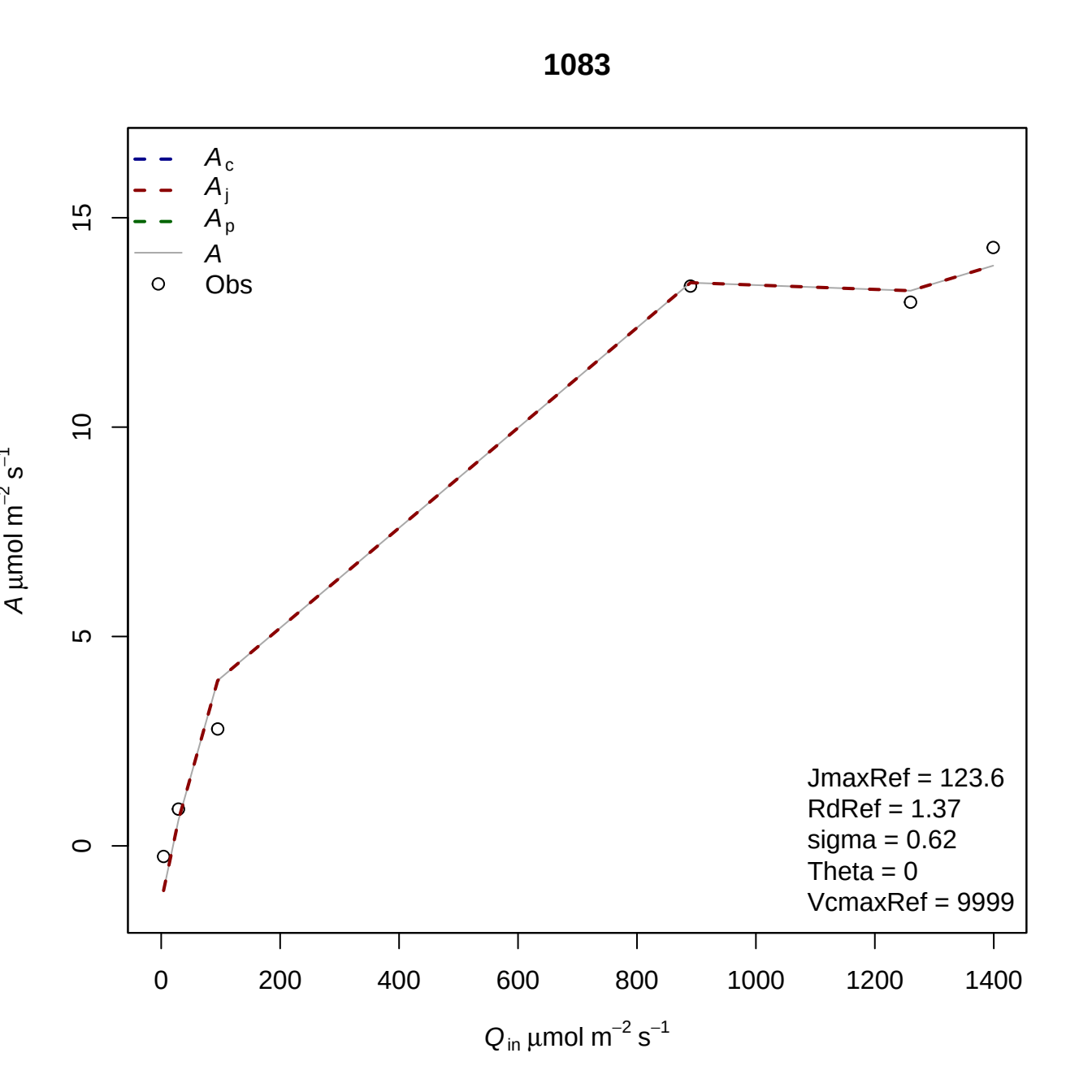

1053
A_c
A_j
A_p
A
Obs
JmaxRef = 145.66
RdRef = 2.05
sigma = 0.52
Theta = 0
VcmaxRef = 84.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

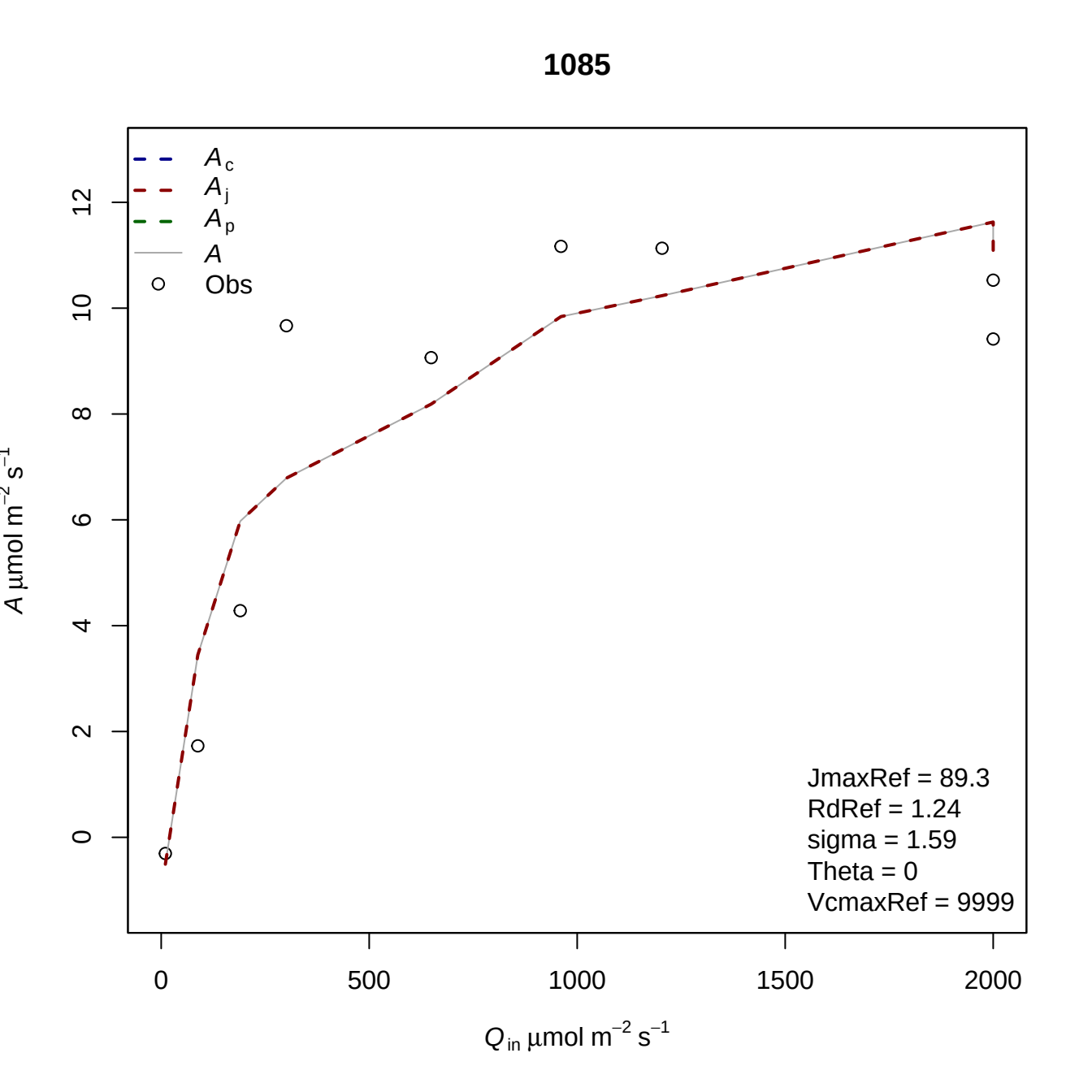

1054
A_c
A_j
A_p
A
Obs
JmaxRef = 103.48
RdRef = 0.94
sigma = 0.24
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

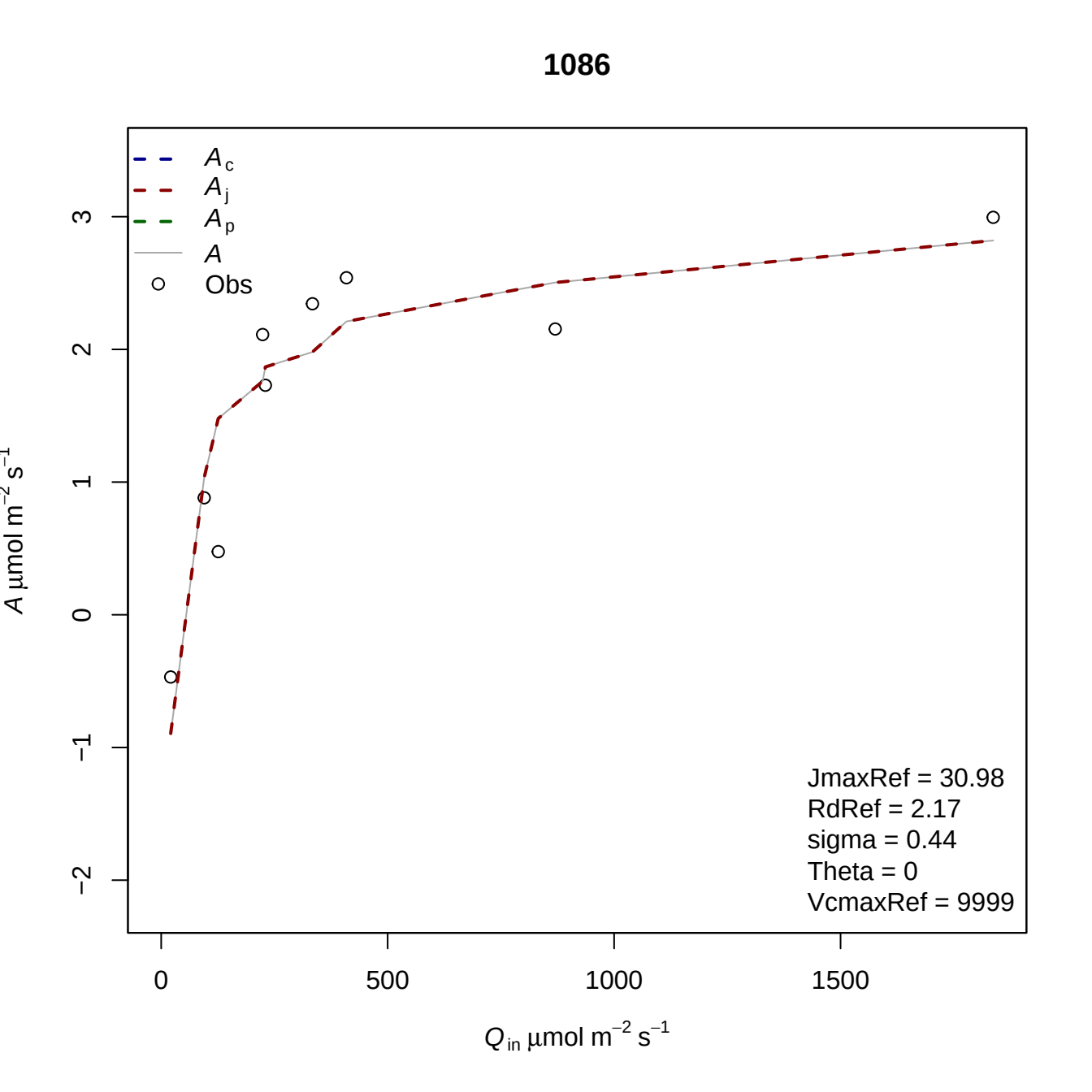

1055
A_c
A_j
A_p
A
Obs
JmaxRef = 112.38
RdRef = 1.45
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

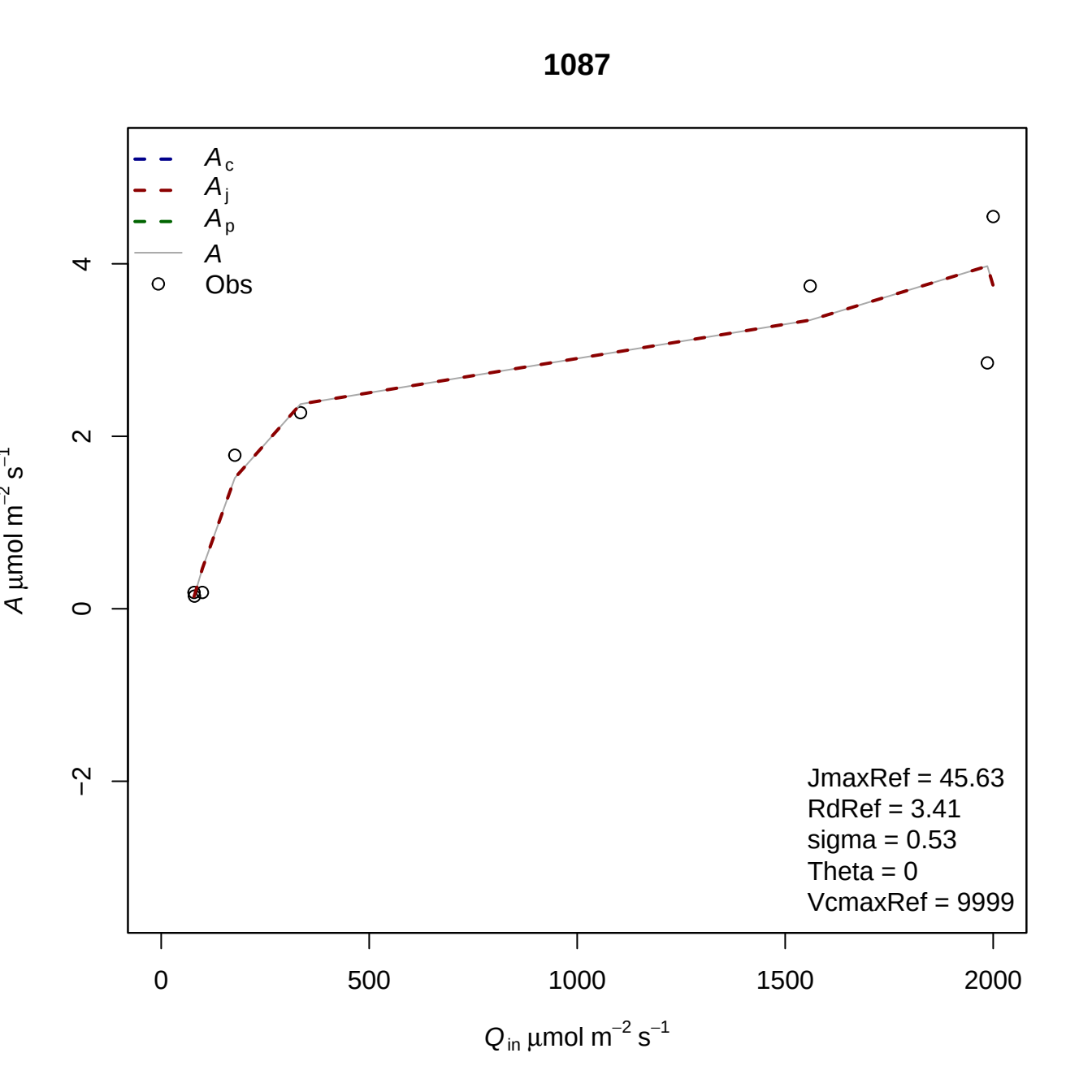

1056
A_c
A_j
A_p
A
Obs
JmaxRef = 157.87
RdRef = 0.78
sigma = 0.18
Theta = 0
VcmaxRef = 72.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

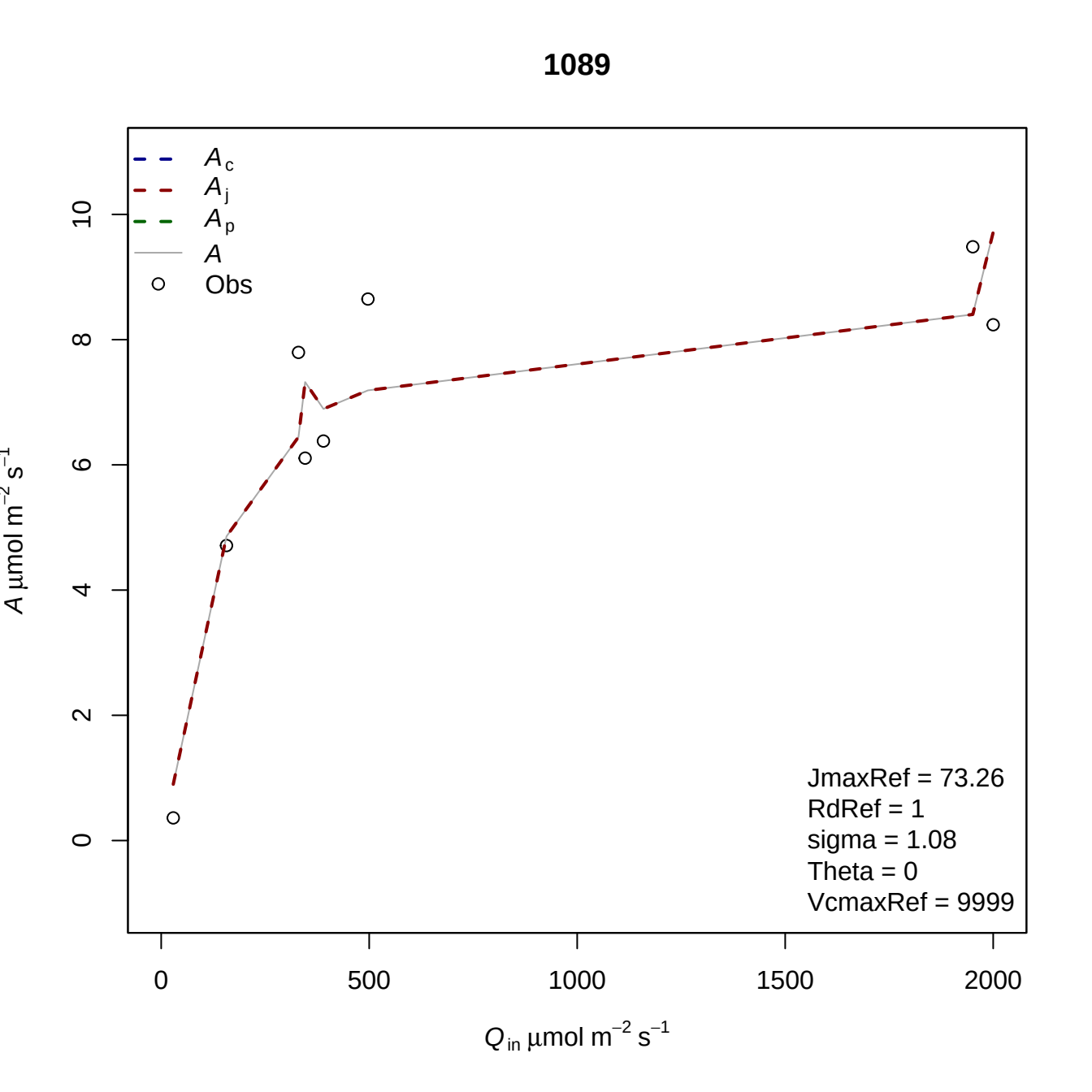

1082
A_c
A_j
A_p
A
Obs
JmaxRef = 60.32
RdRef = 1.64
sigma = 1.26
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

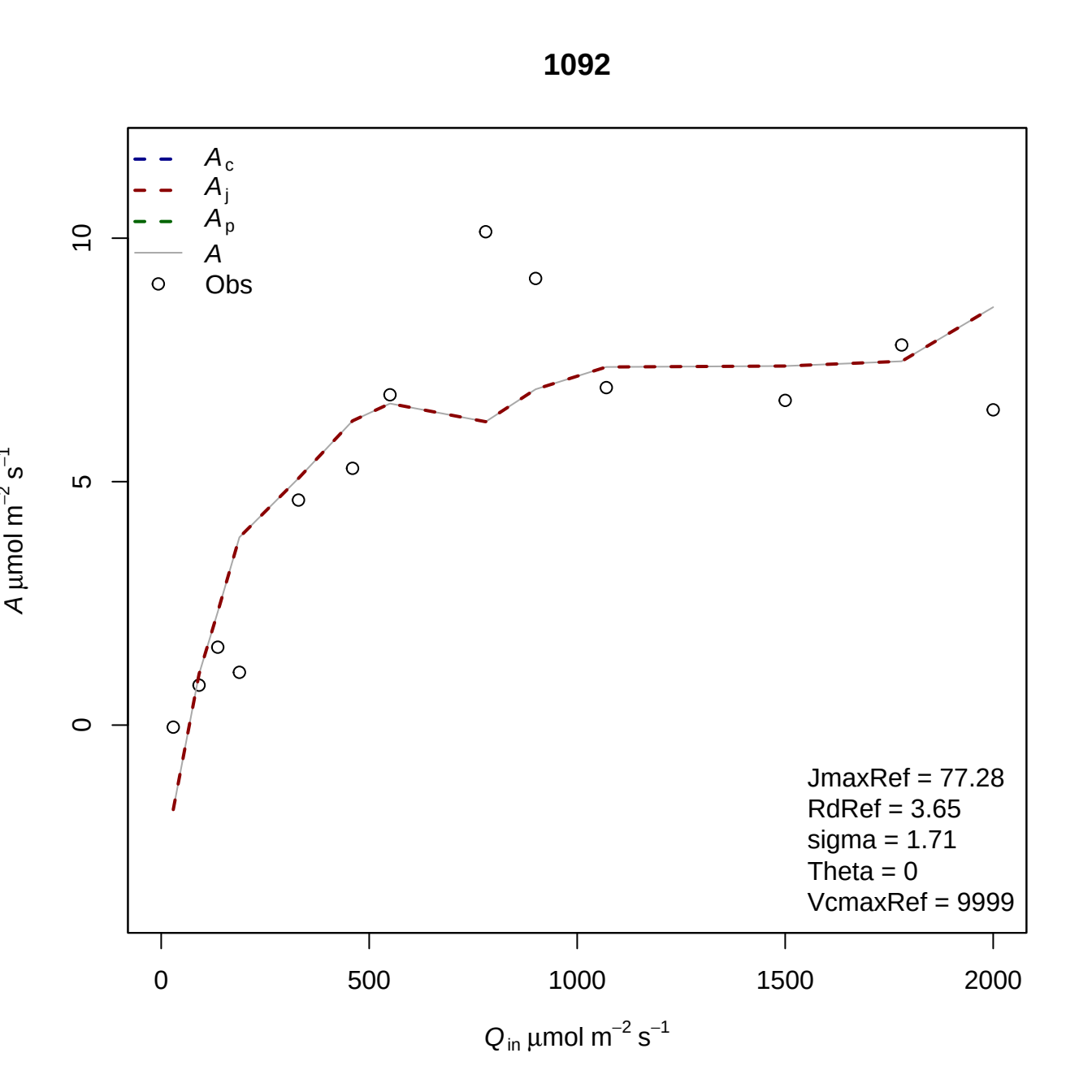

1083
A_c
A_j
A_p
A
Obs
JmaxRef = 123.6
RdRef = 1.37
sigma = 0.62
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

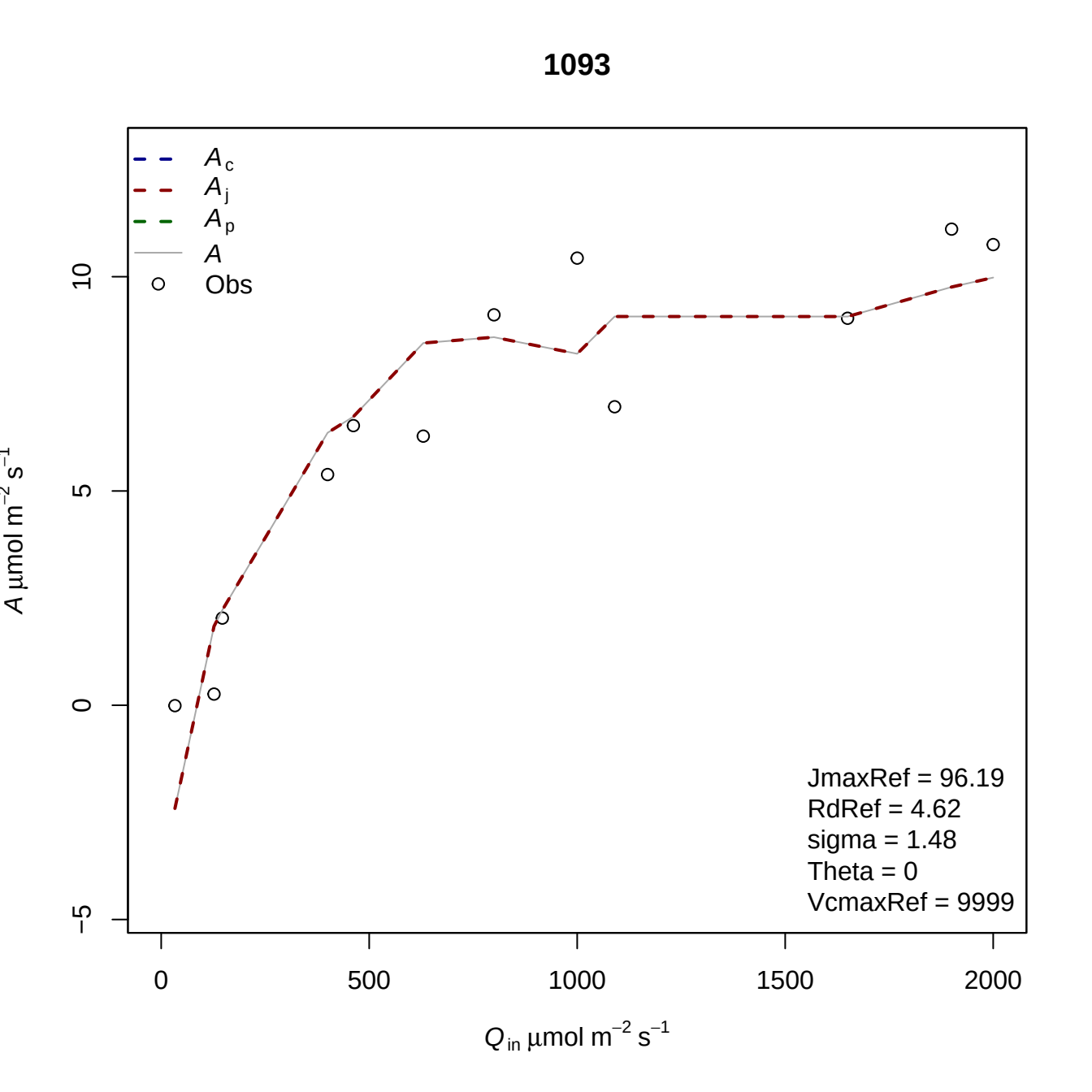

1085
A_c
A_j
A_p
A
Obs
JmaxRef = 89.3
RdRef = 1.24
sigma = 1.59
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

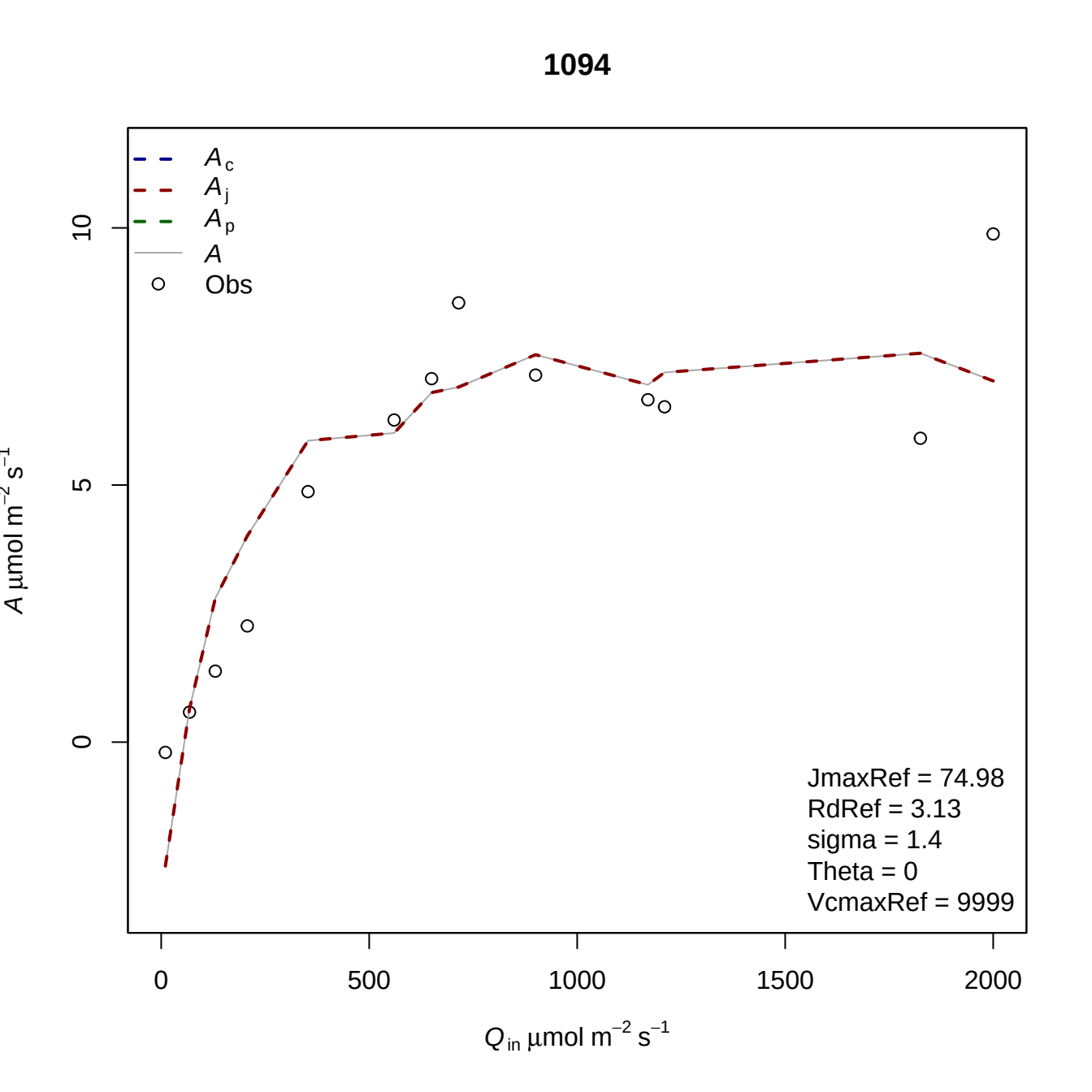

1086
A_c
A_j
A_p
A
Obs
JmaxRef = 30.98
RdRef = 2.17
sigma = 0.44
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
500
1000
1500

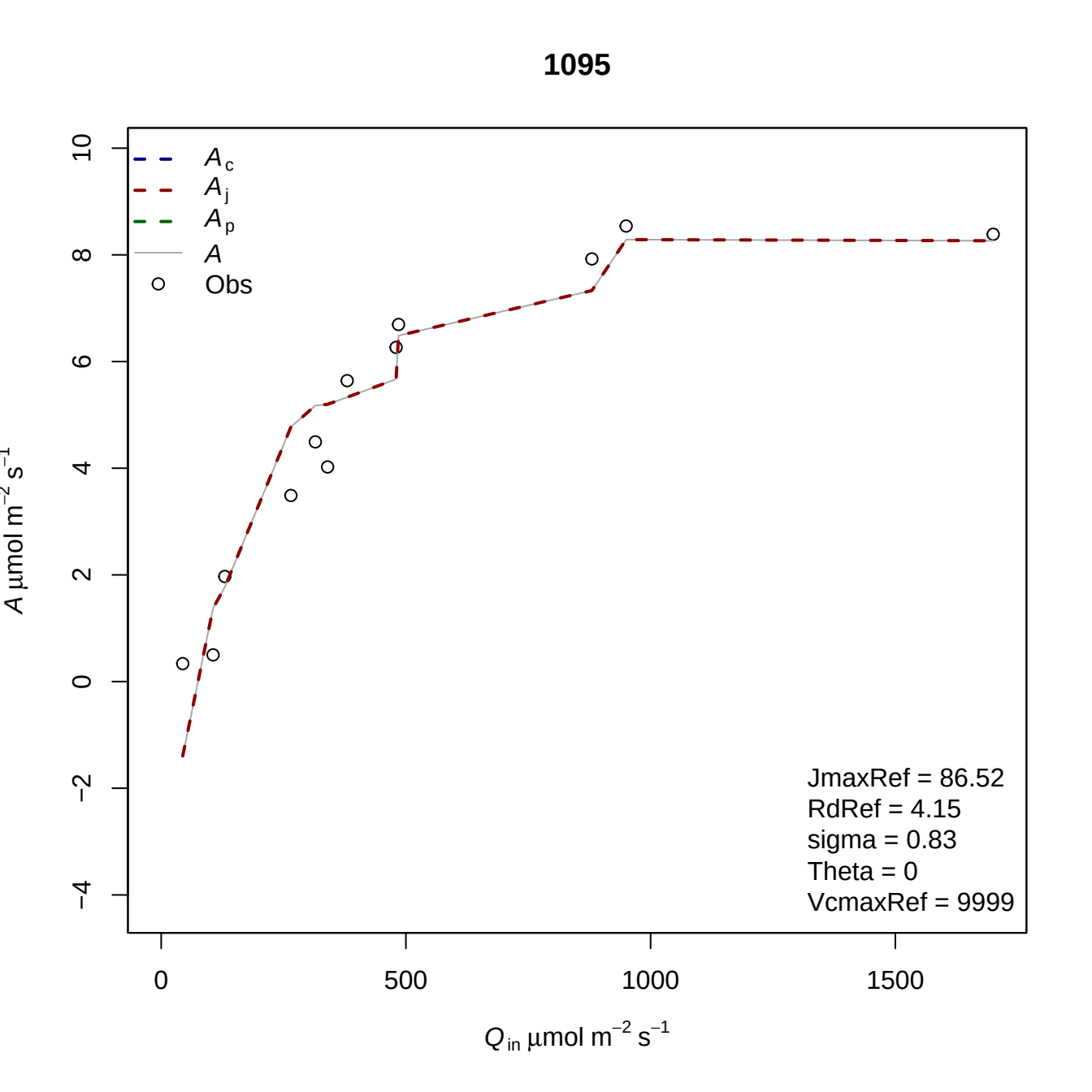

1087
A_c
A_j
A_p
A
Obs
JmaxRef = 45.63
RdRef = 3.41
sigma = 0.53
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

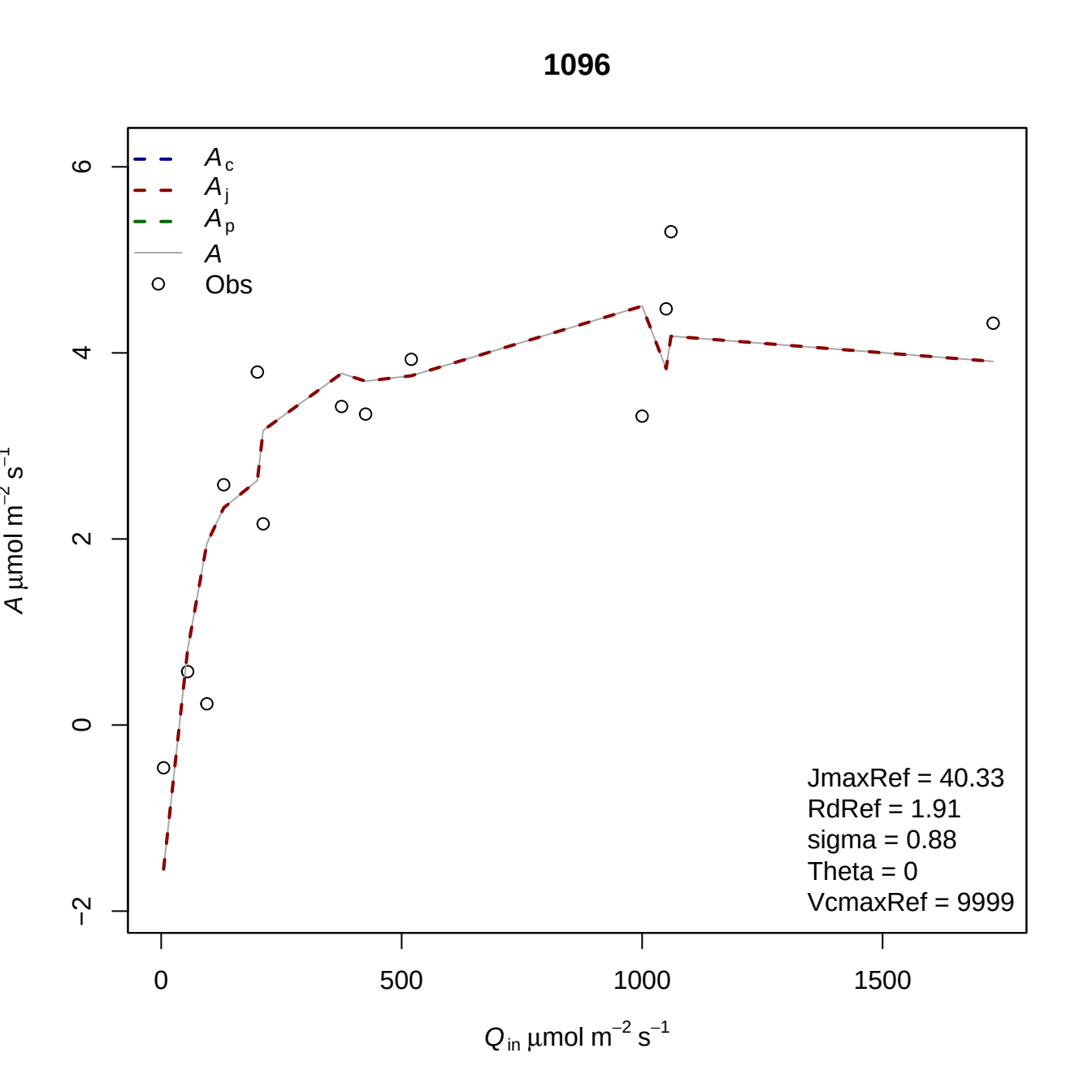

1089
A_c
A_j
A_p
A
Obs
JmaxRef = 73.26
RdRef = 1
sigma = 1.08
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

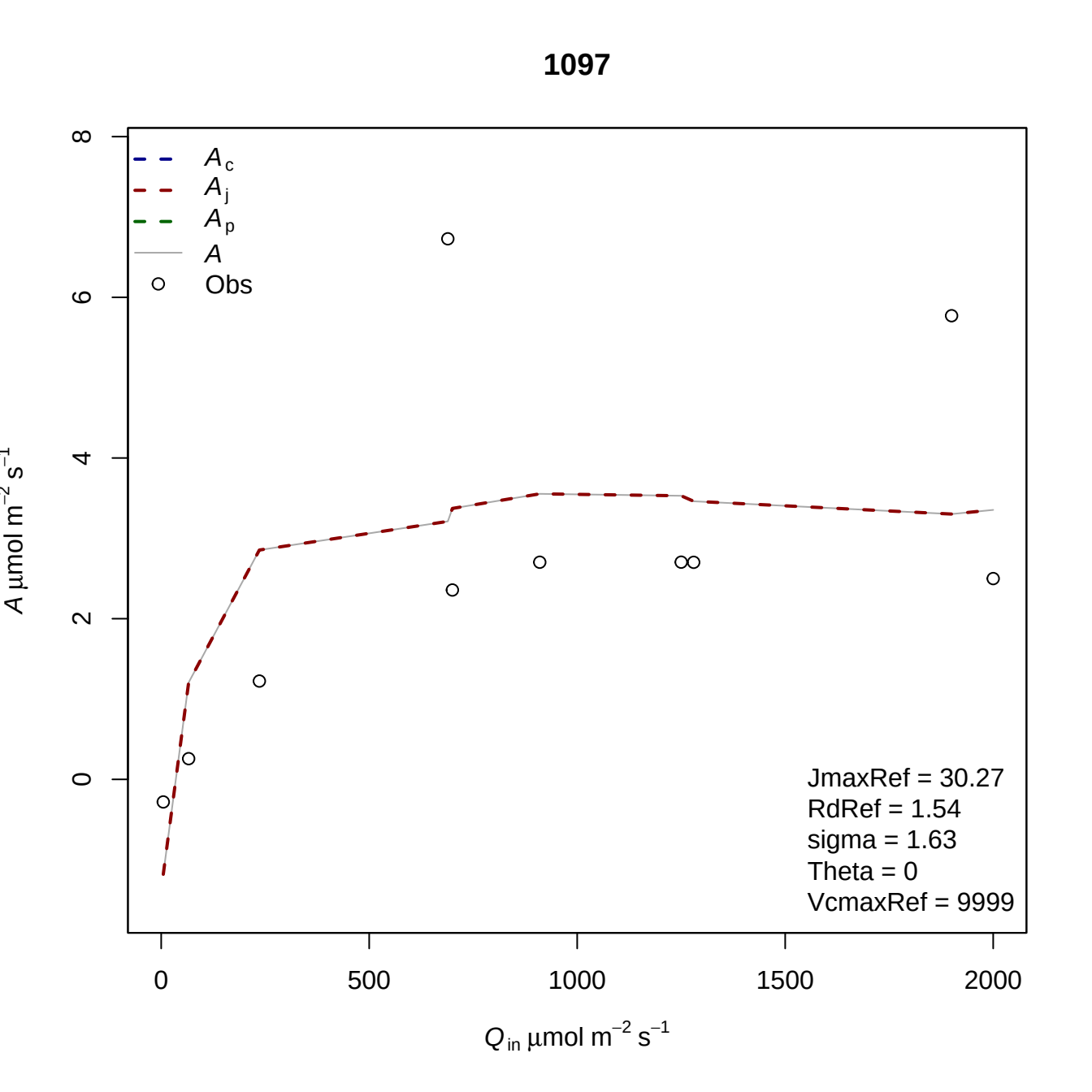

1092
A_c
A_j
A_p
A
Obs
JmaxRef = 77.28
RdRef = 3.65
sigma = 1.71
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

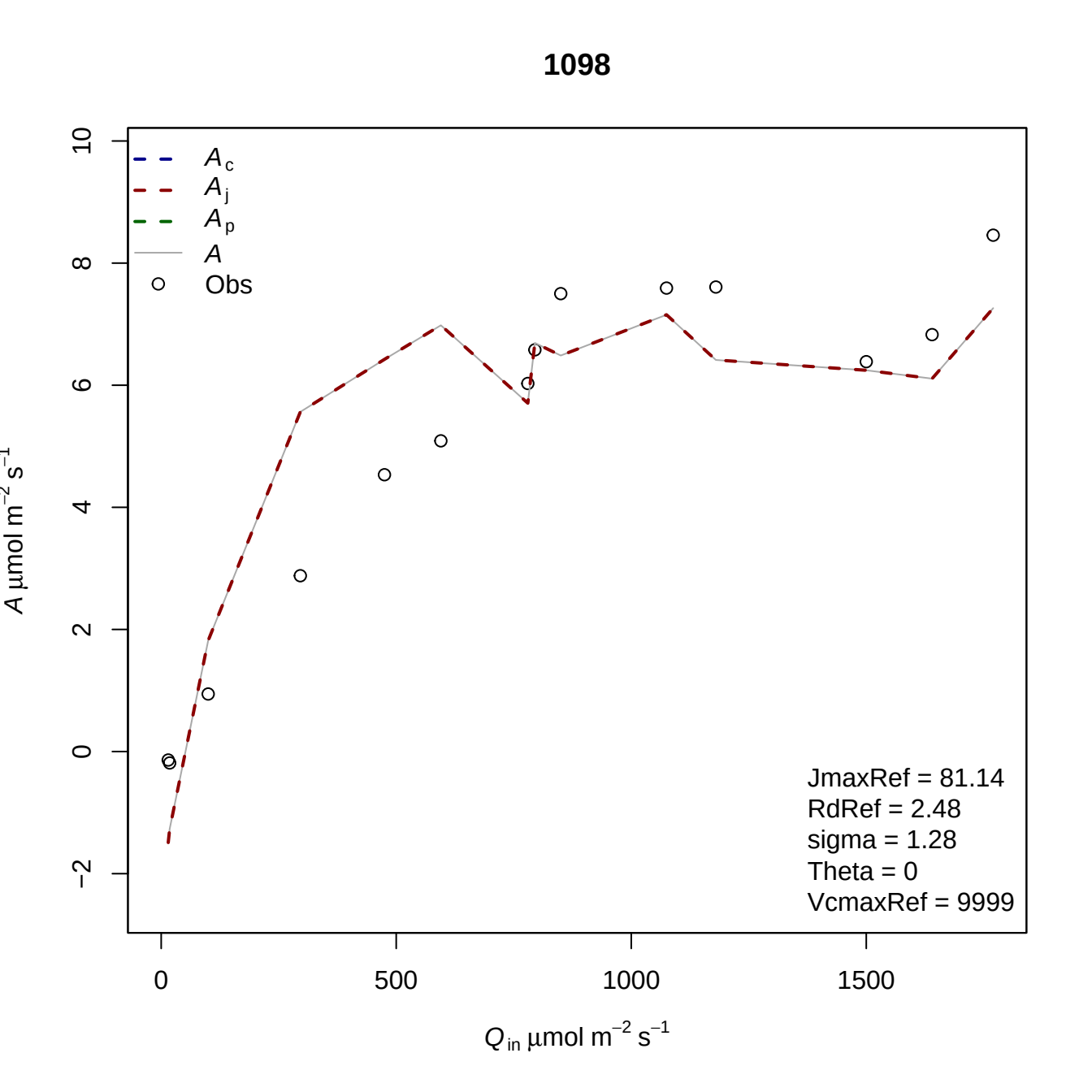

1093
A_c
A_j
A_p
A
Obs
JmaxRef = 96.19
RdRef = 4.62
sigma = 1.48
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

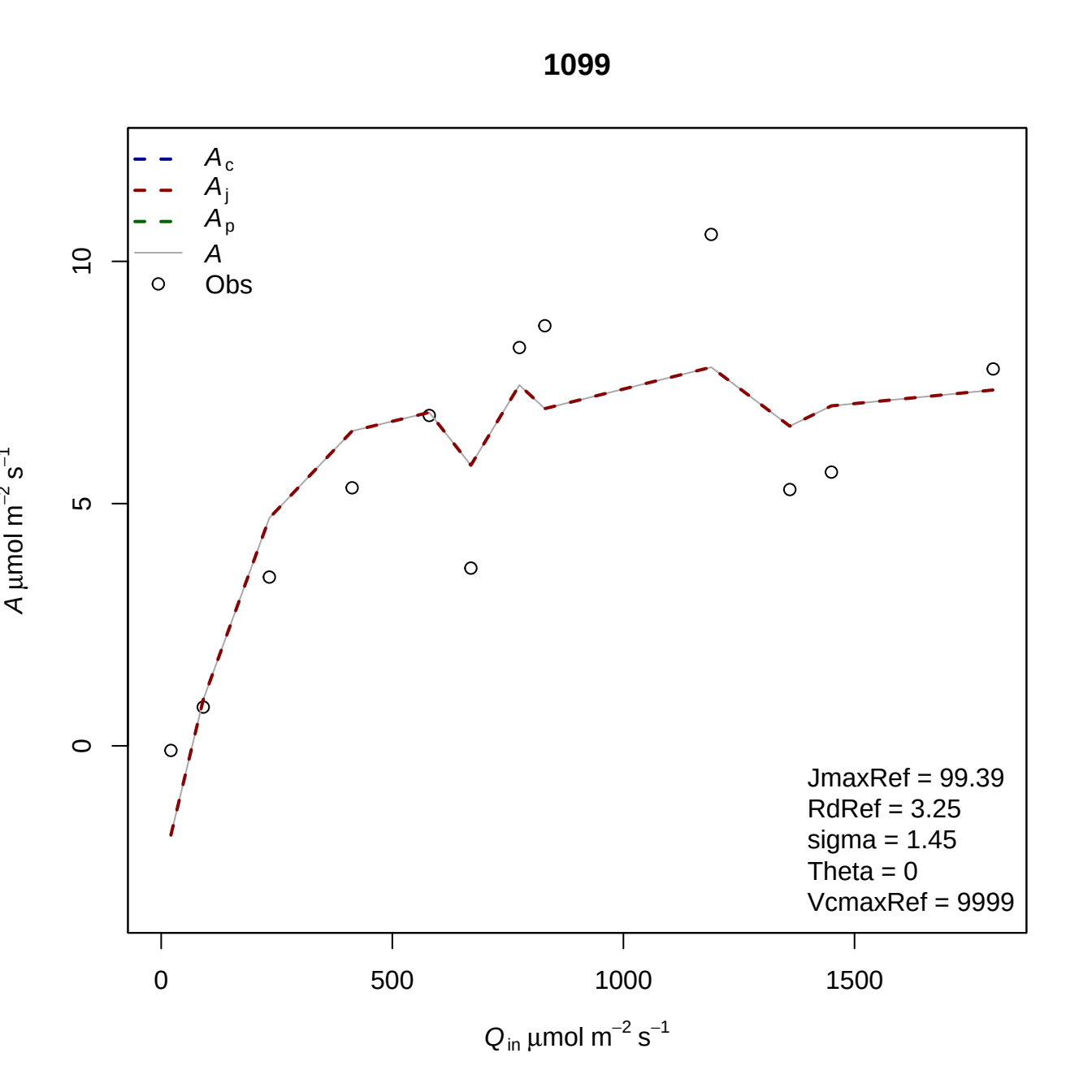

1094
A_c
A_j
A_p
A
Obs
JmaxRef = 74.98
RdRef = 3.13
sigma = 1.4
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

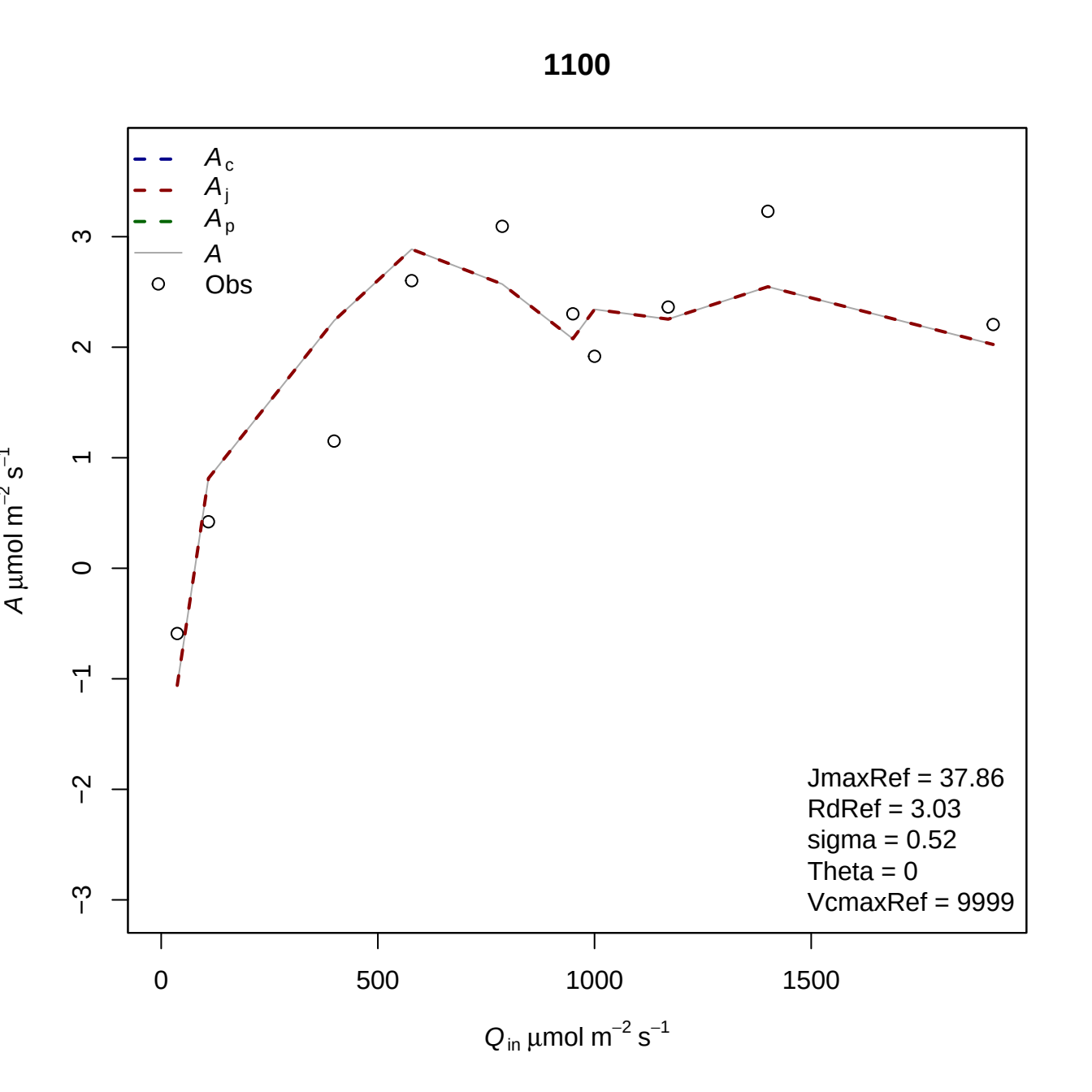

1095
A_c
A_j
A_p
A
Obs
JmaxRef = 86.52
RdRef = 4.15
sigma = 0.83
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

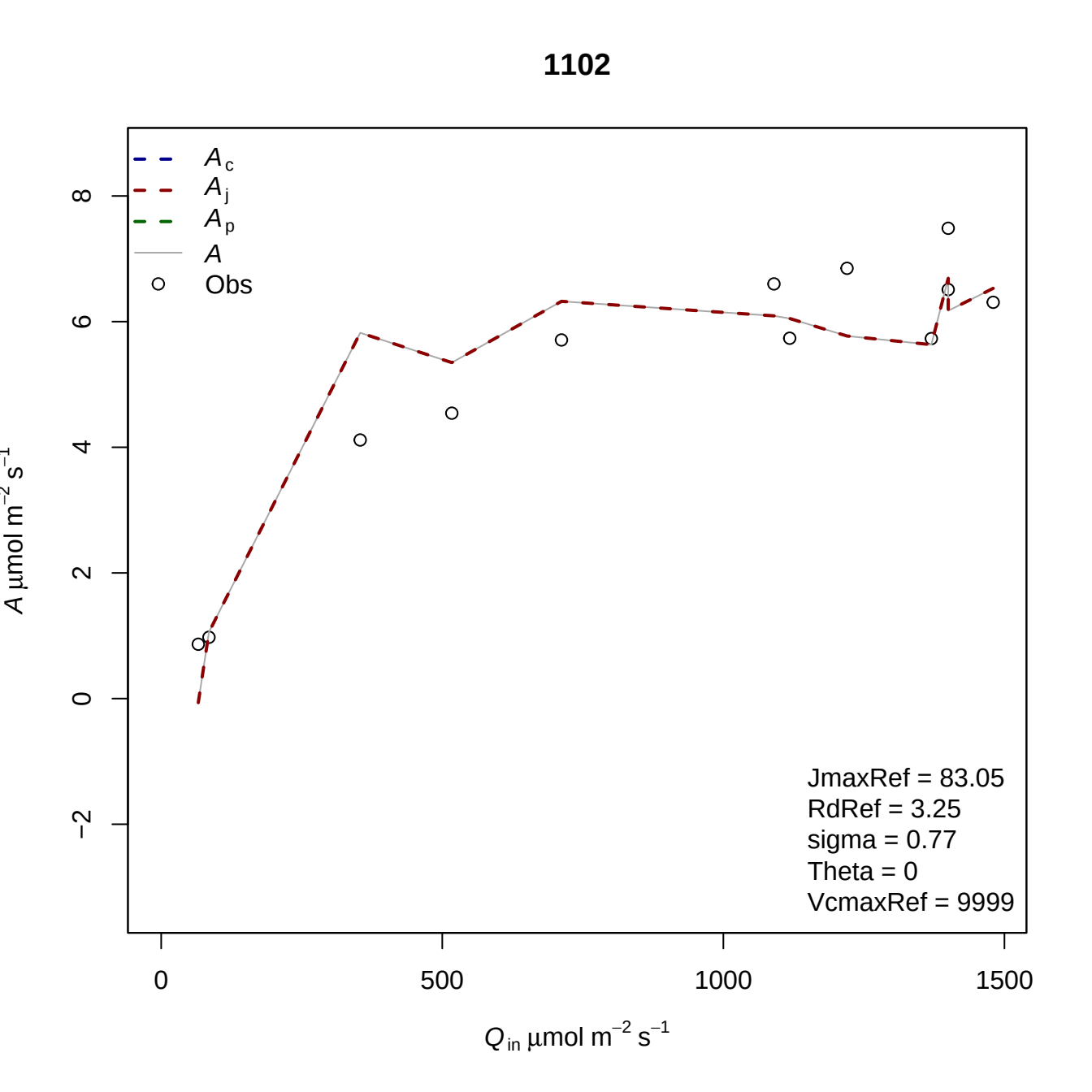

1096
A_c
A_j
A_p
A
Obs
JmaxRef = 40.33
RdRef = 1.91
sigma = 0.88
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

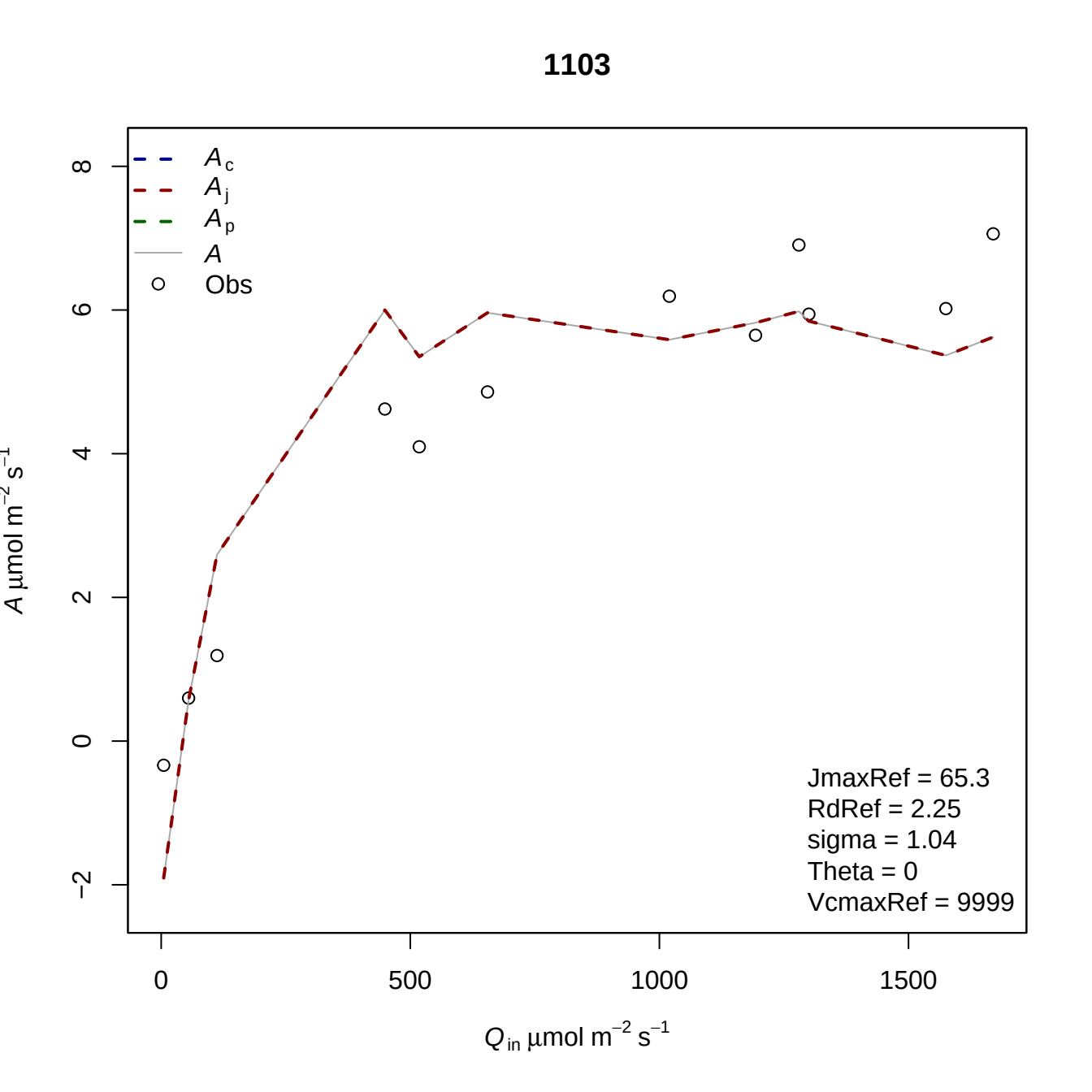

1097
A_c
A_j
A_p
A
Obs
JmaxRef = 30.27
RdRef = 1.54
sigma = 1.63
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

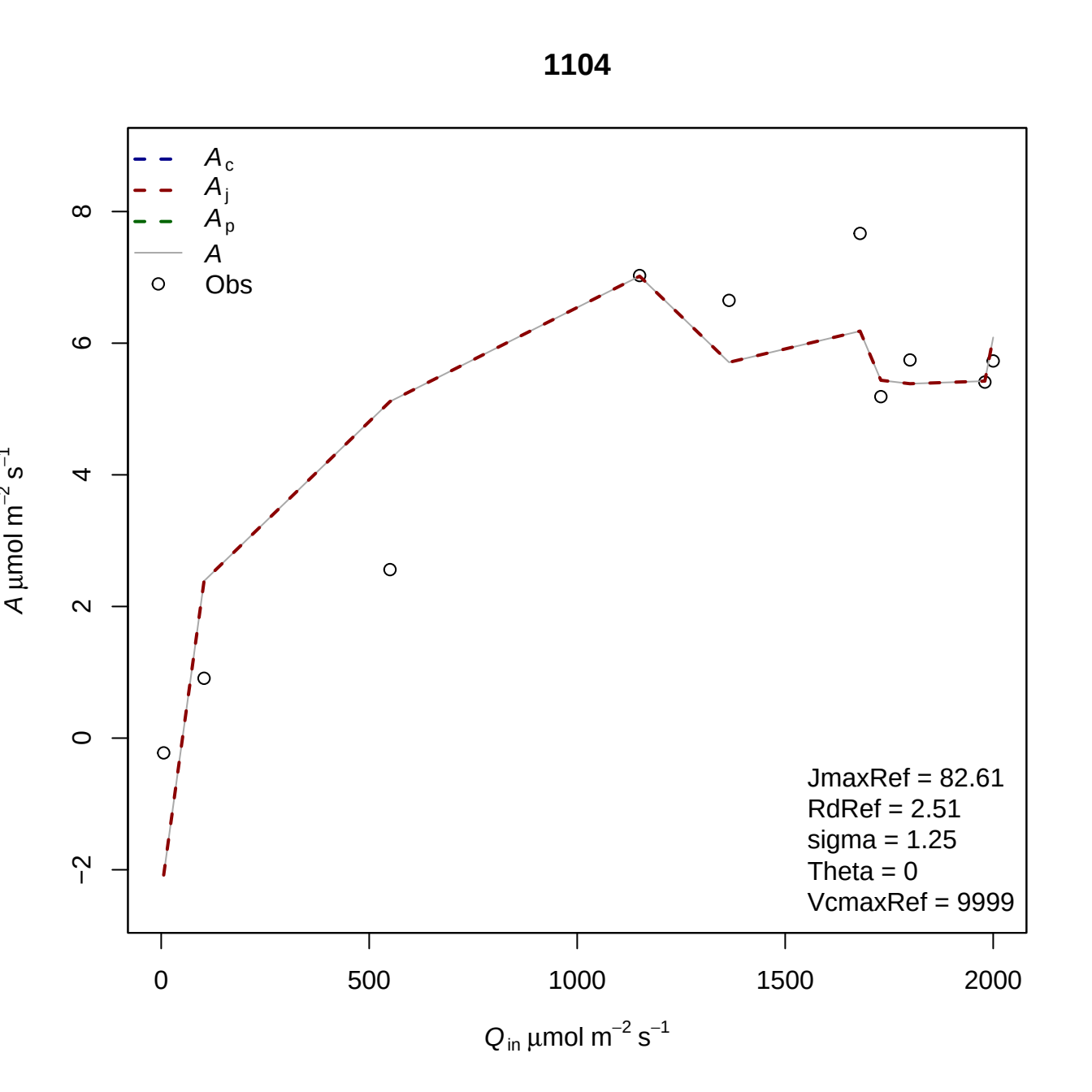

1098
A_c
A_j
A_p
A
Obs
JmaxRef = 81.14
RdRef = 2.48
sigma = 1.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

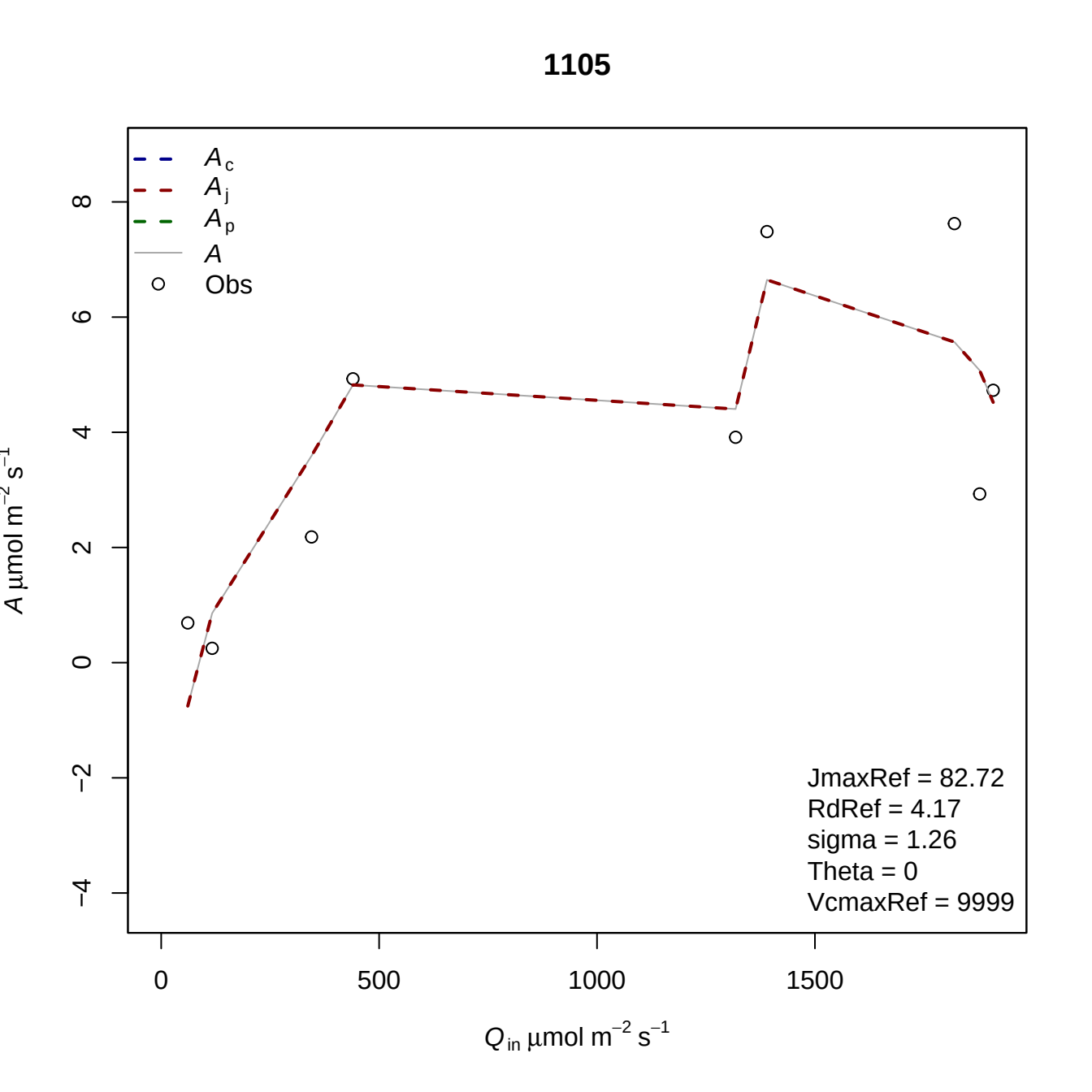

1099
A_c
A_j
A_p
A
Obs
JmaxRef = 99.39
RdRef = 3.25
sigma = 1.45
Theta = 0
VcmaxRef = 9999
0
5
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

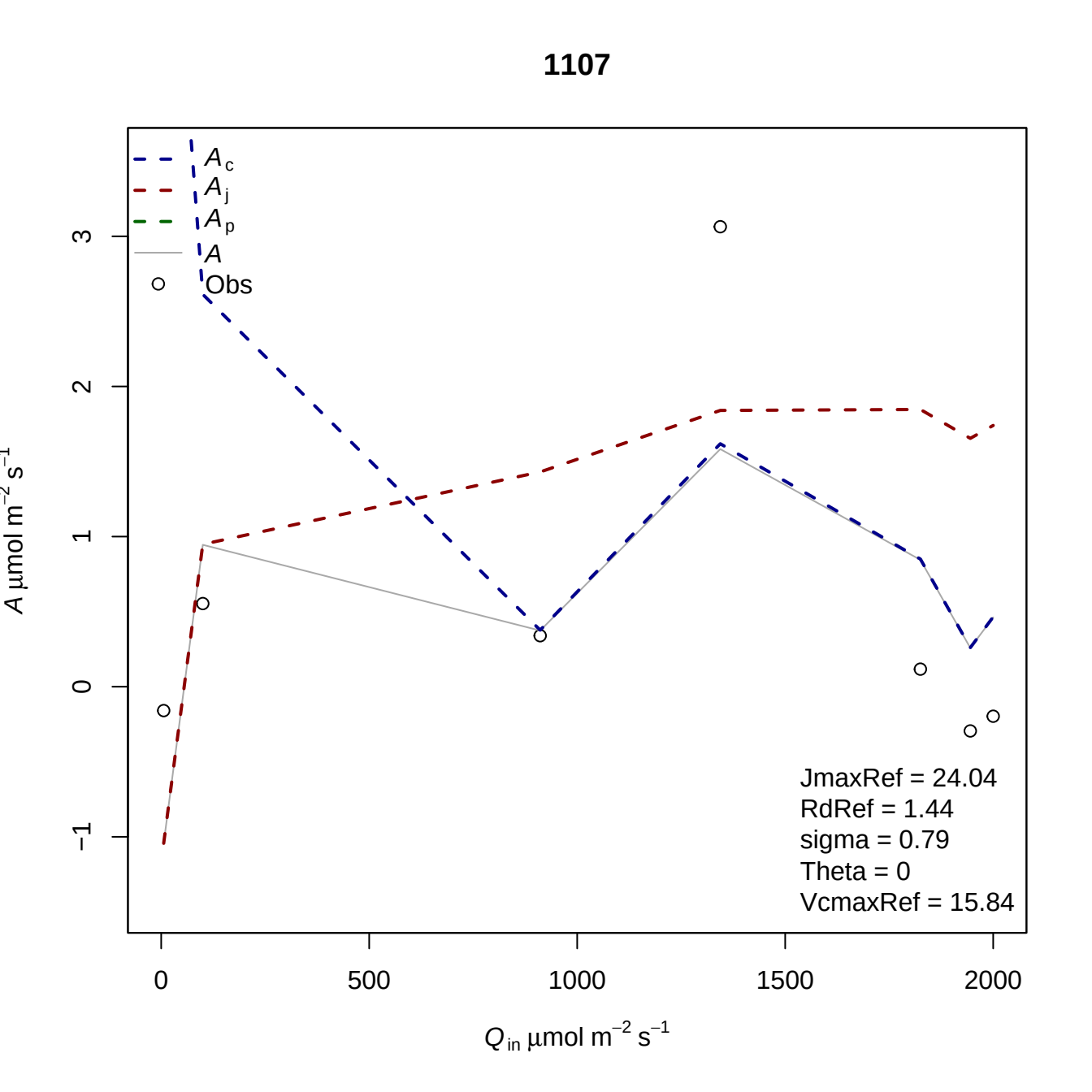

1100
A_c
A_j
A_p
A
Obs
JmaxRef = 37.86
RdRef = 3.03
sigma = 0.52
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
−3
0
500
1000
1500

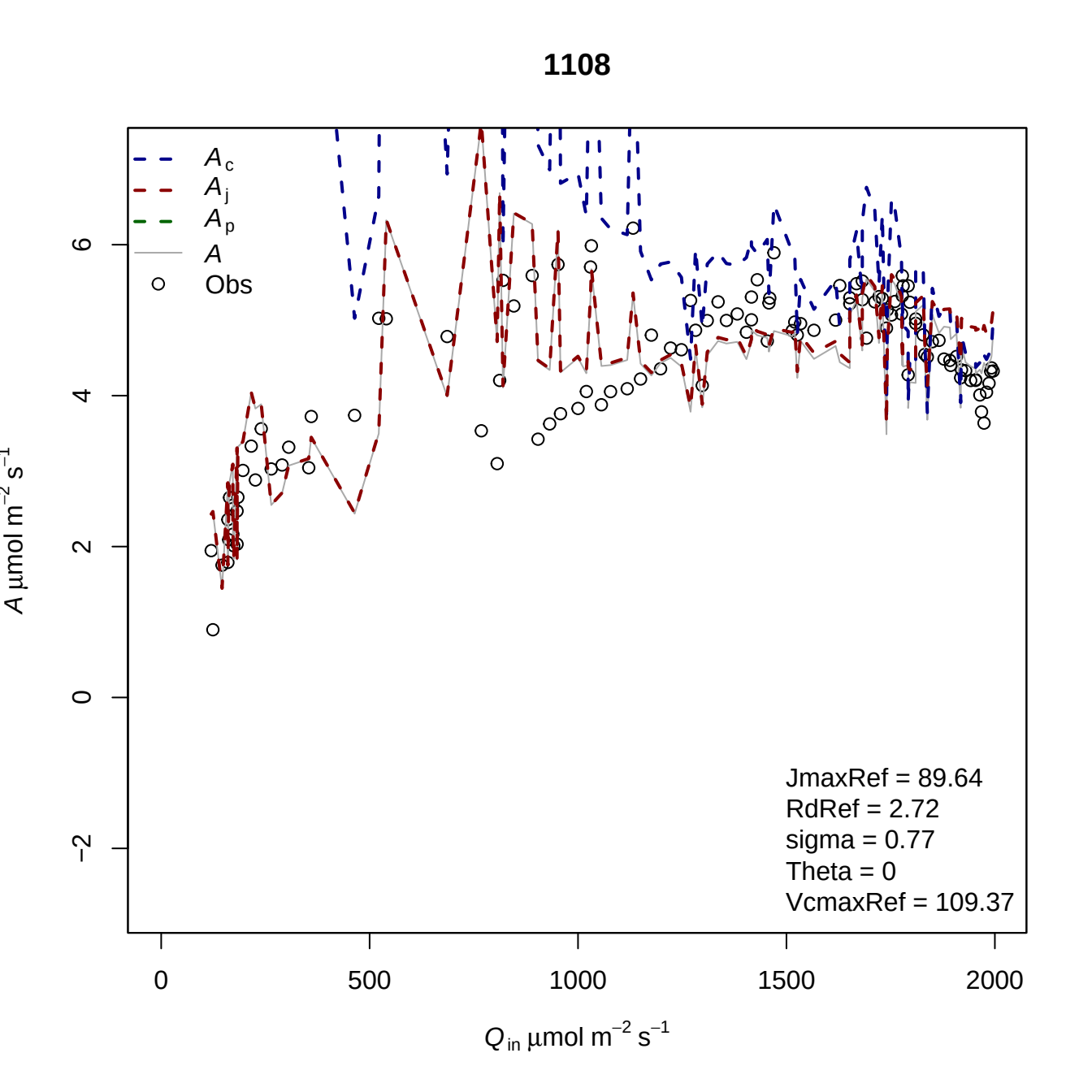

1102
A_c
A_j
A_p
A
Obs
JmaxRef = 83.05
RdRef = 3.25
sigma = 0.77
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

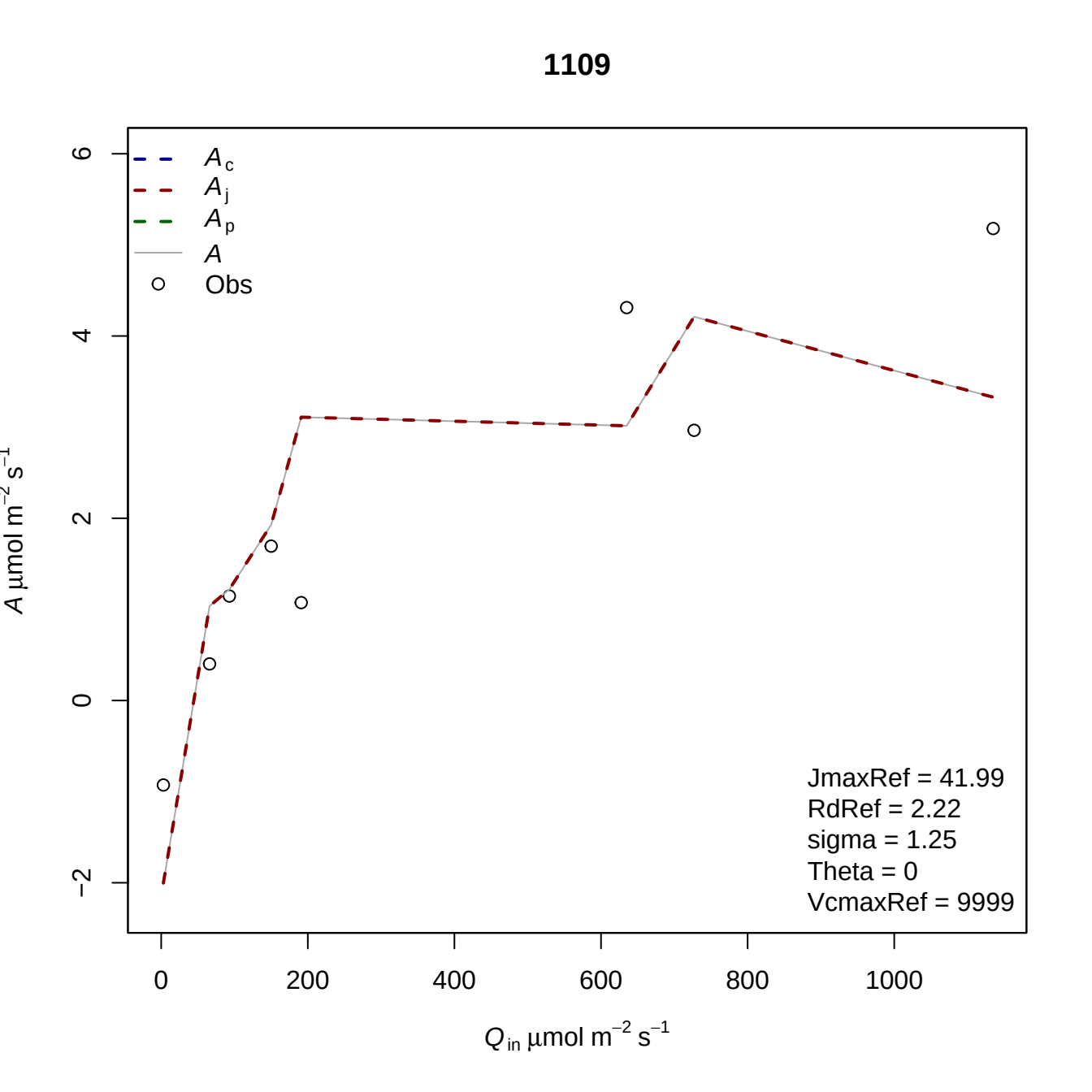

1103
A_c
A_j
A_p
A
Obs
JmaxRef = 65.3
RdRef = 2.25
sigma = 1.04
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

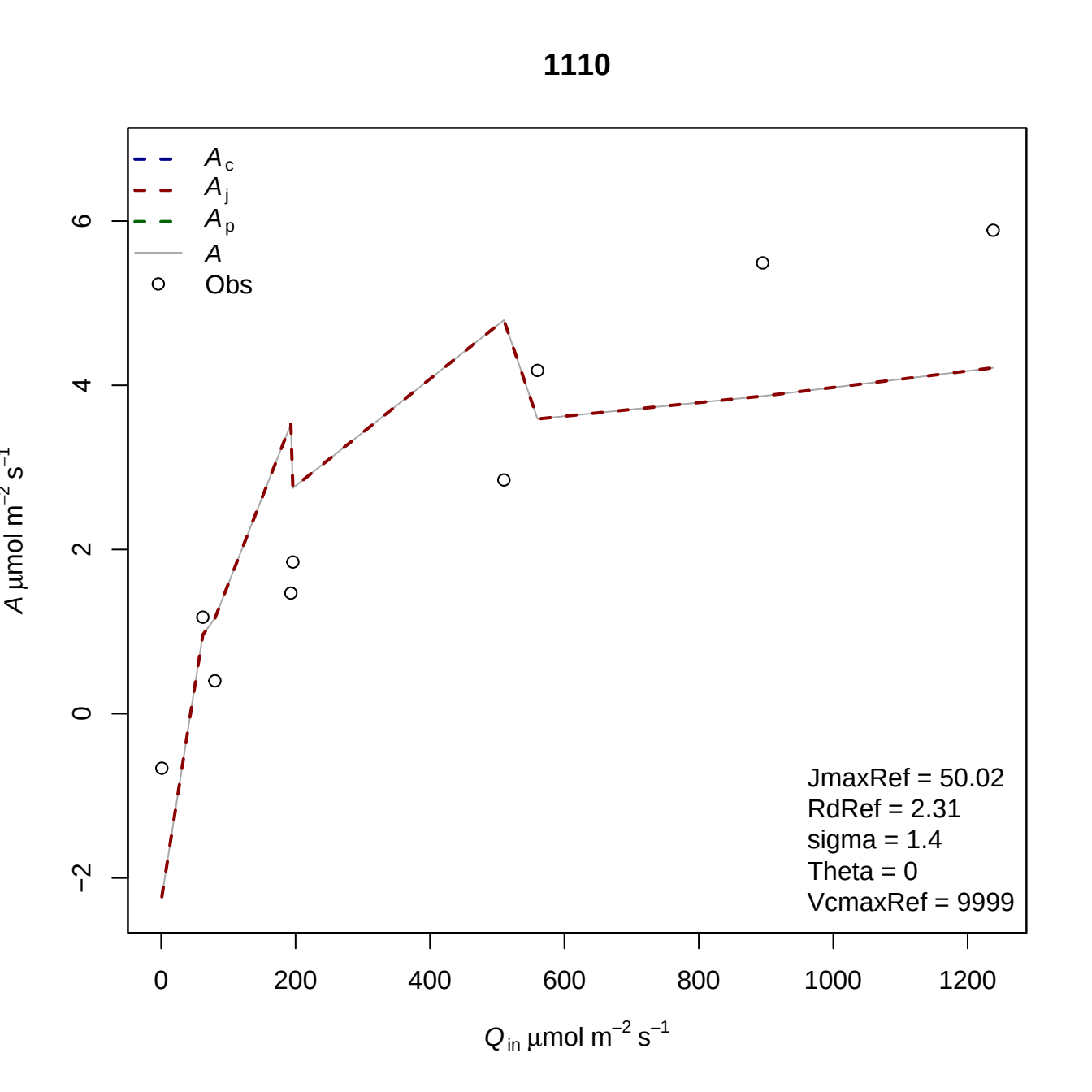

1104
A_c
A_j
A_p
A
Obs
JmaxRef = 82.61
RdRef = 2.51
sigma = 1.25
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

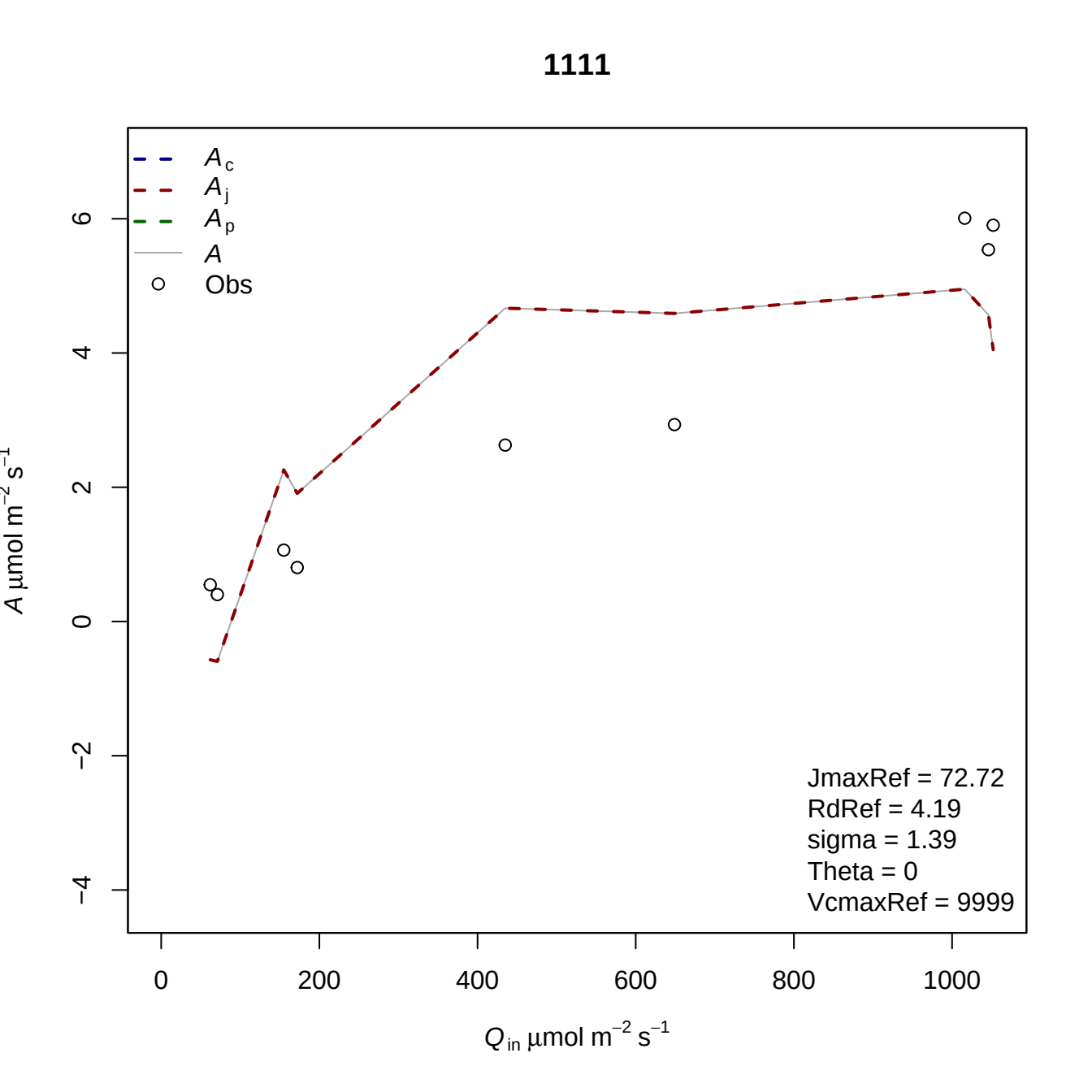

1105
A_c
A_j
A_p
A
Obs
JmaxRef = 82.72
RdRef = 4.17
sigma = 1.26
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

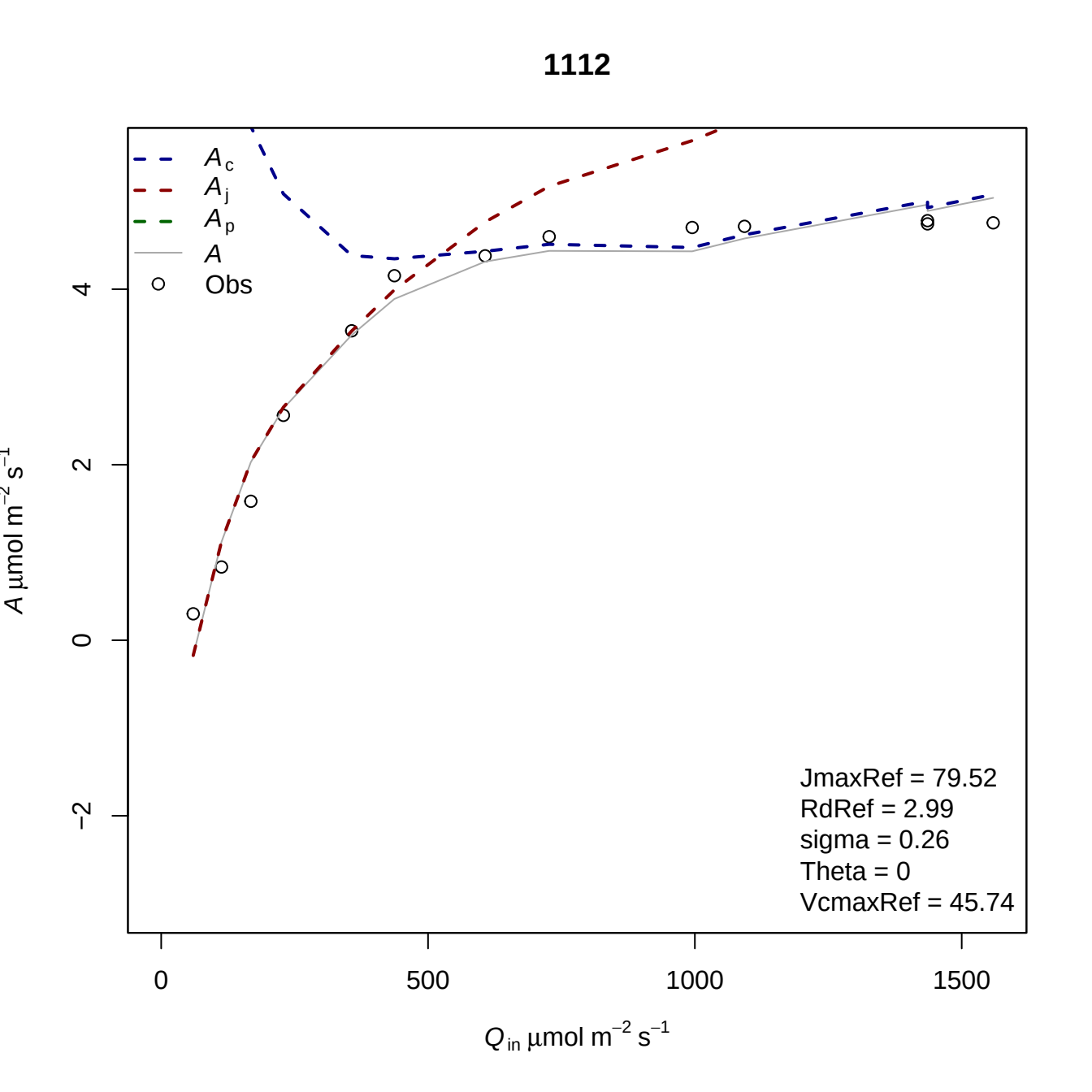

1107
A_c
A_j
A_p
A
Obs
JmaxRef = 24.04
RdRef = 1.44
sigma = 0.79
Theta = 0
VcmaxRef = 15.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

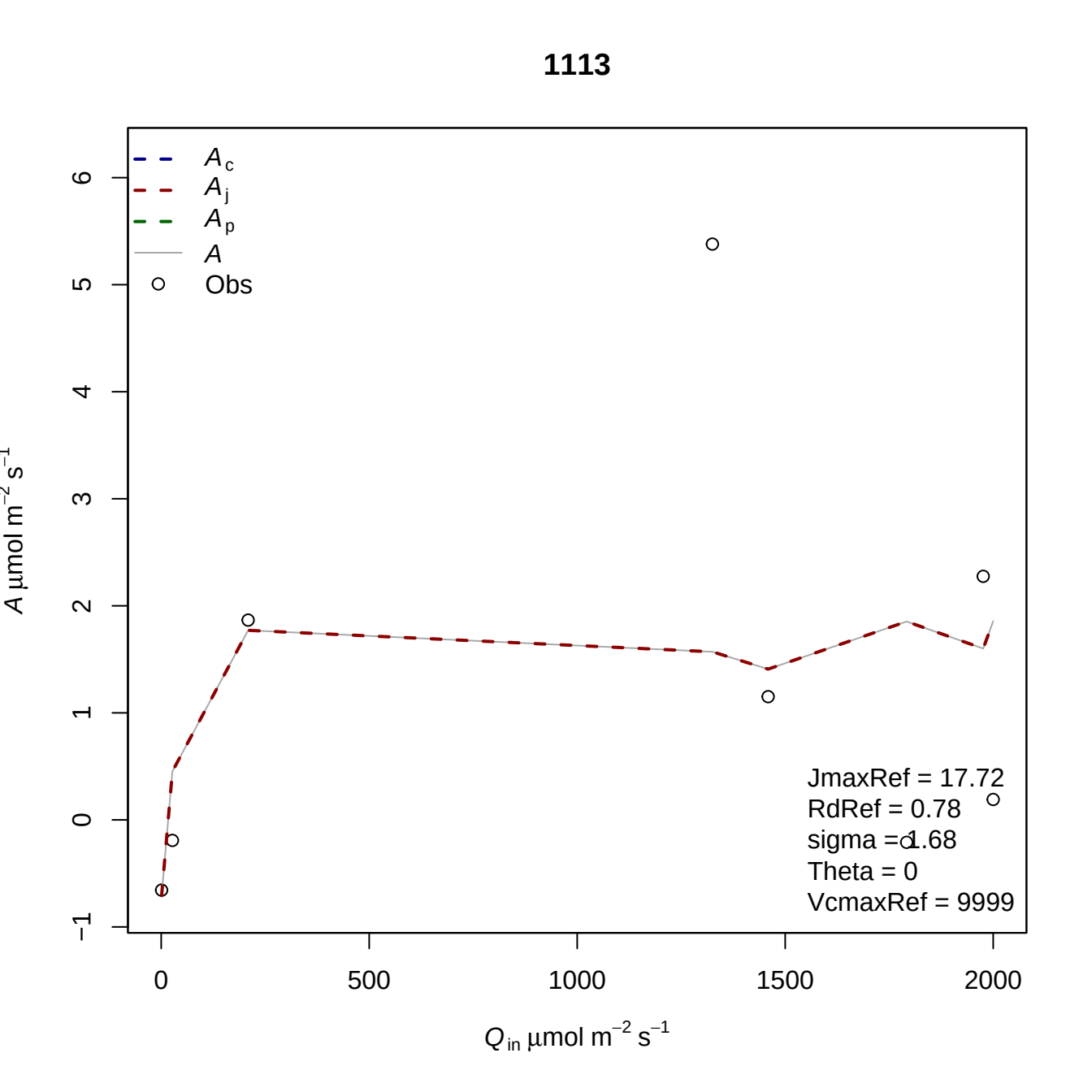

1108
A_c
A_j
A_p
A
Obs
JmaxRef = 89.64
RdRef = 2.72
sigma = 0.77
Theta = 0
VcmaxRef = 109.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

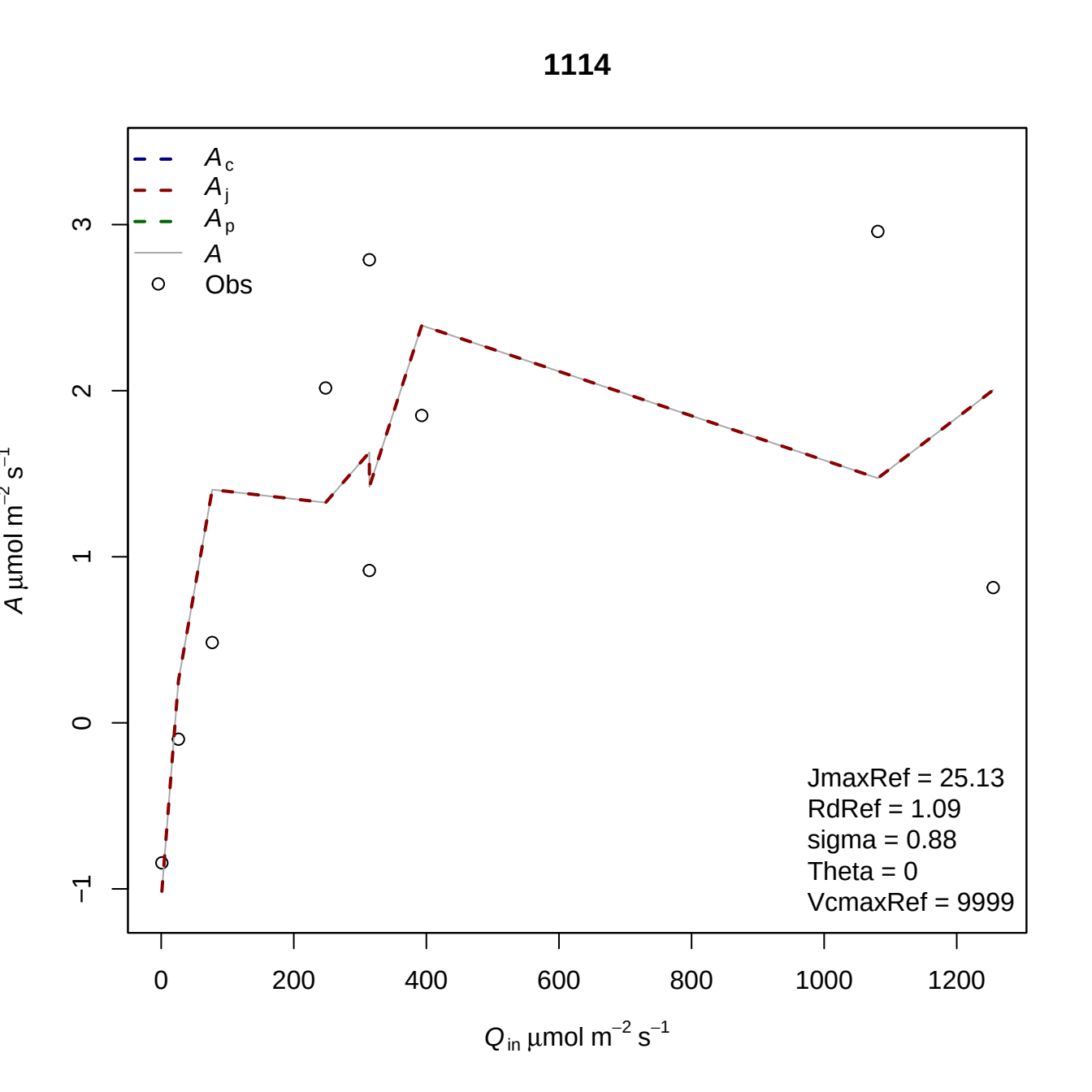

1109
A_c
A_j
A_p
A
Obs
JmaxRef = 41.99
RdRef = 2.22
sigma = 1.25
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

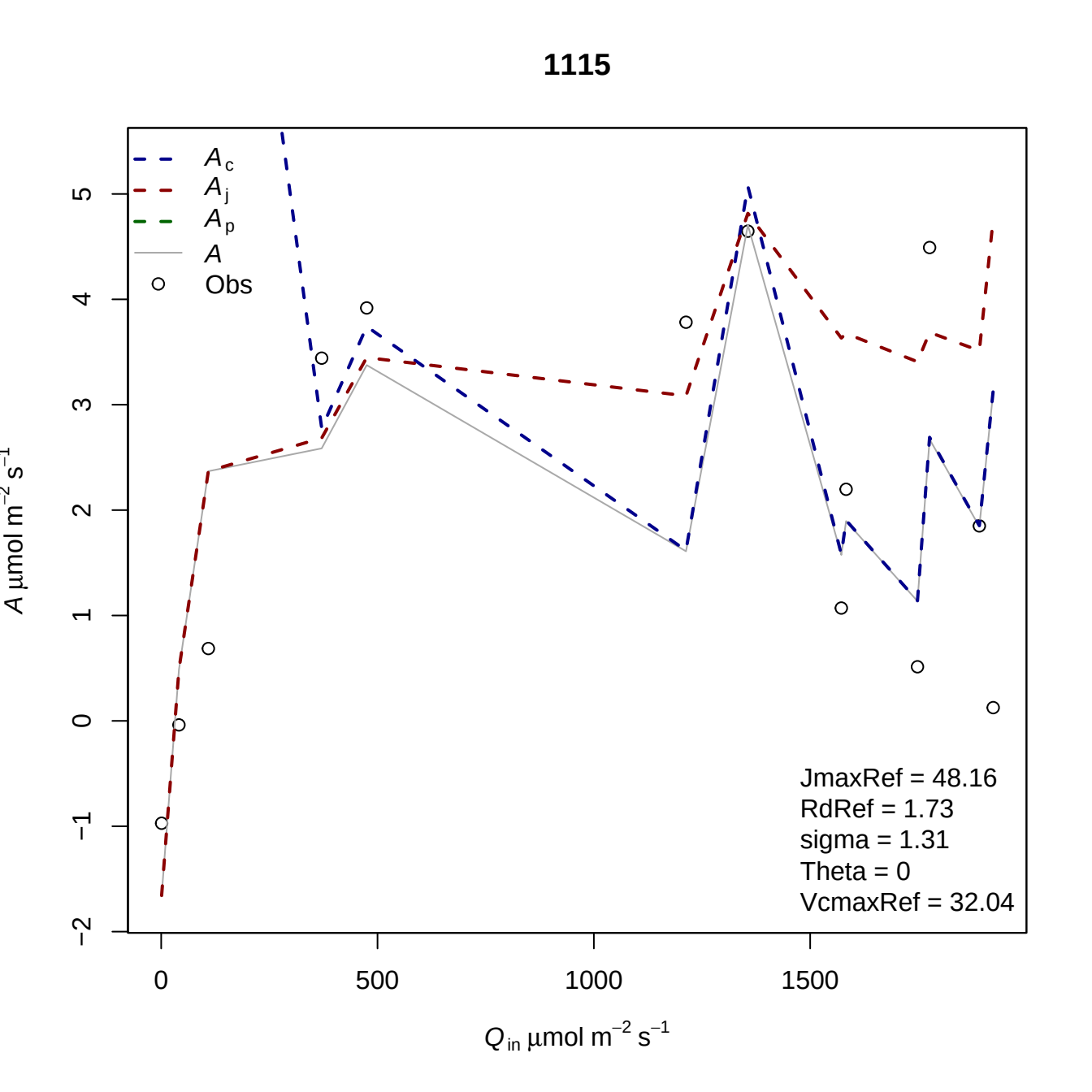

1110
A_c
A_j
A_p
A
Obs
JmaxRef = 50.02
RdRef = 2.31
sigma = 1.4
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

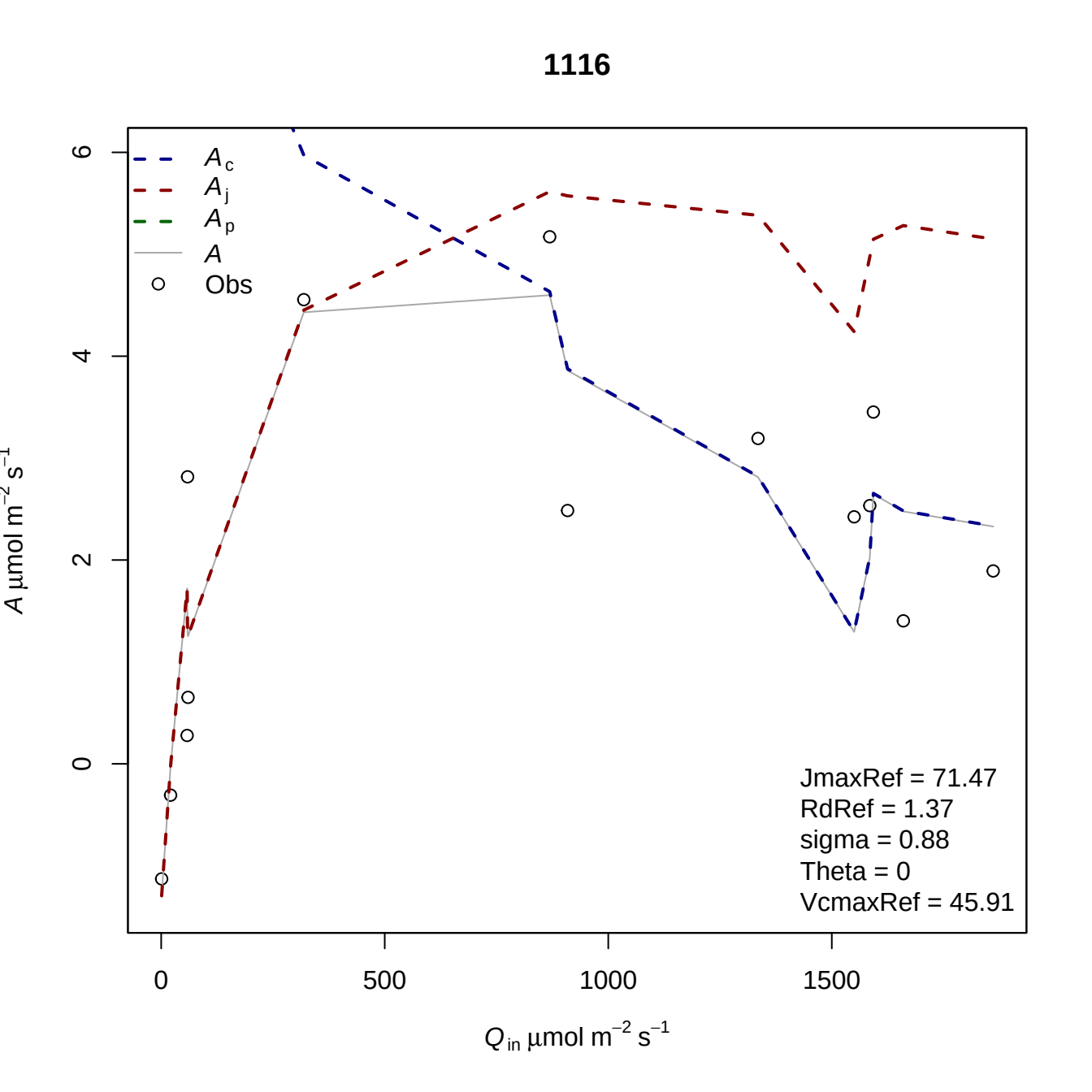

1111
A_c
A_j
A_p
A
Obs
JmaxRef = 72.72
RdRef = 4.19
sigma = 1.39
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000

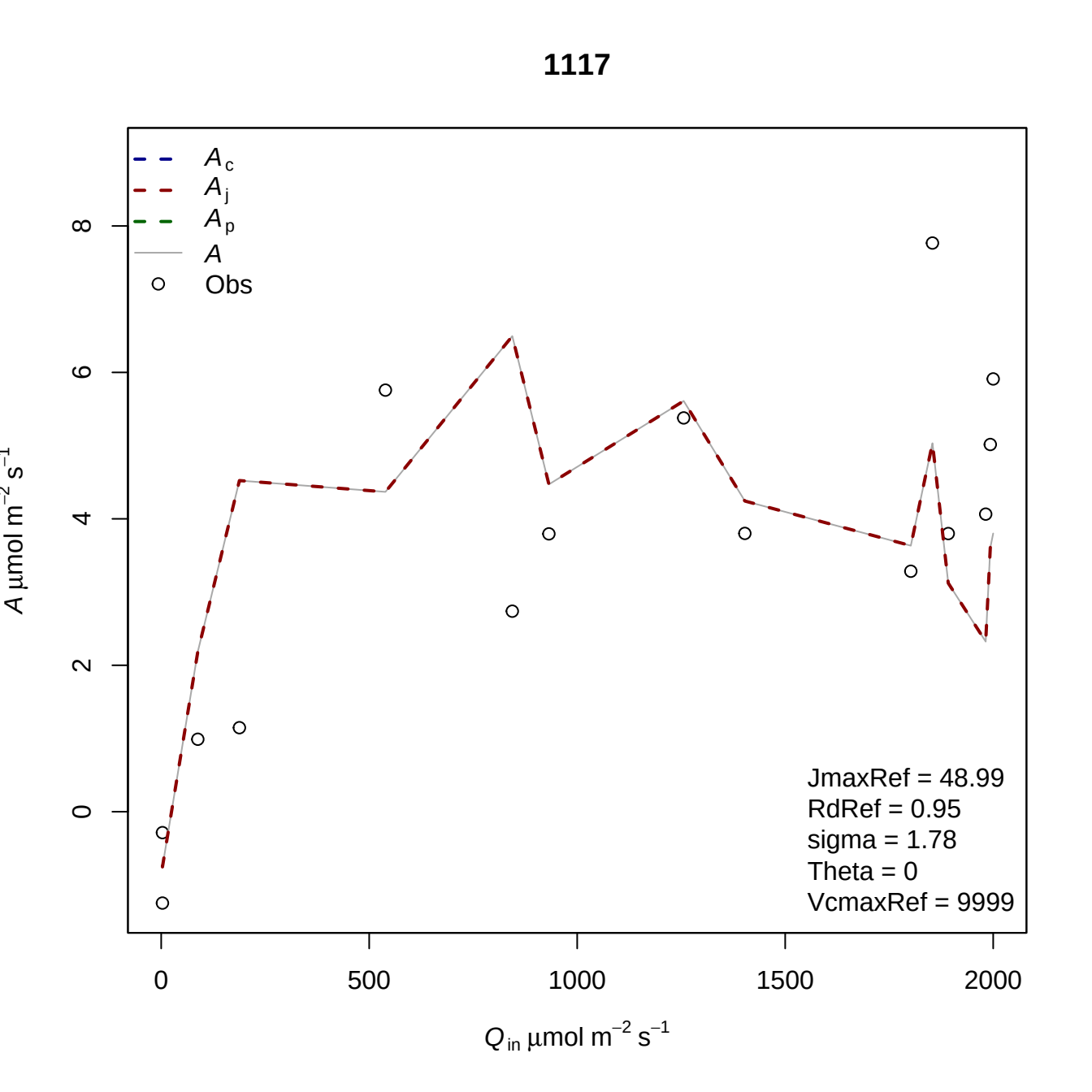

1112
A_c
A_j
A_p
A
Obs
JmaxRef = 79.52
RdRef = 2.99
sigma = 0.26
Theta = 0
VcmaxRef = 45.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

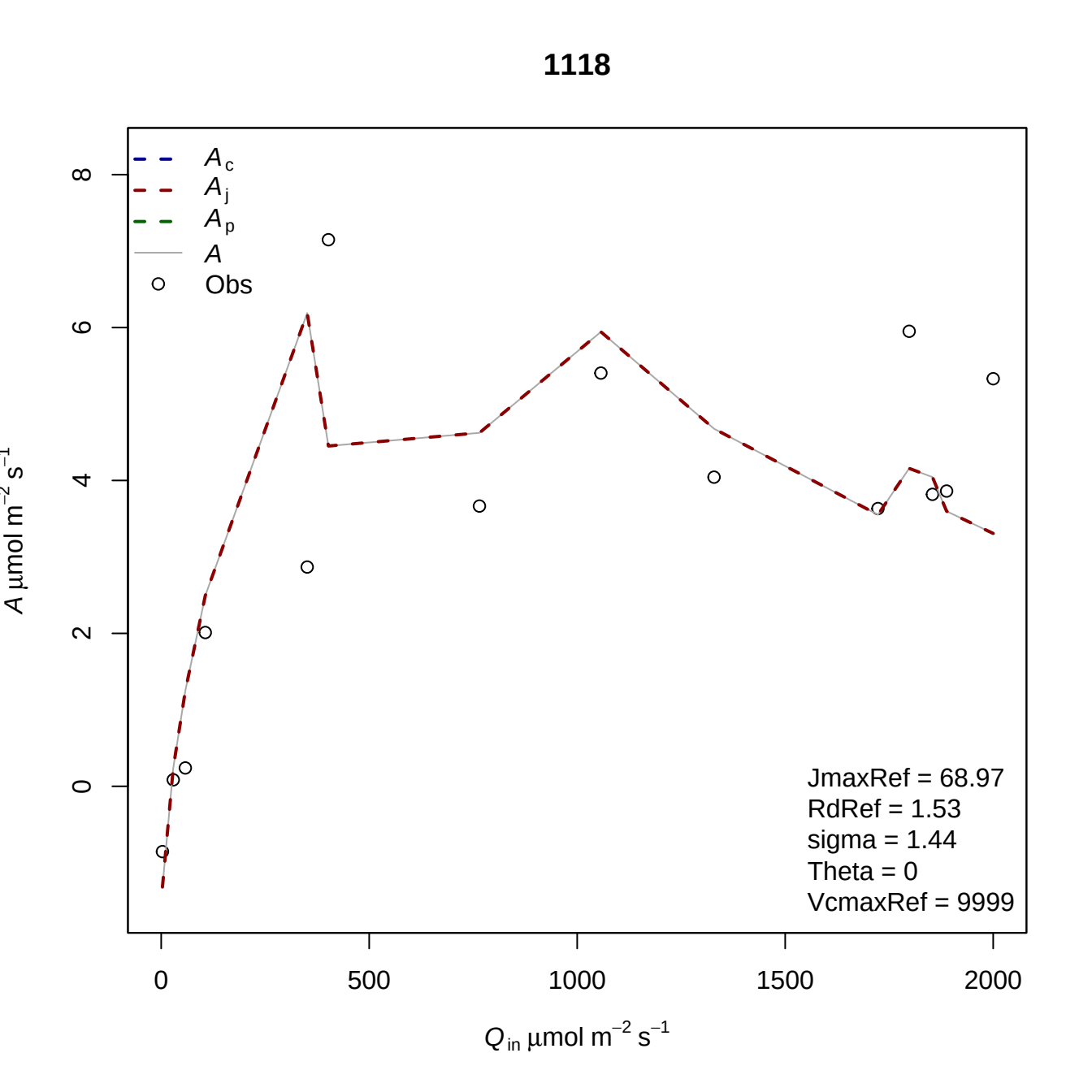

1113
A_c
A_j
A_p
A
Obs
JmaxRef = 17.72
RdRef = 0.78
sigma = 1.68
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

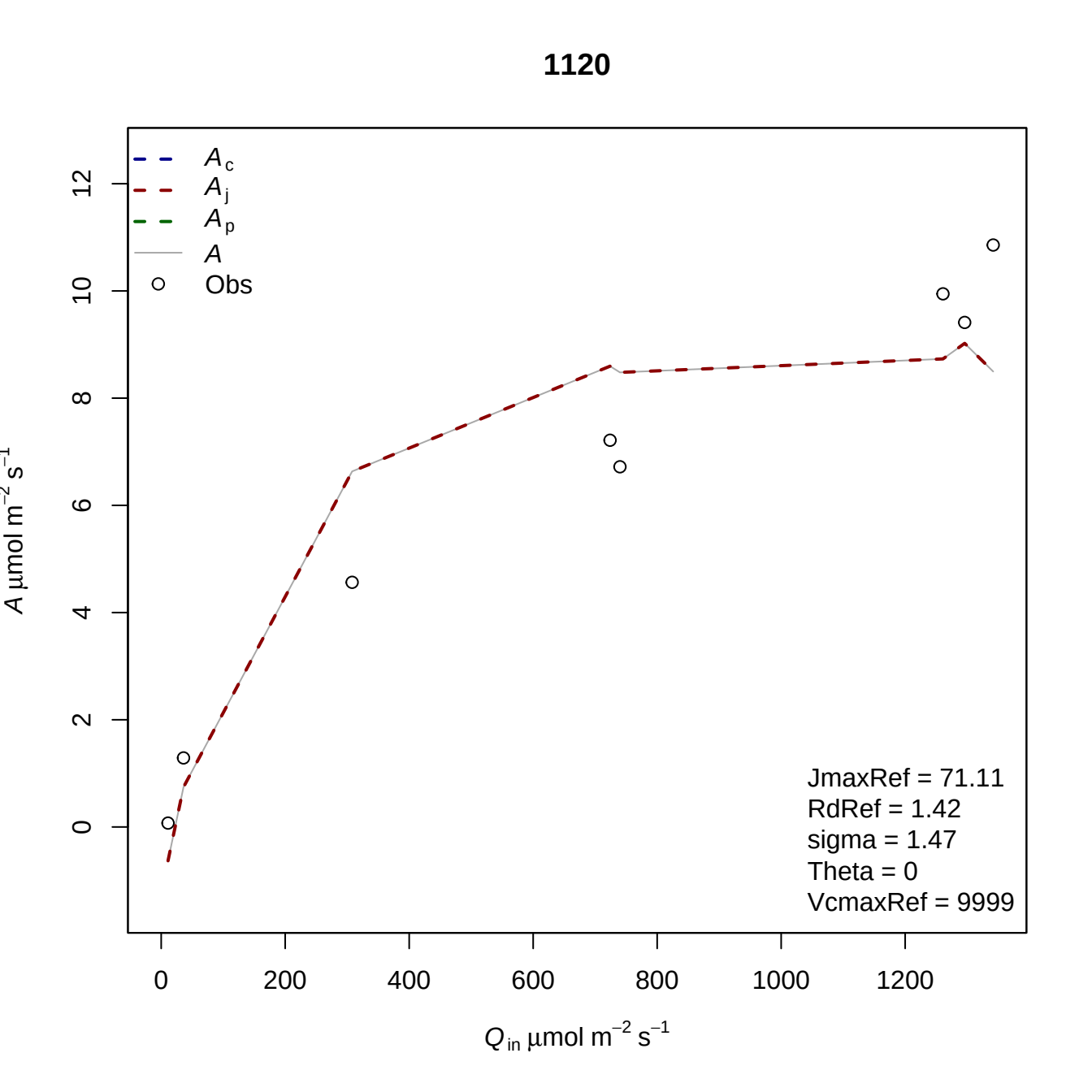

1114
A_c
A_j
A_p
A
Obs
JmaxRef = 25.13
RdRef = 1.09
sigma = 0.88
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200

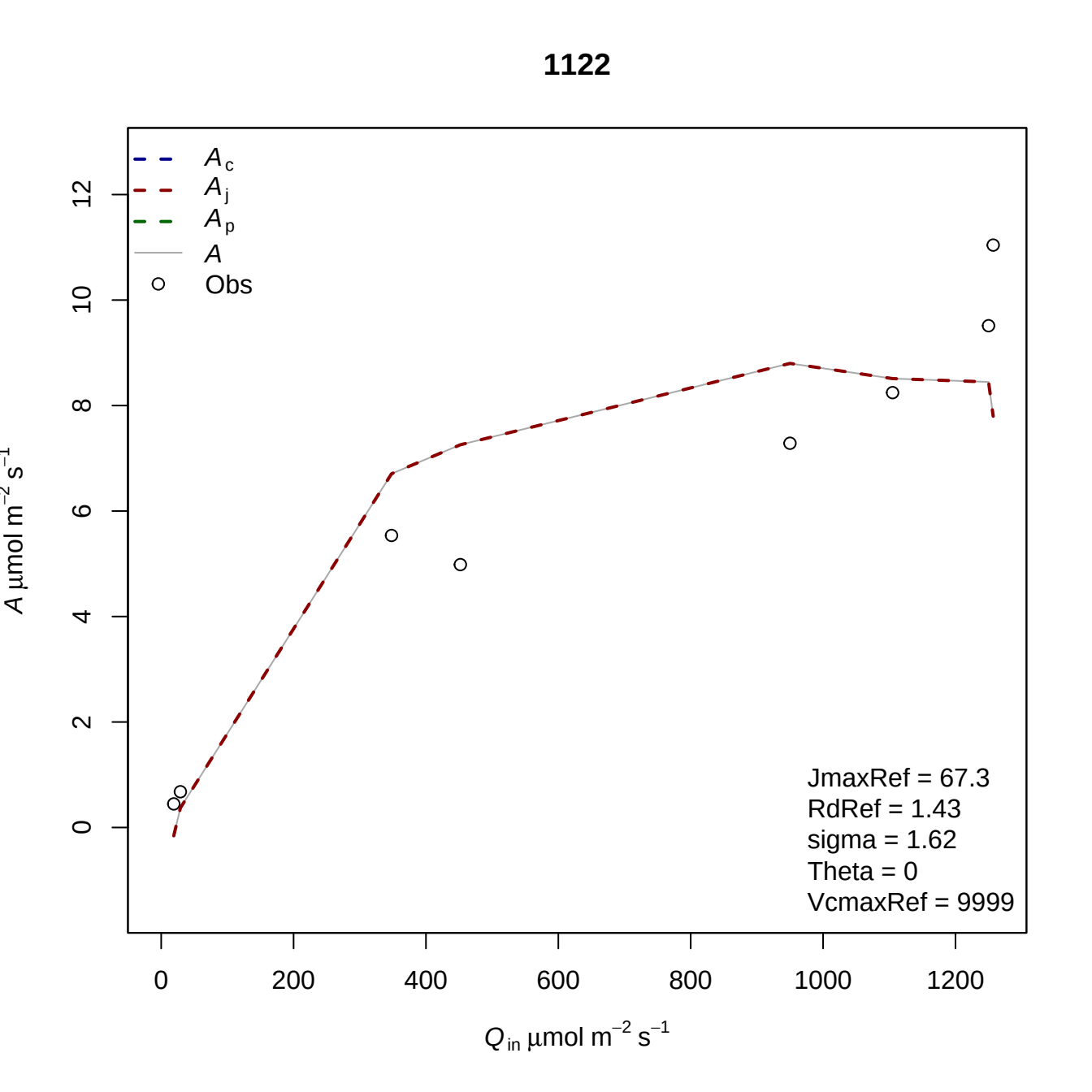

1115
A_c
A_j
A_p
A
Obs
JmaxRef = 48.16
RdRef = 1.73
sigma = 1.31
Theta = 0
VcmaxRef = 32.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
5
0
500
1000
1500

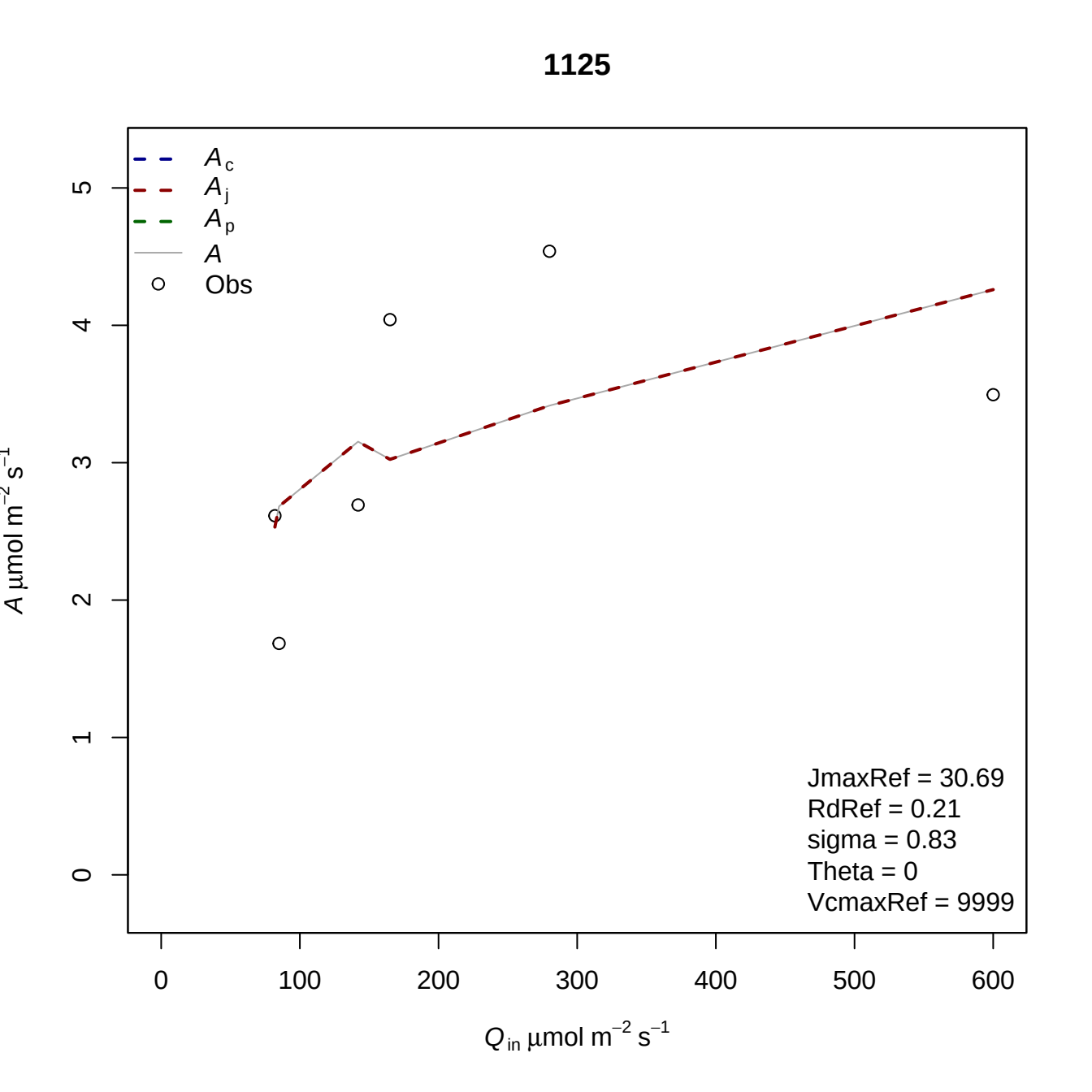

1116
A_c
A_j
A_p
A
Obs
JmaxRef = 71.47
RdRef = 1.37
sigma = 0.88
Theta = 0
VcmaxRef = 45.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

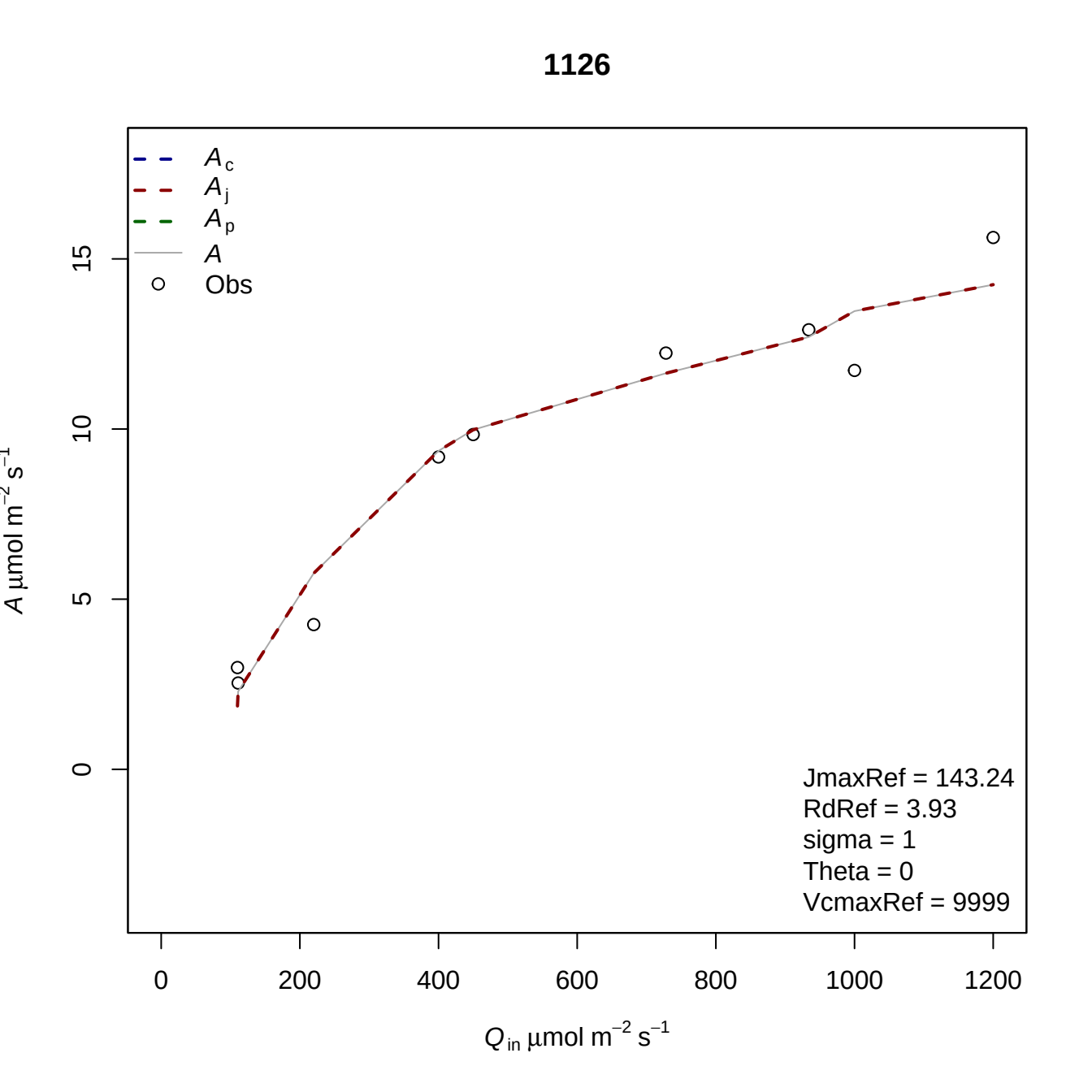

1117
A_c
A_j
A_p
A
Obs
JmaxRef = 48.99
RdRef = 0.95
sigma = 1.78
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

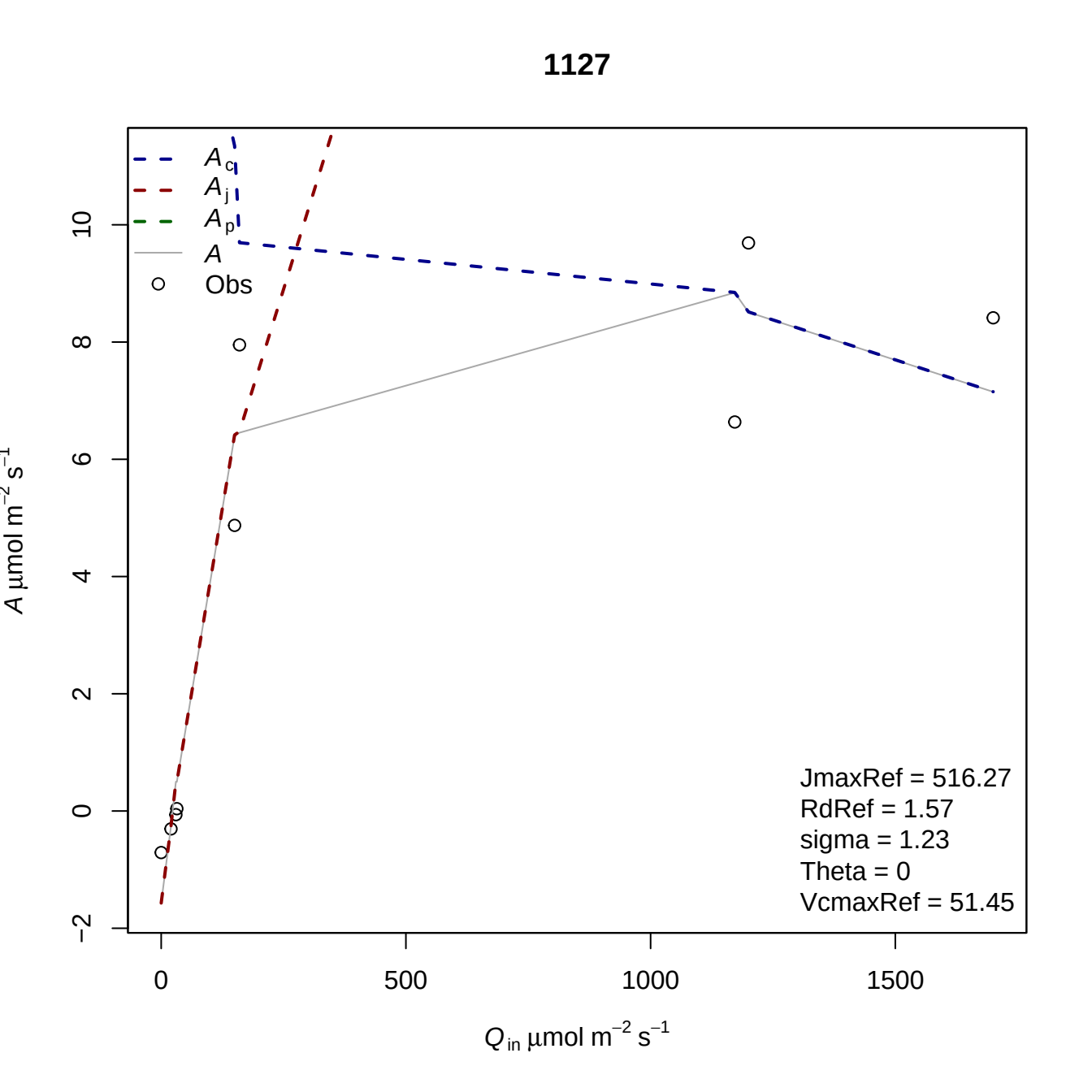

1118
A_c
A_j
A_p
A
Obs
JmaxRef = 68.97
RdRef = 1.53
sigma = 1.44
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

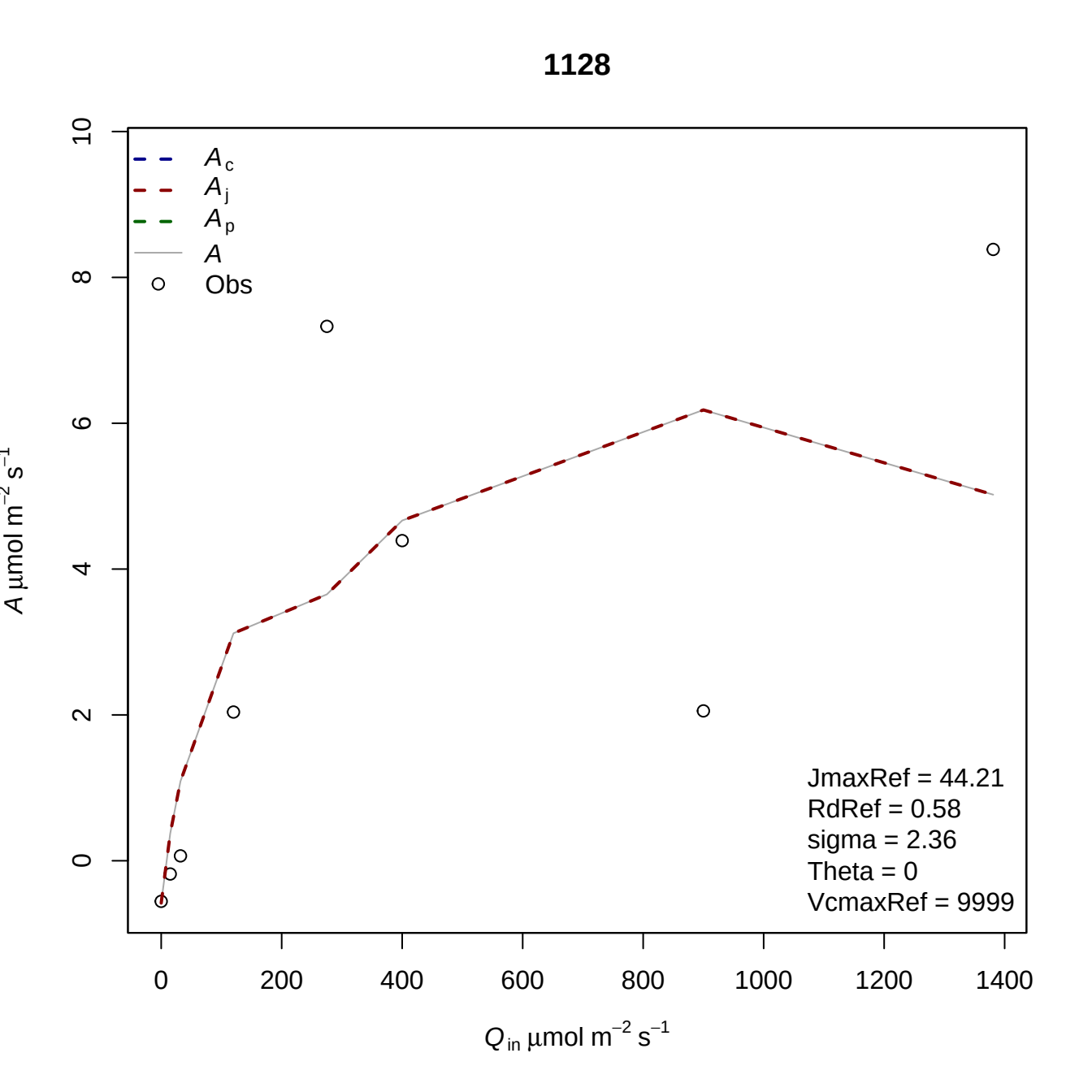

1120
A_c
A_j
A_p
A
Obs
JmaxRef = 71.11
RdRef = 1.42
sigma = 1.47
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

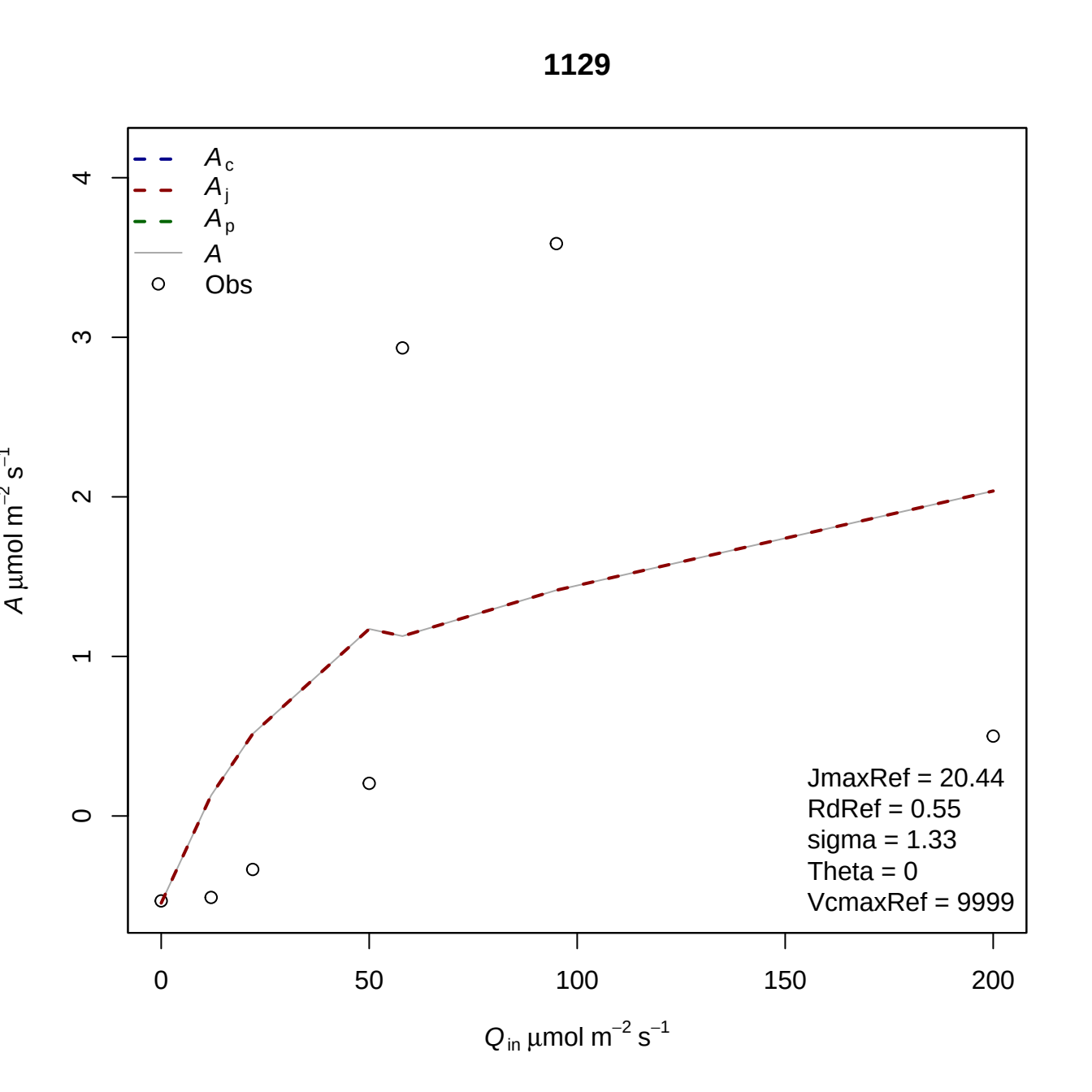

1122
A_c
A_j
A_p
A
Obs
JmaxRef = 67.3
RdRef = 1.43
sigma = 1.62
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

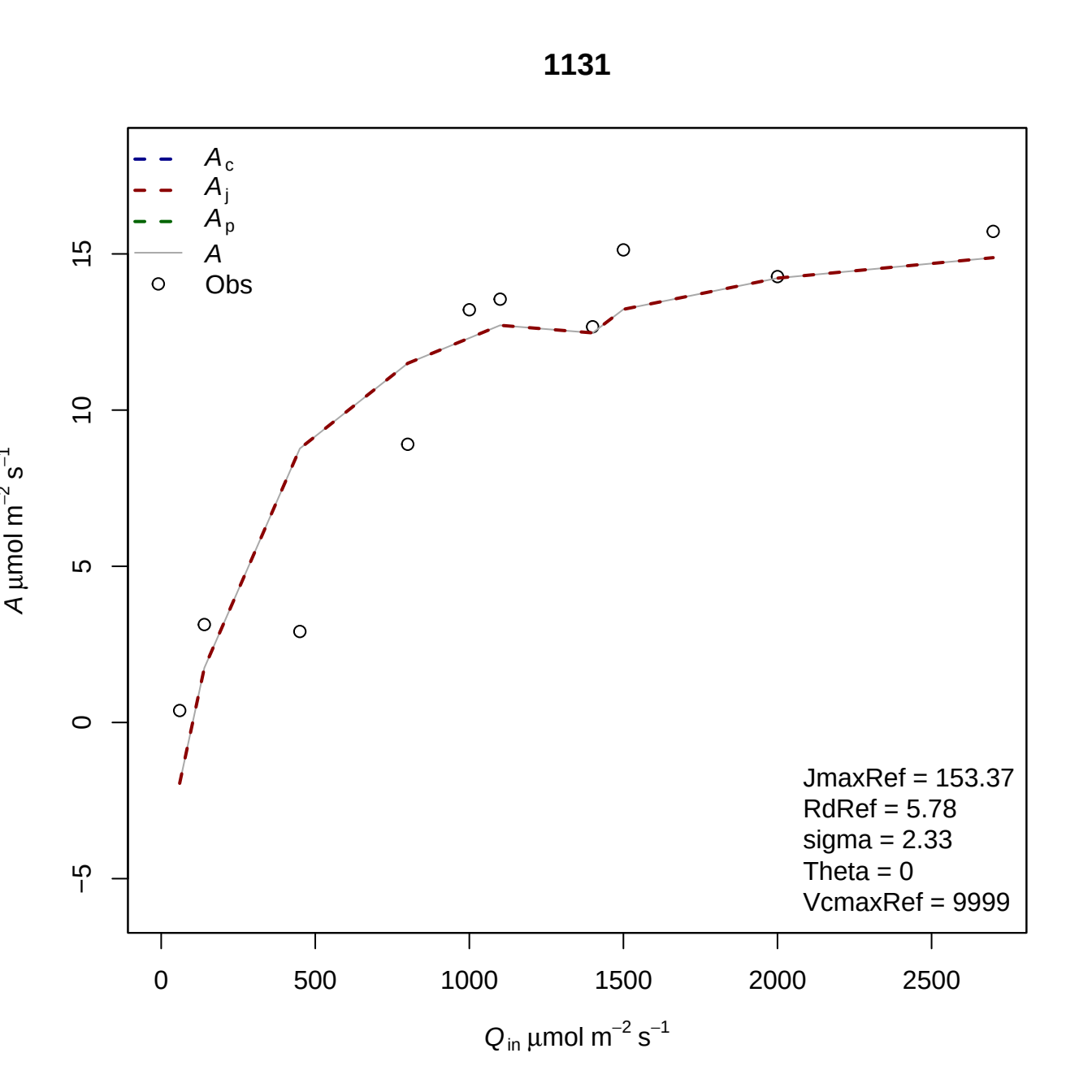

1125
A_c
A_j
A_p
A
Obs
JmaxRef = 30.69
RdRef = 0.21
sigma = 0.83
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
100
200
300
400
500
600

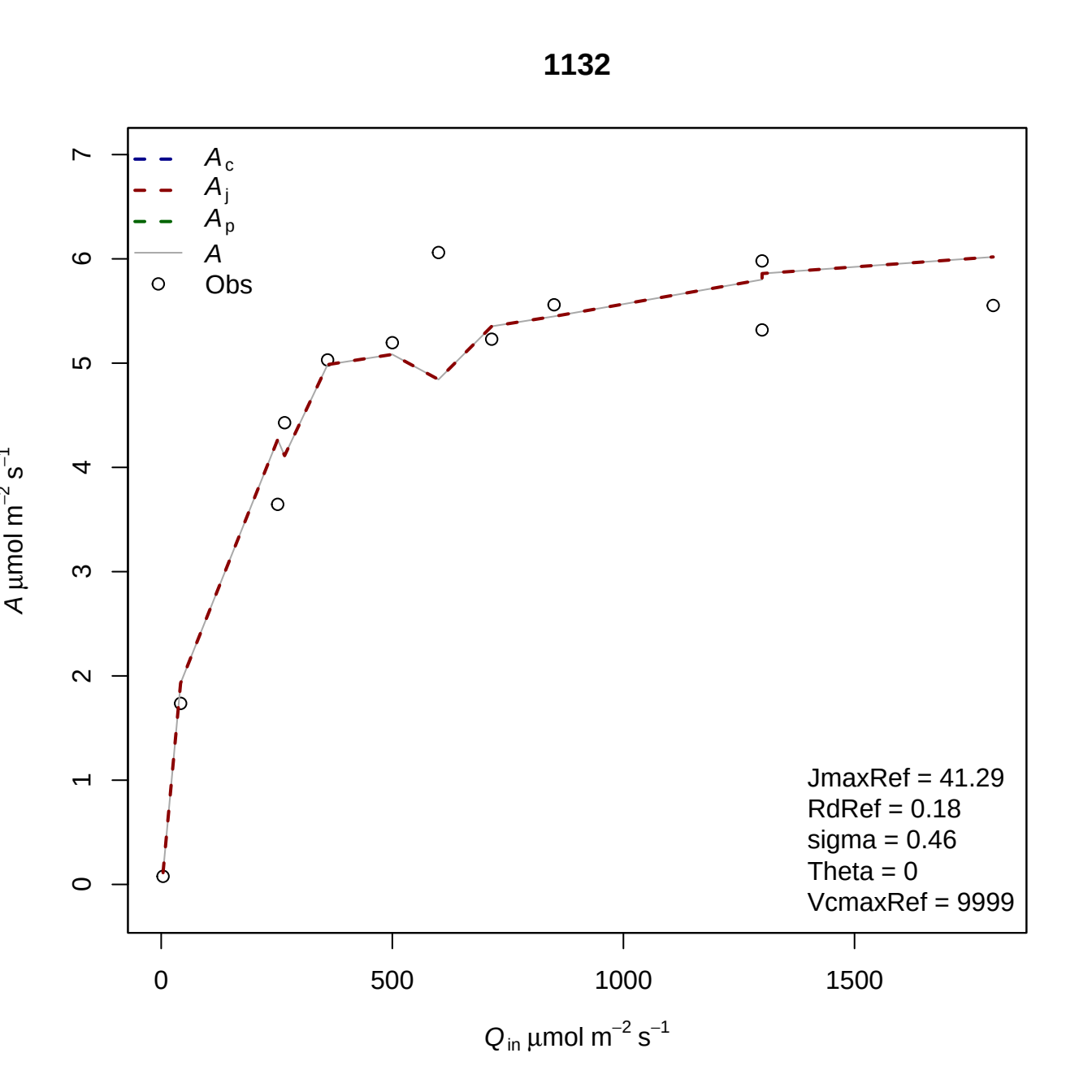

1126
A_c
A_j
A_p
A
Obs
JmaxRef = 143.24
RdRef = 3.93
sigma = 1
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

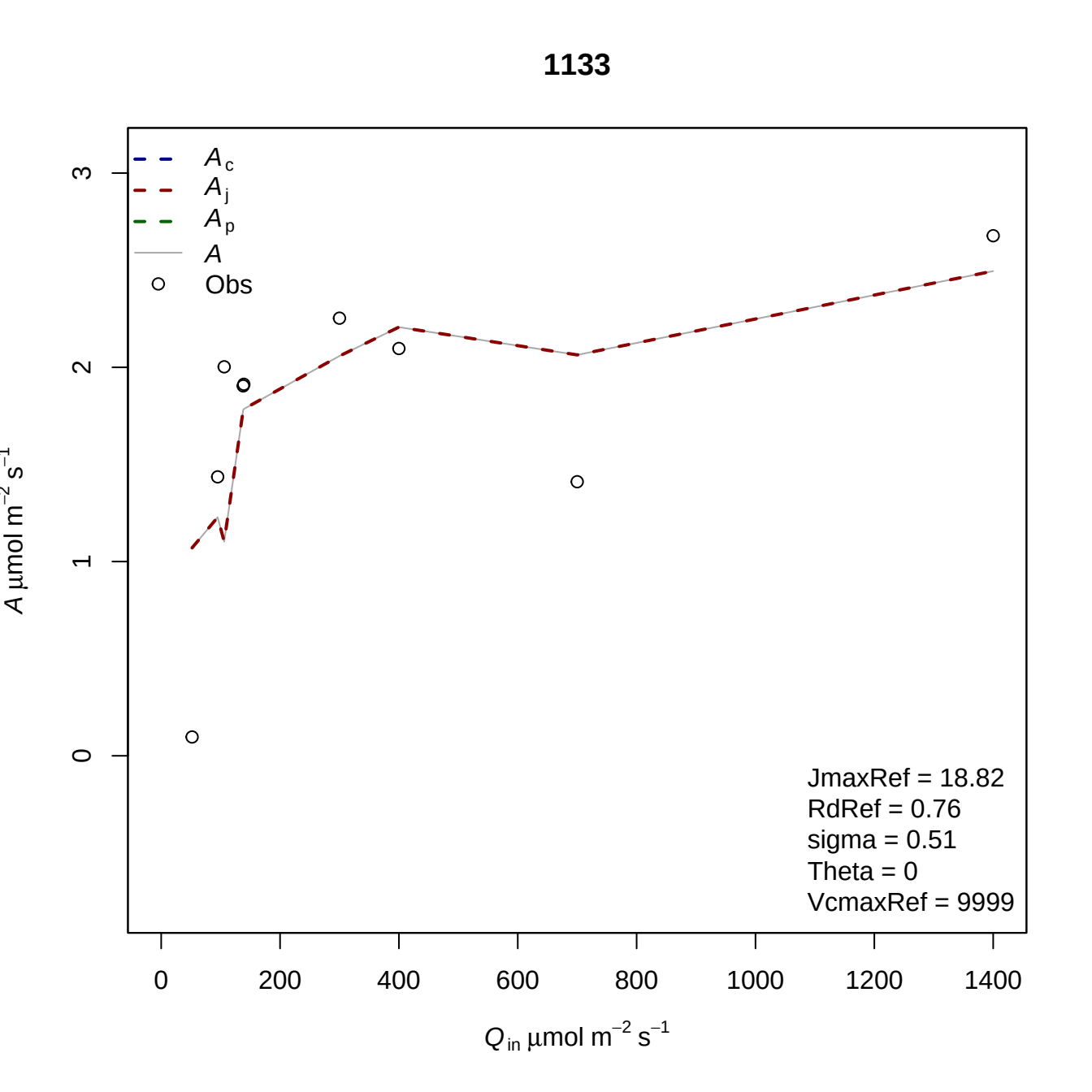

1127
A_c
A_j
A_p
A
Obs
JmaxRef = 516.27
RdRef = 1.57
sigma = 1.23
Theta = 0
VcmaxRef = 51.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

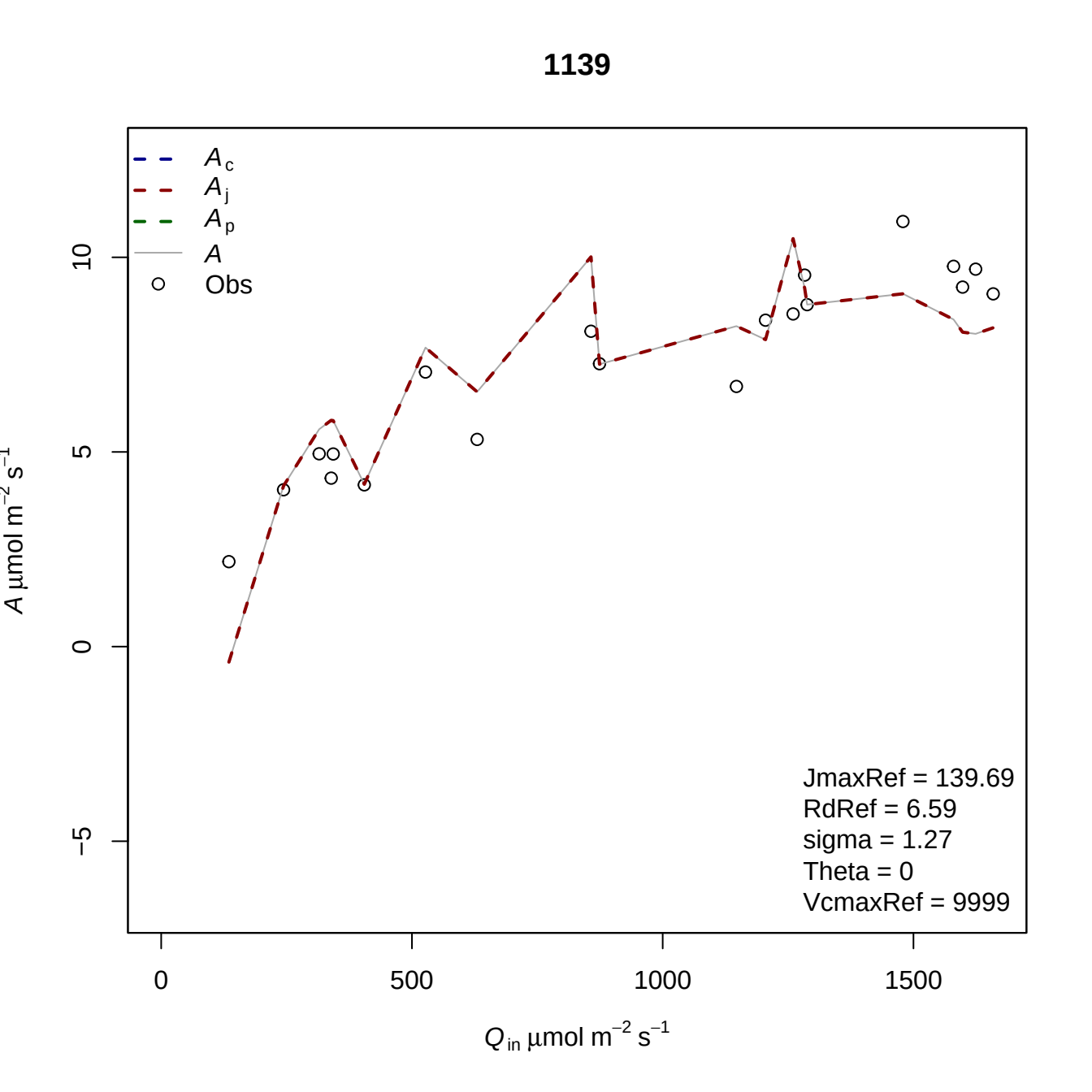

1128
A_c
A_j
A_p
A
Obs
JmaxRef = 44.21
RdRef = 0.58
sigma = 2.36
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

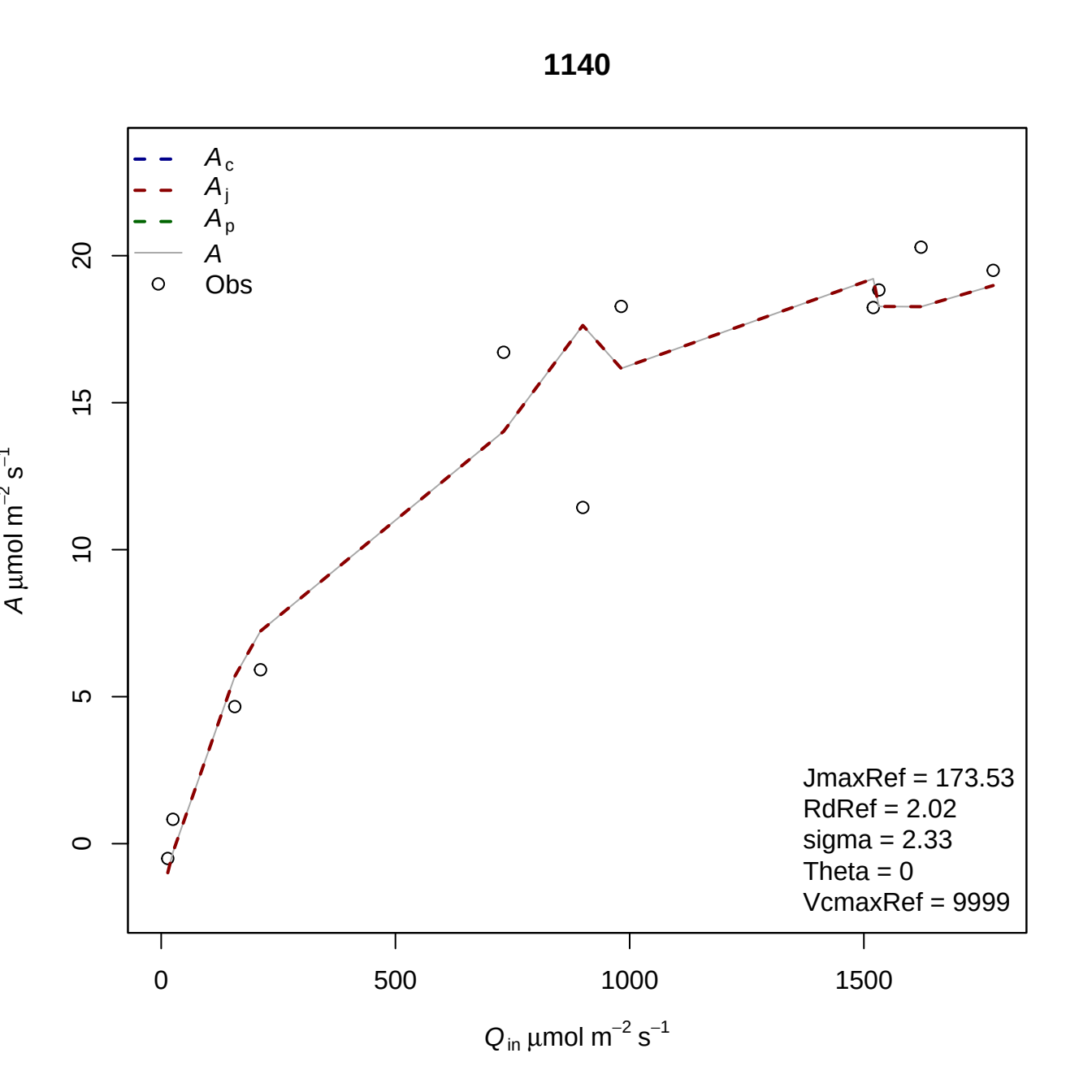

1129
A_c
A_j
A_p
A
Obs
JmaxRef = 20.44
RdRef = 0.55
sigma = 1.33
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
50
100
150
200

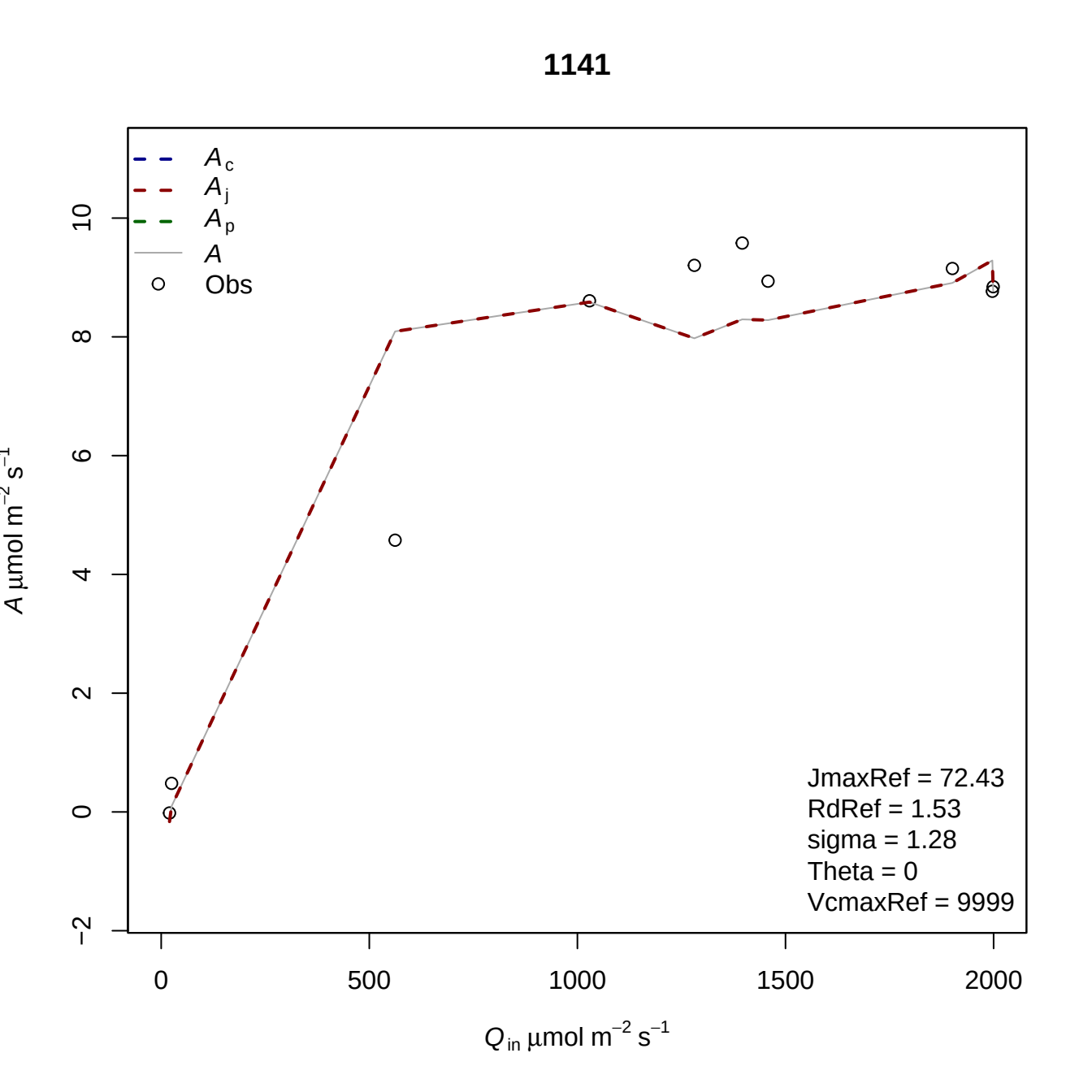

1131
A_c
A_j
A_p
A
Obs
JmaxRef = 153.37
RdRef = 5.78
sigma = 2.33
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000
2500

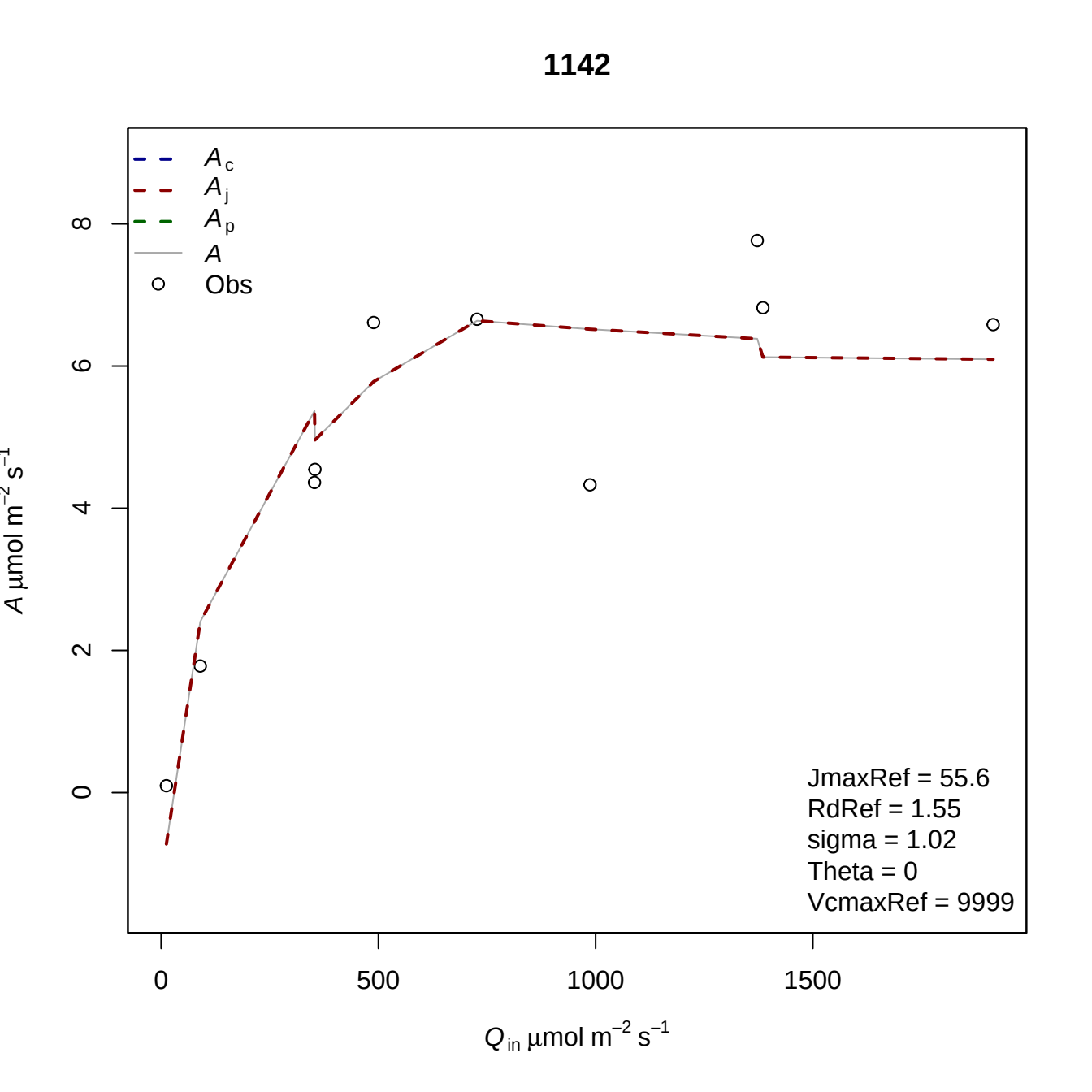

1132
A_c
A_j
A_p
A
Obs
JmaxRef = 41.29
RdRef = 0.18
sigma = 0.46
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
7
0
500
1000
1500

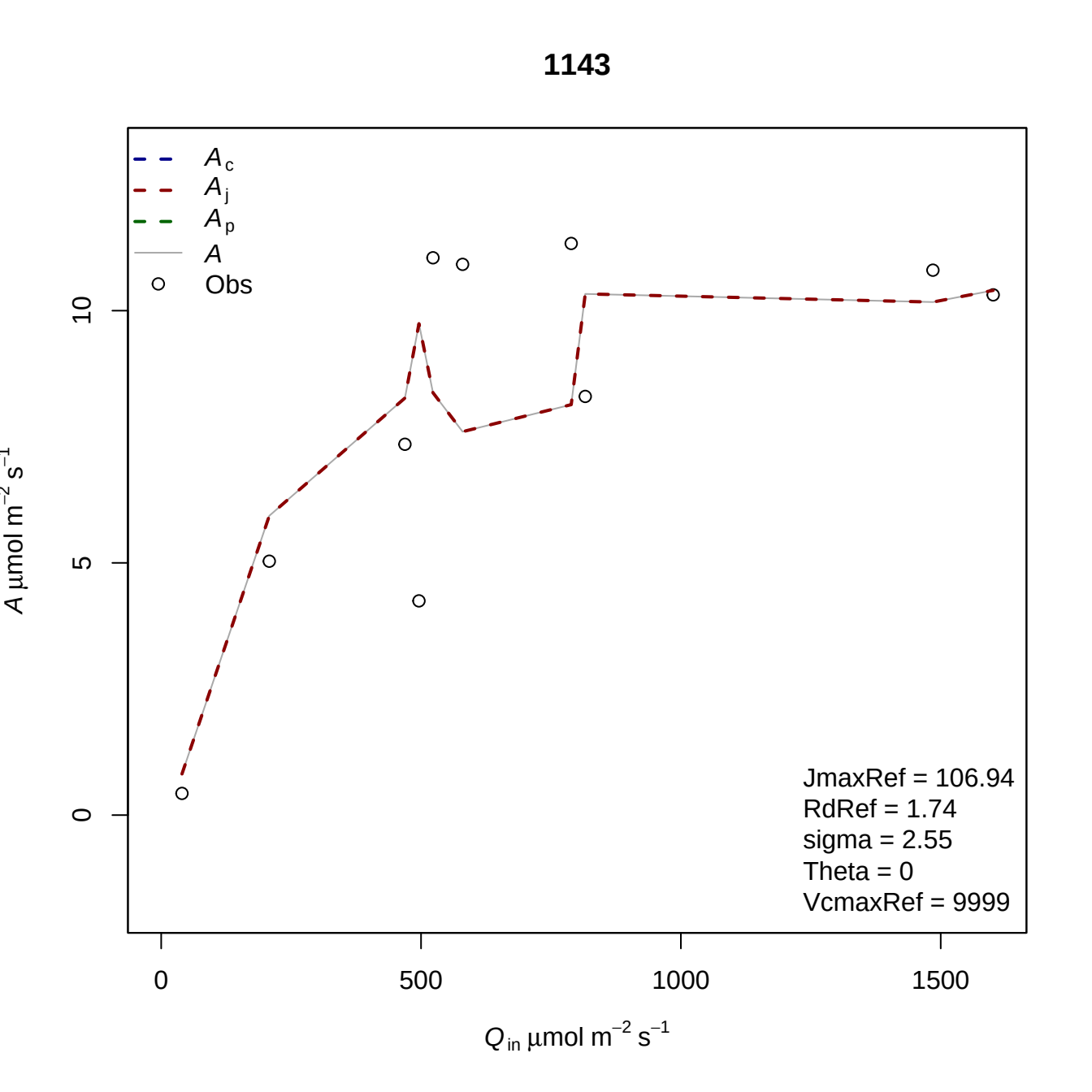

1133
A_c
A_j
A_p
A
Obs
JmaxRef = 18.82
RdRef = 0.76
sigma = 0.51
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000
1200
1400

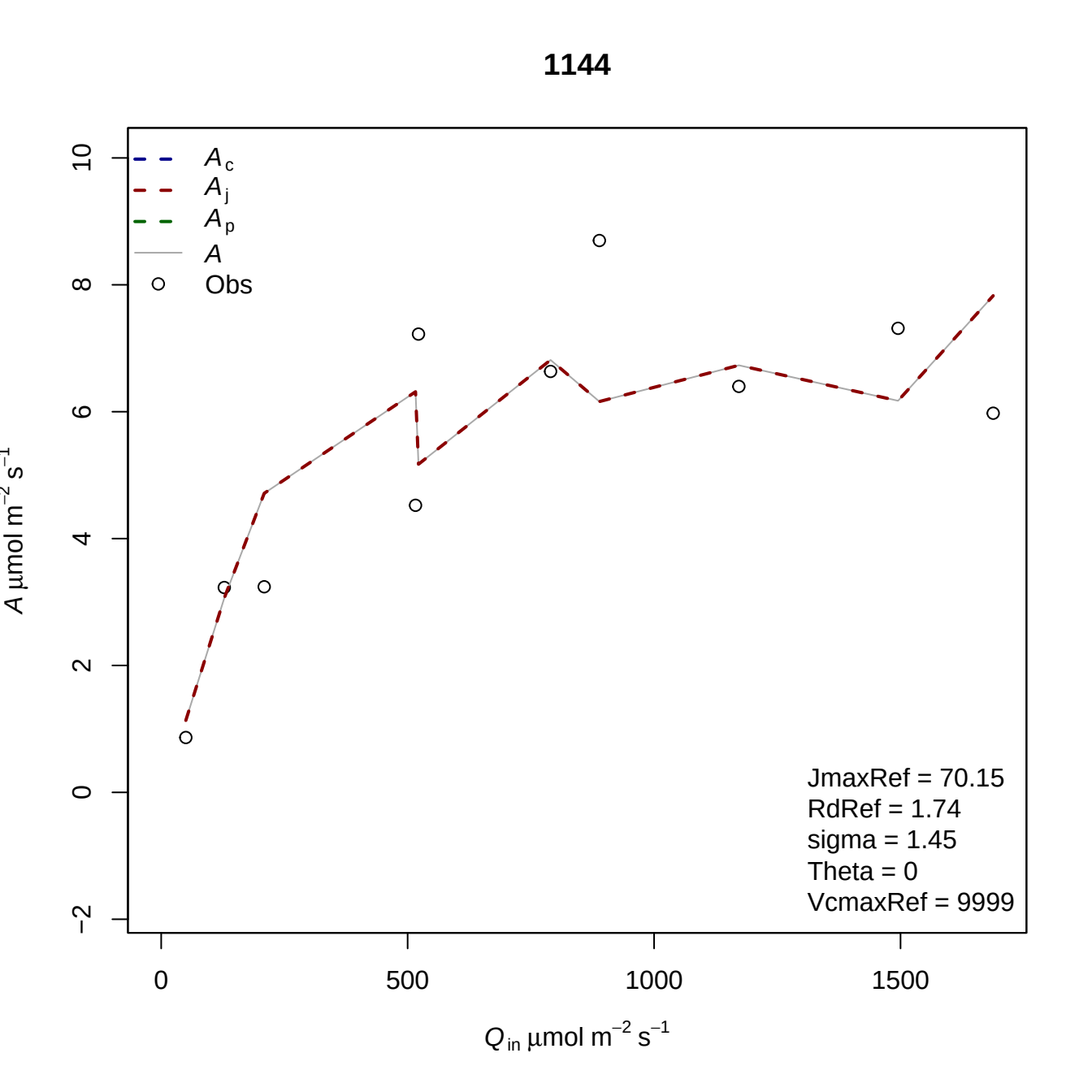

1139
A_c
A_j
A_p
A
Obs
JmaxRef = 139.69
RdRef = 6.59
sigma = 1.27
Theta = 0
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

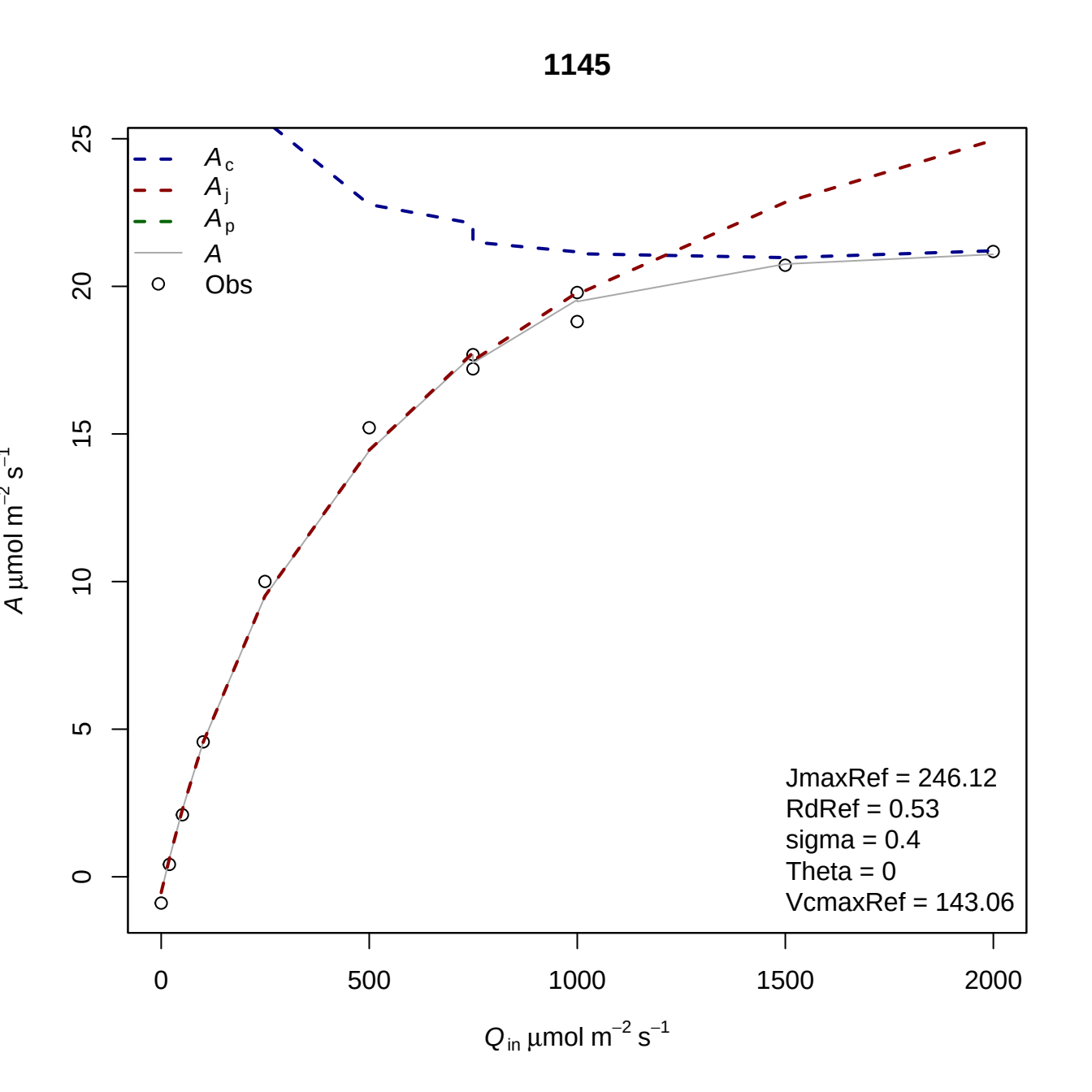

1140
A_c
A_j
A_p
A
Obs
JmaxRef = 173.53
RdRef = 2.02
sigma = 2.33
Theta = 0
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

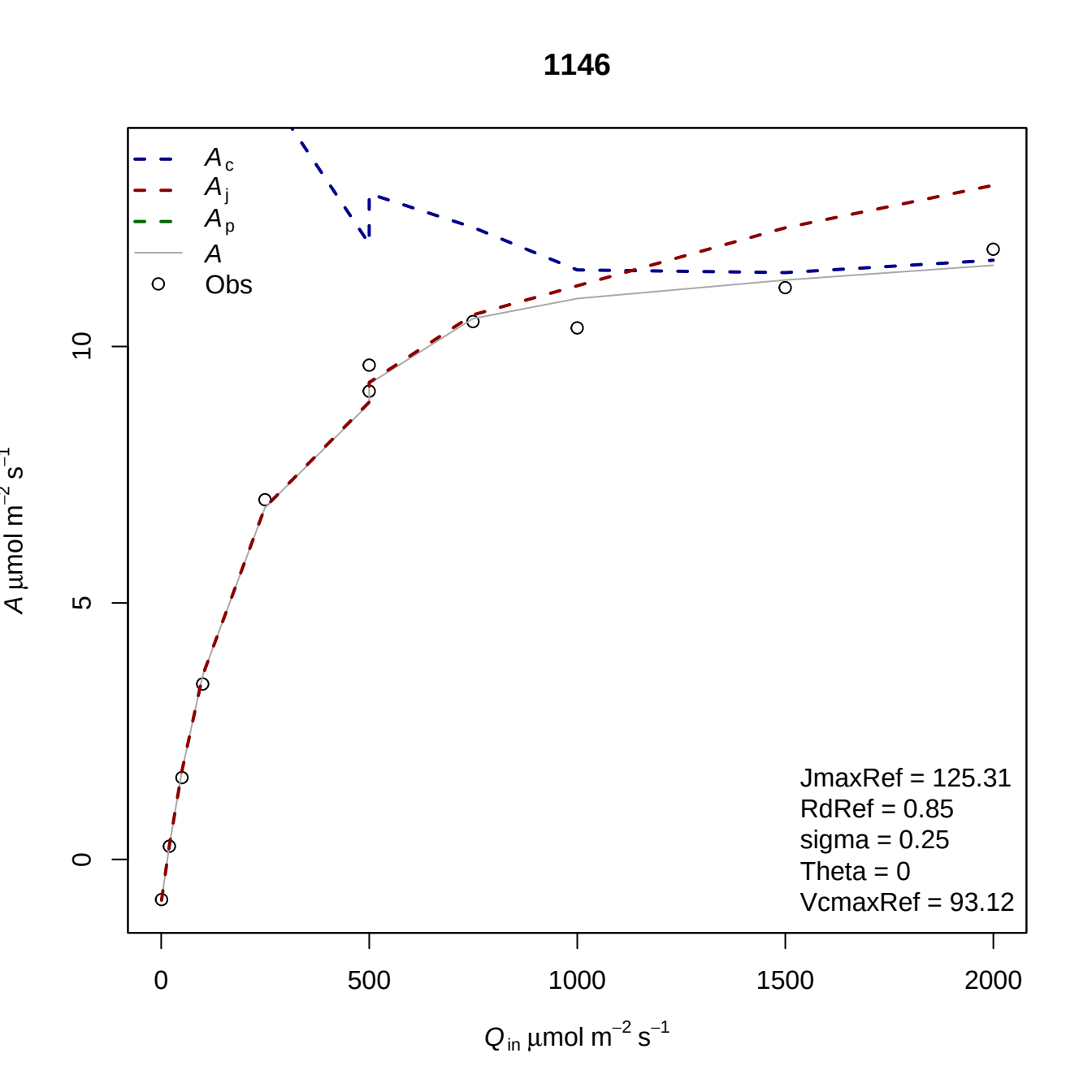

1141
A_c
A_j
A_p
A
Obs
JmaxRef = 72.43
RdRef = 1.53
sigma = 1.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

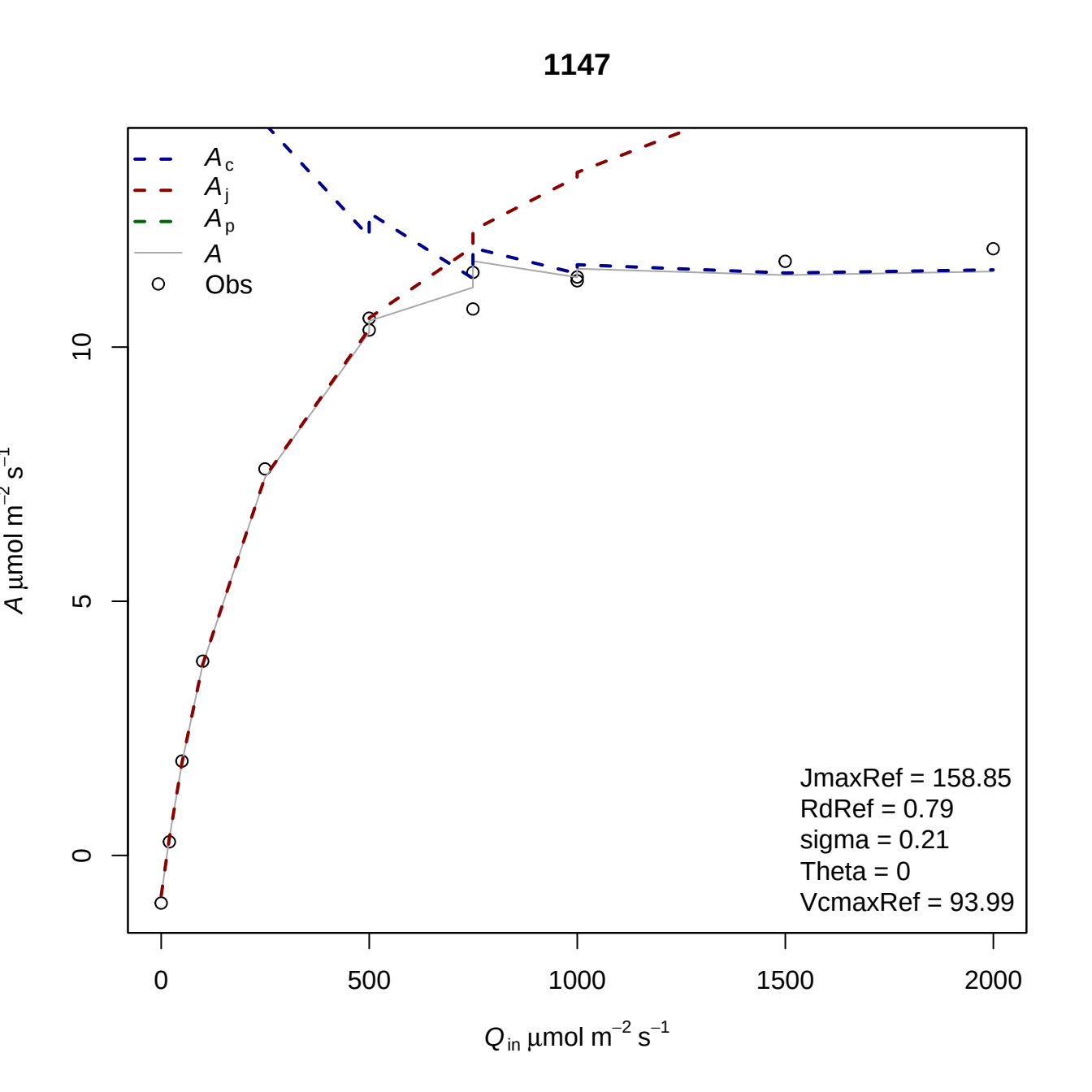

1142
A_c
A_j
A_p
A
Obs
JmaxRef = 55.6
RdRef = 1.55
sigma = 1.02
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

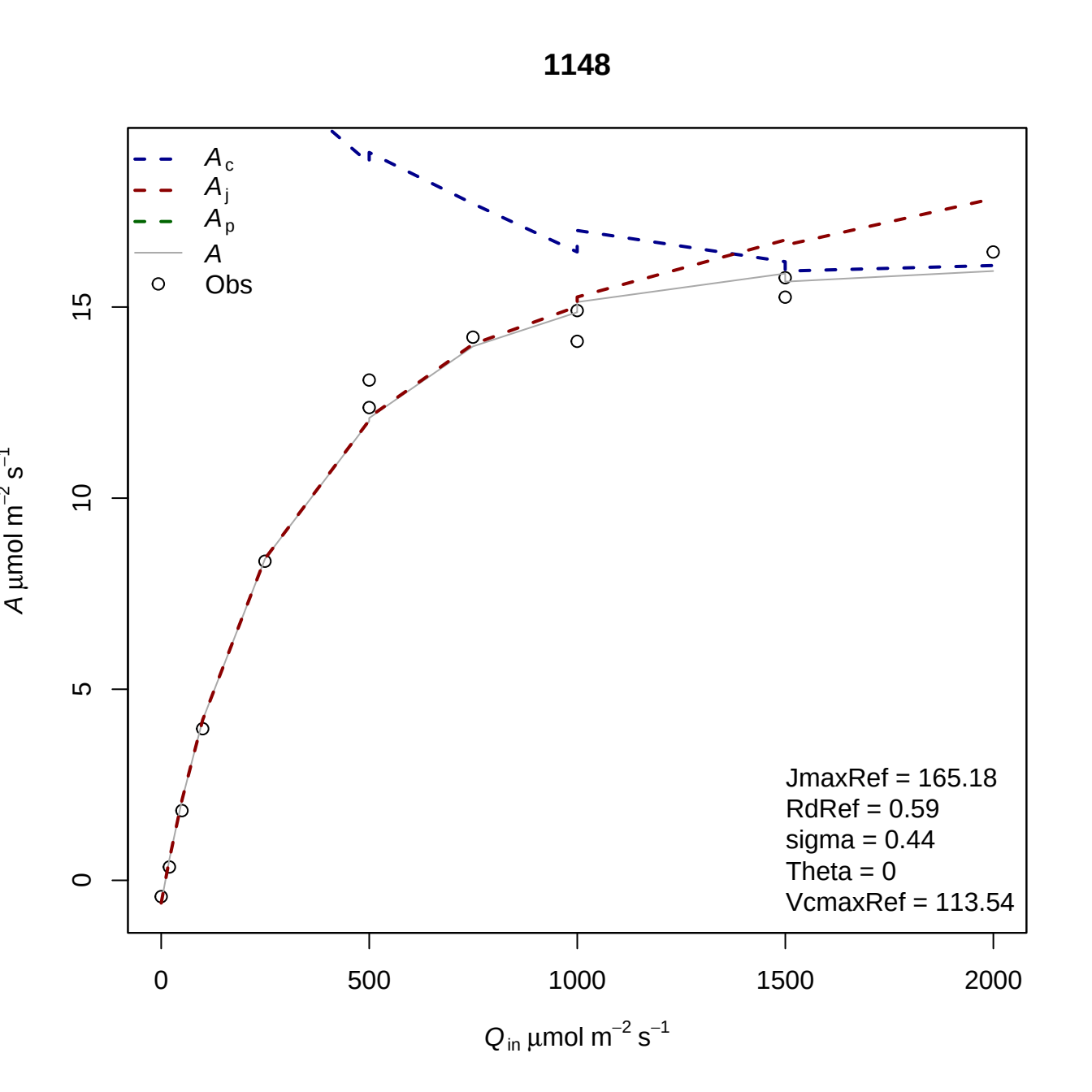

1143
A_c
A_j
A_p
A
Obs
JmaxRef = 106.94
RdRef = 1.74
sigma = 2.55
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

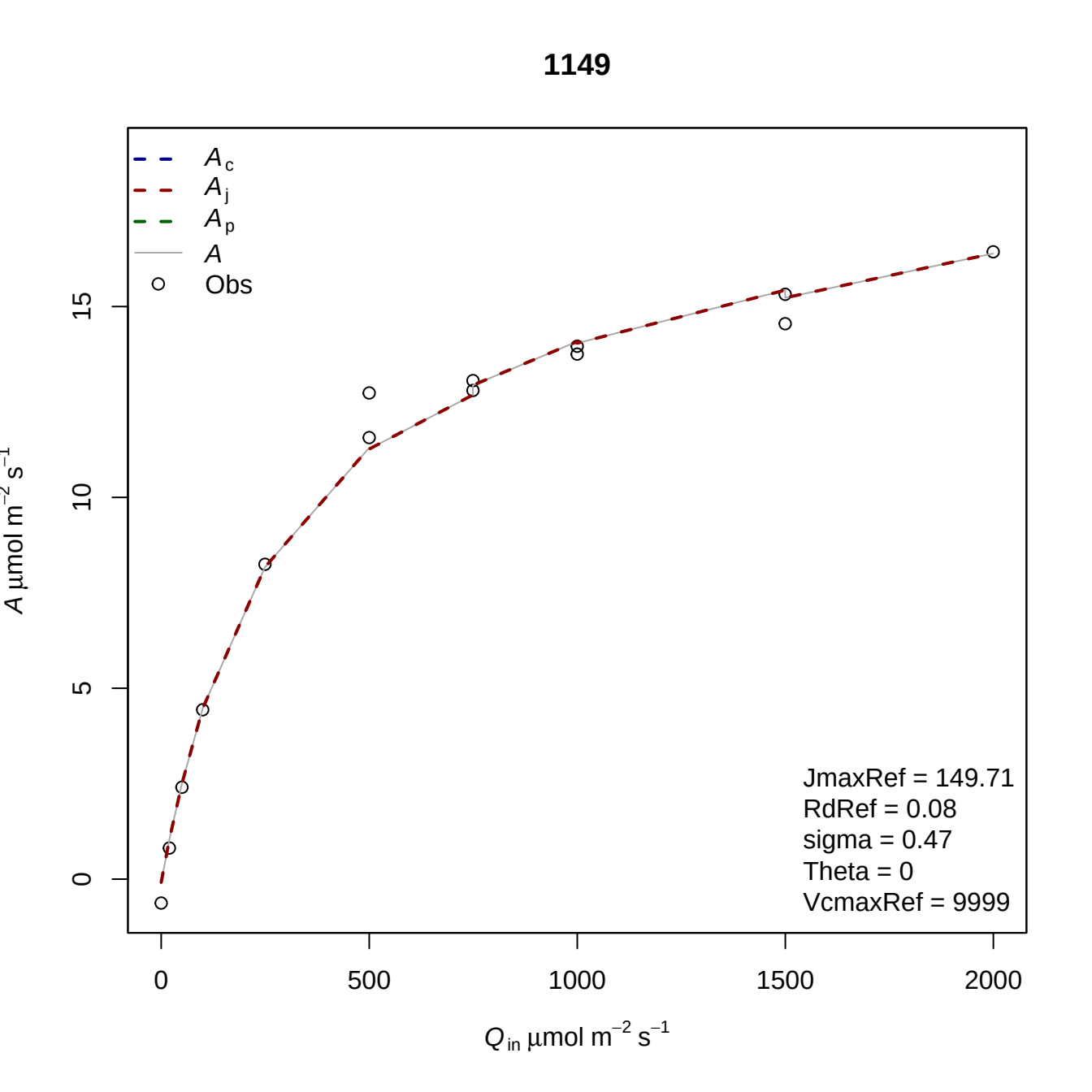

1144
A_c
A_j
A_p
A
Obs
JmaxRef = 70.15
RdRef = 1.74
sigma = 1.45
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

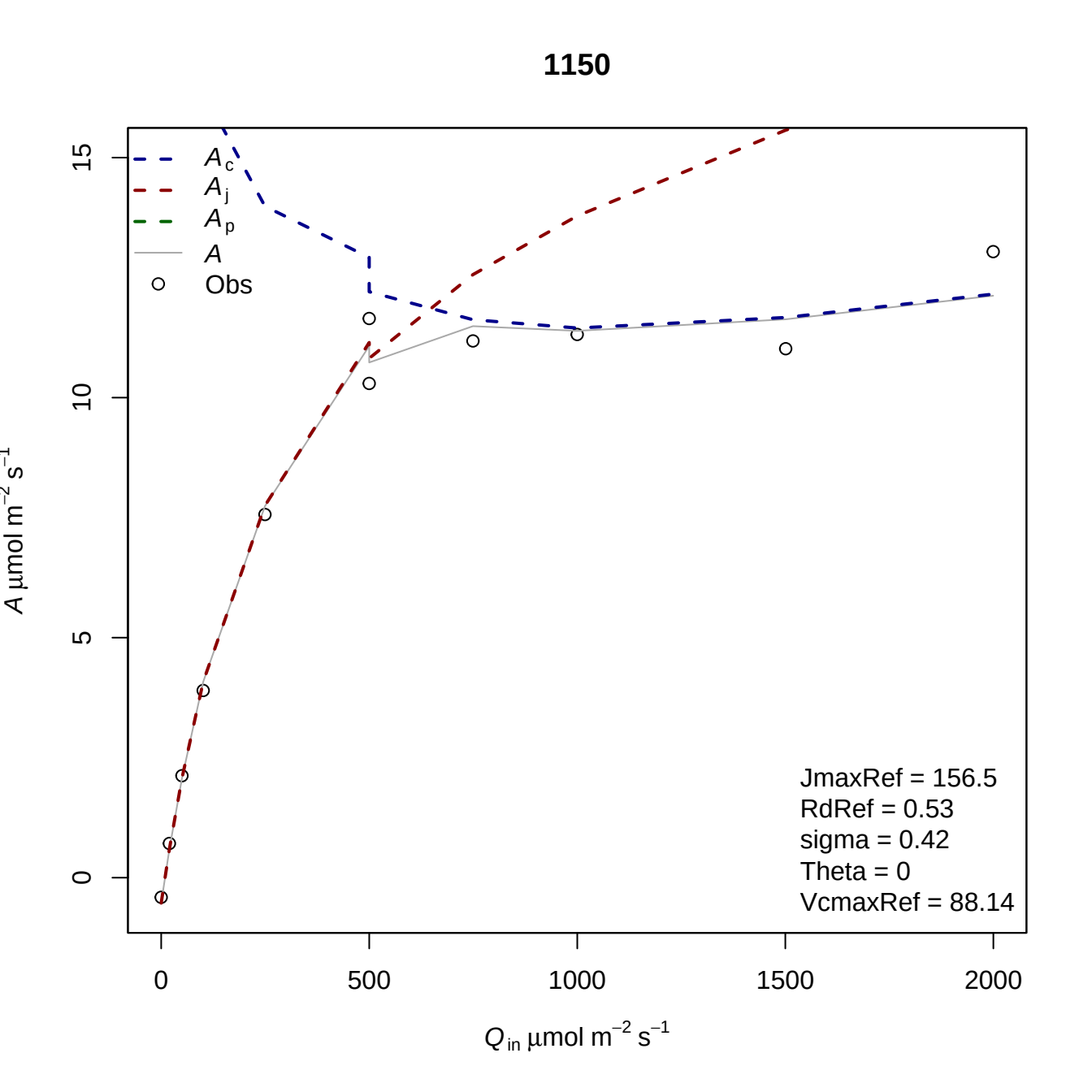

1145
A_c
A_j
A_p
A
Obs
JmaxRef = 246.12
RdRef = 0.53
sigma = 0.4
Theta = 0
VcmaxRef = 143.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

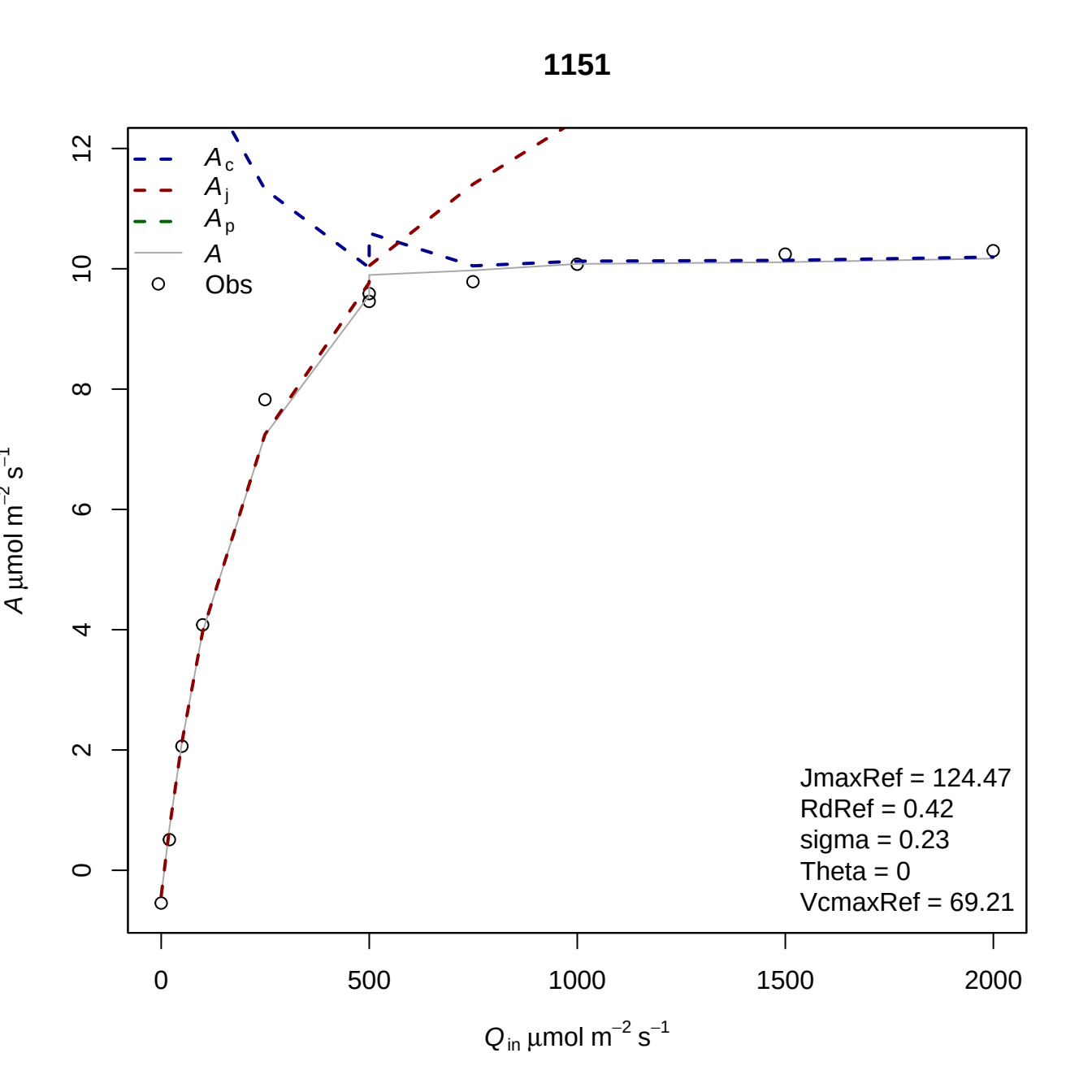

1146
A_c
A_j
A_p
A
Obs
JmaxRef = 125.31
RdRef = 0.85
sigma = 0.25
Theta = 0
VcmaxRef = 93.12
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

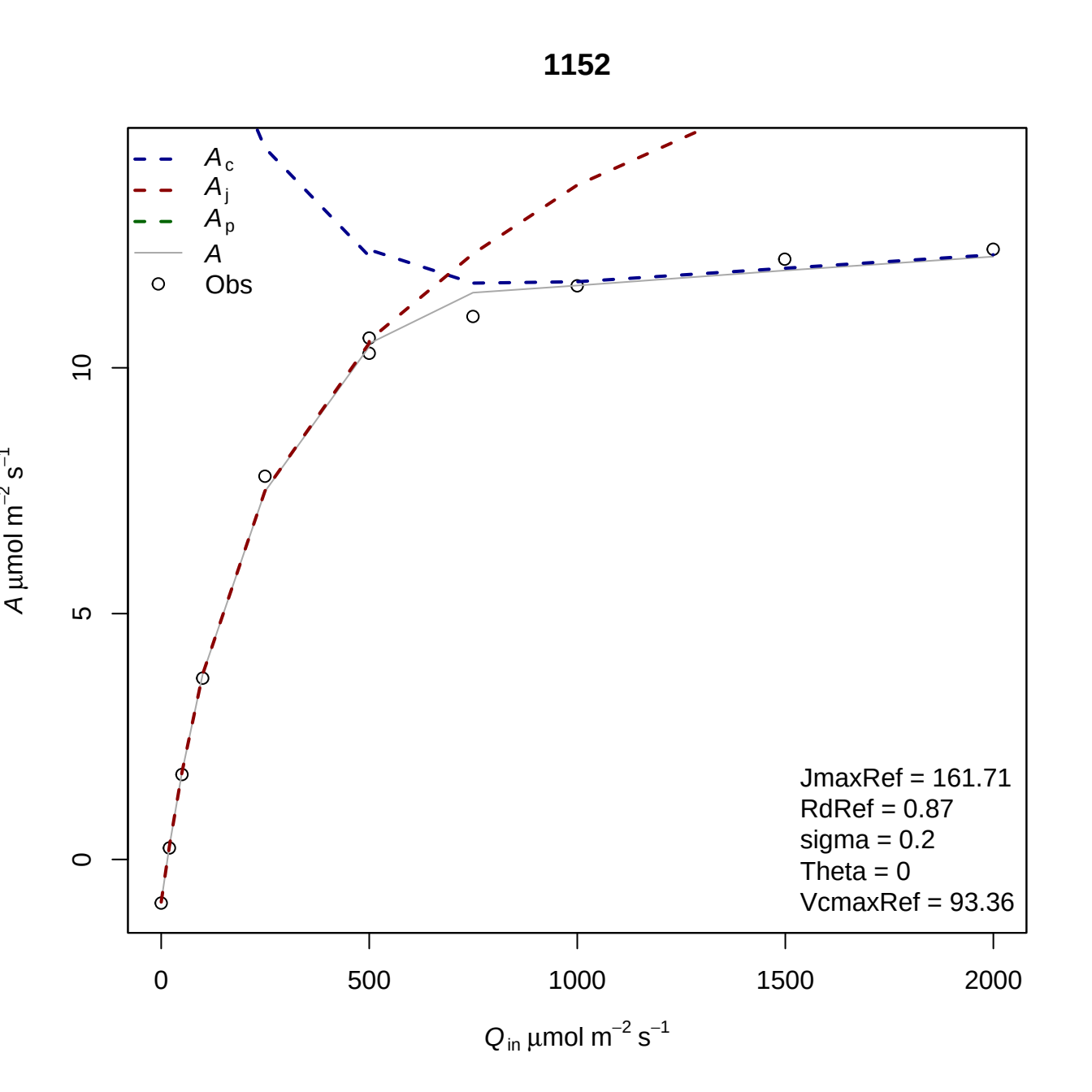

1147
A_c
A_j
A_p
A
Obs
JmaxRef = 158.85
RdRef = 0.79
sigma = 0.21
Theta = 0
VcmaxRef = 93.99
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

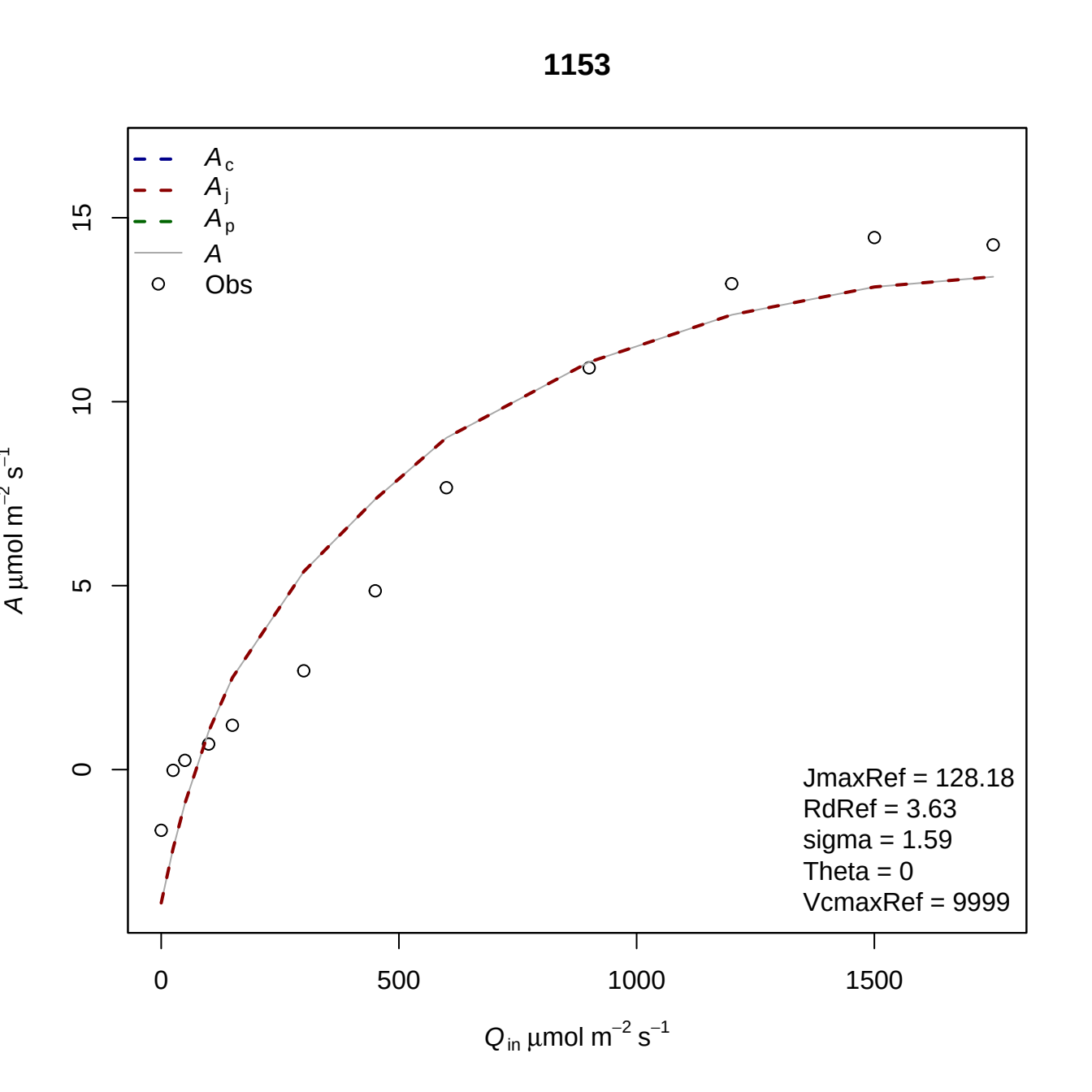

1148
A_c
A_j
A_p
A
Obs
JmaxRef = 165.18
RdRef = 0.59
sigma = 0.44
Theta = 0
VcmaxRef = 113.54
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

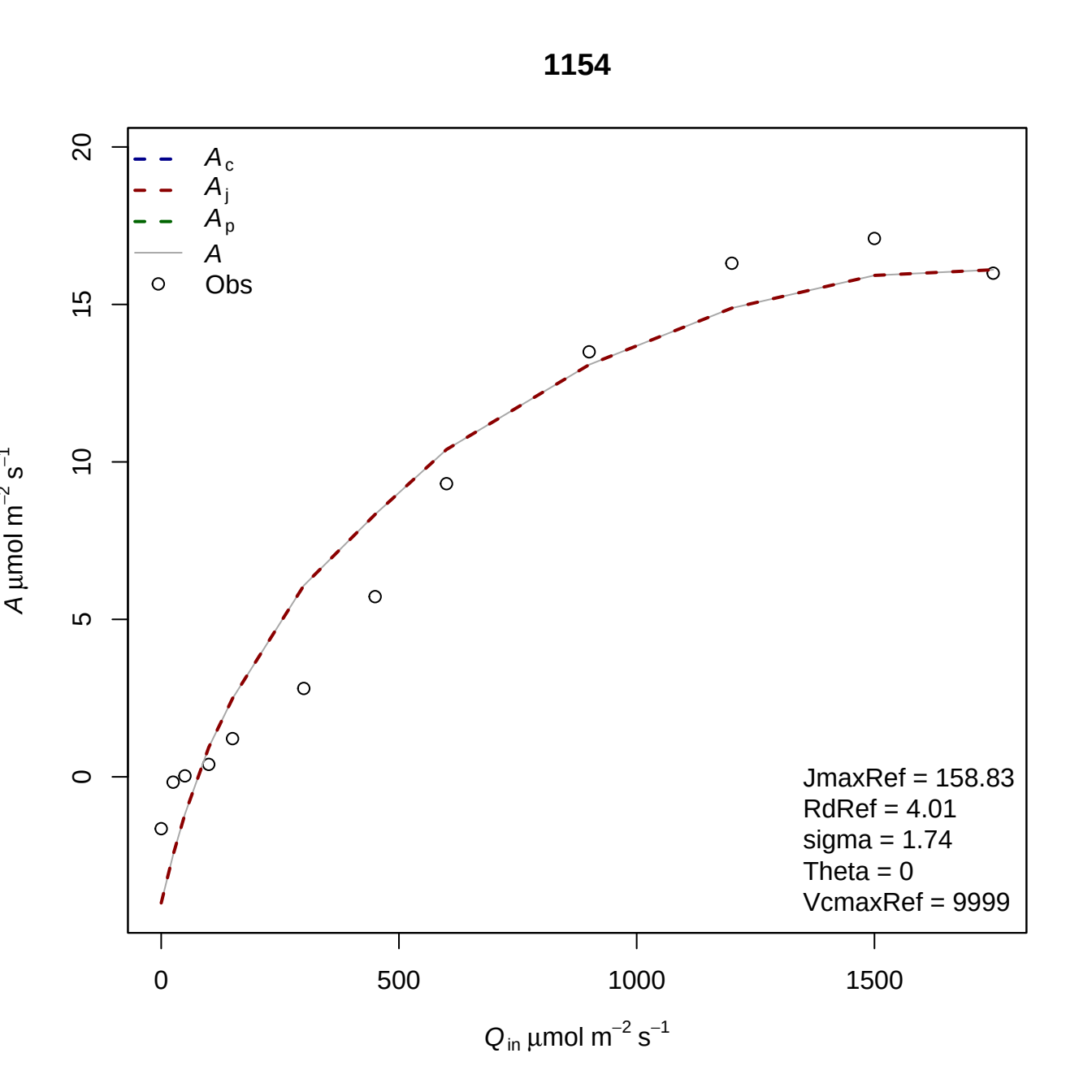

1149
A_c
A_j
A_p
A
Obs
JmaxRef = 149.71
RdRef = 0.08
sigma = 0.47
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

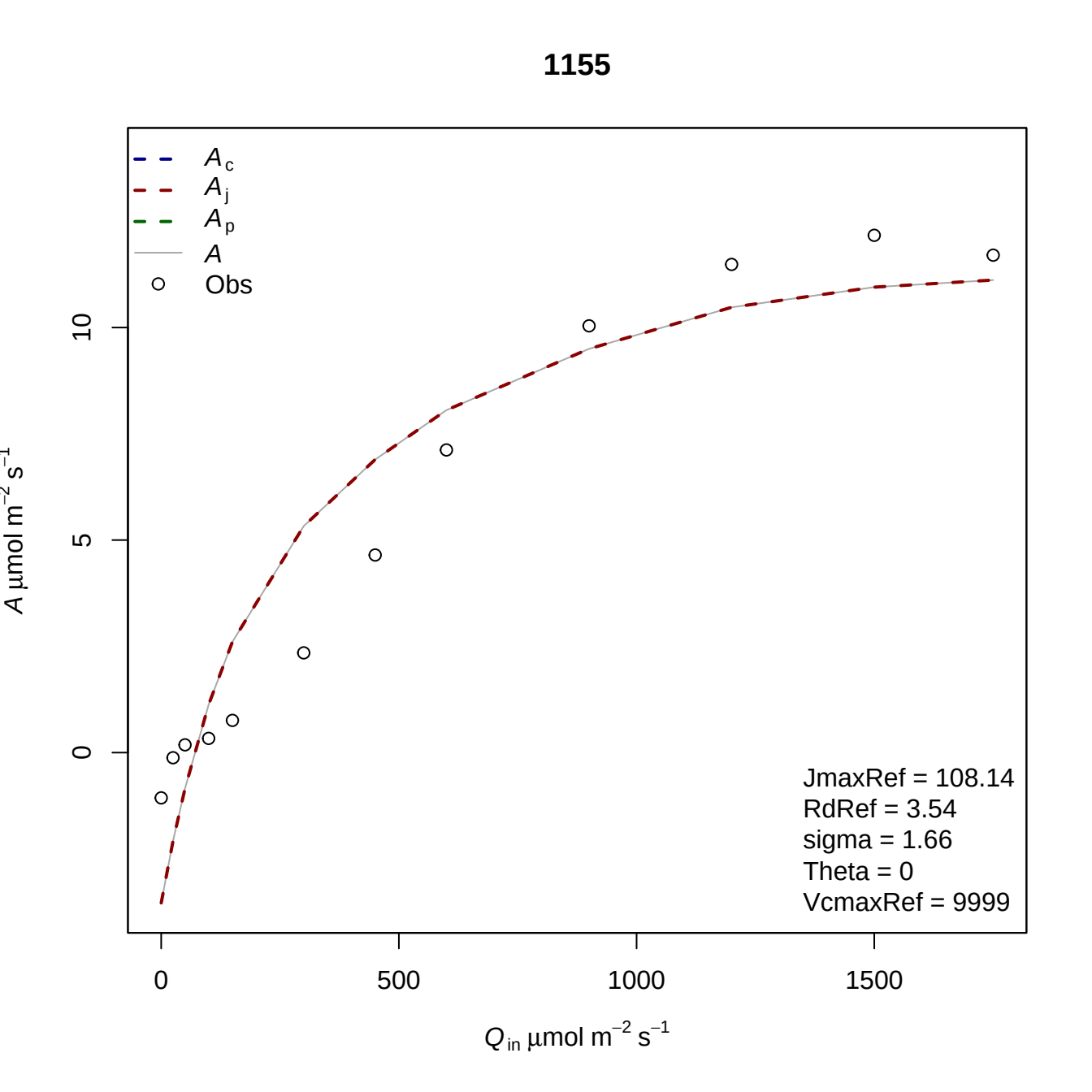

1150
A_c
A_j
A_p
A
Obs
JmaxRef = 156.5
RdRef = 0.53
sigma = 0.42
Theta = 0
VcmaxRef = 88.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

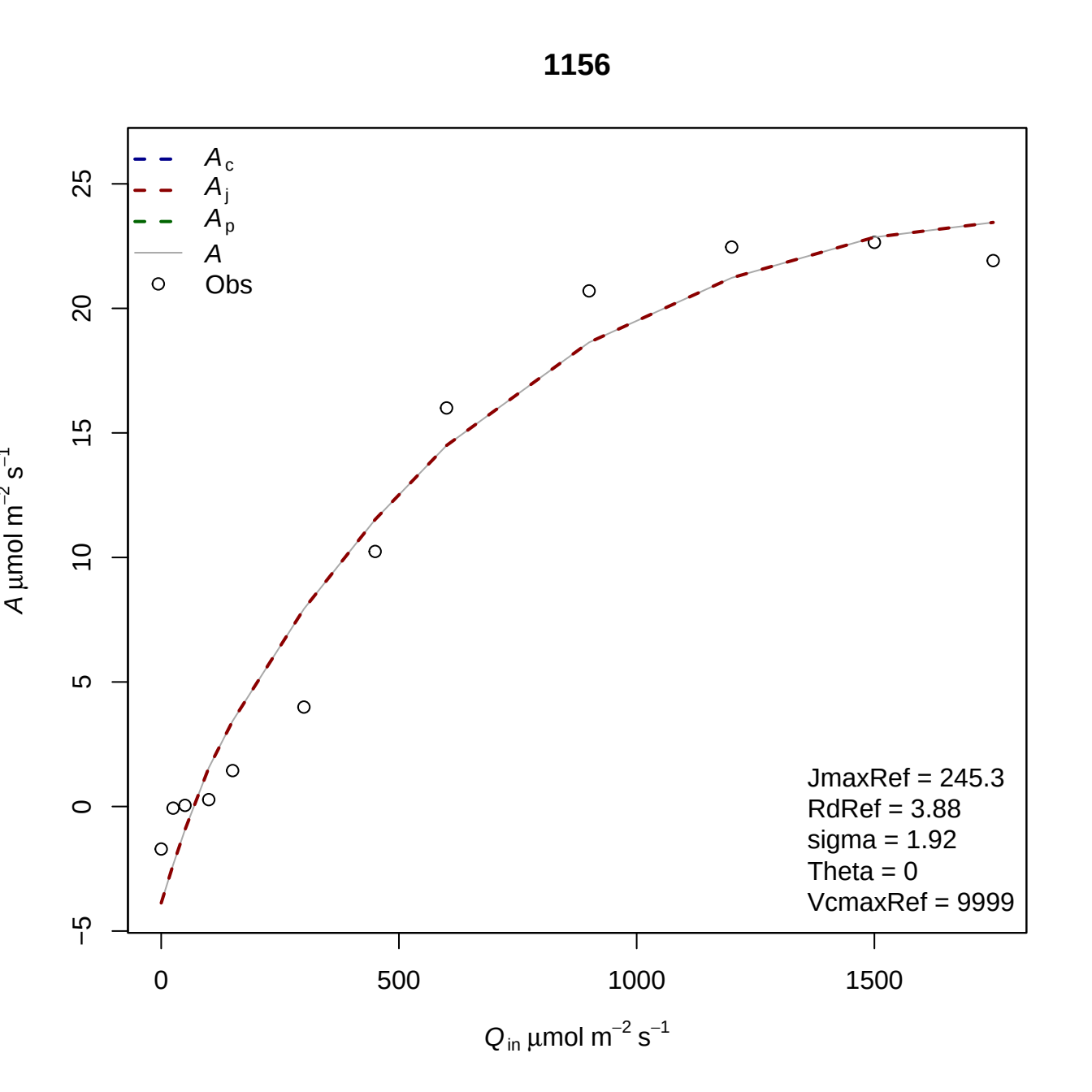

1151
A_c
A_j
A_p
A
Obs
JmaxRef = 124.47
RdRef = 0.42
sigma = 0.23
Theta = 0
VcmaxRef = 69.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

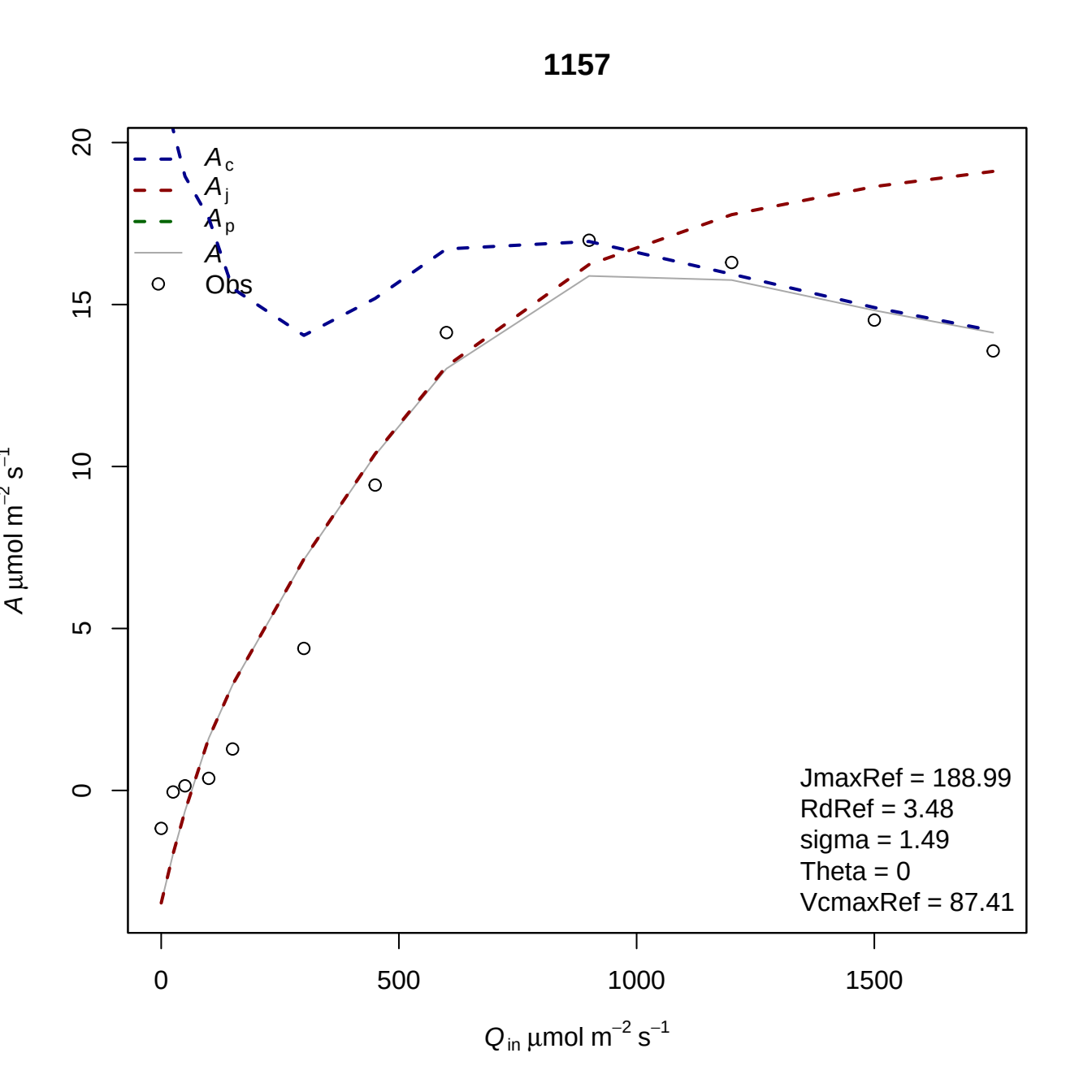

1152
A_c
A_j
A_p
A
Obs
JmaxRef = 161.71
RdRef = 0.87
sigma = 0.2
Theta = 0
VcmaxRef = 93.36
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

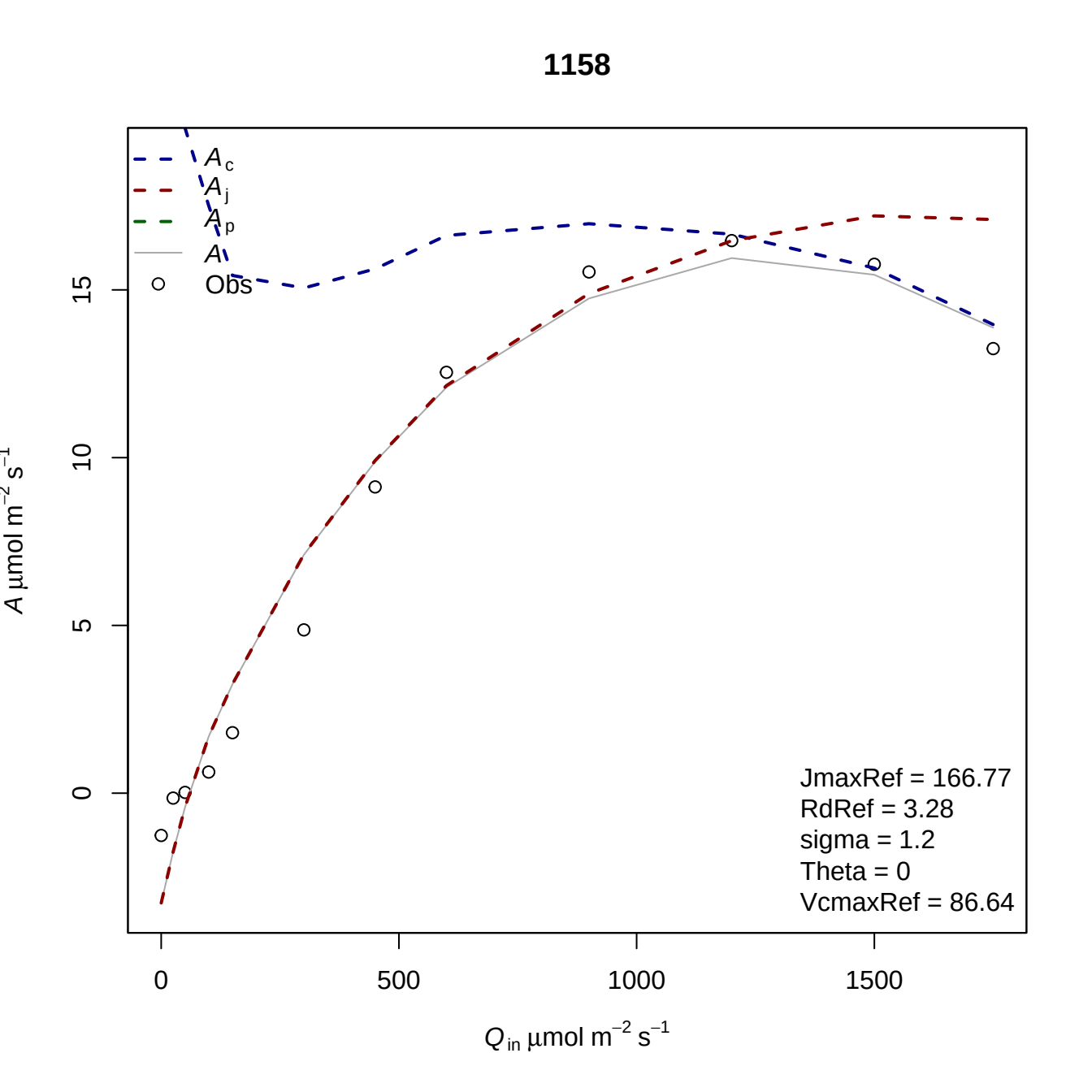

1153
A_c
A_j
A_p
A
Obs
JmaxRef = 128.18
RdRef = 3.63
sigma = 1.59
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

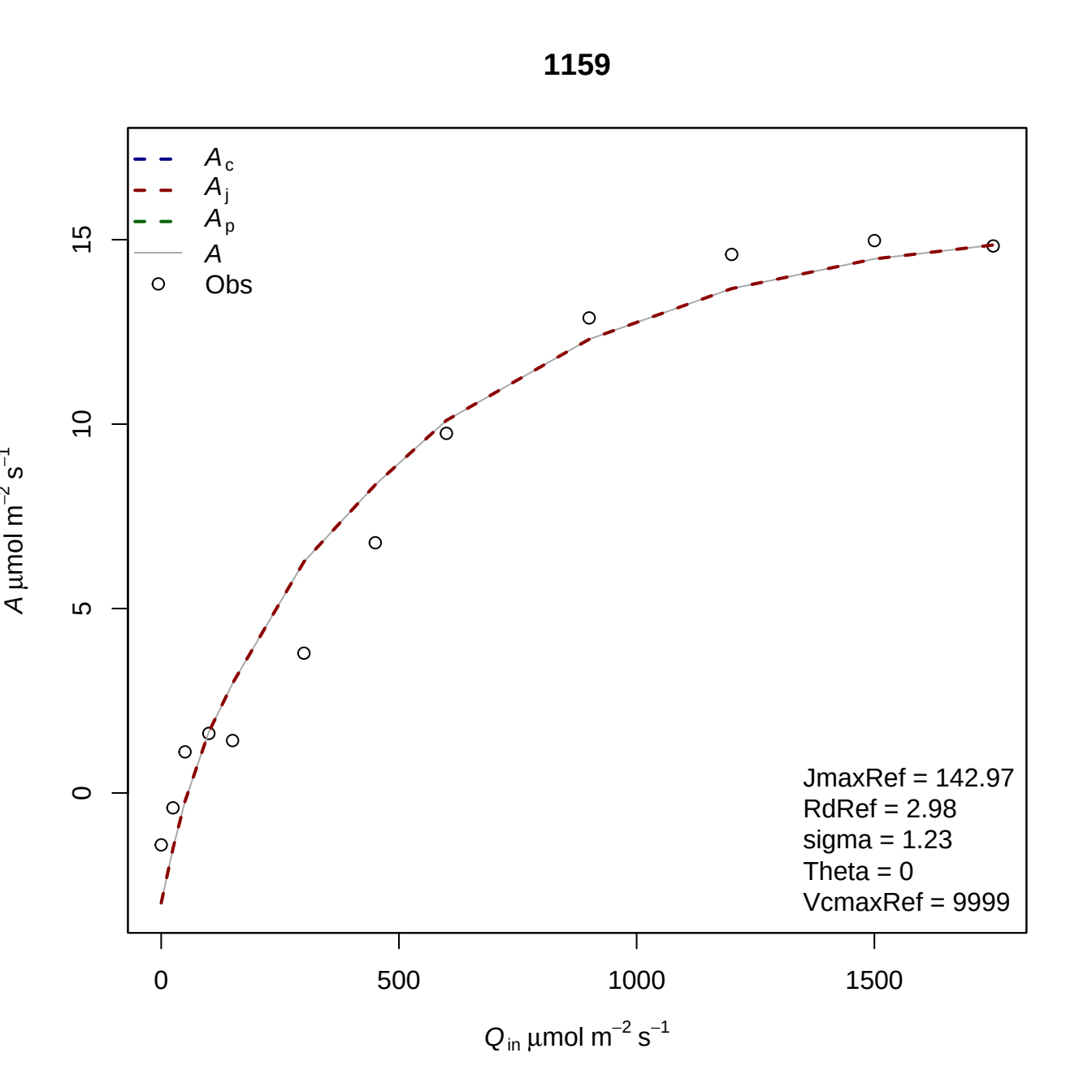

1154
A_c
A_j
A_p
A
Obs
JmaxRef = 158.83
RdRef = 4.01
sigma = 1.74
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

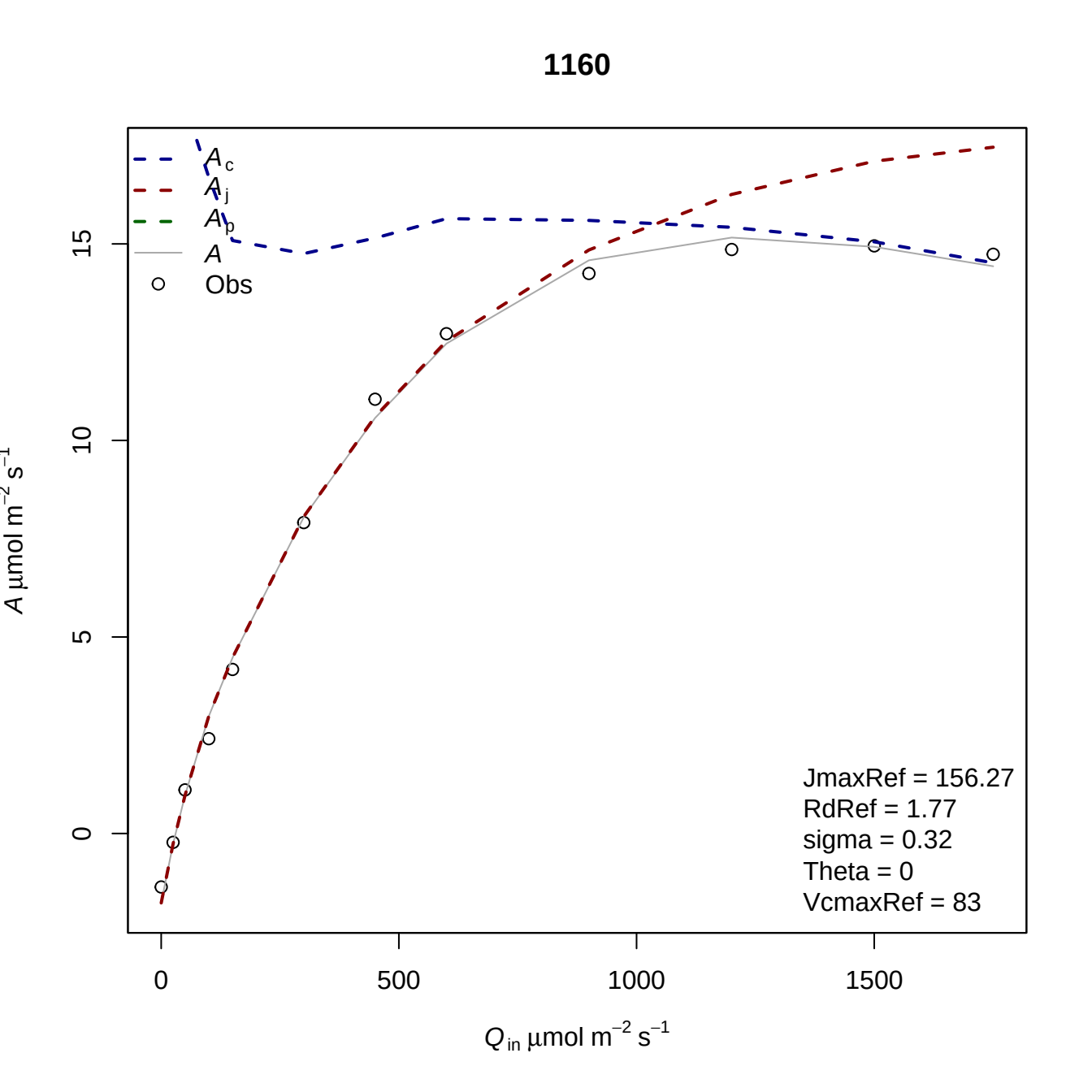

1155
A_c
A_j
A_p
A
Obs
JmaxRef = 108.14
RdRef = 3.54
sigma = 1.66
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

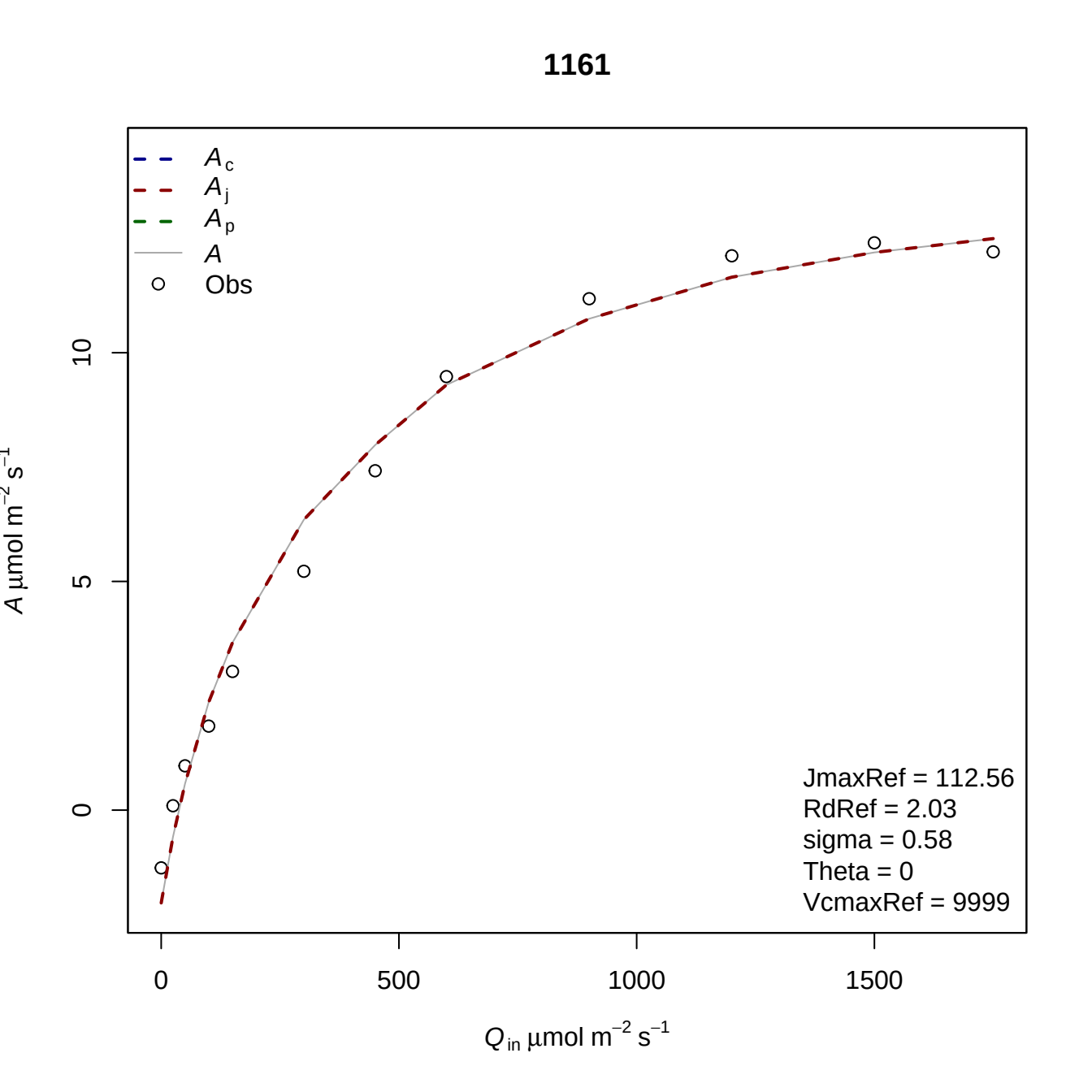

1156
A_c
A_j
A_p
A
Obs
JmaxRef = 245.3
RdRef = 3.88
sigma = 1.92
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500

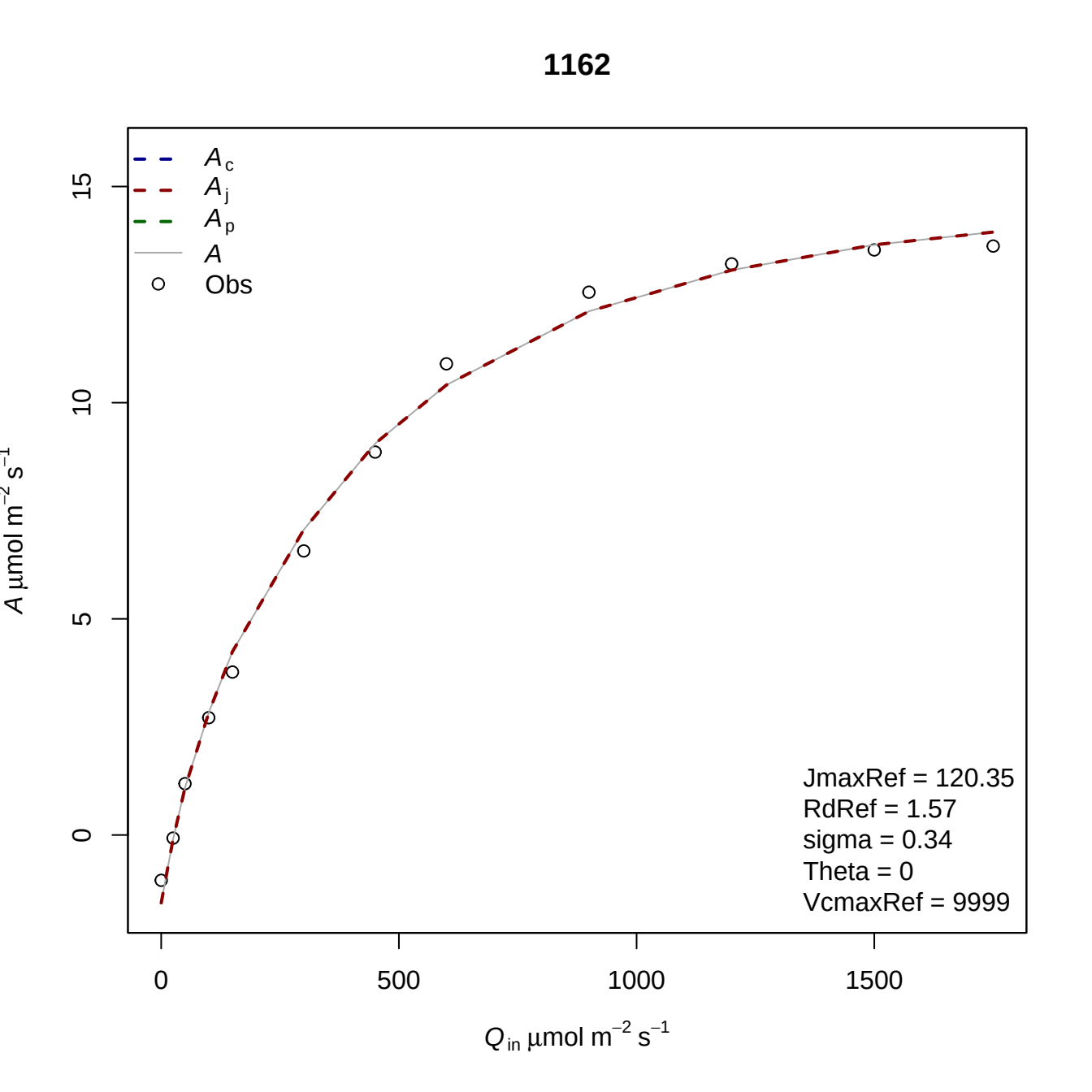

1157
A_c
A_j
A_p
A
Obs
JmaxRef = 188.99
RdRef = 3.48
sigma = 1.49
Theta = 0
VcmaxRef = 87.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

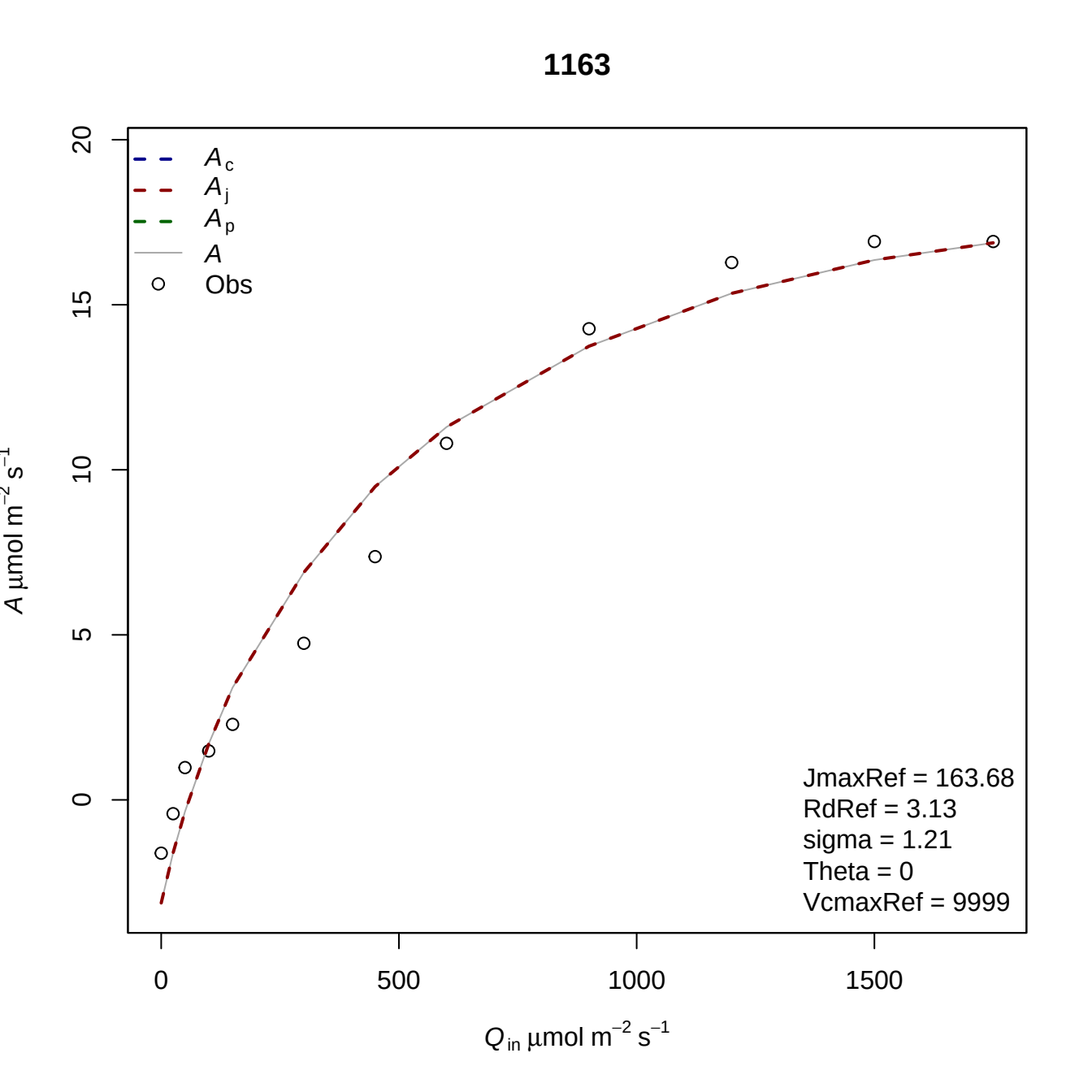

1158
A_c
A_j
A_p
A
Obs
JmaxRef = 166.77
RdRef = 3.28
sigma = 1.2
Theta = 0
VcmaxRef = 86.64
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

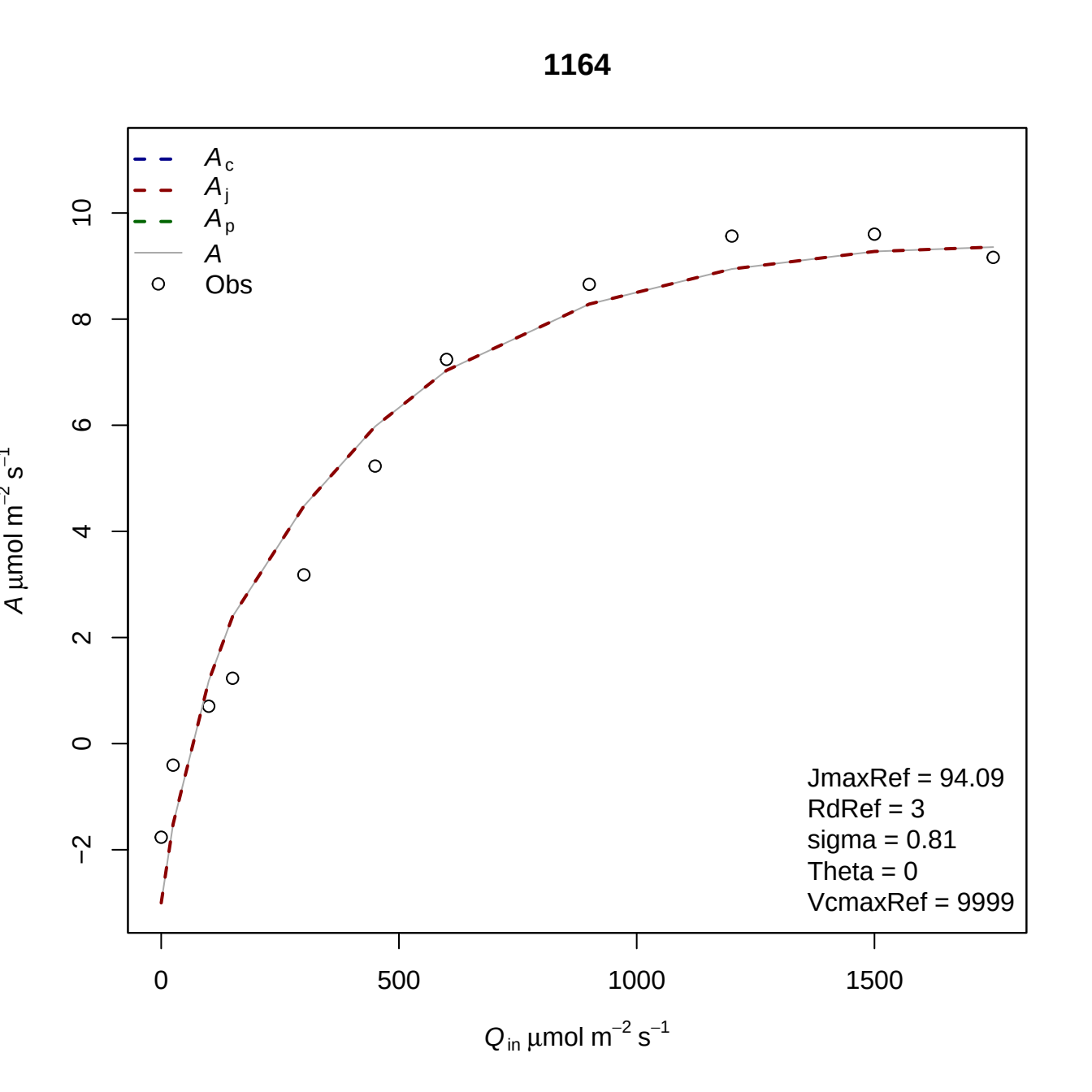

1159
A_c
A_j
A_p
A
Obs
JmaxRef = 142.97
RdRef = 2.98
sigma = 1.23
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

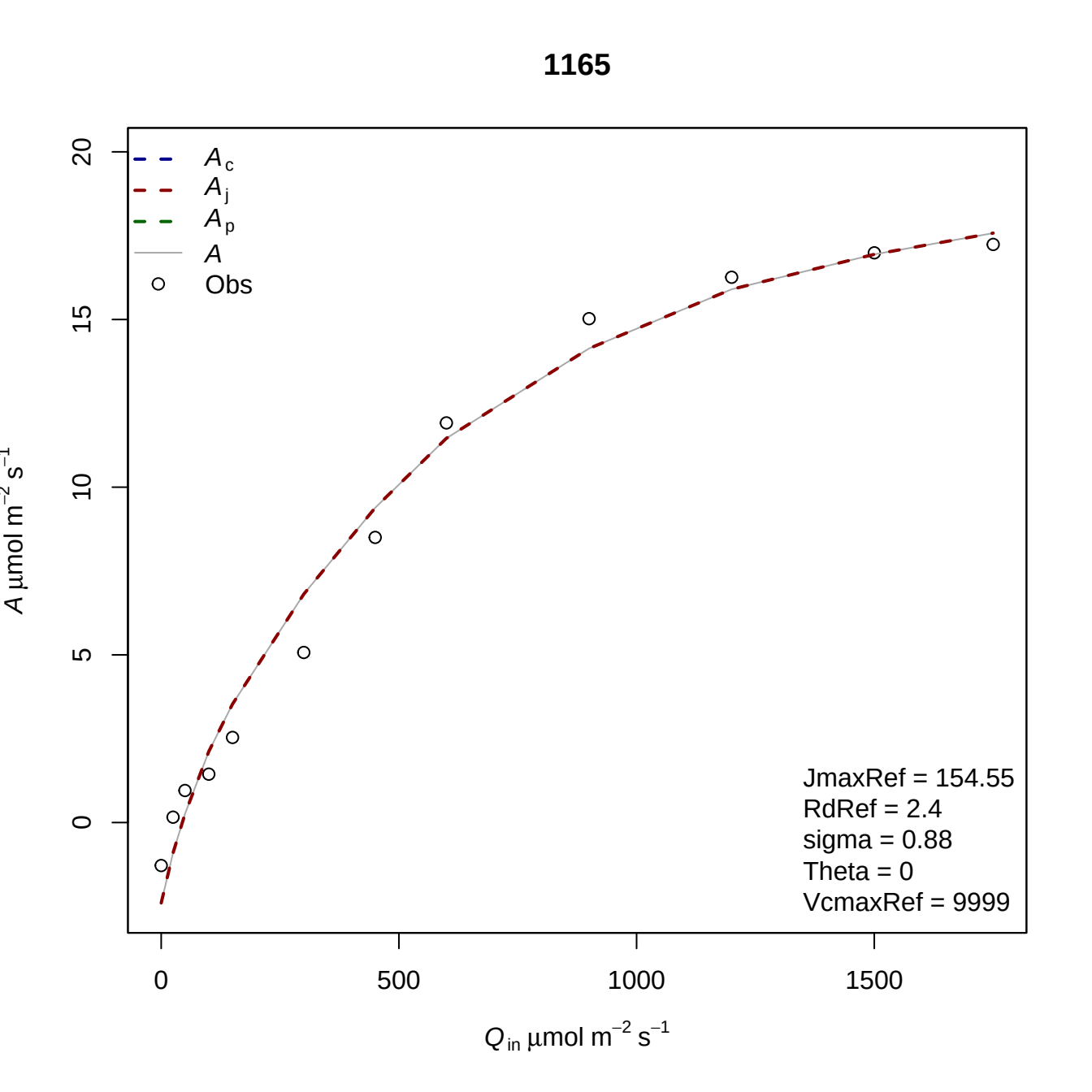

1160
A_c
A_j
A_p
A
Obs
JmaxRef = 156.27
RdRef = 1.77
sigma = 0.32
Theta = 0
VcmaxRef = 83
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

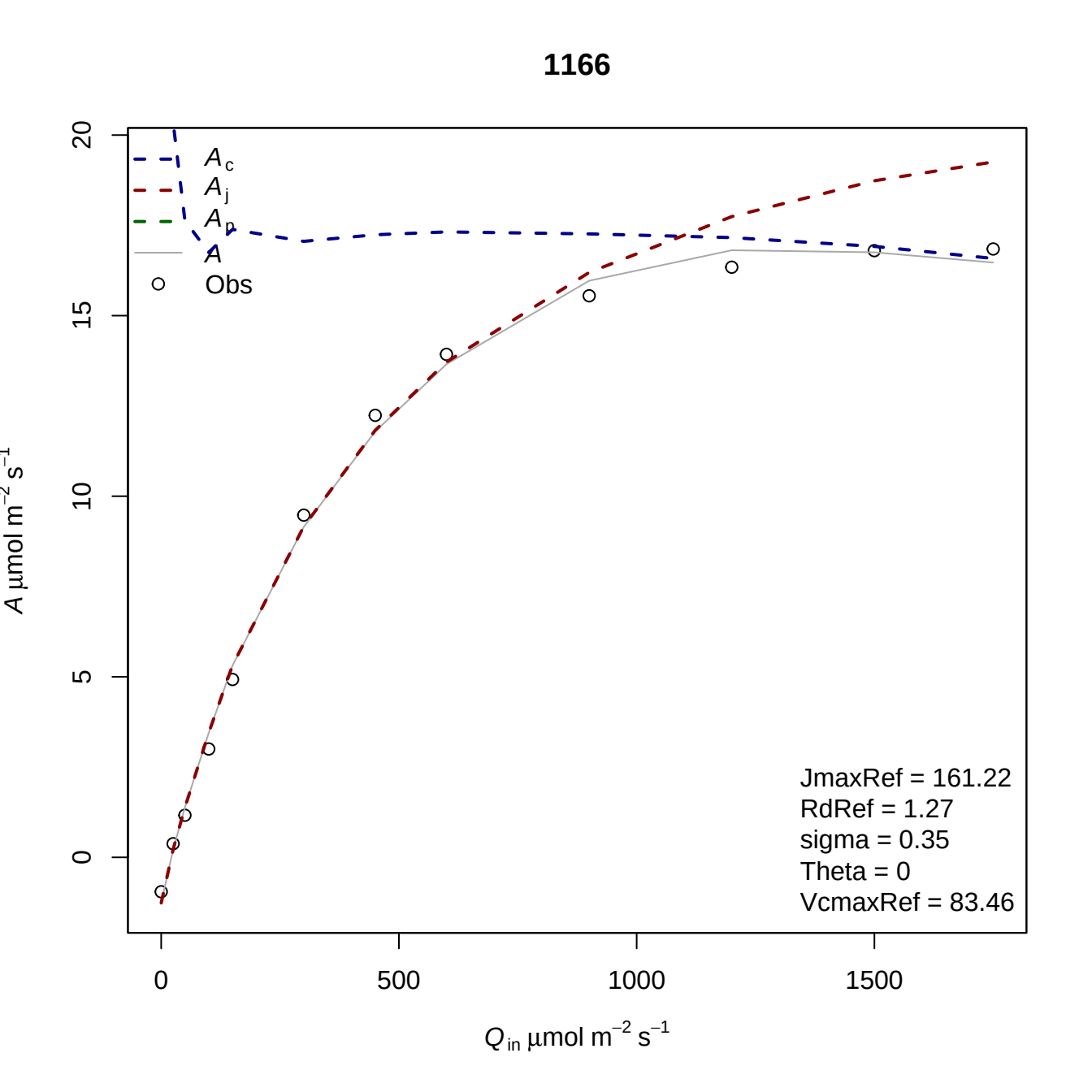

1161
A_c
A_j
A_p
A
Obs
JmaxRef = 112.56
RdRef = 2.03
sigma = 0.58
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

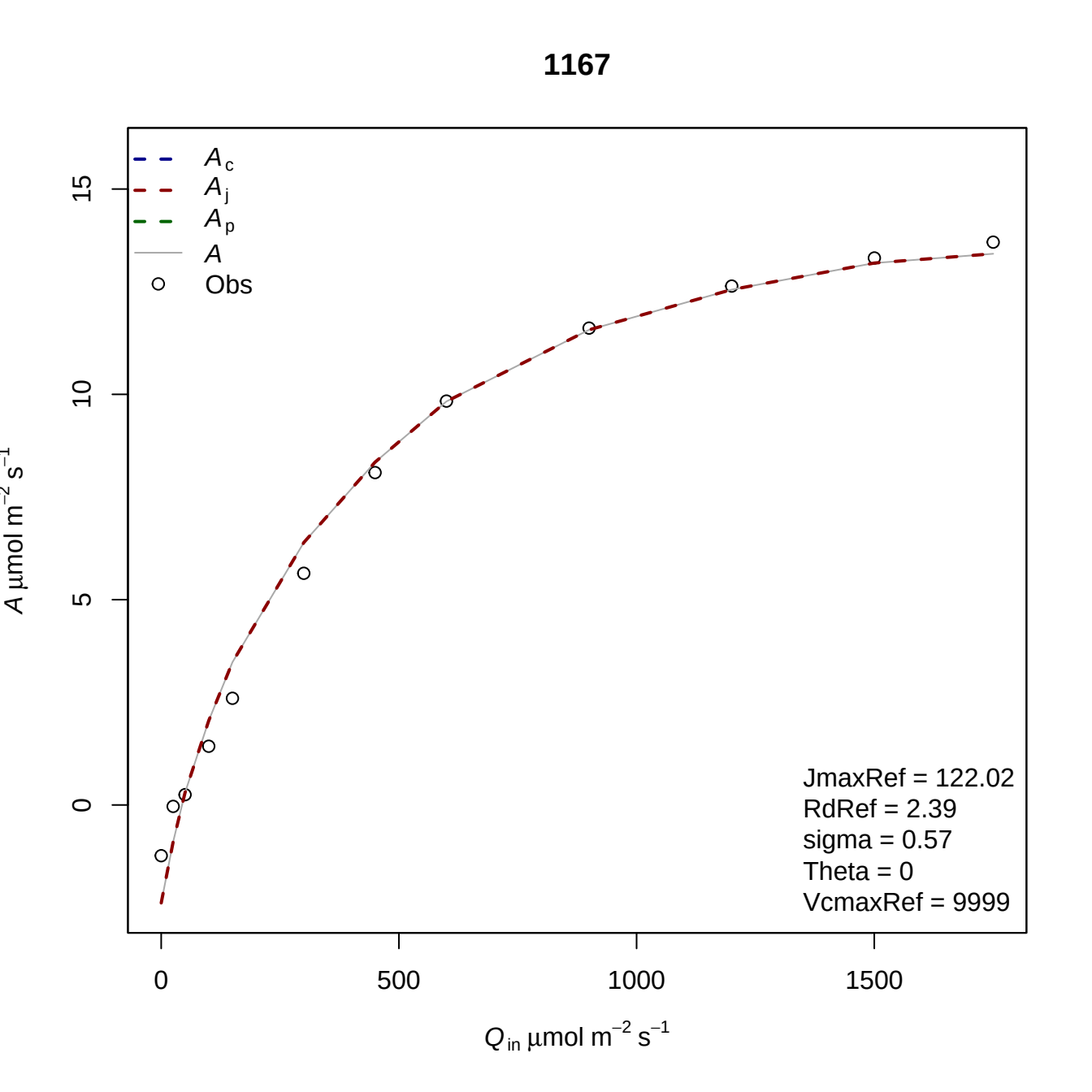

1162
A_c
A_j
A_p
A
Obs
JmaxRef = 120.35
RdRef = 1.57
sigma = 0.34
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

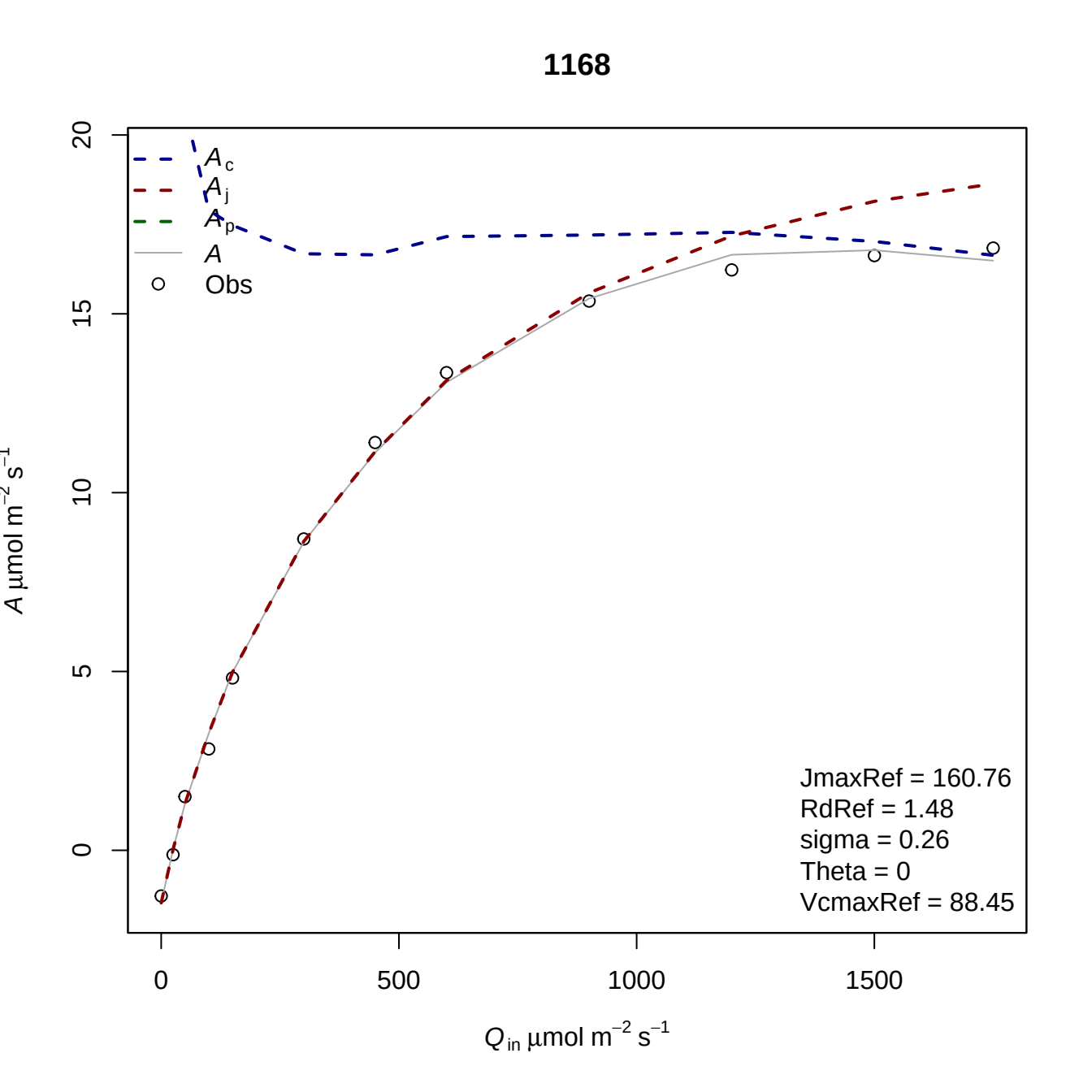

1163
A_c
A_j
A_p
A
Obs
JmaxRef = 163.68
RdRef = 3.13
sigma = 1.21
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

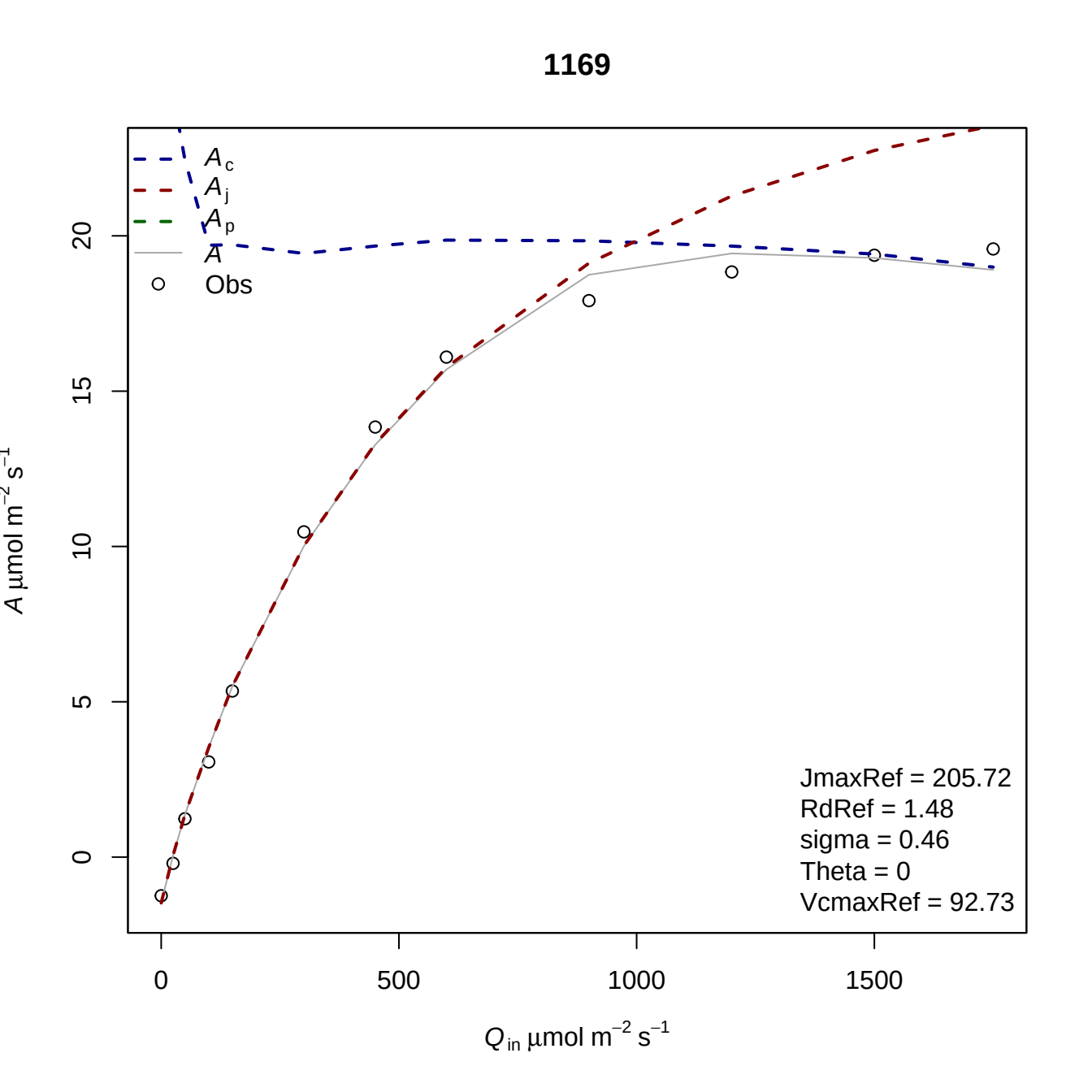

1164
A_c
A_j
A_p
A
Obs
JmaxRef = 94.09
RdRef = 3
sigma = 0.81
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

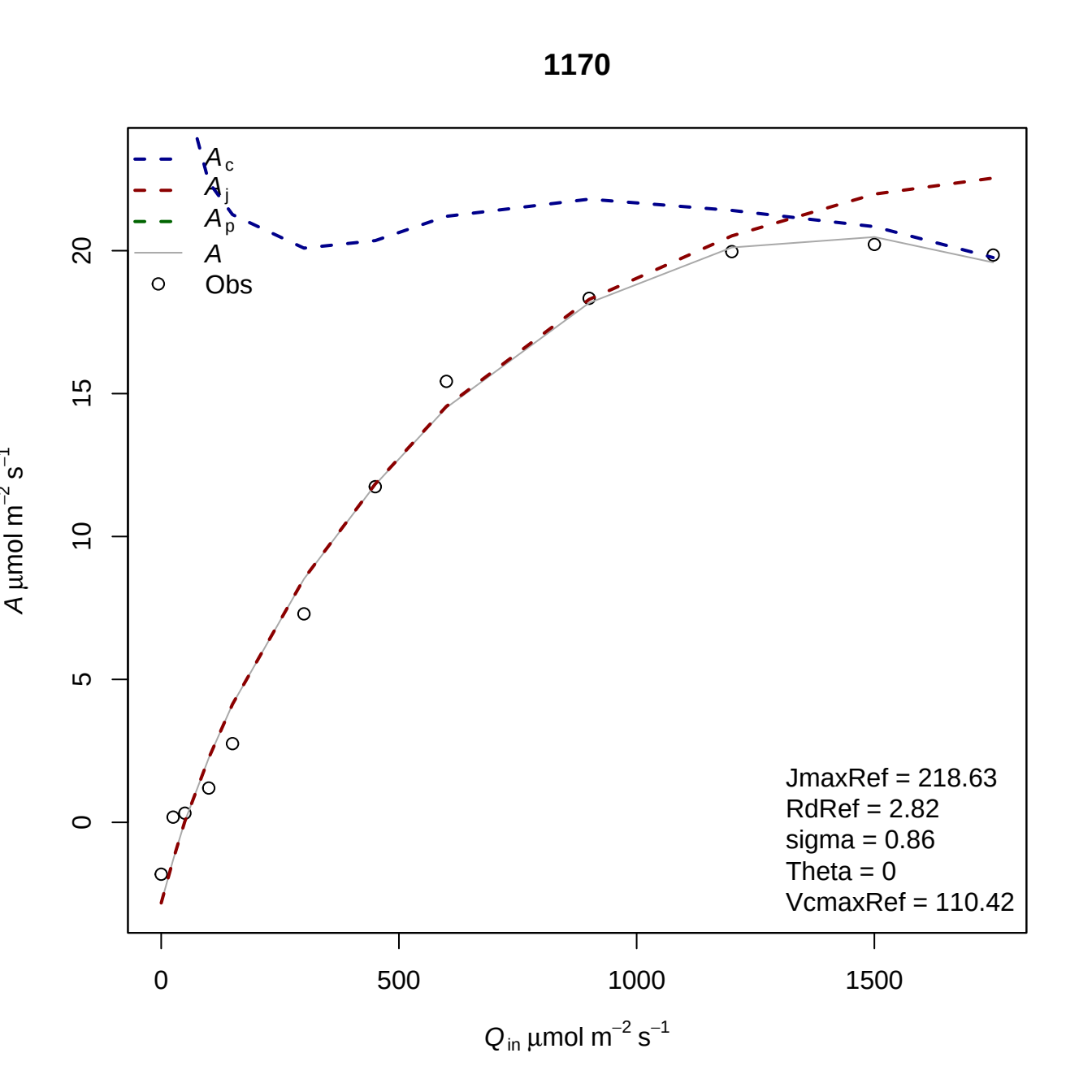

1165
A_c
A_j
A_p
A
Obs
JmaxRef = 154.55
RdRef = 2.4
sigma = 0.88
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

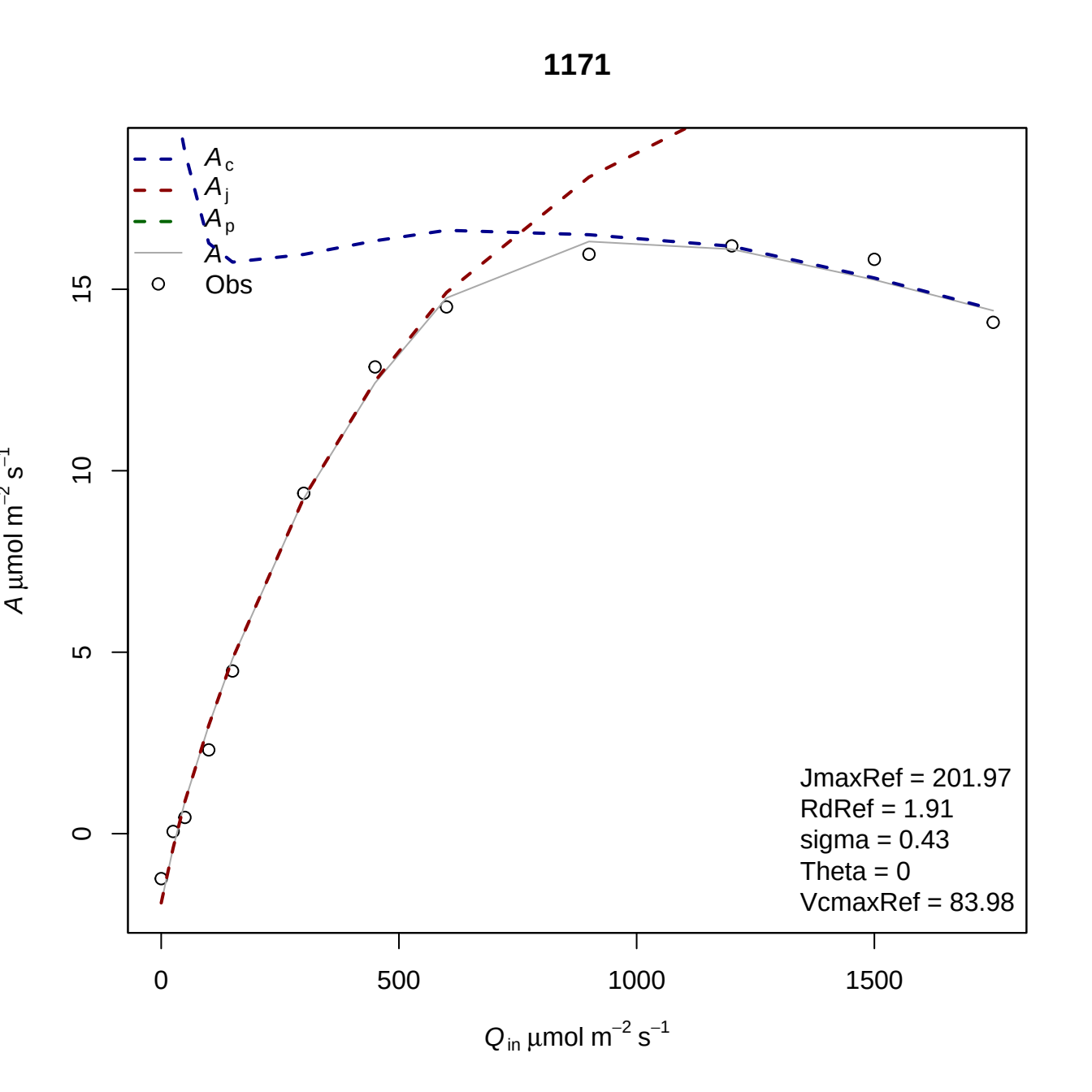

1166
A_c
A_j
A_p
A
Obs
JmaxRef = 161.22
RdRef = 1.27
sigma = 0.35
Theta = 0
VcmaxRef = 83.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

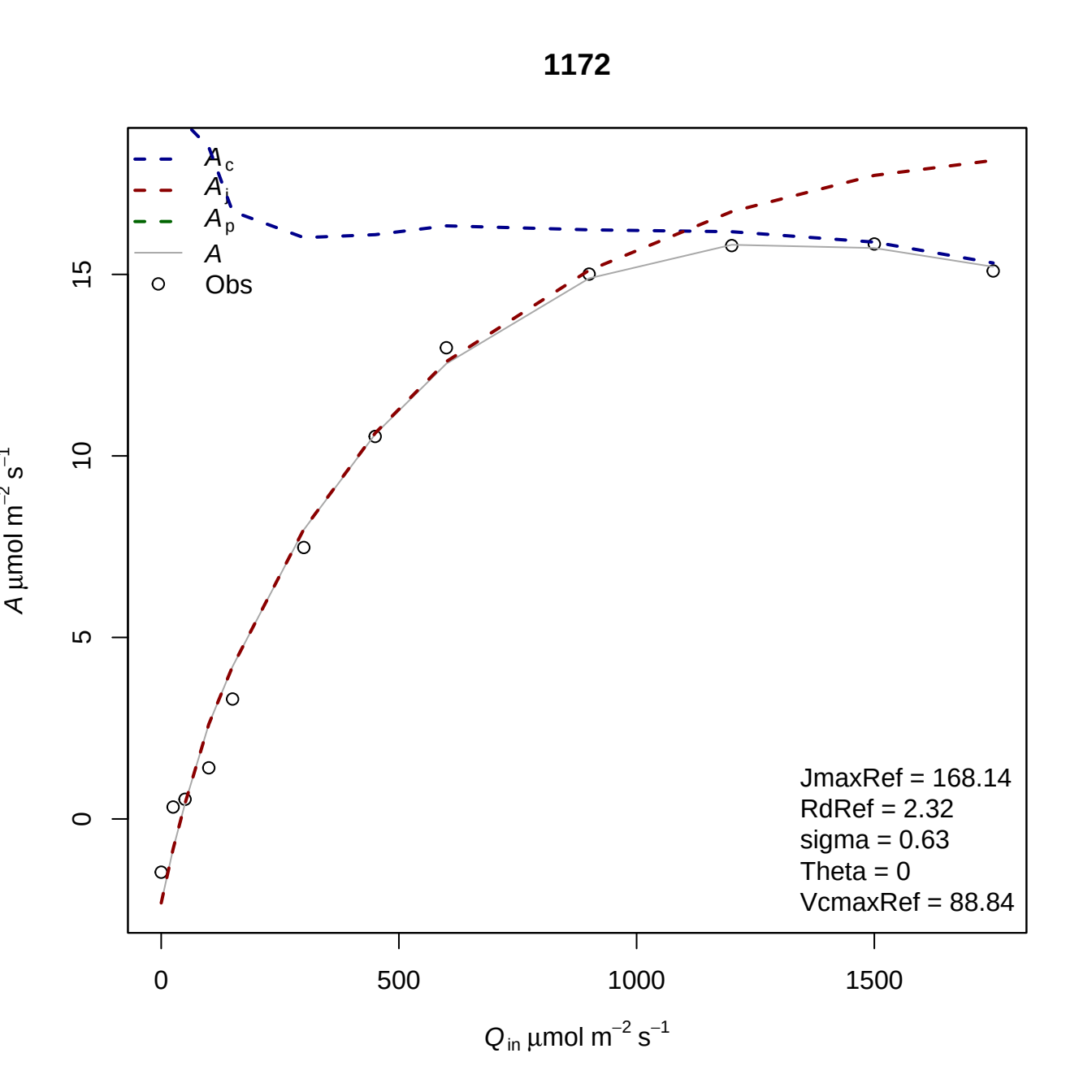

1167
A_c
A_j
A_p
A
Obs
JmaxRef = 122.02
RdRef = 2.39
sigma = 0.57
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

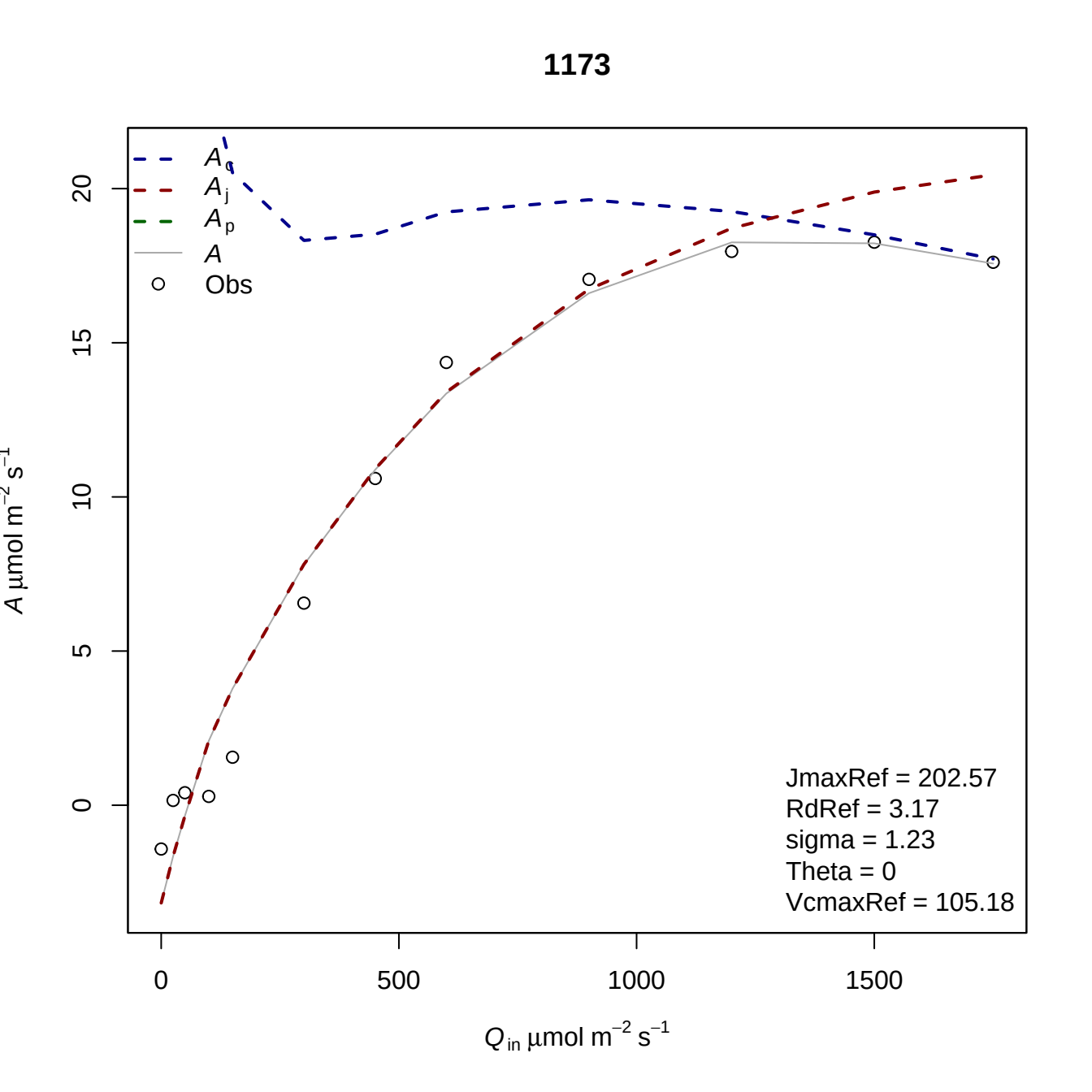

1168
A_c
A_j
A_p
A
Obs
JmaxRef = 160.76
RdRef = 1.48
sigma = 0.26
Theta = 0
VcmaxRef = 88.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

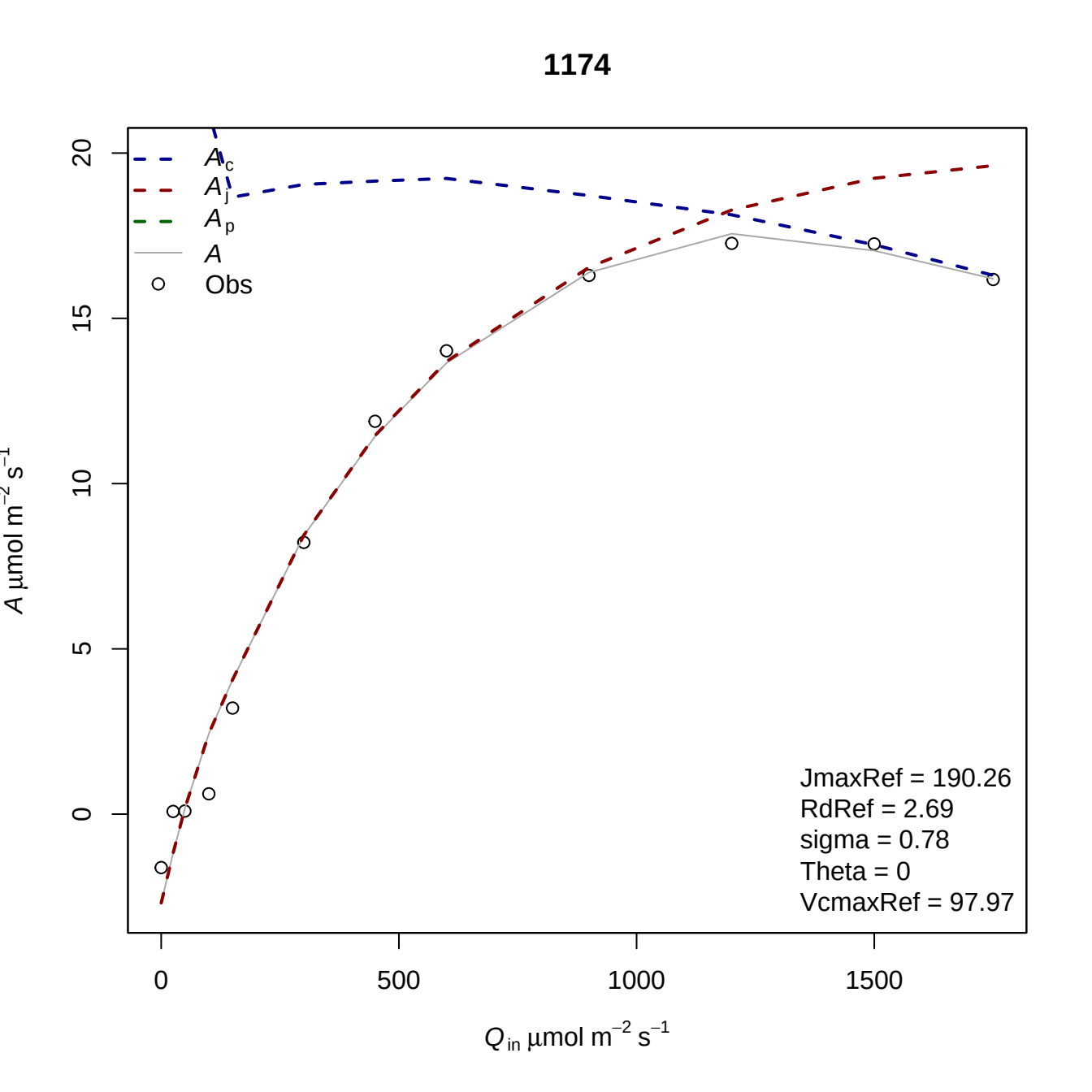

1169
A_c
A_j
A_p
A
Obs
JmaxRef = 205.72
RdRef = 1.48
sigma = 0.46
Theta = 0
VcmaxRef = 92.73
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

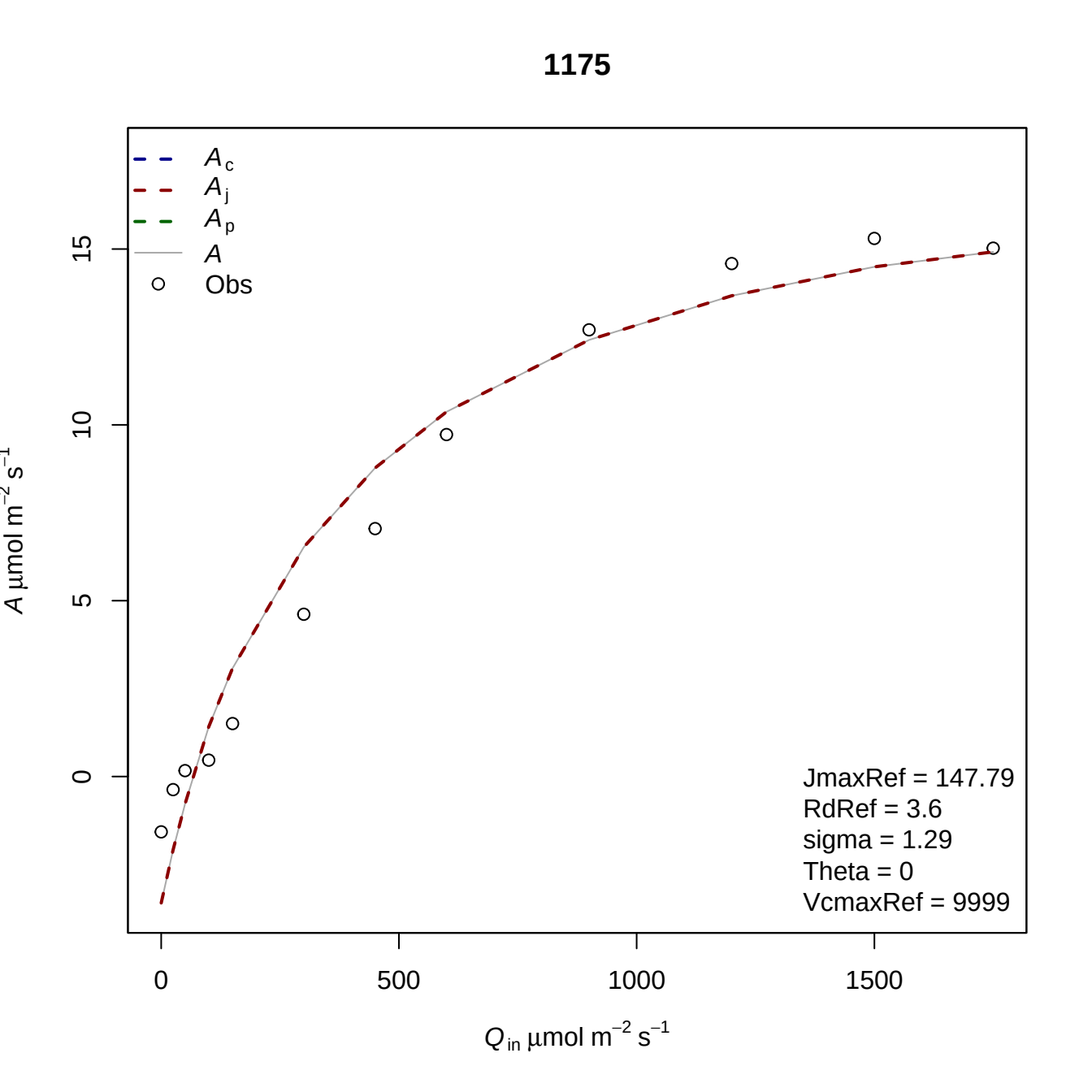

1170
A_c
A_j
A_p
A
Obs
JmaxRef = 218.63
RdRef = 2.82
sigma = 0.86
Theta = 0
VcmaxRef = 110.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

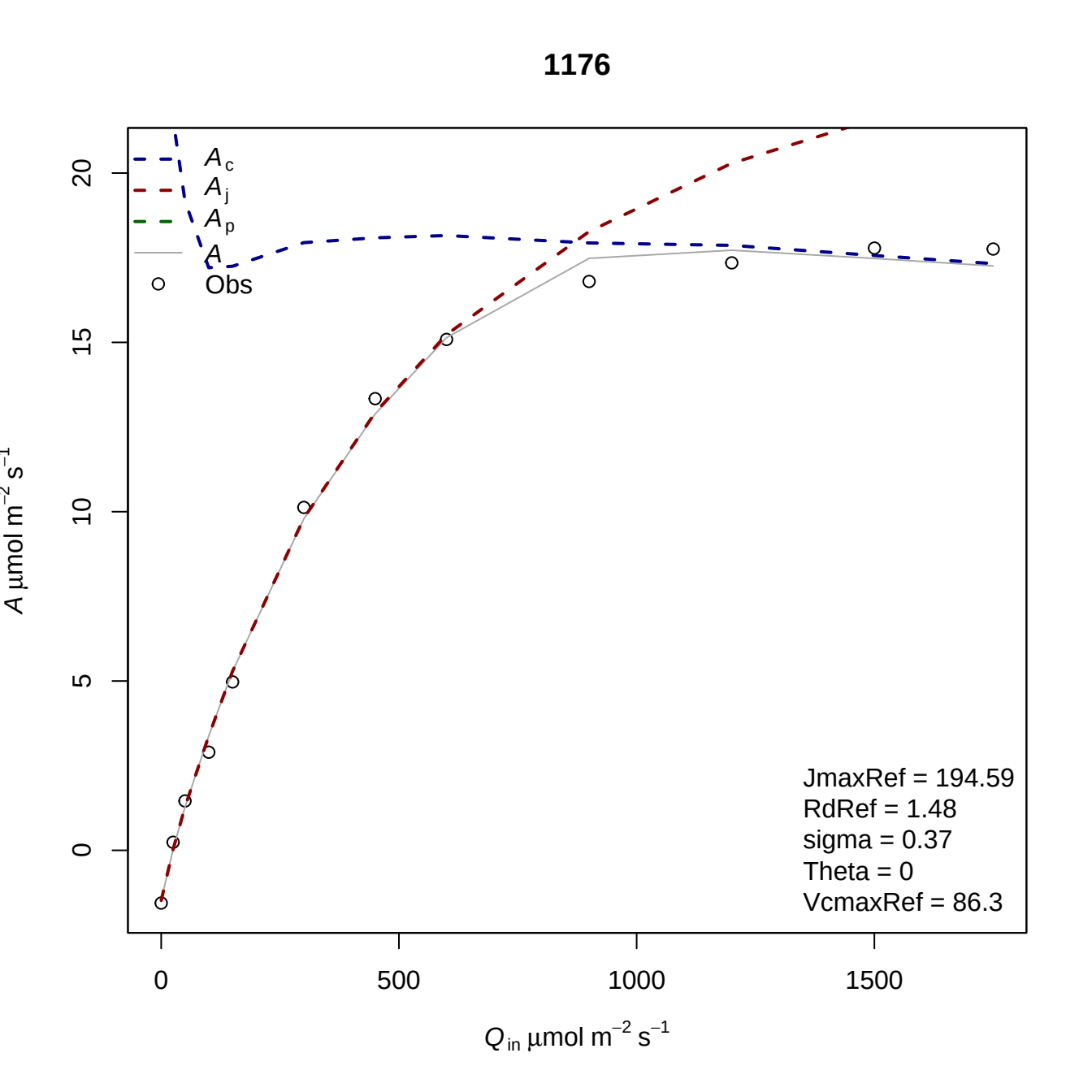

1171
A_c
A_j
A_p
A
Obs
JmaxRef = 201.97
RdRef = 1.91
sigma = 0.43
Theta = 0
VcmaxRef = 83.98
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

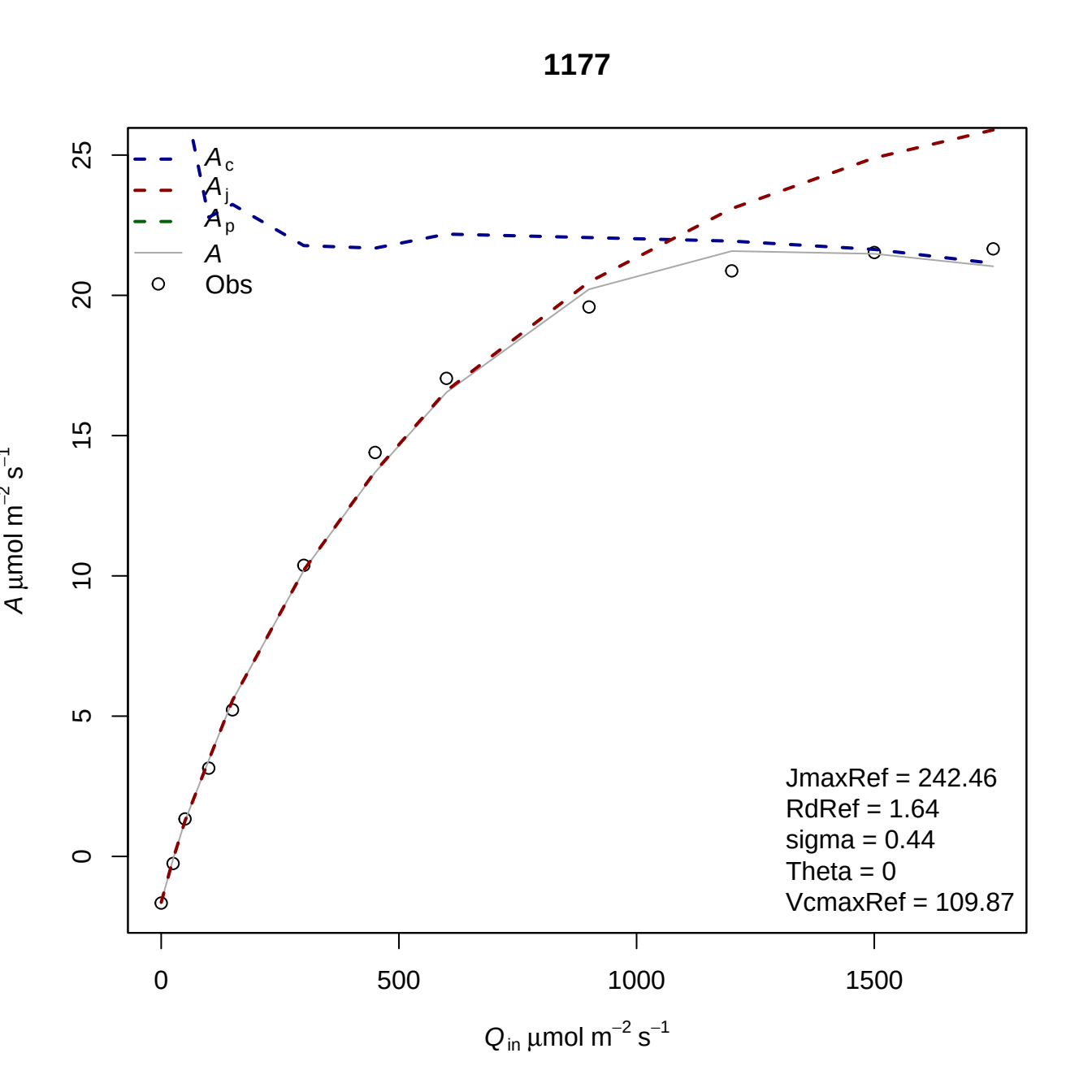

1172
A_c
A_j
A_p
A
Obs
JmaxRef = 168.14
RdRef = 2.32
sigma = 0.63
Theta = 0
VcmaxRef = 88.84
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

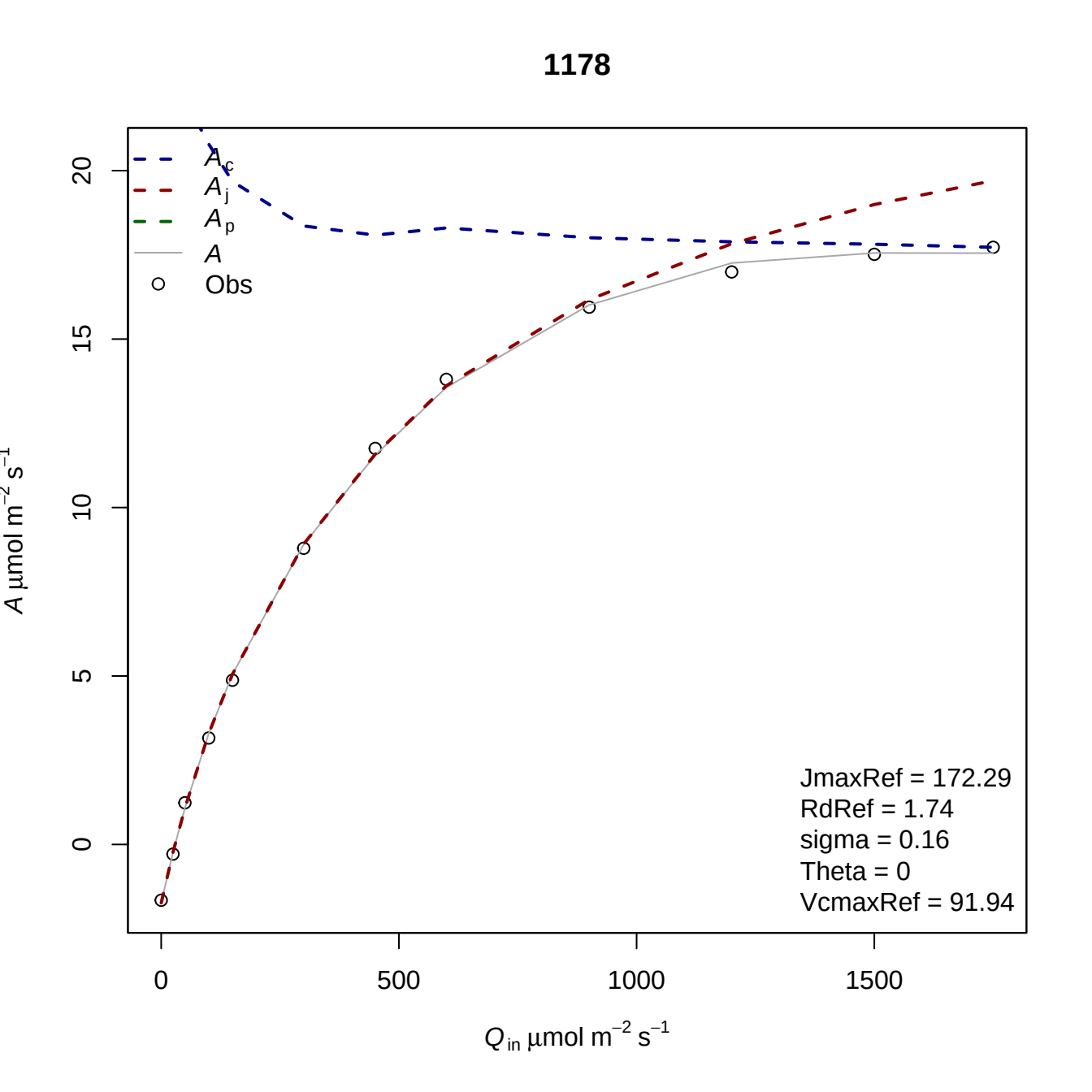

1173
A_c
A_j
A_p
A
Obs
JmaxRef = 202.57
RdRef = 3.17
sigma = 1.23
Theta = 0
VcmaxRef = 105.18
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

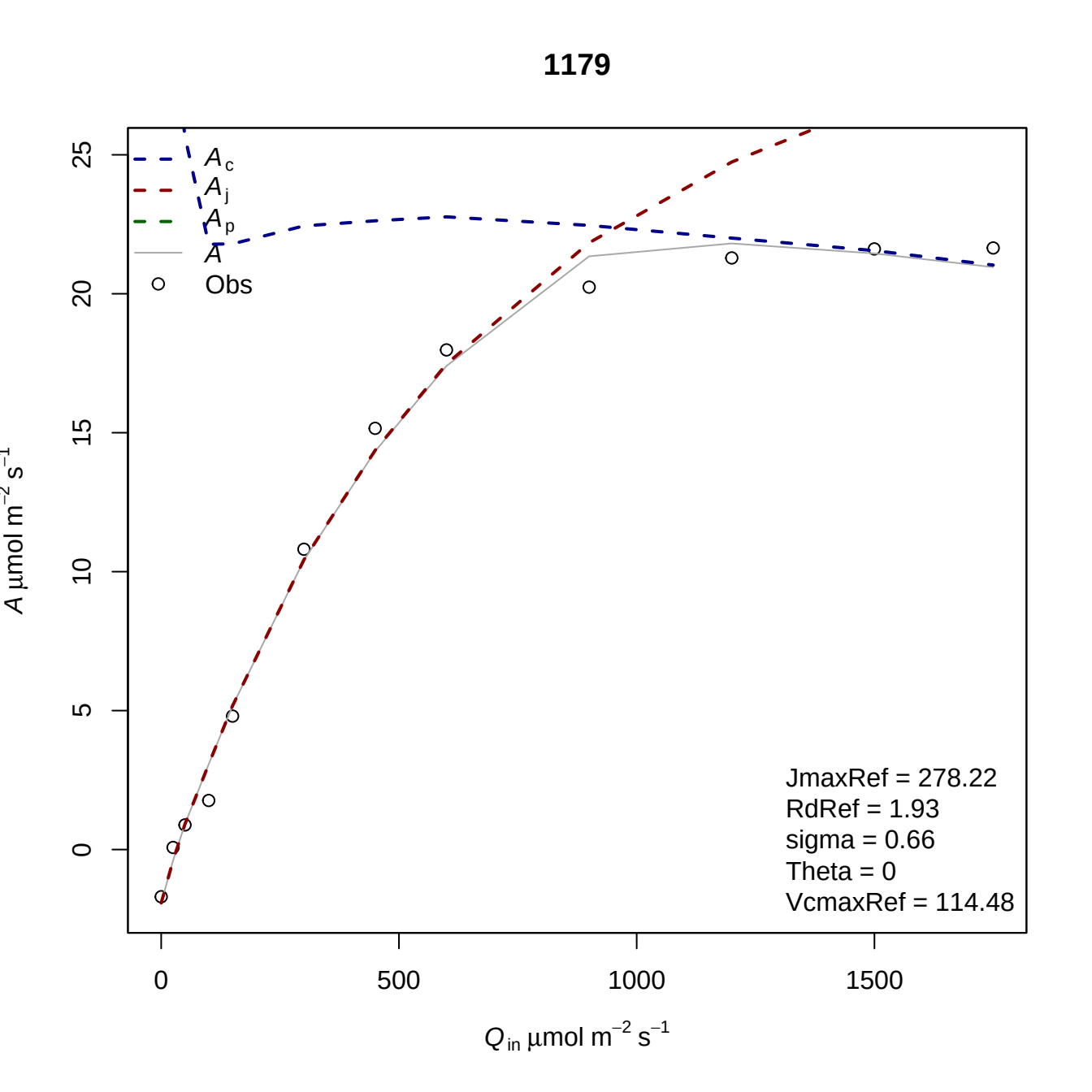

1174
A_c
A_j
A_p
A
Obs
JmaxRef = 190.26
RdRef = 2.69
sigma = 0.78
Theta = 0
VcmaxRef = 97.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

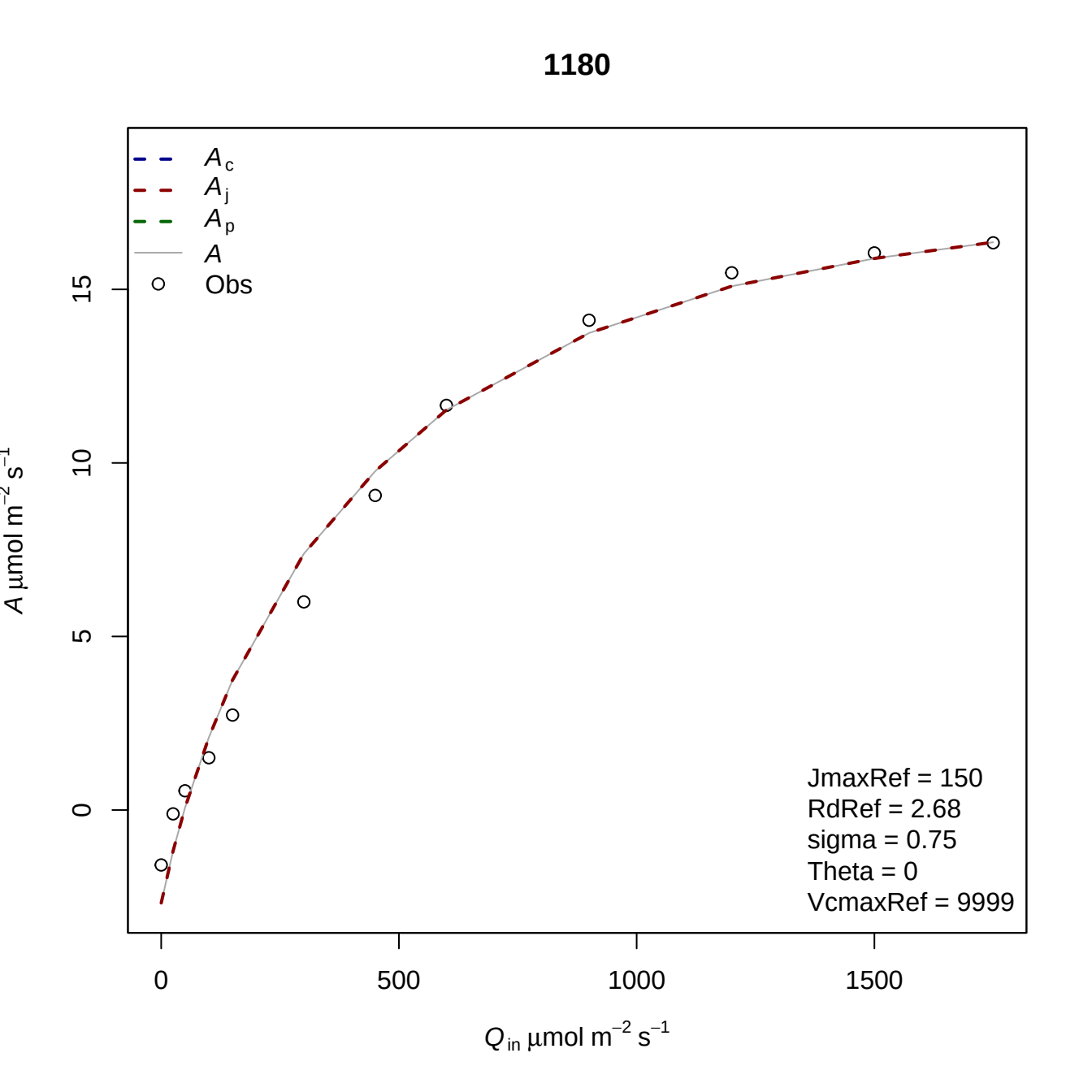

1175
A_c
A_j
A_p
A
Obs
JmaxRef = 147.79
RdRef = 3.6
sigma = 1.29
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

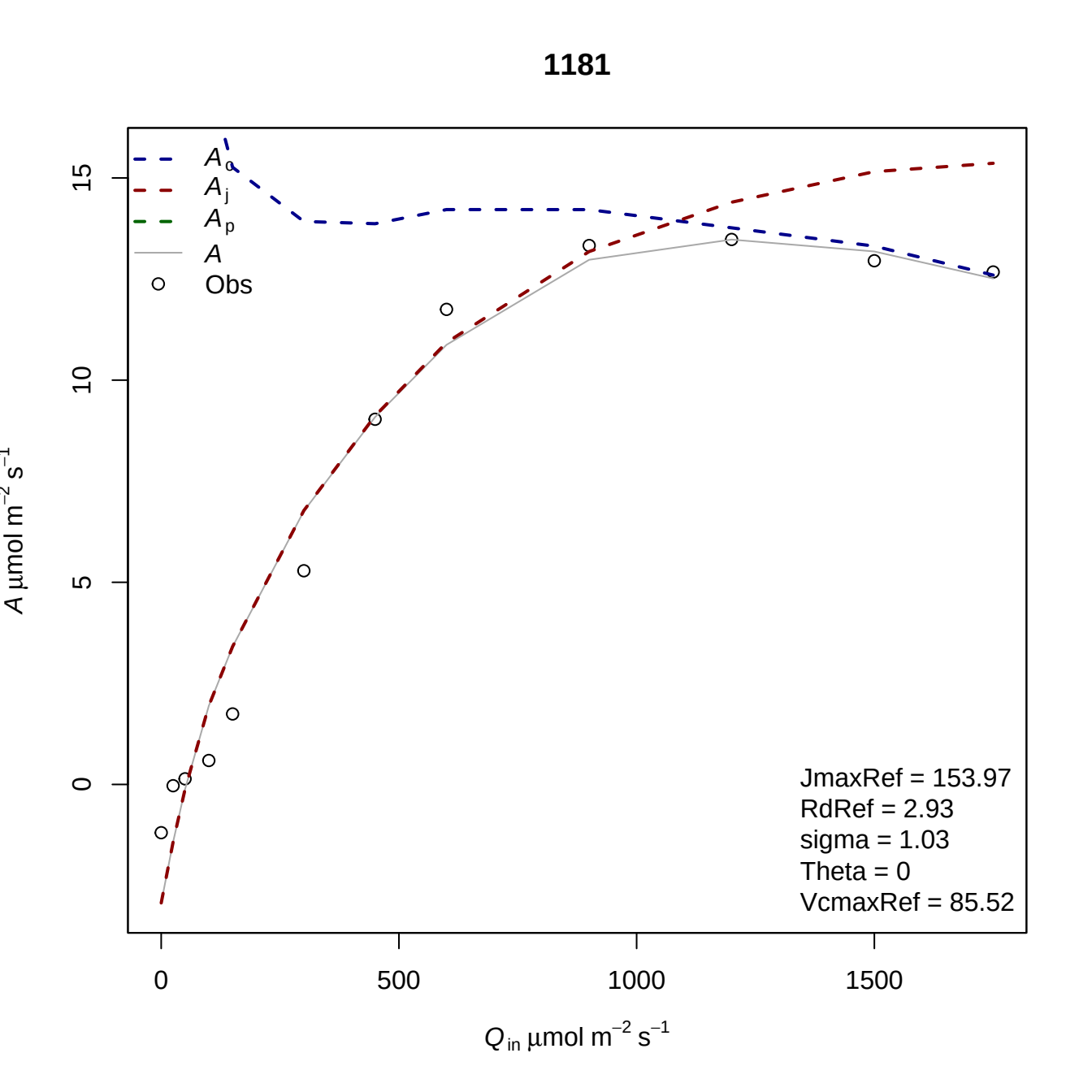

1176
A_c
A_j
A_p
A
Obs
JmaxRef = 194.59
RdRef = 1.48
sigma = 0.37
Theta = 0
VcmaxRef = 86.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

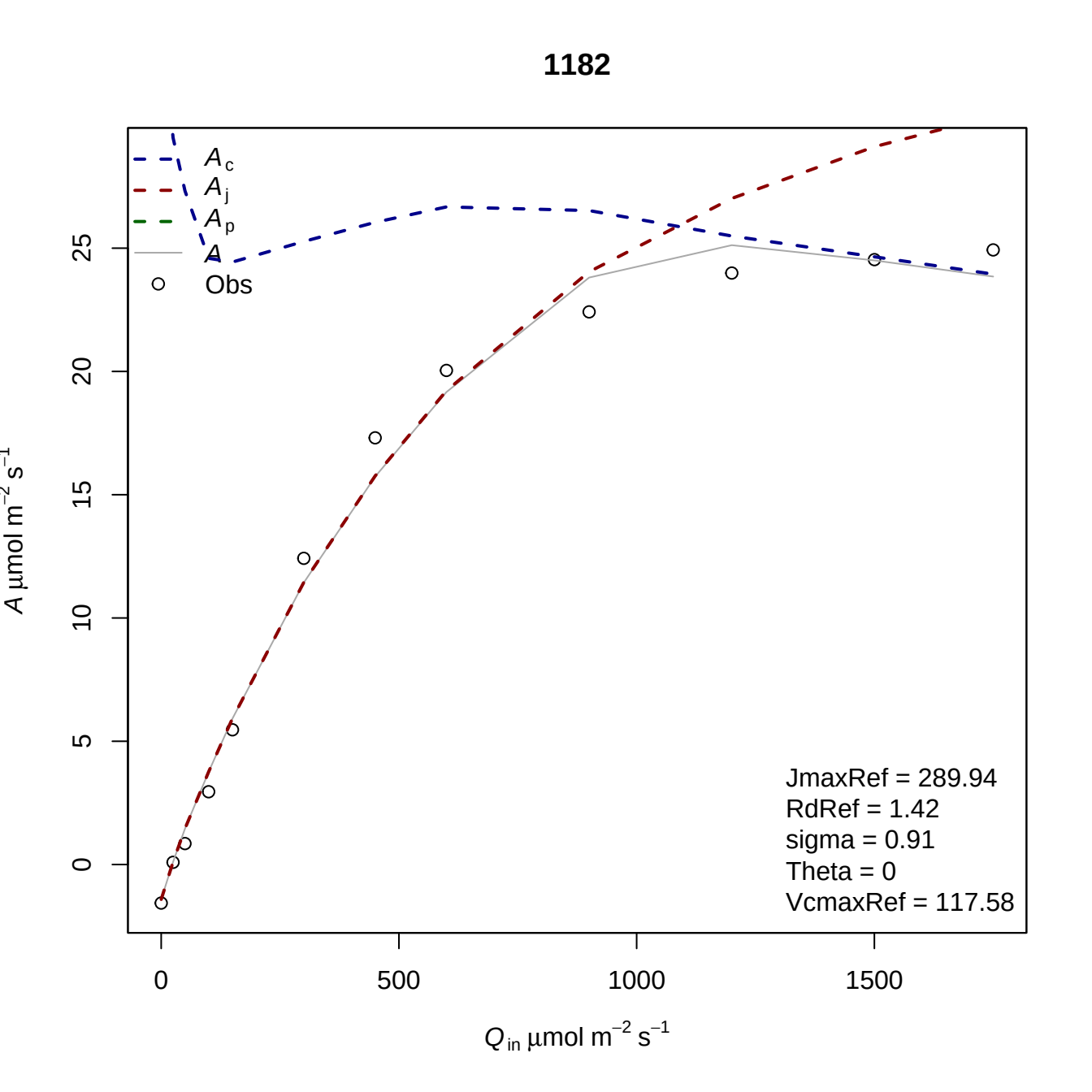

1177
A_c
A_j
A_p
A
Obs
JmaxRef = 242.46
RdRef = 1.64
sigma = 0.44
Theta = 0
VcmaxRef = 109.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

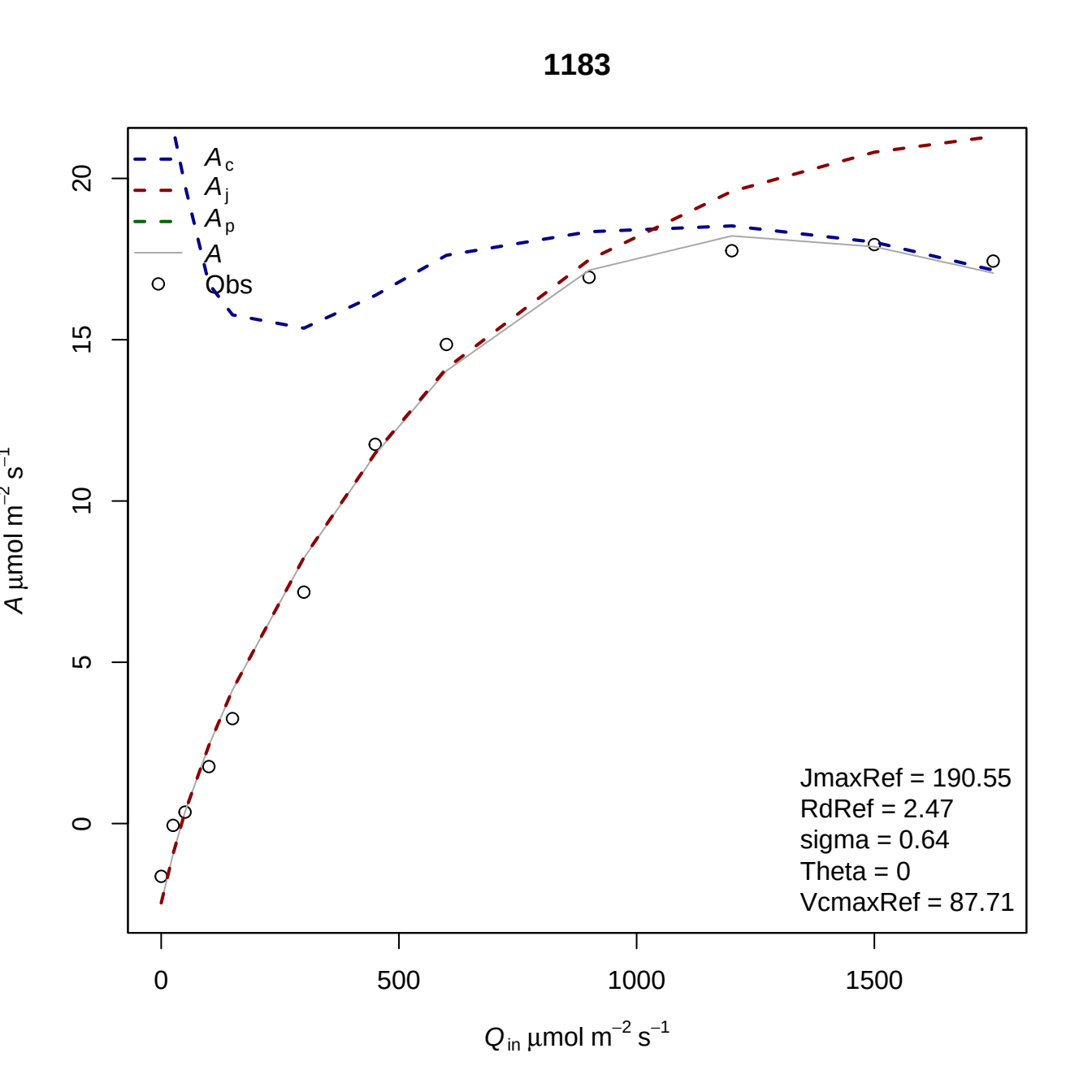

1178
A_c
A_j
A_p
A
Obs
JmaxRef = 172.29
RdRef = 1.74
sigma = 0.16
Theta = 0
VcmaxRef = 91.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

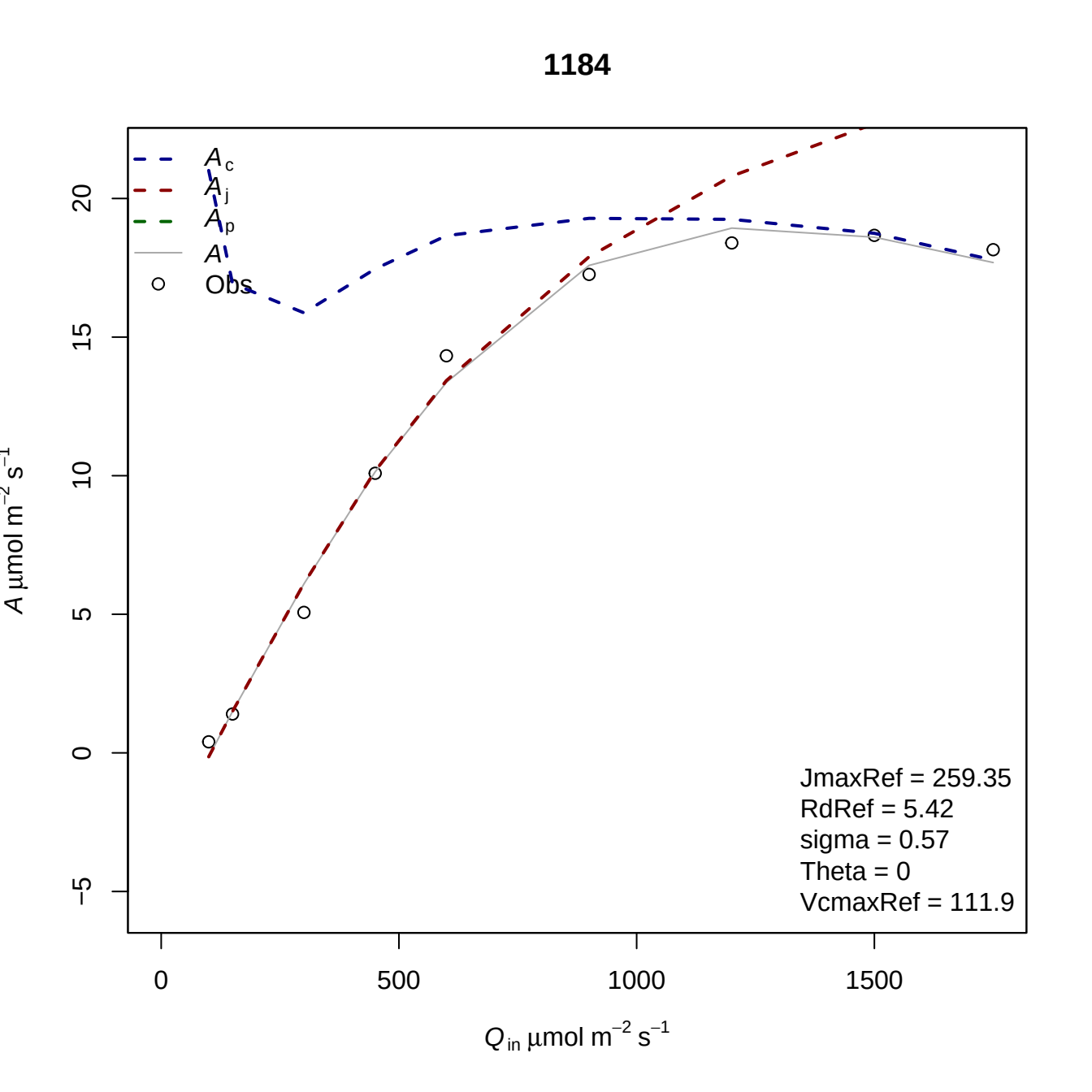

1179
A_c
A_j
A_p
A
Obs
JmaxRef = 278.22
RdRef = 1.93
sigma = 0.66
Theta = 0
VcmaxRef = 114.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

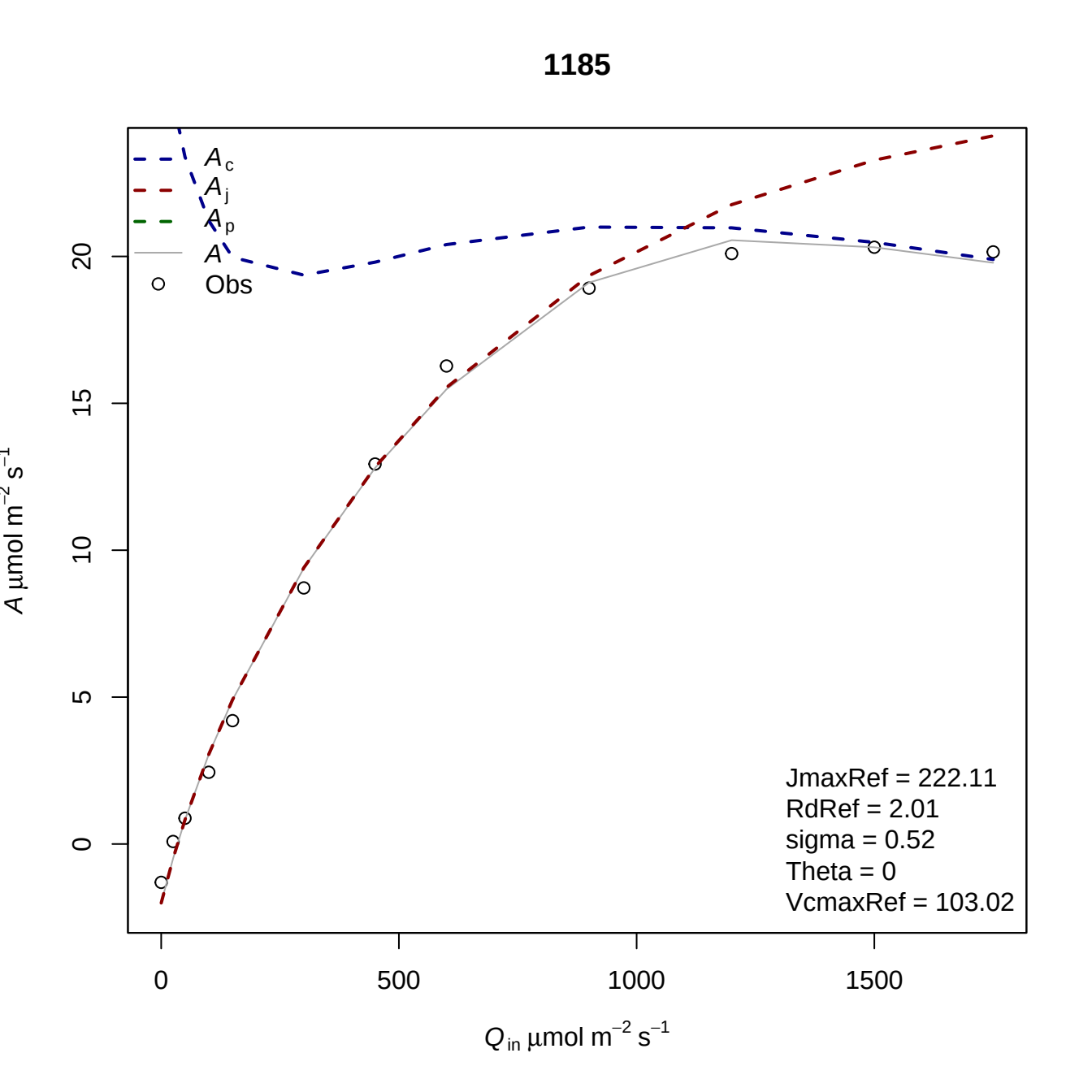

1180
A_c
A_j
A_p
A
Obs
JmaxRef = 150
RdRef = 2.68
sigma = 0.75
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

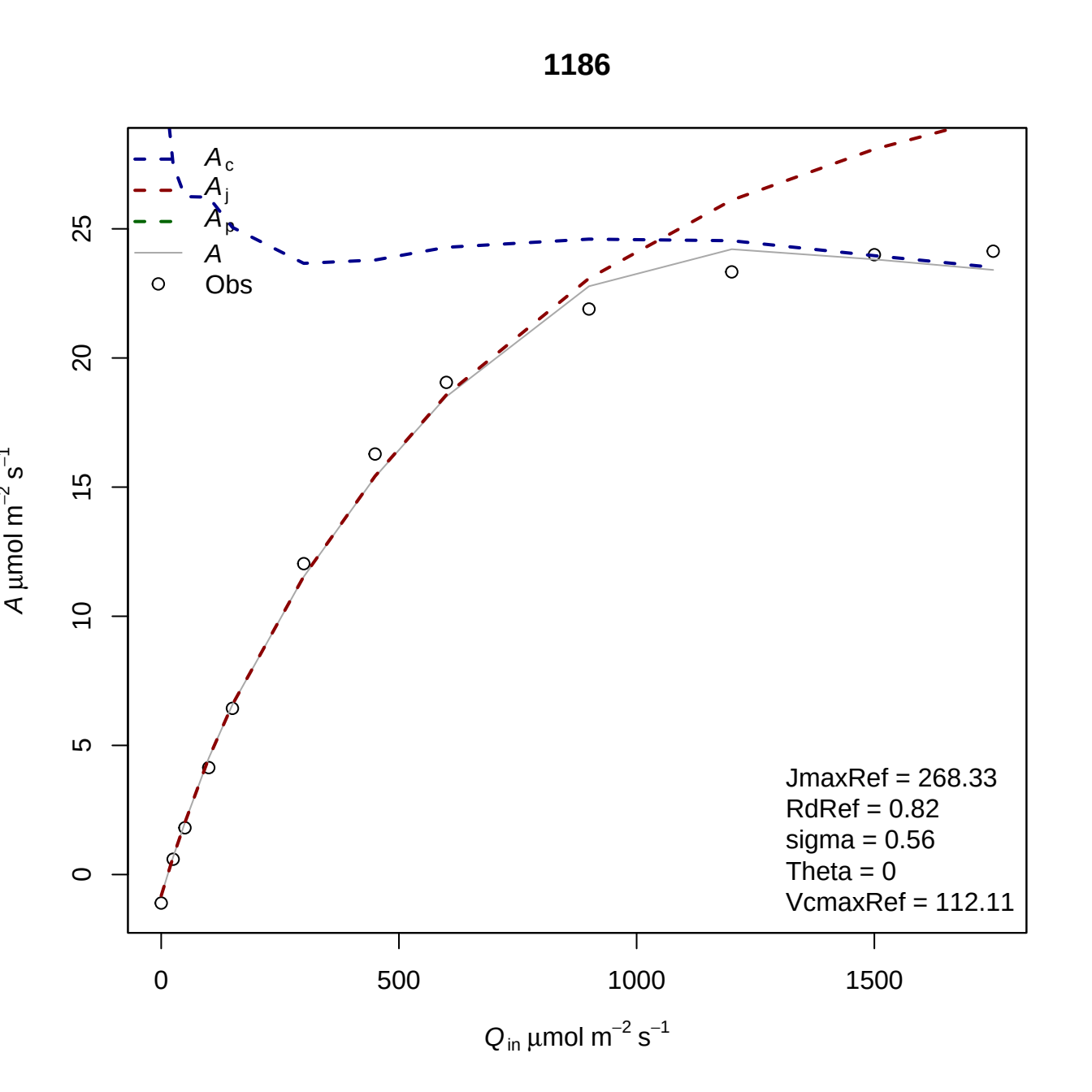

1181
A_c
A_j
A_p
A
Obs
JmaxRef = 153.97
RdRef = 2.93
sigma = 1.03
Theta = 0
VcmaxRef = 85.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

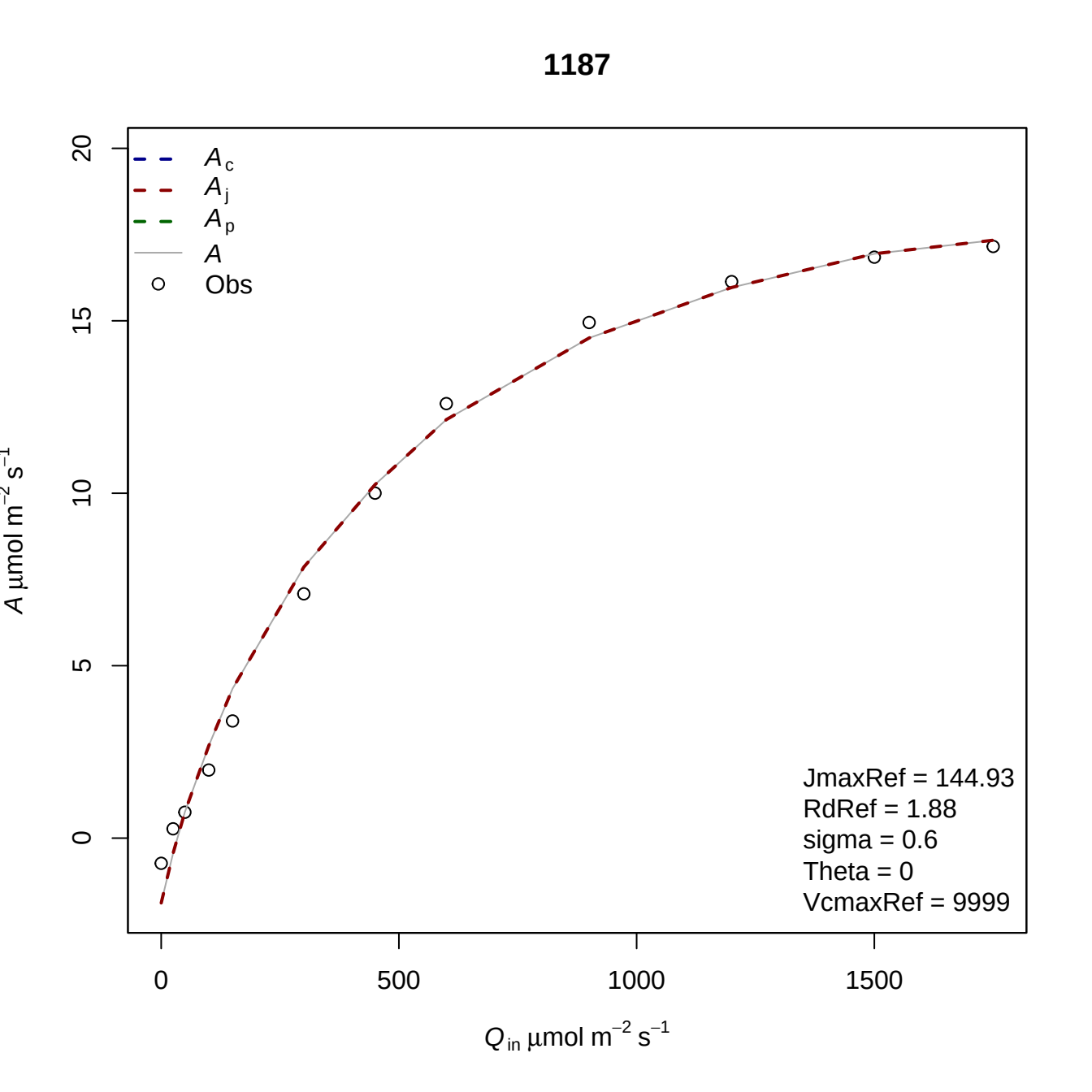

1182
A_c
A_j
A_p
A
Obs
JmaxRef = 289.94
RdRef = 1.42
sigma = 0.91
Theta = 0
VcmaxRef = 117.58
0
5
10
15
20
25
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

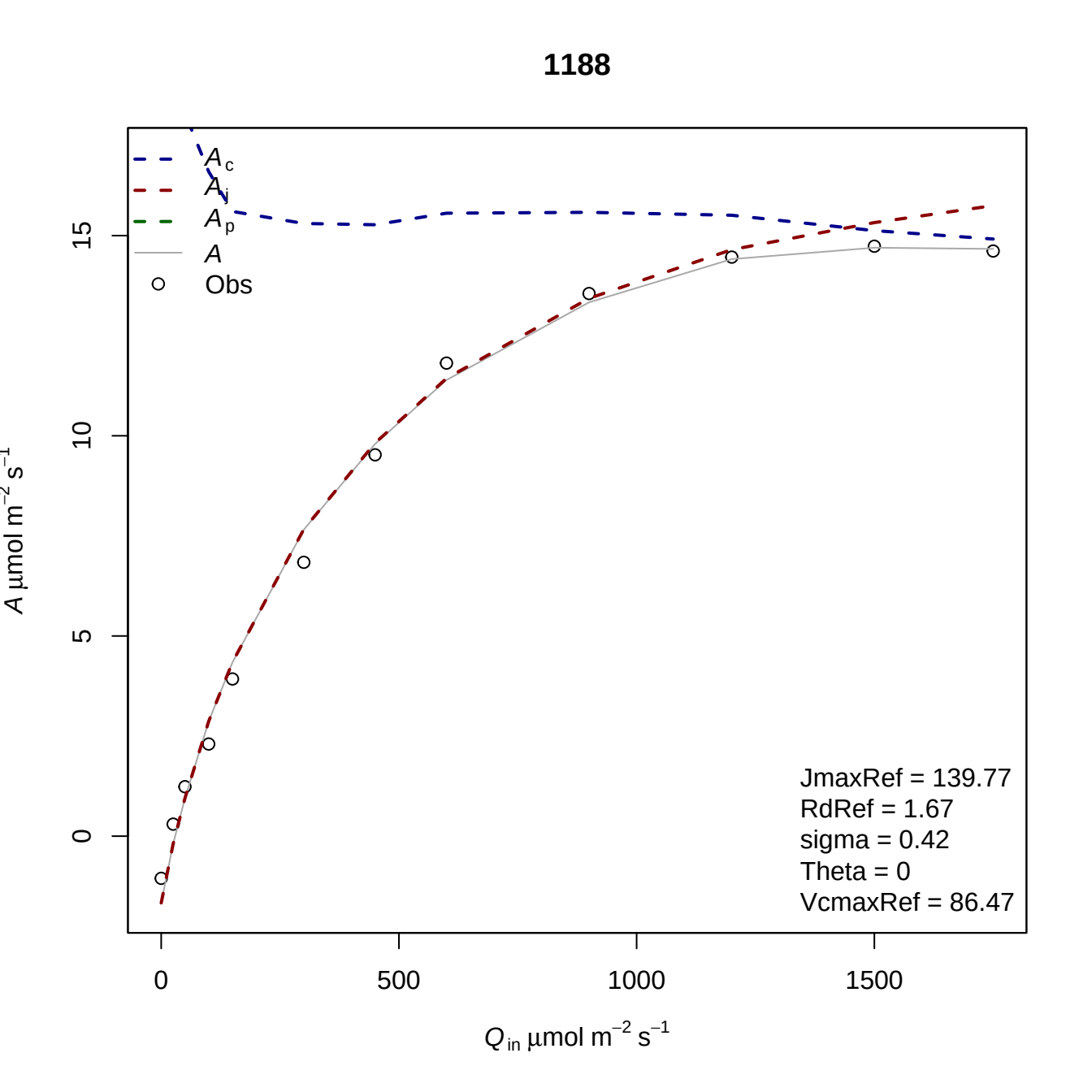

1183
A_c
A_j
A_p
A
Qbs
JmaxRef = 190.55
RdRef = 2.47
sigma = 0.64
Theta = 0
VcmaxRef = 87.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

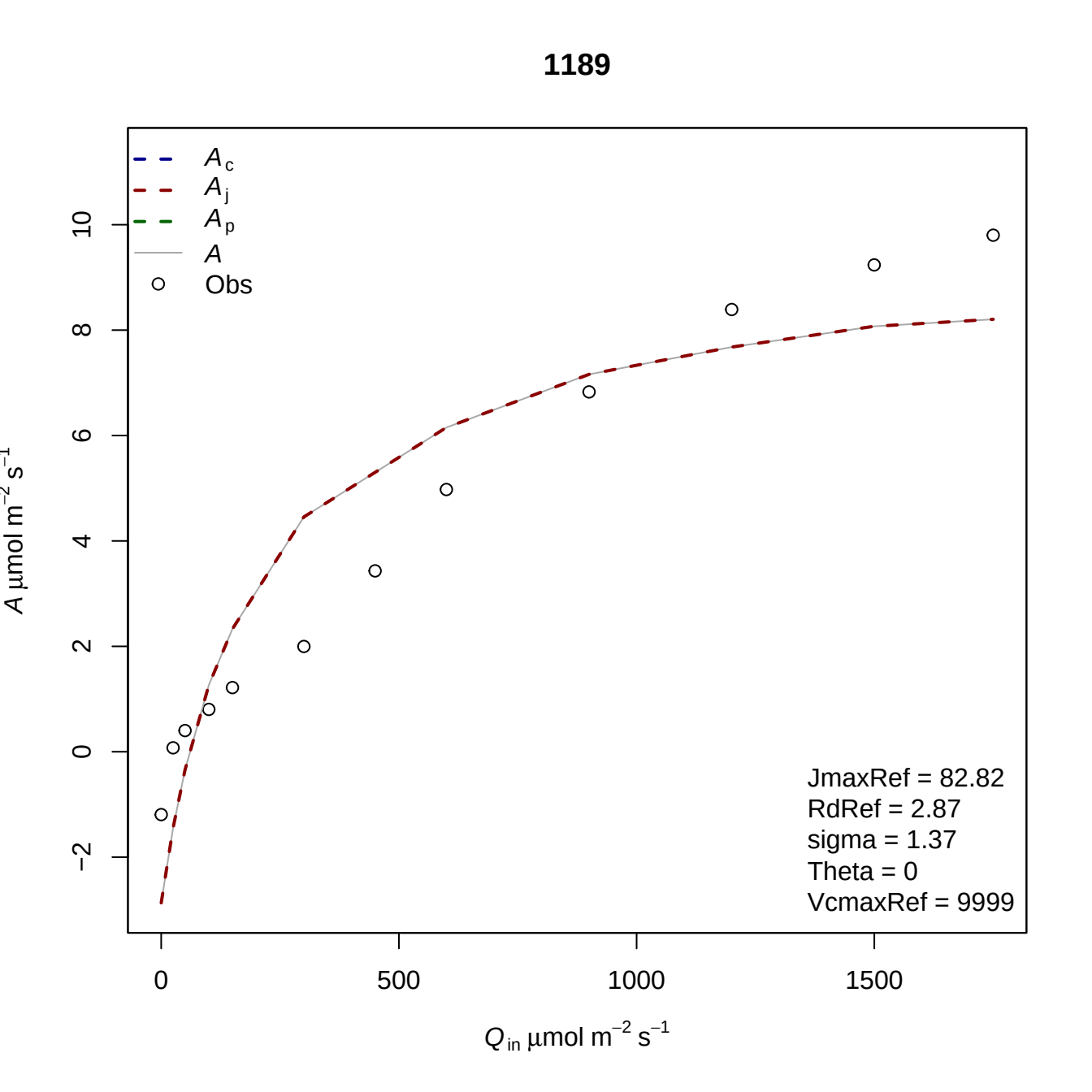

1184
A_c
A_j
A_p
A
Obs
JmaxRef = 259.35
RdRef = 5.42
sigma = 0.57
Theta = 0
VcmaxRef = 111.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500

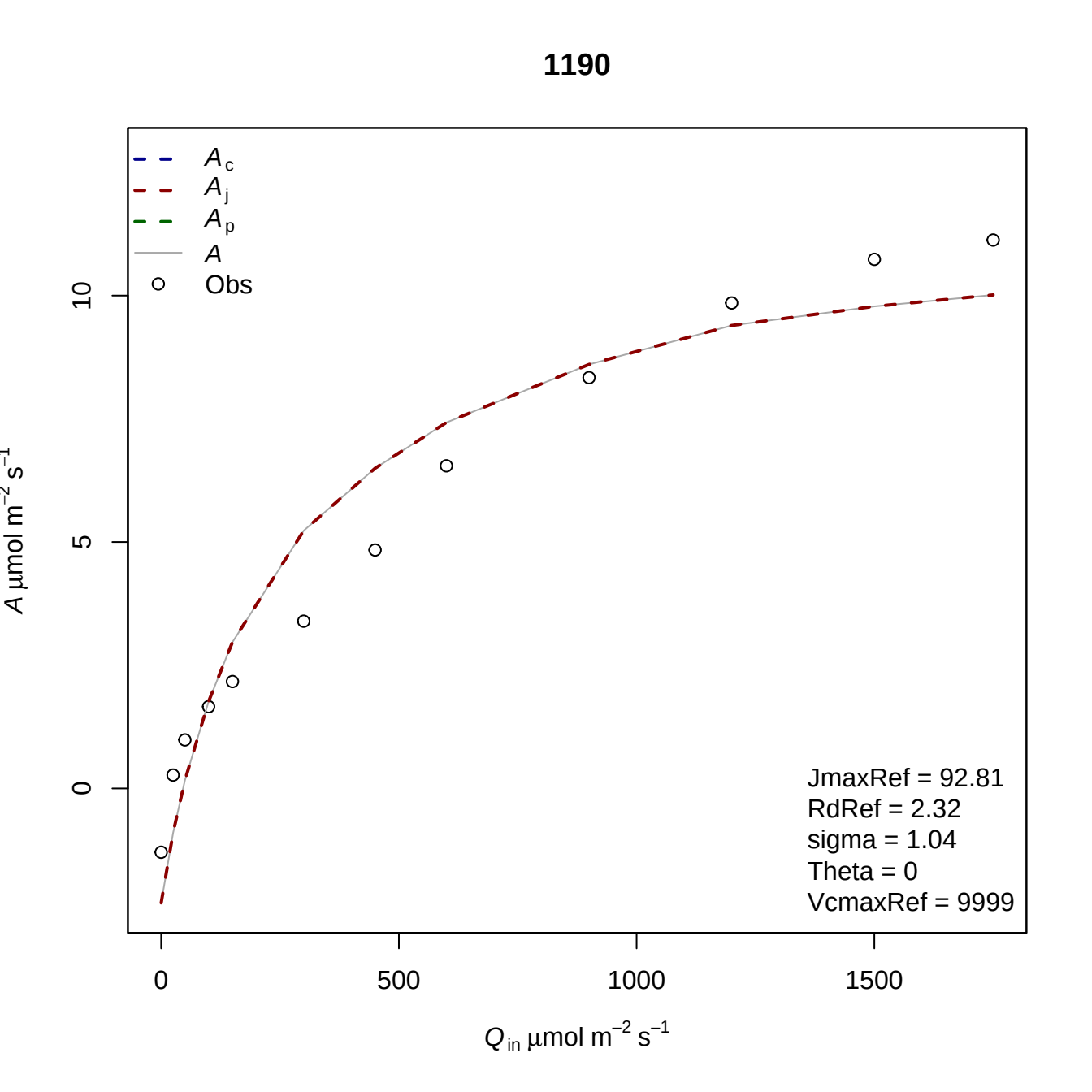

1185
A_c
A_j
A_p
A
Obs
JmaxRef = 222.11
RdRef = 2.01
sigma = 0.52
Theta = 0
VcmaxRef = 103.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

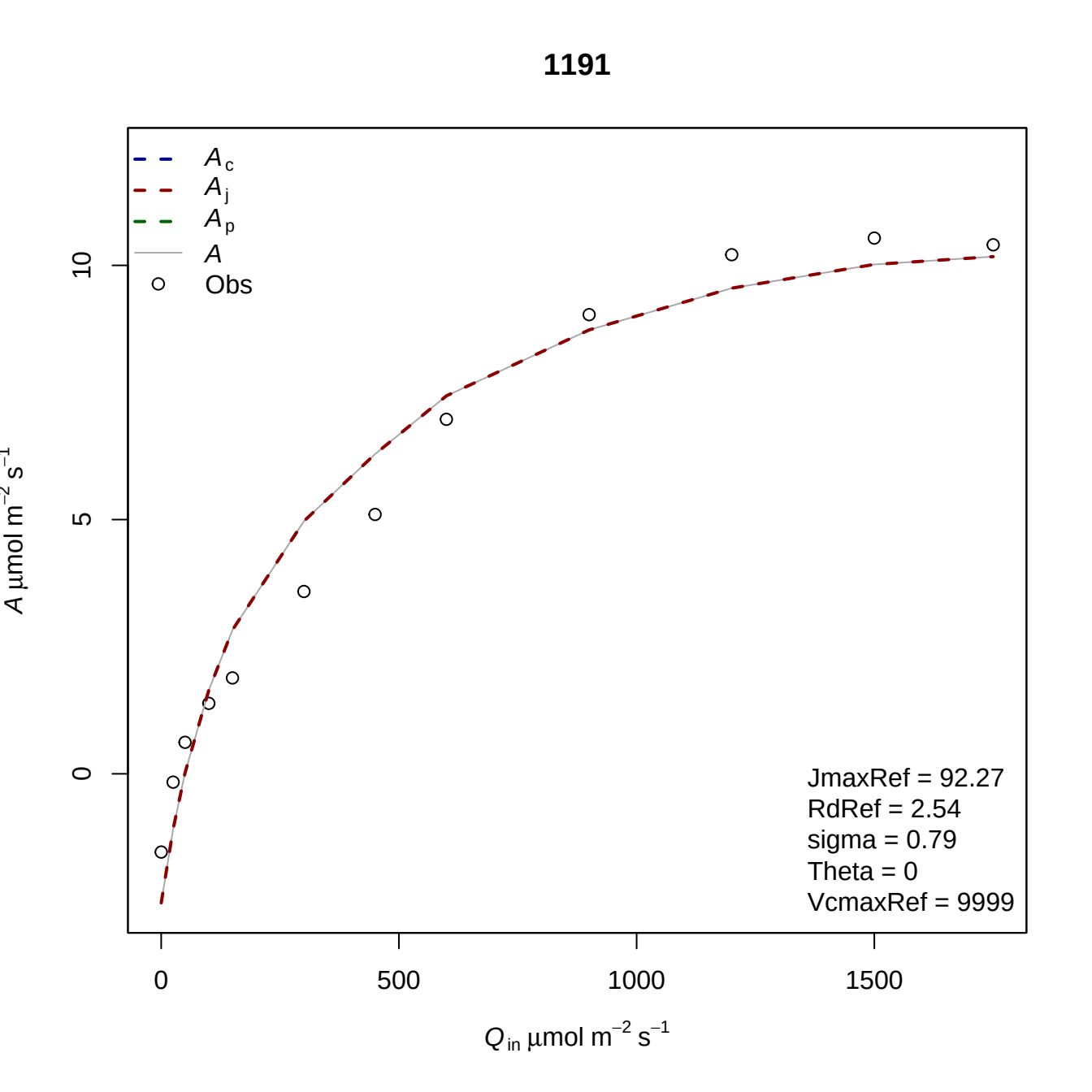

1186
A_c
A_j
A_p
A
Obs
JmaxRef = 268.33
RdRef = 0.82
sigma = 0.56
Theta = 0
VcmaxRef = 112.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

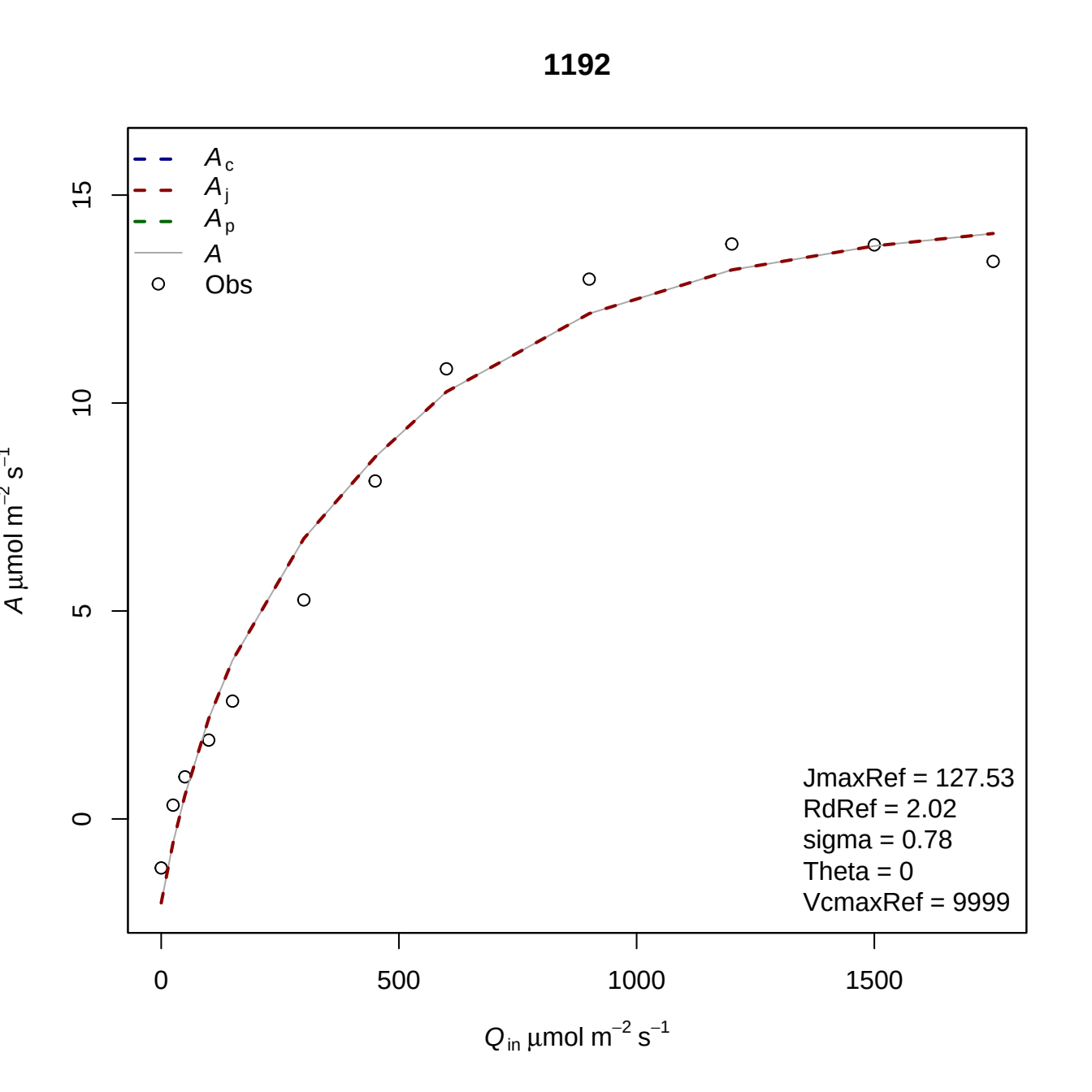

1187
A_c
A_j
A_p
A
Obs
JmaxRef = 144.93
RdRef = 1.88
sigma = 0.6
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

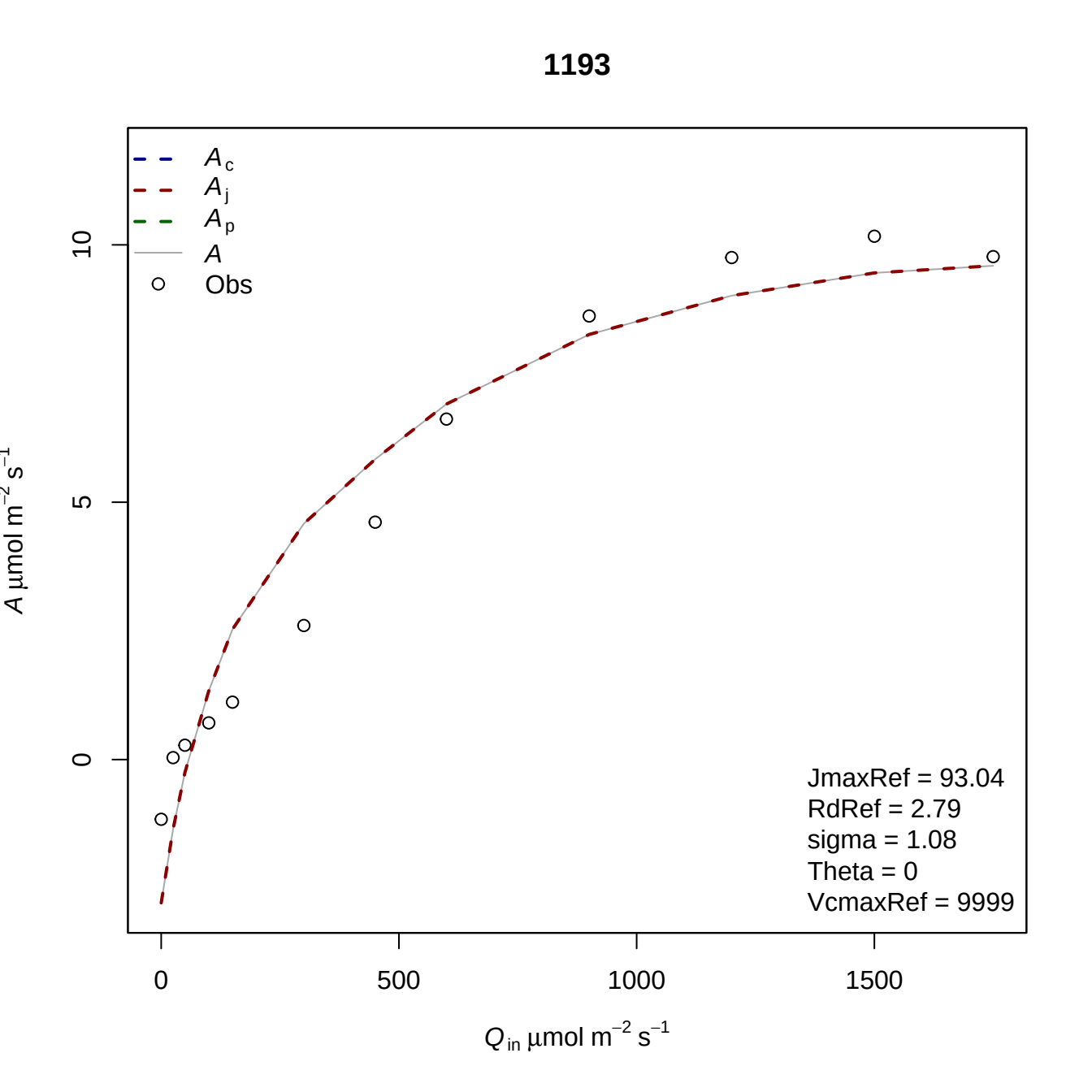

1188
A_c
A_j
A_p
A
Obs
JmaxRef = 139.77
RdRef = 1.67
sigma = 0.42
Theta = 0
VcmaxRef = 86.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

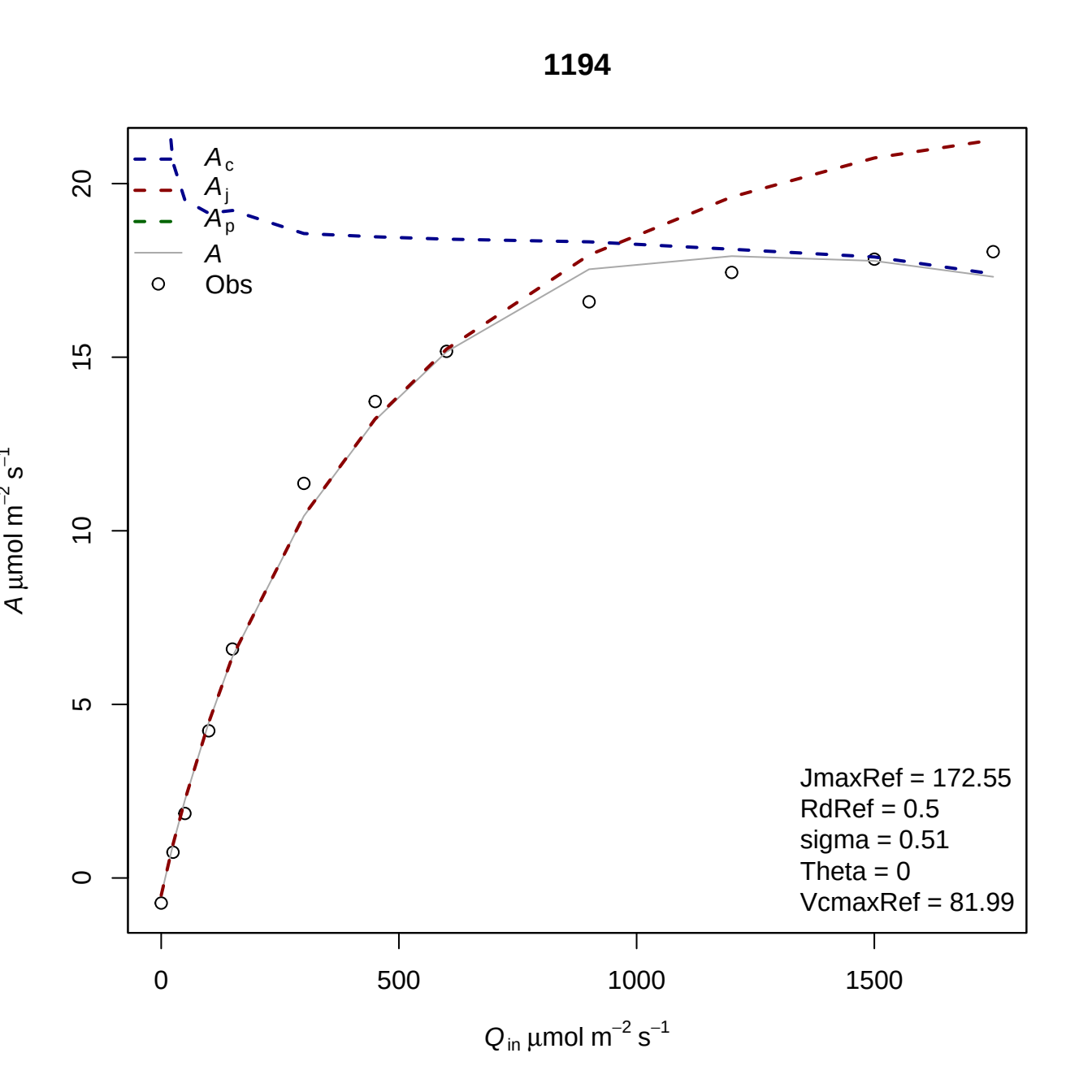

1189
A_c
A_j
A_p
A
Obs
JmaxRef = 82.82
RdRef = 2.87
sigma = 1.37
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

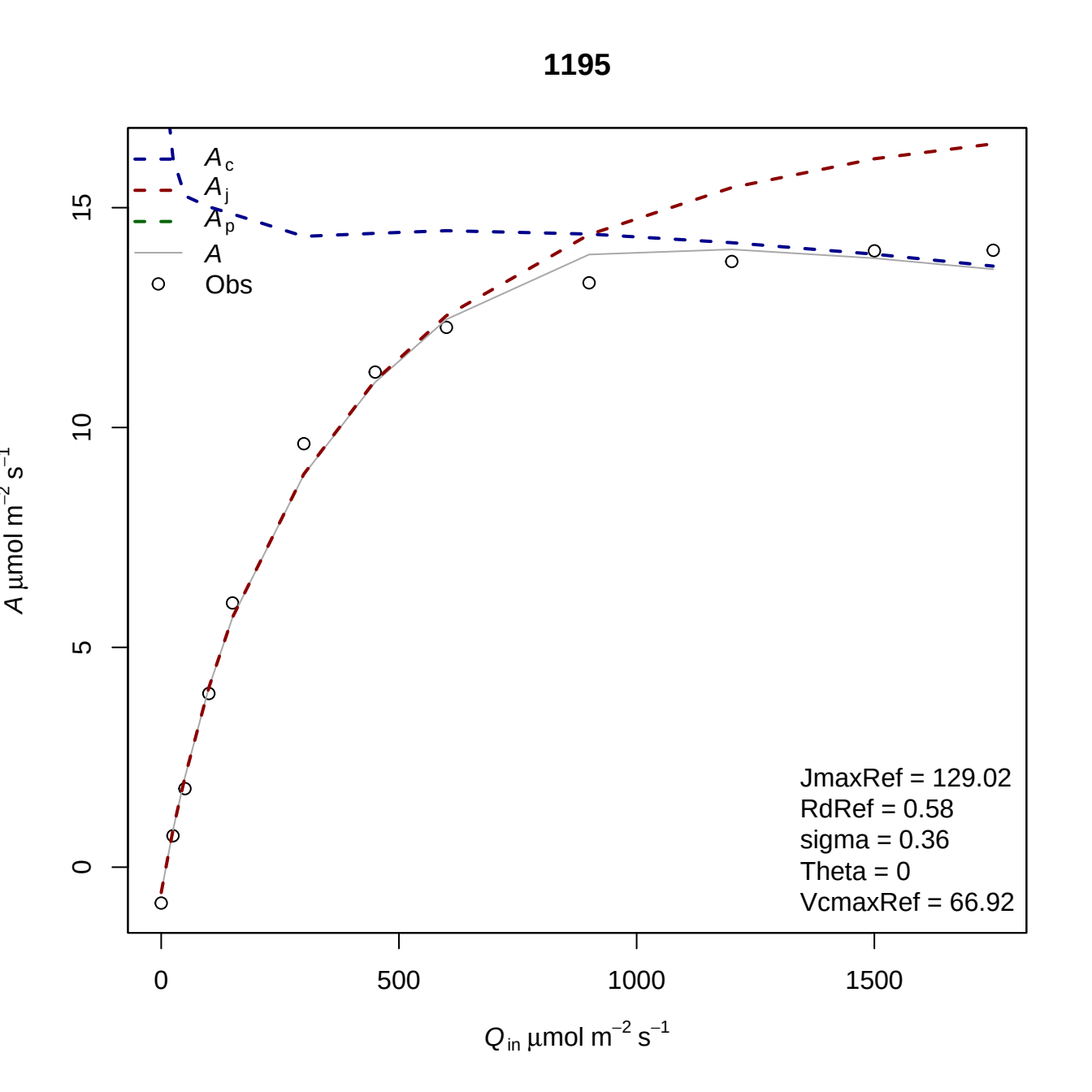

1190
A_c
A_j
A_p
A
Obs
JmaxRef = 92.81
RdRef = 2.32
sigma = 1.04
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

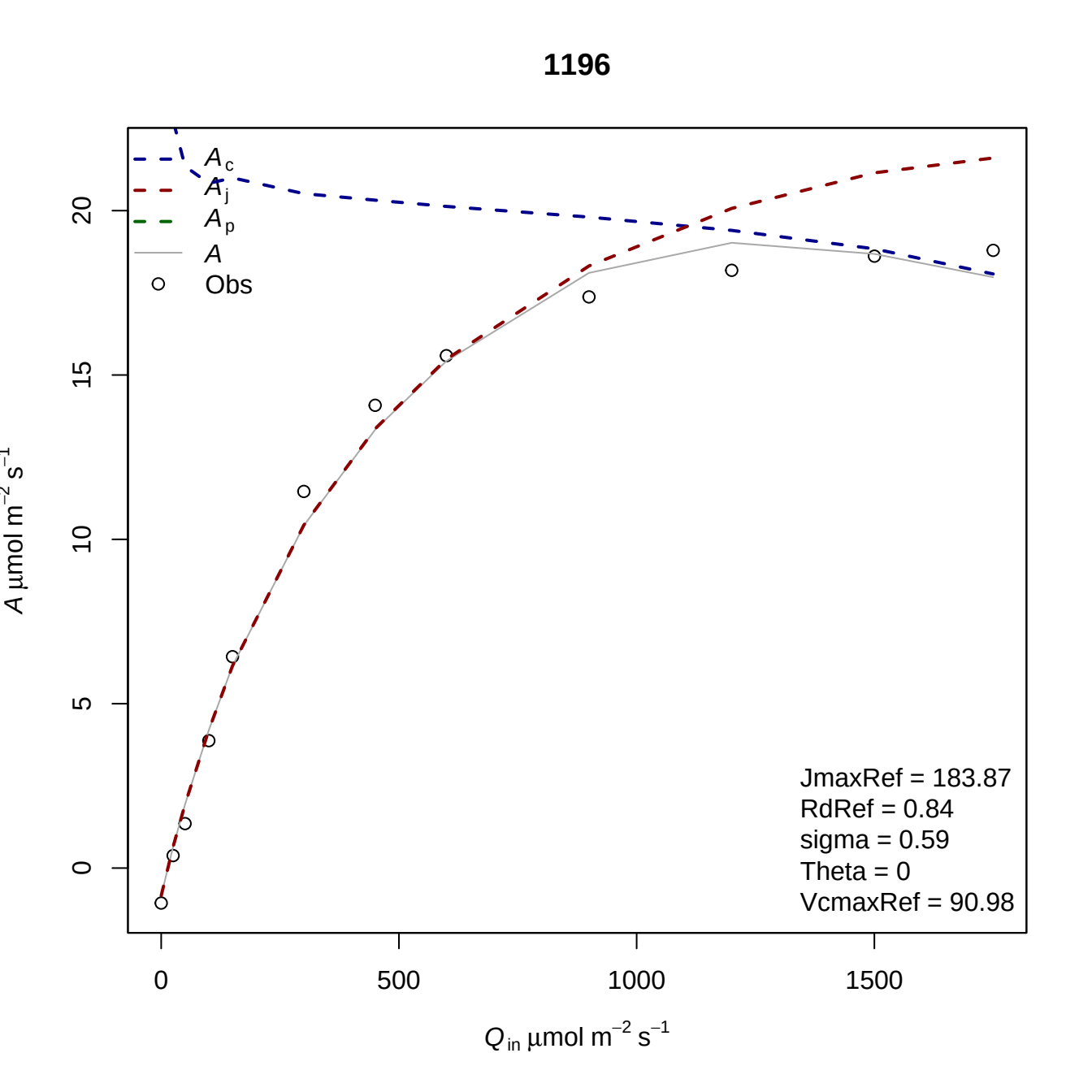

1191
A_c
A_j
A_p
A
Obs
JmaxRef = 92.27
RdRef = 2.54
sigma = 0.79
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

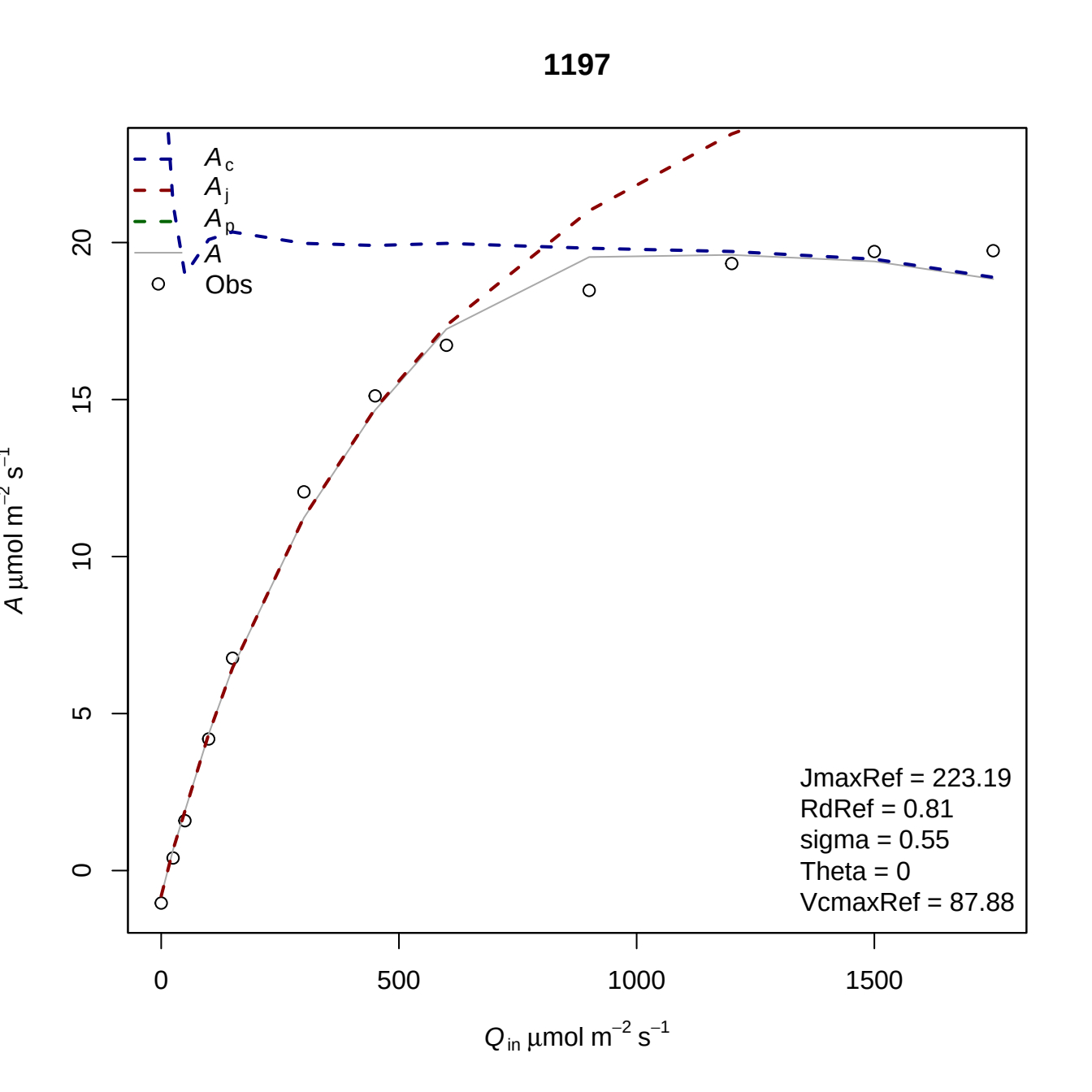

1192
A_c
A_j
A_p
A
Obs
JmaxRef = 127.53
RdRef = 2.02
sigma = 0.78
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

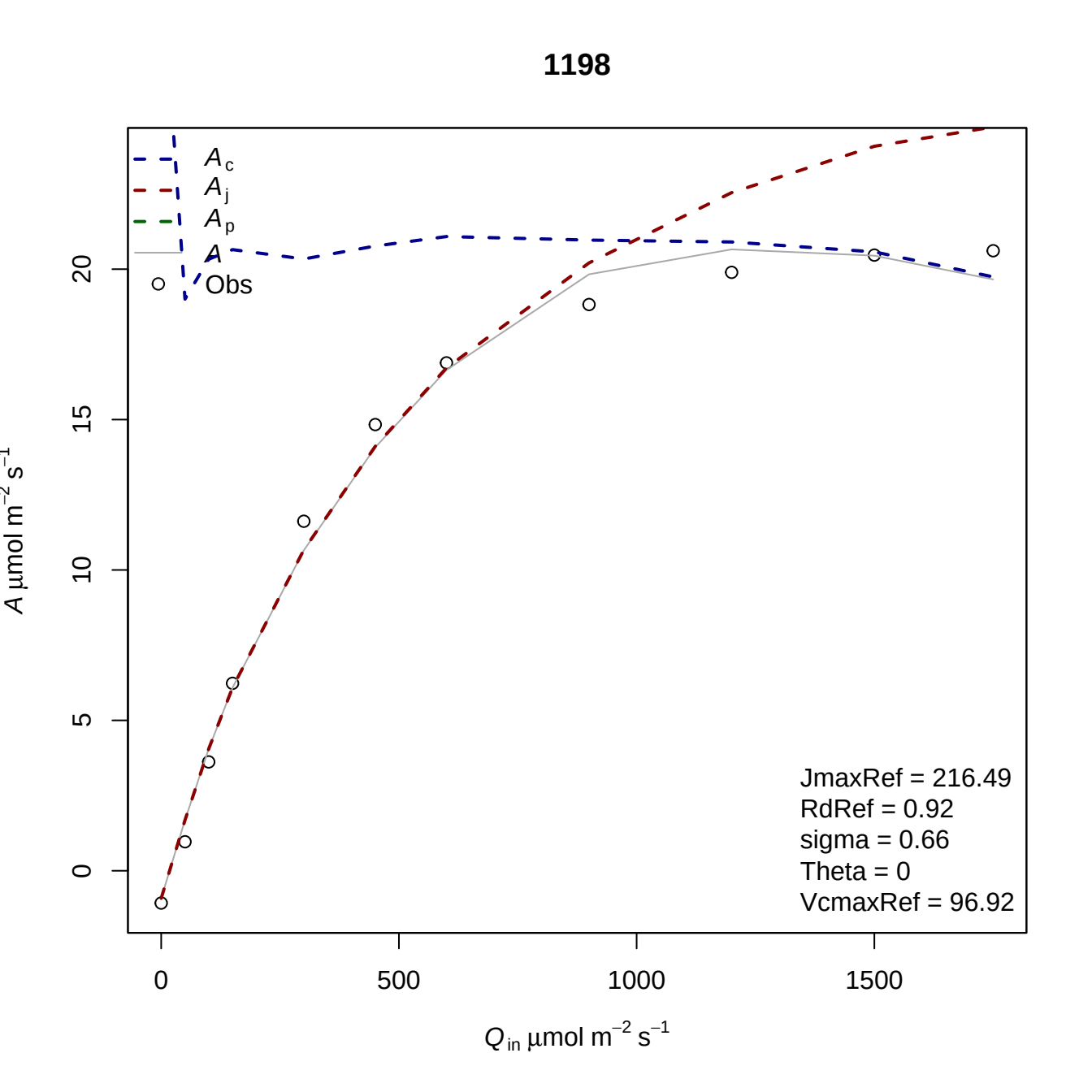

1193
A_c
A_j
A_p
A
Obs
JmaxRef = 93.04
RdRef = 2.79
sigma = 1.08
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

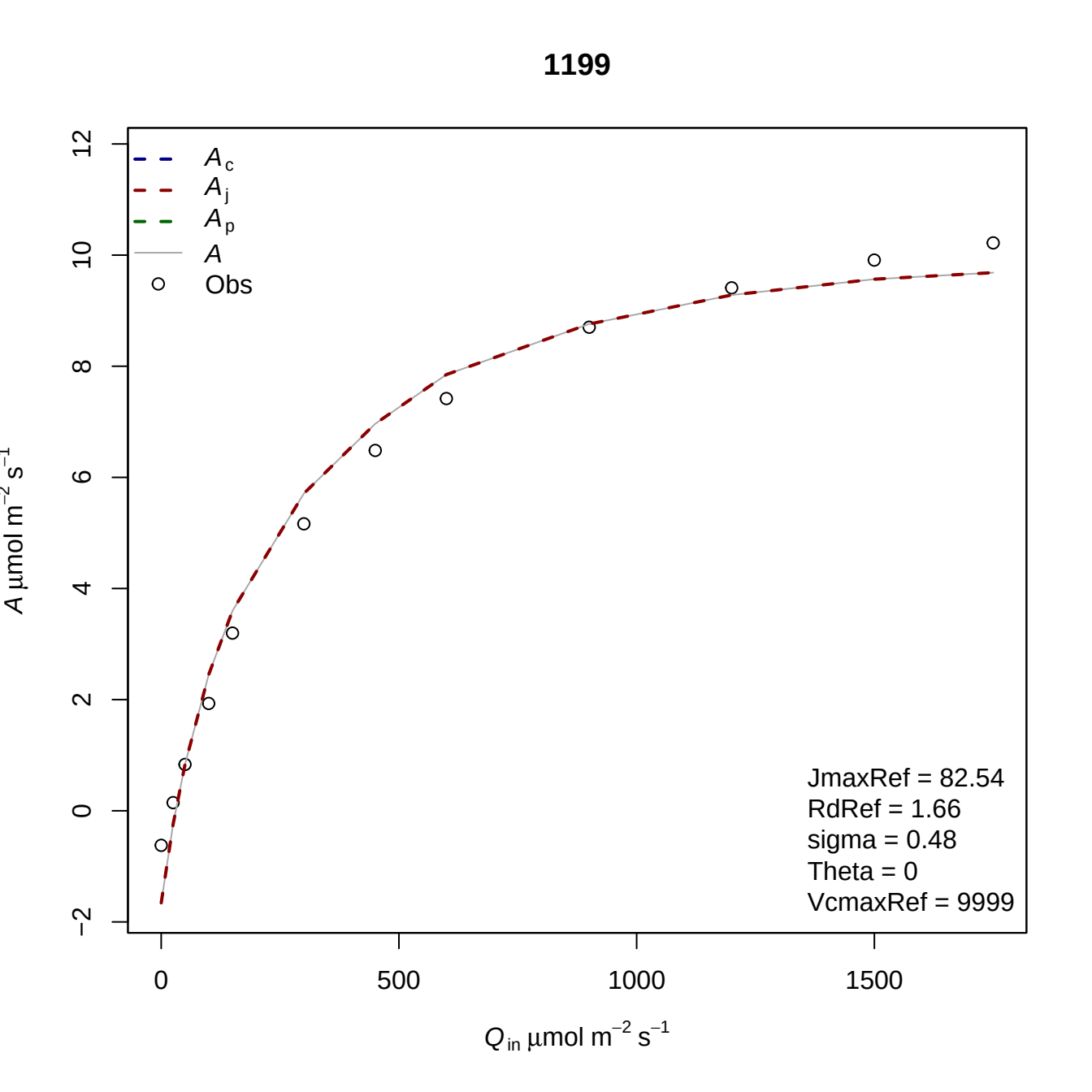

1194
A_c
A_j
A_p
A
Obs
JmaxRef = 172.55
RdRef = 0.5
sigma = 0.51
Theta = 0
VcmaxRef = 81.99
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

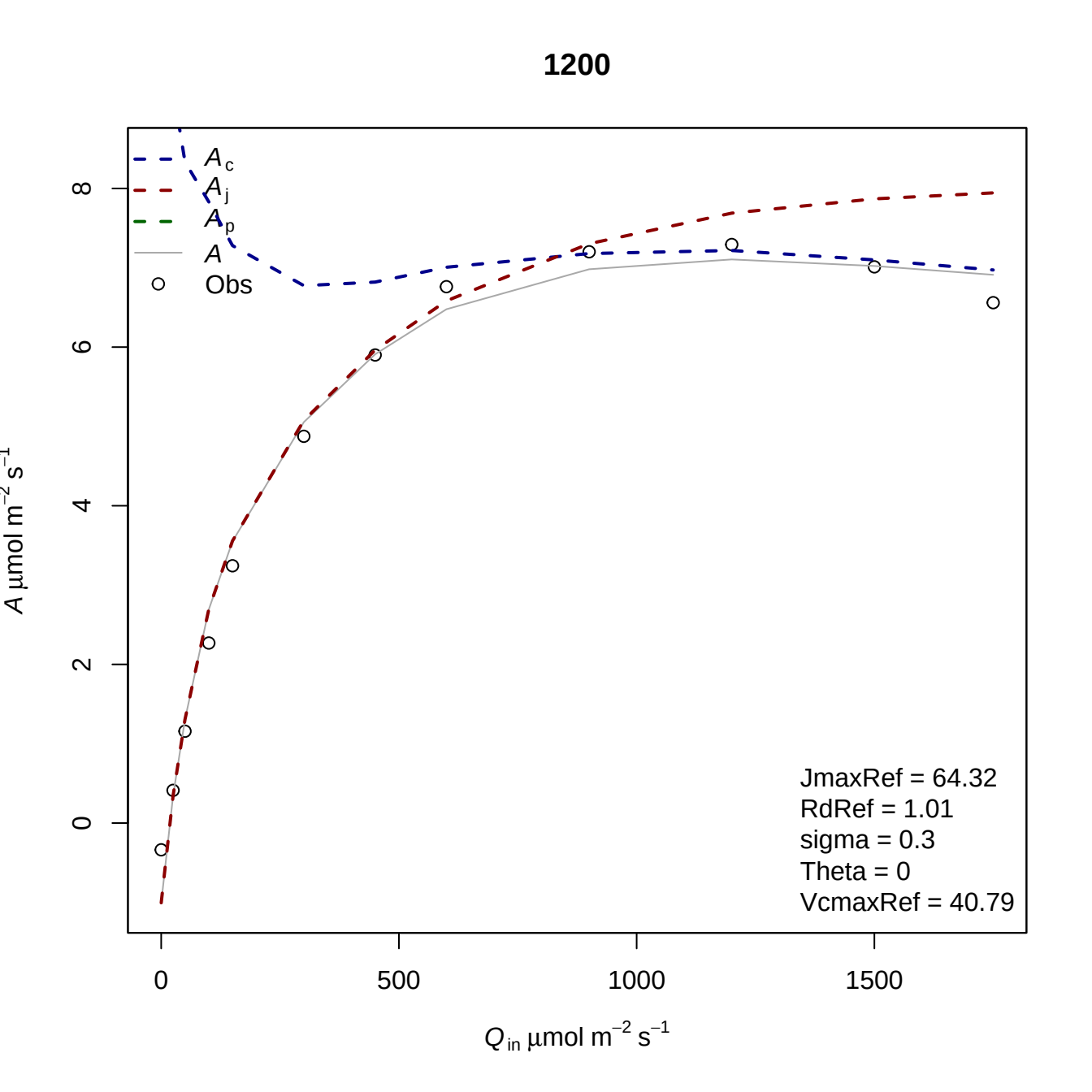

1195
A_c
A_j
A_p
A
Obs
JmaxRef = 129.02
RdRef = 0.58
sigma = 0.36
Theta = 0
VcmaxRef = 66.92
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

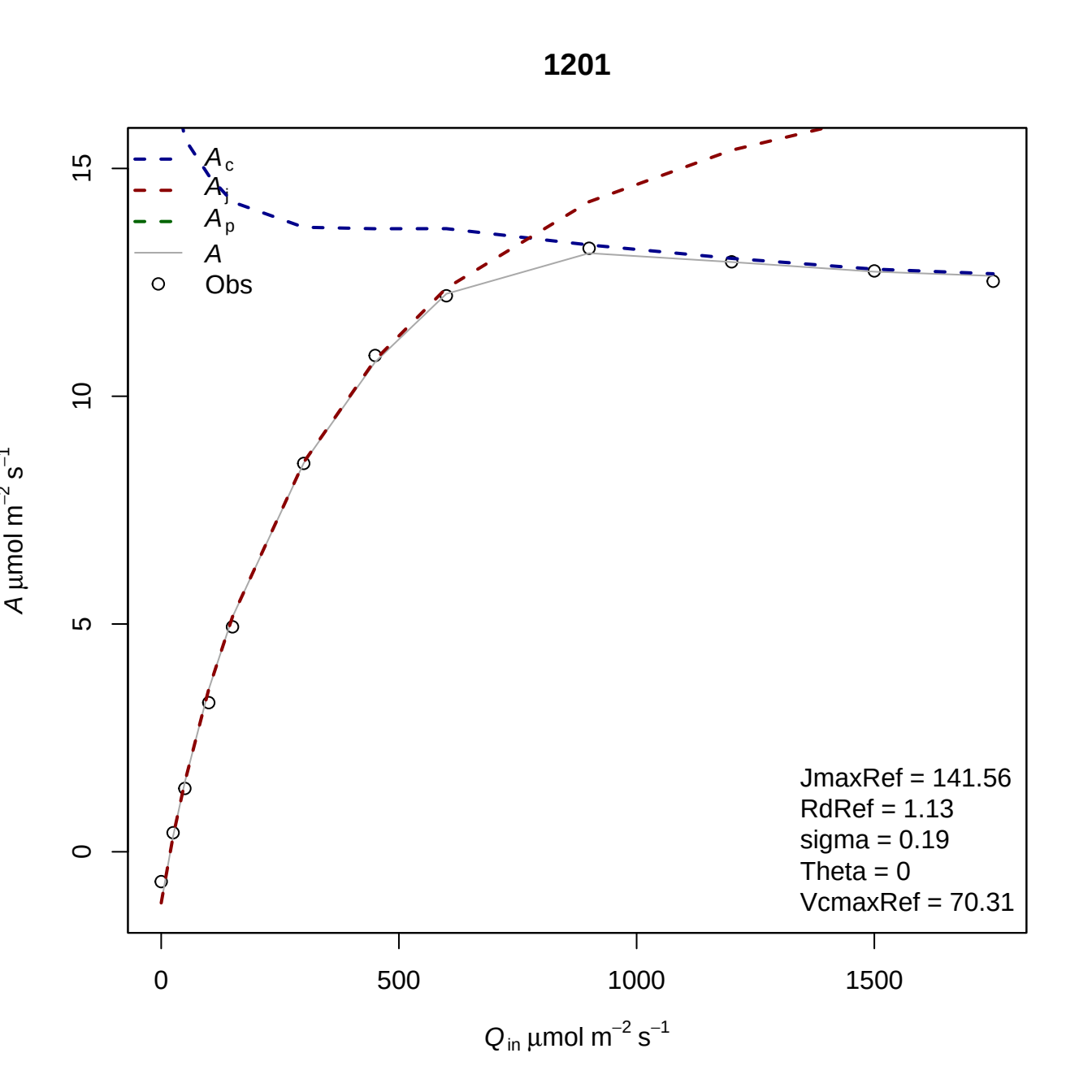

1196
A_c
A_j
A_p
A
Obs
JmaxRef = 183.87
RdRef = 0.84
sigma = 0.59
Theta = 0
VcmaxRef = 90.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

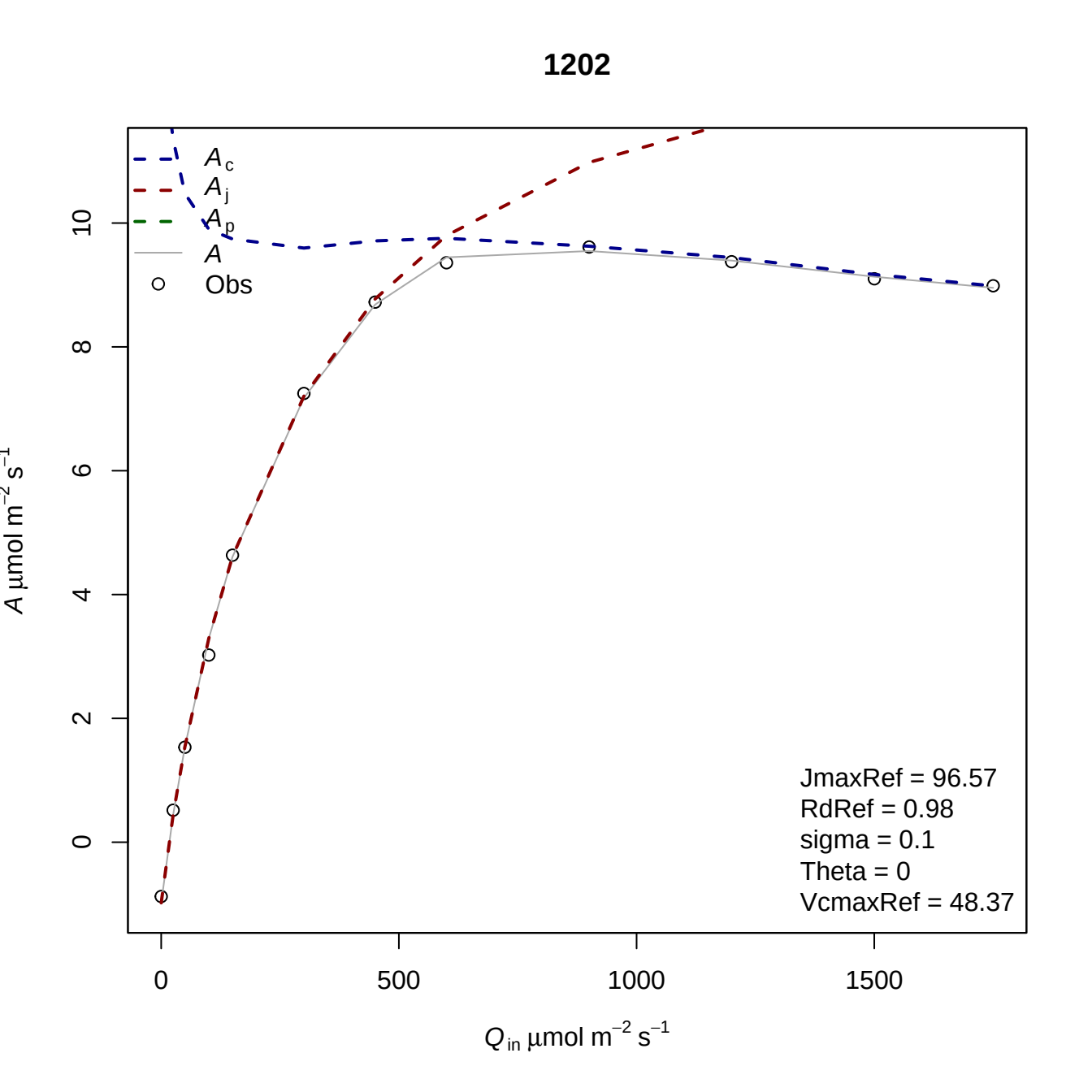

1197
A_c
A_j
A_p
A
Obs
JmaxRef = 223.19
RdRef = 0.81
sigma = 0.55
Theta = 0
VcmaxRef = 87.88
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

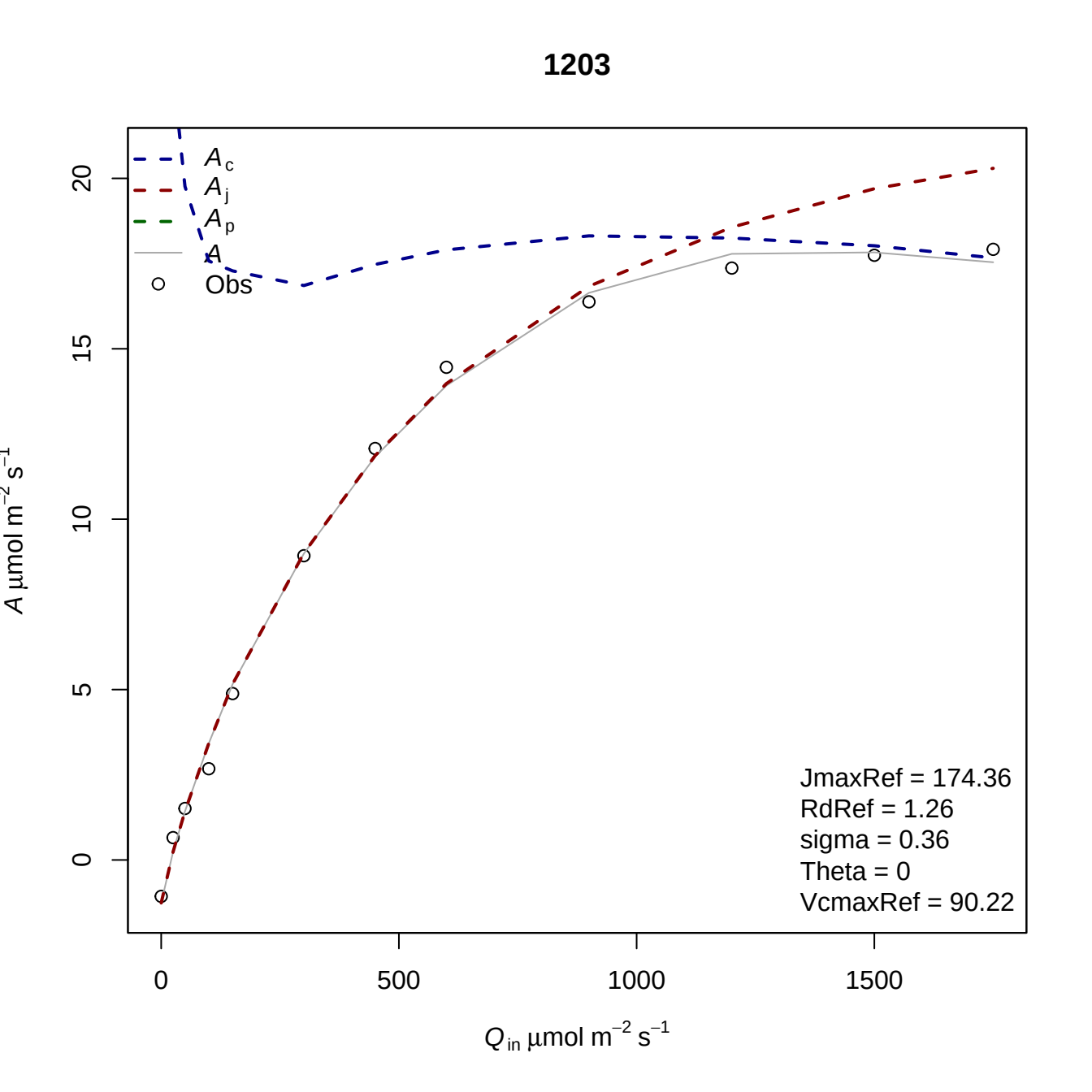

1198
A_c
A_j
A_p
A
Obs
JmaxRef = 216.49
RdRef = 0.92
sigma = 0.66
Theta = 0
VcmaxRef = 96.92
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

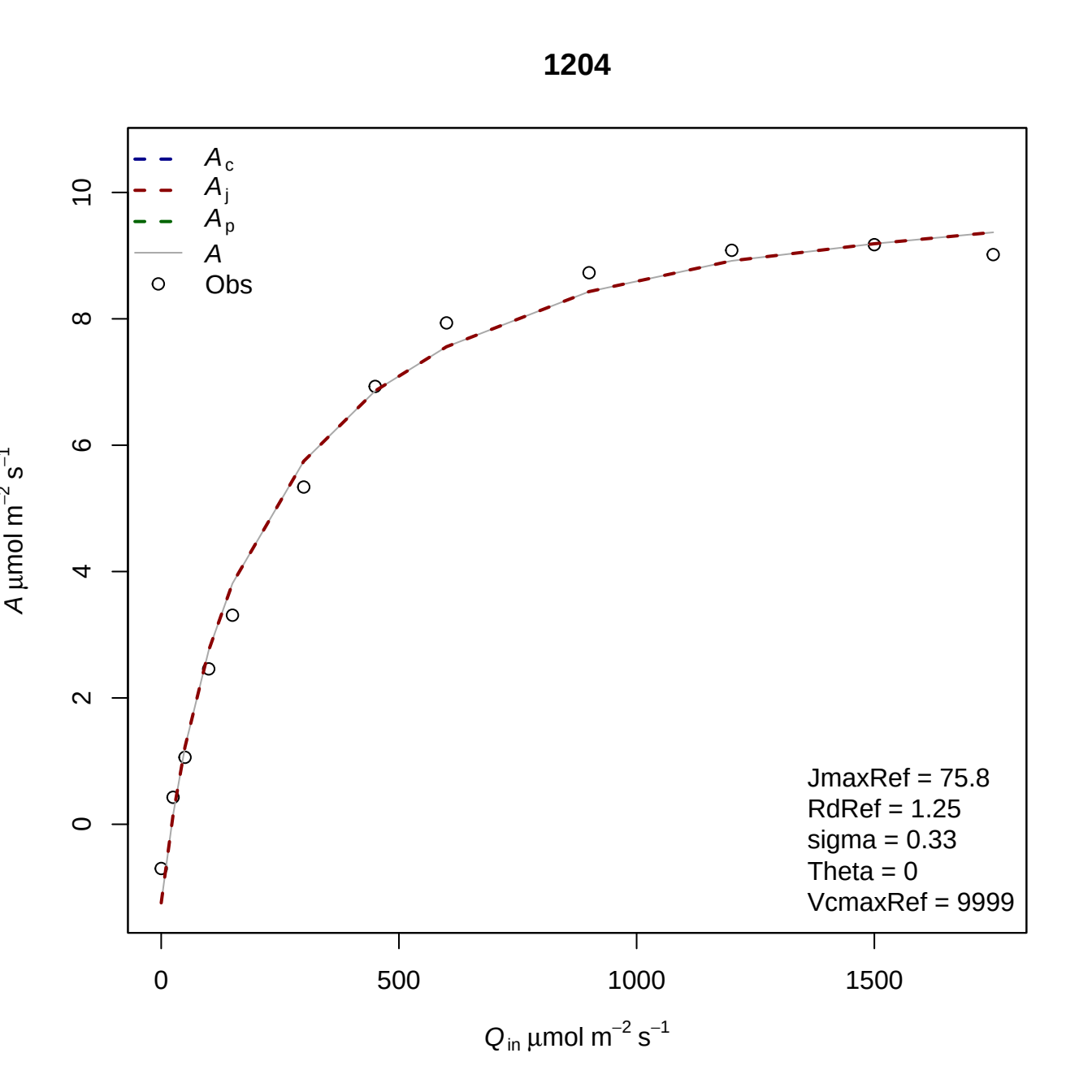

1199
A_c
A_j
A_p
A
Obs
JmaxRef = 82.54
RdRef = 1.66
sigma = 0.48
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

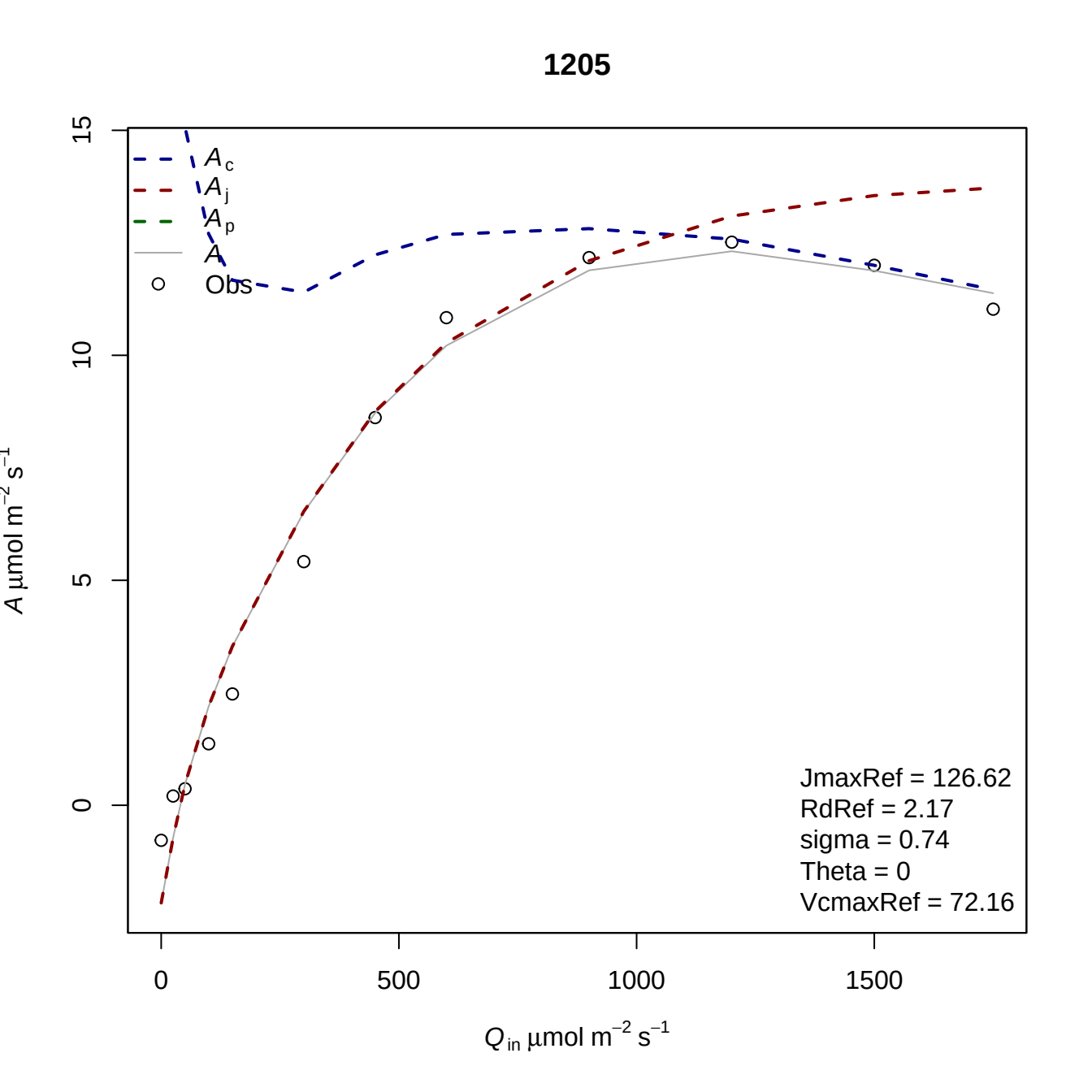

1200
A_c
A_j
A_p
A
Obs
JmaxRef = 64.32
RdRef = 1.01
sigma = 0.3
Theta = 0
VcmaxRef = 40.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

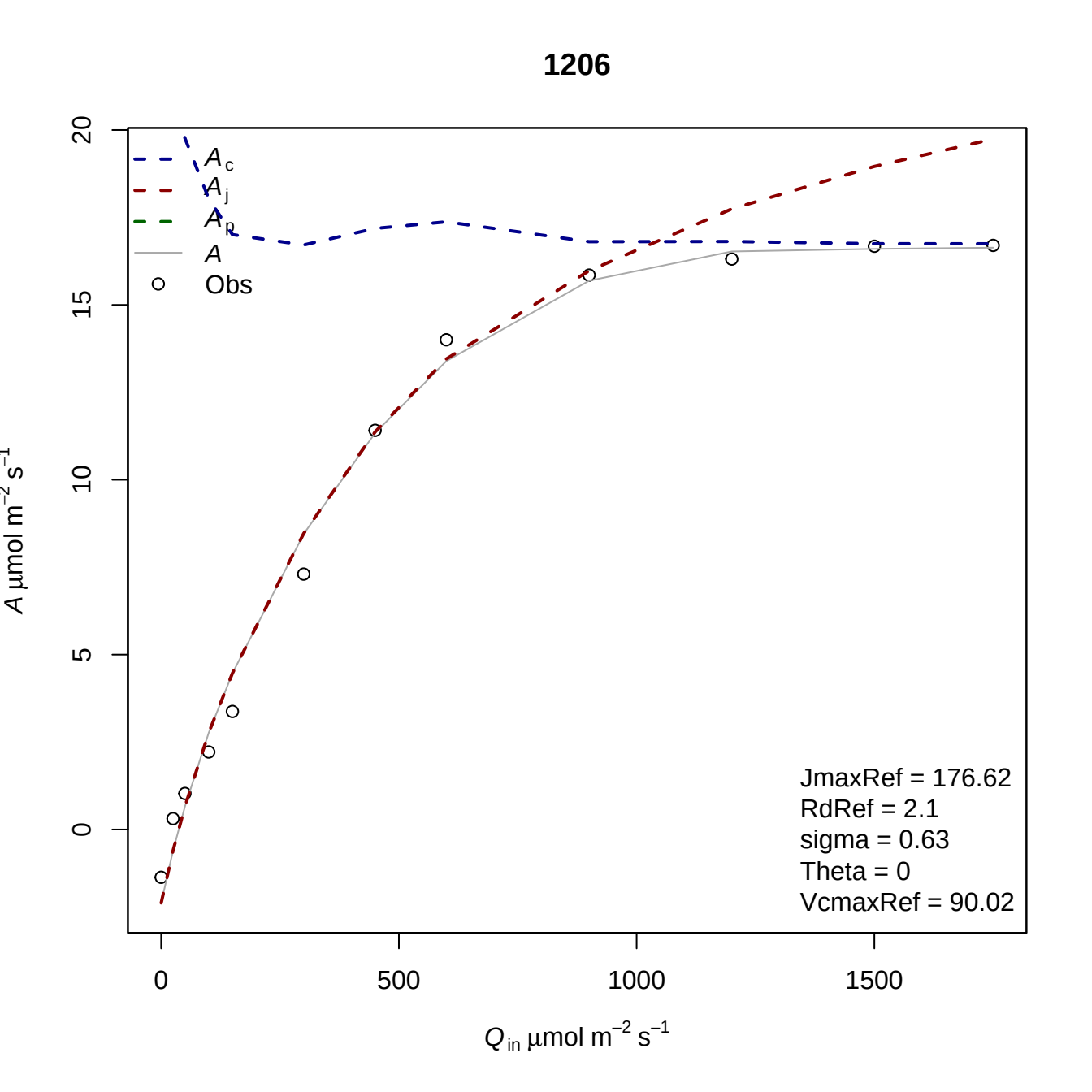

1201
A_c
A_j
A_p
A
Obs
JmaxRef = 141.56
RdRef = 1.13
sigma = 0.19
Theta = 0
VcmaxRef = 70.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

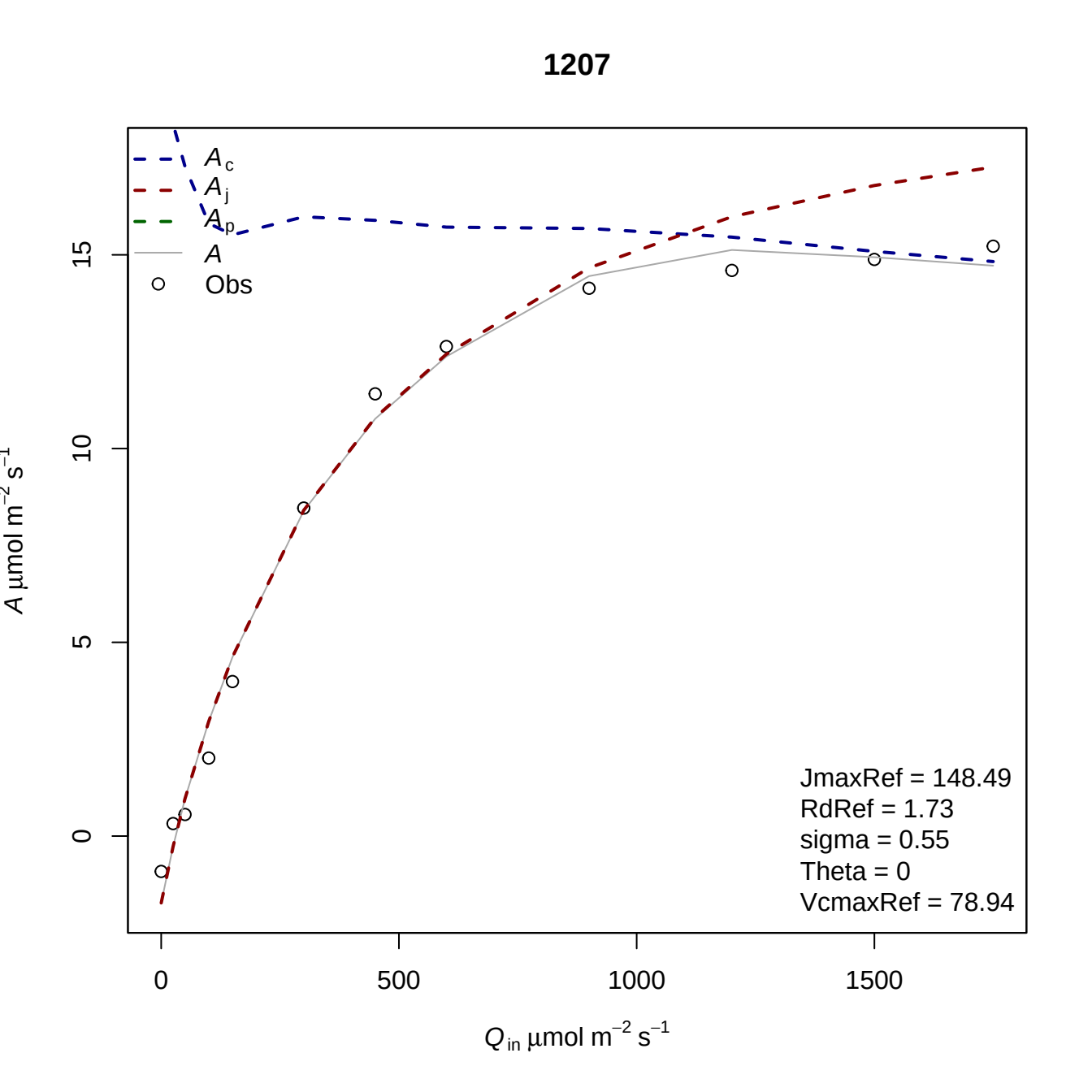

1202
A_c
A_j
A_p
A
Obs
JmaxRef = 96.57
RdRef = 0.98
sigma = 0.1
Theta = 0
VcmaxRef = 48.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

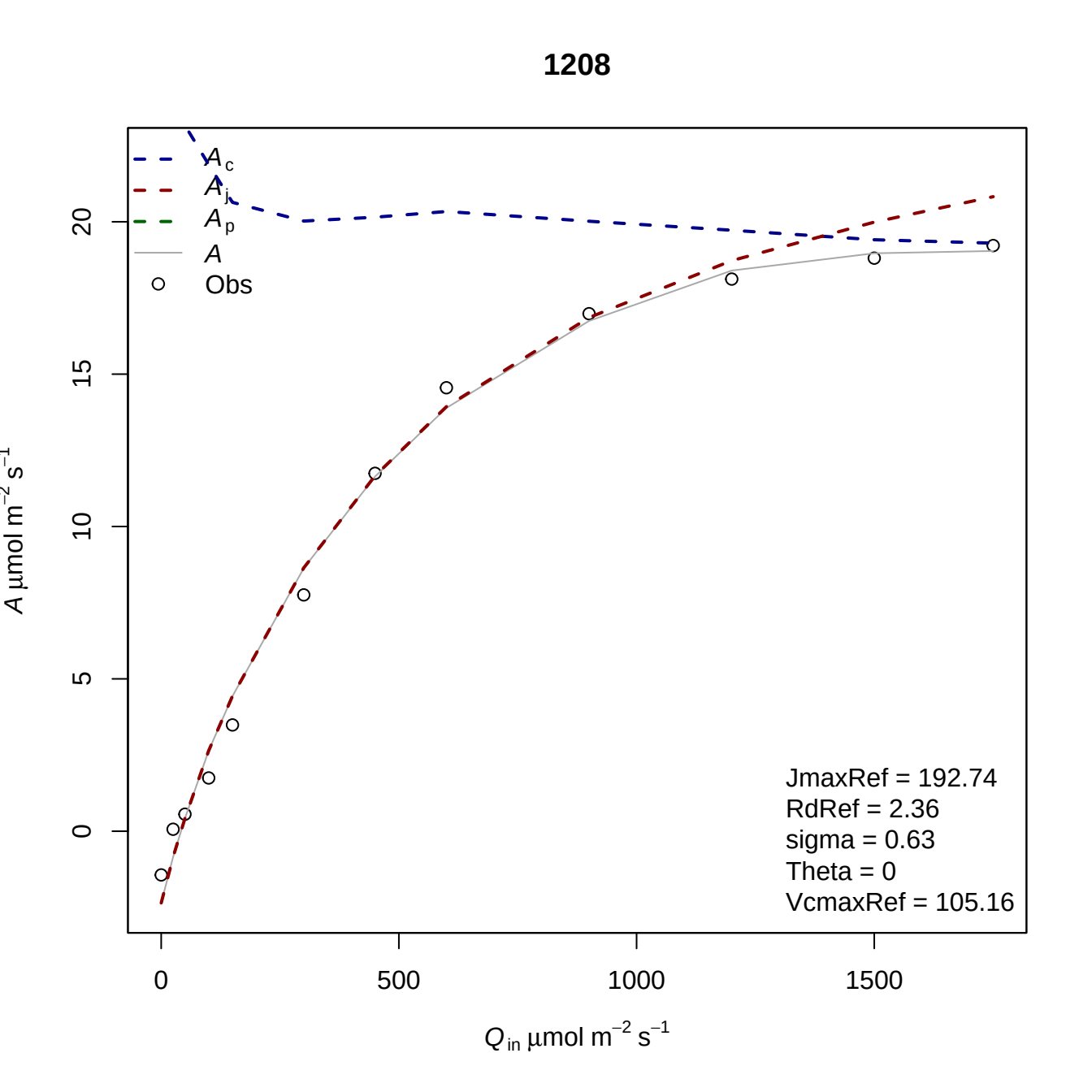

1203
A_c
A_j
A_p
A
Obs
JmaxRef = 174.36
RdRef = 1.26
sigma = 0.36
Theta = 0
VcmaxRef = 90.22
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

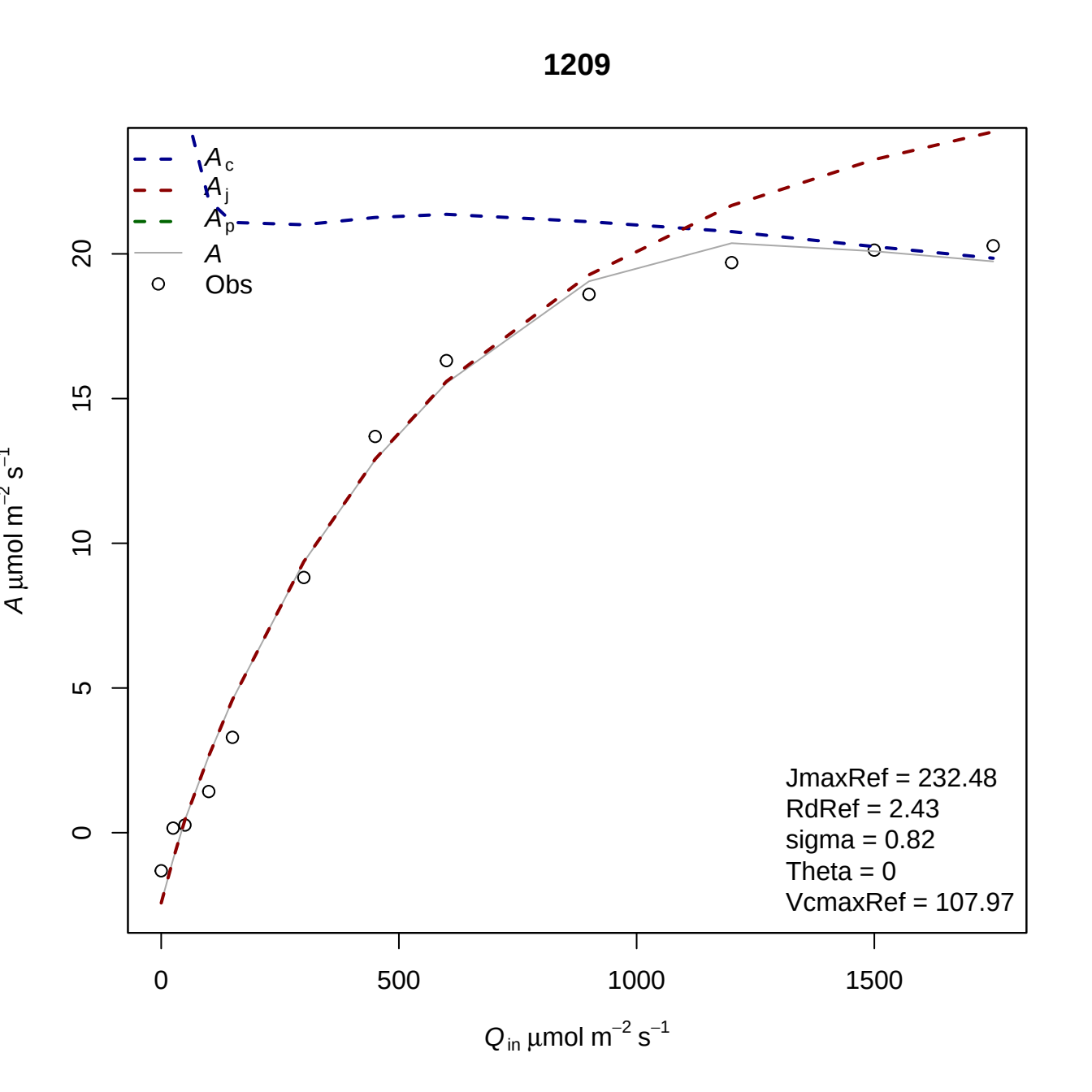

1204
A_c
A_j
A_p
A
Obs
JmaxRef = 75.8
RdRef = 1.25
sigma = 0.33
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

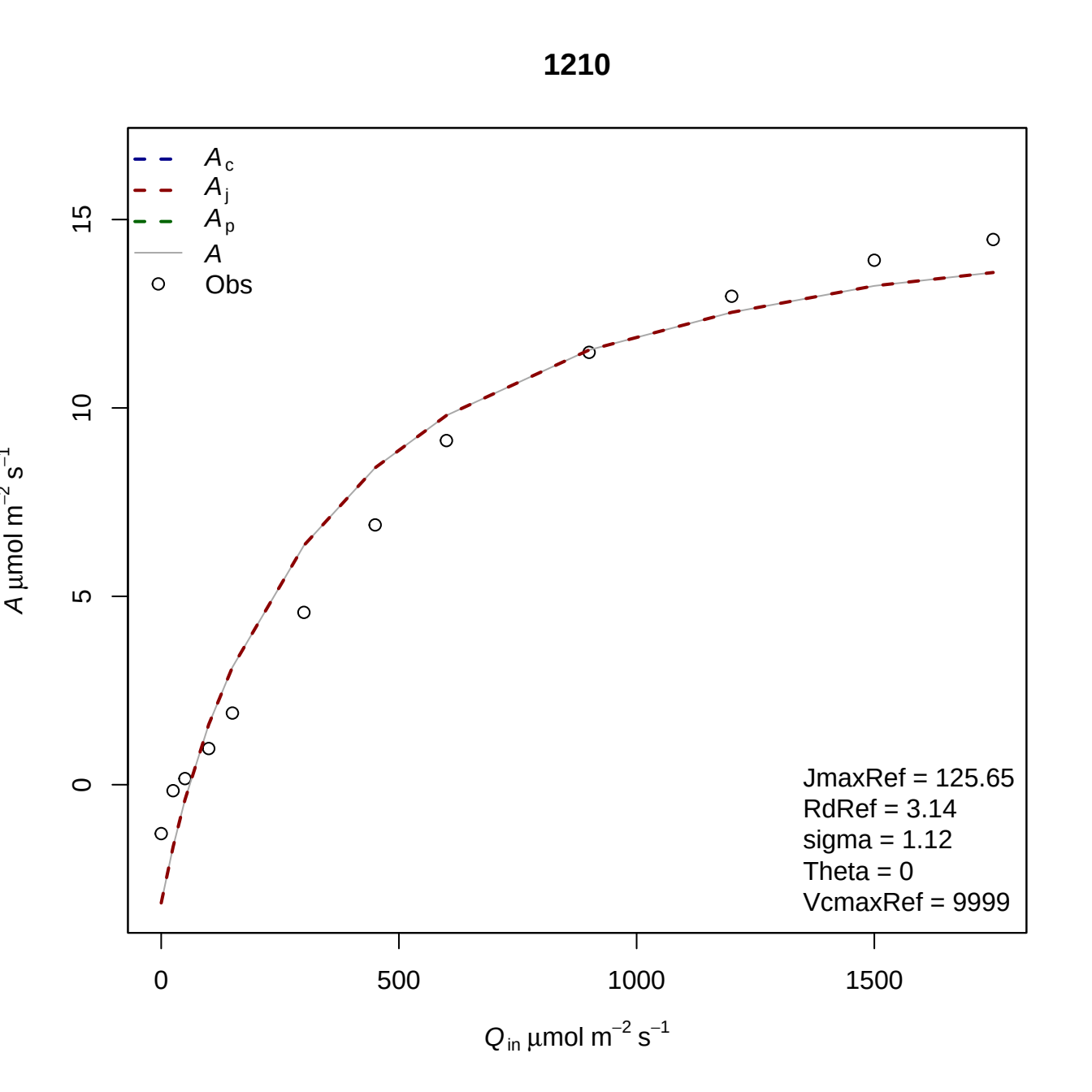

1205
A_c
A_j
A_p
A
Obs
JmaxRef = 126.62
RdRef = 2.17
sigma = 0.74
Theta = 0
VcmaxRef = 72.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

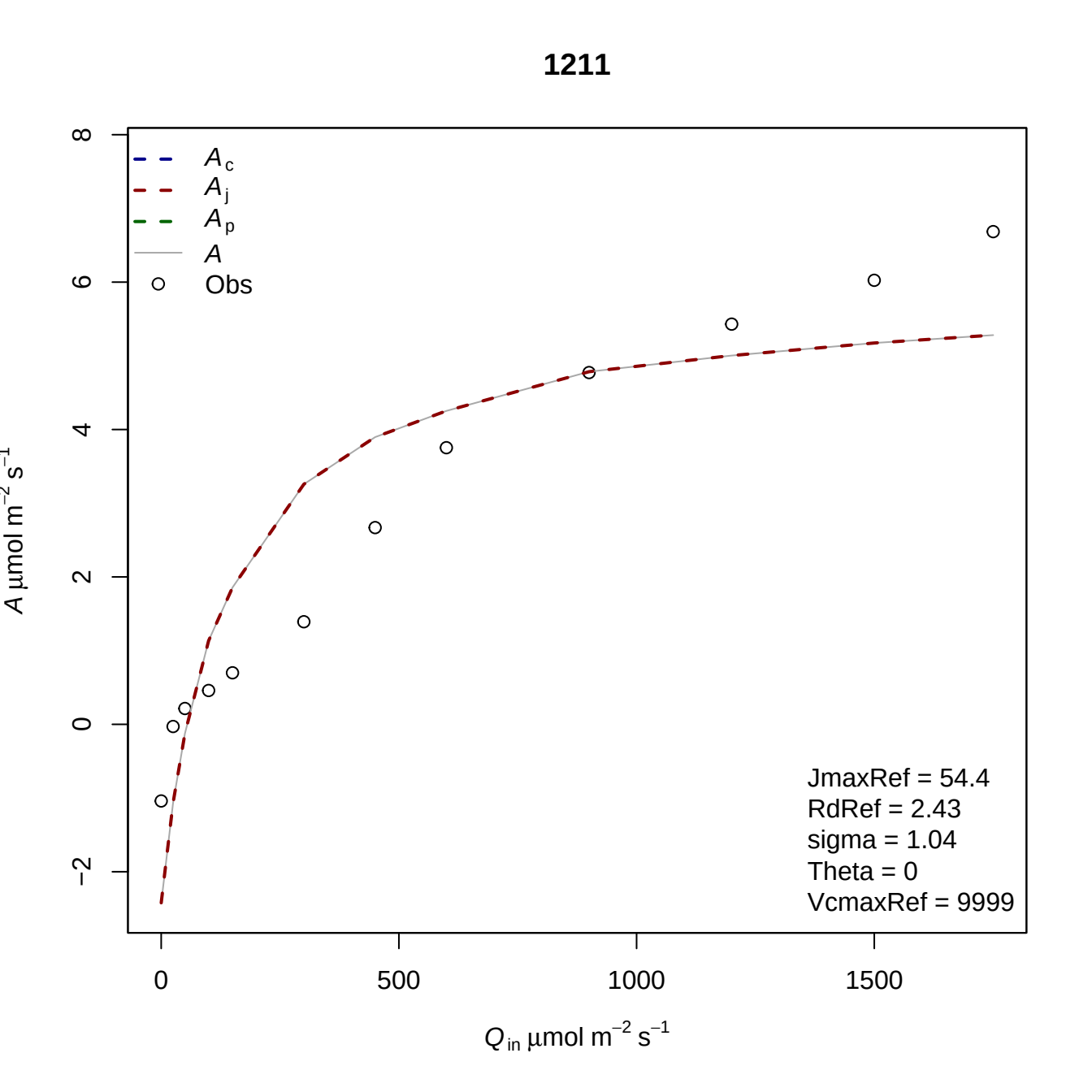

1206
A_c
A_j
A_p
A
Obs
JmaxRef = 176.62
RdRef = 2.1
sigma = 0.63
Theta = 0
VcmaxRef = 90.02
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

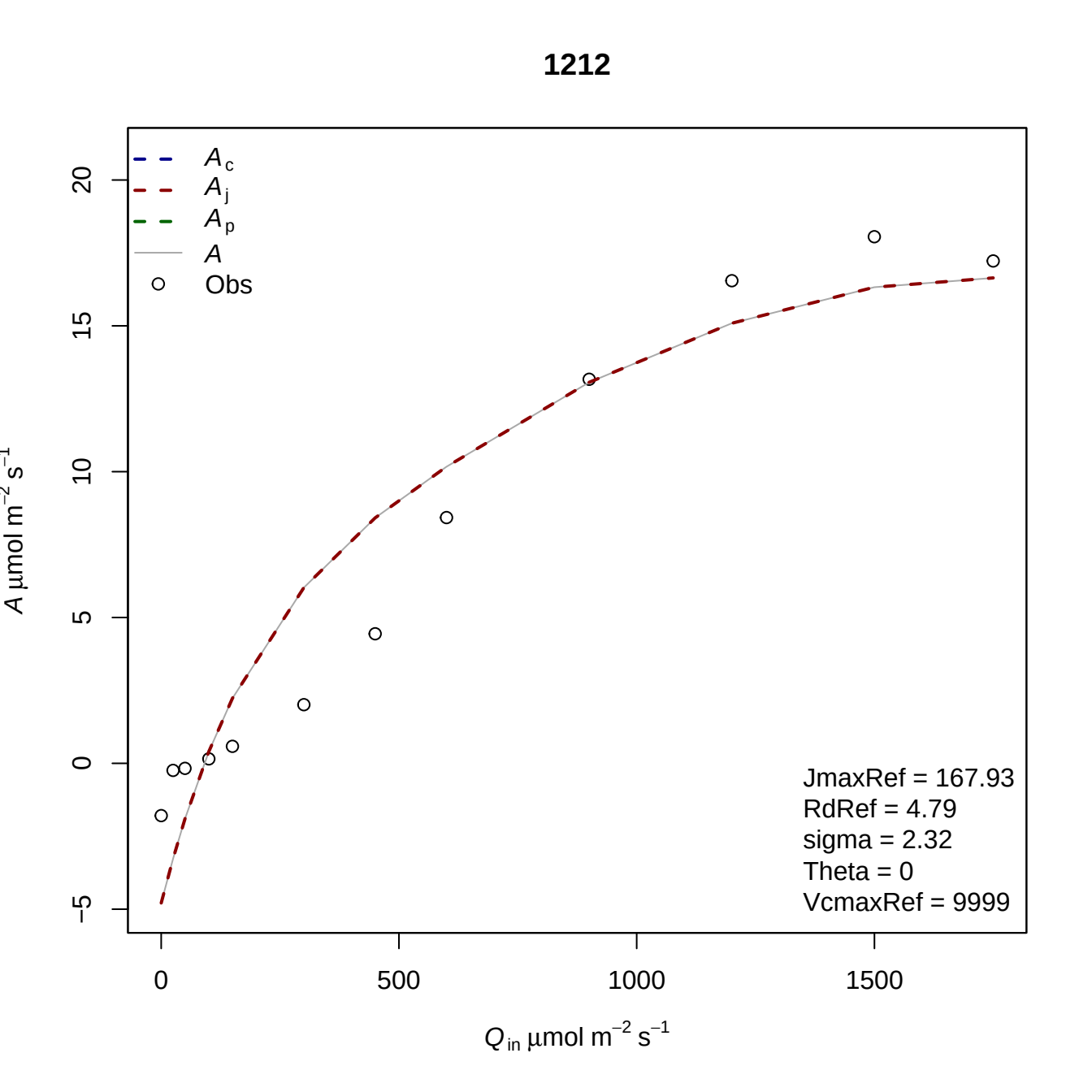

1207
A_c
A_j
A_p
A
Obs
JmaxRef = 148.49
RdRef = 1.73
sigma = 0.55
Theta = 0
VcmaxRef = 78.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

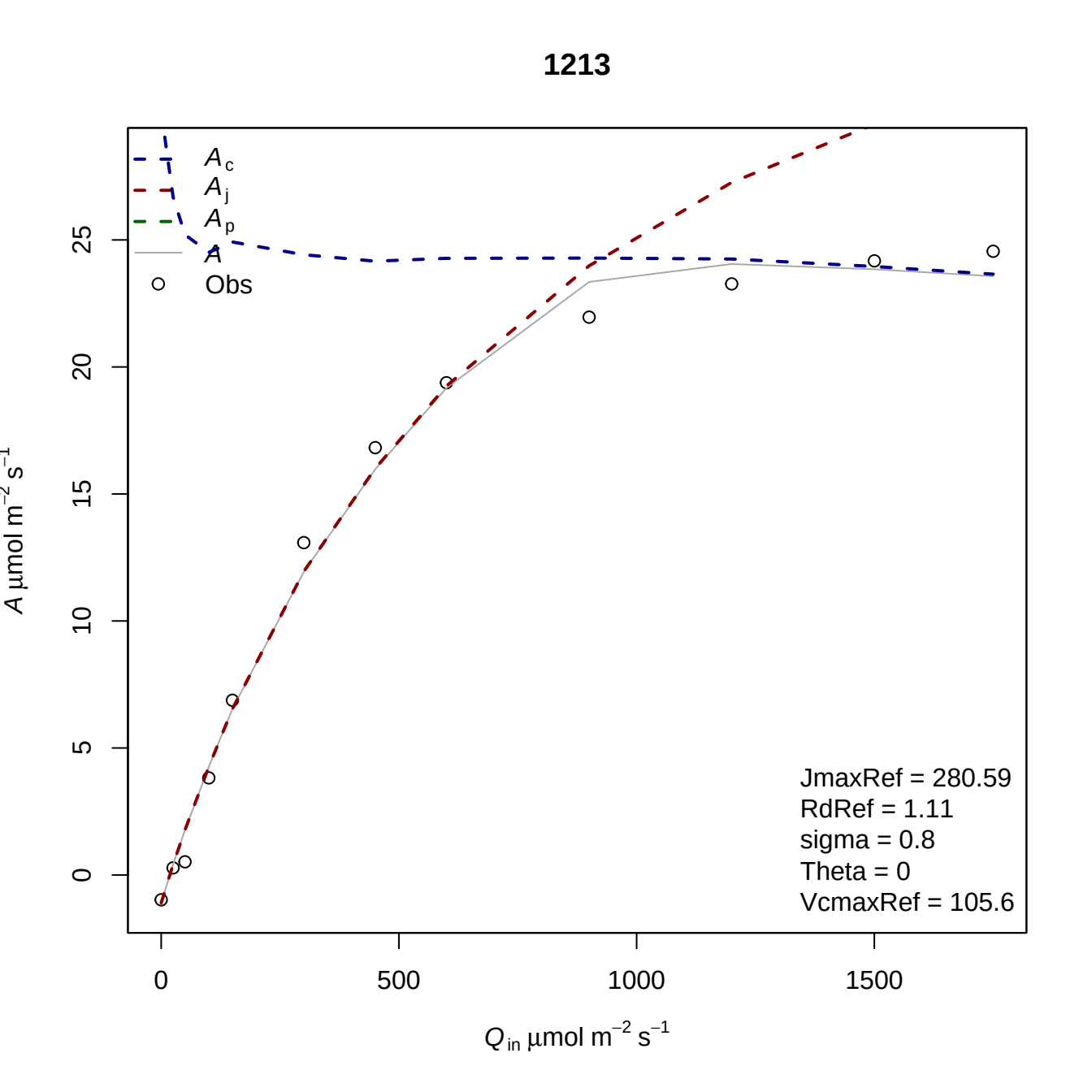

1208
A_c
A_j
A_p
A
Obs
JmaxRef = 192.74
RdRef = 2.36
sigma = 0.63
Theta = 0
VcmaxRef = 105.16
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

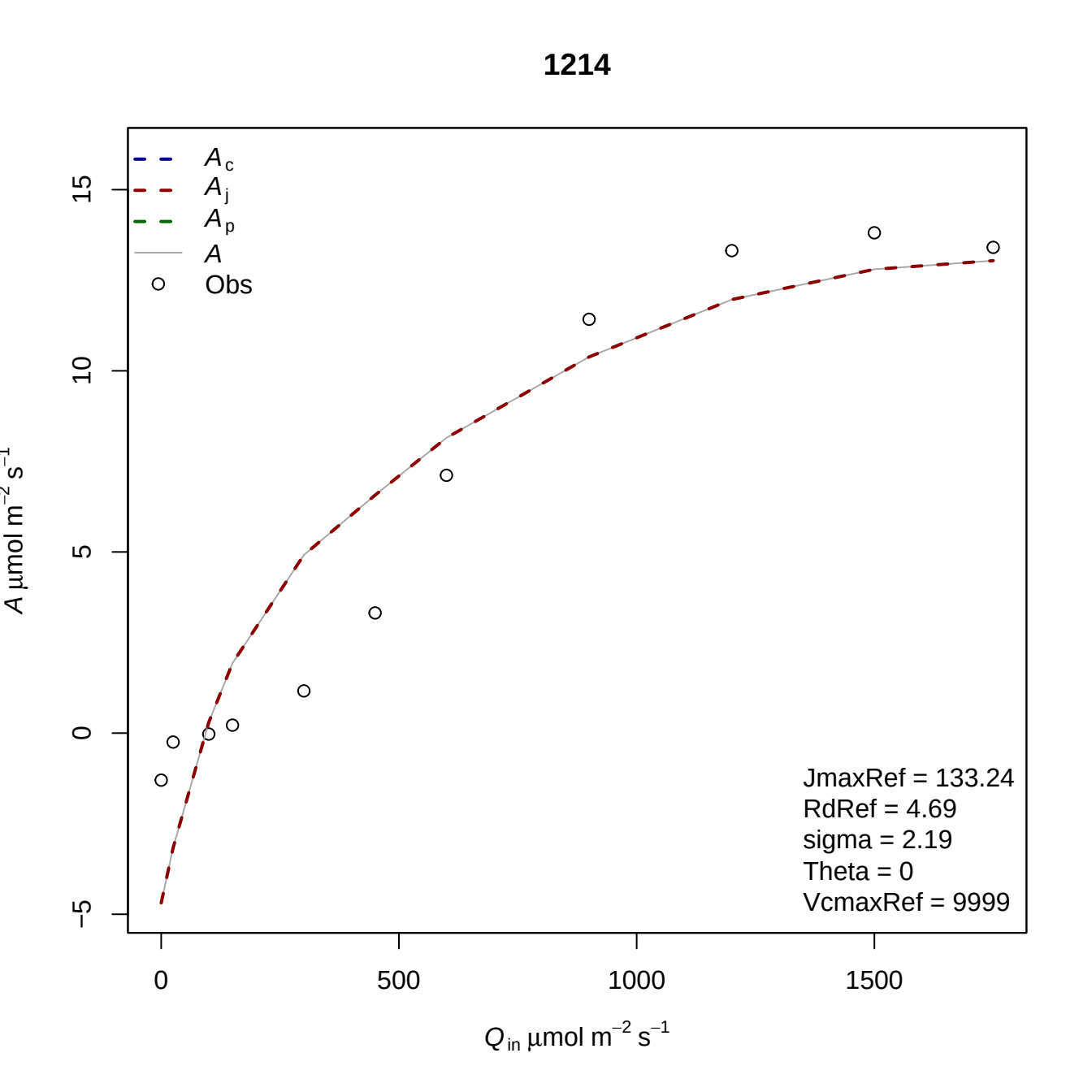

1209
A_c
A_j
A_p
A
Obs
JmaxRef = 232.48
RdRef = 2.43
sigma = 0.82
Theta = 0
VcmaxRef = 107.97
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

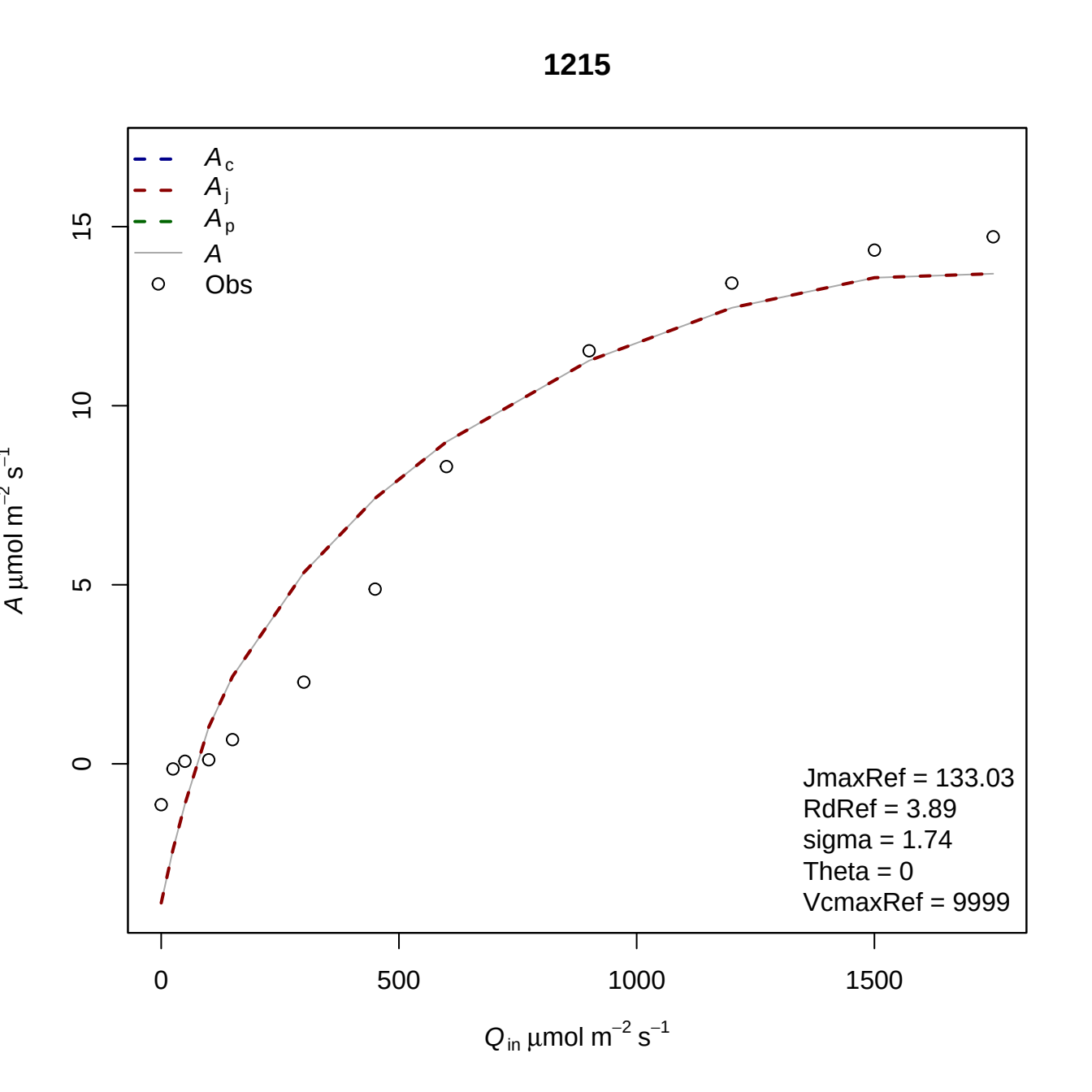

1210
A_c
A_j
A_p
A
Obs
JmaxRef = 125.65
RdRef = 3.14
sigma = 1.12
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

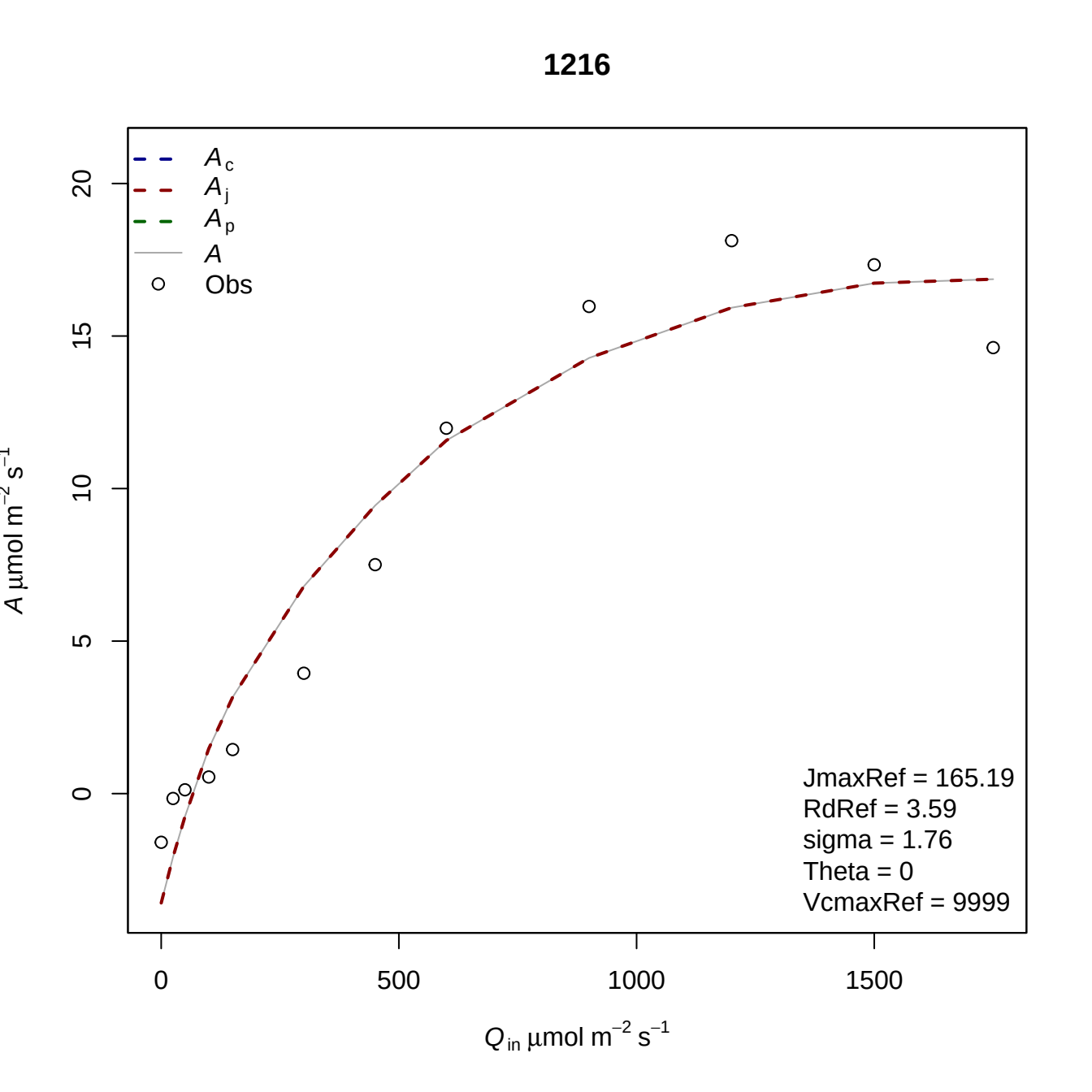

1211
A_c
A_j
A_p
A
Obs
JmaxRef = 54.4
RdRef = 2.43
sigma = 1.04
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

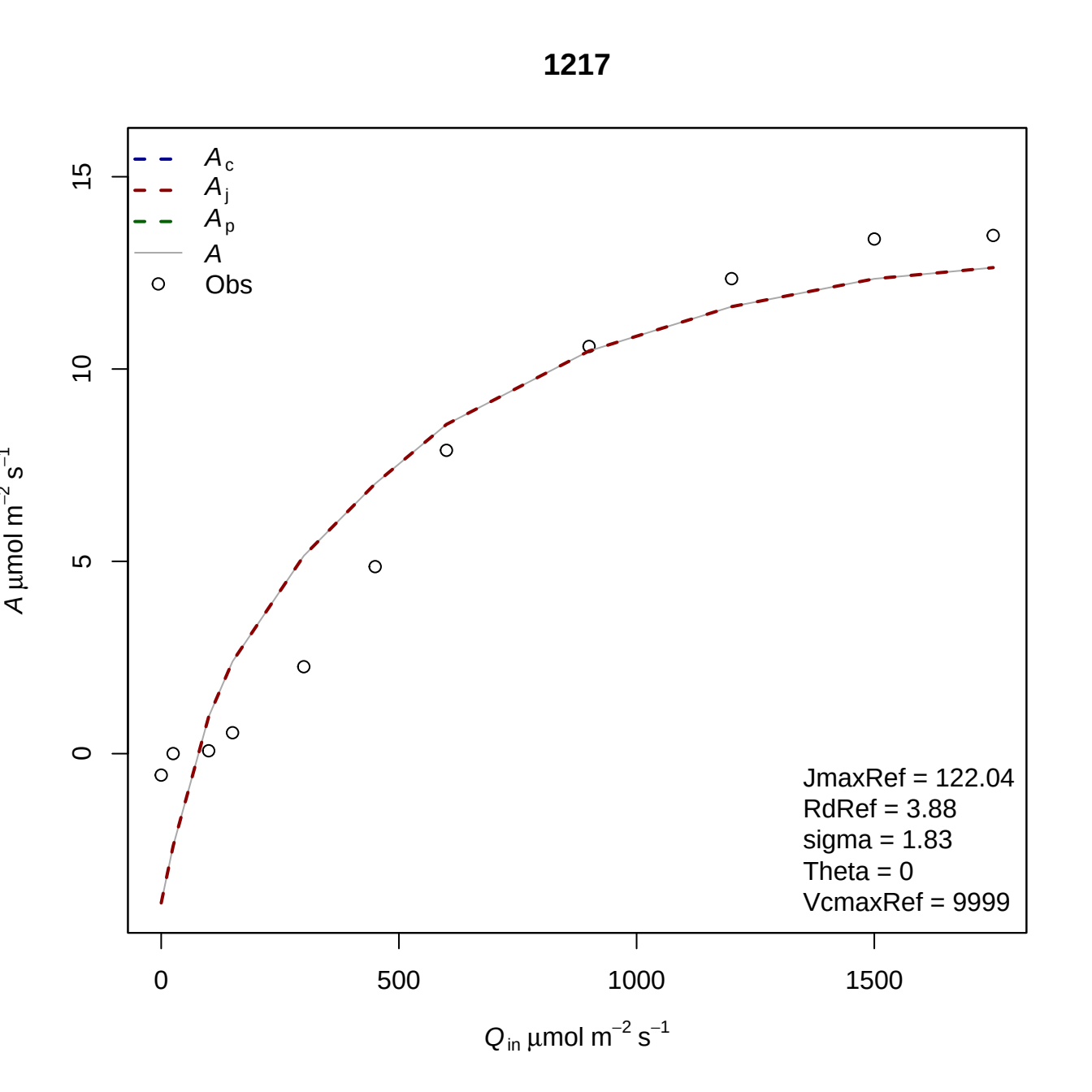

1212
A_c
A_j
A_p
A
Obs
JmaxRef = 167.93
RdRef = 4.79
sigma = 2.32
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500

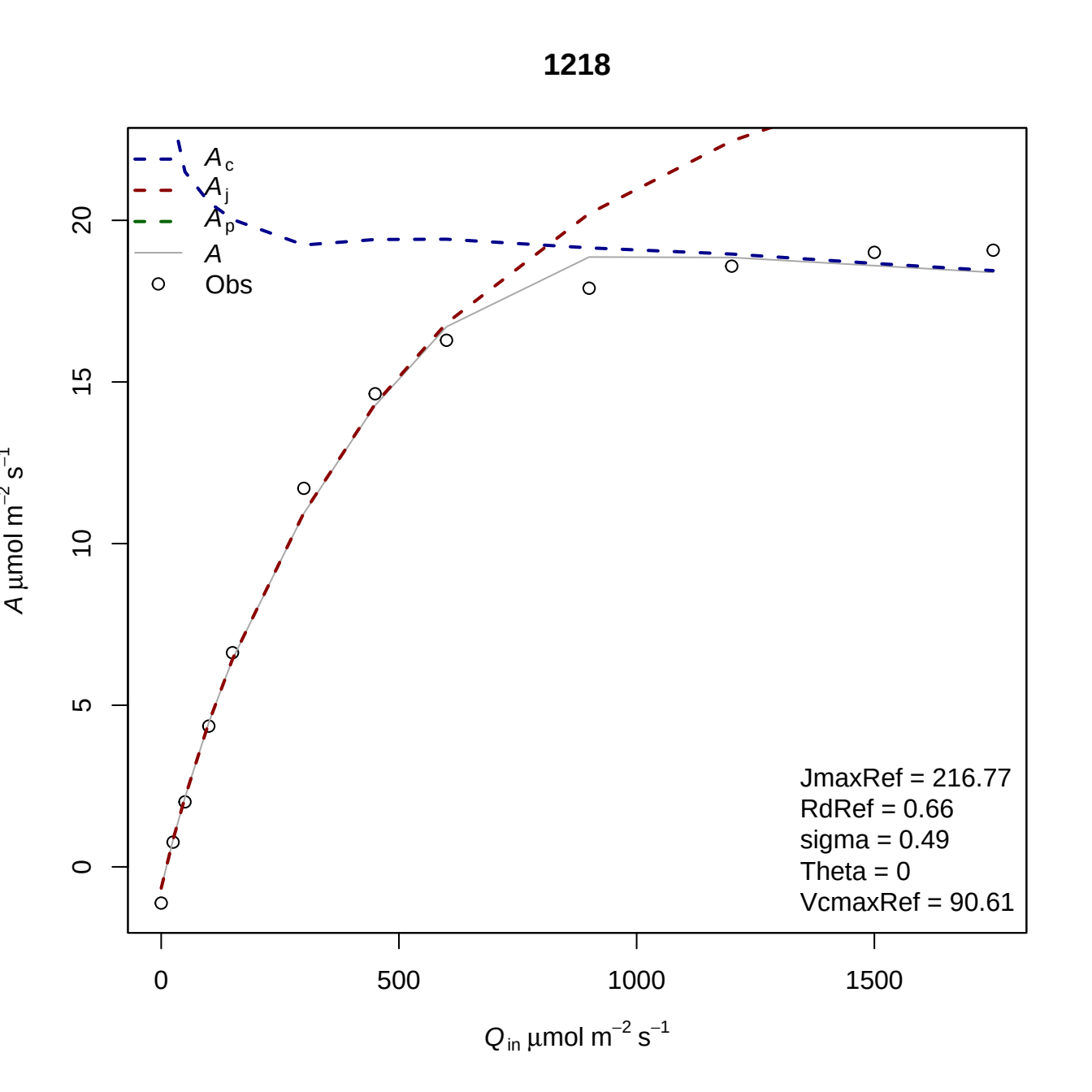

1213
A_c
A_j
A_p
A
Obs
JmaxRef = 280.59
RdRef = 1.11
sigma = 0.8
Theta = 0
VcmaxRef = 105.6
0
5
10
15
20
25
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

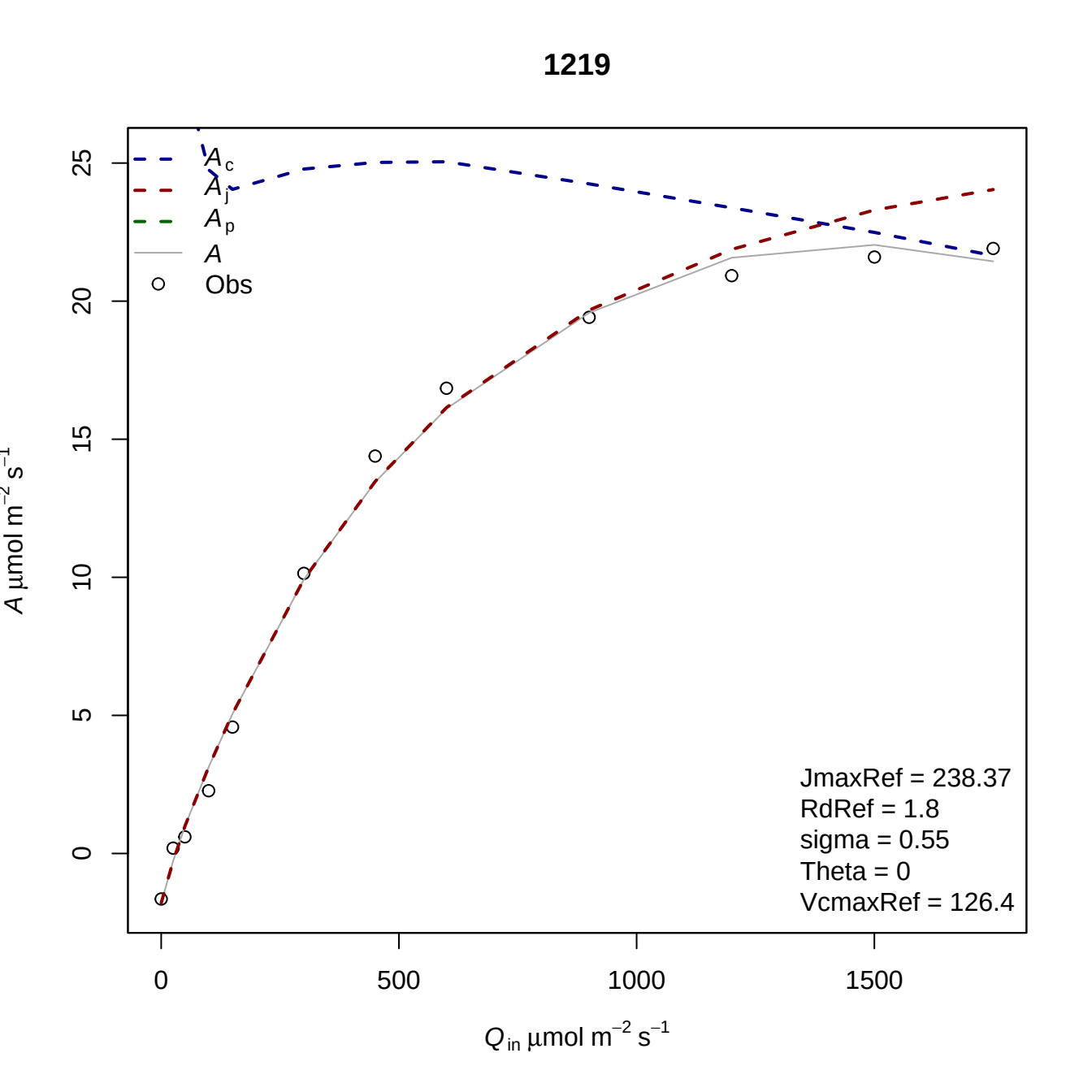

1214
A_c
A_j
A_p
A
Obs
JmaxRef = 133.24
RdRef = 4.69
sigma = 2.19
Theta = 0
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

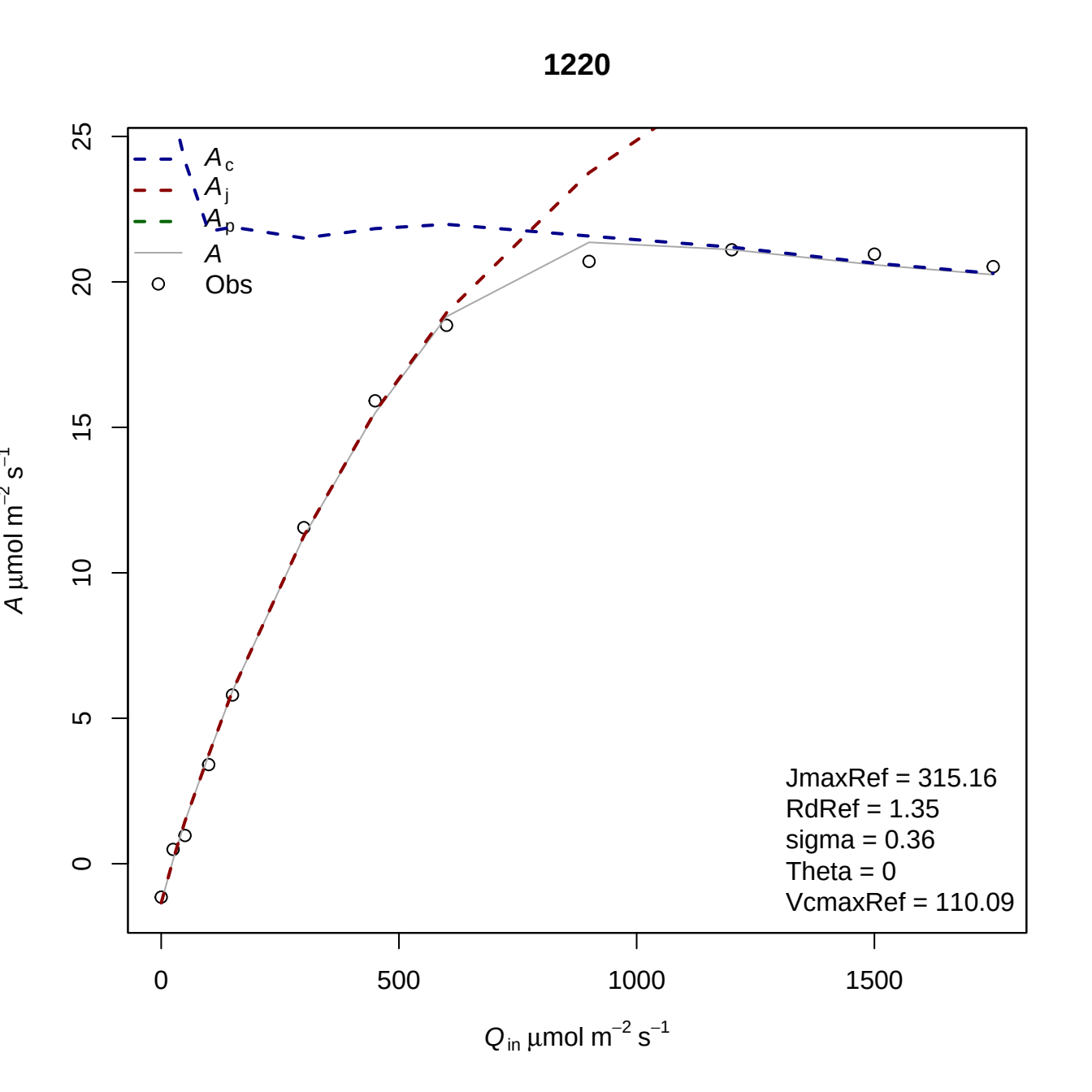

1215
A_c
A_j
A_p
A
Obs
JmaxRef = 133.03
RdRef = 3.89
sigma = 1.74
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

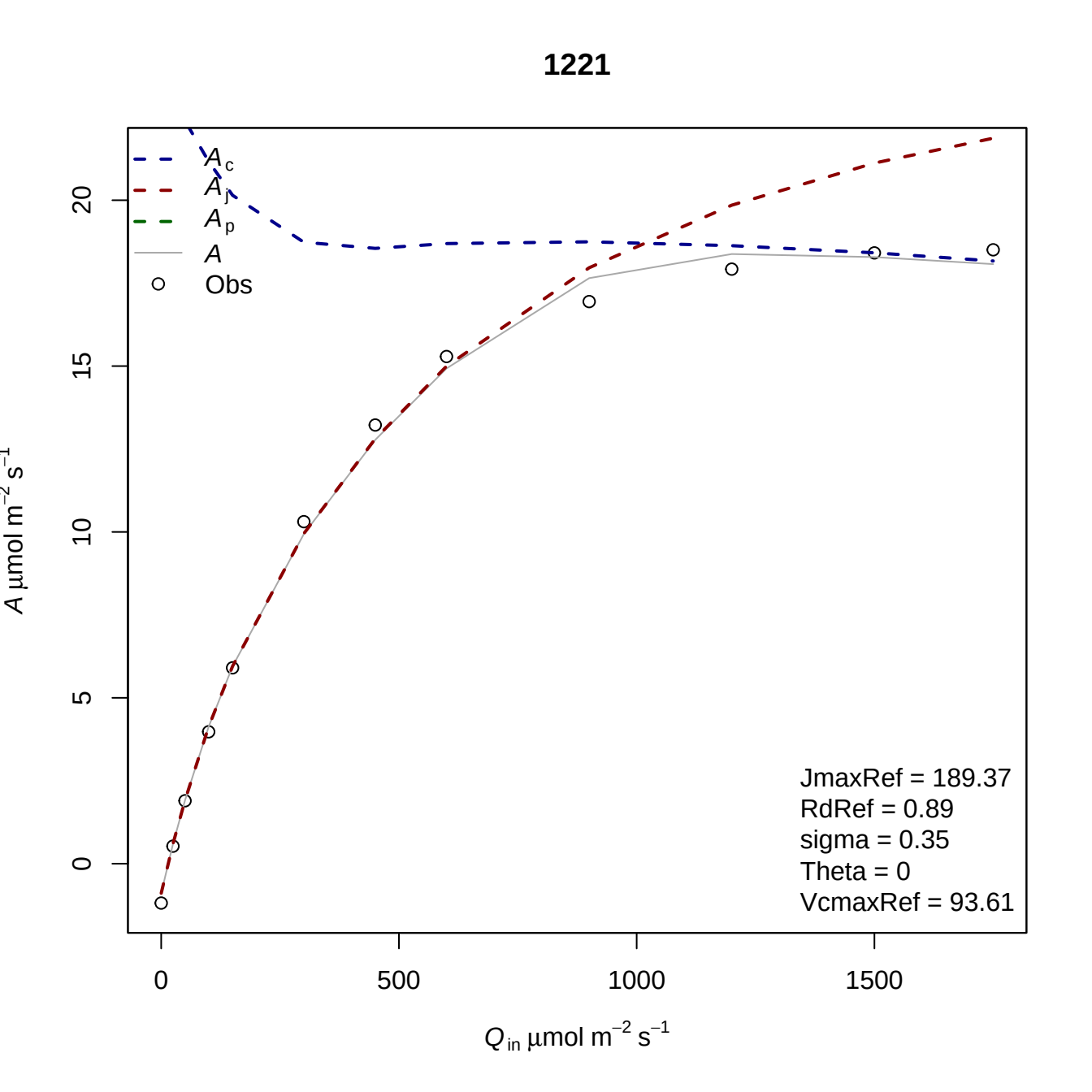

1216
A_c
A_j
A_p
A
Obs
JmaxRef = 165.19
RdRef = 3.59
sigma = 1.76
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

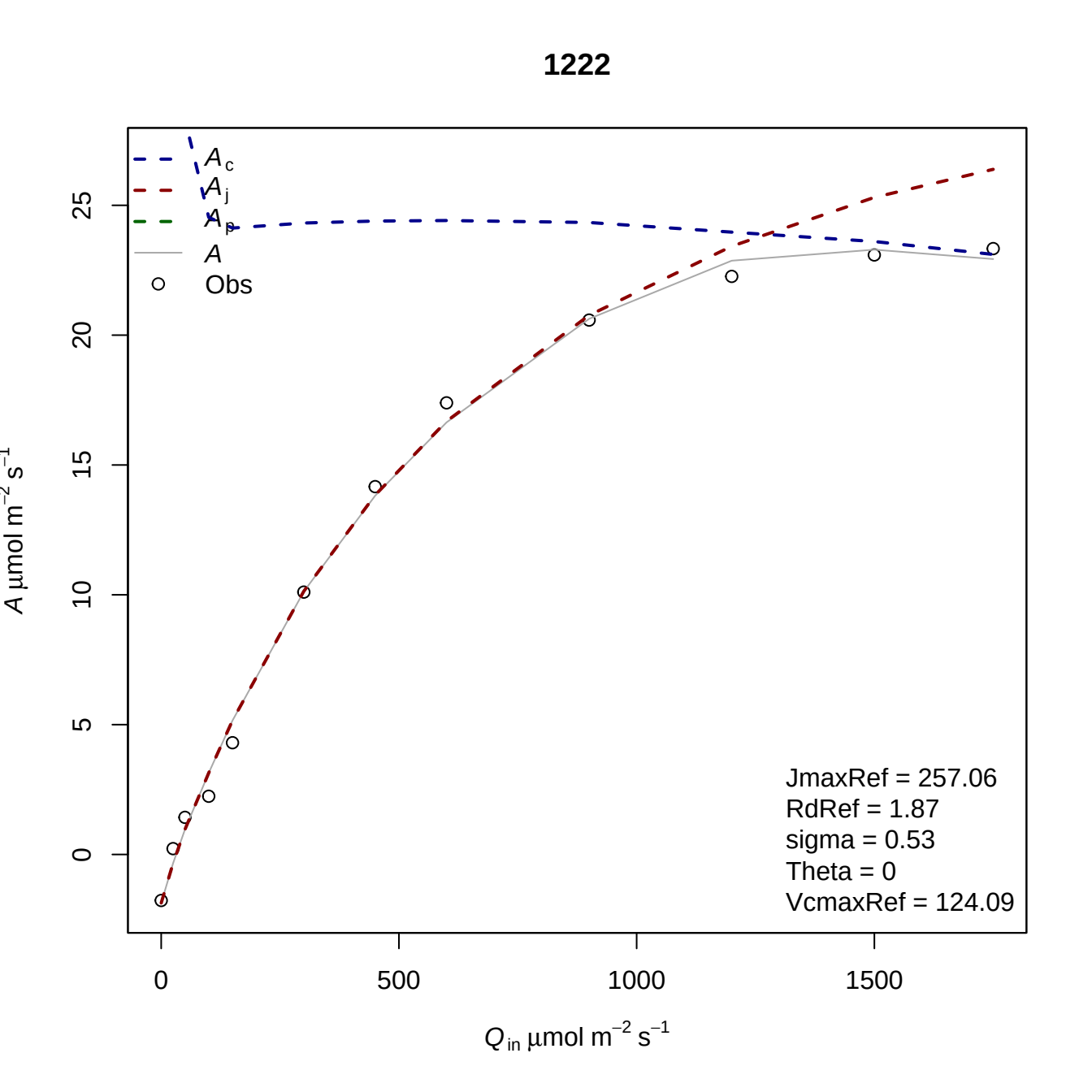

1217
A_c
A_j
A_p
A
Obs
JmaxRef = 122.04
RdRef = 3.88
sigma = 1.83
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

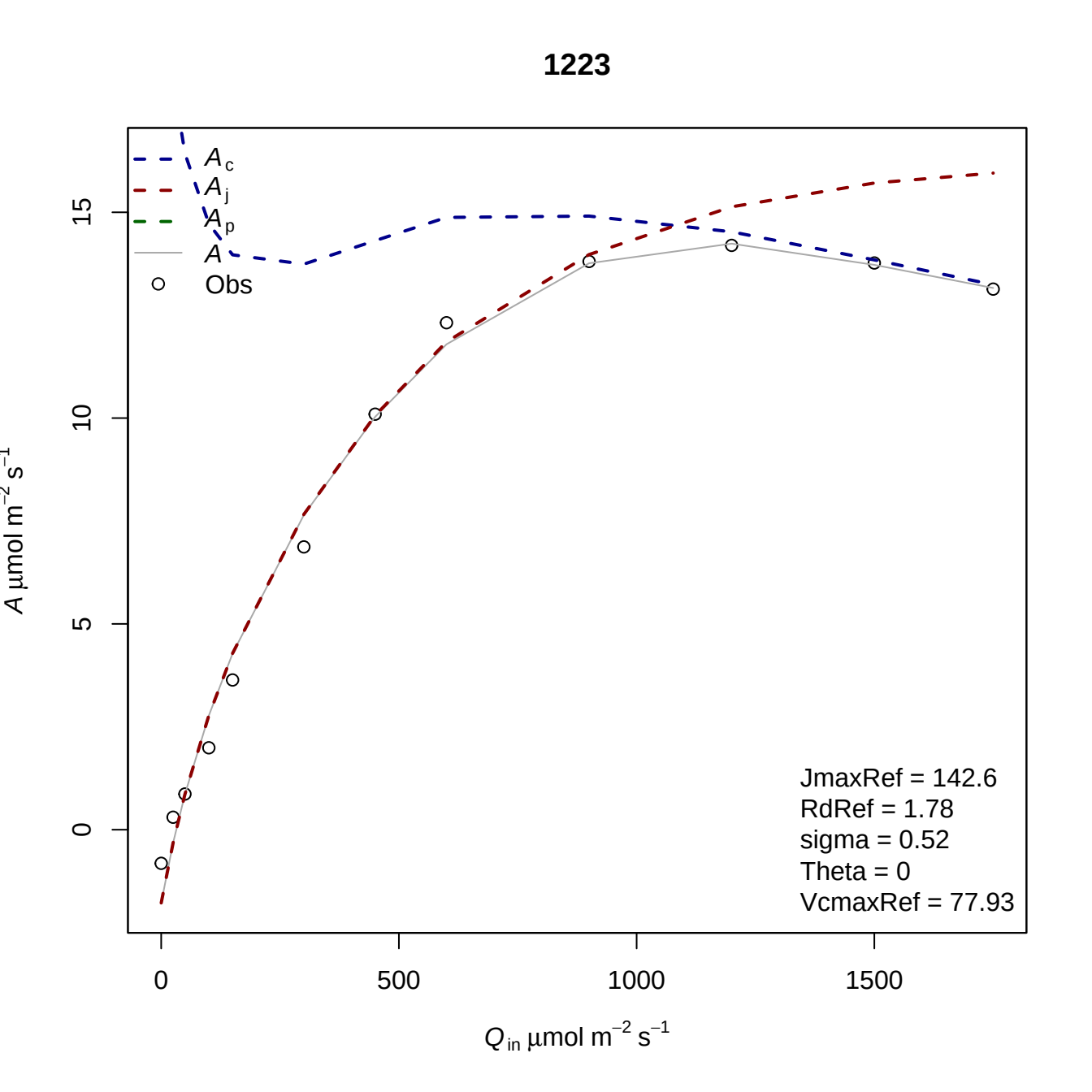

1218
A_c
A_j
A_p
A
Obs
JmaxRef = 216.77
RdRef = 0.66
sigma = 0.49
Theta = 0
VcmaxRef = 90.61
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

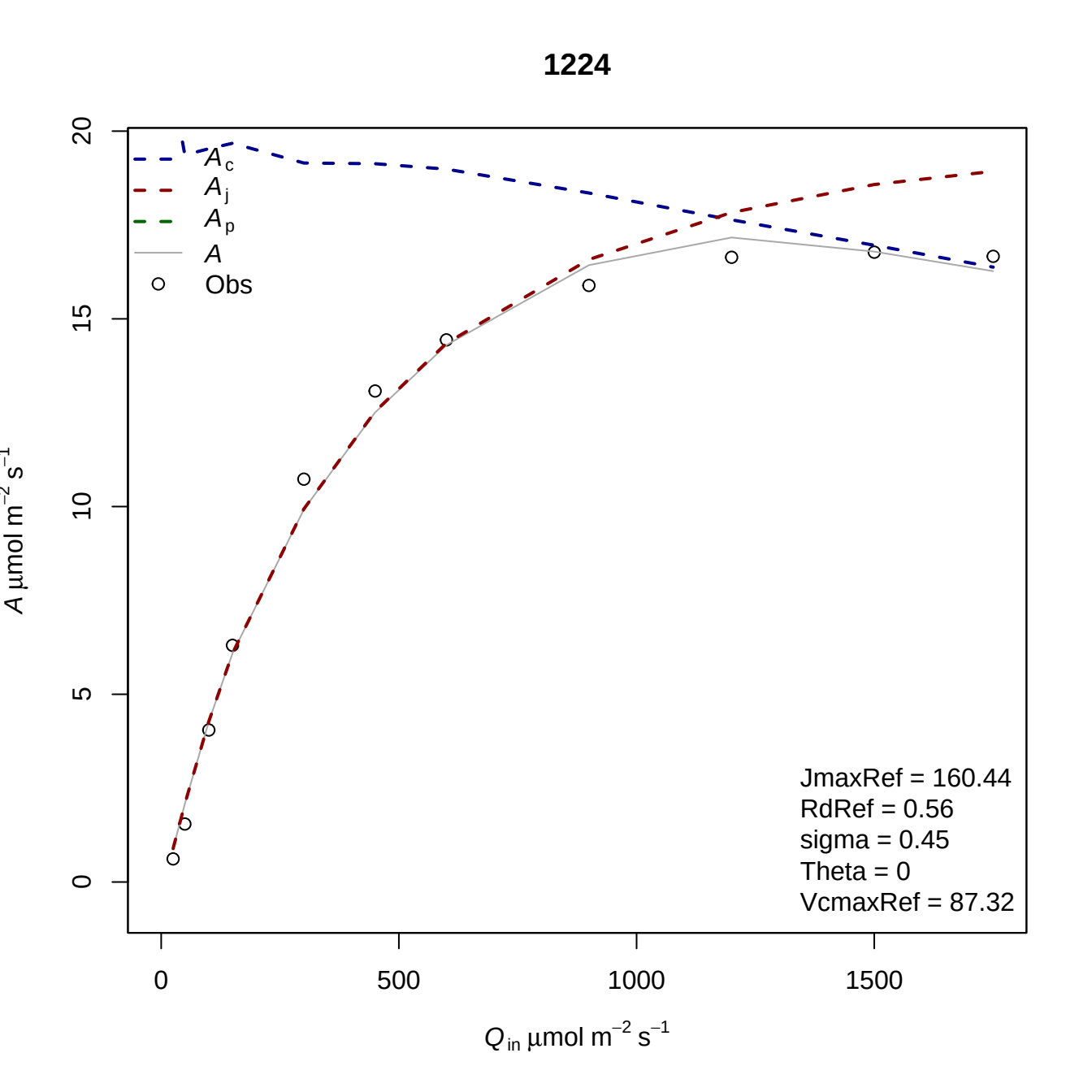

1219
A_c
A_j
A_p
A
Obs
JmaxRef = 238.37
RdRef = 1.8
sigma = 0.55
Theta = 0
VcmaxRef = 126.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

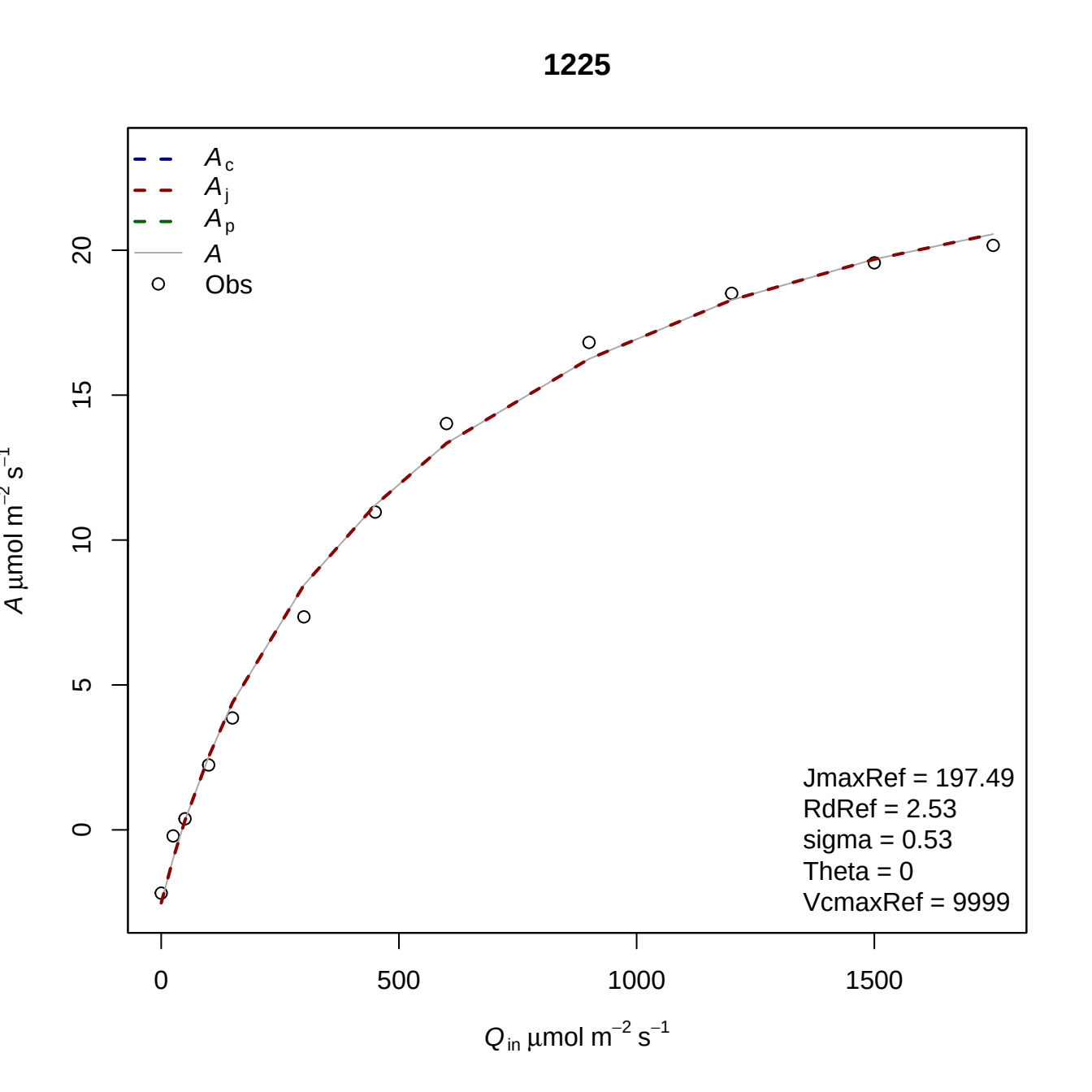

1220
A_c
A_j
A_p
A
Obs
JmaxRef = 315.16
RdRef = 1.35
sigma = 0.36
Theta = 0
VcmaxRef = 110.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

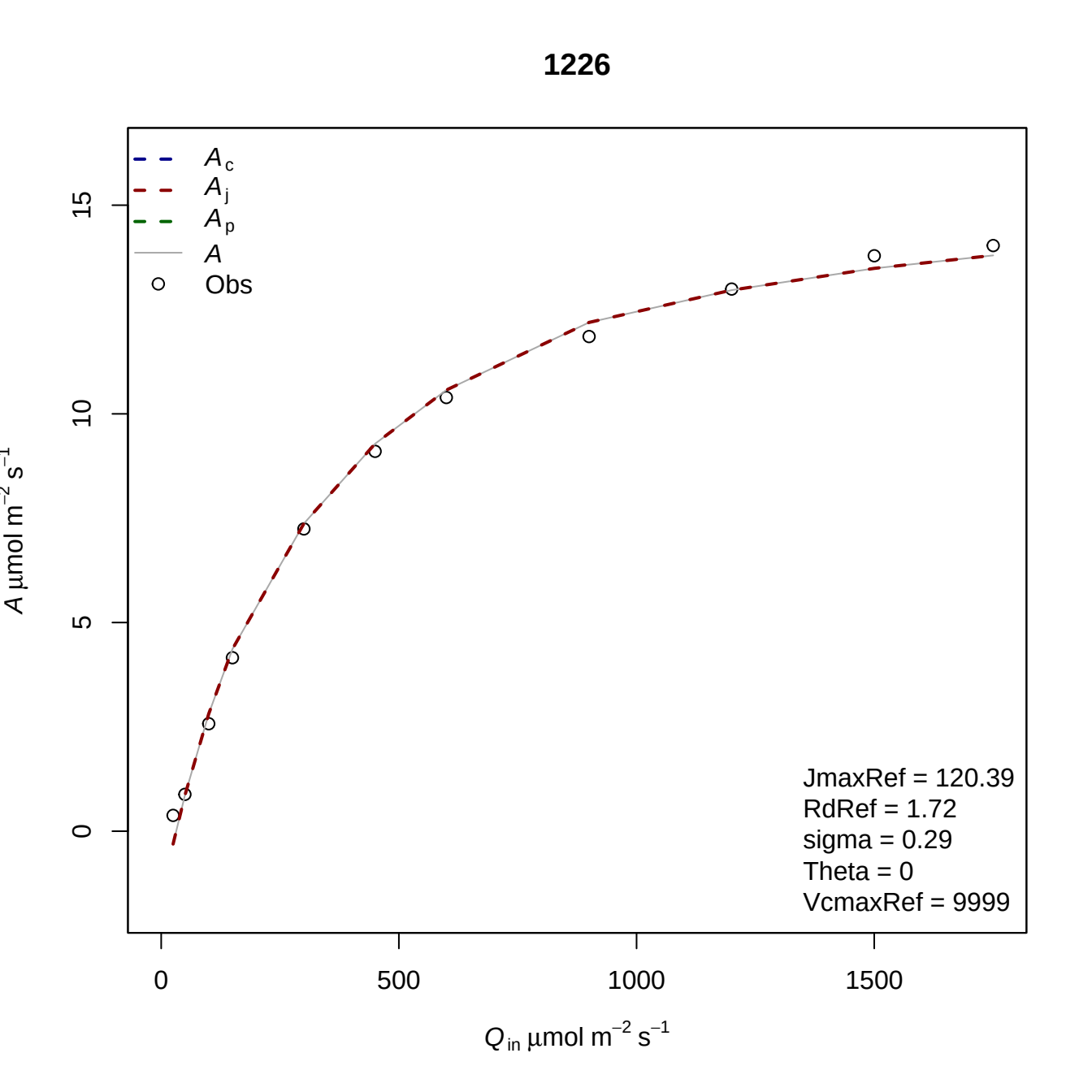

1221
A_c
A_j
A_p
A
Obs
JmaxRef = 189.37
RdRef = 0.89
sigma = 0.35
Theta = 0
VcmaxRef = 93.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

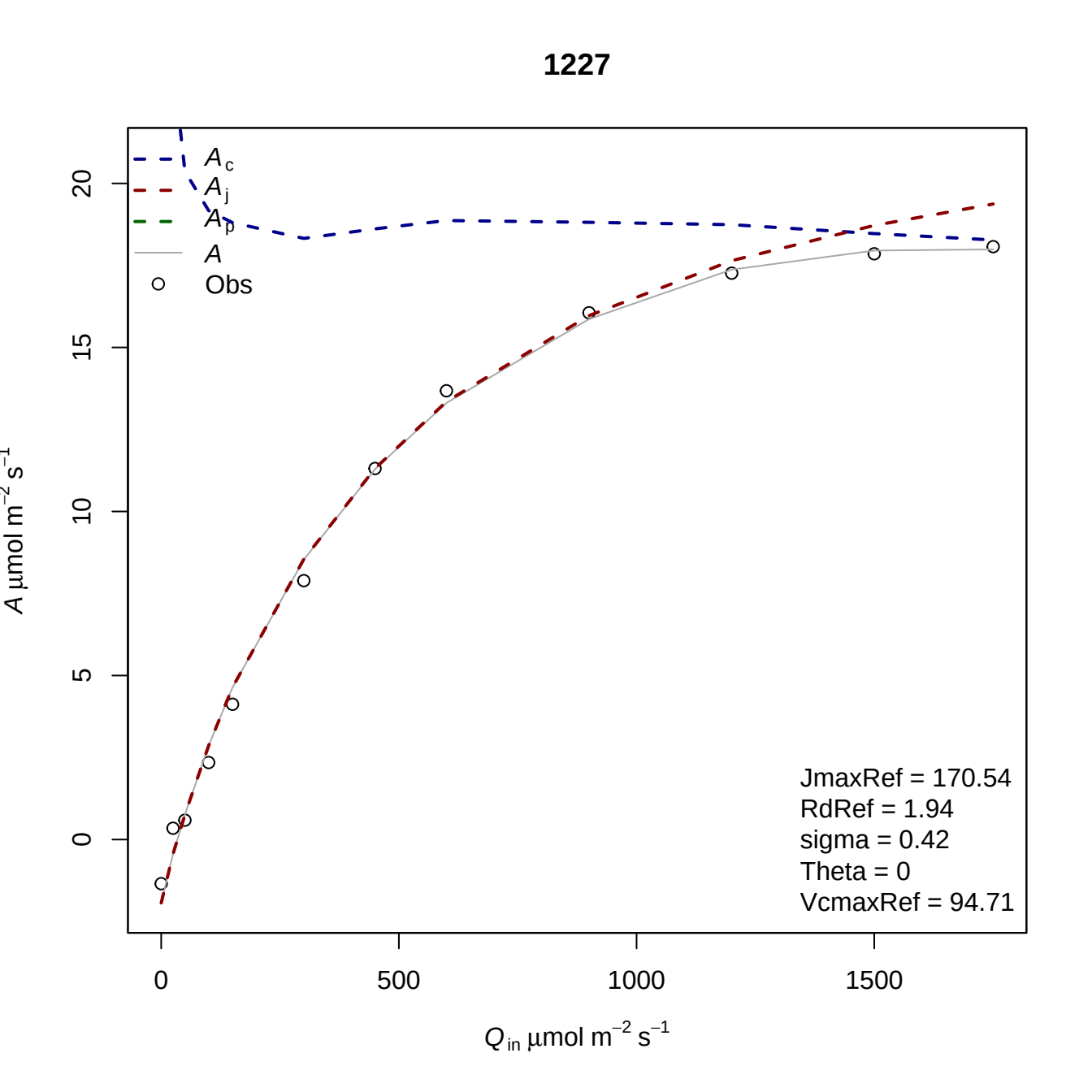

1222
A_c
A_j
A_p
A
Obs
JmaxRef = 257.06
RdRef = 1.87
sigma = 0.53
Theta = 0
VcmaxRef = 124.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

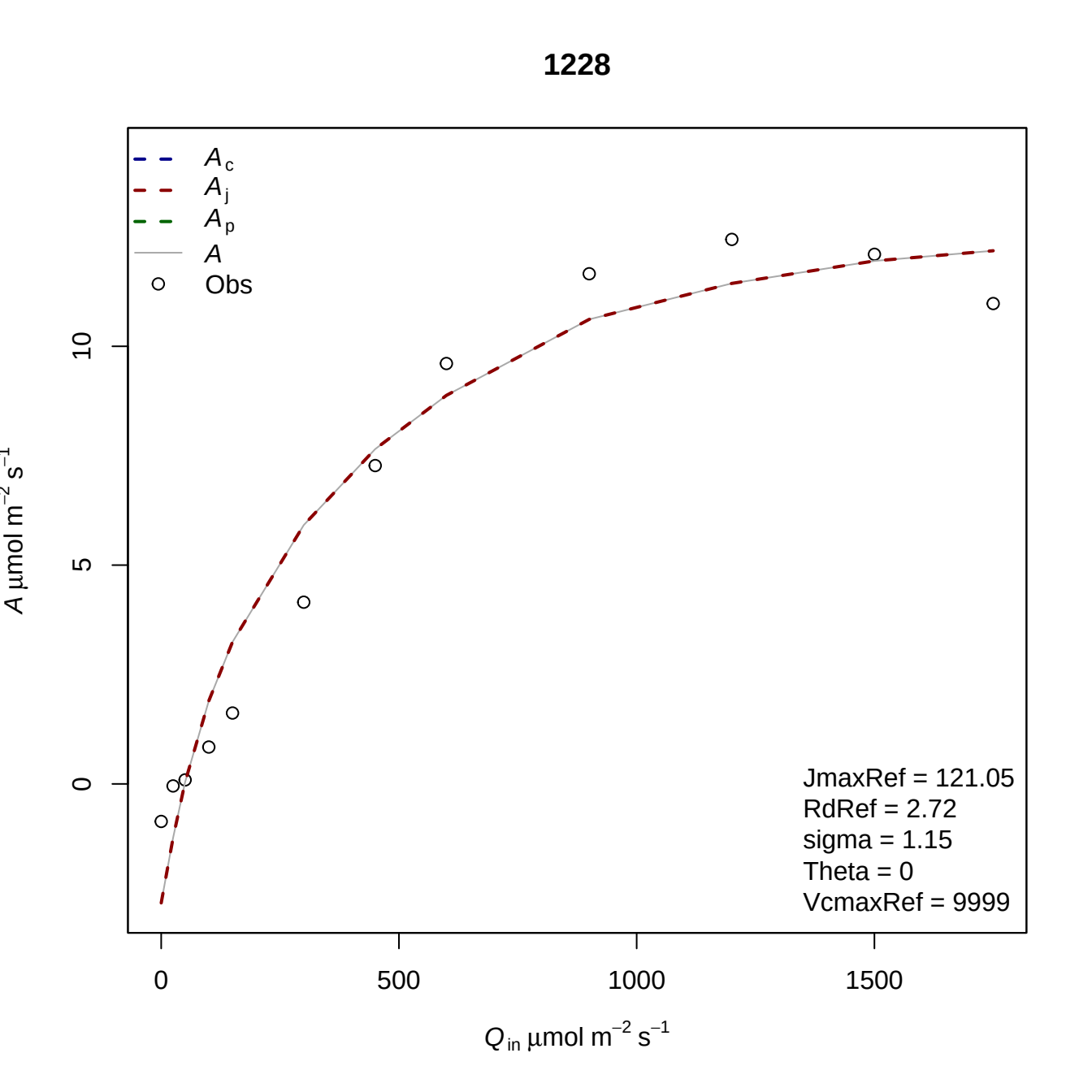

1223
A_c
A_j
A_p
A
Obs
JmaxRef = 142.6
RdRef = 1.78
sigma = 0.52
Theta = 0
VcmaxRef = 77.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

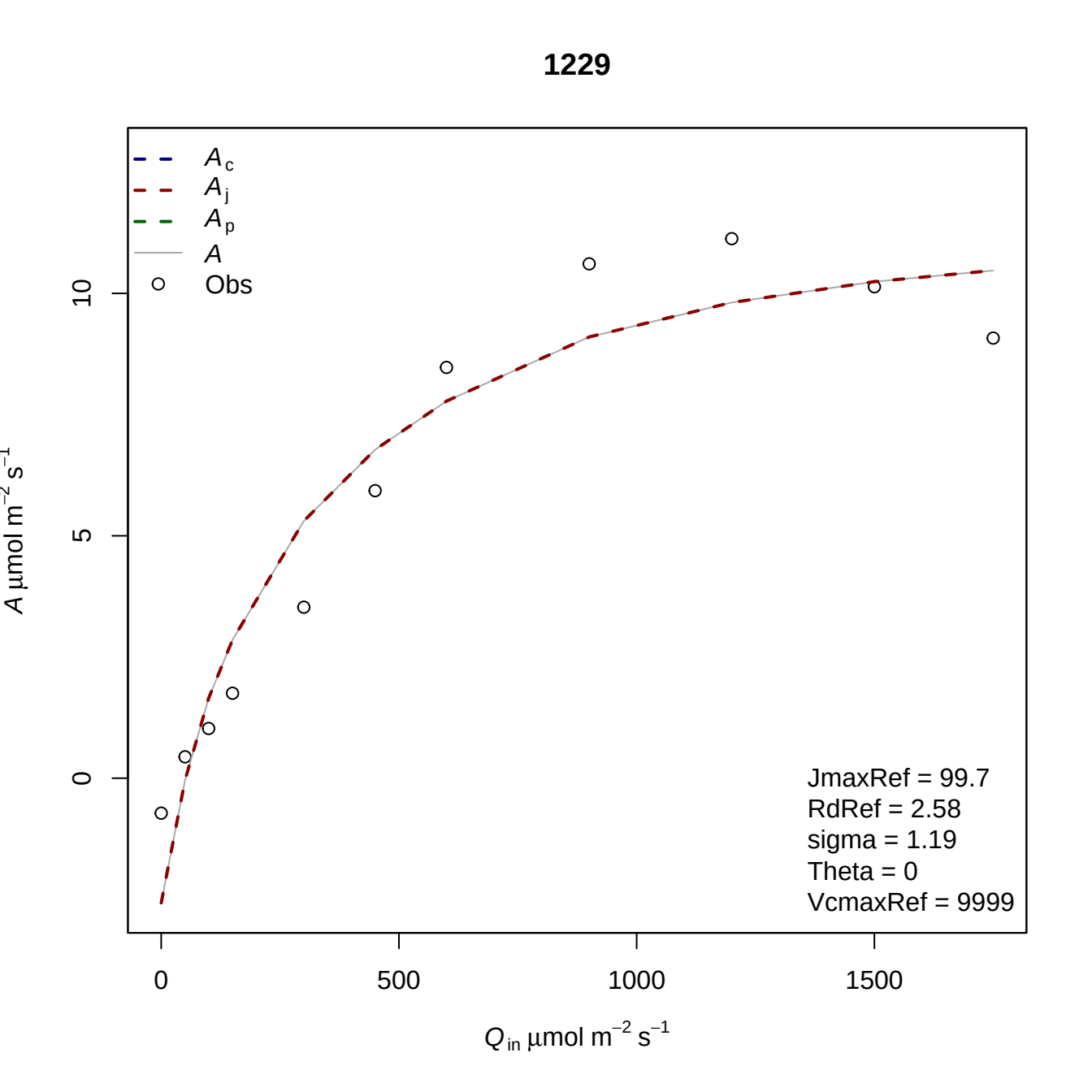

1224
A_c
A_j
A_p
A
Obs
JmaxRef = 160.44
RdRef = 0.56
sigma = 0.45
Theta = 0
VcmaxRef = 87.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

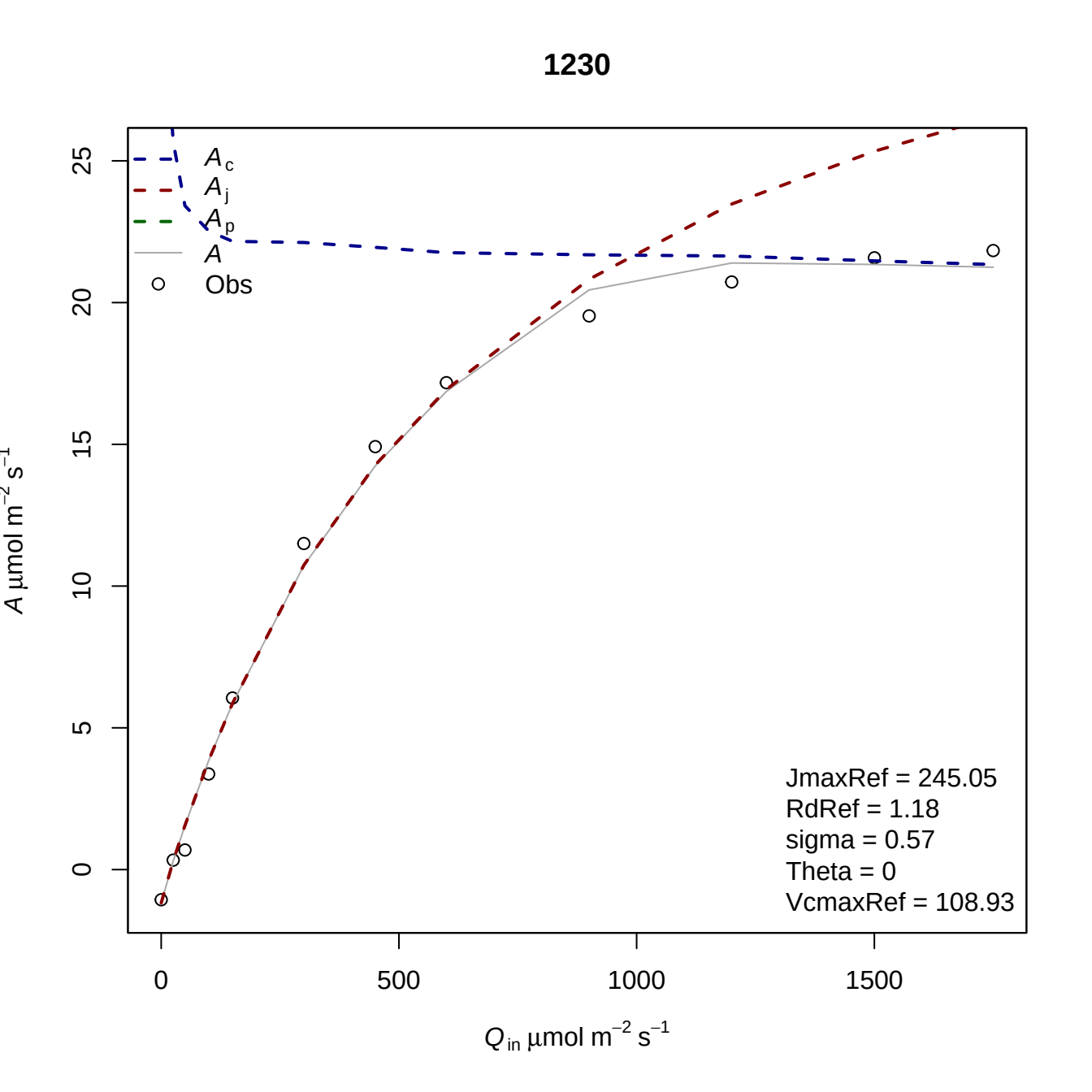

1225
A_c
A_j
A_p
A
Obs
JmaxRef = 197.49
RdRef = 2.53
sigma = 0.53
Theta = 0
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

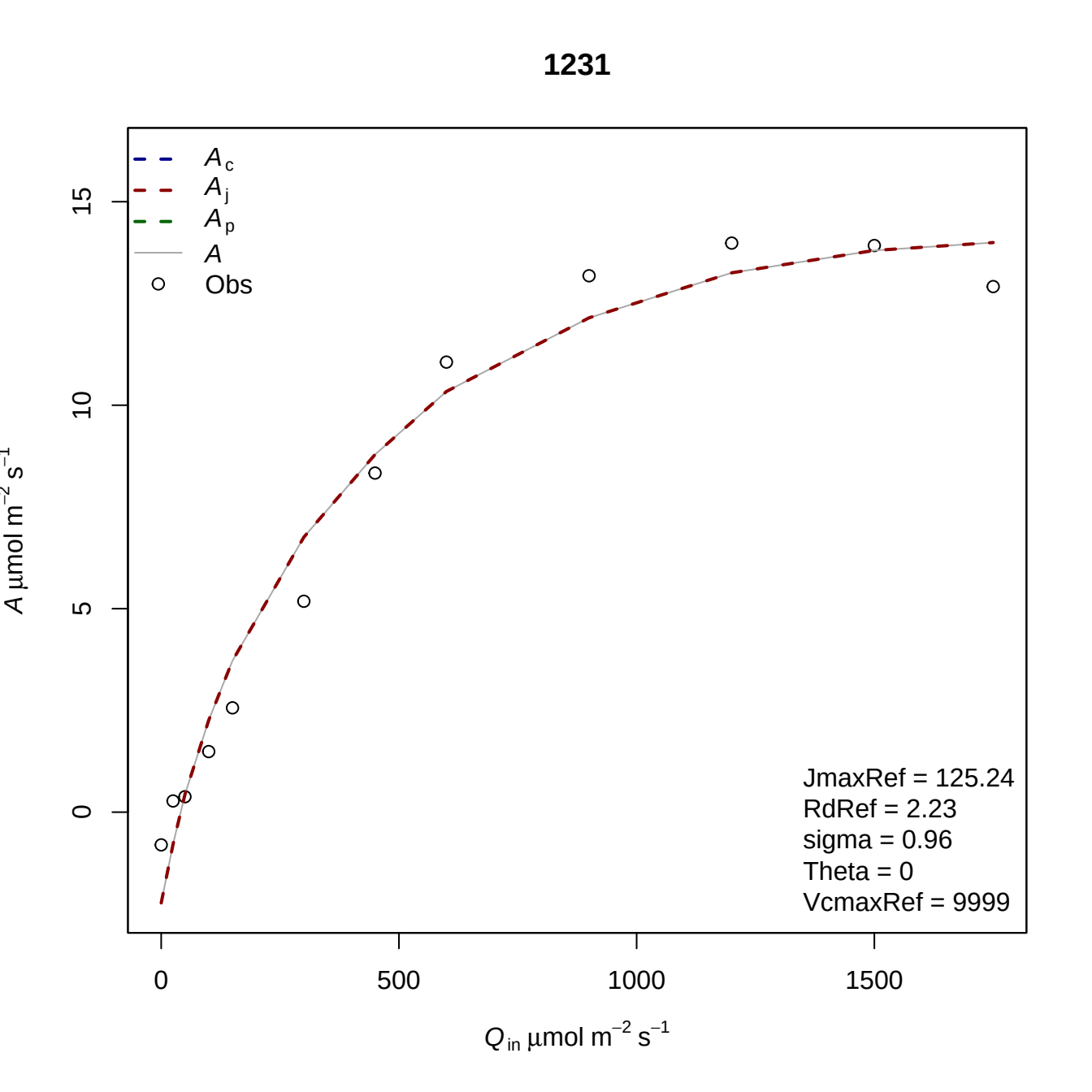

1226
A_c
A_j
A_p
A
Obs
JmaxRef = 120.39
RdRef = 1.72
sigma = 0.29
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

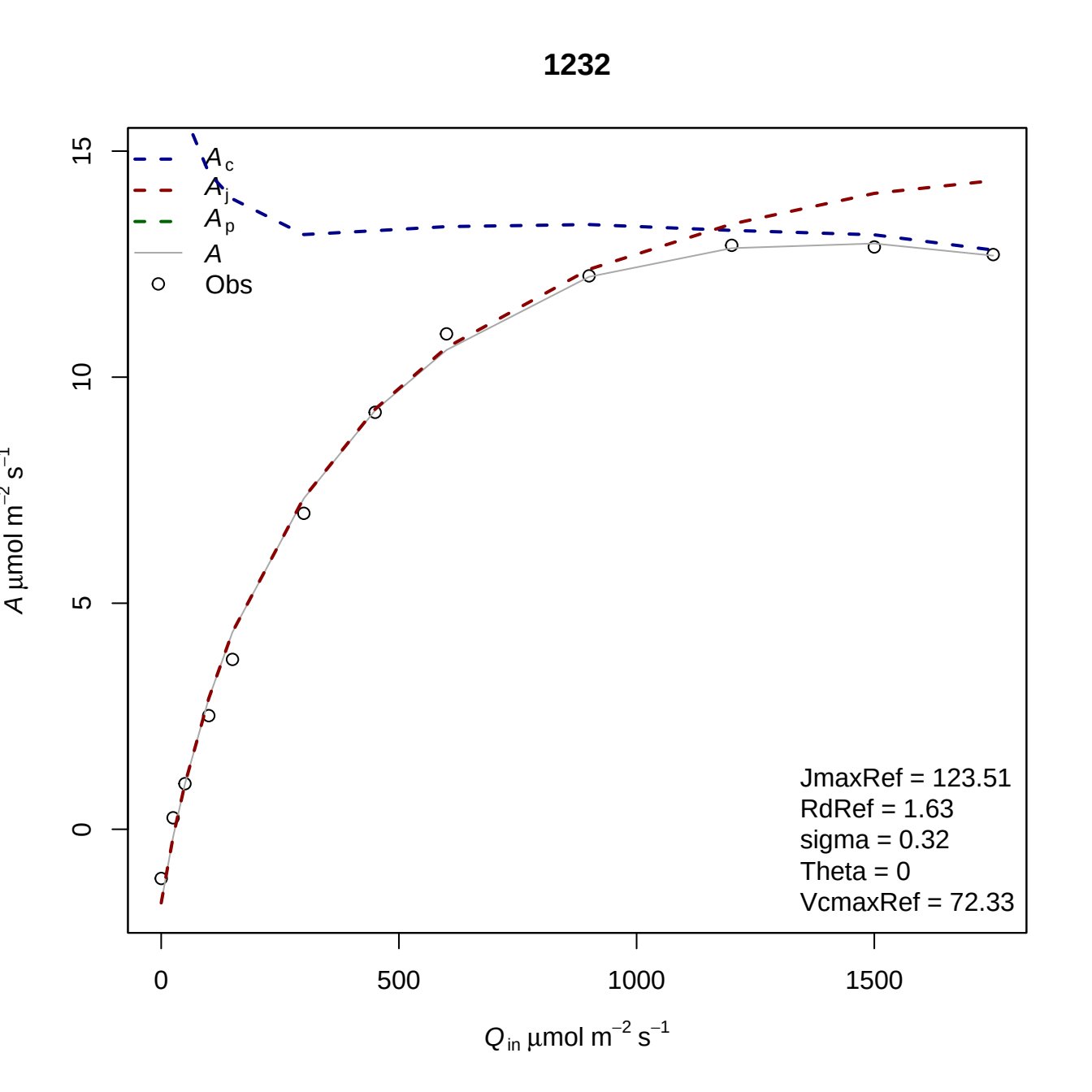

1227
A_c
A_j
A_p
A
Obs
JmaxRef = 170.54
RdRef = 1.94
sigma = 0.42
Theta = 0
VcmaxRef = 94.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

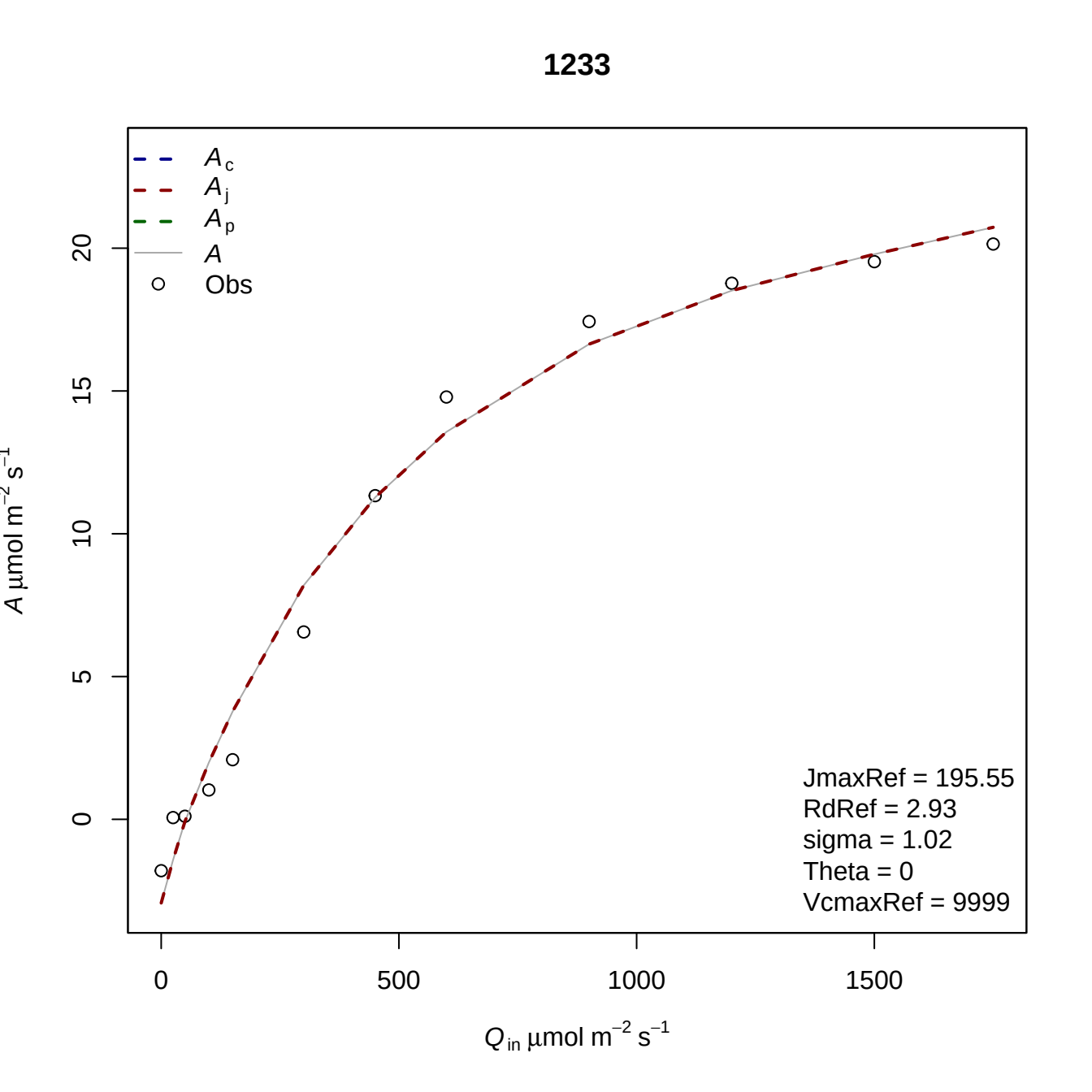

1228
A_c
A_j
A_p
A
Obs
JmaxRef = 121.05
RdRef = 2.72
sigma = 1.15
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

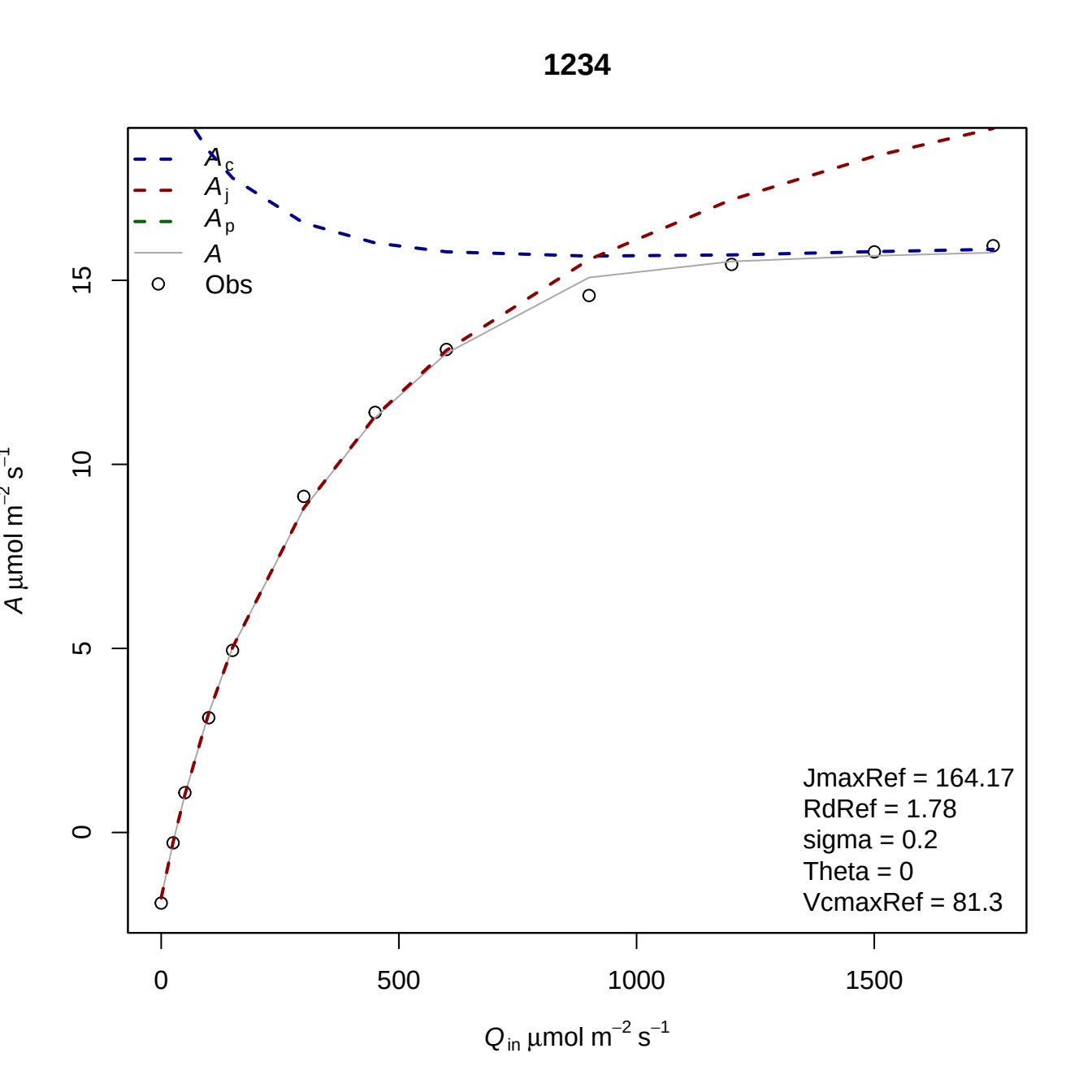

1229
A_c
A_j
A_p
A
Obs
JmaxRef = 99.7
RdRef = 2.58
sigma = 1.19
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

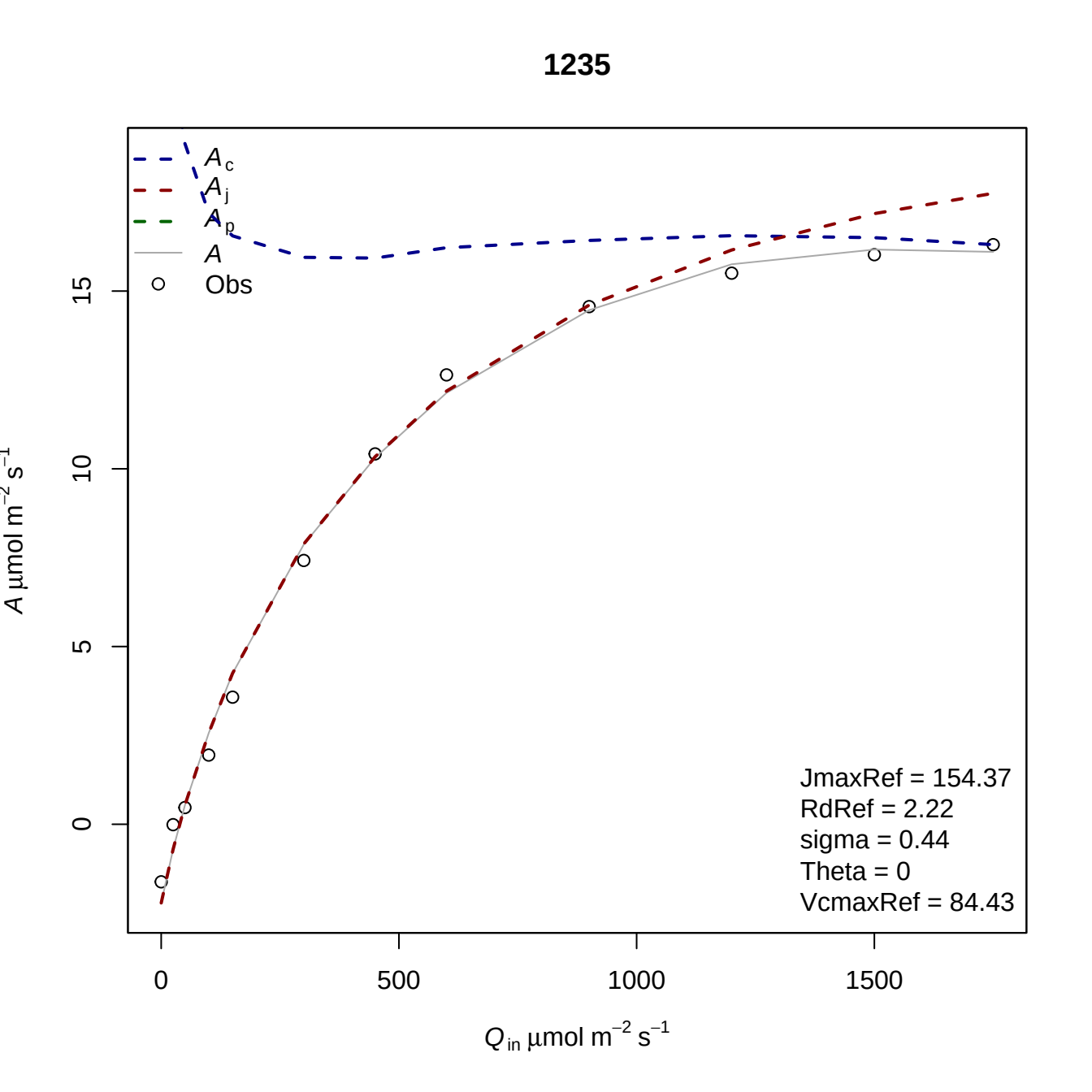

1230
A_c
A_j
A_p
A
Obs
JmaxRef = 245.05
RdRef = 1.18
sigma = 0.57
Theta = 0
VcmaxRef = 108.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

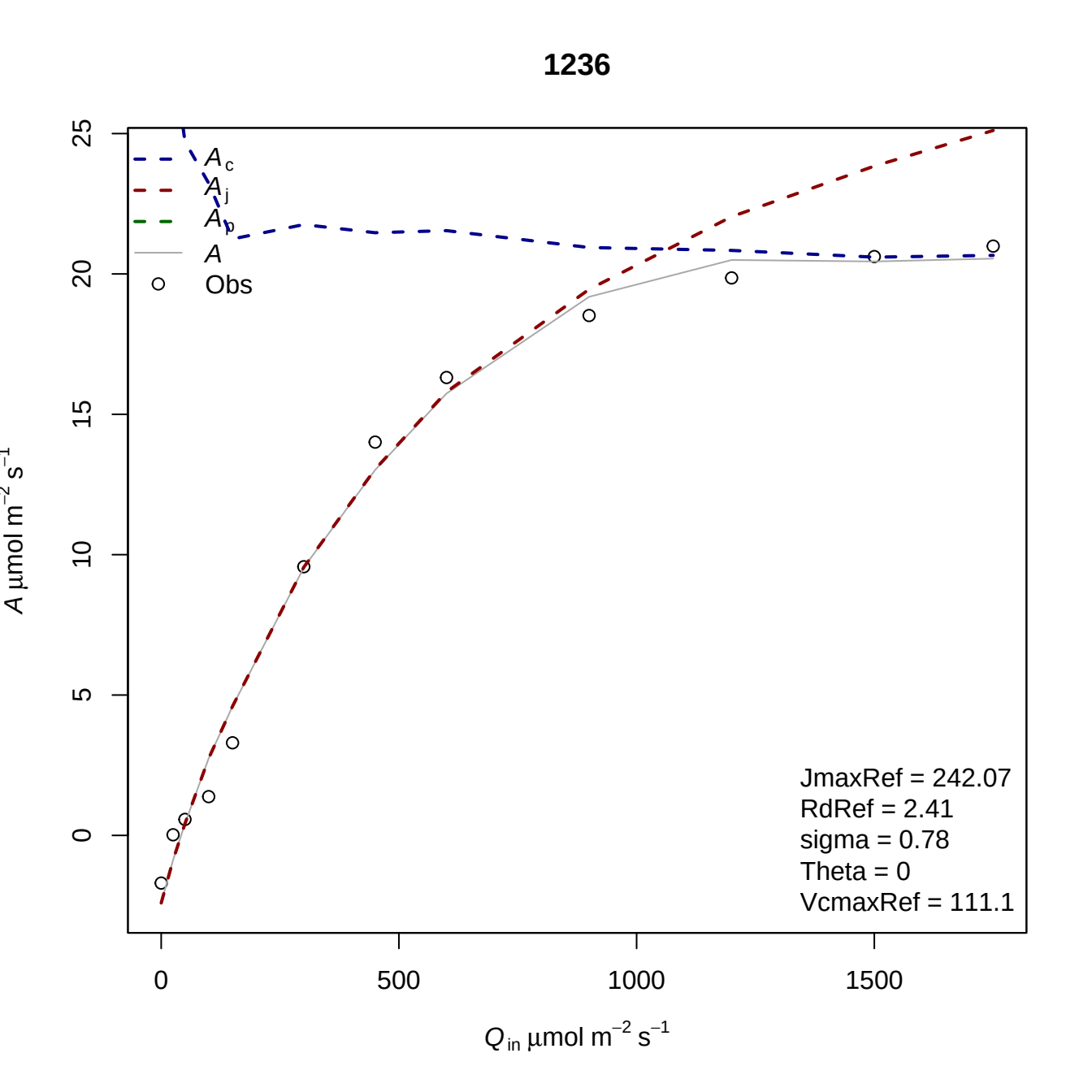

1231
A_c
A_j
A_p
A
Obs
JmaxRef = 125.24
RdRef = 2.23
sigma = 0.96
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

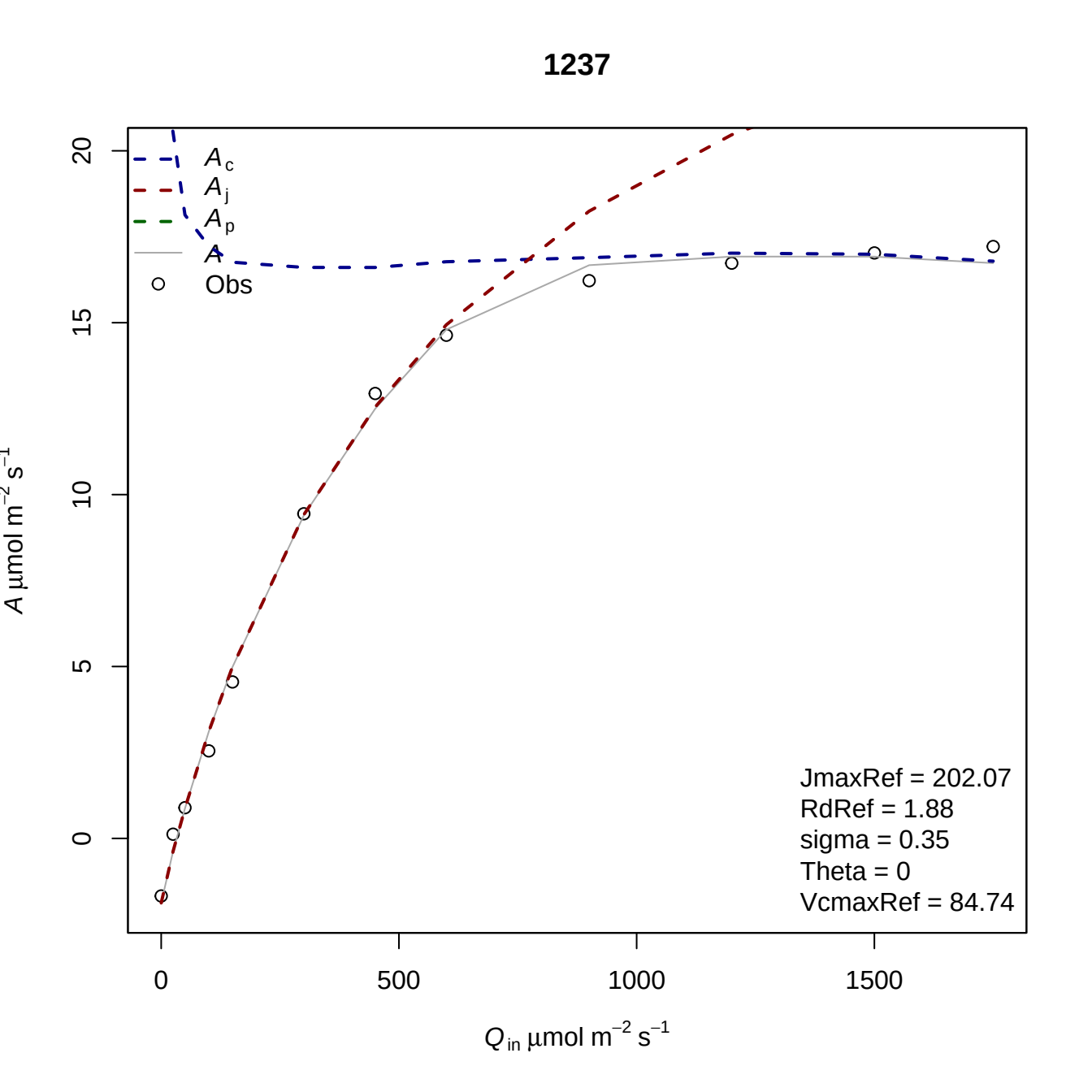

1232
A_c
A_j
A_p
A
Obs
JmaxRef = 123.51
RdRef = 1.63
sigma = 0.32
Theta = 0
VcmaxRef = 72.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

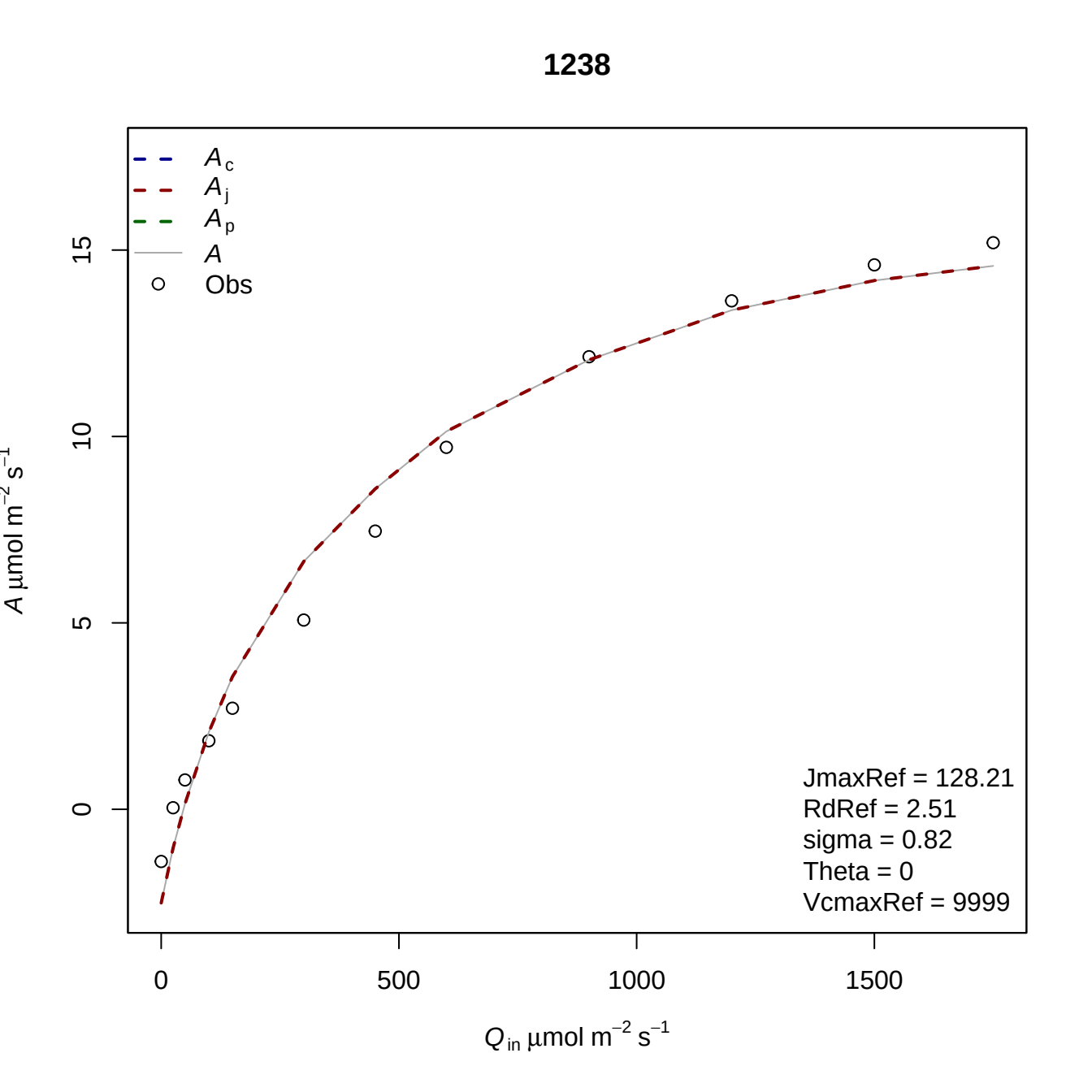

1233
A_c
A_j
A_p
A
Obs
JmaxRef = 195.55
RdRef = 2.93
sigma = 1.02
Theta = 0
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

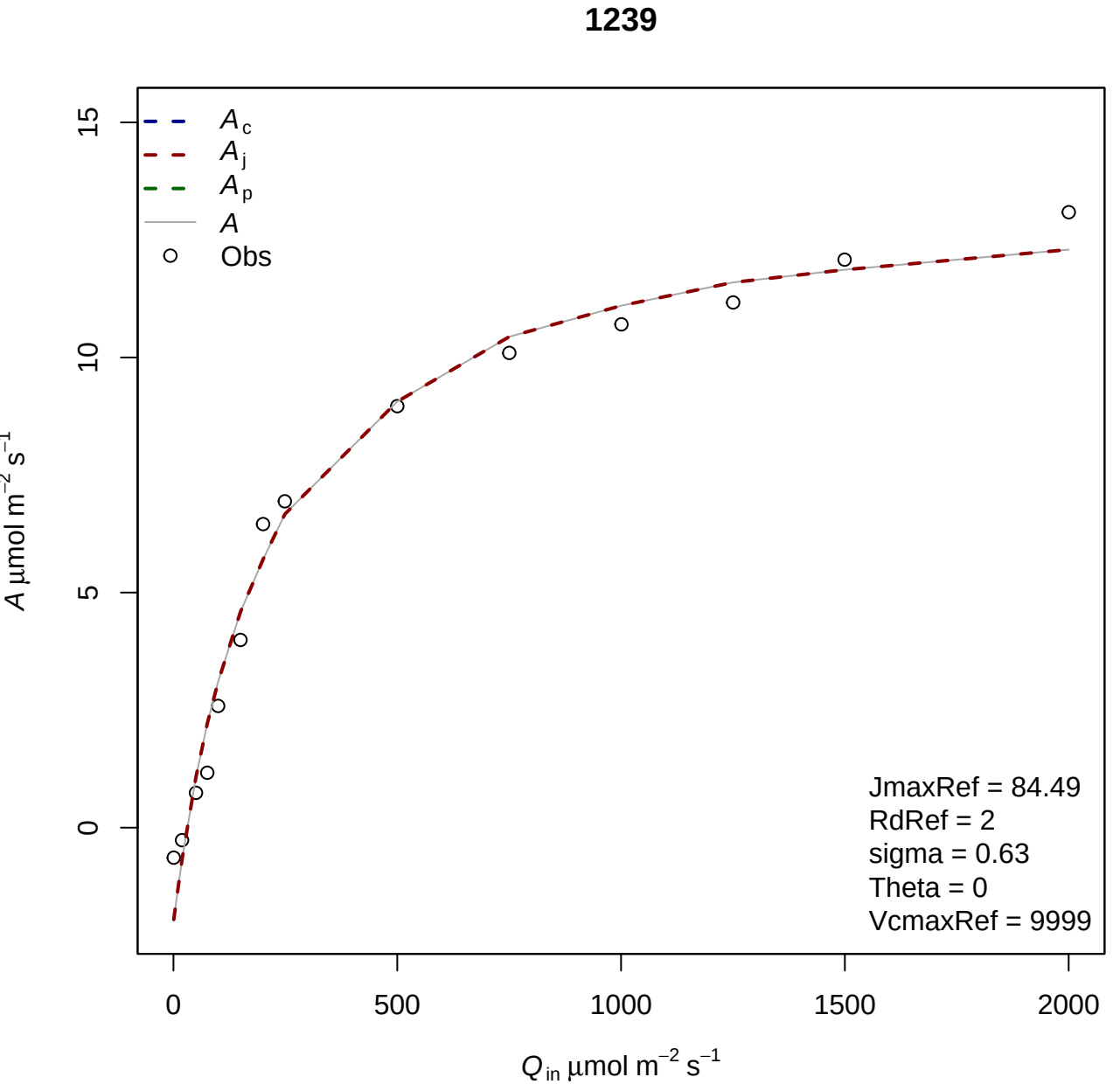

1234
A_c
A_j
A_p
A
Obs
JmaxRef = 164.17
RdRef = 1.78
sigma = 0.2
Theta = 0
VcmaxRef = 81.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

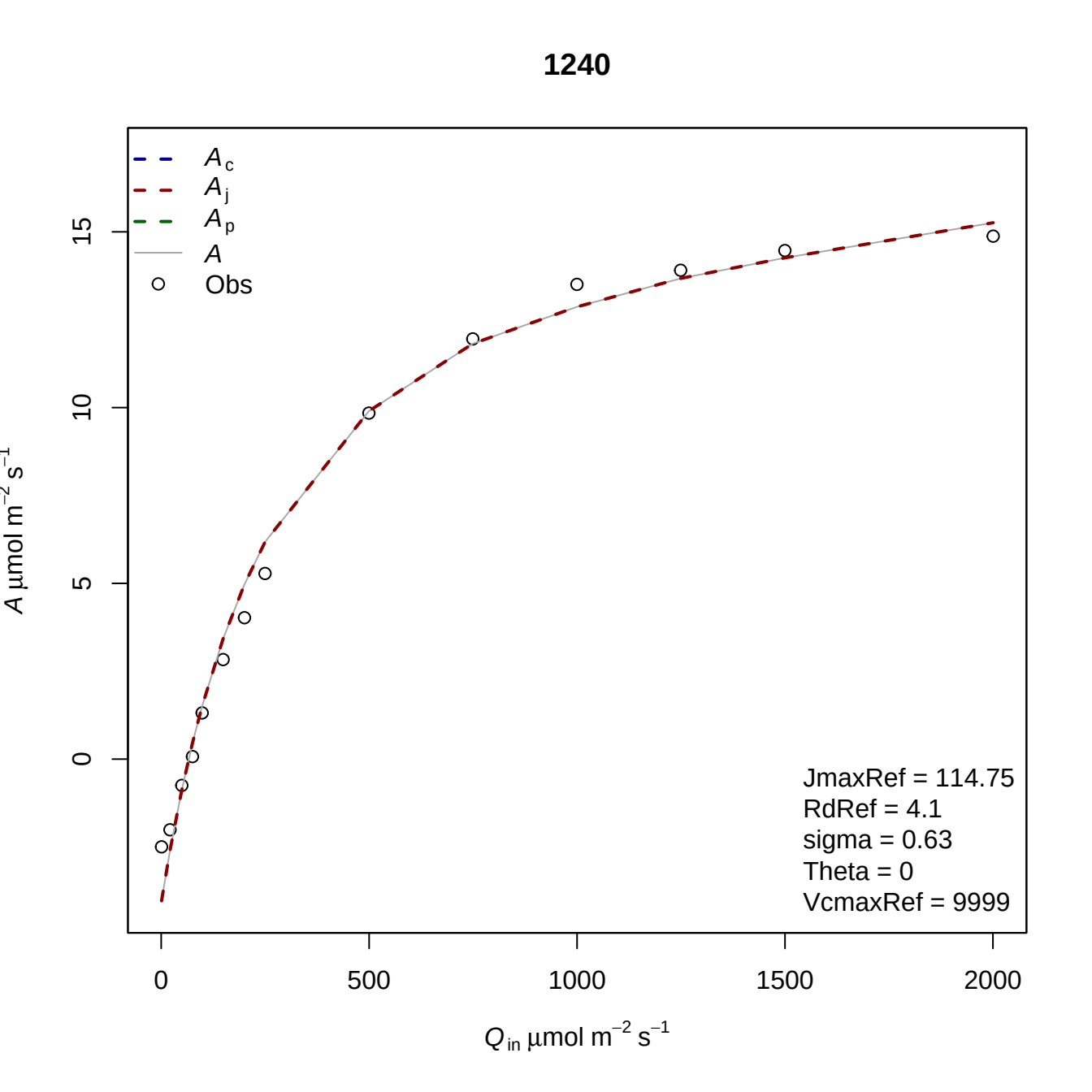

1235
A_c
A_j
A_p
A
Obs
JmaxRef = 154.37
RdRef = 2.22
sigma = 0.44
Theta = 0
VcmaxRef = 84.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

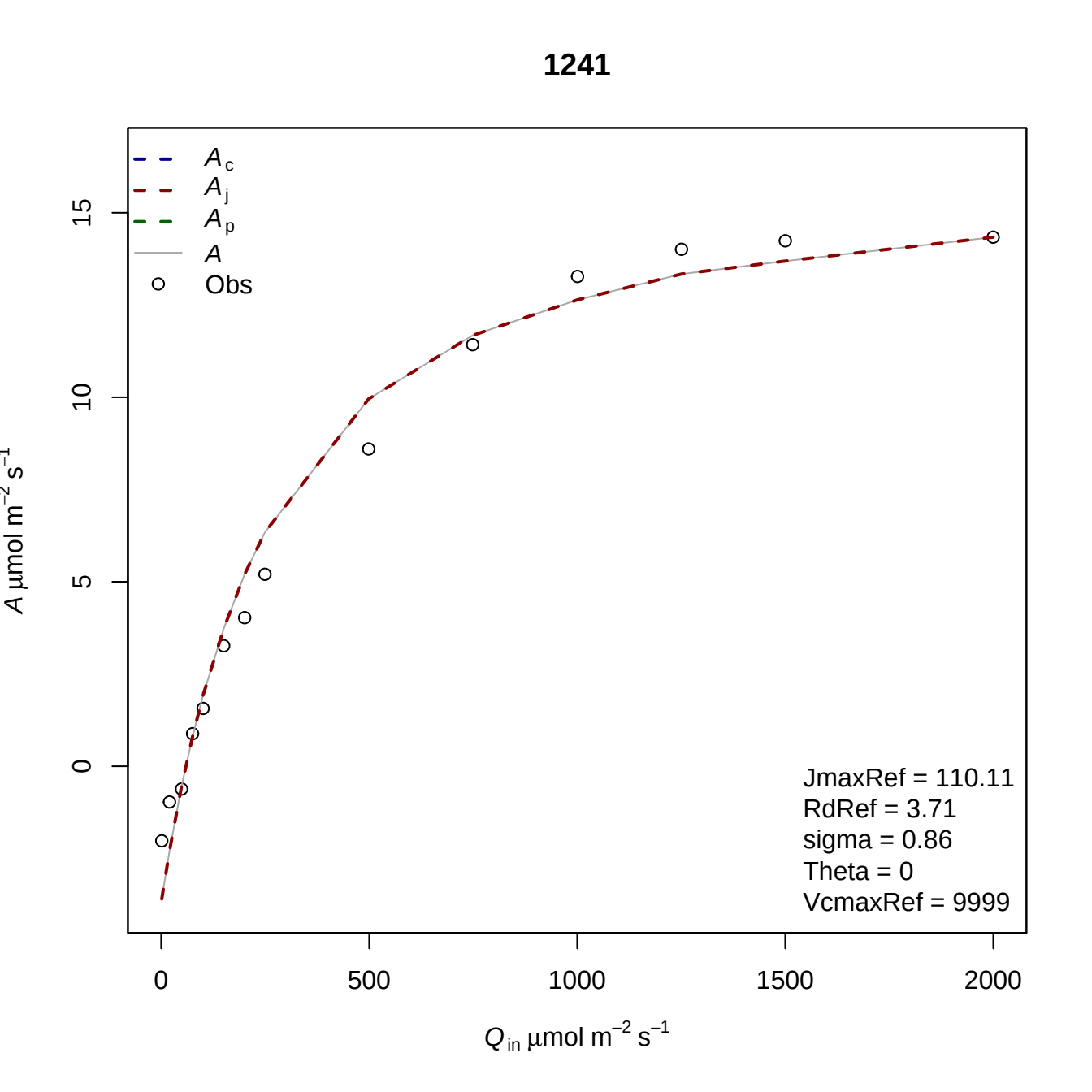

1236
A_c
A_j
A_p
A
Obs
JmaxRef = 242.07
RdRef = 2.41
sigma = 0.78
Theta = 0
VcmaxRef = 111.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

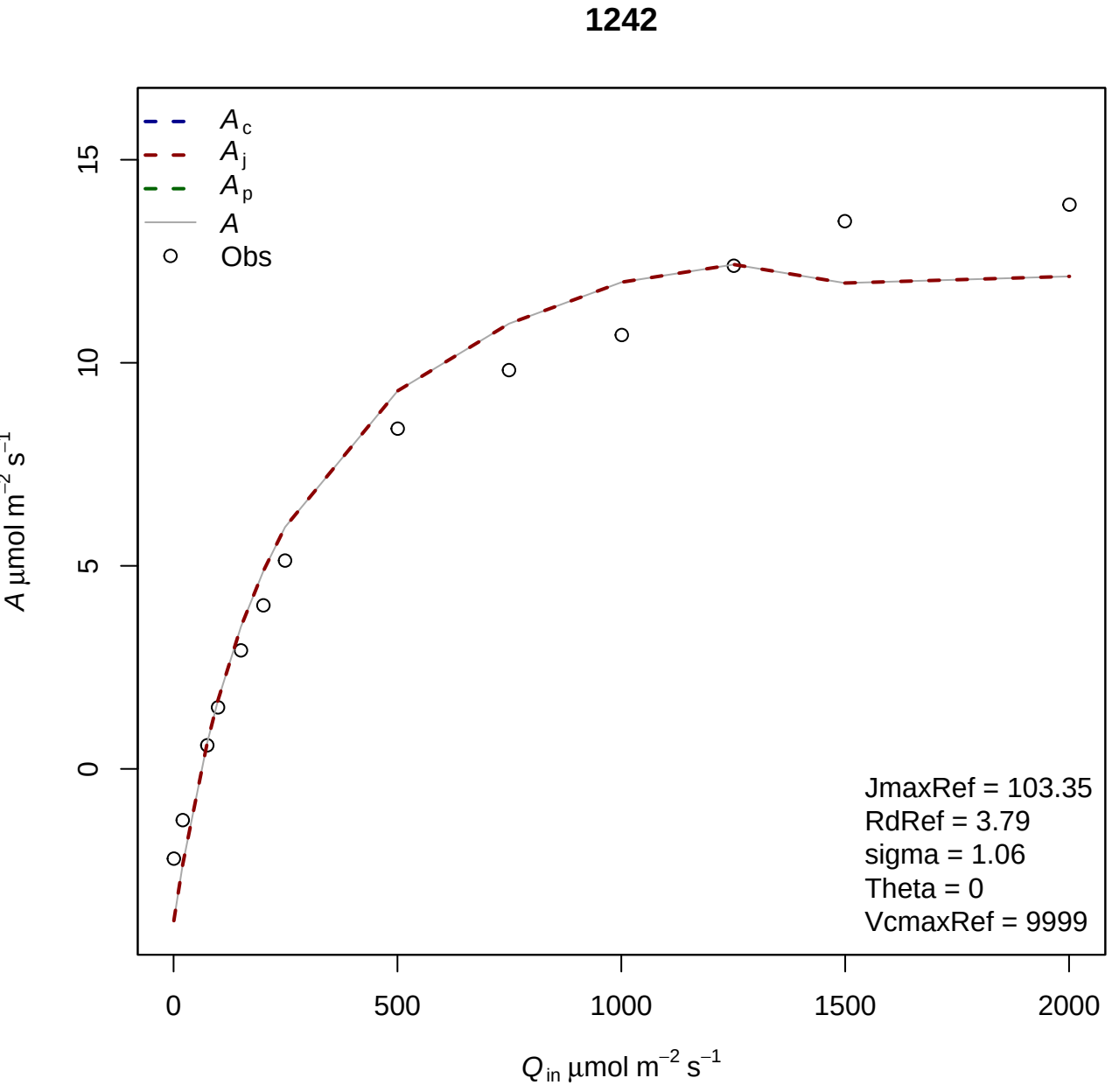

1237
A_c
A_j
A_p
A
Obs
JmaxRef = 202.07
RdRef = 1.88
sigma = 0.35
Theta = 0
VcmaxRef = 84.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

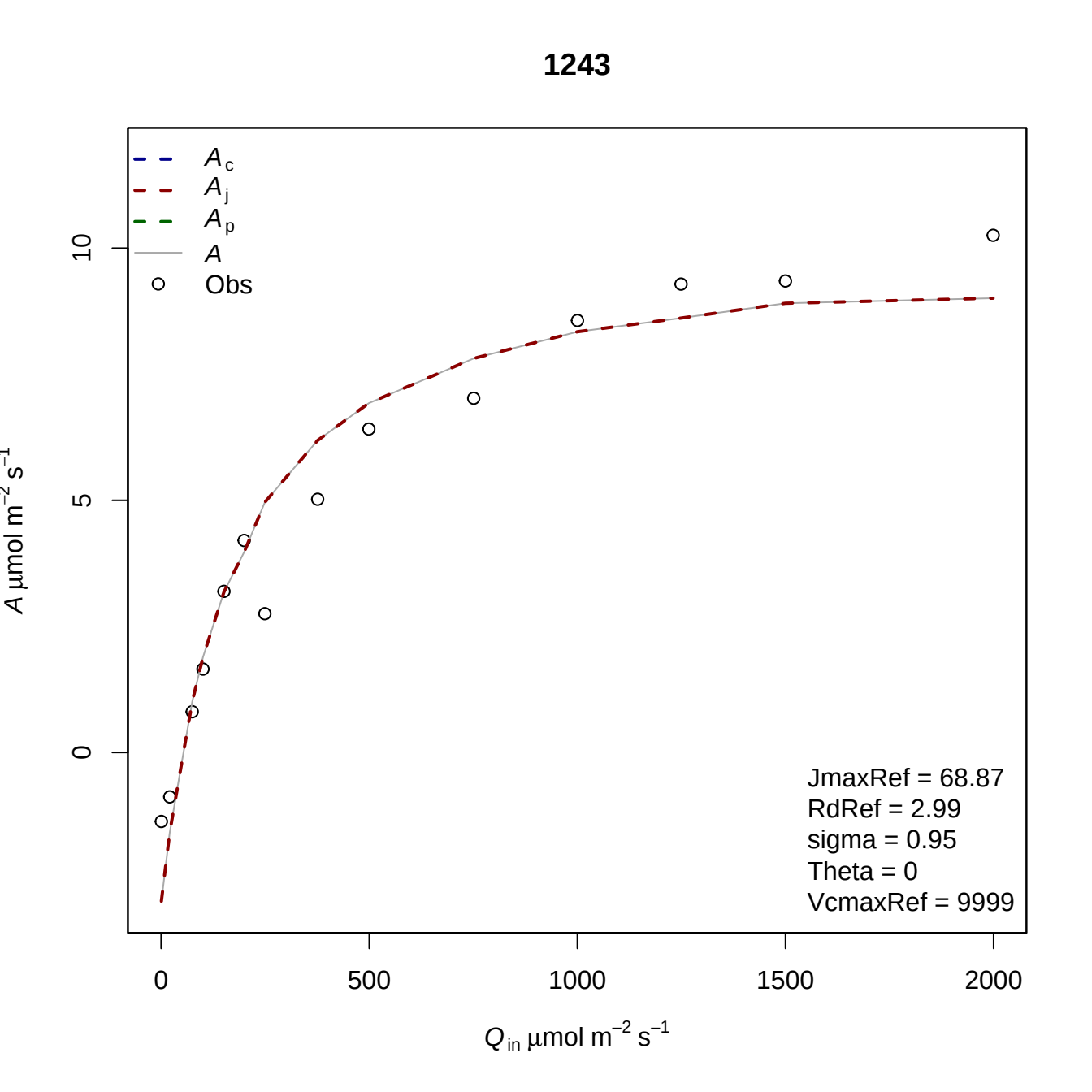

1238
A_c
A_j
A_p
A
Obs
JmaxRef = 128.21
RdRef = 2.51
sigma = 0.82
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

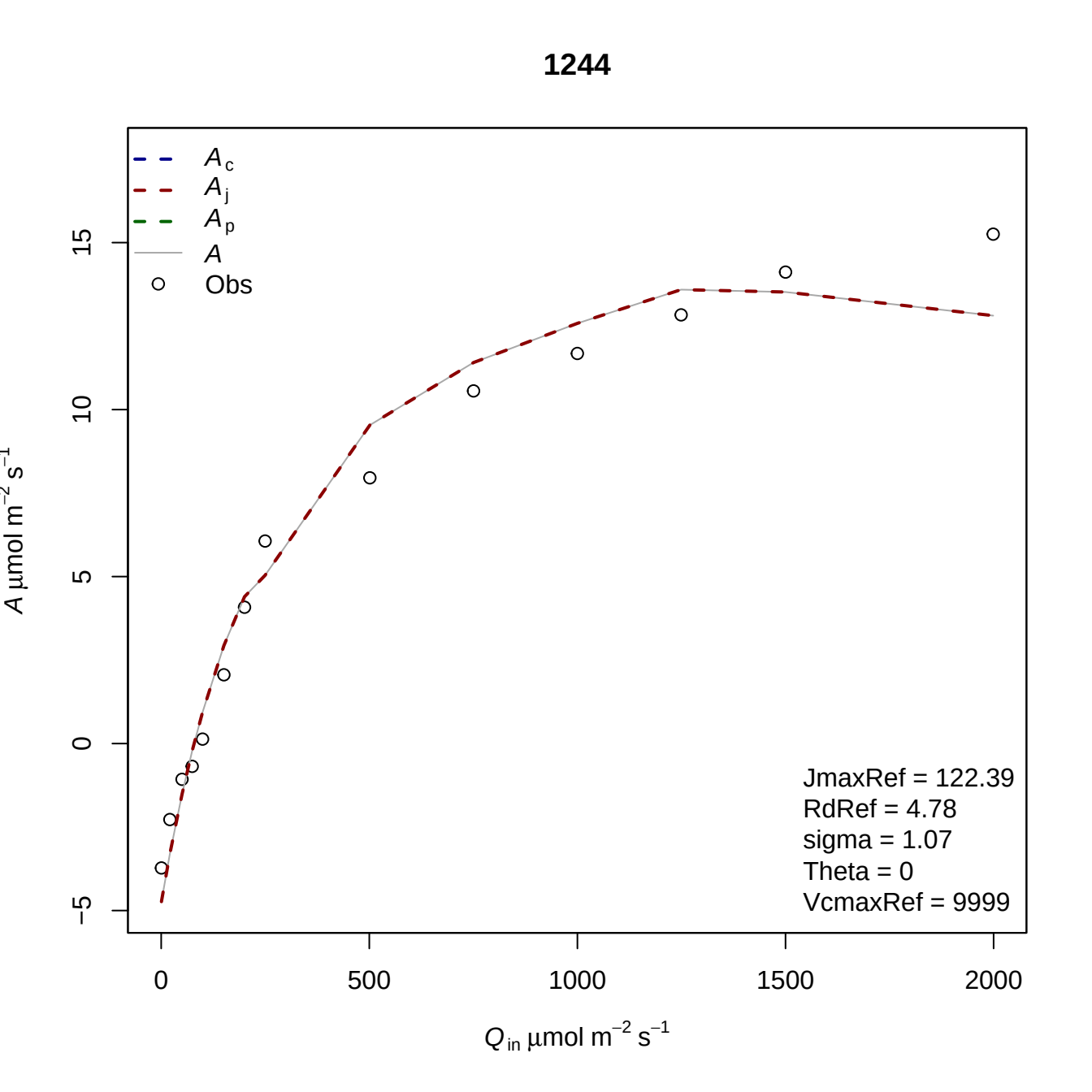

1239
A_c
A_j
A_p
A
Obs
JmaxRef = 84.49
RdRef = 2
sigma = 0.63
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

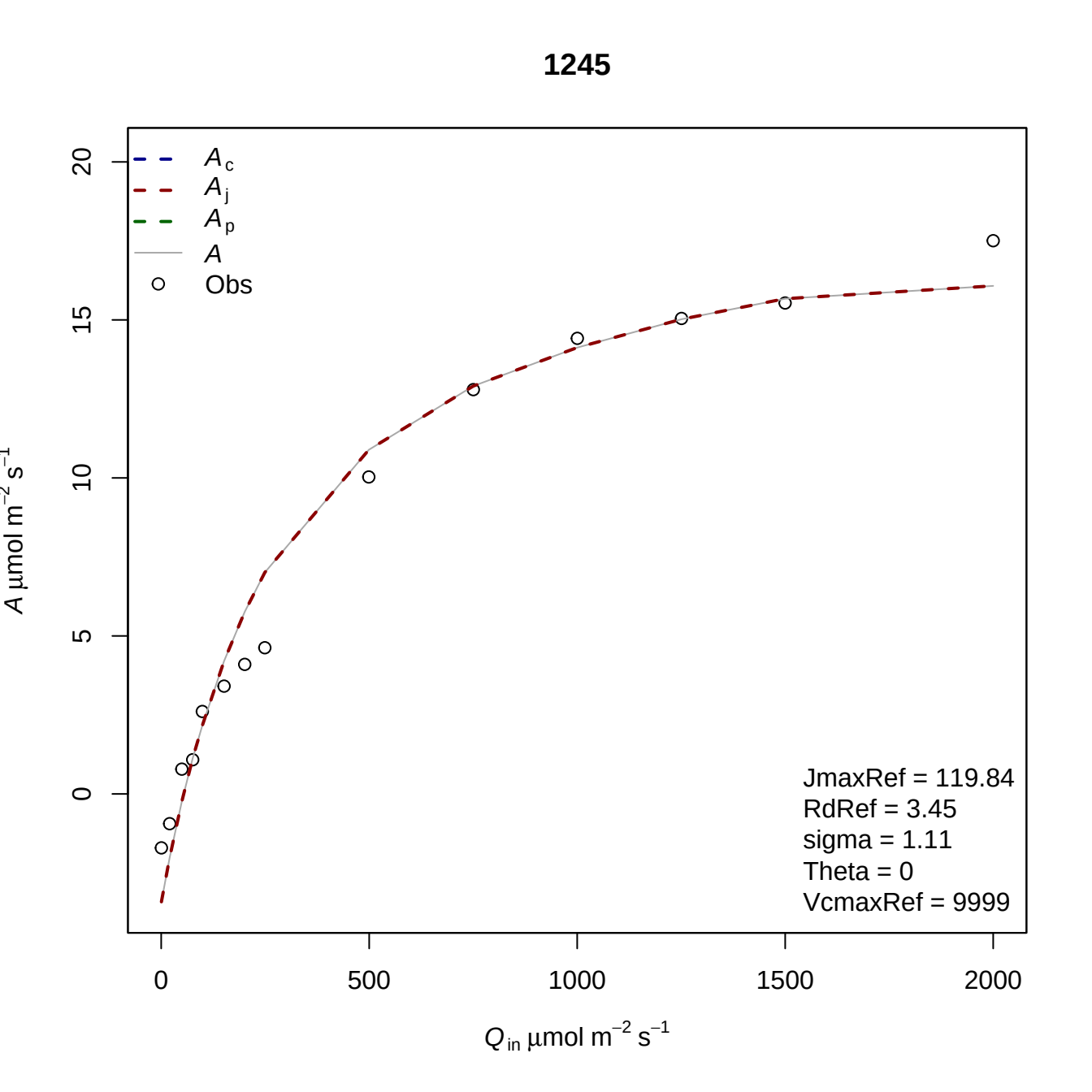

1240
A_c
A_j
A_p
A
Obs
JmaxRef = 114.75
RdRef = 4.1
sigma = 0.63
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

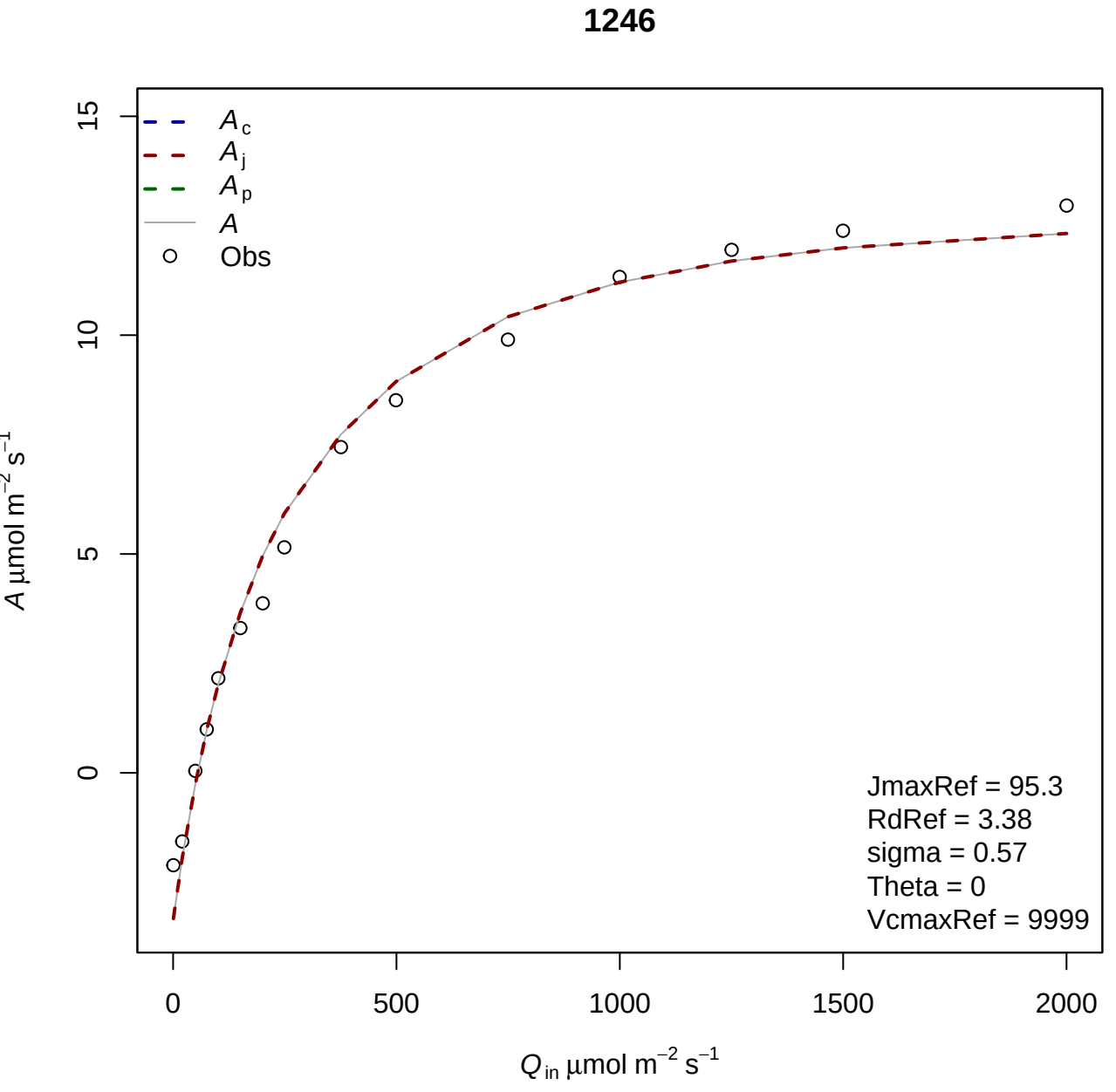

1241
A_c
A_j
A_p
A
Obs
JmaxRef = 110.11
RdRef = 3.71
sigma = 0.86
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

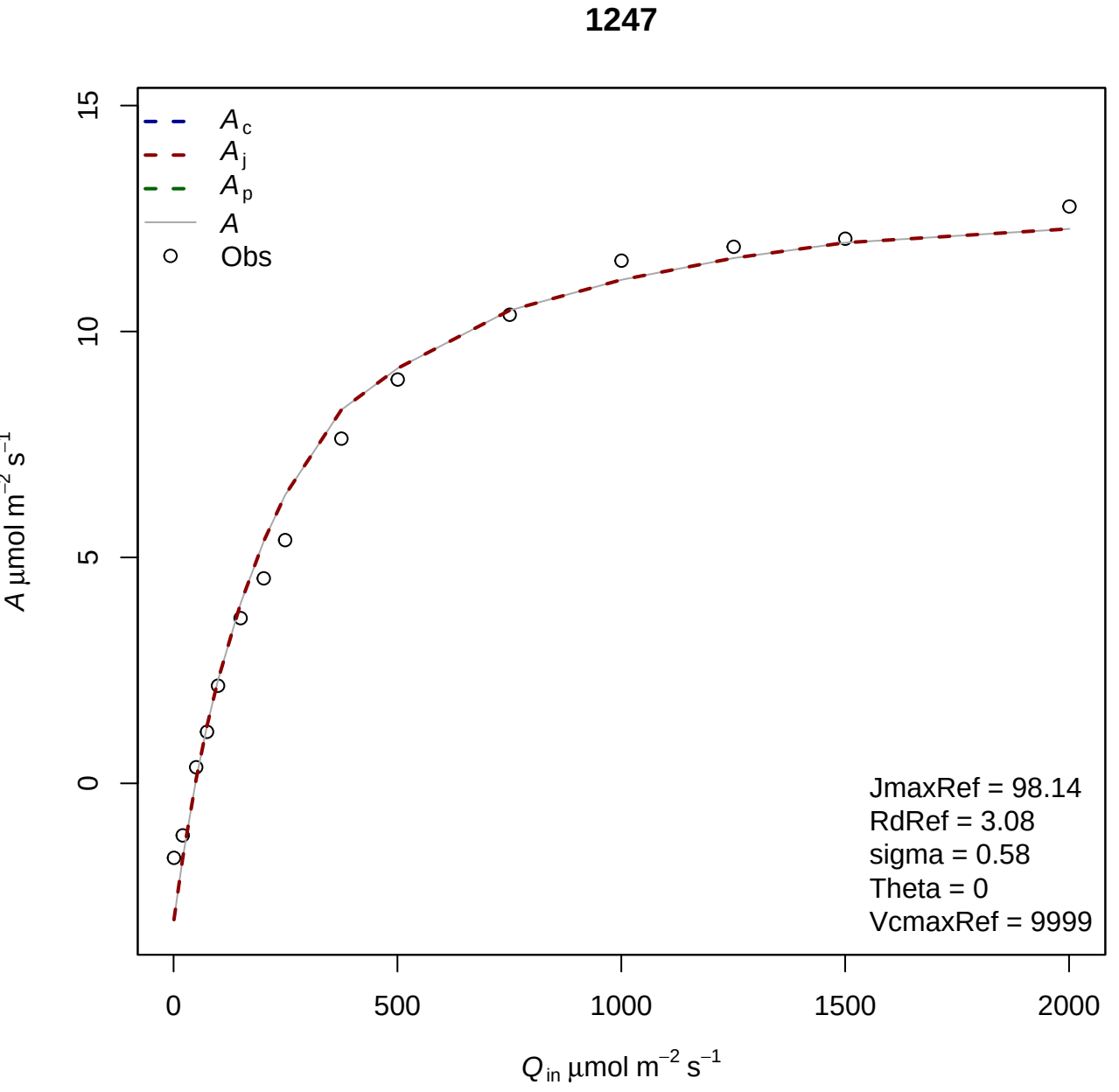

1242
A_c
A_j
A_p
A
Obs
JmaxRef = 103.35
RdRef = 3.79
sigma = 1.06
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

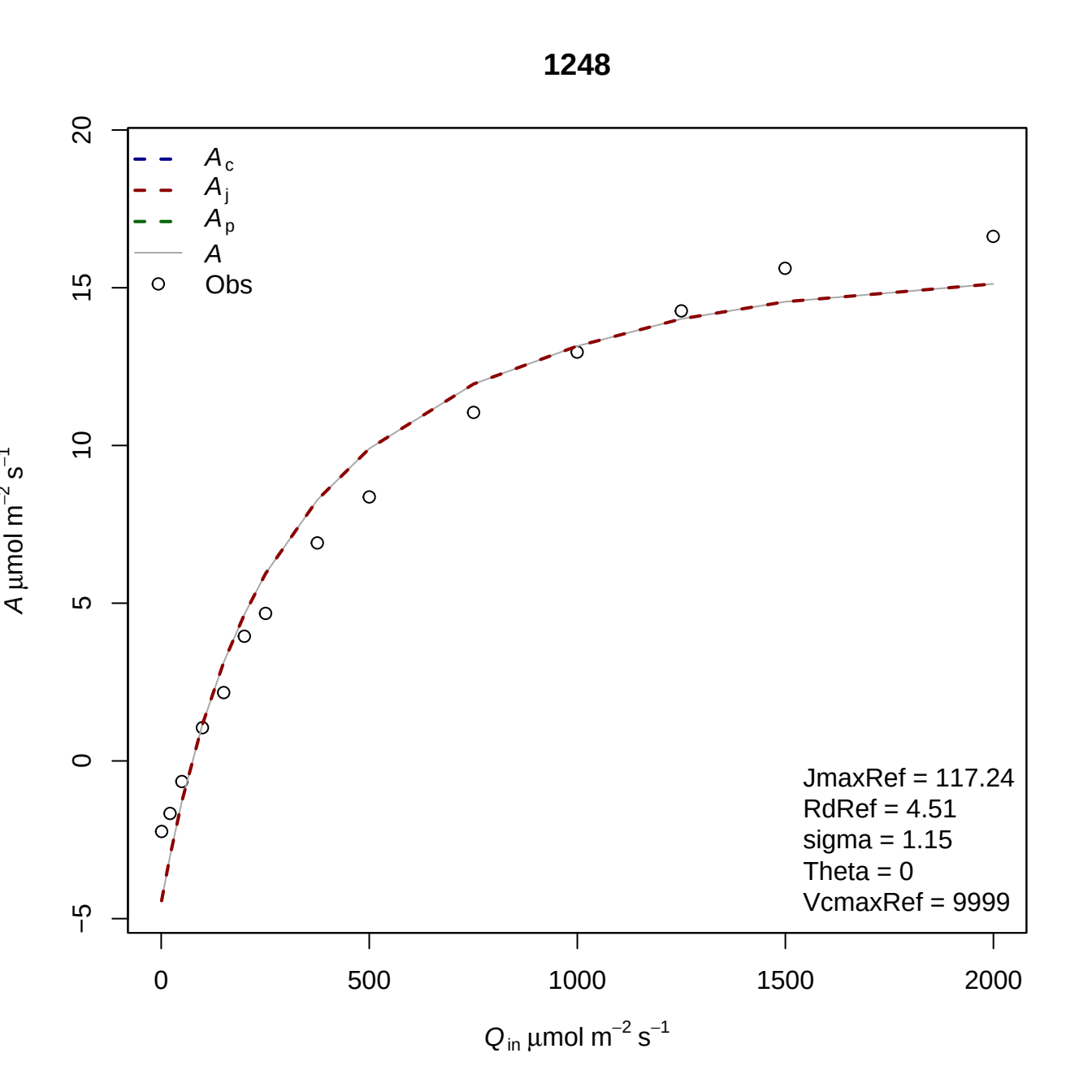

1243
A_c
A_j
A_p
A
Obs
JmaxRef = 68.87
RdRef = 2.99
sigma = 0.95
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

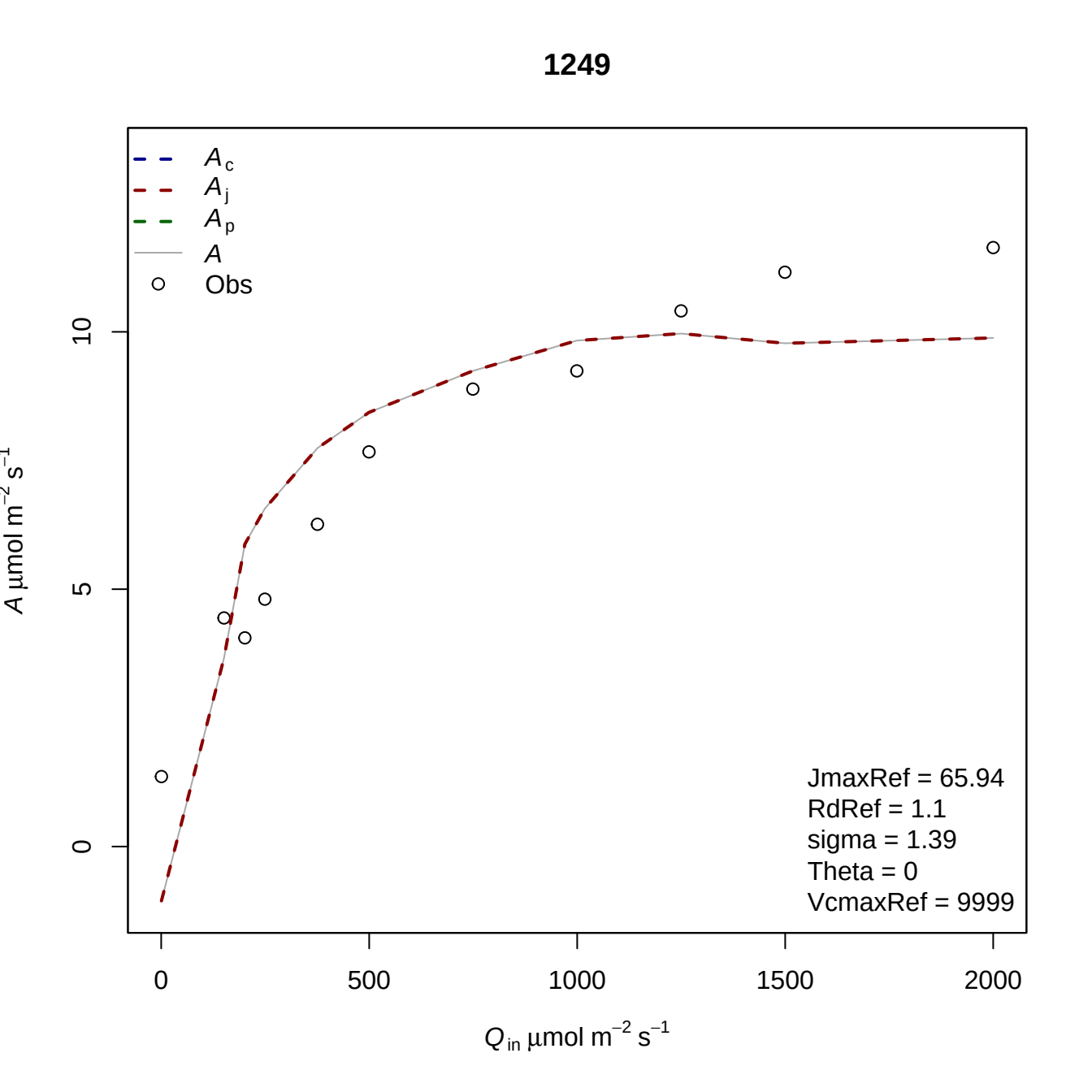

1244
A_c
A_j
A_p
A
Obs
JmaxRef = 122.39
RdRef = 4.78
sigma = 1.07
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

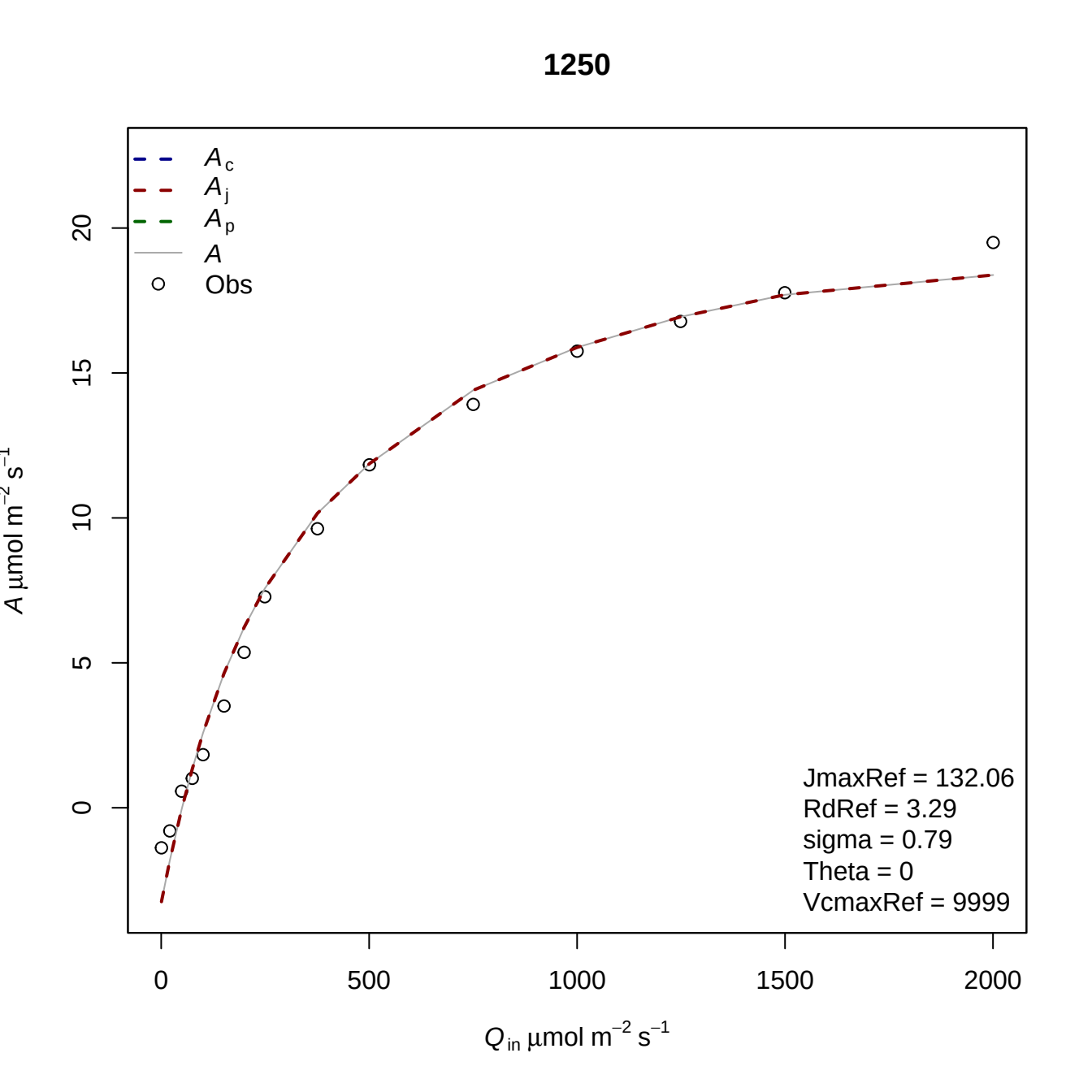

1245
A_c
A_j
A_p
A
Obs
JmaxRef = 119.84
RdRef = 3.45
sigma = 1.11
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

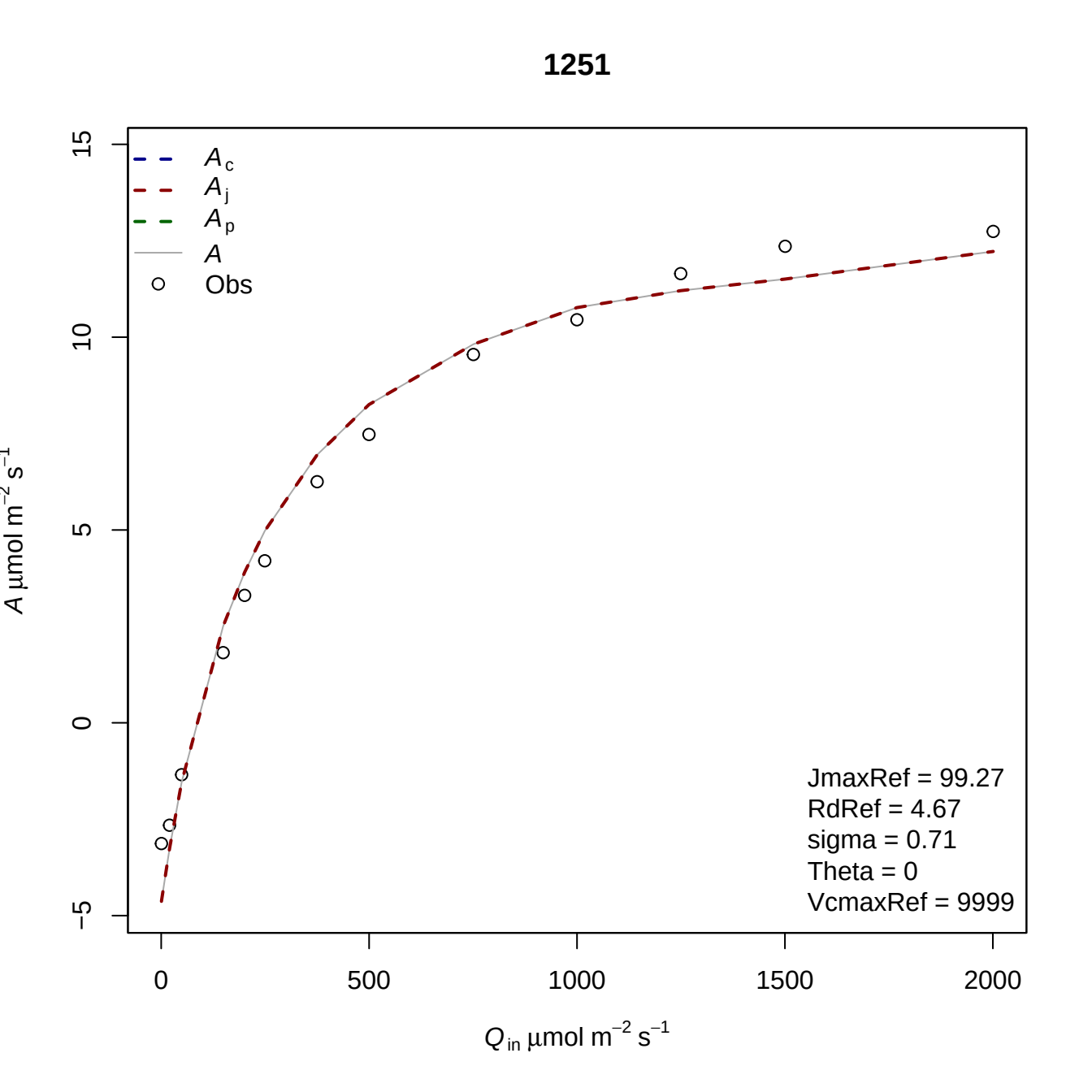

1246
A_c
A_j
A_p
A
Obs
JmaxRef = 95.3
RdRef = 3.38
sigma = 0.57
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

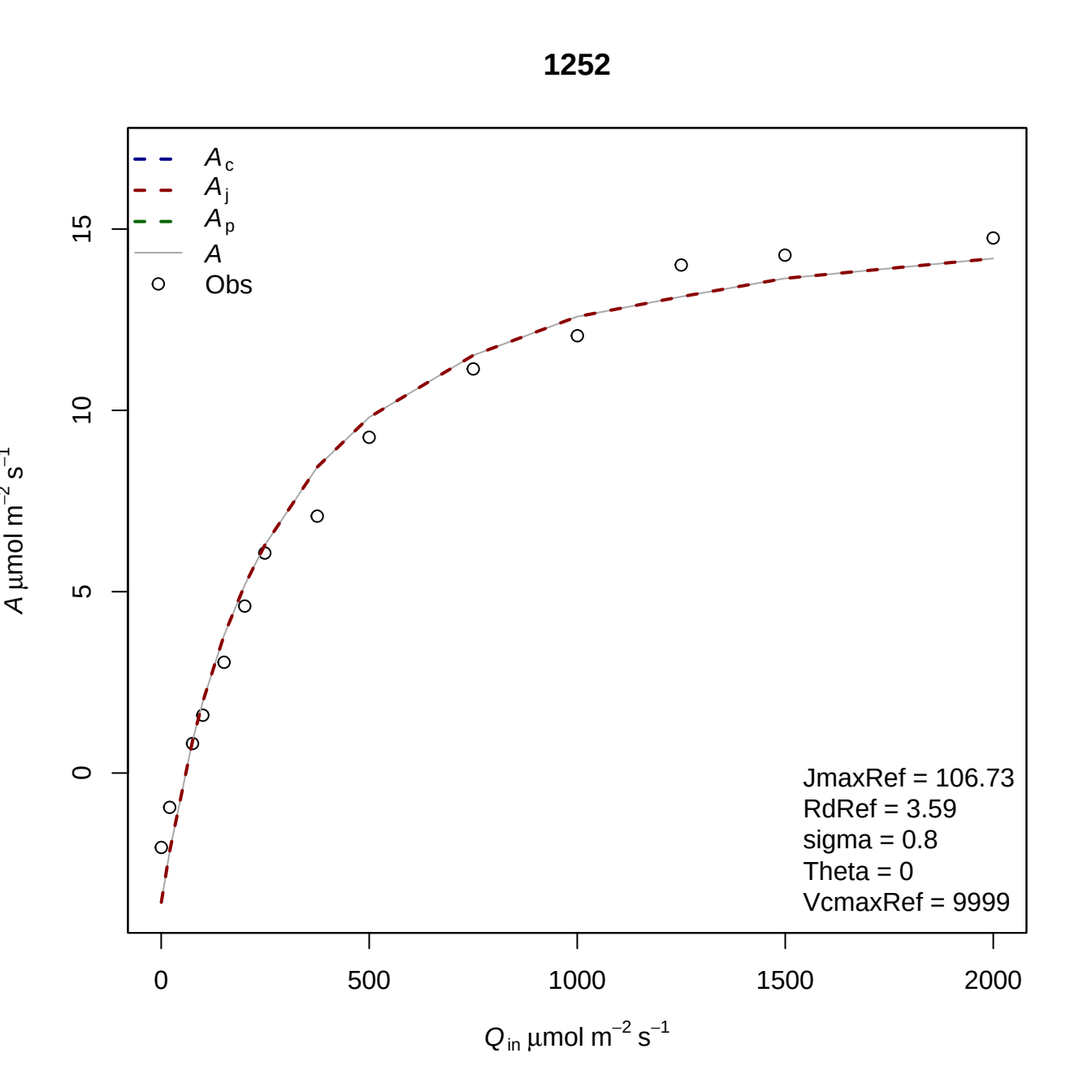

1247
A_c
A_j
A_p
A
Obs
JmaxRef = 98.14
RdRef = 3.08
sigma = 0.58
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

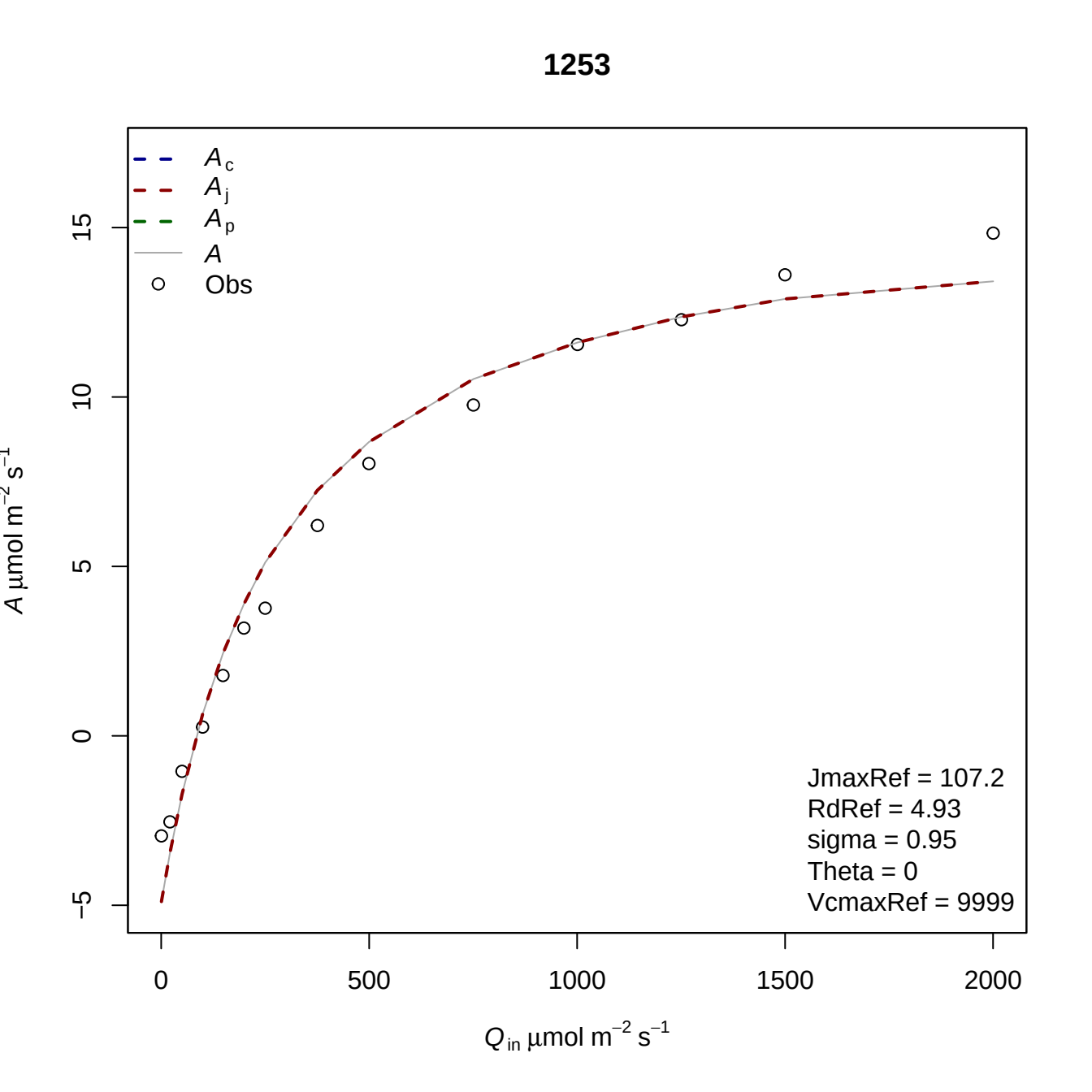

1248
A_c
A_j
A_p
A
Obs
JmaxRef = 117.24
RdRef = 4.51
sigma = 1.15
Theta = 0
VcmaxRef = 9999
20
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

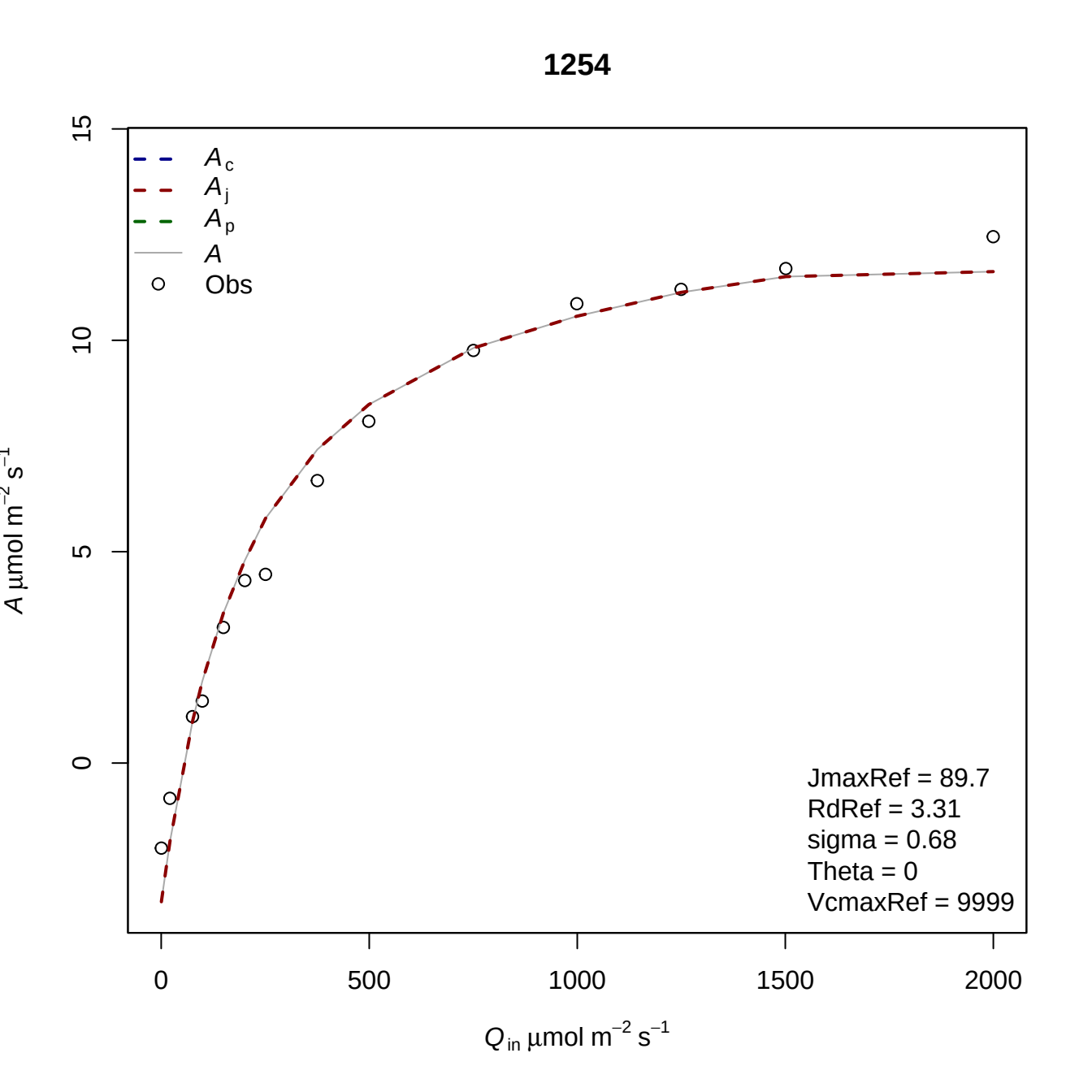

1249
A_c
A_j
A_p
A
Obs
JmaxRef = 65.94
RdRef = 1.1
sigma = 1.39
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

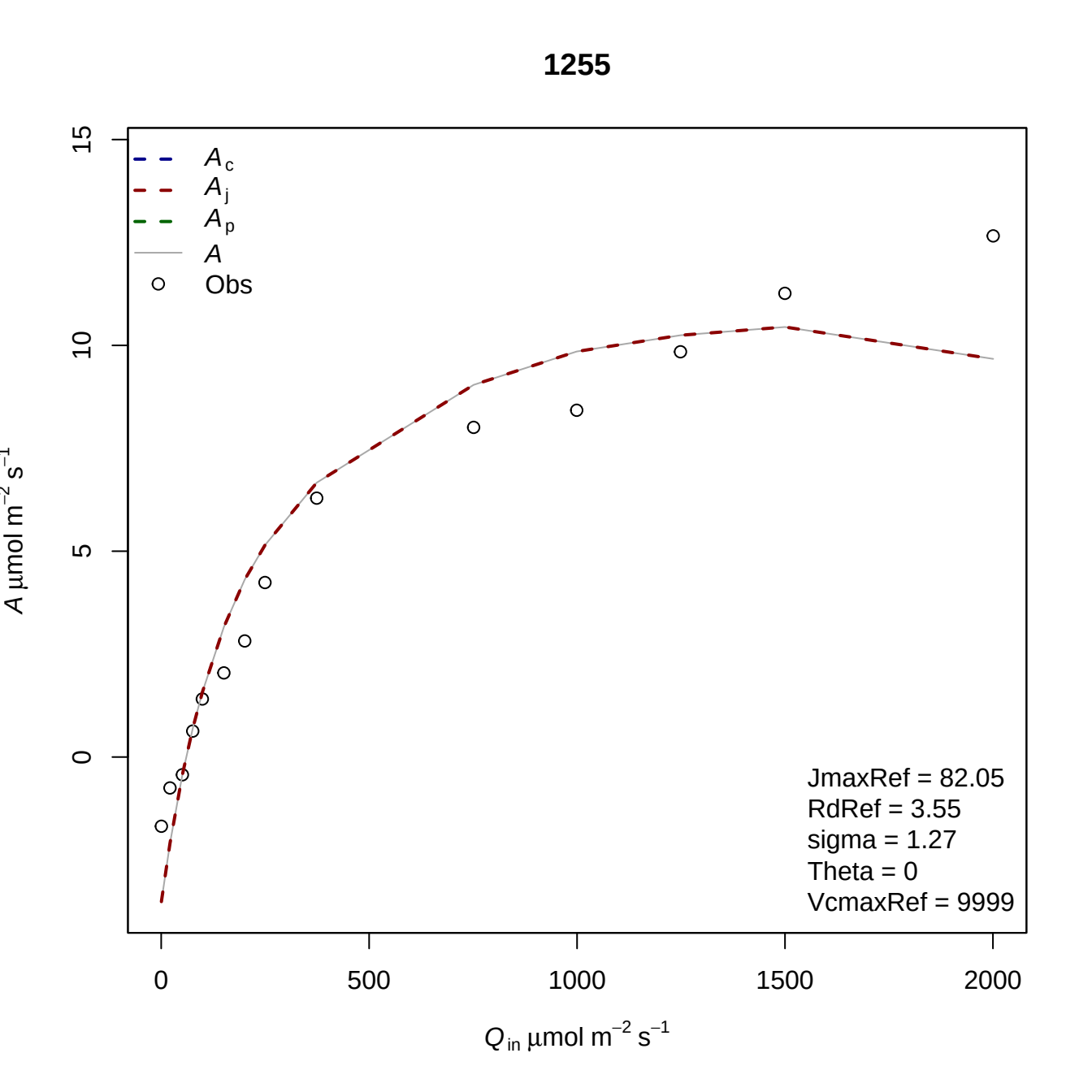

1250
A_c
A_j
A_p
A
Obs
JmaxRef = 132.06
RdRef = 3.29
sigma = 0.79
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

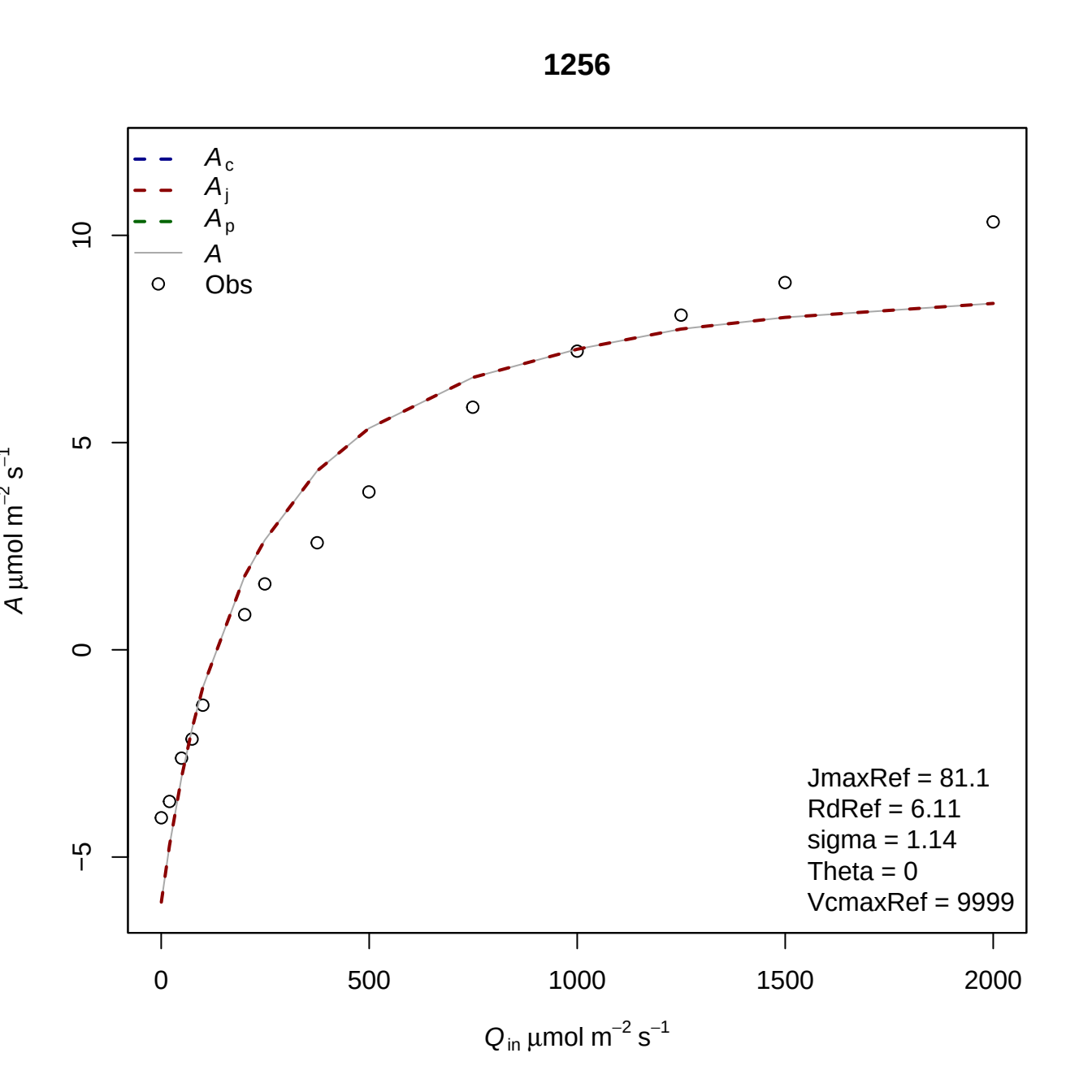

1251
A_c
A_j
A_p
A
Obs
JmaxRef = 99.27
RdRef = 4.67
sigma = 0.71
Theta = 0
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

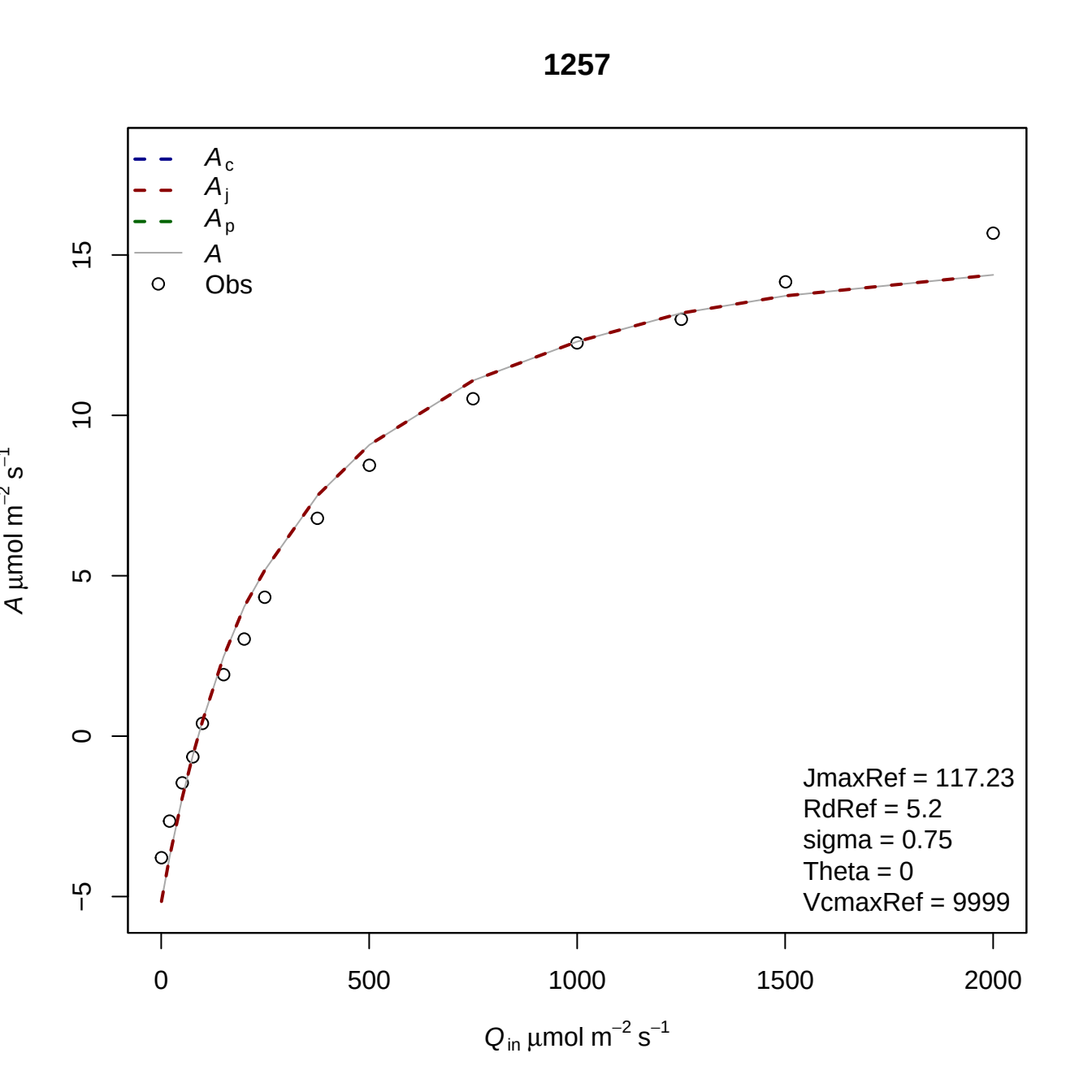

1252
A_c
A_j
A_p
A
Obs
JmaxRef = 106.73
RdRef = 3.59
sigma = 0.8
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

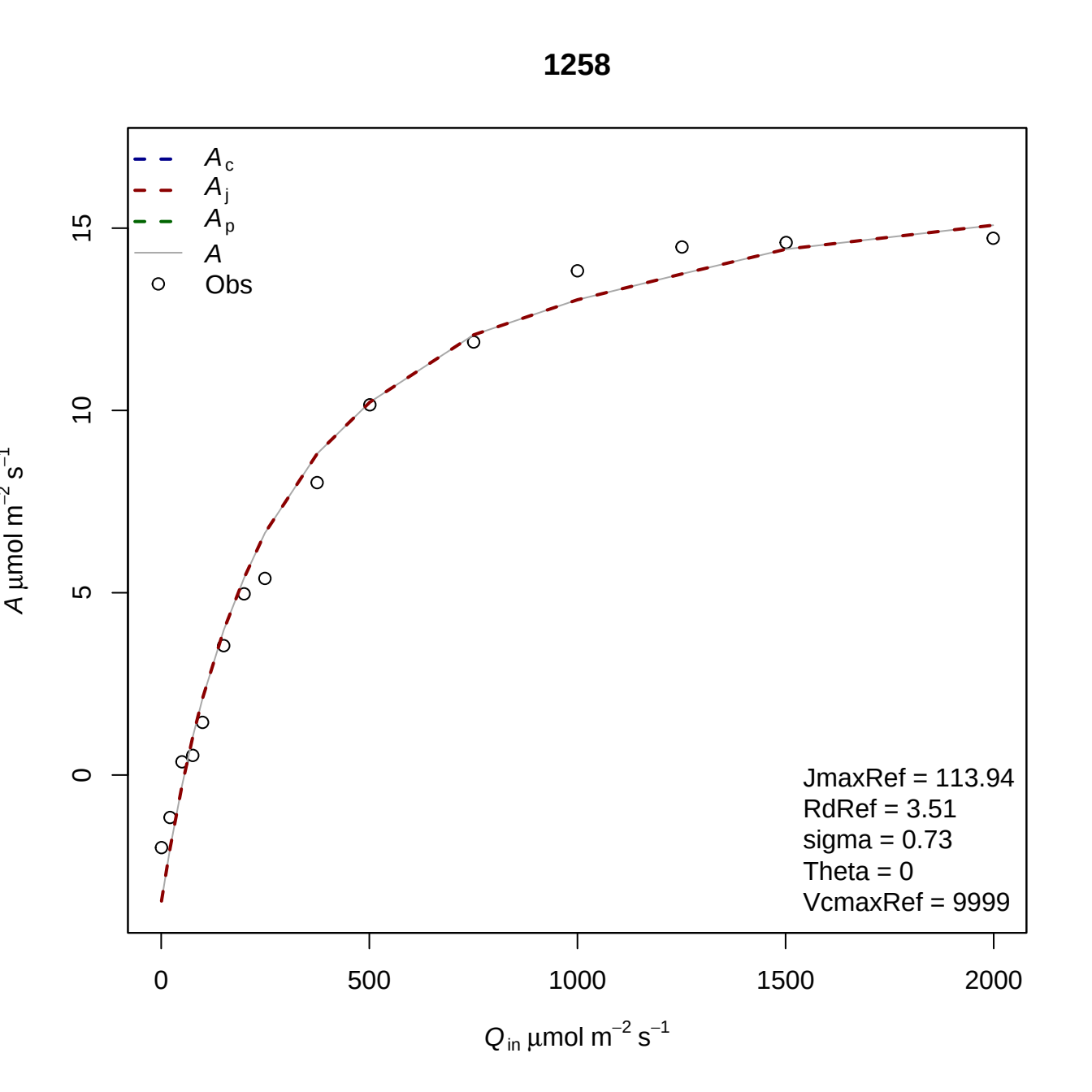

1253
A_c
A_j
A_p
A
Obs
JmaxRef = 107.2
RdRef = 4.93
sigma = 0.95
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

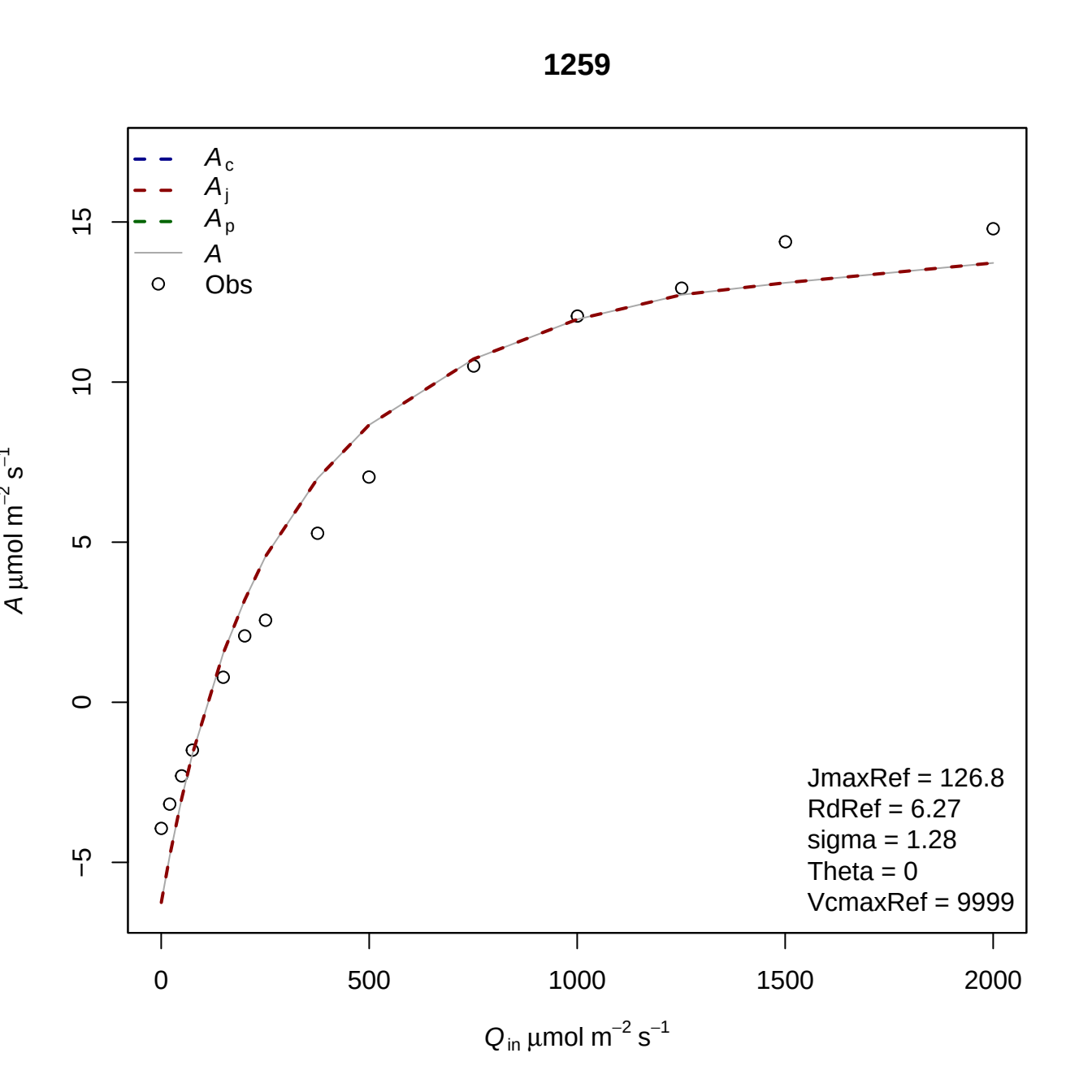

1254
A_c
A_j
A_p
A
Obs
JmaxRef = 89.7
RdRef = 3.31
sigma = 0.68
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

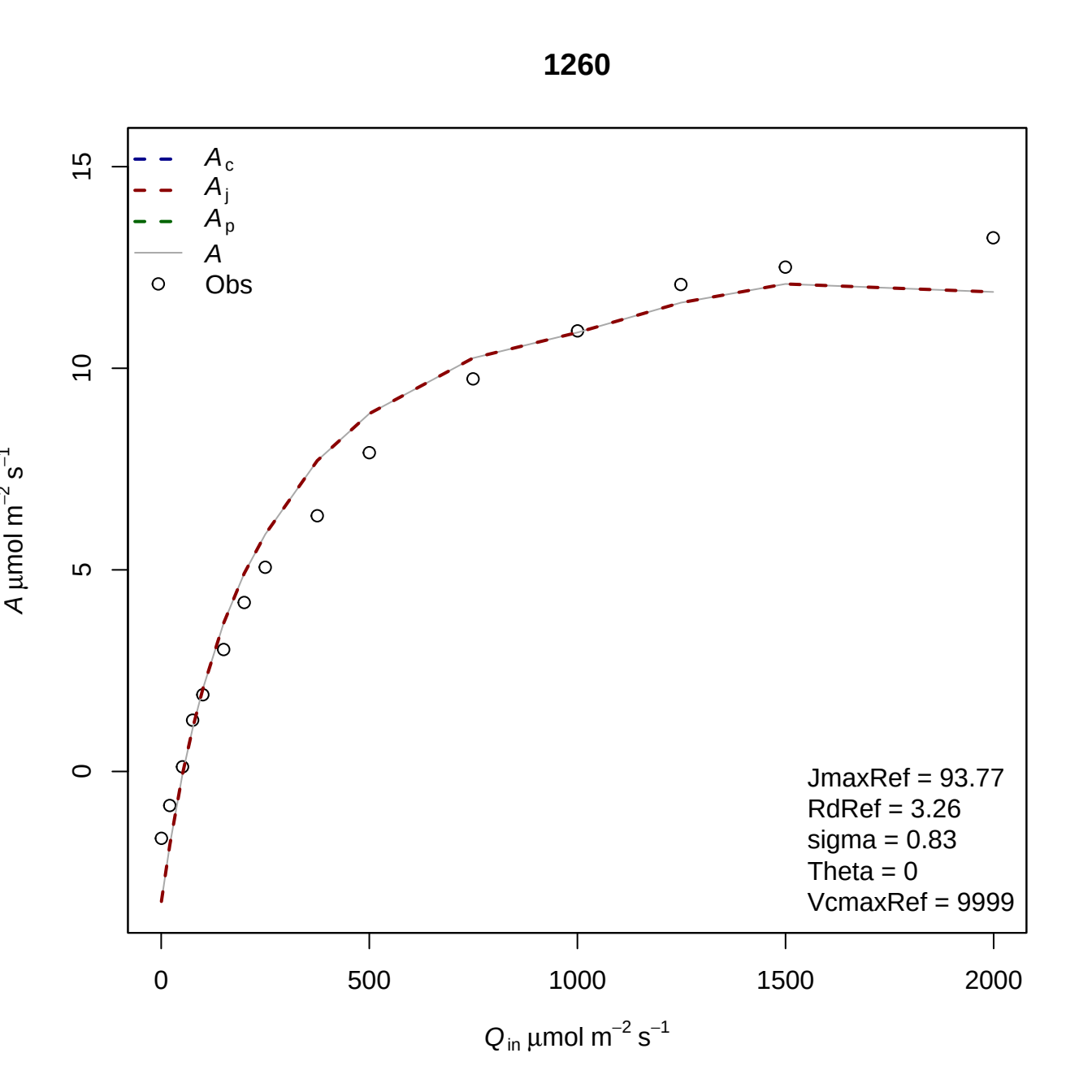

1255
A_c
A_j
A_p
A
Obs
JmaxRef = 82.05
RdRef = 3.55
sigma = 1.27
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

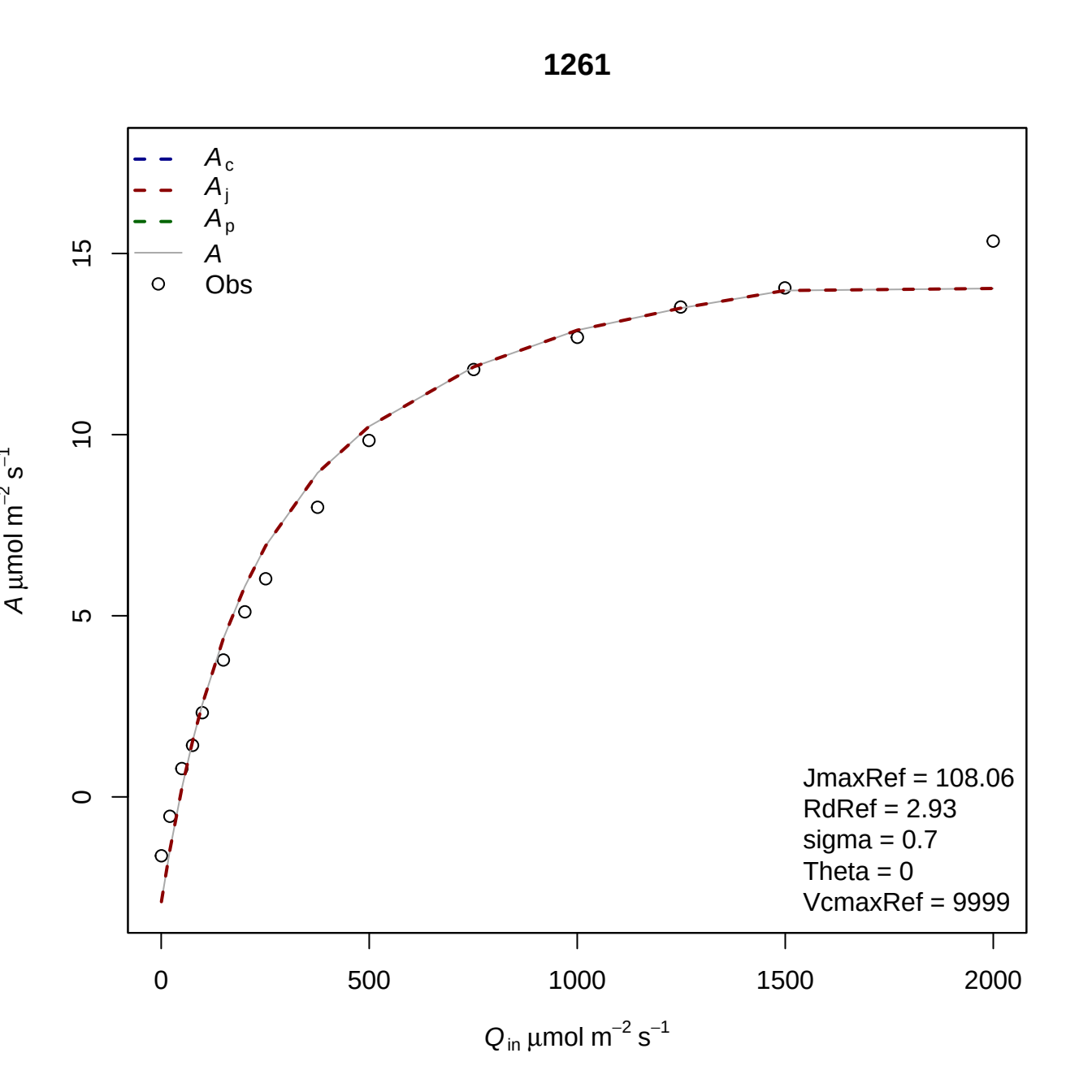

1256
A_c
A_j
A_p
A
Obs
JmaxRef = 81.1
RdRef = 6.11
sigma = 1.14
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

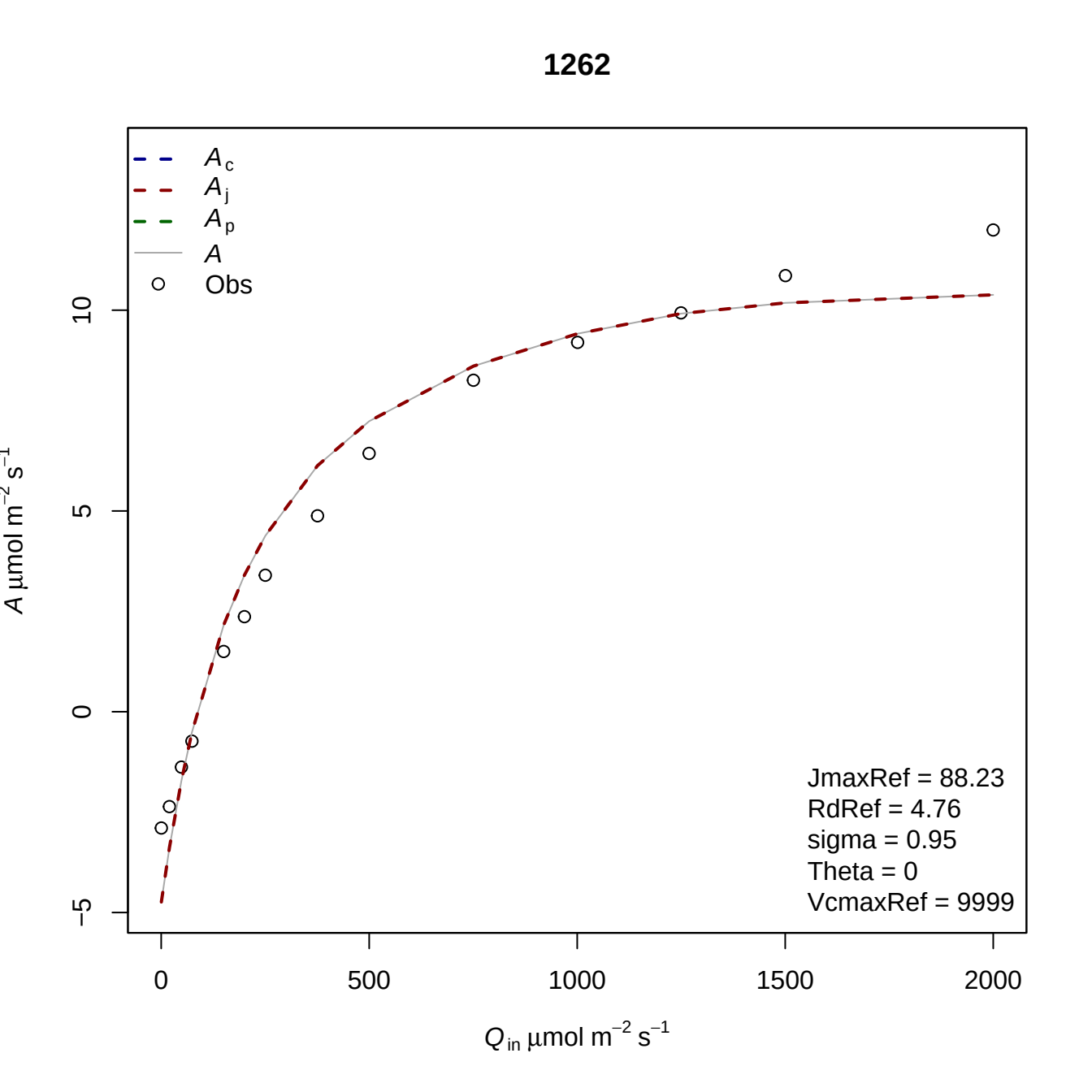

1257
A_c
A_j
A_p
A
Obs
JmaxRef = 117.23
RdRef = 5.2
sigma = 0.75
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

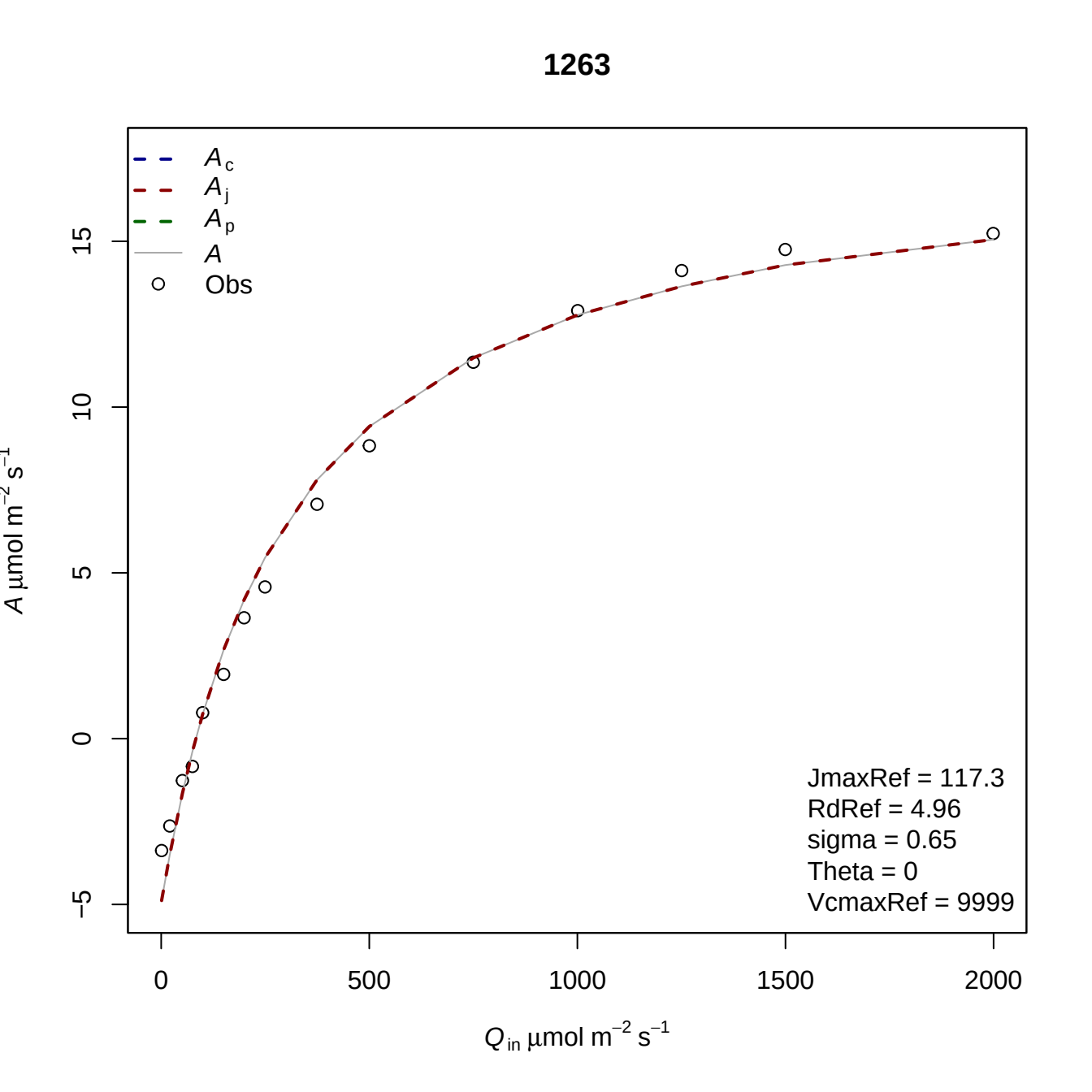

1258
A_c
A_j
A_p
A
Obs
JmaxRef = 113.94
RdRef = 3.51
sigma = 0.73
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

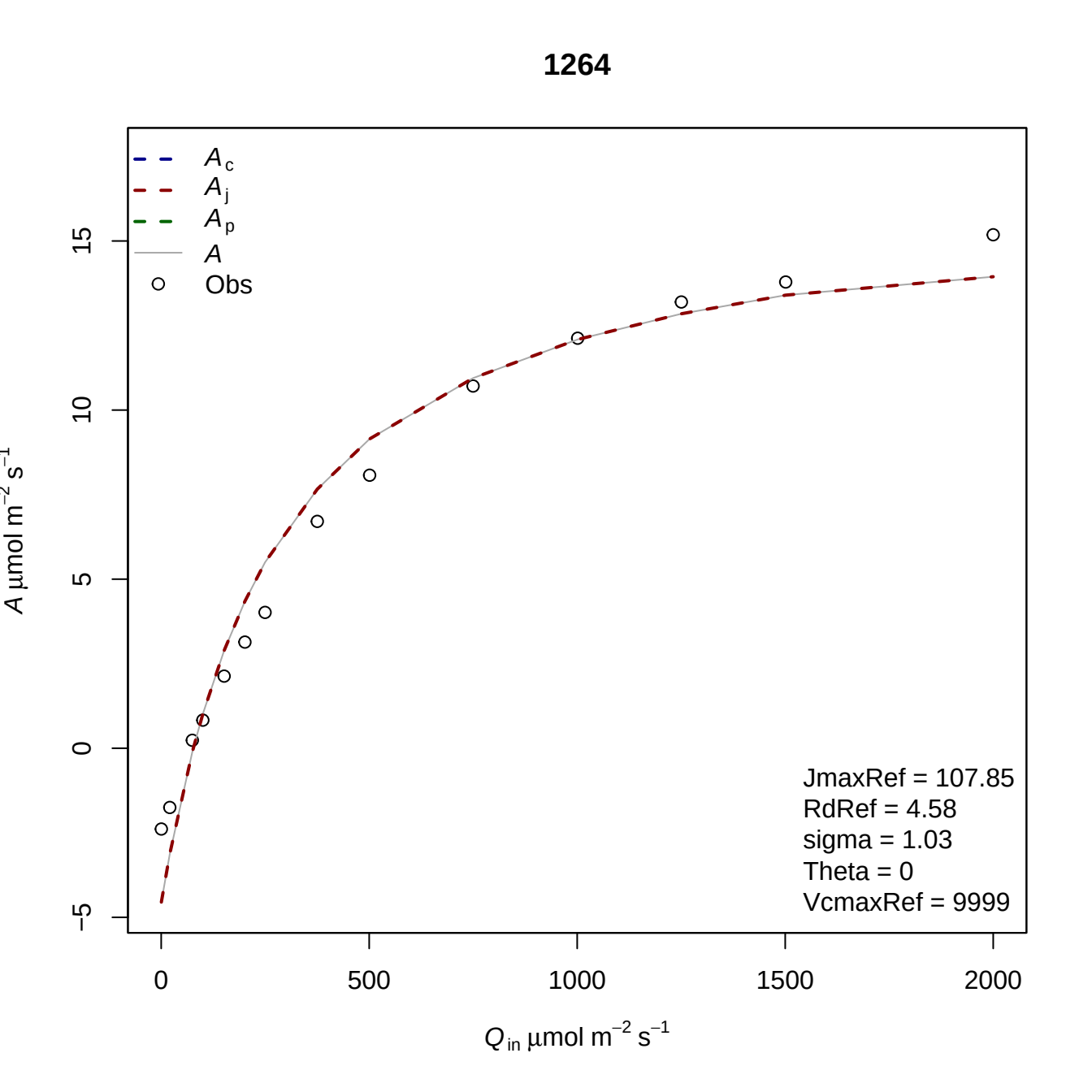

1259
A_c
A_j
A_p
A
Obs
JmaxRef = 126.8
RdRef = 6.27
sigma = 1.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

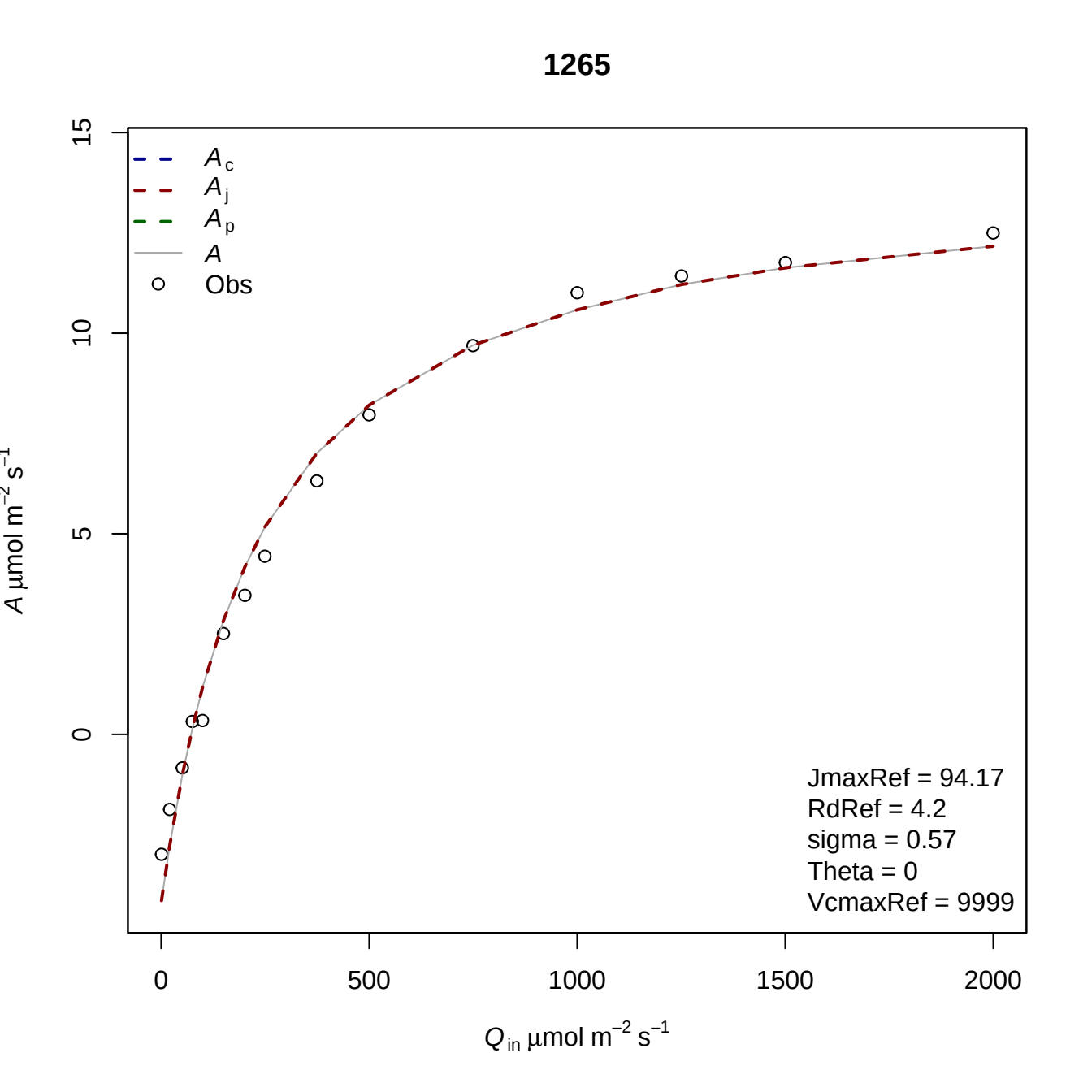

1260
A_c
A_j
A_p
A
Obs
JmaxRef = 93.77
RdRef = 3.26
sigma = 0.83
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

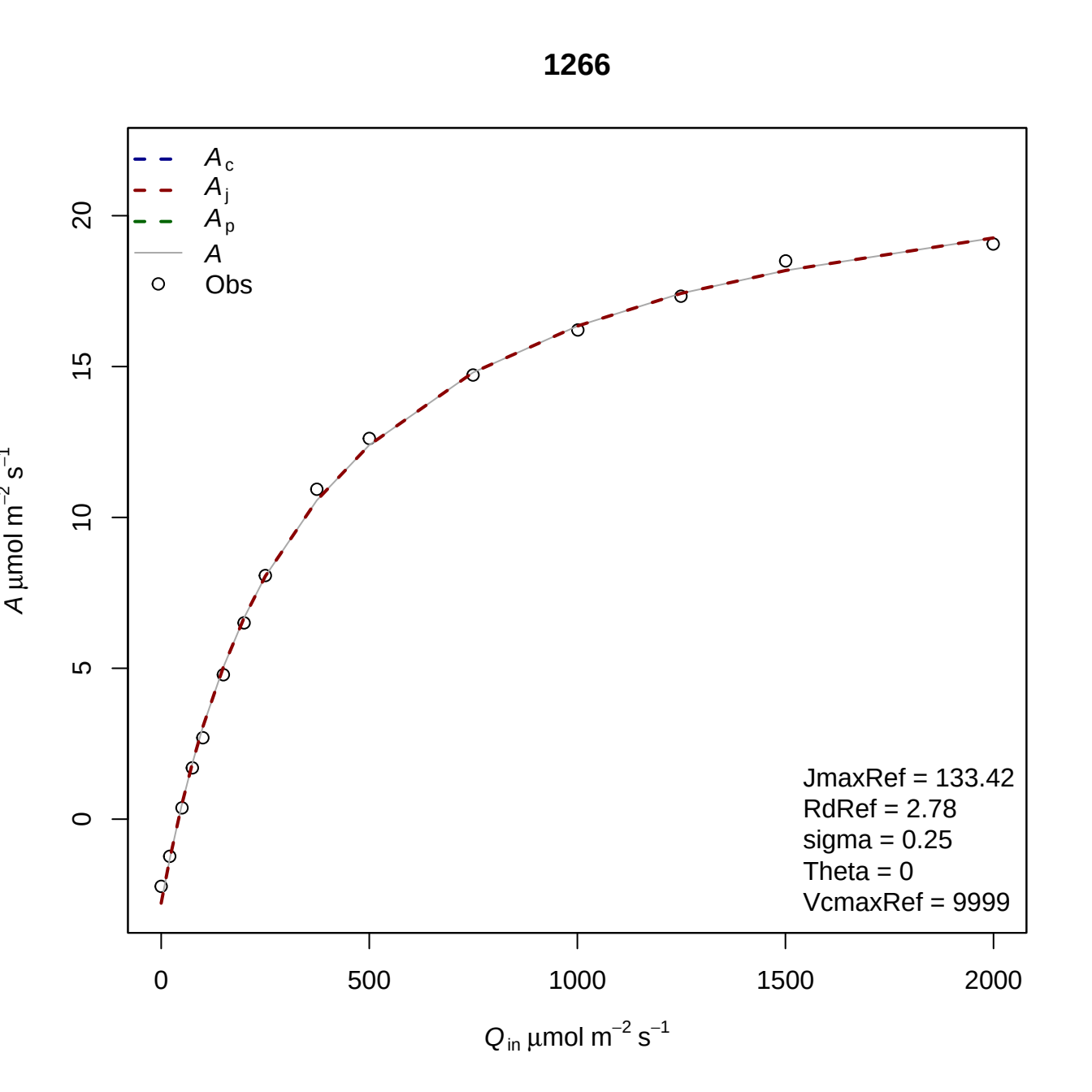

1261
A_c
A_j
A_p
A
Obs
JmaxRef = 108.06
RdRef = 2.93
sigma = 0.7
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

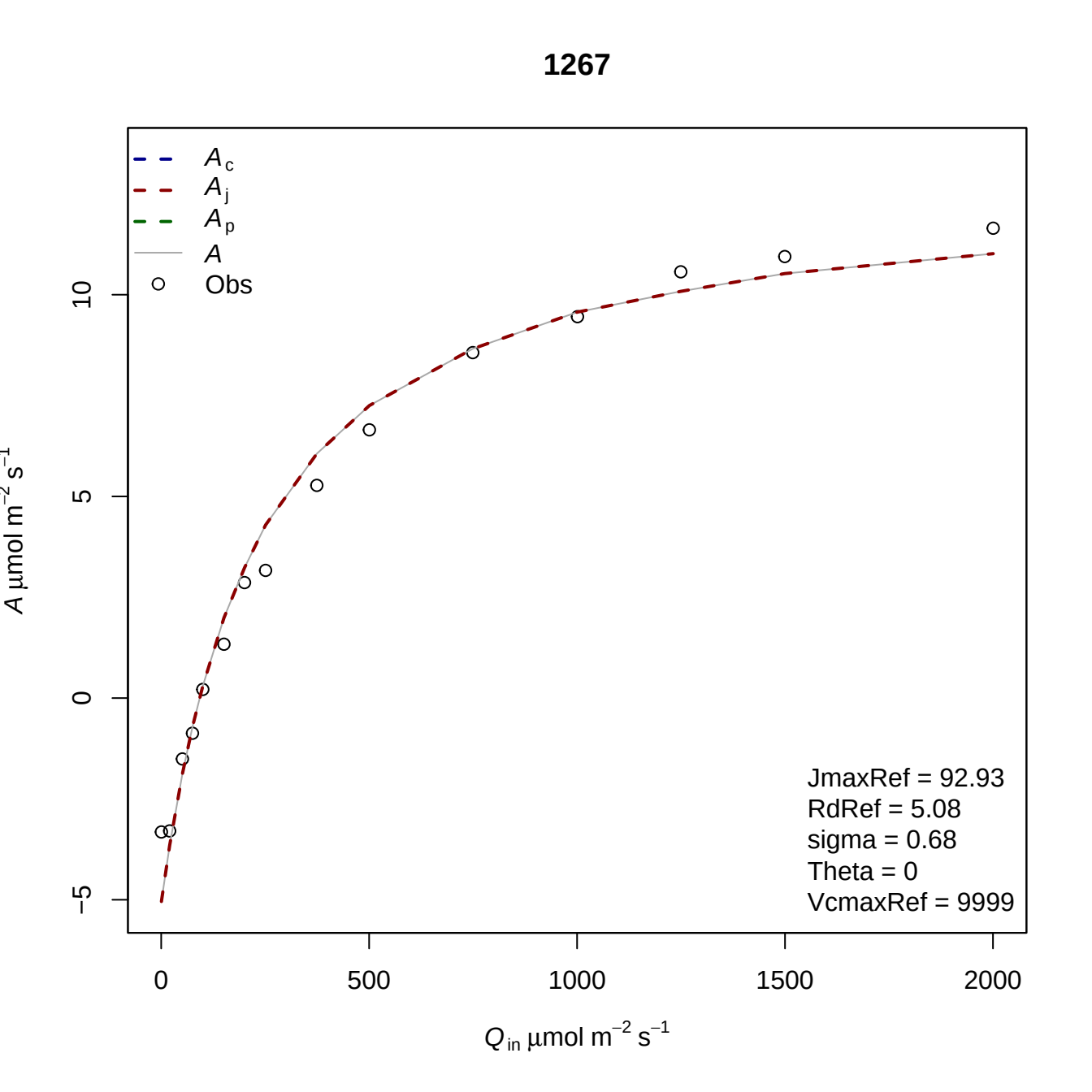

1262
A_c
A_j
A_p
A
Obs
JmaxRef = 88.23
RdRef = 4.76
sigma = 0.95
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

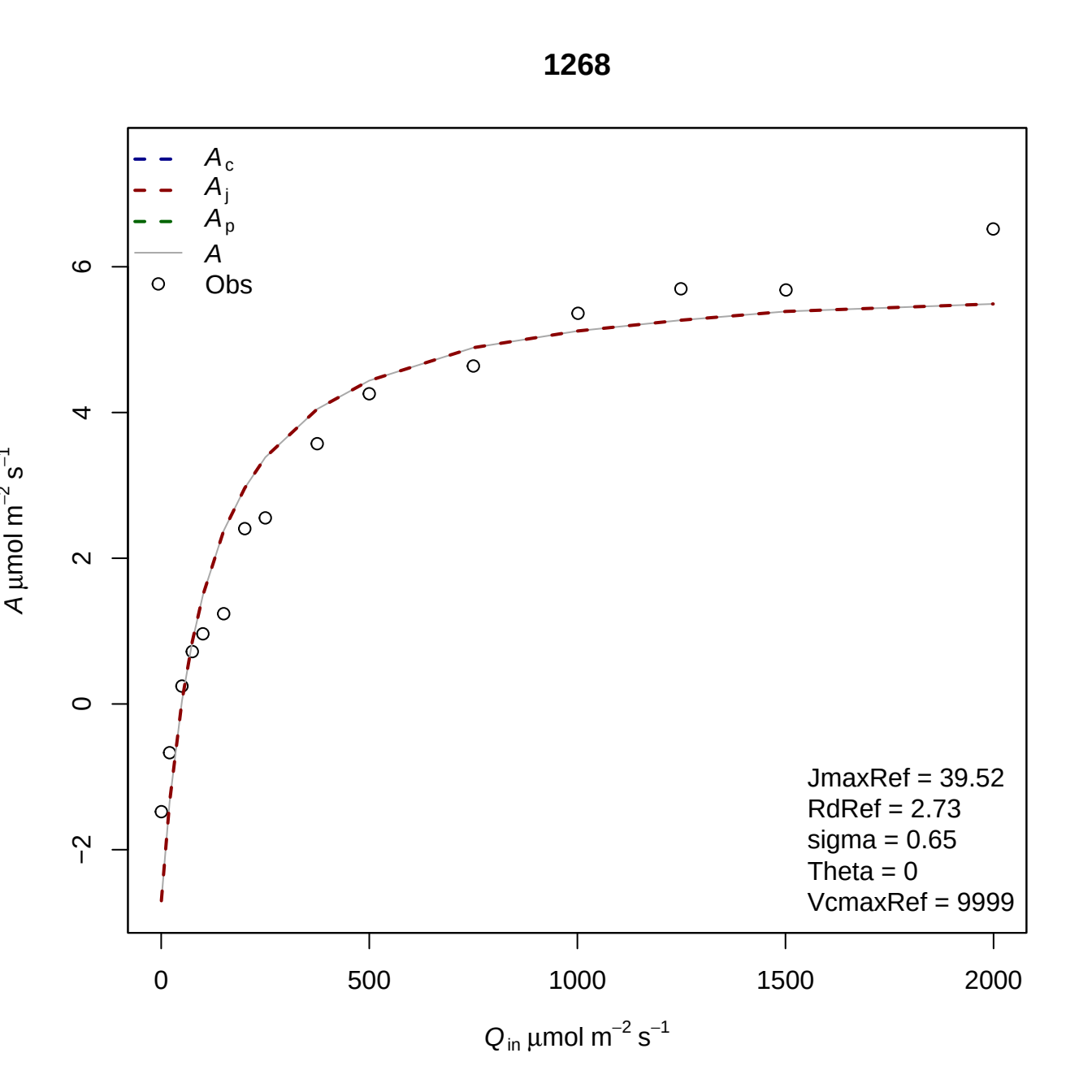

1263
A_c
A_j
A_p
A
Obs
JmaxRef = 117.3
RdRef = 4.96
sigma = 0.65
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

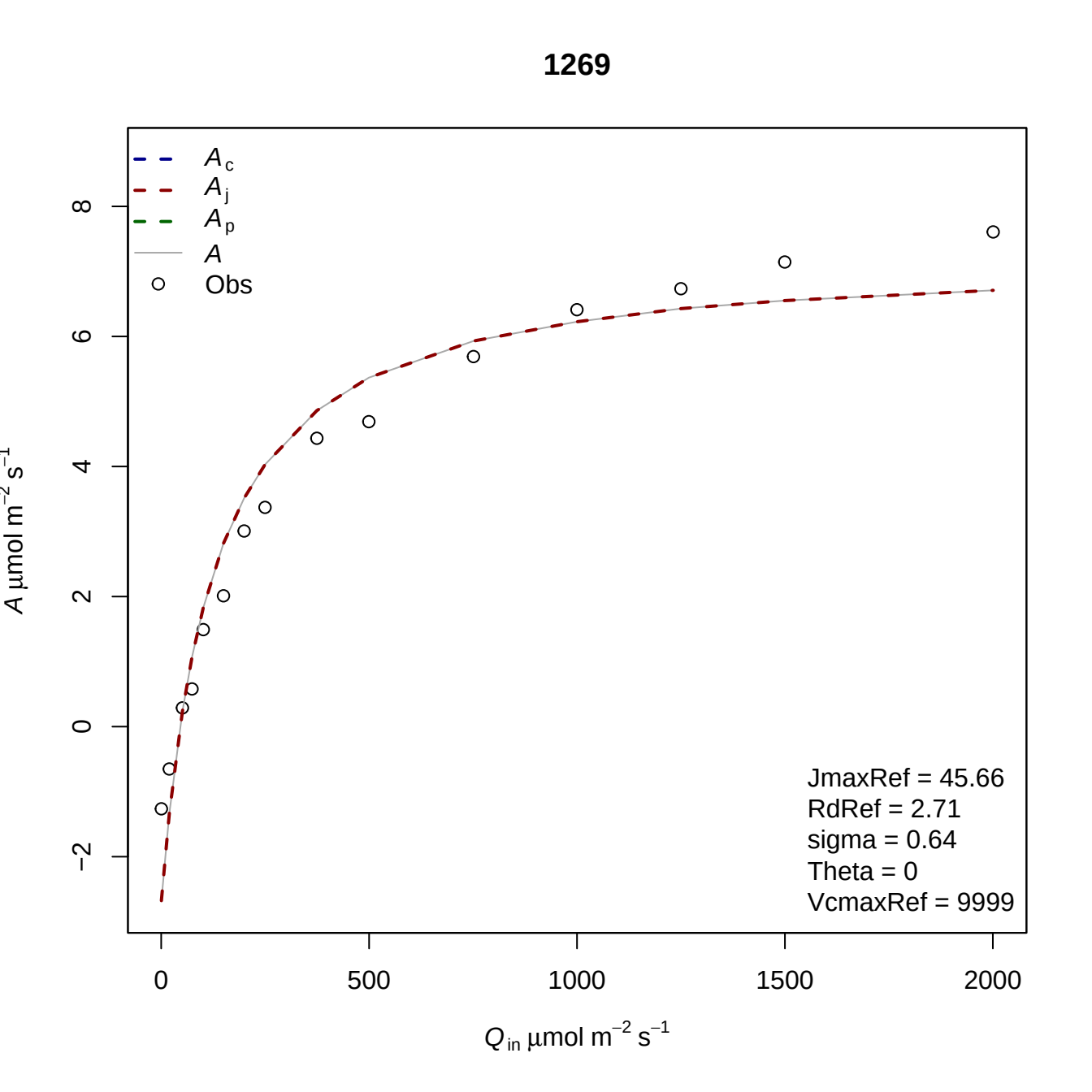

1264
A_c
A_j
A_p
A
Obs
JmaxRef = 107.85
RdRef = 4.58
sigma = 1.03
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

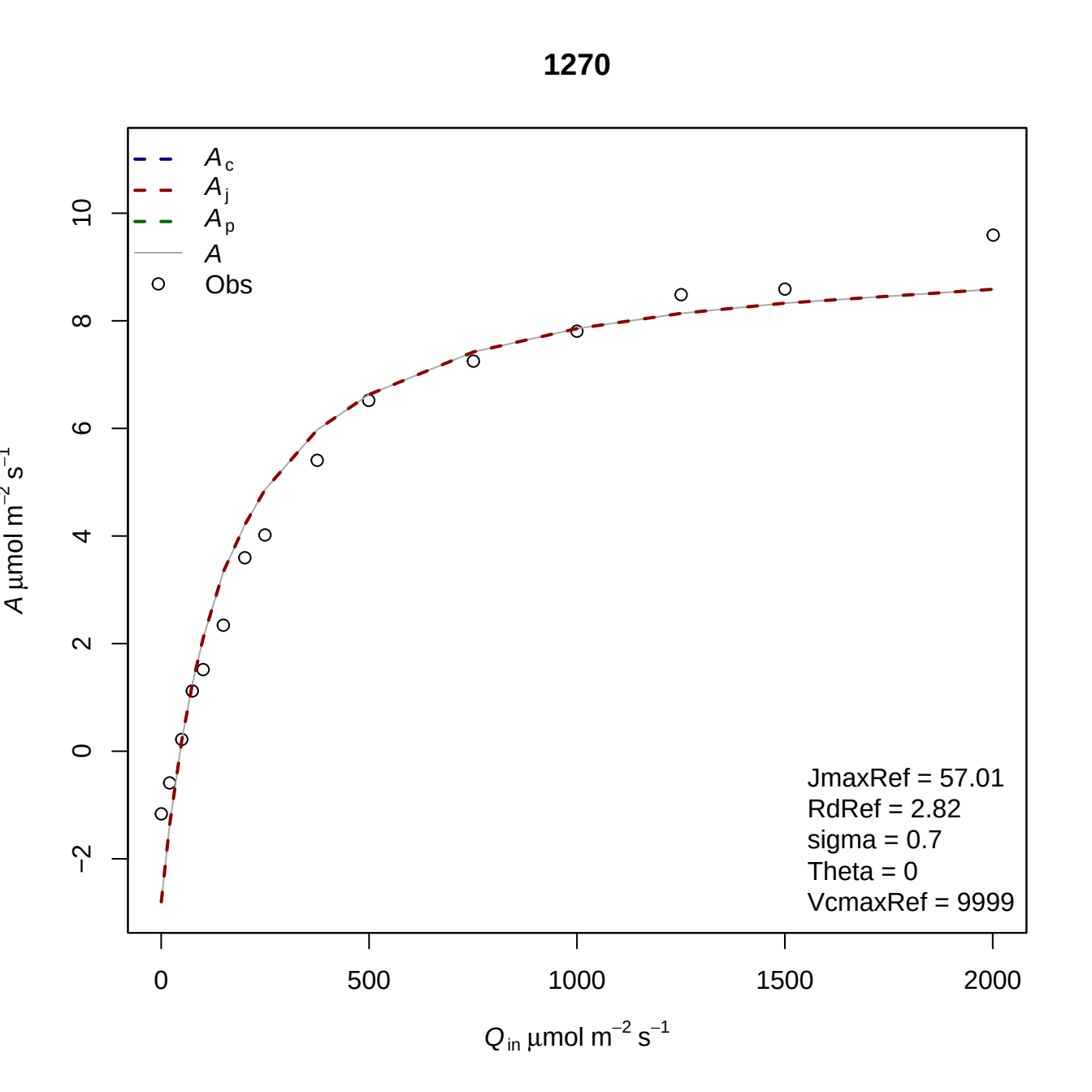

1265
A_c
A_j
A_p
A
Obs
JmaxRef = 94.17
RdRef = 4.2
sigma = 0.57
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

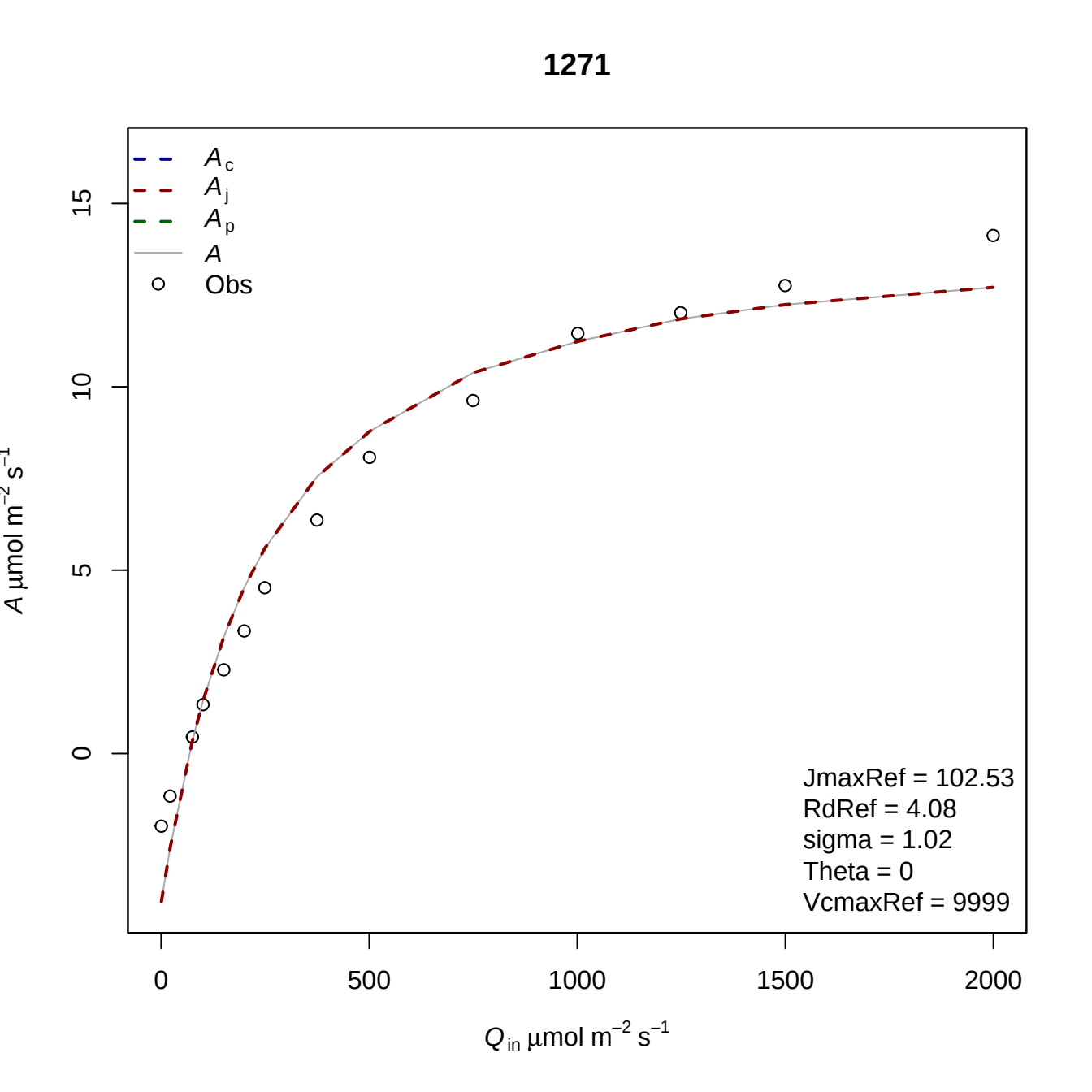

1266
A_c
A_j
A_p
A
Obs
JmaxRef = 133.42
RdRef = 2.78
sigma = 0.25
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

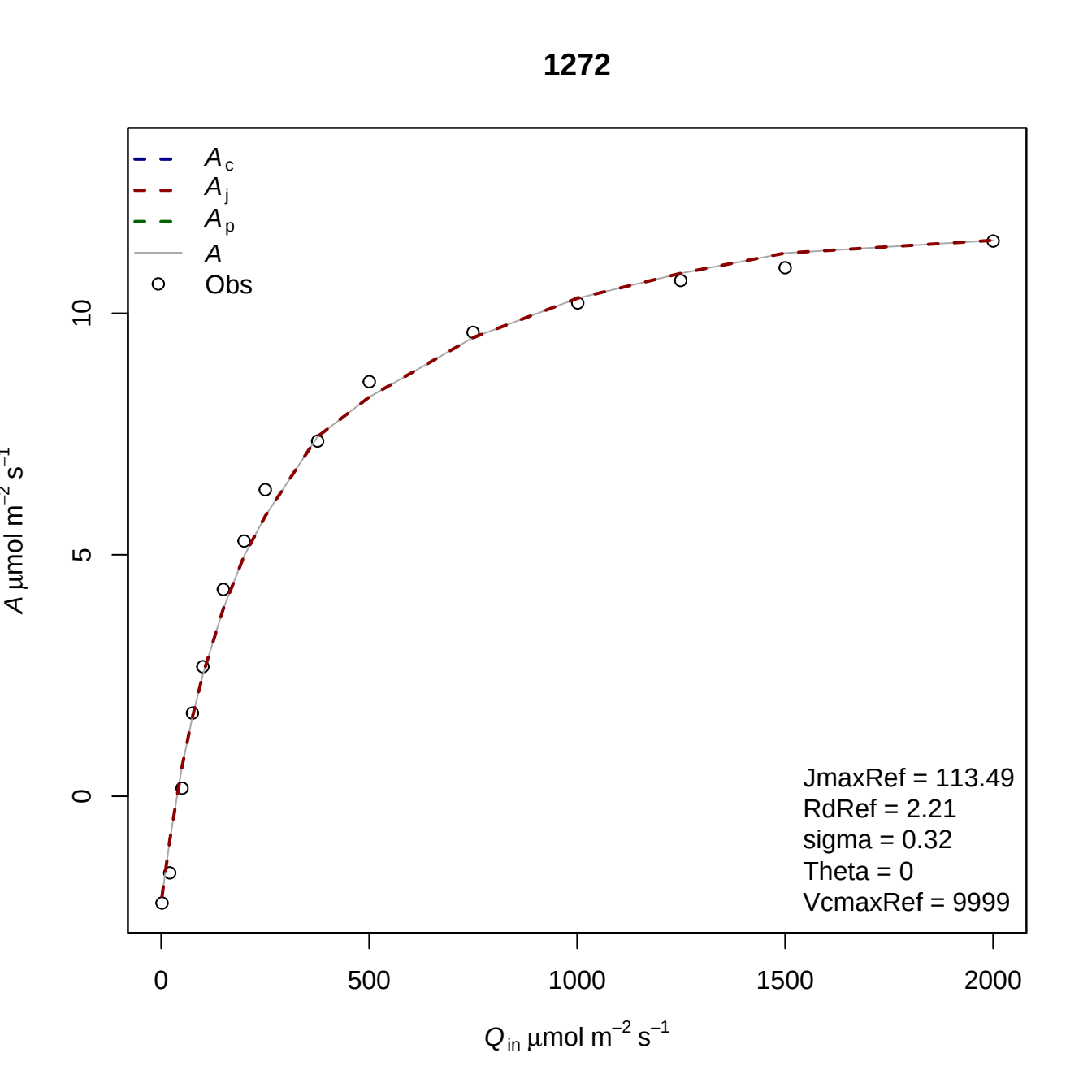

1267
A_c
A_j
A_p
A
Obs
JmaxRef = 92.93
RdRef = 5.08
sigma = 0.68
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

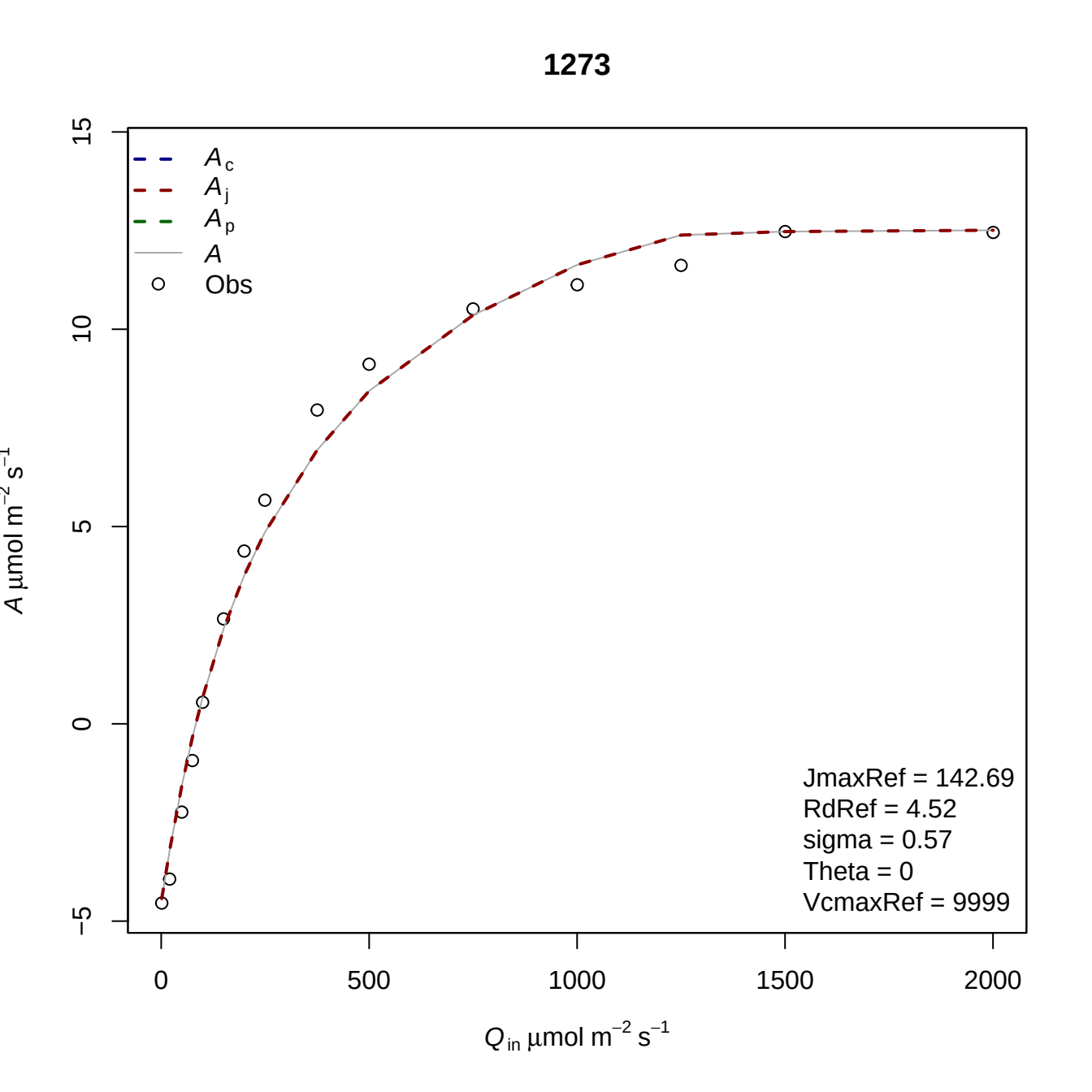

1268
A_c
A_j
A_p
A
Obs
JmaxRef = 39.52
RdRef = 2.73
sigma = 0.65
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

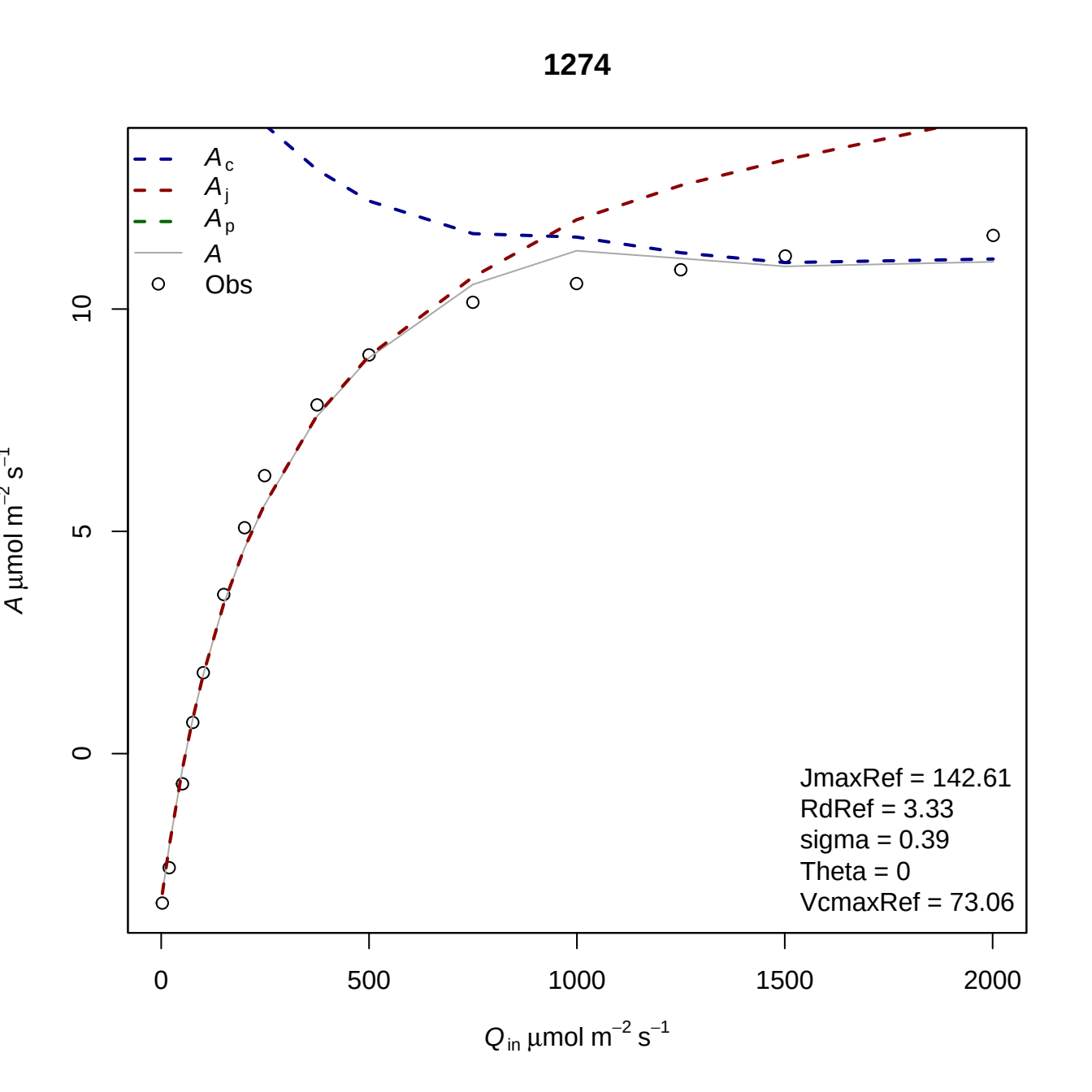

1269
A_c
A_j
A_p
A
Obs
JmaxRef = 45.66
RdRef = 2.71
sigma = 0.64
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

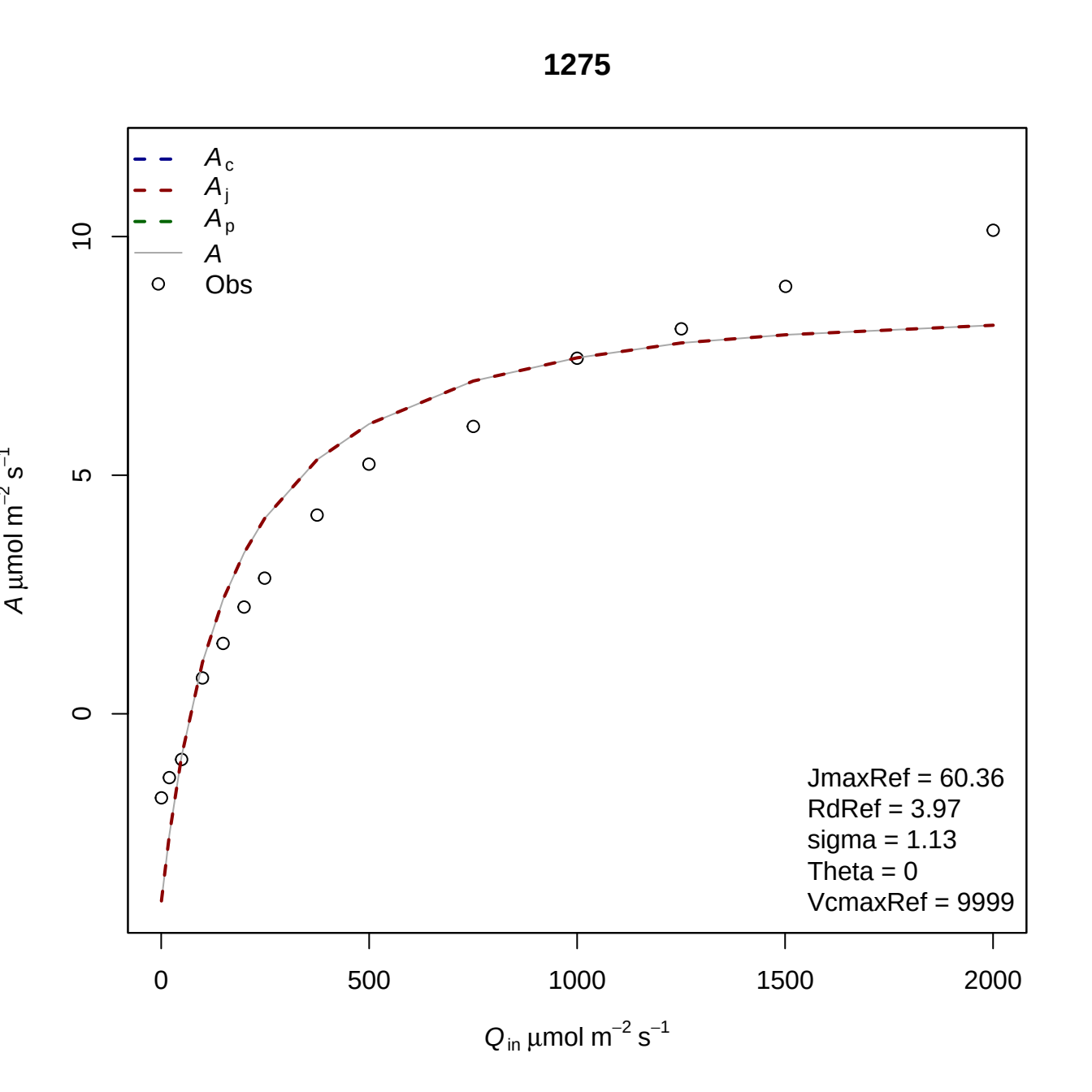

1270
A_c
A_j
A_p
A
Obs
JmaxRef = 57.01
RdRef = 2.82
sigma = 0.7
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

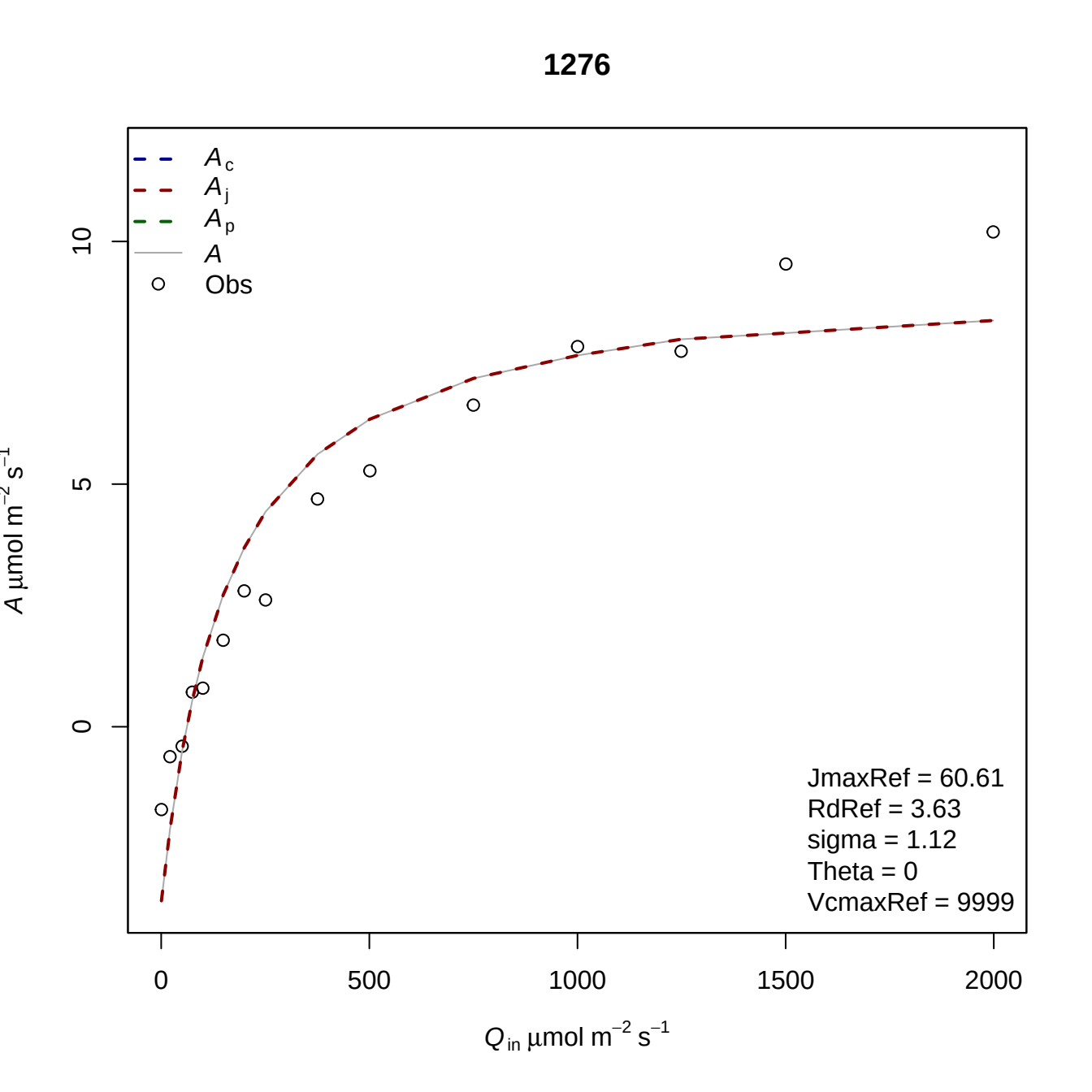

1271
A_c
A_j
A_p
A
Obs
JmaxRef = 102.53
RdRef = 4.08
sigma = 1.02
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

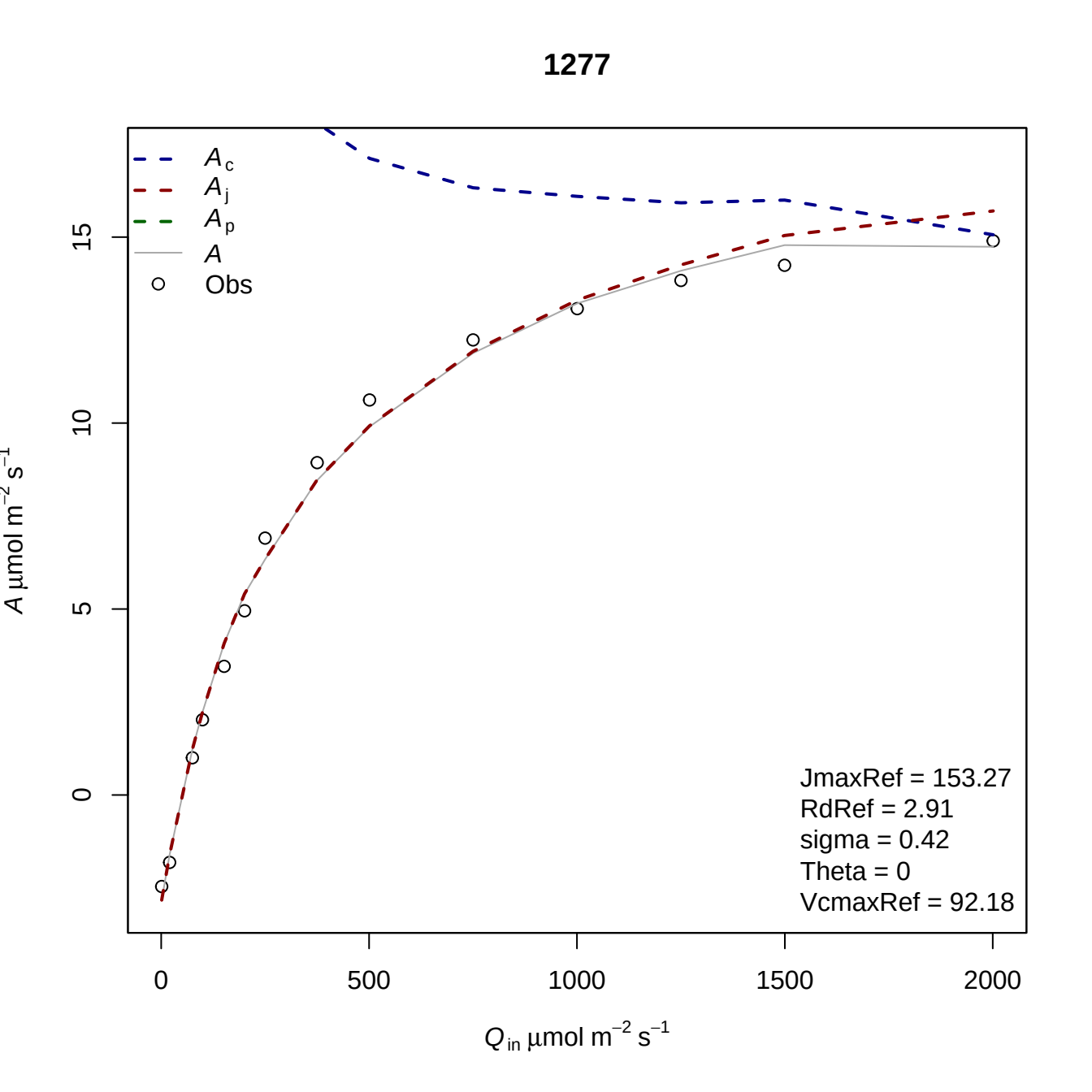

1272
A_c
A_j
A_p
A
Obs
JmaxRef = 113.49
RdRef = 2.21
sigma = 0.32
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

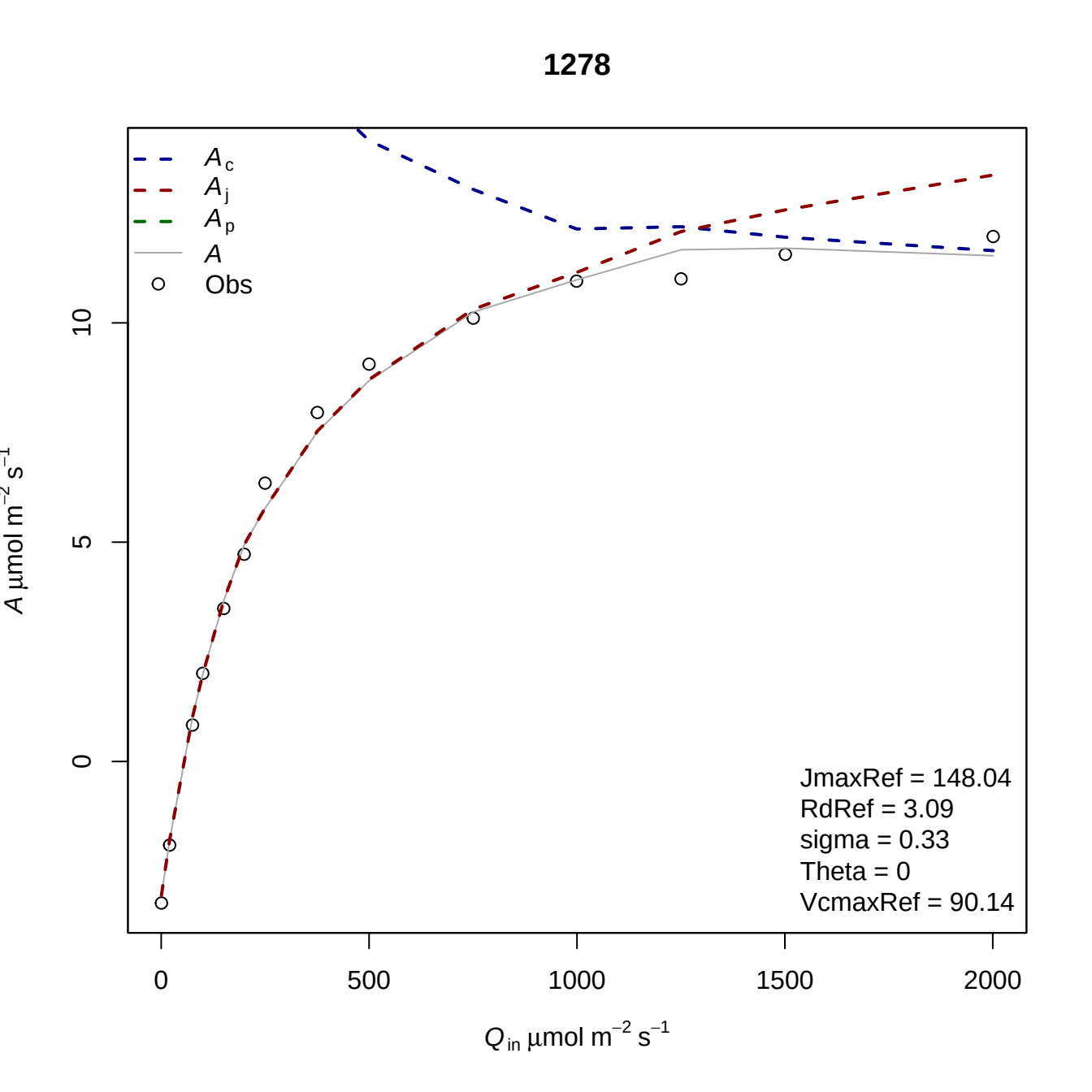

1273
A_c
A_j
A_p
A
Obs
JmaxRef = 142.69
RdRef = 4.52
sigma = 0.57
Theta = 0
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

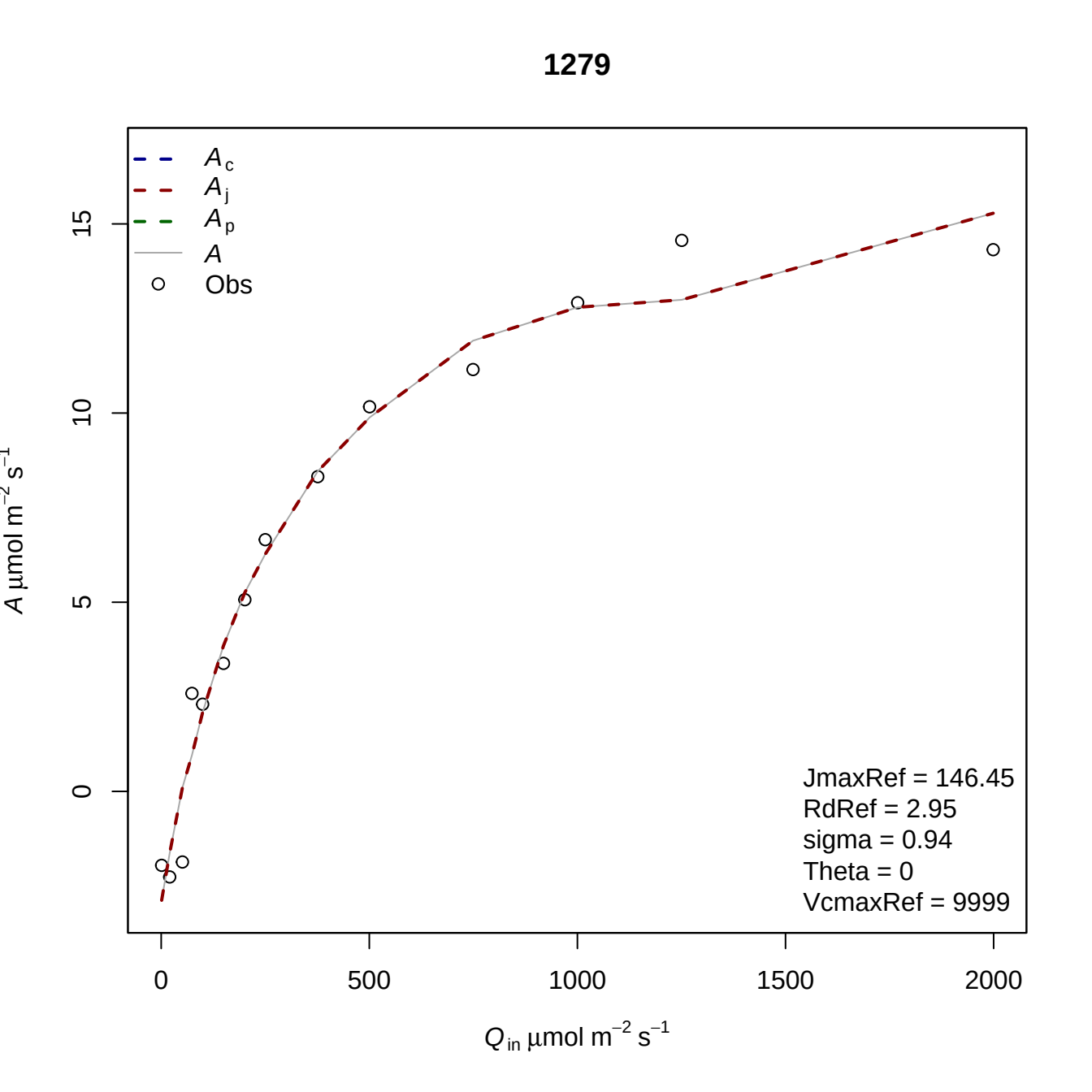

1274
A_c
A_j
A_p
A
Obs
JmaxRef = 142.61
RdRef = 3.33
sigma = 0.39
Theta = 0
VcmaxRef = 73.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

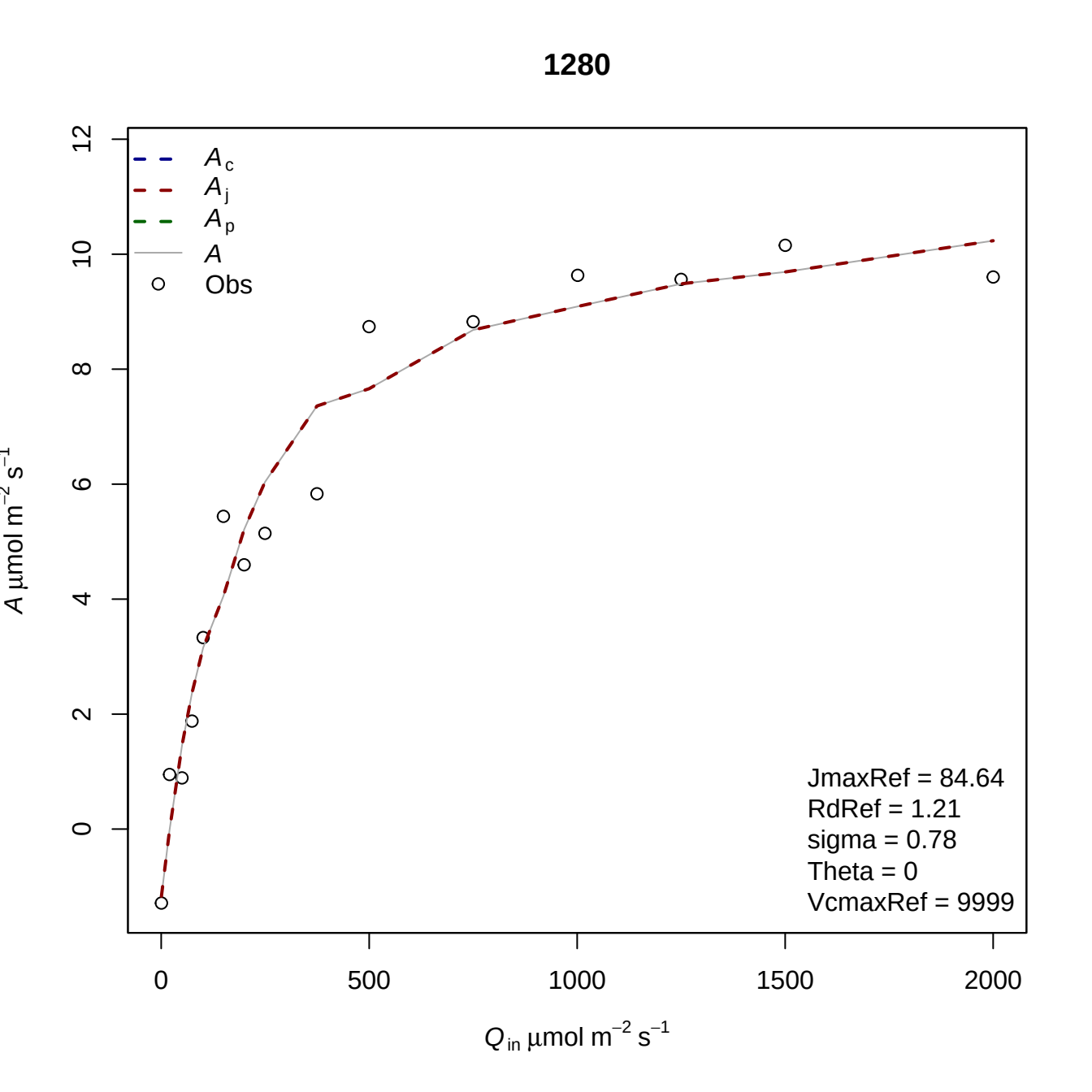

1275
A_c
A_j
A_p
A
Obs
JmaxRef = 60.36
RdRef = 3.97
sigma = 1.13
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

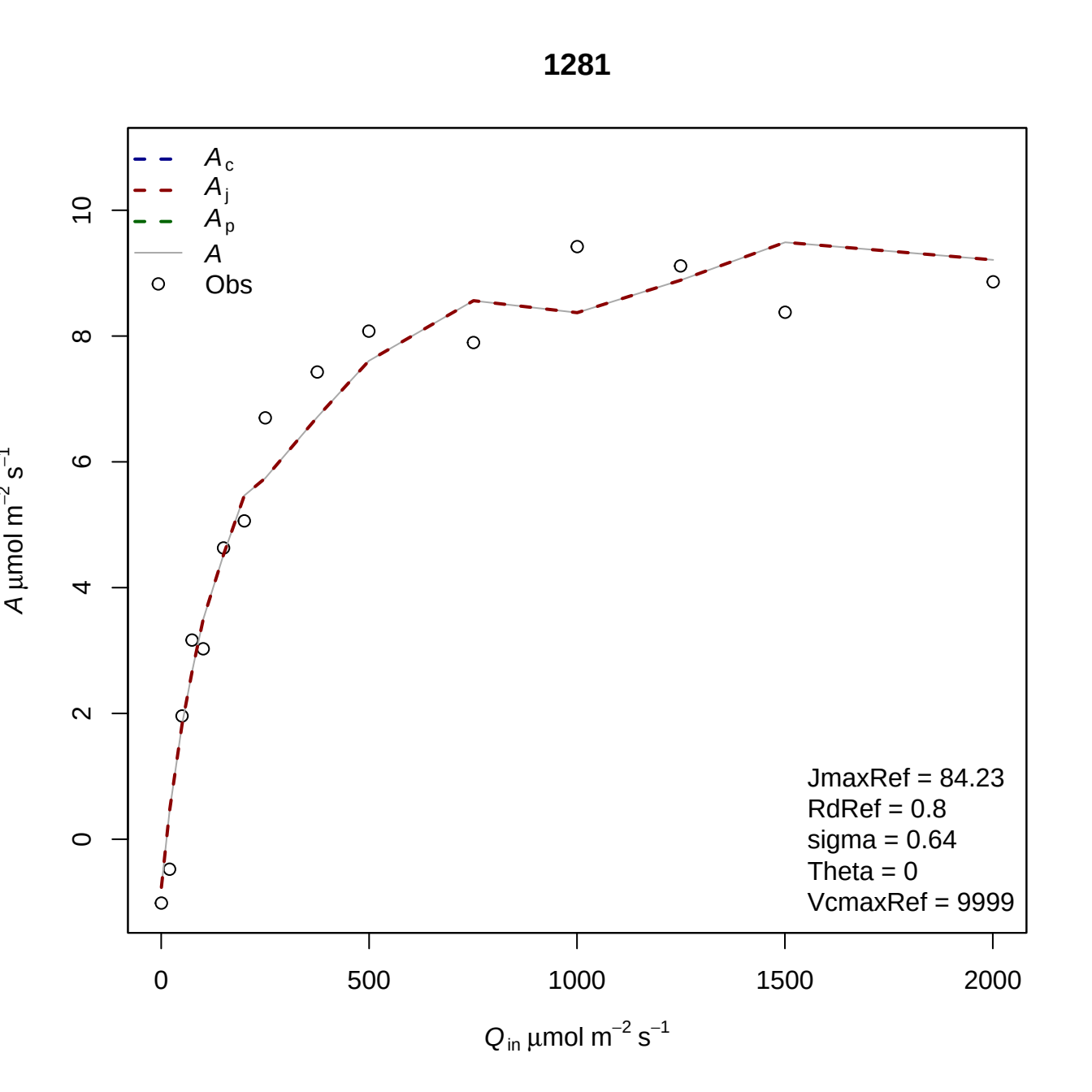

1276
A_c
A_j
A_p
A
Obs
JmaxRef = 60.61
RdRef = 3.63
sigma = 1.12
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

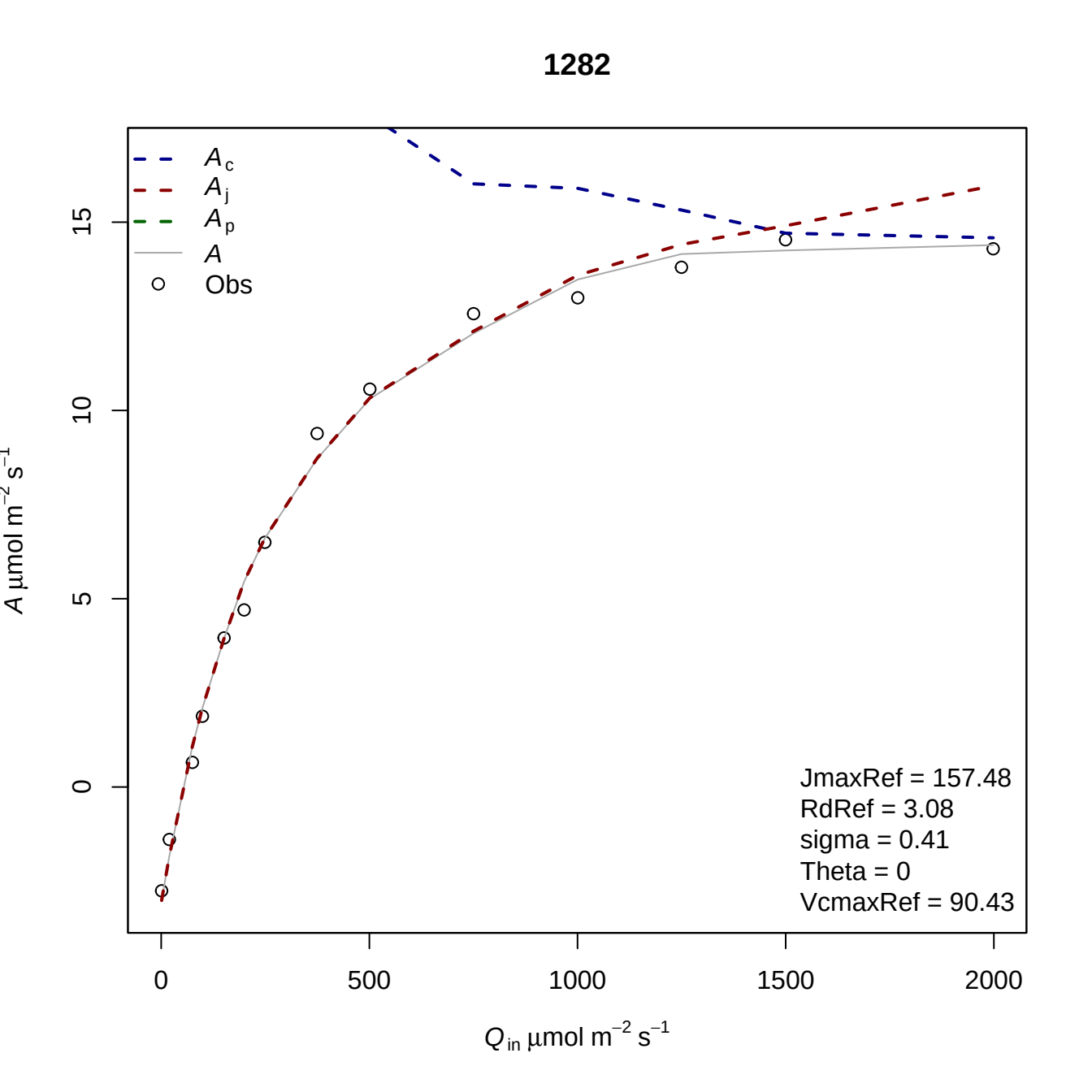

1277
A_c
A_j
A_p
A
Obs
JmaxRef = 153.27
RdRef = 2.91
sigma = 0.42
Theta = 0
VcmaxRef = 92.18
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

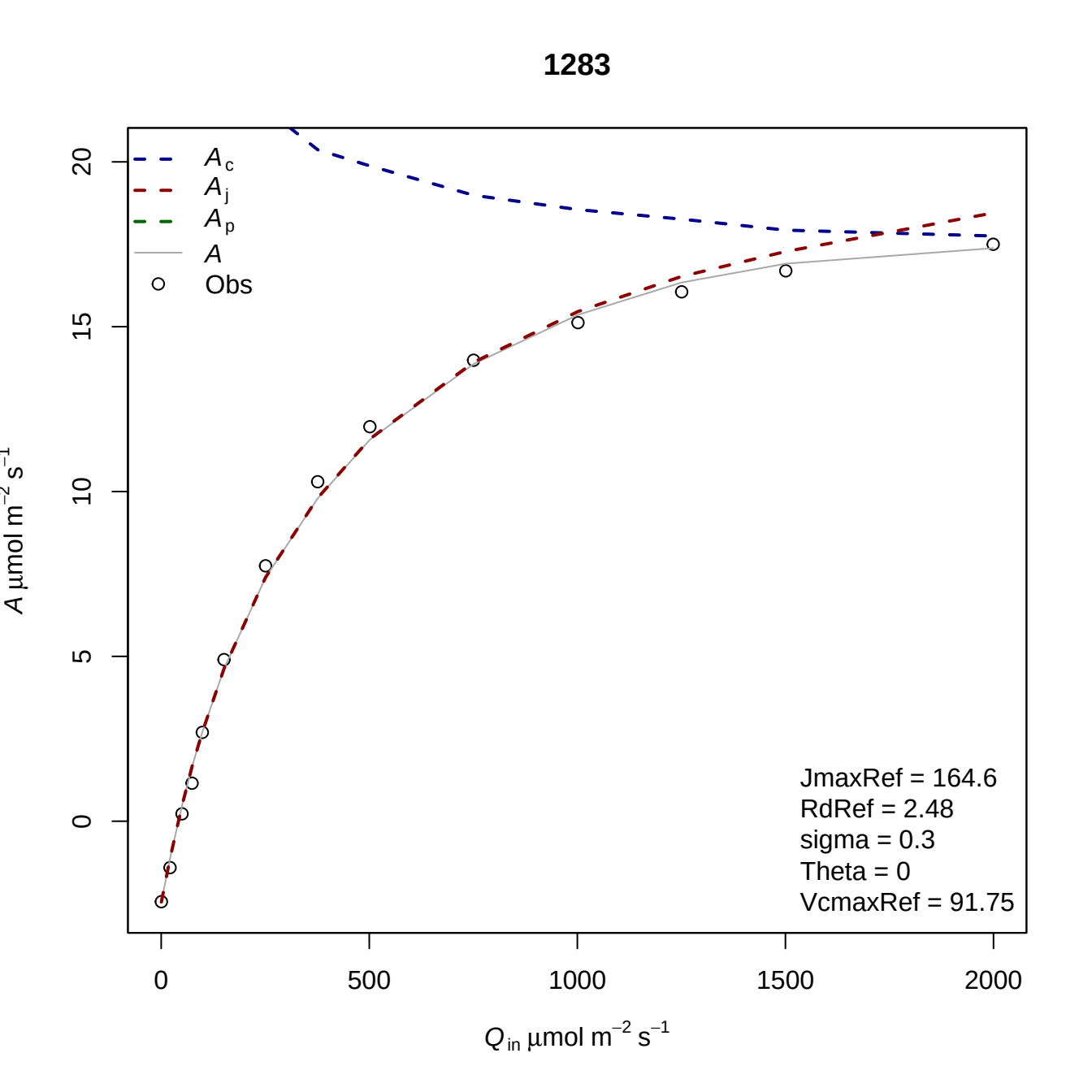

1278
A_c
A_j
A_p
A
Obs
JmaxRef = 148.04
RdRef = 3.09
sigma = 0.33
Theta = 0
VcmaxRef = 90.14
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

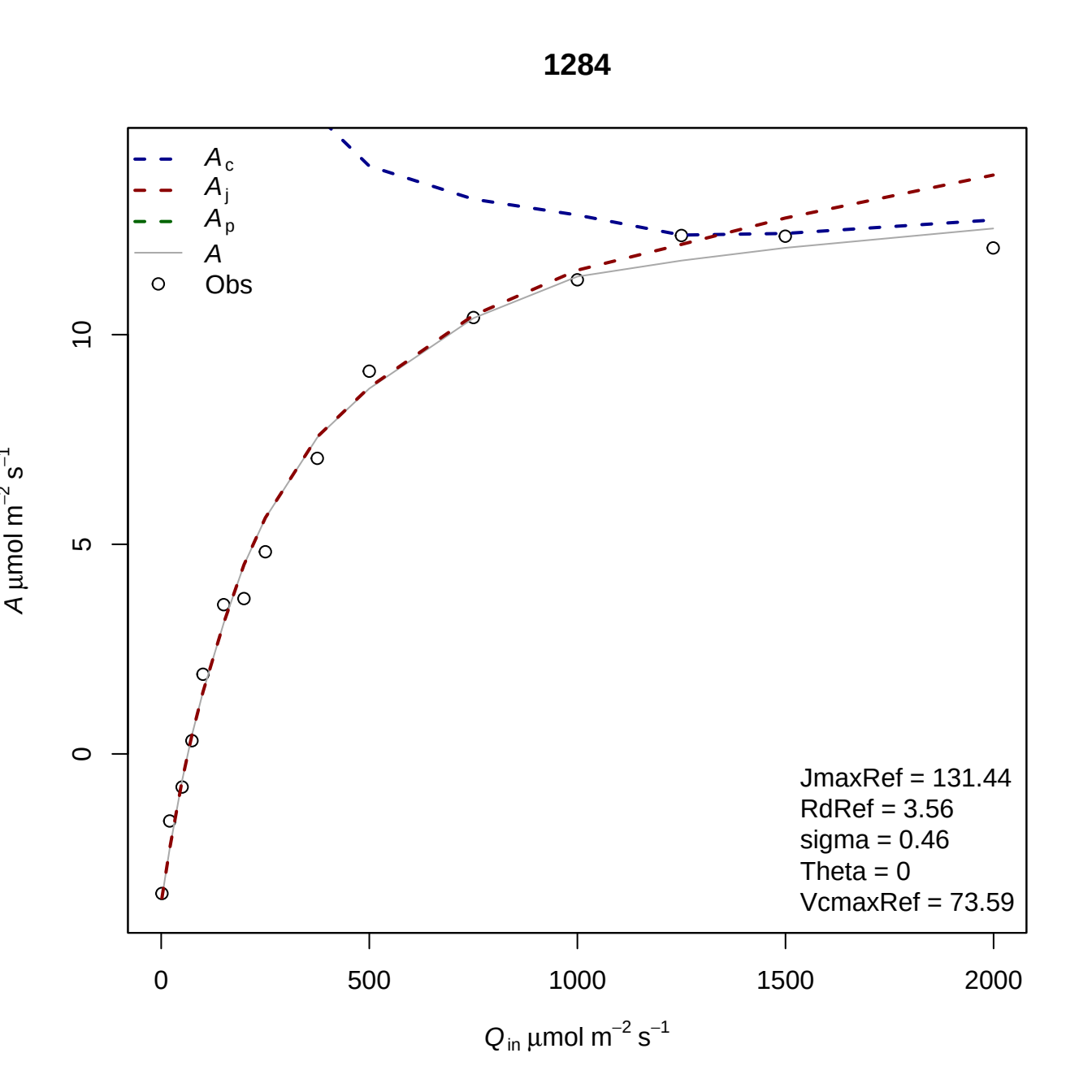

1279
A_c
A_j
A_p
A
Obs
JmaxRef = 146.45
RdRef = 2.95
sigma = 0.94
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

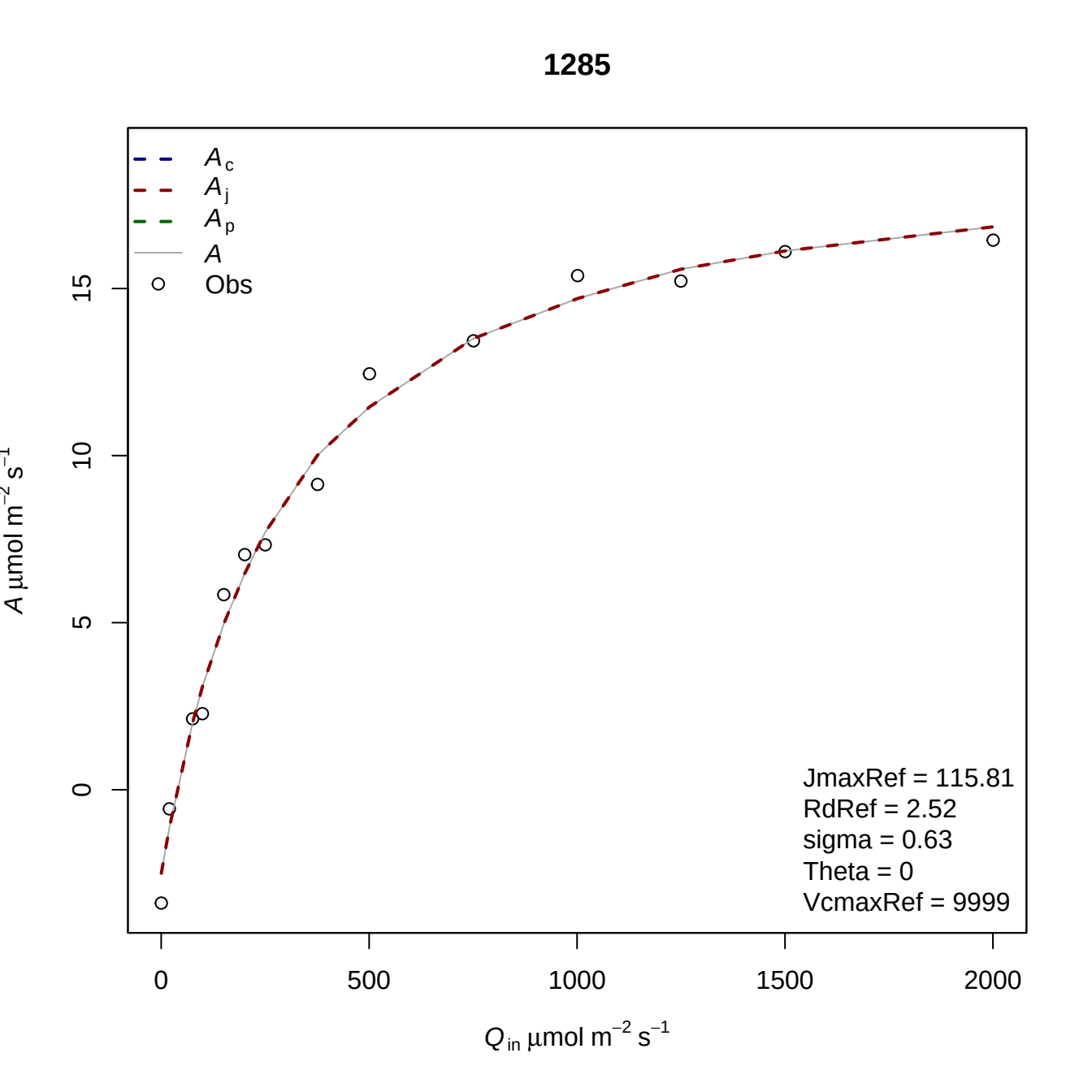

1280
A_c
A_j
A_p
A
Obs
JmaxRef = 84.64
RdRef = 1.21
sigma = 0.78
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

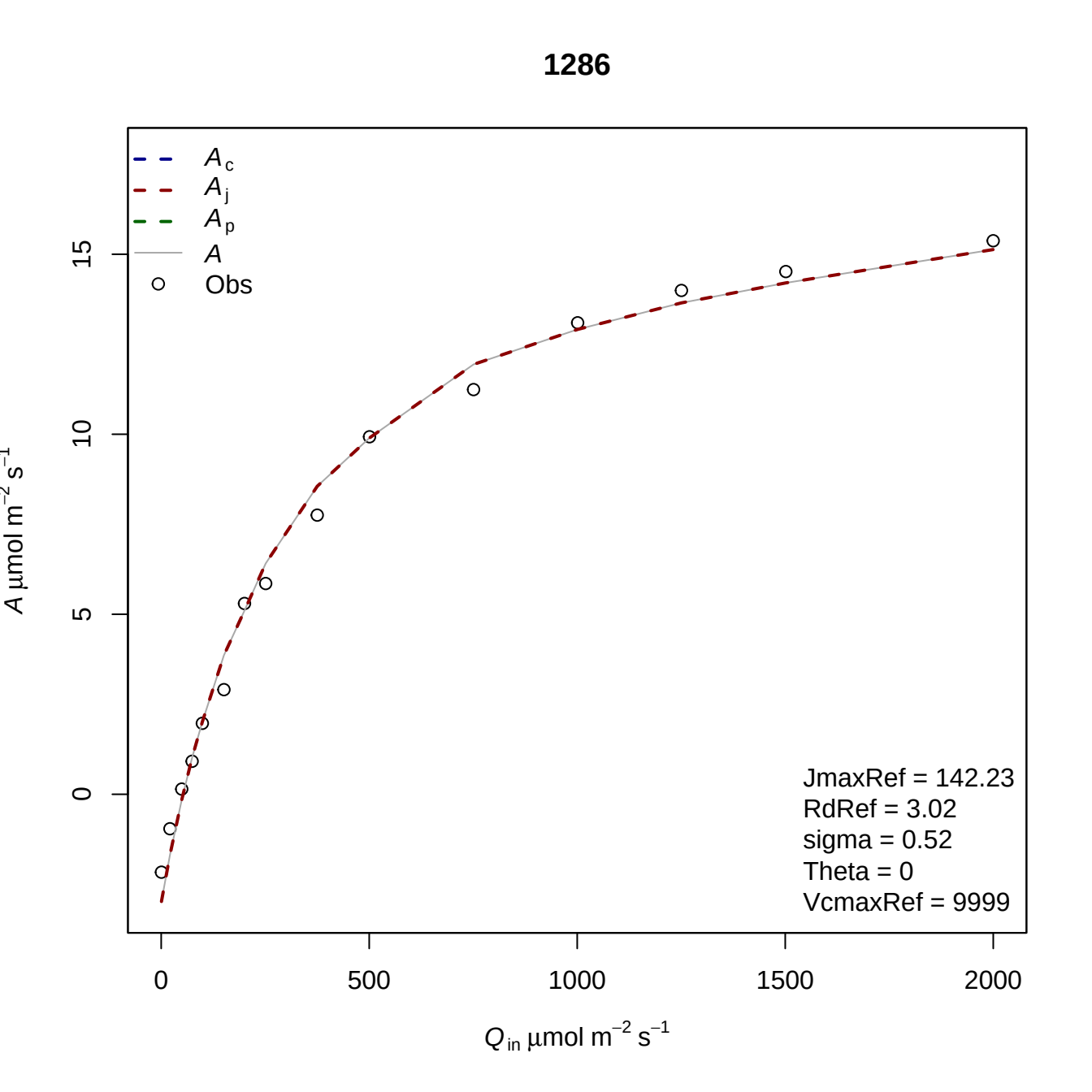

1281
A_c
A_j
A_p
A
Obs
JmaxRef = 84.23
RdRef = 0.8
sigma = 0.64
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

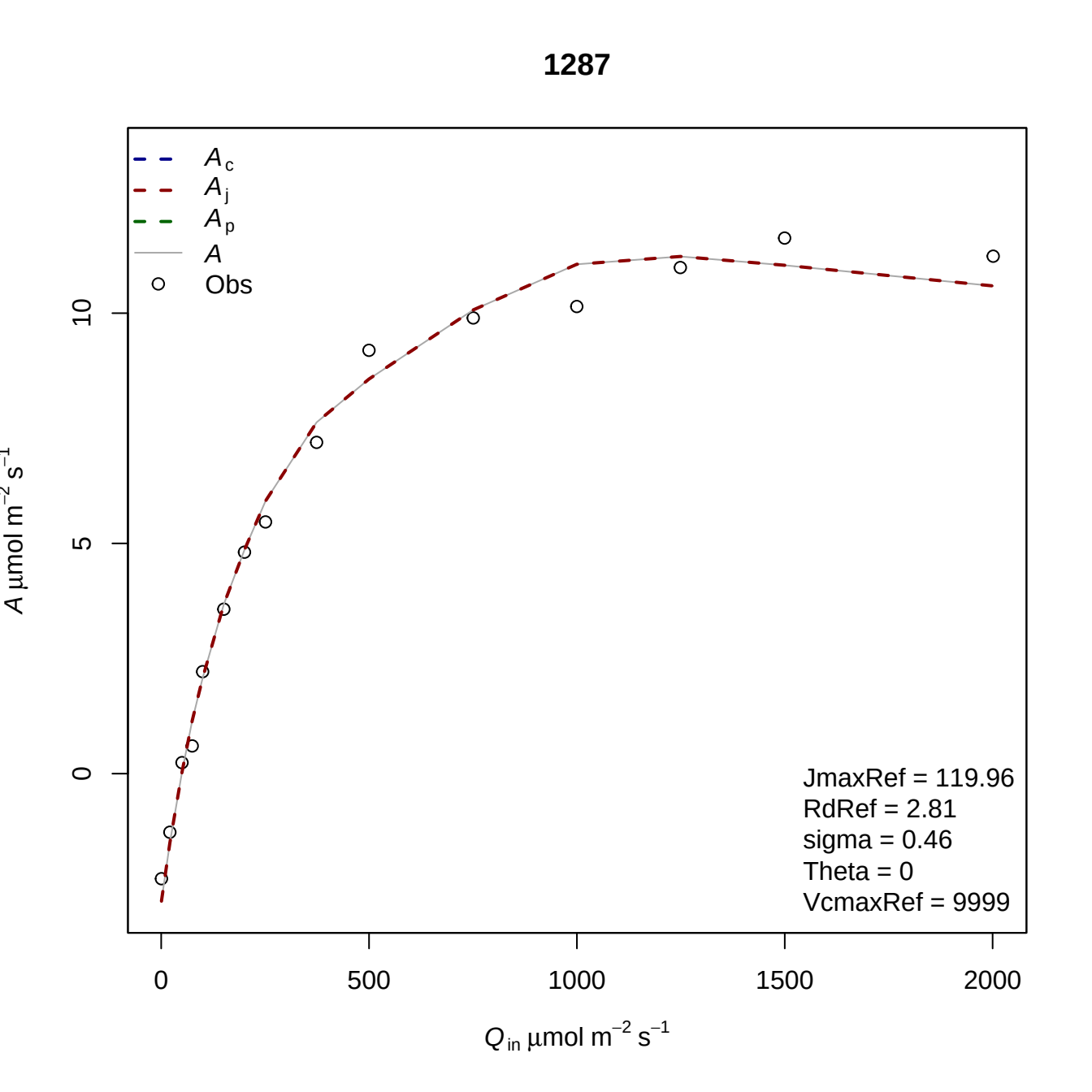

1282
A_c
A_j
A_p
A
Obs
JmaxRef = 157.48
RdRef = 3.08
sigma = 0.41
Theta = 0
VcmaxRef = 90.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

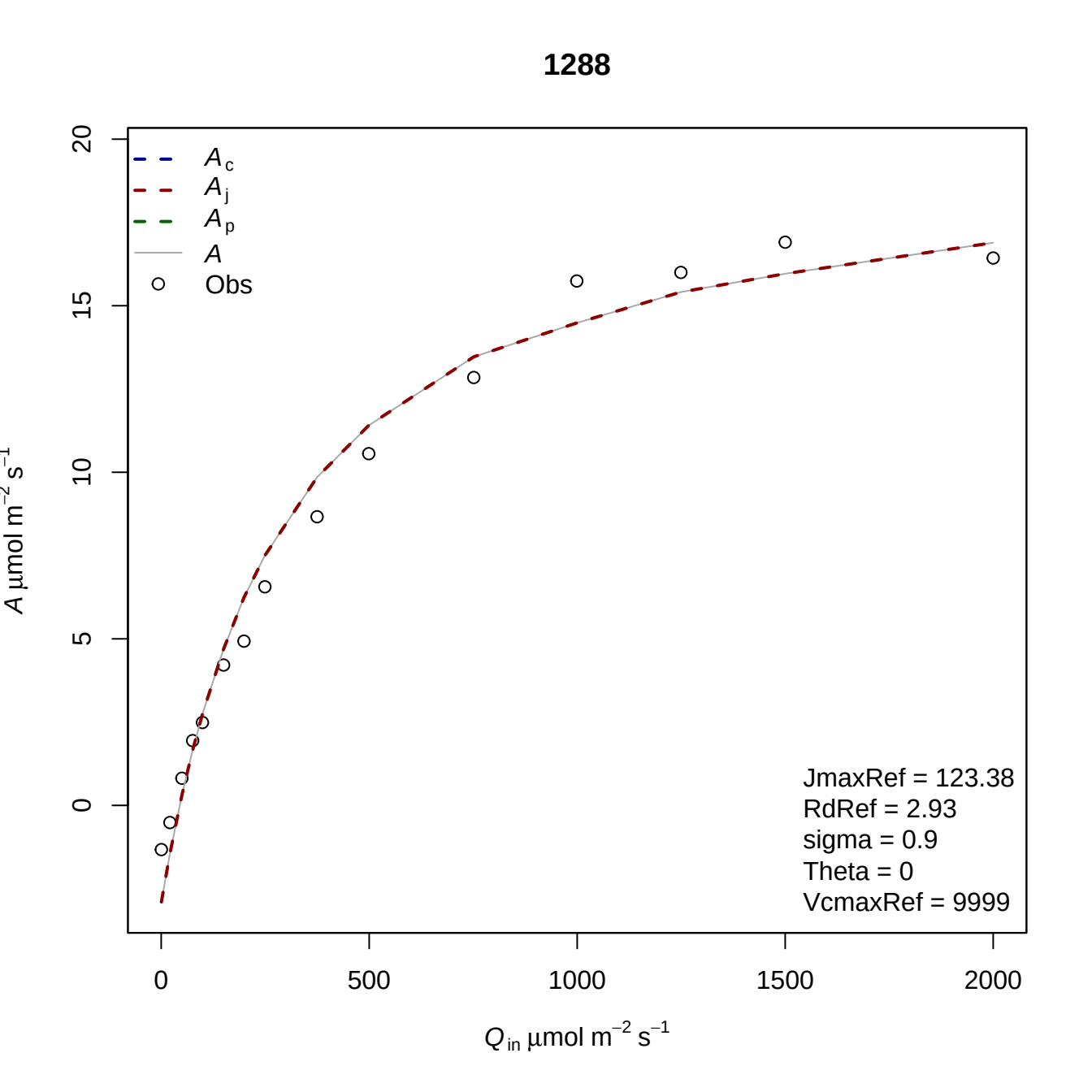

1283
A_c
A_j
A_p
A
Obs
JmaxRef = 164.6
RdRef = 2.48
sigma = 0.3
Theta = 0
VcmaxRef = 91.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

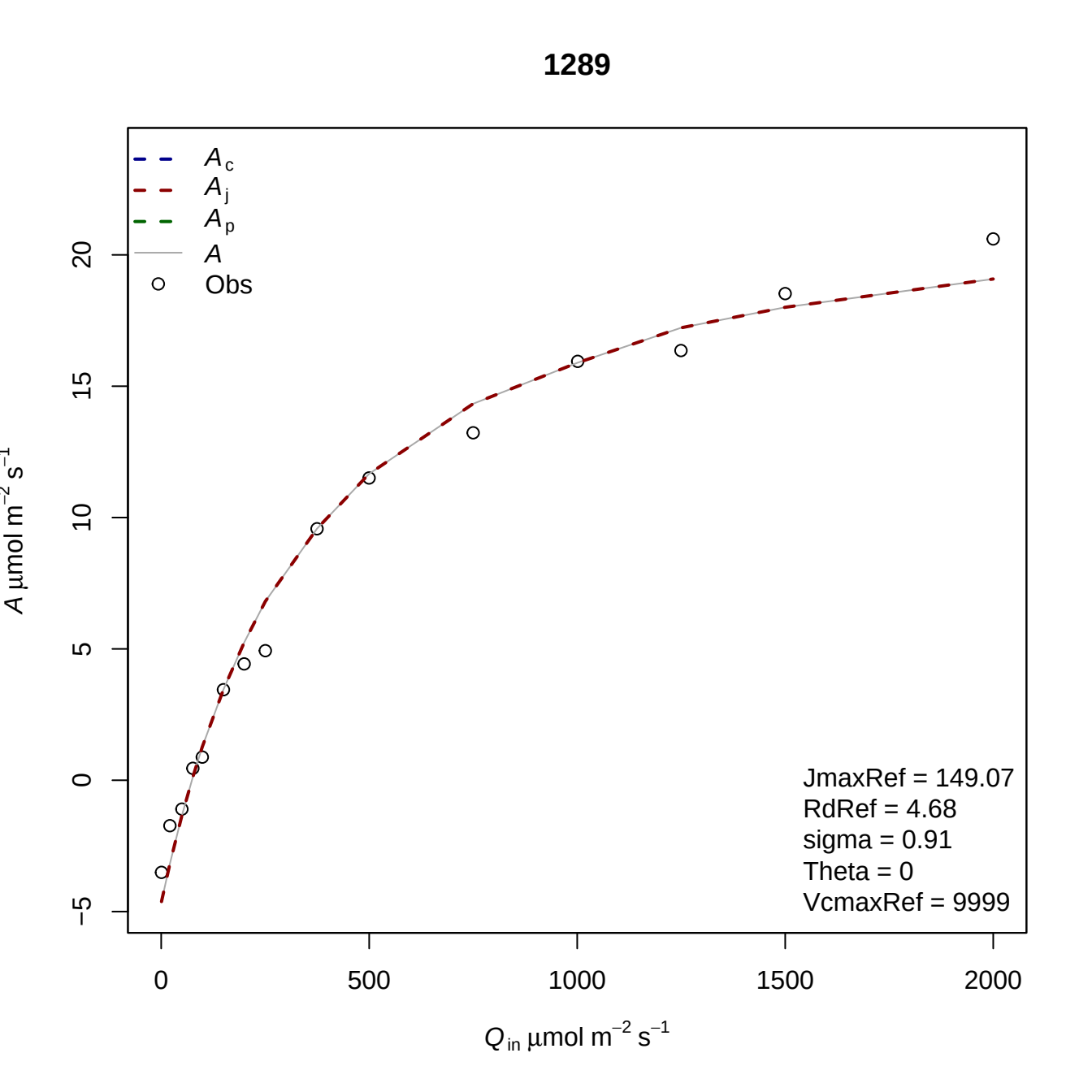

1284
A_c
A_j
A_p
A
Obs
JmaxRef = 131.44
RdRef = 3.56
sigma = 0.46
Theta = 0
VcmaxRef = 73.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

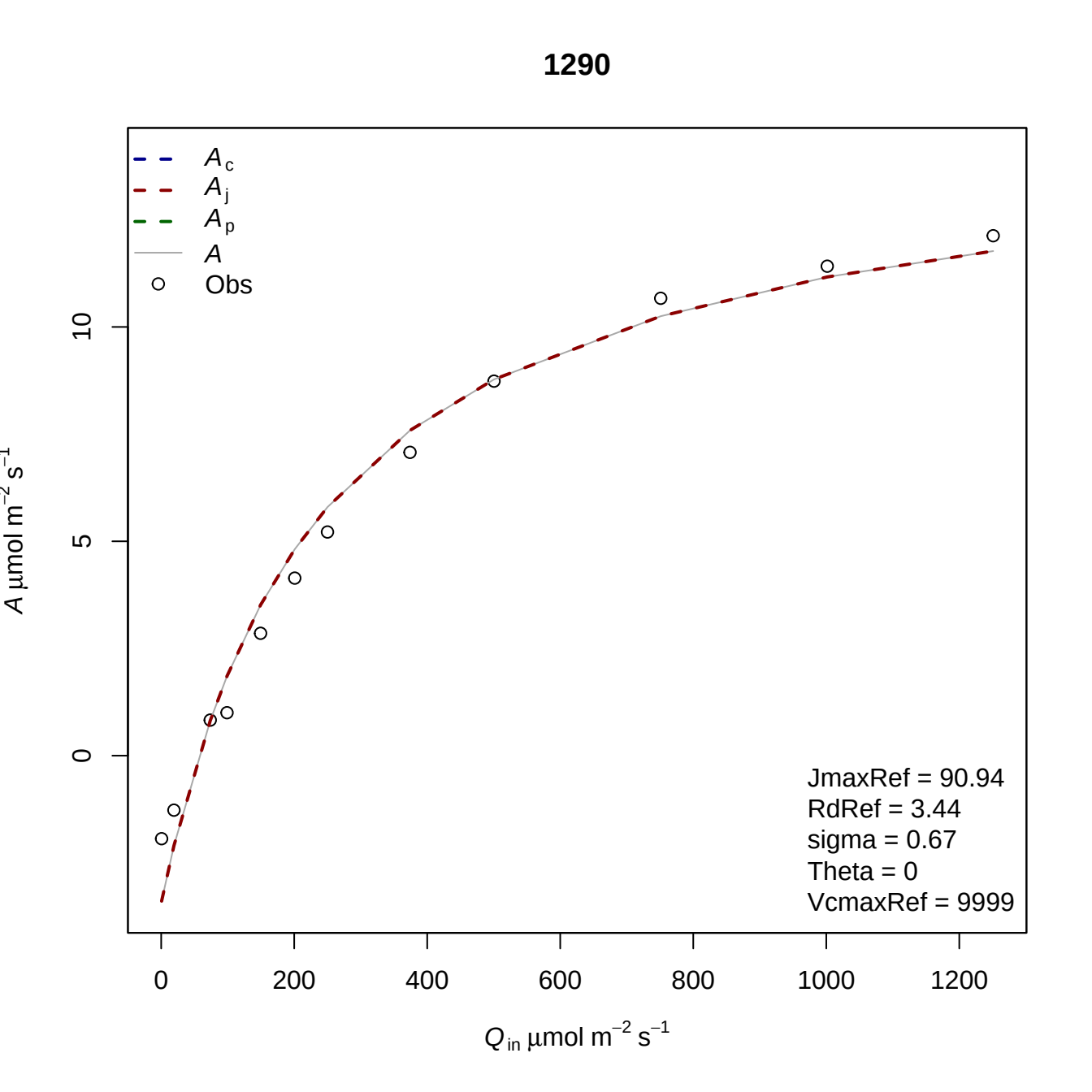

1285
A_c
A_j
A_p
A
Obs
JmaxRef = 115.81
RdRef = 2.52
sigma = 0.63
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

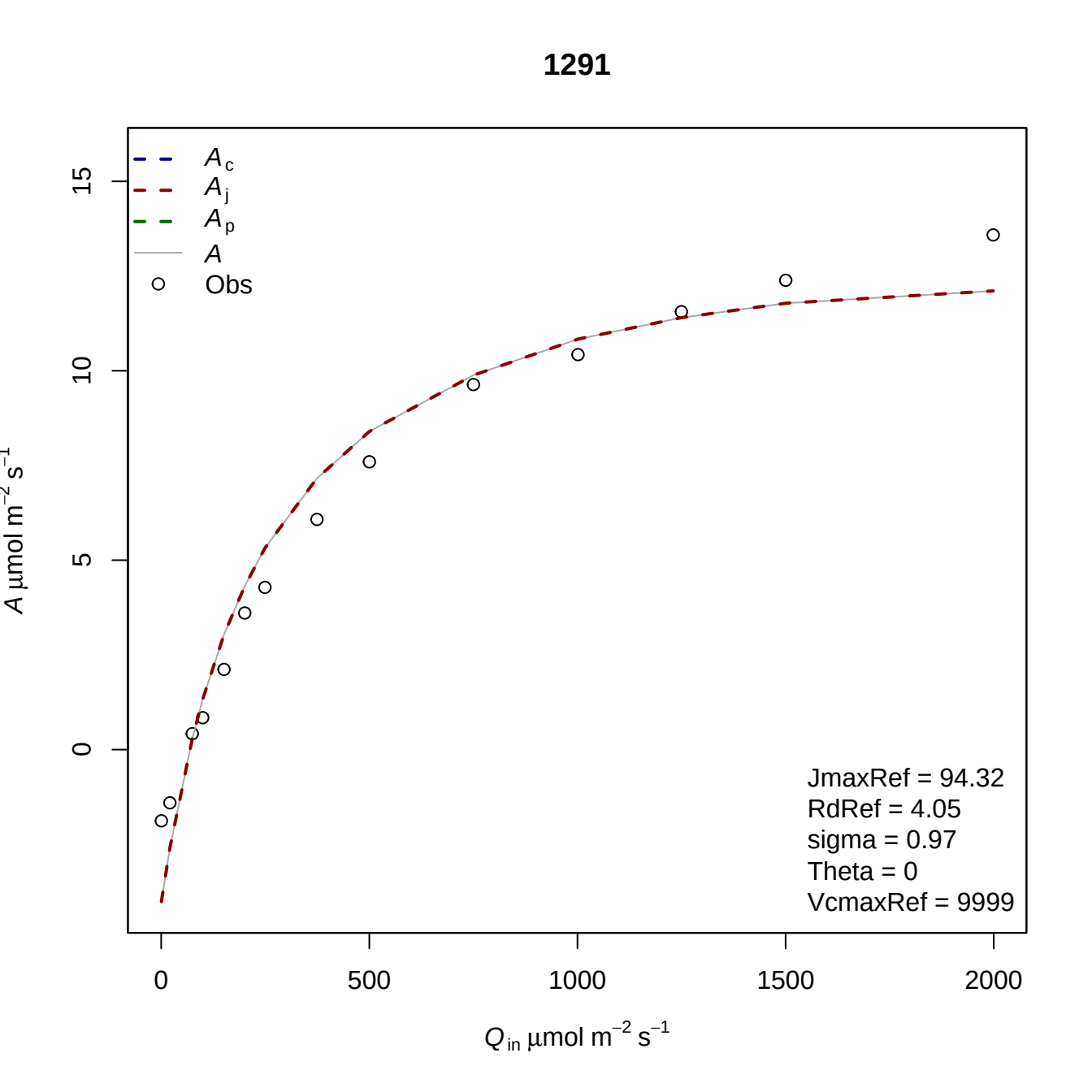

1286
A_c
A_j
A_p
A
Obs
JmaxRef = 142.23
RdRef = 3.02
sigma = 0.52
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

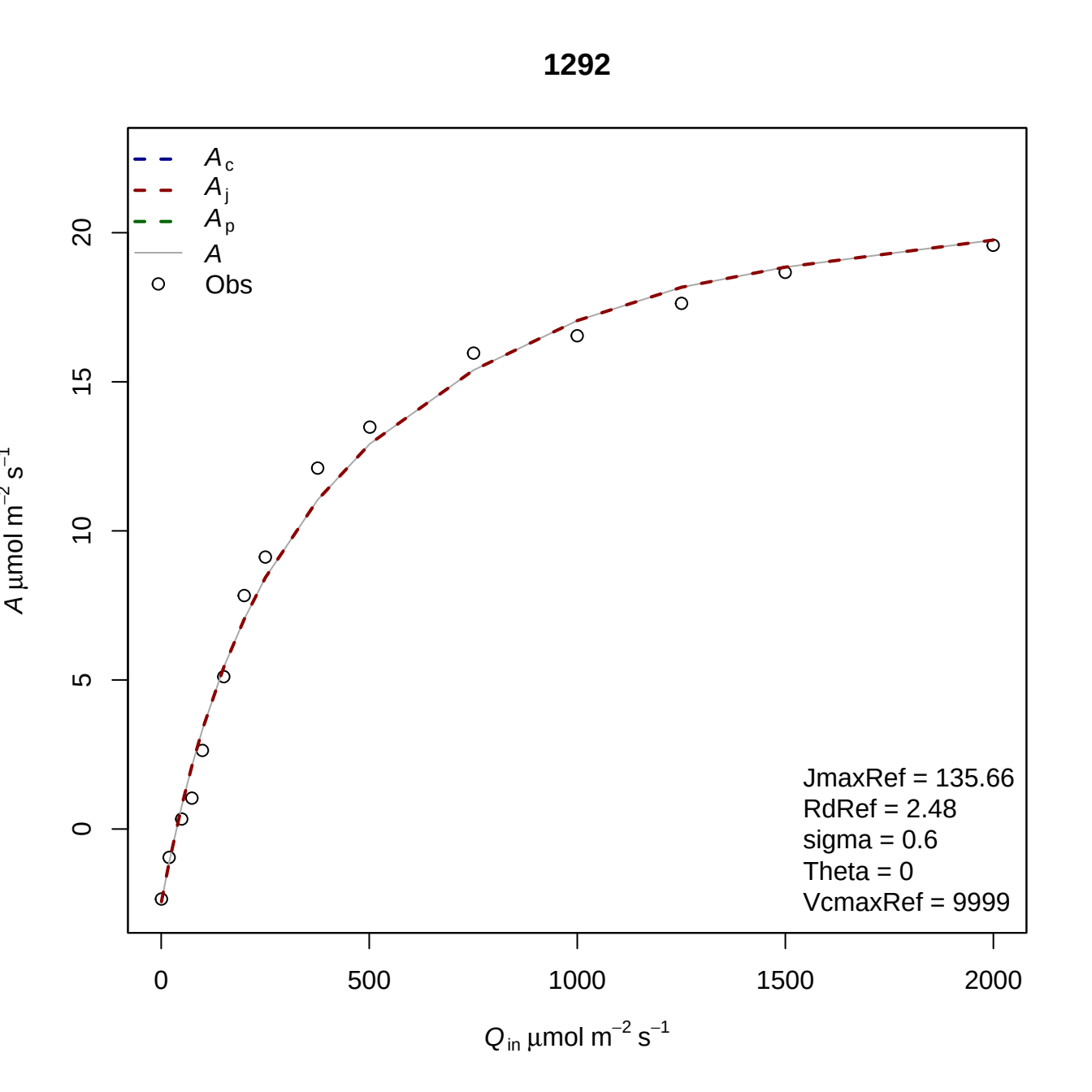

1287
A_c
A_j
A_p
A
Obs
JmaxRef = 119.96
RdRef = 2.81
sigma = 0.46
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

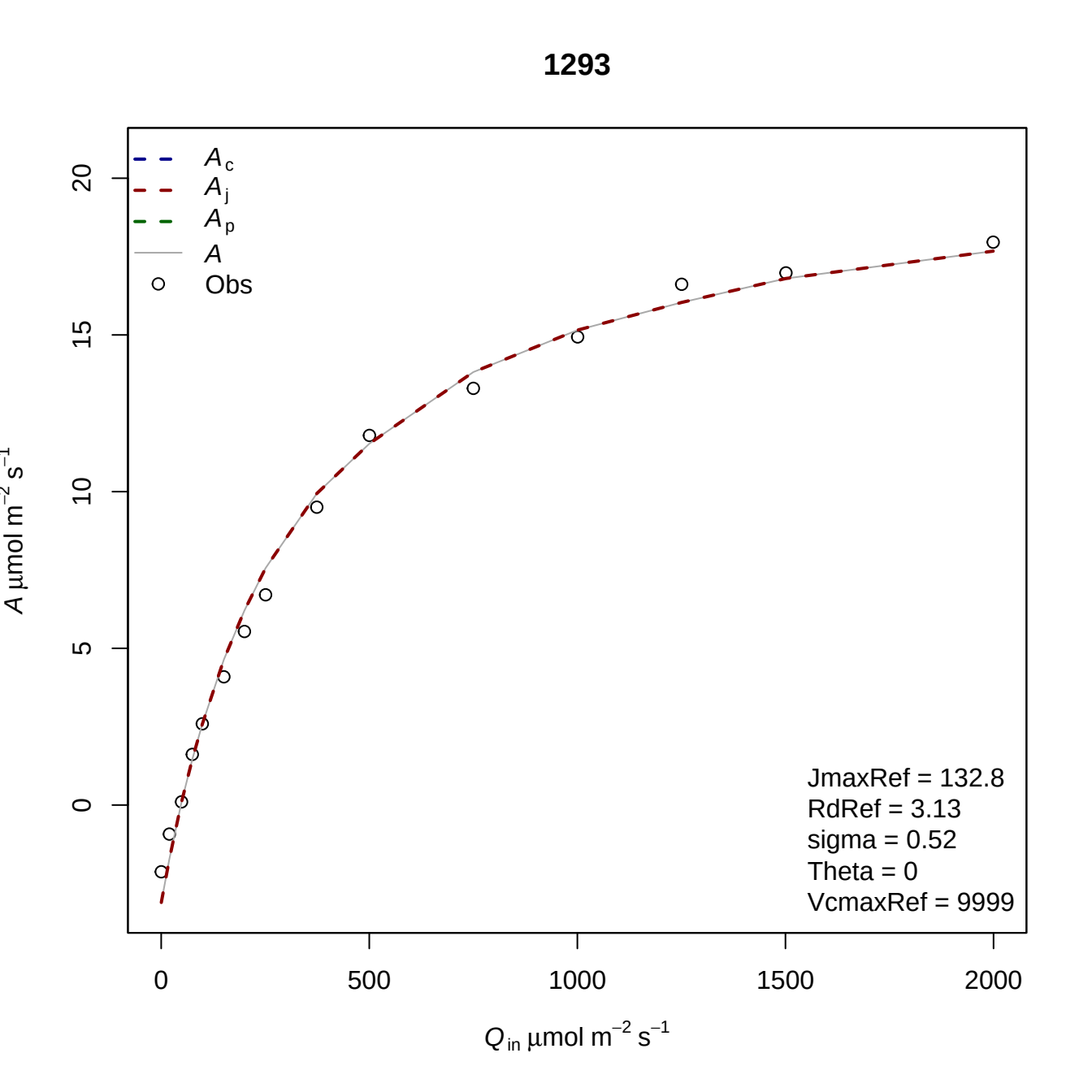

1288
A_c
A_j
A_p
A
Obs
JmaxRef = 123.38
RdRef = 2.93
sigma = 0.9
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

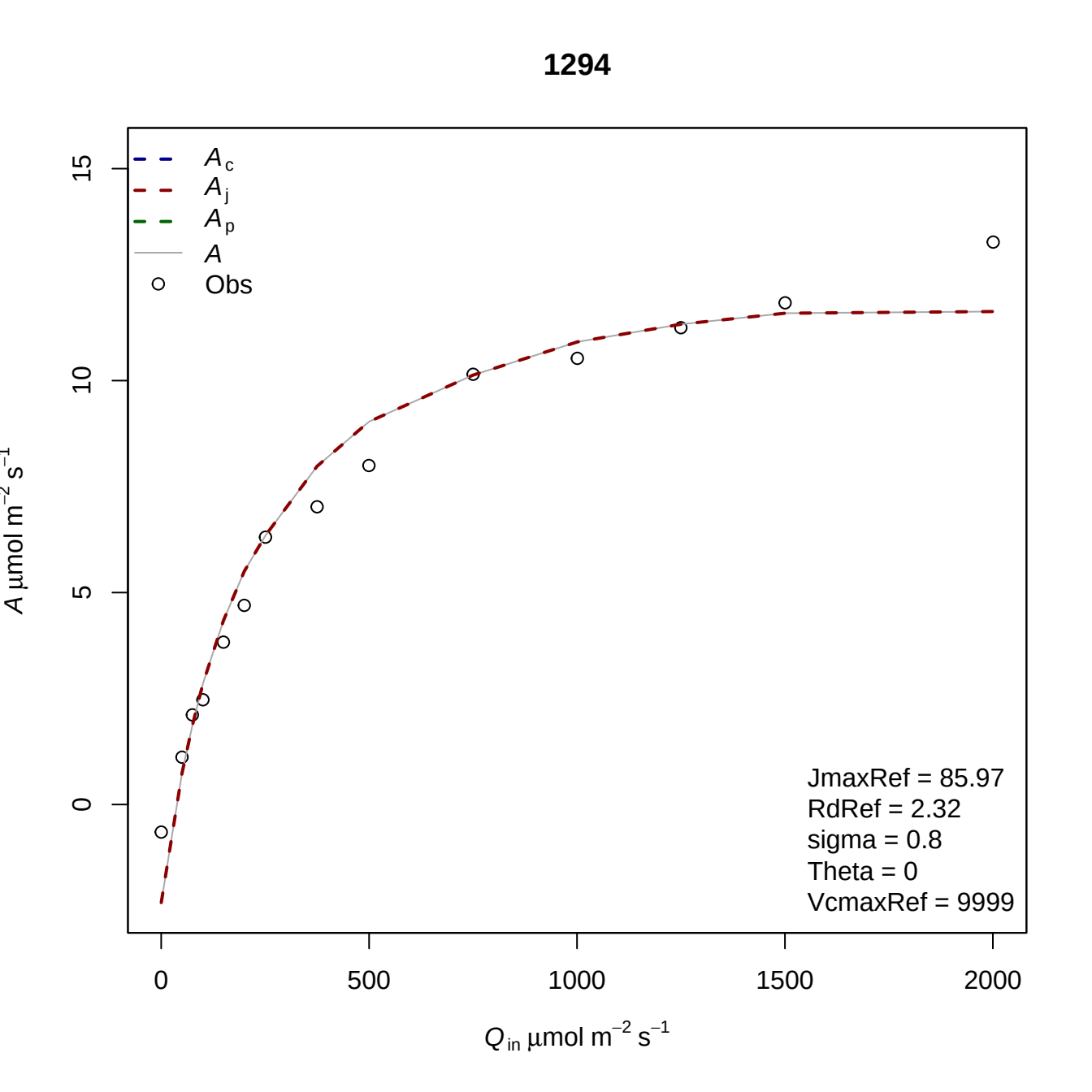

1289
A_c
A_j
A_p
A
Obs
JmaxRef = 149.07
RdRef = 4.68
sigma = 0.91
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

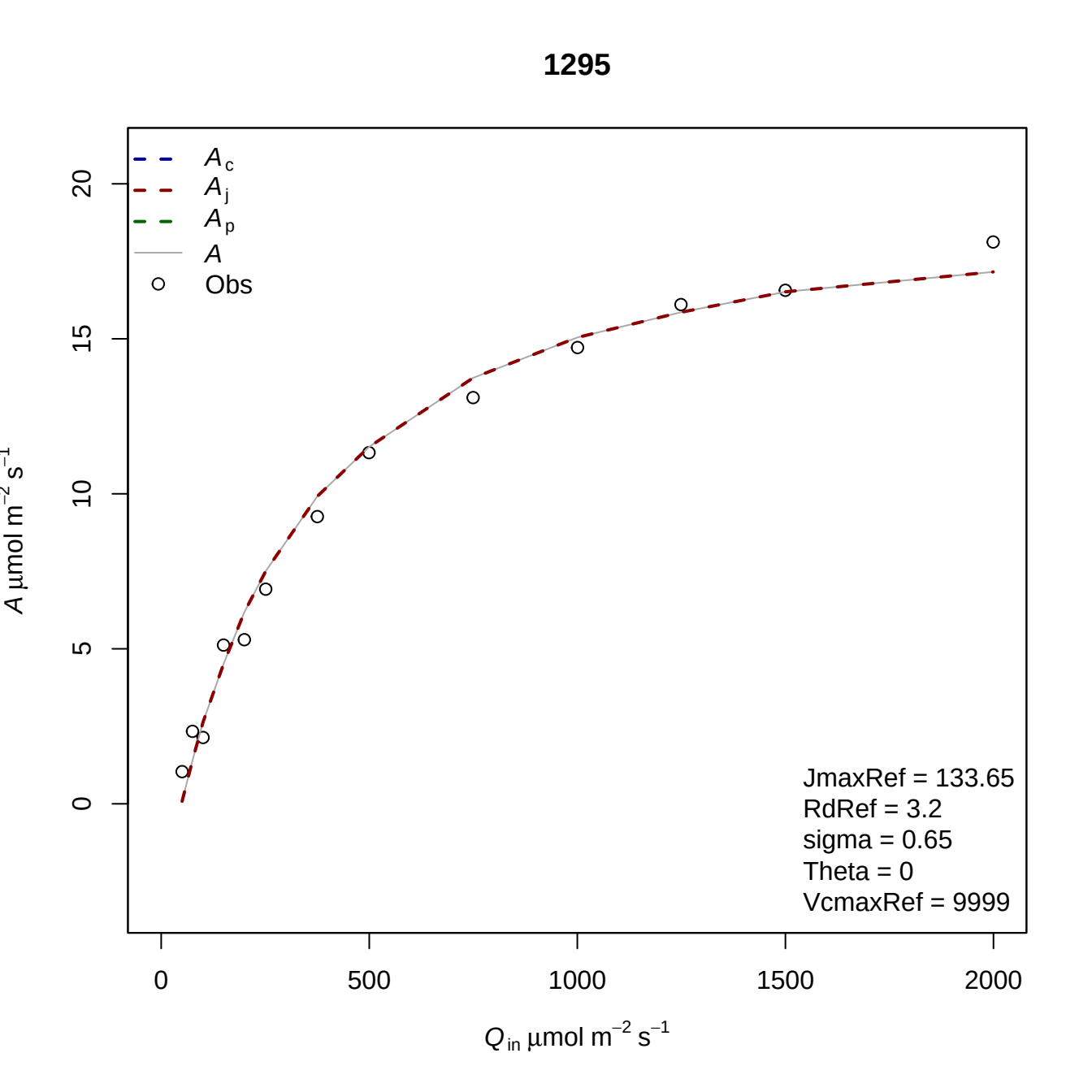

1290
A_c
A_j
A_p
A
Obs
JmaxRef = 90.94
RdRef = 3.44
sigma = 0.67
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

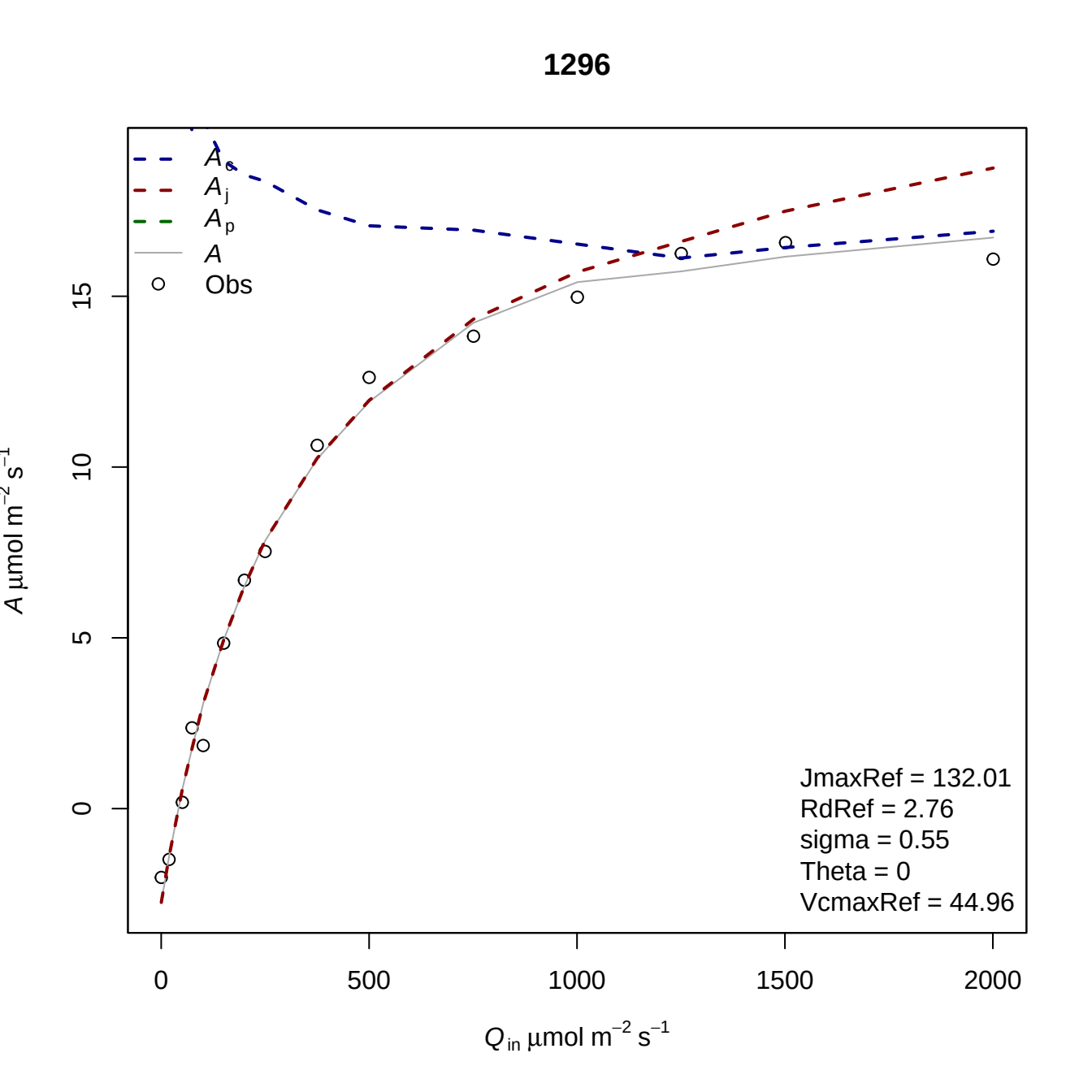

1291
A_c
A_j
A_p
A
Obs
JmaxRef = 94.32
RdRef = 4.05
sigma = 0.97
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

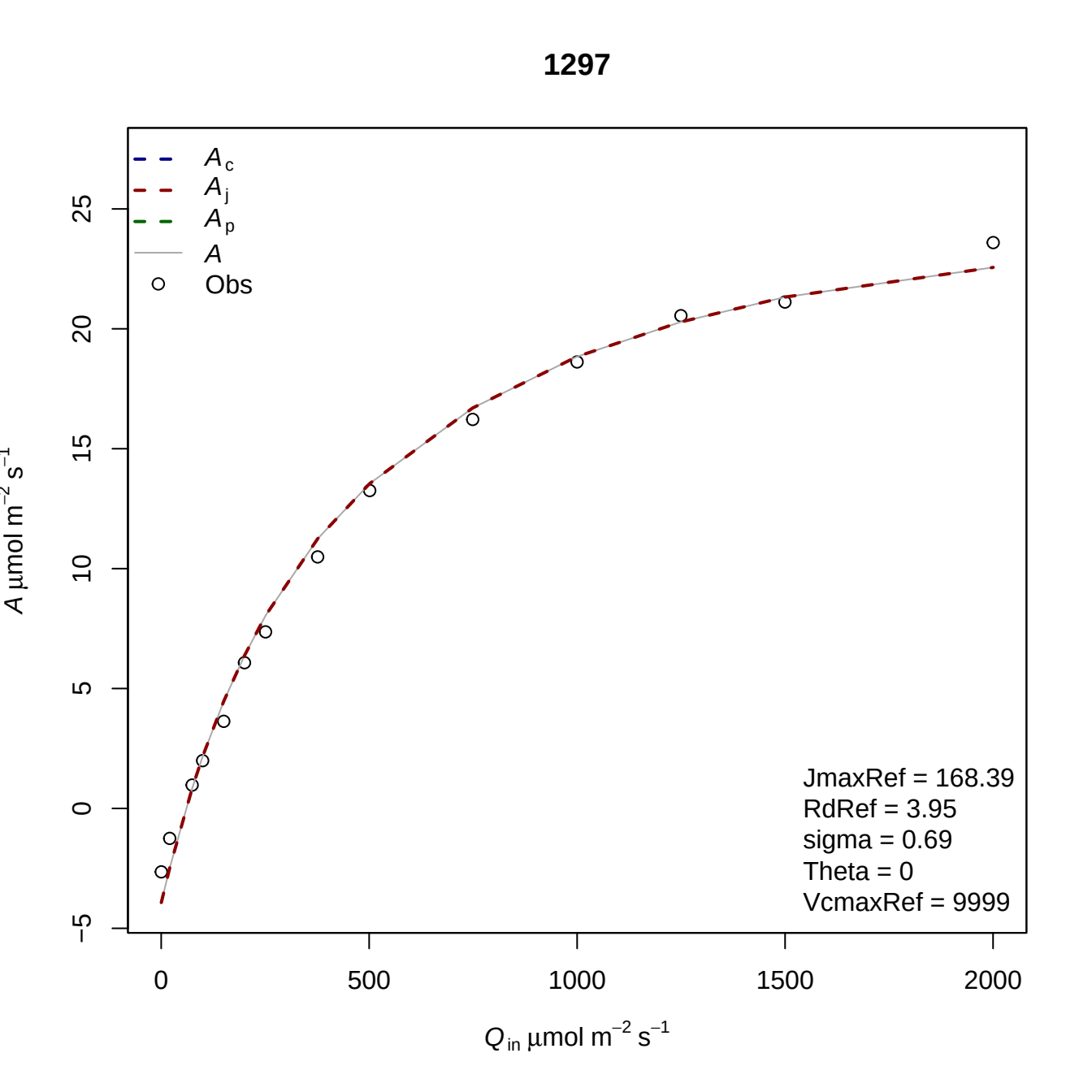

1292
A_c
A_j
A_p
A
Obs
JmaxRef = 135.66
RdRef = 2.48
sigma = 0.6
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

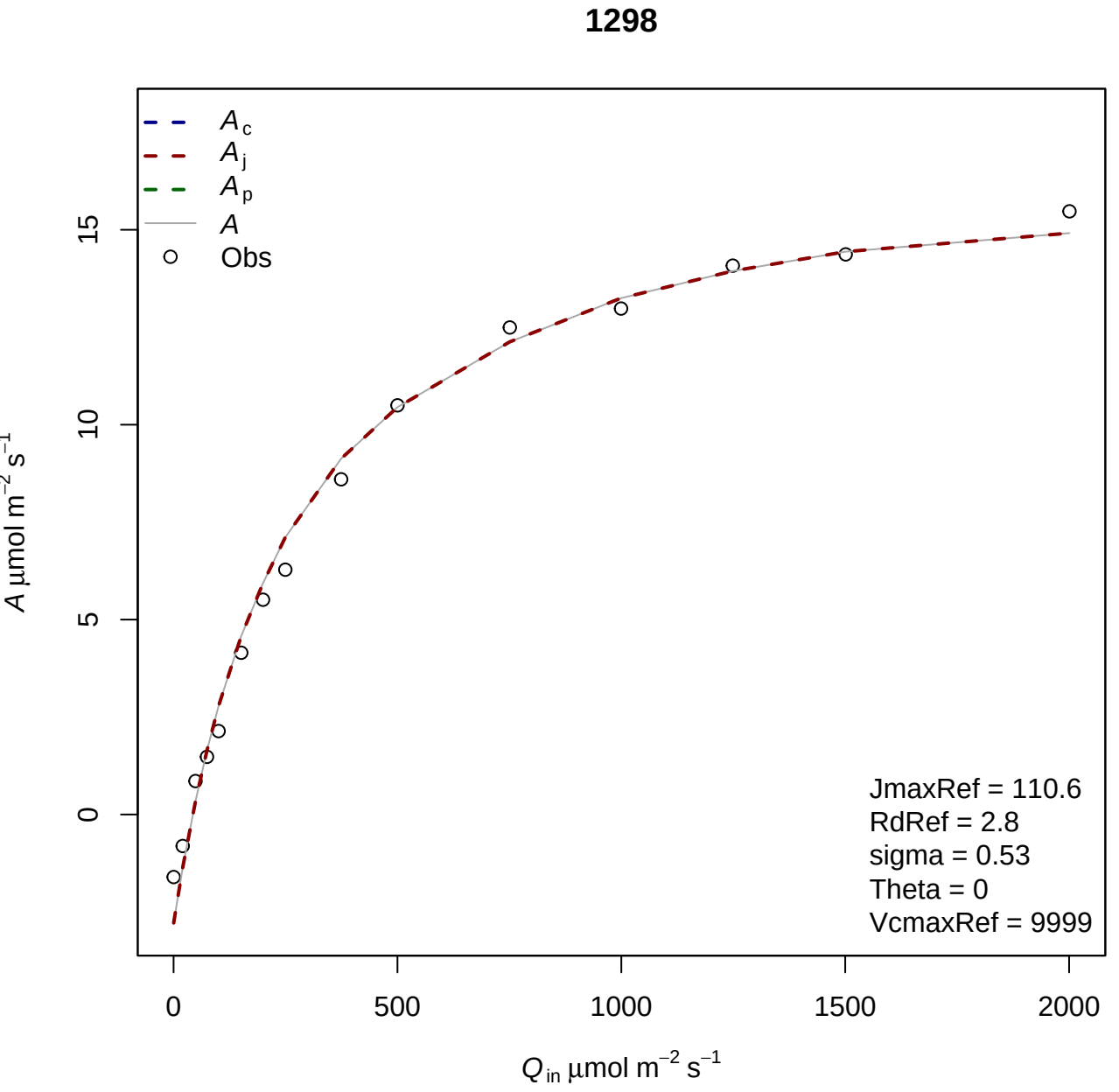

1293
A_c
A_j
A_p
A
Obs
JmaxRef = 132.8
RdRef = 3.13
sigma = 0.52
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

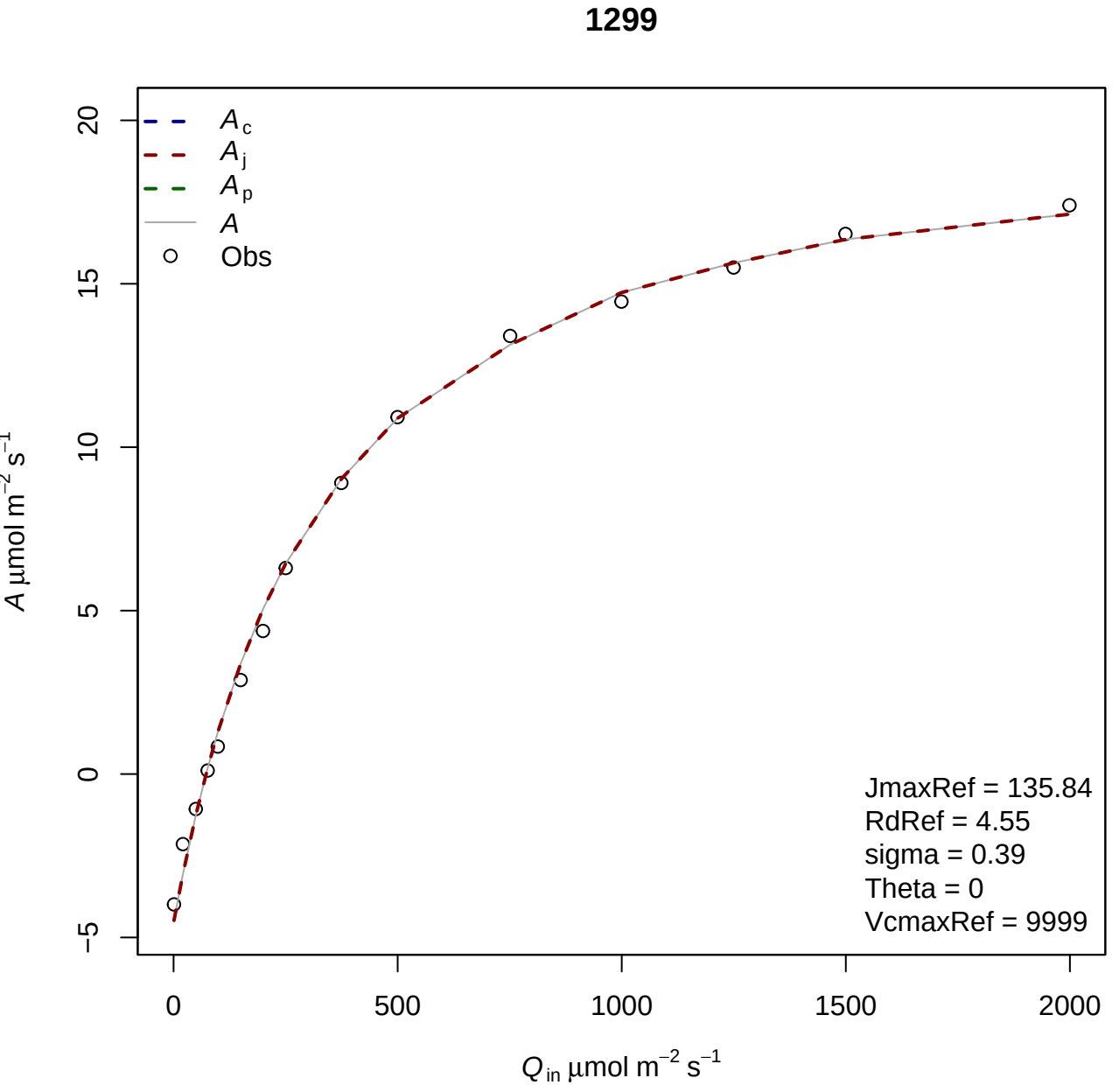

1294
A_c
A_j
A_p
A
Obs
JmaxRef = 85.97
RdRef = 2.32
sigma = 0.8
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

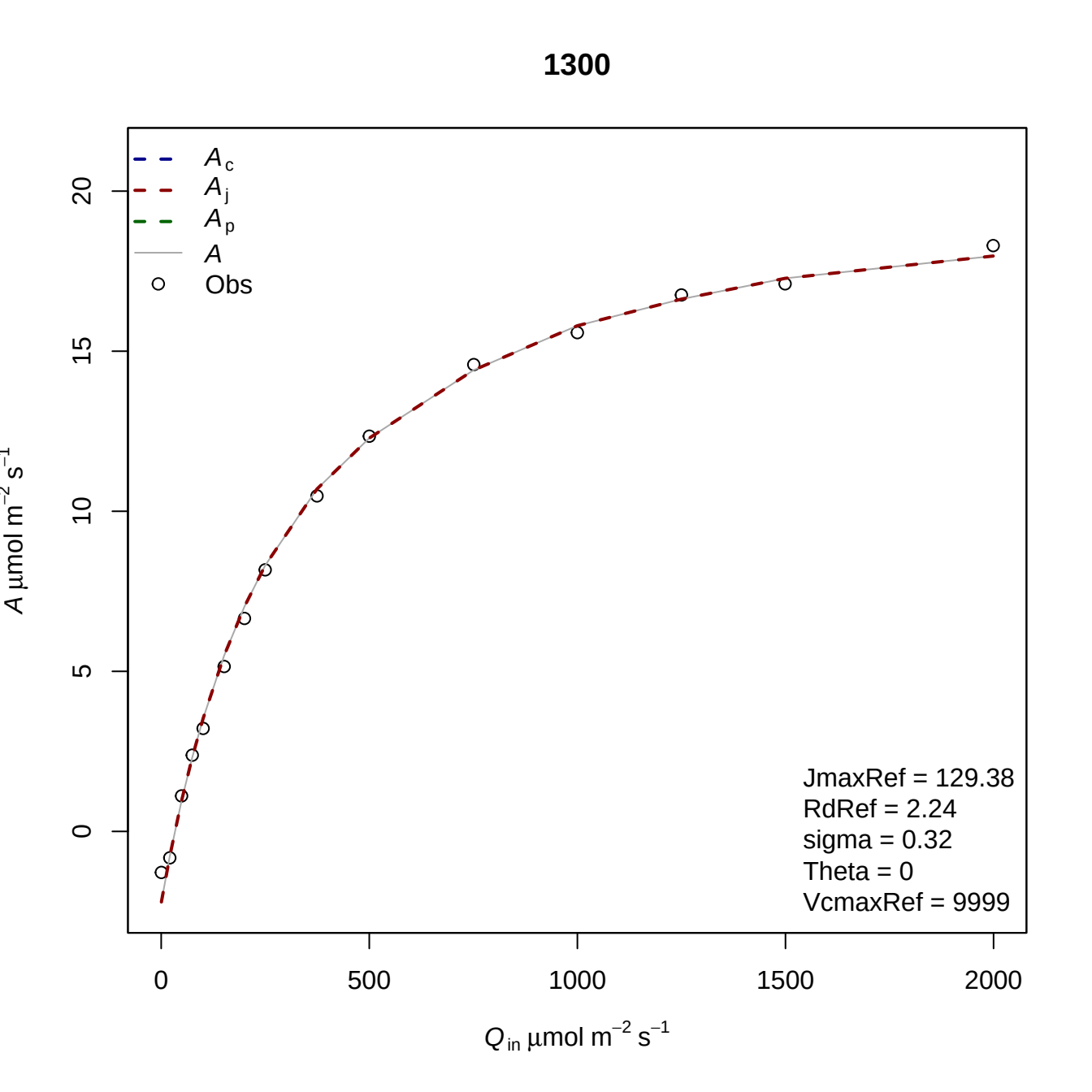

1295
A_c
A_j
A_p
A
Obs
JmaxRef = 133.65
RdRef = 3.2
sigma = 0.65
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

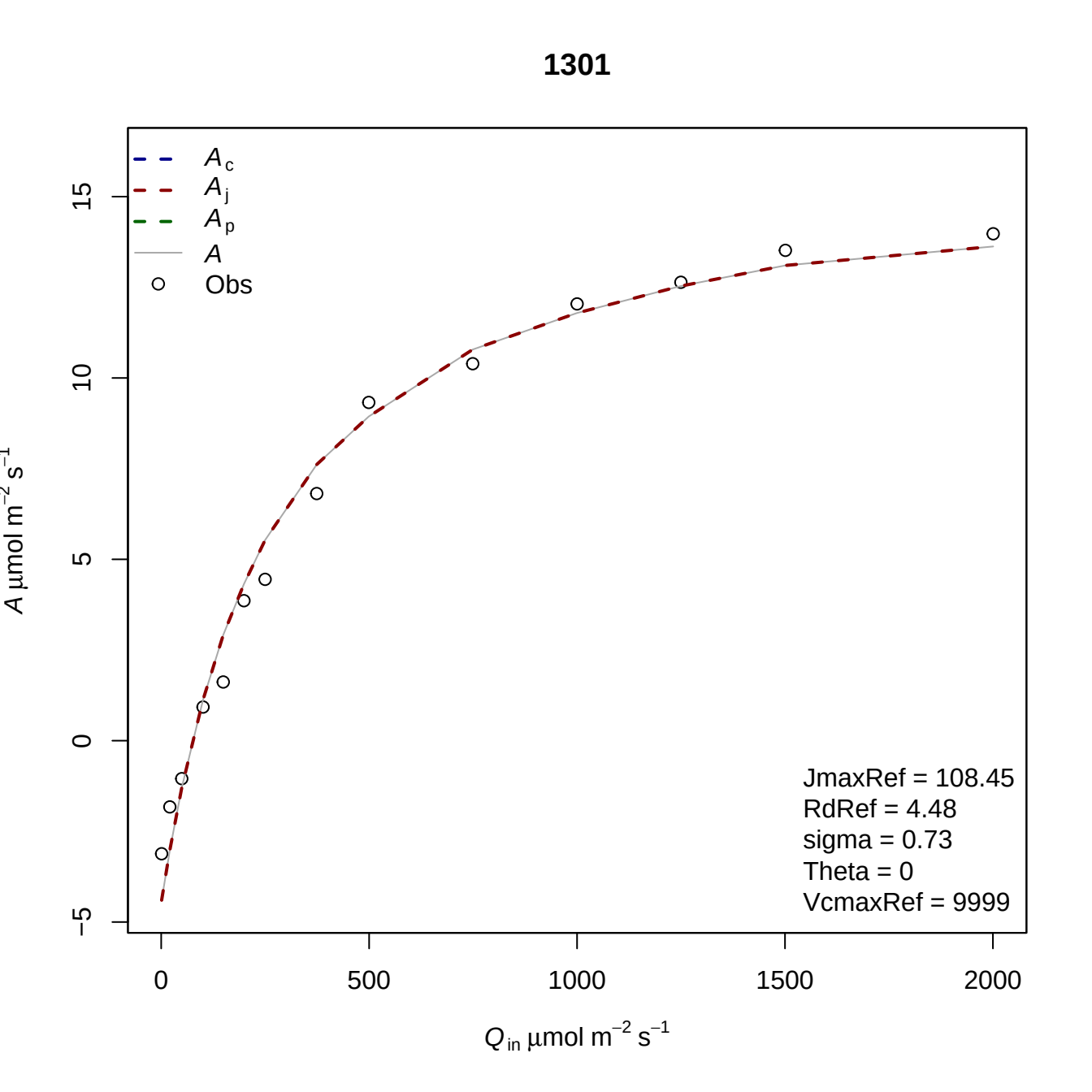

1296
A_c
A_j
A_p
A
Obs
JmaxRef = 132.01
RdRef = 2.76
sigma = 0.55
Theta = 0
VcmaxRef = 44.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

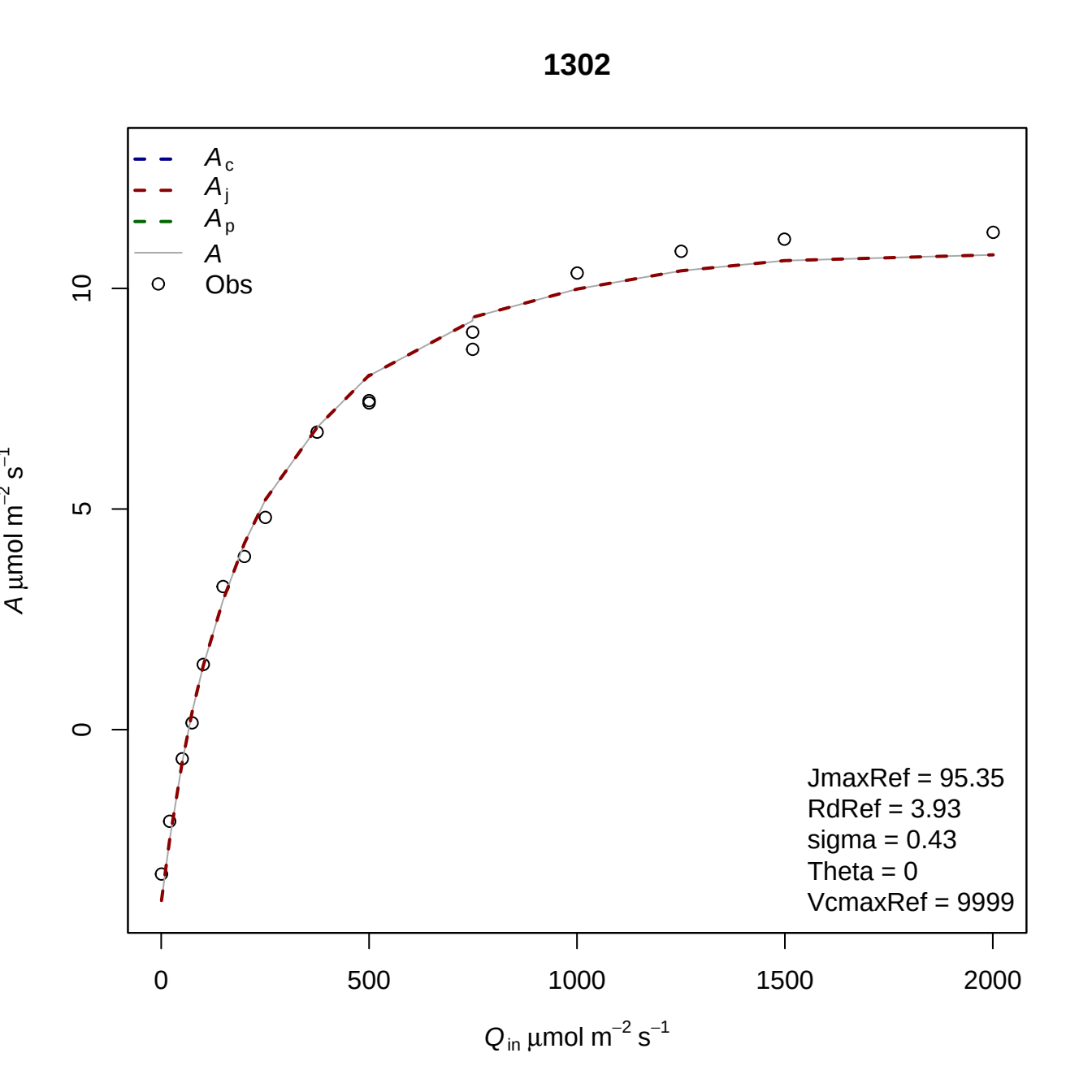

1297
A_c
A_j
A_p
A
Obs
JmaxRef = 168.39
RdRef = 3.95
sigma = 0.69
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500
2000

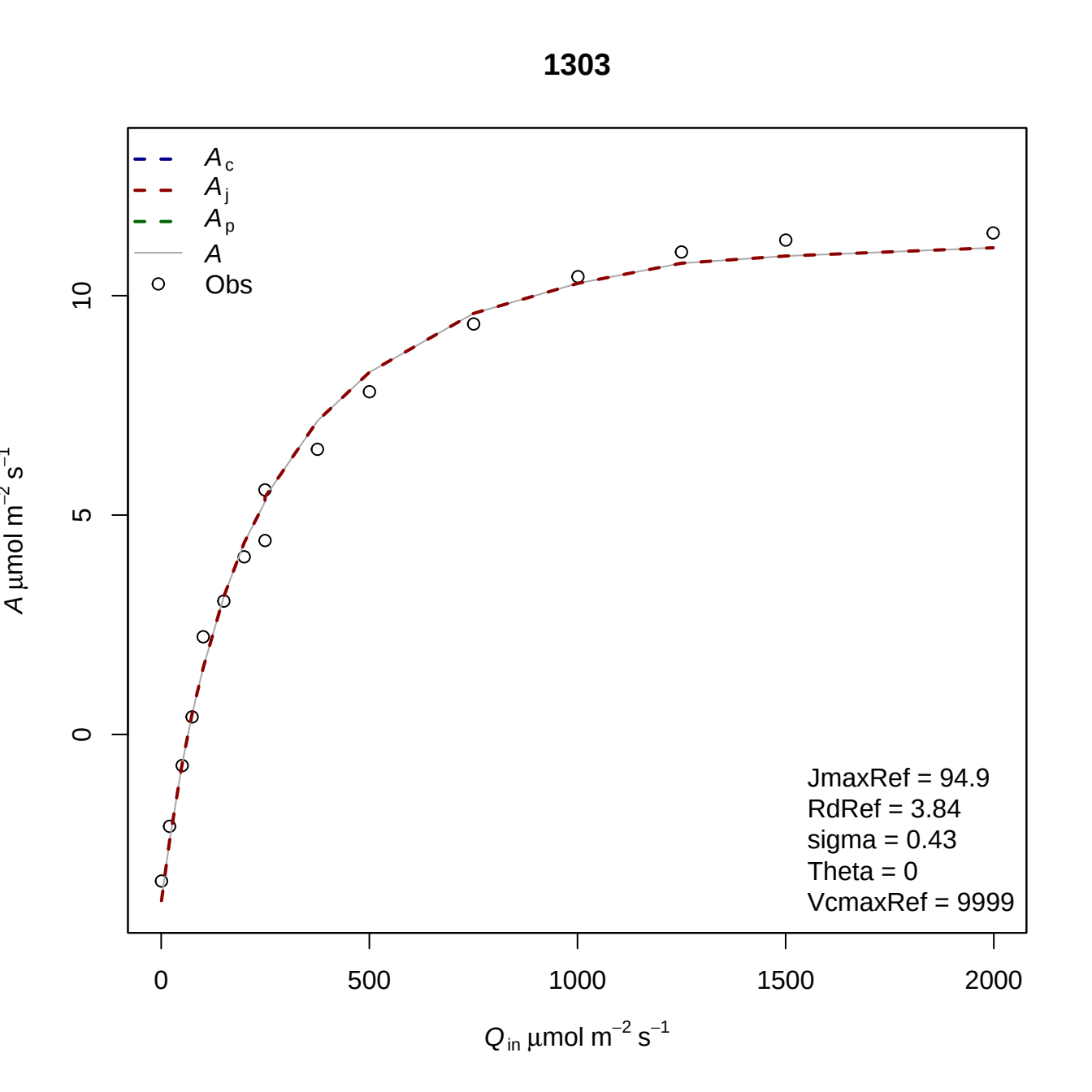

1298
A_c
A_j
A_p
A
Obs
JmaxRef = 110.6
RdRef = 2.8
sigma = 0.53
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

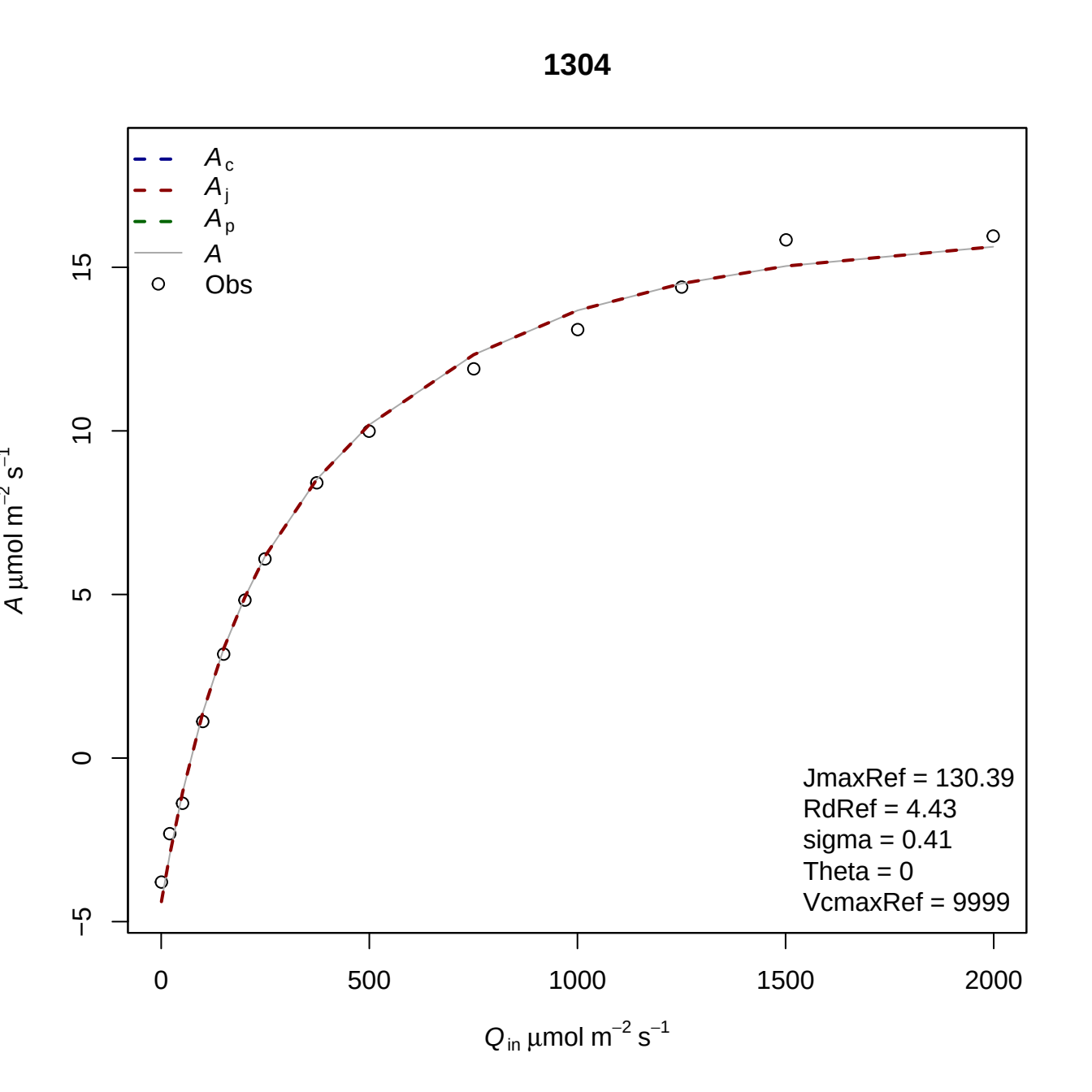

1299
A_c
A_j
A_p
A
Obs
JmaxRef = 135.84
RdRef = 4.55
sigma = 0.39
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

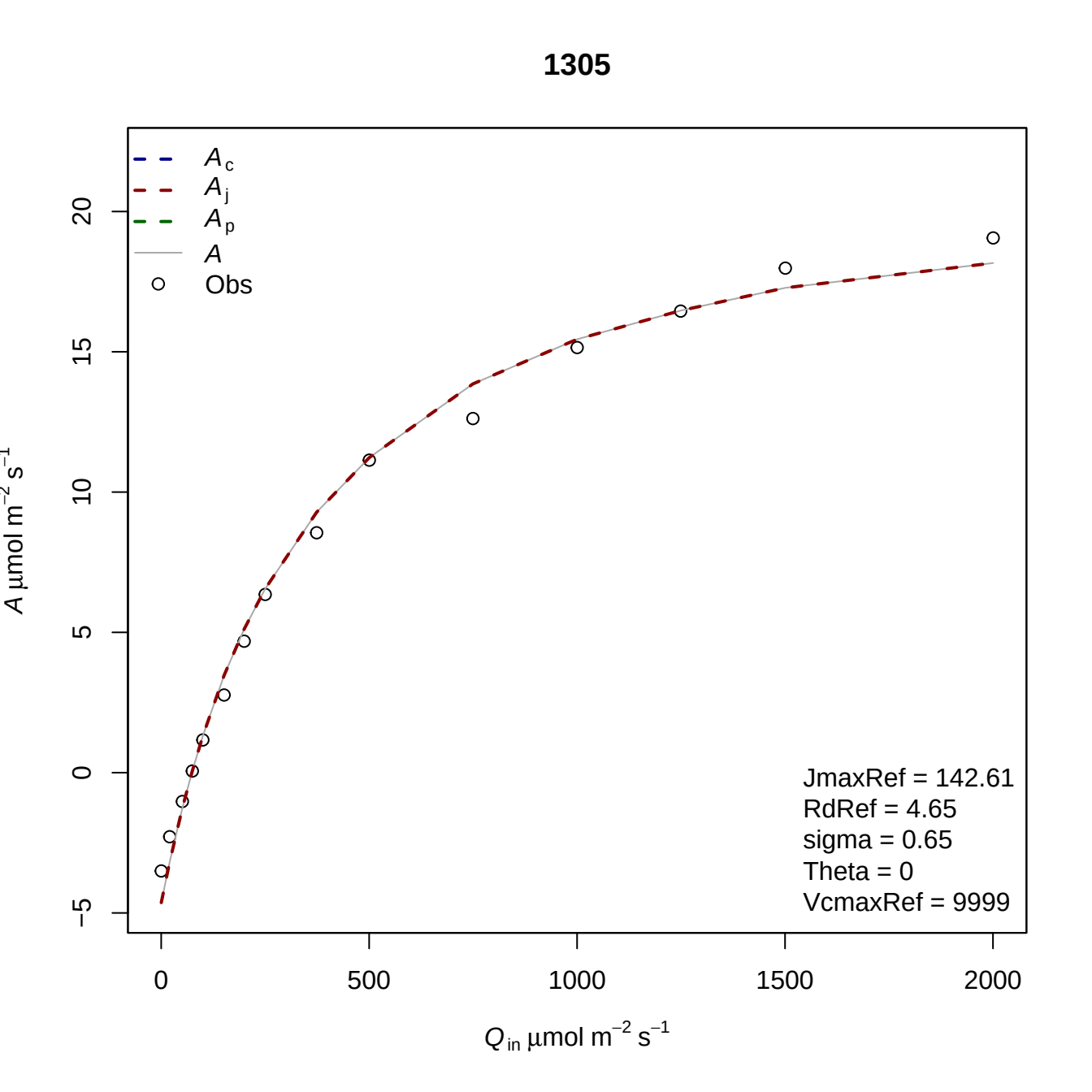

1300
A_c
A_j
A_p
A
Obs
JmaxRef = 129.38
RdRef = 2.24
sigma = 0.32
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

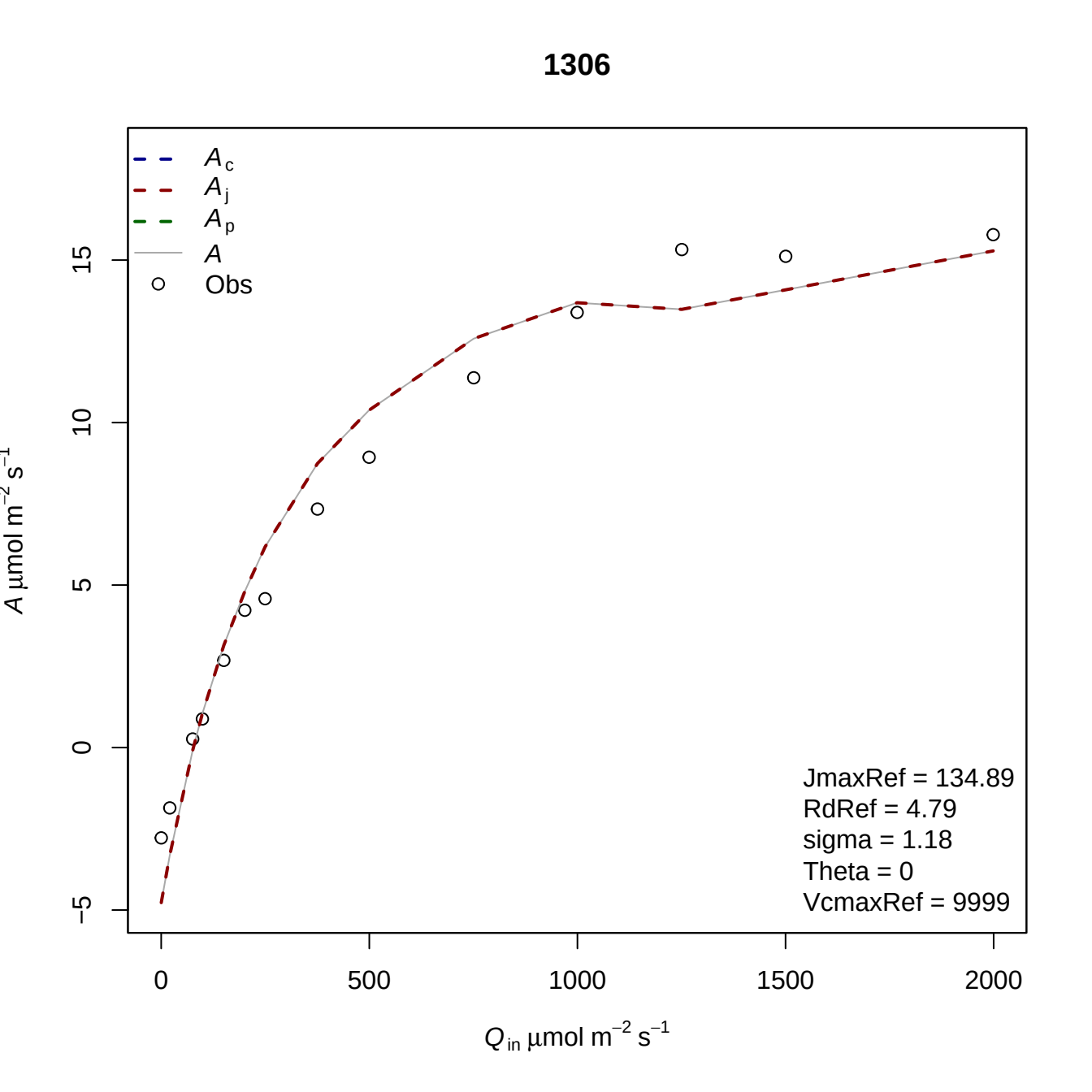

1301
A_c
A_j
A_p
A
Obs
JmaxRef = 108.45
RdRef = 4.48
sigma = 0.73
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

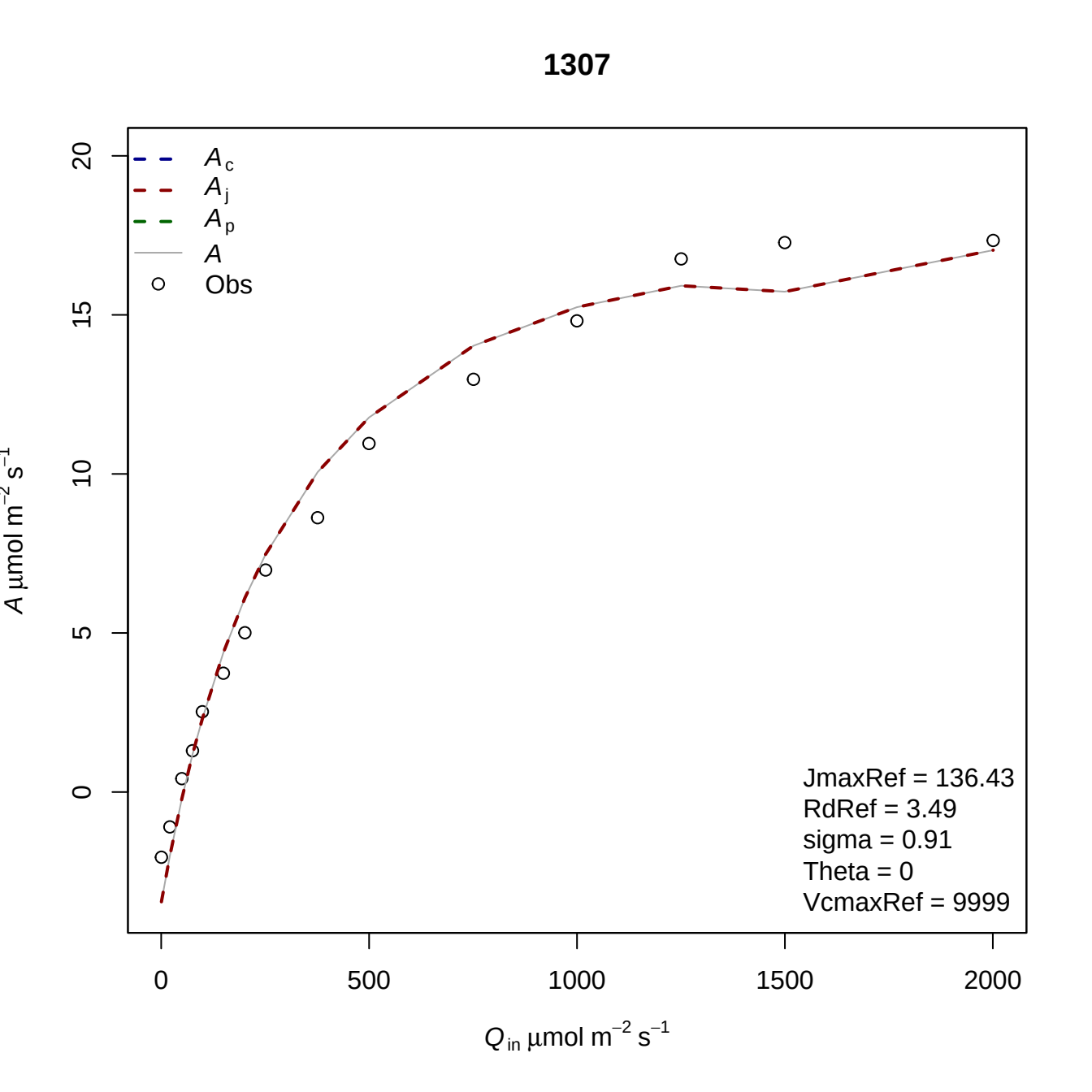

1302
A_c
A_j
A_p
A
Obs
JmaxRef = 95.35
RdRef = 3.93
sigma = 0.43
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

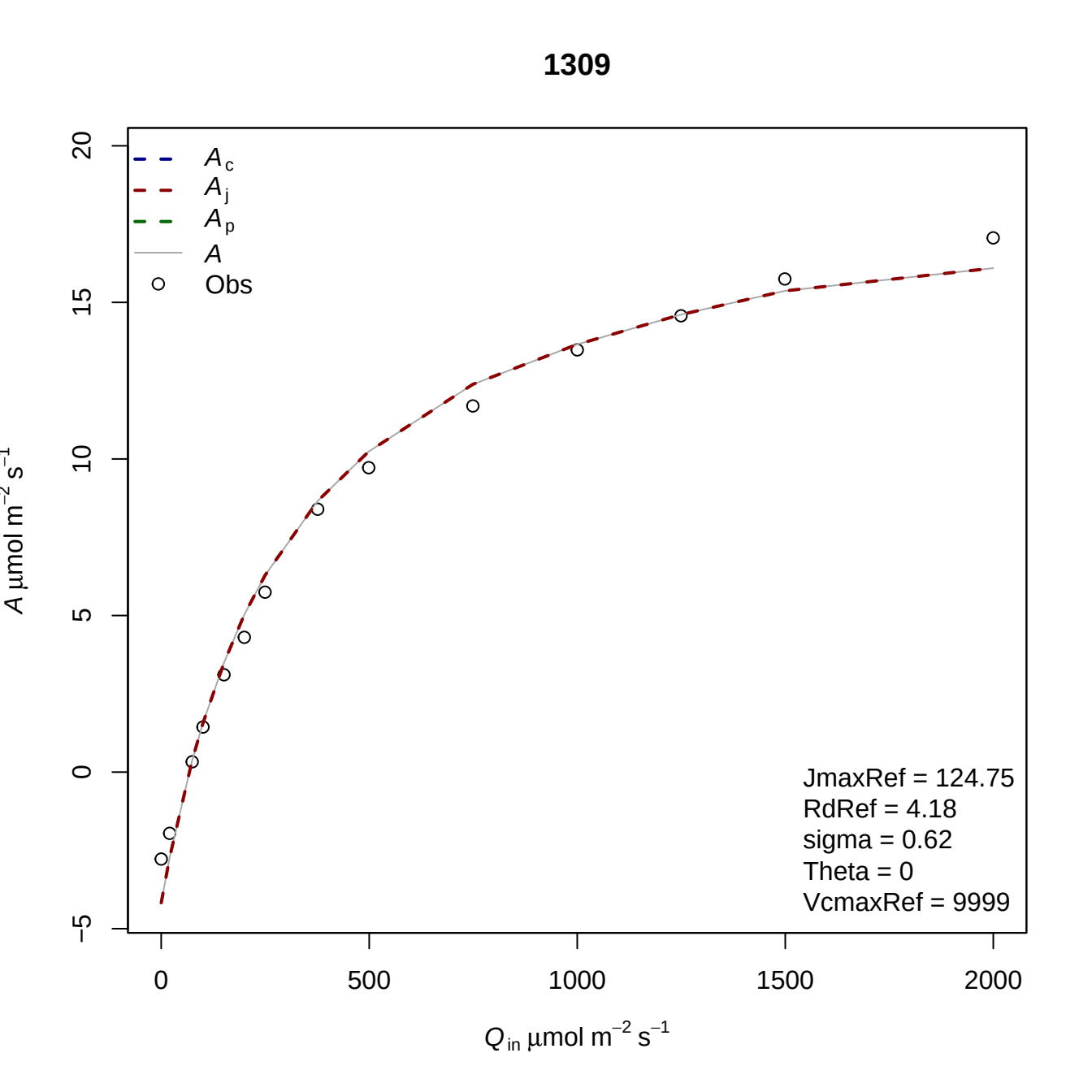

1303
A_c
A_j
A_p
A
Obs
JmaxRef = 94.9
RdRef = 3.84
sigma = 0.43
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

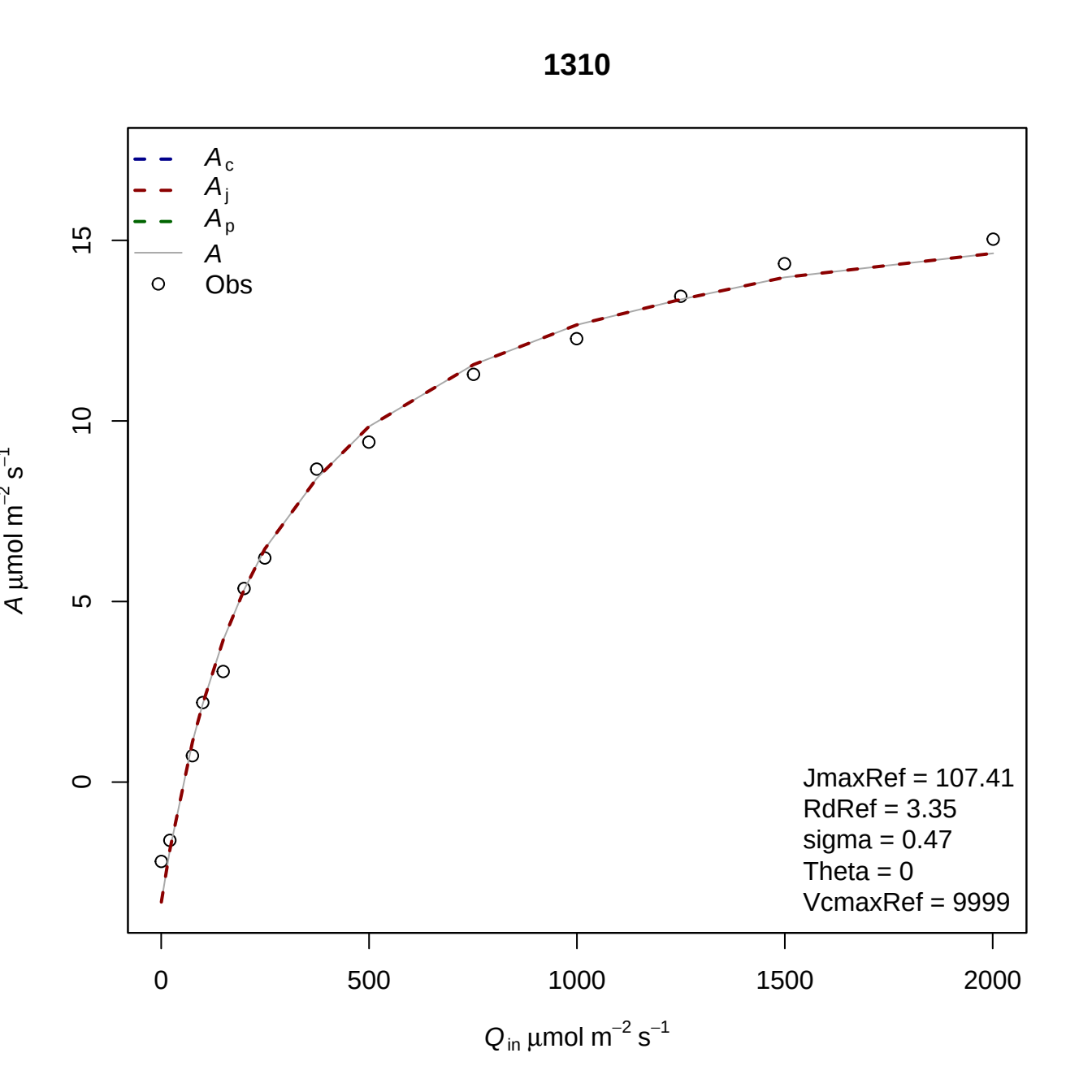

1304
A_c
A_j
A_p
A
Obs
JmaxRef = 130.39
RdRef = 4.43
sigma = 0.41
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

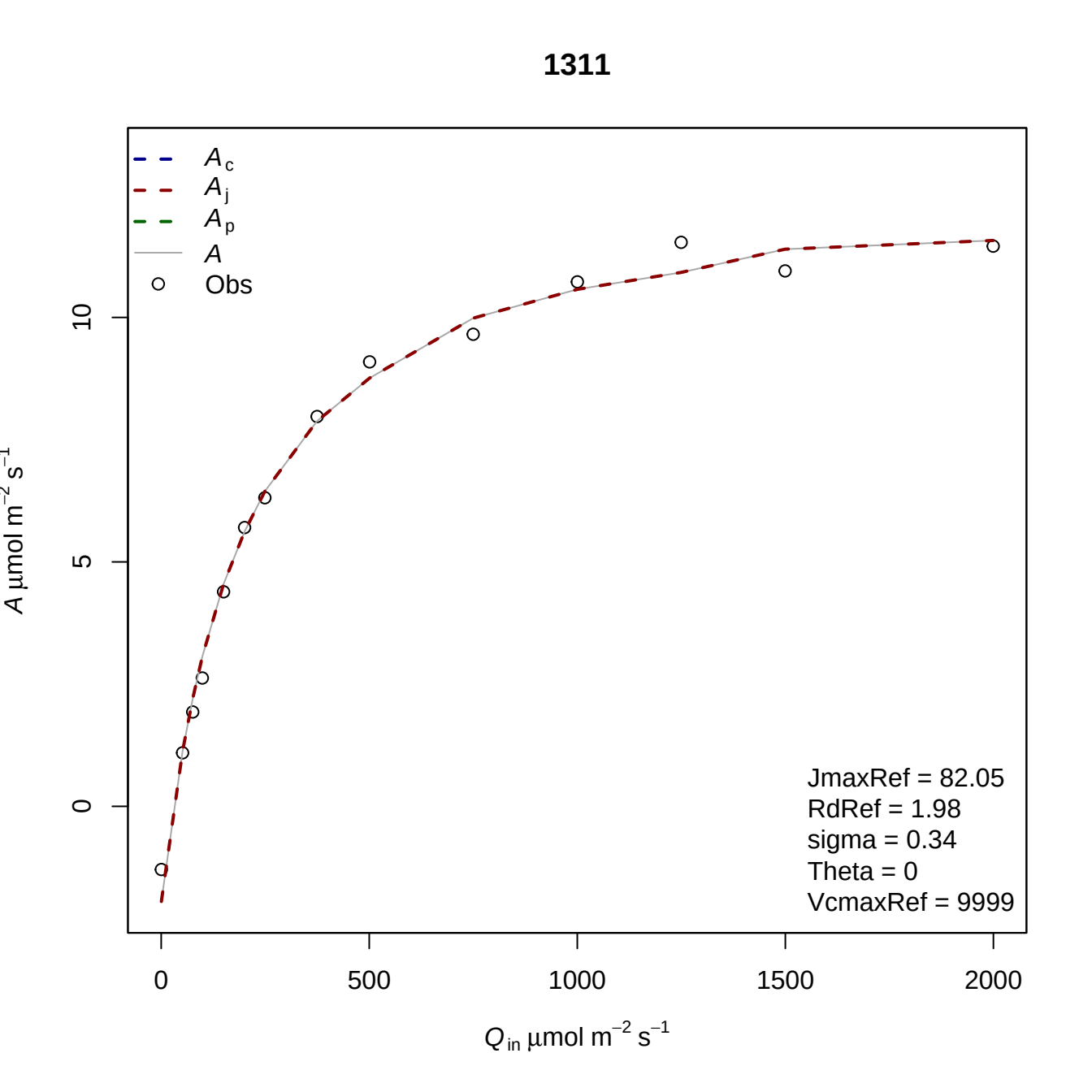

1305
A_c
A_j
A_p
A
Obs
JmaxRef = 142.61
RdRef = 4.65
sigma = 0.65
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

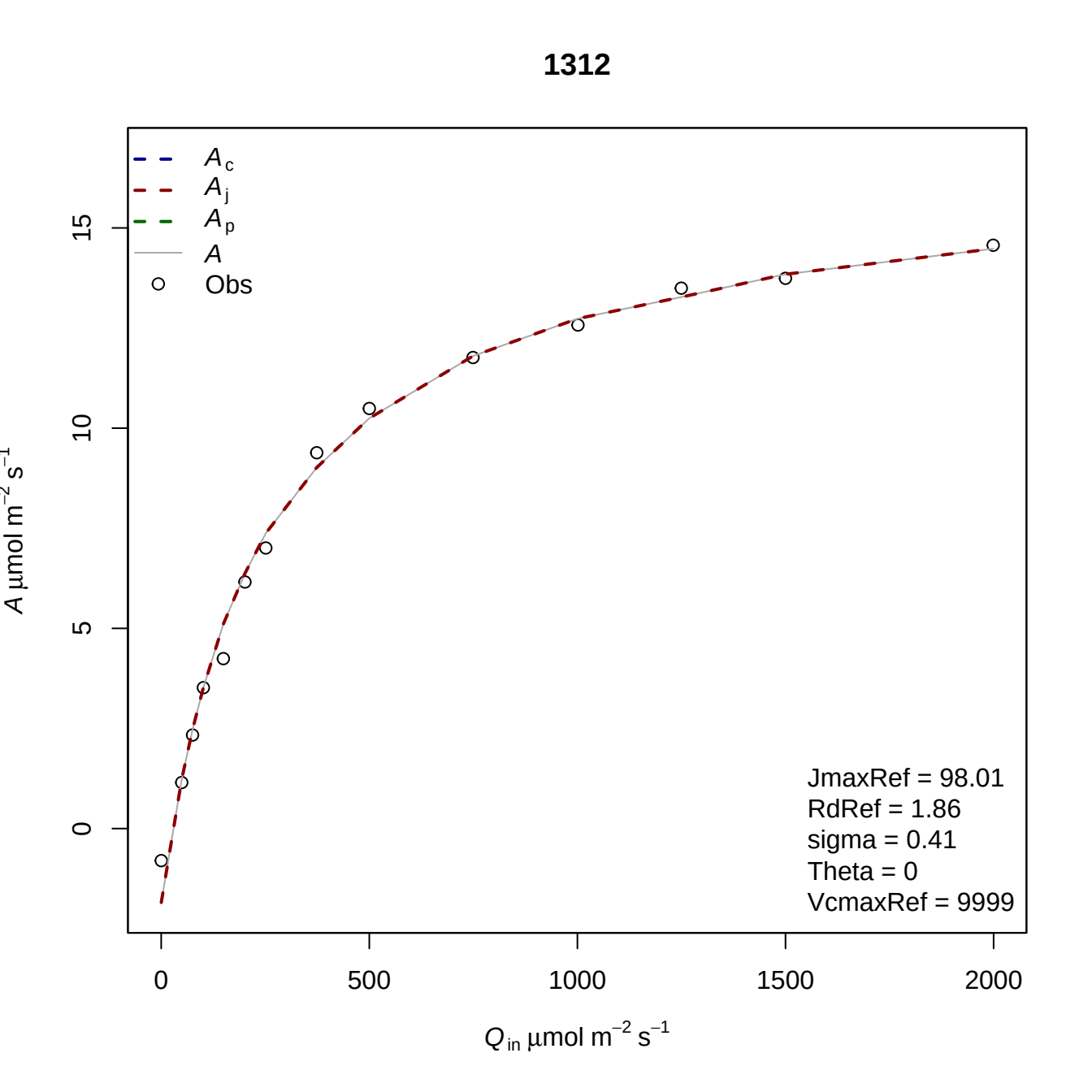

1306
A_c
A_j
A_p
A
Obs
JmaxRef = 134.89
RdRef = 4.79
sigma = 1.18
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

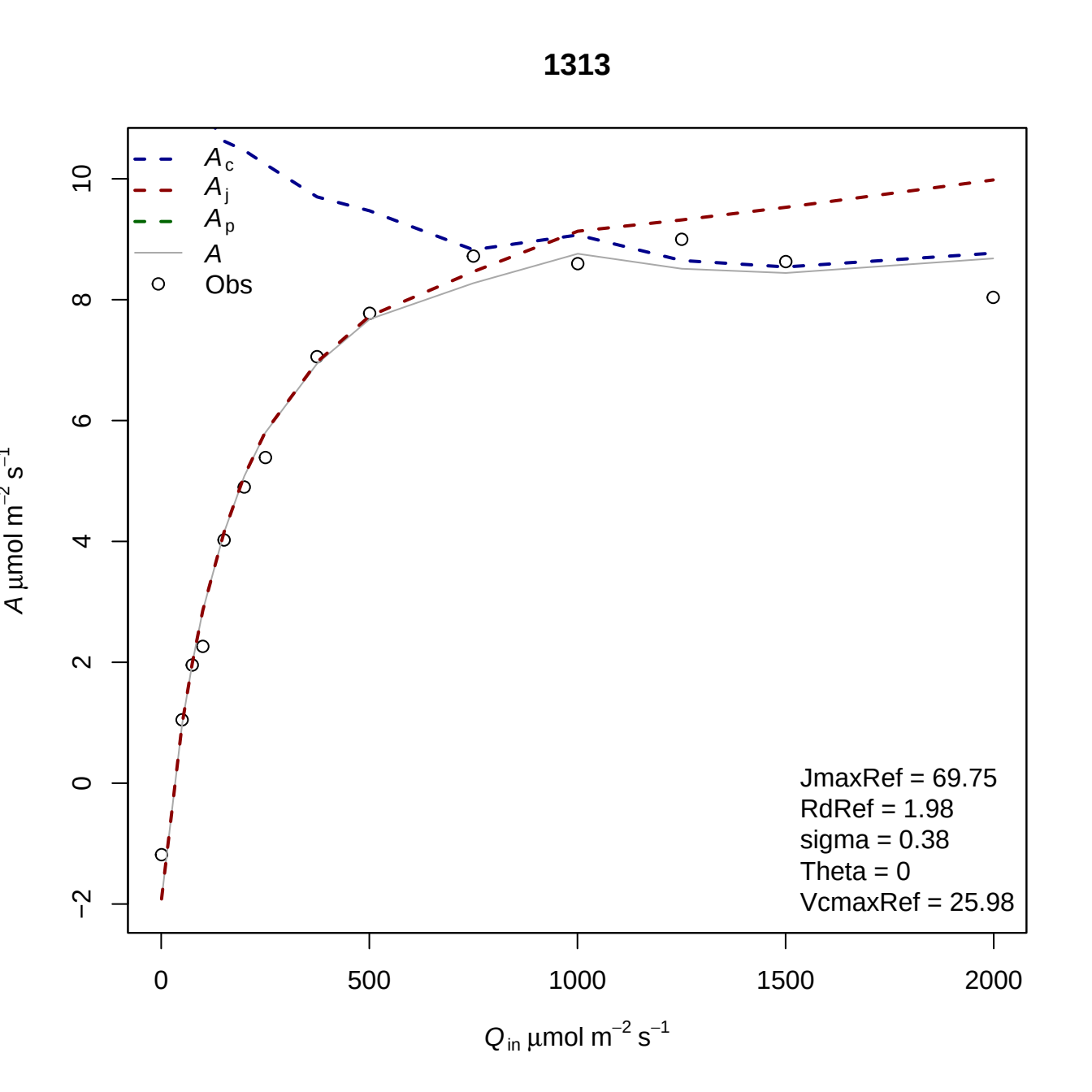

1307
A_c
A_j
A_p
A
Obs
JmaxRef = 136.43
RdRef = 3.49
sigma = 0.91
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

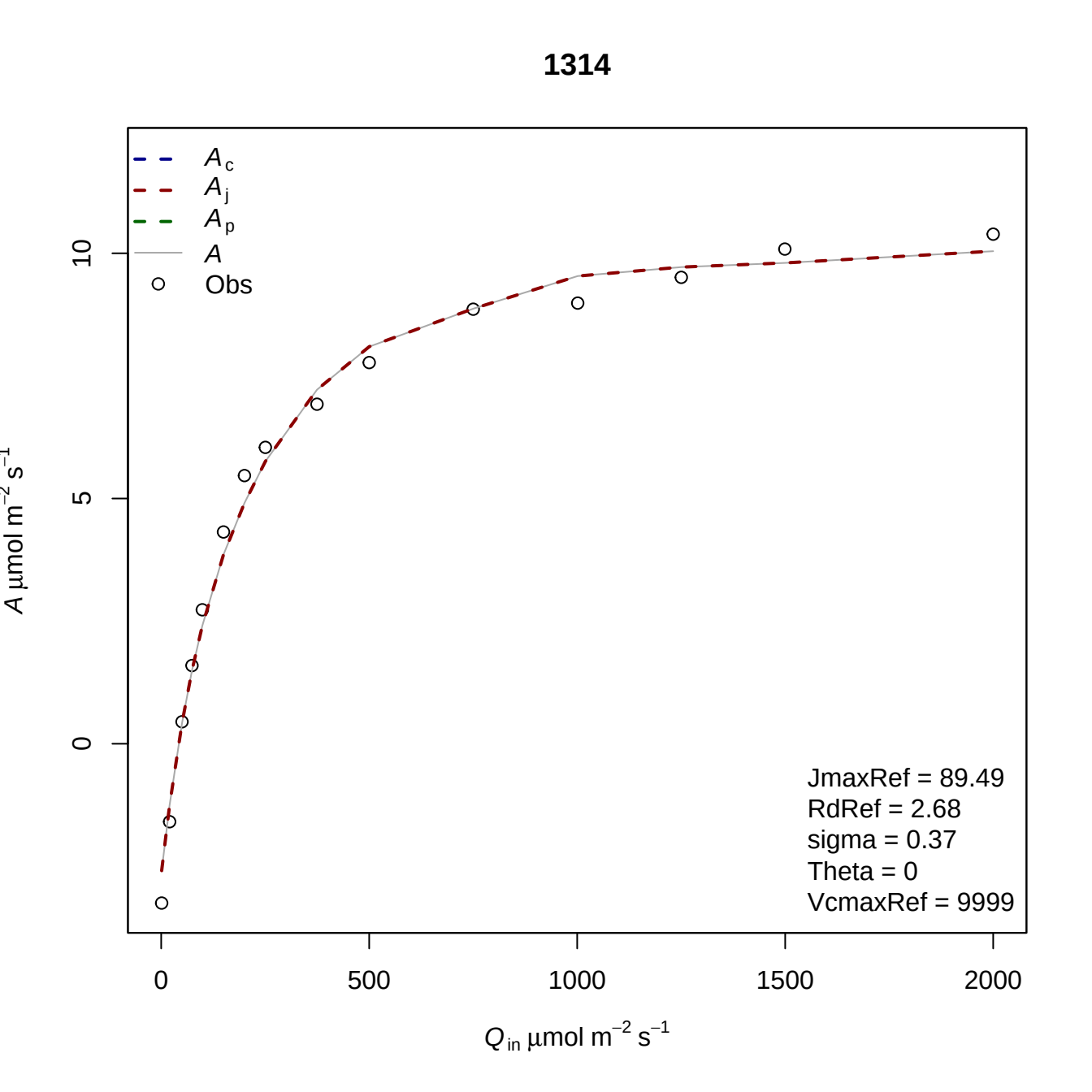

1309
A_c
A_j
A_p
A
Obs
JmaxRef = 124.75
RdRef = 4.18
sigma = 0.62
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

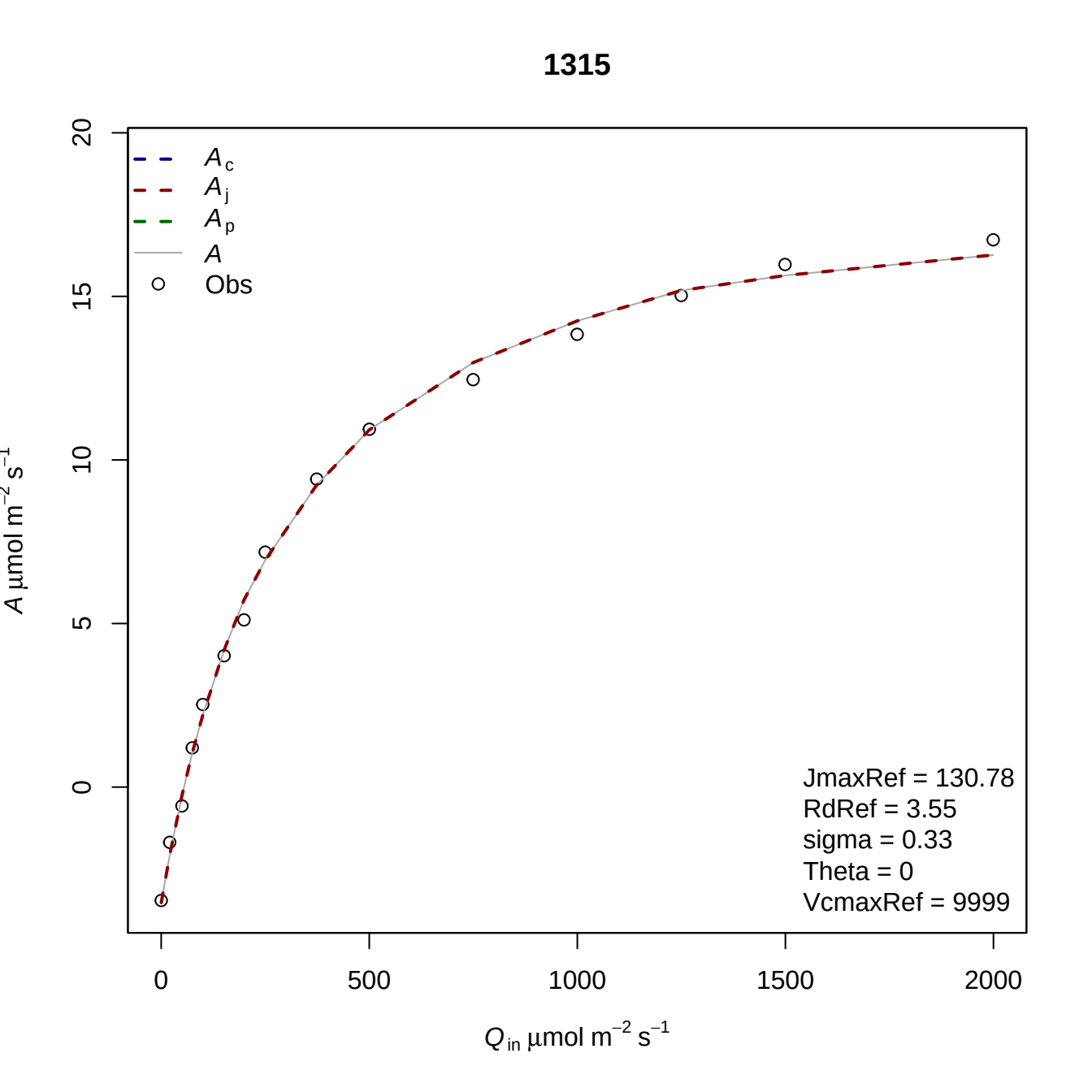

1310
A_c
A_j
A_p
A
Obs
JmaxRef = 107.41
RdRef = 3.35
sigma = 0.47
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

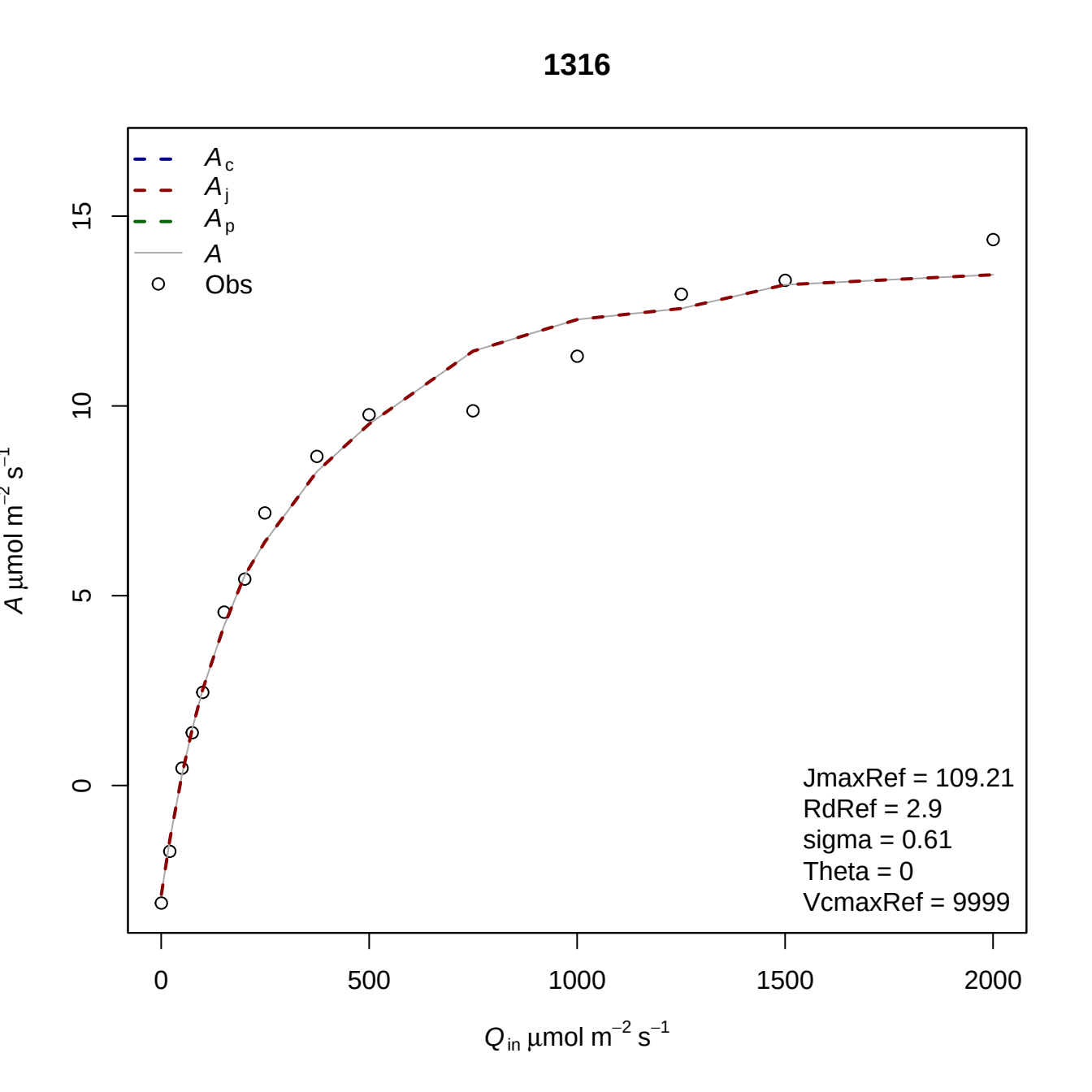

1311
A_c
A_j
A_p
A
Obs
JmaxRef = 82.05
RdRef = 1.98
sigma = 0.34
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

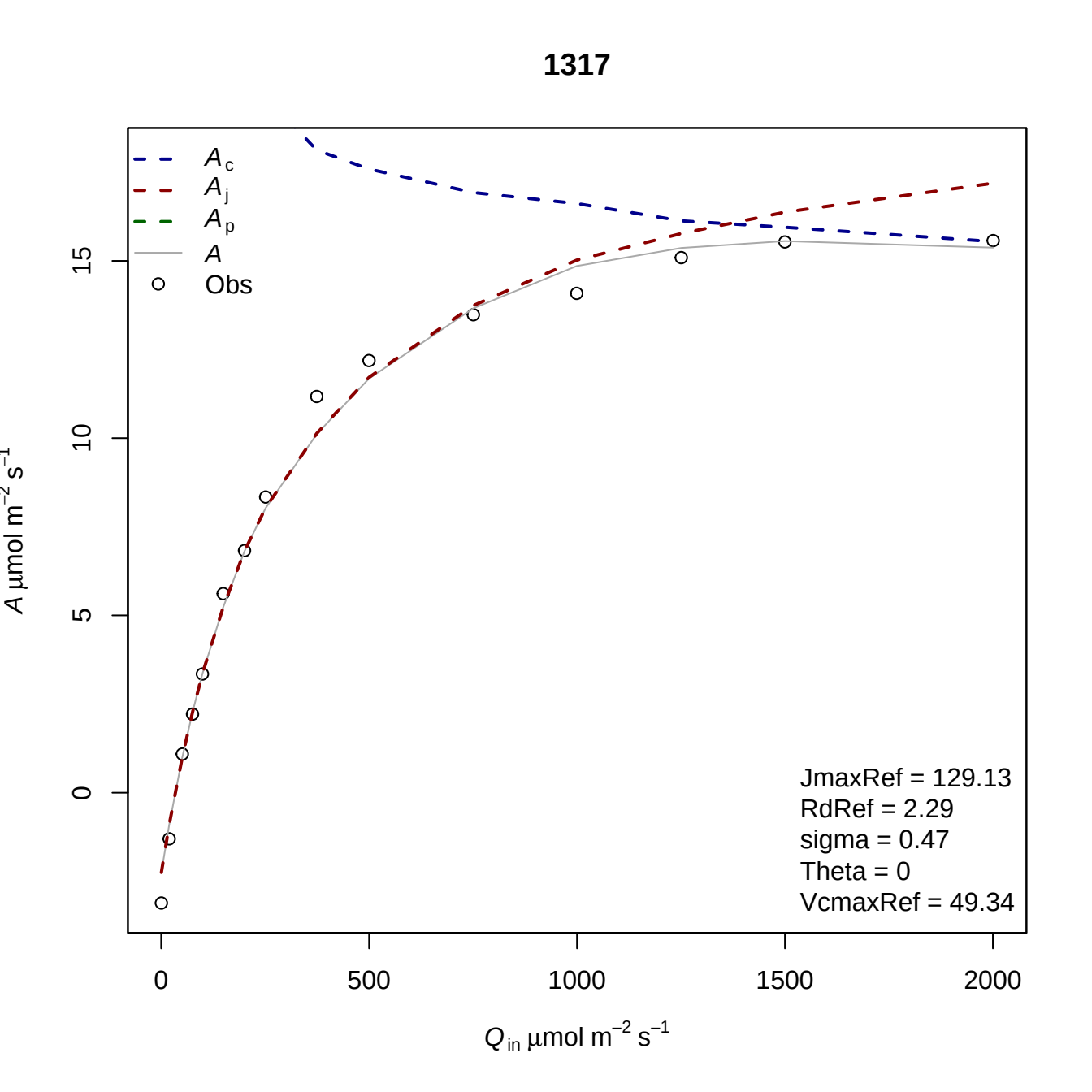

1312
A_c
A_j
A_p
A
Obs
JmaxRef = 98.01
RdRef = 1.86
sigma = 0.41
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

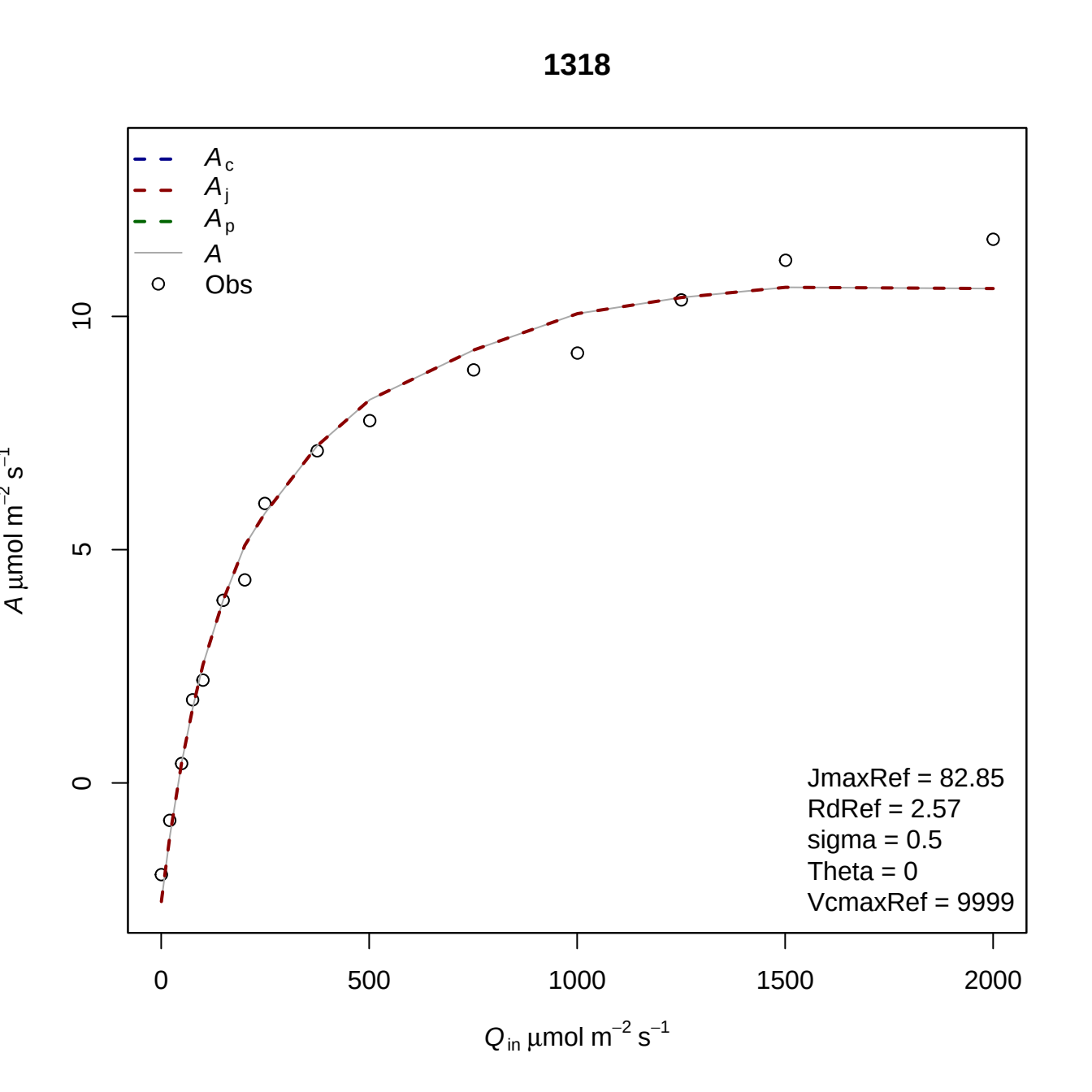

1313
A_c
A_j
A_p
A
Obs
JmaxRef = 69.75
RdRef = 1.98
sigma = 0.38
Theta = 0
VcmaxRef = 25.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

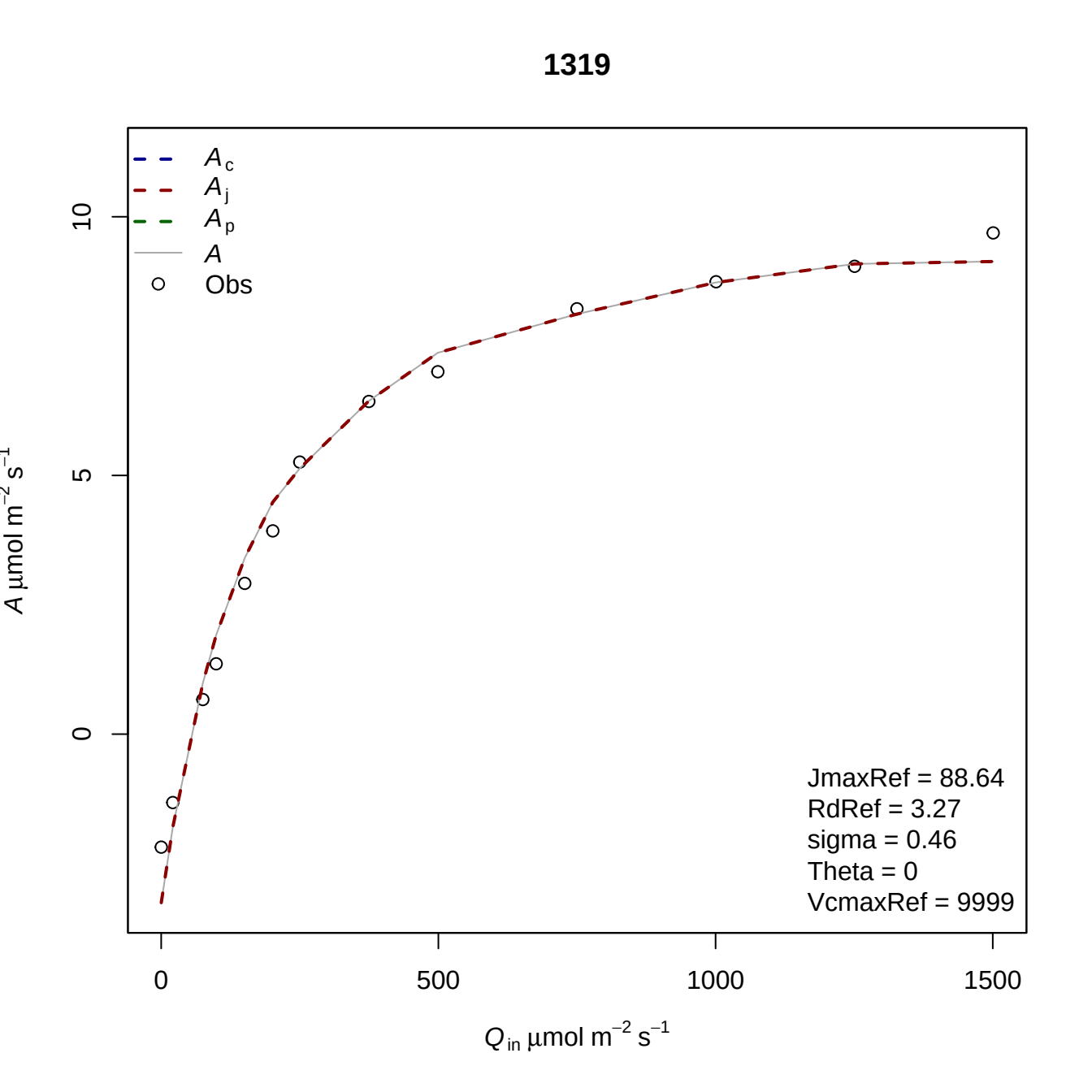

1314
A_c
A_j
A_p
A
Obs
JmaxRef = 89.49
RdRef = 2.68
sigma = 0.37
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

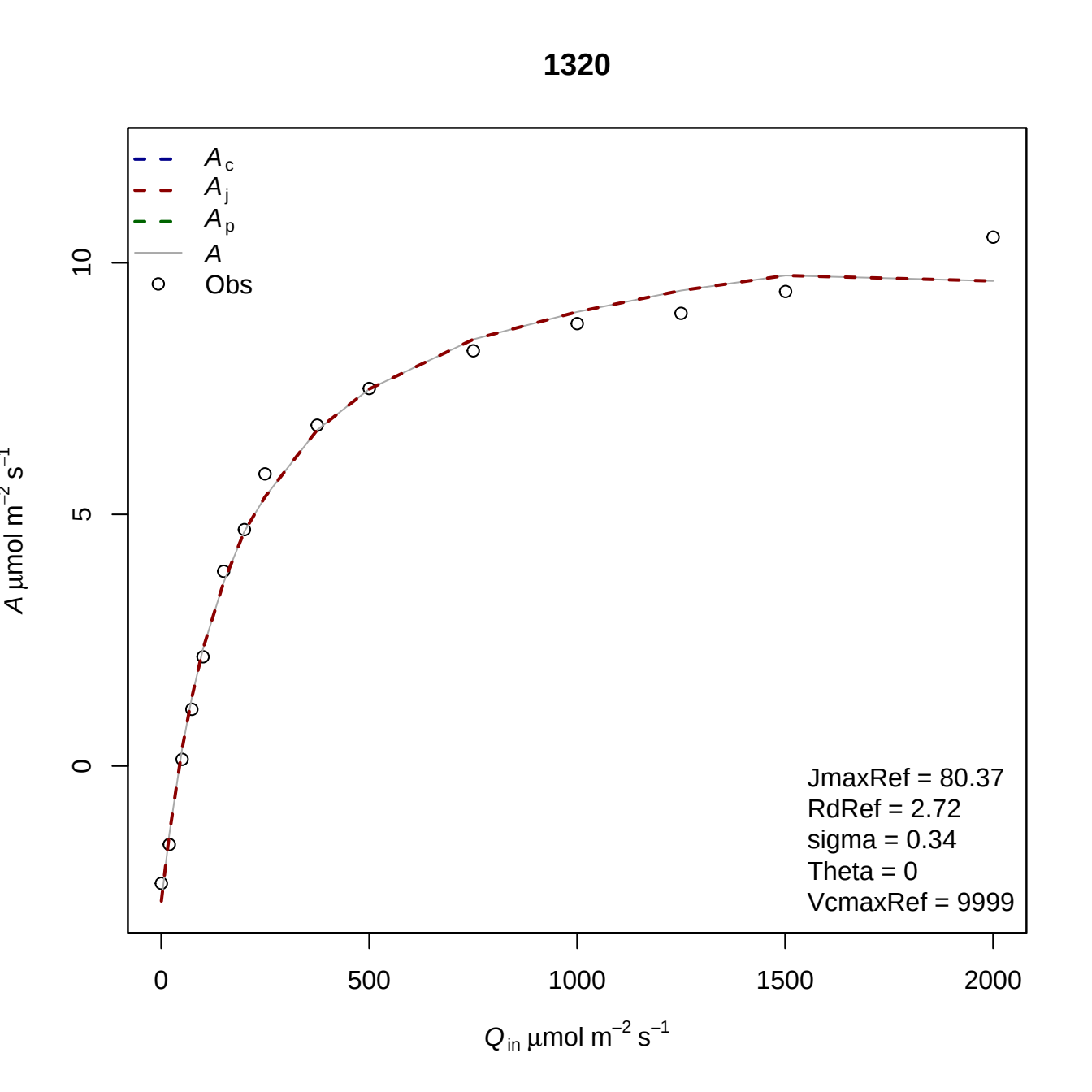

1315
A_c
A_j
A_p
A
Obs
JmaxRef = 130.78
RdRef = 3.55
sigma = 0.33
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

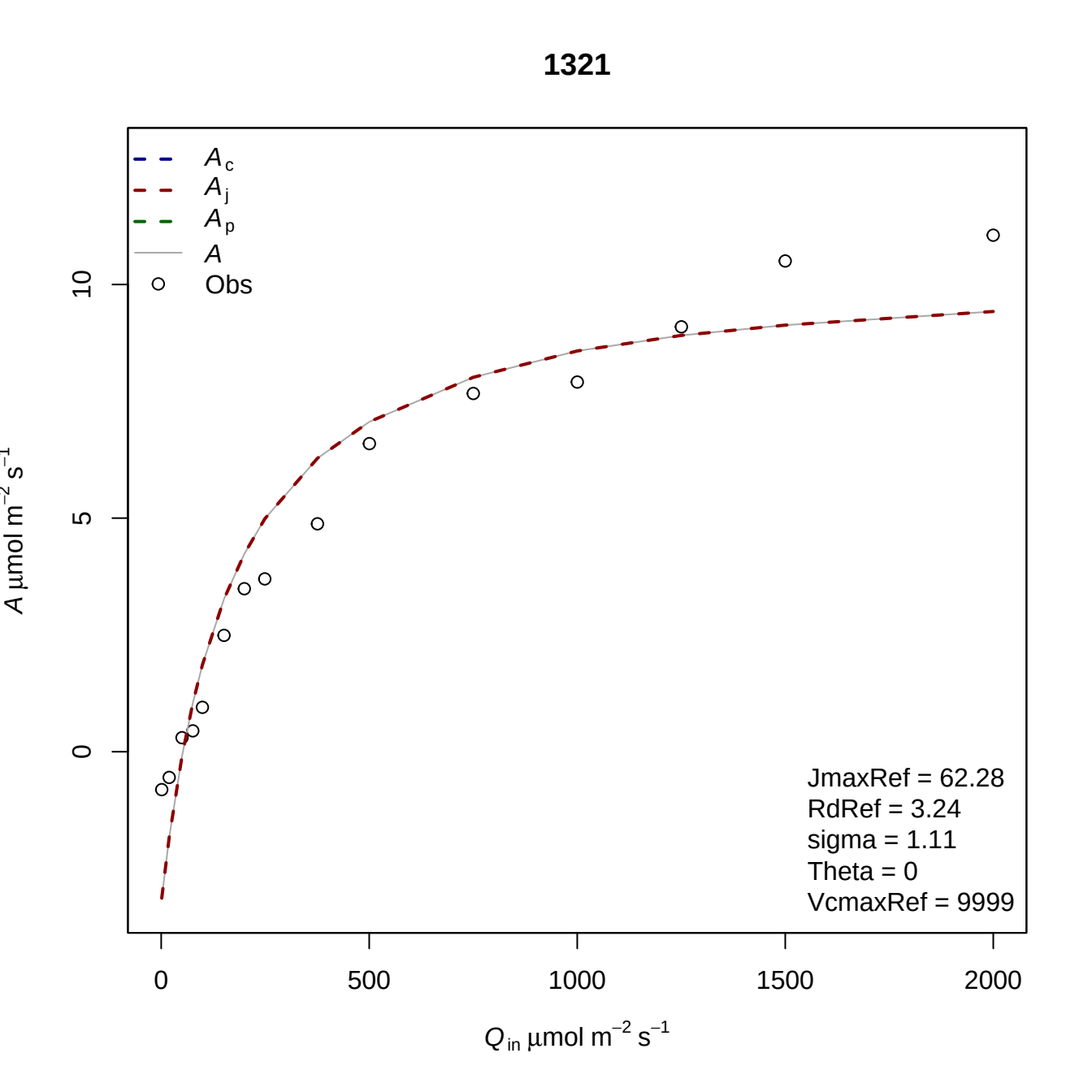

1316
A_c
A_j
A_p
A
Obs
JmaxRef = 109.21
RdRef = 2.9
sigma = 0.61
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

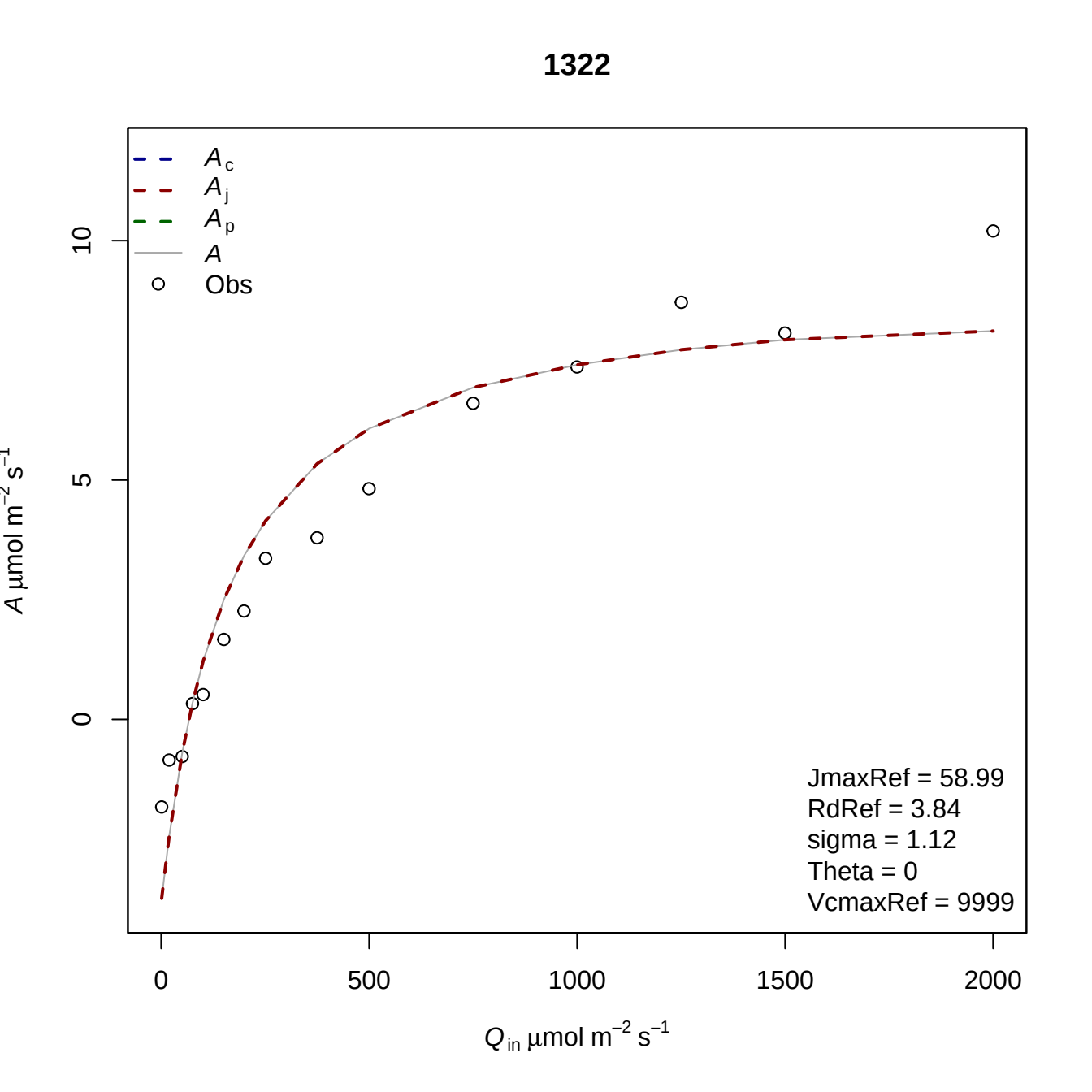

1317
A_c
A_j
A_p
A
Obs
JmaxRef = 129.13
RdRef = 2.29
sigma = 0.47
Theta = 0
VcmaxRef = 49.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

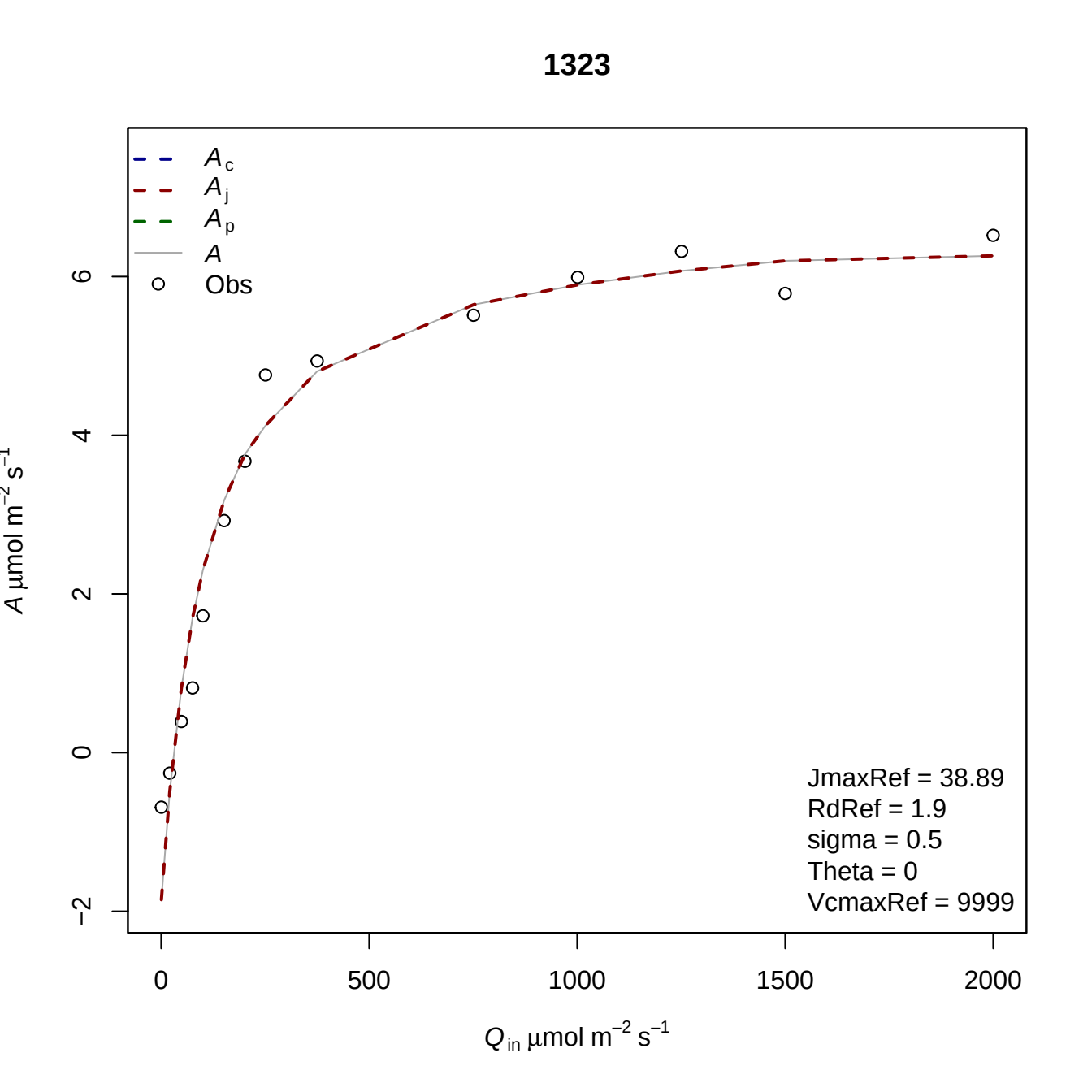

1318
A_c
A_j
A_p
A
Obs
JmaxRef = 82.85
RdRef = 2.57
sigma = 0.5
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

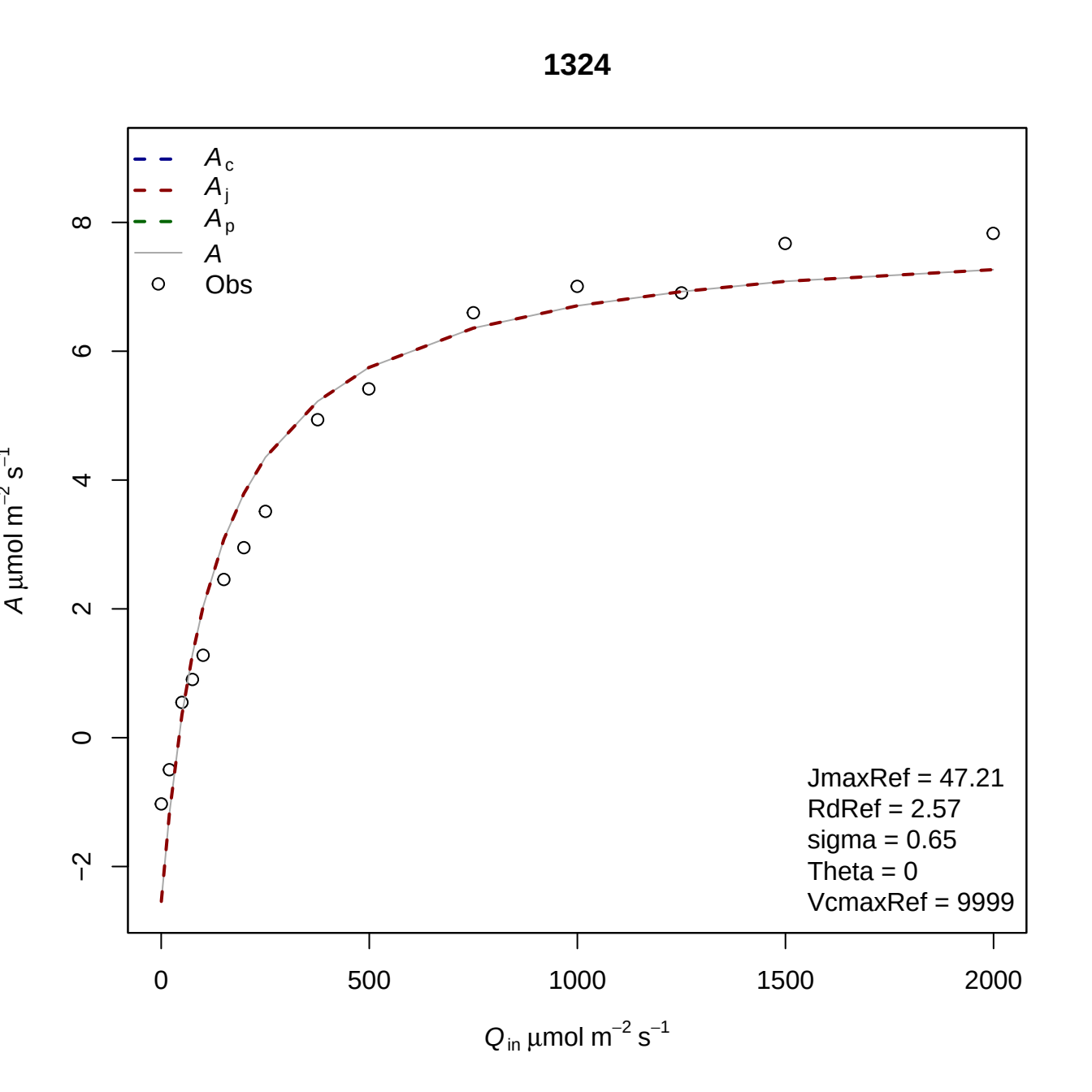

1319
A_c
A_j
A_p
A
Obs
JmaxRef = 88.64
RdRef = 3.27
sigma = 0.46
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

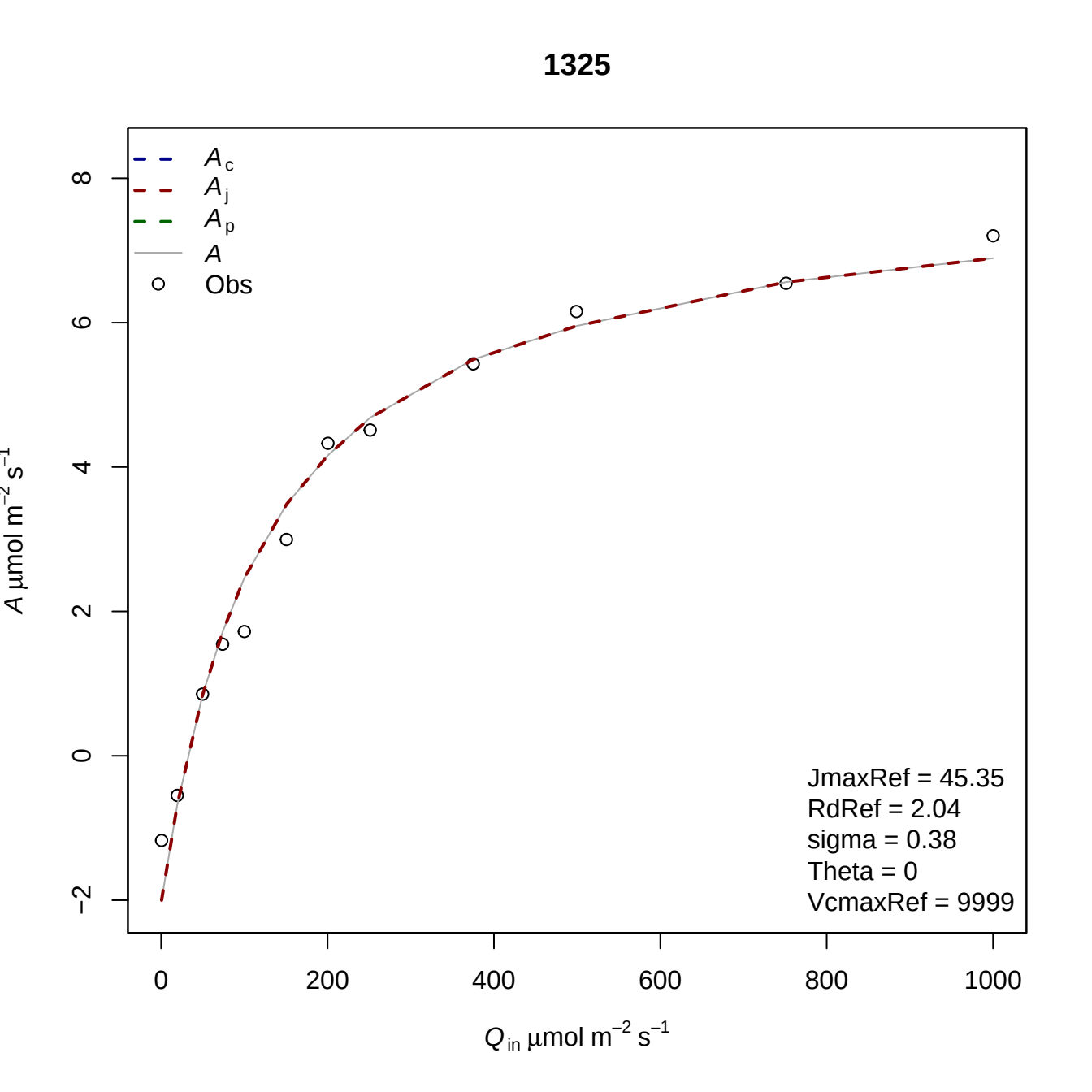

1320
A_c
A_j
A_p
A
Obs
JmaxRef = 80.37
RdRef = 2.72
sigma = 0.34
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

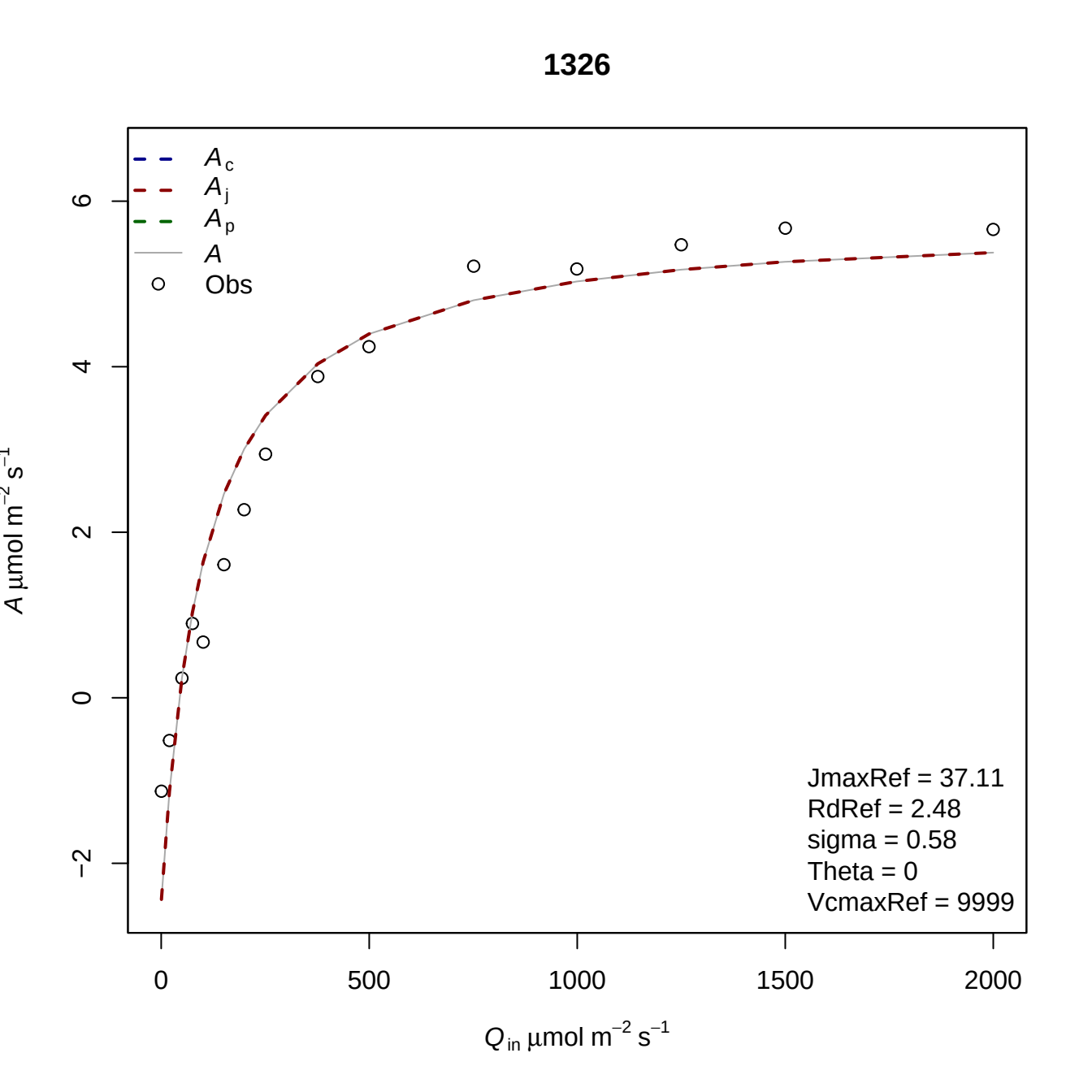

1321
A_c
A_j
A_p
A
Obs
JmaxRef = 62.28
RdRef = 3.24
sigma = 1.11
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

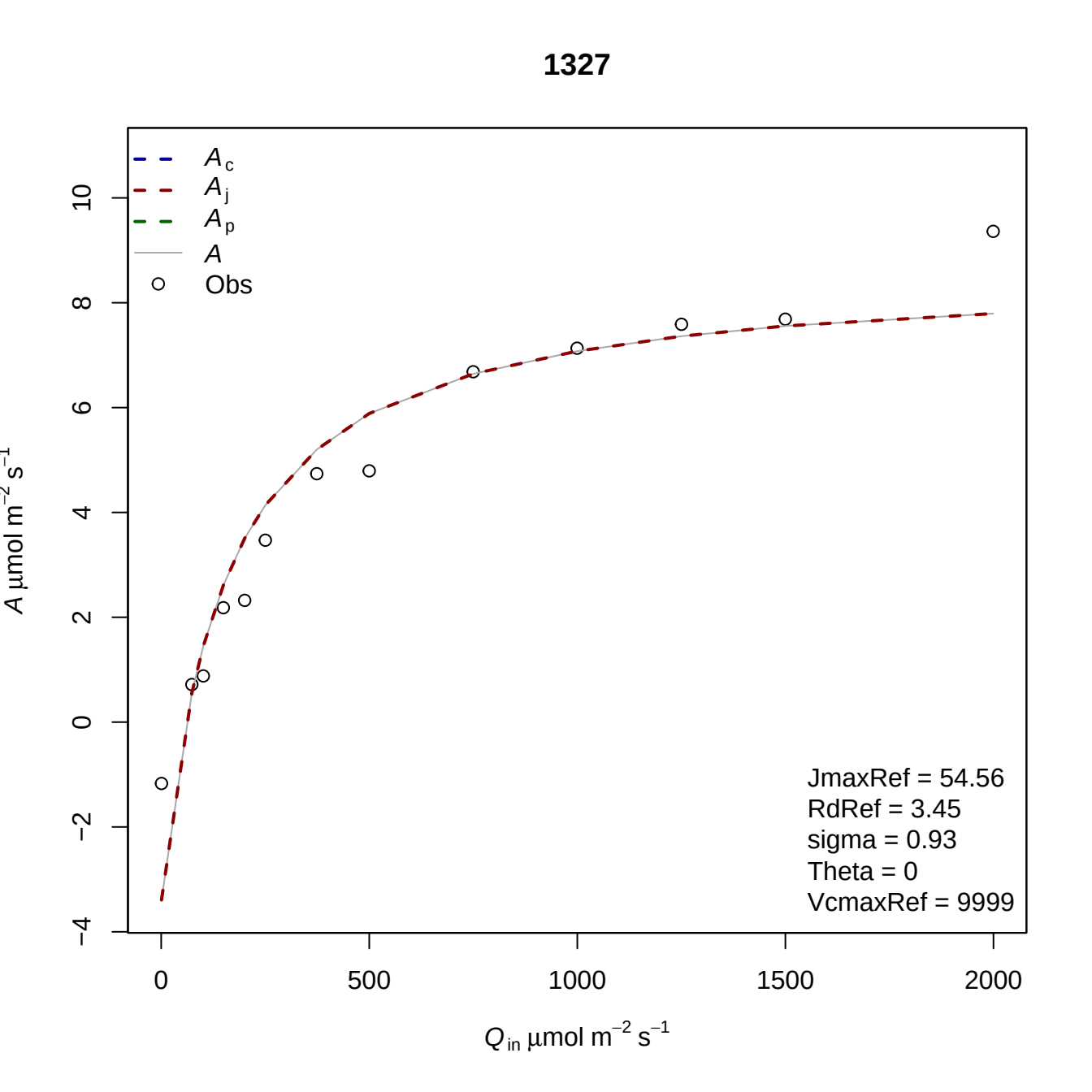

1322
A_c
A_j
A_p
A
Obs
JmaxRef = 58.99
RdRef = 3.84
sigma = 1.12
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

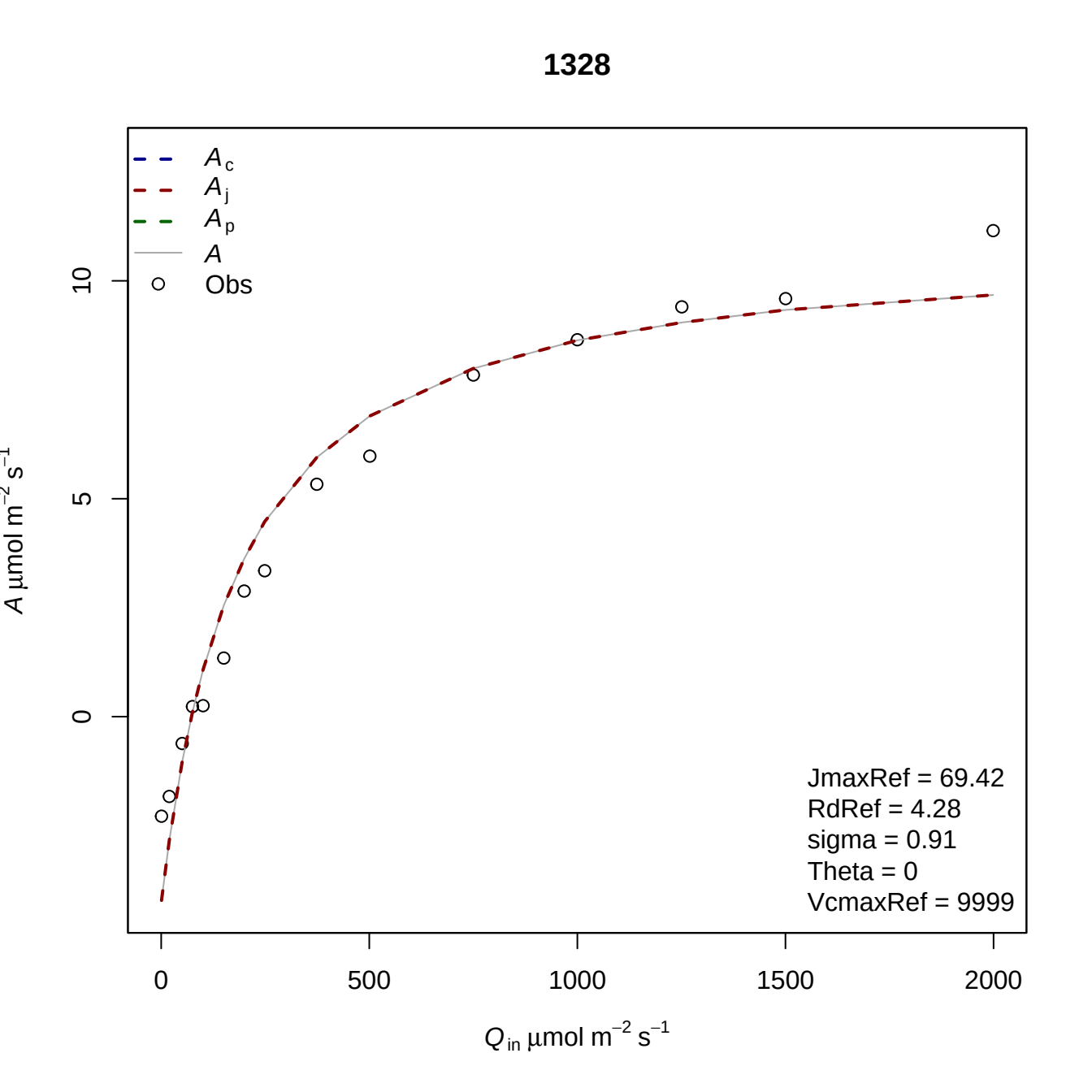

1323
A_c
A_j
A_p
A
Obs
JmaxRef = 38.89
RdRef = 1.9
sigma = 0.5
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

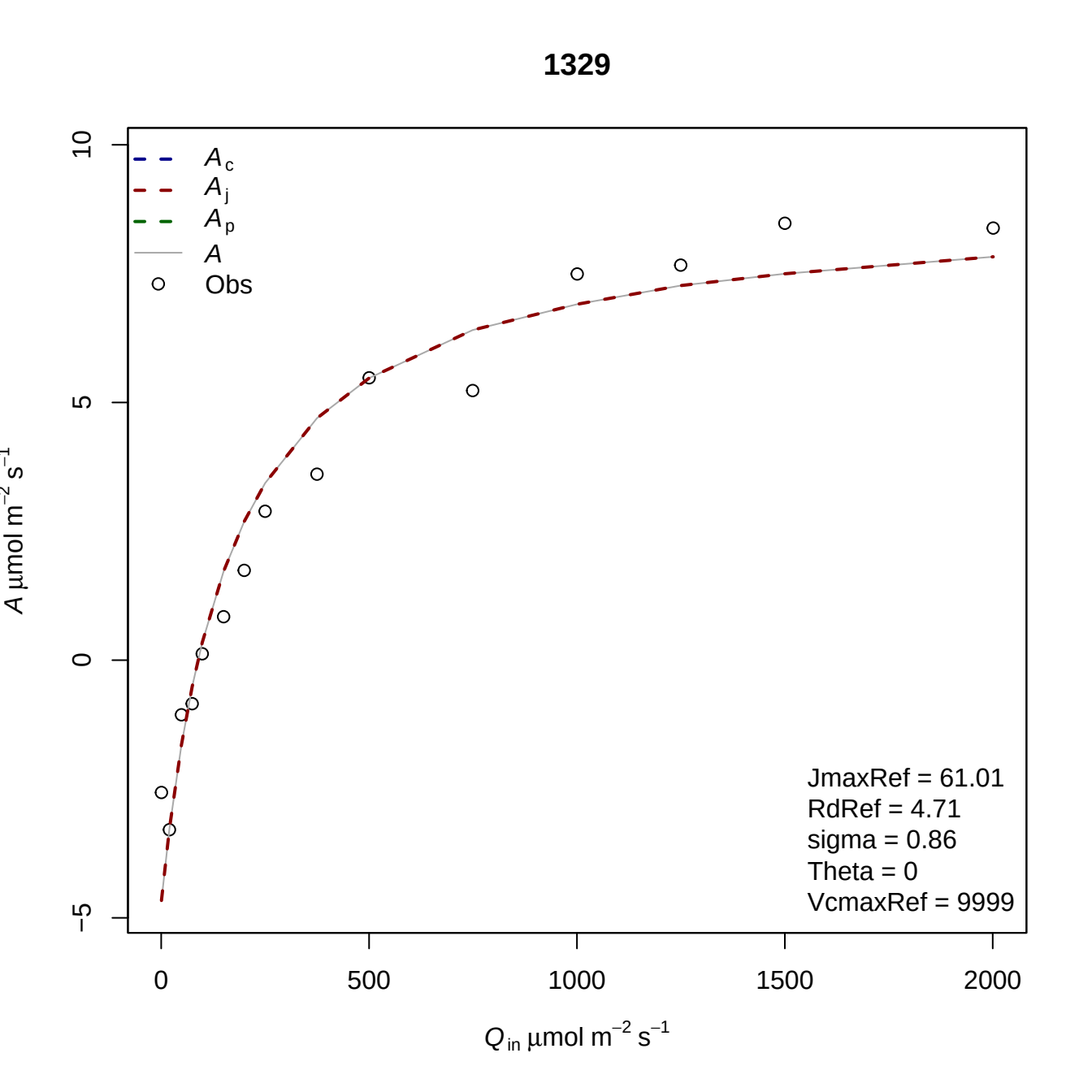

1324
A_c
A_j
A_p
A
Obs
JmaxRef = 47.21
RdRef = 2.57
sigma = 0.65
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

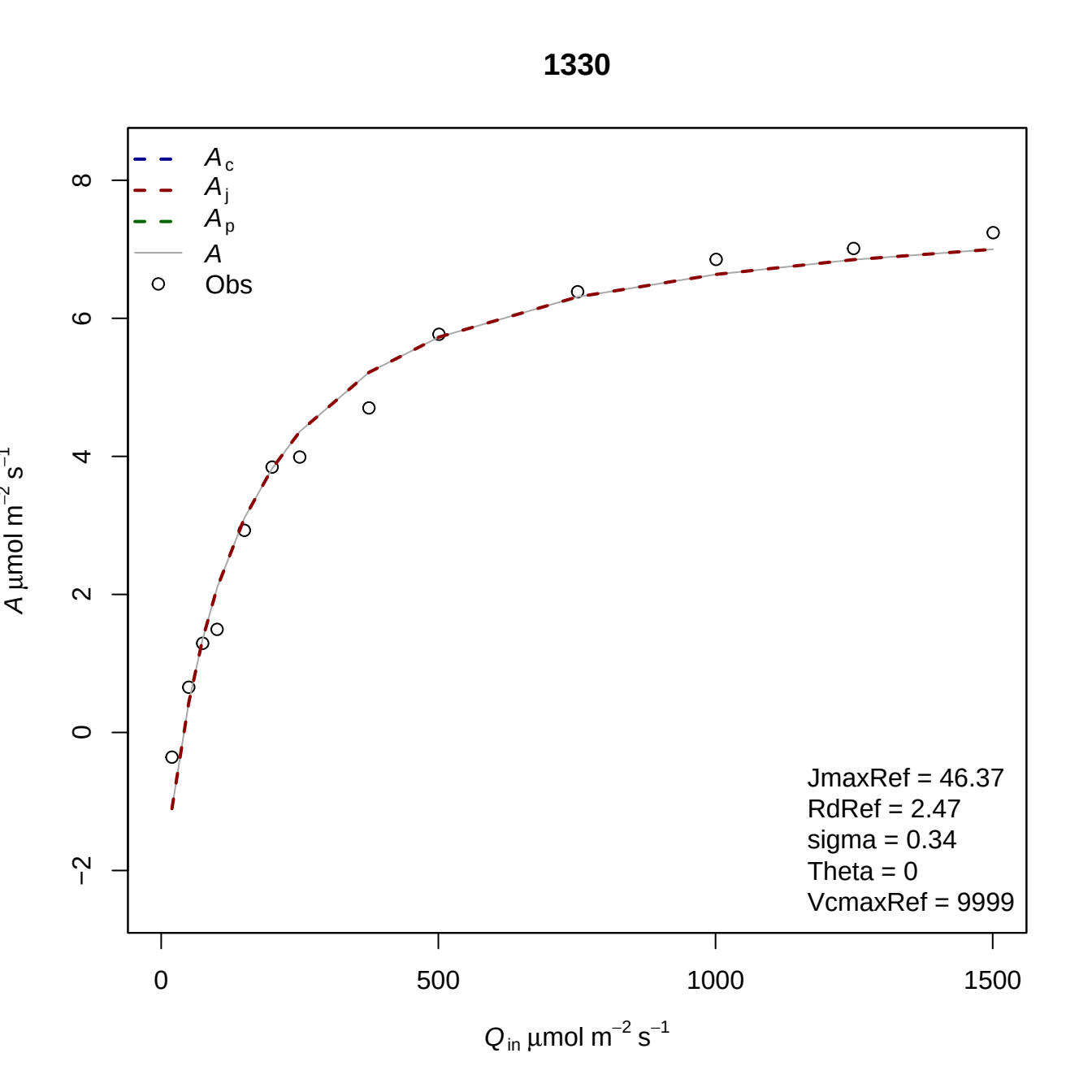

1325
A_c
A_j
A_p
A
Obs
JmaxRef = 45.35
RdRef = 2.04
sigma = 0.38
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

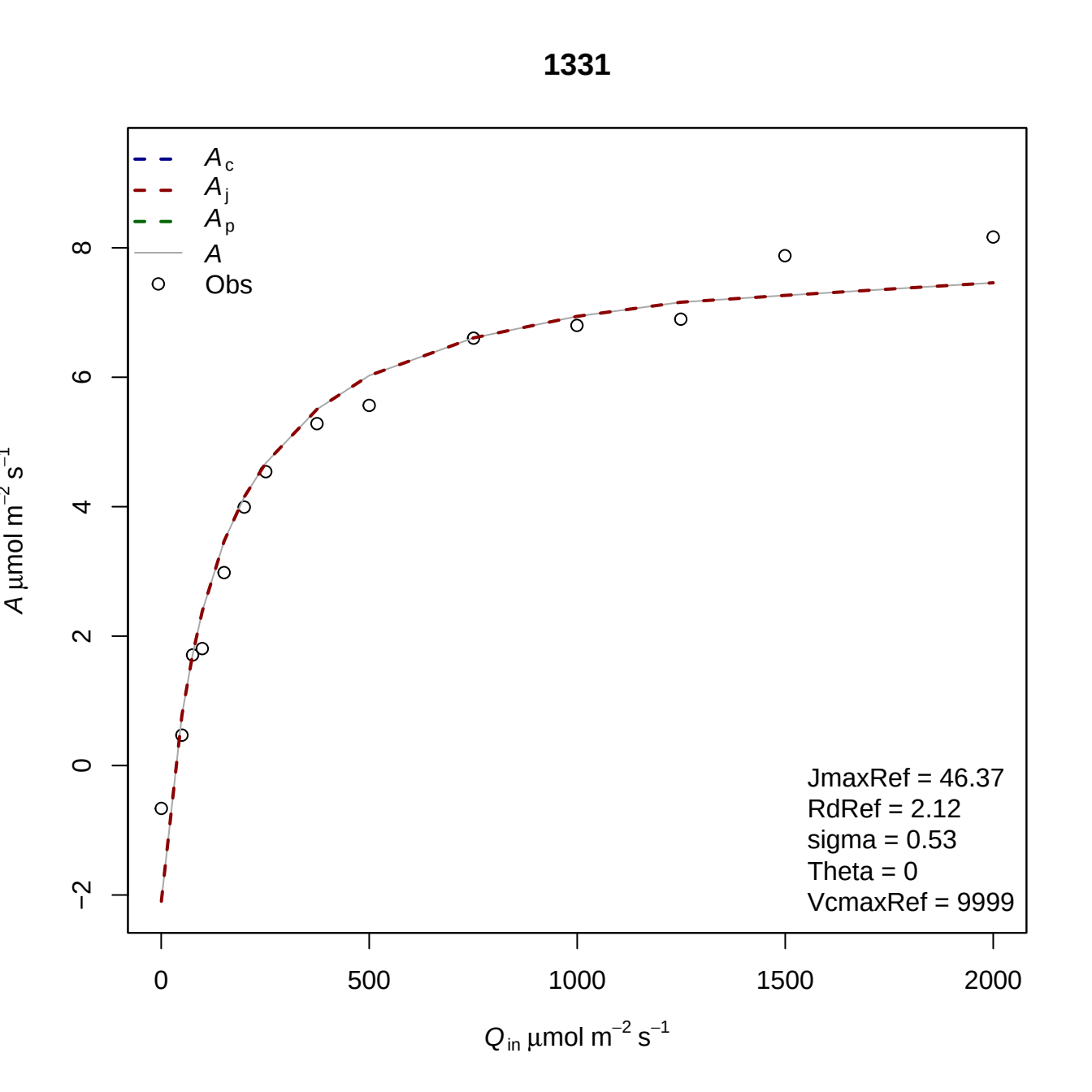

1326
A_c
A_j
A_p
A
Obs
JmaxRef = 37.11
RdRef = 2.48
sigma = 0.58
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

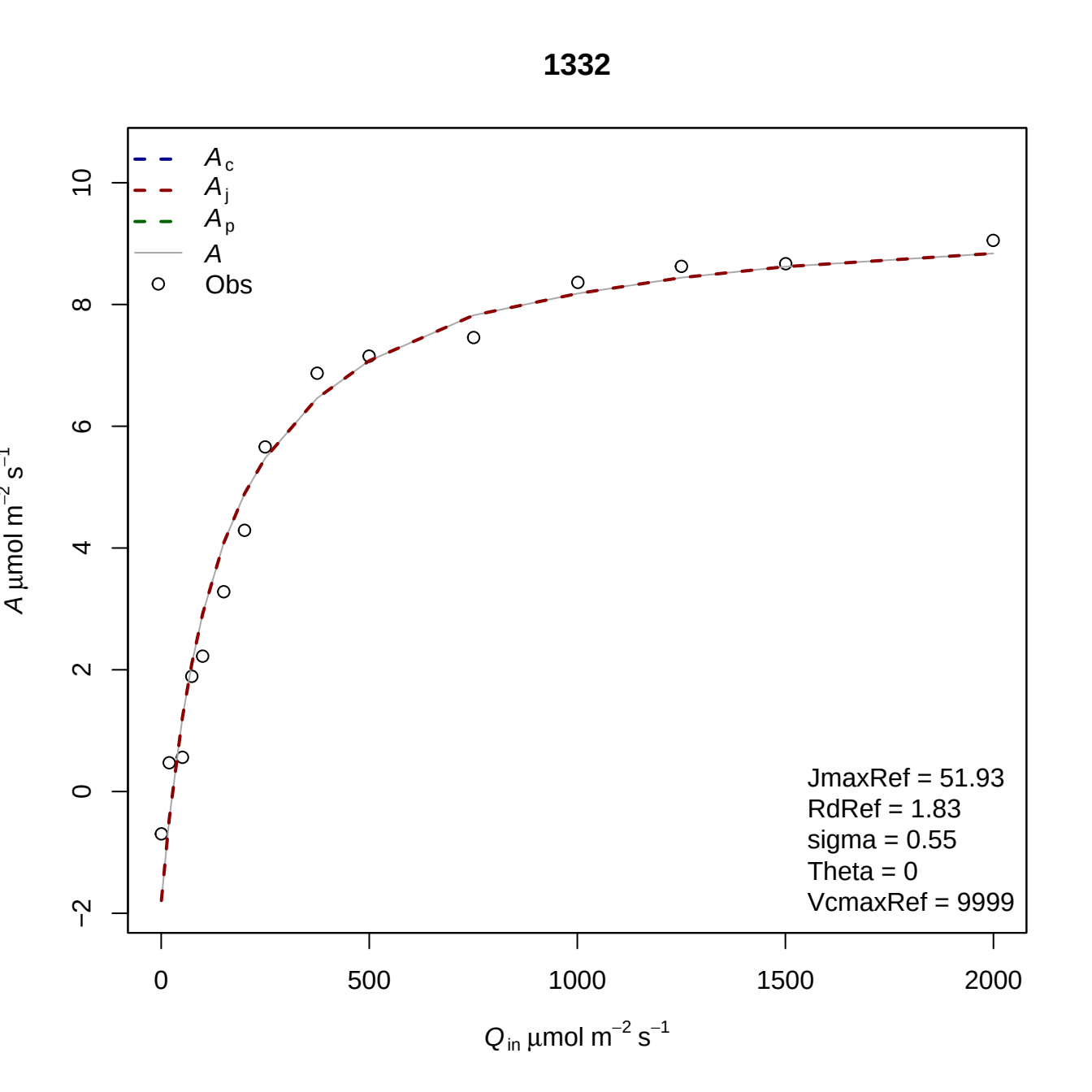

1327
A_c
A_j
A_p
A
Obs
JmaxRef = 54.56
RdRef = 3.45
sigma = 0.93
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

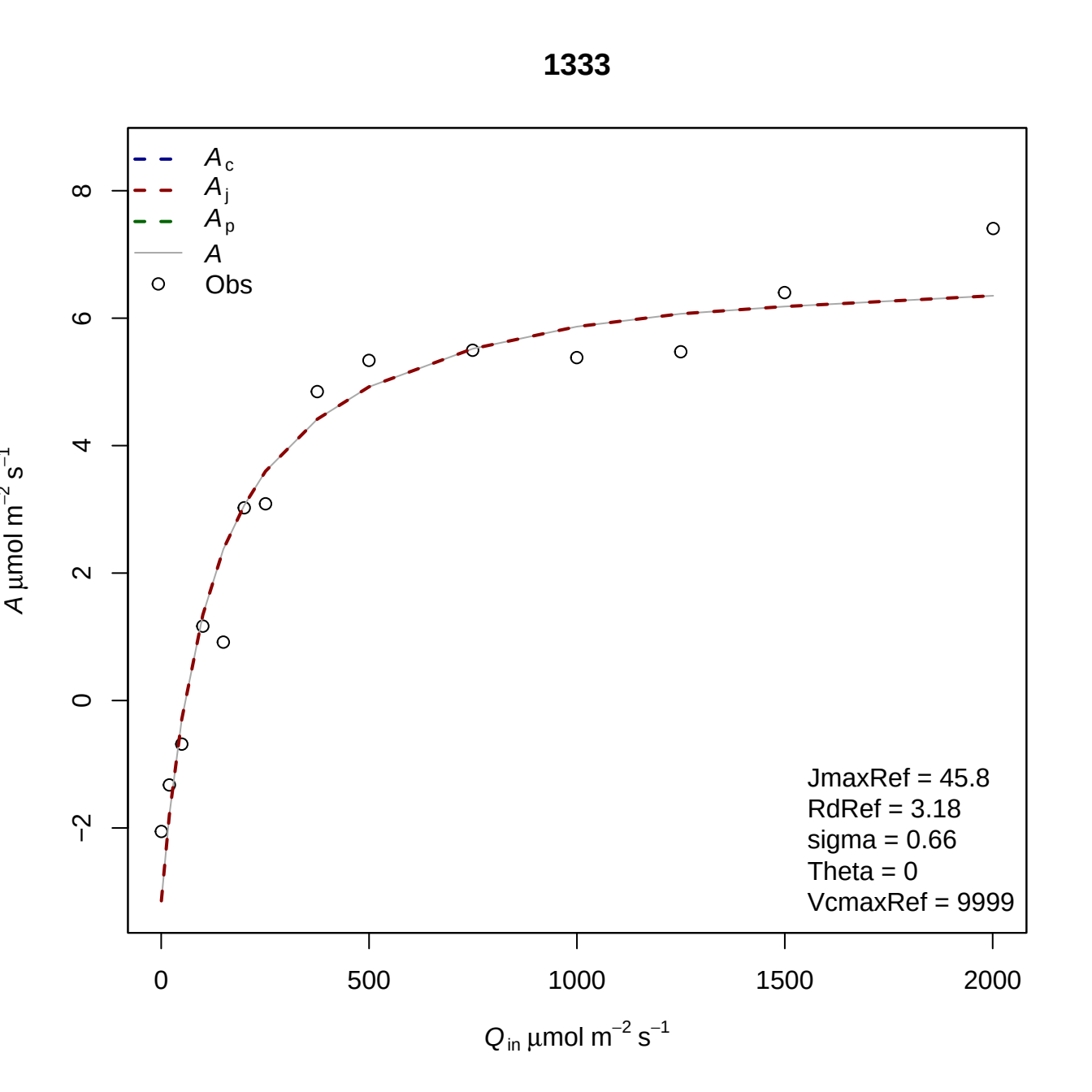

1328
A_c
A_j
A_p
A
Obs
JmaxRef = 69.42
RdRef = 4.28
sigma = 0.91
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

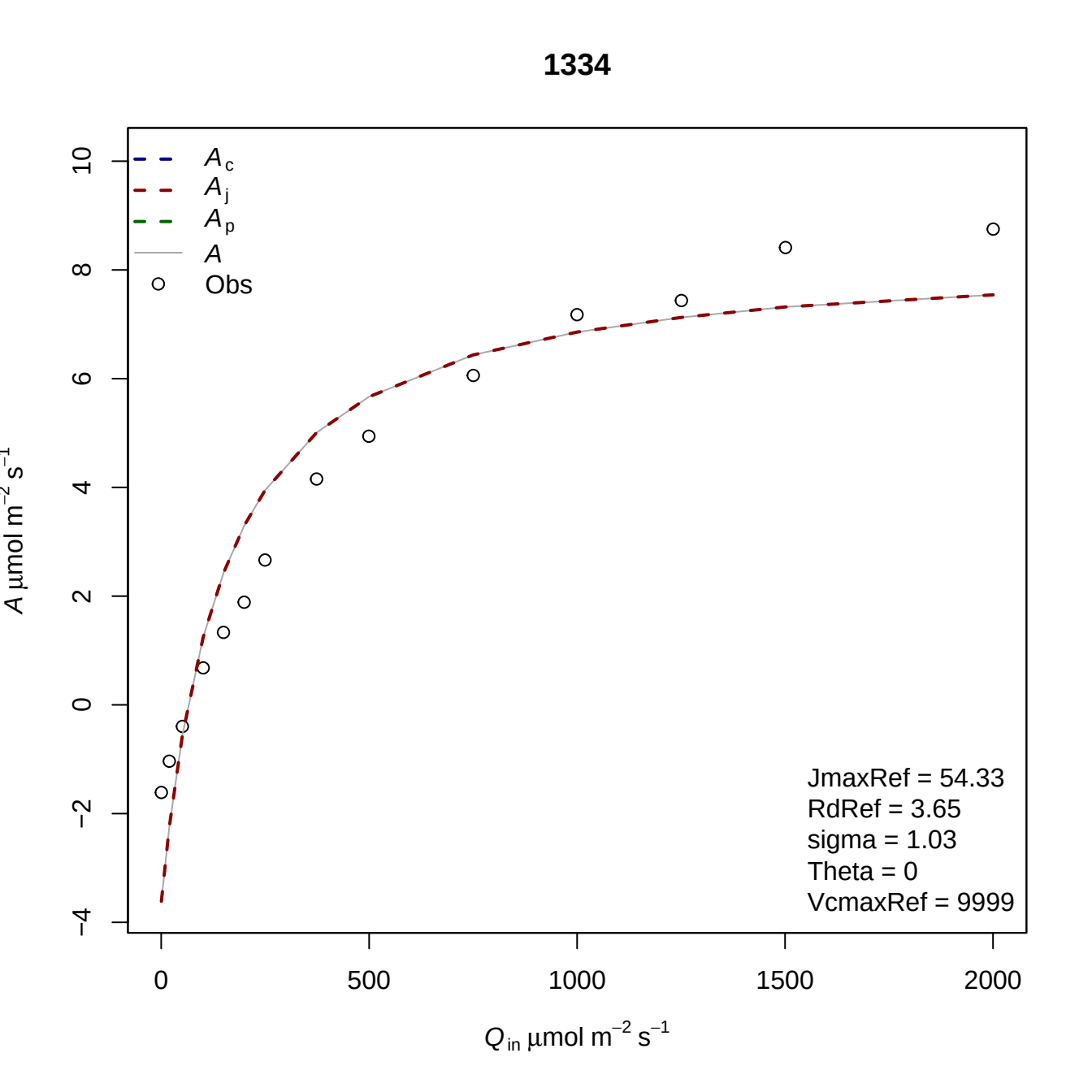

1329
A_c
A_j
A_p
A
Obs
JmaxRef = 61.01
RdRef = 4.71
sigma = 0.86
Theta = 0
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

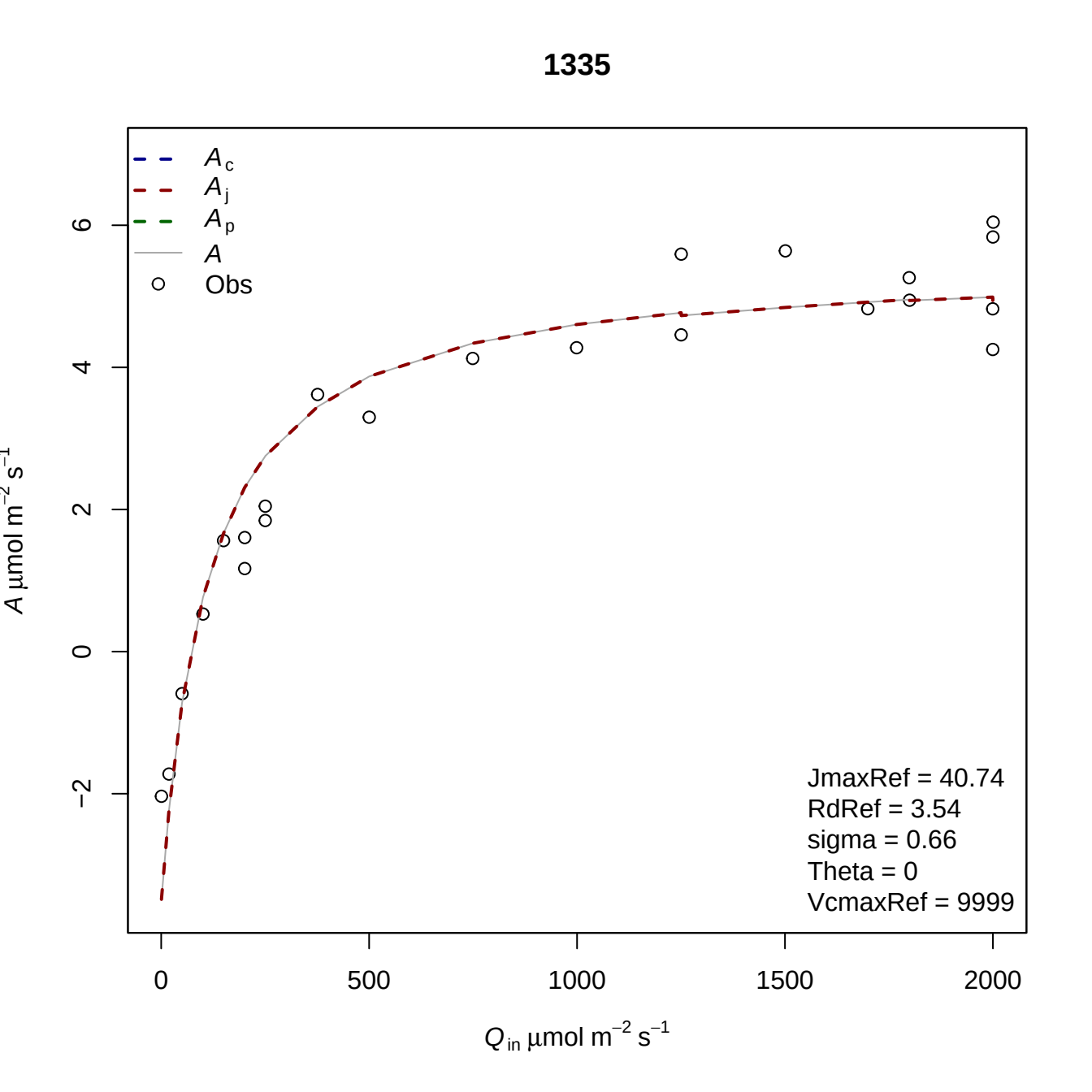

1330
A_c
A_j
A_p
A
Obs
JmaxRef = 46.37
RdRef = 2.47
sigma = 0.34
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

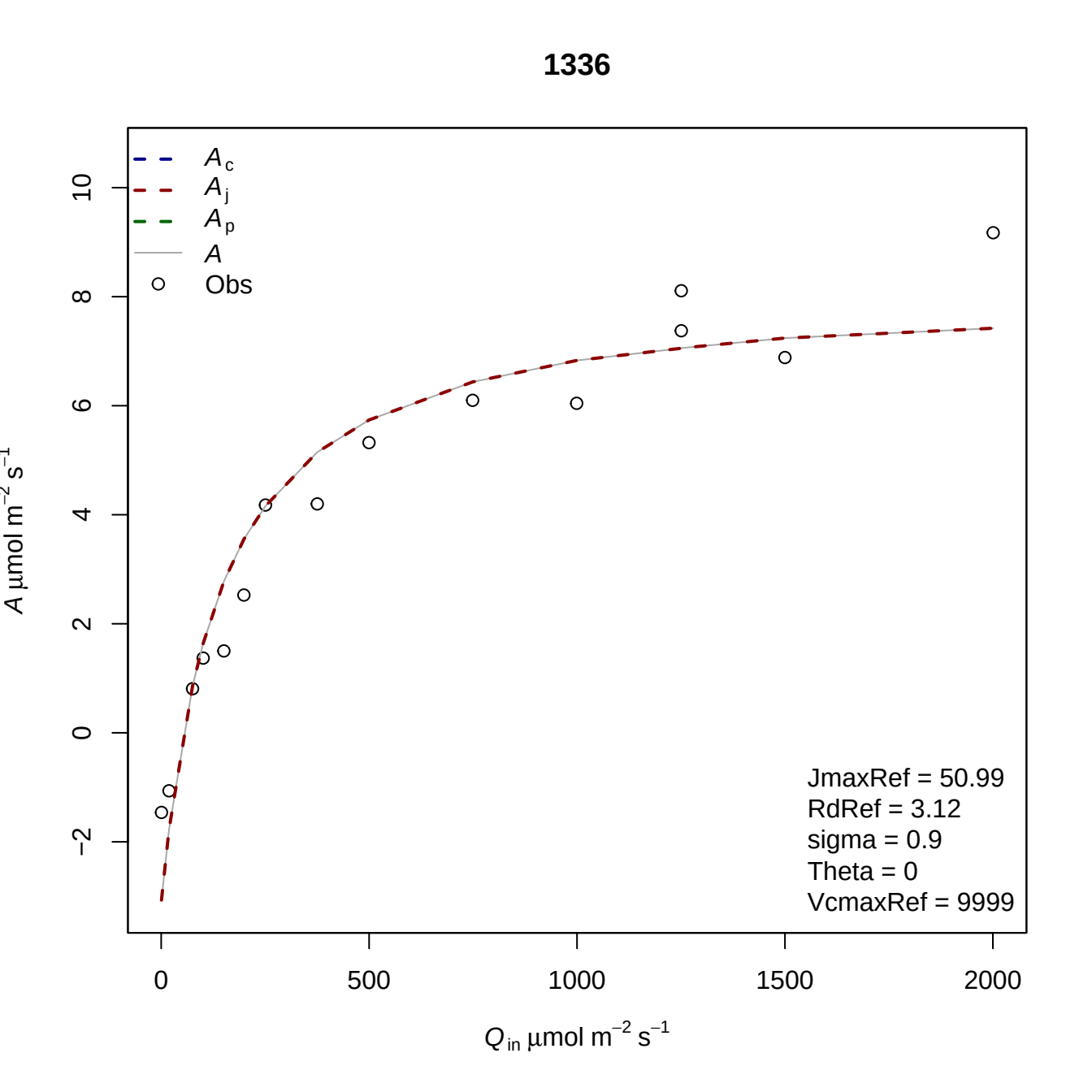

1331
A_c
A_j
A_p
A
Obs
JmaxRef = 46.37
RdRef = 2.12
sigma = 0.53
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

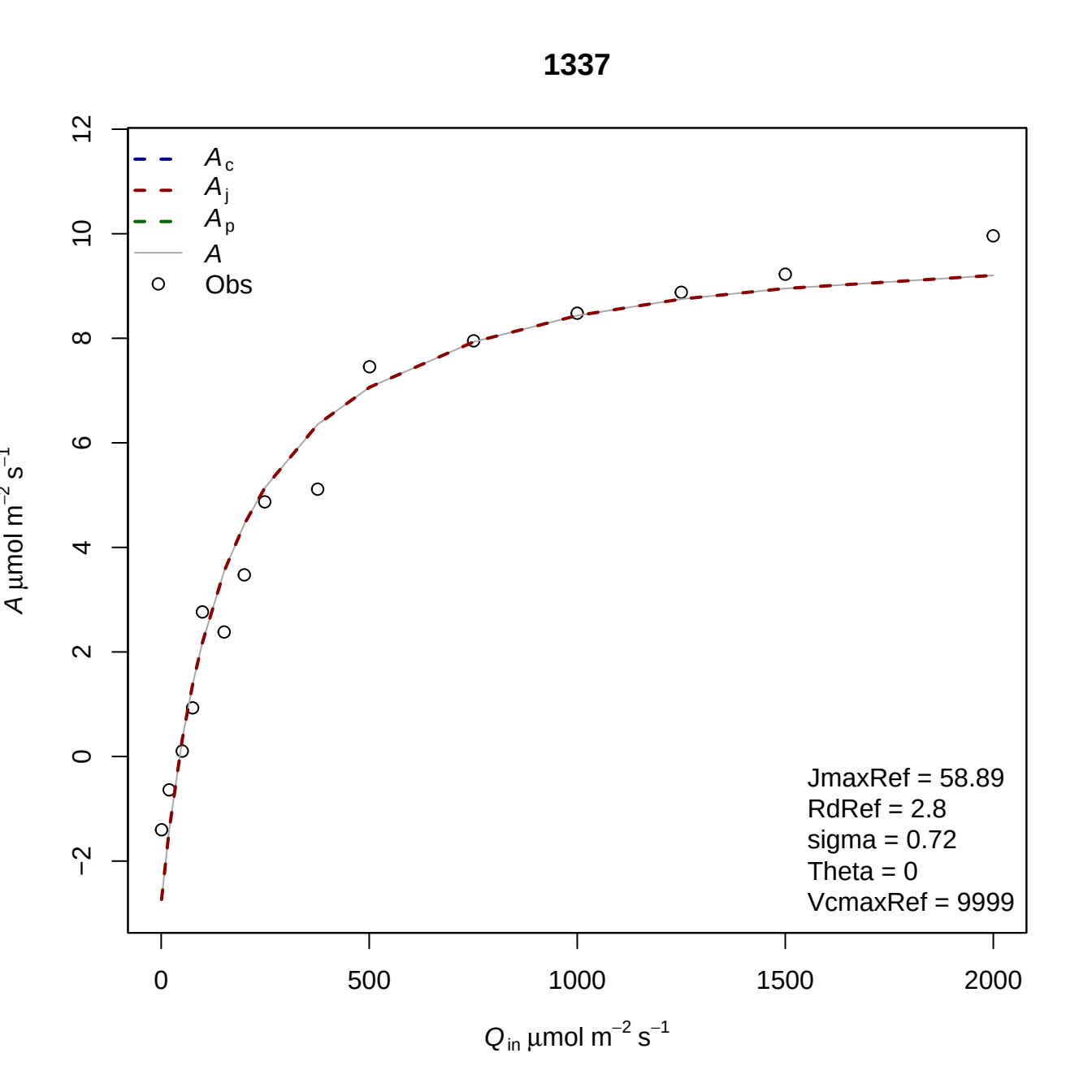

1332
A_c
A_j
A_p
A
Obs
JmaxRef = 51.93
RdRef = 1.83
sigma = 0.55
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

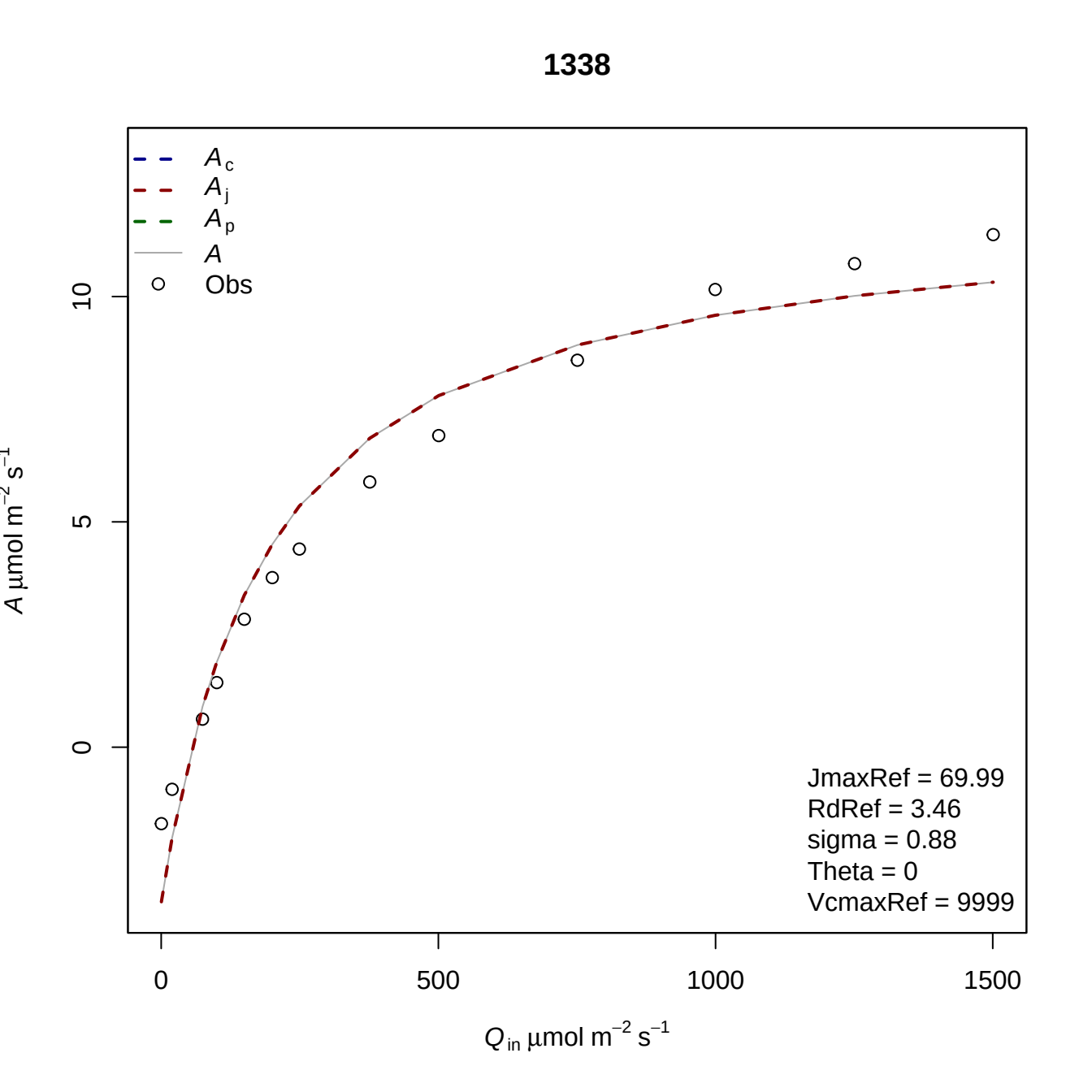

1333
A_c
A_j
A_p
A
Obs
JmaxRef = 45.8
RdRef = 3.18
sigma = 0.66
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

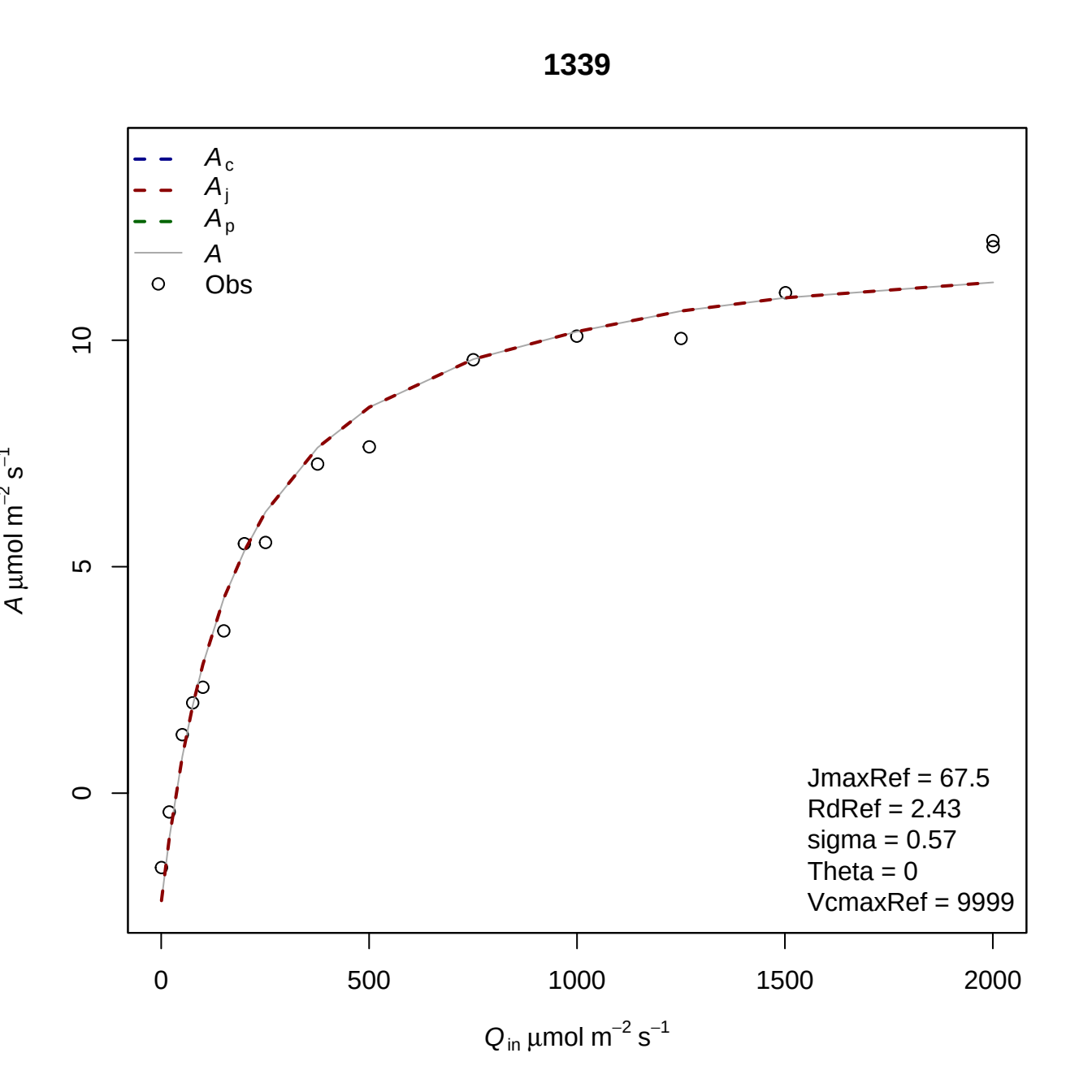

1334
A_c
A_j
A_p
A
Obs
JmaxRef = 54.33
RdRef = 3.65
sigma = 1.03
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

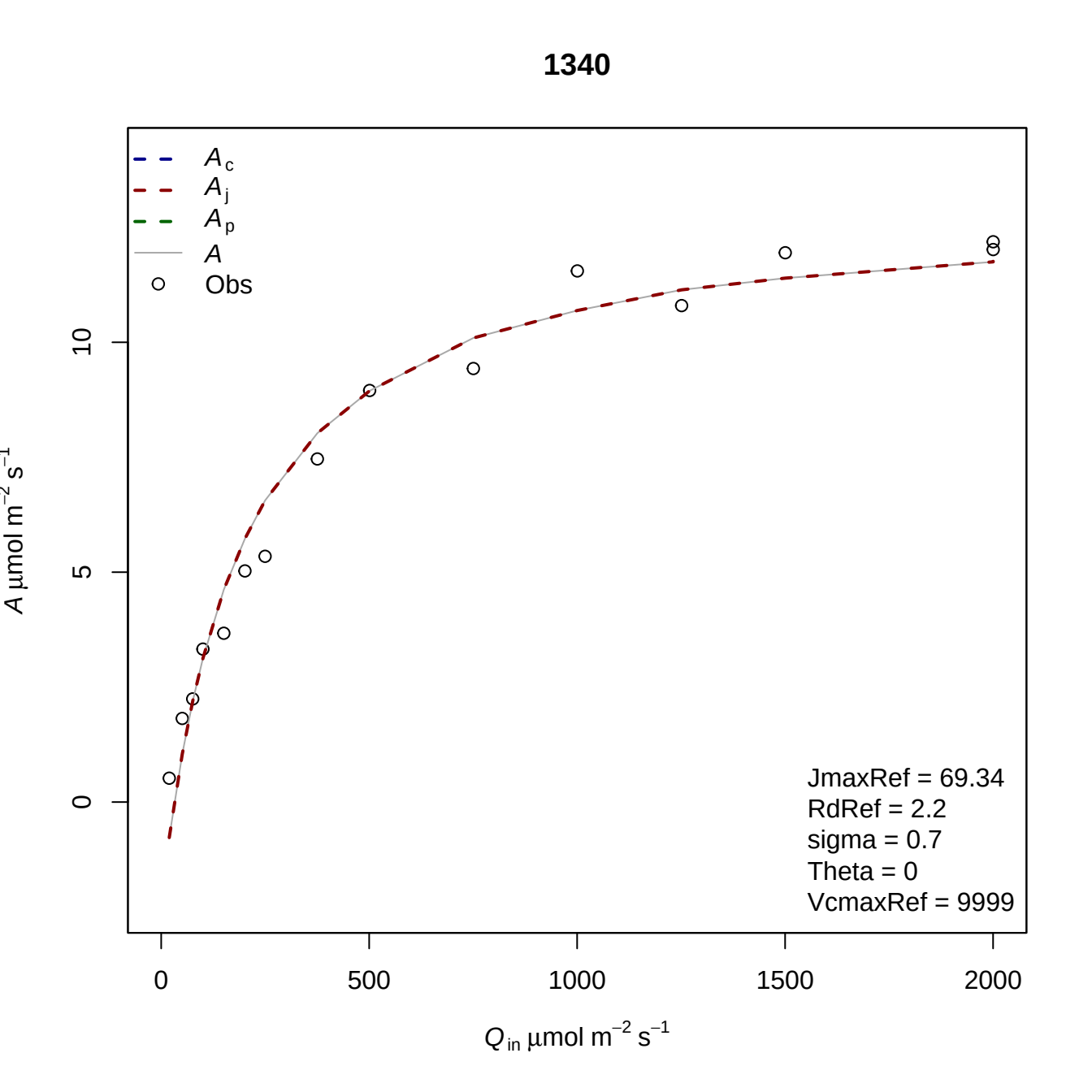

1335
A_c
A_j
A_p
A
Obs
JmaxRef = 40.74
RdRef = 3.54
sigma = 0.66
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

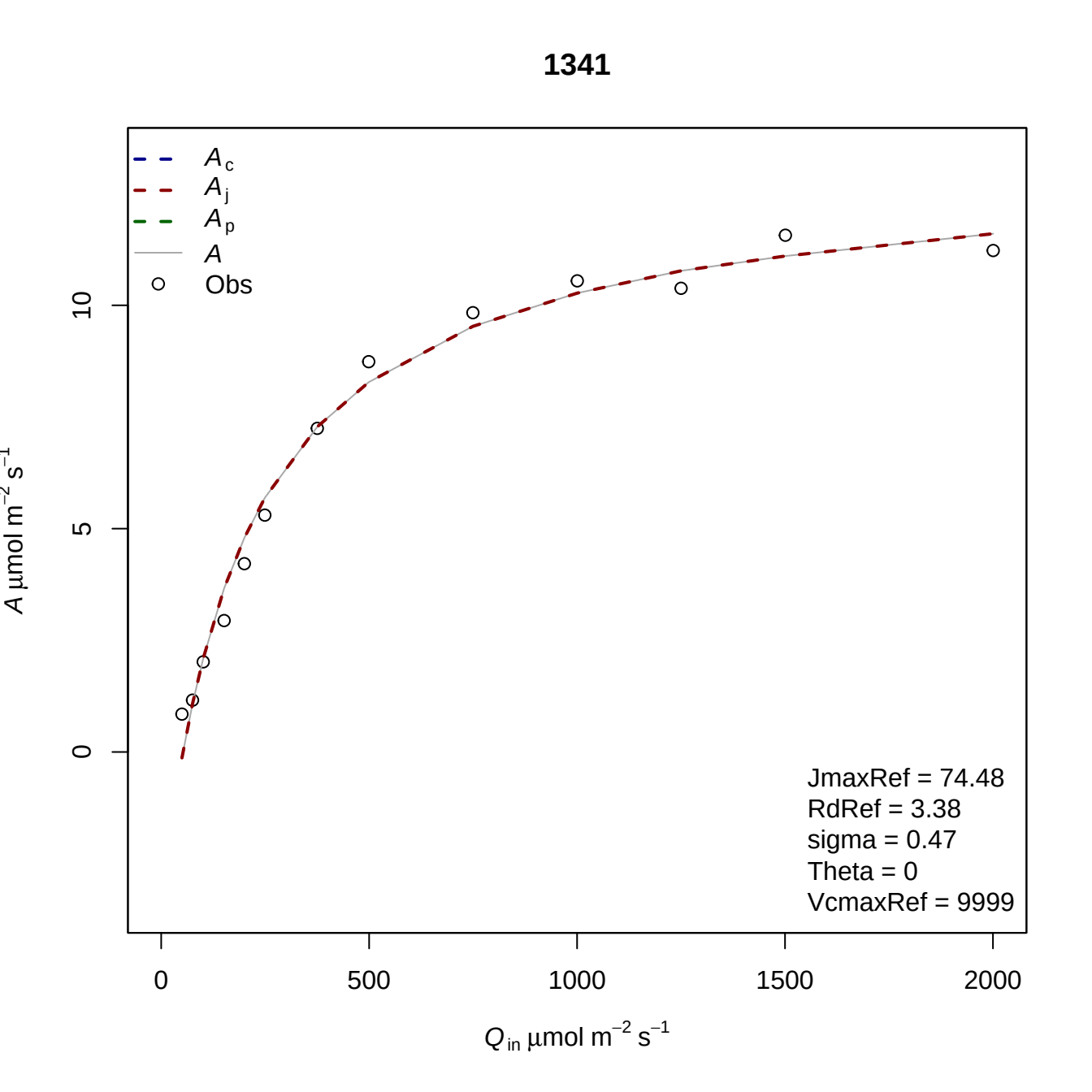

1336
A_c
A_j
A_p
A
Obs
JmaxRef = 50.99
RdRef = 3.12
sigma = 0.9
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

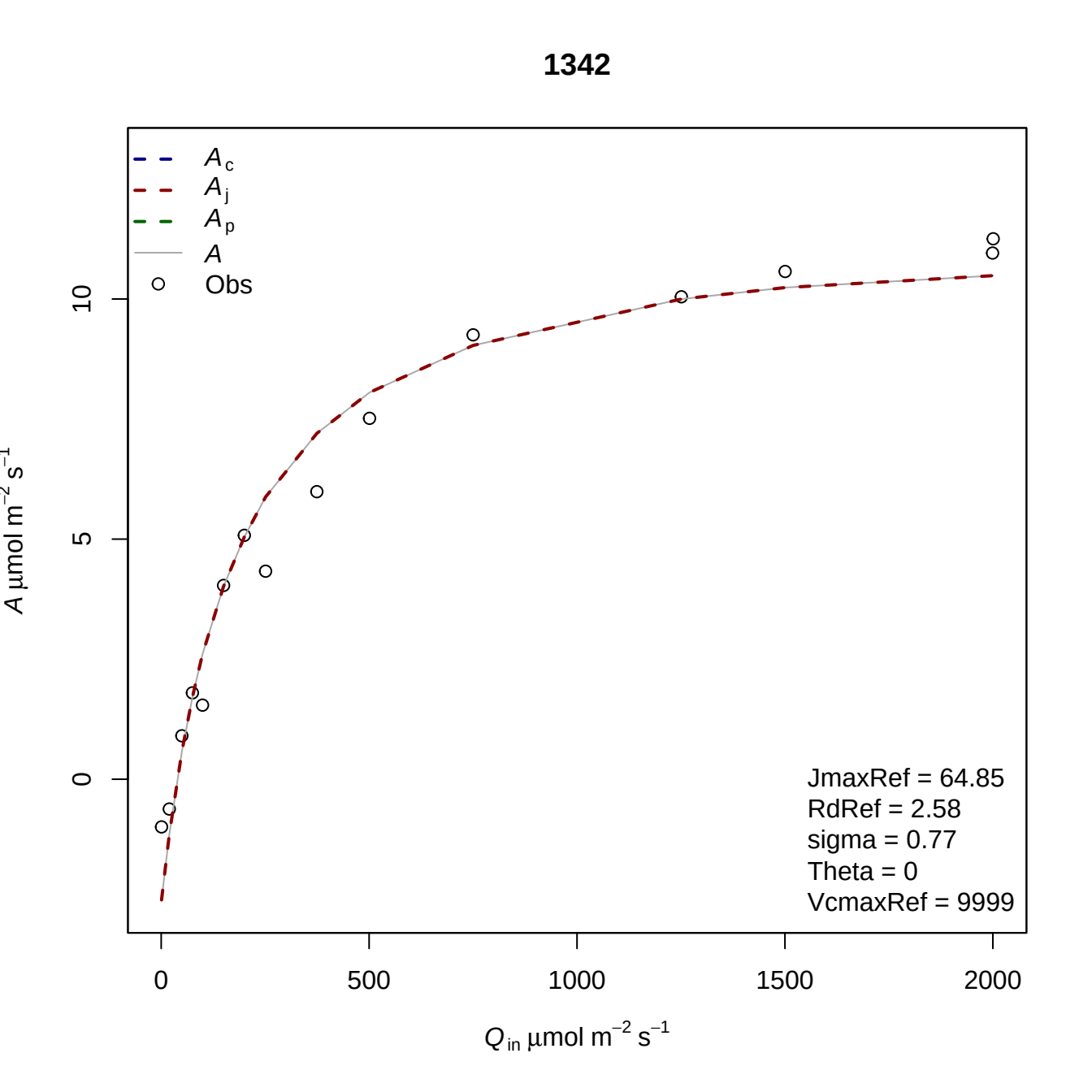

1337
A_c
A_j
A_p
A
Obs
JmaxRef = 58.89
RdRef = 2.8
sigma = 0.72
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

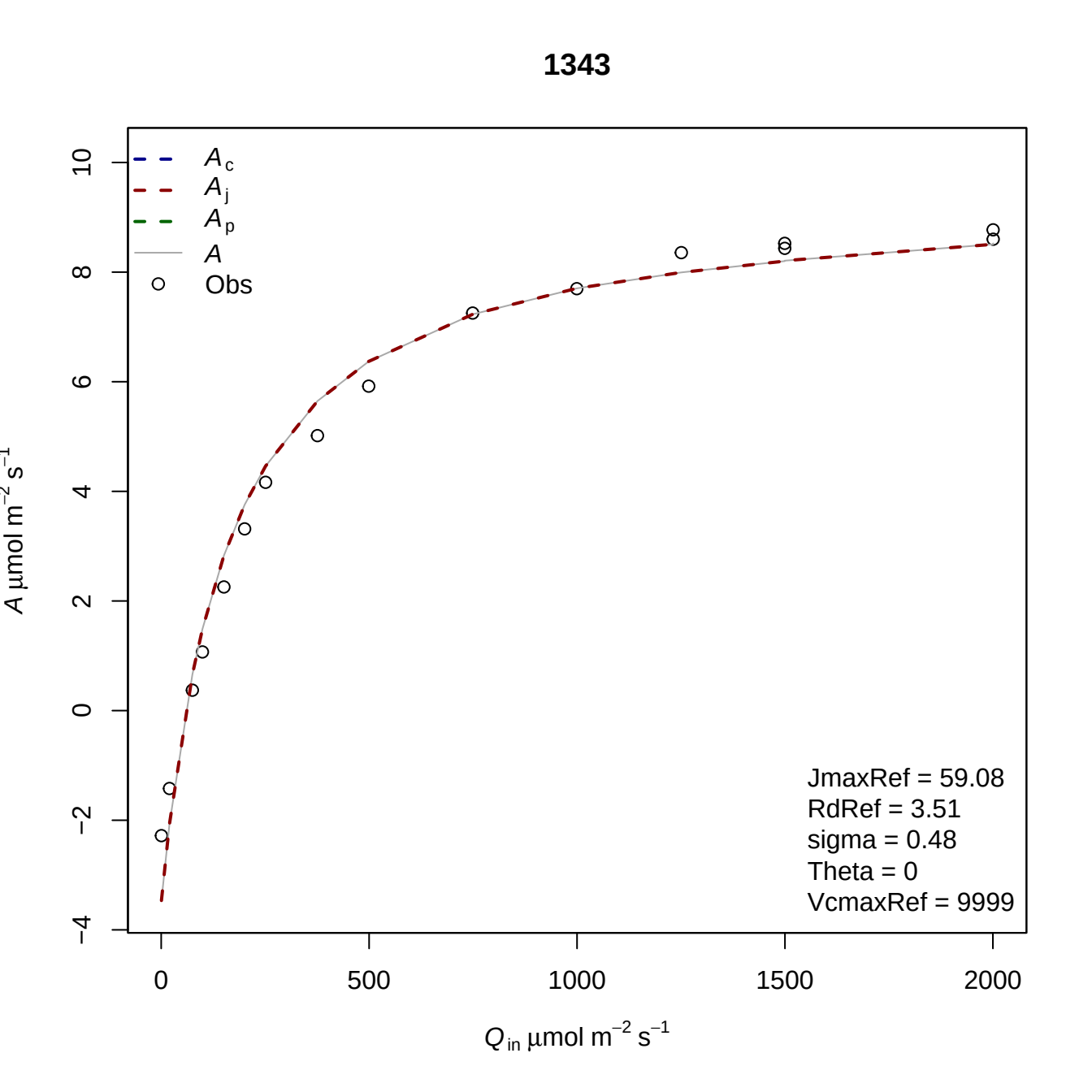

1338
A_c
A_j
A_p
A
Obs
JmaxRef = 69.99
RdRef = 3.46
sigma = 0.88
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

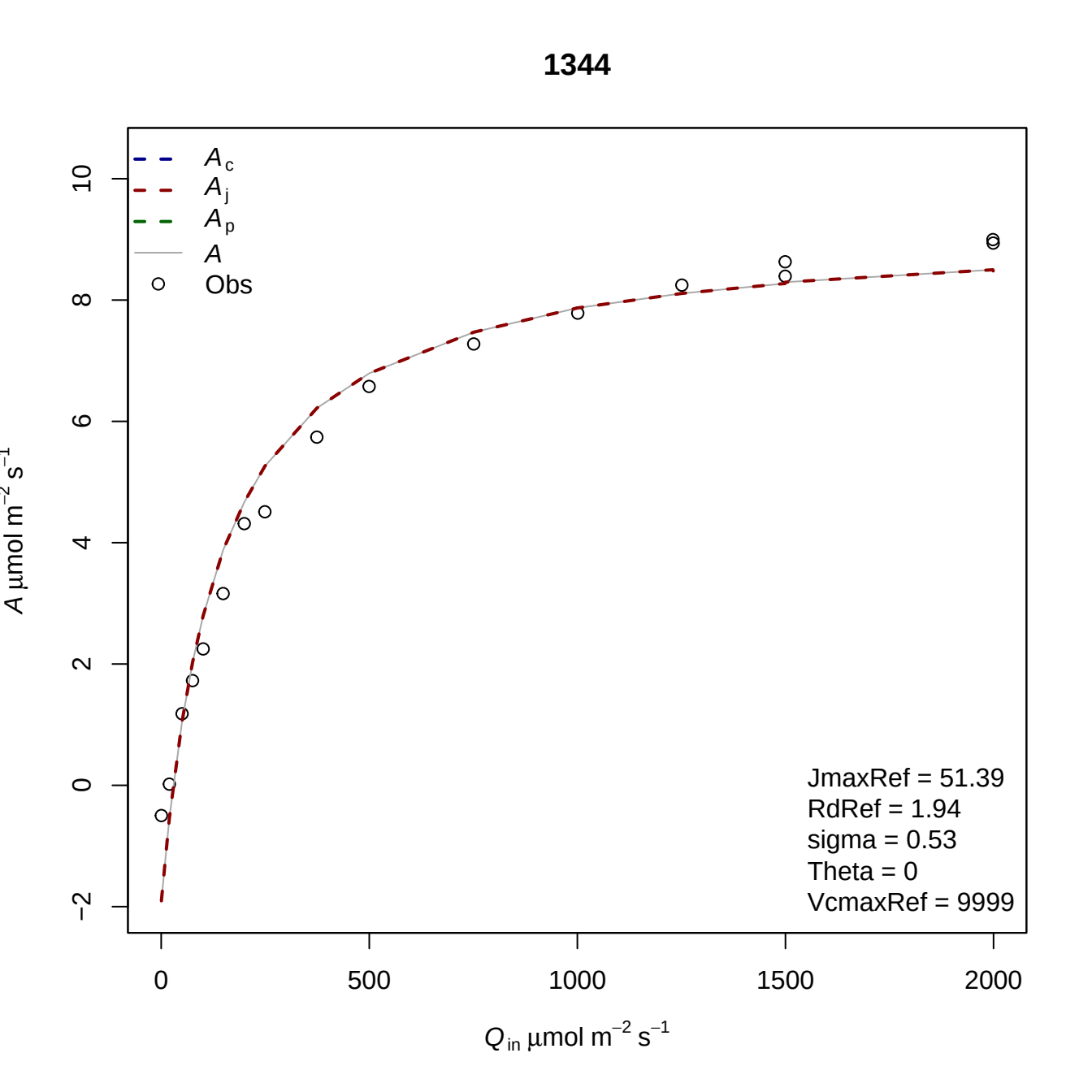

1339
A_c
A_j
A_p
A
Obs
JmaxRef = 67.5
RdRef = 2.43
sigma = 0.57
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

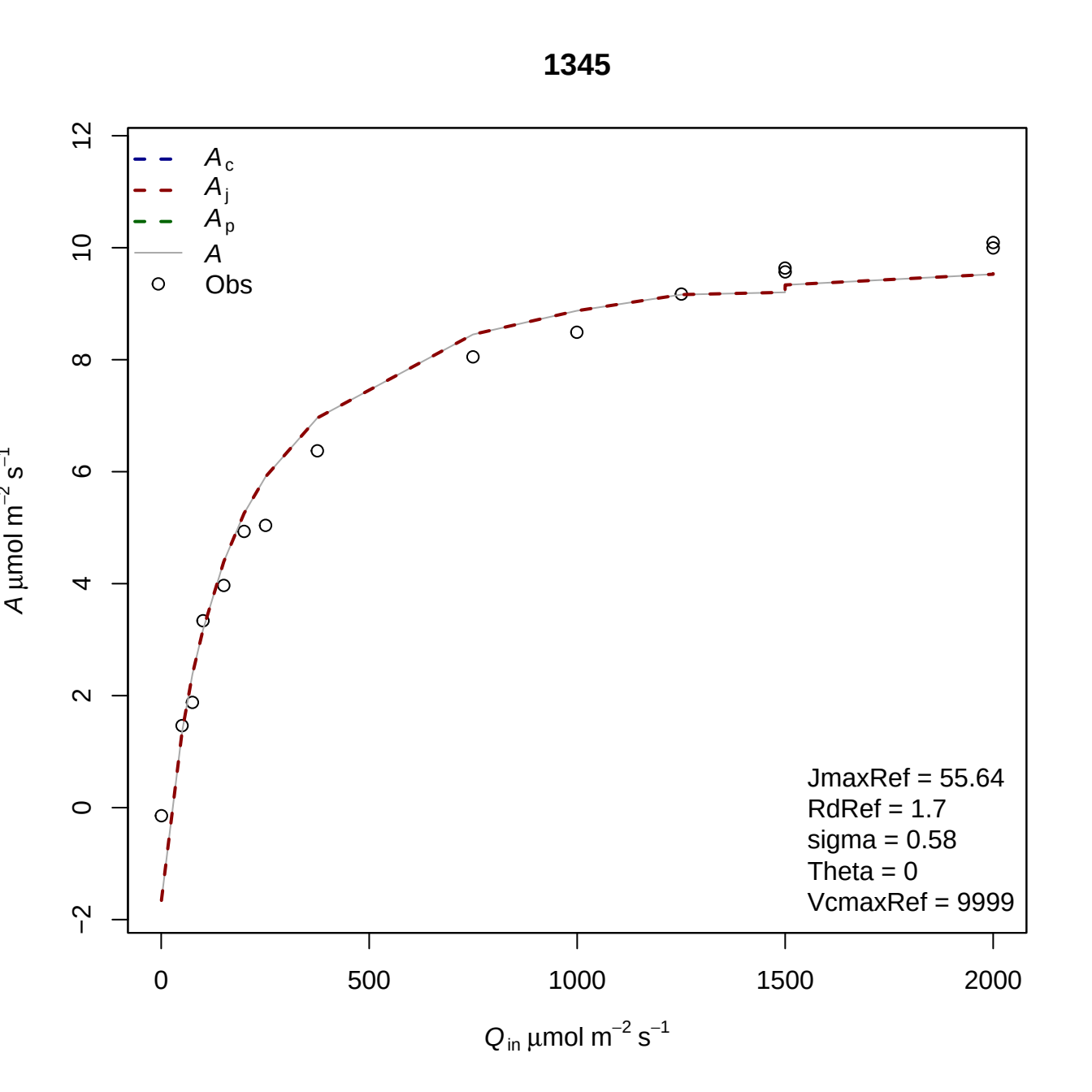

1340
A_c
A_j
A_p
A
Obs
JmaxRef = 69.34
RdRef = 2.2
sigma = 0.7
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

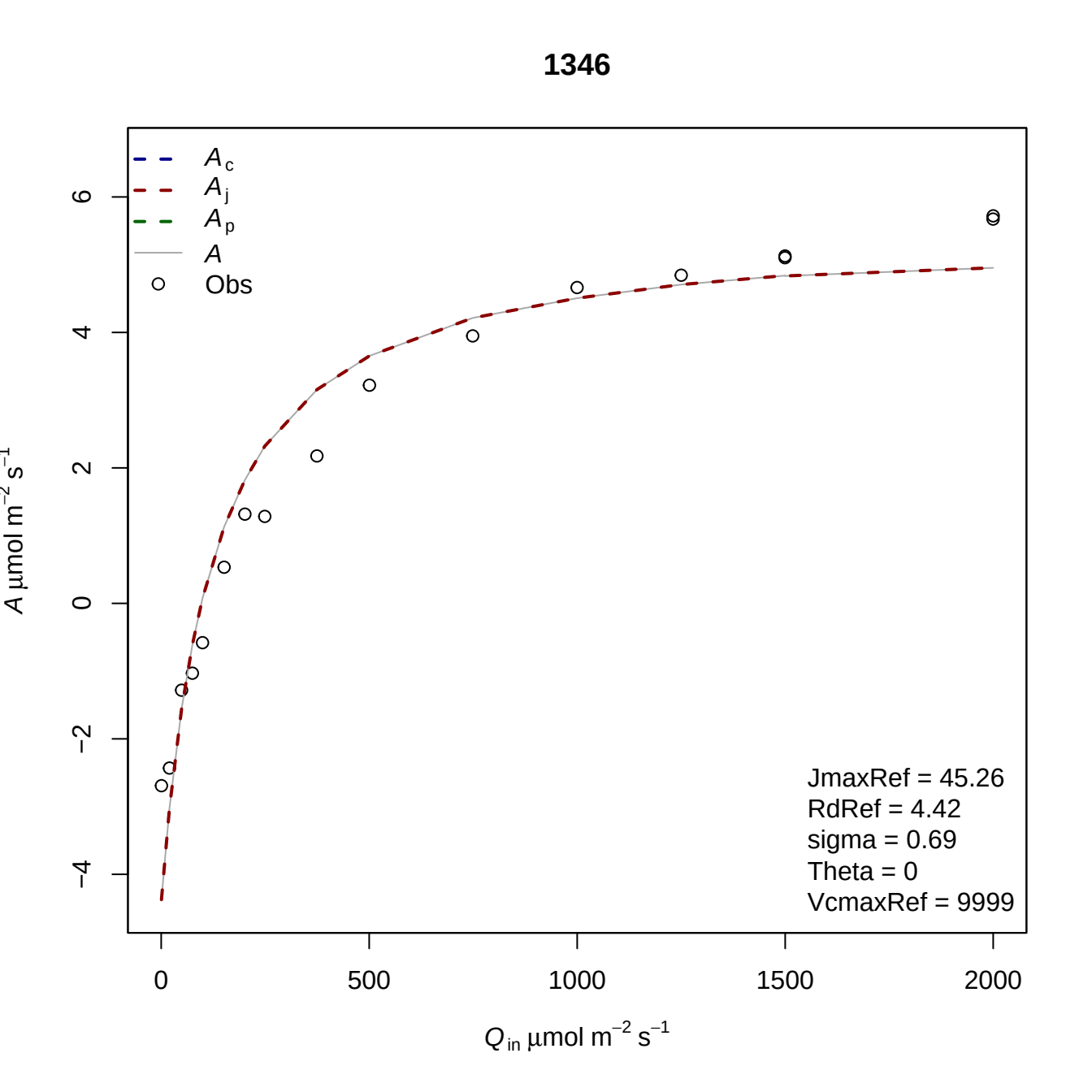

1341
A_c
A_j
A_p
A
Obs
JmaxRef = 74.48
RdRef = 3.38
sigma = 0.47
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

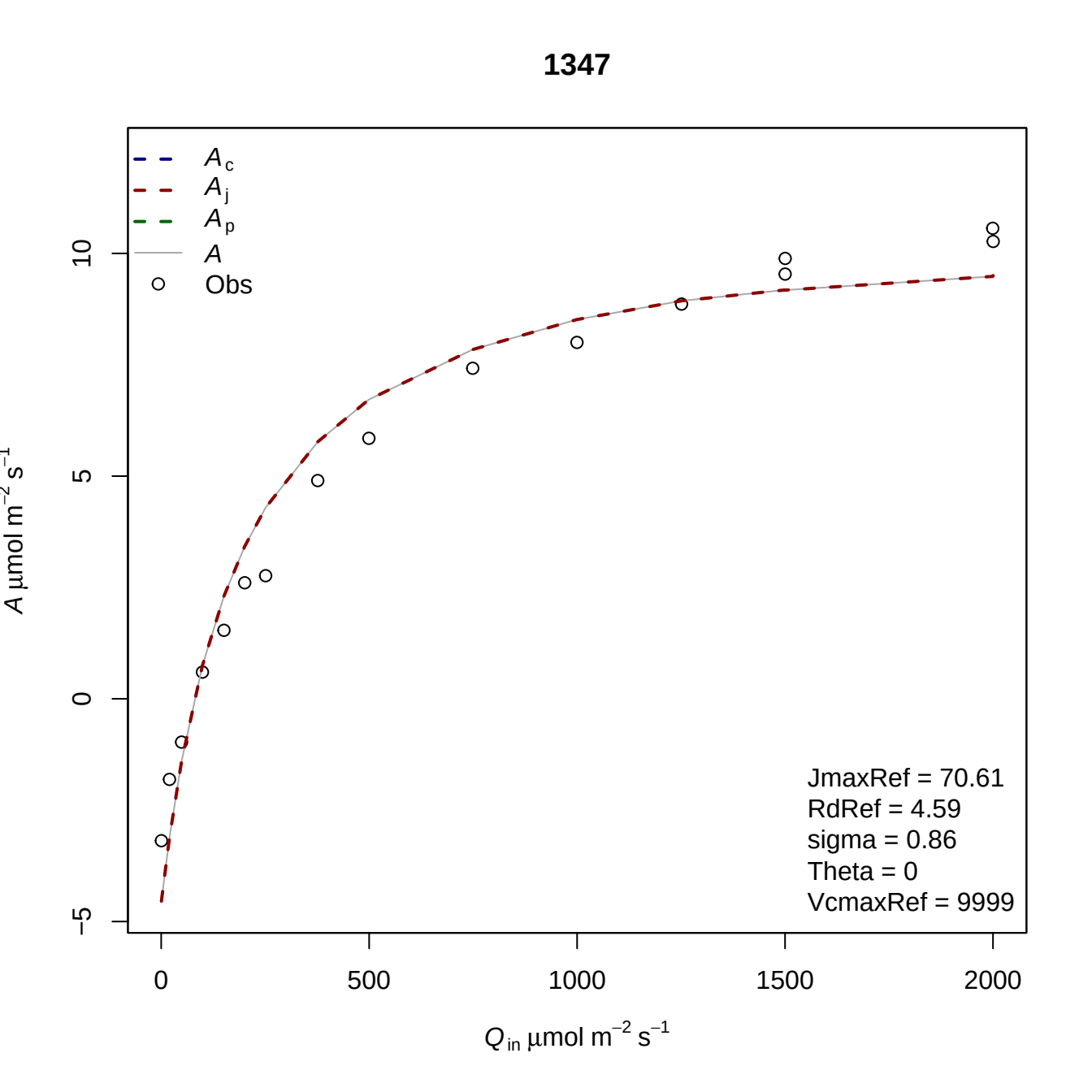

1342
A_c
A_j
A_p
A
Obs
JmaxRef = 64.85
RdRef = 2.58
sigma = 0.77
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

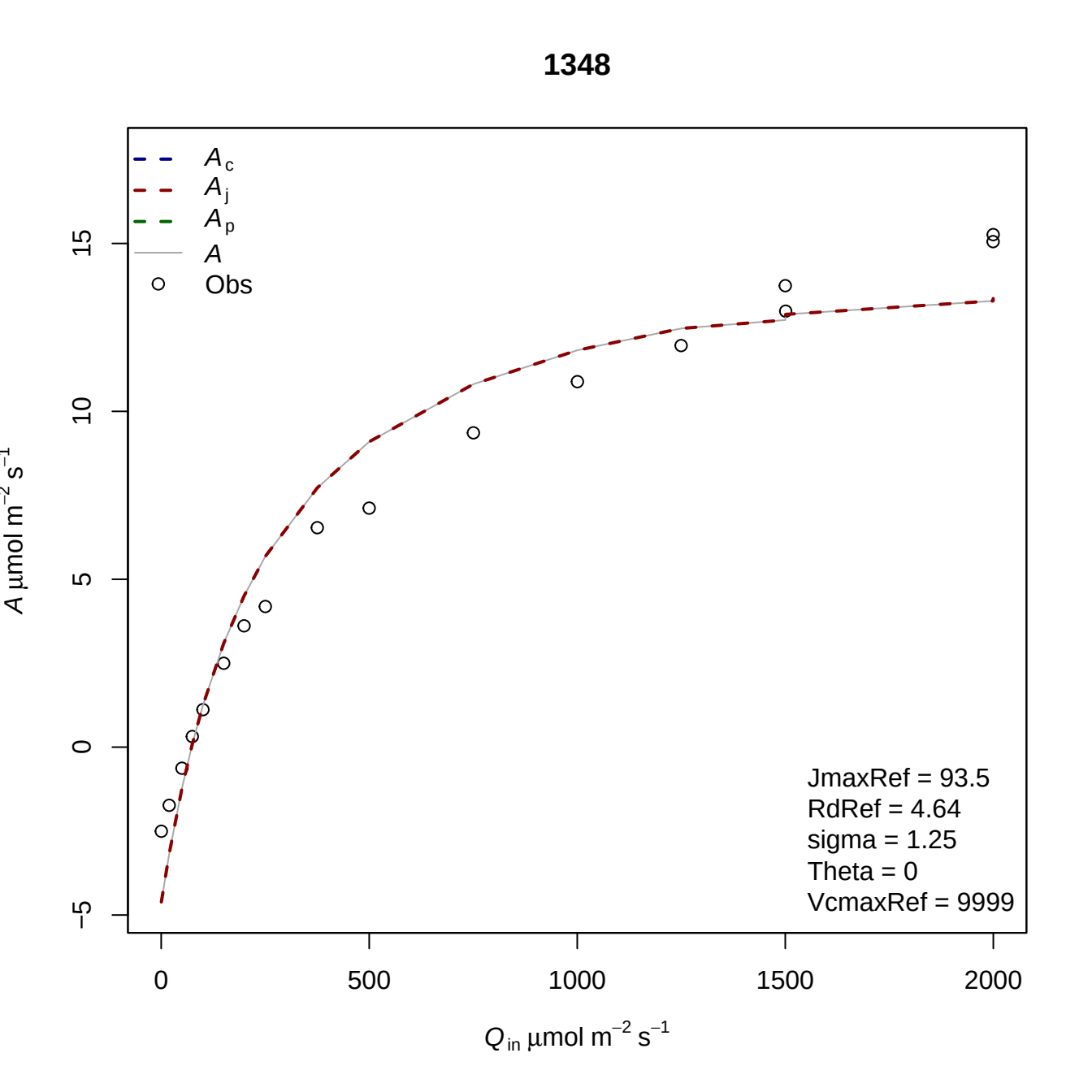

1343
A_c
A_j
A_p
A
Obs
JmaxRef = 59.08
RdRef = 3.51
sigma = 0.48
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

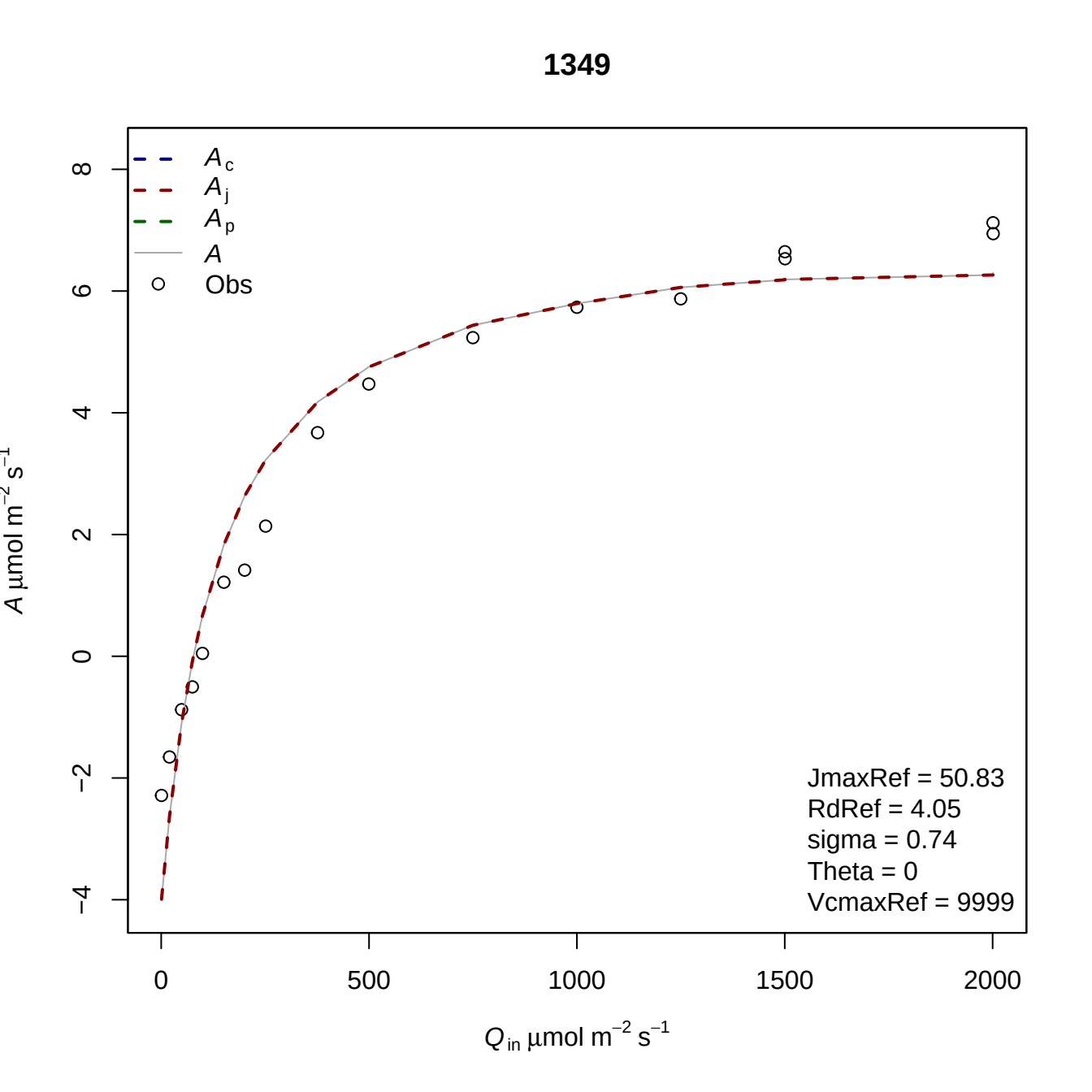

1344
A_c
A_j
A_p
A
Obs
JmaxRef = 51.39
RdRef = 1.94
sigma = 0.53
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

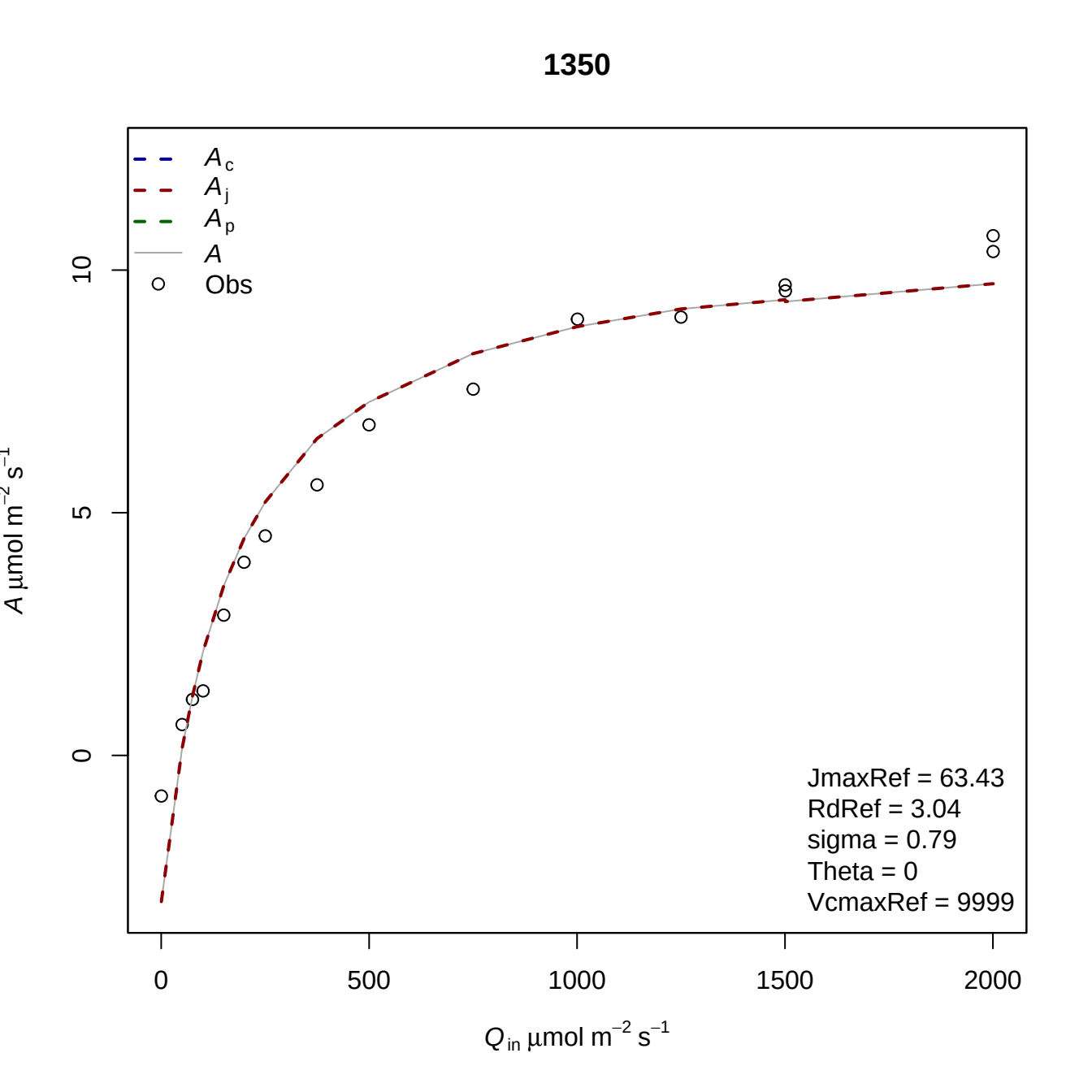

1345
A_c
A_j
A_p
A
Obs
JmaxRef = 55.64
RdRef = 1.7
sigma = 0.58
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

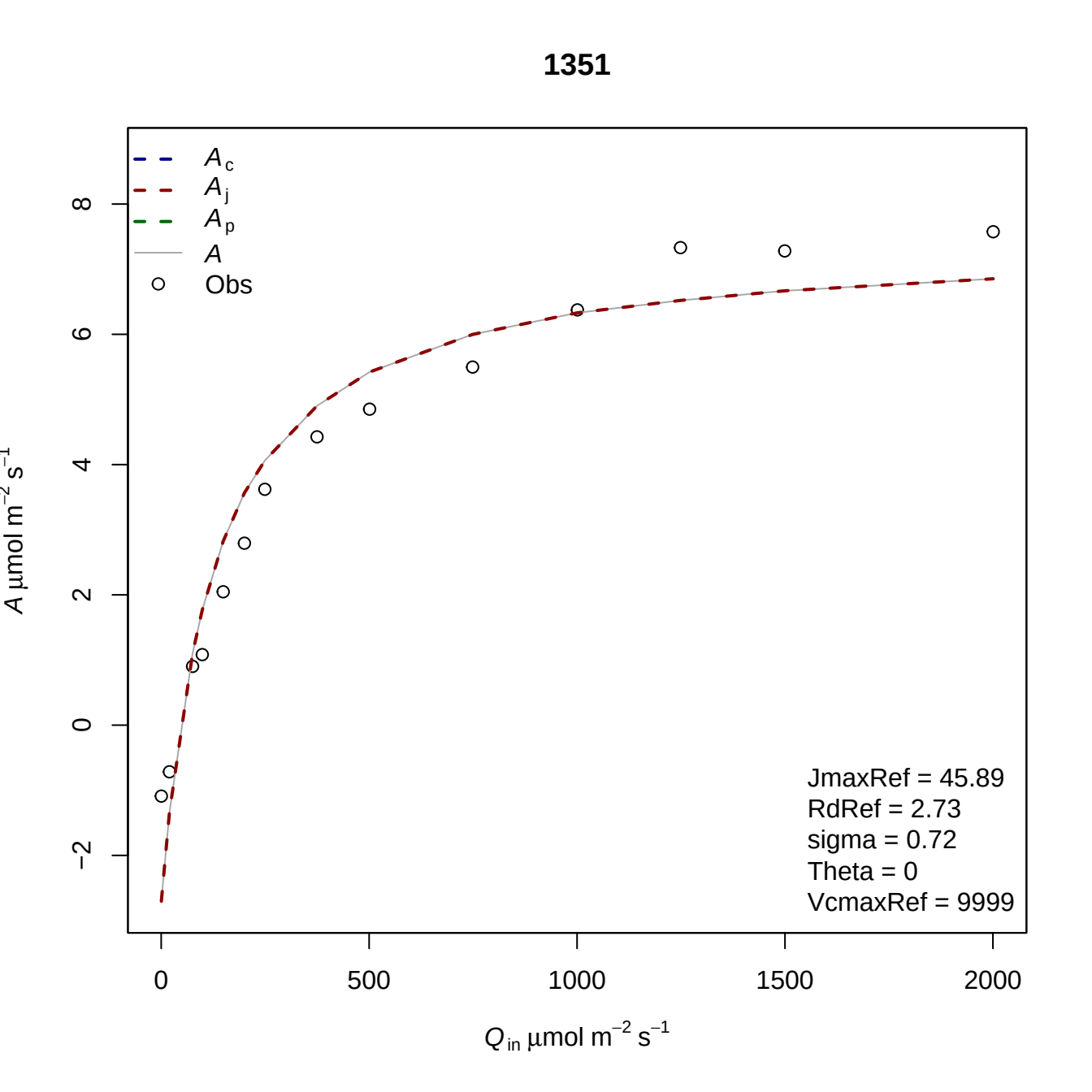

1346
A_c
A_j
A_p
A
Obs
JmaxRef = 45.26
RdRef = 4.42
sigma = 0.69
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

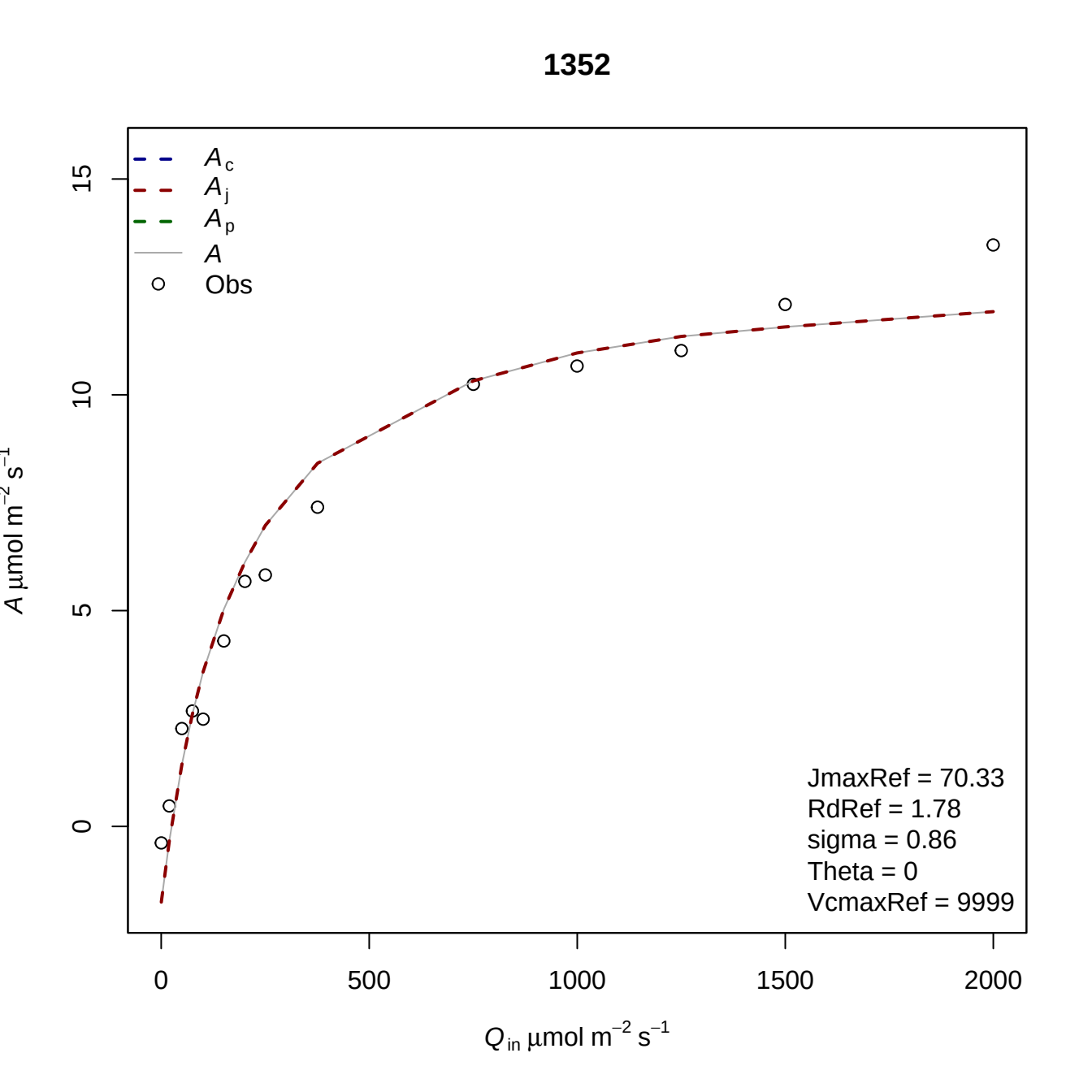

1347
A_c
A_j
A_p
A
Obs
JmaxRef = 70.61
RdRef = 4.59
sigma = 0.86
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

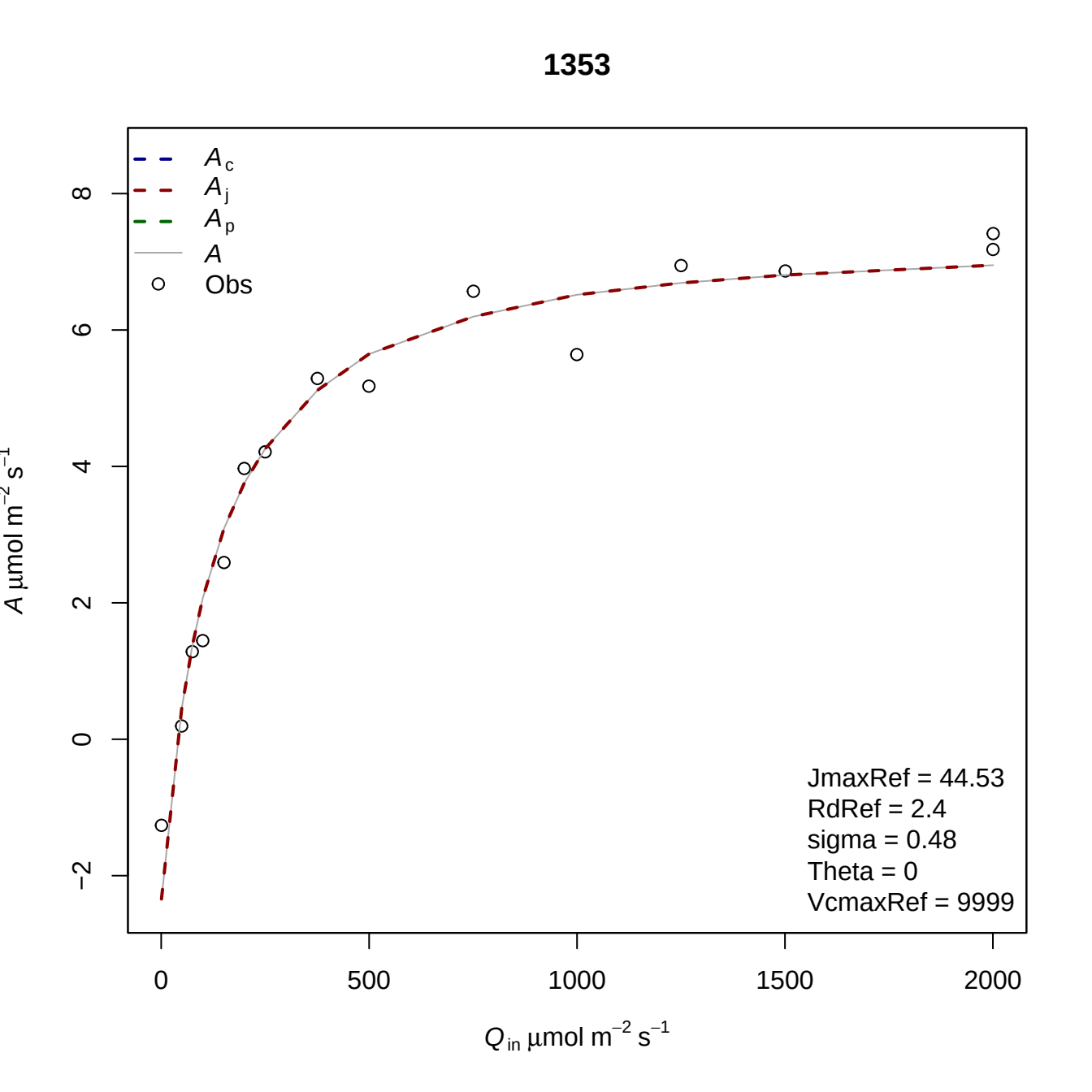

1348
A_c
A_j
A_p
A
Obs
JmaxRef = 93.5
RdRef = 4.64
sigma = 1.25
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

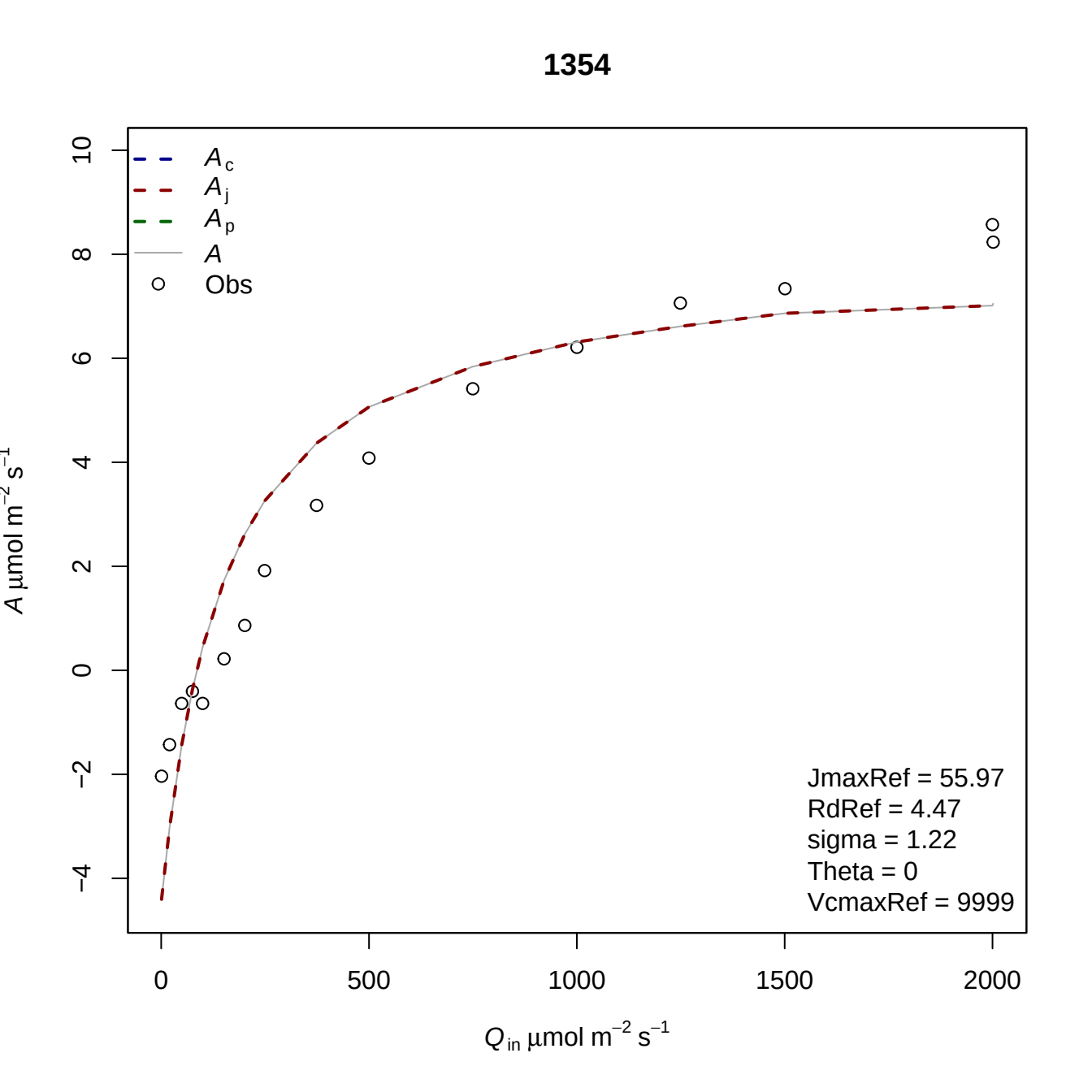

1349
A_c
A_j
A_p
A
Obs
JmaxRef = 50.83
RdRef = 4.05
sigma = 0.74
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

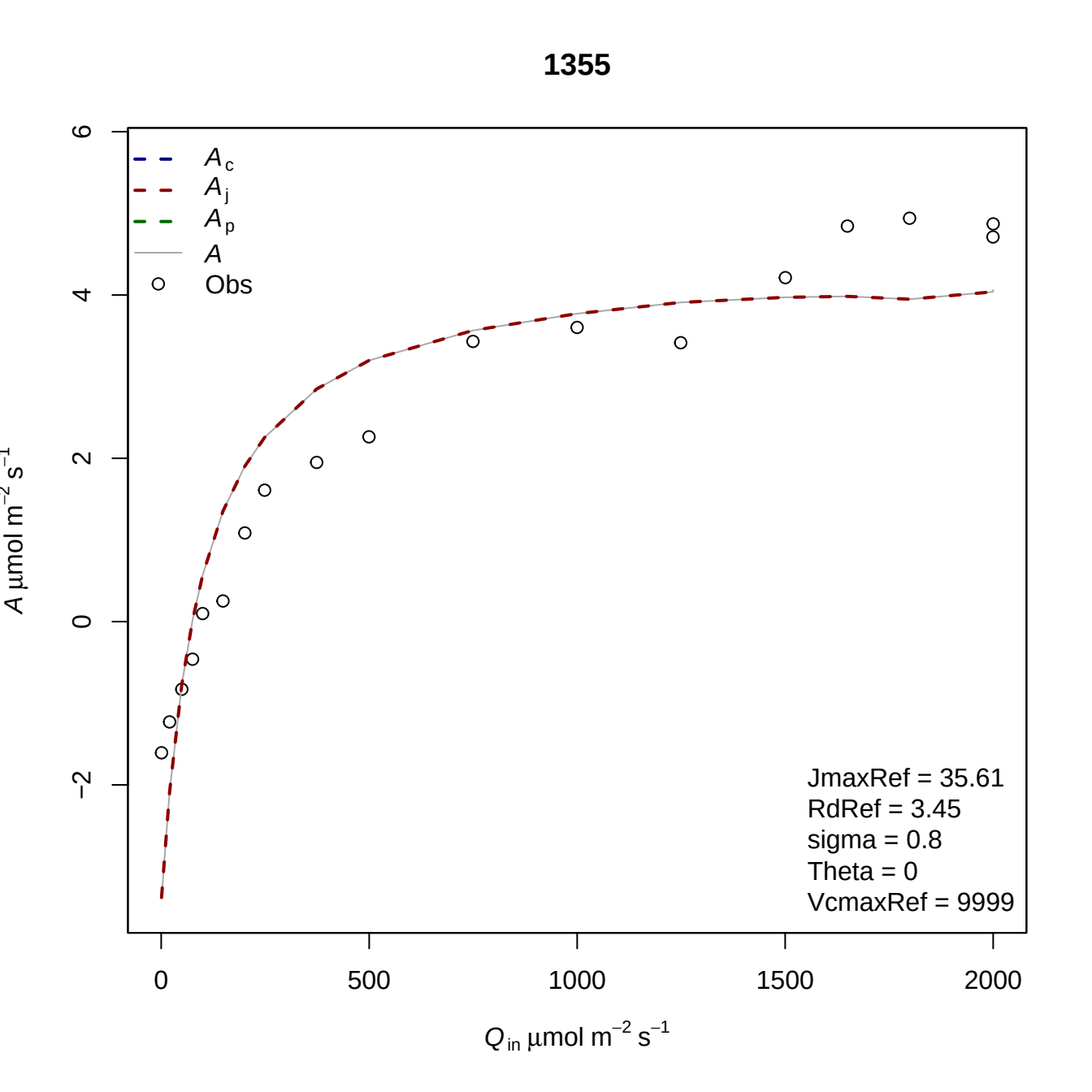

1350
A_c
A_j
A_p
A
Obs
JmaxRef = 63.43
RdRef = 3.04
sigma = 0.79
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

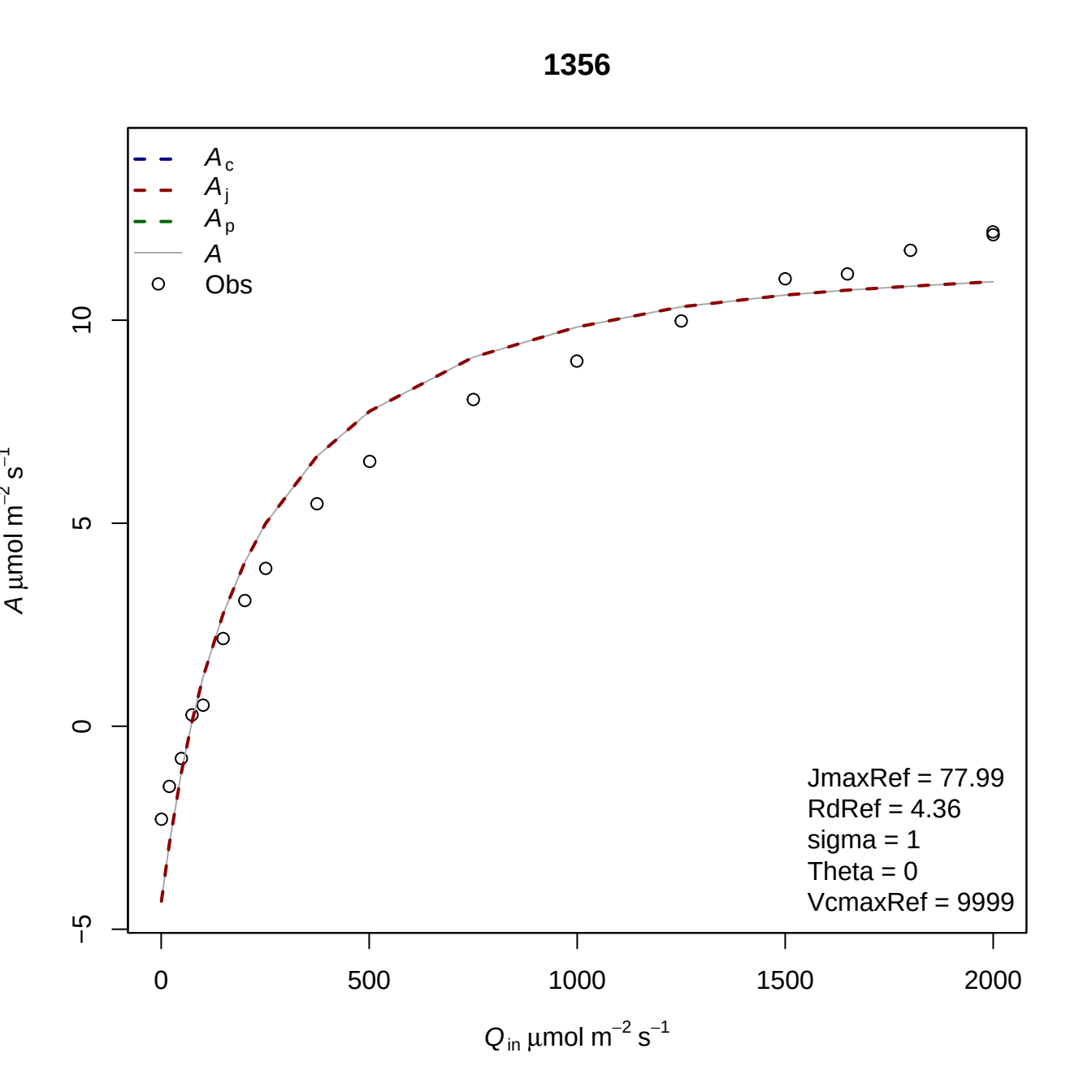

1351
A_c
A_j
A_p
A
Obs
JmaxRef = 45.89
RdRef = 2.73
sigma = 0.72
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

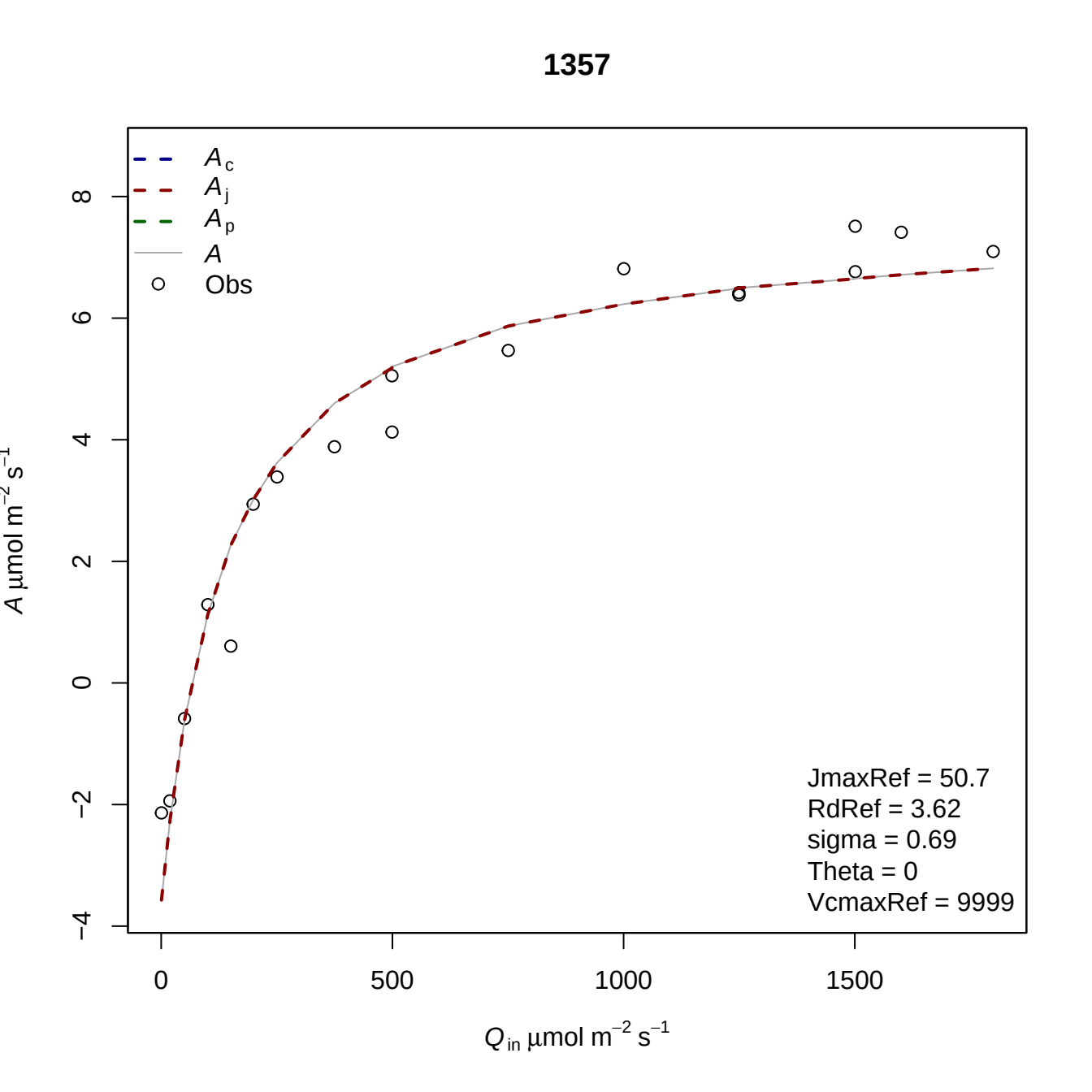

1352
A_c
A_j
A_p
A
Obs
JmaxRef = 70.33
RdRef = 1.78
sigma = 0.86
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

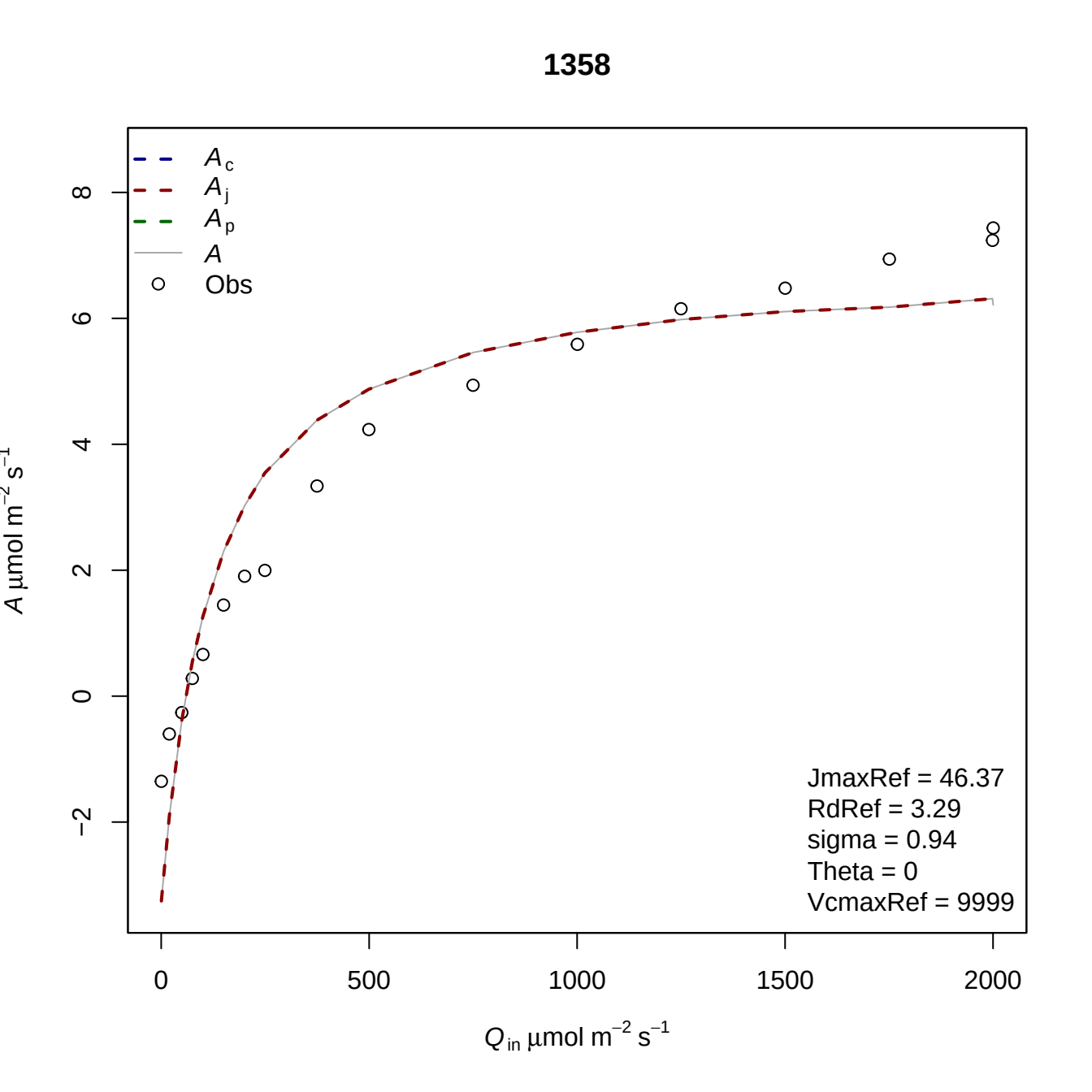

1353
A_c
A_j
A_p
A
Obs
JmaxRef = 44.53
RdRef = 2.4
sigma = 0.48
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

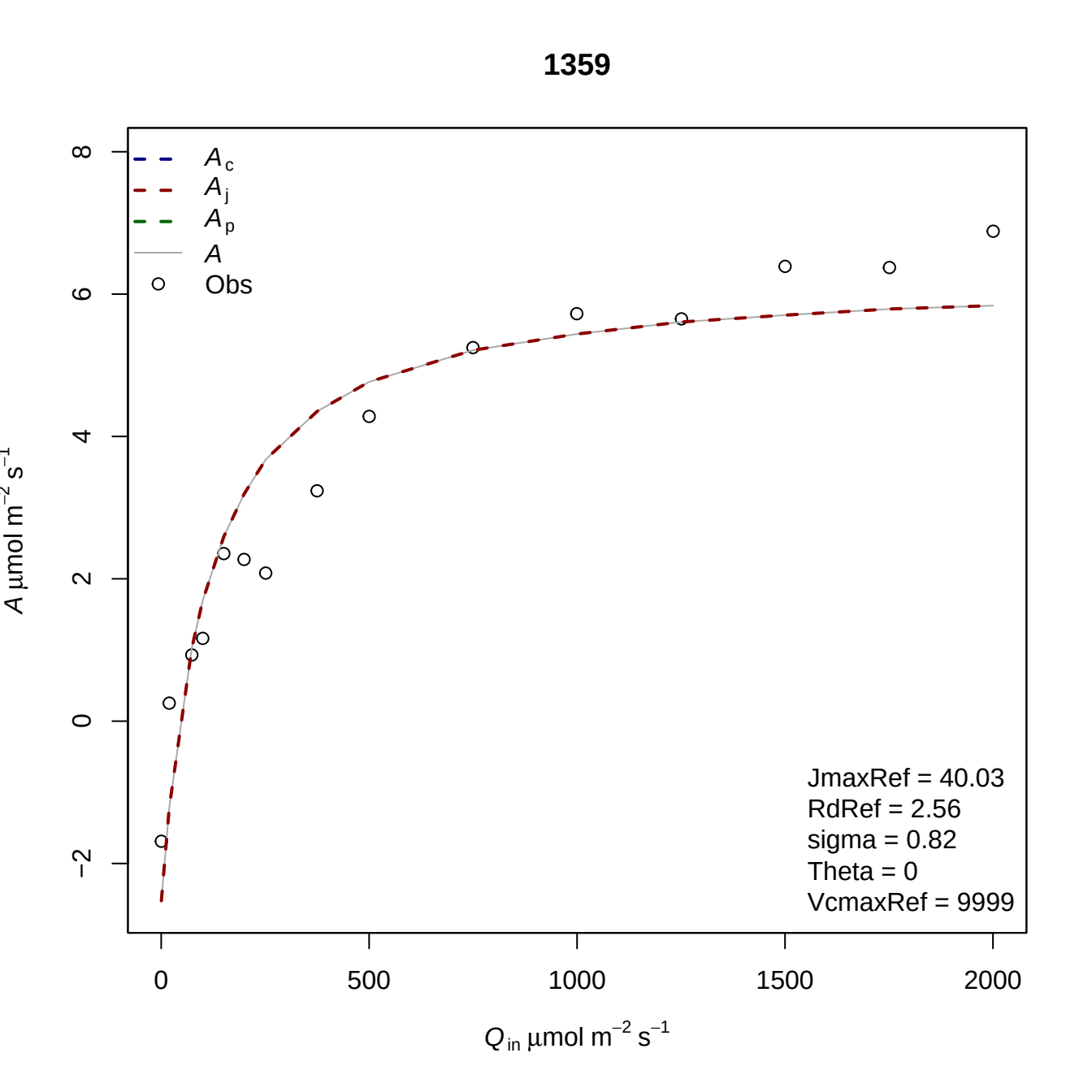

1354
A_c
A_j
A_p
A
Obs
JmaxRef = 55.97
RdRef = 4.47
sigma = 1.22
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

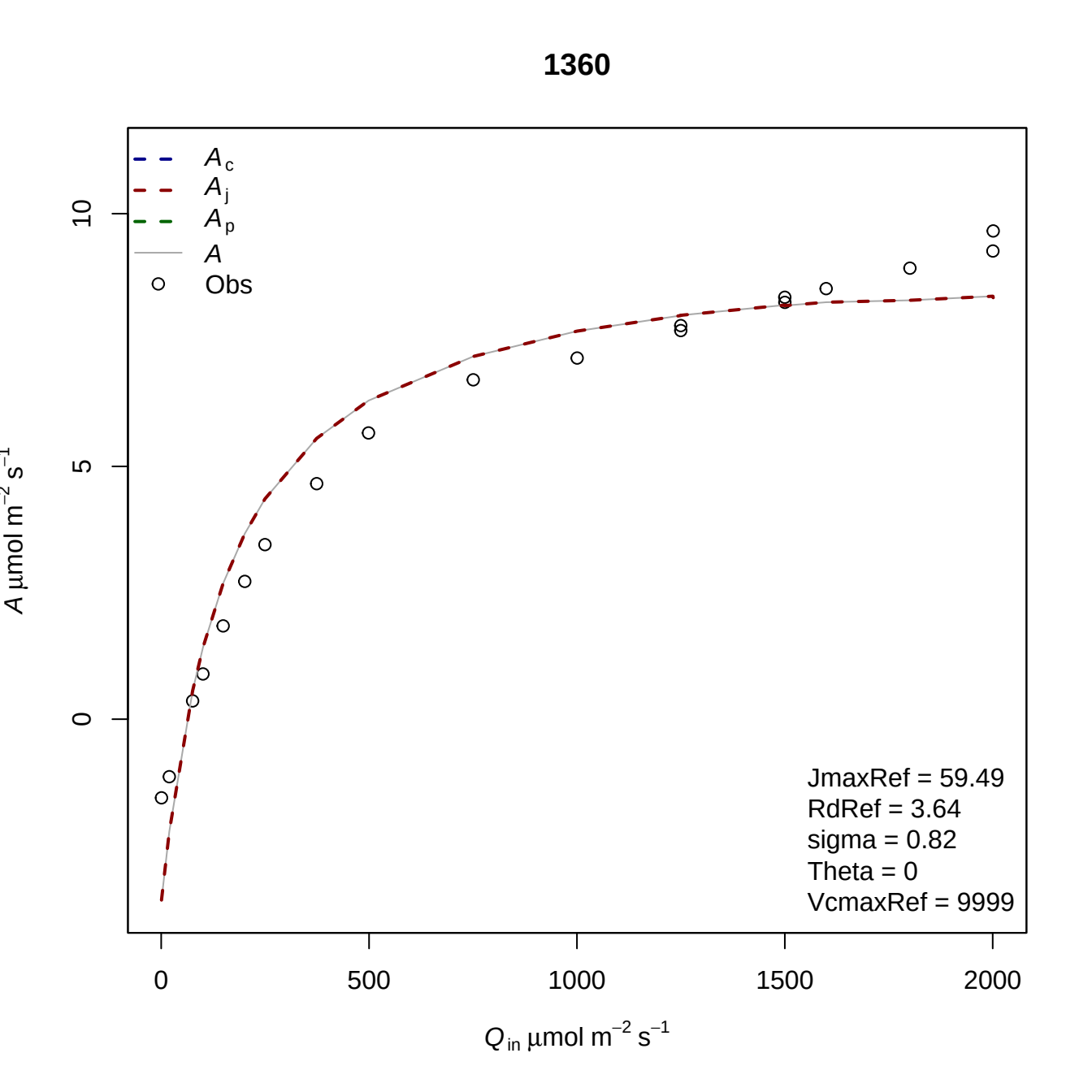

1355
A_c
A_j
A_p
A
Obs
JmaxRef = 35.61
RdRef = 3.45
sigma = 0.8
Theta = 0
VcmaxRef = 9999
6
4
2
0
−2
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

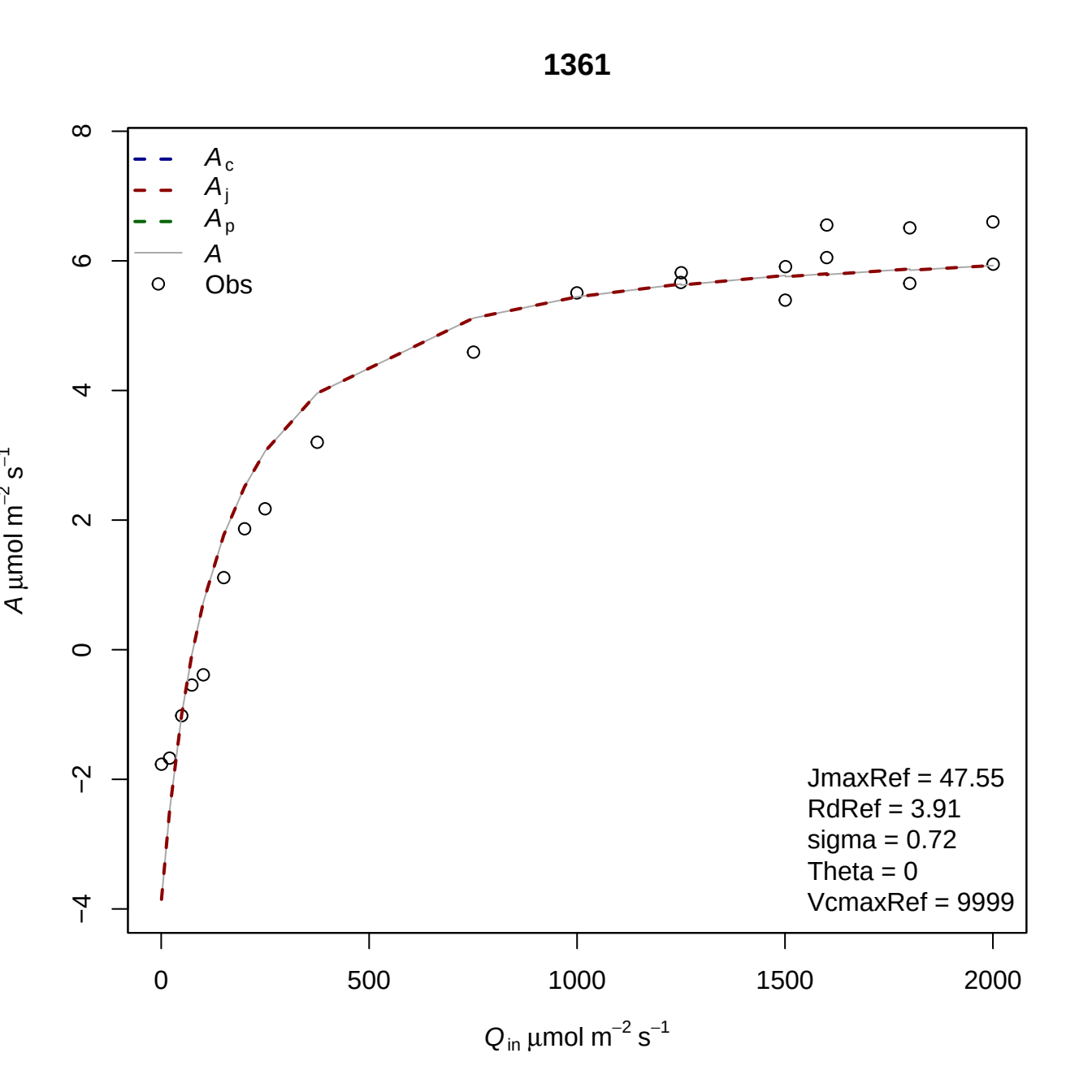

1356
A_c
A_j
A_p
A
Obs
JmaxRef = 77.99
RdRef = 4.36
sigma = 1
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

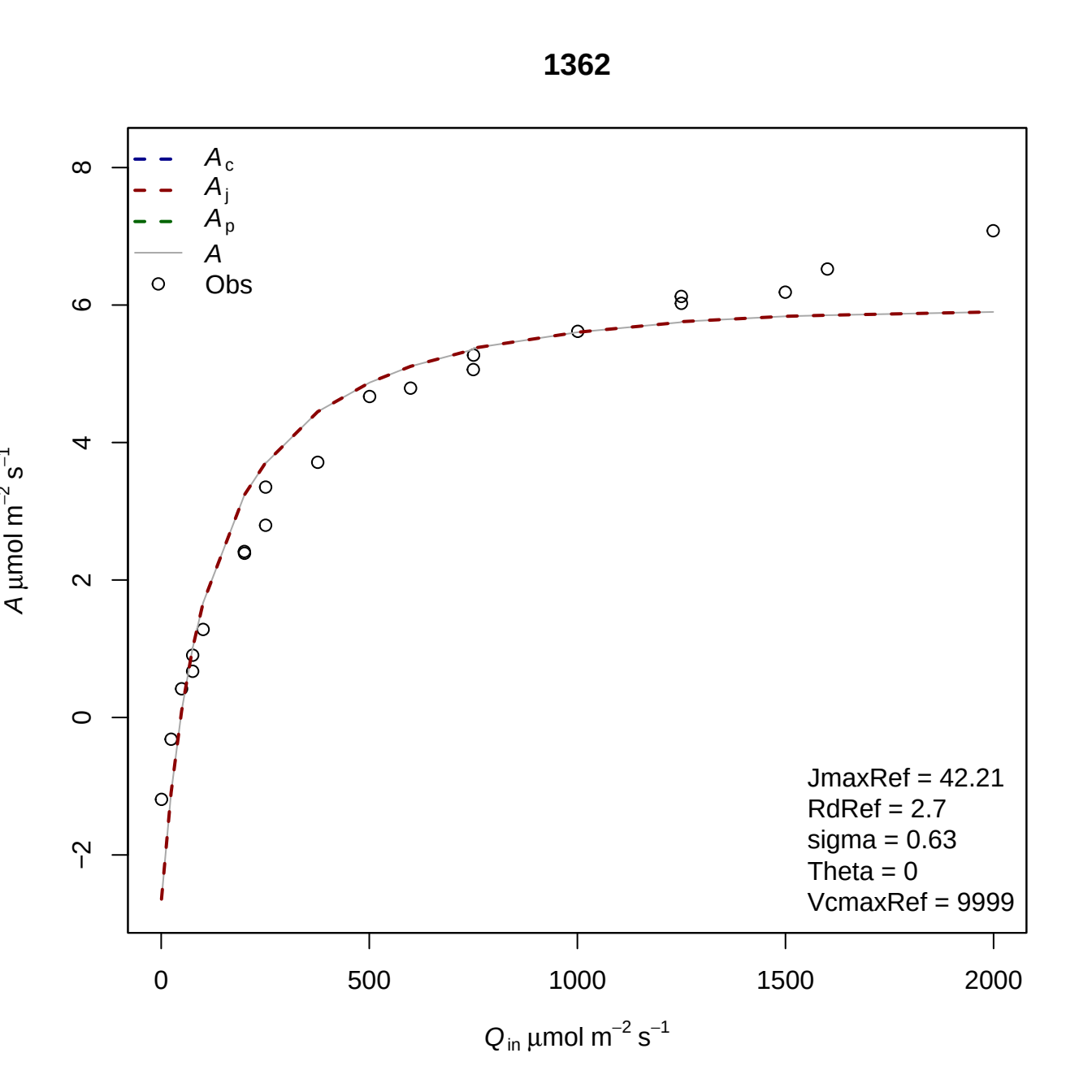

1357
A_c
A_j
A_p
A
Obs
JmaxRef = 50.7
RdRef = 3.62
sigma = 0.69
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

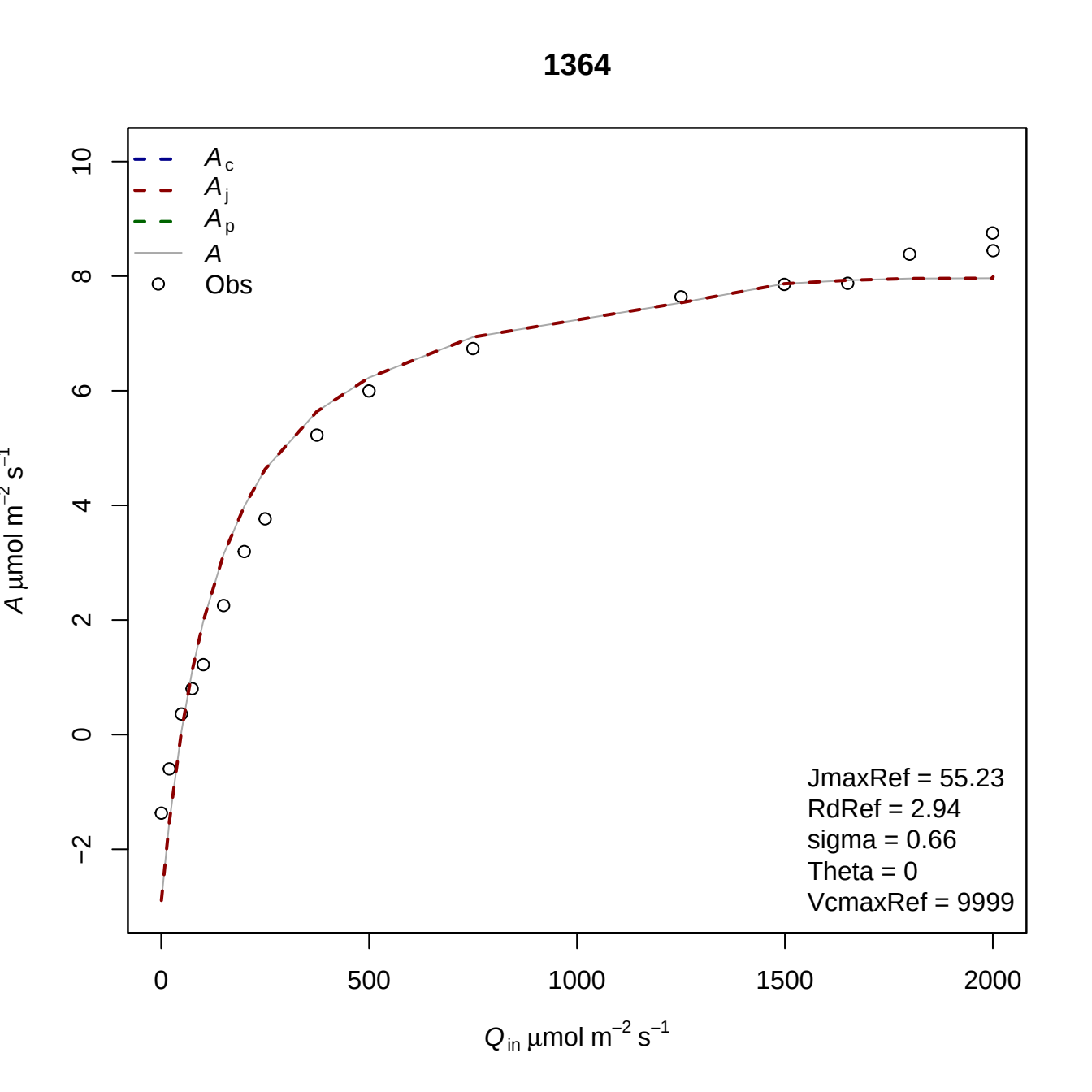

1358
A_c
A_j
A_p
A
Obs
JmaxRef = 46.37
RdRef = 3.29
sigma = 0.94
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

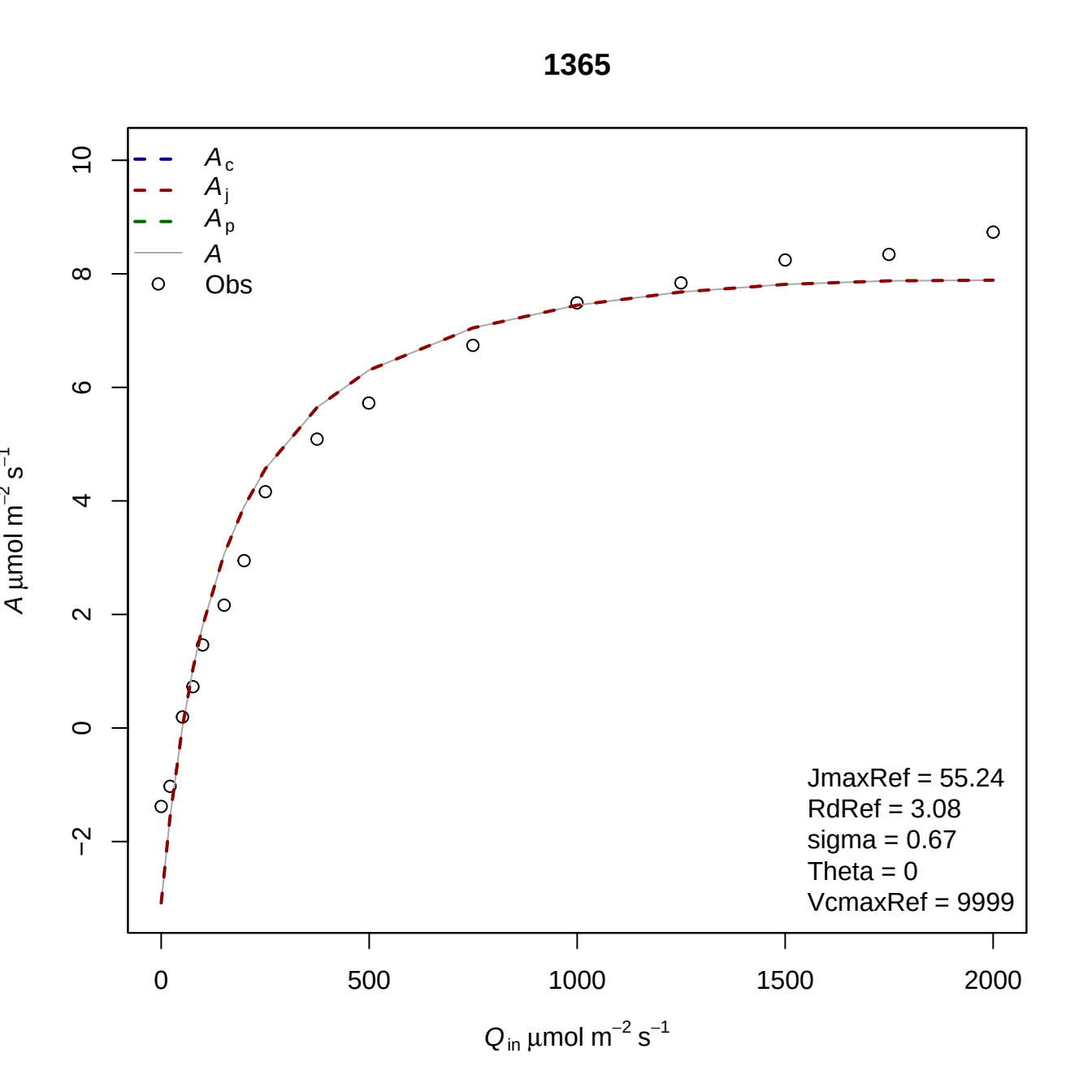

1359
A_c
A_j
A_p
A
Obs
JmaxRef = 40.03
RdRef = 2.56
sigma = 0.82
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

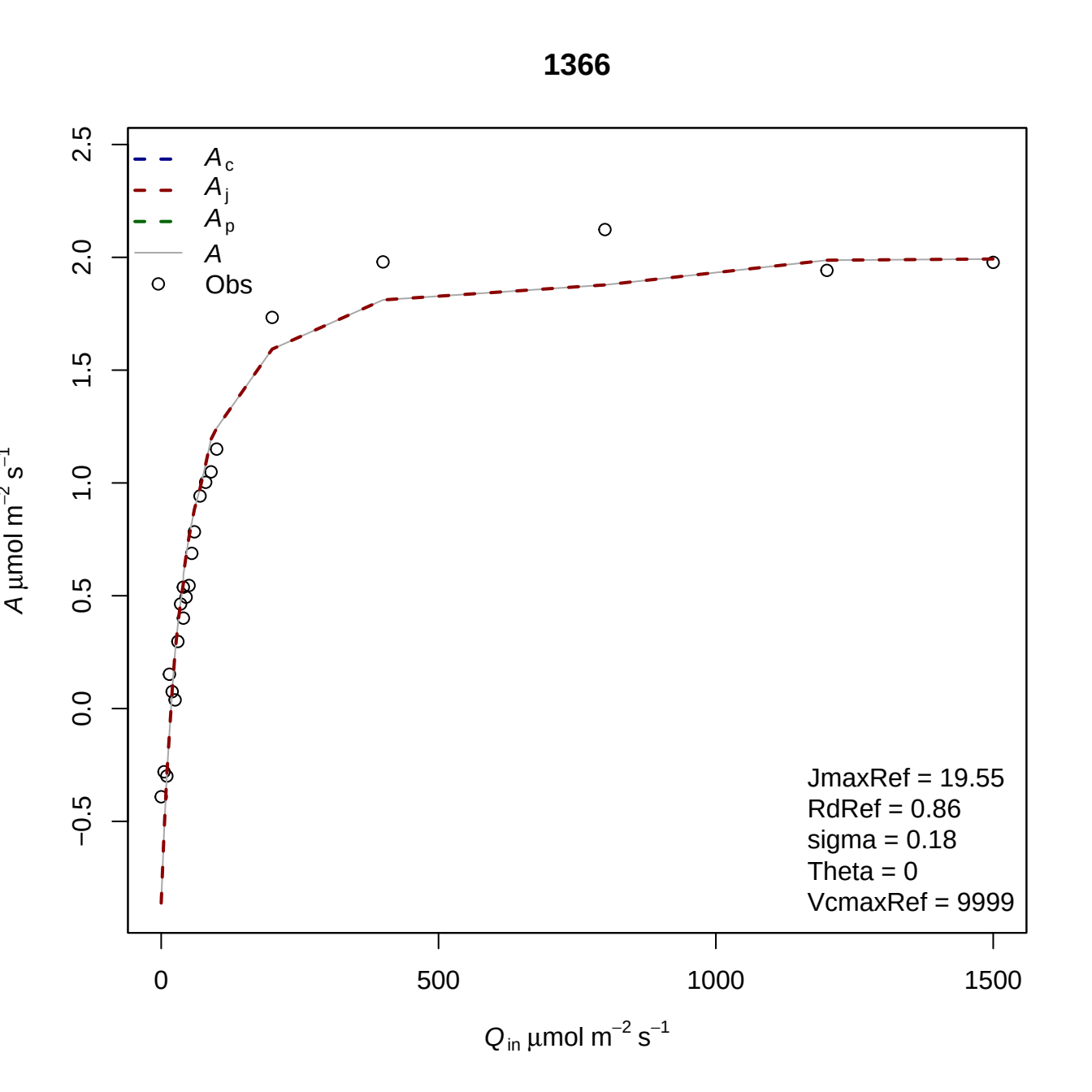

1360
A_c
A_j
A_p
A
Obs
JmaxRef = 59.49
RdRef = 3.64
sigma = 0.82
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

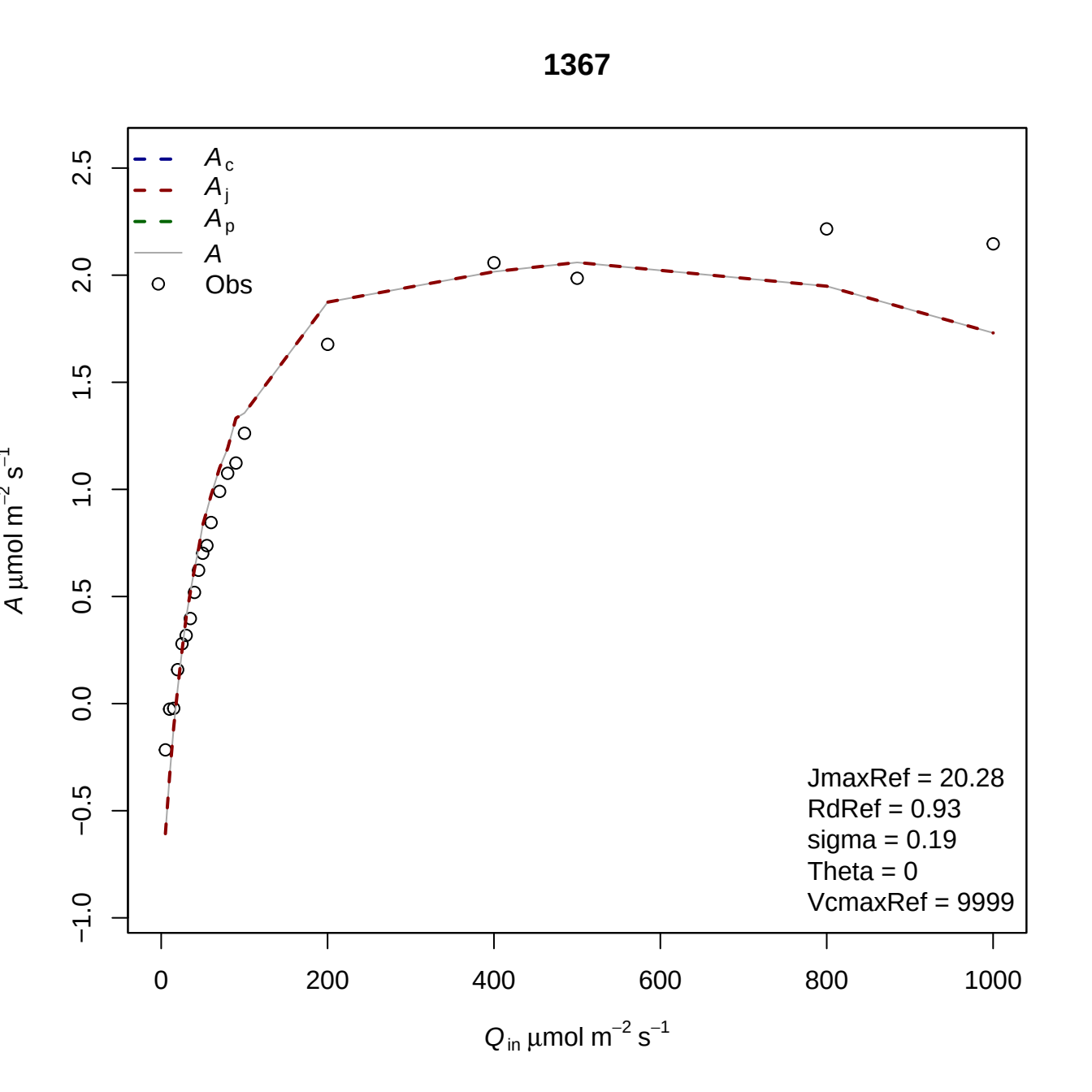

1361
A_c
A_j
A_p
A
Obs
JmaxRef = 47.55
RdRef = 3.91
sigma = 0.72
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

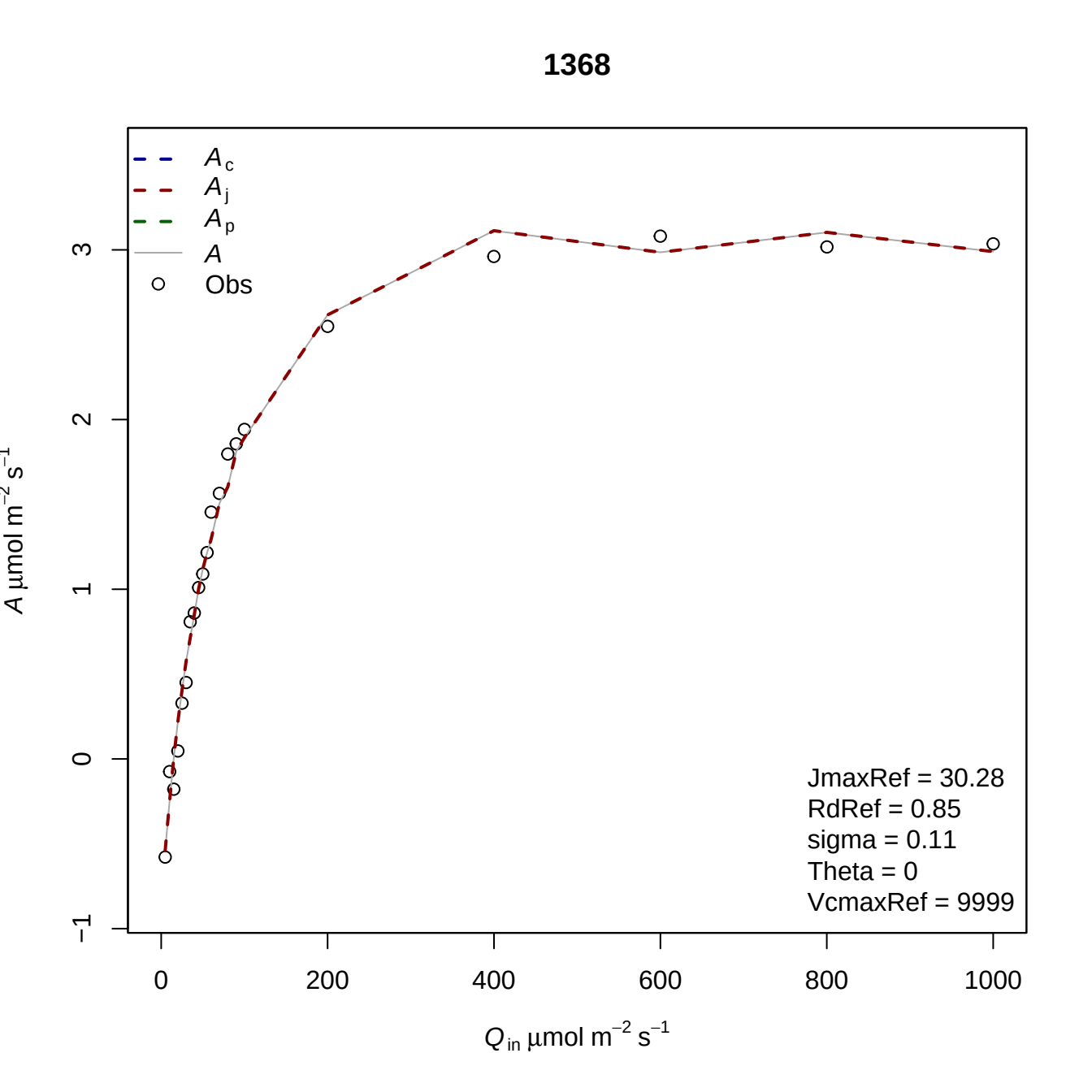

1362
A_c
A_j
A_p
A
Obs
JmaxRef = 42.21
RdRef = 2.7
sigma = 0.63
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

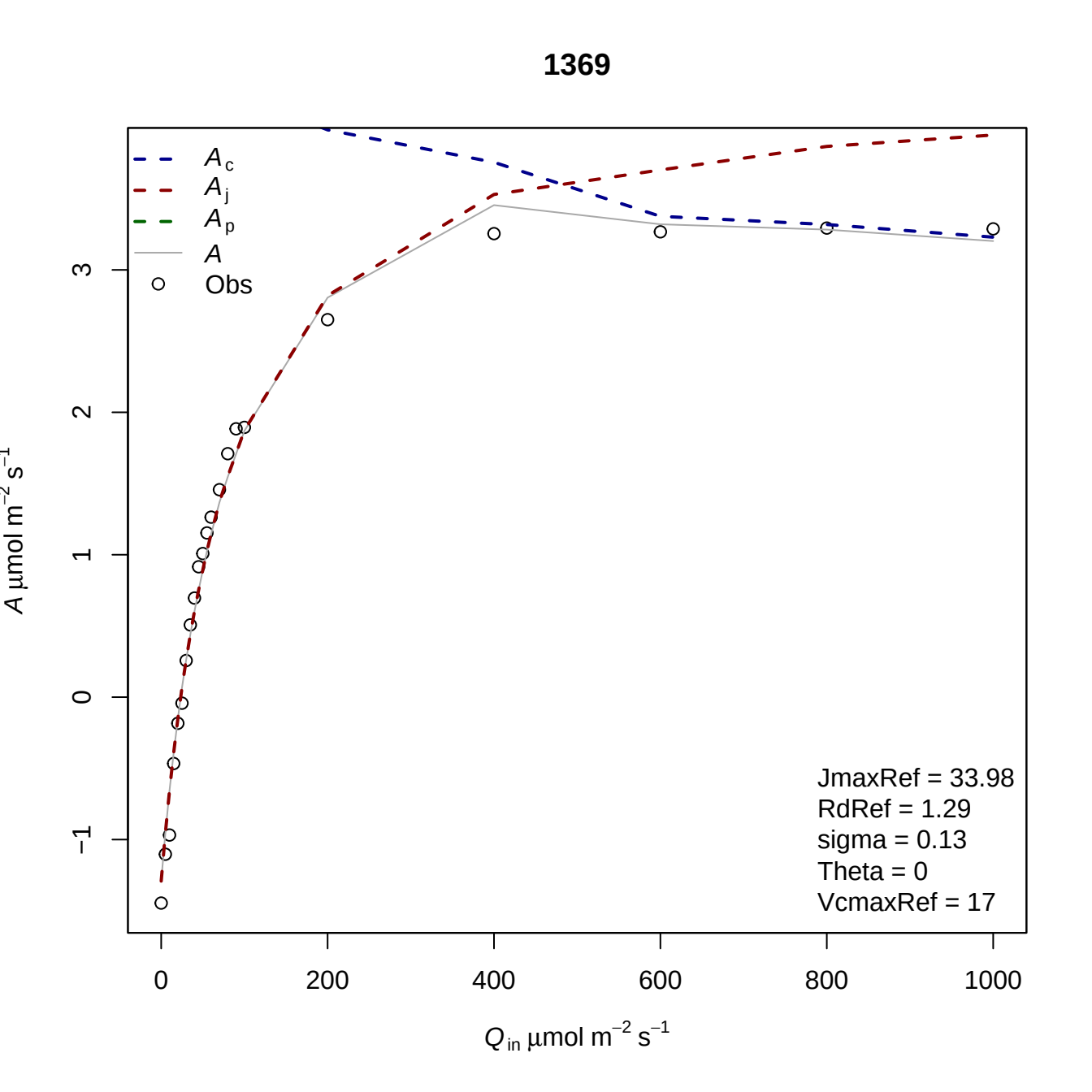

1364
A_c
A_j
A_p
A
Obs
JmaxRef = 55.23
RdRef = 2.94
sigma = 0.66
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

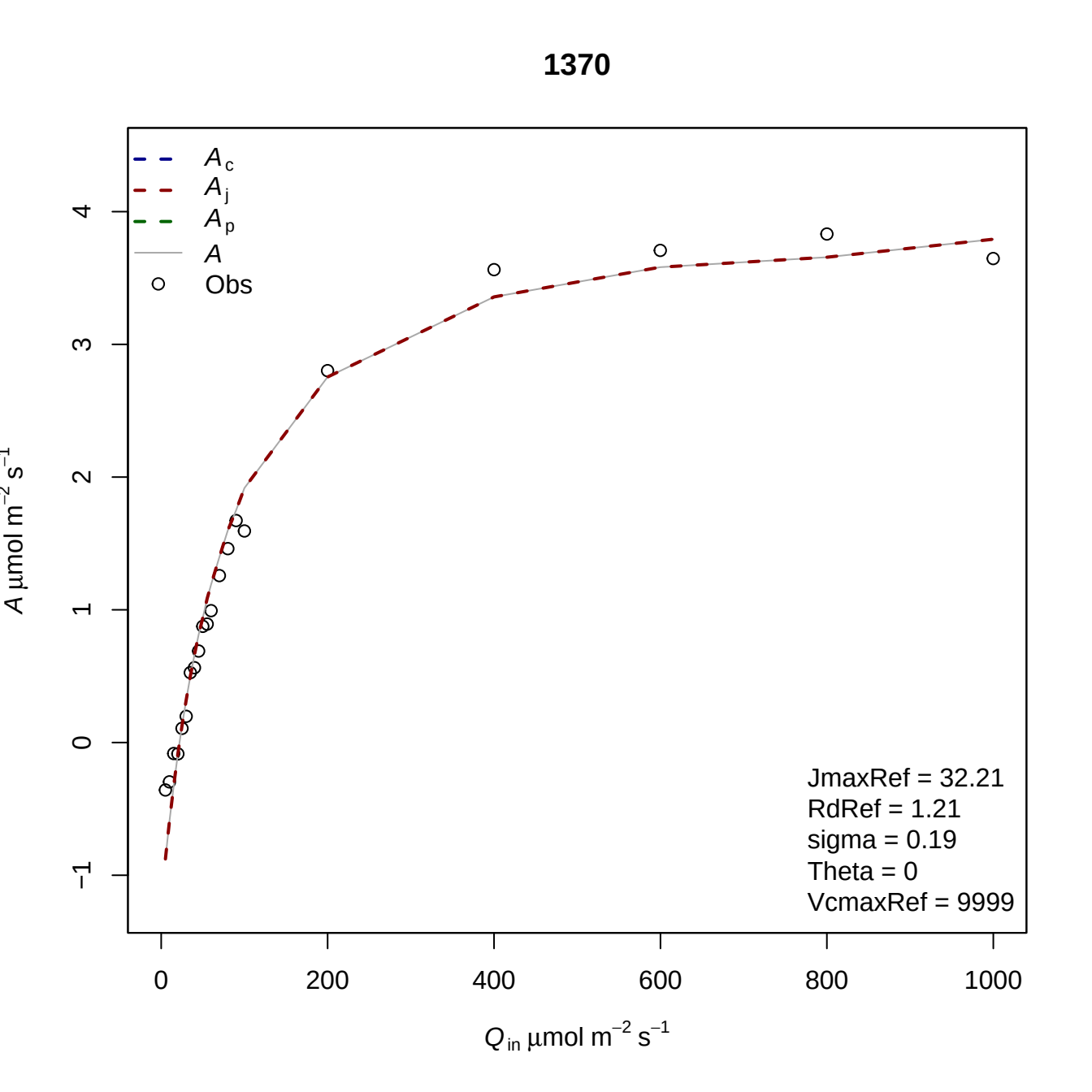

1365
A_c
A_j
A_p
A
Obs
JmaxRef = 55.24
RdRef = 3.08
sigma = 0.67
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

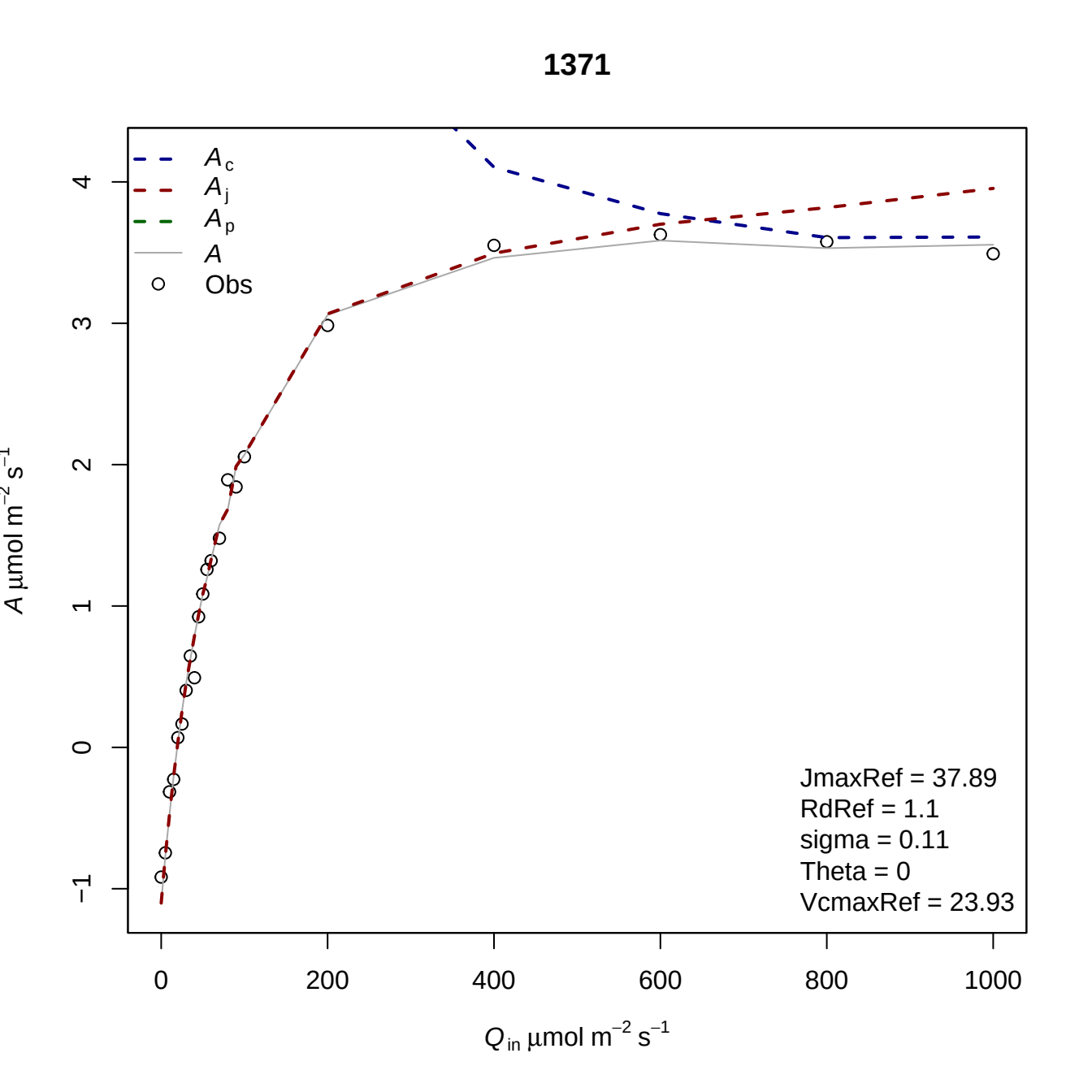

1366
A_c
A_j
A_p
A
Obs
JmaxRef = 19.55
RdRef = 0.86
sigma = 0.18
Theta = 0
VcmaxRef = 9999
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

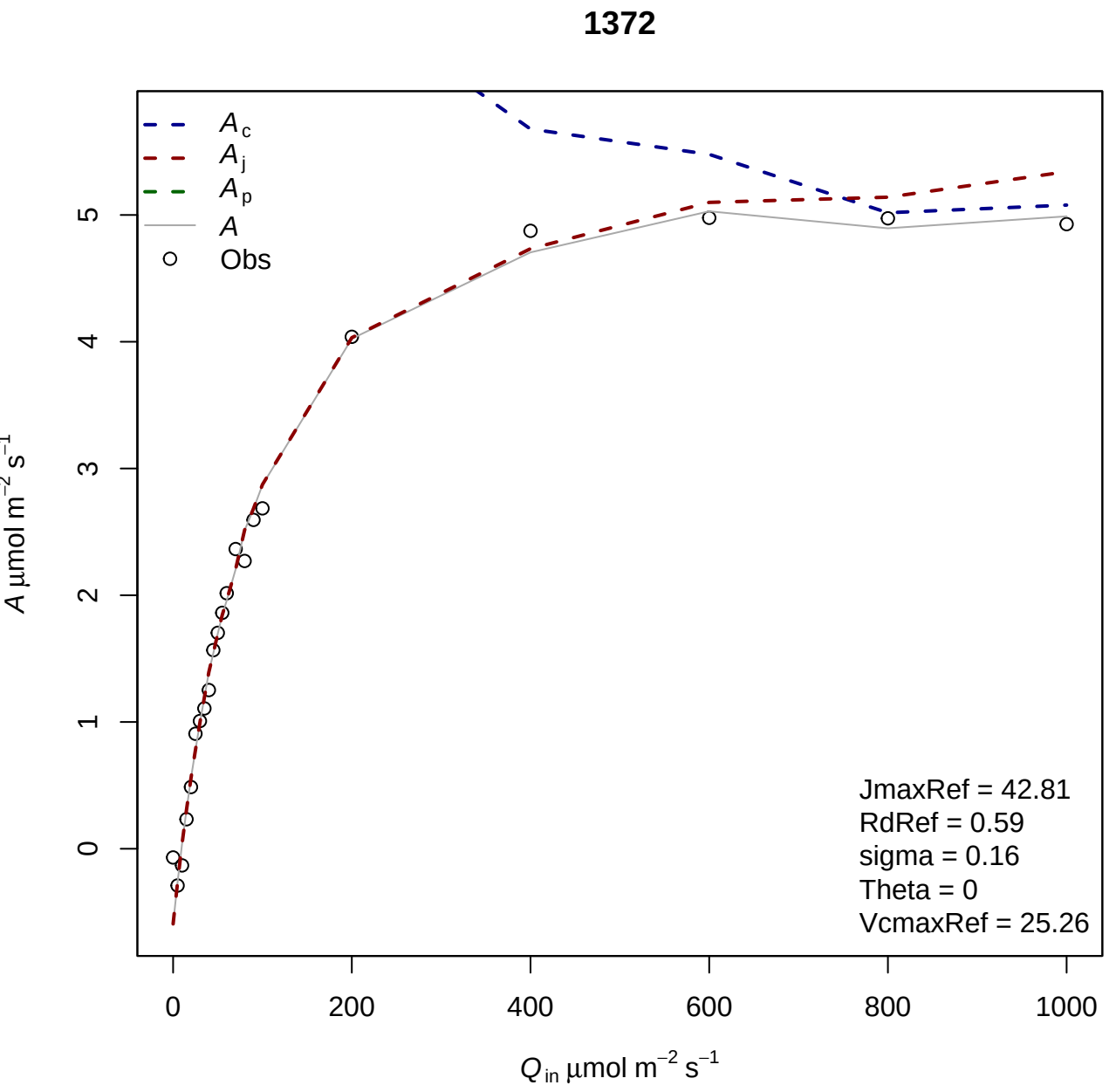

1367
A_c
A_j
A_p
A
Obs
JmaxRef = 20.28
RdRef = 0.93
sigma = 0.19
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
200
400
600
800
1000

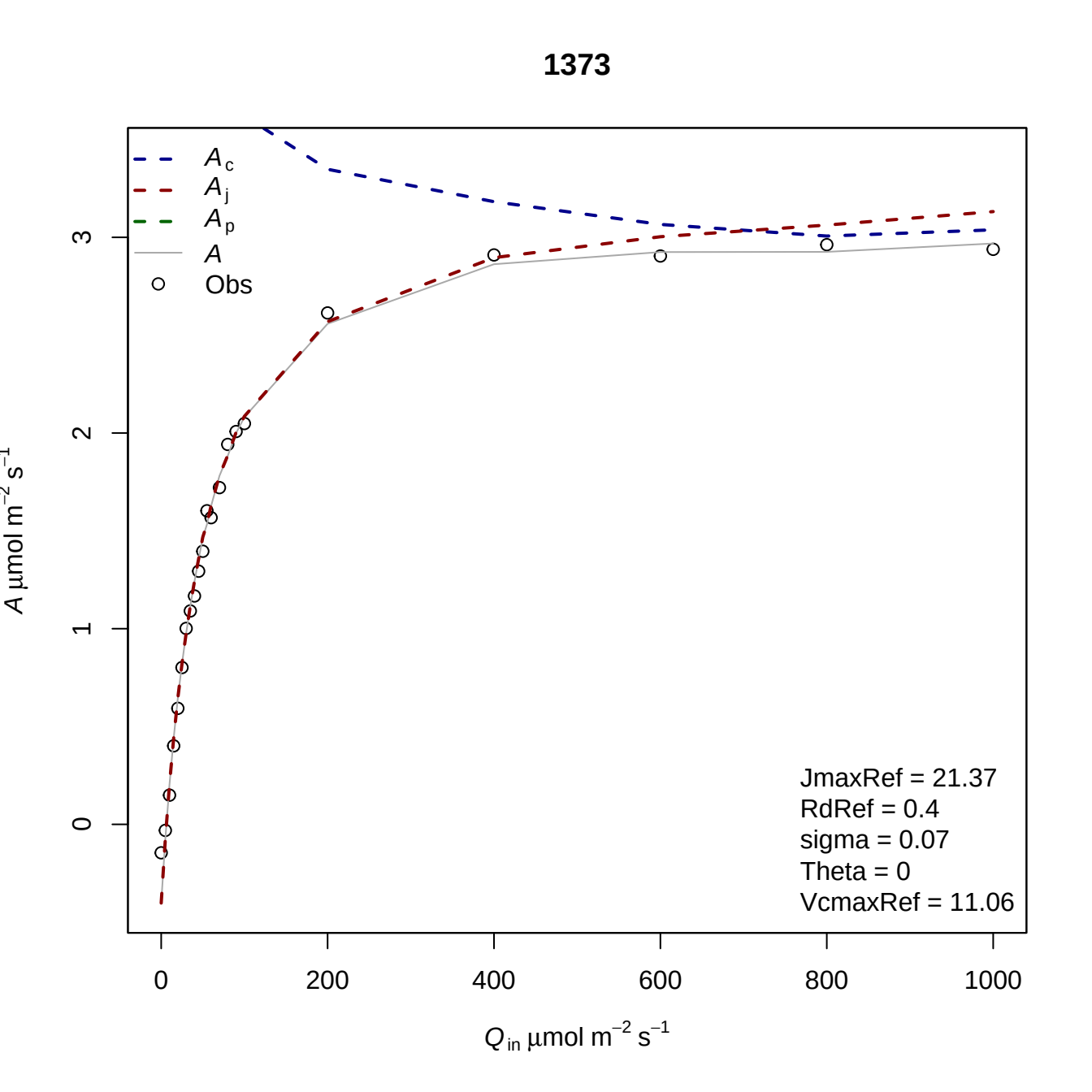

1368
A_c
A_j
A_p
A
Obs
JmaxRef = 30.28
RdRef = 0.85
sigma = 0.11
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

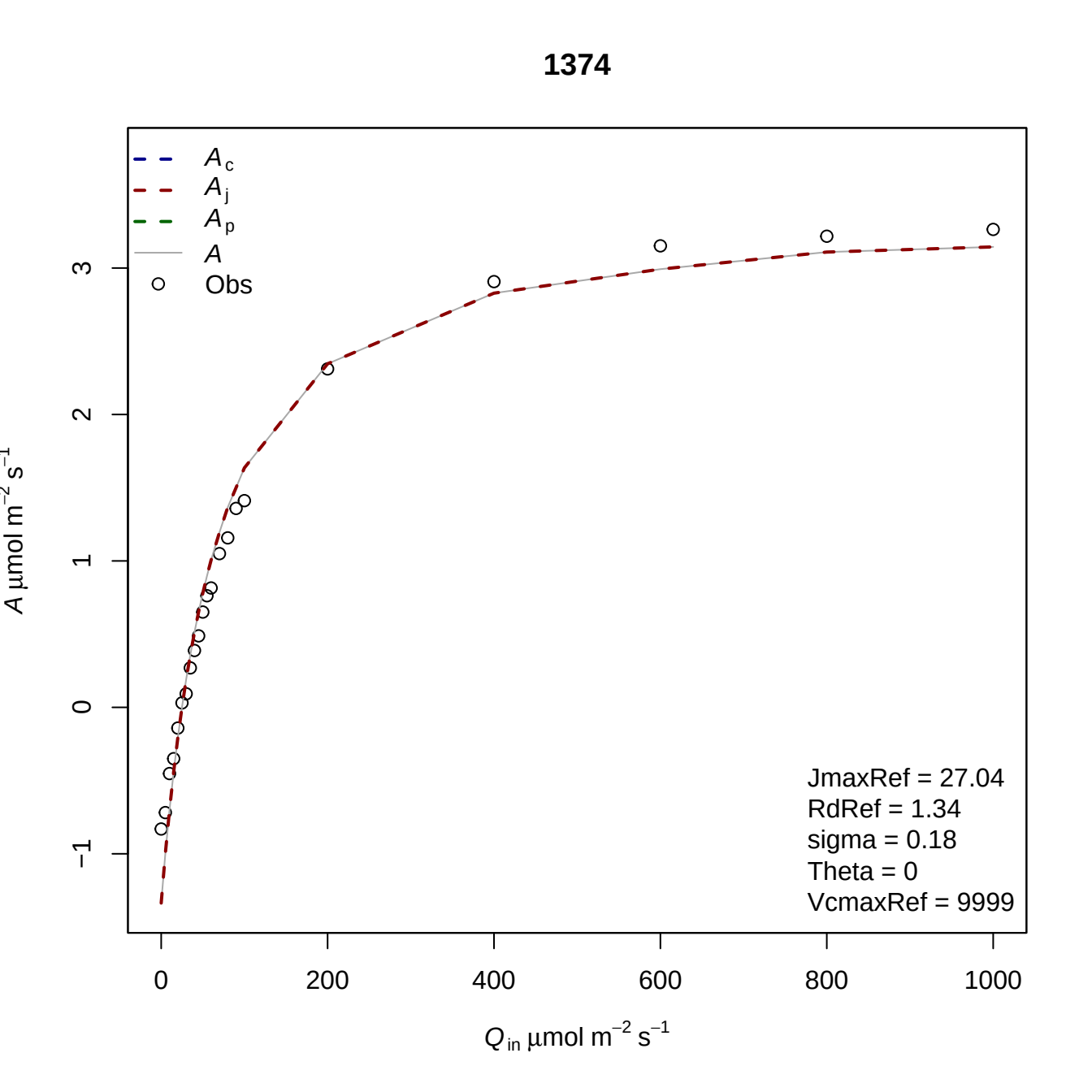

1369
A_c
A_j
A_p
A
Obs
JmaxRef = 33.98
RdRef = 1.29
sigma = 0.13
Theta = 0
VcmaxRef = 17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

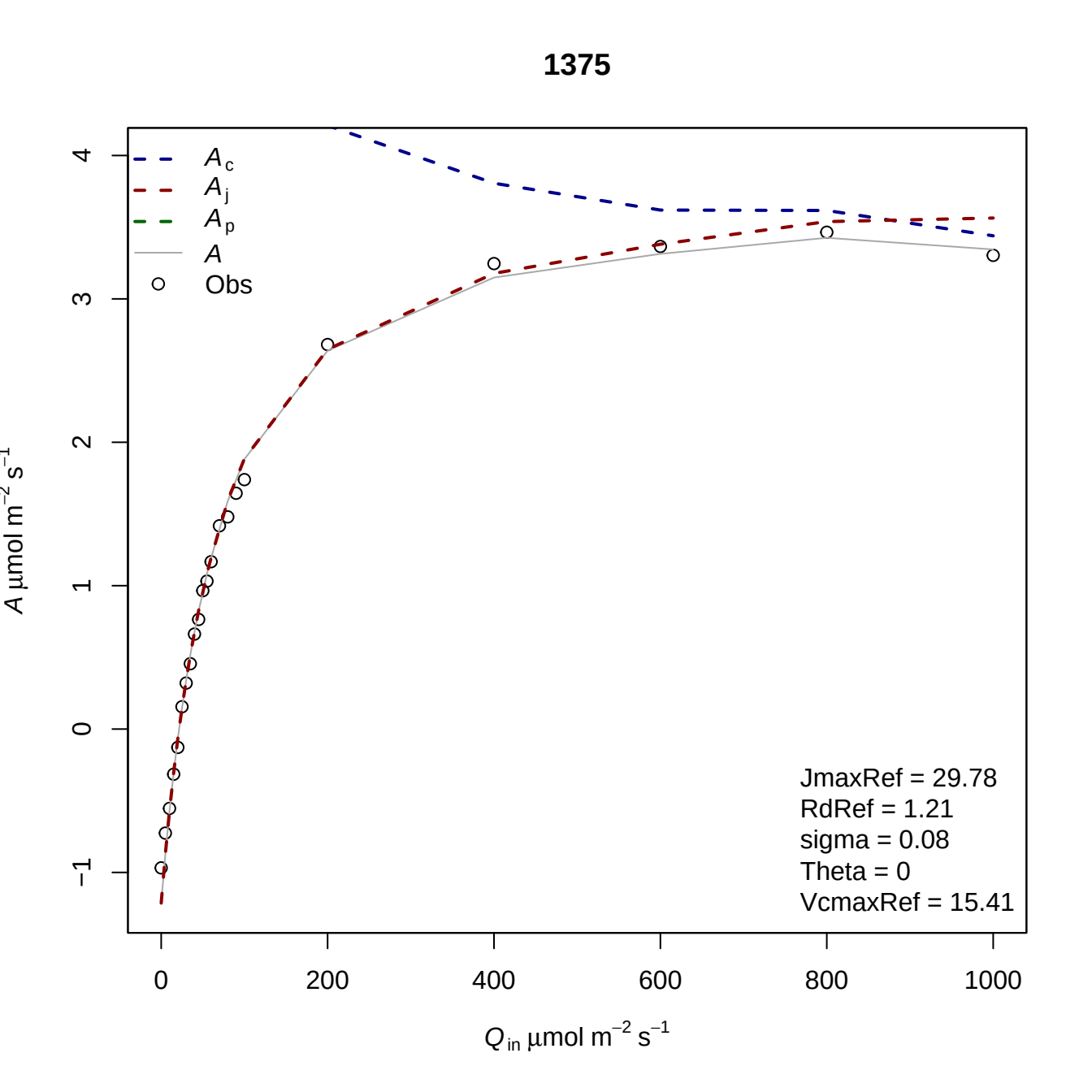

1370
A_c
A_j
A_p
A
Obs
JmaxRef = 32.21
RdRef = 1.21
sigma = 0.19
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

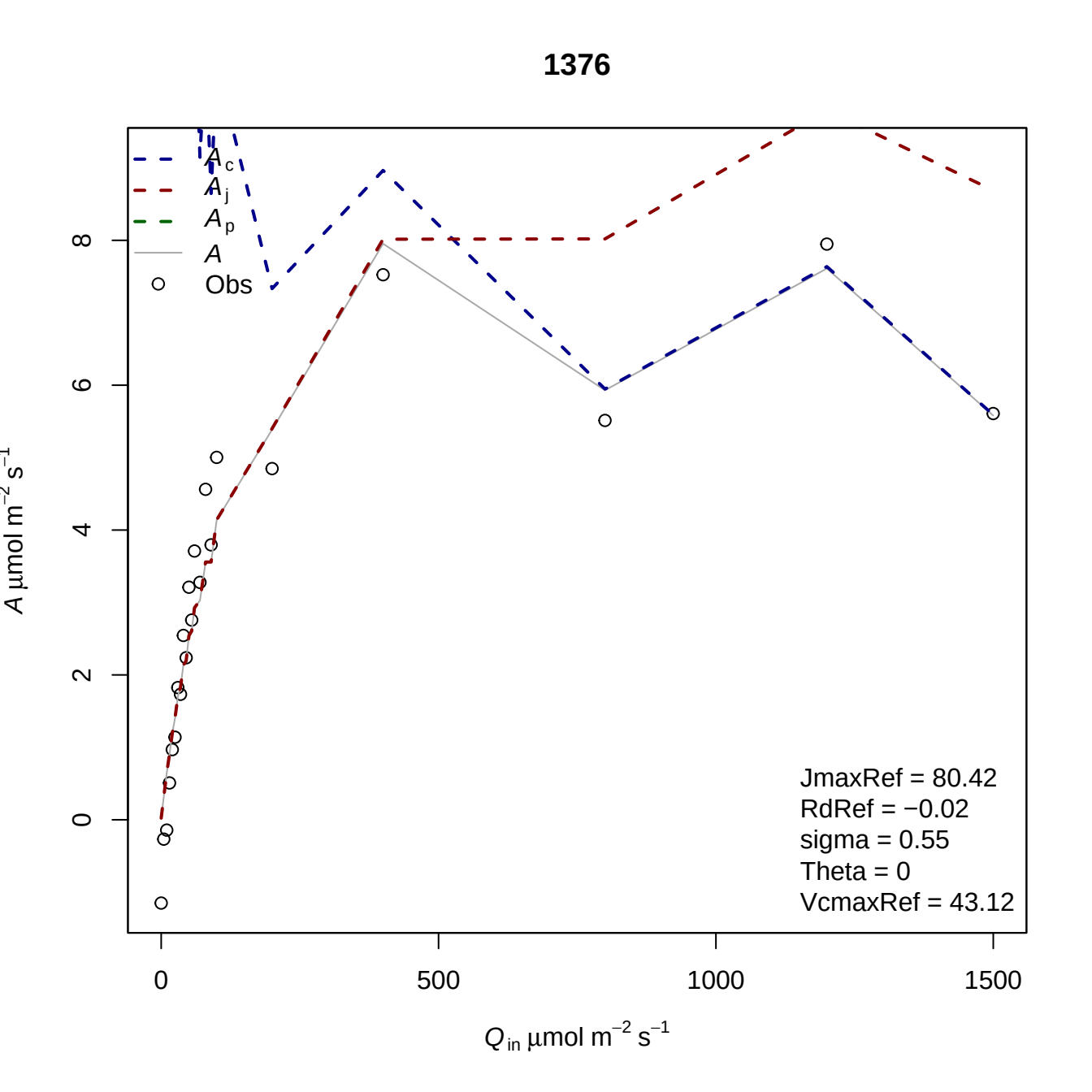

1371
A_c
A_j
A_p
A
Obs
JmaxRef = 37.89
RdRef = 1.1
sigma = 0.11
Theta = 0
VcmaxRef = 23.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

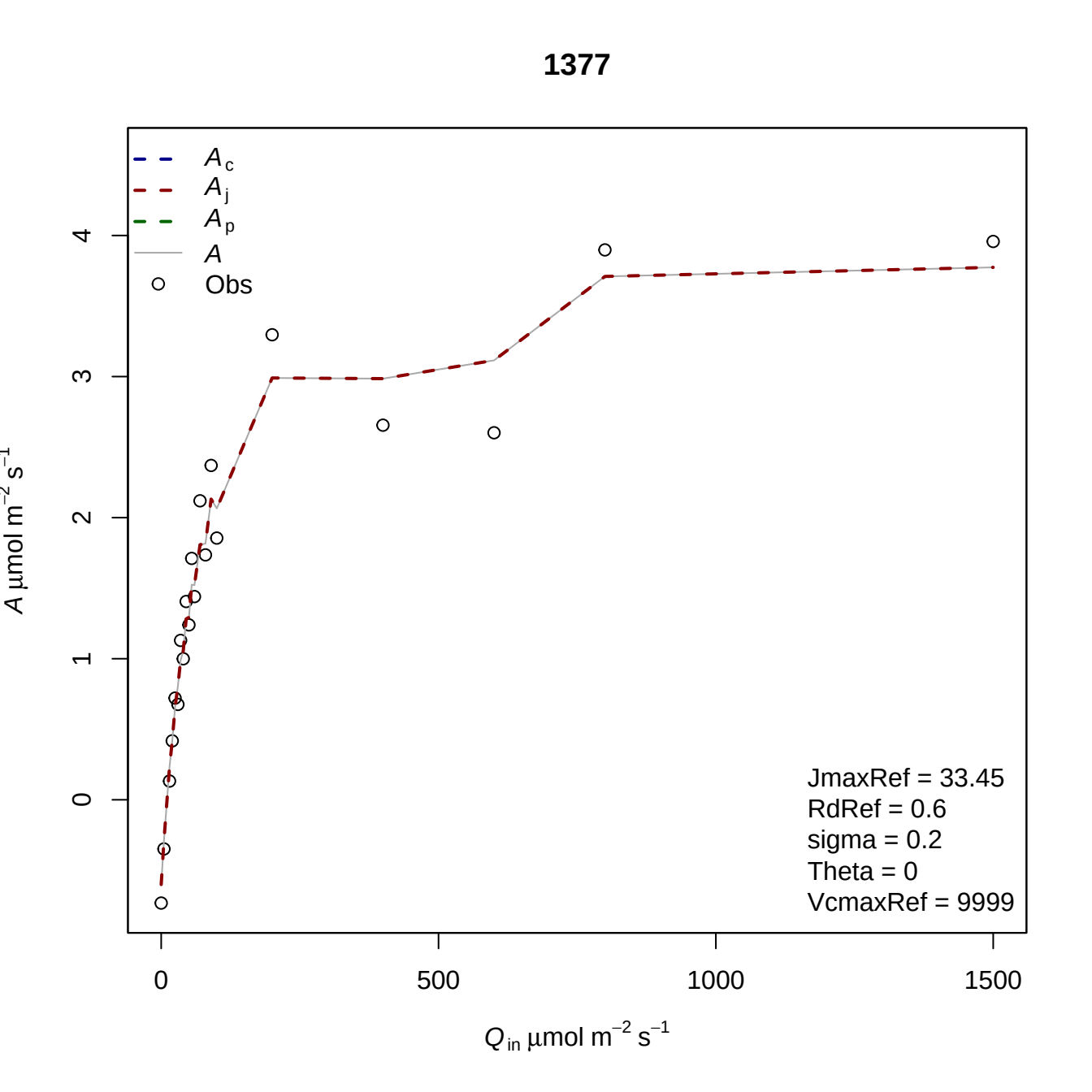

1372
A_c
A_j
A_p
A
Obs
JmaxRef = 42.81
RdRef = 0.59
sigma = 0.16
Theta = 0
VcmaxRef = 25.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000

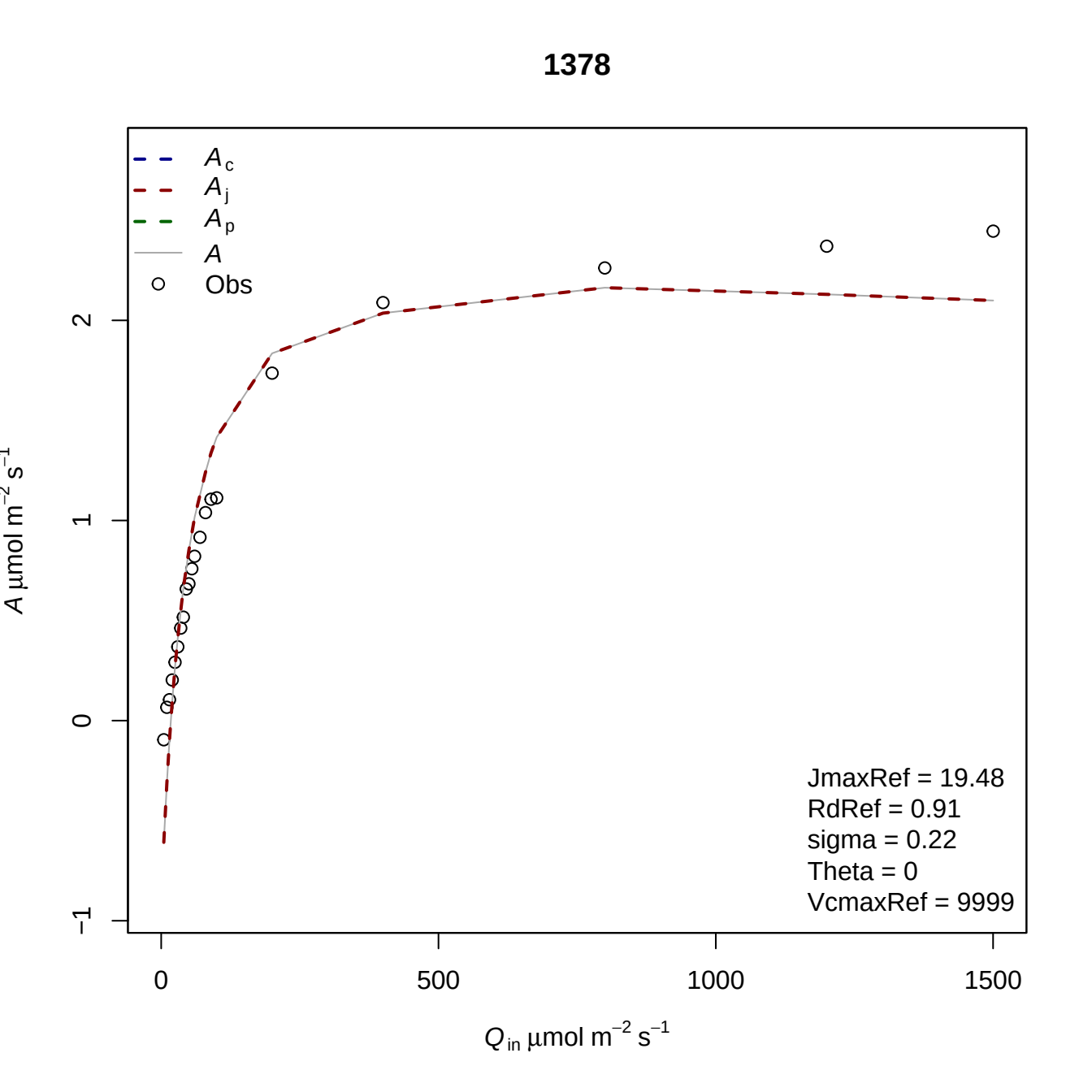

1373
A_c
A_j
A_p
A
Obs
JmaxRef = 21.37
RdRef = 0.4
sigma = 0.07
Theta = 0
VcmaxRef = 11.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

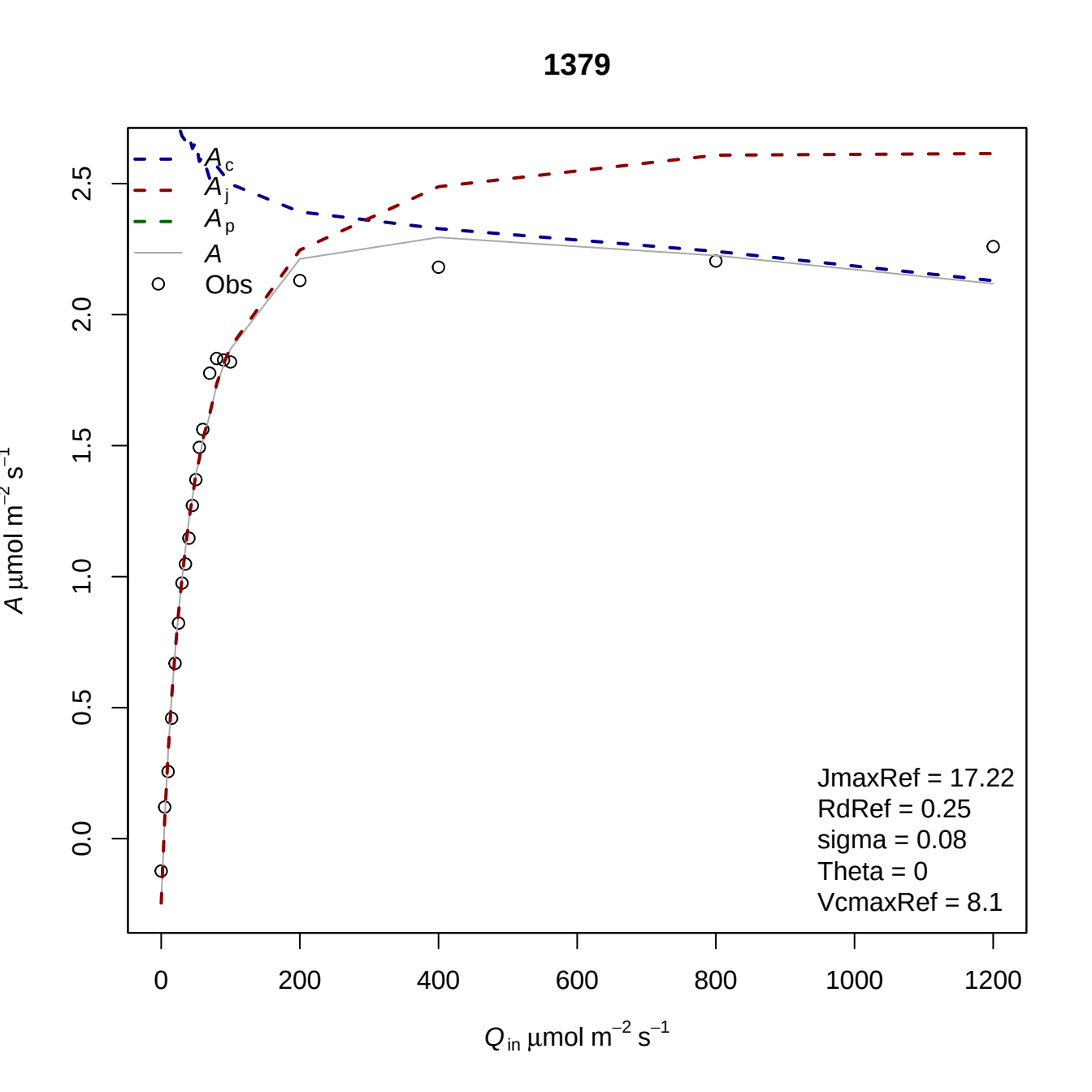

1374
A_c
A_j
A_p
A
Obs
JmaxRef = 27.04
RdRef = 1.34
sigma = 0.18
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

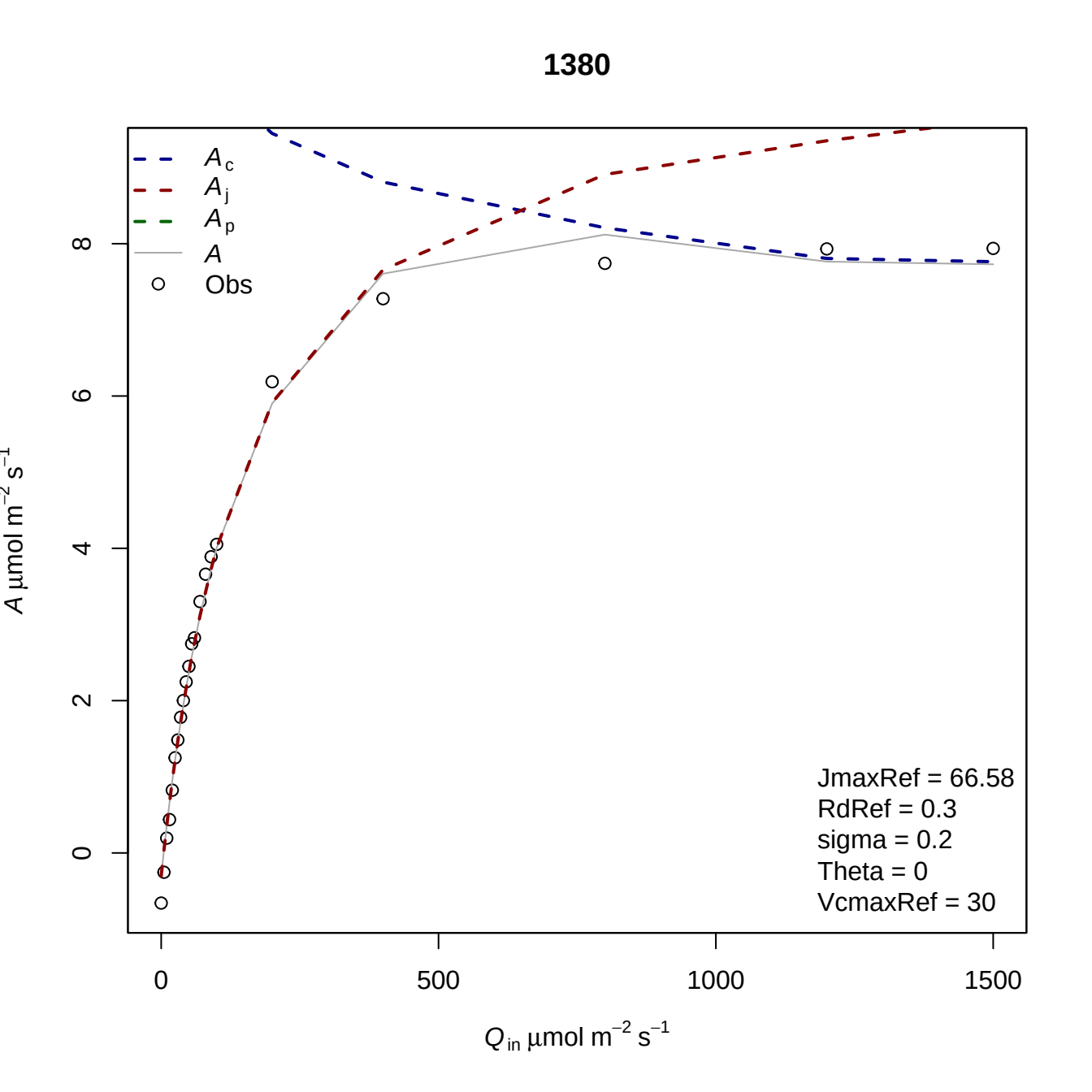

1375
A_c
A_j
A_p
A
Obs
JmaxRef = 29.78
RdRef = 1.21
sigma = 0.08
Theta = 0
VcmaxRef = 15.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

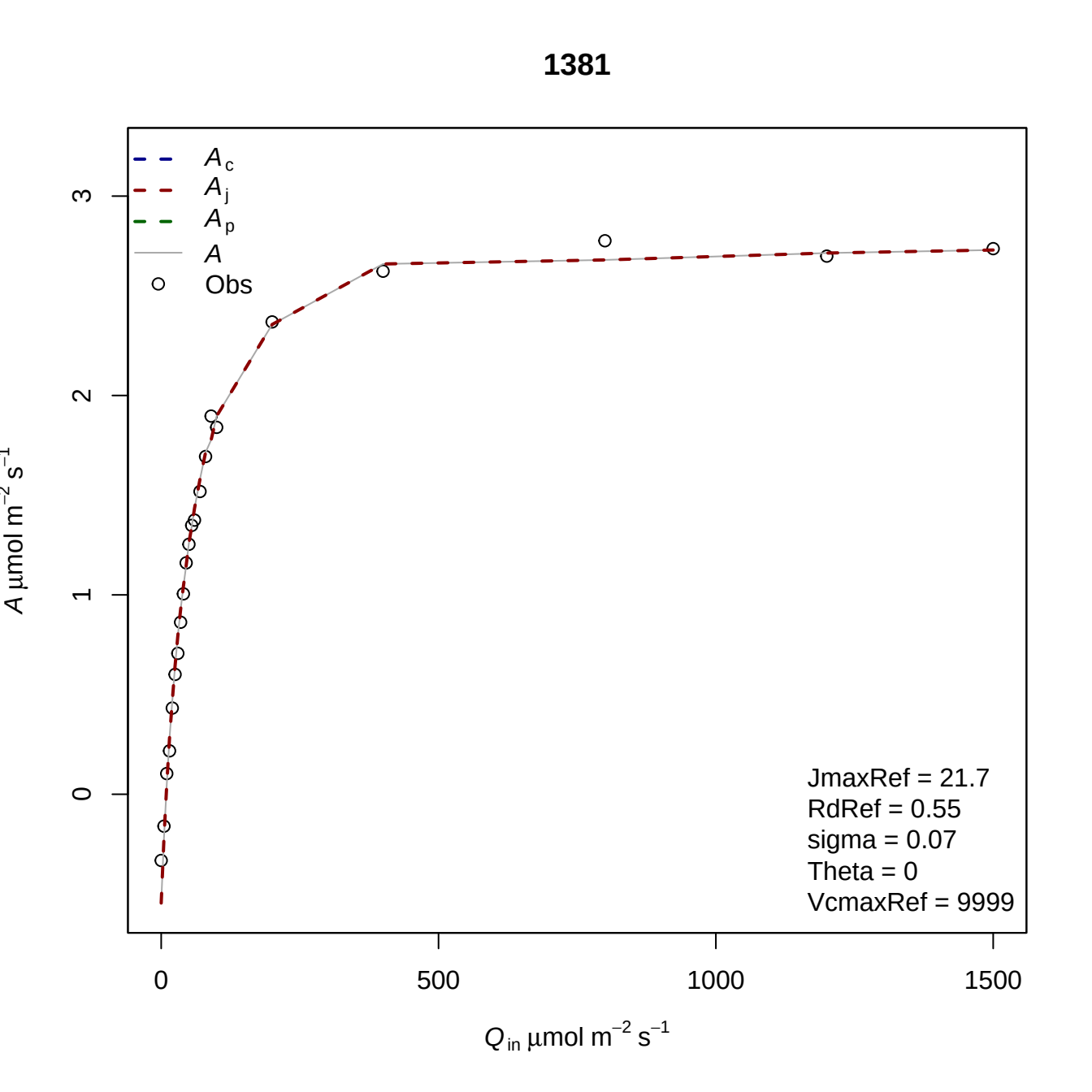

1376
A_c
A_j
A_p
A
Obs
JmaxRef = 80.42
RdRef = −0.02
sigma = 0.55
Theta = 0
VcmaxRef = 43.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

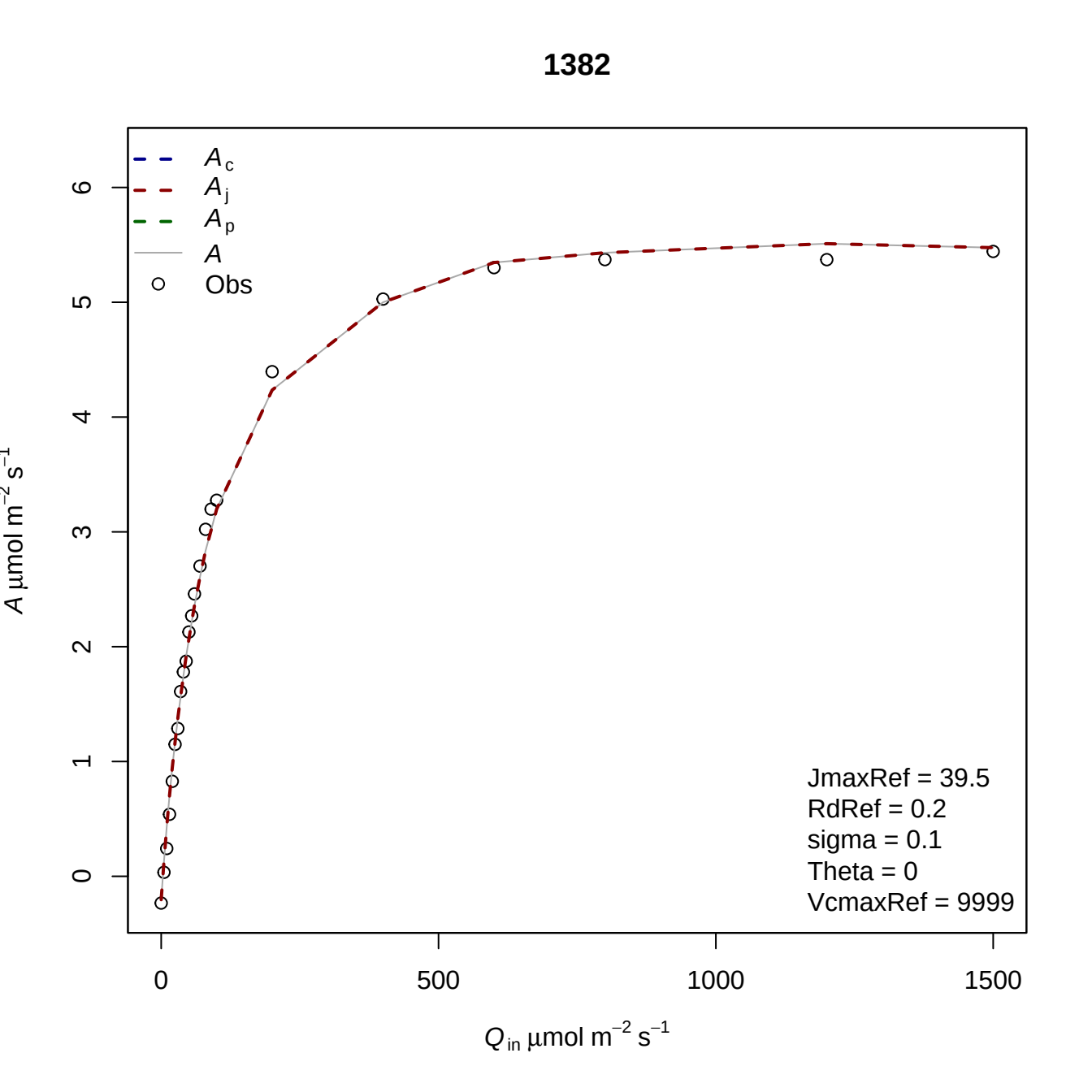

1377
A_c
A_j
A_p
A
Obs
JmaxRef = 33.45
RdRef = 0.6
sigma = 0.2
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

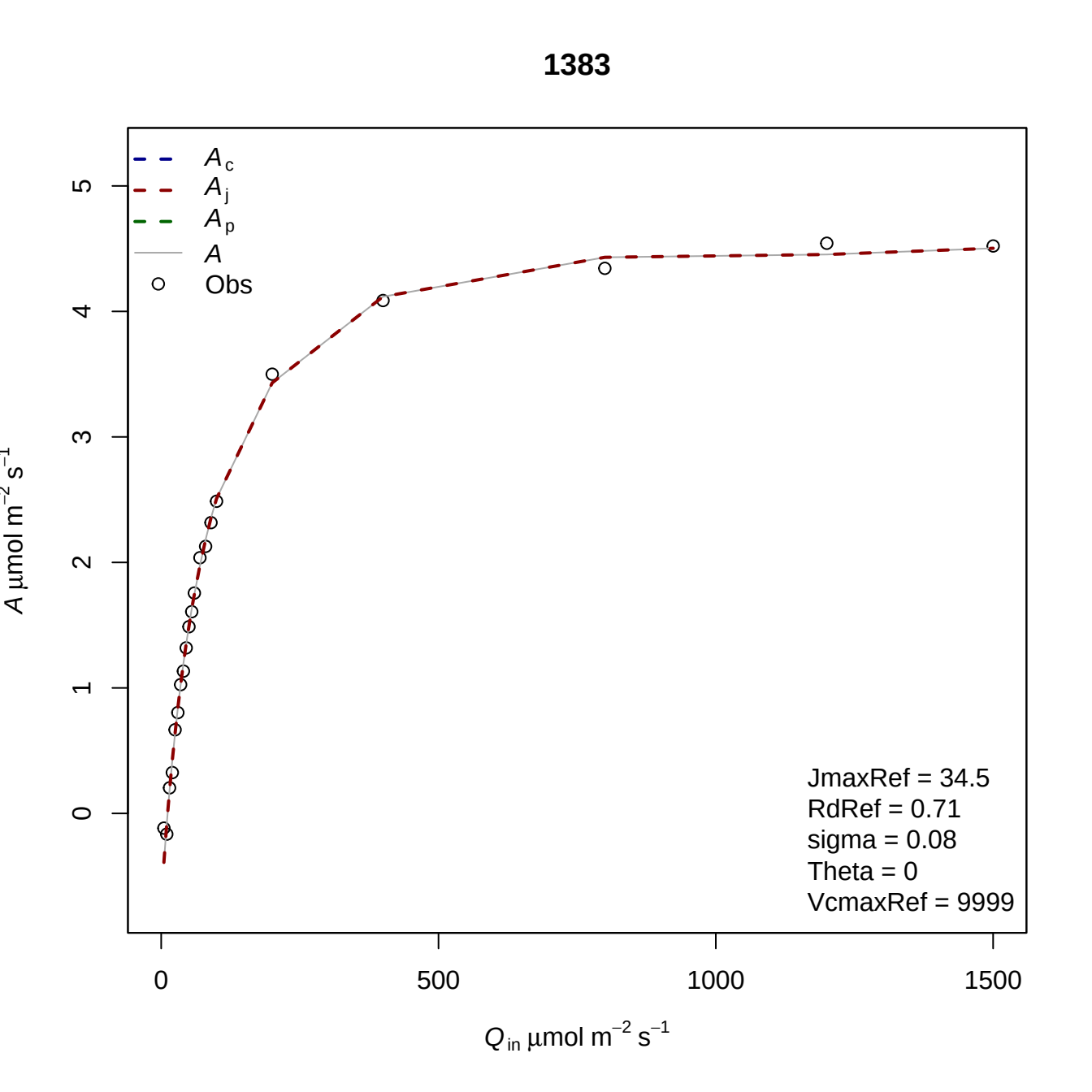

1378
A_c
A_j
A_p
A
Obs
JmaxRef = 19.48
RdRef = 0.91
sigma = 0.22
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

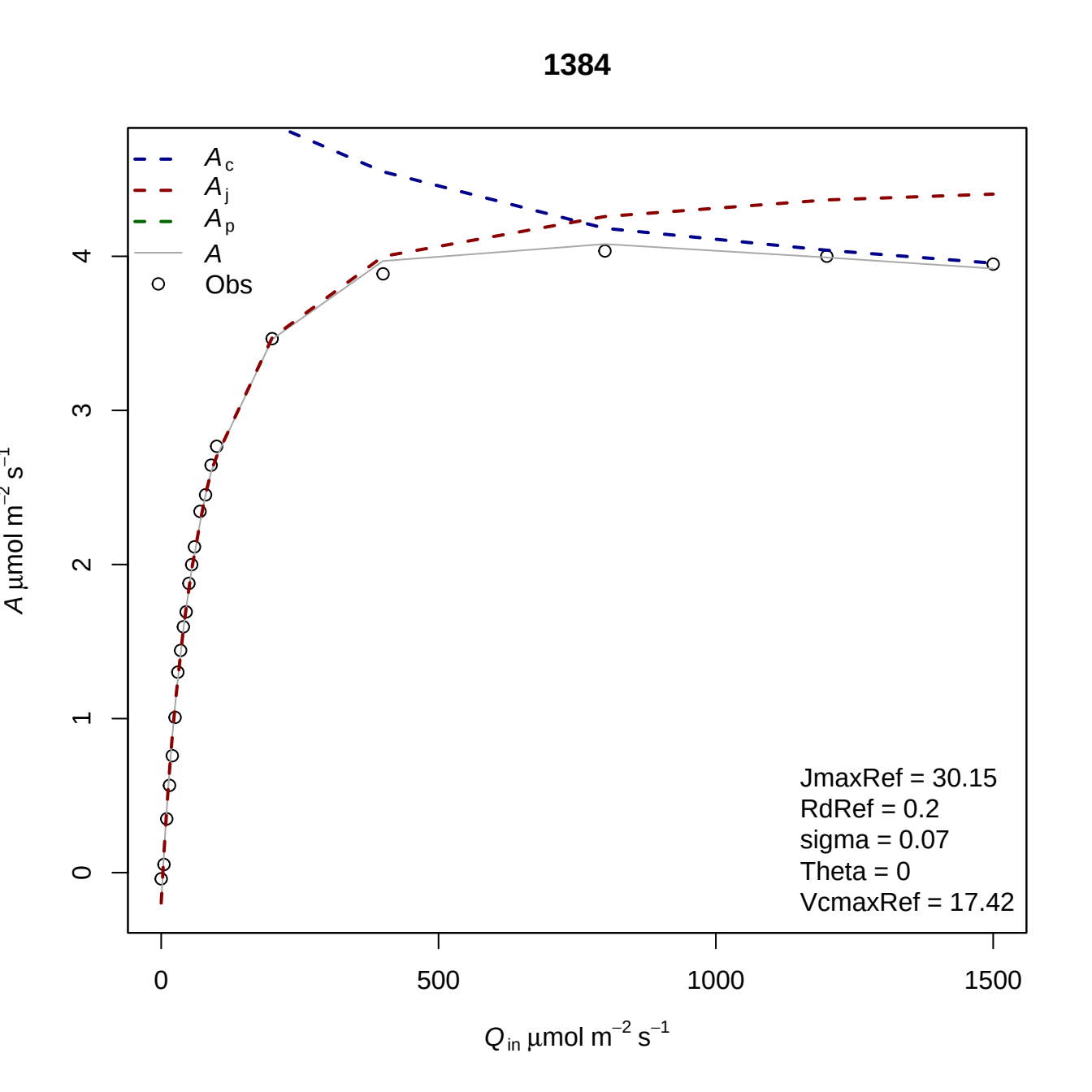

1379
A_c
A_j
A_p
A
Obs
JmaxRef = 17.22
RdRef = 0.25
sigma = 0.08
Theta = 0
VcmaxRef = 8.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
0
200
400
600
800
1000
1200

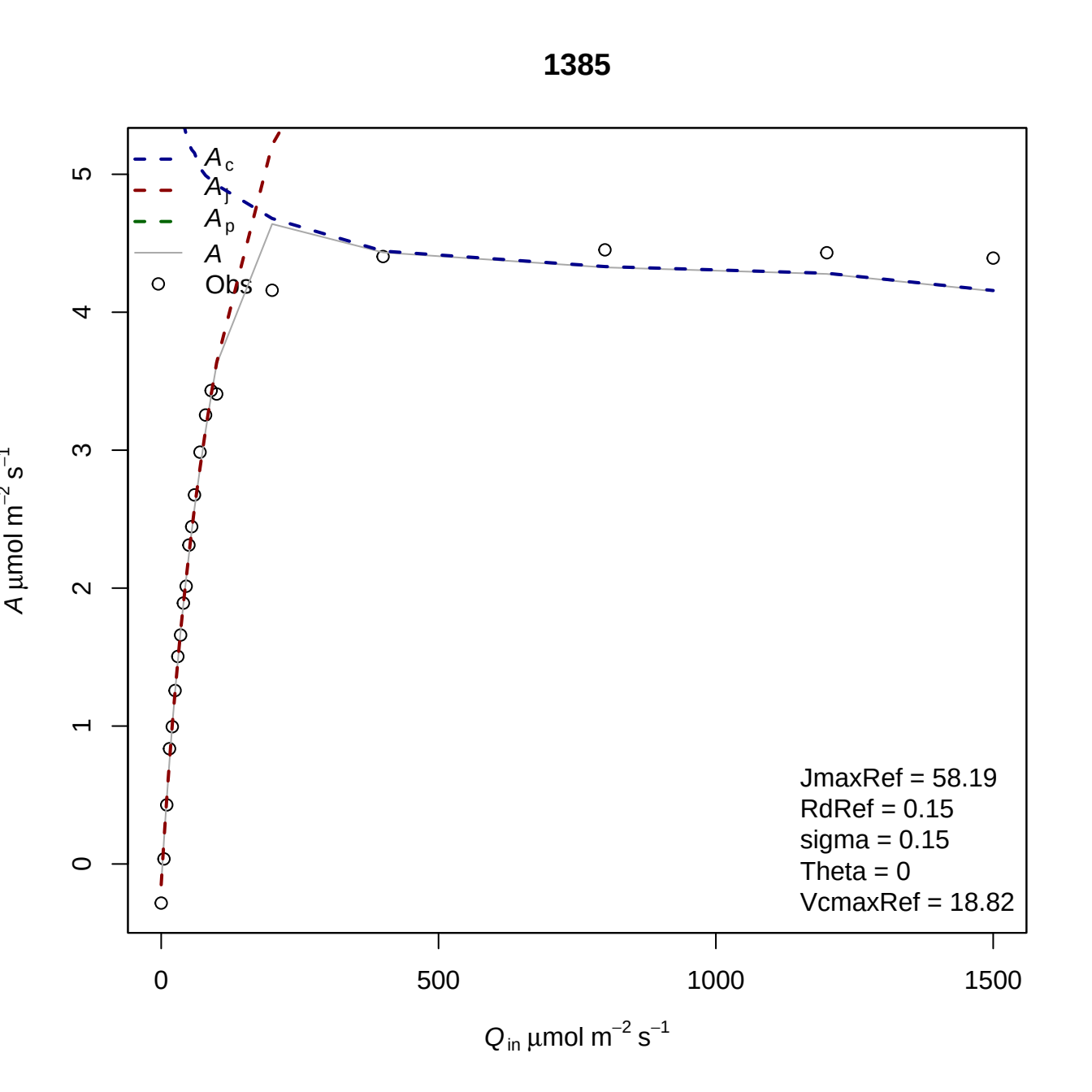

1380
A_c
A_j
A_p
A
Obs
JmaxRef = 66.58
RdRef = 0.3
sigma = 0.2
Theta = 0
VcmaxRef = 30
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

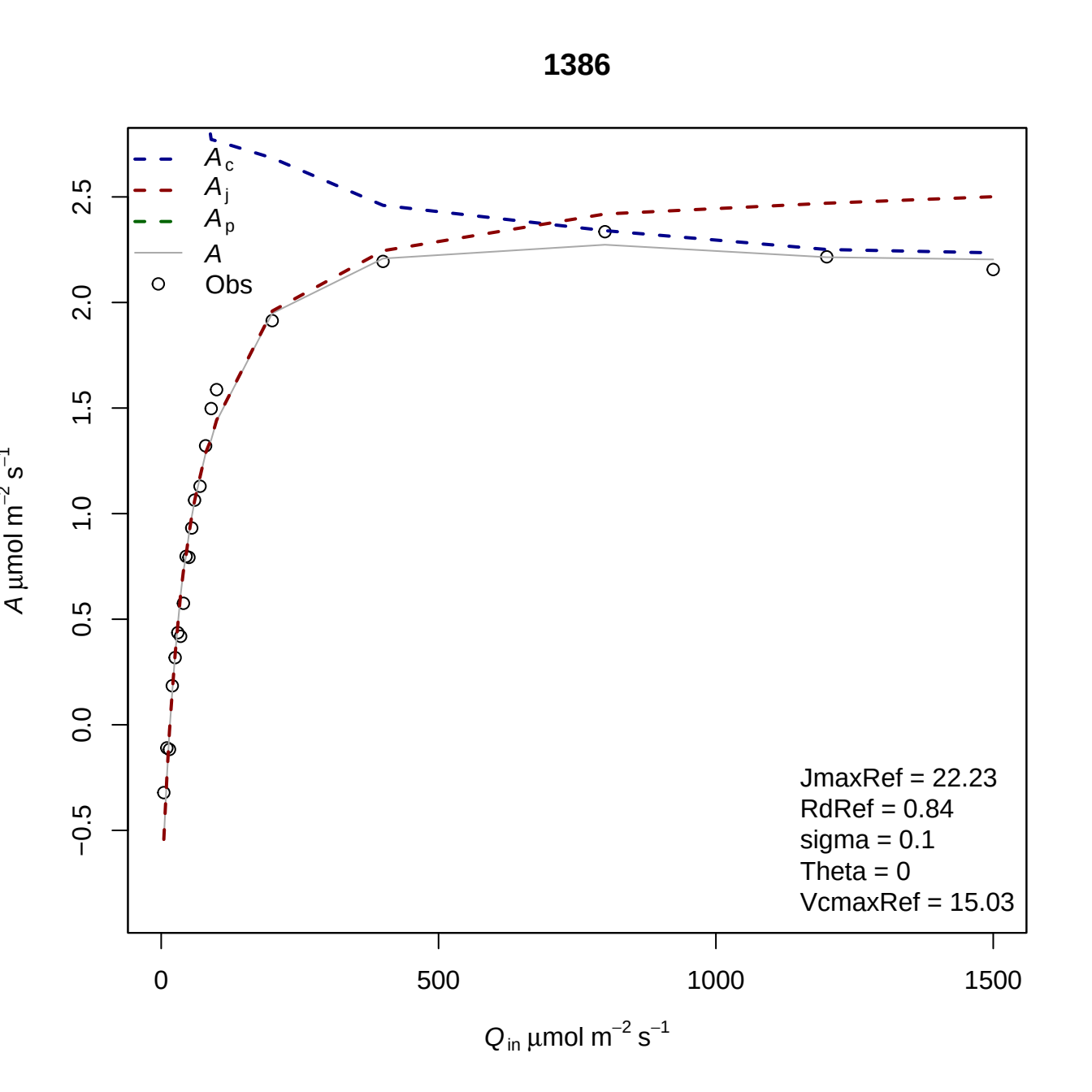

1381
A_c
A_j
A_p
A
Obs
JmaxRef = 21.7
RdRef = 0.55
sigma = 0.07
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

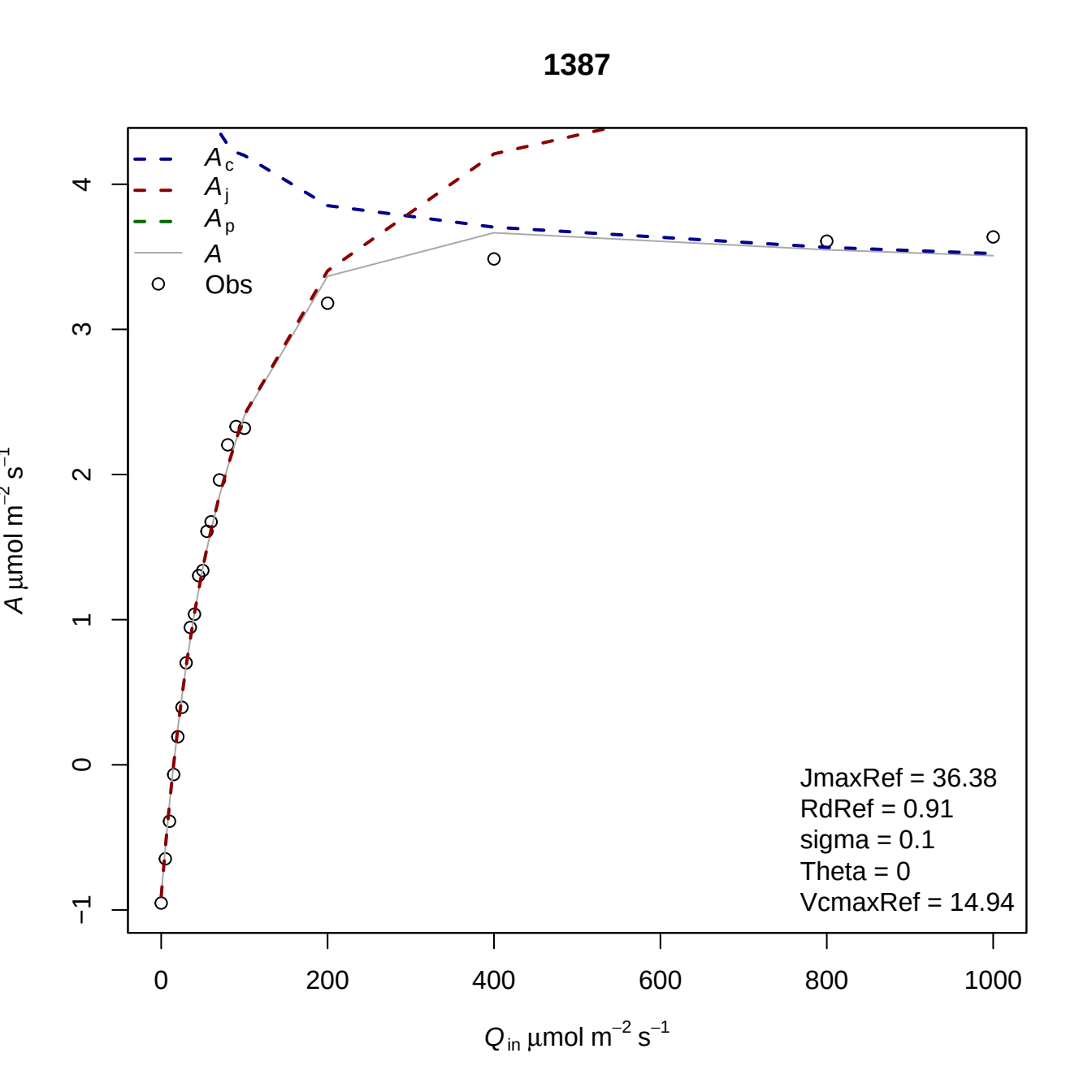

1382
A_c
A_j
A_p
A
Obs
JmaxRef = 39.5
RdRef = 0.2
sigma = 0.1
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

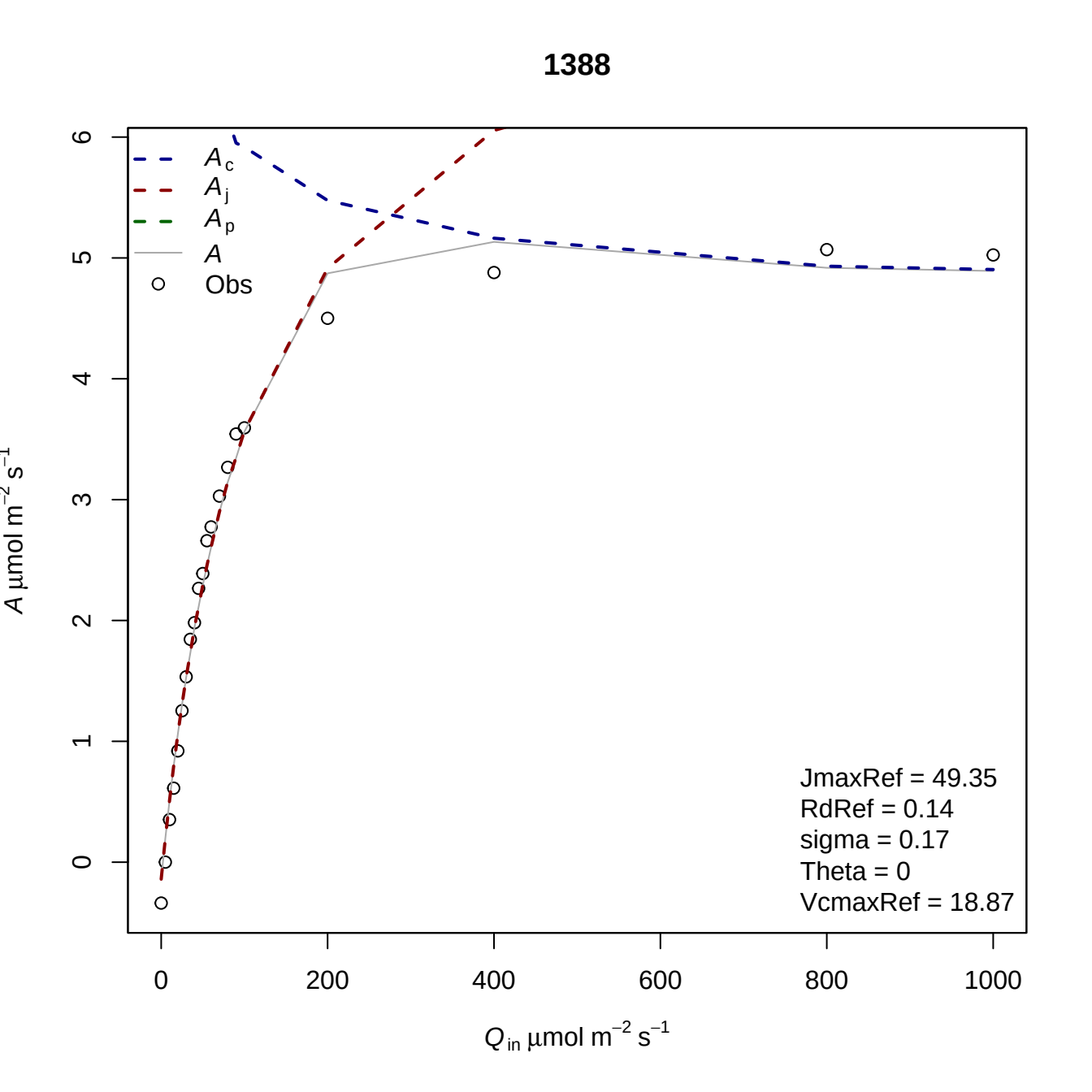

1383
A_c
A_j
A_p
A
Obs
JmaxRef = 34.5
RdRef = 0.71
sigma = 0.08
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

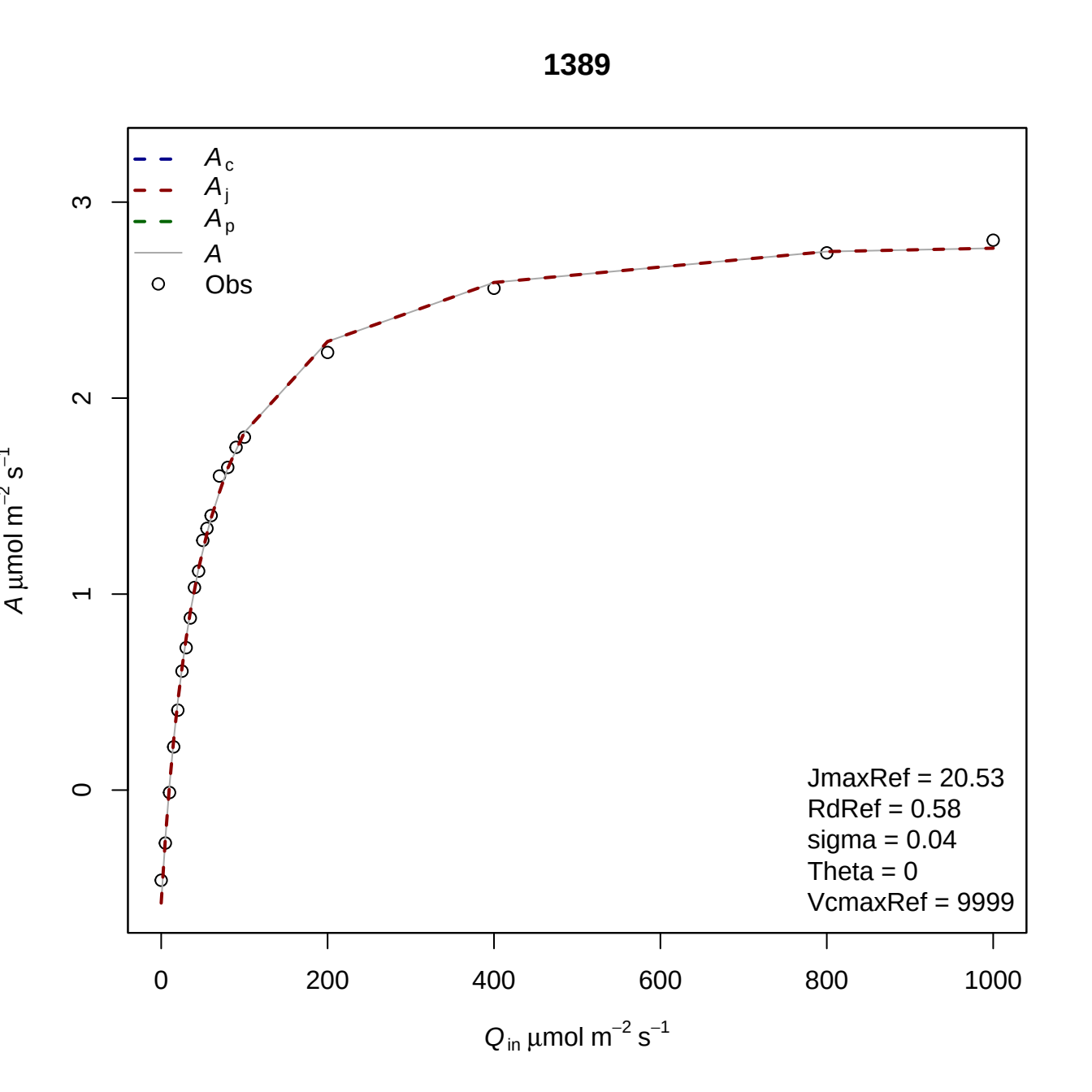

1384
A_c
A_j
A_p
A
Obs
JmaxRef = 30.15
RdRef = 0.2
sigma = 0.07
Theta = 0
VcmaxRef = 17.42
0
1
2
3
4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

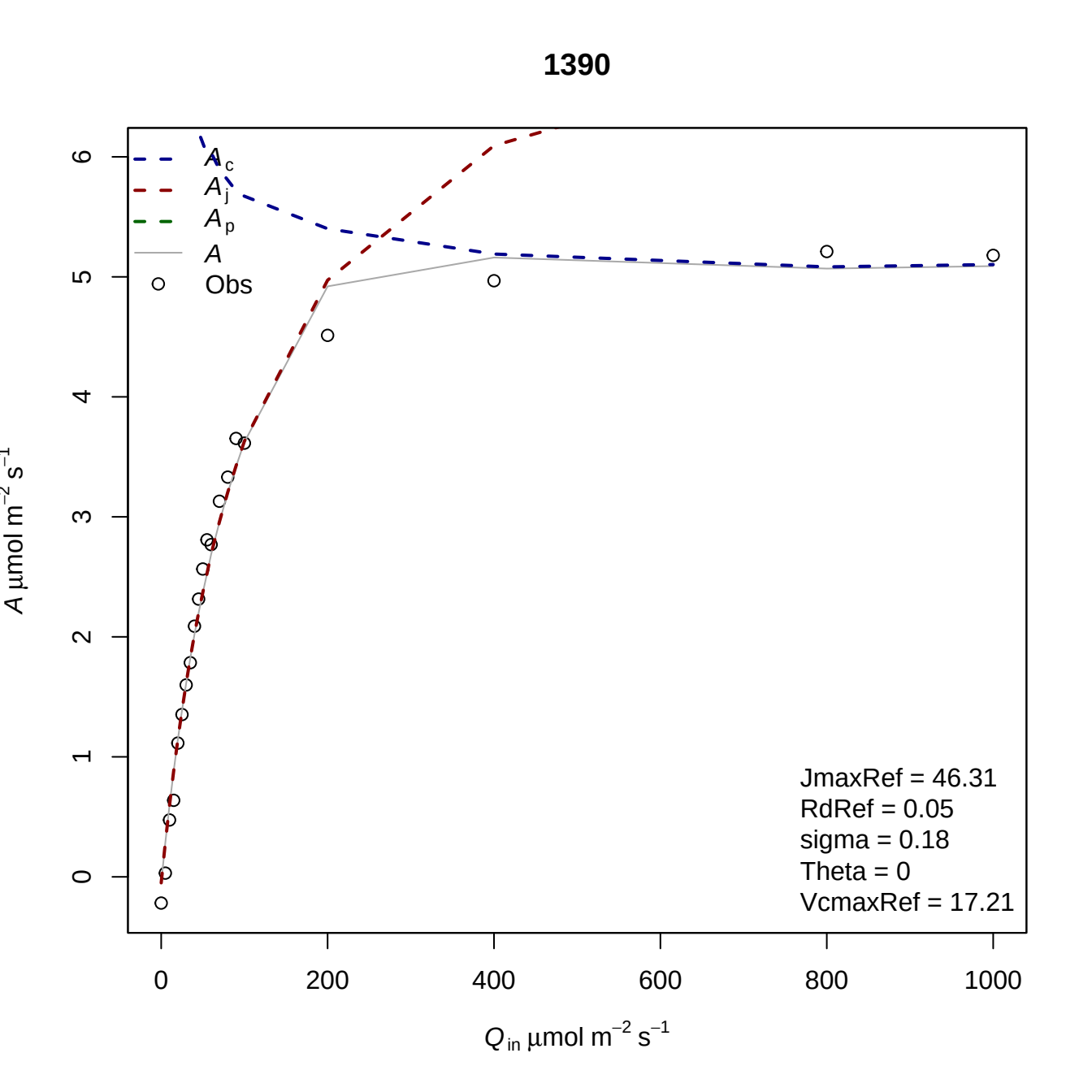

1385
A_c
A_j
A_p
A
Obs
JmaxRef = 58.19
RdRef = 0.15
sigma = 0.15
Theta = 0
VcmaxRef = 18.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

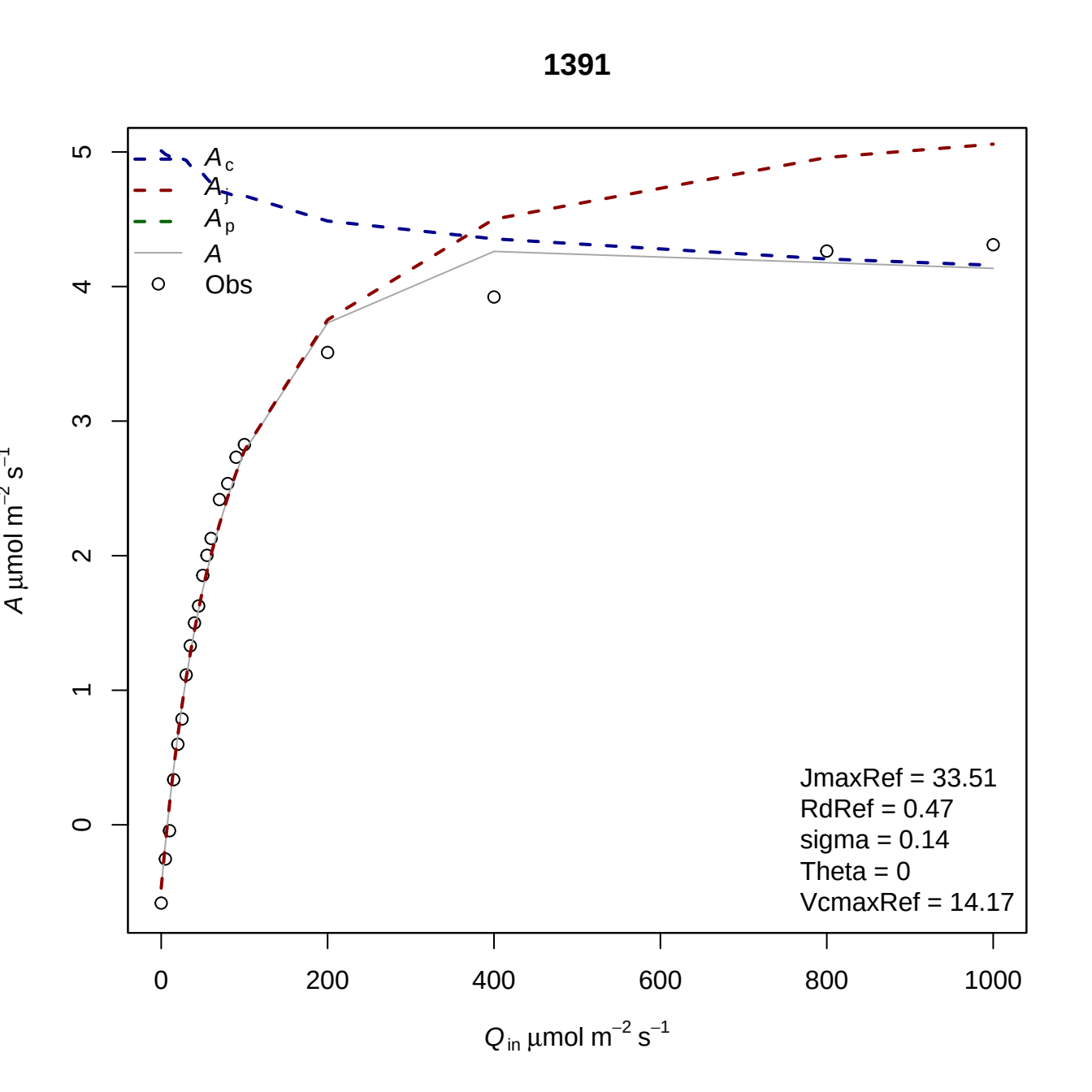

1386
A_c
A_j
A_p
A
Obs
JmaxRef = 22.23
RdRef = 0.84
sigma = 0.1
Theta = 0
VcmaxRef = 15.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

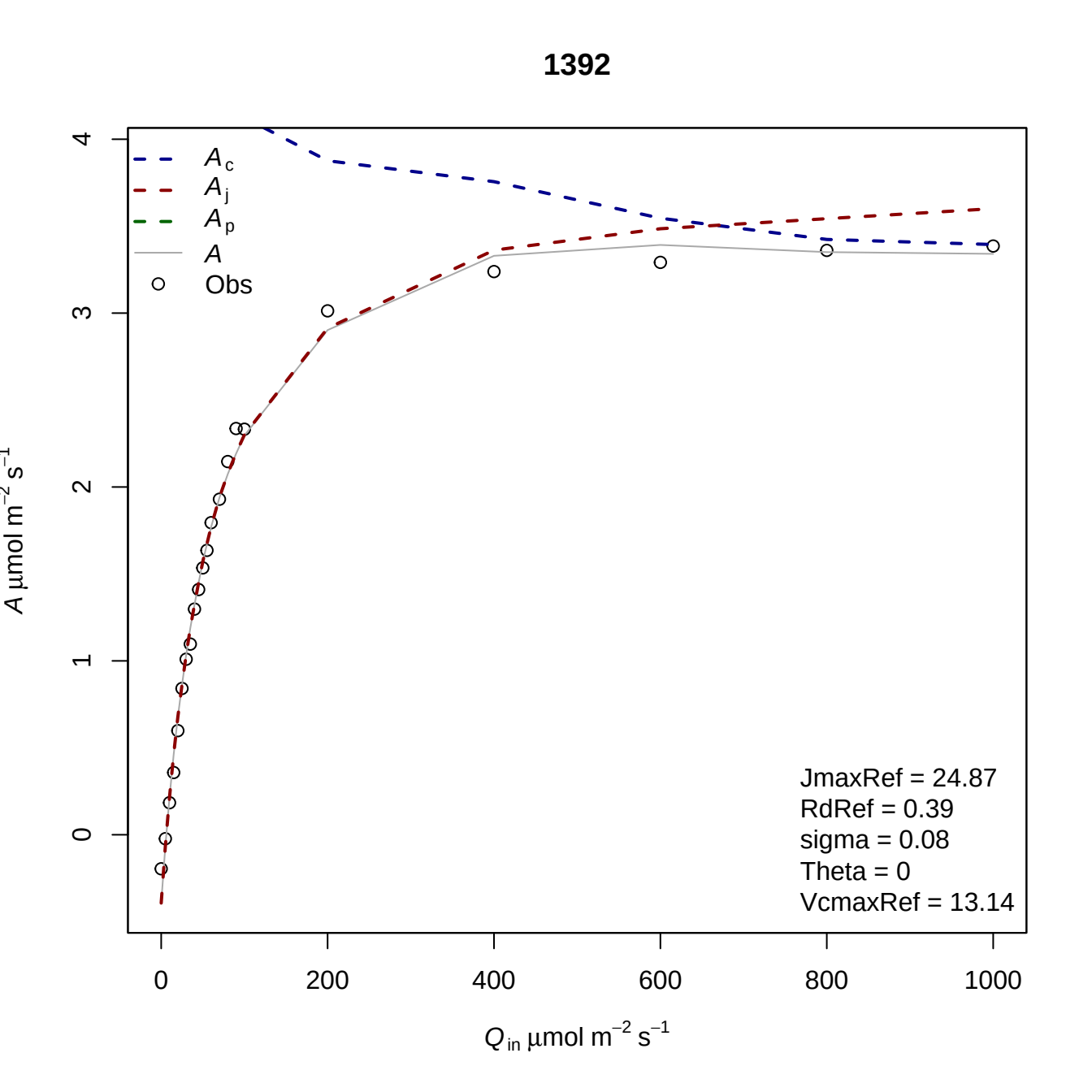

1387
A_c
A_j
A_p
A
Obs
JmaxRef = 36.38
RdRef = 0.91
sigma = 0.1
Theta = 0
VcmaxRef = 14.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

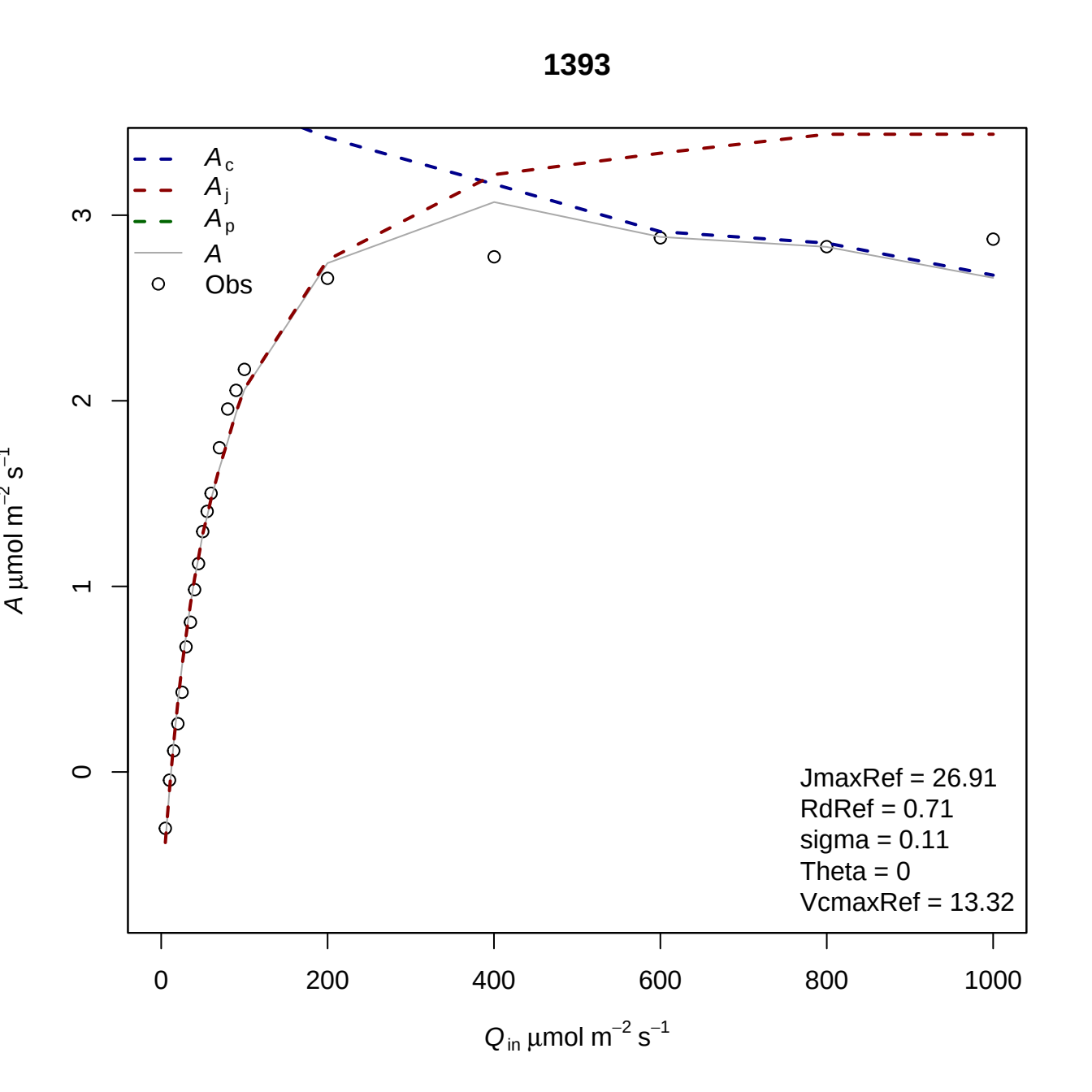

1388
A_c
A_j
A_p
A
Obs
JmaxRef = 49.35
RdRef = 0.14
sigma = 0.17
Theta = 0
VcmaxRef = 18.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

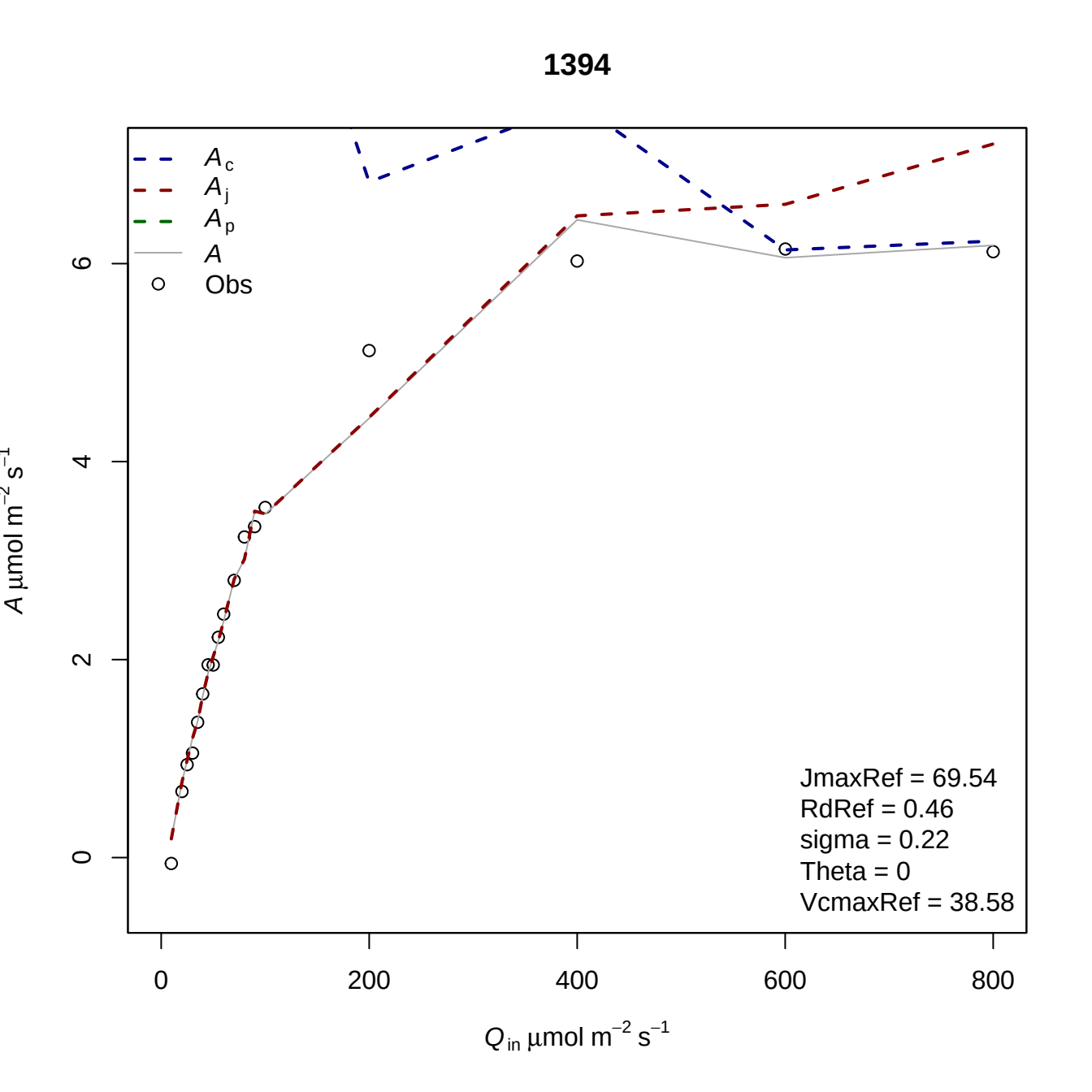

1389
A_c
A_j
A_p
A
Obs
JmaxRef = 20.53
RdRef = 0.58
sigma = 0.04
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

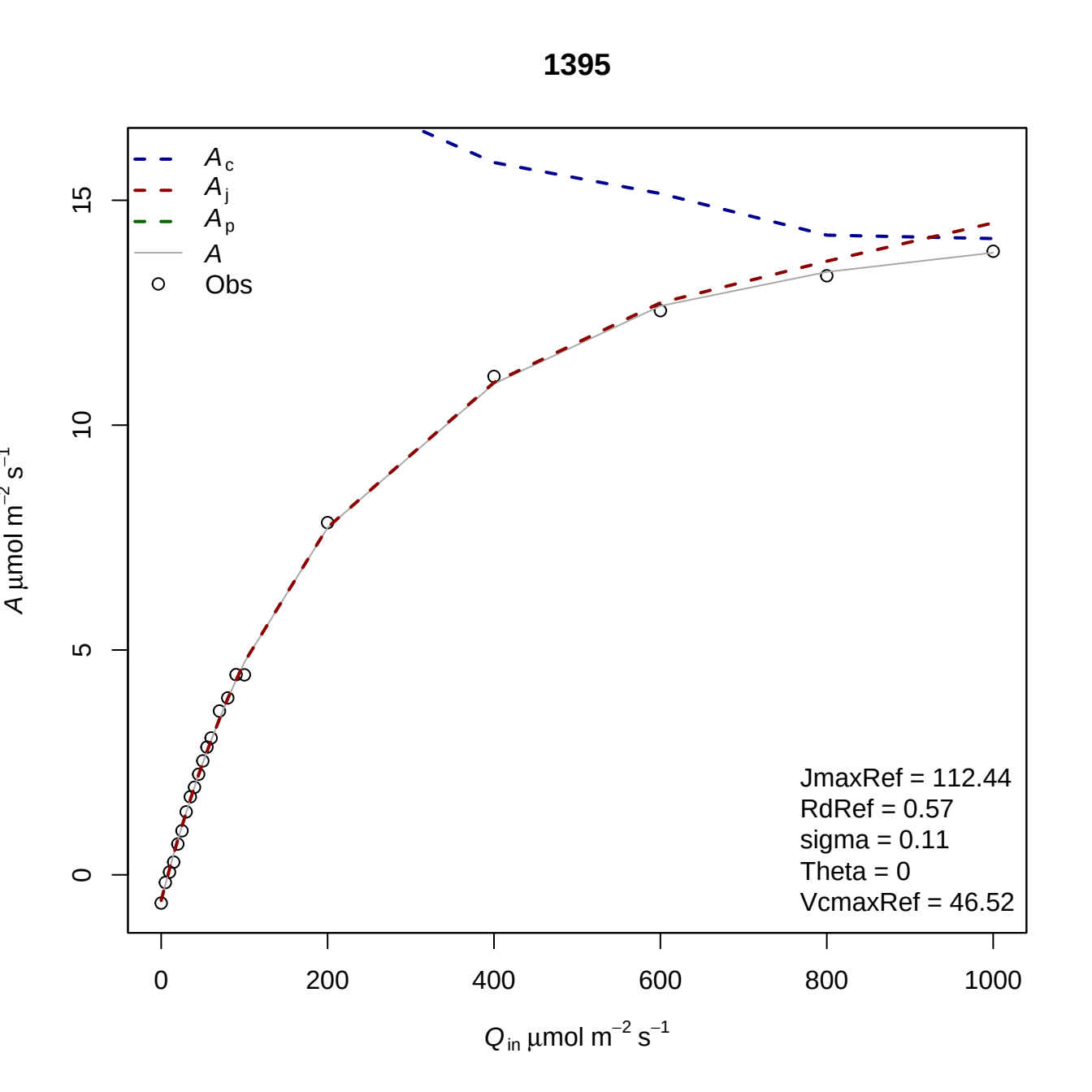

1390
A_c
A_j
A_p
A
Obs
JmaxRef = 46.31
RdRef = 0.05
sigma = 0.18
Theta = 0
VcmaxRef = 17.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

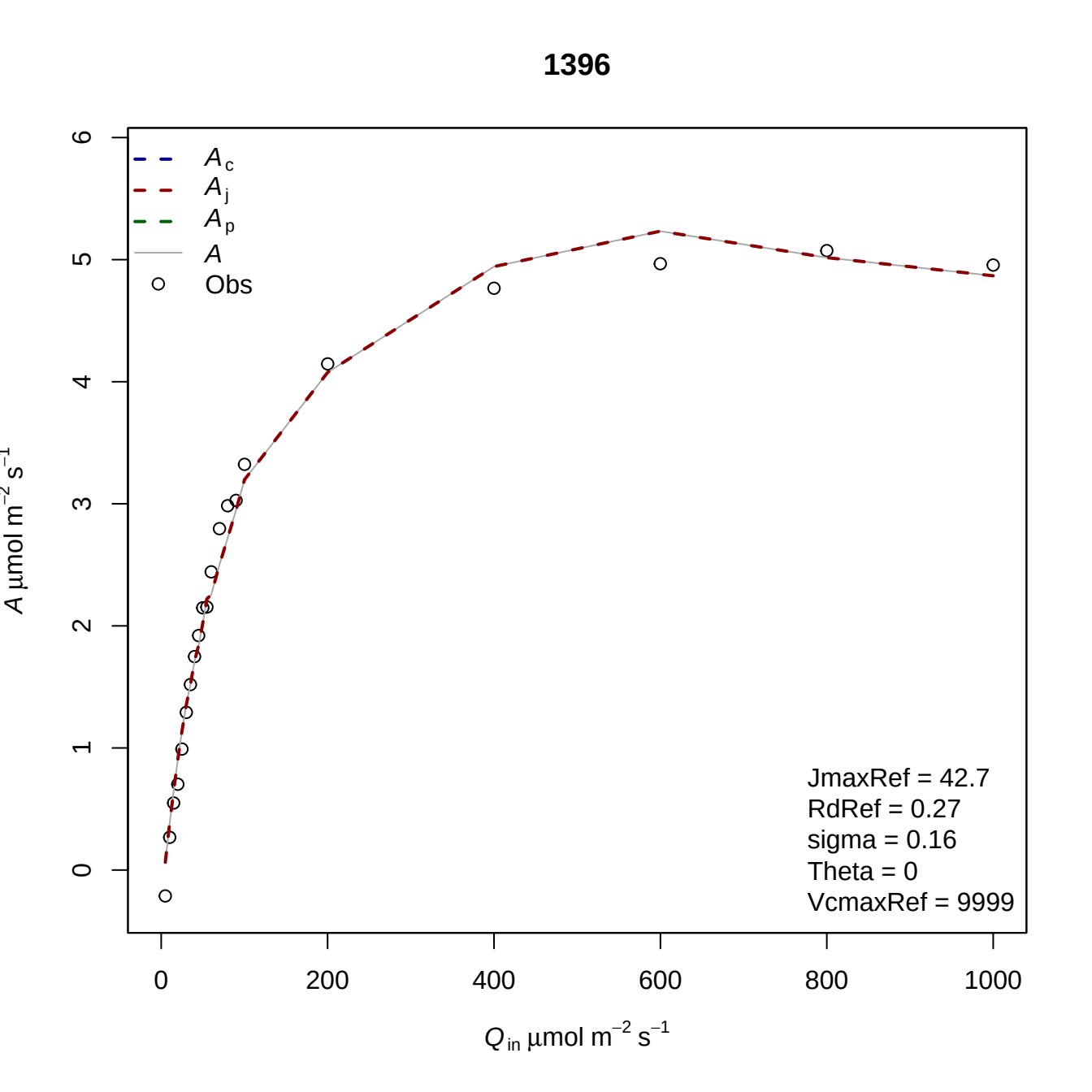

1391
A_c
A_j
A_p
A
Obs
JmaxRef = 33.51
RdRef = 0.47
sigma = 0.14
Theta = 0
VcmaxRef = 14.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000

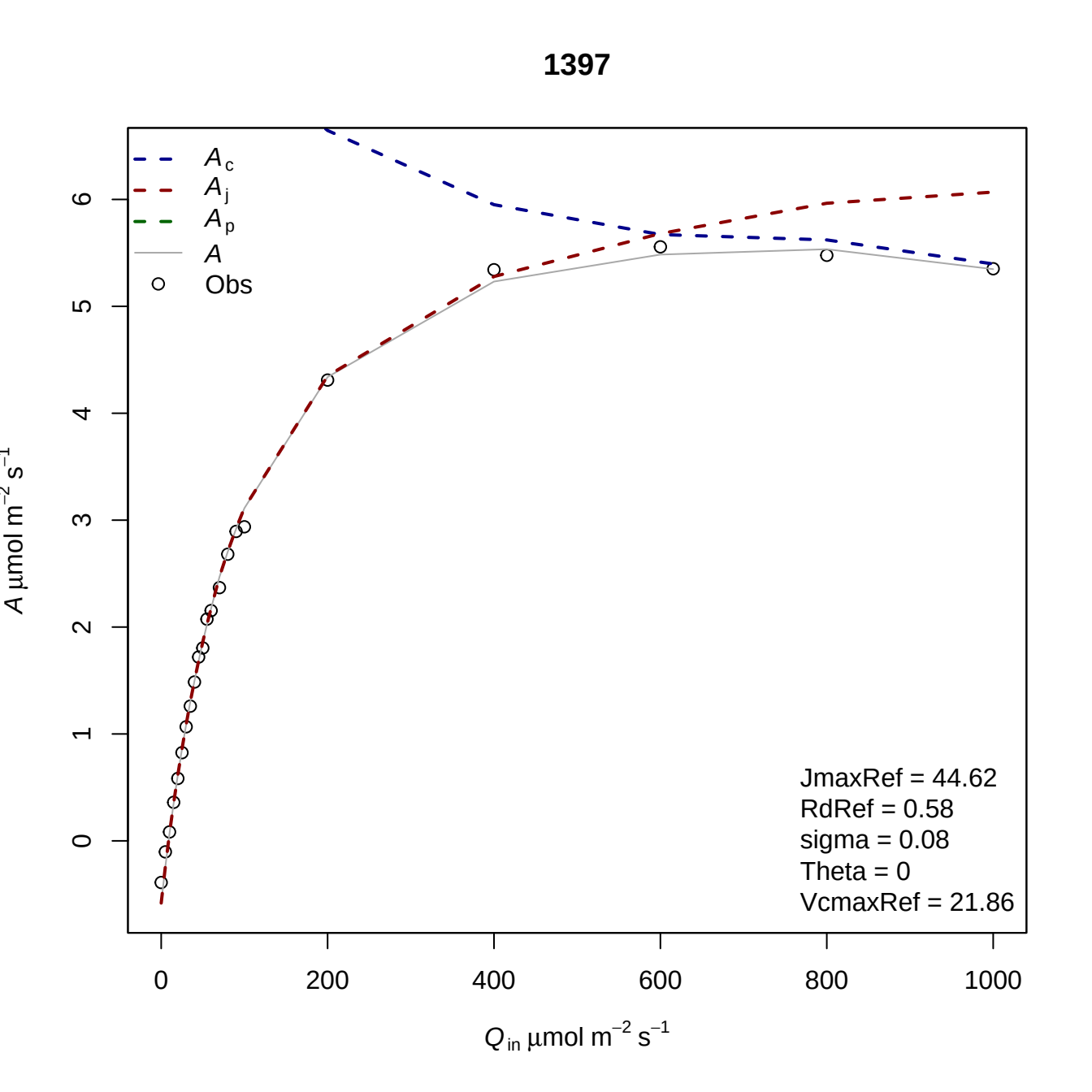

1392
A_c
A_j
A_p
A
Obs
JmaxRef = 24.87
RdRef = 0.39
sigma = 0.08
Theta = 0
VcmaxRef = 13.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
200
400
600
800
1000

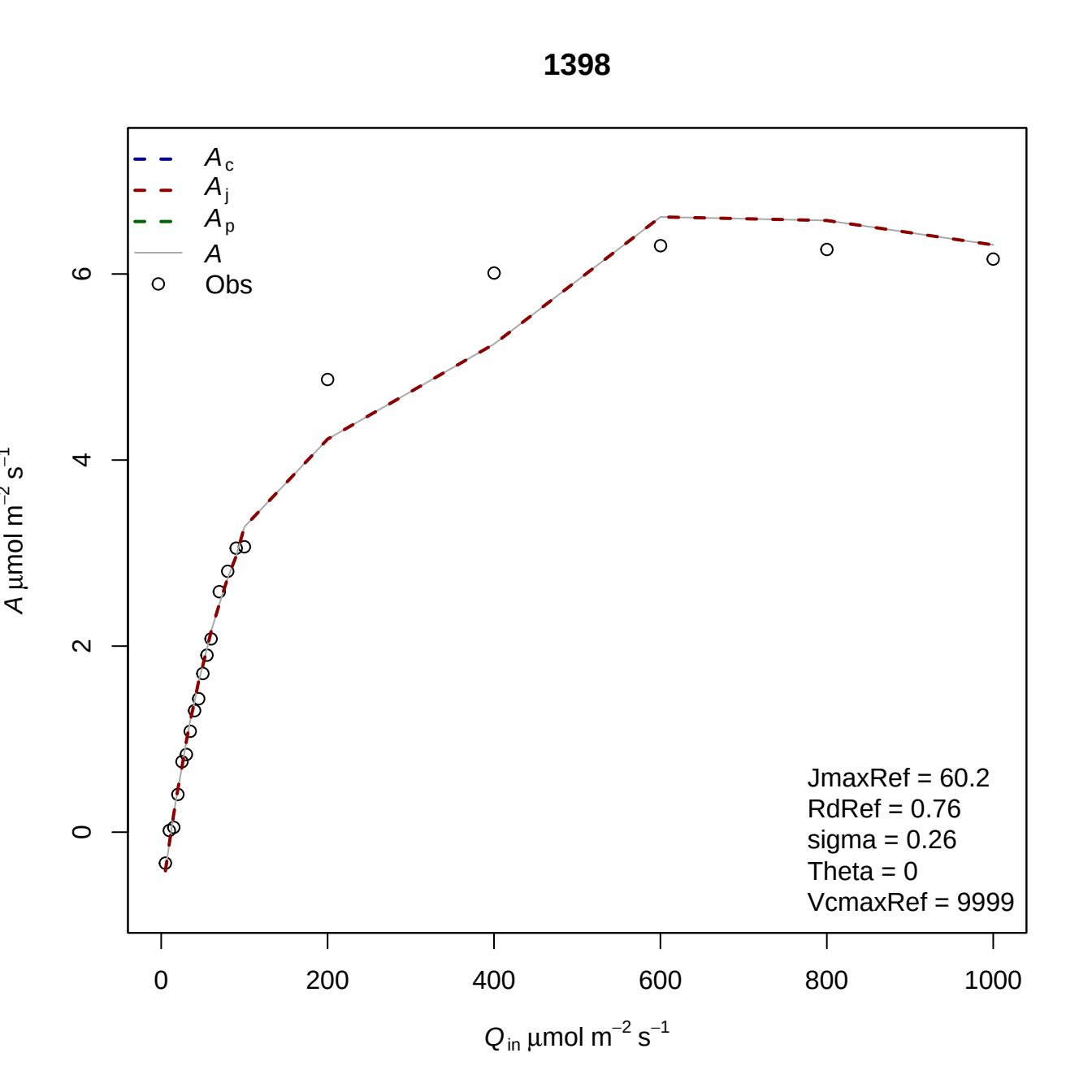

1393
A_c
A_j
A_p
A
Obs
JmaxRef = 26.91
RdRef = 0.71
sigma = 0.11
Theta = 0
VcmaxRef = 13.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

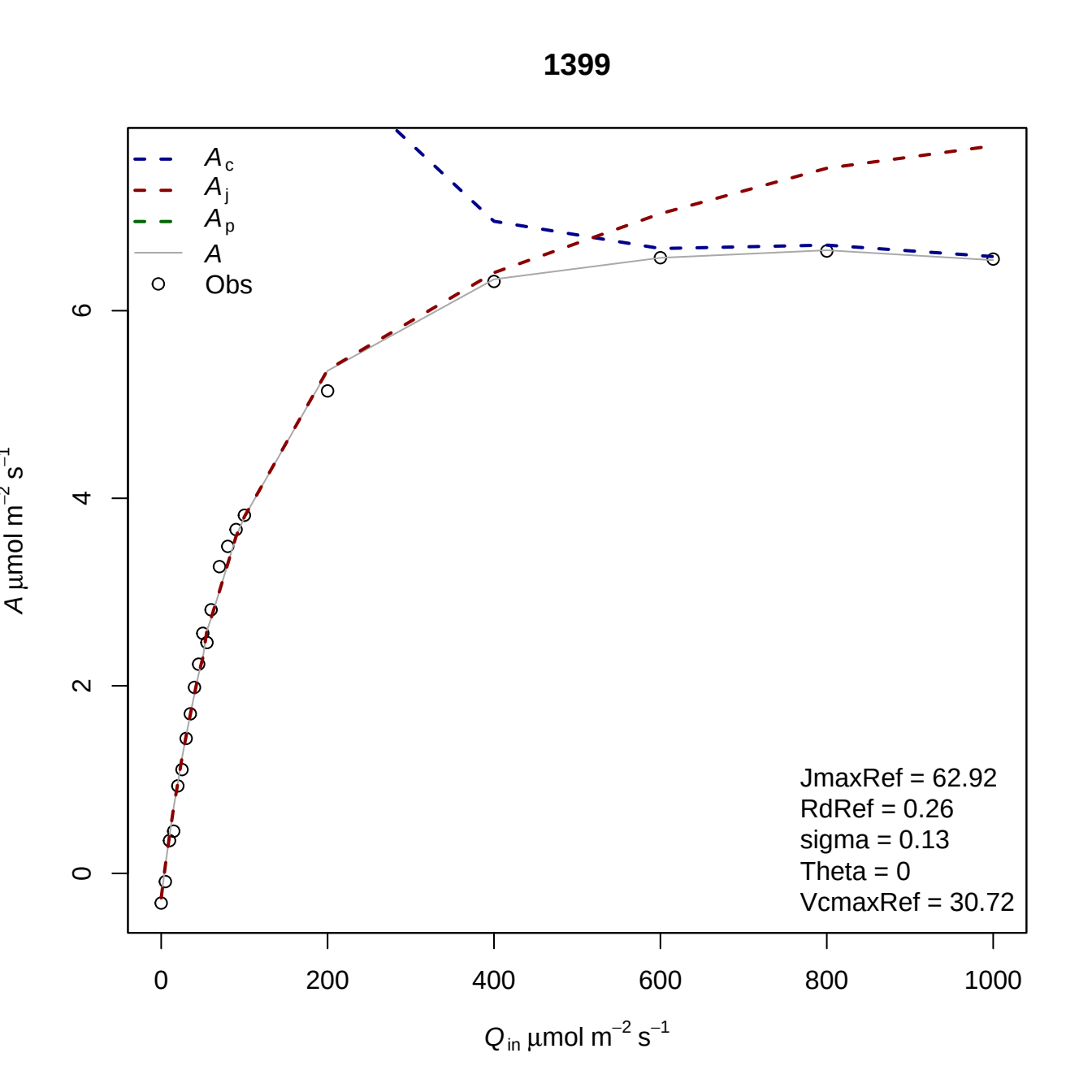

1394
A_c
A_j
A_p
A
Obs
JmaxRef = 69.54
RdRef = 0.46
sigma = 0.22
Theta = 0
VcmaxRef = 38.58
0
2
4
6
A µmol m^{-2} s^{-1}
0
200
400
600
800
Q_{in} µmol m^{-2} s^{-1}

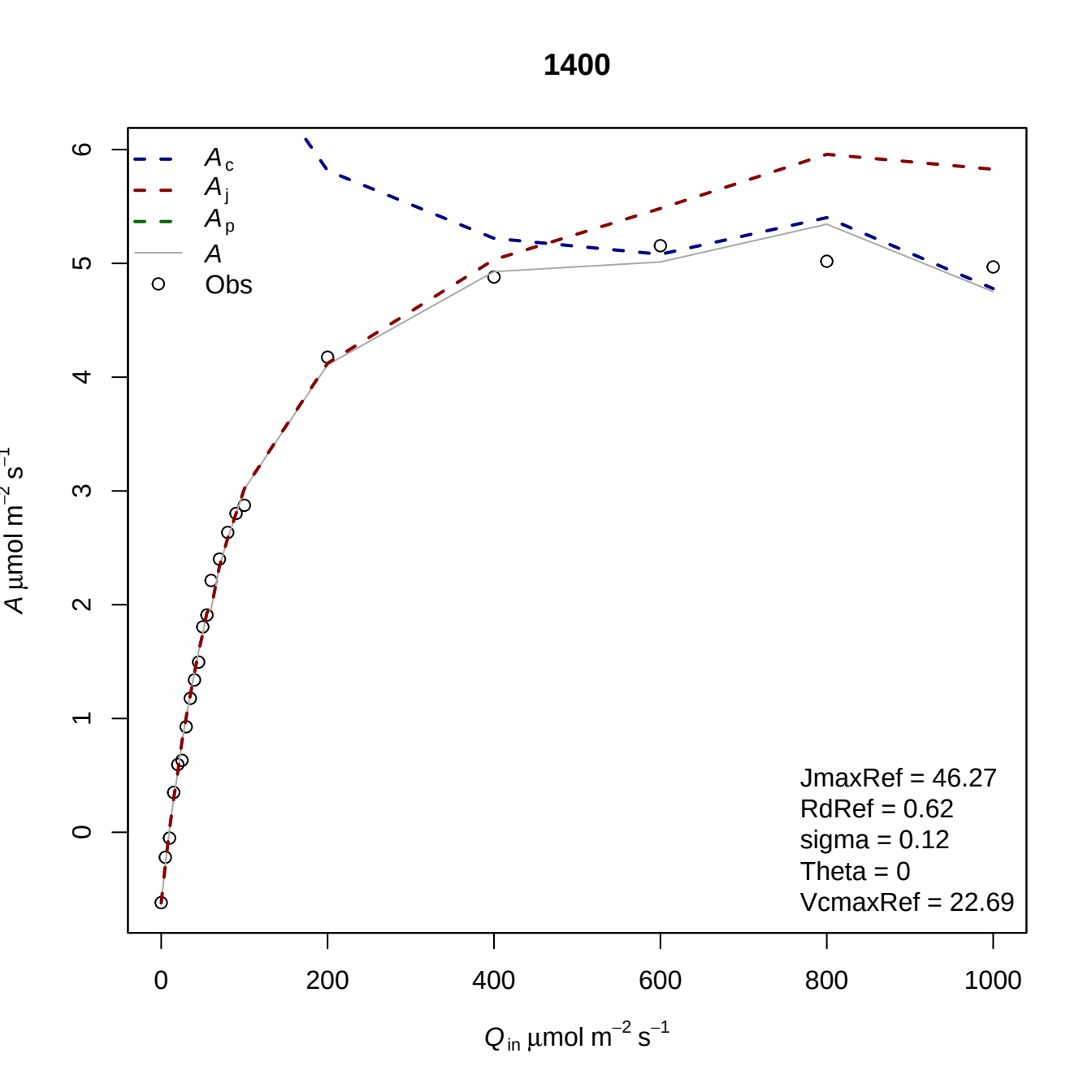

1395
A_c
A_j
A_p
A
Obs
JmaxRef = 112.44
RdRef = 0.57
sigma = 0.11
Theta = 0
VcmaxRef = 46.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000

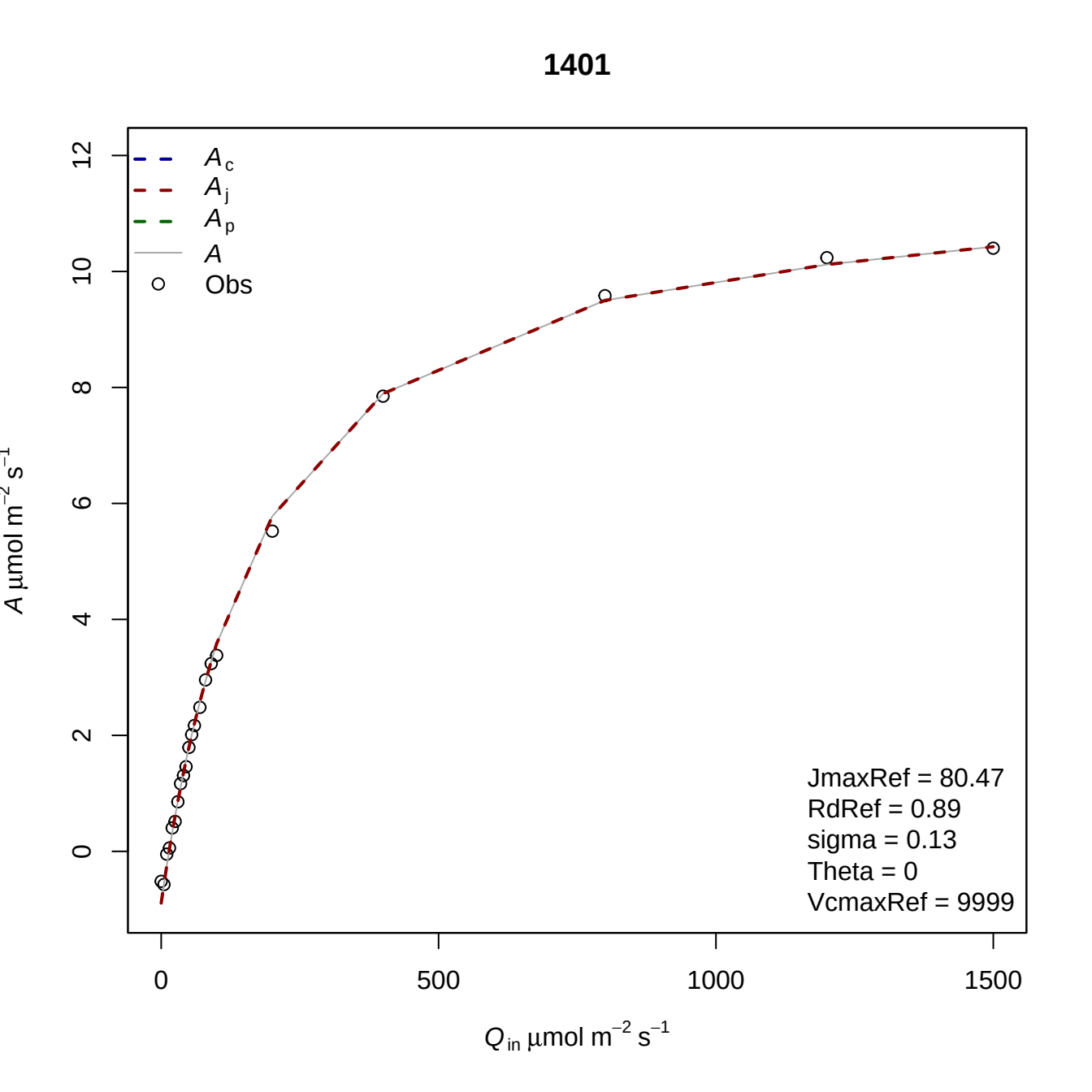

1396
A_c
A_j
A_p
A
Obs
JmaxRef = 42.7
RdRef = 0.27
sigma = 0.16
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

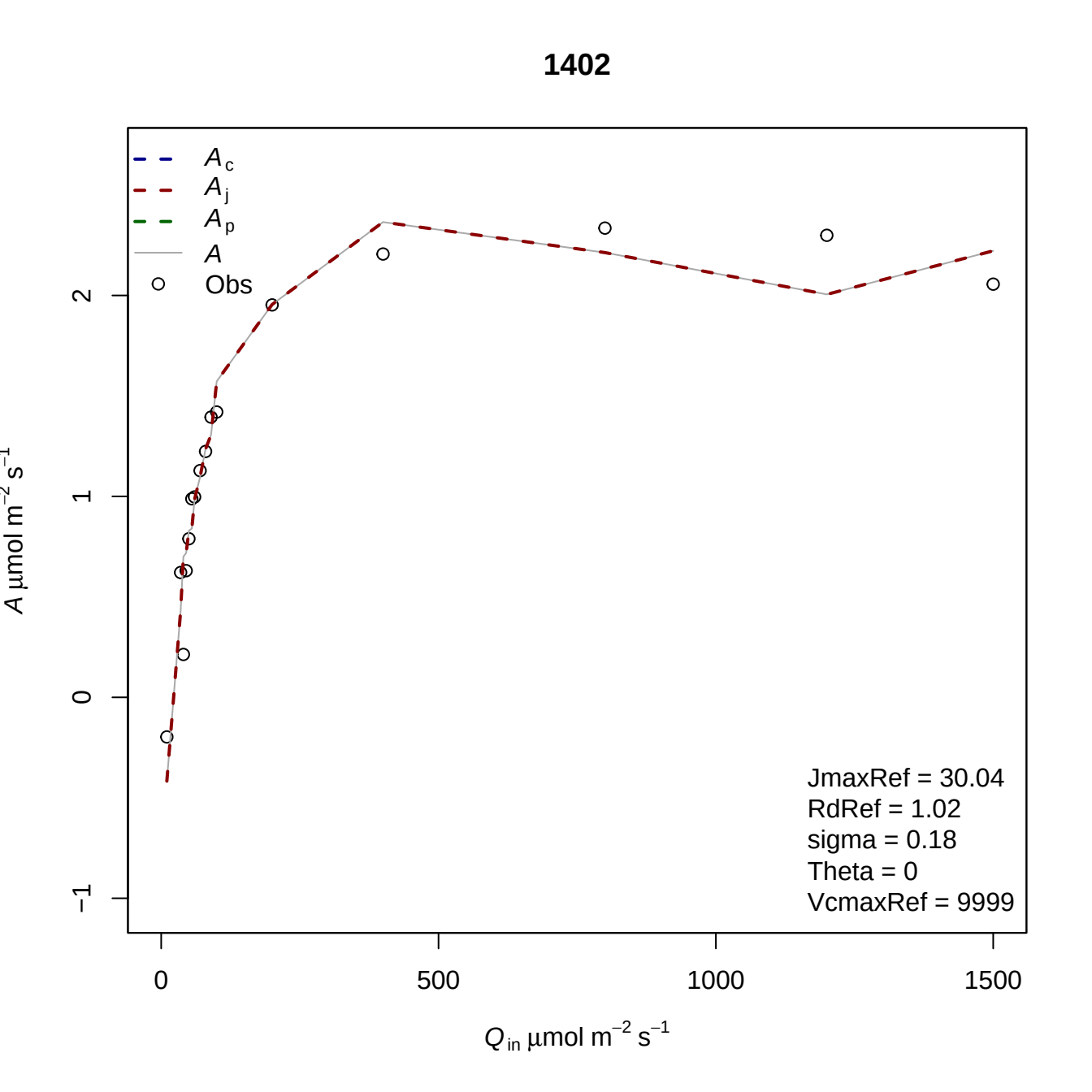

1397
A_c
A_j
A_p
A
Obs
JmaxRef = 44.62
RdRef = 0.58
sigma = 0.08
Theta = 0
VcmaxRef = 21.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

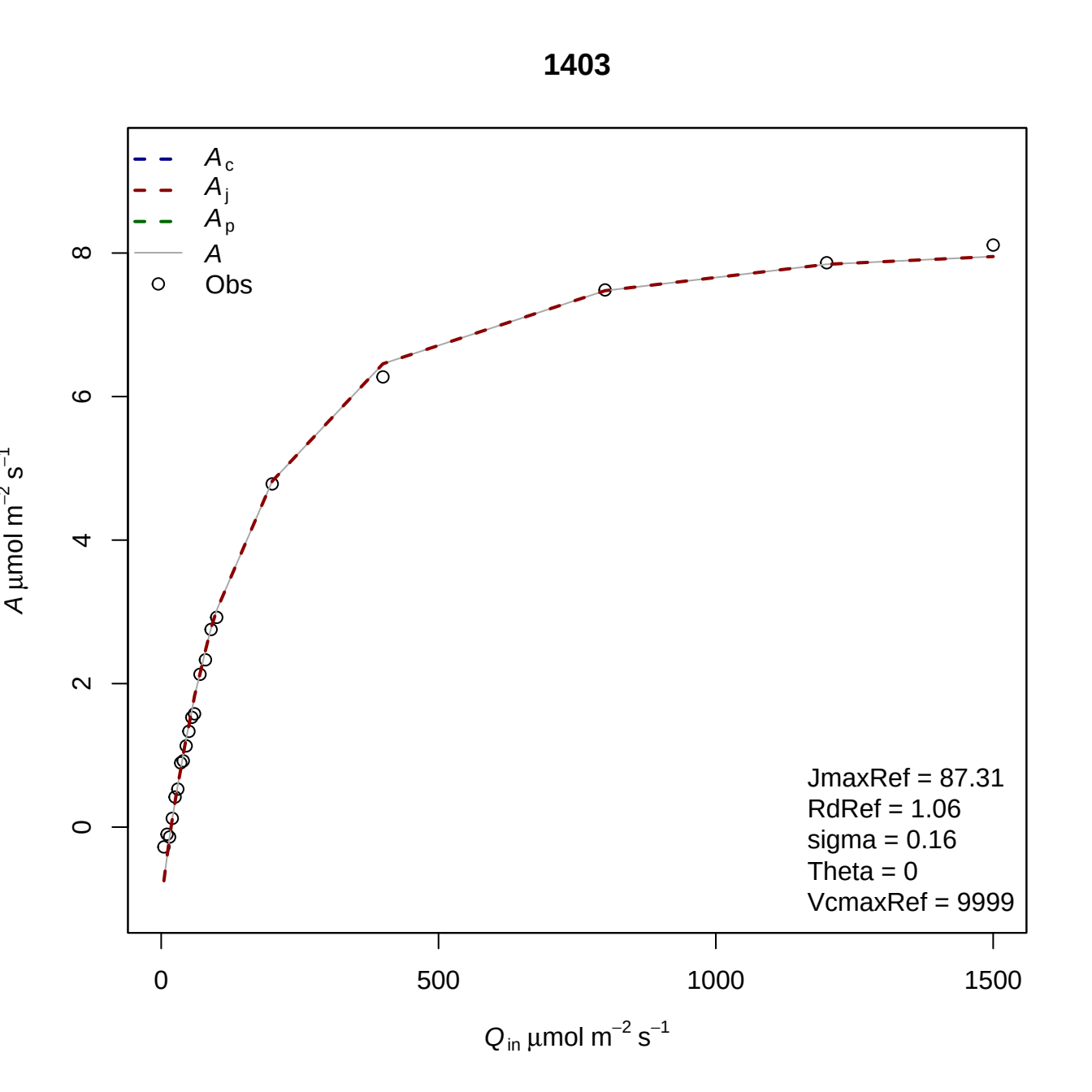

1398
A_c
A_j
A_p
A
Obs
JmaxRef = 60.2
RdRef = 0.76
sigma = 0.26
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

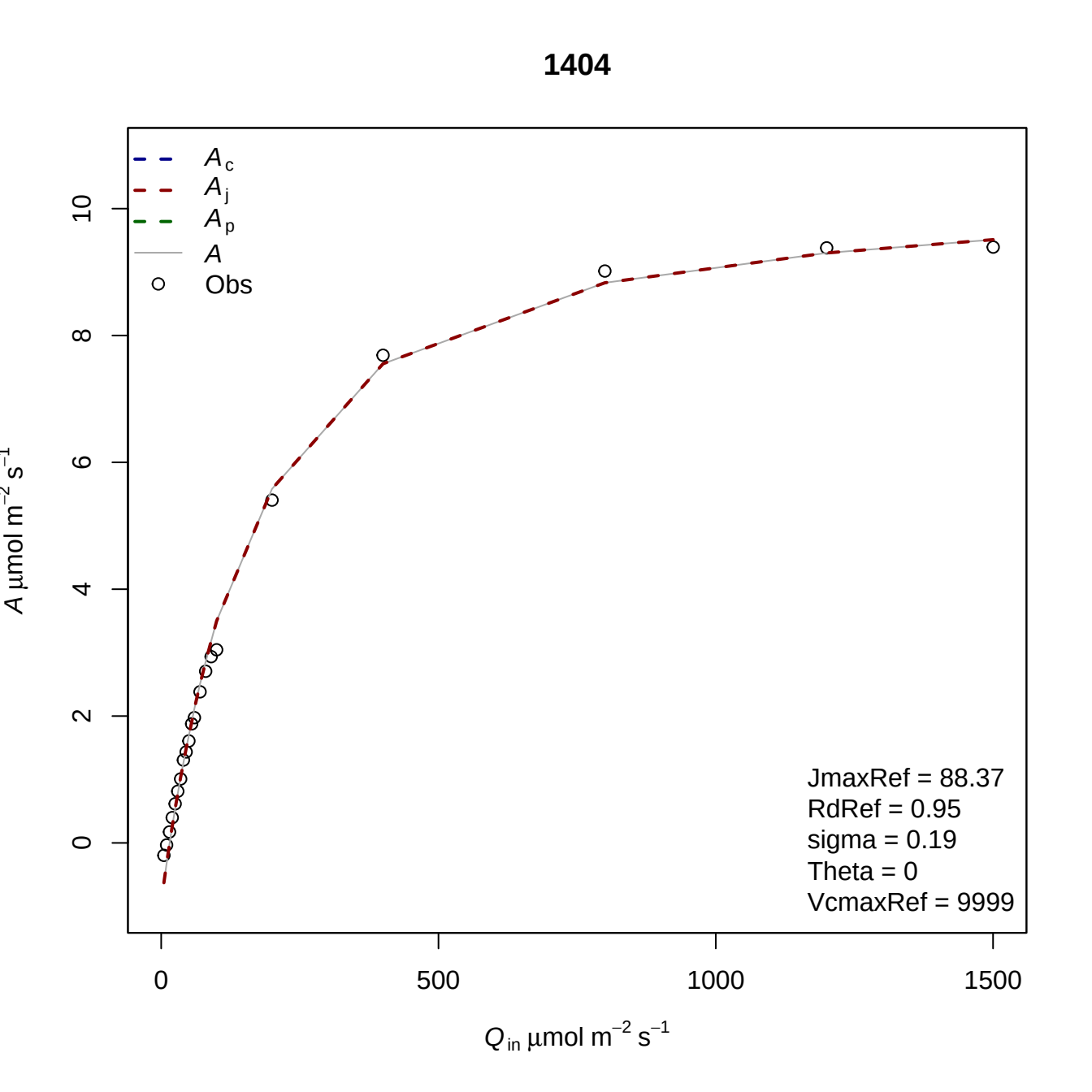

1399
A_c
A_j
A_p
A
Obs
JmaxRef = 62.92
RdRef = 0.26
sigma = 0.13
Theta = 0
VcmaxRef = 30.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

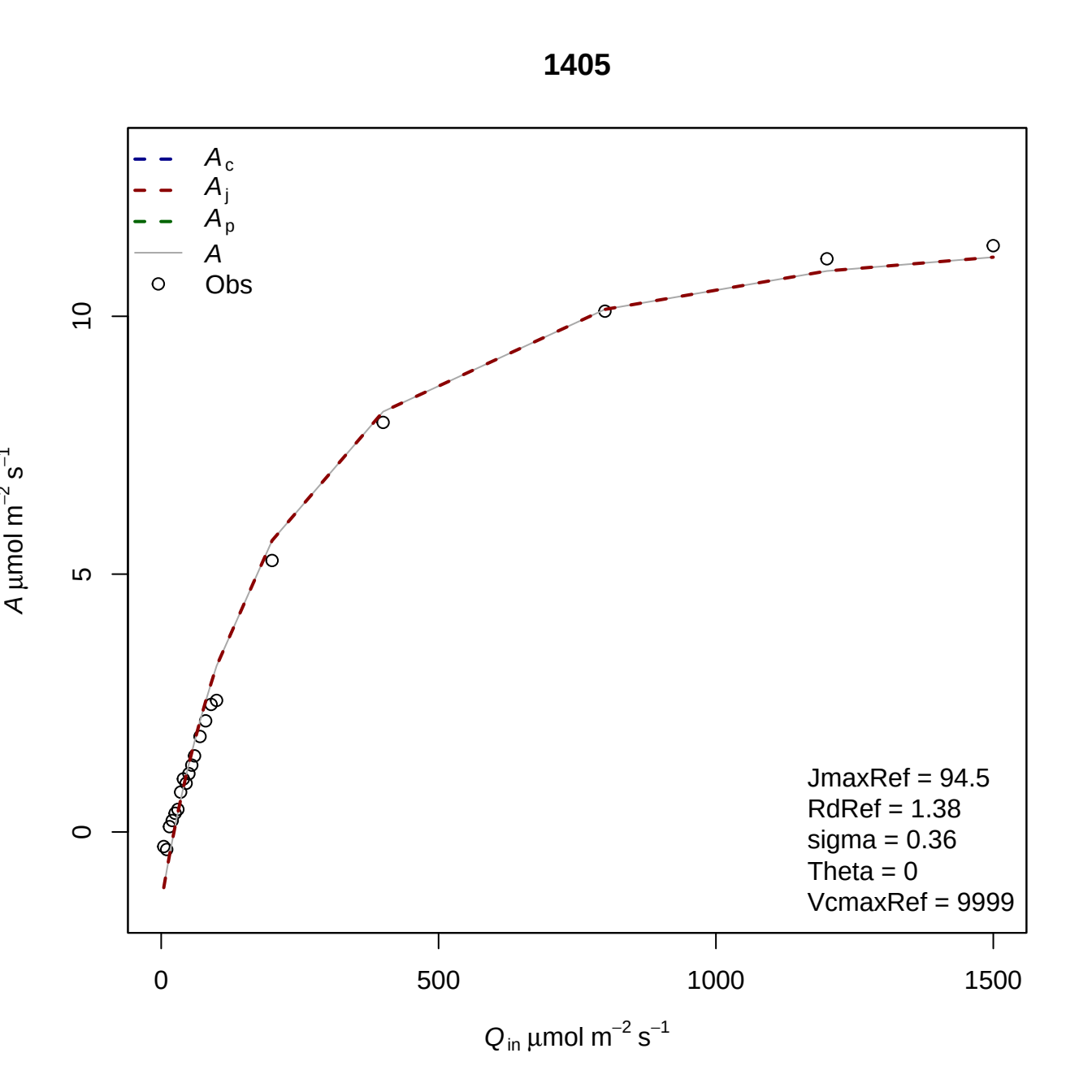

1400
A_c
A_j
A_p
A
Obs
JmaxRef = 46.27
RdRef = 0.62
sigma = 0.12
Theta = 0
VcmaxRef = 22.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

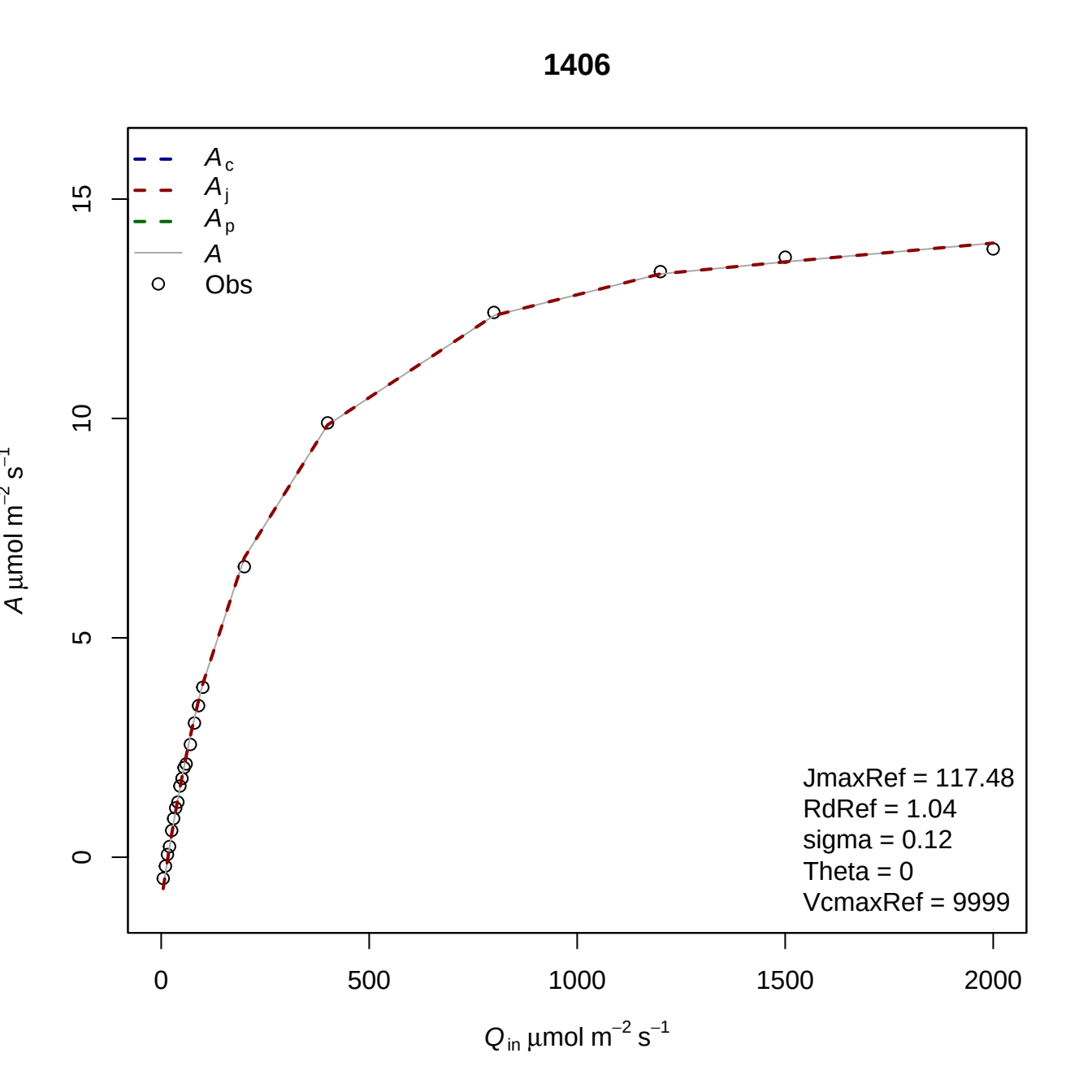

1401
A_c
A_j
A_p
A
Obs
JmaxRef = 80.47
RdRef = 0.89
sigma = 0.13
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

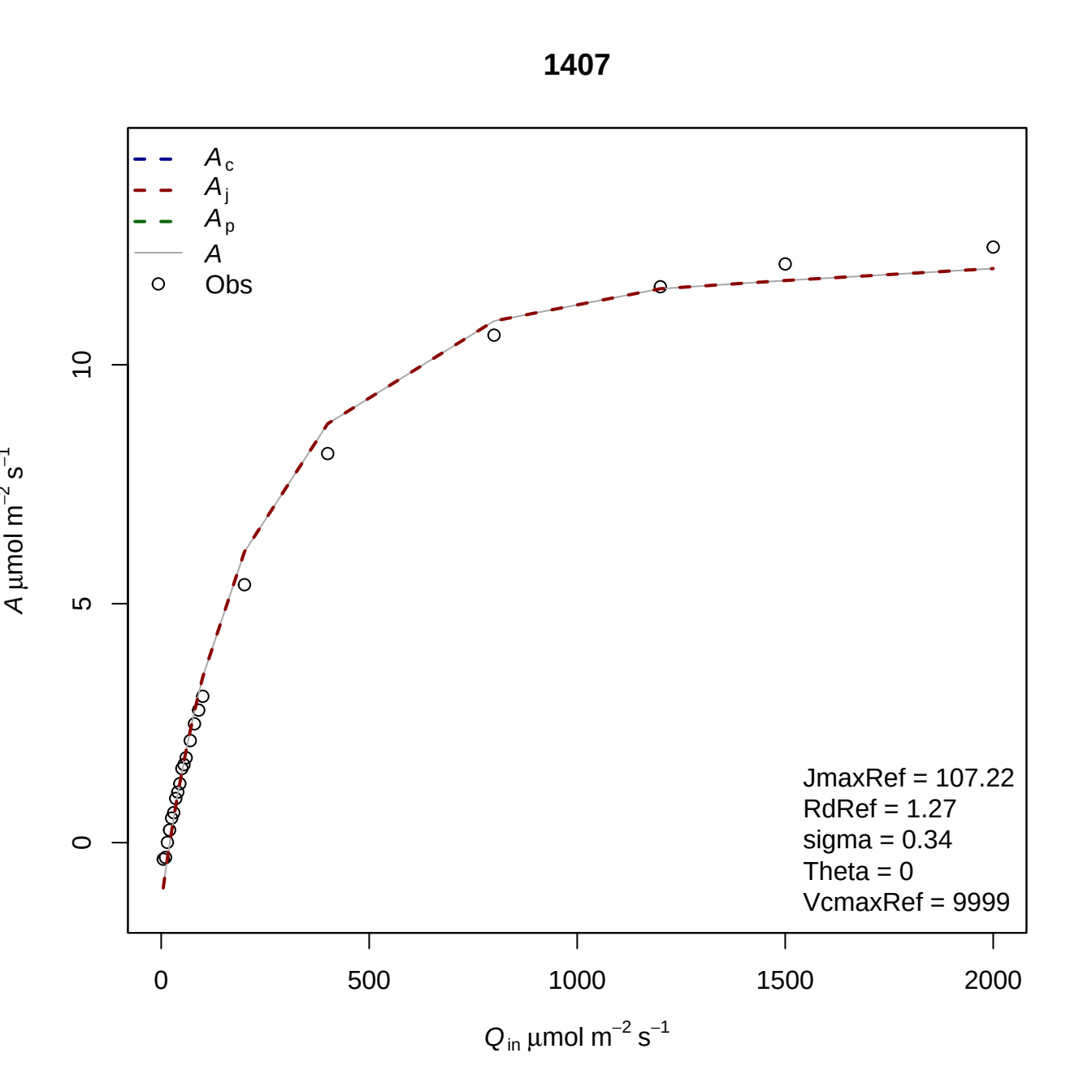

1402
A_c
A_j
A_p
A
Obs
JmaxRef = 30.04
RdRef = 1.02
sigma = 0.18
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

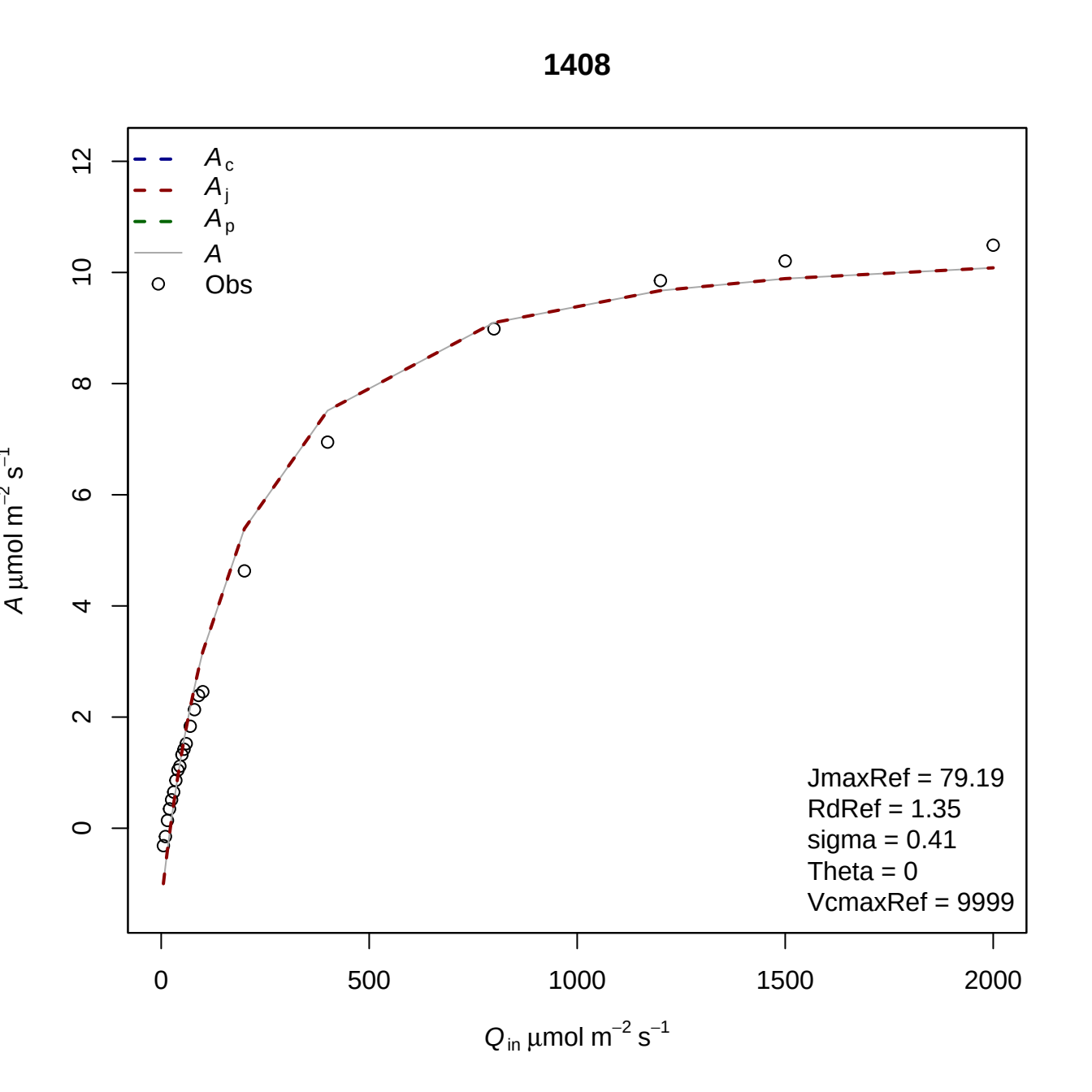

1403
A_c
A_j
A_p
A
Obs
JmaxRef = 87.31
RdRef = 1.06
sigma = 0.16
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

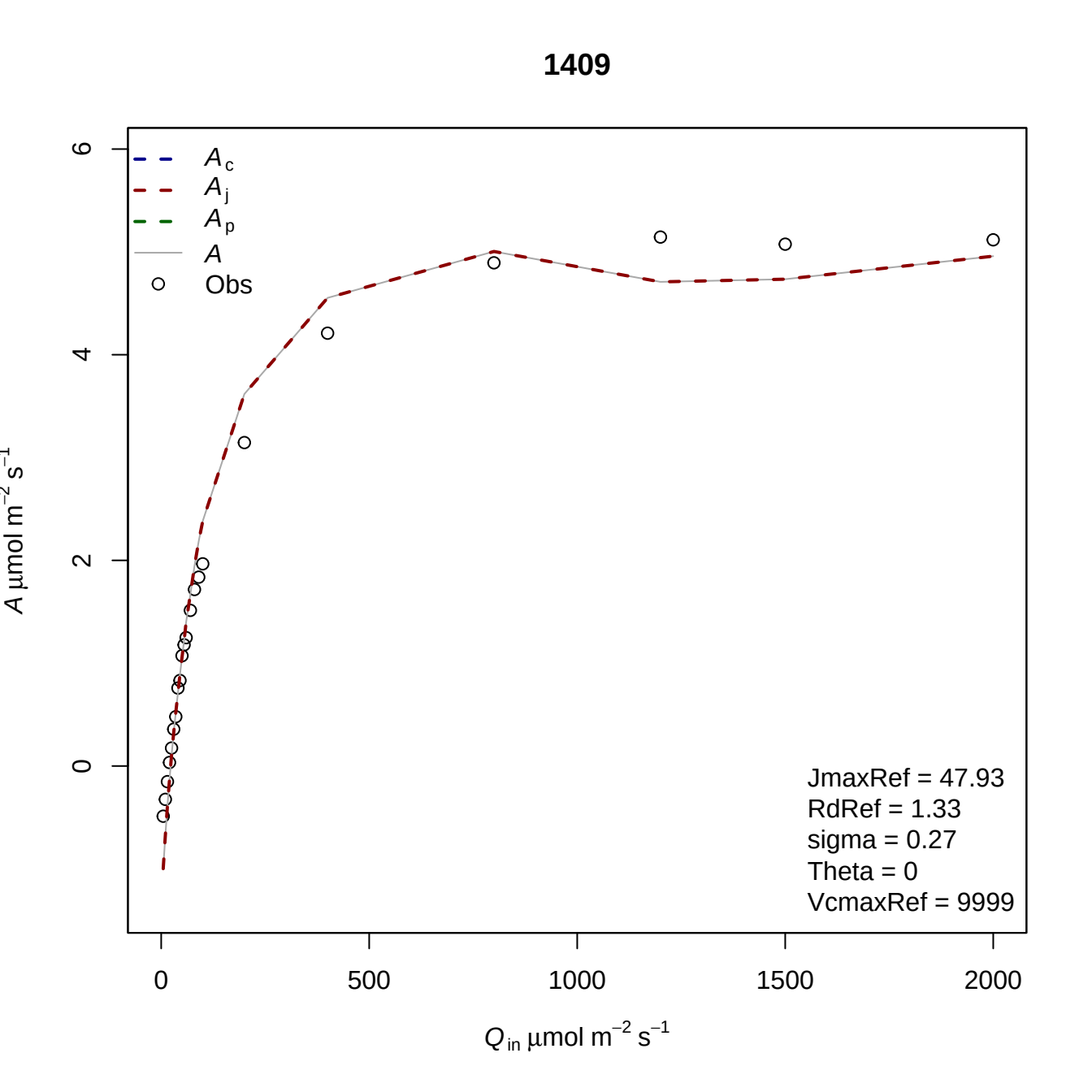

1404
A_c
A_j
A_p
A
Obs
JmaxRef = 88.37
RdRef = 0.95
sigma = 0.19
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

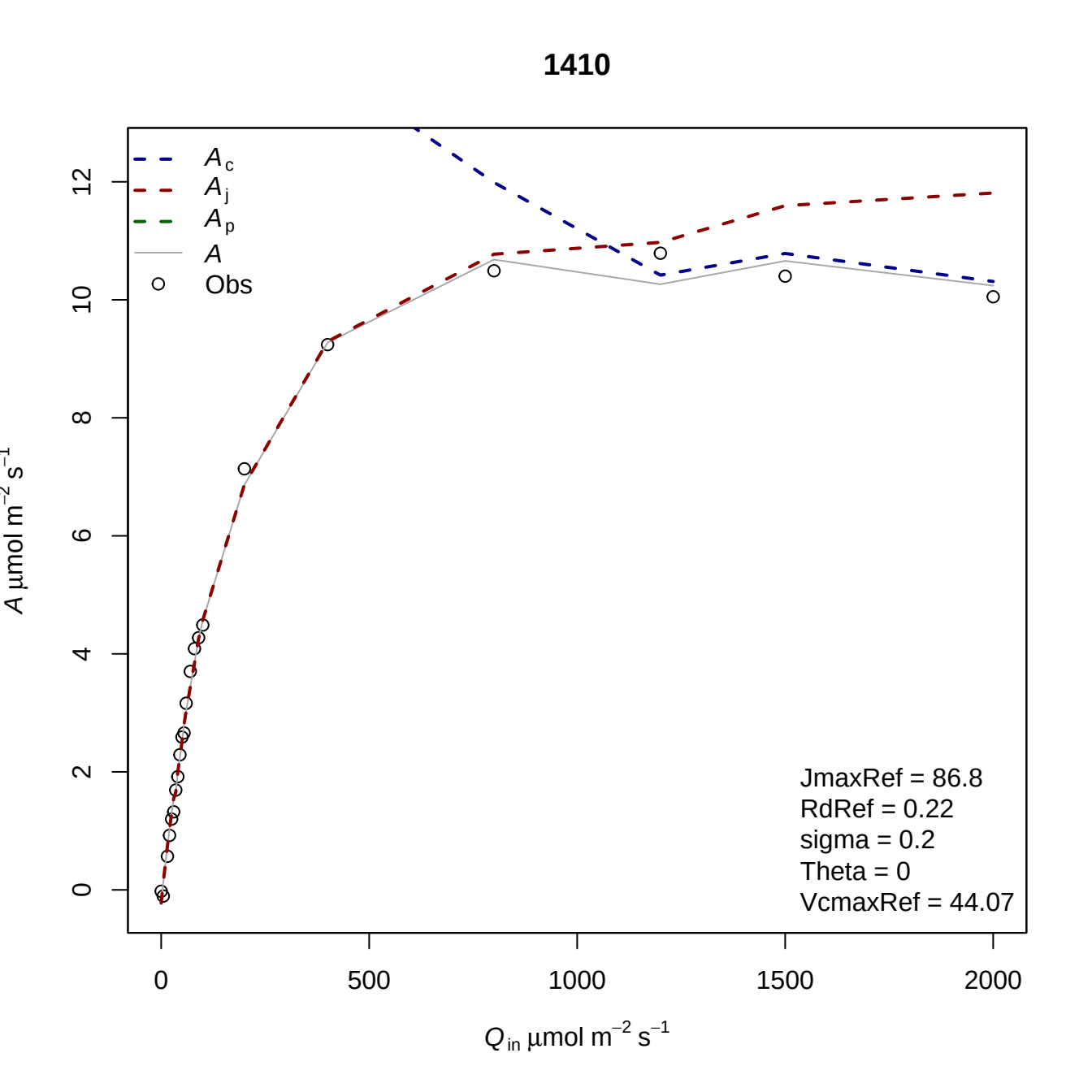

1405
A_c
A_j
A_p
A
Obs
JmaxRef = 94.5
RdRef = 1.38
sigma = 0.36
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

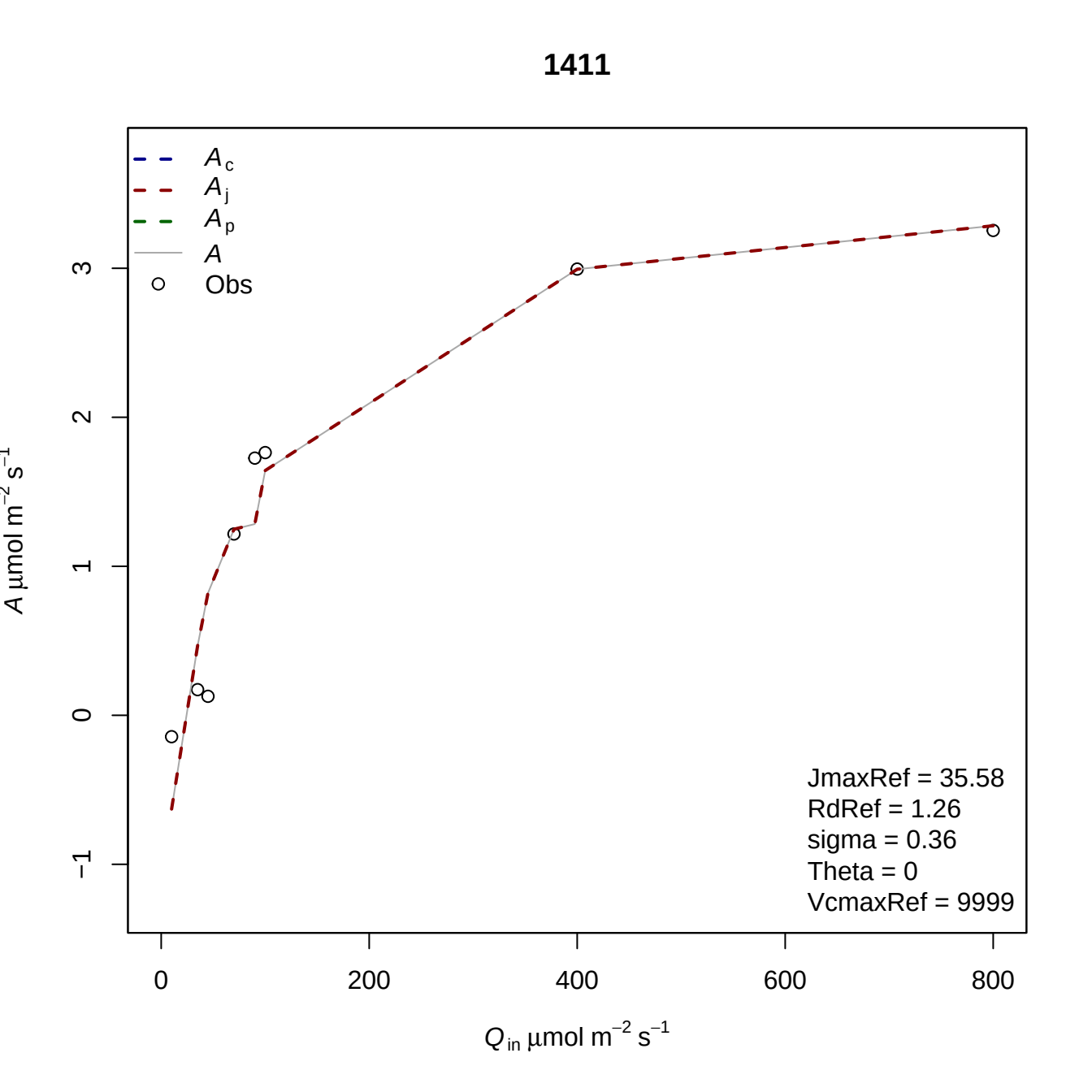

1406
A_c
A_j
A_p
A
Obs
JmaxRef = 117.48
RdRef = 1.04
sigma = 0.12
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

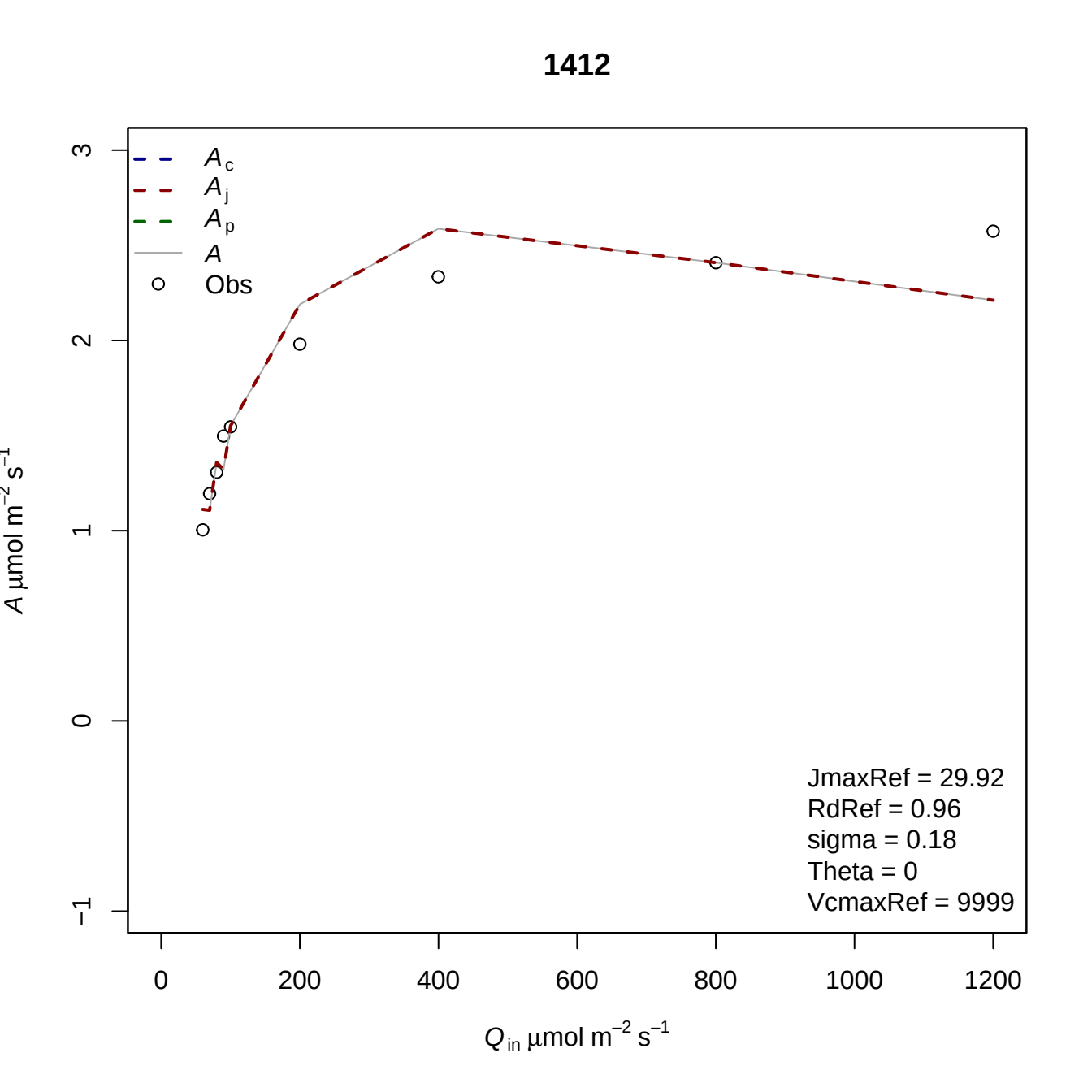

1407
A_c
A_j
A_p
A
Obs
JmaxRef = 107.22
RdRef = 1.27
sigma = 0.34
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

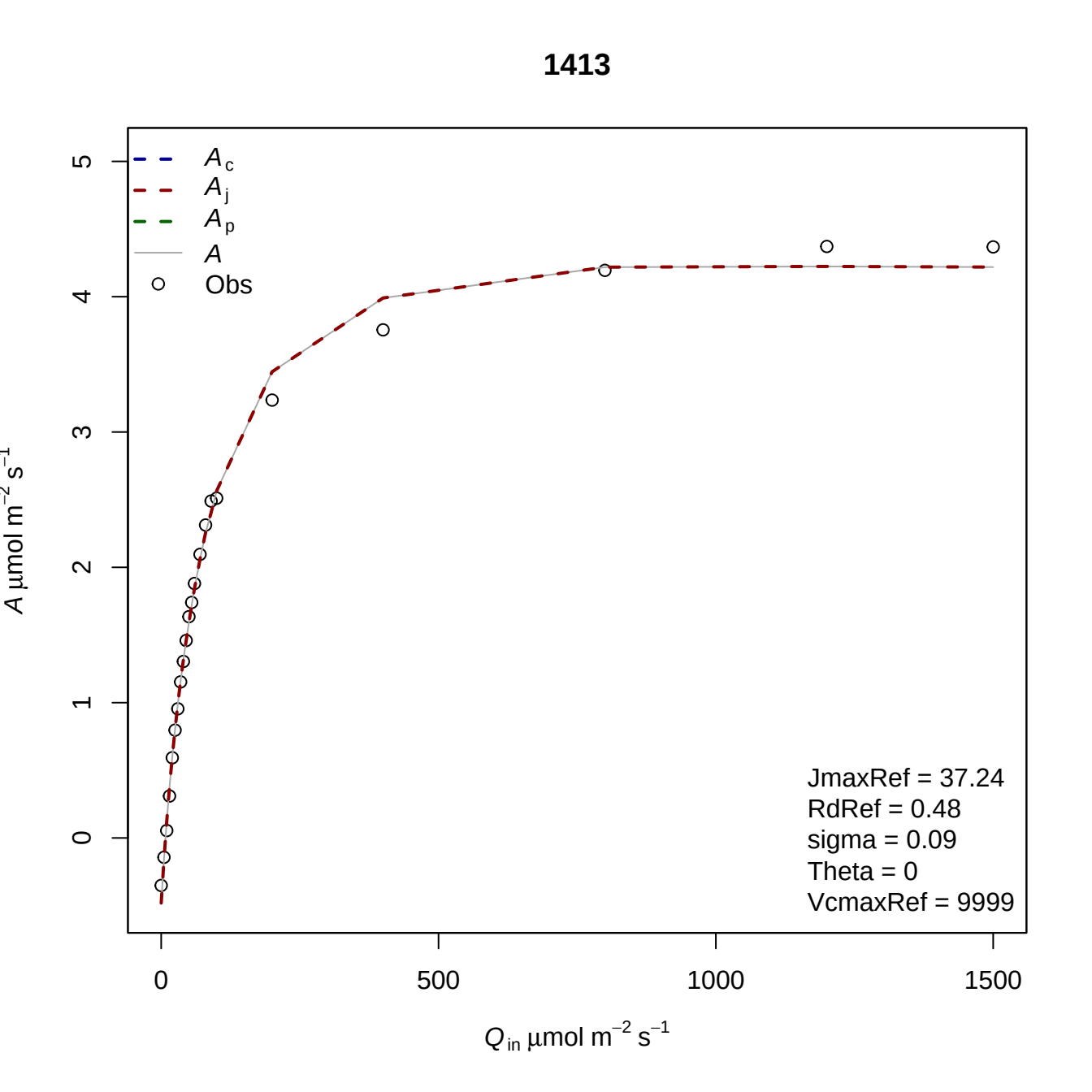

1408
A_c
A_j
A_p
A
Obs
JmaxRef = 79.19
RdRef = 1.35
sigma = 0.41
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

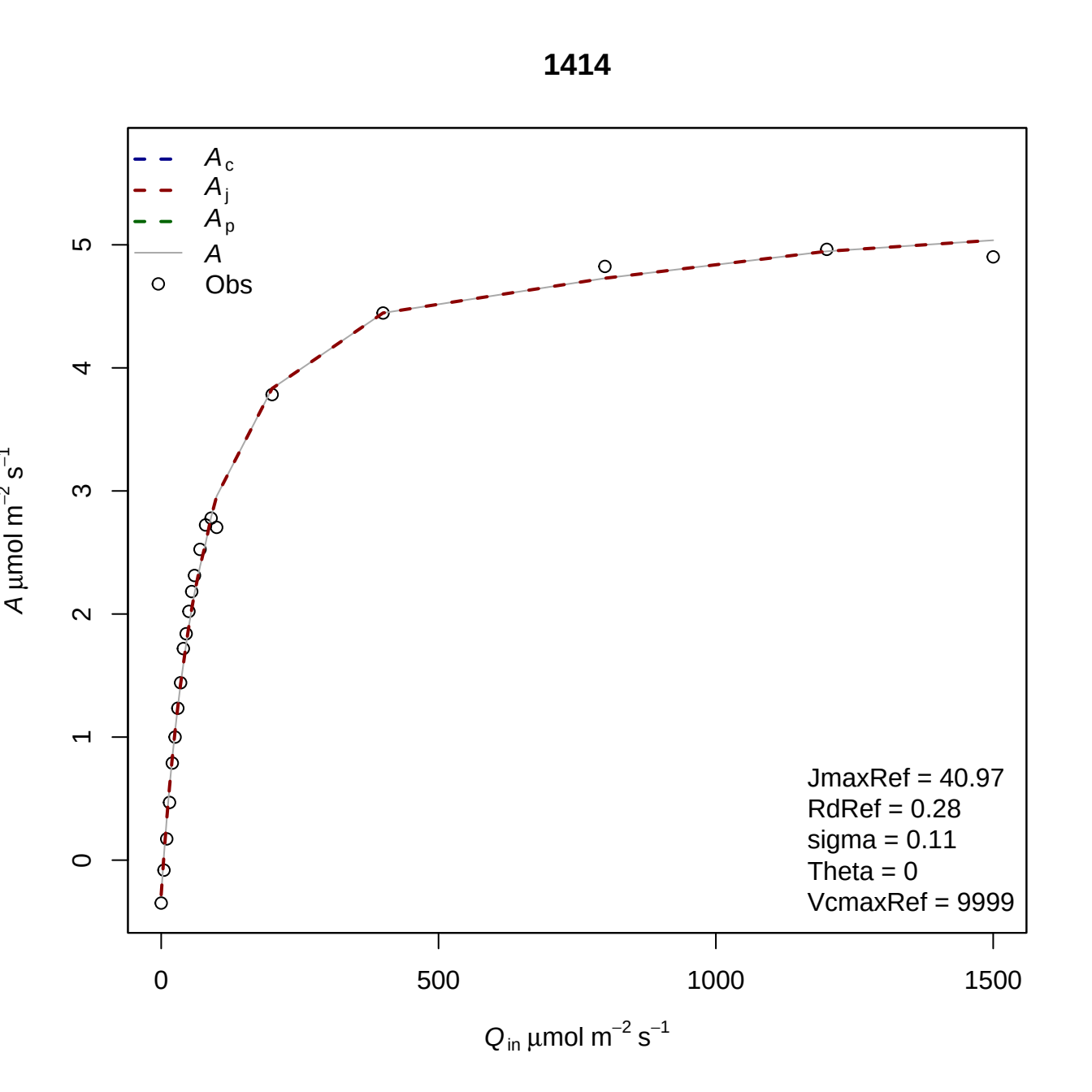

1409
A_c
A_j
A_p
A
Obs
JmaxRef = 47.93
RdRef = 1.33
sigma = 0.27
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

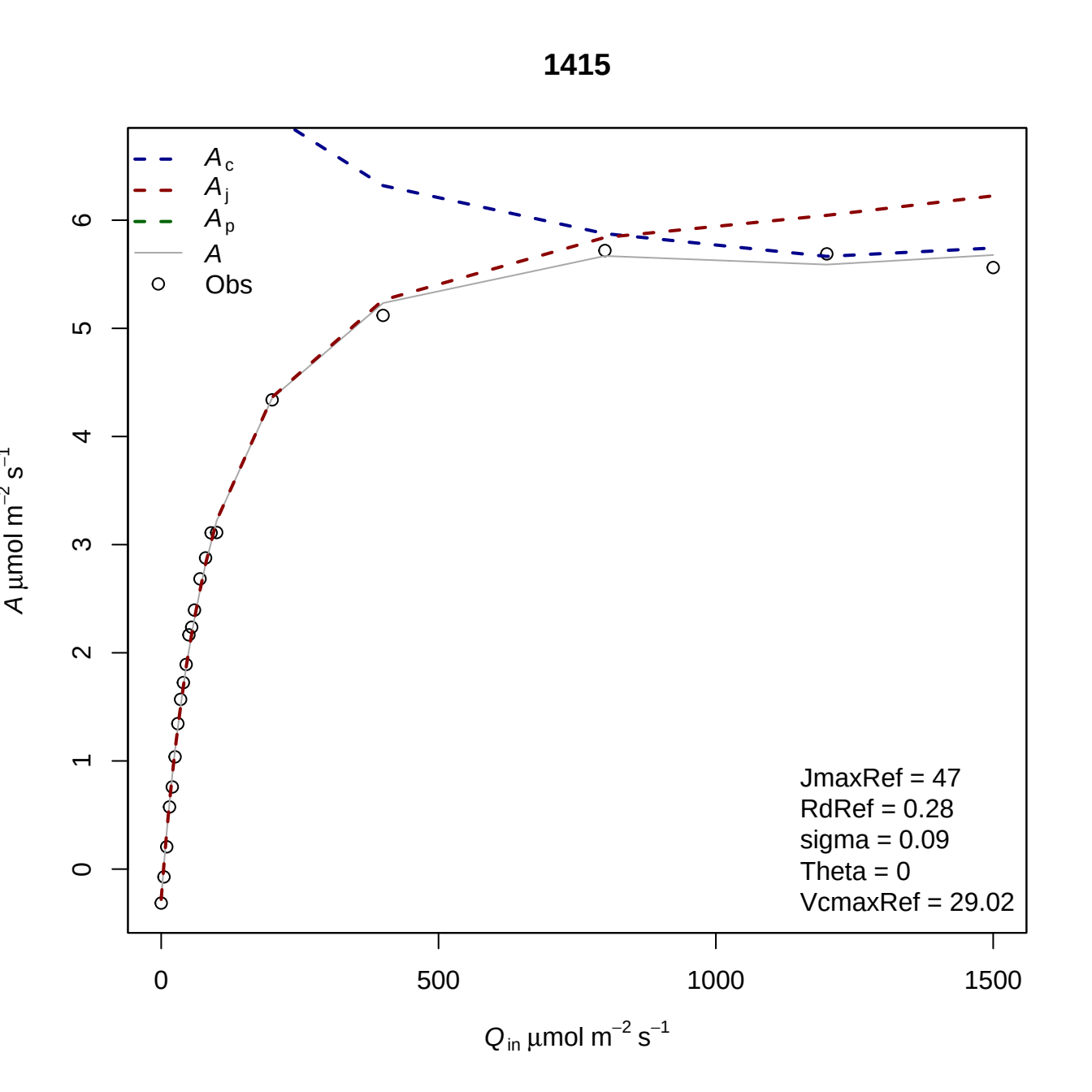

1410
A_c
A_j
A_p
A
Obs
JmaxRef = 86.8
RdRef = 0.22
sigma = 0.2
Theta = 0
VcmaxRef = 44.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

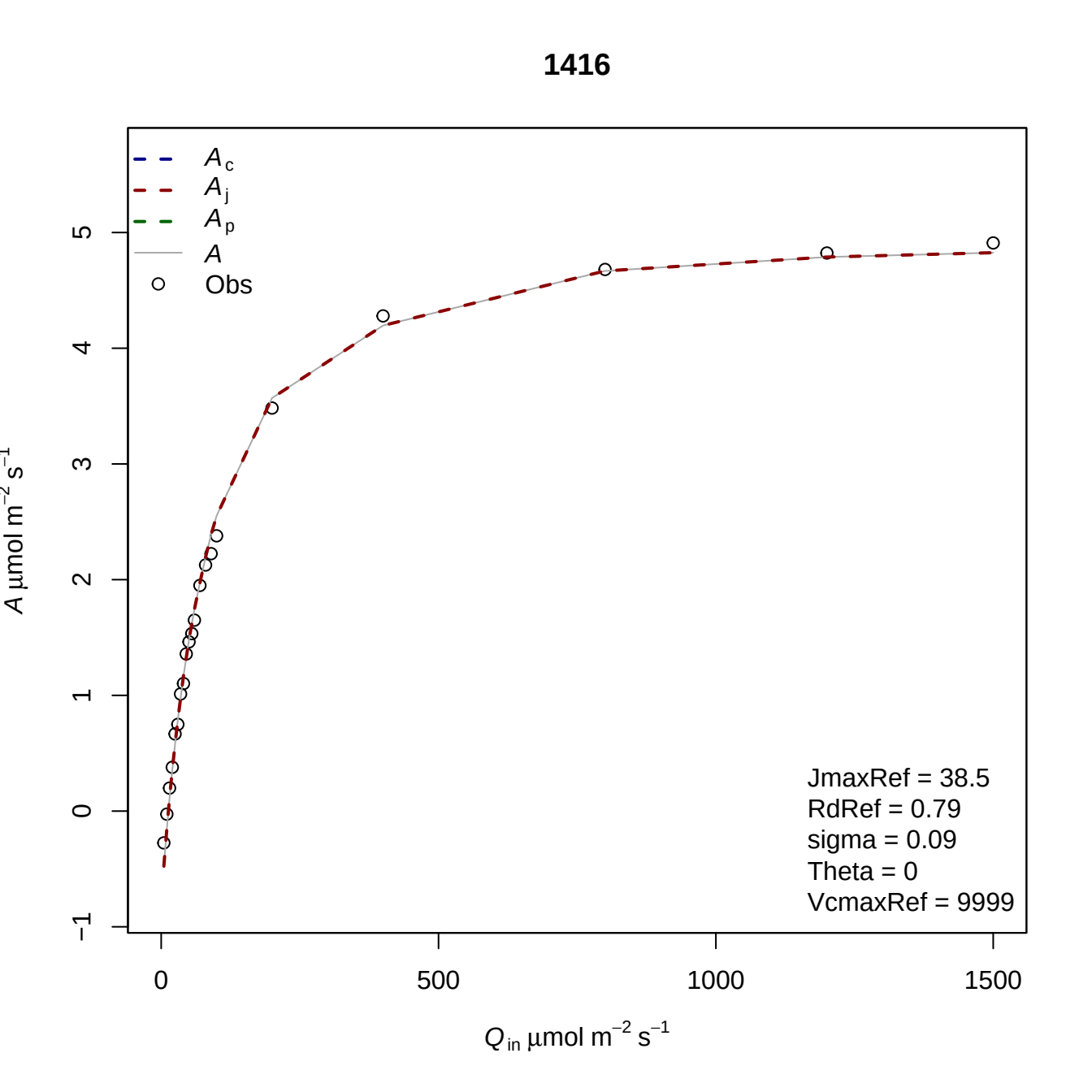

1411
A_c
A_j
A_p
A
Obs
JmaxRef = 35.58
RdRef = 1.26
sigma = 0.36
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800

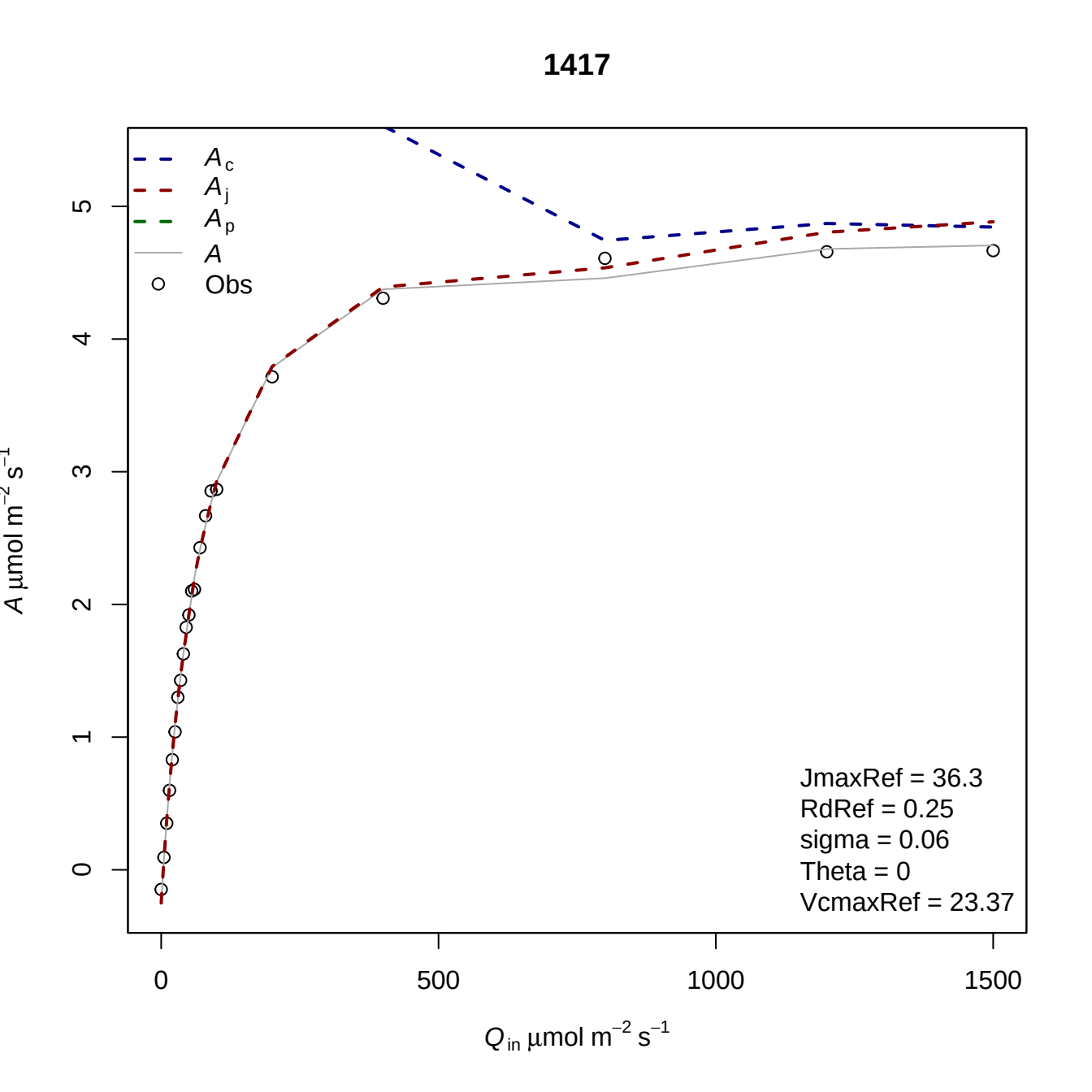

1412
A_c
A_j
A_p
A
Obs
JmaxRef = 29.92
RdRef = 0.96
sigma = 0.18
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
200
400
600
800
1000
1200

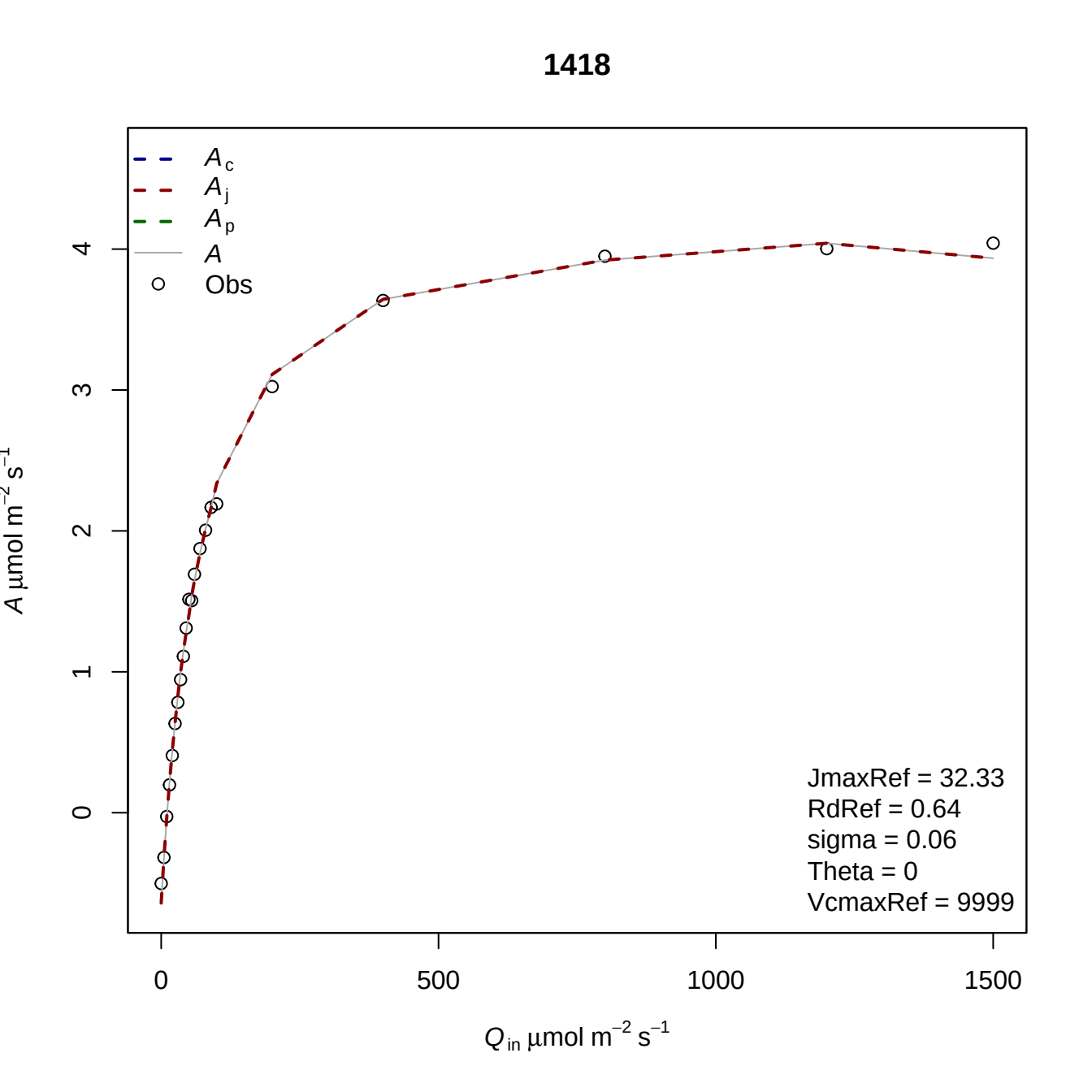

1413
A_c
A_j
A_p
A
Obs
JmaxRef = 37.24
RdRef = 0.48
sigma = 0.09
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

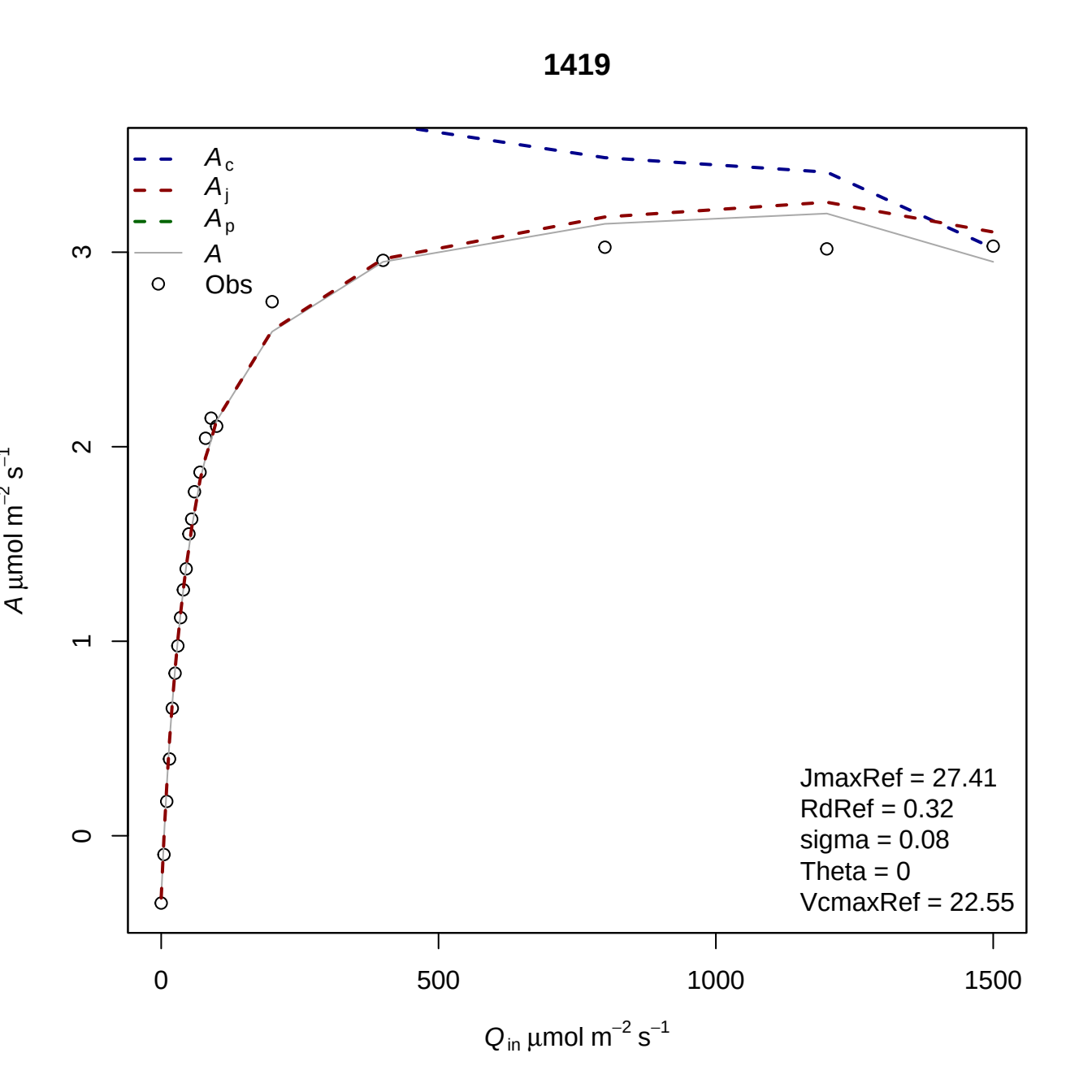

1414
A_c
A_j
A_p
A
Obs
JmaxRef = 40.97
RdRef = 0.28
sigma = 0.11
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

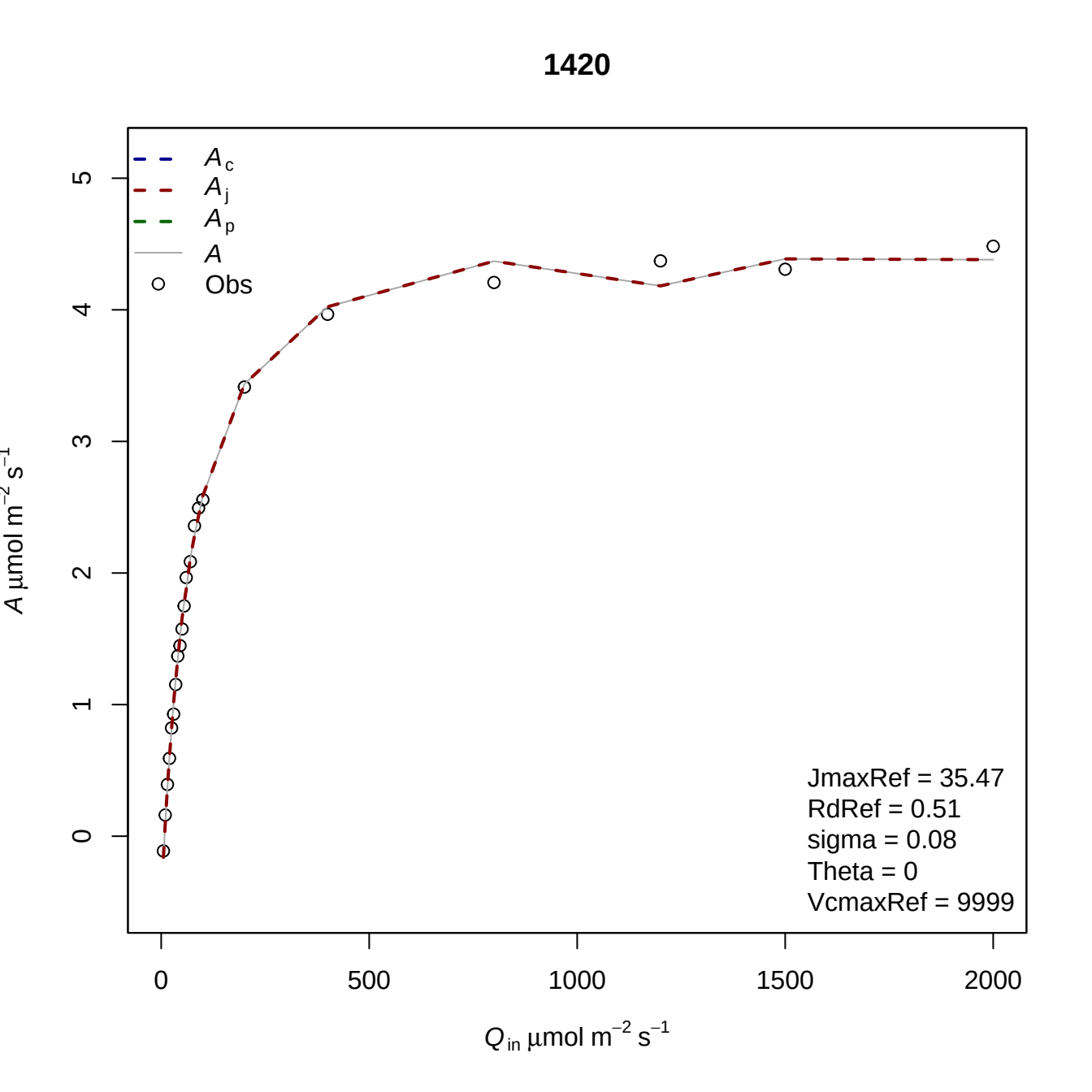

1415
A_c
A_j
A_p
A
Obs
JmaxRef = 47
RdRef = 0.28
sigma = 0.09
Theta = 0
VcmaxRef = 29.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

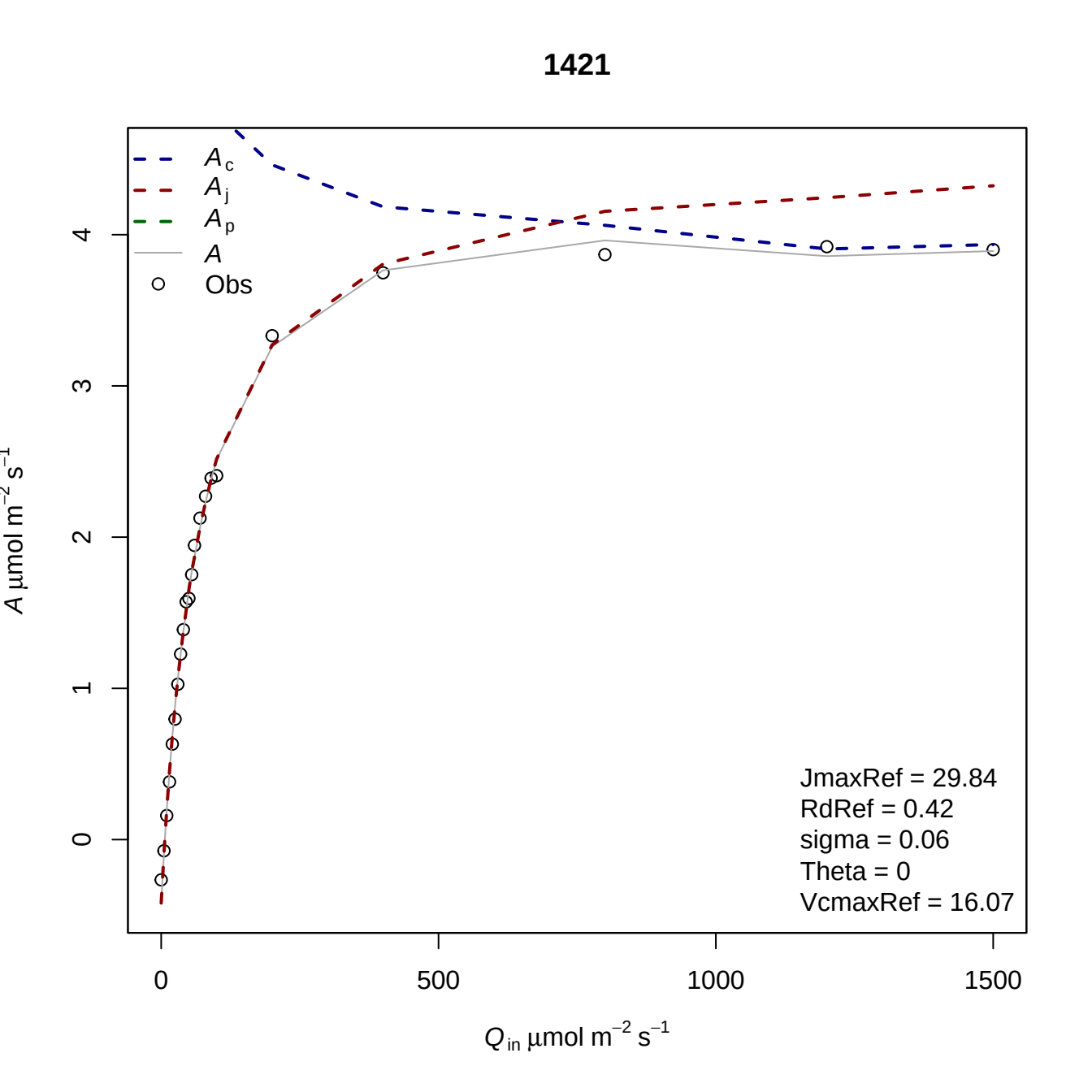

1416
A_c
A_j
A_p
A
Obs
JmaxRef = 38.5
RdRef = 0.79
sigma = 0.09
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

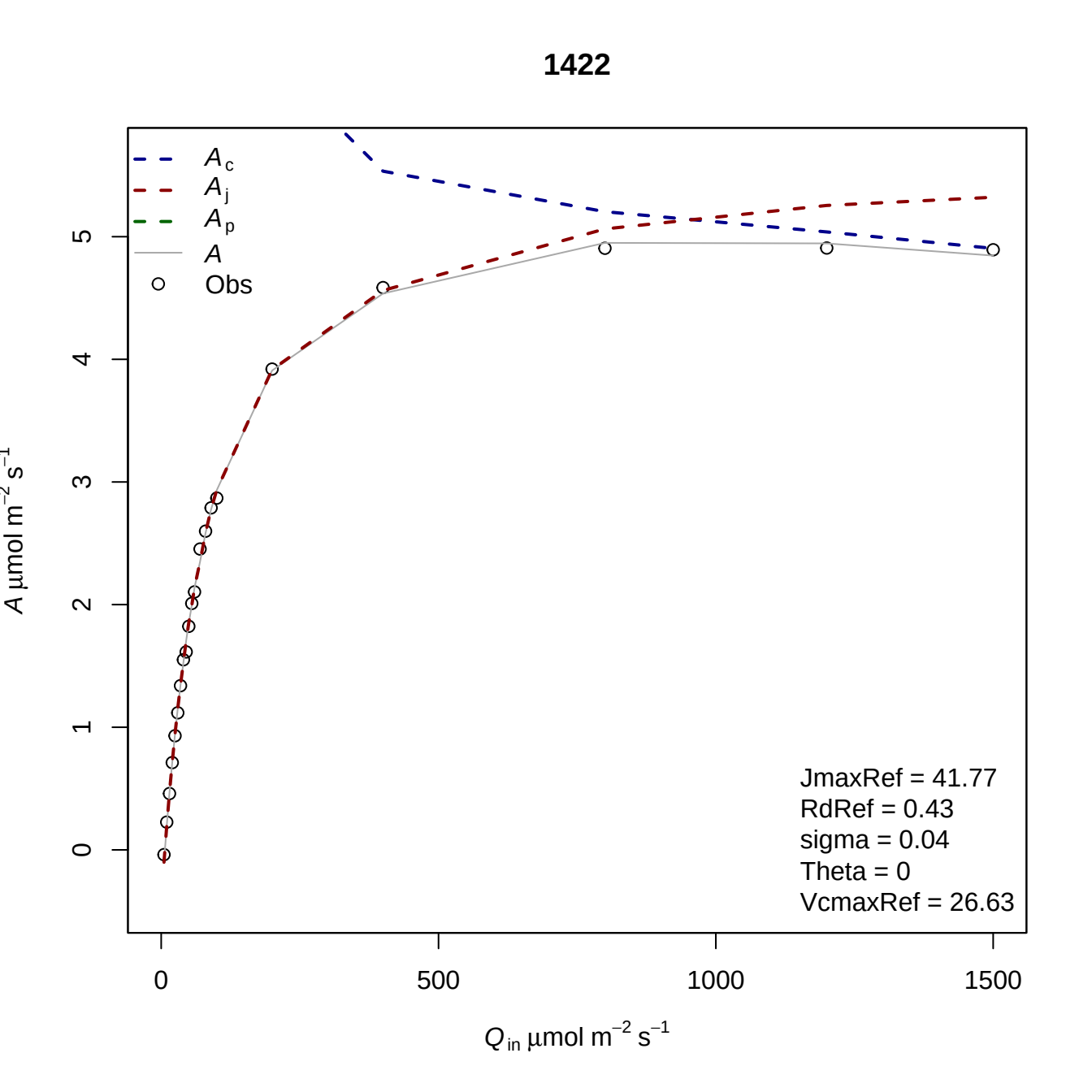

1417
A_c
A_j
A_p
A
Obs
JmaxRef = 36.3
RdRef = 0.25
sigma = 0.06
Theta = 0
VcmaxRef = 23.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

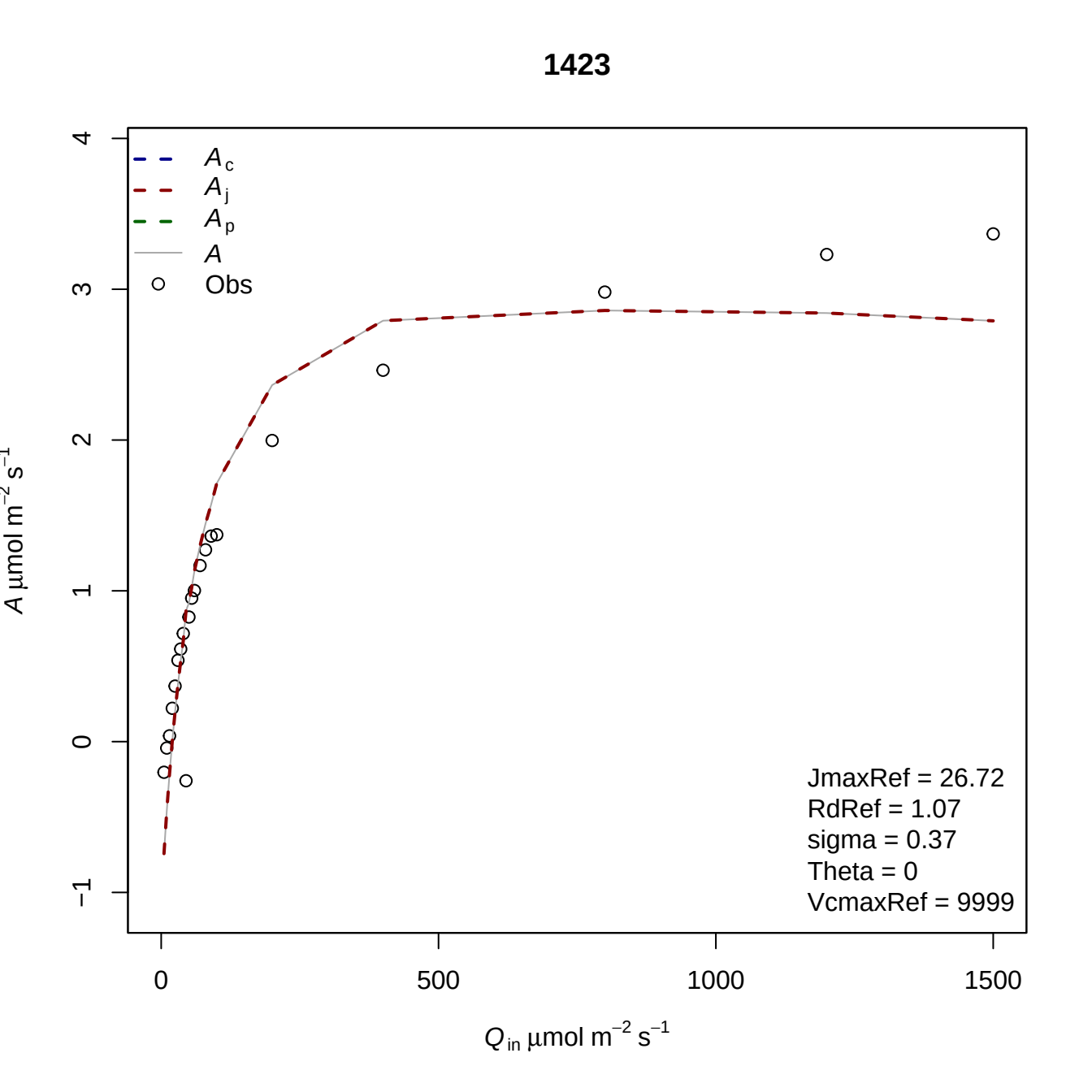

1418
A_c
A_j
A_p
A
Obs
JmaxRef = 32.33
RdRef = 0.64
sigma = 0.06
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

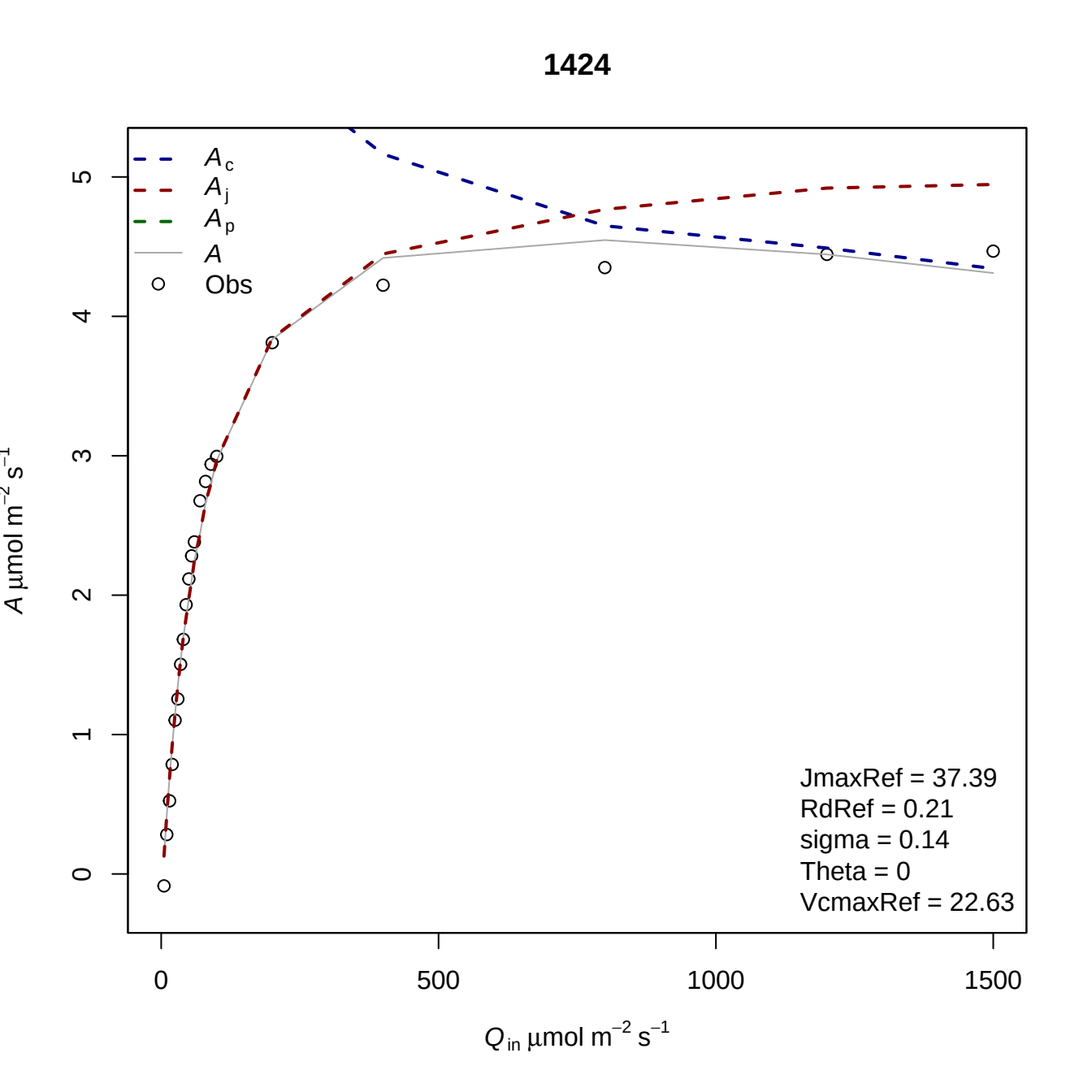

1419
A_c
A_j
A_p
A
Obs
JmaxRef = 27.41
RdRef = 0.32
sigma = 0.08
Theta = 0
VcmaxRef = 22.55
0
1
2
3
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

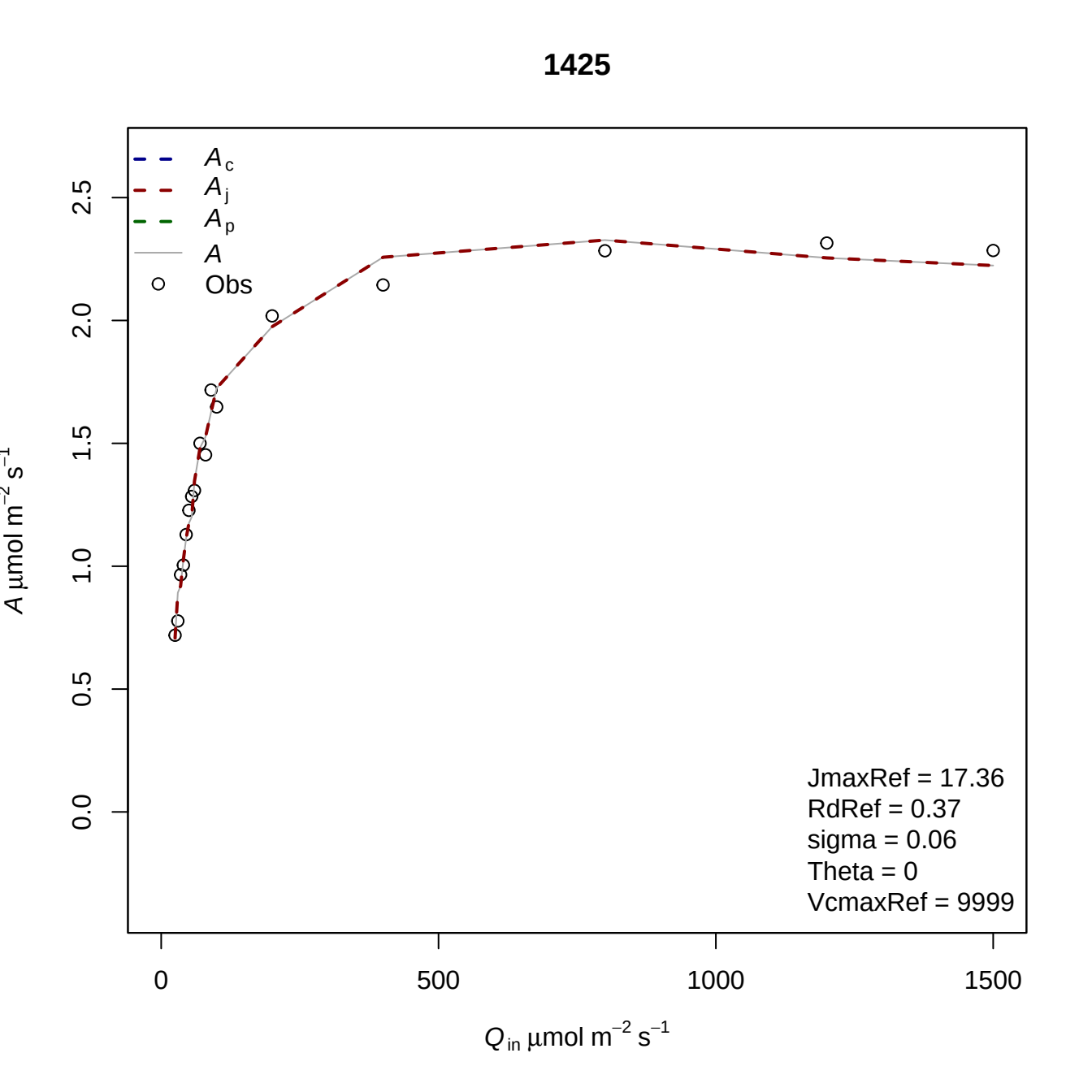

1420
A_c
A_j
A_p
A
Obs
JmaxRef = 35.47
RdRef = 0.51
sigma = 0.08
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500
2000

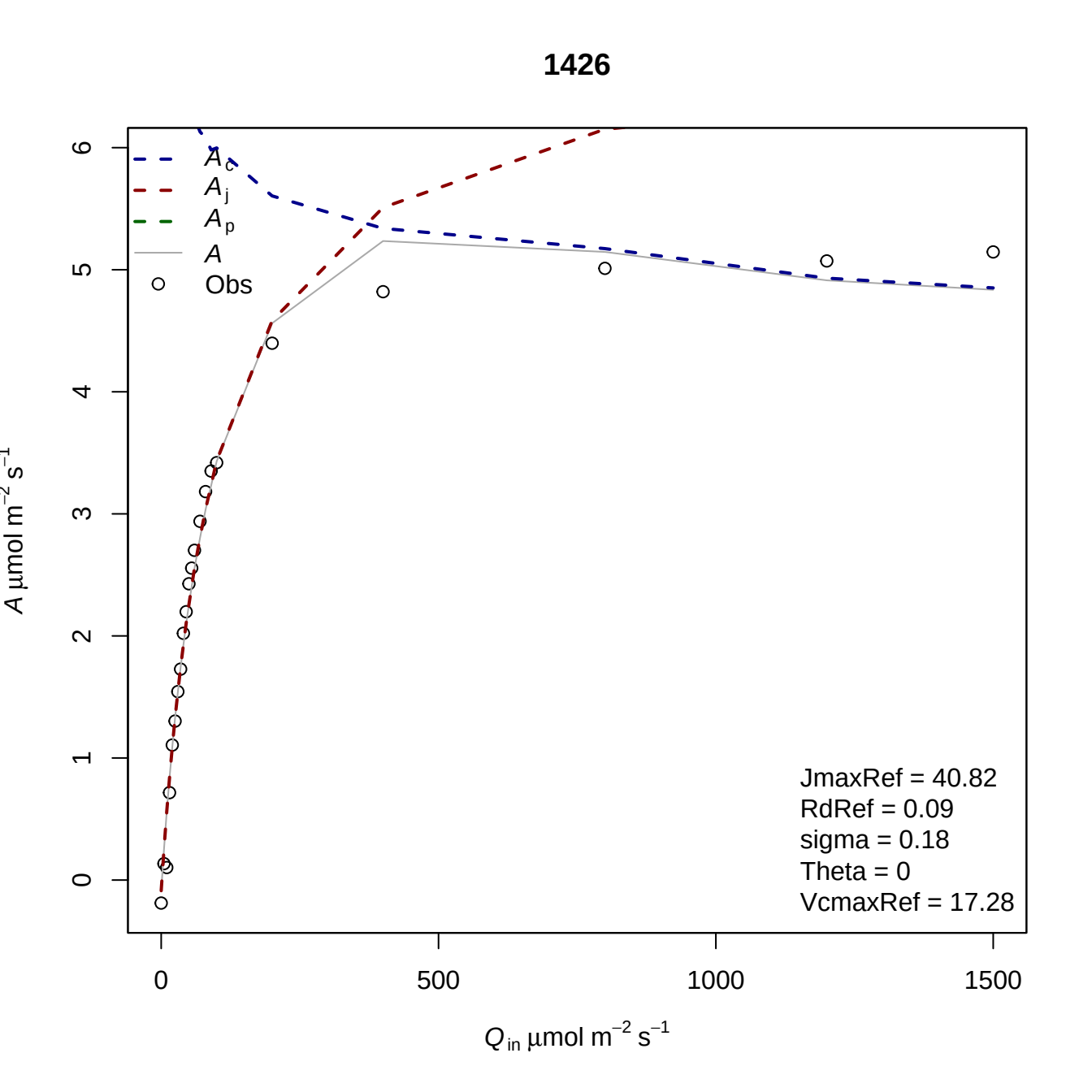

1421
A_c
A_j
A_p
A
Obs
JmaxRef = 29.84
RdRef = 0.42
sigma = 0.06
Theta = 0
VcmaxRef = 16.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

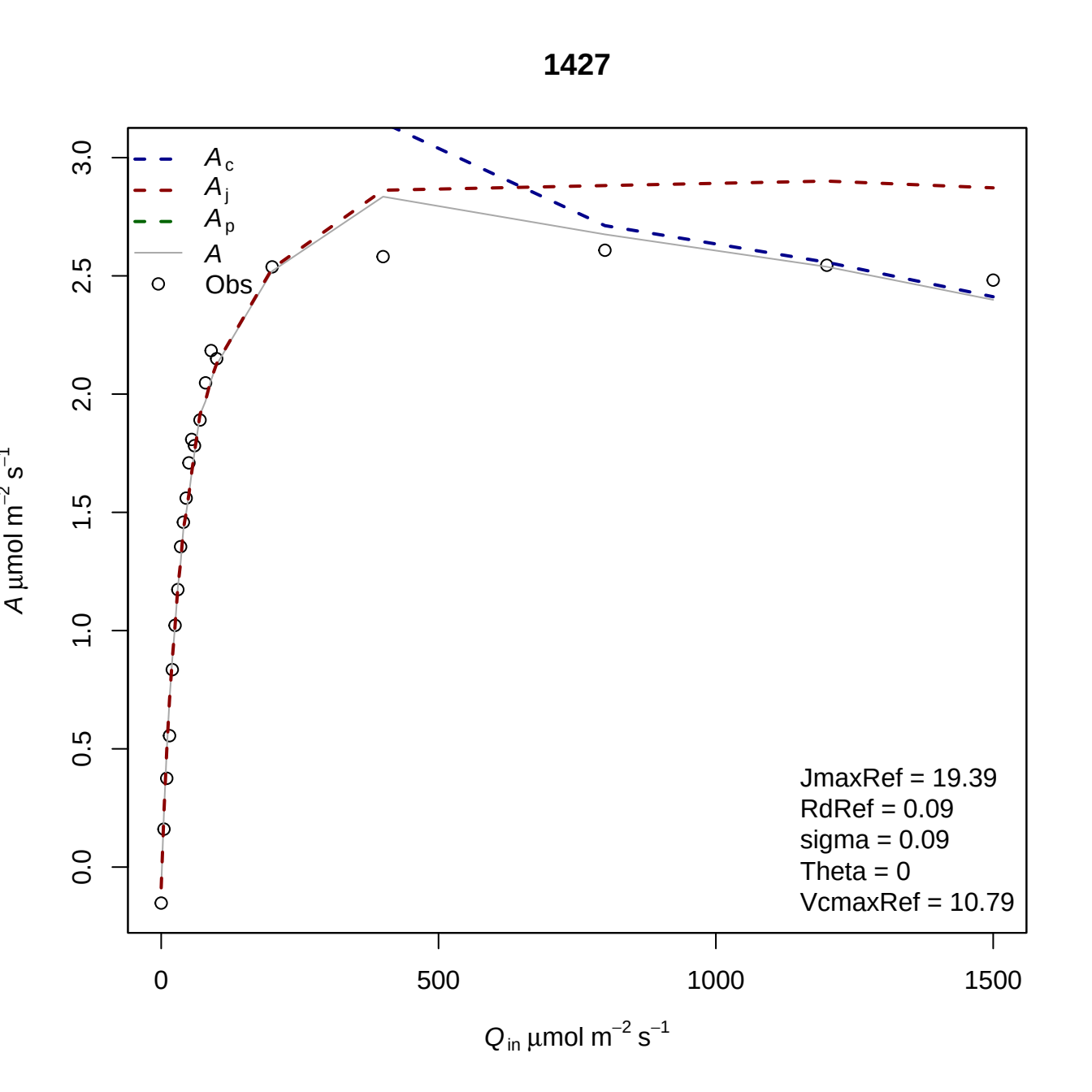

1422
A_c
A_j
A_p
A
Obs
JmaxRef = 41.77
RdRef = 0.43
sigma = 0.04
Theta = 0
VcmaxRef = 26.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

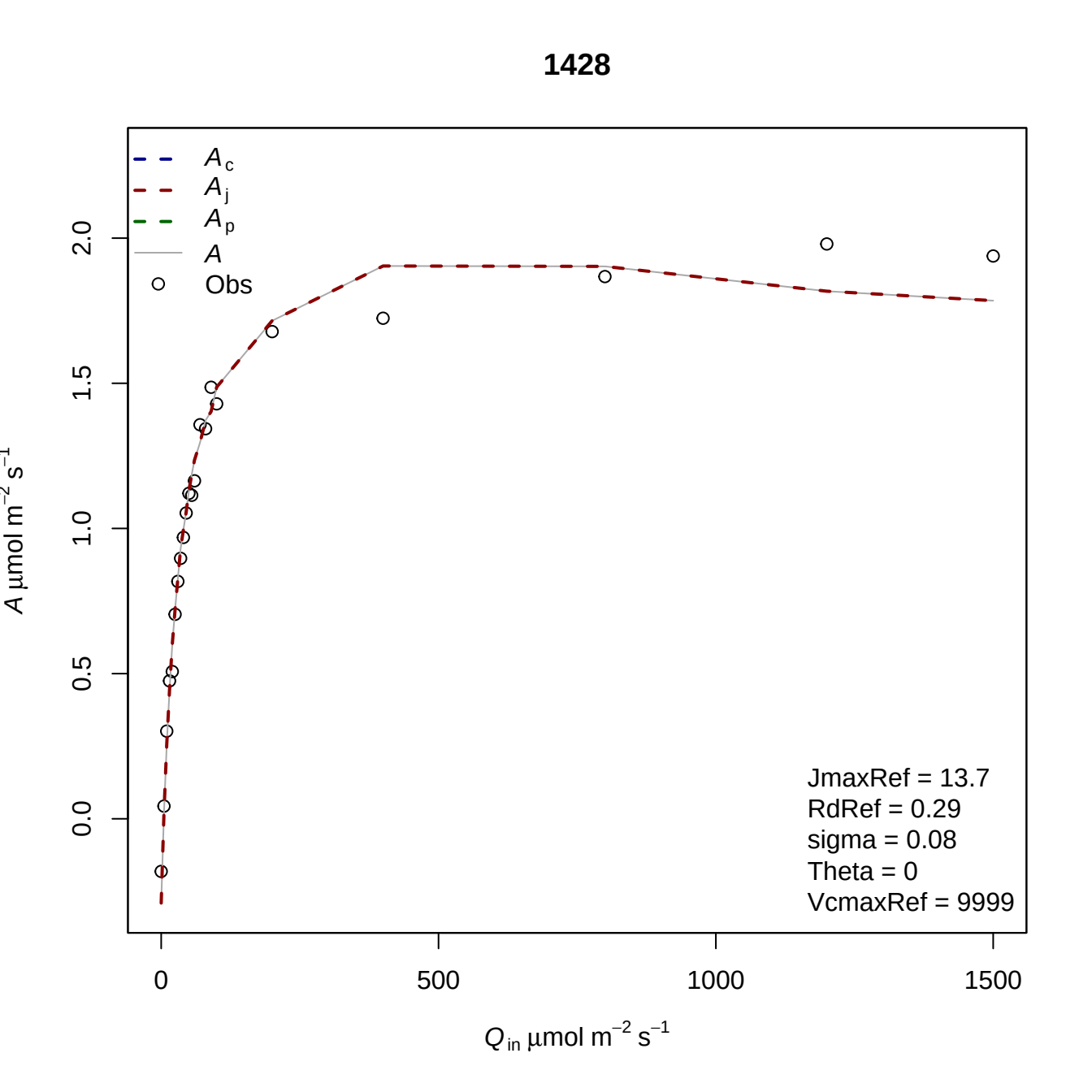

1423
A_c
A_j
A_p
A
Obs
JmaxRef = 26.72
RdRef = 1.07
sigma = 0.37
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500

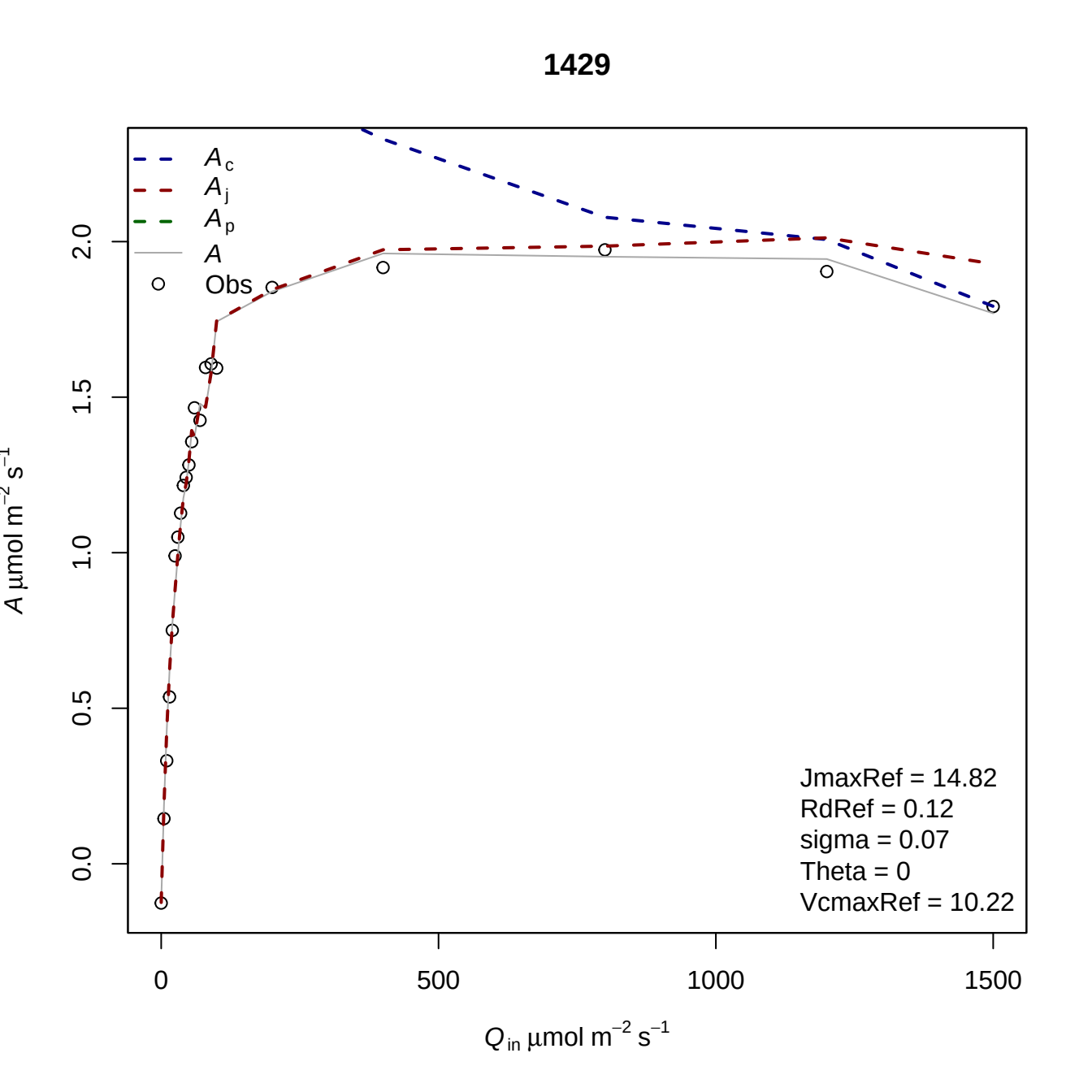

1424
A_c
A_j
A_p
A
Obs
JmaxRef = 37.39
RdRef = 0.21
sigma = 0.14
Theta = 0
VcmaxRef = 22.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

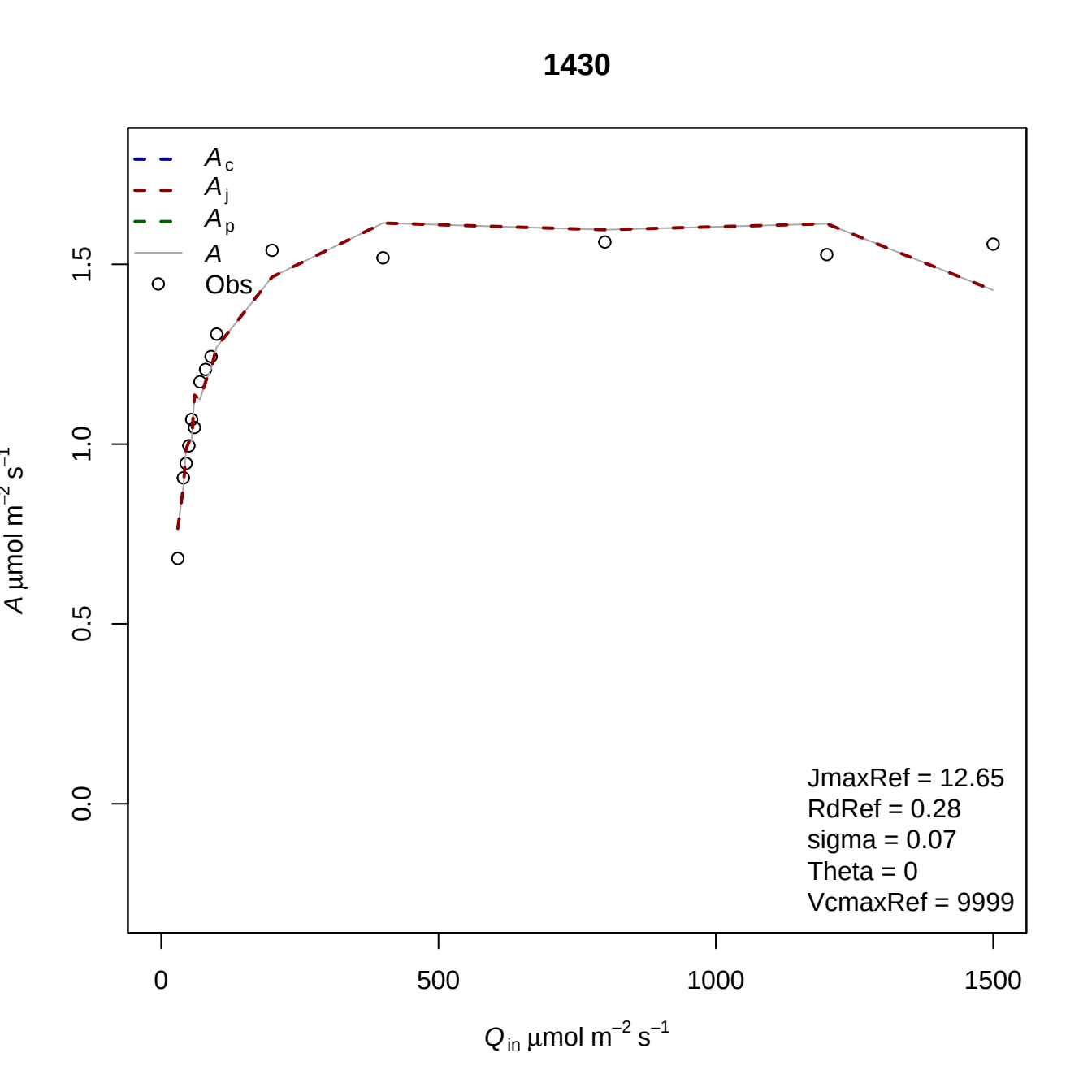

1425
A_c
A_j
A_p
A
Obs
JmaxRef = 17.36
RdRef = 0.37
sigma = 0.06
Theta = 0
VcmaxRef = 9999
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

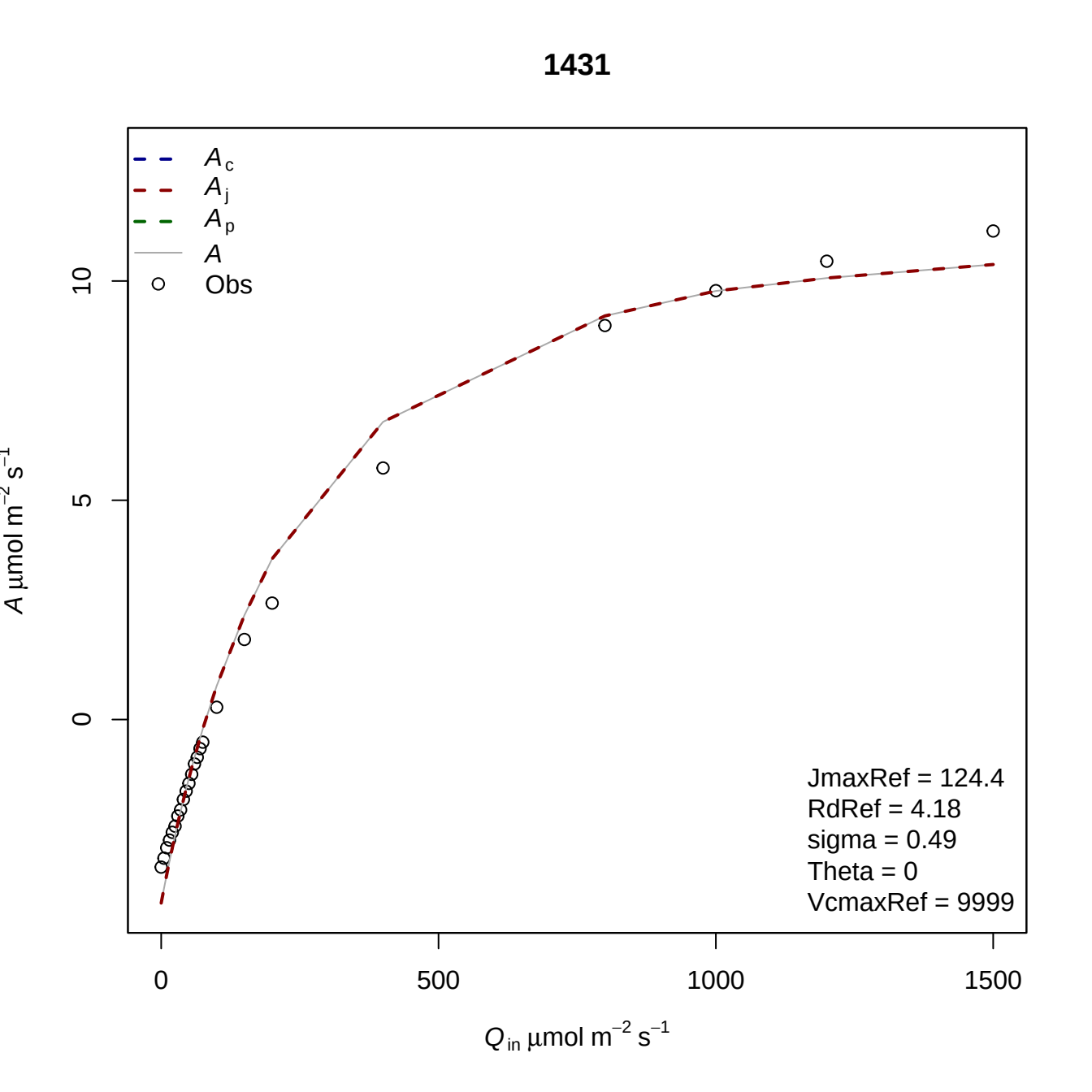

1426
A_c
A_j
A_p
A
Obs
JmaxRef = 40.82
RdRef = 0.09
sigma = 0.18
Theta = 0
VcmaxRef = 17.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

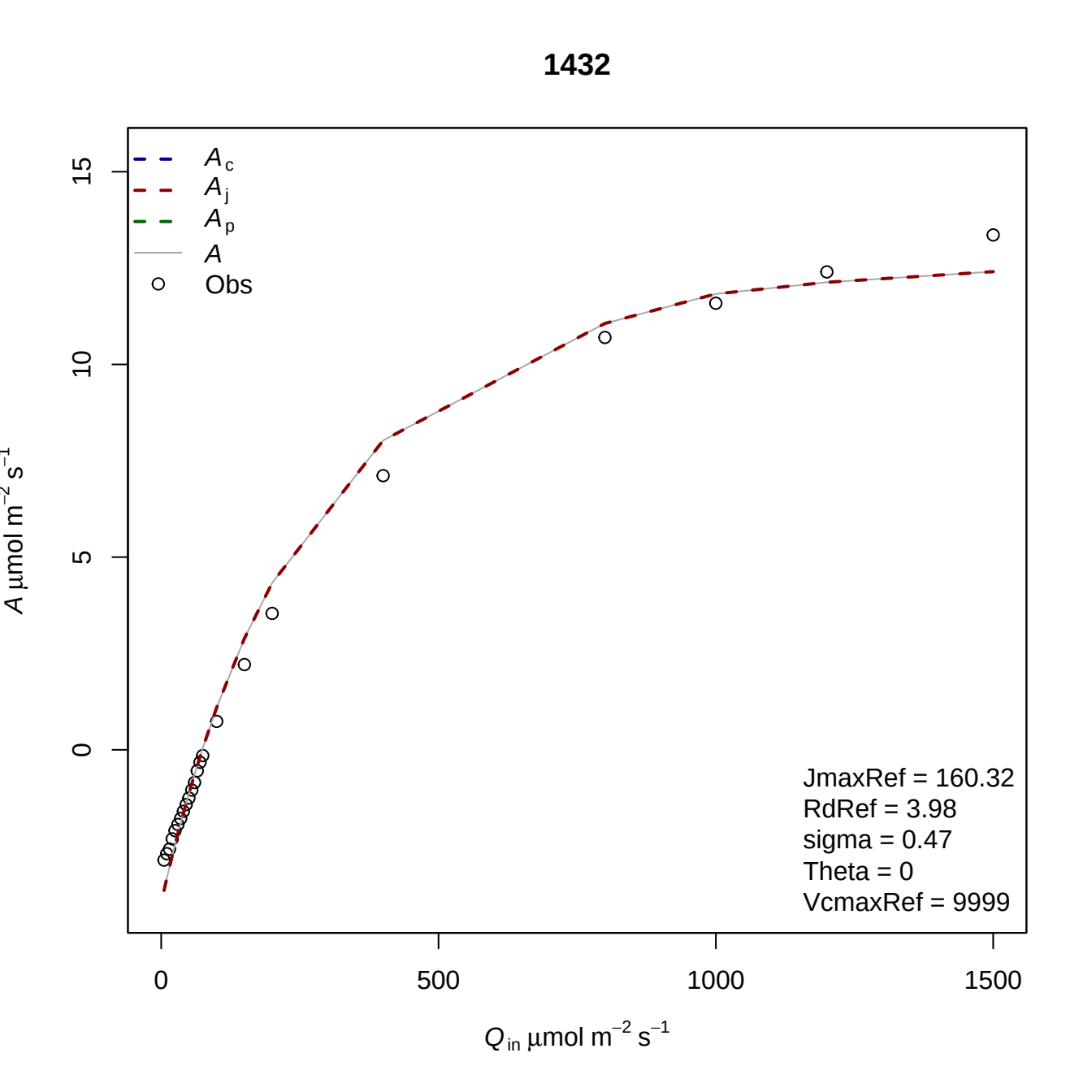

1427
A_c
A_j
A_p
A
Obs
JmaxRef = 19.39
RdRef = 0.09
sigma = 0.09
Theta = 0
VcmaxRef = 10.79
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

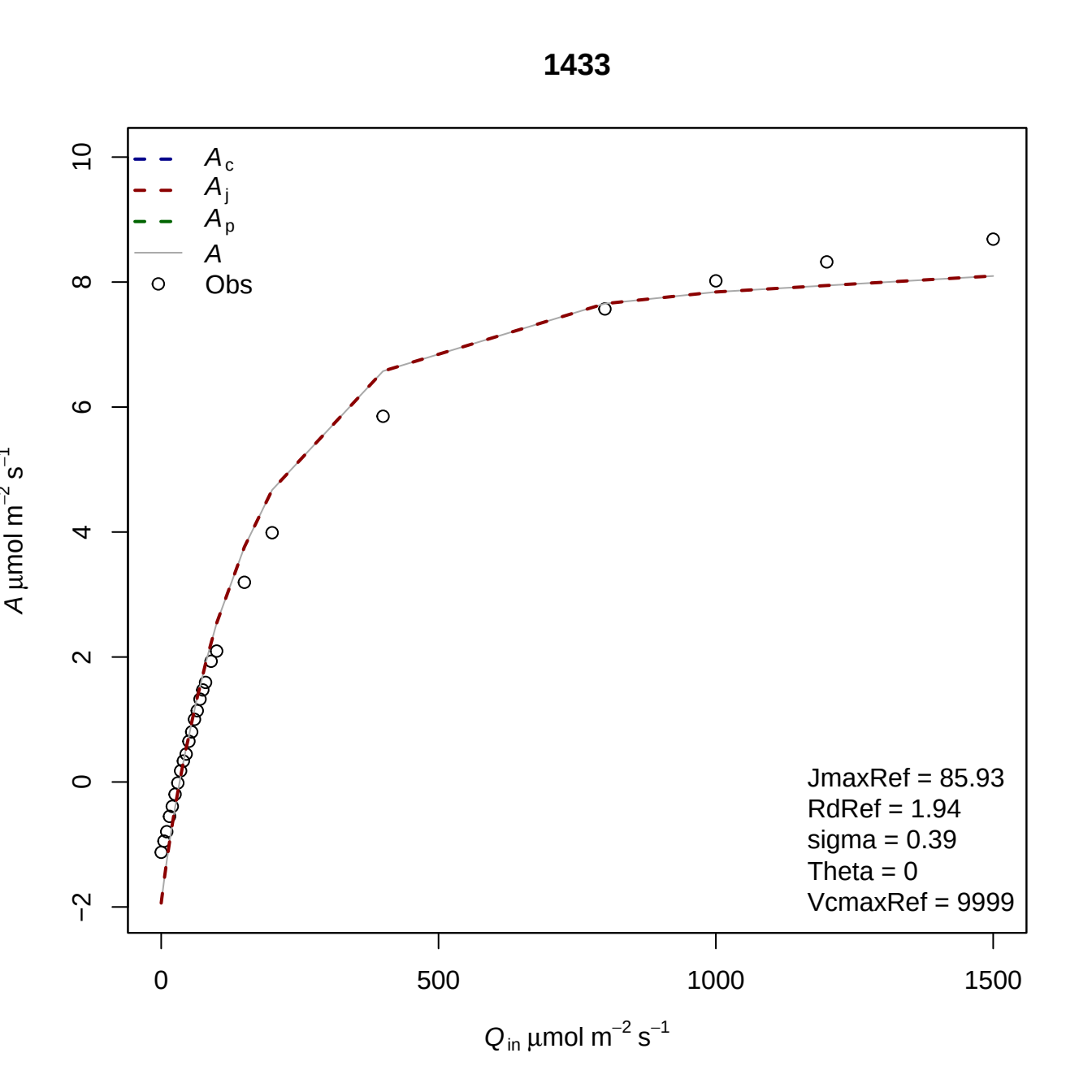

1428
A_c
A_j
A_p
A
Obs
JmaxRef = 13.7
RdRef = 0.29
sigma = 0.08
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500

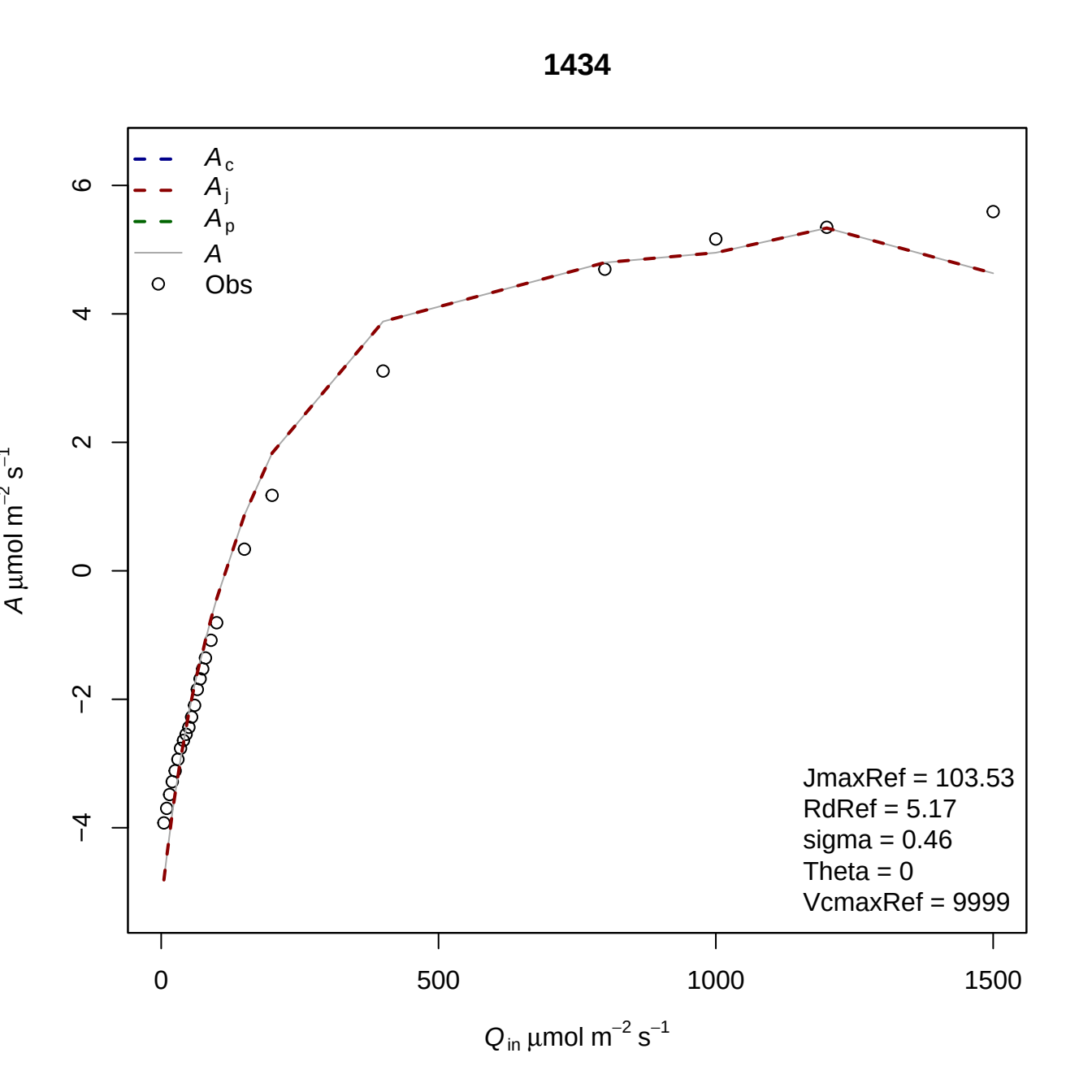

1429
A_c
A_j
A_p
A
Obs
JmaxRef = 14.82
RdRef = 0.12
sigma = 0.07
Theta = 0
VcmaxRef = 10.22
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

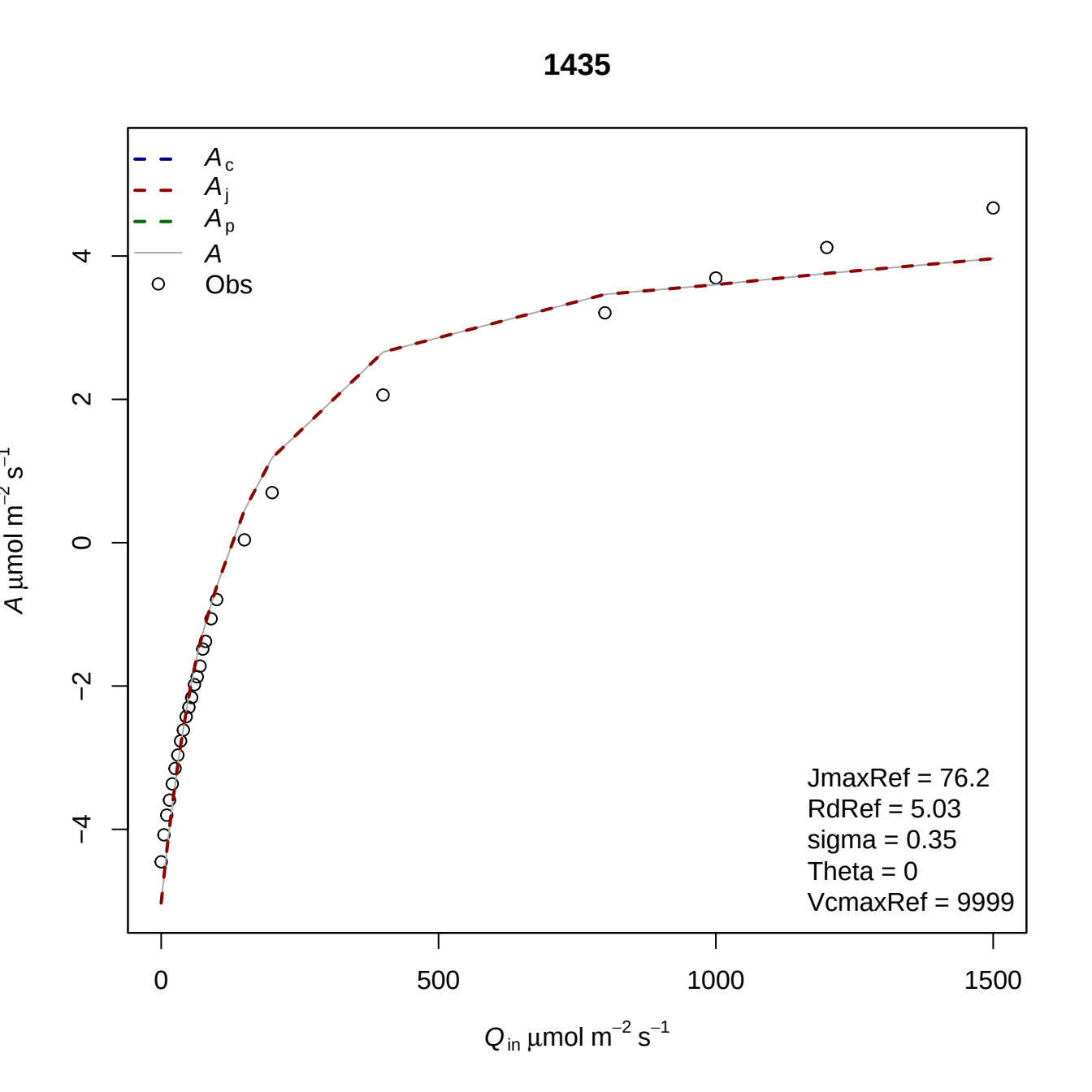

1430
A_c
A_j
A_p
A
Obs
JmaxRef = 12.65
RdRef = 0.28
sigma = 0.07
Theta = 0
VcmaxRef = 9999
0.0
0.5
1.0
1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

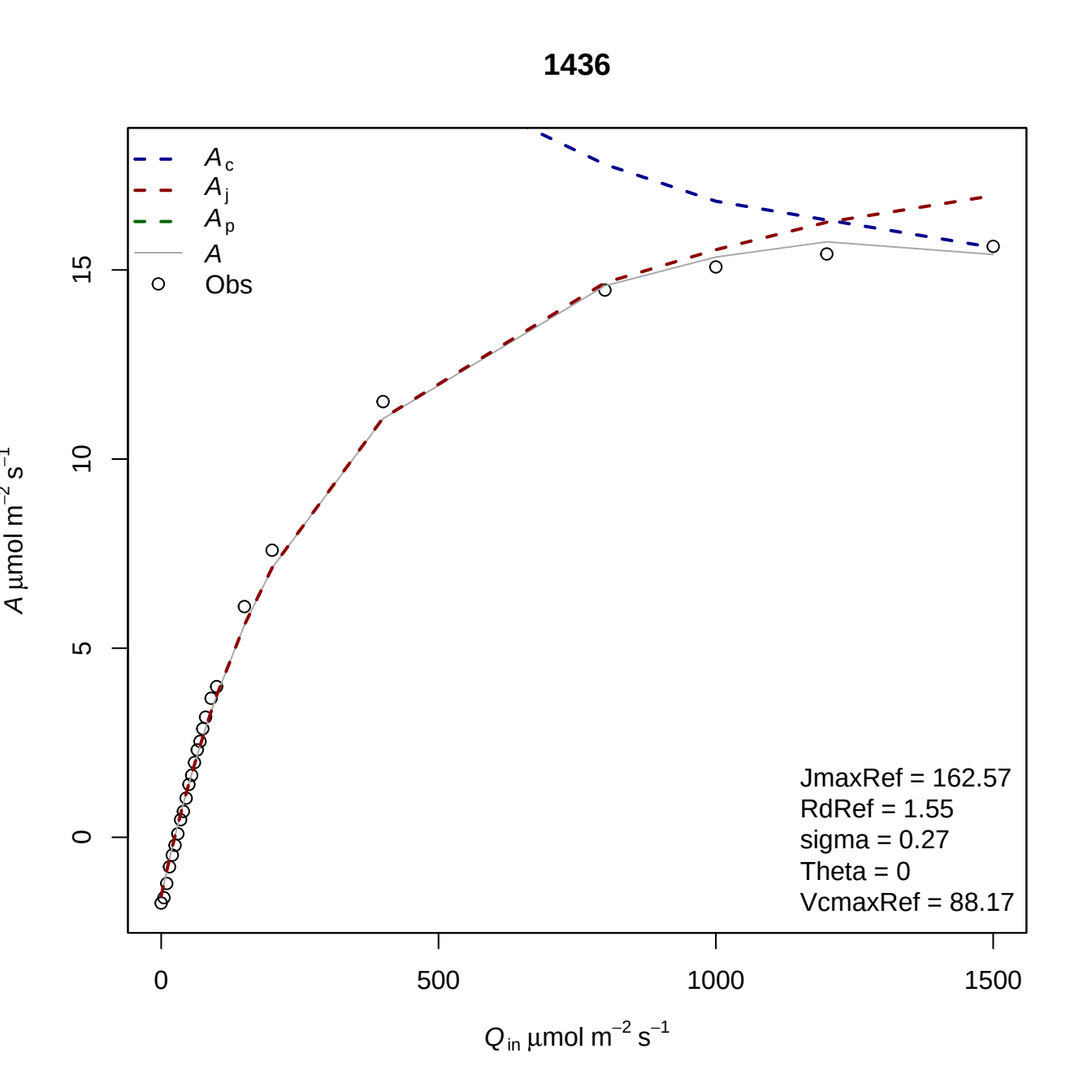

1431
A_c
A_j
A_p
A
Obs
JmaxRef = 124.4
RdRef = 4.18
sigma = 0.49
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

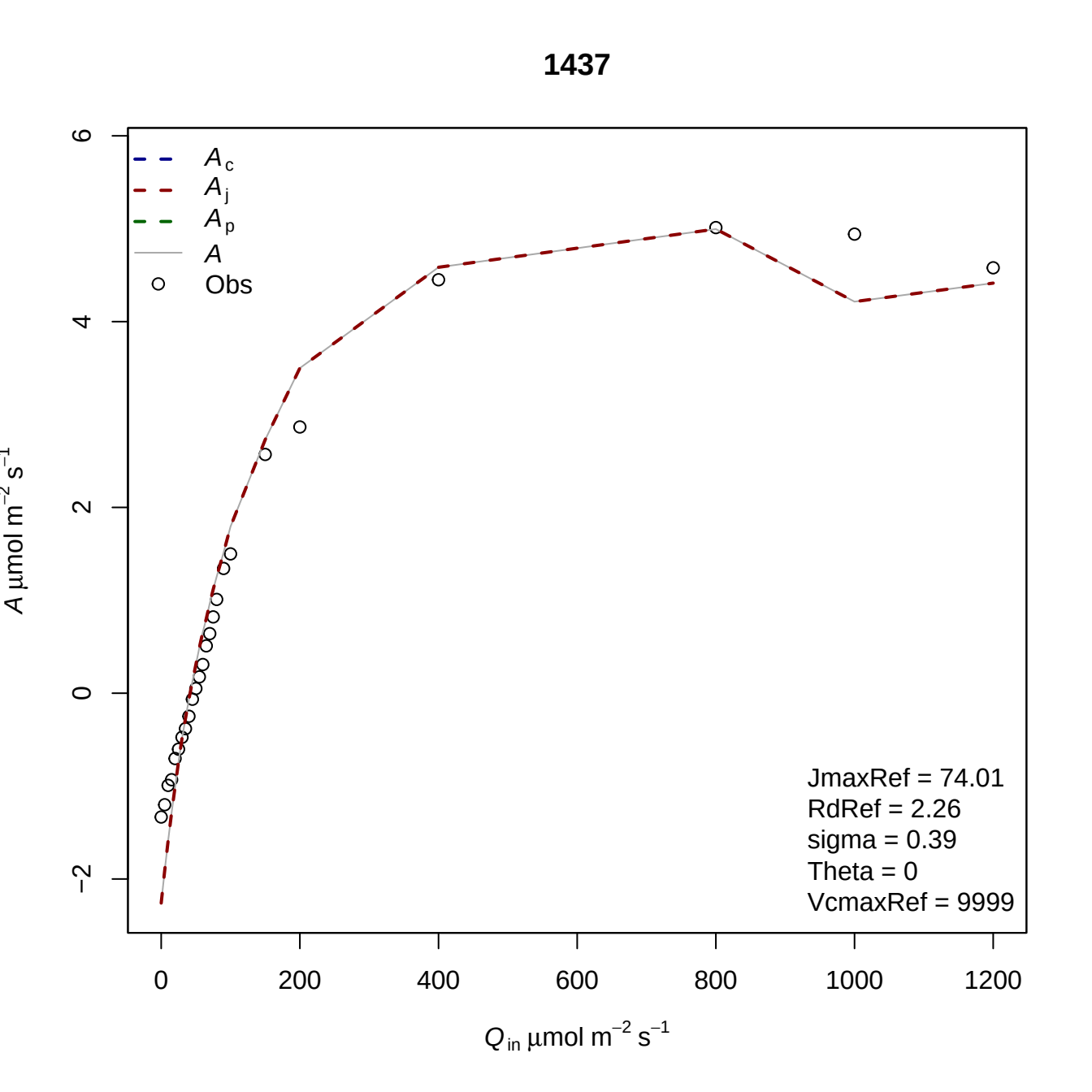

1432
A_c
A_j
A_p
A
Obs
JmaxRef = 160.32
RdRef = 3.98
sigma = 0.47
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

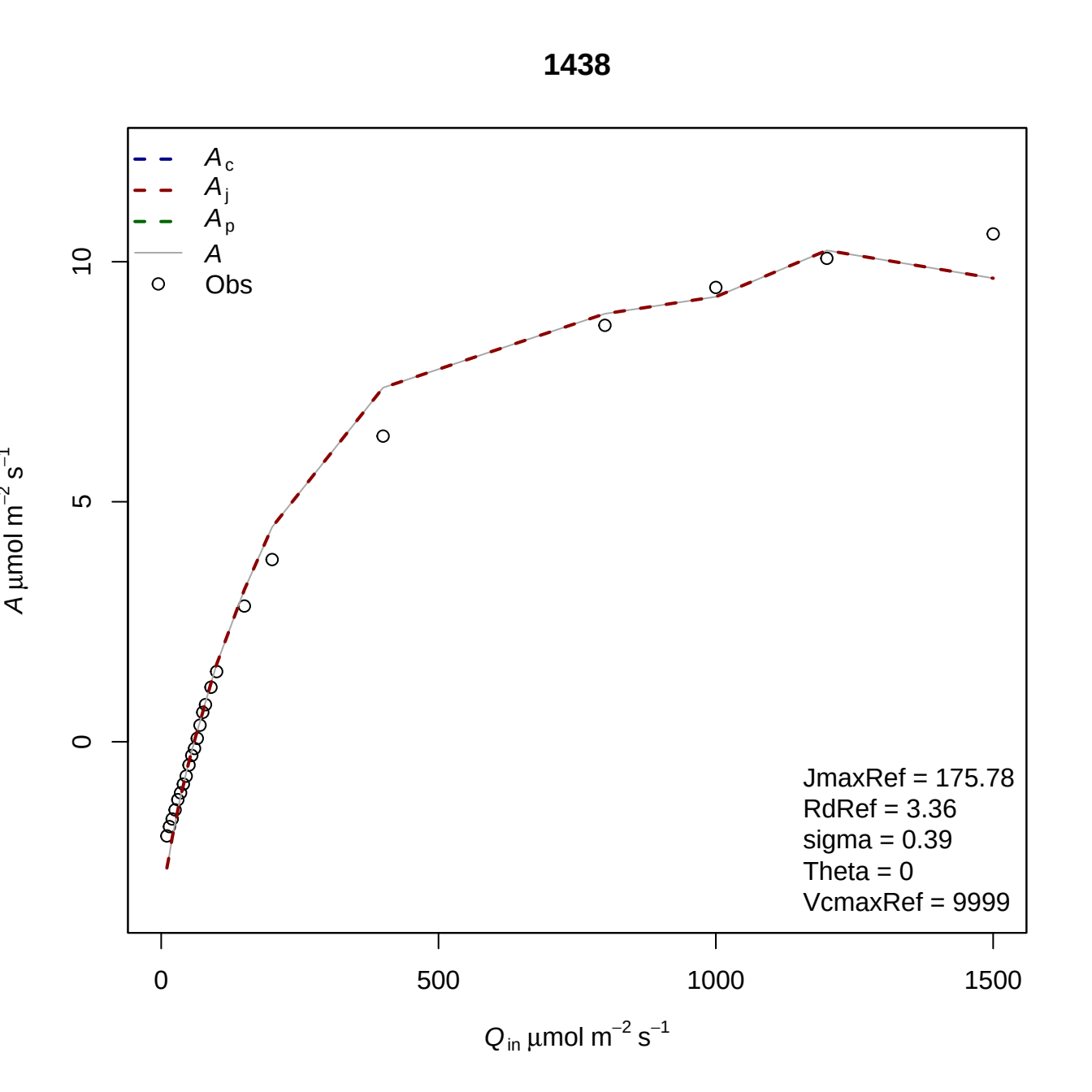

1433
A_c
A_j
A_p
A
Obs
JmaxRef = 85.93
RdRef = 1.94
sigma = 0.39
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

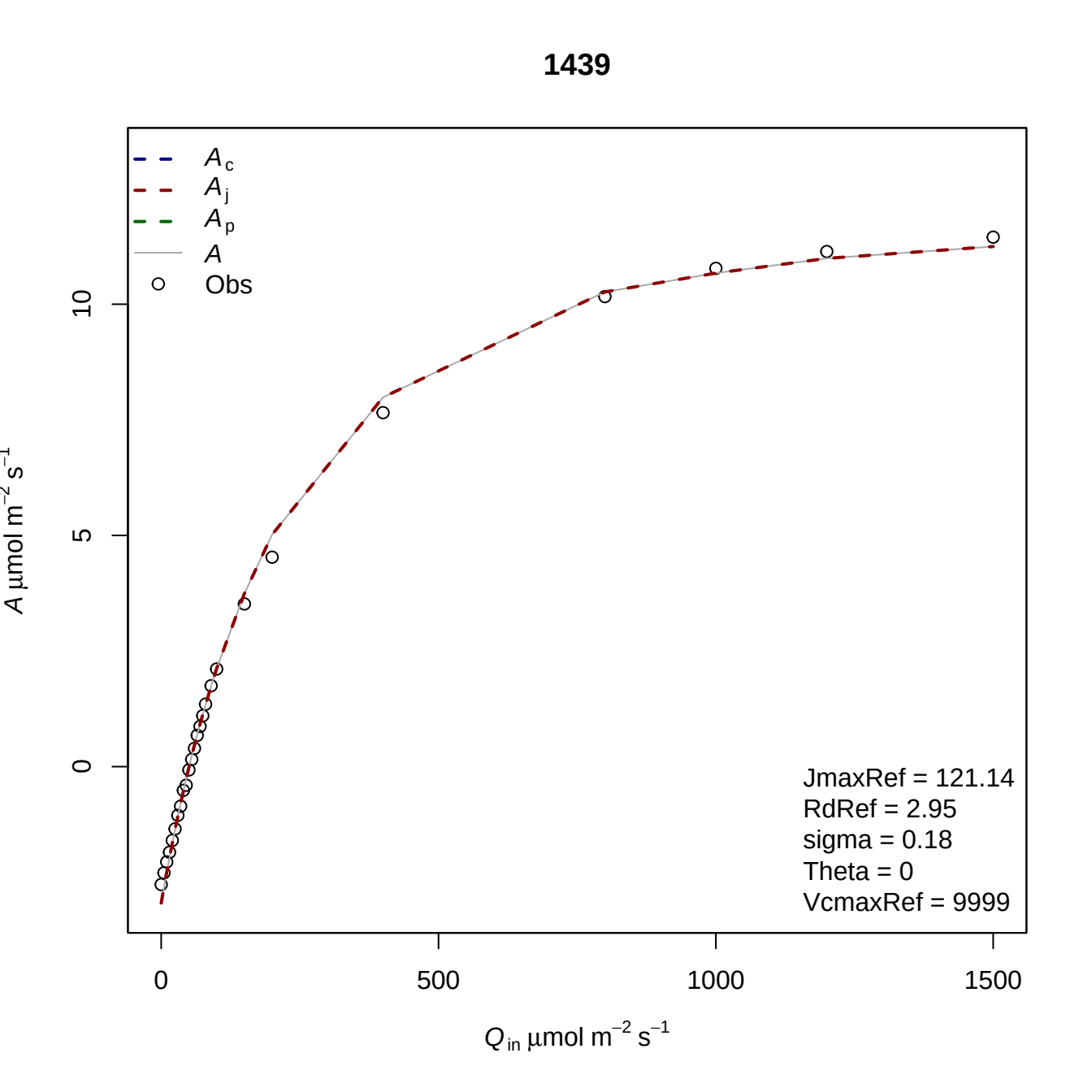

1434
A_c
A_j
A_p
A
Obs
JmaxRef = 103.53
RdRef = 5.17
sigma = 0.46
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

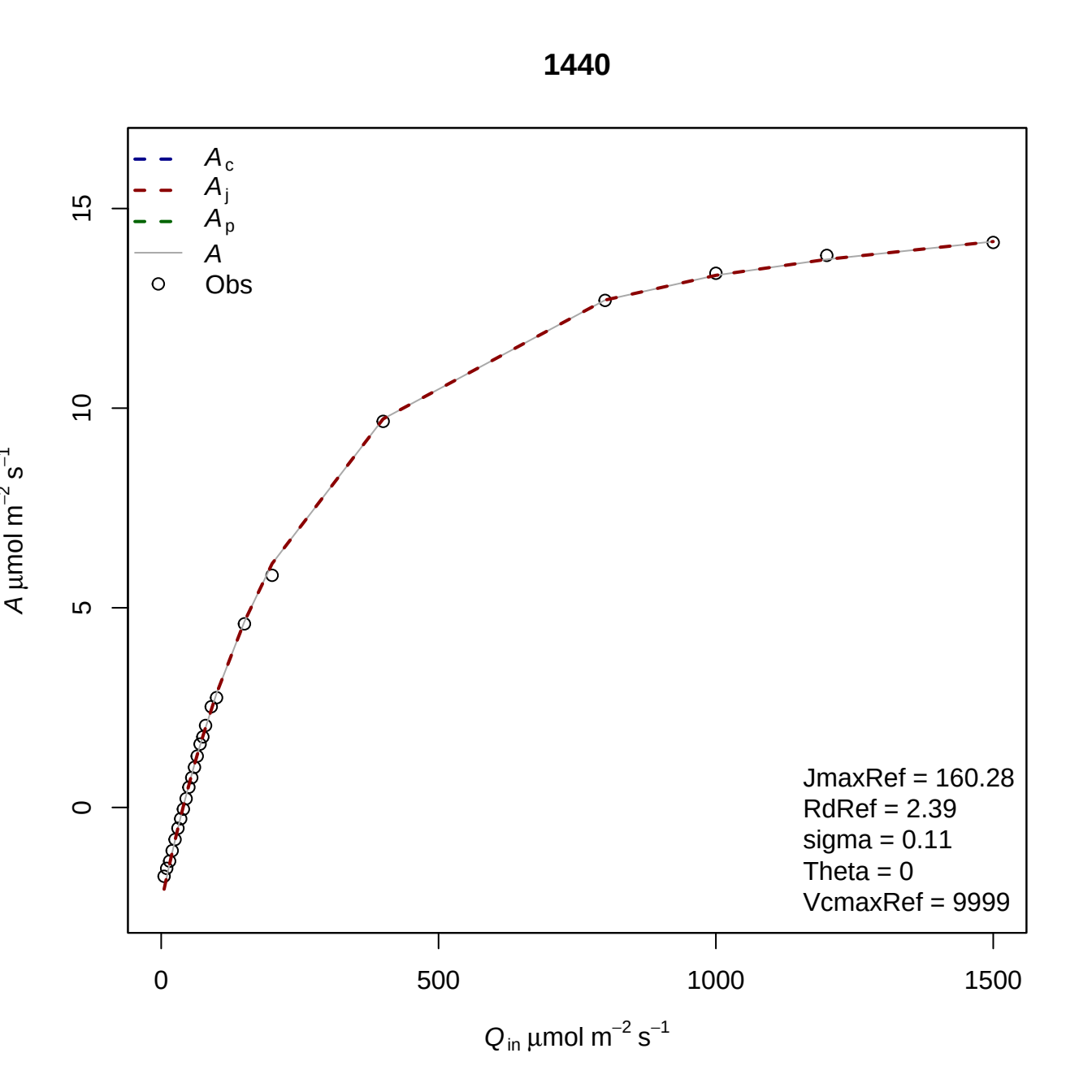

1435
A_c
A_j
A_p
A
Obs
JmaxRef = 76.2
RdRef = 5.03
sigma = 0.35
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

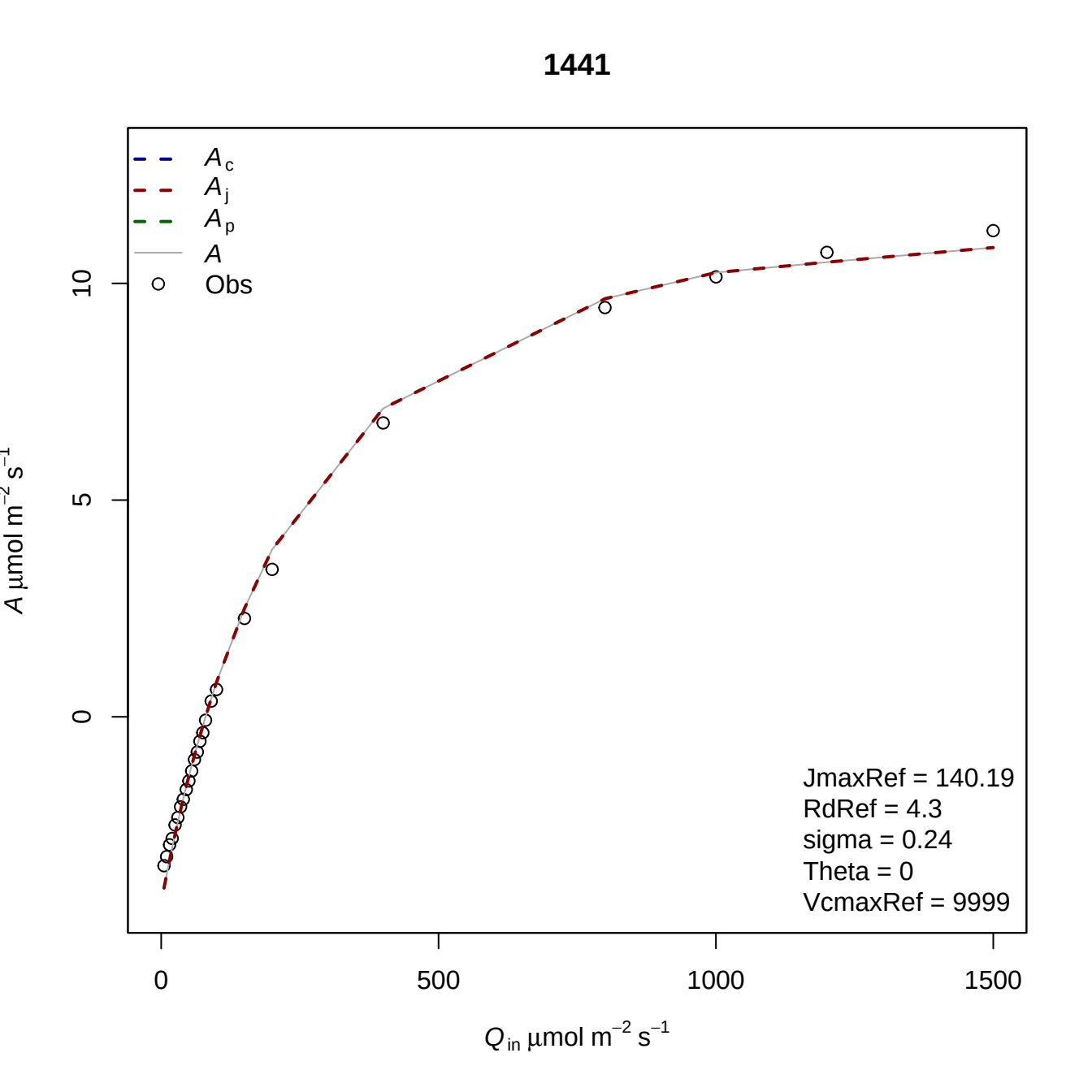

1436
A_c
A_j
A_p
A
Obs
JmaxRef = 162.57
RdRef = 1.55
sigma = 0.27
Theta = 0
VcmaxRef = 88.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

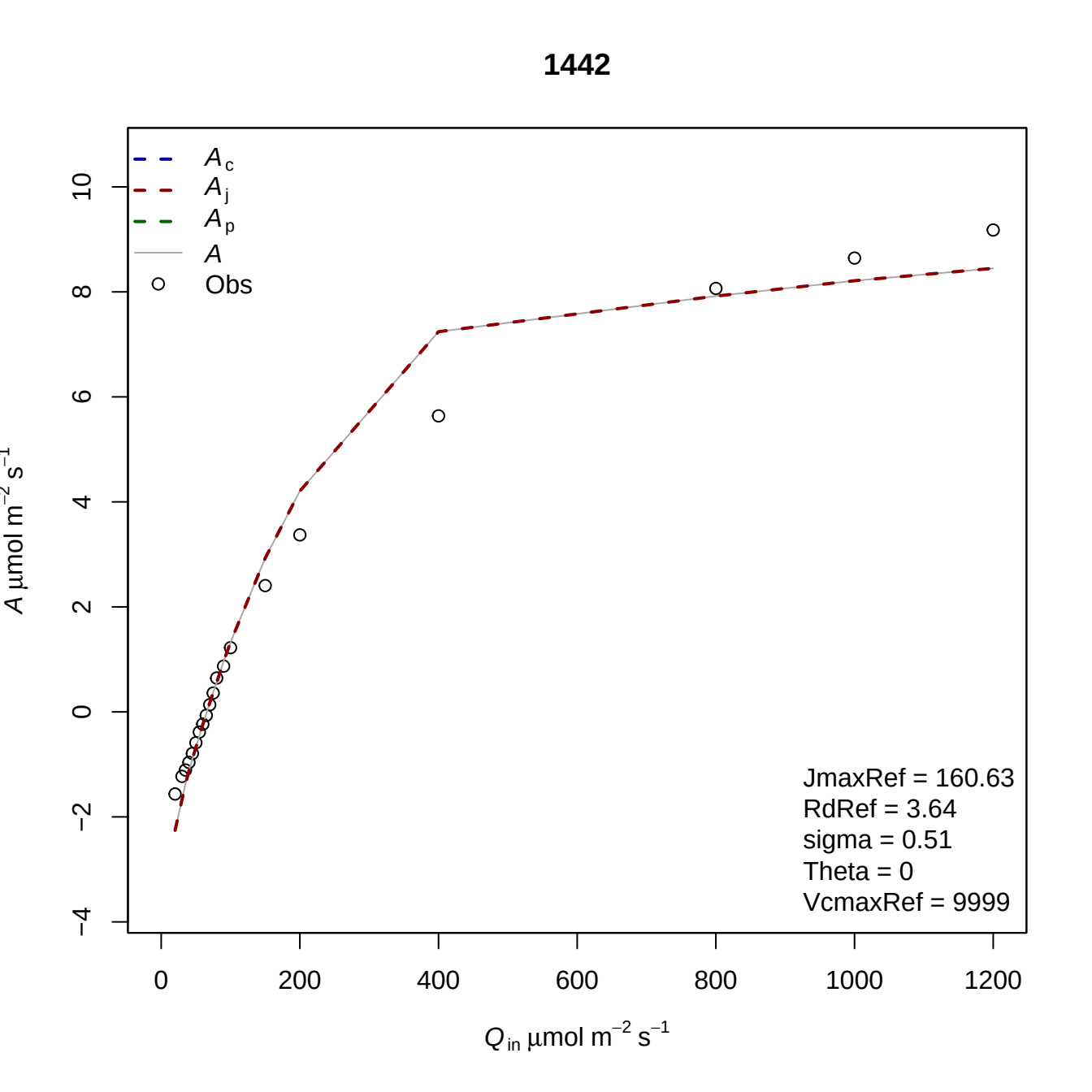

1437
A_c
A_j
A_p
A
Obs
JmaxRef = 74.01
RdRef = 2.26
sigma = 0.39
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

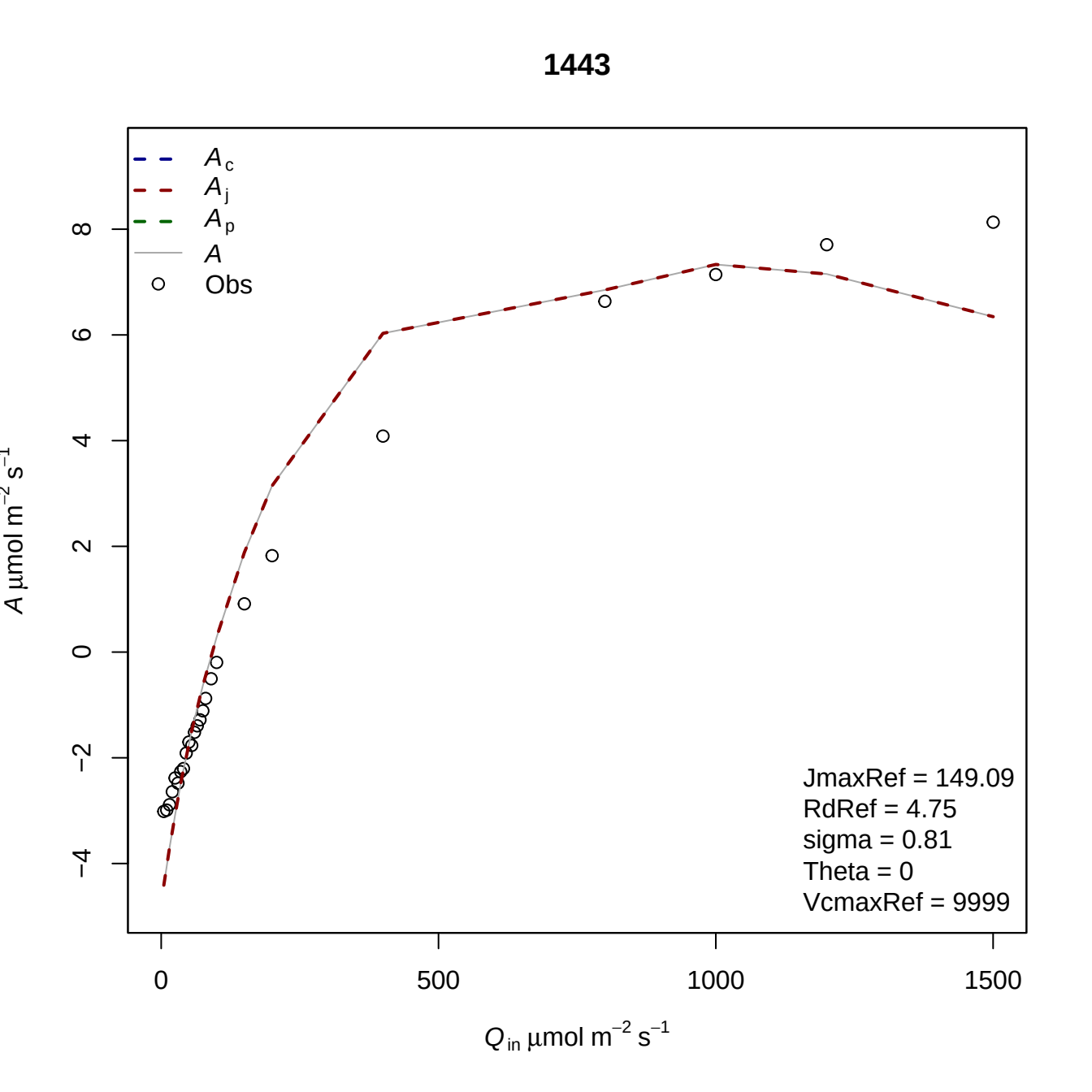

1438
A_c
A_j
A_p
A
Obs
JmaxRef = 175.78
RdRef = 3.36
sigma = 0.39
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

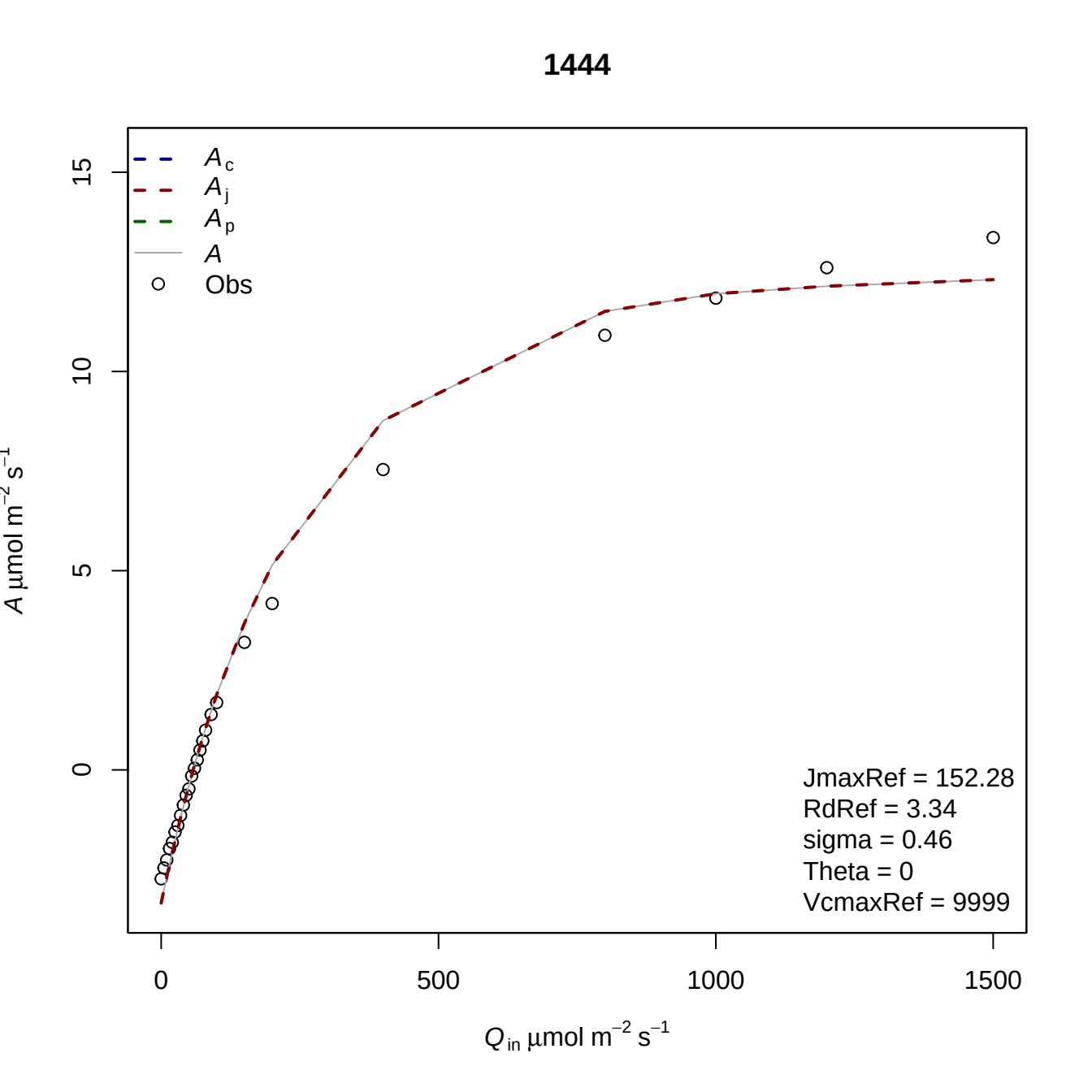

1439
A_c
A_j
A_p
A
Obs
JmaxRef = 121.14
RdRef = 2.95
sigma = 0.18
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

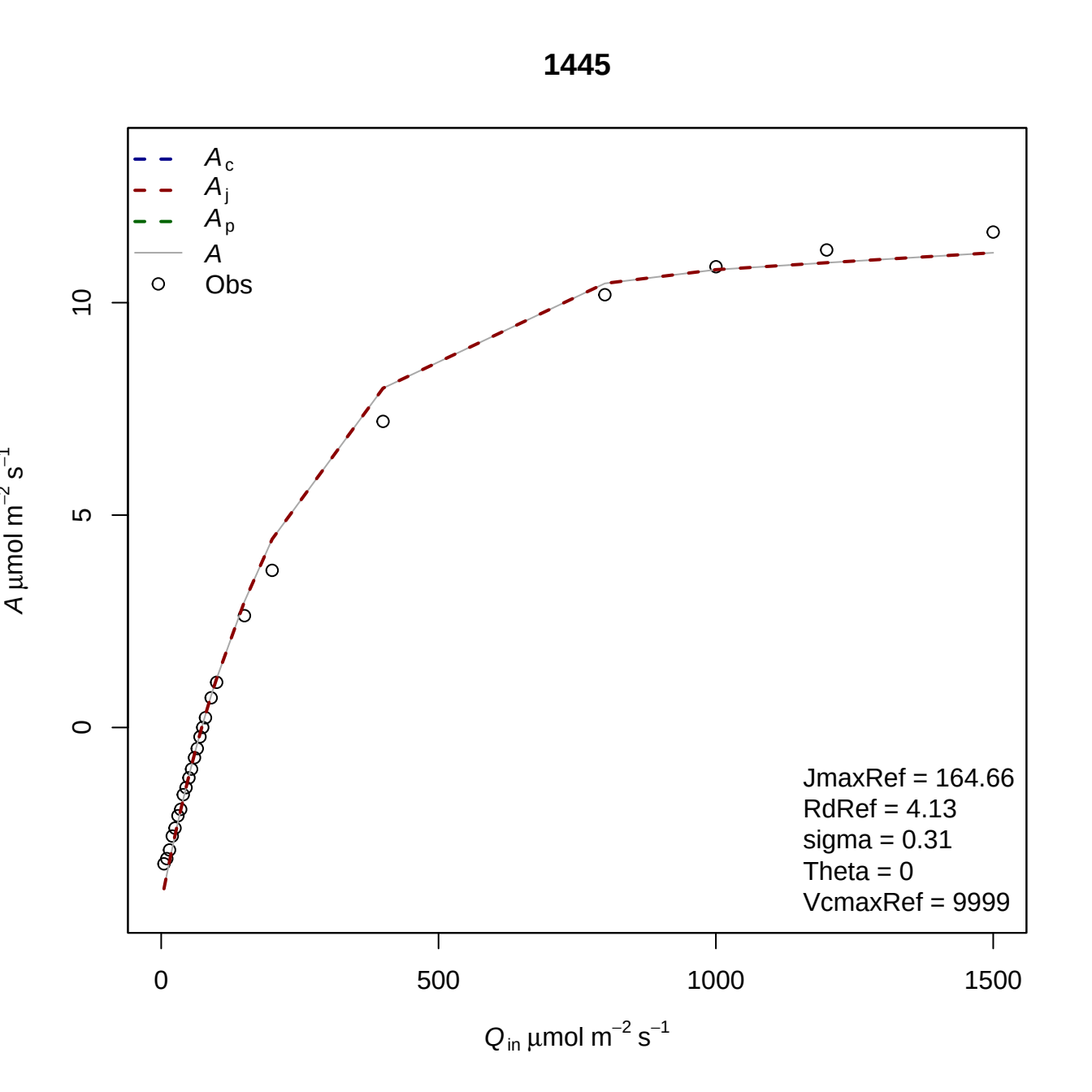

1440
A_c
A_j
A_p
A
Obs
JmaxRef = 160.28
RdRef = 2.39
sigma = 0.11
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

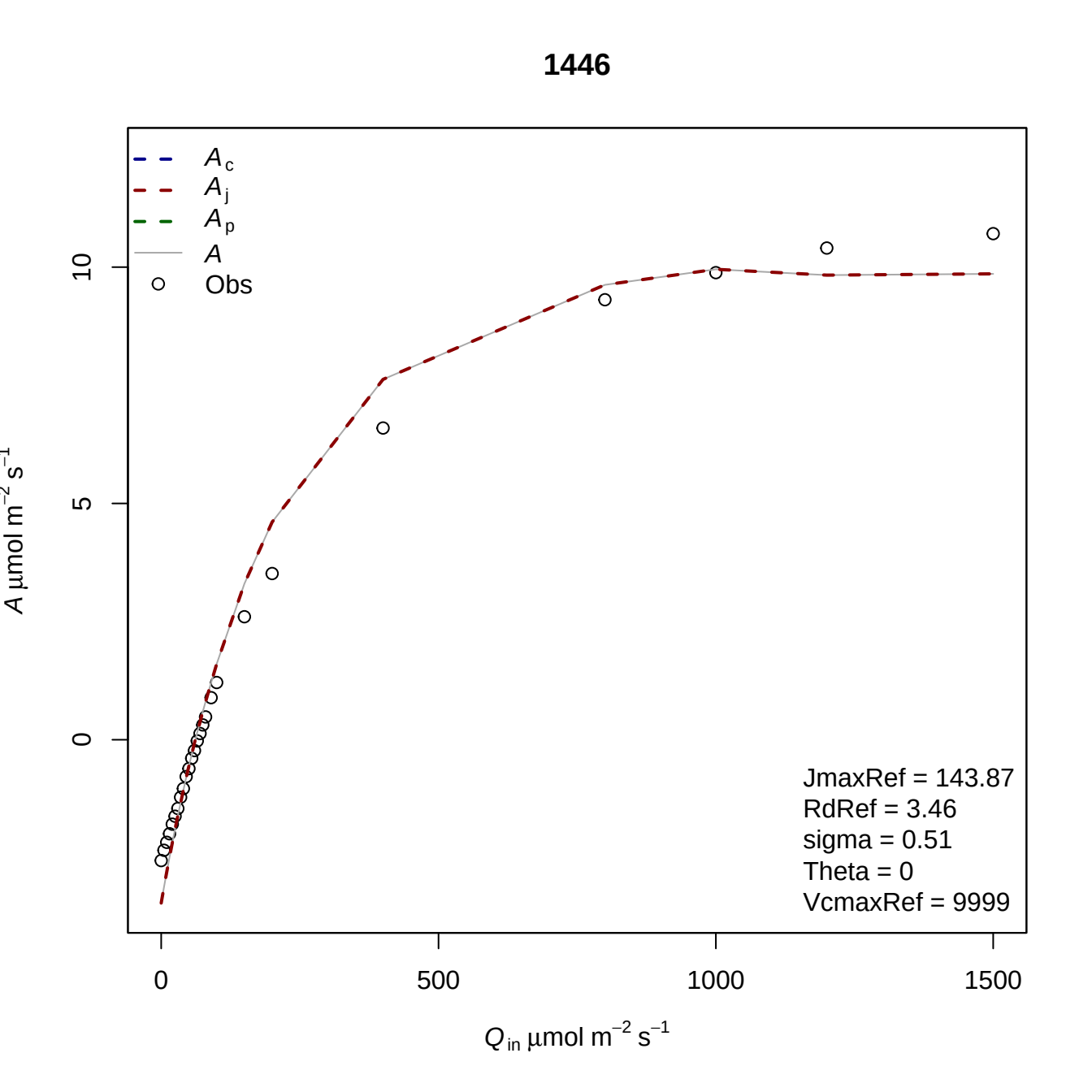

1441
A_c
A_j
A_p
A
Obs
JmaxRef = 140.19
RdRef = 4.3
sigma = 0.24
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

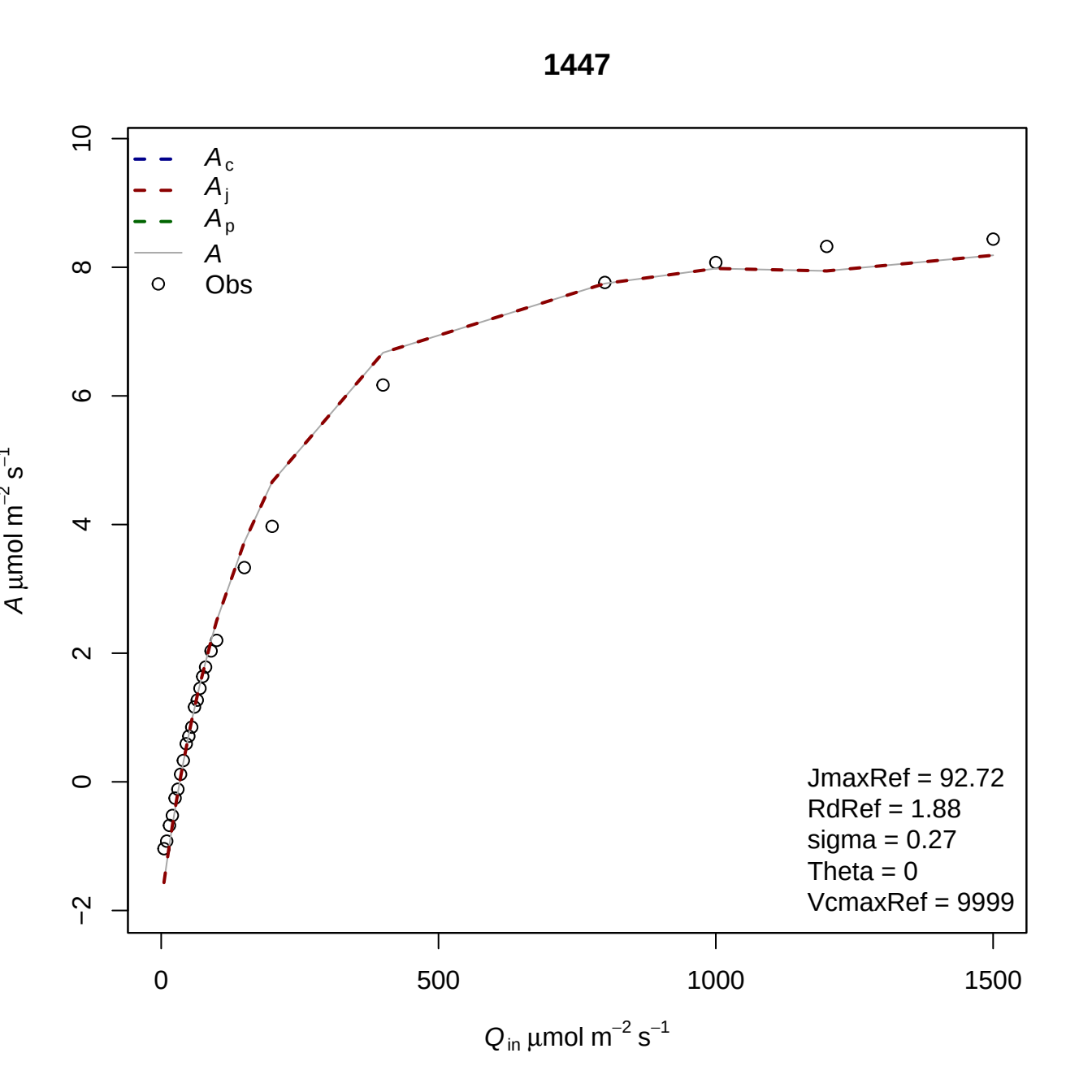

1442
A_c
A_j
A_p
A
Obs
JmaxRef = 160.63
RdRef = 3.64
sigma = 0.51
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

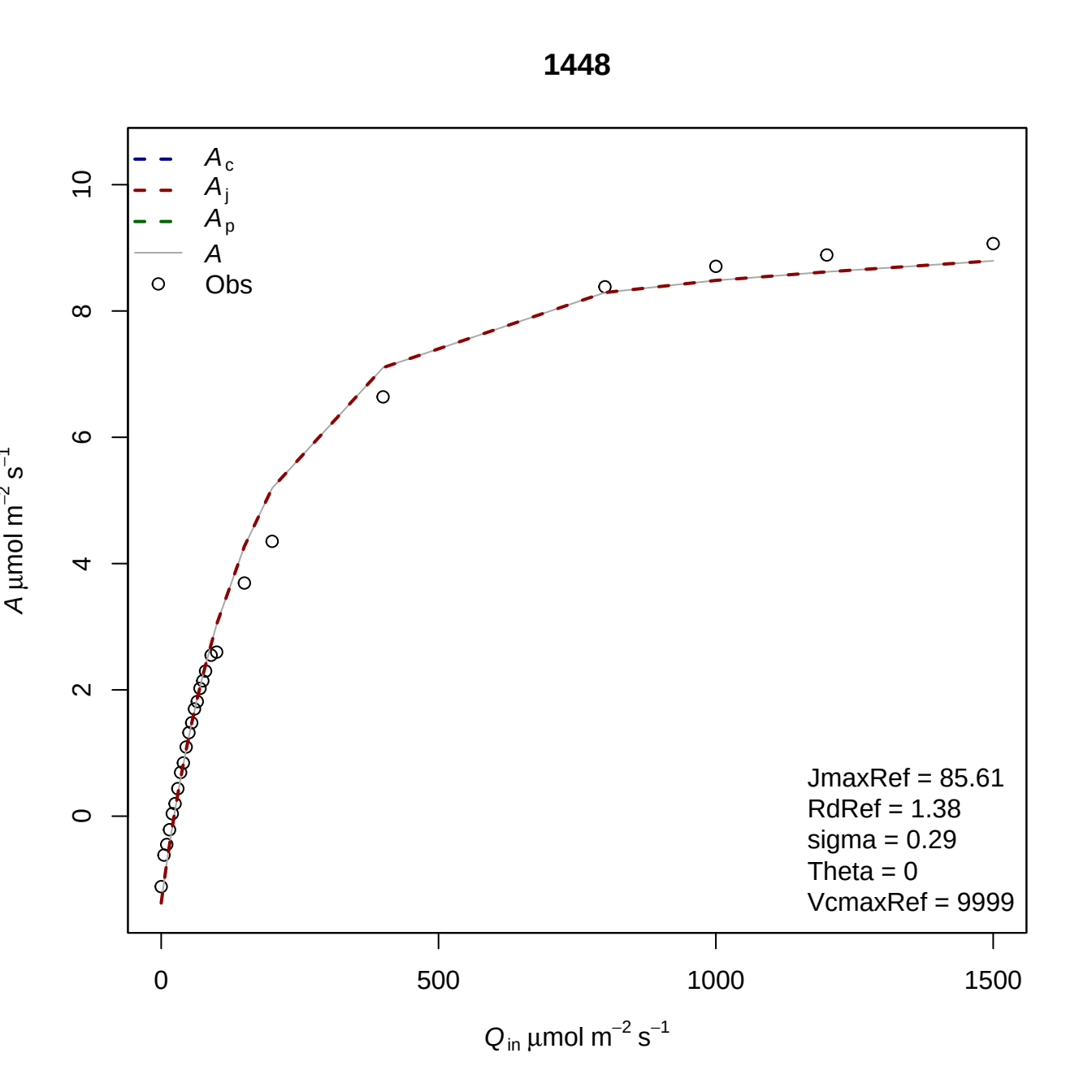

1443
A_c
A_j
A_p
A
Obs
JmaxRef = 149.09
RdRef = 4.75
sigma = 0.81
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

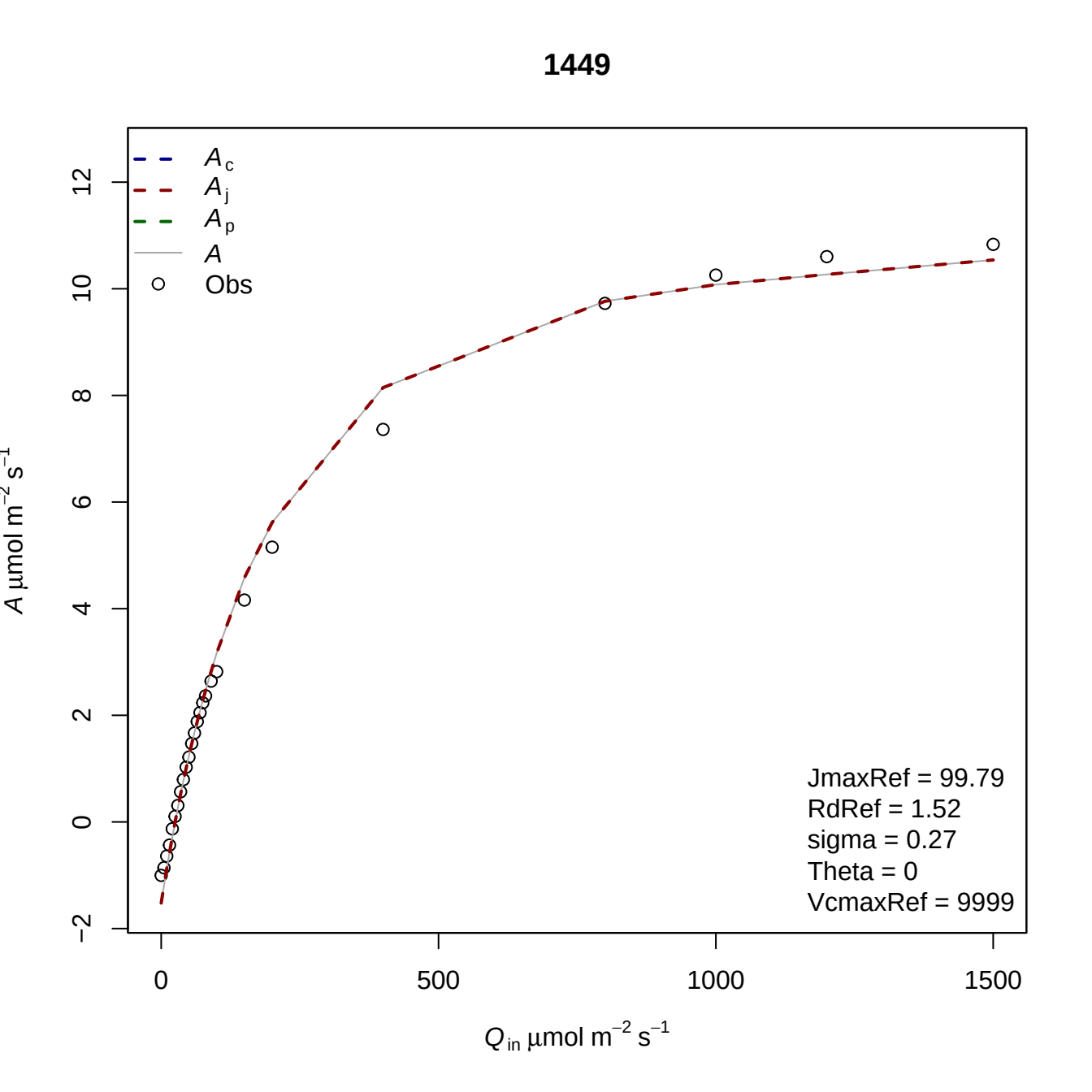

1444
A_c
A_j
A_p
A
Obs
JmaxRef = 152.28
RdRef = 3.34
sigma = 0.46
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

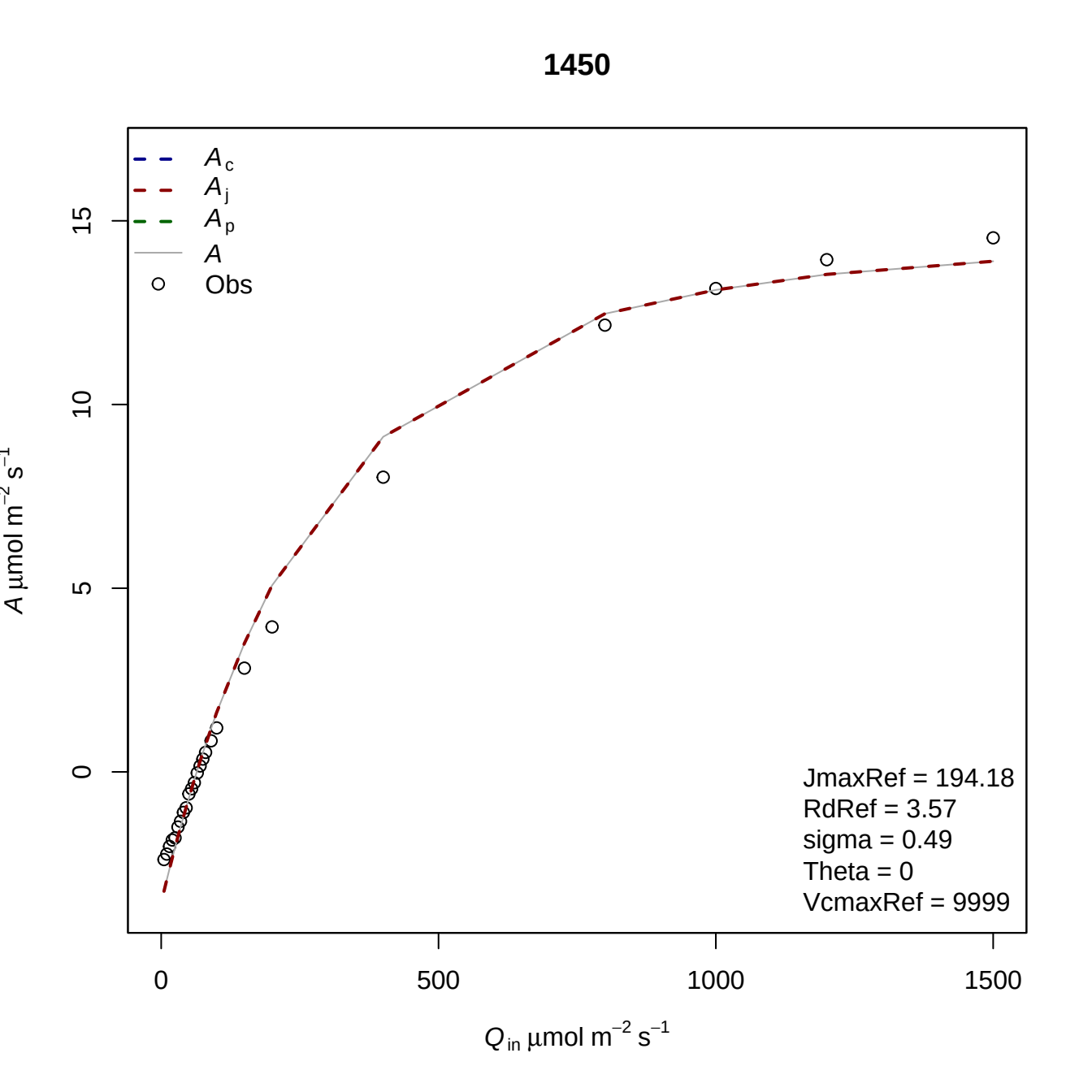

1445
A_c
A_j
A_p
A
Obs
JmaxRef = 164.66
RdRef = 4.13
sigma = 0.31
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

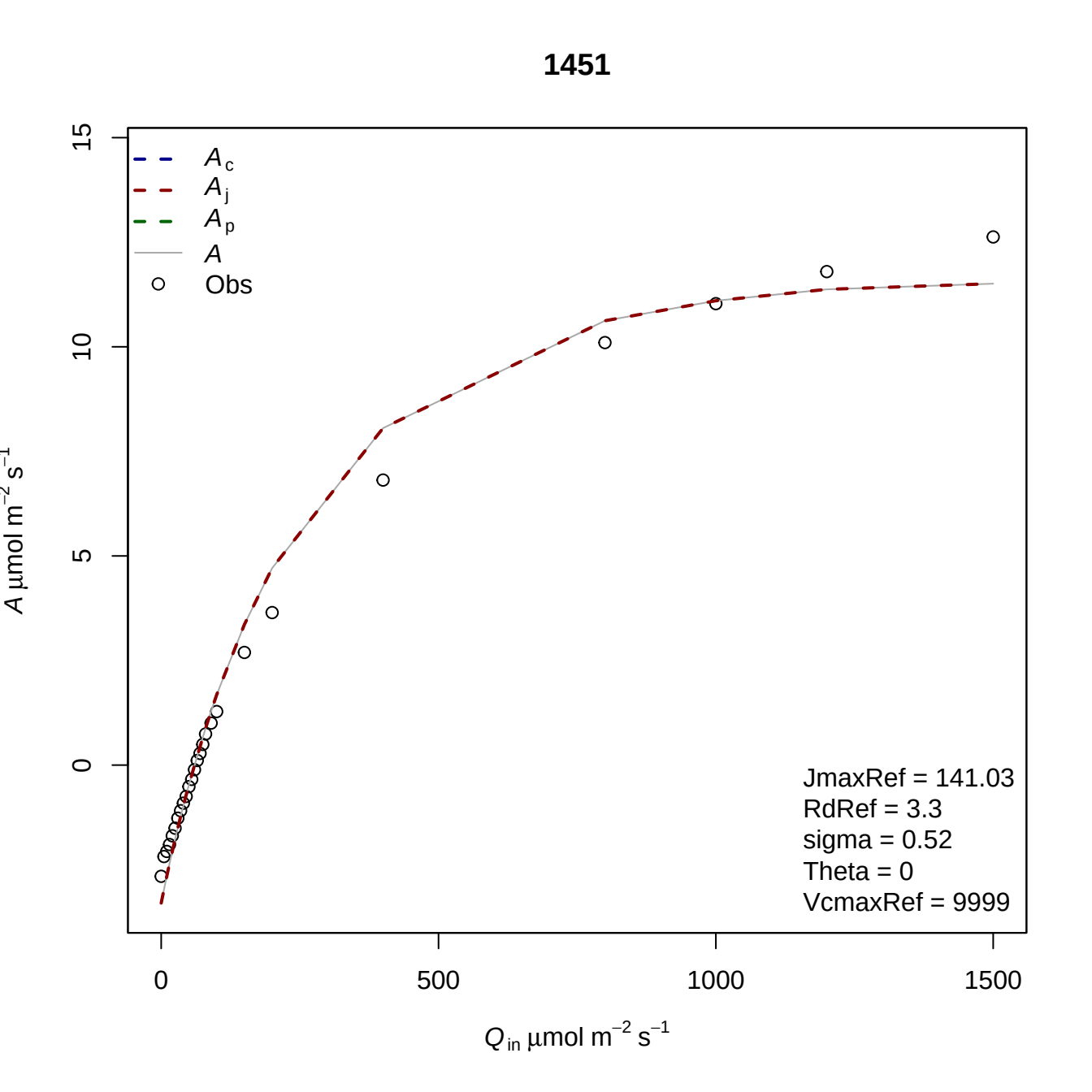

1446
A_c
A_j
A_p
A
Obs
JmaxRef = 143.87
RdRef = 3.46
sigma = 0.51
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

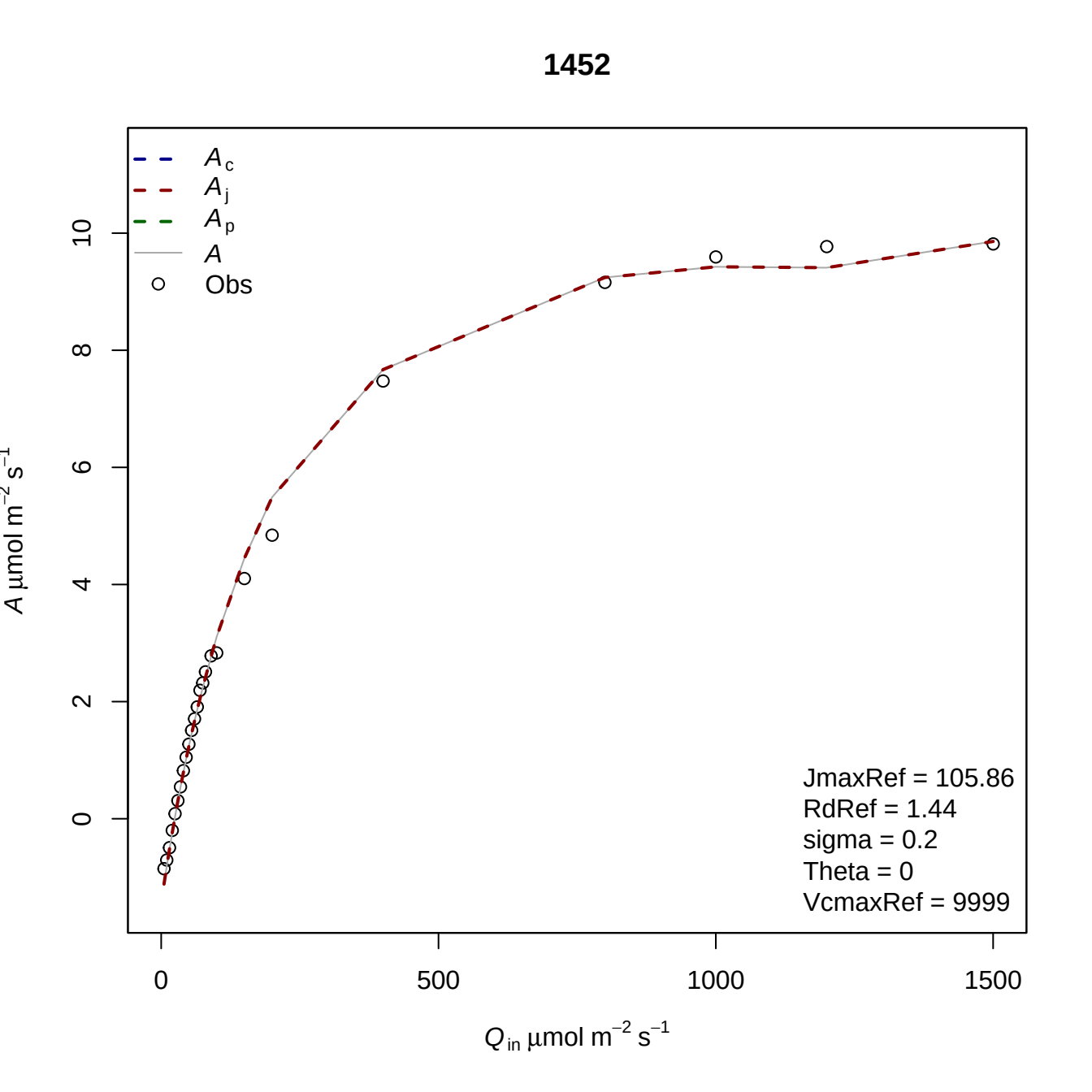

1447
A_c
A_j
A_p
A
Obs
JmaxRef = 92.72
RdRef = 1.88
sigma = 0.27
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

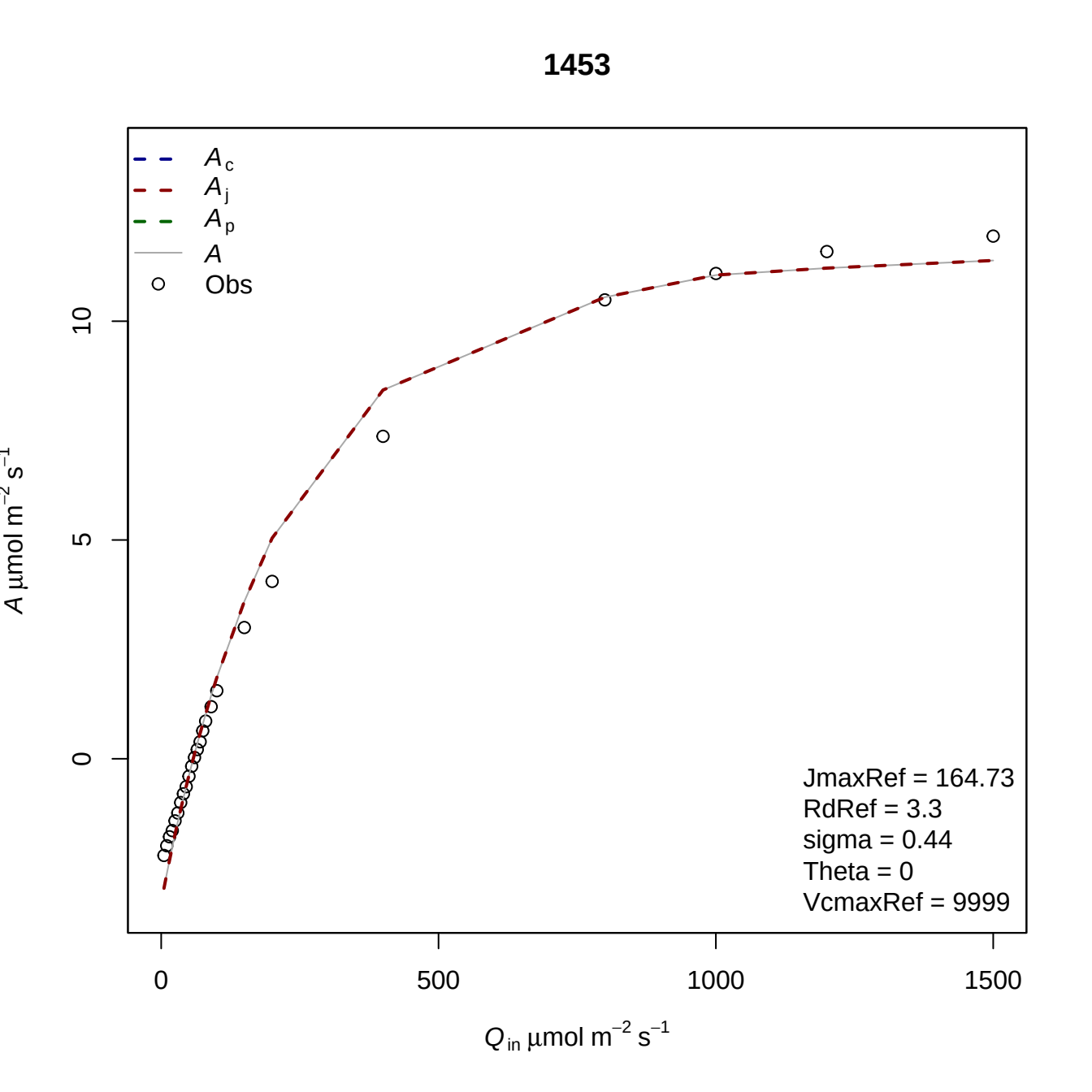

1448
A_c
A_j
A_p
A
Obs
JmaxRef = 85.61
RdRef = 1.38
sigma = 0.29
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

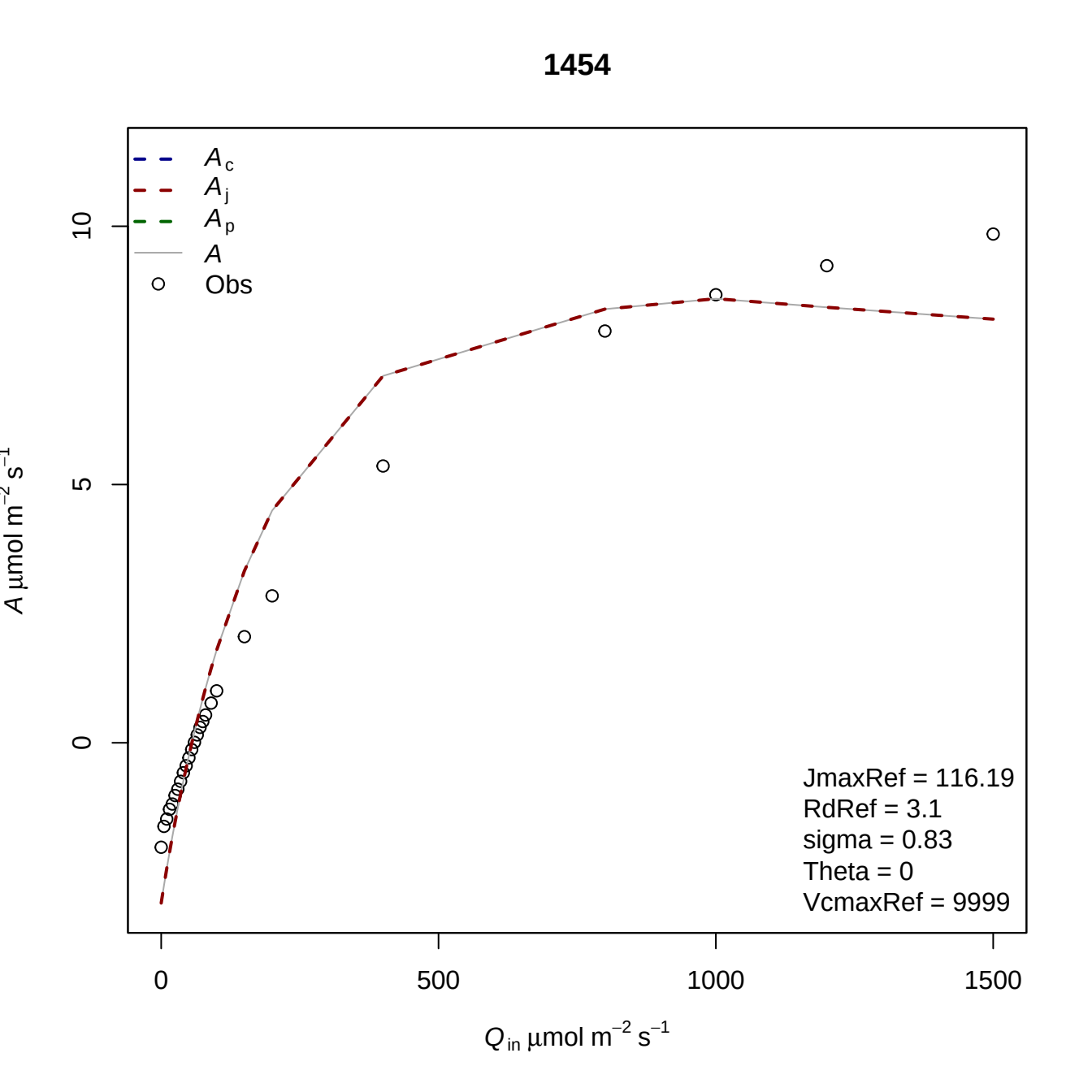

1449
A_c
A_j
A_p
A
Obs
JmaxRef = 99.79
RdRef = 1.52
sigma = 0.27
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

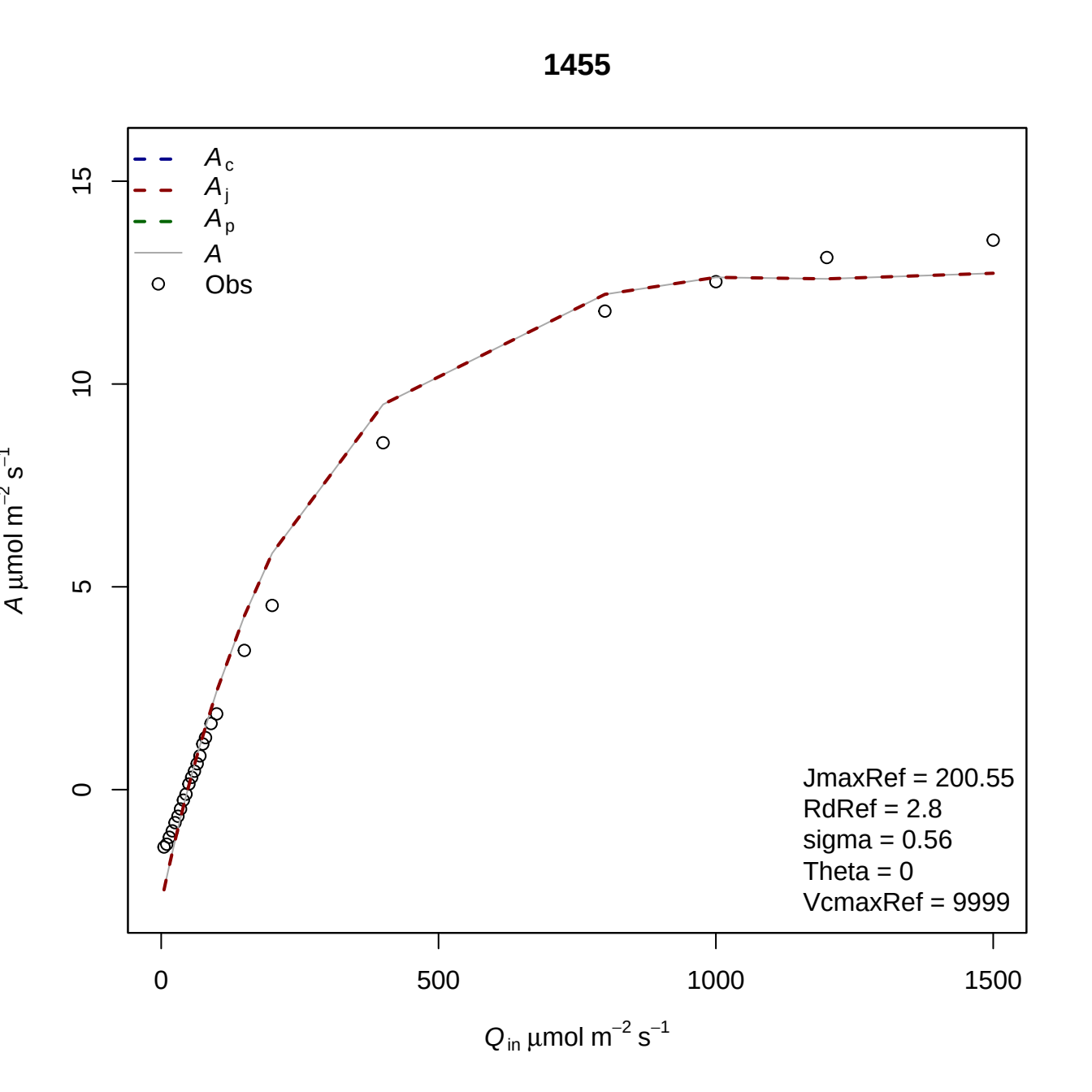

1450
A_c
A_j
A_p
A
Obs
JmaxRef = 194.18
RdRef = 3.57
sigma = 0.49
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

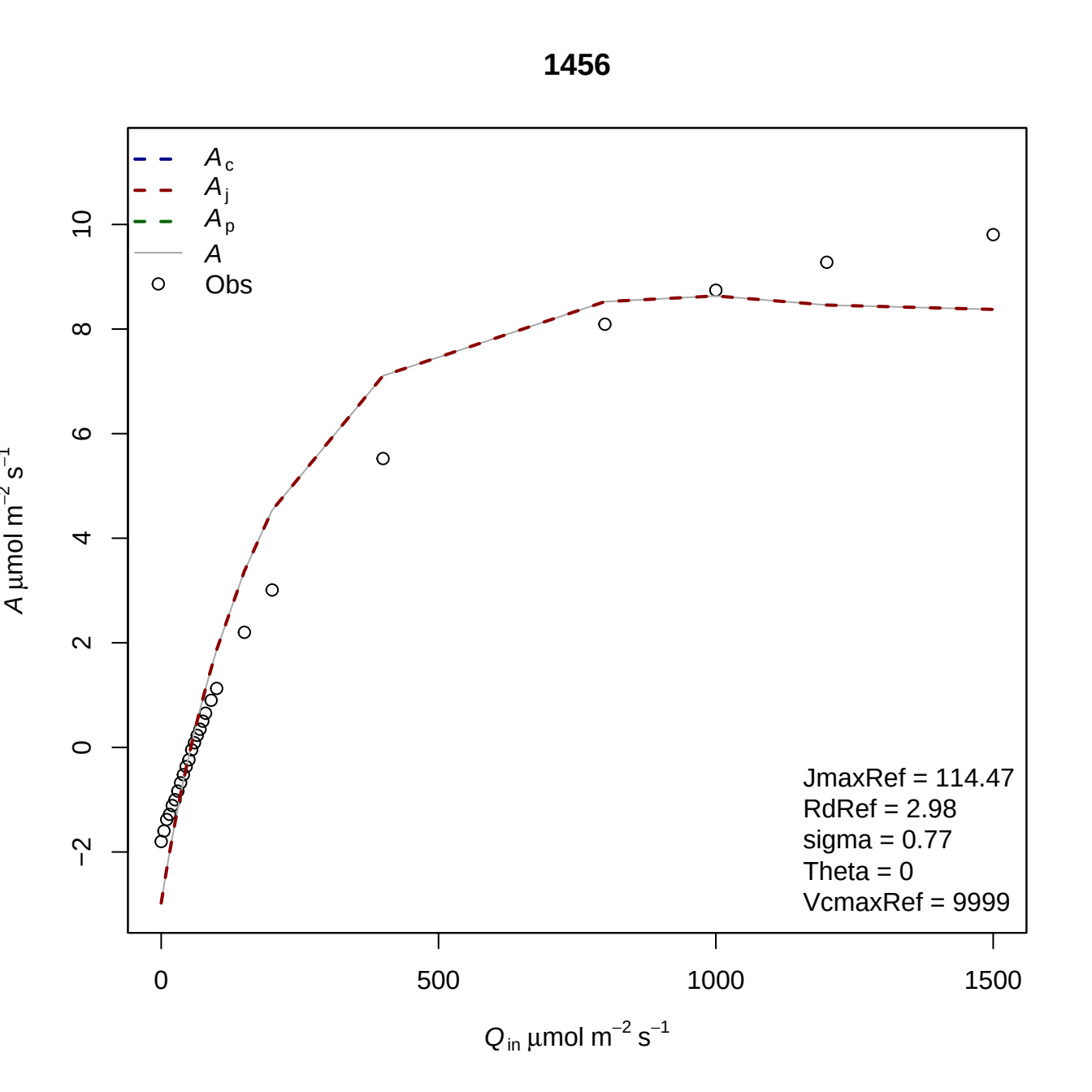

1451
A_c
A_j
A_p
A
Obs
JmaxRef = 141.03
RdRef = 3.3
sigma = 0.52
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

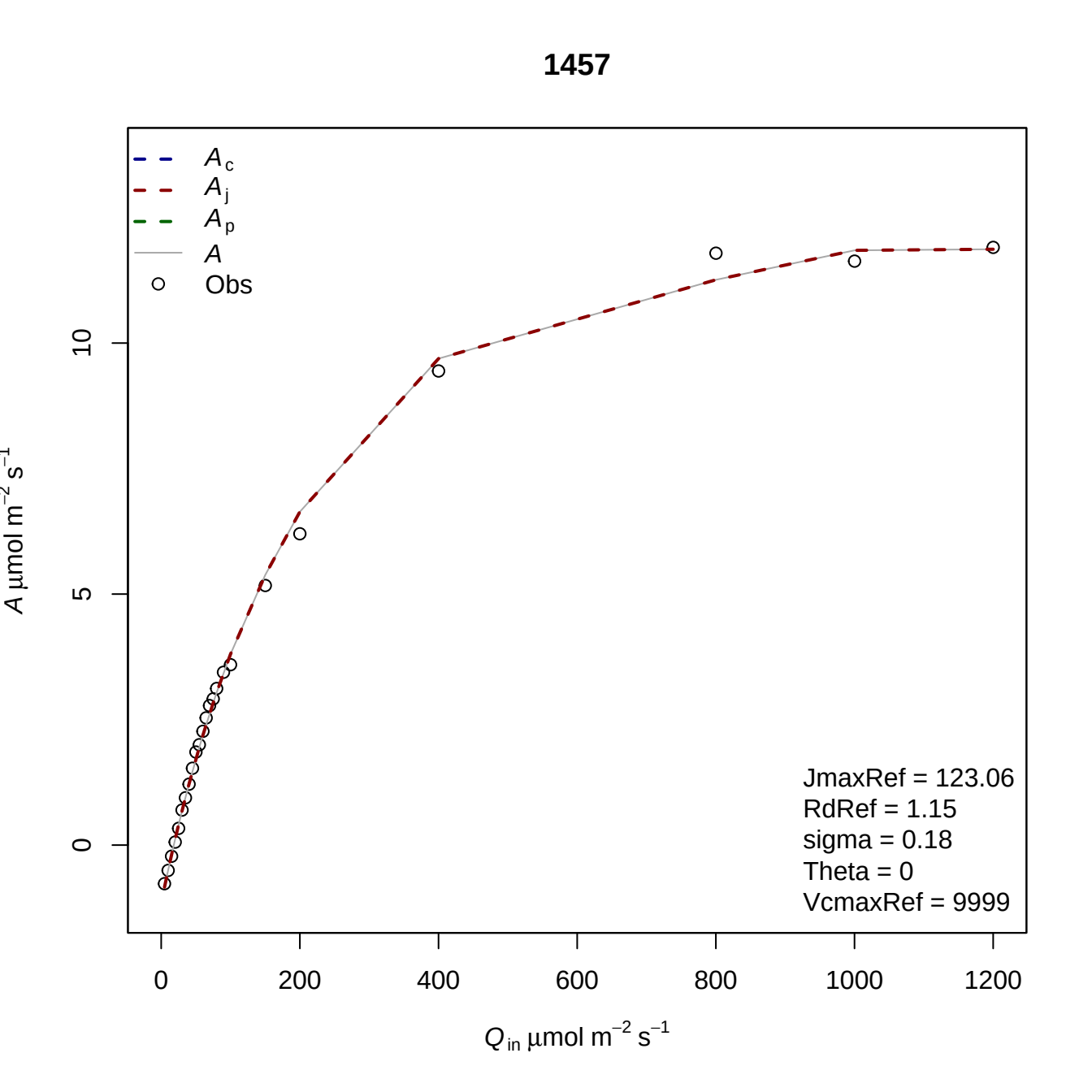

1452
A_c
A_j
A_p
A
Obs
JmaxRef = 105.86
RdRef = 1.44
sigma = 0.2
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

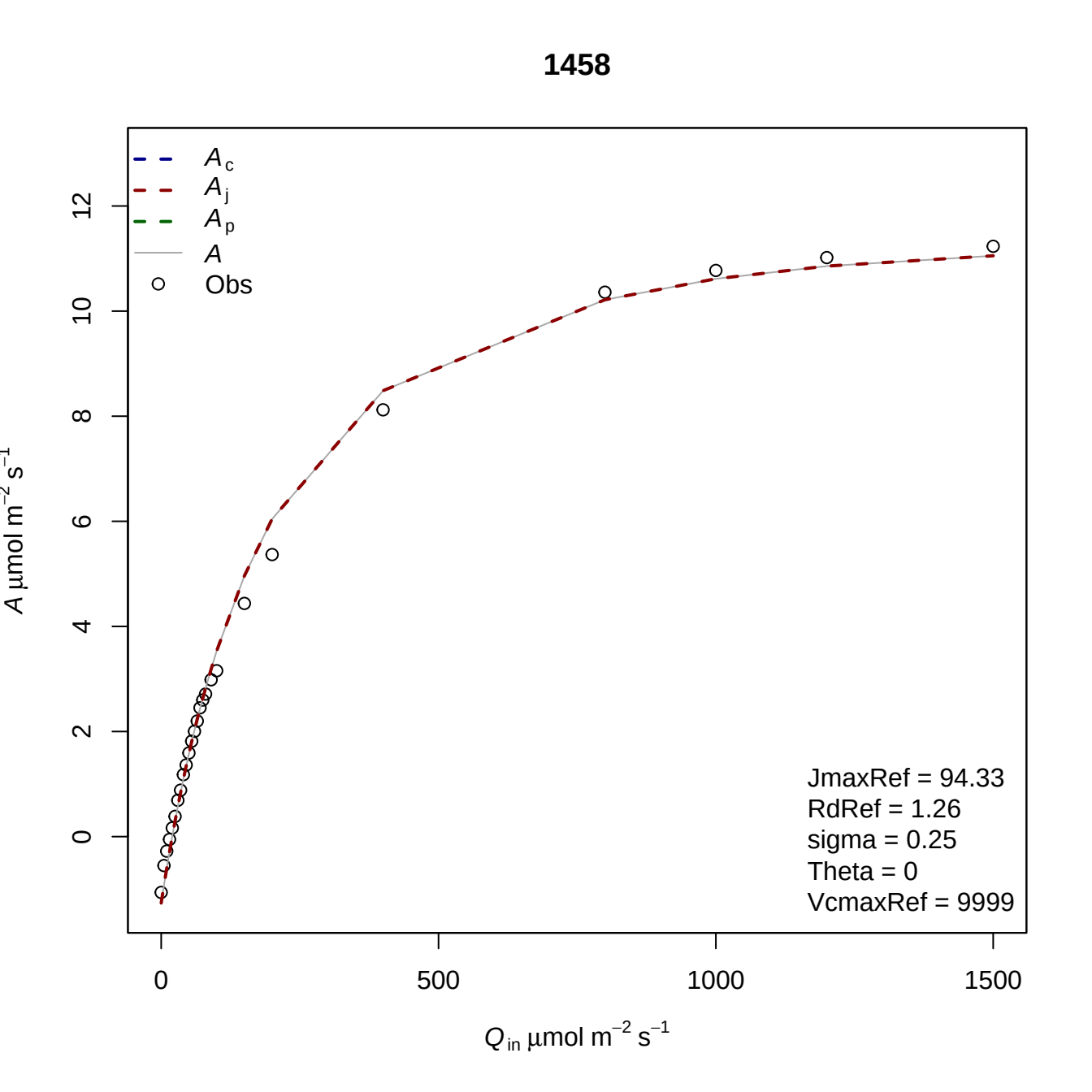

1453
A_c
A_j
A_p
A
Obs
JmaxRef = 164.73
RdRef = 3.3
sigma = 0.44
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

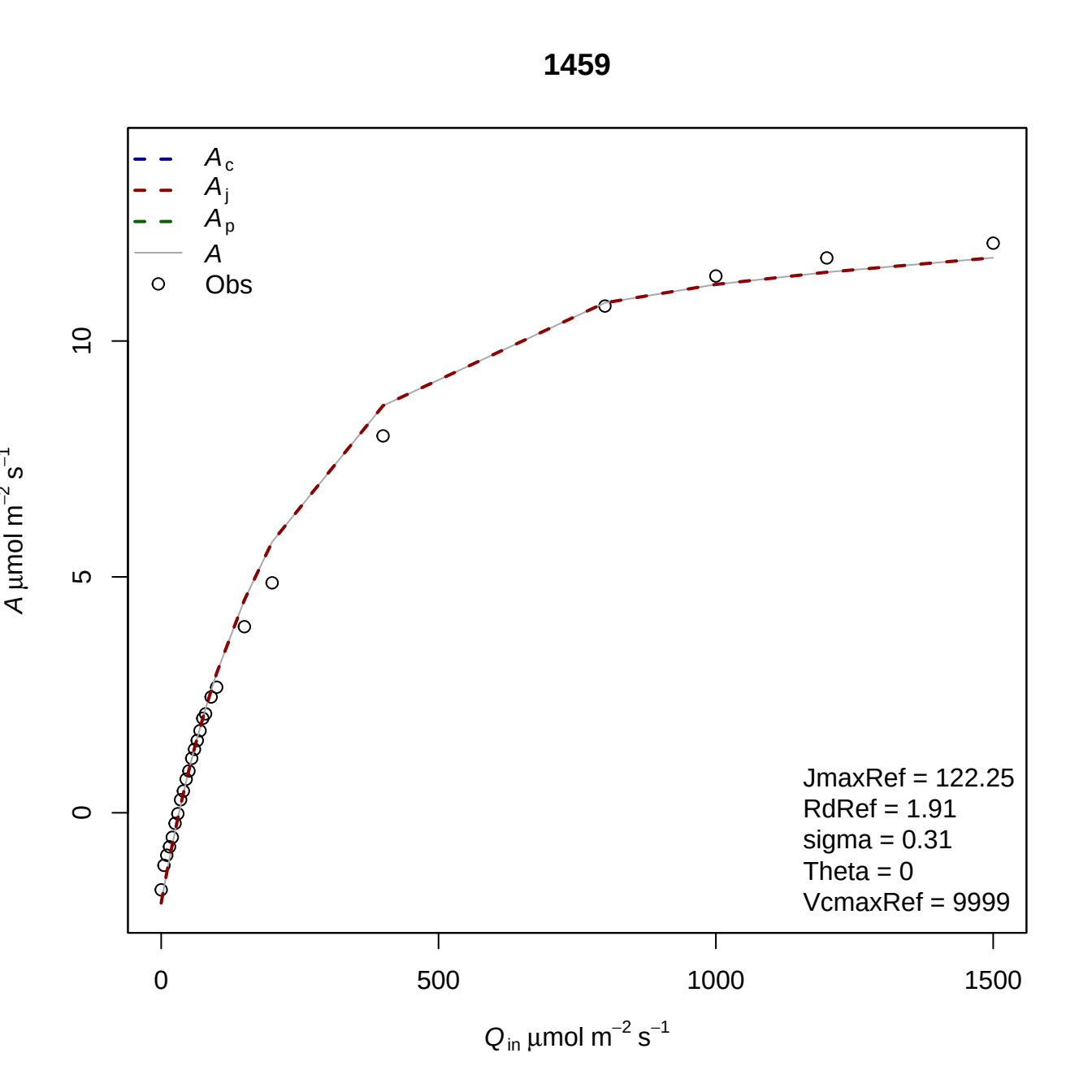

1454
A_c
A_j
A_p
A
Obs
JmaxRef = 116.19
RdRef = 3.1
sigma = 0.83
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

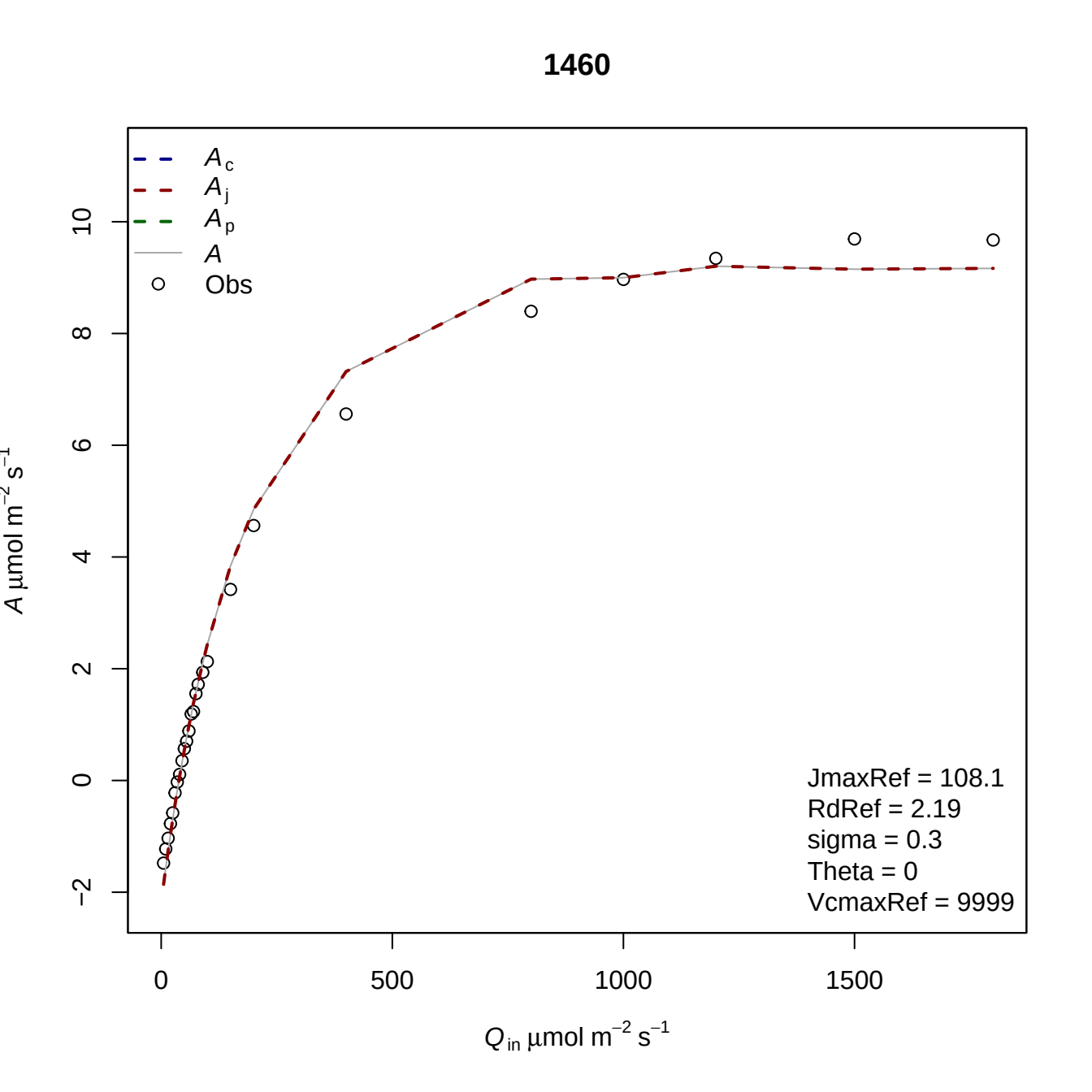

1455
A_c
A_j
A_p
A
Obs
JmaxRef = 200.55
RdRef = 2.8
sigma = 0.56
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

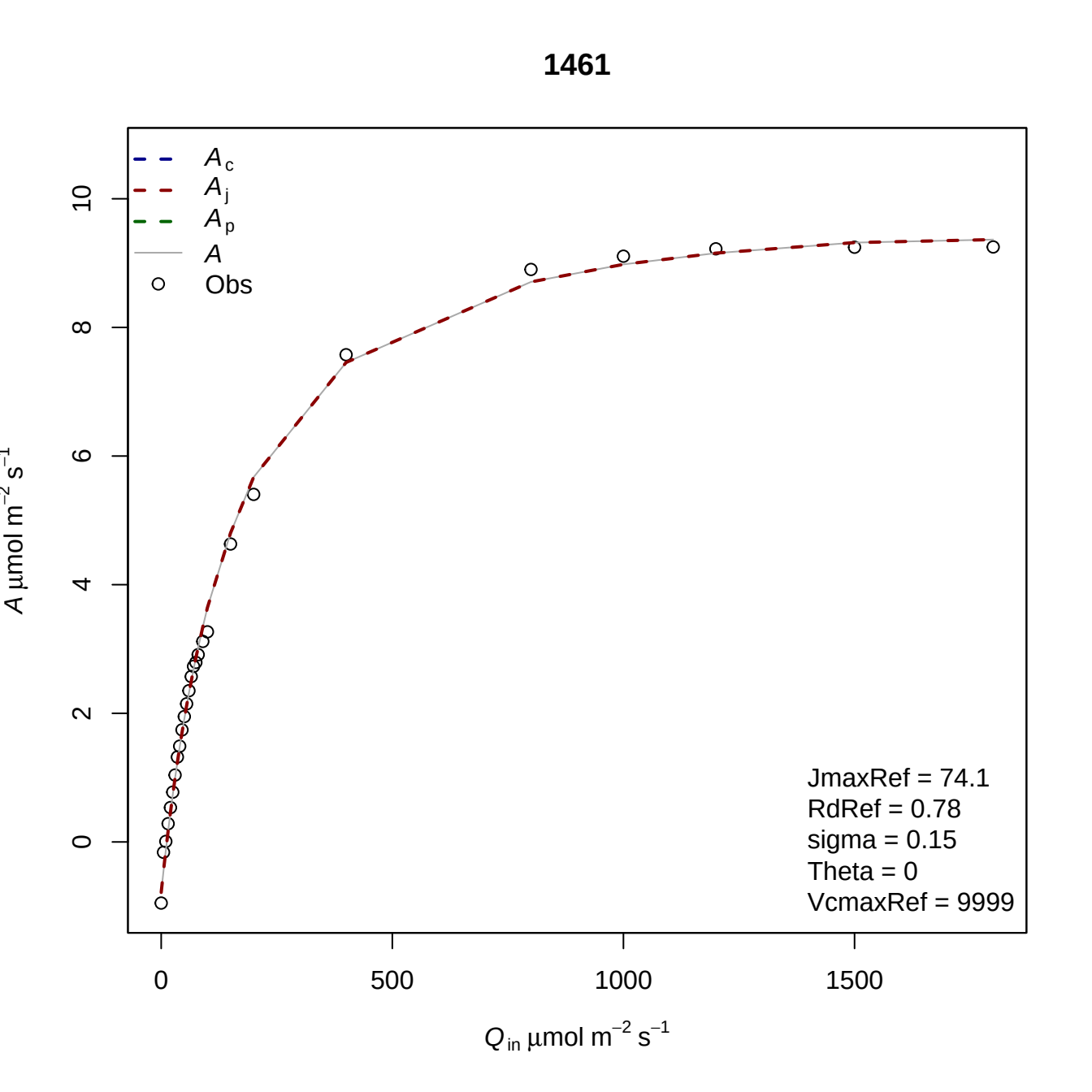

1456
A_c
A_j
A_p
A
Obs
JmaxRef = 114.47
RdRef = 2.98
sigma = 0.77
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

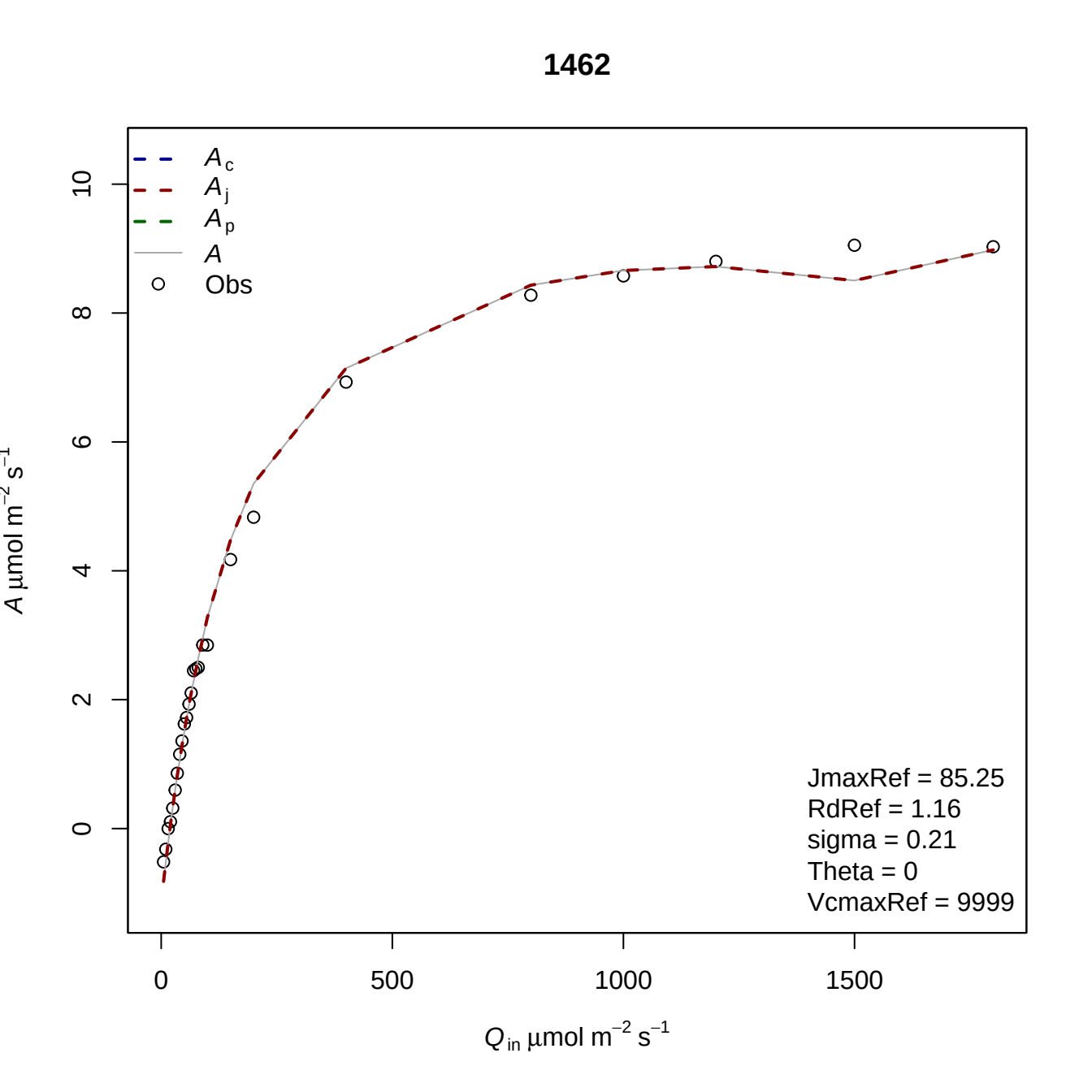

1457
A_c
A_j
A_p
A
Obs
JmaxRef = 123.06
RdRef = 1.15
sigma = 0.18
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

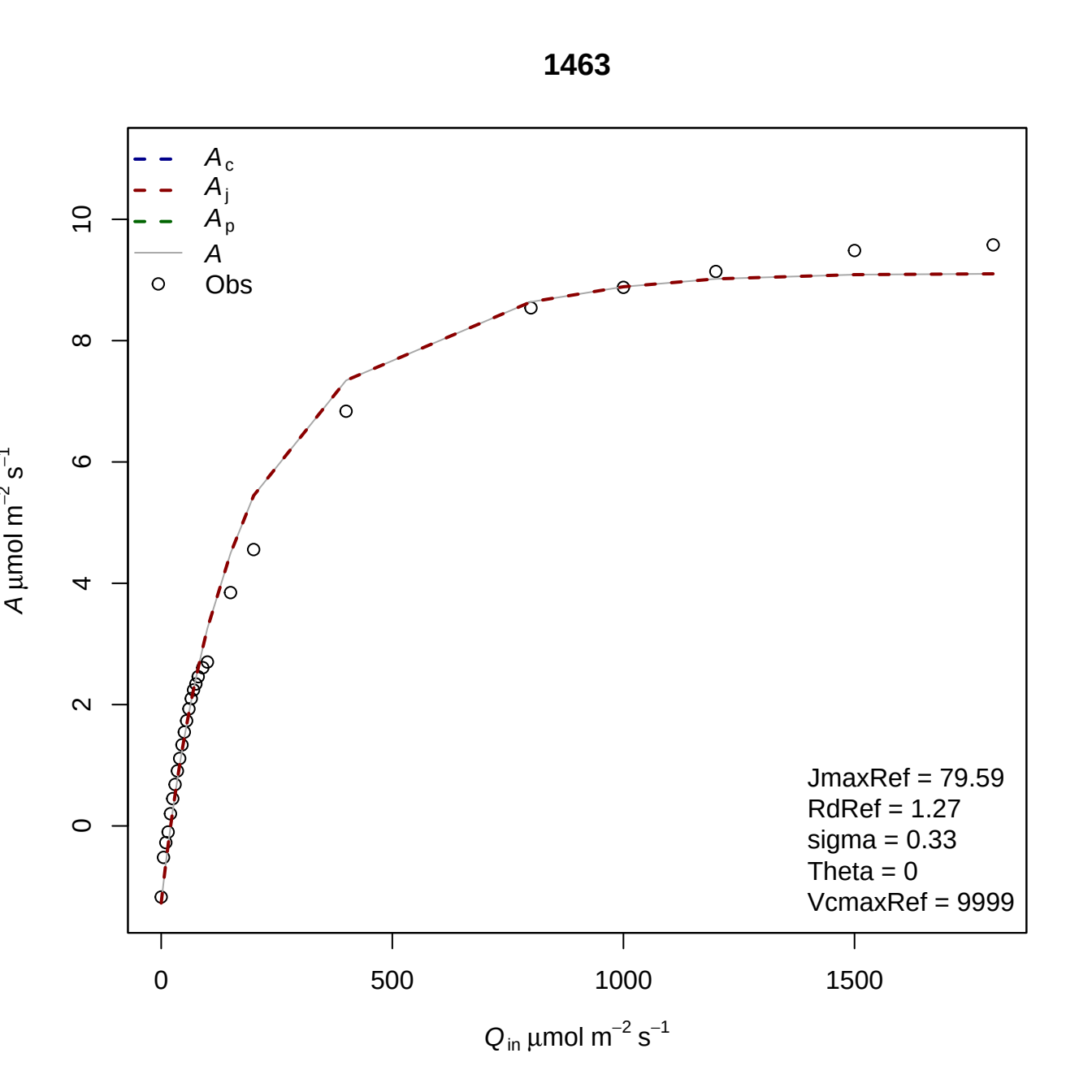

1458
A_c
A_j
A_p
A
Obs
JmaxRef = 94.33
RdRef = 1.26
sigma = 0.25
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

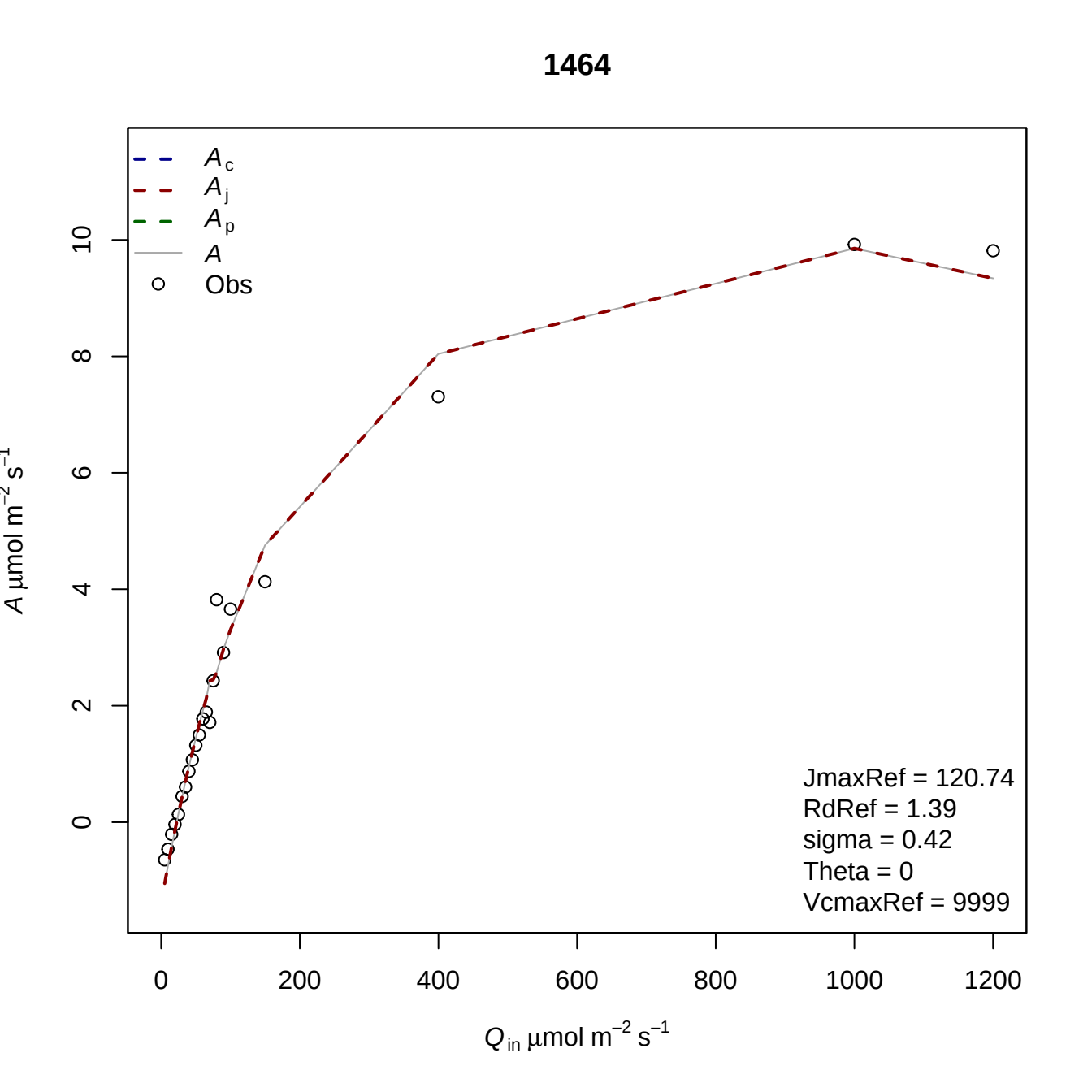

1459
A_c
A_j
A_p
A
Obs
JmaxRef = 122.25
RdRef = 1.91
sigma = 0.31
Theta = 0
VcmaxRef = 9999
0
5
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

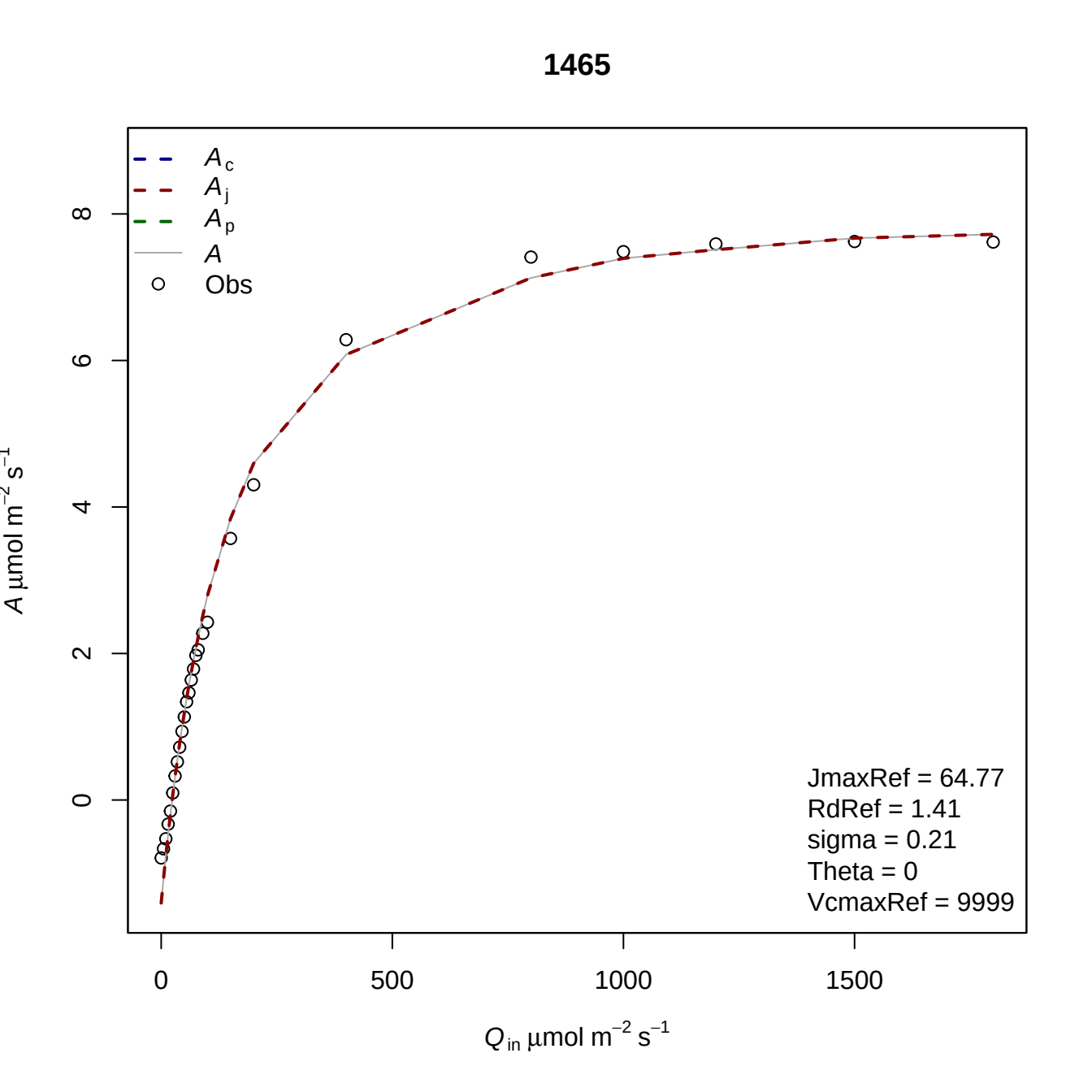

1460
A_c
A_j
A_p
A
Obs
JmaxRef = 108.1
RdRef = 2.19
sigma = 0.3
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

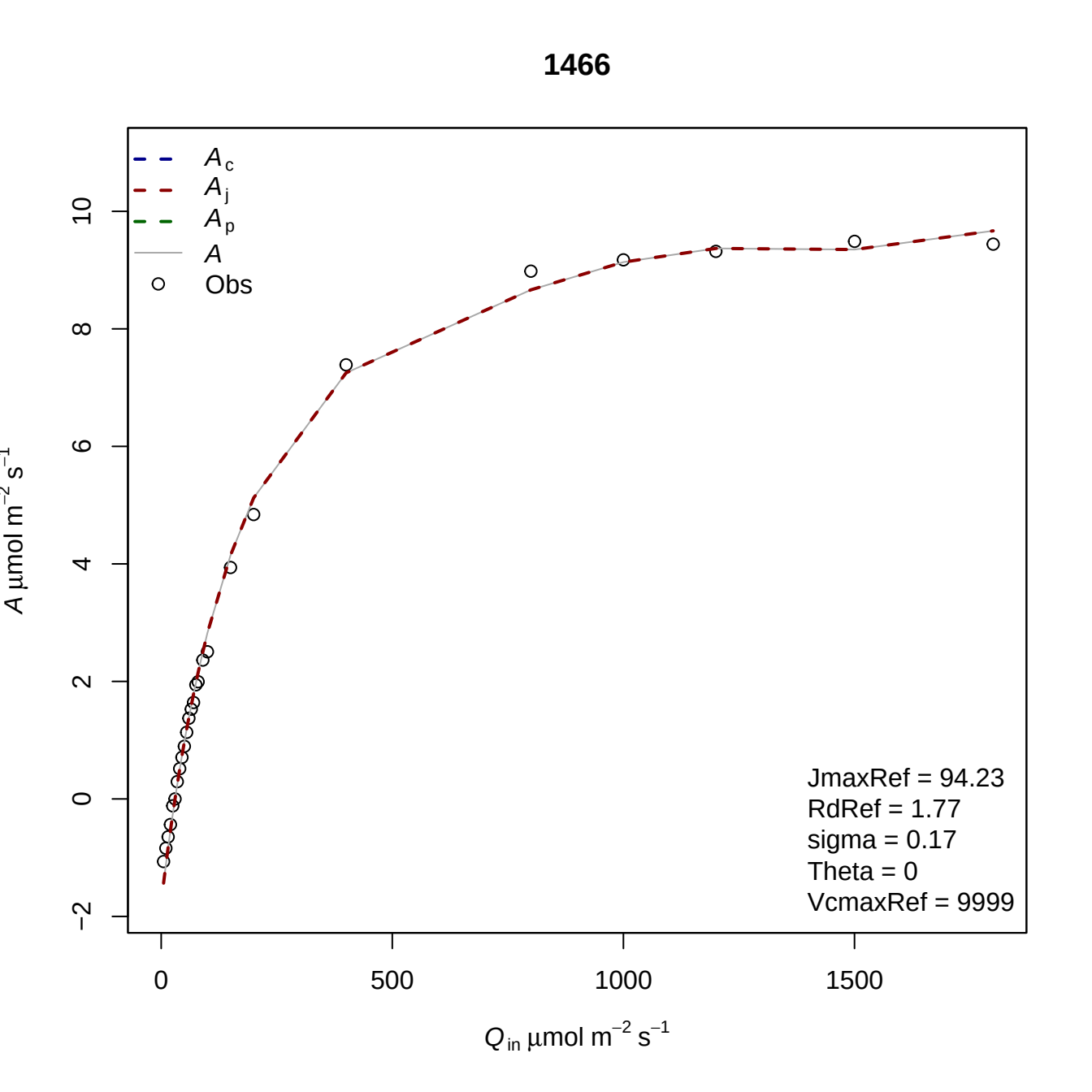

1461
A_c
A_j
A_p
A
Obs
JmaxRef = 74.1
RdRef = 0.78
sigma = 0.15
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

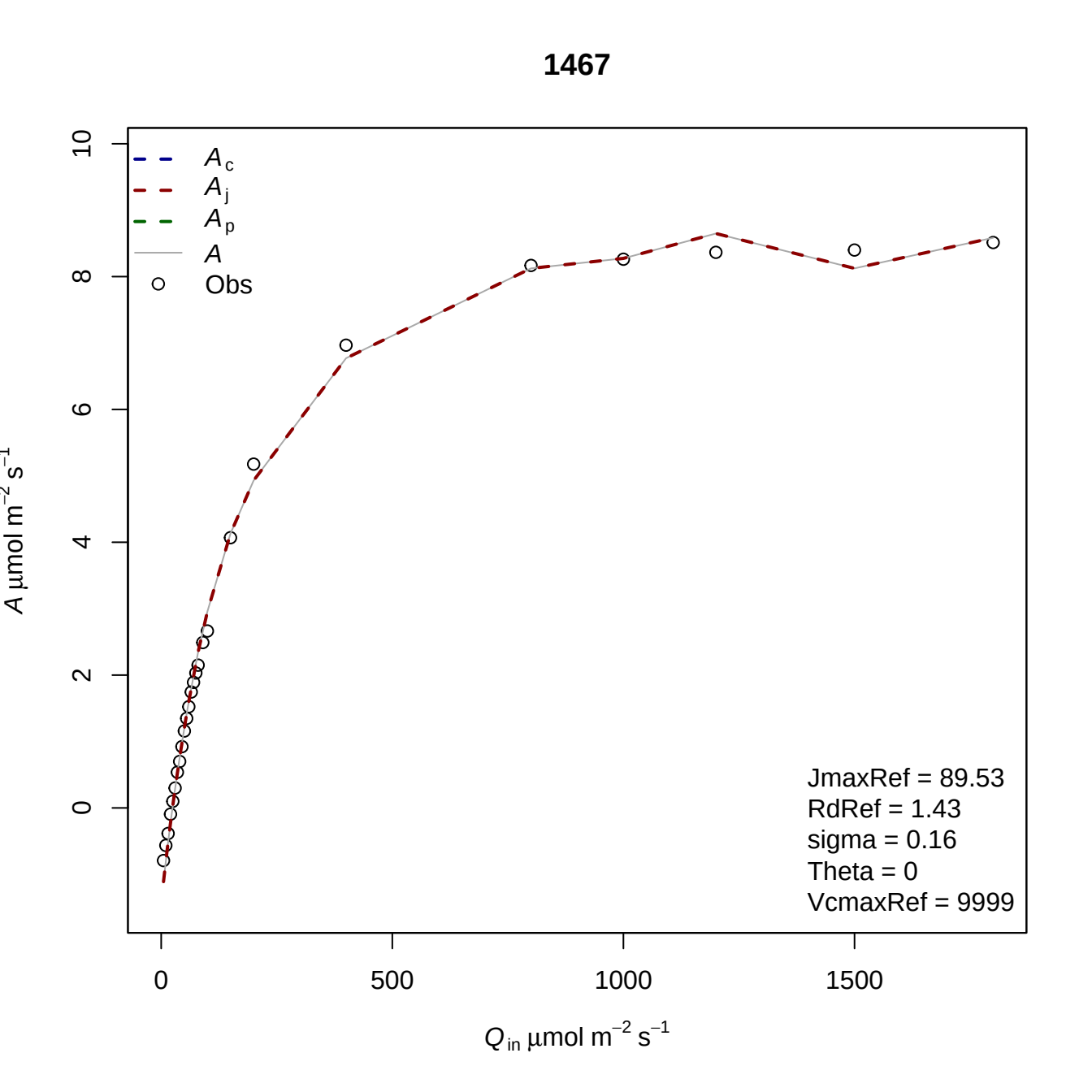

1462
A_c
A_j
A_p
A
Obs
JmaxRef = 85.25
RdRef = 1.16
sigma = 0.21
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

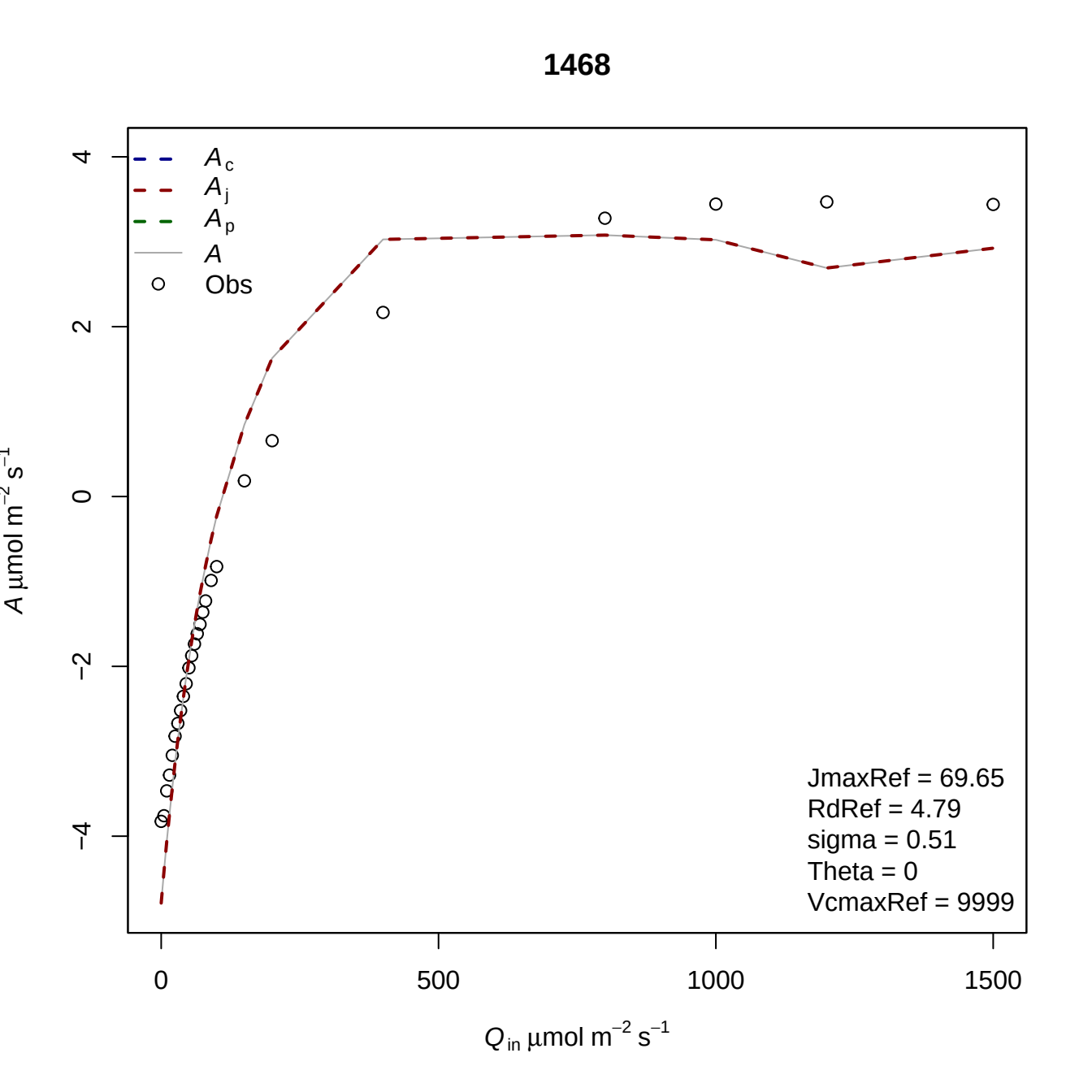

1463
A_c
A_j
A_p
A
Obs
JmaxRef = 79.59
RdRef = 1.27
sigma = 0.33
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

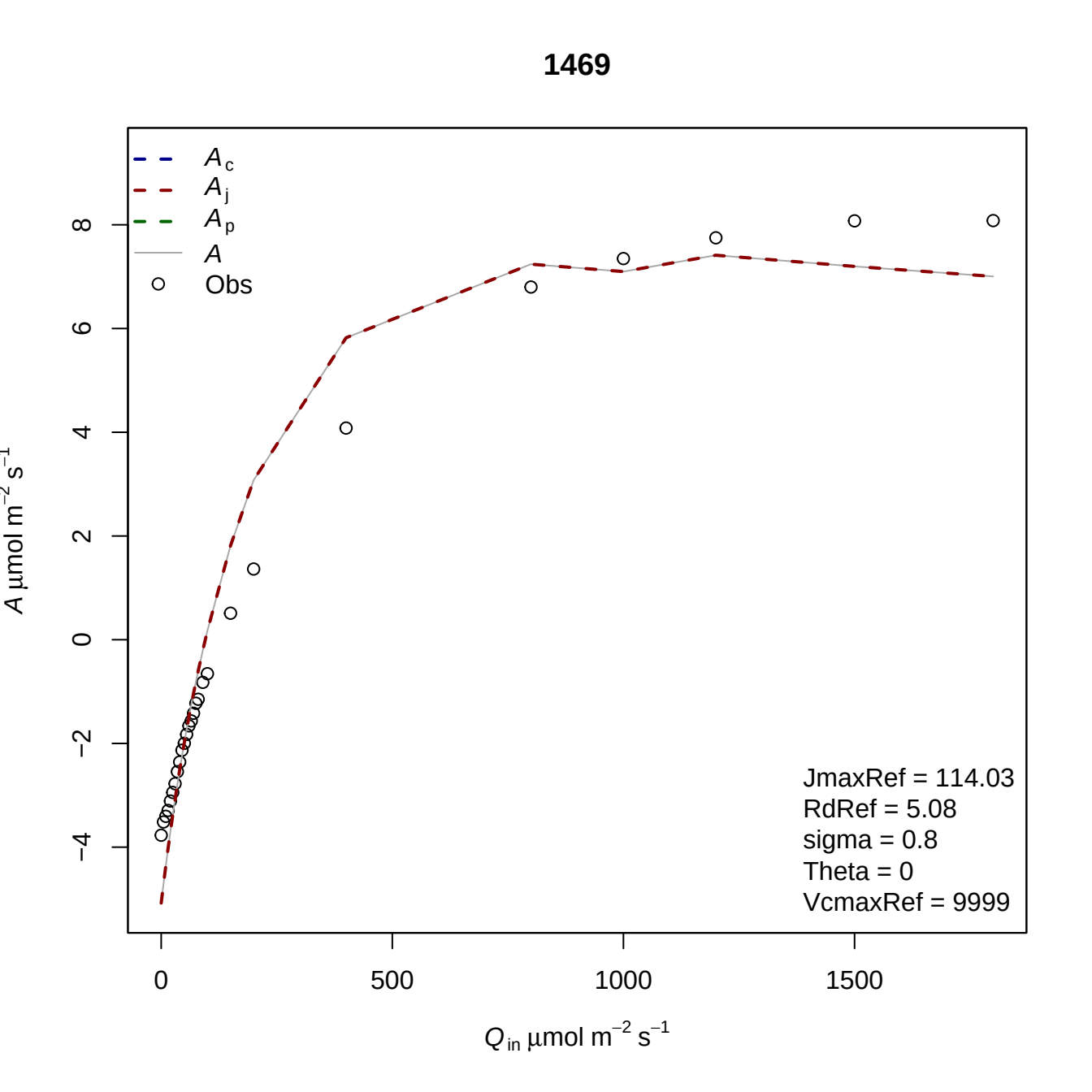

1464
A_c
A_j
A_p
A
Obs
JmaxRef = 120.74
RdRef = 1.39
sigma = 0.42
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

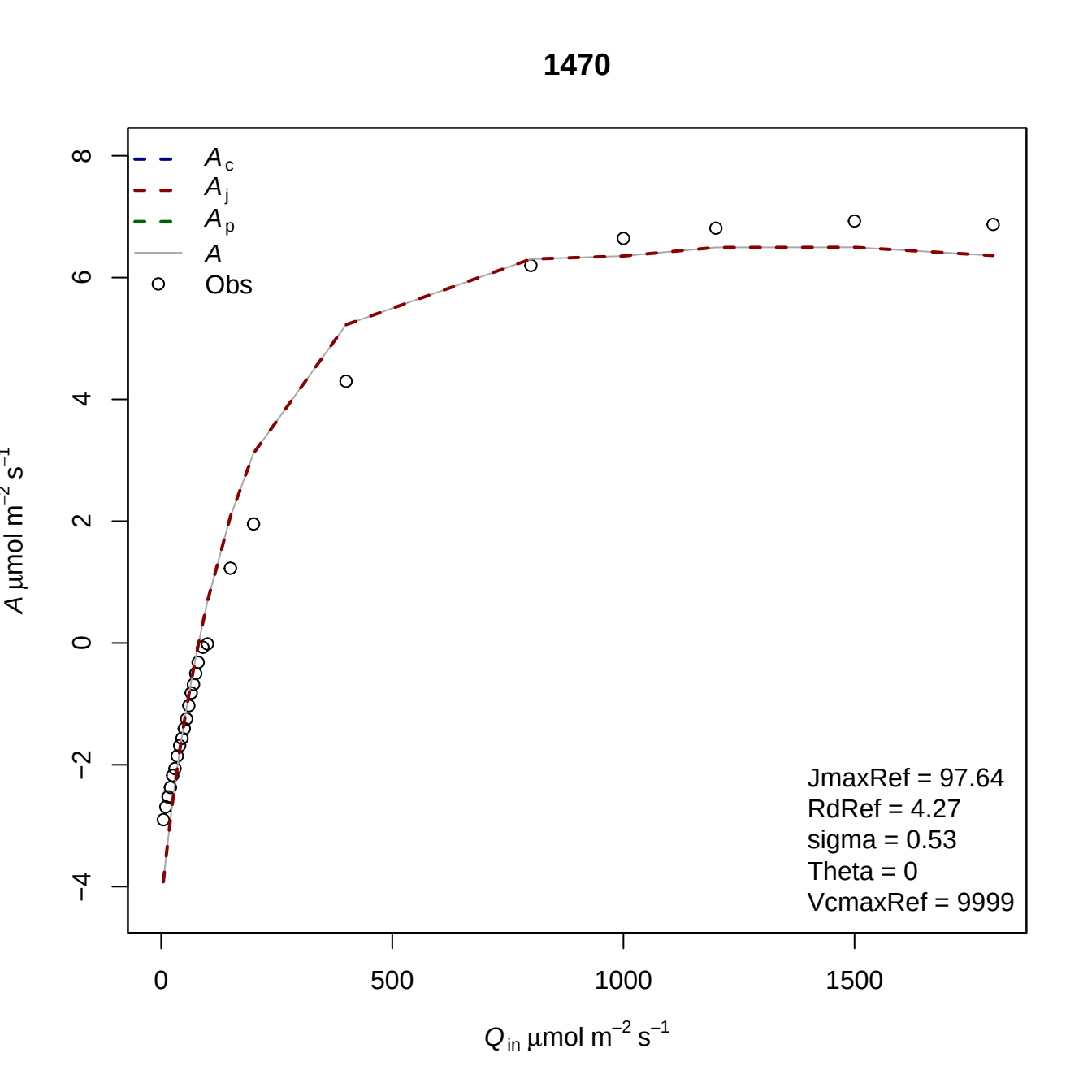

1465
A_c
A_j
A_p
A
Obs
JmaxRef = 64.77
RdRef = 1.41
sigma = 0.21
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

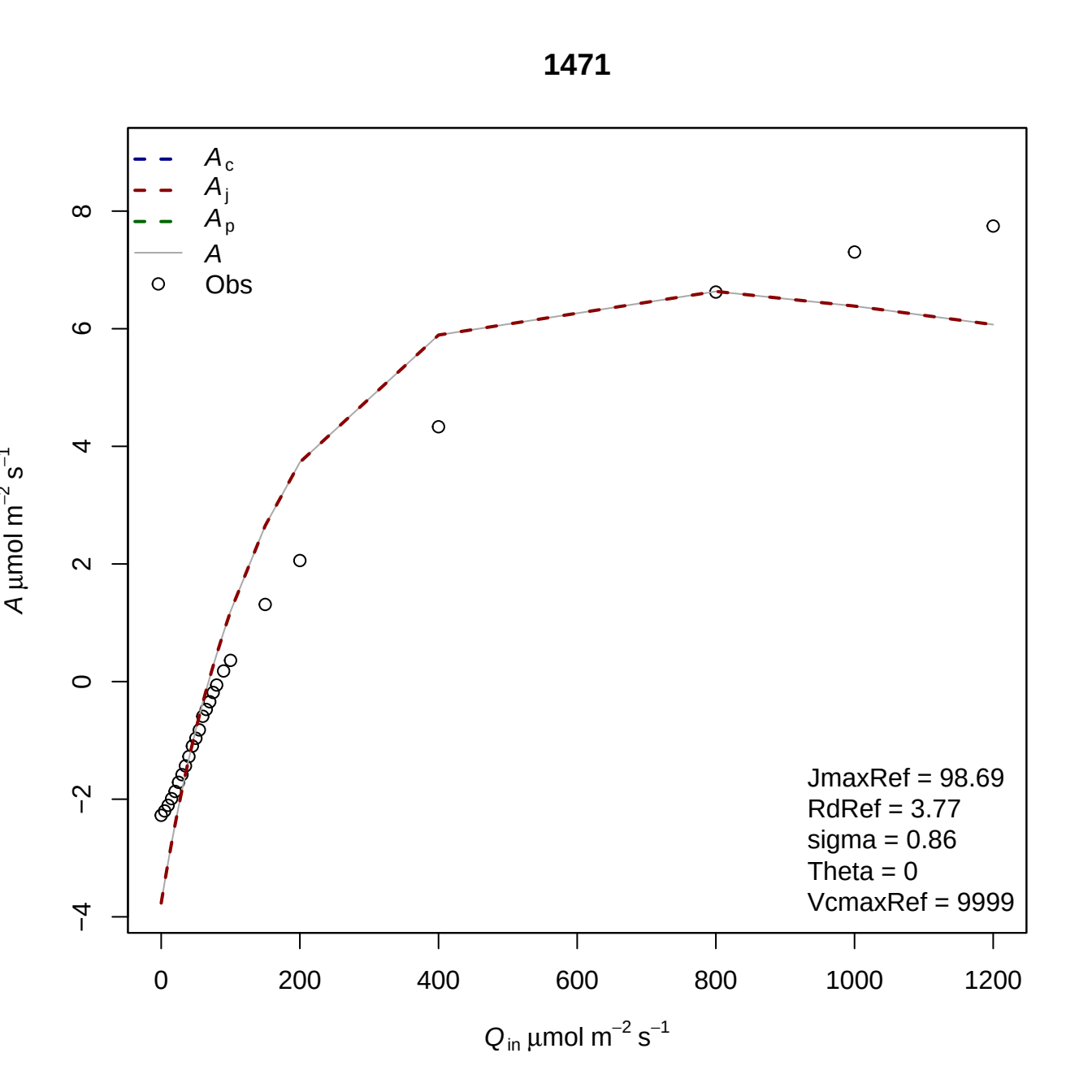

1466
A_c
A_j
A_p
A
Obs
JmaxRef = 94.23
RdRef = 1.77
sigma = 0.17
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

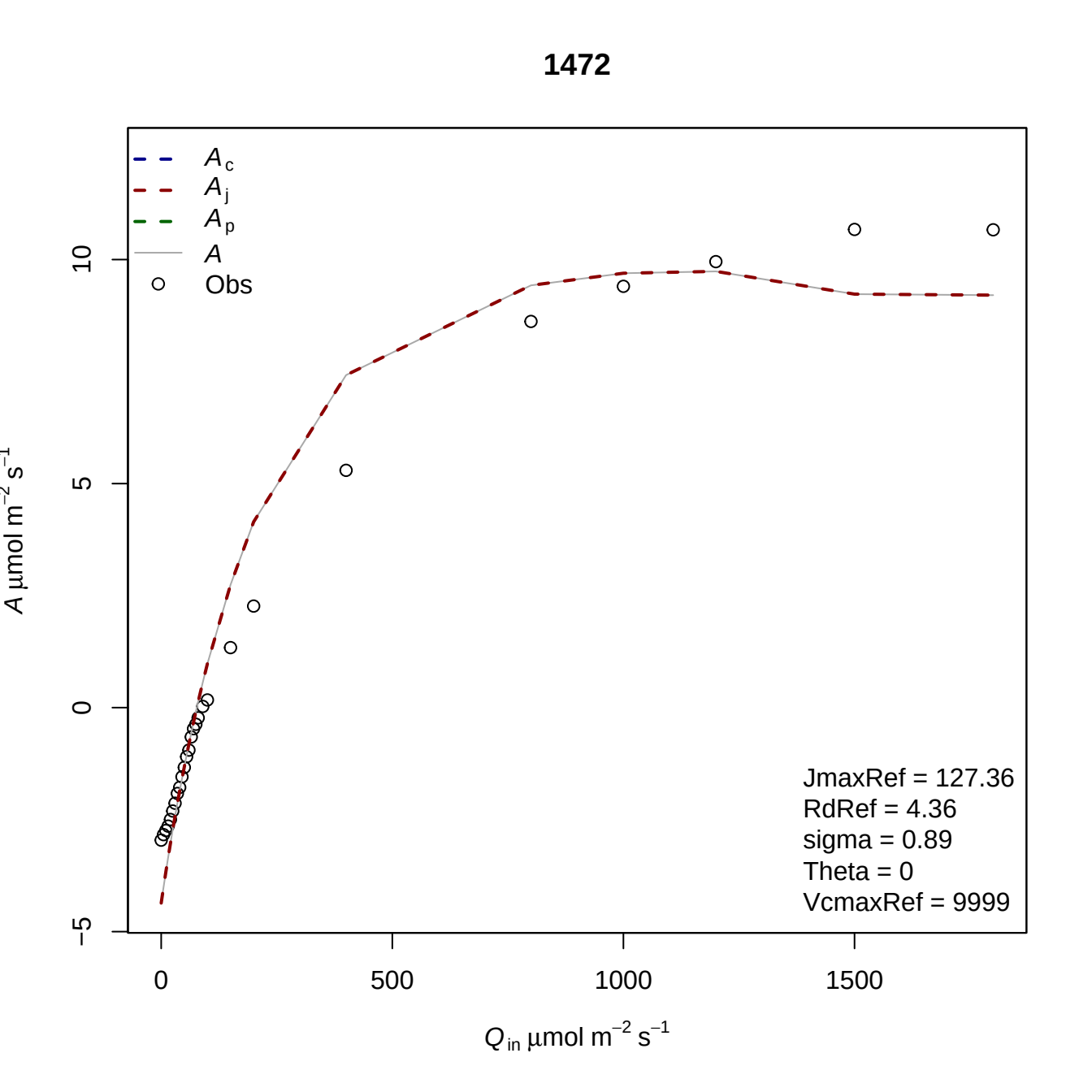

1467
A_c
A_j
A_p
A
Obs
JmaxRef = 89.53
RdRef = 1.43
sigma = 0.16
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

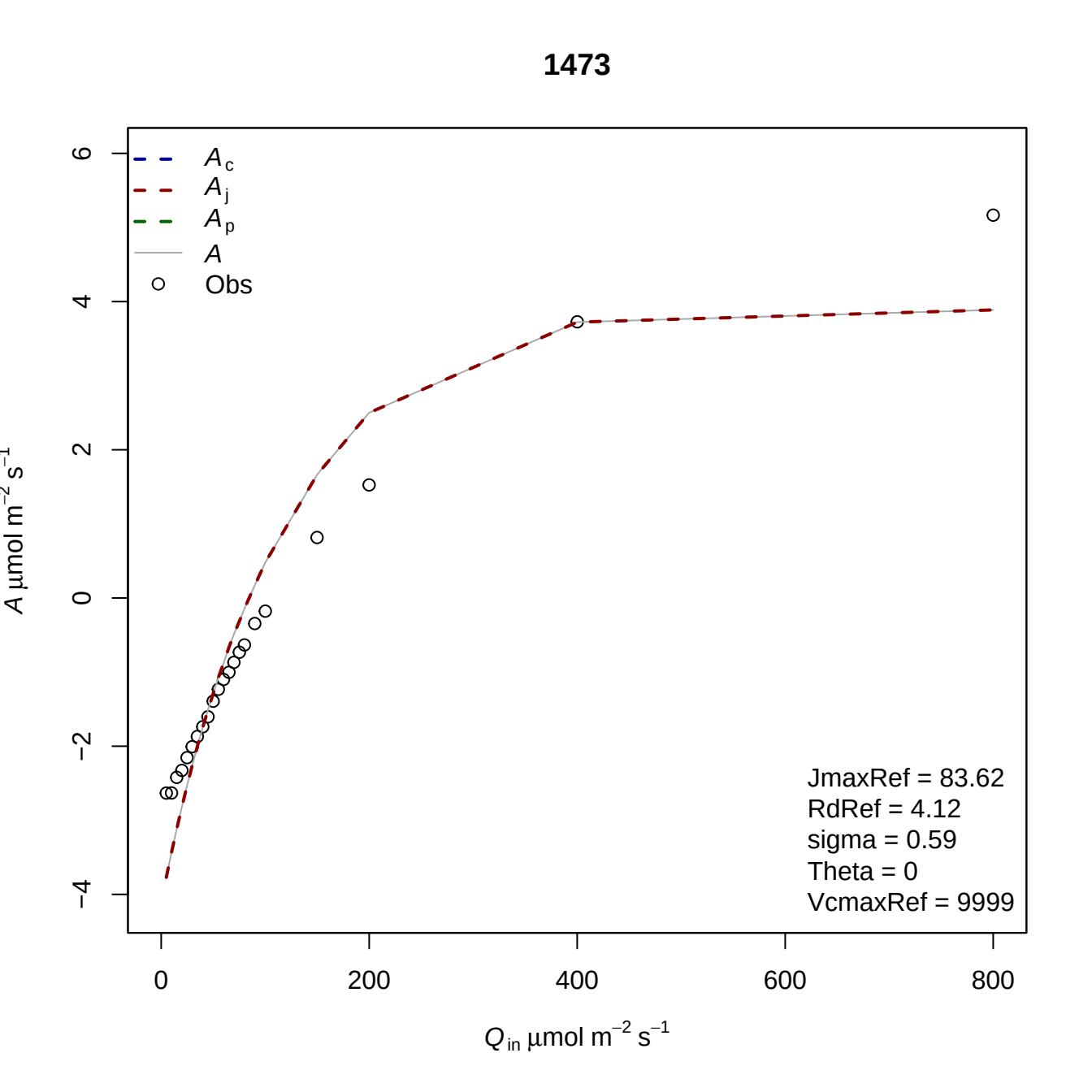

1468
A_c
A_j
A_p
A
Obs
JmaxRef = 69.65
RdRef = 4.79
sigma = 0.51
Theta = 0
VcmaxRef = 9999
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

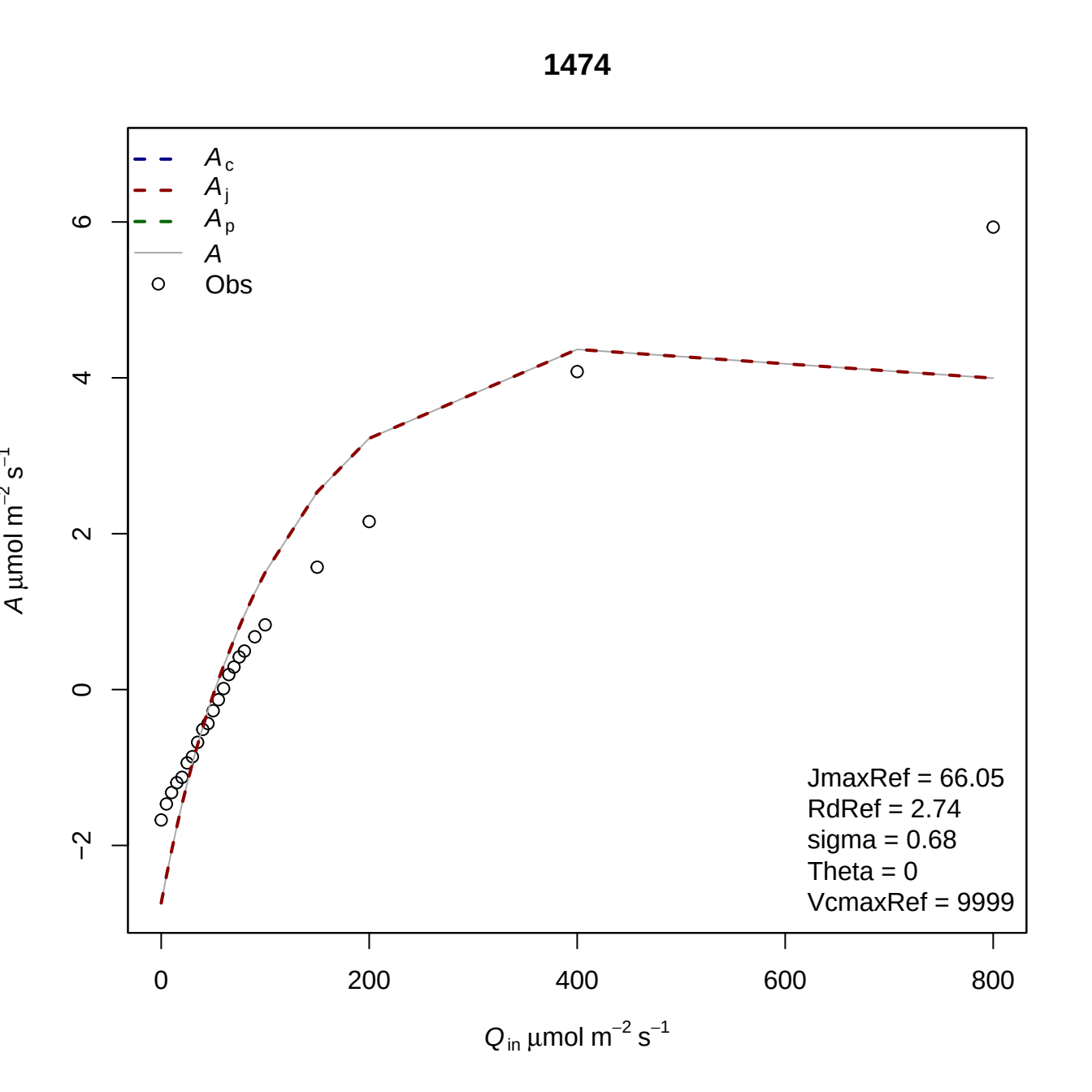

1469
A_c
A_j
A_p
A
Obs
JmaxRef = 114.03
RdRef = 5.08
sigma = 0.8
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

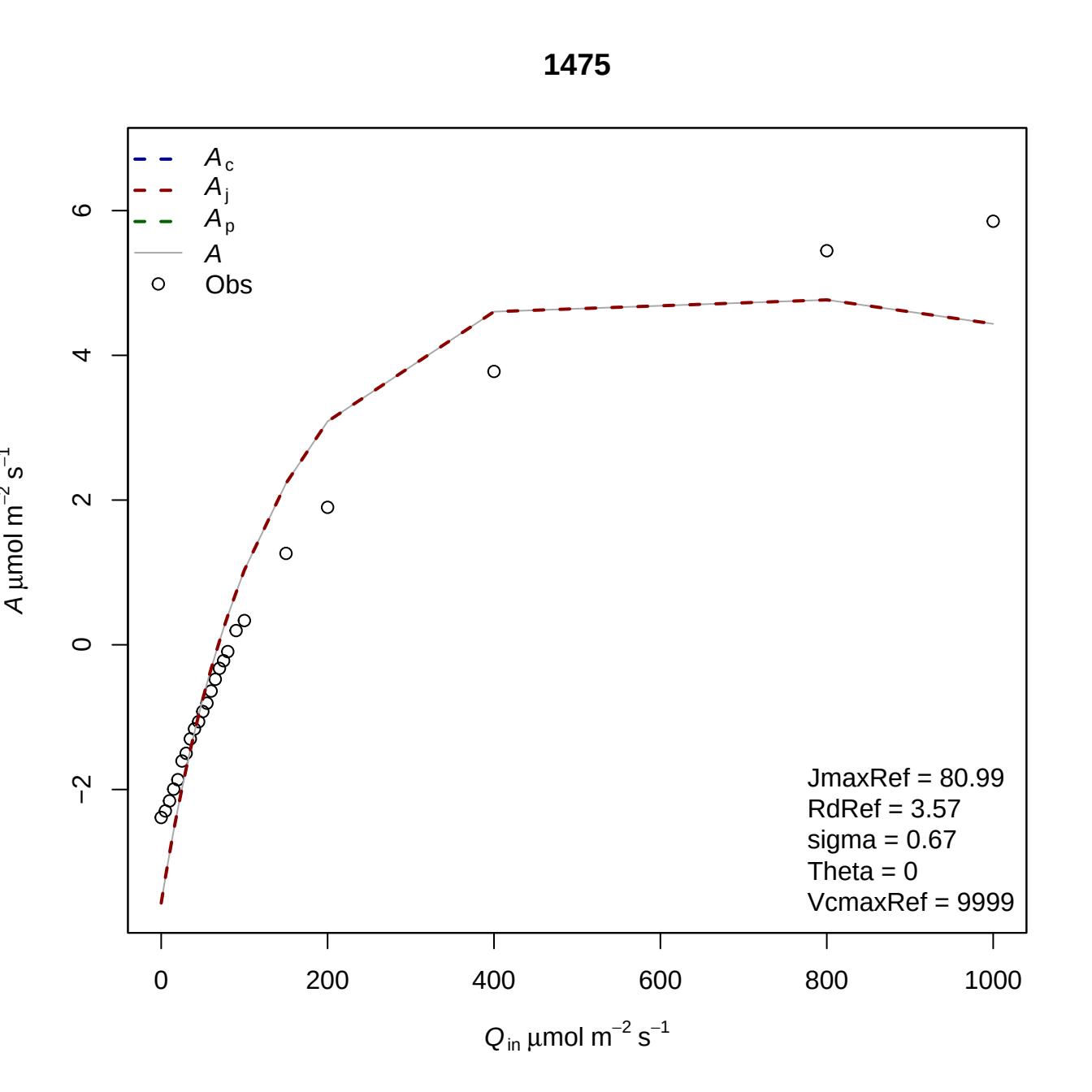

1470
A_c
A_j
A_p
A
Obs
JmaxRef = 97.64
RdRef = 4.27
sigma = 0.53
Theta = 0
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

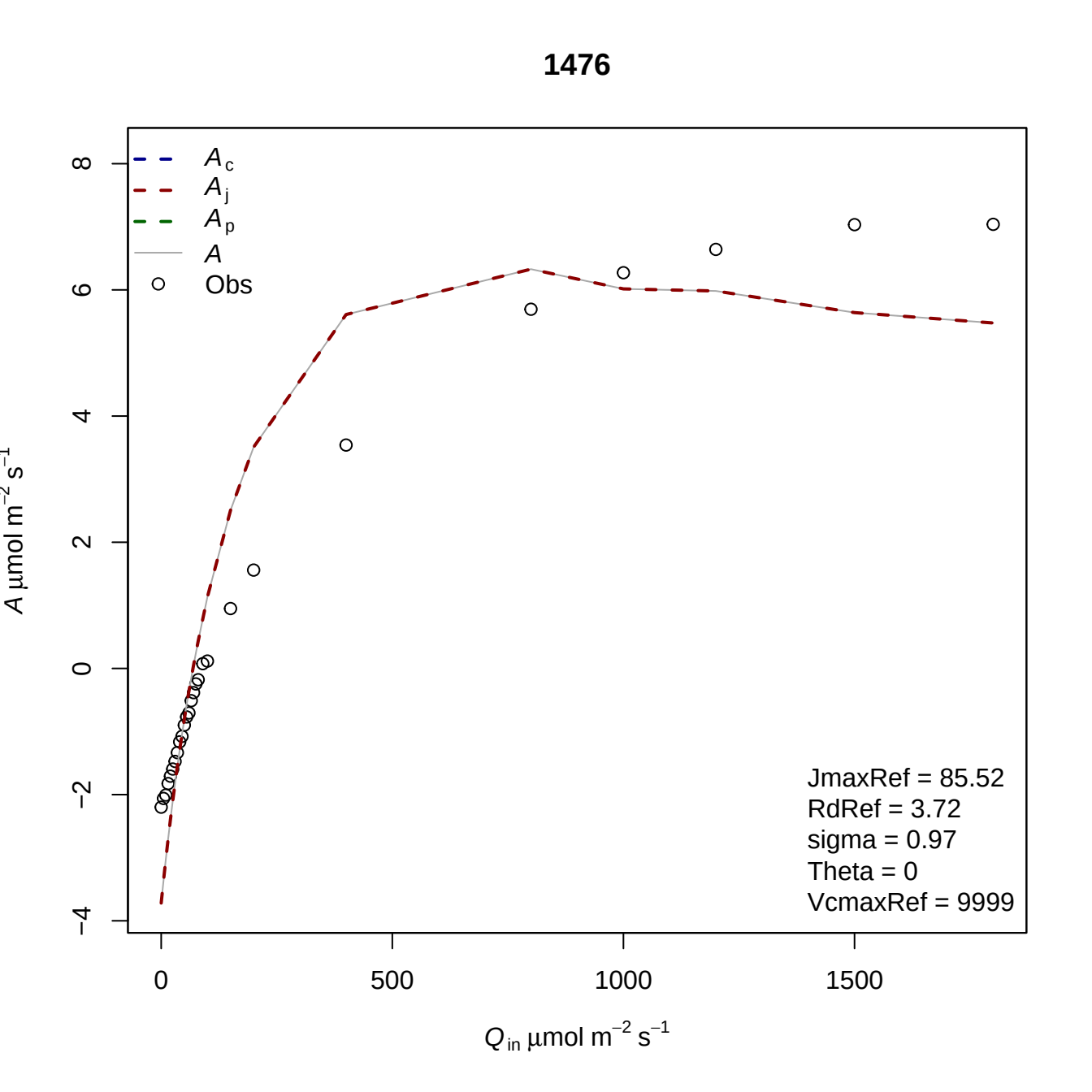

1471
A_c
A_j
A_p
A
Obs
JmaxRef = 98.69
RdRef = 3.77
sigma = 0.86
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

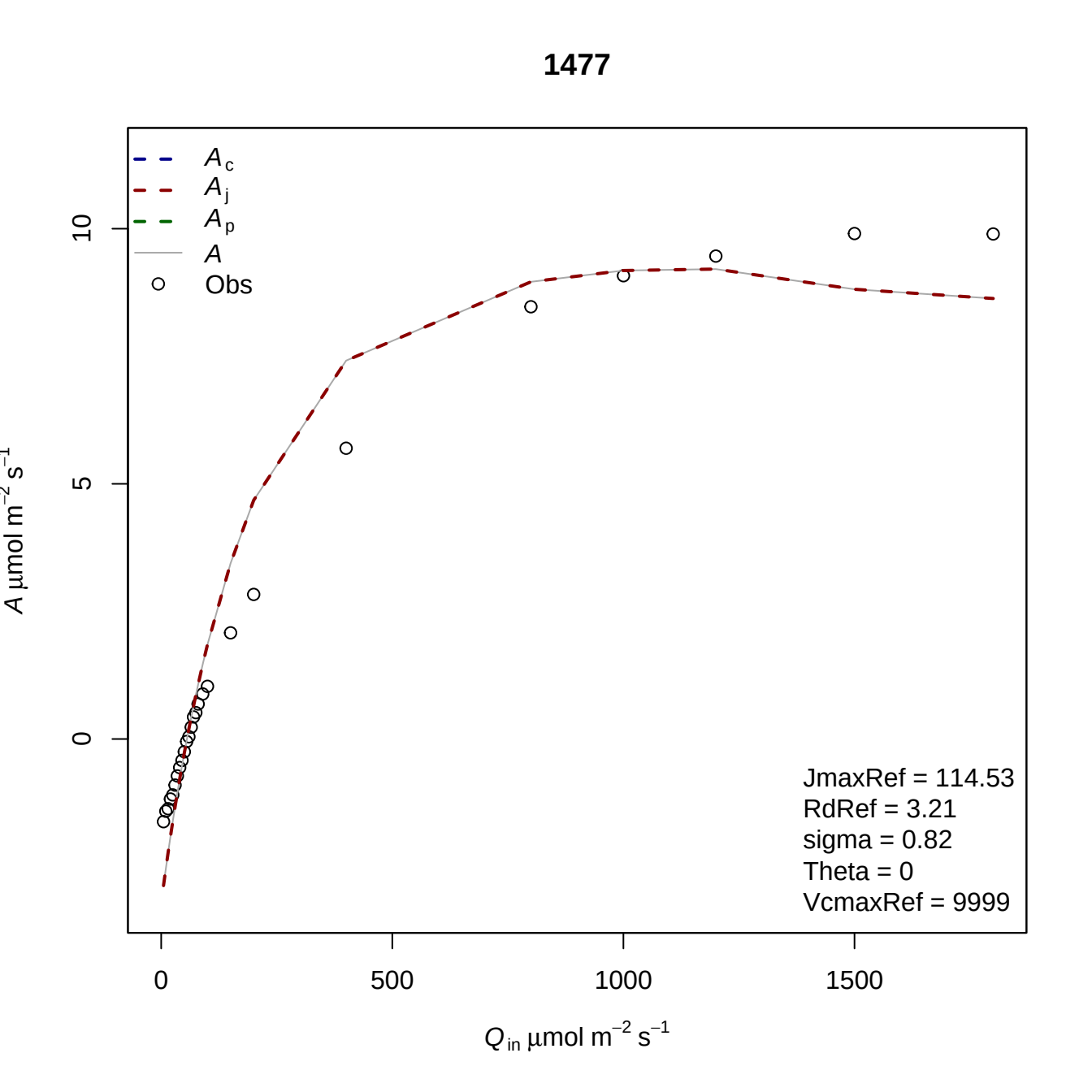

1472
A_c
A_j
A_p
A
Obs
JmaxRef = 127.36
RdRef = 4.36
sigma = 0.89
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

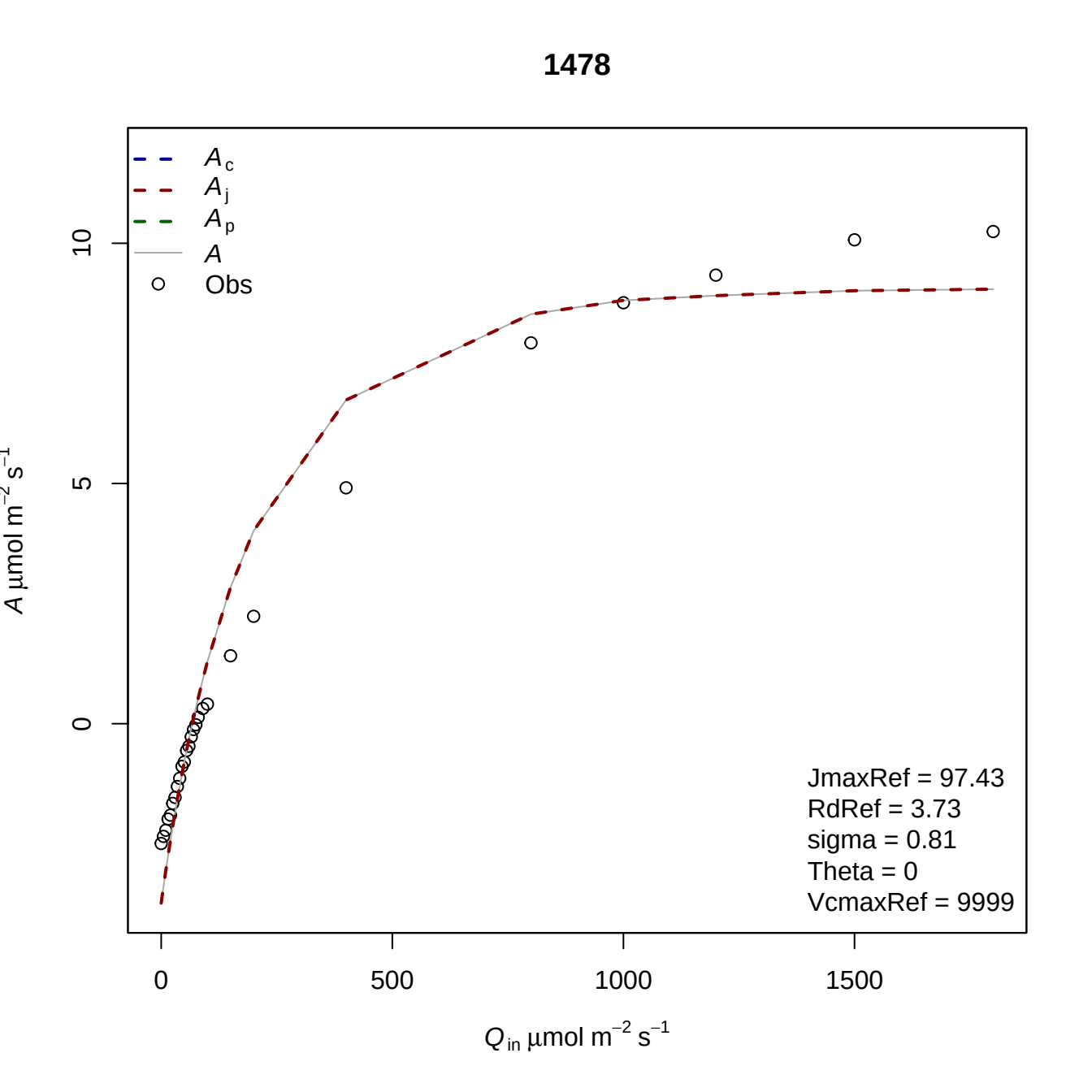

1473
A_c
A_j
A_p
A
Obs
JmaxRef = 83.62
RdRef = 4.12
sigma = 0.59
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800

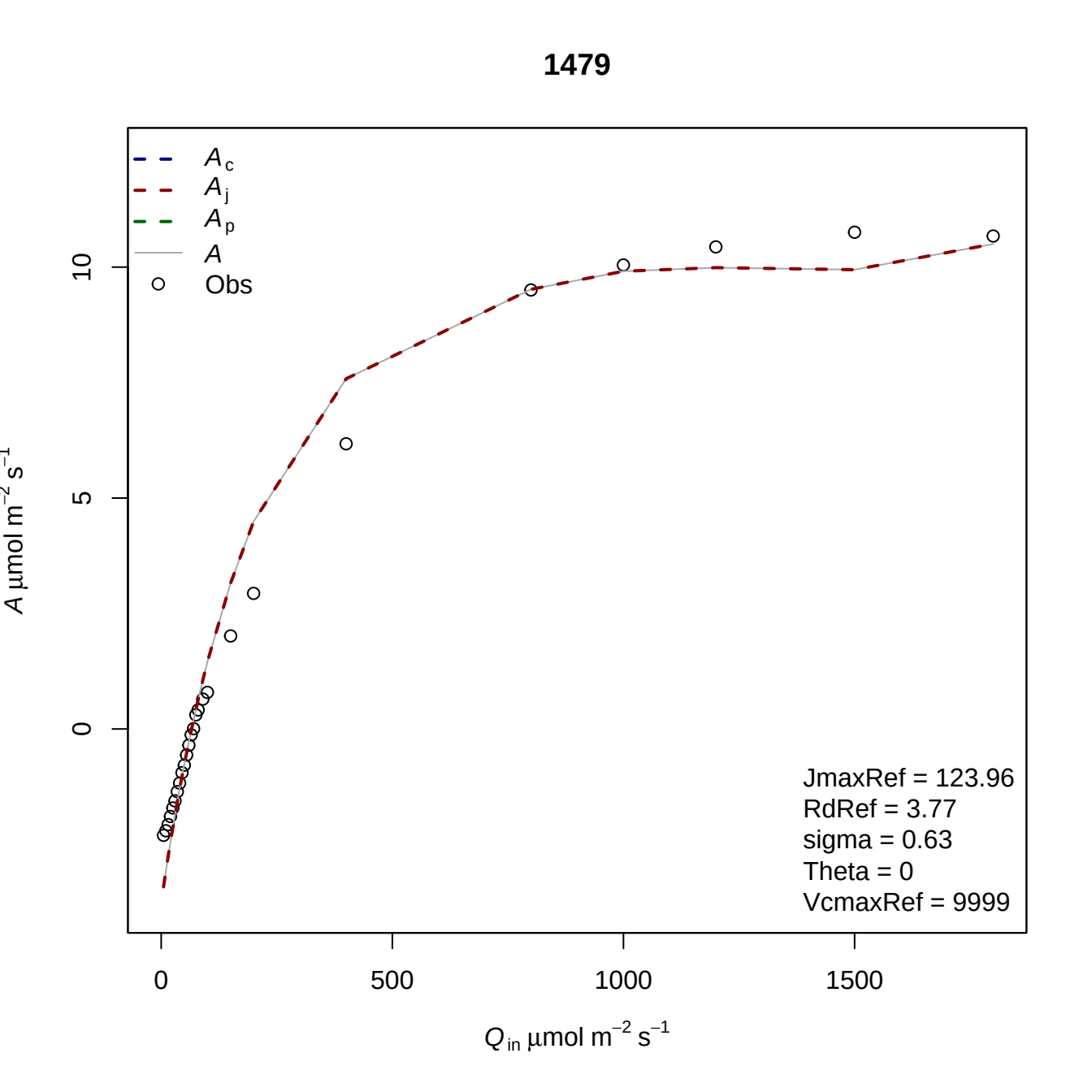

1474
A_c
A_j
A_p
A
Obs
JmaxRef = 66.05
RdRef = 2.74
sigma = 0.68
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

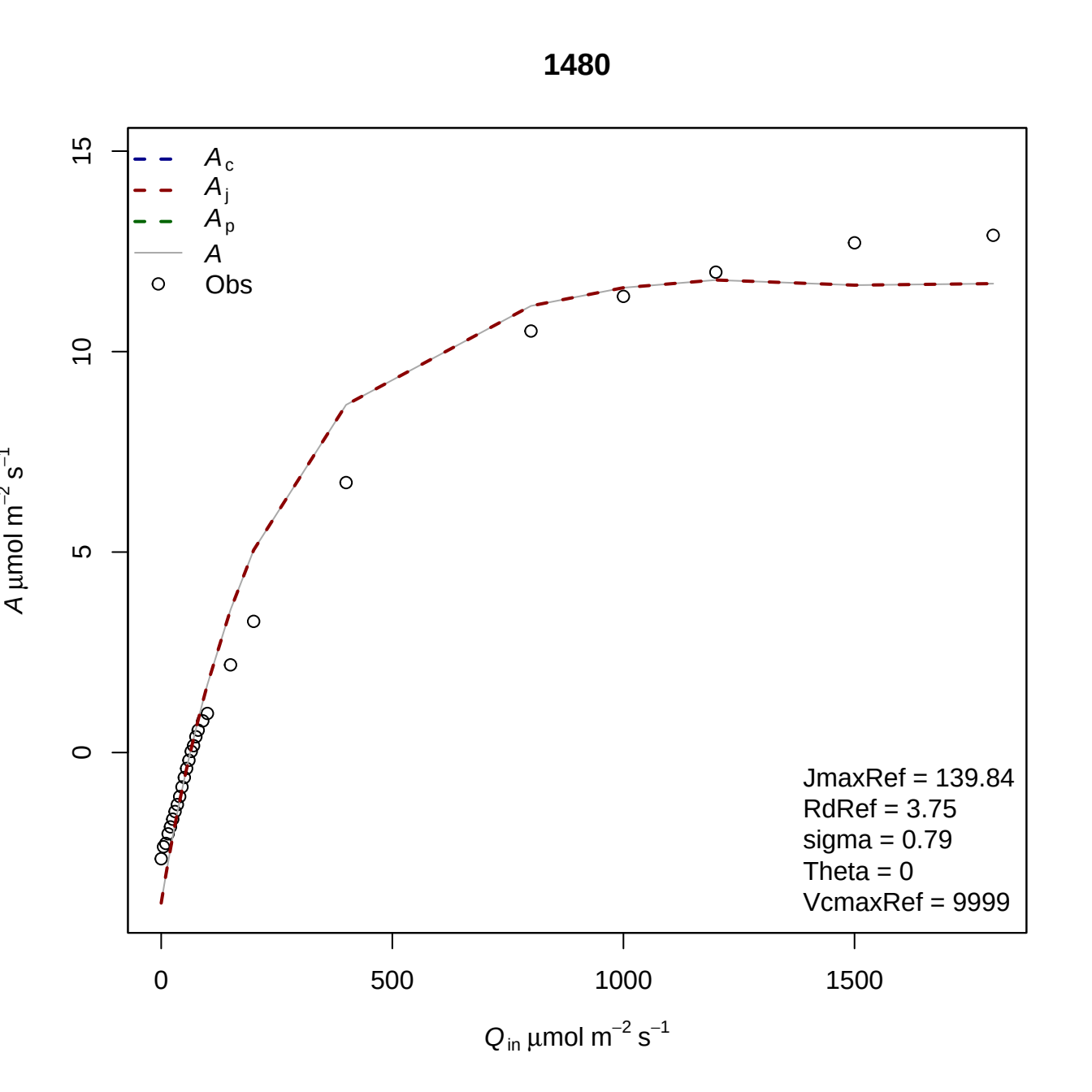

1475
A_c
A_j
A_p
A
Obs
JmaxRef = 80.99
RdRef = 3.57
sigma = 0.67
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

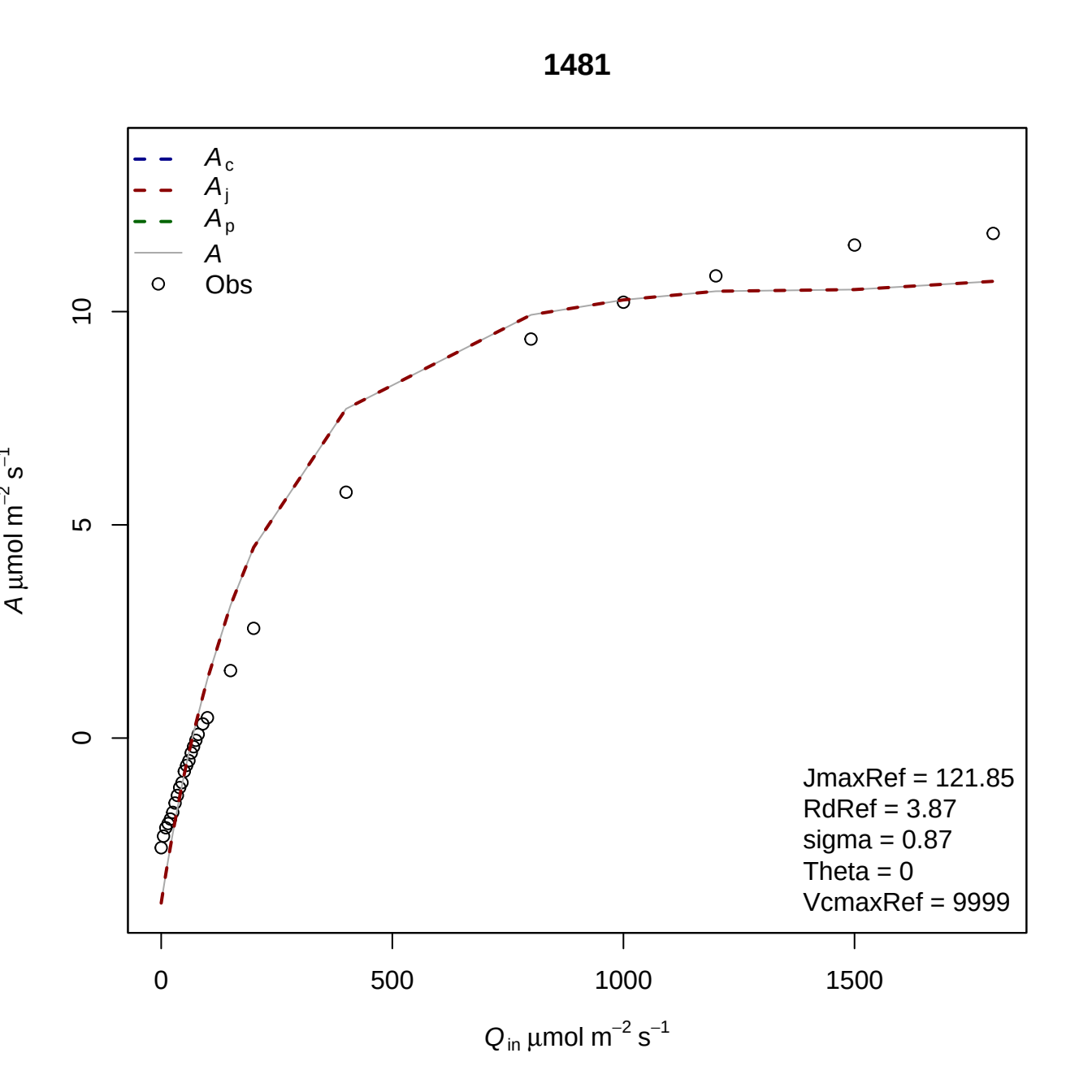

1476
A_c
A_j
A_p
A
Obs
JmaxRef = 85.52
RdRef = 3.72
sigma = 0.97
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

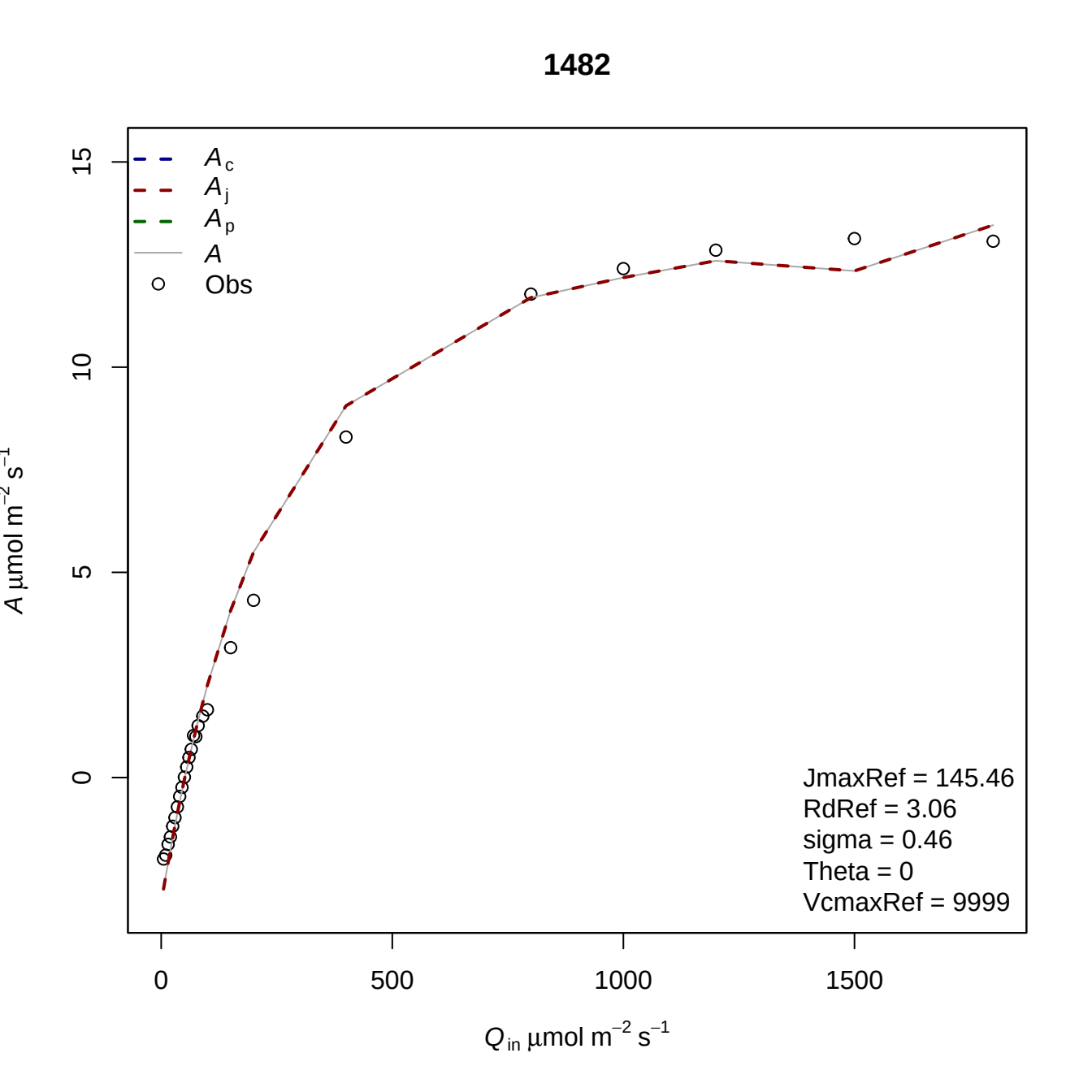

1477
A_c
A_j
A_p
A
Obs
JmaxRef = 114.53
RdRef = 3.21
sigma = 0.82
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

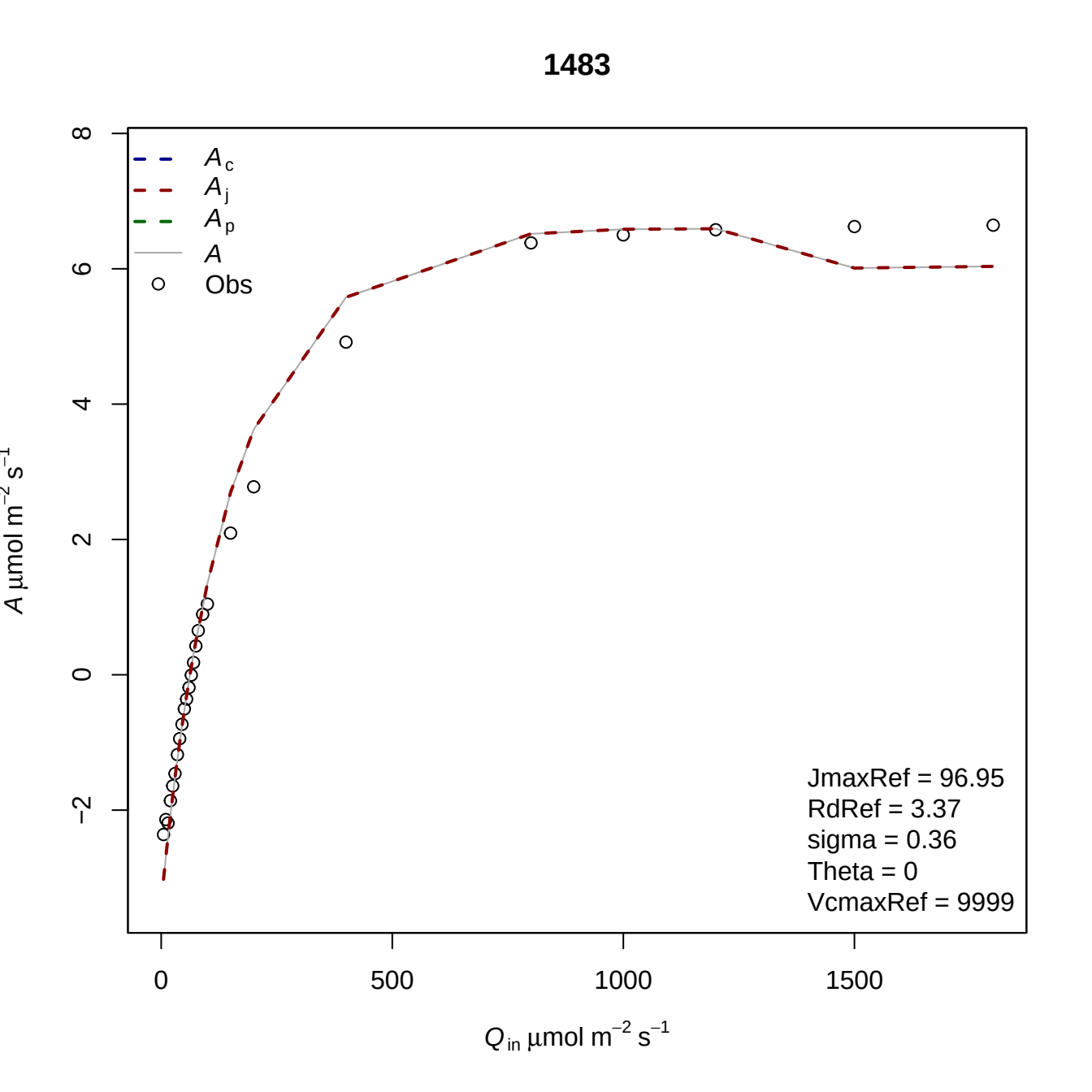

1478
A_c
A_j
A_p
A
Obs
JmaxRef = 97.43
RdRef = 3.73
sigma = 0.81
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

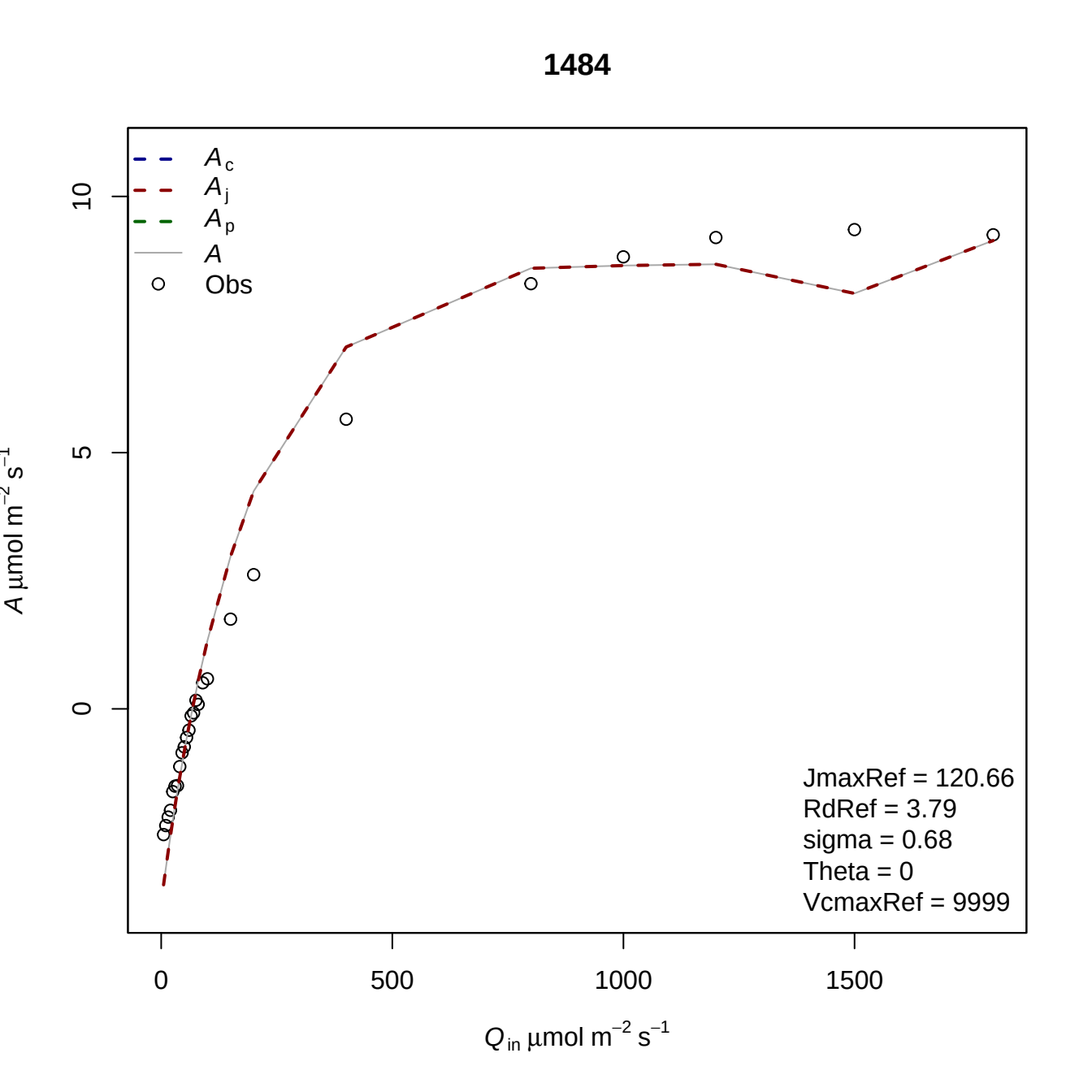

1479
A_c
A_j
A_p
A
Obs
JmaxRef = 123.96
RdRef = 3.77
sigma = 0.63
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

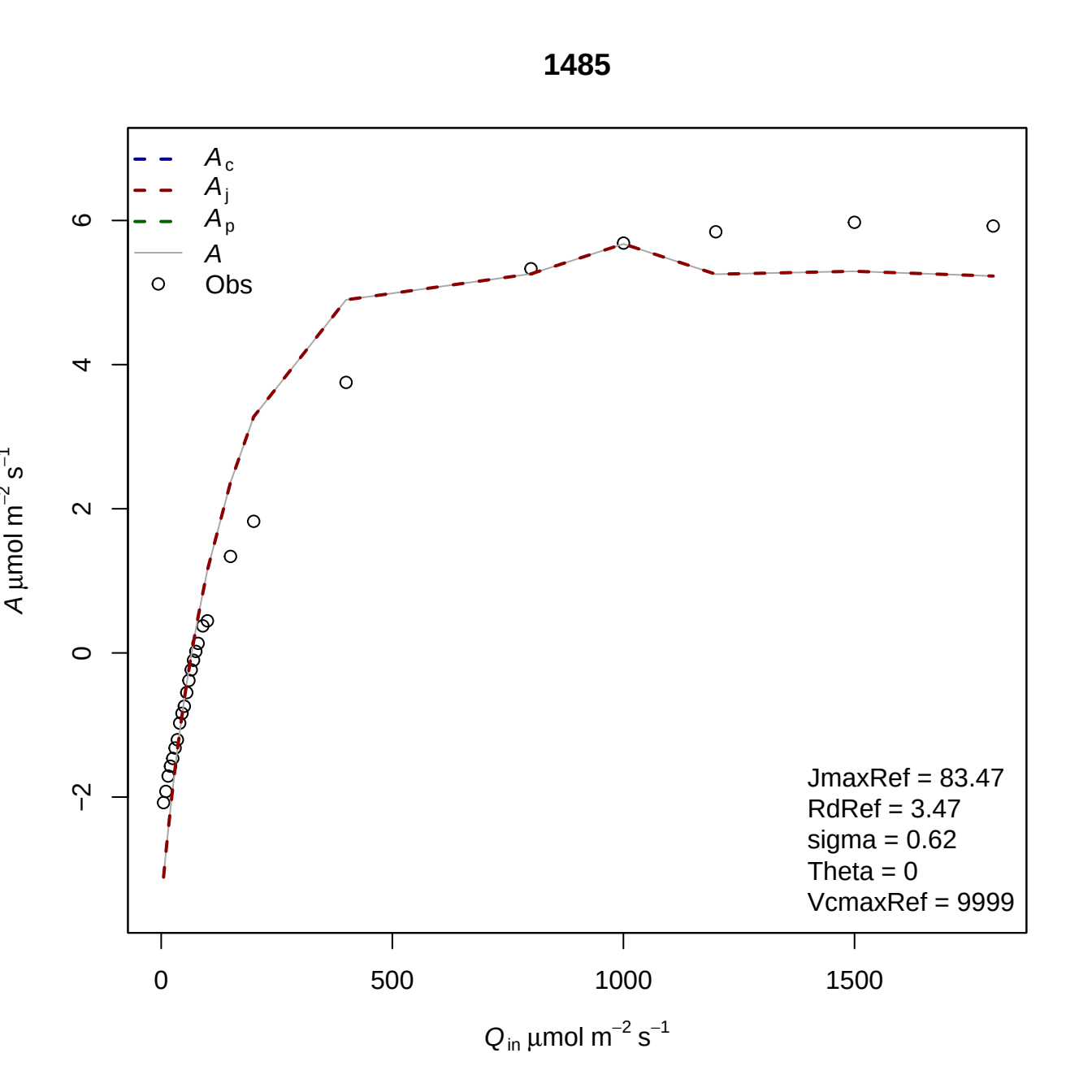

1480
A_c
A_j
A_p
A
Obs
JmaxRef = 139.84
RdRef = 3.75
sigma = 0.79
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

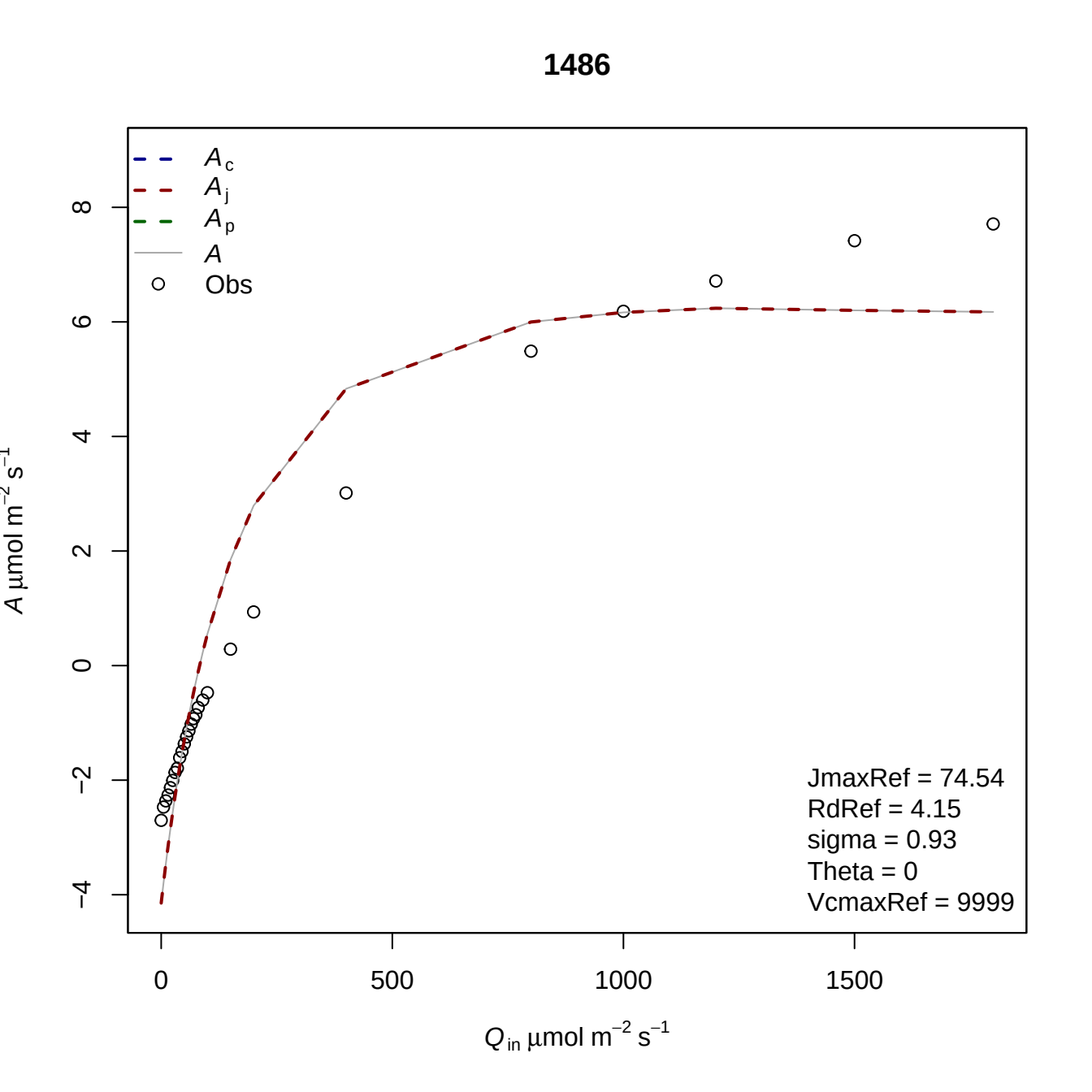

1481
A_c
A_j
A_p
A
Obs
JmaxRef = 121.85
RdRef = 3.87
sigma = 0.87
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

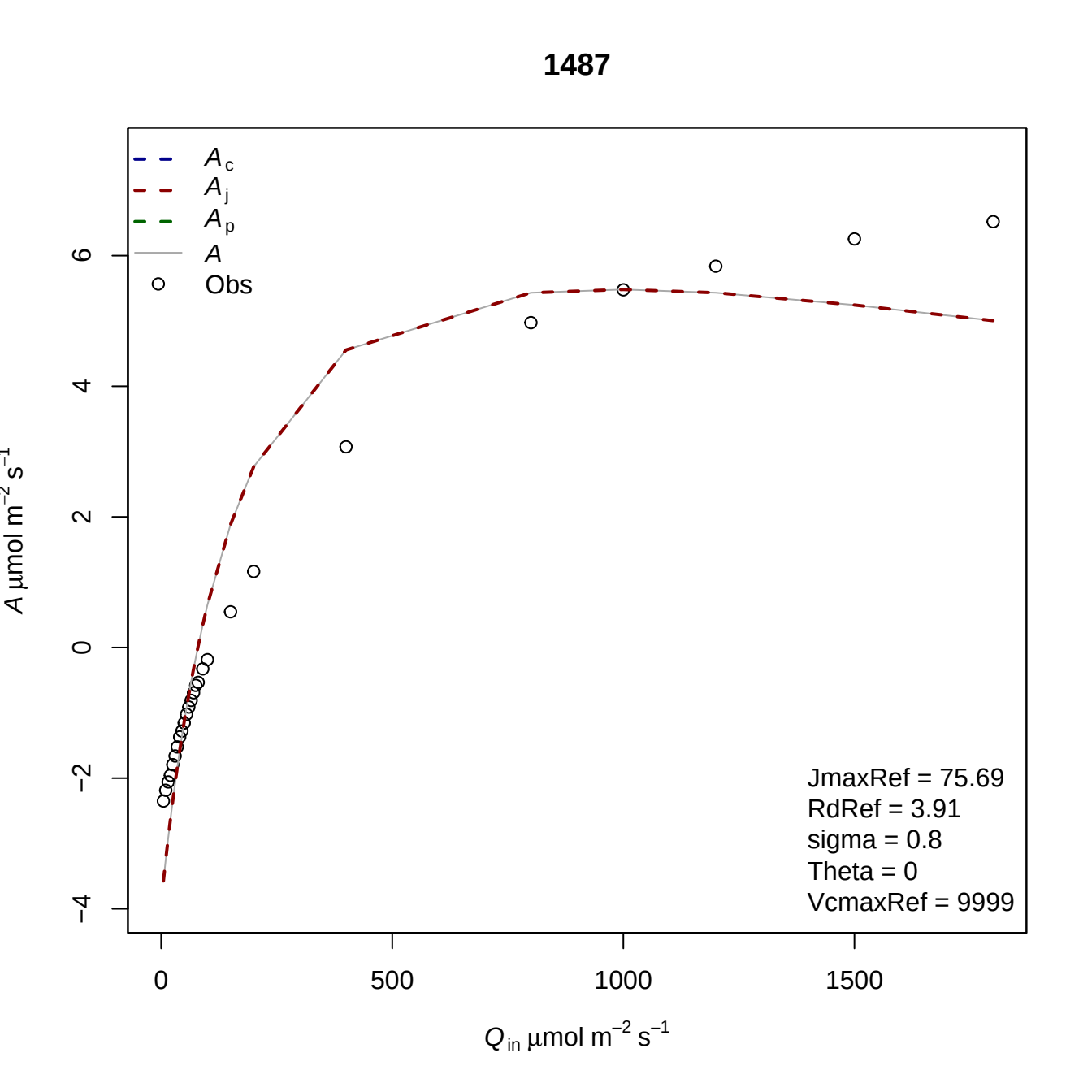

1482
A_c
A_j
A_p
A
Obs
JmaxRef = 145.46
RdRef = 3.06
sigma = 0.46
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

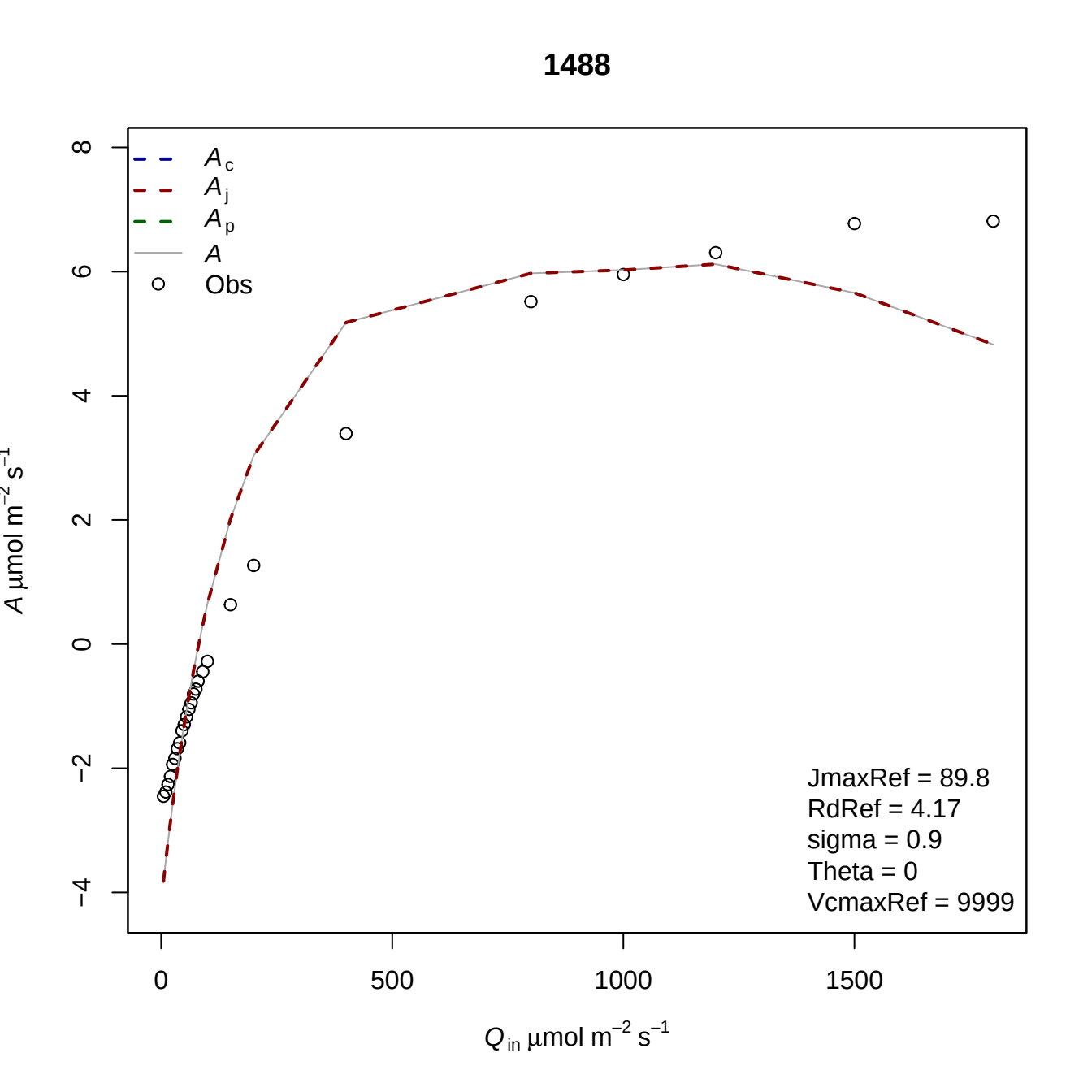

1483
A_c
A_j
A_p
A
Obs
JmaxRef = 96.95
RdRef = 3.37
sigma = 0.36
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

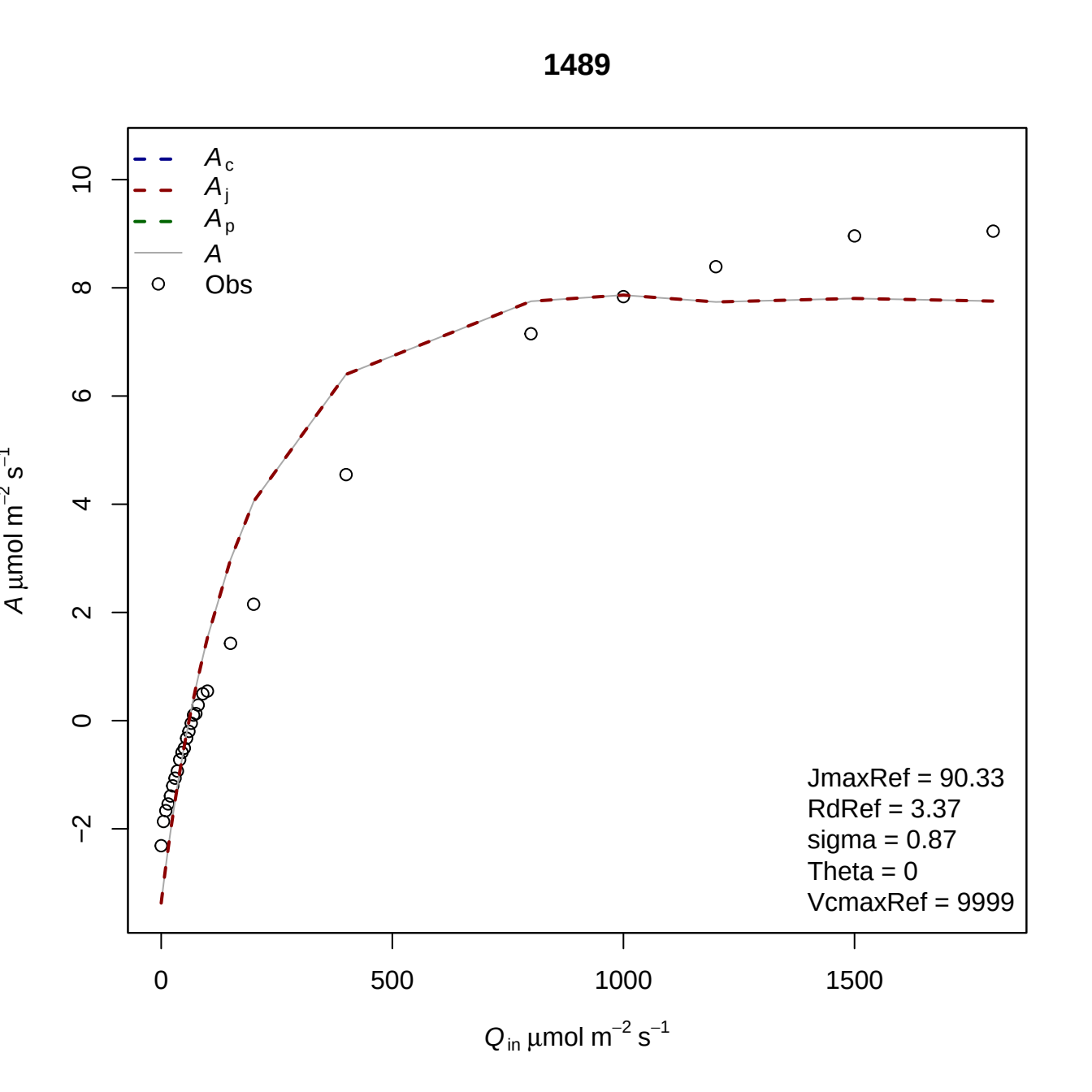

1484
A_c
A_j
A_p
A
Obs
JmaxRef = 120.66
RdRef = 3.79
sigma = 0.68
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

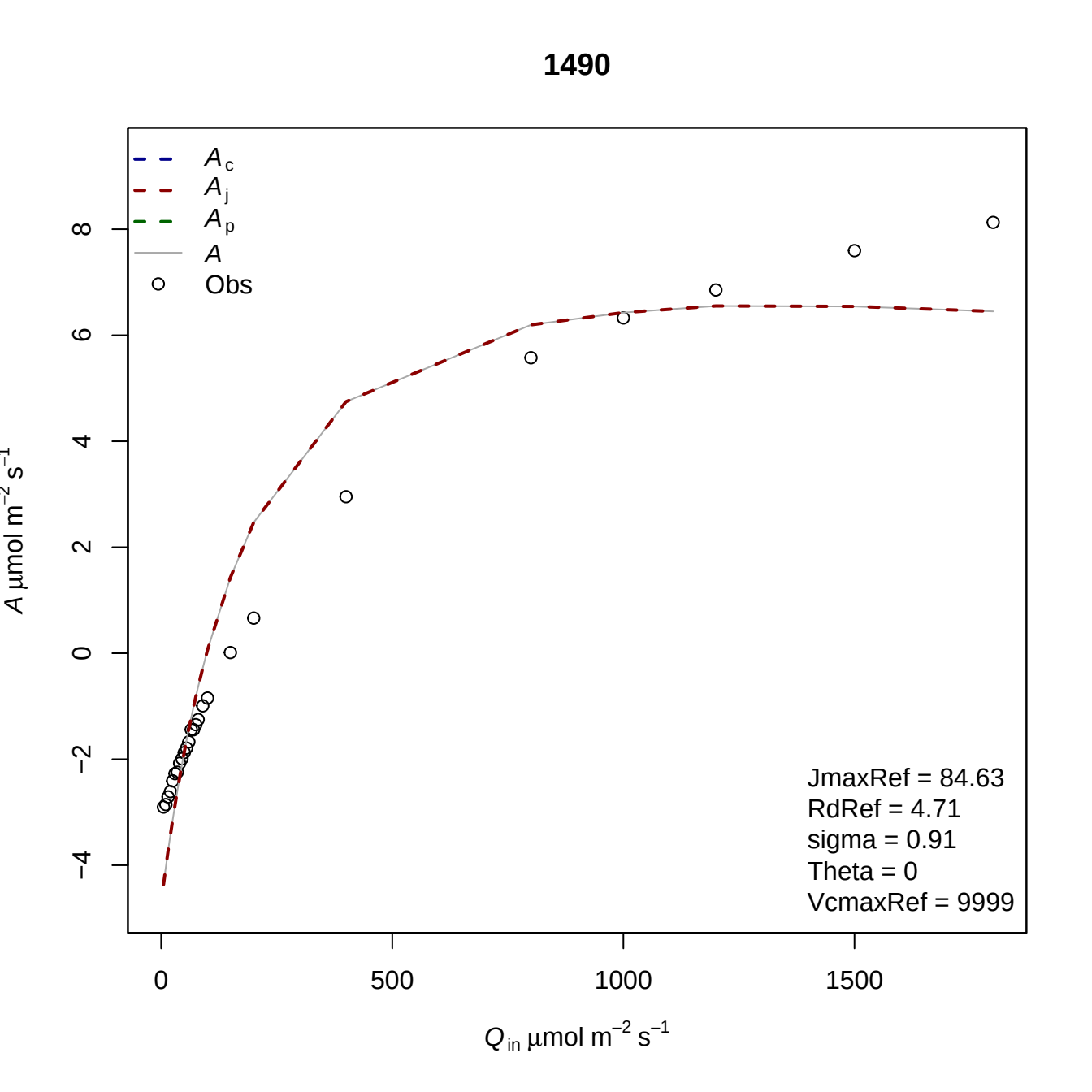

1485
A_c
A_j
A_p
A
Obs
JmaxRef = 83.47
RdRef = 3.47
sigma = 0.62
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

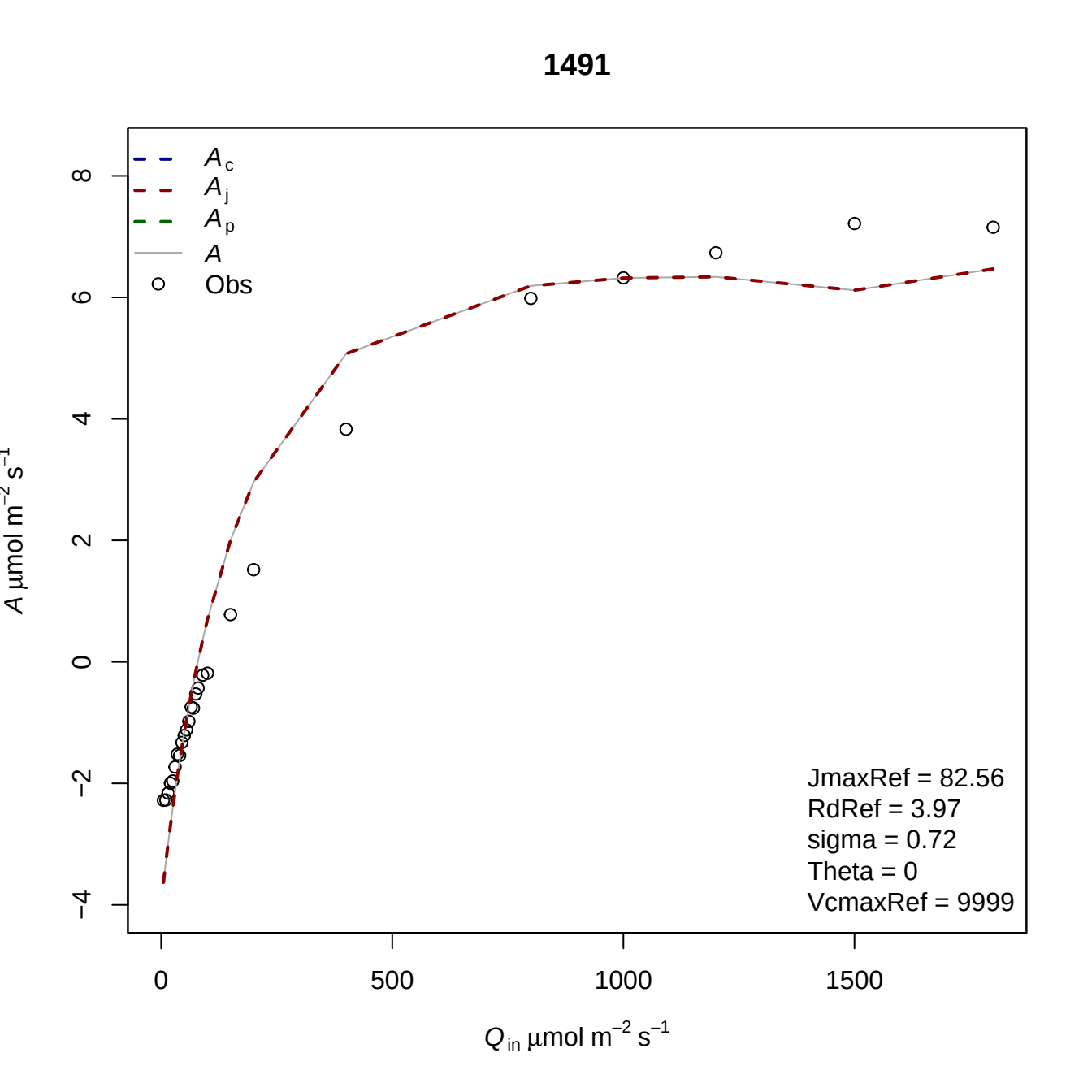

1486
A_c
A_j
A_p
A
Obs
JmaxRef = 74.54
RdRef = 4.15
sigma = 0.93
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

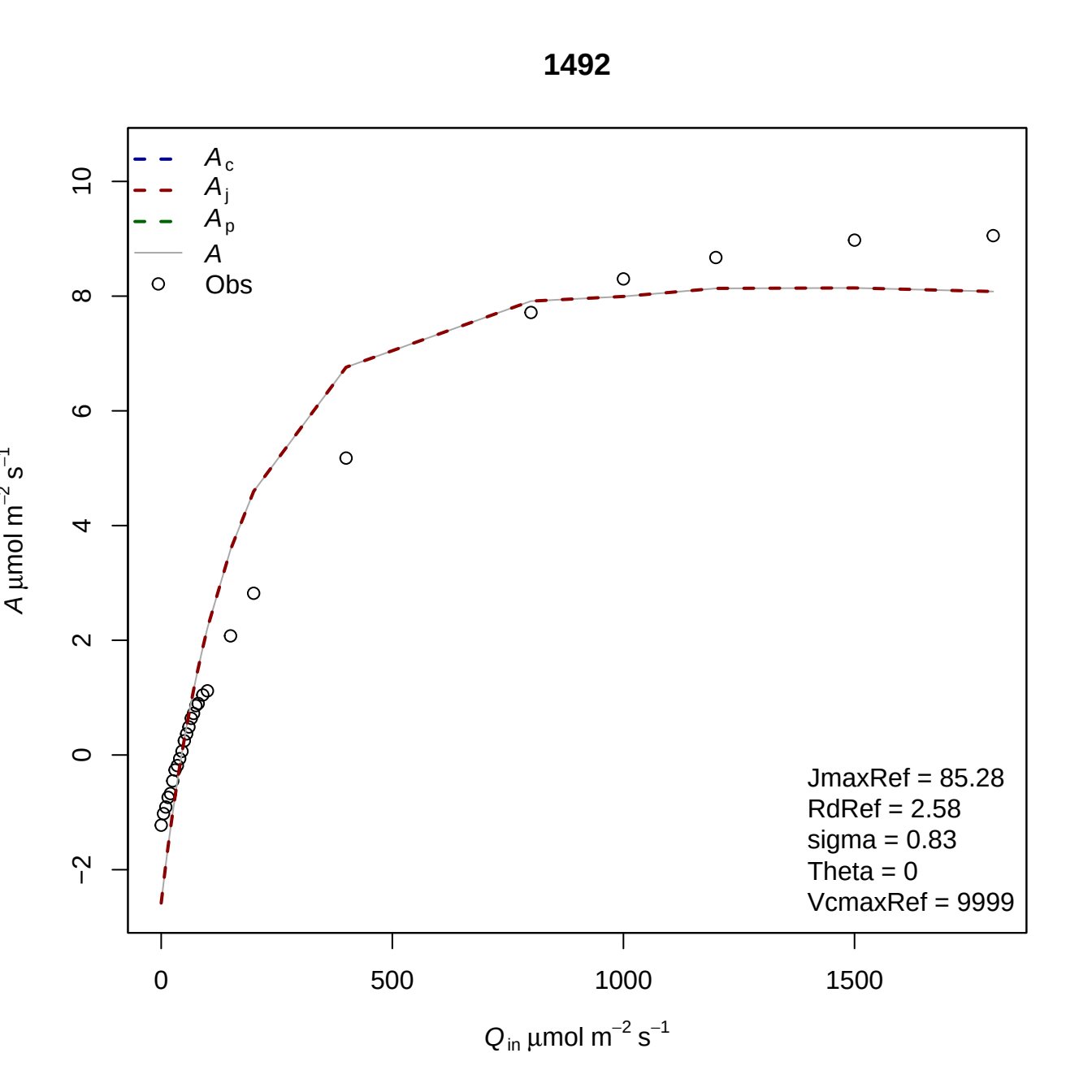

1487
A_c
A_j
A_p
A
Obs
JmaxRef = 75.69
RdRef = 3.91
sigma = 0.8
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

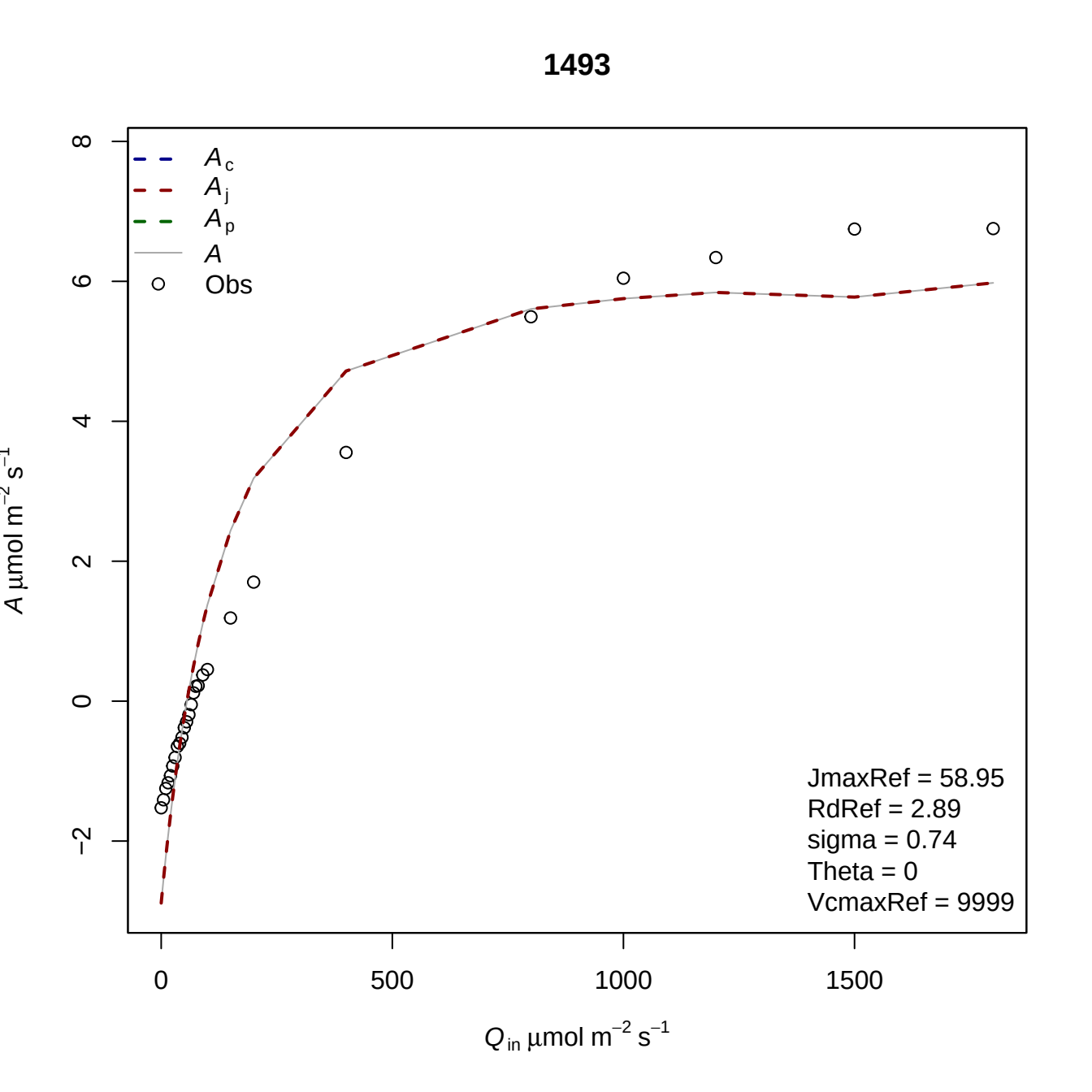

1488
A_c
A_j
A_p
A
Obs
JmaxRef = 89.8
RdRef = 4.17
sigma = 0.9
Theta = 0
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

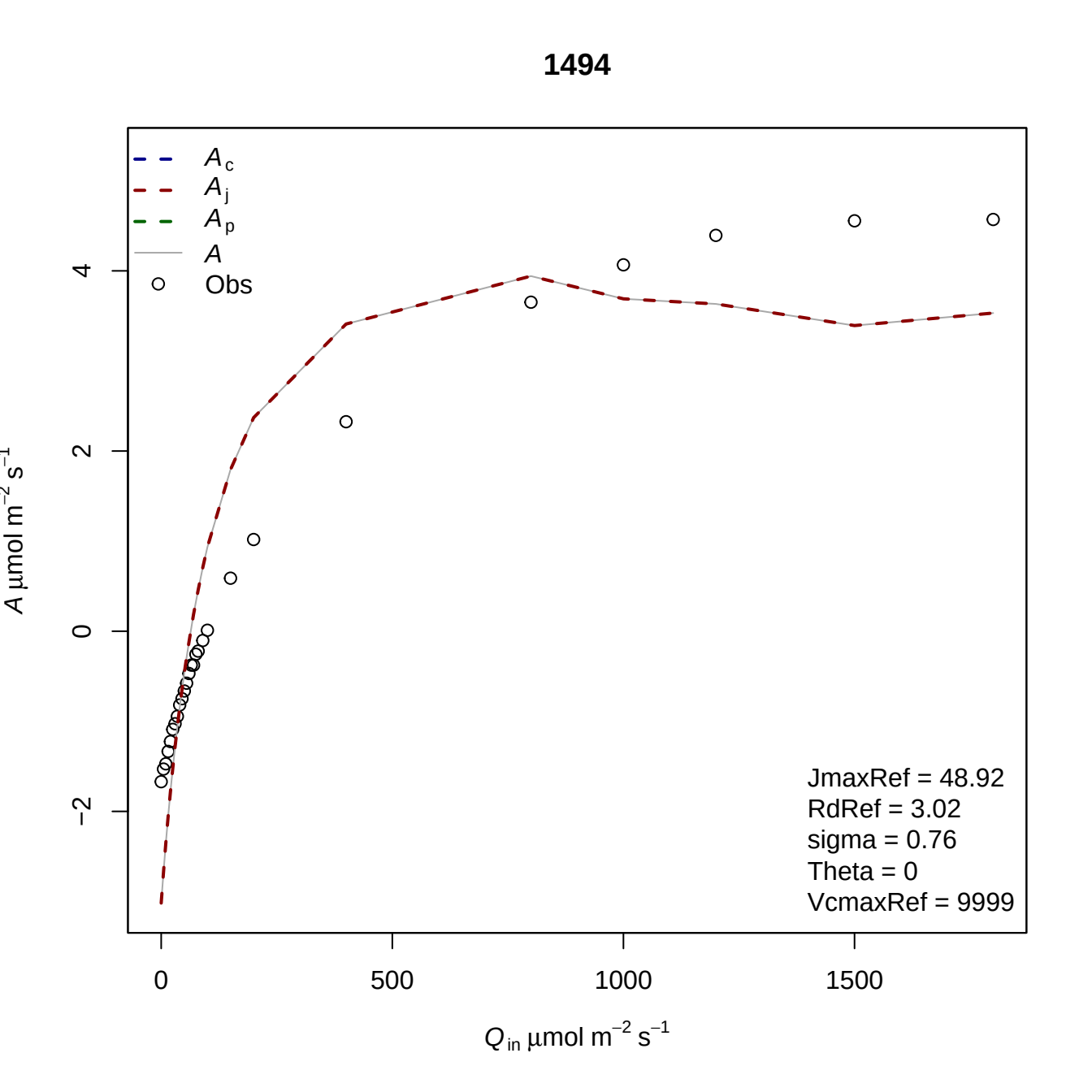

1489
A_c
A_j
A_p
A
Obs
JmaxRef = 90.33
RdRef = 3.37
sigma = 0.87
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

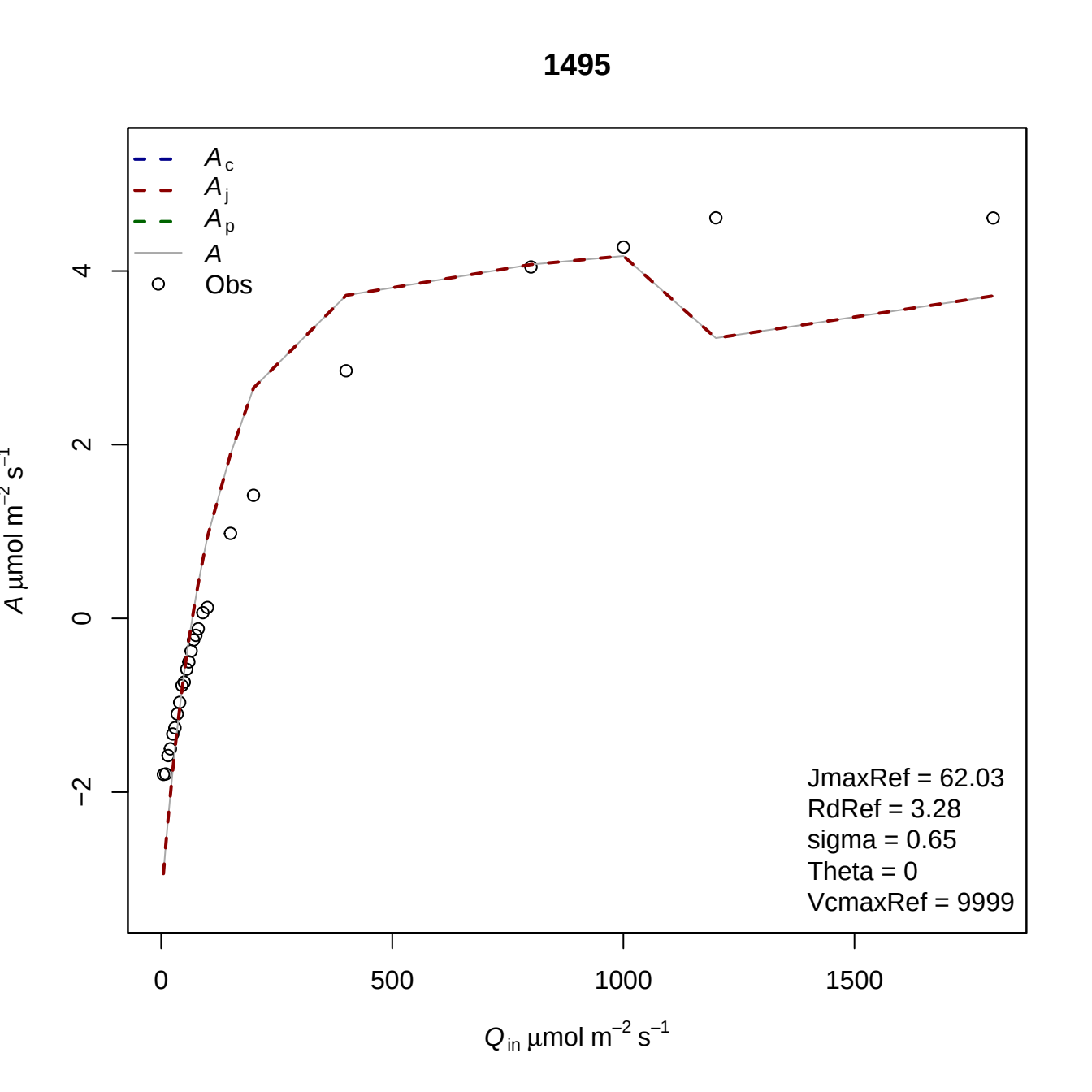

1490
A_c
A_j
A_p
A
Obs
JmaxRef = 84.63
RdRef = 4.71
sigma = 0.91
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

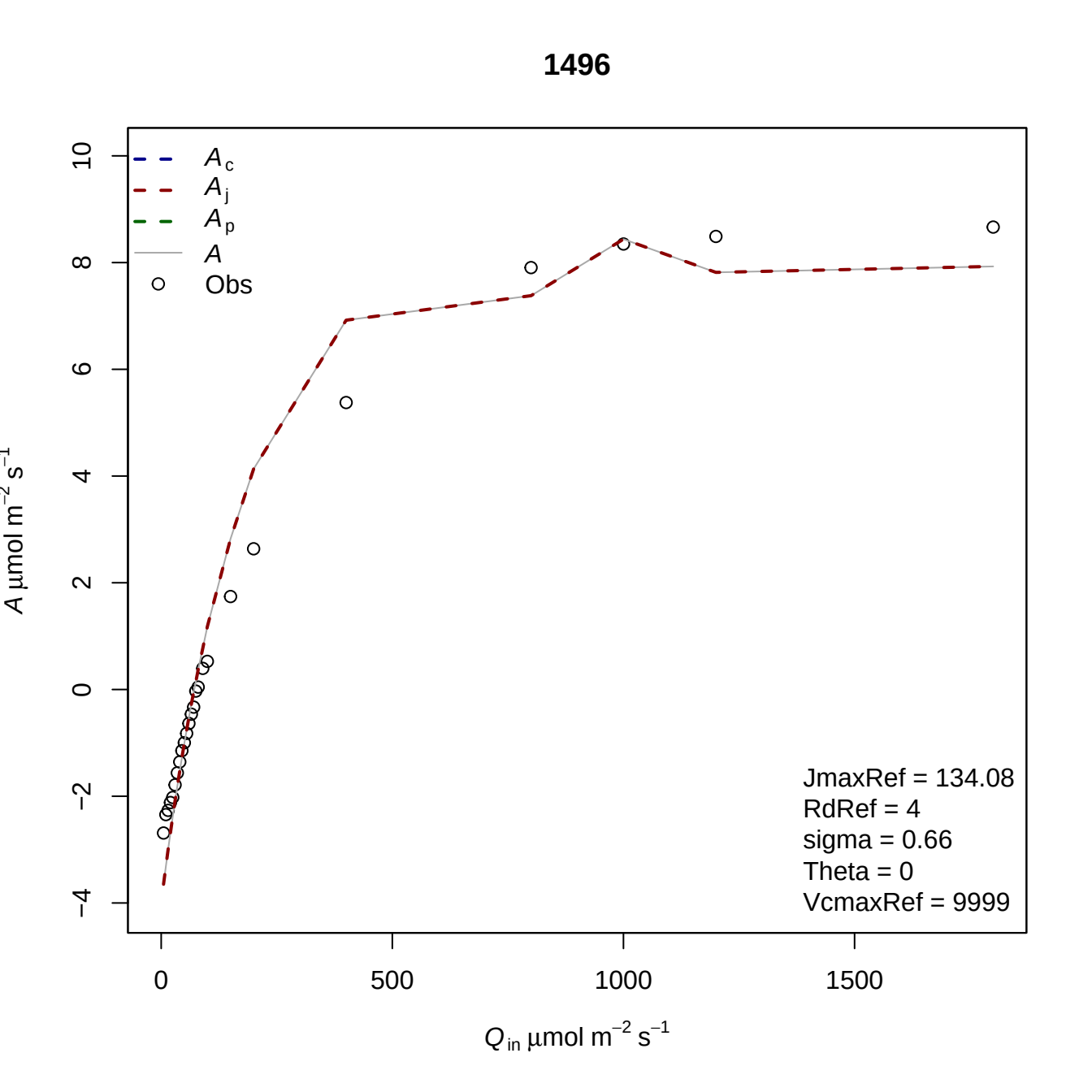

1491
A_c
A_j
A_p
A
Obs
JmaxRef = 82.56
RdRef = 3.97
sigma = 0.72
Theta = 0
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

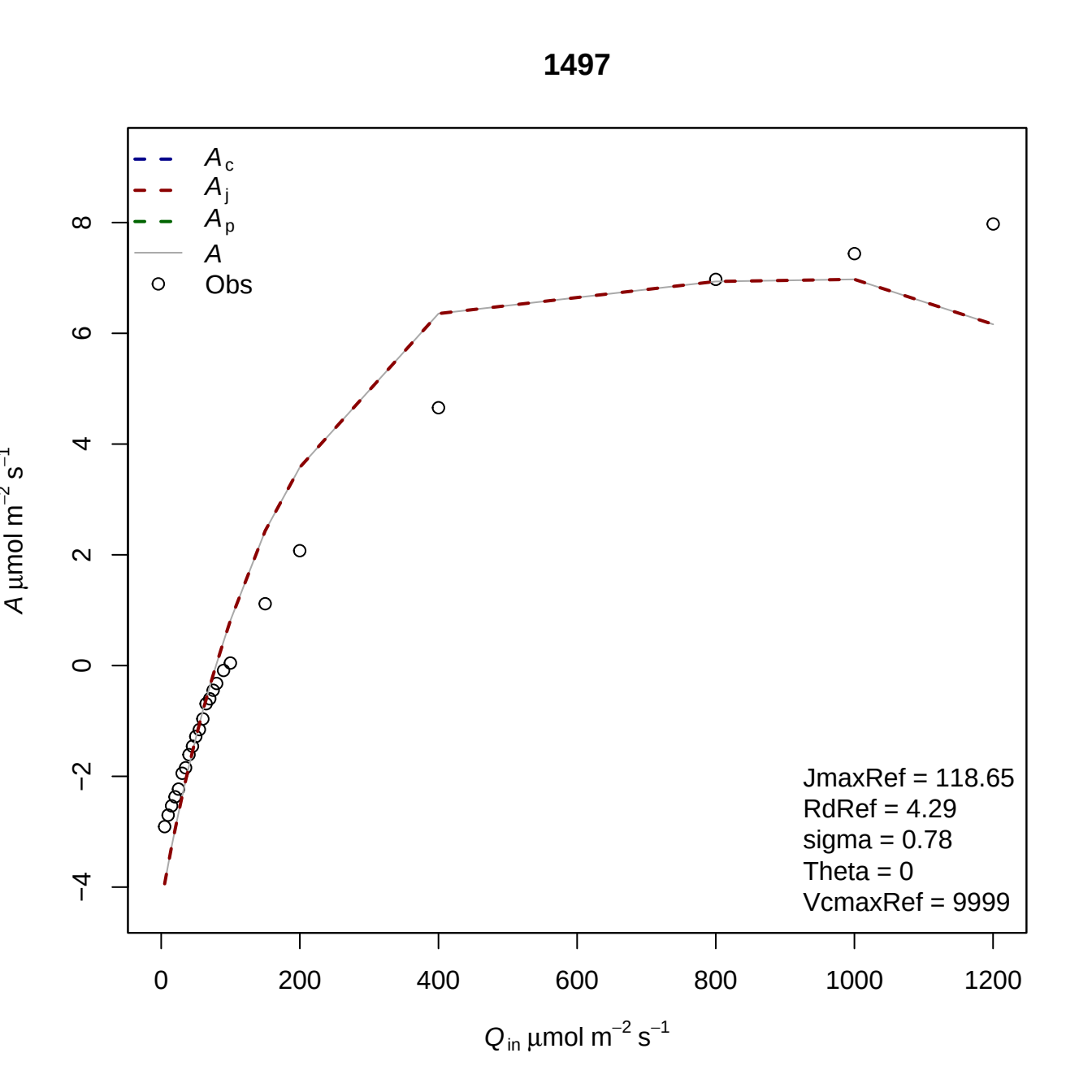

1492
A_c
A_j
A_p
A
Obs
JmaxRef = 85.28
RdRef = 2.58
sigma = 0.83
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

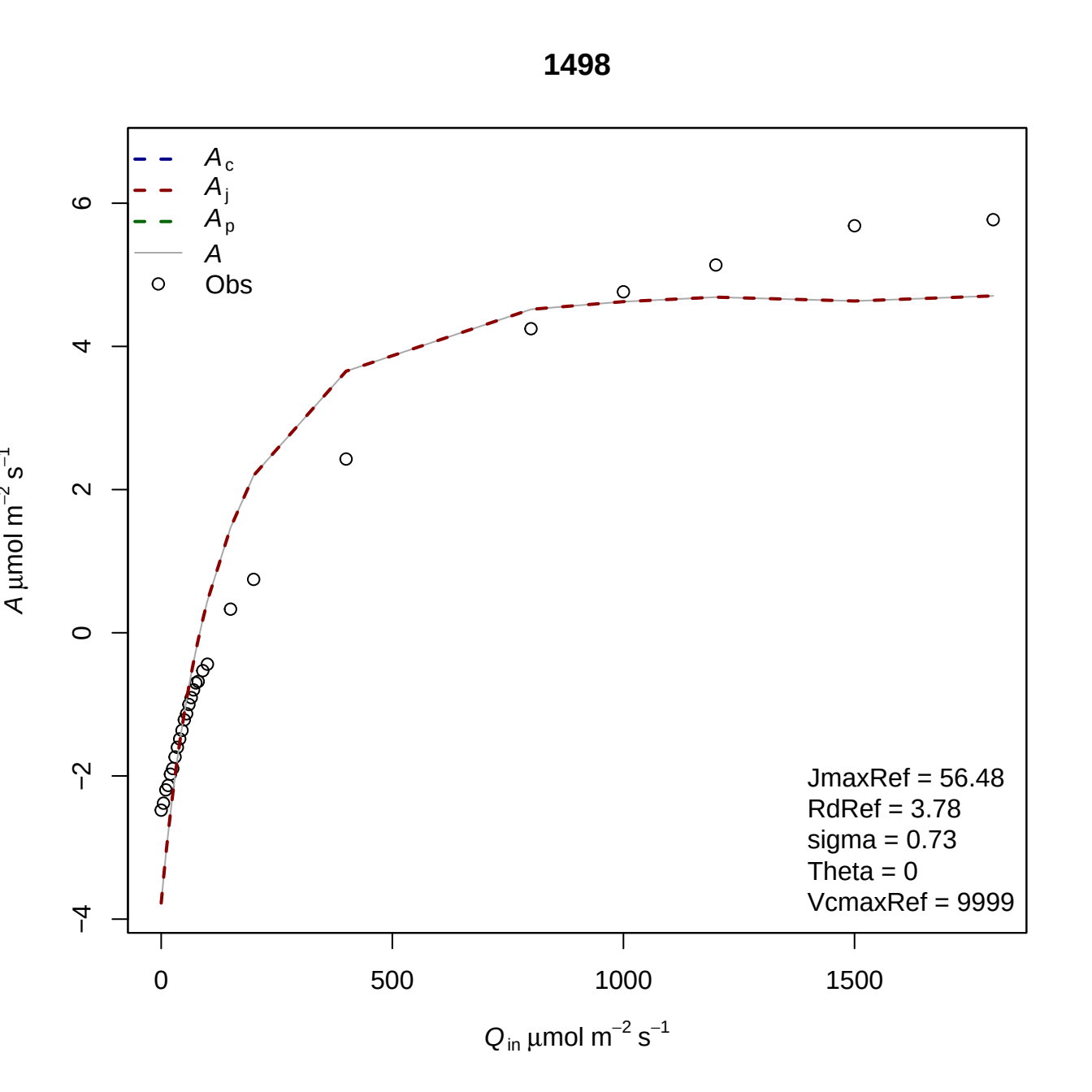

1493
A_c
A_j
A_p
A
Obs
JmaxRef = 58.95
RdRef = 2.89
sigma = 0.74
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

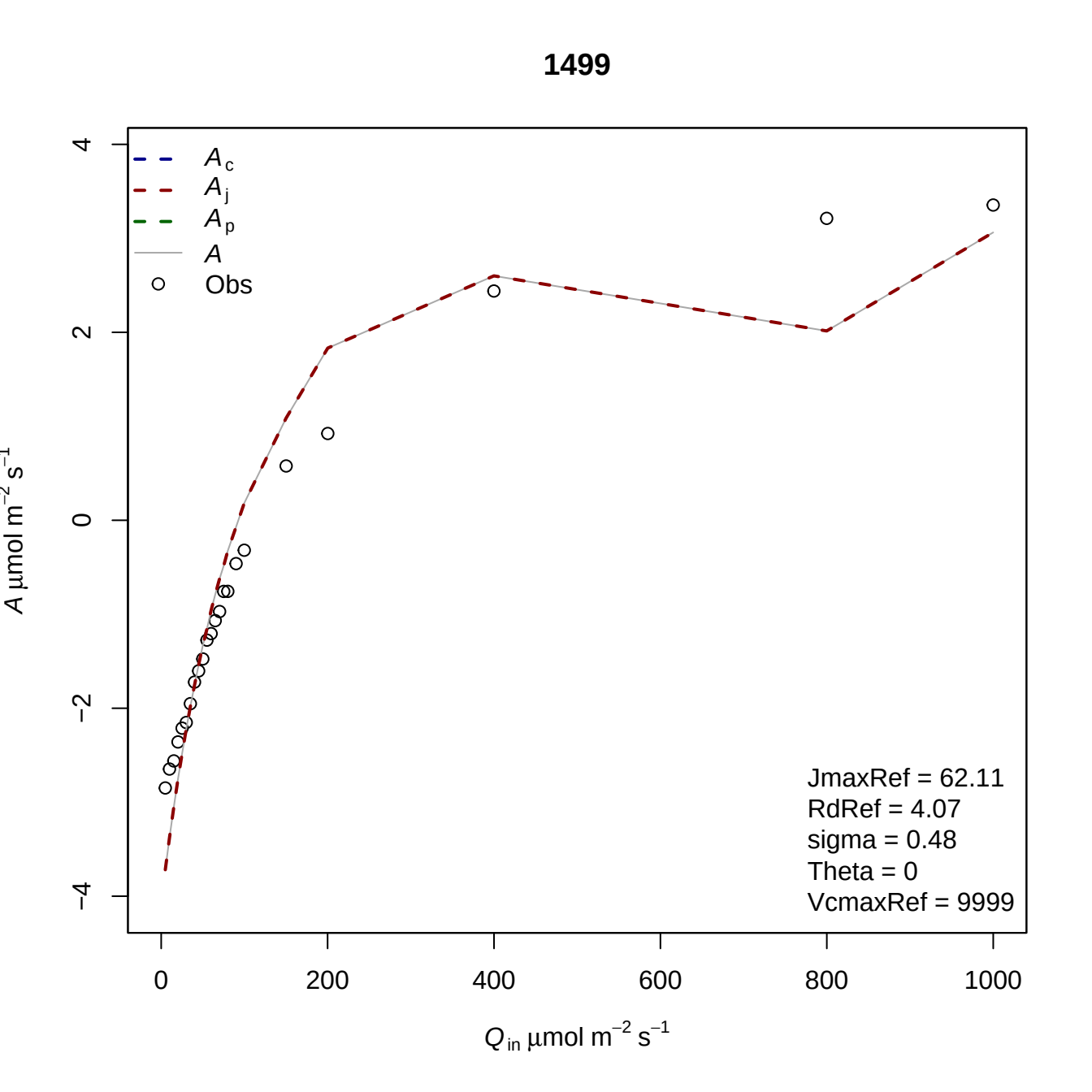

1494
A_c
A_j
A_p
A
Obs
JmaxRef = 48.92
RdRef = 3.02
sigma = 0.76
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

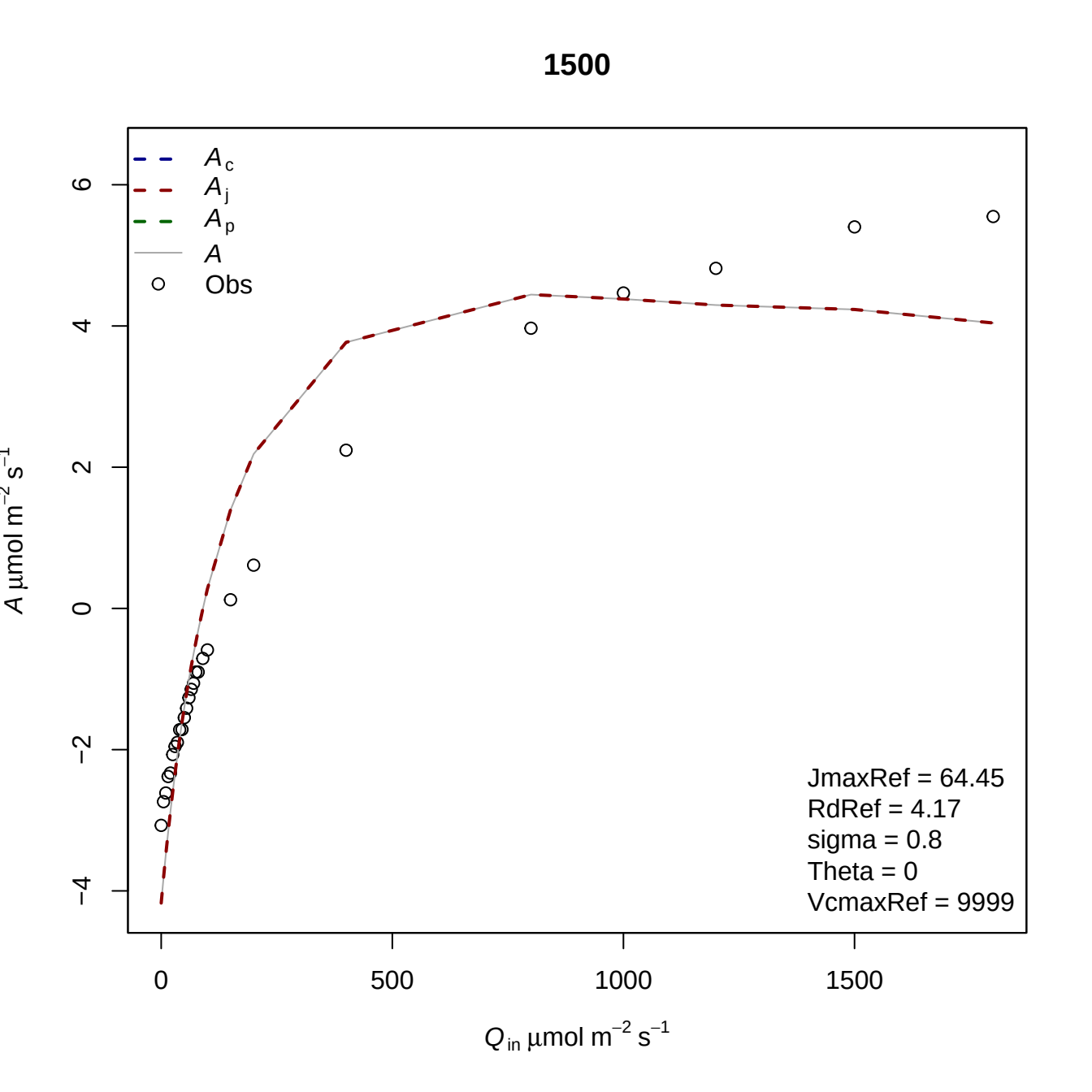

1495
A_c
A_j
A_p
A
Obs
JmaxRef = 62.03
RdRef = 3.28
sigma = 0.65
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

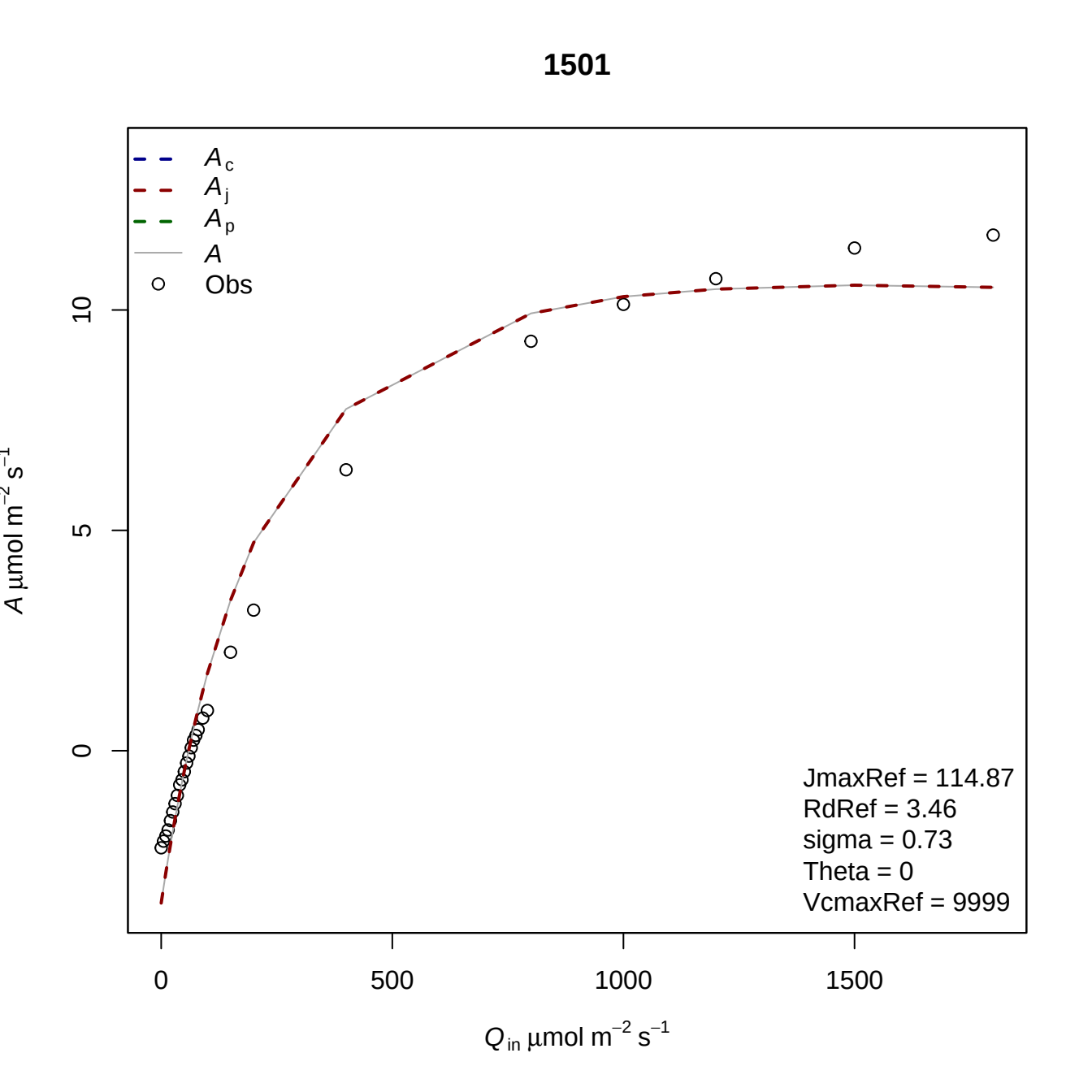

1496
A_c
A_j
A_p
A
Obs
JmaxRef = 134.08
RdRef = 4
sigma = 0.66
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

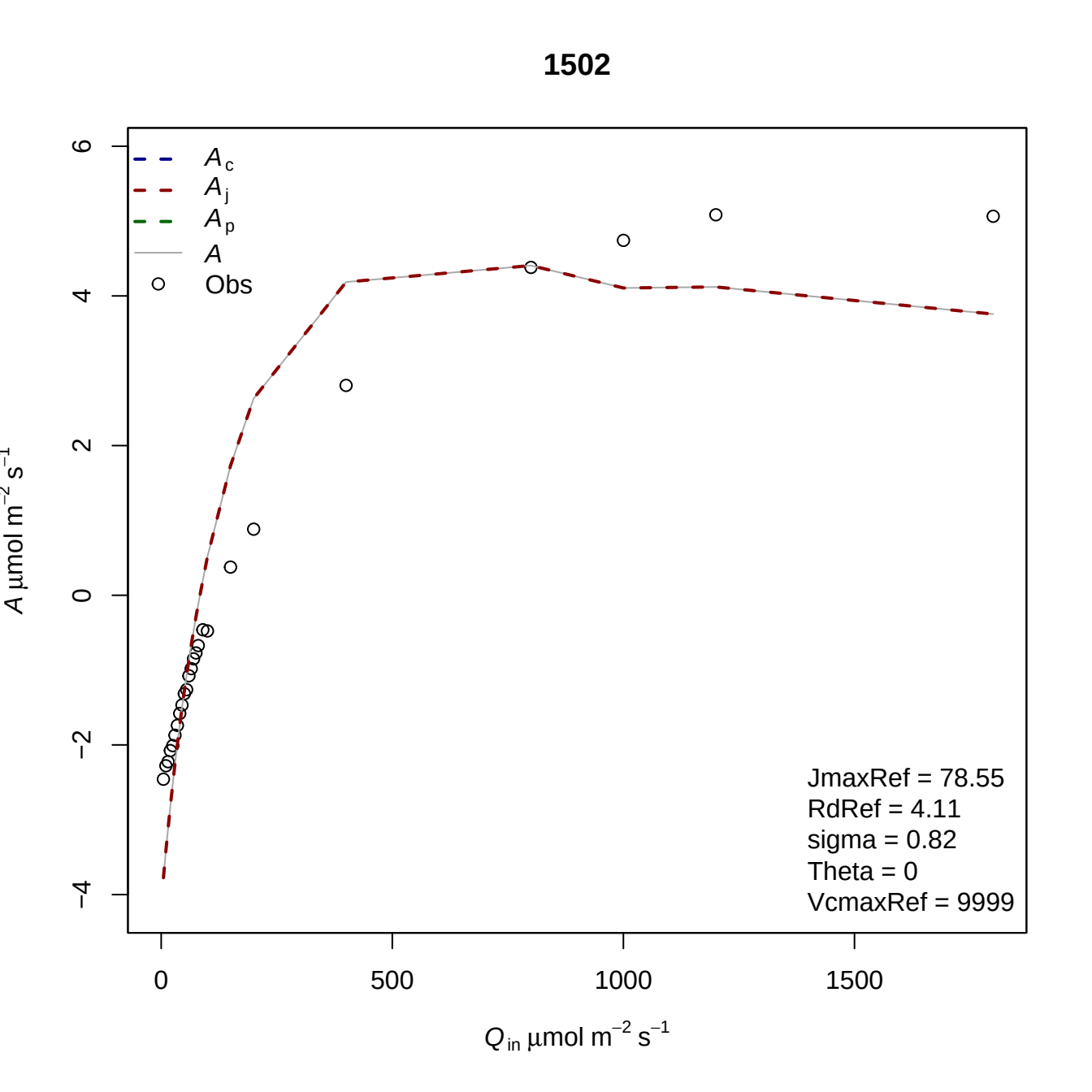

1497
A_c
A_j
A_p
A
Obs
JmaxRef = 118.65
RdRef = 4.29
sigma = 0.78
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

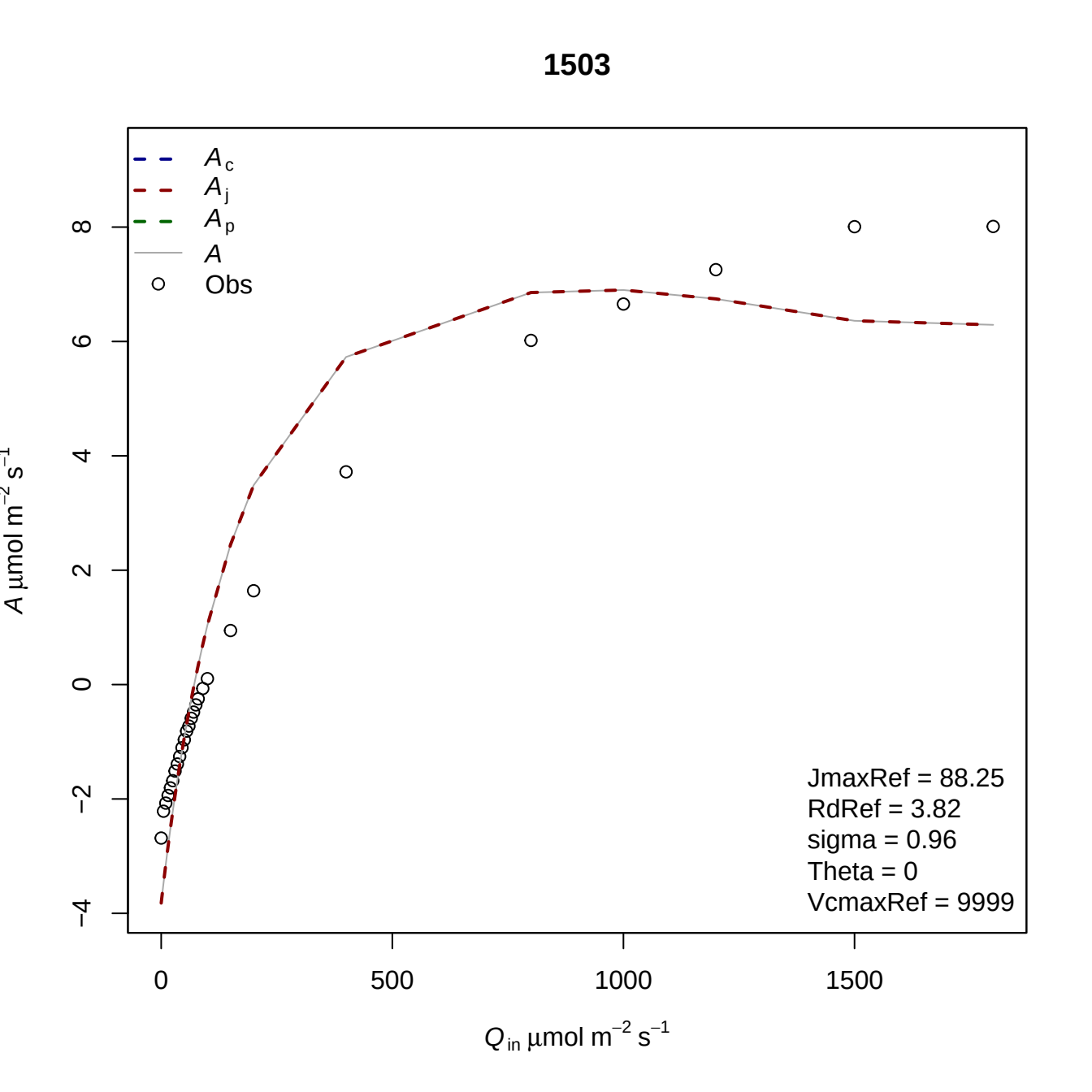

1498
A_c
A_j
A_p
A
Obs
JmaxRef = 56.48
RdRef = 3.78
sigma = 0.73
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

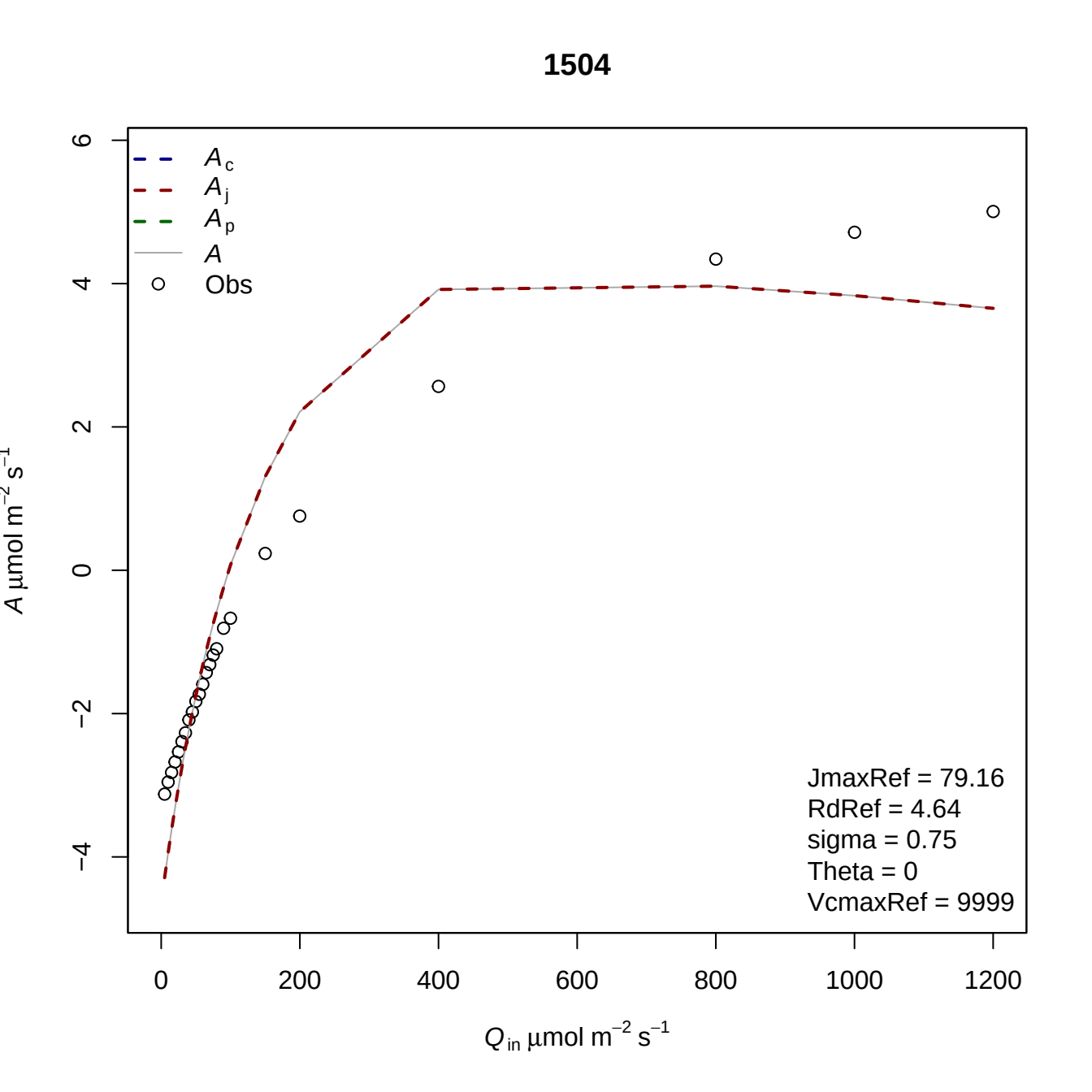

1499
A_c
A_j
A_p
A
Obs
JmaxRef = 62.11
RdRef = 4.07
sigma = 0.48
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
200
400
600
800
1000

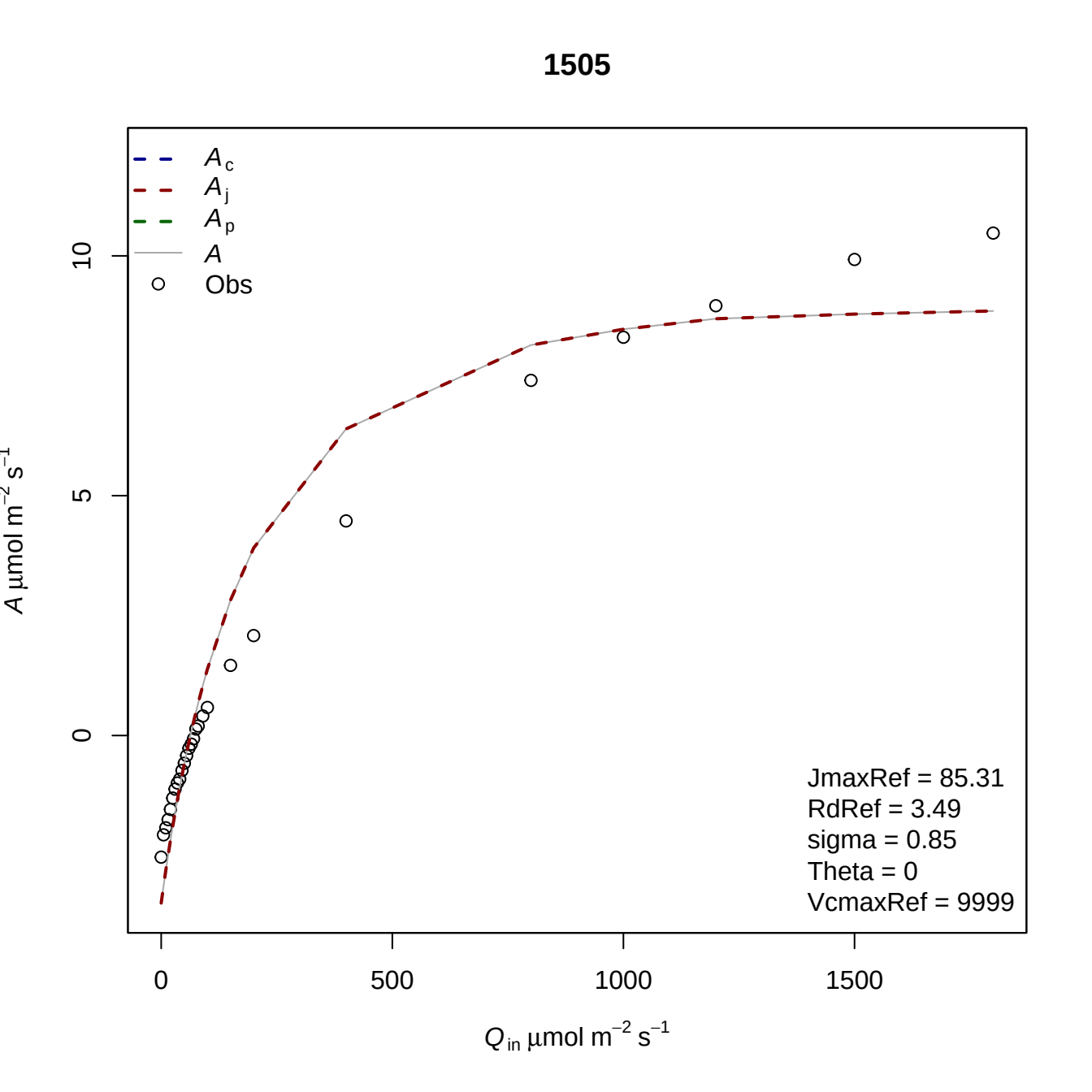

1500
A_c
A_j
A_p
A
Obs
JmaxRef = 64.45
RdRef = 4.17
sigma = 0.8
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

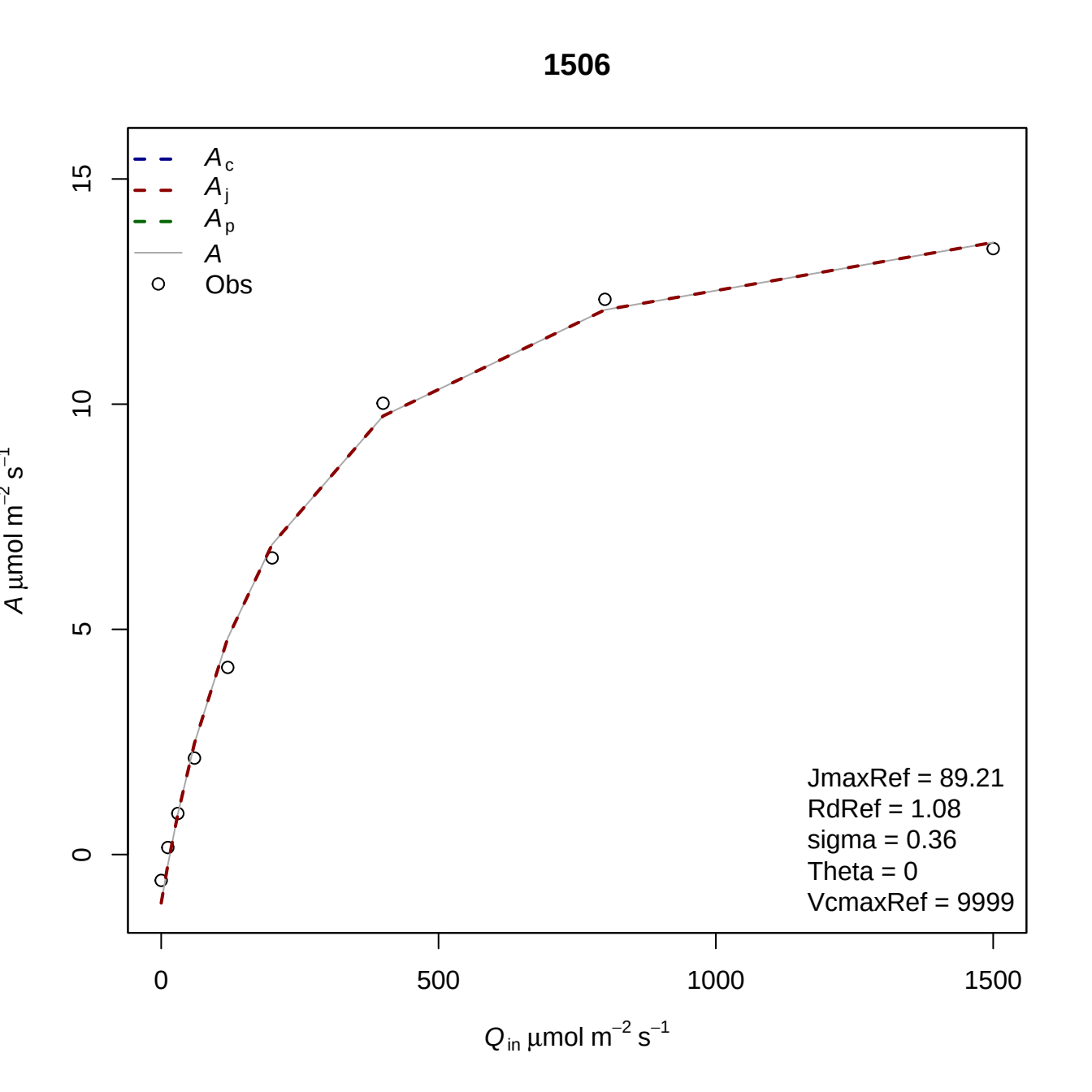

1501
A_c
A_j
A_p
A
Obs
JmaxRef = 114.87
RdRef = 3.46
sigma = 0.73
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

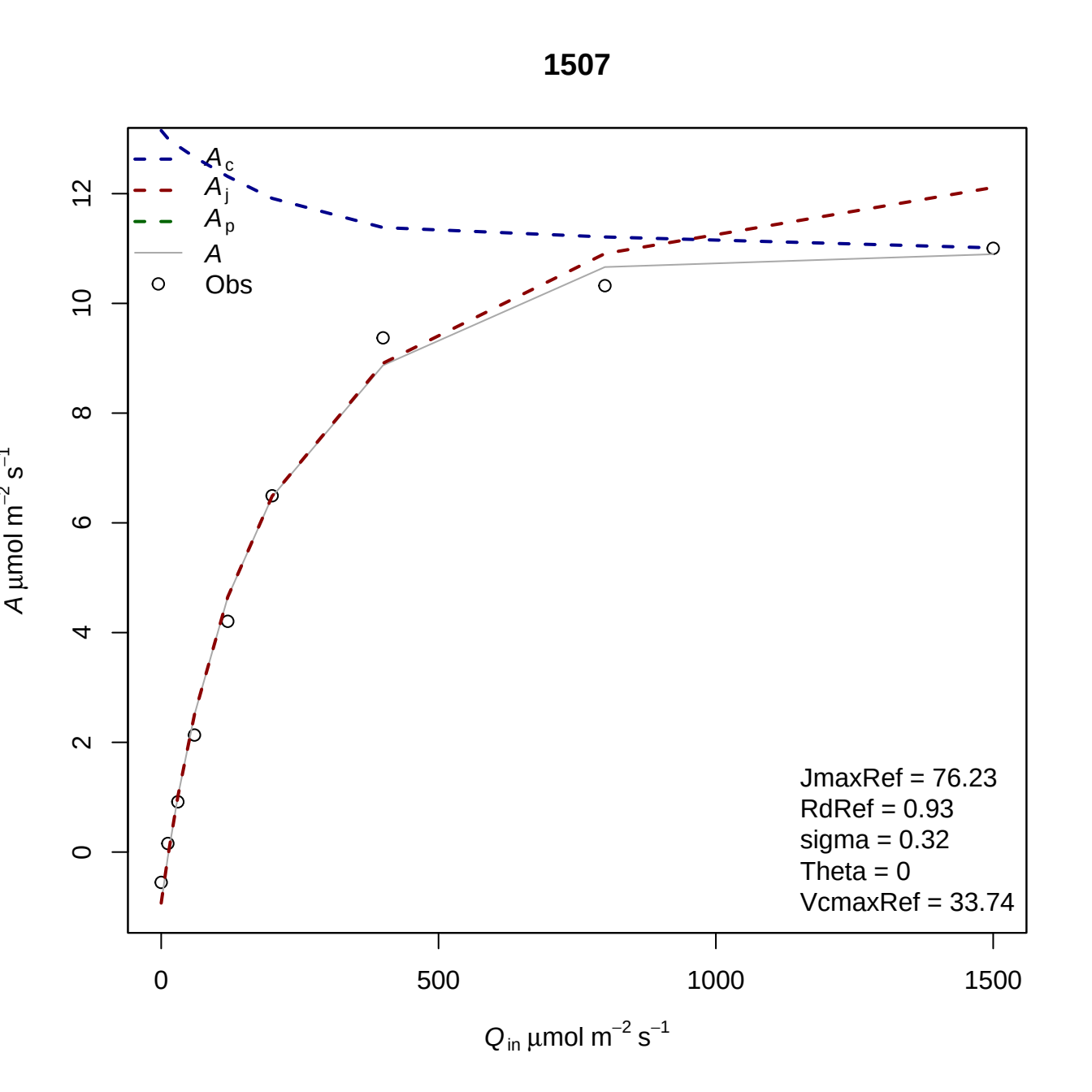

1502
A_c
A_j
A_p
A
Obs
JmaxRef = 78.55
RdRef = 4.11
sigma = 0.82
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

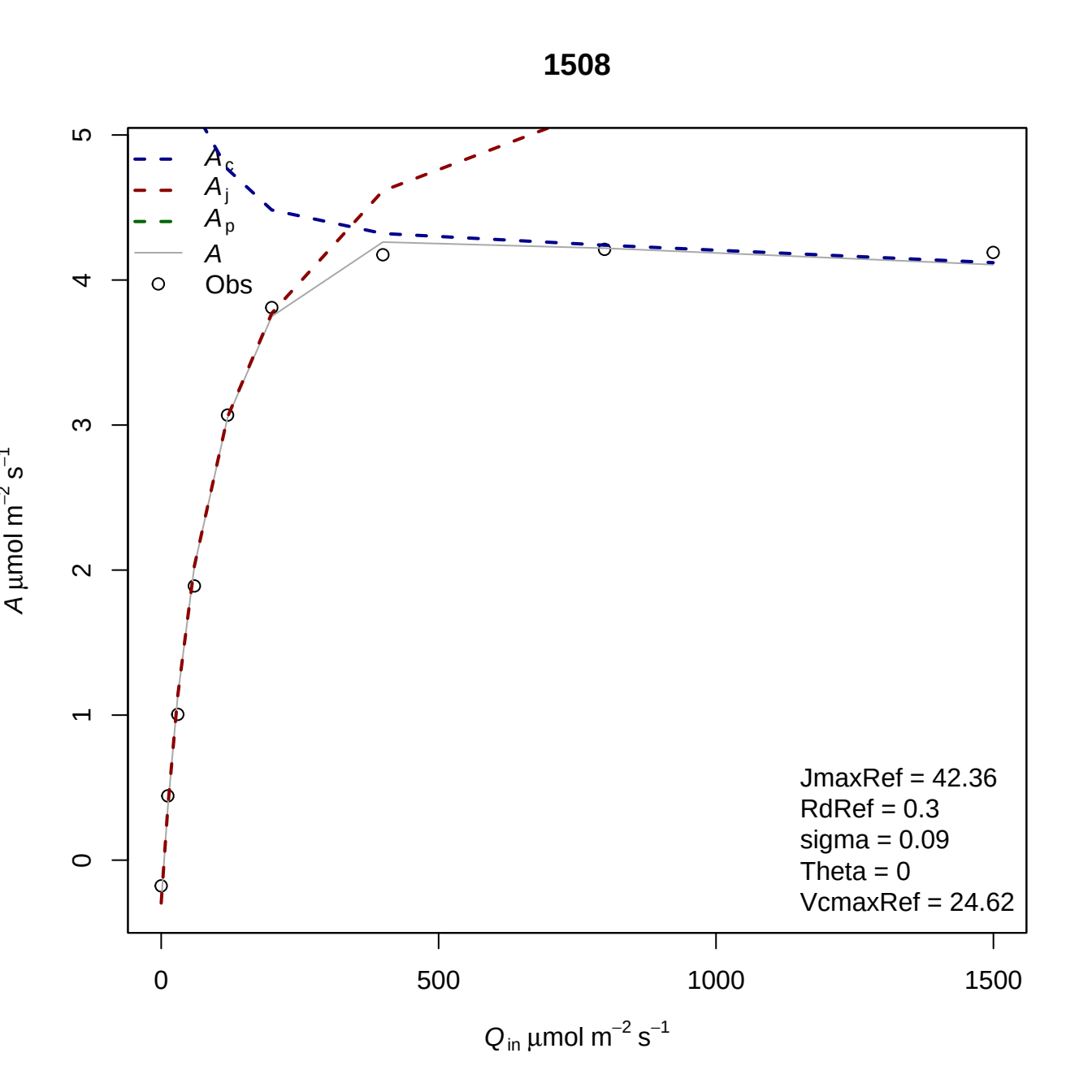

1503
A_c
A_j
A_p
A
Obs
JmaxRef = 88.25
RdRef = 3.82
sigma = 0.96
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

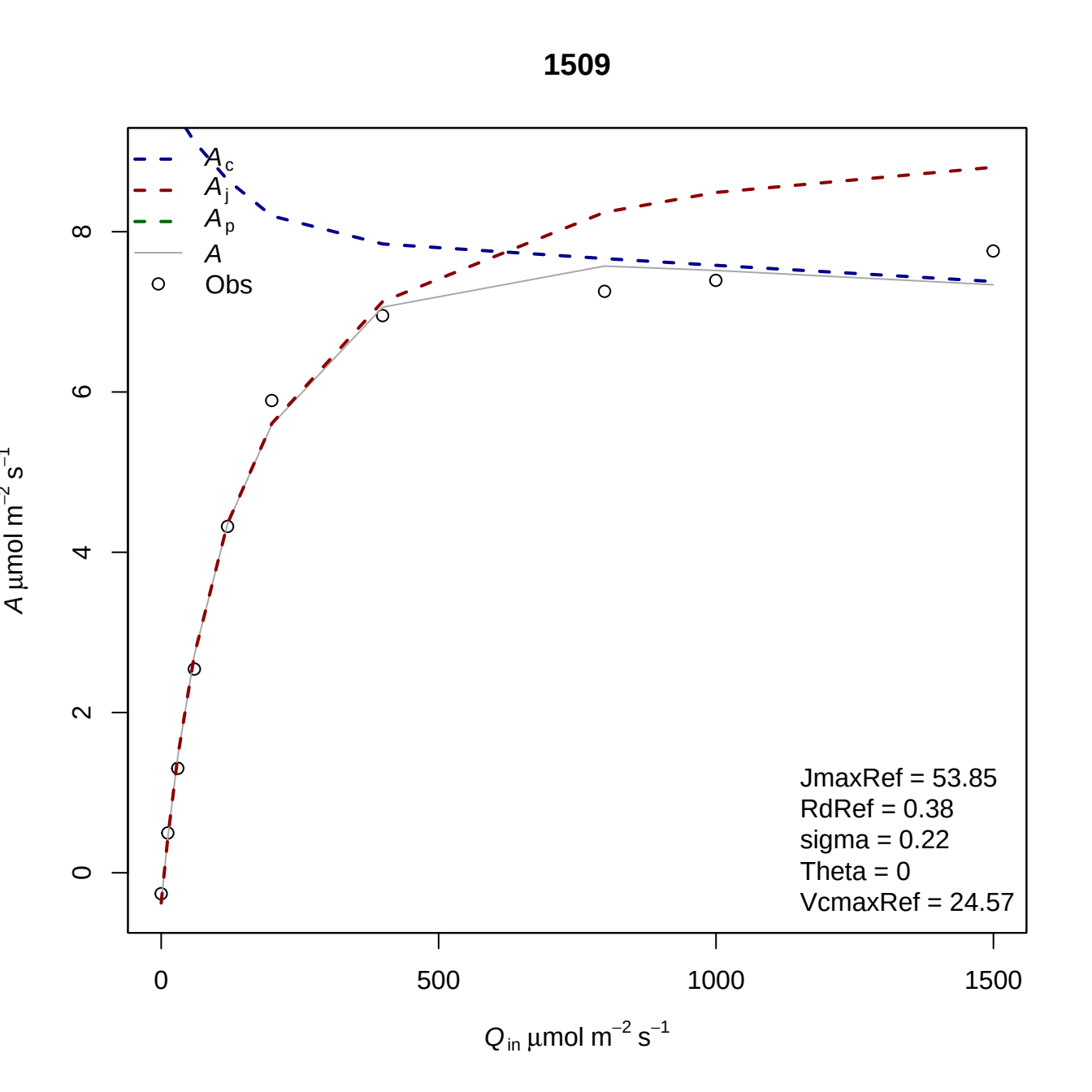

1504
A_c
A_j
A_p
A
Obs
JmaxRef = 79.16
RdRef = 4.64
sigma = 0.75
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

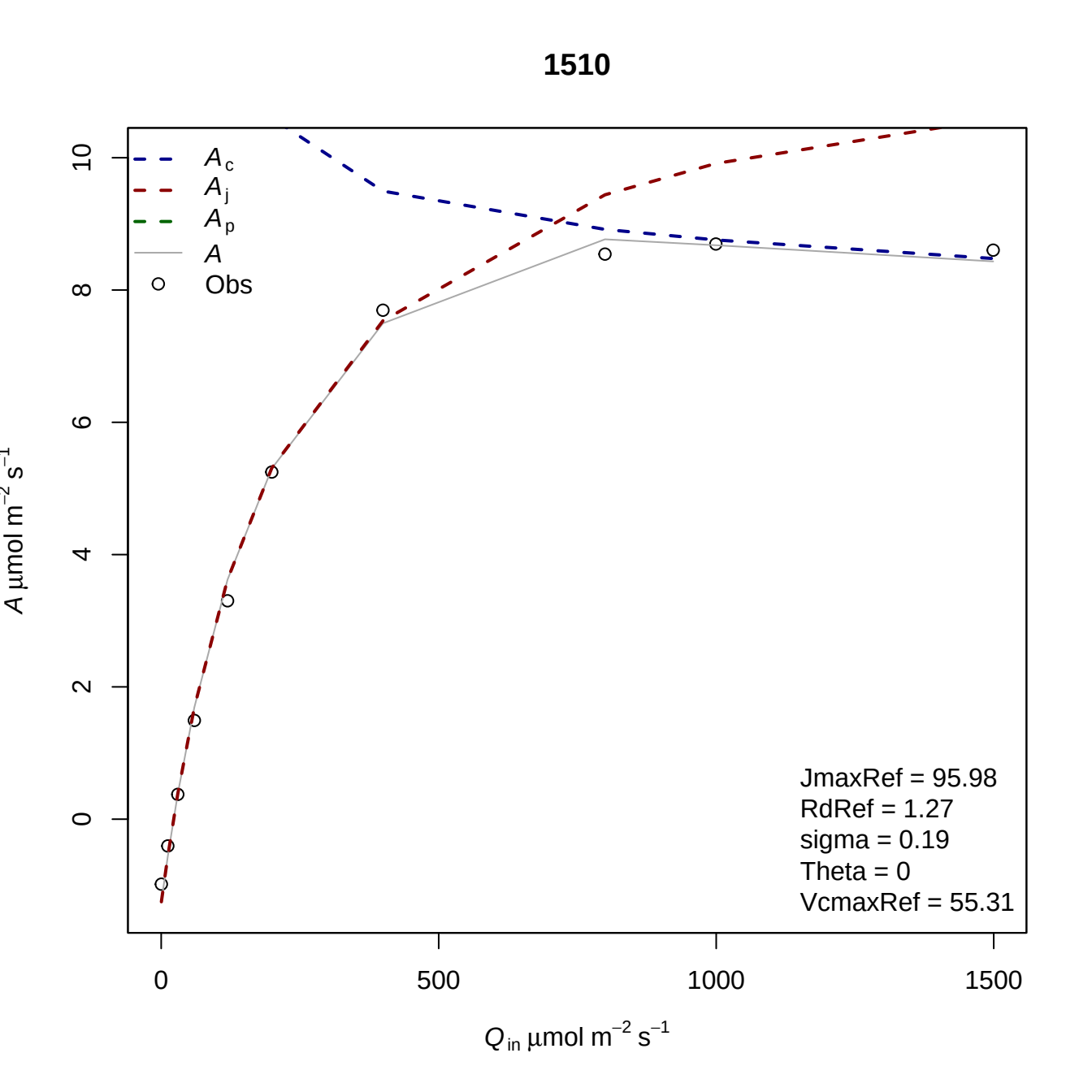

1505
A_c
A_j
A_p
A
Obs
JmaxRef = 85.31
RdRef = 3.49
sigma = 0.85
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

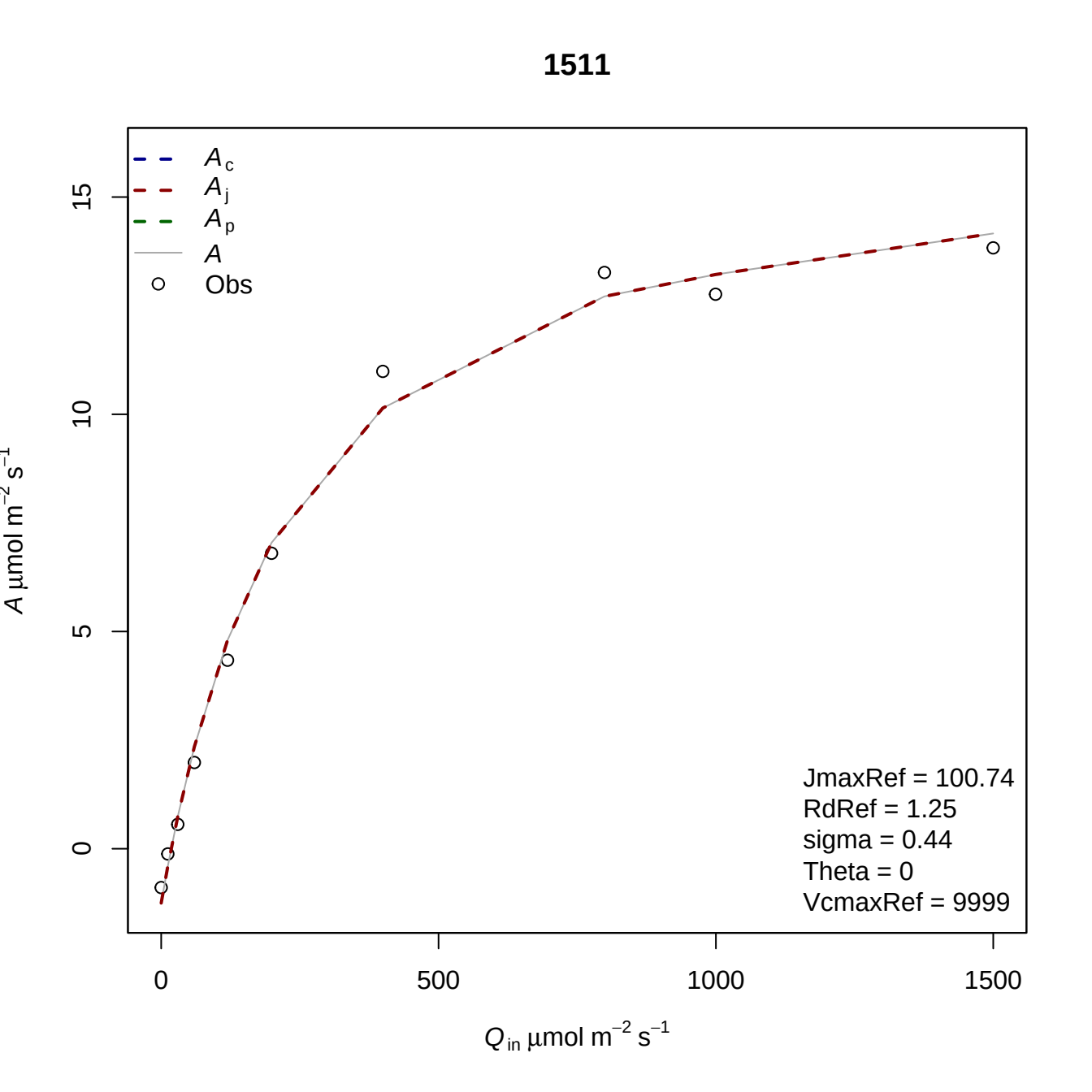

1506
A_c
A_j
A_p
A
Obs
JmaxRef = 89.21
RdRef = 1.08
sigma = 0.36
Theta = 0
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

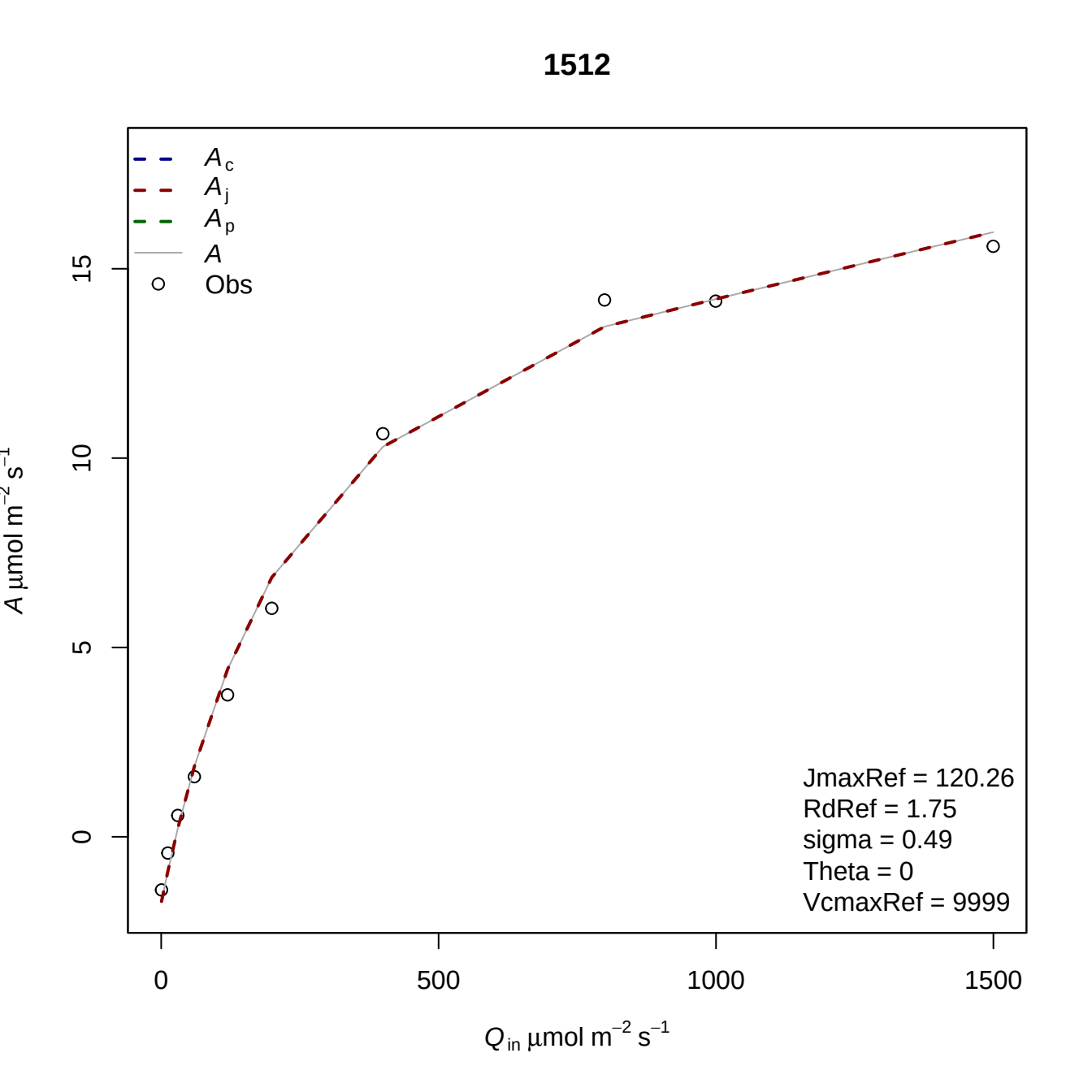

1507
A_c
A_j
A_p
A
Obs
JmaxRef = 76.23
RdRef = 0.93
sigma = 0.32
Theta = 0
VcmaxRef = 33.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

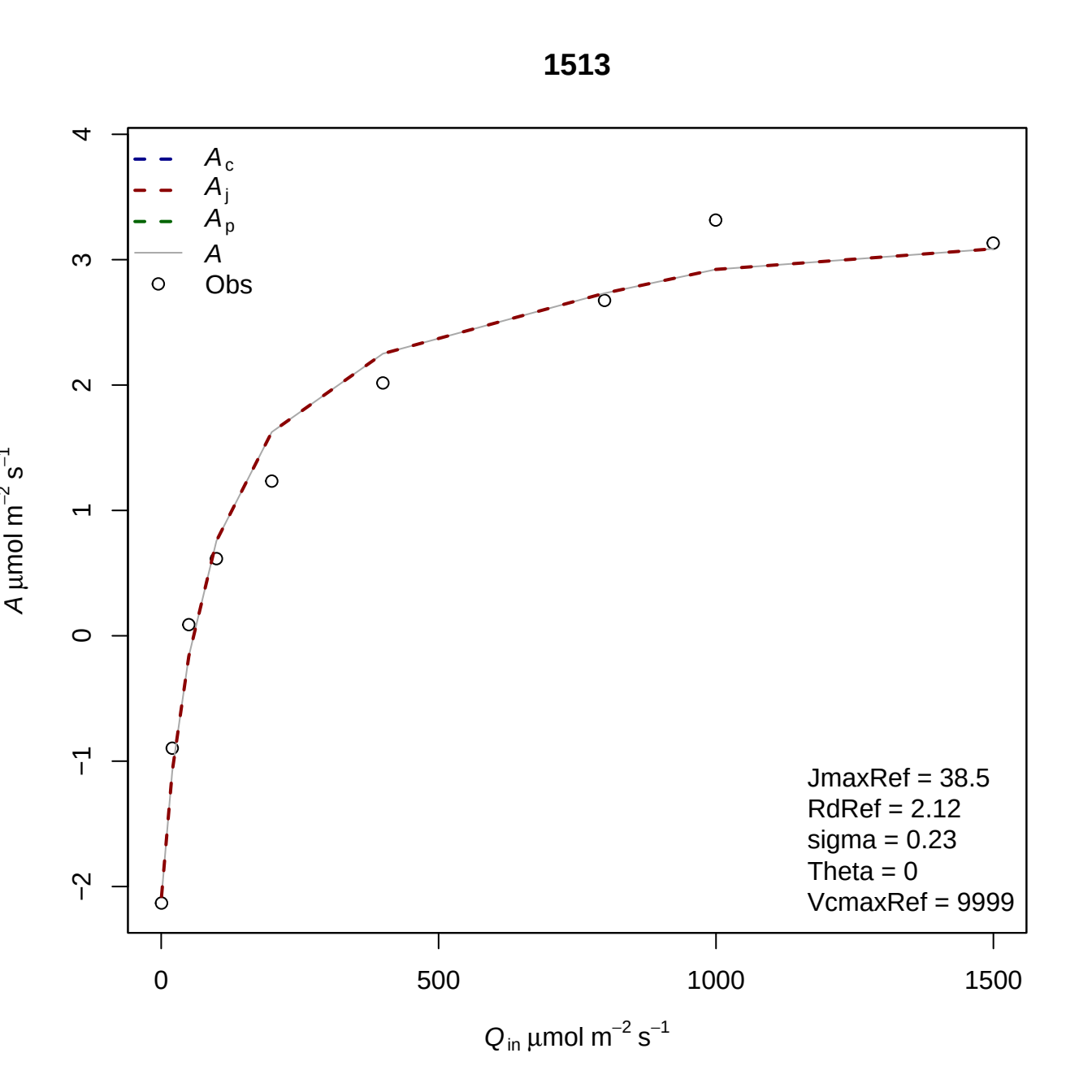

1508
A_c
A_j
A_p
A
Obs
JmaxRef = 42.36
RdRef = 0.3
sigma = 0.09
Theta = 0
VcmaxRef = 24.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

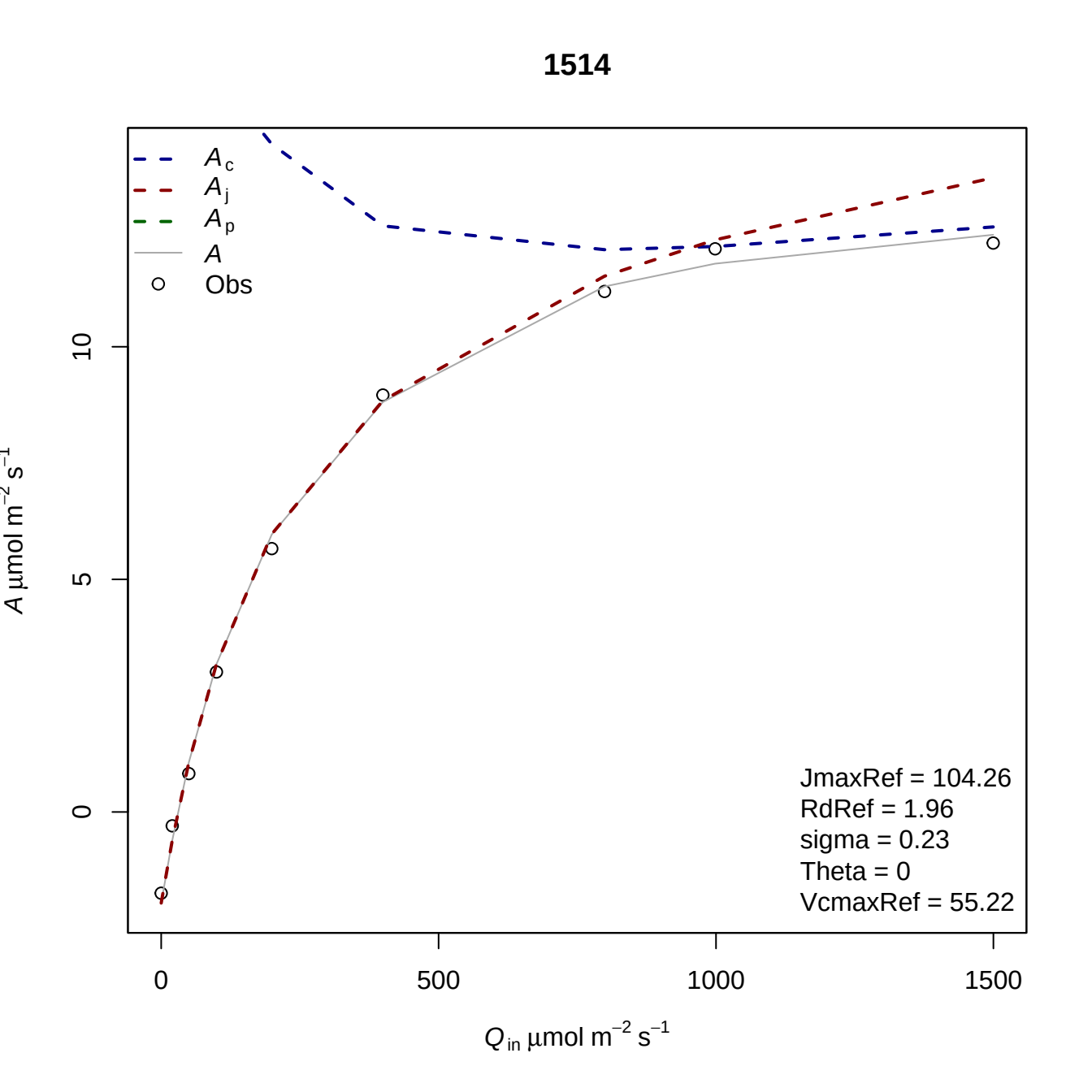

1509
A_c
A_j
A_p
A
Obs
JmaxRef = 53.85
RdRef = 0.38
sigma = 0.22
Theta = 0
VcmaxRef = 24.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

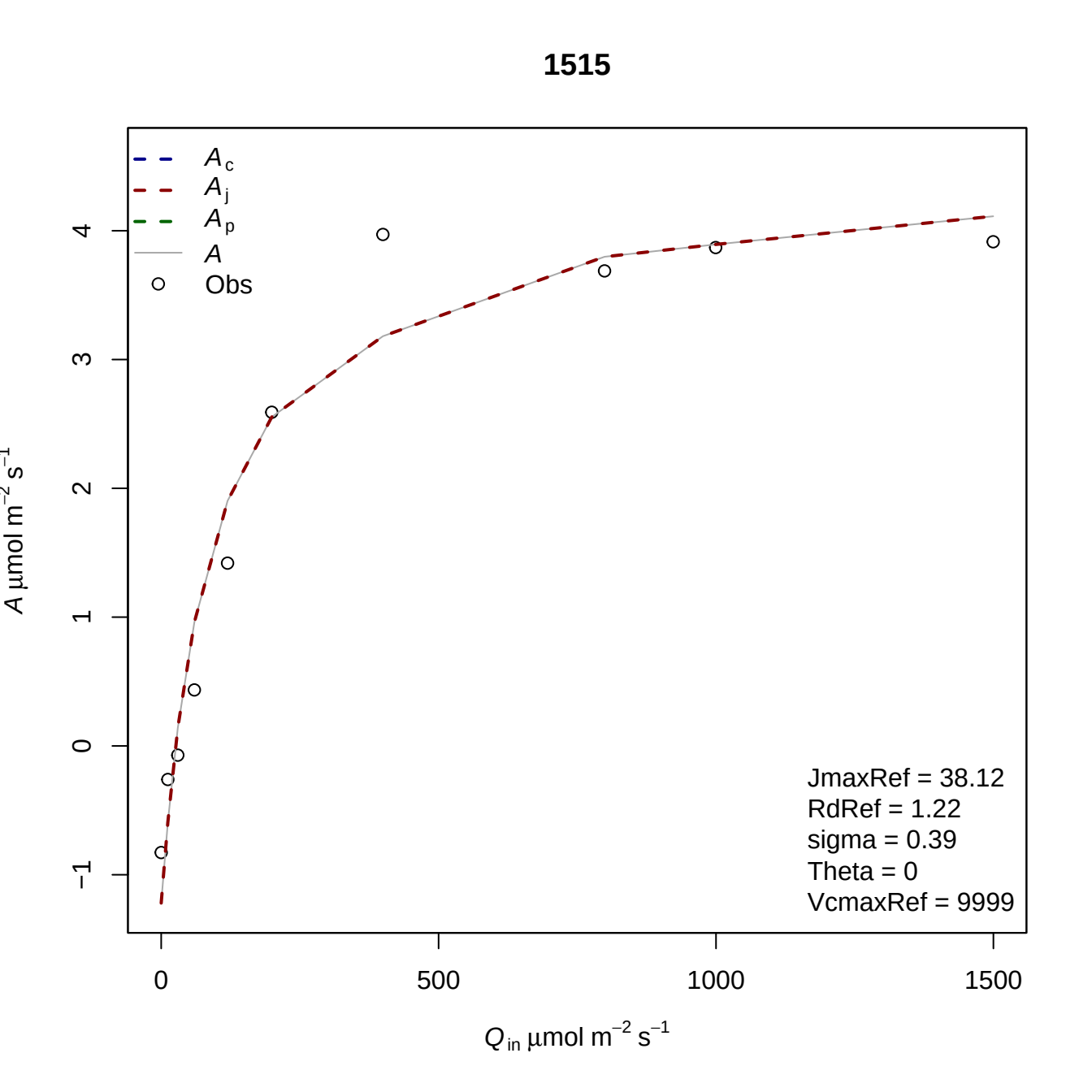

1510
A_c
A_j
A_p
A
Obs
JmaxRef = 95.98
RdRef = 1.27
sigma = 0.19
Theta = 0
VcmaxRef = 55.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

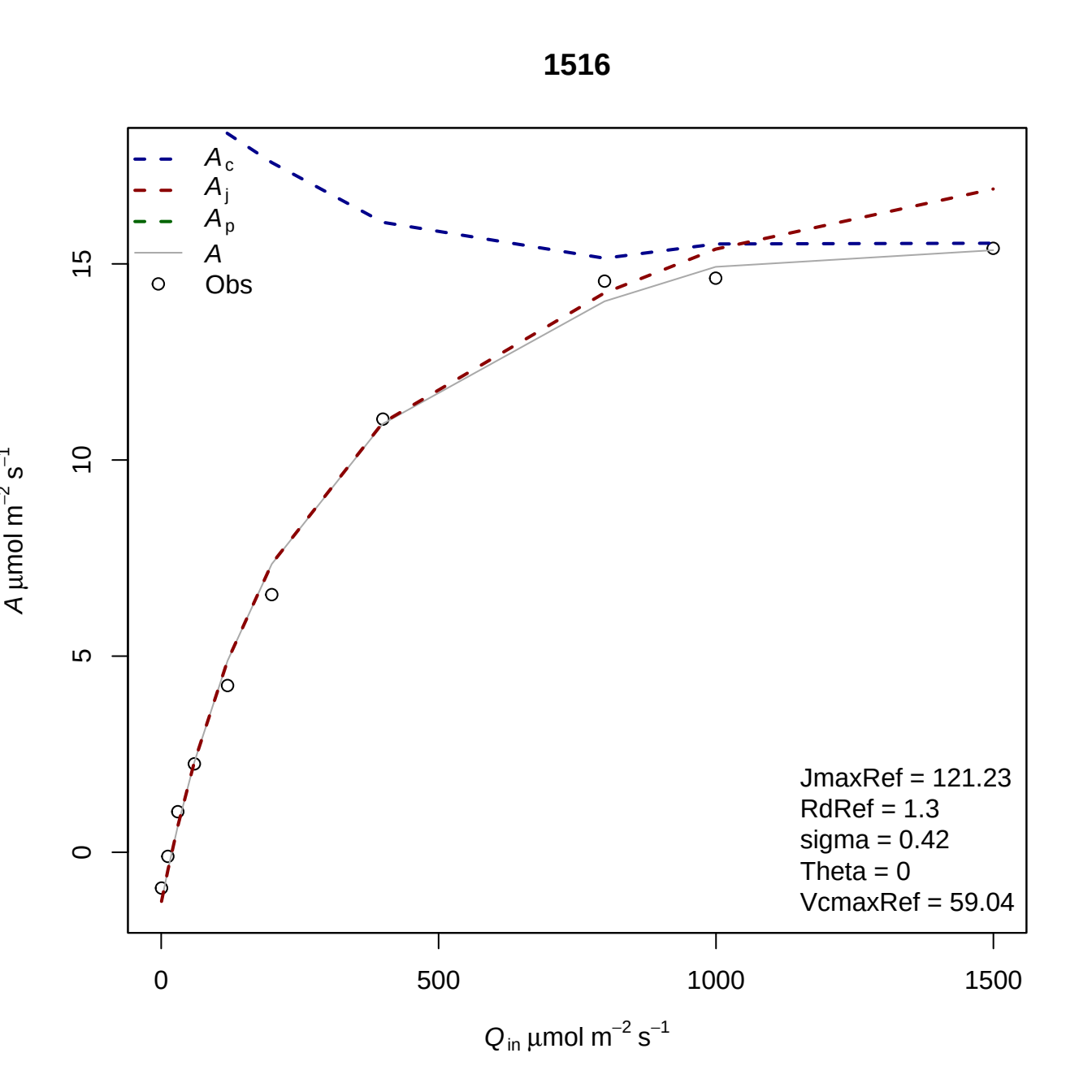

1511
A_c
A_j
A_p
A
Obs
JmaxRef = 100.74
RdRef = 1.25
sigma = 0.44
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

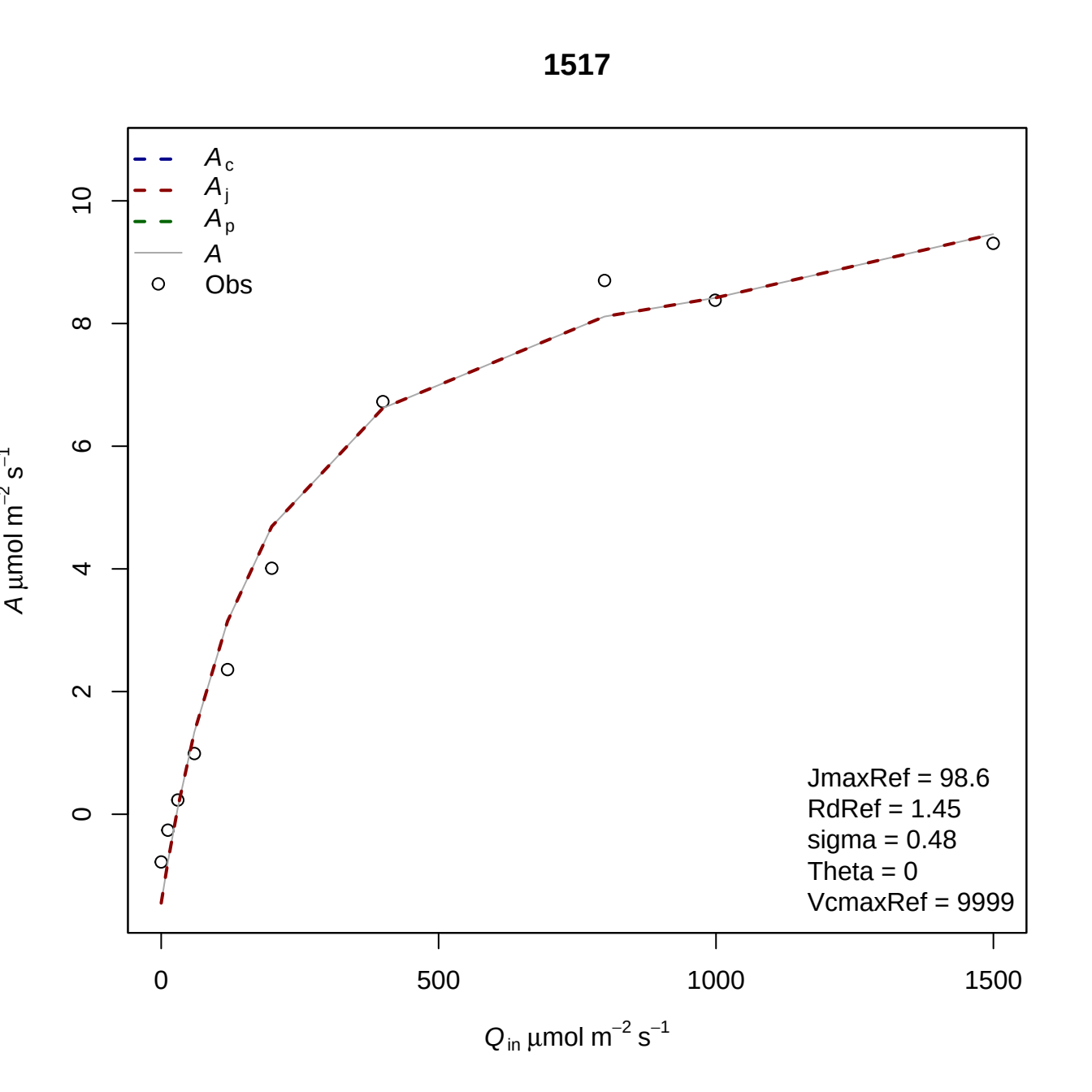

1512
A_c
A_j
A_p
A
Obs
JmaxRef = 120.26
RdRef = 1.75
sigma = 0.49
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

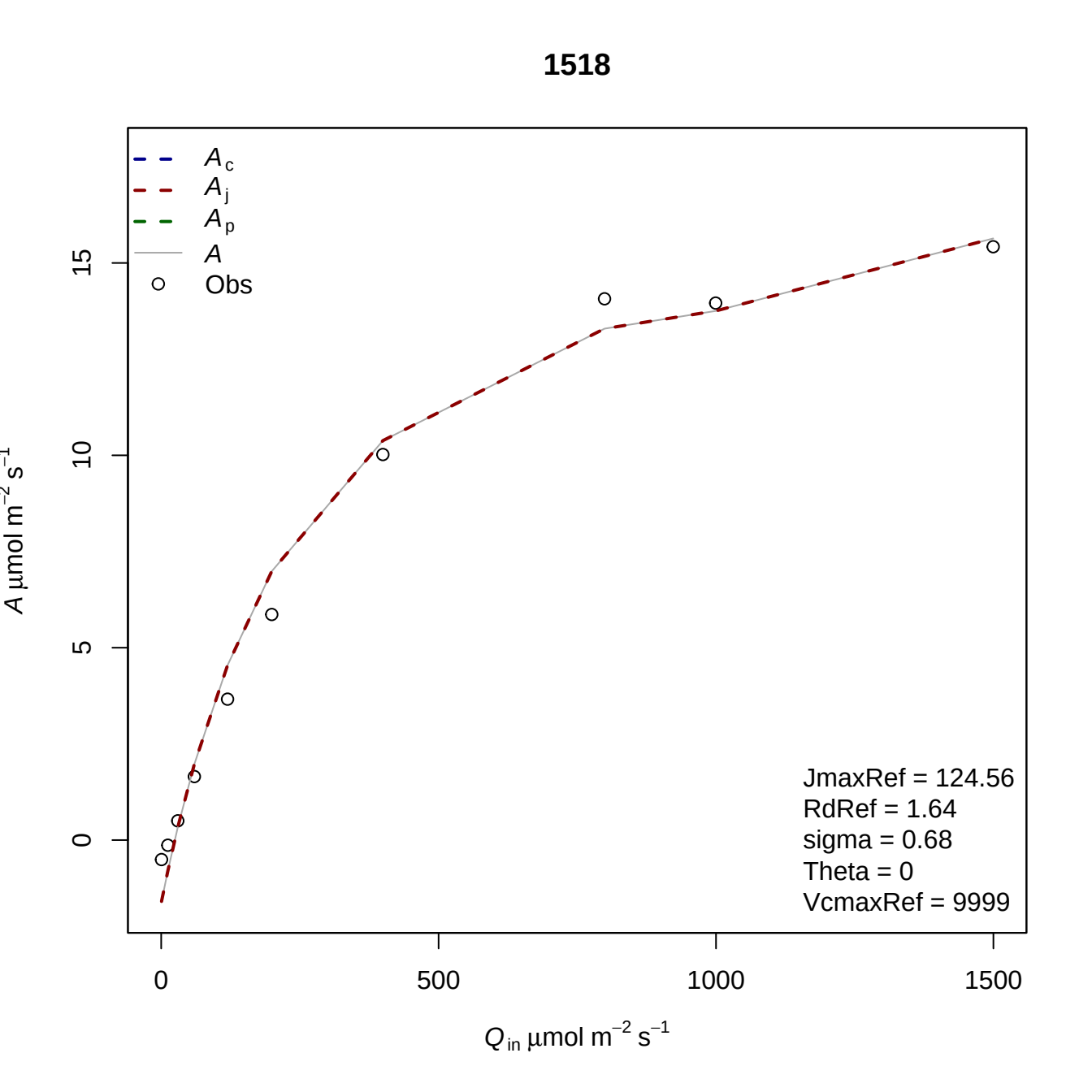

1513
A_c
A_j
A_p
A
Obs
JmaxRef = 38.5
RdRef = 2.12
sigma = 0.23
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

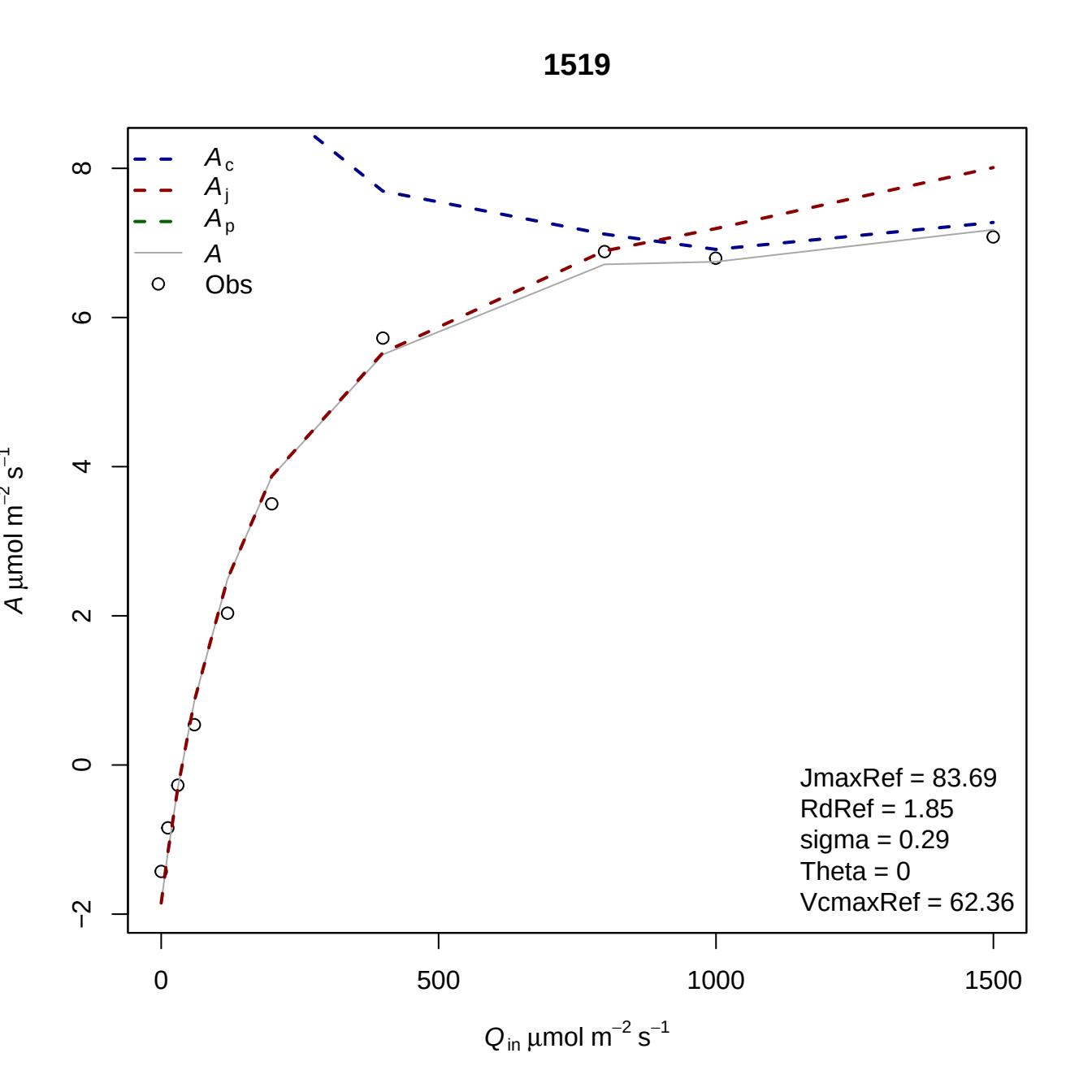

1514
A_c
A_j
A_p
A
Obs
JmaxRef = 104.26
RdRef = 1.96
sigma = 0.23
Theta = 0
VcmaxRef = 55.22
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

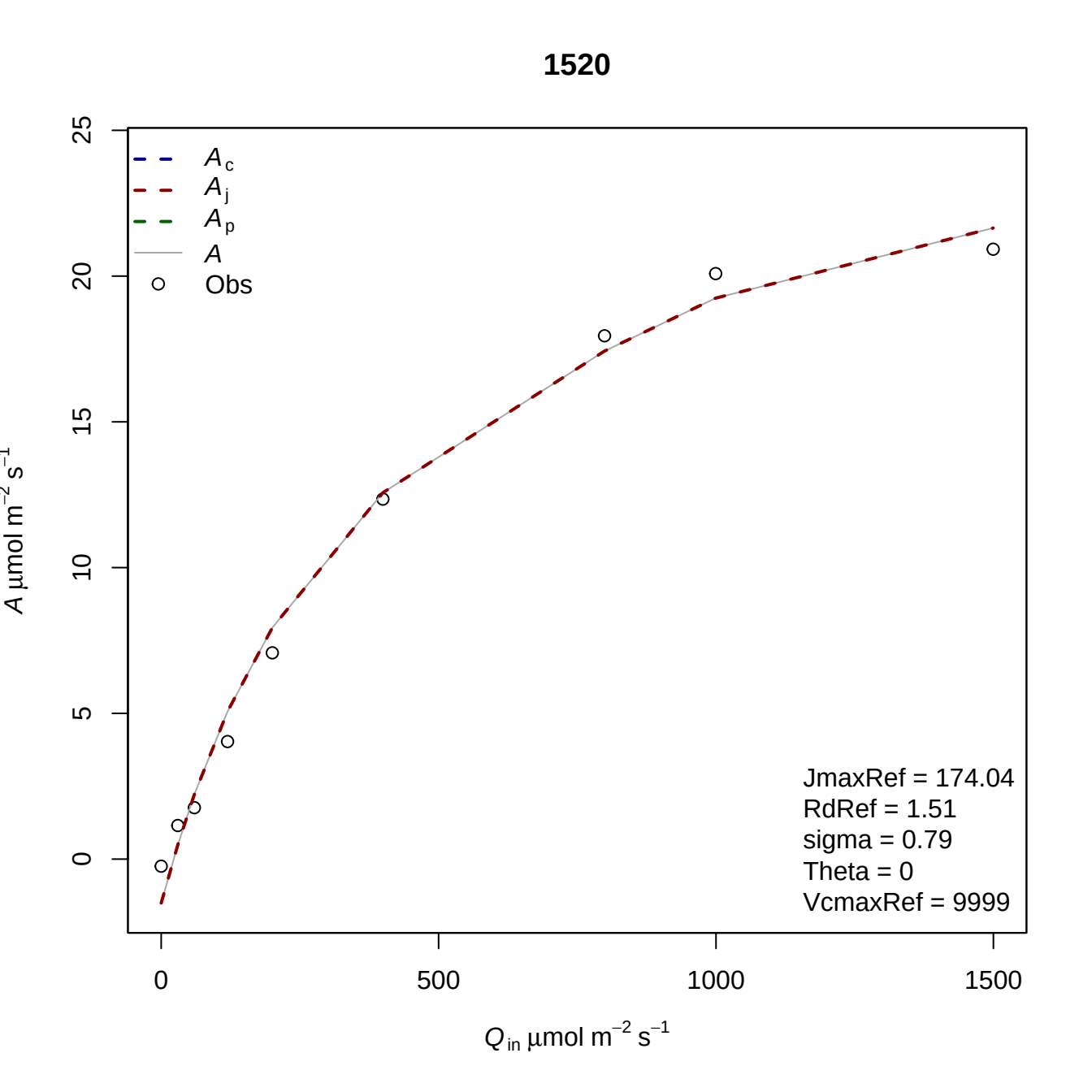

1515
A_c
A_j
A_p
A
Obs
JmaxRef = 38.12
RdRef = 1.22
sigma = 0.39
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

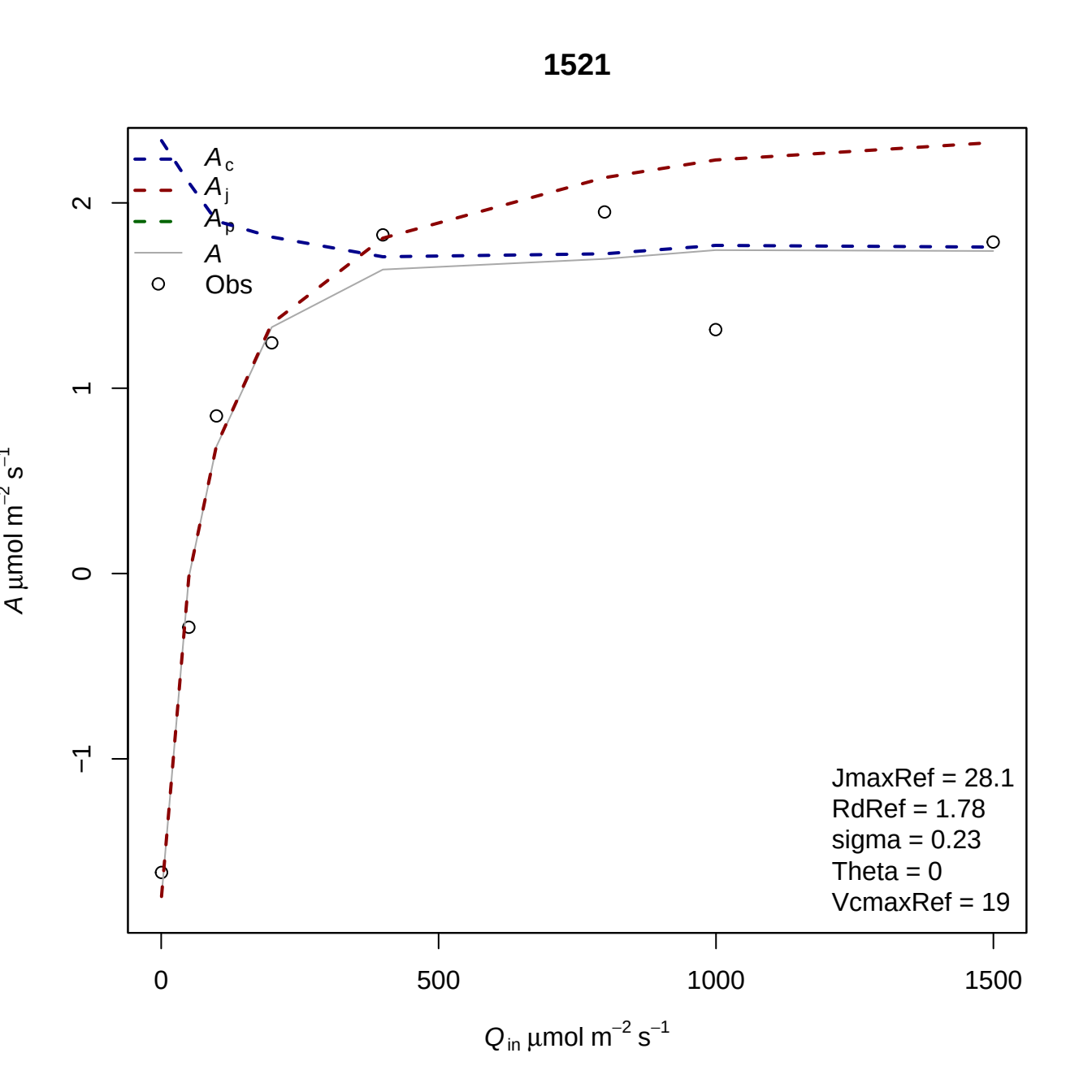

1516
A_c
A_j
A_p
A
Obs
JmaxRef = 121.23
RdRef = 1.3
sigma = 0.42
Theta = 0
VcmaxRef = 59.04
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

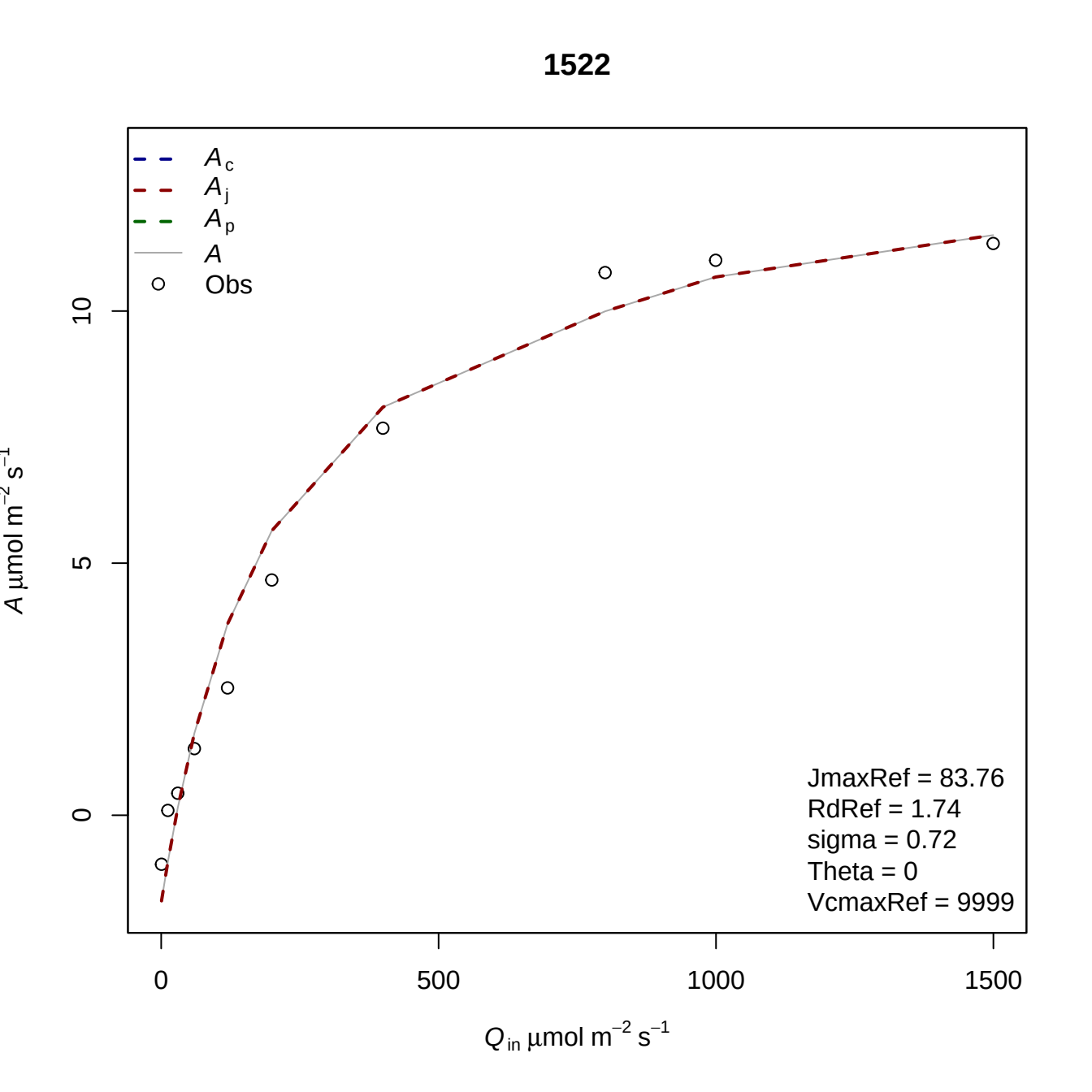

1517
A_c
A_j
A_p
A
Obs
JmaxRef = 98.6
RdRef = 1.45
sigma = 0.48
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

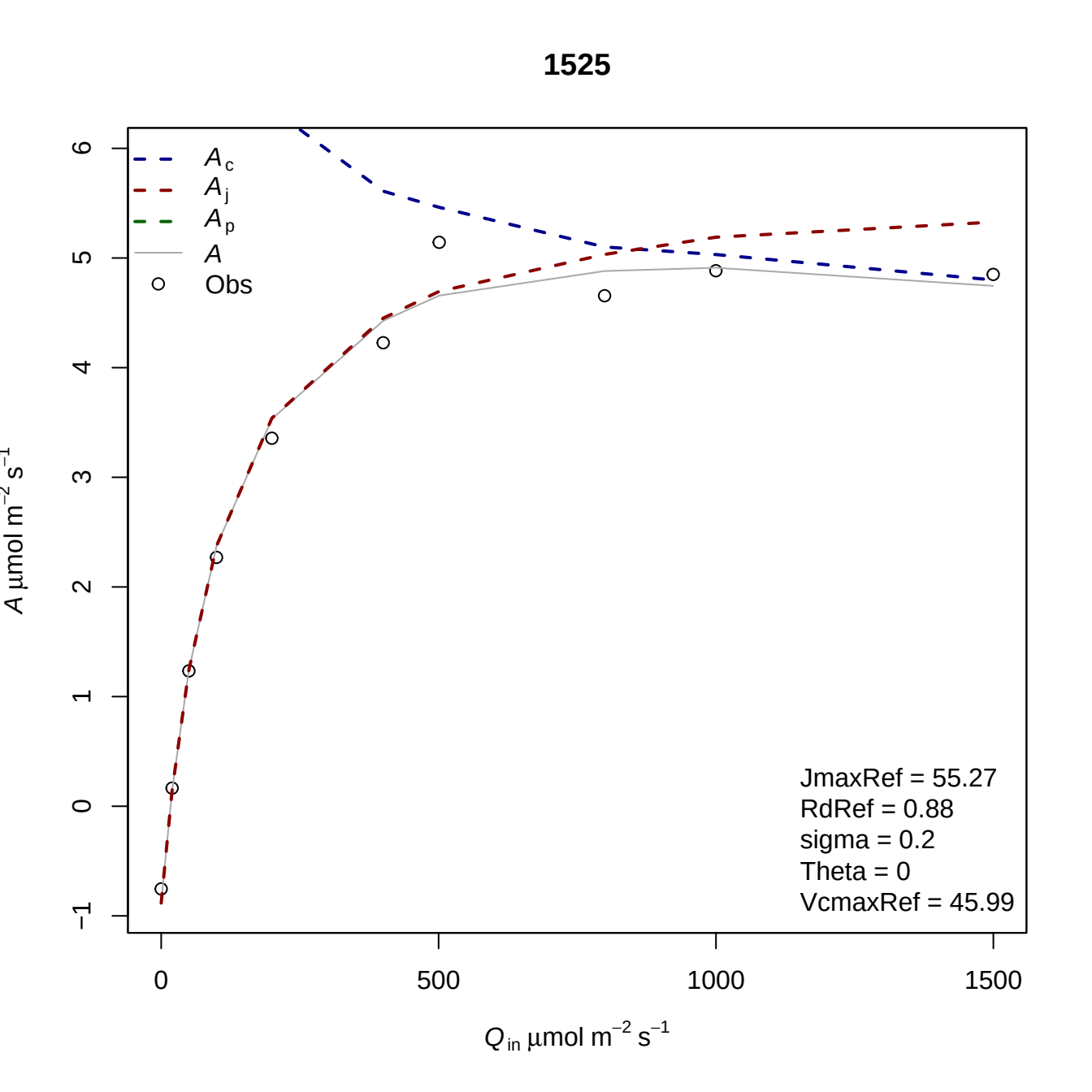

1518
A_c
A_j
A_p
A
Obs
JmaxRef = 124.56
RdRef = 1.64
sigma = 0.68
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

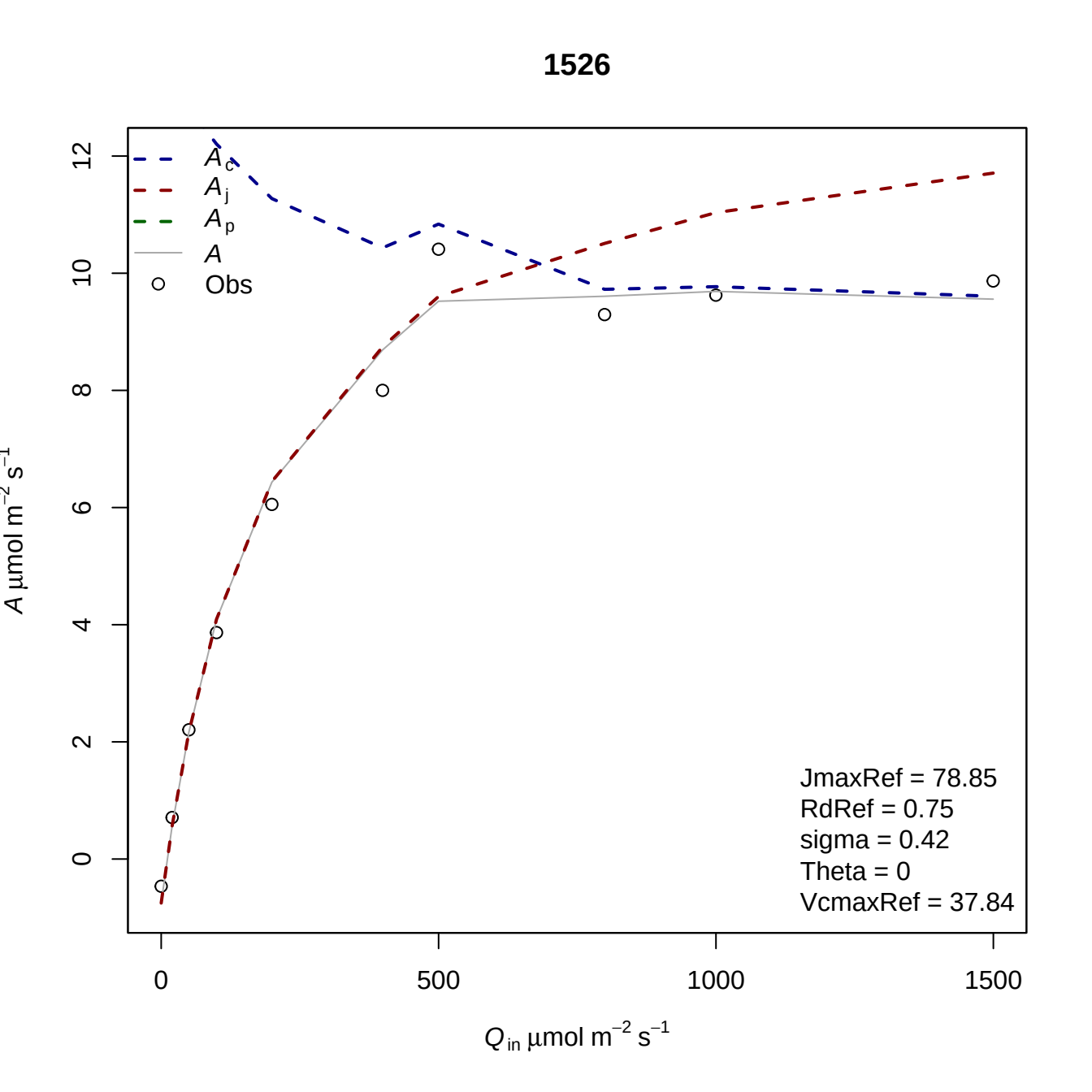

1519
A_c
A_j
A_p
A
Obs
JmaxRef = 83.69
RdRef = 1.85
sigma = 0.29
Theta = 0
VcmaxRef = 62.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

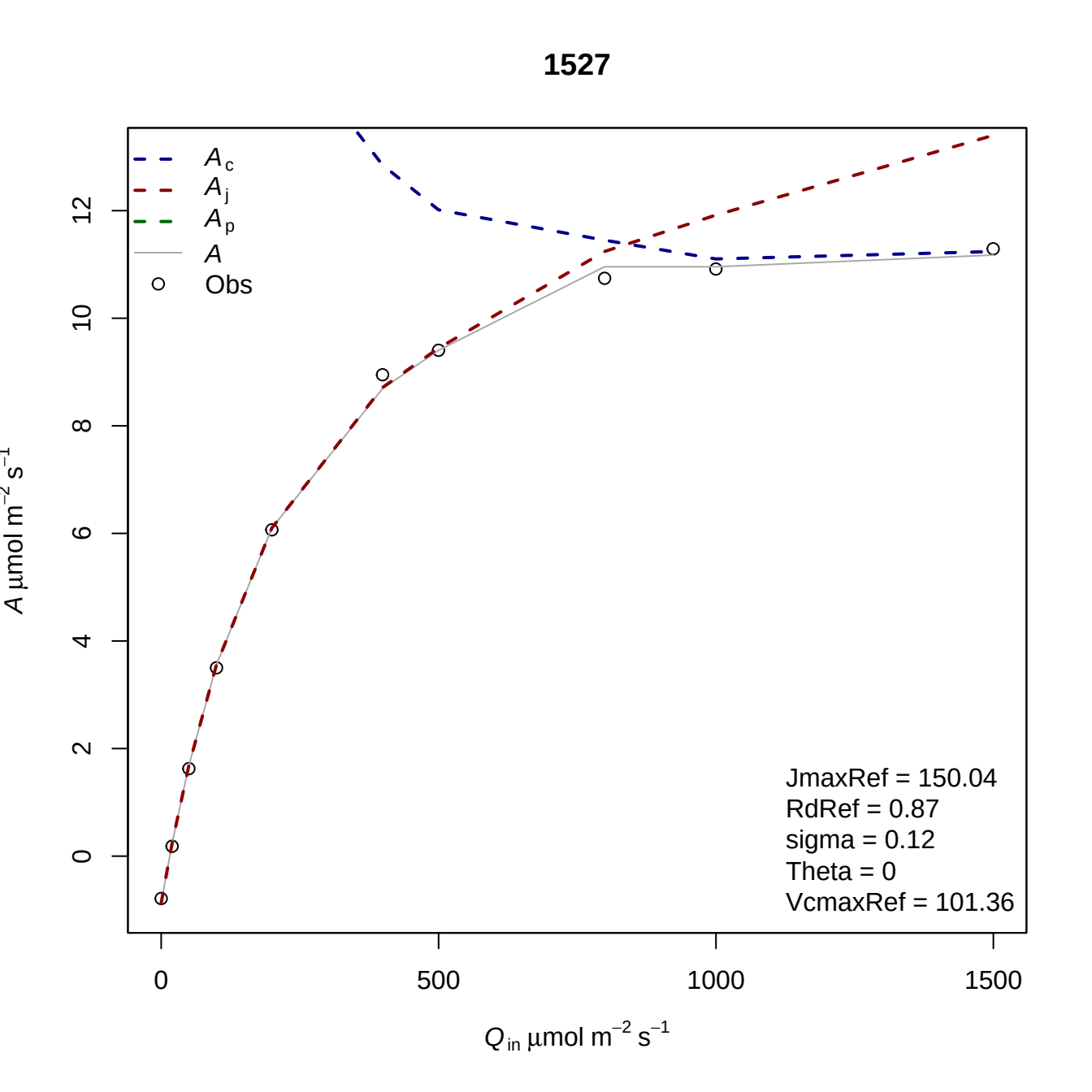

1520
A_c
A_j
A_p
A
Obs
JmaxRef = 174.04
RdRef = 1.51
sigma = 0.79
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

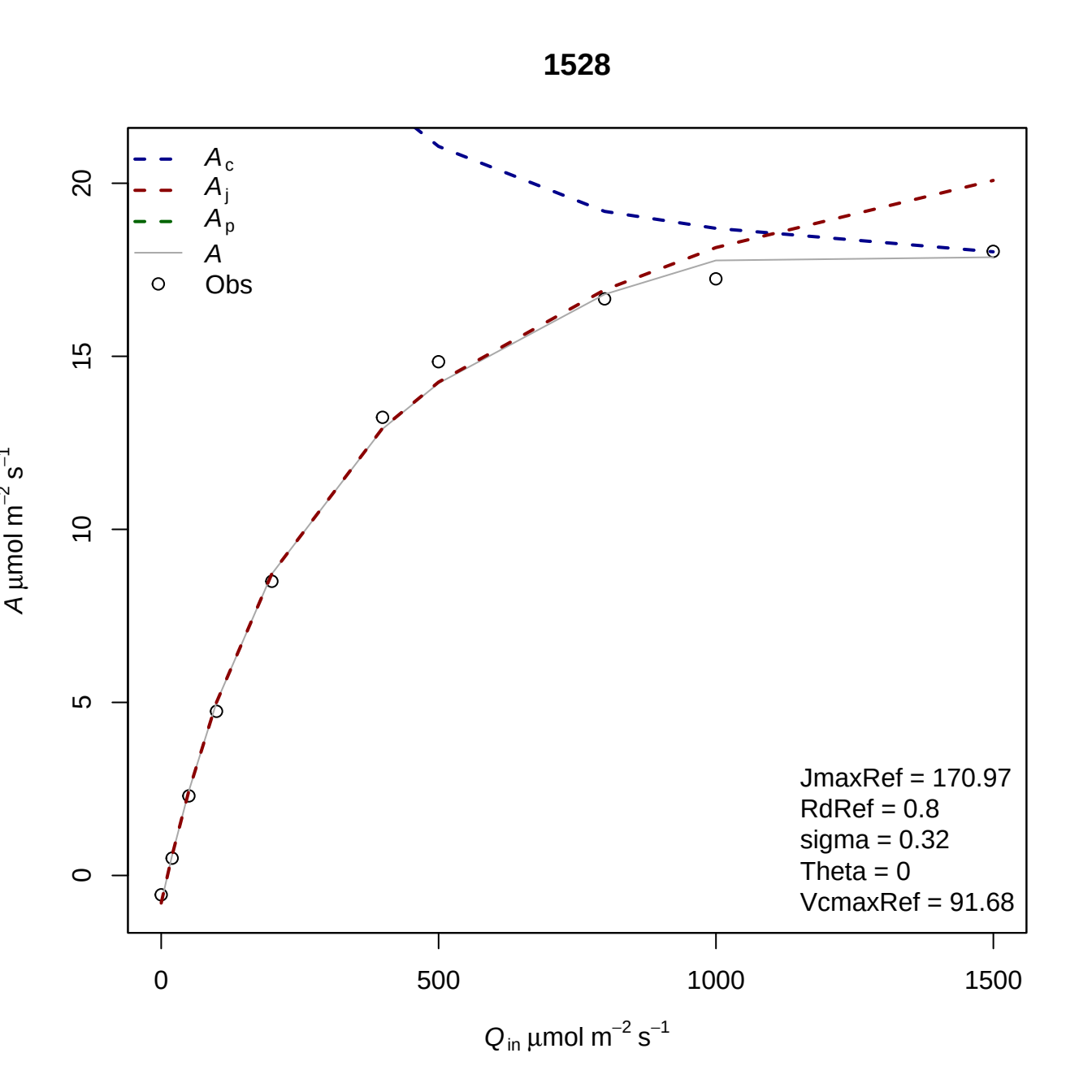

1521
A_c
A_j
A_p
A
Obs
JmaxRef = 28.1
RdRef = 1.78
sigma = 0.23
Theta = 0
VcmaxRef = 19
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

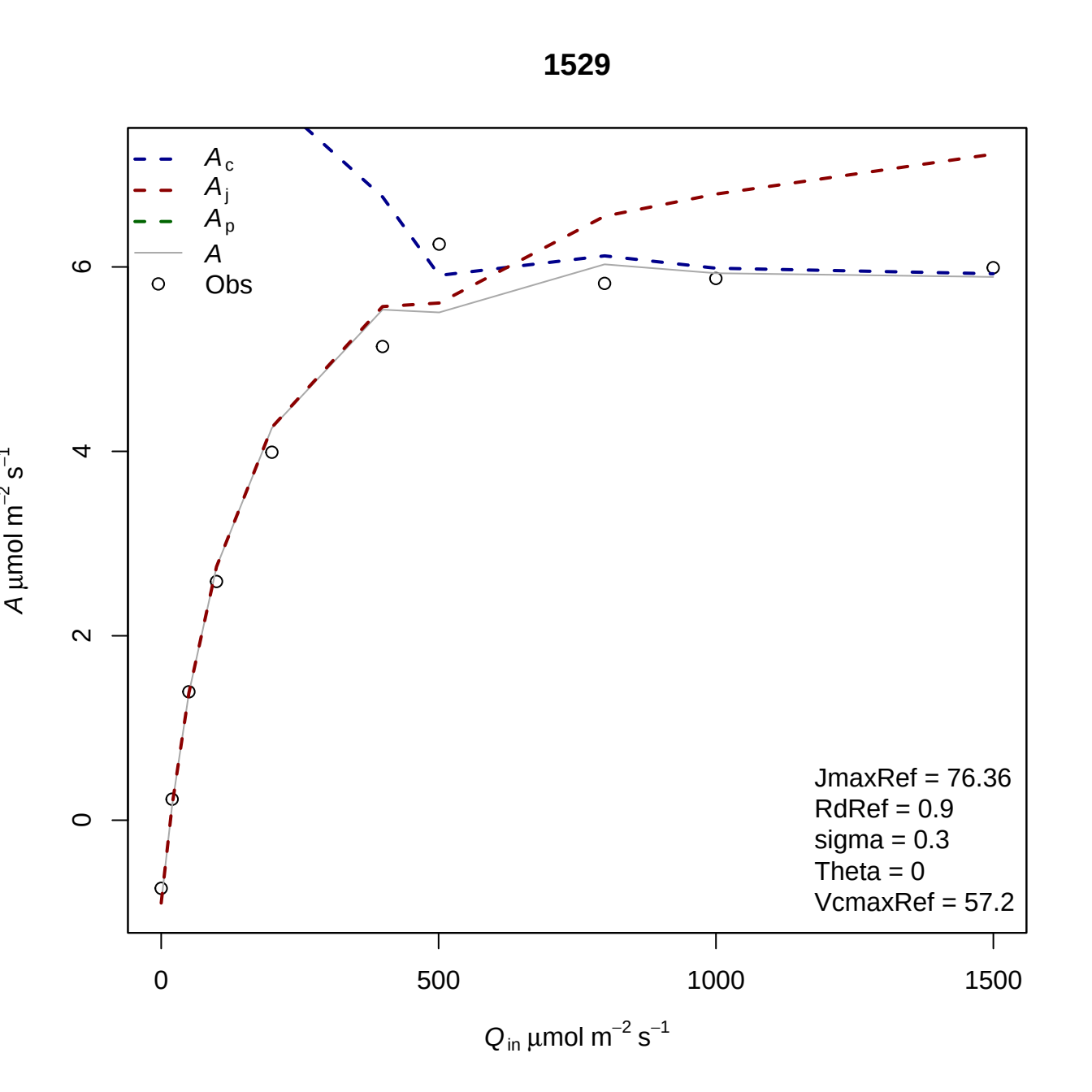

1522
A_c
A_j
A_p
A
Obs
JmaxRef = 83.76
RdRef = 1.74
sigma = 0.72
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

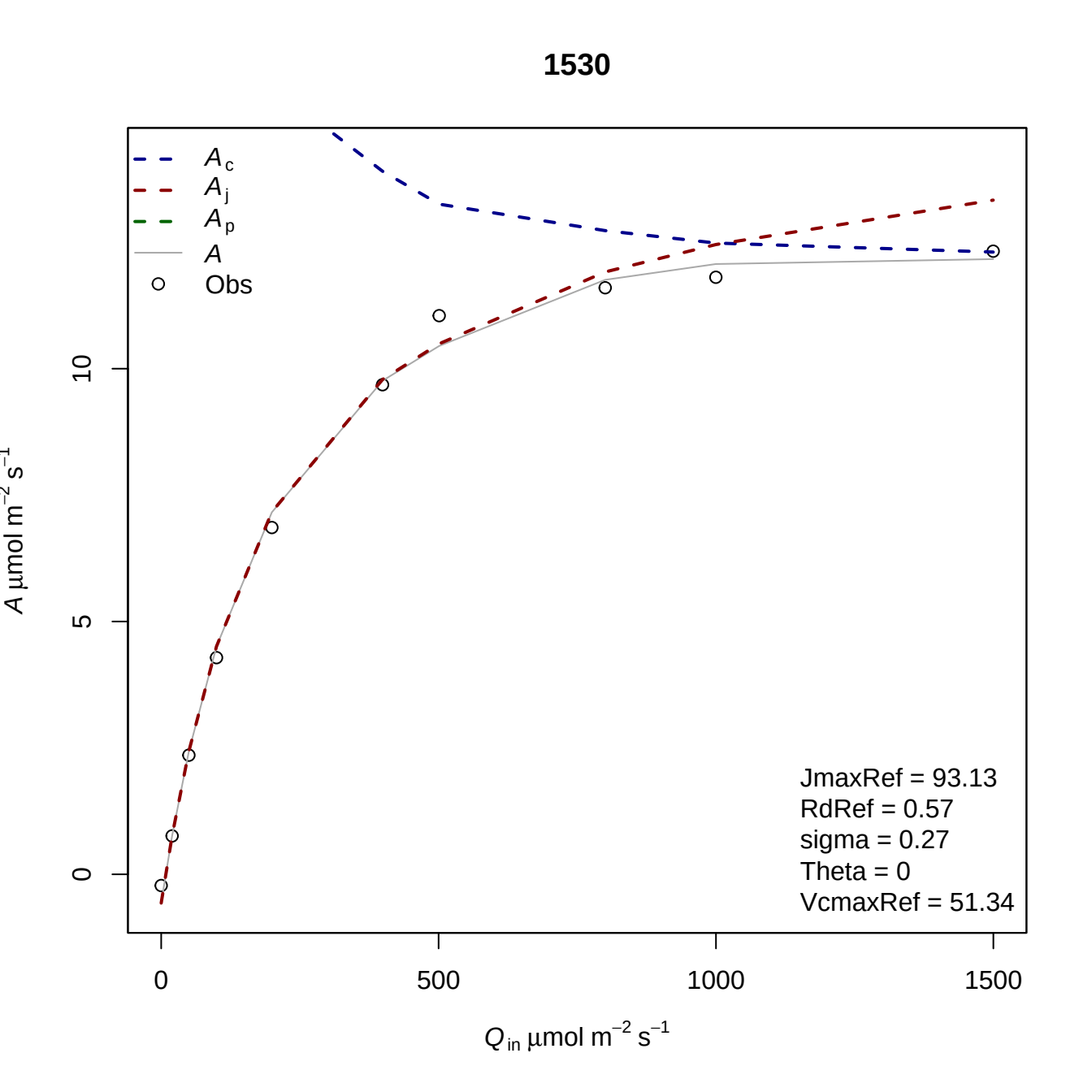

1525
A_c
A_j
A_p
A
Obs
JmaxRef = 55.27
RdRef = 0.88
sigma = 0.2
Theta = 0
VcmaxRef = 45.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

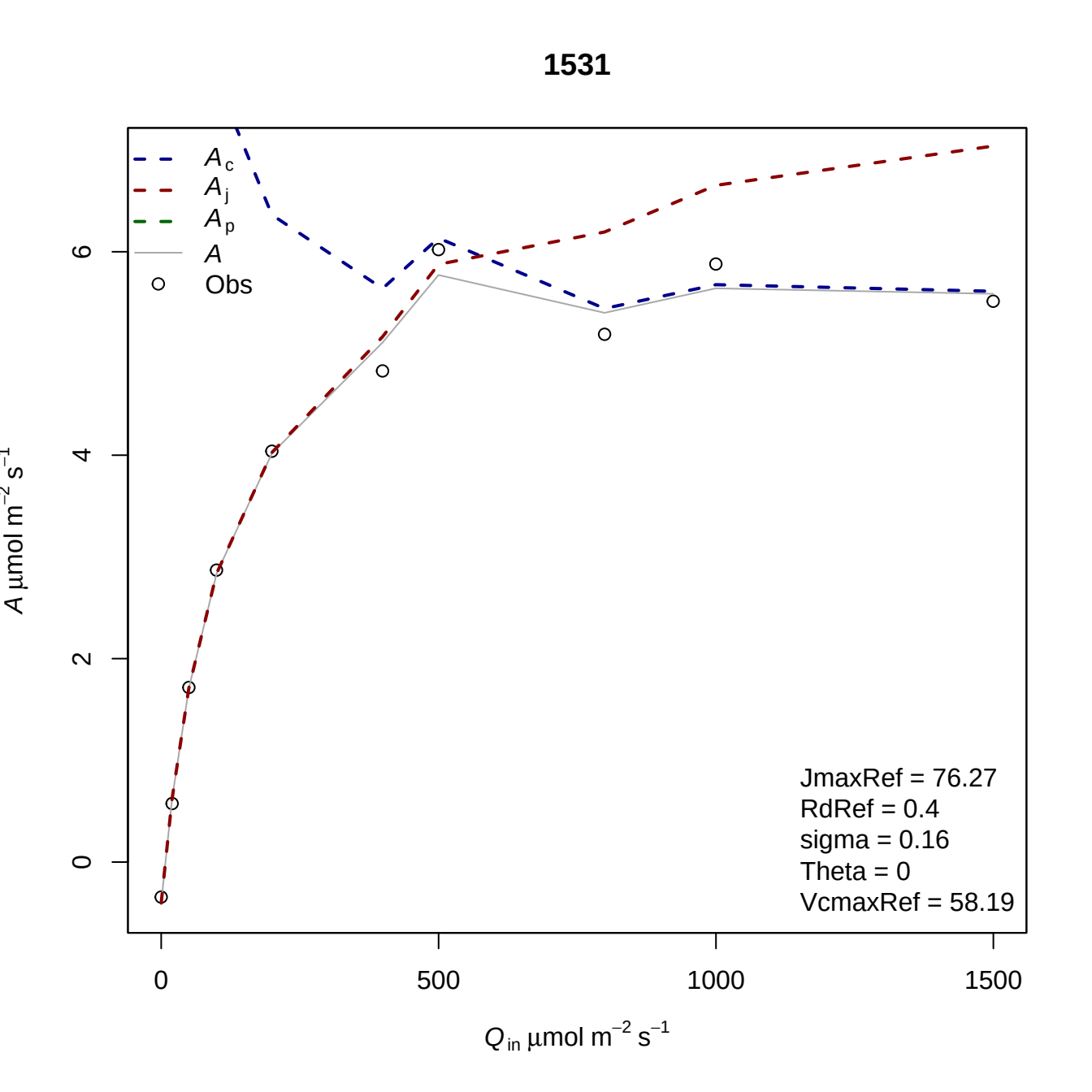

1526
A_c
A_j
A_p
A
Obs
JmaxRef = 78.85
RdRef = 0.75
sigma = 0.42
Theta = 0
VcmaxRef = 37.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

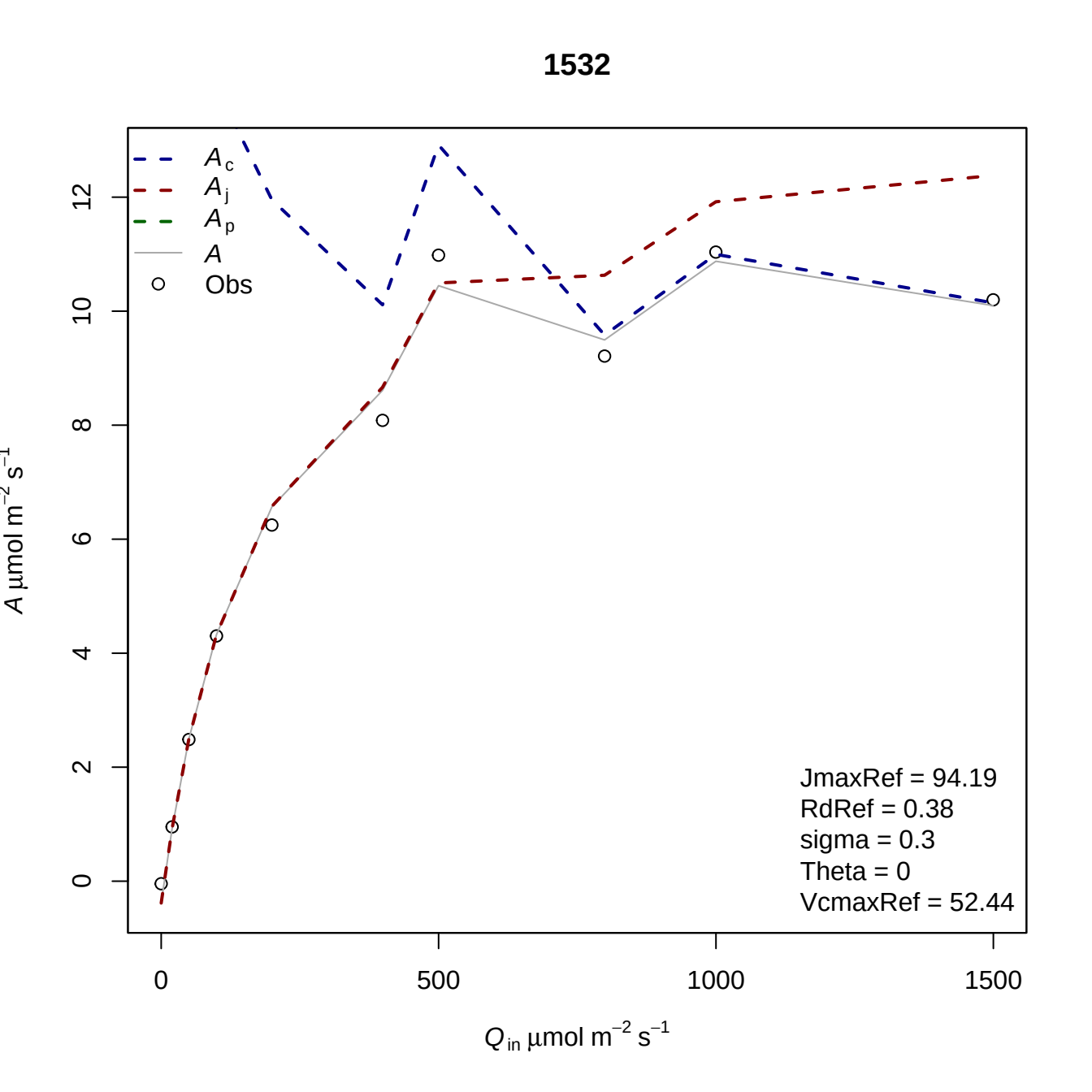

1527
A_c
A_j
A_p
A
Obs
JmaxRef = 150.04
RdRef = 0.87
sigma = 0.12
Theta = 0
VcmaxRef = 101.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

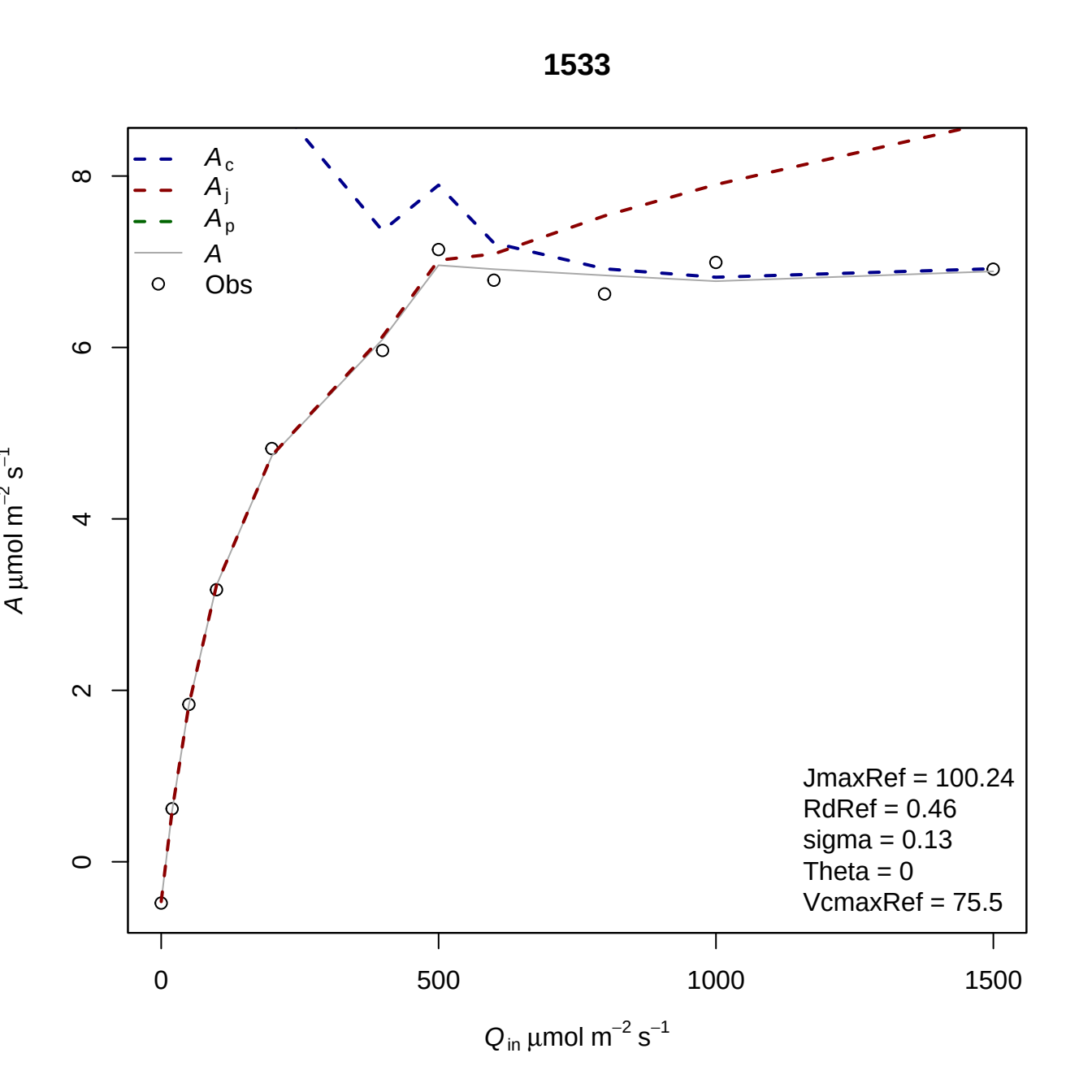

1528
A_c
A_j
A_p
A
Obs
JmaxRef = 170.97
RdRef = 0.8
sigma = 0.32
Theta = 0
VcmaxRef = 91.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

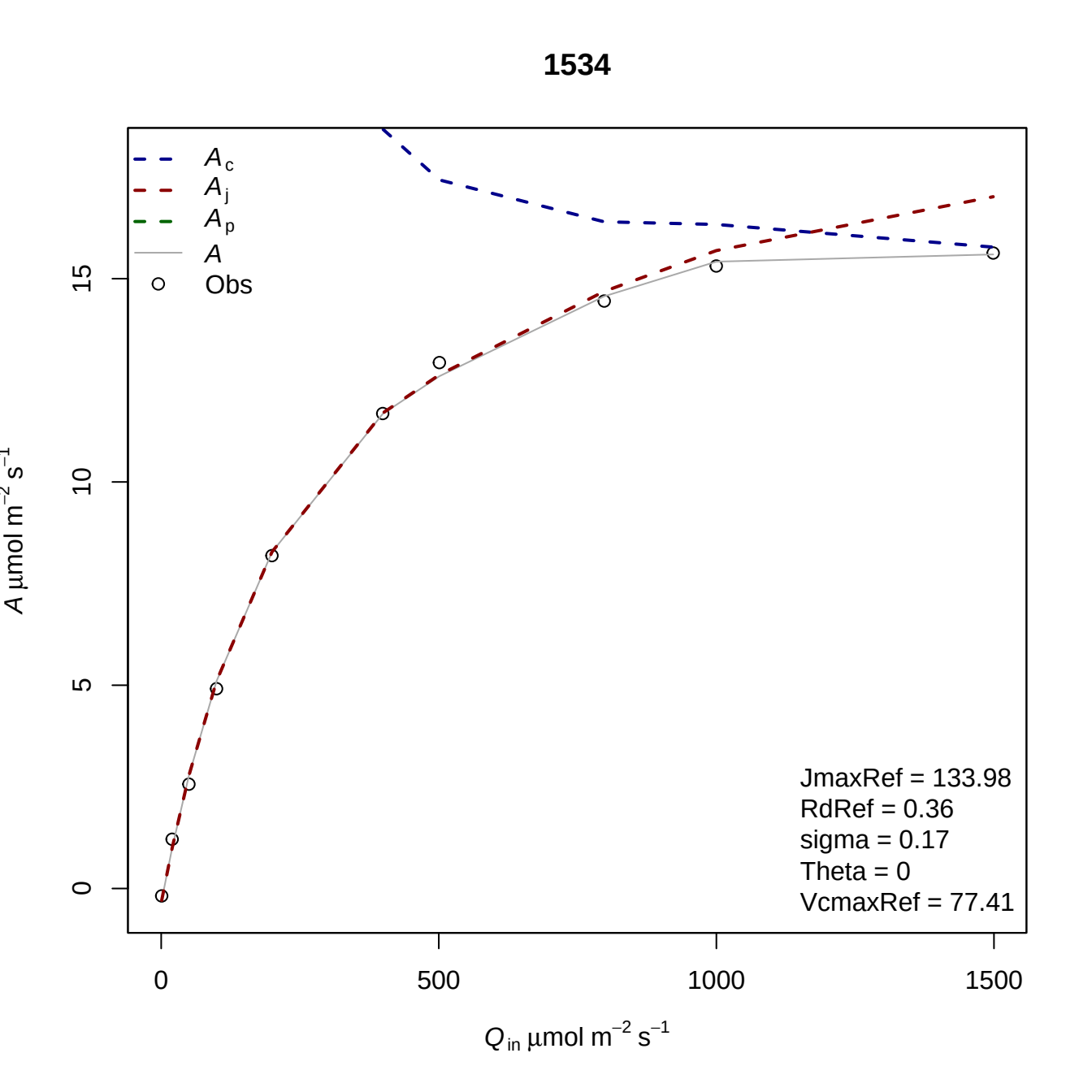

1529
A_c
A_j
A_p
A
Obs
JmaxRef = 76.36
RdRef = 0.9
sigma = 0.3
Theta = 0
VcmaxRef = 57.2
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

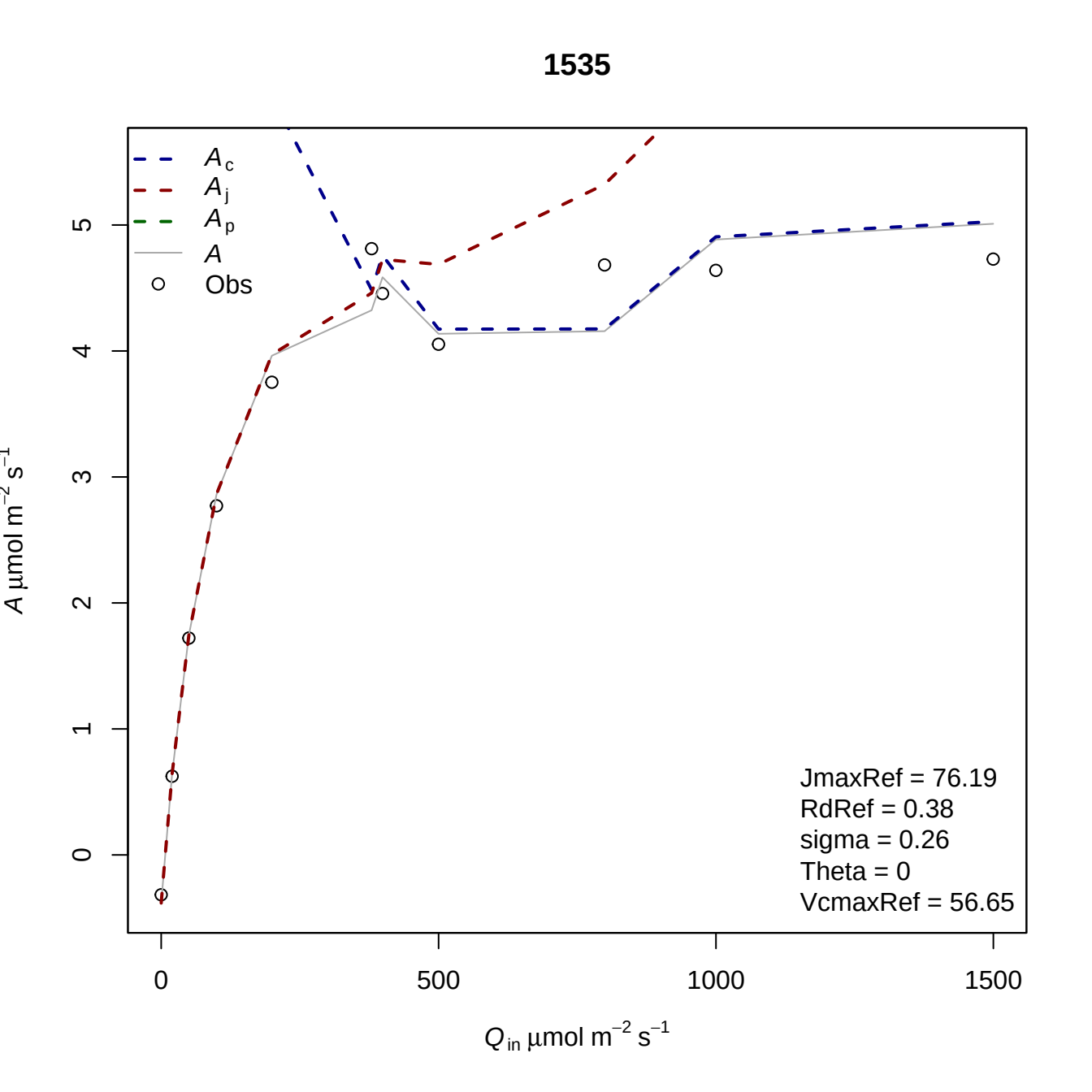

1530
A_c
A_j
A_p
A
Obs
JmaxRef = 93.13
RdRef = 0.57
sigma = 0.27
Theta = 0
VcmaxRef = 51.34
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

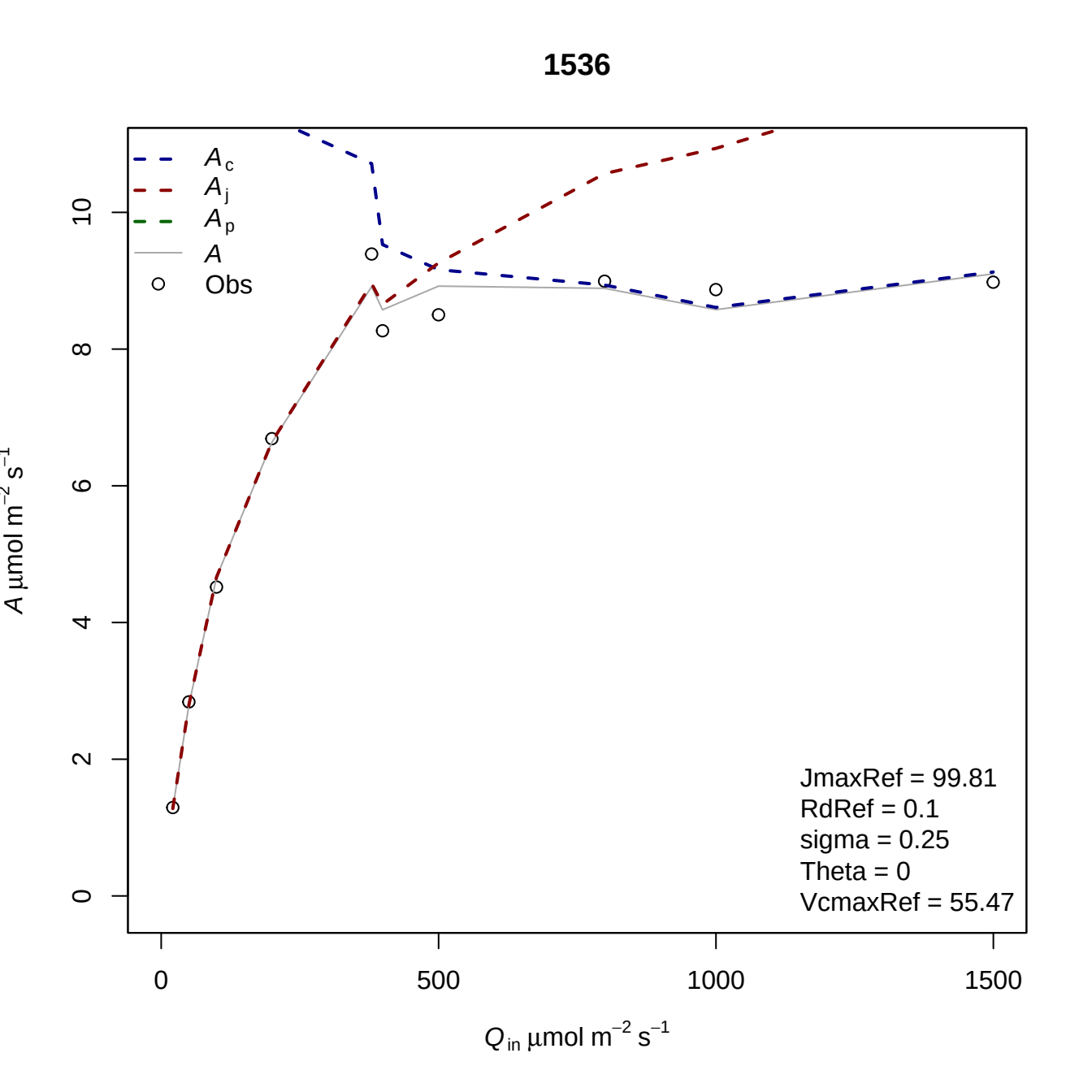

1531
A_c
A_j
A_p
A
Obs
JmaxRef = 76.27
RdRef = 0.4
sigma = 0.16
Theta = 0
VcmaxRef = 58.19
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

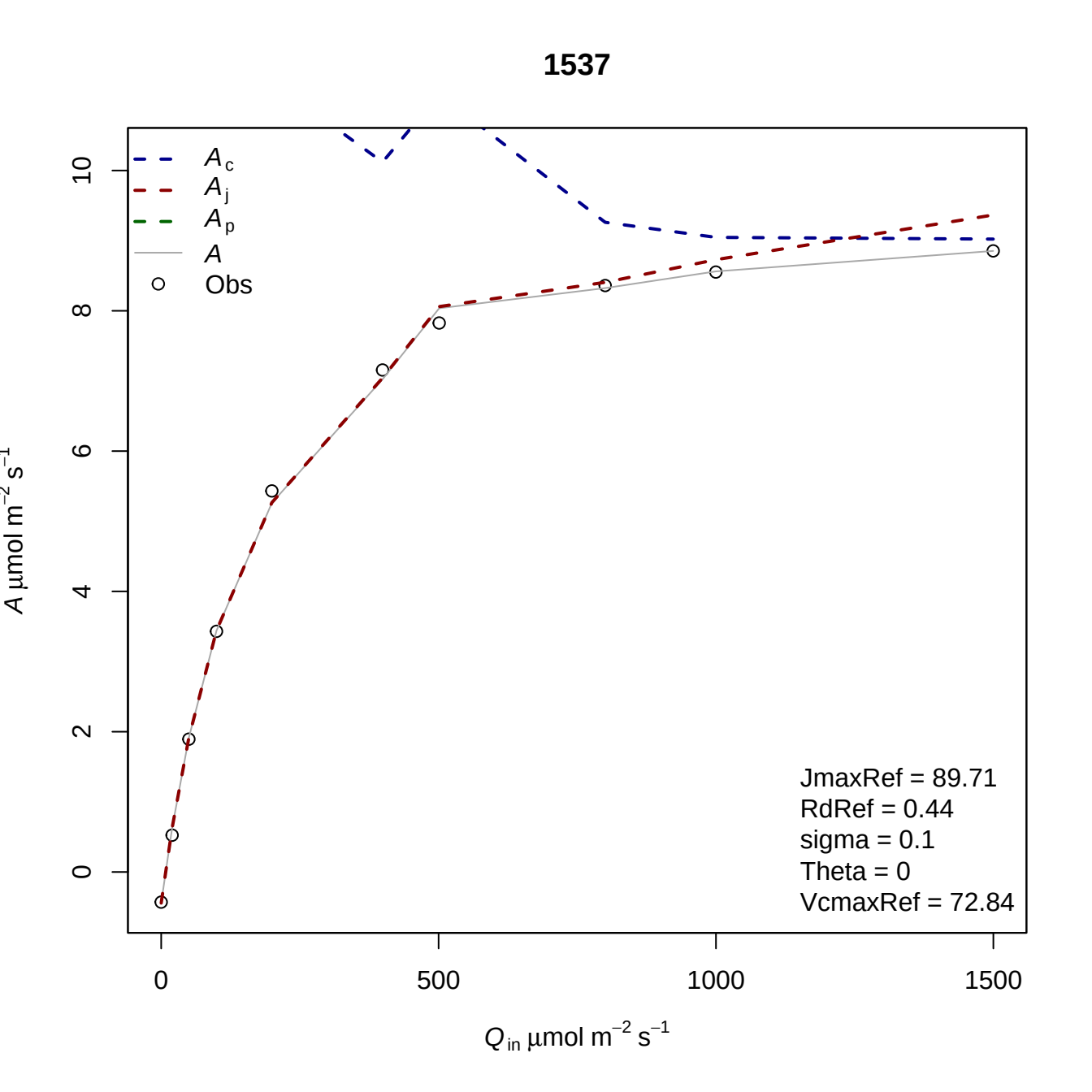

1532
A_c
A_j
A_p
A
Obs
JmaxRef = 94.19
RdRef = 0.38
sigma = 0.3
Theta = 0
VcmaxRef = 52.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

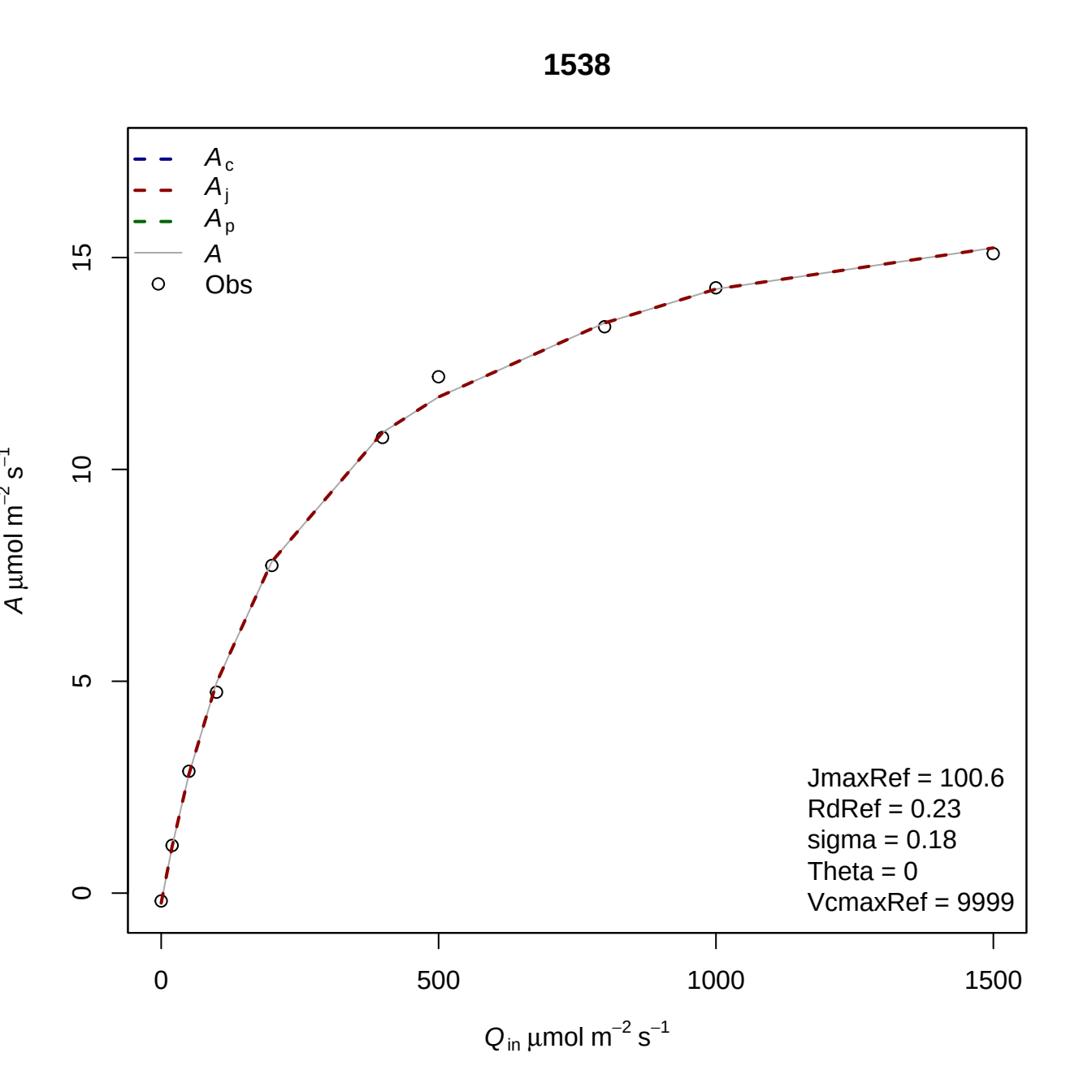

1533
A_c
A_j
A_p
A
Obs
JmaxRef = 100.24
RdRef = 0.46
sigma = 0.13
Theta = 0
VcmaxRef = 75.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

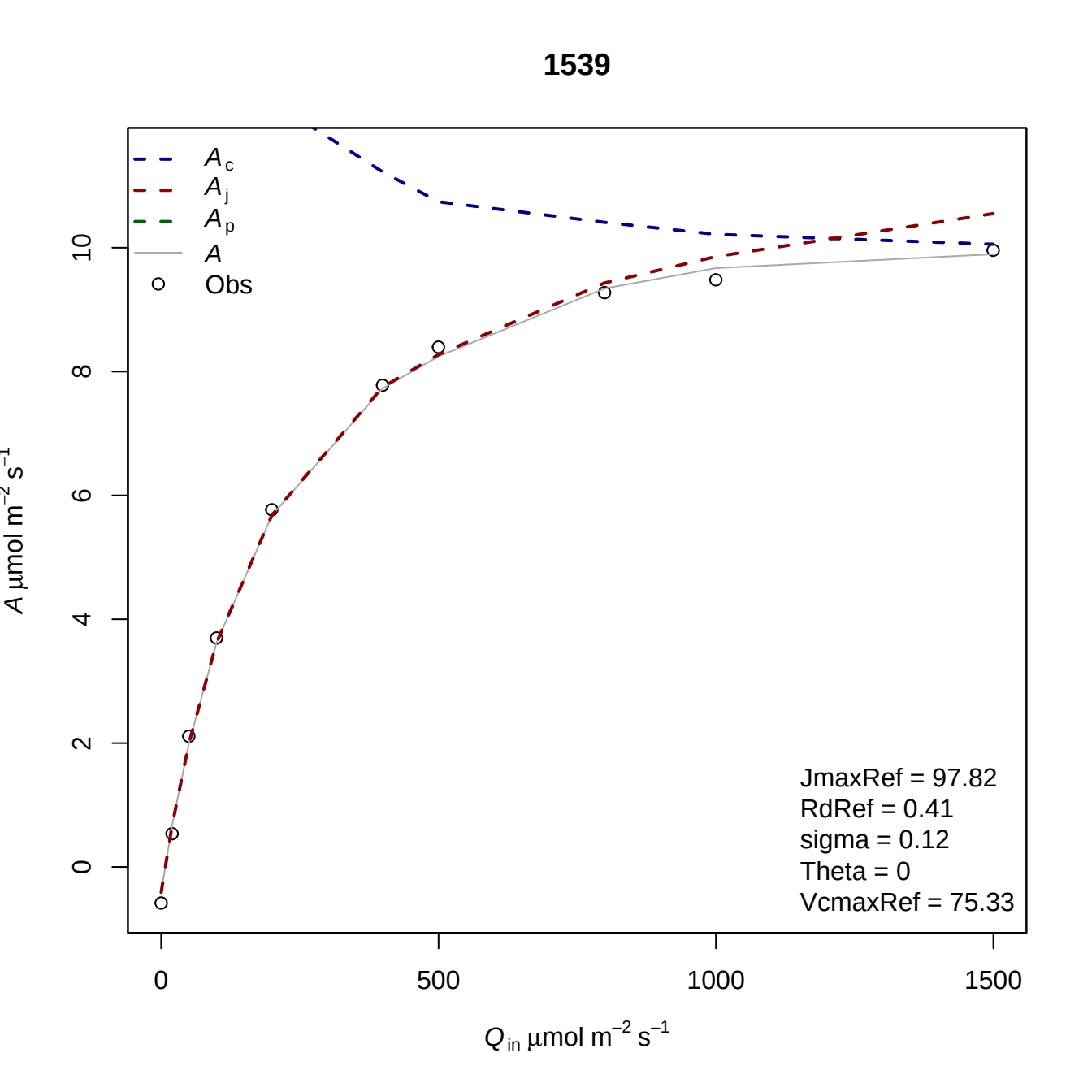

1534
A_c
A_j
A_p
A
Obs
JmaxRef = 133.98
RdRef = 0.36
sigma = 0.17
Theta = 0
VcmaxRef = 77.41
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

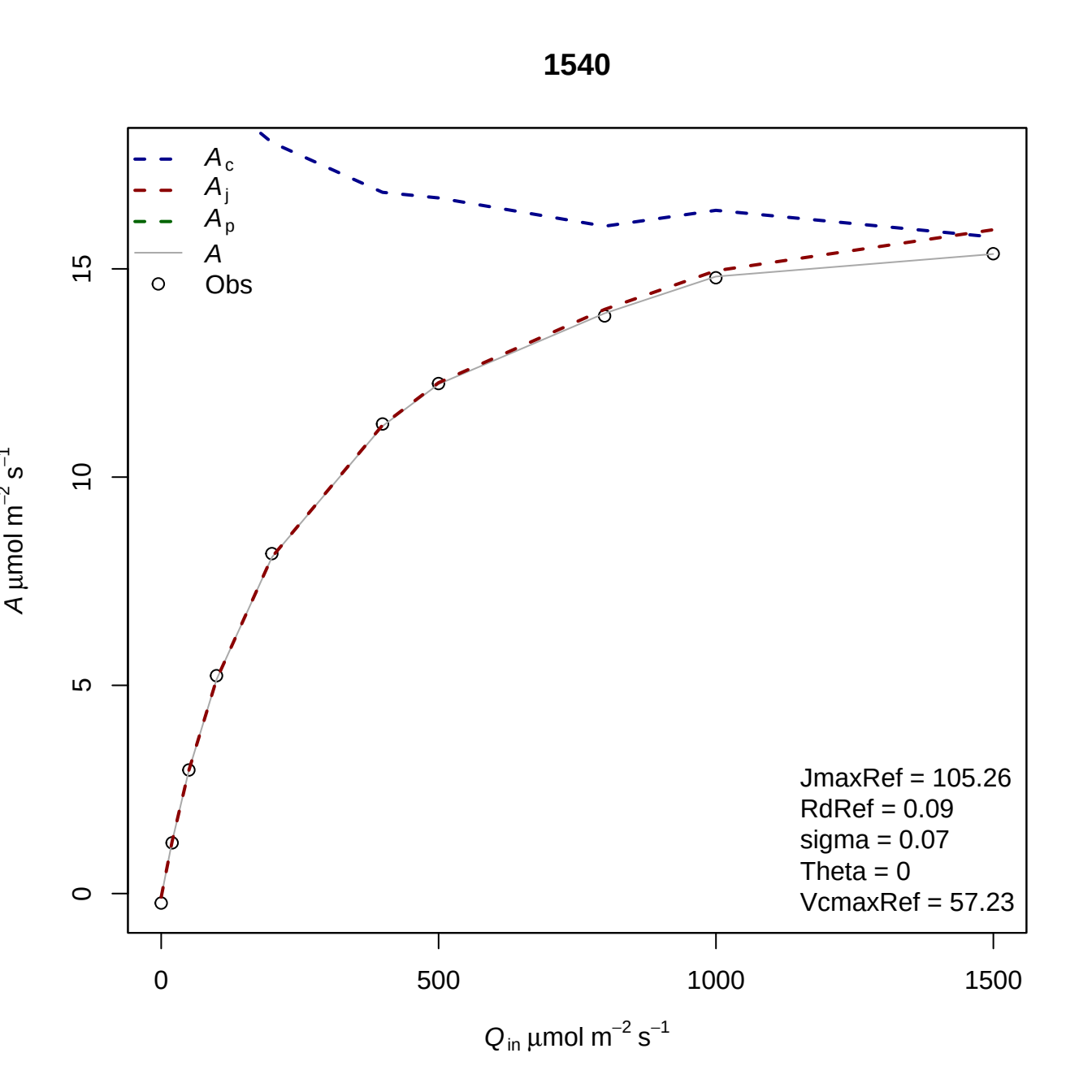

1535
A_c
A_j
A_p
A
Obs
JmaxRef = 76.19
RdRef = 0.38
sigma = 0.26
Theta = 0
VcmaxRef = 56.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

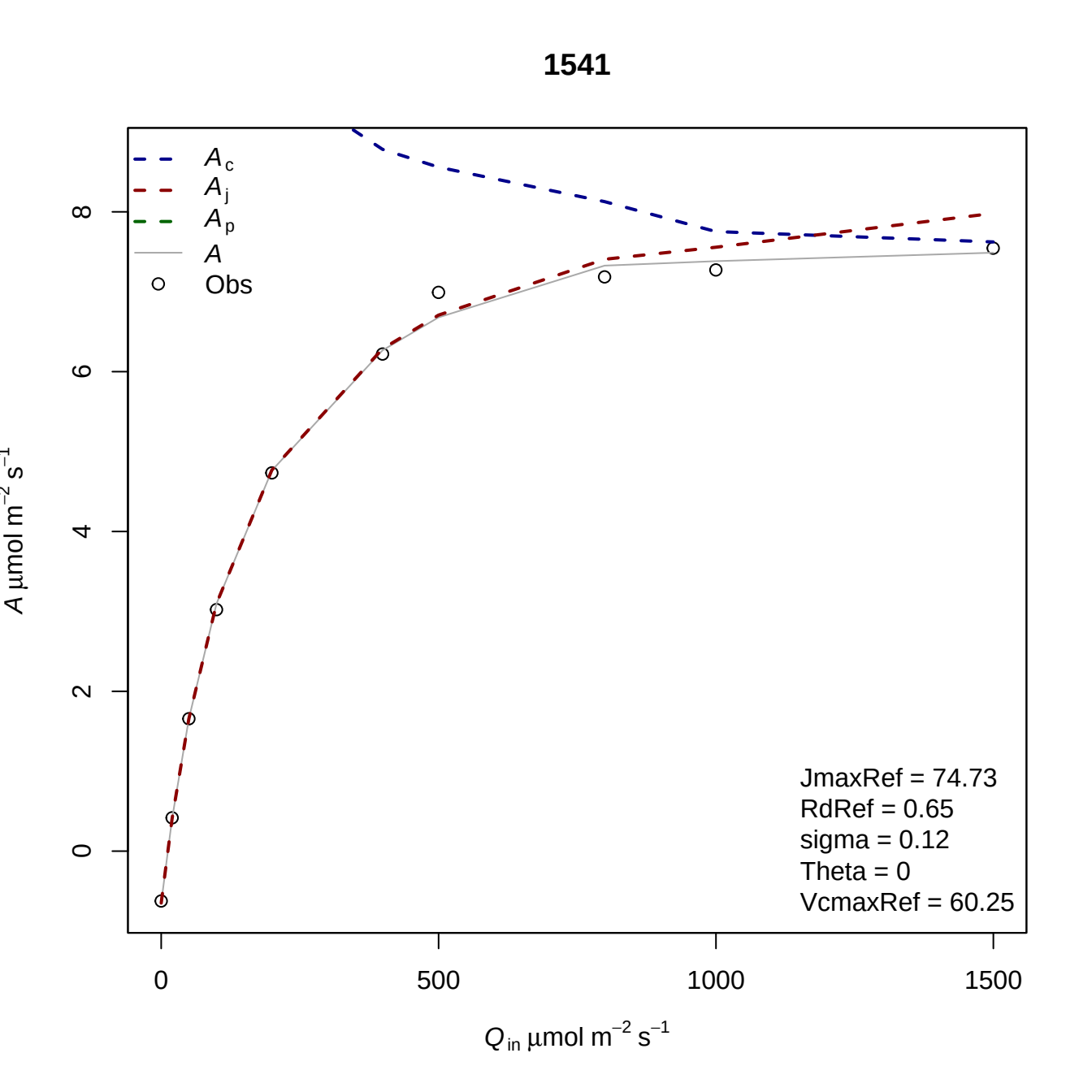

1536
A_c
A_j
A_p
A
Obs
JmaxRef = 99.81
RdRef = 0.1
sigma = 0.25
Theta = 0
VcmaxRef = 55.47
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

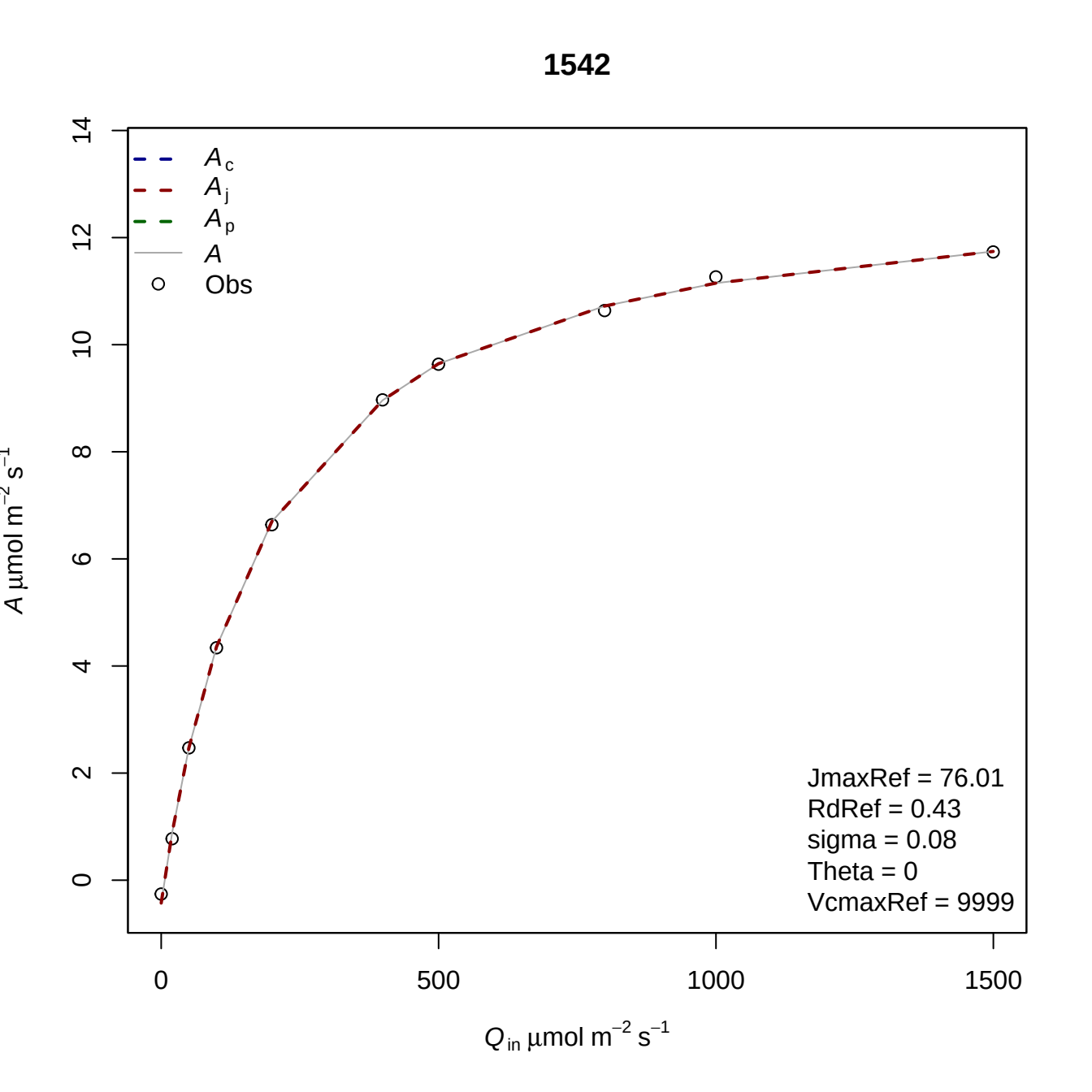

1537
A_c
A_j
A_p
A
Obs
JmaxRef = 89.71
RdRef = 0.44
sigma = 0.1
Theta = 0
VcmaxRef = 72.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

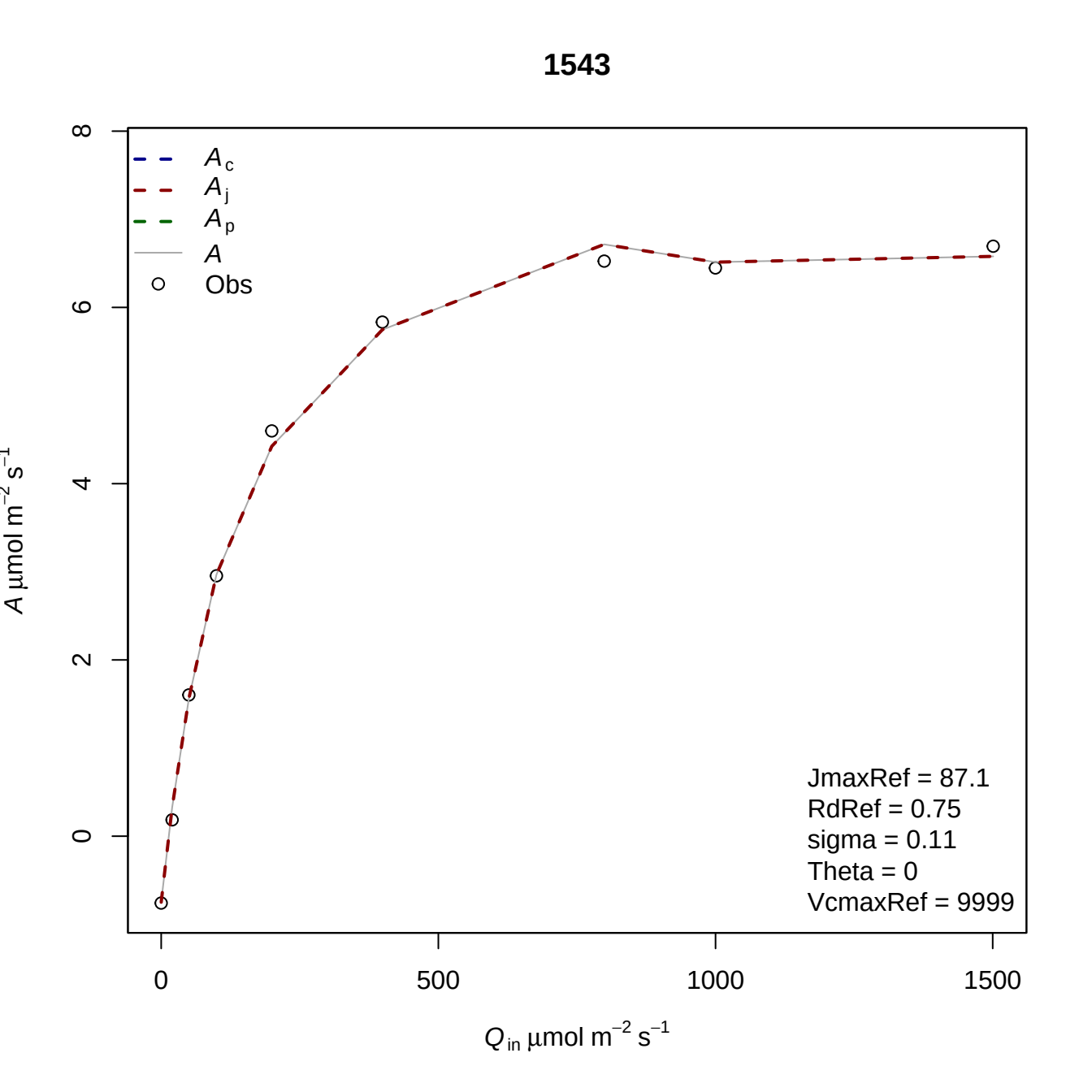

1538
A_c
A_j
A_p
A
Obs
JmaxRef = 100.6
RdRef = 0.23
sigma = 0.18
Theta = 0
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

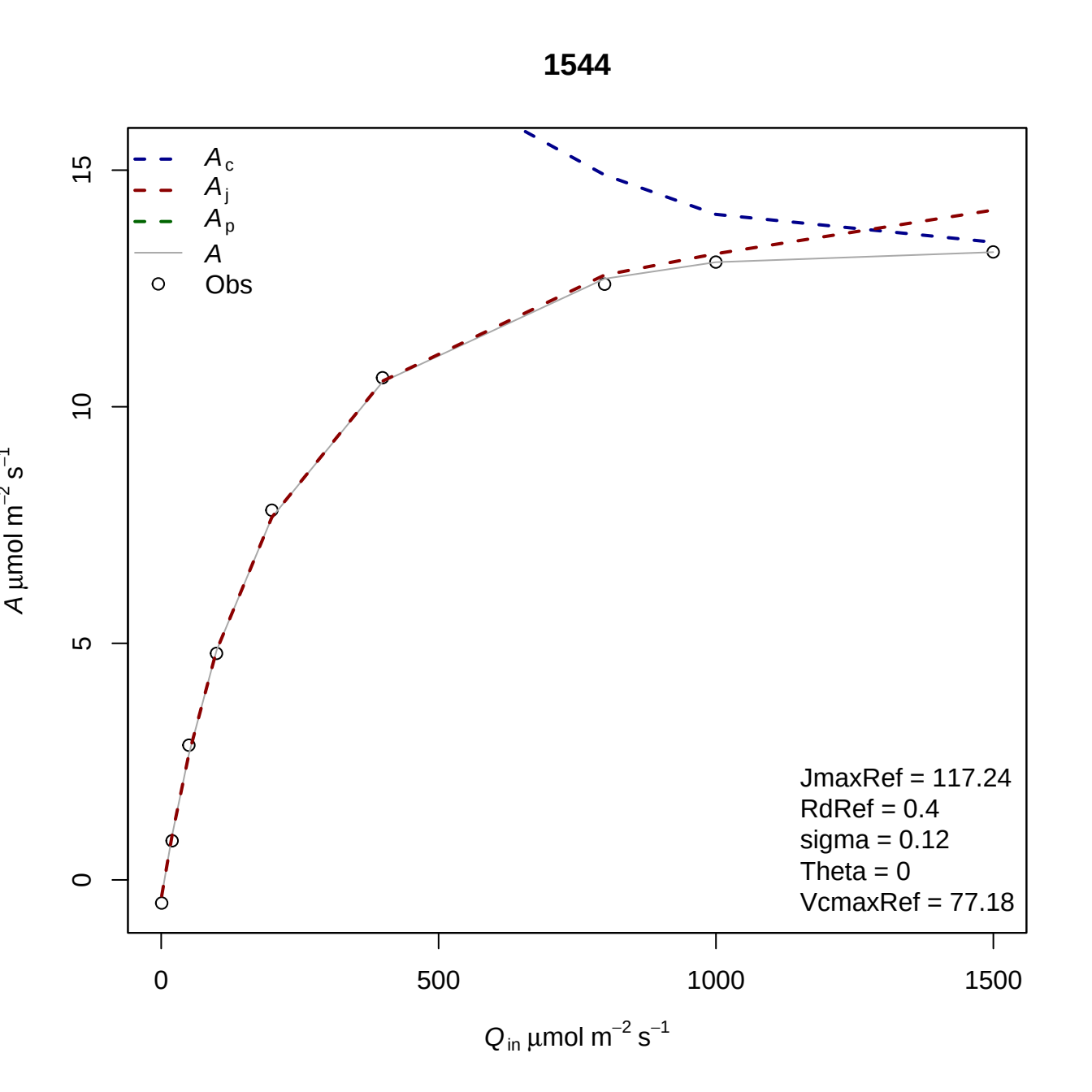

1539
A_c
A_j
A_p
A
Obs
JmaxRef = 97.82
RdRef = 0.41
sigma = 0.12
Theta = 0
VcmaxRef = 75.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

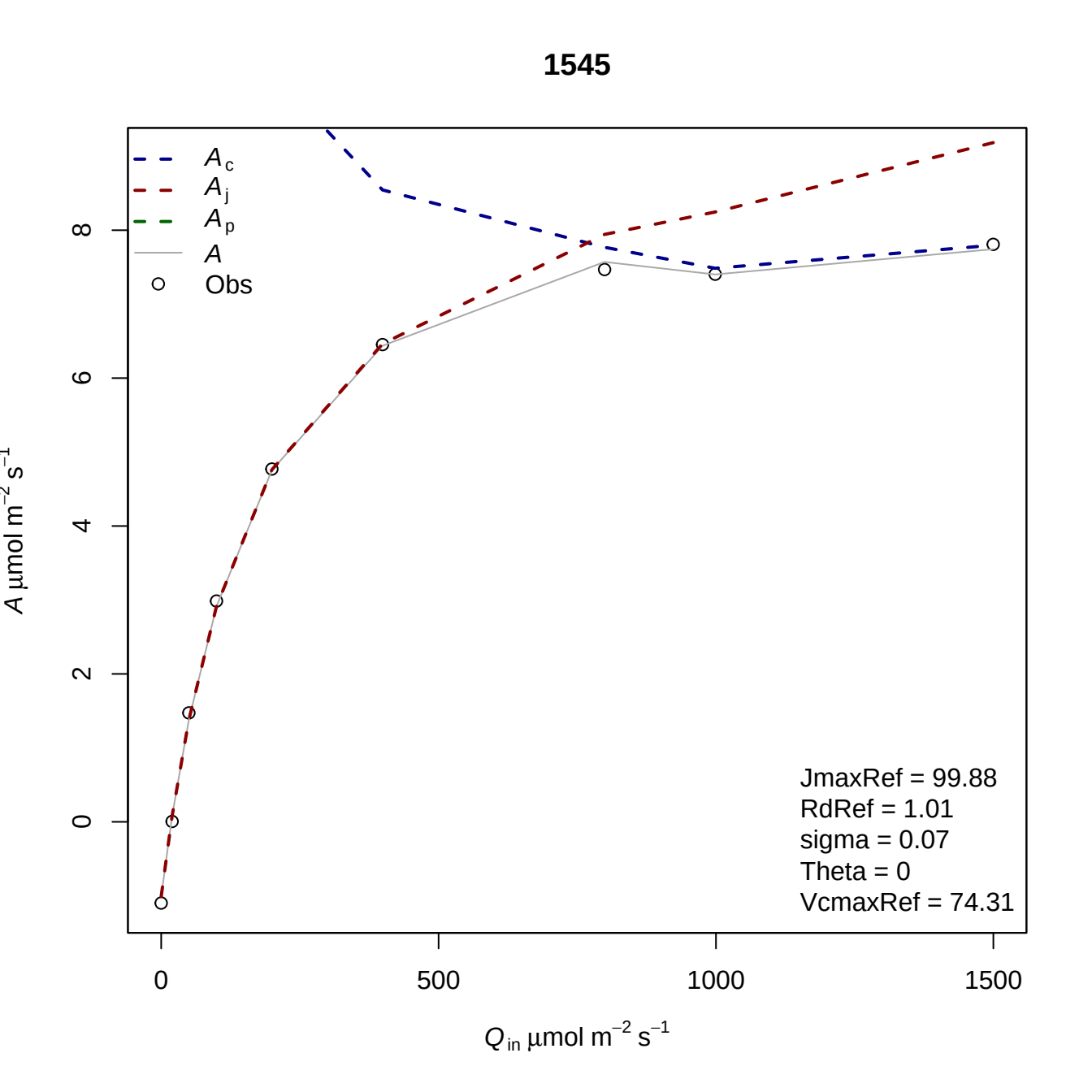

1540
A_c
A_j
A_p
A
Obs
JmaxRef = 105.26
RdRef = 0.09
sigma = 0.07
Theta = 0
VcmaxRef = 57.23
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

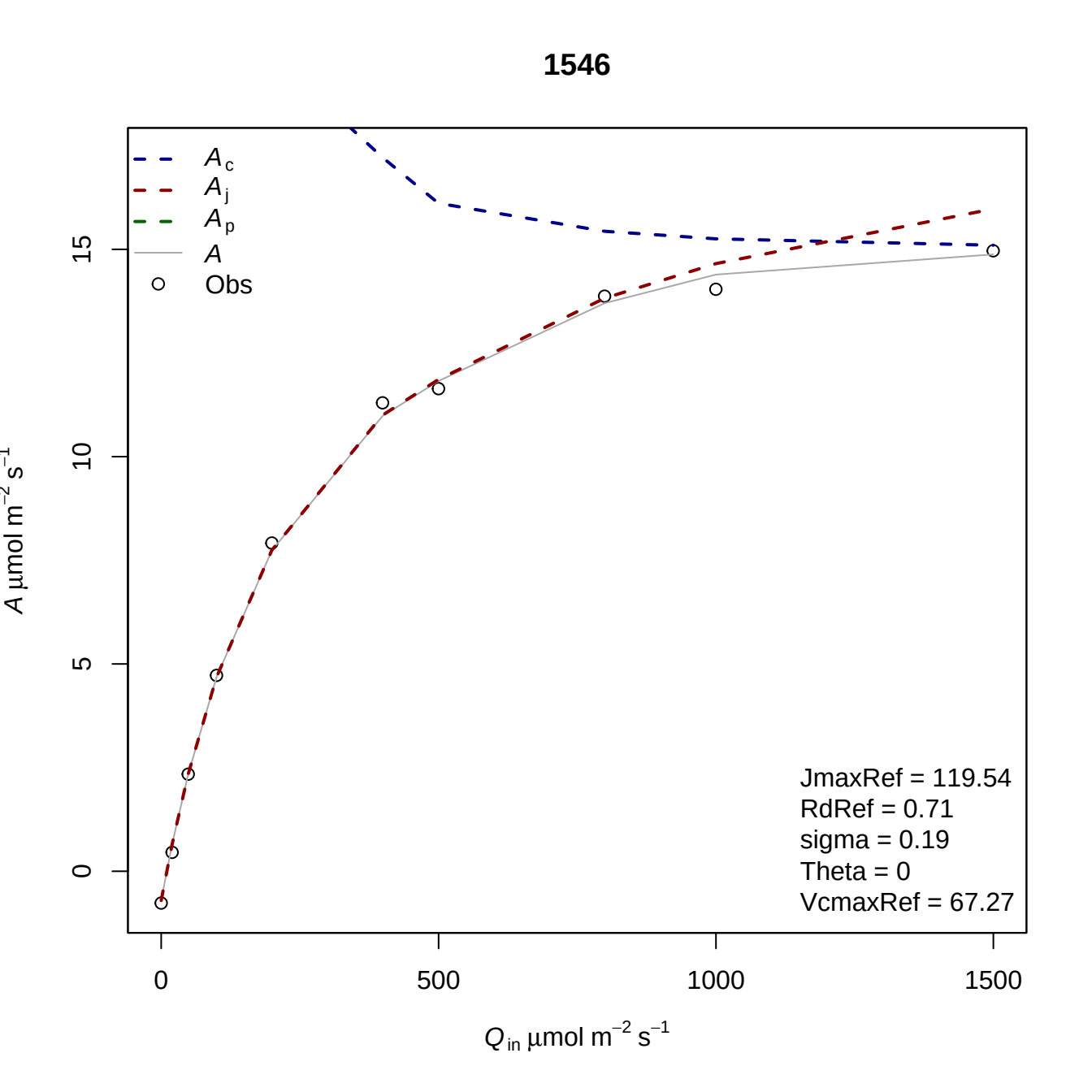

1541
A_c
A_j
A_p
A
Obs
JmaxRef = 74.73
RdRef = 0.65
sigma = 0.12
Theta = 0
VcmaxRef = 60.25
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

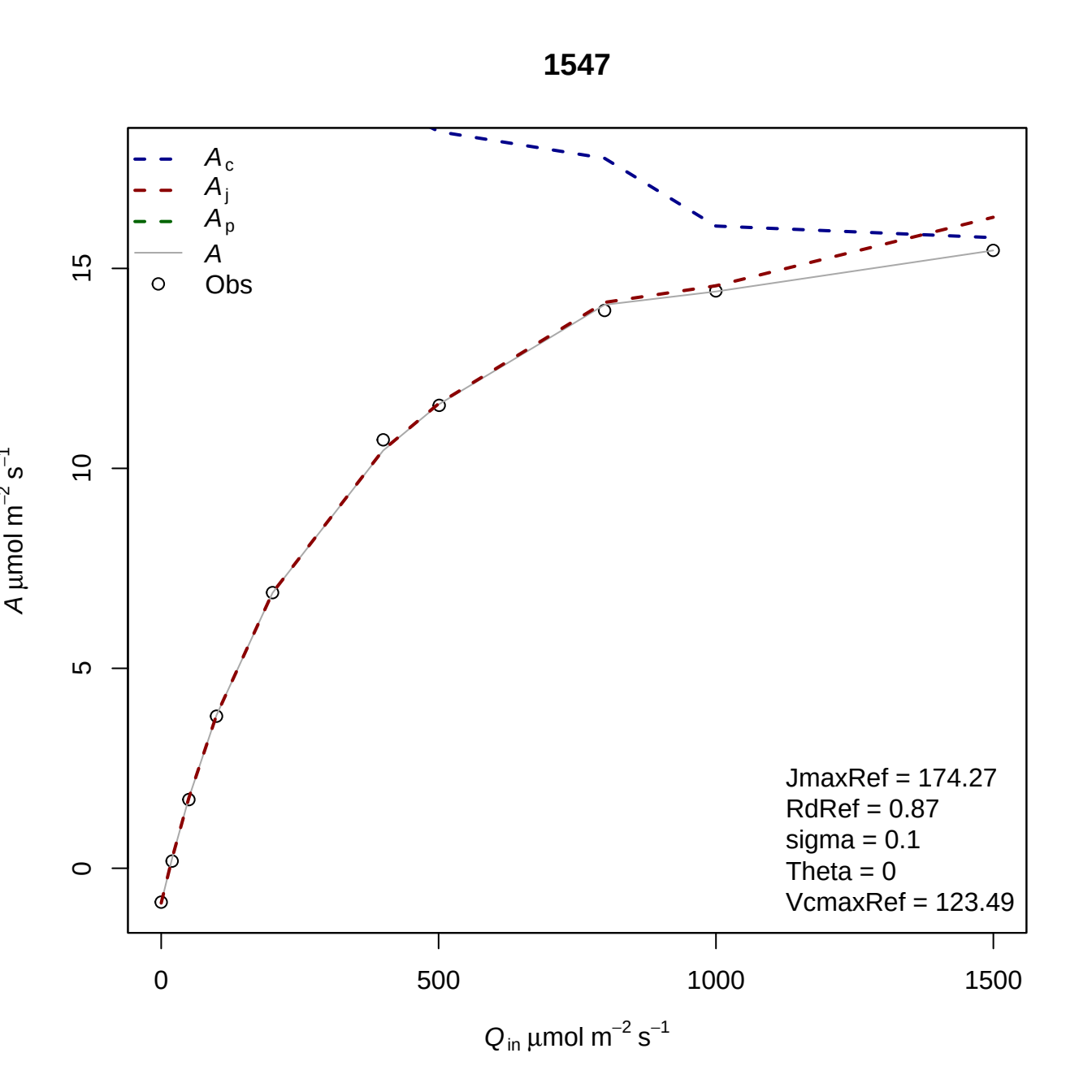

1542
A_c
A_j
A_p
A
Obs
JmaxRef = 76.01
RdRef = 0.43
sigma = 0.08
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

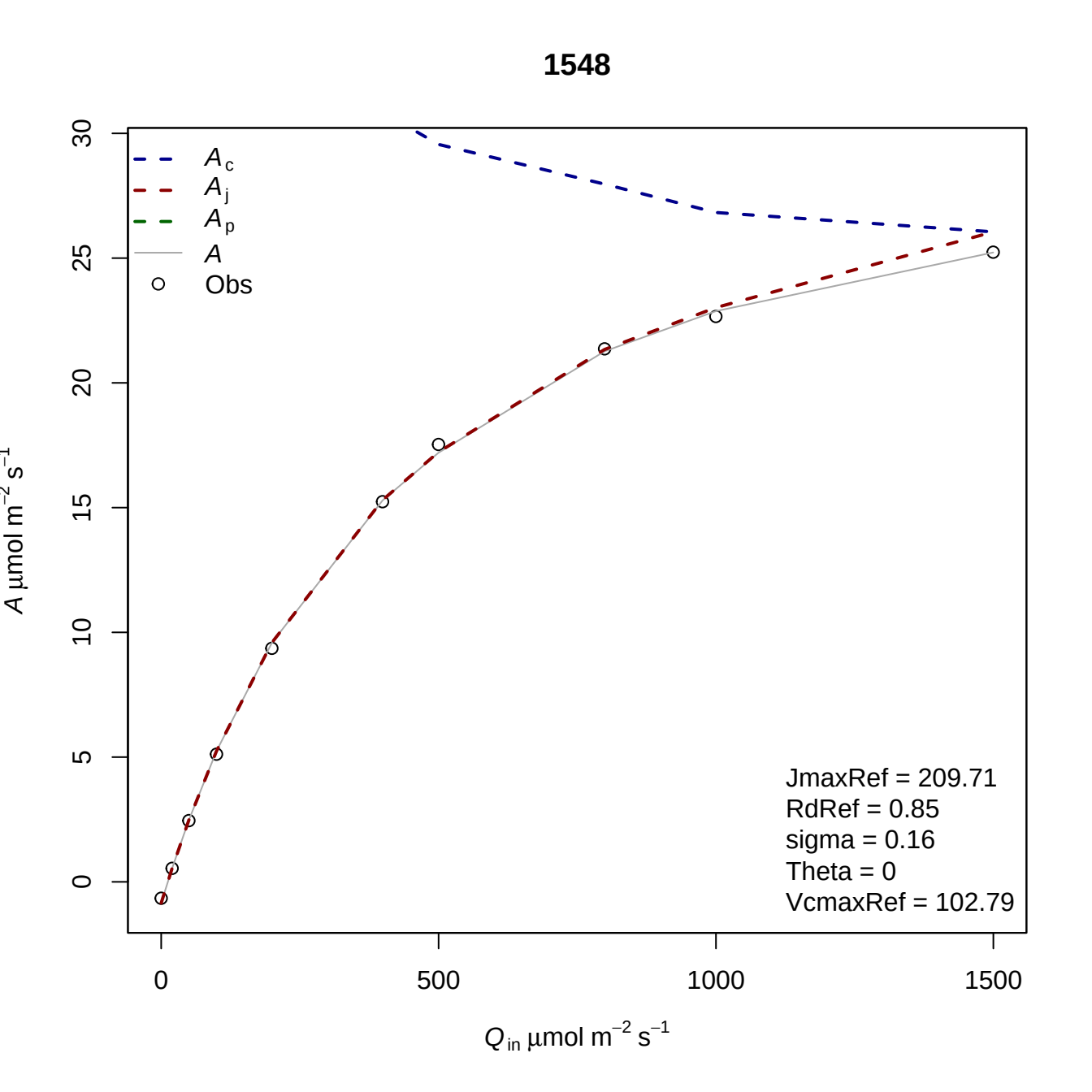

1543
A_c
A_j
A_p
A
Obs
JmaxRef = 87.1
RdRef = 0.75
sigma = 0.11
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

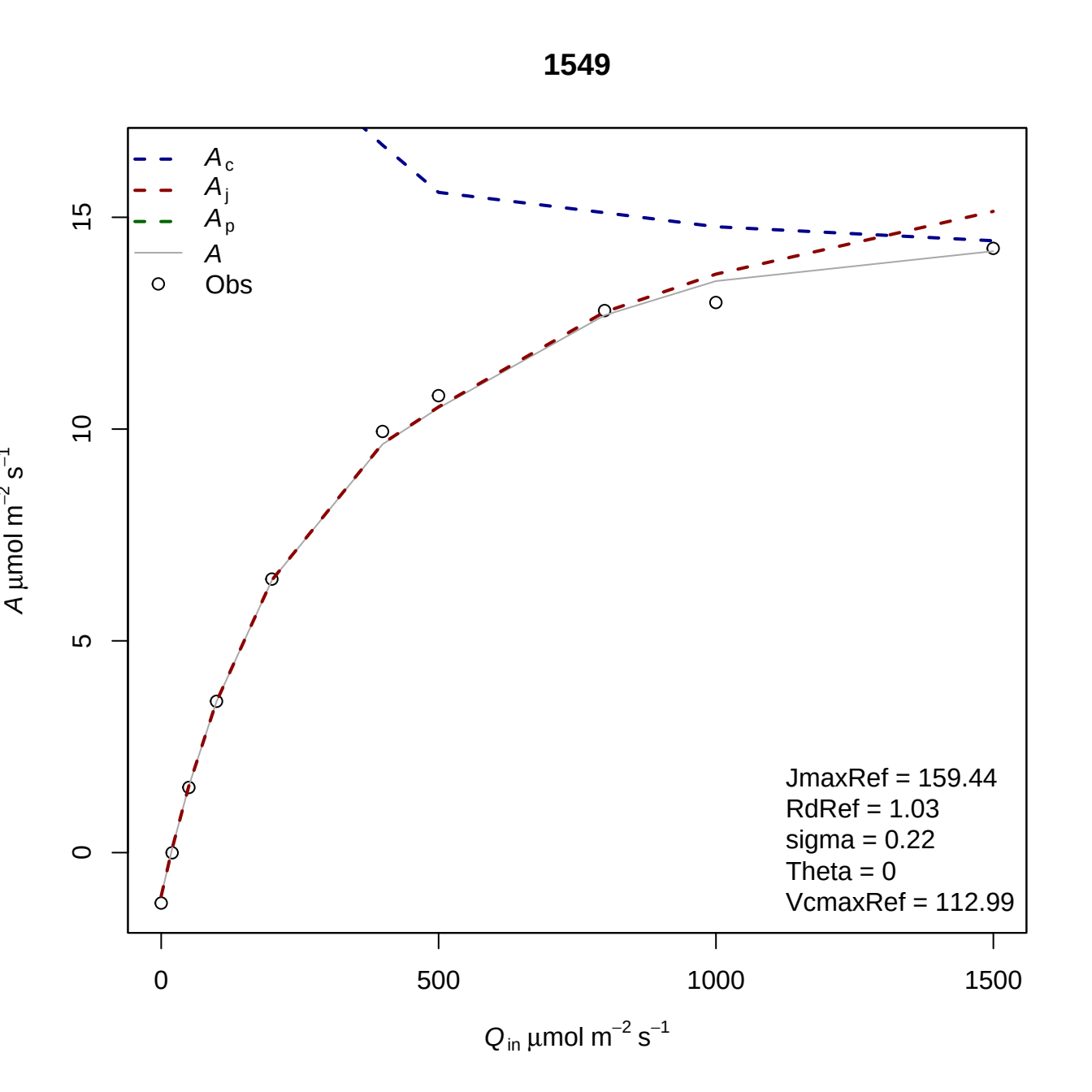

1544
A_c
A_j
A_p
A
Obs
JmaxRef = 117.24
RdRef = 0.4
sigma = 0.12
Theta = 0
VcmaxRef = 77.18
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

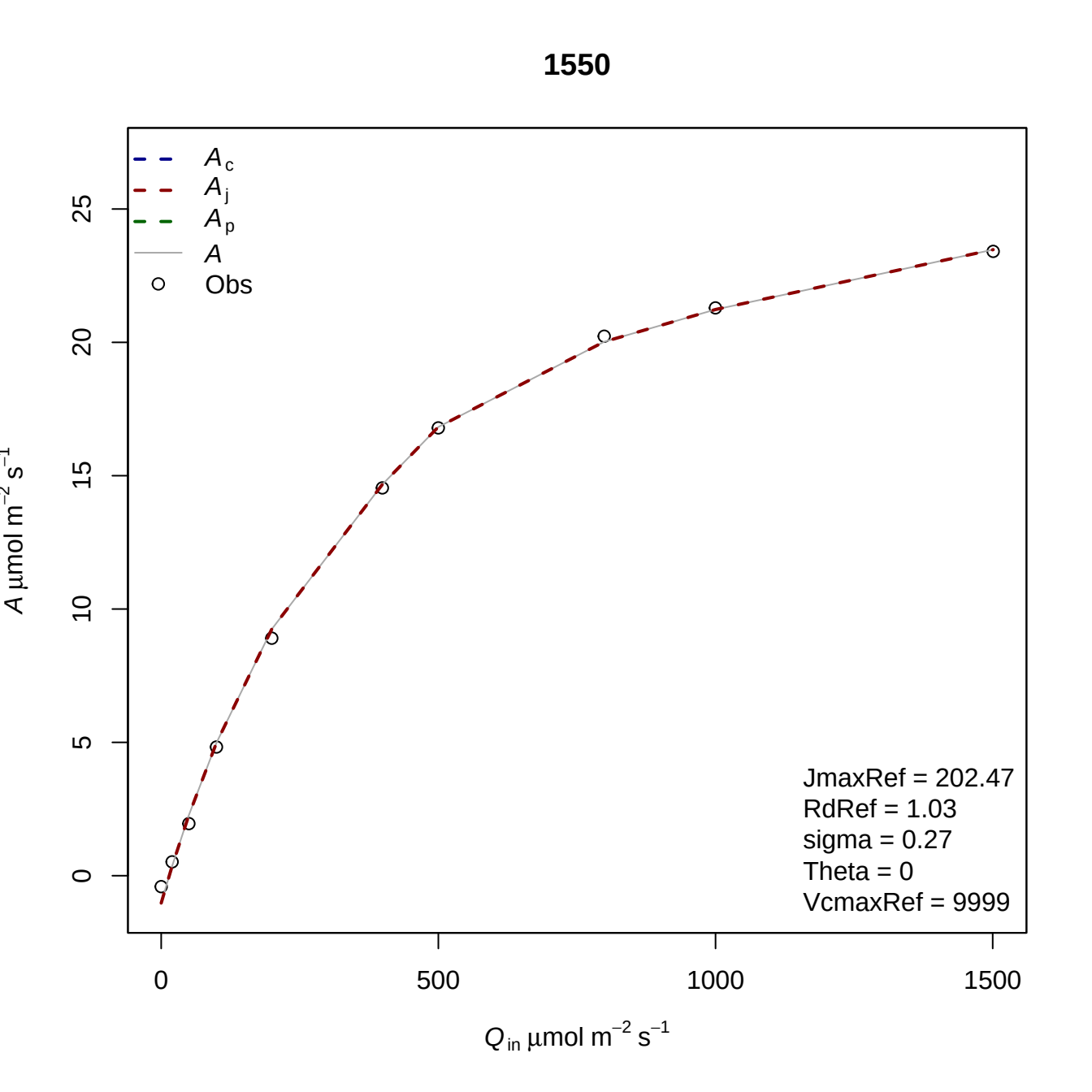

1545
A_c
A_j
A_p
A
Obs
JmaxRef = 99.88
RdRef = 1.01
sigma = 0.07
Theta = 0
VcmaxRef = 74.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

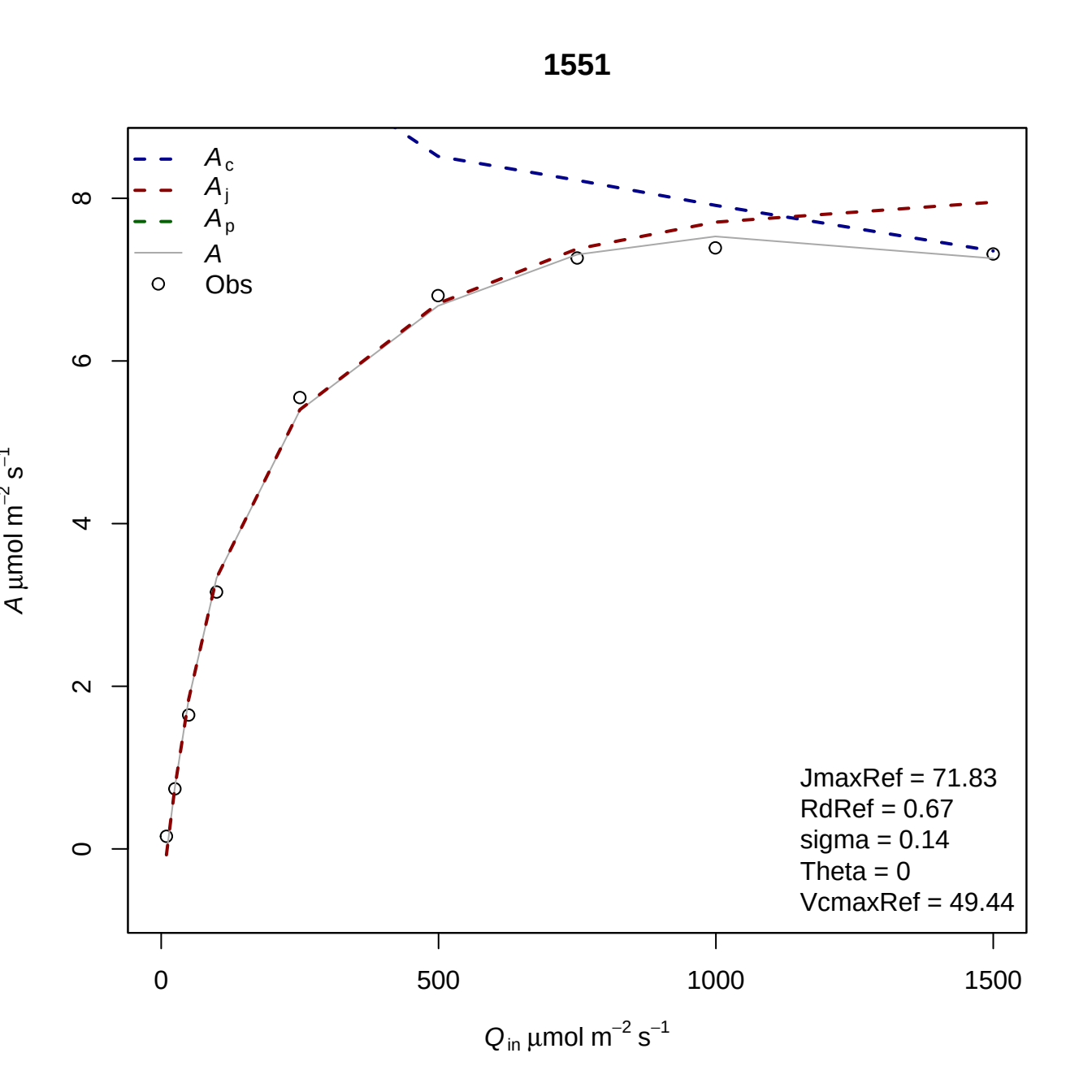

1546
A_c
A_j
A_p
A
Obs
JmaxRef = 119.54
RdRef = 0.71
sigma = 0.19
Theta = 0
VcmaxRef = 67.27
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

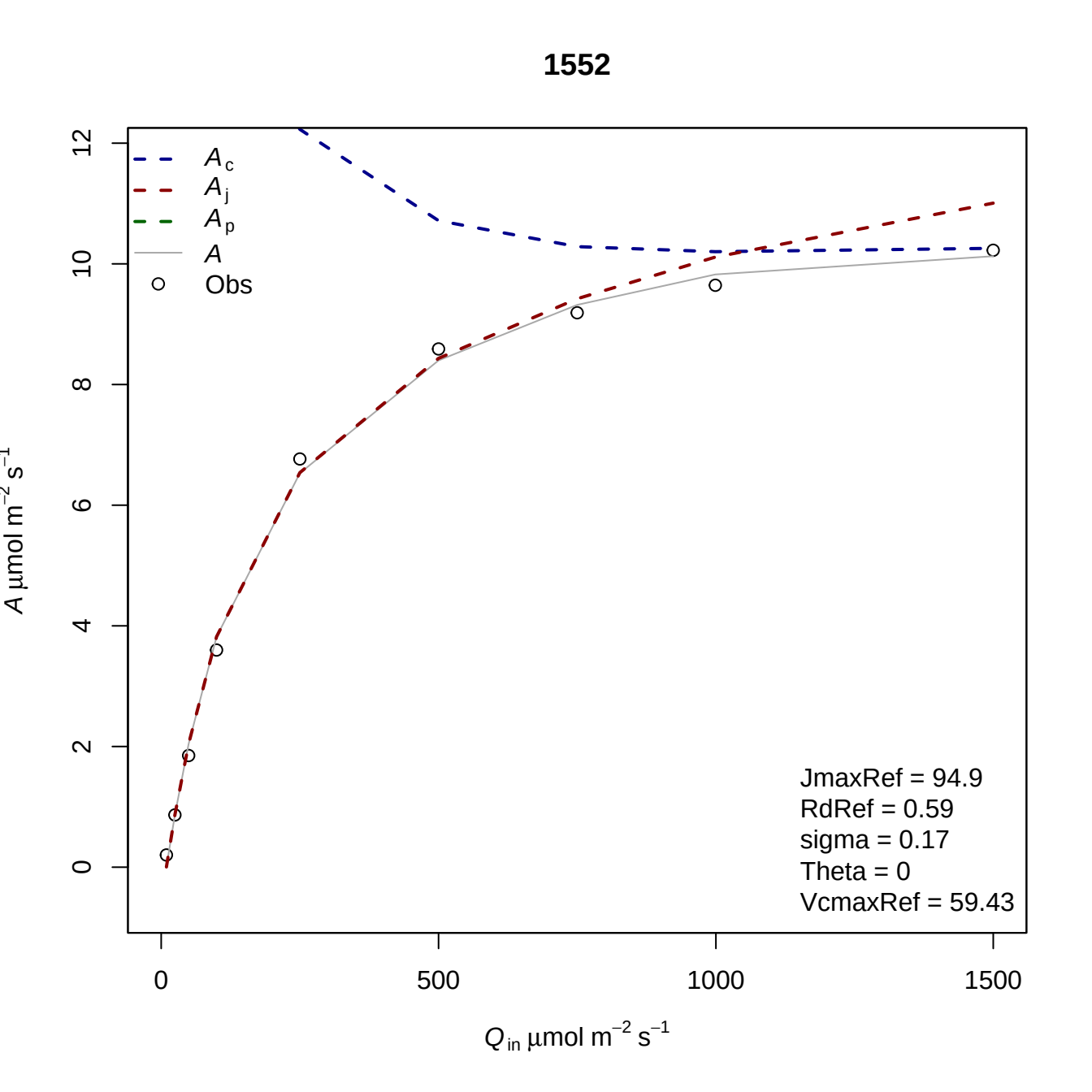

1547
A_c
A_j
A_p
A
Obs
JmaxRef = 174.27
RdRef = 0.87
sigma = 0.1
Theta = 0
VcmaxRef = 123.49
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

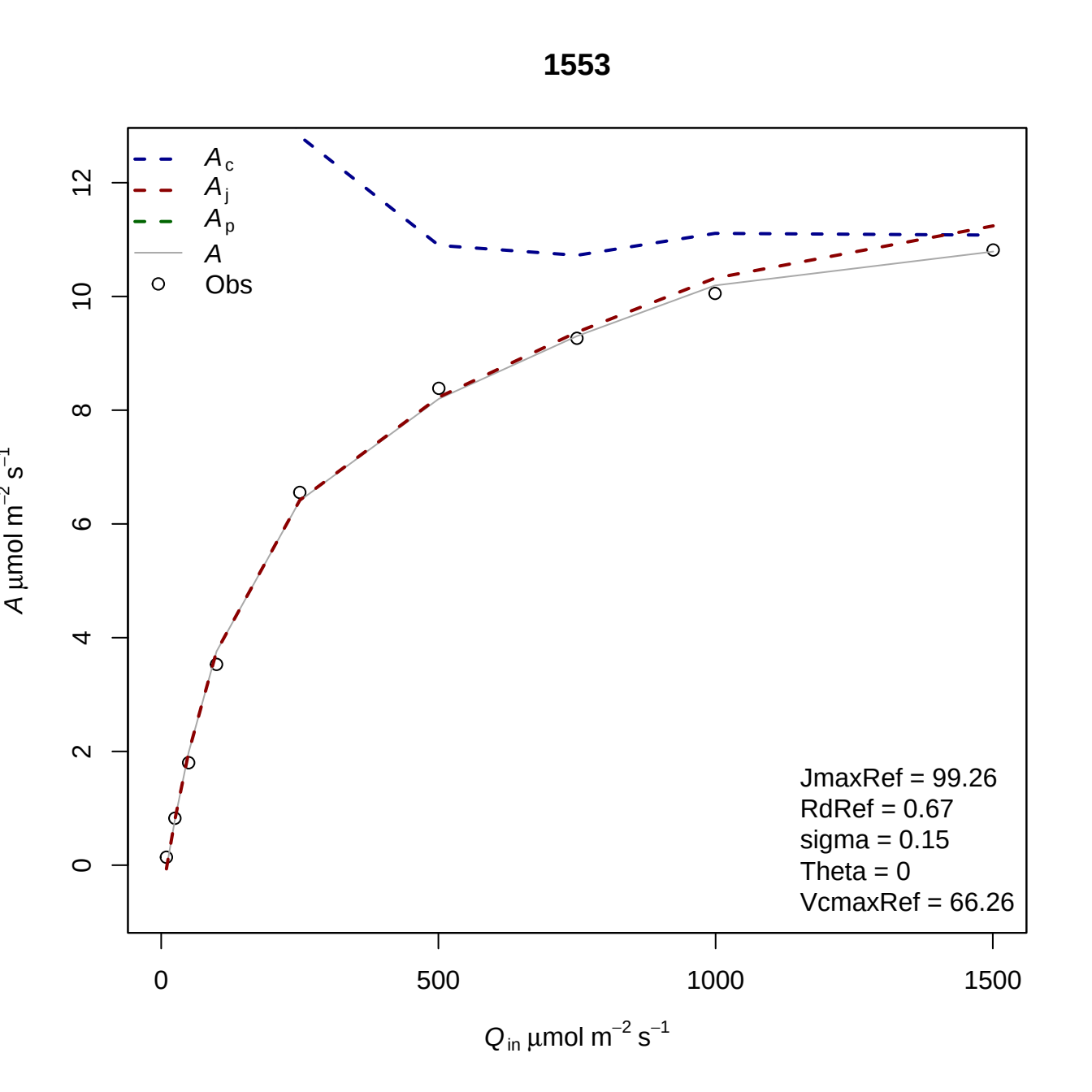

1548
A_c
A_j
A_p
A
Obs
JmaxRef = 209.71
RdRef = 0.85
sigma = 0.16
Theta = 0
VcmaxRef = 102.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500

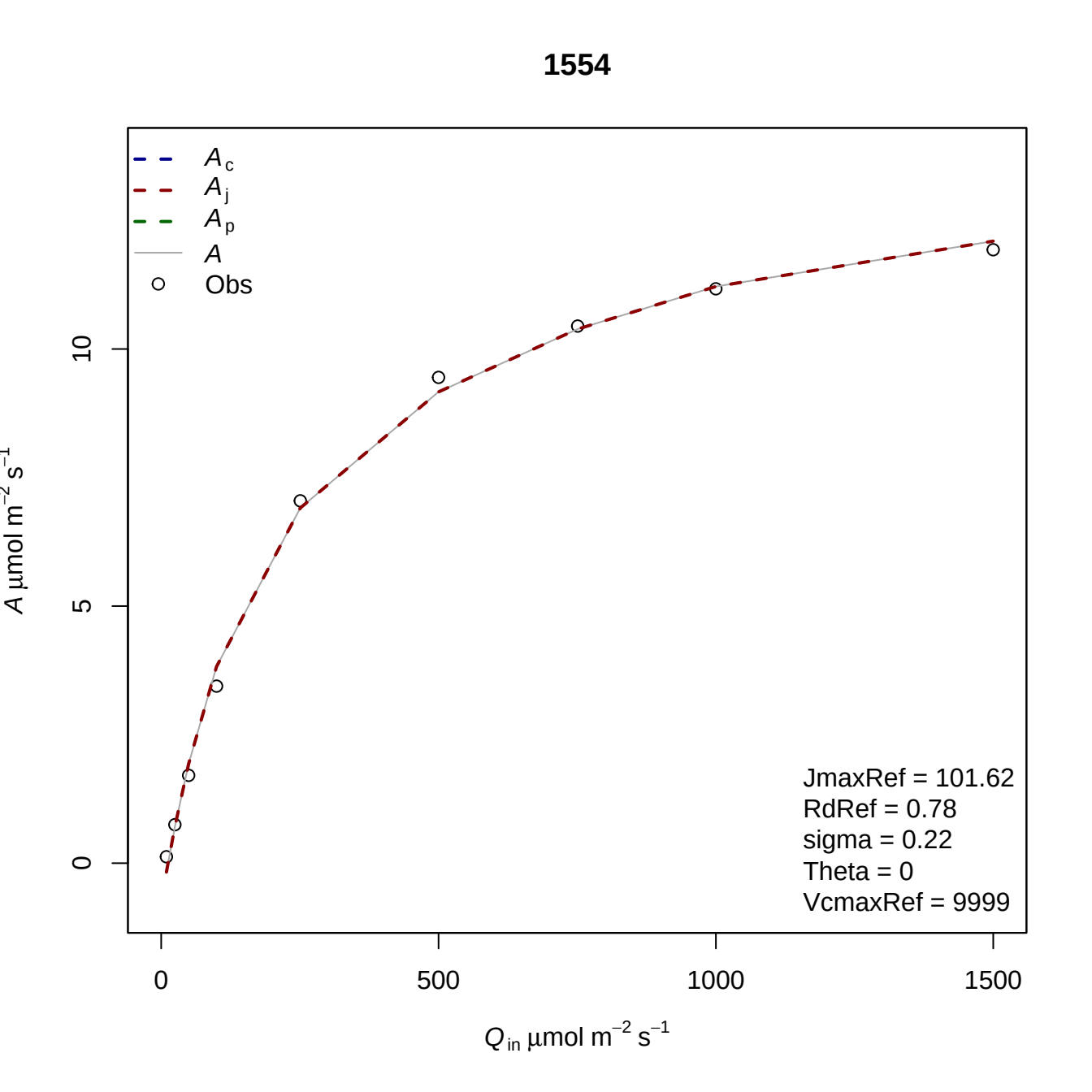

1549
A_c
A_j
A_p
A
Obs
JmaxRef = 159.44
RdRef = 1.03
sigma = 0.22
Theta = 0
VcmaxRef = 112.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

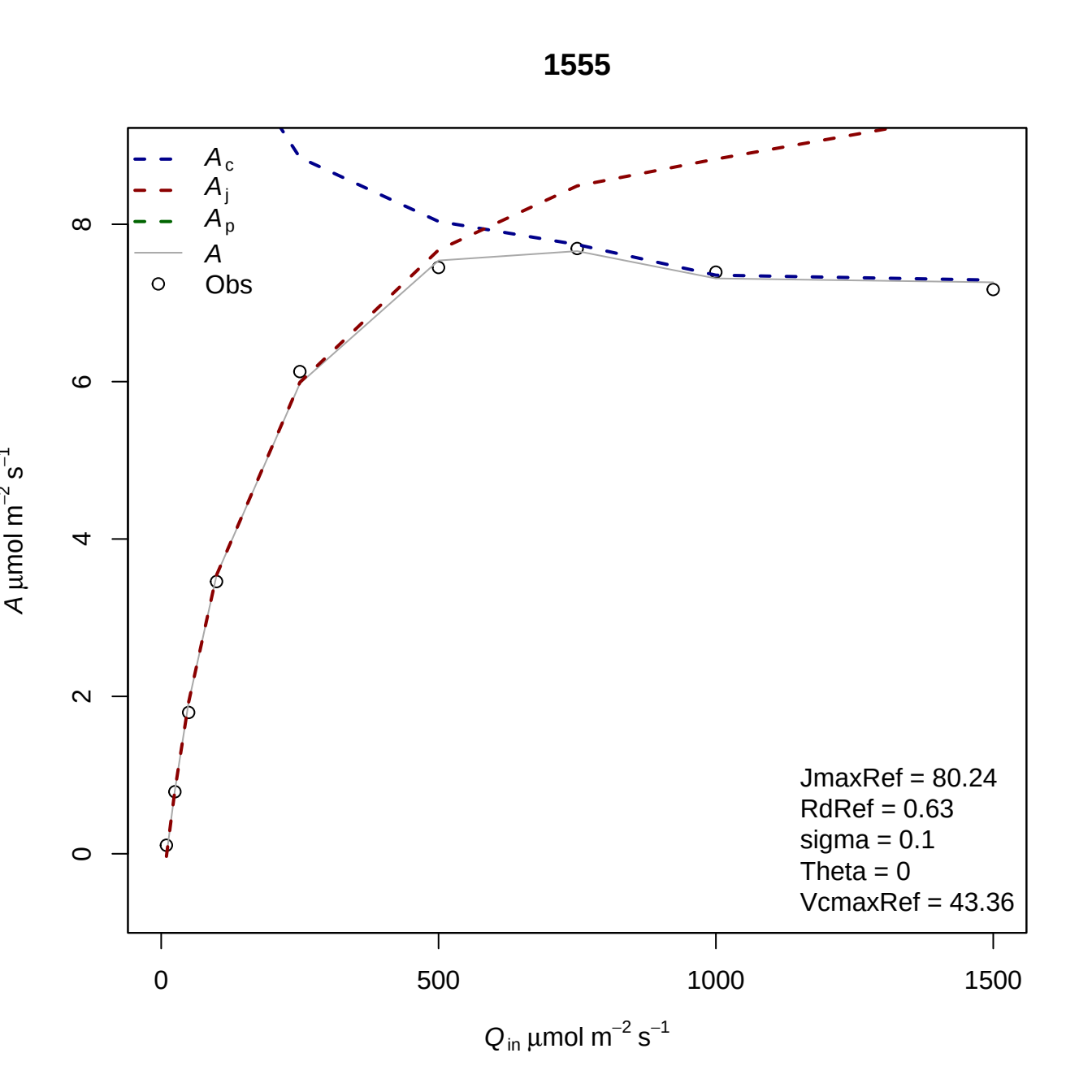

1550
A_c
A_j
A_p
A
Obs
JmaxRef = 202.47
RdRef = 1.03
sigma = 0.27
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

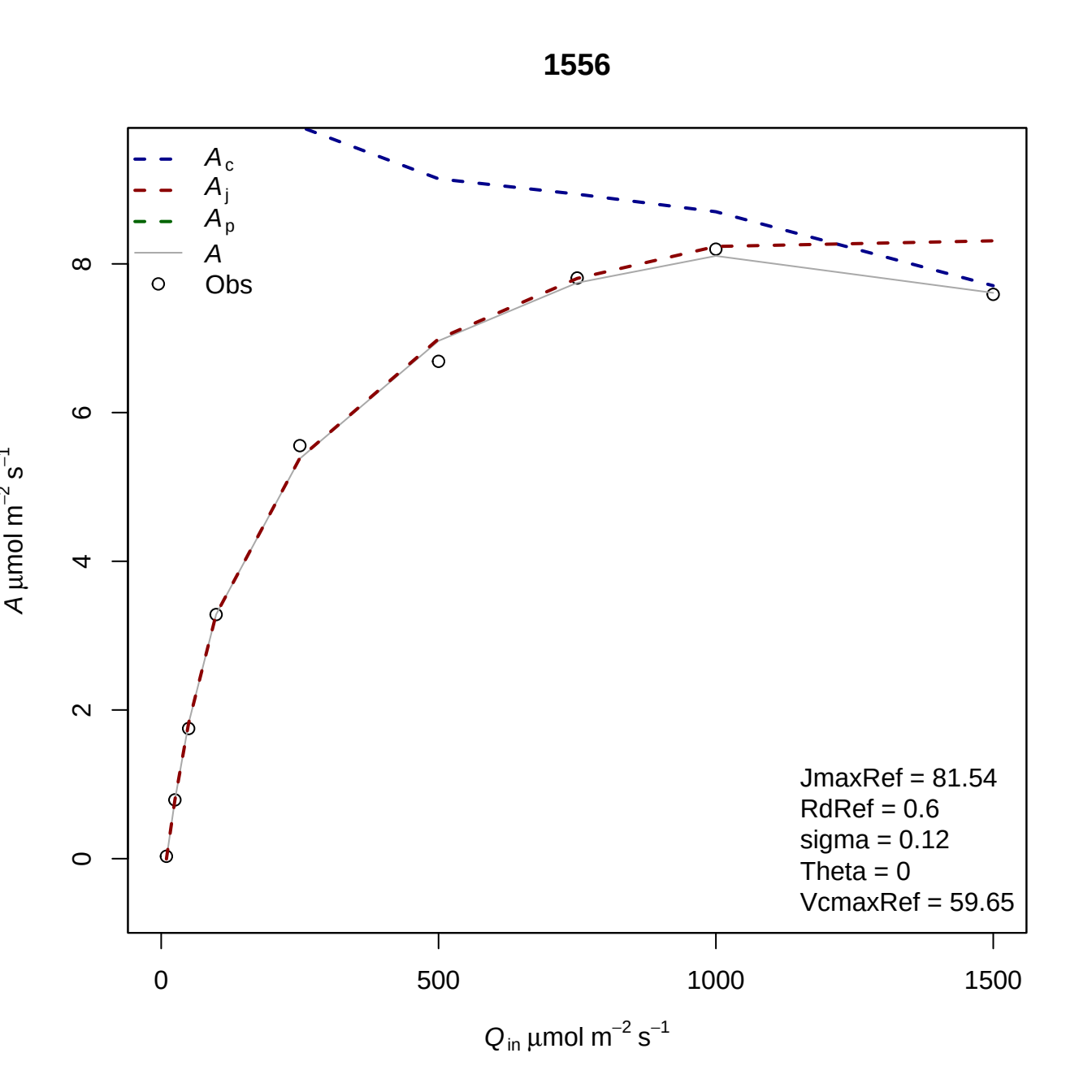

1551
A_c
A_j
A_p
A
Obs
JmaxRef = 71.83
RdRef = 0.67
sigma = 0.14
Theta = 0
VcmaxRef = 49.44
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

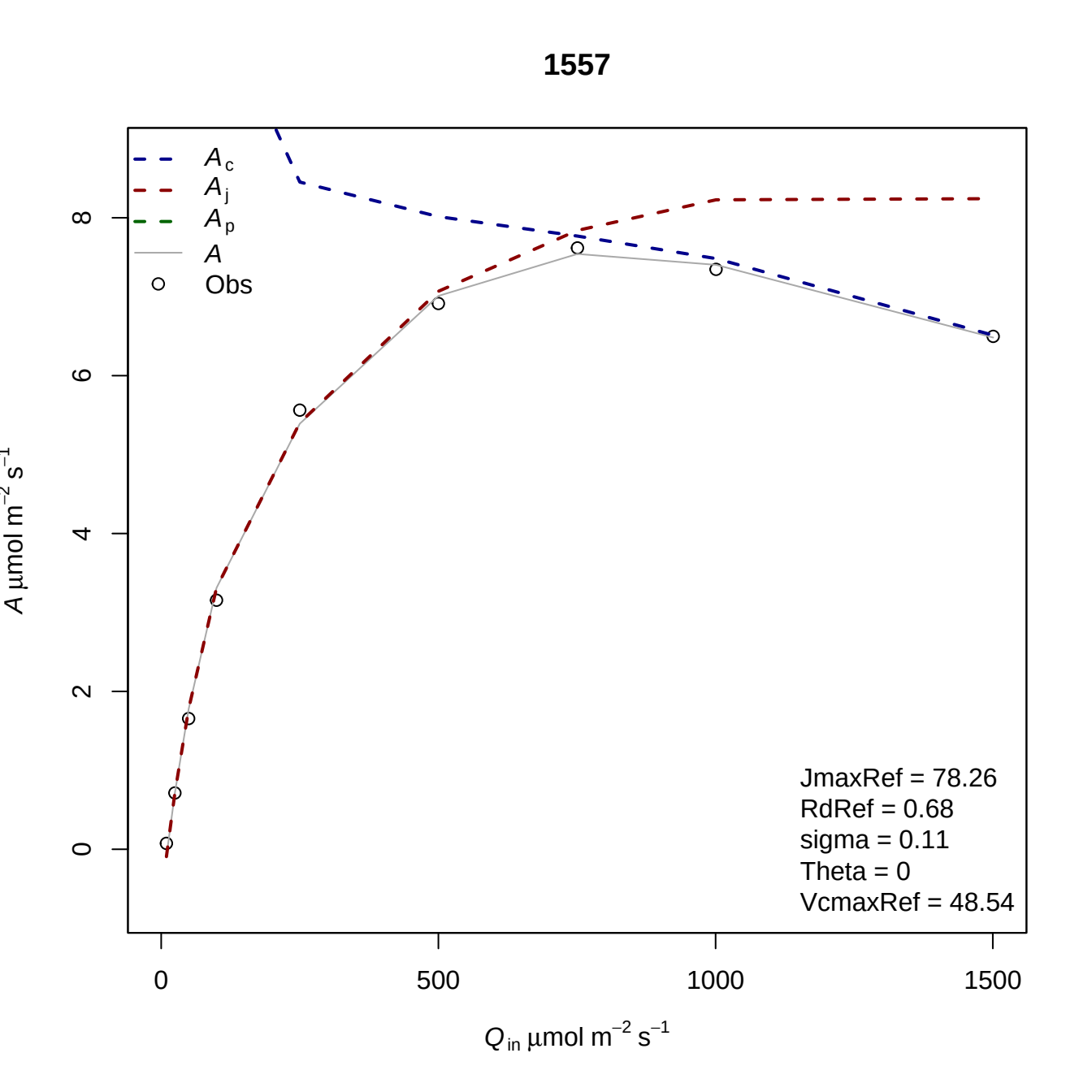

1552
A_c
A_j
A_p
A
Obs
JmaxRef = 94.9
RdRef = 0.59
sigma = 0.17
Theta = 0
VcmaxRef = 59.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

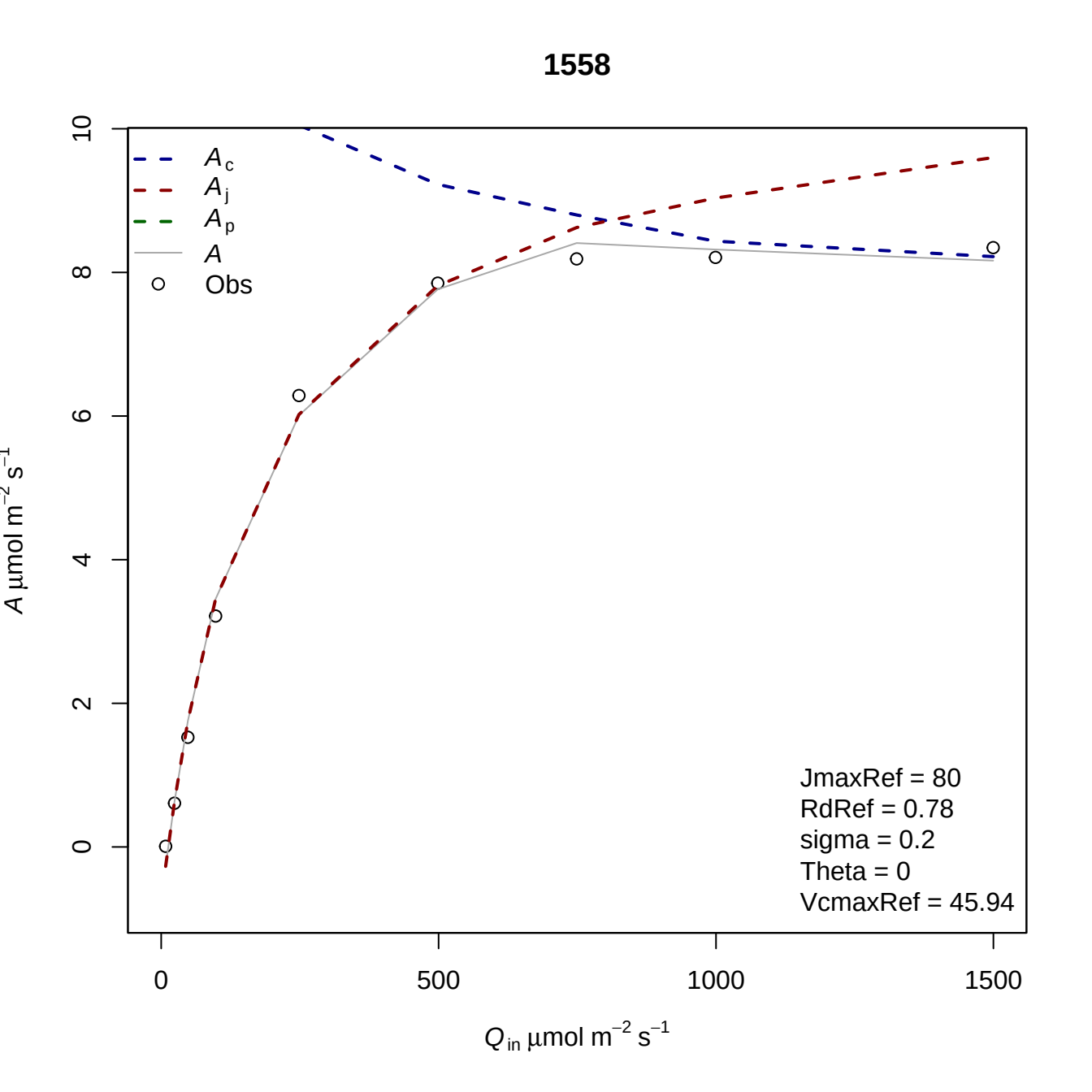

1553
A_c
A_j
A_p
A
Obs
JmaxRef = 99.26
RdRef = 0.67
sigma = 0.15
Theta = 0
VcmaxRef = 66.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

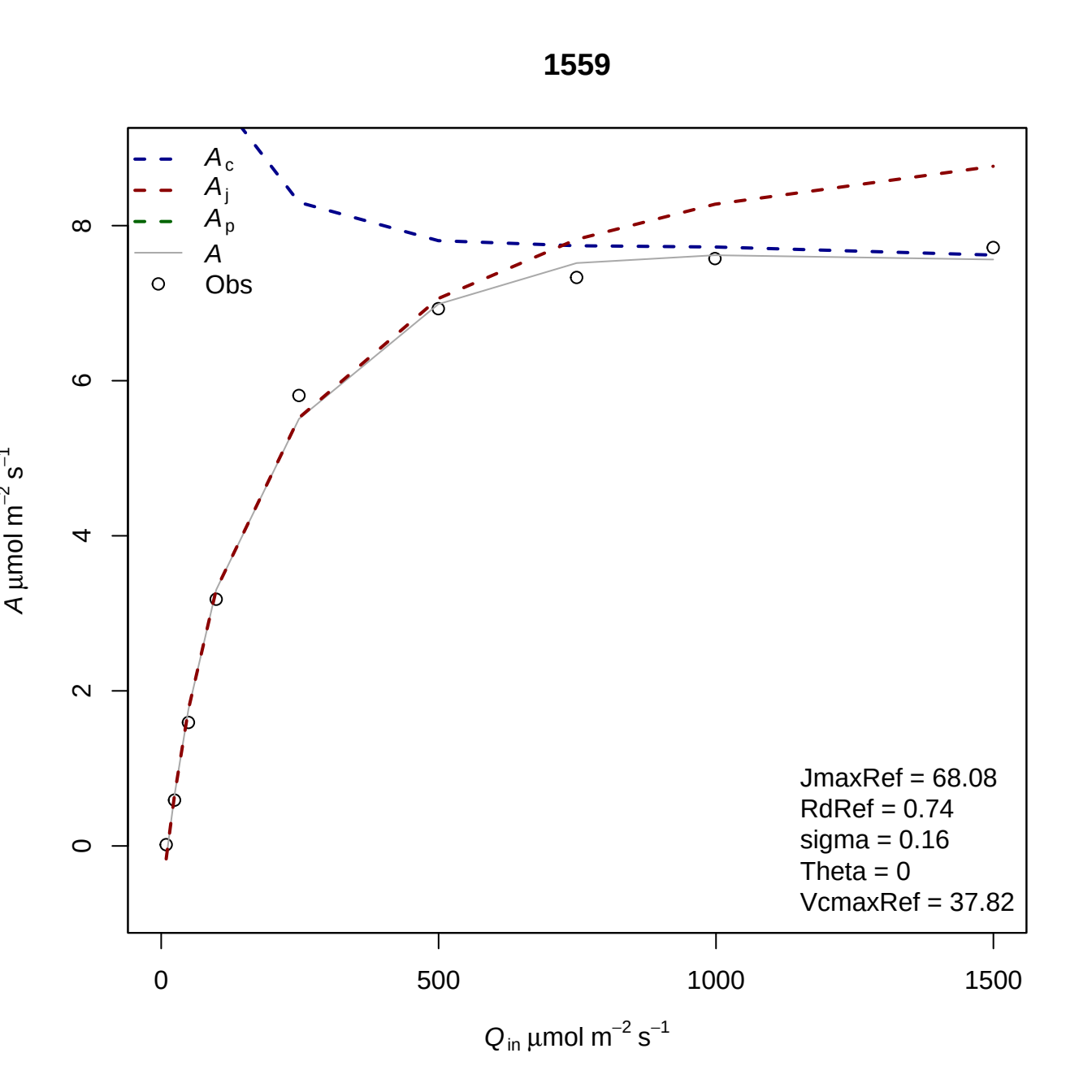

1554
A_c
A_j
A_p
A
Obs
JmaxRef = 101.62
RdRef = 0.78
sigma = 0.22
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

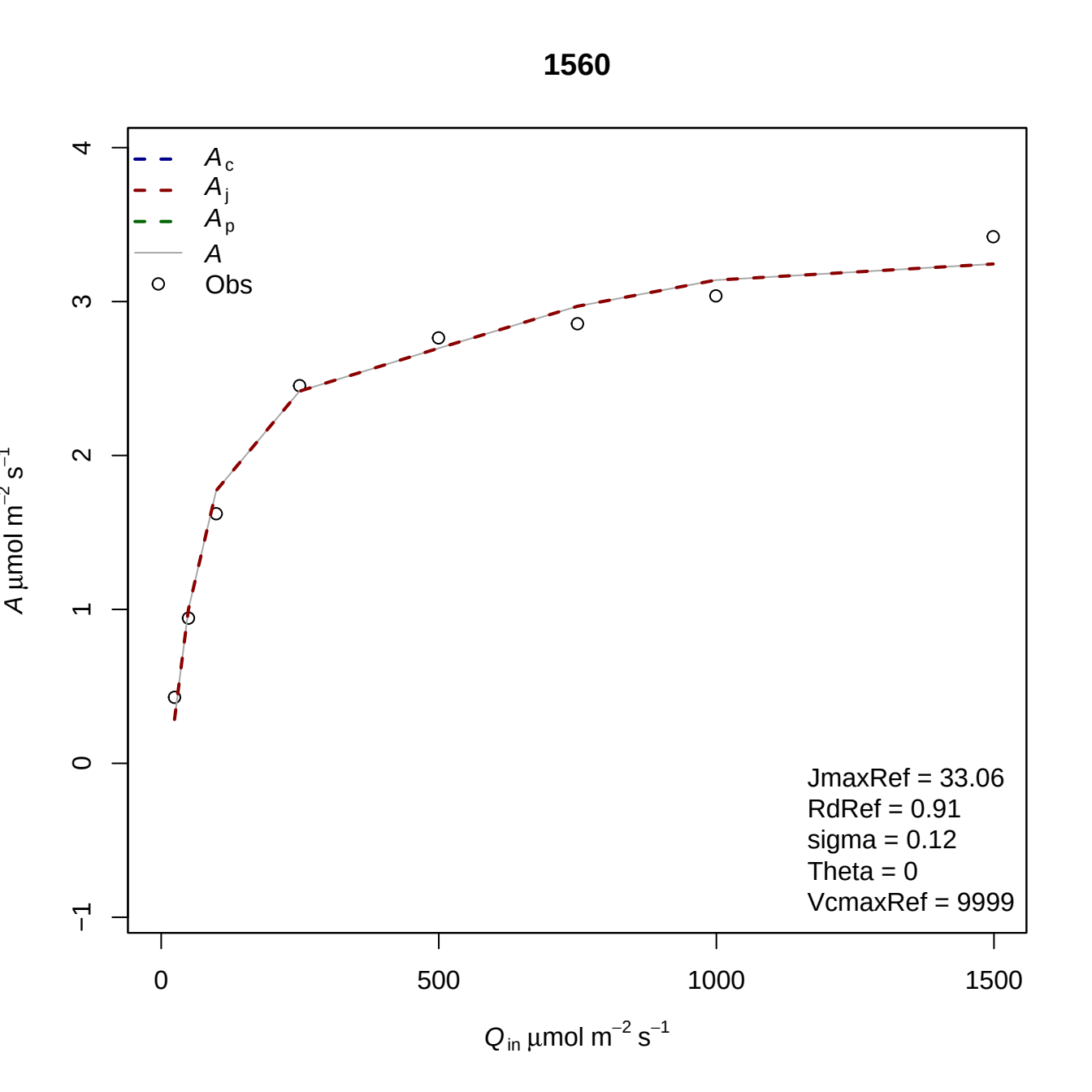

1555
A_c
A_j
A_p
A
Obs
JmaxRef = 80.24
RdRef = 0.63
sigma = 0.1
Theta = 0
VcmaxRef = 43.36
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

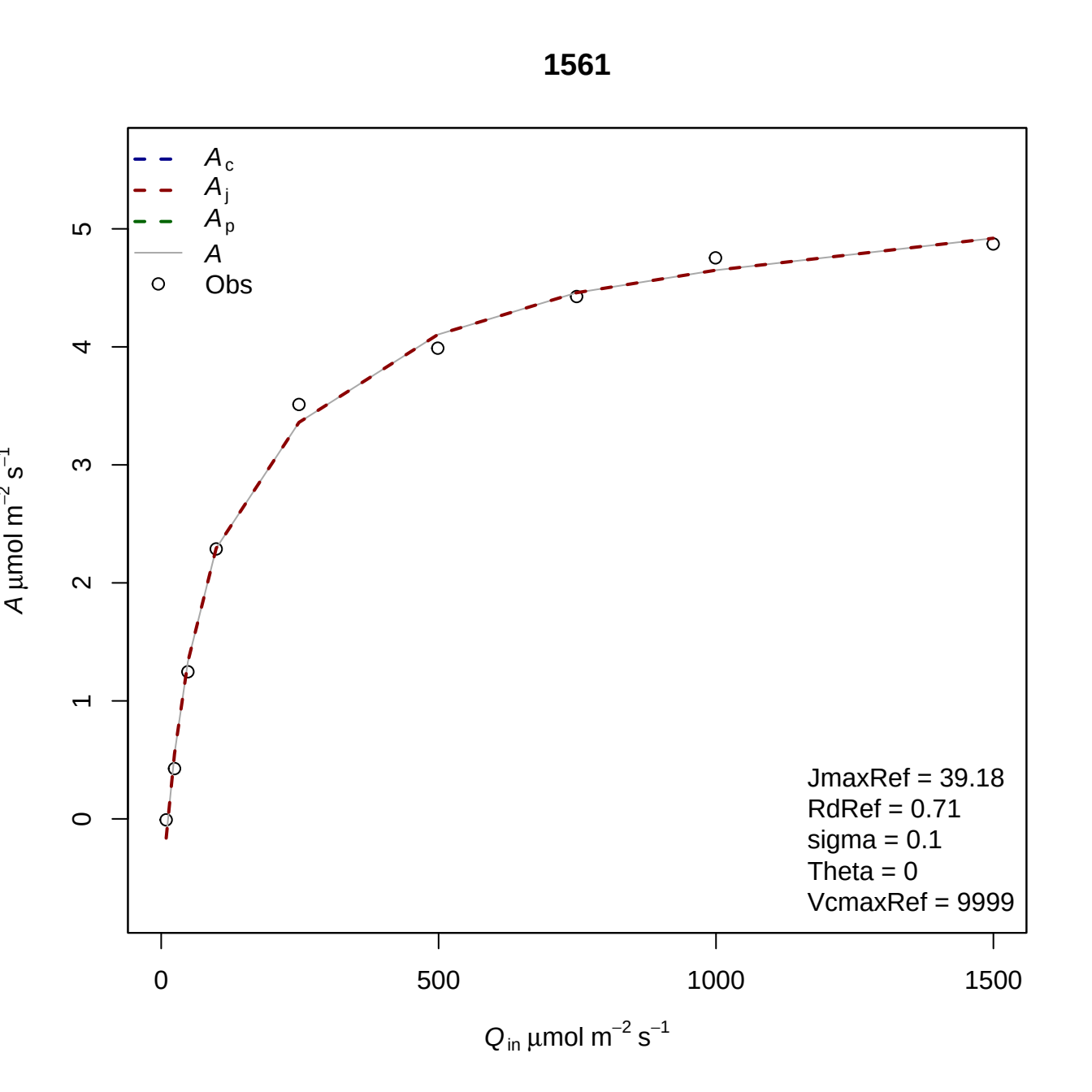

1556
A_c
A_j
A_p
A
Obs
JmaxRef = 81.54
RdRef = 0.6
sigma = 0.12
Theta = 0
VcmaxRef = 59.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

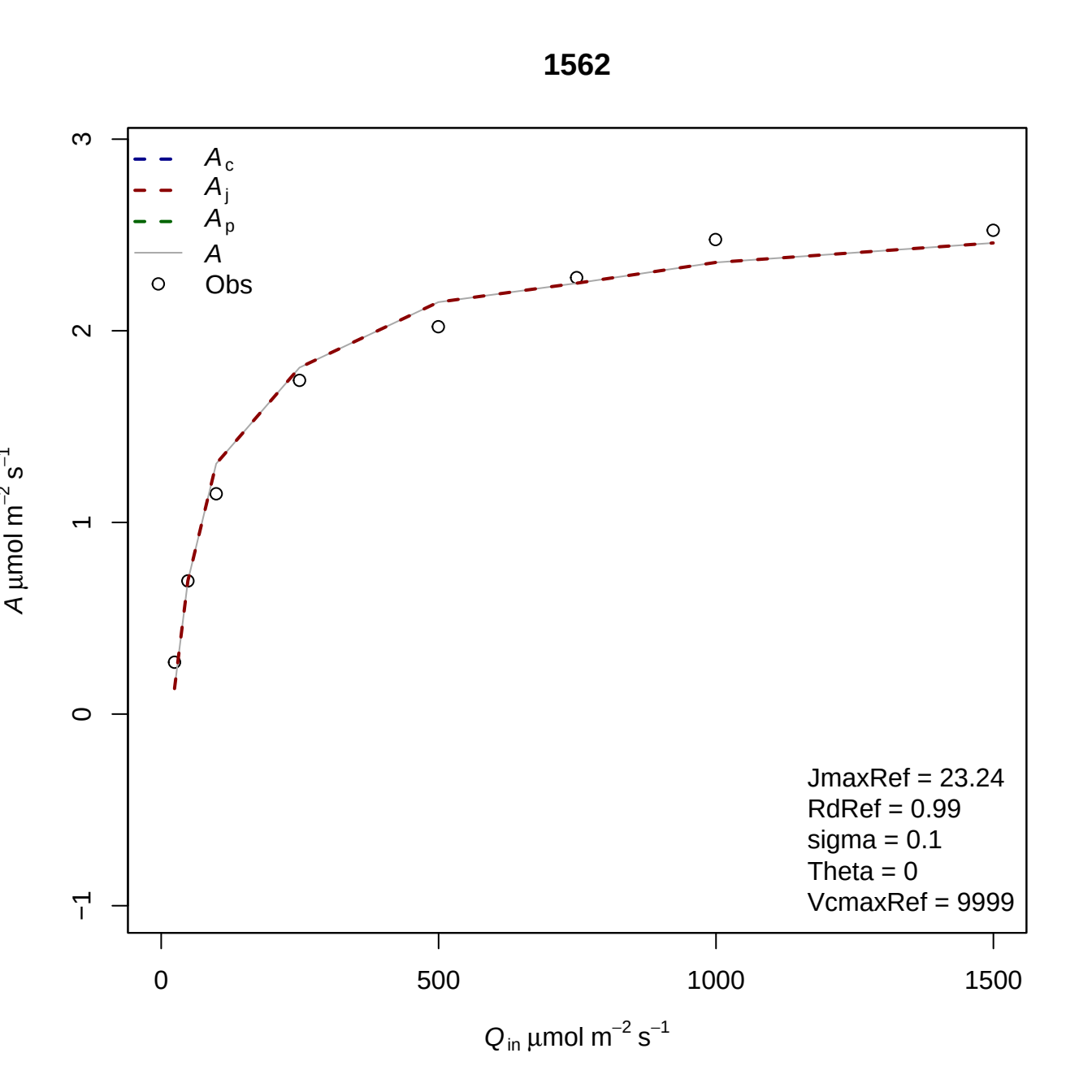

1557
A_c
A_j
A_p
A
Obs
JmaxRef = 78.26
RdRef = 0.68
sigma = 0.11
Theta = 0
VcmaxRef = 48.54
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

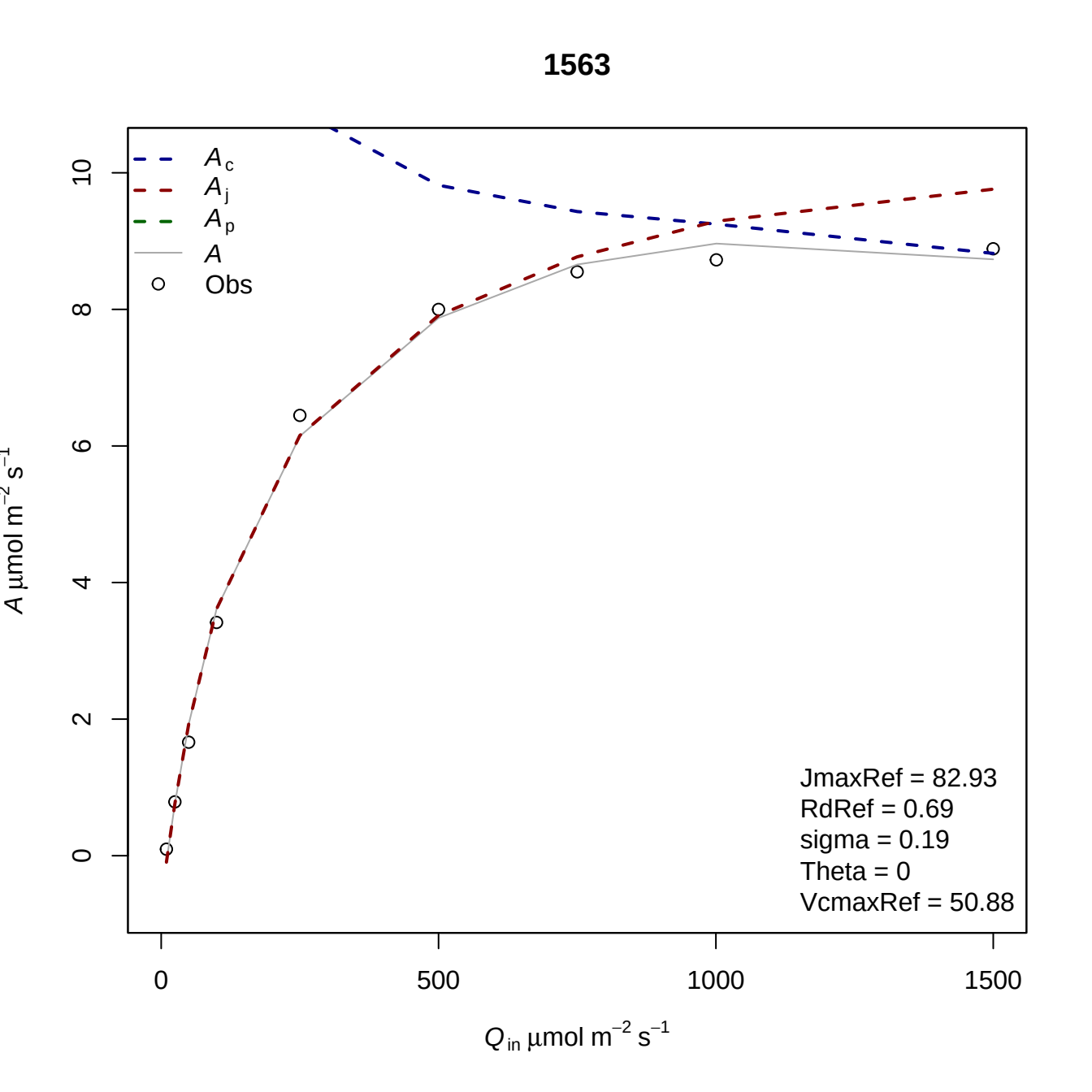

1558
A_c
A_j
A_p
A
Obs
JmaxRef = 80
RdRef = 0.78
sigma = 0.2
Theta = 0
VcmaxRef = 45.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

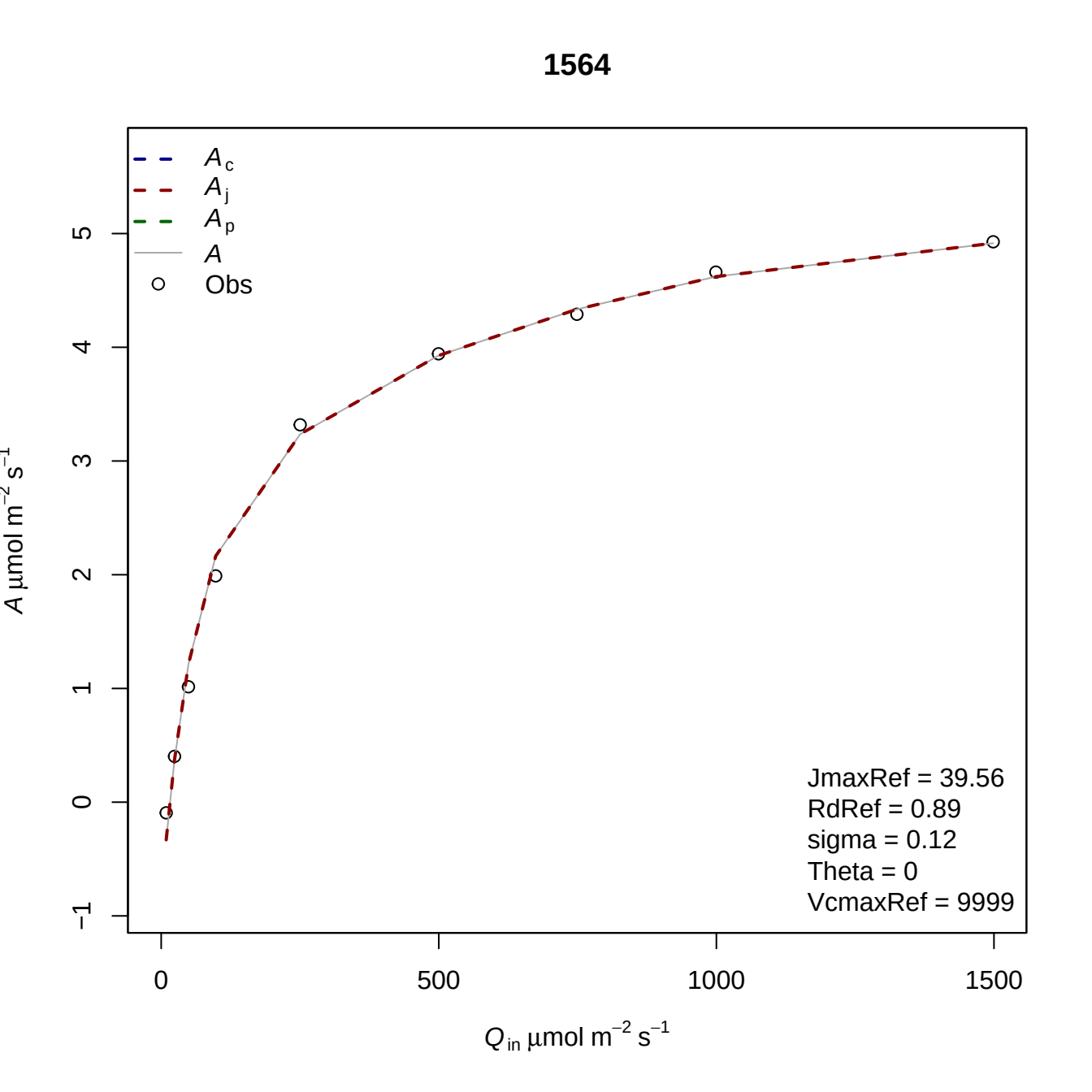

1559
A_c
A_j
A_p
A
Obs
JmaxRef = 68.08
RdRef = 0.74
sigma = 0.16
Theta = 0
VcmaxRef = 37.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

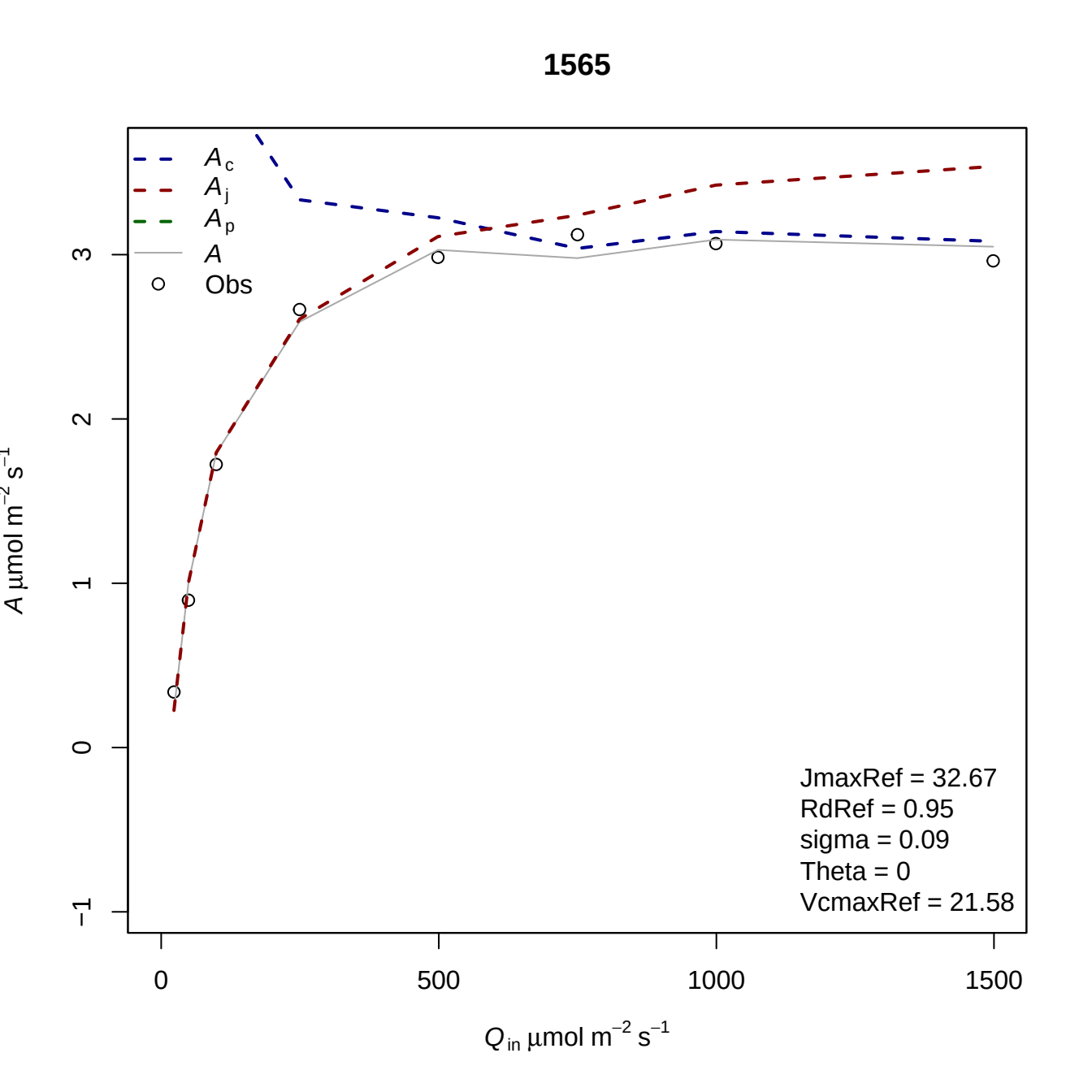

1560
A_c
A_j
A_p
A
Obs
JmaxRef = 33.06
RdRef = 0.91
sigma = 0.12
Theta = 0
VcmaxRef = 9999
4
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

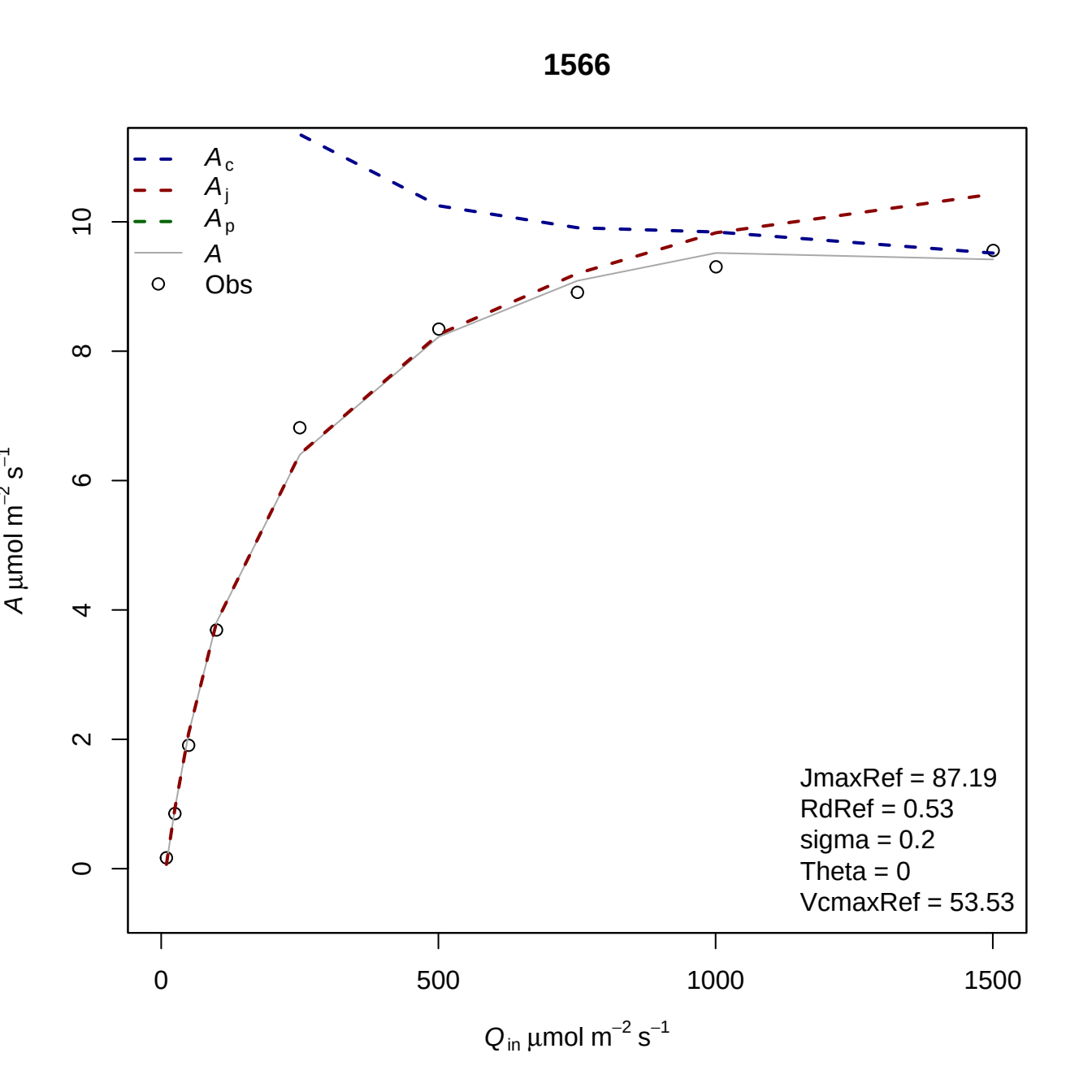

1561
A_c
A_j
A_p
A
Obs
JmaxRef = 39.18
RdRef = 0.71
sigma = 0.1
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

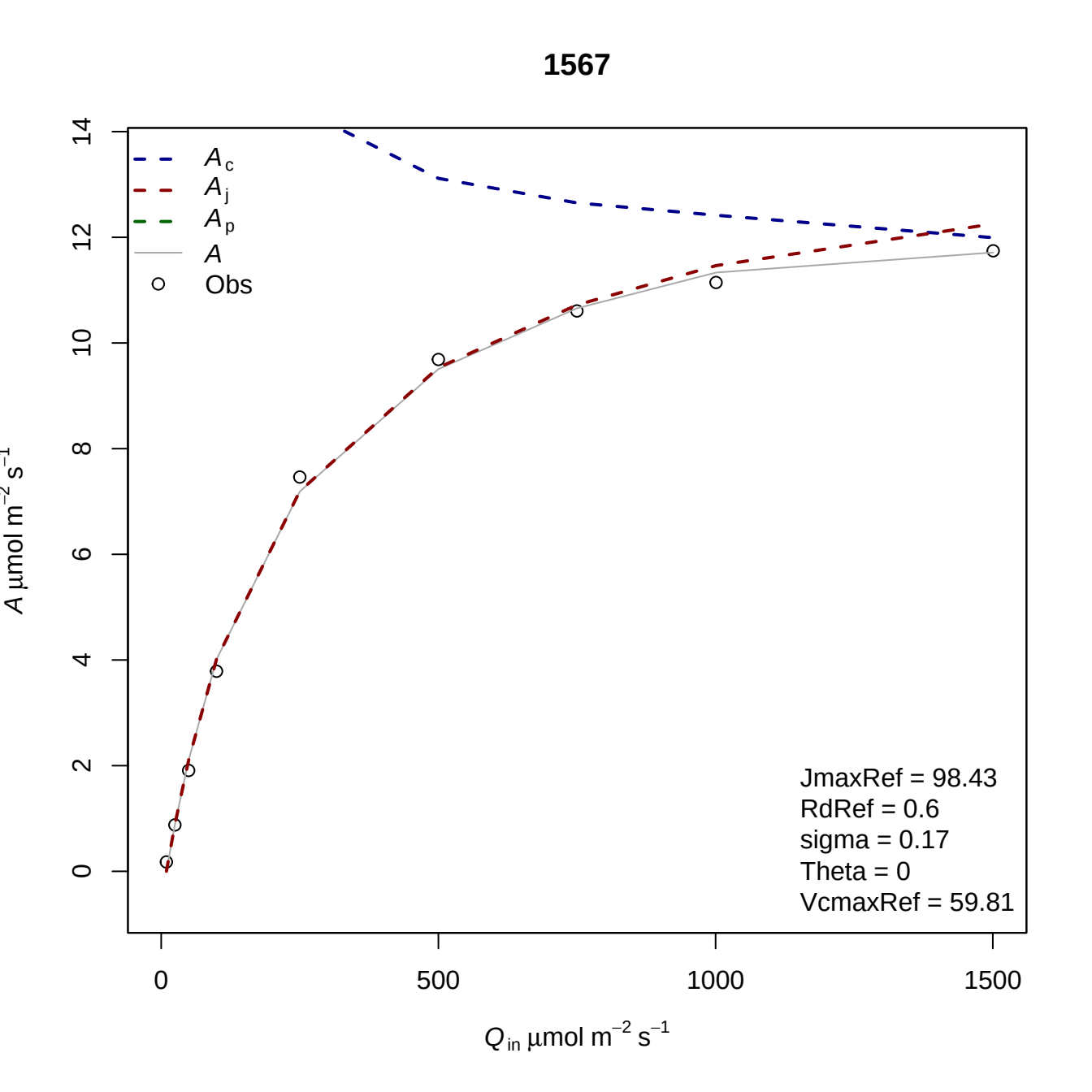

1562
A_c
A_j
A_p
A
Obs
JmaxRef = 23.24
RdRef = 0.99
sigma = 0.1
Theta = 0
VcmaxRef = 9999
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

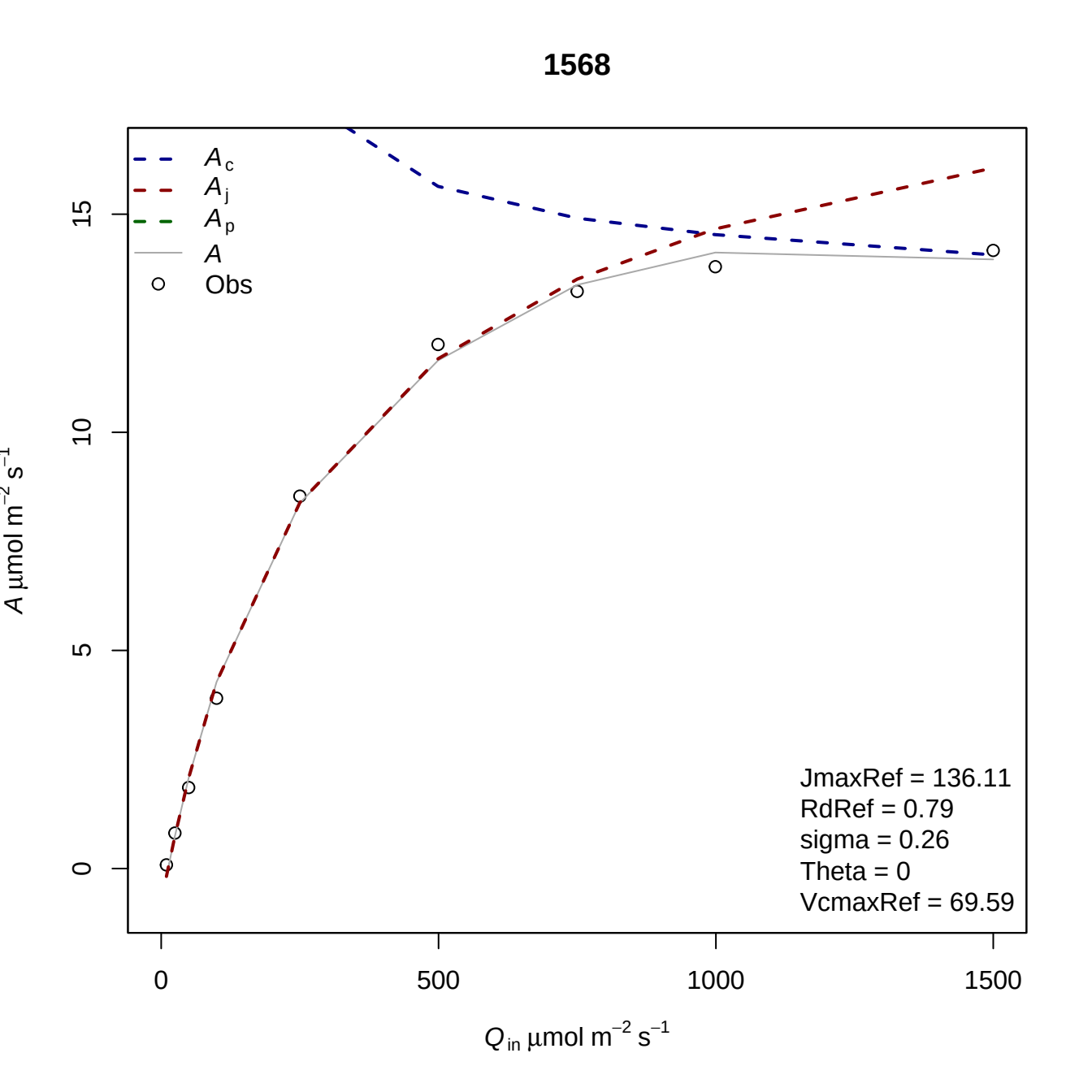

1563
A_c
A_j
A_p
A
Obs
JmaxRef = 82.93
RdRef = 0.69
sigma = 0.19
Theta = 0
VcmaxRef = 50.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

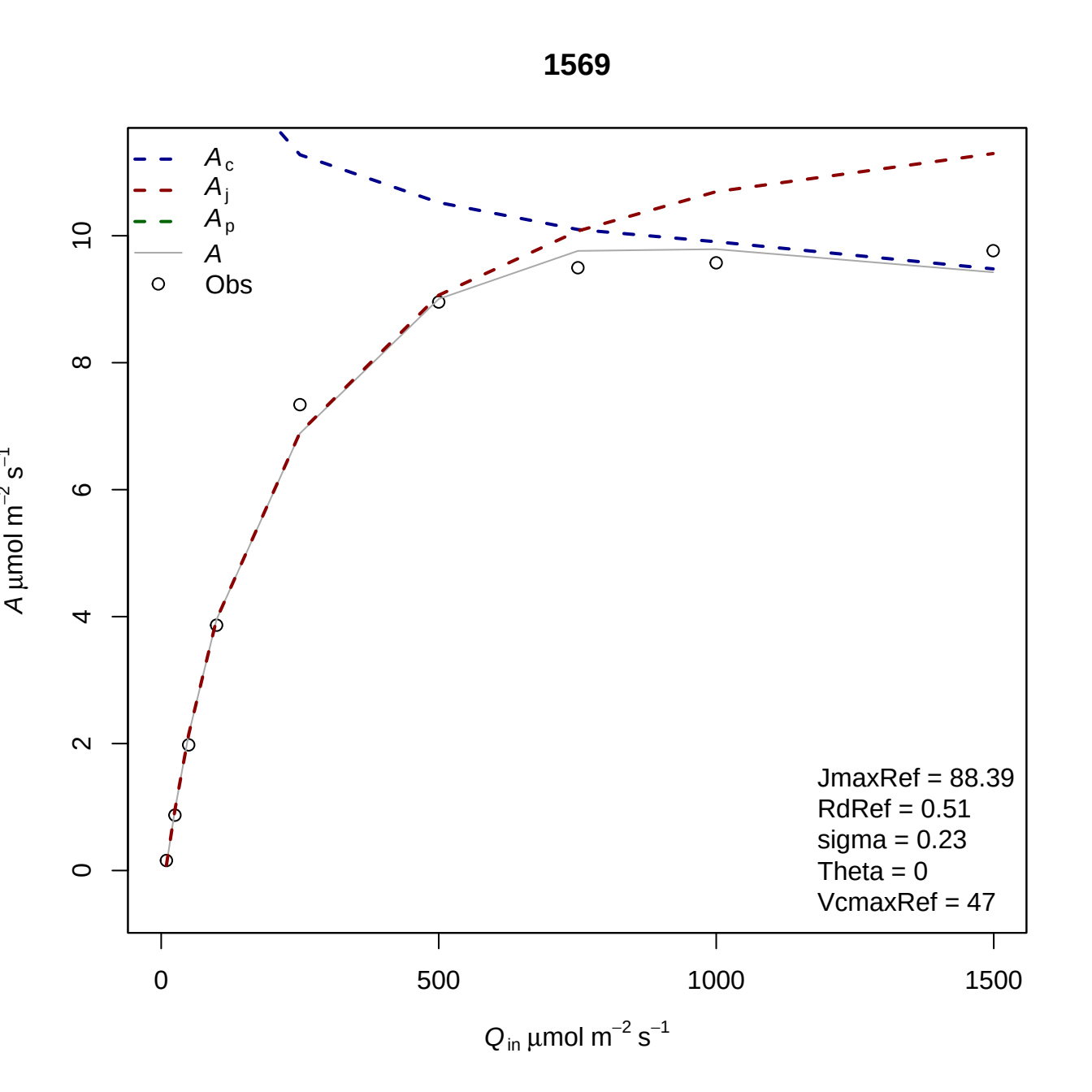

1564
A_c
A_j
A_p
A
Obs
JmaxRef = 39.56
RdRef = 0.89
sigma = 0.12
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

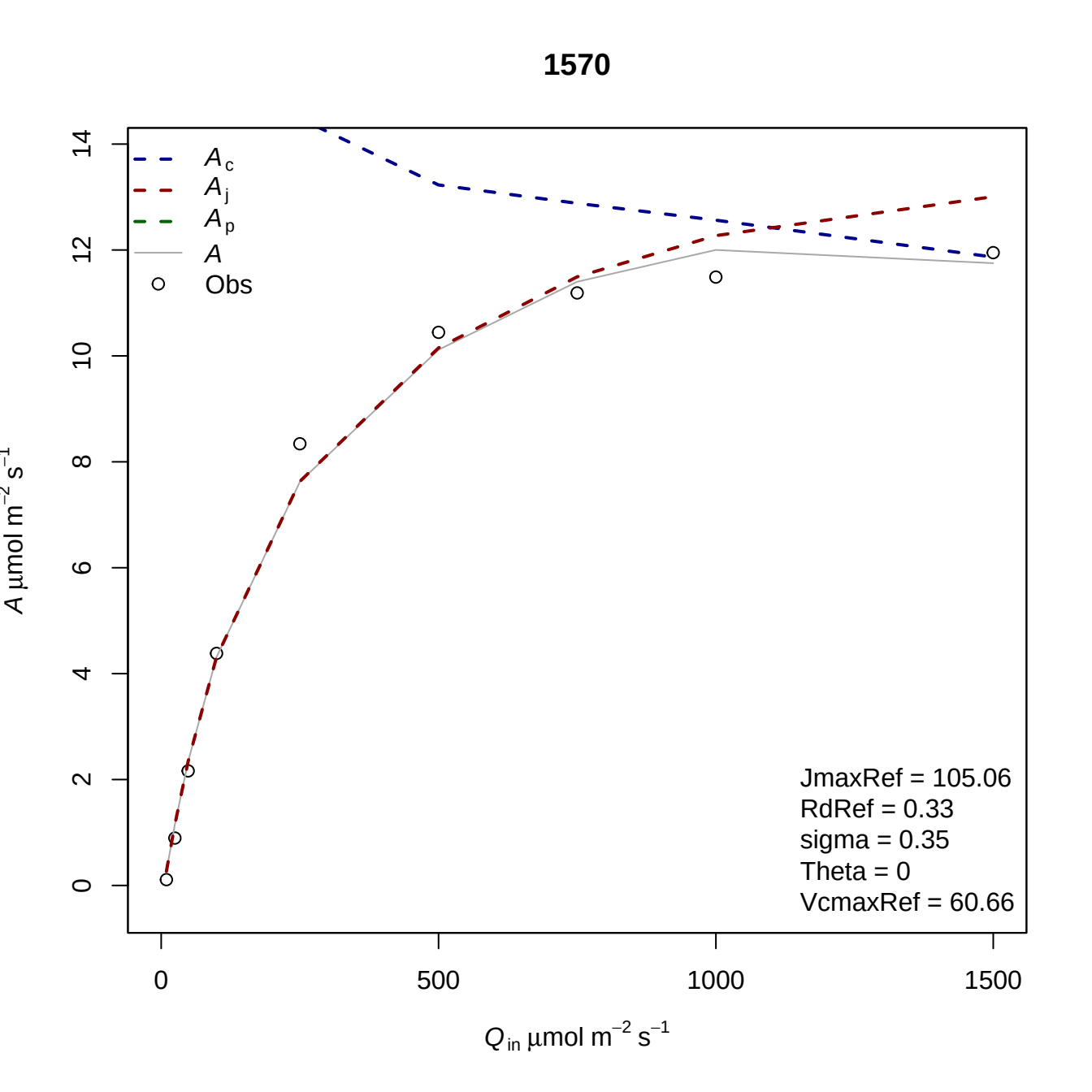

1565
A_c
A_j
A_p
A
Obs
JmaxRef = 32.67
RdRef = 0.95
sigma = 0.09
Theta = 0
VcmaxRef = 21.58
−1
0
1
2
3
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

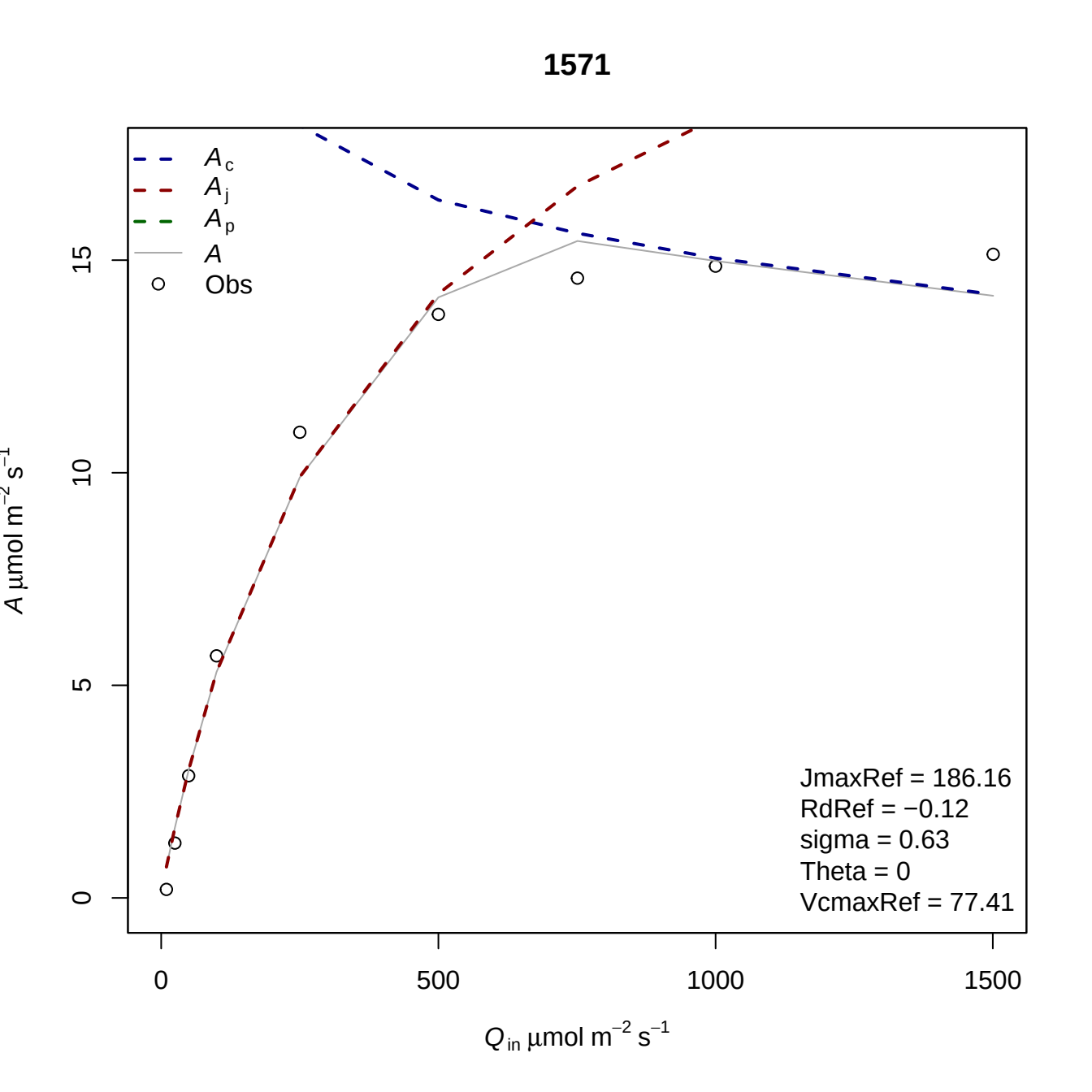

1566
A_c
A_j
A_p
A
Obs
JmaxRef = 87.19
RdRef = 0.53
sigma = 0.2
Theta = 0
VcmaxRef = 53.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

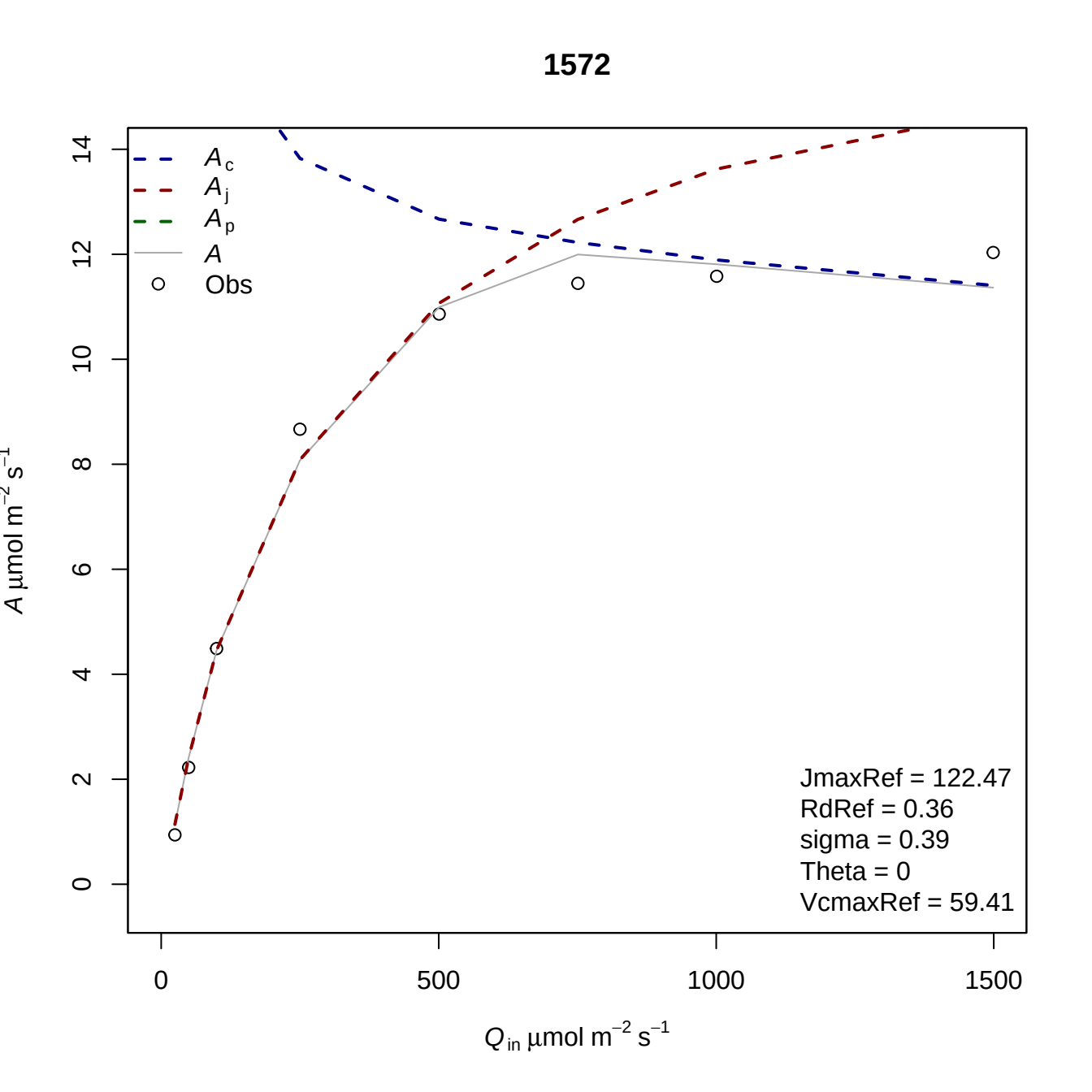

1567
A_c
A_j
A_p
A
Obs
JmaxRef = 98.43
RdRef = 0.6
sigma = 0.17
Theta = 0
VcmaxRef = 59.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

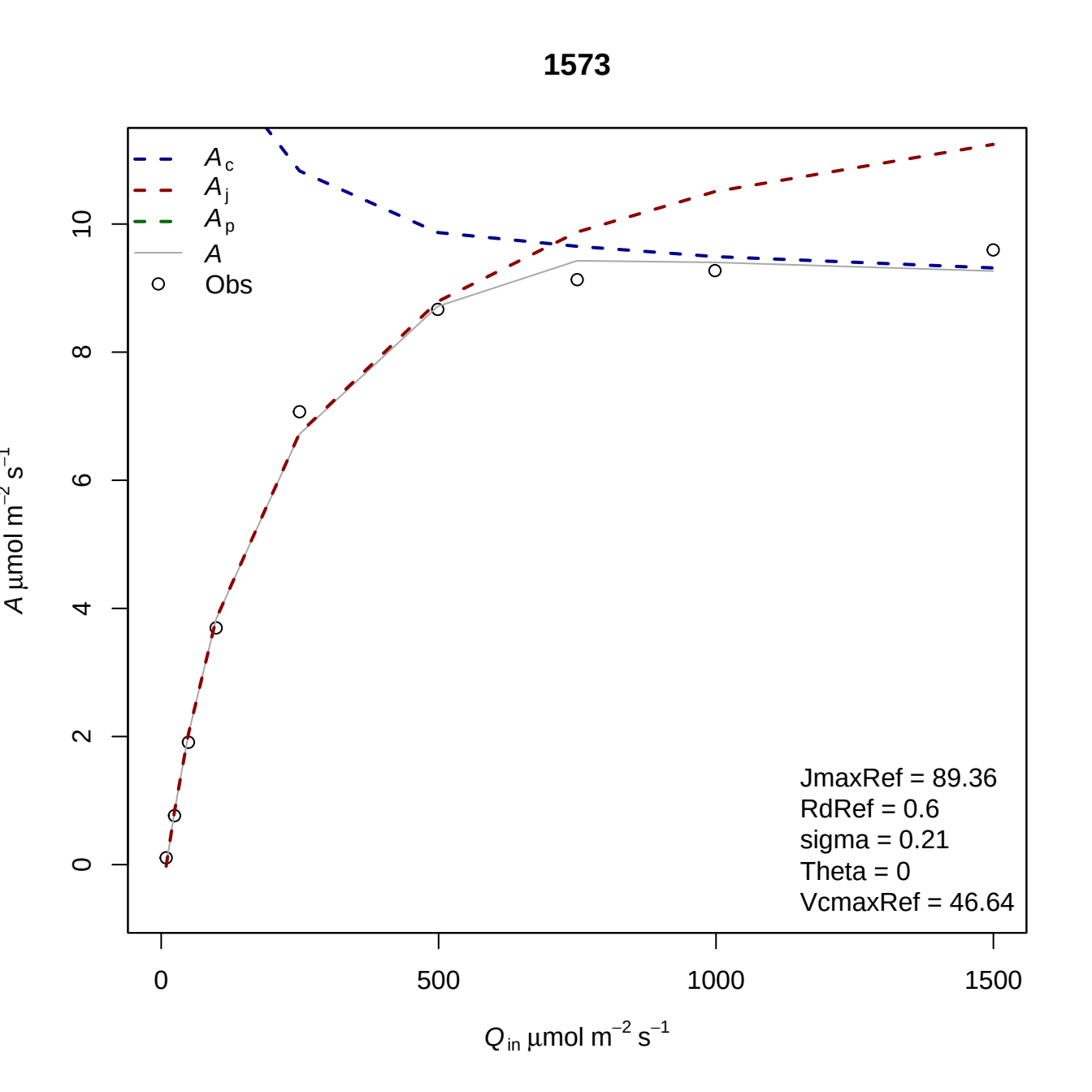

1568
A_c
A_j
A_p
A
Obs
JmaxRef = 136.11
RdRef = 0.79
sigma = 0.26
Theta = 0
VcmaxRef = 69.59
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

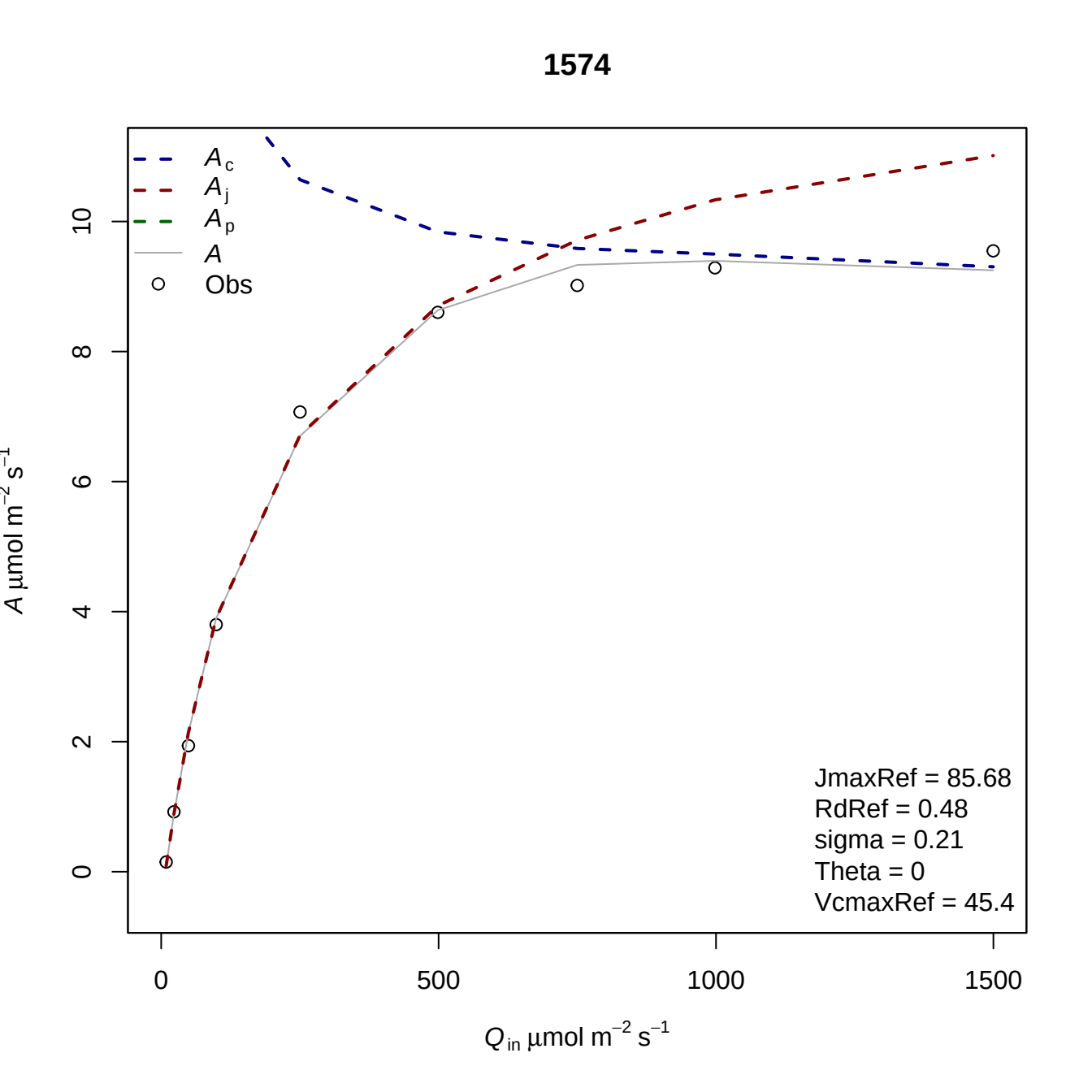

1569
A_c
A_j
A_p
A
Obs
JmaxRef = 88.39
RdRef = 0.51
sigma = 0.23
Theta = 0
VcmaxRef = 47
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

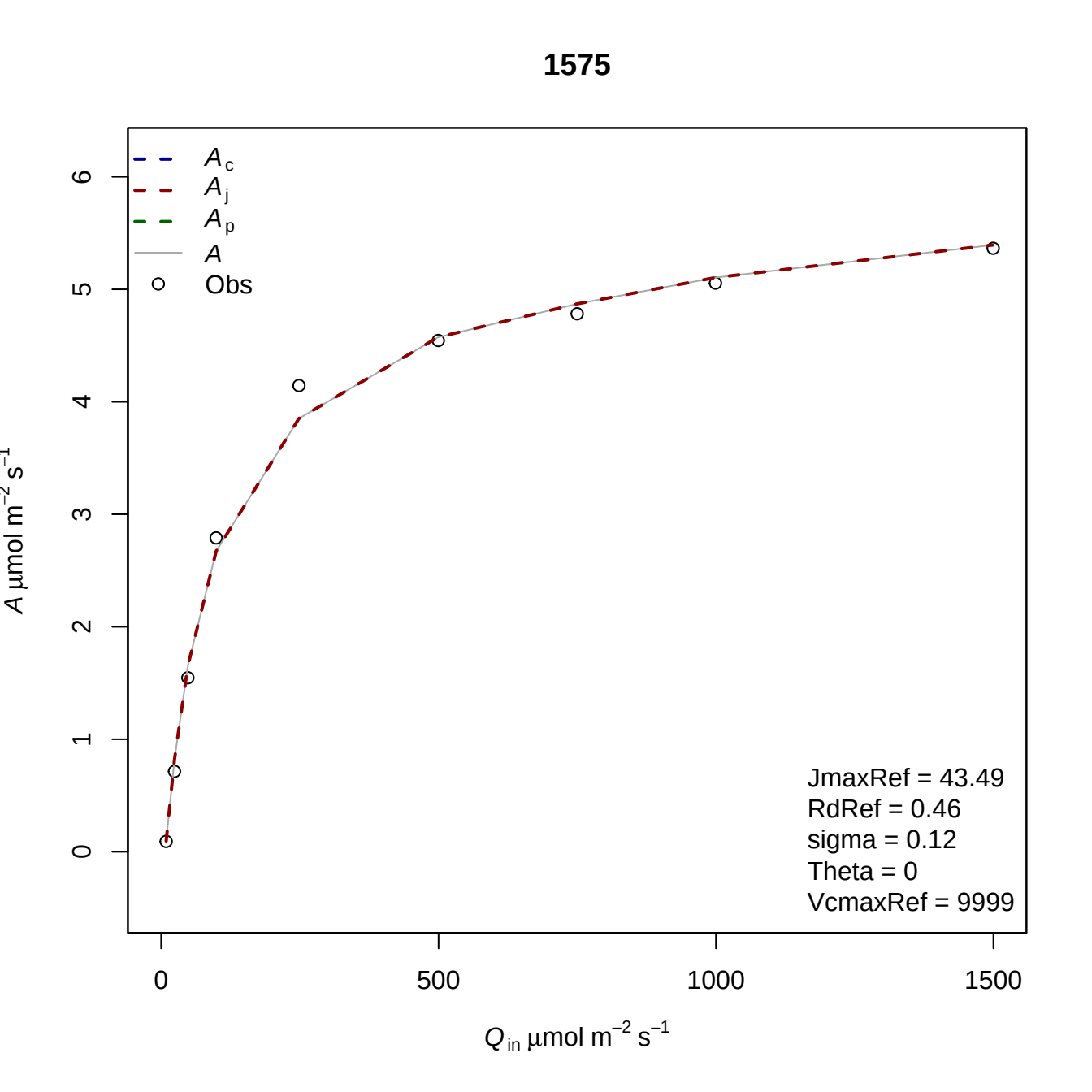

1570
A_c
A_j
A_p
A
Obs
JmaxRef = 105.06
RdRef = 0.33
sigma = 0.35
Theta = 0
VcmaxRef = 60.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

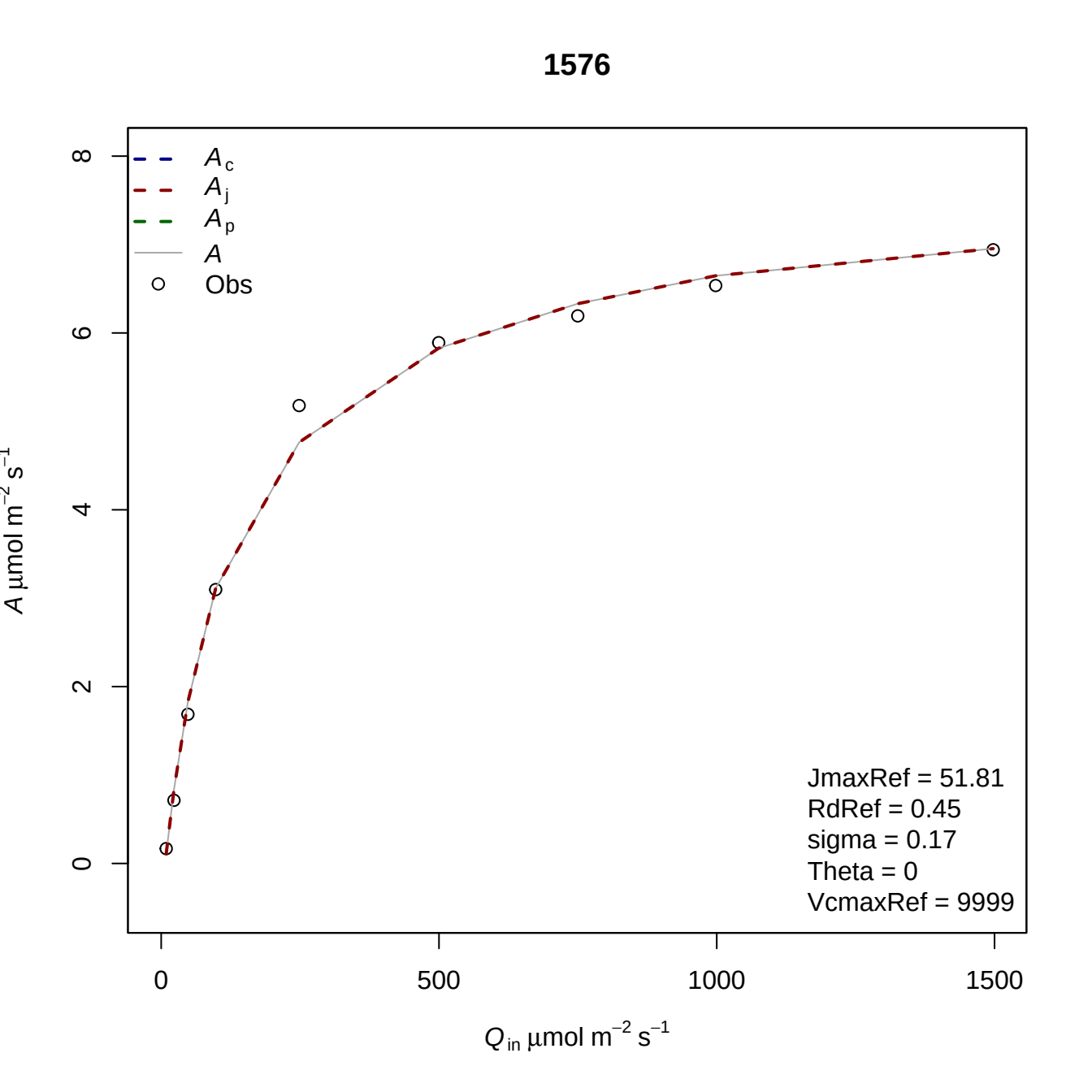

1571
A_c
A_j
A_p
A
Obs
JmaxRef = 186.16
RdRef = −0.12
sigma = 0.63
Theta = 0
VcmaxRef = 77.41
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

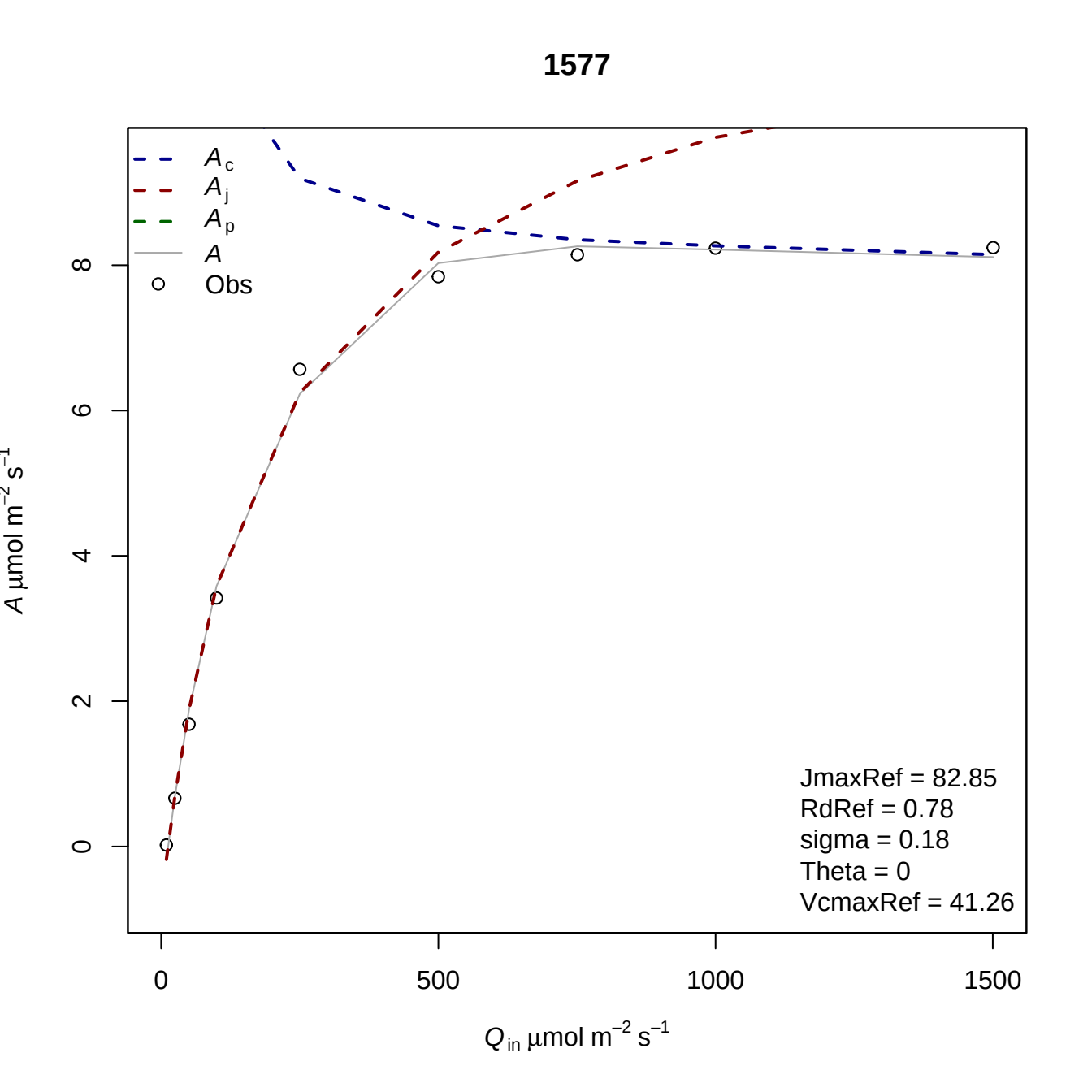

1572
A_c
A_j
A_p
A
Obs
JmaxRef = 122.47
RdRef = 0.36
sigma = 0.39
Theta = 0
VcmaxRef = 59.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

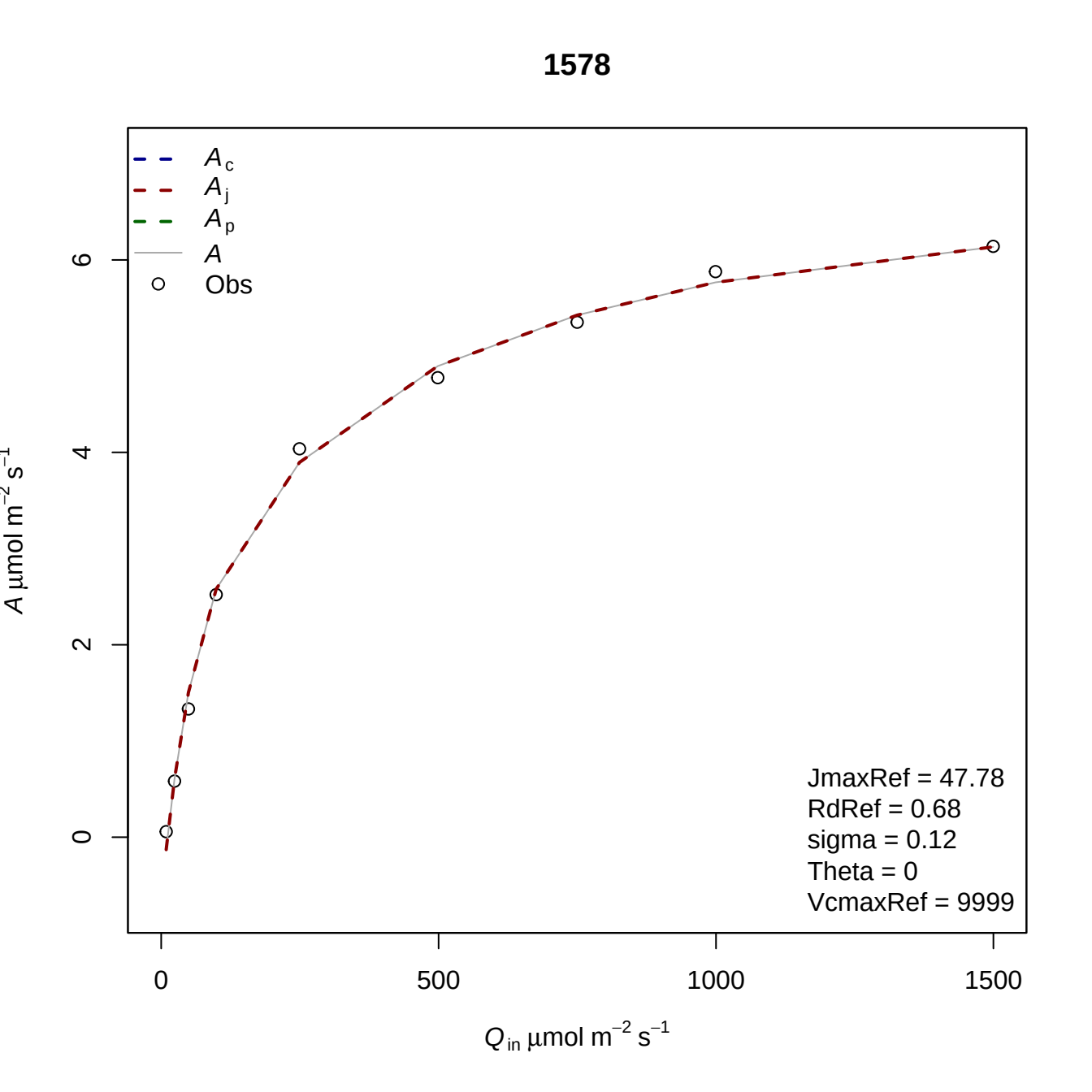

1573
A_c
A_j
A_p
A
Obs
JmaxRef = 89.36
RdRef = 0.6
sigma = 0.21
Theta = 0
VcmaxRef = 46.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

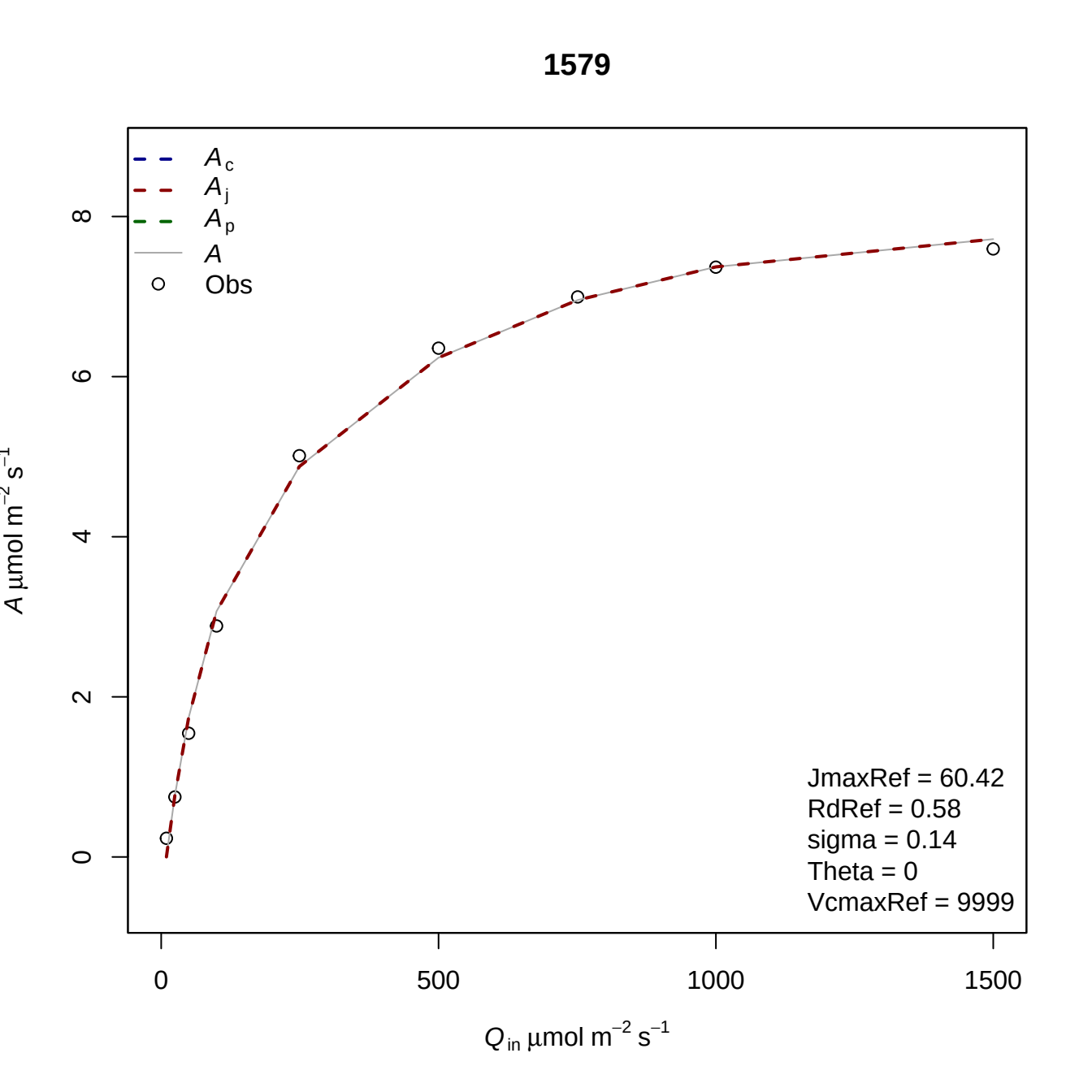

1574
A_c
A_j
A_p
A
Obs
JmaxRef = 85.68
RdRef = 0.48
sigma = 0.21
Theta = 0
VcmaxRef = 45.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

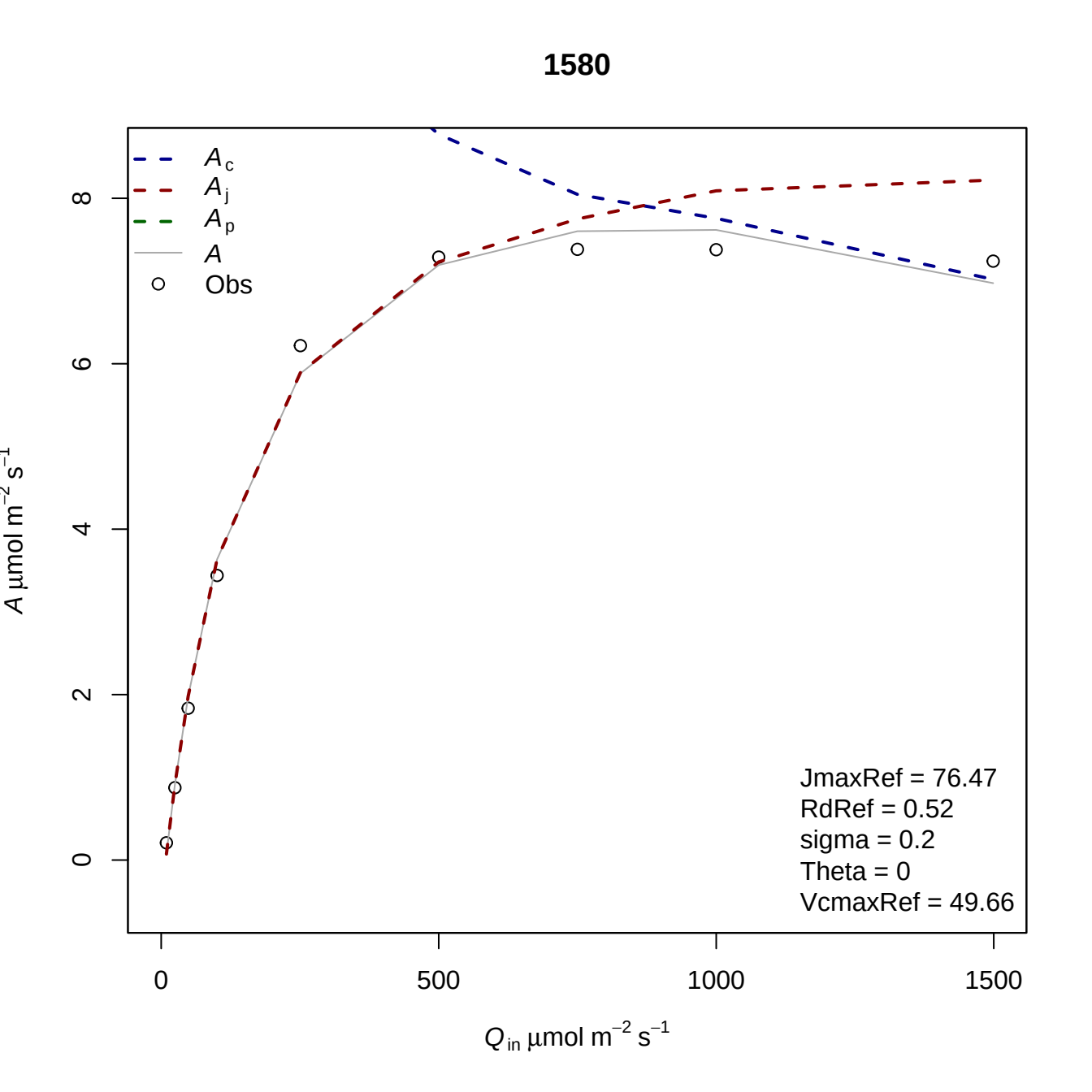

1575
A_c
A_j
A_p
A
Obs
JmaxRef = 43.49
RdRef = 0.46
sigma = 0.12
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

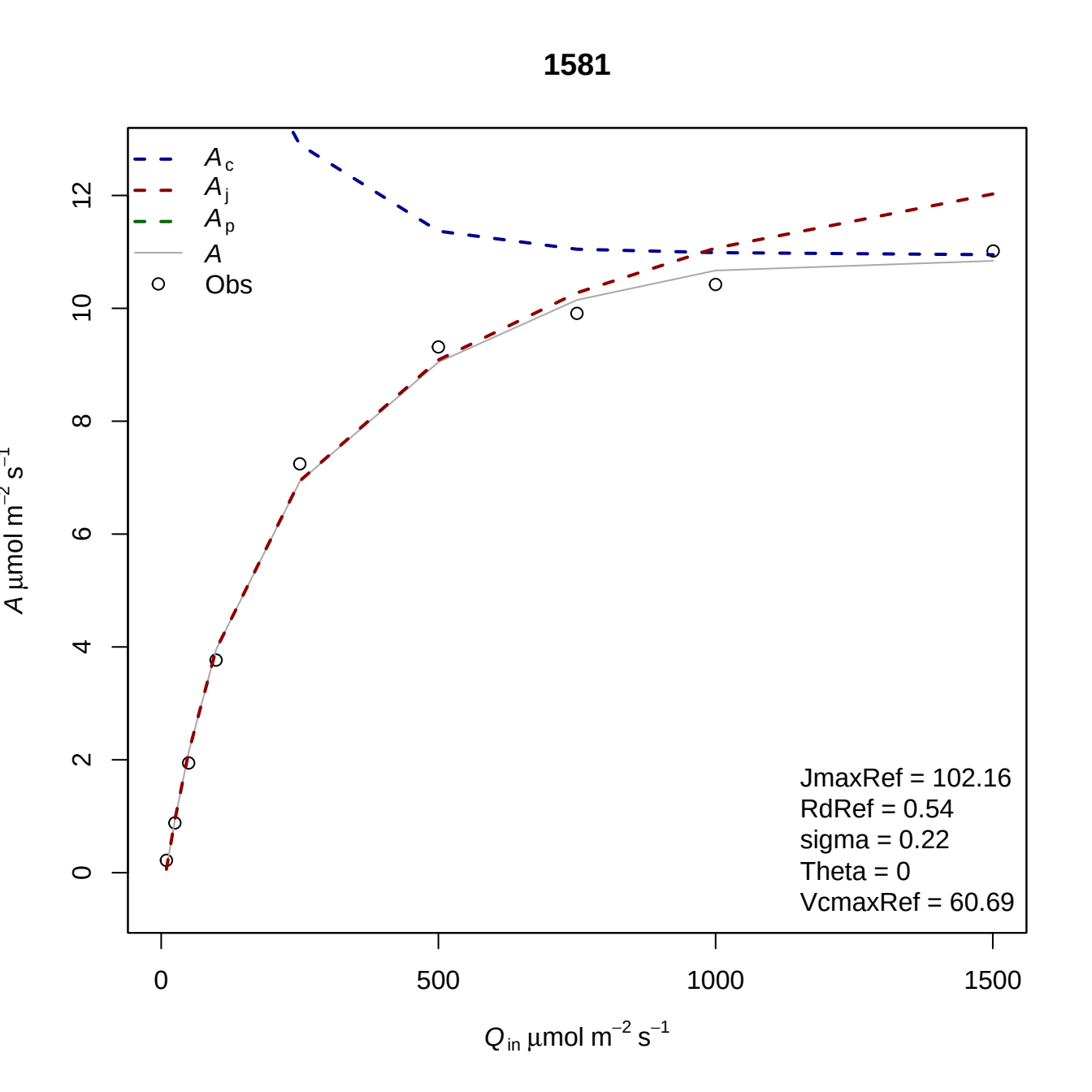

1576
A_c
A_j
A_p
A
Obs
JmaxRef = 51.81
RdRef = 0.45
sigma = 0.17
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

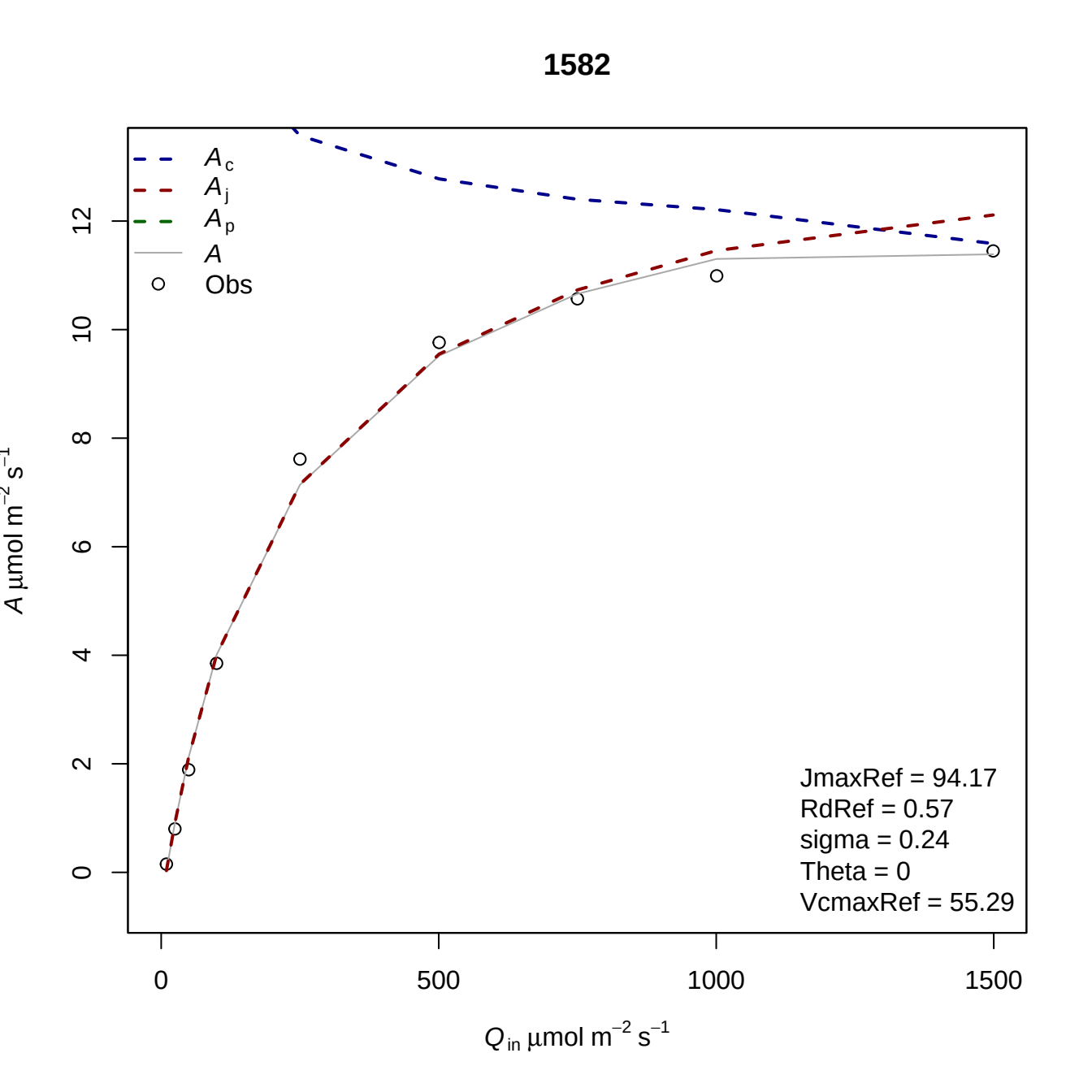

1577
A_c
A_j
A_p
A
Obs
JmaxRef = 82.85
RdRef = 0.78
sigma = 0.18
Theta = 0
VcmaxRef = 41.26
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

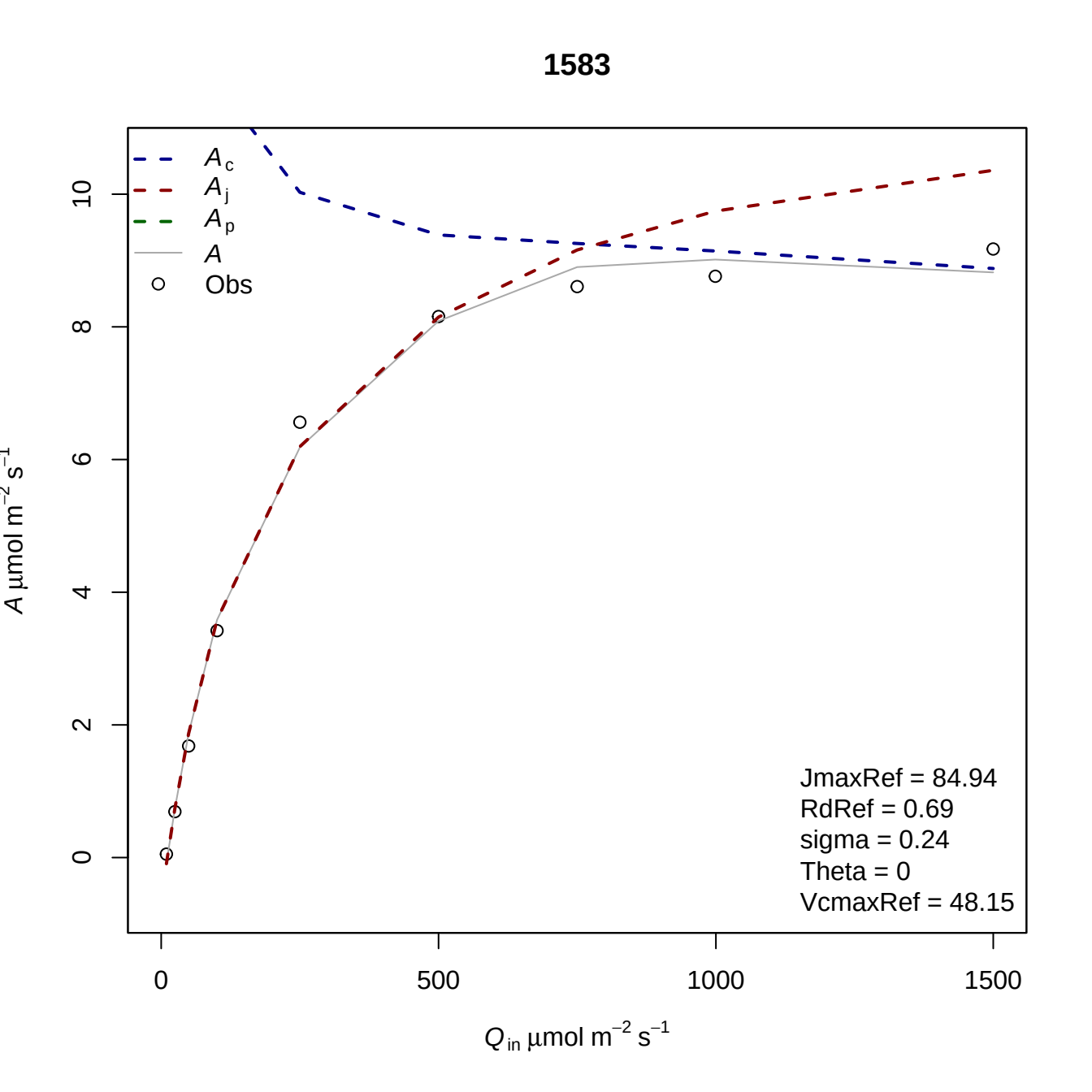

1578
A_c
A_j
A_p
A
Obs
JmaxRef = 47.78
RdRef = 0.68
sigma = 0.12
Theta = 0
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

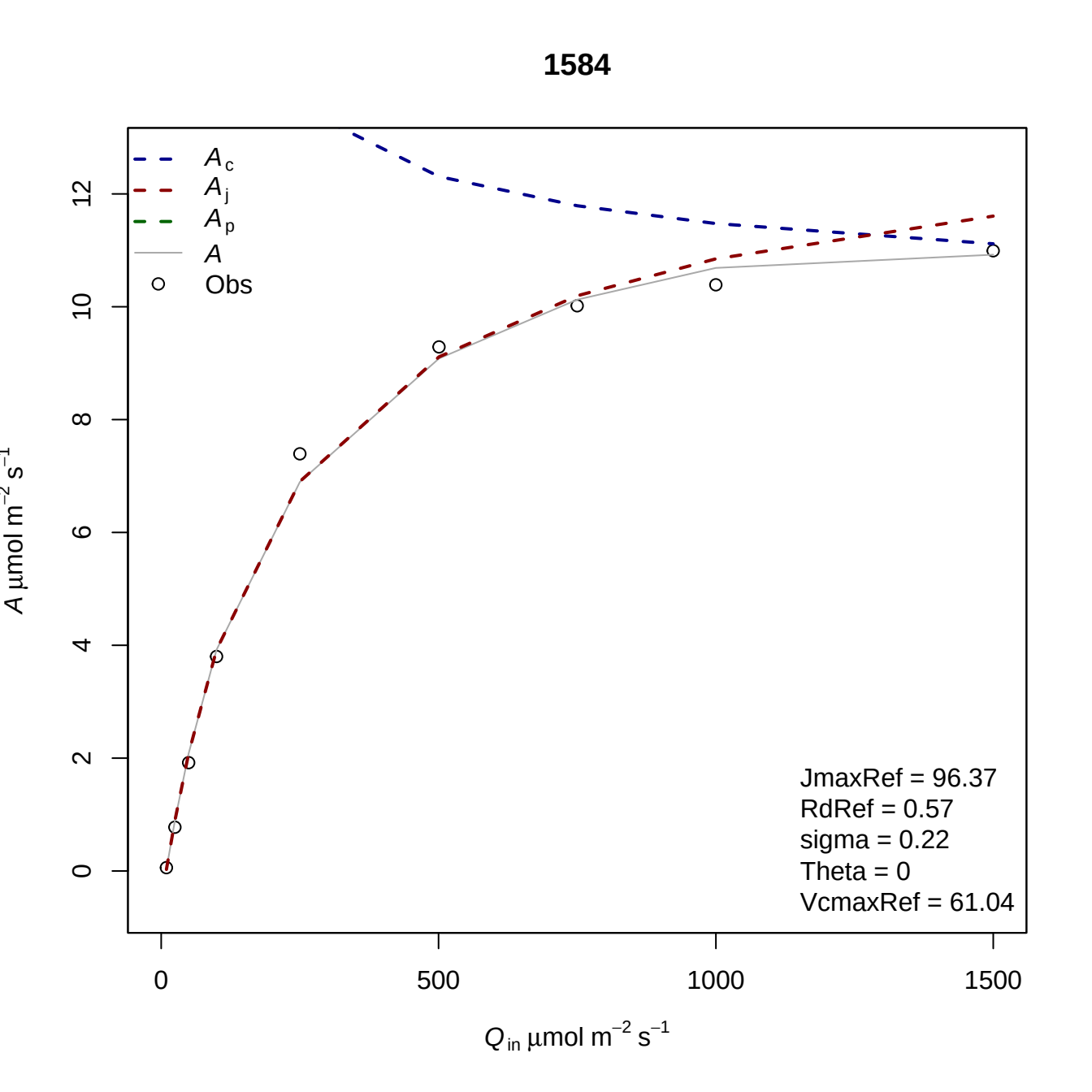

1579
A_c
A_j
A_p
A
Obs
JmaxRef = 60.42
RdRef = 0.58
sigma = 0.14
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

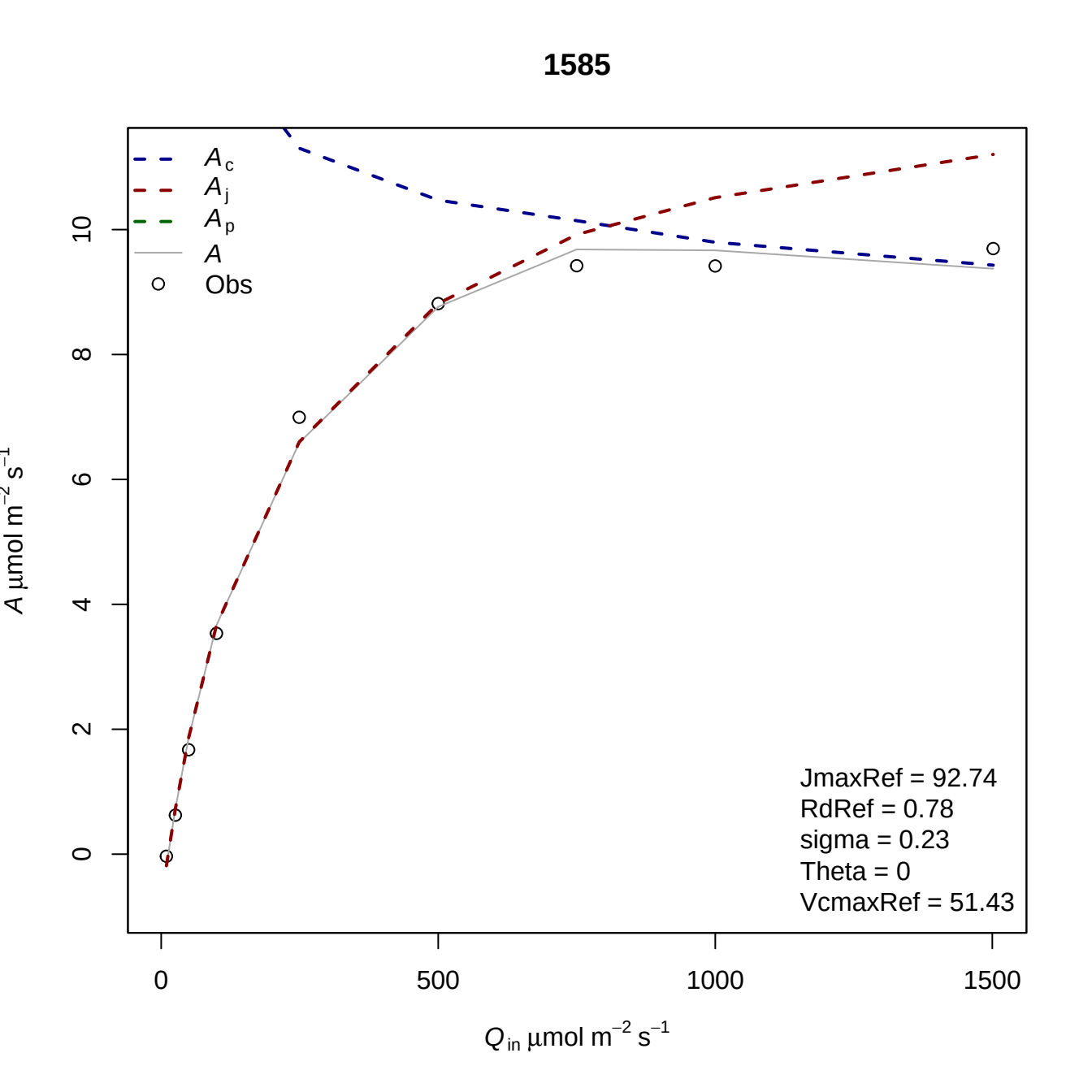

1580
A_c
A_j
A_p
A
Obs
JmaxRef = 76.47
RdRef = 0.52
sigma = 0.2
Theta = 0
VcmaxRef = 49.66
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

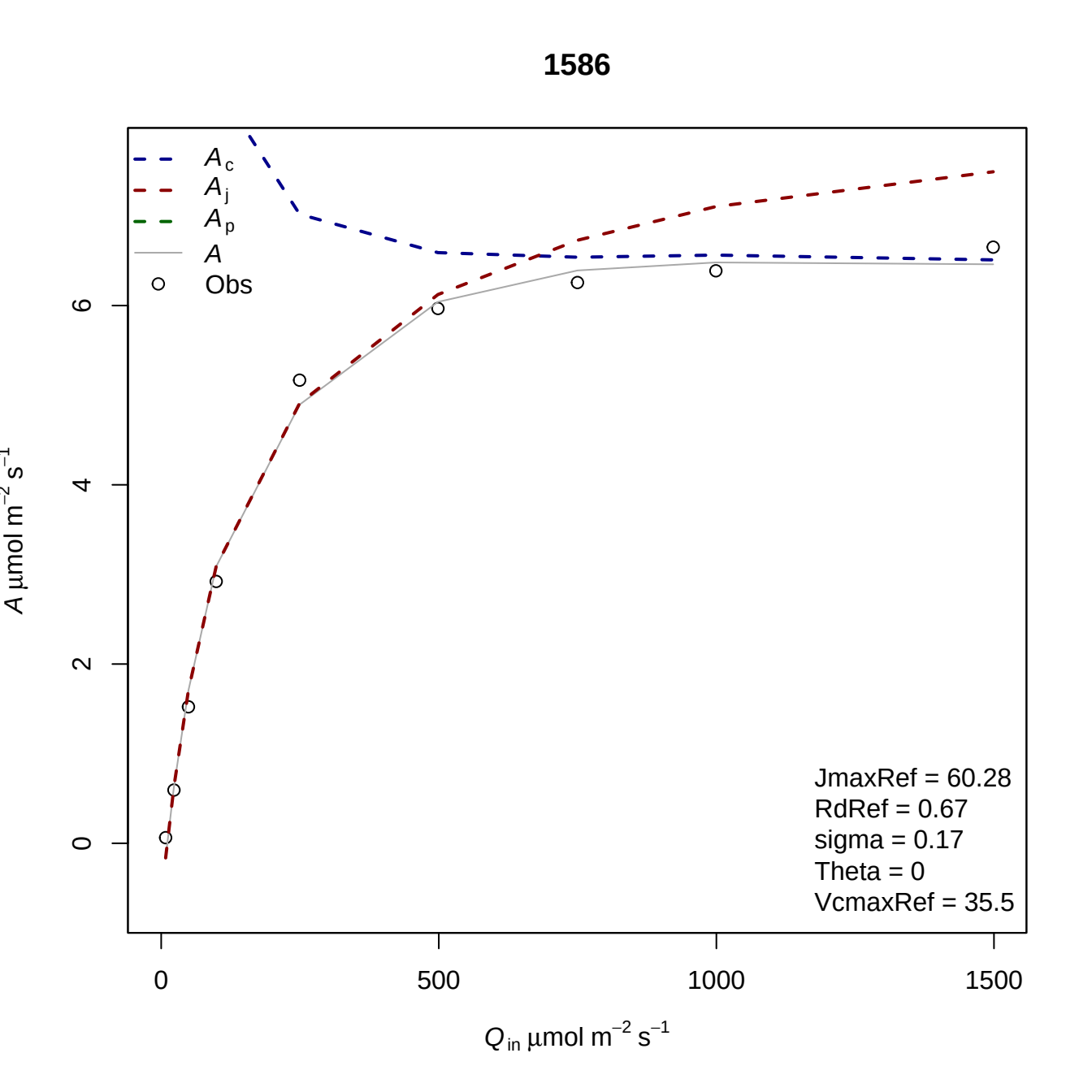

1581
A_c
A_j
A_p
A
Obs
JmaxRef = 102.16
RdRef = 0.54
sigma = 0.22
Theta = 0
VcmaxRef = 60.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

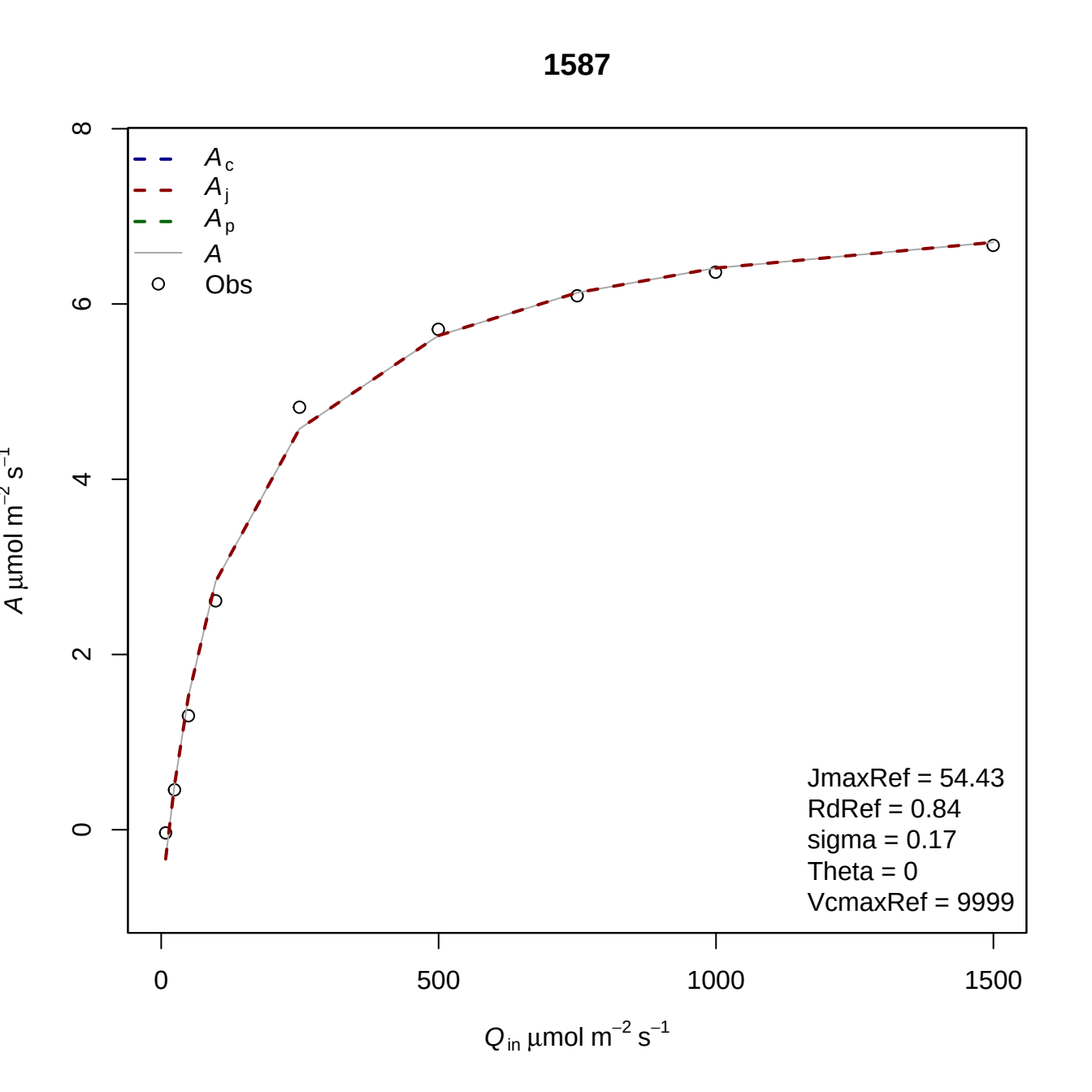

1582
A_c
A_j
A_p
A
Obs
JmaxRef = 94.17
RdRef = 0.57
sigma = 0.24
Theta = 0
VcmaxRef = 55.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

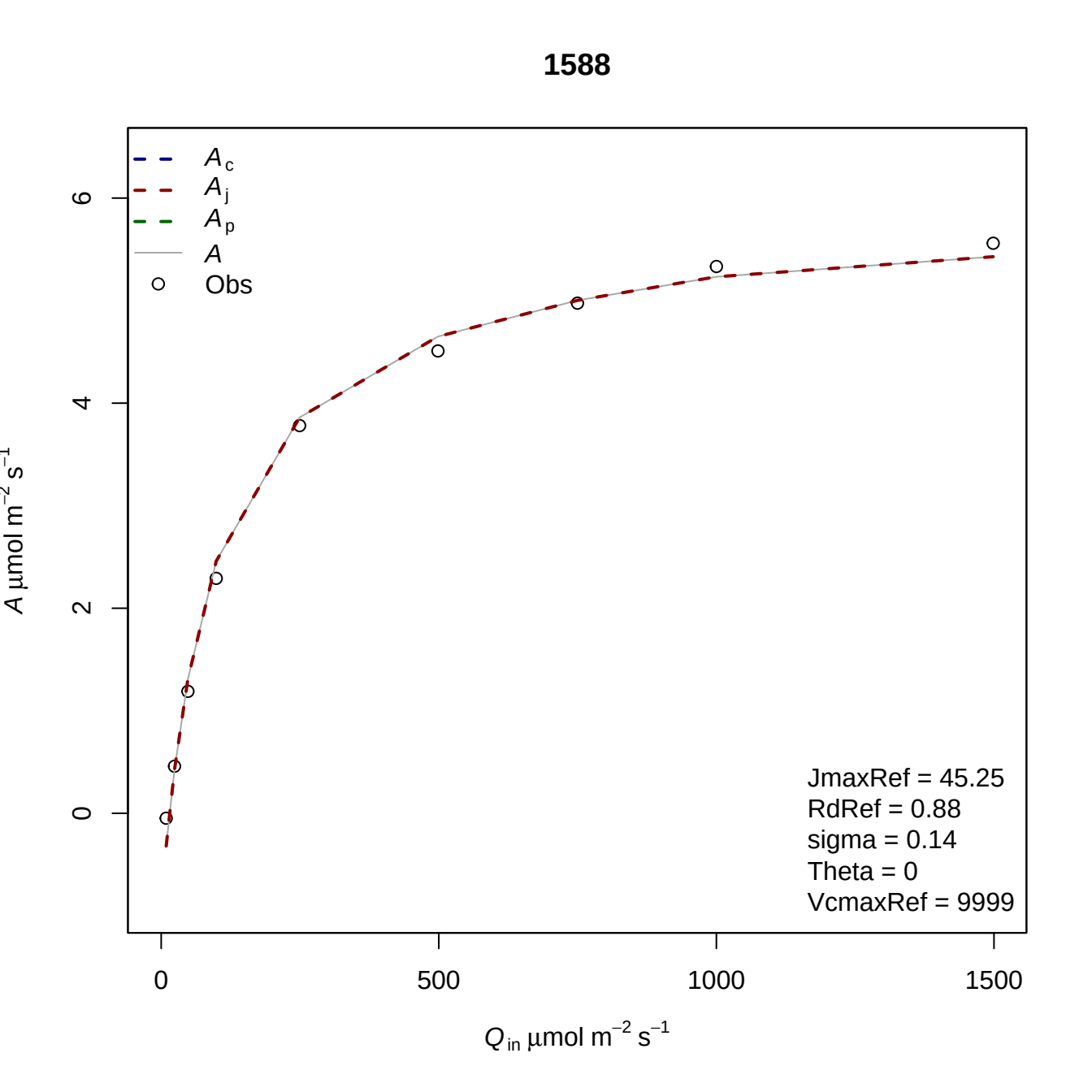

1583
A_c
A_j
A_p
A
Obs
JmaxRef = 84.94
RdRef = 0.69
sigma = 0.24
Theta = 0
VcmaxRef = 48.15
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

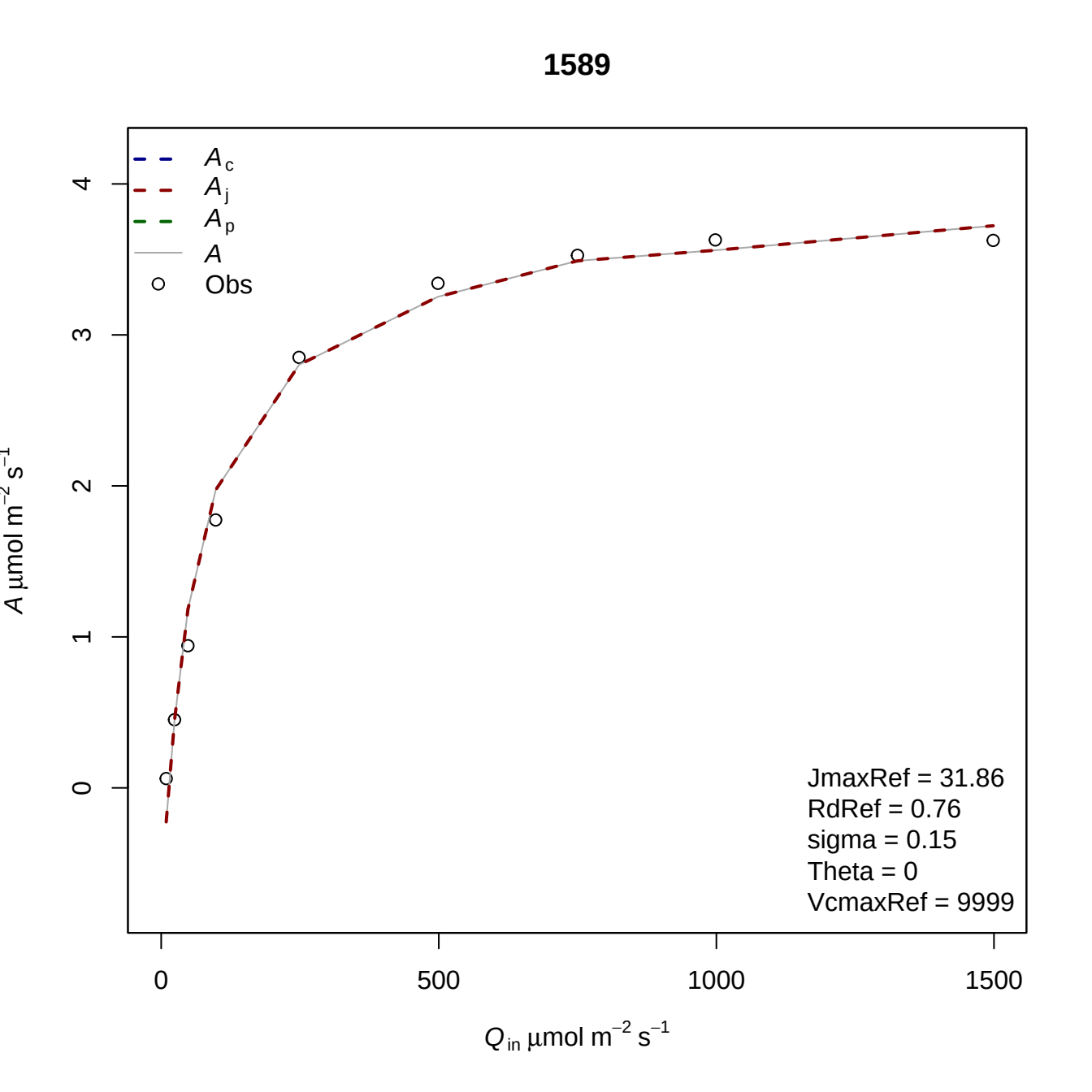

1584
A_c
A_j
A_p
A
Obs
JmaxRef = 96.37
RdRef = 0.57
sigma = 0.22
Theta = 0
VcmaxRef = 61.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

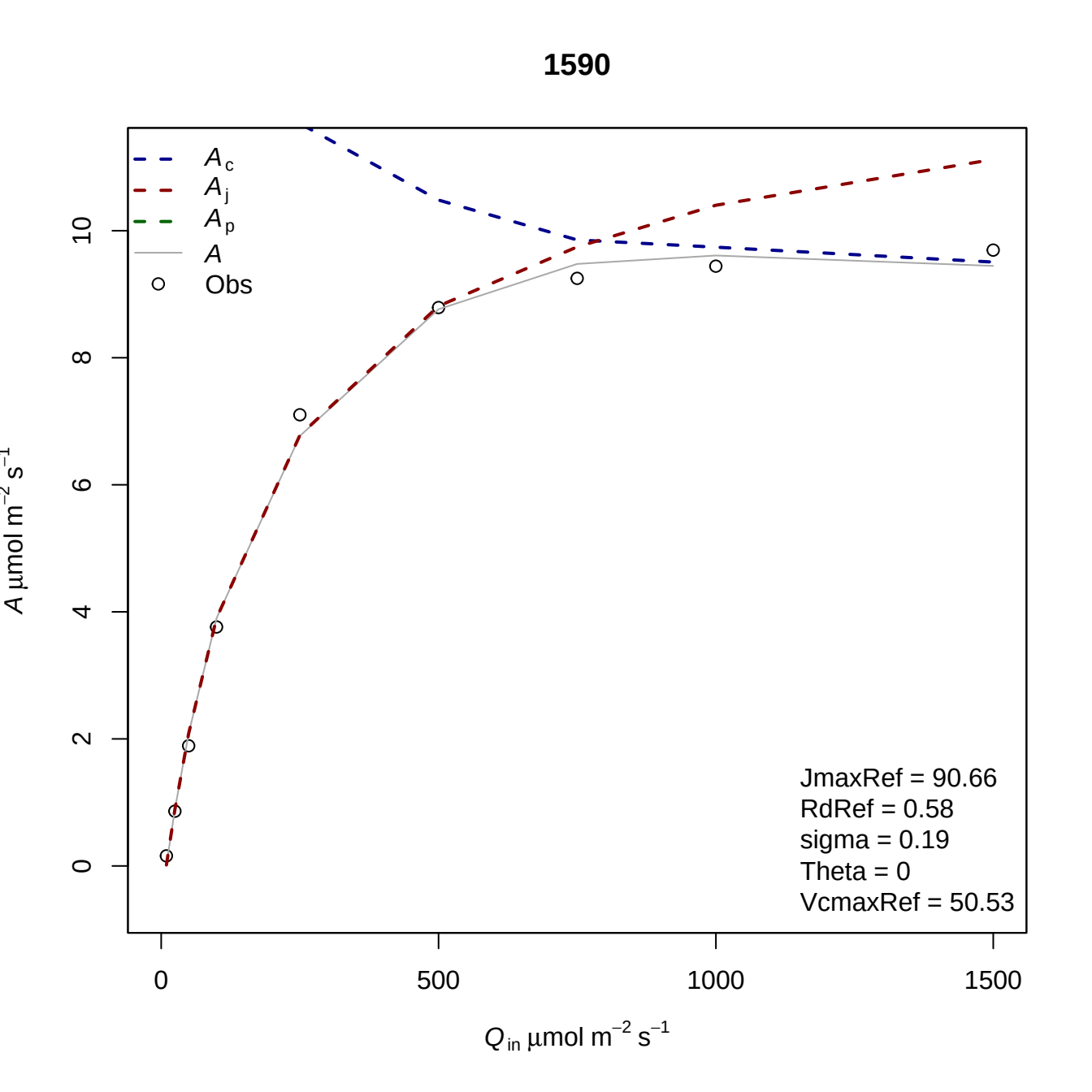

1585
A_c
A_j
A_p
A
Obs
JmaxRef = 92.74
RdRef = 0.78
sigma = 0.23
Theta = 0
VcmaxRef = 51.43
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

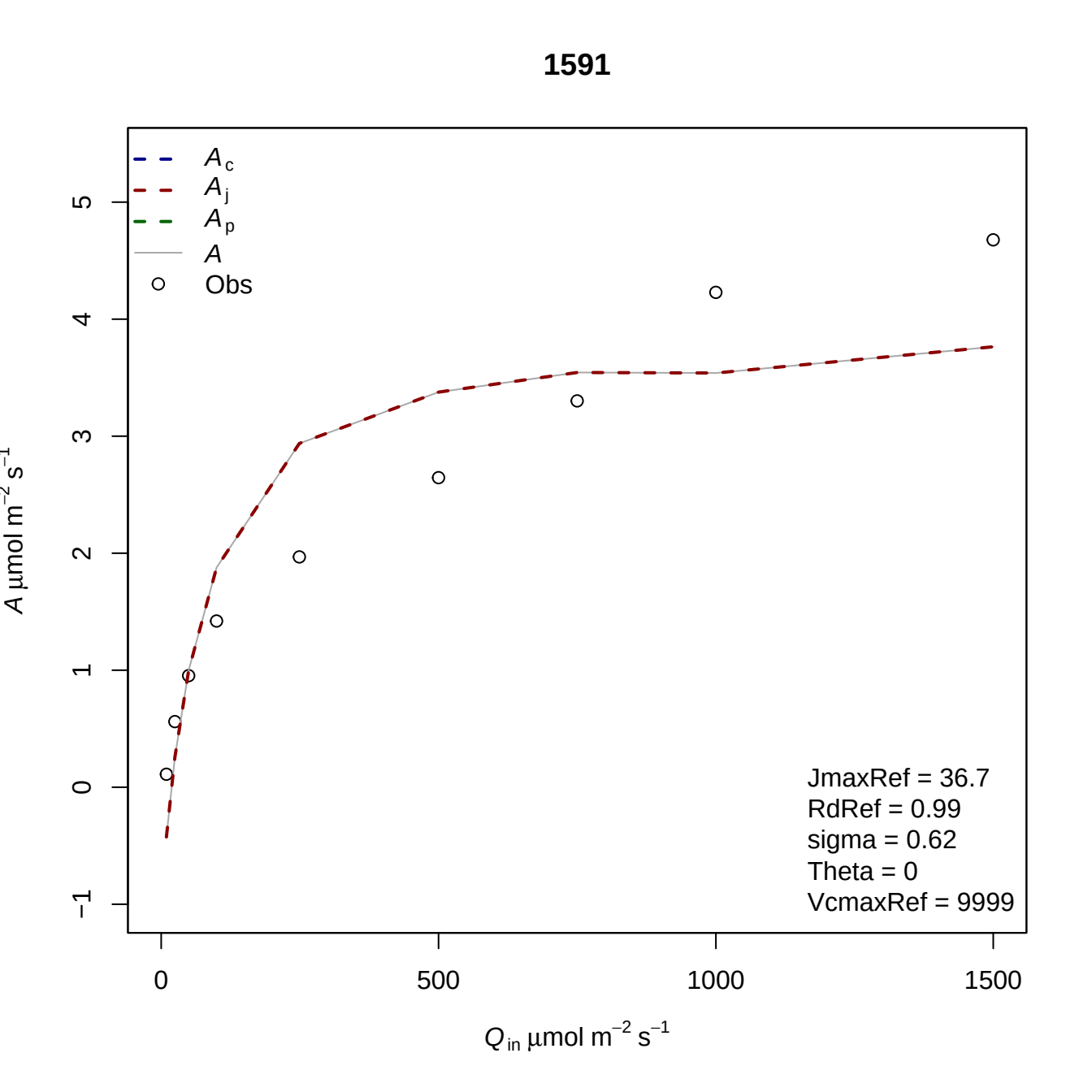

1586
A_c
A_j
A_p
A
Obs
JmaxRef = 60.28
RdRef = 0.67
sigma = 0.17
Theta = 0
VcmaxRef = 35.5
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

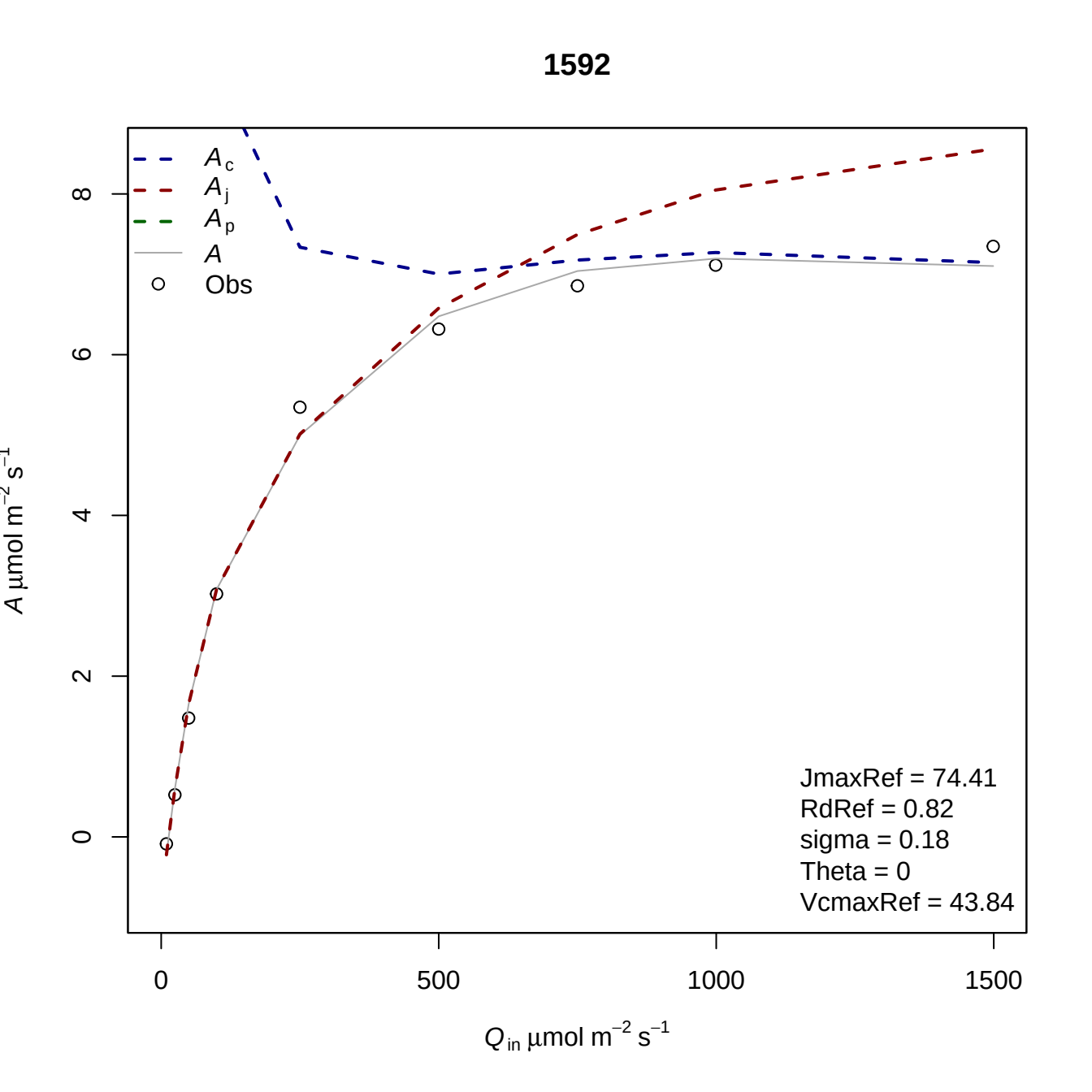

1587
A_c
A_j
A_p
A
Obs
JmaxRef = 54.43
RdRef = 0.84
sigma = 0.17
Theta = 0
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

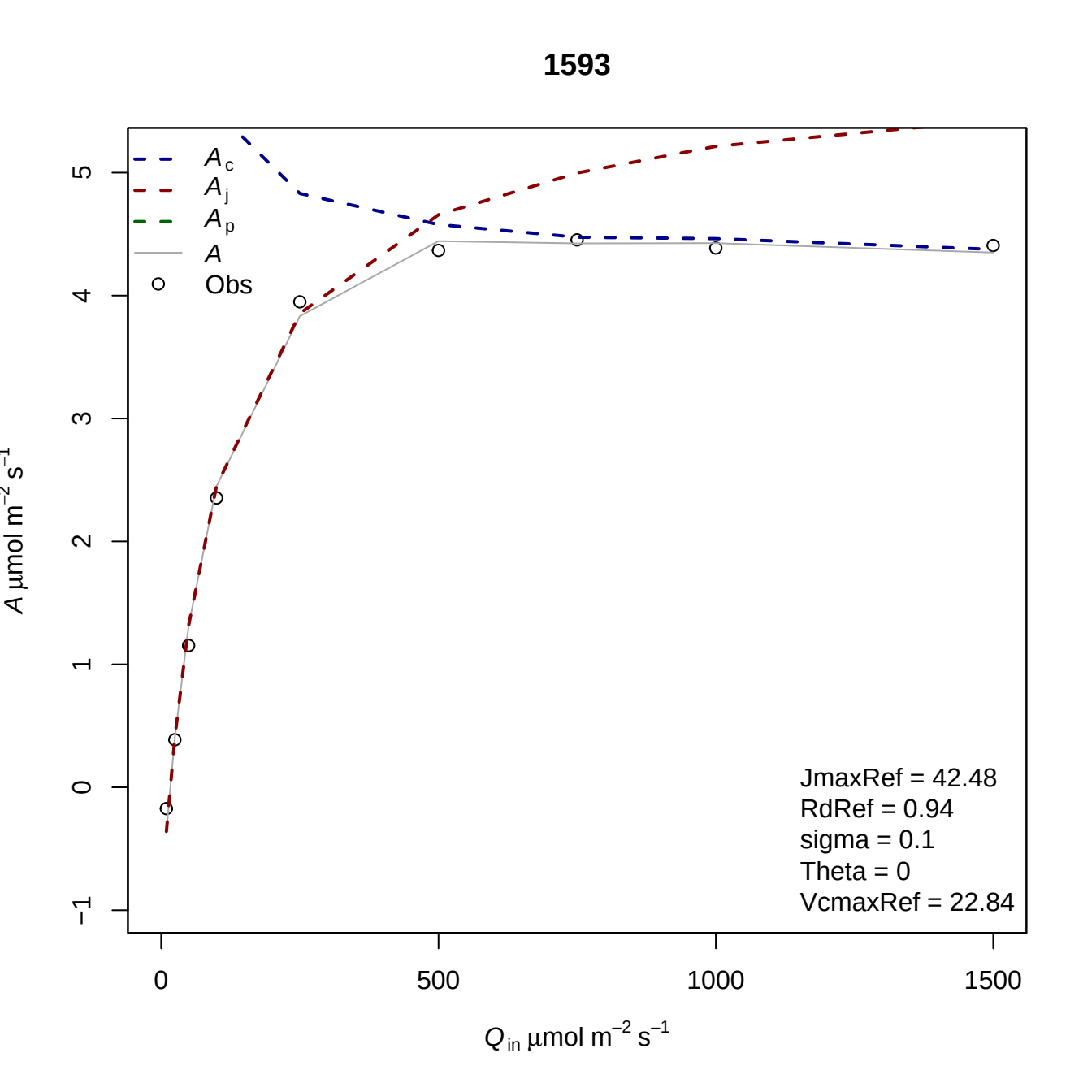

1588
A_c
A_j
A_p
A
Obs
JmaxRef = 45.25
RdRef = 0.88
sigma = 0.14
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

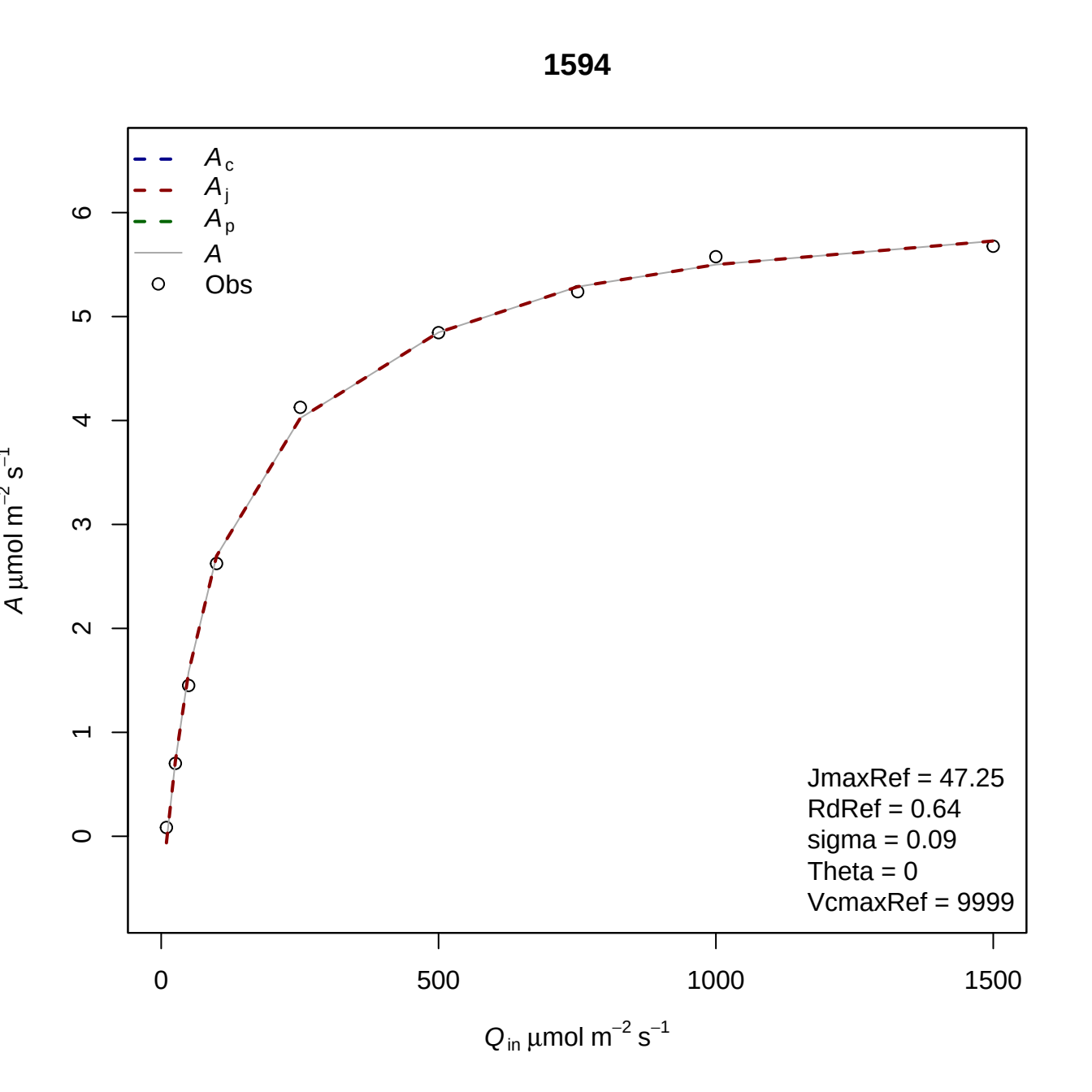

1589
A_c
A_j
A_p
A
Obs
JmaxRef = 31.86
RdRef = 0.76
sigma = 0.15
Theta = 0
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

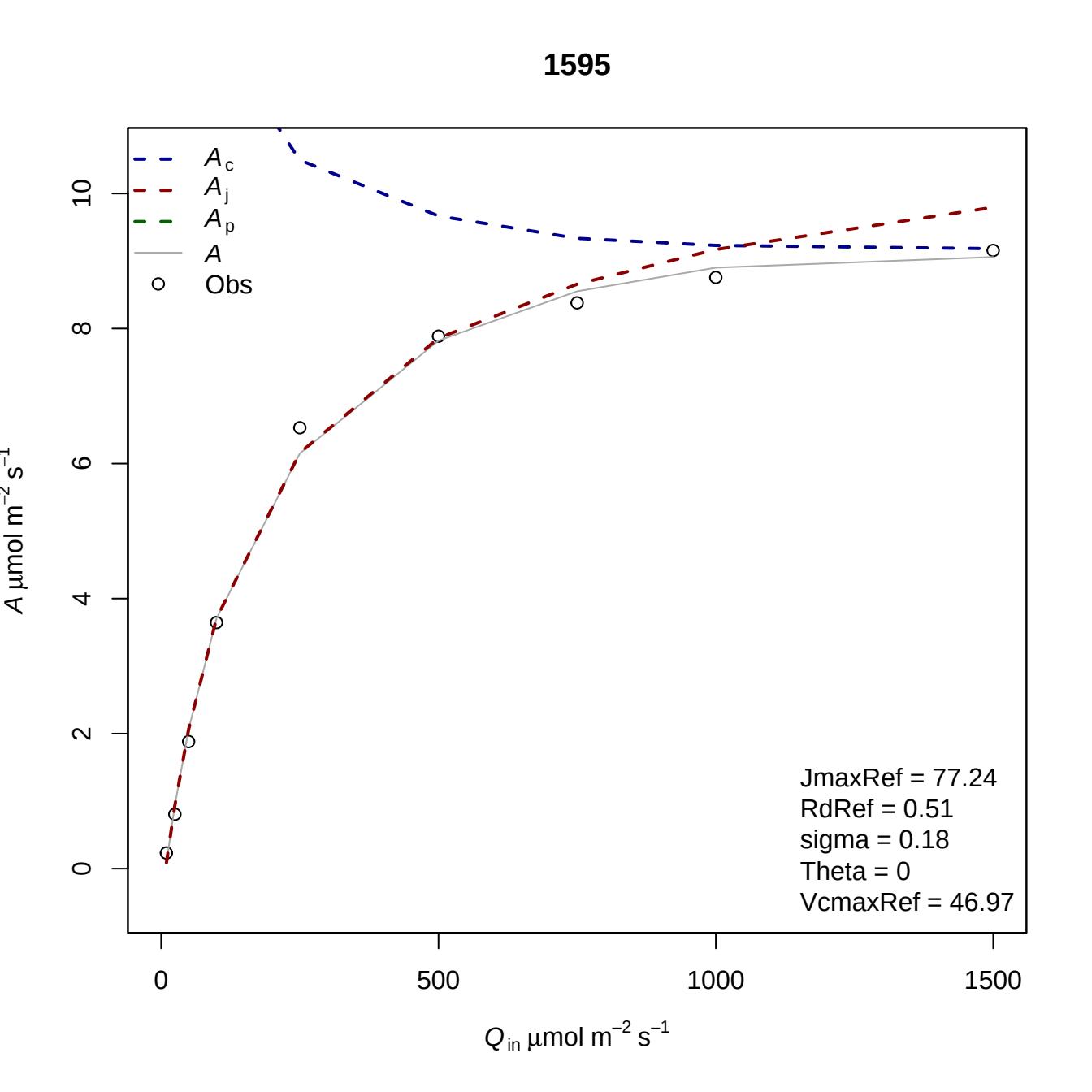

1590
A_c
A_j
A_p
A
Obs
JmaxRef = 90.66
RdRef = 0.58
sigma = 0.19
Theta = 0
VcmaxRef = 50.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

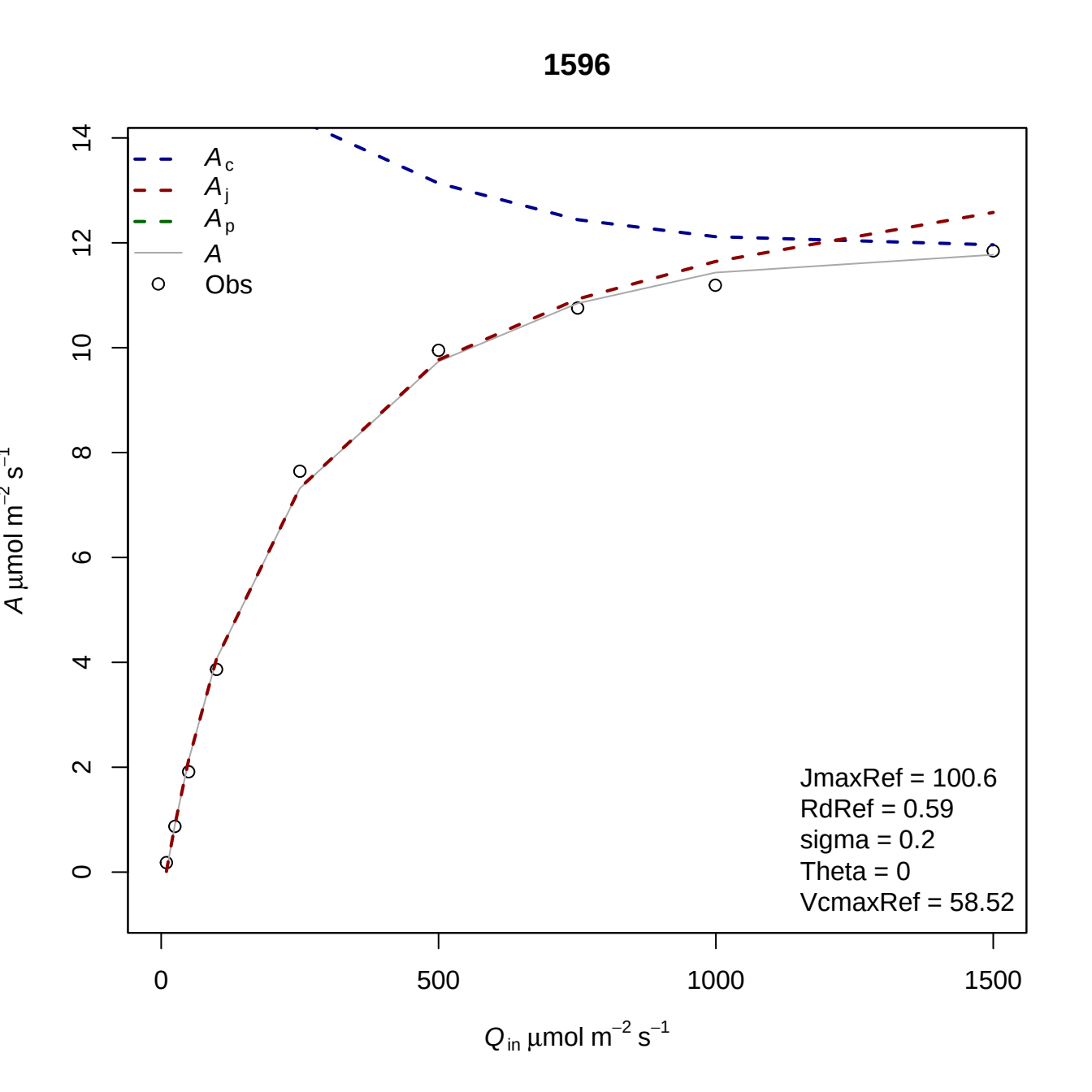

1591
A_c
A_j
A_p
A
Obs
JmaxRef = 36.7
RdRef = 0.99
sigma = 0.62
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

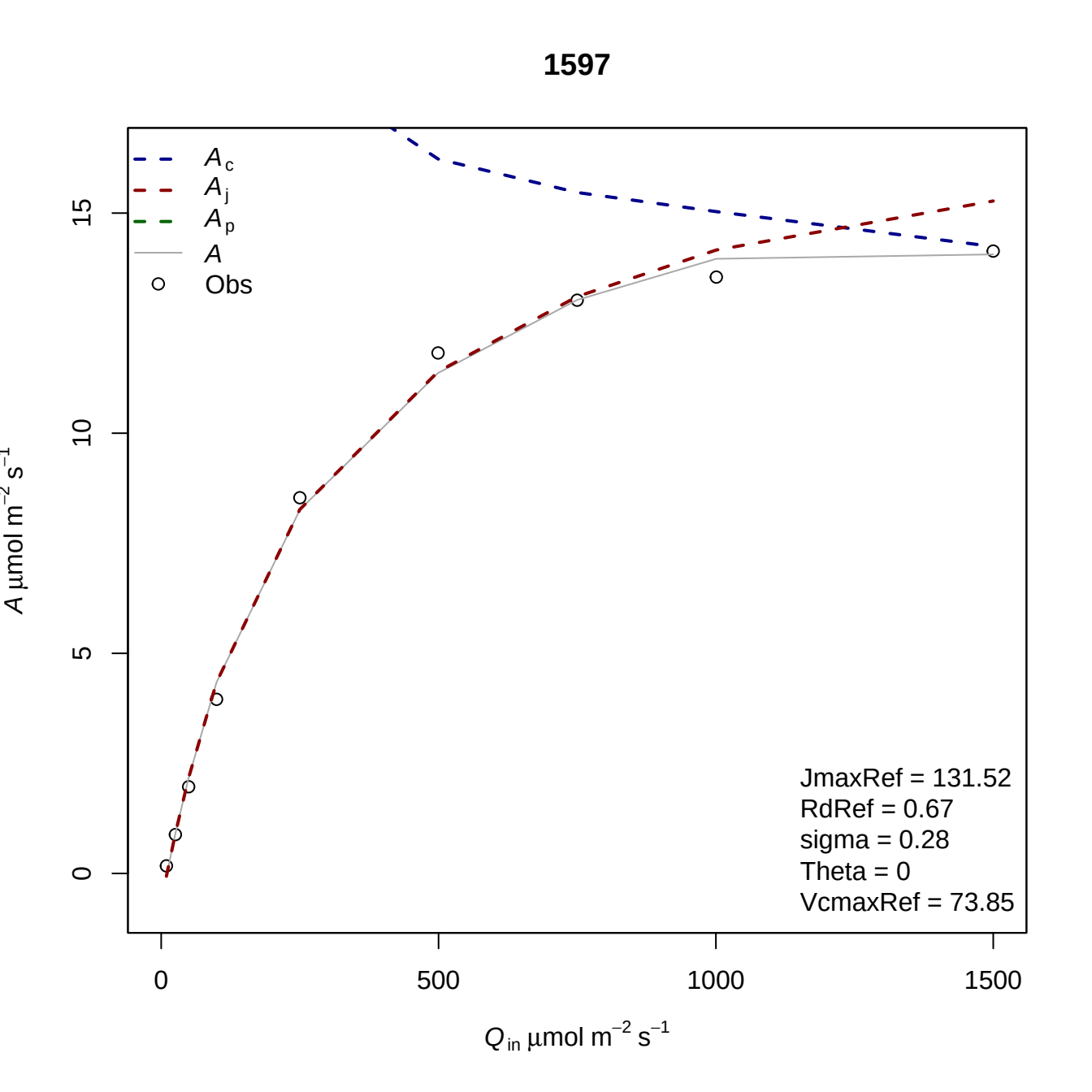

1592
A_c
A_j
A_p
A
Obs
JmaxRef = 74.41
RdRef = 0.82
sigma = 0.18
Theta = 0
VcmaxRef = 43.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

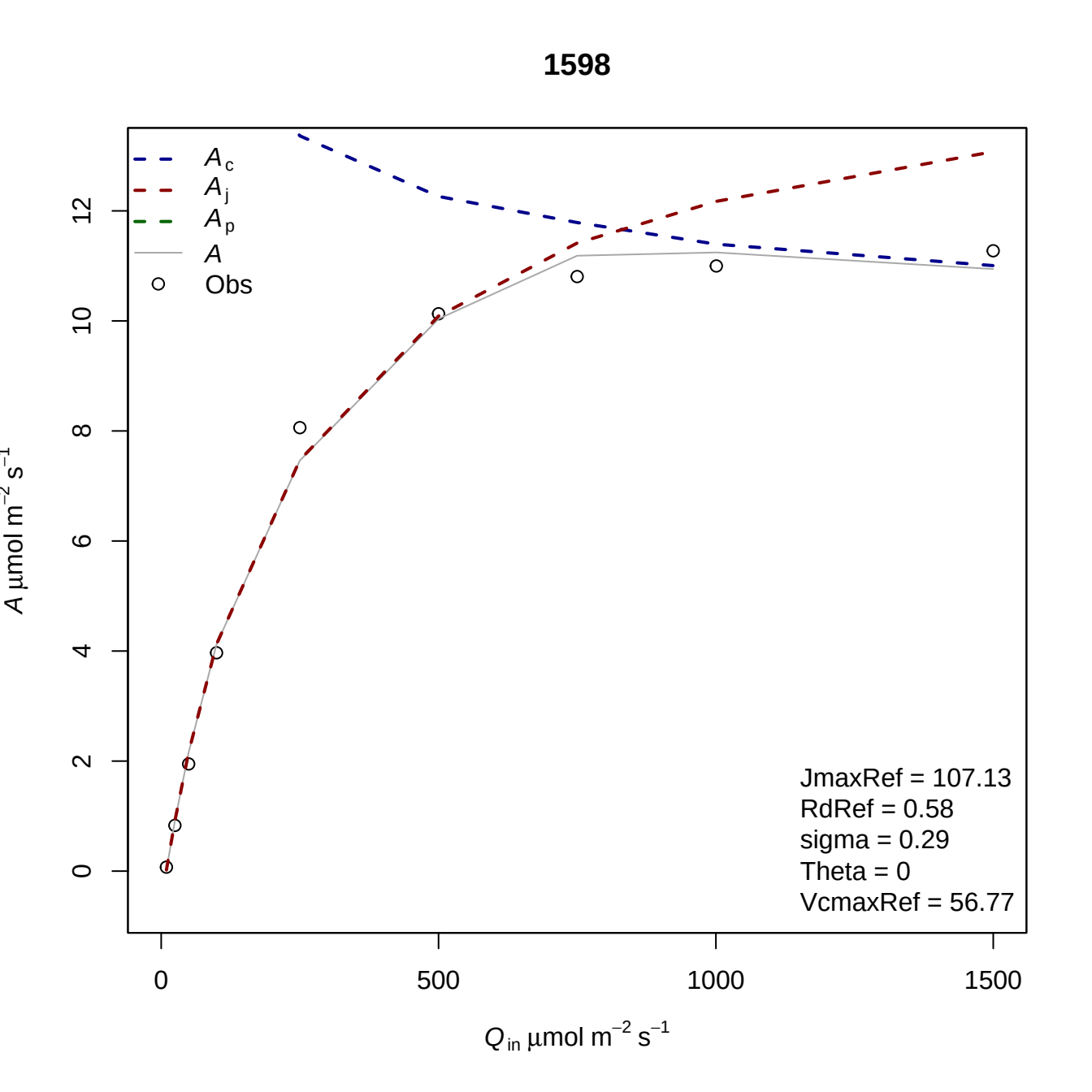

1593
A_c
A_j
A_p
A
Obs
JmaxRef = 42.48
RdRef = 0.94
sigma = 0.1
Theta = 0
VcmaxRef = 22.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

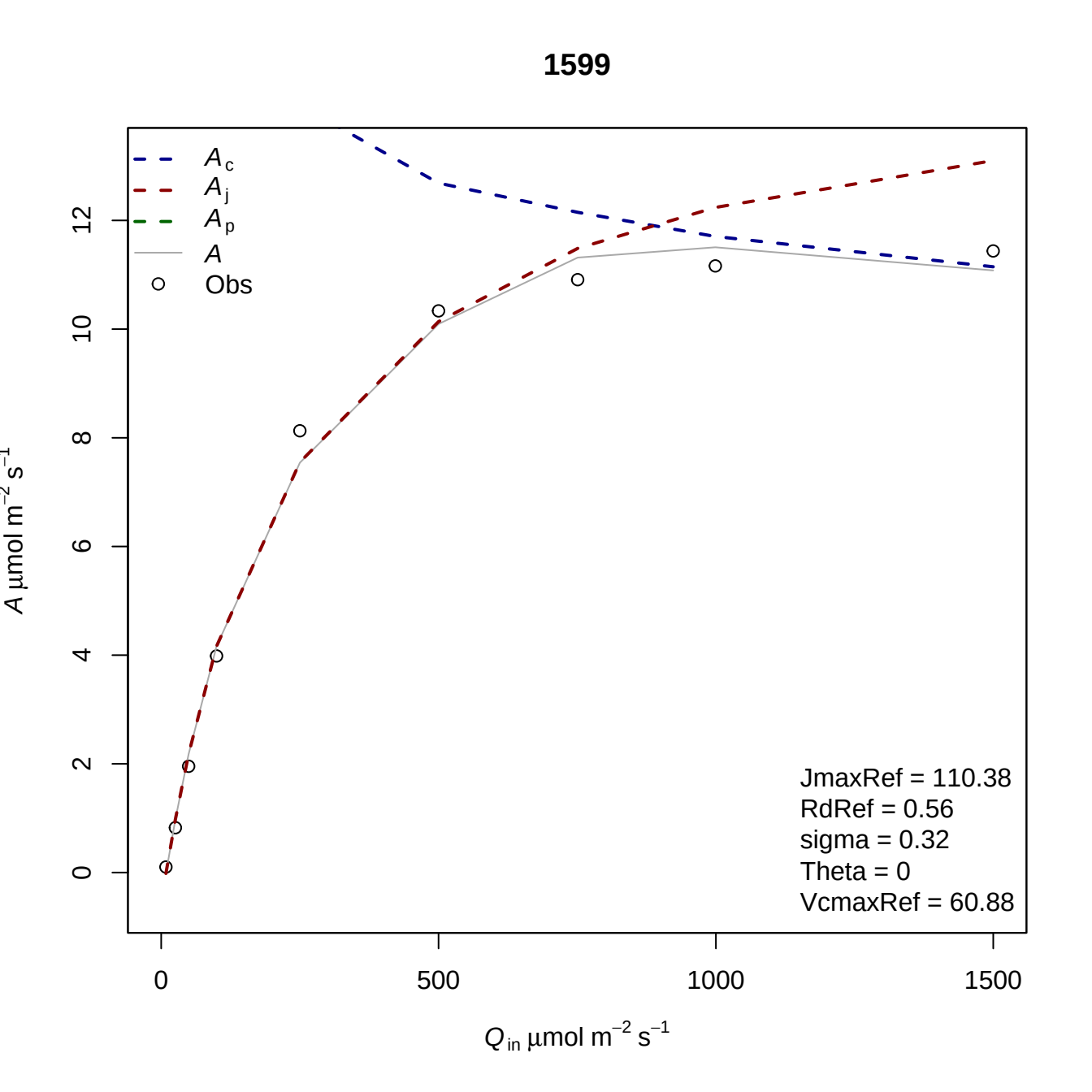

1594
A_c
A_j
A_p
A
Obs
JmaxRef = 47.25
RdRef = 0.64
sigma = 0.09
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

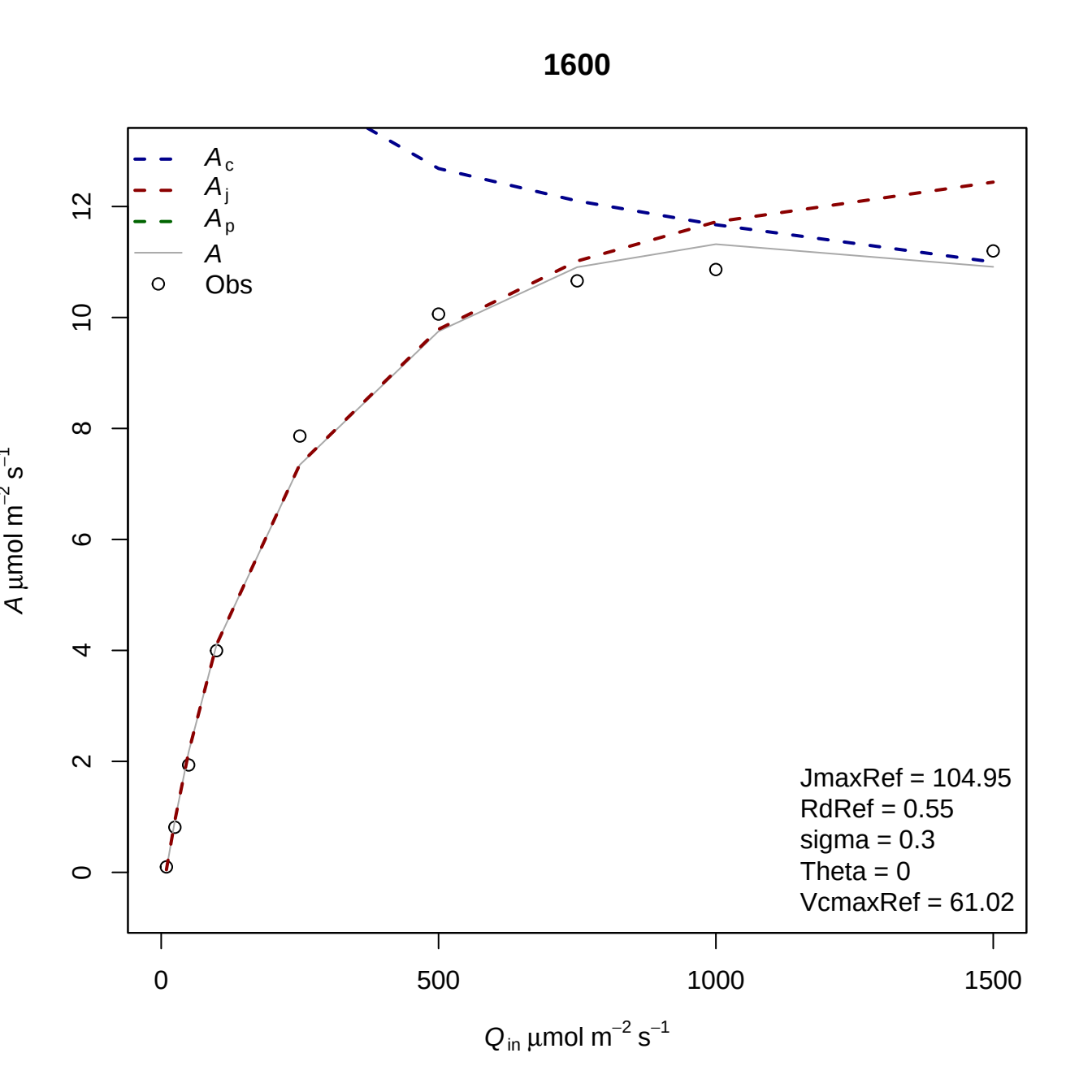

1595
A_c
A_j
A_p
A
Obs
JmaxRef = 77.24
RdRef = 0.51
sigma = 0.18
Theta = 0
VcmaxRef = 46.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

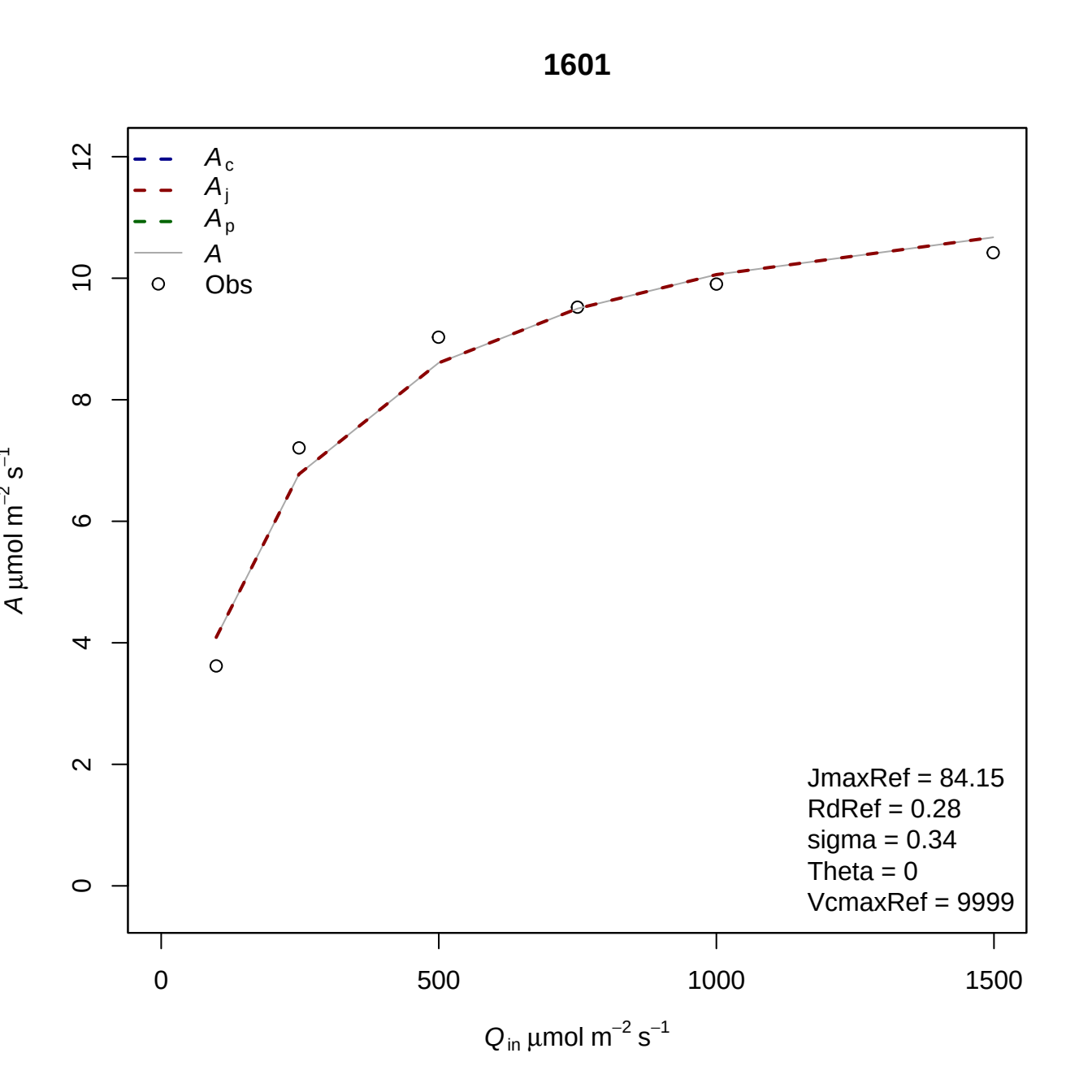

1596
A_c
A_j
A_p
A
Obs
JmaxRef = 100.6
RdRef = 0.59
sigma = 0.2
Theta = 0
VcmaxRef = 58.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

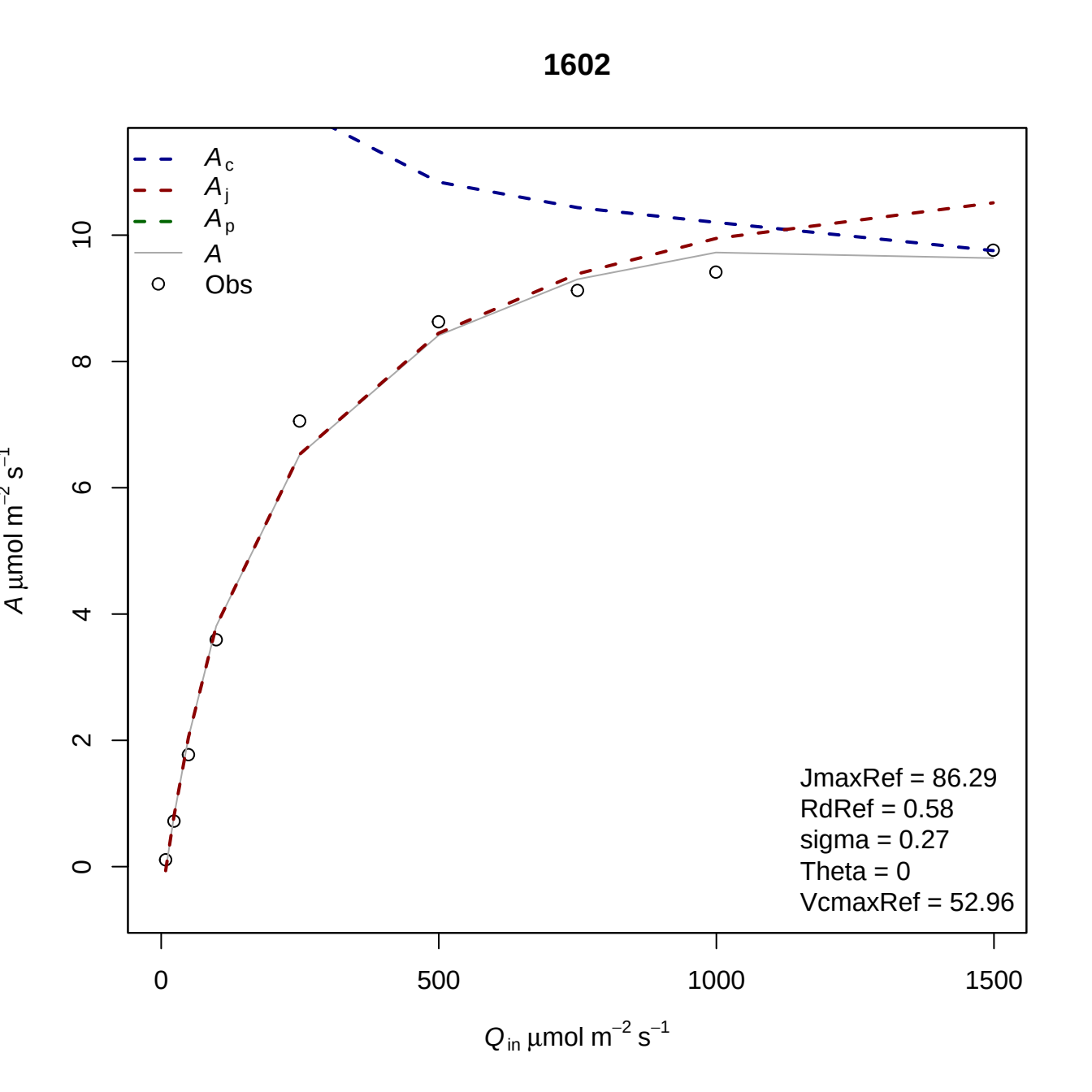

1597
A_c
A_j
A_p
A
Obs
JmaxRef = 131.52
RdRef = 0.67
sigma = 0.28
Theta = 0
VcmaxRef = 73.85
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

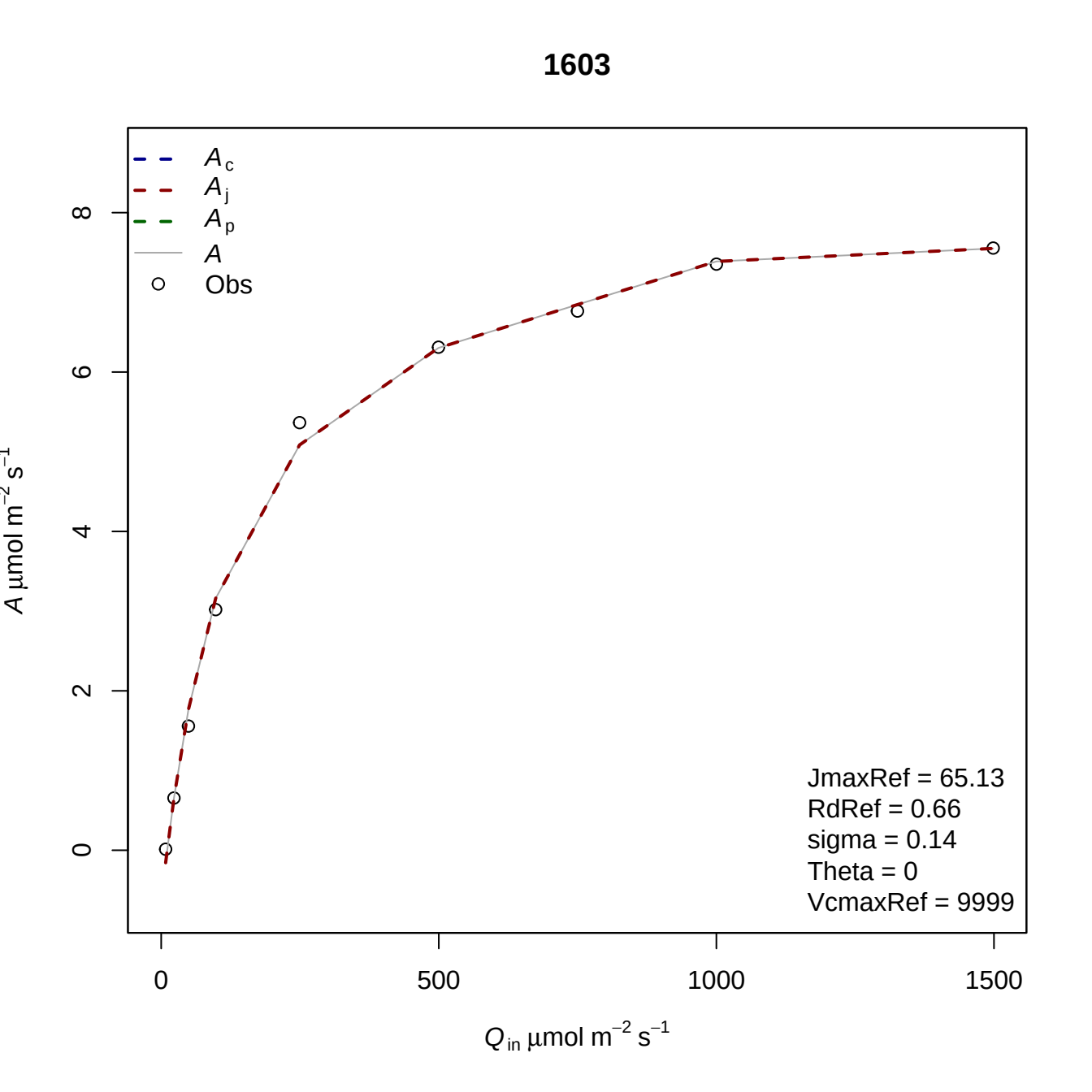

1598
A_c
A_j
A_p
A
Obs
JmaxRef = 107.13
RdRef = 0.58
sigma = 0.29
Theta = 0
VcmaxRef = 56.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

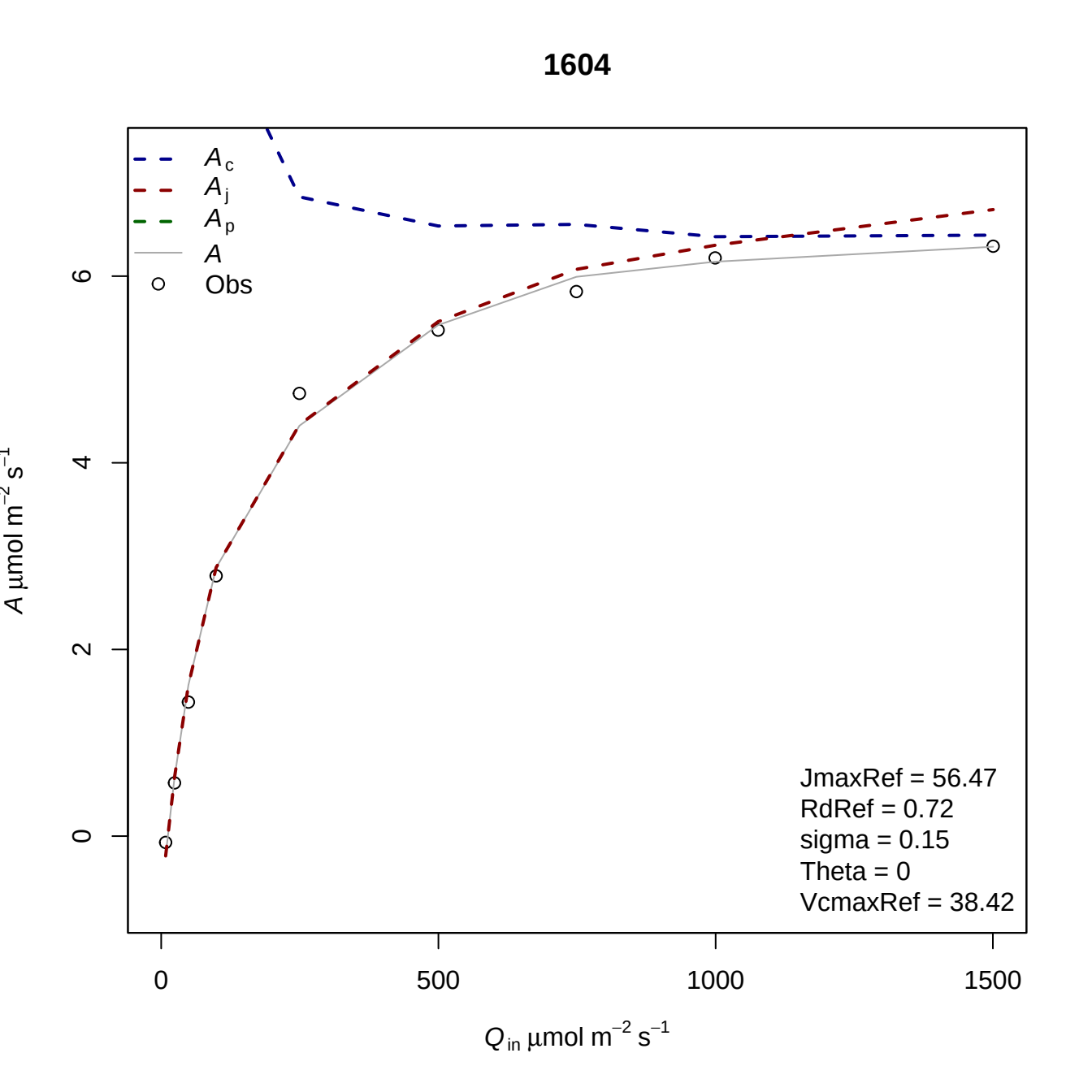

1599
A_c
A_j
A_p
A
Obs
JmaxRef = 110.38
RdRef = 0.56
sigma = 0.32
Theta = 0
VcmaxRef = 60.88
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

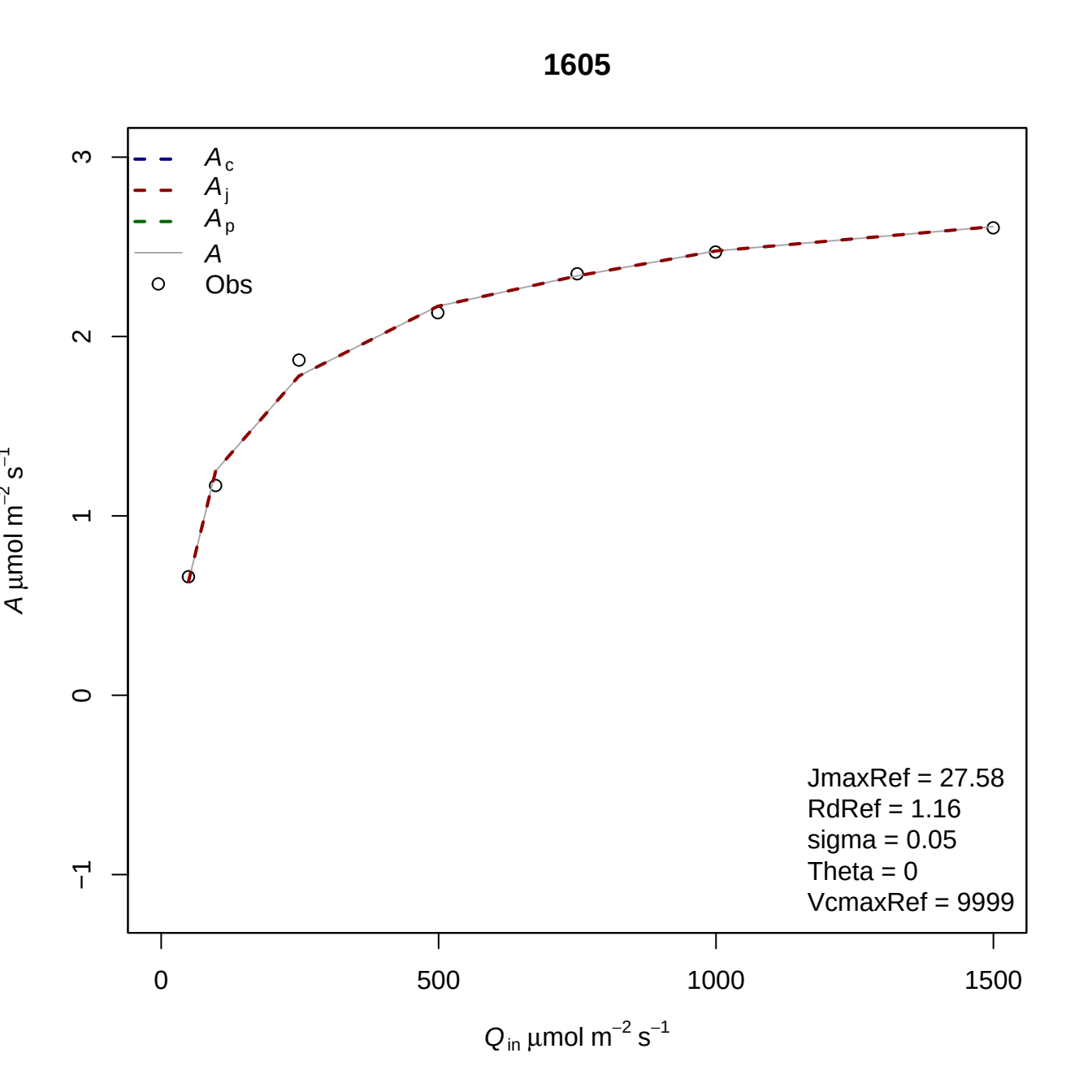

1600
A_c
A_j
A_p
A
Obs
JmaxRef = 104.95
RdRef = 0.55
sigma = 0.3
Theta = 0
VcmaxRef = 61.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

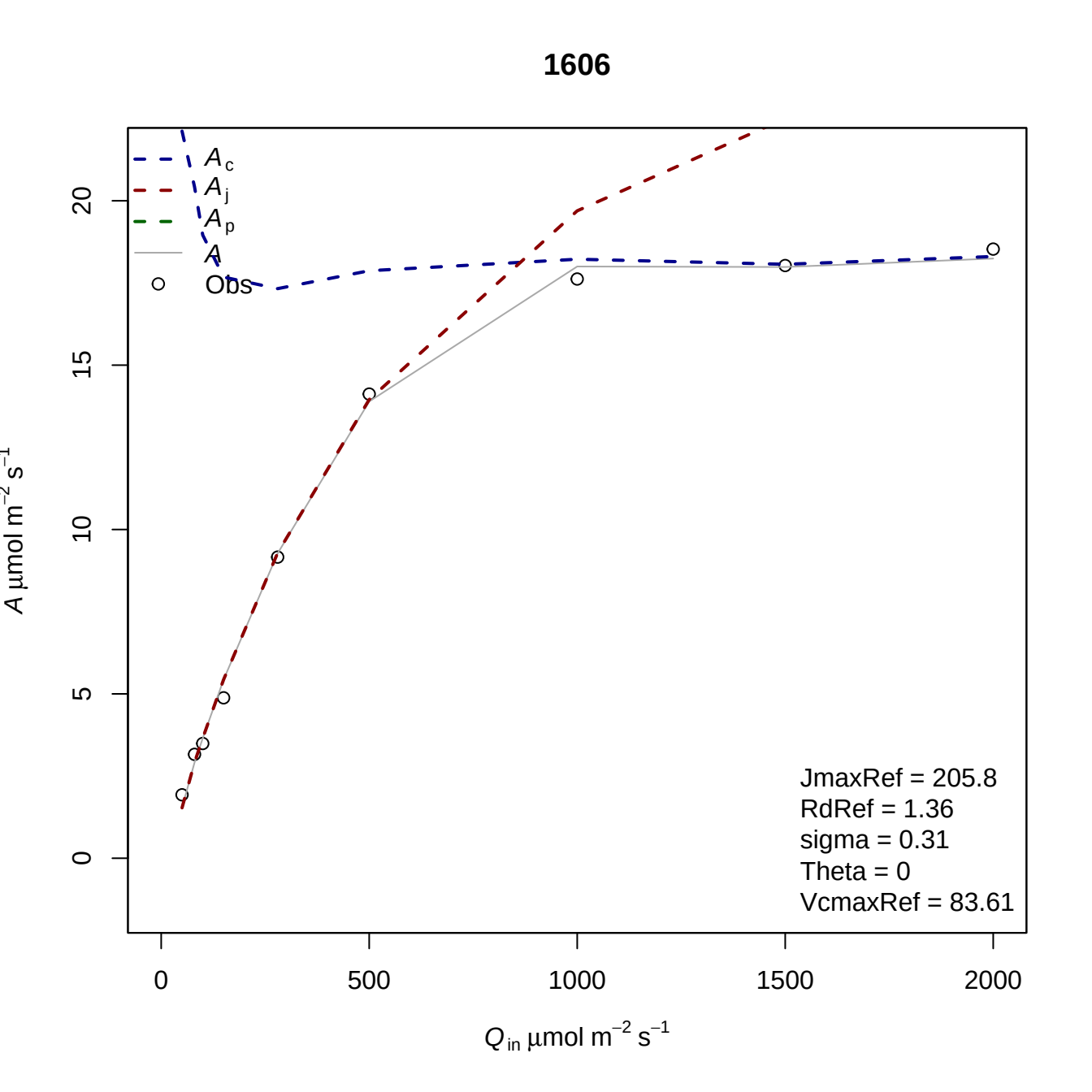

1601
A_c
A_j
A_p
A
Obs
JmaxRef = 84.15
RdRef = 0.28
sigma = 0.34
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

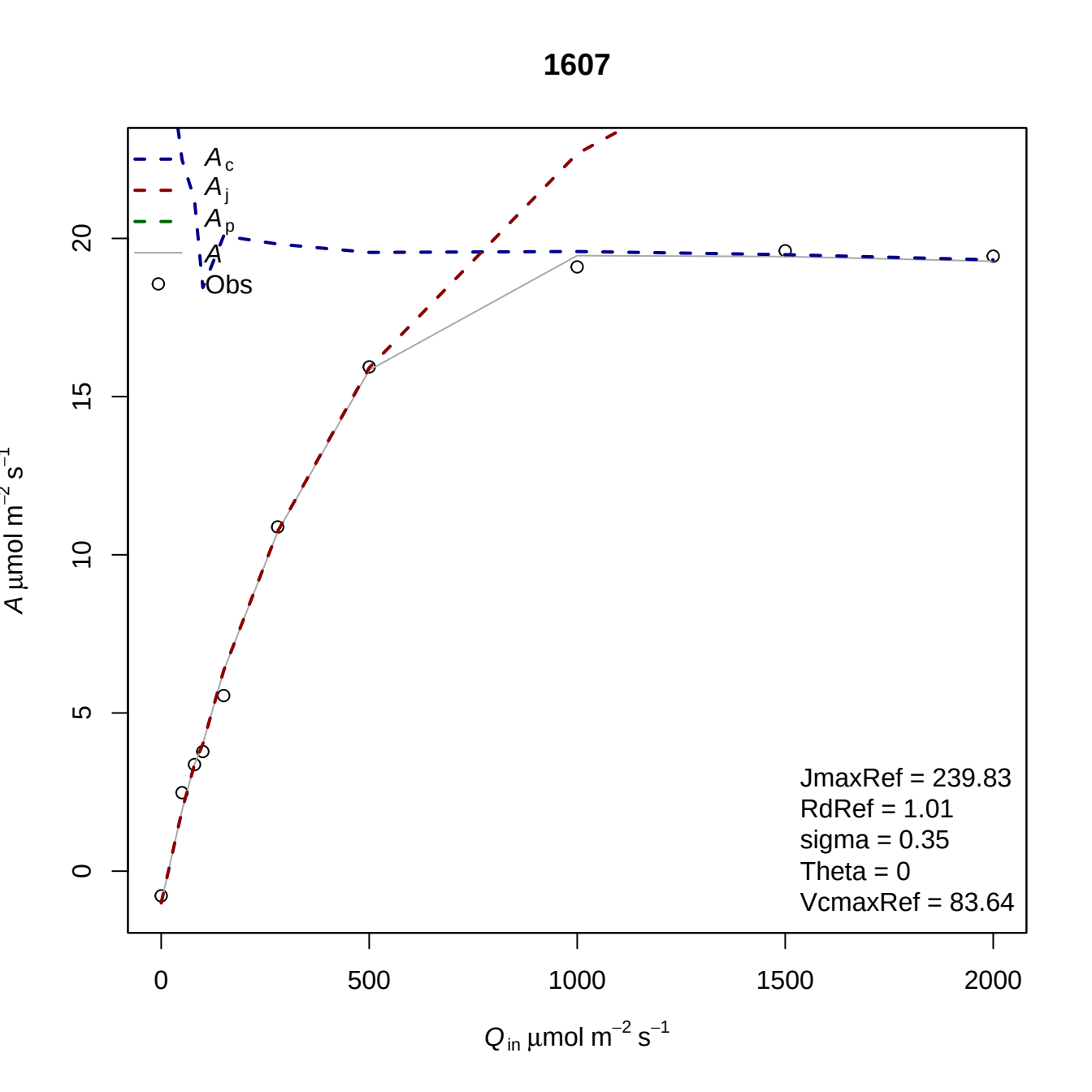

1602
A_c
A_j
A_p
A
Obs
JmaxRef = 86.29
RdRef = 0.58
sigma = 0.27
Theta = 0
VcmaxRef = 52.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

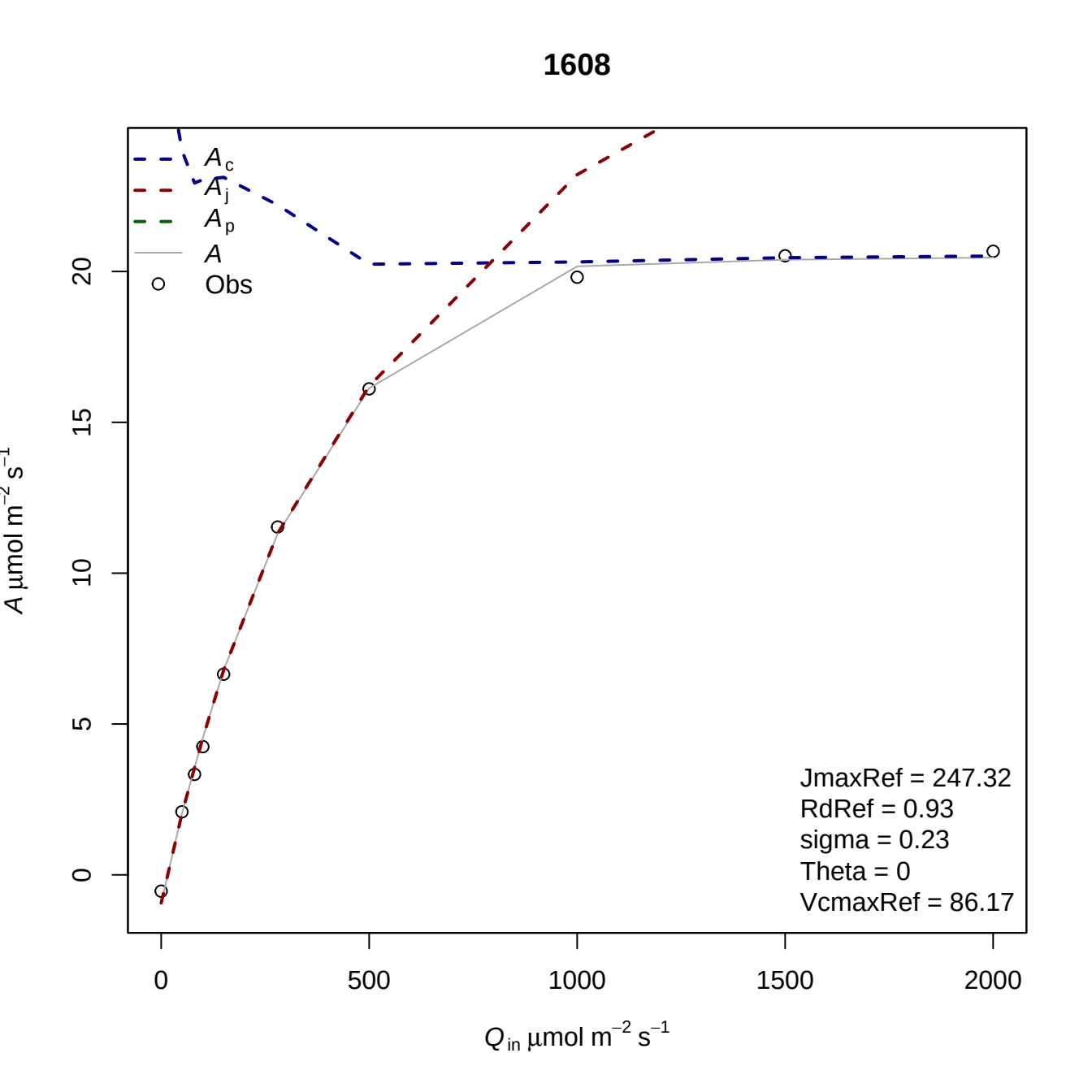

1603
A_c
A_j
A_p
A
Obs
JmaxRef = 65.13
RdRef = 0.66
sigma = 0.14
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

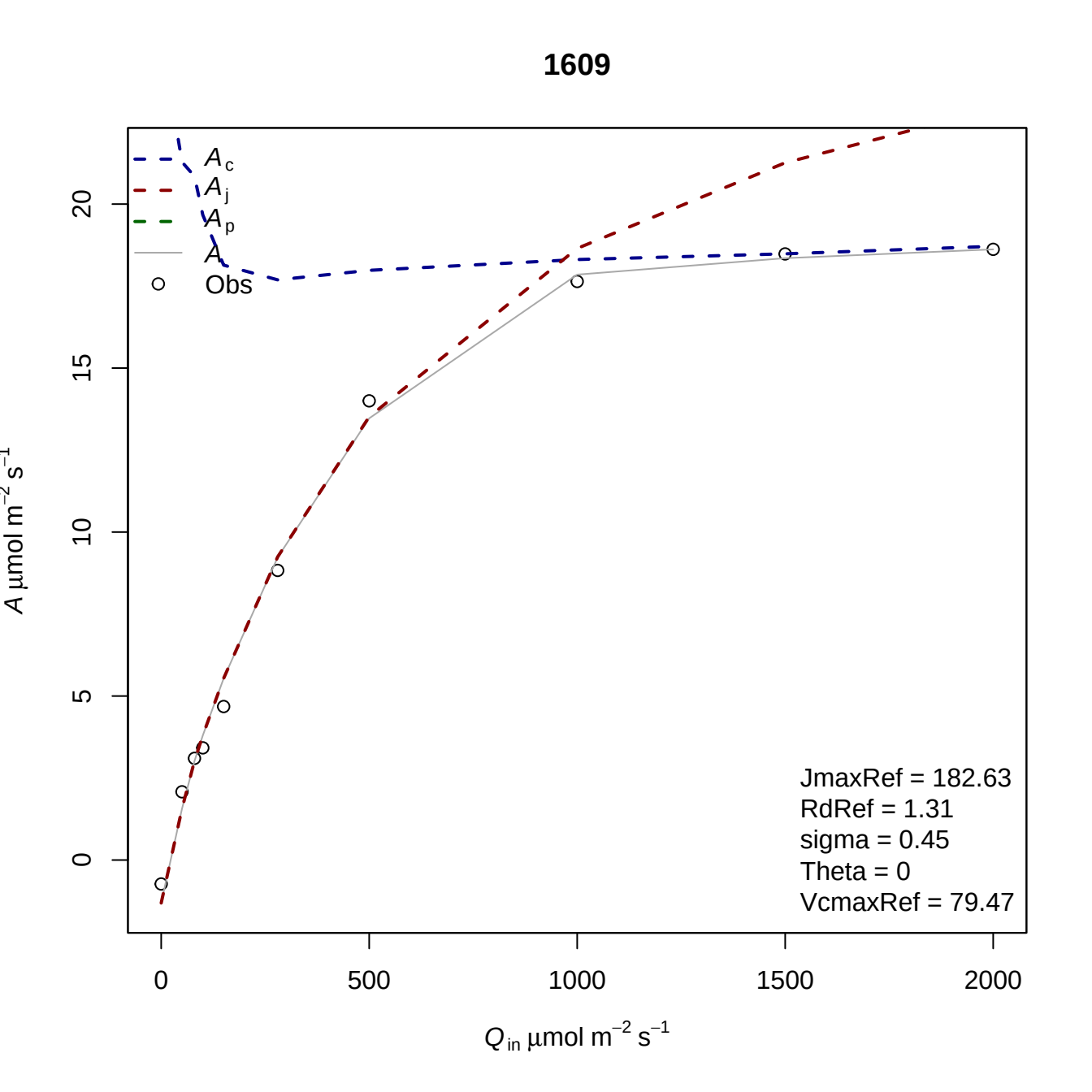

1604
A_c
A_j
A_p
A
Obs
JmaxRef = 56.47
RdRef = 0.72
sigma = 0.15
Theta = 0
VcmaxRef = 38.42
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

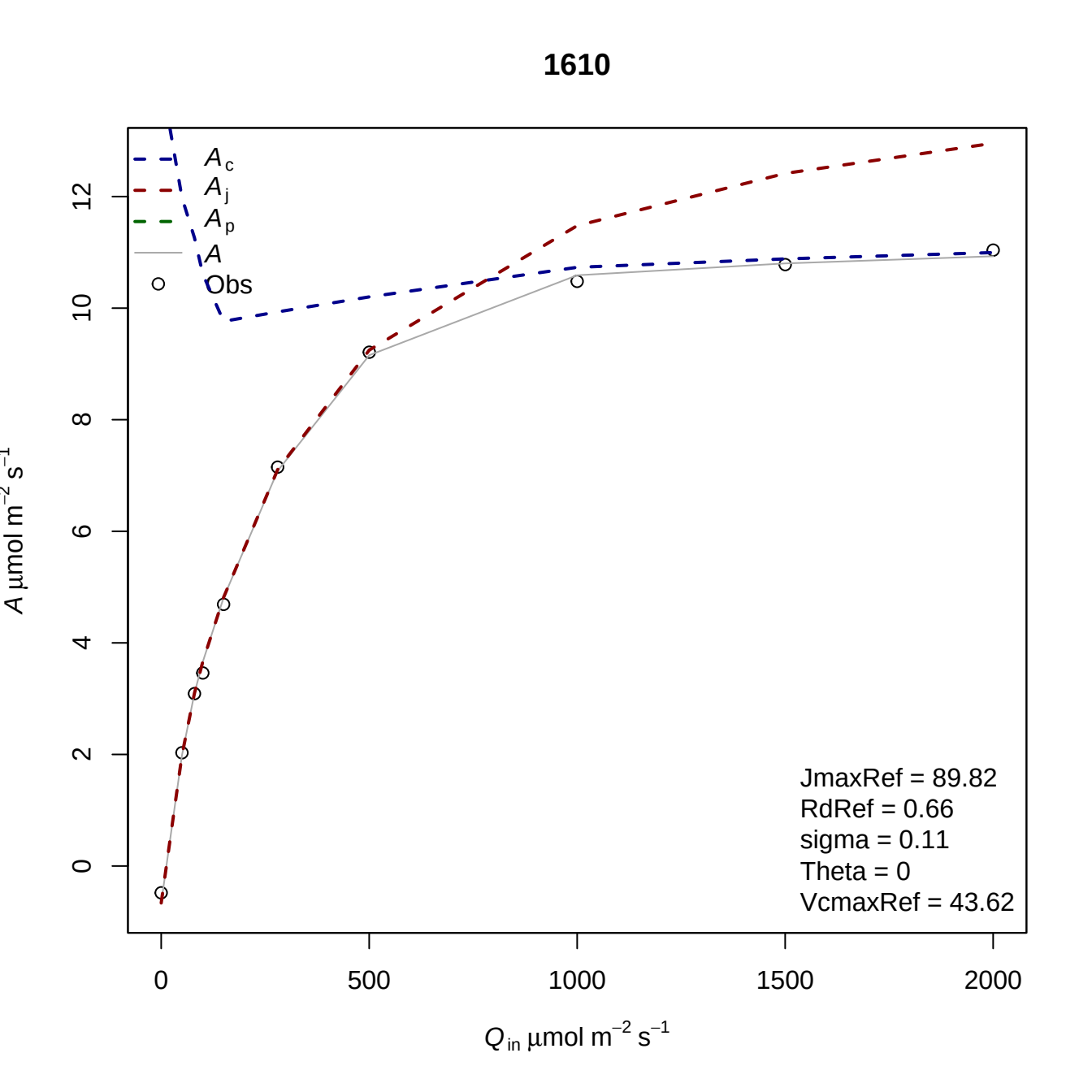

1605
A_c
A_j
A_p
A
Obs
JmaxRef = 27.58
RdRef = 1.16
sigma = 0.05
Theta = 0
VcmaxRef = 9999
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

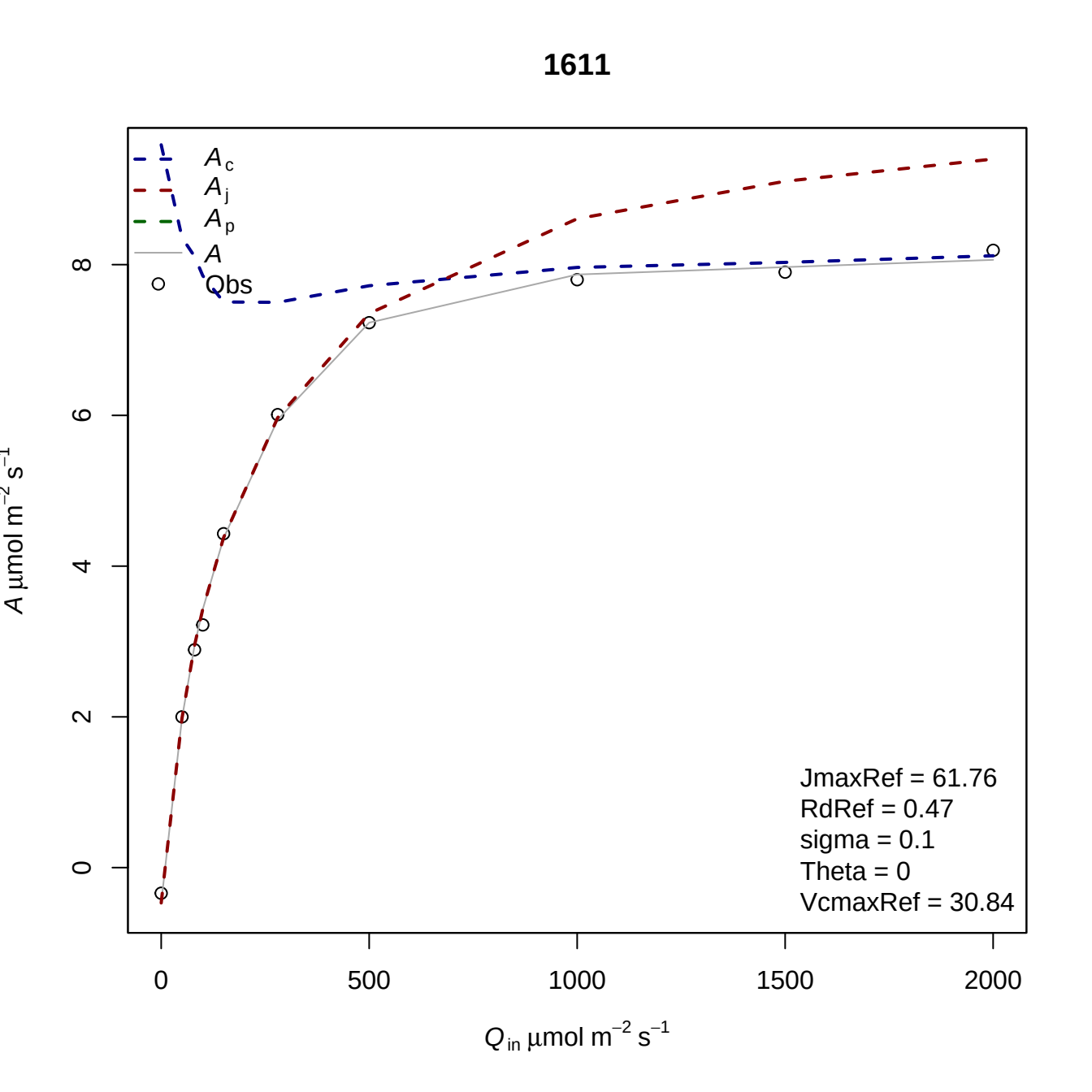

1606
A_c
A_j
A_p
A
Obs
JmaxRef = 205.8
RdRef = 1.36
sigma = 0.31
Theta = 0
VcmaxRef = 83.61
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

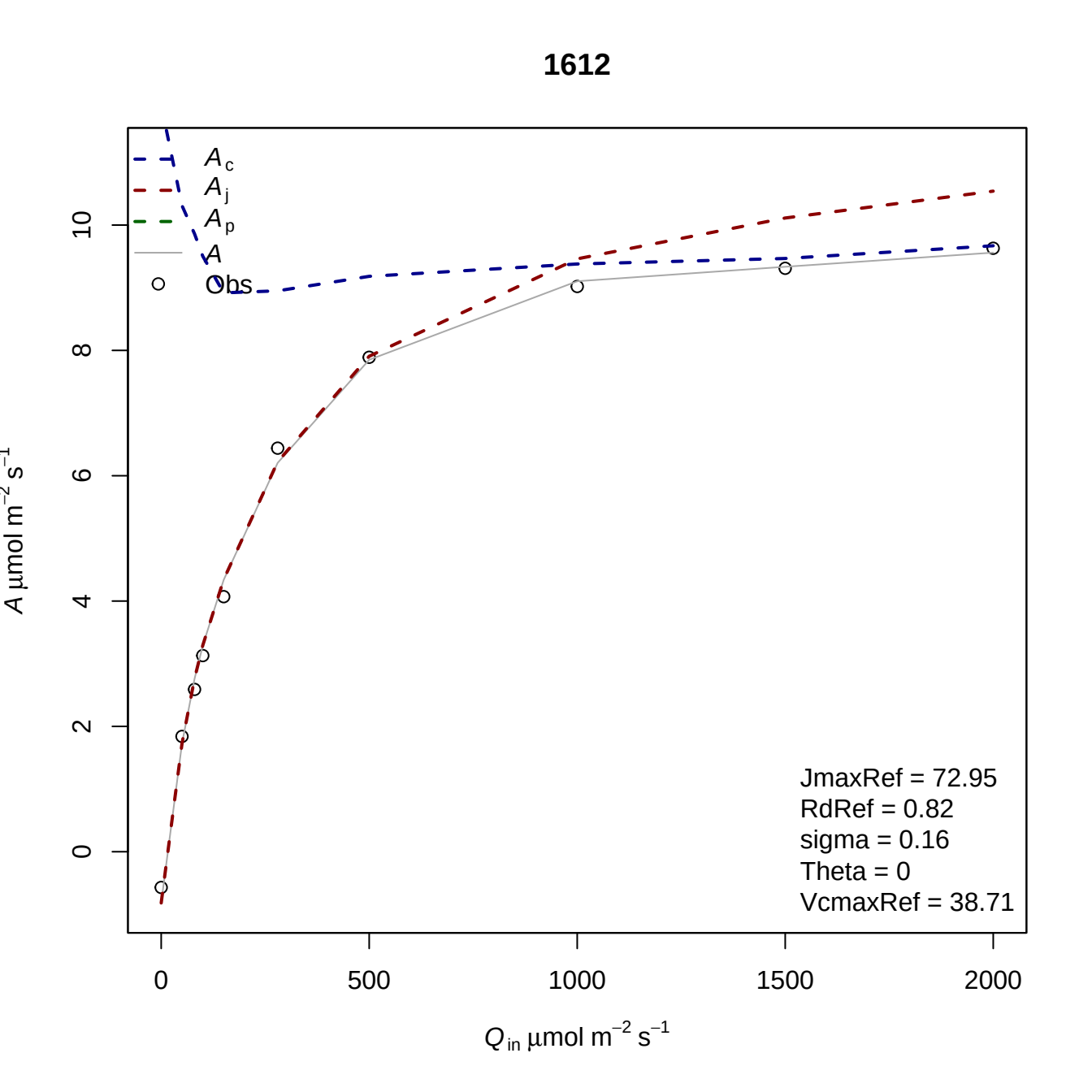

1607
A_c
A_j
A_p
A
Obs
JmaxRef = 239.83
RdRef = 1.01
sigma = 0.35
Theta = 0
VcmaxRef = 83.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

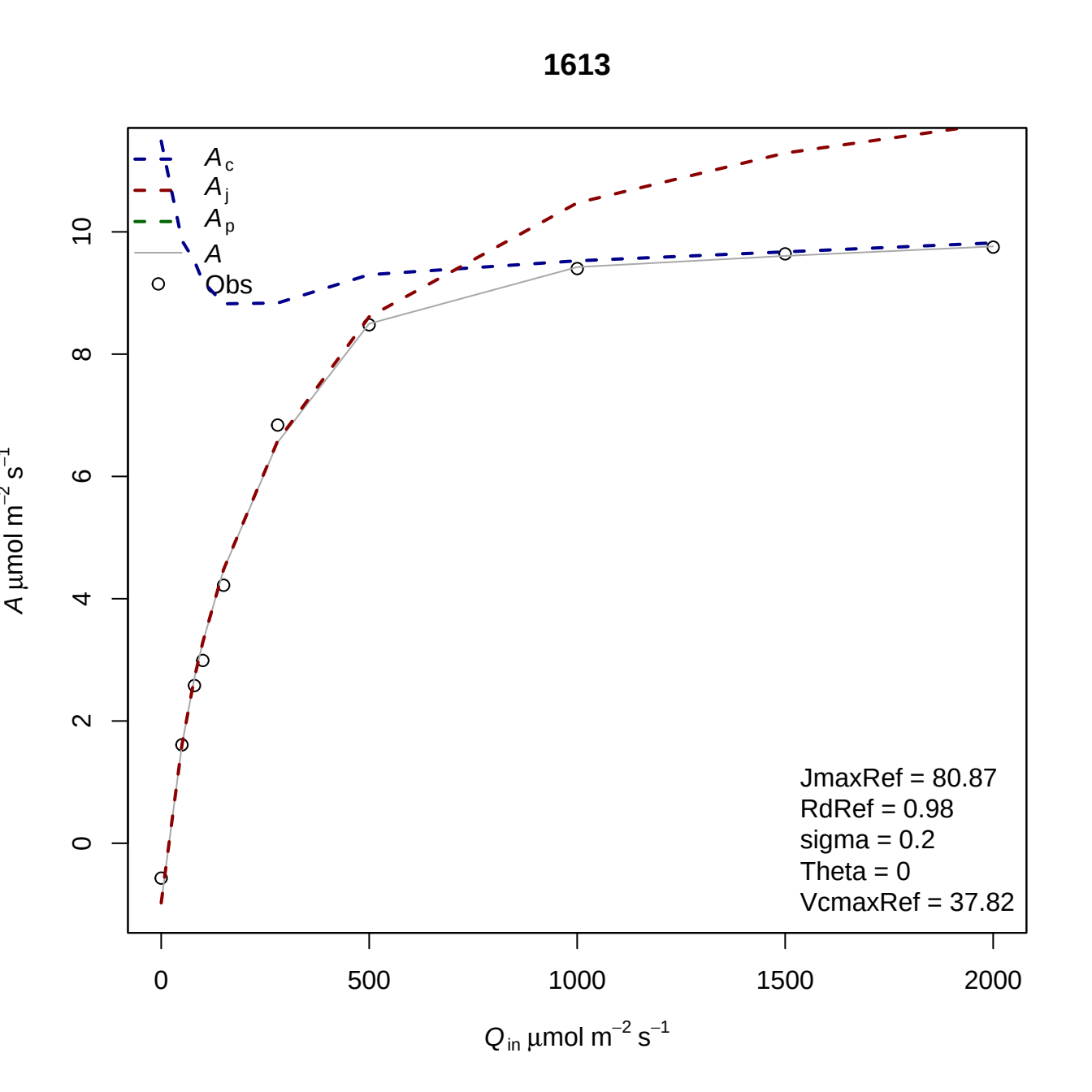

1608
A_c
A_j
A_p
A
Obs
JmaxRef = 247.32
RdRef = 0.93
sigma = 0.23
Theta = 0
VcmaxRef = 86.17
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

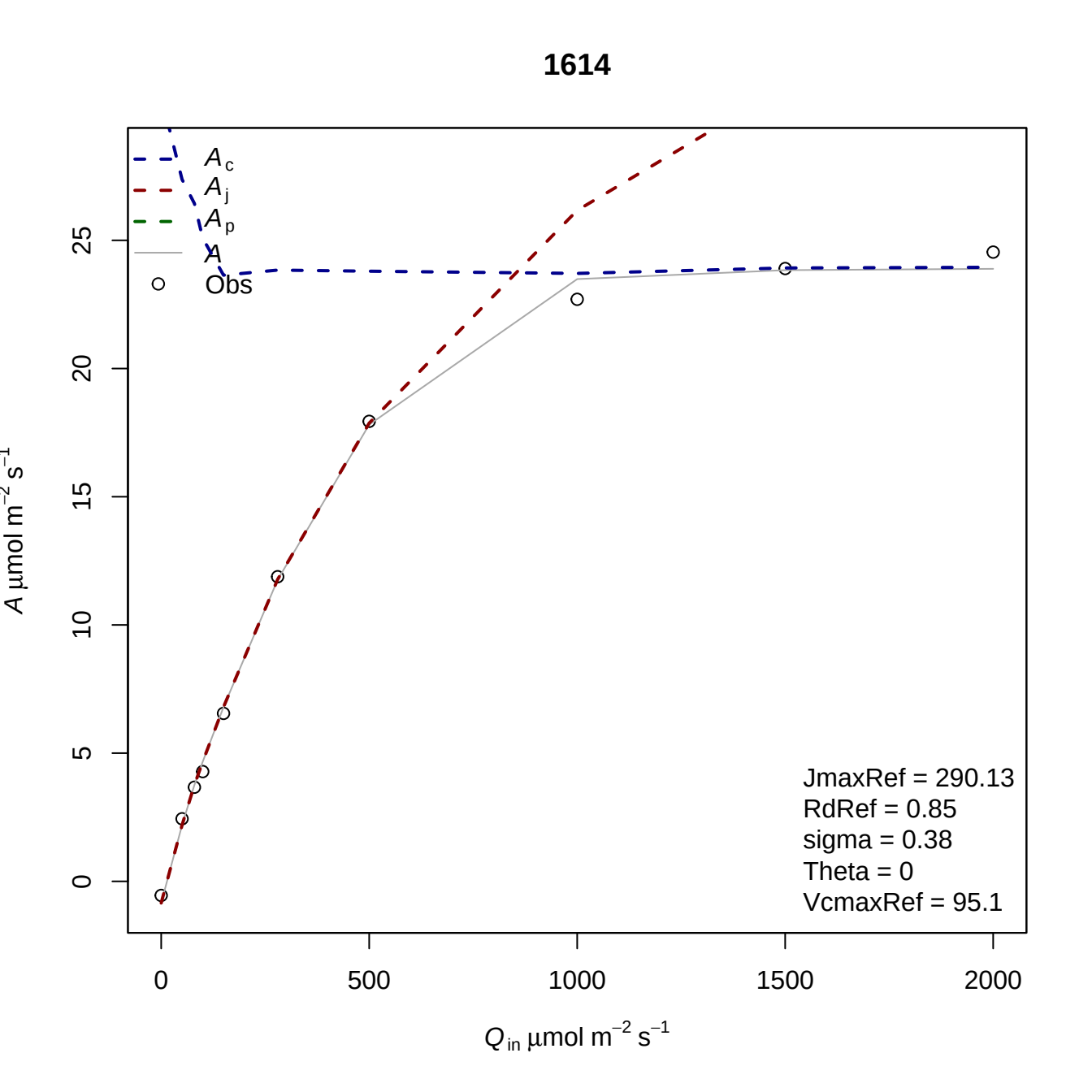

1609
A_c
A_j
A_p
A
Obs
JmaxRef = 182.63
RdRef = 1.31
sigma = 0.45
Theta = 0
VcmaxRef = 79.47
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

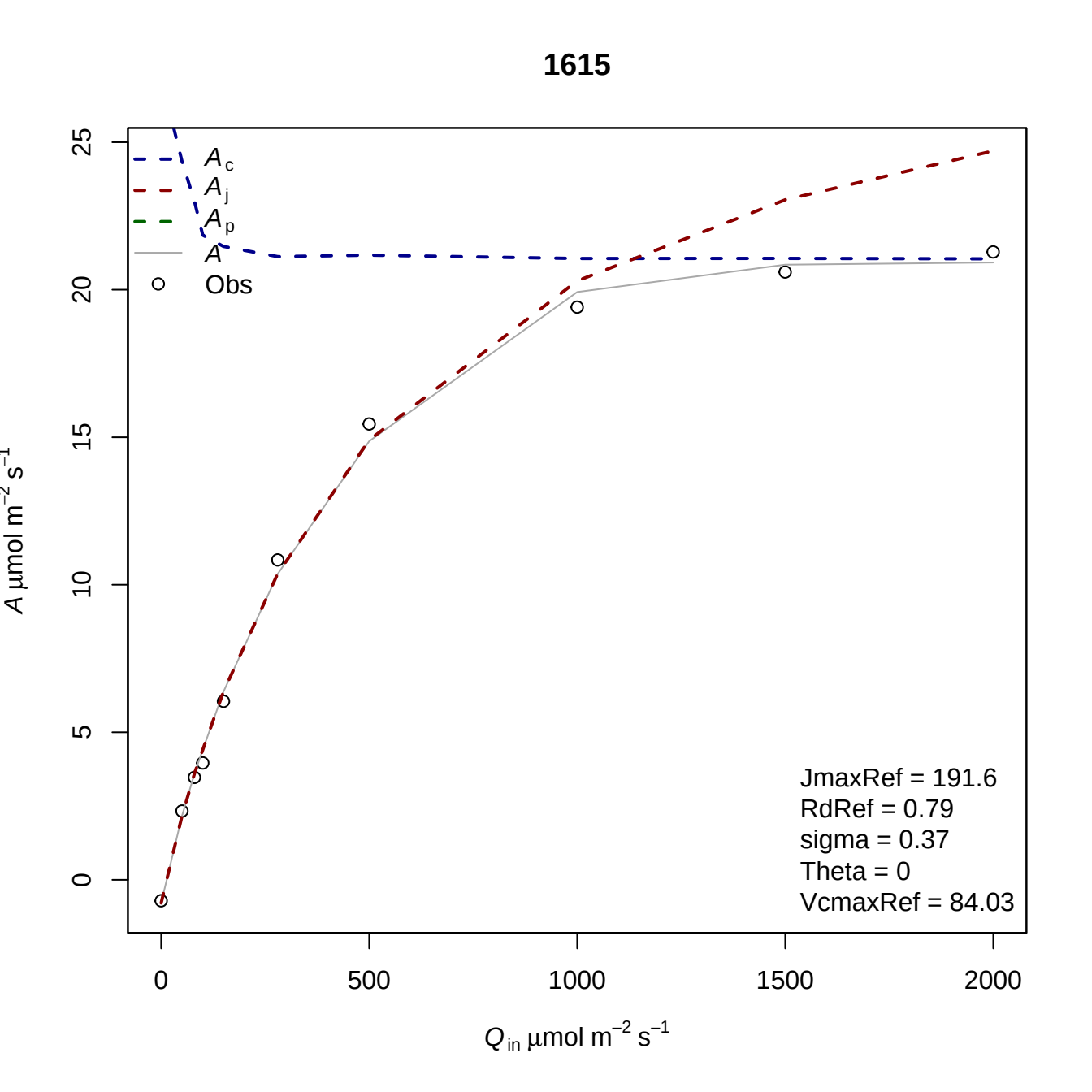

1610
A_c
A_j
A_p
A
Obs
JmaxRef = 89.82
RdRef = 0.66
sigma = 0.11
Theta = 0
VcmaxRef = 43.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

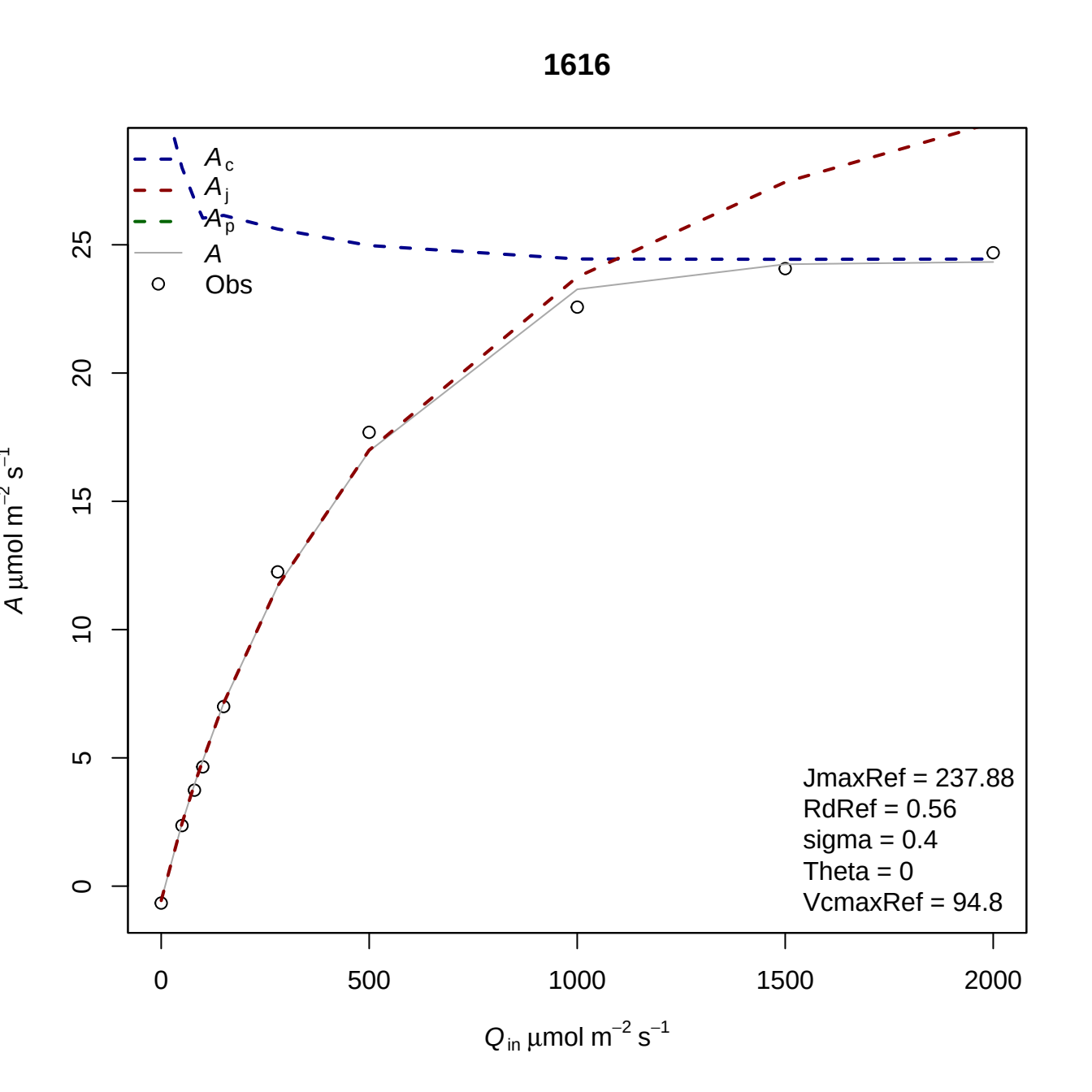

1611
A_c
A_j
A_p
A
Qbs
JmaxRef = 61.76
RdRef = 0.47
sigma = 0.1
Theta = 0
VcmaxRef = 30.84
0
2
4
6
8
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

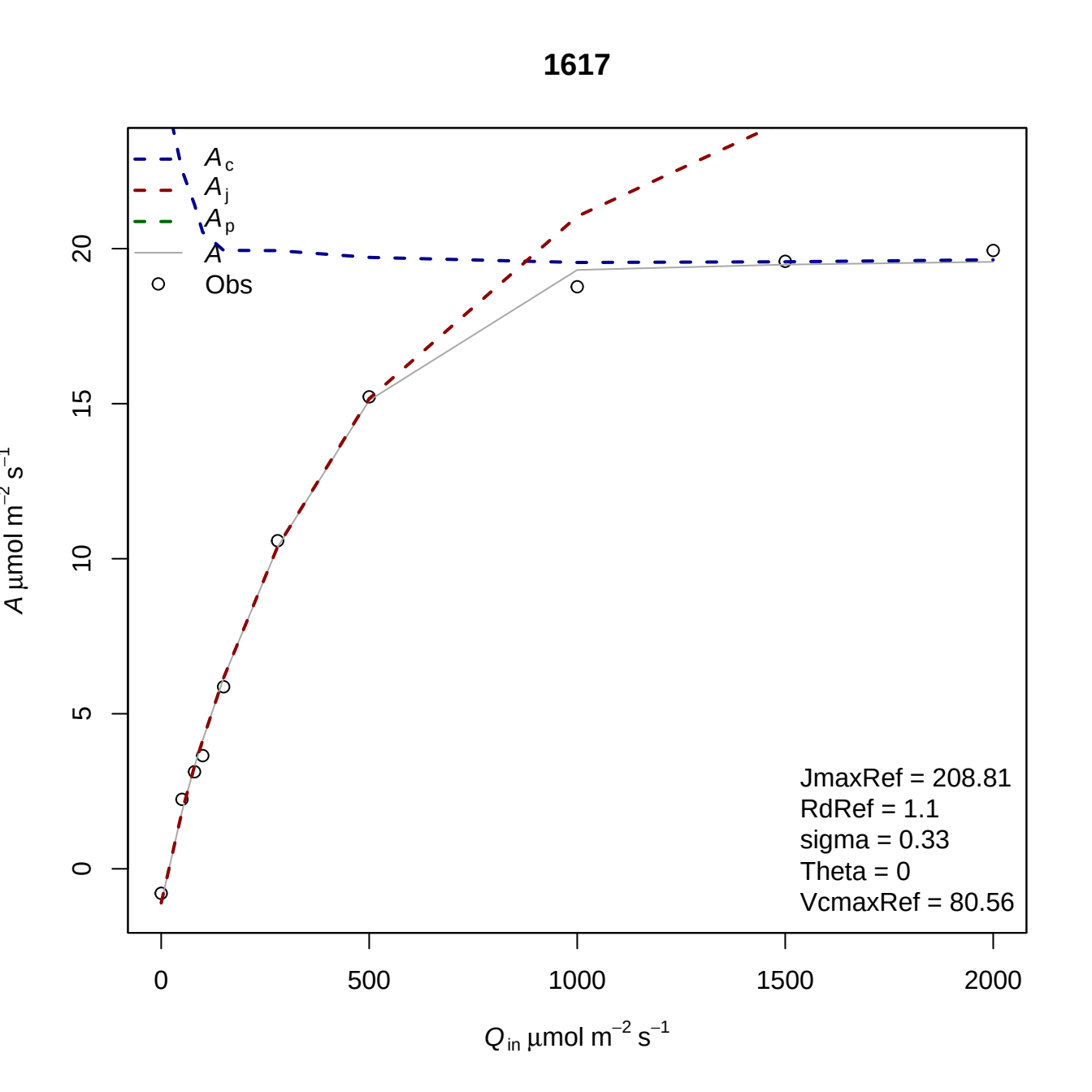

1612
A_c
A_j
A_p
A
Obs
JmaxRef = 72.95
RdRef = 0.82
sigma = 0.16
Theta = 0
VcmaxRef = 38.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

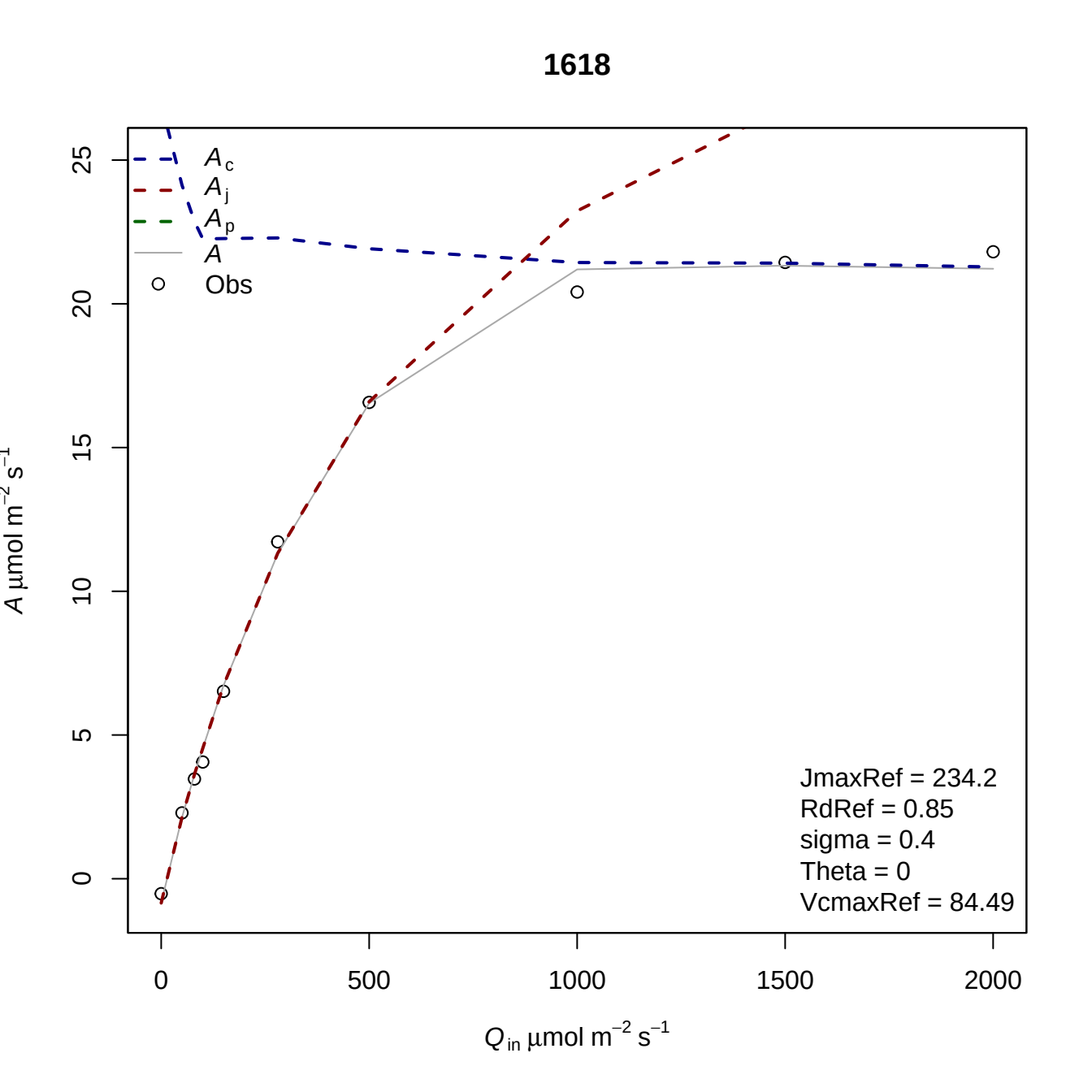

1613
A_c
A_j
A_p
A
Obs
JmaxRef = 80.87
RdRef = 0.98
sigma = 0.2
Theta = 0
VcmaxRef = 37.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

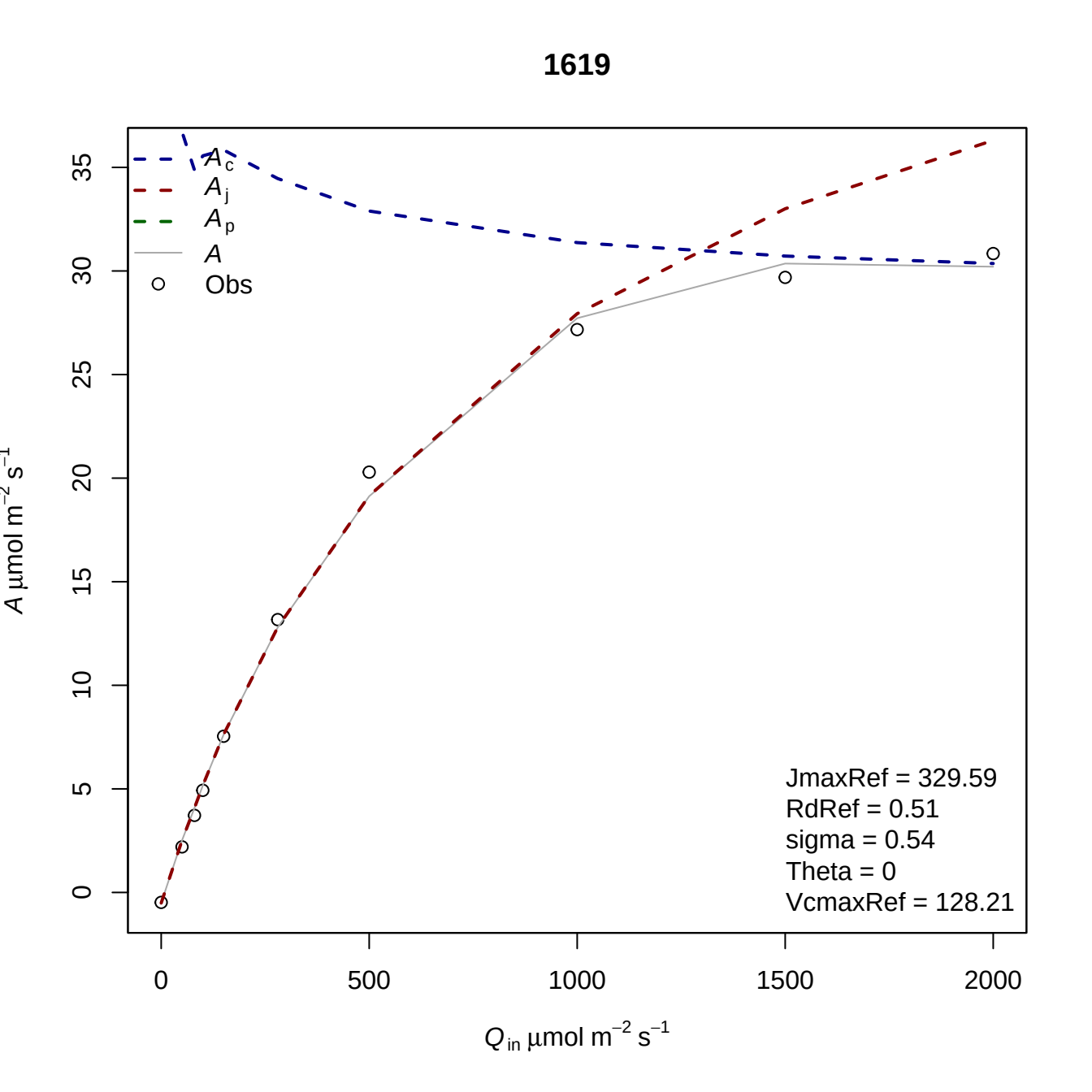

1614
A_c
A_j
A_p
A
Obs
JmaxRef = 290.13
RdRef = 0.85
sigma = 0.38
Theta = 0
VcmaxRef = 95.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

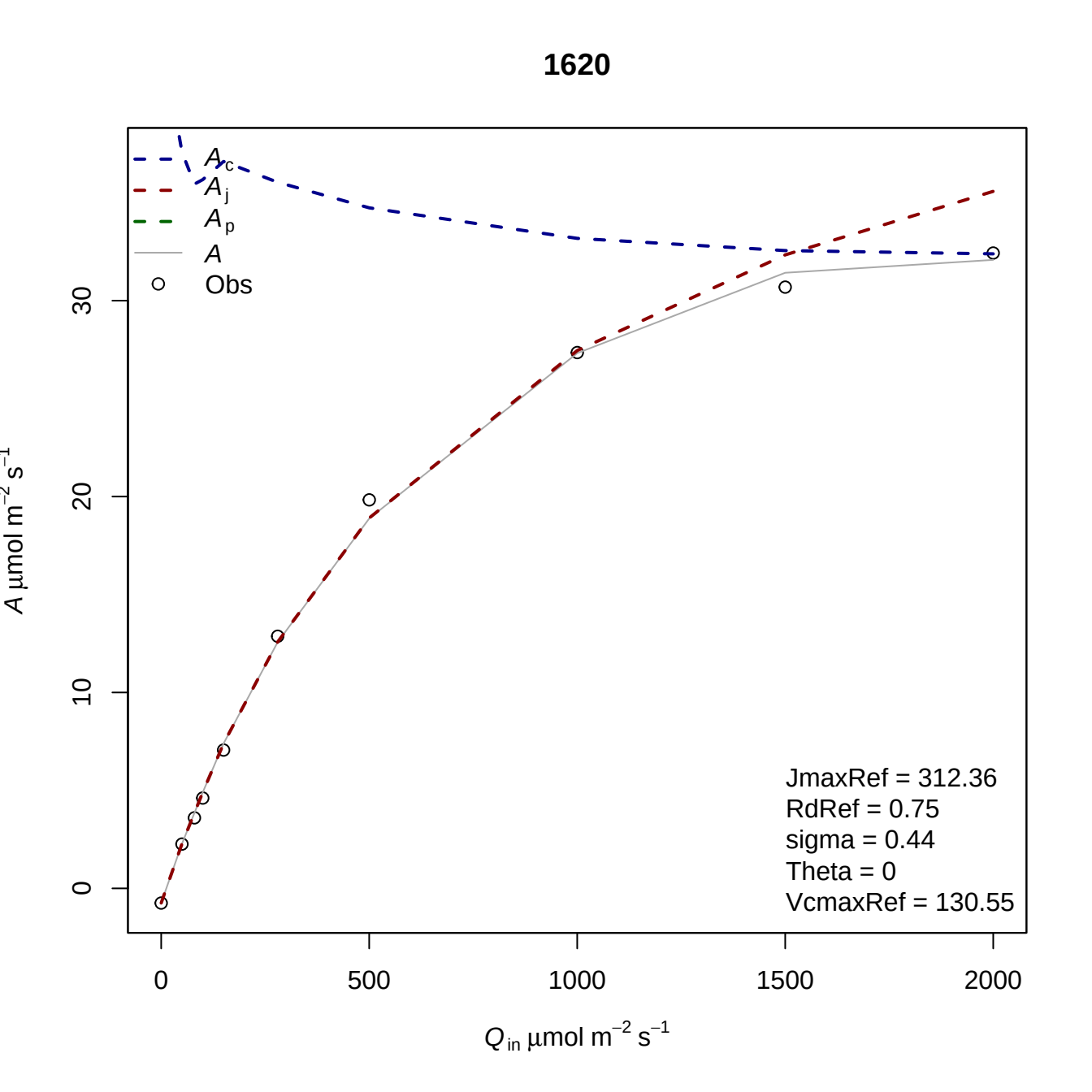

1615
A_c
A_j
A_p
A
Obs
JmaxRef = 191.6
RdRef = 0.79
sigma = 0.37
Theta = 0
VcmaxRef = 84.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

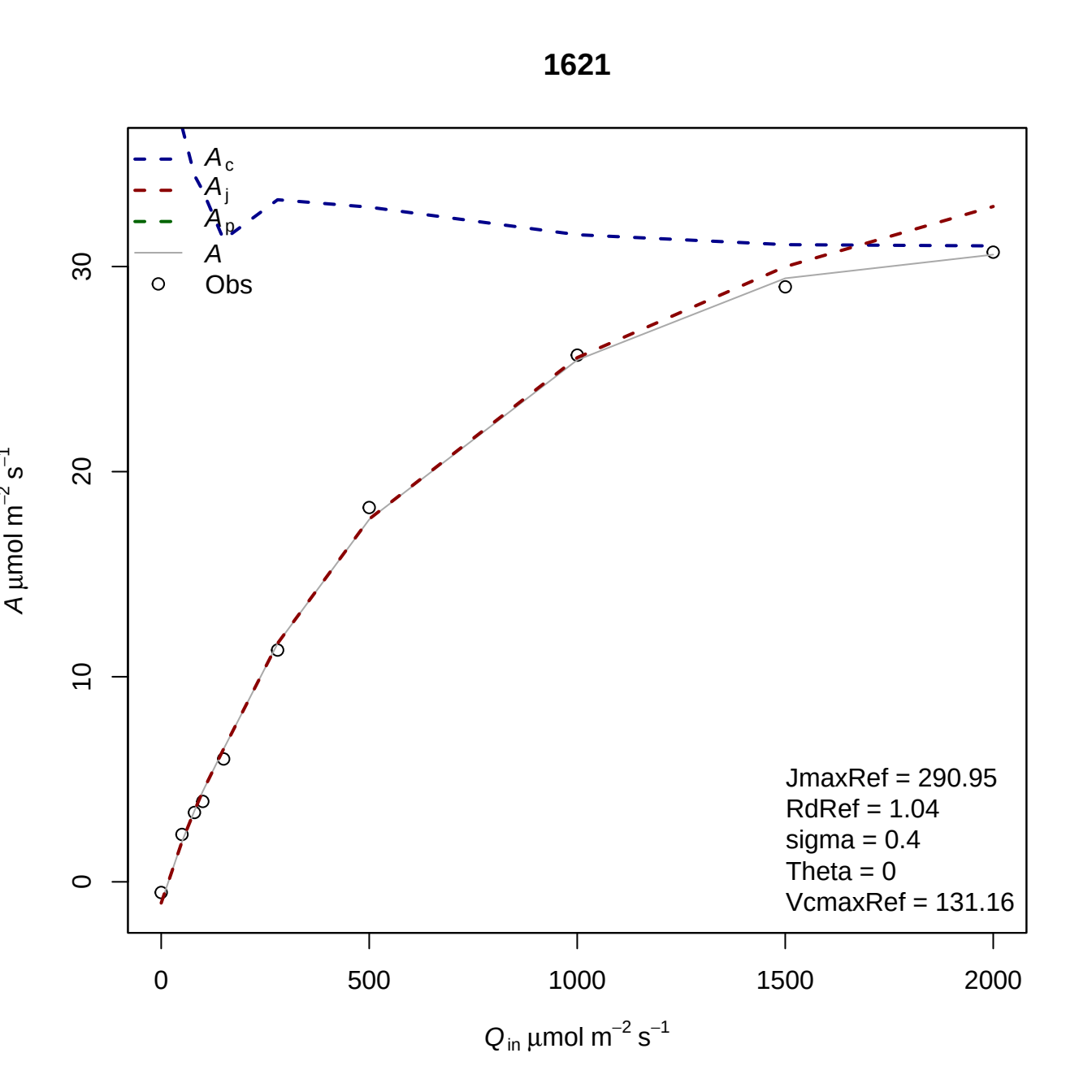

1616
A_c
A_j
A_p
A
Obs
JmaxRef = 237.88
RdRef = 0.56
sigma = 0.4
Theta = 0
VcmaxRef = 94.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

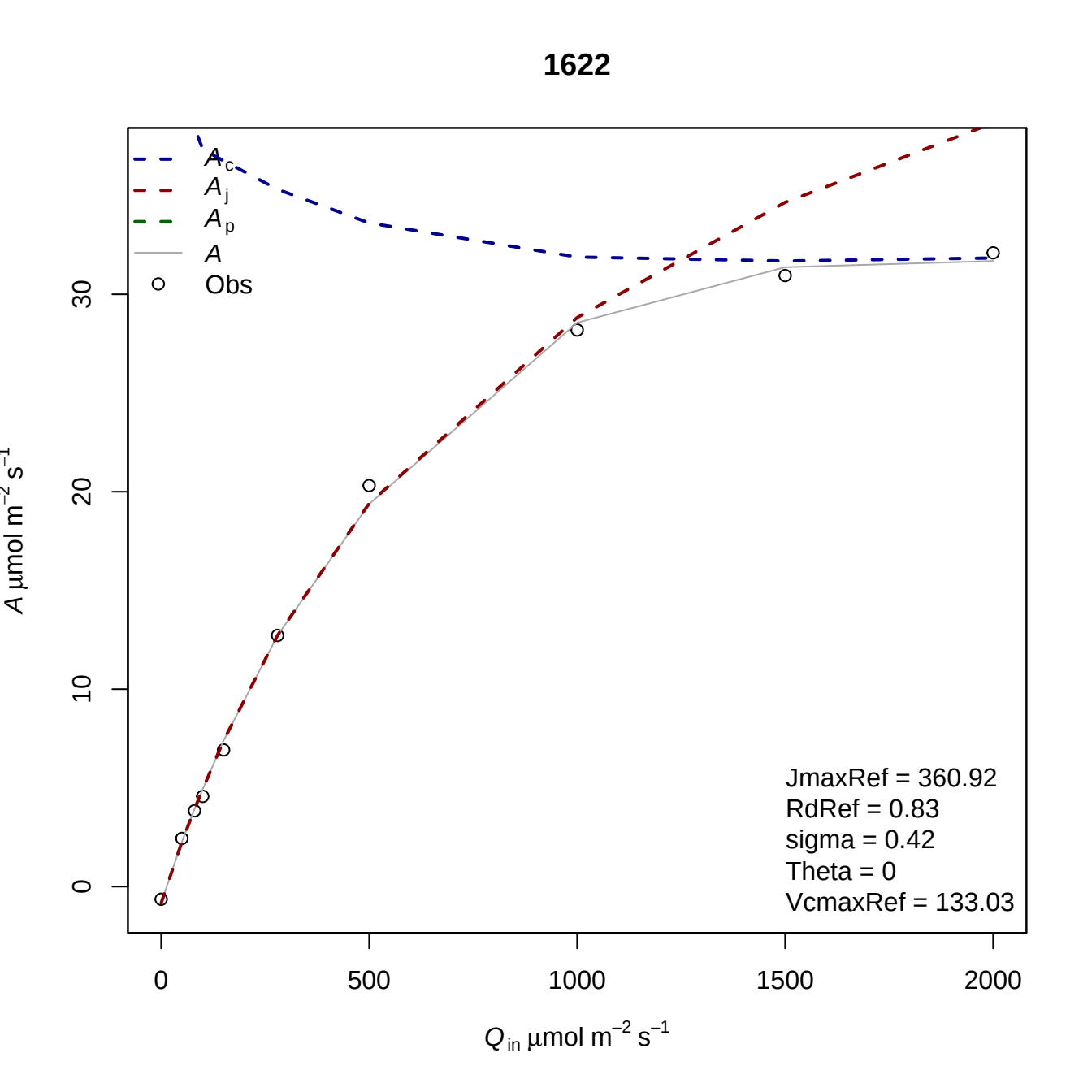

1617
A_c
A_j
A_p
A
Obs
JmaxRef = 208.81
RdRef = 1.1
sigma = 0.33
Theta = 0
VcmaxRef = 80.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

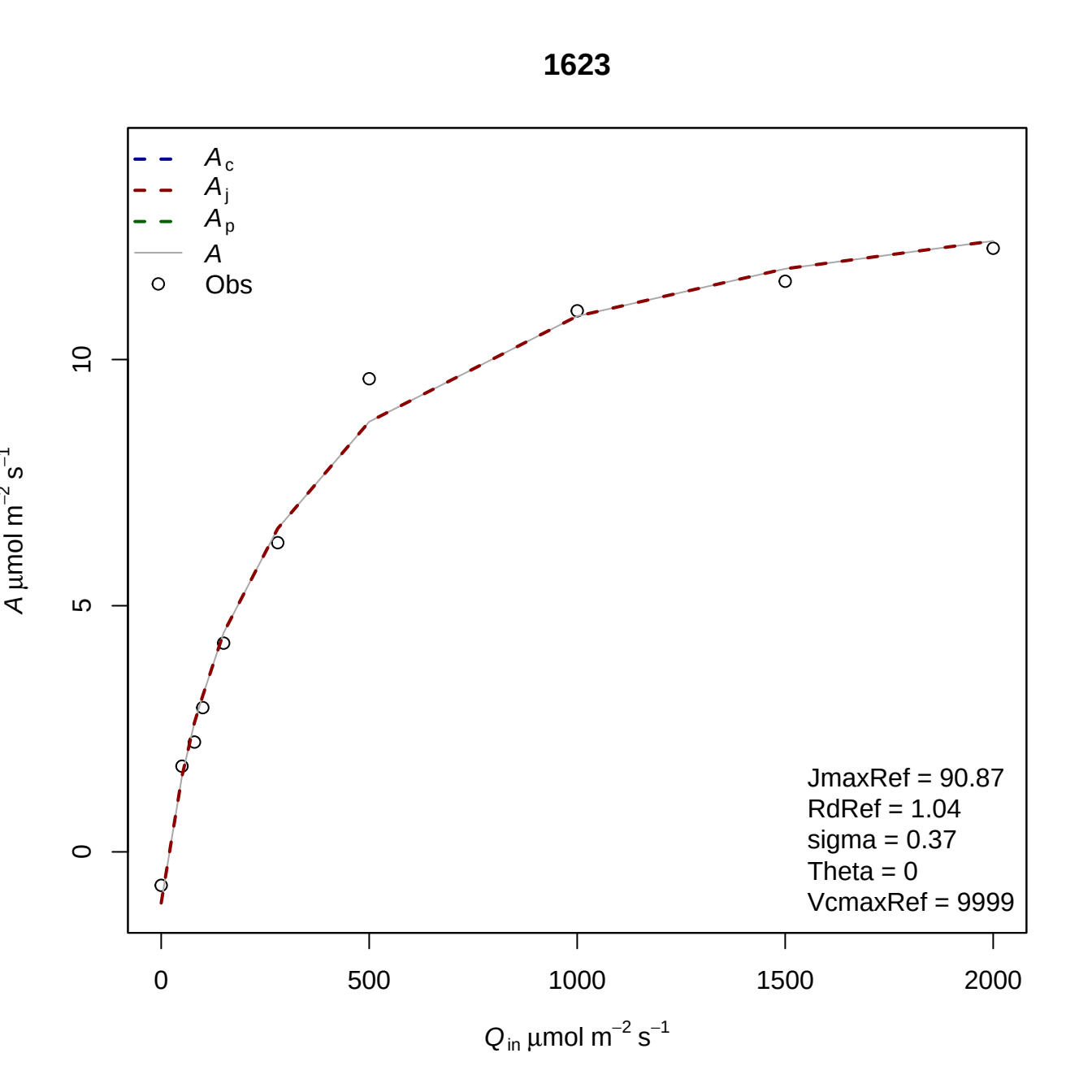

1618
A_c
A_j
A_p
A
Obs
JmaxRef = 234.2
RdRef = 0.85
sigma = 0.4
Theta = 0
VcmaxRef = 84.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

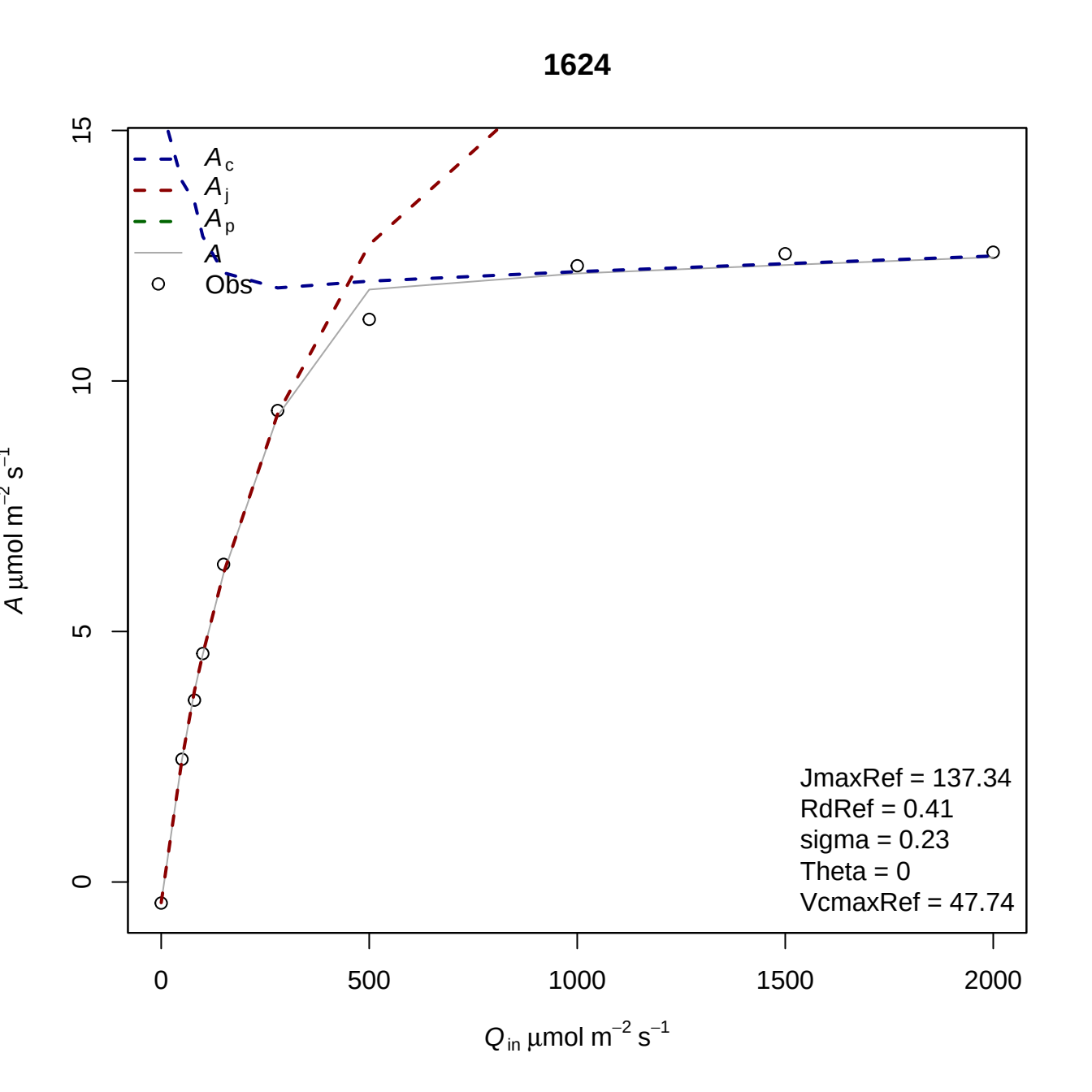

1619
A_c
A_j
A_p
A
Obs
JmaxRef = 329.59
RdRef = 0.51
sigma = 0.54
Theta = 0
VcmaxRef = 128.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

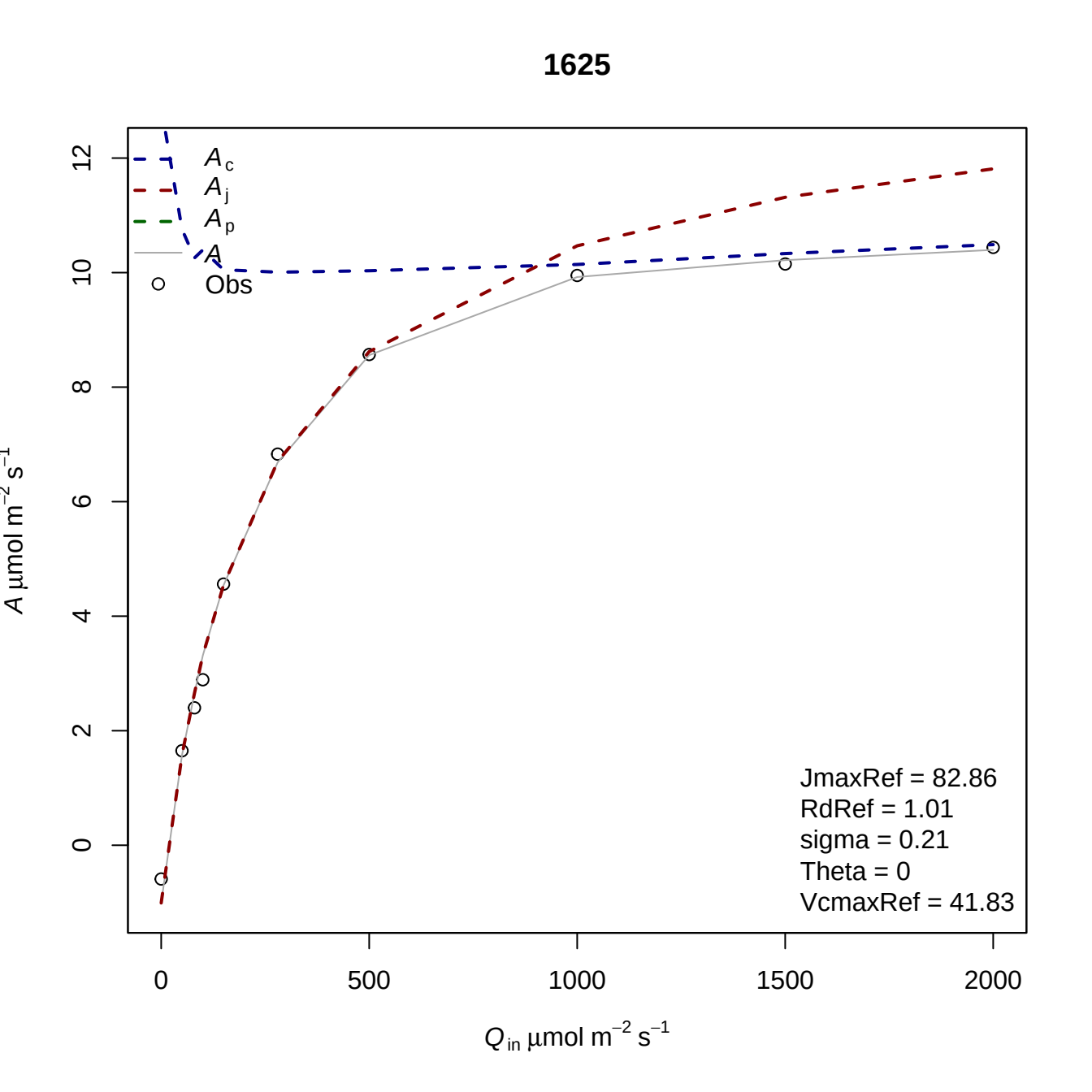

1620
A_c
A_j
A_p
A
Obs
JmaxRef = 312.36
RdRef = 0.75
sigma = 0.44
Theta = 0
VcmaxRef = 130.55
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

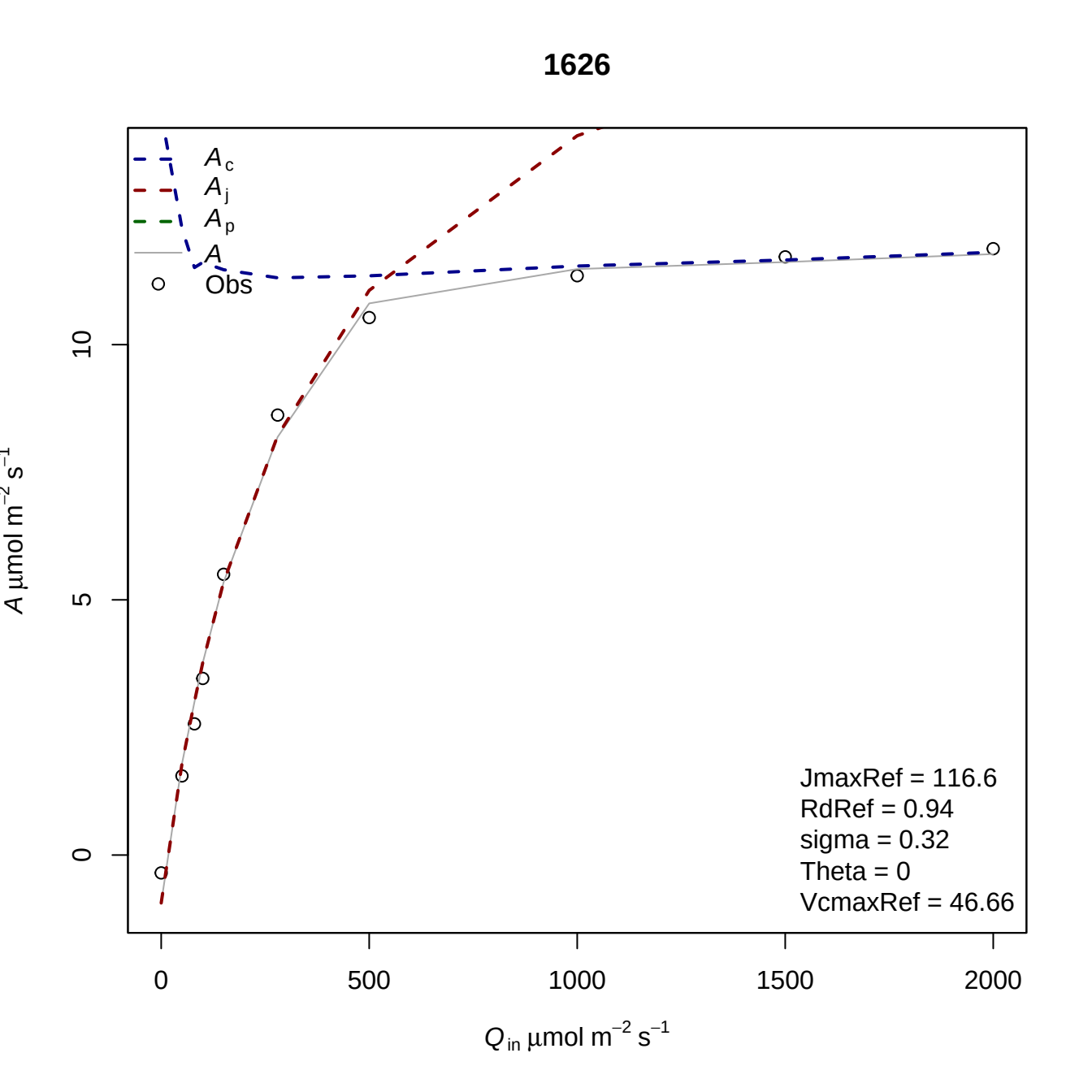

1621
A_c
A_j
A_p
A
Obs
JmaxRef = 290.95
RdRef = 1.04
sigma = 0.4
Theta = 0
VcmaxRef = 131.16
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

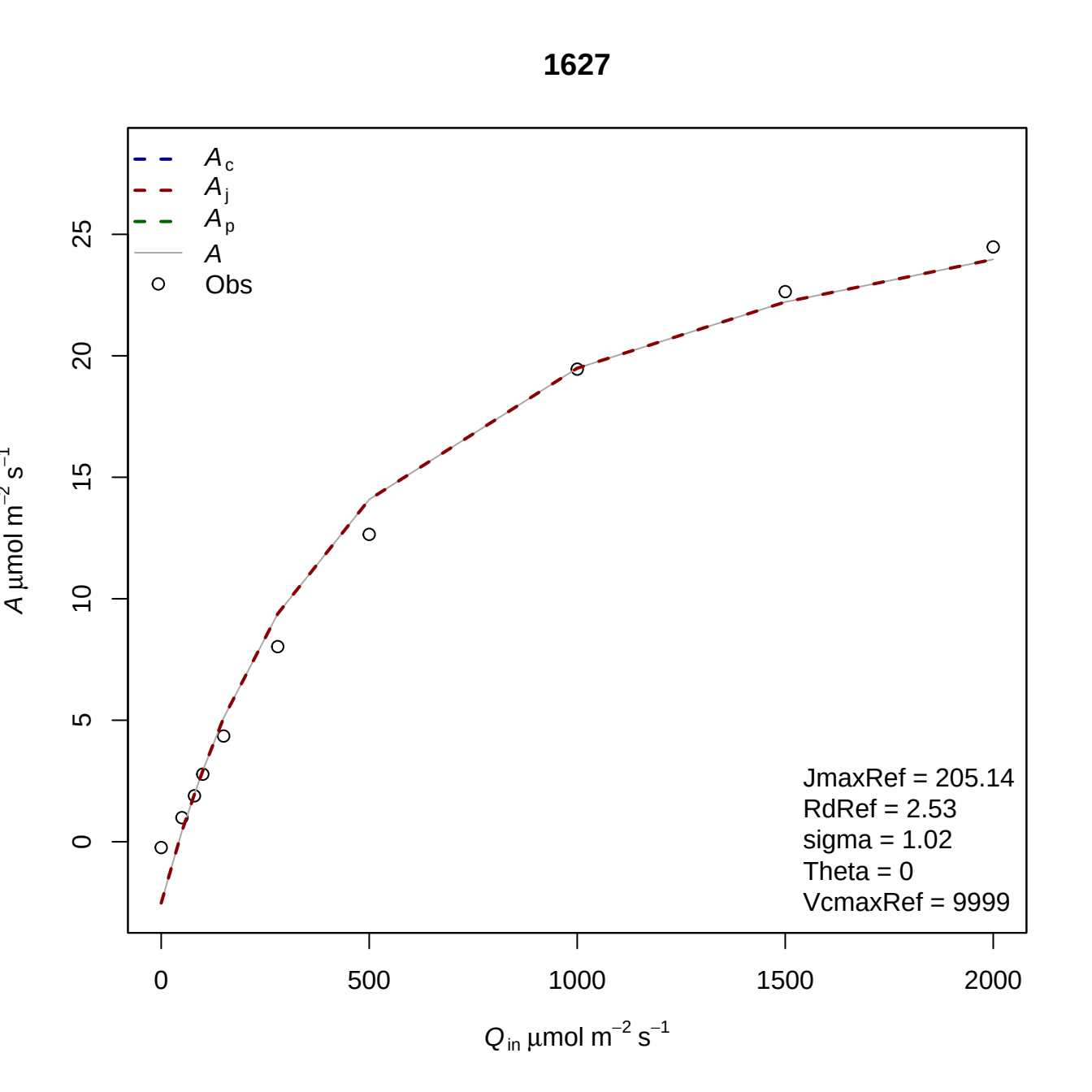

1622
A_c
A_j
A_p
A
Obs
JmaxRef = 360.92
RdRef = 0.83
sigma = 0.42
Theta = 0
VcmaxRef = 133.03
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

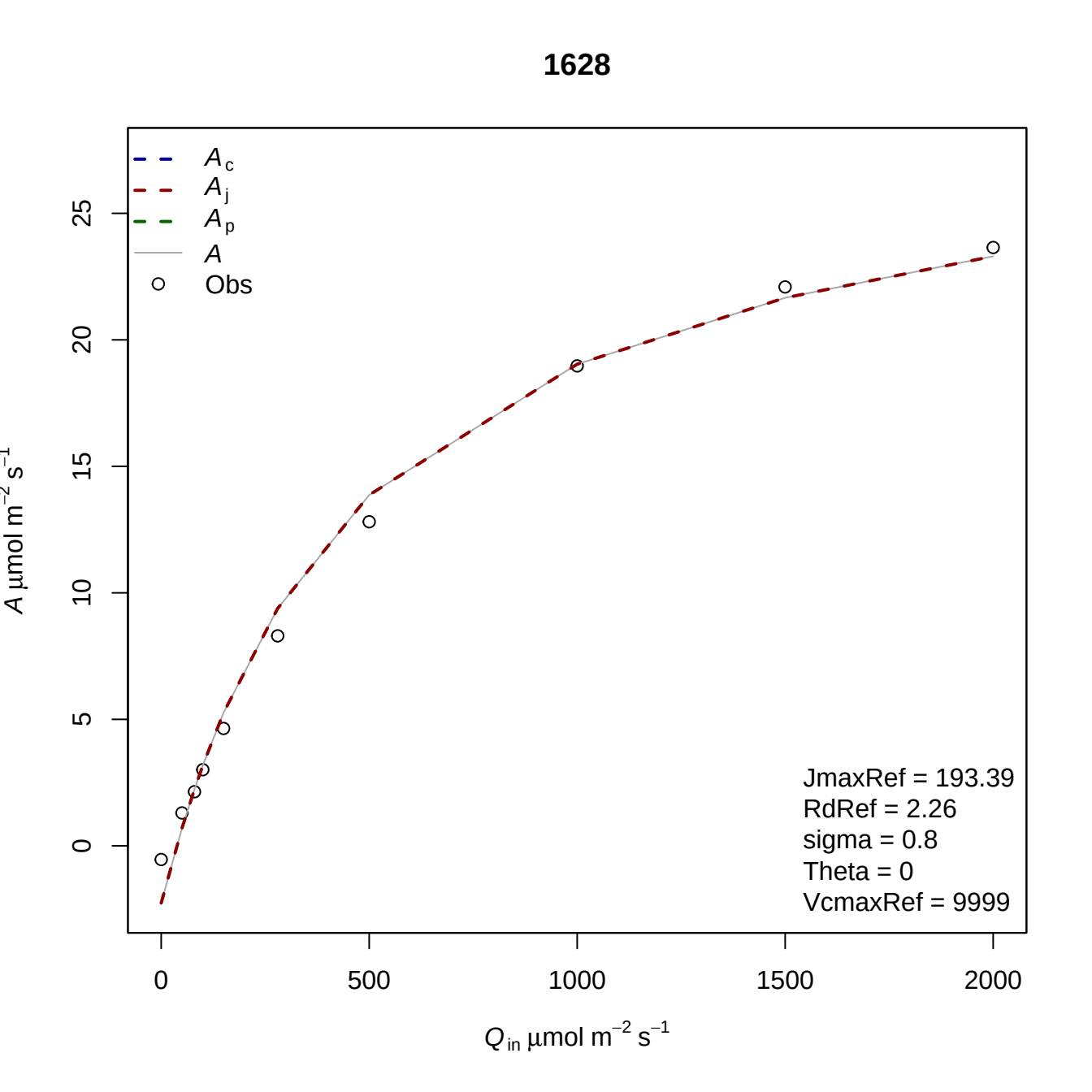

1623
A_c
A_j
A_p
A
Obs
JmaxRef = 90.87
RdRef = 1.04
sigma = 0.37
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

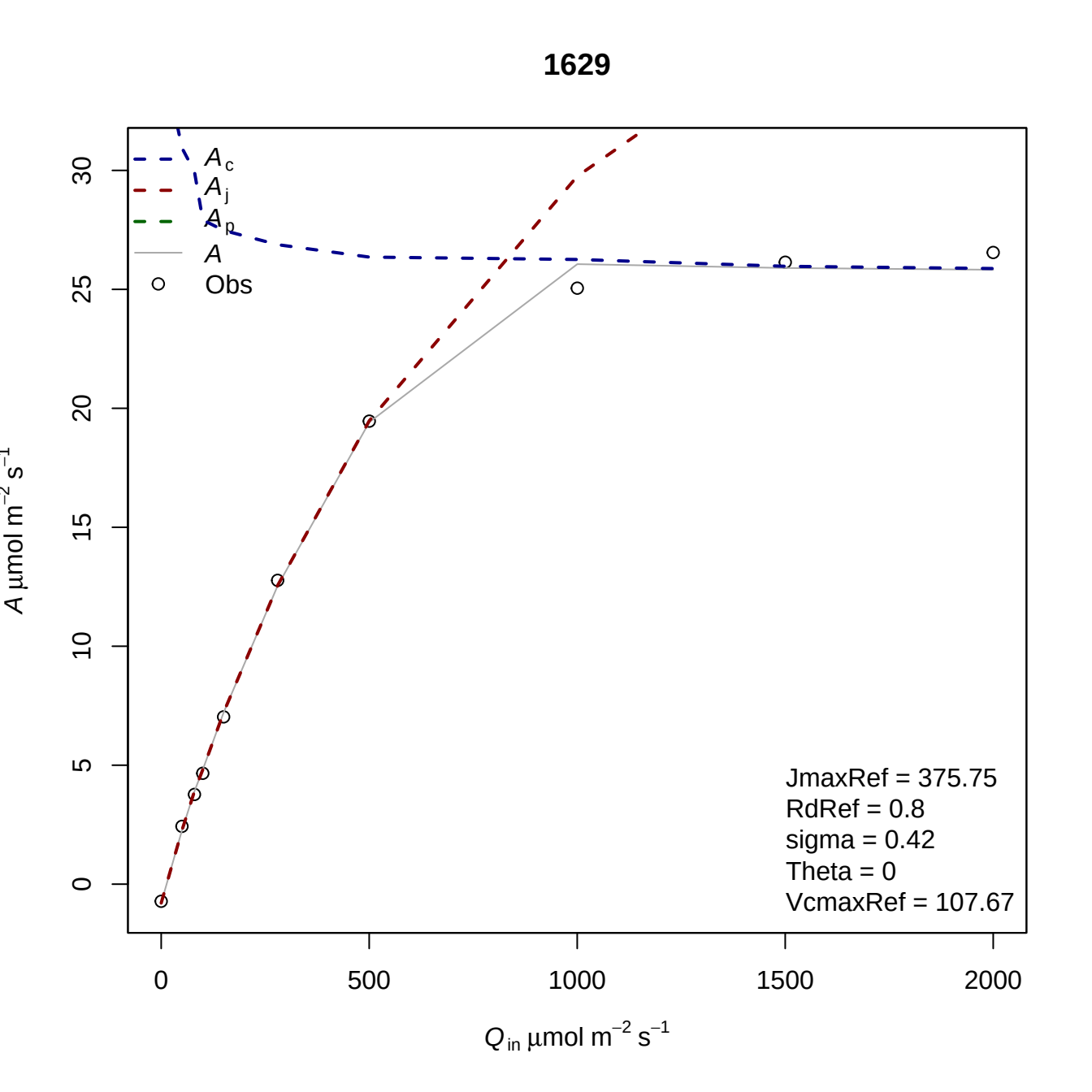

1624
A_c
A_j
A_p
A
Obs
JmaxRef = 137.34
RdRef = 0.41
sigma = 0.23
Theta = 0
VcmaxRef = 47.74
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

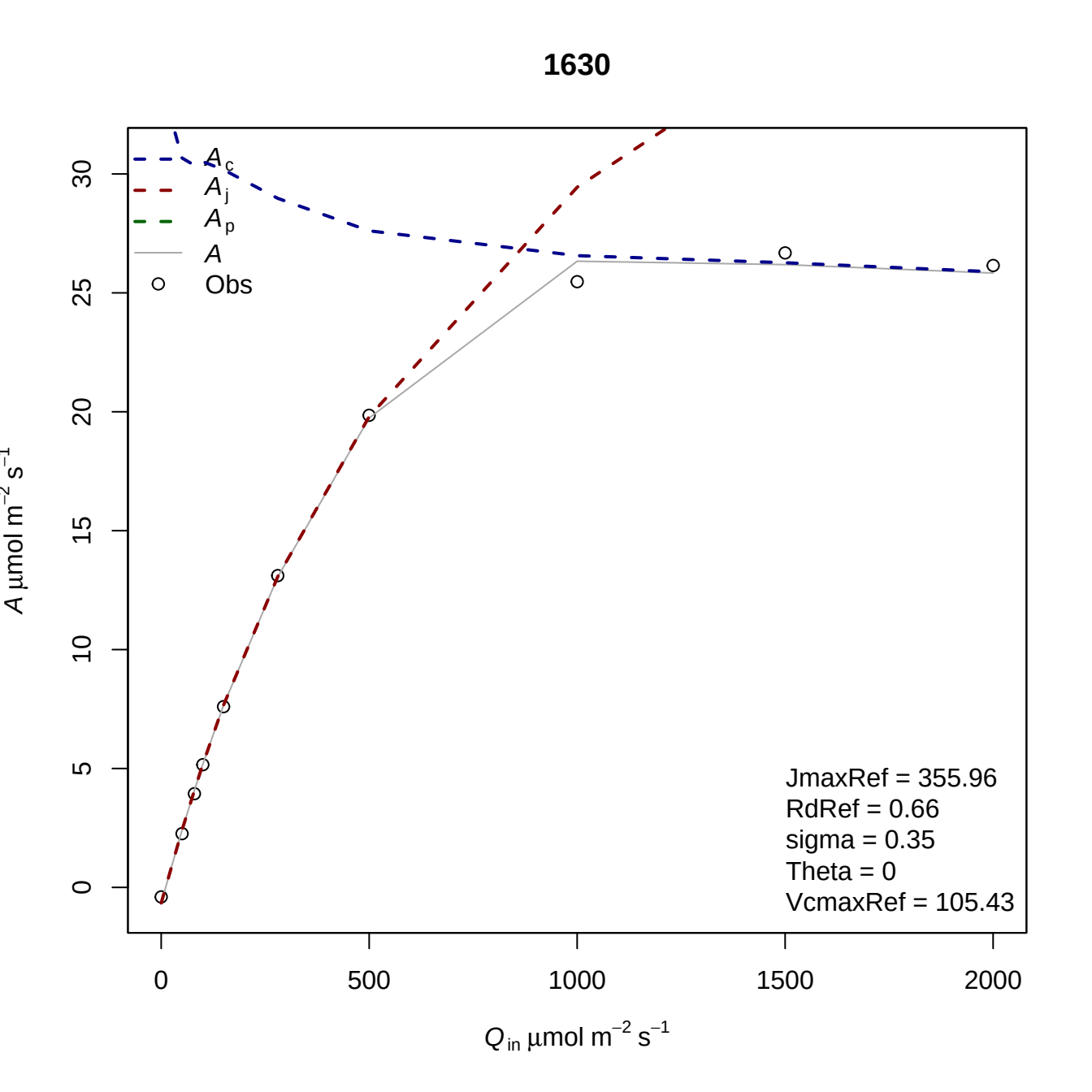

1625
A_c
A_j
A_p
A
Obs
JmaxRef = 82.86
RdRef = 1.01
sigma = 0.21
Theta = 0
VcmaxRef = 41.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

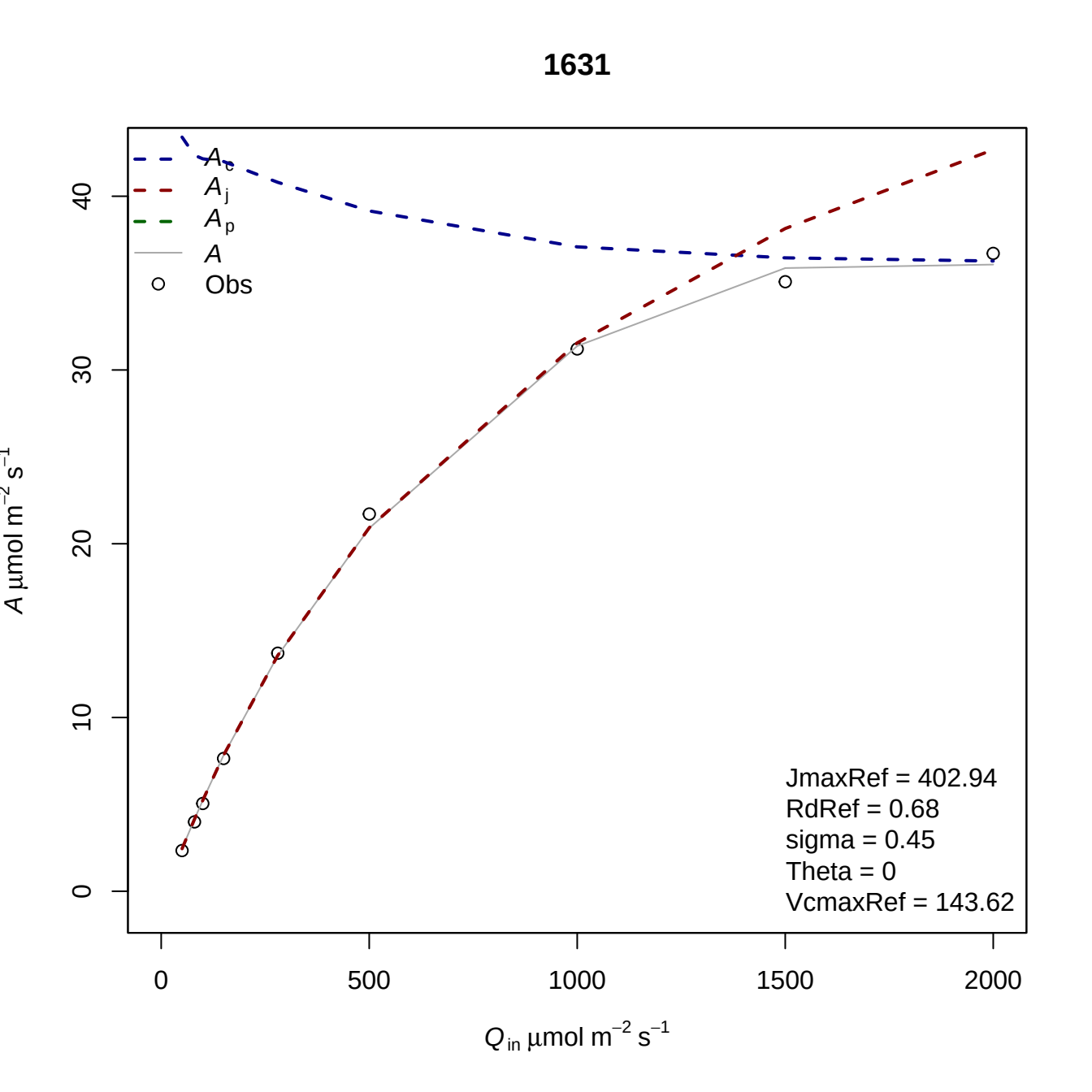

1626
A_c
A_j
A_p
A
Obs
JmaxRef = 116.6
RdRef = 0.94
sigma = 0.32
Theta = 0
VcmaxRef = 46.66
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

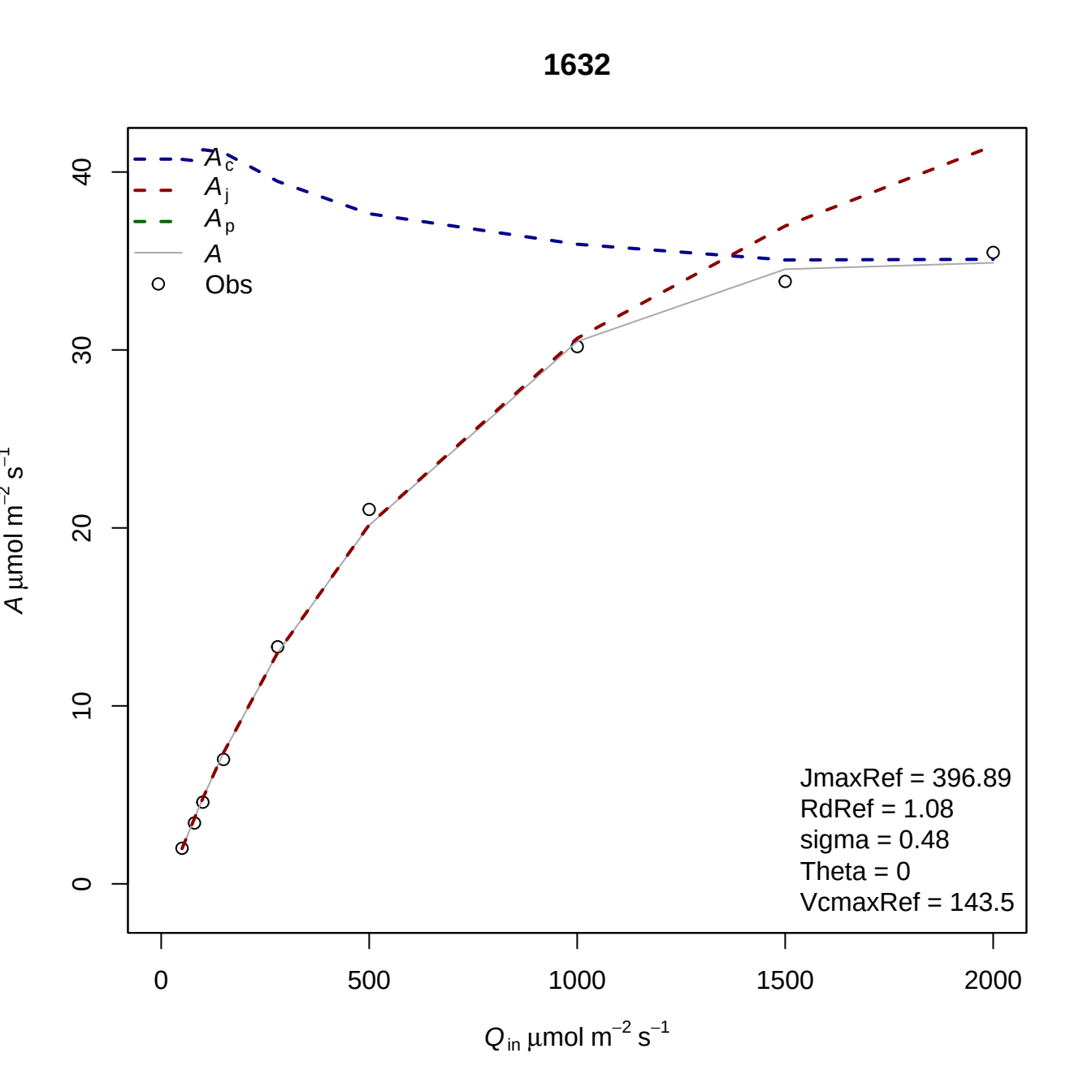

1627
A_c
A_j
A_p
A
Obs
JmaxRef = 205.14
RdRef = 2.53
sigma = 1.02
Theta = 0
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

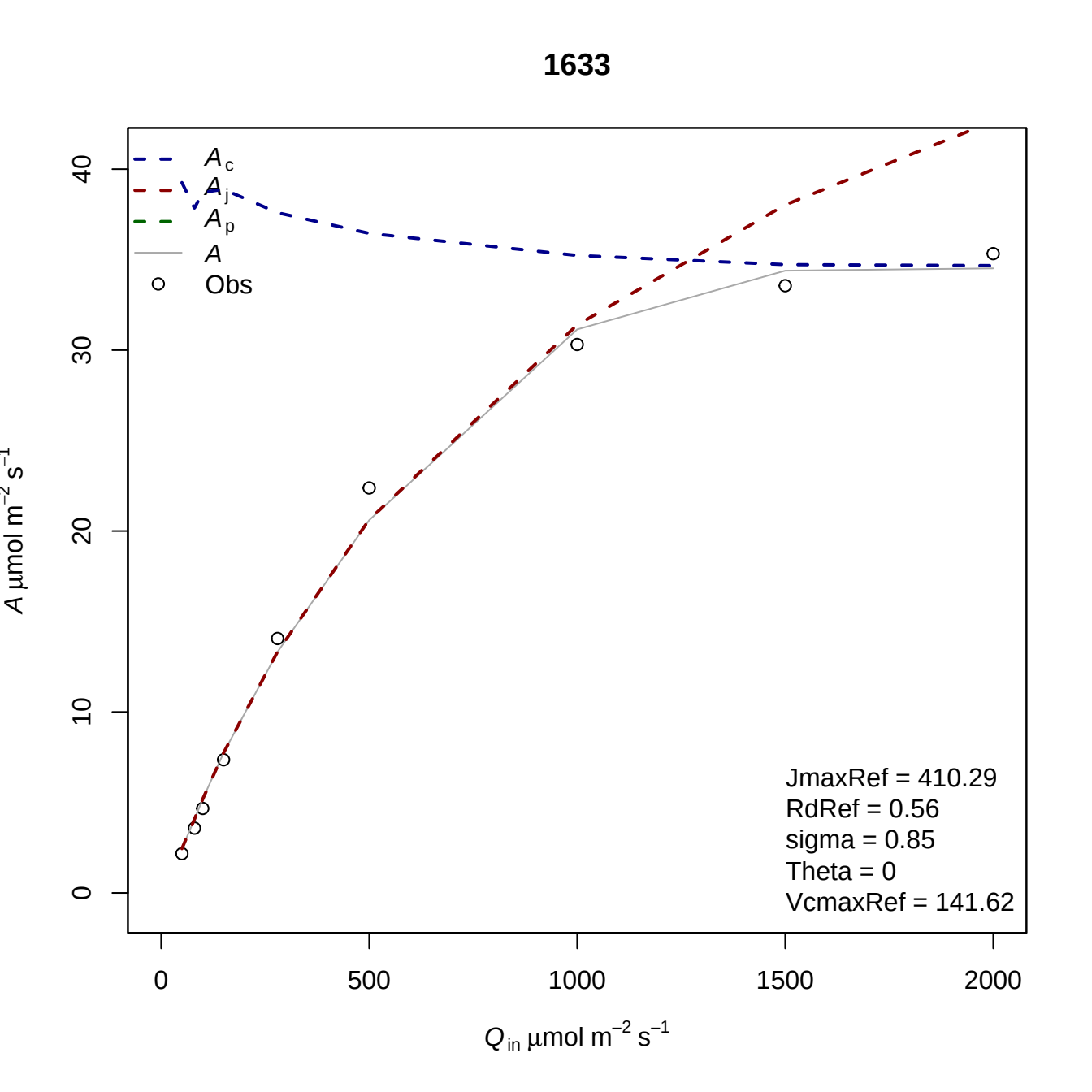

1628
A_c
A_j
A_p
A
Obs
JmaxRef = 193.39
RdRef = 2.26
sigma = 0.8
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

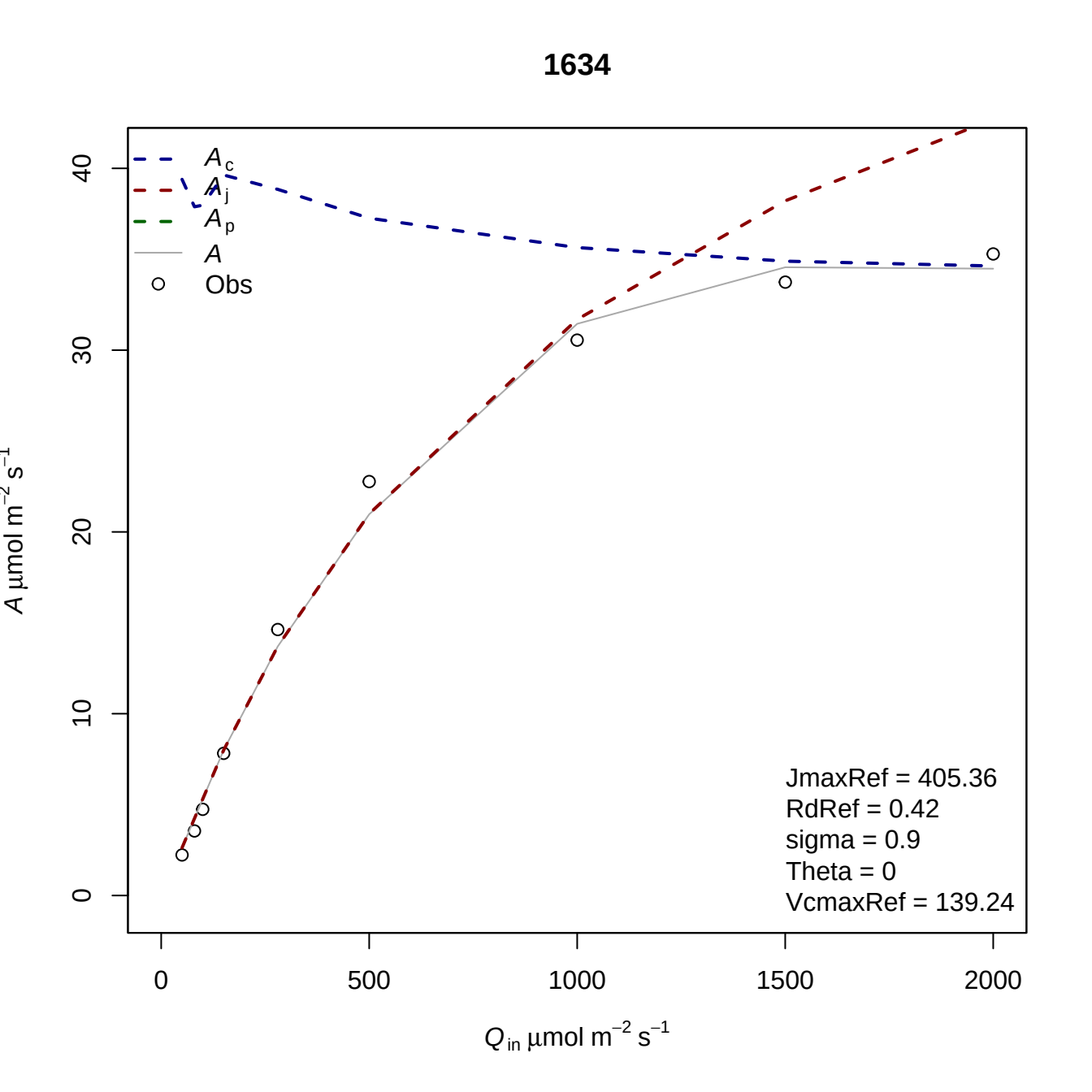

1629
A_c
A_j
A_p
A
Obs
JmaxRef = 375.75
RdRef = 0.8
sigma = 0.42
Theta = 0
VcmaxRef = 107.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

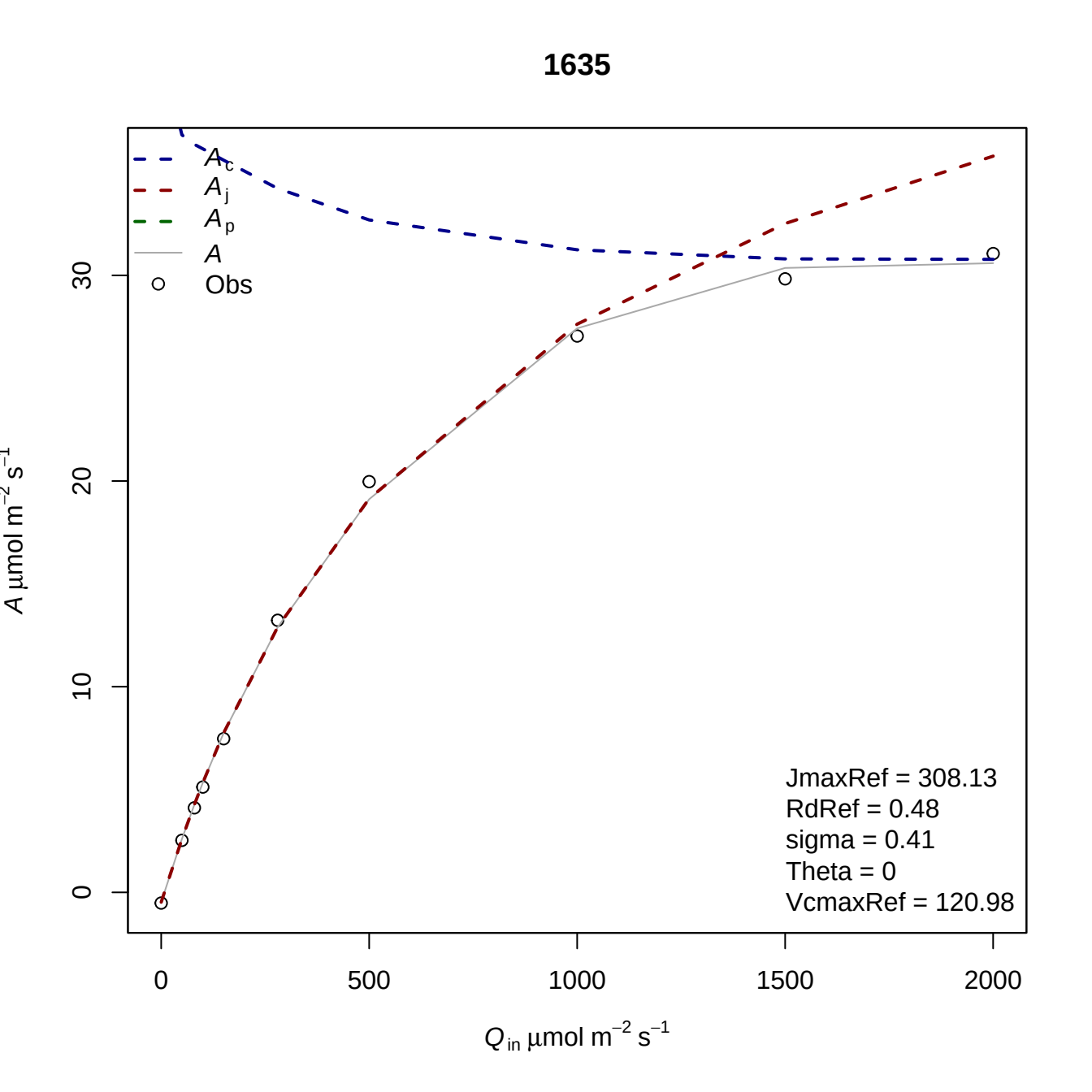

1630
A_c
A_j
A_p
A
Obs
JmaxRef = 355.96
RdRef = 0.66
sigma = 0.35
Theta = 0
VcmaxRef = 105.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

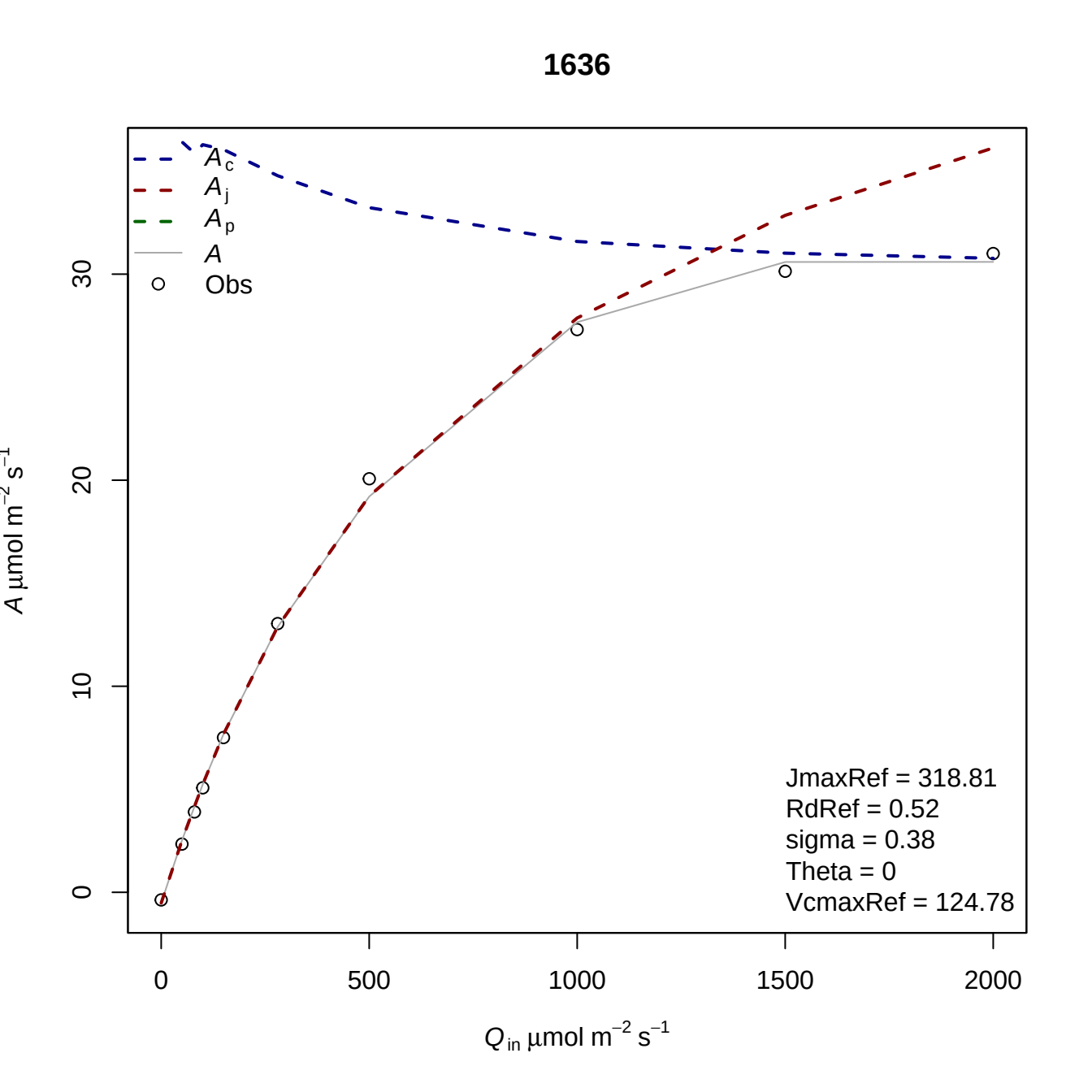

1631
A_c
A_j
A_p
A
Obs
JmaxRef = 402.94
RdRef = 0.68
sigma = 0.45
Theta = 0
VcmaxRef = 143.62
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

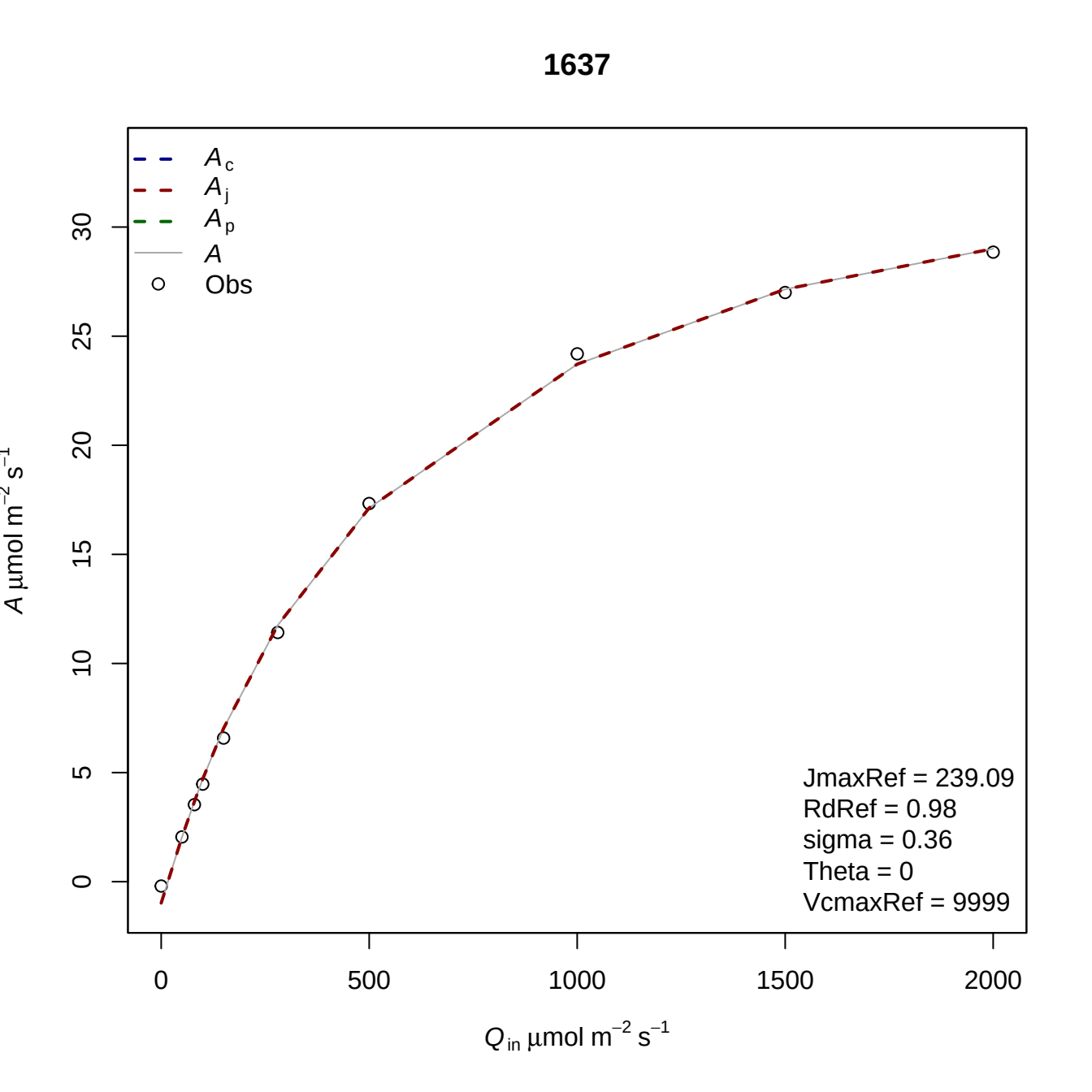

1632
A_c
A_j
A_p
A
Obs
JmaxRef = 396.89
RdRef = 1.08
sigma = 0.48
Theta = 0
VcmaxRef = 143.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

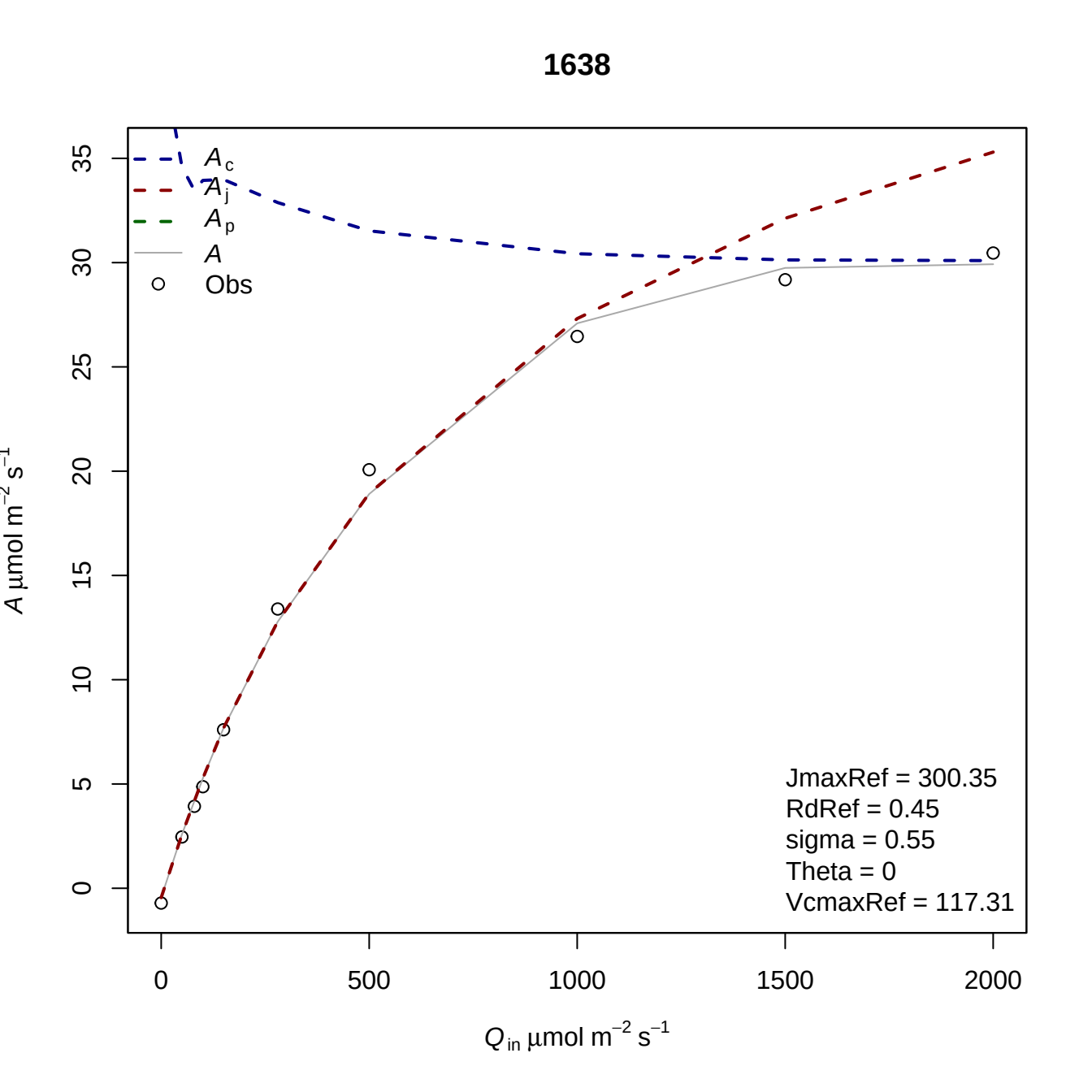

1633
A_c
A_j
A_p
A
Obs
JmaxRef = 410.29
RdRef = 0.56
sigma = 0.85
Theta = 0
VcmaxRef = 141.62
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

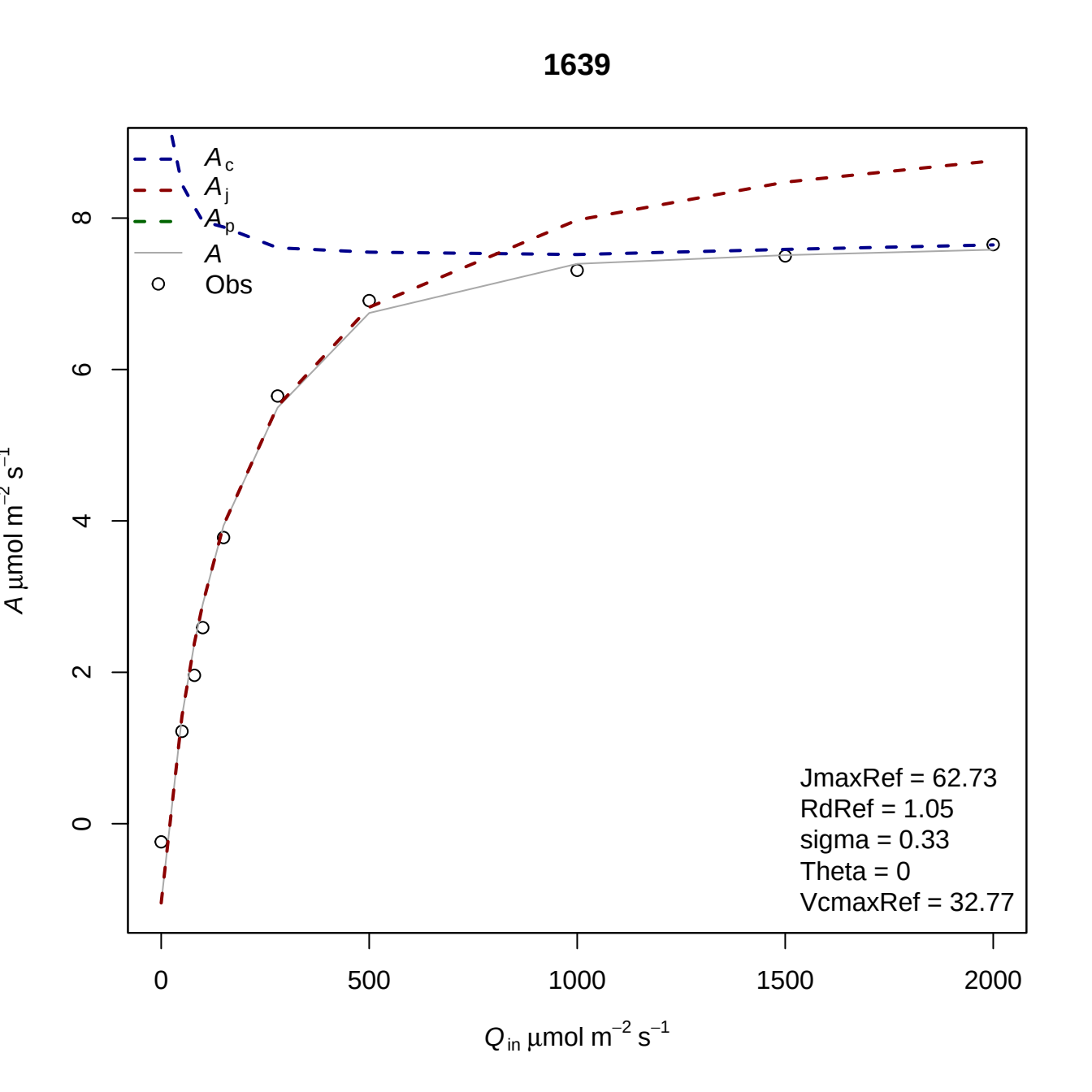

1634
A_c
A_j
A_p
A
Obs
JmaxRef = 405.36
RdRef = 0.42
sigma = 0.9
Theta = 0
VcmaxRef = 139.24
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

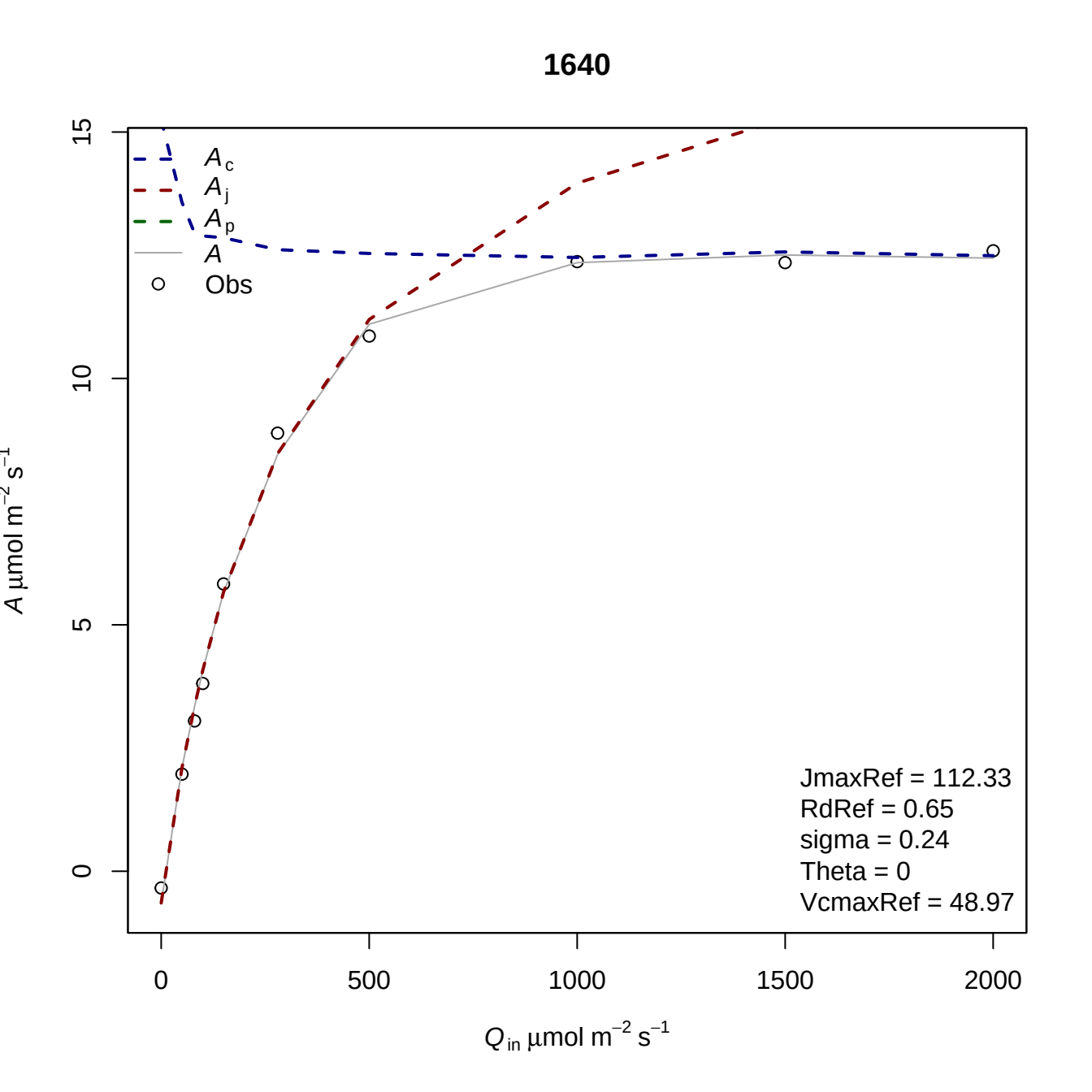

1635
A_c
A_j
A_p
A
Obs
JmaxRef = 308.13
RdRef = 0.48
sigma = 0.41
Theta = 0
VcmaxRef = 120.98
0
10
20
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

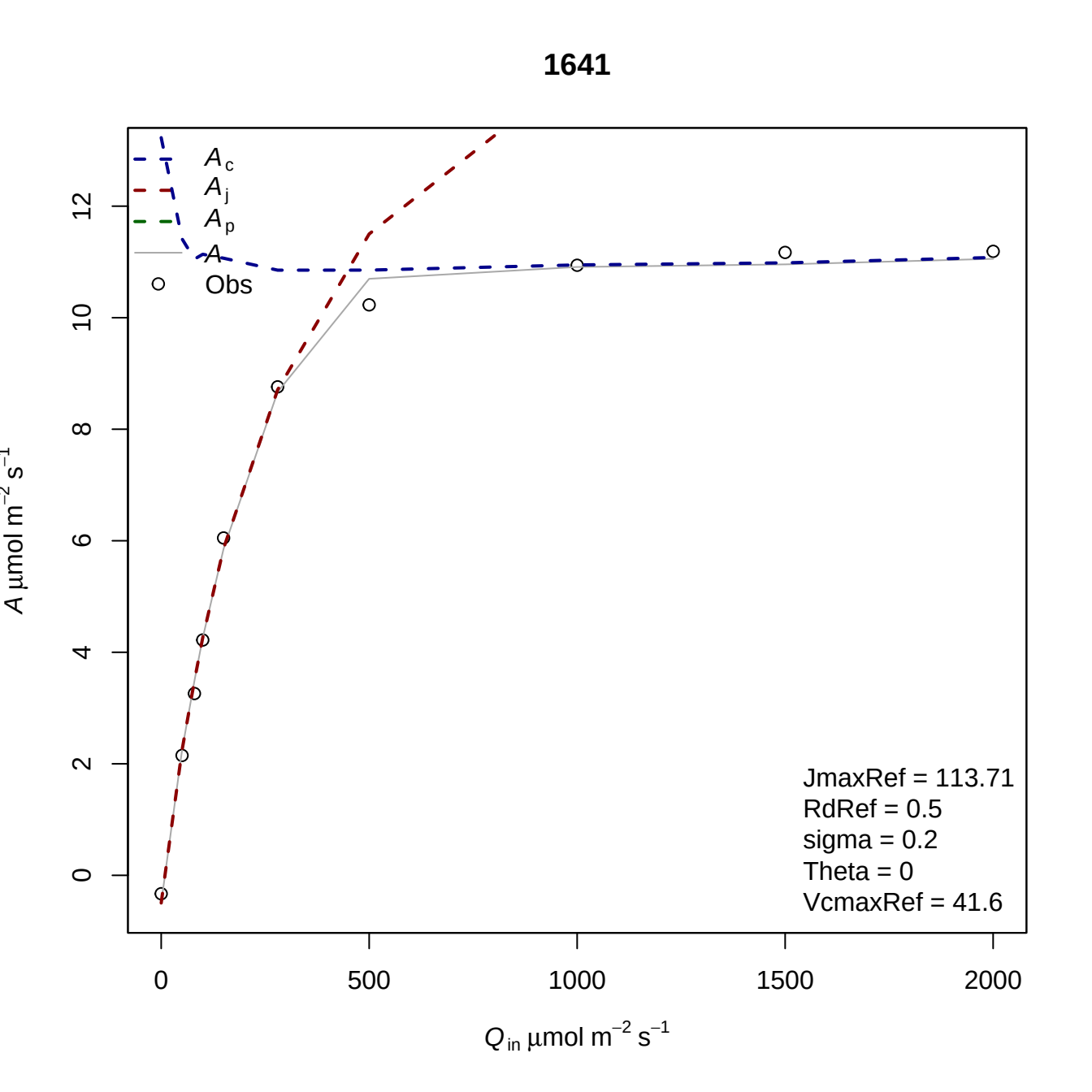

1636
A_c
A_j
A_p
A
Obs
JmaxRef = 318.81
RdRef = 0.52
sigma = 0.38
Theta = 0
VcmaxRef = 124.78
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

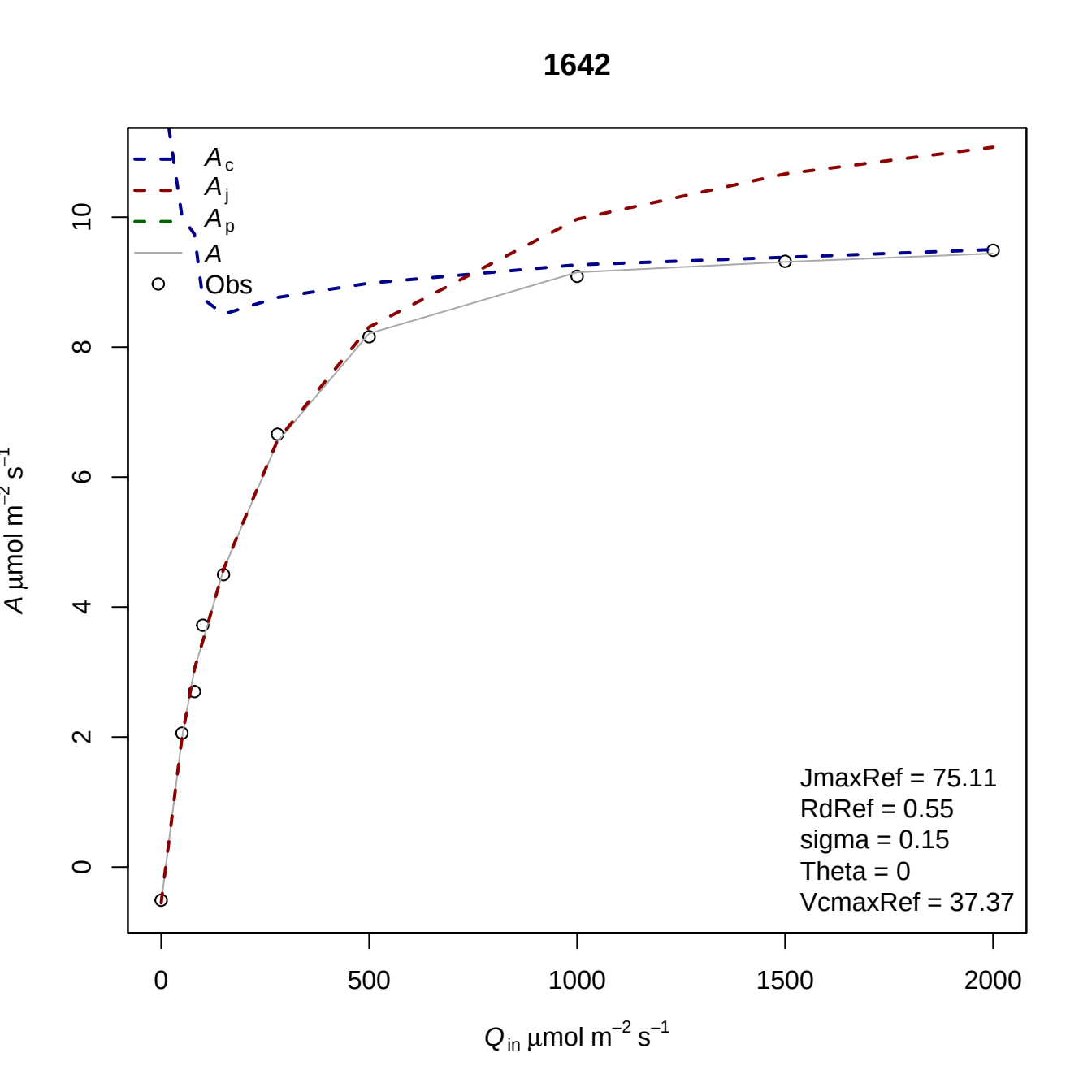

1637
A_c
A_j
A_p
A
Obs
JmaxRef = 239.09
RdRef = 0.98
sigma = 0.36
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

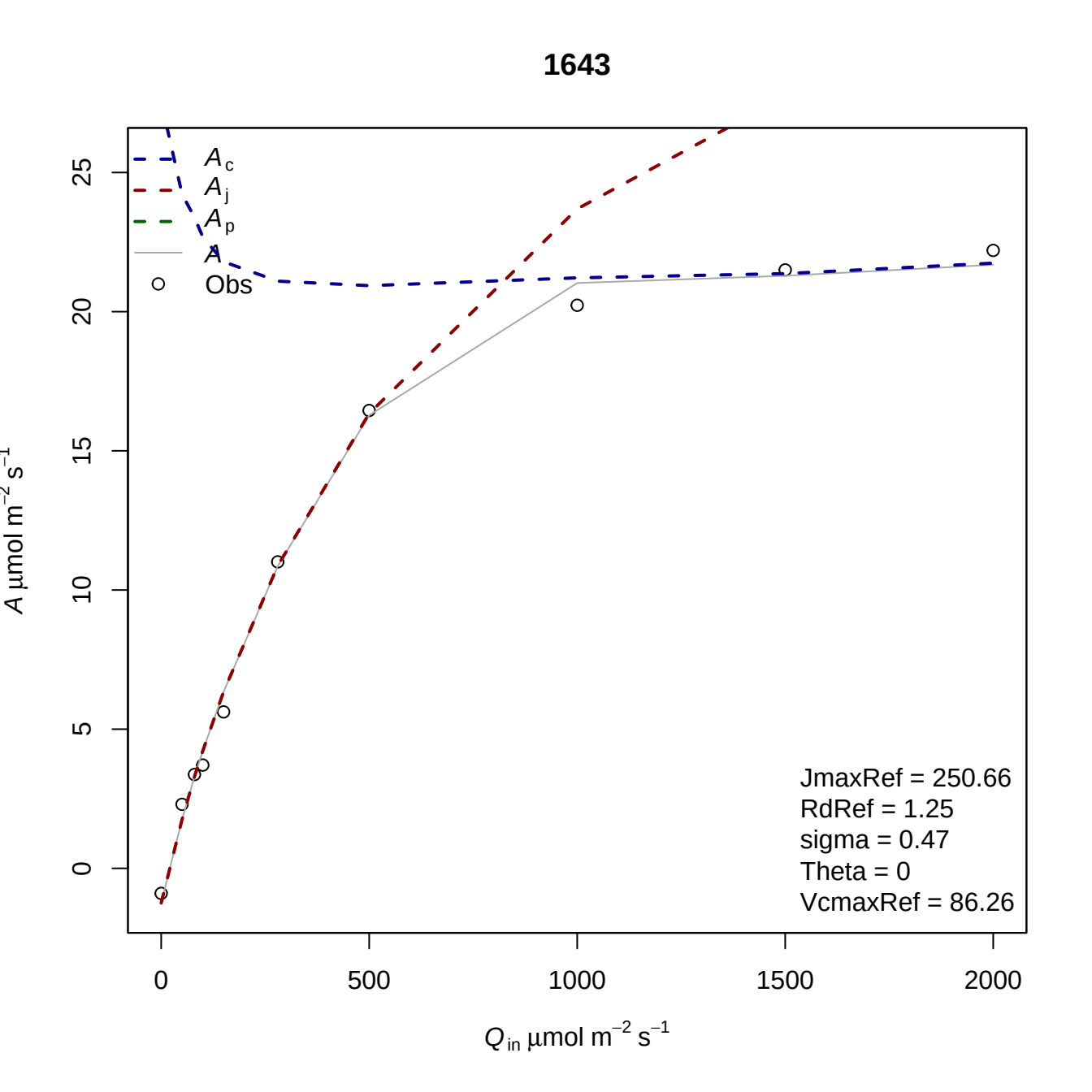

1638
A_c
A_j
A_p
A
Obs
JmaxRef = 300.35
RdRef = 0.45
sigma = 0.55
Theta = 0
VcmaxRef = 117.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

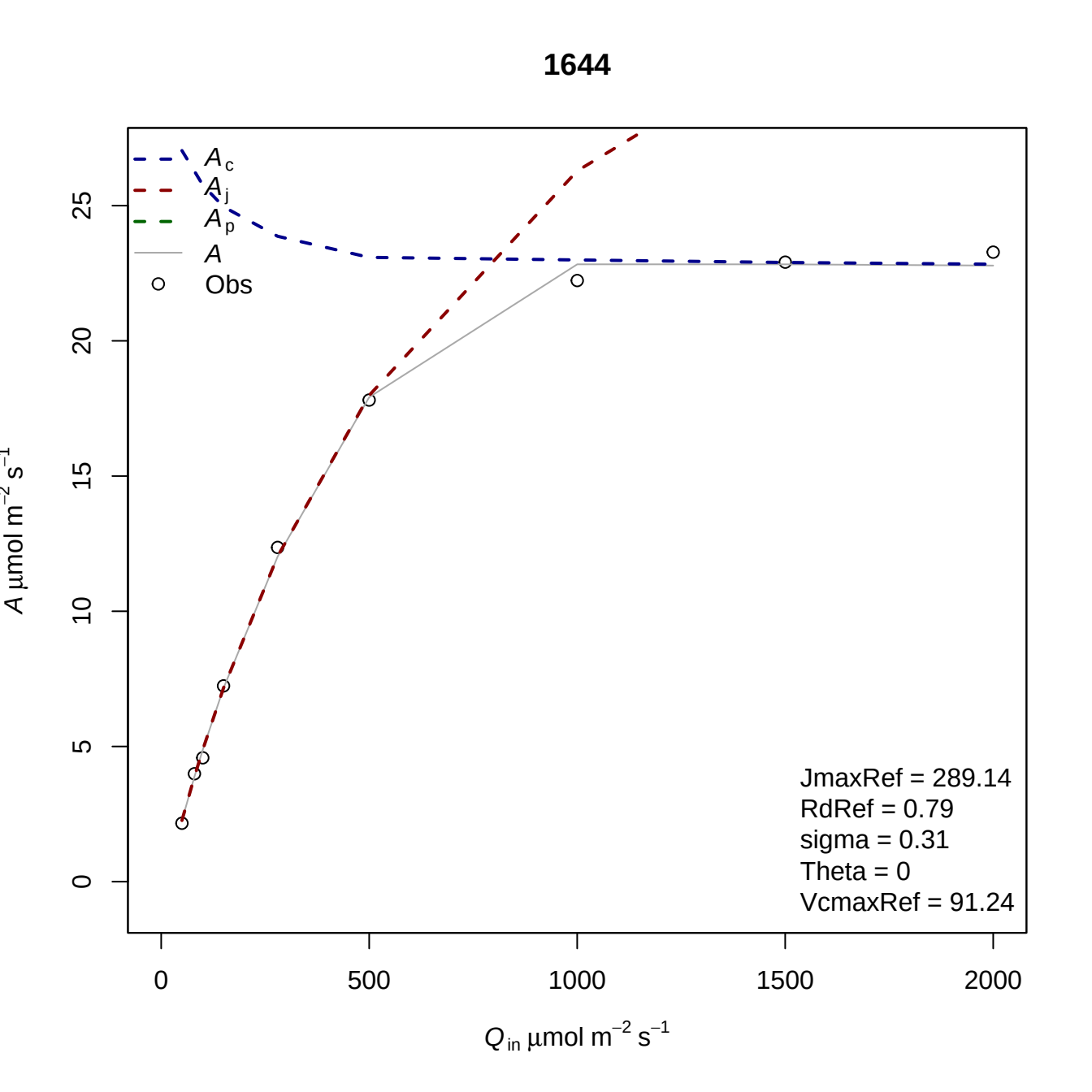

1639
A_c
A_j
A_p
A
Obs
JmaxRef = 62.73
RdRef = 1.05
sigma = 0.33
Theta = 0
VcmaxRef = 32.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

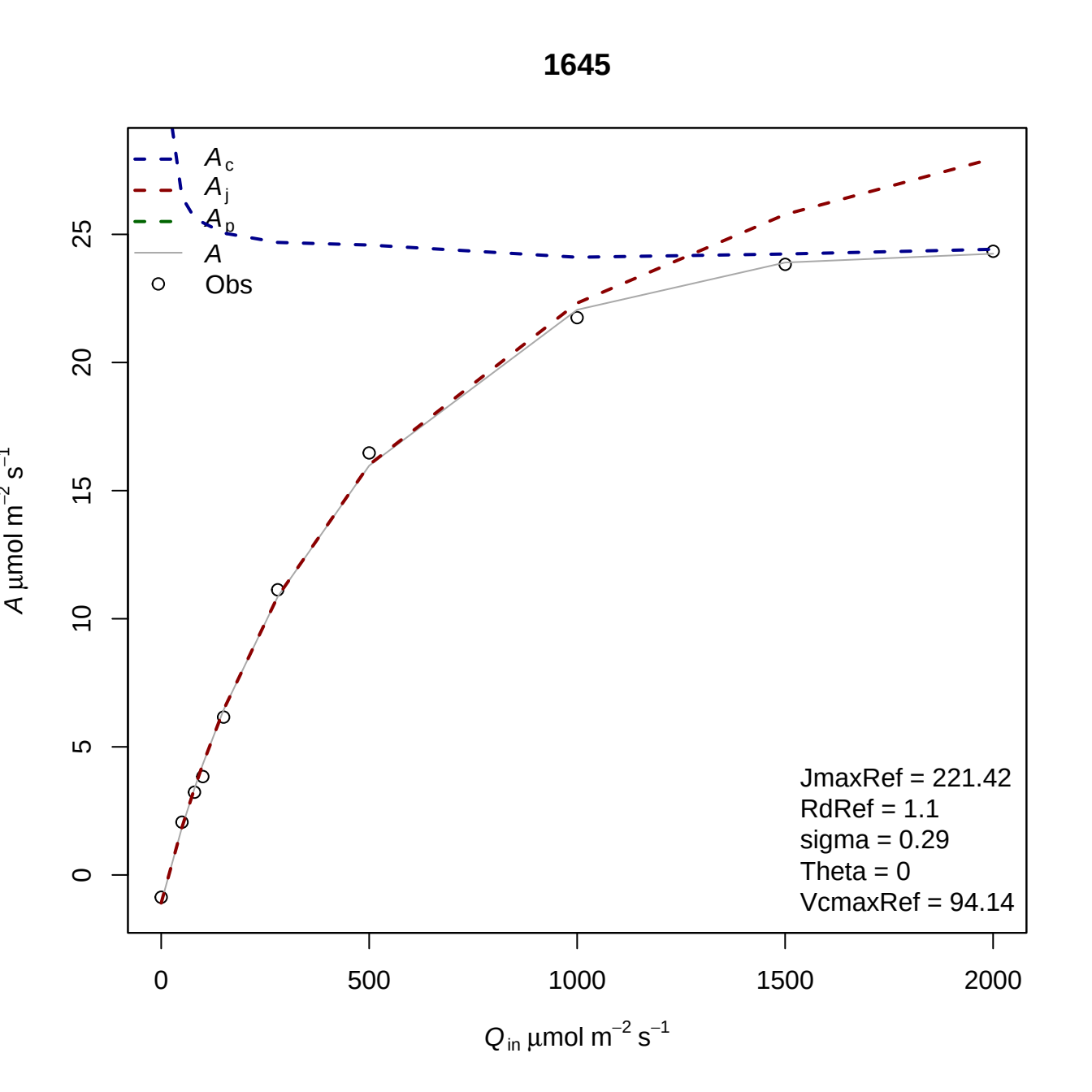

1640
A_c
A_j
A_p
A
Obs
JmaxRef = 112.33
RdRef = 0.65
sigma = 0.24
Theta = 0
VcmaxRef = 48.97
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

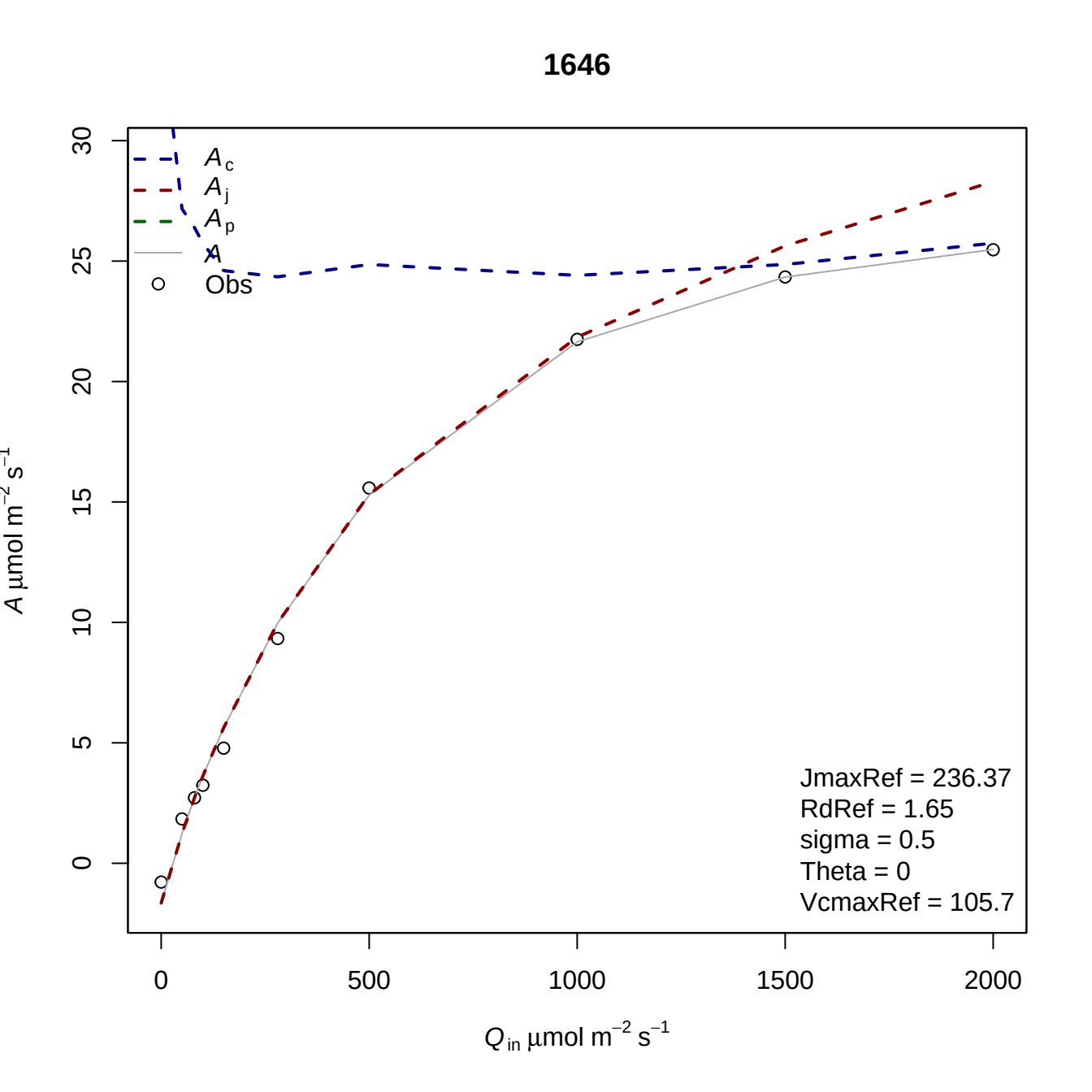

1641
A_c
A_j
A_p
A
Obs
JmaxRef = 113.71
RdRef = 0.5
sigma = 0.2
Theta = 0
VcmaxRef = 41.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

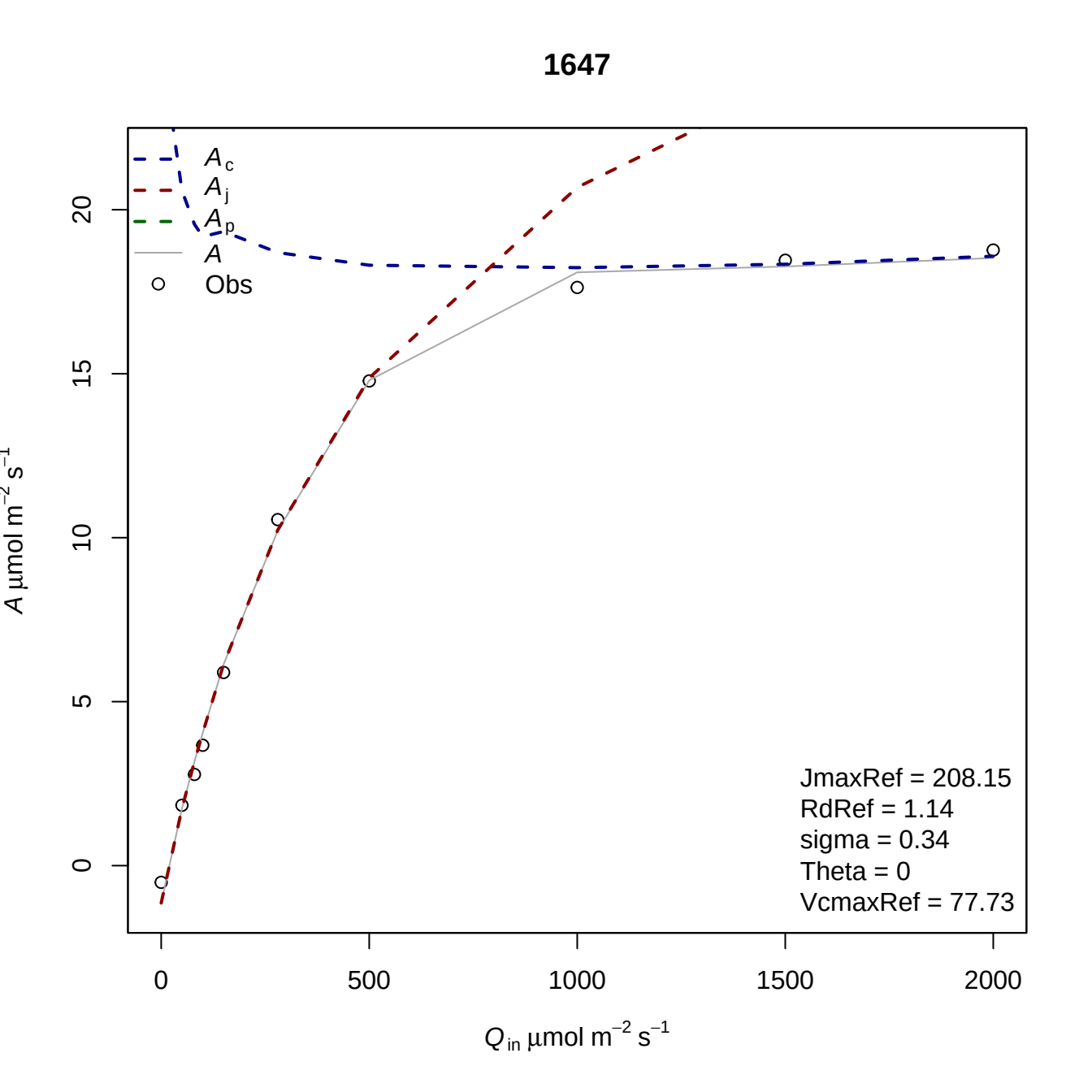

1642
A_c
A_j
A_p
A
Obs
JmaxRef = 75.11
RdRef = 0.55
sigma = 0.15
Theta = 0
VcmaxRef = 37.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

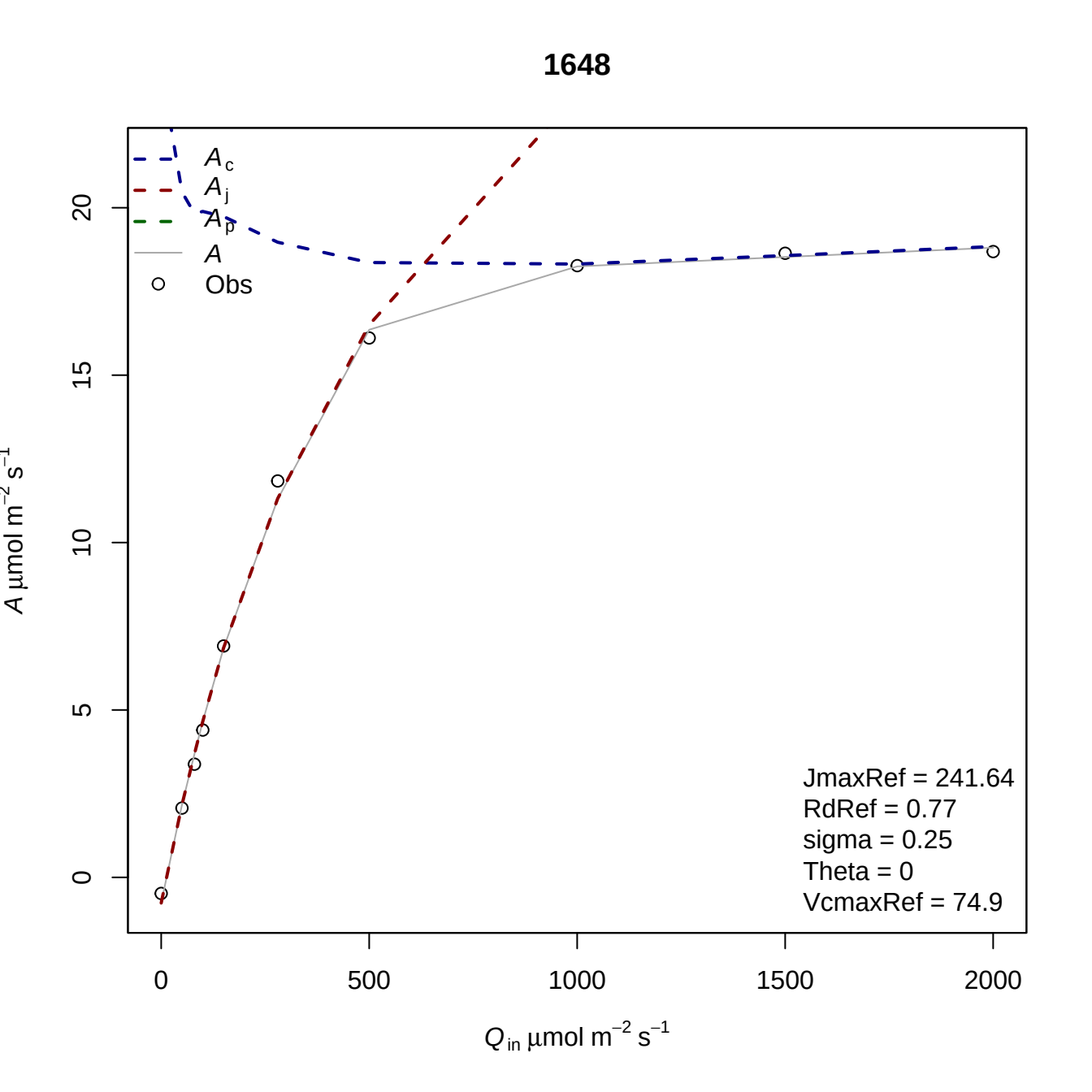

1643
A_c
A_j
A_p
A
Obs
JmaxRef = 250.66
RdRef = 1.25
sigma = 0.47
Theta = 0
VcmaxRef = 86.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

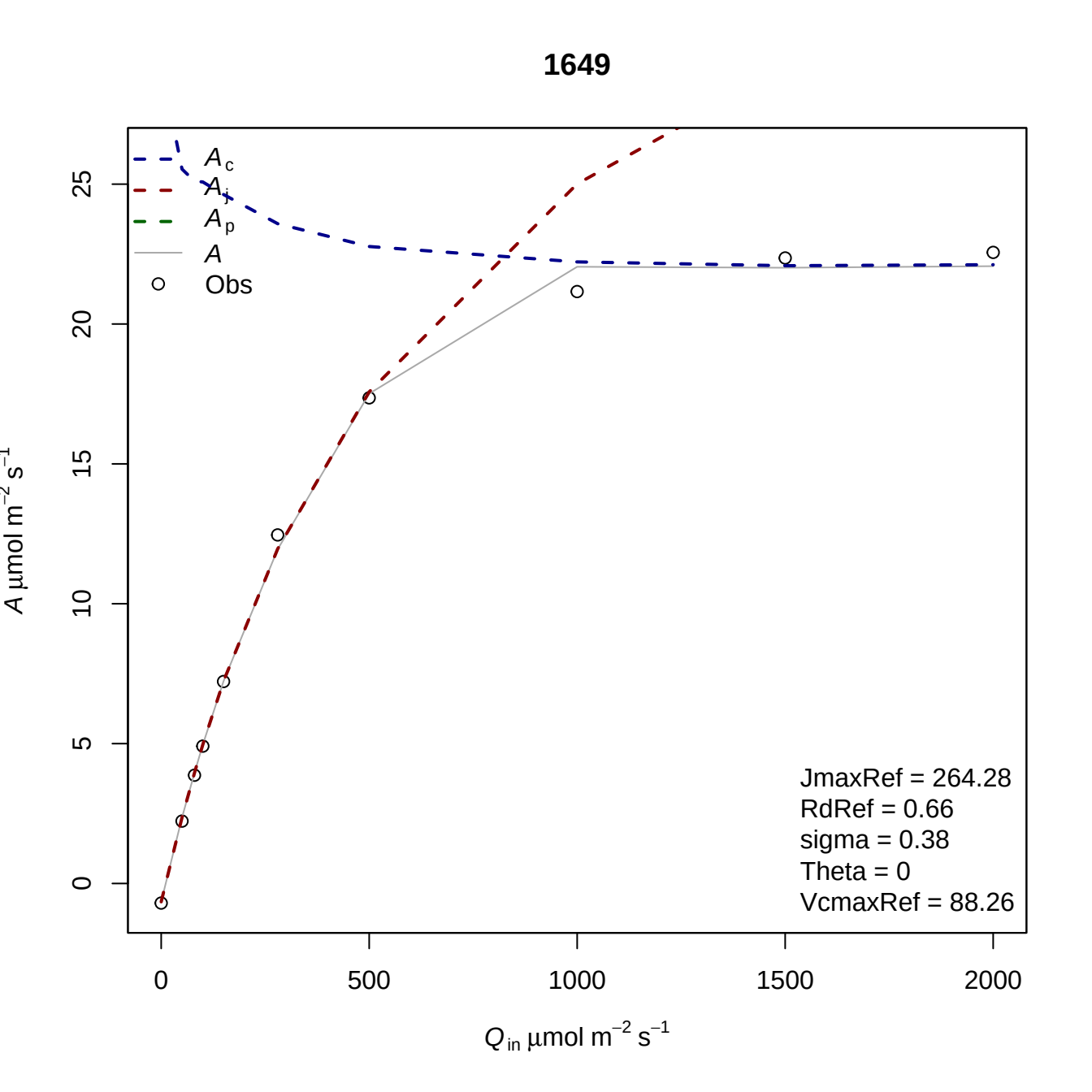

1644
A_c
A_j
A_p
A
Obs
JmaxRef = 289.14
RdRef = 0.79
sigma = 0.31
Theta = 0
VcmaxRef = 91.24
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

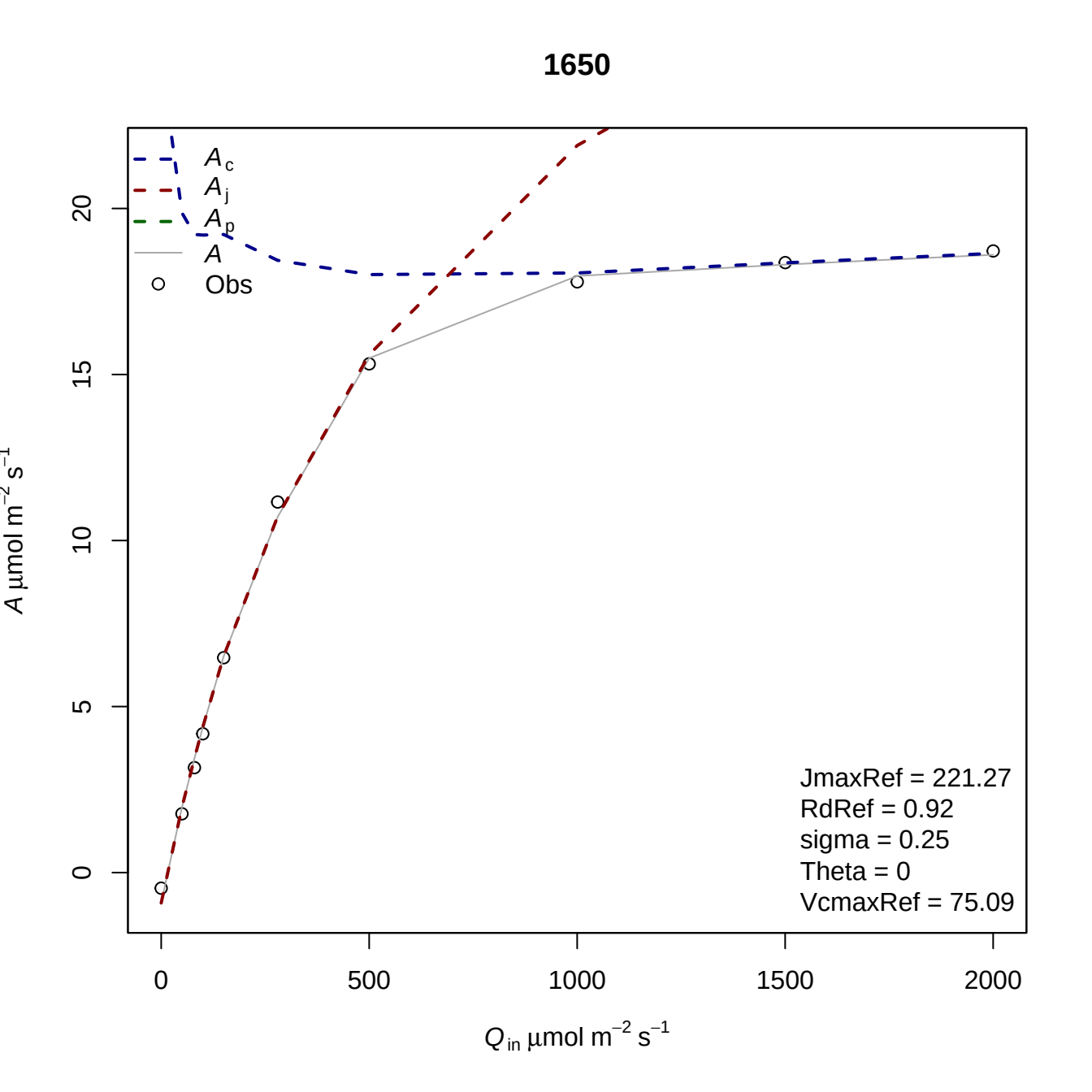

1645
A_c
A_j
A_p
A
Obs
JmaxRef = 221.42
RdRef = 1.1
sigma = 0.29
Theta = 0
VcmaxRef = 94.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

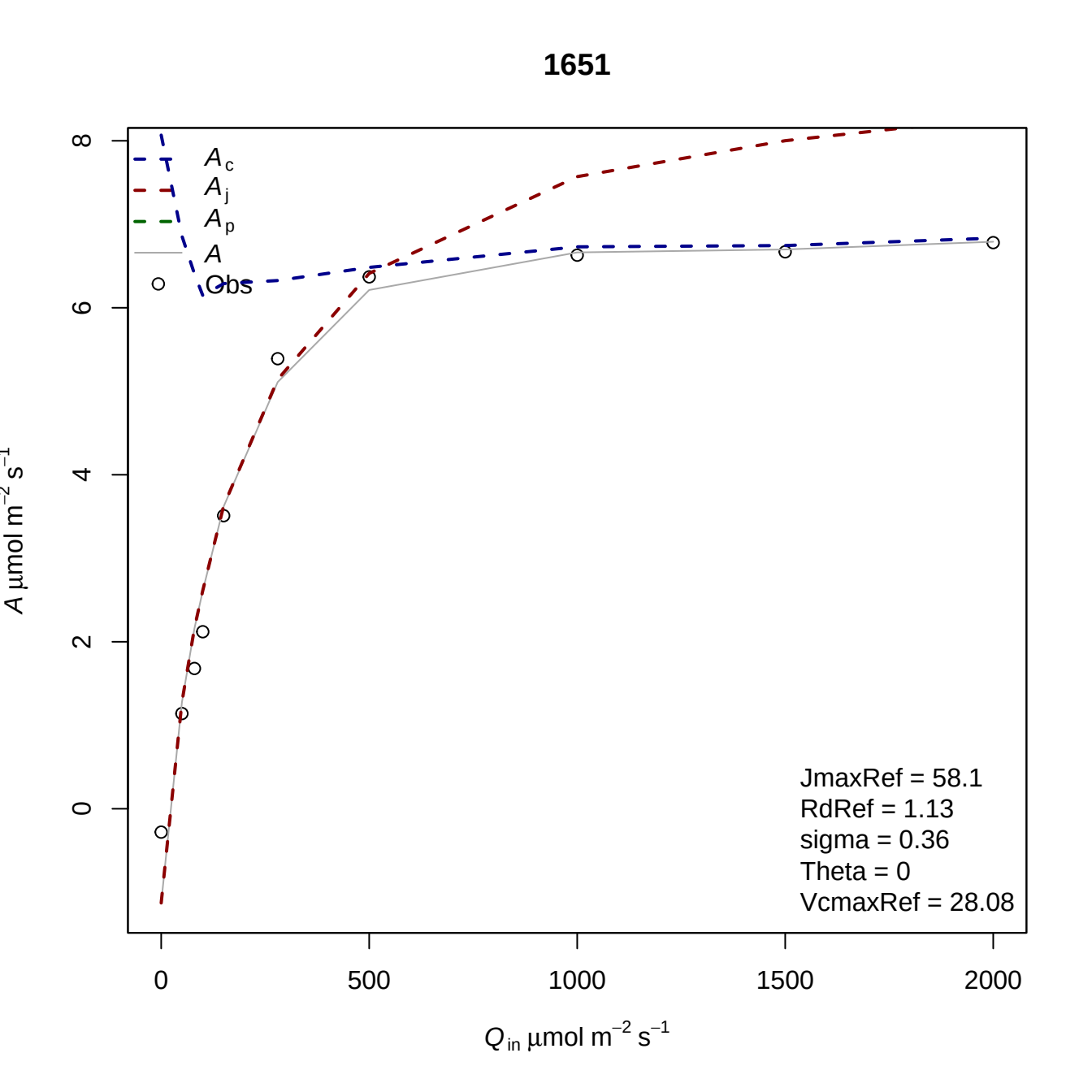

1646
A_c
A_j
A_p
A
Obs
JmaxRef = 236.37
RdRef = 1.65
sigma = 0.5
Theta = 0
VcmaxRef = 105.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

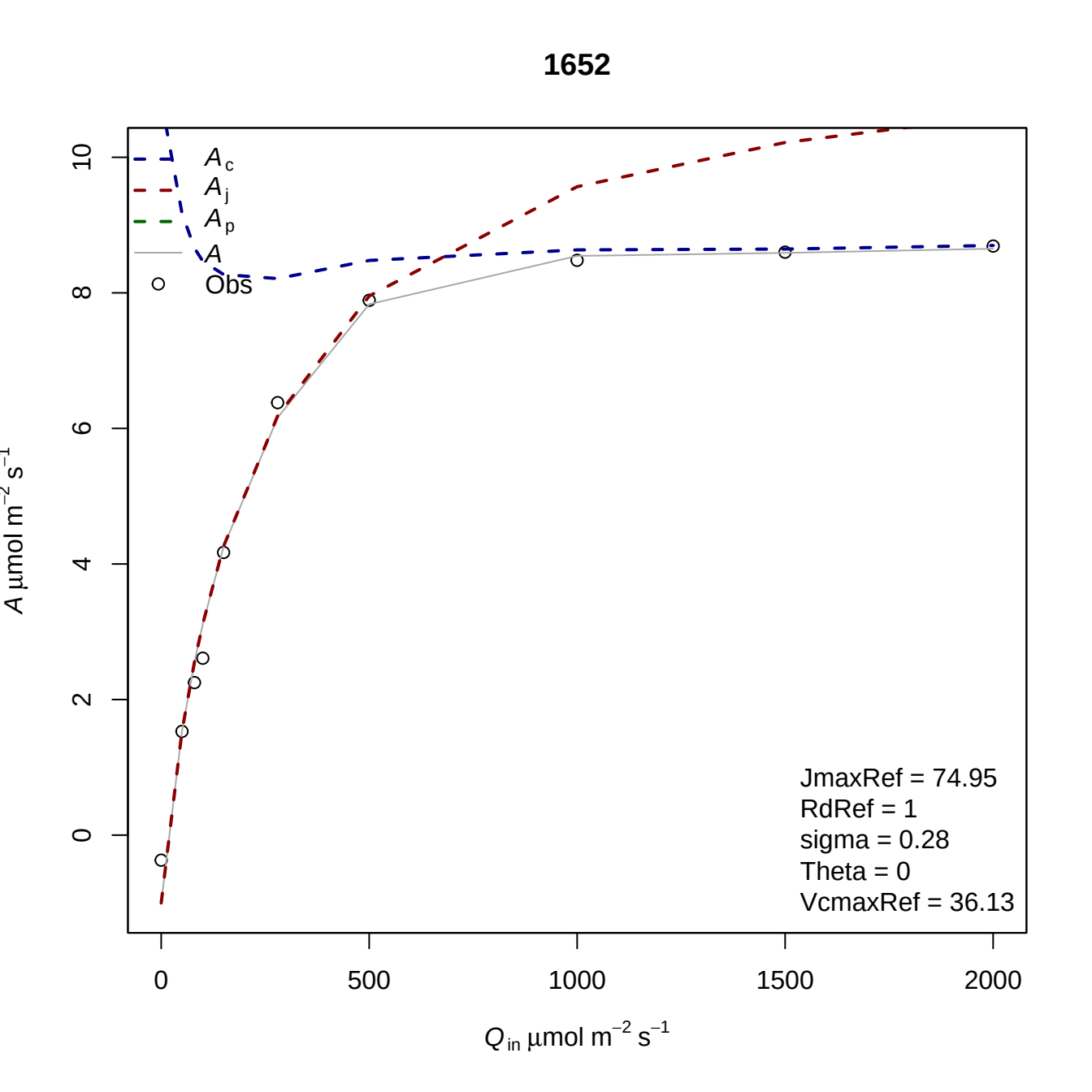

1647
A_c
A_j
A_p
A
Obs
JmaxRef = 208.15
RdRef = 1.14
sigma = 0.34
Theta = 0
VcmaxRef = 77.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

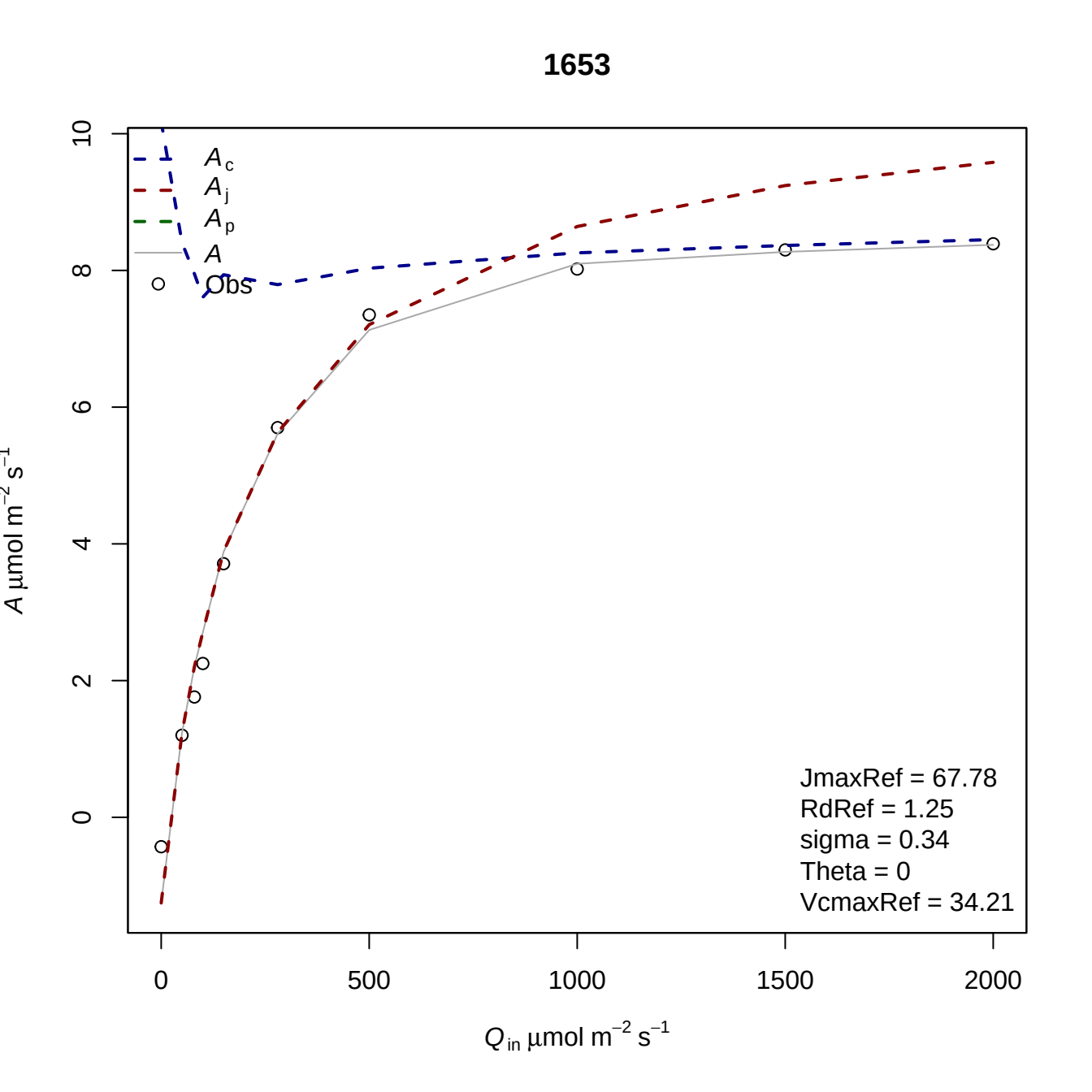

1648
A_c
A_j
A_p
A
Obs
JmaxRef = 241.64
RdRef = 0.77
sigma = 0.25
Theta = 0
VcmaxRef = 74.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

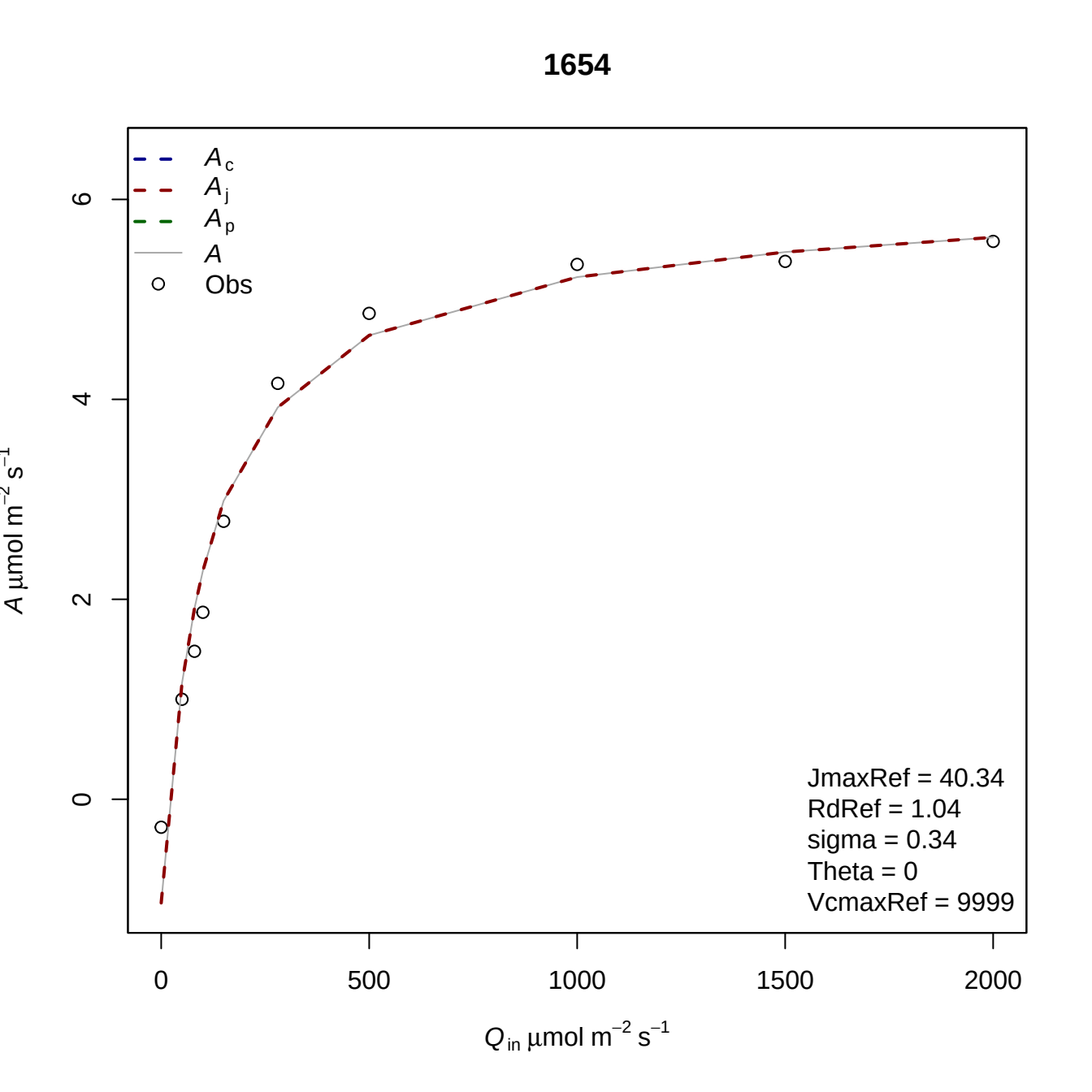

1649
A_c
A_j
A_p
A
Obs
JmaxRef = 264.28
RdRef = 0.66
sigma = 0.38
Theta = 0
VcmaxRef = 88.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

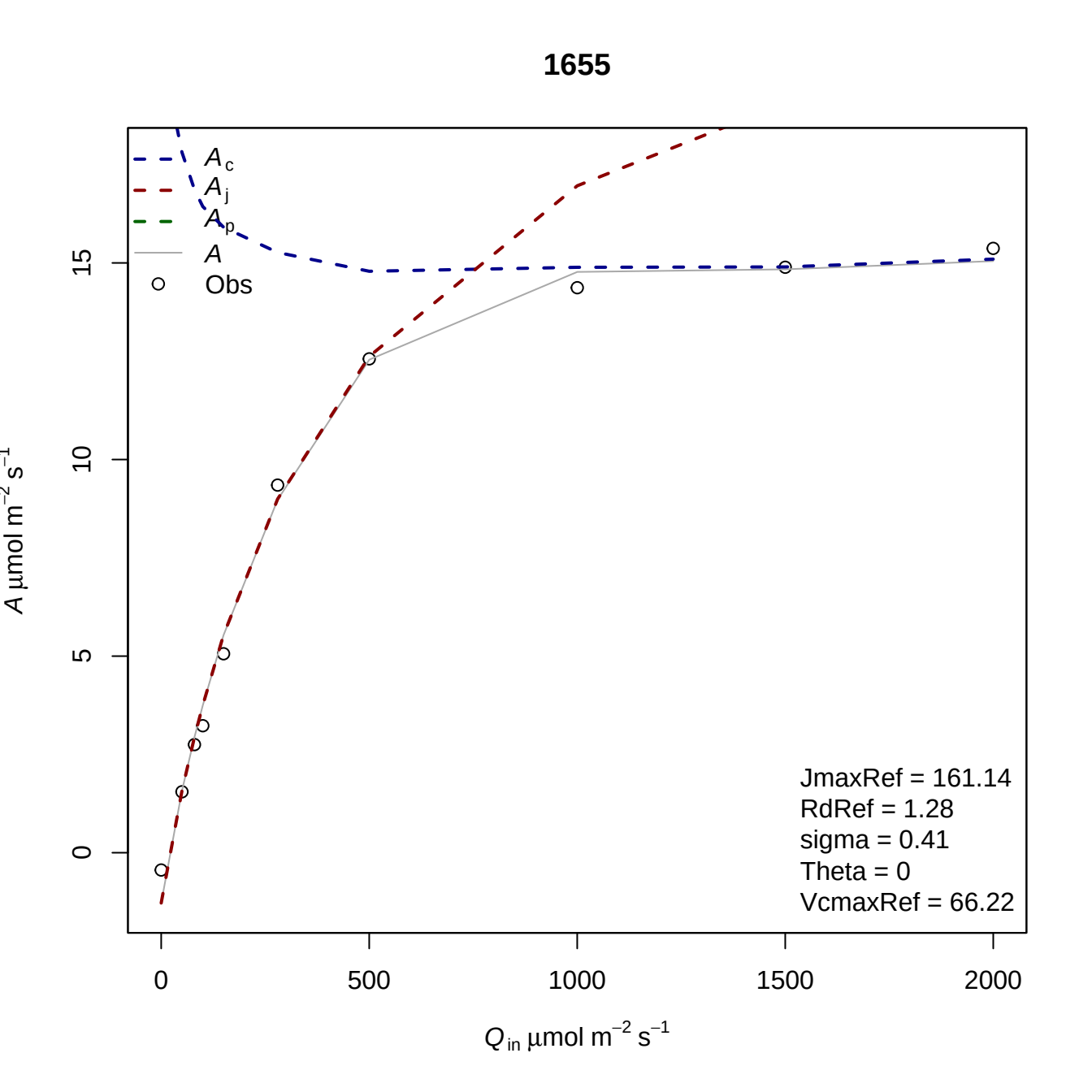

1650
A_c
A_j
A_p
A
Obs
JmaxRef = 221.27
RdRef = 0.92
sigma = 0.25
Theta = 0
VcmaxRef = 75.09
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

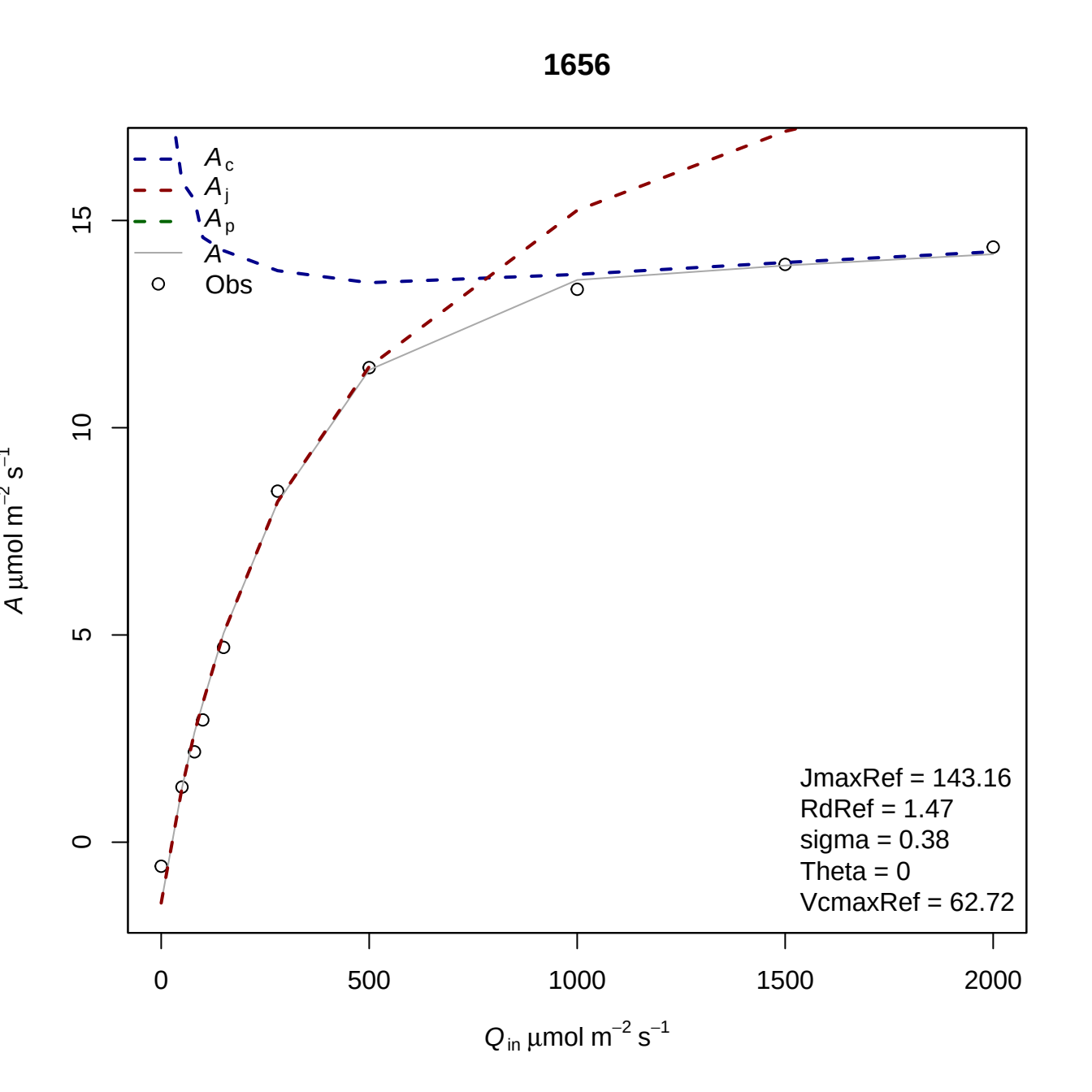

1651
A_c
A_j
A_p
A
Obs
JmaxRef = 58.1
RdRef = 1.13
sigma = 0.36
Theta = 0
VcmaxRef = 28.08
0
2
4
6
8
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

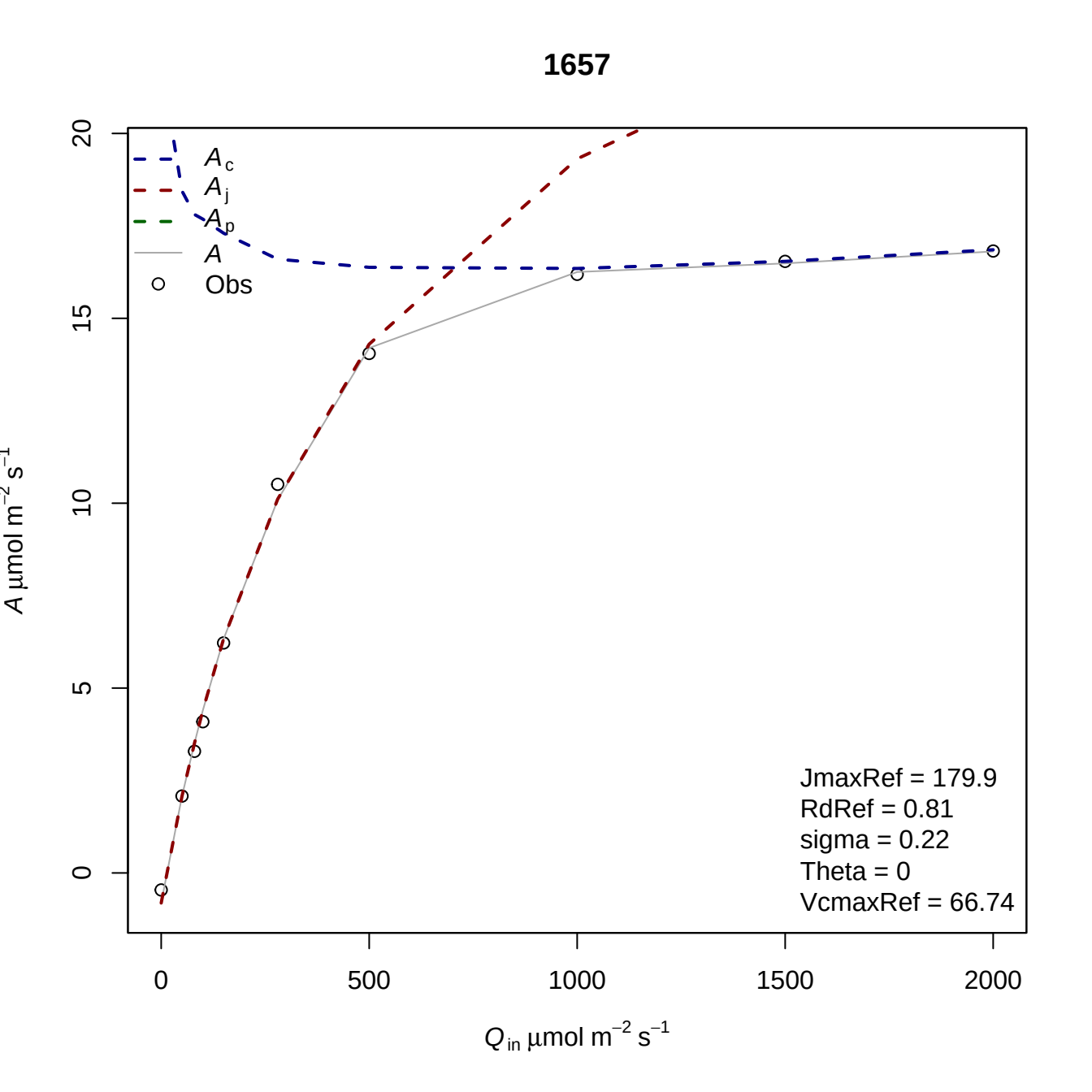

1652
A_c
A_j
A_p
A
Obs
JmaxRef = 74.95
RdRef = 1
sigma = 0.28
Theta = 0
VcmaxRef = 36.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

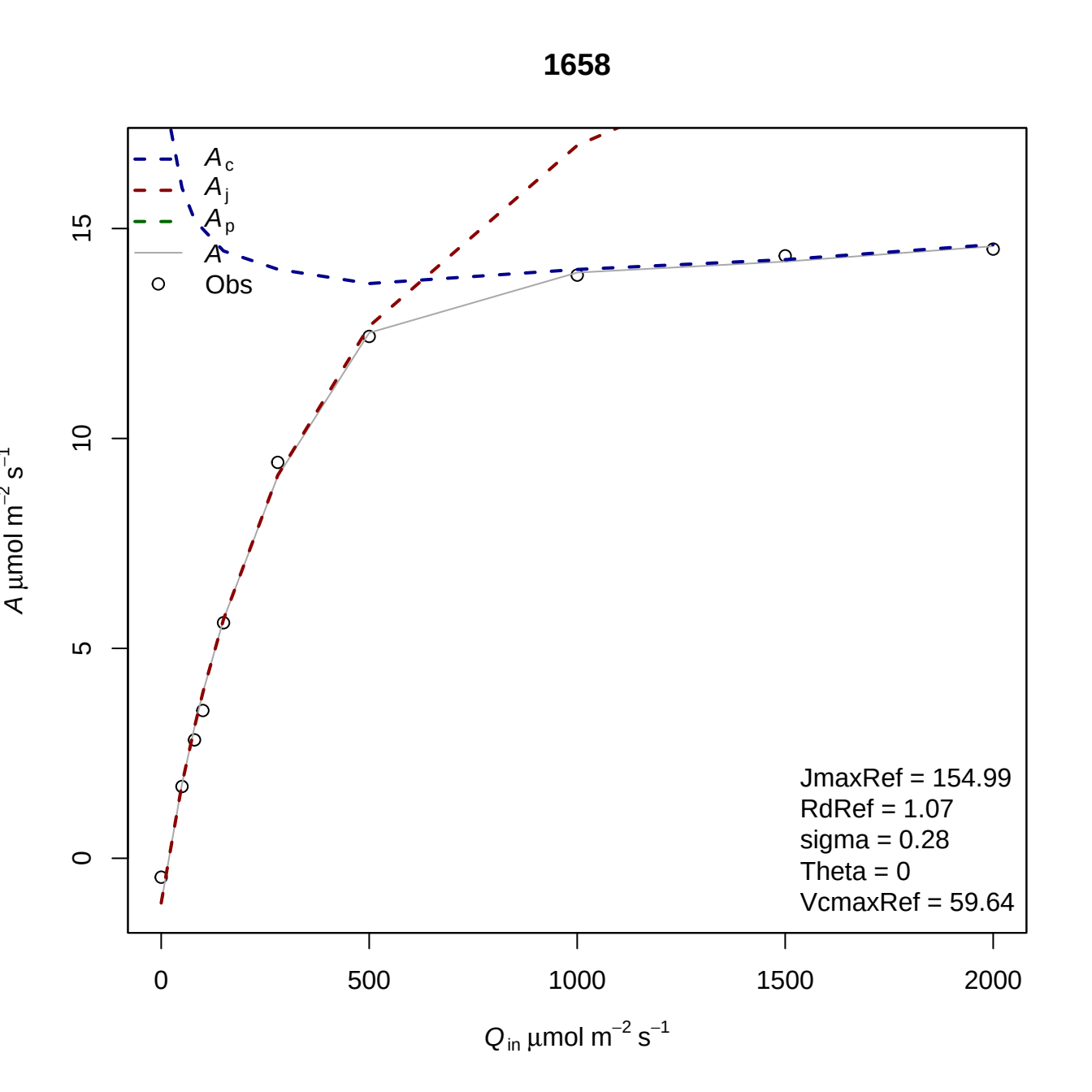

1653
A_c
A_j
A_p
A
Obs
JmaxRef = 67.78
RdRef = 1.25
sigma = 0.34
Theta = 0
VcmaxRef = 34.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

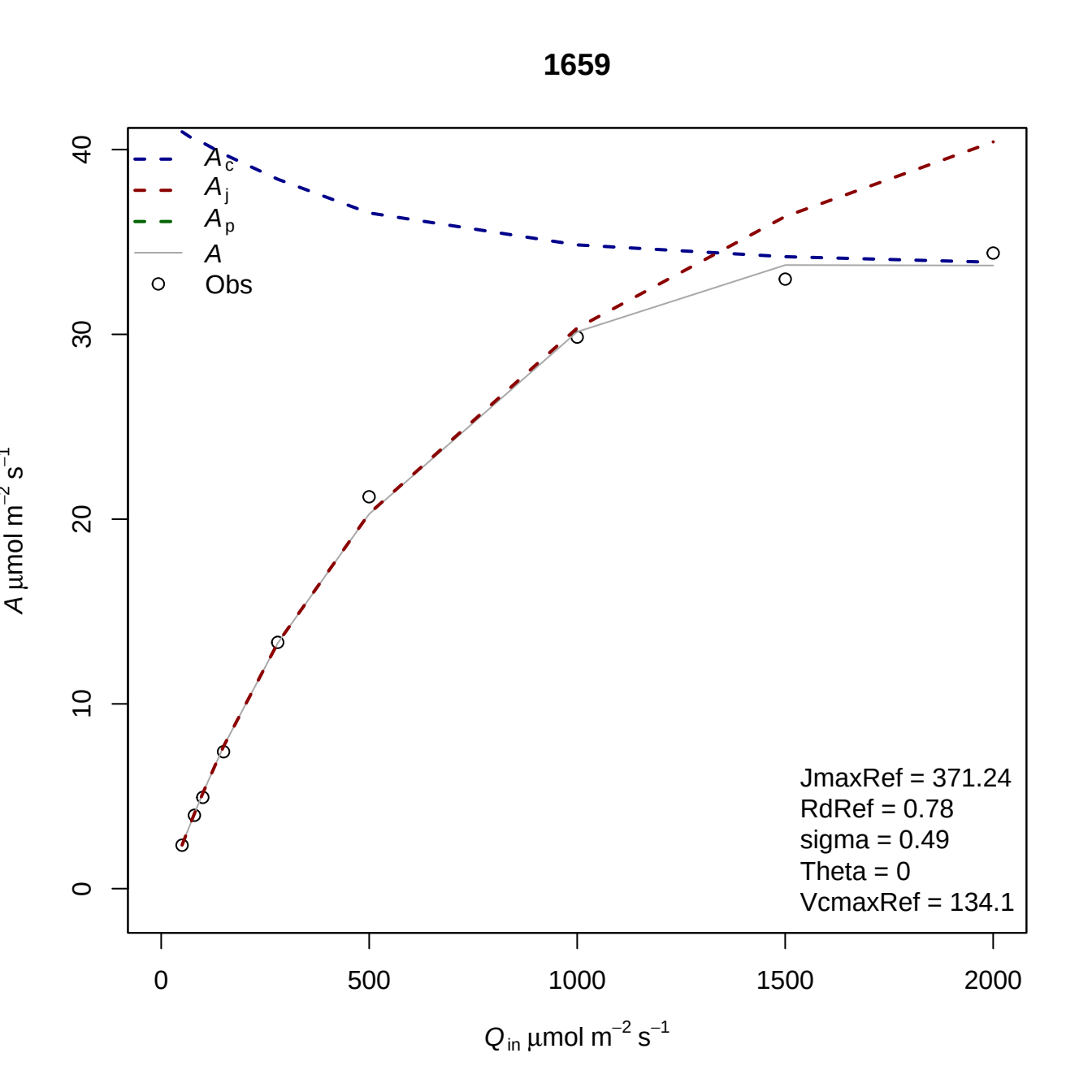

1654
A_c
A_j
A_p
A
Obs
JmaxRef = 40.34
RdRef = 1.04
sigma = 0.34
Theta = 0
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

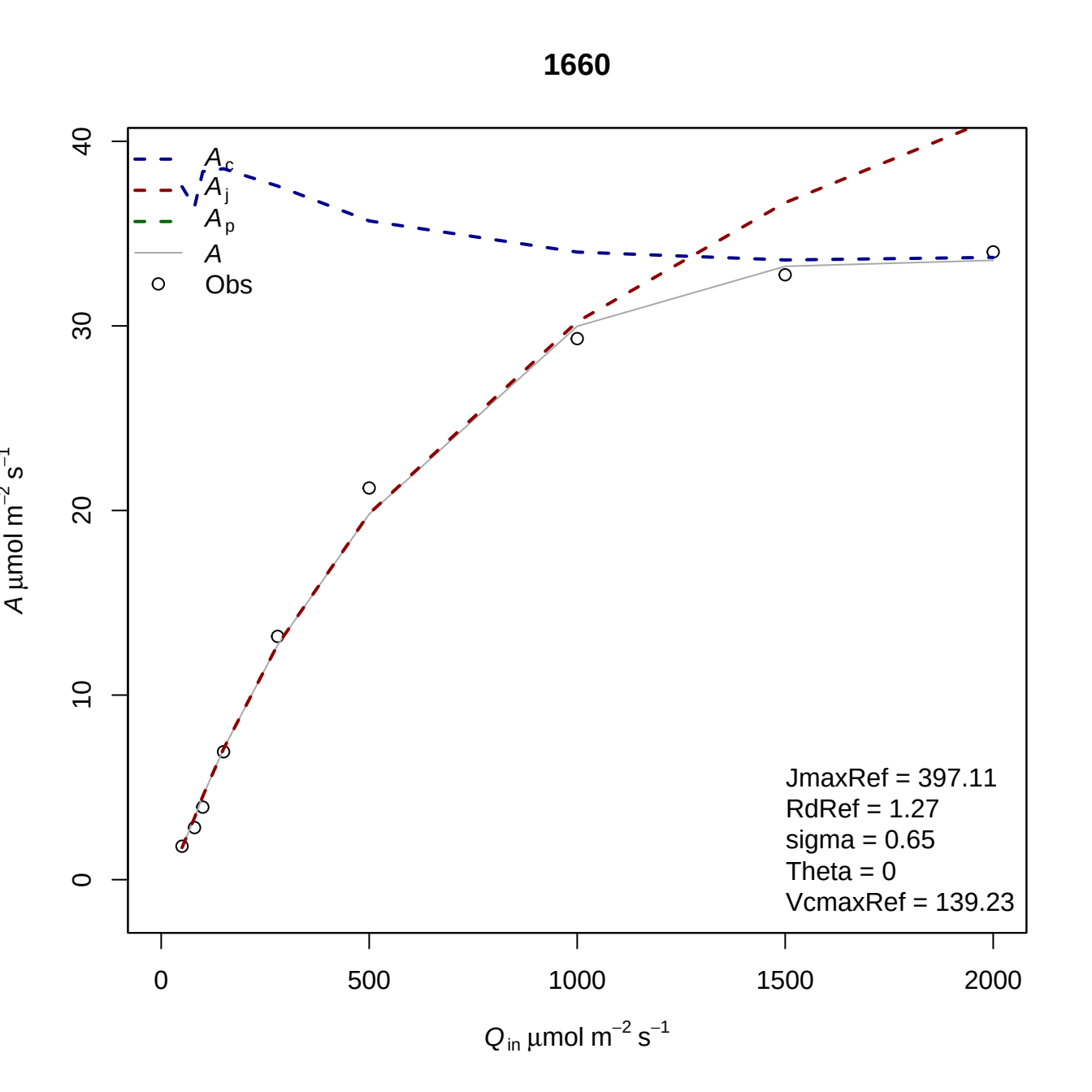

1655
A_c
A_j
A_p
A
Obs
JmaxRef = 161.14
RdRef = 1.28
sigma = 0.41
Theta = 0
VcmaxRef = 66.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

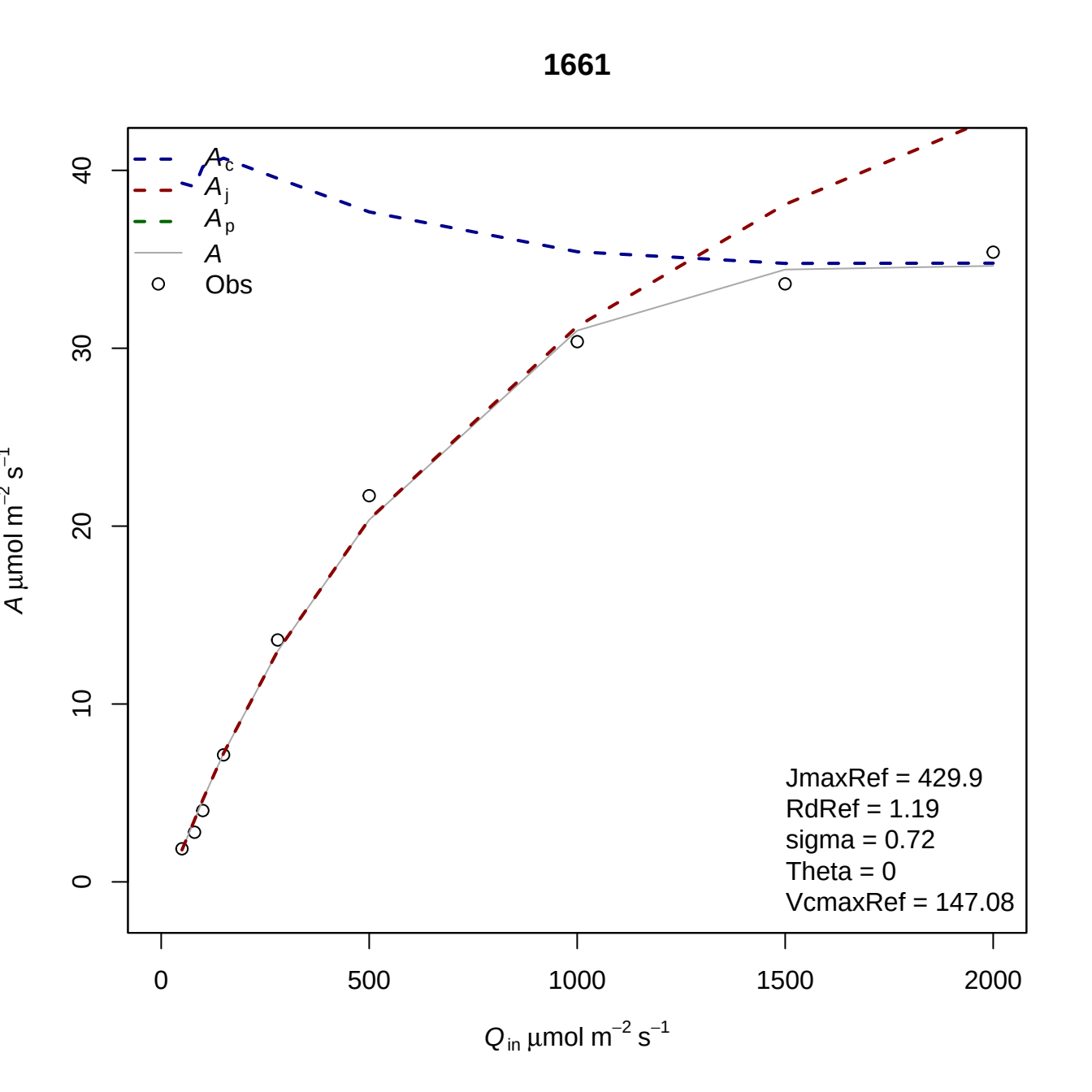

1656
A_c
A_j
A_p
A
Obs
JmaxRef = 143.16
RdRef = 1.47
sigma = 0.38
Theta = 0
VcmaxRef = 62.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

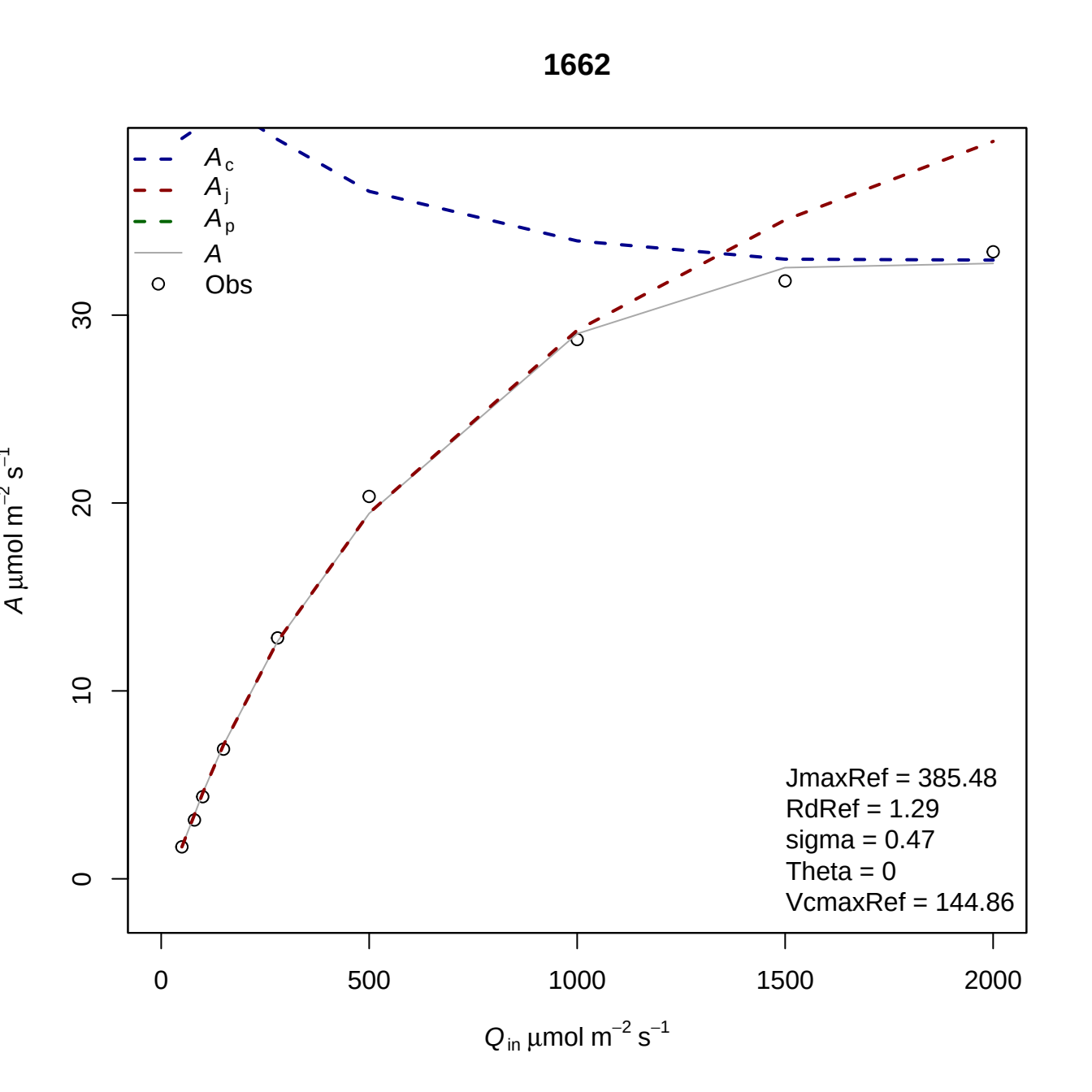

1657
A_c
A_j
A_p
A
Obs
JmaxRef = 179.9
RdRef = 0.81
sigma = 0.22
Theta = 0
VcmaxRef = 66.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000


1658
A_c
A_j
A_p
A
Obs
JmaxRef = 154.99
RdRef = 1.07
sigma = 0.28
Theta = 0
VcmaxRef = 59.64
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1659
A_c
A_j
A_p
A
Obs
JmaxRef = 371.24
RdRef = 0.78
sigma = 0.49
Theta = 0
VcmaxRef = 134.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000


1660
A_c
A_j
A_p
A
Obs
JmaxRef = 397.11
RdRef = 1.27
sigma = 0.65
Theta = 0
VcmaxRef = 139.23
0
10
20
30
40
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}


1661
A_c
A_j
A_p
A
Obs
JmaxRef = 429.9
RdRef = 1.19
sigma = 0.72
Theta = 0
VcmaxRef = 147.08
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1662
A_c
A_j
A_p
A
Obs
JmaxRef = 385.48
RdRef = 1.29
sigma = 0.47
Theta = 0
VcmaxRef = 144.86
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}